Tournament anti-Sidorenko orientations of trees with at most five branch vertices

Yanbin Qiao* Liwei Yao

School of Mathematics, China University of Mining and Technology
Xuzhou, Jiangsu 221116, China

Yanbin Qiao: a15024953277@qq.com
Liwei Yao: 10235419@cumt.edu.cn
ORCID: 0009-0005-2476-208X 0009-0005-6990-9372

Abstract

A finite tree with at most five vertices of degree at least three admits an orientation whose homomorphism density into any tournament is at most the random-tournament benchmark. Degrees, arm multiplicities and positive subdivision lengths are unrestricted, and a single orientation, chosen from the tree alone, satisfies the stronger kernel inequality for every finite probability space and every real skew kernel of absolute value at most one. For four branch vertices we convert terminal fourth moments into weighted second moments across a connector; for five we allocate reference arms in the star and fork skeletons and join the two halves of the path skeleton through compatible fractional moments. Separate arguments cover a central singleton arm and a unit outer connector. The kernel inequality also yields an entropy bound that holds for arbitrarily correlated vertex labels.

Keywords: tournaments; anti-Sidorenko inequality; oriented trees; skew kernels; weighted moments.
MSC 2020: 05C20, 05C05, 05C35, 47A30.

Contents

1 Introduction **9**

2 Kernel preliminaries and rooted messages **10**

3 The four-branch foundation **12**

4 The two branching skeletons and a reduction **12**

5 The complete terminal fourth-moment catalogue **13**
5.1 Zero-step directions 13
5.2 Positive steps: even residual lists 14
5.3 Positive steps: odd residual lists 14

*Yanbin Qiao and Liwei Yao contributed equally and should be considered joint first authors. Both authors are corresponding authors.

5.4 Restoration, reversal, and interface length . . . 15
5.5 The two internal side-table roles . . . 15

6 The chain skeleton **15**
6.1 Four half-tree inequalities . . . 15
6.2 Compatible directions for all four role combinations . . . 16
6.3 The complete central path . . . 17

7 The star skeleton **17**
7.1 The center catalogue . . . 17
7.2 Cubed-weight centers: two incoming paths and one outgoing path . . . 18
7.3 Ninth-weight centers: all three paths incoming . . . 19
7.4 Empty residual center . . . 19

8 Five branch vertices: reduction and geometry **20**

9 The five-branch boundary estimates **20**
9.1 Simple-reference and ordinary moments . . . 20
9.2 A reference second moment for a compound branch . . . 21
9.3 The common-reference outlets for the path skeleton . . . 22

10 The five-vertex star and fork **22**
10.1 Nonempty central residuals . . . 23
10.2 Empty central residual . . . 24

11 The five-vertex path **24**
11.1 A central singleton: the maximal S spine . . . 24
11.2 A unit off-center arm with a unit receiving connector . . . 25
11.3 Two compatible outlets . . . 25
11.4 A worked five-branch example . . . 26

12 The fixed orientation and its consequences **28**
12.1 All maps into a tournament . . . 29
12.2 Correlated vertex labels . . . 29
12.3 Boundary cases . . . 29

13 Further directions **29**

References **31**

A Shared calculus for local moments **32**
A.1 Scalar and propagation tools . . . 33
A.2 A fourth-power coefficient . . . 34
A.3 A common fourth-moment comparison . . . 35
A.4 Five short-connector coefficient profiles . . . 36
A.5 A positive cycle form . . . 36
A.6 Integer reference powers . . . 37

A.7 Convex reference transport . . . 37
A.8 Constants and arithmetic certificates . . . 38
A.9 Polynomial coefficient certificates . . . 39
A.10 Two reference curvature bounds . . . 39

B A worked terminal certificate 40
B.1 The propagated residual pair (1,2) . . . 40

C Trees with at most three branch vertices 42
C.1 Root moments and connector estimates . . . 42
C.2 The lower-branch assembly . . . 44
C.3 The middle cubic construction . . . 44
C.4 Assembly and the tournament inequality . . . 45

D Path and ordered-pair estimates 45
D.1 Spectral bound and positive convolution . . . 45
D.2 Weighted propagation and a weighted path cube . . . 46
D.3 Ordered pairs and exponential moments . . . 47
D.4 Short-pair cubic estimates . . . 49
D.5 Every tree with at most two branch vertices . . . 50

E All positive-length connector estimates 50
E.1 Schur certificates and normalized propagation . . . 50
E.2 An ordinary one-step certificate . . . 51
E.3 Half-step certificates for lengths two to four . . . 52
E.4 Half-step certificates for every length at least five . . . 54
E.5 Mixed half-step certificates . . . 56
E.6 Opposite types and every positive connector length . . . 59

F Rooted cubic estimates 59
F.1 Canonical cubic root certificates . . . 59
F.2 Odd roots with a long anchor or a separated second arm . . . 61
F.3 Short anchors followed by an arm of length three . . . 63
F.4 Residual prefixes of lengths one and two with a nonempty tail . . . 68
F.5 The single residual triples with third length at least four . . . 71
F.6 Residual triples with third length at least six . . . 75
F.7 The critical residual triple of lengths one two and three . . . 77

G Path and singleton moments 82
G.1 Oppositely weighted fourth moments of paths . . . 82
G.2 The fifth-weight estimate at length two . . . 83
G.3 Singletons with the fifth boundary weight . . . 84
G.4 Singletons with the cubed boundary weight . . . 89

H Unpropagated rooted moments 90
H.1 Sixth moments for multiple canonical pairs . . . 90

H.2 The residual pair (2,4) . . . 93
H.3 The residual pair (2,5) . . . 94
H.4 The residual pairs (2,b), b >= 6 . . . 95
H.5 The residual pairs (1,b), b >= 6 . . . 97
H.6 The residual pairs (1,4) and (1,5) . . . 98
H.7 The pair (1,3): a sixth moment . . . 100
H.8 The pair (1,2) and the complete even catalogue . . . 102
H.9 Zero-step triples (1,2,c) . . . 104
H.10 Odd tables of minimum two: compatible moments . . . 105
H.11 Odd tables with a longer outward anchor . . . 106
H.12 All zero-step odd tables with a unit anchor . . . 108

I Positive propagation of even terminal tables 109
I.1 Propagated pairs (1,b), b >= 3 . . . 109
I.2 The propagated pair (2,3) . . . 111
I.3 Propagated pairs (2,b), b >= 4 . . . 113
I.4 Even terminals of minimum at least four . . . 116
I.5 The propagated pair (3,b) . . . 118
I.6 Even terminals of minimum one with multiple pairs . . . 120
I.7 All even terminals of minimum three . . . 122
I.8 Even terminals of minimum two with multiple pairs . . . 123

J Positive propagation of odd terminal tables 125
J.1 The propagated triple (1,2,3) . . . 125
J.2 Propagated triples (1,2,c), c >= 4 . . . 128
J.3 Propagated triples (1,b,c), 3 <= b<c . . . 132
J.4 Odd terminals with a nonunit anchor . . . 138
J.5 Odd terminals beginning with (2,3) . . . 139
J.6 Odd terminals with a unit anchor and multiple pairs . . . 141

K Cubed-weight center estimates 145
K.1 Odd centers with the cubed boundary weight . . . 145
K.2 Even centers of minimum at least three . . . 146
K.3 Even centers of minimum two and unit-arm reductions . . . 148

L Index to the five-branch direction catalogues 151
L.1 Path directions and weights . . . 151
L.2 The inputs to the assembly . . . 151

M Finite calculus and shared local estimates 153
M.1 The full-mean exponential lemma and the minimum-two root catalogue . . . 153
M.2 Finite calculus for the odd-root estimates . . . 167
M.3 One finite packet for short-one terminals . . . 169
M.4 The receiving weight and its two mean endpoints . . . 173
M.5 One retained vector for the two short-arm sums . . . 174

M.6 Five continuous payments for a retained reference square . . . 179
M.7 Propagation of moment bounds . . . 180
M.8 Finite packets at a unit root . . . 180
M.9 The three length-three pair envelopes . . . 186
M.10 Fourth-moment certificates and root replacement . . . 187
M.11 An anchored packet and its reversed unit row . . . 190
M.12 Finite rows with a two-edge root arm . . . 191
M.13 A two-arm terminal at roots of minimum one and two . . . 192
M.14 The actual three-step inverse-q fourth moment . . . 198
M.15 Finite path rows and ordinary operator power bounds . . . 200
M.16 Root multipliers and all outer connectors . . . 212
M.17 Actual unequal-pair envelopes and finite masks . . . 221
M.18 The sharpened losses for actual pairs starting at four . . . 222
M.19 Unanchored propagation and negative odd-root unit bridges . . . 223
M.20 The same-row inverse fourth moment and its endpoint payment . . . 229
M.21 Finite path caps and the sharper fourth-moment outlet . . . 235
M.22 Bare path eighth moments at every length at least six . . . 240

N Reference fourth moments **242**
N.1 The full pointwise coefficient for the one-connector genuine reference . . . 242
N.2 One connector from the terminal $(1, 2)$ . . . 245
N.3 The genuine length-two reference after every connector of length at least six . . . 247
N.4 The one-step genuine short-one, long-three reference . . . 251
N.5 The genuine length-two reference for connectors two through five . . . 254
N.6 Every one-step short-one terminal with long arm at least four . . . 260
N.7 The genuine f_2 reference for every short-one terminal . . . 263
N.8 The actual squares for unit terminals of lengths $(2, 4)$ and $(2, 5)$ . . . 270
N.9 The retained actual square for every unit terminal $(2, b)$ with $b \geq 6$ . . . 273
N.10 The genuine-f_2 reference for every short-two connector . . . 276
N.11 Short-terminal fourth moments with an arbitrary receiving arm . . . 283
N.12 All receiving arms at the unit terminal pair . . . 301
N.13 Bare inward paths of length at least six under the reference f_2 . . . 306
N.14 The short bare inward paths under the reference f_2 . . . 309
N.15 The genuine f_2 reference for terminal $(1, 2, 3)$, $L \geq 2$ . . . 313
N.16 The genuine f_2 reference for terminal $(1, 2, 4)$ at every length . . . 318
N.17 The genuine f_2 reference at the reversed unit terminal $(1, 2, 3)$ . . . 327
N.18 The unit reference terminal $(1, 2, 3)$ . . . 333
N.19 The reference terminal $(1, 2, 4)$ at two connector edges . . . 339
N.20 Reference fourth moments for terminal $(1, 3, c)$ . . . 343
N.21 The reference terminal $(1, 2, 3)$ at every $L \geq 2$ . . . 352
N.22 Reference fourth moments for terminal $(1, b, c)$, $4 \leq b < c$ . . . 360
N.23 A retained reference square for terminals $(1, 2, c)$ . . . 371
N.24 The short connectors for the reference terminal pair $(1, 2)$ . . . 377

N.25 The complete two-connector coefficient certificate . . . 389
N.26 The remaining short-two unit and long-connector rows . . . 393
N.27 The unit connector with terminal arms of lengths two and three . . . 396
N.28 The remaining unit connectors with a terminal arm of length two . . . 400
N.29 The three-step bare reference estimate . . . 404
N.30 A unit connector with a nonunit odd terminal starting with two and three . . . 407
N.31 Four bare steps and two steps from an odd terminal . . . 409
N.32 Every bare input path of length at least five . . . 412
N.33 All positive connectors from an odd terminal starting with two and three . . . 415
N.34 All odd terminals with nonunit shortest arm . . . 418
N.35 Unit shortest arm and second shortest arm at least four . . . 422
N.36 Even terminals whose shortest residual arm is at least four . . . 424
N.37 Unit shortest arm and second shortest arm three . . . 428
N.38 Unit shortest arm and second shortest arm two . . . 431
N.39 Even terminal lists of minimum two or three . . . 438
N.40 Even terminal lists of minimum one, including the global third class . . . 451
N.41 Every single pair of minimum three: the reference proof . . . 468
N.42 Every genuine length-two reference arm: the complete directory . . . 480
N.43 The short-terminal reference catalogue and its row proofs . . . 501

O Ordinary fourth and eighth moments 501
O.1 The shortest-two root: one finite packet and nine payments . . . 502
O.2 The nonunit-inner ordinary eighth moment . . . 506
O.3 Unit receivers with negative root pairs . . . 515
O.4 Negative odd roots at every nonunit inner length . . . 522
O.5 The unit ordinary terminal catalogue . . . 532
O.6 The ordinary eighth moment of the unit short-one connector . . . 535
O.7 Unit-root eighth moments for the terminal $(1, 2)$ . . . 539
O.8 Unit connectors for every short-one terminal . . . 542
O.9 The unit eighth moment for terminal pairs $(2, b)$ on the low domain . . . 546
O.10 High-domain unit eighth moments for every single pair $(2, b)$ . . . 549
O.11 Unit-connector ordinary eighth moments for even terminals of minimum one . . . 554
O.12 The unit ordinary eighth moment for terminal $(1, 2, 3)$ . . . 559
O.13 Unit ordinary eighth moments for terminals $(1, 2, c)$ . . . 564
O.14 The unit-connector eighth moment for every triple $(1, b, c)$, $3 \leq b < c$ . . . 573
O.15 The unit-connector eighth moment for odd terminals with a unit arm and multiple pairs . . . 581
O.16 The unit-connector eighth moment for every triple $(2, b, c)$, $2 < b < c$ . . . 586
O.17 Short connectors for every terminal pair $(2, b)$ . . . 592
O.18 Long connectors for every terminal pair $(2, b)$ . . . 603
O.19 Unit-root eighth moments for terminal pairs $(3, b)$ . . . 615
O.20 Ordinary eighth moments for even terminals of minimum one with multiple pairs 622

O.21 Ordinary eighth moments for odd terminals with an inward arm of length at least two . . . 631
O.22 Short-first-pair ordinary eighth moments for odd terminals of minimum two . . . 636
O.23 The remaining odd triple and two short bare-path eighth moments . . . 643
O.24 Ordinary moments at an actual unit root . . . 651
O.25 Three further unit-terminal ordinary-eighth families . . . 671
O.26 Unit-root ordinary eighth moments for 123 and 124 . . . 672
O.27 One reference arm at a nonempty even center . . . 673
O.28 The unit eighth moments for minimum-three even lists and the pair $(2,3)$ . . . 674
O.29 Ordinary eighth moments for every short-one terminal . . . 677
O.30 Ordinary eighth moments for even terminals with multiple pairs at every connector length . . . 682
O.31 Four unit-root ordinary-eighth families . . . 689
O.32 All nonunit odd terminals: choosing the actual eighth-moment direction . . . 690
O.33 Higher even terminal lists with a nonunit ordinary anchor . . . 696
O.34 Short even multipairs with a nonunit ordinary anchor . . . 701
O.35 Higher even lists: both ordinary moments and outer outlets . . . 707
O.36 Ordinary moments with a unit terminal arm and a nonunit receiver . . . 714
O.37 Short even lists: the fourth moment and all outer lengths . . . 731
O.38 Short terminal pairs and unit-minimum even masks . . . 738
O.39 Ordinary eighth moments and the nonunit-pair outlet . . . 746

P Unit-arm two-step outlets 749
P.1 Compatible square moments at a unit root . . . 749
P.2 Whole-domain retained square moments . . . 751
P.3 Parameters for the unit-root catalogue . . . 752
P.4 All terminal pairs with minimum one at a unit root and a two-edge outlet . . . 753
P.5 All even terminal lists with at least two unequal pairs at a unit root . . . 771
P.6 Single terminal pairs with minimum at least two . . . 785
P.7 Odd terminal triples $(1,2,c)$ at a unit root . . . 789
P.8 The terminal pairs $(2,b)$ for every $b \geq 4$ at a unit root . . . 807
P.9 Odd terminal lists at a unit root with a two-edge outlet . . . 818

Q Odd-root fourth moments and outlets 819
Q.1 Minimum-one odd-root bridges and the complete fourth transfer . . . 819
Q.2 Short terminal pairs at a root beginning (1,2,v) . . . 834
Q.3 The actual unit-arm curvature for the short-two connector . . . 846
Q.4 The terminal pair (2,3) at a root (1,2,v) . . . 852
Q.5 Terminal triples (1,2,w) at a root (1,2,v) . . . 854
Q.6 The exceptional terminal pair (3,4) at a root (1,2,v) . . . 861
Q.7 Terminal triples (1,3,w) at a root (1,2,v) . . . 864
Q.8 All terminal triples (1,a,b) with a at least four . . . 870
Q.9 All odd multipair terminals at a root (1,2,v) . . . 877
Q.10 Large-minimum even terminal packets at a root (1,2,v) . . . 886

Q.11 All nonunit-minimum odd terminals with their actual inward anchors 893
Q.12 Five even multipair packets with the actual two-step root exception 900
Q.13 Odd root lists beginning (1,2,v): lifting all later pairs 907
Q.14 Comparing complete certificates at the remaining odd roots 913
Q.15 The remaining short terminal directions 916
Q.16 All terminals beginning at three, at both remaining roots 917
Q.17 The odd terminal packets with a unit arm 920
Q.18 Even terminal packets at the two remaining root types 923
Q.19 Odd terminals with their actual inward path 926
Q.20 Restoring the complete odd root 930
Q.21 Odd internal roots of minimum one 931
Q.22 The minimum-two odd root: unit and minimum-one odd terminals 933
Q.23 Odd internal roots of minimum at least three 943
Q.24 The terminal pair at a minimum-two odd root 944
Q.25 Minimum-two odd roots with every large-minimum even terminal 944
Q.26 Minimum-two odd roots with every even multipair terminal 950
Q.27 Minimum-two odd roots with every terminal pair (3,t) 959
Q.28 Minimum-two odd roots with every nonunit-minimum odd terminal 963

R Remaining outlets and mixed estimates **970**
R.1 Weighted transfer for a nonunit singleton 970
R.2 Ordinary fourth moments for genuine singleton-root inputs 972
R.3 The retained multiplier and every even internal root 985
R.4 Path second moments and the mixed central estimate 989
R.5 Finite geometric cancellation and every nonunit singleton at three outer edges . . 996
R.6 A genuine two-edge singleton at two outer edges 1005
R.7 Every longer singleton at two outer edges 1013

1 Introduction

For an oriented graph D and a finite tournament $\mathcal{T}$, let $\hom(D,\mathcal{T})$ be the number of arc-preserving vertex maps from D to $\mathcal{T}$; the maps need not be injective. An orientation D of a tree U is *tournament anti-Sidorenko* if

$$\hom(D,\mathcal{T}) \le 2^{-e(U)}|V(\mathcal{T})|^{|V(U)|} \tag{1.1}$$

for every nonempty finite loopless tournament. The factor $2^{-e(U)}$ is the limiting density of D in a uniformly random tournament. Thus (1.1) bounds the number of maps from D into any tournament by this random benchmark. Fox, Himwich, Mani and Zhou conjectured that every tree admits such an orientation [3, Conjecture 1.12]. We prove their conjecture for trees with at most five branch vertices.

Here a *branch vertex* of U is a vertex of degree at least three, and

$$b(U) = |\{v \in V(U) : \deg_U(v) \ge 3\}|, \qquad e(U) = |E(U)|.$$

All degrees refer to U.

Theorem 1.1. *Every nonempty finite tree U with $b(U) \le 5$ has an orientation $D = D(U)$ satisfying (1.1) for every nonempty finite loopless tournament $\mathcal{T}$.*

We prove a kernel form of the theorem, in which one orientation serves every host. For a nonempty finite probability space (X,μ) and a real skew kernel $K : X^2 \to [-1,1]$, put

$$Z_D(K,\mu) = \mathbb{E}_{\mu^{V(U)}} \prod_{u\to v\in E(D)} (1 + K(x_u,x_v)). \tag{1.2}$$

All labels are independently sampled from the same measure; coincident labels are included.

Theorem 1.2. *For every nonempty finite tree U with $b(U) \le 5$, there is an orientation $D = D(U)$ of all its edges such that*

$$Z_D(K,\mu) \le 1 \tag{1.3}$$

for every nonempty finite probability space and every real skew kernel bounded in absolute value by one.

Related results.

Sidorenko's conjecture bounds the homomorphism density of a bipartite graph from below in terms of its edge density [9]; the tournament problem seeks an orientation with a universal upper bound at the random benchmark. Table 1 lists the tree classes covered by the relevant results. These classes overlap only partly: arbitrary subdivisions produce many vertices of degree two, whereas a bound on $b(U)$ restricts neither the degrees nor the multiplicities of equal branches.

Table 1: Tree classes covered by the closest prior results and by this paper.

Work	Scope
Fox–Himwich–Mani–Zhou [3, Proposition 1.13]	Trees with at most one vertex of even degree.
Chen–Clemen–Noel [2]	Every three-legged spider, with arbitrary leg lengths.
He–Mani–Nie–Tung–Wei [6, Theorems 1.3 and 1.4]	Isomorphic-branch attachment and every caterpillar.
Theorem 1.1	Every tree with at most five original vertices of degree at least three, with unrestricted degrees, multiplicities and positive subdivision lengths.

Chen and Lin prove path anti-Sidorenko inequalities when the length is large relative to the number of direction flips [1]. Paths enter our argument in a different role, as connectors that carry rooted messages between branch vertices; their lengths are arbitrary, including one.

Proof strategy.

The proof separates the assembly of a tree from the local moment estimates. Table 2 gives the route through the assembly; the technical appendix proves the inputs listed there.

Table 2: The proof of Theorems 1.1 and 1.2. Each row uses the preceding structural reductions and the indicated local estimates.

Stage	Argument or required input	Location
Path interface	Weighted second moments at two roots control every positive connector length.	Lemma 2.1
Four branches	Pair equal arms; convert terminal fourth moments to the second moments needed by the interface.	Sections 4–7
Five-branch reduction	Reduce to four branches or to P_5, $K_{1,4}$ and the fork T_5.	Lemma 8.1
Star and fork	Allocate disjoint arms to simple-reference fourth, ordinary eighth and compound-reference second moments.	Section 10; Propositions 9.1–9.3
Path alternatives	First treat a central singleton and a unit off-center arm whose receiving connector has length one.	Section 11
Remaining path	Give both sides the same inverse-weighted boundary estimate; close by Hölder at the center.	Proposition 9.4; (11.1)
Conclusion	Fix all choices from the tree, restore equal pairs, and pass from kernels to tournaments.	Section 12

For four branch vertices the terminal fourth moments close the path P_4 and the star $K_{1,3}$. For five, the star and fork need one existing central arm as a reference, while the path uses $I_q(V) = \mathbb{E}[q^{-1/3}|V|^{8/3}]$, with $q = 1 - K\mathbf{1}$. After the two structural path alternatives, both side messages satisfy this bound with the same q; the three weight powers cancel in the central Hölder inequality. The arm allocation, rather than a restriction on the connector lengths, distinguishes these assemblies.

Appendix A gives the shared scalar calculus, and Appendix B proves one terminal estimate in full. Appendix L is the entry point to the remaining direction tables: it separates reference, ordinary and outlet inputs, whose directions and weights differ. The local proofs retain the short-length exceptions before treating the finite tails.

2 Kernel preliminaries and rooted messages

Delete zero-mass atoms and use the real inner product $\langle u, v\rangle = \mathbb{E}(uv)$ on $L^2(\mu)$. Define

$$Ju = (\mathbb{E}u)\mathbf{1}, \quad A = J+K, \quad A^* = J-K, \quad g = K\mathbf{1}, \quad d = \mathbb{E}g^2, \quad p = 1+g, \quad q = 1-g. \tag{2.1}$$

All operators and adjoints are taken with respect to this measure. We also write $r = Kg$. Write $T = A^*$. Both A and T have nonnegative kernels. Since $K(x,x) = 0$ and $\mu(x) > 0$, we have $p(x), q(x) \geq \mu(x) > 0$.

The row constraint is

$$|r(x)| \leq [1 - g(x)^2]/2 = p(x)q(x)/2. \tag{2.2}$$

For $a_y = K(x,y)$, antisymmetrization gives $r(x) = \frac{1}{2}\mathbb{E}_{y,z}[(a_y - a_z)K(y,z)]$. The scalar inequality $|u-v| \leq 1-uv$ on $[-1,1]^2$ follows by squaring, since the difference is $(1-u^2)(1-v^2) \geq 0$. Taking expectations proves (2.2).

For an orientation D of a rooted tree, its message is the conditional integral of the product in (1.2), with the root label fixed and all other labels integrated independently.

For $n \geq 0$ put $f_n = A^n\mathbf{1}$, $\bar{f}_n = (A^*)^n\mathbf{1}$, $\mu_n = \mathbb{E}f_n$ and, for $n \geq 1$, $h_n = A^* f_{n-1}$. At a rooted arm of length n, these are the actual messages of, respectively, all edges pointing away from the root, all edges pointing toward the root, and the first edge pointing toward the root with every remaining edge pointing away along the arm. In particular, $f_1 = p$ and $h_1 = q$.

We use the constants and moments

$$\theta = \frac{13}{32}, \qquad C_* = \frac{101}{50}, \qquad \delta_j = \left\|K^j\mathbf{1}\right\|_2^2, \qquad S_n = \sum_{j=0}^{n} K^j\mathbf{1}. \tag{2.3}$$

Appendix D proves $\|K\|_{2\to 2} \leq 2/\pi$, $d \leq 1/3$, $\delta_{j+1} \leq \theta\delta_j$, $\mu_n \leq 1$, and the positive convolution identity

$$S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad d - \delta_2 \leq \beta_j \leq d \quad (j \geq 3). \tag{2.4}$$

The elementary identity $A + A^* = 2J$ gives

$$f_n + h_n = 2\mu_{n-1}, \qquad 0 < f_n h_n \leq \mu_{n-1}^2 \leq 1. \tag{2.5}$$

At a root, pair equal arm lengths maximally and use f_n, h_n on each pair; let E_0 be the actual product of the pair messages, so $0 < E_0 \leq 1$. The remaining lengths are distinct and are exactly those of odd multiplicity. Since $E_0 \leq 1$, an upper bound may drop E_0.

For an even arm list, its *positive canonical orientation* pairs the remaining distinct lengths adjacently, assigning $h_a f_b$ to a pair $a < b$. For an odd list, retain the shortest residual arm as an outward anchor f_a and pair the rest in this way. Write the odd message as $f_a R$, where R includes all equal and distinct pairs. The *negative canonical orientation* reverses all the corresponding edges. A universal assertion at $-K$ proves the reversed one.

For later use set

$$N_+(H) = \mathbb{E}[pH^2], \qquad N_-(H) = \mathbb{E}[qH^2].$$

Lemma 2.1 (Weighted path interface)**.** *For every integer $L \geq 1$ and all real H, G,*

$$|\langle H, A^L G\rangle| \leq \sqrt{N_+(H)N_-(G)}, \qquad N_-(AG) \leq N_-(G).$$

The reversed statements hold after interchanging A, T and p, q.

Proof. The nonnegative edge measure $(1 + K(x,y))\, d\mu(x)d\mu(y)$ has marginals $p\mu, q\mu$. Cauchy–Schwarz on this measure proves the first inequality for $L = 1$. Row Cauchy–Schwarz gives $(AG)^2 \leq pA(G^2)$, and hence

$$N_-(AG) \leq \mathbb{E}[pqA(G^2)] = \mathbb{E}[T(pq)G^2] \leq \mathbb{E}[qG^2],$$

because $pq = 1 - g^2 \leq 1$ and T preserves order. Iterate $L-1$ times and apply the one-edge inequality. □

Lemma 2.2 (Rooted spider boundary)**.** *Every finite arm multiset admits an orientation, fixed independently of the kernel and measure, with actual message F satisfying $N_-(F) \leq 1$.*

The direction for each residual arm list, and the proof, are given in Appendix D.5. Reversal gives the corresponding N_+ statement.

Table 3: Notation used in the moment calculations.

Symbol	Meaning
g, r	$K\mathbf{1}$ and Kg, both with the original measure.
r_0	The centered function $QKg = Kg + d\mathbf{1}$, used only in Appendix E.
d, δ_j	$d = \mathbb{E}g^2 = \delta_1$ and $\delta_j = \|K^j\mathbf{1}\|_2^2$. Ordinary powers such as d^2 remain powers of d.
$f_n, \bar{f}_n, h_n$	Outward, inward, and first-edge-inward arm messages, respectively.
β_j, ω_j	Positive convolution coefficients, and local envelope or propagation losses; neither is a path function.
ξ	A scalar score coordinate in a polynomial comparison: $p(x) = \xi$, $q(x) = 2 - \xi$.
n_T	Number of vertices of the host tournament.

3 The four-branch foundation

The five-branch argument uses the following four-branch result whenever a reduction removes one branch vertex.

Theorem 3.1. *For every nonempty finite tree U with $b(U) \le 4$, there is an orientation $D = D(U)$ of all its edges such that (1.1) holds for every nonempty finite loopless tournament.*

Theorem 3.2. *The orientation in Theorem 3.1 can be chosen so that $Z_D(K, \mu) \le 1$ for every finite probability space and every real skew kernel bounded in absolute value by one.*

Appendix C proves the assertion for at most three branch vertices. Section 4 reduces to the two skeletons, Section 5 supplies the terminal estimates and the internal side tables; Sections 6 and 7 then close the chain and the star. These inputs also serve the five-branch case.

4 The two branching skeletons and a reduction

Theorem C.2 gives every tree with at most three branch vertices one orientation satisfying (1.3) for every finite model, so it remains to treat $b(U) = 4$.

Let B be the four original branch vertices, and let S be the minimal connected subtree of U containing B. Every vertex of degree at least three in S belongs to B, since it also has original degree at least three. Every leaf of S belongs to B, by minimality. Suppressing the degree-two vertices of S outside B leaves a tree on exactly four vertices, which is either a path P_4 or a star $K_{1,3}$ (Figure 1).

Each skeleton edge represents a positive-length path in the original tree, whose interior vertices have degree two and carry no side arms. Every component outside the core is a path attached at a branch vertex, since another branch point there would belong to B, contrary to the choice of S. These components are the *side arms*, and the core paths together with the side arms partition the edges of U.

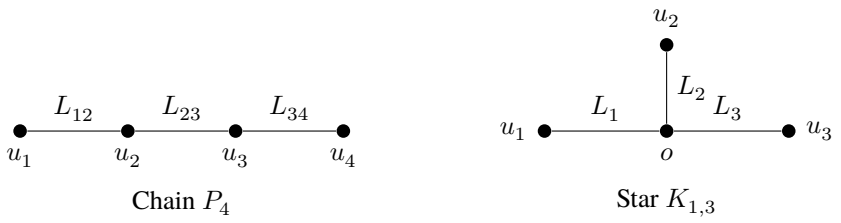


Figure 1: The two skeletons on four branch vertices. Each edge represents its original positive-length connector; side arms are omitted from the drawing.

A leaf of the skeleton is called a *terminal*. It has at least two actual side arms in U. Maximally pair its equal-length arms using (2.5).

Lemma 4.1 (Reducible terminal)**.** *If a terminal has zero or one residual side arm after equal-length pairing, then U has a fixed orientation satisfying* (1.3)*.*

Proof. Orient the equal pairs so that their product is $E_i \le 1$. Removing those paired arms for the comparison leaves the terminal vertex, its remaining arm if any, and all core edges; the terminal then has degree at most two and no other degree has increased. The comparison tree is nonempty, has at most three branch vertices, and carries a fixed orientation from Theorem C.2. Restoring those arms multiplies the nonnegative integrand at the terminal label by $E_i \le 1$, preserving the density bound. □

The reduction applies to both skeletons. We may therefore assume that every terminal has at least two distinct residual lengths. Singleton internal side tables and singleton star-center tables survive the reduction and are estimated in Sections 6 and 7.

5 The complete terminal fourth-moment catalogue

Let Σ be a terminal arm multiset whose distinct residual list has at least two entries, with arbitrary original multiplicities. We use the following local estimate.

Theorem 5.1 (Terminal supply)**.** *For each integer $n \ge 0$ there is an orientation of all the arms, with actual message $F^{+}_{\Sigma,n}$, determined by Σ and n, such that*

$$\mathbb{E}\big[p(T^n F^{+}_{\Sigma,n})^4\big] \le 1. \tag{5.1}$$

Its complete reversal $F^{-}_{\Sigma,n} = \bar{F}^{+}_{\Sigma,n}$ satisfies

$$\mathbb{E}\big[q(A^n F^{-}_{\Sigma,n})^4\big] \le 1. \tag{5.2}$$

The parameter n is a prescribed integer determined by the original connector length.

The catalogue below proves the theorem, with analytic estimates in the indicated sections of Appendices H–J.

5.1 Zero-step directions

Every direction in this subsection includes the equal-pair multiplier E_0. For an even residual list, put

$$P = \prod_{i=1}^{t} h_{a_i} f_{b_i}, \qquad a_1 < b_1 < \cdots < a_t < b_t.$$

This direction $F^{+}_{\Sigma,0} = E_0 P$ satisfies $\mathbb{E}[p(F^{+}_{\Sigma,0})^4] \le 1$. The pair $(1,2)$ is treated directly in Section H.8. Every other even residual list also satisfies a weighted sixth-moment bound, which yields the fourth moment by Hölder against $p\mu$; the complete even catalogue, with its component ranges, is collected at the end of Section H.8, with proofs in Appendix D and Sections H.1–H.7.

For an odd residual list $a < b_1 < c_1 < \cdots < b_t < c_t$, $t \ge 1$, use

$$F^{+}_{\Sigma,0} = \begin{cases} E_0 q f_{b_1} h_{c_1} \displaystyle\prod_{i=2}^{t} h_{b_i} f_{c_i}, & a = 1, \\ E_0 f_a \displaystyle\prod_{i=1}^{t} h_{b_i} f_{c_i}, & a \ge 2. \end{cases} \tag{5.3}$$

The first line is proved in Section H.12, for every number of pairs. The second line is the full reversal of the q-weighted fourth estimates for positive canonical odd messages, with minimum two in Section H.10 and minimum at least three in Section H.11; there the equal-pair arms are reversed together with the residual arms, which is what converts the q weight into the p weight.

5.2 Positive steps: even residual lists

Write the residual list as $a_1 < b_1 < \cdots < a_t < b_t$, $t \geq 1$. The following table gives the residual part of $F^{+}_{\Sigma,n}$ for every $n \geq 1$; multiply each row by its actual E_0.

Table 4: Positive-step terminal directions for even residual lists.

Residual condition	Residual message	Complete proof
$t = 1$, $(a_1, b_1) = (1, 2)$	qf_2	Section B.1
$t = 1$, $a_1 = 1$, $b_1 \geq 3$	qf_{b_1}	Section I.1
$t \geq 2$, $a_1 = 1$	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section I.6
$t = 1$, $(a_1, b_1) = (2, 3)$	$h_2 f_3$	Section I.2
$t = 1$, $a_1 = 2$, $b_1 \geq 4$	$h_2 f_{b_1}$	Section I.3
$t \geq 2$, $a_1 = 2$	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section I.8
$a_1 = 3$, any $t \geq 1$	$f_3 h_{b_1} \prod_{i=2}^{t} h_{a_i} f_{b_i}$	Sections I.5, I.7
$a_1 \geq 4$, any $t \geq 1$	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section I.4

The row of minimum three uses $f_3 h_{b_1}$, with the first pair reversed.

5.3 Positive steps: odd residual lists

Write

$$a < b_1 < c_1 < b_2 < c_2 < \cdots < b_t < c_t, \qquad t \geq 1.$$

Again the table gives only the residual part and covers all $n \geq 1$. Here $\bar{f}_a = T^a \mathbf{1}$ is a path function, whereas b_i in the ordered list is an arm length.

Table 5: Positive-step terminal directions for odd residual lists.

Residual condition	Residual message	Complete proof
$a = 1, t = 1, b_1 = 2, c_1 = 3$	$qf_2 h_3$	Section J.1
$a = 1, t = 1, b_1 = 2, c_1 \geq 4$	$qf_2 h_{c_1}$	Section J.2
$a = 1, t = 1, 3 \leq b_1 < c_1$	$qf_{b_1} h_{c_1}$	Section J.3
$a = 1, t \geq 2$	$qf_{b_1} h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$	Section J.6
$a = 2, b_1 = 3$, any $t \geq 1$	$\bar{f}_2 f_3 h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$	Section J.5
$a \geq 2, b_1 = 4$	$\bar{f}_a f_4 h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$	Section J.4
$a \geq 2, b_1 \geq 5$	$\bar{f}_a \prod_{i=1}^{t} h_{b_i} f_{c_i}$	Section J.4

Completeness of the terminal classification.

For an even list the minimum is $1, 2, 3$, or at least four; the first two cases split into one pair and multiple pairs. For an odd list of minimum one, the cases split by the number of pairs and then by the second length. At minimum two the second length is three, four, or at least five; at larger minimum it is at least four. The inward anchor $\bar{f}_a$ is an actual arm with all of its edges directed toward the root.

5.4 Restoration, reversal, and interface length

The proofs in the tables are for the residual message and already allow arbitrary equal-pair factors. Since $0 \le E_0 \le 1$ and $P \ge 0$ give $T^n(E_0P) \le T^nP$, taking fourth powers and integrating against the nonnegative weight p restores the paired arms without weakening the bound.

Applying (5.1) at $-K$ interchanges T with A, p with q, and the selected arm message with its complete reversal, which proves (5.2).

For a connector of original length L, keeping the edge nearest the internal vertex leaves $n = L-1$ edges to integrate, so $L = 1$ uses the zero-step catalogue and $L \ge 2$ the positive-step one. In the empty-center star construction of Section 7.4 one terminal is propagated through all L edges.

5.5 The two internal side-table roles

After equal pairing, an internal side table in a chain is of one of two kinds. Call it type R if its residual list is empty or has at least two entries. For an empty list take $V^+ = E_0$, and otherwise use the zero-step catalogue. In either case

$$\mathbb{E}[p(V^+)^4] \le 1, \qquad V^- = \bar{V}^+, \qquad \mathbb{E}[q(V^-)^4] \le 1. \tag{5.4}$$

Call the table type S if it has exactly one residual arm, of length $a \ge 1$. Put $C^+ = E_0 f_a$ and $C^- = \bar{C}^+$. The complete singleton estimate is

$$\mathbb{E}[q^5(C^+)^4] \le 1, \qquad \mathbb{E}[p^5(C^-)^4] \le 1. \tag{5.5}$$

For $a = 1$, $q^5p^4 = q(pq)^4 \le q$ proves this directly. The length-two proof is Section G.2, and all lengths $a \ge 3$ are proved in Section G.3.

6 The chain skeleton

Let u_1, u_2, u_3, u_4 be the branch vertices in path order. Write $L_{12}, L_{23}, L_{34} \ge 1$ for their original connector lengths. By Lemma 4.1, the two terminals may be assumed irreducible. Every central-path edge will point from u_2 toward u_3.

6.1 Four half-tree inequalities

The following inequalities account for the weights in the internal catalogue; all factors in each product are evaluated at the same root label.

Lemma 6.1. *For nonnegative functions C, F, G,*

$$N_+(CTF) \le \sqrt{\mathbb{E}[qC^4]\,\mathbb{E}[pF^4]}, \tag{6.1}$$

$$N_-(CTF) \le \sqrt{\mathbb{E}[q^5C^4]\,\mathbb{E}[pF^4]}, \tag{6.2}$$

$$N_-(CAG) \le \sqrt{\mathbb{E}[pC^4]\,\mathbb{E}[qG^4]}, \tag{6.3}$$

$$N_+(CAG) \le \sqrt{\mathbb{E}[p^5C^4]\,\mathbb{E}[qG^4]}. \tag{6.4}$$

Proof. Row Cauchy–Schwarz gives $(TF)^2 \le qT(F^2)$. Hence

$$N_+(CTF) \le \mathbb{E}[pqC^2T(F^2)] \le \mathbb{E}[C^2T(F^2)].$$

The edge measure for T has root marginal $q\mu$ and other marginal $p\mu$. Cauchy–Schwarz on that measure proves (6.1). For (6.2), retain the extra factors:

$$N_-(CTF) \le \mathbb{E}[q^2C^2T(F^2)] \le \sqrt{\mathbb{E}[q(q^2C^2)^2]\,\mathbb{E}[pF^4]}.$$

The first term is exactly $\mathbb{E}[q^5C^4]$. Complete reversal gives (6.3) and (6.4). □

The merger converts fourth moments into the second moments required by Lemma 2.1.

6.2 Compatible directions for all four role combinations

At the left terminal set $n_1 = L_{12} - 1$ and use Theorem 5.1 to define

$$G_1^+ = T^{n_1}F_{1,n_1}^+, \qquad G_1^- = A^{n_1}F_{1,n_1}^-.$$

Then $\mathbb{E}[p(G_1^+)^4] \le 1$ and $\mathbb{E}[q(G_1^-)^4] \le 1$. Use $n_4 = L_{34} - 1$ at the right terminal to obtain $G_4^\pm$ with the same bounds. The superscript at a terminal is chosen according to its adjacent internal role in the following table.

Table 6: Compatible outer directions and half-tree bounds.

Role	Side message	Entire outer path	Half-tree message	Required output
Left R	V_2^-	$u_1 \to u_2$	$H_\ell = V_2^- T G_1^+$	$N_+(H_\ell) \le 1$
Left S	C_2^-	$u_2 \to u_1$	$H_\ell = C_2^- A G_1^-$	$N_+(H_\ell) \le 1$
Right R	V_3^+	$u_3 \to u_4$	$H_r = V_3^+ A G_4^-$	$N_-(H_r) \le 1$
Right S	C_3^+	$u_4 \to u_3$	$H_r = C_3^+ T G_4^+$	$N_-(H_r) \le 1$

For example, a left type-R vertex has side message V_2^- with q-weighted fourth moment at most one. The retained incoming edge contributes T, so (6.1) gives $N_+(H_\ell) \le 1$. A left type-S vertex has the fully reversed side message C_2^- with p^5-weighted fourth moment at most one; (6.4) gives the same N_+ output, using an outgoing outer path. The right type-R and type-S cases follow from (6.3) and (6.2), respectively.

The factor TG or AG integrates the final edge of the outer connector; its preceding $L - 1$ edges are integrated in G. For an outer connector of length one, G is the zero-step terminal message.

The two side tables and the two terminal arm systems are disjoint, so their directions can be chosen independently. For each of the four role combinations RR, RS, SR and SS, the left half produces an N_+ bound and the right half an N_- bound.

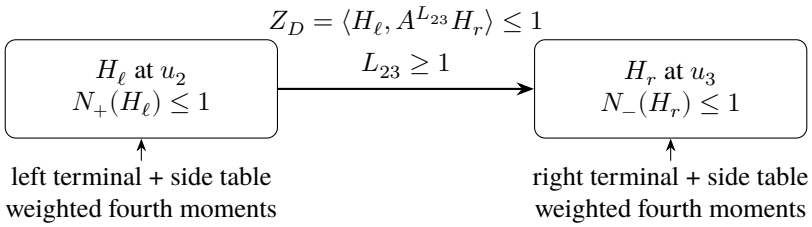


Figure 2: The chain merger. Terminal and side-table fourth moments give the two indicated half-tree bounds. The outer directions depend on the R/S roles in Table 6; the central connector always points from u_2 to u_3.

6.3 The complete central path

After integration over every arm and both outer connector interiors, the original density is exactly

$$Z_D(K,\mu) = \langle H_\ell, A^{L_{23}} H_r \rangle . \tag{6.5}$$

Lemma 2.1 gives

$$Z_D(K,\mu) \le \sqrt{N_+(H_\ell) N_-(H_r)} \le 1.$$

Together with the reducible-terminal case, this proves the theorem for the entire P_4 skeleton.

7 The star skeleton

Let o be the central branch vertex and u_1, u_2, u_3 the three terminal branch vertices, with connector lengths $L_1, L_2, L_3 \ge 1$. The center may have no side arms. Write E_c for its product of equal-length arm pairs, so $0 \le E_c \le 1$.

7.1 The center catalogue

The residual center tables fall into three classes. In the first, a fixed message C satisfies

$$\mathbb{E}[q^3 C^4] \le 1. \tag{7.1}$$

The table gives the direction for each row. The factor E_c multiplies every row, and the products P and V^+ involve only the indicated residual arms.

Table 7: Center directions with a cubed-weight fourth moment.

Residual center table	Message C/E_c	Complete proof
Singleton $a \ge 1$	f_a	Section G.4
Odd, at least three, minimum one	pP, P canonical on the rest	Section K.1; (7.2)
Odd, at least three, minimum $a \ge 2$	$f_a P$, P canonical on the rest	Section K.1
Even, minimum two	$f_2 h_{b_1} \prod_{i=2}^{t} h_{a_i} f_{b_i}$	Section K.3
Even, minimum three	$f_3 h_{b_1} \prod_{i=2}^{t} h_{a_i} f_{b_i}$	Section K.2
Even, minimum at least four	$\prod_{i=1}^{t} h_{a_i} f_{b_i}$	Section K.2
Even, at least four, minimum one	pV^+	Section K.3; (7.2)

The unit-arm reductions in the second and last rows are direct. If $V \ge 0$ is the message of the row in question, with $\mathbb{E}[pV^4] \le 1$, then

$$\mathbb{E}[q^3 (pV)^4] = \mathbb{E}[p(pq)^3 V^4] \le \mathbb{E}[pV^4] \le 1. \tag{7.2}$$

For an odd center of minimum one, $V = P$ is the canonical even product. For an even center with at least four entries and minimum one, the remaining odd list has at least three entries, and V^+ is exactly its zero-step direction in (5.3). The leading p comes from the unit arm.

The singleton row directs its residual arm outward for every length and is proved in Section G.4, with a separate certificate for the two-edge arm.

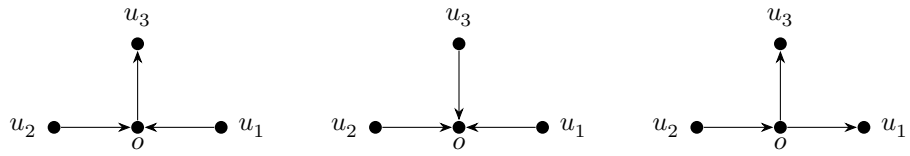
The second center class consists of a residual list exactly $(1, b)$, $b \ge 2$. Direct both residual arms outward and take

$$C = E_c p f_b, \qquad \mathbb{E}[q^9 C^4] \le \mathbb{E}[q^5 f_b^4] \le 1. \tag{7.3}$$

Indeed $q^9C^4 = E_c^4(pq)^4q^5f_b^4$. The last inequality is the complete singleton estimate in Sections G.2, G.3. This class uses the ninth power rather than the cubed weight, including the case $b = 2$.

The third class has an empty residual list and message $C = E_c \le 1$; Section 7.4 handles it with a different merger.

The first class contains every odd nonempty list and every even list whose minimum is two, three, or at least four. An even list with minimum one has either two entries, which gives the second class, or at least four, which gives the first. The empty list is the third class.



Cubed weight
$\mathbb{E}[q^3C^4] \le 1$

Ninth weight
$\mathbb{E}[q^9C^4] \le 1$

Empty residual center
$C = E_c \le 1$

Figure 3: Core directions for the three star-center constructions. The empty-center case uses a full L_1-step terminal on the right-hand path and a second-moment terminal on the upper path. All connector lengths remain positive and arbitrary.

7.2 Cubed-weight centers: two incoming paths and one outgoing path

Assume (7.1). Reducible terminals have already been covered by Lemma 4.1. Fix the core directions

$$u_1 \longrightarrow o, \qquad u_2 \longrightarrow o, \qquad o \longrightarrow u_3,$$

with every original edge of each path pointing consistently. Put

$$G_i = T^{L_i-1}F^+_{i,L_i-1} \quad (i = 1, 2), \qquad H_3 = A^{L_3-1}F^-_{3,L_3-1}.$$

The terminal theorem gives $\mathbb{E}[pG_i^4] \le 1$ and $\mathbb{E}[qH_3^4] \le 1$. Retaining only the final edge near o on each path, the density is

$$Z_D = \mathbb{E}[C(TG_1)(TG_2)(AH_3)]. \tag{7.4}$$

We control two groups at the same center label. First,

$$\begin{aligned} \|CTG_1\|_2^2 &\le \mathbb{E}[qC^2T(G_1^2)] \\ &\le \sqrt{\mathbb{E}[q^3C^4]\,\mathbb{E}[pG_1^4]} \le 1. \end{aligned} \tag{7.5}$$

The first step is row Cauchy–Schwarz; the second applies Cauchy–Schwarz to the T edge measure, whose root marginal is $q\mu$. For the other two paths, row Cauchy–Schwarz and $pq \le 1$ give

$$\begin{aligned} \|(TG_2)(AH_3)\|_2^2 &\le \mathbb{E}[pq\,T(G_2^2)A(H_3^2)] \\ &\le \langle G_2^2, A^2(H_3^2)\rangle \\ &\le \sqrt{\mathbb{E}[pG_2^4]\,\mathbb{E}[qH_3^4]} \le 1. \end{aligned} \tag{7.6}$$

The middle expression uses $T^* = A$, and the last step is Lemma 2.1 with length two. The joint estimate integrates the two adjacent edges at their common label. Applying ordinary Cauchy–Schwarz to the two groups in (7.4) proves $Z_D \le 1$.

7.3 Ninth-weight centers: all three paths incoming

Now use (7.3) and direct every connector from u_i toward o. Set $G_i = T^{L_i-1}F^+_{i,L_i-1}$, so $\mathbb{E}[pG_i^4] \le 1$. The exact density is

$$Z_D = \mathbb{E}\left[C\prod_{i=1}^{3} TG_i\right].$$

Hölder on a row of the nonnegative T kernel gives

$$(TG_i)^4 \le q^3 T(G_i^4). \tag{7.7}$$

The factor q^3 reflects the mass q of the row measure. Apply (7.7), followed by four-factor Hölder on the original probability measure:

$$\begin{aligned} Z_D &\le \mathbb{E}\left[Cq^{9/4}\prod_{i=1}^{3} T(G_i^4)^{1/4}\right] \\ &\le \left(\mathbb{E}[q^9C^4]\prod_{i=1}^{3}\mathbb{E}[T(G_i^4)]\right)^{1/4} \\ &= \left(\mathbb{E}[q^9C^4]\prod_{i=1}^{3}\mathbb{E}[pG_i^4]\right)^{1/4} \le 1. \end{aligned}$$

The equality uses $\mathbb{E}[Tv] = \mathbb{E}[pv]$.

7.4 Empty residual center

Here $C = E_c \le 1$. Choose the core directions $o \to u_1$, $u_2 \to o$, and $o \to u_3$. At the first terminal use the *positive-step* supply through the entire first path:

$$X = A^{L_1}F^-_{1,L_1}, \qquad \mathbb{E}[qX^4] \le 1.$$

At the second terminal retain its last edge and put

$$G = T^{L_2-1}F^+_{2,L_2-1}, \qquad \mathbb{E}[pG^4] \le 1.$$

At the third terminal choose the actual spider direction of Lemma 2.2, with message F_3 and $N_-(F_3) \le 1$; a second-moment bound suffices here.

Let $H = X(TG)$. Row Cauchy–Schwarz, followed by edge Cauchy–Schwarz with the correct marginals, yields

$$\begin{aligned} N_+(H) &\le \mathbb{E}[pqX^2T(G^2)] \\ &\le \sqrt{\mathbb{E}[p^2q^3X^4]\,\mathbb{E}[pG^4]} \\ &\le \sqrt{\mathbb{E}[qX^4]\,\mathbb{E}[pG^4]} \le 1. \end{aligned}$$

In the last comparison $p^2q^2 \le 1$ is used. Multiplying by $C \le 1$ gives $N_+(CH) \le 1$. The exact density is

$$Z_D = \langle CH, A^{L_3}F_3\rangle \le \sqrt{N_+(CH)N_-(F_3)} \le 1.$$

The three center constructions, the terminal catalogue, and Lemma 4.1 prove the assertion for the $K_{1,3}$ skeleton.

8 Five branch vertices: reduction and geometry

Let B be the five original branch vertices and let S be their minimal connecting subtree. Its leaves belong to B, and every vertex of degree at least three in S belongs to B. Suppressing the degree-two vertices outside B reveals the skeleton. The resulting tree on five vertices is P_5, $K_{1,4}$ or T_5: maximum degree four gives the star, maximum degree three gives degree sequence $(3, 2, 1, 1, 1)$, and maximum degree two gives the path. Each skeleton edge retains its original positive path length, and every off-core component is a side arm.

At each skeleton vertex v, pair equal-length side arms maximally. Let c_v be its skeleton degree and t_v the number of remaining distinct arm lengths.

Lemma 8.1. *If $c_v + t_v \le 2$ at any skeleton vertex, then U has an orientation satisfying (1.3) for every finite model.*

Proof. Orient each equal pair as in (2.5), then bound their product at v by one in the integrand. The comparison tree retains v, all core paths and the unpaired arms, and has at most four branch vertices; Theorem 3.2 gives it a fixed orientation. Restoring the pairs at their original attachment multiplies the integrand at v by their actual product, which is at most one, so the original density is also at most one. □

Apply this test at every skeleton vertex. In P_5 it covers an empty internal residual and a terminal with at most one residual arm; an empty residual at the star or fork center needs the separate argument of Section 10.

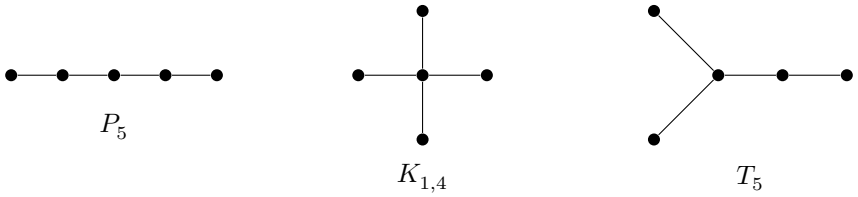


Figure 4: The three skeletons on five branch vertices. Every drawn edge denotes an independently chosen positive-length path in the original tree.

9 The five-branch boundary estimates

Besides $f_n, \bar{f}_n, h_n$, write

$$W_n = A\bar{f}_{n-1} \quad (n \ge 1), \qquad I_q(V) = \mathbb{E}[q^{-1/3}|V|^{8/3}].$$

The arm message W_n directs its first edge away from the root and its remaining edges toward the first neighbor. Reversal exchanges f_n with $\bar{f}_n$ and h_n with W_n.

9.1 Simple-reference and ordinary moments

The terminal residuals below contain at least two distinct lengths. All messages include their actual equal-pair factors.

Proposition 9.1 (Simple-reference fourth moments)**.** *For every existing outward arm f_s, $s \geq 1$, every terminal arm multiset as above, and every connector length $L \geq 1$, there is a structurally chosen terminal direction with actual received message $X = T^L F$ such that*

$$\mathbb{E}[f_s X^4] \leq 1.$$

For $s \geq 3$, the direction in the genuine-reference table satisfies the stronger bound $1 - d/3000$; for $s = 2$ only the bound 1 is used.

For $s = 1$ the assertion is Theorem 5.1 with step count L. For $s \geq 2$ the direction table and its proof are in Sections N.41 and N.42, indexed in Appendix L; the short-arm directions there are specific to this reference catalogue.

For ordinary moments, a *post block* means actual unequal central pairs with negative canonical message

$$P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}, \qquad 4 \leq u_1 < v_1 < \cdots < u_k < v_k.$$

When nonempty, it is required that the designated anchor length satisfy $m < u_1$. This order is automatic in the central allocations below.

Proposition 9.2 (Ordinary eighth moments)**.** *Given an existing arm of length $m \geq 3$, a terminal residual of size at least two, a connector of length $L \geq 1$ and a post block in the stated order, there is a direction of exactly these resources with actual message H satisfying*

$$\mathbb{E}H^8 \leq e^{-d/1024}.$$

Given instead one existing unit arm and no unequal post block, there is an actual message H_1 with $\mathbb{E}H_1^8 \leq 1 - d/20000$.

The complete ordinary direction table, including the first-edge choices, is in Section O.39, and Appendix L indexes its cases. The nonunit proof draws on Sections O.32, O.34, O.35, O.36 and O.38; the unit input is the ordinary catalogue of Section O.5. For the terminal $(1, 2, 3)$, the special direction occurs at $L = 2$ in this ordinary catalogue and at $L = 1$ in the genuine-reference catalogue.

We also use the following path and mixed central bounds:

$$\mathbb{E}f_s^2 \leq 1 \quad (s \geq 2), \qquad \mathbb{E}[f_s^2 Q^8] \leq e^{-17d/64}, \tag{9.1}$$

where the second bound has $Q = W_a \bar{f}_b \prod_i W_{u_i} \bar{f}_{v_i}$ and $2 \leq s < a < b < u_1 < v_1 < \cdots$. The mixed proof, including $s = 2$ and $a = 3$, is in Section R.4. When $Q = 1$ only the first bound is needed. For $s = 2$ its exact value is $1 - d + \delta_2$; for $s \geq 3$ the finite path estimate gives $\mathbb{E}f_s^2 \leq 1 - d + 2\delta_2 \leq 1 - 3d/16$.

9.2 A reference second moment for a compound branch

A fork has a two-segment branch $c - r - t$ with independent lengths $M, N \geq 1$ and arbitrary side-arm tables at r, t.

Proposition 9.3. *For each $s \geq 1$, this actual compound branch admits a direction with received message Y satisfying $\mathbb{E}[f_s Y^2] \leq 1$.*

Proof. If t is reducible, its retained connector and possible residual arm form one comparison arm at r. Choose the canonical endpoint direction used in (C.9), apply that inequality, and restore the equal pairs. Otherwise put $G = T^{N-1} F^+_{t,N-1}$, so $\mathbb{E}[pG^4] \leq 1$.

If the residual at r is empty or has at least two entries, select its negative zero-step factor C_r^- with $\mathbb{E}[q(C_r^-)^4] \le 1$ and set $H = C_r^- TG$, $Y = T^M H$. Row Hölder and Cauchy–Schwarz give

$$(TG)^4 \le q^3 T(G^4), \qquad \mathbb{E}[H^2/q] \le \sqrt{\mathbb{E}[q(C_r^-)^4]\,\mathbb{E}[(TG)^4/q^3]} \le 1. \tag{9.2}$$

Thus $H = \sqrt{q}\,h$ with $\|h\|_2 \le 1$. The operator bounds in Proposition C.4, together with normalized propagation, give

$$\sqrt{f_s} T^M \sqrt{q} = (\sqrt{f_s} T \sqrt{q})(q^{-1/2} T \sqrt{q})^{M-1}, \qquad \|\sqrt{f_s} T^M \sqrt{q}\| \le 1. \tag{9.3}$$

This proves the claim in the R case, including $M = N = 1$.

If r has one residual arm of length a, orient it inward as $\bar{f}_a$. At t, choose the positive canonical direction for an even residual list and the negative canonical direction for an odd residual list, and denote the resulting message by S. Let S^- be its full reversal and put $V = A^N S^-$, $H = \bar{f}_a V$ and $Y = T^M H$. The full reversal of (C.9) gives $\mathbb{E}[\bar{f}_a V^2] \le 1$ for every $a, N \ge 1$. The cross-path contraction $\|\sqrt{f_s} T^M \sqrt{\bar{f}_a}\| \le 1$ proves the assertion. Equal pairs at r are restored by positivity. □

9.3 The common-reference outlets for the path skeleton

Consider one side $c - r - t$ of P_5. Its terminal connector $r - t$ has length $L \ge 1$, and its receiving connector $r - c$ has length $M \ge 1$. The terminal residual has size at least two, and the residual at r is nonempty. The following catalogue is used after the two whole-tree alternatives of Section 11.

Proposition 9.4 (Common-reference outlet)**.** *Except for a unit singleton at r with $M = 1$, every such side has an actual message U at c, determined by the structure, satisfying*

$$I_q(U)^3 \le e^{-\eta d}, \qquad \eta \ge 1/500. \tag{9.4}$$

Both sides of the path use the same q.

Table 8: The path outlet catalogue. The entries locate the proofs in the technical appendix, and Appendix L indexes the direction tables.

Root residual	Structural subdivision	Where proved
Unit singleton	$M = 2$; $M \ge 3$	P.4, P.5, P.6, P.7, P.8, P.9; M.19, O.5
Singleton $a \ge 2$	$M = 1, 3, \ge 4$; $a = 2, M = 2$; $a \ge 3, M = 2$	N.41, N.42, R.5, R.1; R.6; R.7
Exactly two arms	Minimum one; minimum at least two	M.16, O.5; O.39
Odd, at least three	Minimum one, two, or at least three	Q.13, Q.21, Q.22, Q.23, Q.24; Q.25–Q.28
Even, at least four	Minimum one or at least two, with one actual inward multiplier retained	R.3

The unit-singleton row $M = 1$ is handled by the whole-tree construction before this catalogue is consulted.

10 The five-vertex star and fork

Let c be the degree-four star center or the degree-three fork center. There are four simple terminal branches in the star, and two simple terminal branches plus one compound branch in the fork. A reducible simple terminal is covered by Lemma 8.1. Write $E_c \le 1$ for the product of central equal pairs.

10.1 Nonempty central residuals

Table 9 allocates the central arms along the increasing residual list. One actual outward arm f_s serves as the common reference; an ordinary message consumes its anchor, one simple terminal and the connecting path; the central equal pairs belong to one indicated factor. In every nonsingleton row $s \geq 2$,

Table 9: Central arm allocation for $K_{1,4}$ and T_5. Later pairs always retain their displayed negative canonical directions.

Residual list	Reference	Ordinary factors	Other central factors
Singleton s	f_s	None	E_c in one reference message
$(s, m, \text{later pairs}), 2 \leq s < m$	f_s	H: anchor m, later pairs, E_c	None
$(s, a, b, \text{later pairs}), \ 2 \leq s < a < b$	f_s	H_a and H_b; latter holds later pairs and E_c	None
$(1, s, m, \text{later pairs})$	f_s	Unit H_1; H_m holds later pairs and E_c	None
$(1, s)$ or $(1, s, a, b, \text{later pairs})$	f_s	Unit H_1 holds E_c	$Q = 1$ or $W_a \bar{f}_b$ times later pairs

so (9.1) applies, and the anchor preceding a nonempty post block is shorter than all its post lengths, as Proposition 9.2 requires.

For a singleton center the densities are

$$Z_\star = \mathbb{E}[f_s X_1 X_2 X_3 X_4], \qquad Z_{T_5} = \mathbb{E}[f_s X_1 X_2 Y].$$

Use Hölder exponents $(4, 4, 4, 4)$ or $(4, 4, 2)$ and the reference estimates. Both densities are at most one, including $s = 1, 2$.

For a nonunit even center the products are $f_s H X_2 X_3 X_4$ and $f_s H X Y$. Factor the first into $H, f_s^{1/4}, f_s^{1/4} X_2, f_s^{1/4} X_3, f_s^{1/4} X_4$, and the second into $H, f_s^{1/4}, f_s^{1/4} X, f_s^{1/2} Y$. Hölder with exponents $(8, 8, 4, 4, 4)$ or $(8, 8, 4, 2)$ gives, in both cases, $Z_D \leq e^{-d/8192}$.

For a nonunit odd center there are two distinct ordinary factors. The star and fork factors are, respectively,

$$(H_a, H_b, f_s^{1/2}, f_s^{1/4} X_3, f_s^{1/4} X_4), \qquad (H_a, H_b, f_s^{1/2}, f_s^{1/2} Y).$$

The same exponent lists give $Z_D \leq e^{-d/4096}$. For a unit-minimum odd center replace the first ordinary input by $\mathbb{E} H_1^8 \leq 1 - d/20000$. This gives

$$Z_D \leq (1 - d/20000)^{1/8} e^{-d/8192} \leq e^{-(1/160000 + 1/8192)d}.$$

For a unit-minimum even center use the factors $H_1, f_s^{1/4} Q$ and the remaining reference factors. Equations (9.1) and the same Hölder exponents give

$$Z_D \leq \begin{cases} e^{-d/160000}, & Q = 1, \\ e^{-(1/160000 + 17/512)d}, & Q \neq 1. \end{cases} \tag{10.1}$$

The reference arm carries total analytic power one, and the ordinary factors use distinct terminal branches. Hölder applies to the resulting functions of the central label.

10.2 Empty central residual

For the star, choose two paths toward c and two away from it. Put $G_i = T^{L_i-1}F^{+}_{i,L_i-1}$ for $i = 1, 2$ and $H_i = A^{L_i-1}F^{-}_{i,L_i-1}$ for $i = 3, 4$. Then

$$Z_D = \mathbb{E}[E_c(TG_1)(TG_2)(AH_3)(AH_4)].$$

For each pair, row Cauchy–Schwarz, $pq \le 1$, and the path interface give

$$\mathbb{E}[(TG)^2(AH)^2] \le \langle G^2, A^2(H^2)\rangle \le \sqrt{\mathbb{E}[pG^4]\,\mathbb{E}[qH^4]} \le 1.$$

Cauchy–Schwarz on $(TG_1)(AH_3)$ and $(TG_2)(AH_4)$ proves $Z_D \le 1$.

For the fork label its simple ends a, b and its compound $c - r - t$, with lengths L_a, L_b, M, N. Set

$$X = A^{L_a}F^{-}_{a,L_a}, \quad G = T^{L_b-1}F^{+}_{b,L_b-1}, \quad H_c = E_cX(TG).$$

The first terminal uses the L_a-step supply. The edge measure for T has marginals $q\mu, p\mu$, so

$$N_+(H_c) \le \sqrt{\mathbb{E}[p^2q^3X^4]\,\mathbb{E}[pG^4]} \le \sqrt{\mathbb{E}[qX^4]\,\mathbb{E}[pG^4]} \le 1.$$

If r is type R, choose its positive zero-step factor V_r^+ and put $V = A^{N-1}F^{-}_{t,N-1}$, $H_r = V_r^+AV$. If r is a singleton, put $C_r^+ = E_rf_a$, $V = T^{N-1}F^{+}_{t,N-1}$ and $H_r = C_r^+TV$. Inequality (6.3) gives $N_-(H_r) \le 1$ in the R case, and (6.2) in the singleton case, where the q^5 weight enters. Direct the whole M-path from c to r. Then

$$Z_D = \langle H_c, A^M H_r\rangle \le \sqrt{N_+(H_c)N_-(H_r)} \le 1.$$

This proves both skeletons. Only the nonsingleton central allocations give the strict loss $e^{-d/160000}$; the comparison, empty-center and singleton-center directions give the bound 1.

11 The five-vertex path

Write the core in path order as

$$t \;{-}{-}_{L_-}\; r \;{-}{-}_{M_-}\; c \;{-}{-}_{M_+}\; w \;{-}{-}_{L_+}\; z, \qquad L_-, M_-, M_+, L_+ \ge 1.$$

Here $L_\pm$ counts a terminal connector and $M_\pm$ the connector from an off-center root to c. In a local outlet formula, L denotes the terminal-connector length and M the receiving-connector length. After Lemma 8.1, the two terminal residuals have at least two entries and the three internal residuals are nonempty. Call an internal residual S if it is a singleton and R otherwise.

11.1 A central singleton: the maximal S spine

Suppose c is S. Take the maximal consecutive S sequence among r, c, w that contains c. Orient its singleton arms alternately as f_a and $\bar{f}_b$, starting with an outward arm at its left end. Every connector between consecutive S vertices points from the negative end toward the positive end. The corresponding operators between square roots of the actual arm messages are

$$B_i = \sqrt{f_a}T^{M_i}\sqrt{\bar{f}_b} \quad \text{or their adjoints,} \qquad \|B_i\| \le 1$$

by Proposition C.4.

At a positive S end whose arm has message f_s, an ordinary terminal supplies

$$y = \sqrt{f_s}\, T^L F_t, \qquad \|y\|_2 \le 1,$$

where F_t is positive canonical for an even residual list and negative canonical for an odd residual list; the norm bound is (C.9). An R half-tree supplies instead

$$y = \sqrt{f_s}\, T^M H, \qquad H = C^- T G = \sqrt{q}\, h, \qquad \|h\|_2 \le 1.$$

Equations (9.2) and (9.3) give $\|y\|_2 \le 1$ for every $M \ge 1$. At a negative S end reverse the entire outside configuration: the two endpoint functions become $\sqrt{\bar f_s} A^L F_t^-$ and $\sqrt{\bar f_s} A^M H^-$, respectively, with the same norm bound.

Bounding the genuine equal-pair factors on the spine by one gives

$$\langle y_L, B_1 \cdots B_{j-1} y_R \rangle \le 1, \qquad \|y_L\|_2, \|y_R\|_2 \le 1.$$

For a one-vertex spine this is $\langle y_L, y_R \rangle$. This covers the patterns SSS, RSS, SSR and RSR. The two square roots of each arm contribute one arm factor, and each original connector appears once.

11.2 A unit off-center arm with a unit receiving connector

Now suppose the center is R and, say, r has one unit residual arm and its receiving connector $r - c$ has $M_- = 1$. Put $L = L_-$ and choose the terminal direction with step count $L - 1$. Its composite message at r is

$$F_*^+ = E_r^- q A^L F_{t,L-1}^-.$$

The actual unit arm is the factor q. Row Hölder and $pq \le 1$ give, for every real G,

$$\mathbb{E}[q|pTG|^4] \le \mathbb{E}[(pq)^4 T(|G|^4)] \le \mathbb{E}[p|G|^4].$$

Full reversal proves $\mathbb{E}[p(F_*^+)^4] \le 1$. The one edge from r to c has not yet been integrated. Choose the central negative R factor C^- with $\mathbb{E}[q(C^-)^4] \le 1$ and put $H_c = C^- T F_*^+$. Lemma 6.1 gives $N_+(H_c) \le 1$.

At w, use the right R or S construction in Table 6, with its actual terminal and terminal-connector length L_+. It gives a message H_w with $N_-(H_w) \le 1$. Direct the entire M_+-path from c to w. The exact density is

$$Z_D = \langle H_c, A^{M_+} H_w \rangle \le 1.$$

This whole-tree construction replaces the I_q outlet at r.

11.3 Two compatible outlets

In the remaining structural branch c is R, and any unit singleton off-center has receiving-connector length at least two. Choose the central negative zero-step factor C_c as follows. An even residual uses negative canonical pairs. An odd residual beginning with one uses $E_c^- p \bar f_a W_b$ followed by negative canonical pairs. A nonunit odd residual uses $E_c^+ f_a$ followed by positive canonical pairs. These are the full reversals of the corresponding zero-step directions, so

$$\mathbb{E}[q C_c^4] \le 1.$$

Proposition 9.4 supplies actual outputs U_-, U_+ for the two sides, both using this same q. Integration over their disjoint interiors gives $Z_D = \mathbb{E}[C_c U_- U_+]$. Hölder at exponents $4, 8/3, 8/3$ yields

$$\begin{aligned} Z_D &\le (\mathbb{E}[qC_c^4])^{1/4} I_q(U_-)^{3/8} I_q(U_+)^{3/8} \\ &\le e^{-(\eta_- + \eta_+)d/8} \le e^{-d/2000} \le 1. \end{aligned} \tag{11.1}$$

The weight powers $1/4, -1/8, -1/8$ cancel, and both side functions are evaluated at the same central label.

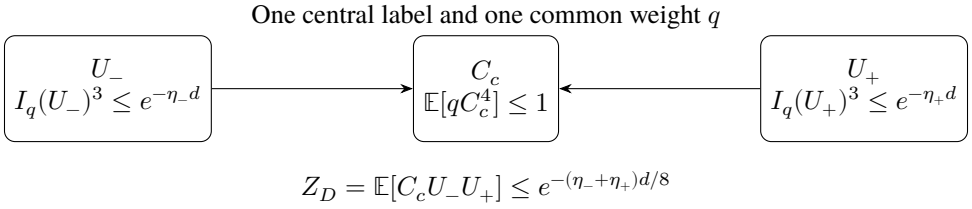


Figure 5: The remaining path branch. The two structural alternatives are applied first; only then are the two local outlets joined with the same central reference weight.

The root catalogue divides nonempty residual lists into singletons, pairs, odd lists of at least three entries, and even lists of at least four. Singletons split according to their arm and receiving-connector lengths: the two-step construction treats an arm of length two differently from one of length at least three, and the choice depends on the length; d enters in the estimates. The minimum-two odd row with a terminal $(2, t)$ has its own first-edge choices, notably $W_2 \bar{f}_t$ at $L = 1$ and $\bar{f}_2 W_t$ at $L \ge 2$; the table in the technical appendix lists them all. Together with the two whole-tree alternatives above, (11.1) proves every P_5.

11.4 A worked five-branch example

Consider the path core in Figure 6, with connector lengths $(L_-, M_-, M_+, L_+) = (2, 1, 2, 1)$ and pendant-arm lists

vertex	t	r	c	w	z
arm lengths	$(2, 4)$	$(1, 3, 4)$	$(4, 5)$	$(2, 4, 5)$	$(2, 3)$.

(11.2)

This tree has 45 edges and five branch vertices, of degrees $3, 5, 4, 5, 3$. It falls in the two-outlet case above. Direct every core edge toward c, and choose the pendant directions specified by the following messages:

$$\begin{aligned} F_- &= \bar{f}_2 W_4, \quad H_- = p W_3 \bar{f}_4 T^2 F_-, \quad U_- = T H_-, \\ F_+ &= W_2 \bar{f}_3, \quad H_+ = f_2 h_4 f_5 T F_+, \quad U_+ = T^2 H_+, \\ &\qquad C = W_4 \bar{f}_5. \end{aligned} \tag{11.3}$$

Integration over the two sides gives $Z_D = \mathbb{E}[C U_- U_+]$. We verify the three moment inputs to this identity directly. When $d = 0$, leaf integration gives $Z_D = 1$; henceforth assume $d > 0$.

Put $B = -K$ and $u = -g = B\mathbf{1}$, so that $T = J + B$, $A = J - B$, $p = 1 - u$, and $q = 1 + u$. We use $d \le 1/3$, $\delta_{j+1} \le \theta \delta_j$, $\theta = 13/32$, and the spectral measure ν of $-B^2$ at u, for which $\int x^j \, d\nu = \delta_{j+1}$. The row bound $|BV| \le \|V\|_2$ and (M.43) give the common fourth-moment rule

$$0 \le H \le e^{a + BV} \quad \Longrightarrow \quad \log \mathbb{E} H^4 \le 4(a + \mathbb{E} BV) + \tfrac{17}{4} \|V\|_2^2. \tag{11.4}$$

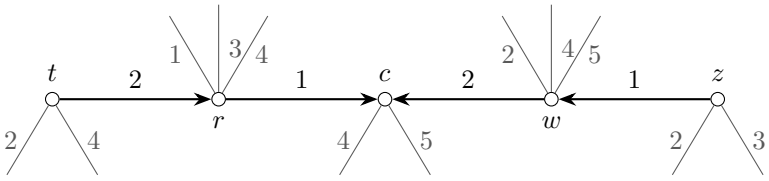


Figure 6: The five-branch example. Each segment represents a path of the indicated length; pendant directions are specified in (11.3).

The arm recurrences $f_{n+1} = Af_n$, $\bar{f}_{n+1} = T\bar{f}_n$, $h_n = Tf_{n-1}$, and $W_n = A\bar{f}_{n-1}$ give

$$\begin{aligned} \bar{f}_2 + W_4 &= 2 - 2d + du - B^2u - B^3u, \\ W_3 + \bar{f}_4 &= 2 - 3d - du + B^3u, \\ W_2 + \bar{f}_3 &= 2 - d + B^2u, \\ h_4 + f_5 &= 2 - 4d + \delta_2 - df_2 - B^4u, \\ W_4 + \bar{f}_5 &= 2 - 4d + \delta_2 - d\bar{f}_2 + B^4u. \end{aligned} \tag{11.5}$$

These identities and $ab \le e^{a+b-2}$ for nonnegative a, b provide all the envelopes below.

The two-step side.

Set

$$R = du - B^2u - B^3u, \quad s = \mathbb{E}R^2 = d^3 + 2d\delta_2 + \delta_3 + \delta_4, \quad \Delta = -\delta_2 + \tfrac{3}{2}s, \quad \epsilon = \Delta - d^2 - \delta_2.$$

The first identity gives $F_- \le e^{-2d+R}$. Since $|R| \le d + \sqrt{\delta_2 + \delta_3} < 1$, the elementary bound $e^y \le 1 + y + 3y^2/4$, $|y| \le 1$, and positivity give

$$TF_- \le e^{-2d}(q + BR + \Delta), \qquad T^2F_- \le e^{-2d}(\bar{f}_2 + B^2R + \Delta q - d^2 - \delta_2).$$

Adding the factors in the last upper bound to p, and then using the second identity in (11.5), yields

$$H_- \le e^{-5d+\epsilon+BV_-}, \qquad V_- = u + BR + B^2u + (\Delta - d)\mathbf{1}.$$

Here

$$-5d + \epsilon + \mathbb{E}BV_- = -6d + \epsilon + \delta_2 + \delta_3 < -\tfrac{27}{5}d.$$

Indeed

$$\epsilon = d^2(-1 + 3d/2) + (-2 + 3d)\delta_2 + \tfrac{3}{2}(\delta_3 + \delta_4) \le 0,$$

because $3\theta(1 + \theta)/2 < 1$.

To bound the complete vector, write $c_0 = \Delta - d$. Its nonconstant part has spectral coordinates $a(x) = 1 - x - x^2$, $b(x) = d + x$. On $[0, \theta]$, $a^2 \le 1 - x$ and $0 \le b \le 1$, so that part has squared norm at most d and mean $-d^2 - \delta_2$. Moreover

$$s/d \le \tfrac{1}{9} + \tfrac{2\theta}{3} + \theta^2 + \theta^3 < \tfrac{5}{8}, \qquad -(d + \delta_2) \le c_0 < 0.$$

Keeping its constant cross term therefore gives

$$\|V_-\|_2^2 \le d + (d + \delta_2)^2 + 2(d + \delta_2)(d^2 + \delta_2) < \tfrac{12}{5}d.$$

The rule (11.4) now proves $\log \mathbb{E}H_-^4 < -57d/5 \le -6d$.

The one-step side.

The third identity gives $F_+ \le e^{-d+B^2u}$. The same Taylor bound gives

$$TF_+ \le e^{-d}(q + B^3u + \tfrac{3}{2}\delta_3).$$

Adding its upper factor to $f_2 = 1 - u + Bu$ leaves $2 + B(u + B^2u) + 3\delta_3/2$. The fourth identity, together with $f_2 \ge 0$ and $4 - \theta > 3$, gives $h_4 f_5 \le e^{-3d-B^4u}$. Consequently

$$H_+ \le e^{-4d+(3/2)\delta_3+BV_+}, \qquad V_+ = u + B^2u - B^3u.$$

Its spectral square is $(1-x)^2 + x^3 \le 1$. Thus

$$\|V_+\|_2^2 \le d, \qquad -4d + \tfrac{3}{2}\delta_3 + \mathbb{E}BV_+ = -5d + \delta_2 + \tfrac{3}{2}\delta_3 < -4d.$$

Equation (11.4) proves $\log \mathbb{E}H_+^4 < -47d/4 \le -6d$.

The center and the receiving connectors.

The last identity gives $C \le e^{-3d+B^4u}$. Put $X = u + 4B^4u$. Skew-adjointness gives $\mathbb{E}X = 0$, and

$$|X| \le 1 + 4\theta^{3/2}\sqrt{d} < 2, \qquad \mathbb{E}X^2 \le (1 + 4\theta^2)^2 d < 3d.$$

Since $q \le e^u$ and $e^y \le 1 + y + 3y^2/2$ for $|y| \le 2$,

$$\mathbb{E}[qC^4] \le e^{-12d}\mathbb{E}e^X \le e^{-12d}(1 + 9d/2) \le e^{-15d/2} \le 1.$$

For a positive operator P, row Cauchy–Schwarz followed by Hölder gives

$$I_q(PH)^3 \le \mathbb{E}[(P\mathbf{1})^4/q]\,\|P\|_{2\to 2}^4(\mathbb{E}H^4)^2.$$

The short-path row and operator bounds proved in Section Q.1 are

$$\mathbb{E}[q^3] \le e^{4d}, \quad \|T\|^4 \le e^{128d/19}, \qquad \mathbb{E}[\bar{f}_2^4/q] \le e^{6d}, \quad \|T^2\|^4 \le e^{4d}.$$

Using $\mathbb{E}H_-^4, \mathbb{E}H_+^4 \le e^{-6d}$ gives

$$I_q(U_-)^3 \le e^{-24d/19}, \qquad I_q(U_+)^3 \le e^{-2d}.$$

Finally, the three-factor Hölder inequality at c yields

$$Z_D \le (\mathbb{E}[qC^4])^{1/4} I_q(U_-)^{3/8} I_q(U_+)^{3/8} \le e^{-(24/19+2)d/8} \le 1.$$

The powers $q^{1/4}, q^{-1/8}, q^{-1/8}$ cancel at the central label.

12 The fixed orientation and its consequences

Fix an ordering of the original vertices and arms to resolve ties. For $b(U) \le 4$ use Theorem 3.2. Otherwise recognize the five-vertex skeleton and apply Lemma 8.1 at the first qualifying vertex, if one exists. For $K_{1,4}$ or T_5, use the corresponding central allocation or empty-center construction of Section 10. For P_5, first use the maximal S spine when the center is S, then the construction for a unit arm with a receiving connector of length one when applicable, and otherwise select the two outlet rows from the original integer data. Restore every previously oriented equal pair at its original attachment.

This rule fixes one orientation of U before the host is chosen and proves Theorem 1.2.

12.1 All maps into a tournament

For a loopless tournament $\mathcal{T}$ on $n_T \geq 1$ vertices, take the uniform measure and its skew sign kernel, with $K(x,x) = 0$. For every original edge and every vertex map ϕ,

$$\mathbf{1}_{\{\phi(u)\to\phi(v) \text{ in } \mathcal{T}\}} \leq \frac{1 + K(\phi(u),\phi(v))}{2}.$$

For distinct images this is equality; for equal images it is the inequality $0 \leq 1/2$. Multiplying over the original edges and summing over all $n_T^{|V(U)|}$ maps gives

$$\hom(D,\mathcal{T}) \leq 2^{-e(U)} n_T^{|V(U)|} Z_D(K,\mu) \leq 2^{-e(U)} n_T^{e(U)+1},$$

which proves Theorem 1.1.

12.2 Correlated vertex labels

Let W_D be the product in (1.2) and let $P = \mu^{V(U)}$. A constant-label assignment has positive mass and weight one, so $Z_D > 0$.

Corollary 12.1. *For the orientation in Theorem 1.2 and every joint law ρ on the original vertex labels,*

$$\mathbb{E}_\rho \log W_D \leq D_{\mathrm{KL}}(\rho\|P) + \log Z_D \leq D_{\mathrm{KL}}(\rho\|P).$$

Proof. When the reference entropy is finite and ρ gives no mass to $W_D = 0$, use the probability law $dP_D = (W_D/Z_D)dP$. Nonnegativity of $D_{\mathrm{KL}}(\rho\|P_D)$ gives the first inequality, and the kernel theorem gives the second. If ρ gives positive mass to $W_D = 0$, the left side is $-\infty$. A law outside the reference support has infinite reference entropy and is covered by the extended-value form of the inequality. □

12.3 Boundary cases

Zero-mass atoms can be omitted without changing the integral. At $d = 0$ we have $K\mathbf{1} = 0$, hence $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, and successive leaf integration gives $Z_D = 1$ for every oriented tree. Reversing every original edge amounts to replacing K by $-K$, which preserves the universal bound.

13 Further directions

The four-branch argument converts terminal fourth moments into weighted second moments. For five branch vertices, the proof allocates central arms to different moment estimates and introduces additional boundary quantities, among them I_q. Each skeleton has a separate assembly. An extension to more branch vertices requires estimates that remain valid for the composite messages produced by further mergers.

Known attachment constructions already produce families with unbounded branch count [6, Theorem 1.3 and Proposition 5.2]. Attaching two copies with the same internal orientation and rooted message $m \geq 0$, where $\mathbb{E}m \leq 1$, through oppositely directed attachment edges contributes $(Am)(Tm) = (\mathbb{E}m)^2 - (Km)^2 \leq 1$, the kernel form of those constructions. Asymmetric composite branches require further boundary estimates.

Two branches give explicit losses, $e^{-d/2000}$ for the two-outlet path construction and $e^{-d/160000}$ for the nonsingleton central allocations. The other constructions yield $Z_D \leq 1$.

Author contributions

Both authors contributed equally, share first authorship, and are corresponding authors.

Acknowledgments

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflicts of interest

The authors declare no conflicts of interest.

Data and code availability

No experimental data were collected. Proof sources, exact-arithmetic verification scripts and rational certificate data are publicly available in the research compendium [7], GitHub release v2026.10.30-r2.

AI assistance

AI assistance was used during symbolic exploration and manuscript preparation: OpenAI Codex assisted with merging, translation and typesetting; DeepSeek assisted with language editing and with language polishing and typesetting of the final manuscript. The authors are responsible for the content of this manuscript.

References

[1] Hao Chen and Yupeng Lin. Oriented paths with few direction flips are tournament anti-Sidorenko, 2026. Preprint, arXiv:2609.14655v1. https://arxiv.org/abs/2609.14655v1.

[2] Hao Chen, Felix Christian Clemen, and Jonathan A. Noel. On tournament anti-Sidorenko orientations of trees, 2026. Preprint, arXiv:2605.14138v2. https://arxiv.org/abs/2605.14138v2.

[3] Jacob Fox, Zoe Himwich, Nitya Mani, and Yunkun Zhou. Variations on Sidorenko's conjecture in tournaments, 2024. Preprint, arXiv:2402.08418v1. https://arxiv.org/abs/2402.08418v1.

[4] Andrzej Grzesik, Daniel Il'kovič, Bartłomiej Kielak, and Daniel Král'. Quasirandom-forcing orientations of cycles. *SIAM Journal on Discrete Mathematics*, 37(4):2689–2716, 2023. doi: 10.1137/23M1548700.

[5] Andrzej Grzesik, Daniel Král', László Miklós Lovász, and Jan Volec. Cycles of a given length in tournaments. *Journal of Combinatorial Theory, Series B*, 158:117–145, 2023. doi: 10.1016/j.jctb.2022.07.007.

[6] Xiaoyu He, Nitya Mani, Jiaxi Nie, Nathan Tung, and Fan Wei. New Sidorenko-type inequalities in tournaments, 2025. Preprint, arXiv:2512.11222. https://arxiv.org/abs/2512.11222.

[7] Yanbin Qiao and Liwei Yao. Tournament anti-Sidorenko orientations of trees with at most five branch vertices: research compendium. GitHub, version 2026.10.30-r2, 2026. https://github.com/E123456829/tournament-anti-sidorenko-five-branch-trees-v2/releases/tag/v2026.10.30-r2.

[8] Ashwin Sah, Mehtaab Sawhney, and Yufei Zhao. Paths of given length in tournaments. *Combinatorial Theory*, 3(2):Paper No. 5, 2023. doi: 10.5070/C63261981.

[9] Alexander Sidorenko. A correlation inequality for bipartite graphs. *Graphs and Combinatorics*, 9 (2):201–204, 1993. doi: 10.1007/BF02988307.

A Shared calculus for local moments

This appendix collects the scalar tools shared by the local estimates and records where each one is used. In every statement, the direction and the weight are part of the hypotheses. References to Appendices C–R point to the technical appendix that accompanies this article. The common spectral and

Table 10: Local estimates and their uses in the assembly.

Location	Estimates	Used in
Appendix G	Path fourth moments; singleton fifth and cubed weights	terminal and center bounds
Appendix H	Unpropagated even and odd messages	zero-step supply; canonical roots
Appendix I	Even residual lists after positive propagation	terminal supply
Appendix J	Odd residual lists after positive propagation	terminal supply
Appendix K	Cubed-weight estimates for nonsingleton centers	four-branch star assembly
Appendix L	Index to direction tables and short-length exceptions	all five-branch constructions
Appendices M–R	Reference fourths, ordinary eighths and common-reference outlets	five-branch boundary estimates

rooted-spider foundations are in Appendix D. The scalar tools below carry no arm parameters; each application supplies its own messages, signed means and parameter interval. A finite coefficient table records a polynomial identity or a bound on a continuous domain.

Common scalar calculations.

For a nonnegative weight w with $\mathbb{E}w = 1$, write $\mathbb{E}_w H = \mathbb{E}[wH]$; the measure and the adjoint of K are unchanged. Each application specifies its weight and message.

If $|R| \le M$, then, for $j \ge 2$, $|\mathbb{E}_w R^j| \le M^{j-2}\mathbb{E}_w R^2$. In particular, set

$$B_r(m, M) = \sum_{j=2}^{r} \binom{r}{j} m^{r-j} M^{j-2} 2^{-j} \qquad (r = 6, 8, 12). \tag{A.1}$$

For $m \ge 0$ and $0 \le P \le (m + R/2)^2$, binomial expansion gives

$$\mathbb{E}_w P^{r/2} \le m^r + \frac{r}{2} m^{r-1} \mathbb{E}_w R + B_r(m, M) \mathbb{E}_w R^2. \tag{A.2}$$

Every coefficient of B_r is nonnegative. Thus upper bounds for m, M can be substituted term by term, retaining the signed linear mean separately. The fourth-power expansions below specify their center and weight.

If instead $P \le \exp(R - \Delta)$, let $\Lambda(t) = \log \mathbb{E}_w e^{tR}$. Its second derivative is the variance of R under the tilted probability measure, and is at most M^2. Hence

$$\mathbb{E}_w P^s \le \exp\{s(\mathbb{E}_w R - \Delta) + s^2 M^2/2\} \qquad (s \ge 0). \tag{A.3}$$

In particular $\Delta - \mathbb{E}_w R \ge 3M^2$ suffices for the sixth moment. For the centered square envelopes, Cauchy–Schwarz also gives

$$V = \mathbb{E}R^2, \quad \|R\|_\infty \le M \quad \Longrightarrow \quad \mathbb{E}[pR^2],\ \mathbb{E}[qR^2] \le V + M\sqrt{dV}. \tag{A.4}$$

Reading the interval tables.

The tables below specify upper bounds on closed intervals. When an upper bound has the form $H(d) + ad$ and $H'' \geq 0$, the two listed endpoint bounds prove the assertion throughout that interval. A decreasing bound needs only the left endpoint. Entries are rational bounds from the displayed finite polynomials or their geometric tails; intermediate integer cross-products are suppressed. The case $d = 0$ is settled by leaf integration: every actual rooted message equals one, so divisions by d below are confined to $d > 0$. Equal-pair restoration and complete reversal follow Section 2.

A.1 Scalar and propagation tools

The scalar inequalities used for odd-root and propagation estimates are

$$\begin{aligned} \log(1-x) &\leq -x - 3x^2/10 \quad (-1 \leq x < 1), \\ \log(1+x) &\leq x - 3x^2/10 \quad (-1 < x \leq 1), \\ e^{y-y^2/7} \leq 1 + y + 3y^2/8, &\qquad e^{y-y^2/8} \leq 1 + y + 5y^2/12. \end{aligned} \tag{A.5}$$

For the logarithm, differentiate $\log(1-x) + x + 3x^2/10$; its derivative is $-x(2+3x)/[5(1-x)]$, and its only possible maxima are $-1, 0$, with $\log 2 < 7/10$. For the last two inequalities the positive quadratic polynomials Q_7, Q_8 have logarithmic differences with derivatives $y(2-5y+6y^2)/(56Q_7)$ and $y(4-8y+5y^2)/(48Q_8)$, respectively. Each quadratic factor is positive, proving the global bounds.

We also need the sharper global bound, for every real x,

$$e^{x-x^2/8} \leq Q(x) := 1 + x + 13x^2/32, \tag{A.6}$$

because $[\log Q - x + x^2/8]' = x(13x^2 - 20x + 8)/(128Q)$ has the sign of x.

Shared propagation identities.

Write $\mathrm{Alt}_h = \sum_{i=1}^{h} (-1)^{i-1}\delta_i$ for each finite $h \geq 0$. For the propagation catalogue write $\tau = d - \delta_2$ and $S'_n = \sum_{j=0}^{n} K'^j \mathbf{1}$. Here and below $K' = -K$, $g' = -g$, $\bar{f}_n = T^n \mathbf{1}$ and $\|F\|_{4,p} = (\mathbb{E}[p|F|^4])^{1/4}$. The finite Duhamel identity, obtained by iterating $T = J + K'$, is

$$T^L u = K'^L u + \sum_{j=0}^{L-1} \mathbb{E}[K'^j u]\bar{f}_{L-1-j}. \tag{A.7}$$

The actual path means also satisfy the stronger uniform loss

$$\mu_n = \mathbb{E} f_n = \mathbb{E}\bar{f}_n \leq 1 - 83d/32 \qquad (n \geq 4). \tag{A.8}$$

For $n = 4, 5, 6$ use their exact formulas; the losses divided by d are at least $83/32, 137/48, 89/32$. For $\mu_7 = 1 - 6d + 4\delta_2 + 6d^2 - 2\delta_3 - 2d\delta_2$ the loss is at least $127/48$. For $n \geq 8$ retain $f_0, \ldots, f_5$ in convolution. The three preceding beta indices are at least seven, each contributing an additional $c_0\delta_3$, $c_0 = 19/32$. Hence

$$1 - \mu_n \geq \tau(7 - 10d + 3\delta_2 + d^2) + 3c_0\delta_3 \geq c_0 d(7 - 10d + 7d^2) \geq 95d/36 > 83d/32.$$

This uses integral moment inequalities $\delta_2 \geq d^2$, $\delta_3 \geq d^3$, without imposing a lower bound on individual spectral frequencies.

A finite perturbation of G_n.

If a spectral polynomial has the form $G_n(z) + E_n(z)$ with

$$|E_n(i\sqrt{x})| \le (4/5)x^{(n-1)/2}, \qquad 0 \le x \le \theta,$$

then its modulus is at most one for every $n \ge 4$. For odd $n \ge 5$, use $1 - |G_n| \ge (x - x^n)/[2(1+x)]$; the required worst-index comparison is $17/18 + 625/20736 = 20209/20736 < 1$. For $n = 4$, $|G_4| \le 1 - x/2 - x^2/2$ and $(4/5)x^{3/2} \le x(1+x)/2$ by a positive quadratic in $\sqrt{x}$. For $n \equiv 0 \pmod 4$, $n \ge 8$, compare against $1 - 1/\sqrt{1+x} \ge x/[2(1+x)]$; the worst bound is $3/(5\sqrt{2}) < 1$. For $n \equiv 2 \pmod 4$, $n \ge 6$, the sufficient inequality is

$$(8/5)(1+x)x^{(n-3)/2} + 2x^{n/2-1} + x^{n-1} \le 1.$$

At the worst index $n = 6$, use $x \le 5/12$, $\sqrt{x} \le 2/3$ and $x^5 < 1/64$ to obtain $17/27 + 25/72 + 1/64 = 1715/1728 < 1$. All higher powers improve these bounds; at $x = 0$ there is equality.

Fourth-moment endpoint comparison.

For $Z \ge 0$, put $\mu = \mathbb{E}Z$, $w = Z - \mu$ and $h = \mathbb{E}[-gw]$. The original p-weighted expansion is exactly

$$\mathbb{E}[pZ^4] = \mu^4 - 4\mu^3 h + \mathbb{E}[pw^2(Z^2 + 2\mu Z + 3\mu^2)].$$

If $0 \le \mu \le \bar{\mu}$, $h \ge h_0 \ge 0$, $\operatorname{Var} Z \le V$ and the pointwise coefficient is at most M, then

$$\mathbb{E}[pZ^4] \le MV + \max\{0, \bar{\mu}^3(\bar{\mu} - 4h_0)\}. \tag{A.9}$$

This treats the two opposite-sign mean terms together: the maximum of $u^3(u - 4h_0)$ on $[0, \bar{\mu}]$ is at an endpoint.

For later error estimates,

$$\mathbb{E}[pq^4] \le 1 + 12d/5, \qquad \|q\|_{4,p} \le e^{3d/5}. \tag{A.10}$$

Indeed $pq^4 = 1 - 3g + 2g^2 + g^3(1-g)(2-g)$; its last term is nonpositive when $g \le 0$, and for $g \ge 0$ use $g(1-g)(2-g) \le 2/(3\sqrt{3}) < 2/5$. The elementary exponential remainder used with these bounds is

$$e^x \le 1 + x + \frac{x^2}{2(1 - x/3)} \qquad (0 \le x < 3). \tag{A.11}$$

Its proof bounds $j! \ge 2\,3^{j-2}$ for every $j \ge 2$ and sums the positive geometric tail.

A.2 A fourth-power coefficient

Set $c = 19/32$, $x = \sqrt{d}$, $x_* = 2/\sqrt{19}$, $m = 1 - 3cx^2$, $b_0 = 1 - cx^2$ and $a_0 = 1 + m + x$. For $B_x(q) = a_0 - b_0 q$ put

$$H_x(q) = q[B_x(q)^2 + 2mB_x(q) + 3m^2] \le 7 - 10x^2 \quad (0 \le x \le x_*,\ 0 \le q \le 2). \tag{A.12}$$

Here $B_x \ge 0$, and $H_x'' = 6b_0^2 q - 4b_0(a_0 + m) \le 4b_0 x(3cx - 1) \le 0$. At $q = 3/2$ the value and derivative are

$$H_0(x) = 51m^2/8 + 9mx/2 + 3x^2/2,$$
$$D(x) = -1/4 - 221x^2/64 - 57x^3/16 + 35739x^4/4096.$$

The derivative $(-D)' = x[221/32 + 171x/16 - 35739x^2/1024]$ is positive: the bracket is concave, and its endpoint lower bound at x_* is $93233/24320 > 0$. Thus $D \le -1/4$ and the maximum in q lies in $[0, 3/2]$. On that interval $H_x'' \le -1813/704 < -5/2$, using $b_0 \ge 7/8$ and the endpoint bound $-3 - 4x + 15cx^2 \le -259/88$. The tangent bound is therefore $\max H_x \le H_0 + D^2/5$. The two derivative certificates are $-D(1/3) < 2/3$ and $-D(x_*) < 89205/92416 < 1$. Write

$$Q = H_0 + 10x^2 - 7 = -5/8 + 9x/2 - 1435x^2/128 - 513x^3/64 + 165699x^4/8192.$$

For $x \le 1/3$ its last two terms have nonpositive sum, so $Q + D^2/5 < -5/8 + 1/2 + 4/45 = -13/360$. For $1/3 \le x \le x_*$ its quadratic part is at most $-427/1152$; the last two terms sum to less than $2/15$, using $x_* < 6/13$. Hence $Q + D^2/5 < -427/1152 + 2/15 + 1/5 = -43/1152$. This proves the full continuous domain of (A.12).

A.3 A common fourth-moment comparison

Set $c = 19/32$, $\tau = d - \delta_2$ and $g' = -g$. Suppose $Z = S - a - bq \ge 0$, with

$$\mathbb{E}S = 1 - \alpha, \quad \mathbb{E}[g'S] = \eta, \quad a, b, \alpha, \eta \in [\tau, d], \quad \|S - 1\|^2 \le d,$$
$$|S - 1 - g'| \le \sqrt{d}.$$

For $\mu = \mathbb{E}Z$, $w = Z - \mu$ the exact identities imply

$$0 \le \mu \le m = 1 - 3cd, \quad \mathbb{E}[g'w] \ge h = cd - d^2,$$
$$\mathbb{E}w^2 = \|S - 1\|^2 - \alpha^2 + b^2 d - 2b\eta \le d - 3c^2d^2 + d^3,$$
$$Z \le (1 - cd)q - cd + \sqrt{d}.$$

More generally, the four displayed output bounds on the mean, covariance, variance and pointwise envelope alone suffice for the conclusion below; the coefficient conditions $a, b, \alpha, \eta \in [\tau, d]$ are one sufficient route to those outputs. In particular, later applications may have $b > d$ if they verify the outputs directly. The coefficient $p(Z^2 + 2\mu Z + 3\mu^2)$ is bounded by $7 - 10d$ on the whole $0 \le d \le 1/3$. For $d \le 4/19$ this is (A.12). For the extension put $x = \sqrt{d}$ and use the same polynomials H_0, D, Q from its proof. On $[2/\sqrt{19}, 1/\sqrt{3}]$, $-D$ increases to less than $7/6$, $H_x'(2) < 0$, and $H_x'' \le -3/2$ on $[0, 3/2]$. The latter follows from $b_0 \ge 77/96$ and $-3 - 4x + 15cx^2 \le -519/224$. Also $Q' < 0$: it is convex and has negative endpoint values

$$Q'(2/\sqrt{19}) = -171/304 - (52421/9728)(2/\sqrt{19}), \quad Q'(1/\sqrt{3}) = -225/64 + (9313/2048)/\sqrt{3}.$$

Since $Q(2/\sqrt{19}) < -3/4$, the tangent remainder is at most $-3/4 + 49/108 = -8/27 < 0$. In the fourth expansion the combined function $\Phi(u) = u^3(u - 4h)$ is maximized on $[0, m]$ at m: $m - 4h = 1 - 7cd + 4d^2 \ge 17/288 > 0$. It follows that

$$\mathbb{E}[pZ^4] \le R_0 := m^3(1 - 7cd + 4d^2) + (7 - 10d)(d - 3c^2d^2 + d^3) \le 1 - d/9. \tag{A.13}$$

The divided coefficients of $R_0 - 1$ are bounded by $-5/2, 37/2, -49, 52, -22$. The resulting polynomial is at most $-5/2 + (37/2)d - 36d^2 \le -71/576$ for $d \le 1/4$. For $d = 1/4 + h$, $h \le 1/12$, its exact expansion is $-27/128 + 19h/8 - 73h^2/4 + 30h^3 - 22h^4$; using $30h^3 \le 5h^2/2$ bounds it by $-979/8064 < -1/9$.

An enlarged short tail. The same comparison holds if the row tail is $(13/12)\sqrt{d}$ and $d \le 1/10$. At $p = 3/2$ the modified derivative and remainder are

$$D_\lambda = -1/4 - 1889x^2/576 - 247x^3/64 + 35739x^4/4096,$$
$$Q_\lambda = -5/8 + 39x/8 - 4205x^2/384 - 2223x^3/256 + 165699x^4/8192.$$

Here $|D_\lambda| \le 2/3$, the maximum is in $p \le 3/2$, and $H'' \le -14/5$. The cubic and quartic sum in Q_λ is nonpositive; the remaining quadratic and tangent gain leave $-5/8 + 4563/8410 + 5/63 = -6499/2119320 < 0$.

A.4 Five short-connector coefficient profiles

For each row below put $x = \sqrt{d}$, u equal to its mean bound, B equal to its row envelope, and $H(\xi) = \xi(B^2 + 2uB + 3u^2)$, with $q = 2 - \xi$. All rows have $H \le 7$. This follows from concavity, $H'(3/2) < 0$, and the tangent estimate $H \le H(3/2) + D^2/(2\kappa)$ when $H'' \le -\kappa$ below $3/2$. The displayed formulas define every coefficient in the table; bounds are on $D = -|D|$ and $Q = H(3/2) - 7$. Differentiation

Table 11: Scalar coefficient profiles for short connectors.

	(u, B)	range of x	$(\lvert D\rvert, \kappa)$	upper bound for Q
A	$(1 - d, (1 - d/3)q - d/3 + x)$	all	$(9/10, 8/3)$	$-7/40$
B	$(1 - 3d/2, (1 - d/2)q - d/2 + 13x/12)$	all	$(11/10, 5/2)$	$-53/200$
C	$(1 - 9d/8, (1 - 3d/10)q + x)$	$x \le 1/3$	$(21/40, 27/10)$	$-1/8$
		$x \ge 1/3$	$(1, 27/10)$	$-2/5$
D	$(1 - 5d/8, q + 4x/5 - 3d/8)$	all	$(1, 3)$	$-13/72$
E	$(1 - d, q + x - 2d/3)$	$x \le 1/3$	$(2/3, 3)$	$-13/64$
		$x \ge 1/3$	$(7/5, 3)$	$-1/2$

gives $H'' = 6b^2\xi - 4b(A + u)$ when $B = A - b\xi$. In each row $-D$ increases; its derivative divided by x is an endpoint-positive concave quadratic. On the upper intervals of C and E, Q' is convex with negative endpoints, giving the tabulated bounds; for A, D and the lower intervals of C and E, discard the nonpositive cubic/quartic sum and maximize the remaining quadratic. For B instead use $x \le 3/5$ to bound that sum by $(207/160)x^3$; then $Q_B \le -5/8 + 39x/8 - (9953/600)x^2 < -53/200$.

A.5 A positive cycle form

Let $r = Kg$, $m_j = \mathbb{E}g^j$ and $C(x, y, z) = (1 + K(x, y))(1 + K(y, z))(1 + K(z, x)) \ge 0$. For every real f the following exact form is nonnegative:

$$\mathbb{E}[(1 + 2r)f^2] + (\mathbb{E}f)^2 - \|Kf + gf\|_2^2 = \tfrac{1}{2}\mathbb{E}[C(x, y, z)(f(x) + f(y))^2] \ge 0. \tag{A.14}$$

To verify it, integrate C first in one or two coordinates:

$$\mathbb{E}_{y,z}C(x, y, z) = 1 - g(x)^2 + 2r(x),$$
$$\mathbb{E}_z C(x, y, z) = (1 + K(x, y))[1 + g(y) - g(x) + K^2(x, y)].$$

The symmetric part of the second kernel is $1 + K^2(x, y) + K(x, y)(g(y) - g(x))$. Thus

$$\mathbb{E}[Cf(x)f(y)] = (\mathbb{E}f)^2 - \|Kf\|_2^2 - 2\mathbb{E}[gfKf].$$

Combining this with the first marginal proves (A.14); in particular $\mathbb{E}C = 1 - 3d$. Taking $f = g + t$ gives, with $\Delta = d - \delta_2 - m_4$,

$$\Delta - 4tm_3 + 2(1-3d)t^2 \geq 0 \quad (t \in \mathbb{R}), \qquad \Delta \geq 0, \qquad 2m_3^2 \leq (1-3d)\Delta.$$

The last inequality also follows without division by Cauchy–Schwarz on the nonnegative measure $C\,d\mu^3$: its quadratic form has values $Q(g) = \Delta$, $Q(1) = 2(1-3d)$, and $Q(g,1) = -2m_3$. Repeated atoms are retained. Their diagonal contribution gives $1 - 3d \geq \sum_x \mu(x)^3 > 0$ and $\Delta \geq 2\sum_x \mu(x)^3 g(x)^2$, so $\Delta > 0$ when $d > 0$. The quantities δ_2, m_3, m_4 satisfy the joint budgets above.

A.6 Integer reference powers

Put $c = pq + 2Kg \geq 0$ and $\sigma_m(z) = \sum_{j=0}^{m-1} 2^{m-1-j} z^j$, with $\sigma_0 = 0$. For every integer $m \geq 0$,

$$A(p^m) \leq \frac{p^{m+1}}{m+1} + \frac{c}{2}\sigma_m(p), \qquad T(q^m) \leq \frac{q^{m+1}}{m+1} + \frac{c}{2}\sigma_m(q). \tag{A.15}$$

These follow from the convex transport proved below: take $\phi(t) = t^m$ and use $(2^m - p^m)/q = \sigma_m(p)$. The cases $m = 0, 1$ are equalities; in particular $f_2 = (p^2 + c)/2$ and $A(p^2) \leq p^3/3 + (p+2)c/2$. Multiplication by a specified nonnegative function and integration preserve these pointwise inequalities.

A.7 Convex reference transport

For every continuous convex $\phi : [0,2] \to \mathbb{R}$,

$$A[\phi(p)](x) \leq \int_0^{p(x)} \phi(t)\,dt + \frac{c(x)}{2q(x)}[\phi(2) - \phi(p(x))]. \tag{A.16}$$

Complete reversal gives the corresponding $T[\phi(q)]$ bound. Affine tests give equality.

First, if s is the score of any finite skew kernel bounded in absolute value by one and $P = 1 + s$, then

$$\mathbb{E}(P-a)_+ \leq (2-a)^2/4 \quad (0 \leq a \leq 2), \qquad \mathbb{E}\psi(P) \leq \tfrac{1}{2}\int_0^2 \psi(t)\,dt$$

for every continuous convex ψ. Indeed, for $S = \{P > a\}$ and $z = \mu(S)$, skewness gives $\int_S s\,d\mu \leq z(1-z)$, and hence $\mathbb{E}(P-a)_+ \leq z(2-a-z) \leq (2-a)^2/4$. Express a convex piecewise linear function as an affine function plus a nonnegative combination of $(t-a)_+$, then take uniform polygonal approximations. The constants and linear terms agree because $\mathbb{E}P = 1$. In particular, for a scalar $0 < P_0 \leq 2$,

$$\mathbb{E}\phi(P_0 P/2) \leq P_0^{-1}\int_0^{P_0} \phi(t)\,dt. \tag{A.17}$$

Write $B = T^2$. Its kernel is $B(x,y) = 1 - g(x) + g(y) + K^2(x,y)$; nonnegative two-step convolution gives $0 \leq B(x,y) \leq 2q(x)$. The cycle marginal just proved is

$$\mathbb{E}_y[A(x,y)B(x,y)] = c(x) \geq 0.$$

At a fixed x, set $p_x = p(x) > 0$. For this scalar comparison alone use $d\nu_x(y) = A(x,y)d\mu(y)/p_x$. Its score is $\gamma_x(y) = [g(y) - K^2(x,y)]/p_x$. Put $a_y = p_x(1+\gamma_x(y))/2$, $b_y = B(x,y)/2$. Then

$$0 \le a_y \le p_x, \quad 0 \le b_y \le q_x, \quad a_y + b_y = p(y), \quad \mathbb{E}_y[A(x,y)b_y] = c(x)/2.$$

For $0 \le a \le p$, $0 \le b \le q$, convexity gives

$$\phi(a+b) - \phi(a) \le \frac{b}{q}[\phi(a+q) - \phi(a)] \le \frac{b}{q}[\phi(2) - \phi(p)].$$

The second inequality is the monotonicity of a convex increment of fixed length, also valid for negative increments. Integrate against $A(x,y)d\mu(y)$ and apply (A.17) to ν_x. This proves (A.16); all target operators and adjoints still use the original μ.

For $\alpha \ge 1$, the tests t^α and $(2-t)^\alpha$ give

$$A(p^\alpha) \le \frac{p^{\alpha+1}}{\alpha+1} + \frac{c(2^\alpha - p^\alpha)}{2q},$$
$$A(q^\alpha) \le \frac{2^{\alpha+1} - q^{\alpha+1}}{\alpha+1} - \frac{c}{2}q^{\alpha-1}.$$

Their full reversals give the other two real-power inequalities. The test $(t-h)_+$, $0 \le h \le 2$, gives

$$A[(p-h)_+](x) \le \frac{(p(x)-h)_+^2}{2} + \frac{c(x)}{2q(x)}[(2-h) - (p(x)-h)_+].$$

The negative cycle term in the opposite-direction power bound is retained.

A.8 Constants and arithmetic certificates

Table 12: Fixed constants and their roles. The rational relaxations are sufficient bounds.

Constant	Origin	Use
$2/\pi$	Skew-kernel norm bound, Appendix D	Spectral estimates.
$\theta = 13/32$	Rational upper bound for $(2/\pi)^2$	Geometric control of δ_{j+1}/δ_j.
$\alpha = 19/32$	$1-\theta$	Centered resolvent and fourth-moment losses.
$\rho = 5/12$	Convenient relaxation of θ	Geometric tails in Appendix E.
$C_* = 101/50$	Relaxed spectral mask bound in the four-branch estimates	Pair and multiple-arm envelopes.
$C_* = 32768/16245$	Sharper finite-mask bound used in the five-branch estimates	Joint terminal and root-mask estimates.
$\eta_* = 1/500$	Common sufficient outlet loss, Appendix L	Two compatible outlets give loss $d/2000$.
$7/15$	Residue-class geometric deficit, (G.1)	Finite spectral polynomials.
625	Negative-part estimate, Section J.6	Signed odd-table auxiliaries.

Each certificate follows the same pattern: a nonnegative message envelope, a spectral polynomial, signed mean and variance bounds, and a scalar moment comparison. Rational endpoints keep the last stage finite while covering the whole continuous parameter domain. Appendix B works out one example in full.

The full local arguments are in the technical appendix.

A.9 Polynomial coefficient certificates

The polynomial comparisons below use rational coefficients on closed intervals or rectangles. The following elementary criterion makes their finite coefficient tables sufficient for the whole domain.

Lemma A.1 (Bernstein coefficient bounds)**.** *Let P be a polynomial on $[a,b]\times[c,d]$, and write*

$$Q(s,t)=P(a+(b-a)s,c+(d-c)t)=\sum_{p=0}^{n}\sum_{q=0}^{m}A_{pq}s^pt^q.$$

Define

$$B_{ij}=\sum_{p=0}^{i}\sum_{q=0}^{j}A_{pq}\frac{\binom{i}{p}}{\binom{n}{p}}\frac{\binom{j}{q}}{\binom{m}{q}},\qquad 0\le i\le n,\quad 0\le j\le m. \tag{A.18}$$

Then

$$\min_{i,j}B_{ij}\ \le P(x,y)\le\ \max_{i,j}B_{ij}\qquad((x,y)\in[a,b]\times[c,d]). \tag{A.19}$$

For a polynomial on one interval, take $m=0$ and omit the second variable. Degrees n,m may exceed the actual degrees. The coefficients and both bounds are rational when the data are rational.

Proof. The binomial theorem gives

$$s^p=\sum_{i=p}^{n}\frac{\binom{i}{p}}{\binom{n}{p}}\binom{n}{i}s^i(1-s)^{n-i}.$$

Apply this identity to both variables. It yields

$$Q(s,t)=\sum_{i=0}^{n}\sum_{j=0}^{m}B_{ij}\binom{n}{i}s^i(1-s)^{n-i}\binom{m}{j}t^j(1-t)^{m-j}.$$

For $s,t\in[0,1]$, the weights are nonnegative and sum to one. This proves (A.19), including the boundary. □

A.10 Two reference curvature bounds

Lemma A.2. *Let $0\le d\le 1/3$, $t=\sqrt{d}$, and let q,C,Z be nonnegative functions on a probability space, with $q\le 2$ and $\mu=\mathbb{E}Z$. Define the following scalar functions:*

$$\begin{aligned}
\mathcal{C}_w&=2-q+6t/5, & \mathcal{M}_w&=1-3t^2/2,\\
\mathcal{S}_w&=q+1+13t/12-(2+5q/8)t^2,\\
\mathcal{C}_s&=2-q+6t/5-19t^2/32, & \mathcal{M}_s&=1-11t^2/8,\\
\mathcal{S}_s&=q+1+t-(13/8+3q/8)t^2.
\end{aligned} \tag{A.20}$$

If, for one $i\in\{w,s\}$,

$$C\le\mathcal{C}_i,\qquad Z+\mu\le\mathcal{S}_i,\qquad \mu\le\mathcal{M}_i,$$

then

$$C\{(Z+\mu)^2+2\mu^2\}\le 9. \tag{A.21}$$

Proof. Enlarge the scalar domain to $0 \le t \le 3/5$, $0 \le q \le 2$. Both $\mathcal{M}_i$ are at least $23/50$, and both $\mathcal{S}_i$ are at least $93/100$: their coefficient of q is positive, and their remaining concave quadratics have these endpoint lower bounds. The two $\mathcal{C}_i$ are nonnegative as well, since $6/5 - 19t/32 > 0$. Thus the asserted coefficient is at most $\mathcal{C}_i(\mathcal{S}_i^2 + 2\mathcal{M}_i^2)$.

Put $D_i = 9 - \mathcal{C}_i(\mathcal{S}_i^2 + 2\mathcal{M}_i^2)$. Divide the enlarged domain at $t = 3/10$ and $q = 1$. On each of the four rectangles, map both variables affinely to $[0, 1]$ and apply (A.18). The complete coefficient lists have the following bounds:

Table 13: Complete coefficient bounds for the reference curvature comparisons.

Polynomial	Bidegree	Rectangles	Total coefficients	Lower bound
D_w	$(5, 3)$	4	96	$1/2$
D_s	$(6, 3)$	4	112	$1/2$

Lemma A.1 gives $D_i \ge 1/2$ throughout the enlarged domain, including its boundary. This proves (A.21) on the original domain $t \le 1/\sqrt{3} < 3/5$. □

B A worked terminal certificate

The residual pair $(1, 2)$ illustrates the pattern used throughout the technical catalogue: choose one nonnegative envelope, compute its propagated mean, covariance and variance, and finish with a scalar fourth-moment comparison. The one-step, two-step and longer-connector calculations below use the same orientation, and the displayed scalar inequalities are part of the proof.

B.1 The propagated residual pair $(1, 2)$

Throughout this proof $\tau = d - \delta_2$ and $D_j = \delta_j/(1 - \theta^2)$. Let $P = qf_2$. We prove $\mathbb{E}[p(T^L P)^4] \le 1$ for every $L \ge 1$. Use the single nonnegative input envelope $P \le F_0 = (1 + r/2)^2$ and $Y_L = T^L F_0 \ge T^L P$.

One step. Put $s = K^2 g$, $t = K^3 g$, $G = \mathbb{E}[gr^2]$, $R_3 = \mathbb{E}r^3$, $V_3 = \mathbb{E}[tr^2]$ and $b = -d + \delta_2/4$. Direct expansion gives

$$\begin{aligned} Y_1 &= q - d - s + \delta_2/4 - K(r^2)/4, \\ \mathbb{E}[p(Y_1 - 1)] &= -2d + 5\delta_2/4 + (G + R_3)/4, \\ \mathbb{E}(Y_1 - 1)^2 &= d - 2\delta_2 + \delta_3 + b^2 - (R_3 + V_3)/2 + \|K(r^2)\|^2/16 + bG/2. \end{aligned}$$

The row bound gives $-1 \le Y_1 - 1 \le -g + e$, $e = 13/102$. Writing $\xi = p(x) \in [0, 2]$, the fourth coefficient is bounded by

$$\xi^3 - (6 + 2e)\xi^2 + (11 + 6e + e^2)\xi < 7 \quad (0 \le \xi \le 2).$$

Indeed its derivative at $3/2$ is $-1/4 + e^2 < 0$; its maximum lies in $[0, 3/2]$, where its second derivative is at most -3. The quadratic tangent leaves margin $463/27744$ below 7. For $d \le 1/4$, $b \in [-1/4, 0]$ and $0 \le 1 + 7b/2 \le 1$. Using $|R_3| \le \sqrt{d}\,\delta_2$, $|V_3| \le \theta\sqrt{d}\,\delta_2$, $|G| \le d\sqrt{\delta_2}$, $\|K(r^2)\|^2 \le \delta_2^2$ gives

$$\mathbb{E}[pY_1^4] - 1 \le -d - 197\delta_2/32 + 251\sqrt{d}\,\delta_2/64 + d\sqrt{\delta_2} + 7d^2 + 7\delta_2^2/16.$$

With $z = \sqrt{d}$ and $v = \delta_2/d \in [z^2, \theta]$, the divided right side decreases in v: its derivative is at most $-3693/1024$. At $v = z^2$ it is $-1 + 59z^2/32 + 251z^3/64 + 7z^6/16 \le -43/1024$. Thus $\mathbb{E}[p(TP)^4] \le 1 - 43d/1024$ there. For $d \ge 1/4$ use the actual mean $\mathbb{E}P = 1 - 2d$ to get $TP \le 2\mathbb{E}P \le 1$.

All lengths at least three. Let $K' = -K$, $S'_L = \sum_{j=0}^{L} K'^j \mathbf{1}$. Finite Duhamel and positive convolution give

$$Y_L \le S'_L + K'^{L+2}\mathbf{1} - \gamma \bar{f}_{L-1} - \sum_{j=3}^{L} \omega_j \bar{f}_{L-j},$$

$$\gamma = d - \delta_2/2 \ge 0, \qquad \omega_j = \begin{cases} \beta_j, & j \text{ even}, \\ \beta_{j+2}, & j \text{ odd}. \end{cases}$$

For $L \ge 4$ retain the last two terms:

$$Z_L = S'_L + K'^{L+2}\mathbf{1} - \omega_L - \omega_{L-1} q \ge Y_L.$$

Here $a = \omega_L$, $b = \omega_{L-1}$, and the mean coefficients are $(\alpha, \eta) = (\beta_{L+3}, \beta_{L+2})$ for even L and $(\beta_{L+1}, \beta_{L+4})$ for odd L. All four lie in $[\tau, d]$. The variance polynomial is $P_L(K')g'$, and the row-tail polynomial is $K' P_{L-1}(K')g'$, so the common comparison applies. Its finite spectral polynomial is

$$P_m(z) = G_m(z) + z^{m+1} = \frac{1 - z^m(1 - z + z^2)}{1 - z}.$$

Its squared numerator is $1 + x^m(1 - x + x^2) - 2\,\mathrm{Re}[(i\sqrt{x})^m(1 - x - i\sqrt{x})]$. For residues $0, 1 \pmod 4$ it is at most 1; for residues $2, 3$, with $m \ge 4$, its deficit divided by x is at least $(1 - 2x^2 - x^5)/(1 + x) \ge 7/15$. For $m = 3$ separately the deficit numerator is $1 - 2x - x^2 + x^3 - x^4$, decreasing to $65295/1048576 > 0$. Thus $\|P_m(K')\| \le 1$ for every required index, proving both the variance and row-tail conditions. For $L = 3$ retain the γr term:

$$Y_3 \le \widehat{S} - \tau - \gamma q, \quad \widehat{S} = S'_3 + K'^5 \mathbf{1} - \gamma r.$$

Its four coefficients are $d(1 - \gamma), \tau, \gamma, \tau + \delta_3 \in [\tau, d]$. Moreover $\widehat{S} - 1 = [P_3(K') - \gamma K']g'$. Subtracting γz changes the imaginary coefficient from one to $1 - \gamma \in [0, 1]$, so its modulus does not exceed that of P_3 and $\|\widehat{S} - 1\|_2^2 \le d$. The row-tail factor is $1 - \gamma + z + z^3$, whose squared spectral modulus is $(1 - \gamma)^2 + x(1 - x)^2 \le (1 - \gamma)^2 + 4/27$. The original K' row bound multiplies its norm by $\sqrt{d}$. For $d \ge 1/10$, $\gamma \ge 51/640$ makes this at most 1; for $d \le 1/10$ it is at most $31/27 < (13/12)^2$. Thus the two comparison versions cover $L = 3$.

Two steps. The same envelope gives $Y_2 \le Z = aq + r + K^3 g \ge 0$, $a = 1 - d + \delta_2/2$. With $g' = -g$ its exact outputs are

$$\mu = 1 - 2d + 3\delta_2/2, \quad \mathbb{E}[g'(Z - \mu)] = ad, \quad \mathrm{Var}\, Z = a^2 d + \delta_2 - 2\delta_3 + \delta_4 - (d - \delta_2)^2.$$

They imply $\mu \le m = 1 - 89d/64$, covariance at least $h = (1 - d)d$, and variance at most

$$V = d[59/51 - 1993d/1024 + 2601d^2/4096],$$

using $v - (2 - \theta)v^2 \le 8/51$ for $v = \delta_2/d$. The row envelope is $Z \le (1 - 51d/64)q + \sqrt{d}$. The fourth coefficient is at most $7 - 7d$. Put $x = \sqrt{d}$, $A = 1 - 51d/64$ and define

$$H_x(\xi) = \xi\{[A(2 - \xi) + x]^2 + 2m[A(2 - \xi) + x] + 3m^2\}.$$

Its derivative $D = \partial_\xi H_x(3/2)$ and comparison remainder $Q = H_x(3/2) + 7x^2 - 7$ are

$$D = -1/4 - 125x^2/128 - 19x^3/16 + 45735x^4/16384,$$
$$Q = -5/8 + 9x/2 - 2021x^2/256 - 687x^3/128 + 347427x^4/32768.$$

The maximum is in $\xi \le 3/2$; curvature is at most -2 there and at most -3 for $x \le 1/3$. On $x \le 1/3$, $Q'' < -12$, $Q(1/4) < -9/256$, $Q'(1/4) < 7/32$ and $|D(1/3)| < 3/8$ leave remainder $-239/24576$. On $x \ge 1/3$, $Q' < 0$, $Q(1/3) + 1/16 = -20061/2654208 < 0$ and $|D| < 1/2$ suffice. The fourth expansion is therefore bounded by $\max\{G, \Phi(m) + G\}$, $G = (7 - 7d)V$, where here $\Phi(u) = u^3(u - 4h)$ with $h = (1 - d)d$. For the first branch, $G \le 7d(1 - d)(59/51 - 7105d/4096) < 31/32$: this cubic has second derivative less than -16 and at $d = 1/4$ has value less than $77/80$ and derivative in $(0, 7/24)$. For the second branch the divided coefficients of $\Phi(m) + G - 1$ are bounded by $-7/5, 11, -32, 34, -10$. The corresponding polynomial is at most $-53/470$ on $d \le 1/4$. At $d = 1/4 + h$ it is $-101/640 + 3h/4 - 41h^2/4 + 24h^3 - 10h^4 \le -991/7040$. Combining both branches yields $\mathbb{E}[p(T^2P)^4] \le 1 - 3d/32$. This proves the bound for every positive integer connector length.

C Trees with at most three branch vertices

We first treat up to three branch vertices using a joint moment estimate for endpoint messages at a common label.

Theorem C.1. *For every nonempty finite tree U with $b(U) \le 3$, there is an orientation $D = D(U)$ of all its original edges such that (1.1) holds for every nonempty finite loopless tournament.*

Theorem C.2. *The orientation in Theorem C.1 may be chosen so that $Z_D(K, \mu) \le 1$ for every finite probability space and every real skew kernel bounded in absolute value by one.*

C.1 Root moments and connector estimates

Proposition C.3. *The same positive canonical even message F satisfies*

$$\mathbb{E}F^3 \le 1, \qquad \mathbb{E}[pF^3] \le 1. \tag{C.1}$$

A positive canonical odd message $f_a R$ satisfies

$$\mathbb{E}[f_a R^3] \le 1. \tag{C.2}$$

All equal-pair factors are included. The complete reversals hold as well.

Section F.1 proves these statements. The anchored estimate (C.2) uses exponent one on the leading path factor in $f_a R$.

Proposition C.4. *For all positive indices in the stated ranges,*

$$\left\|\sqrt{f_n}A^*\right\|_{2\to 2} \le 1 \qquad (n \ge 2), \tag{C.3}$$
$$\left\|\sqrt{f_n}A^*\sqrt{q}\right\|_{2\to 2} \le 1 \qquad (n \ge 1), \tag{C.4}$$
$$\left\|\sqrt{f_i}(A^*)^j\sqrt{\bar{f}_a}\right\|_{2\to 2} \le 1 \qquad (i, a, j \ge 1). \tag{C.5}$$

The same bounds hold after replacing K by $-K$.

The proof, including all short indices, is in Appendix E. Multiplication operators occur in the displayed order.

Lemma C.5. *If a real function z satisfies $\mathbb{E}z^2 \le 1$ and $\mathbb{E}[pz^2] \le 1$, then*

$$\mathbb{E}[f_i((A^*)^j z)^2] \le 1 \qquad (i, j \ge 1). \tag{C.6}$$

Proof. Row Cauchy–Schwarz and positivity give

$$\mathbb{E}[p(A^*z)^2] \le \mathbb{E}[A(pq)z^2] \le \mathbb{E}[pz^2].$$

This iterates and proves the case $i = 1$. For $i \ge 2, j = 1$, use (C.3). Put

$$X_i = \sqrt{f_i}A^*\sqrt{q}, \qquad Y_t = q^{-1/2}(A^*)^t p^{-1/2} \quad (t \ge 1).$$

Row Cauchy–Schwarz proves $\|Y_1\| \le 1$; weighted square propagation proves $\left\|q^{-1/2}A^*\sqrt{q}\right\| \le 1$. Thus $\|Y_t\| \le 1$ for $t \ge 1$. For $j \ge 2$,

$$\sqrt{f_i}(A^*)^j z = X_i Y_{j-1}\sqrt{p}\, z.$$

Apply (C.4). Only positive indices of Y occur. □

Theorem C.6 (Joint endpoint moment)**.** *Let F, G be the complete messages of two arbitrary endpoint arm lists, each of size at least two. Use positive canonical directions at an even left endpoint and negative canonical directions at an odd left endpoint; reverse these choices on the right. For all $L, M \ge 1$,*

$$\mathbb{E}\left[\left(((A^*)^L F)(A^M G)\right)^{3/2}\right] \le 1. \tag{C.7}$$

This statement has no assumption on the middle arm list.

Proof. Write $W = (A^*)^L$, $V = A^M$. At an even left endpoint put $\sigma = L$, $X = F^{3/2}$, $B_{\mathrm{L}} = W$. At an odd left endpoint write $F = \bar{f}_a R_-$ and put

$$\sigma = L + a, \quad X = \sqrt{\bar{f}_a}\, R_-^{3/2}, \quad B_{\mathrm{L}} = W\sqrt{\bar{f}_a}.$$

The even X has both ordinary and p-weighted square norm at most one by (C.1); the odd X has ordinary square norm at most one by (C.2). On the right, use $\tau = M$, $Y = G^{3/2}$, $B_{\mathrm{R}} = V$ in the even case, and

$$\tau = M + b, \quad Y = \sqrt{f_b}\, R_+^{3/2}, \quad B_{\mathrm{R}} = V\sqrt{f_b}$$

when $G = f_b R_+$ is odd. The even Y has ordinary and q-weighted square norms at most one, and the odd Y has ordinary square norm at most one.

For every nonnegative kernel B and nonnegative u, z, row Hölder gives

$$[B(uz)]^{3/2} \le [Bu]^{1/2} B(uz^{3/2}).$$

Choose the reference $u = 1$ at even endpoints and the actual anchor at odd endpoints. Since $W\bar{f}_a = \bar{f}_{L+a}$ and $Vf_b = f_{M+b}$,

$$(WF)^{3/2} \le \sqrt{\bar{f}_\sigma} B_{\mathrm{L}} X, \qquad (VG)^{3/2} \le \sqrt{f_\tau} B_{\mathrm{R}} Y.$$

Multiplication and Cauchy–Schwarz, with the reference weights crossed, give

$$\mathbb{E}[((WF)(VG))^{3/2}] \le \{\mathbb{E}[f_\tau (B_{\mathrm{L}} X)^2]\}^{1/2} \{\mathbb{E}[\bar{f}_\sigma (B_{\mathrm{R}} Y)^2]\}^{1/2} \le 1.$$

For an even endpoint use Lemma C.5; for an odd endpoint use (C.5). Apply the reversed versions on the right. This includes all four endpoint parities and $L = M = 1$. □

C.2 The lower-branch assembly

C.2.1 The three-branch geometry

The case $b(U) \le 2$ is proved in Appendix D. Suppose $b(U) = 3$. The three branch vertices lie on one path: otherwise their connecting subtree has a distinct tripod junction, yielding a fourth branch vertex. Call them u, w, v in path order. Their connector lengths L, M are positive. Every component attached off the core is a path to a leaf. Hence the data are arbitrary positive arm lengths $\alpha_1, \dots, \alpha_r$ at u, $\beta_1, \dots, \beta_s$ at v, and $c_1, \dots, c_k$ at w, with $r, s \ge 2$ and $k \ge 1$. In particular,

$$e(U) = L + M + \sum_i \alpha_i + \sum_j \beta_j + \sum_\ell c_\ell. \tag{C.8}$$

Pair equal middle arms as in Section 2, and let t be the number of distinct residual lengths.

C.2.2 One residual middle arm

Suppose $t = 1$ and let n be its length. Give it message f_n, so the middle message is $E_0 f_n$. Direct both connectors toward w. At each endpoint use positive canonical directions for an even arm count and negative canonical directions for an odd count.

For an even endpoint message F, (C.1) and Hölder, on the probability measures μ and $p\mu$, imply $\mathbb{E}F^2 \le 1$ and $\mathbb{E}[pF^2] \le 1$. Lemma C.5 yields

$$\mathbb{E}[f_n((A^*)^j F)^2] \le 1 \qquad (n, j \ge 1). \tag{C.9}$$

For an odd endpoint $F = \bar{f}_a R_-$, Hölder with the finite measure $\bar{f}_a \mu$ gives

$$\mathbb{E}[\bar{f}_a R_-^2] \le (\mathbb{E}[\bar{f}_a R_-^3])^{2/3} (\mathbb{E}\bar{f}_a)^{1/3} \le 1.$$

Now apply (C.5) to $\sqrt{\bar{f}_a} R_-$ to obtain (C.9). Thus, for the two actual endpoint messages F, G,

$$Z_D = \mathbb{E}[E_0 f_n((A^*)^L F)((A^*)^M G)] \le \{\mathbb{E}[f_n((A^*)^L F)^2]\}^{1/2} \{\mathbb{E}[f_n((A^*)^M G)^2]\}^{1/2} \le 1.$$

This includes $n = 1, 2$ and every positive connector length.

C.3 The middle cubic construction

If $t = 0$, the middle message is $E_0 \le 1$. If $t \ge 2$ is even, use the positive canonical even direction and (C.1). For odd $t \ge 3$, write the distinct residual list as

$$a < b < c < b_2 < c_2 < \cdots < b_m < c_m, \qquad t = 2m + 1, \quad m \ge 1,$$

and put $P_{\text{tail}} = \prod_{i=2}^{m} h_{b_i} f_{c_i}$, with empty product one. The following table specifies a residual message C. The complete middle message is always $C_{\text{mid}} = E_0 C$.

Each cited subsection proves the ordinary cubic bound for the direction in its row. Restoring $E_0 \le 1$ gives

$$\mathbb{E}C_{\text{mid}}^3 \le 1. \tag{C.10}$$

To read the cases, if $a < 3$, then $a = 1$ or 2; next $b \ge 4$, $b = 3$, or $b = 2$. The first two alternatives are listed, while $b = 2$ forces $a = 1$ and is divided by whether the tail is empty. An empty tail is exhausted by $c = 3, 4, 5$ and $c \ge 6$.

Table 14: Middle-root directions for odd residual lists.

Residual condition	Residual message C	Proof
$a \geq 3$	$f_a h_b f_c P_{\text{tail}}$	Section F.2
$a \in \{1,2\},\ b = 4$	$f_a h_4 f_c P_{\text{tail}}$	Section F.2
$a \in \{1,2\},\ b \geq 5$	$f_a h_b f_c P_{\text{tail}}$	Section F.2
$a \in \{1,2\},\ b = 3,\ c = 4$	$f_a h_3 f_4 P_{\text{tail}}$	Section F.3
$a \in \{1,2\},\ b = 3,\ c \geq 5$	$f_a h_3 f_c P_{\text{tail}}$	Section F.3
$a = 1, b = 2, m \geq 2$	$p h_2 f_c P_{\text{tail}}$	Section F.4
$a = 1, b = 2, m = 1, c = 3$	$q f_2 h_3$	Section F.7
$a = 1, b = 2, m = 1, c = 4, 5$	$q f_2 h_c$	Section F.5
$a = 1, b = 2, m = 1, c \geq 6$	$q f_2 h_c$	Section F.6

C.4 Assembly and the tournament inequality

Use the cubic construction when $t = 0$ or $t \geq 2$. Direct the core $u \to w \to v$ consistently and take the endpoint directions in Theorem C.6. Exact integration over arms and connector interiors gives

$$Z_D = \mathbb{E}[C_{\text{mid}}((A^*)^L F)(A^M G)].$$

Hölder with conjugate exponents 3 and $3/2$, (C.10) and (C.7) imply $Z_D \leq 1$. Together with the singleton construction and the lower-branch cases this proves Theorem C.2.

The choices depend only on arm lengths, multiplicities and endpoint parities. Applying the kernel-to-tournament comparison in Section 12.1 to this orientation proves Theorem C.1.

D Path and ordered-pair estimates

This appendix proves the preliminary estimates used throughout. All functions and operators are on the space in Section 2. Put $u_j = \mathbb{E}K^j\mathbf{1}$ and $G_t(z) = \sum_{j=0}^{t-1} z^j$, with $G_0 = 0$.

D.1 Spectral bound and positive convolution

Lemma D.1. *We have*

$$\|K\|_{2\to 2} \leq \frac{2}{\pi}, \quad \|K\|_{2\to 2}^2 \leq \theta, \quad \|Kv\|_\infty \leq \|v\|_2, \quad u_{2j} = (-1)^j \delta_j, \quad u_{2j+1} = 0, \quad \delta_{j+1} \leq \theta\delta_j.$$

Moreover $0 \leq d \leq 1/3$ *and*

$$\|G_t(K)\| \leq 1 \quad (t = 0, 1 \text{ or } t \geq 3), \qquad \|G_2(K)\|^2 \leq 1 + \theta.$$

Proof. For a real unit vector v with $Kv \neq 0$, set $w = Kv/\|Kv\|_2$. Then v, w are orthogonal unit vectors and $|\langle v, Kw\rangle| = \|Kv\|_2$. Write $v(x) + iw(x) = r(x)e^{i\varphi(x)}$, so $\mathbb{E}r^2 = 2$. Antisymmetrization gives

$$|\langle v, Kw\rangle| \leq \frac{1}{2}\mathbb{E}_{x,y} r(x)r(y)|\sin(\varphi(y) - \varphi(x))|.$$

Use the absolutely convergent Fourier series

$$|\sin t| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{n\geq 1} \frac{\cos(2nt)}{4n^2 - 1}.$$

Each nonconstant average is the nonnegative square $|\mathbb{E} r e^{2in\varphi}|^2$ and has a negative coefficient. Hence the last upper bound is at most $(\mathbb{E} r)^2/\pi \le 2/\pi$. The case $Kv = 0$ is immediate. Also $4/\pi^2 < 13/32$; for instance Machin's identity and alternating arctangent truncation give $\pi > 157/50$, and $320000 < 13 \cdot 24649$.

The row bound follows from Cauchy–Schwarz, and the moment identities from skew-adjointness. Expanding the nonnegative cyclic product on three independent points yields

$$0 \le \mathbb{E}_{x,y,z}(1 + K(x,y))(1 + K(y,z))(1 + K(z,x)) = 1 - 3d.$$

The cubic expectation vanishes by a transposition. Thus $d \le 1/3$.

For a rotation frequency $i\lambda$ put $x = \lambda^2 \le \theta < 1/2$. If t is odd, $|G_t(i\lambda)|^2 = (1 + x^t)/(1 + x) \le 1$. If $4 \mid t$, the numerator is $(1 - x^{t/2})^2 \le 1$. If $t \equiv 2 \pmod 4$ and $t \ge 6$, use $2x^2 + x^5 \le 17/32 < 1$ to obtain $(1 + x^{t/2})^2 \le 1 + x$. For $t = 2$ the squared modulus is $1 + x$. Orthogonal rotation blocks prove the assertions. □

Define $\beta_1 = \beta_2 = 0$ and

$$\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1} \delta_j \quad (n \ge 3).$$

The identity $S_n = AS_{n-1} + \beta_n \mathbf{1}$ proves (2.4) by iteration. Alternating sums give $d - \delta_2 \le \beta_n \le d$, while $\mathbb{E} S_n = 1 - \beta_{n+1}$ for $n \ge 1$. Consequently $0 < f_n \le S_n$, $\mu_n \le 1$, and $f_n, h_n \le 2$. Also $\|S_n\|_2 \le 1$ for $n \ge 2$ by the geometric-sum bound, so $\mathbb{E} f_n^2 \le 1$ for $n \ge 2$.

The initial means are

$$\mu_0 = \mu_1 = 1, \quad \mu_2 = 1 - d, \quad \mu_3 = 1 - 2d, \quad \mu_4 = 1 - 3d + \delta_2, \quad \mu_5 = 1 - 4d + 2\delta_2 + d^2.$$

They imply $\mu_n \le 1 - 2d$ for $n = 3, 4, 5$. For $n \ge 6$, retain the four convolution terms with f_0, f_1, f_2, f_3 and use $\beta_j \ge d/2$. Then

$$\mu_n \le 1 - \frac{d}{2} - \frac{d}{2}(4 - 3d) \le 1 - 2d.$$

If $d = 0$, $K\mathbf{1} = 0$ and every rooted tree message equals one by leaf integration. Subsequent divisions by d are therefore confined to $d > 0$.

D.2 Weighted propagation and a weighted path cube

Put $N_{\pm}(F) = \mathbb{E}[(1 \pm g)F^2]$. Row Cauchy–Schwarz gives $(AF)^2 \le pA(F^2)$ and hence

$$N_-(AF) \le \mathbb{E}[A^*(pq)F^2] \le N_-(F). \tag{D.1}$$

Reversal gives $N_+(A^*F) \le N_+(F)$. The edge measure with density $1 + K(x,y)$ has total mass one and marginals p, q. Cauchy–Schwarz and (D.1) thus imply

$$|\langle F, A^L G \rangle| \le \sqrt{N_+(F) N_-(G)} \qquad (L \ge 1). \tag{D.2}$$

In particular $N_-(f_n) \le 1$ and $\mathbb{E}(f_n^2/p) \le 1$: the latter follows by dividing the row square inequality for Af_{n-1} by p and integrating.

Lemma D.2. *For every $n \geq 2$,*

$$\mathbb{E}[qf_n^3] \leq \mathbb{E}[qS_n^3] \leq 1 - d/8.$$

Proof. Write $x = g$, $y = S_n - 1 - g$. Skewness gives $\mathbb{E}y + \mathbb{E}(gy) = -d - u_{n+1}$ and

$$\|y\|_2^2 - u_{n+1} \leq \delta_2. \tag{D.3}$$

For even n, this follows from the geometric-sum bound. For $n = 2h + 1$, integrate against the spectral measure of g. The integrand of the left side of (D.3) minus the right side is

$$\frac{-t^2 + (-1)^h t^h (1-t) + t^{2h+1}}{1+t} \leq 0.$$

For odd h the middle term is nonpositive and $t^{2h+1} \leq t^2$. For even $h \geq 2$, division by t^2 bounds the numerator by $-1 + (1-t) + t = 0$.

The pointwise identity

$$-2x^3 - x^4 - 3x^2(1+x)y - 3x^2y^2 = x^2\left[1 - \frac{(1+x)^2}{4} - 3\left(y + \frac{1+x}{2}\right)^2\right] \leq x^2$$

then gives

$$\mathbb{E}[qS_n^3] \leq 1 - 2d + 3(\|y\|_2^2 - u_{n+1}) + \mathbb{E}[qy^3].$$

For $n = 2$ or $n \geq 4$, $\|y\|_\infty \leq \sqrt{d}$ and $\|y\|_2^2 \leq \delta_2$, so $\mathbb{E}[qy^3] \leq (6/5)\delta_2$. The resulting bound is $1 - (47/160)d \leq 1 - d/8$. For $n = 3$, $\|y\|_2^2 = \delta_2 + \delta_3$, $u_4 = \delta_2$, $\|y\|_\infty^2 \leq 3d/2$, and $\mathbb{E}[qy^3] \leq 9d/8$. Using $\delta_3 \leq d/4$ proves the remaining case. □

D.3 Ordered pairs and exponential moments

For $1 \leq a < b$, set $P_{ab} = h_a f_b$, $R_{ab} = S_b - S_a$. Then

$$P_{ab} \leq e^{R_{ab} - \omega_{ab}}, \qquad \omega_{ab} = \begin{cases} \beta_b, & a = 1, 2, \\ 2d - \delta_2, & a = 3, \\ 2d, & a \geq 4. \end{cases} \tag{D.4}$$

For $a = 1, 2$, use $f_a = S_a$, $h_a = 2 - S_a$ and $f_b \leq S_b - \beta_b$. For $b > a \geq 3$, subtraction of the positive convolutions gives

$$f_b - f_a = R_{ab} + \sum_{i=0}^{a-3} (\beta_{a-i} - \beta_{b-i}) f_i - \sum_{i=a-2}^{b-3} \beta_{b-i} f_i.$$

The positive contribution in the middle sum is at most $4\sum_{j \geq 1} \delta_{2j} \leq 4\delta_2/(1-\theta^2) \leq 2d$; for $a = 3$ it is at most δ_2. Combine this with $h_a + f_b = 2\mu_{a-1} + f_b - f_a$. Finally, for nonnegative h, f with $h + f \leq 2 + C$, AM–GM gives $hf \leq (1 + C/2)^2 \leq e^C$. This proves (D.4).

If a finite polynomial Q has coefficients zero or one, grouping powers modulo four gives

$$|Q(i\lambda)|^2 \leq \frac{1+\lambda^2}{(1-\lambda^4)^2} \leq \frac{32768}{16245} < C_*.$$

For ordered disjoint pairs $a_1 < b_1 < \cdots < a_t < b_t$, put

$$P = \prod_i P_{a_i b_i}, \quad R = \sum_i R_{a_i b_i}, \quad \Delta = \sum_i \omega_{a_i b_i}, \quad a = a_1.$$

Then $R = K^a Q(K) g$ for a zero-one polynomial and

$$P \le e^{R-\Delta}, \qquad \|R\|_\infty^2 \le C_* \theta^{a-1} d, \qquad \|R\|_2^2 \le C_* \theta^a d. \tag{D.5}$$

The same estimates are uniform in the number and size of the pairs. Also

$$\mathbb{E}[pR_{ab}] = u_{a+1} - u_{b+1}. \tag{D.6}$$

For disjoint pairs this implies $\mathbb{E}[pR] \le 0$ if $a = 1$, and $\mathbb{E}[pR] \le d$ if $a \ge 2$. Indeed all boundary indices are distinct and $\sum_{j \ge 2} \delta_j \le 13d/19$. Under the original measure, $\mathbb{E}R \le d/2$ for $a \ge 2$, and $\mathbb{E}R \le -d/2$ for $a = 1$, because $\sum_{j \ge 1} \delta_{2j} \le (416/855)d < d/2$.

For any probability measure and $|R| \le M$,

$$\mathbb{E}e^{sR} \le \exp\{s\mathbb{E}R + s^2 M^2/2\} \quad (s \ge 0). \tag{D.7}$$

To see this, the second derivative of $\log \mathbb{E}e^{sR}$ is the tilted variance, at most M^2; integrate twice. Thus

$$\Delta - \mathbb{E}R \ge (s/2)M^2 \quad \Longrightarrow \quad \mathbb{E}P^s \le 1. \tag{D.8}$$

One may use a scalar weighted probability measure in this criterion without changing any underlying operator or adjoint.

Lemma D.3. *For every disjoint ordered pair list,* $\mathbb{E}[pP^2] \le 1$. *If* $a_1 \ge 3$, *also* $\mathbb{E}[pP^6] \le 1$ *and* $\mathbb{E}P^6 \le 1$.

Proof. For one pair with $a \ge 3$, (D.6) gives $\mathbb{E}[pR_{ab}] \le \delta_2 + \delta_3$. Thus $\Delta - \mathbb{E}[pR] \ge (1047/1024)d$, while $M^2 \le (1+\theta)\theta^2 d = (7605/32768)d$ and $3M^2 \le (1047/1024)d$. For two or more pairs with $a_1 \ge 3$, $\Delta \ge (4-\theta)d$, $\mathbb{E}[pR] \le d$, and $M^2 \le C_* \theta^2 d$; these again satisfy (D.8) for $s = 6$. The ordinary sixth moment follows by the same argument: for one pair $\mathbb{E}R \le \delta_2$ and for several pairs $\mathbb{E}R \le d/2$, which give stronger deficit bounds. Hölder gives the corresponding second moments.

For the weighted second moment with shortest length two and one pair, the following upper and lower bounds are valid, respectively:

b	M^2 upper bound	$\Delta - \mathbb{E}[pR]$ lower bound
3	δ_2	$d + \delta_2$
4	$\delta_2 + \delta_3$	d
$b \ge 5$	δ_2	$d - \delta_2 - \delta_3$.

The last comparison uses $1 - 2\theta - \theta^2 = 23/1024 > 0$. With at least two pairs, the deficit is at least $(2-\theta)d$ and $M^2 \le C_* \theta d$.

For shortest length one and one pair, $b = 2, 3, b \ge 4$ give $(M^2, \Delta - \mathbb{E}[pR])$ bounded by (d, d), $(d + \delta_2, 2d + \delta_2)$ and $(d, 2d - \delta_2 - \delta_3)$. If there are exactly two pairs and the first is $(1, 2)$, the deficit is at least $(2071/1024)d > C_* d \ge M^2$. For at least three pairs beginning with $(1, 2)$ it is at least $(4-\theta)d$. For at least two pairs with first long arm at least three it is at least $(3-\theta)d$. These exhaust the cases and satisfy (D.8) with $s = 2$ under $p\mu$. □

D.4 Short-pair cubic estimates

Lemma D.4. *For $n \geq 2$ and $m \geq 3$,*

$$\mathbb{E}(qf_n)^3 \leq 1 - d/6, \qquad \mathbb{E}(h_2 f_m)^3 \leq 1 - d/10.$$

In particular,

$$\mathbb{E}(qf_2)^3 \leq 1 - \alpha_* d, \qquad \alpha_* = \frac{66751}{92160} > \frac{18}{25}. \tag{D.9}$$

Proof. If nonnegative u, v satisfy $u + v \leq 2m + R$, then $(uv)^3 \leq (m + R/2)^6$. For $m \geq 0$, $|R| \leq M$, expansion gives

$$\mathbb{E}(m + R/2)^6 \leq m^6 + 3m^5 \mathbb{E}R + \mathcal{C}(m, M)\mathbb{E}R^2,$$

$$\mathcal{C}(m, M) = \frac{15}{4}m^4 + \frac{5}{2}m^3 M + \frac{15}{16}m^2 M^2 + \frac{3}{16}mM^3 + \frac{1}{64}M^4.$$

We will use $\mathcal{C} \leq 16133/2880$ for $m \leq 1, M \leq 3/5, M^2 \leq 1/3$; $\mathcal{C} \leq 1579/256$ for $m \leq 1, M \leq 3/4, M^2 \leq 1/2$; and $\mathcal{C} < 197/40$ for $m \leq 15/16$ under the latter two bounds on M. These follow term by term from the displayed polynomial.

For qf_2, use $q + f_2 = 2 + Kg$, $\mathbb{E}Kg = -d$, $\mathbb{E}(Kg)^2 = \delta_2$, $M \leq \sqrt{d}$. The bound is $1 - 3d + (16133/2880)\delta_2 \leq 1 - \alpha_* d$. For qf_3, take $m = 1 - d/2$, $R = Kg + K^2 g$, so $\mathbb{E}R = -d$, $\mathbb{E}R^2 = \delta_2 + \delta_3 \leq 3d/5$, $M \leq 3/4$. The identity

$$(1 - t/2)^6 - 3t(1 - t/2)^5 = 1 - 6t + \frac{45}{4}t^2 - 10t^3 + \frac{75}{16}t^4 - \frac{9}{8}t^5 + \frac{7}{64}t^6$$

is at most $1 - 6t + 45t^2/4 - 9t^3$ on $[0, 1/5]$ and at most the analogous expression with $-8t^3$ on $[0, 1/3]$. Splitting at $d = 1/6$ gives deficits $3d/8$ and $331d/1800$, both at least $d/6$. For $n \geq 4$, set $\tau = (19/32)d$, $m = 1 - \tau/2$ and $R = S_n - 1 - g$. Positive convolution gives $q + f_n \leq 2m + R$, while $\mathbb{E}R \leq -\tau$, $\mathbb{E}R^2 \leq \delta_2$, $M \leq \sqrt{d}$. The preceding polynomial bounds yield a deficit coefficient at least

$$\frac{80275}{32768} - \frac{209729}{92160} = \frac{256711}{1474560} > \frac{1}{6}.$$

For $h_2 f_3$, use $h_2 + f_3 = 2 - d + K^2 g$, $M \leq 3/8$, $\mathbb{E}R = 0$, $\mathbb{E}R^2 = \delta_3$. Since $\mathcal{C} \leq 5$ and $(1 - d/2)^6 \leq 1 - 7d/4$, the deficit is $(947/1024)d$. For $h_2 f_4$, use

$$m = 1 - d, \quad R = K^2 g + K^3 g - dg, \quad \mathbb{E}R = \delta_2, \quad \mathbb{E}R^2 = \delta_3 + \delta_4 + 2d\delta_2 + d^3.$$

When $d \leq 1/6$, $M \leq 1/2$, $\mathbb{E}R^2 < 2d/5$, $\mathcal{C} < 16/3$, and the constant and mean terms are at most $1 - 73d/32$. The deficit is at least $71d/480$. When $d \geq 1/6$, $M < 4/5$, $\mathcal{C} < 7/2$, $\mathbb{E}R^2 < 5d/8$. The constant and mean terms are bounded by $(1 - d)^5(1 + 7d/32) \leq 1 - 5d/2$: this function is convex and its value at $1/3$ is $103/729 < 1/6$. The deficit is then at least $5d/16$.

For $h_2 f_n$, $n \geq 5$, positive convolution gives $h_2 + f_n \leq 2 - 2\tau + R$ with

$$R = S_n - S_2 - \tau g, \quad \mathbb{E}R \leq \delta_2, \quad \mathbb{E}R^2 \leq (\theta + (19/32)d)^2 d, \quad M \leq 23/40, \quad M^2 < 1/3.$$

For $d \leq 2/19$, the constant and mean terms are at most $1 - 915d/512$, and the variance term at most $80665d/65536$; their difference is $36455d/65536$. For $d \geq 2/19$, $m = 1 - \tau \leq 15/16$, $\mathcal{C} < 9/2$, and the variance term is at most $841d/512$. Convexity of $W(d) = (1 - (19/32)d)^5(1 + 5d/8)$ and $W(1/3) < 5/12$ give $W(d) \leq 1 - 7d/4$. The remaining deficit is $55d/512 > d/10$. All interval endpoints are included, and $d = 0$ gives equality. □

D.5 Every tree with at most two branch vertices

Every rooted spider has an orientation with $N_-(F) \le 1$. To prove this, first restore equal pairs only after estimating the residual distinct list $\ell_1 < \cdots < \ell_t$. For $t = 0$ or 1, use $F = 1$ or f_{ℓ_1} and (D.1). For positive even t, use the reversal of the canonical pair product and Lemma D.3. For odd $t \ge 3$ with $\ell_1 = 1$, put $F = pP$, with the remaining pairs positive canonical. Then

$$N_-(F) = \mathbb{E}[p(1-g^2)P^2] \le \mathbb{E}[pP^2] \le 1.$$

For odd $t \ge 3$ with $\ell_1 \ge 2$, put $F = f_{\ell_1} P(-K)$, where P pairs the remaining arms. Their shortest length is at least three, so

$$N_-(F) \le (\mathbb{E}[q f_{\ell_1}^3])^{2/3} (\mathbb{E}[qP(-K)^6])^{1/3} \le 1.$$

Use Lemmas D.2 and D.3. Complete reversal gives an N_+ certificate.

For two branch vertices choose the left endpoint with $N_+(F) \le 1$, the right with $N_-(G) \le 1$, and direct the entire connector left to right. Exact integration and (D.2) give $Z_D = \langle F, A^L G \rangle \le 1$. For one branch vertex, designate one actual arm as an inward path and use the remaining root certificate in the same inequality. A path uses $\mu_n \le 1$ and a single vertex has density one. This proves the lower-branch cases used in Section C.2.

E All positive-length connector estimates

We give the operator certificates needed for Proposition C.4. In this appendix put

$$Q = I - J, \quad C = QKQ, \quad r_0 = Cg = Kg + d\mathbf{1}, \quad s = \|r_0\|_2^2, \quad \alpha = 19/32, \quad \rho = 5/12.$$

Thus $r_0 = QKg$ is centered; the uncentered $r = Kg$ retains its meaning from Section 2. We have $s + d^2 \le \theta d$, $\|K\|^2 \le \theta < \rho < 1/2$ and $Kg = -d\mathbf{1} + r_0$.

E.1 Schur certificates and normalized propagation

If P_0 is a nonnegative symmetric kernel and $v > 0$ satisfies $P_0 v \le \lambda v$, then $\langle z, P_0 z \rangle \le \lambda \|z\|_2^2$: apply $2ab \le a^2 + b^2$ after inserting $v(x)/v(y)$ in the two terms. We use this only with positive semidefinite symmetric kernels.

Write $P = A^*A$, so $Pz = q\mathbb{E}z - \mathbb{E}(gz) - K^2 z$. Weighted propagation (D.1), its reversal and ordinary adjoints give

$$\|B_q\| \le 1, \quad B_q = q^{-1/2} A^* \sqrt{q}, \qquad \|Y\| \le 1, \quad Y = q^{-1/2} A^* p^{-1/2}.$$

For the last assertion, integrate $(A^*v)^2 \le qA^*(v^2)$ after setting $v = z/\sqrt{p}$. Consequently

$$Y_j = q^{-1/2} (A^*)^j p^{-1/2} = B_q^{j-1} Y, \qquad \|Y_j\| \le 1 \quad (j \ge 1). \tag{E.1}$$

The index range starts at one.

E.2 An ordinary one-step certificate

We first prove

$$\|f_n A^*\|^2 \le 1 - d/72 \qquad (n \ge 2). \tag{E.2}$$

For $n \ge 3$ put $H = S_n$, $a = H - 1$, $R = (I + K^2)^{-1}$,

$$e = -R(K^{n+1} + K^{n+2})\mathbf{1}, \quad u = 1 + e, \quad t = 1 - \mathbb{E}H,$$
$$v = \mathbb{E}(gH), \quad b = \mathbb{E}(He), \quad c_0 = \mathbb{E}(ge), \quad a_0 = \mathbb{E}|ae|.$$

Direct expansion, using $(I - K^2)a - g - Kg = (I + K^2)e$, yields

$$M_H P M_H u - u = -a^2 + ae + H\{q(b - t) - (v + c_0) + A^* K(ae)\}. \tag{E.3}$$

The geometric-sum bounds give $\|a\|_2 \le \sqrt{d}$, $|e| \le 6\sqrt{d}/7 \le 18/35$, and hence $17/35 \le u \le 53/35$. The exact formula

$$b = -\mathbb{E}RK^{n+1}\mathbf{1} + (-1)^{n+1}\mathbb{E}RK^{2n+2}\mathbf{1}$$

and $\|e\|_2^2/d \le \rho^n(1 + \rho)/(1 - \rho)^2$ give

n	b upper bound	a_0 upper bound	$a_0 + b$ upper bound
$n \equiv 3 \pmod 4$	0	$4d/7$	$4d/7$
$n \ge 4$ even	$3d/56$	$5d/14$	$23d/56$
$n \equiv 1 \pmod 4,\ n \ge 5$	$9d/28$	$5d/21$	$47d/84$.

All last entries are at most $7d/12 \le t$. For $n = 2m - 1$ or $2m$ the full spectral measure ν at $\mathbf{1}$ gives

$$v + c_0 = \int \left[\frac{x}{1 + x} + (-1)^{m+1}\frac{2x^{m+1}}{1 - x^2}\right] d\nu(x) \ge 2d/7.$$

In the negative branch the weakest case is $m = 2$, where the integrand is $x(1 - 2x)/(1 - x) \ge 2x/7$. Since $A^*K(ae) \le qa_0$, (E.3) is at most $-a^2 + a(e - 2d/7) - 2d/7$. The bound $|e - 2d/7| \le 36\sqrt{d}/35$ and a completed square give a residual at most $-26d/1225$. Schur's argument gives $\|HA^*\|^2 \le 1 - 26d/1855$. Since $0 < f_n/H \le 1$, this proves (E.2) for $n \ge 3$.

For $n = 2$, let $F = f_2 = 1 - d + g + r_0$, $a = F - 1$, $R_C = (I + C^2)^{-1}$ and $e = -C^2 R_C(g + r_0)$, $u = 1 + e$. With the spectral measure η of $-C^2$ at g, put $\kappa = \int x^2/(1 - x)\, d\eta$. Then

$$\langle r_0, e\rangle = \kappa, \quad \langle g, e\rangle = s + \kappa, \quad \kappa \le 13s/19, \quad \|e\|_2^2 \le 585s/361.$$

The row bound gives $|e| \le 2(1 + \sqrt{d})\sqrt{s} \le 13/20 < 312/475$, so $163/475 < u < 787/475$. Exact expansion yields

$$M_F P M_F u - u = -a^2 + ae - F[2d + s - d(g + a) + \kappa g] + FA^* K(ae).$$

Here $\|a\|_2^2 \le 45d/32$, $\mathbb{E}|ae| \le 585d/608$, $\kappa \le 169d/608$, and $\mathbb{E}|ae| + |d - \mathbb{E}|ae| - \kappa| \le 5d/4$. The residual is therefore at most

$$-(1 - d)a^2 + a(e + d/4 - s) - 3d/4 - s.$$

Completing the square leaves, after division by d, the positive quantity

$$3(1-d) - 8(\sqrt{d}+d)s/d - (1+\sqrt{d})\sqrt{s} - d/16.$$

Split $8(\sqrt{d}+d)s/d$ into $8s/\sqrt{d}$ and $8s$. These and the next two losses are at most $8/9$, $25/72$, $1/3$ and $1/48$, respectively, whereas $3(1-d) \geq 2$. Their sum is $229/144$, leaving $59/144$. Thus the residual is at most $-59d/576$. The same u gives $\|FA^*\|^2 \leq 1-(28025/453312)d$, which implies (E.2). Reversal and adjoints preserve the displayed operator order.

E.3 Half-step certificates for lengths two to four

We prove $\left\|\sqrt{H}A^*\right\| \leq 1$ by finding $u > 0$ with $M_H P u \leq u$ and applying Schur to $\sqrt{H}P\sqrt{H}$ with vector $u/\sqrt{H}$. For $H = f_2$, $h_2 = A^*p = 2 - f_2$ and $f_2 h_2 \leq 1$; hence $AM_{f_2}A^*p = A(f_2 h_2) \leq p$, which proves the length-two case.

For the other short lengths, set

$$a_0 = \theta - d, \quad z = \alpha + d, \quad Z_3 = R_C r_0, \quad \beta = \langle r_0, Z_3 \rangle, \quad L = \|Z_3\|_2^2.$$

The Schur complement of $\theta I + K^2$ gives

$$-C^2 + g \otimes g + (r_0 \otimes r_0)/a_0 \leq \theta I.$$

Since $g \perp Z_3$, it follows that

$$\beta \geq \alpha L + \beta^2/a_0, \quad \beta^2 \leq sL, \quad \beta \leq da_0/z, \quad L \leq da_0/z^2, \quad \beta/d \leq 13/19. \tag{E.4}$$

If $\beta = 0$, then $r_0 = Z_3 = 0$ and no division by β is made.

For length three and $0 < d \leq 1/5$, put $c = 1-d$, $G = f_3$, $x = Kf_2 = G - c$, $e = C^2 R_C g = KZ_3$. Then $\mathbb{E}e = 0$, $\langle g, e \rangle = -\beta$, $|e| \leq 13/38$ and

$$P(1+e) = 2 - G + e - dp + \beta q.$$

Cauchy–Schwarz gives $L \geq \beta + \beta^2/d$. For $\beta > 0$ define

$$\lambda = dL/\beta - d - \beta, \quad u_2 = dL/\beta, \quad b = u_2/c, \quad u = 1 + e + \lambda, \quad V = (b-\lambda)g + br_0,$$

$$\mathcal{A} = (1+b)x^2 + cd(1+b) - \beta, \qquad \Psi = (x-d)Z_3 + (c+x)V.$$

Then $\lambda \geq 0$, $u > 0$, and the exact row residual is

$$-(M_G P u - u) = \mathcal{A} - K\Psi, \qquad \|\Psi\|_2^2 = Lx^2/c^2 + E'(c+x)^2,$$

where

$$c^2 E' = d^2 u_2 + s u_2^2 - d(1-2d)\lambda u_2 + dc^2\lambda^2.$$

Since $u_2 \leq d/\alpha$ and $\lambda \leq \theta d/\alpha$, the last two terms together are nonpositive; thus $E' \leq d^3/(\alpha^2 c) < 10d^3/(3c)$. Set

$$j = 1 - ca_0/z, \qquad \eta = a_0/(cz^2) + 30d^2.$$

Using $(c+x)^2 \leq 9c^2/8 + 9x^2$ gives $\mathcal{A} \geq x^2/c + dj$ and $\|\Psi\|_2^2 \leq (d\eta/c)x^2 + (15/4)cd^3$. Their squared comparison reduces to

$$\eta(4j - \eta) - 15dc > 0. \tag{E.5}$$

Here is an exact certificate over the entire interval. Put $t = d/(1-d) \in [0, 1/4]$,

$$W = 19+51t, \quad J_0 = 6+89t+51t^2, \quad U = 32(13-19t)(1+t)^4+30t^2W^2, \quad D_0 = 4J_0(1+t)W.$$

The left side of (E.5) is $P_{10}/[(1+t)^4W^4]$, where $P_{10} = U(D_0-U)-15t(1+t)^2W^4$. Its coefficients from degree zero upward are

$$(16640, 1160833, -8704266, -35377737, 232713992, 819741071,$$
$$-950959922, -5593930311, -4938713932, 86107392, -369664).$$

If these coefficients are p_i, choose the integers ℓ_i in the next table and set $r_i = p_i - 4^i\ell_i$, $B_i = \sum_{j\le i}\binom{10-j}{i-j}\ell_j$:

i	ℓ_i	r_i	B_i
0	16640	0	16640
1	290208	1	456608
2	−544017	6	2816655
3	−552778	55	7539374
4	909039	8	9678989
5	800528	399	4940960
6	−232168	206	38245
7	−341427	9657	1478051
8	−75359	13492	3503691
9	328	124160	1910221
10	−1	678912	270993

Every $r_i \ge 0$ and $B_i \ge 180\binom{10}{i}$. The exact Bernstein transformation $P_{10}(t) = \sum_i b_i(4t)^i(1-4t)^{10-i}$, with $b_i = \sum_{j\le i}\binom{10-j}{i-j}p_j/4^j \ge B_i$, proves $P_{10} \ge 180$. Hence $\mathcal{A} > \|\Psi\|_2$ and the residual is nonnegative. If $\beta = 0$, $u = 1$ has residual $-(x-d)^2 - Gdp \le 0$. For $d \ge 1/5$, $\mu_4 \le 1-(83/32)d \le 77/160$ and $Pf_3 = 2\mu_4 - f_5 \le 77/80$; the Schur vector is the actual f_3. This proves the full length-three case.

For length four and $0 < d \le 1/6$, put

$$c = 1-2d, \quad c_1 = 1-d, \quad Z = (1+C)Z_3, \quad e = CZ,$$
$$L_4 = \|Z\|_2^2 = 2L-\beta, \quad H = f_4, \quad x = Kf_3 = H-c.$$

Then $\langle g, Z\rangle = -\beta$, $\langle r_0, Z\rangle = \beta$, $e = KZ-\beta$ and

$$P(1+e) = 2-H+e-[2(d-\beta)+(2d+\beta)g+dKg].$$

For $\beta > 0$, set $T_0 = r_0 + C^2g$, $D_1 = (2-d)\beta - s \ge c_1\beta$ and

$$b = \frac{2dL_4/c + d\beta + \beta^2}{D_1}, \quad \lambda = cb-2d-\beta, \quad t_0 = d(1+b)+\beta,$$
$$V = t_0g + bT_0, \quad \Psi = (x-2d)Z + (c+x)V.$$

$$\mathcal{A} = (1+b)x^2 + 2d - 2c_1\beta + 2d\lambda.$$

The exact residual is $\mathcal{A} - K\Psi$. The identity

$$\lambda D_1 = 4dL - 5d\beta - (1+d)\beta^2 + (2d+\beta)s \ge (5/57)d\beta$$

uses $L+s \ge 2\beta$ and $L \ge \beta$. Thus $b, t_0, \lambda > 0$, and $u = 1+e+\lambda \ge 1-\sqrt{195}/19-13/57 > 2/57$. Since $\langle Z, V\rangle = 2dL_4/c$,

$$\|\Psi\|_2^2 = L_4 x^2/c^2 + E'(c+x)^2, \qquad E' = \|V\|_2^2 - 4d^2L_4/c^2 \ge 0.$$

Projection onto $Z^\perp$ gives $\langle g_\perp, (T_0)_\perp\rangle = 2(\beta^2 - Ls)/L_4 \le 0$ and $\|(T_0)_\perp\|_2^2 \le \theta^2 L_4/4$. Substitution of $\alpha \le s/\beta \le 1$ and $L/\beta \le (1+\alpha-s/\beta)/\alpha$ yields

$$cb/d \le 11712/2261 < 21/4, \quad ct_0/d \le 389/152 < 13/5,$$
$$c^2E'/d^3 < 169/25 + 55125/27648 < 9.$$

Also $L_4 \le da_0(2-z)/z^2$. Put $j = 1-c_1a_0/z$ and $\eta = [a_0(2-z)/z^2 + 45d^2]/c$. Then $\mathcal{A} \ge x^2/c + 2dj$ and $\|\Psi\|_2^2 \le (d\eta/c)x^2 + 45d^3/4$. The squared comparison is $\eta(8j-\eta) - 45d > 0$.

Put $t = d/(1-2d) \in [0, 1/4]$, $W = 19+70t$, $J_0 = 6+101t+146t^2$, $V_0 = 585+484t-348t^2$, $U = (1+2t)^2V_0 + 45t^2W^2$, $D_0 = 8J_0W$. The comparison equals $P_8/[(1+2t)^2W^4]$, with $P_8 = U(D_0-U) - 45t(1+2t)W^4$. Its coefficients are

$$(191295, 4353483, -12419292, -147707432, 798676159,$$
$$5703292456, 1626691752, -34778574624, -48008315664).$$

The same Bernstein transformation is certified by

i	ℓ_i	r_i	B_i
0	191295	0	191295
1	1088370	3	2618730
2	−776206	4	12198644
3	−2307929	24	26603125
4	3119828	191	31420693
5	5569621	552	28250993
6	397141	2216	29314622
7	−2122716	4320	20811810
8	−732549	15600	4426855

Here $B_i \ge 100000\binom{8}{i}$, so $P_8 \ge 100000$. If $\beta = 0$, then $r_0 = C^2 g = 0$ and $H = cp - dc_1 > 0$ implies $2dp - d^2 > d^2/c > 0$; the residual at $u = 1$ is $-(x-2d)^2 - H(2dp-d^2) \le 0$. For $d \ge 1/6$, the exact mean gives $\mu_5 \le 1 - (51/16)d + d^2 \le 143/288$ and $Pf_4 = 2\mu_5 - f_6 \le 143/144$. This completes length four.

E.4 Half-step certificates for every length at least five

For $n \ge 5$, define

$$D_0 = K^2 \sum_{j=0}^{n-3} K^j \mathbf{1}, \quad Y = (I+K^2)^{-1} D_0, \quad e_S = KY,$$

$$b = \mathbb{E} e_S, \quad c_0 = -\mathbb{E}(g e_S), \quad a_j = \int \frac{x^j}{1-x}\, d\nu(x).$$

Alternating sums give

$$b=\sum_{j=2}^{\lfloor n/2\rfloor}(-1)^j a_j,\quad c_0=\sum_{j=2}^{\lfloor (n+1)/2\rfloor}(-1)^j a_j,\quad 0\le b\le 13d/19,\quad 0\le c_0\le \delta_2+a_4,$$

$$b+c_0\le a_2+\delta_2,\qquad c_0\le 10101d/19456<19d/32.$$

The geometric sum satisfies $x(1+x^3)^2\le(1-x)^2(1+x)$ on $[0,5/12]$; the difference is decreasing and at $5/12$ equals $105043/35831808>0$. Thus $\|Y\|_2^2\le d$ and $|e_S|\le\sqrt{d}$.

For $n=5$, $-\mathbb{E}Y=d$, $b=a_2$, $c_0=\delta_2$, and $\|Y\|_2^2\le\kappa d$ with $\kappa=10101/11552<7/8$. Take $u=1+e_S-d$, so $0<u\le 5/4$. With $\delta=f_5-1$ and $C_0=3d-b-2\delta_2$, exact convolution gives

$$M_{f_5}Pu-u=-\delta^2+\delta(e_S-d)-f_5[C_0+(d+b)g].$$

Center Y by $Y_c=Y+d$, write $v^2=\|Y_c\|_2^2\le\kappa d-d^2$, and use the row constraint $e_S=-dg+\zeta$, $|\zeta|\le v\sqrt{1-g^2}$. With $T=(2d+b)\delta+d+b$, the residual is at most

$$-\delta^2-d\delta-C_0(1+\delta)-Tg+|\delta|v\sqrt{1-g^2}.$$

If $T\ge 0$, positivity gives $f_5\le 2p$, hence $g\ge(\delta-1)/2$. Put $C_0'=[C_0-(d+b)/2]/d\ge 201/304>13/20$ and $a=1+b/(2d)\le 51/38<27/20$. The completed-square margin

$$\Phi=4C_0'(1+ad)-[\sqrt{v^2/d}+\sqrt{d}(C_0'+a)]^2$$

is increasing in C_0'. Lower C_0' to $13/20$ before raising a to $27/20$; the resulting bound is $69/40+51d/100-4\sqrt{d(7/8-d)}\ge 39/200$. Since $1+ad\le 29/20$, the residual is at most $-39d/1160$. If $T<0$, set $x=-\delta\in(0,1)$. The row constraint and $|T|\le dx$ bound the residual by $-x^2+x[(5/2)d+\sqrt{\kappa d}]-3d/2\le -15d/1024$. Thus $\left\|\sqrt{f_5}A^*\right\|^2\le 1-3d/256$.

For $n\ge 6$ and $d\le 1/6$, put $m_0=-\mathbb{E}Y$. Write the frequency sum as $\sum_{j=0}^{n-3}(i\sqrt{x})^j=U+i\sqrt{x}V$. There are two types:

type	m_0/d	correlations
I : n even or $n\equiv 3 \pmod 4$	$[1,777/608]\subset[1,41/32]$	$b\le c_0\le\theta m_0$
II : $n\equiv 1 \pmod 4$, $n\ge 9$	$[1,1024/855]\subset[1,6/5]$	$c_0\le\theta m_0$, $b\le 9c_0/8$.

For type I, $U\ge V$ and $1-x\le U\le 1-x+x^2$. For type II, $(1-x)(1+x^2)\le U\le 1/(1+x)$ and $1\le V/U\le 9/8$; the latter follows from $9x^3-x^2+x-1\le 0$ on $[0,5/12]$. The ratio of the density of $\|Y\|_2^2$ to that of m_0 is at most $3/4$: in type I it is bounded by $x(1+x^3)/(1-x)$, and in type II by $x/(1-x^2)+(9/8)x^2(1-x+x^2)/(1-x)$. Consequently $v^2=\|Y+m_0\|_2^2\le 3m_0/4-m_0^2$, and $u=1+e_S-m_0$ satisfies $71/192\le u\le 19/16$.

Set $\tau=19d/32$, $\delta=f_n-1$ and

$$W_0=(2\tau-m_0)g+\tau r_0,\quad w_0=\|W_0\|_2,\quad C_4=(4-2d)\tau-b-c_0,\quad C_1=C_4-w_0,$$

$$B=\tau(3-d)+b-m_0,\quad A_0=m_0+B,\quad T=A_0\delta+B.$$

Retaining f_0, f_1, f_2, f_3 in the positive convolution and using the same row constraint bounds the residual by

$$-\delta^2 - m_0\delta - C_1(1+\delta) - Tg + |\delta| v\sqrt{1-g^2}. \tag{E.6}$$

For $d > 0$, put $m = m_0/d$, $\eta = 19/32$, and

$$F(m,d) = \sqrt{d[(m-2\eta)^2 + \eta^2(\theta - d)]}.$$

Then $w_0/d \le F(m,d)$. For type I the correlated bounds are

$$C_1/d \ge (4-2d)\eta - 2\theta m - F(m,d),$$
$$(C_1 - B/2)/d \ge \eta(5/2 - 3d/2) + (1/2 - 5\theta/2)m - F(m,d).$$

For type II they are

$$C_1/d \ge (4-2d)\eta - (17\theta/8)m - F(m,d),$$
$$(C_1 - B/2)/d \ge \eta(5/2 - 3d/2) + (1/2 - 43\theta/16)m - F(m,d).$$

The derivative of F in m has magnitude at most $\sqrt{d}$. All four linear slopes are negative with magnitude at least $33/64$, whereas $\sqrt{d} \le \sqrt{1/6} < 33/64$. Thus each complete lower bound decreases in m. Evaluate these bounds at $m = 41/32$ for type I and $m = 6/5$ for type II. At these respective endpoints, $F(m,d) < 1/8$ throughout $0 \le d \le 1/6$; indeed $d[(m-19/16)^2 + (361/1024)(\theta - d)]$ is increasing on this interval and is less than $1/64$ at $d = 1/6$. Using $A_0/d = \eta(3-d) + b/d$ gives the following table:

	C_1/d lower	$(C_1 - B/2)/d$ lower	$A_0/(2d)$ upper
I	$1553/1536$	$1127/2048 > 35/64$	$2357/2048 < 37/32$
II	$1951/1920$	$641/1280 > 1/2$	$1491/1280 < 75/64$.

When $T \ge 0$, again use $g \ge (\delta - 1)/2$. The same Φ calculation has, in type I, $v^2/d \le 123/128 - (1681/1024)d$ and $\Phi \ge 57/1024$. In type II, $v^2/d \le 9/10 - 36d/25$; splitting at $d = 1/8$ gives margins $31/320$ and $197/1920$. Since $1 + ad \le 153/128$, the residual is at most $-19d/1632$. When $T < 0$, $B > 0$, $x = -\delta \in (0,1)$ and $|T| \le m_0 x$. The joint row bound is $x\|Y\|_2 \le x\sqrt{d}$, and (E.6) is at most $-x^2 + x[(73/32)d + \sqrt{d}] - d \le -28823d/589824$. This is stronger than $-19d/1632$. Schur's argument proves $\left\|\sqrt{f_n}A^*\right\|^2 \le 1 - d/102$ in the low range.

For $d \ge 1/6$, retain five final convolution terms. For every $j \ge 7$,

$$\mu_j \le 1 - (d - \delta_2)(6 - 6d + \delta_2) \le 1 - \alpha d[6 - (179/32)d] < 1/2.$$

The last loss is increasing in $d \in [1/6, 1/3]$ and at the left endpoint is $18487/36864 > 1/2$. Thus $Pf_n = 2\mu_{n+1} - f_{n+2} < 1$ and the actual vector $u = f_n$ is a Schur certificate. This completes (C.3) for every $n \ge 2$.

E.5 Mixed half-step certificates

For a positive H put $X_H = \sqrt{H}A^*\sqrt{q}$ and $T = M_q A M_H A^*$. The operator T is similar to $X_H^* X_H$ via multiplication by $\sqrt{q}$. If $u > 0$ and $Tu \le \lambda u$, apply Schur with $u/\sqrt{q}$.

First take a full-space correction

$$Z = (I + K^2)^{-1}(H - 1 - g), \quad e = KZ, \quad b = \mathbb{E}e, \quad D = -\mathbb{E}Z, \quad L = \|Z\|_2^2, \quad u = 1 + e.$$

Writing $\delta = H - 1$, $t = g + Ke$, $m = \mathbb{E}(\delta t)$, exact expansion gives

$$Tu - u = -g^2 - ge + q[-D + bAH - m - K(\delta t)].$$

Since $\delta - t = Z$ and $|vw| - vw \le (v-w)^2/2$,

$$Tu - u \le -g^2 - ge - q[D - L/2 - bAH]. \tag{E.7}$$

For $H = f_2$, $Z = (I + K^2)^{-1}K^2\mathbf{1}$, $b = 0$, and

$$D = \int \frac{x}{1-x}\, d\nu, \quad L = \int \frac{x^2}{(1-x)^2}\, d\nu, \quad d \le D \le 32/57, \quad L \le (13/19)D < 25/64.$$

Hence $|e| < 5/8$, $3/8 < u < 13/8$. With $a = D - L/2$, $4a \ge 50D/19$, whereas $(\sqrt{L} + a)^2 \le (569/228)D$. Completing the square in (E.7) leaves $-31D/912$, so $\left\|X_{f_2}\right\|^2 \le 1 - d/48$.

For $n \ge 3$ use the centered correction

$$Z = R_C(QH - g), \quad e = CZ, \quad b_0 = \langle g, Z\rangle, \quad L = \|Z\|_2^2,$$
$$D = 1 - \mathbb{E}H + \langle r_0, Z\rangle, \quad B = (L + D^2)/2, \quad u = 1 + e.$$

Now Z is centered, $e = KZ + b_0$, and $\delta - t = Z - D$. The exact residual and its row upper bound are

$$Tu - u = -g^2 - ge - q[D + b_0 + \mathbb{E}(\delta t) + Dg + K(\delta t)],$$

$$Tu - u \le -(1-D)g^2 - g(KZ + B) - D - b_0 + B. \tag{E.8}$$

If $D < 1$, the completed-square margin is

$$\Phi = 4(1-D)(D + b_0 - B) - (\sqrt{L} + B)^2.$$

When $0 \le D < 1$, $0 < u < 2$ and $\Phi \ge d/100$, (E.8) implies $\|X_H\|^2 \le 1 - d/800$.

The following uniform spectral bounds supply this margin. Write $R(x) = 1/(1-x)$, $a = \int xR\, d\nu$, $W = 1 + a = \int R\, d\nu$, $U_j(x) = \sum_{\ell=0}^{j-1}(-x)^\ell$, $F_j = xU_j$, $F_0 = 0$. For $H = S_n - c$, let $m = \lfloor n/2 \rfloor$, $\ell = \lfloor (n-1)/2 \rfloor$, $D_n = \int RF_m\, d\nu$, $\mathfrak{b}_n = \int xRF_\ell\, d\nu$, $d_s = D_n/W$, $\mathfrak{b}_s = \mathfrak{b}_n/W$. Direct centering of the full resolvent gives

$$Z = (I + K^2)^{-1}(S_n - 1 - g) + \mathfrak{b}_s(I + K^2)^{-1}g + d_s(I + K^2)^{-1}\mathbf{1}, \quad D = c + d_s, \quad b_0 = -\mathfrak{b}_s,$$

$$L = \int R^2(F_m - d_s)^2\, d\nu + \int xR^2(F_\ell - \mathfrak{b}_s)^2\, d\nu.$$

Each F_j is nondecreasing and 1-Lipschitz, with $0 \le F_j \le x$. To compare variances, put $\pi = R\nu/W$ and use

$$\mathbb{E}_\pi[R(F - \mathbb{E}_\pi F)^2] = \frac{1}{2}\mathbb{E}_{\pi,\pi}[(R_x + R_y - \mathbb{E}_\pi R)(F_x - F_y)^2].$$

Its coefficient is positive because $2 - 1/\alpha > 0$. For the odd part, put $\rho_0 = xR\nu/a$, $\kappa = a/W$ and use

$$\mathbb{E}_{\rho_0}[R(F - \kappa\mathbb{E}F)^2] = \frac{\kappa}{2}\mathbb{E}_{\rho_0,\rho_0}[(R_x + R_y - \kappa\mathbb{E}R)(F_x - F_y)^2] + (1-\kappa)\mathbb{E}_{\rho_0}[(R - \kappa\mathbb{E}R)F^2].$$

Here $\kappa\mathbb{E}R \le \theta/\alpha < 1$, so both coefficients are nonnegative. Replacing F by x in these two formulas proves $L(S_n - c) \le L(S_3)$. From (E.4),

$$L(S_3) = (2 + 2d + d^2)L_{r_0} + (d^2 - 1)\beta + d^3, \qquad L_{r_0} = \|R_C r_0\|_2^2 \le (\beta - \beta^2/a_0)/\alpha.$$

The resulting concave quadratic increases up to $\beta = da_0/z$; its endpoint derivative, after multiplication by αz, is at least $6472/27648$. Thus

$$L \le d\,\ell(d), \qquad \ell(d) = \frac{585/1024 - d}{(19/32 + d)^2}.$$

This convex function has the upper chords

$$\ell(d) \le \begin{cases} 13/8 - 6d, & 0 \le d \le 1/10, \\ 7/5 - 4d, & 1/10 \le d \le 1/5, \\ 4/3 - 11d/3, & 1/8 \le d \le 1/5. \end{cases}$$

Independently, centering the full spectral even and odd parts does not increase their norms, giving

$$L \le \int \frac{x^2[U_m^2 + xU_\ell^2]}{(1-x)^2}\, d\nu.$$

For $n \ge 5$ this is less than $(15/16)d$, since $U_j \le 1 - x + x^2$ for $j \ge 2$; for $n = 4$ it is at most $(10101/11552)d < (7/8)d$.

For $n \ge 5$ and $d \le 1/5$, take $H = S_n - \alpha d \ge f_n$. Then

$$\frac{4}{3}d \le D \le \frac{15}{8}d, \qquad D + b_0 \ge (11/20)D + (171/640)d, \qquad 0 < u < 2.$$

The ratio bound $\mathfrak{b}_n \le (9/20)D_n$ follows pointwise from the finite alternating sums: the only additional even-index ratio is $(x+x^4)/(1-x^4)$, at most $454545/1020015 < 9/20$. The margin is bounded below by

$$\mathcal{F}(D, L) = 4(1 - D)[(11/20)D + (171/640)d - (L + D^2)/2] - [\sqrt{L} + (L + D^2)/2]^2.$$

It decreases in L and is concave in $D \le 3/8$. At $D = td$, $L \le ld$, write $c_t = 11t/20 + 171/640$, $b_t = l/2 + t^2 d/2$; then

$$\mathcal{F}(td, ld)/d = 4(1 - td)(c_t - b_t) - (\sqrt{l} + b_t\sqrt{d})^2.$$

The two endpoints $t = 4/3, 15/8$ suffice. On $d \le 1/10$ use $l = 15/16$: the first term is greater than $3/2$, and the square is at most $(311/256)^2$, leaving $1583/65536 > 1/100$. On $1/10 \le d \le 1/5$, use $l = 7/5 - 4d$. The respective bounds are

t	first term lower	square upper	margin lower
$4/3$	$57/40$	$(1337/1125)^2$	$637865/50625000$
$15/8$	$3/2$	$(973/800)^2$	$13271/640000.$

The first terms are concave quadratics and the square-root sums decrease; the table follows by their interval endpoints. Both margins exceed $1/100$.

For $n = 4$ choose $H = S_4 - d = f_4 + dp$. Then $(5/3)d \le D \le 2d$, $D + b_0 \ge (6/19)D + (13/19)d$ and $0 < u < 2$. Use the same concavity argument with $t = 5/3, 2$ and $c_t = (6t+13)/19$. On $d \le 1/8$, $l = 7/8$ gives first terms at least $205/108, 573/304 > 15/8$ and square at most $(161/128)^2$, leaving $4799/16384$. On $1/8 \le d \le 1/5$, take $l = 4/3 - 11d/3$: the first terms are at least $3/2$ and $702/475$ respectively, while the square is below $23/16$. The uniform margin is at least $307/7600 > 1/100$.

For $n = 3$ use the actual $H = f_3 = S_3 - d$. With $D_2 = a/W$,

$$D = d + D_2, \quad b_0 = -[(1+d)D_2 - d], \quad D + b_0 = d(2 + d - D), \quad D \le d(43/16 - 2d).$$

On $d \le 1/10$, $1 - D > 7/10$, $(D + b_0)/d > 9/5$, $L/d \le 13/8 - 6d$ and $B/d < 7/8$. Thus the first term of Φ/d is at least $259/100$, and the square at most $(191/120)^2$, leaving $815/14400 > 1/100$. On $1/10 \le d \le 1/5$, $1 - D \ge 217/400$, $(D + b_0)/d \ge 697/400$, $L/d \le 7/5 - 4d$ and $B/d < 5/6$. The corresponding bounds are $236747/120000$ and $(11/8)^2 = 226875/120000$, leaving $9872/120000 > 1/100$. In both cases $L \le 1/5$ gives $0 < u < 2$.

Finally for $d \ge 1/5$ a Hilbert–Schmidt bound on the actual message gives $\left\|X_{f_n}\right\|^2 \le 2\mu_{n+2}$. Positive convolution bounds this by $191/200$ for $n \ge 4$, and the exact μ_5 bounds it by $161/200$ for $n = 3$. Both are below $1 - d/800$. This proves (C.4) for $n \ge 2$. For $n = 1$, the factorization $X_1 = [\sqrt{p}A^*p^{-1/2}]\sqrt{pq}$ and weighted propagation prove the assertion.

E.6 Opposite types and every positive connector length

Let $X_i = \sqrt{f_i}A^*\sqrt{q}$ and let $X_j^- = \sqrt{\bar{f}_j}A\sqrt{p}$. Every case of (C.5) is covered by the following factorizations:

indices	factorization or estimate
$i = j = 1$	$[\sqrt{p}A^*p^{-1/2}]^L\sqrt{pq}$
$i \ge 2, j = 1$	$X_iB_q^{L-1}$
$i = 1, j \ge 2$	reversal and adjoint of the preceding case
$i, j \ge 2, L = 2$	$(\sqrt{f_i}A^*)(\sqrt{\bar{f}_j}A)^*$
$i, j \ge 2, L \ge 3$	$X_iY_{L-2}(X_j^-)^*$.

For $i, j \ge 2, L = 1$, Cauchy–Schwarz on the nonnegative edge kernel yields

$$|\left\langle u, \sqrt{f_i}A^*\sqrt{\bar{f}_j}v\right\rangle|^2 \le \left\langle |u|, M_{f_i}A^*|v|\right\rangle \left\langle |u|, A^*M_{\bar{f}_j}|v|\right\rangle \le \|u\|_2^2\,\|v\|_2^2$$

by (E.2) and its reversed adjoint. All factors have already been proved contractions. The connector has L steps; at $L = 3$ the factor is Y_1, while Y_0 lies outside the positive-index range. This finishes Proposition C.4.

F Rooted cubic estimates

These estimates supply the endpoint and middle-root inputs of Section C. The canonical rooted estimates come first; the remaining cases follow the residual-length partition in its middle-root table. The anchor direction is fixed within each case.

F.1 Canonical cubic root certificates

We prove Proposition C.3. Throughout this subsection $P = \prod_i h_{a_i}f_{b_i}$, where $a_1 < b_1 < \cdots < a_t < b_t$. Equal pairs are restored at the end using (2.5).

F.1.1 The ordinary cubic

If $t = 0$ there is nothing to prove. If $a_1 \geq 3$, use the ordinary sixth moment in Lemma D.3. A single pair with $a_1 = 1$ or 2 is covered by Lemma D.4.

For the remaining cases write $P \leq e^{R-\Delta}$ as in (D.5), and let $D_0 = \Delta - \mathbb{E}R$. The following bounds prove $D_0 \geq 3M^2/2$:

condition	D_0 lower bound	M^2 upper bound
$a_1 = 2,\ t \geq 2$	$(67/32)d$	$C_*\theta d$
$a_1 = 1,\ b_1 \geq 3,\ t \geq 2$	$(99/32)d$	C_*d
$(a_1, b_1) = (1, 2),\ t \geq 3$	$(131/32)d$	C_*d.

Indeed the sums Δ in the first two rows are at least $(3-\theta)d$, and in the last row at least $(4-\theta)d$; subtract the ordinary mean estimates after (D.6). The comparisons are $3939/3200 < 67/32$ and $303/100 < 99/32 < 131/32$.

It remains to consider exactly two pairs $(1, 2), (a, b)$, $3 \leq a < b$. If $a = 3$, write

$$R = KQ(K)g, \qquad Q(z) = 1 + z^2 G_{b-3}(z).$$

Then $\|Q(K)\| \leq 1$. For G_1 this is immediate. For $m = b - 3 \geq 2$,

$$Q(z) = \frac{1 - z + z^2 - z^{m+2}}{1 - z}.$$

At $z = i\lambda$, $x = \lambda^2 \leq 1/2$, its numerator has squared modulus at most $1-x+3x^2+x^4 \leq 1+x$, because $3x+x^3 < 2$. Thus $M^2 = d$ is valid. Here $\mathbb{E}R \leq -d+\delta_2$ and $\Delta = 2d-\delta_2$, so $D_0 \geq (35/16)d > 3d/2$.

If $a \geq 4$, the polynomial $Q(z) = 1 + z^{a-1}G_{b-a}(z)$ has no terms of degrees one or two. Grouping powers modulo four yields

$$|Q(i\lambda)|^2 \leq \frac{1 + x^3}{(1 - x^2)^2} \leq \frac{1 + (3/7)^3}{(1 - (3/7)^2)^2} = \frac{259}{160}.$$

Hence $M^2 \leq 259d/160$. Since $\Delta = 2d$ and $\mathbb{E}R \leq -d/2$, $D_0 \geq 5d/2 > 777d/320 = 3M^2/2$. Criterion (D.8) proves $\mathbb{E}P^3 \leq 1$ in all cases.

F.1.2 The compatible weighted cubic

The complete even catalogue proved in Appendix D and Sections H.1–H.7 and collected in Section H.8 gives, for this same canonical P, $\mathbb{E}[pP^6] \leq 1$ unless the sole distinct pair is $(1, 2)$. For that pair Section H.8 proves $\mathbb{E}[pP^4] \leq 1$. Since $\mathbb{E}p = 1$, Hölder under $p\mu$ yields in both cases

$$\mathbb{E}[pP^3] \leq 1. \tag{F.1}$$

Multiplication by the actual factor $E_0 \leq 1$ restores all equal pairs. The foundational estimates in Appendix D supply the inputs to the even catalogue.

F.1.3 An odd anchor to power one

Pair equal lengths, retain the smallest residual arm a, and write $R = E_0 P$ for the remaining pairs. If P is empty, $\mathbb{E}[f_a R^3] \leq \mu_a \leq 1$. If $a = 1$, then $f_a = p$ and all paired shorter lengths are at least two,

so (F.1) applies. If $a \geq 2$, all paired shorter lengths are at least three and

$$\mathbb{E}[f_a R^3] \leq \left(\mathbb{E}\frac{f_a^2}{p}\right)^{1/2} \left(\mathbb{E}[pP^6]\right)^{1/2} \leq 1.$$

Here the inverse-weight path square and weighted sixth moment were proved in Appendix D. Every anchor is an existing arm. Replacing K by $-K$ proves the complete reversed statements and finishes Proposition C.3.

F.2 Odd roots with a long anchor or a separated second arm

F.2.1 Ordinary cubic envelopes for every path of length at least three

For $n \geq 3$ put $t_j = 1 - \mathbb{E}S_j$, $a = t_n$, $b = t_{n-1}$ and $y = S_n - b$. Positive convolution gives $0 < f_n \leq y$ and $\mathbb{E}y = 1 - a - b$, with $(1-\theta)d \leq a, b \leq d$. For odd n, $a = b$; for even $n = 2m$, $a - b = (-1)^{m-1}\delta_m$.

Let $C = \|S_n - 1\|_2^2$ and $v = S_n - \mathbb{E}S_n$. Since $S_n - 1 = KS_{n-1}$ and $\|S_{n-1}\|_2 \leq 1$, we have $|S_n - 1| \leq 1$. Thus $\mathbb{E}v^2 = C - a^2$ and $v^3 \leq (1+a)v^2$, giving

$$\mathbb{E}f_n^3 \leq \mathbb{E}y^3 \leq (1-a-b)^3 + (4-2a-3b)(C-a^2). \tag{F.2}$$

The coefficient is positive. The spectral formulas for G_n give

n mod 4	relation of deficits	C upper bound
3	$a = b$	b
0	$a - b = -\delta_{n/2} \leq 0$	a
1	$a = b$	$b + \delta_3$
2	$a = b + c,\ 0 \leq c = \delta_{n/2} \leq \delta_3$	$a + \theta c.$

For example, for odd $n = 2m+1$, $C = \int x(1+x^{2m+1})/(1+x)\,d\nu$ and $a = b = \int x[1-(-x)^m]/(1+x)\,d\nu$; subtracting yields the two odd rows. For even $n = 2m$, the two integrands are $x[1-(-x)^m]^2/(1+x)$ and $x[1-(-x)^m]/(1+x)$, which prove the even rows.

Write $H(a,b) = (1-a-b)^3 + (4-2a-3b)(a-a^2)$. Direct expansion gives

$$H(b+\delta, b) = 1 - 2b + 3b^2 - 3b^3 + \delta(1-3b) - 3(1-b)\delta^2 + \delta^3.$$

For $0 \leq b \leq 1/3$, $2 - 3b + 3b^2 \geq 4/3$. In the first two rows all additional δ terms are nonpositive. In the third row the excess is at most $4\delta_3$; in the fourth it is at most $(1+4\theta)\delta_3$. Consequently the respective deficit coefficients, relative to d, are

$$19/24, \quad 19/24, \quad 101/768, \quad 8809/24576.$$

We have proved, for the same positive envelope y,

$$\mathbb{E}y^3 \leq 1 - (101/768)d, \qquad \mathbb{E}y \geq 1 - 2d, \qquad \|S_n - 1\|_2^2 \leq d. \tag{F.3}$$

F.2.2 A scalar estimate for an anchored cubic

Lemma F.1. *Suppose $y, P > 0$, $P \leq e^{R-\Delta}$ and $|R| \leq M$. Put $x = \mathbb{E}y^3$ and let $B \geq \mathbb{E}[y^3 R]$. Then*

$$\log \mathbb{E}[y^3 P^3] \leq \log x + \frac{3B}{x} + \frac{9}{2}M^2 - 3\Delta.$$

If $x > 3B$ and $x \leq u$, the first two terms are at most $\log u + 3B/u$. If also $B \geq 0$ and $u \geq 1$, they are at most $\log u + 3B$.

Proof. Apply (D.7) to the probability measure $y^3\mu/x$. The derivative of $\log x + 3B/x$ is $(x-3B)/x^2$. All operator estimates remain on μ. □

We also use, for $z \geq 0$,

$$\log(1+z) \leq z - \frac{z^2}{2(1+z)}. \tag{F.4}$$

The derivative of the difference is $z^2/[2(1+z)^2] \geq 0$.

F.2.3 The smallest residual arm has length at least three

Let $F = E_0 f_a P$, where $a \geq 3$ is smaller than every paired residual length, and let there be ℓ pairs. If $\ell = 0$, use (F.3). Otherwise their shortest length is at least four, so

$$P \leq e^{R-\Delta}, \quad \Delta = 2\ell d \geq 2d, \quad \|R\|_\infty^2 \leq d/7, \quad \mathbb{E}R^2 \leq d/18, \quad \mathbb{E}R \leq d/12.$$

The first two norm bounds follow from $C_*\theta^3 < 1/7$ and $C_*\theta^4 < 1/18$; the mean bound follows by retaining positive means from degree eight onward, using $\delta_4/(1-\theta^2) \leq d/12$. If $d \geq 1/28$, then $\|R\|_\infty \leq 2d \leq \Delta$, hence $P \leq 1$ and the path cubic suffices.

For $0 < d \leq 1/28$, use the actual envelope y in (F.3), put $t = y - 1$ and $Z_0 = \mathbb{E}y^3$. Then $|t| \leq 1$, $\mathbb{E}t^2 \leq d + 3d^2 \leq 31d/28$ and $Z_0 \geq (1-2d)^3 \geq (13/14)^3 > 4/5$. Expanding the cubic weight gives

$$\mathbb{E}[y^3 R] = \mathbb{E}R + 3\mathbb{E}(tR) + \mathbb{E}[t^2(3+t)R] \leq \left(\frac{1}{12} + \frac{3}{4} + \frac{1}{3}\right) d = \frac{7}{6}d.$$

Here $3|\mathbb{E}(tR)| \leq 3\sqrt{31/504}\, d \leq 3d/4$ and the last term is at most $4(1/14)(31/28)d \leq d/3$. Thus its normalized tilted mean is at most $35d/24$. Applying (D.7) under $y^3\mu/Z_0$ yields

$$\mathbb{E}[y^3 P^3] \leq Z_0 \exp[-3\Delta + 3(35/24)d + (9/14)d] \leq Z_0 e^{-(55/56)d} \leq 1 - (101/768)d.$$

Restoring $E_0 \leq 1$ proves the required full root cubic.

F.2.4 A short anchor followed by a residual arm of length at least four

Suppose the residual list is $a < b_1 < c_1 < \cdots < b_t < c_t$ with $a = 1$ or 2. Write $b = b_1$ and $P = \prod_i h_{b_i} f_{c_i}$. For $b = 4$ choose the inward anchor $y = \bar{f}_a$; for $b \geq 5$ choose $y = f_a$. We prove $\mathbb{E}(yP)^3 \leq 1$.

For $z = y - 1$ and $x = \mathbb{E}y^3$ the elementary short-path bounds are

$$|z| \leq 1, \quad \mathbb{E}z^2 \leq 3d/2, \quad (1-d)^3 \leq x \leq 1 + 4d.$$

For a unit anchor $x \geq 1$; for a two-edge anchor $x \leq 1 + (21/8)d$. Indeed $\mathbb{E}z = -d$, $\mathbb{E}z^2 = d + \delta_2$, and $x = 1 + 3\delta_2 + \mathbb{E}z^3$ in the latter case.

If $b \geq 5$, the disjoint-mask bounds give

$$M^2 \leq d/18, \quad \mathbb{E}R^2 \leq d/40, \quad \mathbb{E}R \leq d/12, \quad \mathbb{E}(gR) \leq d/30, \quad \Delta = 2td.$$

For $y = p$, expansion gives $\mathbb{E}[y^3 R] \leq d/5 + 4Md =: B$. If $d \leq 1/8$, then $M \leq 1/12$, $3B \leq 1/5 < x$, and Lemma F.1 gives a logarithmic bound at most $-(3/20)d$. If $d \geq 1/8$, $M \leq 2d/3$ and the pointwise bound gives $\mathbb{E}[y^3 P^3] \leq (1+4d)e^{-4d} \leq 1$.

For $y = f_2$, ordinary Cauchy–Schwarz gives $\mathbb{E}[y^3 R] \le (2/3)d + 6Md =: B$. Put $D = 32/729$. If $d \le D$, then $M \le 4/81$, $3B \le 832/6561 < 1/2 < x$, and the logarithmic bound is $[-9/8 + 18M]d \le -(17/72)d$. For $d \ge D$, $M \le 9d/8$ and $\mathbb{E}[y^3 P^3] \le [1 + (21/8)d]e^{-(21/8)d} \le 1$.

Now take $b = 4$ and one pair. The contiguous remainder is $R = S_c - S_4$, $c \ge 5$, with $M^2 \le (1+\theta)\theta^3 d < d/10$, $\Delta = 2d$. For the inward anchors, the linear cubic-weight term is

$$\mathbb{E}[(1+3z)R] = \begin{cases} \sum_{j=5}^{c}(u_j + 3u_{j+1}), & a = 1, \\ \sum_{j=5}^{c}(u_j + 3u_{j+1} + 3u_{j+2}), & a = 2. \end{cases}$$

For $c = 5$ these equal $-3\delta_3$. For $c \ge 6$ they are at most $-4\delta_3 + 7\delta_4/(1-\theta^2) = -(508/855)\delta_3 \le 0$. Hence $B = 4Md$ or $6Md$ is valid. For the unit anchor, $d \ge 9/40$ gives the pointwise estimate as before. In the other range $M \le 3/20$, $3B < x$, and (F.4) gives

$$\log \mathbb{E}[y^3 P^3] \le d[-31/20 + (12/\sqrt{10})\sqrt{d} - (80/19)d] \le -(139/200)d.$$

For the two-edge anchor use the threshold $D = 32/405$. Above it, $M \le 9d/8$ gives the pointwise estimate. Below it, $M \le 4/45$, $3B < 1/2 < x$, and the logarithmic bound is at most $[-117/40 + 18M]d \le -(53/40)d$.

Finally suppose $b = 4$ and $t \ge 2$. Then $\Delta \ge 4d$, $M^2 \le d/7$, and counting the positive linear-weight coefficients gives $\mathbb{E}[(1+3z)R] \le 7\delta_4/(1-\theta^2) \le 7d/12$. Thus $B = 7d/12 + 6Md$ works for both anchors. For $d \ge 1/16$, $M \le 2d$ and $(1+4d)e^{-6d} \le 1$. For $d \le 1/16$, $M < 1/10$, $3B \le 71/320 < 1/2 < x$, and Lemma F.1 bounds the logarithm by

$$[4 + 7/4 + 9/5 + 9/14 - 12]d = -(533/140)d.$$

All short-anchor configurations with $b \ge 4$ are covered, in the fixed directions stated at the beginning. Restoring E_0 proves the complete middle-root statement in the coverage table.

F.3 Short anchors followed by an arm of length three

We treat every residual list beginning with $a, 3, c$, where $a = 1$ or 2 and $c \ge 4$. The anchor is always inward: $y = \bar{f}_a$. All remaining ordered pairs have their positive canonical directions. Put $z = y - 1$ and $x = \mathbb{E}y^3$. As in Section F.2, $|z| \le 1$; for $a = 1$, $\mathbb{E}z^2 = d$ and $1 \le x \le 1 + 4d$; for $a = 2$, $\mathbb{E}z^2 = d + \delta_2 \le 3d/2$ and $(1-d)^3 \le x \le 1 + (21/8)d$.

F.3.1 The first pair has lengths three and four

Writing $h = Kg$, direct recursion gives

$$h_3 + f_4 = 2 - 3d + R_0, \qquad R_0 = K^3 g - dg.$$

Thus $h_3 f_4 \le e^{R_0 - 3d}$ and

$$M^2 = \|R_0\|_\infty^2 \le \delta_3 + d^2 \le \theta^2 d + d^2.$$

The row vector here is $K^2 g - d\mathbf{1}$, whose mean cross term is zero. Skewness gives $\mathbb{E}R_0 = \delta_2$, $\mathbb{E}(gR_0) = -d^2$, $\mathbb{E}(hR_0) = -\delta_3$. Hence

$$\mathbb{E}[(1+3z)R_0] \le \delta_2 + 3d^2,$$

and valid budgets for Lemma F.1 are $B_1 = \delta_2 + 3d^2 + 4Md$ and $B_2 = \delta_2 + 3d^2 + 6Md$.

For the unit anchor, $d \geq 3/32$ implies $M \leq 5d/3$ and the pointwise estimate gives $(1+4d)e^{-4d} \leq 1$. Below this threshold, $M \leq 5/32$, $3B_1 < x$, and (F.4) gives

$$\log \mathbb{E}[y^3(h_3 f_4)^3] \leq d[-6223/2048 + (169/22)d + 12M] \leq -(847/2048)d.$$

For the two-edge anchor, $d \geq 1/20$ implies $M \leq 17d/8$ and the pointwise bound pays $1 + (21/8)d$. Below the threshold, $M < 1/9$, $3B_2 < 1/2 < x$, and

$$\log \mathbb{E}[y^3(h_3 f_4)^3] \leq d[-9039/2048 + (27/2)d + 18M] \leq -(3407/2048)d.$$

If there are further ordered pairs, put

$$\bar{R} = K^3 g + \sum_{i=2}^{t} (S_{c_i} - S_{b_i}), \quad R = \bar{R} - dg, \quad \Delta = 3d + 2(t-1)d \geq 5d.$$

Then $P \leq e^{R-\Delta}$. Since $\bar{R} = K^3 Q(K) g$ with a zero-one mask, setting $V = K^2 Q(K) g$ gives $\|V\|_2^2 \leq 17d/50$ and $\mathbb{E}V \geq -d/5$. Therefore

$$M^2 \leq (17/50)d + (7/5)d^2.$$

Counting positive moments in $\mathbb{E}[(1+3z)\bar{R}]$ gives a coefficient one at $u_4 = \delta_2$ and at most seven at each later positive moment. Since $\delta_4/(1-\theta^2) \leq d/12$ and $\mathbb{E}(zg) = -d$,

$$\mathbb{E}[y^3 R] \leq d + 3d^2 + 6Md =: B.$$

For $d \geq 1/16$, $M \leq 3d$ and $(1+4d)e^{-6d} \leq 1$. For $d \leq 1/16$, $M < 1/6$, $(9/2)M^2 < 2d$, $3B \leq 105/256 < 1/2 < x$, and Lemma F.1 gives a logarithmic bound at most $-(39/16)d$. This proves all residual lists starting with $a, 3, 4$.

F.3.2 A nonempty tail after the first pair

Suppose the residual list is $a, 3, c_1, b_2, c_2, \ldots, b_t, c_t$, with $t \geq 2$ and $c_1 \geq 5$. The general pair envelope gives

$$\Delta = 2td - \delta_2 \geq (115/32)d, \qquad M^2 \leq C_* \theta^2 d \leq 17d/50.$$

For the inward unit anchor the linear cubic-weight mean is at most $\delta_2 + 4\delta_4/(1-\theta^2) < 3d/4$. For the inward two-edge anchor it is at most $\delta_2 + 7\delta_4/(1-\theta^2) < d$. Thus

$$B_1 = 3d/4 + 4Md, \qquad B_2 = d + 6Md$$

are valid. For $a = 1$, split at $d = 1/14$. Above it $M \leq 9d/4$ and $\Delta - M \geq 43d/32 > 4d/3$. Below it $M < 1/6$ and $3B_1 \leq 17/56 < x$; Lemma F.1 gives

$$\log \mathbb{E}[y^3 P^3] \leq [4 + 9/4 + 153/100 - 345/32 + 2]d = -(801/800)d.$$

For $a = 2$, split at $d = 1/16$. Above it $M \leq 5d/2$ and $\Delta - M \geq 35d/32 > 7d/8$. Below it $M < 3/20$, $3B_2 \leq 57/160 < 1/2 < x$, and the logarithmic bound is

$$[21/8 + 3 + 153/100 - 345/32 + 27/10]d = -(741/800)d.$$

This proves all nonempty-tail cases.

F.3.3 An endpoint version of the scalar estimate

For the remaining three-arm lists we retain the denominator in Lemma F.1. If $0 < L \le x \le U$ and $B \ge 0$, then

$$3B \le \frac{2LU}{L+U} \quad \Longrightarrow \quad \log x + 3B/x \le \log U + 3B/U. \tag{F.5}$$

Indeed the function has only a minimum, so its maximum on an interval is at an endpoint. The inequality $\log t \ge 2(t-1)/(t+1)$ for $t \ge 1$ proves that the upper endpoint is larger under the displayed condition. We will also use

$$\log(1+s) \le s - \frac{s^2}{2(1+s)} - \frac{s^3}{6(1+s)^2} \qquad (s \ge 0). \tag{F.6}$$

The derivative of the difference for the first two terms is $s^2/[2(1+s)^2]$; its integral is at least $s^3/[6(1+s)^2]$.

F.3.4 The single pair of lengths three and five

Set $h = Kg$, $j = K^2 g$, $k = K^3 g$, $\ell = K^4 g$. Exact recursion gives

$$h_3 + f_5 = 2 - \Delta + R, \qquad \Delta = 4d - \delta_2, \qquad R = k + \ell - d(2g + h).$$

Thus $P = h_3 f_5 \le e^{R-\Delta}$ and $\Delta \ge 115d/32$. The two linear cubic-weight means are

$$\mathbb{E}[(1+3z)R] = \begin{cases} \delta_2 - 3\delta_3 + 7d^2 \le d/12 + 7d^2, & a = 1, \\ \delta_2 - 6\delta_3 + 7d^2 - 3d\delta_2 \le d/24 + 7d^2, & a = 2. \end{cases}$$

Here the spectral inequalities are $x - 3x^2 \le 1/12$ and $x - 6x^2 \le 1/24$. The full row vector gives

$$M^2 \le \delta_3 + \delta_4 - 2d\delta_2 + 4d^2 + d^3 \le \max\{(4+d)d^2, S(d)\},$$

$$S(d) = C_0 d + (51/16)d^2 + d^3, \qquad C_0 = (1+\theta)\theta^2 = 7605/32768.$$

The last bound follows by applying the endpoint chord of the convex polynomial $x^2 + x^3 - 2dx$ on $[0, \theta]$ to the spectral measure of g. For $d \le 2/15$ the second expression is larger. Valid cubic-weight budgets are $B_1 = d/12 + 7d^2 + 4Md$ and $B_2 = d/24 + 7d^2 + 6Md$.

For $a = 1$ and $d \ge 2/15$, both row bounds are at most $(81/16)d^2$: use $C_0 < 209/900 \le d(15/8 - d)$. Hence $M \le 9d/4$ and the pointwise estimate proves the cubic. For $d \le 2/15$, $M^2 \le S(d)$, $M \le 3/10$ and $3B_1 < 1 \le x$. Put $U = 1 + 4d$ and retain $3B_1/U$. Lemma F.1 and (F.4) give

$$\frac{1}{d}\log \mathbb{E}[y^3 P^3] \le 4 - \frac{345}{32} + \frac{9}{2}C_0 + \frac{1/4 + 13d + 12\sqrt{S(d)}}{1+4d} + \frac{459}{32}d + \frac{9}{2}d^2.$$

Every varying term increases. For the square-root quotient, differentiate its square; the sign numerator is $C_0 + (2(51/16) - 4C_0)d + 3d^2 + 4d^3 > 0$. At $d = 2/15$ the quotient is at most $335/92 < 73/20$, and the other terms give $-917/160 + 73/20 + 153/80 + 2/25 = -71/800 < 0$.

For $a = 2$ and $d \ge 1/16$, use $C_0 < 139/576 \le d(565/144 - d)$ to get $M \le 8d/3$; hence $\Delta - M \ge 89d/96 > 7d/8$ and the pointwise estimate pays $1 + (21/8)d$. For $d \le 1/16$, $S(d) \le 14261/524288 < 1/36$, $M \le 1/6$ and $3B_2 < 1/2 < x$. The logarithmic bound is

$$[11/4 + (9/2)C_0 - 345/32] + (1131/32)d + 18M + (9/2)d^2 \le -1369/800.$$

This proves both single-pair cases at length five.

F.3.5 The single pair with every longer length

Assume $c \geq 6$. Put

$$r = \delta_2/d, \quad \tau = d - \delta_2 = d(1-r), \quad d \leq r \leq \theta.$$

Here r is a scalar ratio. Retaining f_0, f_1, f_2 in positive convolution gives

$$h_3 + f_c \leq 2 - \Delta + R, \quad \Delta = (4-3r)d, \quad R = R_0 - \tau(2g + Kg), \quad R_0 = S_c - S_3.$$

For $m = c - 3 \geq 3$, the exact geometric formulas imply

$$|G_m(i\sqrt{x})|^2 \leq H(x) = \frac{(1+x^3)^2}{1+x}.$$

The function $xH(x)$ increases and $H(\theta) \leq 13/16$. Let $\Gamma = 169/512$ and $U_0 = K^2 G_{c-3}(K)g$. Then

$$\|U_0\|_2^2 \leq \Gamma\delta_2, \quad \mathbb{E}U_0 \geq \delta_2 - \delta_3, \quad \langle g, U_0\rangle \geq -\delta_2.$$

Since $R = K[U_0 - \tau g - 2\tau\mathbf{1}]$, the complete row expansion yields

$$M^2 \leq dV(d,r), \quad V(d,r) = \Gamma r + d[(1-r)^2(4+d) - (3/8)(1-r)r]. \tag{F.7}$$

The cross terms have been retained. The convex quadratic V in r has coefficients

$$A_r = d(35/8 + d), \quad B_r = \Gamma - d(67/8 + 2d), \quad C_r = d(4+d).$$

Its endpoint formulas are

$$V(d,\theta) = v_0 + v_1 d + v_2 d^2, \quad (v_0, v_1, v_2) = (2197/16384, 10811/8192, 361/1024),$$

$$V(d,d) = dW(d), \qquad W(d) = 4 + \Gamma - (59/8)d + (19/8)d^2 + d^3.$$

On $[0, 1/5]$, $W > 5/2$, $-59/8 \leq W' < 0$, $(dW)' > 0$ and $2W + dW' > 0$. Also $v_0 < 1/7$, $1 < v_1 < 4/3$, $v_2 < 3/8$, $V(1/5,\theta) < 33/80$, $V(1/16,\theta) < 7/32$, $W(1/5) < 3$ and $W(1/16) < 4$.

For the linear cubic weights, finite alternating sums give

$$\mathbb{E}[(1-3g)R_0] \leq \delta_2 - 4\delta_3 + 4\delta_4 \leq d/10,$$

$$\mathbb{E}[(1-3g+3Kg)R_0] \leq \delta_2 - 7\delta_3 + 7\delta_4 \leq d/20.$$

For example, when $c = 2m$, the first expression is $\delta_2 + 4\sum_{j=3}^{m}(-1)^j\delta_j$; for odd c add $3(-1)^{m+1}\delta_{m+1}$. Grouping each later positive term with its preceding negative proves the bound. The spectral scalar estimates are $x(1-2x)^2 \leq 2/27 < 1/10$ and $x - 7x^2 + 7x^3 \leq 1/21 < 1/20$ on $[0, 1/2]$. The correction $-\tau(2g + Kg)$ contributes at most $7\tau d$, so valid budgets are

$$B_1 = d/10 + 7\tau d + 4Md, \qquad B_2 = d/20 + 7\tau d + 6Md. \tag{F.8}$$

For $a = 2, d \geq 1/16$, we claim $M \leq [(11 - 13r)/3]d$. By (F.7), it suffices that

$$d[(85/9 - d) + (-1685/72 + 2d)r + (1037/72 - d)r^2] - \Gamma r \geq 0.$$

This decreases in $r \leq \theta$, so set $r = \theta$. The result is $d[((61/32)^2 - v_1) - v_2 d] - v_0$, increasing on $[1/16, 1/3]$ and positive at $1/16$ because $V(1/16,\theta)/(1/16) < 7/2 < (61/32)^2$. Thus $\Delta - M \geq$

$(1+4r)d/3$, which pays the sharper short-path bound $x \le 1+(1+4r)d$. For $d \le 1/16$, convexity and the endpoint formulas give $V \le 1/4$, so $M \le 1/8$. The budget has $3B_2 \le 297/1280 < 1/2 < x$; Lemma F.1 gives

$$\frac{1}{d}\log \mathbb{E}[y^3P^3] \le 21/8 + 3/20 + 21/16 + 18/8 + 9/8 - 267/32 = -141/160.$$

For $a = 1, d \ge 1/5$, the analogous bound is $M \le (8/3 - 3r)d$. Its squared difference is

$$d[(28/9 - d) + (-61/8 + 2d)r + (37/8 - d)r^2] - \Gamma r,$$

again decreasing in r. At $r = \theta$ it is increasing in $d \le 1/3$ and positive at $1/5$, because $V(1/5,\theta)/(1/5) < 33/16 < (139/96)^2$. Thus $\Delta - M \ge 4d/3$ and the pointwise estimate proves the result.

It remains to handle $a = 1, d \le 1/5$, where $1 + 2d \le x \le U = 1 + 4d$. We justify the upper-endpoint substitution before using it. For $d \le 1/10$, (F.7) gives $M \le 1/4$, so $3B_1 \le 27/50 < x$. For $1/10 \le d \le 1/5$, $\partial_r V \le \Gamma - (617/128)d < 0$, and hence

$$B_1 \le \bar{B}(d) = d[1/10 + 7d(1-d) + 4d\sqrt{W(d)}].$$

Both $\bar{B}/d$ and d/H increase, where $H = (1+2d)(1+4d)/(1+3d)$. At $d = 1/5$, $\bar{B} \le 131/250$ and $3\bar{B} \le 393/250 < 63/40 = H$. Criterion (F.5) therefore proves $\log x + 3B_1/x \le \log U + 3B_1/U$ throughout this range.

For $d \le 1/48$, $V < 1/4$ and $M \le 1/12$, so the logarithmic bound divided by d is at most $4 + 139/80 + 9/8 - 267/32 = -237/160$. For $1/48 \le d \le 1/5$, (F.6) bounds it by

$$\Psi(d,r) = -8 + 9r + \frac{9}{2}V(d,r) + \frac{3/10 + (13 - 21r)d + 12\sqrt{dV(d,r)}}{1+4d} - \frac{32d^2}{3(1+4d)^2}.$$

This is convex in r: the only nonautomatic term is the square root, whose second derivative has the sign of

$$4A_rC_r - B_r^2 = -\Gamma^2 + (67\Gamma/4)d + (151/128)d^2 > 0$$

because $(67/4)d - \Gamma \ge 29/1536$. It is enough to check $r = d, \theta$. Both endpoint functions increase in $d \le 1/5$. For $r = \theta$, the polynomial $V(d,\theta)$ increases and $\sqrt{dV(d,\theta)}/(1+4d)$ increases since the derivative of its square has numerator $v_0 + (2v_1 - 4v_0)d + 3v_2d^2 + 4v_2d^3 > 0$. The derivative of $(9/2)V$ exceeds $9/2$, while the magnitude of the last negative derivative is at most $64/15$. The remaining linear quotient also increases. For $r = d$, the ordinary quotient has derivative numerator $59/5 - 42d - 84d^2 \ge 1/25$, and the square-root quotient increases because $2W + d(1+4d)W' \ge 5 - (9/25)(59/8) > 0$. The derivative of $9d$ pays the same last negative term. Finally, at $d = 1/5$,

$$\Psi(1/5,\theta) \le -199/80 + 743/288 - 32/243 = -1529/38880 < 0,$$

$$\Psi(1/5,1/5) \le -7/2 + 313/90 - 32/243 = -1/45 - 32/243 < 0.$$

This completes the single-pair proof for every $c \ge 6$ and both short anchors.

Every orientation in this subsection depends only on the residual lengths. The estimates include shared interval endpoints, $d = 0$ gives equality, and multiplying by the actual equal-pair product preserves the full cubic.

F.4 Residual prefixes of lengths one and two with a nonempty tail

Consider distinct residual lengths $1, 2, c, b_2, c_2, \dots, b_t, c_t$, where $t \geq 2$ and $c < b_2 < c_2 < \cdots < b_t < c_t$. We prove the ordinary cubic for the fixed message

$$F = pP, \qquad P = h_2 f_c \prod_{i=2}^{t} h_{b_i} f_{c_i}.$$

Throughout $x = \mathbb{E}p^3$ satisfies $1 \leq x \leq 1 + 4d$. We will also use $x \leq 2$. To prove this last bound, sample four independent labels from μ and, conditionally on the labels, orient each pair of the four distinct indices independently with probability $(1 + K(X_i, X_j))/2$. At most one index points to the other three. The expected number of such indices is $4\mathbb{E}(p/2)^3 = x/2$, proving $x \leq 2$, even when some sampled labels coincide.

Write $R_{\text{tail}} = \sum_{i=2}^{t} (S_{c_i} - S_{b_i})$. The later pairs have shorter length at least four, so their product is bounded by $e^{R_{\text{tail}} - 2(t-1)d}$. All estimates below retain the signed denominator in Lemma F.1.

F.4.1 The third residual length is three

Since $h_2 + f_3 = 2 - d + K^2 g$,

$$P \leq e^{R - \Delta}, \quad R = K^2 g + R_{\text{tail}}, \quad \Delta = (2t-1)d \geq 3d.$$

Write $R = K^2 Q(K) g$. The polynomial Q has constant coefficient one and no linear term. Grouping its real and imaginary parts at $i\sqrt{z}$ gives

$$|Q(i\sqrt{z})|^2 \leq \frac{1 + z^3}{(1 - z^2)^2} \leq \frac{1118880}{731025} < \lambda, \qquad \lambda = 25/16.$$

Its real part is nonnegative because $1 - z/(1 - z^2) > 0$; the imaginary sum starts at degree three. Therefore $M^2 = \|R\|_\infty^2 \leq \lambda \delta_2$.

The mandatory degree-three term contributes $-3\delta_2$ to $\mathbb{E}[(1 + 3g)R]$, and the absent degree-four term delays positive ordinary means. Discarding other negative terms gives

$$\mathbb{E}[(1 + 3g)R] \leq -3\delta_2 + \frac{3\delta_3 + \delta_4}{1 - \theta^2} = -\frac{1148}{855}\delta_2 \leq -\frac{4}{3}\delta_2.$$

Thus $B = -(4/3)\delta_2 + 4Md$ bounds $\mathbb{E}[p^3 R]$. This budget may be negative.

For $d \leq 1/5$, completing the square in $\sqrt{\delta_2}$ gives $3B \leq 9\lambda d^2 \leq 9/16 < 1 \leq x$. Put $u = 1 + 4d$. Lemma F.1 and $\log u \leq 4d$ give

$$\log \mathbb{E}[p^3 P^3] \leq -5d + \left[\frac{9}{2}\lambda - \frac{4}{u}\right] \delta_2 + \frac{12Md}{u}.$$

The coefficient of δ_2 is positive. Substituting $\delta_2 \leq \theta d$ and $M^2 \leq \lambda \theta d$ bounds this, after division by d, by

$$-5 + \theta \left[\frac{9}{2}\lambda - \frac{4}{u}\right] + \frac{12\sqrt{\lambda \theta d}}{1 + 4d}.$$

On this interval the middle term is at most $18005/9216$ and the last is less than $12/5$, since $\sqrt{d}/(1+4d)$ increases up to $1/4$ and $\lambda\theta/5 < (9/25)^2$. The result is at most $-29783/46080$. For $d \geq 1/5$, $M \leq 9d/5$, so $P^3 \leq e^{-(18/5)d}$. The bound $x \leq 2$ then gives $\mathbb{E}[p^3 P^3] \leq 2e^{-18/25} < 1$; the first four Taylor terms of $e^{18/25}$ already sum to $31897/15625 > 2$.

F.4.2 The third residual length is four

Exact recursion gives

$$h_2 + f_4 = 2 - 2d + K^2 g + K^3 g - dg.$$

Put $\bar{R} = K^2 g + K^3 g + R_{\text{tail}}$, $R = \bar{R} - dg$, $\Delta = 2td \geq 4d$. Writing $\bar{R} = K^2 Q(K) g$ and $V = KQ(K)g$ gives

$$R = K[V - d\mathbf{1}], \quad \|V\|_2^2 \leq C_* \delta_2, \quad \mathbb{E}V \geq -d/(1-\theta^2),$$

$$M^2 \leq C_* \delta_2 + (17/5) d^2.$$

The mandatory degrees three and four give $-2\delta_2$ in the linear cubic weight; degree five is absent. The remaining positive sum is at most $(\delta_4 + 3\delta_5)/(1-\theta^2) = (11999/27360)\delta_2 < \delta_2/2$. The correction $-dg$ contributes $-3d^2$, so

$$B = -\frac{3}{2}\delta_2 - 3d^2 + 4Md$$

is valid. If $d \leq 1/6$, then $M^2 \leq 6659/28800 < (29/60)^2 < 1/4$, and $3B \leq 6d - 9d^2 \leq 3/4 < x$. With $u = 1 + 4d$ the logarithmic bound becomes

$$-8d + [909/100 - 9/(2u)]\delta_2 + [153/10 - 9/u]d^2 + 12Md/u.$$

Both bracketed coefficients are positive. After replacing δ_2 by θd, their bounds divided by d are $8307/3200$ and $33/20$. The square of $\sqrt{C_* \theta d + (17/5)d^2}/(1+4d)$ has positive derivative numerator $C_*\theta + [34/5 - 4C_*\theta]d$, so the last term is at most $87/25$. The full bound is $-(877/3200)d$. If $d \geq 1/6$, then $M \leq 29d/10$ and $P^3 \leq e^{-(33/10)d}$. Since $d/(1+4d) \geq 1/10$, (F.4) gives $\log x \leq (16/5)d$, leaving $-d/10$.

F.4.3 The third residual length is five

We need a sharper polynomial bound. If the first three coefficients of a zero-one polynomial Q are one, then $\|Q(K)\|^2 \leq 5/4$. For $z \in [0, 5/12]$,

$$|Q(i\sqrt{z})|^2 \leq [1 - z + (6/5)z^2]^2 + (36/25)z = 1 - \frac{14}{25}z + \frac{17}{5}z^2 - \frac{12}{5}z^3 + \frac{36}{25}z^4.$$

The last polynomial is convex, since its second derivative is at least $4/5$; its endpoint values are 1 and $3533/2880 < 5/4$.

Direct recursion gives

$$h_2 + f_5 = 2 - (3d - \delta_2) + K^2 g + K^3 g + K^4 g - 2dg - dKg.$$

Put $\bar{R} = K^2 g + K^3 g + K^4 g + R_{\text{tail}}$, $R = \bar{R} - 2dg - dKg$, and $\Delta = (2t+1)d - \delta_2 \geq 5d - \delta_2$. The polynomial in $\bar{R} = K^2 Q(K) g$ has its first three coefficients one and the next zero. With $V = KQ(K)g$,

$$\mathbb{E}V \geq -d + \delta_2 - \frac{\delta_3}{1-\theta^2}, \qquad \mathbb{E}(gV) \geq -\delta_2 - \frac{\delta_4}{1-\theta^2}.$$

Expansion of $R = K[V - 2d\mathbf{1} - dg]$ gives

$$M^2 \leq (5/4)\delta_2 + 8d^2 + d^3 + (292/855)d\delta_2 \leq (5/4)\delta_2 + (17/2)d^2.$$

The linear cubic-weight mean of $\bar{R}$ is at most $[-2+3\theta+\theta^2(1+3\theta)/(1-\theta^2)]\delta_2 = -(293/855)\delta_2 \le -\delta_2/3$. The two corrections contribute $-5d^2$, so $B = -\delta_2/3 - 5d^2 + 4Md$.

For $d \le 1/7$, $M^2 \le 1543/6272 < 1/4$ and $3B \le 27/49 < x$. Retain the improvement (F.4) in $\log u$. The resulting bound is

$$\log \mathbb{E}[p^3 P^3] \le -11d + [69/8 - 1/u]\delta_2 + [153/4 - 23/u]d^2 + 12Md/u.$$

Both coefficients are positive. The upper bounds after substituting $\delta_2 \le \theta d$ increase with d. The last quotient does too, since its squared derivative numerator is $(5/4)\theta + [17 - 5\theta]d > 0$. At $d = 1/7$, the bound divided by d is at most

$$-11 + 9139/2816 + 1039/308 + 42/11 = -11099/19712.$$

For $d \ge 1/7$, $M \le 7d/2$ and $\Delta \ge 147d/32$, so $P^3 \le e^{-(105/32)d}$. Here $\log x \le (36/11)d$, leaving the negative logarithmic bound $-3d/352$.

F.4.4 Every third residual length at least six

Put $r = \delta_2/d$ and $\tau = d(1-r)$. Retaining four final terms in positive convolution yields

$$P \le e^{R-\Delta}, \quad \bar{R} = S_c - S_2 + R_{\text{tail}}, \quad R = \bar{R} - \tau(3g + 2Kg + K^2 g),$$

$$\Delta \ge \delta = \tau(4-d) + 2d.$$

The first four coefficients of the polynomial Q in $\bar{R} = K^2 Q(K)g$ are one. Such a polynomial satisfies $\|Q(K)\| \le 1$. Indeed both its real part and its imaginary part divided by the frequency are nonnegative and at most $1 - z + (6/5)z^2$; thus

$$|Q(i\sqrt{z})|^2 \le (1+z)[1 - z + (6/5)z^2]^2 = 1 - z + \frac{7}{5}z^2 + z^3 - \frac{24}{25}z^4 + \frac{36}{25}z^5 \le 1.$$

For the last comparison use $(7/5)z + z^2 + (36/25)z^4 \le 461/576 < 1$ on $[0, 5/12]$.

Set $V = KQ(K)g$, $Z = 3\mathbf{1} + 2g + Kg$. The mandatory first four degrees give

$$\mathbb{E}V \ge -d + \delta_2 - \delta_3/(1-\theta^2),$$
$$\mathbb{E}(gV) \ge -\delta_2 + \delta_3 - \delta_4/(1-\theta^2),$$
$$\mathbb{E}[(Kg)V] \ge \delta_2 - \delta_3 - \delta_5/(1-\theta^2).$$

Consequently $\langle V, Z\rangle \ge -3d + (2/5)\delta_2$ and $\|Z\|_2^2 = 9 - 2d + \delta_2 \le 9$. The full row expansion gives

$$M^2 \le \|V - \tau Z\|_2^2 \le \delta_2 + 9\tau^2 + 6\tau d - (4/5)\tau\delta_2 = rd + a(r)d^2,$$

$$a(r) = 15 - (124/5)r + (49/5)r^2 > 0 \quad (0 \le r \le \theta).$$

The mandatory degrees three through six contribute $-2\delta_2 + 2\delta_3$ to the linear cubic weight; later positive contributions are at most $(\delta_4 + 3\delta_5)/(1-\theta^2)$. Thus $\mathbb{E}[(1+3g)\bar{R}] \le -(7/10)\delta_2$. Also $\mathbb{E}[(1+3g)(3g + 2Kg + K^2 g)] = 7d - 3\delta_2$, so

$$B = -\frac{7}{10}\delta_2 - \tau(7d - 3\delta_2) + 4Md$$

is a valid signed budget.

For $d \le 1/7$, convexity in r gives $M^2 \le \max\{15d^2, \theta d + (33497/5120)d^2\} \le 15/49$. Both associated quotients by $(1+4d)^2$ increase, so $M/u < 4/11$. Also $3B < 48/49 < x$. Substitution into Lemma F.1, retaining its signed denominator and using (F.4), gives

$$\frac{1}{d}\log \mathbb{E}[p^3P^3] \le -14 + \frac{33}{2}r - \frac{21r}{10u} + d\left[3(1-r) + \frac{9}{2}a(r) - \frac{29-30r+9r^2}{u}\right] + \frac{12M}{u}.$$

Except for the last term, already bounded by $48/11$, the expression increases with d: the square bracket is increasing in u and at $u=1$ is at least $83/2 - (423/5)\theta = 1141/160 > 0$. Put $d = 1/7, u = 11/7$. The resulting function is convex in r, its quadratic coefficient being $(1/7)(441/10 - 63/11) > 0$. At $r=0$ it is $-339/154 < -1/6$. At $r=\theta$ it is

$$-467/64 - 1911/3520 + 57/224 + 301473/71680 - 18737/11264 + 48/11 \le -1/6,$$

using the respective bounds $-29/4, -1/2, 1/3, 17/4, -3/2, 9/2$. This proves the low range.

For $d \ge 1/7$,

$$M^2/d^2 \le 15 - (89/5)r + (49/5)r^2, \qquad \delta/d \ge 17/3 - (11/3)r.$$

The positive function $s(r) = 137/30 - (11/3)r$ satisfies

$$s(r)^2 - [15 - (89/5)r + (49/5)r^2] = 5269/900 - (706/45)r + (164/45)r^2 > 0.$$

The last polynomial decreases on $[0,\theta]$ and has endpoint value $4741/57600 > 0$. Thus $M \le \delta - (11/10)d$, so $P^3 \le e^{-(33/10)d}$. Since $\log x \le (36/11)d$, the resulting logarithmic bound is at most $-3d/110$.

All four ranges $c = 3, 4, 5, c \ge 6$ have now been proved for every d. At $d = 0$ there is equality. Multiplication by the actual equal-pair factors preserves the cubic without changing any edge or orientation.

F.5 The single residual triples with third length at least four

Within this subsection write

$$r = Kg, \quad k = K^2 g, \quad \ell = K^3 g, \quad u = 1 + r/2, \quad H_c = q f_2 h_c.$$

The direction is the same for every parameter value: the unit arm is inward, the two-edge arm is outward throughout, and the length-c arm is inward on its first edge and outward thereafter. The exact identity

$$q f_2 = u^2 - (g + r/2)^2 \le u^2 \tag{F.9}$$

will be used only as an upper bound on the actual product.

F.5.1 Two scalar estimates for a cubic expansion

Suppose $h_c \le m - r + w$ and $|w| \le M$. Set $Y = (m - r + w)u^2$ and

$$B(r) = (m-r)^2 u^6, \qquad C(r) = [3(m-r) + M]u^6.$$

Since $w^3 \le Mw^2$,

$$h_c^3 u^6 \le Y^3 \le [(m-r)u^2]^3 + 3B(r)w + C(r)w^2. \tag{F.10}$$

If $m = 1-\delta$, $0 \le \delta \le 3/4$ and $|r| \le 1/2$, then

$$(m-r)u^2 = 1 - \delta - \delta r - \frac{3+\delta+r}{4} r^2 \le A_0(\delta) := 1 - \delta + \frac{2}{5}\delta^2. \tag{F.11}$$

The real cube is increasing, so this remains valid when $m - r < 0$. The function A_0 is positive and decreasing on $[0, 3/4]$.

For $1/4 \le m \le 1$ and $|r| \le 1/2$,

$$|B(r) - m^2 - m(3m-2)r| \le (9/4)r^2. \tag{F.12}$$

Here is a proof over the full rectangle. Put $v = 1 + r/2$ and

$$A_1(v) = \frac{v^6 - 6v + 5}{4(v-1)^2}, \quad B_1(v) = \frac{v^6 - 1}{2(v-1)}, \quad C_1(v) = v^6.$$

The remainder divided by r^2 is $A_1 m^2 - 2B_1 m + C_1$, with continuous values at $v = 1$. For its lower bound, complete the square in m. The condition $(C_1 + 9/4)A_1 - B_1^2 \ge 0$ is equivalent to

$$(37 - 24v)v^6 + 41 - 54v = (v-1)^2 Q_1(v) \ge 0,$$

$$Q_1(v) = -24v^5 - 11v^4 + 2v^3 + 15v^2 + 28v + 41.$$

On $[3/4, 5/4]$, $Q_1'' < 0$, $Q_1'(3/4) = -85/32 < 0$ and $Q_1(5/4) = 831/256 > 0$. For the upper bound use convexity in m. At $m = 1$, the logarithmic derivative of $(1-r)^2(1+r/2)^6/(1+r)$ is $-r(5+7r)/[2(1-r)(1+r/2)(1+r)]$, so the remainder is at most zero. At $m = 1/4$ its exact polynomial is

$$-17/64 + (41/32)r + (655/256)r^2 + (523/256)r^3 + (865/1024)r^4 + (23/128)r^5 + r^6/64.$$

Replacing r by $|r| \le 1/2$ bounds it by $21769/16384 < 3/2$. This proves (F.12).

If $r^2 \le d$ and $3m + M \le 4 - (123/16)d$, then

$$C(r) \le 45/8. \tag{F.13}$$

For $r \le 0$, use $(4-3r)(1+r/2)^6 \le 4$. For $r \ge 0$, bound by $c_0(r) = (4-3r-(123/16)r^2)(1+r/2)^6$. Its coefficients after the linear term are all negative:

$$\begin{aligned} c_0(t) = 4 &+ 9t - (27/16)t^2 - (389/16)t^3 - (2085/64)t^4 \\ &- (681/32)t^5 - (1973/256)t^6 - (381/256)t^7 - (123/1024)t^8. \end{aligned}$$

At $t = 1/4$, $c_0 = 376791669/67108864$ and $1 < c_0' = 2302911/2097152 < 9/8$. The function is concave, increases up to $1/4$, and has $c_0'' < -64$ for $t \ge 1/4$. Its quadratic tangent therefore bounds it by $376791669/67108864 + 81/8192 < 45/8$.

F.5.2 The third length is four

Assume $d \le 1/4$ and put

$$m = 1 - 2d, \quad w = -(1-d)g - k - \ell, \quad M_0 = 1 - (27/16)d.$$

Then $h_4 = m - r + w$, and skewness gives

$$\begin{aligned} \mathbb{E}w &= -\delta_2, \quad \mathbb{E}(rw) = \delta_3, \\ \mathbb{E}w^2 &= (1-d)^2 d - 2(1-d)\delta_2 + \delta_3 + \delta_4 \\ &\le (1-d)^2 d - (1463/1024 - 2d)\delta_2. \end{aligned}$$

Writing $w = -K[(1-d)\mathbf{1} + r + k]$, the squared reference norm is $1 - 4d + 3d^2 + \delta_2 + \delta_3 \le 1 - (3511/1024)d + 3d^2$. Its difference from M_0^2 is $(55/1024)d - (39/256)d^2 \ge 0$, so $|w| \le M_0$.

We need the sharper bound

$$|B(r) - m^2 - m(3m-2)r| \le (171/80)r^2. \tag{F.14}$$

For $m = 1 - 2d$, the coefficients $b_2, \dots, b_8$ of B are

2	$-5/4 - 3d + 15d^2$
3	$-2 + 5d + 10d^2$
4	$-5/16 + (25/4)d + (15/4)d^2$
5	$13/16 + 3d + (3/4)d^2$
6	$37/64 + (11/16)d + d^2/16$
7	$5/32 + d/16$
8	$1/64$.

For $d \le 1/16$, $|r| \le 1/4$ and these coefficients bound the absolute remainder divided by r^2 by $19/10 + 10027/262144 < 2$. For $1/16 \le d \le 1/4$, $b_4 \ge 0$. Bounding the terms with $d^{3/2}$ and $d^{5/2}$ using $\sqrt{d} \le 1/2$, all terms except $-b_3\sqrt{d}$ are at most

$$P(d) = 5/4 + (99/32)d - (211/32)d^2 + (311/64)d^3 + d^4/16.$$

Here $P'' < 0$, $P'(1/4) = 729/1024 > 0$ and $P(1/4) = 27/16$. The remaining term $(2 - 5d - 10d^2)\sqrt{d}$ has derivative in $\sqrt{d}$ equal to $-50(d - 1/10)(d + 2/5)$ and maximum $7/(5\sqrt{10}) \le 9/20$. Their sum is $171/80$, proving (F.14).

The identity $3m + M_0 = 4 - (123/16)d$ permits (F.13). Also (F.11) gives $a(d) = 1 - 2d + (8/5)d^2$ for the constant factor. Retaining the exact first two moments of w in (F.10) yields

$$\mathbb{E}Y^3 \le a(d)^3 + (45/8)(1-d)^2 d + K(d)\delta_2,$$

$$K = -3m^2 + 3\theta[m(3m-2)]_+ + (513/80)M_0 - (45/8)(1463/1024 - 2d).$$

For $d \le 1/6$ this equals $-139479/40960 + (3429/1280)d + (21/8)d^2 \le -29/12$. For $1/6 \le d \le 1/4$ it is $-189399/40960 + (15909/1280)d - 12d^2$, increasing with value $-92847/40960 < -2$ at $1/4$. Thus $K \le -2$. Use $\delta_2 \ge d^2$ to get

$$\mathbb{E}Y^3 \le a(d)^3 + (45/8)(1-d)^2 d - 2d^2.$$

After subtracting one and dividing by d, the polynomial is

$$-3/8 + (71/20)d - (863/40)d^2 + (672/25)d^3 - (384/25)d^4 + (512/125)d^5.$$

For $d \le 1/4$ it is at most $-3/8 + (18/5)d - 14d^2 \le -201/1400$. By (F.9), $\mathbb{E}H_4^3 \le 1 - (201/1400)d$.

For $d \ge 15/64$, $h_4 \le 2\mu_3 = 2(1-2d)$ and (D.9) give

$$\mathbb{E}H_4^3 \le 8(1-2d)^3(1-\alpha_* d) \le (17/16)^3(133/160) = 653429/655360 < 1.$$

The factors decrease and the two parameter ranges overlap, proving the full length-four case.

F.5.3 The third length is five

For $d \le 3/16$ put

$$\delta = 3d - \delta_2, \quad m = 1 - \delta, \quad a = 1 - 2d, \quad w = -ag + dr - k - \ell - K^4 g.$$

Then $h_5 = m - r + w$, $m \ge 7/16$ and $\gamma d \le \delta \le 3d$, where $\gamma = 83/32$. Exact moments are

$$\mathbb{E}w = -\delta_2 - d^2, \quad \mathbb{E}(rw) = d\delta_2 + \delta_3,$$

$$\mathbb{E}w^2 = a^2 d + (-2 + 4d + d^2)\delta_2 + (3 - 2d)\delta_3 - \delta_4 + \delta_5 \le a^2 d - L(d)\delta_2,$$

$$L(d) = 25/32 - (51/16)d - d^2 \ge 19/128.$$

The reference $w = -K[a\mathbf{1} - dg + r + k + \ell]$ has squared norm $1 - 6d + 8d^2 + d^3 + (3-2d)\delta_2 - \delta_3 + \delta_4 \le a^2 - L(d)d$, so $|w| \le a$.

Use (F.12) with the weaker constant three. The C estimate here is sharper: $3m + a \le 4 - (313/32)d$, whence $C(r) \le 27/5$. For verification, the positive-r comparison polynomial is

$$\begin{aligned}\Phi(t) &= (4 - 3t - (313/32)t^2)(1 + t/2)^6 \\ &= 4 + 9t - (121/32)t^2 - (979/32)t^3 - (5175/128)t^4 \\ &\quad - (1697/64)t^5 - (4951/512)t^6 - (963/512)t^7 - (313/2048)t^8.\end{aligned}$$

For $t \le 1/8$ it is at most $41/8$. For $t \ge 1/8$, $\Phi'' < -39$. At $t = 1/4$, $\Phi = 717976791/134217728 < 107/20$ and $|\Phi'| = 7263027/4194304 < 7/4$. Thus $\Phi \le 107/20 + 49/1248 < 27/5$. Negative r gives the bound four as before.

Since $[m(3m-2)]_+ \le m^2$ and $m \ge 1 - 3d$, the linear contribution satisfies

$$3\mathbb{E}[B(r)w] \le [9a - 3(1 - \theta - d)(1 - 3d)^2]\delta_2 - 3(1 - 3d)^2 d^2.$$

Combining with the second moment and (F.11),

$$\mathbb{E}(h_5 u^2)^3 \le A_0(\gamma d)^3 + (27/5)(1-2d)^2 d + B_*(d)\delta_2 - 3(1-3d)^2 d^2,$$

$$B_*(d) = 3 + (129/10)d - (4581/160)d^2 + 27d^3 \ge 3.$$

Thus δ_2 may be replaced by θd. After subtracting one and dividing by d, the first four coefficients are $-93/80$, $45549/5120$, $-5138011/163840$, $1572999882/26214400$; the last two are $-(12/25)\gamma^5$, $(8/125)\gamma^6$. This is at most

$$-93/80 + 9d - 31d^2 + 61d^3 + 20d^5 \le -93/80 + 9d - 19d^2 \le -183/1280$$

for $d \le 3/16$. It follows that $\mathbb{E}H_5^3 \le 1 - (183/1280)d$ in that range. For $d \ge 3/16$, $h_5 \le 2\mu_4 \le 2 - (83/16)d$, and

$$\mathbb{E}H_5^3 \le [2 - (83/16)d]^3(1 - \alpha_* d) \le (263/256)^3(139/160) < 1. \tag{F.15}$$

Both factors decrease, proving the full length-five case.

F.5.4 A uniform path mean for the large parameter range

We record the all-length estimate

$$\mu_n \le 1-(83/32)d \qquad (n \ge 4). \tag{F.16}$$

This is the common path-mean estimate (A.8), proved from the convolution and the initial means in Appendix A.

F.6 Residual triples with third length at least six

This continues the single-triple calculation of Section F.5 for all remaining third lengths.

F.6.1 Every third length at least six

Fix $c \ge 6$ and first assume $d \le 3/16$. Let

$$\lambda = 19/32, \quad \tau = d-\delta_2, \quad R_0 = S_c - S_2, \quad \delta = (4-d)\tau, \quad m = 1-\delta, \quad a = 1-3\tau,$$

$$w = -ag + 2\tau r + \tau k - R_0, \qquad M = 1-(5/2)\tau.$$

Apply A^* to the positive convolution for f_{c-1}. The identity $A^*S_{c-1} = 2-\beta_c-S_c$, retaining h_1, h_2, h_3, gives

$$h_c \le 2-\tau-S_c-\tau(h_1+h_2+h_3) = m-r+w.$$

Here $h_1+h_2+h_3 = 3-d-3g-2r-k$. We have $\lambda d \le \tau \le d$, $\delta \le 3/4$, $1/4 \le m \le 1$ and $7/16 \le a \le 1$.

Put $V = KG_{c-2}(K)g$ and $U = a\mathbf{1} - 2\tau g - \tau r + V$. Then $R_0 = KV$, $w = -KU$, $\|V\|_2^2 \le \delta_2$ and $\|R_0\|_2^2 \le \delta_3$. Finite alternating sums give

$$\mathbb{E}R_0 \ge \delta_2-\delta_3, \quad -\delta_3 \le \mathbb{E}(rR_0) \le 0, \quad \mathbb{E}(gR_0) \le -\delta_2+\delta_3, \quad \mathbb{E}(kR_0) \ge \delta_3-\delta_4,$$
$$\mathbb{E}V \le -d+\delta_2, \quad \mathbb{E}(gV) \ge -\delta_2, \quad \mathbb{E}(rV) \ge \delta_2-\delta_3.$$

The indicated initial terms occur for every $c \ge 6$. Therefore

$$\mathbb{E}w \le -2\tau d - \delta_2 + \delta_3, \qquad 2\tau\delta_2 \le \mathbb{E}(rw) \le 2\tau\delta_2+\delta_3,$$

$$\mathbb{E}w^2 \le a^2 d + [-2a+2a\tau+4\tau^2]\delta_2 + [1+2a+\tau^2+2\tau]\delta_3 + 2\tau\delta_4 \le a^2 d - L(\tau)\delta_2,$$
$$L(\tau) = 25/32 - (3433/512)\tau + (51/32)\tau^2.$$

The full row expansion has the same coefficient pattern one moment lower: $\|U\|_2^2 \le a^2 - L(\tau)d$. This proves $|w| \le M$. If $L \ge 0$, use $\|U\|_2 \le a \le M$. If $L < 0$, use $d \le 3/16$ to obtain

$$M^2 - [a^2 - Ld] \ge 75/512 - (2107/8192)\tau - (1255/512)\tau^2 \ge 99/8192 > 0.$$

The last quadratic decreases on $[0, 3/16]$.

Apply (F.10), (F.11) and (F.12). Since $3m+M \le 4-(223/16)\tau \le 4-(4237/512)d \le 4-(123/16)d$, (F.13) also applies. Write $b_1 = m(3m-2)$. Keeping the lower covariance when $b_1 < 0$ yields

$$\mathbb{E}[B(r)w] \le -2m^2\tau d + [-\lambda m^2 + 2\tau b_1 + \theta(b_1)_+ + (9/4)M]\delta_2.$$

Indeed $\mathbb{E}(rw) = 2\tau\delta_2 + t$ with $0 \le t \le \delta_3$. Thus its linear contribution is at most $2\tau b_1\delta_2 + (b_1)_+\delta_3$. Combining terms gives

$$\mathbb{E}[h_c^3 u^6] \le A_0(\delta)^3 + (45/8)a^2 d - 6m^2\tau d + K(d,\tau)\delta_2,$$

$$K = -3\lambda m^2 + 6\tau b_1 + 3\theta(b_1)_+ + (27/4)M - (45/8)L(\tau).$$

Put $Q_0 = (27/4)M - (45/8)L = 603/256 + (85365/4096)\tau - (2295/256)\tau^2$. Then $Q_0 \ge 603/256$, $m^2 \le 1$ and $b_1 \ge -1/3$ imply $K \ge 603/256 - 57/32 - 2\tau \ge 51/256 > 0$. We may replace δ_2 by θd, defining the upper bound

$$P(d,\tau) = A_0(\delta)^3 + (45/8)a^2 d - 6m^2\tau d + \theta d K(d,\tau).$$

For fixed d this function decreases on $\lambda d \le \tau \le d$. Put $b = 4 - d$, so $m = 1 - b\tau$. At the only corner $m = 2/3$, K is continuous. For $m \ge 2/3$,

$$K = (15/8 + 18\tau)m^2 - (39/16 + 12\tau)m + Q_0,$$

$$K_\tau - Q_0' = 54m^2 - 60m + 12 + b[39/16 - (15/4)m] \le 6.$$

For $m \le 2/3$,

$$K = (-57/32 + 18\tau)m^2 - 12\tau m + Q_0, \quad K_\tau - Q_0' = 54m^2 - 60m + 12 + (57/16)bm \le 11/2.$$

These follow by convexity in m and the endpoints $1/4, 2/3, 1$, using $b \le 4$. Since $Q_0' < 21$, $K_\tau \le 27$. Also $A_0'(\delta) < 0$ and $\partial_\tau(-6m^2\tau d) = -6db_1$. Hence

$$P_\tau \le d[-6(45/8)(7/16) + 2 + 27\theta] = -(115/64)d < 0.$$

Continuity at the corner proves $P(d,\tau) \le P(d,\lambda d)$.

At $\tau = \lambda d$, $\delta = \lambda d(4-d) \le 3477/8192 < 3/7$. The two scalar inequalities required there are

$$A_0(z)^3 \le 1 - 3z + (21/5)z^2 - (11/4)z^3, \qquad (1 - 4z + 3z^2)_+ \le 1 - (7/3)z.$$

For the first, the residual coefficient $17/5 - (42/25)z + (12/25)z^2 - (8/125)z^3$ decreases and at $7/16$ is $11/4 + 97/64000$; the second follows by subtraction and nonnegativity of its right side on $[0, 3/7]$. Substitution and collection now give

$$[\mathbb{E}(h_c^3 u^6) - 1]/d \le Q(d),$$

$$\begin{aligned} Q(d) = &-6321/8192 + (189543791/20971520)d \\ &- (1392699791/41943040)d^2 + (22776573/655360)d^3 \\ &- (9815229/1048576)d^4 + (445835/524288)d^5. \end{aligned}$$

The coefficients are bounded by $-77/100$, $226/25$, $-166/5$, $174/5$, 0, 1. For $d \le 3/16$, $(174/5)d^3 + d^5 \le (33/5)d^2$, so

$$Q(d) \le -77/100 + (226/25)d - (133/5)d^2 \le -129/66500.$$

Together with (F.9), this proves $\mathbb{E}H_c^3 \le 1 - (129/66500)d$ for $d \le 3/16$ and every $c \ge 6$.

For $d \ge 3/16$, (F.16) at $n = c - 1$ gives $h_c \le 2\mu_{c-1} \le 2 - (83/16)d$. The decreasing-factor estimate (F.15) therefore applies without change, proving the same fixed-direction cubic for every remaining d. All equal-length pairs may be restored using (2.5), without changing any original edge count.

F.7 The critical residual triple of lengths one two and three

We prove the last root inequality

$$\mathbb{E}[(qf_2h_3)^3] \le 1. \tag{F.17}$$

Only the bounded skew-kernel model and the spectral bound of Lemma D.1 are needed for this root proof. Within this subsection

$$r = Kg, \quad k = K^2 g, \quad D = \mathbb{E}r^2, \quad E_3 = \mathbb{E}k^2.$$

We will prove the quantitative bounds

$$\begin{aligned} &\mathbb{E}[(qf_2h_3)^3] \le 1 - (99/1280)d, && 0 \le d \le 1/8, \\ &\mathbb{E}[(qf_2h_3)^3] \le 1 - (67/6000)d, && 1/8 \le d \le 8/29, \\ &\mathbb{E}[(qf_2h_3)^3] \le 3943/4000, && 8/29 \le d \le 1/3. \end{aligned} \tag{F.18}$$

Each line concerns the same actual orientation.

F.7.1 A geometric bound and a universal reversal

Skewness gives

$$\mathbb{E}g = \mathbb{E}k = \mathbb{E}(gr) = \mathbb{E}(rk) = 0, \quad \mathbb{E}r = -d, \quad \mathbb{E}(gk) = -D, \quad d^2 \le D \le \theta d, \quad E_3 \le \theta D.$$

Use the pointwise constraint (2.2). Thus $c = r + (1 - g^2)/2 \ge 0$ and $\mathbb{E}c = (1 - 3d)/2$. Also $|k| \le \sqrt{D} \le \sqrt{13/96} < 3/8$.

The actual short messages are $f_2 = 1 + g + r$ and $h_3 = 1 - d - g - r - k$. The target at the universally reversed kernel is

$$F = (qf_2h_3)(-K) = p(1 - g + r)(1 - d + g - r + k).$$

It suffices to prove $\mathbb{E}F^3 \le 1$ for every legal K, then substitute $-K$. Set

$$a = 1 - d/2, \quad m = a + k/2, \quad w = -r + (k - d)/2, \quad z = g + w.$$

The last two actual factors of F are $m - z, m + z$, so $F = p(m^2 - z^2)$ and $|z| \le m$. The geometric bounds give

$$w \ge -(1 - g^2)/2 - 17/48, \quad m \ge 31/48, \quad m \le 1 + \theta/8 = 269/256, \quad m^2 < 10/9.$$

Consequently

$$F^3 \le m^6 p^3 e^{-27z^2/10}. \tag{F.19}$$

F.7.2 A scalar quadratic majorant

For $-1 \le g \le 1$ and $w \ge -(1 - g^2)/2 - 17/48$, put $z = g + w$. We claim

$$\frac{p^3}{1 + 27z^2/10} \le Q = 1 + 3g + (12/5)g^2 + (3/2)w^2. \tag{F.20}$$

Let

$$Q_0 = 1 + 3g + (12/5)g^2 = (12/5)(g + 5/8)^2 + 1/16 > 0,$$

$$B = 3/2 + (27/10)Q_0 = 21/5 + (81/10)g + (162/25)g^2 > 0.$$

Expanding and using $Q \geq Q_0$ bounds the difference between the two sides after multiplication by the denominator below by

$$R_g(z) = g^2(9/10 - g) - 3gz + Bz^2.$$

The following four cases prove nonnegativity. For $g \leq -3/5$, directly $Q - p^3 = g^2(-3/5 - g) + (3/2)w^2 \geq 0$. For $-3/5 \leq g \leq 0$, put

$$\Delta(g) = (9/10 - g)B - 9/4 = 153/100 + (309/100)g - (567/250)g^2 - (162/25)g^3.$$

Writing $-g = 1/2 + y$, $-1/2 \leq y \leq 1/10$, gives $\Delta = 57/250 - (249/500)y + (1863/250)y^2 + (162/25)y^3 \geq 1/5 - |y|/2 + 4y^2 \geq 59/320$. Then $R_g(z) = B[z - 3g/(2B)]^2 + (g^2/B)\Delta \geq 0$. For $0 \leq g \leq 7/10$, $\Delta'' < 0$ and its endpoint values are $153/100$ and $1122/3125$, so the same square completion works.

For $g = 7/10 + y$, $y \geq 0$, the assumed lower bound on w gives

$$z \geq T = 109/1200 + (17/10)y + y^2/2, \quad B \geq 13 + 17y, \quad T \geq 9/100 + (17/10)y + y^2/2.$$

Thus $BT - 3g/2 \geq 3/25 + (103/5)y > 0$ and R_g increases for $z \geq T$. The part of $R_g(T)$ outside BT^2 equals $-371/4000 - (1621/400)y - (147/20)y^2 - (5/2)y^3$. Multiplication of the positive lower bounds yields

$$BT^2 \geq 1053/10000 + (41157/10000)y + (3757/100)y^2 + (221/10)y^3.$$

Their sum is at least $251/20000 + (79/1250)y + (1511/50)y^2 + (98/5)y^3 > 0$. This proves (F.20). Since $e^{-u} \leq 1/(1+u)$ for $u \geq 0$, (F.19) implies

$$\mathbb{E}F^3 \leq \mathbb{E}[m^6 Q]. \tag{F.21}$$

F.7.3 Centered moments and the Taylor remainder

Introduce

$$S = \mathbb{E}g^3, \quad M = \mathbb{E}(g^2 k), \quad N = \mathbb{E}(gk^2), \quad B_{22} = \mathbb{E}(g^2 k^2), \quad \rho_0 = D - d^2,$$

$$s = k + dg = K(r + d\mathbf{1}), \quad \sigma = \mathbb{E}s^2, \quad G = \mathbb{E}(gs^2).$$

Since $k = s - dg$, the exact relation is

$$N = -2dM - d^2 S + G. \tag{F.22}$$

The row and spectral bounds give $|s|^2 \leq \rho_0$ and $\sigma \leq \theta\rho_0$. Centering s^2, and using $\mathbb{E}g = 0$, gives

$$|G|^2 \leq d(\mathbb{E}s^4 - \sigma^2) \leq d(\rho_0\sigma - \sigma^2).$$

Since $\theta < 1/2$ and $u(1-u)$ increases up to $1/2$,

$$|G| \leq \sqrt{d\theta(1-\theta)}(D - d^2) \leq (\sqrt{d}/2)(D - d^2). \tag{F.23}$$

If $\rho_0 = 0$, then $s = 0$ and this holds directly.

The remaining bounds are

$$|M| \le d\sqrt{D}, \quad |S| \le d - d^2, \quad B_{22} \le dD, \quad |N| \le D\sqrt{\theta d},$$

$$\mathbb{E}w^2 = D + E_3/4 - 3d^2/4 \le (1 + \theta/4)D - 3d^2/4. \tag{F.24}$$

The score bound for S follows from the expectations of $(1-g)(g+d)^2$ and $(1+g)(g-d)^2$. The nonnegative cycle remainder gives stronger large-d estimates:

$$|S| \le 1 - 3d, \qquad |M| \le (1-d)(1-3d)/2. \tag{F.25}$$

Indeed $\mathbb{E}(gc) = -S/2$; with $b = 1 - g^2$, skewness gives $\mathbb{E}[cKb] = M$, and $|Kb| \le \mathbb{E}b = 1 - d$.

Expand $T = m^6 = (a + k/2)^6$:

$$T = a^6 + 3a^5 k + (15/4)a^4 k^2 + R_3, \quad |R_3| \le C_3 |k|^3,$$

$$C_3 = (5/2)a^3 + (15/16)a^2\sqrt{D} + (3/16)aD + D^{3/2}/64.$$

The bounds $\sqrt{D} < 3/8$ and $a = 1 - d/2$ give

$$C_3 < 3 \ (d \ge 0), \qquad C_3 < 12/5 \ (d \ge 1/8), \qquad C_3 < 2 \ (d \ge 1/4).$$

For example, with denominator 32768 the three upper numerators are 94331, 78462 and 64483. Also $T \le 27/20$ always, since $(269/256)^6 < 27/20$. For $d \ge 8/29$, $\sqrt{\theta d} \le d + 3/50$, so $m \le 103/100$ and $T < 6/5$. The squared comparison is the increasing polynomial $d^2 - (229/800)d + 9/2500$, positive at $11/40 < 8/29$.

Because $Q_0 > 0$, the absolute remainder is bounded with its sign intact by

$$\mathbb{E}(R_3 Q_0) \le C_3\sqrt{D}\,\mathbb{E}(k^2 Q_0) \le R(d)D,$$

where, for a valid constant $\bar{C} \ge C_3$,

$$t = \sqrt{d}, \qquad R(d) = \bar{C}(16/25)t[\theta + 3(16/25)t + (12/5)d].$$

The three choices $\bar{C} = 3, 12/5, 2$ give respectively

$$\begin{array}{c|l} R_S & (39/50)t + (2304/625)t^2 + (576/125)t^3 \\ R_M & (78/125)t + (9216/3125)t^2 + (2304/625)t^3 \\ R_L & (13/25)t + (1536/625)t^2 + (384/125)t^3. \end{array}$$

Using (F.22) in the principal quadratic term yields the master estimate

$$\begin{aligned} \mathbb{E}F^3 \le\ & a^6[1 + (12/5)d] - 9a^5 D + (15/4)a^4 E_3 \\ & + A_M M - (45/4)a^4 d^2 S + (45/4)a^4 G \\ & + 9a^4 B_{22} + R(d)D + (3/2)U[(1 + \theta/4)D - 3d^2/4], \end{aligned} \tag{F.26}$$

where $A_M = (9/10)a^4(8 - 29d)$ and U is the applicable bound on T. Before substituting (F.22), the quadratic term is $(15/4)a^4[E_3 + 3N + (12/5)B_{22}]$ and the linear term is $-9a^5 D + (36/5)a^5 M$. This verifies every sign in (F.26).

F.7.4 A common polynomial for the small and middle ranges

For $d \le 8/29$, $A_M \ge 0$. Use $M \le d\sqrt{D} \le D/4 + d^2$, (F.23), $|S| \le d - d^2$, $B_{22} \le dD$, $E_3 \le \theta D$, and $U = 27/20$. Then

$$\mathbb{E}F^3 \le A_0(d) + V(d)D, \tag{F.27}$$

$$A_0 = a^6[1 + (12/5)d] + a^4d^2[36/5 - (297/20)d - (45/8)t - (45/4)d^2] - (243/160)d^2,$$

$$V = a^4[-3633/640 + (45/8)t + (279/40)d] + R(d) + 11421/5120.$$

The negative term $-(45/8)a^4td^2$ is the centered contribution of G and is retained. With $R = R_M$ and $H_M = A_0 + \theta dV$, direct collection gives

$$\begin{aligned} H_M = a^4[1 - (18557/20480)t^2 + (585/256)t^3 + (10091/1280)t^4 - (45/8)t^5 \\ - (57/4)t^6 - (45/4)t^8] + (148473/163840)t^2 + (507/2000)t^3 \\ - (32067/100000)t^4 + (936/625)t^5. \end{aligned}$$

Write $(H_M - 1)/t^2 = \sum_{j=0}^{14} q_j t^j$. All coefficients and the upper bounds used are

j	q_j	upper bound
0	$-327663/163840$	$-199/100$
1	$81237/32000$	$51/20$
2	$69600837/6400000$	$109/10$
3	$-695817/80000$	$-43/5$
4	$-261131/8192$	$-1593/50$
5	$7515/512$	$147/10$
6	$242409/8192$	$148/5$
7	$-4905/512$	0
8	$-188313/65536$	0
9	$12105/4096$	3
10	$-189589/20480$	0
11	$-45/128$	0
12	$303/64$	$19/4$
13	0	0
14	$-45/64$	$0.$

For $t \le 8/15$, the positive terms of degree at least five are bounded by

$$t^4[(147/10)(8/15) + (148/5)(8/15)^2 + 3(8/15)^5 + (19/4)(8/15)^8] \le (8211/500)t^4.$$

Combining with q_4 and discarding the remaining negative terms gives

$$(H_M - 1)/t^2 \le L_M(t) = -199/100 + (51/20)t + (109/10)t^2 - (43/5)t^3 - (77/5)t^4.$$

With $R = R_S$, only q_1, q_2, q_3 change, to $16653/6400$, $71517765/6400000$, $-133173/16000$. Thus the corresponding $H_S = A_0 + \theta dV$ satisfies

$$(H_S - 1)/t^2 \le L_S(t) = -199/100 + (261/100)t + (56/5)t^2 - (83/10)t^3 - (77/5)t^4.$$

F.7.5 The small range

Suppose $d \le 1/8$ and use R_S. Since $d^2 \le D \le \theta d$, (F.27) is bounded by the larger of $A_0 + Vd^2$ and H_S. First $V \le 1$ on this interval. Indeed $B(t) = -3633/640 + (45/8)t + (279/40)t^2$ is negative and increasing, so $a^4 B$ increases, as does R_S. At $d = 1/8$, use $t \le 99/280$:

$$B \le -563/200, \quad a^4 \ge 193/250, \quad R_S < 941/1000, \quad 11421/5120 < 2231/1000.$$

These give $V < 49941/50000 < 1$. Moreover

$$\begin{aligned} a^6[1 + (12/5)d] &= 1 - (3/5)d - (69/20)d^2 + (13/2)d^3 - (81/16)d^4 \\ &\quad + (33/16)d^5 - (139/320)d^6 + (3/80)d^7 \\ &\le 1 - (3/5)d - (5/2)d^2. \end{aligned}$$

The last comparison follows by dividing the difference by d^2 and bounding it by $-19/20 + 13/16 + 33/8192 + 3/2621440 < 0$. Dropping negative terms in the bracket of A_0 now gives

$$A_0 + Vd^2 \le 1 - (3/5)d + (669/160)d^2 \le 1 - (99/1280)d.$$

For the upper endpoint, $t \le \sqrt{1/8} < 3/8$ and $L_S'(t) \ge 261/100 + (22/5)t > 0$ on $[0, 3/8]$. Thus $L_S(t) \le L_S(3/8) = -18277/102400 < -99/1280$. Both endpoints prove the first range of (F.18).

F.7.6 The middle range

For $1/8 \le d \le 8/29$ use R_M. The coefficient V is positive: $B(t) \ge -375/128$ and $a^4 \le (15/16)^4$ imply $a^4 B > -23/10$; $R_M \ge 26/125$ and $11421/5120 > 11/5$ give $V > 27/250$. Hence D may be replaced by θd. Here $1/3 \le t \le \sqrt{8/29} < 8/15$ and

$$L_M''(t) = 109/5 - (258/5)t - (924/5)t^2 < -15.$$

At $t = 1/2$, $L_M = -11/400$ and $L_M' = -7/10$. Strong concavity gives

$$L_M(t) \le -11/400 - (7/10)(t - 1/2) - (15/2)(t - 1/2)^2 \le -67/6000.$$

This proves the second range of (F.18).

F.7.7 The large range and completion

For $d \ge 8/29$, $A_M \le 0$. Use the cycle bounds (F.25), R_L and $U = 6/5$ in (F.26). The coefficient of D is

$$V_L = a^4[-957/128 + (27/2)d + (45/8)t] + R_L + 1269/640 > 0.$$

Indeed the bracket is at least $-165/128$ for $d \ge 1/4, t \ge 1/2$, and $1269/640 - 165/128 = 111/160 > 0$. Replacing D by θd and collecting gives

$$\mathbb{E}F^3 \le a^4 I(d) + O(d),$$

$$I = P(d) + (45/8)d^{3/2}(\theta - d), \quad P(d) = -13/5 + (528643/20480)d - (15493/320)d^2 + 6d^3,$$

$$O = (16497/20480)d + (169/800)d^{3/2} - (879/2500)d^2 + (156/125)d^{5/2}.$$

For verification, the cycle M term expands as $-18/5 + (549/20)d - 63d^2 + (783/20)d^3$; the two terms containing $\sqrt{d}$ inside a^4 combine to the displayed one in I.

Since $P'' < -84$, $P(2/7) = 3381267/3512320 < 97/100$ and $|P'(2/7)| = 385389/1003520 < 1/2$, a quadratic tangent gives $P(d) \le 97/100 + 1/672 < 39/40$. Also $d(\theta - d)$ decreases on this interval and has maximum $121/3364$ at $8/29$. Using $\sqrt{d} \le 7/12$ bounds the remaining term of I by $38115/322944 < 1/8$. Thus $I \le 11/10$ and $a^4 I \le (25/29)^4(11/10) \le 99/160$. For O, use $d \le 1/3$ and $d^2 \ge 64/841$ to bound its four terms respectively by $27/100$, $21/500$, $-13/500$, $81/1000$. Hence $O \le 367/1000$ and

$$\mathbb{E}F^3 \le 99/160 + 367/1000 = 3943/4000 < 1.$$

This proves the third range. If $d = 0$, all arm messages equal one. The three closed ranges exhaust $[0, 1/3]$. The argument is universal in K, so substituting $-K$ proves (F.17) for the original fixed message qf_2h_3. Multiplying by any actual equal-pair product $E_0 \le 1$ preserves the cubic. Theorem C.6 and the exact assembly in Section C.2 therefore apply to the last residual family as well, completing the proof of the three-branch theorem.

G Path and singleton moments

These same-direction path and singleton estimates provide the boundary weights used by both skeletons. The fifth and cubed weights are kept distinct.

G.1 Oppositely weighted fourth moments of paths

Throughout this proof set $c = 19/32$, $s = K^2 g$, $\tau = d - \delta_2$ and $D_j = \delta_j/(1-\theta^2)$. Write ν for the spectral measure of $-K^2$ at $\mathbf{1}$. For every $n \ge 3$ we prove $\mathbb{E}[qf_n^4] \le 1$ and its full reversal $\mathbb{E}[p\bar{f}_n^4] \le 1$. In particular, if an actual terminal message satisfies $0 \le F \le 1$, positivity gives $T^L F \le \bar{f}_L$ and hence $\mathbb{E}[p(T^L F)^4] \le 1$ for $L \ge 3$.

Improved finite geometric sums. For all $n \ge 3$ and $0 \le x \le \theta$,

$$|G_n(i\sqrt{x})|^2 \le 1 - 7x/15, \qquad \|S_n - 1\|_2^2 \le d - 7\delta_2/15. \tag{G.1}$$

For odd n the squared modulus is $(1+x^n)/(1+x)$ and its loss divided by x is at least $1 - x \ge 19/32$. For $n \equiv 0 \pmod 4$ it is $(1-x^{n/2})^2/(1+x)$, with loss at least $x/(1+x)$. For $n \equiv 2 \pmod 4$, $n \ge 6$, the loss divided by x is at least $(1-2x^2-x^5)/(1+x)$; comparison at θ gives $8 - 7\theta - 30\theta^2 - 15\theta^5 = 1311885/33554432 > 0$. Integrate at the vector g for the second assertion.

The three-edge path. Put $z = f_3 - 1 = g + r + s - d$. The exact moments and row bound give

$$\mathbb{E}[qz] = -3d + \delta_2, \quad \mathbb{E}z^2 = d - \delta_2 + \delta_3 + 3d^2, \quad -1 \le z \le g + \sqrt{45d/32} - d.$$

On $d \le 1/6$ the final correction is increasing and less than $8/25$. Thus $q(6 + 4z + z^2) \le M(q)$, where $M(q) = q^3 - 166q^2/25 + 8139q/625$. On $[0, 2]$, $M'' \le -32/25$, $M(3/2) = 39843/5000$ and $M'(3/2) = -369/2500$, so its quadratic tangent gives $M < 8$. Expansion of $(1+z)^4$ yields

$$\mathbb{E}[qf_3^4] \le 1 - 4d - 4\delta_2 + 8\delta_3 + 24d^2 \le 1 - 4d + 93d^2/4 \le 1 - d/8.$$

For the other range put $\mu = 1 - 2d$ and $v = f_3 - \mu$. Since $\|S_2\|_2^2 = 1 - d + \delta_2 \le 1$, the row bound gives $|S_3 - 1| \le 1$, so $v \le 1 + d$ and $\mathbb{E}v^2 = d - \delta_2 + \delta_3 - d^2$. The inequality $v^3 \le (1+d)v^2$ gives

$$\mathbb{E}f_3^3 \le (1-2d)^3 + (4-5d)(d - \delta_2 + \delta_3 - d^2) \le 1 - 43d/24.$$

Also $\sqrt{qf_3} \le 1 - d/2 + \sqrt{45d/32}/2 < 113/96$, so $qf_3 < 7/5$. For $d \ge 1/6$ it follows that $\mathbb{E}[qf_3^4] \le (7/5)(1 - 43d/24) \le 707/720$.

Paths of length at least four, $d \le 4/19$. For fixed $n \ge 4$, let $a = \beta_n$, $b = \beta_{n-1}$, $\alpha = \beta_{n+1}$, $\eta = \beta_{n+2}$ and $Y = S_n - a - bp \ge f_n > 0$. Set $\mu = \mathbb{E}Y$ and $w = Y - \mu$. All four coefficients lie in $[\tau, d]$, so

$$1 - 3d \le \mu \le m = 1 - 3cd, \quad \mathbb{E}[gw] = \eta - bd \ge h_0 = cd - d^2,$$
$$\mathbb{E}w^2 \le d - 7\delta_2/15 - 3\tau^2 + d^3, \qquad Y \le p + \sqrt{d} - cd - cdp.$$

Equation (A.12) bounds the fourth coefficient by $7 - 10d$. Thus

$$\mathbb{E}[qY^4] \le \mu^4 - 4h_0\mu^3 + (7 - 10d)\mathbb{E}w^2.$$

The negative first-order term must be kept with the fourth power. The function $u^4 - 4h_0u^3$ increases for $\mu \le u \le m$, since $u - 3h_0 \ge 1 - 153d/32 + 3d^2 \ge 365/2888 > 0$. With $a_1 = 7/15 + 3c^2$ this gives the polynomial upper bound

$$R(d) = m^3(1 - 7cd + 4d^2) + (7 - 10d)(d - a_1d^2 + d^3).$$

Its expansion is

$$R - 1 = -5d/2 + C_2d^2 + C_3d^3 + C_4d^4 - 108c^3d^5,$$
$$C_2 = 69c^2 - 139/15 < 16, \quad C_3 = 35/3 + 30c^2 - 36c - 216c^3 = -544915/12288 < -44,$$
$$C_4 = 108c^2 + 189c^4 - 10 < 52.$$

Hence $(R-1)/d \le -5/2 + 16d - (628/19)d^2$. This increases on $[0, 4/19]$ to $-8183/13718 < -1/2$, proving $\mathbb{E}[qf_n^4] \le 1 - d/2$.

The remaining large parameters. For $n = 4$, $R_4 = r + s + K^3g - dg$ satisfies $\|R_4\|_\infty^2 \le d + 77d^2/32$. The positive quadratic $(1/6 + 2d)^2 - (d + 77d^2/32) = 1/36 - d/3 + 51d^2/32$ gives $qf_4 \le 169/144$. For the ordinary cubic use $\alpha = d - \delta_2$, $\beta = d$, $Y = S_4 - \beta$. The spectral integral

$$C = \|S_4 - 1\|_2^2 = \int x(1 - x^2)^2/(1 + x)\, d\nu \le \alpha$$

and $|S_4 - 1| \le 1$ give

$$\mathbb{E}f_4^3 \le (1 - \alpha - \beta)^3 + (4 - 2\alpha - 3\beta)(\alpha - \alpha^2) \le 1 - 4d/3.$$

For the last step write $\delta = \alpha - \beta = -\delta_2 \le 0$; the extra terms are $\delta(1 - 3\beta) - 3(1 - \beta)\delta^2 + \delta^3 \le 0$. Consequently $\mathbb{E}[qf_4^4] \le 6929/8208$ for $d \ge 4/19$. For every $k \ge 4$ the convolution gives $\mu_k \le 1 - 19d/8$: check $k = 4, 5$ from their exact formulas, and for $k \ge 6$ retain f_0, f_1, f_2, f_3 to obtain $1 - \mu_k \ge \tau(5 - 3d) \ge 4\tau$. Thus $f_n \le 2\mu_{n-1} \le 1$ whenever $n \ge 5$ and $d \ge 4/19$. These ranges complete the path theorem and its stated propagation corollary.

G.2 The fifth-weight estimate at length two

Put $P = qf_2$ and $r = Kg$. Since the actual nonnegative factors have $q + f_2 = 2 + r$, $P \le (1 + r/2)^2$. For $0 < d \le 1/4$, the intermediate Hermite bound proved in Section H.8 is

$$\mathbb{E}[p(1 + r/2)^8] \le 1 - d/360.$$

This is the bound on the auxiliary envelope itself, preceding the final bound on P. Under $K \mapsto -K$, $r = K^2\mathbf{1}$ is unchanged and p becomes q. Thus $\mathbb{E}[qP^4] \le 1 - d/360$ in this range.

For $1/4 \le d \le 1/3$, let $c = pq + 2r \in [0, 2pq]$. The function $q^5(p^2 + c)^4/16$ is convex in c; its endpoint chord gives

$$qP^4 \le B(g) + L(g)c, \quad B = p^8q^5/16, \quad L = p^3q^4[(p + 2q)^4 - p^4]/32.$$

The intercept maximum occurs at $(p, q) = (16/13, 10/13)$, giving $B < 9/100$. For the slope discard the negative fourth power and put $G(q) = q^4(2-q)^3(2+q)^4/32$. The logarithm is strictly concave and has derivative $(16 + 2q - 11q^2)/[q(2 - q)(2 + q)]$. Its unique positive zero lies in $(13/10, 131/100)$, where the numerator has endpoint values $1/100, -2571/10000$. The tangent at $13/10$, whose logarithmic derivative is $10/3003 < 1/300$, gives

$$G(q_*) \le G(13/10)\frac{30000}{29999} < \frac{3631}{1000}\frac{30000}{29999} < 91/25.$$

Here $G(13/10) = 11617783760583/3200000000000$. Since $\mathbb{E}c = 1 - 3d$, the full large interval satisfies $\mathbb{E}[qP^4] \le 9/100 + (91/25)(1 - 3d) \le 1$. We have proved $\mathbb{E}[q^5 f_2^4] \le 1$ with the actual outward path direction. The length-one case is $q^5p^4 \le q$; multiplication by E_0 and full reversal supply the complete decorated and reversed statements.

G.3 Singletons with the fifth boundary weight

We prove $\mathbb{E}[q^5 f_a^4] = \mathbb{E}[q(qf_a)^4] \le 1$ for $a \ge 3$. Together with Section G.2 and $q^5p^4 = q(pq)^4 \le q$, this gives (5.5). The auxiliary pair integral has weight q throughout.

Lengths at least six. Put $\alpha = 19/32$, $\epsilon = \alpha d$, $P = qf_a$ and

$$T_0 = S_a - S_1 = KG_{a-1}(K)g, \quad V_0 = 3g + 2r + s, \quad R = T_0 - \epsilon V_0, \quad m_0 = 1 - 2\alpha d + \alpha d^2/2.$$

The actual convolution and the unweighted envelopes of Section H.5 give

$$q + f_a \le 2m_0 + R, \quad P \le (m_0 + R/2)^2,$$
$$\|R\|_\infty^2 \le U = d + 4d^2, \quad V := \mathbb{E}R^2 \le dW, \quad W = 11/32 + 33d/80 + 3d^2,$$
$$P \le (13/12)^2, \qquad P \le 1 \quad (d \ge 4/19).$$

The last bound includes $a = 6$, whose separate threshold $16/83$ is smaller. The alternating tails give $\beta_{a+1}, \beta_{a+2} \ge d - \delta_2 + \delta_3 - \delta_4$. Consequently

$$\mathbb{E}[qT_0] = d - \beta_{a+1} - \beta_{a+2} \le -d + 2\delta_2 - 2\delta_3 + 2\delta_4 \le -3d/8,$$
$$\mathbb{E}[qV_0] = -5d + \delta_2, \qquad \mathbb{E}[qR] \le -3d/8 + 95d^2/32.$$

Indeed $-1 + 2x - 2x^2 + 2x^3$ increases on $[0, \theta]$ and its endpoint value is $-6283/16384 < -3/8$. For $V_q = \mathbb{E}[qR^2]$, Cauchy–Schwarz gives $V_q/d \le W + \sqrt{UW}$. On $d \le 4/19$ we have $3/4 \le m_0 \le m = 1 - 9d/8$. Apply the finite binomial rule (A.2) with $r = 8$ and $r = 6$, using B_r as defined in (A.1). When the mean is nonpositive, discard its linear contribution before increasing the center. When $\mathbb{E}[qR] \le \ell d$ and $\ell \ge 0$, the cubic main term is bounded by $(1 - cd)^5(1 - bd)$, where $c = 9/8$, $b = c - 3\ell$. Its second derivative is $5c(1 - cd)^3(4c + 2b - 6bcd) > 0$ on the intervals used. For

$d \le 1/16$, take $M = 9/32$, $V_q/d < 9/16$, $B_8 < 75/8$. The endpoint certificate $(119/128)^8 < 3/5$ gives

$$\mathbb{E}[qP^4] \le (1 - 9d/8)^8 + 675d/128 \le 1 - 721d/640,$$

since $3/5 + 675/2048 = 9519/10240 < 1$ and the function is convex. For the cubic intervals the complete rational inputs are

I	$\bar{m}$	M	V_q/d	B_6	ℓ
$[1/16, 1/8]$	$15/16$	$7/16$	$3/4$	4	0
$[1/8, 5/32]$	$7/8$	$17/32$	$7/8$	$10/3$	$3/32$
$[5/32, 3/16]$	$5/6$	$23/40$	$19/20$	3	$3/16$
$[3/16, 4/19]$	$4/5$	$5/8$	$26/25$	$21/8$	$1/4$

The endpoint upper bounds for the convex main term plus $dB_6(V_q/d)$ are, in row order,

$$(17/20, 19/24), \quad (1247/1536, 6167/7680),$$
$$(1037/1280, 3197/3840), \quad (7889/9600, 1169/1425).$$

Thus $\mathbb{E}[qP^3] \le 17/20$ and $\mathbb{E}[qP^4] \le (169/144)(17/20) = 2873/2880 < 1$ there. For $d \ge 4/19$ use $P \le 1$.

Length four. For $P_4 = qf_4$, $R_4 = r + s + K^3 g - dg$ and the exact center $m = 1 - d$, Section H.6 supplies the unweighted bounds, while the actual mean is

$$q + f_4 = 2m + R_4, \quad \mathbb{E}[qR_4] = -d + 2\delta_2 + d^2 \le -3d/16 + d^2,$$
$$\|R_4\|_\infty^2 \le U_4 = d + 77d^2/32, \quad \mathbb{E}R_4^2 \le dW_4, \quad W_4 = 10101/32768 + 13d/16 + d^2,$$
$$P_4 \le (13/12)^2.$$

For any $|R| \le M$, $V = \mathbb{E}R^2$, centering R^2 proves

$$|\mathbb{E}[gR^2]| \le \sqrt{dV(M^2 - V)}. \tag{G.2}$$

The resulting $V + \sqrt{dV(M^2 - V)}$ increases for $V \le M^2/2$. Here this condition follows from $1 - 2(10101/32768) + 25d/32 - 2d^2 > 0$. Hence

$$\mathbb{E}[qR_4^2]/d \le W_4 + \sqrt{dW_4 V_4}, \qquad V_4 = 1 - 10101/32768 + 51d/32 - d^2.$$

Both polynomials increase on $[0, 1/3]$. For $d \le 1/16$, the same $M, V_q/d, B_8$ bounds as above, with $(15/16)^8 < 3/5$, give $\mathbb{E}[qP_4^4] \le 1 - 721d/640$. For the other intervals use the following data for B_6 and the main function $(1 - d)^5(1 - bd)$. Its convexity follows from $4 + 2b - 6bd > 0$. In the first row the negative mean is retained at the exact center; the remaining rows use nonnegative mean upper bounds.

I	b	$\bar{m}$	M	V_q/d	B_6
$[1/16, 1/8]$	$19/16$	$15/16$	$13/32$	$7/10$	4
$[1/8, 3/16]$	1	$7/8$	$21/40$	$7/8$	$10/3$
$[3/16, 7/32]$	$29/32$	$13/16$	$7/12$	1	$27/10$
$[7/32, 1/4]$	$13/16$	$25/32$	$41/64$	$17/16$	$99/40$
$[1/4, 9/32]$	$23/32$	$3/4$	$11/16$	$21/20$	$9/4$
$[9/32, 5/16]$	$5/8$	$23/32$	$3/4$	$8/7$	$65/32$
$[5/16, 1/3]$	$9/16$	$11/16$	$7/9$	$6/5$	$29/16$

Its endpoint upper bounds, after adding each row's remainder, are

$$(1493/1760, 2537/3200), \quad (391/480, 271/320),$$
$$(2061/2560, 8511/10240), \quad (16833/20480, 277389/327680),$$
$$(25515/131072 + 189/320, 4219/5120), \quad (7327/8960, 2473/2912), \quad (671/832, 8087/9720).$$

All are at most $17/20$. The pointwise cap converts this cubic bound to $\mathbb{E}[qP_4^4] \le 2873/2880 < 1$.

Length five. Set $P_5 = qf_5$, $R_5 = r + s + K^3 g + K^4 g - d(2g + r)$ and $m_0 = 1 - (3d - \delta_2)/2 \le m = 1 - 5d/4$. The data are

$$q + f_5 = 2m_0 + R_5, \quad P_5 \le (m_0 + R_5/2)^2, \quad P_5 \le (16/15)^2,$$
$$\|R_5\|_\infty^2 \le U_5 = d + 5d^2, \quad \mathbb{E}R_5^2 \le dW_5, \quad W_5 = 1/4 + d/2 + 9d^2/2,$$
$$\mathbb{E}[qR_5] = -d + 2\delta_2 - \delta_3 + 3d^2 \le -11d/32 + 3d^2.$$

The last step uses the increasing polynomial $-1+2x-x^2$ and $(1-\theta)^2 > 11/32$. Since $1+5d-2W_5 = 1/2 + 4d - 9d^2 > 0$,

$$\mathbb{E}[qR_5^2]/d \le W_5 + \sqrt{dW_5 V_5}, \qquad V_5 = 3/4 + 9d/2 - 9d^2/2.$$

For $d \le 1/16$, $M = 3/10$, $V_q/d < 1/2$, $B_8(1, 3/10) < 10$ and $(59/64)^8 < 3/5$ give $\mathbb{E}[qP_5^4] \le 1 - 7d/5$ by convexity. For the subsequent cubic estimates, the main functions in the first row, second row, and remaining rows respectively are

$$(1 - 5d/4)^5(1 - 37d/32), \qquad (1 - 5d/4)^5(1 - 19d/32),$$
$$H = (1 - 5d/4)^5(1 - 73d/32 + 9d^2).$$

All mean upper bounds here are nonnegative, so the center substitution is valid. The first two functions are convex by the preceding formula. For H, the bracket in $H''/(1 - 5d/4)^3$ is $4977/64 - 96495d/256 + 4725d^2/8$; it exceeds $77 - 378d + 590d^2 > 0$, whose discriminant is -38836.

I	$\bar{m}$	M	V_q/d	B_6
$[1/16, 1/8]$	$12/13$	$11/24$	$27/40$	4
$[1/8, 3/16]$	$17/20$	$5/8$	$24/25$	$13/4$
$[3/16, 7/32]$	$10/13$	$11/16$	$65/64$	$49/20$
$[7/32, 1/4]$	$3/4$	$3/4$	$75/64$	$75/32$
$[1/4, 9/32]$	$11/16$	$5/6$	$43/32$	$31/16$
$[9/32, 19/64]$	$13/20$	$55/64$	$10/7$	$27/16$
$[19/64, 5/16]$	$19/30$	$9/10$	$61/40$	$8/5$
$[5/16, 31/96]$	$61/100$	$23/25$	$8/5$	$29/20$
$[31/96, 1/3]$	$3/5$	$19/20$	$317/192$	$17/12$

The endpoint upper bounds for the main functions plus their remainders are

$$(3023/3840, 14577/20480), \quad (81411/102400, 7891/9600),$$
$$(8645/12288, 6367/8192), \quad (54639/65536, 89381/106496),$$
$$(21393/26624, 14189/16384), \quad (5819/7168, 237197/286720),$$
$$(171207/204800, 15455/17920), \quad (14783/17920, 8115/9600), \quad (7/8, 149719/172800).$$

They are at most $7/8$, proving $\mathbb{E}[qP_5^4] \le (256/225)(7/8) = 224/225 < 1$. Every interval table above specifies a finite binomial sum, an increasing variance envelope, and both rational endpoint bounds. Convexity extends these bounds throughout each interval.

Length three: Hermite reduction. For $P = qf_3$, $R = r + s$ and $m = 1 - d/2$, the exact data are

$$P \le (m + R/2)^2, \quad \mathbb{E}[qR] = -d + \delta_2, \quad \mathbb{E}R^2 = \delta_2 + \delta_3, \quad \|R\|_\infty^2 \le d + \delta_2.$$

Writing $a = \delta_2/d \le \theta$, $c_0 = 45/32$, the centered variance bound gives

$$\mathbb{E}[qR^2] \le c_0 ad + d^{3/2}\sqrt{c_0 a(1 - \theta a)}.$$

Here $c_0 a/(1 + a) \le \theta < 1/2$, and both terms increase with a. Put

$$t = \sqrt{45d/32}, \quad k = 16/45, \quad \kappa = 19/45, \quad b = 13/32, \quad c = 29/70, \quad v = b + ct,$$
$$D = 1 + b + (2\kappa + c)t, \quad s_0 = -t(v + \kappa t)/(1 + \kappa t),$$
$$w = (v - \kappa^2 t^2)/D, \quad U = 1 + t/2 - kt^2, \quad W = 1 - kt^2 - t(v + \kappa t)/[2(1 + \kappa t)].$$

On $[0, 11/16]$, $v < 1$, $v - \kappa^2 t^2 > 0$, $-t \le s_0 \le 0$, $0 < w < 1$, and $1 - kt^2 - t/2 \ge 703/1440 > 0$. Our moment bounds are $|R| \le t$, $\mathbb{E}[qR] \le -\kappa t^2$, $\mathbb{E}[qR^2] \le t^2 v$, using $\sqrt{247/1440} < 29/70$. For $\phi(x) = (1 - kt^2 + x/2)^8$, the quadratic Hermite interpolant at the double node s_0 and upper node t majorizes ϕ on $[-t, t]$: the remainder has the sign of $(x - s_0)^2(x - t)$ since $\phi''' \ge 0$. Its quadratic and linear coefficients are positive ($\beta = \phi'(s_0) - 2\alpha s_0 > 0$). The two-node law of masses $w, 1 - w$ has exactly the substituted moments. Thus

$$\mathbb{E}[qP^4] \le H(t) := wU^8 + (1 - w)W^8.$$

The actual range $t \le \sqrt{15/32}$ lies in $[0, 11/16]$.

Low-interval coefficient certificate. For $t \le 11/20$, Hermite interpolation at $-2t/3$ (double) and t gives $H \le Q$, where

$$Q = \omega\phi(t) + (1 - \omega)\phi(-2t/3) + h\phi'(-2t/3),$$
$$\omega = 49/160 - 281t/5250, \qquad h = 5t/32 - 1049t^2/3150.$$

This is an expectation of an upper interpolant, so h need not be positive. Set $z = 1 - kt^2$ and

$$A_j = 49/160 + (222 - 75j)(-2/3)^j/320, \quad B_j = -562/10500 + (562 + 5245j)(-2/3)^j/10500.$$

An exact generator retaining every coefficient is

$$Q = \sum_{j=0}^{8} \binom{8}{j} 2^{-j} z^{8-j} t^j (A_j + B_j t) = 1 + \sum_{j=2}^{16} C_j t^j.$$

Its rational coefficient upper bounds are

j	2	3	4	5	6	7	8
$C_j \le$	$-811/480$	$7291/1440$	$-197/250$	$-196/25$	$13/5$	$137/25$	$-9/5$

j	9	10	11	12	13	14	15	16
$C_j \le$	$-11/5$	$3/5$	$29/50$	0	0	0	$3/500$	$3/2000$

For example $C_4 = -81709/103680$, $C_5 = -6102127/777600$, $C_6 = 40432717/15552000$; the same finite generator supplies every other rational comparison. Therefore $Q \le 1 + t^2 S(t)$, with

$$\begin{aligned} S = & -811/480 + 7291t/1440 - 197t^2/250 - 196t^3/25 \\ & + 13t^4/5 + 137t^5/25 - 9t^6/5 - 11t^7/5 \\ & + 3t^8/5 + 29t^9/50 + 3t^{13}/500 + 3t^{14}/2000. \end{aligned}$$

Using $t^2 \le 121/400$ gives $S' \ge P_0$ and

$$P_0 = 7291/1440 - 197t/125 - 588t^2/25 + 7133t^3/1000 + 45483t^4/2000,$$

$$P_0' \le -197/125 - 1550667t/200000 < 0, \qquad P_0(11/20) > 349/1000 > 1/3.$$

Rational substitution gives $S(11/20) < -944/100000 < -1/200$; its last two positive terms sum to less than $1/100000$. Consequently $H \le 1 - t^2/200$ on the whole low interval.

Curvature and a critical-point bracket. Write $H = A_0 + B_0$, with $A_0 = wU^8$, $B_0 = (1-w)W^8$. Set $A = 45/32$, $B = 793/630$, $e = 6703/10080$, $h = 527/630$. The exact derivatives are

$$w' = [c - 2\kappa b - 2\kappa^2 At - \kappa^2 Bt^2]/D^2, \qquad w'' = -2e^2/D^3,$$

$$U' = 1/2 - 2kt, \qquad U'' = -32/45,$$

$$W' = -2kt - [b + 2ht + \kappa h t^2]/[2(1+\kappa t)^2], \qquad W'' = -32/45 - e/(1+\kappa t)^3.$$

On $[11/20, 5/8]$ the rational endpoint bounds are

$$27/100 \le w \le 7/25, \quad w' \le -3/50, \quad -1/10 \le w'' \le -1/12,$$

$$7/6 \le U \le 47/40, \quad 1/18 \le U' \le 11/100,$$

$$5/8 \le W \le 7/10, \quad |W'| \ge 17/20, \quad |W''| \le 11/10, \quad W'' \le -32/45,$$

and $-W'(11/20) < 7/8$. Substitution in the exact second derivatives gives

$$A_0'' = w''U^8 + 16w'U^7U' + w[56U^6(U')^2 + 8U^7U''] \le -71/16,$$

$$B_0'' = -w''W^8 - 16w'W^7W' + (1-w)[56W^6(W')^2 + 8W^7W''] \le 889/250.$$

For the latter bound, $W^6(W')^2$ decreases because $3(W')^2 + WW'' \ge 559/400 > 0$ and $W' < 0$. Thus $H'' < -4/5$. At $t_0 = 3/5$,

$$U = 293/250,\ U' = 11/150,\ w = 744211/2723475,\ W = 3445933/5264000,$$

$$w' = -19710732160/296692643025, \qquad W' = -32/75 - 1936883/3958528.$$

These give $U^8 < 89/25$, $W^8 < 27/800$, $273/1000 \le w \le 2733/10000$, $-67/1000 \le w' \le -33/500$, and $-23/25 \le W' \le -9/10$. Using also $3 < U^7 < 31/10$ and $1/20 < W^7 < 13/250$ gives $6/25 < A_0' < 27/100$ and $1/4 < -B_0' < 7/25$ there. Hence $H(t_0) < 399/400$, $|H'(t_0)| \le 1/25$, and

$$H(t) \le 399/400 + |t - t_0|/25 - 2(t-t_0)^2/5 \le 1997/2000 < 1$$

on the middle interval.

On $[5/8, 11/16]$, $w', w'' < 0$, $U' > 0$, and $56(U')^2 + 8UU'' \le 56/324 - (256/45)(7/6) < 0$, so $A_0'' < 0$. Its unique maximum lies in $[81/125, 13/20]$. Indeed at $13/20$,

$$w = 6806449/25224975, \quad -w' = 39948064688/565599434445,$$

giving $269/1000 < w < 27/100$ and $7/100 < -w' < 71/1000$. The derivative signs follow from $(-w')/w < 71/269 < 4/15 < 8U'/U$ at $81/125$, and $(-w')/w > 7/27 > 2720/10573 = 8U'/U$ at $13/20$. Since w decreases and U increases,

$$A_0 \le w(81/125)U(13/20)^8 < (27/100)(98/27) = 49/50.$$

The two certificates are

$$w(81/125) < 27/100 \quad \text{and} \quad (10573/9000)^8 < 98/27.$$

The first is equivalently $-22661/3500000 < 0$. Finally $-w' \le 1/10, w \le 7/25, W \le 1, W' \le -4/9$ imply $B_0'/B_0 \le 5/36 - 32/9 < 0$. At $5/8$, $W < 19/30$, $w \ge 27/100$ and $(19/30)^8 < 2/75$, so $B_0 < 73/3750 < 1/50$. Thus $H < 1$ also on the high interval. This completes the three closed intervals, the length-three case, and the singleton supply.

G.4 Singletons with the cubed boundary weight

We prove $\mathbb{E}[q^3 f_a^4] \le 1$ for all $a \ge 1$ in the outward direction. For $a = 1$, $q^3p^4 = p(pq)^3 \le p$. For $a \ge 3$ use Cauchy–Schwarz with the same outward estimates from Sections G.3 and G.1:

$$\mathbb{E}[q^3 f_a^4] \le \sqrt{\mathbb{E}[q f_a^4]\mathbb{E}[q^5 f_a^4]} \le 1.$$

It remains to prove the stronger bound $1 - 3d/160$ for $a = 2$.

A scalar inequality on the full row domain. The row domain is (2.2). We prove on the whole closed domain $-1 \le g \le 1$, $|r| \le pq/2$ that

$$q^3(p + r)^4 \le p + (4 + 3g)r + 49r^2/5. \tag{G.3}$$

Put $\gamma = 49/5$, $z = pq$, $S = 1 + z + z^2$. The right side minus the left is exactly

$$D = pg^2S + (4g^2S + 3g)r + (\gamma - 6p^2q^3)r^2 - 4pq^3r^3 - q^3r^4.$$

Set $A = \gamma - C(q)$, $C(q) = p^2q^3(6 + 2q + q^2/4)$ and $A_- = \gamma - 6p^2q^3$. For $r \ge 0$, $D \ge pg^2S + (4g^2S + 3g)r + Ar^2$. For $r = -v \le 0$, the cubic and quartic remainder is $q^3v^3(4p - v) \ge 0$, so $D \ge pg^2S - (4g^2S + 3g)v + A_-v^2$.

Two shared scalar budgets. The derivative of $C = q^3(2 - q)^2(q^2 + 8q + 24)/4$ has sign $144 - 56q - 38q^2 - 7q^3$, which decreases and has its unique zero in $(5/4, 63/50)$; its endpoint values are $61/64, -111429/125000$. Since $\log C$ is strictly concave, $C(5/4) = 640125/65536 < 489/50$ and $(\log C)'(5/4) = 244/8535 < 1/34$ give $C < 33252/3399 < 49/5$. Also $\max p^2q^3 = 3456/3125$, whence $A_- \ge 9889/3125$. Write $S(u) = 3 - 3u^2 + u^4$, $R(u) = uS(u)$ for $0 \le u \le 1$. Its unique interior maximum lies below $2/3$; there $R \le 3u - 23u^3/9 \le 6/\sqrt{23} < 63/50$. The minimum on $[2/7, 1]$ occurs at an endpoint, giving $3 - 4R(u) \le 0$ for $u \ge 2/7$.

Nonnegative r. If $g \ge 0$ the lower quadratic for D is already nonnegative. For $g = -u$, let $B = 3 - 4R(u)$. Only $u < 2/7$, $B > 0$ needs attention. There $pS \ge 3 - 4u$, $B \le 3 - 21u/2$, and

$$A(1 + u) \ge A_0 = 31/20 - 43u/4 + 55u^2/4 + 85u^3/4 - 6u^4.$$

The discriminant comparison is

$$4pSA - B^2 \ge P_0 = 48/5 - 454u/5 + 907u^2/4 + 35u^3 - 412u^4 + 96u^5 > 0.$$

For its sign put $v = u - 1/4 \in [-1/4, 1/28]$: $P_0 = 33/320 + 421v/80 + 227v^2/2 - 317v^3 - 292v^4 + 96v^5$. For $v = -w \le 0$ the last three terms are nonnegative, and the remaining quadratic has minimum $122399/2905600$. For $v \ge 0$ absorb the negative higher powers into the quadratic; its coefficient remains greater than 100.

Negative r and nonnegative g. For $g = u \ge 0$ the lower quadratic is $pSu^2 - u[3 + 4R(u)]v + A_-(1-u)v^2$. The following intervals certify its discriminant, using $3 + 4R \le 201/25$:

I	$pS \ge$	$A_- \ge$	comparison
$[0, 1/4]$	3	$19/5$	$228/5 > 6^2$
$[1/4, 1/2]$	3	$14963/2560$	$44889/640 > (201/25)^2$
$[1/2, 1]$	2	$649/80$	$649/10 > (201/25)^2$.

Here $pS \ge 3$ on $u \le 1/2$ by expansion, and $pS \ge 2$ globally from $pS - 2 = (1-u)(1 + 4u + u^2 - 2u^3 - u^4) \ge 0$. The A_- bounds use the decreasing function $(1-u)(1-u^2)^2$.

Negative r and negative g. For $g = -u$, only $B_- = 4R - 3 > 0$ needs attention. If $u \le 3/4$, $pS \ge 417/1024$, $B_- \le 51/25$, and $4pSA_- - B_-^2 \ge 794433/800000 > 0$. For $u \ge 3/4$ write $p = 1 - u \in [0, 1/4]$. Then $S \ge 1 + 2p$, $R \le 1 + p + p^2$, $B_- \le (1+2p)^2$. For $p \ge 1/20$, $B_-^2/(4pS) \le (1+2p)^3/(4p) \le 1331/200 < A_-$, using $A_- \ge 19943/2560$. For $0 < p \le 1/20$, $R \ge 1 + p$, so $uB_- \ge 1$ whereas $A_-pq \le 49/50$. Thus the lower quadratic $Q(v)$ decreases up to $v_{\max} = pq/2$. Its endpoint is

$$Q(v_{\max})/p = u[3q/2 - uS(1+2u)] + A_-pq^2/4.$$

The bracket equals $-5p/2 - p^2 + 23p^3 - 29p^4 + 13p^5 - 2p^6 > -13p/5$. Since $A_- > 19/2$, $q > 19/10$, it follows that $Q(v_{\max})/p > (4779/800)p > 0$. At $p = 0$ one has $r = 0$, $D = 0$ directly. All closed sign regions are now covered, proving (G.3). Taking the original expectation, $\mathbb{E}p = 1$, $\mathbb{E}r = -d$, $\mathbb{E}[gr] = 0$ and $\delta_2 \le 13d/32$ yield $\mathbb{E}[q^3 f_2^4] \le 1 - 4d + 49\delta_2/5 \le 1 - 3d/160$. This completes the singleton catalogue, including equal-pair restoration and full reversal by the common rules.

H Unpropagated rooted moments

We first prove the even catalogue, including its exceptional pair, and then the odd catalogue. These bounds concern zero propagation steps. Shared scalar inequalities are in Appendix A.

H.1 Sixth moments for multiple canonical pairs

Here ν_g is the spectral measure of $-K^2$ at g, of total mass d. Let $P = \prod_i h_{a_i} f_{b_i}$, with $a_1 < b_1 < \cdots < a_t < b_t$. We prove $\mathbb{E}[pP^6] \le 1$ for all $t \ge 2$ and for the sole pair $(2, 3)$. The empty table and $a_1 \ge 3$ are already covered by Appendix D.

Deficits and finite masks. Write $R_{ab} = S_b - S_a = K^a G_{b-a}(K) g$. The actual pair envelope is $h_a f_b \le e^{R_{ab} - \omega_{ab}}$, with

$$\omega_{ab} = \begin{cases} \beta_b, & a = 1, 2, \\ 2d - \delta_2, & a = 3, \\ 2d, & a \ge 4. \end{cases}$$

For the block use $R = \sum_i R_{a_i b_i}$, $\Delta = \sum_i \omega_{a_i b_i}$. The common sixth-moment criterion is $D := \Delta - \mathbb{E}[pR] \geq 3M^2$. Here

$$\|R\|_\infty^2 \leq C_* \theta^{a_1 - 1} d, \quad \mathbb{E}[pR] = \sum_i (u_{a_i+1} - u_{b_i+1}), \quad u_j = \mathbb{E} K^j \mathbf{1}.$$

Distinct boundaries let each positive δ_j occur at most once. The mean is at most d if $a_1 \geq 2$, at most zero if $a_1 = 1$, and more precisely at most $-d + \sum_{j \geq 3} \delta_j$ if $a_1 = 1, b_1 \geq 3$, or $-d + \sum_{j \geq 2} \delta_j$ for the first pair $(1, 2)$. Use $\sum_{j \geq 2} \delta_j \leq 13d/19$ and both bounds $\sum_{j \geq 3} \delta_j \leq 169d/608$ and $\sum_{j \geq 3} \delta_j \leq (13/19)\delta_2$ in their respective forms. For $R = KQ(K)g$ starting with $(1, 2)$, Q has constant coefficient one and no linear term. Separating residue signs gives

$$|Q(i\sqrt{x})|^2 \leq (1 + x^3)/(1 - x^2)^2 < 31/20, \qquad M^2 \leq 31d/20.$$

For a general zero-one mask the increasing convex function $(1 + x)/(1 - x^2)^2$ lies below its endpoint chord $1 + (816/325)x$; therefore also $M^2 \leq d + (816/325)\delta_2$.

At least three pairs and minimum two. The table lists the ranges, with admissible lower and upper bounds for D, M^2; each row satisfies $D \geq 3M^2$.

condition	$D \geq$	$M^2 \leq$
$a_1 = 2, t \geq 3$	$115d/32$	$C_* \theta d$
$a_1 = 2, t = 2, b_1 = 3, 4$	$1655d/608$	$C_* \theta d$
$a_1 = 2, t = 2, b_1 \geq 5$	$44d/19$	$117d/200$
sole $(2, 3)$	$d + \delta_2$	δ_2
$a_1 = 1, t \geq 4, b_1 \geq 3$	$211d/32$	$C_* d$
$a_1 = 1, t \geq 4, b_1 = 2$	$179d/32$	$31d/20$
$a_1 = 1, t = 3, b_1 \geq 4$	$101d/19$	$25d/16$
$a_1 = 1, t = 3, b_1 = 3$	$6d + 6\delta_2/19$	$d + 816\delta_2/325$
$a_1 = 1, t = 3, b_1 = 2, a_2 \geq 4$	$2871d/608$	$31d/20$
first $(1, 2), (3, 4)$, $t = 3$	$2377d/608$	d
first $(1, 2), (3, b_2 \geq 5)$, $t = 3$	$2377d/608$	$289d/225$

Here are the prefix norm comparisons needed beyond the mask bound. For minimum two with $b_1 \geq 5$ separate the two intervals: the first G contracts and the second starts at degree at least five. This gives $M^2 \leq \theta[1 + \sqrt{1 + \theta}\theta^2]^2 d < 117d/200$. For $t = 3, b_1 \geq 4$, the first G contracts and the two-interval tail starts at degree four, giving $(1 + \theta^2\sqrt{C_*})^2 d < 25d/16$. For the last two rows use $Q_0(z) = 1 + z^2 G_m(z)$, $m \geq 1$. It contracts: for $m = 1$ this is $1 - x$, and for $m \geq 2$ its numerator $1 - z + z^2 - z^{m+2}$ has squared modulus at most $1 - x + 3x^2 + x^4 \leq 1 + x$. The third segment then contributes less than $2/15$ in norm. For $(1, 2), (3, 4)$ its sharper bound is $|Q| \leq 1 - x + x^2\sqrt{1 + x} \leq 1$. The row with $b_1 = 3$ has margin $D - 3M^2 \geq 519d/7600$; the sole-pair row uses $2\delta_2 \leq d$.

Two pairs $(1, 2), (3, b)$**.** Set $Q = 1 + z^2 G_{b-3}(z)$. Then $\Delta = 2d - \delta_2$, $\mathbb{E}[pR] = -d + \delta_2 - u_{b+1}$ and $D = 3d - 2\delta_2 + u_{b+1}$. Compare the spectral integrands of D and $3\int |Q(i\sqrt{x})|^2 \, d\nu_g$. For $b = 2k$, put $s = (-x)^{k-1}$; after multiplying their difference by $1 + x$ one gets $x(4 - 5x - 6s - 3s^2)$. If $s \leq 0$ this is nonnegative; if $s > 0$, use $s \leq x^2$ and $4 - 5x - 6x^2 - 3x^4 \geq 1570/2401 > 0$ for $x \leq 3/7$. For $b = 2k + 1$, put $s = (-x)^k$; the difference is $x(4 - 5x) + s(5 - 7x) - 3s^2$. If $s \geq 0$, $s \leq x^2$ gives $5 - 7x - 3s \geq 71/49$. If $s < 0$, $|s| \leq x^3$ leaves $x(4 - 5x - 5x^2 - 3x^5) \geq 5x/32$. Thus the integrated criterion holds for every $b \geq 4$.

Sharper pair deficits. Retaining real nonnegative convolution terms yields

$$h_4 + f_b \le 2 + R_{4b} - 4d + 3\delta_2, \qquad h_5 + f_b \le 2 + R_{5b} - 6d + 6\delta_2,$$
$$h_a + f_b \le 2 + R_{ab} - \omega_a, \qquad \omega_6 = 89d/24, \quad \omega_a = 11d/4 \ (a \ge 7).$$

For the first use $h_4 = 2 - S_4 - 2d + dg$ and retain $1, p$ in f_b. For the second use $h_5 = 2 - S_5 - 5d + \delta_2 + d(f_2 + p)$ and retain $1, p, f_2$. For the last use $f_b - f_a \le R_{ab} + 2d$ and $h_a + f_b = 2\mu_{a-1} + f_b - f_a$. The required means follow from $\mu_5 \le 1 - 137d/48$, $\mu_6 \le 1 - 89d/32$ and $\mu_n \le 1 - 19d/8$ for $n \ge 7$; for the last bound retain $f_0, \dots, f_4$ in convolution and use $\sum_{j=0}^{4} \mu_j = 5 - 6d + \delta_2 \ge 3$. AM–GM converts each displayed actual sum to the exponential envelope at the same R_{ab}.

Two pairs $(1, m), (4, b)$**.** For $m = 2$ use the sharper remainder

$$B = \beta_{b-2}, \quad T = d - \beta_{b-1} - B \in [-d, 0], \quad \delta = 2d + \beta_b + \beta_{b-1} + B,$$
$$R' = (1 - B)r + R_{4b} + Tg, \qquad P \le e^{R' - \delta}.$$

Its exact mean is $-d + d^2 - \beta_{b-1}d - u_{b+1}$, so $D \ge 6d - 3\delta_2 - \delta_3 - d\delta_2 \ge 13765d/3072$. Write $R' = K[(1-B)g + U + T\mathbf{1}], U = K^3 G_{b-4}(K)g$. Then $\mathbb{E}U \ge 0$, $\langle g, U\rangle \le \delta_3$ and $\|U\|^2 \le (1+\theta)\theta^3 d$. The nonpositive $2T\mathbb{E}U$ may be discarded, giving $M^2 \le [1 + 7/144 + 2\theta^2 + (1+\theta)\theta^3]d < 1067d/720$. Since $13765/3072 > 1067/240$, the criterion follows. For $m = 3$, use $\Delta = 5d - 3\delta_2$ and exact mean $-d - \delta_2 - u_{b+1}$, so $D \ge 5143d/1024 > 5d$. The polynomial $Q = 1 + z + z^3 G_{b-4}(z)$ has squared modulus at most $5/3$. The tail lengths one and two give $1 + x(1-x)^2$ and $(1+x)(1+x^3)$. For longer tails use $Q = (1 - z^2 + z^3 - z^{j+3})/(1 - z)$ and the bound $|Q|^2 \le 1 + x + 3x^3 + 9x^6/4 < 196523/120000 < 5/3$.

All remaining two-pair blocks. For $(1, m), (a, b)$ the norm polynomial is $G_{m-1}(z) + z^{a-1}G_{b-a}(z)$. The first factor contracts except when $m = 3$, where its modulus is $\sqrt{1+x}$. Combining it with the enhanced deficit gives the table:

$(a; m)$	$D \ge$	$M^2 \le$
$(5; 2)$	$73d/16$	$36d/25$
$(5; 4)$	$89d/16$	$36d/25$
$(5; 3)$	$191d/32$	$245d/128$
$(6; 2)$	$547d/120$	$289d/225$
$(6; 3)$	$137d/24$	$125d/72$
$(6; 4, 5)$	$413d/96$	$289d/225$
$(\ge 7; 2)$	$18d/5$	$6d/5$ (strict)
$(\ge 7; \ge 4)$	$309d/76$	$6d/5$ (strict)
$(\ge 7; 3)$	$19d/4 + 6\delta_2/19$	$d + 3\delta_2/2$

For $a = 5$, the second pair has nonpositive weighted mean; its gap-two factor is retained, giving $(1+\theta)(1+\theta^2)^2 \le 245/128$ when $m = 3$. For $a = 6$, later positive boundaries start at δ_4, with sum less than $3d/20$; also $\theta^{5/2} < 1/9$ gives the $m = 3$ norm. For $a \ge 7$, the tail norm is at most $\sqrt{1+\theta}\theta^3$. In its final row retain the spectral correlation:

$$(1 + x)(1 + x^3)^2 \le 1 + 3x/2,$$

because the bracket $1 + 2x^2 + 2x^3 + x^5 + x^6$ is at most $44697/30000 < 3/2$. Thus $D - 3M^2 \ge 61d/1216$ in that row. The other displayed rows satisfy $D > 3M^2$ directly. All two-pair blocks of minimum one, every larger pair count, and the minimum-two cases have now been covered. Equal-pair restoration and full reversal yield the stated complete canonical sixth moments.

H.2 The residual pair $(2, 4)$

We prove $\mathbb{E}[pP^6] \le 1$ for $P = h_2 f_4$. Use the common notation and put $s = K^2 g, t = K^3 g, m = 1-d$ and $R = s + t - dg$. The exact path formulas give $h_2 + f_4 = 2m + R$, so AM–GM gives

$$0 \le P \le (m + R/2)^2 \le e^{R-2d}.$$

Skew symmetry and the row bound yield

$$\mathbb{E}R = \delta_2, \qquad \mathbb{E}[gR] = -\delta_2 - d^2, \qquad \mathbb{E}[pR] = -d^2,$$
$$\|R\|_\infty^2 \le M_0^2 = \delta_2 + \delta_3 + 3d^2, \qquad V = \mathbb{E}R^2 = \delta_3 + \delta_4 + 2d\delta_2 + d^3.$$

Indeed $R = K[(K + K^2)g - d\mathbf{1}]$, $\langle Kg, K^2 g\rangle = 0$, and $\mathbb{E}[(K + K^2)g] = -d$. With $c = \theta + \theta^2 = 585/1024$, these formulas imply

$$M_0^2 \le cd + 3d^2, \qquad V/d \le v(d) = 7605/32768 + (13/16)d + d^2,$$
$$V_p := \mathbb{E}[pR^2] \le V + M_0\sqrt{dV}.$$

The two polynomial bounds (A.2) are

$$\mathbb{E}[pP^6] \le H(d) + B_{12}(m, M)V_p, \qquad \mathbb{E}[pP^3] \le F(d) + B_6(m, M)V_p,$$

where

$$H = (1-d)^{12} - 6d^2(1-d)^{11}, \qquad F = (1-d)^6 - 3d^2(1-d)^5.$$

Both are convex on $[0, 1/3]$, since

$$H'' = 12(1-d)^9(10 + 13d - 78d^2) > 0, \qquad F'' = 6(1-d)^3(4 + 7d - 21d^2) > 0.$$

For $0 < d \le 1/32$, the exponential criterion applies because

$$(2d + d^2) - 3M_0^2 \ge d(2 - 3\theta - 3\theta^2 - 8d) \ge 37d/1024.$$

For $1/32 \le d \le 1/8$, the following table proves the sixth moment. In each row $m \le 1 - \inf I$; M bounds M_0, and B bounds $B_{12}(1-\inf I, M)$. The last column bounds the resulting convex function at the two endpoints, except in the first row, where $(1-d)^{12} \le 1-8d$ gives the displayed bound directly.

I	M	V_p/d	B	final bound
$[1/32, 1/16]$	$7/32$	$5/12$	19	$1 - d/12$
$[1/16, 3/32]$	$2/7$	$12/25$	$31/2$	$193/200,\ 399/400$
$[3/32, 7/64]$	$8/25$	$13/25$	13	$747/800,\ 1583/1600$
$[7/64, 1/8]$	$7/20$	$14/25$	$45/4$	$601/640,\ 396/400$

These entries use $v(d)$ and (A.4). For the remainder sums, the first five terms are $\frac{33}{2}m^{10}$, $\frac{55}{2}m^9 M$, $\frac{495}{16}m^8 M^2$, $\frac{99}{4}m^7 M^3$, $\frac{231}{16}m^6 M^4$; consecutive tail terms have ratio at most $\frac{5}{8}(M/2)$. This supplies every omitted term.

For the remaining range, completing a square gives $\sqrt{cd + 3d^2} \le 2d + c/6$, and consequently

$$P \le (1 + c/12)^2 = (4291/4096)^2 < (22/21)^2, \quad (22/21)^6 < 4/3, \quad (22/21)^{10} < 8/5.$$

On $[1/8, 1/4]$, apply the third-moment bound with the following data; B now bounds B_6. Convexity of F gives $\mathbb{E}[pP^3] < 3/4$.

I	M	V_p/d	B	endpoint bounds
$[1/8, 3/16]$	$1/2$	$91/120$	$13/4$	$3/4,\ 11/15$
$[3/16, 1/4]$	$7/12$	$11/12$	$8/3$	$29/40,\ 3/4$

Thus $\mathbb{E}[pP^6] < (4/3)(3/4) = 1$ there. Finally, the global bounds $v(d) < 5/8$, $M_0 < 3/4$ give $V_p/d < 49/40$ and

$$\mathbb{E}[pP] \le 1 - 2d + d^3 + V_p/4 < 1 - 2279d/1440 \le 1 - 3d/2.$$

For $d \ge 1/4$ this is less than $5/8$, and therefore $\mathbb{E}[pP^6] < (8/5)(5/8) = 1$. The intervals exhaust $[0, 1/3]$.

H.3 The residual pair $(2, 5)$

Put $P = h_2 f_5$, $r = Kg$, $s = K^2 g$, $t = K^3 g$, $u = K^4 g$, and

$$m = 1 - (3d - \delta_2)/2, \qquad R = s + t + u - 2dg - dr.$$

The path expansion gives $h_2 + f_5 = 2m + R$ and hence $0 \le P \le (m + R/2)^2$. Its exact moment data are

$$\begin{aligned} \mathbb{E}[pR] &= \delta_3 - d^2, \\ M_0^2 &= \delta_2 - \delta_3 + \delta_4 + 8d^2 - 2d\delta_2 + d^3, \\ V = \mathbb{E}R^2 &= \delta_3 - \delta_4 + \delta_5 + 4d\delta_2 - 2d\delta_3 + 4d^3 + d^2\delta_2. \end{aligned}$$

Here $\|R\|_\infty^2 \le M_0^2$ follows by applying the row bound to $K[(K + K^2 + K^3)g - 2d\mathbf{1} - dg]$; the three cross terms are $\|(K+K^2+K^3)g\|^2 = \delta_2 - \delta_3 + \delta_4$, $\mathbb{E}[(K+K^2+K^3)g] = -d + \delta_2$ and $\langle g, (K+K^2+K^3)g\rangle = -\delta_2$. The spectral polynomials $x - x^2 + x^3 - 2dx$ and $x^2 - x^3 + x^4 - 2dx^2 + (4d + d^2)x$ increase on $[0, \theta]$: their derivatives are $3(x-1/3)^2 + 2/3 - 2d$ and $x(2 - 3x + 4x^2) + 4d(1-x) + d^2$. Consequently

$$\begin{aligned} M_0^2 \le U(d) &= \frac{10101}{32768}d + \frac{115}{16}d^2 + d^3, \\ V/d \le W(d) &= \frac{131313}{1048576} + \frac{663}{512}d + \frac{141}{32}d^2, \qquad V_p \le V + M_0\sqrt{dV}. \end{aligned}$$

Set $a = 83/64$, $\tau = 169/1024$, $\bar m = 1 - ad$, $\bar L = \tau d - d^2$ and $\delta_* = 24/83$. Then $m \le \bar m$ and $\mathbb{E}[pR] \le \bar L$. For $d \le \delta_*$ and $k = 1, 3, 6$, the expression $m^{2k} + k\bar L m^{2k-1}$ increases on the permitted range of m: when $\bar L < 0$, its derivative has sign determined by $2m + (2k-1)\bar L \ge 2 - 3d - 11d^2 > 0$. Thus the three base functions in the first-, third- and sixth-moment estimates are

$$G = \bar m^2 + \bar m \bar L, \qquad F = \bar m^5[1 - (821/1024)d - 3d^2], \qquad H = \bar m^{11}[1 - (157/512)d - 6d^2].$$

All are convex on $[0, \delta_*]$. For $b = 157/512$ and $c = 821/1024$,

$$\begin{aligned} H'' &= \bar m^9[(110a^2 + 22ab - 12) + (288a - 132a^2 b)d - 936a^2 d^2], \\ F'' &= \bar m^3[(20a^2 + 10ac - 6) + (72a - 30a^2 c)d - 126a^2 d^2]. \end{aligned}$$

The brackets are positive: use $ab < 1$, $ac < 4/3$, $a > 5/4$ and $d^2 \le 1/9$ to bound them below by $6a^2+22ab-12$ and $6a^2+10ac-6$, respectively. Also $G = 1-(2a-\tau)d+(30661/65536)d^2+ad^3$.

For $d \le 1/16$, the exponential criterion uses $\Delta = 3d-\delta_2$ and $D = \Delta-\mathbb{E}[pR] = 3d-\delta_2-\delta_3+d^2$. Writing $\rho = \delta_2/d$, the bounds $\delta_3 \ge \delta_2^2/d$, $\delta_4 \le \theta\delta_3$ give

$$\begin{aligned}(D-3M_0^2)/d &\ge 3-(4-6d)\rho+(2-3\theta)\rho^2-23d-3d^2\\ &\ge 49281/32768-(329/16)d-3d^2 \ge 6785/32768.\end{aligned}$$

The intermediate polynomial decreases in $0 \le \rho \le \theta$.

For larger d, the following rational data, together with the common binomial bounds and convexity, suffice. The column B refers to B_{12} in the first two rows and B_6 in the third; in each row use $\bar m \le 1 - a\inf I$. The first-moment rows use $G + V_p/4$ directly.

I	moment	M	V_p/d	B	endpoint bounds
$[1/16, 3/32]$	6	$5/16$	$1/2$	14	$15/16,\ 95/96$
$[3/32, 1/8]$	6	$2/5$	$3/5$	11	$1/3+99/160,\ 1$
$[1/8, 3/16]$	3	$4/7$	$19/20$	3	$117/160,\ 235/320$
$[3/16, 7/32]$	1	$13/20$	$229/200$	—	$5/8,\ 5/8$
$[7/32, 1/4]$	1	$3/4$	$11/8$	—	$5/8,\ 5/8$

All upper endpoint bounds in this table are strict. On $[1/4, \delta_*]$, $M < 6/7$, $W < 7/8$ and $\sqrt{W} < 15/16$ give $V_p < (47/28)d \le 282/581 < 1/2$. Since $\bar L \le 0$,

$$\mathbb{E}[pP] \le \bar m^2 + V_p/4 < 121/256 + 1/8 = 153/256 < 5/8.$$

The pointwise bound used to pass from lower moments to the sixth is $P < (22/21)^2$. If $d \ge \delta_*$ then $P \le 4\mu_4 \le 4[1-(3-\theta)d] \le 1$. For $d \le \delta_*$, put $B(d) = 1 - ad + \sqrt{U(d)}/2$. The polynomial $(U')^2 - 16a^2U$ has positive coefficients, since $115/16 - 4a^2 = 471/1024 > 0$; hence B increases. At δ_*, $1-a\delta_* = 5/8$ and $U(\delta_*) < 2042113/2858935 < (71/84)^2$, giving $B(\delta_*) < 22/21$. The preceding third-moment bound is $< 3/4$, while the first-moment bounds are $< 5/8$. Finally use $(22/21)^6 < 4/3$ and $(22/21)^{10} < 8/5$, as in the $(2,4)$ case. This covers every d and proves $\mathbb{E}[p(h_2f_5)^6] \le 1$.

H.4 The residual pairs $(2, b)$, $b \ge 6$

Let $P = h_2f_b$, $m = b-2 \ge 4$, $R_0 = K^2G_m(K)g$. For $0 \le x \le \theta$ the finite residue formulas give

$$|G_m(i\sqrt{x})|^2 \le (1+x^3)^2/(1+x) \le 1-2x/5.$$

For the second inequality it suffices that $2x^2+x^5+2x/5 \le 3/5$; its left side is at most $25/72+1/32+1/6 = 157/288$. Consequently

$$\|KG_m(K)g\|^2 \le \delta_2 - 2\delta_3/5, \quad \|R_0\|^2 \le \delta_3 - 2\delta_4/5,$$
$$\mathbb{E}[KG_m(K)g] \ge -d+\delta_2-\delta_3, \quad \langle g, R_0\rangle \ge -\delta_2+\delta_3-\delta_4, \quad \mathbb{E}[pR_0] = -u_{b+1} \le \delta_5 \le \theta^4 d.$$

Here $u_j = \mathbb{E}K^j\mathbf{1}$. The lower bounds retain one further negative term in each finite alternating sum. The last identity is exact telescoping; its first possible positive value occurs at index $b+1 = 10$.

Set $a = 19/32$, $\epsilon = ad$, $\tau = \theta^4 = 28561/1048576$, $m_0 = 1 - ad$ and $R = R_0 - \epsilon g$. Retaining f_0, f_1 in convolution gives $h_2 + f_b \le 2m_0 + R$, so $P \le (m_0 + R/2)^2$ and $\mathbb{E}[pR] \le L = \tau d - ad^2$. The exact row and variance budgets are

$$\begin{aligned}\|R\|_\infty^2 &\le \delta_2 - 2\delta_3/5 + \epsilon^2 + 2\epsilon(d - \delta_2 + \delta_3) \le U = 871d/2560 + 20539d^2/16384,\\ \mathbb{E}R^2 &\le \delta_3 - 2\delta_4/5 + \epsilon^2 d + 2\epsilon(\delta_2 - \delta_3 + \delta_4) \le dW,\\ W &= 11323/81920 + 191919d/524288 + 361d^2/1024.\end{aligned}$$

For the two spectral substitutions the derivatives of $(1 - 2\epsilon)x + (2\epsilon - 2/5)x^2$ and $x^2 - 2x^3/5 + 2\epsilon(x - x^2 + x^3)$ are nonnegative on $[0, \theta]$. Thus U, W increase, and $V_p/d \le W + \sqrt{UW}$. The cap is $P \le (19/18)^2$: the square of $2ad + 1/9$ minus U is

$$2565d^2/16384 - 1759d/23040 + 1/81 > 0,$$

whose discriminant is $-1009919/530841600$. In particular $(19/18)^6 < 7/5$.

The common binomial estimates have main terms

$$H = (1 - ad)^{11}[1 - (a - 6\tau)d - 6ad^2], \qquad F = (1 - ad)^5[1 - (a - 3\tau)d - 3ad^2].$$

They are convex on $[0, 1/3]$. For H, put $b = a - 6\tau$, $c = 6a$; the bracket in $H''/(1 - ad)^9$ is $110a^2 + 22ab - 2c + 12a(4c - 11ab)d - 156a^2cd^2$. Its linear term is positive and the remainder is at least $a(110a - 104a^2 - 12) + 22ab > 0$. For F, put $b = a - 3\tau$, $c = 3a$; the corresponding bracket is $20a^2 + 10ab - 2c + 6a(4c - 5ab)d - 42a^2cd^2$, bounded below by $2a(10a - 7a^2 - 3) + 10ab > 0$. These signs use $1/2 < a < 3/5$ and $d^2 \le 1/9$. For $d \le 1/32$ the exponential criterion gives $D - 3U > d[1/8 - 13d/4] \ge 3d/128$, with $D \ge (2a - \tau)d + ad^2$.

The remaining sixth-moment intervals use $H + B_{12}V_p$ and the following complete data; all bounds are strict at positive endpoints.

I	$\bar{m}$	M	V_p/d	B_{12}
$[1/32, 1/16]$	$53/54$	$1/6$	$1/4$	19
$[1/16, 3/32]$	$26/27$	$5/24$	$11/40$	17
$[3/32, 1/8]$	$17/18$	$1/4$	$3/10$	$31/2$
$[1/8, 5/32]$	$25/27$	$29/100$	$47/140$	$27/2$
$[5/32, 11/64]$	$49/54$	$5/16$	$9/25$	12

The endpoint upper bounds are respectively

$$(377/384, 185/192), \quad (1841/1920, 1265/1280), \quad (631/640, 157/160),$$
$$(433/448, 5375/5376), \quad (29/30, 799/800).$$

For the cubic intervals use $F + B_6V_p$:

I	$\bar{m}$	M	V_p/d	B_6
$[11/64, 3/16]$	$44/49$	$1/3$	$19/50$	$63/20$
$[3/16, 7/32]$	$8/9$	$11/30$	$5/12$	$31/10$
$[7/32, 1/4]$	$67/77$	$13/32$	$23/50$	3
$[1/4, 2/7]$	$109/128$	$9/20$	$41/80$	$113/40$

The respective endpoint upper bounds are $(228203/320000, 11091/16000)$, $(91/128, 81/200 + 217/768)$, $(1131/1600, 139/200)$ and $(9113/12800, 7993/11200)$. All are less than $5/7$, so the pointwise cap converts the cubic to the sixth moment. Each table uses the increasing U, W, the full finite sum B_r, and convex interpolation of the displayed main polynomial.

Finally, for $d \geq 2/7$, the row bound $|r| \leq pq/2$ gives $ph_2 \leq pq(1 + p/2) \leq 37/24$; for $g \geq 0$ use $g - 3g^2 \leq 1/12$, and for $g \leq 0$ use $3/2$ directly. Positivity gives $f_b \leq 2\mu_{b-2}p$, hence $P \leq 37\mu_{b-2}/12$. For $b \geq 7$ the proved mean bound $\mu_{b-2} \leq 1 - 19d/8$ gives $111/112$. For $b = 6$ retain the exact $\mu_4 \leq 1 - 83d/32$, giving $1073/1344$. Both are less than one. This covers all d and proves $\mathbb{E}[pP^6] \leq 1$.

H.5 The residual pairs $(1, b)$, $b \geq 6$

Put $P = qf_b$, $U_0 = G_{b-1}(K)g$, $T_0 = KU_0 = S_b - S_1$, $a = 19/32$, $\epsilon = ad$, $\tau = \theta^4$ and

$$m_0 = 1 - 2ad + ad^2/2, \qquad R = T_0 - \epsilon(3g + 2r + s).$$

The finite-G comparison in Section H.4 gives $\|U_0\|^2 \leq d - 2\delta_2/5$, $\|T_0\|^2 \leq \delta_2 - 2\delta_3/5$. Every finite alternating truncation also satisfies

$$\mathbb{E}U_0 \geq -d + \delta_2 - \delta_3, \quad \langle g, U_0\rangle \geq d - \delta_2, \quad \langle r, U_0\rangle \geq \delta_2 - \delta_3,$$
$$\langle g, T_0\rangle \geq -\delta_2 + \delta_3 - \delta_4, \quad \langle r, T_0\rangle \geq \delta_2 - \delta_3, \quad \langle s, T_0\rangle \geq \delta_3 - \delta_4.$$

These retain one negative tail term and bound the remaining decreasing alternating sum by zero. Telescoping gives $\mathbb{E}[pT_0] = -d - u_{b+1} \leq -(1 - \tau)d$, where $u_j = \mathbb{E}K^j\mathbf{1}$. Retain f_0, f_1, f_2, f_3 in convolution, whose sum is $4 - d + 3g + 2r + s$. The actual envelope and mean are

$$q + f_b \leq 2m_0 + R, \quad P \leq (m_0 + R/2)^2, \quad \mathbb{E}[pR] \leq -(1 - \tau)d - a^2d^2 \leq -31d/32.$$

Using $R = K[U_0 - \epsilon(3\mathbf{1} + 2g + r)]$ gives

$$\|R\|_\infty^2 \leq d - 2\delta_2/5 + \epsilon^2(9 - 2d + \delta_2) + 2\epsilon(d - 2\delta_2 + 4\delta_3) \leq U = d + 4d^2,$$
$$\mathbb{E}R^2 \leq \delta_2 - 2\delta_3/5 + \epsilon^2(9d - 2\delta_2 + \delta_3) + 2\epsilon(\delta_2 - 2\delta_3 + 4\delta_4) \leq dW,$$
$$W = 11/32 + 33d/80 + 3d^2.$$

For the first reduction the spectral part is bounded by $-2\delta_2/5$, since $4 - 8\theta = 3/4 > \epsilon$; then use $\delta_2 \geq d^2$. For the second, the derivative of $x - 2x^2/5 + \epsilon^2(9 - 2x + x^2) + 2\epsilon(x - 2x^2 + 4x^3)$ is positive on $[0, \theta]$; its first two parts exceed $119/200$. Substitution at θ gives coefficients $871/2560, 53599/131072, 3087633/1048576$, bounded by those in W. As usual $V_p/d \leq W + \sqrt{UW}$.

For $d \geq 4/19$, positivity gives $f_b \leq 2\mu_{b-2}p$, so $P \leq 2\mu_{b-2}$. Use the path mean $\mu_n \leq 1 - 19d/8$ for $b \geq 7$. For $b = 6$, the exact $\mu_4 \leq 1 - 83d/32$ gives the threshold $16/83 < 4/19$. Thus $P \leq 1$ in the entire large range. On $d \leq 4/19$, comparison of U with $(1/6 + 4ad - ad^2)^2$ leaves at least $1/36 - 5d/24 + 5d^2/6 > 0$, whose discriminant is $-85/1728$. Therefore the global cap is $P \leq (13/12)^2$, and $(13/12)^6 < 5/3$.

Put $m = 1 - 9d/8$. The sixth- and third-moment main terms are

$$H = m^{11}(1 - 111d/16), \qquad F = m^5(1 - 129d/32).$$

The negative mean substitution is valid: for the sixth bound on $d \leq 3/32$, $2m_0 - 11(31/32)d \geq 2 - 417d/32 > 0$; for the cubic on $d \leq 4/19$, use $2 - 231d/32 > 0$. Convexity follows from

$$[(1 - cd)^n(1 - bd)]'' = nc(1 - cd)^{n-2}[(n - 1)c + 2b - (n + 1)bcd],$$

with $(n,b)=(11,111/16)$ or $(5,129/32)$, $c=9/8$. For $d\le 1/36$ the exponential criterion applies with $D\ge(4a+1-\tau)d+(a^2-a)d^2$: $D-3U>d[11/32-49d/4]\ge d/288$.

The remaining sixth-moment data are

I	$\bar m$	M	V_p/d	B_{12}	endpoints
$[1/36,1/16]$	$31/32$	$9/32$	$9/16$	21	$191/192, 255/256$
$[1/16,5/64]$	$119/128$	$13/36$	$3/5$	17	$573/640, 319/320$
$[5/64,3/32]$	$11/12$	$13/36$	$13/20$	$29/2$	$2397/2560, 1271/1280$

Here the second row may enlarge its row-norm bound $13/40$ to $13/36$ for the remainder. The displayed endpoints bound $H+dB_{12}(V_p/d)$. For the cubic ranges the table gives a rounded coefficient k with $B_6(V_p/d)\le k$; the final columns bound $F+kd$.

I	$\bar m$	M	V_p/d	k	endpoints
$[3/32,1/8]$	$17/19$	$7/16$	$59/80$	$5/2$	$19/32, 9/16$
$[1/8,5/32]$	$13/15$	$51/100$	$17/20$	$11/4$	$19/32, 371/640$
$[5/32,3/16]$	$33/40$	$23/40$	$19/20$	$8/3$	$17/30, 7/12$
$[3/16,4/19]$	$15/19$	$5/8$	$25/24$	$125/48$	$439/768, 269/456$

The complete finite B_6 bounds used here are $27/8, 16/5, 14/5, 5/2$. All cubic endpoints are less than $3/5$, so the cap converts them to the sixth moment. The finite sums B_r, increasing U, W, and convexity certify all closed intervals. Together with the two outer ranges this proves $\mathbb{E}[p(qf_b)^6]\le 1$ for every $b\ge 6$.

H.6 The residual pairs $(1,4)$ and $(1,5)$

Put $P_j=qf_j$, $r=Kg$, $s=K^2g$, $t=K^3g$, $u=K^4g$. The actual path expansions give

$$R_4=r+s+t-dg, \qquad m_4=1-d, \qquad \mathbb{E}[pR_4]=-d-d^2,$$
$$R_5=r+s+t+u-d(2g+r), \qquad m_5^0=1-(3d-\delta_2)/2, \qquad \mathbb{E}[pR_5]=-d+\delta_3-d^2,$$

and $q+f_j=2m_j+R_j$, with $m_5=m_5^0$ in this identity. Thus $P_j\le(m_j+R_j/2)^2$. The unweighted budgets, also used in Section G.3, are

$$\|R_4\|_\infty^2\le U_4=d+77d^2/32, \qquad \mathbb{E}R_4^2\le dW_4, \quad W_4=10101/32768+13d/16+d^2,$$
$$\|R_5\|_\infty^2\le U_5=d+5d^2, \qquad \mathbb{E}R_5^2\le dW_5, \quad W_5=1/4+d/2+9d^2/2.$$

For the first row use $\|g+r+s\|^2=d-\delta_2+\delta_3$ and $\mathbb{E}R_4^2=\delta_2-\delta_3+\delta_4+2d\delta_2+d^3$; the increasing spectral polynomial is $x-x^2+x^3$. For the second, write $R_5=K[U_0-dW_0]$, $U_0=g+r+s+t$, $W_0=2\mathbf{1}+g$. Then

$$\|U_0\|^2=d-\delta_2-\delta_3+\delta_4, \quad \langle U_0,W_0\rangle=-d+\delta_2, \quad \|W_0\|^2=4+d,$$
$$\mathbb{E}R_5^2=\delta_2-\delta_3-\delta_4+\delta_5+2d(\delta_2-\delta_3)+d^2(4d+\delta_2).$$

Discard $-\delta_4+\delta_5\le 0$ and use the increasing $x-x^2$ on $[0,\theta]$ to obtain the stated W_5. All U_j, W_j increase; use $V_{p,j}/d\le W_j+\sqrt{U_jW_j}$.

Pointwise caps and the outer intervals. Comparing squares gives

$$(1/6+2d)^2-U_4=1/36-d/3+51d^2/32>0,$$
$$(2/15+83d/32)^2-U_5=4/225-37d/120+1769d^2/1024>0.$$

Their discriminants are $-19/288$ and $-1/36$, respectively. Hence $P_4\le(13/12)^2$, $P_5\le(16/15)^2$. The powers used below are $(13/12)^6<5/3$, $(13/12)^{10}<9/4$, $(16/15)^6<3/2$, $(16/15)^{10}<2$. Since $\mathbb{E}[pP_j]\le\mu_j$, the large ranges $d\ge7/32$ for $j=4$ and $d\ge3/16$ for $j=5$ give sixth moments less than $3987/4096$ and $119/128$, respectively, using $\mu_4\le1-83d/32$ and $\mu_5\le1-137d/48$.

Convex main terms. The sixth and cubic main terms are

$$H_4=(1-d)^{11}(1-7d-6d^2),\qquad F_4=(1-d)^5(1-4d-3d^2),$$
$$H_5=(1-5d/4)^{11}(1-121d/20),\qquad F_5=(1-5d/4)^5(1-73d/20).$$

The center for length four is exact. For length five use $m_5^0\le1-5d/4$ and $\mathbb{E}[pR_5]\le-4d/5$. The negative mean substitution is valid since its derivative factors are $2m_5^0-11(4/5)d\ge2-59d/5>0$ on $d\le1/16$ and $2m_5^0-5(4/5)d\ge2-7d>0$ on $d\le3/16$. For length four the second-derivative brackets are $252-636d-936d^2>0$ on $d\le3/32$ and $54-48d-126d^2>0$ on $d\le1/3$. For length five use the derivative formula from Section H.5 with $cd\le1/8$ or $1/4$. Thus all main functions are convex on their table domains.

Complete sixth-moment certificates. Use $H_j+dB_{12}(V_{p,j}/d)$, with the following exact finite-sum bounds.

$j;I$	$\bar m$	M	$V_{p,j}/d$	B_{12}	endpoint bounds
$4;[0,1/32]$	1	$3/16$	$9/20$	23	$1,567/640$
$4;[1/32,1/16]$	$31/32$	$9/32$	$17/32$	21	$933/1024,501/512$
$4;[1/16,5/64]$	$15/16$	$5/16$	$29/50$	16	$689/800,73/80$
$4;[5/64,3/32]$	$12/13$	$11/32$	$5/8$	15	$471/512,1253/1280$
$5;[0,1/32]$	1	$1/5$	$19/50$	$47/2$	$1,2813/3200$
$5;[1/32,1/16]$	$25/26$	$7/24$	$15/32$	20	$1143/1280,111/128$

In the second length-four row the actual $M=27/100$ may be enlarged to $9/32$ for B_{12}, while its recorded variance bound is unchanged. Convexity gives the complete sixth ranges including zero.

Complete cubic certificates. Here k is a rounded upper bound for $B_6(V_{p,j}/d)$, and the endpoint columns bound F_j+kd.

$j;I$	$\bar m$	M	$V_{p,j}/d$	k	endpoint bounds
$4;[3/32,1/8]$	$29/32$	$13/32$	$7/10$	$49/20$	$191/320,173/320$
$4;[1/8,5/32]$	$7/8$	$15/32$	$39/50$	$5/2$	$35/64,35/64$
$4;[5/32,3/16]$	$27/32$	$21/40$	$7/8$	$21/8$	$145/256,71/128$
$4;[3/16,7/32]$	$13/16$	$7/12$	1	$27/10$	$91/160,189/320$
$5;[1/16,3/32]$	$12/13$	$3/8$	$4/7$	$72/35$	$139/210,159/280$
$5;[3/32,1/8]$	$8/9$	$11/24$	$27/40$	$9/4$	$75/128,17/32$
$5;[1/8,5/32]$	$17/20$	$8/15$	$4/5$	$5/2$	$9/16,107/192$
$5;[5/32,3/16]$	$13/16$	$5/8$	$24/25$	$11/4$	$229/384,39/64$

The complete B_6 upper bounds in row order are $7/2,16/5,3,27/10,18/5,10/3,25/8,14/5$. The length-four endpoints are less than $3/5$ and the length-five endpoints less than $2/3$. Their respective pointwise caps convert each cubic estimate into the sixth moment. Together with the outer ranges these prove $\mathbb{E}[p(qf_4)^6],\mathbb{E}[p(qf_5)^6]\le1$ on all $d\in[0,1/3]$.

H.7 The pair $(1,3)$: a sixth moment

Let $P = qf_3$, $R = r + s$ and $m = 1 - d/2$. Exact recursion gives

$$P \le (m + R/2)^2, \quad \mathbb{E}[pR] = -d - \delta_2, \quad \mathbb{E}R^2 = \delta_2 + \delta_3, \quad \|R\|_\infty^2 \le d + \delta_2.$$

The large-variance range. Let $c = pq + 2r$ be the nonnegative cycle marginal and $\mathcal{T}(x) = \mathbb{E}_{y,z}[C(x,y,z)(g(y) + g(z))/2]$. Symmetrizing gives $\mathcal{T} = K(g^2) + s - gr$ and $|\mathcal{T}| \le c$. The exact path decomposition is

$$f_3 = p^3/6 + (p+2)c/6 + (Ac + \mathcal{T})/3.$$

Its remainder after $p^3/6$ is nonnegative, since it is at least $pc/6 + Ac/3$. Put $a_0 = qp^3/6 \le 9/32$. Using $\mathcal{T} \le c$ and $T(pq) = q - d + K(g^2) \le q$ in the adjoint integral gives

$$\mathbb{E}[p(P - a_0)] \le (113/96)(1 - 3d).$$

Here the scalar coefficient is $(7 - g - 5g^2 - g^3)/6 \le 113/96$. The row bound gives $P < 7/5$ globally. The chord of x^6 on $[a_0, 7/5]$ has slope at most $(7/5)^6/(7/5 - 9/32) < 27/4$. Thus $\mathbb{E}[pP^6] < 1/1000 + (3051/384)(1 - 3d) < 1$ for $d \ge 7/24$.

Centered variance and a Hermite upper bound. Now $0 < d \le 7/24$. Write $a = \delta_2/d \in [d, \theta]$, $c_0 = 45/32$. Centering R^2 gives

$$\mathbb{E}[pR^2] \le B_a = c_0 ad + d^{3/2}\sqrt{c_0 a(1 - \theta a)}.$$

The variance substitution is valid because $c_0 a/(1+a) \le \theta < 1/2$; also $B_\theta - B_a \ge c_0 d(\theta - a)$. Set

$$t = \sqrt{c_0 d}, \quad k = 16/45, \quad b = 13/32, \quad c = 5/12, \quad v = b + ct, \quad D = 1 + 2t + v,$$
$$s_0 = -t(t+v)/(1+t), \quad w = (v - t^2)/D, \quad U = 1 + t/2 - kt^2, \quad W = 1 - kt^2 + s_0/2.$$

The actual t range is contained in $[0, 13/20]$. Here $v < 1$, $v - t^2 > 0$, $-9/16 \le s_0 \le 0$, $m - t/2 \ge 61/120 > 0$ and $m + s_0/2 \ge 53/96$. The quadratic Hermite interpolant of $\phi(x) = (m + x/2)^{12}$ at the double node s_0 and upper node t majorizes ϕ on $[-t, t]$, because $\phi''' \ge 0$ and the remainder has sign $(x - s_0)^2(x - t) \le 0$. Write it as $\alpha x^2 + \beta x + \gamma$. Then $\alpha \ge \phi''(s_0)/2 > 0$ and

$$\beta/\alpha \le 4m/11 - 20s_0/11 \le 61/44 < c_0.$$

Hence the combined mean/variance substitution, including a possibly negative β, has the correct sign:

$$\alpha B_a - \beta(1+a)d + \gamma \le \alpha t^2 v - \beta t^2 + \gamma.$$

The two-node law on t, s_0 of masses $w, 1 - w$ has mean $-t^2$ and second moment $t^2 v$. Consequently $\mathbb{E}[pP^6] \le H(t) = wU^{12} + (1-w)W^{12}$.

The full small interval. For $t \le 1/3$, another Hermite interpolant, at $-t/2$ (double) and t, gives $H \le Q$ with

$$Q = \omega\phi(t) + (1 - \omega)\phi(-t/2) + h\phi'(-t/2), \quad \omega = 7/24 - 7t/27, \quad h = t/16 - 11t^2/18.$$

For x^j its expectation is $t^j(A_j + B_j t)$, where

$$A_j = 7/24 + (17 - 3j)(-1/2)^j/24, \qquad B_j = -7/27 + (7 + 33j)(-1/2)^j/27.$$

For $j \geq 5$, $A_j \leq 3/10$, $B_j \leq 0$. For even $j \geq 6$ the latter follows from $7 + 33j \leq 7\,2^j$; odd j are immediate. Let $x = kt^2$, and let $E_{n,r}(x)$ be the degree-r Taylor polynomial of $(1-x)^n$ at zero. Even truncations are upper bounds and odd truncations lower bounds. The following finite table specifies all retained terms, including their required truncation direction:

term in Q	bound for its factor m^n
m^{12}	$E_{12,4}$
$-6t^2m^{11}$	$E_{11,3}$ (lower)
$(33/2)t^2(b+ct)m^{10}$	$E_{10,2}$
$(55/8)t^3m^9 - (165/8)t^4m^9$	$E_{9,2}$; $E_{9,1}$ (lower)
$\widehat{A}_4t^4m^8 + \widehat{B}_4t^5m^8$	$E_{8,2}$
$\widehat{A}_5t^5m^7 - \widehat{N}_5t^6m^7$	$E_{7,2}$; $E_{7,1}$ (lower)
$\widehat{A}_6t^6m^6 - \widehat{N}_6t^7m^6$	1; $E_{6,1}$ (lower)
$j = 7$	discard $B_7 \leq 0$, then $m^5 \leq 1$
$j \geq 8$	$A_j \leq 3/10$, $B_j \leq 0$, $m^{12-j} \leq 1$

The coefficient symbols here are scalar constants:

$$\widehat{A}_4 = 19305/2048,\ \widehat{B}_4 = 495/256,\ \widehat{A}_5 = 3663/512,\ \widehat{N}_5 = 363/32,$$
$$\widehat{A}_6 = 34419/8192,\ \widehat{N}_6 = 2079/1024,\ \widehat{A}_7 = 7425/4096.$$

Substitution gives coefficient upper bounds for powers $t^2, \ldots, t^9$ equal, respectively, to

$$-57/16, \quad 55/4, \quad -3, \quad -37, \quad 19, \quad 50, \quad 116, \quad 32.$$

For example the first exact coefficient is $-3421/960 < -57/16$; the other entries follow from the finite table and the displayed constants. The entire $j = 10, 11, 12$ positive tail is at most $2449t^{10}/122880 < t^{10}/50$. Thus

$$\begin{aligned} Q &\leq 1 + t^2[-57/16 + 55t/4 - 3t^2 - 37t^3 + 19t^4 + 50t^5 + 116t^6 + 32t^7 + t^8/50] \\ &\leq 1 + t^2[-57/16 + 55t/4 - 3t^2 - 20t^3] \leq 1 - 23t^2/432. \end{aligned}$$

The second line uses $t \leq 1/3$ to bound the last five positive terms by $17t^3$; the final bracket increases to $-23/432$ at $1/3$.

A rational maximum bracket on the remaining interval. The function U increases and w decreases on $[0, 13/20]$. For $A = wU^{12}$,

$$A' = U^{11}N(t)/D^2, \quad N = 4657/1536 + 73t/48 - 84049t^2/4320 - 2441t^3/216 + 580t^4/27.$$

The exact integer polynomial $69120N$ is $209565 + 105120t - 1344784t^2 - 781120t^3 + 1484800t^4$. It is positive up to $1/10$; thereafter $N' < 73/48 - (4623/200)t < 0$. At $17/40$, $-1/200 < N < 0$; on $[53/125, 17/40]$ we have $N' < -7$, so $N(53/125) > 1/500$. Thus the unique maximum of A lies in that rational bracket. Its value satisfies

$$A \leq w(53/125)U(17/40)^{12} < (83/500)(263/50) = 21829/25000 < 7/8.$$

The certificates are $w(53/125) < 83/500$ (the reduced numerator is $-783/4000$), $U(17/40) = 20669/18000$ and $(20669/18000)^3 < 53/35$, $(53/35)^4 < 263/50$. For $B = (1-w)W^{12}$, $W' \leq -b/2$ and $W \leq 1$ give $B'/B \leq 2 - 6b = -7/16 < 0$. At $1/3$, $w = 125/637$, $W = 88199/103680 < 23/27$, and $(23/27)^3 < 5/8$, whence $B(1/3) < 625/5096 < 1/8$. Therefore $H < 1$ on $[1/3, 13/20]$. The two scalar intervals cover $0 < d \leq 7/24$; the cycle argument covers $d \geq 7/24$. Hence $\mathbb{E}[p(qf_3)^6] \leq 1$ in the full finite model.

H.8 The pair $(1, 2)$ and the complete even catalogue

For $P = qf_2 = q(p + r)$ we prove

$$\mathbb{E}[pP^4] \le 1 - d/360. \tag{H.1}$$

The argument also retains an intermediate envelope estimate used in Section G.2. The row geometry is $|r| \le \min\{\sqrt{d}, pq/2\}$, $c = pq+2r \in [0, 2pq]$, $\mathbb{E}c = 1-3d$, and $f_2 = (p^2+c)/2$. Since $q+f_2 = 2+r$, $P \le (1 + r/2)^2$.

Orthogonal residual variance. For $0 < d \le 1/4$,

$$|\mathbb{E}[gr^2]|^2 \le d(d - \delta_2)(\delta_2 - d^2)/(1 - d). \tag{H.2}$$

If $V = \delta_2 - d^2 = 0$ then $r = -d$ identically and the left side vanishes. Otherwise put $U = d-\delta_2 > 0$, $H = \mathbb{E}r^3 + d\delta_2$. The function g is orthogonal to $1, r + d$, and the residual of r^2 from their span has squared norm $\mathbb{E}r^4 - \delta_2^2 - H^2/V$. For every real k,

$$0 \le \mathbb{E}[(d - r^2)(r - k)^2] = d\delta_2 - \mathbb{E}r^4 - 2k(d\mathbb{E}r - \mathbb{E}r^3) + k^2U.$$

Complete the square first in k, then in H, to obtain

$$\mathbb{E}r^4 \le d\delta_2 - (H + dU)^2/U, \quad \mathbb{E}r^4 - \delta_2^2 - H^2/V \le UV/(1 - d).$$

Cauchy–Schwarz against g proves (H.2). With $t = \sqrt{d}$, $a = \delta_2/d \in [d, \theta]$, the resulting bound is

$$\mathbb{E}[pr^2]/d \le a + t\sqrt{(1 - a)(a - t^2)/(1 - t^2)}.$$

It increases with a, since $1 + t^2 - 2a > 0$. Thus

$$\mathbb{E}[pr] = -t^2, \quad \mathbb{E}[pr^2] \le t^2 v(t), \quad v = \theta + ts(t), \quad s(t) = \sqrt{(1 - \theta)(\theta - t^2)/(1 - t^2)}.$$

Fixed Hermite envelope. Interpolate $\phi(y) = (1 + y/2)^8$ quadratically at $-2t/3$ (double) and t. Its cubic Rolle remainder is nonpositive on $[-t, t]$ and its quadratic coefficient is nonnegative. Substituting the preceding moments gives

$$\mathbb{E}[pP^4] \le \mathbb{E}[p\phi(r)] \le 1 + t^2\mathcal{F}(t), \qquad \mathcal{F} = Av + B, \tag{H.3}$$

where the complete polynomials are

$$A = 7 - 7t/3 + 35t^2/8 - 35t^3/108 + 455t^4/1296 + t^5/1296 + 457t^6/186624,$$

$$B = -4 + 28t/9 - 371t^2/54 + 7t^3/54 - 1295t^4/972 - 73t^5/729 - 613t^6/15552 - 67t^7/46656.$$

For reconstruction, the coefficients of ϕ are $1, 4, 7, 7, 35/8, 7/4, 7/16, 1/16, 1/256$. For y^j, put $u = -2/3$ and $\eta_j = \sum_{k=0}^{j-2}(k + 1)u^k$. Its interpolant expectation is $t^j[\alpha_j + \eta_j v + \beta_j t]$, where $\alpha_j = (1 - j)u^j + 4\eta_j/9$, $\beta_j = -ju^{j-1} - 4\eta_j/3$. This finite formula supplies every coefficient in A, B.

The small interval. On $[0, 1/2]$, $A \ge 7 - 7t/3 + 455t^2/108 > 13/2$ and $v \le \theta + t/2$. The exact coefficients of $B + A(\theta + t/2)$ are

$$-37/32, \quad 1631/288, \quad -43267/6912, \quad 7553/3456,$$

$$-168175/124416, \quad 28261/373248, \quad -227147/5971968, \quad -79/373248.$$

For $t \le 1/4$ these give $\mathcal{F} \le -37/32 + 1631t/288 - 17t^2/3 \le -109/1152$. The last quadratic increases up to $1/4$, so the bound covers the full interval.

Curvature on the middle interval. Set $c_0 = 19/32$. Direct differentiation gives

$$v' = s - c_0^2 t^2/[s(1-t^2)^2],$$
$$v'' = -3c_0^2 t/[s(1-t^2)^2] - 4c_0^2 t^3/[s(1-t^2)^3] - c_0^4 t^3/[s^3(1-t^2)^4].$$

On $[1/4, 1/2]$ all denominators are positive. The full derivative bounds are

$$v < 3/5, \quad v' \le 1/2, \quad v'' \le -361/600, \quad A'' < 28/3, \quad A' < 9/4, \quad B'' \le -721/54.$$

For $v < 3/5$ use $t^2 s^2 < 9/256$, equivalent to the positive quadratic $(19/32)D^2 - (283/1024)D + 9/256$ in $D = t^2$, whose discriminant is $-7463/1048576$. For v'' retain only its first negative term and use $s \le 1/2$. The polynomial derivative bounds follow termwise at the appropriate endpoints; their rational upper values are $A'' \le 4640045/497664 < 28/3$ and $A' \le 2207545/995328 < 9/4$. If $A' < 0$ then $t < 1/3$, where $v' \ge 0$ because $\theta - 2t^2 + t^4 \ge 53/288 > 0$. Otherwise $2A'v' \le A' < 9/4$. Together with $A > 13/2$ this gives

$$\mathcal{F}'' = A''v + 2A'v' + Av'' + B'' \le 28/5 + 9/4 - 4693/1200 - 721/54 < -9.$$

At the single rational center $t_0 = 3/8$, direct substitution gives

$$673/100 \le A \le 6731/1000, \quad B \le -19099/5000,$$
$$22/25 \le A' \le 9/10, \quad -23/10 \le B' \le -9/4,$$
$$s^2 = 323/1760, \quad 107/250 < s < 3/7,$$
$$113/200 < v \le 127/224 < 567/1000, \quad 27/100 \le v' \le 34/125.$$

In particular $\mathcal{F}(t_0) \le -3991/1120000 < -1/300$ and $143/10000 \le \mathcal{F}'(t_0) \le 22783/250000 < 1/10$. Strong concavity therefore proves

$$\mathcal{F}(t) \le -1/300 + |t - t_0|/10 - 9(t - t_0)^2/2 \le -1/360$$

throughout $[1/4, 1/2]$. This also proves the auxiliary assertion $\mathbb{E}[p(1 + r/2)^8] \le 1 - d/360$ for every $d \le 1/4$.

Large parameters and the full even catalogue. For fixed g, chord convexity in $c \in [0, 2pq]$ gives

$$pP^4 \le p(qp^2/2)^4 + L_4(g)c, \quad L_4 \le q^3(4 - q^2)^4/32 < 8/3.$$

The maximum in the last inequality occurs at $q^2 = 12/11$; there $q^3 < 231/200$, $(4 - q^2)^4 < 72$, yielding $2079/800 < 8/3$. Also $(qp^2/2)^4 \le (16/27)^4 < 1/8$. Thus $\mathbb{E}[pP^4] \le 1/8 + (8/3)(1 - 3d) \le 19/24$ for $d \ge 1/4$, which is stronger than (H.1). Together with the two Hermite ranges and $d = 0$ this proves the stated bound.

Consequently every even canonical message, including its actual E_0, has $\mathbb{E}[pF^4] \le 1$. Indeed all cases other than the sole $(1, 2)$ have the compatible sixth moment: Appendix D covers minimum at least three and empty residuals, Section H.1 covers multiple pairs and $(2, 3)$, Sections H.2–H.4 cover $(2, b \ge 4)$, and Sections H.5–H.7 cover $(1, b \ge 3)$. Hölder under $p\mu$ supplies the fourth moment; full reversal supplies the corresponding q-weighted catalogue.

H.9 Zero-step triples $(1, 2, c)$

For $H_c = qf_2h_c$, $c \geq 3$, we prove

$$\mathbb{E}[pH_c^4] \leq \begin{cases} e^{-22\delta_2/9}, & c = 3, \\ e^{-319d/96}, & c = 4, \\ e^{-7103d/20736}, & c \geq 5. \end{cases}$$

The directions are exactly those of (5.3). Two full-domain scalar inequalities suffice:

$$(1+x)(1-x)^4 \leq e^{-3x-9x^2/4} \quad (|x| < 1), \qquad e^{y-y^2/7} \leq 1 + y + 4y^2/9.$$

The logarithmic differences have derivatives $-x(1+3x)^2/[2(1-x^2)]$ and $y(11-10y+8y^2)/(63Q(y))$, respectively, where $Q = 1 + y + 4y^2/9 > 0$. Their extrema at zero give the asserted signs.

The exact inward-first path identity is

$$h_c = 2 - \beta_c - S_c - \sum_{j=3}^{c-1} \beta_j h_{c-j}.$$

Set $B = \beta_{c-1}$, $\alpha = \beta_c$, $\Delta = \alpha + B$, $W = K^2 G_{c-2}(K)g$, $R = Bg - W$. Retaining the last Bq subtraction gives $f_2 + h_c \leq 2 - \Delta + R$; when $c = 3$, $B = 0$ and the same identity applies. Thus $f_2 h_c \leq e^{R-\Delta}$. Put $I = \|KG_{c-2}(K)g\|^2$, $J_1 = \|W\|^2$, $V = 3g - 4R$. Since $3 - 4B \in [5/3, 3]$, weighted Cauchy gives $V^2/7 \leq 9g^2/4 + 16W^2/3 \leq 9g^2/4 + 16I/3$. The scalar inequalities and exact skew moments therefore yield

$$\begin{aligned} \mathbb{E}[pH_c^4] &\leq e^{\Xi_c}, \\ \Xi_c &= -4\Delta + 4\beta_{c+1} - 32(Bd + d - \beta_{c+2})/3 \\ &\quad + 64[B^2 d + 2B(d - \beta_{c+2}) + J_1]/9 + 16I/3. \end{aligned}$$

Here $\mathbb{E}R = \beta_{c+1} - d$, $\mathbb{E}[gR] = Bd + d - \beta_{c+2}$ and $\mathbb{E}R^2 = B^2 d + 2B(d - \beta_{c+2}) + J_1$; all are taken in the original measure.

For $c = 3$, $B = 0$, $I = \delta_2$, $J_1 = \delta_3$, and the exponent is $-16\delta_2/3 + 64\delta_3/9 \leq -22\delta_2/9$. For $c = 4$, retain the exceptional G_2 data $I = \delta_2 + \delta_3$, $J_1 = \delta_3 + \delta_4$, $\beta_5 = \beta_6 = d - \delta_2$. The exponent becomes

$$-4d - 32d^2/3 + 64d^3/9 + (-28/3 + 128d/9)\delta_2 + 112\delta_3/9 + 64\delta_4/9.$$

After $\delta_4 \leq \theta\delta_3 \leq \theta^2\delta_2$, its δ_2 coefficient is at most $707/432 < 5/3$ and the pure quadratic/cubic terms are nonpositive. Hence $\Xi_4 \leq -319d/96$. For $c \geq 5$, use $I \leq \delta_2$, $J_1 \leq \delta_3$ and the finite beta intervals:

$$\begin{aligned} -4\Delta + 4\beta_{c+1} &\leq -4d + 4\delta_2 + 4\delta_3, \\ Bd + d - \beta_{c+2} &\geq d^2 - d\delta_2 + \delta_2 - \delta_3, \\ B^2 d + 2B(d - \beta_{c+2}) + J_1 &\leq d^3 + 2d\delta_2 + \delta_3. \end{aligned}$$

Combine the $\delta_3 \leq \theta\delta_2$ terms before replacing δ_2 by θd; the combined coefficient is $541/72 + 224d/9 > 0$. This gives

$$\Xi_c/d \leq -2183/2304 - 5d/9 + 64d^2/9 \leq -2183/2304 + 49/81 = -7103/20736.$$

The convex quadratic comparison includes every $d \in [0, 1/3]$. Restoration and full reversal give the complete decorated statements; the ordinary cubic in the same direction remains the separate Q3 estimate.

H.10 Odd tables of minimum two: compatible moments

Let $F = E_0 f_2 P$, $P = \prod_{i=1}^{t} h_{b_i} f_{c_i}$, $3 \le b_1 < c_1 < \cdots < b_t < c_t$, $t \ge 1$. We prove, for this same direction,

$$\mathbb{E}[qF^4] \le e^{-259d/600}, \qquad \mathbb{E}F^2 \le e^{-53d/72}.$$

Shared scalar and packet estimates. The actual ordered pair has envelope $h_b f_c \le e^{R_{bc}-\omega_b}$, $R_{bc} = \sum_{j=b}^{c-1} K^j g$. The base deficits are $\omega_3 = 2d - \delta_2$, $\omega_b = 2d$ for $b \ge 4$. At the same R_{bc}, Section H.1 gives enhanced deficits

$$\omega_4 = 4d - 3\delta_2, \quad \omega_5 = 6d - 6\delta_2, \quad \omega_b \ge 11d/4 \quad (b \ge 6).$$

Write

$$R = \sum_i R_{b_i c_i} = Kv, \quad v = \sum_i \sum_{j=b_i}^{c_i-1} K^{j-1} g, \quad I = \|v\|_2^2, \quad J_1 = \|R\|_2^2, \quad P \le e^{R-\Delta}.$$

The disjoint intervals give a finite zero-one polynomial after factoring K^{b_1-1} out of v. Separating the two signs in each residue class modulo four proves

$$\|Q_0(K)\|^2 \le (1+\theta)/(1-\theta^2)^2 = 32768/16245 < C_* = 101/50.$$

Thus $I \le C_* \delta_{b_1}$, $J_1 \le C_* \delta_{b_1+1}$. For one interval the factor is one except at gap two, where the exact values are $I = \delta_b + \delta_{b+1}$, $J_1 = \delta_{b+1} + \delta_{b+2}$; in all cases the factor $3/2$ suffices. The signed means are exactly

$$\mathbb{E}R_{bc} = \beta_{b+1} - \beta_{c+1}, \quad \mathbb{E}[gR_{bc}] = \beta_{c+2} - \beta_{b+2}, \quad \mathbb{E}[rR_{bc}] = \beta_{b+3} - \beta_{c+3}.$$

For later masks apply the corresponding termwise skew identities. Put $D_j = \delta_j/(1-\theta^2)$. The rational bounds used below are

$$\delta_2 \le 5d/12,\ \delta_3 \le d/6,\ \delta_4 \le d/12,\ \delta_5 \le d/32,\ \delta_6 \le d/64,$$
$$D_2 \le d/2, \quad D_3 \le d/5, \quad D_4 \le d/12, \quad D_5 \le d/24.$$

Use the scalar bounds in (A.5). Positive masses give $|g| < 1$ and $u = g + r = f_2 - 1 \in (-1, 1)$.

Fourth moment. Set $Y = 3g+4r+4R$. The exact decomposition $Y = 3g/5+12u/5+K(8g/5+4v)$ has weighted Cauchy constant $(3/5)^2/(3/10) + (12/5)^2/(6/5) + 1 = 7$. Thus

$$Y^2/7 \le 3g^2/10 + 6u^2/5 + \eta, \qquad \eta = 64d/25 + 16I - 64\mathbb{E}R/5.$$

Combine this with $qf_2^4 \le e^{3g+4r-3g^2/10-6u^2/5}$ and (A.5). Since $\mathbb{E}Y = -4d + 4\mathbb{E}R$ and $\mathbb{E}Y^2 = 9d + 16\delta_2 + 16J_1 + 24\mathbb{E}[gR] + 32\mathbb{E}[rR]$,

$$\mathbb{E}[q(f_2P)^4] \le e^{\Xi}, \quad \Xi = -4\Delta + 387d/200 + 6\delta_2 + 16I + 6J_1 + 9\mathbb{E}[gR] + 12\mathbb{E}[rR] - 44\mathbb{E}R/5. \quad \text{(H.4)}$$

The following table specifies every prefix budget. All entries are upper bounds except the two explicitly marked lower-bound columns; later tails are already included in each multiple-pair row.

$(b,c);t$	$\Delta\ge$	$I\le$	$J_1\le$	$\mathbb{E}R\ge$	$\mathbb{E}[gR]\le$	$\mathbb{E}[rR]\le$
$(3,4);1$	$2d-\delta_2$	δ_3	δ_4	δ_2	0	$-\delta_3$
$(3,5);1$	$2d-\delta_2$	$\delta_3+\delta_4$	$\delta_4+\delta_5$	δ_2	δ_3	$-\delta_3$
$(3,\ge 6);1$	$2d-\delta_2$	δ_3	δ_4	$\delta_2-\delta_3$	δ_3	$-\delta_3+\delta_4$
$(3,c);\ge 2$	$4d-\delta_2$	$C_*\delta_3$	$C_*\delta_4$	$\delta_2-\delta_3-D_3$	δ_3+D_5	$-\delta_3+\delta_4+D_4$
$(4,5);1$	$4d-3\delta_2$	δ_4	δ_5	0	δ_3	0
$(4,6);1$	$4d-3\delta_2$	$\delta_4+\delta_5$	$\delta_5+\delta_6$	$-\delta_3$	δ_3	δ_4
$(4,\ge 7);1$	$4d-3\delta_2$	δ_4	δ_5	$-\delta_3$	δ_3	δ_4
$(4,c);\ge 2$	$6d-3\delta_2$	$C_*\delta_4$	$C_*\delta_5$	$-\delta_3-D_5$	δ_3+D_5	δ_4+D_4
$(\ge 5,c);\ge 1$	$11d/4$	$C_*\delta_5$	$C_*\delta_6$	$-D_3$	D_5	D_4

Substitution in (H.4), followed by the displayed rational spectral bounds, gives the respective upper bounds for Ξ/d

$$-2639/600,\quad -551/400,\quad -259/600,\quad -31/15,\quad -4253/1200,$$
$$-387/800,\quad -431/400,\quad -13883/2400,\quad -3569/1600.$$

The largest is $-259/600$. The two gap-two rows explicitly retain G_2.

Second moment in the identical direction. Use only the base deficits. With $Y_2 = 2u + 2R$, $Y_2^2/8 \le 3u^2/5 + 3I$. The denominator-eight scalar bound yields

$$\mathbb{E}(f_2P)^2 \le e^{\Omega},\quad \Omega = -2\Delta - d/3 + 5\delta_2/3 + 3I + 5J_1/3 + 10(\mathbb{E}[gR] + \mathbb{E}[rR])/3 + 2\mathbb{E}R. \tag{H.5}$$

For one $b = 3$ pair use $\Delta = 2d - \delta_2$, $\mathbb{E}R \le \delta_2$, $\mathbb{E}[gR] + \mathbb{E}[rR] \le \delta_4$, $I \le 3\delta_3/2$, $J_1 \le 3\delta_4/2$; this gives $\Omega \le -53d/72$. For multiple pairs starting at three use $\Delta \ge 4d - \delta_2$, $\mathbb{E}R \le D_2$, $\mathbb{E}[gR] \le D_3$, $\mathbb{E}[rR] \le D_4$; the mask bounds give $\Omega \le -6427d/1800$. For $b \ge 4$, use $\Delta \ge 2d$, $\mathbb{E}R \le D_4$, $\mathbb{E}[gR] \le D_3$, $\mathbb{E}[rR] \le D_4$, obtaining $-27613d/14400$. Thus the required second and fourth bounds hold simultaneously. The residual singleton is excluded. Restore E_0 and reverse all edges only through the common rules; the reversed message supplies the p-fourth and ordinary second moments together.

H.11 Odd tables with a longer outward anchor

Throughout this proof $\tau = d - \delta_2$ and $D_j = \delta_j/(1-\theta^2)$. Let $F = E_0 f_a P$, where $3 \le a < b_1 < c_1 < \cdots < b_t < c_t$, $t \ge 1$, and P is the canonical pair product. We prove $\mathbb{E}[qF^4] \le e^{-59d/200}$ for this direction. The ordinary cubic in Section F.2 additionally gives

$$\mathbb{E}F^3 \le 1 - 101d/768,\quad \mathbb{E}F^2 \le (1 - 101d/768)^{2/3},\quad \mathbb{E}[qF^3] \le e^{-177d/800}.$$

These are compatible moments of one message; the fourth estimate excludes an empty residual tail.

Packet data. Use $R = Kv$, I, J_1 from Section H.10 with $\Delta = 2td$. Write $t_0 = \mathbb{E}[gv]$, $L = \mathbb{E}[rv]$, $W = \mathbb{E}[sv]$, so $\mathbb{E}R = -t_0$, $\mathbb{E}[gR] = -L$, $\mathbb{E}[rR] = -W$. For $K^{j-1}g$ the nonzero contributions to these three moments are

	j parity	value
t_0	odd	$(-1)^{(j-1)/2}\delta_{(j+1)/2}$
L	even	$(-1)^{(j-2)/2}\delta_{j/2+1}$
W	odd	$(-1)^{(j-3)/2}\delta_{(j+3)/2}$.

For one interval use factor $3/2$ in I, J_1; for any number of intervals the signed-mask bound $(1+x)/(1-x)^2 \le 1440/361 < 4$ gives factor 4. For a contiguous interval starting at least at four, $|t_0|, |L| \le \delta_3$,

$|W| \le \delta_4$. For any positive mask starting there, $t_0 \le D_3$, $-D_3 \le L \le D_4$, $W \ge -D_4$. If it starts at least at six these improve to $t_0 \le \delta_5$, $L \ge -\delta_5$, $W \ge -\delta_6$ for one interval, and the corresponding D_j bounds for multiple intervals.

One anchor increment. Set

$$A_a = \beta_a, \quad z_a = KG_{a-2}(K)g, \quad T_a = Kz_a, \quad \hat{f}_a = S_a - \beta_a = 1 + g + r + T_a - A_a.$$

Convolution gives $0 < f_a \le \hat{f}_a = AS_{a-1} \le 2$. Thus $u = \hat{f}_a - 1 \in (-1, 1]$, as required by the logarithmic bound. For $Y_0 = 3g + 4r + 4R$ put $X = Y_0 + 4T_a - 12A_a/5$. The exact decomposition is

$$X = 3g/5 + 12u/5 + KZ, \qquad Z = 8(g + z_a)/5 + 4v.$$

The Cauchy constant is again 7, so $\mathbb{E}[q\hat{f}_a^4 P^4] \le e^{\Xi}$, with $\Xi = -4\Delta - 8A_a/5 + \|Z\|^2 + \mathbb{E}X + 3\mathbb{E}X^2/8$. Define the exact anchor data

$$e_a = \mathbb{E}T_a = d - \beta_{a+1}, \quad I_a = \|z_a\|^2, \quad J_a = \|T_a\|^2,$$
$$L_a = \mathbb{E}[rz_a] = d - \beta_{a+2}, \quad W_a = \mathbb{E}[sz_a] = \beta_{a+3} - \tau,$$
$$C_a = \langle z_a, v\rangle, \qquad D_a = \langle Kz_a, Kv\rangle.$$

They satisfy $\tau \le A_a \le d$, $0 \le e_a \le \delta_2$, $\delta_2 - \delta_3 \le L_a \le \delta_2$, $0 \le W_a \le \delta_3$. If

$$\Xi_0 = -4\Delta + 387d/200 + 6\delta_2 + 16I + 6J_1 - 9L - 12W + 44t_0/5,$$

the complete increment, before discarding any signed term, is

$$\begin{aligned}\Xi - \Xi_0 = {} & -4A_a - 28e_a/25 - 36A_a e_a/5 + 64I_a/25 + 6J_a - 9L_a - 12W_a \\ & + 64C_a/5 + 12D_a + 36A_a d/5 + 36A_a t_0/5 + 54A_a^2/25.\end{aligned}$$

It follows by expanding the two squared norms and using $\mathbb{E}z_a = -A_a$, $\mathbb{E}[gz_a] = -e_a$, $\mathbb{E}[gT_a] = -L_a$, $\mathbb{E}[rT_a] = -W_a$.

The six anchor budgets. In the following substitutions retain the sharper rational bounds $\delta_4 \le 7d/100$, $\delta_5 \le 3d/100$, $\delta_6 \le d/64$, $\delta_7 \le d/128$, $D_3 \le d/5$, $D_4 \le d/12$, $D_5 \le d/24$, $D_6 \le d/48$. For $a = 3$,

$$A_a = d,\ e_a = W_a = 0,\ z_a = r,\ I_a = L_a = \delta_2,\ J_a = \delta_3,\ C_a = L.$$

For one pair use $I \le 3\delta_4/2$, $J_1 \le 3\delta_5/2$, $|t_0|, |L| \le \delta_3$, $|W| \le \delta_4$, $D_a \le \delta_5$. For multiple pairs use $I \le 4\delta_4$, $J_1 \le 4\delta_5$, $t_0 \le D_3$, $L \le D_4$, $W \ge -D_4$, $D_a \le D_5$.

For $a = 4$,

$$A_a = d, \quad e_a = L_a = \delta_2, \quad z_a = r + s, \quad I_a = \delta_2 + \delta_3, \quad J_a = \delta_3 + \delta_4, \quad W_a = \delta_3.$$

For one pair use $I \le 3\delta_5/2$, $J_1 \le 3\delta_6/2$, $t_0 \le \delta_3$, $L, W \ge -\delta_4$, $C_a \le \delta_4 + \delta_5$, $D_a \le \delta_5 + \delta_6$. For multiple pairs use $I \le 4\delta_5$, $J_1 \le 4\delta_6$, $t_0 \le D_3$, $L \ge -D_3$, $W \ge -D_4$, $|C_a| \le 2\delta_3$, $|D_a| \le 2\delta_4$. The cross bounds follow from the two factors $zG_2(z)$ and $z^{b-1}Q(z)$, whose product is bounded by $(12/5)x^{b/2} < 2x^2$ when $b \ge 5$.

For $a \ge 5$, $I_a \le \delta_2$, $J_a \le \delta_3$. The increment is bounded by

$$\begin{aligned}\Xi \le {} & -8td + 387d/200 - 11\delta_2/25 + 15\delta_3 + 16I + 6J_1 - 9L - 12W \\ & + 44t_0/5 + 64C_a/5 + 12D_a - A_a[4 - 36d/5 - 54A_a/25 - 36t_0/5].\end{aligned}$$

For one pair use $I \le 3\delta_6/2$, $J_1 \le 3\delta_7/2$, $t_0 \le \delta_5$, $L \ge -\delta_5$, $W \ge -\delta_6$, $|C_a| \le 6\delta_4/5$, $|D_a| \le 6\delta_5/5$. For multiple pairs use $I \le 4\delta_6$, $J_1 \le 4\delta_7$ and the corresponding D_5, D_6 mean bounds, with $|C_a| \le 2\delta_4$, $|D_a| \le 2\delta_5$. The bracket is respectively at least $101/125$ and $39/50$, so its whole contribution is nonpositive. The cross estimates follow from $|G_{a-2}| \le 1$ and the one-interval or signed-mask bound, rather than independent estimates of all summands.

These substitutions into the exact increment give the six rational bounds

	$t = 1$	$t \ge 2$
$a = 3$	$\Xi/d \le -59/200$	$\Xi/d \le -2813/600$
$a = 4$	$\Xi/d \le -2401/4800$	$\Xi/d \le -373/300$
$a \ge 5$	$\Xi/d \le -71279/80000$	$\Xi/d \le -41243/6000.$

Extra pairs contribute unused deficit $8d$. The largest entry is $-59/200$, proving the fourth estimate. Hölder gives the stated lower moments under the original probability and $q\mu$; reversal preserves their compatibility for the same fully reversed message.

H.12 All zero-step odd tables with a unit anchor

Throughout this proof $\tau = d - \delta_2$ and $D_j = \delta_j/(1-\theta^2)$. Let $F = E_0 q f_b h_c \prod_{i=2}^{t} h_{b_i} f_{c_i}$, $1 < b < c < b_2 < c_2 < \cdots < b_t < c_t$. For $t = 1, b = 2$ apply Section H.9. The remaining cases satisfy

$$\mathbb{E}[pF^4] \le \begin{cases} e^{-5d/2}, & t = 1, b = 3, \\ e^{-2d/3}, & t = 1, b \ge 4, \\ e^{-3d}, & t \ge 2. \end{cases}$$

Only the first residual pair is reversed relative to the canonical pair rule.

A signed packet master. Set $B = \beta_{c-1}$, $C = \beta_b$, $T_0 = \sum_{j=b}^{c-1} K^j g$, $S = \sum_{i=2}^{t} \sum_{j=b_i}^{c_i-1} K^j g$, and

$$W = T_0 - S = Kv, \quad R = Bg - W, \quad \Delta = \beta_c + B + C + 2(t-1)d, \quad I = \|v\|^2, \quad J_1 = \|W\|^2.$$

The exact inward-path identity and $f_b \le S_b - \beta_b$ give $F/E_0 \le q e^{R-\Delta}$. Use the score inequality from Section H.9 and the sharper $3/8$ quadratic from (A.5). The same weighted Cauchy decomposition with $V = 3g - 4R$ yields

$$\mathbb{E}[p(F/E_0)^4] \le e^{\Xi}, \quad \Xi = -4\Delta + 16I/3 + 4\mathbb{E}R + 27d/8 - 9\mathbb{E}[gR] + 6\mathbb{E}R^2. \tag{H.6}$$

The actual signed outputs are

$$\mathbb{E}R = \beta_{c+1} - \beta_{b+1} + \mathbb{E}S,$$
$$\mathbb{E}[gR] = Bd + \beta_{b+2} - \beta_{c+2} + \mathbb{E}[gS],$$
$$\mathbb{E}R^2 = B^2 d + 2B[\beta_{b+2} - \beta_{c+2} + \mathbb{E}[gS]] + J_1.$$

Finite masks and exceptional prefixes. The signed-mask estimate gives $I \le 4\delta_b$, $J_1 \le 4\delta_{b+1}$. For one pair replace 4 by $1 + \theta < 3/2$. For later positive masks, the skew sign sequence gives $\mathbb{E}S \le D_4$, $\mathbb{E}[gS] \ge -D_4$. If $b_2 \ge 5$, also $\mathbb{E}[gS] \le D_5$. The positive terms begin at exponent seven for $\mathbb{E}S$, the negative mixed terms at six, and the positive mixed terms at eight; summing their finite geometric tails proves these bounds, with $D_4 < d/12$, $D_5 < d/24$. For $b = 2$ separate the first interval from the later

mask: $W = K^2[G_{c-2}(K) - K^N Q_0(K)]g$, $N \geq c-1$. Since $\|Q_0\| < 10/7$, the norm factor is at most $177/112$ when $c = 3$, giving $I \leq 5\delta_2/2$, $J_1 \leq 5\delta_3/2$. For $c \geq 5$, it is at most $1 + (10/7)\theta^2 < 5/4$, so factor 2 suffices. For $c = 4$ retain factor 4, including G_2.

Coefficient certificate. The beta intervals and the preceding signed outputs give the following upper polynomials for (H.6):

$$\begin{aligned}
t = 1, b = 3: \quad & \Xi \leq -69d/8 + 4\delta_2 + 21\delta_3 + 9\delta_4 + Q_1, \\
t = 1, b \geq 4: \quad & \Xi \leq -69d/8 + 12\delta_2 + 13\delta_3 + 8\delta_4 + 9\delta_5 + Q_1, \\
t \geq 2, b \geq 3: \quad & \Xi \leq -133d/8 + 12\delta_2 + 103\delta_3/3 + 24\delta_4 + 13D_4 + Q_2, \\
t \geq 2, b = 2, c = 3: \quad & \Xi \leq -69d/8 + 13\delta_2/3 + 15\delta_3 + 13D_4, \\
t \geq 2, b = 2, c = 4: \quad & \Xi \leq -101d/8 + 25\delta_2/3 + 24\delta_3 + 13D_4 + Q_3, \\
t \geq 2, b = 2, c \geq 5: \quad & \Xi \leq -101d/8 + 17\delta_2/3 + 25\delta_3 + 13D_4 + Q_4,
\end{aligned}$$

where the exact remaining terms are

$$\begin{aligned}
Q_1 &= -9d^2 + 9d\delta_2 + 12d\delta_3 + 6d^3, \\
Q_2 &= Q_1 + 12dD_4, \\
Q_3 &= -9d^2 + 12d\delta_2 + 12dD_5 + 6d^3, \\
Q_4 &= -9d^2 + 21d\delta_2 + 12dD_5 + 6d^3.
\end{aligned}$$

The first three are nonpositive by the rational spectral bounds; $Q_4/d \leq 65/96 < 1$. For clarity the one-pair rows use respectively $\Delta \geq d + 2\tau$ or 3τ, $\mathbb{E}R \leq -\delta_2 + \delta_3$ or δ_3, $\mathbb{E}[gR] \geq \tau d - \delta_3$ and $\mathbb{E}R^2 \leq d^3 + 2d\delta_3 + J_1$. For $t \geq 2, b \geq 3$ add $2d$ to the loss and D_4 to the mean/tail budget. For the final three rows the respective loss bounds are $3d, 4d, 4d - 2\delta_2$; retain $B = 0$ in the $c = 3$ row. Their mean upper bounds are $D_4, -\delta_2 + D_4, -\delta_2 + \delta_3 + D_4$; the mixed lower bounds are $\delta_2 - D_4, d^2 + \delta_2 - D_4, d^2 - d\delta_2 + \delta_2 - \delta_3 - D_4$. For the last two, the square-moment upper bound is $d^3 + 2d(\delta_2 + D_5) + J_1$; for $c = 3$ it is simply J_1. Thus all signs and the exceptional zero coefficient are retained.

Substitution gives the respective strict linear bounds $-65d/24$, $-1781d/2400$, $-1469d/450$, $-233d/72$, $-293d/72$, and $(-361/72 + 65/96)d$. They imply the announced losses $5/2, 2/3, 3$ in all branches. These six rows, together with Section H.9, exhaust $t \geq 1$. Restore the original equal pairs and reverse every edge through the common rules.

I Positive propagation of even terminal tables

Every bound in this appendix uses the residual direction in the even table of Section 5 and covers all positive integer propagation lengths. First we settle single short pairs, then products of pairs. The moment comparison, including its short-tail variant, is proved in Section A.3.

I.1 Propagated pairs $(1, b)$, $b \geq 3$

In this proof $\tau = d - \delta_2$. We prove $\mathbb{E}[p(T^L(qf_b))^4] \leq 1$ for every $b \geq 3, L \geq 1$. Use Lemma M.4 with $K' = -K$, $g' = -g$. Thus $N = b - 1$, $R = KG_N(K)g$, $m = 1 - \beta_b/2$, $k = m^{-1}$, $V = \mathbb{E}R^2$, $U = \|KR\|^2$, $\rho = \mathbb{E}(rR)$, $D_j = \mathbb{E}K'^j R$; the positive terminal upper is $F_0 = (m + R/2)^2$. Put

$Y_L = T^L F_0$, $\gamma = k\beta_{b+1} - k^2 V/2$, and $B_0 = \gamma + k(d - \beta_{b+2})$. The common finite bounds give

$$3d/10 \le \gamma \le 6d/5, \quad 5d/8 \le B_0 \le 3d/2, \quad B_0 \ge d - 29\delta_2/32. \tag{I.1}$$

They also give $U \le \theta V$, $\kappa\delta_2 \le \rho \le \delta_2$, $V \le \delta_2$ for $b \ge 4$, and $V = \delta_2 + \delta_3$, $\rho = \delta_2$ at $b = 3$. For $P_n(z) = G_n(z) + kz^{n+1} G_N(-z)$,

$$\|P_n(K')\| \le 1 \ (n \ge 3), \quad \|P_2(K')\|^2 \le 71/63, \quad \|P_1(K')\| \le 1.$$

With $V_L = \sum_{j=0}^{L} K'^j \mathbf{1} + kK'^L R$, $L \ge 3$, the exact identities (M.13) give $1 - \mathbb{E}V_L, \mathbb{E}(g'V_L) \in [\tau, d]$.

The coefficient profiles A–E used below are proved in Section A.4.

We shall compare with the positive Taylor bound

$$m^{-8} \ge 1 + 4\beta_b + 9\beta_b^2 + 15\beta_b^3 \ge 1 + 7d/3 + 49d^2/16 + 35d^3/12. \tag{I.2}$$

Every $L \ge 2$. Duhamel gives

$$Y_L/m^2 \le V_L - \gamma \bar{f}_{L-1} - k(d - \beta_{b+2})\bar{f}_{L-2} - \sum_{j=3}^{L} \varepsilon_j \bar{f}_{L-j}, \quad \varepsilon_j = \beta_j - kD_{j-1}.$$

$d - k\delta_2 \le \varepsilon_3 \le d$ and $\varepsilon_j \in [\tau, d]$ for $j \ge 4$; thus all discarded terms are nonnegative. For $L \ge 4$ retain $Z = V_L - \varepsilon_L - \varepsilon_{L-1} q$. The mean, covariance and variance bounds are $1 - 3d/2$, $d/2 - d^2$, and $d - 3d^2/4 + d^3$. Its row bound permits certificate A, so the fourth moment is at most

$$1 - d + 85d^2/4 - 38d^3 + 621d^4/16 - 27d^5/2.$$

Subtracting (I.2) and dividing by d leaves at most $-10/3 + 291d/16 - 401d^2/16 \le -2597/76992$.

For $L = 3$ retain $Z = \widehat{V} - a - B_0 q$, $\widehat{V} = V_3 - \gamma r$, $a = d - kD_2$. Subtracting γ from the nonnegative imaginary part of P_3 and real part of P_2 decreases their moduli; hence $\|\widehat{V} - 1\|^2 \le d$, with row tail at most $13\sqrt{d}/12$. The exact mean loss of $\widehat{V}$ is at least $2d/5$. Together with (I.1) this gives $\mu \le 1 - 3d/2$, covariance at least $3d/32$, and $\operatorname{Var} Z \le d - 3d^2/4$. For the last claim maximize $t^2 d - (19/16)t$ over $t = B_0/d \in [5/8, 3/2]$, obtaining at most $-235/384$. Certificate B applies. After discarding the favorable covariance, the Taylor difference divided by d is at least $4/3 - 83d/16 + 189d^2/16 \ge 9239/12096$.

For $L = 2$, use $Z = V_2 - \gamma q - k(d - \beta_{b+2})$. It has $\mu \le 1 - 9d/8$, covariance at least $4d/5 - 6d^2/5$, variance at most $71d/63 - 7d^2/10$, and the row bound of certificate C. The positive endpoint branch is

$$1 + 17d/90 + 2927d^2/160 - 21789d^3/640 + 499365d^4/20480 - 2187d^5/320.$$

Its difference above the Taylor bound, divided by d, is at most $-193/90 + 61d/4 - 57d^2/2 \le -2857/27360$.

One step and its coupled spectral budget. Exactly $Y_1 = m^2 Z$, where

$$Z = q - k\beta_{b+1} - kKR + e, \quad e = (V - K(R^2))/(4m^2), \quad 0 \le e \le V/(2m^2).$$

The original mean, covariance and variance bounds are

$$\mu \le 1 - d + 29\delta_2/32, \quad h \ge d - 25\delta_2/16, \quad \operatorname{Var} Z \le d - 5\delta_2/16.$$

The covariance error coefficient is at most $1203/800 < 25/16$. Before adding e, the variance loss is at least $25\delta_2/32$ for $b \ge 4$ and $7\delta_2/5$ for $b = 3$. Centering e adds at most $(k^2V/2)\sqrt{d} + k^4V^2/16$; the respective added coefficients are below $1971/4375 < 15/32$ and $243/400 + 1/20$, leaving the claimed loss. For $b \ge 4$ the row bound is $q + 4\sqrt{d}/5 - 25d/64$; for $b = 3$ it is $q + \sqrt{d} - 5d/7$. Certificates D and E apply.

Put $v = \delta_2/d \in [d, \theta]$, $A_v = 1 - 29v/32$, $B_v = 5 - 229v/32$. The positive endpoint in (A.9) is valid because $B_v d \le 251/288 < 1$. Comparing with the sharper Taylor bound using $\tau = (1 - v)d$ gives the complete polynomial

$$F(v, d) = P_0(v) + Q(v)d - H(v)d^2 + J(v)d^3,$$
$$P_0 = -5 + 187v/16, \quad Q = 9 - 45v/2 + 6615v^2/512,$$
$$H = A_v^3 + 3A_v^2 B_v + 15(1 - v)^3, \quad J = A_v^3 B_v.$$

For $v \le 1/3$, the last two terms sum to a nonpositive quantity; $F \le P_0 + Qv$ increases to $-193/1536$. For $1/3 \le v \le 3/8$, decreasing Q, H, J give $F \le -79/128 + 3d - 5d^2 \le -107/640$. For $3/8 \le v \le \theta$, use $P_0 \le -129/512$, $Q < 119/50$, $H > 589/100$ and $J < 2/3$:

$$F \le -129/512 + (119/50)d - (5101/900)d^2 \le -129/512 + 127449/510100 < 0.$$

This covers all continuous parameters and all finite b. Positivity restores equal pairs; full reversal and the $d = 0$ case follow from the common rules.

I.2 The propagated pair $(2, 3)$

In this proof $\tau = d - \delta_2$. Here $P = h_2 f_3$. We prove $\mathbb{E}[p(T^L P)^4] \le 1$ for every $L \ge 1$, with the same actual directions at every length and parameter. Put $m = 1 - d/2$, $k = 1/m$ and $s = K^2 g$. Since $h_2 + f_3 = 2 - d + s$, positivity and AM–GM give

$$P \le F_0 = (m + s/2)^2, \qquad Y_L = T^L F_0 \ge T^L P \ge 0.$$

We shall compare all moments with

$$m^{-8} \ge 1 + 4d + 9d^2, \qquad dm^8 < 1/9. \tag{I.3}$$

The first is the positive Taylor remainder of $(1 - d/2)^{-8}$. The second follows by differentiation: its maximum is $(2/9)(8/9)^8 < 1/9$ at $d = 2/9$.

Finite spectra and every $L \ge 4$. Let $Q_n(z) = G_n(z) - kz^{n+2}$. For every $n \ge 3$ its spectral norm is at most one. To see this write

$$Q_n = \frac{1 - z^n(1 + kz^2 - kz^3)}{1 - z}, \quad A_x = 1 - kx \ge 41/80, \quad C_x = A_x^2 + k^2 x^3 \le 1.$$

For $n \equiv 0, 3 \pmod 4$ the numerator of the squared modulus is at most one. For residues $1, 2$ the losses after division by x are at least $1 - (12/5)x^3 - x^4 \ge 51/80$ and $1 - 2x^2 - x^5 \ge 15/32$, respectively. This includes every finite length. For $V_L = S'_L - kK'^{L+3}\mathbf{1}$, $L \ge 3$, finite alternating means give $\alpha = 1 - \mathbb{E}V_L, \eta = \mathbb{E}[g'V_L] \in [\tau, d]$. Even added powers are $\beta_{2j-2} + k(-1)^j\delta_j$; pair consecutive terms, using $k\theta^2 < 1$, with the first case $d - k\delta_3 \ge \tau$. Hence $\|V_L - 1\|^2 \le d$ and, for $L \ge 4$, $|V_L - q| \le \sqrt{d}$.

Duhamel and $T(s^2) \le 2\delta_3$ give

$$Y_L \le m^2 V_L - m\delta_2 \bar{f}_{L-2} + (\delta_3/2)\bar{f}_{L-1} - \sum_{j=3}^{L} \omega_j \bar{f}_{L-j},$$

where $\omega_j = m^2\beta_j$ for odd j, and $\omega_j = m^2\beta_j - m(-1)^{j/2}\delta_{j/2+1}$ for even j. All are at least $m^2\tau$: for $j \equiv 0 \pmod 4$ pair the final positive beta term with $\delta_{j/2+1}$, using $m > \theta^2$; the boundary $j = 4$ uses $m > \theta$. For $L \ge 4$ this implies $Y_L \le m^2 Z$ with

$$Z = V_L - a - \tau q, \quad a = \tau - \delta_3/m^2 \ge 2279d/6400 > d/3.$$

Then $\mu = \mathbb{E}Z \le 1 - d$, covariance $\mathbb{E}[g'(Z-\mu)] \ge 0$, $\operatorname{Var} Z \le d$, and $Z \le (1-d/3)q - d/3 + \sqrt{d}$. Profile A of Section A.4 bounds the fourth coefficient by seven for exactly this mean and row envelope. Thus $\mathbb{E}[pY_L^4] \le m^8[(1-d)^4 + 7d] \le 1$; the Taylor difference is $d + 3d^2 + 4d^3 - d^4 \ge 0$.

Three and two steps. For each short case below define $H = p(B^2 + 2uB + 3u^2)$ using that case's stated mean upper bound u and pointwise upper bound B. For $L = 3$, retain the actual term $-m\delta_2 q$:

$$Y_3 \le m^2 Z, \quad Z = V_3 - a - bq, \quad a = d - \delta_3/m^2 \ge 4879d/6400, \quad b = k\delta_2.$$

The mean is at most $1-d$, covariance at least $3d/10$, and variance at most d. The tail Q_2 has squared modulus $(1-kx^2)^2 + x \le 233/200 < (13/12)^2$. Thus $Z \le q - 3d/4 + 13\sqrt{d}/12$. The coefficient certificate is

$$D = -1/4 - 407x^2/144 - 91x^3/24 + 81x^4/16,$$
$$Q = -5/8 + 39x/8 - 1163x^2/96 - 91x^3/16 + 243x^4/32.$$

Here $D = H'(3/2)$, $Q = H(3/2) - 7$ and $H'' \le -3$ on the maximizing interval. For $x \le 1/3$, $-D < 2/3$, $Q < -1/8$, giving margin $-11/216$. For $x \ge 1/3$, convexity and negative endpoints of Q' give $Q \le -25/54$, $-D < 7/5$, giving margin $-92/675$. The same Taylor comparison proves the three-step case.

For $L = 2$, set $\varepsilon = \delta_3/(2m^2)$ and $Z = V_2 - k\delta_2 + \varepsilon q \ge Y_2/m^2$. Then

$$\mu \le 1 - d, \qquad h \ge 4d/5, \qquad \operatorname{Var} Z \le 25d/22, \qquad Z \le q + 21\sqrt{d}/20.$$

For the variance, use $v = \delta_2/d$ and

$$d + \delta_2 - 2k\delta_3 + k^2\delta_5 \le d[1 + v - (2-\theta^2)v^2] < 25d/22;$$

the remaining $-d^2 + 2\varepsilon(d - k\delta_3) + \varepsilon^2 d$ is nonpositive since $\varepsilon < 3d/25$. The row bound is $\sqrt{d + k^2\delta_4} \le 21\sqrt{d}/20$. For $\lambda = 21/20$ the corresponding coefficient function is concave, has maximum below $3/2$ and curvature at most -3; its derivative there has absolute value below $4/3$, while its value is at most $57/8$. The tangent bound $57/8 + 8/27 < 15/2$ proves $M \le 15/2$. The endpoint comparison is therefore

$$\mathbb{E}[pY_2^4] \le m^8 \max\{Cd, (1-d)^3(1 - 21d/5) + Cd\}, \quad C = 375/44.$$

The first branch is below $125/132$ by (I.3). For the second, its difference below $1 + 4d + 9d^2$ is at least $21d/44 + 61d^3/5 \ge 0$.

One step. Exactly $Y_1 = m^2 Z$, where

$$Z = q - kK^3 g + e, \quad e = (\delta_3 - K(s^2))/(4m^2), \quad 0 \le e \le \delta_3/(2m^2).$$

The mean, covariance and centered variance obey

$$\mu \le 1 - 3\delta_2/4, \quad h \ge d - \delta_2/10, \quad \operatorname{Var} Z \le d + 7\delta_2/16.$$

Indeed the mean loss and covariance error are bounded by

$$(1 - \theta/2)\delta_2 = 51\delta_2/64, \qquad (117/800)\sqrt{d}\,\delta_2 < \delta_2/10,$$

respectively. For the variance, use the square of

$$\sqrt{d + k^2\delta_4} + \delta_3/(4m^2).$$

The three added coefficients are $1521/6400$, $2457\sqrt{d}/8000$, $13689\delta_2/640000$, whose sum is below $751/1750 < 7/16$. Put $u = \sqrt{\delta_2} < 3/8$. The row envelope is $Z \le q + 4u/5 + 3u^2/10$. For this envelope and mean $1 - 3u^2/4$, the concave fourth coefficient has curvature at most -3 below $3/2$ and

$$D = -1/4 - 59u^2/25 - 18u^3/25 + 531u^4/400,$$
$$H(3/2) = 51/8 + 18u/5 - 1113u^2/200 - 27u^3/25 + 1593u^4/800.$$

$-D < 5/8$; the cubic and quartic sum is nonpositive, and the quadratic gain is below $3/5$. Hence $M < 51/8 + 3/5 + 25/384 < 57/8$. The zero mean endpoint is below $247/256$ by $\delta_2 \le d/2$ and (I.3). For the other endpoint, expand with $\delta = 3\delta_2/4$:

$$(1 - \delta)^3(1 - \delta - 4h) \le 1 - 4d - 13\delta_2/5 + 9d\delta_2 + 99\delta_2^2/40.$$

Adding $M(d + 7\delta_2/16)$ and subtracting from the Taylor lower bound leaves at least $789d/1280 + 621d^2/160 \ge 0$. This completes every positive length. Equal pairs are restored by positivity, full reversal exchanges p, T with q, A, and $d = 0$ gives equality.

I.3 Propagated pairs $(2, b)$, $b \ge 4$

In this proof $\tau = d - \delta_2$. For $F = h_2 f_b$ we prove $\mathbb{E}[p(T^L F)^4] \le 1$ for all $b \ge 4, L \ge 1$. First assume $0 < d \le 1/4$ and put $m = 1 - d$, $k = m^{-1}$, $u_0 = 1/2 + d$, $z_0 = \theta + 11d/16$.

A retained quadratic envelope. In $h_2 + f_b = 2 + \sum_{j=3}^{b} K^j \mathbf{1} - \sum_{j=3}^{b} \beta_j f_{b-j}$ retain coefficients C_1, C_2, C_3 as follows:

$$(C_1, C_2, C_3) = \begin{cases} (d, 0, 0), & b = 4, \\ (d, \delta_2, 0), & b = 5, \\ (\beta_{b-1}, \lambda\beta_{b-2}, \lambda\beta_{b-3}), & b \ge 6, \end{cases}$$

where $\lambda = (2d - \beta_b - \beta_{b-1})/[\beta_{b-2} + (1 - d)\beta_{b-3}] \in [0, 1]$. Indeed the numerator is at most $2\delta_2$, whereas the full retained constant is at least $(4 - d)\tau \ge 209d/96 > 2d$. Then $0 \le C_i \le d$, $C_1 \ge \tau$, and $\beta_b + C_1 + C_2 + (1 - d)C_3 = 2d$. Set $C = C_1 + C_2 + C_3$, $D = C_2 + C_3$, $E = C_3$ and

$$R = \sum_{j=3}^{b} K^j \mathbf{1} - Cg - Dr - Es.$$

The actual nonnegative sum is at most $2m+R$, so $F \le F_0 = (m+R/2)^2$. At spectral frequency $i\sqrt{x}$, write $G_{b-2} = a + i\sqrt{x}\, b_x$, $1-x \le a, b_x \le 1$. The multiplier of R at g is $-A - i\sqrt{x}B$, with

$$A = C + x(a-E), \quad B = D + xb_x,$$

$$V = \mathbb{E}R^2 = \int (A^2 + xB^2), \quad D_0 = \mathbb{E}R = \int B,$$

$$a_1 = -\mathbb{E}[gR] = \int A, \quad \rho_0 = -\mathbb{E}[rR] = \int xB.$$

All integrals use the spectral measure of g. Since $(1-x)B - A \le -C_1 - C_2 x$,

$$\mathbb{E}[f_2 R] \le -\tau d, \quad a_1 \ge D_0 - \rho_0 + \tau d, \quad D_0^2 - 2D_0 a_1 \le 0. \tag{I.4}$$

The pointwise spectral bounds are

$$\sqrt{A^2 + xB^2} \le u_0, \quad B \le z_0; \qquad \sqrt{A^2 + xB^2} \le 3/8 + 3d/2 \quad (b \ge 5).$$

For $b=4$ use $A = d+x$, $B = x$ and $\theta^2(1+\theta) < 1/4$. For $b=5$ the constant and d vectors have squared norms $131313/1048576 < 9/64$ and $66997/32768 < 9/4$. For $b \ge 6$ use $u = \theta - \theta^2 + \theta^3 < 5/16$ and maximize

$$[u + (1+\theta)d + E(d-\theta)]^2 + \theta[u + 2\theta d + dE]^2$$

at $E = 0, d$. Their square-root bounds are respectively $3/8 + 3d/2$ and $3/8 + 8d/7 + 6d^2/5 \le 3/8 + 3d/2$. The bound for B follows from $u + 2\theta d + d^2 - z_0 \le 5/16 - 13/32 + d/8 + d^2 \le 0$.

Let $D_j = \mathbb{E}[K'^j R]$. The repeating signs are

$$D_{4j+1} = -\int x^{2j} A, \quad D_{4j+2} = -\int x^{2j+1} B, \quad D_{4j+3} = \int x^{2j+1} A, \quad D_{4j+4} = \int x^{2j+2} B.$$

Thus $D_3 \le \theta u_0 d$ and every positive D_j, $j \ge 4$, is at most $\theta^2 z_0 d$. Also $a_1 - D_0 \le d^2 + \theta^3 d$, since $A - B = C_1 - Ex + x(a - b_x)$ and the last possible positive term has integral at most δ_4.

One step. Exactly $Y = TF_0 = m^2 q + m(D_0 - KR) + e$, $0 \le e \le V/2$. Its fourth coefficient relative to m^2 is at most $7m^2$. Here is a direct certificate. Set $x = \sqrt{d} \le 1/2$, $y = q+1$, $A_y = y^2 + 2$ and $p = 3 - y$. Then $pA_y \le 32/5$ and $py^2 \le 4A_y/5$. The row errors divided by m are at most $x/2 + 21d/16$ for $b=4$, and $3x/8 + 7d/4$ for $b \ge 5$. After absorbing cubic and quartic terms using $x \le 1/2$, the coefficient divided by m^2 is bounded respectively by

$$p[A_y + yx - D_y x^2], \quad D_y = 2y^2 - 21y/8 + 2439/1024 \ge A_y/2,$$

and

$$p[A_y + 3yx/4 - D_y x^2], \quad D_y = 2y^2 - 7y/2 + 43/16 \ge 3A_y/8.$$

Completing the square gives $32/5 + 2/5 < 7$ and $32/5 + 3/10 < 7$.

For $w = Y - m^2$, (I.4) yields

$$\mathbb{E}[pw] \le -m^2 d - m\tau d + V/2, \quad \mathbb{E}w^2 \le m^2 dH + (6/5)md^{3/2}v + d^2 v^2/4,$$

where $v = (1/2 + d)^2$ and $H = (733 - 699d + 434d^2)/512 < 36/25$. The fourth expansion is at most

$$U = m^8(1-4d) - 19m^7 d^2/8 + 2m^6 dv + 7m^4 dH + (42/5)m^3 d^{3/2} v + (7/4)m^2 d^2 v^2.$$

Positive Taylor terms of m^{-4}, followed by $m^{-1} \le 1 + 4d/3$, $m^{-2} \le 16/9$, show that $(1-U)/(m^4 d)$ is at least

$$Q(x) = 757/512 - 21x/10 - 13x^2/10 - 56x^3/5 + 25x^4.$$

The discarded even-power coefficients are positive: the remaining fourth and sixth coefficients are $288989/11520 > 25$ and $2023/120$. For $x \le 2/5$, completing the square gives $Q \ge 1838493/8000000 > 0$. For $x = 1/2 - y$, $0 \le y \le 1/10$, its exact expansion is $681/2560 - 7y/10 + 97y^2/5 - 194y^3/5 + 25y^4$, at least $251545/1600000 > 0$.

The common propagation calculation. Put $\delta = kD_0 + k^2 V/2$, $\varepsilon = \delta - ka_1$, $\nu = 1 + \varepsilon$. Duhamel gives the positive upper comparison

$$T^L F_0/m^2 \le Z_L^* = S'_L + \delta S'_{L-1} - ka_1 S'_{L-2} + kK'^L R.$$

The dropped coefficients are $\gamma_3 = d + k\rho_0$, $\gamma_4 = d + \delta d - kD_3$ and $\gamma_j = \beta_j + \delta\beta_{j-1} - ka_1\beta_{j-2} - kD_{j-1}$. They are nonnegative; for $j \ge 5$, $\gamma_j \ge (1/2 - 6d/5)d$. The comparison follows from the endpoint-positive concave polynomial $875/32768 + 40369d/81920 - 11d^2/5$ after multiplying by m. We also have

$$|\varepsilon| \le \frac{297 + 21d + 834d^2}{1024m^2} d \le 315d/512, \qquad 0 \le \delta \le \frac{544 + 800d - 192d^2}{1024m^2} d \le 61d/48.$$

For the lower ε bound use $a_1 - D_0 \le d^2 + \theta^3 d$; the numerator difference is $7307 - 29899d + 59456d^2 > 0$.

With $R = (A - Bz)g'$ define $P_n = G_n + kz^n(A - Bz)$. Then $\|P_n\| \le 1$ for $n \ge 3$, $|P_2|^2 \le 1 + x$ and $\|P_1\| \le 10/7$. For $n = 3$ both real and imaginary parts decrease those of G_3. For $n = 4, 5$ the respective upper bounds are $1 - x/2 + x^2/2$ and $1 - x/2 + 23x^2/20$. For $n \ge 6$ use $(1 + x^3)/\sqrt{1+x} + x^3 \le 1$, which follows from $x(1 + \sqrt{1+x}) \le 143/160 < 1$ (with equality in the original comparison at $x = 0$). For P_2 it suffices that $2 - 4x - kx[u_0(1-x) + xz_0] \ge 323/49152 > 0$; for P_1 the squared upper bound is $1561/768 < 100/49$.

For $L \ge 4$, write $Z_L^* = \nu + Q_L(K')g'$ with $Q_L = P_L + \varepsilon G_{L-2} + \delta z^{L-2}$. Both its norm and row-tail bound are at most $N = 1 + 29d/20$. The boundary $L = 4$ uses $1 + |\varepsilon| + \sqrt{\theta}\delta \le 1 + 11069d/7680 \le N$; the other lengths use $(6/5)|\varepsilon| + \theta\delta$. With $\alpha = \nu - \mathbb{E}Z_L^*$ and $\eta = \mathbb{E}[g' Z_L^*]$, the exact mean losses are

$$\alpha = \beta_{L+1} + \delta\beta_L - ka_1\beta_{L-1} - kD_L,$$
$$\eta = \beta_{L+2} + \delta\beta_{L+1} - ka_1\beta_L - kD_{L+1}.$$

Both are at least $(1/2 - 6d/5)d$. The expansion around ν is exactly

$$\mathbb{E}[pZ^4] = \nu^4 + 4\nu^3\mathbb{E}[p(Z - \nu)] + \mathbb{E}[p(Z - \nu)^2(Z^2 + 2\nu Z + 3\nu^2)],$$
$$\mathbb{E}[p(Z - \nu)] = -\alpha - \eta.$$

Using $\nu \le u = 1 + 2d/3$, the fourth comparison is bounded by

$$u^4 - 4u^3 d + (48/5)u^3 d^2 + (32/5)u^2 N^2 d + (9/4)uN^3 d + 2N^4 d^2.$$

This uses $\sqrt{d} \le 1/2$ and the joint endpoint maximum of $t^3(t - 8(1/2 - 6d/5)d)$; its upper endpoint is positive. The polynomial coefficients are $1, 439/60, 17859/400$, followed by values below $89, 84, 39, 9$. The positive Taylor coefficients of m^{-8} dominate them, leaving $d[41/60 - 3459d/400 + 31d^2] > 0$ by completing the square.

For $L = 3$ retain $\ell = d + k\rho_0$ and use $Z = Z_3^* - \ell$. Its reference is $\nu_0 = 1 + \varepsilon - \ell \in [5/12, 1 - 3d/8]$ and $\|Z - \nu_0\| \le N\sqrt{d}$. Here $\alpha = \nu_0 - \mathbb{E}Z = d + \delta d - kD_3 \ge (19/32)d$ and $\eta = \mathbb{E}[g'Z] = \tau + \varepsilon d - kD_4 \ge 15371d/49152$, so the combined loss is $\alpha + \eta \ge 7d/8$. These bounds and the row bound $q + 11\sqrt{d}/6$ give the coefficient $32/5 + 33\sqrt{d}/4 + 121d/18$. The fourth polynomial after $\sqrt{d} \le 1/2$ has coefficients $1, 221/40, < 421/10, < 42, < 15$. The Taylor difference is at least $d[99/40 - 61d/10] + 78d^3 + 315d^4 \ge 19d/20$.

For $L = 2$, $Z = Z_2^*$ has

$$\mu \le u = 1 - 2d/3, \quad h \ge 19d/32, \quad \operatorname{Var} Z \le 45d/32 + 5d^2/3 + 16d^3/9.$$

The mean estimate uses (I.4); the variance uses the real-shift bound

$$|P_2 + \delta|^2 \le (1 + \delta)^2 + x,$$

and subtracts the squared mean loss. Its fourth coefficient is at most $32/5 + 9\sqrt{d} + 8d$. For $q \le 1$ the row bound is $q + 2\sqrt{d}$. For $1 \le q \le 2$ compare instead $q + 10x/7 + E_q x^2$, $E_q = 5/8 + 9(q - 1)/7$, with that same coefficient. The difference divided by x is at most

$$-8(q + 1)/7 + x[2(q + 1)E_q - 4q/3 - 292/49] + (20/7)E_q x^2 + (E_q^2 + 4/3)x^3.$$

Convexity in $x \in [0, 1/2]$, then in q at $x = 1/2$, reduces it to two negative endpoints, $-565/98 + 95/56 + 25/512$ and $-1621/75264$. Using $\sqrt{d} \le 1/8 + 2d$ gives the fourth polynomial

$$u^4 - (19/8)du^3 + (301/40 + 26d)(45d/32 + 5d^2/3 + 16d^3/9).$$

The upper mean endpoint is positive because $u - 4(19/32)d = 1 - 73d/24 \ge 23/96 > 0$. Its coefficients are $1, 4255/768, 2713/48, 14137/270, 3817/81$. The Taylor difference is at least $d[1889/768 - 985d/48 + 67d^2] + 282d^4 > 0$.

The large range, using the actual message. For $d \ge 1/4$ put $t_b = \mathbb{E}[h_2 f_b]$. The first four means are $1 + d, 1, 1 - d - \delta_2, 1 - 2d - d^2$. Weighted positive convolution gives

$$t_4 = 1 - 3d + \delta_2 - d^2 + \delta_3 \le 1353/4096 < 2/5, \quad t_5 = 1 - 4d + 2(1 + d)\delta_2 \le 65/256 < 2/5.$$

For $b \ge 6$ retain those four actual means to obtain

$$t_b \le 1 - \tau(5 - 2d - d^2 - \delta_2) + \delta_3 \le 1 - d(2871 - 1463d - 608d^2)/1024 \le 2/5.$$

The last loss increases and equals $9869/16384 > 3/5$ at $d = 1/4$. Therefore $TF \le 4/5$. Positivity and the actual path moments $\mathbb{E}[pq^4] \le 11/6$, $\mathbb{E}[p\bar{f}_n^4] \le 2$ for $n \ge 2$ yield, respectively, $256/625$, $1408/1875$, $512/625$ at $L = 1, 2, \ge 3$. Both parameter ranges include $1/4$. The common rules cover $d = 0$, all equal pairs, and full reversal without changing any actual direction.

I.4 Even terminals of minimum at least four

Let $P = \prod_{i=1}^{t} h_{a_i} f_{b_i}$, where $4 \le a_1 < b_1 < \cdots < a_t < b_t$. We prove, for every $L \ge 1$,

$$\mathbb{E}[p(T^L P)^4] \le e^{-d/20}; \qquad \mathbb{E}[p(T^L P)^4] \le e^{-[8(t-2)+481/1000]d} \quad (t \ge 2).$$

The reference weight. Define $U_L = p\bar{f}_L^3$, $W_L = A^L U_L$. Rowwise Hölder for the positive kernel T^L and adjointness give

$$\mathbb{E}[p(T^L F)^4] \le \mathbb{E}[W_L F^4], \qquad \mathbb{E}W_L = \mathbb{E}[p\bar{f}_L^4]. \tag{I.5}$$

The following bounds will also be used for minimum three:

L	$\mathbb{E}W_L$	$\|W_L - \mathbb{E}W_L\|_2/\sqrt{d}$	$\|W_L\|_\infty$
1	$1 + 12d/5$	$23/10$	$8/3$
2	$e^{27d/8}$	$291/80$	4
≥ 3	1	$10093/1280 < 8$	4

We prove these bounds from the path recursion. With $P_0 = I - J$ and $B = P_0 K P_0$, a finite polynomial with coefficients in $[0, M]$ has L^2 norm at most $3M/2$ on B. Indeed at $x = \lambda^2 \le \theta$ its squared modulus is at most $M^2(1+x)/(1-x^2)^2 \le 32768M^2/16245 < 9M^2/4$. This is a spectral norm estimate on B. The centered recursion

$$\bar{f}_n - \mathbb{E}\bar{f}_n = -\sum_{j=0}^{n-1} \mathbb{E}\bar{f}_{n-1-j}(-B)^j g$$

therefore has norm at most $3\sqrt{d}/2$. For $L \ge 2$ the actual cubic path estimate gives $\mathbb{E}U_L \le 1$; positivity implies $W_L \le 4$. Since $|x^3 - y^3| \le 7|x-y|$ for $0 \le x \le 2$, $0 \le y \le 1$, $\|U_L - (\mathbb{E}\bar{f}_L)^3\| \le 37\sqrt{d}/2$. Centering the A recursion yields

$$\|W_L - \mathbb{E}W_L\| \le [3 + (37/2)\theta^{L/2}]\sqrt{d}.$$

For $L \ge 3$ use Section G.1 for its mass. For $L = 1$, $U_1 = pq^3 = 1 + g(-2 + 2g^2 - g^3)$; the last polynomial has absolute value at most two. Thus its centered image has norm at most $(1 + 2\sqrt{\theta})\sqrt{d} \le 23\sqrt{d}/10$, and $\mathbb{E}U_1 \le 1 + d$. The mass bound is (A.10). For $L = 2$ use the scalar inequalities in (A.5) and (A.6) with $X = -3g + 4r$:

$$\mathbb{E}[p\bar{f}_2^4] \le e^{26d/25}[1 - 11d/32 + 13\delta_2/2] \le e^{5339d/1600} \le e^{27d/8}.$$

For centering, $\max pq^2 = 32/27$ gives

$$|U_2 - U_1| \le |r|(32/9 + 3\sqrt{d} + 2d), \qquad \|U_2 - 1\| < 6\sqrt{d}.$$

Expand the actual A^2 before centering:

$$P_0 W_2 = (\mathbb{E}U_2)g + (r + d) + P_0 K^2 (U_2 - 1).$$

The first two terms are orthogonal; hence the norm is at most $(6/5 + 6\theta)\sqrt{d} = 291\sqrt{d}/80$. Uniformly the table implies $\mathbb{E}W_L \le e^{27d/8}$ and $\mathbb{E}W_L \le 10/3$; for the latter use (A.11) at $9/8$, obtaining $251/80 < 10/3$.

Finite exponential tails. Taylor's integral remainder for negative arguments and the positive series ratio for nonnegative arguments give

$$e^u \le 1 + u + 2u^2/3 \ (|u| \le 3/4), \qquad e^u \le 1 + u + 3u^2/4 \ (|u| \le 1). \tag{I.6}$$

The positive remainder coefficient is at most $1/2+u/[6(1-u/4)]$, which is $17/26<2/3$ or $13/18<3/4$. For $W\ge 0$, $M=\mathbb{E}W$, retain the mean through

$$\mathbb{E}[WR]\le M\mathbb{E}R+\|W-M\|_2\|R\|_2. \tag{I.7}$$

For one pair, the actual envelope is $h_af_b\le e^{R-2d}$, $R=K^aG_{b-a}(K)g$. If $a=4,5$, $\mathbb{E}R\le 0$, $\|R\|\le(507/2560)\sqrt{d}<\sqrt{d}/5$, and $|4R|<3/4$. Applying the first inequality in (I.6) with the weight table gives the following upper exponents for $\mathbb{E}[W_Le^{4R}]$:

L	1	2	≥ 3
cost/d	$267/50<6$	$404817/51200<159/20$	$206590332/26214400<79/10$.

The terms are respectively the mass, $4\|W_L-\mathbb{E}W_L\|\|R\|$, and $(32/3)\|W_L\|_\infty\|R\|^2$; thus every entry is specified by the preceding table. Multiplication by e^{-8d} gives losses $2d, d/20, d/10$. For $a\ge 6$, use $\mathbb{E}R\le\theta^3 d$, $\|R\|\le 21\sqrt{d}/250$. The mean, covariance and square costs are at most $14d/15$, $336d/125$, $147d/500$. Together with $27d/8$ their sum is $21871d/3000<15d/2$, leaving loss $d/2$.

For $t\ge 2$, $P\le e^{R-2td}$ where the exponent intervals in $R=\sum_i K^{a_i}G_{b_i-a_i}(K)g$ are disjoint. The finite coefficient bound yields

$$\|R\|\le\sqrt{d}/4,\quad \|R\|^2<63d/1000,\quad \|R\|_\infty^2\le(9/4)\theta^3 d,\quad \mathbb{E}R\le\delta_4/(1-\theta^2)<21d/250.$$

Now $|4R|<1$; the second inequality of (I.6) gives mean, covariance and square costs $28d/25$, $8d$, $378d/125$. Including the mass cost gives $15519d/1000$. The actual loss $8td$ proves the stated stronger bound.

I.5 The propagated pair $(3,b)$

The actual message here is $P=f_3h_b$, $b\ge 4$. We prove $\mathbb{E}[p(T^LP)^4]\le e^{-d/1000}$ for every $L\ge 1$. The length-three arm is outward; the first edge of the longer arm is inward. These directions are kept in the multiple-pair argument below.

The actual retained envelope. Use the coefficients C_1,C_2,C_3 from Section I.3; their construction and $0\le\lambda\le 1$ remain valid up to $d=1/3$. Write $C=C_1+C_2+C_3$, $D=C_2+C_3$, $E=C_3$ and

$$R=-\sum_{j=4}^{b}K^j\mathbf{1}+Cg+Dr+Es.$$

The positive convolution identity $h_b=2-\beta_b-S_b-\sum_{j=3}^{b-1}\beta_jh_{b-j}$ gives $0\le f_3+h_b\le 2-3d+R$. Hence

$$P\le(1-3d/2+R/2)^2\le e^{R-3d}. \tag{I.8}$$

At frequency $i\sqrt{x}$, write $G_{b-2}=a+i\sqrt{x}\,b_x$ and put

$$H=1-a+E,\quad B=D+xb_x,\quad Q=C-xH+i\sqrt{x}B,\quad R=Q(K)g=KV.$$

Since $1-x\le a,b_x\le 1$, we have $\mathbb{E}R=-\int B\,d\nu_g\le-(1-\theta)\delta_2$. For $b\ge 6$, $x-x^2+E\le H\le x+E$ and $B\le D+x-x^2+x^3$. The expression $|Q|^2=C^2+x[-2CH+xH^2+B^2]$ is convex in H; at either endpoint the bracket is at most $x^2(1+x)+D^2$. For the upper endpoint expand it as

$x^2(1+x)+2x(xE-C_1)+(D-E)^2-(1-x)E^2-2C_1E$. For the lower endpoint use $xE\le C_1$, $xE\le 2C$ and $2D\le 4/3<2-x+x^2-x^3$. Integrating against the spectral measure at $\mathbf{1}$ gives

$$\|V\|^2\le C^2+(1+\theta)\delta_3+D^2d\qquad(b\ge 6).\tag{I.9}$$

Only the original row bound is then used for $|R|\le\|V\|$. For $b=4$, $R=dg-K^3g$ and $\|R\|^2=d^3+\delta_4$, $\|V\|^2=d^2+\delta_3$. For $b=5$,

$$\|V\|^2=d^2+(1+d)\delta_2^2+\delta_3+\delta_4,\quad |Q|^2=(d+\delta_2)^2+\delta_2^2x-2dx^2+x^3+x^4.$$

Uniform small and large parameter bounds. For $d\le 1/9$ the preceding formulas imply

$$\|R\|\le 31\sqrt{d}/100,\qquad |R|\le 1/4.\tag{I.10}$$

Here are the finite certificates covering all b. For $b=4$, the normalized squared norm is below $1/12$, and $\|V\|^2\le 5/162<1/16$. For $b=5$, convexity in $d\in[0,1/9]$ reduces its frequency square to $d=0,1/9$; at $d=1/9$ split at $x=1/4$. The two bounds are $109/2304$ and $157/1800$, both below $19/200$; also $\|V\|^2\le 32/2187+1/36<1/16$. For $b\ge 6$ use $C<7/40$, $D<103/1000$ and $D<1/9$. When $x\le d$, $|Q|^2\le 49/1600+4/729<1/16$. When $x\ge d$, $|C-xH|\le\theta^2$ and $B<823/2000$, so $|Q|^2<12293277/128000000<961/10000$. The row-factor bound (I.9) is $49/1600+1/729+1/36<1/16$. These two frequency intervals impose no spectral lower cutoff.

For $1/9\le d\le 1/3$,

$$|R|\le(69/32-1/4000)d.\tag{I.11}$$

For $b=4,5$ the squared row-factor ratios are below $5/2, 9/2$. For $b\ge 6$, (I.9) gives the convex function

$$\|V\|^2/d^2\le(1+\theta+d)^2+d(2\theta+d)^2+(1+\theta)\theta^2/d.$$

Its endpoint bounds, relaxing $\theta<5/12$, are $216799/46656$ and $7351/1728$. The larger is below $(69/32)^2-1/500$, which is below $(69/32-1/4000)^2$. This proves the full large interval.

Propagation along the connector. Use W_L from (I.5). For $d\le 1/9$ its centered constants improve to

$$\frac{\|W_L-\mathbb{E}W_L\|}{\sqrt{d}}\le 23/10\ (L=1),\quad 10/3\ (L=2),\quad 5\ (L\ge 3).\tag{I.12}$$

For $L=2$ the difference-of-cubes bracket is at most $43/9$, giving $\|U_2-1\|\le 919\sqrt{d}/180$; the exact centered A^2 formula yields $18859\sqrt{d}/5760<10\sqrt{d}/3$. For longer paths write $\bar f_L=mq+u$, $m=\mathbb{E}\bar f_{L-1}$, $u=-K(\bar f_{L-1}-m)$. The centered original row bound gives $|u|^2\le(9/4)pqd$, and $\|u\|\le 39\sqrt{d}/40$. Thus $\|U_L-m^3\|\le[2+(39/40)(289/54)]\sqrt{d}<29\sqrt{d}/4$. The centered recursion from Section I.4 now gives $[3+(29/4)\theta^{L/2}]\sqrt{d}\le 12581\sqrt{d}/2560<5\sqrt{d}$.

For $|u|\le 1$, the series proof of (I.6) also gives $e^u\le 1+u+13u^2/18$. Since $\mathbb{E}R\le 0$, (I.10) yields

$$\mathbb{E}[W_Le^{4R}]\le\mathbb{E}W_L+4\|W_L-\mathbb{E}W_L\|\|R\|+(104/9)\|W_L\|_\infty\|R\|^2.$$

The exponent costs are below $9d$, $537763d/45000$, and $11d$ for $L=1,2,\ge 3$. The second consists of $27/8+62/15+24986/5625$; the last linear cost is $59861/5625$. Combining with (I.8) gives losses $3d$, $2237d/45000$, and d. On the large interval, (I.11) and $\mathbb{E}W_L\le e^{27d/8}$ leave $[-12+4(69/32-1/4000)+27/8]d=-d/1000$.

I.6 Even terminals of minimum one with multiple pairs

Let $F = \prod_{i=1}^{t} h_{a_i} f_{b_i}$, where $1 = a_1 < b_1 < a_2 < \cdots < a_t < b_t$, $t \geq 2$. We prove $\mathbb{E}[p(T^L F)^4] \leq 1$ for every $L \geq 1$.

Reduction of the pair count. Put $R_i = \sum_{j=a_i+1}^{b_i} K^j \mathbf{1}$ and $R = \sum_i R_i = KQ(K)g$; Q is a finite $0/1$ mask. Positive convolution gives $h_{a_i} + f_{b_i} \leq 2 - \omega_i + R_i$, with

$$\omega_i = \beta_{b_i} \ (a_i = 1, 2), \quad \omega_i = 2d - \delta_2 \ (a_i = 3), \quad \omega_i = 2d \ (a_i \geq 4).$$

If $\sum \omega_i \geq 2(t-1)d + \kappa_0$, $-\delta_2 \leq \kappa_0 \leq d$, AM–GM on actual positive roots gives $\sqrt{F} \leq [m + C/t]^t$, $m = 1 - d$, $C = R/2 + d - \kappa_0/2$. This decreases for real $t \geq 2$. Indeed its logarithmic derivative is $\log m + h(y)$, $h(y) = \log(1+y) - y/(1+y)$, $y = C/(tm)$. The mask norm gives $|R| \leq 10\sqrt{d}/7$. For $C \geq 0$, $h(y) \leq y^2/2 \leq 225d/392 < d$; for $C < 0$, $tm + C \geq 13/12$ and $h(y) \leq y^2/[2(1+y)^2] \leq 1800d/8281 < d$. Thus with $m_4 = 1 - d/2 - \kappa_0/4$,

$$F \leq (m_4 + R/4)^4 \leq [m_4^2 + M^2 d/16 + m_4 R/2]^2 \quad \text{if } R^2 \leq M^2 d.$$

All quantities squared are nonnegative upper bounds of actual roots. The three types give $F \leq F_0 = m_0^2(1 + kR/2)^2$:

type	first pairs	κ_0	(M^2, m_0)	upper $k = m_4/m_0$
I	$(1,2), (3,b_2)$	$-\delta_2$	$(1, 1 - 13d/20)$	$9/8$
II	$(1,2),\ a_2 \geq 4$	0	$(259/160, 1 - 13d/16)$	$8/7$
III	$(1,b_1),\ b_1 \geq 3$	β_{b_1}	$(101/50, 1 - 31d/32)$	$6/5$

For I the prefix of Q is $(1, 0, 1)$ and the squared-norm denominator margin is $x[2 - 3x - 3x^2 + 3x^3 - x^5] \geq 2501x/16807$. For II the prefix is $(1, 0, 0)$ and $|Q|^2 \leq (1 + x^3)/(1 - x^2)^2 \leq 259/160$. For III use the general mask bound $101/50$. The upper bounds on $m_4^2 + M^2 d/16$ before relaxing to m_0 are $1 - 33495d/49152$, $1 - 6263d/7680$, and $1 - 4769d/4800$.

One signed spectral calculation for all types. Write $Q(i\sqrt{x}) = u + i\sqrt{x}v$ and integrate against ν_g. Let $a = 439/855$, $B = 1024/855$, $a_{\text{II}} = 25163/27360$. The complete prefix bounds are

	u	v
I	$a \leq u \leq 1$	$-Bx \leq v \leq Bx^2$
II	$a_{\text{II}} \leq u \leq B$	$-Bx \leq v \leq Bx^2$
III	$a \leq u \leq B$	$a \leq v \leq B.$

For III the sharper lower function for both is $(1 - x - x^2)/(1 - x^2)$; for II it is $1 - x^3/(1 - x^2)$ for u. Set $V = \mathbb{E}R^2$, $D_j = \mathbb{E}[K'^j R]$. All signs and means are fixed by

$$V = \int x(u^2 + xv^2), \quad D_{4j} = -\int x^{2j} u, \quad D_{4j+1} = -\int x^{2j+1} v,$$

$$D_{4j+2} = \int x^{2j+1} u, \quad D_{4j+3} = \int x^{2j+2} v,$$

$$\mathbb{E}[pR] = -\int (u + xv), \quad \mathbb{E}[f_2 R] = -\int [(1 - x)u + xv].$$

Define $\gamma = -kD_0 - k^2 V/2 \geq 0$, $\varepsilon_j = \beta_j - kD_{j-1}$ and $\delta = k(D_1)_+$. For III retain instead $\omega = -kD_1 \geq 0$. The following bounds follow from the displayed integrals and $\delta_{j+1} \leq \theta \delta_j$:

	γ/d	δ/d	ε_3/d	ε_j/d, $j \ge 4$
I	$[9/32, 9/8]$	$\le 9/40$	$[1/2, 6/5]$	$[1/2, 6/5]$
II	$[27/50, 7/5]$	$\le 23/100$	$[2/5, 5/4]$	$[1/2, 5/4]$
III	$[0, 29/20]$	0	$[1/3, 5/4]$	$[1/3, 5/4]$

For III $\gamma \ge 0$ uses $a - 3939/8000 > 0$; for I and II the lower constants use $a - 117/512 > 9/32$ and $a_{\mathrm{II}} - 3367/8960 > 27/50$. Let $V_n = S_n' + kK'^n R = 1 + P_n(K')g'$, with $P_n = G_n + kz^{n+1}Q(-z)$. The norm bounds $(\|P_1\|, \|P_2\|^2, \|P_3\|)$ are $(1, 3/2, 11/10)$, $(1, 45/32, 9/8)$ and $(1, 45/32, 6/5)$ for I, II and III. For all types $\|P_n\| \le 1$ when $n \ge 4$. For the short indices expand the real and imaginary parts using the prefix table. For III the $n = 3$ square is bounded by $1 - x + 17x^2/4 \le 5305/4096 < 36/25$; for I and II the triangle bounds are $1 - x/2 + 13x^2/8 < 11/10$ and $1 - x/2 + 79x^2/40 < 9/8$. For the long indices put $U = k\sqrt{\theta}\|Q\| \le 11/10$. At $n = 4, 5$ the bounds are $1 - x/2 + 3x^2/5$ and $1 - x/2 + 243x^2/200$. For $n \ge 6$, $(1+x^3)/\sqrt{1+x} + Ux^3 \le 1$, since $x^2(1 + U\sqrt{1+x})(1 + \sqrt{1+x}) \le 107822/128000 < 1$.

Duhamel gives the same comparison at every $L \ge 2$:

$$T^L F_0/m_0^2 \le V_L - \gamma \bar{f}_{L-1} + kD_1 \bar{f}_{L-2} - \sum_{j=3}^{L} \varepsilon_j \bar{f}_{L-j}.$$

Write $\alpha_n = 1 - \mathbb{E}V_n$, $\eta_n = \mathbb{E}[g'V_n]$. For $n \ge 4$ both have the same interval bounds as ε_j in the table. The short bounds needed are $\alpha_3 = \eta_2 \ge 19d/20$ and $\eta_3 \ge \tau$ in I,II; in III they are $\alpha_3 = \eta_2 \ge 3d/4$ and $\eta_3 \ge \tau$. These are the same finite means $\beta_{n+1} - kD_n$, $\beta_{n+2} - kD_{n+1}$ for the specified direction.

All $L \ge 2$: finite coefficient certificates. For $L \ge 4$ retain the last two ε terms and add 2δ for I,II. For $L = 3$ retain ε_3 and add 2δ; for $L = 2$ keep exactly $V_2 - \gamma q + kD_1$. These give positive upper envelopes Z. For $y = q + 1$, $A_y = y^2 + 2$, the elementary scalar bounds are $pA_y \le 32/5$, $py^2 \le 4A_y/5$. Each coefficient certificate below is obtained by expanding $(Z + \mu)^2 + 2\mu^2$, bounding it by $A_y + \ell yx - D_y x^2$, $x = \sqrt{d}$, and using $D_y \ge cA_y$. Its maximum is at most $32/5 + \ell^2/(5c)$ by completing the square.

type	L	ℓ	D_y	coefficient bound
I	≥ 4	$9/4$	$1067/1600 + 11y/10 + y^2$	8
I	3	$5/2$	$421/300 + 21y/10$	$331/40$
I	2	2	$8589/6400 + 47y/80 + 9y^2/16$	$38/5$
II	≥ 4	$9/4$	$31407/40000 + 27y/25 + 4y^2/5$	8
II	3	$12/5$	$53127/40000 + 163y/100$	9
II	2	2	$671/600 + 9y/25 + 27y^2/25$	$37/5$
III	≥ 4	$12/5$	$14/25 + 2y + 2y^2/3$	8
III	3	$12/5$	$878/675 + 8y/3$	8
III	2	2	$2 + 2y$	$15/2$

The corresponding lower factors c are $7/8, 2/3, 2/3$; $4/5, 1/2, 4/5$; $1, 4/5, 8/11$. They follow by checking endpoints of a linear or concave quadratic on $1 \le y \le 3$. Cubic terms of the expansion are nonpositive; quartic terms are absorbed using $x^2 \le 1/3$.

The original centered fourth expansion now gives the following complete upper polynomials; neg-

ative higher terms have only been discarded:

	$L \geq 4$	$L = 3$	$L = 2$
I	$1 + 4d + 6d^2$	$1 + 20051d/4000 + 9d^2$	$1 + 23d/5 + 849d^2/50$
II	$1 + 22d/5 + 243d^2/50$	$1 + 377d/64 + 315d^2/32$	$1 + 801d/160 + 399d^2/40$
III	$1 + 19d/3$	$(1-d)^4 + 288d/25$	$(1-d)^4 + 675d/64.$

For I,II the respective mean bounds at these lengths are $(1-d, 1-d, 1-4d/5)$ and $(1-9d/10, 1-7d/8, 1-19d/20)$. Their positive covariance bounds are respectively $(d/10, d/4, 9d/10 - 9d^2/8)$ and $(d/12, d/2, 2d/5)$. For III the first covariance can be $-d/12$, contributing at most $d/3$; the others are positive. Variances are respectively $(d, 121d/100, 3d/2)$, $(d, 81d/64, 45d/32)$ and $(d, 36d/25, 45d/32)$. The joint mean endpoint in (A.9) is positive whenever used; for example I at $L = 2$ has $1-22d/5+9d^2/2 \geq 1/30$. Compare with the positive Taylor polynomials of m_0^{-8}. Only the I two-step comparison has a negative quadratic difference; its difference is $3d/5-177d^2/100 \geq d/100$. All other displayed comparisons have positive linear and quadratic margins (and the remaining $(1-d)^4$ terms are nonpositive after degree two).

One step. The actual quadratic remainder must be retained:

$$Y = TF_0/m_0^2 = q + kD_0 - kKR + e, \qquad e = (k^2/4)(V - K(R^2)).$$

For the three types the mean, covariance, variance and row budgets are

	$\mathbb{E}Y$	h lower bound	Var Y upper bound	Y upper bound
I	$1 - d/12$	$27d/50 - 13d^{3/2}/100$	$d + 13d^{3/2}/50 + 17d^2/1000$	$q + 3\sqrt{d}/4$
II	$1 - d/3$	$2d/5 - 11d^{3/2}/50$	$d + 11d^{3/2}/25 + d^2/20$	$q + 19\sqrt{d}/20$
III	$1 - d/5$	$2d/5 - 3d^{3/2}/10$	$d + 3d^{3/2}/5 + 9d^2/100$	$q + 11\sqrt{d}/10.$

The mean uses $\mathbb{E}[pR] \leq -6d/19, -439d/608, -439d/608$, respectively. The centered linear square is at most d, because $2 \inf u \geq k\theta\|Q\|^2$; the centered remainder norm is at most $k^2V/4$. The row bounds follow from $\gamma \geq 0$ and $k\sqrt{V}$. For $Y \leq q + \lambda\sqrt{d}$, $\mu \leq 1$, its fourth coefficient is at most $32/5 + (9\lambda/2)\sqrt{d} + 2\lambda^2 d$. Multiplying the table entries in the centered expansion gives, respectively,

$$\begin{aligned}
\mathbb{E}[pY^4] &\leq 1 + 4d + 6d^{3/2} + 3d^2 + 2d^{5/2}/5 + d^3/50, \quad (I),\\
&\leq 1 + 7d/2 + 8d^{3/2} + 13d^2/2 + 21d^{5/2}/20 + d^3/10, \quad (II),\\
&\leq 1 + 4d + 999d^{3/2}/100 + 3583d^2/500 + 759d^{5/2}/400 + 1089d^3/5000, \quad (III).
\end{aligned}$$

The upper mean endpoint is positive after subtracting four times its covariance lower bound, so both possible mean endpoints are paid. With $x = \sqrt{d} \leq 7/12$, comparison with m_0^{-8}, retaining its cubic Taylor term, gives lower differences divided by d:

$$6/5 - 6x + 11x^2 \geq 21/55, \qquad 3 - 8x + 16x^2 \geq 2, \qquad 15/4 - 999x/100 + 25x^2 > 0.$$

These cover I,II,III in order. Every positive L and every min-one multiple-pair table is therefore covered. The common decoration, zero-parameter and full-reversal rules complete the result.

I.7 All even terminals of minimum three

For $t \geq 2$ use the fixed actual message $P = f_3 h_{b_1} \prod_{i=2}^{t} h_{a_i} f_{b_i}$, where $3 < b_1 < a_2 < \cdots < a_t < b_t$. We prove the stronger bound

$$\mathbb{E}[p(T^L P)^4] \leq e^{-[8(t-2)+257/1000]d} \quad (L \geq 1).$$

Let R_0 be the preceding first-pair exponent and $S = \sum_{i=2}^{t} K^{a_i} G_{b_i - a_i}(K) g$, $R = R_0 + S$. Disjoint exponent intervals give a finite $0/1$ mask, and hence

$$\|S\| \le (3/2)\theta^{a_2/2}\sqrt{d}, \quad |S| \le (3/2)\theta^{(a_2-1)/2}\sqrt{d}, \quad \mathbb{E}S \le \delta_4/(1-\theta^2) < \delta_2/5.$$

Thus $\mathbb{E}R \le -63\delta_2/160 \le 0$ and the actual product satisfies $P \le e^{R-(2t+1)d}$. For $d \le 1/9$, $\|R\| \le 9\sqrt{d}/20$ and $|4R| \le 5/4$. If $b_1 = 4$, use $\|R_0\| < 2\sqrt{d}/7$ and $\|S\| < 13\sqrt{d}/80$, while $|R_0| < 3/16$, $|S| < 1/12$. If $b_1 \ge 5$, $a_2 \ge 6$ gives $\|R\| < 83\sqrt{d}/200$ and $|R| \le 73/240 < 5/16$. For $|u| \le 5/4$, Taylor through degree four and series ratio $u/5$ give $e^u \le 1 + u + 4u^2/5$; its positive remainder coefficient is at most $229/288 < 4/5$. Using the masses and pointwise bounds from Section I.4, and (I.12), the total exponent costs of $\mathbb{E}[W_L e^{4R}]$ are respectively

$$3363/250 < 14, \quad 19743/1000, \quad 2421/125 \qquad \text{times } d \quad (L = 1, 2, \ge 3).$$

The first is $12/5 + 207/50 + 864/125$; the second is $27/8 + 6 + 1296/125$; the last is $9 + 1296/125$. Subtracting the actual loss $(8t+4)d$ leaves margins $8(t-2)+6$, $8(t-2)+257/1000$, and $8(t-2)+79/125$. For $d \ge 1/9$, $|S| \le \sqrt{d}/4 \le 3d/4$ gives $|R| \le 93d/32$ and total cost at most $15d$; the resulting margin is $8(t-2)+5$. For $t = 1$ use Section I.5. The common restoration and reversal rules complete all stated ranges, including $d = 0$.

I.8 Even terminals of minimum two with multiple pairs

Let $F = \prod_i h_{a_i} f_{b_i}$ with $2 = a_1 < b_1 < a_2 < \cdots < a_t < b_t$, $t \ge 2$. We prove $\mathbb{E}[p(T^L F)^4] \le 1$ for all $L \ge 1$. The pair-count reduction of Section I.6 applies with $\kappa_0 = \beta_{b_1}$ and $R = K^2 Q(K) g$, Q a finite $0/1$ mask with constant term one. Here $R^2 \le 1313d/1600$. Thus, writing $m_4 = 1 - d/2 - \beta_{b_1}/4$, two positive envelopes of the same actual message are

$$F \le F_0 = m_0^2(1 + kR/2)^2, \quad m_0 = 1 - d, \quad 1 \le k = m_4/m_0 \le 6/5,$$

and, for the one-step comparison,

$$F \le F_1 = m_1^2(1 + kR/2)^2, \quad m_1 = 1 - 6d/5 + d^2/3, \quad 1 \le k = m_4/m_1 \le 5/4.$$

Indeed $m_4^2 + 1313d/25600 \le 1 - 82061d/76800 \le m_0$, and more sharply it is at most $1 - 31d/25 + 7d^2/16 \le m_1$. The upper k bound follows from $(5/4)m_1 - (1 - 83d/128) \ge 43/3456$.

Signed mask and propagation coefficients. At $i\sqrt{x}$ write $Q = u + i\sqrt{x}v$. Then $a = 439/855 \le u \le B = 1024/855$, $-Bx \le v \le B$, $|Q|^2 \le 101/50$. If its linear coefficient is one, both u, v are at least $1 - x/(1-x^2)$; if it is zero, $v \le Bx^2$. For $V = \mathbb{E}R^2$ and $D_j = \mathbb{E}[K'^j R]$, the exact formulas are

$$V = \int x^2(u^2 + xv^2) \le 17069d/51200,$$

$$D_{4j} = \int x^{2j+1} v, \quad D_{4j+1} = -\int x^{2j+1} u, \quad D_{4j+2} = -\int x^{2j+2} v, \quad D_{4j+3} = \int x^{2j+2} u,$$

$$\mathbb{E}[pR] = \int x(v - u), \qquad \mathbb{E}[f_2 R] = \int x[(1-x)v - u].$$

For $k \le 6/5$ put $V_n = S'_n + kK'^n R = 1 + P_n(K')g'$, $P_n = G_n - kz^{n+2}Q(-z)$. The norms are $\|P_1\| \le 13/10$, $\|P_2\|^2 \le 3/2$, $\|P_3\| \le 11/10$, $\|P_n\| \le 1$ for $n \ge 4$. For the first two square directly:

$|P_1|^2 = 1+2kx^2v+k^2x^3|Q|^2 < 169/100$ and $|P_2|^2 = 1+x-2kx^2(u-xv)+k^2x^4|Q|^2 < 3/2$, using $u - xv \geq 0$. For $n = 3$ use $1 - x/2 + 8x^2/5 \leq 679/640 < 11/10$. For $n \geq 4$ the long-polynomial proof of Section I.6 applies with the smaller perturbation budget $U = k\theta\sqrt{101/50} < 7/10$.

Set $\sigma = kD_0 + k^2V/2$, $\delta = k(D_0)_+ + k^2V/2$, $\omega = -kD_1$, $\varepsilon = \delta - \omega$. The prefix distinction gives

$$0 \leq \delta \leq 33d/40, \quad 0 \leq \omega \leq 117d/200, \quad \varepsilon \leq 23d/40, \quad |\varepsilon| \leq 117d/200.$$

For the linear coefficient one use $D_0 + D_1 \leq \delta_3/(1-\theta)$; for coefficient zero use $v < u$, retaining the sign of D_0. Duhamel gives, for $L \geq 3$,

$$T^L F_0/m_0^2 \leq V_L^* - \sum_{j=3}^{L} c_j \bar{f}_{L-j}, \quad V_n^* = S_n' + \delta S_{n-1}' - \omega S_{n-2}' + kK'^n R,$$

where $c_j = \beta_j + \delta\beta_{j-1} - \omega\beta_{j-2} - kD_{j-1}$. The coefficient bounds are $c_3 \geq 9d/10$, $c_4 \geq 3d/4$, $3d/10 \leq c_j \leq 7d/5$. For $j \geq 5$ the lower ratio is at least $19/32 - 39/200 - 2197/22800 = 3/10 + 109/45600$. Put $\nu = 1 + \varepsilon$; then $V_n^* = \nu + N_n(K')g'$. The mean losses of V_n^* are $\alpha_n = \nu - \mathbb{E}V_n^* = c_{n+1}$ and $\eta_n = \mathbb{E}[g'V_n^*] = c_{n+2}$. Its centered polynomial is $N_n = P_n + \varepsilon G_{n-2} + \delta z^{n-2}$ and satisfies

$$\|N_2\| \leq 31/20, \quad \|N_3\| \leq 11/10 + 9d/8, \quad \|N_n\| \leq 1 + 21d/20 \quad (n \geq 4).$$

The exceptional G_2 at $n = 4$ is retained: its correction ratio is $(117/200)(6/5) + (33/40)\theta = 33189/32000 < 21/20$. For $n \geq 5$ the correction is smaller.

All lengths at least two. For $L \geq 4$ retain $Z = V_L^* - c_L - c_{L-1}q$. It has mean at most $1 - 13d/40$, covariance at least $-d/6$, variance at most $(1+21d/20)^2 d$, and $Z + \mathbb{E}Z \leq q + 1 + 3\sqrt{d}/2$. For $L = 3$ use $Z = V_3^* - c_3$; its corresponding bounds are $1-d$, $3d/10$, $(11/10+9d/8)^2 d$, and $q+1+31\sqrt{d}/20$ for the last sum. For $L = 2$ retain the actual signed quantity $Z = V_2 + \sigma q - \omega$, rather than replacing σ by δ. Since $\mathbb{E}[f_2 R] \leq B\delta_4$, its mean is at most $1 - 13d/20$, covariance at least $d/2$, and variance at most $3d/2 + 33d^2/20 + 1089d^3/1600$. For its fourth coefficient, keeping $C_y = (33y - 59)/40$, $y = q + 1$, gives

$$M \leq 32/5 + 117\sqrt{d}/20 + 20d/11 + 26d^{3/2}/33 + 13d^2/5 \leq 32/5 + 117\sqrt{d}/20 + 16d/5.$$

Here $pC_y \leq 10/33$, $pC_y^2 \leq 169/200$; this retains the dependence on y before bounding. For the other lengths the fourth coefficient for a row sum $y + \lambda\sqrt{d}$ is $32/5 + (9\lambda/2)\sqrt{d} + 2\lambda^2 d$. Expanding the original centered fourth identity yields:

L	$[d]$	$[d^{3/2}]$	$[d^2]$	$[d^{5/2}]$	$[d^3]$	$[d^{7/2}]$	$[d^4]$
≥ 4	$173/30$	$27/4$	19	$29/2$	17	$15/2$	5
3	$11/4$	$17/2$	31	$35/2$	20	9	$25/4$
2	5	$351/40$	22	10	10	4	$11/5$

Each row bounds the coefficients of $\mathbb{E}[pZ^4] - 1$ from above. The favorable covariance is retained at $L = 3, 2$ via $d/4, d/2$; the corresponding upper mean endpoints remain positive. Compare with the positive Taylor polynomial of $(1-d)^{-8}$ through degree four. With $x = \sqrt{d} \leq 7/12$, the difference divided by d is at least, for $L \geq 4$, $67/30 - 27x/4 + 8x^2 \geq 6217/7680$; for $L = 2$, $3 - 9x + 8x^2 \geq 15/32$. For $L = 3$ it is at least $21/4 - 17x/2 + 5x^2 - 35x^3/2 + 100x^4 - 9x^5 + 1295x^6/4$. The first two terms are at least $7/24$; after absorbing $-9x^5$ the remaining $x^2(5 - 35x/2 + 90x^2)$ is positive by its discriminant.

One step: a joint spectral budget. Use the stronger envelope F_1, $k \le 5/4$ and $Y = TF_1/m_1^2 = q + kD_0 - kKR + e$, $e = (k^2/4)(V - K(R^2))$. Put

$$\varepsilon_0 = k(D_0 + D_1) + k^2 V/2, \quad h_{\text{lin}} = d - kD_2, \quad V_{\text{lin}} = d + 2k \int x^2 v + k^2 \int x^3(u^2 + xv^2).$$

The essential joint bound is

$$4\varepsilon_0 - 4h_{\text{lin}} + (32/5)V_{\text{lin}} \le 143d/20.$$

Its excess over $12d/5$ is the integral of

$$\Phi = 4kx(v - u) + (44/5)kx^2 v + k^2(2x^2 + 32x^3/5)(u^2 + xv^2).$$

This is separately convex on $1/2 \le u \le 6/5$, $-1/2 \le v \le 6/5$. At $v = -1/2$ discard the two negative linear terms and obtain $\Phi \le (19/16)(36/25{+}13/128) < 19/4$. At $v = 6/5$ increase x, k to $\theta, 5/4$; the two u endpoints give $14723/3200$ and $2939/640$, both below $19/4$. This is a full continuous-domain certificate.

The remaining original-row budgets are

$$V_{\text{lin}} \le 171d/100, \quad k^2 V/4 \le 131d/1000, \quad k\sqrt{V} \le 73\sqrt{d}/100, \quad \sigma \le 87d/100.$$

Centering the actual remainder and retaining the mean square yields

$$\mathbb{E}(Y - 1)^2 \le V_{\text{lin}} + 7d^{3/2}/20 + 3d^2/2, \quad \mathbb{E}[p(Y - 1)] \le \varepsilon_0 - h_{\text{lin}} + 131d^{3/2}/1000.$$

For the exact expansion around the fixed reference one, $\mathbb{E}[pY^4] = 1{+}4\mathbb{E}[p(Y{-}1)]{+}\mathbb{E}[p(Y{-}1)^2((Y{+}1)^2 + 2)]$, its coefficient is at most $32/5 + 33\sqrt{d}/10 + 5d + 13d^{3/2}/5 + 8d^2/5$. Use the joint bound only for its constant term; multiplication gives

$$\mathbb{E}[pY^4] \le 1 + 143d/20 + 17d^{3/2}/2 + 39d^2/2 + 23d^{5/2}/2 + 23d^3/2 + 9d^{7/2}/2 + 12d^4/5.$$

Positive Taylor terms give $m_1^{-8} \ge 1 + 48d/5 + 40d^2 + 150d^3 + 330d^4$, using $6d/5 - d^2/3 \ge 49d/45$. After absorbing higher odd powers with $x \le 7/12$, the difference divided by d is at least $49/20 - 17x/2 + 13x^2 \ge 1103/1040$. This completes the one-step case and hence all lengths.

J Positive propagation of odd terminal tables

A unit anchor requires a signed packet and explicit short-connector bounds. We prove the three-arm inputs first and then the multi-pair estimates. Nonunit inward anchors use the weighted path integral under the fixed measure.

J.1 The propagated triple $(1, 2, 3)$

In this proof $\tau = d - \delta_2$. For $P = qf_2h_3$, we prove

$$\|T^L P\|_{4,p} \le \begin{cases} 1 - 4837d/76800, & L = 1, \\ 1 - 287d/76800, & L = 2, \\ 1 - 589d/1536, & L \ge 3. \end{cases}$$

In particular the fourth moment is at most $e^{-287d/19200}$. The unit arm is inward, the second outward, and the third has its first edge inward; every connector edge points toward the new root. Set $t = 1-d/2$, $a = t^2$, $k = t^{-1}$, $c_0 = 19/32$ and $s = K^2 g$.

A positive comparison and the first step. The actual identity $f_2 + h_3 = 2 - d - s$ gives $P \le q(t - s/2)^2$. Put $H = q - ks$, $B_L = T^L H$. The two positive paths $h_3 = q - d - r - s$ and $\bar{f}_3 = q - d + r - s$ imply $q - s \ge d + |r|$. Since $|s| < 3/8$ and $k - 1 = d/(2t)$, $H \ge 31d/40 \ge 0$, $\mathbb{E}H = 1$, and $0 \le B_1 \le 2$. The exact first-step calculation is

$$TP \le aB_1 + t(|K(gs)| - \delta_2) + \tfrac{1}{2}\mathbb{E}[qs^2] \le aB_1 + 273d/1024.$$

Indeed $|K(gs)| \le \sqrt{d\delta_3}$ and $\sqrt{\theta\rho} - \rho \le \theta/4$ for $\rho = \delta_2/d$; also $\mathbb{E}[qs^2]/2 \le \delta_3 \le \theta^2 d$. Positivity consequently gives, for every L,

$$T^L P \le aB_L + (273/1024)d\,\bar{f}_{L-1}. \tag{J.1}$$

Apply (A.5) and (A.6). Write $B_1 = q + R$, $R = r + kK^3 g$ and

$$X = -3g + 4R, \qquad J = d - 2k\delta_2 + k^2\delta_3, \qquad I = \delta_2 - 2k\delta_3 + k^2\delta_4.$$

The two logarithmic estimates and the original row bound give

$$X^2/8 - 3g^2/10 - 6(-g + R)^2/5 = -3(g + 4R/5)^2/8 + 26R^2/25 \le 26J/25.$$

Since $\mathbb{E}X = 4(-d + k\delta_2)$ and $\mathbb{E}X^2 = 9d + 16I$,

$$\mathbb{E}[pB_1^4] \le \exp\{26J/25 - 11d/32 + 4k\delta_2 + 13I/2\} \le e^{13d/5}.$$

For the last inequality the spectral integrand is

$$P_k(x) = 557/800 + (48k/25 + 13/2)x + (26k^2/25 - 13k)x^2 + (13k^2/2)x^3.$$

It is convex in $k \in [1, 6/5]$ and increasing in $x \in [0, \theta]$; the derivative discriminant divided by $4k^2$ is at most $-2063/100$. Its endpoint values are $4225129/1638400$ and $52694818/20480000$, both below $13/5$. Minkowski in $p\mu$ now yields

$$\|TP\|_{4,p} \le ae^{13d/20} + 273d/1024 \le 1 - 791d/2400 + 273d/1024 = 1 - 4837d/76800.$$

The second step. Use $K' = -K$, $u_j = K'^j\mathbf{1}$ and $S'_n = \sum_{j=0}^{n} u_j$. Exactly $B_2 = V - D$ with $V = S'_3 + ku_5$ and $D = d - k\delta_2 \ge 41d/80$. Thus

$$\mu := \mathbb{E}B_2 \le v = 1 - 3d/2, \quad \mathbb{E}[g'(B_2 - \mu)] \ge h = 19d/32, \quad \operatorname{Var} B_2 \le V_0 = d - 19d^2/25.$$

For the variance use $|1 + z + z^2 + kz^4|^2 = 1 - x + (1 + 2k)x^2 - 2kx^3 + k^2x^4$ at $z = i\sqrt{x}$. At $k = 1$ this is at most one; its k derivative is below $2/5$, and $k - 1 \le 3d/5$. Finally subtract the exact squared mean d^2. The row tail is at most $13\sqrt{d}/12$, because its squared polynomial norm is $1 + x(1 - kx)^2 \le 1 + 4/27$. Hence $B_2 \le q - d/2 + 13\sqrt{d}/12$.

Here is the finite certificate for the fourth coefficient $M \le 7$. Put $x = \sqrt{d}$, $\lambda = 13/12$, $E = \lambda x - x^2/2$, $W = 2 - p + E$ and $H(p) = p(W^2 + 2vW + 3v^2)$. At $p = 3/2$,

$$H_0 = 51/8 + 39x/8 - 1559x^2/96 - 13x^3/2 + 51x^4/4,$$
$$D_0 = -1/4 - 695x^2/144 - 13x^3/3 + 17x^4/2.$$

$D_0 < 0$ and $H'(2) < 0$, so the maximum lies below $3/2$. For $x \le 1/3$, $H'' \le -3$, $H_0 < 55/8$, $-D_0 < 6/7$; the tangent bound is $55/8 + 6/49 < 7$. For $x \ge 1/3$, $H'' \le -14/5$, $H_0 \le 5281/864$ and $-D_0 < 9/5$; the bound is $5281/864 + 81/140 < 7$. These claims follow by differentiating the displayed polynomials: $-D_0$ increases, and H_0' is convex with two negative endpoints on the second interval. Equation (A.9) gives

$$\mathbb{E}[pB_2^4] \le 7V_0 + \max\{0, v^3(v-4h)\} \le 1 + 13d/5.$$

For the zero branch the difference $(22/5)d - 133d^2/25 \le 197/225 < 1$. For the other branch the difference divided by d is

$$159/40 - 7547d/400 + 945d^2/32 - 837d^3/64;$$

its derivative increases but is still $-5663/1600$ at $1/3$, and its terminal value is $2317/4800 > 0$. Using (A.10) and (A.11), $\|q\|_{4,p} \le 1 + 2d/3$. Thus (J.1) gives

$$\|T^2 P\|_{4,p} \le 1 - 427d/5120 + 3059d^2/12800 \le 1 - 287d/76800.$$

Every later step. Put $P_j(k;z) = G_j(z) + kz^{j+1}$. For $j \ge 4$, $\|P_j(k;K')\| \le 1$, and for $j = 4$, $\|P_4(k;K')g'\|^2 \le d - 32\delta_2/45$. Indeed

$$P_j = \frac{1 - z^j(1 - kz + kz^2)}{1 - z}, \qquad C(x) = |1 - kx - ik\sqrt{x}|^2 \le 1.$$

For $j \equiv 0, 1 \pmod 4$ the numerator of the squared modulus is at most one; $1/(1+x) \le 1 - 32x/45$ gives the extra $j = 4$ loss. The other two residues reduce to $2x^2 + x^5 < 1$ and $(12/5)x^3 + x^6 < 1$, respectively. For $j = 3$ the weaker bound is $\|P_3(k;K')\| \le 1 + (k-1)\theta^2$.

For $n = L + 1$, Duhamel gives

$$B_L = V_n - \sum_{j=3}^{n} \omega_j \bar{f}_{n-j}, \quad V_n = S_n' + k u_{n+2}, \quad \omega_j = \begin{cases} \beta_j, & j \text{ even}, \\ \beta_j - k(-1)^{(j-3)/2}\delta_{(j+1)/2}, & j \text{ odd}. \end{cases}$$

$\omega_3 = d - k\delta_2 \ge 41d/80$, and all subsequent coefficients lie in $[\tau, d]$. The same is true of $\alpha = 1 - \mathbb{E}V_n$ and $\eta = \mathbb{E}[g'V_n]$ for $n \ge 5$: pair consecutive alternating terms, using $k\theta < 1$ for the final pair. For $L \ge 4$ retain $Z = V_n - \omega_n - \omega_{n-1}q \ge B_L$. The norm identities $V_n - 1 = P_n(K')g'$ and $V_n - q = K'P_{n-1}(K')g'$ give the four output conditions of Section A.3:

$$\mathbb{E}Z \le 1 - 3c_0 d, \quad \mathbb{E}[g'(Z - \mathbb{E}Z)] \ge c_0 d - d^2, \quad \operatorname{Var} Z \le d - 3c_0^2 d^2 + d^3, \quad Z \le (1 - c_0 d)q - c_0 d + \sqrt{d}.$$

Consequently $\mathbb{E}[pB_L^4] \le 1 - d/9$. For $L = 3$, $B_3 = V_4 - d - bq$ with $b = d - k\delta_2$, $\alpha = \tau + k\delta_3$, $\eta = \tau$. Although b can be below τ, the required mean loss holds because $\alpha + d + b \ge 3\tau$. The improved P_4 norm gives the variance condition using $\delta_2 \ge d^2$ and $32/45 \ge 2c_0(c_0 - 41/80)$. The excess of the row envelope over the required one is at most

$$\theta[-d + (k-1)(2d + \theta\sqrt{d})] \le 0,$$

by $k - 1 = d/(2-d)$ and $3d + \theta\sqrt{d} < 2$. Thus $\|B_L\|_{4,p} \le 1$ for all $L \ge 3$, and

$$\|T^L P\|_{4,p} \le a + 273d/512 \le 1 - 589d/1536.$$

The common decoration and full-reversal rules restore all equal pairs and give the reversed statement. At $d = 0$ every message is one.

J.2 Propagated triples $(1,2,c)$, $c \geq 4$

Let $P_c = qf_2h_c$. We prove $\mathbb{E}[p(T^L P_c)^4] \leq 1$ for every $L \geq 1$. The common path mean (A.8) gives $h_c \leq 77/80$ for $c \geq 5, d \geq 1/5$; for $c = 4, d \geq 1/4$, $h_4 \leq 2\mu_3 = 2(1-2d) \leq 1$. Positivity and Section B.1 therefore cover all these large parameters in the same actual direction. Below $d \leq 1/5$ for $c \geq 5$, or $d \leq 1/4$ for $c = 4$.

Positive input and the actual error. Put

$$\alpha = \beta_c, \quad B = \beta_{c-1}, \quad m = 1-(\alpha+B)/2, \quad k = 1/m, \quad a_0 = \beta_{c-2},$$
$$W_c = K^2 G_{c-2}(K)g, \quad U = Bg - W_c, \quad H = q + kU, \quad B_L = T^L H,$$
$$D_1 = d - \beta_{c+1}, \quad D_2 = d - \beta_{c+2}, \quad E_3 = \beta_{c+3} - \tau, \quad E_4 = \beta_{c+4} - \tau,$$
$$v = 1 - kB, \quad \epsilon = kD_1.$$

The inward-path convolution gives $h_c \leq (1-B)q - r - W_c - \alpha$, and thus $P_c \leq q(m + U/2)^2$. We next verify $H \geq 0$. Take $z = S_{c-3} \geq 0$; then

$$T^3 z = (1 - a_0 + B)q + (1-a_0)r - \alpha - W_c \geq 0.$$

Add $(1-a_0)[(1-B)q - r - \alpha - W_c] \geq 0$ and divide by $2 - a_0$. This proves $q - W_c \geq \alpha + a_0(1-B)q/(2-a_0) \geq \alpha$. It follows that $H \geq k[(\alpha - B) + (\alpha + 3B)g]/2$. For $g \geq 1/4$ the right side is at least $k(d - 2\delta_2)/2 \geq 0$. For $g \leq 1/4$, $q + kBg \geq 3/4$ and $k|W_c| < 3/4$, since $|W_c|^2 \leq \theta(1+\theta)d \leq 195/1024 < 1/4$. Thus $H \geq 0$ and $\mathbb{E}H = 1 + k(\beta_{c+1} - d) \leq 1$.

Expansion of the actual first step gives

$$TP_c \leq m^2 B_1 - m\mathbb{E}[gU] + mK(gU) + T(qU^2)/4.$$

For $J_U = \mathbb{E}U^2$, $v_0 = \mathbb{E}[gU]$, its error is at most $m(\sqrt{dJ_U} - v_0) + J_U$. For $c \geq 5$, $v_0 \geq Bd + c_0\delta_2$, $\sqrt{J_U} \leq B\sqrt{d} + \sqrt{\delta_3}$, where $c_0 = 19/32$. Hence, uniformly on $d \leq 1/5$,

$$T^L P_c \leq m^2 B_L + C_E d\, \bar{f}_{L-1}, \qquad C_E = \theta/(4c_0) + (1/5+\theta)^2. \tag{J.2}$$

This completes a square in $\rho = \delta_2/d$, without a spectral-support assumption. The sharper error retained for short lengths is

$$d[m(y - c_0\rho) + (B+y)^2], \qquad y = \sqrt{\theta\rho}. \tag{J.3}$$

For $c = 4$, $J_U = d^3 + 2d\delta_2 + \delta_3 + \delta_4$, $v_0 = d^2 + \delta_2$; the error is at most $C_4 d$, $C_4 = 2889/4096$, using $\sqrt{dJ_U} - v_0 \leq \theta(1+\theta)d/4$ and $J_U/d \leq 9/16$.

All long connectors for $c \geq 5$. Let $n = L+1$ and $s_0 = c - 2 \geq 3$. Define

$$\Pi_n(z) = G_n(z) + kz^{n-1}[-B + z^2 G_{s_0}(-z)],$$
$$V_n = S'_n - kBK'^n\mathbf{1} + kG_{s_0}(-K')K'^{n+2}\mathbf{1}.$$

The perturbation budget is $k(B+\theta) \leq 97/128 < 4/5$; the shared finite-$G$ lemma therefore gives $\|\Pi_n(K')\| \leq 1$ for $n \geq 4$. Duhamel and convolution give the exact finite coefficient identity

$$B_L = V_n - \sum_{j=2}^{n} \omega_j \bar{f}_{n-j},$$
$$\omega_j = v\beta_j + kB\beta_{j-1} + k(-1)^j[\beta_{j+1} - \beta_{c+j-1}],$$
$$1 - \mathbb{E}V_n = \omega_{n+1}, \qquad \mathbb{E}[g'V_n] = \omega_{n+2}.$$

Here $\omega_2 = kD_1 \geq 0$, $\omega_3 = vd - kD_2 \geq 0$, and $\tau \leq \omega_j \leq d$ for all $j \geq 4$. To see the full index range, expand each beta as its finite alternating sum. For even $j = 2h$, the initial terms are Alt_{h-1} and the next retained tail starts at $k(-1)^{h+1}\delta_{h+1}$, skipping δ_h. The transition is decreasing since $k\delta_{h+1} \leq \delta_{h-1}$; for $h = 2$ check $d - k\delta_3 \geq \tau$ directly. For odd $j = 2h + 1 \geq 5$ the sequence is

$$\mathrm{Alt}_{h-1} + (-1)^{h-1} v\delta_h + k[(-1)^h \delta_{h+1} + (-1)^{h+1}\delta_{h+2} + \cdots].$$

Both transitions decrease because $\delta_{h-1} \geq v\delta_h \geq k\delta_{h+1}$. Every tail ends at the highest index of β_{c+j-1}, so the argument applies to finite paths of either parity. For $L \geq 4$ retain the last two losses: $0 \leq B_L \leq Z = V_n - \omega_n - \omega_{n-1}q$. Since $V_n - 1 = \Pi_n(K')g'$ and $V_n - 1 - g' = K'\Pi_{n-1}(K')g'$, all four beta coefficients, variance and row-tail conditions of (A.13) hold. Thus $\mathbb{E}[pB_L^4] \leq 1 - d/9$. Minkowski in the original probability $p\mu$ and $\bar{f}_{L-1} \leq 2$ now give

$$\|T^L P_c\|_{4,p} \leq m^2 + 2C_E d \leq 1 - 19363d/486400.$$

The exact margin is $5719/5120 - 13/38 - 9409/12800 = 19363/486400$.

The distinct $c = 4$ spectral family. Here $m = 1 - d$, $k \leq 4/3$, $v = 2 - k$, and $\Pi_n = G_n + kz^{n-1}(-d + z^2 - z^3)$. The same finite coefficient formulas apply and $k(d + \theta) \leq 7/8 < 1$ preserves their alternating transitions. For $z = i\sqrt{x}$ write

$$\Pi_n = \frac{1 - z^{n-1}W}{1 - z}, \quad W = a + i\sqrt{x}\, b, \quad a = k(d + x - x^2), \quad b = 1 - kd - 2kx.$$

Then $0 \leq a \leq 2/3$, $-5/12 \leq b \leq 1$, $|W|^2 \leq 245/288 < 1$ and $a \geq x(1 - x)$. For $n \equiv 1 \pmod 4$, $n \geq 5$, the numerator is at most one because $x^{(n-1)/2}|W|^2 \leq 2a$. For the other three residue classes the sufficient upper bounds are

$n \equiv 3,\ n \geq 7$	$(4/3)x^2 + x^5 < 35/96$
$n \equiv 0,\ n \geq 4$	$(5/6)x + x^2 < 2/3$
$n \equiv 2,\ n \geq 6$	$2x^2 + x^4 < 9/16.$

Each is less than one, proving $\|\Pi_n\| \leq 1$ for every $n \geq 4$. The same long-connector comparison gives $\mathbb{E}[pB_L^4] \leq 1 - d/9$ and

$$\|T^L P_4\|_{4,p} \leq (1 - d)^2 + 2C_4 d \leq 1 - 695d/2048.$$

At large parameters the reduction to qf_2 gives norm at most $1 - d/36$. Together with the $c \geq 5$ margin, this proves the optional uniform loss $\mathbb{E}[p(T^L P_c)^4] \leq e^{-d/9}$ for $L \geq 4, c \geq 4$.

Three short connectors for $c = 4$. For $d \leq 1/4$ the following output conditions imply $\mathbb{E}[pZ^4] \leq e^{4d}$:

$$Z \geq 0, \quad \mathbb{E}Z \leq 1 - d/2, \quad \mathbb{E}[-gZ] \geq 3d/4, \quad \mathrm{Var}\, Z \leq 9d/8, \quad Z \leq q + (13/12)\sqrt{d}.$$

To certify the coefficient, put $x = \sqrt{d}$, $u = 1 - x^2/2$, $W = 2 - p + 13x/12$, and $H_x(p) = p(W^2 + 2uW + 3u^2)$. At $p = 3/2$,

$$H_0 = 51/8 + 39x/8 - 335x^2/96 - 13x^3/8 + 9x^4/8,$$
$$D_0 = -1/4 - 119x^2/144 - 13x^3/12 + 3x^4/4.$$

Here $H_0 \le 1499/192$, $|D_0| \le 157/288 < 3/5$; $H_x'(2) < 0$, and on the maximizing interval $p \le 3/2$, $H_x'' \le -3$. Thus the coefficient is at most $1499/192 + 3/50 < 63/8 < 8$. The endpoint expansion (A.9) is bounded by $\max\{9d, (1-d/2)^3(1-7d/2)+9d\} \le e^{4d}$, by the quadratic Taylor lower bound of the exponential.

For B_1, the exact outputs are

$$B_1 = (1-\epsilon)q + Kz_0, \quad z_0 = vg + k(s + K^3 g) - \epsilon,$$
$$\mathbb{E}B_1 = 1 - d + kd^2, \quad \mathbb{E}[-gB_1] = d - k\delta_3,$$
$$\operatorname{Var} B_1 = d + v^2\delta_2 - 2k(v+1)\delta_3 + k^2(\delta_4 + \delta_5) - (vd - k\delta_2)^2.$$

Use $2k(v+1) - k^2\theta(1+\theta) > 9/4$ and $\delta_3 \ge \delta_2^2/d$ to get variance at most $10d/9 < 9d/8$. Also $\|z_0\|^2 \le v^2 d - 2vk\delta_2 + k^2(\delta_3 + \delta_4) \le d$. The mean and covariance satisfy $\mathbb{E}B_1 \le 1 - 2d/3$, $\mathbb{E}[-gB_1] \ge 599d/768 > 3d/4$.

For $B_2 = V_3 - \epsilon q - \omega_3$, $\omega_3 = vd - k\delta_2 \ge d/8$, $\alpha_3 = d - k\delta_3$ and $\eta_3 = \tau + kd\delta_2 + k\delta_3$. Its mean is at most $1 - d/2$ and covariance at least $3d/4$. The squared spectral polynomial is

$$F_k(x) = [1 - 2x + kx(1+x)]^2 + x(1-kx^2)^2.$$

At $k = 1$, $F_1 - 1 = xQ(x)$, $Q = -1 + 3x - 4x^2 + x^3 + x^4 \le -1/3$: Q increases and $Q(\theta) = -363983/1048576$. Also $\partial_k F_k \le 45x/16$, $k - 1 \le 4d/3$, so $F_k \le 1 + [-1/3 + 15d/4]x$. After centering, $\alpha_3 \ge 3d/4$ and $\epsilon^2 d - 2\epsilon\eta_3 \le 0$ give variance at most $9d/8$; for $d \ge 4/45$ its divided bound is $1 + 161/1536 < 9/8$. The row-tail square is $(1-kx^2)^2 + x(v-kx)^2 \le 1 + v^2 x - 9x^2/4 \le 10/9 < 169/144$. Both B_1, B_2 therefore satisfy the displayed fourth comparison.

For B_3 retain ϵr:

$$B_3 = \widehat{V} - a - bq, \quad \widehat{V} = V_4 - \epsilon r, \quad a = d - k\delta_3, \quad b = vd,$$

with $1 - \mathbb{E}\widehat{V} = \tau + k\delta_3$ and $\mathbb{E}[g'\widehat{V}] = \tau + k\delta_4$. These four coefficients lie in $[c_0 d, d]$. Subtracting ϵz lowers the nonnegative imaginary part of Π_4, so $\|\widehat{V} - 1\|^2 \le d$. For its row tail $\|\Pi_3 - \epsilon\| \le 1$ follows by multiplying its squared-modulus difference by $(1-d)^2$ and bounding it by $xQ + A(x)d - C(x)d^2$, where

$$A = 4x - 6x^2 + 6x^3, \quad C = 23/16 + 2x - 2x^2.$$

Completing the square reduces this to $-4CxQ - A^2 = xD(x)/4 \ge 0$, with

$$D = 23 - 101x + 156x^2 - 135x^3 + 105x^4 - 144x^5 + 32x^6.$$

Here $D' \le -105931/8192 < 0$ and $D(\theta) = 2479497/33554432 > 0$. Thus (A.13) gives $\mathbb{E}[pB_3^4] \le 1 - d/9$. Restoring the actual error, for $L = 1, 2$ use $m^2 e^d + C_4 d e^d \le 1 - 2671d/28672$, since $m^2 e^d \le 1 - d$ and $e^d \le 9/7$. For $L = 3$ use $1 - 695d/2048$ as above. This completes all short $c = 4$ cases.

The first two steps for $c \ge 5$. For B_1 the exact mean and covariance are

$$\mathbb{E}B_1 = 1 - d + kBd + k(D_2 - D_1), \qquad \mathbb{E}[-gB_1] = d - kE_3.$$

They are at most $1 - 2d/3$ and at least $3d/4$, respectively. For the mean, $D_2 - D_1 \le 0$ at $c = 5, 6$; for $c \ge 7$ use $D_2 - D_1 \le \delta_4$, $B \le d - c_0\delta_2$ and $kB + k\delta_4/d < 5/16 < 1/3$. Write $G_{s_0}(-i\sqrt{x}) =$

$a - i\sqrt{x}\,b$, so $1 - x \le a, b \le 1$. The squared polynomial for $V_2 - 1$ is $(1 - kx^2 b)^2 + x(v - kxa)^2$. Both brackets are positive, and it is at most

$$F_0 = (1 - x^2 + x^3)^2 + x(1 - x + x^2)^2 \le 1 + x - 4x^2 + 5x^3 \le 13/12.$$

The last cubic has possible maxima at $1/5$ and θ, with values $2/25, 2665/32768 < 1/12$. The row-tail square is at most $v^2 + kx[-2v + (2v + k)x] \le v^2 \le 1$. Thus B_1 has variance at most $13d/12$ and row bound $q + \sqrt{d}$. Using the coefficient $63/8$ proved above for the short $c = 4$ connectors in the endpoint expansion gives

$$\mathbb{E}[pB_1^4] \le \max\{273d/32, 1 + 275d/96 + 26d^2/3 - 140d^3/27 + 88d^4/81\} \le e^{13d/4}.$$

For the two branches, comparison with exponential Taylor polynomials leaves respectively $31/200$ and the positive quadratic $37/96 - 325d/96 + 10d^2$, whose minimum is $36455/368640$.

For B_2 the exact outputs are

$$B_2 = V_3 - \epsilon q - \omega_3, \quad \omega_3 = vd - kD_2, \quad \alpha_3 = d - kE_3, \quad \eta_3 = \tau + kB\delta_2 + kE_4,$$
$$\mu = 1 - (1 + v)d + k(E_3 + D_2 - D_1), \qquad h = \tau + kB\delta_2 + kE_4 - kD_1 d.$$

Finite alternating sums give $0 \le E_3 + D_2 - D_1 \le \delta_3$, hence $\mu \le 1 - 3d/2$. Retain $E_4 \ge c_0\delta_3$ in the covariance. With $\rho = \delta_2/d$,

$$h/d \ge 1 - \rho + \frac{c_0 - d}{1 - d + \rho d}\rho^2 \ge 32125/48128 > 2/3.$$

The first expression decreases in both d and ρ on $[0, 1/5] \times [0, \theta]$, giving the endpoint certificate. For variance use $1 - x \le a \le 1 - x + x^2$, $1 - x \le b \le 1$ in $[1 - vx + kx^2 a]^2 + x[1 - kx^2 b]^2$. Writing $R_0 = 1 - x + x^2 - x^3 + x^4$, the increase over $k = 1$ is at most $(10/3)dx$; the unperturbed square is $1 + xQ_*$, where

$$Q_* = -1 + 3x - 6x^2 + 7x^3 - 3x^4 + x^5 - x^6 + x^7 \le Q \le -1/3.$$

Indeed $Q_* - Q = x^2[-2(1 - x)(1 - 2x) + x^3 - x^4 + x^5] \le 0$ on $[0, \theta]$. Consequently $\|V_3 - 1\|^2 \le d + [-1/3 + 10d/3]\delta_2$. Centering with $\alpha_3 \ge 3d/4$ and $\epsilon^2 d - 2\epsilon\eta_3 \le 0$ gives $\operatorname{Var} B_2 \le 33d/32$; on $d \ge 1/10$ the divided upper bound is $491/480$. The preceding Π_2 row tail gives $B_2 \le q + (13/12)\sqrt{d}$. Thus

$$\mathbb{E}[pB_2^4] \le \max\{2079d/256, 1 - 419d/768 + 51d^2/2 - 63d^3/2 + 225d^4/16\} \le e^{21d/8}.$$

Comparison with the quadratic and cubic exponential Taylor lower bounds has endpoint margins $247/6400$ and $2129/76800$ after their decreasing remainder polynomials are evaluated at $d = 1/5$.

The actual ratio errors. By (J.3), the two norm bounds have the common upper function

$$F = m^2 e^{\lambda d} + \gamma d[m(y - c_0\rho) + (B + y)^2].$$

Use $(\lambda, \gamma) = (13/16, 1)$ for one step and $(21/32, 8/7)$ for two steps; (A.10) gives $\|q\|_{4,p} < 8/7$ on $d \le 1/5$. The function decreases in α and B; its B derivative is at most $-m + 2\gamma d(B + \theta) < 0$. Hence put $\alpha = B = (1 - \rho)d$. Writing $a = 1 - \rho$, use $e^{\lambda d} \le 1 + \lambda d + Cd^2$, with $C = 7/20$ or $1/4$. Then $(F - 1)/d$ is bounded by $L + Qd + Rd^2 + Sd^3$, where

$$L = \lambda - 2 + [2 + \gamma(2\theta - 1)]\rho + \gamma y,$$
$$Q = C - 2a\lambda + a^2 + \gamma a(y + c_0\rho), \quad R = a^2(\lambda + \gamma) - 2aC, \quad S = Ca^2.$$

These full polynomials increase in ρ. In the first row the derivative bounds are $L' \ge 29/16$, $Q' > 0$, $R' \ge -117/40$, $S' \ge -7/10$; the total is at least $16899/10000$. In the second they are $25/14, -129/112, -347/112, -1/2$, yielding at least $1961/1400$. Values at $\rho = 0$ follow by continuity. The complete endpoint coefficients are

L	$L(\theta)$	$Q(\theta)$	$R(\theta)$	$S(\theta)$	$(F-1)/d \le$
1	$-23/512$	$20009/163840$	$18297/81920$	$2527/20480$	$-216589/20480000$
2	$-69/448$	$7529/28672$	$77387/229376$	$361/4096$	$-2503049/28672000.$

The last three coefficients are positive, so evaluating at $d = 1/5$ bounds the polynomial throughout the rectangle.

Three steps for $c \ge 5$. Retain $\widehat{V} = V_4 - \epsilon r$ and write $B_3 = \widehat{V} - a - bq$, where

$$a = d - kE_3, \quad b = d - kBd + k(D_1 - D_2), \quad \alpha = 1 - \mathbb{E}\widehat{V} = \omega_5 - \epsilon d, \quad \eta = \omega_6.$$

Here $a, b \ge 3d/4$, $\alpha \ge 2d/3$, $b \le (4941/4096)d < 1/4$, and $\eta = \tau + kF_5$ with $F_5 = \tau + \delta_3 - \beta_{c+5} \ge c_0\delta_4$. For the lower bound on b, only $c \equiv 3 \pmod 4$ has a negative $D_1 - D_2$. At $c = 7$ use $B = \tau$, $|D_1 - D_2| = \delta_4$; for $c \ge 11$ use $B \le d - c_0\delta_2$ and $|D_1 - D_2| \le \delta_6$. Both give $B + |D_1 - D_2|/d \le 1/5$. Subtracting ϵz decreases the nonnegative imaginary part of Π_4, so $\|\widehat{V} - 1\|^2 \le d$. The exact centered outputs satisfy

$$\mu \le 1 - 13d/6 \le 1 - 3c_0 d, \quad h = \eta - bd \ge \tau - d^2,$$
$$\operatorname{Var} B_3 \le d - 769d^2/576 + 9d^3/16 \le d - 3c_0^2 d^2 + d^3.$$

For the covariance when $b > d$, necessarily $c \equiv 1 \pmod 4$. At $c = 5$ use $\delta_4 \ge d\delta_3$, the integral covariance inequality $\frac{1}{2}\iint (x-y)(x^3 - y^3)\, d\nu(x) d\nu(y) \ge 0$. For $c \ge 9$ use $b - d \le k\delta_5$ and $c_0\delta_4 \ge d\delta_5$. For variance the exact expression is $\|\widehat{V} - 1\|^2 - \alpha^2 + b^2 d - 2b\eta$; its last two terms decrease in the actual b interval. The row tail obeys

$$\|\Pi_3 - \epsilon\|^2 \le 1 + \max\{0, \theta(10d - 1)/3\}.$$

For $d \le 1/10$ it contracts. Otherwise the extra row term is at most $\theta(10d - 1)\sqrt{d}/6 \le 5d/32$. The extra losses $a, b \ge 3d/4$ absorb it, giving $0 \le B_3 \le (1 - c_0 d)q - c_0 d + \sqrt{d}$. All four outputs of (A.13) hold, so $\mathbb{E}[pB_3^4] \le 1 - d/9$. Restoring (J.2) gives the same actual norm $1 - 19363d/486400$ as the long case. The large-parameter reductions and the two short-step margins all join their small ranges. Every $c \ge 4, L \ge 1$ is therefore covered with the original arm and connector directions unchanged.

J.3 Propagated triples $(1, b, c)$, $3 \le b < c$

We prove $\mathbb{E}[p(T^L(qf_b h_c))^4] \le 1$ for every $L \ge 1$. For $c \ge 5, d \ge 1/5$ use $h_c \le 77/80$ and the pair result in Section I.1; for $(b, c) = (3, 4), d \ge 1/4$ use $h_4 \le 1$. Only the complementary small ranges are considered below.

The common positive envelope and finite coefficients. Put

$$\Delta = \beta_b + \beta_c + B, \quad B = \beta_{c-1}, \quad m = 1 - \Delta/2, \quad k = 1/m, \quad v = 1 - kB,$$
$$W = S_c - S_b = K^b G_{c-b}(K) g, \quad U = Bg - W, \quad H = q + kU, \quad B_L = T^L H.$$

Retaining the three indicated beta subtractions gives $qf_bh_c \le q(m+U/2)^2$. For $c \ge 5, d \le 1/5$, $m \ge 7/10$, $k \le 10/7$. Writing $w = K^{b-1}G_{c-b}(K)g$ gives $|w|^2 \le (1+\theta)\delta_{b-1} \le 117/1024 < 1/4$, $\mathbb{E}w = \beta_b - \beta_c \le \delta_2 \le B$ and $m - B \ge 1/2$. Thus $z = w - B + m \ge 0$ and $mH = Tz + B - \mathbb{E}w \ge 0$. The exact actual-error calculation from Section J.2 gives

$$T^L(qf_bh_c) \le m^2 B_L + Cd\,\bar{f}_{L-1} \tag{J.4}$$

whenever $Cd \ge m(\sqrt{dJ} - v_0) + J$, $J = \mathbb{E}U^2$, $v_0 = \mathbb{E}[gU]$.

Let $n = L+1$, $s = c-b$, and define

$$\begin{aligned}
\Pi_n &= G_n + kz^{n-1}[-B + (-1)^b z^b G_s(-z)],\\
V_n &= S_n' - kBK'^n\mathbf{1} + k(-1)^b G_s(-K')K'^{n+b}\mathbf{1},\\
\omega_j &= v\beta_j + kB\beta_{j-1} + k(-1)^j[\beta_{b+j-1} - \beta_{c+j-1}].
\end{aligned}$$

Duhamel gives exactly

$$\begin{gathered}
B_L = V_n - \sum_{j=2}^{n} \omega_j \bar{f}_{n-j}, \quad 1 - \mathbb{E}V_n = \omega_{n+1}, \quad \mathbb{E}[g'V_n] = \omega_{n+2},\\
V_n - 1 = \Pi_n(K')g', \qquad V_n - q = K'\Pi_{n-1}(K')g'.
\end{gathered}$$

For $c \ge 5$ the perturbation budget is less than $3/4 < 4/5$:

$$k[B + \theta^{b/2}\|G_s\|] \le (10/7)[1/5 + \theta\sqrt{\theta(1+\theta)}] < 3/4.$$

The shared finite perturbation lemma proves $\|\Pi_n\| \le 1$ for all $n \ge 4$. Let $t = \tau - k\delta_4$. The finite beta intervals give $|\omega_2| \le k\delta_3$ for $b \ge 4$, $\omega_2 \ge 0$ for $b = 3$, $\omega_3 \ge 0$, and

$$t \le \omega_j \le d + k\delta_4 \quad (j \ge 4), \qquad t/d \ge 57111/114688 > 49/100.$$

For $b \ge 4$ all shifted beta indices are at least seven and their differences have width δ_4; for $b = 3$ check $j = 4$ separately as $d - k(\beta_{c+3} - \tau) \in [\tau, d]$.

Every $L \ge 4$, $c \ge 5$. Let $e = (-\omega_2)_+$ and retain $Z = V_n + 2e - t - tq \ge B_L \ge 0$. For $b \ge 4$ its outputs are

$$\mu \le 1 - d, \quad h \ge 3d/8, \quad \operatorname{Var} Z \le d, \quad Z \le q + \sqrt{d}.$$

The mean and row losses use $3c_0 - (10/7)(2\theta^2 + 3\theta^3) > 1$ and $c_0 - (10/7)(2\theta^2 + \theta^3) > 0$. Coefficient $63/8$ in the fourth expansion gives $\mathbb{E}[pB_L^4] \le e^{13d/4}$. For $b = 3$, $e = 0$ and the sharper outputs are

$$\mu \le 1 - 7d/5, \quad h \ge 3d/8, \quad \operatorname{Var} Z \le d - 2d^2/3, \quad Z \le (1 - d/3)q - d/3 + \sqrt{d}.$$

Here the fourth coefficient is at most 7. The weaker mean bound $1 - d$ and the same row envelope are profile A of Section A.4, so the coefficient is at most seven. The endpoint fourth expansion now gives $\mathbb{E}[pB_L^4] \le e^{2d}$. Its two branch comparisons have margins $4/15$ and $141292/234375$ at $d = 1/5$ after decreasing Taylor remainders. The corresponding actual errors, retaining their dependence on m, are

$$C_b(m,d) = \begin{cases} (143/300)m + d^2 + 19/200, & b = 3,\\ (11/30)m + (d + 1/5)^2, & b \ge 4. \end{cases}$$

For $b \geq 4$ use $\|W\|/\sqrt{d} < 1/5$, $\mathbb{E}[gW] \leq \delta_3$; for $b = 3$ use $\mathbb{E}[gW] \geq 0$, $\|W\|^2 \leq (1+\theta)\delta_4$ and $(1+\theta)\theta^3 < 19/200$. The norm expression is $m^2 e^{\lambda d} + 2dC_b(m,d)$, with $\lambda = 1/2$ or $13/16$. It increases in m; use $m \leq 1 - 57d/64$. The divided polynomial coefficient bounds are

b	constant	d	d^2	d^3	final bound
3	$-331/2400$	< 0	$5/2$	$1/7$	$-3089/84000$
≥ 4	$-373/2400$	< 0	$45/16$	$7/20$	$-2407/60000$.

The exponential upper coefficients are $1/7$ and $7/20$ in (A.11); evaluating the positive terms at $d = 1/5$ proves both norm deficits.

Finite short-connector certificates. The finite convex hull makes all remaining arm indices explicit. For $z = i\sqrt{x}$, every finite $G_j(z)$ lies in the real convex hull of G_1, G_2, G_3, G_4, since

$$G_\infty = (G_3 + xG_1)/(1+x), \qquad G_{4h+r} = (1-x^{2h})G_\infty + x^{2h}G_r \quad (r = 1,2,3,4).$$

Thus $|A + BG_j|^2$ is at most the largest of its four vertex values. Write $\epsilon = k\mathbb{E}W$ and

$$D = \beta_{b+2} - \beta_{c+2}, \quad F = \beta_{b+3} - \beta_{c+3}, \quad G = \beta_{b+4} - \beta_{c+4}.$$

The exact first-step outputs are

$$\mu_1 = 1 - vd + kD - \epsilon, \quad h_1 = d + kF,$$
$$\operatorname{Var} B_1 = d + v^2\delta_2 + 2kF + 2vkG + k^2\|KW\|^2 - (-vd + kD)^2,$$
$$B_1 = q - \epsilon + Kz_0, \quad z_0 = vg + kW, \quad \|z_0\|^2 = v^2 d - 2vkD + k^2\|W\|^2.$$

For $b = 3, 4, 5$ the polynomial $R_b(z) = -1 + vz + kz^{b+1}G_{c-b}(z)$ bounds the variance. Its maximizing hull vertices are respectively $3, 2, 1$; the corresponding z_0 vertices are $2, 2, 1$. Use $k \leq 1 + (15/7)d$, $k^2 \leq 1 + (255/49)d$ and the rational inputs $\theta^3 < 27/400$, $\theta^5 < 9/800$. The resulting complete one-step outputs are

b	$\mu_1 \leq$	$h_1 \geq$	$\operatorname{Var} B_1 \leq$	row tail coefficient
3	$1 - 9d/10$	$3d/4$	$45d/32$	$4/3$
4	$1 - 93d/100 + 221d^2/140$	d	$17d/10 + 7d^2/10$	$11/9$
5	$1 - 23d/30 + 27d^2/14$	d	$31d/20 + d^2/3$	$25/24$
≥ 6	$1 - 3d/5$	$19d/20$	$3d/2$	$13/12$

A row tail coefficient λ means $B_1 \leq q + \lambda\sqrt{d}$. For $b = 4, 5$ the displayed means imply the weaker losses $3d/5, 3d/8$. Their refined row bounds retain the negative kB term:

$$b = 4: \quad 7/6 + 3/175 + k(x/6 - x^2/3) \leq 1699/1400 < 11/9,$$

and for $b = 5$ use $1 + 3/175 + k[x/6 - (19/32)x^2] < 25/24$. For $b \geq 6$, finite beta parity gives

$$(-\mathbb{E}W)_+, (-D)_+, (-F)_+ \leq \delta_5, \quad F_+, G_+ \leq \delta_6, \quad D - \mathbb{E}W \leq \delta_4.$$

The last difference is exactly $u_{c+1} - u_{b+1}$. These signs and $\|W\|/\sqrt{d} < 2/25$ give the fourth row and error below.

The next two steps use the same coefficients:

$$B_2 = V_3 - \epsilon q - \omega_3, \quad \mu_2 = 1 - \omega_4 - \omega_3 - \epsilon, \quad h_2 = \omega_5 - \epsilon d,$$
$$\operatorname{Var} B_2 = \|V_3 - 1\|^2 - \omega_4^2 + \epsilon^2 d - 2\epsilon\omega_5,$$
$$Z_3 = V_4 + 2e - \omega_4 - \omega_3 q \geq B_3, \quad e = (-\epsilon)_+,$$
$$\mu_3 = 1 - \omega_5 - \omega_4 - \omega_3 + 2e, \quad h_3 = \omega_6 - \omega_3 d,$$
$$\operatorname{Var} Z_3 = \|V_4 - 1\|^2 - \omega_5^2 + \omega_3^2 d - 2\omega_3\omega_6.$$

The finite beta intervals give the following coefficient certificate; entries are divided by d:

b	$e/d \leq$	ω_3/d	$\omega_4/d \geq$	ω_5/d	$\omega_6/d \geq$
3	0	$[5/7, 26/21]$	$16/21$	$\geq 79/160$	$19/32$
4	$5/21$	$[5/7, 26/21]$	1	$[79/160, 1]$	$11/20$
5	$5/21$	$[43/70, 1]$	1	$[19/32, 73/70]$	$11/20$
≥ 6	$3/70$	$[43/70, 73/70]$	$19/20$	$[4/7, 11/10]$	$4/7$

For $b = 3$, additionally $\epsilon \geq k(\delta_2 - \delta_3)$ and $\omega_3 + \omega_4 + \epsilon \geq 32d/21$; the covariance retains $h_2 \geq \tau - k\delta_4 - k\delta_2^2 \geq 1499d/3360$. For $b \geq 6$, $\epsilon_+ \leq d/10$. Every coefficient range follows from the actual finite alternating sums; the possible negative ϵ is retained in both variance identities.

The two short polynomials, in the original K coordinate, are

$$P_2 = 1 - vz - kz^{b+1}G_{c-b}(z), \qquad P_3 = 1 - z + vz^2 + kz^{b+2}G_{c-b}(z).$$

The vertex certificates are

b	$\|P_2\| \leq$	$\|P_3\|^2 \leq$
3	$19/16$	1
4	$11/8$	1
5	$13/10$	$11/10$
≥ 6	$19/16 + 13/175$	$101/100$.

For $b = 3$, $|P_3|^2 \leq (1-vx)^2 + x \leq 1 - (347/1568)x$. For $b = 4$, vertex three adds $-2kx^3[1 - 19x/7 + 5x^2/7] + k^2x^6(1 - x + x^2)$; its bracket is at least $109/7168$, and the positive part is at most $(6/245)x$, below the base loss. For $b = 5$, vertex two gives $|P_3|^2 \leq 1 + [-347/1568 + 2/5 + 45/3136]x < 1 + x/5 < 11/10$. For $b \geq 6$, the tail is at most $4x/35$ in modulus, giving $\|P_3\| \leq 501/500$ and the stated squared bound.

Substitution into the exact centered formulas gives the complete outputs:

step;b	mean upper	$h \geq$	variance upper	coefficient M	$\mathbb{E}[pZ^4] \leq$
$2;3$	$1 - 3d/2$	$7d/16$	d	8	e^{4d}
$2;4$	$1 - 4d/3$	$12d/25$	d	8	e^{4d}
$2;5$	$1 - 4d/3$	$7d/12$	$11d/10$	8	e^{4d}
$2;\geq 6$	$1 - 3d/2$	$d/2$	$9d/8$	8	e^{4d}
$3;3$	$1 - 9d/5$	$d/3$	d	7	e^{3d}
$3;4$	$1 - 17d/10$	$3d/10$	d	7	e^{3d}
$3;5$	$1 - 17d/10$	$3d/10$	d	$63/8$	e^{4d}
$3;\geq 6$	$1 - 2d$	$d/3$	d	$63/8$	e^{3d}

For two steps the row bounds are $q + \lambda\sqrt{d}$, with $\lambda = 19/16, 11/8, 13/10, 4/3$ in row order. For three steps and $b = 3, 4$ the row bound is $(1 - d/3)q - d/3 + \sqrt{d}$, so coefficient 7 applies. For $b \geq 5$ it is

$q + (13/12)\sqrt{d}$. In the variance calculation the terms caused by negative ϵ are at most d^2 for $b = 4$, $3d^2/5$ for $b = 5$, and $d^2/10$ for $b \geq 6$; the retained ω_4^2 pays them.

Finite scalar coefficients and exponential certificates. The coefficient-8 entries use the following complete profiles (a, λ) for mean $\mu \leq 1 - ad$ and row bound $q + \lambda\sqrt{d}$:

$$(9/10, 4/3),\ (3/5, 11/9),\ (3/8, 25/24),\ (3/2, 4/3),\ (4/3, 3/2).$$

The first four have $d \leq 1/5$; the last permits $d \leq 1/4$. For $x = \sqrt{d}$ the common value and derivative at $p = 3/2$ are

$$H_0 = 51/8 + (9\lambda/2)x + [(3/2)\lambda^2 - (21/2)a]x^2 - 3a\lambda x^3 + (9/2)a^2x^4,$$
$$D_0 = -1/4 + (\lambda^2 - 4a)x^2 - 2a\lambda x^3 + 3a^2x^4.$$

For all profiles $D_0, H'(2) < 0$ and $H'' \leq -3$ on $p \leq 3/2$, so the maximum is at most $H_0 + D_0^2/6$. The respective rational bounds $(H_0, |D_0|)$ are

$$(51/8 + 540/407, 1),\ (79/10, 4/7),\ (159/20, 1/2),\ (51/8 + 108/157, 3/2),\ (633/85, 8/5).$$

Each gives a bound below 8. For the first and fourth profiles discard the nonpositive cubic/quartic sum and maximize the remaining quadratic; for the second and third H_0 increases to the rational endpoint $x = 9/20$. For the last profile its cubic/quartic sum is again nonpositive. The $63/8$ profile was proved in Section J.2. For one step the four moment bounds from the first output table are $e^{5d}, e^{7d}, e^{6d}, e^{6d}$, respectively. They and the last output table follow from (A.9) and the quadratic/cubic Taylor lower bounds of the exponential. The second endpoint branches may be bounded by

$1; 3$	$1 + 93d/20 + 324d^2/25 - 5103d^3/500 + 28431d^4/10000$
$1; 4$	$1 + 147d/25 + 29d^2$
$1; 5$	$1 + 16d/3 + 24d^2$
$1; \geq 6$	$1 + 449d/80 + 9d^2 - 621d^3/125 + 594d^4/625$
$2; 3$	$1 + d/4 + 171d^2/8$
$2; 4$	$1 + 56d/75 + 1376d^2/75$
$2; 5$	$1 + 17d/15 + 20d^2$
$2; \geq 6$	$1 + d + 45d^2/2 - 27d^3 + 189d^4/16$
$3; 3$	$1 - 23d/15 + 666d^2/25$
$3; 4$	$1 - d + 1173d^2/50$
$3; 5$	$1 - d/8 + 1173d^2/50$
$3; \geq 6$	$1 - 35d/24 + 32d^2 - 48d^3 + 80d^4/3.$

The first endpoint branches are exactly M times the variance bound in the corresponding output table. These formulas specify both branches and the polynomial signs establish them over each interval in d.

Actual short-step error budget. The four error functions in (J.4) are

$$C_b(m, d) = \begin{cases} (143/300)m + d^2 + 19/200, & b = 3, \\ (11/30)m + (d + 1/5)^2, & b = 4, \\ m/8 + (d + 1/8)^2, & b = 5, \\ (3/20)m + (d + 2/25)^2, & b \geq 6. \end{cases}$$

For $b = 5$ retain the favorable $\mathbb{E}[gW] = -D \leq 0$ and $\|W\|/\sqrt{d} < 1/8$; for $b \geq 6$ use the finite sign bounds already given. For each step the norm is bounded by $m^2 e^{\lambda d} + \gamma d C_b(m, d)$, where λ is one

quarter of the moment exponent, and $\gamma = 1, 8/7, 2$ for steps one, two, three. It increases with m; use $m \le 1 - ad$ with $a = 35/32$ for $b = 3, 4$ and $a = 57/64$ for $b \ge 5$. Expansion using (A.11) gives the following upper coefficients for the divided norm minus one. The final column bounds the entire quotient on $0 \le d \le 1/5$.

step;b	constant	d	d^2	d^3	final
$1;3$	$-439/1200$	< 0	3	2	$< -1/5$
$1;4$	$-37/1200$	$-7/15$	< 0	$35/16$	$-1/75$
$1;5$	$-9/64$	$-23/64$	≤ 0	$5/4$	$-209/1600$
$1;\ge 6$	$-2497/20000$	< 0	≤ 0	$5/4$	$-2297/20000$
$2;3$	$-641/1200$	$-5/28$	$5/2$	$5/7$	$< -2/5$
$2;4$	$-6071/8400$	$-59/420$	$5/2$	$5/7$	$< -3/5$
$2;5$	$-139/224$	$-1/7$	$15/7$	$4/7$	$< -1/2$
$2;\ge 6$	$-3/5$	< 0	$15/7$	$4/7$	$-446/875$
$3;3$	$-353/1200$	$-271/300$	3	$3/8$	$< -1/6$
$3;4$	$-749/1200$	$-1/120$	3	$3/8$	$< -1/2$
$3;5$	$-1/2$	$-5/224$	2	$4/7$	$< -2/5$
$3;\ge 6$	$-7/10$	$-2/25$	$11/4$	$3/10$	$-1469/2500$

All entries are substitutions into one explicit polynomial expansion; negative linear terms may be discarded and positive higher terms maximized at $d = 1/5$. The factor $8/7$ follows from (A.10). This completes all $c \ge 5$ short connectors.

The corner $(b, c) = (3, 4)$, every connector length. For $d \le 1/4$, $\Delta = 3d$, $m = 1 - 3d/2$, $k \le 8/5$, $B = d$, $W = K^4\mathbf{1}$. The positive-input construction remains valid since $|K^3\mathbf{1}| \le \sqrt{\theta d} < 3/8 \le m - B$. Now $\Pi_n = G_n + kz^{n-1}(-d - z^3)$ has perturbation budget $329/400 < 5/6$. This larger budget is essential. The four residue comparisons for $n \ge 4$ give respectively: the $n = 4$ quadratic $y^2 - (5/3)y + 1 > 0$; for odd $n \ge 5$, $1026961/1048576 < 1$; for $n \equiv 0$, $n \ge 8$, $5/(8\sqrt{2}) < 1$; and for $n \equiv 2$, $n \ge 6$, $3951/4096 < 1$. Thus $\|\Pi_n\| \le 1$ for the full index range. Here $\omega_2 = k\delta_2$, $\omega_3 = (1 - kd)d$ and every ω_j, $j \ge 4$, lies in $[\tau, d]$: odd indices are convex combinations of adjacent betas, and even indices have the finite form $\mathrm{Alt}_{h-1} + k(-1)^{h+1}\delta_{h+1}$. Long connectors satisfy (A.13). The actual error is less than $2d/5$, since $J = d^3 + \delta_4$, $v_0 = d^2$, $\sqrt{dJ} - d^2 < (13/50)d$ and $J/d < 13/100$. For $L \ge 4$ this gives norm at most $1 - 131d/80$.

For the three short steps the exact coefficients are

$$\epsilon = k\delta_2, \quad \omega_3 = vd, \quad \omega_4 = d - k\delta_3, \quad \omega_5 = \tau + kd\delta_2, \quad \omega_6 = \tau + k\delta_4.$$

At step one, $\mu \le 1 - d$, $h \ge 7d/10$, variance $\le 45d/32$, and row tail $\le q + (13/12)\sqrt{d}$. The coefficient $63/8$ gives moment at most e^{5d}. At step two, $\mu \le 1 - 3d/2$, $h = \tau$, variance $\le d$ and row tail $q + (19/16)\sqrt{d}$ give moment at most e^{4d}. At step three use $Z = V_4 - \omega_4 - \omega_3 q$, with $\mu \le 1 - 9d/5$, $h \ge d/3$, variance $\le d$ and the coefficient-7 row envelope, giving moment at most e^{3d}. The short spectra here are $P_3 = 1 - z + vz^2 + kz^5$, whose square is at most $(1 - 3x/5)^2 + x \le 1$, and $P_2 = 1 - vz - kz^4$, with square at most $1 + x$. The actual norm comparisons use $\gamma = 1, 7/6, 2$ respectively, $C = 2/5$, and the same $m = 1 - 3d/2$. Their divided polynomials are bounded by

$$-27/20 - 5d/8 + 3d^2/16 + 63d^3/32,$$

$$-23/15 - 5d/28 + 15d^2/28 + 9d^3/7,$$

$$-29/20 + 3d/10 + 63d^2/80 + 27d^3/40.$$

All are negative on $[0,1/4]$ by their endpoint upper bounds. For the last moment comparison retain the full second branch $1-23d/15+666d^2/25-4536d^3/125+11421d^4/625$; its exponential Taylor remainder decreases to $117697/120000>0$. For two steps the corresponding margin is $21/64$. Large parameters use $h_4\le 1$, completing the corner and the full catalogue. Equal-pair restoration and full reversal are the shared operations; the half-tree application is exactly Lemma 6.1.

J.4 Odd terminals with a nonunit anchor

Let $2\le a<b_1<c_1<\cdots<b_t<c_t$, $t\ge 1$, $b_1\ge 4$. Use the actual inward anchor $\bar f_a$ and

$$P=\begin{cases}\prod_i h_{b_i}f_{c_i}, & b_1\ge 5,\\ f_4h_{c_1}\prod_{i\ge 2}h_{b_i}f_{c_i}, & b_1=4.\end{cases}$$

We prove $\mathbb{E}[p(T^L(\bar f_aP))^4]\le e^{-d/30}$ for every $L\ge 1$.

The actual anchor occurs once. Put $n=a+L$, $U=p\bar f_n^3$, $Z_j=A^jU$, $e_j=\mathbb{E}Z_j$ and $W_{a,L}=\bar f_aZ_L$. Rowwise Hölder with density $\bar f_a$ gives

$$(T^L(\bar f_aP))^4\le \bar f_{a+L}^3T^L(\bar f_aP^4),\qquad \mathbb{E}[p(T^L(\bar f_aP))^4]\le \mathbb{E}[W_{a,L}P^4]. \tag{J.5}$$

Its exact mass is $\mathbb{E}W_{a,L}=\mathbb{E}[p\bar f_{a+L}^4]\le 1$ by Section G.1. Also $e_0,e_1\le 1$, $e_j\le 2$ for $j\ge 2$. For e_1, use $pq\le 1$ and the ordinary actual cubic bound $\mathbb{E}\bar f_n^3\le 1-101d/768$ proved in Section F.2. It follows that $W_{a,L}\le 4$ for $L=1,2$ and $W_{a,L}\le 8$ otherwise.

For $d\le 1/9$, the centered actual input satisfies $\|U-\mathbb{E}U\|\le 29\sqrt{d}/4$, as derived in Section I.5. The centered A recursion with $B_0=(I-J)K(I-J)$ gives

$$Z_L-e_L=B_0^L(U-\mathbb{E}U)+\sum_{j=0}^{L-1}e_{L-1-j}B_0^jg.$$

The finite coefficient bound from Section I.4 yields centered constants $457/80$, $569/128$, $12581/2560$ at $L=1,2,\ge 3$. Since $\|\bar f_a-\mathbb{E}\bar f_a\|\le 3\sqrt{d}/2$, center the actual product by comparison with the constant $(\mathbb{E}\bar f_a)e_L$ to obtain constants $517/40$, $761/64$, $16421/1280$, all below 13. Consequently

$$\|W_{a,L}-\mathbb{E}W_{a,L}\|\le 13\sqrt{d}. \tag{J.6}$$

For an exponent R with $\mathbb{E}R\le m_Rd$ and $\|R\|\le r_R\sqrt{d}$, the scalar bound $e^{4R}\le 1+4R+16cR^2$ therefore gives

$$\mathbb{E}[W_{a,L}e^{4R}]\le 1+[4\max(m_R,0)+52r_R+128cr_R^2]d.$$

The sharper squared norm may be retained in its final term. The coefficients $c=2/3,13/18,4/5$ apply on $|4R|\le 3/4,1,5/4$, respectively; their series proofs are in Sections I.4–I.7.

First paired length at least five. The actual product is $P\le e^{R-2td}$, $R=\sum_i K^{b_i}G_{c_i-b_i}(K)g$. For one pair $b_1=5$, the finite consecutive sum gives $\mathbb{E}R\le 0$, $\|R\|^2<d/64$ and $|4R|\le 3/4$. The cost is $47d/6$, leaving loss $d/6$ after $8d$. For one pair $b_1\ge 6$, use $\|R\|\le 21\sqrt{d}/250$, $\mathbb{E}R\le 7d/100$; the cost is at most $7/25+546/125+112896/187500<6$, times d. For $t\ge 2$, disjoint masks give $\|R\|\le 13\sqrt{d}/80$, $|R|\le\sqrt{d}/4$ and $\mathbb{E}R\le 21d/250$. With $c=2/3$ the cost is $42/125+169/20+$

$169/75 = 16559/1500 < 12$. It leaves loss $[8(t-2)+4]d$. These bounds apply for $d \le 1/9$. For $d \ge 1/9$, the same row bound is $|R| \le 3d/4$; the exact mass $\mathbb{E}W_{a,L} \le 1$ gives loss $(8t-3)d$ directly.

First paired length four. Use the coefficients C, D, E of Section I.5, with $c = c_1$. Retaining the extra constant in $f_4 = S_4 - 2d - dg$ gives

$$f_4 h_c \le e^{R_4 - 4d}, \qquad R_4 = -\sum_{j=5}^{c} K^j \mathbf{1} + (C-d)g + Dr + Es.$$

For $c = 5$, $R_4 = -K^4 g + \delta_2 g + \delta_2 r$. For $c \ge 6$, its multiplier at g is

$$Q_4 = \alpha + (d-x)E - x(1-u) + i\sqrt{x}[D - x(1-v)], \quad \alpha = d - \beta_c \in [0, \delta_2], \quad G_{c-2} = u + i\sqrt{x}v.$$

Thus $\mathbb{E}R_4 \le \delta_3$. For $d \le 1/9$ the full frequency bound is $|Q_4|^2 \le 9/200$, and for $c = 5$ it improves to $|Q_4| \le 17/100$. For the latter square $(\delta_2 - x^2)^2 + x\delta_2^2$ and use $\theta^4 + \theta^3/81 < 17/600$. For $c \ge 6$ set $D_{\max} = d(2\theta + d) < 103/1000$. On $x \le d$, the real modulus is below $3/50$ and the imaginary coefficient below $103/1000$; on $d \le x \le 1/3$, they are below $3/20$ and $1/9$. The resulting square bounds are $9/2500 + (103/1000)^2/9$ and $9/400 + 1/243$, both below $9/200$. On $1/3 \le x \le \theta$ maximize the two convex quadratics in $D \in [0, D_{\max}]$, using imaginary choices $x^2 - D$ and x^3. The common component bounds are

$$|\Re Q_4| \le x^2 + (x-d)D, \qquad |\text{imaginary coefficient}| \le \max(x^2 - D, x^3).$$

Thus the two quadratics are $[x^2 + (x-d)D]^2 + x(x^2 - D)^2$ and $[x^2 + (x-d)D]^2 + x^7$. At $D = 0$ the square is at most $\theta^4(1+\theta) < 1/25$. At $D = D_{\max}$ the first is at most $1/25 + 1/324$: its excess over $x^4(1+x)$ is $d^2 k[-2x^2 + k((x-d)^2 + x)] \le 1/324$, $k = 2\theta + d < 1$. For the other choice the real modulus increases to below $5927/30000 < 1/5$, and $\theta^7 < 1/500$ gives square below $21/500$. This exhausts the frequency interval. Therefore

$$\|R_4\|^2 \le 9d/200, \quad \|R_4\| \le 3\sqrt{d}/14, \quad |R_4| \le \|Q_4(K)\mathbf{1}\| \le 3/14.$$

For one pair use $c = 13/18$ and retain the sharper squared norm. The cost is $169/256 + 78/7 + 104/25 = 16 - 1657/44800$, leaving loss $1657d/44800 > d/30$. For large d, $R_4 = R_0 + K^4\mathbf{1} - dg$ and $|K^4\mathbf{1} - dg| \le \sqrt{\delta_3 + d^2} < 8d/5$ give $|R_4| \le 601d/160$; the loss is $39d/40$.

For at least two pairs put $R = R_4 + S$, with later minimum at least six. If $c_1 = 5$, the norm constants add as $17/100 + 21/200 < 9/32$; if $c_1 \ge 6$, they add as $3/14 + 13/200 < 9/32$. Also $\mathbb{E}R \le d/4$ and $|R| \le 3/14 + 13/240 < 5/16$. Use $c = 4/5$ in the common budget: the cost is $1 + 117/8 + 81/10 = 949/40$. The actual loss $(8t+8)d$ leaves $[8(t-2) + 11/40]d$. For $d \ge 1/9$, $|R| \le 679d/160$ leaves $[8(t-2) + 281/40]d$. Both parameter ranges include $1/9$; all margins imply the claimed $d/30$ loss. The common rules restore equal pairs, $d = 0$ and full reversal.

J.5 Odd terminals beginning with $(2, 3)$

Let $P = f_3 h_c \prod_{i=2}^{t} h_{a_i} f_{b_i}$, $4 \le c < a_2 < b_2 < \cdots < a_t < b_t$, and retain the actual inward length-two anchor: $F = \bar{f}_2 P$. We prove $\mathbb{E}[p(T^L F)^4] \le e^{-2d/3}$ for every $t \ge 1, L \ge 1$. Write R_0 for the exponent in Section I.5, $S = \sum_{i=2}^{t} K^{a_i} G_{b_i - a_i}(K) g$ and $R = R_0 + S$.

Large parameters and the small-range inputs. For $d \ge 1/24$, the exact row-factor squares from Section I.5 give $|R_0| \le 17d/6$. For $c = 4, 5$ the squared ratios are below 5 and $62/9$; for $c \ge 6$ the

convex ratio $(1+\theta+d)^2+d(2\theta+d)^2+(1+\theta)\theta^2/d$ has endpoint bounds $191/24$ and $69/16$, both below eight. For $t\ge 2$, the disjoint later mask has $|S|<\sqrt{d}/4\le 5d/4$. The actual product satisfies $P\le e^{R-(2t+1)d}\le e^{-d/6}$. Thus $T^LF\le e^{-d/6}\bar f_{L+2}$, and the actual fourth path bound proves the claimed large-range loss.

For $d\le 1/24$ the needed packages are

	$\|R\|/\sqrt{d}$	row control	$\mathbb{E}R$
$t=1$	$31/100$	$\|R\|\le 1/8$	≤ 0
$t\ge 2$	$9/20$	$R=KV,\ \|V\|\le 3\sqrt{d}/4$	$\le 0.$

For the first row, the three row-factor ratios for $c=4,5,\ge 6$ are below $1/4, 81/256, 9/25$. For multiple pairs add later row-factor constants $1/4, 13/80, 21/200$, respectively; the sums are at most $3/4$. The norm and nonpositive-mean proof is the same signed first-pair/tail calculation in Section I.7, with the stronger current domain. In particular $|\mathbb{E}R|\le 3d/4$ follows from the row factor, rather than from a square-root-order estimate.

For one pair AM–GM gives $F\le \bar f_2 m^2(1+kR/2)^2$, $m=1-3d/2$, $k=1/m\le 16/15$. For $t\ge 2$, the positive root $[1-d+(R-d)/(2t)]^{2t}$ decreases with real $t\ge 2$: its logarithmic derivative uses $|u|=|(R-d)/(2t(1-d))|\le\sqrt{d}/4<1/16$, and $\log(1+u)-u/(1+u)\le 8d/225<d$. At $t=2$ use $R^2\le 9d/16$ to get the same form with $m=1-19d/8$, $k=(1-5d/4)/m\le 16/15$. The constant inequality is $(1-5d/4)^2+9d/256\le 1-1843d/768\le m$.

One row comparison and a signed auxiliary. Set $H=\bar f_2+kR$, $H_L=T^LH$. These functions may be signed. Expansion of the actual quadratic envelope gives

$$0\le T^LF\le m^2H_L+\epsilon d\,\bar f_{L-1},\quad \epsilon=17/20\ (t=1),\quad 13/10\ (t\ge 2). \tag{J.7}$$

Indeed the first-step error is at most $[(12/5)\rho+\rho^2]d$ when $\|R\|\le\rho\sqrt{d}$; use $\|\bar f_2-1\|<6\sqrt{d}/5$ and the original nonnegative rows. Both envelopes have $m^2k, m^2k^2\le 1$. Every actual path satisfies $\|\bar f_n\|_{4,p}\le 6/5$ here, including $n=0,1$; for $n\ge 2$ use the actual cubic bound and $\bar f_n\le 2$.

Put $K'=-K$, $V_n=S'_n+kK'^{n-2}R=1+v_n$ and $\omega_j=\beta_{j+2}-k\mathbb{E}[K'^{j-1}R]$. Exact recursion gives

$$H_L=V_{L+2}-\sum_{j=1}^{L}\omega_j\bar f_{L-j},\quad \mathbb{E}V_n=1-\omega_{n-1},\quad \mathbb{E}[g'V_n]=\omega_n,\quad V_n=q+K'v_{n-1}.$$

With $\eta=k\rho$, $\|v_2\|\le(6/5+\eta)\sqrt{d}$ and $\|v_n\|\le[1+\eta\theta^{(n-2)/2}]\sqrt{d}$ for $n\ge 3$. For $j\ge 2$, $|k\mathbb{E}[K'^{j-1}R]|\le\eta\theta^{(j-2)/2}d$; the first coefficient instead uses the exact row-factor mean. The resulting complete bounds are

	$\|v_2\|/\sqrt{d}$	$\|v_3\|/\sqrt{d}$	$\|v_4\|/\sqrt{d}$	$\|v_n\|/\sqrt{d}\ (n\ge 5)$	$(\omega_1/d,\omega_2/d)$
1	$8/5$	$\sqrt{3/2}$	$227/200$	$109/100$	$([1,3/2],[14/15,4/3])$
≥ 2	$42/25$	$4/3$	$239/200$	$113/100$	$([1,9/5],[13/25,37/25]).$

For $t=1$, $\omega_3\ge 39d/100$ and $9d/20\le\omega_j\le 6d/5$ for $j\ge 4$. Also $\mathbb{E}H_1\le 1-199d/100$. To retain this stronger one-step sign, write $M=\int B$, $N=\int(C-xH)$ in the first-pair notation. Then $\omega_1=d+kM, \omega_2=d-kN$ and $M-N\ge-\theta^2d^2$; for $c=4,5$ it is respectively δ_2-d^2, $\delta_2+\delta_3-d^2$. For $t\ge 2$, all $j\ge 3$ have $7d/25\le\omega_j\le 4d/3$, improving for $j\ge 4$ to $319d/800\le\omega_j\le 6d/5$.

Signed fourth coefficients and every length. For real Z with $0 \le \mu = \mathbb{E}Z \le 1$, the centered fourth identity remains exact. The coefficient $p(3\mu^2 + 2\mu Z + Z^2)$ is at most $17/2$, 9, $33/4$ on the respective intervals

$$[-1/3, q + 1/3], \quad [-1/3, q + 7/20], \quad [-1/2, q + 1/4].$$

Convexity in Z reduces this to the endpoints. At the positive endpoint, enlarge μ to one and complete the square in $(2 - v)[3 + 2(v + a) + (v + a)^2]$, $v = q$. The bounds are $457/54$, $1529/200 + (631/400)^2/(14/5)$ and $4177/512$, respectively; the negative endpoints are smaller.

For $L = 1, 2$ take $Z = H_L$. For $L \ge 3$ take $Z = V_{L+2} - \gamma d(1 + q + \bar{f}_2)$, with $\gamma = 3/8$ for one pair and $7/25$ otherwise. The preceding positive ω_j show $H_L \le Z$. All these envelopes may be signed, but have positive means. Their full fourth budgets are:

t	L	mean upper bound	variance bound	M	$[\mathbb{E}(pZ^4) - 1]/d$ upper bound
1	1	$1 - 199d/100$	$3d/2$	$17/2$	$231201/40000 < 6$
1	2	$1 - 9d/4$	$36d/25$	$17/2$	$7209/1600 < 6$
1	≥ 3	$1 - 3d/2$	$81d/64$	$33/4$	$1281/256 < 6$
≥ 2	1	$1 - 38d/25$	$16d/9$	9	$12247/1250 < 10$
≥ 2	2	$1 - 9d/5$	$169d/100$	$17/2$	$319/40 < 10$
≥ 2	≥ 3	$1 - 6d/5$	$49d/36$	$33/4$	$8147/1200 < 10$

The covariance is nonnegative in every row. In the fourth row retain $4\mu^3 h \ge 7d/10$, using $\mu \ge 259/300 > 6/7$ and $h \ge 7d/25$. All other rows may discard that favorable term. For $L \ge 3$ the variance uses $\|v_{L+2} - \gamma d(2g' + K'g')\|$ and $\sqrt{4 + \theta} < 21/10$. Its upper endpoint is at most $q + 1/4$ and its lower endpoint at least $-1/2$, since $S'_n \ge 0$ and $1 + q + \bar{f}_2 \le 5$. For $L = 1, 2$ the lower endpoint is $-1/3$ by the original row mass and the small-range row package. These facts also give all positive mean lower endpoints directly from the displayed δ bounds. The table follows by $(1 - ad)^4 \le 1 - 4ad + 6a^2d^2$ and $d \le 1/24$.

Finally use the nonnegative composite comparison (J.7) with $H_L \le Z$ and Minkowski on absolute fourth norms. For one pair,

$$\|T^L F\|_{4,p} \le e^{-3d/2} + 51d/50 \le 1 - 693d/1600 \le 1 - 2d/5;$$

for multiple pairs it is at most $e^{-9d/4} + 39d/25 \le 1 - 3741d/6400 \le 1 - d/2$. Their fourth powers improve the large-range loss $2d/3$. Both ranges include $1/24$; decorations, full reversal and $d = 0$ are covered by the common rules.

J.6 Odd terminals with a unit anchor and multiple pairs

Let $2 \le b_1 < c_1 < b_2 < c_2 < \cdots < b_t < c_t$, $t \ge 2$, and $F = q f_{b_1} h_{c_1} \prod_{i=2}^{t} h_{b_i} f_{c_i}$. We prove $\mathbb{E}[p(T^L F)^4] \le e^{-2d/5}$ for every $L \ge 1$. Use $b = b_1$, $c = c_1$, $B = \beta_{c-1}$ and the finite sums

$$W = \sum_{j=b}^{c-1} K^j g, \quad S = \sum_{i=2}^{t} \sum_{j=b_i}^{c_i - 1} K^j g, \quad R = Bg - W + S = Kv, \quad \Delta_0 = \beta_b + \beta_c + B.$$

Here the specific row factor is

$$v = B\mathbf{1} - \sum_{j=b}^{c-1} K^{j-1} g + \sum_{i=2}^{t} \sum_{j=b_i}^{c_i - 1} K^{j-1} g.$$

All original directions, including the inward unit arm, are fixed.

A common square and the actual error. For $d \ge 1/24$, the later mask starts at length at least four and $|S| < (3/8)\sqrt{d} \le 15d/8$. Its actual product is at most $e^{-d/8}$. Therefore $F \le e^{-d/8} q f_b h_c$, and the complete three-arm results in Sections J.1, J.2, J.3 prove the bound.

For $0 < d \le 1/24$, the actual arm sums are bounded by $2 - \Delta_0 + Bg - W$ and $2 - 2d+$ the corresponding later sum. AM–GM reduces the positive $2t$-th root to $t = 2$:

$$f_b h_c \prod_{i=2}^{t} h_{b_i} f_{c_i} \le (m_4 + R/4)^4, \quad m_4 = 1 - (\Delta_0 + 2d)/4.$$

To justify the continuous reduction, $d \le \Delta_0 \le 3d$ and the signed-mask bound give $|R| \le d + 13\sqrt{d}/10$. With $u = (R - (\Delta_0 - 2d))/(2t(1-d))$, $|u| \le 103\sqrt{d}/230 < 1/10$. Then $\log(1-d) + \log(1+u) - u/(1+u) < -7d/8$. The inequality follows by integration of $u/(1+u)^2$; its positive correction is at most $530450d/4284900 < d/8$. Since $m_4^2 + \|R\|_\infty^2/16 \le 1 - 307439d/230400$, we have the common positive square envelope

$$F \le q m^2 (1 + kR/2)^2, \quad m = 1 - 4d/3, \quad k = m_4/m < 16/15.$$

Below m is improved for each prefix and k is updated accordingly. Set $H = q + kR$, $H_L = T^L H$. One exact original-row comparison gives

$$0 \le T^L F \le m^2 H_L + \epsilon d\, \bar{f}_{L-1}, \tag{J.8}$$

whenever $\epsilon d \ge \sqrt{dJ} - \mathbb{E}[gR] + J$, $J = \mathbb{E}R^2$. This uses $m^2 k, m^2 k^2 \le 1$, Cauchy, and $T(qR^2) \le 4J$. As in Section J.5, every actual $\bar{f}_j$ has fourth norm at most $6/5$ on this small range.

The finite expansion common to all three prefixes is, for $n = L + 1$,

$$V_n = S'_n + kK'^{n-1}R, \quad \omega_j = \beta_j - k\mathbb{E}[K'^{j-2}R], \quad H_L = V_n - \sum_{j=2}^{n} \omega_j \bar{f}_{n-j}. \tag{J.9}$$

Its two mean coefficients are ω_{n+1} and ω_{n+2}. To distinguish polynomials from their values at K, write $W_{b,c}(z) = \sum_{j=b}^{c-1} z^j$ and $U(z) = \sum_{i=2}^{t} \sum_{j=b_i}^{c_i - 1} z^j$. The general vector polynomial, in the original z convention, is

$$\Pi_n(z) = G_n(-z) + k(-z)^{n-1}[-B + W_{b,c}(z) - U(z)].$$

The formulas below specialize this same finite family. Write $D_j = \delta_j/(1-\theta^2)$ for finite same-sign tail bounds. Only ω_2 may be negative; set $e = k(\mathbb{E}R)_+$ to control it.

The first pair $(2,3)$**.** Here $R = -s + S$, $S = K^4 Q_0(K) g$, $B = 0$. The finite frequency bound $|1 - z^2 Q_0(z)| \le 177/112$ gives $\|v\|^2 \le 5\delta_2/2$, $J \le 5\delta_3/2$, $\mathbb{E}[gR] \ge \delta_2 - D_4$. The improved constant is $m = 1 - 45d/32$, $k = m_4/m \le 16/15$; $m_4^2 + \|R\|_\infty^2/16 \le 1 - 1447d/1024 \le m$. Optimizing $\sqrt{(5/2)\theta\rho} - \rho$ at $\rho = \delta_2/d$ shows that (J.8) holds with $\epsilon = 3/4$: $5\theta/8 + D_4/d + 5\theta^2/2 < 3/4$.

Here $H \ge 0$ by the original row-covariance inequality. In general, if $k\|v\|^2 \le 2\mathbb{E}v$, then

$$q + kKv \ge (a/2)p + (1 - a/2)q - \sqrt{a(2-a)pq} \ge 0, \quad a = k\mathbb{E}v \in [0,2].$$

For this prefix $\mathbb{E}v \ge d - D_3$ and $k\|v\|^2 \le 13d/12 < 2(1 - 2\theta^2)d/(1-\theta^2)$. The finite coefficient ranges are

$$e \le 4d/45, \quad 7d/20 \le \omega_3 \le 11d/10,$$

$$41d/45 \le \omega_4 \le 11d/10, \quad 11d/20 \le \omega_j \le 11d/10 \ (j \ge 5).$$

For the later mask the first positive/negative mean bounds at indices $j = 3, 4, 5, 6, 7$ are (D_3, D_4), (D_4, D_5), (D_5, D_4), (D_6, D_5), (D_5, D_6). Each four-step shift raises both moment indices by two. Combining these with the alternating beta sums gives every stated index, including the first boundary $d - k(\delta_2 + D_3) \geq 7d/20$.

Let $U(z)$ be the finite S mask. The vector polynomial is $\Pi_n = G_n(-z) + k(-z)^{n-1}(z^2 - U(z))$. Its required squared vector norms are

$$\|(1 + kK^2 - kU(K))g\|^2 \leq d, \quad \|\Pi_2 g\|^2 \leq d + \delta_2, \quad \|\Pi_3 g\|^2 \leq 3d/2, \quad \|\Pi_n g\|^2 \leq d \ (n \geq 4).$$

For the first three use $-x^3/(1-x^2) \leq \Re U \leq x^2/(1-x^2)$ and $-\sqrt{x}x^3/(1-x^2) \leq \Im U \leq \sqrt{x}x^2/(1-x^2)$. The first loss divided by kx is at most $-8/5 + (16/25)kx + kx^4/(1-x^2)^2 < 0$. For Π_2 the sufficient comparison is $2x^2/(1-x^2) + kx^6/(1-x^2)^2 < 1/2 < 8/5 - (16/25)kx$. For Π_3 the real part lies in $[0, 1]$ and its imaginary modulus is below $(21/20)\sqrt{x}$. For all $n \geq 4$, $|\Pi_n - G_n(-z)| \leq k[x + (10/7)x^2]x^{(n-1)/2} \leq (4/5)x^{(n-1)/2}$; use the shared perturbation proof.

For $L = 1$, the actual combined mean, covariance and variance bounds are $1 - 4d/15$, $41d/45$, $45d/32$. The first uses the joint positive terms of $\mathbb{E}S + \mathbb{E}[gS]$, with the joint upper bound $\delta_3/(1-\theta)$. For $L = 2$ use $Z = V_3 - \omega_3 + eq \geq H_2 \geq 0$; for $L \geq 3$ use $Z = V_n - \omega_n - \omega_{n-1}q + e\bar{f}_{n-2} \geq H_L \geq 0$. Their fourth certificates are in the first three rows of the table below.

The first pair $(2, c)$, $c \geq 4$. The finite prefix/tail bound is $|G_{c-2}(z) - z^{c-1}Q_0(z)|^2 \leq 5/2$. At $c = 4$ the norm is at most $701/448$, whose square is below $5/2$; for $c \geq 5$ it is at most $1 + (10/7)\theta^2$. Thus $\|v - B\mathbf{1}\|^2 \leq 5\delta_2/2$ and $\|W - S\|^2 \leq 5\delta_3/2$. Now $m = 1 - 23d/16$, $k \leq 16/15$ and $H \geq 0$ by the same row lemma: $\mathbb{E}v \geq 79d/80$, $\|v\|^2 \leq 25d/16$. The improved square constant before relaxation is $1 - 144471d/98304$. The actual error is $\epsilon = 1$. To check this retain $\rho = \delta_2/d$:

$$J \leq d[d + \sqrt{(5/2)\theta\rho}]^2, \qquad \mathbb{E}[gR] \geq (1-\theta)\rho d - D_4.$$

The error increases in ρ and is at most $263/384 + 1/12 - 247/1024 + (263/384)^2 = 146881/147456 < 1$. The coefficient bounds improve to

$$7d/15 \leq \omega_3 \leq d, \quad 11d/15 \leq \omega_4 \leq d, \quad 11d/20 \leq \omega_j \leq 11d/10 \ (j \geq 5), \quad e \leq 4d/45.$$

For the first two use respectively $0 \leq \mathbb{E}[gR] \leq d^2 + \delta_2 + D_5$ and $0 \leq \mathbb{E}[rR] \leq \delta_3 + D_4$. For later indices pair the last modified beta term with its predecessor; the sufficient bounds are $k(B + \theta) < 1$ and $k(B\theta + \theta^2) \leq 53/270 < 1 - \theta$. Restoring the S mask adds at most D_5 to a positive tail and D_4 to a negative tail, as above.

Here $Q(z) = 1 - kB + kW_{2,c}(z) - kU(z)$ satisfies $\|Q(K)\| \leq 1$. Its real part lies between zero and $1 - 3kx/8$ and its imaginary modulus is at most $(3/2)kx\sqrt{x}$; the square loss divided by kx is at most $-3/4 + (9/64)kx + (9/4)kx^2 \leq -23/80$. Also $\langle g, KQg \rangle = k\mathbb{E}[rR] \geq 0$, so the short vector bounds are again $45d/32, 3d/2$. For $n \geq 4$ the perturbation budget is $263/360 < 4/5$. The combined one-step mean is at most $1 - 5d/6$: its positive finite terms, after the Bd term, are disjoint and total at most $\delta_4/(1-\theta)$. The fourth certificates occupy the next three rows of the table.

first pair	L	mean loss a	covariance ratio h_0	variance ratio	M
$(2, 3)$	1	$4/15$	$41/45$	$45/32$	$15/2$
	2	$7/6$	$11/20$	$3/2$	8
	≥ 3	$4/3$	$9/20$	$6/5$	8
$(2, c \geq 4)$	1	$5/6$	$7/10$	$45/32$	$15/2$
	2	$10/9$	$11/20$	$3/2$	8
	≥ 3	$4/3$	$9/20$	$6/5$	8

The mean is at most $1 - ad$ and covariance at least $h_0 d$. Each row uses both endpoints from (A.9). For the coefficients, if $0 \le Z \le q + \alpha$, $\mu \le 1$, the tangent at $p = 3/2$ is $51/8 + 9\alpha/2 + 3\alpha^2/2 + (-1/4 + \alpha^2)^2/6$. It is below $15/2$ at $\alpha = 7/32$ and below eight at $\alpha = 1/4$. The one-step row bounds use $q + \sqrt{d} + kD_4 \le q + 7/32$. For $L = 2$, $\omega_3 - 2e > 0$ gives the endpoint $q + 1/4$. For $L \ge 3$ the first boundary uses $q + 5\sqrt{d}/4 - 67d/180 \le q + 1/4$; all later tails use $\|\Pi_{n-1}\| \le 1$. The two-step variance follows from $-\omega_4^2 + 2e\omega_5 + e^2 d \le 0$; the later variance is bounded by $[1 + (11/10)d + (4/45)\sqrt{2d}]^2 d \le 6d/5$. The positive endpoint's linear and quadratic coefficients in the centered fourth identity are $MV/d - 4(a + h_0)$ and $6a^2 + 12ah_0$; all higher terms sum to a nonpositive quantity on $d \le 1/24$. These formulas and the table give $\mathbb{E}[pH_L^4] \le e^{6d}$ in all six rows. The zero endpoints are below one; the closest second endpoint is $1 + 241d/45 + 398d^2/27$ at $(2, c \ge 4)$, $L = 2$, whose increment ratio is at most $9671/1620 < 6$. Thus the actual norms from (J.8) are at most $e^{-21d/16} + 9d/10 \le 1 - 3d/8$ and $e^{-11d/8} + 6d/5 \le 1 - d/8$, respectively.

All remaining first pairs, $b \ge 3$. The signed-mask bounds give $\|v - B\mathbf{1}\|^2 \le 4\delta_3$, $\|W - S\|^2 \le 4\delta_4$. Use $m = 1 - 7d/4$, $k \le 16/15$; the square constant before relaxation is at most $1 - 107d/60$. The actual error is $\epsilon = 9/8$, since

$$d + 2\theta^{3/2} + \theta^2 + D_4/d + [d + 2\theta^{3/2}]^2 \le 6470929/5760000 < 9/8.$$

Here H may be signed. The original centered row estimate gives $|\mathbb{E}v| \le 13d/8$, $\|v\|^2 \le 21d/20$, and

$$H \ge q - k|\mathbb{E}v| - k\sqrt{2q\|v\|^2} \ge -5d/2, \qquad H_L \ge -5d.$$

For any proxy $Z \ge H_L$, it is therefore sufficient to add $625d^4$ to its fourth moment when bounding $\mathbb{E}[p|H_L|^4]$. The finite coefficient ranges are

$$e \le 4d/45 \ (b = 3), \quad e \le 4d/15 \ (b \ge 4),$$
$$5d/6 \le \omega_3 \le 4d/3, \quad 11d/15 \le \omega_4 \le 4d/3,$$
$$9d/20 \le \omega_j \le 4d/3 \ (j \ge 5).$$

These follow from $\mathbb{E}R = \beta_{c+1} - \beta_{b+1} + \mathbb{E}S$, $\mathbb{E}[gR] = Bd + \beta_{b+2} - \beta_{c+2} + \mathbb{E}[gS]$, and the alternating beta ranges, retaining the first $b = 3, 4$ boundaries. The long perturbation budget is $674/1125 < 4/5$; the short norms are $\|\Pi_2\| < 8/5$, $\|\Pi_3\| < 5/4$.

For $L = 1$ the exact mean satisfies $1 - 3d \le \mu$ and $\mu \le 1 - 5d/6$ at $b = 3, 4$, $\mu \le 1 - 13d/20$ at $b \ge 5$. The positive terms are at most $\delta_4/(1-\theta)$ for $b = 3, 4$ and $\delta_3/(1-\theta)$ otherwise. The covariance is at least $11d/15$ at $b = 3$, and $19d/20$ otherwise; the latter uses $k\theta^4/(1-\theta) < 1/20$. The row bounds are $[-5d, q + 1/3]$ and $[-5d, q + 7/24]$, with fourth coefficients $17/2$ and eight. Convexity includes the negative endpoint $-5/24$. For clarity, the finite-arm variance calculation is

$$\operatorname{Var} H_1 \le d + a_0^2\delta_2 + 2k\mathbb{E}[rP] - 2ka_0\mathbb{E}[sP] + k^2\|KP\|^2, \quad a_0 = 1 - kB, \quad P = W - S.$$

For $b = 3$ the excess over $d + \delta_2$ is at most $2k[-\delta_3 + 2\delta_4 + 2D_5] + 4k^2\delta_5 < 3d/40 + d/8$. For $b = 4$ split $c = 5, 6, \ge 7$; the cross-term bounds are $3d/10$, $7d/18$, $d/3$, and each square term is below $d/30$. These are the finite G_1, G_2 exceptions plus their later masks. For $b \ge 5$ the cross and square bounds are $3d/10, d/40$. The resulting triples $(M, V/d, a, h_0)$ are

$$(17/2, 13/8, 5/6, 11/15), \quad (8, 11/6, 5/6, 19/20), \quad (8, 7/4, 13/20, 19/20).$$

Their positive-endpoint coefficient pairs are $(1811/240, 23/2)$, $(113/15, 41/3)$, $(38/5, 1989/200)$; each is bounded by the corresponding coefficients $8, 32$ of e^{8d}. All zero endpoints are below one. Thus $\mathbb{E}[p|H_1|^4] \le e^{8d}$.

For $L = 2$ use $Z = V_3 - \omega_3 + eq \ge H_2$. It has mean at most $1 - 13d/10$, covariance at least $9d/20$, variance at most $8d/5$, and row interval $[-5d, q + 1/3]$. Its positive endpoint is at most $1 + 33d/5 + 429d^2/25$. Add $625d^4 \le 2d^2$ to control the actual negative part; the result is below e^{8d}, and the zero endpoint is below $821/1440$. For $L \ge 3$ use $Z = V_n - \omega_n - \omega_{n-1}q + e\bar{f}_{n-2}$. Its corresponding bounds are $1 - d$, $d/4$, $4d/3$ and $[-5d, q + 7/24]$. The positive endpoint after adding the negative-part cost is at most $1 + 17d/3 + 11d^2 < e^{8d}$; the zero endpoint is below one. This proves the absolute fourth norm at every length. Finally (J.8) gives

$$\|T^L F\|_{4,p} \le e^{-3d/2} + 27d/20 \le 1 - d/10.$$

The three small-range cases all imply the stated loss $2d/5$. The large range gives loss $d/2$ with the same directions. The common rules restore decorations, full reversal and $d = 0$; the half-tree application is already proved in Section 6.

K Cubed-weight center estimates

The remaining estimates have weight q^3 and the exact center directions of Section 7. Singleton centers were treated in Section G.4; the residual pair $(1, b)$ uses the fifth-weight reduction in (7.3).

K.1 Odd centers with the cubed boundary weight

For $C = E_0 f_a P$, $2 \le a < b_1 < c_1 < \cdots < b_t < c_t$, $t \ge 1$, we prove $\mathbb{E}[q^3 C^4] \le e^{-4d}$. Use the shared packet and anchor data from Sections H.10–H.11; for $a = 2$ set all anchor data $A_a, z_a, T_a, e_a, I_a, J_a, L_a, W_a, C_a, D_a$ equal to zero. The packet deficit is $\Delta = \omega_{b_1} + 2(t-1)d$, using the base pair deficits.

Cubed-weight master. The actual anchor envelope is $\hat{f}_a = 1 + g + r + T_a - A_a \in (0, 2]$. With $u = \hat{f}_a - 1$, the logarithmic inequalities give

$$q^3 \hat{f}_a^4 \le e^{g+4r+4T_a-4A_a-9g^2/10-6u^2/5}.$$

Put $Y_0 = g + 4r + 4R$, $X = Y_0 + 4T_a - 2A_a$, $Z = 2(g + z_a) + 4v$. Then $X = -g + 2u + KZ$ and the reciprocal-weight sum is $10/9 + 10/3 + 23/9 = 7$. Thus $X^2/7 \le 9g^2/10 + 6u^2/5 + (9/23)\|Z\|^2$. The global scalar bound proves

$$\mathbb{E}[q^3 \hat{f}_a^4 P^4] \le e^{\Xi}, \quad \Xi = -4\Delta - 2A_a + (9/23)\|Z\|^2 + \mathbb{E}X + 3\mathbb{E}X^2/8.$$

The base and the complete anchor increment are

$$\begin{aligned}
\Xi_0 &= -4\Delta - 379d/184 + 6\delta_2 + 144I/23 + 6J_1 - 3L - 12W + 52t_0/23, \\
\Xi - \Xi_0 &= -4A_a + (20/23 - 6A_a)e_a + 36I_a/23 + 6J_a - 3L_a - 12W_a \\
&\quad + 144C_a/23 + 12D_a + 6A_a d + 6A_a t_0 + 3A_a^2/2.
\end{aligned}$$

These follow from $\mathbb{E}Y_0 = -4d - 4t_0$, $\mathbb{E}Y_0^2 = d + 16\delta_2 + 16J_1 - 8L - 32W$ and the exact anchor identities. In particular the positive term $20e_a/23$ is retained. The expansion integrates against μ, with $q^3\mu$ kept as a weighted measure.

Complete budget certificate. The packet norm bounds and signed-moment lists are those proved in Section H.11. One extra short-prefix bound is needed: for a single pair starting at three, $t_0 \le -\delta_2 + \delta_3$, $L \ge -\delta_3$, $W \ge \delta_3 - \delta_4$. For masks starting at least at three use $t_0 \le D_3$, $L \ge -D_3$, $W \ge -D_4$. With the exact anchor-three/four data already displayed there, the master has the following upper polynomials. Put $\mathcal{D}(c_1, \dots, c_m) = \sum_{j=1}^{m} c_j \delta_j$. For example, $\mathcal{D}(0, 1, -1) = \delta_2 - \delta_3$; powers such as d^2 are written separately. The seven upper bounds for Ξ are:

$$
\begin{aligned}
a = 2, b = 3, t = 1 &: \quad \mathcal{D}(-1851/184, 178/23, 61/23, 21),\\
a = 2, b \ge 4, t = 1 &: \quad \mathcal{D}(-1851/184, 6, 121/23, 492/23, 9),\\
a = 2, t \ge 2 &: \quad \mathcal{D}(-3323/184, 10, 576/23, 24) + 121D_3/23 + 12D_4,\\
a = 3, t = 1 &: \quad \mathcal{D}(-2587/184, 105/23, 311/23, 492/23, 21) + 15d^2/2,\\
a = 4, t = 1 &: \quad \mathcal{D}(-2587/184, 125/23, 98/23, 627/23, 636/23, 21) + 15d^2/2,\\
a \ge 5, t = 1 &: \quad \mathcal{D}(-1851/184, 125/23, 9, 864/115, 2261/115, 492/23, 9),\\
a \ge 3, t \ge 2 &: \quad \mathcal{D}(-3323/184, 143/23, 3108/115, 6192/115, 24) + 121D_3/23 + 12D_4.
\end{aligned}
$$

For the first three rows use the packet bounds directly. For $a = 3, t = 1$ use $C_a = L$, $D_a \le \delta_5$. For $a = 4, t = 1$ use $C_a \le \delta_4 + \delta_5$, $D_a \le \delta_5 + \delta_6$, as obtained by the two contiguous alternating cross lists. For $a \ge 5, t = 1$ use $|C_a| \le 6\delta_4/5$, $|D_a| \le 6\delta_5/5$; the remaining anchor term is $-A_a[4 - 6d - 3A_a/2 - 6t_0]$, whose bracket is at least $36/25$. For $a \ge 3, t \ge 2$ use $I_a \le 3\delta_2/2$, $J_a \le 3\delta_3/2$, $|C_a| \le 12\delta_3/5$, $|D_a| \le 12\delta_4/5$. Here the same bracket is at least $11/10$. These cross bounds use respectively the finite geometric and signed-mask spectral products, with minimum exponent six or four as appropriate. The spectral products retain the joint dependence of the pair messages.

Using $d^2 \le d/3$ and the rational bounds in Section H.11, the seven rows give respectively

$$\Xi/d \le -29/6,\ -14/3,\ -347/60,\ -5,\ -21/4,\ -541/120,\ -21/5.$$

Each is at most -4, completing the estimate for every $a \ge 2, t \ge 1$. All additional pairs supply unused nonnegative deficit.

A genuine unit anchor. The actual unit-arm reduction is (7.2); it uses the canonical even fourth moment and includes a singleton unit center.

K.2 Even centers of minimum at least three

In this proof $D_j = \delta_j/(1 - \theta^2)$ bounds finite same-sign tails. For an even center use $\prod h_{a_i} f_{b_i}$ if $a_1 \ge 4$; if $a_1 = 3$, use $f_3 h_{b_1} \prod_{i \ge 2} h_{a_i} f_{b_i}$. We prove its original cubed-weight fourth moment is at most $e^{-d/1000}$. The stronger bounds below are retained because they supply the later-pair budgets. Equal-pair multipliers are restored at the end.

A cubed-weight exponential comparison. If $P \le e^{R - \Delta}$, $R = KV$, put $I = \|V\|^2$, $J = \|R\|^2$, $t_0 = \mathbb{E}[gV]$, $L_0 = \mathbb{E}[rV]$. Then

$$\mathbb{E}[q^3 P^4] \le e^{\Xi}, \qquad \Xi = -4\Delta + 6d + 16I + (32/3)J - 4t_0 + 16L_0. \tag{K.1}$$

To prove this use $\log(1 - g) \le -g - 3g^2/10$ and $X = -3g + 4R$. The original row bound and exact square completion give $X^2/11 - 9g^2/10 = 16R^2 - (9/110)(g + 40R/3)^2 \le 16I$. Also $e^{X - X^2/11} \le Q(X) = 1 + X + 2X^2/3$: the logarithmic difference has derivative $X[4(X-2)^2 + 1]/[33Q(X)]$ and

minimum zero. Finally $\mathbb{E}X = -4t_0$ and $\mathbb{E}X^2 = 9d+16J+24L_0$. Here $q^3\mu$ denotes the unnormalized weighted measure.

For a single pair with $a \geq 4$, use $R = K^a G_{b-a}(K)g$, $\Delta = 2d$. The squared norms are at most $3\delta_a/2, 3\delta_{a+1}/2$. Finite alternating signs give the following complete budgets:

a	t_0 lower bound	L_0 upper bound	Ξ upper bound
4	0	$-\delta_3 + \delta_4$	$-3d/2$
5	$\delta_3 - \delta_4$	δ_4	$-5d/16$
≥ 6	$-\delta_4$	δ_4	$-d/10.$

The positive costs are, respectively, $40\delta_4 + 16\delta_5 - 16\delta_3 \leq 31265d/65536$, $3d/4 + d/4 + 11d/16$, and $3d/8 + d/8 + 7d/5$, against the common loss $2d$. For $t \geq 2$, disjoint masks give $I \leq 4\delta_4$, $J \leq 4\delta_5$, $-t_0, L_0 \leq D_4 = \delta_4/(1-\theta^2)$. Equation (K.1) gives $\Xi \leq -[8(t-2) + 63/25]d$.

The first pair of minimum three. Retain $P_0 = f_3 h_b \leq e^{R_0 - 3d}$ and the coefficients C, D, E and row factor V_0 from Section I.5. Write $M = \mathbb{E}[gV_0]$, $N_0 = \mathbb{E}[gR_0]$. The exponent is

$$\Xi_0 = -6d + 16\|V_0\|^2 + (32/3)\|R_0\|^2 - 4M - 16N_0.$$

For $b = 4$, $M = \delta_2$, $N_0 = d^2$ and $\Xi_0 = -6d + 16\delta_3 + (32/3)(d^3 + \delta_4) - 4\delta_2 \leq -d$. For $b = 5$, $M = (1+d)\delta_2$, $N_0 = d^2 + d\delta_2 - \delta_3$, giving

$$\Xi_0 = -6d - 4\delta_2 + 32\delta_3 + 80\delta_4/3 + 32\delta_5/3 + H,$$

where $H = -20d\delta_2 + 16(1+d)\delta_2^2 + (32/3)(d^3 + 2d^2\delta_2 + d\delta_2^2 + \delta_2^3 - 2d\delta_3) \leq 0$. For $u = \delta_2/d \leq 1/3$ use $d \leq u$ and $\delta_3 \geq \delta_2^2/d$; $H/d^2 \leq (4u/3)(-7 + 12u + 20u^2 + 8u^3) \leq 0$. For $u \geq 1/3$ use $d \leq 1/3$ to obtain $(4/9)(8 - 29u + 8u^2 + 8u^3) < 0$. The remaining spectral polynomial is convex and has endpoints $0, 563745/98304 < 23/4$, so $\Xi_0 \leq -d/4$.

For $d \geq 1/9$, use the full row bound $|R_0| \leq (69/32 - 1/4000)d$ from Section I.5 and $\mathbb{E}q^3 \leq 1 + 4d \leq e^{27d/8}$. This proves the uniform loss $d/1000$ for every $b \geq 4$. It remains to treat $b \geq 6, d \leq 1/9$.

All remaining finite lengths in the small range. Put $\alpha = d - \beta_b$, $\alpha' = d - \beta_{b-1}$, $\rho = \beta_{b-3}/(\beta_{b-2} + \beta_{b-3})$ and $S_0 = \alpha + \alpha'$. Then $19/51 \leq \rho \leq 32/51$ and $E = \rho S_0/(1 - d\rho)$, $D = S_0 + dE$. The important joint bound is $\rho S_0 \leq \delta_2$. It is immediate if $\rho \leq 1/2$. For $b = 7$ it is $d\delta_3 \geq \delta_2^2$. For $b \geq 11$ in the remaining residue $b \equiv 3 \pmod 4$, write $\beta_{b-2} = \beta - e$, $S_0 = 2(d - \beta) + 2e - f$, $e = \delta_{(b-3)/2}$, $f = \delta_{(b-1)/2}$. The sufficient inequality is $2(19/32) - (2 + \theta/(19/32))\theta = 59/608 > 0$. Thus every finite residue is included and

$$7\delta_2/16 \leq E \leq 13\delta_2/12, \quad 19\delta_2/16 \leq D < 17\delta_2/8,$$
$$C \geq d + (19/32)\delta_2 + dE, \quad C < 3d/2, \quad D < 7d/8.$$

Let $W = -K^2 G_{b-3}(K)g$ and $V_0 = W + C\mathbf{1} + Dg + Er$. Apply the finite geometric loss (G.1) with $N = b - 3 \geq 3$. Consequently the base norms are $I_b \leq \delta_3 - 7\delta_4/15$, $J_b \leq \delta_4 - 7\delta_5/15$. With $G_{b-2} = a + i\sqrt{x}b_x$, define $M_1 = \int x b_x$, $M_2 = \int x^2 b_x$, $L_1 = \int x(1-a)$, $L_2 = \int x^2(1-a)$. Their ranges are $\delta_2 - \delta_3 \leq M_1 \leq \delta_2$, $0 \leq M_2 \leq \delta_3$, $\delta_3 - \delta_4 \leq L_1 \leq \delta_3$, $0 \leq L_2 \leq \delta_4$. The full dependent quantities are

$$I = I_b + 2Cd - C^2 + D^2 d + E^2\delta_2 + 2DM_1 + 2EL_1,$$
$$J = J_b + C^2 d + D^2\delta_2 + E^2\delta_3 - 2CL_1 + 2DM_2 + 2EL_2 - 2CE\delta_2,$$
$$M = Dd + M_1, \qquad N_0 = Cd - E\delta_2 - L_1.$$

In particular the negative terms $-C^2$ and $-2CE\delta_2$ are retained. Substitution gives

$$\Xi_0 \le -6d - 4\delta_2 + 36\delta_3 + 16\delta_4/5 - 224\delta_5/45 + H,$$

where

$$\begin{aligned} H = & -(16-32d/3)C^2 + 16Cd + D^2(16d + 32\delta_2/3) - 4Dd \\ & + 32D\delta_2 + 64D\delta_3/3 + E^2(16\delta_2 + 32\delta_3/3) \\ & + 16E\delta_2 + 32E\delta_3 + 64E\delta_4/3 - 64CE\delta_2/3 - 64C(\delta_3 - \delta_4)/3. \end{aligned}$$

Using the preceding upper and lower coefficient bounds with their indicated signs yields $H \le d^2H_2(u) + d^3H_3(u)$, $u = \delta_2/d$, with the complete polynomials

$$H_2 = -57u/4 + 14881u^2/144 - 361u^3/48,$$
$$H_3 = (32/3)(1 + 19u/32)^2 - 161u/27 + 755u^2/12 + 28723u^3/432.$$

For example the C term decreases for $C \ge d$, and its bound is $-16(19/32)d\delta_2 - 16(19/32)^2\delta_2^2 + (32/3)d^3(1 + 19u/32)^2 - 161d^2\delta_2/27$. The negative last term uses $\delta_3 - \delta_4 \ge (19/32)\delta_2^2/d$. Jensen gives $\delta_5 \ge du^4$, so

$$\Xi_0/d \le -6 + B(u) + dH_2(u) + d^2H_3(u), \quad B(u) = 3569u/320 - 224u^4/45.$$

B, $\max(H_2, 0)$ and H_3 increase on $[0, \theta]$: $H_2 \le 0$ up to $1/8$, then its derivative is positive and increasing; the other derivatives have the indicated positive coefficient bounds. The endpoint certificates are $B(\theta) < 22/5$, $H_2(\theta) < 54/5$, $H_3(\theta) < 29$. For the last polynomial its four terms are respectively below $33/2, -12/5, 52/5, 9/2$. Hence $\Xi_0/d < -6 + 22/5 + (54/5)/9 + 29/81 = -17/405$.

All later pairs and the original star interface. For $t \ge 2$ put $R = R_0 + R_T$, $V = V_0 + V_T$ and $\Delta = (2t+1)d$. The exact additional cost is

$$16(2\langle V_0, V_T\rangle + \|V_T\|^2) + (32/3)(2\langle R_0, R_T\rangle + \|R_T\|^2) - 4\mathbb{E}[gV_T] + 16\mathbb{E}[rV_T].$$

For $b = 4$, the two signed cross terms are at most $(1+d)D_4$ and $dD_4 + D_5$; $\|V_T\|^2 \le d/16$, $\|R_T\|^2 \le d/36$. Thus the additional cost is at most $568d/81$, leaving $[8(t-2) + 161/81]d$. For $b \ge 5$, use $\|V_0\|^2 \le d/2$, $\|R_0\| \le 31\sqrt{d}/100$, $\|V_T\| \le \sqrt{d}/6$, $\|R_T\| \le \sqrt{d}/9$, and $-\mathbb{E}[gV_T], \mathbb{E}[rV_T] \le D_4$. The cost ratio is at most $268/63 + 5264/6075 + 5/3 < 7$, leaving $[8(t-2)+1]d$. For $d \ge 1/9$, $R_T \le 3d/4$ and the already proved same-direction first-pair bound give the stronger loss $[8(t-2)+5+1/1000]d$. The common rules restore decorations, $d = 0$ and full reversal. The original star assembly using these center moments is proved in Section 7.

K.3 Even centers of minimum two and unit-arm reductions

In this proof $D_j = \delta_j/(1-\theta^2)$ bounds finite same-sign tails. For $2 = a_1 < b_1 < a_2 < \cdots < a_t < b_t$ use $F = f_2 h_{b_1} \prod_{i=2}^{t} h_{a_i} f_{b_i}$. We prove $\mathbb{E}[q^3F^4] \le e^{-d/20}$, with the stronger loss $[8(t-2)+1]d$ when $t \ge 2$. In particular the first pair is f_2h_b, rather than the terminal direction h_2f_b.

A logarithmic comparison retaining both arms. Suppose $h_b \le \hat{h} \le 2$ and $f_2 + \hat{h} = 2 - \Delta + R$, $R = KV$. Write $I = \|V\|^2$, $J = \|R\|^2$, $t_0 = \mathbb{E}[gV]$, $L_0 = \mathbb{E}[rV]$. Then

$$\mathbb{E}[q^3(f_2h_b)^4] \le e^{\Xi}, \tag{K.2}$$

where the complete exponent is

$$\Xi = -4\Delta + 8091d/2056 + 900I/257 + 6J + 9L_0 + (-308/257 + 2\Delta)t_0 + \Delta^2/6.$$

To prove it put $u = f_2 - 1 = g + r$, $v = \widehat{h} - 1$, $w = (u - v)/2 = g + r - R/2 + \Delta/2$. The quadratic logarithmic bound gives

$$q^3(f_2\widehat{h})^4 \le e^{-4\Delta - 3g + 4R - 9g^2/10 - 12w^2/5}.$$

Take $X = -3g + 4R - 2\Delta/3 = -5g/3 - 4w/3 + KZ$, $Z = (4g + 10V)/3$. Weighted Cauchy and the original row bound give $X^2/7 \le 9g^2/10 + 12w^2/5 + \eta$, $\eta = (9/257)\|4g + 10V\|^2$. The scalar inequality $e^{X - X^2/7} \le Q(X) = 1 + X + 3X^2/8$ follows from derivative $X(6X^2 - 5X + 2)/(56Q)$. Finally

$$\eta = 144d/257 + 900I/257 + 720t_0/257, \quad \mathbb{E}X = -4t_0 - 2\Delta/3,$$

and $\mathbb{E}X^2 = 9d + 16J + 24L_0 + (16\Delta/3)t_0 + 4\Delta^2/9$. Taking expectation under the fixed probability measure gives the exponent.

Every single pair. For $b = 3$ take $\widehat{h} = h_3$, $\Delta = d$, $R = -s$, $V = -r$. Thus $(I, J, t_0, L_0) = (\delta_2, \delta_3, 0, -\delta_2)$ and

$$\Xi = -133d/2056 - 1413\delta_2/257 + 6\delta_3 + d^2/6 \le -d/16.$$

For $b \ge 4$, use the retained coefficients C_1, C_2, C_3 from Section I.5 and $\widehat{h} = 2 - \beta_b - S_b - C_1h_1 - C_2h_2 - C_3h_3$. It lies between the actual h_b and two. Let $R_0 = KV_0$ be the same first-pair polynomial as there. Then $\Delta = 2d$, $R = R_0 - s$, $V = V_0 - r$. Put $M = \mathbb{E}[gV_0]$, $N_0 = \mathbb{E}[gR_0]$, $N_1 = -\mathbb{E}[sR_0]$. The quantities entering the exponent are exactly

$$t_0 = M, \quad L_0 = -N_0 - \delta_2, \quad I = \|V_0\|^2 + \delta_2 + 2N_0, \quad J = \|R_0\|^2 + \delta_3 + 2N_1.$$

Hence

$$\begin{aligned}\Xi = {} & -8357d/2056 + (900/257)\|V_0\|^2 + 6\|R_0\|^2 - 1413\delta_2/257 - 513N_0/257 \\ & + 6\delta_3 + 12N_1 + (-308/257 + 4d)M + 2d^2/3.\end{aligned}$$

For $b = 4$, the exact inputs are

$$I = 3d^2 + \delta_2 + \delta_3, \quad J = d^3 + \delta_3 + \delta_4 + 2d\delta_2, \quad t_0 = \delta_2, \quad L_0 = -d^2 - \delta_2.$$

The linear spectral terms are at most $-5\delta_2/3$; using $d \le 1/3$, $\delta_2 \le 5d/12$ gives $\Xi \le -19d/18$. For $b = 5$, they are

$$\begin{aligned}I &= 3d^2 + \delta_2 + 2d\delta_2 + (1 + d)\delta_2^2 - \delta_3 + \delta_4, \\ J &= d^3 + 2d^2\delta_2 + d\delta_2^2 + \delta_2^3 + (1 - 2d)\delta_3 - \delta_4 + \delta_5 + 2d\delta_2 + 2\delta_2^2, \\ t_0 &= (1 + d)\delta_2, \qquad L_0 = -d^2 - d\delta_2 + \delta_3 - \delta_2.\end{aligned}$$

Substitution bounds the exponent by

$$-4d + 9d^2/4 + 6d^3 - 2\delta_2 + 8d\delta_2 + 16d^2\delta_2 + (16 + 10d)\delta_2^2 + 6\delta_2^3.$$

The positive costs divided by d are at most $3/4 + 2/3 + 55/54 + 29/27 + 7/150$, giving $\Xi \le -1199d/2700 \le -2d/5$.

For every $b \geq 6$, the common finite-frequency formulas give

$$\|V_0\|^2 \leq C^2 + (1+\theta)\delta_3 + D^2 d, \quad \|R_0\|^2 \leq C^2 d + \delta_4 + \delta_5 + D^2\delta_2,$$
$$N_0 \geq Cd - \delta_3 - E\delta_2, \quad N_1 \leq C\delta_2 - \delta_4 + \delta_5 - E\delta_3.$$

For $d \leq 1/5$, the M coefficient is negative and $M \geq 0$. Let $A = 900/257$, $B = 513/257$. The higher spectral terms after substitution have nonpositive sum: their coefficient per δ_2 is at most $(41395/3200)(13/32) + 7/32 - 549/100 = -1641/102400$. The remaining positive terms are bounded as follows, using $d \leq C \leq 257d/160$, $D \leq 81d/80$, $E \leq D$:

term divided by d	upper bound
$[(A+6d)C^2 - BCd]/d$	$9/5$
$D^2(Ad + 6\delta_2)/d$	$1/4$
$12C\delta_2/d$	$8/5$
$BE\delta_2/d$	$1/6$
$2d/3$	$2/15$

Their sum is $79/20$, while $8357/2056 > 4$. Thus $\Xi \leq -d/20$ on the full small interval, with every finite b. For $d \geq 1/5$, the stronger actual path means (A.8) give $h_b \leq 77/80$. The same-direction singleton result of Section G.4 gives $\mathbb{E}[q^3 f_2^4] \leq 1$, so the actual pair fourth moment is at most $(77/80)^4 \leq e^{-3/20} \leq e^{-d/20}$.

All later pairs in the same integral. Let $S = KW = \sum_{i=2}^{t} \sum_{j=a_i}^{b_i-1} K^j g$ and put $I_W = \|W\|^2$, $J_W = \|S\|^2$, $t_W = \mathbb{E}[gW]$, $L_W = \mathbb{E}[rW]$. In the preceding logarithmic argument replace X by $X + 4S$ and the row vector Z by $Z + 4W$. The exact additional cost is

$$\text{Extra} = 2160\langle V, W\rangle/257 + 1296 I_W/257 + 12\langle R, S\rangle + 6J_W + 9L_W + (-164/257 + 2\Delta)t_W.$$

The total exponent is $\Xi_{\text{first}} - 8(t-1)d + \text{Extra}$. On $d \leq 1/5$, $|-164/257 + 2\Delta| < 2/3$, $|t_W| \leq D_3 < d/5$ and $L_W \leq D_4 < 21d/250$. For $b = 3$ use $\|V\| \leq 13\sqrt{d}/20$, $\|R\| \leq 5\sqrt{d}/12$, $\|W\| < 2\sqrt{d}/5$, $\|S\| < \sqrt{d}/4$. The six term cost ratios sum to $221/100 + 102/125 + 5/4 + 3/8 + 189/250 + 2/15 = 16621/3000 < 7$. For $b \geq 4$, the exact norms above imply $\|V\| \leq 7\sqrt{d}/5$, $\|R\| \leq 5\sqrt{d}/6$. Indeed $C \leq 13d/8$, $D \leq 21d/20$ give squared ratios below $23/12 < 49/25$ and $1333/2000 < 25/36$. Later minimum at least five gives $\|W\| < \sqrt{d}/4$, $\|S\| < 13\sqrt{d}/80$; the six costs are at most $119/40 + 51/160 + 13/8 + 507/3200 + 189/250 + 2/15 < 7$. This proves the stronger multiple-pair loss $[8(t-2)+1]d$. For $d \geq 1/5$, the actual tail is bounded by $S < 2\sqrt{d}/5 \leq d$. Multiply the same-direction first-pair bound by its actual exponential tail, giving the still stronger loss $[8(t-2)+4+1/20]d$.

A residual unit arm. Except for the sole residual pair $(1,2)$, direct that unit arm outward. If G is the remaining actual message with $\mathbb{E}[pG^4] \leq 1$, then

$$\mathbb{E}[q^3(pG)^4] = \mathbb{E}[p(pq)^3 G^4] \leq \mathbb{E}[pG^4] \leq 1.$$

For sole $(1,b)$, $b \geq 3$, take $G = \bar{f}_b$ and use Section G.1. If at least three arms remain after removing the unit arm, use the fully reversed positive odd canonical direction from Sections H.10–H.11. These are the original zero-step fourth directions. The exception $(1,2)$ has length outside the path theorem and is handled separately in the main star catalogue. All equal pairs, zero parameters and full reversal follow from the common rules.

L Index to the five-branch direction catalogues

This appendix locates the direction tables and their proofs in Appendices M–R. All lengths below are original positive integers. A terminal residual has at least two distinct entries after maximal equal-length pairing. Each actual equal-pair factor remains at its original vertex.

L.1 Path directions and weights

An f_n arm points outward throughout; a $\bar{f}_n$ arm points inward throughout. An h_n arm has its first edge toward the root and all later edges toward its leaf. Its full reversal is W_n. A connector T^L has all L edges pointing toward its receiver. In AT^{L-1} only the edge adjacent to the receiver is reversed. These products include every original subdivision edge.

The quantities below have different roles. A simple branch supplies $\mathbb{E}[f_s X^4] \le 1$ with its actual existing reference arm. An ordinary slot supplies an eighth moment and consumes its stated anchor. A compound reference supplies a second moment. A path outlet supplies $I_q(U)^3 \le e^{-\eta d}$ with the common original $I_q(U) = \mathbb{E}[q^{-1/3}|U|^{8/3}]$. All entries use the fixed measure.

L.2 The inputs to the assembly

For a path side c–r–t, L is the length of the terminal connector r–t and M is the length of the receiving connector r–c. The two whole-tree alternatives of Section 11 are applied before the local outlet catalogue: a central singleton and a unit off-center singleton with $M = 1$. Table 15 gives the adopted inputs, their parameter ranges and the locations of their proofs.

Table 15: The adopted inputs and their proof locations.

Input	Parameter range	Statement/tool and proof location
Finite calculus and connector operators	Finite μ, real skew $\lvert K\rvert \le 1$; original $L^2(\mu)$; crossed lengths positive.	D.1, C.4, (E.2); proof: D, E.
Canonical rooted-spider boundary	Every finite residual list; positive path/anchor lengths; canonical parity direction.	C.3, (C.9); proof: F.1, C.
Four-branch comparison and terminal supply	$b \le 4$; genuine step zero and every positive step; all equal pairs.	3.2, 5.1; proof: 5, 6.1.
Reduction to the three five-vertex skeletons	Original $b = 5$; reducible roles use the four-branch comparison.	8.1; proof: 8.
Unit reference fourth moment	$s = 1$, terminal residual size at least two, every $L \ge 1$.	5.1, (5.1); proof: 5.
Reference arm of length two	$s = 2$, every terminal residual and $L \ge 1$; the $(1,2,3)$ exception occurs at $L = 1$.	9.1; proof: N.42.
Terminal pair with minimum one	$F = qf_b$, $b \ge 2$, all $L \ge 1$; reference weight f_2.	(L58.16.1); proof: N.7.
Terminal pair with minimum two	$F = h_2 f_b$, $b \ge 3$, $L \ge 1$; coefficient construction on $0 \le d \le 1/3$.	(L77.0.4), (L77.4.1); proof: N.10, I.3.
Short terminal reference rows	$a \ge 3$ short directory; $a = 2$ with terminal $(1,2,3)$ or $(1,2,4)$ treated separately.	(R9.1.ref), (Q114.0.3), (Q104.1.1), (Q105.0.2); proof: N.43, N.17, N.15, N.16.
Remaining reference estimates	All remaining parameter ranges of the reference catalogue.	9.1; proof: N, N.41, N.42.
Reference arms of length at least three	$s \ge 3$, all terminal lists and $L \ge 1$; deficit at least $d/3000$.	9.1; proof: N.41, N.43.

Input	Parameter range	Statement/tool and proof location
Unit-arm ordinary eighth moment	One unit anchor, no unequal root posts, all terminal lists and $L \geq 1$.	9.2; proof: O.5.
Nonunit ordinary eighth moment	$m \geq 3$, all $L \geq 1$; ordered negative posts with $m < u_1$ when nonempty.	9.2; proof: O.39.
Finite masks, scalar payments and propagation	Actual finite supports; coupled means/norms; full continuous d and moment intervals.	M.15, M.23, M.2; proof: O.1, Q.11, M.7.
The mixed central moment	$2 \leq s < a < b < u_1 < v_1 < \cdots$; same-label $f_s^2 Q^8$, including $s = 2, a = 3$.	(9.1); proof: R.4.
Compound reference second moment	$s \geq 1$; independent $M, N \geq 1$; arbitrary side residuals; includes the zero-step case $N = 1$.	9.3, (9.2), (9.3); proof: 9.3.
Star/fork with nonempty center	Each residual parity/minimum class; distinct anchor resources; four positive connectors.	9; proof: 10.
Star/fork with empty center	Empty central residual, arbitrary other side lists, all positive connectors.	2.1, (6.3), (6.2); proof: 10.
A central singleton: the maximal S spine	Any singleton arm length; all SSS/RSS/SSR/RSR patterns and positive connectors.	(C.5), (9.3); proof: 11.
Unit off-center arm with a unit outer edge	Central R; one off-center unit singleton with $M = 1$; actual terminal step $L - 1$.	6.1; proof: 11.
Unit singleton, $M = 2$	Unit residual, $M = 2$, every terminal list and inner $L \geq 1$.	9.4; proof: P.
Unit singleton, $M \geq 3$	Unit residual, every $M \geq 3$ and inner $L \geq 1$; no unequal root posts.	9.4; proof: O.5.
Nonunit singleton, $M \neq 2$	$a \geq 2$; $M = 1, 3$ or $M \geq 4$; all terminals and inner $L \geq 1$.	9.4; proof: R.1, R.5.
Nonunit singleton, $M = 2$	$M = 2$; $a = 2$ and $a \geq 3$ have separate constructions.	9.4; proof: R.6, R.7.
Exactly two root arms	Root $(1, a)$ or $2 \leq s < m$; all independent $L, M \geq 1$.	9.4; proof: O.5, O.39.
Odd root of minimum one	Root $(1, 2, v, ...)$ or $(1, a, b, ...)$ with $a \geq 3$; all terminals, $L, M \geq 1$.	9.4, M.31; proof: Q.13, Q.21.
Odd root of minimum two	All odd lists of minimum two; all terminal parity/minimum classes; $L, M \geq 1$.	9.4; proof: Q.22, Q.24, Q.25, Q.26, Q.27, Q.28.
Odd root of minimum at least three	All strict odd root lists of minimum at least three; every terminal class and $L, M \geq 1$.	9.4, M.30; proof: Q.23.
Even root with at least four residual arms	Any strict even root list of size at least four; all terminals; actual retained inward multiplier.	9.4; proof: R.3.
Two compatible outlets	Same original q on both sides; central negative zero-step R factor; exponents $4, 8/3, 8/3$.	(11.1); proof: 11.
Kernel, tournament and entropy conclusions	Original edges and counts; $d = 0$; all tournament maps; arbitrary correlated joint labels.	1.2, 1.1, 12.1; proof: 12, 12.1.

The reference direction for terminal residual $(1, 2, 3)$ has its exception at $L = 1$. The ordinary

unit-arm direction has its exception at $L = 2$. The ordinary nonunit catalogue permits a post block

$$P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}, \qquad 4 \le u_1 < v_1 < \cdots < u_k < v_k, \quad m < u_1 \quad \text{if } k \ge 1.$$

The post lengths and the terminal lengths may be interleaved. The direction compares the integer lengths specified in each row; the parameter d enters its estimates.

Each local outlet row gives $\eta \ge 1/500$. Two compatible outlets therefore yield loss at least

$$\frac{1}{8}\left(\frac{1}{500} + \frac{1}{500}\right) d = \frac{d}{2000}.$$

The preceding whole-tree alternatives have their respective bounds. In the star and fork, Table 9 allocates each central arm once. An ordinary slot consumes its anchor, terminal and connector; unused central pairs keep the prescribed negative direction. The compound second-moment construction is proved in Proposition 9.3, including its zero-step terminal case. Complete reversal reverses the arms, equal pairs, connector edges and reference weight together.

M Finite calculus and shared local estimates

The five-branch proof uses the local estimates in Appendices M–R. Table 16 locates the six families; the direction tables are indexed in Appendix L. Each family begins with its adopted input statements and keeps the short-length cases next to the corresponding general case.

Table 16: The local proof families for five branch vertices.

Location	Inputs	Role in the article
M	Finite convolution, path envelopes and moment propagation	Tools shared by the local catalogues
N	Genuine reference fourth moments	Proposition 9.1
O	Ordinary fourth and eighth moments	Proposition 9.2; nonsingleton outlets
P	Unit-arm two-step outlets	Unit singleton row of Proposition 9.4
Q	Odd-root fourth moments and outlets	Odd nonsingleton row of Proposition 9.4
R	Remaining singleton, even-root and mixed estimates	Remaining rows of Proposition 9.4 and (9.1)

All expectations and adjoints use the original finite measure. The notation $B = -K$ keeps that measure and adjoint. Complete reversal reverses every arm, connector, decoration and reference weight together. For a side c–r–t, L is the length of r–t and M is the length of r–c. The four-branch foundations are in Appendices C–K.

M.1 The full-mean exponential lemma and the minimum-two root catalogue

Q81.0 Local message bounds

Use the finite skew-kernel model of Section M.15 and put $W_n = A\bar{f}_{n-1}$. The root has distinct residual arm lengths $2, u_1, v_1, \dots, u_j, v_j$, where

$$2 < u_1 < v_1 < \cdots < u_j < v_j, \qquad j \ge 1.$$

The terminal has residual arm lengths $(1, b)$, with $b \ge 2$.

For a unit connector, set $P_{\text{unit}} = \prod_{i=1}^{j} W_{u_i} \bar{f}_{v_i}$, except that when $b = 2, j = 1, u_1 = 5$, take $P_{\text{unit}} = h_5 f_{v_1}$. The message and bound are

$$Y_{\text{unit}} = f_2 P_{\text{unit}} T(p \bar{f}_b), \qquad \mathbb{E} Y_{\text{unit}}^8 \le e^{-191d/24}. \tag{Q81.0.1}$$

For a connector of length $L \ge 2$, restrict to $u_1 = 3$ and write $v = v_1$. The message is

$$Y_{\text{nonunit}} = W_2 \bar{f}_3 W_v \prod_{i=2}^{j} W_{u_i} \bar{f}_{v_i} \, T^L(q f_b). \tag{Q81.0.2}$$

It satisfies (Q81.9.3). The first pair is $\bar{f}_3 W_v$, and its direction is fixed by the arm lengths and L.

These bounds include arbitrary genuine equal-pair products at the terminal and root. Insert the terminal product inside T or T^L and multiply the root product outside. Each product lies in $[0, 1]$, so positivity bounds the decorated message by the displayed undecorated message.

Q81.1 Original finite calculus and finite masks

Delete zero-mass atoms and use the measure on the remaining atoms. Since diagonal kernel entries vanish, $p(x), q(x) \ge \mu(x) > 0$. All finite tree messages are positive: the all-equal labeling has positive mass and diagonal edge factors one. Both A and T are positive operators.

Set

$$r = Kg, \quad \sigma = Kr, \quad t = K\sigma, \quad \delta_i = \|K^i \mathbf{1}\|_2^2, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The C014 calculus supplies

$$\|K\|^2 \le \theta, \quad 0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta \delta_i, \quad |KV(x)| \le \|V\|_2, \tag{Q81.1.1}$$

$$Er = -d, \quad E\sigma = 0, \quad Et = \delta_2, \quad Egr = 0, \qquad \kappa d \le \tau \le d - d^2 \le 2/9. \tag{Q81.1.2}$$

The original two-row bound $|K^2 V(x)|^2 \le \|V\|^2/3$ is used only in quoting specified already-paid original scalar certificates. It is not a spectral bound for an arbitrary compression.

Define the finite quantities

$$G_n(z) = \sum_{i=0}^{n-1} z^i, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1} \delta_i.$$

For either sign the exact finite positive convolution is

$$\sum_{i=0}^{n} (\pm K)^i 1 = f_n^{\pm} + \sum_{i=3}^{n} \beta_i f_{n-i}^{\pm}, \qquad \tau \le \beta_n \le d \quad (n \ge 3), \tag{Q81.1.3}$$

where $f_n^+ = f_n, f_n^- = \bar{f}_n$. This finite identity applies at every integer length.

For a rotation frequency write $z = i\sqrt{x}$, $0 \le x \le \theta$, and $G_n(z) = a_n + z c_n$. For $n \ge 2$,

$$1 - x \le a_n, c_n \le 1; \qquad a_1 = 1, \; c_1 = 0. \tag{Q81.1.4}$$

For $n \ge 3$, $a_n \le 1 - x + x^2$ and $|G_n(z)| \le 1$. For $n \ge 4$, both coordinates are at most $1 - x + x^2$. The finite geometric formula proves these statements: odd indices have squared numerator $1 + x^n$,

and even indices have $(1 \pm x^{n/2})^2$, all divided by $1 + x$. In the positive-ending even case $n \geq 6$, use $2x^2 + x^5 \leq 1$. We will also use

$$|G_n(z)|^2 \leq 1 - 7x/15 \qquad (n \geq 3). \tag{Q81.1.5}$$

For $n = 3$, the left side is $1 - x + x^2$, and $1 - x \geq 7/15$. For $n = 4$, it is $(1 - x^2)^2/(1 + x) \leq 1/(1+x) \leq 1-7x/15$. For every $n \geq 5$, the finite numerator is bounded by $(1+x^3)^2$. The remaining comparison is equivalent to $8 - 7x - 30x^2 - 15x^5 \geq 0$; this decreases on the interval and its value at $13/32$ is $1311885/33554432 > 0$. These arguments cover every integer tail.

The spectral measure of g is denoted by ν. It has mass d and $\int x^i \, d\nu = \delta_{i+1}$. Apply the frequency inequalities through the spectral theorem for this kernel.

Q81.1.1 Actual extra pairs and their finite masks

C067 gives $h_u f_v \leq e^{R_{uv} - 3d}$ for every $4 \leq u < v$, where $R_{uv} = \sum_{a=u}^{v-1} K^a g$. C014 gives the exact exceptional loss $2d - \delta_2$ at $u = 3$. Full reversal gives, for negative pairs,

$$\prod_i W_{u_i} \bar{f}_{v_i} \leq e^{S - \Gamma}, \quad S = KU(-K)g, \quad U(z) = \sum_i \sum_{a=u_i}^{v_i - 1} z^{a-1}, \tag{Q81.1.6}$$

$$\Gamma = 3jd \ (u_1 \geq 4), \qquad \Gamma = 2d - \delta_2 + 3(j-1)d \ (u_1 = 3). \tag{Q81.1.7}$$

The intervals are disjoint, so U is a finite zero-one mask. Separating the four residue classes at $i\sqrt{x}$ gives, for every finite zero-one or [0, 1]-coefficient mask Q,

$$|Q(i\sqrt{x})|^2 \leq \frac{1 + x}{(1 - x^2)^2}, \qquad \|Q(K)\|^2 \leq \frac{32768}{16245} < \frac{100}{49}. \tag{Q81.1.8}$$

Geometric series bound the finite positive subcollections in these sums. Consequently

$$\|U(-K)g\| \leq \begin{cases} (3/8)\sqrt{d}, & u_1 \geq 4, \\ (1/4)\sqrt{d}, & u_1 \geq 5, \\ (3/5)\sqrt{d}, & u_1 = 3, \end{cases} \tag{Q81.1.9}$$

$$ES \leq d/12 \ (u_1 \geq 4), \qquad ES \leq \delta_2 + d/12 \ (u_1 = 3). \tag{Q81.1.10}$$

The positive mean terms after excluding the first quadratic term are at most $\delta_4/(1 - \theta^2) < d/12$. For a single pair starting at three, $U = z^2 G_h(z)$, $h = v - 3 \geq 1$, and $ES \leq \delta_2$.

When $u_1 = 3$ the mandatory quadratic coefficient gives

$$U = u - i\sqrt{x}\, w, \quad -6x/5 \leq u \leq -x/2, \qquad -x/2 \leq w \leq 6x/5. \tag{Q81.1.11}$$

For a single such pair, the stronger form is $U = -x(a_h + z c_h)$, where $a_h \in [1 - x, 1]$, $c_h \in [0, 1]$, and $c_h \geq 1 - x$ if $h \geq 2$.

At $d = 0, g = 0$ on every retained atom. K need not be zero. Successive leaf integration nevertheless makes every actual message in this manuscript equal to one. All endpoint assertions follow directly. Every subsequent division by d assumes $d > 0$.

Q81.2 A signed-mean exponential lemma

Theorem T086.1. Let W be any finite real variable with $|W| \le U$ and $EW^2 \le \theta U^2$, where $\theta = 13/32$. For every $s \ge 0$,

$$\boxed{Ee^{sW} \le \exp\{sEW + (17/64)s^2U^2\}.} \tag{Q81.2.1}$$

Consequently, for every real vector V in the original kernel model,

$$\boxed{Ee^{8KV} \le e^{8E(KV)+17\|V\|_2^2}.} \tag{Q81.2.2}$$

The case $U = 0$ is immediate. For $U > 0$, put $m = EW$, $v = E(W-m)^2$, $M = U - m$. The cases $M = 0$ or $v = 0$ follow directly. Otherwise the quadratic Hermite interpolant tangent to e^{sw} at $m - v/M$ and agreeing at U majorizes it on $[-U, U]$: its remainder has the sign of $(w - m + v/M)^2(w-U) \le 0$. The lower contact lies in the interval since $v \le U^2 - m^2$. Matching the first two moments therefore gives

$$Ee^{sW} \le e^{sm}Ee^{sL(B-\pi)}, \qquad L = M + v/M, \qquad \pi = \frac{v}{M^2+v}, \tag{Q81.2.3}$$

where B is Bernoulli with mean pi. For $m \ge 0$, $L \le (1+\theta)U$. The centered two-point log moment has second derivative at most $L^2/4$; integrating twice gives the coefficient $(1+\theta)^2/8 < 17/64$.

For $m \le 0$, put $u = -m/U \in [0, \sqrt{\theta}]$. Increasing the raw second moment to θU^2 enlarges L and decreases $y = 1 - 2\pi$, giving

$$L/U \le \ell(u) = \frac{1+\theta+2u}{1+u}, \qquad y \ge y(u) = \frac{\kappa + 2u + 2u^2}{1+\theta+2u} \ge 0. \tag{Q81.2.4}$$

The Bernoulli bound is

$$Ee^{t(B-\pi)} \le \exp\left\{\frac{1-2\pi}{4\log((1-\pi)/\pi)}t^2\right\}. \tag{Q81.2.5}$$

Indeed, put $a = \frac{1}{2}\log((1-\pi)/\pi)$. The function $z \mapsto \log\cosh\sqrt{z}$ is concave on $z \ge 0$, since $\tanh a/a$ decreases; the derivative test is $(\tanh a - a\,\mathrm{sech}^2\, a)' = 2a\,\mathrm{sech}^2\, a\tanh a \ge 0$. Apply the tangent inequality at a^2 to $(a - t/2)^2$ in $Ee^{t(B-\pi)} = e^{t\tanh(a)/2}\cosh(a - t/2)/\cosh a$. The endpoints follow by continuity.

The positive-term integral for arctanh gives

$$\log\frac{1+y}{1-y} \ge 2(y + y^3/3), \qquad 0 \le y < 1.$$

Thus the coefficient in (Q81.2.3) is at most $L^2/[8(1+y^2/3)]$. It remains to prove

$$8\ell(u)^2 \le 17(1 + y(u)^2/3) \qquad (0 \le u \le \sqrt{\theta}). \tag{Q81.2.6}$$

Set $A_0 = 45 + 64u$ and $B_0 = 19 + 64u + 64u^2$. Then $\ell = A_0/[32(1+u)]$ and $y(u) = B_0/A_0$. Clearing positive denominators reduces (Q81.2.6) to $\mathcal{R}(u) \ge 0$, where

$$\mathcal{R}(u) = 2176(1+u)^2(3A_0^2 + B_0^2) - 3A_0^4.$$

Expansion gives the certificate

$$\begin{aligned}\mathcal{R}(u) &= 1167597 + 918784u + 1048576u^3 + 919552u^4 \\ &\quad + 35651584u^5 + 8912896u^6 + 8564224(2u^2 - 1/4)^2.\end{aligned} \tag{Q81.2.7}$$

Every term is nonnegative for $u \geq 0$, proving (Q81.2.1). Applying the row and spectral bounds in (Q81.1.1) with $W = KV$, $U = \|V\|_2$, and $s = 8$ gives (Q81.2.2).

Q81.3 The unit terminal and a complete correlated frequency estimate

For (Q81.0.1), use only the terminal comparisons of Lemma M.38. For $b \geq 3$, take $\bar{R}, m_b, k_b, t_0, e, V_b, \gamma$ from (M.85), writing $m = m_b$, $k = k_b$, and $V = V_b$. For $b = 2$, take $m = k = 1$, $\gamma = d - \delta_2/2$, and $\sigma = K^2 g$ in (M.84). The terminal lemma has no restriction on the receiving length; the receiving factor here remains the actual f_2.

Since $f_2 = p + r$, the arithmetic mean of the two positive factors yields

$$f_2 T(p\bar{f}_b) \leq \begin{cases} [1 - \gamma/2 + K(1+z)g/2]^2, & b = 2, \\ m^2[1 - \gamma/2 + KQ(z)g/2]^2, & b \geq 3, \end{cases} \qquad z = -K, \quad Q(z) = 1 + kzG_{b-1}(z). \tag{Q81.3.5}$$

The square bases are nonnegative actual arithmetic means. Thus their sixteenth powers are bounded by the corresponding exponentials, including any zero boundary by continuity.

Q81.3.1 Single pair starting at three, $b \geq 3$

For every $N \geq 2, h \geq 1$, k in [1, 6/5] and x in [0, theta],

$$\boxed{|1 + kzG_N(z) + z^2 G_h(z)|^2 \leq 1 - 159x/160, \qquad z = i\sqrt{x}.} \tag{Q81.3.6}$$

The real and normalized imaginary coordinates are $1 - kxc_N - xa_h$ and $ka_N - xc_h$. They are positive: their lower bounds are $17/160$ and $3/16$. It is therefore legitimate to upper-bound each coordinate before squaring.

For $h = 1, N = 2$, the squared norm minus one, divided by x, is

$$k^2 - 2k - 2 + (k+1)^2 x.$$

It is increasing in x and convex in k; at $x = \theta$ its k endpoints are -11/8 and -159/160. For $h = 1$, $N \geq 3$, the coordinate upper bounds are $1 - (k+1)x + kx^2$, $k(1 - x + x^2)$. The same difference quotient is

$$k^2 - 2k - 2 + (-k^2 + 4k + 1)x + (k^2 - 2k)x^2 - k^2 x^3(1 - x).$$

The final two terms are nonpositive. The remaining endpoint values are -11/8 and -951/800, both at most -159/160. For $h \geq 2, N = 2$ use coordinates $1 - (k+1)x + x^2$, $k - x + x^2$. The quotient is

$$k^2 - 2k - 2 + (k^2 + 3)x - x^2 - x^3 + x^4.$$

Its last three terms are nonpositive. The remaining endpoints are -11/8 and -37/32. Finally, for $h \geq 2, N \geq 3$, use $1 - (k+1)x + (k+1)x^2$, $k - (k+1)x + (k+1)x^2$. The quotient is

$$k^2-2k-2+(-k^2+2k+3)x+(k^2-1)x^2-(k+1)^2x^3(1-x).$$

Drop only its last nonpositive term. The remainder increases in x and is convex in k; its endpoints are -11/8 and -32733/25600, both at most -159/160. This covers the four cases.

Q81.3.2 General masks for the other $b \geq 3$ unit rows

If $u_1 = 3$ and $j \geq 2$, (Q81.1.11) gives

$$|Q+U|^2 \leq (1-35x/32)^2 + x(6/5+x/2)^2 < 3/2. \tag{Q81.3.7}$$

The coordinates remain positive. The excess over one is $-299x/400 + (1225/1024+6/5)x^2 + x^3/4$, at most $5/12+1/56 < 1/2$, using $x^2 < 1/6, x^3 < 1/14$.

For every mask starting in degree at least three,

$$\boxed{|Q+U|^2 \leq 4/3.} \tag{Q81.3.8}$$

To prove it, put $\delta = x^2/(1-x^2) \leq 6x^2/5$. The positive upper tail in each coordinate is at most delta. Both total coordinates remain positive; the possible negative tails have magnitudes at most $x^3/(1-x^2)$ and $x/(1-x^2)$, respectively. For $N = 2$ the upper coordinates are $1-kx+6x^2/5$, $k+6x^2/5$. Their squared norm is

$$1+(k^2-2k)x+(k^2+12/5)x^2+(36/25)(x^4+x^5).$$

For $N \geq 3$ they are $1-kx+(k+6/5)x^2$ and $k-kx+(k+6/5)x^2$. Their squared norm is

$$1+(k^2-2k)x+(2k+12/5-k^2)x^2+k^2x^3+(k+6/5)(6/5-k)x^4+(k+6/5)^2x^5.$$

Both are convex in $x \geq 0$ and in k on the specified interval. Thus check $x = 0$, theta and $k = 1, 6/5$. At $x = \theta$, with denominator 838860800, the four numerators are respectively 1015054740, 1089601940, 1080009045, 1111336016. Each is below four thirds of that denominator; for the largest, the cross-multiplication difference is $21435152 > 0$. Convexity extends these endpoint bounds to the interval.

Q81.4 The unit-connector estimate

For $b \geq 3$ combine (Q81.3.5) and (Q81.1.6) before using (Q81.2.2). In every case,

$$\log EY^8 \leq -8(\beta_b+\gamma-ke+d+\Gamma-ES)+17\|(Q+U)g\|^2. \tag{Q81.4.1}$$

For $u_1 = 3$, $j = 1$, the last line of (M.86) and (Q81.3.6) make the right side at most

$$-23d+(23093/800)\delta_2 \leq -10d.$$

For $u_1 = 3, j \geq 2$ use $\Gamma \geq 5d-\delta_2$, $ES \leq \delta_2 + d/12$, and (Q81.3.7). The upper divided by d is at most $-64+(1144/25)\theta+2/3+51/2 < -10$. For $u_1 \geq 4$ use $\Gamma \geq 3d$, $ES \leq d/12$, and (Q81.3.8). The upper is $(-48+(744/25)\theta+2/3+68/3)d < -10d$. Further pairs only strengthen these losses. Thus every $b \geq 3$ row has $EY^8 \leq e^{-10d}$.

For $b = 2$ the common logarithmic expression is

$$-8\gamma - 8d - 8\Gamma + 8ES + 17\|(1+z+U)g\|^2. \tag{Q81.4.2}$$

For a single $u_1 = 3$ pair, $1+z+U = G_{h+2}$. By (Q81.1.5), (Q81.4.2) is at most $-15d + (181/15)\delta_2 \le -(4847/480)d < -10d$. For $u_1 = 3, j \ge 2$, (Q81.1.11) gives

$$|1+z+U|^2 \le (1-x/2)^2 + x(1+x/2)^2 < 5/4.$$

Use $\Gamma \ge 5d - \delta_2$ and $ES \le \delta_2 + d/12$ in (Q81.4.2); the result is below -10d. For $u_1 \ge 4, j \ge 2$, (Q81.1.9) and $\sqrt{1+\theta} < 6/5$ give $\|(1+z+U)g\|^2 < (6/5+3/8)^2 d < 5d/2$. $\Gamma \ge 6d$ and $ES \le d/12$ again make (Q81.4.2) below -10d.

It remains to pay every single pair with $u \ge 4$. At $u = 4$ keep its negative direction, so $U = z^3 G_h(z)$. At $u = 5$ use the positive exception in paragraph Q81.0; its primitive is $U = z^4 G_h(-z)$. In both cases the real coordinate of 1+z+U lies in [0, $1 + x^2$], its normalized imaginary coordinate in [0, 1], and $ES \le 0$. Therefore

$$|1+z+U|^2 \le 1 + x + 2x^2 + x^4.$$

With $\Gamma = 3d$, (Q81.4.2) is at most $-23d + 21\delta_2 + 34\delta_3 + 17\delta_5 \le -8d$. The last comparison has surplus $413535/1048576$ at $x = \theta$.

For every $u \ge 6$ keep the negative direction. The single interval has $|U| \le x^{5/2}\sqrt{1+x}$, so

$$|1+z+U|^2 \le (1+x)(1+(2/3)x^2)^2 < 7/4. \tag{Q81.4.3}$$

The upper is increasing; at theta the numerator and denominator are 130816125 and 75497472. Its comparison with $7/4$ has positive difference 1304451 in that denominator. Using $ES \le d/12$, (Q81.4.2) is now at most

$$[-40 + 119/4 + 2/3 + 4\theta]\, d = -191d/24.$$

The cases above prove (Q81.0.1) for $b \ge 2$, every first short arm, every finite number of pairs, and every finite last arm.

Q81.5 The exchanged first pair for the nonunit family

For (Q81.0.2), exchange the first pair of arms; the preceding direction has factor $W_3 \bar{f}_v$.

Apply A, a positive operator, to the negative finite convolution for $\bar{f}_{v-1}$. Since

$$A\sum_{i=0}^{v-1}(-K)^i 1 = 2 - \beta_v - \sum_{i=0}^{v}(-K)^i 1,$$

retaining the real final term $\beta_{v-1}W_1 = \beta_{v-1}p$ gives

$$W_v \le 2 - \beta_v - S_v^- - \beta_{v-1}p, \qquad S_v^- = \sum_{i=0}^{v}(-K)^i 1. \tag{Q81.5.1}$$

Also $\bar{f}_3 = S_3^- - d$. Set

$$h = v-3 \ge 1, \quad \alpha = \beta_v, \quad \beta = \beta_{v-1}, \quad U(z) = z^2 G_h(z), \quad S = KU(-K)g, \quad \Gamma = d + \alpha + \beta. \tag{Q81.5.2}$$

Here $S_v^- - S_3^- = S$. Adding (Q81.5.1) to the b3 identity and using AM–GM on two nonnegative actual factors proves

$$\boxed{\bar{f}_3 W_v \le \exp\{-S - \beta g - \Gamma\}, \qquad \Gamma \ge 3d - 2\delta_2 \ge 35d/16.} \tag{Q81.5.3}$$

Exchanging the first pair changes the sign of S and gives the displayed loss.

Write $G_h = a_h + z c_h$, with the conventions of (Q81.1.4), and put $\zeta = \delta_2/d$ when $d > 0$. Direct skew integration gives

$$S = KUg, \quad ES = \int x a_h \, d\nu \in [\kappa\delta_2, \delta_2], \quad E(Ug) = -\int x c_h \, d\nu. \tag{Q81.5.4}$$

For $h = 1, \alpha = \beta = d$, $ES = \delta_2$, $E(Ug) = 0$. For $h = 2, \beta = d$, $ES = \delta_2$, $E(Ug) = -\delta_2$. For $h \ge 3$,

$$\tau \le \beta \le d - \kappa\delta_2, \quad |G_h| \le 1, \quad E(Ug) \le -\kappa\delta_2. \tag{Q81.5.5}$$

The bound on beta is the finite alternating comparison $\beta_n \le d - \delta_2 + \delta_3 \le d - \kappa\delta_2$ for $n \ge 5$. For all $h \ge 2, E(Ug) \le -\kappa\delta_2$, while $\tau \le \beta \le d$. We retain these same-variable correlations.

Q81.6 The base table and its full-mean conversion

Put $X = T^L(q f_b)$, $Y_0 = W_2 X = (p - r)X$, $L \ge 2$. The C071/C075/C076 actual positive masks have the form

$$0 < X \le C[Bq + Br + Z - a], \qquad Y_0 \le CB\left(1 + \frac{Z - a}{2B}\right)^2, \quad Z = KV. \tag{Q81.6.1}$$

The base of the square is the arithmetic mean of actual nonnegative upper factors. The table below records the finite envelopes and scalar bounds used in this proof.

For $b = 2$ set $\gamma = d - \delta_2/2$, $B_2 = 1 - \gamma$. For $b \ge 3$ use the same actual positive terminal remainder

$$R = KG_{b-1}(K)g, \quad m = 1 - \beta_b/2, \quad k = 1/m, \quad e = d - \beta_{b+2}, \quad \rho = E(rR),$$
$$\gamma = k\beta_{b+1} - k^2 E R^2/2, \quad a_1 = ke, \quad B_* = \gamma + a_1. \tag{Q81.6.2}$$

Here $1 \le k \le 6/5$, $0 \le e \le \delta_2$, $\kappa\delta_2 \le e, \rho \le \delta_2$, $\gamma \le kd$, $a_1 \le 39d/80$, and $1 - \gamma \ge 17/25$. At a frequency, writing $G_{b-1} = a_N + z c_N$, $e = \int x c_N \, d\nu$ and $\rho = \int x a_N \, d\nu$. Define

$$P_n(z) = G_n(z) + z^{n+1} \quad (b = 2), \qquad P_n(z) = G_n(z) + k z^{n+1} G_{b-1}(-z) \quad (b \ge 3). \tag{Q81.6.3}$$

Table 17: Finite masks for $X = T^L(q f_b)$, indexed by (b, L).

(b, L)	C	B	a	Actual Z
2, 2	1;	B2;	0;	$K^2(\gamma 1 + r)$.
2, 3	1;	B2;	τ;	$-K^2(g + \sigma)$.
2, 4	1;	$B2 - \tau$;	d B2;	$K^2[\tau 1 - B2g + r + t]$.
$2, L \ge 5$	1;	$1 - 2\tau$;	τ;	$K[\tau + z P_{L-2}(z)]g$.

(b,L)	C	B	a	Actual Z
$b\ge 3,2$	m^2;	$1-\gamma$;	a1;	$K^2(\gamma 1+kR)$.
$b\ge 3,3$	m^2;	$1-B*$;	$d-k\rho$;	$K[a_1+zP1(z)]g$.
$b\ge 3,4$	m^2;	$1-41d/80$;	κd;	$K[41d/80+zP2(z)]g$.
$b\ge 3,L\ge 5$	$1-d/2$;	$1-d/2$;	$d/2$;	$K[d/2+zP_{L-2}(z)]g$.

In this table $z=-K$; displayed strings such as P1 designate the finite polynomials in (Q81.6.3), not an unspecified path input.

The complete old scalar expressions are

$$8\log(CB)-8a/B+8M/B+\begin{cases}6D_2/B^2, & Z=K^2V_2,\\ (2025/128)D_1/B^2, & Z=KV_1,\end{cases} \tag{Q81.6.4}$$

with $M\ge \max(EZ,0)$ and the original specified vector bounds D_i. Every expression (Q81.6.4) is at most -d/10. For $b\ge 3,L=2$ it is at most -d/8. The improved b2L2 calculation in Section O.2, paragraph Q75.3.2, explicitly pays that original scalar, with $D_2=\delta_2$ and $M=\delta_2-\gamma d$; it is not merely a moment theorem. The finite positive-end-sign family $b=2$ mod 4 is included by C071's same-remainder d3 payment. Its sign is not discarded here.

For reference, the whole-domain normalizations for the four two-row rows, in their order above, are

$$\frac{D_2}{B^2d}\le \theta\frac{324}{169},\quad \frac{324}{169},\quad \frac{217}{81}<\frac{25}{9},\quad \frac{540}{289}<2. \tag{Q81.6.5}$$

The one-row normalizations, in their order, are

$$\frac{D_1}{B^2d}\le \frac{289}{256},\quad \frac{481}{400},\quad \frac{27}{32},\quad \frac{18}{25}; \tag{Q81.6.6}$$

all are at most $5/4$. The lower B bounds are respectively $13/18, 13/18, 1/2, 5/9, 17/25, 1/2, 4/5, 5/6$. For example the b2L4 normalization is $(1+4/9)^2+16/27=217/81$, using $B_2=B+\tau$. For $b\ge 3$, L2, use $\theta(1+\theta)<3/5$, $k\le 6/5$, and $B\ge 17/25$. The b2 tail quotient $(\theta-2\theta\kappa\tau+\tau^2)/(1-2\tau)^2$ increases for $0\le\tau\le 2/9$ and has endpoint $289/256$. These are the full original normalizations from Q5-C076.

Set $V_0=V/B$ in (Q81.6.1). Changing (Q81.6.4) to the exact full-mean 17 scalar (Q81.2.2) costs at most $(17\theta-6)D_2/B^2=(29/32)D_2/B^2$ in a two-row case, and $(17-2025/128)D_1/B^2=(151/128)D_1/B^2$ in a one-row case. Actual $EZ\le M$ is retained. Therefore

$$\mathcal{E}_0:=8\log(CB)-8a/B+8E(KV_0)+17\|V_0\|^2\le c_0 d,\qquad \|V_0\|^2\le 5d/4, \tag{Q81.6.7}$$

$$c_0=5/2\ (b=2),\qquad c_0=8/5\ (b\ge 3,L=2),\qquad c_0=3/2\ (b\ge 3,L\ge 3). \tag{Q81.6.8}$$

The arithmetic is $-1/10+(29/32)(25/9)=3481/1440<5/2$, $-1/8+(29/32)(540/289)=1813/1156<8/5$, and $-1/10+(151/128)(5/4)=3519/2560<3/2$. The norm bound in (Q81.6.7) follows from (Q81.6.5) after multiplication by theta and from (Q81.6.6).

Q81.6.1 Finite coefficient facts used below

For every finite $n \geq 3$ in (Q81.6.3), writing $P_n = A_P + zB_P$,

$$1 - x \leq A_P, B_P \leq 1. \tag{Q81.6.9}$$

This is the exact C075 finite coefficient lemma. One direct proof separates n modulo four in the finite alternating sums $S_a = \sum_{i=0}^{a}(-x)^i$. For $n = 4a$ the two additions have signs +,+ and magnitudes $kx^{2a+1}c_N, kx^{2a}a_N$; for $n = 4a + 1$ they have signs -,+ and magnitudes $kx^{2a+1}a_N, kx^{2a+1}c_N$; for $n = 4a + 2$ they have signs -,- and magnitudes $kx^{2a+2}c_N, kx^{2a+1}a_N$; and for $n = 4a + 3$ they have signs +,- and magnitudes $kx^{2a+2}a_N, kx^{2a+2}c_N$. Each fits inside the final alternating interval because $kx < 1$ and $kx^2 < 1$. At $n = 3$ its second coordinate is directly at least 1-x. The four residue classes cover every integer n. The b2 polynomial is included by $k = 1, N = 1$. For b2 also $A_P \leq 1 - x + x^2$.

The two short positive-terminal polynomials have exact coordinates

$$P_1 : \ A_P = 1 - kxa_N, \ B_P = kxc_N; \qquad P_2 : \ A_P = 1 - kx^2c_N, \ B_P = 1 - kxa_N. \tag{Q81.6.10}$$

In particular $A_{P_1} \geq 41/80$, $A_{P_2} \geq 4/5$. In the later tail we may use the weaker bounds $A_P \geq 1/2$ and $B_P \geq 3/8$, as well as their upper bound one.

Q81.7 Single exchanged pair: terminal $b = 2$, every $L \geq 2$

Let $Y_1 = Y_0\bar{f}_3W_v$. Combine the two actual remainders before applying (Q81.2.2). Define

$$\Delta = \|V_0 - Ug - \beta\mathbf{1}\|^2 - \|V_0\|^2. \tag{Q81.7.1}$$

Equations (Q81.5.3),(Q81.6.1) give the exact scalar certificate

$$\log EY_1^8 \leq \mathcal{E}_0 - 8\Gamma + 17\Delta - 8ES. \tag{Q81.7.2}$$

We prove, throughout the full continuous original domain,

$$\boxed{17\Delta - 8ES \leq 7d \qquad (b = 2, L \geq 2).} \tag{Q81.7.3}$$

Together with (Q81.6.8) and $\Gamma \geq 35d/16$ this proves $EY_1^8 \leq e^{-8d}$. In the computations below $\zeta = \delta_2/d \in [d, \theta]$. Spectral Cauchy also gives $\delta_3 \geq \delta_2^2/d$.

Q81.7.1 $L = 2$

Here $V_0 = (\gamma + z^2)g/B$, $B = B2$. Put $\lambda = \gamma/B$. Then $0 \leq \lambda \leq 5/13$, because $\gamma \leq d - d^2/2 \leq 5/18$. Since $EV0 = 0$, the beta cross term is $\beta^2 + 2\beta E(Ug) \leq \beta^2$. The frequency increment is

$$x^2|G_h|^2 - (2/B)x^2a_h + 2\lambda xa_h. \tag{Q81.7.4}$$

For $h = 1$, this gives $\Delta \leq d^2 + 2\lambda\delta_2 - \delta_3$. For $h = 2$, use $|G_2|^2 = 1 + x, a_2 = 1$ to get $\Delta \leq d^2 + 2\lambda\delta_2 - \kappa\delta_3$. As $ES = \delta_2$ in both cases, their cost divided by d is at most

$$17/3 + (66/13)\zeta - 17\kappa\zeta^2 \leq 17/3 + \frac{(66/13)^2}{68\kappa} < 20/3 < 7. \tag{Q81.7.5}$$

For the strict comparison use $34848 < 54587$. The h1 quadratic is stronger. The concave quadratic bounds apply throughout the stated interval.

For $h \geq 3, |G_h|^2 \leq 1$ and $a_h \geq \kappa$ make the d3 term in (Q81.7.4) nonpositive. By (Q81.5.5),

$$\Delta \leq 2\lambda\delta_2 + d^2(1-\kappa\zeta)^2.$$

Thus its cost divided by d is at most

$$(170/13 - 8\kappa)\zeta + (17/3)(1-\kappa\zeta)^2.$$

This increases on [0, theta]: its derivative is at least $170/13 - 8\kappa - 34\kappa/3 > 0$. At theta it is $20904801/3145728 < 7$. This proves the full L2 row.

Q81.7.2 $L = 3$

Here $V_0 = (z + z^3)g/B$, $B = B2$, and $EV_0 = \tau/B$. Because $\beta \leq d$ and $2\tau/B \geq 2\kappa d > d$, the entire beta cross $\beta^2 - 2\beta EV_0 + 2\beta E(Ug)$ is nonpositive. The remaining increment is at most

$$\int x^2\{1 + x + (2/B)(1-x)\}\, d\nu \leq \int \frac{x^2(49-23x)}{13}\, d\nu. \tag{Q81.7.6}$$

Use $B \geq 13/18$. After subtracting $8ES \geq 8\kappa$ d2, the frequency upper has its maximum at $x = \theta$ or zero: its quotient by x is increasing, since $49 - 46x > 0$. At theta the value is $217217/32768 < 7$. This proves L3.

Q81.7.3 $L = 4$

Now $B = B2 - \tau$ and $V_0 = [\tau + B_2 z + z^2 + z^4]g/B$, so $EV_0 = B_2 d/B$. The frequency increment excluding beta is

$$x^2|G_h|^2 - (2/B)x^2(1-x)a_h + 2x^2c_h + (2\tau/B)(xa_h + x^2c_h). \tag{Q81.7.7}$$

The beta term is at most $\tau^2 - 2\tau B_2 d/B$: the function $\beta^2 - 2\beta B_2 d/B$ decreases on [tau, d], and $EUg \leq 0$. Combining it with the last term of (Q81.7.7) gives a nonpositive quantity. Indeed it suffices that

$$2[B_2 - (1+\theta)\zeta] - B(1-\zeta) \geq 0.$$

Its exact left side is $1 - (1+2\theta)\zeta - \zeta(5-3\zeta)d/2$. It decreases in both variables on the original rectangle, and at $\zeta = \theta, d = 1/3$ equals $47/6144 > 0$.

In the remainder of (Q81.7.7), use $B \leq 1, a_h \geq 1 - x, c_h \leq 1$, and $|G_h|^2 \leq 1 + x$. This gives

$$\Delta \leq \int x^2(1 + 5x - 2x^2)\, d\nu.$$

After subtraction of 8kappa d2 the frequency quotient by x increases, and its endpoint value is $2961647/524288 < 7$. This proves L4.

Q81.7.4 Every $L \geq 5$

Here $V_0 = [\tau + zP_n]g/B$, $n = L - 2 \geq 3, B = 1 - 2\tau \geq 5/9$. Write $Pn = A + zC$. In this b2 family, $1 - x \leq A, C \leq 1$, $A \leq 1 - x + x^2$, and $EV_0 \geq \kappa d/B$. For $h = 1$, the frequency term is

at most $(1-2\kappa/B)\delta_3+(2\tau/B)\delta_2$. With $\beta=d$, the constant term is at most $d^2-2\kappa d^2/B$. Since $\tau\delta_2\le\kappa d^2$, these give $\Delta\le d^2$. Thus $17\Delta-8ES\le 9d^2\le 3d$.

For $h\ge 2$, the frequency cross satisfies

$$Ac_h-Ca_h\le(1-x+x^2)-(1-x)^2=x. \tag{Q81.7.8}$$

The terms involving tau and beta together are at most

$$\tau d\{1-2\kappa\zeta-(2/B)(\kappa-\zeta)\}\le(73/512)\tau d\le(73/2304)d. \tag{Q81.7.9}$$

To see this, use $\beta^2\le\beta$ d, $\beta\ge\tau, EUg\le-\kappa$ d2, $EV0\ge\kappa$ d/B, $ES\le\delta_2$, and $B\le 1$. The coefficient of beta before this substitution is negative. Its final upper is $1-2\kappa+2\theta\zeta\le 73/512$.

Equations (Q81.7.8)–(Q81.7.9) therefore give

$$\Delta\le\delta_3+(23/5)\delta_4+(73/2304)d.$$

Consequently

$$17\Delta-8ES\le\left[17\theta^2+(391/5)\theta^3+1241/2304-8\kappa\theta\right]d=\frac{9817163}{1474560}d<7d. \tag{Q81.7.10}$$

Use $d3\le\theta$ d2, $d4\le\theta^2\delta_2$, followed by $\delta_2\le\theta$ d; the coefficient being multiplied by d2 is positive. This proves every integer $L\ge 5$ and completes (Q81.7.3).

Q81.8 Single exchanged pair: every $b\ge 3$ and every $L\ge 2$

For these rows we prove

$$\boxed{17\Delta-8ES\le 8d.} \tag{Q81.8.1}$$

The same exact definition (Q81.7.1) and the same actual terminal parameters are used. Together with (Q81.6.8), this yields exp(-79d/10) at L2 and exp(-8d) at every $L\ge 3$.

Q81.8.1 $L=2$

Put $c=k/B$ and $\lambda=\gamma/B$. Then $c\le 30/17$, $\lambda\le cd$, and

$$V_0=[\gamma+kz^2G_N(-z)]g/B,\quad N=b-1\ge 2,\qquad EV_0=ce\ge c\kappa\delta_2. \tag{Q81.8.2}$$

Its frequency increment is

$$x^2|G_h|^2+2\lambda x a_h-2cx^2(a_Na_h-xc_Nc_h). \tag{Q81.8.3}$$

For $h=1,\beta=d$ and $EUg=0$. Combining the gamma and beta terms gives at most $d^2+2c\theta d\delta_2$, since $e\ge\kappa$ d2. The remaining frequency contribution is at most $-3\delta_3/16$. Thus

$$17\Delta-8ES\le 17d^2+60\theta d\delta_2-8\delta_2-(51/16)\delta_3\le(17/3)d+\delta_2/8<6d. \tag{Q81.8.4}$$

For $h=2,\beta=d$ and $EUg=-\delta_2$. The gamma plus constant part is at most d^2, because $ctheta<1$. The cross in (Q81.8.3) is nonnegative before its minus sign: $a_N-xc_N\ge 1-2x>0$. Hence $\Delta\le d^2+\delta_3+\delta_4$, and

$$17\Delta - 8ES \le [17/3 + \{17\theta(1+\theta) - 8\}\theta]d < 7d. \tag{Q81.8.5}$$

For $h \ge 3, |Gh| \le 1$ and $a_N a_h - x c_N c_h \ge 1 - 3x + x^2 \ge -55/1024$. The part of (Q81.8.3) not involving gamma is at most $(5177/4352)\delta_3 < (6/5)\delta_3$. The gamma and beta terms together are at most d^2. For this bound, Write $\beta = t$ d, $1 - \zeta \le t \le 1 - \kappa\zeta$. After using $\lambda \le cd, EUg \le -\kappa$ d2, and (Q81.8.2), their upper divided by d^2 is

$$t^2 + (60/17)\zeta - (893/272)t\zeta. \tag{Q81.8.6}$$

It is convex in t. At $t = 1 - \zeta$ its excess over one is $\zeta[-477 + 1165\zeta]/272 \le 0$. At $t = 1 - \kappa$ zeta that excess is $\zeta[-16/17 + (40071/17408)\zeta] \le 0$. At theta the two brackets are strictly negative, with numerators -119/32 and -3365/557056, respectively. Thus $(Q81.8.6) \le 1$ throughout. It follows that

$$17\Delta - 8ES \le [17/3 + (102/5)\theta^2 - 8\kappa\theta]d = \frac{54557}{7680}d < 8d. \tag{Q81.8.7}$$

This proves every $b \ge 3$ terminal, including all finite end signs, at L2.

Q81.8.2 $L = 3$

Here $V_0 = [\delta + zP_1]g/B$, $\delta = ke, B \ge 1/2$. Use (Q81.6.10). The general frequency increment, before beta, is

$$x^2|G_h|^2 + (2\delta/B)xa_h + (2/B)x^2(A_P c_h - B_P a_h). \tag{Q81.8.8}$$

For $h \ge 2$, all terms involving delta and beta have nonpositive sum. To verify this fully, write $\beta = t$ d, $1 - \zeta \le t \le 1$. Since $EV_0 = (d - k\rho)/B \ge (d - k\delta_2)/B$ and $\delta \le k$ d2, their upper divided by d^2 is

$$t^2 - 2\kappa t\zeta - (2/B)[t(1 - k\zeta) - k\zeta^2].$$

The bracket is at least $1 - (1+k)\zeta \ge 17/160 > 0$. Since $B \le 1$, the last display is at most

$$t^2 - 2t + 2(k - \kappa)t\zeta + 2k\zeta^2. \tag{Q81.8.9}$$

At $t = 1$ it is $-1 + 2(k-\kappa)\zeta + 2k\zeta^2$; at $t = 1 - \zeta$ it is $-1 + 2(k-\kappa)\zeta + (1+2\kappa)\zeta^2$. Both increase in k, zeta and are negative at $k = 6/5, \zeta = \theta$. Convexity in t proves the assertion on the full interval.

For $h = 1$ the same terms are at most d^2, since $1 - k\zeta(1+\zeta) > 0$; the remaining cross in (Q81.8.8) is nonpositive. Thus $\Delta \le d^2 + \delta_3$, which pays (Q81.8.1).

For $h = 2, A_P - B_P \le 1 - 2\kappa x$. Using $2/B \le 4$ gives $\Delta \le 5\delta_3 - 15\delta_4/4$. As $ES = \delta_2$,

$$17\Delta - 8ES \le (852501/131072)d < 8d. \tag{Q81.8.10}$$

For $h \ge 3, |G_h|^2 \le 1$, and $A_P c_h - B_P a_h \le 1 - \kappa(1+\kappa)x$. Hence $\Delta \le 5\delta_3 - (969/256)\delta_4$, and

$$17\Delta - 8ES \le (65299507/8388608)d < 8d. \tag{Q81.8.11}$$

For both endpoint comparisons, the frequency upper divided by x is increasing on [0, theta]. Its derivative is positive there: respectively $85 - (255/2)x > 0$ and $85 - (16473/128)x > 0$. The estimate follows for every frequency and h.

Q81.8.3 $L = 4$

Now $V_0 = [\delta + zP_2]g/B$, $\delta = 41d/80, B = 1 - 41d/80 \geq 199/240 > 4/5$. The delta and beta terms in (Q81.8.8) have nonpositive sum for every h. For $h = 1$, their upper divided by d^2 is

$$1 - (2/B)[1 - (k\theta + 41/80)\zeta] \leq -1 + 2\zeta < 0.$$

For $h \geq 2$ use $t = \beta/d$ in [1-zeta, 1], $EUg \leq -\kappa$ d2, and $EV_0 \geq (d - k\delta_3)/B$. The bracket $t(1 - k\theta\zeta) - (41/80)\zeta$ is positive: at its smallest permitted t it is at least $1 - 2\zeta + (39/80)\zeta^2 > 0$. Replace 1/B by one only in its negative contribution. The resulting upper is

$$t^2 - 2t + 2(k\theta - \kappa)t\zeta + (41/40)\zeta \leq -\kappa + (41/40)\theta < 0. \tag{Q81.8.12}$$

For $h = 1$ the remaining frequency cross is nonpositive, so (Q81.8.1) is immediate. For $h \geq 2$, the exact coordinates in (Q81.6.10) give

$$A_P c_h - B_P a_h \leq x[11/5 - (153/80)x]. \tag{Q81.8.13}$$

Indeed its upper is $A_P - B_P(1 - x)$, then use $a_N \leq 1, c_N \geq \kappa$, and maximize the resulting expression in k. The right side is nonnegative. With $2/B \leq 5/2$ and $|Gh|^2 \leq 1 + x$,

$$\Delta \leq \delta_3 + (13/2)\delta_4 - (153/32)\delta_5.$$

Subtracting $8ES \geq 8\kappa$ d2 gives an increasing frequency quotient by x and endpoint value

$$17[\theta^2 + (13/2)\theta^3 - (153/32)\theta^4] - 8\kappa\theta = 203700679/33554432 < 8. \tag{Q81.8.14}$$

The derivative test is $1 + 13x - (459/32)x^2 > 0$. This proves L4.

Q81.8.4 Every $L \geq 5$

Here $B = 1 - d/2 \geq 5/6, \delta = d/2$, and $V_0 = [\delta + zP_n]g/B$, $n = L - 2 \geq 3$. Use $A_P \geq 1/2$, $B_P \geq 3/8$, and $A_P, B_P \leq 1$. For $h \geq 2$ the delta and beta terms have nonpositive sum. Their upper divided by d^2 is

$$t^2 - 2\kappa t\zeta - (t - \zeta)/B \leq t^2 - t + \zeta - 2\kappa t\zeta. \tag{Q81.8.15}$$

Here $t \geq 1 - \zeta > \zeta$. At $t = 1$ its value is $(1 - 2\kappa)\zeta \leq 0$; at $t = 1 - \zeta$ it is $\zeta[(1 + 2\kappa)\zeta - 2\kappa] \leq 0$. Convexity gives the bound between these endpoints.

For $h = 1$ that sum is at most $(6/5)\zeta d^2$, and the frequency part is at most d3. Therefore its cost is at most $(913/160)\zeta d < 8d$.

For $h \geq 2, A_P c_h - B_P a_h \leq 5/8 + 3x/8$. Since $2/B \leq 12/5$,

$$\Delta \leq (5/2)\delta_3 + (19/10)\delta_4,$$

$$17\Delta - 8ES \leq [(85/2)\theta^2 + (323/10)\theta^3 - 8\kappa\theta]d = 2375711d/327680 < 8d. \tag{Q81.8.16}$$

The finite coefficient lemma gives these bounds for every $n \geq 3$. This completes (Q81.8.1) and every terminal $b \geq 3$, inner $L \geq 2$ single-pair case.

Q81.9 Further pairs in the nonunit message

The single-pair proof gives a scalar bound for the combined remainder. Its primitive is $V_* = V_0 - Ug - \beta\mathbf{1}$. From (Q81.6.7), (Q81.1.4), and $\beta \le d$,

$$\|V_*\| \le (9/8 + 1/2 + 3/5)\sqrt{d} = (89/40)\sqrt{d} < (9/4)\sqrt{d}. \tag{Q81.9.1}$$

For the middle term use $\|Ug\| \le \theta\sqrt{1+\theta}\sqrt{d} < \sqrt{d}/2$; for the last use $\sqrt{d} < 3/5$. These are original norm bounds.

All remaining pairs satisfy $u_2 \ge v + 1 \ge 5$. Their complete disjoint mask has primitive Vtail of norm at most $\sqrt{d}/4$, mean $E(KV_{tail}) \le d/12$, and actual total loss $3(j-1)d$. Combining it with V* before (Q81.2.2) adds at most

$$\left[\frac{2}{3} + 17\left(2 \cdot \frac{9}{4} \cdot \frac{1}{4} + \frac{1}{16}\right) - 24(j-1)\right] d = \left[\frac{1001}{48} - 24(j-1)\right] d. \tag{Q81.9.2}$$

For $j \ge 2$ this is at most -151d/48. The bound is uniform in the number, gaps, and largest length of the remaining arms. Combining (Q81.7.2)–(Q81.8.16) with (Q81.9.2) yields

$$\boxed{E\left[W_2 \bar{f}_3 W_v \prod_{i=2}^{j} W_{u_i} \bar{f}_{v_i} T^L(q f_b)\right]^8 \le e^{-79d/10} \quad (b \ge 2, L \ge 2, j \ge 1).} \tag{Q81.9.3}$$

This proves (Q81.9.3), including every finite short connector and both infinite integer tails. All continuous payments used only the original domain $d \le 1/3, d \le \zeta \le 13/32$, and the stated same-kernel spectral moments.

M.2 Finite calculus for the odd-root estimates

Work on a finite probability space after deleting zero-mass atoms. Let $B = -K$, where K is a real antisymmetric kernel with $|K| \le 1$, and write

$$T = J + B, \quad A = J - B = T^*, \quad u = B\mathbf{1}, \quad p = 1 - u, \quad q = 1 + u, \qquad d = \mathbb{E}u^2, \quad \delta_j = \|B^j\mathbf{1}\|_2^2.$$

The four arm messages are $f_n = A^n\mathbf{1}$, $\bar{f}_n = T^n\mathbf{1}$, $h_n = Tf_{n-1}$ and $W_n = A\bar{f}_{n-1}$. Thus f_n points outward, $\bar{f}_n$ inward, h_n first inward and then outward, and W_n first outward and then inward. The path calculus in Section D gives

$$\begin{gathered} 0 \le d \le \tfrac{1}{3}, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j, \quad \|B\|^2 \le \theta = \tfrac{13}{32}, \quad \kappa = \tfrac{19}{32}, \\ |BV(x)| \le \|V\|_2, \qquad 0 \le f_n, \bar{f}_n, h_n, W_n \le 2, \qquad \mathbb{E}f_n = \mathbb{E}\bar{f}_n \le 1, \\ 0 \le f_n h_n, \bar{f}_n W_n \le 1. \end{gathered} \tag{M.1}$$

Here the path bounds concern $n \ge 1$, and the mean bound also holds at $n = 0$. The last line permits only genuine equal-length pair decorations. All operators and expectations below use the original measure. Since $p(x), q(x) \ge \mu(x) > 0$, actual tree messages and the upper bounds dominating them are positive. When $d = 0$, leaf integration gives density one; subsequent estimates take $d > 0$.

Lemma M.1 (Finite spectral coordinates)**.** *Let $G_n(z) = \sum_{j=0}^{n-1} z^j$, $G_0 = 0$, and*

$$\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j \quad (n \geq 1).$$

Empty sums are zero. With $\tau = d - \delta_2$,

$$S_n := \mathbf{1} + G_n(B)u = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j}, \qquad \tau \leq \beta_j \leq d \quad (j \geq 3), \qquad \kappa d \leq \tau \leq \tfrac{2}{9}. \tag{M.2}$$

There is a positive measure ν on $[0, \theta]$ of mass d such that $\int x^j \, d\nu = \delta_{j+1}$. If $P(i\sqrt{x}) = a(x) + i\sqrt{x}\, b(x)$, then

$$\begin{aligned} \mathbb{E}[P(B)u] &= -\int b \, d\nu, & \mathbb{E}[BP(B)u] &= -\int a \, d\nu, \\ \|P(B)u + c\mathbf{1}\|_2^2 &= \int (a^2 + xb^2)\, d\nu - 2c \int b \, d\nu + c^2. \end{aligned} \tag{M.3}$$

For $G_m = E + zC$, $(E, C) = (1, 0)$ at $m = 1$. For $m \geq 2$, $1 - x \leq E, C \leq 1$; even m has $E = C$, and odd $m \geq 3$ has $E = 1 - xC$. In particular $G_2 = 1 + z$ and $|G_2|^2 = 1 + x$; no contraction bound is imposed at this index. For $m \geq 3$, $E \leq 1 - x + x^2$ and $|G_m| \leq 1$.

Proof. Skew-adjointness gives $\mathbb{E}B^{2j+1}\mathbf{1} = 0$ and $\mathbb{E}B^{2j}\mathbf{1} = (-1)^j\delta_j$. Hence $TS_n = S_{n+1} - \beta_{n+1}\mathbf{1}$, which proves the finite convolution by induction. The alternating sums lie between $d - \delta_2$ and d. Also $\tau \leq d - d^2 \leq 2/9$. The spectral measure of $-B^2$ at u gives ν and the norm identity. Skew-adjointness then gives the two means and the displayed constant cross term. The coordinate relations follow directly by separating the finite even and odd powers of G_m. For odd $m \geq 3$, $|G_m|^2 = (1 - xC)^2 + xC^2 \leq 1$. For even $m \geq 4$, use $C \leq 1 - x + x^2$ and

$$1 - (1 + x)(1 - x + x^2)^2 = x(1 - x - x^2 + x^3 - x^4) \geq 3x/16 \geq 0 \quad (0 \leq x \leq 1/2).$$

□

Lemma M.2 (One finite mask)**.** *For every finite polynomial $P(z) = \sum_j \epsilon_j z^j$, $\epsilon_j \in \{0, 1\}$, and every $0 \leq x \leq \theta$,*

$$|P(i\sqrt{x})|^2 \leq \frac{1 + x}{(1 - x^2)^2} \leq \frac{32768}{16245} < \frac{101}{50}. \tag{M.4}$$

Consequently, for every integer $r \geq 0$, $\|B^r P(\pm B)u\|^2 \leq (101/50)\theta^r d$. The constants are independent of the number of nonzero coefficients.

Proof. In each real or imaginary coordinate, separate the two alternating signs and take the larger same-sign subseries. Each such series is bounded by $(1 - x^2)^{-1}$. This proves the first inequality; its two positive factors increase on $[0, \theta]$. Integrate against the original spectral measure for the norm bound. The geometric series bound finite sums only; no path or message is replaced by an infinite one. □

Lemma M.3 (A joint finite path bound)**.** *For every $n \geq 3$, $m \geq 1$ and $0 \leq x \leq \theta$,*

$$|G_n(z) - zG_m(z)| \leq \tfrac{6}{5}, \qquad z = i\sqrt{x}. \tag{M.5}$$

When $m = 1$, the bound improves to one.

Proof. The difference is $1 + z^{m+1}G_{n-1-m}$ if $n-1 \geq m$, and $1 - z^n G_{m-n+1}$ otherwise. The empty tail gives one. For a nonempty tail, the finite coordinate bounds above give the following bounds for the square:

tail term	squared upper bound
$+z^2G_l$	$(1-x+x^2)^2 + x^3 \leq 1$
$-z^3G_l$	$1 + x^3$
$+z^3G_l$	$1 + 2x^2 + x^3 + x^4$
$\pm z^k G_l,\ k \geq 4$	$1 + 2x^2 + x^4(1+x)$.

The third row is at most $1493553/1048576 < 36/25$ at $x = \theta$; the last is at most $345857/248832 < 7/5 < 36/25$ even at $x = 5/12$. The second row is smaller still. For the first row, the difference from one is $x[-2+3x-x^2+x^3] \leq 0$ on $[0, 1/2]$. When $m = 1$, use this first row on $G_n - z = 1 + z^2 G_{n-2}$. These bounds include G_1 and G_2. □

M.3 One finite packet for short-one terminals

Use the original finite-space conventions of Lemma M.1, with $B_0 = -K$, $u = -g$, $g = K\mathbf{1}$, $r = K^2\mathbf{1}$, $\theta = 13/32$, $\kappa = 19/32$, and $\tau = d - \delta_2$. All statements at $d = 0$ follow by leaf integration.

Lemma M.4 (Finite short-one packet)**.** *For an integer $b \geq 2$, put*

$$\begin{gathered} F_b = qf_b, \quad R = \sum_{j=2}^{b} K^j\mathbf{1} = B_0 G_{b-1}(-B_0)u, \quad m = 1 - \beta_b/2, \quad k = m^{-1}, \\ t_0 = \beta_{b+1}, \quad e = d - \beta_{b+2}, \quad V = \mathbb{E}R^2, \quad \rho = \mathbb{E}(rR), \quad D_j = \mathbb{E}B_0^j R, \\ \gamma = kt_0 - k^2V/2, \quad a_1 = ke, \quad B_* = \gamma + a_1, \quad \varepsilon_j = \beta_j - kD_{j-1}. \end{gathered} \tag{M.6}$$

Then $F_b \leq F_0 := (m + R/2)^2$, $5/6 \leq m \leq 1$, $1 \leq k \leq 6/5$, and $D_0 = -t_0, D_1 = -e, D_2 = \rho, 0 \leq D_3 \leq \delta_3$. For $L \geq 1$, set

$$V_L = \sum_{j=0}^{L} B_0^j\mathbf{1} + kB_0^L R, \qquad P_n(z) = G_n(z) + kz^{n+1}G_{b-1}(-z).$$

The positive finite upper bound is

$$0 < X_{b,L} := T^L F_b \leq T^L F_0 \leq m^2 \left[V_L - \gamma \bar{f}_{L-1} - a_1 \bar{f}_{L-2} - \sum_{j=3}^{L} \varepsilon_j \bar{f}_{L-j} \right]. \tag{M.7}$$

Terms with negative path index are absent. The parameters satisfy

$$\begin{gathered} \tfrac{3}{10}d \leq \gamma \leq \tfrac{6}{5}d, \quad \tfrac{5}{8}d \leq B_* \leq \tfrac{3}{2}d, \quad B_* \geq d - \tfrac{29}{32}\delta_2, \quad 1 - \gamma \geq \tfrac{17}{25}, \\ \varepsilon_3 = d - k\rho \geq \tfrac{41}{80}d, \qquad \tau \leq \varepsilon_j \leq d \quad (j \geq 4), \\ \kappa\delta_2 \leq \rho \leq \delta_2, \qquad 0 \leq a_1 \leq \tfrac{39}{80}d, \qquad \mathbb{E}(KR)^2 \leq \theta V. \end{gathered} \tag{M.8}$$

For $b = 2$, $m = k = 1, R = r, e = 0, V = \rho = \delta_2$, $\gamma = d - \delta_2/2$, and $\varepsilon_j \geq \tau$ also at $j = 3$. For $b = 3$, $V = \delta_2 + \delta_3, \rho = e = \delta_2$. For $b \geq 4$, $V \leq \delta_2$. For $b \geq 3$, $\kappa\delta_2 \leq e \leq \delta_2$ and

$$m^2 \leq 1 - \tfrac{209}{384}d. \tag{M.9}$$

The full finite spectral catalogue is

$$\begin{gathered}|P_1(i\sqrt{x})|^2 \le 1-\tfrac{4}{5}x, \quad |P_2(i\sqrt{x})|^2 \le 1+x-\tfrac{63}{32}x^2 \le \tfrac{71}{63} \quad (b\ge 3),\\ |P_3(i\sqrt{x})|^2 \le 1-\tfrac{1}{8}x \quad (b\ge 3),\\ |P_n(i\sqrt{x})| \le 1 \quad (n\ge 3), \qquad |P_4(i\sqrt{x})|^2 \le 1-\tfrac{7}{10}x,\\ |P_n(i\sqrt{x})|^2 \le 1-\tfrac{7}{15}x \quad (n\ge 6,\ \textit{or } b=2,\ n\ge 4).\end{gathered} \tag{M.10}$$

Writing $P_n = A_n + zC_n$, $z = i\sqrt{x}$, its coordinates obey

n	A_n	C_n
1	$[1-6x/5, 1]$	$[0, 6x/5]$
2	$[1-x, 1]$	$[1-6x/5, 1]$
$n\ge 3$	$[1-x, 1]$	$[1-x, 1]$.

(M.11)

For $b = 2, n \ge 3$, the real-coordinate interval improves to

$$1-x \le \Re P_n(i\sqrt{x}) \le 1-x+x^2, \qquad \Re P_4 = 1-x. \tag{M.12}$$

For $L \ge 3$ the exact mean identities are

$$\begin{aligned}\alpha_L &:= 1-\mathbb{E}V_L = \varepsilon_{L+1} \in [\tau, d], \qquad \mathbb{E}(uV_L) = \alpha_{L+1},\\ R_L &:= V_L - q = B_0 P_{L-1}(B_0)u, \qquad \mathbb{E}[r(V_L-1)] = d-\alpha_{L+2}.\end{aligned} \tag{M.13}$$

Finally the finite terminal end sign is

$$\rho - e = \begin{cases} 0, & b \textit{ odd},\\ -\delta_{b/2+1}, & b\equiv 0 \pmod 4,\\ +\delta_{b/2+1}, & b\equiv 2 \pmod 4.\end{cases} \tag{M.14}$$

Proof. The outward version of (M.2), retaining its constant loss β_b, gives $q + f_b \le 2m + R$. Both sides are positive; AM–GM proves the square bound. Write $h = b-1$ and $G_h(-z) = a - zc$. At $h = 1$, $(a, c) = (1, 0)$; at $h \ge 2$, $1-x \le a, c \le 1$. In the spectral measure ν of u,

$$t_0 = \int a\,d\nu, \quad e = \int xc\,d\nu, \quad \rho = \int xa\,d\nu, \quad V = \int x(a^2+xc^2)\,d\nu.$$

More generally,

$$\begin{aligned}D_{4j} &= -\int x^{2j}a\,d\nu, \quad D_{4j+1} = -\int x^{2j+1}c\,d\nu,\\ D_{4j+2} &= \int x^{2j+1}a\,d\nu, \quad D_{4j+3} = \int x^{2j+2}c\,d\nu.\end{aligned}$$

These formulas give the signed means, ρ's bounds, the exceptional $b = 3$ values and $V \le \delta_2$ for $b \ge 4$. The two alternating sums for ρ, e differ only in their last term, proving (M.14).

As $T(R^2) \le 2V$, positivity gives $m^{-2}TF_0 \le q + kB_0R - \gamma$. Apply finite Duhamel and (M.2) to obtain (M.7). Also $V_L - 1 = P_L(B_0)u$. Skew adjunction gives $\alpha_L = \beta_{L+1} - kD_L = \varepsilon_{L+1}$; one and two further adjunctions give the other identities in (M.13).

Here are the coefficient bounds at every finite index. At $j = 3$, $\varepsilon_3 = d - k\rho \ge (1-6\theta/5)d$. At $j = 4$, $D_3 \le \delta_3$ and $k\theta < 1$ give $\tau \le \varepsilon_4 \le d$. For $j \ge 5$, $\beta_j \le d - \kappa\delta_2$; for $j \ge 7$, $\beta_j \ge \tau + \kappa\delta_3$. If

$D_{j-1} \le 0$, its magnitude is at most δ_3; if $D_{j-1} > 0$ and $j \ge 7$, it is at most $\delta_4 \le \theta\delta_3$. Since $k\theta < \kappa$, these prove all ε_j bounds. The $b = 2$ value $\varepsilon_3 = \tau$ follows directly.

For γ's lower bound use $V \le \theta(1+\theta)d$ and $t_0 \ge \kappa d$. The concave quadratic $k\kappa - k^2\theta(1+\theta)/2$ has endpoint values $631/2048, 771/2560$, both above $3/10$. For $b = 3$, $B_* \ge kd$, $B_* \le 449d/320$. For $b \ge 4$, $t_0 + e \in [d - \delta_3, d + \delta_4]$, so

$$B_*/d \ge k(1-\theta^2) - k^2\theta/2 \ge 647/1024 > 5/8, \qquad B_*/d \le k(1+\theta^3) < 3/2.$$

Moreover

$$B_* \ge k(d - \theta\delta_2) - k^2\delta_2/2 \ge d - (\theta + 1/2)\delta_2,$$

because $(\theta + 11/10)\theta < 1$. At $b = 2$, the same stated bounds follow from $\gamma = d - \delta_2/2$, $a_1 = 0$. Set $v_0 = kt_0$. Since $V \ge t_0^2$, $\gamma \le v_0 - v_0^2/2$, where $v_0 \le d/(1 - d/2) \le 2/5$; hence $1 - \gamma \ge 17/25$. Finally $m^2 \le 1 - 11\beta_b/12$ gives (M.9) when $b \ge 3$.

The norm bounds reduce to a few scalar inequalities. All coordinate upper bounds used below are nonnegative. For $b \ge 3$, the loss of $|P_1|^2$ is at least

$$kx\{2(1-x) - kx[(1-x)^2 + x]\}.$$

After division by x, this is concave in k. Its endpoint values decrease in x; at $x = \theta$ they are $28811/32768, 200931/204800$, both above $4/5$. For $b \ge 3$, the two coordinates of P_2 are $1 - kx^2c, 1 - kxa$. They decrease in k, giving

$$|P_2|^2 \le (1 - x^2 + x^3)^2 + x(1 - x + x^2)^2 \le 1 + x - \tfrac{63}{32}x^2.$$

The second difference divided by x^2 is $65/32 - 5x + x^2 + x^3 - x^4 \ge 0$. For P_3, bound its coordinates by $1 - x + kx^2, 1 - kx^2(1-x)$. Their squared sum is convex in k. At $k = 1, 6/5$, its loss divided by x is

$$Q_k = 1 - (1+2k)x + 4kx^2 - (2k + k^2)x^3 - k^2x^4 + 2k^2x^5 - k^2x^6.$$

Put $C_k = 3(2k + k^2) - 10k^2\theta^2$. Then $Q_k' \le -(1+2k) + 8kx - C_kx^2 < 0$, since $4(1+2k)C_k > 64k^2$ at both endpoints. Their values at $x = \theta$ are $247671943/1073741824$ and $937169087/6710886400$, both above $1/8$. Convexity in k proves the claimed uniform P_3 loss. The case $b = 2, n = 3$ is $|P_3|^2 = 1 - x + 3x^2 - 2x^3 + x^4 \le 1$; the loss divided by x decreases to $1451/32768 > 0$. For P_4,

$$|P_4|^2 \le (1 - x + 6x^3/5)^2 + x(1 - x + 6x^2/5)^2 \le 1 - \tfrac{7}{10}x.$$

Indeed its loss divided by x is at least $1 + x[\kappa - (24/5)\theta\kappa - (36/25)\theta^3(1+\theta)]$; the smaller endpoint value exceeds $7/10$.

For all remaining lengths use the exact finite identity

$$P_n = \frac{1 + z^n S}{1 - z}, \qquad S = -1 + kz(1-z)G_h(-z).$$

For $h \ge 2$, $|S| \le 1$ follows from

$$k(1+x)(a^2 + xc^2) \le \tfrac{6}{5}(1+\theta)^2 = 1215/512 < 1216/512 = 4\kappa \le 2(a+c).$$

At $h = 1$, it follows from $1 - x + x^2 \le 1$. Thus $n = 5$ is contractive because $2x^{3/2} + x^4 \le 4\theta/3 + \theta^4 < 1$; for $n \ge 6$,

$$|P_n|^2 \le \frac{(1+x^3)^2}{1+x} \le 1 - \tfrac{7}{15}x.$$

The last difference, after multiplication by $15(1+x)/x$, is $8-7x-30x^2-15x^5$, which decreases to $1311885/33554432>0$. The same bound holds for G_n, $n\ge 5$, by its finite geometric formula. At $b=2, n=5$, P_5 is obtained from G_5 by reducing its positive real coordinate, so the stronger bound also holds there.

For the coordinate bounds at $n=3,4,5$, the exact pairs are

$$\begin{array}{c|cc} 3 & 1-x+kx^2a & 1-kx^2c \\ 4 & 1-x+kx^3c & 1-x+kx^2a \\ 5 & 1-x+x^2-kx^3a & 1-x+kx^3c. \end{array}$$

They lie in $[1-x,1]^2$, using $kx<1$; this also holds at $(a,c)=(1,0)$. The rows $n=1,2$ follow from their explicit coordinates above. For $n\ge 6$, the geometric identity places P_n within distance $x^{n/2}/\sqrt{1+x}$ of $(1+z)/(1+x)$. The error in either real coefficient is at most $x^{(n-1)/2}/\sqrt{1+x}\le x^2/(1+x)$, since $\theta(1+\theta)<1$. This proves (M.11), including $x=0$ by continuity. At $b=2$, separate the finite even powers of G_n+z^{n+1}. Their real part is always a truncation of $1-x+x^2-\cdots$ ending at or after the x-term, proving (M.12). □

Lemma M.5 (A two-row bound and an exponential comparison)**.** *For every real vector H,*

$$|K^2H(x)|^2\le \|H\|_2^2/3. \tag{M.15}$$

If a finite real variable Z satisfies $|Z|\le U$, $\mathbb{E}Z^2\le hU^2$, $0\le h<1$, then

$$\mathbb{E}e^{sZ}\le \exp\{s\max(\mathbb{E}Z,0)+s^2(1+h)^2U^2/8\}\qquad (s\ge 0). \tag{M.16}$$

Consequently, if $C,B>0$, $M\ge\max(\mathbb{E}Z,0)$, and the actual nonnegative message has an upper bound

$$0\le Y\le CB[1+(Z-a)/(2B)]^2 \tag{M.17}$$

whose square base is nonnegative, then

$$\begin{aligned}\log\mathbb{E}Y^8\le\mathcal{E}:=8\log C+8\log B-\frac{8a}{B}+\frac{8M}{B}\\ +\begin{cases}6D/B^2, & Z=K^2H,\ \|H\|_2^2\le D,\\ 8c_*D/B^2, & Z=KH,\ \|H\|_2^2\le D,\end{cases}\qquad c_*=(1+\theta)^2.\end{aligned} \tag{M.18}$$

Proof. For any legal skew kernel L, integrating

$$1+ab+bc+ca=\{(1+a)(1+b)(1+c)+(1-a)(1-b)(1-c)\}/2\ge 0$$

at $a=L(x,y), b=L(y,z), c=L(z,x)$ gives $\|L\mathbf{1}\|_2^2\le 1/3$. Conjugating K by any coordinate sign vector, then taking convex combinations, gives $\|Kv\|_2^2\le 1/3$ for $|v|\le 1$. Apply this to $v(y)=K(x,y)$ and use skew adjunction and Cauchy–Schwarz to prove (M.15).

For the exponential inequality let $\mu=\mathbb{E}Z\ge 0$, $v=\operatorname{Var}Z$. Unless $U=0$, $U-\mu>0$. The quadratic tangent to e^{sz} at $\mu-v/(U-\mu)$ and passing through U majorizes it on $[-U,U]$: its Hermite remainder has sign $(z-\mu+v/(U-\mu))^2(z-U)$. The lower contact lies in $[-U,U]$ because $v\le U^2-\mu^2$. The two-point law with these contacts and the same first two moments has range

$$U-\mu+\frac{v}{U-\mu}\le U\left(2-\frac{1-h}{1-\mu/U}\right)\le(1+h)U.$$

The tilted log-moment has second derivative at most one quarter of the squared range. Integrating twice proves the claim. If $\mathbb{E}Z < 0$, take instead the contacts $-hU, U$. The interpolating quadratic has nonnegative quadratic coefficient C_2 and linear coefficient $se^{-shU} + 2hUC_2 \geq 0$, hence permits replacing the mean by zero and the raw second moment by hU^2. The resulting mean-zero law has range $(1+h)U$. Degenerate cases follow by continuity.

For $Z = KH$ take $U^2 = D, h = \theta$; for $Z = K^2H$ take $U^2 = D/3, h = 1/2$, using $\theta^2 < 1/6$. Apply (M.16) at $s = 8/B$. Since the base in (M.17) is nonnegative, its sixteenth power is at most $e^{8(Z-a)/B}$, including at a zero base. This proves the complete expression (M.18). □

M.4 The receiving weight and its two mean endpoints

Work on the original finite probability space, with zero-mass atoms deleted, and let K be the real antisymmetric kernel with $|K| \leq 1$ used in Lemma D.1. Write

$$B = -K, \quad u = B\mathbf{1}, \quad A = J - B, \quad T = J + B,$$
$$p = 1 - u, \quad q = 1 + u, \quad f_n = A^n\mathbf{1}, \quad d = \mathbb{E}u^2, \quad \delta_j = \|B^j\mathbf{1}\|_2^2.$$

Put $G_0 = 0$, $G_n(z) = \sum_{j=0}^{n-1} z^j$, and

$$\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j \quad (n \geq 1), \qquad \theta = \tfrac{13}{32}, \quad \kappa = \tfrac{19}{32}, \quad \tau = d - \delta_2.$$

In particular $\beta_1 = \beta_2 = 0$. The finite path and spectral calculus of Lemma M.1 and (M.1) gives $0 \leq d \leq 1/3$, $\kappa d \leq \tau \leq \beta_n \leq d$ for $n \geq 3$, and a positive measure ν on $[0, \theta]$ of mass d such that $\int x^j \, d\nu = \delta_{j+1}$ for every integer $j \geq 0$. Thus the measure in each integral below is the actual spectral measure of $-B^2$ at u, not a separately chosen moment distribution. The finite convolution (M.2) also holds with B replaced by $-B$, since this leaves every δ_j and β_n unchanged. All expectations and inner products use the original measure, with spectral means and norms as in (M.3). When $d = 0$, leaf integration gives all the tree densities below directly; divisions by d in individual parameter tables concern $d > 0$.

Lemma M.6 (A finite receiving arm)**.** *For an integer $a \geq 2$, let*

$$S_a = 1 - G_a(-B)u, \qquad C_{a,t} = S_a - \beta_a - tp, \qquad 0 \leq t \leq \beta_{a-1}.$$

The case $a = 2$ has $t = \beta_1 = 0$, so $C_{2,0} = f_2$. In every case $C_{a,t} \geq f_a \geq 0$, and

$$\mathbb{E}C_{a,t} = 1 - \beta_a - \beta_{a+1} - t. \tag{M.19}$$

For a real finite polynomial Q, put $Z = c + Q(B)u$ and $\alpha = -\mathbb{E}[Q(B)u]$. Then

$$\operatorname{Var} Z = \int |Q(i\sqrt{x})|^2 \, d\nu - \alpha^2, \tag{M.20}$$

$$\operatorname{Cov}(C_{a,t}, Z) = -\int \operatorname{Re}(G_a(i\sqrt{x})Q(i\sqrt{x})) \, d\nu - \beta_{a+1}\alpha + t\int \operatorname{Re} Q(i\sqrt{x}) \, d\nu. \tag{M.21}$$

Proof. The reversed finite convolution (M.2) gives

$$S_a = f_a + \sum_{j=3}^{a} \beta_j f_{a-j}.$$

When $a \ge 4$, remove its terms $\beta_a f_0$ and $t f_1 = tp$; the latter is at most the available term $\beta_{a-1} f_1$. For $a = 2, 3$, the displayed assertions follow from $\beta_1 = \beta_2 = 0$. This proves nonnegativity without introducing any arm. The finite spectral mean is $\mathbb{E}[G_a(-B)u] = \beta_{a+1}$, proving (M.19). Skew-adjointness gives

$$\langle G_a(-B)u, Q(B)u\rangle = \int \mathrm{Re}(G_a(i\sqrt{x})Q(i\sqrt{x}))\, d\nu, \qquad \langle u, Q(B)u\rangle = \int \mathrm{Re}\, Q\, d\nu.$$

The norm identity gives (M.20). Subtracting the two actual means gives (M.21); its signed term $-\beta_{a+1}\alpha$ is retained. Constants in Z do not change its covariance or variance. □

Lemma M.7 (A reference fourth-moment payment)**.** *Suppose* $0 \le X \le sZ$, $s > 0$, $Z \ge 0$, $C \ge f_a \ge 0$, *and* $\mu = \mathbb{E}Z \in [0, M]$. *Assume*

$$\mathbb{E}C \le h, \quad h \ge 0, \qquad \mathrm{Cov}(C, Z) \le b, \qquad \mathrm{Var}\, Z \le V,$$

where b *can have either sign. If* $C[(Z+\mu)^2 + 2\mu^2] \le K_0$, *then*

$$\mathbb{E}[f_a X^4] \le s^4[K_0 V + \max\{0, hM^4 + 4bM^3\}]. \tag{M.22}$$

If instead $s = e^{-\Gamma}$ *and the pointwise coefficient* K_0 *has been proved on* sZ, *then the corresponding bound is*

$$\mathbb{E}[f_a X^4] \le e^{-4\Gamma} \max\{0, hM^4 + 4bM^3\} + K_0 e^{-2\Gamma} V. \tag{M.23}$$

In the scaled assertion, the pointwise hypothesis is $C[(sZ + s\mu)^2 + 2s^2\mu^2] \le K_0$.

Proof. Taylor's exact fourth identity at the actual mean gives

$$\mathbb{E}[CZ^4] = (\mathbb{E}C)\mu^4 + 4\mu^3\, \mathrm{Cov}(C, Z) + \mathbb{E}[C(Z-\mu)^2\{(Z+\mu)^2 + 2\mu^2\}].$$

The derivative of $hy^4 + 4by^3$ is $4y^2(hy + 3b)$; on $[0, M]$, its maximum is at an endpoint, including when $b < 0$ or $h = 0$. This proves (M.22). Applying the same identity to sZ gives (M.23). In particular a pointwise coefficient proved on the scaled message multiplies the variance with its second-power scale. □

The actual f_2 row has mass $1 - d$. The weak rows $a \ge 3, t = 0$ have mass $1 - \beta_a - \beta_{a+1}$, while the strengthened rows retain the additional term $-t$. These weights share the displayed algebra but retain different finite covariance and pointwise hypotheses in their row proofs.

Lemma M.8 (Changing the receiving mass)**.** *Suppose* $1 \le \eta \le 2$, $0 \le \mu \le M$, *and the scalar comparison bracket is* $(1 - \eta d)\mu^4 + 4b_0\mu^3$. *For a receiving factor of mass* $1 - d$, *this same bracket remains an upper if its covariance with the unscaled positive upper is at most* $b_0 - (\eta - 1)Md/4$.

Proof. The new bracket minus the comparison bracket is at most $(\eta - 1)d\mu^3(\mu - M) \le 0$. No endpoint substitution for μ or for a spectral moment was used in this comparison. □

M.5 One retained vector for the two short-arm sums

Use the original finite kernel, with $B = -K$, $u = B\mathbf{1}$, and $\mathsf{X} = -B^2$, and write $g = K\mathbf{1} = -u$. The path coefficients β_j are those in (M.2); put $\tau = d - \delta_2$, $\theta = 13/32$, and $\kappa = 1 - \theta$.

Lemma M.9 (A retained pair of arm sums)**.** *Let $n \geq 5$. Take either $\sigma = 2$, $0 \leq d \leq 1/3$, or $\sigma = 9/4$, $0 \leq d \leq 1/5$, and set $m = 1 - \sigma d/2$. For $n = 5$ put*

$$a_1 = d, \qquad a_2 = (\sigma - 2)d + \delta_2, \qquad a_3 = 0.$$

For $n \geq 6$ put

$$\lambda = \frac{\sigma d - \beta_n - \beta_{n-1}}{\beta_{n-2} + (1-d)\beta_{n-3}}, \qquad a_1 = \beta_{n-1}, \quad a_2 = \lambda\beta_{n-2}, \quad a_3 = \lambda\beta_{n-3}.$$

At $d = 0$ all a_i are zero. These are genuine retained coefficients: $0 \leq a_i \leq \beta_{n-i}$. Define

$$C_0 = a_1 + a_2 + a_3, \quad D_0 = a_2 + a_3, \quad E_0 = a_3, \qquad U = \sum_{j=3}^{n} K^j\mathbf{1} - C_0 g - D_0 K g - E_0 K^2 g.$$

The same actual vector gives both comparisons

$$h_2 + f_n \leq 2m + U, \qquad f_2 + h_n \leq 2m - U. \tag{M.24}$$

Each right side is nonnegative. In particular $h_2 f_n \leq (m + U/2)^2$ and $f_2 h_n \leq (m - U/2)^2$.

Write $U = [a(\mathsf{X}) - b(\mathsf{X})B]u$. At $n = 5$ its exact coordinates are

$$a = (\sigma - 1)d + \delta_2 + x - x^2, \qquad b = (\sigma - 2)d + \delta_2 + x.$$

For $n \geq 6$, writing $G_{n-2}(i\sqrt{x}) = a_N + i\sqrt{x}\, b_N$, they are

$$a = C_0 + x(a_N - E_0), \qquad b = D_0 + x b_N, \qquad a - (1-x)b \geq a_1 + a_2 x \geq \tau. \tag{M.25}$$

All statements retain the global d, δ_2, E_0 while x varies.

Proof. For $d > 0$ the denominator defining λ is at least $(2-d)\tau > 0$. Its numerator is nonnegative. The full available constant loss is at least $(4-d)\tau$: it exceeds $2d$ on $[0, 1/3]$, since $209/96 > 2$, and exceeds $9d/4$ on $[0, 1/5]$, since $361/160 > 9/4$. Thus $0 \leq \lambda \leq 1$. For $n = 5$, $a_2 \leq (21/32)d \leq \beta_3$ even at $\sigma = 9/4$; $a_3 = 0$ takes nothing from $\beta_2 = 0$. In both cases

$$\beta_n + a_1 + a_2 + (1-d)a_3 = \sigma d, \quad C_0 = \sigma d - \beta_n + dE_0, \quad D_0 = \sigma d - \beta_n - \beta_{n-1} + dE_0.$$

Retain $\beta_n\mathbf{1}, a_1 f_1, a_2 f_2, a_3 f_3$ in the positive convolution for f_n. Since $h_2 + \sum_{j=0}^{n} K^j\mathbf{1} = 2 + \sum_{j=3}^{n} K^j\mathbf{1}$, this proves the first sum. Applying T to the convolution for f_{n-1} gives exactly

$$f_2 + h_n = 2 - \beta_n - \sum_{j=3}^{n} K^j\mathbf{1} - \sum_{j=1}^{n-3} \beta_{n-j} h_j.$$

Retaining $a_1 h_1, a_2 h_2, a_3 h_3$ proves the second sum, with the opposite sign of the same U. The finite geometric identity gives the displayed coordinates. For $N \geq 4$, alternating sums give $1 - x \leq a_N, b_N \leq 1 - x + x^2$ and $a_N \geq (1-x)b_N$. The last inequality in (M.25) follows after inserting $C_0 - D_0 = a_1$ and $D_0 - E_0 = a_2$; the $n = 5$ formula verifies the same assertion directly. □

Lemma M.10 (The two retained first-row packets)**.** *Take the actual vector $U = [a(\mathsf{X}) - b(\mathsf{X})B]u$ of Lemma M.9, with $n = c \geq 5$. The following two rows hold for $0 < d \leq d_{\max}$:*

σ	$d_{\max}$	ℓ	*a upper*	*b upper*	D
2	$1/4$	2	$7/10$	$3/5$	$d/2$
$9/4$	$1/5$	$15/7$	$16/25$	$9/16$	$2d/5$.

In each row, with $m = 1 - \sigma d/2$, $k = m^{-1}$ and $R = -kU$, one has

$$\ell(a^2 + xb^2) \leq a + b, \qquad T(qf_2h_c) \leq m^2(\bar{f}_2 + BR + D\mathbf{1}). \tag{M.26}$$

The same coefficients satisfy $a, b \geq x(1-x)$ and $ka \geq k\tau + (1-x)kb$. Let $D_i = \mathbb{E}B^iR$ and $\gamma_j = \beta_j - D_{j-2}$. Their exact signs and the bounds needed at every later finite index are

$$\begin{aligned}
D_{4j+1} &= k\int x^{2j}a\,d\nu, & D_{4j+2} &= k\int x^{2j+1}b\,d\nu,\\
D_{4j+3} &= -k\int x^{2j+1}a\,d\nu, & D_{4j+4} &= -k\int x^{2j+2}b\,d\nu,\\
D_1 &< d, & D_2 &< d/3,\\
\gamma_3 &\geq 0, & \gamma_4 &\geq 2d/3,\\
\gamma_5 &\geq \tau, & \gamma_j &\geq 41d/96 \quad (j \geq 5).
\end{aligned} \tag{M.27}$$

Every positive D_i with $i \geq 3$ is less than $d/6$. In the $9/4$ row, also $D_1 + D_2 < 5d/4$. The $2d$ row retains its own original vector; it is available through $d = 1/4$ independently of the $9d/4$ row. In the $9d/4$ row, the displayed coordinate cap further gives $D_1 < 128d/155$, hence $\gamma_3 > 27d/155 > d/8$. At $d = 0$, $U = R = 0$ and all non-strict bounds extend directly.

Proof. Write $E = a_3$, $C = a_1 + a_2 + a_3$ and $v(x) = x - x^2$, $w(x) = x - x^2 + x^3$. The actual retained coefficients give

$$(\sigma-1)d + dE \leq C \leq (\sigma-\kappa)d + dE, \quad 0 \leq E \leq d, \quad a \leq C + w(x) - xE, \quad b \geq C - d + v(x) \geq 0.$$

The last two comparisons also hold at $c = 5$, using its explicit coordinates. For $c = 5$, its explicit coordinates are increasing in x and give

$$a \leq \theta - \theta^2 + (\sigma - 1 + \theta)d, \qquad b \leq \theta + (\sigma - 2 + \theta)d.$$

At the respective $d_{\max}$ these bounds are $(607/1024, 65/128)$ for $\sigma = 2$, and $(2931/5120, 43/80)$ for $\sigma = 9/4$; they are strictly below the four coordinate caps in the table. For $c \geq 6$, first increase x to θ, keeping the original d, E fixed. This is legitimate since $w'(x) - E \geq 2/3 - E > 0$. Because $d < \theta$,

$$a \leq w(\theta) + (\sigma - \kappa)d, \qquad b \leq w(\theta) + (\sigma - 2\kappa)d + d^2.$$

At $d = d_{\max}$ the respective bounding pairs are

$$\begin{aligned}
\sigma = 2: &\quad (21621/32768, 18805/32768),\\
\sigma = 9/4: &\quad (104777/163840, 459373/819200).
\end{aligned}$$

They imply the displayed coordinate bounds and $0 \leq ka, kb < 1$. Also $b \geq x(1-x)$, while the lower bound for a exceeds $x(1-x) + d(\sigma - 1) + E(d - x) \geq x(1-x)$. The relation between ka and kb follows from (M.25).

Put $\Phi(a,b)=\ell(a^2+xb^2)-a-b$. The displayed bounds imply $2\ell xb<1$, so Φ decreases in b and is convex in a. The endpoint $a=0$ is nonpositive. At the other endpoint put $A=C+w(x)-xE$, and decrease b to $B_*=C-d+v(x)$. For fixed original C,d,E, the exact second derivative is

$$2\ell\big[(1-E-2x+3x^2)^2+(6x-2)(d-Ex+x^3)+x(1-2x)^2\big]+4-6x.$$

For $x\le 1/3$, the square is at least $25/144$, the d term at least $-1/2$, and the $-Ex$ term is nonnegative. The remaining terms combine as $x(1-4x+2x^2+6x^3)\ge 0$; its parenthesis decreases to $1/9$. Thus this derivative is at least $101/168>0$ for both rows. For $x\ge 1/3$ the bracket is nonnegative. It suffices to check $x=0,\theta$.

At zero, $\Phi=\ell C^2-2C+d$ decreases in C, since $C\le 53/128$ and $(15/7)(53/128)<1$. Substituting $C\ge(\sigma-1)d$ proves the required sign. At $x=\theta$, convexity reduces C to $\alpha d+dE$, where $\alpha=\sigma-1,\sigma-\kappa$, and then E to $0,d$. The $E=0$ expressions are convex in d. For $E=d$ the coordinates are

$$A=w(\theta)+(\alpha-\theta)d+d^2,\qquad B_*=v(\theta)+(\alpha-1)d+d^2.$$

Their second d derivative is $2\ell[(A')^2+2A+\theta((B_*')^2+2B_*)]-4>0$, because $\kappa^2+2w(\theta)+2\theta v(\theta)>1$ and $\alpha\ge 1$. Only $d=0,d_{\max}$ remain. At zero use $w(\theta)<5/16$, $v(\theta)<1/4$ and $w(\theta)+v(\theta)>1/2$. For $d=d_{\max}$, in the order $(\alpha,E)=(\sigma-1,0),(\sigma-\kappa,0),(\sigma-1,d),(\sigma-\kappa,d)$, exact substitution gives

$$\begin{aligned}-536870912\,\Phi\big|_{\sigma=2}&\in\{69198055,19544295,112116455,65444583\},\\ -187904819200\,\Phi\big|_{\sigma=9/4}&\in\{20259598945,2787865185,33811712609,17975757409\}.\end{aligned}$$

This proves the coefficient inequality on the full continuous domains.

Let $V=\mathbb{E}U^2=\int(a^2+xb^2)\,d\nu$. Expanding the original positive square gives

$$\frac{T(qf_2h_c)}{m^2}\le \bar f_2+BR+\delta\mathbf{1},\qquad \delta=-k\int(a+b)\,d\nu+k\sqrt{dV}+k^2V.$$

Here the linear row error is bounded by $\mathbb{E}|uR|$, and the quadratic term by $\mathbb{E}R^2$, using the original kernel row bound and $q\le 2$. For $\sigma=2$, use $\sqrt{dV}\le 3d/8+(2/3)V$ and $k+2/3\le 2$ to get $\delta\le 3kd/8\le d/2$. For $\sigma=9/4$, use $\sqrt{dV}\le 3d/10+(5/6)V$ and $k+5/6\le 15/7$ to get $\delta\le 3kd/10\le 12d/31<2d/5$.

Skew adjunction gives the four exact sign classes in (M.27). The coordinate caps in the table imply $D_1<d$, $D_2<d/3$ and, in the second row, $D_1+D_2<5d/4$. The later positive classes cost at most $k\theta^2(\sup a)d$ and $k\theta^3(\sup b)d$, both less than $d/6$. Combining them with $\beta_j\ge\tau\ge\kappa d$ proves all the γ bounds. At $d=0$ the retained vectors vanish. □

Lemma M.11 (Positive continuation of a retained square)**.** *For either row above, let $F=qf_2h_c$ and use its same actual R,D. For every finite $L\ge 1$,*

$$\begin{aligned}&T^LF\le m^2Y_L,\\ &Y_L=\bar f_{L+1}+B^LR+D\bar f_{L-1}+\sum_{i=1}^{L-1}D_i\bar f_{L-1-i}\\ &\quad=\mathbf{1}+G_{L+1}(B)u+B^LR+D\bar f_{L-1}-\sum_{j=3}^{L+1}\gamma_j\bar f_{L+1-j}.\end{aligned}\tag{M.28}$$

Every loss in the last sum is nonnegative.

Proof. Apply T^{L-1} to the original first row and use $T = J + B$. The identity $T^{L-1}(BR) = B^L R + \sum_{i=1}^{L-1} (\mathbb{E}B^i R)\bar{f}_{L-1-i}$ follows by finite induction. Substituting the positive path convolution for $\bar{f}_{L+1}$ gives the second expression. □

Lemma M.12 (Strong norms for the $9d/4$ packet)**.** *For the actual $\sigma = 9/4$ coefficients of Lemma M.9, for every finite $c \geq 5$ and $0 \leq d \leq 1/5$, one has*

$$\sqrt{a^2 + xb^2} \leq u_0 = 3/8 + 9d/5 \leq 147/200, \qquad b \leq z_0 = \theta + 13d/16 \leq 91/160. \tag{M.29}$$

Proof. For $c = 5$ the coordinates increase in x. First set $x = \theta$ with the original d, δ_2 fixed, then use $\delta_2 \leq \theta d$. The resulting vector is bounded coordinatewise by

$$(\theta - \theta^2, \sqrt{\theta}\,\theta) + d(53/32, \sqrt{\theta}\,21/32).$$

Its constant and linear squared norms are $131313/1048576 < 9/64$ and $95621/32768 < 81/25$. The triangle inequality proves the norm bound; $b \leq \theta + 21d/32 \leq z_0$ gives the second assertion.

For $c \geq 6$ put $w(x) = x - x^2 + x^3$. The actual global moment gives

$$a \leq w(x) + 5d/4 + \delta_2 + E(d - x), \qquad b \leq w(x) + d/4 + 2\delta_2 + dE.$$

Both upper functions increase in x: their derivatives are at least $2/3 - E \geq 7/15$ and $2/3$. Set $x = \theta$, then $\delta_2 \leq \theta d$. With $w_\theta = 10101/32768 < 5/16$, the bounding coordinates are

$$a_* = w_\theta + (53/32)d + E(d - \theta), \qquad b_* = w_\theta + (17/16)d + dE.$$

Their squared norm is convex in $E \in [0, d]$. At $E = 0$ the constant norm is below $3/8$ and the linear squared norm is $26229/8192 < 81/25$. At $E = d$ the linear squared norm is $16557/8192 < 100/49$ and the quadratic vector has norm $\sqrt{45/32} < 6/5$. Thus

$$\sqrt{a_*^2 + \theta b_*^2} \leq 3/8 + (10/7)d + (6/5)d^2 \leq 3/8 + (10/7 + 6/25)d < 3/8 + 9d/5.$$

Finally $b \leq 5/16 + (17/16)d + d^2$, whose excess above z_0 is $-3/32 + d/4 + d^2 \leq -3/800$. All frequency maximizations kept the original d, δ_2, E fixed. □

Lemma M.13 (The two short retained polynomials)**.** *For the same $\sigma = 9/4$ packet, put $t = ka$, $s = kb$ and*

$$Q_2 = 1 - sx + z(1 - t), \qquad Q_3 = 1 - x + tx + z(1 - sx).$$

For every actual finite $c \geq 5$ and every $x \in [0, \theta]$,

$$|Q_2|^2 \leq 1 + 8/63 < 8/7, \tag{M.30}$$

$$|Q_3|^2 \leq 8/7. \tag{M.31}$$

Proof. Both normalized imaginary coordinates are nonnegative since $0 \leq t, s \leq 1$. Using $a, b \geq x(1 - x)$ and $k \geq 1$ gives

$$|Q_2|^2 \leq 1 + x - 4x^2 + 5x^3 - x^4 - x^5 + x^6 \leq 1 + x - (63/32)x^2 \leq 1 + 8/63.$$

Indeed $4 - 5x + x^2 + x^3 - x^4$ decreases to $2279183/1048576 > 63/32$.

For Q_3, first suppose $x \le d$. Then $a \le (53/32)d + dE + x(1-E) \le 85d/32$, hence $ka \le 85/124 < 3/4$. Thus $|Q_3|^2 \le (1-x/4)^2 + x \le 441/400 < 8/7$. For $x \ge d$, increase a to $C + w(x) - xE$ and decrease b to $C - d + v(x)$, with $v = x - x^2$, $w = x - x^2 + x^3$. This increases both nonnegative coordinates. Their squared norm increases in C for $x > 0$, because its derivative is $2kx[\operatorname{Re} Q_3 - x \operatorname{Im} Q_3/\sqrt{x}] > 0$; the parenthesis is at least $1 - 2x > 0$. The case $x = 0$ is direct. Take $C = 53d/32 + dE$. Increasing E now decreases both coordinates, so take $E = 0$. At fixed x the coordinates are affine in $d/m \in [0, 8/31]$, and the squared norm is convex. It suffices to use $d = 0, 1/5$, allowing the whole rectangle for this last bound. At zero it is at most $(1 - x + x^2)^2 + x \le 1$. At $1/5$, the coordinates are

$$1 - (71/124)x + (40/31)x^2 - (40/31)x^3 + (40/31)x^4, \quad 1 - (21/124)x - (40/31)x^2 + (40/31)x^3.$$

In their squared norm $P(x) = 1 + \sum_{j=1}^{8} a_j x^j$,

$$\begin{aligned}(a_1, \dots, a_5) &= (-9/62, 39513/15376, -101639/15376,\\ &\qquad 134400/15376, -55040/15376),\\ (a_6, a_7, a_8) &= (k^2, -k^2, k^2), \qquad k = 40/31.\end{aligned}$$

Thus $P'' \ge 5 - 40x + 104x^2 - 72x^3 > 0$; the remaining term is $k^2x^4(30 - 42x + 56x^2) > 0$. For $y = x - 1/4 \le 5/32$, the cubic lower bound is $3/8 - (3/2)y + 50y^2 - 72y^3 \ge 3/8 - (3/2)y + (155/4)y^2 > 0$. At $x = 0$ the norm is one; at θ the two coordinates are less than $93/100, 81/100$, giving $P(\theta) < 362061/320000 < 8/7$. Convexity proves the bound on the entire frequency interval. □

M.6 Five continuous payments for a retained reference square

Lemma M.14. *For the functions $\mathcal{U}_i$ below, one has $\mathcal{U}_i(d) \le 1 - \varepsilon_i d$ throughout $0 \le d \le D_i$.*

$$\begin{aligned}\mathcal{U}_1(d) &= e^{-37d/4} + \frac{39}{4} d e^{-4d - 2d^2} - \frac{3}{4} d e^{-851d/96} - 9d^2(\kappa - 2d)^2 e^{-7d/3},\\ \mathcal{U}_2(d) &= e^{-159d/16} + \frac{261}{26} d e^{-9d/2 - 81d^2/32} - \frac{3}{4} d e^{-10d},\\ \mathcal{U}_3(d) &= e^{-203d/16} + \frac{72}{7} d e^{-15d/4 - 81d^2/32},\\ \mathcal{U}_4(d) &= e^{-209d/16} + 9d e^{-25d/8 - 2d^2} + (d/4 + 4d^2) e^{-349d/32},\\ \mathcal{U}_5(d) &= e^{-25d/2} + \frac{91}{10} d e^{-7d/2}.\end{aligned}$$

Here $\kappa = 19/32$, and the parameters are

i	D_i	ε_i	$\deg P_i$	$\min_j b_{i,j}$ *lower bound*
1	$1/4$	$1/800$	32	$1/100$
2	$1/5$	$1/16$	32	$1/5$
3	$1/5$	$1/160$	32	$1/10$
4	$1/4$	$1/600$	32	$1/20$
5	$1/4$	$1/60$	16	$1/100$.

The polynomials P_i and their Bernstein coefficients $b_{i,j}$ are defined in the proof, on one interval for each row.

Proof. Put

$$E(z)=\sum_{j=0}^{16}\frac{(-z)^j}{j!},\qquad O(z)=\sum_{j=0}^{15}\frac{(-z)^j}{j!}.$$

The signed Taylor remainder gives $O(z)\le e^{-z}\le E(z)$ for every $z\ge 0$. In the displayed $\mathcal{U}_i$, replace an exponential with a positive prefactor by E, and one with a negative prefactor by O; call the resulting polynomial $\widehat{\mathcal{U}}_i$. All prefactors have the indicated sign on their stated intervals. The polynomial

$$P_i(d)=\frac{1-\varepsilon_i d-\widehat{\mathcal{U}}_i(d)}{d}$$

has a removable zero in the numerator at $d=0$. If $P_i(d)=\sum_{k=0}^{n_i}a_{i,k}d^k$, its coefficients in the Bernstein basis on $[0,D_i]$ are exactly

$$b_{i,j}=\sum_{k=0}^{j}a_{i,k}D_i^k\frac{\binom{j}{k}}{\binom{n_i}{k}},\qquad 0\le j\le n_i.$$

Substitution of the five displayed formulas gives the 149 positive coefficients with the rational lower bounds in the table. Applying Lemma A.1 gives $P_i\ge 0$ on the entire interval. Therefore $\mathcal{U}_i\le\widehat{\mathcal{U}}_i\le 1-\varepsilon_i d$, including the endpoint $d=0$. □

M.7 Propagation of moment bounds

M.8 Finite packets at a unit root

Work on the original finite probability space. Put

$$A=J+K,\quad T=J-K=J+B,\quad u=B\mathbf{1}=-g,\quad p=1-u,\quad q=1+u,\quad d=\|g\|^2,$$

where $K^*=-K$, $|K(x,y)|\le 1$, and $\|K\|^2\le 13/32$. Zero-mass atoms are omitted. Thus $p,q>0$ on the remaining support. The notation $\bar f_0=\mathbf{1}$ includes the constant vector. Write $G_n(z)=\sum_{j=0}^{n-1}z^j$ and use the positive-convolution coefficients β_j of (M.2). In particular $\beta_j\ge d-\delta_2\ge\kappa d$ for $j\ge 3$, where $\delta_2=\|K^2\mathbf{1}\|^2$ and $\kappa=19/32$. Paragraph M.1 supplies the signed-mean bound

$$\mathbb{E}e^{sBW}\le e^{s\mathbb{E}BW+(17/64)s^2\|W\|^2}\qquad (W\text{ real},\ s\ge 0). \tag{M.32}$$

Lemma M.15 (Finite continuation)**.** *For a real vector Z and an integer $m\ge 0$,*

$$T^mZ=B^mZ+\sum_{i=0}^{m-1}(\mathbb{E}B^iZ)\bar f_{m-1-i}. \tag{M.33}$$

Proof. The empty sum gives $m=0$. Apply T, use $TB^mZ=B^{m+1}Z+(\mathbb{E}B^mZ)\mathbf{1}$ and $T\bar f_j=\bar f_{j+1}$, and induct on m. □

Lemma M.16 (A finite exponential envelope)**.** *If $|S|\le b<3$ and $c\ge(1-b/3)^{-1}$, then $e^S\le 1+S+cS^2/2$.*

Proof. For $S\ge 0$, use $n!\ge 2\cdot 3^{n-2}$ for $n\ge 2$ in the exponential series. For $S\le 0$, its integral remainder is at most $S^2/2$. Both give the displayed bound. □

Lemma M.17 (The unit-arm logarithm)**.** *For $-1 \le u < 1$,*

$$\log(1-u) \le -u - 3u^2/10.$$

Proof. The derivative of $\log(1-u) + u + 3u^2/10$ is $-u(2+3u)/[5(1-u)]$. Its maxima on $[-1,1)$ occur at -1 and 0. Their values are negative and zero, respectively, since the first four terms of $e^{7/10}$ sum to $12013/6000 > 2$. □

Theorem M.18 (Unit-root finite-packet transport)**.** *The positive first row and its continuation. Suppose $0 \le w \le W$, $Tw = \bar{f}_r$, $r \ge 1$, and*

$$0 < F \le w(s_0 + s_1 Z + s_2 Z^2), \qquad s_0 > 0, \quad s_2 \ge 0.$$

For any real λ, set

$$R = \frac{s_1 \lambda}{s_0} Z, \qquad D \ge \frac{s_1}{s_0}\mathbb{E}(wZ) + \frac{|s_1|}{s_0}\|w - \lambda \mathbf{1}\|\|Z\| + \frac{2W s_2}{s_0}\|Z\|^2.$$

For every integer $L \ge 1$, with $D_i = \mathbb{E}B^i R$,

$$0 < T^L F \le s_0 Y_L, \qquad Y_L = \bar{f}_{r+L-1} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{M.34}$$

In particular the scalar D propagates to $D\bar{f}_{L-1}$. For $n = r + L - 1$, put $m_L = \max(n-3, L-1)$ and, for $0 \le k \le m_L$, define the finite coefficient

$$e_k = D\,\mathbf{1}_{\{k=L-1\}} + D_{L-1-k}\,\mathbf{1}_{\{0\le k\le L-2\}} - \beta_{n-k}\,\mathbf{1}_{\{0\le k\le n-3\}}. \tag{M.35}$$

An indexed term outside its indicated range is absent. For any subset $\mathcal{J} \subseteq \{0, \dots, m_L\}$ the same comparison gives

$$Y_L \le 1 + G_n(B)u + B^L R + \sum_{k\in\mathcal{J}} e_k \bar{f}_k + 2\sum_{k\notin\mathcal{J}} (e_k)_+. \tag{M.36}$$

Pairing with the actual unit arm. If $0 < G \le Cq + S$, $C > 0$, then

$$pG \le C\left(1 + \frac{S}{2C}\right)^2, \qquad 1 + \frac{S}{2C} \ge 0. \tag{M.37}$$

The complete-vector moment payment. Suppose an actual positive message obeys

$$0 < G \le sY, \quad s > 0, \qquad Y \le Cq + c + tu + BV, \quad C > 0.$$

For $\nu \in \{4, 8\}$,

$$\log \mathbb{E}(pG)^\nu \le \nu \log(sC) + \frac{\nu}{C}(c + \mathbb{E}BV) + \frac{1}{C^2}\left\{\frac{17\nu^2}{64}\|V\|^2 + \frac{5\nu}{6}t^2\right\}. \tag{M.38}$$

For one fixed comparison, use the same norm upper bound $N_0 \ge \|V\|^2/C^2$ in both right-hand sides. Their values satisfy $\mathcal{B}_4 = \mathcal{B}_8/2 - 17N_0/4$.

If instead the actual positive message has the square form

$$0 < H \le \eta(c + BV/2)^2, \qquad c + BV/2 \ge 0, \quad \eta > 0,$$

then

$$\log \mathbb{E}H^4 \le 4\log\eta + 8(c-1) + 4\mathbb{E}BV + \tfrac{17}{4}\|V\|^2. \tag{M.39}$$

The last term may instead be $2\|V\|^2 + 6\|BV\|^2$. *The underlying exponential estimate, valid for every real* V, *is*

$$\mathbb{E}e^{4BV} \le e^{4\mathbb{E}BV + 2\|V\|^2 + 6\|BV\|^2}. \tag{M.40}$$

Proof. Write $wZ = \lambda Z + (w-\lambda)Z$. The identity $T(wZ) = \mathbb{E}(wZ) + \lambda BZ + B((w-\lambda)Z)$ and the row bound give

$$|B((w-\lambda)Z)| \le \mathbb{E}|(w-\lambda)Z| \le \|w - \lambda\mathbf{1}\|\|Z\|, \qquad T(wZ^2) \le 2W\|Z\|^2.$$

This proves the first positive comparison. Apply the finite identity (M.33) to BR and then use positivity to obtain (M.34). Subtract the exact positive path convolution from $\bar{f}_n$ in (M.34). Its term of index k is $-\beta_{n-k}\bar{f}_k$, and the other two terms have exactly the indices in (M.35). On unretained indices use $0 \le \bar{f}_k \le 2$ according to the sign of e_k, proving (M.36). For (M.37), the upper bound $Cq+S$ is positive because it dominates $G > 0$. Apply AM–GM to the two nonnegative functions p and $(Cq+S)/C$, whose sum is $2 + S/C$.

For the moment payment use Lemma M.17, $\log p \le -u - 3u^2/10$ and $\log Y \le \log C + (Y-C)/C$. Completing the scalar square gives

$$\nu tu/C - 3\nu u^2/10 \le \frac{5\nu}{6C^2}t^2.$$

The signed-mean exponential estimate at ν/C now proves (M.38). Using the same norm upper N_0 in both formulas, the other terms are linear in ν, proving their stated relation.

For (M.39), first keep the positive factor η outside the fourth moment, and use $\log(c + BV/2) \le c + BV/2 - 1$. This retains the exact $4\log\eta$. Here is the variance-sensitive alternative, retaining the same mean. For a real W with $|W| \le U$, put $h = \mathbb{E}W^2/U^2 \le \theta$ and $m = \mathbb{E}W$. The moment-matching two-point comparison in paragraph M.1 applies. If $m \ge 0$, its separation is at most $(1+h)U$, so its centered log moment is at most $s^2U^2(1+3h)/8$, using $(1+h)^2 \le 1+3h$. If $m \le 0$, write $w = -m/U$, $a = (h-w^2)/(1+w)^2$. The same Bernoulli estimate gives the desired coefficient precisely when

$$\frac{3(1+a)^4}{4(1+a+a^2)} \le \frac{1+3w^2}{(1+w)^2} + 3a.$$

After subtracting $3a$, the left side is $3(1+a^2)^2/[4(1+a+a^2)] \le 3/4$, since $a^3 + a \le \theta^3 + \theta < 1$; the other side differs from $3/4$ by $(3w-1)^2/[4(1+w)^2]$. Consequently $\mathbb{E}e^{sW} \le e^{s\mathbb{E}W + s^2(U^2+3\mathbb{E}W^2)/8}$. Apply this with $W = BV$, $U = \|V\|$ and $s = 4$; the zero and degenerate moment cases follow directly. □

Corollary M.19 (The original two-edge output)**.** *For an actual nonnegative* H, *put* $U = T^2H$ *and* $I_q(U) = \mathbb{E}[q^{-1/3}|U|^{8/3}]$. *Then*

$$I_q(U)^3 \le \min\left\{[\mathbb{E}(U^4/q)]^2, e^{(24427/3420)d - 5d^2}[\mathbb{E}H^4]^2\right\}.$$

Proof. Hölder and $\mathbb{E}q = 1$ give the first bound. For the second, apply the generic transfer (R29.11.4), the inverse row (R29.11.3), and its cost (R29.11.6). The ordinary norm input is proved in paragraph M.15. □

Corollary M.20 (The first connector of a square)**.** *Let $m > 0$ and use the same original kernel throughout. If $0 < F \le (m + Z/2)^2$, then*

$$pTF \le (c + BZ/2)^2, \qquad c = m + \tfrac{1}{2}\mathbb{E}Z + \frac{\|Z\|^2}{4m}. \tag{M.41}$$

If instead $0 < F \le q(m + Z/2)^2$, $Z = ag - W$, and $e = \mathbb{E}W$, put

$$N = K((m - a)g + W), \qquad \Delta \ge m\{\sqrt{d\|Z\|^2} - \mathbb{E}(gZ)\} + \|Z\|^2.$$

Then

$$0 < TF \le m^2 q + mN - me + \Delta, \qquad pTF \le \left(m - e/2 + \frac{\Delta}{2m} + N/2\right)^2. \tag{M.42}$$

Both square bases are nonnegative. The assertions hold for the actual signed $e, \mathbb{E}(gZ)$ and need no separate sign condition on $q + Z/m$.

Proof. Use the first row of Theorem M.18 with $s_0 = m^2$, $s_1 = m$, $s_2 = 1/4$, and $\lambda = 1$. For $w = 1$, $r = 1$, it gives $TF \le m^2 q + mBZ + m\mathbb{E}Z + \|Z\|^2/2$. Equation (M.37) gives (M.41). For $w = q$, $r = 2$, use $\bar{f}_2 = q + Kg$ and $\mathbb{E}Z = -e$ to obtain

$$m^2 \bar{f}_2 + mBZ = m^2 q + mK((m - a)g + W).$$

The first-row scalar error is at most $-me + \Delta$. The complete upper bound in (M.42) is therefore positive. Apply the same unit pairing with $C = m^2$. □

Corollary M.21 (Bare inward paths)**.** *For the actual paths on the same measure,*

$$\mathbb{E}(p\bar{f}_2)^4 \le e^{d/4}, \qquad \mathbb{E}(p\bar{f}_3)^4 \le e^{-259d/128}, \qquad \mathbb{E}(p\bar{f}_n)^4 \le e^{-d/2} \quad (n \ge 4).$$

Proof. At $n = 2$, $p + \bar{f}_2 = 2 + Bu$ and $\mathbb{E}Bu = -d$. For $n \ge 3$, retain $\beta_n \bar{f}_0$ in the positive convolution:

$$p + \bar{f}_n \le 2 - \beta_n + BG_{n-1}(B)u, \qquad \mathbb{E}BG_{n-1}(B)u = -\beta_{n+1}.$$

The squared norm of $G_2(B)u$ is $d + \delta_2 \le 45d/32$; for $n \ge 4$, Lemma M.1 gives $\|G_{n-1}(B)u\|^2 \le d$. Apply (M.39) to the positive arithmetic-mean square. Its fourth exponent is $-4\beta_n - 4\beta_{n+1} + (17/4)\|G_{n-1}(B)u\|^2$. At $n = 3$, $\beta_3 = \beta_4 = d$, so this is at most $-259d/128$; thereafter use $\beta_n, \beta_{n+1} \ge \kappa d$ and $8\kappa - 17/4 = 1/2$. □

A common propagation lemma

Write $B = -K$, $u = B\mathbf{1} = -g$, $T = J + B$, and $p = 1 - u$. All expectations and norms use the original finite measure. The path convolution, the bounds $0 < \bar{f}_n \le 2$, and $\beta_n \in [\tau, d]$ for $n \ge 3$ are given by (M.2). The signed-mean inequality of Section M.1, paragraph M.1, gives

$$\mathbb{E}e^{4BV} \le e^{4\mathbb{E}BV + (17/4)\|V\|^2}, \qquad \mathbb{E}e^{8BV} \le e^{8\mathbb{E}BV + 17\|V\|^2} \quad (V \text{ real}). \tag{M.43}$$

The continuation identity (M.33) and Theorem M.18 give the following two specializations.

Lemma M.22 (Ordinary fourth-moment propagation)**.** *Suppose $F > 0$, $F \le e^{R-\Gamma}$, and $|R| \le t$, where $0 \le t < 3$. For $c \ge (1 - t/3)^{-1}$, put*

$$e^R \le 1 + R + \frac{c}{2}R^2, \tag{R29.5.1}$$

$$\sigma = \mathbb{E}R^2, \quad D_i = \mathbb{E}B^i R, \quad \delta = D_0 + c\sigma, \quad \epsilon = \delta + D_1. \tag{R29.5.2}$$

Let $H = pT^L F$, $L \ge 1$. Then

$$\log \mathbb{E}H^4 \le -4\Gamma + 4\epsilon + \tfrac{17}{4}\sigma, \qquad L = 1, \tag{R29.5.6}$$

$$\log \mathbb{E}H^4 \le -4\Gamma + 4\epsilon - 4d + 4D_2 + \tfrac{17}{4}\|u + BR\|^2 + \tfrac{10}{3}\delta^2, \qquad L = 2, \tag{R29.5.7}$$

$$\log \mathbb{E}H^4 \le -4\Gamma + 4\epsilon + 4D_2 + 4D_3 - 4\delta d - 8d + \tfrac{17}{4}\|(1+B)u + \delta u + B^2 R\|^2 + \tfrac{10}{3}\epsilon^2, \qquad L = 3. \tag{R29.5.8}$$

For $L \ge 4$, assume additionally that $h \ge 0$ and

$$\gamma_k := \beta_k - D_{k-1} \ge hd \quad (k \ge 3). \tag{R29.5.9}$$

With $V_L = G_{L-1}(B)u + B^{L-1}R$, one has

$$\log \mathbb{E}H^4 \le -4\Gamma + 8\delta_+ + 8(D_1)_+ - 12hd + \tfrac{17}{4}\|V_L\|^2 + \tfrac{10}{3}h^2 d^2. \tag{R29.5.10}$$

Proof. Use the E row of Table 80, that is, $w = 1, r = 1, \lambda = 1, s_0 = e^{-\Gamma}$ and R as stated. Lemma M.16 gives (R29.5.1); the first row and continuation give

$$0 < e^{\Gamma} T^L F \le Y_L = \bar{f}_L + \delta \bar{f}_{L-1} + B^L R + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{R29.5.3}$$

The scalar inequalities in the common proof are

$$\log p \le -u - \tfrac{3}{10}u^2 \qquad (-1 \le u < 1), \tag{R29.5.4}$$

$$4a_0 u - \tfrac{6}{5}u^2 \le \tfrac{10}{3}a_0^2 \qquad (a_0 \in \mathbb{R}). \tag{R29.5.5}$$

For the complete-vector payment (M.38), take $s = e^{-\Gamma}$, $C = 1$, and the following actual (c, t, V):

L	c	t	V
1	δ	0	R
2	ϵ	δ	$u + BR$
3	$\epsilon + D_2 - d$	ϵ	$(1+\delta)u + Bu + B^2 R.$

The signed means of BV are D_1, $-d + D_2$, and $-(1+\delta)d + D_3$. This gives the three short formulas. For $L \ge 4$, the finite ledger retains $\bar{f}_0 = 1$ and $\bar{f}_1 = 1 + u$, giving

$$Y_L \le q + 2\delta_+ + 2(D_1)_+ - hd(2 + u) + BV_L.$$

Here $\mathbb{E}BV_L = -\gamma_{L+1} \le -hd$. The same payment with $t = -hd$ proves (R29.5.10). □

Lemma M.23 (Ordinary eighth moments with an inward path arm)**.** *Let $a \ge 1$, $F > 0$, and $F \le \bar{f}_a e^{R-\Gamma}$. Suppose*

$$|R| \le \tfrac{1}{2}, \qquad \|R\| \le \varrho\sqrt{d}, \qquad \|\bar{f}_a - \mu_a\| \le s\sqrt{d}, \quad \mu_a = \mathbb{E}\bar{f}_a \in [0,1].$$

Here $s, \varrho \ge 0$. Write $\mu = \mu_a$, $D_i = \mathbb{E}B^i R$, and $c_0 = 2s\varrho + (12/5)\varrho^2$. Set $\delta = \mu D_0 + c_0 d$. The positive first step obeys

$$\begin{aligned} T(\bar{f}_a e^R) &\le \bar{f}_{a+1} + T(\bar{f}_a R) + \tfrac{3}{5}T(\bar{f}_a R^2) \\ &\le \bar{f}_{a+1} + \mu BR + \delta. \end{aligned} \tag{Q93.4.1}$$

For $L \ge 1$, $n = a + L$, define

$$0 < e^{\Gamma} T^L F \le Y_L = \bar{f}_n + \mu B^L R + \delta \bar{f}_{L-1} + \mu \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \tag{Q93.4.3}$$

$$V = G_{n-1}(B)u + \mu B^{L-1} R, \quad N = BV, \quad \mathbb{E}N = -\beta_{n+1} + \mu D_L, \tag{Q93.4.4}$$

$$c_1 = \delta - \beta_n, \qquad c_L = 2\delta_+ + 2\mu \sum_{i=1}^{L-1} (D_i)_+ - \beta_n \quad (L \ge 2). \tag{Q93.4.5}$$

Then

$$\log \mathbb{E}[(pT^L F)^8] \le -8\Gamma + 8c_L + 8\mathbb{E}N + 17\|V\|^2. \tag{Q93.4.6}$$

Proof. Use the A row of Table 80, with $w = \bar{f}_a$, $r = a+1$, $\lambda = \mu$, $c = 6/5$ and normalized seed μR. The scalar error is at most $\mu D_0 + 2s\varrho d + (12/5)\varrho^2 d = \delta$. This proves (Q93.4.1) and the exact continuation (Q93.4.3). The finite ledger retains $\beta_n \bar{f}_0$; the remaining positive paths are bounded by two, giving $p + Y_L \le 2 + c_L + N$ with (Q93.4.4)–(Q93.4.5). Apply (M.38) at $\nu = 8$, $s = e^{-\Gamma}$, $C = 1$, $t = 0$ to obtain (Q93.4.6). □

Scalar comparisons for ordinary moments

Lemma M.24 (Quadratic Hermite comparison)**.** *Let W be a finite real variable with $|W| \le U$, and let $f \in C^3([-U, U])$ satisfy $f''' \ge 0$. For $-U \le c < U$, put*

$$Q_c(w) = f(c) + f'(c)(w-c) + b_c(w-c)^2, \qquad b_c = \frac{f(U) - f(c) - f'(c)(U-c)}{(U-c)^2}. \tag{M.44}$$

Then $f(w) \le Q_c(w)$ throughout $[-U, U]$. If also $f', f'' \ge 0$, $0 \le a \le U$, $\mathbb{E}W \le -a$ and $\mathbb{E}W^2 \le v_0$, then

$$\mathbb{E}f(W) \le f(-a) + b_{-a}(v_0 - a^2). \tag{M.45}$$

Here the case $U = a = 0$ is read directly. If in addition $f \ge 0$, then $0 \le b_{-a} \le f(U)/(U+a)^2$ when $U + a > 0$. In particular, for $0 \le h \le 1$,

$$\mathbb{E}W \le 0, \quad \mathbb{E}W^2 \le hU^2 \quad \Longrightarrow \quad \mathbb{E}f(W) \le \frac{f(-hU) + hf(U)}{1+h}, \qquad (f', f'', f''' \ge 0). \tag{M.46}$$

For moment matching, write $m = \mathbb{E}W$, $v = \mathbb{E}(W-m)^2$, $M = U - m > 0$. With

$$c = m - v/M, \qquad \ell = M + v/M, \qquad \pi = \frac{v}{M^2 + v}, \tag{M.47}$$

one has $-U \le c \le m < U$, and the Bernoulli variable Z of mean π gives

$$\mathbb{E}f(W) \le \mathbb{E}f(m + \ell(Z - \pi)). \tag{M.48}$$

If $M = 0$, then $W = U$; if $v = 0$, both sides equal $f(m)$. The matching statement needs only $f''' \ge 0$, not a restriction on the sign of m or a bound involving θ.

Proof. The Hermite remainder has sign $f'''(\xi)(w-c)^2(w-U)/6 \le 0$. Under the additional hypothesis $f'' \ge 0$, convexity gives $b_c \ge 0$; at $c = -a$, the coefficient of $\mathbb{E}W$ in $\mathbb{E}Q_c(W)$ is $f'(c) + 2ab_c \ge 0$. Substituting the two moment bounds proves (M.45); the upper bound for b_{-a} follows by dropping the nonnegative terms in its numerator. For (M.46), take $c = -hU$: the enlarged moments $0, hU^2$ are realized at $-hU, U$ with probabilities $1/(1+h), h/(1+h)$. For matching, $v \le U^2 - m^2 = M(U + m)$ places the lower contact in the interval. The two-point variable in (M.48) has the same first two moments as W and lies at the contact points, so its expectation of Q_c is $\mathbb{E}Q_c(W)$. □

The centered Bernoulli estimate used with these comparisons is (Q81.2.5), proved in paragraph M.1. Moment matching retains the actual mean of W, of either sign. The continuous balanced coefficient is $1/8$; at a deterministic endpoint the centered variable is zero.

Lemma M.25 (Ordinary moments from a positive sum or square)**.** *Use the original finite kernel and $B = -K$. For $r \in \{4, 8\}$, write $C_r = 17r^2/64$. If $H \ge 0$, $a > 0$, and $H \le a^2(c + BV/2)^2$ with $c + BV/2 \ge 0$, then*

$$\log \mathbb{E}H^r \le 2r \log a + 2r(c-1) + r\mathbb{E}BV + C_r\|V\|_2^2. \tag{M.49}$$

If $X, Y \ge 0$, $X + Y \le 2 + c + BV$, and $0 \le H \le a^2 e^{-\Gamma} XY$, then

$$\log \mathbb{E}H^r \le r(2 \log a - \Gamma + c + \mathbb{E}BV) + C_r\|V\|_2^2. \tag{M.50}$$

Both formulas allow either sign of $\mathbb{E}BV$ and a constant coordinate in V. A zero moment is read with $\log 0 = -\infty$.

Proof. For $y \ge 0$, $y^{2r} \le e^{2r(y-1)}$. Apply (M.43) to the resulting exponential of rBV to obtain (M.49). For the sum form, AM–GM gives $XY \le (1 + (c + BV)/2)^2$, whose base is nonnegative since it dominates $(X + Y)/2$; use the square form with $ae^{-\Gamma/2}$ and $1 + c/2$. □

M.9 The three length-three pair envelopes

Lemma M.26 (Length-three pairs, with their full constants)**.** *Let $c \ge 4$, $m = c - 3$, $G_m = E + zD$, and put $\Gamma_+ = 3d - 2\delta_2$, $\beta = \beta_{c-1}$ and $\Gamma_- = d + \beta_c + \beta$. The three literal arm products have the following envelopes:*

$$h_3 f_c \le e^{-\Gamma_+ + BV_+}, \qquad V_+ = (-xE + zxD)u + \tau\mathbf{1}; \tag{M.51}$$

$$W_3 \bar{f}_c \le e^{-\Gamma_+ + BV_{\text{rev}}}, \qquad V_{\text{rev}} = (-xE - zxD)u - \tau\mathbf{1}; \tag{M.52}$$

$$\bar{f}_3 W_c \le e^{-\Gamma_- + B(U_- + \beta\mathbf{1})}, \qquad U_- = (xE + zxD)u, \quad \Gamma_- \ge 3d - 2\delta_2. \tag{M.53}$$

The nonconstant parts of all three vectors have norm less than $\sqrt{d}/2$. In addition

$$\begin{gathered} \mathbb{E}B(V_+ - \tau\mathbf{1}) \le \delta_2, \qquad \mathbb{E}BU_- \le 0, \quad -\delta_2 \le \mathbb{E}U_- \le 0, \\ \|Bu + U_-\| \le \sqrt{d}, \qquad \|-u + U_-\| \le \sqrt{d}. \end{gathered} \tag{M.54}$$

At $m = 1$, $(E, D) = (1, 0)$. No lower bound on D is imposed there. Constants $\tau\mathbf{1}$, $-\tau\mathbf{1}$ and $\beta\mathbf{1}$ belong to their respective complete vectors.

Proof. The exact arm identity is $h_3 + f_3 = 2 - 2d$. In the finite convolution for f_c, retain $\beta_c f_0 + \beta_{c-1} f_1 \geq \tau(1 + p)$. With $R_{3c} = \sum_{j=4}^{c} K^j \mathbf{1} = B^3 G_m(-B)u$, this gives

$$h_3 + f_c \leq 2 - (3d - 2\delta_2) + R_{3c} + \tau u.$$

The actual factors are nonnegative, so $xy \leq e^{x+y-2}$ proves (M.51). Complete reversal gives (M.52). For the exchanged pair, retain $\beta \bar{f}_0$ before applying A to $\bar{f}_{c-1}$; then add the exact $\bar{f}_3$. Using $G_{c-1} = 1 + B + B^2 G_m$ gives

$$\bar{f}_3 + W_c \leq 2 - d - \beta_c - \beta + B[-B^2 G_m(B)u + \beta\mathbf{1}].$$

This proves (M.53). The common nonconstant square is $\int x^2(E^2 + xD^2)\, d\nu \leq \theta^2(1 + \theta)d < d/4$. The means are given by (M.3). Finally

$$\|Bu + U_-\|^2 \leq \int [x^2 + x(1 + x)^2]\, d\nu \leq d, \quad \| - u + U_-\|^2 \leq \int [(1 - x + x^2)^2 + x^3]\, d\nu \leq d.$$

The first bound uses $\theta + 3\theta^2 + \theta^3 < 1$, and the second is the positive-exponent-two case of Lemma M.3. These norm estimates concern the displayed nonconstant vectors; a constant-square payment in a later certificate is retained separately. □

M.10 Fourth-moment certificates and root replacement

Use the signed-mean estimate (M.43), whose full proof is in Section M.1. It applies to vectors with nonzero means and constant components. For a terminal envelope $F \leq e^{R-\Gamma}$, write $\sigma, D_j, \delta, \epsilon$ as in (R29.5.2), using the row's pointwise Taylor coefficient $c = \eta$. The finite positive majorant is (R29.5.3). Once that majorant is known, an additional root pair is handled by combining its envelope with this same majorant and recomputing the complete vector and signed mean.

Lemma M.27 (Two fourth-moment conventions)**.** *If* $\log H \leq a + BQ$, *then* $\log \mathbb{E}H^4 \leq 4(a + \mathbb{E}BQ) + (17/4)\|Q\|_2^2$. *If an actual unit arm supplies p and the combined bound is* $\log H \leq a + BQ + cu - 3u^2/10$, *then*

$$\log \mathbb{E}H^4 \leq 4(a + \mathbb{E}BQ) + \tfrac{17}{4}\|Q\|_2^2 + \tfrac{10}{3}c^2, \qquad c \in \mathbb{R}. \tag{M.55}$$

The first convention includes every constant in Q; the second uses the displayed unit-arm curvature to pay cu.

Proof. The first bound is (M.43). The second combines that bound with (R29.5.4) and (R29.5.5), taking $a_0 = c$. This permits negative c and makes no orthogonality assumption about the remaining vector. □

For a long connector, the positive-majorant step in the proof of Lemma M.22 also gives the following useful form before any root-pair envelope is added. If $h \geq 0$ and $\gamma_j = \beta_j - D_{j-1} \geq hd$ for all $j \geq 3$, then for $L \geq 4$, $n = L - 1$,

$$\begin{gathered} Y_L \leq 1 + u + 2\delta_+ + 2(D_1)_+ - 2hd + B\{G_n(B)u + B^n R - hd\mathbf{1}\}, \\ \mathbb{E}B\{G_n(B)u + B^n R - hd\mathbf{1}\} = -\gamma_{L+1} \leq -hd. \end{gathered} \tag{M.56}$$

Its terminal scalar plus mean is at most $-\Gamma + 2\delta_+ + 2(D_1)_+ - 3hd$. The two retained terms are $\gamma_L \bar{f}_0$ and $\gamma_{L-1}\bar{f}_1$, already present at $L = 4$. The ordinary propagation proof is shared; the root pair and its constant are added using Lemma M.27.

Lemma M.28 (The two complete root pairs)**.** *Let $v \geq 3$, $m = v - 2$, $G_m = E + zC$, $\beta = \beta_{v-1} = \int C\, d\nu$ and $\Gamma_v = \beta_v + \beta = \int (E + C)\, d\nu$. The two specified arm directions have envelopes*

$$\bar{f}_2 W_v \leq \exp\{-\Gamma_v + B[-BG_m(B)u + \beta \mathbf{1}]\}, \tag{M.57}$$

$$h_2 f_v \leq \exp\{-\Gamma_v + B[-BG_m(-B)u + \beta \mathbf{1}]\}. \tag{M.58}$$

Their scalar-plus-mean losses are

$$\begin{gathered} \ell_s = \Gamma_v + \int xC\, d\nu, \quad \ell_a = \Gamma_v - \int xC\, d\nu, \\ \ell_s = \ell_a = d \quad (v = 3), \qquad \ell_s \geq \tfrac{1463}{1024} d, \quad \ell_a \geq \tfrac{969}{1024} d \quad (v \geq 4). \end{gathered} \tag{M.59}$$

Dropping the nonpositive $\beta(u - 1)$ instead gives the weaker envelopes

$$\bar{f}_2 W_v \leq e^{-\beta_v - B^2 G_m(B)u}, \qquad h_2 f_v \leq e^{-\beta_v - B^2 G_m(-B)u}. \tag{M.60}$$

Their scalar-plus-mean losses are at least $5d/6$ and $3d/16$, respectively. These versions keep the same arm directions and omit the constant only by the displayed pointwise comparison.

Proof. Retain $\beta_{v-1}\bar{f}_0$ in the positive convolution for $\bar{f}_{v-1}$, then apply A. This gives

$$W_v \leq 1 - \beta_v - u - BG_{v-1}(B)u - \beta p.$$

Adding $\bar{f}_2 = 1 + u + Bu$, using $G_{v-1} = 1 + BG_m$, and applying $xy \leq e^{x+y-2}$ proves (M.57). For (M.58), retain both $\beta_v f_0$ and $\beta_{v-1} f_1$ in the reversed convolution for f_v, and add h_2. The means follow from (M.3). At $m = 1$, $C = \beta = 0$. For even $m \geq 2$,

$$\ell_s = \int (2 + x)C \geq (2 - \theta - \theta^2)d, \qquad \ell_a = \int (2 - x)C \geq (2 - 3\theta + \theta^2)d.$$

For odd $m \geq 3$, the corresponding bounds are $\ell_s \geq (2-\theta)d$ and $\ell_a \geq (2-3\theta+2\theta^2)d$. These prove the stated constants while retaining the full $\beta\mathbf{1}$ component in both root envelopes. For the weak standard envelope, the two parities give $\int (1 + x)C \geq (1 - \theta^2)d > 5d/6$ and d. For the weak alternative, they give $\int (1 - x)C \geq \kappa^2 d$ and $d - 2\int xC \geq (1 - 2\theta)d = 3d/16$. The $v = 3$ endpoint has loss d in both cases. □

Lemma M.29 (A root pair and a positive comparison)**.** *Suppose $0 \leq X \leq e^{\alpha}[C_0 q + c_0 + BV]$, with the bracket positive, and take either root pair of Lemma M.28. Write $\varepsilon = 1$, $\ell = \ell_s$ for $\bar{f}_2 W_v$, and $\varepsilon = -1$, $\ell = \ell_a$ for $h_2 f_v$. For the full actual message $H = pPX$,*

$$\begin{aligned} \log \mathbb{E} H^4 \leq 4[\alpha + C_0 + c_0 - 1 + \mathbb{E} BV - \ell] \\ + \tfrac{17}{4} \|V - BG_m(\varepsilon B)u\|^2 + \tfrac{10}{3}(\beta + C_0 - 1)^2. \end{aligned} \tag{M.61}$$

The vector V retains any already displayed constant components. At a zero of the message, interpret $\log 0 = -\infty$.

Proof. Use $\log Y \le Y-1$ on the same positive bracket, add the full root envelope and $\log p \le -u-3u^2/10$. The residual coefficient of u is $\beta+C_0-1$, while its remaining vector is $V-BG_m(\varepsilon B)u$. Apply (M.55); the root's signed mean is already included in ℓ. □

Lemma M.30 (Constant terms in a root replacement)**.** *Suppose the same propagated input has a positive upper bound $X \le e^{\alpha}[C_0 q + c_0 + BQ(B)u]$. Put $\gamma = C_0 - 1$, $k_0 = C_0 + c_0 - 1$, and write $Q = U + zV$. Take a new root $f_a h_b f_c$, $3 \le a < b < c$, with the actual pair envelope $h_b f_c \le e^{BV_{bc}-16d/5}$. Write $G_{a-1}(-z) = E_a - zD_a$ and $V_{bc} = (r+zs)u$. Let*

$$V_0 = (Q-B)u, \qquad V_* = (Q + G_{a-1}(-B))u + V_{bc}, \qquad \phi = 1 - D_a + s.$$

Relative to the old root pP_0, $P_0 \le e^{-d-B^2u}$, the complete fourth exponents satisfy

$$\begin{aligned} j_Q = & -\tfrac{44}{5} - 4(E_a + D_a + r) \\ & + \tfrac{17}{4}\{(U + E_a + r)^2 - U^2 + x[(V - D_a + s)^2 - (V-1)^2]\}. \end{aligned} \tag{M.62}$$

If the old certificate retained $\gamma\mathbf{1}$ in its full vector,

$$J_{\text{new}} - J_{\text{old,full}} = \int j_Q \, d\nu - \tfrac{17}{2}\gamma \int \phi \, d\nu. \tag{M.63}$$

If the old certificate paid γu using its actual p,

$$J_{\text{new}} - J_{\text{old,curv}} = \int j_Q \, d\nu + \tfrac{17}{2}\gamma \mathbb{E} V_* + \tfrac{11}{12}\gamma^2. \tag{M.64}$$

Proof. Logarithms of the positive majorant give $\log X \le \alpha + k_0 + u + B(Qu + \gamma\mathbf{1})$. The finite convolution also gives $f_a \le 1 - u + BG_{a-1}(-B)u - \beta_a$, where $\beta_a = \int D_a \, d\nu$. The new complete vector is $V_* + \gamma\mathbf{1}$, with scalar $\alpha + k_0 - \beta_a - 16d/5$. The old scalar is $\alpha + k_0 - d$: its full-vector convention uses $V_0 + \gamma\mathbf{1}$, while its curved convention uses V_0 and pays γu with the actual p. Subtract these complete exponents using (M.3); this gives (M.62). In the first convention the constant squares cancel and $\mathbb{E}(V_* - V_0) = -\int \phi \, d\nu$. In the second the new root must retain $\gamma\mathbf{1}$, while the old root paid $10\gamma^2/3$; thus $17/4 - 10/3 = 11/12$, giving the second identity. These are exponent comparisons for the same input, not divisions by an old root moment. Root-specific coordinate rectangles and their signed overshoot charges are still required. □

Lemma M.31 (Adding all later root pairs at once)**.** *Suppose a complete fourth certificate has rate λ, with its full real vector bounded by $(7/2)\sqrt{d}$ on $0 < d \le 1/64$. Suppose $k \ge 1$ later unequal pairs, in their specified common direction, have disjoint exponent intervals and give the actual bound*

$$E_k \le \exp\{BV_k - (16k/5)d\}, \qquad \|V_k\| \le \tfrac{3}{8}\sqrt{d}, \qquad \mathbb{E} BV_k \le d/12.$$

Keep any already used unit-arm curvature payment separate. Then the augmented certificate has rate at least

$$\lambda + \tfrac{2737}{3840} + \tfrac{64}{5}(k-1). \tag{M.65}$$

On $d \ge 1/64$, the same direction instead gives the stronger pointwise fourth gain $4/5 + 64(k-1)/5$.

Proof. The low-domain scalar gain is $64kd/5$; its mean and square costs total at most

$$\tfrac{d}{3} + \tfrac{17}{4}[2(\tfrac{7}{2})(\tfrac{3}{8}) + \tfrac{9}{64}]d = \tfrac{9283}{768}d.$$

Subtracting gives (M.65). For the high domain the row bound gives $BV_k \le \|V_k\| \le 3d$, so the pointwise fourth gain is $64k/5 - 12$. The required mask estimates, when the original powers start at five, follow from Lemma M.2 and the actual signed even-moment tail:

$$|P(i\sqrt{x})|^2 \le \frac{1+x}{(1-x^2)^2} \le \tfrac{101}{50}, \quad \|V_k\|^2 \le \tfrac{101}{50}\theta^3 d < \tfrac{9}{64}d, \quad \mathbb{E}BV_k \le \frac{\delta_4}{1-\theta^2} < \tfrac{d}{12}.$$

Here P is one finite zero-one mask. Disjoint intervals make the bound independent of k. Reversal preserves the same even moment mean bound. Each pair must retain its true $16d/5$ scalar loss from the pair estimate; the lemma does not make arbitrary unequal pairs free. □

Finally the ordinary, unanchored fourth outlet of Sections Q.1 and M.21 gives, for every real Y and independent $M \ge 1$,

$$\begin{gathered} I_q(T^M Y)^3 \le e^{c(M)d}(\mathbb{E}|Y|^4)^2, \qquad I_q(X) = \mathbb{E}[q^{-1/3}|X|^{8/3}], \\ c(1) = \tfrac{204}{19}, \quad c(2) = 10, \quad c(3) = 0, \quad c(4) = 7, \quad c(M) = 6 \ (M \ge 5). \end{gathered} \tag{M.66}$$

In particular one may use the weaker uniform cost 11. This outlet uses one arm factor and an unweighted input estimate.

M.11 An anchored packet and its reversed unit row

Lemma M.32 (A correlated first row and the W_2 cancellation)**.** *Let F be an actual terminal, $R = r(B)u$, and suppose its actual first row has been proved in the form*

$$TF \le e^{\alpha}(\bar{f}_2 + BR + D), \qquad D \ge 0. \tag{M.67}$$

For a terminal $F \le qe^{R-\Gamma}$, $\alpha = -\Gamma$. A square row with prefactor $m^2 e^{-\Gamma}$ instead has $\alpha = 2\log m - \Gamma$; that scalar is kept throughout. Set $D_i = \mathbb{E}B^i R$ and $\gamma_j = \beta_j - D_{j-2}$. Finite propagation and the positive path convolution give

$$T^L F \le e^{\alpha}\left[1 + G_{L+1}(B)u + B^L R + D\bar{f}_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar{f}_{L+1-j}\right]. \tag{M.68}$$

For $L \ge 2$, if the displayed γ_j are nonnegative,

$$\begin{gathered} T^L F \le e^{\alpha}[1 + u + c_L + BQ_L(B)u], \qquad c_L = 2D - \gamma_{L+1}, \quad Q_L = G_L + z^{L-1} r, \\ \mathbb{E}BQ_L(B)u = -\gamma_{L+2}. \end{gathered} \tag{M.69}$$

The mean identity requires the same actual moment at index $L+2$.

Under complete terminal reversal, $\bar{R} = -r(-B)u$ and

$$\mathbb{E}\bar{R} = \mathbb{E}R, \quad \|\bar{R}\| = \|R\|, \quad \mathbb{E}(u\bar{R}) = -\mathbb{E}(uR). \tag{M.70}$$

If the first-row proof used $D \ge \max\{\mathbb{E}R + \mathbb{E}(uR) + \sqrt{d\mathbb{E}R^2} + c_T\mathbb{E}R^2, 0\}$, the same proof and the same D give $T\bar{F} \le e^{\alpha}[Tp + B\bar{R} + D]$. The coefficient c_T is the one proved for the actual terminal.

Suppose the chosen root pair has the full envelope $P_r \le e^{-\Gamma_r + BV_r}$, including its actual constant. The resulting complete certificates are

actual message	a	*complete V*
$f_2 P_r T\bar{F}$	$\alpha + D - \Gamma_r$	$\bar{R} + V_r$
$W_2 P_r T^L F,\ L \ge 2$	$\alpha + c_L - \Gamma_r$	$(Q_L - 1)u + V_r.$

(M.71)

They mean $\log H \le a + BV$*, with fourth exponent* $4(a + \mathbb{E}BV) + (17/4)\|V\|^2$*. For the second row its scalar-plus-mean, before the actual root mean* $\mathbb{E}BV_r$*, is* $\alpha - \Gamma_r + 2D + d - \gamma_{L+1} - \gamma_{L+2}$*.*

Proof. Apply T^{L-1} to (M.67) and use the bare finite Duhamel identity (M.33). Substituting (M.2) gives (M.68). Retaining the final $\gamma_{L+1}\bar{f}_0$ and using $\bar{f}_{L-1} \le 2$ gives (M.69); the two spectral means give its next-index identity and (M.70). In the first-row calculation $T(qR) = \mathbb{E}R + \mathbb{E}(uR) + BR + B(uR)$. Reversal changes its mean sum to $\mathbb{E}\bar{R} - \mathbb{E}(u\bar{R})$, which is the same number; the row-product bound remains $\sqrt{d\mathbb{E}R^2}$. Finally $f_2 = 1 - u + Bu$ cancels both u and Bu in $Tp = 1 + u - Bu$, whereas $W_2 = 1 - u - Bu$ cancels u and subtracts one u from the primitive in the long row. Apply $\log y \le y - 1$ to these positive factors and their positive majorants. This gives the displayed full vectors, without a new unit-arm curvature reserve. □

M.12 Finite rows with a two-edge root arm

Lemma M.33 (Continuation from an anchored first row)**.** *Let* $s \ge 1$*, and suppose an actual terminal has the positive upper bound* $TF \le e^{\alpha}[\bar{f}_s + \mu BR + D]$*, where* R *is real. Put* $D_i = \mathbb{E}B^i R$*. For every integer* $L \ge 1$*,*

$$0 < e^{-\alpha} T^L F \le \bar{f}_{s+L-1} + \mu B^L R + D\bar{f}_{L-1} + \mu \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{M.72}$$

The constants α, μ, D *are those of the proved first row.*

Proof. Apply T^{L-1} to that row and use (M.33) with the vector BR.

□

For the actual two-edge arm $W_2 = 1 - u - Bu$, any positive upper bound $T^L F \le e^{-\Gamma}[Cq + BV + c]$ gives

$$\log(W_2 T^L F) \le -\Gamma + C + c - 1 + B[V - u + (C-1)\mathbf{1}]. \tag{M.73}$$

This follows by adding $\log W_2 \le -u - Bu$ to $\log(Cq + BV + c) \le Cq + BV + c - 1$. The complete bracket, rather than each summand, is required to be positive.

Lemma M.34 (The ordinary terminal at a W_2 root)**.** *Suppose* $F > 0$*,* $F \le e^{R-\Gamma}$*, and the actual remainder satisfies* $e^R \le 1 + R + (c_T/2)R^2$*. Write* $\sigma, D_i, \delta, \epsilon$ *as in* (R29.5.2)*, with* $c = c_T$*. Then* $\log(W_2 T^L F) \le a + B(V_0 + c_0\mathbf{1})$*, where*

L	a	V_0	c_0
1	$-\Gamma + \delta$	$R - u$	0
2	$-\Gamma + \epsilon$	BR	δ
3	$-\Gamma + \epsilon - d + D_2$	$(B + \delta)u + B^2 R$	ϵ
$L \ge 4$	$-\Gamma + 2\delta_+ + 2(D_1)_+ - 2hd$	$(G_{L-1} - 1)u + B^{L-1}R$	$-hd$

(M.74)

The last row assumes $h \ge 0$ *and* $\gamma_j = \beta_j - D_{j-1} \ge hd$ *for every* $j \ge 3$*. With* $n = L - 1$*, its two means are*

$$\mathbb{E}BV_0 = d - \gamma_{L+1} \le (1 - h)d, \qquad \mathbb{E}V_0 = -\gamma_{n+1} \le -hd. \tag{M.75}$$

At a unit-length row whose actual arm is f_2*, replace only* $R - u$ *by* $R + u$*. If the actual root-pair product has envelope* $P \le e^{-\Gamma_r + BV_r}$*, including its full constant,*

$$\begin{gathered} V = V_0 + c_0\mathbf{1} + V_r, \qquad S = a - \Gamma_r + \mathbb{E}BV, \\ \log \mathbb{E}[(PW_2 T^L F)^4] \le 4S + \tfrac{17}{4}\|V\|^2. \end{gathered} \tag{M.76}$$

For the unit f_2 exception the same conclusion uses that actual arm.

Proof. The finite positive majorant is (R29.5.3); its derivation needs only the displayed remainder bound and $T(R^2) \le 2\sigma$. Expand its first three lengths and add $\log W_2 \le -u - Bu$. For $L \ge 4$, use (M.56); adding $\log W_2$ subtracts one u from its primitive and gives the last row of (M.74). The signed means follow from (M.3). The unit exception uses $\log f_2 \le -u + Bu$ instead. Finally add the actual pair envelope and apply Lemma M.27 to the whole vector V. There is no independent unit-arm curvature payment at this root. □

M.13 A two-arm terminal at roots of minimum one and two

Use the original model of Section M.2 and put $h = 17/32$. All lengths below are positive integers, all ordered lists are finite, and all operators use the same original probability measure.

Lemma M.35 (Transporting a complete constant)**.** *Suppose an actual positive message satisfies*

$$\log H \le a + B(V + c\mathbf{1}) - \tfrac{3}{10}(u + \xi Bu)^2, \qquad \xi \in \{0, 1\}, \quad c \in \mathbb{R}.$$

Its two available fourth certificates are

$$J_f = 4(a + \mathbb{E}BV) + \tfrac{17}{4}\|V + c\mathbf{1}\|^2, \tag{M.77}$$

$$J_c = 4\{a + \mathbb{E}B(V - \xi cu)\} + \tfrac{17}{4}\|V - \xi cu\|^2 + \tfrac{10}{3}c^2. \tag{M.78}$$

Each bounds $\log \mathbb{E}H^4$*. The case* $\xi = 0$ *uses the actual* p *curvature, and* $\xi = 1$ *uses the actual* W_2 *curvature.*

Proof. For $w \in [-1, 1)$, $\log(1-w) \le -w - 3w^2/10$. Its derivative proof is the unit-arm proof applied to w; the endpoint -1 is paid by $\log 2 < 7/10$. Both $w = u$ and $w = u + Bu = 1 - W_2$ have this range. The full certificate follows by discarding the nonpositive square and applying (M.43). For the second use the exact inequality

$$4cu - \tfrac{6}{5}(u + \xi Bu)^2 \le \tfrac{10}{3}c^2 - 4\xi cBu$$

before that same signed-mean bound. In particular no mean or constant-vector cross term is declared zero. □

Lemma M.36 (The terminal packet actually needed)**.** *For* $t \ge 3$*, set* $m = t - 2$*,* $G_m = e + zf$ *at* $z = i\sqrt{x}$*, and*

$$\gamma = \beta_{t-1} = \int f \, d\nu, \quad \Gamma = \beta_t + \gamma = \int (e + f) \, d\nu, \quad R = (\gamma + xe + zxf)u.$$

The actual terminal $F_t = \bar{f}_2 W_t$ *satisfies* $F_t \le e^{R-\Gamma}$*. At* $t = 3$ *the same envelope also holds for the distinct terminal* $h_2 f_3$*. Put* $D_j = \mathbb{E}B^j R$*,* $\sigma = \mathbb{E}R^2$*,* $\delta = D_0 + \eta\sigma$*, and* $\epsilon = \delta + D_1$*, where* $\eta = 3/2$ *at* $t \ge 4$ *and* $\eta = 8/7$ *at* $t = 3$*. Then*

$$\begin{gathered} TF_t \le e^{-\Gamma}(q + BR + \delta), \qquad \Gamma \ge d, \quad 0 \le \gamma \le d, \quad \|R\| \le \tfrac{4}{5}\sqrt{d}, \\ -\delta_2 \le \delta \le \tfrac{7}{12}d, \qquad -\tfrac{13}{16}d \le \epsilon \le 0, \qquad D_1 \le 0, \\ \Gamma - 2\delta_+ \ge \tfrac{3}{5}d, \qquad \Gamma - (1-d)\delta \ge \tfrac{4}{5}d, \qquad \gamma_j := \beta_j - D_{j-1} \ge hd \quad (j \ge 3). \end{gathered} \tag{M.79}$$

The first-row statement also holds for the stated $h_2 f_3$ *choice.*

Proof. Retain $\beta_{t-1}W_1$ in the positive A-image defining W_t. Adding $\bar{f}_2$ gives $\bar{f}_2 + W_t \le 2 - \Gamma + R$; positive AM–GM proves the envelope. At $t = 3$ both actual sums $\bar{f}_2 + W_3$ and $h_2 + f_3$ equal $2 - d - B^2 u$. The exact moments are

$$\begin{aligned} \sigma &= \gamma^2 d + 2\gamma \int xe\, d\nu + \int x^2(e^2 + xf^2)\, d\nu, \\ D_0 &= -\int xf\, d\nu, & D_1 &= -\gamma d - \int xe\, d\nu, \\ D_2 &= \int x^2 f\, d\nu, & D_3 &= \gamma\delta_2 + \int x^2 e\, d\nu. \end{aligned} \tag{M.80}$$

For $m \ge 2$, $1 - x \le e, f \le 1$. The row bound gives $|R| \le d + \sqrt{\delta_2 + \delta_3} < 1$; hence $e^R \le 1 + R + 3R^2/4$ and the positive first row follows. The positive remainder coefficient is at most $e - 2 < 3/4$; the negative remainder is at most $1/2$. Moreover

$$\sigma/d \le \tfrac{1}{9} + \tfrac{2\theta}{3} + \theta^2 + \theta^3 < \tfrac{5}{8} < \tfrac{16}{25}.$$

For fixed actual e, f, ϵ is convex in $\gamma \in [0, 1/3]$. At zero, its two negative linear terms absorb the positive squares. At $\gamma = 1/3$ it is at most

$$-d/6 + \int [-x + (5/2)x^2 + (3/2)x^3]\, d\nu \le 0.$$

Here maximize the actual e-term at one and the f-term at $1 - x$; the latter derivative is negative. The displayed convex cubic has endpoints zero and $7007/65536 < 1/6$. Conversely $\sigma \ge \gamma^2 d + 2\gamma \int xe\, d\nu$ gives

$$\epsilon/d \ge -2\theta + (3\theta - 1)\gamma + (3/2)\gamma^2 \ge -13/16.$$

Here $3\gamma - 1 \le 0$ and both first integrals are at most $\delta_2 \le \theta d$.

We prove the joint scalar bound before using it to bound δ. When $\delta \le 0$, $\Gamma \ge 2\kappa d > 5d/6$. Otherwise consider $\Gamma - 2\delta$, retaining $\gamma = \int f\, d\nu$. If m is even, $e = f$. With $X = \int xf\, d\nu$, $Y = \int x^2(1 + x)f^2\, d\nu$,

$$\Gamma - 2\delta = 2\gamma - 3\gamma^2 d + (2 - 6\gamma)X - 3Y, \qquad Y \le \theta(1 + \theta)X.$$

If $2 - 6\gamma - 3\theta(1 + \theta) \ge 0$, the result is at least $95d/96$. Otherwise use $X \le \theta\gamma$ and $a = \gamma/d \in [\kappa, 1]$. The lower bound divided by d is

$$Ca - \tfrac{55}{48}a^2, \qquad C = 2 + 2\theta - 3\theta^2(1 + \theta) > 21/10.$$

Its two endpoint lower bounds are $229/240$ and $207157/245760$, both exceeding $5/6$. If $m \ge 3$ is odd, $e = 1 - xf$ and

$$\sigma = \gamma^2 d + 2\gamma\delta_2 - 2\gamma \int x^2 f\, d\nu + \delta_3 - \int x^3 f[2 - (1 + x)f]\, d\nu.$$

Using $\int xf\, d\nu \ge \kappa\delta_2$, $\int x^2 f\, d\nu \ge \kappa\delta_3$, and $f[2 - (1 + x)f] \ge \kappa^2$ gives

$$\Gamma - 2\delta \ge d + \gamma - 3\gamma^2 d + \int J_\gamma\, d\nu, \quad J_\gamma = (\kappa - 6\gamma)x + (6\gamma\kappa - 3)x^2 + 3\kappa^2 x^3.$$

Its second derivative is at most $4\kappa-6+18\kappa^2\theta<0$, and its endpoint at θ is negative: the γ-coefficient is negative and the $\gamma=0$ value is negative. After using $d\le 1/3$, its lower bound divided by d is

$$A+Ca-a^2/3,\quad A=1+\kappa\theta-3\theta^2+3\kappa^2\theta^3>13/16,\quad C=1-2\theta+2\kappa\theta^2>3/8.$$

The two endpoint bounds exceed $41/48$ and $233/256$. Thus, for every finite $m\ge 2$,

$$\Gamma-2\delta_+\ge 5d/6\qquad (t\ge 4). \tag{M.81}$$

Since $\Gamma\le 2d$, this already implies the sufficient $\delta\le 7d/12$. Also $\delta\ge D_0\ge-\delta_2$, and $(1-d)\delta\le 2\delta_+$ proves the other scalar bound.

At $m=1$, $(e,f,\gamma,\Gamma)=(1,0,0,d)$, $R=-B^2u$, and $|R|\le\sqrt{\delta_2}<3/8$. The finite factorial-tail bound for the remainder coefficient is $7281/12800<4/7$. Therefore

$$\delta=8\delta_3/7<d/5,\quad \epsilon=\delta-\delta_2,\quad D_2=0,\quad D_3=\delta_3.$$

These values give all the stated scalar bounds. Finally the complete four residue classes, for every $j\ge 0$, are

$$D_{4j}=-\int x^{2j+1}f\,d\nu,\quad D_{4j+1}=-\gamma\delta_{2j+1}-\int x^{2j+1}e\,d\nu,$$
$$D_{4j+2}=\int x^{2j+2}f\,d\nu,\qquad D_{4j+3}=\gamma\delta_{2j+2}+\int x^{2j+2}e\,d\nu.$$

The first two positive classes give $D_2\le\delta_3$, $D_3\le d\delta_2+\delta_3$. Every later positive value is at most $(d+\theta)\delta_4<3\delta_4/4<d/16$, and each subsequent member contracts by θ^2. Combining with the finite β_j bounds proves $\gamma_j\ge 17d/32$, including the next mean index at every connector length. □

Theorem M.37 (The two root minima with the exact terminal $(2,t)$)**.** *Let the root residual list be $(r,a,b,s_1,t_1,\dots,s_k,t_k)$, with $r\in\{1,2\}$, $3\le a<b<s_1<t_1<\cdots<s_k<t_k$, and terminal list $(2,t)$, $t\ge 3$. The directions are as follows:*

Table 18: The two root minima and their terminal directions.

r	L	*Root factor*	*Terminal factor*
1	*all*	$pW_a\bar f_b\prod W_{s_i}\bar f_{t_i}$	h_2f_3 *if* $t=3$*, otherwise* $\bar f_2W_t$
2	*1*	$f_2h_af_b\prod h_{s_i}f_{t_i}$	$W_2\bar f_t$
2	≥ 2*,* $a=3$	$W_2\bar f_3W_b\prod W_{s_i}\bar f_{t_i}$	$\bar f_2W_t$
2	≥ 2*,* $a\ge 4$	$W_2W_a\bar f_b\prod W_{s_i}\bar f_{t_i}$	$\bar f_2W_t$

Every inner edge points terminal-to-root and every outer edge root-to-receiver. Let H_k be the actual root factor times the actual T^L-propagated terminal, with arbitrary compatible equal-pair decorations at both roots. Put $z_0=0$ and, for $k\ge 1$, $z_k=2737/3840+64(k-1)/5$. Then for every original host,

$$\mathbb{E}H_k^4\le e^{-(6+z_k)d},\qquad I_q(T^MH_k)^3\le e^{-[12+2z_k-c(M)]d}\le e^{-24d/19}, \tag{M.82}$$

for every independent $L,M\ge 1$, with $c(1)=204/19, c(2)=10, c(3)=0, c(4)=7, c(M)=6$ for $M\ge 5$.

Proof. First omit decorations and later pairs. The actual pair envelopes are Lemma M.26. Write $G_{b-3} = E + zC$, $U = (xE + zxC)u$, $\beta = \beta_{b-1}$. The reversed pair is bounded by $e^{-g_r + B(-U-\tau\mathbf{1})}$, with $g_r = 3d - 2\delta_2$. The exchanged pair is bounded by $e^{-g_e + B(U+\beta\mathbf{1})}$, where $g_e = d + \beta_b + \beta \geq 3d - 2\delta_2$ and $\kappa d \leq \beta \leq d$. At the finite boundary $b = 4$, $(E, C) = (1, 0)$ and $\beta = d$. For a short root arm at least four, the one-pair envelope is $e^{-16d/5 + BV_*}$, with $\|V_*\| \leq 3\sqrt{d}/8$, $\mathbb{E}BV_* \leq d/12$. The same bounds apply in the positive direction at a unit inner edge. They follow from the actual pair loss and the single finite mask:

$$\|V_*\|^2 \leq (101/50)\theta^3 d < 9d/64, \qquad \mathbb{E}BV_* \leq \delta_4/(1-\theta^2) < d/12.$$

The common finite root calculation.

Use Lemma M.36. The positive finite Duhamel majorant is

$$Y_L = \bar{f}_L + \delta \bar{f}_{L-1} + B^L R + \sum_{j=1}^{L-1} D_j \bar{f}_{L-1-j}.$$

For $L \geq 4$, the two actual final losses give $Y_L \leq 1 + u + 2\delta_+ - 2hd + B[G_n u + B^n R - hd\mathbf{1}]$, where $n = L - 1 \geq 3$. Its next-index mean is $\mathbb{E}B(G_n u + B^n R) = -\gamma_{L+1} \leq -hd$. Before adding a root pair, the following table is exact:

L	a_L	V_L	c_L
1	$-\Gamma + \delta$	R	0
2	$-\Gamma + \epsilon$	$u + BR$	δ
3	$-\Gamma + \epsilon - d + D_2$	$(1 + \delta + B)u + B^2 R$	ϵ
≥ 4	$-\Gamma + 2\delta_+ - 2hd$	$G_n u + B^n R$	$-hd$

For the actual p prefix, $\log(pT^L F) \leq a_L + B(V_L + c_L\mathbf{1}) - 3u^2/10$. For the actual W_2 prefix, at $L \geq 2$, replace only V_L by $V_L - u$, and retain $-3(u + Bu)^2/10$. Thus the same finite input supplies both root minima. Constants are paid by Lemma M.35, not discarded from this table.

The following entries record the complete payments. Here A means $a + \mathbb{E}BV \leq -Ad$; N bounds the square of the final full or transported vector divided by d; C_0 bounds the separate curvature expense divided by d. Each loss is $4A - 17N/4 - C_0$.

Table 19: Complete root certificates for the nonunit packet. Every displayed loss exceeds six.

Root and length	A	N	C_0	Loss
$pW_3\bar{f}_b, 1$	12/5	2/3	0	203/30
$pW_3\bar{f}_b, 2$	18/5	15/8	0	1029/160
$pW_3\bar{f}_b, 3$	179/40	15/8	15/4	989/160
$pW_3\bar{f}_b, \geq 4$	159/40	25/16	8/3	6329/960
$pW_a\bar{f}_b, 1; a \geq 4$	247/60	$(47/40)^2$	0	203501/19200
$pW_a\bar{f}_b, 2; a \geq 4$	76057/15360	$(67/40)^2$	0	75673/9600
$pW_a\bar{f}_b, 3; a \geq 4$	349/60	4	0	94/15
$pW_a\bar{f}_b, \geq 4; a \geq 4$	2549/480	64/25	1/3	6017/600
$W_2\bar{f}_3 W_b, 2$	51/16	227/144	0	3485/576
$W_2\bar{f}_3 W_b, 3$	319/80	3/2	0	383/40
$W_2\bar{f}_3 W_b, \geq 4$	541/160	25/16	0	2203/320
$W_2 W_a\bar{f}_b, 2; a \geq 4$	79/20	$(53/40)^2$	0	53367/6400

Root and length	A	N	C_0	Loss
$W_2 W_a \bar{f}_b, 3; a \geq 4$	$289/60$	$(41/24)^2$	0	$79067/11520$
$W_2 W_a \bar{f}_b, \geq 4; a \geq 4$	$2069/480$	$64/25$	0	$3817/600$

Every entry exceeds six; the ninth has margin $29/576$. We justify the entire finite table next.

The p prefix and its reversed length-three pair.

Add $-U - \tau\mathbf{1}$ to the displayed complete primitive. At $L = 1$, its nonconstant coordinates are $\gamma + x(e - E), x(f - C)$. Because $|e - E| \leq x$ and $|f - C| \leq 1$, its full square, including $-\tau\mathbf{1}$, is at most

$$[d^2 + 2d\theta^2 + \theta^4 + \theta^3 + d]d < 2d/3.$$

The constant cross costs at most $\tau^2 + 2\tau\delta_2 = d^2 - \delta_2^2 \leq d^2$. Its scalar plus mean is at most $-4d + 2\delta_2 + \delta + \delta_3 \leq -12d/5$.

At $L = 2$, the nonconstant coordinates are $1 - x^2 f - xE, \gamma + x(e - C)$, and the constant is $c = \delta - \tau \in [-d, 0]$. The real coordinate is positive and its square is at most $1 - x$; the imaginary absolute value is at most $d + x < 1$. Thus its nonconstant square is at most d. Since its mean is at least $-\gamma d - \delta_2$, the full square is at most $\{1 + d(1 + 2d + 2\theta)\}d \leq 263d/144 < 15d/8$. The actual scalar plus mean is at most $-5d + 3\delta_2 + \delta_3 \leq -18d/5$.

At $L = 3$ the nonconstant coordinates are

$$1 + \delta - \gamma x - x^2 e - xE, \qquad 1 - x^2 f - xC.$$

They are positive: the real lower bound is $1 - 2\theta/3 - \theta^2 - \theta = 485/3072$. Their square is at most $(43/36)^2 + \theta < 15/8$. The actual scalar plus mean, before the constant $c = \epsilon - \tau$, is

$$\begin{aligned} &- \{\Gamma - (1 - d)\delta\} - 2d - g_r - \gamma(d - \delta_2) - \int xe \, d\nu \\ &\quad + \int x^2(e + f) \, d\nu + \int xE \, d\nu \\ &\leq -\tfrac{29}{5}d + 2\delta_2 + 3\delta_3 < -179d/40. \end{aligned}$$

Transport the same c by the actual p curvature; $|c| \leq 29d/16$ and its expense is below $15d/4$.

For $L \geq 4$ the nonconstant primitive is $G_n u + B^n R - U$, and $c = -hd - \tau$. The coordinates of $G_n u - U$ are nonnegative and bounded by $1 - 2x + 2x^2, 1$; their square is at most one, since $-3 + 8x - 8x^2 + 4x^3 < 0$ on $[0, \theta]$. The actual finite tail has norm below $16\sqrt{d}/75$, giving $\|V\| \leq 91\sqrt{d}/75 < 5\sqrt{d}/4$. Its scalar plus mean is at most $-[3/5 + 3h + 3 - 3\theta]d = -159d/40$. The actual p curvature pays c, with expense below $8d/3$. This proves the first four rows.

The p prefix and every higher first pair.

Add the one actual mask V_*. At length one use $\|R + V_*\| \leq 47\sqrt{d}/40$ and $-\Gamma + \epsilon - 16d/5 + \mathbb{E}BV_* \leq -247d/60$. At length two the whole primitive before V_* is $u + BR + \delta\mathbf{1}$. Its square divided by d is at most

$$1 + \theta(d + \theta)^2 + (7/6)d(d + \theta) + (49/144)d < (13/10)^2.$$

This includes the absolute constant cross, using $|\delta| \leq 7d/12$ and $|\mathbb{E}(u + BR)| \leq (d + \theta)d$. The scalar plus mean is at most $-[2 - \theta^2 - 1/12 + 16/5]d = -76057d/15360$. At length three the nonconstant coordinates are $1 + \delta - \gamma x - x^2 e, 1 - x^2 f$. Their square is below $15d/8$, and including $\epsilon\mathbf{1}$ costs at

most $195d/256$. Their full square is therefore below $675d/256 < (13/8)^2 d$. Adding V_* gives norm at most $2\sqrt{d}$. The scalar plus mean is at most $-[4/5 + 2 + 16/5 - 1/10 - 1/12]d = -349d/60$, since $-\int xe + \int x^2(e+f) \le -\delta_2 + 3\delta_3 < d/10$. For the long rows, pay $c = -hd$ by p curvature. The remaining norm is below $(1+16/75+3/8)\sqrt{d} < 8\sqrt{d}/5$, its scalar plus mean is at most $-2549d/480$, and its curvature expense is below $d/3$. This proves the next four rows.

The W_2 prefix and the exchanged length-three pair.

At length two the nonconstant coordinates are $xE - x^2 f, \gamma + xe + xC$, the constant is $c = \delta + \beta$, and the scalar is $-\Gamma + \epsilon - g_e$. The scalar plus mean is at most $-51d/16$, using $D_2 - \int xE \le -\delta_2 + 2\delta_3 \le 0$. The nonconstant square is at most $20d/27$, while $0 < c \le 19d/12$ and its mean is nonpositive. Thus the full square is at most $20d/27 + 361d/432 = 227d/144$.

At length three the coordinates are $\delta - \gamma x - x^2 e + xE, 1 - x^2 f + xC$, and $c = \epsilon + \beta \in [-7d/32, d]$. Their real absolute value is below $5/8$; their imaginary coefficient lies between $5/6$ and $17/12$. The nonconstant square is below $5d/4$. If $c \ge 0$, its constant correction is nonpositive. If $c < 0$, that correction is at most $2051d^2/3072 < d/4$. The full square is therefore below $3d/2$. Its exact scalar plus mean is at most $-[4/5 + 1 + 35/16]d = -319d/80$; the two negative first moments absorb the two positive second moments before this estimate.

For the long rows the nonconstant primitive is $(G_n - 1)u + B^n R + U$, and $c = \beta - hd \in [d/16, 15d/32]$. Before the tail, its coordinate bounds are $[-x^2, x^2]$ and $[1-x, 1+x]$, giving square below one. Its full norm before $c\mathbf{1}$ is below $5\sqrt{d}/4$. The imaginary coefficient after the actual finite tail is at least $1 - 4\theta/3 - \theta^2 = 901/3072 > 15/64$. Consequently $c(c + 2\mathbb{E}V) \le 0$, so the full constant does not increase this square. The scalar plus mean is at most $-541d/160$. This proves all three exchanged-pair rows, including G_1.

The W_2 prefix and every higher first pair.

At length two the complete vector is $BR + \delta\mathbf{1} + V_*$. Its norm is at most $(3/5 + 7/20 + 3/8)\sqrt{d} = 53\sqrt{d}/40$, and its scalar plus mean is at most $-79d/20$. At length three the nonconstant coordinates before V_* are $\delta - \gamma x - x^2 e, 1 - x^2 f$. The real absolute value is below $1/2$: bound its two signs separately by $7/36$ and $2\theta/3 + \theta^2$. The nonconstant norm is below $5\sqrt{d}/6$. The actual $\epsilon\mathbf{1}$ has norm below $\sqrt{d}/2$, so the complete norm is at most $41\sqrt{d}/24$. The scalar plus mean is at most $-289d/60$. For the long rows, keep $c = -hd$ and the actual W_2 curvature. Transporting it replaces the primitive by $(G_n - 1 + hd)u + B^n R + V_*$. Its norm is below $(4/5 + 16/75 + 3/8 + 17/96)\sqrt{d} < 8\sqrt{d}/5$. The extra scalar correction is $-4hd^2 + (10/3)h^2 d^2 \le 0$, while the uncorrected scalar plus mean is at most $-2069d/480$. The last three rows follow.

The distinct unit-length minimum-two direction.

Here the actual terminal is $W_2 \bar{f}_t$, rather than $\bar{f}_2 W_t$. The finite convolution gives

$$W_2 + \bar{f}_t \le 2 - \beta_t + B^2 G_{t-2} u.$$

Thus $F \le e^{R^- - \beta_t}$, with $R^- = (-xe - zxf)u$, and

$$TF \le e^{-\beta_t}(q + BR^- + \delta^-), \qquad \delta^- = \int xf \, d\nu + (3/2) \int x^2(e^2 + xf^2) \, d\nu.$$

The actual f_2 cancels u, leaving primitive $u + R^-$. For $a = 3$, add the positive length-three pair with primitive $(-xE + zxC)u + \tau\mathbf{1}$. The full nonconstant coordinates are $1 - x(e+E), x(C-f)$. The

true constant correction is at most $\tau^2 + 2\tau\delta_2 \le d^2 \le d/3$. Keeping all terms in the same original spectral integral bounds the remaining fourth exponent by $\int P(x)\,d\nu$, where

$$P = -63/4 + 7x + 36x^2 - (95/4)x^3 + 17x^4.$$

Indeed use $-4e+4xe \le -4(1-x)^2$, $f, E \le 1$, $|C-f| \le 1$, and $0 \le 1-x(e+E) \le 1-2x+2x^2$ in the exact fourth expansion. The polynomial increases on $[0, 5/12]$, where its endpoint is $-20189/2592$. The paid constant gives loss $16517/2592 > 6$. Its nonconstant square is at most $(1+\theta^3)d < (7/6)^2 d$, and $\tau \le 3\sqrt{d}/5$; its complete norm is therefore less than $2\sqrt{d}$, as needed for later pairs. For $a \ge 4$, the base primitive has norm at most $\sqrt{d}$, and its scalar plus mean is at most $-71d/128$: its integrand is bounded by the increasing cubic $-2 + 3x + x^2/2 + 3x^3/2$ at $5/12$. The actual root mask then gives complete norm at most $11\sqrt{d}/8$ and loss $25537/3840 > 6$. Both calculations include $(e, f) = (1, 0)$ when $t = 3$.

All later pairs, decorations and the outer length.

Every complete or transported vector above has norm at most $2\sqrt{d}$ on the whole host domain; its already paid curvature constant is unchanged by a later root pair. The entire ordered list of $k \ge 1$ later pairs is one finite mask, with scalar loss $16kd/5$, norm at most $3\sqrt{d}/8$, and signed mean at most $d/12$. Adding it to the same certificate costs at most

$$\left[\tfrac{1}{3} + \tfrac{17}{4}\{2(2)(3/8) + 9/64\}\right] d.$$

The resulting gain is $21097/3840 + 64(k-1)/5 > z_k$, proving the first claim. Equal-pair products lie in $[0, 1]$ in their actual compatible directions, and positivity restores them at both original vertices.

If $\mathcal{M}$ is any original positive operator and $r = \mathcal{M}\mathbf{1}$, row Cauchy and Hölder give

$$I_q(\mathcal{M}H)^3 \le \mathbb{E}(r^4/q)\|\mathcal{M}\|^4(\mathbb{E}|H|^4)^2.$$

This uses the original q, which is positive on every retained atom. The operator bounds and path rows of Sections Q.1 and M.21 give exactly the costs $204/19, 10, 0, 7, 6$ in the five stated length ranges: at $M = 1$ the two costs are 4 and $128/19$; at $M = 2$ they are 6 and 4; at $M = 3$ both are nonpositive; at $M \ge 4$ the operator norm is at most one and the path-row costs are $7, 6$.

Since $12 - 204/19 = 24/19$, it proves the second claim for every independent outer length. When $d = 0$, leaf integration makes every actual message one, even if $B \ne 0$. All finite indices, continuous endpoints and arm directions have been retained. □

M.14 The actual three-step inverse-q fourth moment

We retain Q5-C056, Section 3, with its necessary elementary premises. On original positive-mass support $q(x) \ge \mu(x) > 0$. Use $r = Kg$, $s = Kr$, $d = \mathbb{E}g^2$, $\delta_2 = \mathbb{E}r^2$, $\delta_3 = \mathbb{E}s^2$. The exact original skew identities are $\mathbb{E}g = 0$, $\mathbb{E}r = -d$, $\mathbb{E}s = 0$, $\mathbb{E}gr = \mathbb{E}rs = 0$, $\mathbb{E}gs = -\delta_2$. The actual paths are $\bar{f}_1 = q$, $\bar{f}_2 = q + r$, $\bar{f}_3 = q + r - s - d$. We need only the elementary weaker spectral bound $\|K\|^2 \le 1/2$: the original Hilbert–Schmidt square is at most one; for a maximizing real unit vector v, $u = Kv/\|K\|$ is orthogonal to v and $\|Ku\| \ge \|K\|$. Extending u, v to an orthonormal basis gives $\|K\|_{\mathrm{HS}}^2 \ge 2\|K\|^2$. Thus $\delta_2 \le d/2$, $\delta_3 \le \delta_2/2$. The original three-cycle inequality gives $d \le 1/3$.

Put $v = r - s$, $u_0 = v - d = \bar{f}_3 - q$, $W = 2v - g$. Their exact moments are

$$\mathbb{E}v = -d, \quad \mathbb{E}gv = \delta_2, \quad \mathbb{E}v^2 = \delta_2 + \delta_3, \quad \mathbb{E}u_0 = -2d, \quad \mathbb{E}gu_0 = \delta_2,$$
$$\mathbb{E}u_0^2 = \delta_2 + \delta_3 + 3d^2,$$

$$\mathbb{E}W = -2d, \qquad \mathbb{E}W^2 = d + 4\delta_3.$$

The latter equality keeps the cancellation $4(\delta_2 + \delta_3) - 4\delta_2 + d$. The actual generating identity is $W = K(-1 + 2g - 2r)$, with input square $1 + 4\delta_2$. Therefore the original row bound gives $|W| \le \sqrt{1 + 4\delta_2} \le \sqrt{1 + 2d} \le 1+d =: t$. This includes small-mass atoms where $|g|$ is large. The correlated cubic combination is

$$\mathcal{C} = -\mathbb{E}g^3 + 4\mathbb{E}(g^2 v) - 6\mathbb{E}(gv^2) + 4\mathbb{E}v^3 = \tfrac{1}{2}\{\mathbb{E}W^3 + \mathbb{E}(g^2 W)\}.$$

Since $(W+2d)^2(W-t) \le 0$, exact expansion and $\mathbb{E}W = -2d$ give $\mathbb{E}W^3 \le (t-4d)\mathbb{E}W^2 - 4td^2 + 8d^3$. Also $\mathbb{E}(g^2 W) \le td$, hence

$$\mathcal{C} \le td - 2(1 + t)d^2 + 4d^3 + 2(t - 4d)\delta_3. \tag{L56.3.8}$$

The cubic interpolation applies for every W. For the only inverse-weight quartic term, use the actual identity $u_0 = T(\bar{f}_2 - 1)$. For $w = \bar{f}_2 - 1$, $\mathbb{E}w^2 = d + \delta_2$; row Cauchy–Schwarz and the original kernel bound two give $u_0^2 \le qT(w^2)$, $T(w^2) \le 2(d + \delta_2)$. Thus

$$\mathbb{E}(u_0^4/q) \le 2(d + \delta_2)(\delta_2 + \delta_3 + 3d^2). \tag{L56.3.9}$$

This controls arbitrarily small original q without a uniform bound on $1/q$. The exact fourth expansion is $R_4 := \mathbb{E}(\bar{f}_3^4/q) = \mathbb{E}q^3 + 4\mathbb{E}(q^2 u_0) + 6\mathbb{E}(qu_0^2) + 4\mathbb{E}u_0^3 + \mathbb{E}(u_0^4/q)$, with individual terms

$$\begin{aligned}
\mathbb{E}q^3 &= 1 + 3d - \mathbb{E}g^3, \\
\mathbb{E}(q^2 u_0) &= -2d - 2\delta_2 + \mathbb{E}(g^2 v) - d^2, \\
\mathbb{E}(qu_0^2) &= \delta_2 + \delta_3 + 3d^2 - \mathbb{E}(gv^2) + 2d\delta_2, \\
\mathbb{E}u_0^3 &= \mathbb{E}v^3 - 3d(\delta_2 + \delta_3) - 4d^3.
\end{aligned}$$

The $d\delta_2$ terms cancel exactly, leaving $R_4 = 1 - 5d - 2\delta_2 + 6\delta_3 + 14d^2 - 16d^3 - 12d\delta_3 + \mathcal{C} + \mathbb{E}(u_0^4/q)$. Combining (L56.3.8)–(L56.3.9) gives

$$R_4 - 1 \le -(5 - t)d - 2\delta_2 + (6 + 2t - 20d)\delta_3 + (12 - 2t)d^2 - 12d^3 + 2(d + \delta_2)(\delta_2 + \delta_3 + 3d^2).$$

For $d > 0$ put $x = \delta_2/d \in [0, 1/2]$ and substitute $t = 1 + d$. The separate δ_3 coefficient is $8 - 18d \ge 2$ and its coefficient in the last product is also nonnegative, so $\delta_3 \le \delta_2/2$ can be inserted legally everywhere. The normalized upper bound becomes

$$-4 + 11d + (2 - 6d + 6d^2)x + 3dx^2 - 8d^2.$$

Its x derivative is positive since $2 - 6d + 6d^2 = 6(d - 1/2)^2 + 1/2 > 0$. At $x = 1/2$ it is $-3 + (35/4)d - 5d^2$. This increases throughout $[0, 1/3]$, derivative at least $65/12$, and at $1/3$ is exactly $-23/36$. We obtain the complete original-measure interface

$$\mathbb{E}[\bar{f}_3^4/q] \le 1 - 23d/36. \tag{L56.3.15}$$

At $d = 0$, $g = r = s = 0$, all these actual path messages and weights are one, while K may be nonzero. The proof uses no lower spectral moment inequality or stronger spectral constant.

M.15 Finite path rows and ordinary operator power bounds

Q64.1 Measure, operators, and inputs

Let $X \neq \emptyset$ be finite with a probability measure μ, after deleting zero-mass atoms. Assume

$$K(x,y) = -K(y,x), \qquad |K(x,y)| \le 1. \tag{Q64.1.1}$$

Every expectation and every operator adjoint is with respect to this one original μ. Define

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n \mathbf{1}, \quad \bar{f}_n = T^n \mathbf{1}, \quad f_0 = \bar{f}_0 = 1, \qquad h_n = T f_{n-1} \quad (n \ge 1),$$
$$I_q(X) = E[q^{-1/3}|X|^{8/3}], \qquad \beta(X) = E[X^2/q]. \tag{Q64.1.2}$$

The kernels of A, T are nonnegative. Since $K(x,x) = 0$, $p(x), q(x) \ge \mu(x) > 0$. Every $f_n, \bar{f}_n$ is strictly positive.

Write

$$r = Kg, \quad s = Kr, \quad t = Ks, \quad \delta_j = \|K^j \mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = \frac{13}{32}, \quad c = \frac{19}{32}, \quad k = \frac{32}{19}. \tag{Q64.1.3}$$

We use the kernel spectral bound $\|K\|^2 \le \theta$ from the Q3/Q4 package. We keep p and q as weights. In particular,

$$0 \le d \le \frac{1}{3}, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta \delta_j, \quad \delta_3 d \ge \delta_2^2. \tag{Q64.1.4}$$

Here $Er = -d$ proves $d^2 \le \delta_2$. The last inequality is Cauchy–Schwarz in the finite spectral measure of $-K^2$ at 1. part Q64.9 includes the elementary proof of $d \le 1/3$.

The other inputs are the following exact finite identities and estimates.

Ordinary path multiplier. For every $a \ge 2$,

$$\|M_{f_a} T\|_{2\to 2}^2 \le 1 - d/72. \tag{Q64.1.5}$$

This is the full ordinary multiplier, in this actual order, stated in Q4 C115, original formula (8.3) and reconstructed in the Q3 operator proof in Appendix E. Apply it to $-K$, then take the ordinary μ-adjoint, to obtain

$$\boxed{\|T M_{\bar{f}_a}\|_{2\to 2}^2 \le 1 - d/72, \quad a \ge 2.} \tag{Q64.1.6}$$

The full multiplier and propagation occur in the displayed order.

Inverse path fourth, C056.

$$E[\bar{f}_3^4/q] \le 1 - \frac{23}{36} d. \tag{Q64.1.7}$$

Eighth-moment bounds, C049/C050 and C058. For the same directions at every positive inner length, put

$$Y_{b,L} = pT^L(qf_b), \qquad b \ge 2, \quad L \ge 1.$$

Then

$$EY_{b,L}^8 \le 1 - d/12000. \tag{Q64.1.8}$$

For $b = 2$, the stronger inputs are

$$EY_{2,L}^8 \le 1 - d/500 \quad (L \ge 1), \qquad EY_{2,L}^8 \le 1 - d/50 \quad (L \ge 2). \tag{Q64.1.9}$$

The same terminal message is used for every positive connector length in (Q64.1.8).

The four-branch zero-step R table and equal-arm pairs. The actual negative zero-step table satisfies

$$E[q(C^-)^4] \le 1. \tag{Q64.1.10}$$

The residual arm table is empty or has at least two distinct lengths after maximal equal-length pairing; a singleton S table is not added. An equal positive pair of length ℓ contributes $f_\ell h_\ell \le 1$ pointwise. Reversing every edge gives the negative pair inequality.

If $d = 0$, then $g = 0$, $p = q = 1$, and every $f_n = \bar{f}_n = 1$. The kernel need not vanish. Leaf integration gives value one for actual finite-tree messages with boundary values one. Expressions using δ_2/d below are restricted to $d > 0$.

Q64.2 Row bounds and the full positive path convolution

For $Ev = 0$, row centering gives

$$|Kv(x)|^2 = |E_y[(K(x,y) - g(x))v(y)]|^2 \le p(x)q(x)\|v\|_2^2. \tag{Q64.2.1}$$

Thus $r^2 \le pqd$. Row Cauchy for the nonnegative T-kernel yields

$$(TX)^2 \le qT(X^2), \qquad \beta(TX) \le E[T(X^2)] = E[pX^2] \le \beta(X), \tag{Q64.2.2}$$

where the last step uses $pq \le 1$. Since $\bar{f}_2 = q + r$,

$$\beta(\bar{f}_2) = 1 - 2d + E[r^2/q] \le 1 - d.$$

Consequently

$$\boxed{\beta(\bar{f}_n) \le 1 - d, \qquad n \ge 2.} \tag{Q64.2.3}$$

Define

$$S_n = \sum_{j=0}^{n} K^j \mathbf{1}, \quad G_n(z) = \sum_{j=0}^{n-1} z^j, \quad \tau = d - \delta_2,$$

$$\beta_1 = \beta_2 = 0, \qquad \beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1}\delta_i \quad (j \ge 3). \tag{Q64.2.4}$$

The scalar β_j is distinct from the quadratic functional $\beta(X)$. Skewness gives $ES_n = 1 - \beta_{n+1}$. The recurrence $S_n = AS_{n-1} + \beta_n 1$ and induction give

$$\boxed{S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}.} \tag{Q64.2.5}$$

The sum is empty for $n < 3$. Alternation and (Q64.1.4) imply

$$\tau \le \beta_j \le d \ (j \ge 3), \qquad cd \le \tau \le d - d^2 \le 2/9. \tag{Q64.2.6}$$

This is a positive finite convolution for every n.

We will also use

$$\|G_n(K)\|_{2\to 2} \le 1 \ (n \ge 3), \qquad \|S_j\|_2 \le 1 \ (j \ge 2). \tag{Q64.2.7}$$

For a full proof, diagonalize the real skew operator into rotation blocks and put $z = i\sqrt{x}$, $0 \le x \le \theta < 1/2$. For odd $n \ge 3$,

$$|G_n(z)|^2 = (1 + x^n)/(1 + x) \le 1.$$

For even n, the numerator is $(1 - (-1)^{n/2}x^{n/2})^2$. If $4 \mid n$, the claim is immediate. Otherwise $n \ge 6$, and with $m = n/2 \ge 3$,

$$2x^{m-1} + x^{2m-1} \le 2x^2 + x^5 \le 1/2 + 1/32 < 1.$$

This proves the first inequality in (Q64.2.7); the second follows from $S_j = G_{j+1}(K)1$. Since $|Ku| \le E|u| \le \|u\|_2$,

$$0 \le S_n = 1 + KS_{n-1} \le 2 \qquad (n \ge 3). \tag{Q64.2.8}$$

In particular $0 < f_n \le 2$ for $n \ge 3$, by (Q64.2.5).

For $0 \le Z \le U$, $\mu_Z = EZ$, and $V_Z = \operatorname{Var} Z$, we shall use the exact identity and bound

$$EZ^4 = \mu_Z^4 + E[(Z - \mu_Z)^2(Z^2 + 2\mu_Z Z + 3\mu_Z^2)] \le \mu_Z^4 + (U^2 + 2\mu_Z U + 3\mu_Z^2)V_Z. \tag{Q64.2.9}$$

Expansion and $E(Z - \mu_Z) = 0$ prove the identity. Every use below retains the nonnegativity of the mean and of its majorants.

Q64.3 Length four: a same-kernel fourth moment

Assume $d > 0$ and put $x = \delta_2/d$, so $d \le x \le \theta$. The exact path, mean, and upper bound are

$$f_4 = S_4 - d(1 + p) = 1 - 2d + (1 - d)g + r + s + t,$$

$$m = Ef_4 = 1 - (3 - x)d, \qquad 0 < f_4 \le U = 2 - d. \tag{Q64.3.1}$$

Skewness gives

$$\operatorname{Var} f_4 = d - \delta_2 - \delta_3 + \delta_4 - 3d^2 + 4d\delta_2 - \delta_2^2 + d^3. \tag{Q64.3.2}$$

Indeed $\langle g, s\rangle = -\delta_2$, $\langle r, t\rangle = -\delta_3$, odd total-degree pairings vanish, and the mean of $(1-d)g+r+s+t$ is $-d + \delta_2$. Using $\delta_4 \le \theta\delta_3$ and $\delta_3 \ge \delta_2^2/d$, define

$$V(d, x) = d[1 - x - cx^2 - 3d + 4dx - dx^2 + d^2],$$

$$F(d, x) = m^4 + (U^2 + 2mU + 3m^2)V(d, x). \tag{Q64.3.3}$$

Then $Ef_4^4 \le F(d, x)$.

For the exact x-dependence put

$$m_* = 1 - \frac{83}{32}d, \quad V_0 = d - 4d^2 + \frac{141}{32}d^3 - d^4,$$

$$b(d) = 1 - \frac{45}{16}d + 2d^2, \quad C_{\min} = 11 - 36d + 34d^2. \tag{Q64.3.4}$$

Throughout $0 < d \le 1/3$, $d \le x \le \theta$,

$$1 - 3d \le m \le m_*, \quad V \le V_0, \quad V_x \le -db(d), \quad U^2 + 2mU + 3m^2 \ge C_{\min}. \tag{Q64.3.5}$$

Here $b(d) \ge 41/144 > 0$. All variance majorants are nonnegative: V decreases with x, and

$$V(d, \theta)/d = c(1 - \theta^2) - \frac{1577}{1024}d + d^2 \ge \frac{27581}{294912} > 0. \tag{Q64.3.6}$$

The last quadratic decreases on the whole interval and the displayed constant is its value at $d = 1/3$.

Differentiating (Q64.3.3) at fixed d gives $F_x \le -dD(d)$, where

$$D = bC_{\min} - [4m_*^3 + (2U + 6m_*)V_0] = 7 - \frac{733}{16}d + \frac{34325}{256}d^2 - \frac{1737845}{8192}d^3 + \frac{79557}{512}d^4 - \frac{281}{16}d^5. \tag{Q64.3.7}$$

A coefficientwise lower polynomial is

$$P(d) = 7 - 46d + 134d^2 - 213d^3 + 155d^4 - 18d^5.$$

For $s = 3d \in [0, 1]$, the degree-five Bernstein coefficients of $P(s/3)$ are

$$7, \quad 59/15, \quad 106/45, \quad 133/90, \quad 362/405, \quad 41/81. \tag{Q64.3.8}$$

They are positive, so $D > 0$ and F decreases with x. A degree-n Bernstein list $(B_0, \dots, B_n)$ here always means the exact identity

$$P(s/3) = \sum_{i=0}^{n} B_i \binom{n}{i} s^i (1 - s)^{n-i}. \tag{Q64.3.9}$$

Thus each ledger specifies a finite algebraic certificate and its entire interval, not a finite collection of test points.

At $x = d$, expansion gives

$$F_0(d) := F(d, d) = 1 - d - 22d^2 + \frac{2959}{32}d^3 - \frac{1365}{8}d^4 + \frac{1351}{8}d^5 - \frac{689}{8}d^6 + \frac{679}{32}d^7 - 2d^8. \tag{Q64.3.10}$$

The expression $(F_0 - 1 + d)/d^2$ is at most

$$Q_4(d) = -22 + 93d - 170d^2 + 169d^3 - 86d^4 + 22d^5.$$

The degree-five Bernstein coefficients of $Q_4(s/3)$ are

$$-22, \quad -79/5, \quad -517/45, \quad -2279/270, \quad -2528/405, \quad -1118/243. \tag{Q64.3.11}$$

Every coefficient is at most -4. Therefore

$$\boxed{Ef_4^4 \le F(d, x) \le F_0(d) \le 1 - d - 4d^2.} \tag{Q64.3.12}$$

Changing K to $-K$ gives the same bounds for $\bar{f}_4$ with the same d, δ_2, x. This asserts common bounds, not equality between the two fourth moments for a fixed kernel.

Q64.4 Length five: a global strict fourth budget

The convolution gives

$$f_5 = S_5 - d(f_2 + p + 1) + \delta_2, \quad Ef_5 = 1 - 4d + 2\delta_2 + d^2 \le 1 - \frac{51}{16}d + d^2, \quad 0 < f_5 \le 2. \tag{Q64.4.1}$$

Its nonconstant polynomial is $G_5(K)g - d(2g + r)$. For $z = i\sqrt{x}$, $0 \le x \le \theta < 1/2$,

$$|G_5(z)|^2 = \frac{1 + x^5}{1 + x} \le 1 - \frac{7}{15}x,$$

$$\mathrm{Re}\{G_5(z)\overline{(2+z)}\} = 2 - x + x^2 \geq \frac{7}{4}, \qquad |2+z|^2 \leq \frac{9}{2}. \tag{Q64.4.2}$$

For the first inequality, $(1-x^4)/(1+x) \geq (15/16)/(3/2) = 5/8 > 7/15$. Integrating against the spectral measure at g and dropping the nonnegative square of the mean gives

$$\mathrm{Var}\, f_5 \leq d - \frac{7}{15}\delta_2 - \frac{7}{2}d^2 + \frac{9}{2}d^3 \leq d - \frac{37}{15}d^2 =: V_5. \tag{Q64.4.3}$$

This last majorant is nonnegative on $[0, 1/3]$, and is positive when $d > 0$.

For $m = 1 - ad$, the right side of (Q64.2.9) with $U = 2, V = V_5$ is

$$1 + (11 - 4a)d + C_2 d^2 + C_3 d^3 + C_4 d^4,$$

$$C_2 = 6a^2 - 10a - 407/15, \quad C_3 = -4a^3 + 3a^2 + (74/3)a, \quad C_4 = a^4 - (37/5)a^2. \tag{Q64.4.4}$$

This bound is increasing in a nonnegative mean majorant m. For $d \leq 1/6$, (Q64.4.1) allows $a = 145/48$, for which

$$11 - 4a = -13/12, \quad C_2 = -4971/1920 < -2, \quad C_3 = -926260/110592 < 0, \quad C_4 < 16. \tag{Q64.4.5}$$

The last inequality follows from $a^2 < 913/100$ and $(913/100)(913/100 - 37/5) = 157949/10000 < 16$. Thus (Q64.4.4) is at most

$$1 - \frac{13}{12}d - 2d^2 + 16d^4 \leq 1 - \frac{13}{12}d. \tag{Q64.4.6}$$

For $1/6 \leq d \leq 1/3$, use $a = 137/48$, so

$$11 - 4a = -5/12, \quad C_2 = -13051/1920 < -6, \quad C_3 = 203308/110592 < 3, \quad C_4 < 7. \tag{Q64.4.7}$$

Indeed $a^2 < 49/6$ and $(49/6)(49/6 - 37/5) = 1127/180 < 7$. Hence (Q64.4.4) is at most

$$1 - \frac{5}{12}d - 6d^2 + 3d^3 + 7d^4 \leq 1 - \frac{5}{12}d - \frac{38}{9}d^2 \leq 1 - \frac{121}{108}d \leq 1 - \frac{13}{12}d. \tag{Q64.4.8}$$

The intervals meet and $d = 0$ is immediate. We have proved

$$\boxed{Ef_5^4 \leq 1 - \frac{13}{12}d, \qquad E\bar{f}_5^4 \leq 1 - \frac{13}{12}d.} \tag{Q64.4.9}$$

The second statement is the global sign reversal of the first.

Q64.5 All lengths at least six: one finite mask proof

For every $n \geq 6$, the four final coefficients in (Q64.2.5) are at least τ, so

$$0 < f_n \leq Z := S_n - \tau(1 + p + f_2 + f_3). \tag{Q64.5.1}$$

This uses the full integer convolution, not extrapolation from samples. Since $1 + p + f_2 + f_3 = 4 - d + 3g + 2r + s$, and $\alpha = \beta_{n+1} \in [\tau, d]$,

$$EZ = 1 - \alpha - \tau(4 - 3d) \leq \mu_0 := 1 - \tau(5 - 3d), \qquad Z \leq 2 - \tau. \tag{Q64.5.2}$$

We have $\mu_0 \geq 1/9$, because $(d - d^2)(5 - 3d)$ increases on $[0, 1/3]$ to $8/9$.

The nonconstant polynomial of Z is

$$G_n(K)g - \tau(3g + 2r + s).$$

Put $P(z) = 3 + 2z + z^2$. For $z = i\sqrt{x}$,

$$|P(z)|^2 = 9 - 2x + x^2 \le 9, \qquad \mathrm{Re}\{(1-z)^{-1}\overline{P(z)}\} = \frac{3+x}{1+x} \ge \frac{7}{3}. \tag{Q64.5.3}$$

For $n \ge 6$, the geometric tail satisfies

$$|((1-z)^{-1} - G_n(z))\overline{P(z)}| \le 3\theta^3 < 1/4.$$

Therefore

$$\mathrm{Re}\{G_n(z)\overline{P(z)}\} \ge 2. \tag{Q64.5.4}$$

Together with (Q64.2.7), this gives the squared-norm majorant $d(1 - 4\tau + 9\tau^2) \le d(1 - 2\tau)$, since $\tau \le 2/9$. The mean of the nonconstant polynomial is $-\alpha + 2\tau d$; moreover $\alpha - 2\tau d \ge \tau(1 - 2d) \ge 0$. Thus

$$\mathrm{Var}\, Z \le d(1 - 2\tau) - \tau^2(1 - 2d)^2. \tag{Q64.5.5}$$

In the fourth-moment bound (Q64.2.9), the upper quantities μ_0, $2-\tau$, and the right side of (Q64.5.5) decrease with τ. They are nonnegative where used; the variance bound majorizes an actual nonnegative variance and increases when τ decreases. Hence substitute $\tau = cd$ and put

$$m = 1 - 5cd + 3cd^2, \quad U = 2 - cd, \quad V = d - (2c + c^2)d^2 + 4c^2d^3 - 4c^2d^4. \tag{Q64.5.6}$$

All substituted factors are nonnegative.

The coefficients of $m^4 + (U^2 + 2mU + 3m^2)V - 1$, degrees one through eight, are

$$\begin{array}{c|l} 1 & 11 - 20c = -7/8 \\ 2 & -66c + 139c^2 \\ 3 & 30c + 62c^2 - 444c^3 \\ 4 & -146c^2 + 474c^3 + 539c^4 \\ 5 & 27c^2 - 4c^3 - 1060c^4 \\ 6 & -66c^3 + 595c^4 \\ 7 & -48c^4 \\ 8 & -27c^4 \end{array} \tag{Q64.5.7}$$

At $c = 19/32$, degrees two through six equal respectively

$$\frac{10051}{1024} < 10, \quad -\frac{1745492}{32768} < -53, \quad \frac{120309387}{1048576} < 115, \quad -\frac{129037284}{1048576} < -122, \quad \frac{63054787}{1048576} < 61. \tag{Q64.5.8}$$

Discard the last two negative terms and divide by $d > 0$. The result is at most

$$Q_{\ge 6}(d) = -7/8 + 10d - 53d^2 + 115d^3 - 122d^4 + 61d^5.$$

The degree-five Bernstein coefficients of $Q_{\ge 6}(s/3)$ are

$$-7/8, \quad -5/24, \quad -47/360, \quad -233/1080, \quad -1099/3240, \quad -829/1944. \tag{Q64.5.9}$$

Each is at most $-47/360$. Including $d = 0$, we obtain

$$\boxed{Ef_n^4 \le 1 - \frac{47}{360}d, \quad E\bar{f}_n^4 \le 1 - \frac{47}{360}d, \qquad n \ge 6.} \tag{Q64.5.10}$$

Together with parts Q64.3–4 this yields the weaker common bound $1 - 47d/360$ for every $n \ge 4$.

Q64.6 Positive-kernel transfer from an ordinary norm

Let B be a positive linear operator on this same finite $L^2(\mu)$, with strictly positive row mass $a = B\mathbf{1}$. For real Y, row Jensen gives

$$|BY|^4 \le a^3 B(|Y|^4).$$

Raise this to power $2/3$, multiply by $q^{-1/3}$, and apply Hölder with exponents 3 and $3/2$:

$$I_q(BY) \le \{E[(B(|Y|^4))^2]\}^{1/3}\{E[a^3/\sqrt{q}]\}^{2/3}.$$

Thus, putting $R(a) = E[a^3/\sqrt{q}]$, we have

$$\boxed{I_q(BY)^3 \le \|B\|_{2\to 2}^2 E|Y|^8 R(a)^2.} \tag{Q64.6.1}$$

Cauchy–Schwarz supplies

$$R(a)^2 \le \beta(a) E a^4, \qquad R(\bar f_3)^2 \le (E\bar f_3^4/q)(E\bar f_3^2). \tag{Q64.6.2}$$

For the second, factor $\bar f_3^3/\sqrt{q} = (\bar f_3^2/\sqrt{q})\bar f_3$. The exact second moment is

$$E\bar f_3^2 = 1 - 3d - \delta_2 + \delta_3 + 3d^2 \le 1 - 2d, \tag{Q64.6.3}$$

using $\bar f_3 = q + r - s - d$, $\delta_3 \le \delta_2$, and $3d^2 \le d$. Consequently (Q64.1.7) gives

$$R(\bar f_3)^2 \le (1 - 23d/36)(1 - 2d) \le 1. \tag{Q64.6.4}$$

Q64.7 Two short ordinary operator bounds

Q64.7.1 A T-norm bound retaining δ_2

Put $\Pi = I - J$, $C = \Pi K \Pi$ on $1^\perp$, $U = -C^2$, and $h = Cg$. Then $0 \le U \le \theta I$, $\|h\|^2 = \delta_2 - d^2$, and

$$T = \begin{pmatrix} 1 & g^* \\ -g & -C \end{pmatrix}, \qquad T^*T = \begin{pmatrix} 1+d & (g-h)^* \\ g-h & U + g\otimes g \end{pmatrix}. \tag{Q64.7.1}$$

Our convention is $(u \otimes v)w = u\langle v, w\rangle$. Set

$$t_0 = d + k\delta_2, \quad \Lambda = (1+t_0)^2, \quad R = (\Lambda - U)^{-1}, \quad \sigma = \langle g, Rg\rangle.$$

Commutation with U and skewness of C give

$$\langle g-h, Rg\rangle = \sigma, \qquad \langle g-h, R(g-h)\rangle = (1+\Lambda)\sigma - d. \tag{Q64.7.2}$$

Moreover

$$\sigma \le \frac{d + (\delta_2 - d^2)/(\Lambda - \theta)}{\Lambda} \le \frac{t_0}{(1+t_0)^2} \le 1 - \Lambda^{-1/2} < 1 \tag{Q64.7.3}$$

when $t_0 > 0$. If $t_0 = 0$, then $g = 0, \sigma = 0$, and the same non-strict inequalities are immediate.

The rank-one inverse formula, verifiable by multiplication, gives

$$\langle g-h, (\Lambda - U - g\otimes g)^{-1}(g-h)\rangle = \Lambda\sigma - d + \frac{\sigma}{1-\sigma}. \tag{Q64.7.4}$$

The lower block is positive since $\sigma < 1$. Its Schur complement in $\Lambda I - T^*T$ is nonnegative exactly when $\Lambda(1-\sigma)^2 \ge 1$, which follows from (Q64.7.3). Therefore

$$\boxed{\|T\|_{2\to 2}^2 \le (1 + d + k\delta_2)^2 =: \Lambda(d, x) = (1 + d + kxd)^2.} \tag{Q64.7.5}$$

The same $x = \delta_2/d$ will be preserved in part Q64.8.

Q64.7.2 A T^2-norm bound using cyclic even/odd subspaces

Let $S = -K^2$. The finite cyclic subspace generated by 1 under K, and its orthogonal complement, are invariant. On the latter, $J = 0$, so $T^2 = K^2$ has norm at most θ. Inside the cyclic subspace put

$$E = \operatorname{span}\{S^j 1 : j \geq 0\}, \qquad O = \operatorname{span}\{KS^j 1 : j \geq 0\}. \tag{Q64.7.6}$$

Odd skew moments vanish, so these spaces are orthogonal and span the cyclic subspace. The operator K interchanges them and S preserves them.

Multiplying T^2 on the left by the orthogonal sign operator $P = \operatorname{diag}(I_E, -I_O)$ gives

$$N = PT^2 = \begin{pmatrix} J - S_E & 1 \otimes g \\ g \otimes 1 & S_O \end{pmatrix}. \tag{Q64.7.7}$$

This operator is self-adjoint and has the same norm as T^2. Let ν be the finite spectral measure of S at 1:

$$\operatorname{supp} \nu \subset [0, \theta], \quad \int d\nu = 1, \quad \int x \, d\nu = d, \quad \int x^2 \, d\nu = \delta_2. \tag{Q64.7.8}$$

This scalar spectral measure changes no original adjoint.

Take $\lambda \geq 1 > \theta$. In $\lambda I + N$ the lower block is positive and its Schur complement is

$$\lambda - S_E + \{1 - \langle g, (\lambda + S_O)^{-1} g \rangle\} J > 0,$$

because the bracket is at least $1 - d/\lambda > 0$. For $\lambda I - N$, the rank-one Schur condition is

$$\lambda h_\lambda t_\lambda \leq 1, \quad h_\lambda = \int \frac{d\nu(x)}{\lambda - x}, \quad t_\lambda = \int \frac{d\nu(x)}{\lambda + x}. \tag{Q64.7.9}$$

Indeed

$$1 + \langle g, (\lambda - S_O)^{-1} g \rangle = 1 + \int \frac{x \, d\nu(x)}{\lambda - x} = \lambda h_\lambda.$$

Put

$$\alpha = \int \frac{x^2 \, d\nu(x)}{\lambda^2 - x^2}, \qquad \eta = \int \frac{x \, d\nu(x)}{\lambda^2 - x^2}.$$

Then $h_\lambda = (1 + \alpha)/\lambda + \eta$ and $t_\lambda = (1 + \alpha)/\lambda - \eta$, whence

$$\lambda h_\lambda t_\lambda \leq (1 + \alpha)^2 / \lambda.$$

Choose

$$\kappa = (1 - \theta^2)^{-1} = 1024/855, \qquad \lambda = (1 + \kappa\delta_2)^2.$$

We have $\alpha \leq \kappa\delta_2$, so (Q64.7.9) follows. Both signs of N, and the orthogonal complement, are controlled. Hence

$$\boxed{\|T^2\|_{2\to 2} \leq (1 + \kappa\delta_2)^2, \qquad \|T^2\|_{2\to 2}^2 \leq (1 + d/2)^4.} \tag{Q64.7.10}$$

Here $\kappa\theta = 416/855 < 1/2$. This is an ordinary $L^2(\mu)$ bound, not the separate generic weighted two-step conjecture.

Q64.8 Full continuous payment for outer length two

The expression to pay is

$$(1-d)\Lambda(d,x)F(d,x), \qquad 0<d\le 1/3, \quad d\le x\le \theta. \tag{Q64.8.1}$$

Separately maximizing the factors in x would lose needed same-kernel information. By part Q64.3, $F_x \le -dD$, while

$$\Lambda_x/\Lambda = \frac{2kd}{1+d+kxd} \le 2kd.$$

All factors are nonnegative. Since $F \le F_0$,

$$(\Lambda F)_x \le -\Lambda d\{D(d) - 2kF_0(d)\}. \tag{Q64.8.2}$$

We now certify positivity of this bracket on the whole interval.

The coefficients of $D - 2kF_0$, in degrees zero through eight, are

$$\frac{69}{19},\quad -\frac{12903}{304},\quad \frac{1012623}{4864},\quad -\frac{81499311}{155648},\quad \frac{7102623}{9728},\quad -\frac{178267}{304},\quad \frac{5512}{19},\quad -\frac{1358}{19},\quad \frac{128}{19}. \tag{Q64.8.3}$$

They are coefficientwise at least those of

$$L(d) = 18/5 - (85/2)d + 208d^2 - 524d^3 + 730d^4 - 587d^5 + 290d^6 - 72d^7 + 6d^8. \tag{Q64.8.4}$$

For $s = 3d$, the power coefficients of $65610L(s/3)$ are

$$236196,\ -929475,\ 1516320,\ -1273320,\ 591300, -158490,\ 26100,\ -2160,\ 60. \tag{Q64.8.5}$$

Its degree-eight Bernstein coefficients are $N_i/\binom{8}{i}$, where

$$(N_0,\dots,N_8) = (236196,\ 960093,\ 1623483,\ 1532601,\ 971595, 495261,\ 204543,\ 54333,\ 6531). \tag{Q64.8.6}$$

Thus the Bernstein coefficients of $L(s/3)$ are $N_i/[65610\binom{8}{i}]$, all strictly positive. The finite rational identities (Q64.8.3)–(Q64.8.6) prove $D - 2kF_0 > 0$. Consequently

$$(1-d)\Lambda(d,x)F(d,x) \le (1-d)(1+d+kd^2)^2(1-d-4d^2). \tag{Q64.8.7}$$

The final product is exactly

$$1 - \frac{50}{19}d^2 - \frac{140}{19}d^3 - \frac{3251}{361}d^4 + \frac{612}{361}d^5 + \frac{1792}{361}d^6 + \frac{4096}{361}d^7. \tag{Q64.8.8}$$

The last three terms are at most

$$2d^5 + 5d^6 + 12d^7 \le (5/3)d^4,$$

whereas $3251/361 > 9$. Thus (Q64.8.8) is at most

$$1 - \frac{50}{19}d^2 - \frac{140}{19}d^3 - \frac{22}{3}d^4 \le 1.$$

The zero boundary is direct. We have proved

$$\boxed{(1-d)\Lambda(d,x)F(d,x) \le 1} \tag{Q64.8.9}$$

on the entire permitted region with the same kernel moments.

Q64.9 Ordinary contraction of every power of length at least three

This section proves the long-power bound associated with R157/R158.

Q64.9.1 Actual-kernel budgets and the two-dimensional comparison

For $a, b, c \in [-1, 1]$,

$$(1 + ab)^2 - (a + b)^2 = (1 - a^2)(1 - b^2) \geq 0, \qquad 1 + ab \geq 0.$$

Hence $ab + c(a + b) \geq ab - |a + b| \geq -1$. Apply this to $a = K(x, y), b = K(y, z), c = K(z, x)$ and integrate over three independent original variables. Each product has mean $-d$; therefore $d \leq 1/3$, including all atomic coincidences.

Use $C = \Pi K \Pi$, $U = -C^2$, and put $H = Cg$, $r_C = \|C\|$. In the decomposition $\mathbb{R}1 \oplus 1^{\perp}$,

$$K = \begin{pmatrix} 0 & -g^* \\ g & C \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & -g^* \\ g & C \end{pmatrix}.$$

Orthogonality of Hilbert–Schmidt blocks gives

$$2d + \|C\|_{\mathrm{HS}}^2 = \|K\|_{\mathrm{HS}}^2 = E_{x,y} K(x, y)^2 \leq 1. \tag{Q64.9.1}$$

A real skew operator satisfies $2\|C\|^2 \leq \|C\|_{\mathrm{HS}}^2$. Indeed, for unit u with $Cu \neq 0$, the unit vector $v = Cu/\|Cu\|$ is orthogonal to u, and $|\langle Cv, u\rangle| = \|Cu\|$. The Hilbert–Schmidt square contains at least $\|Cu\|^2 + \|Cv\|^2 \geq 2\|Cu\|^2$. Take the supremum. Consequently

$$r_C^2 \leq 1/2 - d, \qquad r_C \leq 3/4, \qquad \langle g, H\rangle = 0, \qquad \|H\|^2 \leq (1/2 - d)d. \tag{Q64.9.2}$$

For any block operator

$$B = \begin{pmatrix} a & -w^* \\ v & D \end{pmatrix}, \qquad 0 \leq a \leq \bar{a}, \quad \|v\|, \|w\| \leq N, \quad \|D\| \leq e,$$

with $\bar{a}, N, e \geq 0$, componentwise norm estimates imply

$$\|B\| \leq \left\| \begin{pmatrix} \bar{a} & N \\ N & e \end{pmatrix} \right\|. \tag{Q64.9.3}$$

The symmetric comparison matrix has nonnegative entries and trace, so its largest eigenvalue has modulus at least that of the other. To bound it by $\rho = 1 - d/8$, it is therefore enough to show $\rho I - \left(\begin{smallmatrix} \bar{a} & N \\ N & e \end{smallmatrix}\right)$ is positive semidefinite. The original block operator need not be self-adjoint.

Q64.9.2 The cube

Exact multiplication gives

$$A^3 = \begin{pmatrix} 1 - 2d & -w_3^* \\ v_3 & D_3 \end{pmatrix},$$

$$v_3 = (1 - d - U)g + H, \quad w_3 = (1 - d - U)g - H,$$

$$D_3 = C^3 - g \otimes g - H \otimes g + g \otimes H. \tag{Q64.9.4}$$

Commutation and skewness eliminate the cross term in the squared off-diagonal norms. The scalar polynomial $(1 - d - u)^2 + u$ decreases for $0 \leq u \leq 1/2 - d$. Thus

$$\|v_3\|^2 = \|w_3\|^2 \leq d(1 - d)^2. \tag{Q64.9.5}$$

The combined skew rank-two operator $g \otimes H - H \otimes g$ has norm $\|g\|\|H\|$, since $g \perp H$. Hence

$$\|D_3\| \le r_C^3 + d + r_C d \le d + r_C/2 \le d + 3/8. \tag{Q64.9.6}$$

Use (Q64.9.3) with $\bar{a} = 1 - 2d$, $N = \sqrt{d}(1-d)$, $e = d + 3/8$. The two diagonals of $\rho I - M$ are $15d/8$ and $5/8 - 9d/8 \ge 1/4$. Its determinant is

$$d\left(\frac{11}{64} - \frac{7}{64}d - d^2\right) \ge \frac{7}{288}d. \tag{Q64.9.7}$$

The bracket decreases to $7/288$ at $d = 1/3$. This proves $\|A^3\| \le \rho$, including $d = 0$.

Q64.9.3 The fourth power

Multiplication by A gives

$$a_4 = 1 - 3d + d^2 + \|H\|^2, \quad v_4 = (1 - 2d - U)g + (1 - d - U)H,$$

$$w_4 = (1 - 2d - U)g - (1 - d - U)H,$$

$$D_4 = U^2 - (1-d)g \otimes g + (Ug) \otimes g + g \otimes (Ug) + H \otimes H + g \otimes H - H \otimes g. \tag{Q64.9.8}$$

Here

$$1 - 3d + d^2 \le a_4 \le 1 - 5d/2. \tag{Q64.9.9}$$

The lower bound is at least $1/9$, so the constant block is nonnegative. The squared off-diagonal norm is obtained by integrating

$$f_4(u) = (1 - 2d - u)^2 + u(1 - d - u)^2.$$

Its derivative is

$$f_4'(u) = -1 + 2d + d^2 + 2(-1 + 2d)u + 3u^2.$$

This is convex in u; at $u = 0$ it is at most $-2/9$, and at $u = 1/2 - d$ it is $3d - 5/4 \le -1/4$. It is therefore nonpositive throughout the spectral interval. Thus

$$\|v_4\|^2 = \|w_4\|^2 \le d(1 - 2d)^2. \tag{Q64.9.10}$$

Bounding the last two skew terms of D_4 together, and the other terms separately, gives

$$\|D_4\| \le r_C^4 + (1-d)d + 3r_C^2 d + r_C d \le e_4 := 1/4 + (9/4)d - 3d^2 \le 2/3. \tag{Q64.9.11}$$

Use $\bar{a} = 1 - 5d/2$, $N = \sqrt{d}(1 - 2d)$, $e = e_4$ in (Q64.9.3). The diagonals of $\rho I - M$ are $19d/8$ and $3/4 - 19d/8 + 3d^2 \ge 7/24$. The determinant is

$$d\left(\frac{25}{32} - \frac{105}{64}d + \frac{25}{8}d^2\right) \ge \frac{15}{64}d. \tag{Q64.9.12}$$

This proves $\|A^4\| \le \rho$.

Q64.9.4 The fifth power

The next multiplication gives

$$a_5 = 1 - 4d + 3d^2 + 2\|H\|^2,$$

$$v_5 = [a_4 - (1-d)U + U^2]g + (1 - 2d - U)H, \quad w_5 = [a_4 - (1-d)U + U^2]g - (1 - 2d - U)H. \tag{Q64.9.13}$$

The scalar block obeys

$$0 \le (1-d)(1-3d) \le a_5 \le 1 - 3d + d^2. \tag{Q64.9.14}$$

To control the off-diagonal terms, put $b = 1-2d \in [1/3, 1]$, $c_0 = 1-d = (1+b)/2$, and $\vartheta = a_4$. The spectral variable satisfies $0 \le u \le b/2$, and $\vartheta \ge \vartheta_0 = (b^2 + 4b - 1)/4$. For $P(u) = \vartheta - c_0 u + u^2$, we have

$$P(u) \ge \vartheta_0 - b/4 = (b^2 + 3b - 1)/4 \ge 1/36. \tag{Q64.9.15}$$

Indeed P decreases on this interval. The relevant norm polynomial and its derivative are

$$f_5(u) = P(u)^2 + u(b-u)^2, \qquad f_5'(u) = -2P(u)(c_0 - 2u) + (b-u)(b-3u). \tag{Q64.9.16}$$

On $b/3 \le u \le b/2$ both terms are nonpositive. For $0 \le u \le b/3$, the negative derivative increases with ϑ, and its value at ϑ_0 gives

$$\begin{aligned} -f_5'(u) &\ge \frac{b^3 + b^2 + 3b - 1}{4} - \frac{(1+b)(3b-1)}{2}u + u^2(3b - 4u) \\ &\ge \frac{-3b^3 - b^2 + 11b - 3}{12} \ge 1/27. \end{aligned} \tag{Q64.9.17}$$

For the last step the numerator is increasing on $[1/3, 1]$, since its derivative is $(1-b)(9b+11)$, and its value at $1/3$ is $4/9$. Therefore $f_5(u) \le f_5(0) = a_4^2$, and

$$\|v_5\|^2 = \|w_5\|^2 \le d a_4^2 \le d(1 - 5d/2)^2. \tag{Q64.9.18}$$

The lower-right block can be kept as $D_5 = -g \otimes w_4 + C D_4$. Equations (Q64.9.10)–(Q64.9.11) give

$$\|D_5\| \le d(1-2d) + (3/4)e_4 = e_5 := 3/16 + (43/16)d - (17/4)d^2 \le 5/6. \tag{Q64.9.19}$$

It is nonnegative by its first expression.

Use $\bar{a} = 1 - 3d + d^2$, $N = \sqrt{d}(1 - 5d/2)$, and $e = e_5$. The diagonals of $\rho I - M$ are $d(23/8 - d)$ and $13/16 - 45d/16 + 17d^2/4 \ge 1/8$. Its determinant is

$$d\left[\frac{171}{128} - \frac{499}{128}d + \frac{281}{32}d^2 - \frac{17}{4}d^3\right] > 0 \quad (d > 0). \tag{Q64.9.20}$$

The constant and linear terms in the bracket sum to at least $7/192$; the remaining part is $d^2[281/32 - (17/4)d] \ge 0$. The zero boundary is positive semidefinite. Thus $\|A^5\| \le \rho$.

Q64.9.5 Every integer power

For $\ell = 3m$, use m factors A^3. For $\ell = 3m + 1 \ge 4$, use A^4 and $m - 1$ factors A^3. For $\ell = 3m + 2 \ge 5$, use A^5 and $m - 1$ factors A^3. Submultiplicativity and adjunction prove

$$\boxed{\|A^\ell\|,\ \|T^\ell\| \le (1 - d/8)^{\lfloor \ell/3 \rfloor} \le 1, \qquad \ell \ge 3.} \tag{Q64.9.21}$$

For $d = 0$, A is block diagonal with blocks $1, C$, so these powers have norm exactly one. This handles the boundary without dividing by d.

M.16 Root multipliers and all outer connectors

Q66.1 Original model and exact old interfaces

Let X be a nonempty finite probability space (X,μ), with zero-mass atoms deleted. Assume

$$K(x,y)=-K(y,x),\qquad |K(x,y)|\le 1.$$

All expectations, inner products, and adjoints use the same original measure μ. Put

$$Jv=(Ev)1,\quad A=J+K,\quad T=J-K=A^*,\quad g=K1,\quad p=1+g,\quad q=1-g,$$

$$f_n=A^n\mathbf{1},\quad \bar f_n=T^n\mathbf{1},\quad h_n=Tf_{n-1},\quad f_0=\bar f_0=1,\qquad I_q(X)=E[q^{-1/3}|X|^{8/3}]. \tag{Q66.1.1}$$

Since $K(x,x)=0$, $p(x),q(x)\ge\mu(x)>0$. All actual path messages are positive, and A,T preserve order.

Write

$$\delta_j=\|K^j\mathbf{1}\|_2^2,\quad d=\delta_1,\quad r=Kg,\quad s=Kr,\quad t=Ks,$$

$$\theta=13/32,\quad c=19/32,\quad k=32/19,\quad \tau=d-\delta_2. \tag{Q66.1.2}$$

The original spectral estimate and elementary moment relations give

$$\|K\|^2\le\theta,\quad 0\le d\le 1/3,\quad d^2\le\delta_2\le\theta d,\quad \delta_{j+1}\le\theta\delta_j,\quad cd\le\tau\le d-d^2\le 2/9. \tag{Q66.1.3}$$

We use the following exact interfaces.

1. The ordinary path multiplier, from the Q3 operator estimate T061 and Q4 C115 formula (8.3), after global sign reversal and ordinary adjunction:

$$\|TM_{\bar f_a}\|_{2\to2}^2\le 1-d/72,\qquad a\ge 2. \tag{Q66.1.4}$$

Multiplication and propagation are in precisely this order.

2. The C064 positive-kernel transfer. For a positive operator B with row $a=B\mathbf{1}>0$,

$$I_q(BY)^3\le\|B\|_{2\to2}^2E|Y|^8\left(E[a^3/\sqrt{q}]\right)^2. \tag{Q66.1.5}$$

Indeed $|BY|^4\le a^3B(|Y|^4)$, followed by Hölder with exponents $3,3/2$, proves it. In particular, if $\beta(X)=E[X^2/q]$,

$$\left(E[\bar f_n^3/\sqrt{q}]\right)^2\le\beta(\bar f_n)E\bar f_n^4,\qquad \beta(\bar f_n)\le 1-d\quad(n\ge 2). \tag{Q66.1.6}$$

The latter follows from row Cauchy $\beta(TX)\le E[pX^2]\le\beta(X)$, $\bar f_2=q+r$, and $r^2\le pqd$.

3. The two Q5-C064 ordinary short norms:

$$\|T\|^2\le(1+d+k\delta_2)^2\le(1+kd)^2,\qquad \|T^2\|^2\le(1+d/2)^4. \tag{Q66.1.7}$$

The second inequality in the first chain uses $\delta_2\le\theta d$ and $1+k\theta=k$. These are ordinary $L^2(\mu)$ norms, not a generic weighted two-step contraction.

4. C064 already gives (0.2) for $a=2$, all $M\ge1$, and for every $a\ge2$ when $M=1$ or $M\ge4$.

5. C058 gives, with the same actual terminal qf_b at every inner length,

$$E[(pT^L(qf_b))^8]\le 1-d/12000,\qquad b\ge2,L\ge1. \tag{Q66.1.8}$$

C049/C050 give $1 - d/500$ when $b = 2, L \geq 1$, and $1 - d/50$ when $b = 2, L \geq 2$.

6. C143 supplies its exact zero-step R directions with $E[q(C^-)^4] \leq 1$, and actual equal pairs $f_\ell h_\ell \leq 1$, including their complete reversals.

If $d = 0$, then $g = 0$, $p = q = 1$, and all $f_n = \bar{f}_n = 1$. The kernel need not vanish. All actual tree messages are then one by leaf integration. New expressions involving δ_2/d or τ/d below are restricted to $d > 0$; the all-real maps at $d = 0$ follow directly from (Q66.1.4)–(Q66.1.6).

Q66.2 Full path convolution and a sharper geometric-sum estimate

Define

$$S_n = \sum_{j=0}^{n} K^j \mathbf{1}, \qquad G_n(z) = \sum_{j=0}^{n-1} z^j,$$

$$\gamma_1 = \gamma_2 = 0, \qquad \gamma_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1} \delta_i \quad (j \geq 3).$$

Skewness gives $ES_n = 1 - \gamma_{n+1}$, and $S_n = AS_{n-1} + \gamma_n 1$ gives the full finite identity

$$S_n = f_n + \sum_{j=3}^{n} \gamma_j f_{n-j}, \qquad \tau \leq \gamma_j \leq d \quad (j \geq 3). \tag{Q66.2.1}$$

Every summand is nonnegative. This proves statements for every integer n, without assuming ordinary path means decrease. For the same finite skew spectral decomposition,

$$\|G_n(K)\| \leq 1 \quad (n \geq 3), \qquad \|S_j\|_2 \leq 1 \quad (j \geq 2), \qquad 0 \leq S_n \leq 2 \quad (n \geq 3). \tag{Q66.2.2}$$

The scalar verification uses

$$|G_n(i\sqrt{x})|^2 = \begin{cases} (1 + x^n)/(1 + x), & n \text{ odd}, \\ (1 - (-1)^{n/2} x^{n/2})^2/(1 + x), & n \text{ even}, \end{cases} \qquad 0 \leq x \leq \theta.$$

For $n = 2 \bmod 4,\ n \geq 6$, the required inequality follows from $2x^2 + x^5 < 1$; the other cases are immediate. Finally $|KS_{n-1}| \leq \|S_{n-1}\|_2$ proves the pointwise bound in (Q66.2.2). Positivity in (Q66.2.1) gives $0 < f_n \leq 2$ for $n \geq 3$.

We need a uniform improvement which retains δ_2:

$$\boxed{|G_n(i\sqrt{x})|^2 \leq 1 - \frac{7}{15}x, \quad n \geq 5, \quad 0 \leq x \leq \theta.} \tag{Q66.2.3}$$

For every $n \geq 5$, its numerator is at most $(1 + x^3)^2$. For odd n use $x^n \leq 2x^3 + x^6$; for $4 \mid n$ use a numerator at most one; the remaining even case has $n/2 \geq 3$. It therefore suffices to check

$$8 - 7x - 30x^2 - 15x^5 \geq 0. \tag{Q66.2.4}$$

The polynomial decreases on $[0, \theta]$, and its endpoint is

$$\frac{1311885}{33554432} > 0.$$

This proves (Q66.2.3), including $x = 0$. Integration against the spectral measure at g gives

$$\|G_n(K)g\|^2 \leq d - \frac{7}{15}\delta_2 \quad (n \geq 5). \tag{Q66.2.5}$$

A second uniform improvement is needed for $n \ge 6$. Put $P(z) = 3 + 2z + z^2$. Then, for $z = i\sqrt{x}$,

$$|P(z)|^2 = 9 - 2x + x^2 \le 9, \qquad \mathrm{Re}\{(1-z)^{-1}\overline{P(z)}\} = \frac{3+x}{1+x} \ge \frac{109}{45}.$$

The geometric tail is bounded by $3\theta^3 = 6591/32768$. Since

$$\frac{109}{45} - \frac{6591}{32768} - \frac{11}{5} = \frac{6217}{294912} > 0,$$

we obtain the all-length statement

$$\boxed{\mathrm{Re}\{G_n(z)\overline{P(z)}\} \ge 11/5, \qquad n \ge 6.} \tag{Q66.2.6}$$

Q66.3 Every row length at least six: a uniform joint payment

Q66.3.1 The actual finite mask and its variance

For $n \ge 6$, retain the four final terms in (Q66.2.1):

$$0 < f_n \le Z := S_n - \tau(1 + p + f_2 + f_3). \tag{Q66.3.1}$$

Here

$$1 + p + f_2 + f_3 = 4 - d + 3g + 2r + s.$$

With $\alpha = \gamma_{n+1} \in [\tau, d]$,

$$EZ = 1 - \alpha - \tau(4 - 3d) \le m := 1 - (5 - 3d)\tau, \qquad Z \le U := 2 - \tau. \tag{Q66.3.2}$$

Moreover $m \ge 1/9$: the maximum of $(d - d^2)(5 - 3d)$ on $[0, 1/3]$ is $8/9$, since its derivative $5 - 16d + 9d^2$ is positive there.

The nonconstant part of Z is $G_n(K)g - \tau P(K)g$. Equations (Q66.2.5)–(Q66.2.6) give

$$\|G_n(K)g - \tau P(K)g\|^2 \le d - \frac{7}{15}\delta_2 - \frac{22}{5}\tau d + 9\tau^2 d \le d - \frac{7}{15}\delta_2 - \frac{12}{5}\tau d,$$

where $9\tau^2 \le 2\tau$. Its mean is $-\alpha + 2\tau d$, whose absolute value is at least $\tau(1 - 2d)$. Hence

$$\mathrm{Var}\, Z \le V(d, \tau) := d - \frac{7}{15}\delta_2 - \frac{12}{5}\tau d - \tau^2(1 - 2d)^2. \tag{Q66.3.3}$$

Using $\delta_2 = d - \tau$, this is

$$V = \frac{8}{15}d + \left(\frac{7}{15} - \frac{12}{5}d\right)\tau - (1 - 2d)^2\tau^2. \tag{Q66.3.4}$$

This variance majorant is positive over the full continuous region

$$0 < d \le 1/3, \qquad cd \le \tau \le d - d^2. \tag{Q66.3.5}$$

Indeed $\delta_2/d \le \theta$, $\tau \le 2/9$, $\tau \le d$, and $d(1 - 2d)^2 \le 2/27$ imply

$$V/d \ge \frac{389}{480} - \frac{8}{15} - \frac{2}{27} = \frac{877}{4320} > 0. \tag{Q66.3.6}$$

The bound $d(1-2d)^2 \le 2/27$ follows by differentiating on $[0, 1/3]$; the only interior maximum occurs at $d = 1/6$.

For $0 \le W \le U$, with mean μ_W, the exact identity

$$EW^4 = \mu_W^4 + E[(W - \mu_W)^2(W^2 + 2\mu_W W + 3\mu_W^2)]$$

gives

$$EW^4 \le \mu_W^4 + (U^2 + 2\mu_W U + 3\mu_W^2)\operatorname{Var} W. \tag{Q66.3.7}$$

Its right side is increasing in a nonnegative mean majorant. Thus, putting $C = U^2 + 2mU + 3m^2$,

$$Ef_n^4 \le EZ^4 \le F(d,\tau) := m^4 + C\,V(d,\tau). \tag{Q66.3.8}$$

Q66.3.2 A full-region monotonicity proof in the shared moment

We prove that $F(d,\tau)$ decreases with τ throughout (Q66.3.5). This is a statement about an explicit fourth-moment majorant, not monotonicity of a mean-plus-variance functional.

Put $B = 5 - 3d$, so $B \ge 4$, $m = 1 - B\tau$, $U = 2 - \tau$. Since $0 \le m \le 1$, $0 \le U \le 2$,

$$-C_\tau = 2(U + m) + 2B(U + 3m) \ge 4C. \tag{Q66.3.9}$$

For a direct verification, after replacing B by four the difference from $4C$ is

$$10U + 26m - 4U^2 - 8mU - 12m^2 \ge 0.$$

For fixed m, this is concave in U, so its minimum is on $U = 0, 2$. At $U = 0$ it is $26m - 12m^2 \ge 0$. At $U = 2$ it is $4 + 10m - 12m^2$, whose endpoint values on $0 \le m \le 1$ are four and two.

Write $V = v_0 + v_1\tau - w\tau^2$, where

$$v_0 = 8d/15, \quad v_1 = 7/15 - (12/5)d, \quad w = (1-2d)^2.$$

If $v_1 < 0$, then $V_\tau < 0$, and (Q66.3.6), (Q66.3.9) immediately give $F_\tau < 0$. If $v_1 \ge 0$, then

$$V_\tau - 4V = v_1 - 4v_0 - 4v_1\tau - 2w\tau(1-2\tau) \le \frac{7 - 68d}{15}. \tag{Q66.3.10}$$

For $d \ge 7/68$, this again proves $F_\tau < 0$. For $0 \le d \le 7/68$, use $C \le 11$, $m \ge 1 - 5d$, and (Q66.3.9) to obtain

$$F_\tau \le -16(1-5d)^3 + \frac{11}{15}(7 - 68d) =: q_0(d).$$

On this interval

$$q_0'(d) = 240(1-5d)^2 - \frac{748}{15} \ge \frac{461648}{69360} > 0.$$

Thus $q_0(d) \le q_0(7/68) = -16(33/68)^3 < 0$. All cases are covered, proving

$$F(d,\tau) \le F(d, cd) =: F_*(d). \tag{Q66.3.11}$$

The exact same δ_2 was retained through (Q66.3.3)–(Q66.3.11). No independent replacement of its variance contribution by zero was made.

Q66.3.3 The complete finite scalar certificate

At $\tau = cd$, the quantities in (Q66.3.8) are

$$m = 1 - 5cd + 3cd^2, \quad U = 2 - cd,$$

$$V = \frac{389}{480}d - \left(\frac{12}{5}c + c^2\right)d^2 + 4c^2d^3 - 4c^2d^4, \qquad c = 19/32. \tag{Q66.3.12}$$

Expanding $F_* = m^4 + (U^2 + 2mU + 3m^2)V$, its coefficients after the constant one are as follows:

Table 20: Coefficients of d^j in the fourth-moment envelope $F_*(d)$.

j	Exact coefficient	Convenient upper coefficient
1	$-1421/480$;	$-59/20$.
2	$207461/15360$;	$271/20$.
3	$-26785934/491520$;	$-1089/20$.
4	$575254583/5242880$;	$2196/20$.
5	$-1837517436/15728640$;	$-2336/20$.
6	$303421583/5242880$;	$1158/20$.
7	$-48c^4$;	0.
8	$-27c^4$;	0.

Each entry in the last column is at least the exact coefficient. Let $\mathcal{P}(d)$ be the degree-six upper polynomial with constant one and that column's coefficients. It is nonnegative because it majorizes $F_*(d) > 0$.

Since $k = 32/19 < 17/10$,

$$(1-d)(1+kd)^2F_*(d) \le (1-d)(1+17d/10)^2\mathcal{P}(d). \tag{Q66.3.13}$$

The factor on the right outside $\mathcal{P}$ is $1 + (12/5)d - (51/100)d^2 - (289/100)d^3$. The right side of (Q66.3.13), minus one, is $dQ(d)/2000$, where

$$\begin{aligned} Q(d) = \; & -1100 + 11920d - 46631d^2 - 38530d^3 + 270660d^4 \\ & - 242115d^5 - 237588d^6 + 616046d^7 - 334662d^8. \end{aligned} \tag{Q66.3.14}$$

For $d \le 1/3$,

$$-237588d^6 + 616046d^7 \le -\frac{96718}{3}d^6 \le 0.$$

Dropping this pair and the final negative term leaves the degree-five upper polynomial

$$Q_5 = -1100 + 11920d - 46631d^2 - 38530d^3 + 270660d^4 - 242115d^5.$$

For $z = 3d \in [0,1]$, its degree-five Bernstein coefficients are

$$-1100, \quad -\frac{916}{3}, \quad -\frac{2591}{90}, \quad -\frac{111529}{270}, \quad -\frac{755394}{810}, \quad -\frac{112574}{81}. \tag{Q66.3.15}$$

Each is at most $-2591/90$. Here a Bernstein list $(B_i)_{i=0}^n$ means the exact identity

$$Q_5(z/3) = \sum_{i=0}^{n} B_i \binom{n}{i} z^i(1-z)^{n-i}.$$

Consequently, with $\eta = 2591/180000$,

$$\boxed{(1-d)(1+kd)^2 E f_n^4 \le 1-\eta d, \qquad n \ge 6.} \tag{Q66.3.16}$$

Global sign reversal gives the same bound for $E\bar{f}_n^4$. The $d = 0$ boundary is direct. These are common bounds under sign reversal, not an equality of opposite fourth moments for a fixed kernel.

Q66.4 The remaining row length five

We need a stronger global fifth-path fourth budget than the one used in C064:

$$\boxed{E f_5^4,\ E\bar{f}_5^4 \le 1-\frac{3}{2}d.} \tag{Q66.4.1}$$

The exact convolution gives

$$f_5 = S_5 - d(f_2+p+1) + \delta_2 = S_5 - d(f_2+p) - \tau.$$

Therefore

$$E f_5 = 1 - 4d + 2\delta_2 + d^2 \le m := 1 - \frac{51}{16}d + d^2, \qquad 0 < f_5 \le U := 2 - cd. \tag{Q66.4.2}$$

The mean majorant is positive throughout the interval; at $d = 1/3$ it is $7/144$, and it is decreasing on the whole interval.

The nonconstant polynomial is $G_5(K)g - d(2g+r)$. For $z = i\sqrt{x}$,

$$|G_5(z)|^2 \le 1 - \frac{7}{15}x, \quad \mathrm{Re}\{G_5(z)\overline{(2+z)}\} = 2 - x + x^2 \ge 7/4, \quad |2+z|^2 \le 9/2.$$

Thus, dropping the nonnegative square of the mean,

$$\mathrm{Var}\, f_5 \le d - \frac{7}{15}\delta_2 - \frac{7}{2}d^2 + \frac{9}{2}d^3 \le d - \frac{37}{15}d^2 =: V_5. \tag{Q66.4.3}$$

The last quantity is nonnegative, and positive for $d > 0$. Apply (Q66.3.7) with (Q66.4.2)–(Q66.4.3). The coefficients of the resulting polynomial after its constant one are

Table 21: Coefficients of d^j in the fifth-path fourth-moment envelope.

j	Exact coefficient	Convenient upper coefficient
1	$-7/4$;	$-7/4$.
2	$4589/1920$;	$12/5$.
3	$-91549/2560$;	$-143/4$.
4	$99070783/983040$;	$504/5$.
5	$-352321/3072$;	$-573/5$.
6	$36839/640$;	$288/5$.
7	$-51/4$;	retained with degree eight.
8	1;	retained with degree seven.

The final two terms sum to $d^7(-51/4+d) \le 0$. After discarding them, the difference from $1-3d/2$, divided by $d > 0$, is at most

$$Q_{\text{five}}(d) = -\frac{1}{4} + \frac{12}{5}d - \frac{143}{4}d^2 + \frac{504}{5}d^3 - \frac{573}{5}d^4 + \frac{288}{5}d^5. \tag{Q66.4.4}$$

The degree-five Bernstein coefficients of $Q_{\text{five}}(z/3)$ are

$$-\frac{1}{4},\quad -\frac{9}{100},\quad -\frac{589}{1800},\quad -\frac{353}{600},\quad -\frac{1057}{1350},\quad -\frac{13}{15}. \tag{Q66.4.5}$$

They are all negative. This proves (Q66.4.1) on the whole domain, including $d = 0$ separately. The improved pointwise upper bound $f_5 \le 2 - \tau$, which follows from the exact positive convolution, is retained in this proof.

Q66.5 Additional strict ordinary loss for the actual root $\bar{f}_3$

We prove the strengthened ordinary multiplier

$$\boxed{\|TM_{\bar{f}_3}\|^2 \le (1-d/2)^2(1-d/72).} \tag{Q66.5.1}$$

We use the S_3 case of the Q3 T061 Schur argument below.

Set

$$H = S_3 = 1 + g + r + s,\quad a = H - 1,\quad R = (I + K^2)^{-1},\quad e = -R(K^4 + K^5)1,\quad u = 1 + e.$$

The inverse exists by $\|K\|^2 \le 13/32 < 1$. Let

$$P = TA,\quad t_0 = 1 - EH = d,\quad v = E(gH),\quad b = E(He),\quad c_0 = E(ge),\quad a_0 = E|ae|.$$

The elementary exact formulas are

$$Pz = qEz - E(gz) - K^2 z,\qquad (I - K^2)a - g - r = (I + K^2)e. \tag{Q66.5.2}$$

Expanding by (Q66.5.2) gives

$$M_H P M_H u - u = -a^2 + ae + H\{q(b - t_0) - (v + c_0) + TK(ae)\}. \tag{Q66.5.3}$$

For a direct check, write $E(Hu) = 1 - t_0 + b$, $E(gHu) = v + c_0 + E(gae)$, and $K^2(Hu) = K^2 1 + K^2 a + K^2 e + K^2(ae)$. The last two ae contributions form $TK(ae)$; the remaining terms inside $P(Hu)$ reduce to $1 - a + e$. Multiplying by $H = 1 + a$ and subtracting u yields (Q66.5.3).

Let ν be the finite spectral measure of $-K^2$ at 1. Using skewness and the explicit powers in H, e,

$$b = -\delta_2 - \delta_3 \le 0,\qquad v + c_0 = \int \frac{x(1-2x)}{1-x}\, d\nu(x) \ge \frac{2}{7} d. \tag{Q66.5.4}$$

For the second bound use $x \le \theta < \rho := 5/12$ and $(1 - 2\rho)/(1 - \rho) = 2/7$. Also $\|a\|_2 \le \sqrt{d}$, since $a = G_3(K)g$ and $|G_3(i\sqrt{x})|^2 = (1 + x^3)/(1 + x) \le 1$.

The exact spectral formulas for e give

$$\|e\|_2^2 = \int \frac{x^4(1+x)}{(1-x)^2}\, d\nu(x) \le \frac{2125}{7056} d \le \frac{16}{49} d,$$

and, writing $e = -KR(K^3 + K^4)1$,

$$|e|^2 \le \|R(K^3 + K^4)1\|_2^2 \le \frac{425}{588} d \le \frac{36}{49} d. \tag{Q66.5.5}$$

The two rational bounds are obtained by evaluating the increasing scalar factors at $\rho = 5/12$. They imply

$$a_0 \le 4d/7, \quad |e| \le 6\sqrt{d}/7 \le 18/35, \quad 17/35 \le u \le 53/35. \tag{Q66.5.6}$$

Here $\sqrt{d} \le 3/5$ follows from $d \le 1/3$.

Pointwise $|K(ae)| \le a_0$, and positivity of T gives $TK(ae) \le qa_0$. Since $b - d + a_0 \le 0$, (Q66.5.3)–(Q66.5.4) imply

$$M_H P M_H u - u \le -a^2 + a(e - 2d/7) - 2d/7.$$

Moreover $|e - 2d/7| \le (36/35)\sqrt{d}$. Completing the square,

$$M_H P M_H u - u \le \frac{324}{1225} d - \frac{2}{7} d = -\frac{26}{1225} d. \tag{Q66.5.7}$$

Because $u \le 53/35$, this is at most $-26du/1855$.

The operator $M_H P M_H$ is positive semidefinite and has a nonnegative kernel. Indeed $P = TA = A^*A$, and $H = f_3 + d > 0$. The elementary weighted Schur argument therefore gives

$$\|M_H T\|^2 = \|M_H P M_H\| \le 1 - \frac{26}{1855} d. \tag{Q66.5.8}$$

For clarity, if a symmetric nonnegative kernel Q satisfies $Qu \le \lambda u$ with $u > 0$, the inequality $2z(x)z(y) \le (u(y)/u(x))z(x)^2 + (u(x)/u(y))z(y)^2$, integrated against $Q(x,y)\mu(x)\mu(y)$, gives $\langle z, Qz\rangle \le \lambda \|z\|^2$. Positive semidefiniteness then controls the full operator norm.

Finally $f_3 = H - d$, $0 < H \le 2$, so

$$0 < f_3/H \le 1 - d/2.$$

Thus

$$\|M_{f_3} T\|^2 \le (1 - d/2)^2 (1 - 26d/1855).$$

Global sign reversal followed by the ordinary adjoint gives the same estimate for $TM_{\bar{f}_3}$. Since $26/1855 \ge 1/72$, (Q66.5.1) follows.

Q66.6 All short outer lengths for every $a \ge 3$

Put $B_{M,a} = T^M M_{\bar{f}_a}$; its actual row is $\bar{f}_{M+a}$. Using (Q66.1.4)–(Q66.1.6),

$$I_q(B_{M,a} Y)^3 \le (1 - d/72)\|T^{M-1}\|^2 (1 - d) E \bar{f}_{M+a}^4 \, E|Y|^8. \tag{Q66.6.1}$$

If $M = 2, a \ge 4$, the row length is at least six. Equations (Q66.1.7) and (Q66.3.16) show

$$I_q(T^2(\bar{f}_a Y))^3 \le (1 - d/72)(1 - \eta d) E|Y|^8 \le (1 - d/72) E|Y|^8. \tag{Q66.6.2}$$

If $M = 3, a \ge 3$, again the row length is at least six, and

$$(1 + d/2)^2 \le 1 + \frac{13}{12} d \le 1 + kd.$$

Therefore the same payment proves

$$I_q(T^3(\bar{f}_a Y))^3 \le (1 - d/72)(1 - \eta d) E|Y|^8. \tag{Q66.6.3}$$

This covers every such a, not merely sufficiently long roots.

It remains to consider $a = 3, M = 2$. Use (Q66.5.1) in place of the weaker ordinary multiplier in (Q66.6.1), and use (Q66.4.1):

$$I_q(T^2(\bar{f}_3 Y))^3 \le (1 - d/72)(1 - d/2)^2(1 + kd)^2(1 - d)(1 - 3d/2)E|Y|^8. \tag{Q66.6.4}$$

All displayed factors are nonnegative. We have

$$(1 - d)(1 - 3d/2) \le (1 - 5d/4)^2,$$

and $k < 7/4$, so the square root of the extra factor in (Q66.6.4) is at most

$$(1 - d/2)(1 + 7d/4)(1 - 5d/4) = 1 - \frac{39}{16}d^2 + \frac{35}{32}d^3 \le 1. \tag{Q66.6.5}$$

The last step uses $d \le 1/3$. This proves the final short case on the whole continuous domain.

The regular case follows directly from (Q66.1.4)–(Q66.1.6). Equations (Q66.6.2)–(Q66.6.5) prove (0.1). Together with the exact C064 cases, they prove the complete all-a, all-M map (0.2). The argument uses the operator order $T^{M-1}(TM_{\bar{f}_a})$, the row $\bar{f}_{M+a}$, and the fixed measure.

Q66.10.1 A complete outer-length interface for every negative even root

Let an actual even residual root table be

$$1 \le s_1 < t_1 < s_2 < t_2 < \cdots < s_j < t_j, \qquad j \ge 1.$$

The negative canonical pair orientation has factor

$$C_\Sigma^- = E_r \prod_{i=1}^{j} \bar{h}_{s_i} \bar{f}_{t_i}, \qquad \bar{h}_s = A\bar{f}_{s-1}. \tag{Q66.10.1}$$

Thus each shorter arm has its first edge away from the root and all remaining edges toward the root; each longer arm is wholly toward the root. Equal pairs in E_r are the actual reversed pairs. Every original arm and its length is retained.

Choose the first longer arm $t_1 \ge 2$ and set

$$W_\Sigma = E_r \bar{h}_{s_1} \prod_{i=2}^{j} \bar{h}_{s_i} \bar{f}_{t_i}.$$

For any fixed actual terminal message F and any $L, M \ge 1$, the actual half-tree factor and output are

$$H_\Sigma = C_\Sigma^- T^L F = \bar{f}_{t_1}(W_\Sigma T^L F), \qquad T^M H_\Sigma.$$

The full multiplier theorem proves the unconditional analytic bound

$$\boxed{I_q(T^M H_\Sigma)^3 \le (1 - d/72)E[(W_\Sigma T^L F)^8], \qquad M \ge 1.} \tag{Q66.10.2}$$

There is no short-outer exception left in this interface, and L, M are independent. The missing responsibility for these more general roots is now an explicit ordinary eighth input, not an unspecified outer-propagation claim.

For the terminal family $F = E_t q f_b$, $b \geq 2$, the precise sufficient new input is

$$E\left[\left\{E_r(A\bar{f}_{s_1-1})\left(\prod_{i=2}^{j}(A\bar{f}_{s_i-1})\bar{f}_{t_i}\right)T^L(E_t q f_b)\right\}^8\right] \leq 1. \tag{Q66.10.3}$$

When $j = 1, s_1 = 1$, this is exactly the input with factor $p = A1$, and parts Q66.8–9 finish all outer lengths. For $s_1 \geq 2$, or $j \geq 2$, (Q66.10.3) is a sufficient local condition, not a necessary condition for the full mapping or for the existence of an orientation of the original tree.

The following two symbolic examinations show why the missing factors matter in the current positive comparison.

M.17 Actual unequal-pair envelopes and finite masks

We use the pair and finite-mask bounds of Q5-C090, Sections 2.1–2.3. For actual residual arm lengths $a < b$ write $R_{ab} = \sum_{n=a+1}^{b} K^n \mathbf{1} = \sum_{n=a}^{b-1} K^n g$. The exact finite positive-convolution difference is

$$f_b - f_a = R_{ab} + \sum_{i=0}^{a-3}(\beta_{a-i} - \beta_{b-i})f_i - \sum_{i=a-2}^{b-3}\beta_{b-i}f_i. \tag{L90.2.4}$$

The first sum's positive part is at most $4\delta_2/(1-\theta^2) \leq 1664d/855$; this is the finite alternating-beta difference bound from the original Q3 convolution, using $f_i \leq 2$. In particular the short sums give

$$h_2 + f_b \leq 2 + R_{2b} - \beta_b \leq 2 + R_{2b} - \kappa d, \qquad h_3 + f_b \leq 2 + R_{3b} - (2d - \delta_2).$$

The shortest cases $b = 3, 4$ retain the actual $\beta_b f_0$ term.

We include the precise all-integer path-mean input, also recorded in Q5-C067, Section 2, and the Q4 path foundation. For $n = 4, 5, 6, 7$, the exact means are respectively

$$1 - 3d + \delta_2, \quad 1 - 4d + 2\delta_2 + d^2, \quad 1 - 5d + 3\delta_2 + 3d^2 - \delta_3,$$
$$1 - 6d + 4\delta_2 + 6d^2 - 2\delta_3 - 2d\delta_2.$$

Their losses divided by d are at least $83/32, 137/48, 89/32, 127/48$. For every finite $n \geq 8$, retain $f_0, \ldots, f_5$ and three additional convolution indices at least seven. Positive convolution gives

$$1 - \mu_n \geq \tau(7 - 10d + 3\delta_2 + d^2) + 3\kappa\delta_3 \geq \kappa d(7 - 10d + 7d^2) \geq 95d/36 > 83d/32.$$

Here $\delta_2 \geq d^2$, $\delta_3 \geq d^3$ are original global spectral moment inequalities; the bracket is positive and decreasing on $[0, 1/3]$. Thus $\mu_n \leq 1 - 83d/32$ for every $n \geq 4$, without using monotonicity of the sequence μ_n.

For $a = 4$, retain the actual negative f_2 term in (L90.2.4): $f_b - f_4 \leq R_{4b} + \delta_2(1+p) - \tau f_2$. The original pointwise row inequality $|Kg| \leq pq/2$ gives $f_2 = p + Kg \geq p^2/2$. For $d > 0$, completing the square gives

$$\delta_2(1+p) - \tau f_2 \leq \delta_2 + \delta_2 p - \tau p^2/2 \leq \delta_2 + \frac{\delta_2^2}{2\tau} \leq 3d/4.$$

Since $h_4 + f_4 = 2 - 4d$, the sum loss is at least $13d/4 > 16d/5$. For $a \geq 5$, use $h_a + f_a = 2\mu_{a-1} \leq 2 - 83d/16$ and the finite positive-difference bound. The loss is at least $[83/16 - 1664/855]d =$

$(44341/13680)d = [16/5 + 113/2736]d > 16d/5$. Every actual two-arm sum is nonnegative, so AM–GM followed by $1 + t \le e^t$ applies with a nonnegative square base and gives

$$h_a f_b \le \begin{cases} e^{R_{ab}-\kappa d}, & a = 2, \\ e^{R_{ab}-(2-\theta)d}, & a = 3, \\ e^{R_{ab}-16d/5}, & a \ge 4. \end{cases} \tag{L90.2.3}$$

For a finite ordered pair list, its intervals are disjoint; sum them once into R. The actual product F satisfies $F \le e^{R-\lambda d}$, where

$$\lambda = \begin{cases} \kappa + 16(j-1)/5, & a_1 = 2, \\ 2 - \theta + 16(j-1)/5, & a_1 = 3, \\ 16j/5, & a_1 \ge 4. \end{cases}$$

For $j \ge 2$ the first two values are at least $607/160, 767/160$. No expectations of distinct pair factors are multiplied.

If a finite polynomial P has all coefficients in $[0, 1]$, put $z = i\sqrt{x}$, $0 \le x \le \theta$. The positive and negative real-coordinate subsums are each bounded by their geometric sums, so the absolute real coordinate is at most $1/(1 - x^2)$. The imaginary coordinate, divided by $\sqrt{x}$, has the same bound. Consequently

$$|P(i\sqrt{x})|^2 \le \frac{1 + x}{(1 - x^2)^2} \le C_* := \frac{32768}{16245} < \frac{100}{49}. \tag{L90.2.5}$$

Apply this on each original skew-operator rotation block. The actual finite mask is $R = K^{a_1} P(K) g$, with coefficients zero or one and lowest coefficient one. Hence

$$\begin{gathered} \|R\|_2^2 \le \theta^{a_1} C_* d, \quad |R(x_0)|^2 \le \theta^{a_1 - 1} C_* d, \quad |U(i\sqrt{x})|^2 \le x^{a_1} C_*, \\ \|B^r R\|_2^2 \le \theta^{a_1 + r} C_* d. \end{gathered} \tag{L90.2.6}$$

Here U is the actual polynomial representing R in B, u coordinates; it is not pointwise division by u. The pointwise bound removes only the outermost original kernel using its row bound. For $|R| \le t < 3$, the positive Taylor terms obey $n! \ge 2 \cdot 3^{n-2}$ for $n \ge 2$, and negative R is bounded by the integral remainder. Thus

$$e^R \le 1 + R + \frac{R^2}{2(1 - t/3)}.$$

For $a_1 \ge 4$, $|R|^2 < d/7 \le 1/21 < 1/16$, so $t = 1/4$ works; for $a_1 = 3$, $|R|^2 < d/3 \le 1/9$, so $t = 1/3$ works. Both give $e^R \le 1 + R + 3R^2/5$. For $a_1 = 2$, $|R|^2 < 5d/6 \le 5/18 < 9/25$; take $t = 3/5$ to get $e^R \le 1 + R + 5R^2/8$. At $d = 0$ the actual messages are one by original leaf integration.

M.18 The sharpened losses for actual pairs starting at four

Use the original finite calculus and notation of Q5-C090. For $4 \le s < c$ put $R_{sc} = \sum_{r=s+1}^{c} K^r \mathbf{1}$. The finite difference retains, for the shortest case $s = 4$, the actual negative convolution term

$$f_c - f_4 \le R_{4c} + \delta_2(1 + p) - \tau f_2.$$

The original row constraint gives $f_2 \ge p^2/2$. For $d > 0$, $\delta_2/d \le 13/32$, $\tau/d \ge 19/32$, so

$$\delta_2(1 + p) - \tau f_2 \le \delta_2 + \delta_2 p - \tau p^2/2 \le \delta_2 + \frac{\delta_2^2}{2\tau} \le \frac{663}{1216} d.$$

Since $h_4 + f_4 = 2\mu_3 = 2 - 4d$, $h_4 + f_c \le 2 + R_{4c} - (4201/1216)d$ and $4201/1216 - 69/20 = 29/6080 > 0$. AM–GM on the nonnegative actual two-arm sum and $1 + t \le e^t$ therefore prove

$$h_4 f_c \le e^{R_{4c} - 69d/20}. \tag{L96.2.5}$$

For every finite $s \ge 5$, the original difference's positive part is at most $4\delta_2/(1-\theta^2)$, and the all-integer path mean proved in Q5-C090 gives $2\mu_{s-1} \le 2 - 83d/16$. Thus

$$h_s f_c \le e^{R_{sc} - \gamma d}, \qquad \gamma = \frac{83}{16} - \frac{1664}{855} = \frac{16}{5} + \frac{113}{2736} > \frac{16}{5}. \tag{L96.2.6}$$

Every $s \ge 4$ may consequently use the common loss $16d/5$. The finite-mask estimate used with these losses is (L90.2.5), with the stronger rational comparison $C_* < 101/50 < 100/49$. Useful exact original tail comparisons are

$$\sqrt{\theta} < 13/20 < 2/3, \quad \theta^2 < 1/6, \quad \theta^3 < 7/100 < 1/14, \quad \theta^{3/2} < 4/15,$$
$$\theta/(1-\theta^2) = 416/855 < 1/2, \quad \theta^2/(1-\theta^2) = 169/855 < 1/5,$$
$$\theta^3/(1-\theta^2) = 2197/27360 < 1/12, \quad \theta^4/(1-\theta^2) = 28561/875520 < 1/30,$$
$$\theta^5/(1-\theta^2) = 371293/28016640 < 1/72.$$

The last numerator comparison is $26733096 < 28016640$. These bounds hold for every finite mask. At $d = 0$ equality follows by original leaf integration.

M.19 Unanchored propagation and negative odd-root unit bridges

Q79.1 Original finite identities, directions, and inputs

Delete zero-mass atoms. Since K(x, x) = 0, on each retained atom p(x), $q(x) \ge \mu(x) > 0$. Every actual finite rooted-tree message is positive, because the all-equal labeling at the root has positive mass and all diagonal edge factors are one. Write

$$r = Kg, \quad s = Kr, \quad t = Ks, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The original finite skew calculus gives

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{m+1} \le \theta \delta_m,$$
$$Er = -d, \quad Eg = Es = 0, \quad E(gr) = E(rs) = 0, \quad E(gs) = -\delta_2. \tag{Q79.1.1}$$

Original row centering gives $r^2 \le pqd$. The stronger pointwise $|r| \le pq/2$ also follows directly by antisymmetrizing its two inner labels:

$$r(x) = \tfrac{1}{2} E_{y,z}[(K(x,y) - K(x,z))K(y,z)],$$

using $|u - v| \le 1 - uv$ for u, v in [-1, 1]. Consequently

$$|r| \le pq/2 \le 1/2. \tag{Q79.1.2}$$

The exact C064 ordinary norm inputs are

$$\|T\|^2 \le (1 + 32d/19)^2, \qquad \|T^2\|^2 \le (1 + d/2)^4,$$

$$\|T^M\| \le (1-d/8)^{\lfloor M/3 \rfloor} \le 1 \quad (M \ge 3). \tag{Q79.1.3}$$

Their order is the ordinary original L2(mu) order. In particular the last line is an integer-power theorem, not a claim that $\|T\| \le 1$. The exact C064 path fourth and C056 inverse fourth inputs are

$$E\bar{f}_4^4 \le 1-d-4d^2, \qquad E\bar{f}_5^4 \le 1-13d/12, \qquad E\bar{f}_n^4 \le 1-47d/360 \quad (n \ge 6), \tag{Q79.1.4}$$

$$E[\bar{f}_3^4/q] \le 1-23d/36. \tag{Q79.1.5}$$

Only these ranges are used. The convenient weaker common bound $1-47d/360$ also holds for all $n \ge 4$.

The unit-connector directions in Proposition M.39 are as follows:

Terminal unit arm
Away from the terminal, with factor p.

Terminal b-arm
Entirely toward the terminal, with factor $\bar{f}_b$.

Unit inner connector
From the terminal toward the root, with operator T.

Root shortest arm a
Entirely away from the root, with factor f_a.

Root arm u_i
The first edge points away from the root and the remaining edges toward its first neighbor, with factor W_{u_i}.

Root arm v_i
Entirely toward the root, with factor $\bar{f}_{v_i}$.

Outer path of length M
Every edge points from the root toward the receiving root, with operator T^M.

Thus the root has exactly $2k+1$ residual arms, and (Q79.6.8) lists them all. The directions depend solely on the original ordered lengths. The same directions are used in every continuous parameter interval.

If $d = 0$ then $g = 0$, but K need not vanish. $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, so leaf integration gives value one for every actual component. For the all-real theorem, (Q79.1.3) and row mass one give the asserted estimates directly as shown in part Q79.2. Divisions by d below are confined to $d > 0$.

Q79.2 A positive-kernel transfer on the fixed measure

For any positive operator B with row mass $h = B\mathbf{1} > 0$, original row Jensen gives $|BY|^4 \le h^3 B(|Y|^4)$. Raising to power $2/3$, multiplying by $q^{-1/3}$, and applying Holder with exponents 3 and $3/2$ gives

$$I_q(BY) \le \{E[B(|Y|^4)^2]\}^{1/3}\{E[h^3/\sqrt{q}]\}^{2/3}.$$

Here $B(|Y|^4)^2$ means the square of $B(|Y|^4)$. Its L2 norm is bounded by ‖B‖ times the L2 norm of $|Y|^4$. Thus

$$I_q(BY)^3 \le \|B\|^2 E|Y|^8 R(h)^2, \qquad R(h) = E[h^3/\sqrt{q}]. \tag{Q79.2.1}$$

The C064 transfer has the displayed weight, operator order and input exponent. It applies to every real Y.

Taking $B = T^M$, its actual row is $\bar{f}_M$. For $d = 0, q = \bar{f}_M = 1$ and the norm in (Q79.1.3) is at most one, so (Q79.2.1) proves (0.1) for all-real Y also in that boundary case. We now pay the row for every M.

Q79.3 The two shortest rows, with complete scalar certificates

Q79.3.1 A uniform original q fifth-half moment

For $0 \le q \le 2$ put $z = \sqrt{q}$. The exact identity is

$$q^{5/2} - 1 - \tfrac{5}{2}(q-1) = (z-1)^2(z^3 + 2z^2 + 3z + 3/2).$$

The difference between $(9/4)(z+1)^2$ and its last factor is

$$-z^3 + \tfrac{1}{4}z^2 + \tfrac{3}{2}z + \tfrac{3}{4} \ge \tfrac{1}{4}z^2 - \tfrac{1}{2}z + \tfrac{3}{4} = \tfrac{1}{4}(z-1)^2 + \tfrac{1}{2} > 0, \tag{Q79.3.1}$$

because $z^3 \le 2z$. Multiplying by $(z-1)^2$ gives the full-interval pointwise inequality

$$q^{5/2} \le 1 + \tfrac{5}{2}(q-1) + \tfrac{9}{4}(q-1)^2.$$

Since $Eq = 1$ and $E(q-1)^2 = d$,

$$R(\bar{f}_1) = Eq^{5/2} \le 1 + 9d/4. \tag{Q79.3.2}$$

Together with (Q79.1.3) and $\log(1+x) \le x$, this gives

$$\|T\|^2 R(\bar{f}_1)^2 \le (1 + 32d/19)^2(1 + 9d/4)^2 \le e^{(64/19+9/2)d} = e^{(299/38)d}. \tag{Q79.3.3}$$

This proves the $M = 1$ line without asserting that it is a contraction. Indeed for $Y = 1$ and a nonconstant q, $I_q(T1) = Eq^{7/3} > 1$, so a universal cost-free unit map would be false.

Q79.3.2 The actual b2 weighted row

The exact path is $\bar{f}_2 = q + r$. First note that, again with $z = \sqrt{q}$,

$$q^{3/2} - 1 - \tfrac{3}{2}(q-1) = (z-1)^2(z + 1/2),$$

and $0 \le z + 1/2 \le (z+1)^2/2$. The nonnegative remainder H(q) therefore satisfies $0 \le H(q) \le g^2/2$. Using $Egr = 0$ and (Q79.1.2),

$$E[q^{3/2}r] = -d + E[H(q)r] \le -d + \tfrac{1}{2}EH(q) \le -3d/4. \tag{Q79.3.4}$$

The two remaining terms in the cubic expansion have the bounds

$$E[\sqrt{q}\, r^2] \le \tfrac{10}{7}\delta_2, \qquad E[r^3/\sqrt{q}] \le \tfrac{4}{7}\delta_2. \tag{Q79.3.5}$$

For the first use $\sqrt{2} < 10/7$. For the second, (Q79.1.2) gives $r^3/\sqrt{q} \le r^2$ p $\sqrt{q}/2$. The square of the last coefficient is $(2-q)^2 q/4$. Its derivative has sign (2-q)(2-3q), so its maximum on [0, 2] is

$8/27$ at $q = 2/3$. The comparison $8/27 < 16/49$ proves p $\sqrt{q}/2 < 4/7$. This argument needs no host-independent positive lower bound on q.

Expand the same actual cubic row and combine (Q79.3.2), (Q79.3.4), (Q79.3.5):

$$\begin{aligned} R(\bar{f}_2) &= Eq^{5/2} + 3E[q^{3/2}r] + 3E[\sqrt{q}\,r^2] + E[r^3/\sqrt{q}] \\ &\le 1 + (9/4 - 9/4)d + (34/7)\delta_2 \\ &\le 1 + (221/112)d \le 1 + 2d. \end{aligned} \tag{Q79.3.6}$$

The cancellation of the first-order q cost comes from the actual negative mean of r with the same q. The terms were not replaced by independent extremal function budgets. Equation (Q79.1.3) now gives

$$\|T^2\|^2 R(\bar{f}_2)^2 \le (1 + d/2)^4 (1 + 2d)^2 \le e^{6d}. \tag{Q79.3.7}$$

This pays every kernel and the entire continuous domain for $M = 2$.

Q79.4 Every outer length at least three

Let $\beta(U) = E[U^2/q]$. Row Cauchy for the original positive T gives

$$(TU)^2 \le qT(U^2), \qquad \beta(TU) \le E[pU^2] \le \beta(U), \tag{Q79.4.1}$$

where the last inequality uses $pq \le 1$. Since $r^2 \le pqd$,

$$\beta(\bar{f}_2) = 1 - 2d + E[r^2/q] \le 1 - d, \quad \beta(\bar{f}_M) \le 1 - d \quad (M \ge 2). \tag{Q79.4.2}$$

For $M \ge 4$, ordinary Cauchy and the exact (Q79.1.4) give

$$R(\bar{f}_M)^2 \le \beta(\bar{f}_M) E\bar{f}_M^4 \le 1 - d. \tag{Q79.4.3}$$

All integer lengths are included: $M = 4$, 5 use their own fourths and $M \ge 6$ uses the complete finite tail.

The boundary $M = 3$ uses its actual inverse fourth, not an assumed ordinary fourth. The exact path $\bar{f}_3 = 1 - d - g + r - s$ has mean 1-2d, and the odd/even skew orthogonality in (Q79.1.1) gives

$$\mathrm{Var}(\bar{f}_3) = d - \delta_2 + \delta_3 - d^2, \qquad E\bar{f}_3^2 = 1 - 3d + 3d^2 - \delta_2 + \delta_3 \le 1 - 2d. \tag{Q79.4.4}$$

Thus (Q79.1.5) and Cauchy give

$$R(\bar{f}_3)^2 \le E[\bar{f}_3^4/q] E\bar{f}_3^2 \le (1 - 23d/36)(1 - 2d) \le 1 - d. \tag{Q79.4.5}$$

The final factors are nonnegative on $0 \le d \le 1/3$. Use the true ordinary-power contraction (Q79.1.3) in (Q79.2.1) to conclude

$$I_q(T^M Y)^3 \le (1 - d) E|Y|^8 \quad (M \ge 3). \tag{Q79.4.6}$$

The cases $M = 1$, $M = 2$, and $M \ge 3$ prove, for every real input Y, the uniform bound

$$I_q(T^M Y)^3 \le e^{(299/38)d} \mathbb{E}|Y|^8 \qquad (M \ge 1). \tag{M.83}$$

Here $299/38 > 6$; the sharper contraction for $M \ge 3$ remains (Q79.4.6).

M.19.1 A unit connector at a nonunit receiving arm

All lengths below are positive integers, and the operators use the same original finite probability measure. Put $\theta = 13/32$, $\tau = d - \delta_2$, and use the finite G_n and β_n of (M.2), with $K = -B$. The actual terminal is $p\bar{f}_b$; its connector is one edge with operator T. The receiving arm points outward.

Lemma M.38 (The reversed unit terminal)**.** *For every $b \geq 2$, the following positive comparisons hold. At $b = 2$,*

$$T(p\bar{f}_2),\ T(qf_2) \leq q - \gamma - K^2 g, \qquad \gamma = d - \delta_2/2. \tag{M.84}$$

For $b \geq 3$, write

$$\begin{gathered} \bar{R} = KG_{b-1}(-K)g, \quad m_b = 1 - \beta_b/2, \quad k_b = 1/m_b, \\ t_0 = \beta_{b+1}, \quad e = d - \beta_{b+2}, \quad V_b = \mathbb{E}\bar{R}^2, \quad \gamma = k_b t_0 - k_b^2 V_b/2. \end{gathered} \tag{M.85}$$

Then

$$\begin{gathered} T(p\bar{f}_b) \leq m_b^2(q - \gamma - k_b K\bar{R}), \\ 5/6 \leq m_b \leq 1, \quad 1 \leq k_b \leq 6/5, \quad 0 \leq e \leq \delta_2, \quad 0 \leq \gamma \leq k_b d, \\ \gamma - k_b e \geq d - 68\delta_2/25. \end{gathered} \tag{M.86}$$

This lemma is independent of the length of the receiving arm.

Proof. The finite positive convolution gives $p + \bar{f}_b \leq 2m_b + \bar{R}$. The right side is positive, so AM–GM and positivity of T give

$$p\bar{f}_b \leq (m_b + \bar{R}/2)^2, \qquad T\bar{R} = -t_0 - K\bar{R}, \qquad T(\bar{R}^2) \leq 2V_b.$$

These prove the first line of (M.86). Finite skew identities give $\mathbb{E}\bar{R} = -t_0$, $\mathbb{E}[g\bar{R}] = e$, $V_3 = \delta_2 + \delta_3$, and $V_b \leq \delta_2$ for $b \geq 4$, using $\|G_{b-1}\| \leq 1$ in the latter range. For $b \geq 4$, $t_0 - e \geq d - 2\delta_2 \geq 0$, hence

$$\gamma - k_b e = k_b(t_0 - e) - k_b^2 V_b/2 \geq d - 2\delta_2 - 18\delta_2/25.$$

For $b = 3$, $t_0 = d$, $e = \delta_2$ give the stronger $\gamma - k_b e \geq d - (161/80)\delta_2$. Also $\gamma \geq 0$: for $b \geq 4$, use $k_b\tau - k_b^2\delta_2/2 \geq (25/64)d$; for $b = 3$, use $d - (18/25)(\delta_2 + \delta_3) \geq 0$, strictly when $d > 0$. The upper bound $\gamma \leq k_b d$ is immediate. At $b = 2$, both $p + \bar{f}_2$ and $q + f_2$ equal $2 + Kg$, and the same calculation uses $T(Kg) = -d - K^2 g$ and $T((Kg)^2) \leq 2\delta_2$. □

Proposition M.39 (One receiving arm, with all later negative pairs)**.** *Let $a \geq 3$, $b \geq 2$, $k \geq 0$, and put*

$$J_{a,b} = f_a T(p\bar{f}_b), \qquad P = \prod_{i=1}^{k} W_{u_i}\bar{f}_{v_i}, \qquad a < u_1 < v_1 < \cdots < u_k < v_k.$$

For $k = 0$, take $P = 1$. The directions are fixed by these integer lengths alone. Then

$$\mathbb{E}(PJ_{a,b})^8 \leq e^{-\Lambda_k d}, \qquad \Lambda_0 = \frac{9}{4}, \qquad \Lambda_k = \frac{2005}{192} + 24(k-1) \quad (k \geq 1). \tag{M.87}$$

Genuine equal pairs may be retained at either vertex. The terminal length has no ordering restriction relative to the receiving arm or the post lengths. When $b = 2$, the same bounds also hold with the distinct actual terminal qf_2 in place of $p\bar{f}_2$.

Proof. First suppose $d > 0$. We keep the complete positive comparison for the bare input. For $b \geq 3$, the root envelope $f_a \leq p + KG_{a-1}(K)g - \beta_a$, together with Lemma M.38, gives

$$\begin{gathered} J_{a,b} \leq m_b^2[1 - (\beta_a + \gamma)/2 + Z/2]^2, \\ Z = KV_0, \qquad V_0 = [G_{a-1}(K) - k_b KG_{b-1}(-K)]g, \\ \mathbb{E}Z = -\beta_{a+1} + k_b e \leq -17d/160. \end{gathered} \tag{M.88}$$

The square base is half the sum of the same two nonnegative upper factors. For $b = 2$, the formula holds with $m_b = k_b = 1$, $\gamma = d - \delta_2/2$, and $V_0 = [G_{a-1}(K) - K]g$, $\mathbb{E}Z = -\beta_{a+1}$.

For every finite $r, s \geq 2$, $1 \leq k \leq 6/5$, and $z = i\sqrt{x}$, $0 \leq x \leq \theta$,

$$|G_r(z) - kzG_s(-z)|^2 \leq 1 - 17x/20. \tag{M.89}$$

Indeed, the real and normalized imaginary coordinates of each G_n, $n \geq 2$, lie in $[1 - x, 1]$. The real part of the difference is positive, at least $17/160$, and at most $1 - kx(1 - x)$; its imaginary coordinate has absolute value at most $k - 1 + x$. The squared bound is $1 + xP(k, x)$, where

$$P = k^2 - 4k + 1 + (k^2 + 4k - 2)x + (1 - 2k^2)x^2 + k^2x^3.$$

Here $P_x \geq 589/400 > 0$ and $P_{kk} = 2(1 + x - 2x^2 + x^3) > 0$. Its maximum is therefore at $x = \theta$ and $k = 1$ or $6/5$, where its values are $-28811/32768$ and $-174331/204800$, both below $-17/20$. This proves the frequency bound for all finite indices, including zero frequency. Integration gives $\|V_0\|^2 \leq d - 17\delta_2/20$ when $b \geq 3$.

At $b = 2$, the squared primitive bounds are d for $a = 3$ and $d - 51\delta_2/32$ for $a = 4$. For every $a \geq 5$, the finite identity

$$G_{a-1}(z) - z = \frac{1 - z + z^2 - z^{a-1}}{1 - z}$$

gives

$$|G_{a-1}(z) - z|^2 \leq \frac{1 - x + 3x^2 + x^4}{1 + x} \leq 1 - 65x/128. \tag{M.90}$$

The first bound uses $|1 - z + z^2| \leq 1$ and $|z^{a-1}| \leq x^2$. For the second, the decreasing polynomial $191/128 - (449/128)x - x^3$ has value $3/32768 > 0$ at θ. In every case $\|V_0\|^2 \leq d$.

Apply (M.43) to $-V_0$, since $B = -K$. The actual signed mean in (M.88) is retained, and

$$\log \mathbb{E}J_{a,b}^8 \leq 16 \log m_b - 8(\beta_a + \gamma) + 8\mathbb{E}Z + 17\|V_0\|^2. \tag{M.91}$$

For $b \geq 3$, $16 \log m_b \leq -8\beta_b$, and

$$\beta_a + \beta_{a+1} + \beta_b + \gamma - k_b e \geq 4d - 143\delta_2/25.$$

Thus the complete exponent is at most $-15d + (3131/100)\delta_2 \leq -(7297/3200)d < -9d/4$. At $a = 3$, retaining $\beta_3 + \beta_4 = 2d$ improves this by $16\delta_2$. The other short cases give the following complete lower bounds for the bare rate:

(a, b)	rate
$(3, 2)$	$43/8$
$(4, 2)$	7
$(a \geq 5, 2)$	$9757/4096$
$(3, b \geq 3)$	$28097/3200$
$(a \geq 4, b \geq 3)$	$7297/3200$

(M.92)

For the first three rows the respective full exponents are $-7d + 4\delta_2$, $-7d - 483\delta_2/32$, and $-7d + 1455\delta_2/128$. Every listed rate exceeds $9/4$.

Now take $k \geq 1$. Complete reversal of (L90.2.3), with the weaker loss $3d$ per pair, gives

$$P \leq e^{S-3kd}, \quad S = KV_S, \quad V_S = \sum_{i=1}^{k} \sum_{j=u_i}^{v_i-1} (-K)^{j-1} g. \tag{M.93}$$

The intervals are disjoint and start in degree at least $u_1 - 1 \geq a$. The finite-mask bound (L90.2.5) gives

$$\|V_S\|^2 \leq (100/49)\theta^a d, \qquad \|V_S\| \leq \varepsilon_a \sqrt{d}, \qquad \varepsilon_a = \begin{cases} 3/8, & a = 3, \\ 1/4, & a \geq 4. \end{cases}$$

The first possible positive mean in the tail is δ_4, so

$$\mathbb{E}S \leq \frac{\delta_4}{1-\theta^2} \leq \frac{2197}{27360} d < d/12, \qquad \mathbb{E}(Z+S) \leq -11d/480.$$

Keep the original scalar in (M.91) and apply the same signed estimate once to the full primitive $V_0 + V_S$. Its squared norm increases by at most $(2\varepsilon_a + \varepsilon_a^2)d \leq 57d/64$. The additional complete exponent is therefore at most

$$[17(57/64) + 2/3 - 24k]d = -[1573/192 + 24(k-1)]d.$$

Together with the bare rate $9/4$, this proves (M.87). Equal pairs decrease the same positive input. At $d = 0$, original leaf integration gives every actual message value one, even if $K \neq 0$. □

When another genuine inward arm $\bar{f}_s$, $s \geq 2$, is available, Section M.16 gives, for every independent $M \geq 1$,

$$I_q(T^M[\bar{f}_s P J_{a,b}])^3 \leq (1 - d/72)e^{-\Lambda_k d}. \tag{M.94}$$

Its multiplication by the actual inward arm precedes T^M. For an unanchored output and $k \geq 1$, the independently proved all-real bound (M.83) instead gives

$$I_q(T^M[PJ_{a,b}])^3 \leq e^{-[9391/3648 + 24(k-1)]d} \leq e^{-25d/912} \qquad (M \geq 1). \tag{Q79.6.8}$$

The unanchored conclusion here uses a nonempty post block.

M.20 The same-row inverse fourth moment and its endpoint payment

Q100.2 Kernel notation and inherited estimates

For finite trees with at most five branch vertices, the orientation problem requires the following bound on every finite probability space and every real skew $|K| \leq 1$:

$$Z_D = E_{\mu^{V(U)}} \prod_{v \to w \in D} (1 + K(x_v, x_w)) \leq 1.$$

Delete zero-mass atoms only. On the unchanged original L2(mu), use

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}, \quad \delta_j = \|K^j\mathbf{1}\|^2, \quad d = \delta_1.$$

The finite calculus gives $\theta = 13/32, \kappa = 19/32, d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{j+1} \le \theta\delta_j$, and $\|K\|^2 \le \theta$. Skewness gives $K(x,x) = 0$, so p(x), $q(x) \ge \mu(x) > 0$ on retained atoms. Every actual rooted tree message is positive by the all-equal labeling.

The proof uses the finite calculus and the four-branch pairing and zero-step interfaces already established above.

The genuine equal-pair comparison is $f_l h_l \le 1$, or its full reverse $\bar{f}_l W_l \le 1$. Each such factor stays at its original vertex and is removed only from a nonnegative upper comparison. The path-mean bound used here applies to f_n and $\bar{f}_n$. In contrast, $\mathbb{E}h_2 = \mathbb{E}W_2 = 1 + d$. The f2 case uses the non-strict bound.

Q100.3 Independent finite-G reconstruction

Put $G_m(z) = \sum_{j=0}^{m-1} z^j$ and $E_r(x) = \sum_{j=0}^{r-1}(-x)^j$. For $0 \le x \le \theta < 1$, the finite alternating sum satisfies $1 - x \le E_r \le 1$. At $z = i\sqrt{x}$, write $G_m(z) = A_m(x) + zD_m(x)$. For the actual

$$R = KG_m(-K)g, \qquad h = E(gR), \qquad V = ER^2, \qquad m = n - 1 \ge 3,$$

the block expression is $R = (zA_m + xD_m)g$. Hence the genuine finite spectral measure ν_g of -K^2 at g gives

$$h = \int xD_m \, d\nu_g, \qquad V = \int x(A_m^2 + xD_m^2) \, d\nu_g.$$

Its mass is d and its first moment is δ_2. This proves the signs and identifies the same R as in the original correlated cubic calculation.

Q100.3.1 Even and odd finite lengths

For $m = 2r, r \ge 2, A_m = D_m = E_r$, and the FINITE identity $(1 + x)E_r = 1 - (-x)^r$ yields

$$x[(1 + x)E_r^2 - E_r] = (-1)^{r+1}x^{r+1}E_r.$$

Positive cases have r odd, so $r \ge 3$. They are bounded by x^4, because $E_r \le 1$ and $x \le 1$. Furthermore $x^4 \le x^3(1 - x + x^2)$, since their difference is $x^3(1 - x)^2$. Negative cases automatically satisfy the same upper bound.

For $m = 2r + 1, r \ge 1, D_m = E_r$ and $A_m = 1 - xE_r$. Substitution, or replacing $(1 + x)E_r$ by its finite expression, gives

$$A_m^2 + xD_m^2 - D_m = (-x)^r A_m.$$

Here $A_m > 0$. Only even $r \ge 2$ gives a positive term. In that case

$$A_m = \frac{1 + x^{r+1}}{1 + x} \le \frac{1 + x^3}{1 + x} = 1 - x + x^2, \qquad x^{r+1} \le x^3.$$

Multiplication by x proves the same uniform upper polynomial. The two short endpoints are explicit:

$$m = 3 : \ -x^2(1 - x), \qquad m = 4 : \ -x^3(1 - x).$$

They are negative, not missing positive cases. At $m = 5$ the expression is exactly $x^3(1 - x + x^2)$, so that polynomial is a real finite expression. These calculations cover every integer $m \geq 3$, with no infinite-tail substitution.

Q100.3.2 Uniform spectral payment and exact constants

The derivative of $x^2(1 - x + x^2)$ is $x(2 - 3x + 4x^2)$, nonnegative; the quadratic has discriminant -23 and positive leading coefficient. Therefore

$$\begin{aligned} V - h &\leq \int x^3(1 - x + x^2)\, d\nu_g \\ &\leq \theta^2(1 - \theta + \theta^2) \int x\, d\nu_g = r_* \delta_2. \end{aligned}$$

Since $1 - \theta + \theta^2 = 777/1024$ and $\theta^2 = 169/1024, r_* = 131313/1048576$. The comparisons in (2.6)-(2.7) check exactly:

- $r_* > 1/8$ because $131313 > 131072$.
- $r_* < 13/100$ because $13131300 < 13631488$.
- $\theta r_* = 1707069/33554432 < 51/1000$ because $1707069000 < 1711276032$.
- $\theta^2 - 51/1000 = (21125 - 6528)/128000 = 14597/128000$. This is >$9/80$ because $14597 > 14400$.

Thus $\theta(\theta - r_*) > \delta_* = 14597/128000 > 0$. No step assumes $x \geq d$. In particular this is a correlated V-h bound, not independent worst-case budgets for V and h.

Q100.4 Exact same-row calculation

This section reconstructs the calculation through (2.10). Write $\alpha = \kappa d$, $C = 1 - \alpha, t = 1 + d/2$ and $\beta = \beta_{n+1}$.

The finite convolution is

$$S'_n = \sum_{j=0}^{n} (-K)^j 1 = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j} = q + R.$$

For $n \geq 4$ retain its actual b0 and b1 terms. Since β_n, β_{n-1} are at least alpha, $\bar{f}_n \leq Cq + R - \alpha$. Also T $v \leq 2Ev$ and $T^2 v \leq 2(Ev)q$ for $v \geq 0$. The same finite convolution gives $E\bar{f}_{n-2} \leq 1 - \beta_{n-1} \leq C$, hence $\bar{f}_n \leq 2Cq$. At $n = 4$ the latter uses the actual b2 mean 1-d; no short term is lost.

For $s \geq -1$,

$$\min(1 + s, 2)^4 \leq 1 + 4s + 7s^2 + 4s^3.$$

On [-1, 1] the difference from $(1 + s)^4$ is $s^2(1 - s^2)$. On [1, infinity), the cubic starts at 16 and has positive derivative. The upper Cq+R-alpha is positive, so $s = (R - \alpha)/(Cq) > -1$. Applying the inequality to the same actual row gives

$$B_n \leq C^4 E q^3 + 4C^3 E[q^2(R - \alpha)] + 7C^2 E[q(R - \alpha)^2] + 4CE(R - \alpha)^3. \qquad \text{(Q100.A24.1)}$$

The exact finite quantities satisfy $ER = -\beta, \alpha \le \beta \le d, \kappa\delta_2 \le h \le \delta_2, V \le \delta_2, |R| \le \sqrt{d}$, and E[$G_m(-K)g$]=$\beta_n \ge \alpha$. The h and V bounds follow from the actual finite coordinates $D_m \ge 1 - x$ and $|G_m| \le 1$ for $m \ge 3$.

Put $W = 2R - Cg$. The original row estimate gives

$$|W|^2 \le C^2 + 4d - 4C\alpha = 1 + (4 - 6\kappa)d + 5\kappa^2 d^2 \le t^2.$$

The last difference is $d[9/16 - (1549/1024)d] \ge 179d/3072$. Also

$$EW = -2\beta, \qquad D_W = EW^2 = C^2 d + 4V - 4Ch \ge 0.$$

The pointwise inequality $(W + 2\beta)^2(W - t) \le 0$ gives

$$EW^3 \le (t - 4\beta)D_W - 4t\beta^2 + 8\beta^3.$$

The full correlated identity is

$$-C^4 Eg^3 + 4C^3 E(g^2 R) - 6C^2 E(gR^2) + 4CER^3 = \frac{C}{2}[EW^3 + C^2 E(g^2 W)].$$

Use $E(g^2 W) \le td$. Of the seven copies of -$C^2 E(gR^2)$ in the expansion (Q100.A24.1), six are in this identity and one remains. This is where a separate absolute cubic budget would lose the intended gain.

At fixed TRUE $D_W \ge 0$, the beta-linear coefficient from the rest of (Q100.A24.1) is

$$-4C^3 + 14C^2\alpha - 12C\alpha^2 = -2C(2C - 3\alpha)(C - 2\alpha) < 0.$$

Indeed $\alpha \le 19/96 < 1/5$. The derivative of the cubic bound, after its C/2 factor, is $C[-2D_W - 4t\beta + 12\beta^2] \le 0$ because $3\beta \le 1 \le t$. Thus one first lowers beta to alpha, while holding D_W fixed. Only afterwards $t - 4\alpha = 1 - 15d/8 \ge 3/8$ allows its substitution.

Before grouping V-h, the coefficients are precisely

$$C[7 + 2t - 27\alpha]V + C^2[-8 - 2t + 30\alpha]h.$$

Writing $V = (V - h) + h$ makes the h coefficient

$$\begin{aligned} C\{7 + 2t - 27\alpha + (1 - \alpha)(-8 - 2t + 30\alpha)\} &= C[-1 + (11 + 2t)\alpha - 30\alpha^2] \\ &= C[-1 + (13 + d)\alpha - 30\alpha^2]. \end{aligned} \tag{Q100.A24.2}$$

This independently proves (2.10), including the disputed coefficient. Together with its unchanged constants it yields

$$\begin{aligned} B_n \le\ & C^4(1 + 3d) + C^3[td - 6\alpha d - 8\alpha] + C\alpha^2[19 - d - 33\alpha] \\ & + CB(d)(V - h) + C[-1 + (13 + d)\alpha - 30\alpha^2]h - C^2 E(gR^2), \end{aligned} \tag{Q100.A24.3}$$

where $B(d) = 7 + 2t - 27\alpha = 9 - 481d/32$ and $B \ge 383/96 > 0$.

Q100.4.2 Complete signed substitution

Split the actual h coefficient term by term, using the lower h bound on its negative terms and the upper h bound on its positive term:

$$[-1+(13+d)\alpha-30\alpha^2]h \le [-\kappa+13\kappa d+\kappa(1-30\kappa^2)d^2]\delta_2 = H(d)\delta_2.$$

After the proved V-h improvement, the combined coefficient is $S_new = r_*B + H$. Its constant is >$17/32$, using $r_* > 1/8$. Its linear coefficient is

$$13\kappa - \frac{481}{32}r_* > \frac{24700-6253}{3200} = \frac{18447}{3200} > 5.$$

Its quadratic coefficient is -$93157/16384 > -6$. Thus

$$S_{\text{new}} > 17/32 + 5d - 6d^2 > 0, \qquad 0 \le d \le 1/3.$$

This sign is proved BEFORE multiplying the upper bound $\delta_2 \le \theta d$.

Q100.5 Old polynomial, exact difference and continuous row

For the coefficient P, let F0 be the first three terms of (Q100.A24.3). The combined coefficient is

$$S_{\text{old}} = \theta B + H = \frac{49}{16} + \frac{1651}{1024}d - \frac{93157}{16384}d^2.$$

For example, its linear coefficient is -$6253/1024 + 7904/1024 = 1651/1024$.

The old polynomial is $P = F_0 + C\theta d S_{\text{old}}$. Its coefficient ledger in kappa is

Table 22: Coefficients of d^j in the comparison polynomial $P(d)$.

j	Coefficient before setting $\kappa = 19/32$	Exact value
1	$13 - 31\kappa + 10\kappa^2$;	$-963/512$.
2	$3/2 - 46\kappa + 110\kappa^2 - 37\kappa^3$;	$171113/32768$.
3	$-(3/2)\kappa + 53\kappa^2 - 152\kappa^3 + 57\kappa^4$;	$-7275575/1048576$.
4	$(1/2)\kappa^2 - 29\kappa^3 + 72\kappa^4 - 30\kappa^5$;	$14103187/16777216$.
5	$9\kappa^4 - (1/2)\kappa^3$;	$1063145/1048576$.

Expanding the finite products gives these coefficients. For instance the degree-two numerator is $49152 - 894976 + 1270720 - 253783 = 171113$; the degree-three numerator is -$933888 + 19592192 - 33362176 + 7428297 = -7275575$. At degree four, the numerator over 33554432 is $5914624 - 203684864 + 300259584 - 74282970 = 28206374$, giving the printed fraction after division by two. At degree five it is $1172889 - 109744 = 1063145$ over 1048576.

The correct new polynomial is consequently EXACTLY

$$F0 + C\theta d(r_*B + H) = P - CB\theta(\theta - r_*)d. \tag{Q100.A24.4}$$

The correlation satisfies

$$|E(gR^2)| \le \sqrt{d\,ER^4} \le d\sqrt{V} \le \sqrt{\theta}\,d^{3/2} \le \frac{2}{3}d^{3/2}.$$

Here $|R|^2 \le d$ and $V \le \theta$ d are the actual same-R bounds. Combining them with (Q100.A24.4) proves (3.4) exactly.

The rational majorization of P also checks: $A2 < 21/4$ follows from $171113 < 172032$; $A3 < -69/10$ from $72755750 > 72351744$; $A4 < 1$ and $A5 < 33/32$ from $14103187 < 16777216$ and $1063145 < 1081344$. Since

$$d^3 + \frac{33}{32}d^4 \le \left(\frac{1}{3} + \frac{33}{288}\right) d^2 = \frac{43}{96}d^2,$$

and $69/10 - 43/96 = 3097/480 > 32/5$, one gets (3.5) for $d > 0$.

Using $\theta(\theta - r_*) > \delta_*$ proves

$$B_n \le 1 - s(d)d,$$

$$s(d) = \frac{963}{512} - \frac{21}{4}d + \frac{32}{5}d^2 - \frac{2}{3}(1-\kappa d)^2\sqrt{d} + \delta_*(1-\kappa d)(9 - 481d/32).$$

All factors in its last product are positive. The derivative of its first three terms is at most -$21/4 + 64/15 = -59/60$. The product (1-kappa d)(9-481d/32) decreases, since both positive factors decrease. For $d > 0$,

$$\frac{d}{dd}\{(1-\kappa d)^2\sqrt{d}\} = \frac{(1-\kappa d)(1-5\kappa d)}{2\sqrt{d}} > 0;$$

at $d = 1/3$, 1-5kappa $d = 1/96$. The negative of this term therefore decreases too. These exact signs prove decreasing s on the whole interval, with its continuous extension at zero.

Q100.6 Completing the inverse-row bound.

The final scalar payment for $B_n = \mathbb{E}[\bar{f}_n^4/q]$, $n \ge 4$, is as follows. The derivative above is positive on $0 < d \le 1/3$, and $1/\sqrt{3} < 11/19$, since $361 < 363$. Hence

$$\frac{2}{3}(1-\kappa d)^2\sqrt{d} \le \frac{2}{3}\left(\frac{77}{96}\right)^2 \frac{11}{19} = \frac{130438}{525312} < \frac{1}{4}. \tag{Q100.3.6}$$

The last comparison is $521752 < 525312$. Dropping the nonnegative improvement in $s(d)$, the preceding coefficient bound therefore gives, for $d > 0$,

$$\frac{B_n - 1}{d} \le -\frac{835}{512} + \frac{21}{4}d - \frac{32}{5}d^2. \tag{Q100.3.7}$$

This quadratic increases on $[0, 1/3]$: its derivative is at least $21/4 - 64/15 = 59/60 > 0$. Its value at the right endpoint is

$$-\frac{835}{512} + \frac{7}{4} - \frac{32}{45} = -\frac{13639}{23040} < -\frac{1}{2}. \tag{Q100.3.8}$$

At $d = 0$, $\bar{f}_n = q = 1$. Thus every finite $n \ge 4$ satisfies

$$\boxed{\mathbb{E}[\bar{f}_n^4/q] \le 1 - d/2.} \tag{Q100.3.9}$$

The separate $n = 3$ estimate in Section M.14 is $1 - 23d/36$; since $23/36 > 13639/23040 > 1/2$, it implies the same weaker bound. These are the actual inward paths for every length throughout the score interval.

M.21 Finite path caps and the sharper fourth-moment outlet

Q164.0 Original goal and exact new scope

For every nonempty finite original tree U with at most five original vertices of degree at least three, we seek an orientation fixed from U alone before the host, for which every original finite probability measure and every real skew pointwise kernel $|K| \le 1$ satisfy $Z_D \le 1$. All original arms, equal multiplicities, subdivisions, four independent positive core lengths, original e(U), arbitrary joint-label KL laws and noninjective maps must remain. The operator outlet gives the conditional tree application below.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. Zero-mass atoms alone may be removed. On positive-mass atoms p, q are strictly positive, since K(x, x) $= 0$ and p(x), $q(x) \ge \mu(x)$.

The new actual path-row bounds are

$$\boxed{E[\bar{f}_4^4/q] \le e^{7d}, \qquad E[\bar{f}_M^4/q] \le e^{6d} \quad (M \ge 5).} \tag{Q164.0.1}$$

Combined with the exact short-length outlet and the original operator estimate specified in part Q164.1, this gives for every real H and every original integer $M \ge 1$

$$\boxed{I_q(T^M H)^3 \le e^{c(M)d}(E|H|^4)^2, \qquad I_q(Y) = E[q^{-1/3}|Y|^{8/3}],} \tag{Q164.0.2}$$

where the structure-only cost table is

$$c(1) = 204/19, \quad c(2) = 10, \quad c(3) = 0, \quad c(4) = 7, \quad c(M) = 6 \ (M \ge 5). \tag{Q164.0.3}$$

In particular, the uniform coefficient $204/19$ is valid for every $M \ge 1$. No reference arm is inserted into (Q164.0.2). It is an ordinary fourth-moment outlet, not a same-reference fourth-moment theorem.

Q164.1 Exact inputs and finite path identities

Use Section M.2 with $K = -B$ and $g = -u$. The reversed form of (M.2) is

$$1 + G_n(K)g = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad \tau \le \beta_j \le d \quad (j \ge 3). \tag{Q164.1.1}$$

The same formula with B, u gives $\bar{f}_n$. Besides $0 \le f_n, \bar{f}_n \le 2$ and $0 \le \mathbb{E}f_n = \mathbb{E}\bar{f}_n \le 1$, the path means give

$$\mathbb{E}f_2 = \mathbb{E}\bar{f}_2 = 1 - d, \qquad \mathbb{E}f_n = \mathbb{E}\bar{f}_n \le 1 - 2d \quad (n \ge 3). \tag{Q164.1.2}$$

Also $\bar{f}_n \le 2\mathbb{E}\bar{f}_{n-1}$, since the kernel of T is at most two.

Section Q.1 supplies

$$\|T^M\|_{2\to 2} \le 1 \quad (M \ge 3), \tag{Q164.1.3}$$

and its real-input outlet costs $204/19, 10, 0$ at $M = 1, 2, 3$. The following argument proves the new costs $7, 6$ for $M = 4$ and $M \ge 5$, respectively. The finite spectral coordinates and moments are Lemma M.1, retaining $|G_2|^2 = 1 + x$.

Q164.2 The exact path caps

For $M = 4$, (Q164.1.2) and positivity give

$$\bar{f}_3 = T\bar{f}_2 \le 2E\bar{f}_2 = 2(1-d), \qquad \bar{f}_4 = T\bar{f}_3 \le 2(1-d)q. \tag{Q164.2.1}$$

For every $M \ge 5, M-2 \ge 3$ and hence

$$\bar{f}_{M-1} \le 2E\bar{f}_{M-2} \le 2(1-2d), \qquad \bar{f}_M \le 2(1-2d)q. \tag{Q164.2.2}$$

Thus define the actual upper cap

$$c_4 = 2(1-d) \in [4/3, 2], \qquad c_{\ge 5} = 2(1-2d) \in [2/3, 2].$$

The endpoint $2/3$ is an allowed closed-domain upper-bound calculation; no positive-mass host is required to attain $d = 1/3$.

For all c in [$2/3$, 2] and all $s \ge -1$,

$$\boxed{\min\{1+s, c\}^4 \le 1 + 4s + (c^2 + 2c + 3)s^2.} \tag{Q164.2.3}$$

Indeed, on [-1, c-1] expand $(1+s)^4$ and use that $s^2 + 4s + 6$ is increasing for $s \ge -1$. At $s = c-1$ its value is $c^2 + 2c + 3$. On [c-1, infinity), the quadratic on the right has derivative at its left endpoint

$$4 + 2(c^2 + 2c + 3)(c-1) = 2(c^3 + c^2 + c - 1) > 0.$$

The last polynomial increases for $c \ge 0$ and at $c = 2/3$ is $11/27$. The quadratic derivative then only increases. Its value at c-1 equals c^4, proving the second range. Both boundary points are included.

Consequently the relevant quadratic coefficients are

$$a_4(d) = 11 - 12d + 4d^2, \qquad a_{\ge 5}(d) = 11 - 24d + 16d^2. \tag{Q164.2.4}$$

They are positive throughout the entire host domain.

Q164.3 Same-kernel finite row moments

Fix $M \ge 4$. The finite convolution (Q164.1.1), with the actual sign reversal, gives

$$0 < \bar{f}_M \le U = q + R - \beta_M, \qquad R = KG_{M-1}(-K)g. \tag{Q164.3.1}$$

U is nonnegative because it dominates $\bar{f}_M$. Put $\beta = \beta_M$. The polynomial index M-1 is at least three, so all uses of $|G| \le 1$ are within the valid finite range.

The exact same-measure identities imply

$$-d \le ER \le -\tau, \quad \kappa\delta_2 \le E(gR) \le \delta_2, \quad ER^2 \le \delta_2, \quad |R| \le \sqrt{d}. \tag{Q164.3.2}$$

To check the signs directly, at $z = -K$ and $G(z) = a + zc, R = -zG(z)g$ has real coefficient xc and imaginary coefficient -a. Thus $\mathbb{E}R = -\int a\, d\nu$ and $\mathbb{E}(gR) = \int xc\, d\nu$. The bounds $1 - x \le a, c \le 1$ give the first two inequalities, using $\delta_3 \le \theta\delta_2$ for the lower bound on E(gR). Also $ER^2 = \int x|G|^2 d\nu \le \delta_2$. The original row bound applied to $KG(-K)g$ gives $|R| \le \|G(-K)g\|_2 \le \sqrt{d}$.

Because $\tau \le \beta \le d$ and $Eg = 0$,

$$E[q^2(R - \beta)] \le -2\tau - 2\kappa\delta_2 + d\sqrt{d} - \tau d. \tag{Q164.3.3}$$

Expanding $q^2 = 1 - 2g + g^2$, the first term satisfies $\mathbb{E}R - \beta \le -2\tau$, the next is $-2\mathbb{E}(gR) \le -2\kappa\delta_2$, and $\mathbb{E}(g^2R) - \beta d \le d\sqrt{d} - \tau d$.

The same actual cross terms give

$$E[q(R-\beta)^2] \le \delta_2 + 3d^2 + d\sqrt{\delta_2} + 2d\delta_2. \tag{Q164.3.4}$$

The expansion is

$$ER^2 - 2\beta ER + \beta^2 - E(gR^2) + 2\beta E(gR).$$

The five upper bounds are respectively $\delta_2, 2d^2, d^2$, d $\sqrt{\delta_2}$, and $2d\delta_2$. For the fourth use

$$|E(gR^2)| \le \sqrt{d\,ER^4} \le d\sqrt{ER^2} \le d\sqrt{\delta_2},$$

where $R^2 \le d$ is the same pointwise row bound. No independent copies of R or of the spectral measure have been introduced. Finally

$$Eq^3 = 1 + 3d - Eg^3 \le 1 + 4d \tag{Q164.3.5}$$

because $|g| \le 1$ and $|Eg^3| \le d$.

Q164.4 The common scalar row upper bound

Put $s = (R-\beta)/q$. By (Q164.3.1), $s \ge -1$. The two simultaneous actual pointwise bounds are $\bar{f}_M \le q(1+s)$ and $\bar{f}_M \le c$ q. Since these quantities are nonnegative, (Q164.2.3) gives

$$\bar{f}_M^4/q \le q^3 + 4q^2(R-\beta) + a(d)q(R-\beta)^2. \tag{Q164.4.1}$$

Apply the scalar majorant at the actual s.

Combining (Q164.3.3)–(Q164.3.5), and only then using $\delta_2 \le \theta$ d, yields

$$E[\bar{f}_M^4/q] \le 1 + d\,B_a(t), \qquad t = \sqrt{d},$$

$$B_a(t) = -4 + 8\theta^2 + a\theta + (4 + 2a/3)t + (-19/8 + 61a/16)t^2. \tag{Q164.4.2}$$

Here $\sqrt{\theta} < 2/3$ was used in d $\sqrt{\delta_2}$, and all coefficients multiplying the replaced δ_2 and $d\delta_2$ are positive. Equivalently, before this replacement the nonconstant upper bound is

$$-4d + (8\theta + a)\delta_2 + 4d\sqrt{d} + (3a-4)d^2 + (4+2a)d\delta_2 + ad\sqrt{\delta_2}.$$

This records all constant and cross contributions.

For $M \ge 5$, insertion of $a = 11 - 24t^2 + 16t^4$ gives

$$B_5(t) = \frac{229}{128} + \frac{34}{3}t + \frac{477}{16}t^2 - 16t^3 - 85t^4 + \frac{32}{3}t^5 + 61t^6. \tag{Q164.4.3}$$

For $M = 4, a = 11 - 12t^2 + 4t^4$ instead gives

$$B_4(t) = \frac{229}{128} + \frac{34}{3}t + \frac{555}{16}t^2 - 8t^3 - \frac{353}{8}t^4 + \frac{8}{3}t^5 + \frac{61}{4}t^6. \tag{Q164.4.4}$$

These are finite exact rational expansions, obtained by handwritten multiplication of (Q164.4.2). No polynomial was generated or checked by code.

Q164.5 One continuous quartic certificate

The following explicit quartic is positive on the whole interval $0 \le t \le 1/\sqrt{3}$:

$$R_0(t) = \frac{539}{128} - \frac{34}{3}t - \frac{189}{16}t^2 + \frac{1414}{15}t^4 > 0. \tag{Q164.5.1}$$

On [0, $1/3$], $t^3 \le t/9$, so

$$R_0'(t) \le -\frac{34}{3} + \frac{19733}{1080}t \le -\frac{16987}{3240} < 0.$$

The endpoint is

$$R_0(1/3) = \frac{44237}{155520} > \frac{1}{4}.$$

This proves the first whole interval.

On $[1/3, 1/\sqrt{3}]$,

$$R_0''(t) = -\frac{189}{8} + \frac{5656}{5}t^2 \ge \frac{36743}{360} > 100.$$

Take the fixed rational point $c = 3/8$, which lies in this interval. Direct handwritten substitution gives

$$R_0(c) = \frac{1679}{10240} > \frac{4}{25}, \qquad R_0'(c) = -\frac{37}{120}, \quad |R_0'(c)| < \frac{1}{3}.$$

Taylor's integral formula, or integrating R_0"≥100 in either direction between c and t, gives

$$R_0(t) \ge R_0(c) + R_0'(c)(t-c) + 50(t-c)^2 \ge R_0(c) - R_0'(c)^2/200 > \frac{4}{25} - \frac{1}{1800} = \frac{287}{1800} > \frac{1}{8}.$$

This proves (Q164.5.1) continuously, with no finite sampling of t.

Q164.6 All $M \ge 5$: the coefficient six

For $0 \le t \le 1/\sqrt{3}$, use $t \le 3/5$ and $t^2 \le 1/3$. The exact difference is

$$6 + 18t^2 + 36t^4 - B_5(t) = \frac{539}{128} - \frac{34}{3}t - \frac{189}{16}t^2 + 16t^3 + 121t^4 - \frac{32}{3}t^5 - 61t^6.$$

The last two adverse terms have the rigorous lower bounds

$$-\frac{32}{3}t^5 \ge -\frac{32}{5}t^4, \qquad -61t^6 \ge -\frac{61}{3}t^4.$$

Since $121 - 32/5 - 61/3 = 1414/15$, the whole difference is at least $R_0(t) + 16t^3 > 0$. Therefore

$$1 + dB_5(\sqrt{d}) \le 1 + 6d + 18d^2 + 36d^3 \le e^{6d}.$$

The last comparison is the positive Taylor lower bound for e^{6d}. Together with (Q164.4.2), it proves the $M \ge 5$ part of (Q164.0.1). No d-dependent orientation has been selected.

Q164.7 The exceptional $M = 4$: the coefficient seven

For the same whole interval,

$$7 + \frac{49}{2}t^2 + \frac{343}{6}t^4 - B_4(t) = \frac{667}{128} - \frac{34}{3}t - \frac{163}{16}t^2 + 8t^3 + \frac{2431}{24}t^4 - \frac{8}{3}t^5 - \frac{61}{4}t^6.$$

The fifth and sixth terms are bounded below using $t \le 3/5, t^2 \le 1/3$. Their combined fourth-power coefficient is at least

$$\frac{2431}{24} - \frac{8}{5} - \frac{61}{12} = \frac{11353}{120} = \frac{1414}{15} + \frac{41}{120}.$$

Hence the difference is at least

$$R_0(t) + 1 + \frac{13}{8}t^2 + 8t^3 + \frac{41}{120}t^4 > 0.$$

Thus

$$E[\bar{f}_4^4/q] \le 1 + 7d + \frac{49}{2}d^2 + \frac{343}{6}d^3 \le e^{7d}.$$

This explicitly uses the weaker but valid $M = 4$ cap from (Q164.2.1).

Q164.8 Real inputs, every outer length, and uniform specialization

The positive-kernel outlet is elementary. If B0 is any positive original operator with $h = B0\ 1$, row Cauchy gives $|\mathrm{B0}\ H|^2 \le hB0(|H|^2)$. Hölder with exponents 3 and $3/2$ then gives

$$I_q(B0H)^3 \le E[h^4/q]\,(E[(B0(|H|^2))^2])^2 \le E[h^4/q]\,\|B0\|_{2\to2}^4 (E|H|^4)^2. \tag{Q164.8.1}$$

The expectation in the middle factor is the original second moment. This argument permits arbitrary real H.

For $M \ge 4$ set $B0 = T^M$, apply (Q164.1.3) and parts Q164.6–7. For $M = 1, 2, 3$ use the exact C084 costs stated in part Q164.1. This proves the complete table (Q164.0.3) and (Q164.0.2). Since $10 < 204/19$ and 7, 6, $0 < 204/19$, the uniform version is

$$\boxed{I_q(T^M H)^3 \le e^{204d/19}(E|H|^4)^2 \quad (M \ge 1).} \tag{Q164.8.2}$$

If a separately proved actual nonnegative half-tree has $EH^4 \le \exp(-\lambda\ \mathrm{d})$, the exact outlet is

$$I_q(T^M H)^3 \le \exp\{-[2\lambda - c(M)]d\}. \tag{Q164.8.3}$$

For example the already specified value $\lambda = 45/8$ gives the uniform loss $39/76$, while $\lambda = 705/128$ gives $339/1216$. These are substitutions into a conditional statement; they are not new claims that an arbitrary actual half-tree has either input moment.

All $d = 0$ cases are valid. Then $g = 0, q = 1$ and $\bar{f}_M = 1$, though K can be nonzero. The real-input estimate also follows from (Q164.8.1) and the short operator facts. Actual finite tree messages equal one by original leaf integration.

M.22 Bare path eighth moments at every length at least six

Use $A = J + K$, $T = J - K = A^*$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $f_n = A^n\mathbf{1}$, $\bar{f}_n = T^n\mathbf{1}$, $d = \mathbb{E}g^2$, $\delta_j = \|K^j\mathbf{1}\|_2^2$, $\theta = 13/32$, $\kappa = 19/32$ and $\tau = d - \delta_2$. We prove

$$\mathbb{E}[(p\bar{f}_n)^8] \le 1 - d/50 \qquad (n \ge 6). \tag{Q72.bare}$$

The finite calculus supplies $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{j+1} \le \theta\delta_j$, $\|K\|^2 \le \theta$ and $|Kv(x)| \le \|v\|_2$. Write $G_m(z) = \sum_{j=0}^{m-1} z^j$, with $G_0 = 0$, and $\beta_1 = \beta_2 = 0$, $\beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor}(-1)^{i-1}\delta_i$ for $j \ge 3$. Then

$$\sum_{j=0}^{n}(-K)^j\mathbf{1} = \bar{f}_n + \sum_{j=3}^{n}\beta_j\bar{f}_{n-j}, \qquad \tau \le \beta_j \le d \quad (j \ge 3). \tag{Q72.1.2}$$

In particular $\kappa d \le \tau \le d - d^2 \le 2/9$. The paths lie in $[0, 2]$ and have means $\mu_n = \mathbb{E}f_n = \mathbb{E}\bar{f}_n \in [0, 1]$. The finite geometric bounds are

$$\|G_m(K)\| \le 1 \quad (m = 0, 1 \text{ or } m \ge 3), \qquad \|G_2(K)\|^2 \le 1 + \theta. \tag{Q72.1.4}$$

At $d = 0$ leaf integration gives equality in (Q72.bare); divisions by d below use $d > 0$.

Q72.2 Finite masks and centered paths

For a finite polynomial with coefficients in $[0, 1]$, sign-separated geometric sums at $z = i\sqrt{x}$, $0 \le x \le \theta$, give

$$|\operatorname{Re} P(z)| \le (1 - x^2)^{-1}, \qquad |\operatorname{Im} P(z)| \le \sqrt{x}(1 - x^2)^{-1}.$$

Thus

$$|P(z)|^2 \le \frac{1 + x}{(1 - x^2)^2} \le \frac{32768}{16245} < \frac{100}{49} < \frac{9}{4}. \tag{Q72.2.1}$$

The strict comparison uses $32768 \cdot 49 = 1605632 < 1624500$. Let P_0 be orthogonal projection onto the mean-zero subspace and $C_0 = P_0KP_0$ there. This operator is skew and $\|C_0\|^2 \le \theta$. For $v_n = \bar{f}_n - \mu_n\mathbf{1}$, the path recursion gives

$$v_{n+1} = -\mu_n g - C_0 v_n, \qquad v_n = -\sum_{j=0}^{n-1}\mu_{n-1-j}(-C_0)^j g. \tag{Q72.2.2}$$

Since its coefficients lie in $[0, 1]$, (Q72.2.1) and reversal give

$$\sigma(\bar{f}_n), \sigma(f_n) \le (3/2)\sqrt{d}. \tag{Q72.2.3}$$

Here σ is standard deviation under the given measure. Set $H_n = p\bar{f}_n$. The recursion also yields

$$H_n = \mu_{n-1}(1 - g^2) - pKv_{n-1}, \qquad \mathbb{E}H_n = \mu_{n+1} \le 1.$$

Comparison with the constant μ_{n-1}, using $p \le 2$ and $\sqrt{\theta} < 2/3$, gives

$$\sigma(H_n) \le (1 + 3\sqrt{\theta})\sqrt{d} < 3\sqrt{d}. \tag{Q72.2.5}$$

Likewise, $\bar{f}_a f_L - \mu_a\mu_L = f_L(\bar{f}_a - \mu_a) + \mu_a(f_L - \mu_L)$ gives $\sigma(\bar{f}_a f_L) \le (9/2)\sqrt{d}$.

Q72.3.1 An exponential mass estimate

Put $w_n = \sum_{j=2}^{n} (-K)^j \mathbf{1} = K G_{n-1}(-K) g$. Retaining $\beta_n \bar f_0$ in (Q72.1.2) gives

$$0 \le \bar f_n \le q + w_n - \beta_n, \qquad \mathbb{E} w_n = -\beta_{n+1}. \tag{Q72.3.1}$$

For $n \ge 4$, $|w_n| \le \sqrt{d}$, $\mathbb{E} w_n^2 \le \theta d$ and $|\mathbb{E} w_n| \le d$. Arithmetic–geometric mean gives

$$p \bar f_n \le [1 + (w_n - \beta_n)/2]^2, \qquad \mathbb{E} H_n^8 \le e^{-8\beta_n} \mathbb{E} e^{8 w_n}. \tag{Q72.3.2}$$

Use the negative-mean lemma of Section O.14, whose coefficient is $\lambda(x) = (188/25)(2 - \kappa/(1+x))^2$. It increases with x, and $\lambda(\sqrt{d}) \le \lambda(3/5) = 32691132/1638400 < 20$. Consequently

$$\mathbb{E} H_n^8 \le e^{-8\beta_n - 8\beta_{n+1} + 20 d} \le e^{21 d/2} \qquad (n \ge 4). \tag{Q72.3.4}$$

For $n = 3$, $w_3 = r - s$ with $r = K^2 \mathbf{1}$, $s = K^3 \mathbf{1}$. Its squared norm is $\delta_2 + \delta_3 \le \theta(1+\theta) d$, $|w_3| \le \sqrt{(1+\theta) d}$, and $\mathbb{E} w_3 = -d$. The same lemma with $A_0 = 1 + \theta$ gives

$$\mathbb{E} H_3^8 \le e^{[20(1+\theta) - 16] d} = e^{97 d/8}. \tag{Q72.3.5}$$

Equation (Q72.3.1) and $pq \le 1$ also imply $H_n \le 1 + 2\sqrt{d}$ for $n \ge 4$. Thus, using $\sqrt{1+\theta} < 6/5$,

$$\mathbb{E} H_n^8 \le e^{97 d/8} \le e^{13 d}, \qquad 0 \le H_n \le 1 + (12/5)\sqrt{d} \quad (n \ge 3). \tag{Q72.3.6}$$

Q72.3.2 Four retained convolution losses

Let $B_0 = -K$ and $R_n = \sum_{j=2}^{n} B_0^j \mathbf{1}$. For $n \ge 6$, the four convolution indices $n, n-1, n-2, n-3$ are all at least three. Retaining their τ multiples gives

$$\bar f_n \le q + R_n - \tau(\bar f_0 + \bar f_1 + \bar f_2 + \bar f_3) = (1 - 3\tau) q + \widetilde R_n - \tau(1 - d), \tag{Q72.3.7}$$

where

$$\widetilde R_n = R_n - 2\tau r + \tau s = B_0 Q_{n-1}(B_0)(B_0 \mathbf{1}), \qquad Q_m(z) = G_m(z) - \tau(2 + z). \tag{Q72.3.8}$$

Here $\bar f_0 = 1$, $\bar f_1 = q$, $\bar f_2 = q + r$ and $\bar f_3 = q + r - s - d$. For every $m \ge 5$, $0 \le x \le \theta$ and $z = i\sqrt{x}$,

$$|G_m(z)| \le 1, \qquad x |G_m(z)|^2 \le 1/3. \tag{Q72.3.9}$$

The first bound is (Q72.1.4). If m is odd, $|G_m|^2 = (1 + x^m)/(1 + x)$; the sufficient inequality $2x + 3x^{m+1} \le 1$ follows from $13/16 + 1/72 < 1$, using $\theta^2 < 1/6$. If m is divisible by four, $x|G_m|^2 \le x/(1+x) \le 13/45$. If $m \equiv 2 \pmod 4$, then $m \ge 6$ and $|G_m|^2 \le (1 + x^3)^2/(1+x)$. Now $2x + 6x^4 + 3x^7 \le 13/16 + 1/6 + 13/2304 = 2269/2304 < 1$.

Sign-separated sums also give

$$\operatorname{Re}[G_m(z)(2 - z)] \ge 2 - \frac{3x}{1 - x^2} \ge 154/285.$$

Since $\tau(4 + x) \le 47/48 < 308/285$,

$$|Q_m(z)|^2 = |G_m(z)|^2 - 2\tau \operatorname{Re}[G_m(z)(2 - z)] + \tau^2(4 + x) \le |G_m(z)|^2. \tag{Q72.3.10}$$

The row bound and (Q72.3.9) imply

$$\begin{aligned} |\widetilde R_n| &\le \sqrt{d}, \qquad \mathbb{E} \widetilde R_n^2 \le d/3, \\ \mathbb{E} \widetilde R_n &= -\beta_{n+1} + 2\tau d \le -\tau(1 - 2d). \end{aligned} \tag{Q72.3.11}$$

Q72.3.3 The scalar interface

Apply row V of Table 56. For $0 \le d \le 1/3$, put $\eta = \kappa d$, $B = 1 - 3\eta$, $a_0 = \eta(1-2d)$, and $a_* = \eta(1-d)$. The row proves $\mathbb{E}H^8 \le 1 - d \le 1 - d/50$ for every nonnegative H with $|R| \le \sqrt{d}$, $\mathbb{E}R \le -a_0$, $\mathbb{E}R^2 \le d/3$, and

$$H \le B\left(1 + \frac{R - a_*}{2B}\right)^2. \tag{Q72.3.12}$$

Its full-interval certificate also verifies the compatible moments and the positive interpolation base. Here $\tau \ge \eta$ makes (Q72.3.7) at most $Bq + \widetilde{R}_n - a_*$. Multiplying by p and applying AM–GM to Bp and this nonnegative upper gives the displayed square for H_n. Equation (Q72.3.11), with $1 - 2d \ge 0$, supplies its three actual moment bounds and proves (Q72.bare).

N Reference fourth moments

This family proves the genuine-reference catalogue of Proposition 9.1. Its receiving arm of length two has its own input; the directions for references of length at least three are stated separately.

N.1 The full pointwise coefficient for the one-connector genuine reference

Use the original finite model. Fix the actual terminal $F = qf_2$, $X = TF$, $f_2 = p + r$, with the unit arm inward, the two-edge arm outward, and the connector toward the reference root. For $0 < d \le 1/5$, put $m = 1 - 2d$, $v = \delta_2/d \in [d, \theta]$. The full coefficient bound is

$$f_2(X^2 + 2mX + 3m^2) \le C(d, v) := 255/32 - 13d + 2(v - 3/10)_+. \tag{L29.T}$$

This statement concerns the pointwise coefficient; the covariance payment is treated separately. At $d = 0$ the actual messages are one and the coefficient is six, without defining v.

The actual identity with $u = g(g + r)$ is $F = 1 + r - u$, $\mathbb{E}F = m$, $X = q - 2d - s + Ku$, where $s = Kr$. The original row and Cauchy bounds give

$$\mathbb{E}|u| \le d\sqrt{1+v}, \quad |s| \le \sqrt{dv}, \quad 0 \le X \le q + \alpha,$$
$$\alpha = -2d + \sqrt{dv} + d\sqrt{1+v} \ge 0, \quad X \le 2m.$$

Centering an original row gives $r(x) = \mathbb{E}[(K(x,\cdot) - g(x))g]$ and $r(x)^2 \le p(x)q(x)d$, so $f_2 \le p + \sqrt{pqd} \le p + \sqrt{d}$. The common error bound is

$$\alpha \le e(v) := \min\{1/8, ((1+d)v - d)/2\}. \tag{L29.e}$$

The second branch follows from $\sqrt{dv} \le (d+v)/2$ and $\sqrt{1+v} \le 1 + v/2$. For the first, use $\sqrt{1+v} \le \sqrt{45/32} < 19/16$ and complete the square in $\alpha \le \sqrt{\theta d} - 13d/16 \le 1/8$.

The whole low interval $0 < d \le 1/9$. Let $x = \sqrt{d} \le 1/3$. The actual coefficient is at most the maximum on $0 \le q \le 2$ of

$$F(q) = (2 + x - q)\{(q + 1/8)^2 + 2m(q + 1/8) + 3m^2\}.$$

We prove the stronger bound $\max F \le 255/32 - 13d$. Put $a = 9/8 - 2x^2$, $b = a^2 + 2m^2$, $B_1 = 1 + 2a + b$. One has $F'(0) > 0$ and

$$F'(1/2) = 1/4 + 3x + 8x^2 - 4x^3 - 12x^4 + (1/4)(x + 2x^2) - 1/64 > 0.$$

For example $8x^2 - 4x^3 - 12x^4 \geq 3x^2$ for $x \leq 1/2$. Since F' is a concave quadratic, a maximum lies in $[1/2, 2]$. There $F''(q) \leq -L$, with $L = 7/2 - 2x - 8x^2 \geq 35/18$. The quadratic tangent at one therefore gives

$$\max F \leq F_1 + z^2/(2L),$$
$$F_1 = (1+x)(417/64 - (33/2)x^2 + 12x^4),$$
$$z = -145/64 + (17/4)x + (25/2)x^2 - 4x^3 - 12x^4.$$

Let $D = 255/32 - 13x^2 - F_1$. The complete remainder is

$$R(x) := 4096(2LD - z^2) = 20639 - 131744x + 269888x^2 + 333312x^3 \\ - 1567232x^4 - 401408x^5 + 2146304x^6 + 393216x^7 - 589824x^8.$$

Its derivative is

$$R' = -131744 + 539776x + 999936x^2 - 6268928x^3 \\ - 2007040x^4 + 12877824x^5 + 2752512x^6 - 4718592x^7.$$

Absorb the positive fifth and sixth terms into the cubic, using $-6268928 + 12877824/9 + 2752512/27 = -127875072/27 < -4700000$. Dropping the other nonpositive terms gives $R' \leq h = -131744 + 539776x + 999936x^2 - 4700000x^3$. On $[0, 1/5]$, h' is concave and positive at both endpoints, so $h \leq h(1/5) = -534784/25 < 0$. On $[1/5, 1/3]$, $h'' \leq -3640128 < -3000000$. At $2/7$, $h = -1891936/343 < -5000$ and $h' = -1952768/49$, $|h'| < 40000$. Strong concavity gives $h < -5000 + 800/3 < -4700 < 0$ on this entire interval. Thus R decreases and $R(1/3) = 264973/243 > 0$. Since $L > 0$, this proves the stronger low-interval coefficient.

The entire middle interval $1/10 \leq d \leq 3/16$**.** Now keep the sharper actual row bound. The coefficient is at most

$$\Phi(e) := \max_{0 \leq q \leq 2} \{2 - q + x\sqrt{q(2-q)}\}\{(q+e)^2 + 2m(q+e) + 3m^2\}, \qquad x = \sqrt{d},$$

at $e = e(v)$. As a maximum of increasing convex quadratics with nonnegative coefficients, Φ increases and is convex for $e \geq 0$. Set $e_0 = 3/20 - 7d/20$, $v_1 = (d + 1/4)/(1 + d)$. Here $0 < e_0 < 1/8$ and $3/10 < v_1 \leq 7/19 < \theta$. For $v \leq 3/10$, $e(v) \leq e_0$; on $[3/10, v_1]$ it is affine; after v_1 it equals $1/8$. Thus it suffices to prove

$$\Phi(e_0) \leq 255/32 - 13d, \qquad \Phi(1/8) \leq 255/32 - 13d + 2(v_1 - 3/10).$$

The intermediate segment follows from the convex chord bound and the outer segments from monotonicity and the nonnegative target slope.

For either endpoint, and every $e \in [e_0, 1/8]$, put $a = m + e$, $b = a^2 + 2m^2$, $B_1 = 1 + 2a + b$, $z = 2(1+x)(1+a) - B_1$. One has $5/16 < x < 7/16$, $m \geq 5/8$, $a \geq 227/320 > 7/10$, and $0 < z \leq 5/4$. Indeed $z_e = 2x - 2a < 0$; at $e = 1/8$ the same polynomial z as above increases on $[5/16, 7/16]$, since $17/4 - 12x^2 + x(25 - 48x^2) > 0$, and its left value is $765/16384 > 0$. For the upper bound discard e and use the function $1 + 2x(1+m) - 3m^2$, increasing in x and decreasing in m; its endpoint bound is $5/4$. For $t = 2 + x$, the cubic $F(q) = (t - q)(q^2 + 2aq + b)$ has $F'(0) > 0$, $F'(1) = z > 0$, so it increases on $[0, 1]$. For $q = 1 + w \geq 1$, the original row improvement $\sqrt{q(2-q)} = \sqrt{1 - w^2} \leq 1 - w^2/2$ bounds the coefficient by

$$(1+x)B_1 + zw - A'w^2 - w^3, \qquad A' = 1 - x + 2a + xB_1/2 \geq 101/40 > 5/2.$$

Only additional nonpositive cubic and quartic terms were discarded. The lower bound uses $B_1 \ge 18/5$ and $5/16 \le x \le 7/16$. Since $z \le A'/2$, completing the square gives, also for $q \le 1$,

$$\Phi(e) \le (1+x)B_1 + z^2/(4A') \le (1+x)B_1 + z/8 = H_e := (7/8+x)B_1 + (1+x)(1+a)/4. \quad \text{(L29.II.4)}$$

At $e = e_0$, $a = (23 - 47x^2)/20$, $B_1 = (2649 - 7242x^2 + 5409x^4)/400$. For $D_0 = 255/32 - 13x^2 - H_{e_0}$, the exact polynomial is

$$3200 D_0 = 5237 - 22912x + 10974x^2 + 59816x^3 - 37863x^4 - 43272x^5.$$

On the wider interval $I = [5/16, 7/16]$, $D_0'' = 5487/800 + (22431/200)x - (113589/800)x^2 - (5409/20)x^3$ is concave, since its second derivative is $-113589/400 - (16227/10)x < 0$. Its endpoint values $8105109/409600$, $2499363/409600$ exceed six. Also $D_0(7/16) = 31687275/419430400 > 3/40$ and $D_0'(7/16) = 3509875/26214400 \in (0, 1/4)$. Strong convexity gives $D_0 > 3/40 - 1/192 = 67/960 > 0$ on I.

At $e = 1/8$, $H_{1/8} = 3191/512 + (451/64)x - (239/16)x^2 - 17x^3 + (21/2)x^4 + 12x^5$. Multiply the second endpoint's desired excess by $1 + x^2$ and set

$$P = (1 + x^2)\left[255/32 - 13x^2 + 2\left(\frac{x^2 + 1/4}{1 + x^2} - 3/10\right) - H_{1/8}\right].$$

Its complete polynomial and second derivative are

$$2560 P = 4189 - 18040x + 12989x^2 + 25480x^3 - 21920x^4 + 12800x^5 - 26880x^6 - 30720x^7,$$

$$P'' = 12989/1280 + (1911/32)x - (411/4)x^2 + 100x^3 - 315x^4 - 504x^5.$$

Since $411/4 + 315(7/16)^2 + 504(7/16)^3 < 206$ and $100x^3 \ge (125/4)x^2$, $P'' \ge 12989/1280 + (1911/32)x - (699/4)x^2$. The latter concave quadratic has positive endpoint values $60131/5120$, $14471/5120$, so $P'' > 0$. Finally $P'(7/16) = -9291053/(8192 \cdot 2560) < 0$ and $P(7/16) = 4683285/(131072 \cdot 2560) > 0$. Thus P decreases to a positive endpoint on I, proving the second endpoint certificate and the entire middle interval.

The actual truncation for $3/16 \le d \le 1/5$. Here retain the additional actual bound $X \le 2m$. Let $s_0 = \sqrt{d}$, $t = 2 + s_0$, $e = e(v)$ and $q_0 = 2m - e \in (0, 2)$. The coefficient is at most

$$\max_{0 \le q \le 2} (t - q)\{y^2 + 2my + 3m^2\}, \qquad y = \min(q + e, 2m).$$

Before truncation the derivative of $F(q) = (t - q)\{(q + e)^2 + 2m(q + e) + 3m^2\}$ satisfies $F'(q_0) = m[6(2 + s_0) + 6e - 23m] \ge 0$, since $46d + 6s_0 \ge 11$ ($s_0 \ge 43/100$). Also $F'(0) > 0$: $m \ge 3/5$, $m + e \le 3/4$ give $2t(m + e) \ge 12/5$ whereas $(m + e)^2 + 2m^2 \le 43/32$. Its concave quadratic derivative is nonnegative on $[0, q_0]$; after truncation the expression is $11m^2(t - q)$ and decreases. Consequently $C_{\text{actual}} \le 11m^2(4d + e + s_0)$.

Write $C_* = 8 - 13d + 2(v - 3/10)_+$. For $v \le 3/10$, $e \le e_0 = 3/20 - 7d/20$ and set $B = 3/20 + 73d/20 + s_0$, $G = 11m^2 B$. On the entire interval, $B \ge 2023/1600$, $B' \le 73/20 + 50/43$, $3/5 \le m \le 5/8$. The bracket is negative, so

$$G' = 11m(-4B + mB') \le (33/5)[-2023/400 + (5/8)(73/20 + 50/43)] < -13.$$

At $d = 3/16$, $s_0 = \sqrt{3}/4 < 7/16$ gives $B \le 407/320$, $G \le 111925/20480$, whereas $8 - 13d = 113920/20480$. Hence $G \le 8 - 13d - 1995/20480$ everywhere. For $v \ge 3/10$, $e \le e_0 + (1+d)(v - 3/10)/2$. The additional slope $11m^2(1+d)/2$ decreases in d and is at most $5225/2048$. Compared with the slope two of C_*, its maximum loss is $(5225/2048 - 2)(17/160) = 19193/327680$. The original margin is $1995/20480 = 31920/327680$, so

$$C_{\text{actual}} \le C_* - 12727/327680 < C_* - 1/32 = C(d, v). \tag{L29.III}$$

The three closed score intervals overlap and cover $0 < d \le 1/5$. Their actual terminal, receiving weight and original directions are the same, proving (L29.T) with every local variable defined.

N.2 One connector from the terminal $(1, 2)$

Use the original finite measure and a real skew kernel K, $|K| \le 1$. Write $A = J + K$, $T = J - K$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $f_2 = A^2\mathbf{1}$, $d = \mathbb{E}g^2$, $\delta_j = \|K^j\mathbf{1}\|^2$, and $\theta = 13/32$. Lemma D.1 gives $0 \le d \le 1/3$, $\|K\|^2 \le \theta$, and $\delta_{j+1} \le \theta\delta_j$. Direct the terminal unit arm inward, its two-edge arm outward, the connector toward the receiving root, and that root's length-two arm outward. Thus $F = qf_2$ and $X = TF$. We prove

$$\mathbb{E}[f_2 X^4] \le 1. \tag{L32.T}$$

The inequality is strict for $d > 0$. Both directions and the reference weight are fixed throughout the proof.

The common covariance criterion.

Put $r = Kg$, $s = Kr$, $U = g(g+r)$, $m = 1-2d$, $k = 1-d$, and $h = 4m^3$. Since $f_2 = 1+g+r \in [0, 2]$, $|g + r| \le 1$. Skewness gives

$$\mathbb{E}U = d, \quad \mathbb{E}U^2 \le d, \quad \operatorname{Var} U \le dk, \quad \mathbb{E}|U| \le \sqrt{d(d + \delta_2)}.$$

The last bound is Cauchy–Schwarz; hence $\|KU\|^2 \le d^2(1 + v)$, where $v = \delta_2/d$ when $d > 0$. The actual identities, with $Z = X - m$, are

$$\begin{aligned} F &= 1 + r - U, \qquad X = q - 2d - s + KU, \\ Z &= -g - s + KU, \qquad \mathbb{E}f_2 = k, \\ \mathbb{E}[f_2 Z] &= -d + \delta_2 - \mathbb{E}[(g + r + s)U], \\ \mathbb{E}Z^2 &= d - 2\delta_2 + \delta_3 + \|KU\|^2 \\ &\quad + 2\mathbb{E}[(r + K^2 r)U]. \end{aligned} \tag{L30.2}$$

Here $m = \mathbb{E}F$; it need not equal $\mathbb{E}X$. Let $c \ge 0$ satisfy the actual pointwise bound $f_2(X^2 + 2mX + 3m^2) \le c$. Expanding the fourth power at m, and using $\delta_3 \le \theta\delta_2$, gives

$$\mathbb{E}[f_2 X^4] - 1 \le -d\mathcal{L} + \mathbb{E}[(V - \mathbb{E}V)(U - d)],$$

where

$$\begin{aligned} V &= -h(g + s) + (2c - h)r + 2cK^2 r, \\ \mathcal{B} &= 3m^4 + 4m^3 + 2m^2 + 2m + 2, \\ \mathcal{L} &= \mathcal{B} - kc + [(2 - \theta - 3d)c - h]v, \\ \mathbb{E}V &= d(h - 2c + 2cv). \end{aligned}$$

The uncentered constant before subtracting one is $km^4 - hd + h\delta_2 + c[d - 2\delta_2 + \theta\delta_2 + d^2(1+v)] + d^2(h - 2c + 2cv)$; it equals $-d\mathcal{L} + 1$.

Let σ be the spectral measure of $-K^2$ at g. It has mass d, support $[0, \theta]$ and moments $\int x^j \, d\sigma = \delta_{j+1}$. Thus

$$\|V\|^2 = \int \{h^2(1-x)^2 + x(2c - h - 2cx)^2\} \, d\sigma.$$

If $b = 4c^2(2-\theta) - 4ch - h^2 \geq 0$, use $\delta_4 \leq \theta\delta_3$ and the integrated restriction $\delta_3 \geq \delta_2^2/d$, then subtract $(\mathbb{E}V)^2$:

$$\operatorname{Var} V \leq d\mathcal{Q}, \qquad \mathcal{Q} = h^2 + (4c^2 - 4ch - h^2)v - bv^2 - d(h - 2c + 2cv)^2. \tag{L30.3.3}$$

In particular $\mathcal{Q} \geq 0$ for every actual kernel to which this calculation applies. Original-measure Cauchy–Schwarz now gives

$$\mathbb{E}[f_2 X^4] \leq 1 + d\{-\mathcal{L} + \sqrt{k\mathcal{Q}}\}. \tag{L30.3.5}$$

Consequently it suffices to prove $b \geq 0$, $\mathcal{L} > 0$, and $\mathcal{D} := \mathcal{L}^2 - k\mathcal{Q} > 0$.

One certificate on the whole range $0 < d \leq 1/5$.

The proved pointwise coefficient (L29.T) supplies

$$c = 255/32 - 13d + 2(v - 3/10)_+. \tag{N.1}$$

The actual moment range is $d \leq v \leq \theta$, since $d^2 \leq \delta_2 \leq \theta d$. Here $859/160 \leq c \leq 1309/160$, $0 < h \leq 4$; hence

$$b \geq 4(51/32)(859/160)^2 - 16(859/160) - 16 = 16762811/204800 > 80.$$

Here the lower bound first uses $h \leq 4$; the resulting polynomial $4(51/32)c^2 - 16c - 16$ increases for $c \geq 859/160$. Also $[(2 - \theta - 3d)c - h] \geq 34181/25600 > 4/3$. Writing $m \in [3/5, 1]$, one obtains

$$\mathcal{L} \geq P(m) := 3m^4 + 4m^3 - \tfrac{5}{4}m^2 - \tfrac{669}{320}m + \tfrac{371}{320}.$$

Indeed $c \leq 1309/160 - 13d$. Since $P'' = 36m^2 + 24m - 5/2 > 0$ and $P'(3/5) = 26571/8000 > 0$, $\mathcal{L} \geq P(3/5) = 3539/5000 > 0$.

It remains to check a single polynomial on each side of $v = 3/10$. An expanded form useful for this check is

$$\begin{aligned}\mathcal{Q} = {} & h^2(k - v + v^2) + ch[4d - 4(1+d)v + 4v^2] \\ & + c^2[-4d + 4(1+2d)v - 4(2 - \theta + d)v^2].\end{aligned}$$

For the lower branch put $v = d + (3/10 - d)t$; for the upper branch put $v = 3/10 + (\theta - 3/10)t$, with $0 \leq t \leq 1$. The latter entire rectangle is allowed because $d \leq 1/5 < 3/10$. These domains retain the actual restriction $v \geq d$, but need not consist entirely of realizable kernel moments.

For $i = 0, 1, 2, 3$, $j = 0, 1$, substitute $d = (i+x)/20$, $t = (j+y)/2$, $(x, y) \in [0, 1]^2$. The resulting polynomial $\mathcal{D}_{ij}$ has degree at most $(n, m) = (9, 2)$ on the lower branch and $(8, 4)$ on the upper branch. Apply Lemma A.1 to its power coefficients. The Bernstein coefficients β_{rs} are the explicit rationals given by (A.18). Every such coefficient is at least the corresponding entry below. The two columns in each branch correspond to $j = 0, 1$. This finite table follows by expanding the displayed $\mathcal{L}$ and $\mathcal{Q}$ and applying (A.18). All denominators are positive integers; none depends on d, v, x, y. The Bernstein

Table 23: Lower bounds for every Bernstein coefficient in the sixteen cells.

i	lower branch		upper branch	
0	3	5	13	18
1	$3/2$	$3/2$	4	$13/2$
2	$3/5$	$3/40$	$3/4$	$5/3$
3	$1/2$	$1/200$	$1/25$	$7/40$

basis is nonnegative and sums to one on the unit square. Thus $\mathcal{D} \geq 1/200 > 0$ on both closed branches, including their common boundary. This proves (L32.T) for $0 < d \leq 1/5$ by (L30.3.5). Indeed $\mathcal{L} \leq 13 + (51/32)(1309/160)\theta < 19$, and $\mathcal{L} - \sqrt{k\mathcal{Q}} = \mathcal{D}/(\mathcal{L} + \sqrt{k\mathcal{Q}}) \geq 1/7600$. Thus this range has deficit at least $d/7600$.

The same direction on $1/5 \leq d \leq 1/3$.

The direct actual first moment is

$$\mathbb{E}[f_2 X] = 1 - 4d + \delta_2 + 2d^2 - \mathbb{E}[(g + r + s)U].$$

Since $g + s$ is orthogonal to r, $\|g + r + s\|^2 = d - \delta_2 + \delta_3 \leq d - (1 - \theta)\delta_2$, and $\|U\|^2 \leq d$. Writing $\kappa = 19/32$, this gives

$$\mathbb{E}[f_2 X] \leq 1 - 4d + 2d^2 + d[v + \sqrt{1 - \kappa v}] \leq 1 - \tfrac{87}{32}d + 2d^2 \leq h_*(d) := 1 - \tfrac{81}{32}d + 2d^2.$$

Here $v + \sqrt{1 - \kappa v}$ increases on $[0, \theta]$, and its right endpoint is less than $13/32 + 7/8 = 41/32$. The original positive row gives $0 \leq X \leq m_*(d) = 2(1 - 2d)$. The functions m_* and h_* are nonnegative and decreasing on this interval; therefore

$$\mathbb{E}[f_2 X^4] \leq m_*^3 h_* \leq (6/5)^3(459/800) = 12393/12500 < 1.$$

The two parameter ranges use the same original orientation and meet at $d = 1/5$. At $d = 0$, $K\mathbf{1} = 0$, so every actual tree message is one by leaf integration even if $K \neq 0$. For actual equal-pair factors $0 \leq E_i \leq 1$, positivity gives $T((\prod_i E_i) q f_2) \leq X$, and a receiving equal-pair factor is also at most one. Full reversal sends K to $-K$ and gives $\mathbb{E}[\bar{f}_2\{A(p\bar{f}_2)\}^4] \leq 1$ with the same strictness.

N.3 The genuine length-two reference after every connector of length at least six

Keep $F = qf_2$, $f_2 = p + r$, and $X_n = T^n F$, with every connector edge directed from the terminal toward the receiving root. For every finite integer $n \geq 6$,

$$\mathbb{E}[f_2 X_n^4] \leq 1. \tag{L33.A}$$

More precisely the upper bounds are $1 - 483d/400$ for $0 < d \leq 1/10$, and $1 - 853/124416$ strictly for $1/10 \leq d \leq 1/3$. Actual equal-arm pairs and full reversal are included by positivity, reversing all original edges and weights together.

Write $B = -K$, $u = B\mathbf{1} = -g$, $c = 19/32$, $\tau = d - \delta_2$, $\theta = 13/32$, and $s = K^2 g$. We use the original spectral and row bounds $\|K\|^2 \leq \theta$, $d^2 \leq \delta_2 \leq \theta d$, $\delta_{j+1} \leq \theta\delta_j$, $0 \leq d \leq 1/3$, and $|Kz| \leq \|z\|_2$. The scalar coefficient proved in Section A.3 says that

$$\begin{aligned} &0 \leq Z \leq (1 - cd)q - cd + \sqrt{d}, \qquad 0 \leq \mu \leq 1 - 3cd \\ \Longrightarrow \quad &p(Z^2 + 2\mu Z + 3\mu^2) \leq 7 - 10d \quad (0 \leq d \leq 1/3). \end{aligned} \tag{L33.1.2}$$

This is only a scalar input; the additional r weight is retained below.

Four actual positive-convolution losses. Define

$$S_j = \sum_{i=0}^{j} B^i \mathbf{1}, \quad \beta_1 = \beta_2 = 0,$$
$$\beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1} \delta_i \quad (j \geq 3).$$

Skew-adjointness and decreasing alternating sums give

$$\begin{gathered} \mathbb{E} S_j = 1 - \beta_{j+1}, \quad \mathbb{E}[u S_j] = \beta_{j+2}, \quad \tau \leq \beta_j \leq d \ (j \geq 3), \\ S_j = \bar{f}_j + \sum_{i=3}^{j} \beta_i \bar{f}_{j-i}. \end{gathered} \tag{L33.2.1}$$

The last identity follows recursively from $T S_{j-1} = S_j - \beta_j \mathbf{1}$; every $\bar{f}_j = T^j \mathbf{1}$ is nonnegative. The same actual input satisfies

$$F = (1 + r/2)^2 - (g + r/2)^2 \leq F_0 := (1 + r/2)^2.$$

For $Y_n = T^n F_0$, the inequality $T(r^2) \leq 2\delta_2$ and finite Duhamel expansion yield

$$Y_n \leq \widehat{S}_n - \gamma \bar{f}_{n-1} - \sum_{j=3}^{n} \omega_j \bar{f}_{n-j}, \quad \widehat{S}_n = S_n + B^{n+2} \mathbf{1}, \quad \gamma = d - \delta_2/2, \quad \omega_j = \begin{cases} \beta_j, & j \text{ even}, \\ \beta_{j+2}, & j \text{ odd}. \end{cases} \tag{L33.2.2}$$

Indeed, the signed r contribution is exactly

$$T^n r = B^{n+2} \mathbf{1} - d \bar{f}_{n-1} + \delta_2 \bar{f}_{n-3} - \delta_3 \bar{f}_{n-5} + \cdots,$$

retaining only nonnegative indices; combine this with (L33.2.1). For each $n \geq 6$ retain the last four terms:

$$Z = \widehat{S}_n - \sum_{i=0}^{3} a_i \bar{f}_i, \qquad a_i = \omega_{n-i} \in [\tau, d], \qquad 0 \leq X_n \leq Y_n \leq Z. \tag{L33.2.3}$$

Every discarded term is a nonnegative subtraction. If $\mathbb{E}\widehat{S}_n = 1 - \alpha$ and $\mathbb{E}[u\widehat{S}_n] = \eta$, then

$$\alpha = \begin{cases} \beta_{n+3}, & n \text{ even}, \\ \beta_{n+1}, & n \text{ odd}, \end{cases} \qquad \eta = \begin{cases} \beta_{n+2}, & n \text{ even}, \\ \beta_{n+4}, & n \text{ odd}. \end{cases}$$

Thus $\alpha, \eta, a_0, \ldots, a_3 \in [\tau, d]$. The actual short paths are

$$\bar{f}_0 = 1, \quad \bar{f}_1 = q, \quad \bar{f}_2 = q + r, \quad \bar{f}_3 = q + r - s - d,$$

with means $1, 1, 1 - d, 1 - 2d$. Consequently, for $\mu = \mathbb{E} Z$ and $w = Z - \mu$,

$$0 \leq \mu \leq 1 - \tau(5 - 3d) \leq M(d) := 1 - 5cd + 3cd^2. \tag{L33.2.5}$$

The common finite spectral polynomial. Put $P_n(z) = \sum_{j=0}^{n-1} z^j + z^{n+1}$. Exactly,

$$\widehat{S}_n - 1 = P_n(B) u, \qquad \widehat{S}_n - 1 - u = B P_{n-1}(B) u.$$

For $0 \le x \le \theta$,

$$|P_n(i\sqrt{x})|^2 = \frac{1 + x^n(1 - x + x^2) - 2\Re\{(i\sqrt{x})^n(1 - x - i\sqrt{x})\}}{1 + x}. \tag{L33.3.1}$$

For all $n \ge 4$ this is at most $1 - 7x/15$. If $n \equiv 0, 1 \pmod 4$, the numerator is at most one, and the loss divided by x is at least $1/(1 + x) \ge 32/45$. For $n \equiv 2 \pmod 4$ with $n \ge 6$ and $n \equiv 3 \pmod 4$ with $n \ge 7$, the loss divided by x is at least $(1 - 2x^2 - x^5)/(1 + x) \ge 7/15$: its numerator is positive and decreasing, its denominator increasing, and

$$15(1 - 2\theta^2 - \theta^5) - 7(1 + \theta) = 1311885/33554432 > 0.$$

At $x = 0$ equality is direct. In particular $\|P_j(B)\| \le 1$ for $j \ge 4$, so $|\widehat{S}_n - 1 - u| \le \sqrt{d}$ for $n \ge 6$. Keeping a_0, a_1 in (L33.2.3) gives

$$0 \le Z \le (1 - cd)q - cd + \sqrt{d}. \tag{L33.3.3}$$

Write $P_n(i\sqrt{x}) = R_n(x) + i\sqrt{x}I_n(x)$. For $n \ge 6$ both finite alternating sums satisfy

$$\rho(x) := 1 - x + x^2 - x^3 \le R_n(x), I_n(x) \le 1. \tag{L33.3.4}$$

For $n = 4k, 4k+1, 4k+2, 4k+3$, the respective last exponents in (R_n, I_n) are $(2k-1, 2k), (2k+1, 2k-1), (2k, 2k+1), (2k+2, 2k)$. All needed exponents are at least two; the exponent-two sum also exceeds ρ, and all remaining cases follow from decreasing alternating sums. Let $\Pi z = z - \mathbb{E}z$ and $A_0 = a_1 + a_2 + a_3$, $B_0 = a_2 + a_3$, $C_0 = a_3$. These scalars are distinct from the operators A, B. The actual short paths give

$$w = \Pi\{[P_n(B) - A_0 - B_0 B - C_0 B^2]u\}.$$

At $i\sqrt{x}$ the subtracted real and normalized imaginary parts obey $\tau(3 - x) \le A_0 - C_0 x \le 3d$ and $2\tau \le B_0 \le 2d$. Since $2\rho(x) \ge 2\rho(\theta) > 1$ and $3d \le 1$, each subtraction is at most twice its original component. The componentwise square difference is

$$|P_n - A_0 - B_0 z - C_0 z^2|^2 \le |P_n|^2 - \tau\{(3 - x)[2\rho(x) - 3d] + 2x[2\rho(x) - 2d]\}.$$

All bracketed factors are nonnegative. Moreover $(6 + 2x)\rho(x) = 6 - 4x + 4x^2 - 4x^3 - 2x^4 \ge 6 - 4x + 2x^2$, because $4 - 4\theta - 2\theta^2 > 2$. Thus

$$|P_n - A_0 - B_0 z - C_0 z^2|^2 \le 1 - 7x/15 - \tau[6 - 9d - (4 + d)x + 2x^2].$$

Integrate against the spectral measure of u for $-B^2$, whose mass, first moment and second moment are d, δ_2, δ_3; use $\delta_3 \ge \delta_2^2/d$ and contraction of Π. For $d > 0$,

$$\frac{\mathbb{E}w^2}{d} \le \mathcal{V}(d, v) := 1 - \frac{7}{15}v - d(1 - v)[6 - 9d - (4 + d)v + 2v^2], \qquad v = \delta_2/d \in [d, \theta]. \tag{L33.3.7}$$

The subtracted bracket is nonnegative, since even its crude lower bound $6 - 3 - (13/3)\theta$ is positive; hence also $\mathbb{E}w^2 \le d$.

The additional reference weight has an actual signed correlation. By (L33.3.4), $\rho_r := \mathbb{E}[r(\widehat{S}_n - 1)] = \int x I_n(x)\, d\sigma_u(x) \le \delta_2$. The exact paths and $\mathbb{E}r = -d$, $\mathbb{E}[ru] = \mathbb{E}[rs] = 0$ give

$$\begin{aligned}
\mathbb{E}[uw] &= \eta - a_1 d - a_2 d - a_3 \tau \ge \tau - 2d^2 - d\tau, \\
\mathbb{E}[rw] &= \rho_r - \alpha d - (a_2 + a_3)(\delta_2 - d^2) \\
&\le \delta_2 - \tau d - 2\tau(\delta_2 - d^2).
\end{aligned}$$

Consequently

$$\mathbb{E}[f_2 w] = -\mathbb{E}[uw] + \mathbb{E}[rw] \le d[-1 + 2v + 2d - 2\tau(v-d)] \le dH(d), \quad H(d) = -3/16 + 2d. \tag{L33.3.9}$$

The full length-two fourth coefficient. Put $R_4(Z,\mu) = Z^2 + 2\mu Z + 3\mu^2$. Equations (L33.2.5) and (L33.3.3) allow (L33.1.2). We now retain the extra r term and prove

$$rR_4(Z,\mu) \le 9/4, \qquad f_2 R_4(Z,\mu) \le C(d) := 37/4 - 10d. \tag{L33.4.1}$$

For an original row $k(y) = K(x,y)$, the equality $\mathbb{E}g = 0$ gives $|r(x)| = |\mathbb{E}[(k - g(x))g]| \le \sqrt{p(x)q(x)d}$. Set $x = \sqrt{d}$, $b = 1 - cx^2$, $e = x - cx^2$, $M = 1 - 5cx^2 + 3cx^4$. Then $0 \le Z \le bq + e$, $0 \le \mu \le M$. Direct maximization for $0 \le q \le 2$, $p = 2 - q$ yields

$$\sqrt{pq} \le 1, \quad \sqrt{pq}\, q \le 3\sqrt{3}/4 < 21/16, \quad \sqrt{pq}\, q^2 \le 25\sqrt{5}/27 < 25/12.$$

The additional term is therefore at most

$$\begin{aligned}\mathcal{E}(x) &= x\{(25/12)b^2 + (21/8)b(e+M) + (e+M)^2 + 2M^2\} \\ &= \frac{185}{24}x + \frac{37}{8}x^2 - \frac{24103}{768}x^3 - \frac{2223}{256}x^4 + \frac{319295}{6144}x^5 + \frac{57}{16}x^6 - \frac{299991}{8192}x^7 + \frac{9747}{1024}x^9.\end{aligned}$$

The first row of Table 24 proves $9/4 - \mathcal{E}(3y/5) \ge 1/40$ for $0 \le y \le 1$. Since $x = \sqrt{d} \le 1/\sqrt{3} < 3/5$, this gives $rR_4(Z,\mu) < 9/4$ on the entire actual domain. Together with (L33.1.2), it proves (L33.4.1).

The entire small-score interval. The exact fourth expansion is

$$\mathbb{E}[f_2 Z^4] = (1-d)\mu^4 + 4\mu^3 \mathbb{E}[f_2 w] + \mathbb{E}[f_2 w^2 R_4(Z,\mu)]. \tag{L33.5.1}$$

For $0 < d \le 1/10$, the bracket in (L33.3.7) is at least $51/10 - (41/10)\theta = 1099/320 > 17/5$. Using $\tau \ge cd$ and $v \ge d$ therefore gives

$$\mathbb{E}w^2 \le d(1 - \mathcal{B}d), \quad \mathcal{B} = 1193/480, \quad \mu \le N = 1 - \mathcal{A}d,$$
$$\mathcal{A} = 893/320,$$

the latter because $5 - 3d \ge 47/10$. The maximum of $\phi(t) = (1-d)t^4 + 4dH(d)t^3$ on $[0,N]$ is at an endpoint: for $H \ge 0$ it increases; for $H < 0$ its derivative has one positive root, changing from negative to positive. Furthermore $(1-d)N + 4dH \ge (9/10)(7/10) - 3/40 > 0$, so N is the maximizing endpoint. Equations (L33.3.9)–(L33.5.1) give

$$\mathbb{E}[f_2 Z^4] \le N^3[1 - (\mathcal{A} + 7/4)d + (\mathcal{A} + 8)d^2] + (37/4 - 10d)d(1 - \mathcal{B}d).$$

For $\lambda \in \{1193/480, 2\}$, let

$$\mathcal{S}_\lambda(d) = \frac{1 - \ell_\lambda d - [(1-d)N^4 + 4dH(d)N^3 + C(d)d(1-\lambda d)]}{d},$$
$$\ell_{1193/480} = 483/400, \quad \ell_2 = 303/400.$$

The numerator vanishes at zero, so each quotient is a polynomial, including at that endpoint. The two low-score rows of Table 24 prove $\mathcal{S}_{1193/480} > 0$ and $\mathcal{S}_2 > 0$ on $[0, 1/10]$. The first yields

$$\mathbb{E}[f_2 X_n^4] \le \mathbb{E}[f_2 Z^4] \le 1 - 483d/400. \tag{L33.5.3}$$

The entire remaining score interval. Let $1/10 \le d \le 1/3$. Differentiating (L33.3.7) gives

$$\partial_v \mathcal{V} = -7/15 + d[10 - 8d - (12 + 2d)v + 6v^2],$$

which decreases in v. At $v = \theta$ it is $-7/15 + (3131/512)d - (141/16)d^2$, a concave quadratic with positive endpoint values $4357/76800$ and $4551/7680$. Hence the maximum variance factor is

$$V_\theta(d) = 389/480 - (45771/16384)d + (5719/1024)d^2.$$

Since $H(d) > 0$, the mean part of (L33.5.1) increases in μ. With $M = 1 - 95d/32 + 57d^2/32$, set

$$\begin{gathered} \mathbb{E}[f_2 Z^4] \le P(d) := M^3 B_1 + F(d), \\ B_1 = 1 - (151/32)d + (51/4)d^2 - (57/32)d^3, \quad F = d(37/4 - 10d)V_\theta. \end{gathered} \tag{L33.6.2}$$

Put $P_0(d) = M(d)^3 B_1(d)$. The two remaining rows of Table 24 prove directly that

$$P(d) < 1 - 853/124416 \qquad (1/10 \le d \le 1/3), \tag{N.2}$$

and also the shared monotonicity statement

$$P_0'(d) < 0 \qquad (1/10 \le d \le 1/3). \tag{N.3}$$

The latter is the mean-part estimate used for connector four.

For the table, apply (A.18) to each displayed polynomial in y. All coefficients are at least its listed positive bound. Every clearing denominator is a positive constant integer. Lemma A.1 therefore proves the assertions throughout the stated closed intervals. At $d = 0$, $g = 0$ and all the actual messages are

Table 24: Complete-interval scalar payments for the genuine length-two reference.

Polynomial in $0 \le y \le 1$	degree	coefficient lower bound
$9/4 - \mathcal{E}(3y/5)$	9	$1/40$
$\mathcal{S}_{1193/480}(y/10)$	4	$1/8$
$\mathcal{S}_2(y/10)$	4	$3/16$
$1 - 853/124416 - P(1/10 + 7y/30)$	9	$1/2500$
$-P_0'(1/10 + 7y/30)$	8	$3/20$

one, so equality is direct; the kernel need not vanish.

N.4 The one-step genuine short-one, long-three reference

For the actual terminal $F_{13} = qf_3$ with its unit arm inward and length-three arm outward, set $X = TF_{13}$. The connector points from the terminal toward the receiving root, whose outward length-two arm supplies the weight f_2. Then

$$\mathbb{E}[f_2 X^4] \le 1 - 59d/6000. \tag{L35.T}$$

Write $r = Kg$, $s = Kr$, $t = Ks$, $\theta = 13/32$, and $\delta_j = \|K^j \mathbf{1}\|_2^2$. Here t is a Krylov vector. The exact short paths are $f_2 = p + r$, $f_3 = p - d + r + s$, and

$$\begin{gathered} \mathbb{E}r = -d, \quad \mathbb{E}s = 0, \quad \mathbb{E}t = \delta_2, \quad \mathbb{E}[gr] = \mathbb{E}[rs] = \mathbb{E}[gt] = \mathbb{E}[st] = 0, \\ \mathbb{E}[gs] = -\delta_2, \quad \mathbb{E}[rt] = -\delta_3, \quad \mathbb{E}r^2 = \delta_2, \quad \mathbb{E}s^2 = \delta_3, \quad \mathbb{E}t^2 = \delta_4. \end{gathered}$$

For an original row, centering gives $r(x) = \mathbb{E}_y[(K(x,y) - g(x))g(y)]$ and $r(x)^2 \le p(x)q(x)d$. We use the scalar profile D of Section A.4: if $z, u_0 \ge 0$, $z \le q + (4/5)\sqrt{d} - 3d/8$, $u_0 \le 1 - 5d/8$, then

$$p(z^2 + 2u_0 z + 3u_0^2) \le 7 \quad (0 \le d \le 1/3). \tag{L35.1.5}$$

The C127 scalar profile bounds the p contribution in the f_2-weighted calculation below.

A positive square envelope for the same actual terminal. Put $a = 1 - d/2$, $R = r + s$, $V = \mathbb{E}R^2 = \delta_2 + \delta_3$. Then $5/6 \le a \le 1$ and $q + f_3 = 2a + R$. Hence $0 < F_{13} \le (a + R/2)^2$ and positivity of T, together with $TR = -d - s - t$ and $T(R^2) \le 2V$, gives

$$0 < X \le T\{(a + R/2)^2\} \le Z := a^2 q - ad - a(s + t) + V/2. \tag{L35.2.2}$$

Thus the nonnegativity of Z follows from the actual positive message. For $\mu = \mathbb{E}Z$ and $w = Z - \mu$,

$$\begin{aligned} \mu &= a^2 - ad - a\delta_2 + (\delta_2 + \delta_3)/2 \\ &= 1 - 2d + 3d^2/4 - (1 - d)\delta_2/2 + \delta_3/2, \end{aligned}$$

so $0 < \mu \le 1 - 2d + 3d^2/4 \le N := 1 - 7d/4$, using $\delta_3 \le \theta\delta_2$ and $1 - d - \theta > 0$. Also $N \ge 5/12$ and $w = -a^2 g - as - a(t - \delta_2)$.

The actual weighted covariance and variance. The signed identities give $\mathbb{E}[gw] = -a^2 d + a\delta_2$, $\mathbb{E}[rw] = a\delta_3 - ad\delta_2$. Therefore

$$\begin{gathered} \mathbb{E}[f_2 w] = -a^2 d + a(1 - d)\delta_2 + a\delta_3 \le dH(v), \quad H(v) = -a^2 + a(1 - d + \theta)v, \\ v = \delta_2/d \in [d, \theta]. \end{gathered} \tag{L35.3.1}$$

For $d > 0$ the same v is retained in the exact variance:

$$\begin{gathered} \mathbb{E}w^2 = a^4 d - 2a^3\delta_2 + a^2(\delta_3 + \delta_4 - \delta_2^2) \le dW(v), \quad W(v) = a^2[a^2 - (2a - \eta)v - dv^2], \\ \eta = 585/1024 < 3/5. \end{gathered} \tag{L35.3.3}$$

Here $\eta = \theta(1 + \theta)$, the negative cross term uses $\mathbb{E}[gs] = -\delta_2$, and the square of the nonzero mean $\mathbb{E}t = \delta_2$ is retained.

The full fourth coefficient on the continuous domain. Since $V \le \eta d$, the row bound $|KR| \le \sqrt{V}$ yields

$$Z \le a^2 q + a\sqrt{V} - ad + V/2 \le a^2 q + (4/5)\sqrt{d} - d/2.$$

The rational comparisons used are $\eta < 16/25$ and $5/6 - \eta/2 = 3365/6144 > 1/2$. Together with $\mu \le N$, these imply the hypotheses of (L35.1.5). For the additional r term set $x = \sqrt{d} \in [0, 1/\sqrt{3}]$,

$$b = 1 - 11x^2/12, \quad e = 4x/5 - x^2/2, \quad n = 1 - 7x^2/4, \quad u_0 = e + n = 1 + 4x/5 - 9x^2/4.$$

Since $a^2 \le 1 - 11d/12$, one has $Z \le bq + e$, $\mu \le n$. All right sides are nonnegative on the stated interval. Maximizing the squares over $q \in [0, 2]$, with $p = 2 - q$, gives

$$\sqrt{pq} \le 1, \qquad q\sqrt{pq} \le 3\sqrt{3}/4 < 21/16, \qquad q^2\sqrt{pq} \le 25\sqrt{5}/27 < 25/12.$$

The latter critical points are $q = 3/2, 5/3$, and both endpoint values are zero. Consequently, also when $r < 0$,

$$\begin{aligned} r(Z^2 + 2\mu Z + 3\mu^2) &\le x\sqrt{pq}[(bq + u_0)^2 + 2n^2] \le \mathcal{E}(x), \\ \mathcal{E}(x) &= x[(25/12)b^2 + (21/8)bu_0 + u_0^2 + 2n^2] \\ &= \frac{185}{24}x + \frac{37}{10}x^2 - \frac{82771}{3600}x^3 - \frac{221}{40}x^4 + \frac{63425}{3456}x^5. \end{aligned}$$

As $x^2 \le 1/3$ and $-82771/3600 + 63425/10368 + 67/4 = -32287/259200 < 0$,

$$\mathcal{E}(x) \le G(x) := 185x/24 + 37x^2/10 - 67x^3/4.$$

On $[0, 1/3]$, G' is concave with positive endpoint values $185/24, 551/120$, so $G \le G(1/3) = 2549/1080 < 3$. On $[1/3, 1/\sqrt{3}]$, $G'' = 37/5 - (201/2)x \le -261/10 < -26$. At $1/2$, $G = 1289/480 < 27/10$, $G' = -277/240$, $|G'| < 6/5$. The strong concavity bound gives $G(x) \le G(1/2) + G'(1/2)^2/52 < 27/10 + 9/325 < 3$. The added term is zero at $x = 0$. Thus the full coefficient satisfies

$$f_2(Z^2 + 2\mu Z + 3\mu^2) \le 10. \tag{L35.4.6}$$

A compatible endpoint reduction. The exact fourth expansion and (L35.3.1)–(L35.4.6) give

$$\mathbb{E}[f_2 Z^4] \le k\mu^4 + 4dH(v)\mu^3 + 10dW(v), \qquad k = 1 - d.$$

The derivative of $\Phi(t) = kt^4 + 4dH(v)t^3$ is $4t^2[kt + 3dH(v)]$. On $[0, N]$ the function increases, or first decreases and then increases. Its maximum is therefore at an endpoint:

$$\mathbb{E}[f_2 Z^4] \le \max\{10dW(v),\ kN^4 + 4dH(v)N^3 + 10dW(v)\}. \tag{L35.5.2}$$

Both branches strictly decrease in v. For the first use $2a - \eta > 0$. For the second its v derivative divided by da is $4(1 - d + \theta)N^3 - 10a(2a - \eta + 2dv)$. The first term is at most $45/8$, whereas the second is at least $10(5/6)(5/3 - 3/5) = 80/9 > 45/8$. Thus the same endpoint $v = d$ can be used for both compatible budgets.

For the zero endpoint, $\eta \le 3/5$ gives

$$W(d) \le a^2[1 - (12/5)d + (5/4)d^2 - d^3] \le a^2(1 - 6d/5)^2 \le (1 - 3d/2)^2.$$

The second comparison has difference $19d^2/100 + d^3 \ge 0$. The last uses the nonnegative quantities $a(1 - 6d/5) = 1 - 17d/10 + 3d^2/5 \le 1 - 3d/2$. The cubic $10d(1 - 3d/2)^2$ has derivative $10(1 - 3d/2)(1 - 9d/2)$ and its maximum on $[0, 1/3]$ is $80/81$ at $d = 2/9$. Therefore $10dW(d) \le 80/81$.

For the other endpoint use $\theta \le 5/12$ and $d^3/2 \le d^2/6$:

$$H(d) \le -1 + 29d/12 - 47d^2/24 + d^3/2 \le -1 + 29d/12 - 43d^2/24.$$

That branch is consequently at most

$$P(d) = (1 - 7d/4)^3[1 - 27d/4 + 137d^2/12 - 43d^3/6] + 10d(1 - 3d/2)^2.$$

The bracket can be negative at an endpoint value. The complete divided excess is

$$\frac{P(d) - 1}{d} = F(d) := -2 + \frac{625}{24}d - \frac{5375}{48}d^2 + \frac{45745}{256}d^3 - \frac{97559}{768}d^4 + \frac{14749}{384}d^5. \tag{L35.6.5}$$

The following is a continuous certificate on $[0, 1/3]$, using the polynomial extension at zero:

$$F'' = -5375/24 + (137235/128)d - (97559/64)d^2 + (73745/96)d^3,$$
$$F''' = 137235/128 - (97559/32)d + (73745/32)d^2,$$
$$F'''' = (-97559 + 147490d)/32 < 0.$$

Although F''' decreases, $F'''(1/3) = 359387/1152 > 0$. Hence F'' increases to $F''(1/3) = -77737/10368 < 0$, and $F''(1/4) = -80263/2048 < -30$. At $1/5$, $F = -38711/1200000 < -3/100$, $F' = -34071/32000$, $|F'| < 11/10$. On $[0, 1/4]$ strong concavity therefore gives

$$F(d) \le F(1/5) + F'(1/5)^2/60 < -3/100 + 121/6000 = -59/6000.$$

On $[1/4, 1/3]$, $F'' < 0$ and $F'(1/5) < 0$, so F decreases and the same bound holds. Thus $P(d) < 1 - 59d/6000$ for $d > 0$. Finally $59d/6000 \le 59/18000 < 1/81$, so the zero endpoint also obeys $80/81 < 1 - 59d/6000$. Substitution in (L35.5.2) and $0 < X \le Z$ proves (L35.T). At $d = 0$, $g = 0$ and $f_2 = f_3 = F_{13} = X = 1$; this allows $K \ne 0$ and gives equality directly. Actual equal-arm factors at either end are restored by positivity. Applying the same statement to $-K$, with every edge and weight reversed, gives $\mathbb{E}[\bar{f}_2\{A(p\bar{f}_3)\}^4] \le 1 - 59d/6000$.

N.5 The genuine length-two reference for connectors two through five

For the same terminal $F = qf_2$ and reference arm f_2, let $X_n = T^n F$, with all connector edges directed toward the reference root. We prove $\mathbb{E}[f_2 X_n^4] \le 1$ for $n = 2, 3, 4, 5$. Together with Sections N.2 and N.3 this supplies every positive connector for precisely this terminal and reference. The quantitative bounds proved here are

n	first score interval	remaining interval
4	$1 - 303d/400 \quad (0 < d \le 1/10)$	$1473/1600 \quad (1/10 \le d \le 1/3)$
5	$1 - 483d/400 \quad (0 < d \le 1/10)$	$1 - 853/124416 \quad (1/10 \le d \le 1/3)$
3	$1 - 41d/3328 \quad (0 < d \le 1/4)$	$61/64 \quad (1/4 \le d \le 1/3)$
2		$1 - 259d/128000 \quad (0 < d \le 1/3).$

(L37.0)

The bounds on the remaining interval for $n = 4, 5, 3$, and the $n = 2$ bound, are strict. At $d = 0$ all actual messages are one. Equal-arm factors are restored by positivity, and full reversal applies to every original edge and weight together.

Use $B = -K$, $u = B\mathbf{1} = -g$, $r = Kg$, $s = K^2 g$, $t_3 = K^3 g$, $c = 19/32$, $\theta = 13/32$, $\tau = d - \delta_2$, $\gamma = d - \delta_2/2$, and $\Pi z = z - \mathbb{E}z$. The finite polynomials $S_n, \widehat{S}_n, P_n$ are as in Section N.3. In the following envelope notation $\mathcal{U}_n$ is distinct from the paper's receiving message W_n. The same difference of squares and finite Duhamel identity give

$$0 \le X_n \le T^n\{(1 + r/2)^2\} \le \mathcal{U}_n := \bar{f}_n + T^n r + (\delta_2/2)\bar{f}_{n-1}, \qquad \mathcal{U}_{n+1} = T\mathcal{U}_n.$$

Explicitly,

$$\begin{aligned}
\mathcal{U}_2 &= (1 - \gamma)q + r + t_3, \\
\mathcal{U}_3 &= \widehat{S}_3 - \tau - \gamma\bar{f}_2, \\
\mathcal{U}_4 &= \widehat{S}_4 - d - \tau\bar{f}_1 - \gamma\bar{f}_3, \\
\mathcal{U}_5 &= \widehat{S}_5 - (\tau + \delta_3) - d\bar{f}_1 - \tau\bar{f}_2 - \gamma\bar{f}_4.
\end{aligned} \tag{L37.1.4}$$

All are nonnegative by the preceding actual comparison. The short paths are

$$\bar f_1 = q, \quad \bar f_2 = q + r, \quad \bar f_3 = q + r - s - d, \quad \bar f_4 = 1 - 2d + (1-d)u + r - s + t_3,$$

of means $1, 1-d, 1-2d, 1-3d+\delta_2$.

We use precisely the coefficient and scalar comparisons already proved in Section N.3. Put

$$M = 1 - 95d/32 + 57d^2/32, \quad H = -3/16 + 2d, \quad C = 37/4 - 10d,$$
$$\mathcal{V}(d,v) = 1 - 7v/15 - d(1-v)[6 - 9d - (4+d)v + 2v^2].$$

For $Z \ge 0$, $\mu = \mathbb{E}Z$, the bounds $Z \le (1-cd)q - cd + \sqrt{d}$, $\mu \le M$ give $f_2(Z^2 + 2\mu Z + 3\mu^2) \le C$. If also $\mathbb{E}[f_2(Z-\mu)] \le dH$ and $\operatorname{Var} Z \le d\mathcal{V}(d,v)$, $d \le v \le \theta$, the two scalar conclusions (L33.5.3) and the final high-score bound there apply. In every case the starting identity is

$$\mathbb{E}[f_2 Z^4] = (1-d)\mu^4 + 4\mu^3 \mathbb{E}[f_2(Z-\mu)] + \mathbb{E}\{f_2(Z-\mu)^2[Z^2 + 2\mu Z + 3\mu^2]\}. \tag{L37.2.5}$$

Connector four: exact moments and retained variance. Take $Z = \mathcal{U}_4$, $d > 0$, $v = \delta_2/d \in [d, \theta]$. The finite signed sums give

$$\mathbb{E}\widehat{S}_4 = 1 - \tau - \delta_3, \quad \mathbb{E}[u\widehat{S}_4] = \tau,$$
$$\mathbb{E}[r(\widehat{S}_4 - 1)] = \delta_2 - \delta_3 + \delta_4.$$

Consequently

$$\mu_4 = 1 - 4d + (5/2)\delta_2 + 2d^2 - d\delta_2 - \delta_3 \le 1 - d[4 - (5/2)v + v^2] + d^2(2-v) \le M. \tag{L37.3.1}$$

The last comparison reduces to $4 - (5/2)v + v^2 - 5c - d(2 - v - 3c) \ge 0$. Its left side decreases in v and at θ equals $185/1024 + 3d/16 > 0$. The actual covariances $\operatorname{Cov}(f_2, \bar f_1) = -d$, $\operatorname{Cov}(f_2, \bar f_3) = -d + 2\delta_2 - d^2$ give

$$\mathbb{E}[f_2(Z - \mu_4)] = -d + 2\delta_2 - (1+d)\delta_3 + \delta_4 + \gamma d(1 - 2v + d).$$

Using $\delta_4 \le \theta\delta_3$, $\gamma \le d$ and $1 - 2v + d \le 1 + d \le 4/3$ bounds this divided by d by $-3/16 + (4/3)d \le H$.

On $[0, \theta]$, $|P_3(i\sqrt{x})|^2 = 1 - x + 3x^2 - 2x^3 + x^4 \le 1$, because $j(x) = 1 - 3x + 2x^2 - x^3$ decreases to $j(\theta) = 1451/32768 > 0$. Thus $\widehat{S}_4 \le q + \sqrt{d}$ by the row bound and $\widehat{S}_4 - 1 - u = BP_3(B)u$. Retaining $-d, -\tau q$ in (L37.1.4) yields $0 \le Z \le (1-cd)q - cd + \sqrt{d}$. Hence the full coefficient C applies. Exactly,

$$Z - \mu_4 = \Pi\{[P_4(B) - (\tau + \gamma) - \gamma B - \gamma B^2]u\}.$$

The real and normalized imaginary parts of $P_4(i\sqrt{x})$ are $1 - x$ and $1 - x + x^2$, and $|P_4|^2 = 1 - x - x^2 + 3x^3 - 2x^4 + x^5 \le 1 - x$ by $j \ge 0$. The respective subtracted parts lie in $[\tau(2-x), 2d]$ and $[\tau, d]$. Since $2(1-x) - 2d > 0$ and $2(1-x) - d > 0$, their square difference gives

$$\begin{aligned} |P_4 - (\tau + \gamma) - \gamma z - \gamma z^2|^2 &\le 1 - x - \tau\{(2-x)[2(1-x) - 2d] + x[2(1-x) - d]\} \\ &= 1 - x - \tau[4 - 4d - (4-d)x]. \end{aligned}$$

Integrating against the same spectral measure of u, of mass d and first moment δ_2, and contracting by Π, gives

$$\operatorname{Var}(Z)/d \le V_4(d,v) := (1-v)[(1-2d)^2 + d(4-d)v]. \tag{L37.3.8}$$

For $0 < d \le 1/10$, the preceding spectral bracket is at least $18/5 - 4\theta = 79/40$, whence $\mathrm{Var}(Z)/d \le 1 - d - c(79/40)d \le 1 - 2d$. Set $\mathcal{A} = 893/320$ and $N = 1 - \mathcal{A}d \ge \mu_4$. The function $(1-d)t^4 + 4dHt^3$ has its maximum on $[0, N]$ at N: its derivative either is nonnegative or changes once from negative to positive, and $(1-d)N + 4dH \ge (9/10)(7/10) - 3/40 > 0$. Here the actual variance factor is $1 - 2d$, so the polynomial $\mathcal{S}_2$ in Table 24 is the divided difference between $1 - 303d/400$ and the resulting fourth-moment upper bound. Its strictly positive coefficients prove the first $n = 4$ bound in (L37.0).

For $1/10 \le d \le 1/3$, the mean part is at most $P_0 = M^3 B_1$, where $B_1 = 1 - 151d/32 + 51d^2/4 - 57d^3/32$. Equation (N.3) gives $P_0' < 0$ throughout this actual interval. The endpoint estimates are

d	M upper bound	B_1 upper bound	P_0 upper bound
$1/10$	$3/4$	$2/3$	$9/32$
$3/20$	$3/5$	$23/40$	$621/5000 < 1/8$
$1/5$	$12/25$	$9/16$	$972/15625 < 1/16$
$1/4$	$37/100$	$3/5$	$151959/5000000 < 1/32$.

Writing $A_1 = (1-2d)^2$, $B_2 = d(4-d)$ makes $V_4 = (1-v)(A_1 + B_2 v)$ concave in v. On $[1/10, 1/4]$ it decreases in d, since $\partial_d V_4 = -(1-v)[4 - 8d - (4-2d)v] < 0$, with bracket at least $2 - (7/2)\theta = 37/64$. The allowed v interval also shrinks. The first three endpoint maxima occur at $v = d$; the later ones are bounded by the unrestricted quadratic maximum $(A_1 + B_2)^2/(4B_2)$:

d	upper bound on $\max_v V_4$	simpler bound
$1/10$	$6111/10000$	$5/8$
$1/8$	$2233/4096$	$11/20$
$3/20$	$78421/160000$	$1/2$
$1/5$	$196/475$	$5/12$
$1/4$	$361/960$	$2/5$
$1/3$	$4/11$	$2/5$.

Each simpler bound is strict. On $[1/4, 1/3]$, for each fixed $v \in [0, \theta]$, V_4 is convex in d, so its last two endpoint bounds give $V_4 < 2/5$ on the entire interval. Since $dC(d)$ increases, the following closed intervals cover the full remaining domain:

d interval	P_0	dC	V_4	total upper bound
$[1/10, 1/8]$	$9/32$	1	$5/8$	$29/32$
$[1/8, 3/20]$	$9/32$	$93/80$	$11/20$	$1473/1600$
$[3/20, 1/5]$	$1/8$	$29/20$	$1/2$	$17/20$
$[1/5, 1/4]$	$1/16$	$27/16$	$5/12$	$49/64$
$[1/4, 1/3]$	$1/32$	$71/36$	$2/5$	$1181/1440$.

Thus $\mathbb{E}[f_2 X_4^4] \le \mathbb{E}[f_2 Z^4] < 1473/1600$.

Connector five: verification of all common comparison inputs. Let $Z = \mathcal{U}_5$. The finite signed sums are

$$\mathbb{E}\widehat{S}_5 = 1 - \tau, \quad \mathbb{E}[u\widehat{S}_5] = \tau + \delta_3 - \delta_4,$$

$$\mathbb{E}[r(\widehat{S}_5 - 1)] = \delta_2 - \delta_3.$$

The exact mean is

$$\mu_5 = 1 - 5d + (7/2)\delta_2 + 4d^2 - (7/2)d\delta_2 + \delta_2^2/2 - \delta_3.$$

Since $\mu_5 - \mu_4 = -\tau[1 - (d/2)(4 - v)] \le 0$, one has $0 \le \mu_5 \le M$. The actual covariances are $\mathrm{Cov}(f_2, \bar{f}_2) = -d + \delta_2 - d^2$ and $\mathrm{Cov}(f_2, \bar{f}_4) = -d + 2\delta_2 - \delta_3 + d\delta_2$. Consequently

$$\begin{aligned}\mathbb{E}[f_2(Z - \mu_5)] = &-d + 2\delta_2 - 2\delta_3 + \delta_4 + d^2 - \tau(\delta_2 - d^2)\\ &+ \gamma(d - 2\delta_2 + \delta_3 - d\delta_2).\end{aligned}$$

After $\delta_4 \le \theta\delta_3$ the total coefficient of δ_3 is at most $-(2-\theta-\gamma) < 0$. Discarding this and $-\tau(\delta_2 - d^2)$, using $\gamma \le d$ and $1 - 2v - dv \le 1$, gives $\mathbb{E}[f_2(Z - \mu_5)]/d \le -1 + 2v + d + \gamma \le H$. The bound $|P_4| \le 1$ proved above gives $\widehat{S}_5 \le q + \sqrt{d}$. Retain the constant $\tau + \delta_3$ and dq in (L37.1.4); then $0 \le Z \le (1 - cd)q - cd + \sqrt{d}$.

The real and normalized imaginary parts of P_5 are $\rho = 1 - x + x^2 - x^3$ and $1 - x$, and

$$|P_5(i\sqrt{x})|^2 = 1 - x + x^2 - 3x^3 + 3x^4 - 2x^5 + x^6 \le 1 - 6x/7.$$

Indeed its loss divided by x is at least $1 - x + 3cx^2 \ge 1 - 1/(12c) = 49/57 > 6/7$, using $1 - x \ge c$ and $2x^4 - x^5 \ge 0$; at zero equality is direct. In the centered polynomial the subtracted parts are $r_0 = d + \tau + \gamma(1 - d - x)$ and $r_1 = \tau + \gamma(1 - x)$. As $1 - d - x > 0$, these lie in $[\tau(3 - d - x), 3d]$ and $[\tau(2 - x), 2d]$. The factors $2\rho - 3d$ and $2(1 - x) - 2d$ are positive, so the componentwise square difference is at least τB_5, where

$$\begin{aligned}B_5 &= (3 - d - x)(2\rho - 3d) + x(2 - x)[2(1 - x) - 2d]\\ &= 6 - 11d + 3d^2 + (d - 4)x + 2x^2 - (6 - 2d)x^3 + 2x^4\\ &\ge 6 - 11d + 3d^2 - L(d)x, \qquad L(d) = (2139 - 681d)/512.\end{aligned}$$

The last inequality uses $x^3 \le \theta x^2$, drops $2x^4$, and uses $(6 - 2d)\theta - 2 \ge 1/6$ and $x^2 \le \theta x$. Spectral integration and centering give

$$\mathrm{Var}(Z)/d \le V_5(d, v) := 1 - 6v/7 - \tau[6 - 11d + 3d^2 - L(d)v].$$

Its exact difference from the common variance factor is

$$V_5 - \mathcal{V} = -41v/105 + \tau[2d - 3d^2 + (91 - 1193d)v/512 + 2v^2].$$

For $d \ge 1/10$ the linear coefficient is negative, and

$$\frac{\tau}{v}(2d - 3d^2 + 2v^2) \le d(1 - v)(2 - 3d + 2v) \le d(1 - d)(2 - d) \le 10/27 < 41/105.$$

Here the derivative in v is $3d - 4v < 0$, and the final cubic increases on $[0, 1/3]$; the strict difference is $19/945$. For $0 < d \le 1/10$, retaining the positive part of the linear term yields

$$\begin{aligned}\frac{\tau}{v}[2d - 3d^2 + (91 - 1193d)v/512 + 2v^2] &\le 2d(1 - v^2) + (91/512)d(1 - v)\\ &\le 1115d/512 \le 1115/5120 < 41/105.\end{aligned}$$

Thus $V_5 < \mathcal{V}$ everywhere. All common coefficient, mean, covariance and variance hypotheses have now been verified, so Section N.3 gives both stated $n = 5$ bounds.

Connector three: negative reference covariance retained. Let $Z = \mathcal{U}_3$, $\alpha = 51/64$, $h = c^2 = 361/1024$. Exactly,

$$\mu_3 = 1 - 3d + (3/2)\delta_2 + d^2 - d\delta_2/2 \le M_3(d) := 1 - 3\alpha d + \alpha d^2 \le 1 - 2d.$$

The first upper bound takes $v = \theta$ in an increasing affine expression. From $\mathbb{E}\widehat{S}_3 = 1 - d$, $\mathbb{E}[u\widehat{S}_3] = \tau + \delta_3$, $\mathbb{E}[r(\widehat{S}_3 - 1)] = \delta_2$ and the covariance of $\bar{f}_2$,

$$\mathbb{E}[f_2(Z - \mu_3)] = -d + 2\delta_2 - \delta_3 - d^2 + \gamma d(1 - v + d) \le -d(1 - v)^2 \le -hd.$$

This uses $\delta_3 \ge dv^2$, $v \ge d$ and $\gamma(1-v+d) \le d$. The centered polynomial is $\Pi\{[P_3(B)-\gamma-\gamma B]u\}$. Its norm before centering obeys $|P_3-\gamma-\gamma z|^2 = |P_3|^2-2\gamma(1+x^2)+\gamma^2(1+x) \le 1-2\gamma+\gamma^2(1+x)$. Its exact mean is $-(1-\gamma)d$, so

$$\mathrm{Var}(Z)/d \le 1 - 2\gamma + \gamma^2(1 + v) - (1 - \gamma)^2 d.$$

This decreases in $\gamma \in [\alpha d, d]$, because its derivative divided by two is $-(1 - d) + \gamma(1 + v - d) < 0$: $\gamma(1 + v - d) \le (1 + \theta)/3 = 15/32 < 2/3 \le 1 - d$. It increases in v, hence

$$\begin{aligned}\mathrm{Var}(Z)/d &\le 1 - (83/32)d + (325941/131072)d^2 - (2601/4096)d^3 \\ &\le V_3(d) := 1 - (83/32)d + (5/2)d^2.\end{aligned}$$

For the p coefficient set $S = \widehat{S}_3 - \gamma r$; then $Z = S - \tau - \gamma q$. The four coefficient parameters for Section A.3 are $\alpha_0 = d(1-\gamma)$, $a_0 = \tau$, $b_0 = \gamma$, $\eta_0 = \tau + \delta_3$, all in $[\tau, d]$: $\alpha_0 - \tau = \delta_2 - d\gamma \ge 0$ and $\tau + \delta_3 \le d$. The polynomial $P_3 - \gamma z$ only reduces the normalized imaginary part from one to $1 - \gamma$, so $\|S - 1\|_2^2 \le d$. Its tail polynomial $(1 - \gamma) + z + z^3$ has squared modulus $(1 - \gamma)^2 + x(1 - x)^2 \le (1 - \gamma)^2 + 4/27$, giving row tail at most $(13/12)\sqrt{d}$ everywhere. For $d \ge 1/10$, $\gamma \ge 51/640$ and $2\gamma - \gamma^2 \ge 62679/409600 > 4/27$, improving it to $\sqrt{d}$. The standard and enlarged-tail comparisons of that section cover the whole domain and give $p[Z^2 + 2\mu_3 Z + 3\mu_3^2] \le 7 - 10d$. Throughout, $Z \le (1 - cd)q - cd + (13/12)\sqrt{d}$.

For the additional r weight set $x = \sqrt{d}$, $\lambda = 13/12$, $b = 1 - cx^2$, $e = \lambda x - cx^2$, $m = 1 - 2x^2$. The same original-row bound and extrema of $q^j\sqrt{pq}$ used in Section N.4 bound the added term by $x\{(25/12)b^2 + (21/8)b(e + m) + (e + m)^2 + 2m^2\}$. All substituted terms are nonnegative; taking $b \le 1$ gives

$$\begin{aligned}\mathcal{E}_3(x) &= x\{25/12 + (21/8)U + U^2 + 2m^2\}, \quad U = 1 + (13/12)x - (83/32)x^2 \\ &= (185/24)x + (481/96)x^2 - (43367/2304)x^3 - (1079/192)x^4 + (15081/1024)x^5.\end{aligned}$$

For $0 \le x \le 1/3$, $43367/2304 - 15081/9216 > 17$, so $\mathcal{E}_3 \le (185/24)x + (481/96)x^2 - 17x^3$. Its derivative is concave and positive at both endpoints, giving the upper bound $719/288 < 5/2$. For the remaining interval use $1/\sqrt{3} < 3/5$, and set $\mathcal{T}(y) = 3 - \mathcal{E}_3(1/3 + 4y/15)$. This degree-five polynomial has the six Bernstein coefficients

$$\frac{1}{777600000}\,(450809375, 249628875, 115378375, \\ 62732115, 91480239, 171085851).$$

They are all positive, so Lemma A.1 gives $\mathcal{E}_3 < 3$ on $[1/3, 1/\sqrt{3}]$. Consequently the full coefficient is at most $10 - 10d$.

The pure variance branch is

$$G(d) = 10d(1 - d)V_3(d) = 10d - (575/16)d^2 + (815/16)d^3 - 25d^4.$$

Its second derivative $-575/8 + (2445/8)d - 300d^2$ increases on $[0, 1/3]$ to $-10/3$. As $G(1/4) = 975/1024$ and $G'(1/4) = 5/256$, strong concavity gives $G \le 975/1024 + 15/262144 < 61/64$. The remaining part is $\phi(t) = (1-d)t^4 - 4hdt^3$, whose maximum on any $[0, N]$ is at an endpoint. For $d \le 1/4$, take $N = 1 - \mathcal{A}d \ge \mu_3$, $\mathcal{A} = 561/256$. The divided excess of $\phi(N) + G$ is exactly

$$\begin{aligned} &-301/256 + [6\mathcal{A}^2 + (4+12h)\mathcal{A} - 575/16]d \\ &\quad + [-4\mathcal{A}^3 - (6+12h)\mathcal{A}^2 + 815/16]d^2 \\ &\quad + [\mathcal{A}^4 + (4+4h)\mathcal{A}^3 - 25]d^3 - \mathcal{A}^4 d^4. \end{aligned}$$

The middle coefficients are less than $11, -40, 56$, respectively, using $219/100 < \mathcal{A} < 274/125$ and $44/125 < h < 353/1000$:

$$\begin{gathered} 6(274/125)^2 + (4 + 12 \cdot 353/1000)(274/125) - 575/16 < 11, \\ -4(219/100)^3 - (6 + 12 \cdot 44/125)(219/100)^2 + 815/16 < -40, \\ (274/125)^4 + (4 + 4 \cdot 353/1000)(274/125)^3 - 25 < 56. \end{gathered}$$

The entire excess is therefore at most $-301/256 + 11d - 26d^2 \le -41/3328$, by the maximum of this concave quadratic. The zero endpoint $G < 61/64$ also satisfies the required $1 - 41d/3328$ bound. For $d \ge 1/4$, M_3 and $(1-d)M_3$ decrease, whereas $4hd$ increases. At $1/4$, $(1-d)M_3 - 4hd = -55/4096 < 0$. Hence $\phi(M_3) < 0$ and its maximum on $[0, M_3]$ is zero, so the whole remaining interval is bounded strictly by $61/64$.

Connector two: exact moment budget. Let $Z_2 = \mathcal{U}_2 = aq + R$, $a = 1 - d + \delta_2/2$, $R = r + t_3$. The exact moments are

$$\begin{aligned} \mu_2 &= 1 - 2d + (3/2)\delta_2, \\ \mathbb{E}[f_2(Z_2 - \mu_2)] &= -d + \delta_2 - \delta_3 + d\delta_2/2, \\ \mathrm{Var}(Z_2) &= a^2 d + \delta_2 - 2\delta_3 + \delta_4 - (d - \delta_2)^2. \end{aligned} \tag{L37.7.1}$$

The complete continuous pointwise proof in Section N.25 supplies

$$f_2[Z_2^2 + 2\mu_2 Z_2 + 3\mu_2^2] \le C_2(d) := 1023/128 - 8d. \tag{L37.7.C}$$

Keeping the same δ_3 in both moments and using $\delta_3 \ge dv^2$, $\delta_4 \le \theta\delta_3$ in (L37.2.5) gives

$$\mathbb{E}[f_2 Z_2^4] \le P(d, v) := (1-d)\mu^4 + 4d\mu^3 H_2 + C_2(d) d V_2,$$

where $\mu = 1 - 2d + (3/2)dv$, $a = 1 - d + dv/2$, $H_2 = -1 + v - v^2 + dv/2$ and $V_2 = a^2 + v - (51/32)v^2 - d(1-v)^2$. The following monotonicity works for any v-independent coefficient $21/4 \le C(d) \le 8$, which includes C_2:

$$\frac{\partial_v P}{d} = 6\mu^2 \mathcal{B} + 4\mu^3 H_{2,v} + C V_{2,v}, \quad \mathcal{B} = 1 - 6d + 2d^2 + (9/2)dv - 3dv^2,$$

$$H_{2,v} = 1 - 2v + d/2 > 0, \qquad V_{2,v} \ge W(d) := -151/512 + (35/16)d - (51/64)d^2.$$

W increases on $[0, 1/3]$. Use the following five intervals. For $0 < d \le 1/16$, use $\mu \ge 7/8$, $\mathcal{B} \ge 5/8$, $H_{2,v} \ge 3/16$, $CV_{2,v} \ge -151/64$; the resulting sum is $2077/2048 > 0$. For $1/16 \le d \le 1/8$, use $\mu \ge 3/4$, $\mathcal{B} \ge 9/32$, $H_{2,v} \ge 7/32$ and $8W(1/16) = -2643/2048$, giving $243/256 + 189/512 -$

$2643/2048 = 57/2048 > 0$. For $1/8 \le d \le 3/20$, use $\mu \ge 7/10$, $\mathcal{B} \ge 29/200$, $H_{2,v} \ge 1/4$ and $8W(1/8) = -139/512$; the second term alone is $343/1000 > 139/512$, and the first is positive. For $3/20 \le d \le 1/5$, $W \ge 391/25600 > 0$ and

$$\mathcal{B} \ge \mathcal{B}_{\min}(d) := 1 - 6d + (13/2)d^2 - 3d^3 > 0.$$

Here $v(9/2 - 3v)$ increases for $v \in [0, \theta]$, $\mathcal{B}_{\min}$ decreases on $[0, 1/3]$, and $\mathcal{B}_{\min}(1/5) = 9/250$. For $1/5 \le d \le 1/3$, the second term is at least $23/160$, since $\mu \ge 1/2$ and $H_{2,v} \ge 23/80$; the third exceeds $29/50$, because $(21/4)W(1/5) = 29757/51200 > 29/50$. Only a negative first term needs payment. Set $m = 1 - (89/64)d$. On the subdomain $\mathcal{B}_{\min} < 0$, $6\mu^2\mathcal{B} \ge 6m^2\mathcal{B}_{\min}$. The derivative of $m^2(-\mathcal{B}_{\min})$, divided by positive m, is at least $(103/192)(8/3) - 2(89/64)(7/18) = 67/192 > 0$, using $m \ge 103/192$, $-\mathcal{B}_{\min} \le 7/18$ and $(-\mathcal{B}_{\min})' \ge 8/3$. Thus $6\mu^2\mathcal{B} \ge -74263/110592 > -27/40$. The total exceeds $-27/40 + 23/160 + 29/50 = 39/800 > 0$. If $\mathcal{B}_{\min} \ge 0$ all terms were already nonnegative. Consequently P increases strictly in v, and $v = \theta$ is a compatible common endpoint.

At this endpoint, $\mu = 1 - (89/64)d$, $H_2 = -777/1024 + (13/64)d$ and $V_2 = 37461/32768 - (1993/1024)d + (2601/4096)d^2$. The exact divided excess is

$$\begin{aligned} Q(d) := \frac{P(d,\theta) - 1}{d} = & -1932885/4194304 + (778497/131072)d \\ & - (23816013/1048576)d^2 + (1496032917/67108864)d^3 \\ & - (99400629/16777216)d^4. \end{aligned}$$

Its third derivative exceeds $6 \cdot 22 - 8 \cdot 6 = 84$, since its cubic coefficient exceeds 22 and quartic coefficient exceeds -6. So Q'' increases. The quadratic, cubic and quartic coefficients are less than $-227/10, 223/10, -59/10$, giving $Q''(1/3) < -227/5 + 223/5 - 118/15 = -26/3 < -8$. The anchor values are $Q(1/6) = -278262102/86973087744 < -3/1000$ and $Q'(1/6) = 844767381/7247757312 \in (0, 1/8)$. Strong concavity yields $Q(d) < -3/1000 + 1/1024 = -259/128000$ on the full interval. Using (L37.7.C) and $0 \le X_2 \le Z_2$ proves the claimed $n = 2$ bound. All four connectors have now been treated with the same actual terminal, reference measure and fixed directions.

N.6 Every one-step short-one terminal with long arm at least four

For every finite $b \ge 4$, keep the actual orientation $F_b = qf_b$, one connector toward the receiving root, and its outward length-two arm f_2. We prove

$$\mathbb{E}[f_2\{T(qf_b)\}^4] \le 1 - (47/65000)d. \tag{L38.0.1}$$

All quantities use the original finite measure. Write $c = \kappa = 19/32$, $r = Kg$, and $\delta_j = \|K^j\mathbf{1}\|_2^2$. The original row constraint gives $|r| \le pq/2$ and $0 \le f_2 \le 2 - q^2/2$. Put $R = \sum_{j=2}^{b} K^j\mathbf{1}$, $\beta = \beta_b$, $a = 1 - \beta/2$, $t_0 = \beta_{b+1} = -\mathbb{E}R$, $k_0 = \mathbb{E}KR = d - \beta_{b+2}$, $V = \mathbb{E}R^2$. Actual positive convolution retains its $\beta_b f_0$ term and gives $q + f_b \le 2a + R$. Thus

$$0 < T(qf_b) \le Z := a^2q - at_0 - aKR + V/2. \tag{L38.2.1}$$

For $d > 0$, $v = \delta_2/d \in [d, \theta]$ and $m := 1 - d/2 \le a \le A_0 := 1 - (1 - v)d/2$, $5/6 \le a \le 1$.

For every finite $N \geq 3$ the exact geometric polynomial satisfies

$$|G_N(i\sqrt{x})|^2 \leq 1 - (7/15)x \quad (0 \leq x \leq \theta). \tag{L38.2.3}$$

At $x = 0$ equality holds. For odd N, the finite formula reduces to $x^{N-1} + (7/15)x \leq 8/15$, whose left side is at most $\theta^2 + (7/15)\theta = 5447/15360 < 8/15$. For $N \equiv 0 \pmod 4$, the numerator $(1 - x^{N/2})^2 \leq 1$ and $(7/15)(1+x) \leq 21/32 < 1$. For $N \equiv 2 \pmod 4$, $N \geq 6$, it suffices that $2x^{N/2-1} + x^{N-1} + (7/15)x \leq 8/15$. Use $8/15 - 2\theta^2 - (7/15)\theta = 7/512$ and $\theta^5 = 371293/33554432 < 7/512$ (numerators $371293 < 458752$). This covers every finite integer. Consequently, since $b - 1 \geq 3$,

$$V \leq \delta_2 - (7/15)\delta_3, \quad U := \mathbb{E}(KR)^2 \leq \delta_3 - (7/15)\delta_4 \leq \theta_* \delta_2,$$
$$\theta_* = 5057/15360 < 1/3, \quad \theta - \theta_* = 1183/15360 > 1/13.$$

The last endpoint bound follows from increasing $x - 7x^2/15$ on $[0, \theta]$, and $1183 \cdot 13 - 15360 = 19$. Set $\rho = \mathbb{E}(rR)$ and $\sigma = \mathbb{E}(sR)$. The exact finite signed sums are

$$\rho = \sum_{h=0}^{\lfloor b/2 \rfloor - 1} (-1)^h \delta_{h+2},$$
$$k_0 = \sum_{h=0}^{\lfloor (b-1)/2 \rfloor - 1} (-1)^h \delta_{h+2}, \quad \sigma = \sum_{h=0}^{\lfloor (b-1)/2 \rfloor - 1} (-1)^h \delta_{h+3}.$$

Thus $c\delta_2 \leq \rho, k_0 \leq \delta_2$. For $b = 2h \geq 4$, $\rho + \sigma = \delta_2$, $t_0 + k_0 = d$; for $b = 2h + 1 \geq 5$, $\rho + \sigma = \delta_2 + (-1)^{h-1}\delta_{h+2}$ and $t_0 + k_0 = d + (-1)^{h+1}\delta_{h+1}$. The first positive extra term is at $b = 7$, so with $L_0 = 1 + \theta^3 = 34965/32768$,

$$\rho + \sigma \leq L_0 \delta_2, \qquad t_0 + k_0 \geq d - \delta_3. \tag{L38.2.8}$$

The normalized positive bound $\widehat{Z} = Z/a^2$ satisfies $\widehat{Z} \leq q + (4/5)\sqrt{d} - 3d/8$, $\mathbb{E}\widehat{Z} \leq 1 - 5d/8$. Indeed $k = 1/a \in [1, 6/5]$, $V \leq \delta_2$, $k\sqrt{\theta} \leq 4/5$, and $kt_0 - k^2 V/2 \geq (ck - \theta k^2/2)d$. The last concave quadratic has endpoint values $25/64, 21/50$, both at least $3/8$. The normalized mean loss is at least $[k(1 - \theta^2) - k^2\theta/2]d$, whose endpoint values $647/1024, 9081/12800$ exceed $5/8$. For any nonnegative z, u_0 satisfying these same bounds, the full continuous coefficient is

$$f_2(z^2 + 2u_0 z + 3u_0^2) \leq 29/3. \tag{L38.3.3}$$

To prove it put $x = \sqrt{d}$, $n = 1 - 5x^2/8$, $t = 1 + 4x/5 - x^2$. The bracket is at most $(q + t)^2 + 2n^2$, whose exact expansion is

$$q^2 + 2q + 3 + (8/5)(q + 1)x - (2q + 193/50)x^2 - (8/5)x^3 + (57/32)x^4.$$

Since $x < 3/5$, its cubic and quartic sum is nonpositive. Complete the square in the remaining quadratic to add at most $32(q + 1)^2/(100q + 193)$ to $q^2 + 2q + 3$. The base after multiplication by $2 - q^2/2$ is at most nine. The added part is below $2/3$ because, for $y = q - 1 \in [-1, 1]$,

$$100q + 193 - 24(4 - q^2)(q + 1)^2 = 5 + 4y + 24y^2(y + 3)^2 \geq 1.$$

All denominators are positive, including both q endpoints. By homogeneity the actual Z, $\mu_0 = \mathbb{E}Z$ has coefficient Ca^4, $C = 29/3$.

For $w = Z - \mu_0$, exact original skew calculations give

$$\mu_0 = a^2 - a(t_0 + k_0) + V/2, \quad \mathbb{E}[f_2 w] = -a^2 d + a(\rho + \sigma) - a d k_0, \\ \mathbb{E}w^2 = a^4 d - 2a^3\rho + a^2(U - k_0^2). \tag{L38.4.1}$$

The actual mean square k_0^2 is retained. Equations (L38.2.3)–(L38.2.8) imply

$$\mu_0 \le M(d,v) := 1 - [2 - (1739/960)v]d + [(1-v)(3 - 29v/16)/4]d^2. \tag{L38.4.3}$$

Before taking $a \le A_0$ the bound is $a^2 - ad + (a - 7/30)\delta_3 + \delta_2/2$; $a - 7/30 \ge 3/5$ permits $\delta_3 \le \theta\delta_2$, and its derivative in a is $2a - d + \theta\delta_2 > 0$. The weighted centered mean is at most $dH(d,v)$, where

$$H = -m^2 + m(L_0 - cd)v < 0.$$

The upper expression decreases in a, derivative $-2a + (L_0 - cd)v < 0$, justifying $a \ge m$. The original variance and homogeneous curvature give

$$Ca^4\mathbb{E}w^2 \le B_*(d,v) := CdA_0^6 Q_*, \quad Q_* = A_0^2 - (2A_0 c - \theta_*)v - c^2 d v^2. \tag{L38.4.5}$$

The bracket increases in a and upper-bounds a nonnegative actual variance; on the full domain it is also positive, since $A_0 \ge 5/6$, $2c\theta < 1/2$, $c^2 d v^2 < 1/48$ yield $Q_* > 25/144$. The exact fourth expansion therefore gives the two-endpoint bound

$$\mathbb{E}[f_2 Z^4] \le \max\{B_*, (1-d)N^4 + 4dHN^3 + B_*\} \quad (\mu_0 \le N). \tag{L38.4.7}$$

This follows because $t \mapsto (1-d)t^4 + 4dHt^3$ decreases then increases, or increases throughout, on $[0, N]$.

The actual envelope has $\mu_0 = \mathbb{E}Z \ge 0$, by (L38.2.1). Take $N = M(d,v)$ from (L38.4.3). On the whole moment triangle $0 \le d \le 1/3$, $d \le v \le \theta$, this bound satisfies $M \ge 1 - 2d \ge 1/3$: its remaining displayed terms are nonnegative. Moreover $H \le m[-5/6 + L_0\theta] < 0$, where $m = 1 - d/2$. For $\phi(t) = (1-d)t^4 + 4dHt^3$, $\phi'(t) = 4t^2[(1-d)t + 3dH]$. The bracket increases, so ϕ decreases and then increases, or is monotone, on $[0, M]$. Its maximum is at zero or M. Thus (L38.4.7) reduces the actual integral to

$$\max\{B_*,\ (1-d)M^4 + 4dHM^3 + B_*\}.$$

Put $\lambda = 47/65000$, and define the two polynomial gaps

$$\begin{aligned} \Phi(d,v) &= (1-d)M(d,v)^4 + 4dH(d,v)M(d,v)^3 + B_*(d,v), \\ D_0(d,v) &= 1 - \lambda d - B_*(d,v), \\ D_1(d,v) &= [1 - \lambda d - \Phi(d,v)]/d. \end{aligned} \tag{N.4}$$

The second numerator vanishes identically at $d = 0$; after exact cancellation, D_1 is a polynomial there as well. The substitutions

$$d = x/3, \qquad v = d + (\theta - d)y, \qquad (x,y) \in [0,1]^2$$

cover the full triangle, since $\theta - d \ge 7/96 > 0$. They may include moment pairs that no kernel realizes; proving the polynomial bounds on this larger set is sufficient. Put $\widetilde{D}_i(x,y) = D_i(x/3, x/3 + (\theta - x/3)y)$. These two polynomials have the following complete-matrix Bernstein bounds: Each entry

Table 25: Joint-parameter payments for the genuine one-step reference.

Polynomial	bidegree	number of coefficients	coefficient lower bound
$\widetilde{D}_0$	$(17,8)$	162	$1/8$
$\widetilde{D}_1$	$(16,8)$	153	$1/2$

is obtained from the displayed M, H, B_* by (A.18). All clearing denominators are positive constant integers. Lemma A.1 gives $\widetilde{D}_0 \geq 1/8$ and $\widetilde{D}_1 \geq 1/2$ on the full closed square, hence the same gap bounds throughout the original triangle. For $d > 0$ these inequalities make both actual endpoint budgets strictly smaller than $1 - \lambda d$, proving (L38.0.1). At $d = 0$ original leaf integration gives equality without $K = 0$. Actual equal-pair decorations decrease the positive messages; full reversal simultaneously reverses every original edge and its weight.

N.7 The genuine f_2 reference for every short-one terminal

Let $F_b = qf_b$, $X_{b,L} = T^L(qf_b)$, with the terminal unit arm inward, its b arm outward, and all L connector edges toward the receiving root. The reference-fourth bound from Q5-C058 is

$$\mathbb{E}[f_2 X_{b,L}^4] \leq 1 \quad (b \geq 2, L \geq 1). \tag{L58.16.1}$$

For $b = 2$, the exact base rows are Section N.2 for $L = 1$, Section N.5 for $2 \leq L \leq 5$, and Section N.3 for $L \geq 6$. For $L = 1$, the $b = 3$ row is Section N.4 and every $b \geq 4$ row is Section N.6. The proof below treats every $b \geq 3, L \geq 2$ with deficit at least $d/1000$. Use the finite short-one packet of Lemma M.4, with its actual $R, m, k, t_0, e, V, \rho, \gamma, a_1, B_*, D_j, \varepsilon_j$. Locally write $B = -K$, $u = -g$, $U = \mathbb{E}(KR)^2$, and $M = 71/63$. For $b \geq 3$ the common bounds are

$$\begin{gathered} 3d/10 \leq \gamma \leq 6d/5, \quad 5d/8 \leq B_* \leq 3d/2, \quad \varepsilon_3 = d - k\rho \geq 41d/80, \\ \tau \leq \varepsilon_j \leq d \quad (j \geq 4). \end{gathered} \tag{L58.1.6}$$

In particular $\kappa\delta_2 \leq \rho \leq \delta_2$, $U \leq \theta V$, $V \leq \delta_2$ for $b \geq 4$; at $b = 3$, $V = \delta_2 + \delta_3, \rho = \delta_2$. Set $S'_L = \sum_{j=0}^{L} B^j \mathbf{1}$ and $V_L = S'_L + kB^L R$. The same finite packet gives

$$0 < X_{b,L} \leq m^2 \left[V_L - \gamma \bar{f}_{L-1} - a_1 \bar{f}_{L-2} - \sum_{j=3}^{L} \varepsilon_j \bar{f}_{L-j} \right]. \tag{L58.1.7}$$

Only nonnegative path indices occur. Here $V_L - 1 = P_L(B)u$, $\|P_n\| \leq 1$ for $n \geq 3$, $\|P_2\|^2 \leq M$, and $\|P_1\| \leq 1$. For $L \geq 3$, $\alpha_L = 1 - \mathbb{E}V_L$, $\eta_L = \mathbb{E}(uV_L) = \alpha_{L+1} \in [\tau, d]$. Also $R_L = V_L - q = BP_{L-1}(B)u$, so at $L \geq 4$, $|R_L| \leq \sqrt{d}$ and $\mathbb{E}R_L^2 \leq \theta d$. The exact mixed mean is

$$\mathbb{E}[r(V_L - 1)] = d - \alpha_{L+2}. \tag{L58.1.9}$$

We first give two continuous curvature certificates. The original row constraint yields $0 \leq f_2 \leq 2 - q^2/2$. If $Z \geq 0$, $\mu = \mathbb{E}Z$, and $Z \leq (1 - d/2)q - d/2 + \sqrt{d}$, $\mu \leq 1 - 27d/16$, then

$$f_2[Z^2 + 2\mu Z + 3\mu^2] \leq 19/2. \tag{L58.8.3}$$

Indeed put $x = \sqrt{d}$, $a = 27/16$, $b(q) = 35/16 + q/2$, $n = 1 - ax^2$. Expanding $(Z + \mu)^2 + 2\mu^2$, discarding its negative cubic and using $x^4 \leq x^2/3$, gives $q^2 + 2q + 3 + 2(q + 1)x - A(q)x^2$, $A(q) =$

$(11/12)q^2+(223/48)q+5093/768$. Complete the square in x. The base $(2-q^2/2)(q^2+2q+3) \le 9$ and the gain is at most $1/2$, since

$$A(q) - (4-q^2)(q+1)^2 = q^4 + 2q^3 - (25/12)q^2 - (161/48)q + 2021/768 > 0.$$

For $h = q-7/8 \in [-7/8, 9/8]$, the last polynomial is $113/4096+(35/128)h+h^2[745/96+(11/2)h+h^2]$. The bracket increases from $713/192 > 3$; the expression is at least $113/4096 - (35/128)^2/12 = 4199/196608 > 0$ on the entire interval.

For $L = 4$, (M.10) gives $|P_4(i\sqrt{\xi})|^2 \le 1-(7/10)\xi$. In (L58.1.7) retain the last two losses and take $\delta = \kappa d$ to obtain $X_{b,4} \le m^2 Z$, $Z = V_4 - \delta - (d/2)q \ge 0$. The actual mean is at most $u_0 = 1-27d/16$; the exact mixed identity gives

$$C_f := \mathbb{E}[f_2(Z-\mu)] = -\eta_4 + d^2/2 + d - \alpha_6 - d\alpha_4 \le -3d/16, \quad \operatorname{Var} Z \le d - (13/8)d^2 + d^3/4.$$

Before the last variance simplification the quadratic loss is $(7/10)\delta_2 + (\kappa^2+\kappa)d^2$. The curvature certificate applies. Taylor expansion gives

$$\mathbb{E}[f_2 X_{b,4}^4] \le m^8 W, \quad W = (1-d)u_0^4 - (3/4)du_0^3 + (19/2)[d - (13/8)d^2 + d^3/4].$$

The mean polynomial has maximum at an endpoint; its nonzero endpoint is positive because $(1-d)u_0 - (3/4)d \ge 1/24$. The exact expansion is $W = 1 + d + (1561/128)d^2 - (10327/256)d^3 + (2027349/65536)d^4 - (531441/65536)d^5$. Using $m^{-8} \ge 1+4\kappa d+9\kappa^2 d^2+15\kappa^3 d^3$, the normalized difference is at most $-11/8 + 10d - 32d^2 \le -19/32$. The intermediate exact linear coefficient is $9239/1024 < 10$, negative quadratic magnitude $1424741/32768 > 43$, and positive cubic coefficient below 33; absorb $d^3 \le d^2/3$ and discard the final negative quartic. Since $m^8 \ge (5/6)^8 > 1/5$, this gives deficit $d/10$.

For every $L \ge 5$, the final two coefficients are at least τ. Take $Z = V_L - \delta - \delta q$, $\delta = \kappa d$, so $\mu \le u_0 = 1-3\kappa d$, $C_f \le -3d/16$, and $\operatorname{Var} Z \le d - 3\kappa^2 d^2 + \kappa^2 d^3$. The curvature certificate still applies; the covariance uses the same $\delta \le \alpha_L$ before simplification. The resulting W is $(1-d)u_0^4 - (3/4)du_0^3 + (19/2)[d - 3\kappa^2 d^2 + \kappa^2 d^3]$. Its nonzero mean endpoint is positive since $(1-d)u_0 - (3/4)d \ge 1/48$. Retain the inverse-power expansion through $(165/8)\kappa^4 d^4$. A sufficient normalized difference is

$$Q(d) = -7/4 + (34713/2048)d - Ad^2 + Bd^3 - Cd^4,$$
$$A = 1591649/32768, \quad B = 288139731/8388608, \quad C = 10556001/1048576.$$

$Q(1/4) = -27393775/536870912 < -1/20$, $Q'(1/4) = -204774799/134217728$, $|Q'(1/4)| < 31/20$. Here $A > 97/2$, $34 < B < 69/2$, $10 < C < 11$. $Q''' = 6B - 24Cd > 6\cdot 34 - 8\cdot 11 > 0$, so Q'' increases and $Q''(1/3) = -2A + 2B - (4/3)C < -124/3 < -40$. The full continuous quadratic tangent bound is therefore $Q \le -1/20 + (31/20)^2/80 = -639/32000$. With $m^8 > 1/5$ this gives deficit $d/300$ for all $L \ge 5$.

For $L = 3$ retain the actual term γr: $X_{b,3} \le m^2 Z$, $Z = \widehat{V} - a - B_* q$, $\widehat{V} = V_3 - \gamma r$, $a = d - k\rho$. The exact finite means obey $\hat{\alpha} = 1 - \mathbb{E}\widehat{V} = d - kD_3 - \gamma d \ge (4/5 - 6d/5)d$, $\hat{\eta} = \mathbb{E}(u\widehat{V}) = \alpha_4$. The finite alternating tail gives $D_4 \le -(\delta_3 - \delta_4)$, $D_5 \le -(\delta_4 - \delta_5)$, hence $\alpha_4 \ge \tau + \kappa\delta_3$, $\alpha_5 \ge \tau + \kappa\delta_4$. With the global ratio $v = \delta_2/d$, original moment inequalities give $\delta_3 \ge v^2 d$, $\delta_4 \ge v^3 d$. The exact covariance is $C_f = -\alpha_4 + B_* d + d - \alpha_5 - \gamma\delta_2 - d\hat{\alpha}$, so

$$C_f/d \le -1 + 2v - \kappa(v^2+v^3) + (7/10)d + (9/10)d^2 \le h(d) := -13/40 + (7/10)d + (9/10)d^2.$$

The v polynomial increases up to θ and $\kappa\theta^2(1+\theta) = 144495/1048576 > 11/80$ (cross-difference 25264). The actual mean loss is at least $(31/16)d - (6/5)d^2$, so $\mu \le 1 - 13d/8$ on $d \le 1/4$ and $\mu \le 1 - 3d/2$ throughout. The finite packet gives the stronger bound $|P_3|^2 \le 1 - \xi/8 \le 1 - (3/25)\xi$. Also $\mathbb{E}[r(V_3 - 1)] = \delta_2 - k|D_5| \ge (4/5)\delta_2$. It follows that $\|\widehat{V} - 1\|_2^2 \le d - (3/25 + 9d/25)\delta_2$. The exact variance is $\|\widehat{V} - 1\|_2^2 - \hat{\alpha}^2 + B_*^2 d - 2B_*\hat{\eta}$. For $t = B_*/d \in [5/8, 3/2]$, the convex endpoint comparison of $t^2 d - 2\kappa t$ chooses $5/8$, since the endpoint difference is $119d/64 - 7\kappa/4 < 0$. Thus

$$\operatorname{Var} Z \le d - (4807/3200)d^2 + (3121/1600)d^3 - (36/25)d^4 \le V_3(d) := d - (3/2)d^2 + 2d^3 - (7/5)d^4.$$

The pointwise and mean bounds are $Z \le (1 - 5d/8)q - d/2 + (13/12)\sqrt{d}$, $\mu \le 1 - 3d/2$. Expanding as above gives the curvature coefficient $29/3$. In detail the quadratic loss is $A(q) = (215/192)q^2 + (53/12)q + 863/144$. The difference from $(169/192)(4 - q^2)(q+1)^2$ is $(169/192)q^4 + (169/96)q^3 - (73/48)q^2 - (21/8)q + 89/36$. At $q = 1 + h$, it equals $557/576 + (301/96)h + h^2[217/24 + (169/32)h + (169/192)h^2]$. The bracket is at least $297/64 > 9/2$, and the whole is at least $15/16 - (16/5)^2/18 = 1327/3600 > 0$. Completing the square in $\sqrt{d}$ adds at most $2/3$ to the base nine, proving the full-rectangle coefficient $29/3$. For $d \le 1/4$, use $u_0 = 1 - 13d/8$ and $W = (1-d)u_0^4 + 4dh(d)u_0^3 + (29/3)V_3(d)$. The mean endpoint replacement is legitimate because $(1-d)u_0 \ge 57/128$ and $4dh(d) \ge -13/40$. After the inverse-power expansion through degree four, a sufficient normalized upper polynomial is

$$Q_s = -3/2 + (221/16)d - (297/8)d^2 + (73/4)d^3 + (153/16)d^4 - (123/8)d^5.$$

For exact coefficient traceability the W coefficients of degrees one through six are $13/15, 2717/160, -16331/480, 1278763/61440, 39039/4096, -19773/1280$. After subtraction the constant is $-181/120 < -3/2$, linear $70699/5120 < 221/16$, quadratic magnitude $18266219/491520 > 297/8$ (cross-difference 18539), and the cubic comparison has positive difference 19291 over 125829120. $Q_s(1/4) = -489/8192 < -7/128$, $Q_s'(1/4) = -2111/2048$ of absolute value below $9/8$; Q_s''' is a concave quadratic positive at both endpoints, so $Q_s'' \le Q_s''(1/4) = -5697/128 < -40$. The tangent bound is $Q_s \le -7/128 + (9/8)^2/80 = -199/5120$. For $1/4 \le d \le 1/3$ take $u_0 = 1 - 3d/2$. Here $h \ge -3/16$, so the nonzero mean endpoint remains positive by $(1-d)u_0 + 4dh \ge 1/3 - 1/4 = 1/12$. The W coefficients are $41/30, 273/20, -3053/120, 727/60, 783/80, -243/20$. Retain inverse-power degree five, $(99/4)\kappa^5 d^5$. The normalized difference Q_h has constant $-121/120$, linear coefficient below $21/2$, negative quadratic magnitude above $57/2$, cubic below ten, quartic below eight, and negative quintic magnitude above twelve. Hence $Q_h'' < -57 + 20 + 32/3 < 0$ and $Q_h'(1/4) < 21/2 - 57/4 + 30/16 + 8/16 = -11/8 < 0$. At $1/4$, $W = 113737/61440 < 1157/625$ (cross-difference 455), whereas the inverse-power lower bound exceeds $1 + 19/32 + 1983/10000 + 4905/100000 + 1001/100000 + 1783/1000000 > 1.8528$. Its exact degree-two through degree-five terms are $3249/16384, 102885/2097152, 21502965/2147483648, 245133801/137438953472$; the displayed rational comparison cross-differences are $528, 1969440, 665368352, 80146959424$. Thus $W - m^{-8} < -1/625$ and $Q_h < -4/625$ throughout. With $m^8 > 1/5$, the complete $L = 3$ row has deficit $d/1000$.

It remains to prove $L = 2$ for all finite $b \ge 3$. The actual nonnegative upper bound is

$$X_{b,2} \le m^2 Z, \quad Z = V_2 - \gamma q - a_1 = (1-\gamma)q + R_2 - a_1, \quad R_2 = r + kK^2 R, \quad |R_2| \le \sqrt{d}. \tag{L58.12.1}$$

Set $\alpha = 1 - \mathbb{E}V_2 = d - k\rho$, $\eta = \mathbb{E}(uV_2) = d - kD_3$, and $\rho_3 = -\mathbb{E}(rK^2R)$. Exact centering gives

$$\begin{aligned} \mu &= 1 - \alpha - \gamma - a_1, \qquad \operatorname{Var} Z = \|V_2 - 1\|_2^2 - \alpha^2 + \gamma^2 d - 2\gamma\eta, \\ C_f &= -d + \delta_2 + k(D_3 - \rho_3) + (\gamma - d + k\rho)d. \end{aligned} \tag{L58.12.2}$$

For $G_{b-1}(-z) = a_\xi - zb_\xi$, $D_3 = \int \xi^2 b_\xi \, d\nu$, $\rho_3 = \int \xi^2 a_\xi \, d\nu$. Even finite lengths have $a_\xi = b_\xi$; if $b - 1 = 2j + 1$ then $b_\xi - a_\xi = -(-\xi)^j$. Hence $D_3 - \rho_3 \le \delta_4$ for all $b \ge 3$, and $D_3 - \rho_3 \le \delta_6$ for $b \ge 7$, whose first positive terminal term comes from $b - 1 = 7, j = 3$. For $v = \delta_2/d \in [d, \theta]$ the actual P_2 estimate is

$$\begin{aligned} \|V_2 - 1\|_2^2 &\le d + \delta_2 - 4\delta_3 + 5\delta_4 - \delta_5 - \delta_6 + \delta_7 \le dJ(v), \\ J(v) &= 1 + v - (63/32)v^2 - (855/1024)v^4 - v^5 \le 71/63. \end{aligned}$$

The replacements use global original spectral Jensen and $\delta_{j+1} \le \theta\delta_j$, not a spectral-point lower bound.

The common continuous $L = 2$ curvature is

$$f_2[Z^2 + 2\mu Z + 3\mu^2] \le 29/3. \tag{L58.12.5}$$

It suffices after expanding to have quadratic loss $A_0(q) = (45/64)q^2 + (27/8)q + 125/24$. Indeed $A_0 - (3/4)(4 - q^2)(q + 1)^2 = (3/4)q^4 + (3/2)q^3 - (99/64)q^2 - (21/8)q + 53/24$. At $q = 7/8 + h$ it equals $8437/49152 + (63/512)h + h^2[747/128 + (33/8)h + (3/4)h^2]$. The bracket increases from $717/256 > 11/4$, the constant exceeds $1/6$ and the linear coefficient is below $1/8$, so the difference is at least $1/6 - 1/704 > 0$. Completing the square adds at most $2/3$ to the base nine. For $b = 3, 4, 5/6$ below use $Z \le (1 - cd)q + \sqrt{d}$, $\mu \le 1 - ad$ with $(a, c) = (17/10, 7/10), (8/5, 7/16), (47/32, 15/32)$. Expansion, dropping the nonpositive cubic and using $d^2 \le d/3$, gives respective quadratic-loss coefficients

b	q^2	q	1
3	$371/300$	$601/150$	$631/100$
4	$623/768$	$433/120$	$151/25$
$5, 6$	$885/1024$	$1749/512$	$5791/1024.$

Each dominates the corresponding coefficient of A_0. For $b \ge 7$ below, $\gamma \ge (1 - 3v/2)d$, $a_1 \ge \kappa v d$, and $\mu \le 1 - A(v)d$, $A(v) = 2 - (337/160)v + (57/80)v^2$. The sum $(1 - 3v/2)q + \kappa v + A(v)$ decreases in v and at θ exceeds $3/2 + 3q/8$: $A(\theta) = 103377/81920 > 5/4$, $\kappa\theta + A(\theta) = 123137/81920 > 3/2$, $1 - 3\theta/2 = 25/64 > 3/8$. Thus $Z + \mu \le q + 1 + \sqrt{d} - (3/2 + 3q/8)d$, $\mu \le 1 - 5d/4$, yielding exactly A_0. For every case the exact Taylor expression must pay both mean endpoints. If $\mu \le u_0$, $C_f \le dh \le 0$ and $\operatorname{Var} Z \le V_0$, then

$$\mathbb{E}[f_2 X_{b,2}^4] \le m^8 \max\{(29/3)V_0, (1 - d)u_0^4 + 4dhu_0^3 + (29/3)V_0\}. \tag{L58.12.6}$$

The mean polynomial $t^3[(1 - d)t + 4dh]$ has maximum on $[0, u_0]$ at an endpoint. For $u_0 = 1 - ad$, $h = h_0 + h_1 d + h_2 d^2$, its degree-one through degree-six coefficients are

$$\begin{aligned} c_1 &= -1 - 4a + 4h_0, & c_2 &= 6a^2 + 4a - 12ah_0 + 4h_1, \\ c_3 &= -4a^3 - 6a^2 + 12a^2h_0 - 12ah_1 + 4h_2, & c_4 &= a^4 + 4a^3 - 4a^3h_0 + 12a^2h_1 - 12ah_2, \\ c_5 &= -a^4 - 4a^3h_1 + 12a^2h_2, & c_6 &= -4a^3h_2. \end{aligned} \tag{L58.12.7}$$

For $b = 3$, $\beta_b = t_0 = d$, $e = \rho = \delta_2$, $V = \delta_2 + \delta_3$, $D_3 = \rho_3 = \delta_3$. γ increases with $k \in [1, 6/5]$, so $\gamma \geq [1 - (1+\theta)v/2]d \geq (1463/2048)d > 7d/10$. Since $\alpha + \gamma + a_1 = d + \gamma$, $\mu \leq 1 - 17d/10$. Using $k - k^2/2 \leq 1/2$ in (L58.12.2) gives $C_f \leq d[-\kappa + (13/64)d + (3/5)d^2]$. The actual P_2 square is at most $1 + \xi - 4\xi^2 + \xi^3 + \xi^4$, hence $\|V_2 - 1\|_2^2 \leq d[1 + v - (3511/1024)v^2] \leq 43d/40$. $\alpha \geq 41d/80$, $\eta \geq 4d/5$; the convex endpoint comparison in $\gamma/d \in [7/10, 6/5]$ selects $7/10$, giving $\operatorname{Var} Z \leq (43/40)d - (11/8)d^2 + d^3/2$. The zero-mean endpoint is paid by $m^{-8} \geq 1 + 4d + 9d^2$ with difference at least $1 - (13/2)d + 20d^2 \geq 151/320$. For the nonzero endpoint use (L58.12.7) and inverse degree three. Its normalized coefficients are bounded by $-3, 15, -69, 35, 9$, with nonpositive last coefficient; the exact first three are $-227/60, 1108/75, -208481/3000$ and the next bounds use $689537/20000 < 35$, $677127/80000 < 9$. Absorbing higher powers using $d \leq 1/3$ gives $-3 + 15d - 55d^2 \leq -3 + 225/220 < 0$. For $b = 4$, $\beta_b = d$, $t_0 = \tau$, $e = \delta_2$, $\rho = \delta_2 - \delta_3$, $V = \delta_2 - \delta_3 + \delta_4$, $D_3 - \rho_3 = \delta_4$. Monotonicity in k gives $\gamma \geq [1 - 3v/2 + (\kappa/2)v^2]d \geq 7d/16$ (endpoint $28811/65536$, excess $139/65536$), and $\alpha + \gamma + a_1 \geq [2 - 3v/2 + (83/64)v^2]d \geq 8d/5$ (endpoint $105163/65536$, cross-difference 1527). The exact identity $\gamma - d + k\rho = (k-1)d - k\delta_3 - k^2V/2$ gives $C_f \leq d[-1/2 + (3/5)d^2]$. The finite square bound is $1 + \xi - 4\xi^2 + 3\xi^3 - \xi^4 + \xi^5$, so $\|V_2 - 1\|_2^2 \leq d[1 + v - (89/32)v^2] \leq 12d/11$. With $\alpha \geq 7d/10$, $\eta \geq 4d/5$, the same convex endpoint method gives variance at most $(12/11)d - (19/16)d^2 + d^3/5$. The zero endpoint inverse difference is at least $1 - 7d + 19d^2 \geq 27/76$. The nonzero endpoint normalized coefficients are $-157/55, 13057/1200, -21664/375, 12256/625, 7424/625$, followed by a nonpositive term; bound them by $-14/5, 11, -57, 20, 12$ to get $-14/5 + 11d - 49d^2 \leq -14/5 + 121/196 < 0$. Both short values yield deficit at least $d/1000$.

For $b = 5, 6$ keep the shared $\beta_b = \tau$ and common global ratio v. Define

$$\Gamma(v) = 1 - 3v/2 + v^2/2 + (\kappa/2)v^3, \quad a(v) = 1 - 6v/5 + (57/80)v^2, \quad e(v) = 1 - 3v/10,$$
$$C(v) = a(v)^2 + 2\Gamma(v)e(v), \quad u_0(v) = 1 - [1 + \Gamma(v)]d, \quad h(v) = -1 + v + (3/5)d^2,$$
$$V_0(v) = dJ(v) - C(v)d^2 + \Gamma(v)^2 d^3, \qquad I(v) = [1 - (1-v)d/2]^{-8}. \tag{L58.14.1}$$

The actual bounds are $\gamma \geq \Gamma(v)d$, $\alpha \geq a(v)d$, $\eta \geq e(v)d$, $\mu \leq u_0(v)$, $C_f \leq dh(v)$, $\operatorname{Var} Z \leq V_0(v)$. For $b = 5$ the exact values are $V = \delta_2 - \delta_3 - \delta_4 + \delta_5$, $\rho = e = \delta_2 - \delta_3$; the increasing-$k$ lower bound is $\gamma \geq d - (3/2)\delta_2 + \delta_3/2 + (\delta_4 - \delta_5)/2$, and the mean loss is exactly $d + \gamma$. For $b = 6$, $V = \delta_2 - \delta_3 + \delta_4 - \delta_5 + \delta_6$, $\rho = \delta_2 - \delta_3 + \delta_4$, $e = \delta_2 - \delta_3$; the lower bounds are $[1 - 3v/2 + (83/64)v^2]d$ and $[2 - 3v/2 + (3\kappa/2)v^2]d$ for γ and mean loss. They dominate the shared parameters. Both have $D_3 = \delta_3 - \delta_4$ and $\eta \geq d - (6/5)\theta\kappa\delta_2 \geq e(v)d$, since $(6/5)\theta\kappa = 741/2560 < 3/10$. $\rho \leq \delta_2 - \kappa\delta_3$ gives the α bound. For covariance, the exact $\gamma - d + k\rho$ formulas are $(k-1)d - k\delta_3 - k^2V/2$ and $(k-1)d + k\delta_4 - k^2V/2$. In the second use $V \geq \kappa\delta_2$, $\theta^2 < \kappa/2$; together with $D_3 - \rho_3 \leq 0$, $k - 1 \leq 3d/5$, both yield $h(v)$. The variance's convex endpoint is $\Gamma(v)$ because $[(6/5) + \Gamma(v)]d - 2e(v) < 0$. Γ decreases and $63/128 < \Gamma(\theta) = 1033999/2097152 < 1/2$, which provides the common curvature premises above.

Set $B_0 = 29/3$ and $W(v) = (1-d)u_0^4 + 4dh(v)u_0^3 + B_0V_0(v)$. With $A = 1 + \Gamma$, $A_- = -A'$, the exact normalized derivative is

$$\partial_v[(W - I)/d] = 4u_0^2\{A_-[(1-d)u_0 + 3dh] + u_0\} + B_0[J' - dC' + 2\Gamma\Gamma' d^2] + 4[1 - (1-v)d/2]^{-9}. \tag{L58.14.4}$$

Uniformly $J' > -1$, and for $v \leq 1/3$, $J' > -1/2$: J' decreases, $J'(1/3) = -1/2 + 43/20736$, $J'(\theta) > -1$. Also $-C' > 13/5$, $2\Gamma\Gamma' \geq -3$, improved to $> -3/2$ when $v \geq 1/3$. For $-C'$ use

$a \ge 5/8$, $-a' \ge 3/5$, $-\Gamma' \ge 9/10$, $e \ge 7/8$, $\Gamma \ge 49/100$ to get $3/4+63/40+147/500 > 13/5$. For $v \ge 1/3$, $\Gamma < 3/5$, $-\Gamma' < 11/10$ gives the latter bound. On this interval $9/10 \le A_- \le 11/10$, $u_0 \ge 1-8d/5$, $h \ge -2/3$, so the bracket is at least $I_+ = 19/10-(307/50)d+(36/25)d^2 > 0$, decreasing to $1/75$. The inverse derivative is at least $4 + 18\kappa d$. Expanding gives the sufficient positive lower bound $29/15 - (53/4)d + 62d^2$, whose real minimum is $29/15 - 2809/3968 > 0$. The coefficients of $4(1-8d/5)^2 I_+$ used here are $38/5, -1222/25, 12976/125, -50816/625, 9216/625$; absorb the negative cubic with $d \le 1/3$. For $v \le 1/3$, $1 \le A_- \le 3/2$, $u_0 \ge 1-2d$, $h \ge -1+d$, and the bracket is at least $I_- = 2-(19/2)d+(13/2)d^2$. For $d \le 1/5$ this is positive and the expanded lower bound is $7 - 35d + 129d^2 > 0$: the actual constant is $43/6$, linear coefficient above -35, quadratic 181 before absorbing $-256d^3$. For $d \ge 1/5$, use $I_- \ge -4/9$; the three terms in the derivative are at least $-16/9, -29/30, 4$, totaling $113/90 > 0$. Thus $W - I$ increases in the same global v throughout $[d, \theta]$.

The zero-mean endpoint must be paid separately. For $d \ge 1/5$, $B_0 V_0 - I$ also increases in v. For $v \ge 1/3$ its normalized derivative is at least $B_0[-1 + (13/5)d - (3/2)d^2] + 4 + 18\kappa d$, equal to $367/400 > 0$ at $1/5$ and increasing. For $v \le 1/3$ use $B_0[-1/2 + (13/5)d - 3d^2] + 4 \ge 91/30 > 0$ and increasing. At θ, $J < 21/20$, $C > 5/4$, $\Gamma^2 < 1/4$; the middle comparison is $C > (5/8)^2 + 2(63/128)(7/8) = 641/512 > 5/4$. So $V_0(\theta) \le (21/20)d - (5/4)d^2 + d^3/4$, and inverse degree three gives difference at least $1-(311/40)d+(61/4)d^2 \ge 879/97600 > 0$. For $d \le 1/5$, the full correlated bound is $V_0(v) \le Md - (3/2)d^2$. Indeed $C \ge 3 - 6v + (209/50)v^2$, $\Gamma \le 1 - v$; the required expression is concave in d, so check zero and $1/5$. The second endpoint has sufficient lower polynomial $D(v) = 1219/3150 - (53/25)v + (11059/4000)v^2 + (855/1024)v^4 + v^5$. $D'' > 5$, $D(7/20) > 1/1000$, $|D'(7/20)| < 1/25$, so $D \ge 1/1000 - (1/25)^2/10 > 0$. The exact five terms at $7/20$ are $1219/3150, -371/500, 541891/1600000, 2052855/163840000, 16807/3200000$, whose sum exceeds 0.0014430625; $D' = -0.184675 + 293265/2048000 + 2401/32000 \in (0, 1/25)$. These terminating decimals denote exact rationals. The inverse difference is then at least $1 - (43/5)d + (35/2)d^2 + 3d^3$, decreasing to $1/250$ at $1/5$; use $2059/189 - 19/8 < 43/5$, $29/2 + 3249/1024 > 35/2$, $102885/32768 > 3$. For the nonzero endpoint, monotonicity reduces to $v = \theta$. Relax u_0 to $1 - 149d/100$, retaining the endpoint maximum; take $h = -\kappa + (3/5)d^2$ and $V_0 = (21/20)d - (5/4)d^2 + d^3/4$. Inverse degree five and (L58.12.7) give normalized coefficients $-39/25$, $< 147/10$, $< -203/5$, < 13, < 10, with nonpositive final term. The linear comparison has positive numerator 113923 over 1920000; the quadratic magnitude exceeds $1951/48 > 203/5$; the raw next two coefficients are $3057803777/200000000 < 153/10$ and $1105587599/100000000 < 111/10$, with $\kappa^4 > 3/25$, $\kappa^5 > 3/50$ supplying their inverse costs. Thus the normalized difference is at most $-39/25 + (147/10)d - 35d^2 \le -231/14000$. Both endpoints with $m^8 > 1/5$ give deficit $d/1000$ for $b = 5, 6$.

Finally let every finite $b \ge 7$ retain the extra alternating loss:

$$\beta_b \ge \tau + \kappa\delta_3 \ge t(v)d, \quad t(v) = 1 - v + \kappa v^2, \quad t_0 \le d - \kappa\delta_2. \tag{L58.15.1}$$

The third positive beta term remains, the fourth negative term is at most $\theta\delta_3$, and all subsequent complete pairs are nonnegative. Also $V \ge c_G\delta_2$, $c_G = (1+\theta)\kappa^2 = 16245/32768$, because the finite coordinates are at least $1-\xi$ and $(1+\xi)(1-\xi)^2$ decreases on $[0, \theta]$. The actual covariance is at most $dh(v, d)$, where

$$h = -1 + (31/30)v + d/40 + (3/5)d^2.$$

Indeed $k(D_3 - \rho_3)/d \le (6/5)\theta^4 v < v/30$. The remaining correlated expression $k(\theta v - \kappa v^2) -$

$k^2 c_G v/2$ decreases in $k \ge 1$ because $c_G > \theta$; its upper bound is $v(\theta - c_G/2) - \kappa v^2 \le 1/72 < 1/40$, using $c_G > 23/48$, $\theta - c_G/2 < 1/6$, $\kappa > 1/2$. The residual $k - 1 \le 3d/5$ yields the stated h. Define

$$\begin{aligned} g_0 &= 1 - 3v/2, \quad a = 1 - 6v/5 + (57/80)v^2, \\ e_0 &= 1 - 39v/80, \quad A = 2 - (337/160)v + (57/80)v^2, \\ C &= a^2 + 2g_0 e_0, \quad u_0 = 1 - Ad, \\ V_0 &= dJ - Cd^2 + g_0^2 d^3, \quad I = [1 - t(v)d/2]^{-8}. \end{aligned} \tag{L58.15.4}$$

The actual bounds are $\gamma \ge g_0 d$, $\alpha \ge ad$, $\eta \ge e_0 d$, $\mu \le u_0$, $\operatorname{Var} Z \le V_0$, $m^{-8} \ge I$. The stronger bound $B_* \ge d - (29/32)\delta_2$ follows directly from the finite parameters. Put $s_* = t_0 + e$. Consecutive beta values differ by zero or one signed moment of index at least three, so $s_* \ge d - \delta_3$. Also $V \le \delta_2$ and

$$\frac{d}{dk}\left(k s_* - \frac{k^2 V}{2}\right) \ge d - \delta_3 - \frac{6}{5}\delta_2 > 0,$$

where the last inequality uses $\delta_3 \le \theta \delta_2$, $\delta_2 \le \theta d$ and $1 - (\theta + 6/5)\theta > 0$. Evaluating at $k = 1$ therefore gives

$$B_* \ge s_* - V/2 \ge d - \delta_3 - \delta_2/2 \ge d - (29/32)\delta_2.$$

Furthermore, $k(1 - v) - k^2 v/2$ increases for $1 \le k \le 6/5$, giving the gamma bound. The variance convex endpoint is g_0 since $[(6/5) + g_0]d - 2e_0 < 0$. The curvature premise $a_1 \ge \kappa \delta_2$ was included above. For $W = (1 - d)u_0^4 + 4dhu_0^3 + B_0 V_0$, the exact normalized derivative is

$$\begin{aligned} \partial_v[(W - I)/d] = {} & 4u_0^2\{A_-[(1 - d)u_0 + 3dh] + (31/30)u_0\} \\ & + B_0[J' - dC' - 3g_0 d^2] + 4(1 - 2\kappa v)[1 - t(v)d/2]^{-9}. \end{aligned} \tag{L58.15.6}$$

Here $3/2 \le A_- = -A' \le 17/8$, $-C' > 7/2$, $J' > -1$, and $t(v) \ge t(\theta) = 22667/32768 > 11/16$. For $-C'$ use $a \ge 5/8$, $-a' \ge 3/5$, $e_0 \ge 4/5$, $g_0 \ge 25/64$ to get $3/4 + 12/5 + (39/40)(25/64) > 7/2$. When $v \le 1/3$, $1 - 2\kappa v \ge 29/48$ and $v \ge d$ gives $u_0 \ge 1 - 2d + (21/10)d^2 - (3/4)d^3$, $h \ge -1 + (127/120)d + (3/5)d^2$. The bracket is at least $I_- = 38/15 - 13d + 15d^2 - (5/4)d^3$; the stronger raw coefficients are $38/15, -1553/120, 24107/1600, -49/40, 9/8$. For $d \le 1/4$, $I_- > 0$ and $u_0 \ge 1 - 2d$. Expansion gives lower bound $77/10 - 52d + 165d^2 > 0$, with real minimum $77/10 - 2704/660 > 0$. The coefficients of $4(1 - 2d)^2 I_-$ are $152/15, -1388/15, 4628/15, -453, 260, -20$; add $B_0[-1/2 + (7/2)d - 3d^2]$ and $(29/12)[1 + (99/32)d]$, then absorb negative cubic and quintic using $d \le 1/4$. For $d \ge 1/4$, $I_- \ge -97/540 > -1/5$; the three derivative terms are at least $-4/5, 29/16, 29/12$, with positive sum. When $v \ge 1/3$, $A_- \le 5/3$, $u_0 \ge 1 - 7d/5$, $h \ge -2/3$, $1 - 2\kappa v \ge 265/512$. The bracket is at least $I_+ = 5/2 - (25/3)d + (21/10)d^2$. For $d \le 1/4$ it is nonnegative; expansion gives the lower bound $12/5 - 22d + 70d^2 > 0$, whose real minimum is $12/5 - 484/280 > 0$. The coefficients of $4(1 - 7d/5)^2 I_+$ are $10, -184/3, 364/3, -6664/75, 2058/125$; after adding the other terms the constant is $923/384 > 12/5$, linear coefficient above -22, and, absorbing the negative cubic, quadratic above seventy. For $d \ge 1/4$, use $I_+ \ge -2/45$: the three terms are at least $-8/45, -145/48, (265/128)(1 + 99/128) > 18/5$, again a positive sum. Thus $W - I$ increases throughout the complete correlated interval.

For the zero endpoint with $d \ge 1/5$, $B_0 V_0 - I$ also increases. For $v \le 1/3$, its variance derivative is at least $B_0[-1/2 + (7/2)d - 3d^2]$, positive at $1/5$ and increasing. For $v \ge 1/3$, $g_0 \le 1/2$, so use $B_0[-1 + (7/2)d - (3/2)d^2]$ and inverse derivative at least $2(1 + 3d + 5d^2)$. At $1/5$ their

sum is at least $-87/25+18/5=3/25>0$, increasing. The inverse bounds use $265/128>2$, $99/32>3$, $5445/1024>5$. At θ, $J=1124767513/1073741824<21/20$ (cross-difference 53228044), $C>65/64>1$, $g_0^2=625/4096<5/32$. Thus $V_0(\theta)\le(21/20)d-d^2+(5/32)d^3$ and inverse degree three yields difference at least $1-(37/5)d+(139/10)d^2\ge 21/1390>0$; the actual quadratic is $10691/768>139/10$ and the omitted cubic is positive. For $d\le 1/5$, the full correlated variance is $V_0\le Md-(4/3)d^2$. Expand $C\ge 3-(51/8)v+(465/128)v^2$; the needed expression is concave in d, so check zero and $1/5$. At the latter a sufficient polynomial is $D(v)=662/1575-(431/200)v+(8337/3200)v^2+(855/1024)v^4+v^5$. $D''>5$, $D(3/8)>1/500$, $|D'(3/8)|<1/12$ imply $D\ge 1/500-(1/12)^2/10>0$. The exact terms there are $662/1575$, $-1293/1600$, $75033/204800$, $69255/4194304$, $243/32768$, whose sum exceeds 0.00247; $D'=-2573/12800+23085/131072+405/4096\in(0,1/12)$. Then the inverse difference is at least $1-(33/4)d+17d^2$, decreasing to $3/100$ at $1/5$; use $2059/189-11/4<33/4$ and $1089/256+116/9>17$. For the nonzero endpoint reduce by monotonicity to $v=\theta$, relax u_0 to $1-5d/4$, and take $h=-23/40+d/40+(3/5)d^2$, $V_0=(21/20)d-d^2+(5/32)d^3$. Retaining the zero endpoint justifies any relaxation of a negative mean polynomial. Use $I\ge 1+(11/4)d+(1089/256)d^2+(19965/4096)d^3$. The normalized coefficients from (L58.12.7) are $-9/10$, $403/30-1089/256<46/5$, $-733/30-19965/4096<-29$, $1591/256<25/4$, $2205/256<35/4$, followed by $-75d^5/16$. Therefore the normalized difference is at most $-9/10+(46/5)d-25d^2\le -67/1250<0$. Both endpoints, $m^8>1/5$ and $d\le 1/3$ prove the claimed deficit $d/1000$ for every finite $b\ge 7$. This finishes all $b\ge 3, L\ge 2$; the precise base rows stated above give (L58.16.1). At $d=0$ original leaf integration gives equality. Actual equal pairs $E_t\in[0,1]$ are restored inside T^L by positivity. Full reversal simultaneously reverses the actual reference arm, terminal and all connector and decoration edges.

N.8 The actual squares for unit terminals of lengths $(2,4)$ and $(2,5)$

The following square and moment calculations are used in the reference proofs. Put $r=Kg$, $s=K^2g$, $t_0=K^3g$ and $m=1-d$. For $(2,4)$, the actual positive paths satisfy $h_2=q-r$, $f_4=p+r+s+t_0-d-dp$, so $h_2+f_4=2m+R$, $R=s+t_0-dg$. The actual product is at most $F_0=(m+R/2)^2$. For any such square write

$$\begin{aligned} Y=TF_0=m^2+w_0+e,\quad w_0=-m^2g+mD_0-mKR,\quad D_0=\mathbb{E}R,\\ e=T(R^2)/4,\quad 0\le e\le V/2,\quad V=\mathbb{E}R^2. \end{aligned} \tag{L69.10.square}$$

For $(2,4)$ the exact original moments are $\mathbb{E}R=\delta_2$, $\mathbb{E}KR=d^2+\delta_2$, $\mathbb{E}(rR)=-\delta_3$, $\mathbb{E}(sR)=\delta_3+d\delta_2$. Consequently

$$\mathbb{E}[f_2(w_0+e)]=-m^2d-md^2+\mathbb{E}[f_2e]\le -m^2d-md^2+V/2, \tag{L69.10.2}$$

where the actual adjoint gives $\mathbb{E}[f_2e]=\mathbb{E}[f_3R^2]/4\le V/2$. The complete second moment, using $m+d=1$, is

$$\mathbb{E}w_0^2=m^2[m^2d+2\delta_3+\delta_4+\delta_5-d^2\delta_2-\delta_2^2]. \tag{L69.10.3}$$

On $0<d\le 1/4$ set $v=15/64+(13/16)d+d^2$, $H=m^2+17/40-(4/25)d-(2/5)d^2$. The exact $V=d^3+2d\delta_2+\delta_3+\delta_4$ gives $V\le dv$ and $V\le d(1/2+d)^2$, using $\theta^2+\theta^3=7605/32768<15/64$. For the bracket in (L69.10.3), normalized by d, write $v_1=\delta_2/d$. Its upper bound is $m^2+(2\theta+\theta^2+\theta^3)v_1-d^2v_1-dv_1^2$, increasing on $[0,\theta]$ since its derivative is at least

$2\theta - d^2 - 2d\theta > 0$. At $v_1 = \theta$ use $2\theta^2 + \theta^3 + \theta^4 = 444977/1048576 < 17/40$, $\theta \ge 2/5$, $\theta^2 \ge 4/25$ to obtain $\mathbb{E}w_0^2 \le m^2 dH$. $0 < H \le 57/40 < (6/5)^2$ and Cauchy–Schwarz with $e \le V/2$ give

$$\mathbb{E}(w_0 + e)^2 \le m^2 dH + (6/5)md^{3/2}v + d^2v^2/4. \tag{L69.10.4}$$

We supply the full original-f_2 curvature certificate. With $x = \sqrt{d} \in [0, 1/2]$, the row bound gives $Y \le m^2 q + B_0$, $B_0/m \le x/2 + (41/32)x^2 \le C = x/2 + (21/16)x^2$. Since $f_2 \le p + x = 2 + x - q$, its fourth Taylor coefficient divided by m^2 is at most $Q(q) = (2+x-q)\{[m(q+1)+C]^2 + 2m^2\}$. Put $c = C/m \le 37x/24 < 1$. $Q'(0) > 0$ and $Q'(1/2)/m^2 = 1/4 + 3x + 2xc - c^2 > 0$; Q' is concave, while $Q'(1)/m^2 = -2 + 4x + (-2+2x)c - c^2 \le 0$ and $Q'' < 0$ after one. Thus its maximum is in $[1/2, 1]$, where $Q'' \le -2m^2$. At $q = 3/4$ the normalized derivative is $D = -11/16 + (7/2)x + (2x-1)c - c^2$, with $|D| \le 7/10$. For the lower bound use $c + c^2 \le (3145/1152)x < (7/2)x$. For $x \le 2/5$, use $c \ge x/2$ to get $D \le 269/400 < 7/10$; for $x \ge 2/5$, c/x increases and is at least $205/168 > 61/50$, giving $D \le -11/16 + (57/25)x + (2379/2500)x^2 \le 863/1250 < 7/10$. Strong concavity gives

$$\max Q \le \Phi(x) = (5/4 + x)\{[7/4 + x/2 - 7x^2/16]^2 + 2(1 - x^2)^2\} + (49/400)(1 - x^2)^2.$$

Its coefficients, in ascending degree, are $10321/1600, 29/4, -621/128 - 49/200, -373/64$, $2357/1024 + 49/400, 561/256$. Absorb $x^4 \le x^2/4$, $x^5 \le x^3/4$, using $459827/102400 > 22/5$, $5407/1024 > 21/4$, to get $\Phi \le 13/2 + (29/4)x - (22/5)x^2 - (21/4)x^3$. This increases up to $2/5$ and there equals $209/25 < 17/2$. For $2/5 \le x \le 1/2$, use $x^3 \ge (2/5)x^2$ to get the increasing upper bound $13/2 + (29/4)x - (13/2)x^2$, equal to $17/2$ at $1/2$. Thus

$$f_2(Y^2 + 2m^2Y + 3m^4) \le (17/2)m^2. \tag{L69.10.5}$$

The exact fourth expansion and the above moments bound the target by

$$U = m^8(1 - 5d) - 4m^7d^2 + 2m^6dv + (17/2)m^4dH + (51/5)m^3d^{3/2}v + (17/8)m^2d^2v^2.$$

The positive Taylor series gives $[m^{-4} - m^4(1 - 5d)]/d \ge 13 - 16d + 54d^2 + 14d^3 + 61d^4$. After combining all nonsquare-root terms the normalized deficit is at least

$$67/160 + (2269/400)d + (6029/160)d^2 + (227/8)d^3 + 55d^4 - (51/5)(\sqrt{d}/m)v - (17/8)(d/m^2)v^2.$$

Use $m^{-1} \le 1 + 4d/3$, $m^{-2} \le 16/9$. The first negative cost has x, x^3, x^5, x^7 coefficients $153/64, 459/40, 85/4, 68/5$; the second has d through d^5 coefficients $425/2048, 1105/768$, $4913/1152, 221/36, 34/9$. With $d^5 \le d^4/4$, $x^5 \le x^4/2$, $x^7 \le x^6/2$, the deficit is at least $67/160 - (153/64)x + (27/5)x^2 - (459/40)x^3 + 25x^4$. The retained quadratic and quartic coefficients are at least $279807/51200 > 27/5$, $98371/3840 > 25$; remaining sixth and eighth coefficients are nonnegative. Since $25x^2 - (459/40)x + 27/5 \ge 4$, whose square-completion remainder is $13319/160000 > 0$, the deficit is at least $67/160 - (153/64)x + 4x^2 \ge 20171/327680$. Multiplying by $m^4 \ge 81/256$ proves the low-range bound $\mathbb{E}[f_2\{T(h_2f_4)\}^4] \le 1 - d/100$. The exact actual mean is $\mathbb{E}(h_2f_4) = 1 - 3d + \delta_2 - d^2 + \delta_3$. For $d \ge 1/4$ its decreasing upper bound is $1353/4096 < 2/5$, so $TF < 4/5$ and the target is at most $256/625$.

For $(2, 5)$ keep the same m but use the actual retained vector

$$R = \sum_{j=3}^{5} K^j\mathbf{1} - (d + \delta_2)g - \delta_2 r. \tag{L69.11.vector}$$

Finite positive convolution retaining f_0, f_1 and the actual $\delta_2 f_2$ portion gives $h_2 + f_5 \le 2m + R$. Its multiplier at g is $-A - i\sqrt{\xi}B$, where $A = d + \delta_2 + \xi - \xi^2$, $B = \delta_2 + \xi$. Let $v = 101/800 + (13/16)d + (33/16)d^2$ and $H = m^2 + 49/128 + (3/10)d - (1/2)d^2 - (19/20)d^3$. Here $D_0 = (1+d)\delta_2$ and the exact reference cancellation is

$$\mathbb{E}[f_2(w_0 + e)] = -m^2 d - md^2(1 + \delta_2) + m(\delta_3 - \delta_4) + \mathbb{E}[f_2 e], \quad \mathbb{E}[f_2 e] \le V/2. \tag{L69.11.2}$$

The actual difference $\delta_3 - \delta_4$ is bounded as one spectral term: $\xi^2(1-\xi)$ increases to $\theta^2\kappa$, with $4\theta^2\kappa = 3211/8192 < 2/5$. The exact vector square is

$$V = d(d + \delta_2)^2 + [2(d + \delta_2) + \delta_2^2]\delta_2 + (1 - 2d)\delta_3 - \delta_4 + \delta_5 \le dv.$$

The positive $2\delta_2^2$ cancels with $-2d\delta_3$ using $\delta_3 d \ge \delta_2^2$; the remaining frequency bound gives $V/d \le \theta^2 - \theta^3 + \theta^4 + 2\theta d + [(1+\theta)^2 + \theta^3]d^2$. Use $131313/1048576 < 101/800$ and $66997/32768 < 33/16$ for the two constants. The complete second square, writing $u_2 = \delta_2$, is

$$\begin{aligned}\mathbb{E}w_0^2/m^2 = {} & m^2 d + (1 - 2d - d^2)u_2^2 - d^2 u_2 - 2d^3 u_2 + u_2^3 \\ & + [2 + (4 + 2d)u_2 + u_2^2]\delta_3 + (1 - 2d)\delta_4 - \delta_5 + \delta_6.\end{aligned}$$

The complete high-frequency combination $2\xi^2 + (1 - 2d)\xi^3 - \xi^4 + \xi^5$ increases on $[0, \theta]$. After bounding it, the upper expression increases in u_2/d, with derivative at least $d(4\theta^2 - d - 2d^2) > 0$ on $d \le 1/4$. At $u_2/d = \theta$ it is bounded by

$$m^2 + (2\theta^2 + \theta^3 - \theta^4 + \theta^5) + \theta^2(1 + 2\theta)d - (2\theta^2 + \theta - 3\theta^3 - \theta^4)d^2 - (\theta^2 + 2\theta)d^3 \le H.$$

The four respective coefficient comparisons are $< 49/128$, $< 3/10$, $> 1/2$, $> 19/20$; the first and third positive differences have numerators 62403 over 33554432 and 8335 over 1048576. H is positive and decreasing, $H \le 177/128 < (6/5)^2$, hence

$$\mathbb{E}(w_0 + e)^2 \le m^2 dH + (6/5)md^{3/2}v + d^2 v^2/4. \tag{L69.11.6}$$

The five-arm curvature certificate has, with $x = \sqrt{d}$, $\sqrt{V} \le (3/8 + 3d/2)\sqrt{d}$: the two squared vector constants are $(\theta - \theta^2)^2 + \theta^3 = 131313/1048576 < 9/64$ and $(1+\theta)^2 + \theta^3 = 66997/32768 < 9/4$. $D_0 \le 37d/64$ and $V/(2m) \le 3d/8$ give $Y \le m^2 q + B_0$, $B_0/m \le C = 3x/8 + 109x^2/64$. For $Q = (2 + x - q)\{[m(q+1) + C]^2 + 2m^2\}$ the same derivative argument locates the maximum in $[1/2, 1]$ with $Q'' \le -2m^2$. Now $c = C/m \le 157x/96 < 1$; the normalized derivative at $3/4$ is $D = -11/16 + (7/2)x + (2x-1)c - c^2$, $|D| \le 7/10$. The lower estimate uses $c + c^2 \le (54793/18432)x < (7/2)x$; the upper uses $c \ge 3x/8$ for $x \le 2/5$ to get $69/100$, and $c/x \ge 845/672 > 5/4$ for $x \ge 2/5$ to get $43/64$. The strong-concavity upper polynomial is

$$\Phi = (5/4 + x)\{[7/4 + 3x/8 - 3x^2/64]^2 + 2(1 - x^2)^2\} + (49/400)(1 - x^2)^2.$$

Its ascending coefficients are

$$\begin{gathered}10321/1600, \quad 429/64, \quad -1903/512 - 49/200, \\ -4165/1024, \quad 40429/16384 + 49/400, \quad 8201/4096.\end{gathered}$$

Absorb $x^4 \le x^2/4$, $x^5 \le x^3/4$ to get $\Phi \le 13/2 + (27/4)x - (33/10)x^2 - (7/2)x^3$. This increases to $1056/125 < 35/4$ on $[0, 2/5]$; afterward use $x^3 \ge (2/5)x^2$ to obtain upper bound $13/2 + (27/4)x - (47/10)x^2 \le 87/10 < 35/4$. Thus on the full low rectangle

$$f_2(Y^2 + 2m^2 Y + 3m^4) \le (35/4)m^2. \tag{L69.11.7}$$

The full normalized deficit after Taylor expansion is at least

$$3173/12800 + (591/200)d + (14119/400)d^2 + (3307/80)d^3 + (423/8)d^4$$
$$-(21/2)(\sqrt{d}/m)v - (35/16)(d/m^2)v^2.$$

The same-kernel $\delta_3 - \delta_4$ contribution retains the favorable factor $-(2/5)m^3$ before collecting coefficients. Using $m^{-1} \le 1 + 4d/3$, $m^{-2} \le 16/9$, the first cost has x, x^3, x^5, x^7 coefficients $2121/1600, 8239/800, 1057/32, 231/8$; the second cost has d through d^5 coefficients $357035/5760000, 45955/57600, 132265/28800, 15015/1152, 4235/256$. Use $d^5 \le d^4/4$. On $0 \le x \le 1/3$, absorbing $x^5 \le x^4/3$, $x^7 \le x^6/3$ gives a lower polynomial $3173/12800 - (2121/1600)x + (289/100)x^2 - (8239/800)x^3 + (469/20)x^4$. Its actual quadratic and quartic coefficients are at least $16663765/5760000 > 289/100$ and $1352981/57600 > 469/20$; the sixth and eighth remainders are nonnegative. The quadratic $(469/20)x^2 - (8239/800)x + 289/100$ is at least $7/4$: use $8239/800 < 103/10$ and $(469/5)(57/50) > 10609/100$. The deficit is therefore at least $3173/12800 - (2121/1600)x + (7/4)x^2$, decreasing up to $1/3$, with endpoint $53/115200 > 0$. On $1/3 \le x \le \sqrt{1/6} < 5/12$, absorb instead using $5/12$. The lower polynomial is $Q_* = 3173/12800 - (2121/1600)x + (289/100)x^2 - (8239/800)x^3 + (207/10)x^4$; the actual quartic coefficient is $1194431/57600 > 207/10$. At $1/3$, its value, derivative and second derivative are $431/345600, 1127/4800, 5113/400$, all positive, and $Q_*''' = -24717/400 + (2484/5)x > 0$ afterward. Hence it increases and is at least $431/345600 > 53/115200$. These overlapping continuous intervals cover $d \le 1/6$. Since $m^4 \ge 81/256$, their loss proves $\mathbb{E}[f_2\{T(h_2 f_5)\}^4] \le 1 - d/10000$. For $d \ge 1/6$ the exact actual mean is

$$\mathbb{E}(h_2 f_5) = 1 - 4d + 2(1+d)\delta_2 \le 1 - (51/16)d + (13/16)d^2 \le 283/576 < 1/2. \qquad \text{(L69.11.8)}$$

The quadratic decreases on $[1/6, 1/3]$, so $TF \le 283/288 < 1$ and the reference integral is at most $1 - d$. All stated square inputs and claims use actual fixed directions; at $d = 0$ use original leaf integration, and restore equal pairs and simultaneous full reversal by original positivity.

N.9 The retained actual square for every unit terminal $(2, b)$ with $b \ge 6$

We use the square calculation from Q5-C073, Sections 1.1 and 3.1–3.5, for the direction $h_2 f_b$. For $0 < d \le 1/5$ put $m = 1 - 9d/8$, $N = b - 2 \ge 4$, and

$$\lambda = \frac{9d/4 - \beta_b - \beta_{b-1}}{\beta_{b-2} + (1-d)\beta_{b-3}}, \qquad C_1 = \beta_{b-1},\ C_2 = \lambda\beta_{b-2},\ C_3 = \lambda\beta_{b-3},$$
$$C = C_1 + C_2 + C_3, \quad D = C_2 + C_3, \quad E = C_3, \qquad R = \sum_{j=3}^{b} K^j \mathbf{1} - Cg - Dr - Es.$$

(L73.3.retained)

Here $r = Kg$, $s = K^2 g$. The denominator is at least $(2-d)\tau > 0$, the numerator is nonnegative, and the four available constant portions sum to at least $(4-d)\tau \ge 361d/160 > 9d/4$. Thus $0 \le \lambda \le 1$ and these are actual retained portions of the finite positive convolution. It gives $0 \le h_2 + f_b \le 2m + R$, so $F_b = h_2 f_b \le (m + R/2)^2$ without any change of actual orientation.

On the original g spectral measure put $G_N(i\sqrt{\xi}) = a + i\sqrt{\xi}\, b_0$. The finite alternating sums give $1 - \xi \le a, b_0 \le 1 - \xi + \xi^2$ and $a \ge (1-\xi)b_0$ for every finite $N \ge 4$. The multiplier of R at g is

$-A - i\sqrt{\xi}B$, with $A = C + \xi(a - E)$, $B = D + \xi b_0$, both nonnegative. For $V = \mathbb{E}R^2$, $D_0 = \mathbb{E}R$, $a_1 = \mathbb{E}KR$, $\rho_0 = -\mathbb{E}(rR)$, the exact spectral data are

$$V = \int (A^2 + \xi B^2)\, d\nu, \quad D_0 = \int B\, d\nu, \quad a_1 = \int A\, d\nu, \quad \rho_0 = \int \xi B\, d\nu,$$
$$(1 - \xi)B - A \le -C_1 - C_2\xi, \quad a_1 \ge \kappa D_0 + \tau d, \quad D_0^2 - 2D_0 a_1 \le 0. \tag{L73.3.1}$$

If N is even, $a = b_0$; for $N = 2j + 1 \ge 5$, $a - b_0 = (-\xi)^j$. Thus all finite parities satisfy

$$I := \int (1 - \xi)(B - A)\, d\nu \le -\tau^2 + d(\delta_2 - \delta_3) + \int \xi^4(1 - \xi)\, d\nu$$
$$\le -(57/512)d^2 + d/60.$$

The global relation $d\delta_3 \ge \delta_2^2$ bounds the first two terms by $-(d - \delta_2)(d - 2\delta_2)$; $\xi^4(1 - \xi)$ increases to $\theta^4\kappa = 542659/33554432 < 1/60$.

Put $v = (3/8 + 9d/5)^2$, $z_0 = 5/16 + 17d/16 + d^2$. The full frequency bounds are

$$A^2 + \xi B^2 \le v, \quad B \le z_0, \quad V \le dv, \quad D_0 \le dz_0, \quad \rho_0 \le \theta d z_0. \tag{L73.3.3}$$

To prove them write $u_0 = \theta - \theta^2 + \theta^3 < 5/16$ and use $C = 9d/4 - \beta_b + dE$, $D = 9d/4 - \beta_b - \beta_{b-1} + dE$. The increasing frequency upper bounds for A, B are $u_0 + (5/4 + \theta)d + E(d - \theta)$ and $u_0 + (1/4 + 2\theta)d + dE$. Their frequency derivatives before endpoint substitution are at least $2/3 - E > 0$ and $2/3$, so $\xi = \theta$ is legal. Their squared sum is convex in $E \in [0, d]$. For $E = 0$ the squared linear-d vector norm is $26229/8192 < (9/5)^2$; for $E = d$ it is $16557/8192 < (10/7)^2$, its quadratic-d vector norm is at most $6/5$, and $10/7 + (6/5)(1/5) < 9/5$. The constant vector has norm $u_0\sqrt{1 + \theta} < 3/8$. This proves the square bound; the scalar B bound is exactly z_0.

Use the actual square notation (L69.10.square). The exact second moment is

$$\mathbb{E}w_0^2 = m^4 d + 2m^3\rho_0 + m^2\{\mathbb{E}(KR)^2 + D_0^2 - 2D_0 a_1\} \le m^2 dH,$$

where

$$H = m^2 + 2m\theta z_0 + \theta v = 2685/2048 - (11509/10240)d + (124067/51200)d^2 - (117/128)d^3. \tag{L73.3.4}$$

The exact reference identity $\mathbb{E}[f_2 w_0] = -m^2 d + mI - mdD_0$ gives

$$\mathbb{E}[f_2(w_0 + e)] \le -m^2 d + (m/60)d + V/2. \tag{L73.3.5}$$

Only the nonpositive terms $-(57m/512)d^2 - mdD_0$ were dropped. The cross term with the actual $e = T(R^2)/4$ is

$$2\mathbb{E}(w_0 e) = \frac{mV}{2}(D_0 - a_1) - \frac{m^2}{2}\mathbb{E}(rR^2) + \frac{mD_0}{2}\mathbb{E}(gR^2) - \frac{m}{2}\mathbb{E}[(K^2R)R^2]. \tag{L73.1.6}$$

The second-row inequality is $|K^2 v(x)|^2 \le \|v\|_2^2/3$. For any legal skew kernel L, the three-variable inequality $1 + ab + bc + ca \ge 0$, with $a = L(x, y), b = L(y, z), c = L(z, x)$, follows by expanding the two products of three nonnegative factors and averaging. Independent integration gives $\|L\mathbf{1}\|_2^2 \le 1/3$. For a sign vector v, $v(x)K(x, y)v(y)$ is legal, hence $\|Kv\|_2^2 \le 1/3$; finite convex mixtures of sign vectors give the same for $|v| \le 1$. For the original row $k_x(y) = K(x, y)$, skew-adjointness gives

$K^2v(x) = -\langle Kk_x, v\rangle$, proving the stated inequality. This gives the row bound for the kernel. Use it in (L73.1.6), along with $D_0 - a_1 \le \theta D_0 - \tau d$ and $|g| \le 1$, to get

$$2\mathbb{E}(w_0 e) \le (49/80)md^{3/2}v + (2/5)md^2v.$$

The coefficients follow from $m/2 + \sqrt{v}/(2\sqrt{3}) \le 49/80 - 9d/400 \le 49/80$, using $1/\sqrt{3} < 3/5$, and $(1+\theta)z_0/2 \le 1017/2560 < 2/5$. Thus

$$\mathbb{E}(w_0 + e)^2 \le m^2 dH + (49/80)md^{3/2}v + (2/5)md^2v + d^2v^2/4. \tag{L73.3.6}$$

For the original-f_2 curvature, the row upper bound gives $Y \le m^2 q + B_0$, $B_0/m = D_0 + \sqrt{V} + V/(2m)$. With $x = \sqrt{d} < 9/20$, $z_0 \le 113/200$, $V/(2m) \le 7d/20$, this is at most $3x/8 + (69/40)x^2$. Put $m_0 = 1 - d$, $C_0 = 3x/8 + (109/64)d$. Since $69/40 - 109/64 = 7/320 < 1/8$ and $q + 1 \ge 1$, $m(q+1) + B_0/m \le m_0(q+1) + C_0$, $m \le m_0$. The complete scalar rectangle certificate proved in (L69.11.7), using the displayed derivative and tangent-polynomial coefficients there, is exactly

$$(2 + x - q)\{[m_0(q+1) + C_0]^2 + 2m_0^2\} \le 35/4 \quad (0 \le x \le 1/2, 0 \le q \le 2).$$

Consequently $f_2(Y^2 + 2m^2Y + 3m^4) \le (35/4)m^2$. The exact fourth expansion bounds the reference target by

$$\begin{aligned} U = {} & m^8(1 - 5d) + m^7 d/15 + 2m^6 dv + (35/4)m^4 dH \\ & + (343/64)m^3 d^{3/2}v + (7/2)m^3 d^2 v + (35/16)m^2 d^2 v^2. \end{aligned}$$

The positive Taylor series gives

$$\frac{m^{-4} - m^4(1 - 5d)}{d} \ge 14 - (279/16)d + (4617/64)d^2 + (53217/2048)d^3 + (111537/1024)d^4.$$

Discard the favorable nonconstant part of $-m^3/15$ and combine $-2m^2v - 35H/4$; the resulting constant through quartic coefficients are $267943/122880, -396097/40960, 10276211/204800, 2311461/51200, 2578473/25600$. With $m^{-1} \le 1 + 3d/2$, $m^{-2} \le 5/3$, the square-root cost has x, x^3, x^5, x^7 coefficients at most $3087/4096, 342657/40960, 361179/12800, 83349/3200$. The $-(7/2)(d/m)v$ cost has d through d^4 coefficients $63/128, 6993/1280, 7371/400, 1701/100$. The remaining $-(35/16)(d/m^2)v^2$ cost has d through d^5 coefficients $4725/65536, 2835/2048, 5103/512, 5103/160, 15309/400$. Using $d^5 \le d^4/5$, the full normalized deficit is at least

$$\begin{aligned} & 109/50 - (19/25)x - (41/4)x^2 - (42/5)x^3 + (433/10)x^4 \\ & -(113/4)x^5 + (67/4)x^6 - (261/10)x^7 + 44x^8. \end{aligned}$$

Absorb the fifth and seventh powers using $x \le 9/20$ to get $Q(x) = 109/50 - (19/25)x - (41/4)x^2 - (42/5)x^3 + (61/2)x^4$. Its derivative is at most $-19/25 - (41/2)x + (297/10)x^2$; this convex quadratic is negative at both endpoints of $[0, 9/20]$. Thus Q decreases and $Q(9/20) = 79237/320000 > 6/25$. Since $(6/25)(31/40)^4 > 1/12$, the entire low range gives

$$\mathbb{E}[f_2\{T(h_2 f_b)\}^4] \le 1 - d/12 \quad (b \ge 6, d \le 1/5). \tag{L73.3.low}$$

For the high range the same actual terminal mean, supplied by the finite Q4-C128, Section 8, calculation retained in Section I.3, is

$$\mathbb{E}(h_2 f_b) \le 1 - L(d), \qquad L(d) = d(2871 - 1463d - 608d^2)/1024 \quad (b \ge 6). \tag{L73.3.5mean}$$

L increases on $[0,1/3]$ and $L(1/5) = 15963/32000 > 99/200$. For $d \geq 1/5$, $TF_b \leq 101/100$ and $\mathbb{E}[f_2(TF_b)^4] \leq (4/5)(101/100)^4 < 9/10 < 1 - d/12$. The low and high ranges overlap at $1/5$, with the same actual orientation for every finite b. Equal-pair decorations and full reversal follow by positivity; $d = 0$ by original leaf integration.

N.10 The genuine-f_2 reference for every short-two connector

We use the finite-frequency proof of Sections 1–7. The actual terminal is always $F_b = h_2 f_b$, $b \geq 3$, and every connector edge points from it toward the receiving root. The claim, with decorations restored, is

$$\mathbb{E}[f_2\{T^L(E_t h_2 f_b)\}^4] \leq 1 - d/10000 \quad (b \geq 3, L \geq 1). \tag{L77.0.4}$$

The proof below treats the $L = 2, 3$ rows; the exact unit and $L \geq 4$ rows are the finite square and propagation proofs of Q5-C069 and Q5-C073. All use the same original directions.

Put $B = -K$, $u = -g$, and use the original spectral measure ν, whose mass is d and support is $[0, \theta]$. Suppose an actual nonnegative terminal has $F \leq F_0 = (m + R/2)^2$, $m > 0$, $k = 1/m$. Write

$$V = \mathbb{E}R^2, \quad D_j = \mathbb{E}B^j R, \quad a_1 = \mathbb{E}KR, \quad \rho_0 = -\mathbb{E}(rR),$$
$$\delta = kD_0 + k^2 V/2, \quad a = ka_1, \quad \varepsilon = \delta - a, \quad \nu_0 = 1 + \varepsilon.$$

For $R = (A - B_0 z)(B)u$, $z = i\sqrt{\xi}$, exact spectral integration gives

$$\begin{aligned} V &= \int (A^2 + \xi B_0^2), \quad D_0 = \int B_0, \quad a_1 = \int A, \quad \rho_0 = \int \xi B_0, \\ D_1 &= -a_1,\ D_2 = -\rho_0,\ D_3 = \int \xi A,\ D_4 = \int \xi^2 B_0,\ D_5 = -\int \xi^2 A. \end{aligned} \tag{L77.1.4}$$

Every unspecified integral here uses the same original ν. Define finite polynomials $P_n = G_n + kz^n(A - B_0 z)$, $Q_n = P_n + \delta G_{n-1} - aG_{n-2}$ and $Z_n^* = \nu_0 + Q_n(B)u$, using only nonnegative indices. The bound $T(R^2) \leq 2V$ and finite Duhamel identity give

$$T^L F_0/m^2 \leq \bar{f}_L + kB^L R + \delta \bar{f}_{L-1} - a\bar{f}_{L-2} + k\sum_{j=2}^{L-1} D_j \bar{f}_{L-1-j}. \tag{L77.1.6}$$

Thus the nonnegative actual upper bounds for $L = 2, 3$ are respectively $Z = Z_2^*$ and $Z = Z_3^* - \ell_0$, $\ell_0 = d + k\rho_0$. If $\alpha_n = \nu_0 - \mathbb{E}Z_n^*$, then

$$\alpha_2 = d + k\rho_0, \quad \alpha_3 = d + \delta d - kD_3, \quad \alpha_4 = \tau + \varepsilon d - kD_4,$$
$$\alpha_5 = \tau + \varepsilon d - \delta\delta_2 + k\int \xi^2 A.$$

The exact centered data, writing $C_Z = \mathbb{E}[f_2(Z - \mathbb{E}Z)]$, are

$$\begin{aligned} L = 2: \quad \mu_Z &= 1 - d + k\int [(1-\xi)B_0 - A] + k^2 V/2, \\ C_Z &= -d + \delta_2 + k(D_3 + D_4) - \delta d - d^2 - kd\rho_0, \\ \operatorname{Var} Z &= \int [(1 + \delta - k\xi A)^2 + \xi(1 + k\xi B_0)^2] - (d + k\rho_0)^2; \end{aligned} \tag{L77.1.8}$$

$$L=3:\quad \mu_Z=(1+\varepsilon-d-k\rho_0)-\alpha_3,$$
$$C_Z=-d+2\delta_2-\varepsilon d+\delta\delta_2+k\int\xi^2(B_0-A)-d\alpha_3, \tag{L77.1.9}$$
$$\operatorname{Var} Z=\|Q_3(B)u\|_2^2-\alpha_3^2.$$

The actual row forms are $Z_2^*=(1+\delta)q-a+BP_1(B)u$ and $Z_3^*-\ell_0=\nu_0 q-\ell_0+BQ_2(B)u$. Use the original centered fourth identity

$$\mathbb{E}[f_2Z^4]=(1-d)\mu_Z^4+4\mu_Z^3C_Z+\mathbb{E}[f_2(Z-\mu_Z)^2\{(Z+\mu_Z)^2+2\mu_Z^2\}].$$

When $C_Z\le c\le 0$ and $\mu_Z\le v_0$, the mean polynomial $t^3[(1-d)t+4c]$ has its maximum on $[0,v_0]$ at an endpoint. We explicitly pay both endpoints, or prove the nonzero one nonnegative.

For $(2,3), L=2$, the actual square is $m=1-d/2$, $R=s=K^2g$. Put $a=k\delta_2$, $\delta=k^2\delta_3/2$, $\eta=a-\delta$. The exact data simplify to

$$Z=1-\eta+(1+\delta+B-kB^4)u,\quad \mu_Z=1-d-\eta,$$
$$C_Z=-d+\delta_2+k\delta_3-\delta d-d^2, \tag{L77.2.1}$$
$$\operatorname{Var} Z=(1+\delta)^2d+\delta_2-2k(1+\delta)\delta_3+k^2\delta_5-d^2.$$

$\delta/a\le 39/160$ implies $\delta q-a\le 0$. The actual row tail has norm $\|1-kB^3\|\le 21/20$, since $(36/25)\theta^3<41/400$. Hence $Z\le q+(21/20)\sqrt{d}$, $\mu_Z\le 1-d$. For $x=\sqrt{d}$, $y=q+1$,

$$(Z+\mu_Z)^2+2\mu_Z^2\le y^2+2+(21/10)yx-[2y+759/400]x^2.$$

Here the negative cubic was omitted and $3x^4\le x^2$ used. The exact base identity and auxiliary bound are $9-(2-q^2/2)(y^2+2)=(q-1)^2(q^2+4q+6)/2\ge 0$, $(2-q^2/2)y\le 25/8$; the latter follows from $q^3\ge 3q-2$. Completing the square adds at most $1225/936$ because $y/(2y+759/400)\le 400/1053$. Thus the full reference coefficient is at most $31/3$. On $d\le 1/6$, put $u_2=\delta_2/d$, $v_3=\delta_3/d$. Then $k\le 12/11$, $\eta\ge(7/9)du_2$ since $\eta/(kdu_2)\ge 137/176>7/9$, and

$$\mu_Z\le 1-\sigma d,\quad \sigma=1+7u_2/9,\quad C_Z\le dc-d^2,\quad c=-1+u_2+(12/11)v_3\le -2/5,$$
$$\operatorname{Var} Z\le d(1+u_2-7v_3/4-d).$$

The variance uses $2k-k^2(d+\theta^2)\ge 7/4$; after multiplication by m^2, its decreasing difference has value $287/9216>0$ at $d=1/6$. The covariance comparison uses $\theta+(12/11)\theta^2=1651/2816<3/5$. With coefficient $C=31/3$, the nonzero endpoint polynomial has linear coefficient at most $4/3+(101/9)u_2-(1811/132)u_2^2\le 11/3$ (square-completion increment $112211/48897<7/3$), quadratic at most $23/3+(88/9)u_2-(5582/297)u_2^2-(112/11)u_2^3\le 9$ (increment $85184/66984<4/3$), nonpositive cubic $\sigma[-4\sigma^2-6\sigma+12c\sigma+12]$, quartic at most $\sigma^2(\sigma^2+8\sigma-12)\le 64/81<1$, and quintic $\sigma^3(4-\sigma)\le 64/9<8$. Thus it is at most $1+(11/3)d+9d^2+d^4+8d^5$. Compare with the positive coefficients $1,4,9,15,165/8,99/4$ of $(1-d/2)^{-8}$ to get difference at least $d/3$. For the zero endpoint, $v_3\ge u_2^2$ gives $C\operatorname{Var} Z\le(248/21)d$. The inverse difference is at least $1-(164/21)d+9d^2+15d^3+(165/8)d^4$, decreasing on $[0,1/6]$ since its derivative is at most $-164/21+3+5/4+55/144<0$. At $1/6$ it is $-13/252+5/72+55/3456>1/30\ge d/5$. Both endpoints are paid. For $1/6\le d\le 1/3$, the exact formulas give $\mu_Z\le 1-d$, $C_Z\le -3d/8-d^2$, $\operatorname{Var} Z\le(33/28)d-d^2$. Use $\kappa-(6/5)\theta^2>3/8$ and combine the actual delta terms to get $d+\delta_2-d^2-(7/5)\delta_3$; $1+u_2-7u_2^2/5\le 33/28$. The nonzero

endpoint plus variance expands to $1+(159/28)d+d^2/6-(5/2)d^3-(11/2)d^4+3d^5$. Its inverse difference is at least $d[-47/28+(53/6)d+(35/2)d^2]\ge(47/168)d$. The zero endpoint difference is at least $1-(229/28)d+(58/3)d^2\ge 24565/181888$. Since $m^8>(2/5)$ in the low interval and $m^8>1/5$ globally, these bounds prove

$$\mathbb{E}[f_2\{T^2(h_2f_3)\}^4]\le 1-d/20. \tag{L77.2.8}$$

For $(2,4)$ on $d\le 1/4$, use the actual square of Module Q5-C069: $m=1-d$, $R=s+t_0-dg$, $A=d+\xi$, $B_0=\xi$, $V=d^3+2d\delta_2+\delta_3+\delta_4\le d(1/2+d)^2$. Then $\delta=k\delta_2+k^2V/2$, $\varepsilon=-kd^2+k^2V/2$, and $|\varepsilon|\le d/6$, $\delta\le 25d/24$. For the upper epsilon bound multiply by m^2 to obtain $(1-4d)(1+8d)/24\ge 0$; for its lower bound use $V\ge 3d^3$ and $z=d/m\le 1/3$ so $\varepsilon/d\ge -z+3z^2/2\ge -1/6$. The delta endpoint bound increases to $25/24$. At $L=2$, (L77.1.8) gives

$$\mu_Z=1-d-kd^2-k\delta_3+k^2V/2\le 1-d,\quad C_Z=-d+\delta_2+\delta_3+k\delta_4-d^2-k^2dV/2\le -d/3-d^2.$$

The mean comparison uses $V-2m(d^2+\delta_3)\le d^2(3d-2+2\theta)+(2d-1+\theta)\delta_3\le 0$; the covariance uses $\theta+(37/24)\theta^2=16237/24576<2/3$. The P_2 squared bound is $1+\xi-2kd\xi-(2/3)\xi^2$. Its coefficient follows from $2k\kappa-k^2[(d+\theta)^2+\theta^3]\ge 2/3$; after multiplication by m^2, the decreasing quadratic difference has endpoint $587/32768>0$. The added real delta costs at most $2\delta+\delta^2$ and actual mean-square subtraction gives $\operatorname{Var} Z\le d+\delta_2-(2/3)\delta_3-d^2+k^2dV+\delta^2d\le(13/10)d+(11/10)d^3$. Use $k^2dV\le d^2$, $\delta^2d\le 625d^3/576$, $1+u_2-2u_2^2/3\le 1991/1536<13/10$. The original P_1 row norm is below $27/20$: its square is at most $1+2k\theta^2+k^2\theta[(d+\theta)^2+\theta^3]<1+4/9+13/36=65/36<(27/20)^2$. Therefore

$$Z+\mu_Z\le q+1+(27/20)\sqrt{d}-5d/6+\delta(q-1).$$

For $d\le 1/8$, if $q\le 1$ omit the nonpositive last term to get $Z+\mu_Z\le q+7/5$, $f_2\le 19/8-q$. The resulting cubic has $Q''\le -17/20$, $Q(4/5)=10773/1000$, $Q'(4/5)=9/100$, so its strong-concavity bound is below eleven. If $q\ge 1$ use $\delta\le 25d/24$ and complete the square to get $Z+\mu_Z\le(8/7)q+5/4$: the actual constant is at most $1243/1000<5/4$ and coefficient at most $217/192<8/7$. The cubic $(19/8-q)[((8/7)q+5/4)^2+2]$ decreases after one, with derivative $-161/784$, negative second derivative, and value $66627/6272<11$ at one. For the whole $d\le 1/4$, a coefficient thirteen suffices. On $q\le 1$, use $Z+\mu_Z\le q+22/15$; the cubic $(5/2-q)[(q+22/15)^2+2]$ has $Q''\le -13/15$, value below $25/2$ at $9/10$, derivative $-1/36$, so bound below thirteen. On $q\ge 1$, use $(19/15)q+5/4$; the cubic has $Q''\le -3$, $Q(1)=30001/2400$, $|Q'(1)|=4427/3600<5/4$, giving upper bound $30001/2400+25/96<13$. With coefficient $C=11$ or thirteen, the nonzero endpoint is

$$W_C=(1-d)^5-(4d/3+4d^2)(1-d)^3+C[(13/10)d+(11/10)d^3].$$

For $C=11$ its coefficients are $1, 239/30, 10, 101/10, -17/3, 3$, termwise below $(1-d)^{-8}$ with linear gap $1/30$. For $C=13$, the first three positive coefficients are $317/30, 10, 123/10$ and on $[1/8,1/4]$ the inverse difference is at least $d[-77/30+26d]\ge 41d/60$. The zero endpoint using the larger coefficient thirteen has inverse difference at least $1-(89/10)d+(1297/40)d^2\ge 5049/12970$. Since $m^8\ge(3/4)^8>1/10$, the low range gives deficit $d/300$. For $d\ge 1/4$, the actual mean is at most $1353/4096<1/3$, so $TF_4\le 2/3$. The direct actual bound $\mathbb{E}[f_2q^4]\le 1+4d\le 7/3$ gives target at most $112/243$.

For $(2,4), L=3$, the exact $\alpha_3 = d - k\delta_3 + k^2 dV/2 \geq 3d/4$ gives $\mu_Z \leq 1 - 7d/4$. The exact covariance cancellation is

$$C_Z = -d + 2\delta_2 - d^2 + kd^3 + k\delta_2^2 + (k^2V/2)(\delta_2 - d - d^2) \leq -3d/16.$$

The last term is nonpositive and $k(d+\theta^2) < 1$ pays the others. The original P_3 contraction gives $\|Q_3\| \leq 1 + 27d/32$ and variance at most $d + (9/8)d^2 + (3/4)d^3$. Also $\|Q_2\|^2 \leq (1+\delta)^2 + \theta < 2$, so $Z \leq (1+d/6)q - d + (10/7)\sqrt{d}$ and $Z + \mu_Z \leq q + 1 + (10/7)\sqrt{d} - (29/12)d \leq q + 1 + 3/14$. The pointwise base identity above gives curvature below eleven, since $9 + 2(3/14)(25/8) + 2(3/14)^2 < 11$. The nonzero mean endpoint is nonnegative (bracket at $1/4$ at least $15/64$). Using $(1-7d/4)^4 \leq 1 - 7d + (147/8)d^2$ and $(1-7d/4)^3 \geq 1 - 21d/4$ bounds the target before m^8 by $1 + (9/4)d + (667/16)d^2$. The inverse difference is at least $d[23/4 - (91/16)d + 120d^2] > 5d$, giving low loss $d/2$. For the high range $X_{4,3} \leq (2/3)\bar{f}_2$. The original pointwise inequality

$$f_2 \bar{f}_2 \leq 4p \tag{L77.3.7}$$

follows from $r \leq pq/2$, since the left side is at most $pq(1+q/2)(1+p/2)$ and $q(1+q/2)(2-q/2)$ increases on $[0,2]$ to four. The original reversed path cube gives $\mathbb{E}[p\bar{f}_2^3] \leq 1$, so $\mathbb{E}[f_2\bar{f}_2^4] \leq 4$ and the high target is at most $64/81$. In particular the full $(2,4), L=3$ row has deficit $d/100$.

For every finite $b \geq 5$ on $d \leq 1/5$, take $m = 1 - 9d/8$. For $b=5$, retain $C_1 = d$, $C_2 = d/4 + \delta_2$, $C_3 = 0$; $C_2 \leq 21d/32 \leq \beta_3$ and $\beta_5 + C_1 + C_2 = 9d/4$. For $b \geq 6$, use (L73.3.retained). With $C = C_1 + C_2 + C_3$, $D_* = C_2 + C_3$, $E = C_3$ and $R = \sum_{j=3}^{b} K^j \mathbf{1} - Cg - D_* r - Es$, the actual positive convolution subtracts $\beta_b, C_1 f_1, C_2 f_2, Ef_3$. Its exact constant loss is $\beta_b + C_1 + C_2 + (1-d)E = 9d/4$, so $0 \leq h_2 + f_b \leq 2m + R$. For $b=5$, $A = d + C_2 + \xi - \xi^2$, $B_0 = C_2 + \xi$. For $b \geq 6$, $A = C + \xi(a_f - E)$, $B_0 = D_* + \xi b_f$, $G_{b-2}(i\sqrt{\xi}) = a_f + i\sqrt{\xi}\, b_f$. For even $N = 2j \geq 4$, $a_f = b_f = \sum_{i=0}^{j-1}(-\xi)^i$; for odd $N = 2j+1 \geq 5$, these sums end at j and $j-1$. Thus $1 - \xi \leq a_f, b_f \leq 1 - \xi + \xi^2$ and $a_f - (1-\xi)b_f = [\xi + (-\xi)^j]/(1+\xi) \geq 0$ in the odd case. Every finite case, including $N = 4, 5$, therefore satisfies

$$A, B_0 \geq 0, \qquad A - (1-\xi)B_0 \geq C_1 + C_2\xi, \qquad C_1 \geq \tau. \tag{L77.4.1b}$$

For $b=5$ the latter difference is exactly $d + C_2\xi$.

We use the full-frequency proof of

$$\sqrt{A^2 + \xi B_0^2} \leq u_0 = 3/8 + 9d/5, \qquad B_0 \leq z_0 = \theta + 13d/16. \tag{L77.4.1}$$

For $b = 5$, hold the actual δ_2 fixed while differentiating: $A' = 1 - 2\xi \geq 3/16$, $B_0' = 1$, so take $\xi = \theta$ first and only then use $\delta_2 \leq \theta d$. The endpoint vector is bounded componentwise by $(\theta - \theta^2, \sqrt{\theta}\,\theta) + d(53/32, \sqrt{\theta}\,21/32)$. Their squares are $131313/1048576 < 9/64$ and $95621/32768 < (9/5)^2$, the last difference $263683/819200 > 0$; also $B_0 \leq \theta + (21/32)d \leq z_0$. For $b \geq 6$, exact retained constants give $C = 9d/4 - \beta_b + dE$, $D_* = 9d/4 - \beta_b - \beta_{b-1} + dE$. Put $u(\xi) = \xi - \xi^2 + \xi^3$. The true frequency upper bounds, with original global constants d, δ_2, E held fixed, are

$$\overline{A} = u(\xi) + 5d/4 + \delta_2 + E(d - \xi), \qquad \overline{B} = u(\xi) + d/4 + 2\delta_2 + dE.$$

$u' = 2/3 + 3(\xi - 1/3)^2$, so $\overline{A}' \geq 2/3 - E \geq 7/15 > 0$, $\overline{B}' \geq 2/3$. They start nonnegative; thus the squared vector increases to $\xi = \theta$. Only at this endpoint use $\delta_2 \leq \theta d$, giving $A_* = u +$

$(5/4+\theta)d+E(d-\theta)$, $B_*=u+(1/4+2\theta)d+dE$, $u=10101/32768<5/16$. These remain nonnegative: the minimum A_* on $E\in[0,d]$ is $u+5d/4+d^2$. The square $A_*^2+\theta B_*^2$ is convex in E, with quadratic coefficient $(d-\theta)^2+\theta d^2\ge 0$. At $E=0$ its vector is $u(1,\sqrt{\theta})+d(53/32,\sqrt{\theta}\,17/16)$. The constant norm is below $3/8$ using $1125/8192<1152/8192$; the d-square is $26229/8192<(9/5)^2$, difference $7827/204800$. At $E=d$ the vector is $u(1,\sqrt{\theta})+d(5/4,\sqrt{\theta}\,17/16)+d^2(1,\sqrt{\theta})$. The d-square $16557/8192<(10/7)^2$, difference $7907/401408$; the d^2 norm is below $6/5$, square difference $27/800$. Thus its norm is at most $3/8+(10/7)d+(6/5)d^2<3/8+(9/5)d$; the coefficient difference at $d=1/5$ is $23/175$. Finally the independent B_0 endpoint gives $B_0\le 5/16+(17/16)d+d^2=:z$. $z-z_0=-3/32+d/4+d^2$ increases to $-3/800<0$ on the low interval. This proves (L77.4.1) without ever replacing the global δ_2 by a local frequency multiple ξd.

These inputs imply $|\varepsilon|\le 3d/5$, $\delta\le 6d/5$, $ku_0\le 1$. The delta comparison, multiplied by $2m^2$, is the decreasing quadratic $463/320-(955/128)d+(2601/1600)d^2$, whose endpoint is $1577/80000>0$. For upper epsilon use $a_1\ge\kappa D_0+\tau d$ from (L77.4.1b); the decreasing difference is $1867/2560-(64539/20480)d-(118503/51200)d^2$, endpoint $16519/2560000>0$. For lower epsilon the actual finite mean difference is $a_1-D_0\le d^2+\theta^3 d$, giving $-\varepsilon/d\le k(d+\theta^3)<3/5$. For $b=5$ that difference is $d^2-\delta_3$; for $b\ge 6$ use $A-B_0=C_1-E\xi+\xi(a_f-b_f)$ and its first possible positive terminal power ξ^3. The original row tail has $\|P_1\|^2\le 1+2k\theta z_0+k^2\theta u_0^2<1+3/5+3/8<2<(10/7)^2$; the two endpoint terms are $5915/9920$ and $280917/768800$. The actual real-shifted P_2 satisfies

$$|P_2+\delta|^2\le(1+\delta)^2+\xi. \tag{L77.4.3}$$

For its unshifted square it suffices that $2-4\xi-k\xi[u_0(1-\xi)+\xi z_0]\ge 0$. The subtracted part increases in ξ,d, and its endpoint gap is $79745/3174400>0$; the real shift adds the favorable $-2\delta k\xi A$. P_3 has coordinates $1-\xi-k\xi^2B_0$, $1-k\xi A$, nonnegative and bounded by G_3 coordinates, hence $\|P_3\|\le 1$.

For $L=2$, the exact mean obeys $\mu_Z\le 1-d-k\kappa d^2+k^2du_0^2/2\le 1-7d/10$. The last decreasing difference after multiplication by m^2 is $147/640-(121/160)d-(12213/6400)d^2$, endpoint $337/160000>0$. To retain covariance cancellations, use the genuine coarse frequency bounds $A\le\xi+(15/8)d$, $B_0\le\xi+(101/80)d$, $B_0\ge\xi(1-\xi)$. For $b\ge 6$ these follow from $C\le(5/4+\theta+d)d\le 297d/160<15d/8$, $D_*\le(1/4+2\theta+d)d\le 101d/80$ and the finite coordinates; $b=5$ has stronger constants $53/32, 21/32$. Thus $\delta d+kd\rho_0\ge kd(\delta_2-\delta_4)$, and

$$C_Z\le -d+\delta_2+k(\delta_3+\delta_4)-d^2+kd[(7/8)\delta_2+(101/80)\delta_3+\delta_4]\le -d/3+d^2/5.$$

The bracket coefficient is at most $103363/163840$ times d, whose product with k is below $5/6$; $k-1\le 45d/31$, $(45/31)(\theta^2+\theta^3)<7/20$ give the intermediate bound $-(11851/32768)d+(11/60)d^2$. Equation (L77.4.3), with actual mean-square subtraction, gives Var $Z\le(45/32)d+(7/5)d^2+(36/25)d^3$. For the continuous curvature put $x=\sqrt{d}<9/20$, $y=q+1$. The row bound is $Z+\mu_Z\le y+(10/7)x+e(q)d$, where $e=-1/10$ for $q\le 1$ and $e=(6/5)q-13/10$ for $q\ge 1$. With $f_2\le p+x$, the base $p(y^2+2)=6+q-q^3\le 32/5$, the linear-x coefficient is $61/7+(34/7)q-(13/7)q^2\le 1082/91<12$. For $q\le 1$, discard the negative e; later costs total at most $425d/49<9d$. For $q=1+u$, $0\le u\le 1$, use $pye=-1/5+(5/2)u-(11/10)u^2-(6/5)u^3\le 15/22$, by $u^3\ge 3u/4-1/4$ and square completion. Also $pe\le 121/480$, $e\le 11/10$, $y\le 3$. The complete d,x^3,d^2,x^5 coefficients are at most $765/77<10$, $121/168+100/49+33/5<19/2$, $3047/700<9/2$, $121/100$. Using $x^3\le 9d/20$, $d^2\le d/5$, $(121/100)x^5<d/8$ gives total cost below $153d/10<31d/2$. Finally $x\le 2/15+(15/8)d$ by square completion, proving the full coefficient $8+38d$. Take $v_0=1-7d/10$, $c=-d/3+d^2/5$.

The upper mean bracket is $1-(91/30)d+(3/2)d^2$, at least $34/75$. After the variance payment the exact polynomial coefficients are $1, 367/60, 29591/400, 7096/125$, then a quartic below 59 and a negative quintic. The mean and variance components giving these are

degree	mean term	variance term
1	$-77/15$	$45/4$
2	$467/50$	$56/5+855/16$
3	$-994/125$	$1618/25$
4	$97363/30000$	$1368/25$
5	$-1029/2000$	$0.$

The inverse positive series yields difference at least $d[173/60-(5683/200)d+114d^2]\ge d$; the square-completion integer difference is $123330660>0$. Its quartic inverse coefficient exceeds 500, and all higher terms are nonnegative. Since $m\ge 31/40$, $m^8>1/8$, the complete low $L=2$ row has deficit $d/8$.

For $b\ge 5, L=3$ on the same low interval, $\alpha_3\ge(1-k\theta u_0)d\ge\kappa d$ and $\mu_Z\le 1-(7/10+\kappa)d\le 1-5d/4$. The actual finite tail retains $\int\xi^2(B_0-A)\le\delta_5$. For $b=5$, $B_0-A=-d+\xi^2$. For $b\ge 6$, $B_0-A=-C_1+E\xi-\xi(a_f-b_f)$; the first terms are at most $-(\kappa-\theta)d\le 0$. Even N has no last difference; for odd $N=2j+1\ge 5$, even j is favorable, while odd $j\ge 3$ gives after multiplication at most $\xi^6\le\xi^4$. This covers every finite b. Thus $C_Z\le[-3/16+k\theta^4]d+[3/5+(6/5)\theta-\kappa]d^2\le -211d/4000<-d/20$, using $k\theta^4\le 9/250$ and quadratic coefficient $79/160$. $\|Q_3\|\le 1+69d/50\le 1+7d/5$ gives variance at most $d+(5/2)d^2+2d^3$ after actual α_3^2 subtraction. $\|Q_2\|^2\le 38877/20000<49/25$ gives $Z\le(1+3d/5)q-d+(7/5)\sqrt{d}$ and $Z+\mu_Z\le q+1+(7/5)\sqrt{d}-(21/20)d\le q+22/15$. The cubic $(49/20-q)[(q+22/15)^2+2]$ has $Q''\le -29/30$, $Q(9/10)=212071/18000<59/5$ and $|Q'(9/10)|=119/450<3/10$, so curvature is below twelve. The upper mean bracket is at least $14/25$. The exact Taylor bound is at most $1+(29/5)d+(361/8)d^2+(117/8)d^3$. The inverse first three coefficients $9, 729/16, 10935/64$ are larger, with linear gap $16/5$; $m^8>1/8$ gives deficit $2d/5$.

For $1/5\le d\le 1/4$, keep the original direction but use the other actual C128 square with $m=1-d$. For $b=5$ retain $C_1=d, C_2=\delta_2, C_3=0$; for $b\ge 6$ use the finite coefficient formula above with numerator constant $2d$. The precise Q4-C128, Sections 2,4,6,7, inputs, retained in Section I.3, are

$$\begin{gathered}|\varepsilon|\le(315/512)d<2d/3,\quad \delta\le(61/48)d<4d/3,\quad \mu_{Z_2}\le 1-2d/3,\\ \|P_1\|\le 10/7,\quad |P_2+\delta|^2\le(1+\delta)^2+\xi,\quad \|P_3\|\le 1,\quad \alpha_3\ge\kappa d.\end{gathered}\tag{L77.6.1}$$

The actual coordinates remain nonnegative, $A\ge(1-\xi)B_0$, $B_0\ge\xi(1-\xi)$, and $A\le\xi+(53/32)d$, $B_0\le\xi+(17/16)d$. The same covariance cancellation uses $(21/32)\theta+(17/16)\theta^2+\theta^3=16679/32768$, whose product with k is below $7/10$; $k-1\le 4d/3$, $(4/3)(\theta^2+\theta^3)<1/3$. Hence $C_{Z_2}\le -d/3+d^2/5$ and variance at most $(45/32)d+(5/3)d^2+(16/9)d^3$. Here $\mu\le 13/15$ and the row bound splits at $q=1$: $Z+\mu\le q+12/7$ when $q\le 1$, $Z+\mu\le(4/3)q+29/21$ when $q\ge 1$. The first cubic $(5/2-q)[(q+12/7)^2+338/225]$ has $Q''\le -13/7$, $Q(1)<27/2$, $|Q'(1)|<1$. The second, replacing the square by $((4/3)q+29/21)^2$, has $Q''\le -64/7<-9$, $Q(6/5)<68/5$, $|Q'(6/5)|<1$. Both give curvature below fourteen over their full intervals. With $v_0=1-2d/3$, $c=-d/3+d^2/5$, the upper mean bracket is $1-3d+(22/15)d^2>0$. The full polynomial coefficients are $1, 235/16, 482/15, 2384/135, 128/45, -176/405$. The inverse difference is at least $d[-107/16+3d+102d^2+327d^3]\ge d/2$, since its bracket increases from $1217/2000>1/2$ at

$1/5$. $m^8 > 1/10$ gives middle $L = 2$ deficit $d/20$. For middle $L = 3$, $\mu_Z \le 1 - 5d/4$ and the same finite-tail calculation gives $C_Z \le [-3/16 + k\theta^4]d + (413/768)d^2 \le -(239/15360)d < 0$, using $k\theta^4 \le 3/80$. The Q_3 cost is $315/512 + (13/20)(61/48) = 11069/7680 < 3/2$, so variance is at most $d + (8/3)d^2 + (9/4)d^3$. $\|Q_2\|^2 \le 78985/36864 < 9/4$ yields $Z \le (1+2d/3)q - d + (3/2)\sqrt{d}$ and $Z + \mu \le q + 1 + (3/2)\sqrt{d} - (11/12)d \le q + 31/20$. $\mu \le 3/4$, $f_2 \le 5/2 - q$; the cubic has $Q'' \le -6/5$, $Q(1) = 9153/800 < 23/2$, $Q'(1) = 9/400$, so curvature is below twelve. Discarding only the nonpositive covariance gives $\mathbb{E}[f_2 Z^4] \le 1 + 6d + (371/8)d^2 + (141/8)d^3$. Its inverse difference is at least $d[2 - (83/8)d + (819/8)d^2] > (3/2)d$, since the square-completion loss $6889/26208 < 1/2$. The middle $L = 3$ deficit exceeds $3d/20$.

For $d \ge 1/4$ use actual terminal means $t_j = \mathbb{E}[h_2 f_j]$:

$$\begin{gathered} t_0 = 1 + d,\ t_1 = 1,\ t_2 = 1 - d - \delta_2,\ t_3 = 1 - 2d - d^2, \\ t_4 = 1 - 3d + \delta_2 - d^2 + \delta_3, \quad t_5 = 1 - 4d + 2(1+d)\delta_2, \\ t_6 = 1 - 5d + 2d^2 + d^3 + (3+2d)\delta_2 - \delta_3 - \delta_4. \end{gathered} \tag{L77.7.1}$$

For every finite $b \ge 7$, $\mathbb{E}[h_2 S_b] \le 1 - \tau + \delta_3$; retain actual $t_0, \dots, t_4$ in positive convolution and use $t_4 \ge 1 - 3d$. This gives $t_b \le 1 - \tau(6 - 5d - d^2 - \delta_2) + \delta_3 \le 1 - \mathcal{L}(d)$, $\mathcal{L}(d) = d(3479 - 3287d - 608d^2)/1024$. All retained beta indices are at least three, all actual terms nonnegative, and $\mathcal{L}' \ge 1085/1024 > 0$ on $[0, 1/3]$. The upper bounds for t_4, t_5, t_6 also decrease and at $1/4$ are at most $1353/4096, 65/256, 63/256 < 1/3$. For $b \ge 7$, the bound is $t_b \le 5907/16384 < 13/36$ there. Thus $TF_b \le 13/18$ for every $b \ge 4$. At $d \ge 4/15$, $\mathcal{L}(4/15) = 2303308/3456000 > 13/20$ improves it to $TF_b \le 7/10$. For $L = 2$, use $X_{b,2} \le (13/18)q$ and $\mathbb{E}[f_2 q^4] \le 7/3$ to get target at most $199927/314928 < 99/100$. For $L = 3$ on $1/4 \le d \le 4/15$, the actual bare-two bound is $\mathbb{E}[f_2 \bar{f}_2^4] \le 7/2$. Its independent proof uses mean $1 - d$, covariance $-d + \delta_2 - d^2 \le -\kappa d - d^2$, variance at most $45d/32 - d^2$, and curvature $651/64 < 11$ from the same original square completion as the $(2,3)$ calculation above. The actual mean bracket $1 - (35/8)d - 3d^2$ is negative on this range, so the centered expansion gives target at most $11(45d/32 - d^2) \le 6017/1800 < 7/2$. Hence the $L = 3$ target is at most $(13/18)^4(7/2) = 199927/209952 < 99/100$. For $d \ge 4/15$ instead use (L77.3.7) and the original path cube to get $4(7/10)^4 = 2401/2500 < 99/100$. All high bounds are stronger than the stated linear losses.

The complete exact same-direction union is

terminal/connector	proof	deficit
$b = 3, L = 1$	Q5-C073, Section 2	$d/125$
$b = 4, L = 1$	Q5-C069, Section 10	$d/100$
$b = 5, L = 1$	Q5-C069, Section 11	$d/10000$
$b \ge 6, L = 1$	Q5-C073, Section 3	$d/12$
$b \ge 3, L = 2$	the calculations above	$d/300$
$b = 3, L = 3$	M.20	$d/4$
$b \ge 4, L = 3$	the calculations above	$d/100$
$b \ge 3, L \ge 4$	M.20	$d/50$.

(L77.7.table)

Together with the exact unit rows in Sections N.8 and N.9 and the remaining rows in Section N.26, this proves (L77.0.4).

At $d = 0$ all original messages are one.

N.11 Short-terminal fourth moments with an arbitrary receiving arm

All operators, expectations, and paths use the original finite probability space of Lemma M.6. Let $a \geq 3$ be the receiving-arm length, and put $h_2 = Tp = q - Kg$. The receiving arm and the terminal b-arm point outward; every connector points from terminal to receiver. For h_2, the internal vertex of the two-edge arm points to the terminal and to its leaf. These directions depend only on the arm lengths. Let E_o, E_t denote arbitrary actual equal-pair products at the receiving and terminal vertices, respectively.

Proposition N.1 (The three receiving families)**.** *For every legal original model and all the indicated integers,*

$$\mathbb{E}[E_o f_a \{T^L(E_t q f_b)\}^4] \leq 1 - d/3000, \qquad b \geq 3, \quad L \geq 1, \tag{N.5}$$

$$\mathbb{E}[E_o f_a \{T^L(E_t q f_2)\}^4] \leq 1 - 3d/800, \qquad L \geq 6, \tag{N.6}$$

$$\mathbb{E}[E_o f_a \{T^L(E_t h_2 f_b)\}^4] \leq 1 - d/400, \qquad b \geq 3, \quad L \geq 2. \tag{N.7}$$

The proof uses nine finite positive envelopes. Their actual means, variances, and signed receiving covariances feed the same fourth identity. Tables of finite parameters then cover the short integer prefixes and the unbounded connector tails. The high domains use direct positive path comparisons.

N.11.1 A finite receiving-symbol bound

The reversed positive path convolution gives

$$S_n = 1 + G_n(K)g = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad \tau \leq \beta_j \leq d \quad (j \geq 3). \tag{Q117.1.1}$$

Consequently the weak receiving upper is

$$0 \leq f_a \leq C_{a,0} = S_a - \beta_a \leq 2, \qquad \mathbb{E}C_{a,0} = 1 - \beta_a - \beta_{a+1}. \tag{Q117.1.2}$$

The upper bound also follows directly from $C_{a,0} = AS_{a-1}$, $S_{a-1} \geq 0$, and $\mathbb{E}S_{a-1} = 1 - \beta_a$, using the original kernel bound $A \leq 2$. For $a \geq 4$, the next actual positive term permits $C_{a,\beta_{a-1}} = S_a - \beta_a - \beta_{a-1}p \geq f_a$. These are the common weights of Lemma M.6. The original spectral measure ν has support $[0, \theta]$, mass d, and $\int x^j \, d\nu = \delta_{j+1}$. For $d > 0$, $d \leq v := \delta_2/d \leq \theta$; the lower bound follows from $\mathbb{E}(Kg) = -d$ and Cauchy's inequality. This constraint concerns aggregate moments, rather than each spectral frequency.

Lemma N.2 (All finite short-one receiving symbols)**.** *For $b \geq 3$, put $N = b-1 \geq 2$ and $1 \leq k \leq 6/5$. For $b = 2$, put $N = k = 1$. For every integer $L \geq 1$, let $P_L(z) = G_L(z) + kz^{L+1}G_N(-z)$. Then for every $a \geq 3$, $0 \leq x \leq \theta$, and $z = i\sqrt{x}$,*

$$\mathrm{Re}(G_a(z)P_L(z)) \geq \phi(x) := 1 - 3x + 2x^2 - x^3 \geq 1451/32768 > 1/25. \tag{N.8}$$

Proof. Put $c = 1 - x$. Write $G_a = u_a + zv_a$; finite alternating sums give $c \leq u_a, v_a \leq 1$. For $N \geq 2$, write $G_N(-z) = \alpha_0 - z\beta_0$. The same finite sums give

$$c \leq \alpha_0, \beta_0 \leq 1, \qquad 0 \leq \alpha_0 - c\beta_0 \leq x, \qquad c \leq c\alpha_0 + x\beta_0 \leq 1.$$

Indeed for $N = 2h$, both coefficients are $a_h = (1 - (-x)^h)/(1 + x)$, so their first difference is xa_h. For $N = 2h + 1$, it is $(x + (-x)^h)/(1 + x)$, lying in $[0, x]$ for every $h \geq 1$. The other inequalities are the corresponding finite alternating endpoints. For $N = 1$, use separately $\alpha_0 = 1, \beta_0 = 0$.

Write $P_L = r_L + zs_L$. For $N \ge 2$, the first two rows are $(r_1, s_1) = (1 - kx\alpha_0, kx\beta_0)$ and $(r_2, s_2) = (1 - kx^2\beta_0, 1 - kx\alpha_0)$. Both coordinates are nonnegative. For $L \ge 3$, each unperturbed coordinate is at least c, and each possible negative correction has magnitude at most kx^2. Hence $r_L, s_L \ge c - kx^2 \ge 1013/2560 > 0$. The same argument works for $N = k = 1$, allowing $s_1 = 0$. It follows that

$$\mathrm{Re}(G_a P_L) = u_a r_L - x v_a s_L \ge c r_L - x s_L = \mathrm{Re}((c + z)P_L).$$

Put $B_0 = (1-2x)/(1+x)$, $A_0 = (2-x)/(1+x)$, and $D_0 = A_0 - c = (1-x+x^2)/(1+x) > 0$. Then $D_0 + A_0 x = 1$, $kx \le 1$, and $B_0 - D_0 x^2 = \phi(x)$. Finite geometric division gives

$$\mathrm{Re}((c + z)G_{2m}) = B_0(1 - (-x)^m), \qquad \mathrm{Re}((c + z)G_{2m+1}) = B_0 + (-1)^m A_0 x^{m+1}.$$

When $N \ge 2$, the $L = 1$ value is at least $c - kx$, whose excess over ϕ is at least $x(4/5 - 2x + x^2) \ge (781/5120)x$. The $L = 2$ value is $1 - 2x + kx^2(\alpha_0 - c\beta_0) \ge 1 - 2x \ge \phi$. For even $L = 2m \ge 4$, the correction is $-k(-1)^m x^{m+1}(\alpha_0 - c\beta_0)$. Its negative part costs at most $kx^4 \le x^3$, paid by the unperturbed lower bound $c(1 - 2x)$. For odd $L = 2m + 1 \ge 3$, put $C_0 = c\alpha_0 + x\beta_0 \in [c, 1]$. The exact value is $B_0 + (-1)^m(A_0 - kC_0)x^{m+1}$. For odd m, this is at least $B_0 - D_0 x^{m+1} \ge \phi$. For even $m \ge 2$, a negative correction can occur only if $k > A_0$; then $(k - A_0)x \le D_0$, so it costs at most $D_0 x^2$.

For $N = k = 1$, the even correction is $-(-1)^m x^{m+1}$. At $m = 1$ the result is $(1 - x)^2 \ge \phi$; at $m \ge 2$ its negative part is at most x^3. The odd value, including $L = 1$, is exactly $B_0 + (-1)^m D_0 x^{m+1}$, again at least ϕ. Finally $\phi' = -3 + 4x - 3x^2 < 0$, and $\phi(\theta) = 1451/32768$. □

Q117.4 The wider coefficient-nine entrance

This entrance also applies when the receiving upper is not one of the $a \ge 3$ weights of the proposition. Suppose $C, Z \ge 0$, $u_0 = \mathbb{E}Z \ge 0$, and

$$C \le p + (6/5)\sqrt{d}, \qquad Z \le (1 - 5d/8)q - d/2 + (13/12)\sqrt{d}, \qquad u_0 \le 1 - 3d/2. \quad \text{(Q117.4.1)}$$

Then on $0 \le d \le 1/3$,

$$C[(Z + u_0)^2 + 2u_0^2] \le 9. \quad \text{(Q117.4.2)}$$

Indeed $t = \sqrt{d}$ gives exactly case w of Lemma A.2, with the three scalar uppers in (A.20).

N.11.2 The actual finite envelopes

Use the original variables and receiving weights of Lemma M.6. Every table entry has $Z = z_0 + Q(B)u$, so its exact mean is $z_0 - \alpha$, where $\alpha = -\mathbb{E}[Q(B)u]$. Its mass, variance, and signed covariance are (M.19), (M.20), and (M.21). The fourth-moment bound (M.22) is used with $s = m^2$; in particular its outside scale is m^8.

For the terminal qf_b, write

$$\mathsf{P}_{L,b}(z) = G_L(z) + kz^{L+1}G_{b-1}(-z), \qquad k = m^{-1}, \quad m = 1 - \beta_b/2 \quad (b \ge 3).$$

For $b = 2$, take $m = k = 1$. All these are finite polynomials. Write $V_L = 1 + \mathsf{P}_{L,b}(B)u$, and put

$$R_1 = KG_{b-1}(K)g, \quad \rho = \mathbb{E}[(Kg)R_1], \quad \gamma = k\beta_{b+1} - k^2\|R_1\|^2/2,$$
$$a_1 = k(d - \beta_{b+2}), \quad B_* = \gamma + a_1, \quad a_* = d - k\rho, \quad \lambda = \kappa d, \quad e_4 = 41d/80.$$

These are the precise retained coefficients of (L58.1.6) and (L58.1.7).

For the terminal $h_2 f_b$, use its actual positive square $F \le (m + R/2)^2$. Write $R = \mathcal{R}(B)u$, where on a rotation block $\mathcal{R}(z) = A(x) - zB_0(x)$, $x = -z^2$. With $D_j = \mathbb{E}B^j R$, put

$$\delta = kD_0 + k^2\|R\|^2/2, \quad \varepsilon = \delta - k\mathbb{E}(KR), \quad \nu_0 = 1 + \varepsilon,$$

$$\mathsf{U}_L = G_L + kz^L\mathcal{R}(z), \qquad \mathsf{Q}_L = \mathsf{U}_L + \varepsilon G_{L-2} + \delta z^{L-2}, \quad L \ge 2.$$

The last expression equals $\mathsf{U}_L + \delta G_{L-1} - k\mathbb{E}(KR)G_{L-2}$, by the finite identity $G_{L-1} = G_{L-2} + z^{L-2}$. Also put $\ell = d + k\rho_0$, $\rho_0 = -\mathbb{E}[(Kg)R]$.

The actual square packets have four parameter choices:

b	d range	m and R
3	$(0, 1/3]$	$m = 1 - d/2, \quad R = K^2 g$
4	$(0, 1/4]$	$m = 1 - d, \quad R = K^2 g + K^3 g - dg$
$b \ge 5$	$(0, 1/5]$	$m = 1 - 9d/8, \quad R = R_{9/4,b}$
$b \ge 5$	$[1/5, 1/4]$	$m = 1 - d, \quad R = R_{2,b}$.

To specify the last two finite vectors, let $c = 9/4$ or 2. These quotient definitions concern $d > 0$; the degenerate model is handled directly by leaf integration. For $b = 5$, take $C_1 = d$, $C_2 = (c-2)d + \delta_2$, $C_3 = 0$. For $b \ge 6$, take

$$C_1 = \beta_{b-1}, \qquad \Lambda = \frac{cd - \beta_b - \beta_{b-1}}{\beta_{b-2} + (1-d)\beta_{b-3}}, \qquad C_2 = \Lambda\beta_{b-2}, \quad C_3 = \Lambda\beta_{b-3}.$$

On the indicated domains these are retained portions of the actual positive convolution, with $0 \le \Lambda \le 1$. Put

$$C_* = C_1 + C_2 + C_3, \quad D_* = C_2 + C_3, \quad E_* = C_3,$$

$$R_{c,b} = \sum_{j=3}^{b} K^j 1 - C_* g - D_* Kg - E_* K^2 g.$$

The exact identity $\beta_b + C_1 + C_2 + (1-d)C_3 = cd$ gives $0 \le h_2 + f_b \le 2m + R_{c,b}$. The finite retained-coefficient proofs are (L73.3.retained) for $c = 9/4$ on the low domain and Section I.3 for $c = 2$; the latter has its full retained construction and its stated middle-domain analytic bounds. The finite norm and coordinate inputs are (L77.4.1b), (L77.4.1), and (L77.6.1). Only those actual analytic inputs, not their genuine-f_2 fourth-moment conclusions, are used here.

Table 26: Exact envelopes $Z = z_0 + Q(B)u$. The original comparison is $0 \le T^L F \le m^2 Z$ in every row.

Terminal and range	Scale	z_0	$Q(z)$	Receiver t
$qf_b,\ b \ge 3,\ L \ge 5$	$1 - \beta_b/2$	$1 - 2\lambda$	$\mathsf{P}_{L,b} - \lambda$	0
$qf_b,\ b \ge 3,\ L = 4$	$1 - \beta_b/2$	$1 - \lambda - e_4$	$\mathsf{P}_{4,b} - e_4$	0
$qf_b,\ b \ge 3,\ L = 3$	$1 - \beta_b/2$	$1 - a_* - B_*$	$\mathsf{P}_{3,b} - \gamma z - B_*$	t_a
$qf_b,\ b \ge 3,\ L = 2$	$1 - \beta_b/2$	$1 - \gamma - a_1$	$\mathsf{P}_{2,b} - \gamma$	0
$qf_b,\ b \ge 3,\ L = 1$	$1 - \beta_b/2$	$1 - \gamma$	$\mathsf{P}_{1,b}$	0
$qf_2,\ L \ge 6$	1	$1 - \tau(4-d)$	$\mathsf{P}_{L,2} - \tau(3 + 2z + z^2)$	0
$h_2 f_b,\ b \ge 3,\ L \ge 4$	Square packet	$\nu_0 - 2hd$	$\mathsf{Q}_L - hd$	0

Terminal and range	Scale	z_0	$Q(z)$	Receiver t
$h_2 f_b,\ b \ge 3,\ L = 2$	Square packet	ν_0	Q_2	See below
$h_2 f_b,\ b \ge 3,\ L = 3$	Square packet	$\nu_0 - \ell$	Q_3	0

Here $t_a = 0$ for $a = 3$, and $t_a = \beta_{a-1}$ for $a \ge 4$. The $h_2 f_b, L = 2$ row uses $t = t_a$ at $b = 3$ and $t = 0$ at $b \ge 4$. The square packet is the actual $b = 3$, $b = 4$, low $b \ge 5$, or middle $b \ge 5$ packet, on its stated continuous domain; changing that analytic envelope leaves F and the original direction unchanged.

The rows are algebraic forms of the finite positive propagation proofs (L58.1.7), (L77.1.6), and (L73c.4.2), with their actual finite inputs. In the first two rows the last two retained losses give respectively $Z = V_L - \lambda(1+q)$ and $Z = V_4 - \lambda - e_4 q$. The third row is $Z = V_3 - \gamma K g - a_* - B_* q$. The next two rows are $Z = V_2 - \gamma q - a_1$ and $Z = q - kKR_1 - \gamma$.

For $q f_2$, the four actual retained path losses have sum

$$\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3 = (4 - d) + (3 + 2B + B^2)u.$$

This gives its displayed scalar and polynomial, retaining the entire constant $4 - d$ before multiplying by $-\tau$. The $h_2 f_b$ tail subtracts $hd(1+q)$, with $h = 9/16 - d/2$ at $b = 3$ and $h = 1/2 - 6d/5$ for the other low long-connector packets; its two short rows retain the actual scalar ℓ at $L = 3$. Thus all means, centered variances, and receiving covariances follow from the same finite identities, with their scalar losses and signs retained.

N.11.3 The finite parameter rows

For $d > 0$, put $v = \delta_2/d$; then $d \le v \le \theta$. In Tables 27– 30, use the normalization

$$H_r = 1 - h_r d, \quad U_r = 1 - a_r d, \quad J_r = d j_r, \quad V_r = d v_r, \qquad W_r = H_r U_r^4 + 4 J_r U_r^3 + M_r V_r. \tag{N.9}$$

The preceding finite receiving identities and the proofs below give the two actual mean endpoints $M_r V_r$ and W_r. Rows marked † compare the actual receiving prefix after its own endpoint reduction. The row marked ‡ is formed after the joint covariance–variance reduction in (N.20); its displayed j_r is part of that complete endpoint polynomial.

The two-step short-one functions are

$$\begin{aligned}
A_2 &= 2 - 3v/2 + 9v^2/40, & g_2 &= A_2 - 1,\\
B_2 &= 2 - 2v + v^2, & M_2 &= 1 - 2v + 7v^2/8,\\
M_{2,f} &= 1 - 2v + 2v^2/3, & H_2 &= 5v/4 - 17v^2/20,\\
J_2 &= 1 + v - 63v^2/32 - 855v^4/1024 - v^5, & & \\
a_0 &= 1 - 6v/5 + 57v^2/80, & e_0 &= 1 - 39v/80 + 57v^3/80,\\
D_2 &= a_0^2 + 2 g_2 e_0, & \mathcal{V}_2 &= J_2 - D_2 d + g_2^2 d^2.
\end{aligned} \tag{N.10}$$

For the long $(1,2)$ row write

$$T_6 = (1 - v)[2 - 25v/32 + 2d(1 - v)], \qquad \mathcal{V}_6 = 1 - 7v/15 - d(1 - v)[6 - 9d - (4 + d)v + 2v^2].$$

Table 27: Endpoint polynomials for the short-one terminals.

Row	h_r	a_r	j_r	v_r	M_r
S1	2κ	3κ	$\kappa(1-\kappa)d$	$1-3\kappa^2 d+\kappa^2 d^2$	9
S2	2κ	$27/16$	$(41/80-\kappa^2)d$	$1-13d/8+d^2/4$	9
S3L	$7/4$	$13/8$	$-1/25+81d/40+57d^2/80$	$1-3d/2+2d^2-7d^3/5$	9
S3H	$7/4$	$3/2$	$-1/25+81d/40+57d^2/80$	$1-3d/2+2d^2-7d^3/5$	9
S4†	B_2	A_2	$-M_2+H_2 d+3d^2/5$	$\mathcal{V}_2$	9
S4F†	B_2	A_2	$-M_{2,f}+H_2 d+3d^2/5$	$\mathcal{V}_2$	9
S5	$2(1-v)$	$1-779v/960$	$-1+11v/5-57v^2/80-\kappa v(1-v)d$	$1-41v/48-\kappa^2 v^2 d$	$37/4$
S5F	$2(1-v)$	1	$-1+11v/5-v(1-v)d$	$1-7v/5-v^2 d$	10
S6	$2(1-v)$	$(1-v)(5-3d)$	$-1/10+dT_6$	$\mathcal{V}_6$	9

Table 28: Endpoint polynomials for the short-two long connectors.

Row	h_r	a_r	j_r	v_r	M_r
D4F	$19/16$	$27/16-3d/2$	$-1/10+4d/5$	$\mathcal{W}(1,\eta_3)$	10
D4B	$19/16$	$4/3-18d/5$	$3d$	$\mathcal{W}(1+d,\eta_*)$	13
D4T	$19/16$	$9/10-18d/5$	$3d$	$\mathcal{W}(1+7d/5,\eta_*)$	13

For the short-two long rows put $\mathcal{W}(N,\eta)=N^2-3\eta^2 d+\eta^2 d^2$, $\eta_3=9/16-d/2$, and $\eta_*=1/2-6d/5$.

Let $K_*(d)=124/15+38d$, and put $C_{23}=1-2v+49v^2/100$, $A_{23}=1+51v/64$. The strengthened receiving row D2F covers $a\geq 4$, using the actual $a=4$ comparison proved below. D2F3L and D2F3H are the remaining actual $a=3$ prefix.

Table 29: Endpoint polynomials for the short-two two-step connector.

Row	h_r	a_r	j_r	v_r	M_r
D2F†	$3-3v+v^2$	A_{23}	$-C_{23}+vd$	$1+v-7v^2/5-d$	10
D2F3L‡	2	A_{23}	$-1+2v+12v^2/11-v^3-d$	$1+v-7v^2/4-d$	10
D2F3H	2	1	$-1/20-d$	$33/28-d$	10
D2BL†	$19/16$	1	$-3/16+7d/32$	$13/10+11d^2/10$	K_*
D2BM†	$19/16$	1	$-3/16+7d/32$	$13/10+11d^2/10$	13
D2TL†	$19/16$	$7/10$	$-1/5+3d/4$	$45/32+7d/5+36d^2/25$	K_*
D2TM†	$19/16$	$2/3$	$-1/5+3d/5$	$45/32+5d/3+16d^2/9$	14

N.11.4 Finite data for the short-one rows

Write $G_a(z)=u_a(x)+zv_a(x)$, where $z=i\sqrt{x}$. The finite alternating coefficients satisfy $1-x\leq u_a,v_a\leq 1$ for every $a\geq 3$. The receiving-symbol lemma above gives $\mathrm{Re}(G_a \mathsf{P}_{L,b})\geq\phi(x)>1/25$ for every finite a,b,L in its stated ranges. For $a\geq 3$, positivity and the actual path convolution give

$$0\leq f_a\leq C_{a,t}\leq C_{a,0}\leq p+\frac{6}{5}\sqrt{d}-\kappa d. \tag{N.11}$$

Indeed $S_a-p=KG_{a-1}(K)g$, and $\|G_2(K)\|^2\leq 45/32<36/25$, whereas $\|G_{a-1}(K)\|\leq 1$ for $a\geq 4$. Also $\beta_a\geq\kappa d$. In particular the actual receiving weights are bounded by $5/2-q$ on $d\leq 1/3$ and by $49/20-q$ on $d\leq 1/5$. The increment $6y/5-\kappa y^2$, $y=\sqrt{d}$, is increasing on the relevant interval. At its two right endpoints use respectively $y<26/45$ and $y<9/20$ to obtain the asserted bounds.

Table 30: Endpoint polynomials for the short-two three-step connector.

Row	h_r	a_r	j_r	v_r	M_r
D3F	$2(1-v)$	$1+25v/64$	$-1+3v-v^2-v^3+d/10$	$1-6v/7-16d/25$	11
D3B	$19/16$	$7/4$	$1/10+2d$	$1+9d/8+3d^2/4$	11
D3TL	$19/16$	$5/4$	$1/10+2d$	$1+5d/2+2d^2$	12
D3TM	$19/16$	$5/4$	$1/10+2d$	$1+8d/3+9d^2/4$	12

The longer connectors S1–S3.

For S1, the actual finite propagation (L58.1.7) has nonnegative coefficients $\gamma, a_1, \varepsilon_j$, with $\varepsilon_j \ge \tau$ for $j \ge 4$. Retaining its last two terms gives the first envelope in Table 26. The finite inputs (L58.1.6) give

$$\|\mathsf{P}_{L,b}(B)\| \le 1, \qquad \alpha_L := -\mathbb{E}[\mathsf{P}_{L,b}(B)u] \ge \tau, \qquad \eta_L := \langle u, \mathsf{P}_{L,b}(B)u\rangle \ge \tau, \quad L \ge 5.$$

Thus the exact variance after subtracting λu is at most $d - 3\kappa^2 d^2 + \kappa^2 d^3$. Its mean is at most $1 - 3\kappa d$. The common covariance identity and $\int u_a \, d\nu \le d$ give

$$\mathrm{Cov}(C_{a,0}, Z) = -\int \mathrm{Re}(G_a \mathsf{P}_{L,b}) \, d\nu - \beta_{a+1}\alpha_L + \lambda \int u_a \, d\nu \le \kappa(1-\kappa)d^2.$$

These are the S1 data, including mass $1 - 2\kappa d$.

For S2 retain the actual coefficient $\varepsilon_3 \ge 41d/80$. The finite fourth-connector calculation following (L58.1.6) gives

$$\mu \le 1 - 17d/10 \le 1 - 27d/16, \qquad \mathrm{Var}\, Z \le d - 13d^2/8 + d^3/4.$$

In the variance, increasing the coefficient of q from $d/2$ to $41d/80$ decreases the bound: its derivative is $2ed - 2\eta_4 < 0$ for $e \le 41d/80$, since $\eta_4 \ge \kappa d$. The same covariance calculation gives $J = (41/80 - \kappa^2)d^2$.

For S3 the finite third-connector inputs of Section N.7 give, with $\widehat{V} = V_3 - \gamma K g$,

$$5d/8 \le B_* \le 3d/2, \qquad 1 - \mathbb{E}\widehat{V} \ge (4/5 - 6d/5)d, \qquad \tau \le \alpha_4 \le d,$$
$$\mu \le 1 - 13d/8 \quad (d \le 1/4), \qquad \mu \le 1 - 3d/2 \quad (d \le 1/3),$$
$$\mathrm{Var}\, Z \le d - 3d^2/2 + 2d^3 - 7d^4/5.$$

Here $\alpha_4 = -\mathbb{E}[\mathsf{P}_{4,b}(B)u]$. Use $t = t_a$ from Table 26. Its actual mass is $1 - 2d$ for $a = 3$, and at most $1 - 3\tau$ for $a \ge 4$, hence at most $1 - 7d/4$. Let $M_a = \int \mathrm{Re}(G_a \mathsf{P}_{3,b}) \, d\nu$, $J_a = \int x v_a \, d\nu$, and $D_a = \int u_a \, d\nu$. The covariance, with every receiving correction retained, is

$$\mathrm{Cov}(C_{a,t_a}, Z) = -M_a - \gamma J_a - \beta_{a+1}(1 - \mathbb{E}\widehat{V}) + B_* D_a + t_a \alpha_4 - t_a B_* d. \tag{N.12}$$

Use $M_a \ge d/25$, $J_a \ge 0$, $D_a, t_a \le d$, and the preceding finite bounds. Dropping the nonpositive terms gives

$$\mathrm{Cov}(C_{a,t_a}, Z) \le -d/25 + (81/40)d^2 + (57/80)d^3.$$

This proves S3L and S3H on their respective continuous domains.

The two-step terminal data S4 and S4F.

Put $V_R = \|R_1\|^2$, $\rho = \langle Kg, R_1\rangle$, $D_3 = \mathbb{E}B^3R_1$, $t_0 = \beta_{b+1}$, and $e = d - \beta_{b+2}$. The actual finite identities (L58.12.1)–(L58.12.2) are

$$\begin{aligned} &\alpha = d - k\rho, && \eta = d - kD_3, \\ &\gamma = kt_0 - k^2V_R/2, && a_1 = ke, \\ &\mu = 1 - \alpha - \gamma - a_1, && \operatorname{Var} Z = \|V_2 - 1\|^2 - \alpha^2 + \gamma^2 d - 2\gamma\eta. \end{aligned} \tag{N.13}$$

In particular $\rho = d - \beta_{b+3}$, and the finite alternating sums give

$$t_0 + e - \rho = \begin{cases} \beta_{b+1}, & b \text{ odd}, \\ \beta_{b+3}, & b \text{ even}. \end{cases}$$

Thus $1 - \mu = d + k(t_0 + e - \rho) - k^2V_R/2$. The finite geometric loss (Q81.1.5) implies the weaker $\|G_N(i\sqrt{x})\|^2 \le 1 - 9x/20$ for every $N \ge 3$. For $b \ge 4$, $V_R \le \delta_2 - 9\delta_3/20$, and $k\tau - k^2V_R/2$ increases on $1 \le k \le 6/5$, since its derivative is at least $17d/160$. Jensen's moment inequality therefore gives $\mu \le 1 - A_2 d$ and $\gamma \ge g_2 d$. For $b = 3$, use instead $t_0 + e - \rho = d$ and $V_R = \delta_2 + \delta_3$. The same derivative is positive, and $1 - \mu \ge (2 - 45v/64)d$, $\gamma \ge (1 - 45v/64)d$, which imply these same two bounds.

The finite norm proof following (L58.12.2) gives $\|V_2 - 1\|^2 \le dJ_2(v)$, with the exact J_2 in (N.10). For $b \ge 5$, the two finite alternating coefficients of G_{b-1} are at most $1 - \kappa x$; hence $\rho \le \delta_2 - \kappa\delta_3$ and $D_3 \le \delta_3 - \kappa\delta_4$. Using $\delta_3 \ge v^2 d$, $\delta_3 \le \theta v d$, and $\delta_4 \ge v^3 d$ gives $\alpha \ge a_0 d$ and $\eta \ge e_0 d$. In (N.13), the larger endpoint of the convex expression in $\gamma \in [g_2 d, 6d/5]$ is $\gamma = g_2 d$, because $[(6/5) + g_2]d - 2e_0 < 0$. Consequently

$$\operatorname{Var} Z \le d\mathcal{V}_2(v, d), \qquad b \ge 3. \tag{N.14}$$

The two finite prefixes in this last assertion use the following comparisons. At $b = 4$, put $a_4 = 1 - 6v/5 + 6v^2/5$, $e_* = 1 - 39v/80$. Then

$$a_4^2 - a_0^2 \ge 1599v^2/3200, \qquad 2g_2(e_0 - e_*) \le 57v^2/224.$$

These follow from $a_4 - a_0 = 39v^2/80$, $a_4 + a_0 \ge 41/40$, $g_2 \le 1 - 7v/5$, and $v(1 - 7v/5) \le 5/28$. At $b = 3$, put $a_* = 1 - 6v/5$, $g_* = 1 - 45v/64$. The actual comparison is $dJ_2 - (a_*^2 + 2g_*e_*)d^2 + g_*^2 d^3$. Here $g_* - g_2 \ge 7v/10$, $g_* + g_2 \le 2 - 2v$, and $2e_* - (g_* + g_2)d > 6/5$; the difference from (N.14) is paid by $(21/25)v - (1881/1120)v^2 > 0$.

The two-step receiving symbols and endpoints.

For $a \ge 4$, every finite terminal satisfies

$$\operatorname{Re}(G_a \mathsf{P}_{2,b}) \ge 1 - 2x + 2x^2/3; \quad b \ge 5 \implies \operatorname{Re}(G_a \mathsf{P}_{2,b}) \ge 1 - 2x + 7x^2/8.$$

Here is the finite-index verification. Write $G_{b-1}(-z) = \alpha_0 - z\beta_0$ and $\mathsf{P}_{2,b} = c + ze$, with $c = 1 - kx^2\beta_0$, $e = 1 - kx\alpha_0$. For $b \ge 5$, $\alpha_0 - \beta_0 \ge -x^3$, so $c - xe > 0$. At $a = 4, 5$, use $G_4 = (1 - x)(1 + z)$ and $G_5 = G_4 + x^2$, together with $kx(1 - x) \le 3/10$. For $a \ge 6$, the finite coefficients satisfy $v_a \ge (1 - x)(1 + x^2)$, $u_a - v_a \ge -x^3$. The sufficient excess divided by x^2 is at least

$$9/8 - 3x + 7x^2/10 - 3x^4/10 \ge 142669/10485760 > 0.$$

The polynomial decreases on $[0,\theta]$. For arbitrary $b \ge 3$, use instead $\alpha_0 - \beta_0 \ge -x$, $\alpha_0 \ge (1-x)\beta_0$, and $\beta_0 \ge 1-x$. At $a=4$ the payment is $kx(1-x) \le 3/10 < 1/3$. For $a \ge 6$ it reduces to

$$4/3 - 3x + x^2 - k(1-x)x + k(1-x)x^4 > 0.$$

Its minimum in k occurs at $6/5$; the resulting polynomial decreases, and at θ its first three combined terms are $-151/15360$, while its final term exceeds $1/60$. The remaining receiver $a = 5$ has the stronger bound $\mathrm{Re}(G_5 \mathsf{P}_{2,b}) \ge 1 - 2x + 3x^2/2$ for every b, from $kx < 1/2$.

Let $M_a = \int \mathrm{Re}(G_a \mathsf{P}_{2,b})\, d\nu$, $D_a = \int u_a\, d\nu \le d - \kappa\delta_2$. The exact identity is

$$\mathrm{Cov}(C_{a,0}, Z) = -M_a - \beta_{a+1}\alpha + \gamma D_a.$$

Finite alternating coefficients also give $V_R \ge c_G \delta_2$, $c_G = 16245/32768$. Keeping the same k, the last two terms divided by d^2 are at most

$$(k-1) + v[1 + k(1-\kappa)] - kv^2 - \tfrac{1}{2}k^2(V_R/d)(1-\kappa v) \le H_2(v) + 3d/5.$$

Here $k - 1 \le 3d/5$, $1 + (6/5)\theta - c_G/2 < 5/4$, and $1 - c_G\kappa/2 > 17/20$. For $a = 3$, its actual mass is $1 - 2d$, and $-d\alpha + \gamma\tau \le (k-1)d^2$, so the stronger covariance is $-(1-2v)d + 3d^3/5$.

For $b \ge 5$, the mass $H = 1 - B_2 d$ and covariance in row S4 therefore apply directly at $a = 4$ and $a \ge 6$. The finite mass losses retain $2\tau + \delta_3$: this is exact at $a = 6$; at $a = 4$ use $\delta_2 \ge \delta_3$; and at $a \ge 7$ each sum retains $\tau + \kappa\delta_3$, with $2\kappa > 1$. Compare the remaining receivers only after applying their own two mean endpoints. Their zero endpoints are unchanged. At $a = 5$, the nonzero endpoint difference from S4 is at most $v^2 dU^3(U - 5/2) \le 0$. At $a = 3$, divided by dU^3, it is

$$v[-2 + 9v/2] + dv[-1 - 8v/5 + 39v^2/20 - 9v^3/40] \le 0,$$

since $v \le \theta < 4/9$ and $(39/20)\theta < 8/5$. This proves the complete S4 endpoints for every receiving length. For $b = 3, 4$, replace M_2 by $M_{2,f}$ as in S4F. The same receiving comparisons remain valid. Its nonzero endpoint increases by $(5/6)v^2 dU^3$, while the actual inverse scale obeys

$$(1 - d/2)^{-8} - [1 - (1-v)d/2]^{-8} \ge 4vd \ge (5/6)v^2 dU^3. \tag{N.15}$$

The first inequality follows termwise from the positive inverse binomial series. Thus the S4 certificate also pays S4F.

The unit connector S5 and S5F.

For $b \ge 4$, the actual $Z = q - kKR_1 - \gamma$ satisfies

$$\mu = 1 - k(t_0 + e) + k^2 V_R/2, \qquad \mathrm{Var}\, Z = d - 2k\rho + k^2\|KR_1\|^2 - k^2 e^2.$$

By (Q81.1.5), $V_R \le \delta_2 - 7\delta_3/15$ and $\|KR_1\|^2 \le \delta_3 - 7\delta_4/15 < \delta_2/3$. The finite alternating identities give $\rho, e \ge \kappa\delta_2$ and $t_0 + e \ge d - \delta_3$. The last sum equals d for even b; its first negative odd correction is $-\delta_3$, and later ones are smaller. Both derivatives in k have the favorable sign: $t_0 + e - kV_R > 0$ and $k\|KR_1\|^2 < \rho$. At $k = 1$ these yield exactly the S5 mean and variance. For its receiving symbol write $G_{b-1}(-z) = \alpha_0 - z\beta_0$. Every finite $b \ge 4$ satisfies $(1-x)\alpha_0 + x\beta_0 \le 1 - \kappa x$. Since the two coefficients of $\mathsf{P}_{1,b}$ are nonnegative,

$$\mathrm{Re}(G_a \mathsf{P}_{1,b}) \ge 1 - 11x/5 + 57x^2/80.$$

The common covariance identity, with $\alpha_1 = ke \ge \kappa\delta_2$, gives the S5 value of J. For $b = 3$, the exact data are $R_1 = Kg + K^2 g$, $t_0 = d$, $e = \rho = \delta_2$, $V_R = \delta_2 + \delta_3$, and $\|KR_1\|^2 = \delta_3 + \delta_4$. The same derivative argument gives $\mu \le 1 - d$ and $\mathrm{Var}\, Z \le d(1 - 7v/5) - v^2 d^2$. Here $2 - \theta - \theta^2 > 7/5$, and the finite receiving symbol is $\mathrm{Re}(G_a \mathsf{P}_{1,3}) \ge 1 - 11x/5$. Together with $\alpha_1 = k\delta_2$, these prove S5F.

The four-loss tail S6.

The exact identity $qf_2 = (1 + Kg/2)^2 - (g + Kg/2)^2$, positivity of T, and $T((Kg)^2) \le 2\delta_2$ give the finite propagation

$$0 \le T^L(qf_2) \le V_L - \gamma \bar{f}_{L-1} - \sum_{j=3}^{L} \omega_j \bar{f}_{L-j}, \qquad \gamma = d - \delta_2/2, \quad \omega_j = \begin{cases} \beta_j, & j \text{ even}, \\ \beta_{j+2}, & j \text{ odd}. \end{cases}$$

This is the finite Duhamel expansion of the actual Kg term; every coefficient $\omega_j \ge \tau$. For $L \ge 6$, the last four indices are at least three and give the S6 mask. Its unmodified mean shift is $\alpha_L = \beta_{L+3}$ for even L, β_{L+1} for odd L, hence in $[\tau, d]$. The four retained path means sum to $4 - 3d$, proving $\mu \le 1 - \tau(5 - 3d)$.

For every finite $n \ge 4$, the receiver-independent finite calculation (L33.3.1) gives $|\mathsf{P}_{n,2}|^2 \le 1 - 7x/15$. For $L \ge 6$, (L33.3.4) gives both finite coefficients in $[\rho_*(x), 1]$, where $\rho_*(x) = 1 - x + x^2 - x^3$. Those proofs include the four residue classes and their actual finite last exponents. The actual receiving symbol consequently obeys $\mathrm{Re}(G_a \mathsf{P}_{L,2}) \ge 7/60$. Indeed its finite lower value is at least $B_0 - D_0 x^4$, with $B_0 = (1 - 2x)/(1 + x)$, $D_0 = (1 - x + x^2)/(1 + x)$. This decreases, and at θ exceeds $2/15 - 1/60 = 7/60$, because $259 \cdot 28561 < 8388608$.

In the S6 mask the real and imaginary coefficient subtractions are $\tau(3 - x)$ and 2τ. The componentwise proof of (L33.3.7) permits each of its four actual retained coefficients to lie in $[\tau, d]$. Setting all four equal to τ gives precisely this S6 polynomial. Its same-measure spectral integration, using $\delta_3 \ge v^2 d$ and projection onto mean-zero functions, gives the stated S6 variance. The actual receiving covariance is computed separately below. For the covariance let $D_a = \int u_a \, d\nu$, $J_a = \int x v_a \, d\nu$, $E_a = \int x u_a \, d\nu$. The actual centered mean shift is $\alpha_L - 2\tau d$, so

$$\mathrm{Cov}(C_{a,0}, Z) = -M_a + \tau(3D_a - 2J_a - E_a) - \beta_{a+1}(\alpha_L - 2\tau d).$$

On $d \le 5/24$, this last mean shift is positive. Use $D_a \le d$, $J_a, E_a \ge \kappa\delta_2$, $\beta_{a+1}, \alpha_L \ge \tau$, and $M_a \ge 7d/60$. Since $3\kappa - 1 = 25/32$, the resulting bound is exactly the S6 value of J, after weakening $-7d/60$ to $-d/10$.

N.11.5 Finite data for the short-two rows

The four actual square packets in Table 26 have the following finite analytic bounds. They are the propagation, coordinate, and moment bounds of (L77.1.6), (L77.1.8)–(L77.1.9), (L77.4.1b), (L77.4.1), and (L77.6.1), together with the long-connector finite loss calculation (L73c.4.2). Here N bounds both $\|\mathsf{Q}_L\|$ and $\|\mathsf{Q}_{L-1}\|$ when $L \ge 4$, and hd bounds the retained losses and the three actual mean shifts at those connector lengths.

b	d range	ε	δ	h	N
3	$(0, 1/3]$	$-\eta$	$k^2\delta_3/2$	$9/16 - d/2$	1
4	$(0, 1/4]$	$\lvert\varepsilon\rvert \le d/6$	$\le 25d/24$	$1/2 - 6d/5$	$1 + d$
$b \ge 5$	$(0, 1/5]$	$\lvert\varepsilon\rvert \le 3d/5$	$\le 6d/5$	$1/2 - 6d/5$	$1 + 7d/5$
$b \ge 5$	$[1/5, 1/4]$	$\lvert\varepsilon\rvert \le 2d/3$	$\le 4d/3$	not used	not used

(N.16)

At $b = 3$, $\eta = k\delta_2 - k^2\delta_3/2$, $0 \le \eta \le 39d/80$, and $0 \le \delta \le \eta/3$. For $b = 4$, $A = d + x$, $B_0 = x$. For $b \ge 5$, both coordinates A, B_0 are nonnegative. On the low domain $\sqrt{A^2 + xB_0^2} \le 3/8 + 9d/5$, and $k(3/8 + 9d/5) \le 1$; the $b = 4$ packet also has $k\sqrt{A^2 + xB_0^2} \le 1$. These are bounds at fixed

original moments: in their finite proof one first maximizes the actual functions of x, and only then uses $\delta_2 \le \theta d$. No substitution $\delta_2 = xd$ is made in a frequency polynomial.

For all three low long-connector packets, the positive propagation and the actual mean shifts give

$$\mu \le \nu_0 - 3hd, \qquad \operatorname{Var} Z \le N^2 d - 3h^2 d^2 + h^2 d^3, \qquad Z \le (\nu_0 - hd)q - hd + N\sqrt{d}. \quad \text{(N.17)}$$

The negative square comes from the actual mean of $\mathsf{Q}_L(B)u$, as in (M.20).

Long connectors D4F, D4B, and D4T.

First let $b \ge 4$ in its low packet. The finite polynomial $\mathsf{U}_L = G_L + kz^L(A - zB_0)$ satisfies

$$\operatorname{Re}(G_a \mathsf{U}_L) \ge 1/50, \qquad a \ge 3, \quad L \ge 4.$$

At $L = 4$ the perturbation contributes $kx^2(u_a A + x v_a B_0) \ge 0$, and the unperturbed product is at least $(1-x)(1-2x) \ge 57/512$. At $L = 5$ its unperturbed value is at least $B_0^* + A_0^* x^3 \ge 2/15$, with $B_0^* = (1-2x)/(1+x)$, $A_0^* = (2-x)/(1+x)$. The perturbation costs at most $x^{5/2} \le 1/9$, leaving $1/45 > 1/50$. At $L = 6$ the unperturbed product exceeds B_0^*; at $L = 7$ it is at least $2/15 - (17/15)\theta^4 > 11/108$. For $L \ge 8$, the finite even and odd formulas give, respectively, $(2/15)(1-\theta^4) > 7/54$ and $2/15 - 2\theta^6 > 67/540$, both exceeding $3/25$. The perturbation at $L \ge 6$ costs at most $\theta^3 < 7/100$. These cover every finite connector length.

In $\mathsf{Q}_L = \mathsf{U}_L + \varepsilon G_{L-2} + \delta z^{L-2}$, $|\operatorname{Re}(G_a G_{L-2})| \le 1$; at $L = 4$ this follows directly from $0 \le u_a - x v_a \le 1$, and later from the two finite norm bounds. The last correction is bounded below by $-\delta\theta$. Thus the additional receiving cost is at most $151d/256$ for $b = 4$ and $87d/80$ for $b \ge 5$. Using the exact centered mean $\alpha_L \ge hd$, the covariance is

$$\operatorname{Cov}(C_{a,0}, Z) = -\int \operatorname{Re}(G_a \mathsf{Q}_L)\, d\nu - \beta_{a+1}\alpha_L + hd \int u_a \, d\nu \le 3d^2. \quad \text{(N.18)}$$

For $b = 3$, $\mathsf{U}_L = G_L - kz^{L+2}$ has the stronger finite receiving lower bound $1/10$. The $L = 4$ correction is favorable. At $L = 5$, its cost kx^4 is paid by $A_0^* x^3$, since $A_0^* \ge 17/15 > kx$. At $L = 6$ the lower value is $B_0^* + x^3(B_0^* - kx) \ge 2/15 - (17/48)(7/100) > 1/10$. At $L = 7$ the correction is favorable and the unperturbed value exceeds $11/108$. For $L \ge 8$ the cost is at most $(6/5)\theta^5 < 1/50$, paid by the unperturbed $3/25$. The remaining corrections cost at most $\eta(1 + \theta/3) \le 4251d/7680$. Retaining $-\beta_{a+1}\alpha_L$, rather than discarding it, now gives

$$\operatorname{Cov}(C_{a,0}, Z) \le -d/10 + [4251/7680 + \theta h]d^2 \le -d/10 + 4d^2/5,$$

since $4251/7680 + \theta(9/16) = 1001/1280 < 4/5$. Together with (N.17), these are D4F, D4B, and D4T. In every case the actual receiving mass is at most $1 - 2\tau \le 1 - 19d/16$.

The finite terminal $b = 3, L = 2$.

For this packet the actual polynomial and scalar are

$$\mathsf{Q}_2 = 1 + z + \delta - kz^4, \qquad Z = 1 - \eta + \mathsf{Q}_2(B)u, \qquad \mu = 1 - d - \eta.$$

Since $k\delta_2 - k^2\delta_3/2$ increases with k, $\eta \ge \delta_2 - \delta_3/2 \ge 51\delta_2/64$. The exact variance of this same message is

$$(1+\delta)^2 d + \delta_2 - 2k(1+\delta)\delta_3 + k^2\delta_5 - d^2.$$

It is at most $d(1+v-7w/4-d)$ on $d \le 1/6$, and at most $d(1+v-7v^2/5-d)$ on $d \le 1/3$, where $w = \delta_3/d$. To see the first bound, combine the exact δ_3 terms before estimating them. After multiplication by m^2, the sufficient difference is $1/4-\theta^2-d/4-7d^2/16$, decreasing and equal to $287/9216 > 0$ at $1/6$. The full-range coefficient is $7/5$, as in the finite calculation for (L77.2.1).

For $a \ge 4$, use $t = \beta_{a-1}$ and put $M_a = \int \mathrm{Re}\{G_a(1+z-z^4)\}\, d\nu$. The exact receiving cancellation is

$$\begin{aligned}\mathrm{Cov}(C_{a,\beta_{a-1}}, Z) = &-M_a + d(\beta_{a-1}-\beta_{a+1}) \\ &+ \delta\left(\beta_{a-1}d - \int u_a\, d\nu\right) + (k-1)\int u_a x^2\, d\nu - k\beta_{a-1}\delta_3.\end{aligned} \tag{N.19}$$

The δ term is nonpositive, since $u_a \ge \kappa > d \ge \beta_{a-1}$. The last two terms have nonpositive sum: $k-1 = kd/2$, $u_a \le 1$, and $\beta_{a-1} \ge \tau > d/2$. The beta difference is one signed δ_j, $j \ge 2$. Hence the covariance is at most $-M_a + vd^2$. For $a = 4$, the exact frequency polynomial is $1-2x+x^3$, and the actual receiving mass is $1-(3-v)d$. For every finite $a \ge 5$,

$$M_a \ge (1-2v+49v^2/100)d, \qquad \mathbb{E}C_{a,\beta_{a-1}} \le 1-(3-3v+v^2)d.$$

The finite frequency polynomials at $a = 5, 6, 7$ are respectively $1-2x+x^2+x^3-x^4$, $1-2x+x^2-x^4$, and $1-2x+x^2-x^3-x^4+x^5$. In the last, the coefficient after x^2 is $(1-x)(1-x^2) \ge 16245/32768 > 49/100$. For $a \ge 8$, both coefficients are at least $(1-x)(1+x^2)$, their difference is at least $-x^5$, and the product is at least $1-2x+x^2-x^3+x^7 \ge 1-2x+\kappa x^2$. Integrate and use $\delta_3 \ge v^2 d$. The three actual beta losses at $a = 5, 6, 7$ retain $3\tau+\delta_3$; at $a \ge 8$, each retains $\tau+\delta_3-\delta_4$, and $3\kappa > 1$. This proves the mass bound as well.

The D2F endpoints therefore apply directly at $a \ge 5$. At $a = 4$, after taking its own two mean endpoints, the nonzero endpoint difference from D2F divided by dU^3 is

$$v[-(2-v)U + (49/25)v - 4v^2] \le 0.$$

Indeed $U \ge 3433/6144 > 5/9$, and the bracket is at most $-(2-\theta)(5/9)+(49/25)\theta = -107/1200 < 0$. The zero endpoint is the same. This extends D2F to all $a \ge 4$.

For $a = 3$, the actual mass is $1-2d$, and the covariance is

$$-d + 2\delta_2 + k\delta_3 - k\delta_4 - \delta(d-\delta_2) - d^2.$$

On $d \le 1/6$, $k \le 12/11$. With $w = \delta_3/d$, $w_4 = \delta_4/d$, the actual joint bounds are

$$J \le d[-1+2v+(12/11)w-w_4] - d^2, \qquad V \le d[1+v-7w/4-d].$$

They must be combined before reducing the moments. In the complete nonzero endpoint, the coefficient of w, divided by d, is $(48/11)U^3 - 35/2 \le -289/22 < 0$; the coefficient of w_4 is $-4U^3 \le 0$. Consequently $w \ge v^2$ and $w_4 \ge v^3$ give

$$HU^4 + 4JU^3 + 10V \le W_{\mathrm{D2F3L}}(d, v). \tag{N.20}$$

At the zero endpoint, $10V$ separately decreases with w. Thus D2F3L is a joint endpoint row, not an independent covariance upper obtained by substituting $w \ge v^2$ into J. On $d \ge 1/6$, use $w^2 \le vw_4$, $w \le \theta v$. The quadratic $-1+2v+(6/5)w-w^2/v$ increases in $w \in [0, \theta v]$; its resulting upper increases with v. Thus $-1+2v+(6/5)w-w_4 \le -9257/163840 < -1/20$, which gives D2F3H with $\mu \le 1-d$ and $V \le d(33/28-d)$.

The two-step finite symbols for $b \ge 4$.

The actual centered mean shift is $\alpha_2 = d + k\rho_0$, with $\rho_0 = \int xB_0\,d\nu \ge 0$. For $b = 4$, remove the nonlinear retained constants and set $k = 1$ to obtain the baseline real coefficients $1 - x^2$, $1 + x^2$ of U_2. Its receiving lower bound is $1 - 2x - x^2 > 1/50$ at $a = 3$. For $a = 4$ it is $1 - 2x + x^4 \ge 3/16$, and $a = 5$ is larger. For $a \ge 6$, the finite lower polynomial is $1 - 2x + x^2 - 3x^3 + x^4 + x^5 + x^6$. It decreases on $[0, \theta]$, and its excess over $3/16$ at θ is $\theta^2(42545/1048576) > 0$. The baseline vector norm is at most $1/2$. Since $k - 1 \le 4d/3$, the actual perturbation of U_2 has norm at most $13d/16$. The extra real shift δ has a favorable receiving sign. The common covariance formula therefore gives

$$J_3 \le -d/50 - 3d^2/16, \qquad J_a \le -3d/16 + 7d^2/32 \quad (a \ge 4).$$

For $b \ge 5$, write the actual finite decomposition $A = A_0 + C_* - E_*x$, $B_0 = B_{00} + D_*$. At $b = 5$, $A_0 = x - x^2$, $B_{00} = x$. At $b \ge 6$, $A_0 = xa_f$, $B_{00} = x\bar{a}_f$, with the two finite alternating coefficients $a_f, \bar{a}_f \in [1 - x, 1 - x + x^2]$. The baseline vector norm is at most $3/8$: at $b = 5$ its square is $x^2 - x^3 + x^4$, and later it is at most $(x - x^2 + x^3)^2(1 + x) < (3/8)^2$. The receiving baseline obeys

$$\mathrm{Re}(G_3\mathsf{U}_2^0) \ge 1 - 2x - x^2 + \kappa x^3 > 3/50, \qquad \mathrm{Re}(G_a\mathsf{U}_2^0) \ge 1/5 \quad (a \ge 4).$$

For $a = 4, 5$, use $a_f + x\bar{a}_f \le 1 + x^3$, so the lower bound is $(1 - x)(1 - x - x^2 - x^5) > 1/5$. For $a \ge 6$, the finite coefficients give $1 - 2x + x^2 - 2x^3 + x^6 - x^7 + x^8$; its derivative is negative, and its endpoint exceeds $7/20 - 7/50 - 1/500 = 26/125 > 1/5$. The $a = 3$ polynomial decreases and at θ is $65295/1048576 > 3/50$.

On the low domain the true retained coefficients give $0 \le C_* - E_*x \le 15d/8$, $0 \le D_* \le 101d/80$. Their perturbing vector norm is at most $21d/10$, and $k - 1 \le 45d/31$, $k \le 40/31$ give a total U_2 perturbation $(10491/7936)d$. On the middle domain the corresponding constants are $C_* \le 53d/32$, $D_* \le 17d/16$, vector norm $9d/5$, and total perturbation $(377/320)d$. Including the actual mean product yields

	$a = 3$	$a \ge 4$
$d \le 1/5$	$J \le -3d/50 + d^2/3$	$J \le -d/5 + 3d^2/4$
$1/5 \le d \le 1/4$	$J \le -3d/50 + d^2/5$	$J \le -d/5 + 3d^2/5$.

The unrounded quadratic coefficients are $2555/7936, 5779/7936$ and $57/320, 187/320$, respectively. The exact two-step mean and variance data from (L77.1.8), (L77.4.3), and (L77.6.1) are those in D2BL, D2BM, D2TL, D2TM. For each $a = 3$ receiver take its own two mean endpoints first. The nonzero endpoint differences from the formal $a \ge 4$ rows are

$b = 4$	$-dU^3(57/400 + 13d/16)$
$b \ge 5,\ d \le 1/5$	$-dU^3(101/400 + 527d/480)$
$b \ge 5,\ 1/5 \le d \le 1/4$	$-dU^3(101/400 + 127d/120)$.

They are nonpositive, and the zero endpoints agree. This proves the all-receiver endpoint interpretation of those four rows.

Three-step connectors D3B, D3TL, and D3TM.

The scalar $\ell = d + k\rho_0$ is retained in the mean. It cancels from the covariance, which is exactly

$$J = -\int \mathrm{Re}(G_a\mathsf{U}_3)\,d\nu - \varepsilon\int u_a\,d\nu + \delta\int xv_a\,d\nu - \beta_{a+1}\alpha_3.$$

For the baseline packets just used, the two coefficients are $1-x-x^3\bar{a}_f$ and $1-x^2a_f$. Use $a_f=\bar{a}_f=1$ at $b=4$, and $a_f=1-x,\bar{a}_f=1$ at $b=5$. In every finite case $a_f-(1-x)\bar{a}_f\ge 0$, so

$$\mathrm{Re}(G_a\mathsf{U}_3^0)\ge 1-3x+x^2+x^3[a_f-(1-x)\bar{a}_f]\ge -55/1024>-3/50.$$

The three-step perturbation is at most $\sqrt{\theta}<13/20$ times the two-step perturbation. Thus it is below $9d/10$ on the low $b\ge 5$ domain, $4d/5$ on its middle domain, and $169d/320$ at $b=4$. The actual finite mean shifts satisfy $\alpha_3\ge\kappa d$. The resulting covariance fees, divided by d^2, are bounded by

$$9/10+3/5+(6/5)\theta-\kappa^2,\quad 4/5+2/3+(4/3)\theta-\kappa^2,\quad 169/320+1/6+(25/24)\theta-\kappa^2,$$

respectively; every one is below two. Hence $J\le d/10+2d^2$. The actual means and centered variances of (L77.1.9) and (L77.6.1) give D3B, D3TL, and D3TM.

The finite three-step terminal D3F.

For $b=3$, the exact polynomial is

$$\mathsf{Q}_3=1+z+z^2-\eta+\delta z-kz^5,\qquad Z=1-\eta-d+\mathsf{Q}_3(B)u.$$

Its centered mean shift is $\alpha_3=d-k\delta_3+\delta d\ge 2053d/2560>4d/5$, and

$$\mu=1-2d-k\delta_2+k\delta_3+(1-d)\delta\le 1-2d-25\delta_2/64\le 1-(1+25v/64)d.$$

Indeed $k(1-d)\le 1$, and $k[\delta_2-(3/2)\delta_3]\ge 25\delta_2/64$. Before the η,δ shifts, the squared norm is at most $1-x+x^2-2x^3+x^5\le 1-6x/7$:

$$1/7-x+2x^2-x^4\ge 1/7-x+(11/6)x^2\ge 1/154>0.$$

The actual shifts change the norm by at most

$$-2\eta(1-x)+\eta^2+2\delta x(1-kx^2)+\delta^2x\le -11\eta/12+\eta^2(1+\theta/9)\le 0,$$

using $\delta\le\eta/3,\eta\le 13/80$. Subtracting the true mean square gives $V\le d-(6/7)\delta_2-(16/25)d^2$, as in D3F. The baseline receiving product satisfies

$$\mathrm{Re}\{G_a(1+z+z^2-z^5)\}\ge 1-3x+x^2+x^3.$$

Its two coordinates are nonnegative; increasing k only increases that product. The η,δ fees are at most $(109/80)vd^2$, and the negative mean product is at least $(4\kappa/5)d^2$. Their difference is at most $(201/2560)d^2<d^2/10$. Integrating the baseline with $\delta_3\ge v^2d$ and $\delta_4\ge v^3d$ gives exactly the D3F covariance.

N.11.6 Pointwise fourth coefficients

Write $y=\sqrt{d}$, and put

$$\mathfrak{C}=2-q+6y/5-\kappa y^2,\qquad \mathfrak{C}_\infty=5/2-q,\qquad \mathfrak{C}_\ell=49/20-q.$$

The receiving profiles (N.11) give the indicated upper for every actual $C_{a,t}$. All three profiles are nonnegative on the rectangles used below. We prove the coefficient bounds in the row tables by bounding $Z+\mu$ and its actual mean μ, then certifying the resulting polynomial in (q,y).

The two coefficient-nine entrances.

S1, S2, S3L, S3H, and S6 satisfy the hypotheses (Q117.4.1). For S1 and S6 their actual point upper is at most $(1-\kappa d)q-\kappa d+y$; the difference from the (Q117.4.1) upper is $y/12+(3-q)d/32\ge 0$. For S2 the difference is at least $y/12-21d/160\ge 0$, since $y\le 3/5$. S3 has exactly the required finite point upper from its third-connector propagation. Their actual means are at most $1-3d/2$. Thus (Q117.4.2) gives their coefficient nine.

For clarity retain the stronger two-step entrance with its full original hypotheses. If $C,Z\ge 0$, $\mu=\mathbb{E}Z\ge 0$, and

$$C\le p+(6/5)\sqrt{d}-\kappa d,\qquad Z+\mu\le q+1+\sqrt{d}-(13/8+3q/8)d,\qquad \mu\le 1-11d/8, \tag{Q117.6.14}$$

then

$$C[(Z+\mu)^2+2\mu^2]\le 9. \tag{Q117.6.15}$$

This is Lemma A.2, case s. The actual S4 and S4F messages satisfy these hypotheses: $Z\le(1-\gamma)q+y-a_1$, $\gamma\ge(1-3v/2)d$, $a_1\ge\kappa vd$, and $1-\mu\ge(2-3v/2)d$. At $v\le\theta$, the resulting coefficient of q is at least $25/64>3/8$, the other loss is at least $1671/1024>13/8$, and the mean loss is at least $89/64>11/8$.

The remaining point prerequisites.

For S5, the exact data of the unit connector give $\gamma\ge d-3\delta_2/2>3d/8$ and $k|KR_1|\le(6/5)\sqrt{\theta d}<4y/5$. Its mean is at most $1-5d/8$, so $Z+\mu\le q+1+4y/5-d$. For S5F, $\gamma>7d/10$ and $k|KR_1|\le 11y/12$; the latter follows from $(36/25)\theta(1+\theta)<121/144$. Together with $\mu\le 1-d$, this gives its point upper below. For D2F and its two actual $a=3$ prefixes, $\delta/(k\delta_2)\le 39/160$ and $\|1-kB^3\|^2\le 1+(36/25)\theta^3<(21/20)^2$. The actual row formula therefore gives $Z\le q+21y/20$, with $\mu\le 1-d$.

For the D4 rows, use (N.17). At $b=3$, $h\ge 19/48$, $\nu_0,N\le 1$, whence

$$Z+\mu\le q+1+y-(19/48)(q+4)d\le q+22/19.$$

The last inequality completes the square $y-19y^2/12\le 3/19$. At $b=4$, $\nu_0\le 1+d/6$, and the coefficient of q is at most one. The remaining upper is $1+y-11y^2/6+y^3+24y^4/5$, increasing on $[0,1/2]$ and equal to $22/15<3/2$ at $1/2$. For its derivative, use $1-11y/3>0$ on $[0,1/4]$; on $[1/4,1/2]$ the lower quadratic $1-11y/3+39y^2/5$ is positive. For $b\ge 5$, the q coefficient is at most $1+d/10+6d^2/5\le 267/250<107/100$. The remaining Z term is at most $(1+7d/5)\sqrt{d}-d/2+6d^2/5$, increasing and below $131/250<21/40$ at $d=1/5$, using $\sqrt{1/5}<9/20$. In all three D4 rows $\mu\le 1$.

For low two-step $b\ge 5$, the actual finite row gives

$$Z+\mu\le q+1+10y/7+e(q)d,\qquad \mu\le 1,\quad e(q)=\begin{cases}-1/10, & q\le 1,\\ 6q/5-13/10, & q\ge 1.\end{cases}$$

It uses $\|\mathsf{U}_1\|\le 10/7$, $\varepsilon\le 3d/5$, and $\delta\le 6d/5$. The $b=4$ row has the stronger upper $q+1+27y/20-5d/6+\delta(q-1)$, with $\delta\le 25d/24$. For $q\le 1$, the difference from the preceding common upper is $11y/140+11d/15\ge 0$. For $q\ge 1$, it is $11y/140+(19q/120+23/40)d\ge 0$. On the middle $b\ge 5$ domain, use $\varepsilon\le 2d/3$, $\delta\le 4d/3$, $\|\mathsf{U}_1\|\le 10/7$, and $\mu\le 1-2d/3\le 13/15$. The two point uppers are $q+12/7$ for $q\le 1$, and $4q/3+29/21$ for $q\ge 1$. At $b=4$, the middle mean is at most $4/5$.

The stronger row just displayed is at most $q + 22/15$ when $q \le 1$, since $27y/20 - 5y^2/6 \le 7/15$ for $y \le 1/2$. For $q \ge 1$, it is at most $19q/15 + 5/4$: the difference has nonnegative q coefficient $4/15 - 25y^2/24 \ge 1/160$, and at $q = 1$ is at least $31/60 - 27y/20 + 5y^2/6 \ge 1/20$.

For the three-step low $b \ge 5$ packet, $\ell \ge d$, $\|\mathsf{Q}_2\| \le 7/5$, $\varepsilon \le 3d/5$, and $\mu \le 1 - 5d/4$ give

$$Z + \mu \le q + 1 + 7y/5 - 21d/20 \le q + 22/15.$$

On its middle domain use $\|\mathsf{Q}_2\| \le 3/2$, $\varepsilon \le 2d/3$, and $\mu \le 3/4$; then $Z+\mu \le q+1+3y/2-11d/12 \le q + 31/20$. At $b = 4$, the inputs $\|\mathsf{Q}_2\| < 10/7$, $\varepsilon \le d/6$, $\ell \ge d$, and $\mu \le 1 - 7d/4$ give $Z + \mu \le q+1+10y/7-29d/12 \le q+1+3/14 < q+5/4$. Here the maximum of the last quadratic increment is $300/1421 < 3/14$. At $b = 3$, $\delta \le 13/240$ gives $\|\mathsf{Q}_2\|^2 \le 87409/57600 < 25/16$. The actual row gives $Z \le (1-\eta)q-d+5y/4$; using $\mu \le 1-d$, $Z+\mu \le q+1+5y/4-2d \le q+153/128 < q+5/4$.

One rational certificate table.

In Table 31, let $\mathcal{P} = M - \mathcal{C}(Y^2 + 2m_0^2)$. The entries $\mathcal{C}, Y, m_0$ are actual point and mean uppers proved above. The last row is also used for the high-domain bare path. Each row has $\mathcal{P} \ge 0$ on its full rectangle. Here is an exact coefficient certificate: subdivide the indicated q and y intervals into the stated equal grid, map each cell affinely to $[0,1]^2$, and apply (A.18) with the printed bidegree. Every Bernstein coefficient is nonnegative. By Lemma A.1, this proves

$$C_{a,t}[(Z + \mu)^2 + 2\mu^2] \le M \tag{N.21}$$

for each actual row, including every boundary. For the first three rows, $0 \le y \le 26/45$; for the two low two-step rows, $0 \le y \le 9/20$. The rows without y have degree zero in that variable. Their grids and bidegrees determine every coefficient by a finite rational formula.

For the two low two-step entries abbreviate the already proved upper by $Y_\ell(q, y) = q+1+10y/7+e(q)y^2$. The function K_* is defined with the row data above.

Table 31: Fourth-coefficient certificates. A full q interval is $[0, 2]$; the two symbols $-, +$ denote $[0, 1], [1, 2]$. The grid lists the number of equal subdivisions in (q, y).

Rows	$\mathcal{C}$	Y	m_0	M	Degree	Grid
S5	$\mathfrak{C}$	$q + 1 + 4y/5 - y^2$	$1 - 5y^2/8$	$37/4$	$(3, 6)$	$(2, 2)$
S5F	$\mathfrak{C}$	$q + 1 + 11y/12 - 17y^2/10$	$1 - y^2$	10	$(3, 6)$	$(2, 2)$
D2F, D2F3L/H	$\mathfrak{C}$	$q + 1 + 21y/20 - y^2$	$1 - y^2$	10	$(3, 6)$	$(2, 2)$
D4F	$\mathfrak{C}_\infty$	$q + 22/19$	1	10	$(3, 0)$	$(8, 1)$
D4B	$\mathfrak{C}_\infty$	$q + 3/2$	1	13	$(3, 0)$	$(2, 1)$
D4T	$\mathfrak{C}_\ell$	$107q/100 + 61/40$	1	13	$(3, 0)$	$(2, 1)$
D2BL/TL $-$	$\mathfrak{C}$	$Y_\ell(q, y)$	1	$K_*(y^2)$	$(3, 6)$	$(1, 1)$
D2BL/TL $+$	$\mathfrak{C}$	$Y_\ell(q, y)$	1	$K_*(y^2)$	$(3, 6)$	$(1, 1)$
D2TM $-$	$\mathfrak{C}_\infty$	$q + 12/7$	$13/15$	14	$(3, 0)$	$(1, 1)$
D2TM $+$	$\mathfrak{C}_\infty$	$4q/3 + 29/21$	$13/15$	14	$(3, 0)$	$(1, 1)$
D2BM $-$	$\mathfrak{C}_\infty$	$q + 22/15$	$4/5$	13	$(3, 0)$	$(1, 1)$
D2BM $+$	$\mathfrak{C}_\infty$	$19q/15 + 5/4$	$4/5$	13	$(3, 0)$	$(1, 1)$
D3TL	$\mathfrak{C}_\ell$	$q + 22/15$	1	12	$(3, 0)$	$(4, 1)$
D3TM	$\mathfrak{C}_\infty$	$q + 31/20$	$3/4$	12	$(3, 0)$	$(2, 1)$
D3F, D3B	$\mathfrak{C}_\infty$	$q + 5/4$	1	11	$(3, 0)$	$(2, 1)$
Bare $\bar{f}_2$	$\mathfrak{C}_\infty$	$q + 5/4$	$3/4$	$75/8$	$(3, 0)$	$(4, 1)$

Table 32: Integer scopes, continuous domains, and inverse payments. Every row has $a \geq 3$ unless a smaller receiving range is printed. S rows use qf_b; D rows use $h_2 f_b$.

Row	Integer range	d-interval	σ_r	Δ_r
S1	$b \geq 3,\ L \geq 5$	$[0, 1/3]$	κ	$\frac{1}{600}$
S2	$b \geq 3,\ L = 4$	$[0, 1/3]$	κ	$\frac{1}{600}$
S3L	$b \geq 3,\ L = 3$	$[0, 1/4]$	κ	$\frac{1}{600}$
S3H	$b \geq 3,\ L = 3$	$[1/4, 1/3]$	κ	$\frac{1}{600}$
S4	$b \geq 5,\ L = 2$	$[0, 1/3]$	$1 - v$	$\frac{3}{80}$
S4F	$b = 3, 4,\ L = 2$	$[0, 1/3]$	1	$\frac{3}{80}$
S5	$b \geq 4,\ L = 1$	$[0, 1/3]$	$1 - v$	$\frac{1}{600}$
S5F	$b = 3,\ L = 1$	$[0, 1/3]$	1	$\frac{1}{600}$
S6	$b = 2,\ L \geq 6$	$[0, 5/24]$	0	$\frac{3}{800}$
D4F	$b = 3,\ L \geq 4$	$[0, 1/3]$	1	$\frac{1}{80}$
D4B	$b = 4,\ L \geq 4$	$[0, 1/4]$	2	$\frac{1}{40}$
D4T	$b \geq 5,\ L \geq 4$	$[0, 1/5]$	$\frac{9}{4}$	$\frac{1}{50}$
D2F	$b = 3,\ L = 2,\ a \geq 4$	$[0, 1/3]$	1	$\frac{1}{80}$
D2F3L	$b = 3,\ L = 2,\ a = 3$	$[0, 1/6]$	1	$\frac{1}{80}$
D2F3H	$b = 3,\ L = 2,\ a = 3$	$[1/6, 1/3]$	1	$\frac{1}{80}$
D2BL	$b = 4,\ L = 2$	$[0, 1/5]$	2	$\frac{1}{40}$
D2BM	$b = 4,\ L = 2$	$[1/5, 1/4]$	2	$\frac{1}{40}$
D2TL	$b \geq 5,\ L = 2$	$[0, 1/5]$	$\frac{9}{4}$	$\frac{1}{50}$
D2TM	$b \geq 5,\ L = 2$	$[1/5, 1/4]$	2	$\frac{1}{40}$
D3F	$b = 3,\ L = 3$	$[0, 1/3]$	1	$\frac{1}{80}$
D3B	$b = 4,\ L = 3$	$[0, 1/4]$	2	$\frac{1}{40}$
D3TL	$b \geq 5,\ L = 3$	$[0, 1/5]$	$\frac{9}{4}$	$\frac{1}{50}$
D3TM	$b \geq 5,\ L = 3$	$[1/5, 1/4]$	2	$\frac{1}{40}$

Table 33: Certificates for the two whole payment polynomials. The bidegrees are those of P_r^+, P_r^0 after the substitution $v = d + (\theta - d)t$; a degree-zero second variable is omitted. Every coefficient in every listed cell is at least $1/100$.

Row	Degree $(+, 0)$	Cells (c_+, c_0)	Coefficients	d-cut for $c_+ = 2$
S1	$4, 5$	$2, 1$	16	$1/6$
S2	$4, 5$	$1, 1$	11	$-$
S3L	$5, 5$	$1, 1$	12	$-$
S3H	$5, 5$	$1, 1$	12	$-$
S4	$(14, 10), (10, 5)$	$2, 1$	396	$1/6$
S5	$(9, 5), (10, 5)$	$1, 1$	126	$-$
S5F	$(6, 2), (5, 2)$	$1, 1$	39	$-$
S6	$(13, 5), (5, 3)$	$1, 1$	108	$-$
D4F	$8, 5$	$1, 1$	15	$-$
D4B	$8, 5$	$1, 1$	15	$-$
D4T	$8, 5$	$1, 1$	15	$-$
D2F	$(10, 6), (5, 2)$	$1, 1$	95	$-$
D2F3L	$(9, 6), (5, 2)$	$1, 1$	88	$-$

Row	Degree $(+,0)$	Cells (c_+, c_0)	Coefficients	d-cut for $c_+ = 2$
D2F3H	$4, 5$	$1, 1$	11	–
D2BL	$4, 5$	$1, 1$	11	–
D2BM	$4, 5$	$1, 1$	11	–
D2TL	$4, 5$	$2, 1$	16	$1/10$
D2TM	$4, 5$	$1, 1$	11	–
D3F	$(9, 6), (5, 1)$	$1, 1$	82	–
D3B	$4, 5$	$1, 1$	11	–
D3TL	$4, 5$	$1, 1$	11	–
D3TM	$4, 5$	$1, 1$	11	–

S4F is paid by the S4 row and the exact stronger-prefix scale comparison (N.15).

N.11.7 The common continuous payment

Table 32 specifies the complete integer and continuous ranges of the parameter rows. A closed interval starting at zero includes the polynomial extension at $d = 0$; the actual zero-variance model is treated below. In each row the finite data and point bounds just proved give

$$\mathbb{E}[f_a\{T^L F\}^4] \le m^8 \max\{M_r V_r, W_r\}. \tag{N.22}$$

For the rows with a receiving-endpoint comparison, this means the two actual endpoints followed by the proved comparison, rather than a replacement of the covariance at an interior mean.

The outside scale in every row has $m \le 1 - \sigma_r d/2$, with σ_r in the scope table. Define the finite inverse polynomial and its two exact payment polynomials by

$$\begin{aligned} S_{5,r} &= \sum_{j=0}^{5} \binom{7+j}{7} (\sigma_r d/2)^j, \\ P_r^+ &= \frac{S_{5,r} - W_r}{d} - \Delta_r, \\ P_r^0 &= S_{5,r} - M_r V_r - \Delta_r d. \end{aligned} \tag{N.23}$$

The quotient is a polynomial, since both numerator terms have constant term one in d. The positive inverse-binomial series gives $m^{-8} \ge (1 - \sigma_r d/2)^{-8} \ge S_{5,r}$. All its denominators are positive on the displayed domains.

For each row depending on v, set $v = d + (\theta - d)t$, $0 \le t \le 1$. Since $\theta - d \ge 7/96 > 0$, this parametrizes exactly $d \le v \le \theta$. For the other rows, the polynomial is constant in t. Table 33 gives a rational certificate for both entire polynomials in (N.23). The two bidegrees are in (d, t); c_+, c_0 give the number of closed d-cells. Subdivide only at the indicated midpoint when $c_+ = 2$; in every cell the t-interval is $[0, 1]$. After the affine map, compute the Bernstein coefficients by (A.18). Every coefficient is at least $1/100$. The rational coefficient tables include their complete values and affine power coefficients. Lemma A.1 therefore proves both payment polynomials nonnegative throughout their domains. Consequently

$$\max\{M_r V_r, W_r\} \le S_{5,r} - \Delta_r d \le m^{-8} - \Delta_r d. \tag{N.24}$$

The finite-prefix comparison (N.15) extends the S4 certificate to S4F, paying its entire additional nonzero endpoint.

For S1–S5F, $m \geq 5/6$ and $(5/6)^8 > 1/5$. At S6, $m = 1$. For the short-two packets the respective lower bounds on m^8 are $1/5$ at $b = 3$, $1/10$ at $b = 4$ or on the middle $b \geq 5$ domain, and $1/8$ on the low $b \geq 5$ domain. Multiplying (N.24) by the actual m^8 therefore gives the three required deficits on every low and middle row. The S4 and S4F bound is stronger,

$$\mathbb{E}[f_a\{T^2(qf_b)\}^4] \leq 1 - 3d/400, \qquad a, b \geq 3. \tag{N.25}$$

N.11.8 The high domains

These remaining ranges use the same original terminal and orientation. The needed path estimates are (A.8), (L56.3.15), and (Q64.3.12):

$$\mathbb{E}f_n = \mathbb{E}\bar{f}_n \leq 1 - 83d/32 \quad (n \geq 4), \qquad \mathbb{E}(\bar{f}_3^4/q) \leq 1 - 23d/36, \qquad \mathbb{E}\bar{f}_4^4 \leq 1 - d - 4d^2.$$

In particular $\bar{f}_n \leq 2 - 5d$ for $n \geq 5$, and $\mathbb{E}f_a \leq 1 - 2d$ for $a \geq 3$. From $f_a \leq 5/2 - q$, $f_a q \leq 25/16 < 8/5$.

For $F = qf_2$, $\mathbb{E}F = 1 - 2d$. Positivity gives $T^L F \leq 2(1 - 2d)\bar{f}_{L-1}$. At $L \geq 6, d \geq 5/24$, its fourth moment is at most

$$(1 - 2d)[2(1 - 2d)(2 - 5d)]^4 \leq 45927/49152 < 15/16 < 1 - 3d/800.$$

The factors are positive and decreasing, so it suffices to use their left endpoint; the displayed rational bound also uses $161/144 < 9/8$.

For $F = h_2 f_b, b \geq 5, d \geq 1/5$, the true terminal mean from Section N.26 is at most $91/200$. Thus $T^L F \leq (91/100)\bar{f}_{L-1}$. At $L = 4$, use the inverse q-weighted $\bar{f}_3$ estimate and $f_a q < 8/5$; the resulting budget is below $35/36$. At $L = 5$, use $f_a \leq 2$ and the $\bar{f}_4$ estimate, giving a budget below $8/9$. At every $L \geq 6$, $\bar{f}_{L-1} \leq 1$, and the budget is at most $(91/100)^4$. Each is below $1 - d/400$. For $b = 4, d \geq 1/4$, the same terminal calculation gives $TF \leq 4/5$; the identical three path ranges have common budget $512/625 < 1 - d/400$.

For the two short high-domain connectors use the actual finite terminal estimate (L77.7.1): $T(h_2 f_b) \leq 13/18$ for every $b \geq 4, d \geq 1/4$. At $L = 2$, positivity gives $T^2 F \leq (13/18)q$. Since $\mathbb{E}q^3 = 1 + 3d - \mathbb{E}g^3 \leq 1 + 4d \leq 7/3$,

$$\mathbb{E}[f_a(T^2F)^4] \leq (13/18)^4(25/16)(7/3) = 4998175/5038848 < 1 - d/400. \tag{N.26}$$

For $L = 3$, the actual bare $\bar{f}_2 = q + Kg$ has mean $1 - d$, variance at most $45d/32 - d^2$, and covariance

$$\mathrm{Cov}(C_{a,0}, \bar{f}_2) = -\int \mathrm{Re}\{G_a(1 + z)\}\, d\nu - \beta_{a+1} d < 0,$$

since $\mathrm{Re}\{G_a(1+z)\} \geq 1 - 2x > 0$. Its point upper satisfies $\bar{f}_2 + (1 - d) \leq q + 1 + \sqrt{d} - d \leq q + 5/4$, and $1 - d \leq 3/4$. The last row of Table 31 gives coefficient $75/8$. The exact fourth identity with mass at most one and nonpositive covariance yields

$$\mathbb{E}[f_a \bar{f}_2^4] \leq (3/4)^4 + (75/8)(103/288) = 1409/384. \tag{N.27}$$

Here $45d/32 - d^2$ increases on $[1/4, 1/3]$ and is at most $103/288$. Therefore

$$\mathbb{E}[f_a(T^3F)^4] \leq (13/18)^4(1409/384) = 40242449/40310784 < 1 - d/400. \tag{N.28}$$

The exact gaps in the last two comparisons are $40673/5038848$ and $68335/40310784$, respectively; both exceed $1/1200$.

The degenerate model, decorations, and reversal.

When $d = 0$, $g = 0$ on every positive-mass atom. Successive leaf integration makes every actual arm and terminal message identically one, even if $K \neq 0$. Thus each reference moment is one, with no moment ratio required. For $d > 0$, the low, middle, and high intervals above cover their shared boundaries. Actual equal-pair products lie in $[0, 1]$; positivity of T restores E_t, E_o in the three estimates. This proves Proposition N.1.

Whole reversal gives the same three bounds with receiver $\bar{f}_a$, propagation A^L, and terminals $p\bar{f}_b$, $p\bar{f}_2$, and $(Aq)\bar{f}_b$, respectively. It reverses every actual edge at once. With short terminal-arm length $s = 1$ or 2, an original decorated component has $a + b + L + s + 2S_t + 2S_o$ edges and one more vertex. The fourth verification tree has $a + 4(b + L + s) + 8S_t + 2S_o$ edges and one more vertex; the receiving arm occurs once.

N.12 All receiving arms at the unit terminal pair

Q187.0 The local claim

Work on the original finite probability space with a real skew kernel $|K(x, y)| \leq 1$. Write

$$B = -K, \quad T = J + B, \quad A = J - B = T^*, \quad u = B\mathbf{1}, \quad p = 1 - u, \quad q = 1 + u,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad \delta_j = \|B^j\mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The terminal residual is $(1, 2)$. Its actual message is $F = qf_2$: the unit arm points inward, the two-edge arm outward, and the single connector edge points from terminal to receiver. The receiver has an outward arm f_a of length $a \geq 3$. We prove

$$\mathbb{E}[f_a\{T(qf_2)\}^4] \leq 1 - d/50. \tag{Q187.T198.0}$$

The same bound holds with genuine equal-pair factors E_o, E_t in their original positions, as $\mathbb{E}[E_o f_a\{T(E_t qf_2)\}^4] \leq 1 - d/50$. The claim here concerns one connector edge; the $L \geq 6$ row is supplied separately by Section N.11.

Q187.1 The receiving upper

The finite calculus gives

$$0 \leq d \leq 1/3, \quad d^2 \leq \delta_2 \leq \theta d, \quad \|B\|^2 \leq \theta, \quad |BV(x)| \leq \|V\|_2.$$

The kernels of A, T lie in $[0, 2]$. Their path messages have $\mathbb{E}f_n = \mathbb{E}\bar{f}_n \leq 1$, and $\mathbb{E}f_n \leq 1 - 2d$ for $n \geq 3$. Set

$$G_n(z) = \sum_{j=0}^{n-1} z^j, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j.$$

For $n \geq 3$, $\tau \leq \beta_n \leq d$. Finite positive convolution gives $f_a \leq S_a - \beta_a$, where $S_a = 1 - G_a(-B)u$. For $a \geq 4$ retain one more nonnegative convolution term:

$$f_a \leq C_a := S_a - \beta_a - \beta_{a-1}p. \tag{Q187.T198.1}$$

For $a = 3$ set $C_3 = f_3 = S_3 - d$. These uppers are nonnegative, being f_a plus unused nonnegative convolution terms. With $V_a = G_{a-1}(-B)u$, finite skew identities give

$$\mathbb{E}V_a = \beta_a, \qquad S_a - \beta_a = (1 - \beta_a)p + B(V_a - \beta_a).$$

For $a \ge 4$, $\|V_a\|_2 \le \sqrt{d}$. For $a = 3$, $V_3 = u - Bu$ and $\mathrm{Var}(V_3) = d + \delta_2 - d^2 \le 45d/32 \le 36d/25$. The centered-row inequality for a mean-zero real V is

$$|BV(x)| \le \sqrt{p(x)q(x)}\,\|V\|_2.$$

It follows by subtracting the row mean $u(x)$ and bounding its variance by $1 - u(x)^2 = pq$. Since $\beta_a + \beta_{a-1} \ge 2\tau \ge 19d/16 \ge d$ for $a \ge 4$, both cases give

$$0 \le C_a \le (1-d)p + \tfrac{6}{5}\sqrt{pqd}. \tag{Q187.T198.2}$$

At $a = 3$ this uses $\beta_3 = d$; at larger a it uses the retained second convolution term.

Q187.2 The terminal square and its actual mean

Put $r = Bu$, $s = -B^2u$, agreeing with the $r = Kg, s = K^2g$ notation in Q5-C117. The identity

$$qf_2 = (1 + r/2)^2 - (-u + r/2)^2$$

and $T(r^2) \le 2\mathbb{E}r^2 = 2\delta_2$ give

$$0 \le X := T(qf_2) \le W := q - \gamma + B^2u, \qquad \gamma = d - \delta_2/2. \tag{Q187.T198.3}$$

Here $Tr = -d + B^2u$. In particular $W \ge 0$ and $m := \mathbb{E}W = 1 - d + \delta_2/2$. The spectral measure ν of u for $-B^2$ has mass d, support $[0, \theta]$, and $\int x^j\,d\nu = \delta_{j+1}$. Thus

$$W = m + (1 + B^2)u, \qquad N := \mathrm{Var}(W) = d - 2\delta_2 + \delta_3 = \int (1-x)^2\,d\nu. \tag{Q187.T198.4}$$

Since $\mathbb{E}r = -d$, we have $B^2u = -du + B(r + d)$ and $\mathrm{Var}(r) = \delta_2 - d^2$. The centered-row inequality therefore yields

$$W \le (1-d)q + \delta_2/2 + \sqrt{pq}\sqrt{\delta_2 - d^2}. \tag{Q187.T198.5}$$

All these quantities come from the same kernel and measure.

Q187.3 A coupled third-moment bound

Set $w = r + d$. Then $\mathbb{E}w = 0$, $\|w\|_2^2 = \delta_2 - d^2$, and

$$B^2u = Bw - du, \qquad \langle Bw, u\rangle = -\langle w, Bu\rangle = -\|w\|_2^2.$$

The norm bound gives $\delta_3 \le (\theta + 2d)(\delta_2 - d^2) + d^3$. For $d > 0$ put $v = \delta_2/d \in [d, \theta]$ and $c = 2 - \theta - 2d = 51/32 - 2d$. Then

$$0 \le N \le \overline{N} := d[(1-d)^2 - c(v-d)]. \tag{Q187.T198.6}$$

At $v = d$ this is $d(1-d)^2$. This uses a global Hilbert-space identity, without imposing a pointwise spectral lower cutoff.

Q187.4 Continuous fourth-expansion coefficients

The exact expansion at the actual mean is

$$W^4 = m^4 + 4m^3(W-m) + (W-m)^2[(W+m)^2 + 2m^2].$$

We show

$$C_a[(W+m)^2 + 2m^2] \le k(d), \tag{Q187.T198.7}$$

where $k = 8$ on $[0, 1/8]$, $k = 121/16$ on $[1/8, 1/5]$, and $k = 117/16$ on $[1/5, 1/4]$. In fact the first coefficient works on the whole original domain. Adjacent ranges include their common endpoint and use the same direction.

Let $t = \sqrt{d}$, $\alpha = 1 - t^2$. From $\sqrt{pq} \le 1$, $\sqrt{\theta} < 13/20$, and $\delta_2 \le \theta d$, we obtain

$$C_a \le \alpha p + \tfrac{6}{5}t, \quad W + m \le \alpha(q+1) + \tfrac{13}{20}t + \theta t^2 =: y, \quad m \le 1 - \tfrac{51}{64}t^2 =: \overline{m}.$$

Set $C = 3 + 37t/20 - 83t^2/32$. Since $C - y = \alpha p + 6t/5 \ge 0$,

$$C_a[(W+m)^2 + 2m^2] \le (C-y)(y^2 + 2\overline{m}^2). \tag{Q187.T198.8}$$

For $0 \le t \le 1/4$, C increases and $C \le 8449/2560 < 53/16$, while $\overline{m} \le 1$. For $0 \le y \le 53/16$,

$$8 - (53/16 - y)(y^2 + 2) = y^3 - \tfrac{53}{16}y^2 + 2y + \tfrac{11}{8} > 0.$$

Indeed, with $h = y - 7/4$ this is $h^3 + 31h^2/16 - 13h/32 + 23/256$. For $-7/4 \le h \le 0$, group the cubic and quadratic; all resulting terms are nonnegative. For $h \ge 0$, dropping h^3 and completing the square leaves at least $23/256 - 169/7936 = 17/248 > 0$.

For $t \ge 1/4$, completing the square gives $C \le 3 + 1369/4150 < 10/3$. Also $\overline{m} \le 973/1024 < 61/64$, so $2\overline{m}^2 < 15/8$. For $0 \le y \le 10/3$,

$$8 - (10/3 - y)(y^2 + 15/8) = y^3 - \tfrac{10}{3}y^2 + \tfrac{15}{8}y + \tfrac{7}{4}.$$

Putting $h = y - 2$ gives $h^3 + 8h^2/3 + 13h/24 + 1/6$. It is positive for $h \ge 0$. On $[-2, 0]$, $h^3 + 8h^2/3 \ge 2h^2/3$, and the remaining quadratic is at least $1/6 - 507/4608 = 29/512 > 0$. This proves $k = 8$ everywhere.

If $d \ge 1/8$, then $\overline{m} \le 461/512$ and $2\overline{m}^2 < 13/8$. At $y = 2 + h$,

$$\tfrac{121}{16} - (10/3 - y)(y^2 + 13/8) = h^3 + \tfrac{8}{3}h^2 + \tfrac{7}{24}h + \tfrac{1}{16}.$$

It is positive for $h \ge 0$; the same quadratic bound on $[-2, 0]$ gives at least $47/1536 > 0$. If $d \ge 1/5$, then $\overline{m} \le 269/320$ and $2\overline{m}^2 < 23/16$, and

$$\tfrac{117}{16} - (10/3 - y)(y^2 + 23/16) = h^3 + \tfrac{8}{3}h^2 + \tfrac{5}{48}h + \tfrac{1}{16},$$

whose analogous lower bound is $359/6144 > 0$. These are continuous polynomial certificates for all three coefficients.

Q187.5 Covariance and the two scalar bounds

Write $G_a(z) = E_a(x) + zD_a(x)$ on the spectral block. For $a \geq 3$, $E_a \geq 1 - x$. Since the centered polynomial of W is $1 - x$,

$$\mathrm{Cov}(S_a, W) = -\int E_a(x)(1-x)\, d\nu, \qquad \mathrm{Cov}(p, W) = -\tau.$$

For $a = 3$, $E_3 = 1 - x$, giving

$$\mathbb{E}C_3 = 1 - 2d, \qquad \mathrm{Cov}(C_3, W) = -N. \tag{Q187.T198.9}$$

For $a \geq 4$, (Q187.T198.1) gives

$$\mathbb{E}C_a = 1 - \beta_a - \beta_{a+1} - \beta_{a-1}, \qquad \mathrm{Cov}(C_a, W) \leq -N + \beta_{a-1}\tau. \tag{Q187.T198.10}$$

Integrate the exact fourth expansion, retaining the actual mean m:

$$R_3 = (1-2d)m^4 + (k - 4m^3)N,$$

$$R_a \leq (1-2\tau)m^4 + \beta_{a-1}m^3(4\tau - m) + (k - 4m^3)N \qquad (a \geq 4).$$

For $d \leq 1/4$ and $v \geq d$, $4\tau - m \leq -1 + 5d - 9d^2/2 \leq -1/32 < 0$. Thus inserting $\beta_{a-1} \geq \tau$ in its combined coefficient and $N \leq \overline{N}$ gives

$$\mathbb{E}[f_a X^4] \leq \mathbb{E}[C_a W^4] \leq R_*(d, v) := (1-3\tau)m^4 + 4m^3\tau^2 + (k - 4m^3)\overline{N} \quad (a \geq 4), \tag{Q187.T198.11}$$

$$R_3(d, v) := (1-2d)m^4 + (k - 4m^3)\overline{N}. \tag{Q187.T198.12}$$

Here $m = 1 - d + dv/2$, $\tau = d(1-v)$, and $c = 51/32 - 2d$. The variance coefficient is positive since $k \geq 117/16$ and $m \leq 1$.

Q187.6 Continuous monotonicity in the moment ratio

Differentiate in v, holding d, k fixed within a closed strip:

$$\frac{1}{d}\partial_v R_* = 3m^4 + (2 + 4c)m^3 - 14m^3\tau + 6m^2\tau^2 - 6m^2\overline{N} - ck,$$

$$\frac{1}{d}\partial_v R_3 = 2m^3(1-2d) + 4cm^3 - 6m^2\overline{N} - ck.$$

On $d \leq v \leq \theta$, $d \leq 1/4$, $\overline{N} \geq 0$: its quotient by d decreases in v, and at $v = \theta$ equals $1 - 2\theta + \theta^2 + \theta d - d^2 \geq \kappa^2$. For $0 < d \leq 1/8$, $k = 8$, $m \leq 1$, $\tau \leq d$. The first derivative divided by d is at most

$$5 + 6d^2 - 4c = -11/8 + 8d + 6d^2 \leq -9/32,$$

and the second is at most $2 - 4c < 0$. For $1/8 \leq d \leq 1/5$, $k = 121/16$ and $m \leq 461/512 < 29/32$, hence $m^3 < 3/4$, $m^4 < 11/16$, $\tau^2 \leq 1/25$, $c \geq 1$. The two upper bounds are

$$33/16 + 3/2 + 6/25 - (73/16)c \leq -19/25, \qquad 3/2 - (73/16)c < 0.$$

For $1/5 \leq d \leq 1/4$, $k = 117/16$ and $m \leq 269/320 < 27/32$, so $m^3 < 5/8$, $m^4 < 9/16$, $\tau^2 \leq 1/16$, $c \geq 1$. The corresponding bounds are $53/16 - (77/16)c \leq -3/2$ and $5/4 - (77/16)c < 0$. Both scalar uppers therefore decrease in v. Set

$$m_0 = 1 - d + d^2/2, \quad \tau_0 = d(1-d), \quad N_0 = d(1-d)^2.$$

Their worst allowed values are

$$G_3 = (1-2d)m_0^4 + (k - 4m_0^3)d(1-d)^2,$$

$$G_* = (1 - 3d + 3d^2)m_0^4 - 4d(1-d)^3m_0^3 + kd(1-d)^2.$$

Q187.7 Polynomial payment on the low domain

The finite powers are

$$m_0^3 = 1 - 3d + \tfrac{9}{2}d^2 - 4d^3 + \tfrac{9}{4}d^4 - \tfrac{3}{4}d^5 + d^6/8,$$

$$m_0^4 = 1 - 4d + 8d^2 - 10d^3 + \tfrac{17}{2}d^4 - 5d^5 + 2d^6 - d^7/2 + d^8/16.$$

For $d > 0$ define $P_3 = (1 - G_3)/d$, $P_* = (1 - G_*)/d$. Expansion gives

$$\begin{aligned} P_3 &= 10 - k - (36 - 2k)d + (72 - k)d^2 - \tfrac{185}{2}d^3 + 81d^4 - 49d^5 \\ &\quad + 20d^6 - \tfrac{81}{16}d^7 + \tfrac{5}{8}d^8, \\ P_* &= 11 - k - (47 - 2k)d + (112 - k)d^2 - \tfrac{345}{2}d^3 + \tfrac{367}{2}d^4 - \tfrac{277}{2}d^5 \\ &\quad + 74d^6 - \tfrac{433}{16}d^7 + \tfrac{99}{16}d^8 - \tfrac{11}{16}d^9. \end{aligned} \tag{Q187.T198.13}$$

Pairing successive signed higher terms on $[0, 1/4]$ gives

$$P_3 \geq 10 - k - (36 - 2k)d + (72 - k)d^2 - \tfrac{185}{2}d^3 + \tfrac{275}{4}d^4,$$

$$P_* \geq 11 - k - (47 - 2k)d + (112 - k)d^2 - \tfrac{345}{2}d^3 + \tfrac{1191}{8}d^4.$$

For example pair $81d^4 - 49d^5$ and $(367/2)d^4 - (277/2)d^5$, then the successive signed terms, using $d \leq 1/4$.

On $[0, 1/8]$, $k = 8$. Discard nonnegative terms of degree at least four and use $d^3 \leq d^2/8$. The resulting decreasing quadratics give

$$P_3 \geq 2 - 20d + \tfrac{839}{16}d^2 \geq 327/1024, \qquad P_* \geq 3 - 31d + \tfrac{1319}{16}d^2 \geq 423/1024.$$

Their derivatives are still negative at the right endpoint.

On $[1/8, 1/5]$, $k = 121/16$. The cubic lower bound for P_3 has second derivative at least $1031/8 - 555/5 = 143/8 > 0$, first derivative $-31/5$ at $1/5$, and endpoint value $1/10$. For P_* use

$$Q_* = \frac{55}{16} - \frac{255}{8}d + \frac{1671}{16}d^2 - \frac{345}{2}d^3 + \frac{1191}{8}d^4.$$

Its second derivative is $1671/8 - 1035d + (3573/2)d^2 > 0$: the first two terms alone are at least $15/8$. Its first derivative at $1/5$ is $-1509/250$, and $Q_*(1/5) = 491/5000$. Thus

$$P_3 \geq 1/10, \qquad P_* \geq 491/5000. \tag{Q187.T198.14}$$

On $[1/5, 1/4]$, $k = 117/16$. The fourth-degree lower bound for P_3 has second derivative $1035/8 - 555d + 825d^2 \geq 189/8$, first derivative $-133/64$ at $1/4$, and endpoint value $215/1024$. For P_* use

$$Q_* = \frac{59}{16} - \frac{259}{8}d + \frac{1675}{16}d^2 - \frac{345}{2}d^3 + \frac{1191}{8}d^4.$$

Its second derivative is at least $1675/8 - 1035/4 + 3573/50 = 4417/200 > 0$, its first derivative at $1/4$ is $-393/128$, and $Q_*(1/4) = 47/2048$. Hence

$$P_3 \geq 215/1024, \qquad P_* \geq 47/2048. \tag{Q187.T198.15}$$

The three closed strips cover $[0, 1/4]$. Since $327/1024$, $491/5000$, and $47/2048$ exceed $1/50$, the actual reference expression is at most $1 - d/50$ for $0 < d \leq 1/4$.

Q187.8 The high domain, zero boundary and decorations

On the high domain use X directly:

$$\mathbb{E}(qf_2) = 1 - 2d, \qquad 0 \le X = T(qf_2) \le 2(1-2d).$$

For $1/4 \le d \le 1/3$, $X \le 1$, so $\mathbb{E}[f_a X^4] \le \mathbb{E}f_a \le 1 - 2d \le 1 - d/50$. If $d = 0$, then $u = 0$ on the positive support, though K need not vanish. Leaf integration with $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$ makes every actual tree message one.

Each actual equal pair $f_\ell h_\ell$, or its full reversal, is at most one. Positivity gives $0 \le T(E_t F) \le TF$ and $E_o \le 1$, proving the decorated claim. All pair edges and vertices remain in the original tree and its final edge count. The finite positive support has $p, q \ge \mu(x) > 0$ throughout.

N.13 Bare inward paths of length at least six under the reference f_2

Work on a nonempty finite probability space with a real skew kernel $|K| \le 1$. Use $A = J + K$, $T = J - K$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $f_n = A^n\mathbf{1}$, $\bar{f}_n = T^n\mathbf{1}$ and $h_n = Tf_{n-1}$. We prove

$$\mathbb{E}[f_2 \bar{f}_n^4] \le 1 - d/20 \qquad (n \ge 6). \tag{Q99.0.1}$$

All lengths here are finite. At $d = 0$, leaf integration gives equality. The finite calculus in Section D gives $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{j+1} \le \theta\delta_j$, $\theta = 13/32$, and $0 \le f_n, \bar{f}_n, h_n \le 2$. The paths f_n and $\bar{f}_n$ have means at most one. On positive-mass atoms $p, q \ge \mu(x) > 0$.

Put $z = -K$, $u = z\mathbf{1} = -g$, $\delta_j = \|K^j\mathbf{1}\|_2^2$, $\kappa = 19/32$ and $\tau = d - \delta_2$. Then

$$\kappa d \le \tau \le d - d^2 \le 2/9, \qquad \beta_1 = \beta_2 = 0, \quad \beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1}\delta_i \quad (j \ge 3), \qquad \tau \le \beta_j \le d.$$

For $G_n(z) = \sum_{i=0}^{n-1} z^i$, $G_0 = 0$, finite positive convolution gives

$$S'_n := \sum_{i=0}^{n} z^i\mathbf{1} = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j}.$$

The norm of $G_m(z)$ is at most one for $m = 0, 1$ or $m \ge 3$; $\|G_2(z)\|^2 \le 1 + \theta$. Equal-arm factors satisfy $f_\ell h_\ell \le (\mathbb{E}f_{\ell-1})^2 \le 1$. For $n \ge 6$, retain four terms of the convolution to obtain

$$0 < \bar{f}_n \le Z := S'_n - \tau(\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3).$$

The four retained coefficients all have index at least three, and

$$\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3 = 4 - d + 3u + 2zu + z^2u.$$

Consequently,

$$Z = 1 - \tau(4 - d) + Q_n(z)u, \qquad Q_n(z) = G_n(z) - \tau(3 + 2z + z^2). \tag{Q99.0.2}$$

Q99.1 The finite predecessor family

At $z = i\sqrt{x}$, $0 \le x \le \theta < 5/12$, each real and normalized imaginary coordinate of G_n lies between $1-x$ and $1-x+x^2$. Indeed these are finite alternating sums in $-x$ with at least three terms, including the longer coordinate when n is odd. Put $a = 1-3\tau$ and $b = 1-2\tau$. Both coordinates of Q_n are nonnegative, since

$$a-(1-\tau)x \ge 1/3-(7/9)(5/12) = 1/108 > 0, \qquad b-x \ge 5/9-5/12 > 0.$$

Their upper bounds are $a-(1-\tau)x+x^2$ and $b-x+x^2$. Thus

$$|Q_n|^2 - a^2 \le x\{-1+4\tau-2\tau^2+(1-4\tau+\tau^2)x+(1-2\tau)x^2-x^3+x^4\}.$$

The derivative of the brace with respect to τ is $4(1-x)-\tau(4-2x)-2x^2 \ge 79/72 > 0$ on this rectangle. Taking $\tau = 2/9$ gives

$$\{\cdots\} \le -17/81+(13/81)x+(5/9)x^2-x^3+x^4 \le -17/81+65/972+125/1296 = -181/3888 < 0.$$

Here $-x^3+x^4 \le 0$; at $x=0$ equality holds in the preceding norm comparison. Hence

$$\|Q_n(z)\| \le 1-3\tau \qquad (n \ge 6). \tag{Q99.1.1}$$

Q99.2 Mean, covariance and variance

The finite moments give

$$\mathbb{E}[G_n(z)u] = -\beta_{n+1}, \qquad \mathbb{E}[Q_n(z)u] = -\beta_{n+1}+2\tau d.$$

Writing $m = \mathbb{E}Z$, we have

$$0 \le m = 1-4\tau-\beta_{n+1}+3\tau d \le 1-(5-3d)\tau \le 1-4\kappa d =: u_0, \qquad u_0 \ge 5/24. \tag{Q99.2.1}$$

For the actual reference $f_2 = 1-u+zu$, telescoping gives

$$\mathbb{E}[f_2 S_n'] = 1-\beta_{n+1}-\beta_{n+2}-\beta_{n+3}, \qquad \operatorname{Cov}(f_2, S_n') = d-\beta_{n+2}-\beta_{n+3}-d\beta_{n+1}.$$

The short-path covariances are

$$\operatorname{Cov}(f_2, \bar{f}_1) = -d, \qquad \operatorname{Cov}(f_2, \bar{f}_2) = -d+\delta_2-d^2,$$
$$\operatorname{Cov}(f_2, \bar{f}_3) = -d+2\delta_2-d^2.$$

Their sum is $-3\tau-2d^2$. Therefore

$$\operatorname{Cov}(f_2, Z) \le d-2\tau-d\tau+3\tau^2+2\tau d^2.$$

The right side decreases with τ, because its derivative satisfies $6\tau-2-d+2d^2 \le 5d-4d^2-2 \le -7/9$. At $\tau = \kappa d$ this bound becomes

$$v(d) = -3d/16+(475/1024)d^2+(19/16)d^3.$$

The quotient $v(d)/d$ increases and equals $913/9216 < 1/10$ at $d = 1/3$. Thus

$$\operatorname{Cov}(f_2, Z) \le d/10. \tag{Q99.2.2}$$

By (Q99.1.1),

$$\operatorname{Var} Z \le (1-3\tau)^2 d-(\beta_{n+1}-2\tau d)^2 \le (1-3\tau)^2 d-\tau^2(1-2d)^2.$$

This expression decreases with $\tau \le 2/9$, since its derivative is $-6d(1-3\tau)-2\tau(1-2d)^2 < 0$. Consequently,

$$\operatorname{Var} Z \le w(d) := d-(6\kappa+\kappa^2)d^2+13\kappa^2 d^3-4\kappa^2 d^4. \tag{Q99.2.3}$$

Q99.3 The reference coefficient

We use the following pointwise coefficient with $c = 19/48$:

$$\left.\begin{aligned} Z \ge 0, \quad m = \mathbb{E}Z, \\ Z \le (1-cd)q - cd + \sqrt{d}, \qquad m \le 1 - 3cd \end{aligned}\right\} \implies f_2[(Z+m)^2 + 2m^2] \le 10. \quad \text{(Q99.3.1)}$$

For the four-term Z, positive convolution and $\|G_{n-1}\| \le 1$ give

$$Z \le S_n' - \tau(1+q) \le (1-\tau)q - \tau + \sqrt{d} \le (1-cd)q - cd + \sqrt{d}.$$

Also $m \le 1 - 4\kappa d \le 1 - 3cd$. Here is the coefficient calculation. The row identity

$$2Kg(x) = \mathbb{E}_{y,z}[(K(x,y) - K(x,z))K(y,z)]$$

and $|a-b| \le 1-ab$ for $a, b \in [-1,1]$ imply $|Kg| \le pq/2$. Hence $0 \le f_2 \le w(q) := 2 - q^2/2$. Put $x = \sqrt{d}$ and $y = q+1$. The hypotheses give $Z + m \le y + x - c(y+3)x^2$ and $m \le 1 - 3cx^2$. Expand the squares, drop the negative cubic term and use $x^4 \le x^2/3$:

$$(Z+m)^2 + 2m^2 \le y^2 + 2 + 2yx - A(q)x^2,$$

where

$$A(q) = \frac{5111}{6912}q^2 + \frac{3059}{864}q + \frac{17767}{3456}.$$

Coefficient subtraction gives

$$A(q) - w(q)y^2 \ge q^4/2 + q^3 - (4/5)q^2 - q/2 + 3 > 0.$$

On $0 \le q \le 1$ the lower bound is $17/10$; on $1 \le q \le 2$ use $q^3 \ge q^2$ and $q^4 \ge q$. Moreover,

$$9 - w(q)(y^2+2) = \tfrac{1}{2}(q-1)^2(q^2+4q+6) \ge 0.$$

Completing the square gives $2yx - Ax^2 \le y^2/A$. Multiplication by $w(q) \ge 0$ then bounds the coefficient by $9 + w(q)y^2/A(q) \le 10$. The polynomial comparisons include $q = 0$.

The fourth-order Taylor identity around $m = \mathbb{E}Z$ is

$$\mathbb{E}[f_2 Z^4] = (1-d)m^4 + 4m^3 \operatorname{Cov}(f_2, Z) + \mathbb{E}[f_2(Z-m)^2\{(Z+m)^2 + 2m^2\}].$$

Using the positive covariance upper bound in (Q99.2.2), the resulting terms increase with $m \ge 0$. Thus

$$\mathbb{E}[f_2 \bar{f}_n^4] \le (1-d)u_0^4 + (2/5)du_0^3 + 10w(d). \quad \text{(Q99.3.2)}$$

Q99.4 The scalar estimate

Put $a = 4\kappa = 19/8$, so $u_0 = 1 - ad$. The increment of the right side of (Q99.3.2) over one, divided by d, is

$$-\frac{1}{10} + \frac{3439}{2560}d - \frac{89167}{2560}d^2 + \frac{1350501}{20480}d^3 - \frac{130321}{4096}d^4.$$

For $0 \le d \le 1/3$, the cubic term is at most $(1350501/61440)d^2$. The resulting negative quadratic coefficient is $263169/20480 > 64/5$, and $3439/2560 < 27/20$. Drop the negative quartic term and complete the square:

$$\frac{\mathbb{E}[f_2 \bar{f}_n^4] - 1}{d} \le -1/10 + (27/20)d - (64/5)d^2 \le -1/10 + 729/20480 < -1/20,$$

where the last comparison is $729 < 1024$. This proves (Q99.0.1) for $d > 0$, and the zero case follows from the leaf-integration observation above.

N.14 The short bare inward paths under the reference f_2

Use the finite model and notation of Section N.13, with $z = -K$, $u = z\mathbf{1} = -g$, $\delta_j = \|K^j\mathbf{1}\|_2^2$, $\theta = 13/32$, $\kappa = 19/32$, and $\tau = d - \delta_2$. The following bounds hold for every real skew $|K| \le 1$:

$$\mathbb{E}[f_2\bar{f}_3^4] \le 1 - d/8, \qquad \mathbb{E}[f_2\bar{f}_4^4] \le 1 - 2d/5, \qquad \mathbb{E}[f_2\bar{f}_5^4] \le 1 - 7d/10. \tag{Q101.0.1}$$

Together with (Q99.0.1), these give

$$\mathbb{E}[f_2\bar{f}_n^4] \le 1 - d/20 \qquad (n \ge 3). \tag{Q101.0.2}$$

All adjoints and moments use the same finite measure. In particular, $d^2 \le \delta_2 \le \theta d$, $\delta_3 \ge \delta_2^2/d$ for $d > 0$, and $\delta_{j+1} \le \theta\delta_j$. At $d = 0$, the result follows by leaf integration. Actual equal-arm factors satisfy $f_\ell h_\ell \le 1$.

Q101.1 Finite convolution and the coefficient ten

Write $G_n(z) = \sum_{i=0}^{n-1} z^i$. The finite positive convolution is

$$S_n' = 1 + G_n(z)u = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j}, \qquad \beta_3 = \beta_4 = d, \quad \beta_5 = \beta_6 = \tau, \quad \beta_7 = \beta_8 = \tau + \delta_3.$$

For $n = 4, 5$, positivity and $\|G_{n-1}\| \le 1$ give

$$0 \le \bar{f}_n \le S_n' - \tau(1 + q) \le (1 - \tau)q - \tau + \sqrt{d}.$$

Their means below satisfy $m \le 1 - 3cd$, where $c = 19/48$. Since $\tau \ge \kappa d \ge cd$, the pointwise coefficient (Q99.3.1) applies:

$$f_2[(Z + m)^2 + 2m^2] \le 10, \qquad Z = \bar{f}_4 \text{ or } \bar{f}_5.$$

For clarity, that coefficient uses $f_2 \le w(q) = 2 - q^2/2$, $x = \sqrt{d}$, $y = q + 1$, and

$$(Z + m)^2 + 2m^2 \le y^2 + 2 + 2yx - A(q)x^2, \qquad A(q) = \frac{5111}{6912}q^2 + \frac{3059}{864}q + \frac{17767}{3456}.$$

The full coefficient proof is given in the coefficient calculation of Section N.13. The Taylor identity is

$$\mathbb{E}[f_2 Z^4] = (1 - d)m^4 + 4m^3 \operatorname{Cov}(f_2, Z) + \mathbb{E}[f_2(Z - m)^2\{(Z + m)^2 + 2m^2\}]. \tag{Q101.1.1}$$

For $n = 4, 5$ the covariances are nonpositive, so their middle terms can be discarded.

Q101.2 Length four

The exact message is

$$\bar{f}_4 = S_4' - d(1 + q) = 1 - 2d + P_4(z)u, \qquad P_4 = G_4 - d,$$

and its mean satisfies

$$m_4 = 1 - 3d + \delta_2 \le 1 - (83/32)d \le 1 - 5d/2.$$

Here m_4 denotes a path mean; ordinary powers are written explicitly. At $z = i\sqrt{x}$,

$$|P_4|^2 = (1 - d)^2 + (-1 + 2d)x - x^2 + x^3.$$

Consequently,

$$\mathrm{Var}(\bar{f}_4) = (1-d)^2 d - (1-2d)\delta_2 - \delta_3 + \delta_4 - \tau^2,$$

and

$$\mathrm{Cov}(f_2, \bar{f}_4) = -d + 2\delta_2 - \delta_3 + d\delta_2.$$

Put $t = \delta_2/d \in [d, \theta]$. Since $\delta_3 \geq t^2 d$,

$$\frac{\mathrm{Cov}(f_2, \bar{f}_4)}{d} \leq -1 + 2t - t^2 + td \leq -\kappa^2 + \theta d \leq -667/3072 < 0.$$

The derivative $2 - 2t + d$ is positive, justifying the endpoint $t = \theta$. For the variance, use $\delta_4 \leq \theta\delta_3$ and $\delta_3 \geq t^2 d$:

$$\frac{\mathrm{Var}(\bar{f}_4)}{d} \leq (1-d)^2 - (1-2d)t - \kappa t^2 - d(1-t)^2.$$

Its derivative in t is $-1 + 4d - 2(\kappa + d)t$. For $t \geq d$ this is at most $-1 + (45/16)d - 2d^2$, which increases on $[0, 1/3]$ and equals $-41/144 < 0$ at its right endpoint. Thus the maximum occurs at $t = d$, giving

$$\mathrm{Var}(\bar{f}_4) \leq v_4(d) := d - 4d^2 + (141/32)d^3 - d^4.$$

Apply the coefficient ten and $m_4 \leq 1 - 5d/2$:

$$\mathbb{E}[f_2 \bar{f}_4^4] \leq (1-d)(1-5d/2)^4 + 10 v_4(d).$$

The increment over one, divided by d, expands as

$$-1 + (15/2)d - (895/16)d^2 + (1465/16)d^3 - (625/16)d^4.$$

Use $d^3 \leq d^2/3$ and discard the negative quartic term:

$$\frac{\mathbb{E}[f_2 \bar{f}_4^4] - 1}{d} \leq -1 + (15/2)d - (305/12)d^2 \leq -109/244 < -2/5.$$

The last comparison is $545 > 488$.

Q101.3 Length five

The finite convolution retains its actual coefficients:

$$\bar{f}_5 = S_5' - d(\bar{f}_1 + \bar{f}_2) - \tau = 1 - 2d - \tau + P_5(z)u, \qquad P_5 = G_5 - d(2+z).$$

Its mean satisfies

$$m_5 = 1 - 4d + 2\delta_2 + d^2 \leq 1 - (51/16)d + d^2 \leq 1 - (137/48)d \leq 1 - 14d/5.$$

At $z = i\sqrt{x}$,

$$\frac{|P_5|^2 - (1-2d)^2}{x} = -1 + 2d + d^2 + (1-2d)x - x^2 + x^3.$$

This increases with d, because $2 + 2d - 2x > 0$. At $d = 1/3$ it is at most $-2/9 + x/3 \leq -25/288 < 0$, using $0 \leq x \leq \theta < 1$ and $-x^2 + x^3 \leq 0$. Hence $\|P_5\| \leq 1 - 2d$. The exact mean of $P_5 u$ is $-\tau + d^2$, so

$$\mathrm{Var}(\bar{f}_5) \leq (1-2d)^2 d - (\tau - d^2)^2 \leq v_5(d) := d - (4457/1024)d^2 + (83/16)d^3 - d^4.$$

Here $\tau \geq \kappa d > d^2$. The exact covariance is

$$\mathrm{Cov}(f_2, \bar{f}_5) = -d + 2\delta_2 - 2\delta_3 + d^2 + d^3.$$

With $t = \delta_2/d$,

$$\frac{\mathrm{Cov}(f_2, \bar{f}_5)}{d} \leq -1 + 2t - 2t^2 + d + d^2 \leq -265/512 + d + d^2 \leq -337/4608 < 0.$$

The derivative $2 - 4t$ is positive for $t \leq \theta$. The coefficient ten therefore gives

$$\mathbb{E}[f_2 \bar{f}_5^4] \leq (1-d)(1-14d/5)^4 + 10v_5(d).$$

The increment over one, divided by d, is

$$-11/5 + (188347/12800)d - (82973/1000)d^2 + (87046/625)d^3 - (38416/625)d^4.$$

Absorb the cubic using $d \leq 1/3$. The negative quadratic coefficient becomes $548227/15000 > 73/2$, and $188347/12800 < 59/4$. Thus

$$\frac{\mathbb{E}[f_2 \bar{f}_5^4] - 1}{d} \leq -11/5 + (59/4)d - (73/2)d^2 \leq -11/5 + 3481/2336 < -7/10,$$

with final margin $23/2336$.

Q101.4 Length three

Put $\eta = \sqrt{45/32}$, $x = \sqrt{d}$, and $m = 1 - 2d$. The exact message and row norm are

$$\bar{f}_3 = q - d + K(g - Kg), \qquad \|g - Kg\|^2 = d + \delta_2 \leq 45d/32.$$

Hence $0 \leq \bar{f}_3 \leq q - d + \eta\sqrt{d}$. Using $f_2 \leq w(q) = 2 - q^2/2$ and $c = \eta x - 3x^2$ gives

$$f_2[(\bar{f}_3 + m)^2 + 2m^2] \leq H(q,d) := w(q)[(q + 1 + c)^2 + 2m^2].$$

We establish, for $0 \leq q \leq 2$, $0 \leq d \leq 1/3$,

$$H(q,d) \leq C(d) := 10 - 15d + 8d^2. \tag{Q101.4.1}$$

Concavity of c as a function of x gives $-1/3 \leq c \leq 1/8$: the lower endpoint uses $\sqrt{15/32} > 2/3$, and its maximum is $\eta^2/12 = 15/128 < 1/8$. Differentiation gives

$$H_q = (q + 1 + c)[4 - 2q^2 - q(1 + c)] - 2qm^2.$$

For $0 \leq q \leq 4/5$ the bracket is positive and

$$\begin{aligned} H_q &\geq (q + 2/3)(4 - 2q^2 - 9q/8) - 2q \\ &= 8/3 + 5q/4 - 59q^2/24 - 2q^3 \\ &\geq 8/3 - (599/300)q \geq 1604/1500 > 0. \end{aligned}$$

For $q \geq 3/2$ the derivative is negative, since $1 + c \geq 2/3$. It remains to prove (Q101.4.1) for $4/5 \leq q \leq 3/2$. Put $y = q + 1$. After moving $15x^2 - 8x^4$ to the left, the cubic and quartic terms are

$$-6\eta w x^3 + (17w - 8)x^4 \leq 0.$$

Indeed $x \le 1/\sqrt{3}$, $w \le 42/25$, $\eta\sqrt{3} > 41/20$, and $(17-6\eta\sqrt{3})w < (47/10)(42/25) = 1974/250 < 8$. Thus

$$H + 15d - 8d^2 \le w(y^2+2) + 2\eta wyx - A(q)x^2, \qquad A(q) = w(q)(6q + 403/32) - 15.$$

Put $v = q - 1 \in [-1/5, 1/2]$. Then

$$A = (825 - 614v - 979v^2 - 192v^3)/64 \ge 997/256 > 0,$$
$$D := 10 - w(y^2+2) = 1 + 11v^2/2 + 3v^3 + v^4/2.$$

The bound for A follows from its negative derivative on this interval. Completing the square in x reduces the claim to $AD \ge \eta^2 w^2 y^2$. The exact expansion is

$$\begin{aligned} 128(AD - \eta^2 w^2 y^2) = 30 - 688v + 9232v^2 - 2008v^3 - 14438v^4 \\ - 8960v^5 - 2176v^6 - 192v^7. \end{aligned} \tag{Q101.4.2}$$

For $-1/5 \le v \le 0$, the odd-degree terms are nonnegative, and absorbing the negative even powers leaves quadratic coefficient at least $9232 - 14438/25 - 2176/625 > 0$. For $0 \le v \le 1/4$, absorbing degrees at least three into v^2 leaves coefficient $7678 + 15/16 > 7600$. Hence the right side of (Q101.4.2) is at least $30 - 688v + 7600v^2 > 14$, since $688^2 = 473344 < 486400$. For $1/4 \le v \le 1/2$, the coarser absorption leaves $6713/2$; using $v \le 4v^2$ gives a lower bound $30 + (6713/2 - 2752)v^2 > 0$. This proves the coefficient on the middle interval, and the monotonicity in q covers the two tails.

For the actual $\bar{f}_3$, the exact moments are

$$\mathbb{E}\bar{f}_3 = m = 1 - 2d, \qquad \mathrm{Cov}(f_2, \bar{f}_3) = -d + 2\delta_2 - d^2, \qquad \mathrm{Var}(\bar{f}_3) = d - \delta_2 + \delta_3 - d^2 \le d - d^2 - \kappa\delta_2.$$

Apply (Q101.1.1) with coefficient $C(d)$. The coefficient of δ_2 in the resulting upper bound is

$$B(d) = 8(1-2d)^3 - \kappa C(d) = 33/16 - (1251/32)d + (365/4)d^2 - 64d^3.$$

For $0 \le d \le 1/16$, use $B(d) \le 33/16$ and $\delta_2 \le \theta d$ when $B \ge 0$; when $B < 0$ discard $B\delta_2$. This gives

$$\frac{\mathbb{E}[f_2 \bar{f}_3^4] - 1}{d} \le -1107/512 + 27d - 57d^2 + 24d^3 + 16d^4 \le -1919/4096 < -2/5.$$

For the last step, discard the negative quadratic and use $-1107/512 + 27/16 + 3/512 + 1/4096 = -1919/4096$. For $1/16 \le d \le 1/3$, the function B is convex because $B'' = 365/2 - 384d > 0$. Its endpoint values are $B(1/16) = -41/1024$ and $B(1/3) = -2765/864$, so $B \le 0$. Using $\delta_2 \ge d^2$ gives the quotient

$$P(d) = -3 + (465/16)d - (3075/32)d^2 + (461/4)d^3 - 48d^4.$$

Set $t = d - 1/4 \in [-3/16, 1/12]$. Its exact expansion is

$$P = -65/512 - (3/8)t - (885/32)t^2 + (269/4)t^3 - 48t^4.$$

For $t \le 0$, drop the negative cubic and quartic terms and complete the quadratic to obtain $P \le -65/512 + 3/2360 < -1/8$, where the last comparison is $1536 < 2360$. For $t \ge 0$, use $t^3 \le t^2/12$; every nonconstant term is then nonpositive, giving $P \le -65/512 < -1/8$. This proves the length-three bound and completes (Q101.0.1) and (Q101.0.2).

N.15 The genuine f_2 reference for terminal $(1,2,3)$, $L \geq 2$

Q104.1 Statement and original directions

On a nonempty finite probability space (X,μ) let K be real skew with $|K(x,y)| \leq 1$. Every operator, norm, expectation, adjoint and spectral decomposition uses the same original μ. Put

$$Jv = (\mathbb{E}v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \quad d = \mathbb{E}g^2,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad \delta_j = \|K^j\mathbf{1}\|_2^2, \quad \theta = 13/32, \quad \kappa = 19/32.$$

For the terminal residual $(1,2,3)$, fix at every $L \geq 2$ the same actual message $F = qf_2h_3$, and put $X_L = T^L(E_tF)$. The unit terminal arm points inward, the two-edge arm outward, and the three-edge arm has first edge inward and the other two outward. Every original connector edge points terminal-to-receiver. The receiver has its actual outward two-edge arm, producing the genuine reference f_2. Arbitrary original equal pairs at the terminal and receiver have products E_t, E_o, each pair using $f_\ell h_\ell \leq 1$. For this predetermined direction,

$$\mathbb{E}[E_o f_2 X_L^4] \leq 1 - d/500 \leq 1 \qquad (L \geq 2). \tag{Q104.1.1}$$

The fourth power copies only the terminal branch in the analytic moment; the receiving arm occurs once. The statement excludes $L = 1$, other odd terminal tables and a generic real-input reference-fourth map. Global reversal gives the $\bar{f}_2/A$ version with every edge reversed. The original finite framework gives

$$0 \leq d \leq 1/3, \quad \|K\|^2 \leq \theta, \quad |Kv(x)| \leq \|v\|_2, \quad d^2 \leq \delta_2 \leq \theta d, \quad \delta_{j+1} \leq \theta\delta_j, \quad \delta_3 \geq \delta_2^2/d \quad (d > 0). \tag{Q104.1.2}$$

It also supplies path positivity, means at most one and the equal-pair comparison. The necessary square identities are rederived below. Delete zero-mass atoms. Then $p, q \geq \mu(x) > 0$, and the all-equal labeling makes each actual finite-tree message strictly positive. At $d = 0$, $g = 0$ though K may be nonzero; leaf integration gives all actual messages one. Henceforth take $d > 0$.

Q104.2 A genuine-reference coefficient and exact moment identity

The original row identity $2Kg(x) = \mathbb{E}_{y,z}[(K(x,y) - K(x,z))K(y,z)]$ and $|a - b| \leq 1 - ab$ for $a, b \in [-1,1]$ give $|Kg| \leq pq/2$. Consequently

$$0 \leq f_2 = p + Kg \leq w(q) := 2 - q^2/2, \qquad \mathbb{E}f_2 = 1 - d. \tag{Q104.2.1}$$

Suppose $Y \geq 0$, $u = \mathbb{E}Y$, and

$$0 \leq u \leq 1 - d, \quad Y + u \leq q + 1 + \eta\sqrt{d} - (3/2 - q/4)d, \quad \eta^2 \leq 4/3. \tag{Q104.2.2}$$

Then the pointwise coefficient obeys

$$f_2[(Y + u)^2 + 2u^2] \leq 21/2. \tag{Q104.2.3}$$

Replacing a negative η by its absolute value enlarges the upper, so take $\eta \geq 0$. Set $t = \sqrt{d}$, $y = q + 1$ and $B = 3/2 - q/4 > 0$. The upper for $Y + u$ is nonnegative. Square it, discard the negative cubic and use $t^4 \leq t^2/3$ to get

$$(Y + u)^2 + 2u^2 \leq y^2 + 2 + 2\eta yt - \mathcal{A}(q)t^2,$$

$$\mathcal{A}(q) = 2yB + 10/3 - \eta^2 - B^2/3 \geq -25q^2/48 + 11q/4 + 17/4 =: \mathcal{A}_0(q) > 0.$$

The identity $9 - w(q)(y^2+2) = (q-1)^2(q^2+4q+6)/2 \geq 0$ and completion of the t square reduce the claim to $3\mathcal{A}_0 \geq (8/3)wy^2$. Multiplying by 48 gives

$$P(q) = 64q^4 + 128q^3 - 267q^2 - 116q + 356 \geq 0 \quad (0 \leq q \leq 2).$$

For $q \leq 3/4$, $P \geq 356 - 267(9/16) - 87 > 0$. For $3/4 \leq q \leq 1$, $64q^4 + 128q^3 \geq 132q^2$, so $P \geq 356 - 135 - 116 = 105$. For $q \geq 1$, write $v = q - 1 \geq 0$: $P = 165 - 10v + 501v^2 + 384v^3 + 64v^4 > 0$, since $165 - 10v + 501v^2 \geq 165 - 100/2004 > 0$. The exact fourth identity is

$$\mathbb{E}[f_2 Y^4] = (1-d)u^4 + 4u^3 \operatorname{Cov}(f_2, Y) + \mathbb{E}[f_2 (Y-u)^2((Y+u)^2 + 2u^2)]. \tag{Q104.2.4}$$

Therefore (Q104.2.2) implies

$$\mathbb{E}[f_2 Y^4] \leq (1-d)u^4 + 4u^3 \operatorname{Cov}(f_2, Y) + (21/2) \operatorname{Var} Y. \tag{Q104.2.5}$$

The kernel and adjoint remain at the original measure.

Q104.3 Finite convolution and positive square propagation

Put $z = -K$, $g' = -g = z\mathbf{1}$ and $\mathsf{X} = -z^2$. At a rotation frequency $z = i\sqrt{x}$, $x \in [0, \theta]$. The spectral measure at g' has mass d and moments δ_{j+1}. Write $r = Kg$, $s = K^2 g = \mathsf{X} g'$, so $\mathbb{E}r = -d$, $\mathbb{E}s = 0$, $\mathbb{E}(gs) = -\delta_2$. Let $G_n(z) = \sum_{j=0}^{n-1} z^j$, $G_0 = 0$, $\beta_1 = \beta_2 = 0$, and $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1} \delta_j$ for $n \geq 3$. Finite induction using $T = J + z$ gives

$$S_n' = \sum_{j=0}^{n} z^j \mathbf{1} = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j}, \quad \kappa d \leq d - \delta_2 \leq \beta_j \leq d \quad (j \geq 3). \tag{Q104.3.1}$$

Also $S_n' = \mathbf{1} + G_n g' = q + z G_{n-1} g'$ and $\mathbb{E} z G_n g' = -\beta_{n+2}$. Finite geometric sums have norm at most one for $n = 0, 1$ or $n \geq 3$, while $\|G_2\|^2 \leq 1 + \theta$. For $n \geq 4$, both real coordinates of $G_n = a + zb$ lie in $[1-x, 1-x+x^2]$; use finite alternating sums and $(1-z)G_n = 1 - z^n$, with the positive-ending even case $n \geq 6$ using $2x^2 + x^5 \leq 1$. Since $f_2 + h_3 = 2 - d - s$, let $m = 1 - d/2$, $k = 1/m \in [1, 6/5]$ and $R = -ks$. AM–GM for the actual arms gives $F \leq q(m - s/2)^2$. Expand $T(qs)$ and use $|K(gs)| \leq \sqrt{d\delta_3}$, $T(qs^2) \leq 4\delta_3$:

$$TF/m^2 \leq \bar{f}_2 + zR + \delta \mathbf{1}, \quad \delta = -k\delta_2 + k\sqrt{d\delta_3} + k^2 \delta_3.$$

For $t = \delta_2/d$, $\delta_3 \leq \theta \delta_2$ and completion of a square give

$$\delta/d \leq k\{\sqrt{\theta t} - (1 - k\theta)t\} \leq \frac{k\theta}{4(1-k\theta)} \leq 39/164 < 1/4. \tag{Q104.3.2}$$

Choose $D \geq \delta$, $D \geq 0$. The upper $\bar{f}_2 + zR + D\mathbf{1}$ is nonnegative. Apply T^{L-1} to obtain

$$0 < T^L F/m^2 \leq Y_L = \bar{f}_{L+1} + z^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \quad D_i = \mathbb{E} z^i R. \tag{Q104.3.3}$$

Here $D_1 = k\delta_2$, $D_2 = 0$, $D_3 = -k\delta_3$, $D_4 = 0$. Every positive D_i for $i \geq 3$ is at most $k\delta_4 \leq k\theta^3 d < d/12$. For $\sigma = 49d/96$,

$$\gamma_j := \beta_j - D_{j-2} \geq \sigma \quad (j \geq 3). \tag{Q104.3.4}$$

Indeed $\gamma_3 \ge 41d/80 > \sigma$, $\gamma_4 = d$, and later cases use $\kappa d - d/12$. All discarded losses below are nonnegative. If $Y = (C+c)\mathbf{1} + Q(z)g' = Cq + zP(z)g' + c\mathbf{1}$, then $Q = C + zP$ and

$$\operatorname{Cov}(f_2, Y) = \int [-\operatorname{Re} Q + x \operatorname{Im}_z Q]\, d\nu + d\mathbb{E}[Q(z)g']. \tag{Q104.3.5}$$

Here Im_z is the real coefficient of z in $Q = a + zb$. Also $\operatorname{Var} Y = \|Q(z)g'\|_2^2 - (\mathbb{E}Q(z)g')^2$.

Q104.4 The complete $L = 2$ proof

Choose $D = d/4$. The actual positive comparison is

$$Y = Cq + zPg' + c\mathbf{1}, \quad C = 1 + d/4, \quad P = 1 + z + kz^3, \quad c = k\delta_2 - d, \quad T^2F \le m^2 Y. \tag{Q104.4.1}$$

The terminal remains qf_2h_3. At a frequency,

$$|P|^2 = 1 + x(1-kx)^2 \le 31/27 < 4/3. \tag{Q104.4.2}$$

Differentiate the cubic on $0 \le kx \le 39/80 < 1$ to get $x(1-kx)^2 \le 4/(27k)$; hence $|zPg'| \le \sqrt{31d/27}$. Since $g' = z\mathbf{1}$ and $Q = C + z + z^2 + kz^4$, skewness gives $\mathbb{E}[Q(z)g'] = \mathbb{E}(Cz\mathbf{1} + z^2\mathbf{1} + z^3\mathbf{1} + kz^5\mathbf{1}) = -d$. Thus $u = \mathbb{E}Y = 1 - 7d/4 + k\delta_2$, and

$$u \le 1 - (101/80)d \le 1 - (21/20)d. \tag{Q104.4.3}$$

Moreover $c \le -(41/80)d \le -d/2$. Therefore $u \le 1-d$ and $Y+u \le q+1+\sqrt{31d/27}-(3/2-q/4)d$, verifying the coefficient conditions. The coordinates of Q are $C - x + kx^2$ and one. The identity

$$|Q|^2 - C^2 = x[-2 + 1/k + 3kx - 2kx^2 + k^2x^3] \tag{Q104.4.4}$$

uses $d = 2(1-1/k)$. The bracket is convex in k and increasing in x. At $x = \theta$, its values at $k = 1$, $6/5$ are $-1451/32768$ and $-2281/614400$, so it is negative throughout. Hence $\|Qg'\|_2^2 \le C^2 d$; retain $\mathbb{E}Qg' = -d$ to get

$$\operatorname{Var} Y \le C^2 d - d^2 = d - d^2/2 + d^3/16 \le d. \tag{Q104.4.5}$$

The exact covariance is

$$\operatorname{Cov}(f_2, Y) = -d - 5d^2/4 + 2\delta_2 - k\delta_3 \le -\kappa^2 d - 5d^2/4 \le -(7/20)d - d^2. \tag{Q104.4.6}$$

For the first comparison put $t = \delta_2/d$, use $\delta_3 \ge t^2 d$ and $k \ge 1$, and maximize $2t - t^2$ on $[0, \theta]$. For the second use $\kappa^2 = 361/1024 > 7/20$, $5/4 \ge 1$. Let $a = 21/20$, $v = 1 - ad$, $c_0 = 7/20$. The mean-covariance polynomial is $F_0(u) = (1-d)u^4 - 4(c_0 d + d^2)u^3$. On $[0, v]$ its maximum is zero or $F_0(v)$: the only possible positive critical point is a minimum. If $F_0(v) \ge 0$, then (Q104.2.5) gives $\mathbb{E}[f_2 Y^4] \le B_2(d) := (1-d)v^4 - 4(c_0 d + d^2)v^3 + (21/2)d$, whose exact expansion is

$$B_2 = 1 + (39/10)d + (449/40)d^2 - (819/250)d^3 - (922131/160000)d^4 + (546399/160000)d^5. \tag{Q104.4.7}$$

For $0 \le d \le 1/3$ the positive binomial series gives

$$m^{-8} \ge 1 + 4d + 9d^2 + 15d^3 + (165/8)d^4 + (99/4)d^5. \tag{Q104.4.8}$$

Subtracting (Q104.4.7), the degree-four and degree-five coefficients are positive, while the first three terms satisfy

$$d[1/10 - (89/40)d + (4569/250)d^2] \ge d/50. \tag{Q104.4.9}$$

Indeed $2/25-(89/40)d+(4569/250)d^2>0$ by $(89/40)^2<4(2/25)(4569/250)$. Since $m^8>1/5$, this proves $\mathbb{E}[f_2(T^2F)^4]\le 1-d/250$. If $F_0(v)<0$, the alternative upper is $(21/2)dm^8$. The maximum of $d(1-d/2)^8$ on $[0,1/3]$ is at $2/9$. As $(8/9)^4<5/8$, this is less than $(7/3)(25/64)=175/192<1-d/250$.

Q104.5 The complete $L=3$ proof

For the remaining lengths take $D=3d/8$. The exact positive upper is

$$Y_3=Cq+zPg'-d\mathbf{1},\quad C=1-5d/8+k\delta_2,\quad P=1+D+z+(I-k\mathsf{X})z^2. \tag{Q104.5.1}$$

One has $231/256\le C\le 1-d/8$; the lower bound follows by minimizing $1-5d/8+d^2$. The coordinates of $Q=C+zP$ are $C-x$ and $1+D-x+kx^2$, both nonnegative. Their squared difference is

$$|Q|^2-C^2=x[-2C+x+(1+D-x+kx^2)^2]. \tag{Q104.5.2}$$

The bracket is convex in x, with second derivative $2(-1+2kx)^2+4k(1+D-x+kx^2)\ge 0$. At zero it is at most $-138/256$; at θ, use $1+D-\theta+k\theta^2\le 2347/2560<15/16$ to get $-133/256$. Thus $\|Q\|\le C$ and $\operatorname{Var}Y_3\le d$. Furthermore

$$\mathbb{E}Qg'=-(1+D)d+\delta_2-k\delta_3\le-(3/4+D)d,\quad \mathbb{E}Y_3\le 1-15d/8\le 1-7d/4. \tag{Q104.5.3}$$

The covariance integrand in (Q104.3.5) is $-C+(2+D)x-x^2+kx^3$, increasing in x. With $C\ge 1-5d/8$, $D\le 1/8$, $k\le 6/5$, its upper is $-19/24+(17/8)\theta-\theta^2+(6/5)\theta^3=-3187/245760<0$. The $d\mathbb{E}Qg'$ term is also nonpositive. For the row norm, the quotient of $|P|^2-(1+D)^2$ by x is

$$-5/2+3/(2k)+(7k/2-1/2)x-2kx^2+k^2x^3.$$

It is convex in k, and increasing in x at $k=1,6/5$, with endpoint values $-1451/32768$, $-9507/204800$. Hence $|zPg'|\le(1+D)\sqrt{d}$ and $(1+D)^2\le 81/64<4/3$. The pointwise and mean rows verify (Q104.2.2). Put $v=1-7d/4$. The moment is at most $(1-d)v^4+(21/2)d$. Subtracting it from (Q104.4.8) gives

$$(3/2)d-(131/8)d^2+(877/16)d^3-(2609/256)d^4+(8737/256)d^5.$$

Since $(2609/256)d^4\le(7/2)d^3$, this is at least $d[3/2-(131/8)d+(821/16)d^2]\ge d/10$. After subtracting $1/10$ the quadratic has negative discriminant. Multiplying by $m^8>1/5$ proves deficit at least $d/50$.

Q104.6 The complete $L=4$ proof

The exact positive upper is

$$\begin{aligned}Y_4&=Cq+zPg'+c\mathbf{1},\quad C=1-13d/8+k\delta_2,\quad A_0=C+d,\\ P&=A_0+(1+D)z+z^2+(I-k\mathsf{X})z^3,\quad c=-(d-\delta_2)-Dd-k\delta_3.\end{aligned} \tag{Q104.6.1}$$

Here $1/2<41/72\le C\le 1-9d/8$ and $231/256\le A_0\le 1-d/8$. The positive coordinates of $Q=C+zP$ are $Q_R=C-(1+D)x+(1-kx)x^2$ and $Q_I=C+d-x$. Since $Q_R\le C-x+x^2$,

$$|Q|^2-C^2\le x[(C+d)^2-2C+(1-2d)x-x^2+x^3]. \tag{Q104.6.2}$$

The bracket increases in d for $C \in [1/2, 1]$. At $d = 1/3$, convexity in C places its maximum at $C = 1$, with constant $-2/9$; the remaining $(1/3)x - x^2 + x^3$ increases and reaches $3679/98304 < 1/8$ at θ. Thus $\operatorname{Var} Y_4 \le C^2 d$. One has $\mathbb{E}Qg' = -A_0 d + \delta_2 < 0$. The covariance integrand $-C + (1 + C + D + d)x - 2x^2 + kx^3$ increases in x and is at most $-1/2 + (47/24)\theta - 2\theta^2 + (6/5)\theta^3 = 11293/245760 < 1/16$. Hence $\operatorname{Cov}(f_2, Y_4) \le d/16 \le d/4$. The exact mean correction is $c + \mathbb{E}Qg' = -2d + 2\delta_2 + d^2/4 - kd\delta_2 - k\delta_3 \le (-2 + 2\theta + 1/12)d = -53d/48$, which with $C \le 1 - 9d/8$ gives $u = \mathbb{E}Y_4 \le 1 - 2d$. The direct identity $|P|^2 = A_0^2 + x[(1 + D - x + kx^2)^2 + x - 2A_0]$ has convex bracket, negative at both endpoints: at zero it is at most $-1 + 2d + D^2 < -1/4$, and at θ at most $-229/768$. Thus $\|P\| \le A_0 \le 1$; together with $c \le -\kappa d$, this verifies (Q104.2.2). For $v = 1 - 2d$,

$$\begin{aligned}\mathbb{E}[f_2 Y_4^4] &\le (1 - d)v^4 + dv^3 + (21/2)(1 - 9d/8)^2 d \\ &= 1 + (5/2)d + (19/8)d^2 - (3931/128)d^3 + 40d^4 - 16d^5.\end{aligned}$$

The last three terms are nonpositive, since $40d^4 \le (40/3)d^3 < (3931/128)d^3$. Compare with $m^{-8} \ge 1 + 4d + 9d^2$ to leave at least $(3/2)d$, giving reference deficit at least $3d/10$.

Q104.7 Every finite $L \ge 5$

For $n \ge 5$, $0 \le \sigma \le 49/288$, $0 \le x \le \theta$, one has

$$|G_n(z) - \sigma(3 + 2z + z^2)| \le 1 - 3\sigma, \qquad z = i\sqrt{x}. \tag{Q104.7.1}$$

For $n = 5$ the squared difference from $(1 - 3\sigma)^2$, divided by x, is $-1 + 4\sigma - 2\sigma^2 + (1 - 4\sigma + \sigma^2)x + (-1 + 2\sigma)x^2 + x^3$. It increases in σ up to $3/16$. At $3/16$ discard the negative quadratic term and put $x = \theta$ to obtain $-4503/32768 < 0$. For $n \ge 6$, multiplication by $1 - z$ gives $M - z^n$, where $M = (1 - 3\sigma) + \sigma z + \sigma z^2 + \sigma z^3$. Writing $a = 1 - 3\sigma$, $|M| \le 1$ and

$$a^2(1 + x) - |M|^2 = x[a^2 + 2a\sigma - \sigma^2(1 - x + x^2)] \ge x(1 - 4\sigma + 2\sigma^2) > 3x/8.$$

The minimum is $15649/41472 > 3/8$. The finite tail contributes at most $2x^{n/2} + x^n$; after division by x, this is at most $2\theta^2 + \theta^5 < 1/3 + 1/64 < 7/20 < 3/8$. The zero frequency is direct. The estimate $|M| \le 1$ follows from $|M|^2 \le a^2 + \sigma^2 x \le 1 - 6\sigma + (9 + \theta)\sigma^2 \le 1$. Set $\sigma = 49d/96$ and retain the last four nonnegative losses in (Q104.3.3) using (Q104.3.4). As $\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3 = \mathbf{1} + 3q + z(2 + z)g' - d\mathbf{1}$ and $\bar{f}_{L-1} \le S'_{L-1}$, the actual nonnegative upper is

$$Y = Cq + zPg' + c\mathbf{1}, \quad C = 1 - 37d/32 > 0, \quad c = -\sigma(1 - d), \quad P = G_L - k\mathsf{X}z^{L-1} + DG_{L-2} - \sigma(2 + z), \quad T^L F \le m^2 Y. \tag{Q104.7.2}$$

For $Q = C + zP$,

$$Q = (1 - kx)[G_{L+1} - \sigma S] + kx[G_L - \sigma S] + DG_{L-1}, \qquad S = 3 + 2z + z^2.$$

The convex-combination coefficients are nonnegative and $kx \le 39/80$. By (Q104.7.1) and $\|G_{L-1}\| \le 1$, $\|Q\| \le 1 - 3\sigma + D = C$ and $\operatorname{Var} Y \le C^2 d$. The shorter shift is

$$|G_n - \sigma(2 + z)| \le 1 - 2\sigma \quad (n \ge 3, \sigma \le 3/16). \tag{Q104.7.3}$$

$n = 3, 4$ have squared gaps $x[-1 + 2\sigma + \sigma^2 + x]$ and $x[-1 + 2\sigma + \sigma^2 + (-1 + 2\sigma)x + x^2]$. For $n \ge 5$ the numerator is $1 - 2\sigma + \sigma z + \sigma z^2 - z^n$. Its rational-part squared gap is $x[1 - 2\sigma - \sigma^2(1 + x)] \ge$

$4715x/8192$, while the finite tail divided by x is at most $2\theta^{3/2}+\theta^4<9/16$. This proves (Q104.7.3) and gives $\|P\|\le 1+D-2\sigma=1-31d/48\le 1$. The actual mean term is

$$\mathbb{E}Qg'=-\beta_{L+2}+D_L-D\beta_L+2\sigma d\le -\sigma-D\kappa d+2\sigma d\le 0.$$

Consequently

$$u=\mathbb{E}Y\le 1-(209/96)d+(335/256)d^2\le 1-(17/10)d. \tag{Q104.7.4}$$

Every G polynomial in Q has index at least four. Its finite coordinate bounds give $-\operatorname{Re}G+x\operatorname{Im}_z G\le -9/32$, with endpoint margin $1-2\theta+\theta^2-\theta^3=9355/32768>9/32$. The $-\sigma S$ term costs at most 3σ. Using $\mathbb{E}Qg'\le 0$ in (Q104.3.5) gives

$$\operatorname{Cov}(f_2,Y)\le[-9(1+D)/32+3\sigma]d\le(11/48)d<d/4. \tag{Q104.7.5}$$

The row $\|P\|\le 1$, $c\le -d/3$, the mean bound (Q104.7.4), and $C=1-37d/32$ verify (Q104.2.2), since $37q/32+61/30\ge 3/2-q/4$. Set $a=17/10$, $v=1-ad$. The moment bound is

$$\mathbb{E}[f_2Y^4]\le(1-d)v^4+dv^3+(21/2)(1-37d/32)^2d. \tag{Q104.7.6}$$

Its linear coefficient is $37/10$; its quadratic is $476/25-777/32=-4193/800<-5$; its cubic is $-14161/500+28749/2048<-14$; its quartic is $230911/10000<24$; its quintic is $-83521/10000<0$. Since $d^4\le d^2/9$, all terms of degree at least two sum to at most $(-5+24/9)d^2-14d^3<0$. Hence

$$\mathbb{E}[f_2(T^LF)^4]\le m^8(1+37d/10)\le e^{-3d/10}\le 1-3d/11\le 1-d/500. \tag{Q104.7.7}$$

The penultimate step uses $e^t\ge 1+t$ for $t=3d/10\le 1/10$. The same direction covers every $L\ge 5$, completing all $L\ge 2$.

Q104.8 Original decorations, reversal and edge counts

Original pair products satisfy $0<E_t,E_o\le 1$. Positivity of T gives $\mathbb{E}[E_of_2\{T^L(E_tF)\}^4]\le\mathbb{E}[f_2(T^LF)^4]$. All equal-pair arms remain in the original oriented object. If their half-lengths sum to S, an original reference component has $L+6+2+2S=L+8+2S$ edges and $L+9+2S$ vertices, counting the receiving two-edge arm, the terminal arms, connector subdivisions and pair decorations once. Only in the fourth-moment comparison is the terminal branch copied four times while the reference arm stays single. All directions precede μ,K; indeed the same canonical terminal works for all $L\ge 2$. Replacing the whole kernel by $-K$ reverses the reference, propagation, terminal and all other actual edges together. The ordinary $qf_2\bar f_3$ message at two edges is not substituted for the canonical reference message qf_2h_3.

N.16 The genuine f_2 reference for terminal $(1,2,4)$ at every length

Q105.0 Actual scope

On a nonempty original finite probability space (X,μ) let K be real skew with $|K|\le 1$. Every operator, adjoint, norm, expectation and spectral measure uses this original μ. Put

$$Jv=(\mathbb{E}v)\mathbf{1},\quad A=J+K,\quad T=J-K=A^*,\quad g=K\mathbf{1},\quad p=1+g,\quad q=1-g,$$

$$f_n=A^n\mathbf{1},\quad \bar f_n=T^n\mathbf{1},\quad h_n=Tf_{n-1},\quad \delta_j=\|K^j\mathbf{1}\|_2^2,\quad d=\delta_1,\quad \theta=13/32,\quad \kappa=19/32. \tag{Q105.0.1}$$

For residual terminal $(1,2,4)$ and every positive original L, use the same canonical $F = qf_2h_4$ and $X_L = T^L(E_tF)$. The receiver has its original outward two-edge arm, giving the genuine reference f_2. For arbitrary original equal-pair products E_t, E_o,

$$\mathbb{E}[E_o f_2 X_L^4] \le 1 \qquad (L \ge 1). \tag{Q105.0.2}$$

More precisely,

$$\mathbb{E}[E_o f_2 X_L^4] \le 1 - d/50 \qquad (0 \le d \le 1/4, L \ge 1). \tag{Q105.0.3}$$

The unit proof gives $1 - d/40$; the high domain uses the same F, $h_4 \le 1$, and the same-direction qf_2 reference fourth of C058. The direction is independent of μ, K, d.

The different canonical $(1,2,3)$ unit target is not an input here:

$$\mathbb{E}[f_2\{T(qf_2h_3)\}^4] \overset{?}{\le} 1. \tag{Q105.0.4}$$

Q105.1 Original directions and finite foundations

Along an original arm, f_n points every edge outward, $\bar{f}_n$ every edge inward, and h_n only the first edge inward. Thus the terminal unit arm points inward, its two-edge arm outward, and its four-edge arm only first-edge inward. All L connector edges point terminal-to-receiver; the receiving two-edge arm is outward. Equal pairs use $f_\ell h_\ell \le 1$, fixed before the host, retaining original multiplicities, subdivision vertices and leaves. The finite framework is

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad |Kv(x)| \le \|v\|_2, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j. \tag{Q105.1.1}$$

After zero-mass atoms are deleted, $p, q \ge \mu(x) > 0$. The operators are positive; the all-equal labeling gives strictly positive actual messages. Path means are at most one, and $f_\ell + h_\ell = 2\mathbb{E}f_{\ell-1}$ gives $0 < f_\ell h_\ell \le 1$. At $d = 0$, $g = 0$ with possibly nonzero K; original leaf integration gives all messages one. Every division by d below is restricted to $d > 0$. Put

$$z = -K, \quad g' = -g = z\mathbf{1}, \quad \mathsf{X} = -z^2 = -K^2, \quad r = Kg = zg', \quad s = K^2g = \mathsf{X}g', \quad w = K^3g = -\mathsf{X}zg'. \tag{Q105.1.2}$$

The operator X is nonnegative and self-adjoint. At $z = i\sqrt{x}$, $x \in [0,\theta]$, the original measure ν at g' has

$$\int x^j \, d\nu = \delta_{j+1}, \quad \nu([0,\theta]) = d, \quad \mathbb{E}[z\mathsf{X}^j g'] = -\delta_{j+1}, \quad \mathbb{E}[\mathsf{X}^j g'] = 0. \tag{Q105.1.3}$$

Global moment inequalities are not pointwise $x \ge d$; all polynomials are finite. The row identity $2r(x) = \mathbb{E}_{y,t}[(K(x,y) - K(x,t))K(y,t)]$ and $|a - b| \le 1 - ab$ for $a, b \in [-1,1]$ give

$$|r| \le pq/2, \quad 0 < f_2 = p + r \le W(q) := 2 - q^2/2, \quad \mathbb{E}f_2 = 1 - d. \tag{Q105.1.4}$$

This pointwise fact is specific to the genuine f_2.

Q105.2 Five continuous reference coefficients

For $Y \ge 0$ and $u = \mathbb{E}Y$, the exact identity is

$$\mathbb{E}[f_2Y^4] = (1-d)u^4 + 4u^3 \operatorname{Cov}(f_2, Y) + \mathbb{E}[f_2(Y-u)^2\{(Y+u)^2 + 2u^2\}]. \tag{Q105.2.1}$$

A pointwise bound $f_2[(Y+u)^2 + 2u^2] \le C_*$ therefore yields

$$\mathbb{E}[f_2Y^4] \le (1-d)u^4 + 4u^3 \operatorname{Cov}(f_2, Y) + C_* \operatorname{Var} Y. \tag{Q105.2.2}$$

All five coefficient proofs below cover continuous intervals.

Q105.2.1 Coefficient A: $21/2$

If $0 \le u \le 1-d$ and

$$Y + u \le q + 1 + \eta\sqrt{d} - (3/2 - q/4)d, \qquad 0 \le \eta^2 \le 4/3, \tag{Q105.2.3}$$

then $C_* = 21/2$. This is precisely the genuine-reference lemma proved in part N.15: the same pointwise $f_2 \le W(q)$, the same original measure, and identical u, η, d hypotheses apply. Its finite polynomial certificate $64q^4 + 128q^3 - 267q^2 - 116q + 356 \ge 0$ supplies the proof here.

Q105.2.2 Coefficient B: another $21/2$ on the low domain

If $d \le 1/4$, $u \le 1-d$, and

$$Y + u \le q + 1 + (5/4)\sqrt{d} - (7/4 - q/2)d, \tag{Q105.2.4}$$

then $C_* = 21/2$. Set $t = \sqrt{d}$, $y = q+1$, $B = 7/4 - q/2 \ge 3/4$. As in coefficient A, use $t^4 \le t^2/4$ to get quadratic loss $A_0(q) = -(17/16)q^2 + (47/16)q + 299/64 > 0$. Complete the square using $W(q)(y^2+2) \le 9$; it remains to show $24A_0 \ge 25W(q)y^2$. Eight times the exact difference is $P(q) = 100q^4+200q^3-504q^2-236q+497$. For $v = q-1 \in [-1,1]$ this is $57-244v+696v^2+600v^3+100v^4$. At $v \le 0$, $696v^2 + 600v^3 \ge 96v^2$ and $-244v \ge 0$. At $v \ge 0$, $57 - 244v + 696v^2 > 0$ because $244^2 < 4 \cdot 57 \cdot 696$; the other terms are nonnegative.

Q105.2.3 Coefficient C: twelve for $L = 2$

If $d \le 1/4$, $u \le 1 - 2d/5$, and

$$Y + u \le q + 1 + (21/20)\sqrt{d} - (21/40 - q/2)d, \tag{Q105.2.5}$$

then $C_* = 12$. For fixed q, square this upper and add $2(1 - 2t^2/5)^2$. Its derivative on $0 \le t \le 1/2$ is minimized at $q = 0$, where it is $21/10 - (619/200)t - (1323/400)t^2 + (953/400)t^3$. This decreases to $3/128 > 0$ at $1/2$. Thus take $t = 1/2$, with upper $(9/8)q + 223/160$ for $Y + u$ and $9/10$ for u. Clearing positive denominators reduces the claim to $P(q) = 32400q^4 + 80280q^3 - 38399q^2 - 321120q + 249596 \ge 0$. At $v = q - 1$ it is $2757 - 27478v + 396841v^2 + 209880v^3 + 32400v^4$. For $v \le 0$, the quadratic and cubic combine nonnegatively, as does the linear term. For $v \ge 0$, the quadratic is at least $2757 - 27478^2/(4 \cdot 396841) > 0$, and the remainder nonnegative.

Q105.2.4 The unit low-domain coefficient twelve

If $u \le 1$ and $Y + u \le q + 11/8$, then $C_* = 12$. After using (Q105.1.4) and clearing denominators the polynomial is $64q^4 + 176q^3 - 7q^2 - 704q + 540$. At $v = q - 1 \in [-1,1]$ it is $69 + 66v + 905v^2 + 432v^3 + 64v^4$. For $v \ge 0$ this is immediate; for $v \le 0$ it is at least $69 + 66v + 473v^2 > 0$, since $66^2 < 4 \cdot 69 \cdot 473$.

Q105.2.5 The unit middle-domain coefficient fifteen

If $u \le 35/32$ and $Y + u \le q + 55/32$, then $C_* = 15$. The polynomial after clearing denominators is $1024q^4 + 3520q^3 + 1379q^2 - 14080q + 8820$. At $v = q-1$ it is $663 + 3334v + 18083v^2 + 7616v^3 + 1024v^4$. For $v \ge 0$ it is immediate; for $v \le 0$ it is at least $663 + 3334v + 10467v^2 > 0$, since $3334^2 < 4 \cdot 663 \cdot 10467$.

Q105.3 The actual square and finite propagation

For $0 < d \le 1/4$, $f_2 + h_4 = 2 - 2d - s - w + dg$. Let

$$m = 1 - d, \quad k = 1/m \in [1, 4/3], \quad a = 2 - k = 1 - kd,$$
$$U = s + w - dg = (dI + \mathsf{X} - \mathsf{X}z)g', \quad R = -kU, \quad V = \mathbb{E}U^2 = d^3 + 2d\delta_2 + \delta_3 + \delta_4. \tag{Q105.3.1}$$

Actual-arm AM–GM gives $F \le q(m - U/2)^2$. Expand $T(qU)$, retaining its mean, and use $|K(gU)| \le \mathbb{E}|gU| \le \sqrt{dV}$ and $T(qU^2) \le 4V$ to obtain

$$TF/m^2 \le \bar{f}_2 + zR + \delta\mathbf{1}, \quad \delta = -k(d^2 + 2\delta_2) + k\sqrt{dV} + k^2V. \tag{Q105.3.2}$$

For the full low domain set $v = \delta_2/d$, $b = \theta(1 + \theta) = 585/1024$ and $J_0 = V/d \le d^2 + (2d + b)v$. Using $\sqrt{J_0} \le 3/8 + (2/3)J_0$ gives

$$\delta/(kd) \le 3/8 - d + 2d^2 + v(-2 + 4d + 2b) \le 3/8. \tag{Q105.3.3}$$

If the v coefficient is nonpositive, discard it; otherwise put $v = \theta$. The remaining quadratic increases with d and has endpoint $-1099/16384$ at $1/4$ for the part after $3/8$. Thus $\delta \le d/2$, and nonunit rows use $D = d/2$. For $d \le 1/8$, a stronger choice is $D = d/6$. Let $A_0 = b + 2d$, $B_0 = 2 - b - 4d > 0$, $J = d^2 + A_0v$. Since $J_0 \le J$ and $\sqrt{J_0} + kJ_0$ increases, first enlarge to $\sqrt{J} + kJ$, then substitute the exact $v = (J - d^2)/A_0$:

$$\delta/d \le k[-d - 2v + \sqrt{J} + kJ] = -\frac{bd}{mA_0} + \frac{1}{m}\left[\sqrt{J} - \frac{B_0}{mA_0}J\right] \le -\frac{bd}{mA_0} + \frac{A_0}{4B_0}. \tag{Q105.3.4}$$

The last upper is at most $1/6$ exactly when

$$P(d) = b(4 - 5b) + (8 - 4b - 7b^2)d - (36 + 24b)d^2 + 28d^3 \ge 0. \tag{Q105.3.5}$$

It is concave on $[0, 1/8]$, since $P'' = -2(36 + 24b) + 168d < 0$. At zero it is positive, and $P(1/8) = 63/128 + (25/8)b - (47/8)b^2 > 0$. The latter is concave in $b \in [1/2, 3/5]$, with positive endpoint values. For either valid D, set $D_i = \mathbb{E}z^iR$. Positive propagation gives

$$0 < T^LF/m^2 \le Y_L = \bar{f}_{L+1} + z^LR + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i\bar{f}_{L-1-i}, \tag{Q105.3.6}$$

with the exact coefficient generator

$$D_{2j-1} = (-1)^{j+1}k(d\delta_j + \delta_{j+1}), \qquad D_{2j} = (-1)^{j+1}k\delta_{j+2} \quad (j \ge 1). \tag{Q105.3.7}$$

Let $G_n(z) = \sum_{j=0}^{n-1} z^j$, $G_0 = 0$, $\beta_1 = \beta_2 = 0$ and $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor}(-1)^{j-1}\delta_j$. Then

$$S'_n = \sum_{j=0}^{n} z^j\mathbf{1} = \bar{f}_n + \sum_{j=3}^{n} \beta_j\bar{f}_{n-j}, \quad \kappa d \le \beta_j \le d \quad (j \ge 3). \tag{Q105.3.8}$$

Also $S'_n = q + zG_{n-1}g'$ and $\mathbb{E}zG_ng' = -\beta_{n+2}$, from finite induction with $T = z + J$. At $D = d/2$ the losses $\gamma_j = \beta_j - D_{j-2}$ satisfy

$$\gamma_3 \ge d/8, \quad \gamma_4 \ge 3d/4, \quad \gamma_j \ge 41d/96 \quad (j \ge 5). \tag{Q105.3.9}$$

Indeed $D_1 \le 7d/8$, $D_2 < d/4$; for $i \ge 3$ every positive $D_i < d/6$. The initial positive odd and even values satisfy $D_5 \le 1183d/8192 < d/6$ and $D_6 \le (4/3)\theta^4 d < d/6$, followed every four indices by a further factor at most θ^2. For $Y = Cq + zPg' + c\mathbf{1} = (C+c)\mathbf{1} + Qg'$, $Q = C + zP = Q_R + zQ_I$,

$$u = \mathbb{E}Y = C + c + \mathbb{E}Qg', \quad \operatorname{Var} Y = \|Qg'\|_2^2 - (\mathbb{E}Qg')^2, \quad \operatorname{Cov}(f_2, Y) = \int [-Q_R + xQ_I]\, d\nu + d\mathbb{E}Qg'. \tag{Q105.3.10}$$

This uses the actual $f_2 = 1 - g' + zg'$ with mean $1-d$. For the exact propagated Y_L, before discarding losses,

$$C_{L+1} = C_L + c_L, \quad P_{L+1} = Q_L = C_L + zP_L, \qquad c_{L+1} = \mathbb{E}(Q_L g'). \tag{Q105.3.11}$$

These are the direct expansion of TY_L.

Q105.4 The unit connector on the full low domain

The exact first-row upper is

$$Y_1 = q + zP_1 g' + D\mathbf{1}, \quad P_1 = aI - k\mathsf{X} + k\mathsf{X}z, \quad Q_1 = 1 + zP_1. \tag{Q105.4.1}$$

Its frequency coordinates are

$$(Q_{1,R}, Q_{1,I}) = (1 - kx^2, a - kx). \tag{Q105.4.2}$$

Both are nonnegative. Since $k \ge 1, a \le 1, x(1+x) \le 1$,

$$|Q_1|^2 \le (1-x^2)^2 + x(1-x)^2 \le 1 + x - 3x^2 \le 13/12. \tag{Q105.4.3}$$

Also

$$|P_1|^2 = a^2 + kx[-2a + kx(1+x)] \le a^2 \le 1, \tag{Q105.4.4}$$

since $kx(1+x) \le 195/256 < 4/3 \le 2a$. The exact mean is $u_1 = 1 + D - ad + k\delta_2$, $\mathbb{E}Q_1 g' = -ad + k\delta_2 < 0$, and

$$\operatorname{Cov}(f_2, Y_1) = -d + a\delta_2 + d(-ad + k\delta_2) \le -\kappa d. \tag{Q105.4.5}$$

The mean is negative in its nonconstant part because $a - k\theta \ge 2/3 - (4/3)\theta = 1/8$. The variance is at most $13d/12$. For $0 < d \le 1/8$ take $D = d/6$ from (Q105.3.4)–(Q105.3.5). Since $k \le 8/7$, $u_1 \le 1 - (19/84)d \le 1 - d/5$. Then $Y_1 + u_1 \le q + 1 + \sqrt{d} \le q + 11/8$, so the unit coefficient twelve applies. For the retained mean margin put $\alpha = 41/96$, $\beta = 45/28$, $v = 1 - \alpha d + \beta d^2$. The exact mean and $k = 1 + d/m$ give $0 \le u_1 \le v \le 1$. The polynomial $F_0(u) = (1-d)u^4 - 4\kappa d u^3$ has maximum zero or $F_0(v)$ on $[0, v]$, its only positive critical point being a minimum. Here $F_0(v) > 0$ since $v \ge 1 - \alpha d, d \le 1/8$. Using $\alpha < 7/16$, $\beta < 13/8$, $v^4 \le 1 - 4\alpha d + 8d^2$ and $v^3 \ge 1 - 3\alpha d$. The first follows from the fourth expansion of $e = v - 1 \in [-\alpha d, 0]$ and $4\beta + 6\alpha^2 + \alpha^4/64 < 8$. Thus

$$\mathbb{E}[f_2 Y_1^4] \le 1 + (95/12)d + 13d^2. \tag{Q105.4.6}$$

The linear coefficient is exactly $13 - 1 - 4\alpha - 4\kappa = 95/12$; the quadratic is $8 + 4\alpha + 12\kappa\alpha < 13$. Since $m^{-8} \ge 1 + 8d + 36d^2$ and $m^8 \ge (7/8)^8 > 1/3$,

$$\mathbb{E}[f_2 (TF)^4] \le 1 - m^8 d/12 \le 1 - d/40. \tag{Q105.4.7}$$

For $1/8 \le d \le 1/4$, use $D = d/2$. The mean has $u_1 \le v = 1 + 3d/8 \le 35/32$; $Y_1 + u_1 \le q + 1 + \sqrt{d} + 7d/8 \le q + 55/32$. The coefficient fifteen and the same covariance and variance yield

$$\mathbb{E}[f_2(TF)^4] \le m^8[v^3\{1 - 3d - (3/8)d^2\} + (65/4)d] =: S(d). \tag{Q105.4.8}$$

The first factor in braces is positive. Both m^8v^3 and $1 - 3d - (3/8)d^2$ decrease, and dm^8 decreases for $d \ge 1/8 > 1/9$. Thus S decreases. At $d = 1/8$, use $(7/8)^8 < 9/25$, $(67/64)^3 < 7/6$ and the brace less than $5/8$ to obtain

$$S(d) \le S(1/8) < (9/25)[(7/6)(5/8) + 65/32] = 159/160 \le 1 - d/40. \tag{Q105.4.9}$$

This pays the entire unit low domain with its actual direction.

Q105.5 Finite-frequency contractions for the short nonunit rows

Let $D = d/2$, $C_2 = 1 + D$. The second Q in (Q105.3.11) has

$$Q_{2,R} = C_2 - ax + kx^2, \qquad Q_{2,I} = 1 - kx^2. \tag{Q105.5.1}$$

Its exact squared gap is

$$|Q_2|^2 - C_2^2 = xB(k,x), \quad B = 3k + 2/k - 6 + (k^2 - k + 3)x + (2k^2 - 6k)x^2 + k^2x^3 + k^2x^4. \tag{Q105.5.2}$$

This B is convex in k. At $k = 1$, $B \le -1 + 3x - 2x^2 + x^3$, increasing to $-1451/32768$. At $k = 4/3$, it increases in x, since nine times its derivative is at least $31 - 80x + 48x^2 > 0$; the endpoint is $196657/589824 < 7/20$. The convex chord gives

$$B(k,x) \le (21/20)(k-1) \le (7/5)d, \quad |Q_2|^2 \le C_2^2 + (91/160)d \le (1+d)^2. \tag{Q105.5.3}$$

The third-row constant is

$$C_3 = 1 - d/2 + k(d^2 + \delta_2), \qquad 31/32 \le C_3 \le 1 + 3d/8. \tag{Q105.5.4}$$

The lower bound uses $\delta_2 \ge d^2$ and $1 - d/2 + 2d^2 \ge 31/32$. The coordinates are

$$Q_{3,R} = C_3 - x + kx^3, \quad Q_{3,I} = C_2 - ax + kx^2 =: R_2(x). \tag{Q105.5.5}$$

One has $\|Q_3\| \le C_3$:

$$|Q_3|^2 - C_3^2 = x\{-2C_3(1 - kx^2) + x(1 - kx^2)^2 + R_2(x)^2\}. \tag{Q105.5.6}$$

The bracket's second derivative is at least $4k(C_3 + R_2 - 3x) > 0$, since $C_3 \ge 15/16$, $R_2 \ge 3/4$, $3x \le 39/32$. At zero it is at most $81/64 - 30/16 < 0$. At θ, use $1 - k\theta^2 \ge 599/768 > 7/9$, $\theta(1 - \theta^2)^2 < 29/100$, $R_2(\theta) \le 275/256$ and $(275/256)^2 < 7/6$ to get $-35/24 + 29/100 + 7/6 = -1/600$. Convexity covers the full interval.

Q105.6 Four exact short-connector payments

Each row comes from the same $Y_L = T^{L-1}Y_1$ through the original mean and covariance dictionary. At $L = 2$, $C_2 = 1 + D$, $P_2 = Q_1$, $c_2 = k(d^2 + \delta_2) - d$, and

$$u_2 = 1 - 3d/2 + k(d^2 + \delta_2 + \delta_3) \le 1 - (311/768)d \le 1 - 2d/5. \tag{Q105.6.1}$$

The bounds $c_2 \le -d/8$ and $\|P_2\|^2 \le 13/12 < (21/20)^2$ verify coefficient C, twelve. By (Q105.5.3), $\operatorname{Var} Y_2 \le (1+d)^2 d$. The covariance integrand is $-C_2 + (3-k)x - kx^2 - kx^3 \le -1 + 2x - x^2 - x^3$, increasing to $-13749/32768 < -2/5$ at θ. Also $\mathbb{E} Q_2 g' = -d + k\delta_3 < 0$, so

$$\operatorname{Cov}(f_2, Y_2) \le -2d/5. \tag{Q105.6.2}$$

For $v = 1 - 2d/5$, the same endpoint-minimum argument as at $L = 1$ allows the mean-covariance upper at $u = v$, with positive endpoint value. Thus

$$\mathbb{E}[f_2 Y_2^4] \le W_2(d) := (1-d)v^4 - (8/5)dv^3 + 12d(1+d)^2. \tag{Q105.6.3}$$

At $L = 3$, take C_3 from (Q105.5.4), $P_3 = Q_2$ and $c_3 = -d + k\delta_3 \le -3d/4$. Then

$$u_3 = 1 - 5d/2 + 2\delta_2 + (k - 1/2)d^2 \le 1 - (71/48)d \le 1 - 7d/5. \tag{Q105.6.4}$$

The bounds $\|P_3\| \le 1 + d \le 5/4$, $C_3 \le 1 + d/2$, $c_3 \le -3d/4$ and $u_3 \le 1 - d$ verify coefficient B, $21/2$. The variance is at most $C_3^2 d \le (1 + 3d/8)^2 d$. The covariance integrand simplifies to $-C_3 + (2 + D)x - ax^2$, at most $-15/16 + (17/8)\theta - (2/3)\theta^2 = -283/1536 < 0$. Also $\mathbb{E} Q_3 g' = -(1 + D)d + a\delta_2 - k\delta_3 < 0$. With $v = 1 - 7d/5$,

$$\mathbb{E}[f_2 Y_3^4] \le W_3(d) := (1-d)v^4 + (21/2)(1 + 3d/8)^2 d. \tag{Q105.6.5}$$

At $L = 4$, $C_4 = u_2$, $P_4 = Q_3$, $c_4 = \mathbb{E} Q_3 g' \le -\kappa d$, and

$$1 - 3d/2 \le C_4 \le 1 - 2d/5, \quad Q_{4,R} = C_4 - xR_2(x), \quad Q_{4,I} = C_3 - x + kx^3. \tag{Q105.6.6}$$

Since $R_2 \ge 3/4$ and $\|P_4\| \le C_3 \le C_4 + d$, $|Q_4|^2 - C_4^2 \le x[(C_4 + d)^2 - (3/2)C_4] \le 0$. The opposite bracket is concave in C_4, with endpoint lower bounds $11/64$ and $11/400$ at $1 - 3d/2, 1 - 2d/5$ for $d \le 1/4$. The exact means combine as

$$c_4 + \mathbb{E} Q_4 g' = -2d + 2\delta_2 - kV, \qquad u_4 \le 1 - (127/80)d \le 1 - 3d/2. \tag{Q105.6.7}$$

The covariance integrand is $-C_4 + x(R_2 + C_3 - x + kx^3)$. Discard its nonpositive higher-degree group to bound it by $-C_4 + \theta(1 + D + C_3) \le -(1 - \theta)C_4 + \theta(1 + 3d/2) \le 3/16$. As $\mathbb{E} Q_4 g' = -C_3 d + \delta_2 - k\delta_4 < 0$, $\operatorname{Cov}(f_2, Y_4) \le d/4$. The bounds $\|P_4\| \le 35/32$, $C_4 \le 1$, $c_4 \le -\kappa d$ and $u_4 \le 1 - 3d/2$ verify coefficient A. The variance is at most d. For $v = 1 - 3d/2$,

$$\mathbb{E}[f_2 Y_4^4] \le W_4(d) := (1-d)v^4 + dv^3 + (21/2)d. \tag{Q105.6.8}$$

At $L = 5$, $C_5 = u_3$, $P_5 = Q_4$, $c_5 = \mathbb{E} Q_4 g' \le -9d/16$, and

$$17/32 \le C_5 \le 1 - 7d/5, \qquad \|P_5\| \le 1. \tag{Q105.6.9}$$

The lower bound follows from $u_3 \ge 1 - (5/2)d + (5/2)d^2 \ge 17/32$. Since $Q_5 = C_5 + zQ_4$, $Q_{4,I} \ge 31/32 - x$, and $\|Q_4\| \le 1$, $|Q_5|^2 \le C_5^2 + x[1 - 2C_5(31/32 - x)]$. This is convex in C_5. At $C_5 = 1$ it is at most $1 + x(-15/16 + 2x) \le 1$; at $17/32$ it is at most $289/1024 + (17/16)\theta^2 < 1$. The mean dictionary gives $\mathbb{E} Q_5 g' = -C_4 d + C_2\delta_2 - a\delta_3 + k\delta_4 < 0$ and

$$u_5 = C_5 - (C_3 + C_4)d + (2 + D)\delta_2 - a\delta_3 \le 1 - 2d. \tag{Q105.6.10}$$

The first sign uses $C_4 \ge 5/8$, $C_2 \le 9/8$, $k \le 4/3$; the second uses $C_3 + C_4 \ge 51/32$, $(2 + D)\theta \le 221/256$, $C_5 \le 1 - 7d/5$. The covariance integrand is $-C_5 + (C_3 + C_4)x - (2 + D)x^2 + ax^3$, at

most $-17/32+(67/32)\theta = 327/1024 < 1/2$. Hence the covariance is at most $d/2$ and the variance at most d. The row conditions $C_5 \le 1, c_5 \le -9d/16, u_5 \le 1-2d, \|P_5\| \le 1$ verify coefficient A. For $v = 1-2d$,

$$\mathbb{E}[f_2 Y_5^4] \le W_5(d) := (1-d)v^4 + 2dv^3 + (21/2)d. \tag{Q105.6.11}$$

The four exact polynomials are

$$\begin{aligned}
W_2 &= 1 + (39/5)d + (712/25)d^2 + (1252/125)d^3 + (48/125)d^4 - (16/625)d^5,\\
W_3 &= 1 + (39/10)d + (5047/200)d^2 - (340151/16000)d^3 + (9261/625)d^4 - (2401/625)d^5,\\
W_4 &= 1 + (9/2)d + 15d^2 - (81/4)d^3 + (243/16)d^4 - (81/16)d^5,\\
W_5 &= 1 + (7/2)d + 20d^2 - 32d^3 + 32d^4 - 16d^5.
\end{aligned} \tag{Q105.6.12}$$

Compare coefficients with the positive binomial bound

$$m^{-8} \ge 1 + 8d + 36d^2 + 120d^3 + 330d^4. \tag{Q105.6.13}$$

Each difference is at least $d/5$. Since $m^8 \ge (3/4)^8 > 1/10$,

$$\mathbb{E}[f_2 (T^L F)^4] \le 1 - d/50 \qquad (L = 2,3,4,5). \tag{Q105.6.14}$$

Q105.7 Every finite $L \ge 6$

Set $\sigma = 41d/96 \le 41/384 < 1/8$. The needed finite-symbol facts are

$$|G_n - \sigma(2+z)| \le 1 - 2\sigma \quad (n \ge 3), \qquad |G_n - \sigma(3+2z+z^2)| \le 1 - 3\sigma \quad (n \ge 6). \tag{Q105.7.1}$$

For $H_n = G_{n-1} + z^n$, the same respective bounds hold for $n \ge 6$ and $n \ge 7$. For the two-term G bound let $a_2 = 1 - 2\sigma$. At $n = 3, 4$ the squared gaps are $x[-1 + 2\sigma + \sigma^2 + x]$ and $x[-1 + 2\sigma + \sigma^2 + (-1+2\sigma)x + x^2]$, nonpositive since $\sigma \le 1/8$ and $x \le \theta < 1 - 2\sigma$. For $n \ge 5$, the numerator after multiplication by $1 - z$ is $M_2 - z^n$ with $M_2 = a_2 + \sigma z + \sigma z^2$. Then $|M_2| \le a_2 + \sigma\sqrt{x + x^2} \le 1$ and $a_2^2(1+x) - |M_2|^2 = x[1 - 2\sigma - \sigma^2(1+x)] \ge (4715/8192)x$, even under the looser $\sigma \le 3/16$. The finite tail increases the square by at most $2x^{n/2} + x^n$; divided by x, this is at most $2\theta^{3/2} + \theta^4 < 9/16 < 4715/8192$. An exact comparison uses $\sqrt{\theta} < 13/20$ to bound the left side by $169/320 + 28561/1048576 < 9/16$. For H_n, its numerator tail is $z^{n-1}(1 - z + z^2)$, whose extra factor has squared modulus $1 - x + x^2 \le 1$; $n \ge 6$ suffices. Setting $\sigma = 0$ also gives the later-used $|G_n| \le 1$ for $n \ge 3$. For the three-term bound, the base numerator is $M = (1-3\sigma) + \sigma z + \sigma z^2 + \sigma z^3$, whose norm is at most $(1 - 3\sigma) + \sigma\sqrt{x - x^2 + x^3} \le 1$. Writing $a = 1 - 3\sigma$,

$$a^2(1+x) - |M|^2 = x[a^2 + 2a\sigma - \sigma^2(1 - x + x^2)] \ge x(1 - 4\sigma + 2\sigma^2) \ge 17x/32.$$

For $G_n, n \ge 6$ and $H_n, n \ge 7$ the finite tail increment divided by x is at most $2\theta^2 + \theta^5 < 7/20 < 17/32$. Indeed $\theta^5 < \theta^4/2$ gives the upper $169/512 + 28561/2097152 < 7/20$. The zero frequency is direct, completing all finite integers. At $L \ge 6$, (Q105.3.6), (Q105.3.8) and (Q105.3.9) give coefficient at least σ on the last four actual losses $\bar{f}_0, \bar{f}_1, \bar{f}_2, \bar{f}_3$. Retain them and use $D\bar{f}_{L-1} \le DS'_{L-1}$ to get

$$\begin{aligned}
Y &= Cq + zPg' + c\mathbf{1}, \quad C = 1 + D - 3\sigma = 1 - 25d/32, \quad c = -\sigma(1-d),\\
P &= G_L - k(dI + \mathsf{X})z^{L-1} + k\mathsf{X}z^L + DG_{L-2} - \sigma(2+z), \qquad T^L F \le m^2 Y.
\end{aligned} \tag{Q105.7.2}$$

At a frequency set $A_1 = k(d+x)$, $B_1 = kx$, so $0 \le B_1 \le A_1 \le 7/8$. The first three unshifted terms are $(1-A_1)G_L + (A_1-B_1)G_{L-1} + B_1 H_L$. For $Q = C + zP$ they become $(1-A_1)G_{L+1} + (A_1 - B_1)G_L + B_1 H_{L+1}$, with DG_{L-1} added and $\sigma(3+2z+z^2)$ subtracted. Thus

$$\|P\| \le 1 + D - 2\sigma = 1 - 17d/48 \le 1, \qquad \|Q\| \le 1 + D - 3\sigma = C. \tag{Q105.7.3}$$

Retain the exact mean $\mathbb{E}Qg' = -\beta_{L+2} + D_L - D\beta_L + 2\sigma d \le -\sigma - D\kappa d + 2\sigma d < 0$, so

$$u \le 1 - (157/96)d + (63/64)d^2 \le 1 - 4d/3. \tag{Q105.7.4}$$

For $G_n, n \ge 4$, finite coordinates give $-\operatorname{Re} G_n + x \operatorname{Im}_z G_n \le -9/32$, using $\operatorname{Re} G_n \ge 1-x$, $\operatorname{Im}_z G_n \le 1 - x + x^2$ and $1 - 2\theta + \theta^2 - \theta^3 = 9355/32768 > 9/32$. For $H_n, n \ge 7$, the additional z^n contributes in absolute value less than $\sqrt{1+\theta}\theta^{n/2} < 1/16$, so its upper is $-7/32$. The negative three-term shift costs at most 3σ. Using (Q105.3.10),

$$\operatorname{Cov}(f_2, Y) \le [-7(1+D)/32 + 3\sigma]d \le d/4, \quad \operatorname{Var} Y \le C^2 d \le d. \tag{Q105.7.5}$$

The bounds $C \le 1, c \le -41d/128, u \le 1 - 4d/3, \|P\| \le 1$ verify coefficient A. At $v = 1 - 4d/3$,

$$\begin{aligned} \mathbb{E}[f_2 Y^4] &\le (1-d)v^4 + dv^3 + (21/2)d \\ &= 1 + (31/6)d + 12d^2 - (400/27)d^3 + (832/81)d^4 - (256/81)d^5. \end{aligned} \tag{Q105.7.6}$$

Comparison with (Q105.6.13) leaves at least $(17/6)d$; therefore every finite $L \ge 6$ satisfies $\mathbb{E}[f_2 (T^L F)^4] \le 1 - d/50$.

Q105.8 High domain, genuine decorations and reference use

For $d \ge 1/4$, $h_4 \le 2\mathbb{E}f_3 = 2(1-2d) \le 1$. Positivity gives $T^L(q f_2 h_4) \le T^L(q f_2)$. The same genuine-$f_2$ reference fourth proved in Section N.7 is at most one for every $L \ge 1$. Only the estimate changes across the analytic intervals; the canonical direction is unchanged. For original equal pairs $0 < E_t, E_o \le 1$, $\mathbb{E}[E_o f_2 \{T^L(E_t F)\}^4] \le \mathbb{E}[f_2 (T^L F)^4]$. If the equal-pair half-length sum is S, one actual reference component has

$$e = L + (1+2+4) + 2 + 2S = L + 9 + 2S, \qquad |V| = L + 10 + 2S. \tag{Q105.8.1}$$

The receiving two-edge reference exists in the original tree and is single in the fourth-moment comparison; only the terminal branch is copied. Complete reversal of K reverses every actual edge, changes the reference to $\bar{f}_2$ and propagation to A^L, with corresponding scope. For a genuine four-terminal star whose central side is the existing outward two-edge arm and equal pairs, each branch may independently use this $(1,2,4)$ role at any $L > 0$, or another precisely established same-f_2 role. For actual branch messages X_i, original Hölder gives

$$Z_D \le \mathbb{E}[f_2 \prod_{i=1}^{4} X_i] \le \prod_{i=1}^{4} (\mathbb{E}[f_2 X_i^4])^{1/4} \le 1. \tag{Q105.8.2}$$

This concerns the stated central-side subfamily; other gluing uses the explicit slot requiring this same-direction reference and does not assign an arbitrary compound output the same certificate. The enhanced-edge comparison in the original uniform tournament model is $\hom(D,T) \le 2^{-e}|T|^{e+1} Z_D$, retaining all original noninjective maps, decoration edges and subdivisions.

N.17 The genuine f_2 reference at the reversed unit terminal $(1,2,3)$

Q114.0 The actual unit interface

Let (X,μ) be any nonempty original finite probability space and K a real skew kernel with $|K|\le 1$. Every operator, expectation, inner product, norm, adjoint and spectral measure uses this same original μ. Put

$$Jv=(\mathbb{E}v)\mathbf{1},\quad A=J+K,\quad T=J-K=A^*,\quad g=K\mathbf{1},\quad p=1+g,\quad q=1-g,$$

$$f_n=A^n\mathbf{1},\quad \bar f_n=T^n\mathbf{1},\quad h_n=Tf_{n-1},\quad r=Kg,\quad s=K^2g,\quad \delta_j=\|K^j\mathbf{1}\|_2^2,\quad d=\delta_1. \tag{Q114.0.1}$$

The terminal residual after maximal equal pairing is $(1,2,3)$ and the original connector is one edge. The receiver's original outward two-edge arm gives the genuine reference f_2. Use the fixed actual terminal and output

$$F_{\text{alt}}=qf_2\bar f_3,\qquad X=TF_{\text{alt}}. \tag{Q114.0.2}$$

For arbitrary original equal-pair products E_t,E_o,

$$\mathbb{E}[E_of_2\{T(E_tF_{\text{alt}})\}^4]\le 1-d/256\le 1. \tag{Q114.0.3}$$

The terminal unit arm points inward, the two-edge arm outward throughout, and the three-edge arm inward throughout. The connector points to the receiver, whose two-edge arm is outward. The direction precedes the host. The three-edge arm is $\bar f_3$, rather than canonical h_3. The low and high analyses use this same direction. For $0\le d\le 1/4$ the stronger bound is

$$\|X\|_{L^4(f_2\mu)}\le 1-d/256. \tag{Q114.0.4}$$

The high domain gives $\mathbb{E}[f_2X^4]\le 83521/131072<3/4$.

Q114.1 Original foundations, positivity and the zero boundary

The original Q3-C014 finite framework, with $\theta=13/32$, gives

$$0\le d\le 1/3,\quad \|K\|^2\le\theta,\quad |Kv(x)|\le\|v\|_2,\quad d^2\le\delta_2\le\theta d,\quad \delta_{j+1}\le\theta\delta_j. \tag{Q114.1.1}$$

Skewness gives $\mathbb{E}K^{2j}\mathbf{1}=(-1)^j\delta_j$ and $\mathbb{E}K^{2j+1}\mathbf{1}=0$. In particular $\mathbb{E}r=-d$, $\mathbb{E}s=0$, $\mathbb{E}(gr)=\mathbb{E}(rs)=0$, and $\mathbb{E}(gs)=-\delta_2$. After zero-mass atoms are removed, $p,q\ge\mu(x)>0$, and A,T are positive. Actual tree messages are nonnegative; path means are at most one, $0\le f_n,\bar f_n\le 2$ and $f_\ell h_\ell\le 1$. The genuine reference obeys

$$f_2=p+r,\qquad \mathbb{E}f_2=1-d,\qquad 0<f_2\le 2. \tag{Q114.1.2}$$

At $d=0$, $g=0$ with possibly nonzero K; original leaf integration gives all actual messages one. Below every division by d requires $d>0$. For a mean-zero original vector w, the centered row gives

$$|Kw(x)|^2=|\mathbb{E}_y[(K(x,y)-g(x))w(y)]|^2\le p(x)q(x)\|w\|_2^2. \tag{Q114.1.3}$$

All spectral polynomials are finite. The original spectral measure ν at g is supported in $[0,\theta]$, has mass d and satisfies $\int x^j\,d\nu=\delta_{j+1}$. Only global moment inequalities such as $\delta_2\ge d^2$ are used.

Q114.2 The true five-edge mean and the high domain

The short-path identities are

$$f_2 = p + r, \qquad \bar{f}_3 = q - d\mathbf{1} + r - s. \tag{Q114.2.1}$$

The original adjoint gives

$$\mu_5 := \mathbb{E}[f_2 \bar{f}_3] = \langle A^2\mathbf{1}, T^3\mathbf{1}\rangle = \mathbb{E}[A^5\mathbf{1}] = 1 - 4d + 2\delta_2 + d^2. \tag{Q114.2.2}$$

The same finite expansion follows by applying A once more to $f_3 = p + r + s - d\mathbf{1}$ and $f_4 = \mathbf{1} - 2d\mathbf{1} + (1-d)g + r + s + K^3 g$. In particular $\mu_5 \geq 0$, and the actual mean has

$$u = \mathbb{E}X = \mathbb{E}[pq f_2 \bar{f}_3] \leq \mu_5. \tag{Q114.2.3}$$

Also $\mathbb{E}F_{\text{alt}} \leq 2\mu_5$ and $X \leq 4\mu_5$, so

$$\mathbb{E}[f_2 X^4] \leq 2\mathbb{E}X^4 \leq 2(4\mu_5)^3 \mathbb{E}X \leq 128\mu_5^4. \tag{Q114.2.4}$$

For $d \geq 1/4$, $\mu_5 \leq 1 - (51/16)d + d^2 \leq 17/64$; the scalar decreases on $[1/4, 1/3]$. Therefore

$$\mathbb{E}[f_2 X^4] \leq 83521/131072 < 3/4 \qquad (1/4 \leq d \leq 1/3). \tag{Q114.2.5}$$

Q114.3 The actual square and a nonnegative auxiliary

For $0 < d \leq 1/4$ put

$$m = 1 - 3d/2 \in [5/8, 1], \quad k = 1/m \in [1, 8/5], \quad U = r + d\mathbf{1} - s/2, \quad V_0 = g + (d\mathbf{1} + s)/2. \tag{Q114.3.1}$$

Here m differs from $\mathbb{E}f_2$. The actual arm identities $f_2 + \bar{f}_3 = 2(m\mathbf{1} + U)$ and $f_2 - \bar{f}_3 = 2V_0$ give

$$F_{\text{alt}} = q\{(m+U)^2 - V_0^2\} \leq q(m+U)^2. \tag{Q114.3.2}$$

Define the analytic auxiliary

$$H = q + 2k(r + d\mathbf{1}) - ks, \qquad \mathcal{B} = TH. \tag{Q114.3.3}$$

We prove $H \geq 0$, $\mathbb{E}H = 1$, and $0 \leq \mathcal{B} \leq 2$. Let $w = 2g - r - d\mathbf{1}$, so $\mathbb{E}w = 0$ and $\|w\|_2^2 = 4d + \delta_2 - d^2$. The centered row yields $mH = (1 - 5d/2)q + 3d\mathbf{1} + Kw \geq a_0 q + b_0\mathbf{1} - \sqrt{pqd(4 + b - d)}$, where $a_0 = 1 - 5d/2 > 0$, $b_0 = 3d$, $b = \delta_2/d \leq \theta$. The exact square is $(a_0 q + b_0)^2 - b_0(2a_0 + b_0)pq = ((a_0 + b_0)q - b_0)^2 \geq 0$. Also $b_0(2a_0 + b_0) = 6d(1-d) \geq d(4 + b - d)$, since $b + 5d \leq 13/32 + 5/4 = 53/32 < 2$. Thus $H \geq 0$. Its mean is one, and the original kernel of T lies in $[0, 2]$, so $0 \leq TH \leq 2\mathbb{E}H = 2$. This is an analytic auxiliary, not an added actual tree message. Expanding at the original kernel gives

$$\mathcal{B} = q + r - 2k(s + dg) + kK^3 g. \tag{Q114.3.4}$$

Q114.4 The full genuine-f_2 exponential budget

We prove

$$\mathbb{E}[f_2 \mathcal{B}^4] \leq \exp\{3d + (29/2)d^2\}. \tag{Q114.4.1}$$

For $-1 < x \leq 1$,

$$\log(1 + x) \leq x - (3/10)x^2. \tag{Q114.4.2}$$

The difference has derivative $x(2-3x)/(5(1+x))$; on the negative half and $[0,2/3]$ its sign is direct, while on $[2/3,1]$ use the positive endpoint $7/10-\log 2$. This follows from $e^{7/10} > 1+7/10+49/200+343/6000 > 2$. For every real z,

$$e^{z-z^2/8} \le 1+z+(13/32)z^2. \tag{Q114.4.3}$$

The right side $Q(z) > 0$ and $[\log Q - z + z^2/8]' = z(13z^2-20z+8)/(128Q)$. The quadratic has negative discriminant, making zero the global minimum. Let $e_1 = f_2 - 1$, $e_2 = \mathcal{B} - 1$, and $S = e_1 + 4e_2$. Since $f_2, \mathcal{B} \in [0,2]$, (Q114.4.2) and continuity give $f_2\mathcal{B}^4 \le \exp\{S-(3/10)e_1^2-(6/5)e_2^2\}$. The exact completion is

$$S^2/8-(3/10)e_1^2-(6/5)e_2^2 = -(3/8)\{(9e_1+4e_2)/5\}^2+(26/25)(e_1+e_2)^2. \tag{Q114.4.4}$$

Here $e_1+e_2 = Kw_1$, with $w_1 = 2g-2kr+ks-2kd\mathbf{1}$, and

$$J_1 := \|w_1\|_2^2 = 4d+(4k^2-4k)\delta_2+k^2\delta_3-4k^2d^2. \tag{Q114.4.5}$$

The row bound gives $(e_1+e_2)^2 \le J_1$. Integrating (Q114.4.3) and using $1+y \le e^y$ yields

$$\mathbb{E}[f_2\mathcal{B}^4] \le e^{\mathcal{E}}, \quad \mathcal{E} = (26/25)J_1 + \mathbb{E}S + (13/32)\mathbb{E}S^2. \tag{Q114.4.6}$$

The quantity $\mathbb{E}Q(S)$ is strictly positive. From the actual auxiliary expansion, $S = -(3+8kd)g + 5r - 8ks + 4kK^3g$, $\mathbb{E}S = -5d+4k\delta_2$. Odd and even powers are orthogonal, giving

$$\mathbb{E}S^2 = (3+8kd)^2d+(25-48k-128k^2d)\delta_2+(64k^2-40k)\delta_3+16k^2\delta_4. \tag{Q114.4.7}$$

Thus $\mathcal{E} = \int P(x)\,d\nu$, where

$$\begin{aligned} P &= C_0+C_1x+C_2x^2+C_3x^3, \\ C_0 &= -21/25+(13/32)(3+8kd)^2-(104/25)k^2d, \\ C_1 &= (104/25)k^2-(983/50)k+325/32-52k^2d, \\ C_2 &= (676/25)k^2-(65/4)k > 0, \qquad C_3 = (13/2)k^2 > 0. \end{aligned} \tag{Q114.4.8}$$

The convex chord on $[0,\theta]$ gives

$$\mathcal{E} \le C_0d+S_0\delta_2, \qquad S_0 = C_1+C_2\theta+C_3\theta^2. \tag{Q114.4.9}$$

Set $h = k-1 \in [0,3/5]$, so $d = 2h/[3(1+h)]$. The exact expansion is

$$C_0 = 2253/800+(767/75)h+(1976/225)h^2. \tag{Q114.4.10}$$

As a quadratic in k, S_0 has coefficients $A_* = -2833753/153600$, $B_* = 80689/9600$, $C_* = 325/32$. At $k=1$ it equals $5757/51200 < 1/8$, with derivative $-2188241/76800 < -28$, and $A_* < -18$. Hence $S_0 \le 1/8-28h-18h^2$. Global $d^2 \le \delta_2 \le \theta d$ gives

$$\mathcal{E}/d \le C_0+\theta/8-(28h+18h^2)d. \tag{Q114.4.11}$$

Let $\epsilon = 851/6400 = 3-2253/800-\theta/8$. Subtract $3+(29/2)d$ and multiply by $1+h$ to get exactly

$$-\epsilon+(14/25-\epsilon)h+(77/225)h^2-(724/225)h^3. \tag{Q114.4.12}$$

Use $h^2 \le 3h/5$, $\epsilon > 17/128$, $287/375-\epsilon < 2/3$ and $724/225 > 16/5$ to bound it by $-17/128+(2/3)h-(16/5)h^3 < 0$. The positive cubic combination has maximum $\sqrt{10}/27 < 17/128$, with squared comparison $163840 < 210681$. This proves (Q114.4.1) on the whole low domain.

Q114.5 The true square remainder and its variance

Using $\mathbb{E}U = 0$ and $\mathbb{E}(gU) = \delta_2/2$, the expansion of (Q114.3.2) gives

$$X \le m^2\mathcal{B} - m\delta_2 + 2mK(gU) + T(qU^2). \tag{Q114.5.1}$$

Put $V = \mathbb{E}U^2 = \delta_2 + \delta_3/4 - d^2$. Then

$$|K(gU)| \le \sqrt{dV}, \qquad T(qU^2) \le \min\{4V,\, 2V + 2|\mathbb{E}[gU^2]|\}. \tag{Q114.5.2}$$

Since $\mathbb{E}g = 0$, Cauchy–Schwarz gives $|\mathbb{E}[gU^2]|^2 \le d\,\mathrm{Var}(U^2)$. Also $U = K(g - r/2) + d\mathbf{1}$ and $\mathbb{E}(gr) = 0$, so

$$\|U\|_\infty \le d + \sqrt{d + \delta_2/4} =: M_U, \qquad \mathrm{Var}(U^2) \le M_U^2 V - V^2. \tag{Q114.5.3}$$

The negative term $-V^2$ is retained in the subsequent estimate.

Q114.5.1 Reduction of the global moment parameters

Write $b = \delta_2/d$, $c = \delta_3/d$, $v = V/d = b + c/4 - d$ and $t = \sqrt{d}$. Then $b \in [d, \theta]$, $0 \le c \le \theta b$, and

$$0 \le v \le \alpha - d, \qquad \beta = 1 + \theta/4 = 141/128, \qquad \alpha = \beta\theta = 1833/4096. \tag{Q114.5.4}$$

Equations (Q114.5.1)–(Q114.5.3) give $X \le m^2\mathcal{B} + \rho d$, where either of the following expressions is an upper bound for the same error coefficient:

$$\begin{aligned} &-mb + 2m\sqrt{v} + 4v, \\ &-mb + 2m\sqrt{v} + 2v + 2t\sqrt{v\{(t + \sqrt{1 + b/4})^2 - v\}}. \end{aligned} \tag{Q114.5.5}$$

Both are bounded above by their values at $b = \theta$, $c = \theta b$. For fixed b, $v(W - v)$ increases with v, because $W = (t + \sqrt{1 + b/4})^2 \ge 1 > 2\alpha \ge 2v$. After setting $c = \theta b$, we have $v = \beta b - d$ and W increases with b. The derivative of $-mb + 2m\sqrt{v}$ with respect to b is $-m + m\beta/\sqrt{v} > 0$, since $v \le \alpha < \beta^2$. The other terms also increase; the case $v = 0$ follows by continuity. Thus the two global bounds are

$$\begin{aligned} \rho_4 &= -m\theta + 2m\sqrt{\alpha - d} + 4(\alpha - d), \\ \rho_2 &= -m\theta + 2m\sqrt{\alpha - d} + 2(\alpha - d) \\ &\qquad + 2t\sqrt{(\alpha - d)\{(t + \sqrt{\beta})^2 - (\alpha - d)\}}. \end{aligned} \tag{Q114.5.6}$$

One may use $\min\{\rho_2, \rho_4\}$. Every parameter belongs to the same original kernel.

Q114.5.2 Two finite square-root bounds

The tangent bound from a nonnegative square gives

$$\sqrt{\alpha - d} \le \tfrac{3}{4}(\alpha - d + 4/9) = 32881/49152 - 3d/4 \le 67/100 - 3d/4. \tag{Q114.5.7}$$

Also $\sqrt{\beta} < 21/20$. Put $\gamma = \beta - \alpha = 2679/4096$ and $I(t) = (\alpha - t^2)\{\gamma + 2t^2 + (21/10)t\}$. This bounds the expression inside the second square root in (Q114.5.6). For every $0 \le t \le 1/2$,

$$\sqrt{I(t)} \le Q(t) := 11/20 + (9/10)t - (21/20)t^2. \tag{Q114.5.8}$$

Here is the finite certificate. The polynomial Q is positive on this interval. Put $z = t - 1/4 \in [-1/4, 1/4]$. The exact expansion is

$$Q^2 - I = \frac{478089}{419430400} - \frac{97}{10240}z + \frac{75261}{102400}z^2 + \frac{53}{16}z^3 + \frac{1241}{400}z^4. \tag{Q114.5.9}$$

Its constant term exceeds $1/1000$, its linear coefficient belongs to $(-1/100, 0)$, and its quadratic coefficient exceeds $73/100$. For $z \geq 0$, the first three terms are at least $1/1000 - z/100 + (73/100)z^2 > 0$, and the last two are nonnegative. For $z \leq 0$, the linear term is nonnegative. The bracket multiplying z^2,

$$75261/102400 + (53/16)z + (1241/400)z^2,$$

increases on $[-1/4, 0]$; its left endpoint is at least $73/100 - 53/64 + 1241/6400 = 613/6400 > 0$. Thus (Q114.5.9) is strictly positive throughout and proves (Q114.5.8).

Q114.5.3 The two polynomial error bounds

Substituting (Q114.5.7)–(Q114.5.8) into (Q114.5.6) gives

$$\rho_2 \leq \Delta_2(d) = 747/800 + 2\alpha + (11/10)\sqrt{d} - (4961/1600)d - (21/10)d^{3/2} + (9/4)d^2, \tag{Q114.5.10}$$

$$\rho_4 \leq \Delta_4(d) = 747/800 + 4\alpha - (11041/1600)d + (9/4)d^2. \tag{Q114.5.11}$$

Both are positive on the ranges where they are used: for $d \leq 4/25$, $\Delta_2 \geq 747/800 + 2\alpha - 4961/10000 - 168/1250 > 0$; for $d \leq 1/4$, $\Delta_4 \geq 747/800 + 4\alpha - 11041/6400 > 0$.

Q114.6 Minkowski with the original reference mass

The norm $L^4(f_2\mu)$ uses the original nonnegative finite weighted measure, of total mass $\mathbb{E}f_2 = 1 - d$. Minkowski applied to the positive upper bound in (Q114.5.1), retaining the norm of the constant function, yields

$$\begin{aligned} \|X\|_{L^4(f_2\mu)} &\leq m^2\|\mathcal{B}\|_{L^4(f_2\mu)} + \Delta d(1-d)^{1/4} \\ &\leq m^2 \exp\{(3/4)d + (29/8)d^2\} + \Delta d. \end{aligned} \tag{Q114.6.1}$$

Use $\Delta = \Delta_2$ for $d \leq 4/25$ and $\Delta = \Delta_4$ for $d \geq 4/25$. The measure in this estimate remains $f_2\mu$ for the same kernel. The inequality $\log(1-x) \leq -x - x^2/2$ gives $2\log m \leq -3d - (9/4)d^2$. Consequently the first term of (Q114.6.1) is at most $\exp\{-h(d)\}$, where $h(d) = (9/4)d - (11/8)d^2 > 0$. Using $e^{-h} \leq 1 - h + h^2/2$ and putting $L(d) = 9/4 - (125/32)d + (99/32)d^2 - (121/128)d^3$, we obtain

$$\|X\|_{L^4(f_2\mu)} \leq 1 - d\{L(d) - \Delta(d)\}. \tag{Q114.6.2}$$

Q114.6.1 The interval $0 < d \leq 4/25$

Write $t = \sqrt{d} \leq 2/5$. The exact difference is

$$\begin{aligned} L(d) - \Delta_2(d) = P(t) = 21567/51200 &- (11/10)t - (1289/1600)t^2 \\ &+ (21/10)t^3 + (27/32)t^4 - (121/128)t^6. \end{aligned} \tag{Q114.6.3}$$

The polynomial decreases strictly: using $t^2 \leq (2/5)t$ and $t^3 \leq (4/25)t$ gives $P'(t) \leq -11/10 + (1159/800)t \leq -1041/2000 < 0$. Hence throughout this interval,

$$P(t) \geq P(2/5) = 142671/32000000 > 1/256. \tag{Q114.6.4}$$

Q114.6.2 The interval $4/25 \le d \le 1/4$

Here the exact difference is

$$L(d) - \Delta_4(d) = R(d) = -12129/25600 + (4791/1600)d + (27/32)d^2 - (121/128)d^3. \tag{Q114.6.5}$$

It increases strictly, since $R' \ge 4791/1600 - 363/2048 > 0$. Thus

$$R(d) \ge R(4/25) = 368623/16000000 > 1/256. \tag{Q114.6.6}$$

Equations (Q114.6.2), (Q114.6.4) and (Q114.6.6) prove (Q114.0.4) for all $0 < d \le 1/4$. They imply

$$\mathbb{E}[f_2 X^4] \le (1 - d/256)^4 \le 1 - d/256. \tag{Q114.6.7}$$

Together with the high-domain estimate and the case $d = 0$, this proves the bare interface in (Q114.0.3) for every allowed original kernel.

Q114.7 Decoration, full reversal and the original-edge count

Every genuine equal pair $f_\ell h_\ell$ is at most one in the original integral. Positivity of T gives

$$\mathbb{E}[E_o f_2 \{T(E_t F_{\text{alt}})\}^4] \le \mathbb{E}[f_2 (T F_{\text{alt}})^4].$$

This restores every original equal-pair multiplicity and its original edges and vertices. If the sums of pair half-lengths at the terminal and receiver are S_t, S_o, respectively, a single original component has $9 + 2S_t + 2S_o$ edges and $10 + 2S_t + 2S_o$ vertices. The fourth-moment test tree has $30 + 8S_t + 2S_o$ edges and $31 + 8S_t + 2S_o$ vertices. The reference two-edge arm remains a single original arm; analytic copies add no reference arm to the original tree. Reversing all original edges at once corresponds to $K \mapsto -K$. The terminal becomes $p\bar{f}_2 f_3$, the connector uses A, and the receiving reference becomes $\bar{f}_2$. The true reversed interface is

$$\mathbb{E}[\overline{E}_o \bar{f}_2 \{A(\overline{E}_t p \bar{f}_2 f_3)\}^4] \le 1 - d/256, \tag{Q114.7.1}$$

with the same scope. This is a simultaneous reversal of terminal, propagation and reference.

The genuine f_2 reference catalogue for $(1, 2, 3)$ can now be combined with (Q104.1.1), which supplies canonical $q f_2 h_3$ with the same-direction propagation T^L and bound $1 - d/500$ for every original $L \ge 2$. Define, before the host is chosen,

$$F_L = \begin{cases} q f_2 \bar{f}_3, & L = 1, \\ q f_2 h_3, & L \ge 2. \end{cases}$$

Then for every positive integer original connector length,

$$\mathbb{E}[E_o f_2 \{T^L (E_t F_L)\}^4] \le 1 - d/500. \tag{Q114.7.2}$$

The choice depends on the original integer length and precedes the kernel. A single original component has $L + 8 + 2S_t + 2S_o$ edges and $L + 9 + 2S_t + 2S_o$ vertices. The changed unit direction is used here only for the genuine reference fourth moment. The conclusion concerns this terminal reference interface; its receiving weight is exactly f_2, not a general reference f_a or a compound output.

N.18 The unit reference terminal $(1,2,3)$

Q165.0 The actual unit interface

For every original finite probability space and legal real skew kernel, the terminal residual $(1,2,3)$ with one connector edge has the fixed actual message $F = qf_2\bar{f}_3$. For every outward receiving arm f_a, $a \geq 3$, and original equal-pair factors E_o, E_t,

$$\mathbb{E}\big[E_o f_a\{T(E_t q f_2 \bar{f}_3)\}^4\big] \leq 1 - d/256. \tag{Q165.T172.0}$$

The direction is chosen independently of d, the host and the kernel.

Q165.1 Original space, direction and parameters

All quantities use the same original finite probability measure μ. Let K be real and skew, with $|K(x,y)| \leq 1$, and put

$$Jv = (\mathbb{E}v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad R = Kg, \quad S = K^2 g, \quad W = K^3 g,$$
$$\delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32.$$

The receiving length $a \geq 3$ is distinct from the vector R. At the terminal the unit arm points inward, the two-edge arm outward throughout, and the three-edge arm inward throughout. The connector points terminal-to-receiver; the receiving a-arm points outward. The Q3-C014 finite framework gives

$$0 \leq d \leq 1/3, \quad \|K\|^2 \leq \theta, \quad |Kv(x)| \leq \|v\|_2, \quad d^2 \leq \delta_2 \leq \theta d, \quad \delta_{j+1} \leq \theta\delta_j.$$

Original paths are nonnegative with means at most one, and $f_\ell h_\ell \leq 1$. Skewness gives $\mathbb{E}R = -d$, $\mathbb{E}S = 0$, $\mathbb{E}(gR) = \mathbb{E}(RS) = 0$ and $\mathbb{E}(gS) = -\delta_2$. After zero-mass atoms are deleted, $p, q \geq \mu(x) > 0$. At $d = 0$, K can still be nonzero; integration from the original leaves makes all actual messages one. We use $d > 0$ below; the final bounds have no division by d.

Q165.2 The receiving envelope and its pointwise bound

Define $G_n(z) = \sum_{j=0}^{n-1} z^j$ and $S_n = \mathbf{1} + G_n(K)g$. Finite positive convolution gives

$$S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad \beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j, \qquad \beta_1 = \beta_2 = 0.$$

For $n \geq 3$, $\kappa d \leq \beta_n \leq d$, and hence

$$0 \leq f_a \leq C_a := S_a - \beta_a\mathbf{1}. \tag{Q165.T172.1}$$

A separate pointwise bound on S_a is needed. At $z = i\sqrt{x}$, $0 \leq x \leq \theta$, every $n \geq 3$ satisfies $|G_n(z)| \leq 1$. For odd n the square is $(1 + x^n)/(1 + x) \leq 1$. For even $n = 2j$ it is $(1 - (-x)^j)^2/(1 + x)$: even j is direct, and odd j with $n \geq 6$ uses $2x^2 + x^5 < 1$ for $x \leq \theta < 1/2$. The finite rotation blocks of the original skew operator therefore give $\|G_a(K)\| \leq 1$. The row bound and commutativity yield

$$|S_a(x) - 1| = |KG_a(K)\mathbf{1}(x)| \leq \|G_a(K)\mathbf{1}\|_2 \leq 1.$$

Thus

$$0 \leq S_a \leq 2, \qquad 0 \leq C_a \leq 2. \tag{Q165.T172.2}$$

This uses G_a with $a \geq 3$; the exceptional G_2 is excluded. The later $a = 3$ coordinates use the actual $G_3 = 1 + z + z^2$, retaining its G_2 predecessor cost. Write $\beta = \beta_a$. Where $S_a > 0$, $1 - \beta/S_a \leq e^{-\beta/S_a} \leq e^{-\beta/2}$. If $S_a = 0$, (Q165.T172.1) implies $\beta = 0$ and $f_a = 0$. Consequently,

$$f_a \leq e^{-\beta_a/2} S_a. \tag{Q165.T172.3}$$

This pays the true constant loss $-\beta_a$ without replacing the reference weight by f_2.

Q165.3 The high domain

Let $F = q f_2 \bar{f}_3$ and $X = TF$. The actual short-arm identities are

$$f_2 = p + R, \qquad \bar{f}_3 = q - d\mathbf{1} + R - S, \qquad \mu_5 = \mathbb{E}[f_2 \bar{f}_3] = 1 - 4d + 2\delta_2 + d^2.$$

The adjoint gives $\mathbb{E}X = \mathbb{E}[pq f_2 \bar{f}_3] \leq \mu_5$. Also $\mathbb{E}F \leq 2\mu_5$ and $X \leq 2\mathbb{E}F \leq 4\mu_5$. Hence

$$\mathbb{E}[f_a X^4] \leq 2\mathbb{E}X^4 \leq 128\mu_5^4.$$

For $d \geq 1/4$, $\mu_5 \leq 1 - (51/16)d + d^2 \leq 17/64$, since that scalar decreases on $[1/4, 1/3]$. Thus $\mathbb{E}[f_a X^4] \leq 83521/131072 < 3/4 < 1 - d/256$, with the same direction.

Q165.4 The actual low-domain square and a nonnegative auxiliary

Let $0 < d \leq 1/4$ and set

$$m = 1 - 3d/2, \quad k = m^{-1} \in [1, 8/5], \quad U = R + d\mathbf{1} - S/2, \quad V_0 = g + (d\mathbf{1} + S)/2.$$

The two nonnegative arms have $f_2 + \bar{f}_3 = 2(m\mathbf{1} + U)$ and $f_2 - \bar{f}_3 = 2V_0$. Therefore

$$F = q\{(m + U)^2 - V_0^2\} \leq q(m + U)^2.$$

Define $H = q + 2k(R + d\mathbf{1}) - kS$ and $Z = TH$. To prove $H \geq 0$, put $w = 2g - R - d\mathbf{1}$. Then $\mathbb{E}w = 0$ and $\|w\|_2^2 = 4d + \delta_2 - d^2$. The centered row gives $|Kw|^2 \leq pq\|w\|_2^2$. With $a_0 = 1 - 5d/2 > 0$, $b_0 = 3d$ and $b = \delta_2/d \leq \theta$,

$$mH = a_0 q + b_0 \mathbf{1} + Kw \geq a_0 q + b_0 \mathbf{1} - \sqrt{pq\, d(4 + b - d)}.$$

The identity $(a_0 q + b_0)^2 - b_0(2a_0 + b_0)pq = ((a_0 + b_0)q - b_0)^2 \geq 0$ and $b_0(2a_0 + b_0) = 6d(1 - d) \geq d(4 + b - d)$, since $b + 5d \leq 53/32 < 2$, imply $H \geq 0$. As $\mathbb{E}H = 1$, $0 \leq Z \leq 2$. The actual expansion is

$$Z = q + R - 2k(S + dg) + kW = \mathbf{1} + P(K)g, \quad P(z) = -1 - 2kd + z - 2kz^2 + kz^3. \tag{Q165.T172.4}$$

The auxiliary Z uses the same original kernel; it is an analytic quantity rather than an additional tree message.

Q165.5 An exponential functional and an algebraic f_2 comparison

This paragraph establishes an algebraic comparison without importing an f_2-reference conclusion to change the weight. For $-1 < t \leq 1$, $\log(1 + t) \leq t - (3/10)t^2$. The derivative of the difference is $t(2 - 3t)/(5(1 + t))$; at $t = 1$ use $\log 2 < 7/10$, since $e^{7/10} > 1 + 7/10 + 49/200 + 343/6000 > 2$. For every real z,

$$e^{z - z^2/8} \leq 1 + z + (13/32)z^2. \tag{Q165.T172.5}$$

The right side $Q(z)$ is positive, and $[\log Q - z + z^2/8]' = z(13z^2 - 20z + 8)/(128Q)$. The quadratic has negative discriminant, so zero is the global minimum. For $1 + e, 1 + v \in [0, 2]$, let $s = e + 4v$. The exact completion is

$$s^2/8 - (3/10)e^2 - (6/5)v^2 = -(3/8)\{(9e + 4v)/5\}^2 + (26/25)(e + v)^2.$$

If $e + v = Kw$, the original row bound, (Q165.T172.5), and $1 + y \le e^y$ give

$$\mathbb{E}[(1 + e)(1 + v)^4] \le \exp\{(26/25)\|w\|_2^2 + \mathbb{E}s + (13/32)\mathbb{E}s^2\}. \tag{Q165.T172.6}$$

Here $\mathbb{E}Q(s) > 0$, and zero values are included by continuity. For the algebraic comparison take $e = G_2(K)g = g + R$ and $v = Z - 1$; call its exponent $\mathcal{E}_2$. The original spectral measure ν has support $[0, \theta]$, mass d, and $\int x^j \, d\nu = \delta_{j+1}$. Direct expansion at the same kernel gives

$$J_2 = 4d + (4k^2 - 4k)\delta_2 + k^2\delta_3 - 4k^2d^2,$$

$$\mathbb{E}s_2 = -5d + 4k\delta_2,$$
$$\mathbb{E}s_2^2 = (3 + 8kd)^2 d + (25 - 48k - 128k^2 d)\delta_2 + (64k^2 - 40k)\delta_3 + 16k^2\delta_4.$$

Thus $\mathcal{E}_2 = \int P_2(x) \, d\nu$, where

$$P_2 = C_0 + C_1 x + C_2 x^2 + C_3 x^3,$$
$$C_0 = -21/25 + (13/32)(3 + 8kd)^2 - (104/25)k^2 d,$$
$$C_1 = (104/25)k^2 - (983/50)k + 325/32 - 52k^2 d,$$
$$C_2 = (676/25)k^2 - (65/4)k > 0, \qquad C_3 = (13/2)k^2 > 0.$$

The convex chord gives $\mathcal{E}_2 \le C_0 d + S_0\delta_2$, with $S_0 = C_1 + C_2\theta + C_3\theta^2$. Let $h = k - 1 \in [0, 3/5]$, so $d = 2h/[3(1 + h)]$. The exact coefficients are

$$C_0 = 2253/800 + (767/75)h + (1976/225)h^2, \quad S_0 = 5757/51200 - (2188241/76800)h - (2833753/153600)h^2. \tag{Q165.T172.7}$$

For the full comparison payment use $S_0 \le 1/8 - 28h - 18h^2$ and $d^2 \le \delta_2 \le \theta d$ to get $\mathcal{E}_2/d \le C_0 + \theta/8 - (28h + 18h^2)d$. Subtract $3 + (29/2)d$ and multiply by $1 + h$; the result is exactly

$$-\epsilon + (14/25 - \epsilon)h + (77/225)h^2 - (724/225)h^3, \qquad \epsilon = 851/6400.$$

Using $h^2 \le 3h/5$, $\epsilon > 17/128$, $287/375 - \epsilon < 2/3$ and $724/225 > 16/5$, this is at most $-17/128 + (2/3)h - (16/5)h^3 < 0$. The positive cubic combination has maximum $\sqrt{10}/27 < 17/128$, with squared comparison $163840 < 210681$. Consequently

$$\mathcal{E}_2 \le 3d + (29/2)d^2. \tag{Q165.T172.8}$$

Q165.6 The all-a comparison at the same kernel

Take $e = S_a - 1 = G_a(K)g$ and $v = Z - 1$, whose pointwise ranges were proved above. Set

$$w_a = \{G_{a-1}(K) + I - 2kK + kK^2\}g - 2kd\mathbf{1}.$$

Then $e+v=Kw_a$. Equations (Q165.T172.3) and (Q165.T172.6) imply

$$\begin{aligned}\mathbb{E}[f_aZ^4]&\le e^{\mathcal{E}_a},\\ \mathcal{E}_a&=(26/25)\|w_a\|_2^2+\mathbb{E}s_a+(13/32)\mathbb{E}s_a^2-\beta_a/2,\\ s_a&=G_a(K)g+4P(K)g.\end{aligned}\tag{Q165.T172.9}$$

Every receiving and terminal symbol below is multiplied at the same x and integrated under the same original ν. At $z=i\sqrt{x}$ write

$$G_a(z)=1-xt+zC,\qquad 1-x\le t,C\le 1,\qquad \eta=1-C\in[0,x].$$

These finite alternating coordinates cover every $a\ge 3$, with $t=C=1$ at $a=3$. Skewness gives exactly

$$\beta_a=\int t\,d\nu,\qquad \beta_{a+1}=\int C\,d\nu.\tag{Q165.T172.10}$$

The first identity uses the imaginary coordinate t of G_{a-1}, since $\mathbb{E}G_{a-1}(K)g=-\beta_a$. The polynomial part of w_a has coordinates $C+1-kx$ and $t-2k$. Its mean $2kd-\beta_a$ crosses the actual constant $-2kd\mathbf{1}$, yielding

$$\|w_a\|_2^2-J_2=\int\{-2(2-kx)\eta+\eta^2+x(t^2-4kt)+4kdt\}\,d\nu.\tag{Q165.T172.11}$$

In particular the constant cross term $4kd\beta_a$ is retained. Let $A_0=-3-8kd+8kx$ and $B_0=5-4kx$. The coordinates of s_2 are A_0, zB_0, and those of s_a are $A_0-xt, z(B_0-\eta)$. Therefore

$$\mathbb{E}s_a-\mathbb{E}s_2=\int\eta\,d\nu,$$

$$\mathbb{E}s_a^2-\mathbb{E}s_2^2=\int\{-2A_0xt+x^2t^2+x(-2B_0\eta+\eta^2)\}\,d\nu.\tag{Q165.T172.12}$$

Substitute (Q165.T172.10)–(Q165.T172.12) into the exponent difference. The entire η contribution is at most

$$\eta\{-79/25+[(52/25)k+26/25-65/16]x+[(13/4)k+13/32]x^2\}\le 0.$$

Indeed, at $k\le 8/5$ the linear bracket is at most $611/2000<1/3$, and the quadratic bracket at most $897/160<6$. Since $x<1/2$, the braces are less than $-79/25+1/6+3/2=-112/75$. This first uses $\eta^2\le x\eta$; it bounds the full legal interval without requiring arbitrary independent choices of η,t to come from a receiving path. For the remaining terms use $t^2\le t$ to obtain

$$\mathcal{E}_a-\mathcal{E}_2\le\int t\{\chi-a_*x-b_*x^2\}\,d\nu,\tag{Q165.T172.13}$$

where

$$\chi=-1/2+(104/25)kd=-1/2+(208/75)h,\quad a_*=273/400-(13/75)h>0,\quad b_*=195/32+(13/2)h>0.$$

If $\chi\le 0$, the integral is nonpositive and (Q165.T172.8) applies throughout that domain. If $\chi\ge 0$, $t\le 1$ and $t\ge 1-x$ give

$$t(\chi-a_*x-b_*x^2)\le\chi-a_*x+(a_*-b_*)x^2+b_*x^3.$$

Add this to P_2. The new quadratic coefficient is positive: $C_2+a_*-b_* > 0$, because $C_2 \ge (1079/100)k$ and $b_* < (13/2)k$; the cubic coefficient is also positive. The convex chord gives

$$\mathcal{E}_a \le (C_0+\chi)d + S_*\delta_2, \qquad S_* = S_0 - \kappa(a_* + b_*\theta).$$

The exact expansion is

$$\begin{aligned} S_* &= -1443981/819200 - (4601499/153600)h - (2833753/153600)h^2 \\ &\le -7/4 - (299/10)h - (92/5)h^2 < 0. \end{aligned} \tag{Q165.T172.14}$$

Using only the original global moment inequality $\delta_2 \ge d^2$ gives

$$\mathcal{E}_a/d \le F(h) := 1853/800 + 13h + (1976/225)h^2 + dS_*. \tag{Q165.T172.15}$$

No pointwise lower spectral bound $x \ge d$ is imposed. For $d \le 4/25$, $h \le 6/19$. By (Q165.T172.14), $(1+h)[F(h) - 3 - (29/2)d]$ is at most

$$-547/800 + (3559/2400)h + (416/225)h^2 - (784/225)h^3 \le -547/800 + (3/2)h + (15/8)h^2.$$

The right side increases, with value $-6667/288800 < 0$ at $6/19$. For either sign of χ,

$$\mathbb{E}[f_a Z^4] \le \exp\{3d + (29/2)d^2\} \qquad (0 < d \le 4/25). \tag{Q165.T172.16}$$

For $4/25 \le d \le 1/4$, every $h \le 3/5$ satisfies

$$(1+h)[F(h) - 3 - 15d] \le -547/800 + (23/20)h + (15/8)h^2 - (69/20)h^3 < 0.$$

For a continuous proof, let $P(h) = 547/800 - (23/20)h - (15/8)h^2 + (69/20)h^3$. Its derivative is a convex quadratic on $[0, 1/2]$ with negative endpoint values, so P decreases to $P(1/2) = 57/800$. For $h \ge 1/2$, $P'' \ge 33/5$, $P'(1/2) = -7/16$, and completing the square gives $P \ge 57/800 - 245/16896 > 0$, using $57/800 > 1/16$ and $245/16896 < 1/64$. If $\chi \le 0$, (Q165.T172.8) is stronger. Thus

$$\mathbb{E}[f_a Z^4] \le \exp\{3d + 15d^2\} \qquad (4/25 \le d \le 1/4). \tag{Q165.T172.17}$$

Q165.7 The actual square remainder with its variance term

Let $V = \mathbb{E}U^2 = \delta_2 + \delta_3/4 - d^2$. Since $\mathbb{E}U = 0$ and $\mathbb{E}(gU) = \delta_2/2$, the actual square gives

$$X \le m^2 Z - m\delta_2 \mathbf{1} + 2mK(gU) + T(qU^2).$$

Here $|K(gU)| \le \sqrt{dV}$ and $T(qU^2) \le \min\{4V, 2V + 2|\mathbb{E}(gU^2)|\}$, while $[\mathbb{E}(gU^2)]^2 \le d\,\mathrm{Var}(U^2)$. Moreover,

$$\|U\|_\infty \le d + \sqrt{d + \delta_2/4}, \qquad \mathrm{Var}(U^2) \le [d + \sqrt{d + \delta_2/4}]^2 V - V^2.$$

The negative $-V^2$ is retained. Set $b = \delta_2/d$, $c = \delta_3/d$, $v = V/d = b + c/4 - d$ and $t = \sqrt{d}$. Then $b \in [d, \theta]$, $c \le \theta b$ and $v \le \alpha - d$, where $\beta_0 = 141/128$ and $\alpha = 1833/4096$. We have $X \le m^2 Z + \delta_{\mathrm{err}} d\mathbf{1}$, with either upper

$$-mb + 2m\sqrt{v} + 4v, \quad \text{or} \quad -mb + 2m\sqrt{v} + 2v + 2t\sqrt{v\{(t + \sqrt{1 + b/4})^2 - v\}}.$$

Both can be bounded jointly at $b=\theta, c=\theta b$. First fix b and use the increase of $v(W_0-v)$ when $W_0 \ge 1 > 2\alpha \ge 2v$; then $-mb+2m\sqrt{\beta_0 b-d}$ has derivative $-m+m\beta_0/\sqrt{v}>0$, and the other terms increase. Thus the two corresponding upper bounds are

$$\rho_4 = -m\theta + 2m\sqrt{\alpha-d} + 4(\alpha-d),$$

$$\rho_2 = -m\theta + 2m\sqrt{\alpha-d} + 2(\alpha-d) + 2t\sqrt{(\alpha-d)\{(t+\sqrt{\beta_0})^2-(\alpha-d)\}}.$$

The finite radical comparisons are

$$\sqrt{\alpha-d} \le 32881/49152 - 3d/4 \le 67/100 - 3d/4,$$

$$\sqrt{I(t)} \le Q(t) = 11/20 + (9/10)t - (21/20)t^2, \quad I(t) = (\alpha-t^2)\{2679/4096 + 2t^2 + (21/10)t\}.$$

The second holds for $0 \le t \le 1/2$, using $\sqrt{\beta_0} < 21/20$. For its full-domain certificate write $z = t - 1/4$:

$$Q^2 - I = 478089/419430400 - (97/10240)z + (75261/102400)z^2 + (53/16)z^3 + (1241/400)z^4.$$

At $z \ge 0$ the first three terms are at least $1/1000 - z/100 + (73/100)z^2 > 0$, and the last two are nonnegative. At $z \le 0$ the linear term is nonnegative; the remaining z^2 bracket increases on $[-1/4, 0]$, with left endpoint at least $613/6400 > 0$. Also $Q > 0$. Substitution gives the paid uppers

$$\begin{aligned} \Delta_2 &= 747/800 + 2\alpha + (11/10)\sqrt{d} - (4961/1600)d - (21/10)d^{3/2} + (9/4)d^2, \\ \Delta_4 &= 747/800 + 4\alpha - (11041/1600)d + (9/4)d^2. \end{aligned} \tag{Q165.T172.18}$$

Use Δ_2 in the first range and Δ_4 in the second. They are positive on their respective ranges: for $d \le 4/25$, $\Delta_2 \ge 747/800 + 2\alpha - 4961/10000 - 168/1250 > 0$; for $d \le 1/4$, $\Delta_4 \ge 747/800 + 4\alpha - 11041/6400 > 0$. These remainder bounds concern the same F, U at the original kernel and reference weight.

Q165.8 The complete low-domain payment under $f_a\mu$

The original weight $f_a\mu$ is a nonnegative finite measure of mass $\mathbb{E}f_a \le 1$. Apply Minkowski to $X \le m^2 Z + \Delta d\mathbf{1}$ without renormalizing that weight into another kernel:

$$\|X\|_{L^4(f_a\mu)} \le m^2\|Z\|_{L^4(f_a\mu)} + \Delta d(\mathbb{E}f_a)^{1/4} \le m^2\|Z\|_{L^4(f_a\mu)} + \Delta d.$$

In the first range, (Q165.T172.16) and $2\log m \le -3d - (9/4)d^2$ bound the first term by $e^{-(9/4)d+(11/8)d^2}$. Using $e^{-h} \le 1 - h + h^2/2$ gives

$$\|X\|_{L^4(f_a\mu)} \le 1 - d[L_1(d) - \Delta_2(d)], \quad L_1 = 9/4 - (125/32)d + (99/32)d^2 - (121/128)d^3.$$

For $t = \sqrt{d} \le 2/5$, the gap is

$$21567/51200 - (11/10)t - (1289/1600)t^2 + (21/10)t^3 + (27/32)t^4 - (121/128)t^6.$$

Using $t^2 \le 2t/5$ and $t^3 \le 4t/25$, its derivative is at most $-11/10 + (1159/800)t \le -1041/2000 < 0$. The gap is therefore at least $142671/32000000 > 1/256$. In the second range, (Q165.T172.17) bounds the first term by $e^{-(9/4)d+(3/2)d^2}$. The same finite Taylor comparison gives

$$L_2 = 9/4 - (129/32)d + (27/8)d^2 - (9/8)d^3,$$

$$L_2 - \Delta_4 = -12129/25600 + (4591/1600)d + (9/8)d^2 - (9/8)d^3.$$

This increases on $[4/25, 1/4]$, since its derivative is at least $4591/1600 - 27/128 > 0$; its left endpoint is $152047/16000000 > 1/256$. This pays the extra all-a exponent loss at the actual reference weight. Thus throughout the low domain, $\|X\|_{L^4(f_a\mu)} \le 1 - d/256$, and

$$\mathbb{E}[f_a X^4] \le (1 - d/256)^4 \le 1 - d/256.$$

Together with the high domain and $d = 0$, this proves the bare interface for every $a \ge 3$ and every original legal kernel.

Q165.9 Decorations, complete reversal and original counts

Each genuine equal-pair factor lies in $[0, 1]$. Positivity of T gives

$$\mathbb{E}[E_o f_a \{T(E_t F)\}^4] \le \mathbb{E}[f_a (TF)^4].$$

This restores all original multiplicities and subdivisions. If the sums of the equal-pair half-lengths are S_t, S_o, one original component has $a + 7 + 2S_t + 2S_o$ edges and $a + 8 + 2S_t + 2S_o$ vertices. The fourth-moment test tree has $a + 28 + 8S_t + 2S_o$ edges and $a + 29 + 8S_t + 2S_o$ vertices. The analytic copies add no reference arm to the original component. Reversing every edge, or replacing K by $-K$, gives

$$\mathbb{E}[\overline{E}_o \bar{f}_a \{A(\overline{E}_t p \bar{f}_2 f_3)\}^4] \le 1 - d/256.$$

The reference weight, terminal and connector reverse together. All vertex variables remain under the same original product measure; original edge counts, noninjective maps and arbitrary original joint-label laws retain their meanings. This is the specified reference fourth interface; ordinary eighth moments, arbitrary internal-root closures, compound outputs and other skeleton payments require their own inputs.

N.19 The reference terminal $(1, 2, 4)$ at two connector edges

Q167.0 The fixed local interface

For terminal residual $(1, 2, 4)$, the fixed canonical direction $F = q f_2 h_4$ and original connector length $L = 2$ satisfy, for every outward receiving arm f_a, $a \ge 3$, and every original legal finite kernel,

$$\mathbb{E}[E_o f_a \{T^2(E_t q f_2 h_4)\}^4] \le 1 - d/16. \tag{Q167.T175.0}$$

The same direction is used on both low and high domains.

Q167.1 Original objects and the receiving envelope

Work at the original finite probability measure. Let $K^* = -K$ and $|K(x, y)| \le 1$, and put

$$A = J + K, \quad T = J - K = A^*, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \quad d = \|g\|_2^2, \quad \delta_j = \|K^j \mathbf{1}\|_2^2, \quad \theta = 13/32, \quad \kappa = 19/32.$$

Every mean, adjoint and spectral measure is original. The actual arm messages are $f_n = A^n \mathbf{1}$, $\bar{f}_n = T^n \mathbf{1}$, $h_n = T f_{n-1}$. The Q3-C014 finite framework gives $d \le 1/3$, $\|K\|^2 \le \theta$, $|Kv(x)| \le \|v\|_2$, $d^2 \le \delta_2 \le \theta d$ and $\delta_{j+1} \le \theta \delta_j$. Original path means are at most one, and for $n \ge 3$ at most $1 - 2d$; actual messages are nonnegative and $f_\ell h_\ell \le 1$. Let $G_n(z) = \sum_{j=0}^{n-1} z^j$, $S_n = \mathbf{1} + G_n(K) g$. Finite

convolution gives $f_a \le C_a = S_a - \beta_a \mathbf{1}$, with β_n the finite decreasing alternating moment sum beginning at d, and $d - \delta_2 \le \beta_n \le d$ for $n \ge 3$. Thus

$$\mathbb{E}C_a = 1 - \beta_a - \beta_{a+1} \le 1 - 2d + 2\delta_2 \le 1 - (19/16)d. \tag{Q167.T175.1}$$

For $a \ge 3$, the finite norm $\|G_a\| \le 1$ from part N.18 gives $|S_a - 1| = |KG_a(K)\mathbf{1}| \le 1$, so $0 \le C_a \le S_a \le 2$. The actual $a = 3$ predecessor cost is retained:

$$C_a = p + KG_{a-1}(K)g - \beta_a \mathbf{1} \le p + (6/5)\sqrt{d}\,\mathbf{1}, \tag{Q167.T175.2}$$

since $\|G_2\| \le \sqrt{1+\theta} < 6/5$, while $a \ge 4$ has predecessor norm at most one. The terminal unit arm points inward, its two-edge arm outward, and its four-edge arm has first edge inward and the remaining three outward. Both connector edges point terminal-to-receiver; the receiving arm is outward.

Q167.2 The actual square and a positive two-step upper

Let $0 < d \le 1/4$ and put $R = Kg$, $S = K^2 g$, $W = K^3 g$, $m = 1 - d$ and $k = 1/m \in [1, 4/3]$. The actual identity $h_4 = q - 2d\mathbf{1} - R - S - W + dg$ gives $f_2 + h_4 = 2m\mathbf{1} - U$, where $U = S + W - dg$; hence $F = qf_2h_4 \le q(m - U/2)^2$. One has $\mathbb{E}U^2 = V = d^3 + 2d\delta_2 + \delta_3 + \delta_4$ and $\mathbb{E}U - \mathbb{E}(gU) = d^2 + 2\delta_2$. Expand $T(qU)$ and use $|K(gU)| \le \sqrt{dV}$, $T(qU^2) \le 4V$ to obtain

$$TF/m^2 \le \bar{f}_2 + K'R_0 + \delta\mathbf{1}, \quad K' = -K, \quad g' = -g, \quad R_0 = -kU, \quad \delta = -k(d^2 + 2\delta_2) + k\sqrt{dV} + k^2 V.$$

Put $v = \delta_2/d$ and $J_0 = V/d \le d^2 + [2d + \theta(1+\theta)]v$. With $\sqrt{J_0} \le 3/8 + (2/3)J_0$ and $k \le 4/3$,

$$\delta/(kd) \le 3/8 - d + 2d^2 + v[-2 + 4d + 2\theta(1+\theta)] \le 3/8.$$

If the v coefficient is nonpositive, discard it; otherwise use $v \le \theta$. The resulting quadratic increases in d, with endpoint value $-1099/16384 < 0$ at $1/4$ for the part after $3/8$. Thus $D = d/2 \ge \delta$ and $D \ge 0$. Apply the original positive operator once more. Its exact $D_1 = \mathbb{E}(K'R_0) = k(d^2 + \delta_2)$ and $\bar{f}_3 = S_3' - d\mathbf{1}$ give

$$0 \le X := T^2 F \le \mathcal{W} := m^2 Y, \quad Y = Cq + K'Q(K')g' + c\mathbf{1}. \tag{Q167.T175.3}$$

Here

$$C = 1 + d/2, \quad c = k(d^2 + \delta_2) - d, \quad Q(z) = (1 - kx^2) + [1 - k(d + x)]z, \quad z = i\sqrt{x}.$$

The upper Y is nonnegative by actual positive propagation. Both real coefficients are nonnegative, since $k(d + x) \le 7/8$ and $kx^2 \le 169/768$. Therefore

$$|Q|^2 \le (1 - x^2)^2 + x(1 - x)^2 \le 1 + x - 3x^2 \le 13/12. \tag{Q167.T175.4}$$

All comparisons are on $[0, \theta]$, without a lower condition $x \ge d$.

Q167.3 Every condition for coefficient nine

The coefficient-nine lemma in part Q117.4 of Section N.11 states that nonnegative C_0, Z with

$$C_0 \le p + (6/5)\sqrt{d}, \quad Z \le (1 - 5d/8)q - d/2 + (13/12)\sqrt{d}, \quad 0 \le \mathbb{E}Z \le 1 - 3d/2$$

satisfy $C_0\{(Z + \mathbb{E}Z)^2 + 2(\mathbb{E}Z)^2\} \le 9$. Use the receiving row (Q167.T175.2). Since $c \le -kd(\kappa - 2d)$, (Q167.T175.4) yields

$$\mathcal{W} \le m^2 Cq - dm(\kappa - 2d)\mathbf{1} + m^2\sqrt{13/12}\sqrt{d}\,\mathbf{1}.$$

The coefficient $m^2C = 1 - 3d/2 + d^3/2 \le 1 - 5d/8$. For $t = \sqrt{d} \le 1/2$, use $\sqrt{13/12} < 25/24$. The remaining difference from the lemma's upper is at least

$$t/24 + (3/32)t^2 + (25/12)t^3 - (83/32)t^4 - (25/24)t^5 + 2t^6 > 0.$$

Combine the two negative higher terms with the positive cubic using $t \le 1/2$; its remaining cubic coefficient is $101/192 > 0$. Let $\mu = \mathbb{E}\mathcal{W}$, distinguished from the measure and score. The exact mean $\mathbb{E}(K'Q(K')g') = -d + k\delta_3$ gives

$$\mu = m^2[1 - 3d/2 + k(d^2 + \delta_2 + \delta_3)]. \tag{Q167.T175.5}$$

For the lemma's weak condition, $c + \mathbb{E}(K'Q(K')g') = -2d + k(d^2 + \delta_2 + \delta_3) \le -695d/768 < -7d/8$. Thus $\mu \le m^2(1 - 3d/8) \le 1 - 3d/2$, and

$$C_a\{(\mathcal{W} + \mu)^2 + 2\mu^2\} \le 9. \tag{Q167.T175.6}$$

Q167.4 A stronger bound on the actual mean

Using $\delta_2 + \delta_3 \le \theta(1 + \theta)d = (585/1024)d$ in (Q167.T175.5) gives $\mu \le m[1 - (1975/1024)d + (5/2)d^2]$. The upper equals $1 - dL(d)$ with $L(d) = 2999/1024 - (4535/1024)d + (5/2)d^2$. On $[0, 1/4]$, $L' < 0$, so $L(d) \ge L(1/4) = 8101/4096 > 63/32$. Hence

$$0 \le \mu \le 1 - (8101/4096)d \le 1 - (63/32)d. \tag{Q167.T175.7}$$

This improves the mean estimate for the same positive upper and direction.

Q167.5 Nonpositive covariance at the original kernel

The spectral measure ν at g has support $[0, \theta]$, mass d, and $\int x^j\, d\nu = \delta_{j+1}$. Write $G_a = A_0 + zB_0$ with $1 - x \le A_0, B_0 \le 1$. The coordinates of $C + zQ$ are $P_0 = C - x + kdx + kx^2 > 0$ and $P_1 = 1 - kx^2 \ge 0$. Then

$$\mathrm{Re}\{G_a(z)(C + zQ(z))\} \ge (1 - x)P_0 - xP_1 \ge 1 - 3x + 2x^2 \ge 57/512.$$

Retaining the original mean correction,

$$\mathrm{Cov}(C_a, \mathcal{W}) \le -m^2[(57/512)d + \beta_{a+1}(d - k\delta_3)] \le 0. \tag{Q167.T175.8}$$

Here $d - k\delta_3 > 0$ since $k \le 4/3$, $\delta_3 \le \theta^2 d$. The payment discards this proved nonpositive covariance while retaining the negative square in the variance.

Q167.6 The full variance after clearing a denominator

Set $j = mC = 1 - d/2 - d^2/2$ and $a_0 = 2d - 1$. For $\rho = P_0^2 + xP_1^2$, the exact polynomial is

$$\begin{aligned} m^2\rho(x) = v(x) &= (j + a_0x + x^2)^2 + x(m - x^2)^2 \\ &= j^2 + (-1 + 3d - 2d^3)x + (3 - 5d + 3d^2)x^2 \\ &\quad + (6d - 4)x^3 + x^4 + x^5. \end{aligned} \tag{Q167.T175.9}$$

The actual mean gives

$$\mathrm{Var}(\mathcal{W}) = m^2\left\{\int v(x)\, d\nu - (md - \delta_3)^2\right\}. \tag{Q167.T175.10}$$

Let $F_0(x) = (v(x) - j^2)/x$, extended polynomially at zero. Its second derivative $2(6d-4)+6x+12x^2$ has maximum $-149/256 < 0$ on $d \le 1/4, x \le \theta$. Its first derivative at θ is $4205/8192 - d/8 + 3d^2 > 0$, so F_0 increases throughout the interval. Thus $v(x) \le j^2 + \mathcal{B}(d)x$, where

$$\mathcal{B}(d) = -363983/1048576 + (1003/512)d + (39/32)d^2 - 2d^3. \tag{Q167.T175.11}$$

The derivative of $\mathcal{B}(d) - (49/16)d^2$ is $1003/512 - (59/16)d - 6d^2 \ge 339/512 > 0$ on $[0, 1/4]$; its right endpoint is $-4047/1048576 < 0$. Hence $\mathcal{B}(d) \le (49/16)d^2$. Even though $\mathcal{B}$ can change sign, first multiply this inequality by $\delta_2 \ge 0$, then use the nonnegative upper coefficient: $\mathcal{B}(d)\delta_2 \le (49/16)d^2\delta_2 \le (637/512)d^3$. Also $md - \delta_3 \ge (m - \theta^2)d > 0$. With $e_0 = 1 - \theta^2 = 855/1024$, retain the variance square:

$$\begin{aligned} \operatorname{Var}(\mathcal{W}) &\le m^2\{j^2 d + (637/512)d^3 - (e_0 - d)^2 d^2\} \\ &= m^2\{d - (1 + e_0^2)d^2 + (277/128)d^3 - d^4/2 + d^5/4\}. \end{aligned} \tag{Q167.T175.12}$$

The negative square is part of the bound used in the following payment.

Q167.7 The complete low-domain fourth payment

The original comparisons $X \le \mathcal{W}$ and $f_a \le C_a$ are nonnegative. The exact fourth identity gives

$$\begin{aligned} \mathbb{E}[f_a X^4] \le (\mathbb{E}C_a)\mu^4 + 4\mu^3 \operatorname{Cov}(C_a, \mathcal{W}) \\ + \mathbb{E}[C_a(\mathcal{W} - \mu)^2\{(\mathcal{W} + \mu)^2 + 2\mu^2\}]. \end{aligned}$$

The established mean, coefficient, covariance and variance bounds give

$$Q(d) = (1 - 19d/16)(1 - 63d/32)^4 + 9(1 - d)^2\{d - (1 + e_0^2)d^2 + (277/128)d^3 - d^4/2 + d^5/4\}.$$

The exact finite expansion is

$$Q(d) = 1 - d/16 + q_2 d^2 + q_3 d^3 + q_4 d^4 + q_5 d^5 - 9d^6 + (9/4)d^7,$$

$$\begin{aligned} q_2 &= -699417/1048576 < -2/3, \quad q_3 = 928818/1048576 < 9/10, \\ q_4 &= -7820208/1048576 < -7, \quad q_5 = 216199917/16777216 < 13. \end{aligned} \tag{Q167.T175.13}$$

The comparisons have positive denominators; for example $3 \cdot 699417 - 2 \cdot 1048576 = 1099 > 0$, $7820208 > 7 \cdot 1048576$, and $216199917 < 13 \cdot 16777216$. Group the coefficients by even degrees:

$$Q(d) \le 1 - d/16 + d^2[-2/3 + (9/10)d] + d^4[-7 + 13d] + d^6[-9 + (9/4)d].$$

For $d \le 1/4$, the three brackets are at most $-53/120, -15/4, -135/16$, all nonpositive. Hence

$$\mathbb{E}[f_a\{T^2(q f_2 h_4)\}^4] \le 1 - d/16 \qquad (a \ge 3, 0 \le d \le 1/4). \tag{Q167.T175.14}$$

At $d = 0$ original leaf integration gives equality, without requiring $K = 0$.

Q167.8 A high-domain mean proof, decorations and reversal

For $d \geq 1/4$ retain $F = qf_2h_4$, and let $Y = TF$, $X = TY$. The original adjoint, $pq \leq 1$, and nonnegativity give

$$\mathbb{E}Y = \mathbb{E}[pF] = \mathbb{E}[pqf_2h_4] \leq \mathbb{E}[f_2h_4] = \langle f_2, Tf_3\rangle = \langle Af_2, f_3\rangle = \mathbb{E}f_3^2. \tag{Q167.T175.15}$$

This computes the true one-step mean. The original kernel $1 - K \in [0, 2]$ gives $X \leq 2\mathbb{E}Y$. Use $f_3 = 1 - d + g + R + S$, $\mathbb{E}R = -d$, $\mathbb{E}(gR) = \mathbb{E}(RS) = 0$ and $\mathbb{E}(gS) = -\delta_2$ to expand $\mathbb{E}f_3^2 = 1 - 3d + 3d^2 - \delta_2 + \delta_3 \leq 1 - 3d + (77/32)d^2$. This last quadratic decreases on $[1/4, 1/3]$, with derivative at most $-67/48$ and left endpoint $205/512$. Therefore

$$0 \leq X \leq 205/256 < 7/8, \quad \mathbb{E}[f_aX^4] \leq (205/256)^4 < (7/8)^4 = 2401/4096 < 3/4 < 1 - d/16. \tag{Q167.T175.15a}$$

Only $\mathbb{E}f_a \leq 1$ is used here, simultaneously for all $a \geq 3$. Together with the low-domain result and the $d = 0$ boundary, this proves (Q167.T175.0) at every original finite kernel, without an ordinary-moment or general change-of-weight input. For every proved d, $E_t, E_o \in [0, 1]$ are actual equal-pair products; positivity of T gives the decorated result in place. All multiplicities and subdivisions remain as original directed edges. For equal-pair half-length sums S_t, S_o, an original component has $a + 9 + 2S_t + 2S_o$ edges and $a + 10 + 2S_t + 2S_o$ vertices; its fourth-moment test tree has $a + 36 + 8S_t + 2S_o$ edges and $a + 37 + 8S_t + 2S_o$ vertices. Complete reversal replaces K by $-K$, the terminal by $p\bar{f}_2W_4$ with $W_4 = A\bar{f}_3$, propagation T^2 by A^2, and receiving f_a by $\bar{f}_a$, with all decorations reversed together. Every calculation stays at the same original μ; original joint-label laws, edge counts, vertices and noninjective maps retain their meanings. This reference fourth interface does not itself assign arbitrary compound half-tree or internal-root moments.

N.20 Reference fourth moments for terminal $(1, 3, c)$

Q175.0 The actual local interface

For every $c \geq 4$, $a \geq 2$ and original positive connector length L, the fixed canonical direction $F_c = qf_3h_c$ satisfies

$$\mathbb{E}[E_of_a\{T^L(E_tF_c)\}^4] \leq 1 - d/800. \tag{Q175.T180.0}$$

The result concerns this same-reference local interface. Until paragraph N.20, take $a \geq 3$. The genuine case $a = 2$ is proved there using its own receiving mass and finite G_2 covariance. On the original finite probability space let $A = J + K$, $T = J - K = A^*$, where K is real skew and $|K| \leq 1$. Write

$$g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \quad \delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2,$$

$$f_n = A^n\mathbf{1}, \qquad \bar{f}_n = T^n\mathbf{1}, \qquad h_n = Tf_{n-1}.$$

The Q3-C014 finite framework gives $d \leq 1/3$, $\|K\|^2 \leq \theta$, $|Kv| \leq \|v\|_2$, $d^2 \leq \delta_2 \leq \theta d$ and $\delta_{j+1} \leq \theta\delta_j$. Actual messages are nonnegative; path means are at most one and, for $n \geq 3$, $\mathbb{E}f_n = \mathbb{E}\bar{f}_n \leq 1 - 2d$. Equal pairs satisfy $f_\ell h_\ell \leq 1$. Let $G_n(z) = \sum_{j=0}^{n-1} z^j$, $S_n = \mathbf{1} + G_n(K)g$, and use the finite alternating moments β_n, with $\beta_1 = \beta_2 = 0$. Positive convolution gives

$$0 \leq f_a \leq C_a = S_a - \beta_a\mathbf{1} \leq 2, \quad \mathbb{E}C_a = 1 - \beta_a - \beta_{a+1} \leq 1 - 2\tau, \quad \tau \leq \beta_n \leq d \quad (n \geq 3).$$

The independent bound $S_a \le 2$ follows from $|S_a - 1| = |KG_a(K)\mathbf{1}| \le \|G_a(K)\mathbf{1}\|_2 \le 1$, since $\|G_a\| \le 1$ for $a \ge 3$. The true G_2 predecessor at $a = 3$ has norm $\sqrt{1+\theta} < 6/5$; hence the receiving row is

$$C_a \le p + (6/5)\sqrt{d}\,\mathbf{1} - \beta_a\mathbf{1} \le p + (6/5)\sqrt{d}\,\mathbf{1}. \tag{Q175.T180.1}$$

All quantities use the same original measure. At $d = 0$, K can be nonzero, but integration from the original leaves gives messages one.

Q175.1 Centered rows and actual constants

Put $z = -K$, $g' = z\mathbf{1} = -g$, so $q = 1 + g'$ and $p = 1 - g'$. For $\mathbb{E}w = 0$, subtract the original row mean $g'(x)$ and use Cauchy–Schwarz:

$$|zw(x)|^2 \le p(x)q(x)\mathbb{E}w^2. \tag{Q175.T180.2}$$

The row variance is at most $1 - g'(x)^2 = pq$; this is a bound on the original kernel. For an analytic output $Y = Cq + zV + c\mathbf{1}$, where V is an actual real vector on the original space, set $e = \mathbb{E}V$ and $\sigma^2 = \operatorname{Var} V$. The complete pointwise bound is

$$Y \le (C + e)q + (c - e)\mathbf{1} + \sqrt{pq}\,\sigma. \tag{Q175.T180.3}$$

It retains the true mean e. When finite propagation gives $c = e$, the free constant cancels and $Y \le (C + e)q + \sqrt{pq}\sigma$. The coefficient lemmas below concern original nonnegative variables and their actual means, rather than a conversion of ordinary eighth moments.

Q175.2 The coefficient $17/2$

Suppose $0 \le d \le 1/4$, $t = \sqrt{d}$, $\gamma = 35/16$, $0 < \alpha \le e^{-\gamma d}$, $p + q = 2$, $p, q \ge 0$, $C_0, Z \ge 0$ and $\mu = \mathbb{E}Z \ge 0$. Assume

$$C_0 \le p + (6/5)t, \quad \mu \le \alpha, \quad Z + \mu \le \alpha\{q + 1 + (6/5)\sqrt{pq}\,t + (57/160)t^2\}. \tag{Q175.T180.4}$$

Then $C_0\{(Z+\mu)^2 + 2\mu^2\} \le 17/2$ pointwise. To prove this, use $\sqrt{pq} \le 1$, put $a = 6/5$, $b = 57/160$, $\delta = at + bt^2$, and $H_0 = q^2 + 2q + 3$. The coefficient is at most $e^{-2\gamma t^2}(p + at)\{H_0 + 2(q+1)\delta + \delta^2\}$. The constant coefficient is $pH_0 = 6 + q - q^3 \le 13/2$, since $\max(q - q^3) = 2/(3\sqrt{3}) < 1/2$. The linear coefficient is $a\{H_0 + 2p(q+1)\} = a(-q^2 + 4q + 7) \le 66/5$. The next four coefficients satisfy

$$A_2 \le 6a^2 + (9/2)b < 41/4, \quad A_3 = a^3 + 6ab < 43/10,$$
$$A_4 \le 2b^2 + 2a^2b < 13/10, \quad A_5 = ab^2 < 4/25.$$

Here $p(q+1) \le 9/4$, $0 \le p, q \le 2$, and all coefficient uppers are nonnegative. Since $t \le 1/2$, compression to the quadratic gives $41/4 + 43/20 + 13/40 + 1/50 = 2549/200 < 13$. The polynomial before the exponential is at most $13/2 + (66/5)t + 13t^2$, while $(17/2)e^{2\gamma t^2} \ge 17/2 + (595/16)t^2$. Their difference is at least $2 - (66/5)t + (387/16)t^2$, which is positive on the real line: its leading coefficient is positive and discriminant $-963/50 < 0$. The case $d = 0$ follows directly.

Q175.3 Coefficients nine and eleven for connector tails

Keep $t = \sqrt{d}$, $\alpha \le e^{-35d/16}$, $C_0 \le p + (6/5)t$ and $0 \le \mu \le \alpha$. If $d \le 1/8$ and

$$Z + \mu \le \alpha\{q + 1 + \sqrt{pq}\,t + (3/2)t^2\},$$

then $C_0\{(Z+\mu)^2+2\mu^2\} \le 9$. If $d \le 1/4$ and $3/2$ is replaced by $5/2$, the coefficient is at most 11. For the proof take $\sqrt{pq} \le 1$, $\delta = t + bt^2$ and $a = 6/5$ in the same expansion. The constant is at most $13/2$, the linear coefficient at most $66/5$. At $b = 3/2$, degrees two through five have bounds $319/20, 12, 81/10, 27/10$. Using $t \le 1/\sqrt{8} < 3/8$, their quadratic compression is less than $87/4$. Compare with $9e^{2\gamma t^2} \ge 9 + (315/8)t^2$: the difference is at least $5/2 - (66/5)t + (141/8)t^2$, with discriminant $-201/100 < 0$. At $b = 5/2$, the four bounds are $409/20, 96/5, 37/2, 15/2$. At $t \le 1/2$ their compressed coefficient is $2849/80 < 36$. Compare with $11e^{2\gamma t^2} \ge 11+(385/8)t^2$: the difference is at least $9/2-(66/5)t+(97/8)t^2$, with discriminant $-4401/100 < 0$. Both full continuous domains follow.

Q175.4 The exponential first row for all $c \ge 4$

Let $0 < d \le 1/4$, $m = c - 3 \ge 1$, and let $B_0 = \beta_{c-1}$ be a scalar. Finite positive convolution retains β_c and $B_0 q$, giving

$$f_3 + h_c \le (2-\Gamma)\mathbf{1} + R, \quad \Gamma = d + \beta_c + B_0, \quad R = -B_0 g' - z^3 G_m(-z) g'. \tag{Q175.T180.5}$$

Indeed $h_c = T f_{c-1} \le T(S_{c-1} - B_0\mathbf{1})$, and adding the exact f_3 gives the bound. AM–GM for the actual arms yields $F_c \le q e^{R-\Gamma}$. Also $\Gamma \ge d + 2\tau \ge (35/16)d$, $\Gamma \le 3d$. Set $\alpha = e^{-\Gamma}$. For every m, $\|G_m\| \le 6/5$, and the row bound gives $|R| \le d + (6/5)\theta\sqrt{d} \le 79/160 < 1/2$. The Taylor bound $e^r \le 1 + r + (3/5)r^2$ for $r \le 1/2$ gives

$$T(qe^R) \le \bar{f}_2 + zR + \delta\mathbf{1}, \quad \delta = \mathbb{E}R + \mathbb{E}(g'R) + \sqrt{d\mathbb{E}R^2} + (12/5)\mathbb{E}R^2. \tag{Q175.T180.6}$$

The mixed means are retained jointly; use $|z(g'R)| \le \mathbb{E}|g'R| \le \sqrt{d\mathbb{E}R^2}$ and $T(qR^2) \le 4\mathbb{E}R^2$. The Taylor constant follows from the increase of $(e^r - 1 - r)/r^2$ and the finite positive-series bound $e^{1/2} < 33/20$. We now prove $D = 7d/20 \ge \max(\delta, 0)$, with the direction fixed. Put $v = \delta_2/d$. For $c = 4$, $R = -dg' - z^3 g'$, $\mathbb{E}R + \mathbb{E}(g'R) = -\delta_2 - d^2$, and $\mathbb{E}R^2 = d^3 + \delta_4$. Young's bound $\sqrt{J} \le 1/8 + 2J$ gives

$$\delta/d \le 1/8 - d + (22/5)d^2 + v[-1 + (22/5)\theta^2] \le 3/20.$$

The v coefficient is negative; the remaining convex quadratic has maximum $3/20$ at the endpoints zero and $1/4$. For $c = 5$, the same-kernel expansion gives $\mathbb{E}R + \mathbb{E}(g'R) = -\delta_2 - d^2 + \delta_3$ and $\mathbb{E}R^2/d \le d^2 + [\theta^2(1+\theta) - 2\theta d]v$. Thus

$$\delta/d \le 1/8 - d + (22/5)d^2 + v[-\kappa + (22/5)\theta^2(1+\theta) - (44/5)\theta d].$$

For $d \le 1/8$, discard the negative vd and $-d + (22/5)d^2$, then use $v \le \theta$ to obtain $156575/524288 < 3/10$. For $d \ge 1/8$, the v coefficient is $-1593/81920 < 0$ at the left endpoint and decreases; discarding it gives $3/20$. For $c \ge 6$, one has $\mathbb{E}R + \mathbb{E}(g'R) \le -(3/16)\delta_2 - \kappa d^2$ and $\mathbb{E}R^2/d \le d^2 + [\theta^2 - 2\kappa^2\theta d]v$, whence

$$\delta/d \le 1/8 - \kappa d + (22/5)d^2 + v[-3/16 + (22/5)\theta^2 - (44/5)\kappa^2\theta d].$$

These expansions use the finite alternating moments of $W = K^3 G_m(K) g$: $\mathbb{E}W - \mathbb{E}(gW) \ge \delta_2 - 2\delta_3$, $\mathbb{E}(gW) \ge \kappa\delta_3$, and $\|W\|_2^2 \le \delta_4$. The v bracket is at least $373/2000 > 0$ on the full quarter-domain, using $2/5 \le \theta \le 5/12$, $\kappa \le 3/5$. Set $v = \theta$; the resulting convex quadratic has endpoint values $28167/81920$ and $1795141/5242880$, both less than $7/20$. Thus $D = 7d/20$ is valid in every case.

Q175.5 Finite propagation, signed moments and fourth expansion

Let $D_i = \mathbb{E}z^i R$. Define the finite propagation by

$$Y_L = \bar{f}_{L+1} + z^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \qquad (L \geq 1). \tag{Q175.T180.7}$$

The first Y_1 is a nonnegative upper in (Q175.T180.6); $Y_L = T^{L-1}Y_1 \geq 0$ and $T^L F_c \leq \alpha Y_L$. The finite alternating sums give

$$-\delta_3 \leq D_1 \leq d^2, \quad 0 \leq D_2 \leq \delta_3, \quad D_3 \leq \delta_4, \quad D_4 \leq 0. \tag{Q175.T180.8}$$

More fully, $D_{2r-1} = (-1)^{r+1}(B_0\delta_r - H_{r+2})$ and $D_{2r} = (-1)^{r+1}\widetilde{H}_{r+2}$, where $H, \widetilde{H}$ are the actual finite decreasing alternating sums obtained by taking the means of the displayed finite polynomial R, both in $[0, \delta_{r+2}]$. This includes the empty sum at $m = 1$. Pay the positive sign classes separately:

$$\sum_{i\geq 1}(D_i)_+ \leq \frac{B_0 d + \delta_3 + \delta_4}{1-\theta^2} \leq (169/608)d + (1024/855)d^2. \tag{Q175.T180.9}$$

The geometric sum only bounds the actual finite positive terms. For any nonnegative Z on the original space and $\mu = \mathbb{E}Z$,

$$\mathbb{E}[C_a Z^4] = (\mathbb{E}C_a)\mu^4 + 4\mu^3 \operatorname{Cov}(C_a, Z) + \mathbb{E}[C_a(Z-\mu)^2\{(Z+\mu)^2 + 2\mu^2\}]. \tag{Q175.T180.10}$$

A pointwise coefficient bound K_0 bounds the last term by $K_0 \operatorname{Var} Z$. We apply this to $W_L = \alpha Y_L$ or its actual nonnegative upper. The original spectral measure ν of $-K^2$ at g (or its full reversal g') has support $[0, \theta]$, mass d, and $\int x^j \, d\nu = \delta_{j+1}$; all symbol products use the same frequency and original measure. For the finite exponential denominator set

$$\gamma = 35/16, \quad a_0 = 35/8, \quad b_0 = 1225/128, \quad H(d) = 1 + a_0 d + b_0 d^2. \tag{Q175.T180.11}$$

Then $e^{2\gamma d} \geq H(d)$. For $d \leq 1/4$, $H \leq 5513/2048 < 11/4$ and $H^2 < 8$; for $d \leq 1/8$, $H < 2$ and $H^2 < 4$. The remaining payments after clearing H^2 are finite polynomials.

Q175.6 The unit connector: coupled moments and the variance square

Write $G_m(-z) = U - zV$. For $m \geq 2$, $1 - x \leq U, V \leq 1$; for $m = 1$, $U = 1, V = 0$. Define the finite polynomial P by $R = (P - I)g'$, namely $P = 1 - B_0 + x^2 V + zxU$. Then $Y_1 = \mathbf{1} + D\mathbf{1} + g' + zPg'$. Since $\mathbb{E}R \leq 0$, $|\mathbb{E}R| \leq \delta_2$ and $\|Pg'\|_2^2 \leq (36/25)d$, the centered row gives

$$Y_1 + \mathbb{E}Y_1 \leq q + 1 + (6/5)\sqrt{pqd} + (57/160)d.$$

Indeed, $Y_1 \leq (1 + \mathbb{E}R)q + (D - \mathbb{E}R)\mathbf{1} + \sqrt{pq}\sqrt{\operatorname{Var}(Pg')}$, and $\mathbb{E}Y_1 = 1 - d + D_1 + D \leq M_1 := 1 - (13/20)d + d^2 \leq 1$. The sum of the two constant remainders is at most $2D - d + d^2 + \delta_2 \leq (57/160)d$. The norm estimate follows from $|P|^2 \leq (1 + x^2)^2 + x^3 \leq 1493553/1048576 < 36/25$. Thus $W_1 = \alpha Y_1$ satisfies part N.20, with coefficient $17/2$. The nonconstant symbol of Y_1 is $Q_2 = 1 + zP$, with real coordinate $1 - x^2 U$ and imaginary coordinate $1 - B_0 + x^2 V$. Finite same-kernel coordinates give

$$|Q_2|^2 \leq 1 + (1 - B_0)^2 x - \eta x^2, \qquad \eta = 215/1024. \tag{Q175.T180.12}$$

Using $U \ge 1-x, V \le 1$, expansion gives $1+(1-B_0)^2x-2x^2+(4-2B_0)x^3+x^4-x^5+x^6$. Then use $4x^3 \le 13x^2/8$, $x^4 \le \theta^2 x^2$, and $x^6 \le x^5$. In particular $\|Q_2\|^2 \le 11/8$. The covariance is nonpositive:

$$\mathrm{Re}(G_a Q_2) \ge (1-x)(1-x^2U) - x(1-B_0+x^2V) \ge 1-2x-x^2 \ge 23/1024,$$

and its mean term $-\beta_{a+1}(d-D_1)$ is nonpositive. Let $v=\delta_2/d \in [d,\theta]$. Spectral Cauchy–Schwarz gives $\delta_3 \ge \delta_2^2/d$. Retaining the negative square yields

$$\mathrm{Var}\, Y_1 \le dV(v,d), \quad V = 1+v-\eta v^2 - 2v(1-v)d + v(1-v)^2d^2 - d(1-d)^2. \tag{Q175.T180.13}$$

This uses $B_0 \ge \tau = (1-v)d$ and $(d-D_1)^2 \ge d^2(1-d)^2$. The upper $V>0$ has v derivative at least $9/32>0$, using $\eta<1/4$, $v \le 5/12$, $d \le 1/4$ and $1-4v+3v^2 \ge -1/6$. Thus the receiving upper $\mathbb{E}C_a \le 1-2(1-v)d$, the bound on $e^{-\Gamma}$ and V all increase in v in this upper estimate. At $v=\theta$, relax to $V \le 11/8-(47/32)d+(69/32)d^2$. Consequently

$$\mathbb{E}[f_a W_1^4] \le e^{-4\gamma d}(1-19d/16)M_1^4 + (17/2)de^{-2\gamma d}[11/8-(47/32)d+(69/32)d^2].$$

The reference arm occurs once. Use $M_1^4 \le 1-(13/5)d+(1307/200)d^2-(39/5)d^3+6d^4$. Let A be this polynomial times $1-19d/16$ and $V_0 = 11/8-(47/32)d+(69/32)d^2$. Clearing the denominator gives

$$H^2 - A - (17/2)dV_0H = d[17/20 + g_2d + g_3d^2 + g_4d^3 + g_5d^4],$$

where

$$g_2 = -31967/3200 > -10, \quad g_3 = 1215413/51200 > 237/10,$$
$$g_4 = 9471771/81920, \quad g_5 = -1378557/8192, \quad g_4+g_5d \ge 73.$$

The bracket is at least $17/20-10d+(237/10)d^2+73d^3$. For $d \le 1/12$ it is at least $1/60$. For $d \ge 1/12$ it is at least $17/20-10d+(1787/60)d^2$, whose real-line minimum is $379/35740 > 1/100$. Thus the cleared difference is at least $d/100$; $H^2<8$ gives

$$\mathbb{E}[f_a(TF_c)^4] \le 1-d/800. \tag{Q175.T180.14}$$

Q175.7 The centered structures at $L=2,3,4$

Let $R_0(z) = -B_0 - z^3G_m(-z)$, so $R = R_0(z)g'$; all expressions below use this finite polynomial, without division by g'. Collect the propagation terms as $Y_j = C_jq + zP_j(z)g' + c_j\mathbf{1}$. The polynomials are

$$Q_2 = G_2 + zR_0, \qquad Q_3 = G_3 + z^2R_0, \qquad Q_4 = G_4 + z^3R_0,$$

$$\begin{aligned}
&C_2 = 1+D, && P_2 = Q_2, && c_2 = D_1 - d = \mathbb{E}(P_2g'),\\
&C_3 = 1-d+D+D_1, && P_3 = Q_3 + D, && c_3 = D_2 - d = \mathbb{E}(P_3g'),\\
&C_4 = 1-2d+D+D_1+D_2, && P_4 = Q_4 - \lambda + Dz, && c_4 = -\tau - Dd + D_3 = \mathbb{E}(P_4g'),\\
& && \lambda = d-D-D_1 \in [2d/5, d]. &&
\end{aligned} \tag{Q175.T180.15}$$

Each c_j is the actual mean $\mathbb{E}(P_jg')$, so the free constant in (Q175.T180.3) cancels; $C_j + \mathbb{E}(P_jg') = \mathbb{E}Y_{j-1} \le 1$. Direct finite coordinates give

$$\|Q_2\|^2 \le 11/8, \qquad \|Q_3\| \le 1, \qquad \|Q_4\| \le 1, \qquad \|P_4\| \le 1+d/4 < 6/5. \tag{Q175.T180.16}$$

The Q_3 coordinates are $1-(1-B_0)x-x^3V$ and $1-x^2U$; they are positive and their squared bound is $(1-3x/4)^2+x\le 1$. The Q_4 coordinates are $1-x+x^3U$ and $1-(1-B_0)x-x^3V$. After subtracting one and dividing by x, the square is at most $-1-x/2+(41/16)x^2-2x^3+x^5<0$. Since $\operatorname{Re} Q_4\ge\kappa$ and $0\le\lambda\le d$, $\|Q_4-\lambda\|\le 1$; adding Dz costs $D\sqrt{\theta}\le d/4$. This includes $m=1,a=3$ and the full quarter-domain. The actual means satisfy

$$\mathbb{E}Y_2\le 1-(7/5)d+d^2\le 1-(23/20)d,\quad \mathbb{E}Y_3\le 1-(7/4)d,\quad \mathbb{E}Y_4\le 1-2d. \tag{Q175.T180.17}$$

For Y_2 use the exact mean $-d+D_2$ and $D_1\le d^2$. For Y_3 use $\mathbb{E}\bar f_4=1-3d+\delta_2$, $D_1\le d^2$, $D_2\le\delta_3$, $D_3\le\delta_4$. For Y_4 use $\mathbb{E}\bar f_5=1-4d+2\delta_2+d^2$ and $D_4\le 0$. The rational comparisons are $\theta+\theta^2+\theta^3<13/20$ and $2\theta+\theta^2+\theta^3<21/20$; all means are nonnegative. Every $W_j=\alpha Y_j$ satisfies the stronger version of part N.20, with the $57/160$ constant replaced by zero, and coefficient $17/2$. Retaining the actual mean terms, the covariance and variance bounds are

$$\begin{aligned}
&\operatorname{Cov}(C_a,W_2)\le(55/1024)\alpha d, && \operatorname{Var}Y_2\le d-d^2/10+(49/400)d^3,\\
&\operatorname{Cov}(C_a,W_3)\le\alpha d(1/16+d), && \operatorname{Var}Y_3\le d,\\
&\operatorname{Cov}(C_a,W_4)\le\alpha d(1/16+9d/4), && \operatorname{Var}Y_4\le d[45/32-2d+(27/20)d^2].
\end{aligned} \tag{Q175.T180.18}$$

The full original variance is $\int|C_j+zP_j|^2\,d\nu-(\int\operatorname{Re}P_j\,d\nu)^2$; the covariance is $-\int\operatorname{Re}\{G_a(C_j+zP_j)\}\,d\nu-\beta_{a+1}\int\operatorname{Re}P_j\,d\nu$. For these three short cases, $\operatorname{Re}P_j>0$, so the last covariance term is nonpositive. At $L=2$, $U-(1-x)V\ge 0$ gives $\operatorname{Re}\{G_a(C_2+zQ_2)\}\ge 1-3x+x^2\ge -55/1024$. Also $\operatorname{Im}_z Q_2\ge 1-B_0$, and $\|Q_2\|^2\le 11/8$, so $|C_2+zQ_2|^2\le C_2^2-x/8$. The negative mean square is at least $(1-\theta^2)^2d^2$; use $\delta_2\ge d^2$ and $7/10-1/8-(855/1024)^2<-1/10$ to obtain the variance bound. At $L=3$, $C_3\ge 1-17d/20$, the normalized imaginary coordinate of P_3 is at least $1-\theta^2$, and $\|P_3\|\le 1+D$. Then $2C_3(1-\theta^2)\ge(1+D)^2$, so the squared symbol is at most C_3^2 and the variance at most d. The receiving real part is at least $1-3x+x^2-d$. At $L=4$, $C_4\ge 1-11d/6$, $\operatorname{Re}P_4\le 1$, its normalized imaginary coordinate is at most $1+D$, and $C_4-x\operatorname{Im}_z P_4>0$. The receiving real part is at least $1-3x+x^2-(9/4)d$. The variance symbol is bounded by $C_4^2+\theta\|P_4\|^2$; $C_4\le 1-23d/20$ and $\|P_4\|\le 1+d/4$ give the stated variance. Here Im_z denotes the real coefficient of z in a symbol $R+zI$. These relaxations discard only the displayed nonpositive terms, retaining each actual mean.

Q175.8 Finite payments at the three short connectors

For the positive covariances in (Q175.T180.18), use $\mu\le\alpha M\le\alpha$. The fourth-expansion contributions are at most α^4 times $(55/256)d$, $d/4+4d^2$, and $d/4+9d^2$. For $L=2$ put $M=1-23d/20$. Using $M^4\le 1-(23/5)d+(1587/200)d^2$ and the original receiving mean, the upward quadratic bound is $A_2=1-(7133/1280)d+(5359/400)d^2$. Relax the variance to $d+(49/400)d^3$, and set

$$G_2=H^2-A_2-(17/2)d[1+(49/400)d^2]H.$$

Its coefficients at degrees one through five are at least $29/5,-25/2,0,0,-10$. Thus $G_2\ge 2d$ for $d\le 1/4$, giving $\mathbb{E}[f_a(T^2F_c)^4]\le 1-d/4$. For $L=3$, $M=1-7d/4$ gives the mean-plus-covariance bound $A_3=1-(127/16)d+(491/16)d^2$; relax the variance to d. The coefficients of $G_3=H^2-A_3-(17/2)dH$ at degrees one through four are at least $8,-30,0,0$. Thus $G_3\ge d/2$ and $\mathbb{E}[f_a(T^3F_c)^4]\le 1-d/16$. For $L=4$, set $M=1-2d$, $A_4=1-(143/16)d+(85/2)d^2$ and $V_4=45/32-2d+(27/20)d^2$. The five nonzero coefficients of $G_4=H^2-A_4-(17/2)dV_4H$ are at least

$57/10, -40, 32, 200, -110$. Hence

$$G_4/d \geq 57/10 - 40d + 32d^2 + (345/2)d^3,$$

where $-110d^4$ is paid using $d \leq 1/4$. The cubic on the right at $d_0 = 9/40$ is $36465/128000 > 1/4$; its derivative is $383/640 < 1$, and its second derivative is at least 64. Completing the square gives a lower bound $1/4 - 1/128 = 31/128$, so $\mathbb{E}[f_a(T^4F_c)^4] \leq 1 - d/64$. These are explicit rational coefficient comparisons on continuous domains.

Q175.9 Every finite tail $L \geq 5$

Positive convolution retains $\bar{f}_0, \bar{f}_1, \bar{f}_2, \bar{f}_3$. For $n = L + 1 \geq 6$,

$$\bar{f}_n \leq (1 - 3\tau)q + z[G_L - \tau(2 + z)]g' - \tau(1 - d)\mathbf{1}.$$

Let $P_{\text{cost}} = 2D + 2\sum_{i\geq 1}(D_i)_+$, $C = 1 - 3\tau$, $a_1 = 1 - 2\tau$, and define the actual finite polynomial

$$P_L = G_L - \tau(2 + z) - B_0 z^{L-1} - z^{L+2}G_m(-z).$$

Then $Y_L \leq Z_L = Cq + zP_Lg' + c\mathbf{1} \geq 0$, where $c = P_{\text{cost}} - \tau(1-d)$. We prove $\|P_L\| \leq a_1$ throughout $d \leq 1/4$. For $L = 5$, the nonnegative coordinate uppers are $a_1 - x + x^2 + x^4$ and $1 - \tau - x + x^3$. The squared difference from a_1^2, divided by x, increases with τ; at $\tau = 1/4$ it is at most

$$-7/16 + x/2 - x^2 + (7/2)x^3 - 4x^4 + 2x^5 + x^6 + x^7.$$

Use $x \leq 5/12$ and $2x^3 + x^4 + x^5 \leq 5/24$ to bound it by $-7/16 + x/2 + (2/3)x^2 < 0$. The actual coordinates are nonnegative using $a_1 \geq 1/2, B_0 \leq 1/4$. For $L = 6$, the coordinate uppers are $a_1 - x + x^2$ and $1 - \tau - x + x^2 + x^4$. The analogous difference increases with τ; at $1/4$ it is at most $-7/16 + x/2 + x^2/2 - x^3 + (5/2)x^4$, and hence at most $-7/16 + x/2 + (11/12)x^2 \leq -121/1728$. For $L \geq 7$ use only as a comparison the rational symbol $P_\infty = 1/(1-z) - \tau(2+z)$. The exact identity $a_1^2 - |P_\infty|^2 = x[a_1/(1+x) - \tau^2]$ gives $a_1 - |P_\infty| \geq 211x/720$. For all finite $m \geq 1, L \geq 7$, the finite remainder divided by x is at most $x^{5/2} + x^2/4 + (6/5)x^{7/2} \leq 125/576 < 211/720$, using $\sqrt{x} < 2/3$ and $x < 5/12$. This comparison proves the norm bound for every finite tail. Its normalized imaginary coordinate I is positive and at most $9/8$; its real coordinate satisfies $R \geq \kappa - 1/16 - (2 + \theta^2)d \geq -1/96$. For these bounds use $(6/5)\theta^{7/2} < 1/16$ and $I \geq \kappa - (1 + \theta^2)d - 1/12 \geq 7/32 > 0$.

Q175.10 Long-connector means, centered rows and covariance

By (Q175.T180.9), $P_{\text{cost}} \leq a_{\text{cost}}d + b_{\text{cost}}d^2$, where $a_{\text{cost}} = 7/10 + 169/304 < 63/50$ and $b_{\text{cost}} = 2048/855 < 12/5$. Therefore

$$\mathbb{E}Z_L \leq 1 - (3/2)d + 6d^2 =: M_L \leq 1. \qquad \text{(Q175.T180.19)}$$

A more precise common bound is $\mathbb{E}Z_L \leq 1 - (1317/800)d + (167/30)d^2$. For $d \leq 1/8$ it is at most $1 - 9d/10$, and for $d \leq 1/4$ at most $1 - d/4$. This uses $\mathbb{E}Z_L = C + c - \int R\, d\nu$, $R \geq \kappa - 1/16 - (2+\theta^2)d$, and $\tau \geq \kappa d$. The actual mean $\mathbb{E}P_Lg' = -\int I\, d\nu$ lies in $[-9d/8, 0]$. In the low domain $c \leq 35d/32$; throughout the quarter-domain $c \leq 49d/32$. The centered row (Q175.T180.3) gives

$$\alpha Z_L + \mathbb{E}(\alpha Z_L) \leq \alpha\{q + 1 + \sqrt{pqd} + bd\},$$

with $b = 3/2$ for $d \le 1/8$ and $b = 5/2$ for $d \le 1/4$. Thus the coefficients in part N.20 are nine and eleven, respectively. Since $I \ge 0$, the full original variance satisfies

$$\begin{aligned} \operatorname{Var} Z_L &= \int |C + zP_L|^2 \, d\nu - (\int R \, d\nu)^2 \\ &\le d\{(1 - 3\kappa d)^2 + \theta(1 - 2\kappa d)^2\} =: dV_L. \end{aligned} \tag{Q175.T180.20}$$

The upper is $V_L = 45/32 - (1159/256)d + (30685/8192)d^2$; the negative mean square was displayed before this relaxation. For covariance write $P_L = R + zI$ and $G_a = A_0 + zB_1$. If $C - xI \ge 0$, use $A_0 \ge 1 - x, B_1 \le 1$. Otherwise the extra error is at most $x(xI - C)_+$. The full finite coordinates give

$$\int \operatorname{Re}\{G_a(C + zP_L)\} \, d\nu \ge -d/10 - (7/8)d^2 - (17/16)d^2.$$

Before the last sign correction, the main expression $1 - 3x + x^2 - \tau(3 - 6x + x^2)$ decreases throughout the frequency interval. At θ its constant error is $1219/12288 < 1/10$, its τ coefficient is $745/1024 < 3/4$, and the remaining d error is $\theta^3(1 + \kappa) < 1/8$. The sign correction is $x(xI - C)_+ \le \theta(3 - \theta)\tau < (17/16)d$. The mean term $-\beta_{a+1} \int R \, d\nu$ adds at most $d^2/96$. Combining gives

$$\operatorname{Cov}(C_a, \alpha Z_L) \le \alpha(d/10 + 2d^2). \tag{Q175.T180.21}$$

The actual β and sign terms are retained.

Q175.11 Two complete long-connector payments

The positive covariance contribution in the fourth expansion is at most $\alpha^4[(2/5)d + 8d^2]$. Use $M_L^4 \le 1 - 6d + (75/2)d^2 - 108d^3 + 216d^4$ and $\mathbb{E}C_a \le 1 - 19d/16$. With the covariance contribution this gives

$$A_L = 1 - (543/80)d + (421/8)d^2 - (4881/32)d^3 + (1377/4)d^4 - (513/2)d^5.$$

For $d \le 1/8$ use $K_0 = 9$ and $G = H^2 - A_L - 9dV_L H$. Its five coefficients have lower bounds $14/5$, $-29, 259, -11, -67$. Hence $G/d \ge 14/5 - 29d + 256d^2 > 1$, giving reference fourth at most $1 - d/4$. For $1/8 \le d \le 1/4$ use $K_0 = 11$. The five coefficient bounds are $1/16, -129/4, 264, 43, -138$, so

$$G/d \ge 1/16 - (129/4)d + 264d^2 + 43d^3 - 138d^4.$$

This polynomial increases on the interval: its derivative is at least $201/8 > 0$. The left endpoint is $423/2048 > 1/5$. Thus $G \ge d/5$ and $H^2 < 8$ give reference fourth at most $1 - d/40$. Every $L \ge 5$, $c \ge 4$ uses the same fixed direction and payment.

Q175.12 The high domain at every length

For $d \ge 1/4$ the actual arm satisfies $h_c \le 2\mathbb{E}f_{c-1} \le 2(1 - 2d) \le 1$. Thus $TF_c \le T(qf_3) \le 2\mathbb{E}(qf_3)$, and $\mathbb{E}(qf_3) = 1 - 3d + \delta_2 \le 1 - (83/32)d$ gives $TF_c \le M \le 45/64$. Positivity yields $T^L F_c \le M\bar{f}_{L-1}$. For $n \ge 1$, $\bar{f}_n \le 2$ and $\mathbb{E}[f_a \bar{f}_n] = \mu_{a+n} \le 1 - 2d$, so $\mathbb{E}[f_a \bar{f}_n^4] \le 8\mathbb{E}[f_a \bar{f}_n] \le 4$. For $n = 0$, $\mathbb{E}f_a \le 1 \le 4$. Therefore every $L \ge 1$ obeys

$$\mathbb{E}[f_a (T^L F_c)^4] \le 4(45/64)^4 < 47/48 \le 1 - d/16. \tag{Q175.T180.22}$$

This is a same-reference original path integral; it uses no conversion from an ordinary eighth moment.

The genuine receiving arm of length two

The same actual direction $F_c = qf_3h_c$ also satisfies

$$\mathbb{E}[f_2\{T^L F_c\}^4] \le 1 - d/800 \qquad (c \ge 4,\ L \ge 1). \tag{N.29}$$

Use the finite packet and positive upper functions already constructed above, with $B = -K$, $u = B\mathbf{1}$, and $D = 7d/20$. Here the receiving factor is exactly $C_2 = f_2 = 1 - u + Bu$, with $\mathbb{E}C_2 = 1 - d$ and $C_2 \le p + \sqrt{d}$. Thus the pointwise coefficient proofs in paragraphs N.20 and N.20 apply to it.

For $Y = c + Q(B)u$, write $\alpha = -\mathbb{E}[Q(B)u]$. Lemma M.6, with $a = 2, t = 0$, gives

$$\mathrm{Cov}(C_2, Y) = -\int \mathrm{Re}\{(1+z)Q(z)\}\, d\nu - d\alpha, \qquad z = i\sqrt{x}. \tag{N.30}$$

We use Lemma M.8 at the actual mean, before any mean endpoint or δ_2/d substitution.

On $0 < d \le 1/4$, the same finite polynomials give the following covariances and comparison budgets for the unscaled positive uppers:

L	new covariance upper	v_0	η
1	$-d/3$	0	$2(1-\delta_2/d)$
2	$-d/10$	$55d/1024$	$19/16$
3	$-3d/16 + (127/128)d^2$	$d/16 + d^2$	$19/16$
4	$-3d/16 + (11/6 + 7\theta/20)d^2$	$d/16 + 9d^2/4$	$19/16$
≥ 5	$-11d/40 + (289/96)d^2$	$d/10 + 2d^2$	$19/16.$

(N.31)

All these uppers have nonnegative actual mean at most one.

Here are finite checks of the new column. Write the actual packet polynomial as $R_0 = -\beta - z^3 G_{c-3}(-z)$, $\beta = \beta_{c-1}$. At $L = 1$, the full output symbol is $Q = 1 + z(1 + R_0)$. If $G_{c-3}(-z) = U - zV$, its real and normalized imaginary coordinates are $1 - x^2U, 1 - \beta + x^2V$, with $U, V \in [0,1]$, including $(U,V) = (1,0)$ at $c = 4$. Its genuine product is at least $1 - x - x^2 - x^3 > 1/3$, and $\alpha = d - D_1 > 0$. At $L = 2$, the full symbol is $G_3 + D + z^2R_0$. Its genuine product is at least $1 - 2x - x^3 > 1/10$, and its normalized imaginary coordinate $1 - x^2U$ is positive. These give the first two rows of (N.31) on the whole spectrum.

At $L = 3, 4$, use the actual form $Y_L = C_L q + BP_L(B)u + c_L$ in (Q175.T180.15). Its full symbol is $C_L + zP_L$, and its mean parameter $\int \mathrm{Re}\, P_L\, d\nu$ is positive. At length three, $C_3 \ge 1 - 17d/20$, $\mathrm{Re}\, P_3 \le 1 + D$, and $\mathrm{Im}_z\, P_3 \le 1$; at length four, $C_4 \ge 1 - 11d/6$, $\mathrm{Re}\, P_4 \le 1$, and $\mathrm{Im}_z\, P_4 \le 1 + D$. Since the genuine product is $C_L - x(\mathrm{Re}\, P_L + \mathrm{Im}_z\, P_L)$, these bounds give precisely the third and fourth rows.

For $L \ge 5$, the full symbol of the positive upper in paragraph N.20 is

$$G_{L+1} - \tau(3 + 2z + z^2) - \beta z^L - z^{L+3} G_{c-3}(-z).$$

The finite even/odd formulas give $\mathrm{Re}(G_2 G_n) \ge (1-x)^2 \ge \kappa^2 > 7/20$ for every $n \ge 4$: for even n the two coordinates of G_n agree and are at least $1 - x$; for odd $n \ge 5$, they give the lower $1 - 2x + 2x^2 - 2x^3 \ge (1-x)^2$. The retained subtraction costs at most $3\tau d$, since $\mathrm{Re}[G_2(3 + 2z + z^2)] = 3 - 3x \le 3$. The two finite tails together cost less than $(3/40)d$:

$$\frac{6}{5}\left\{d\theta^{L/2} + \frac{6}{5}\theta^{(L+3)/2}\right\} \le \frac{6}{5}\left\{\frac{1}{36} + \frac{1}{30}\right\} = \frac{11}{150} < \frac{3}{40}.$$

Here the shortest length is $L = 5$; later powers decrease. The actual bound $\operatorname{Re} P_L \geq -1/96$ gives an adverse signed-mean term of at most $d^2/96$, rather than discarding that term. This proves the last covariance row.

The first row improves its comparison budget by at least $d/4$. Rows two through four improve theirs by more than $3d/64$. For the last row the gain after the same fee is

$$\left(\frac{3}{8} - \frac{3}{64}\right) d - \frac{97}{96} d^2 \geq \frac{29}{384} d > 0.$$

Thus every bracket comparison is valid with $M = 1$. The unchanged variance and pointwise coefficients can now be used in the scalar payments of paragraphs N.20, N.20, and N.20. They give the respective deficits $d/800, d/4, d/16, d/64$, and at least $d/40$ for every long length.

For $d \geq 1/4$, the same direction has $h_c \leq 1$ and $TF_c \leq 45/64$, as in paragraph N.20. For $n \geq 1$, the actual adjoint identity gives

$$\mathbb{E}[f_2 \bar{f}_n^4] \leq 8\mathbb{E}[f_2 \bar{f}_n] = 8\mu_{n+2} \leq 8(1 - 2d) \leq 4;$$

for $n = 0$, use $\mathbb{E} f_2 \leq 1$. Hence the high-domain moment is at most $4(45/64)^4 < 47/48 \leq 1 - d/16$. At $d = 0$, leaf integration gives equality. The original equal-pair insertions and full reversal use positivity exactly as above, retaining the same receiving arm and connector.

Q175.13 Completion, decorations and reversal

On the low domain the deficits are $d/800$ at $L = 1$, at least $d/4$ at $L = 2$, $d/16$ at $L = 3$, $d/64$ at $L = 4$, and $d/40$ at $L \geq 5$. The high-domain deficit is at least $d/16$ for every L. This proves (Q175.T180.0) for all stated integers and original kernels. The terminal unit arm points inward, its three-edge arm outward throughout, and its c-arm has first edge inward and the remaining edges outward. Every connector edge points terminal-to-receiver; the receiving arm is outward. The analytic intervals estimate this one fixed message. The factors $E_t, E_o \in [0, 1]$ are restored by positivity of T. If the equal-pair half-length sums are S_t, S_o, an original component has $a + c + L + 4 + 2S_t + 2S_o$ edges and one more vertex. Its fourth-moment test tree has $a + 4c + 4L + 16 + 8S_t + 2S_o$ edges and one more vertex; the receiving arm occurs once. Complete reversal replaces K by $-K$, F_c by $p\bar{f}_3 W_c$ with $W_c = A\bar{f}_{c-1}$, T^L by A^L and f_a by $\bar{f}_a$, with all decorations reversed. The $d = 0$ leaf integration, arbitrary original joint-label laws, original edge counts and noninjective maps retain their original meanings. The proof gives this local reference interface and does not itself assign compound-half-tree moments or other skeleton closures. For comparison, for $4 \leq b < c$ finite positive convolution retains

$$f_b + h_c \leq (2 - \Gamma_{bc})\mathbf{1} + R_{bc}, \quad \Gamma_{bc} = \beta_b + \beta_c + \beta_{b-1} + \beta_{c-1},$$

$$R_{bc} = (\beta_{c-1} - \beta_{b-1})g - K^b G_{c-b}(K)g.$$

This different residual requires its own first-row error and finite signed D_i; the preceding $b = 3$ data are not its input. The $(1, b, c)$ family with $b \geq 4$ is supplied separately in Section N.22.

N.21 The reference terminal $(1, 2, 3)$ at every $L \geq 2$

Q176.0 The original objects and the actual direction

Fix a nonempty original finite probability space (X, μ) and a real kernel $K(x, y) = -K(y, x)$, $|K(x, y)| \leq 1$. All means, norms, adjoints, covariances and spectral measures use this original μ. Put

$$Jv = (\mathbb{E} v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad B = -K, \quad u = B\mathbf{1}, \quad p = 1 - u, \quad q = 1 + u,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}.$$

The message f_n is the outward n-arm, h_n has its first edge inward and the remaining edges outward, and $\bar{f}_n$ is entirely inward. The terminal residual is $(1,2,3)$, the L connector edges point terminal-to-receiver, and the receiver has a true outward arm f_a, $a \geq 3$. For every independently specified $a \geq 3$, $L \geq 2$, the fixed canonical terminal $F = qf_2h_3$ satisfies

$$\mathbb{E}[E_o f_a \{T^L(E_t F)\}^4] \leq 1 - d/16, \qquad d = \mathbb{E}u^2. \tag{Q176.T182.A}$$

The genuine pair products E_o, E_t remain at their original receiver and terminal. The unit terminal arm points inward, the two-edge arm outward, and the three-edge arm has first edge inward and the rest outward. These directions are fixed from the original tree before the host. Together with Section N.18, the structural table is

L	actual terminal	reference fourth, all $a \geq 3$
1	$qf_2\bar{f}_3$	$1 - d/256$
$L \geq 2$	qf_2h_3	$1 - d/16$.

The common bound is $1 - d/256$ with directions chosen from L alone. Delete only zero-mass atoms. Then $p, q \geq \mu(x) > 0$, and A, T are positive. Every actual tree message is strictly positive: the configuration with all labels equal to the root label has positive weight. At $d = 0$, K may be nonzero, but original leaf integration gives all messages one. Below take $d > 0, E_o = E_t = 1$, and restore decorations at the end. The finite framework gives

$$0 < d \leq 1/3, \quad \|B\|^2 \leq \theta = 13/32, \quad |Bv(x)| \leq \|v\|_2, \quad d^2 \leq \delta_2 \leq \theta d, \quad \delta_{j+1} \leq \theta\delta_j, \quad \delta_j = \|B^j\mathbf{1}\|_2^2.$$

Let $\kappa = 19/32$, $\tau = d - \delta_2$, $G_n(z) = \sum_{j=0}^{n-1} z^j$, $G_0 = 0$, $\beta_1 = \beta_2 = 0$, and $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j$. Then $\tau \leq \beta_n \leq d$ for $n \geq 3$, and

$$\mathbf{1} + G_n(B)u = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j}.$$

Complete reversal gives the receiving envelope

$$0 \leq f_a \leq C_a = \mathbf{1} - G_a(-B)u - \beta_a\mathbf{1}, \quad \mathbb{E}C_a = 1 - \beta_a - \beta_{a+1} \leq 1 - 19d/16, \quad C_a \leq p + (6/5)\sqrt{d}\,\mathbf{1}. \tag{Q176.T182.0}$$

The $a = 3$ predecessor retains its true norm $\sqrt{1+\theta} < 6/5$; for $a \geq 4$ the finite predecessor norm is at most one. For every $n \geq 3$, $|G_n(i\sqrt{x})| \leq 1$ on $[0, \theta]$: odd n has squared modulus $(1+x^n)/(1+x)$; even $n = 2j$ has $(1-(-x)^j)^2/(1+x)$, direct for even j, while odd $j \geq 3$ uses $2x^2 + x^5 < 1$. The spectral measure ν of $-B^2$ at u has mass d, support $[0, \theta]$ and $\int x^j\, d\nu = \delta_{j+1}$. The symbols $x, z = i\sqrt{x}$ represent the same original spectral calculation; there is no pointwise assumption $x \geq d$.

Q176.1 An independent coefficient-ten lemma

Suppose $C_0, Z \geq 0$, $\mu = \mathbb{E}Z \geq 0$, $t = \sqrt{d} \leq 1/\sqrt{3}$,

$$C_0 \leq p + (6/5)t, \quad Z \leq q + (13/12)t, \quad \mu \leq 1 - (5/4)t^2, \quad p + q = 2.$$

Then

$$C_0\{(Z+\mu)^2 + 2\mu^2\} \leq 10. \tag{Q176.T182b.1}$$

Use the positive upper $q+1+(13/12)t-(5/4)t^2$ for $Z+\mu$. Expand, use $t^4 \le t^2/3$, retain the negative cubic, and multiply by $p+(6/5)t$ to get the upper

$$\begin{aligned} P(q,t) &= 6+q-q^3+B(q)t-D_0(q)t^2-E_0(q)t^3-(13/4)t^4,\\ B(q) &= 119/15+(137/30)q-(29/30)q^2,\\ D_0(q) &= 1247/180-(851/360)q-(5/2)q^2,\\ E_0(q) &= 167/15+(7/24)q. \end{aligned}$$

The squared bracket before multiplication is

$$q^2+2q+3+(13/6)(q+1)t-[(5/2)q+343/72]t^2-(65/24)t^3.$$

Each substitution is made before multiplying by a nonnegative factor. For $q \le 1/2$, $P_q \ge 1/4+(18/5)t-(7/24)t^3 > 0$. For $q \ge 1/2$, $P_{qq} \le -3-(29/15)t+5t^2 \le -12/5$. The latter quadratic is convex in t, with endpoint values at most -3 and $-4/3-29/(15\sqrt{3}) < -256/105 < -12/5$, using $1/\sqrt{3} > 4/7$. Set

$$F_0(t) = P(1,t) = 6+(173/15)t-(743/360)t^2-(457/40)t^3-(13/4)t^4,$$

$$H_0(t) = P_q(1,t) = -2+(79/30)t+(2651/360)t^2-(7/24)t^3.$$

The function H_0 strictly increases, with

$$H_0(3/10) = -4441/8000, \quad H_0(2/5) = 479/2250, \quad H_0(1/2) = 3229/2880 < 9/8, \quad H_0(1/\sqrt{3}) < 2.$$

For the last value use $1/\sqrt{3} < 3/5$ and its exact expression $491/1080+913/(360\sqrt{3})$. The function F_0 increases on $[0,2/5]$, since $F_0'' < 0$ and $F_0'(2/5) > 0$. On $[0,3/10]$, $F_0 \le 178789/20000 < 9$ and $|H_0| \le 2$; strong concavity gives $P \le F_0+(5/24)H_0^2 < 9+5/6 < 10$. On $[3/10,2/5]$, $F_0 \le 106523/11250 < 19/2$, $|H_0| \le 3/5$, so $P < 19/2+3/40 < 10$. On $[2/5,1/2]$, $F_0'' \le -36$, $F_0(1/2) = 3463/360$ and $F_0'(1/2) = -1043/1440$, of absolute value less than one. The upper tangent quadratic gives $F_0 \le 3463/360+1/72 < 97/10$, and $0 < H_0 \le 9/8$. Thus $P < 97/10+405/1536 < 10$. On $[1/2,1/\sqrt{3}]$, $H_0 > 0$, so for $q \le 1$ the maximum is at most $P(1,t)$. For $q \ge 1$ use $P_{qq} \le -27/5$: the convex quadratic $-6-(29/15)t+5t^2$ has endpoint values $-343/60$ and less than $-571/105$, both below $-27/5$. Since F_0 decreases for $t \ge 1/2$, $P \le 3463/360+(5/54)4 = 10789/1080 < 10$. This covers every q,t in the stated continuous domain.

Q176.2 The canonical first row and the fourth-expansion interface

Let $m = 1-d/2$, $k = 1/m \in [1,6/5]$, and $D = d/4$. The actual arm identity $f_2+h_3 = 2m\mathbf{1}+B^2u$ gives $R = kB^2u$, whose frequency symbol is $-kxu$. Hence

$$F \le m^2q(1+R/2)^2, \qquad TF \le m^2[\bar{f}_2+BR+\delta\mathbf{1}], \qquad \delta = -k\delta_2+k\sqrt{d\delta_3}+k^2\delta_3.$$

With $s = \delta_2/d$, use $\delta_3 \le \theta\delta_2$ and $\sqrt{\theta s}-(1-k\theta)s \le \theta/[4(1-k\theta)]$. Since $k \le 6/5$ and $1-k\theta \ge 41/80$, $\delta/d \le 39/164 < 1/4$. Thus $D = d/4$ is valid throughout the host domain. For $D_i = \mathbb{E}B^iR$, the exact values include $D_1 = k\delta_2$, $D_2 = 0$, $D_3 = -k\delta_3$. Successive application of the same positive T gives

$$0 < T^LF/m^2 \le Y_L = \bar{f}_{L+1}+B^LR+D\bar{f}_{L-1}+\sum_{i=1}^{L-1} D_i\bar{f}_{L-1-i}.$$

The Z_2, Z_3, Z_4 below are exactly the reorganized Y_2, Y_3, Y_4; they are nonnegative before taking fourth powers. The same original envelope C_a satisfies $\mathbb{E}C_a \le 1 - 19d/16$ and $C_a \le p + (6/5)\sqrt{d}$. For $Z = s_0\mathbf{1} + Q(B)u$ with $\alpha = -\mathbb{E}[Q(B)u] > 0$,

$$\operatorname{Var} Z = \|Q(B)u\|_2^2 - \alpha^2, \qquad \operatorname{Cov}(C_a, Z) = -\int \operatorname{Re}(G_a Q)\, d\nu - \beta_{a+1}\alpha. \tag{Q176.T182b.2}$$

Here $a \ge 3$ and $\beta_{a+1} \ge \tau$. With $\mu_Z = \mathbb{E}Z$, the exact fourth identity and the coefficient-ten lemma imply

$$\mathbb{E}[C_a Z^4] \le (\mathbb{E}C_a)\mu_Z^4 + 4\mu_Z^3 \operatorname{Cov}(C_a, Z) + 10 \operatorname{Var} Z$$

when that lemma's pointwise and mean conditions have been proved. Since $T^L F \le m^2 Z$, multiply the entire right side by m^8; variance scales by m^4. For $G_a = E_a + zD_a$, even a has $E_a = D_a$, odd a has $E_a = 1 - xD_a$, and $0 \le D_a \le 1$. We use $\psi(x) = 1 - 3x + 2x^2 - x^3 \ge \psi(\theta) = 1451/32768 > 0$; its derivative is negative throughout the interval.

Q176.3 Two connector edges

Finite propagation gives

$$Z_2 = C_2 q + BP_2(B)u + c_2\mathbf{1}, \quad C_2 = 1 + D, \quad P_2 = 1 + (1 - kx)z, \quad c_2 = k\delta_2 - d,$$

$$Q_2 = C_2 - x + kx^2 + z, \quad \alpha_2 = d, \quad \mu_2 = 1 - 7d/4 + k\delta_2 \le 1 - 101d/80. \tag{Q176.T182b.3}$$

One has $|P_2|^2 = 1 + x(1 - kx)^2 \le 1 + x(1 - x)^2 \le 31/27 < (13/12)^2$. Since $Dq + c_2 \le d/2 - 41d/80 < 0$, $Z_2 \le q + (13/12)\sqrt{d}$; also $\mu_2 \le 1 - 5d/4$. Thus the coefficient-ten lemma applies. To prove $|Q_2| \le C_2$, divide the squared difference by x, put $s = d/2, k = 1/(1 - s)$, and multiply by $(1 - s)^2$. The result is

$$\Phi(s, x) = -1 + s + s^2 - s^3 + (1 - s)(3x - 2x^2) + x^3.$$

It increases in both variables on $0 \le s \le 1/6$, $0 \le x \le \theta$: $\Phi_s = (1 - x)(1 - 2x) + 2s - 3s^2 \ge 0$ and $\Phi_x = (1 - s)(3 - 4x) + 3x^2 > 0$. Its upper-right value is $-2281/884736 < 0$. Hence $\operatorname{Var} Z_2 \le C_2^2 d - d^2 = (1 - d/4)^2 d$. The real coordinate of Q_2 is at least $1 - x + x^2$. Even a gives $\operatorname{Re}(G_a Q_2) \ge D_a(1 - 2x + x^2) \ge 0$; for odd a the worst $D_a = 1$ gives the positive lower bound $\psi(x)$. Therefore $\operatorname{Cov}(C_a, Z_2) \le 0$, and

$$\mathbb{E}[f_a (T^2 F)^4] \le e^{-819d/80} + 10de^{-9d/2}. \tag{Q176.T182b.4}$$

For $d \le 1/6$, $k \le 12/11$ and $\mu_2 \le 1 - 115d/88$. The first exponent improves to $1833/176 > 52/5$, giving

$$\mathbb{E}[f_a (T^2 F)^4] \le e^{-52d/5} + 10de^{-9d/2}. \tag{Q176.T182b.5}$$

For the full continuous payment, first let $0 < d \le 1/8$ and define $L(d) = [e^{9d/2} - e^{-59d/10}]/d$. Degree-four Taylor bounds give

$$L(d) \ge 52/5 - (182/25)d + (37063/750)d^2 - (250523/7500)d^3 \ge 52/5 - (182/25)d + 45d^2.$$

The positive degree-four Taylor sum and a tail with ratios at most $1/8$ give $e^{3/4} < 17/8$. Convexity gives $e^{9d/2} \le 1 + 27d/4$ for $d \le 1/6$. Thus $L - 10 - e^{9d/2}/16$ is at least $27/80 - (12323/1600)d + 45d^2 \ge 3663671/460800000 > 0$. This pays (Q176.T182b.5) by $1 - d/16$. On $1/8 \le d \le 1/6$,

its right side decreases: the derivative test reduces to $e^{(59/10)d}(1-9d/2)$, decreasing there, with left endpoint less than $e^{3/4}7/16 < 119/128 < 26/25$. At $1/8$, $e^{-13/10} < 11/40$ by the positive degree-five Taylor sum, and $e^{-9/16} < 4/7$ by the positive degree-three sum, with gap $3/8192$. The bound is less than $277/280 < 95/96 \le 1-d/16$. On $1/6 \le d \le 1/3$, the right side of (Q176.T182b.4) also decreases. Before $2/9$ the test expression $e^{(459/80)d}(1-9d/2)$ decreases, starting below $e/4 < 3/4 < 819/800$; afterwards both derivatives are negative. At $1/6$, $e^{-273/160} < 3/16$, since the degree-five positive Taylor sum for $e^{17/10}$ exceeds $16/3$; and $e^{-3/4} < 19/40$, since its degree-four positive sum is $4331/2048 > 40/19$. Thus the bound is less than $3/16 + 19/24 = 47/48 \le 1-d/16$. All $a \ge 3$ and the entire d-domain are covered at $L = 2$.

Q176.4 Three connector edges

The finite expression is

$$Z_3 = C_3 q + BP_3(B)u - d\mathbf{1}, \quad C_3 = 1 - d + D + k\delta_2, \quad P_3 = 1 + D - x + kx^2 + z,$$

$$Q_3 = C_3 - x + z(1 + D - x + kx^2), \qquad \alpha_3 = (1+D)d - \delta_2 + k\delta_3 \ge \tau,$$

$$\mu_3 = 1 - 3d + D + (k+1)\delta_2 - Dd - k\delta_3 \le 1 - 297d/160, \qquad C_3 \le 1 - 21d/80. \tag{Q176.T182b.6}$$

Also $C_3 \ge 1 - 3d/4 + d^2 \ge 31/36$; the final difference factors as $(1/3-d)(5/12-d)$. Put $A = 1 + D - x + kx^2$, so $0 < A \le 1 + D \le 13/12$. Then $|Q_3|^2 - C_3^2 = x[-2C_3 + x + A^2] \le -(41/288)x \le 0$. Since $P_3 = Q_2$, $|P_3| \le C_2 \le 13/12$, so $Z_3 \le q + (13/12)\sqrt{d}$. The mean is at most $1 - 5d/4$, and the coefficient-ten lemma applies; $\operatorname{Var} Z_3 \le C_3^2 d$. For $Q_{30} = 1 - x + z(1 - x + x^2)$, both parity classes give $\operatorname{Re}(G_a Q_{30}) \ge 0$, with odd-class lower bound ψ. The actual perturbation is $(-d + D + k\delta_2) + z[D + (k-1)x^2]$. Therefore

$$\operatorname{Cov}(C_a, Z_3) \le d^2 + D\delta_2 + (k-1)\delta_4 - \beta_{a+1}\alpha_3 \le d^2.$$

Use $k - 1 \le 3d/5$, $\theta/4 + 3\theta^3/5 < 37/240 < 1/4 < \kappa^2$ and $\beta_{a+1}\alpha_3 \ge \kappa^2 d^2$ for the last step. The actual negative constant has not been set to zero. Consequently

$$\begin{aligned} \mathbb{E}[f_a (T^3 F)^4] &\le e^{-1009d/80} + 10de^{-181d/40} + 4d^2 e^{-1531d/160} \\ &\le e^{-25d/2} + (103/10)de^{-9d/2}. \end{aligned} \tag{Q176.T182b.7}$$

The normalized covariance cost is at most $240/811 < 3/10$, using $te^{-ct} \le 1/(ce)$, $e > 8/3$ and $c = 811/160$. For $0 < d \le 1/4$, put $L = [e^{9d/2} - e^{-8d}]/d$. The positive and negative degree-four Taylor bounds yield

$$\begin{aligned} L &\ge 25/2 - (175/8)d + (4825/48)d^2 - (58975/384)d^3 \\ &\ge 25/2 - (175/8)d + 62d^2 \ge 25/2 - 30625/15872 > 211/20. \end{aligned}$$

Since $211/20 = 103/10 + 1/4$ and $e^{9/8} < 4$, this gives $1 - d/16$. For $d \ge 1/4$, both terms decrease and at $1/4$ are less than $1/16 + 103/120 = 221/240 < 47/48$. The lower bounds $e^{25/8} > 16$ and $e^{9/8} > 3$ follow from positive Taylor sums through degree four; $e^{9/8} < 4$ follows from $e < 3$ and $e^{1/8} \le 8/7$.

Q176.5 Four connector edges and the actual mean

The finite T^4 contains the actual $D_3 = -k\delta_3$:

$$Z_4 = C_4 q + BP_4(B)u + c_4\mathbf{1}, \quad C_4 = 1 - 2d + D + k\delta_2, \quad A_1 = C_4 + d = C_3,$$
$$P_4 = A_1 - x + z(1 + D - x + kx^2) = Q_3, \quad c_4 = -\tau - Dd - k\delta_3,$$
$$Q_4 = C_4 - xA + z(A_1 - x), \qquad A = 1 + D - x + kx^2.$$

The exact mean loss is $\alpha_4 = A_1 d - \delta_2 \geq (131/288)d > 5d/12$, and the mean is

$$\mu_4 = 1 - 4d + D + (k+2)\delta_2 + d^2 - 2Dd - kd\delta_2 - k\delta_3. \tag{Q176.T182b.8}$$

There is exactly one $-k\delta_3$ term. Since $k(1-d) \leq 1$, $\mu_4 \leq 1 - 15d/4 + 3\delta_2 + d^2/2 \leq 1 - 227d/96$ and $C_4 \leq 1 - 101d/80$. Also $C_4 \geq C_{\text{lo}} := 1 - 7d/4 + d^2 \geq 19/36 > 0$ and $C_4 \leq 1$. Using $1 + D - x \leq A \leq 1 + D$ gives

$$\frac{|Q_4|^2 - C_4^2}{x} \leq C_4^2 + [(3/2)d - 2]C_4 + d^2 + x(1 - 3d/2 + d^2/16) + x^2.$$

The right side increases in x, so set $x = \theta$; it is convex in C_4, so check $C_{\text{lo}}, 1$. At $C_4 = 1$ it is $F_1(d) = -439/1024 + (3\kappa/2)d + (1 + \theta/16)d^2$, increasing up to $d = 1/3$, where it equals $-55/3072 < 0$. The C_{lo} value exceeds this by $d^2(7/4 - d)(1/4 - d)$, nonpositive at $d \geq 1/4$. At $d \leq 1/4$ it is at most $7d^2/16$, giving the endpoint upper $-939/8192 < 0$ at $1/4$. Thus $|Q_4| \leq C_4$ throughout the domain. Since $P_4 = Q_3$, $|P_4| \leq C_3 \leq 1$ and $c_4 < 0$, $Z_4 \leq q + \sqrt{d}$. Its mean is at most $1 - 5d/4$, so the coefficient-ten lemma again applies. For $Q_{40} = 1 - x + x^2 - x^3 + z(1 - x)$, the product with every G_a has nonnegative real part. For even a its lower bound is $D_a(1 - 2x + 2x^2 - x^3)$; for odd a it is $(1 - x)(1 - 2x + x^2 - x^3) > 0$. The actual perturbation has real part $s_C - Dx - (k-1)x^3$ and imaginary coordinate s_A, with $s_C = -7d/4 + k\delta_2 < 0$ and $s_A = -3d/4 + k\delta_2 < 0$. Thus the imaginary contribution $-xD_a s_A$ is nonnegative, and

$$\operatorname{Cov}(C_a, Z_4) \leq (7/4)d^2 + D\delta_2 + (k-1)\delta_4 - \beta_{a+1}\alpha_4 \leq (7/4)d^2.$$

Here $\beta_{a+1}\alpha_4 > (5/24)d^2$, while the extra positive cost is less than $37d^2/240$. With $\operatorname{Var} Z_4 \leq C_4^2 d$, this yields

$$\begin{aligned}\mathbb{E}[f_a(T^4F)^4] &\leq e^{-703d/48} + 10de^{-261d/40} + 7d^2e^{-355d/32} \\ &\leq e^{-703d/48} + (74/7)de^{-13d/2}.\end{aligned} \tag{Q176.T182b.9}$$

The normalized covariance cost is at most $4/7$, using exponent gap $147/32$ and $e > 8/3$. Set $\alpha = 703/48$, $\beta = 13/2$ and $c = 391/48$. For $d \leq 1/10$, the linear positive-exponential lower and quadratic negative-exponential upper give

$$\frac{e^{\beta d} - e^{-cd}}{d} \geq \alpha - c^2 d/2 > 29/2 - 1089/320 = 3551/320 > 74/7 + 1/4.$$

Since $e^{13/20} < 3 < 4$, this pays by $1 - d/16$. For $d \geq 1/10$ the upper decreases: before $2/13$, $e^{cd}(1 - \beta d)$ decreases and starts below $3 \cdot 7/20 = 21/20 < \alpha/(74/7)$; afterwards both derivatives are negative. At $1/10$, the positive quadratic Taylor sums give $e^{-703/480} < 1/3$ and $e^{-13/20} < 5/9$. The upper is less than $58/63 < 47/48$. This completes $L = 4$ for every receiving length on the full domain.

Q176.6 Four actual losses for every $L \ge 5$

Keep $m = 1 - d/2$, $k = 1/m \in [1, 6/5]$. The actual identity $f_2 + h_3 = 2m\mathbf{1} + B^2u$ gives $F \le q(m + B^2u/2)^2$. The first row has $R = -kxu$, $\delta \le 39d/164 < d/4$ and $D = d/4$. Set $D_i = \mathbb{E}B^iR$, $\gamma_j = \beta_j - D_{j-2}$. The exact initial values are $D_1 = k\delta_2, D_2 = 0, D_3 = -k\delta_3, D_4 = 0$; later positive values are at most $k\delta_4 < d/12$. Thus every $j \ge 3$ has $\gamma_j \ge h := 49d/96$: $\gamma_3 = d - k\delta_2 \ge 41d/80 > 49d/96$, $\gamma_4 = d$, and the others have $\gamma_j \ge \kappa d - d/12 = 49d/96$. These are the same finite square coefficients as C102, rather than an application of its ordinary eighth conclusion. Retain the actual last four losses $\bar f_0, \bar f_1, \bar f_2, \bar f_3$ and use $D\bar f_{L-1} \le DS'_{L-1}$. The finite positive upper is

$$\begin{aligned} Y_L &= \bar f_{L+1} + B^LR + D\bar f_{L-1} + \sum_{i=1}^{L-1} D_i \bar f_{L-1-i} \\ &= S'_{L+1} + B^LR + D\bar f_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar f_{L+1-j} \ge T^LF/m^2 > 0. \end{aligned}$$

Since $S'_n = \mathbf{1} + G_n(B)u$ and $\bar f_0 + \bar f_1 + \bar f_2 + \bar f_3 = (4-d)\mathbf{1} + 3u + 2Bu + B^2u$,

$$\begin{aligned} 0 < T^LF \le W = m^2Z, \qquad Z = Cq + BP_L(B)u - h(1-d)\mathbf{1}, \\ C = 1 + D - 3h = 1 - 41d/32, \quad P_L = G_L - kxz^{L-1} + DG_{L-2} - h(2+z). \end{aligned}$$

Here $x = -B^2$ is finite-symbol notation, with frequency $z = i\sqrt{x}$; it is not an identification of a frequency with d. Equivalently $Z = s\mathbf{1} + Q_L(B)u$, where

$$s = 1 + D - 4h + hd, \qquad Q_L = G_{L+1} - kxz^L + DG_{L-1} - h(3 + 2z + z^2).$$

The uppers come from positivity of the original T, so $Z > 0$ before raising it to the fourth power.

Q176.7 A finite three-term shift bound

For $0 \le \sigma \le 49/288$, every $n \ge 4$, and $0 \le x \le \theta$, $z = i\sqrt{x}$, one has

$$|G_n(z) - \sigma(3 + 2z + z^2)| \le 1 - 3\sigma. \tag{Q176.T182.1}$$

Put $A = 1 - 3\sigma$. At $n = 4$ the squared difference from A^2, divided by x, is

$$-1 + 4\sigma - 2\sigma^2 + (-1 + 2\sigma + \sigma^2)x + x^2.$$

The constant is at most $-3/8$, the linear coefficient negative, and $x^2 \le \theta^2 < 1/6$, so it is negative. At $n = 5$ the quotient is

$$\begin{aligned} &-1 + 4\sigma - 2\sigma^2 + (1 - 4\sigma + \sigma^2)x + (-1 + 2\sigma)x^2 + x^3 \\ &\quad \le -3/8 + x - (95/144)x^2 + x^3. \end{aligned}$$

The final polynomial increases, since its derivative is at least $1 - (95/72)\theta > 0$; its value at θ is $-3121/294912 < 0$. Here $1 - 4\sigma + 2\sigma^2 \ge 15649/41472 > 3/8$. For $n \ge 6$, multiplication by $1 - z$ gives numerator $M - z^n$, with $M = A + \sigma z + \sigma z^2 + \sigma z^3$ and

$$A^2(1 + x) - |M|^2 = x[1 - 4\sigma + 2\sigma^2 + \sigma^2 x - \sigma^2 x^2] \ge (3/8)x.$$

Since $|z+z^2+z^3|^2 = x - x^2 + x^3 \le x \le 1$, $|M| \le A + \sigma \le 1$. The finite tail increment divided by x is at most $2\theta^2 + \theta^5 < 169/512 + 1/80 = 877/2560 < 3/8$. The case $x = 0$ is direct, completing all finite integers. Since $0 \le kx \le 39/80$, $G_n - kxz^{n-1} = (1-kx)G_n + kxG_{n-1}$. For Q_L, $n = L+1 \ge 6$; (Q176.T182.1) and $\|G_{L-1}\| \le 1$ give

$$|Q_L| \le 1 - 3h + D = C, \qquad \|Q_L(B)u\|_2^2 \le C^2 d. \tag{Q176.T182.2}$$

Apply the proved two-term shift (Q102.5.6), valid for $n \ge 3, \sigma \le 3/16$, to the same convex combination:

$$|P_L| \le 1 - 2h + D = 1 - 37d/48 =: A_0. \tag{Q176.T182.3}$$

At $L = 5$ the $G_{L-2} = G_3$ and convex combination G_5, G_4 all lie in the actual domain.

Q176.8 Long-connector means, covariance and variance

Let $\alpha = -\mathbb{E}[Q_L(B)u]$. The exact finite mean loss is

$$\alpha = \gamma_{L+2} + D\beta_L - 2hd \ge h + D\tau - 2hd > 0.$$

Therefore

$$\mu_Z = \mathbb{E}Z \le 1 - (221/96)d + (177/128)d^2 \le 1 - (707/384)d. \tag{Q176.T182.4}$$

The exact variance $\|Q_L(B)u\|_2^2 - \alpha^2$ is at most $C^2 d$; the negative squared mean is displayed before being discarded. The required finite symbol fact is $\mathrm{Re}(G_a G_n) \ge 0$ for $a \ge 3, n \ge 4$. If either index is even, the lower bound follows from $(1-x)(1-2x) > 0$. If both are odd, $n \ge 5$ has imaginary coordinate at most $1 - x + x^2$; using the receiving imaginary coordinate at most one gives

$$1 - 3x + 3x^2 - 3x^3 + x^4 \ge 1 - 3x + 2x^2 - x^3 \ge 1451/32768 > 0.$$

The finite coordinate relations are even $E = D$, odd $E = 1 - xD$. Thus both G convex-combination terms and DG_{L-1} in Q_L give nonnegative receiving real parts. Since $\mathrm{Re}[G_a(3 + 2z + z^2)] \le (3 - x)E_a \le 3$,

$$\mathrm{Cov}(C_a, Z) = -\int \mathrm{Re}(G_a Q_L)\, d\nu - \beta_{a+1}\alpha \le 3hd = (49/32)d^2. \tag{Q176.T182.5}$$

The actual constant and mean terms are retained.

Q176.9 Every pointwise condition for coefficient nine

The coefficient-nine lemma proved in part Q117.4 of Section N.11 requires nonnegative C_a, W with

$$C_a \le p + (6/5)\sqrt{d}, \quad W \le (1 - 5d/8)q - d/2 + (13/12)\sqrt{d}, \quad \mathbb{E}W \le 1 - 3d/2.$$

It gives $C_a[(W + \mathbb{E}W)^2 + 2(\mathbb{E}W)^2] \le 9$ at the same original point. From (Q176.T182.3), $Z \le Cq + A_0\sqrt{d} - h(1-d)$. The q coefficient satisfies $m^2 C \le 1 - 5d/8$, and $\mathbb{E}W \le \mu_Z \le 1 - 3d/2$. With $t = \sqrt{d}$, the remaining pointwise difference is exactly

$$\begin{aligned}
&(13/12)t - m^2 A_0 t + m^2 h(1-d) - d/2 \\
&= t/12 + t^2/96 + (85/48)t^3 - (49/48)t^4 - (49/48)t^5 \\
&\quad + (245/384)t^6 + (37/192)t^7 - (49/384)t^8 \ge 0.
\end{aligned}$$

Indeed $t \le 3/5, t^2 \le 1/3$ give $85 - 49t - 49t^2 \ge 589/15 > 0$ and $245 - 49t^2 > 0$; all remaining terms are positive. All the coefficient-nine conditions are now supplied.

Q176.10 The full continuous long-connector payment

The exact fourth identity and the same-kernel data give

$$\begin{aligned}\mathbb{E}[f_a(T^L F)^4] &\le \mathbb{E}[C_a W^4] \\ &\le (\mathbb{E}C_a)(\mathbb{E}W)^4 + 4(\mathbb{E}W)^3 \operatorname{Cov}(C_a, W) + 9 \operatorname{Var} W \\ &\le e^{-1205d/96} + 9de^{-73d/16} + (49/8)d^2 e^{-1219d/128}.\end{aligned}$$

Here $\operatorname{Var} W = m^4 \operatorname{Var} Z$, and the covariance term has scale $m^8 \mu_Z^3 \operatorname{Cov}(C_a, Z)$. Using $e > 8/3$ and $te^{-ct} \le 1/(ce)$, the covariance coefficient relative to $de^{-73d/16}$ is at most $294/635 < 1/2$. Therefore

$$\mathbb{E}[f_a(T^L F)^4] \le e^{-25d/2} + (19/2)de^{-9d/2}. \tag{Q176.T182.6}$$

For $0 < d \le 1/4$, keep the positive cubic and negative quartic Taylor bounds:

$$\begin{aligned}\frac{e^{9d/2} - e^{-8d}}{d} &\ge 25/2 - (175/8)d + (4825/48)d^2 - (512/3)d^3 \\ &\ge 25/2 - 22d + 57d^2 \ge 1183/114 > 39/4.\end{aligned}$$

Thus the moment is at most $1 - (d/4)e^{-9d/2} \le 1 - d/16$, using $e^{9/8} < 4$, which follows from $e < 3, e^{1/8} \le 8/7$. On $[1/4, 1/3]$ both terms in (Q176.T182.6) decrease; at $1/4$ they sum to less than $1/16 + 19/24 = 41/48 < 1 - d/16$. The lower bounds $e^{25/8} > 16$ and $e^{9/8} > 3$ follow from positive Taylor sums through degree four. Hence

$$\mathbb{E}[f_a\{T^L(qf_2h_3)\}^4] \le 1 - d/16 \qquad (a \ge 3, L \ge 5). \tag{Q176.T182.7}$$

Together with the separate $L = 2, 3, 4$ payments this proves (Q176.T182.A).

Q176.11 Genuine decorations, original counts and full reversal

Every original equal-pair message lies in $[0, 1]$, so positivity of the same T gives $0 < T^L(E_t F) \le T^L F$; at the receiver, $E_o \le 1$. This restores all actual multiplicities, equal lengths and subdivisions. If the terminal and receiving pair half-length sums are S_t, S_o, one original component has $a + L + 6 + 2S_t + 2S_o$ edges. The reference-fourth test tree shares only the receiver and its single a-arm, with

$$e_{\text{verification}} = a + 4(L + 6) + 8S_t + 2S_o, \qquad |V| = e_{\text{verification}} + 1. \tag{Q176.T182.8}$$

The analytic copies describe the verification integral, rather than extra edges in the original application. Distinct four-branch messages may be combined under the same f_a through Hölder, with each actual message paid. Complete reversal replaces K by $-K$, f_a by $\bar{f}_a$, T by A, and qf_2h_3 by $p\bar{f}_2W_3$. The unit row $qf_2\bar{f}_3$ becomes $p\bar{f}_2f_3$. Connectors, terminal arms and decorations reverse together, with d unchanged; the reference weight is never reversed alone.

N.22 Reference fourth moments for terminal $(1, b, c)$, $4 \le b < c$

Q178.0 The original local theorem and fixed direction

On a nonempty original finite probability space (X, μ), let K be real skew with $|K(x, y)| \le 1$. All expectations, inner products, spectra, variances, covariances and row bounds use this same original μ. Put

$$Jv = (\mathbb{E}v)\mathbf{1}, \quad B = -K, \quad A = J - B, \quad T = J + B = A^*, \quad u = B\mathbf{1}, \quad p = 1 - u, \quad q = 1 + u,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad \delta_j = \|B^j\mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32.$$

The receiver has a true outward arm of length $a \geq 3$. The terminal residual is $(1, b, c)$, $4 \leq b < c$, and L is any independently specified positive original connector length. All connector edges point terminal-to-receiver. The terminal unit arm points inward, its b-arm outward throughout, and its c-arm has first edge inward and the rest outward. Thus $F = qf_bh_c$. Genuine equal pairs at the two roots have fixed messages $f_\ell h_\ell$ or the complete pair reversal; their products E_o, E_t lie in $[0, 1]$. Every edge belongs to the original tree. For all these independent integers,

$$\mathbb{E}[E_o f_a \{T^L(E_t q f_b h_c)\}^4] \leq 1 - d/3000. \tag{Q178.T186.A}$$

The direction precedes the host, measure and kernel and stays fixed across the analytic intervals. The associated four-terminal applications use the stated receiving arm, rather than an arbitrary compound weight.

Q178.1 Finite tools and the true receiving weight

The finite framework gives $0 \leq d \leq 1/3$, $\|B\|^2 \leq \theta$, $|u| \leq 1$, and positive A, T with kernels in $[0, 2]$. For $n \geq 1$, $0 \leq f_n, \bar{f}_n \leq 2$ and their means are at most one; for $n \geq 3$ the means are at most $1 - 2d$. Equal pairs are in $[0, 1]$. Delete only zero-mass atoms; then $p(x), q(x) \geq \mu(x) > 0$ and the true tree messages are strictly positive. Let $G_n(z) = \sum_{j=0}^{n-1} z^j$, $G_0 = 0$, $\beta_1 = \beta_2 = 0$, and $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j$. For $n \geq 3$, $\tau = d - \delta_2 \leq \beta_n \leq d$, and

$$\mathbf{1} + G_n(B)u = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j}. \tag{Q178.T186.1}$$

The spectral measure ν of $-B^2$ at u has mass d, support $[0, \theta]$, and $\int x^j \, d\nu = \delta_{j+1}$. At $z = i\sqrt{x}$, for a finite symbol $A_0 + zC_0$,

$$\|(A_0 + zC_0)(B)u\|_2^2 = \int (A_0^2 + xC_0^2) \, d\nu, \qquad \mathbb{E}[(A_0 + zC_0)(B)u] = -\int C_0 \, d\nu. \tag{Q178.T186.2}$$

Global Cauchy–Schwarz gives $d^2 \leq \delta_2$ and $\delta_3 \geq \delta_2^2/d$ for $d > 0$; these are not pointwise $x \geq d$. For $n \geq 2$, $G_n = U + zV$ has $1 - x \leq U, V \leq 1$; $n = 1$ has $(U, V) = (1, 0)$. For $n \geq 3$, $|G_n| \leq 1$, while the true G_2 norm is $\sqrt{1 + \theta} < 6/5$. Odd n has squared modulus $(1 + x^n)/(1 + x)$; even $n = 2j$ has $(1 - (-x)^j)^2/(1 + x)$, with odd $j \geq 3$ covered by $2x^2 + x^5 < 1$. The actual nonnegative receiving upper is

$$0 \leq f_a \leq C_a = \mathbf{1} - G_a(-B)u - \beta_a\mathbf{1}, \quad \mathbb{E}C_a = 1 - \beta_a - \beta_{a+1} \leq 1 - 2\tau \leq 1 - 19d/16. \tag{Q178.T186.3}$$

The row bound, including the $a = 3$ predecessor G_2, gives $C_a \leq p + (6/5)\sqrt{d}$. For $Z = s\mathbf{1} + Q(B)u$ and $\zeta = -\mathbb{E}[Q(B)u]$,

$$\operatorname{Var} Z = \|Q(B)u\|_2^2 - \zeta^2, \quad \operatorname{Cov}(C_a, Z) = -\int \operatorname{Re}(G_a Q) \, d\nu - \beta_{a+1}\zeta. \tag{Q178.T186.4}$$

The last index is $a + 1$, and the original mean is retained. For $W \geq 0$ and its actual mean $m = \mathbb{E}W$,

$$W^4 = m^4 + 4m^3(W - m) + (W - m)^2\{(W + m)^2 + 2m^2\}. \tag{Q178.T186.5}$$

If $C_a\{(W+m)^2+2m^2\} \le k$ pointwise, then

$$\mathbb{E}[C_a W^4] \le (\mathbb{E}C_a)m^4 + 4m^3 \operatorname{Cov}(C_a, W) + k \operatorname{Var} W. \tag{Q178.T186.6}$$

For $W = \alpha Z$ this is exactly $\alpha^4[(\mathbb{E}C_a)(\mathbb{E}Z)^4 + 4(\mathbb{E}Z)^3 \operatorname{Cov}(C_a, Z)] + k\alpha^2 \operatorname{Var} Z$. Negative covariance uses a lower mean bound; positive covariance uses an upper mean bound. For a mean-zero original real vector V, $|BV(x)|^2 \le pq\mathbb{E}V^2$. Thus for $Z = Cq + BV + c\mathbf{1}$, $e = \mathbb{E}V$ and $\sigma^2 = \operatorname{Var} V$,

$$Z \le (C+e)q + (c-e)\mathbf{1} + \sqrt{pq}\,\sigma. \tag{Q178.T186.7}$$

Only when $c = e$ has been proved does the free constant cancel; this cancellation does not hold at $L = 1$.

Q178.2 The finite first-row packet and propagation

Use the actual envelope and signed first-row facts proved in parts C173.2–4 of Section Q.8. This input concerns finite envelopes and coefficients, rather than a conversion of its ordinary-fourth conclusion. Let $n = c - b \ge 1$, $c_0 = \beta_{c-1} - \beta_{b-1}$, and

$$\Gamma = \beta_b + \beta_{b-1} + \beta_c + \beta_{c-1}, \quad r(z) = -c_0 + (-1)^b z^b G_n(-z), \quad R = r(B)u.$$

Retain $\beta_b, \beta_{b-1}p$ from f_b and $\beta_{c-1}q$ from h_c:

$$f_b + h_c \le (2-\Gamma)\mathbf{1} + R, \qquad F \le \alpha q e^R, \quad \alpha = e^{-\Gamma}. \tag{Q178.T186.8}$$

This is AM–GM for the two actual nonnegative arms. For $0 < d \le 1/4$, $|R| < 13/50 < 1/2$, and the Taylor bound $e^R \le 1 + R + (3/5)R^2$ gives

$$T(qe^R) \le \bar{f}_2 + BR + \delta\mathbf{1}, \quad \delta = \mathbb{E}R + \mathbb{E}(uR) + \sqrt{d\mathbb{E}R^2} + (12/5)\mathbb{E}R^2. \tag{Q178.T186.9}$$

The same R and its signed means give the complete quarter-domain packet

b	Γ/d lower	$D/d \quad (D \ge \delta_+)$	$\lvert r\rvert$ upper ρ
4	51/16	1	3/10
5	89/32	2/3	1/4
$b \ge 6$	19/8	7/20	1/8.

Write $G_n(-z) = U - zV$. At $b = 4$, $r = -c_0 + x^2U - zx^2V$, $c_0 \le 0$, $|c_0| \le \delta_2$. At $b = 5$, $r = -c_0 - x^3V - zx^2U$ with the same c_0 bounds. For $b \ge 6$, $|c_0| \le \delta_3$; if $r = A_0 + zC_0$ then $|A_0| \le \delta_3 + x^3$ and $|C_0| \le x^3$. These include $n = 1$ and both parities; ρ is a finite-symbol bound, not a pointwise quotient R/u. For $D_i = \mathbb{E}B^iR$, the exact finite coefficients are

$$D_{2j-1} = (-1)^j \int x^{j-1} A_0 \, d\nu, \qquad D_{2j} = (-1)^{j+1} \int x^j C_0 \, d\nu. \tag{Q178.T186.10}$$

At $b = 4$, $D_1, D_2 \le 0$; at $b = 5$, $D_1 \le \delta_4, D_2 \le 0$. For $b \ge 6$, $|D_1| < 67d/600$, $|D_2| < d/36$ and $4(D_1 + D_2) < 3d/5$. Every $\gamma_j = \beta_j - D_{j-2}$, $j \ge 3$, satisfies

$$\gamma_j \ge h := 15d/32. \tag{Q178.T186.11}$$

Specifically $\gamma_3 \ge d/2$, $\gamma_4 \ge 11d/12$; the others use $|D_i| < d/8$ and $\tau \ge 19d/32$. Positive propagation gives

$$T^L F \le \alpha Y_L, \qquad Y_L = \bar{f}_{L+1} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{Q178.T186.12}$$

The first $Y_1 = \bar{f}_2 + BR + D\mathbf{1}$ is a nonnegative actual upper; $Y_L = T^{L-1}Y_1 \geq 0$. Finite convolution rewrites it as

$$Y_L = \mathbf{1} + G_{L+1}(B)u + B^L R + D\bar{f}_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar{f}_{L+1-j}. \tag{Q178.T186.13}$$

At $L = 3, 4, \geq 5$ retain only the existing last two, three, four messages, respectively. The bound $D\bar{f}_{L-1} \leq D[\mathbf{1} + G_{L-1}(B)u]$ uses this same finite convolution.

Q178.3 Finite symbol and expansion-coefficient lemmas

For $n \geq 2$, $G_n(-z) = U - zV$ satisfies $0 \leq U - (1-x)V \leq x$. For even $n = 2j$, $U = V = a_j = (1-(-x)^j)/(1+x)$, giving difference xa_j; for odd $n = 2j+1$ it is $(x + (-x)^j)/(1+x)$. At $j = 1$ it is zero; even $j \geq 2$ gives at most x, and odd $j \geq 3$ lies between zero and x. The $n = 1$ coordinates $(1, 0)$ are exceptional; the unit $(b, c) = (4, 5)$ is treated separately. For receiving $G_a = E_a + zD_a$, even a has $E_a = D_a$ and odd a has $E_a = 1 - xD_a$. Thus $\mathrm{Re}(G_a G_n) \geq 0$ for $a \geq 3, n \geq 4$: if either is even, use $(1-x)(1-2x) > 0$; if both are odd, $n \geq 5$ has $D_n \leq 1 - x + x^2$, so the lower bound is $1 - 3x + 3x^2 - 3x^3 + x^4 \geq \psi(x) = 1 - 3x + 2x^2 - x^3 \geq 1451/32768 > 0$. Here $\psi' = -3 + 4x - 3x^2 < 0$. For $n = 3$ the uniform bound is only

$$\mathrm{Re}(G_a G_3) \geq 1 - 3x + x^2 \geq -55/1024 > -1/16. \tag{Q178.T186.14}$$

For $n = 2$, $\mathrm{Re}(G_a G_2) \geq 0$. Also $\mathrm{Re}[G_a(2+z)] \leq 2$ and $\mathrm{Re}[G_a(3 + 2z + z^2)] \leq (3-x)E_a \leq 3$.

Q178.3.2 The finite two-term shift

For $n \geq 3$, $0 \leq \sigma \leq 3/16$, $|G_n - \sigma(2+z)| \leq 1 - 2\sigma$. Set $A_0 = 1 - 2\sigma$. At $n = 3, 4$ the squared differences from A_0^2, divided by x, are respectively $-1+2\sigma+\sigma^2+x$ and $-1+2\sigma+\sigma^2+(-1+2\sigma)x+x^2$, strictly negative on the rectangle. For $n \geq 5$, multiply by $1 - z$ to obtain numerator $M - z^n$, $M = A_0 + \sigma z + \sigma z^2$, $|M| \leq 1$, and

$$A_0^2(1+x) - |M|^2 = x[1 - 2\sigma - \sigma^2(1+x)] \geq (4715/8192)x.$$

The finite tail increment divided by x is at most $2\theta^{3/2} + \theta^4 < 889/1600 < 9/16 < 4715/8192$. At $x = 0$ it is direct. This is the finite proof also used by C102.

Q178.3.3 The finite three-term shift

For $n \geq 4$, $0 \leq \sigma \leq 49/288$, $|G_n - \sigma(3 + 2z + z^2)| \leq 1 - 3\sigma$. Set $A_0 = 1 - 3\sigma$. At $n = 4$ the quotient of the squared difference is $-1 + 4\sigma - 2\sigma^2 + (-1 + 2\sigma + \sigma^2)x + x^2 < 0$: the constant is at most $-3/8$, linear coefficient negative, and $x^2 < 1/6$. At $n = 5$ it is $-1+4\sigma-2\sigma^2+(1-4\sigma+\sigma^2)x+(-1+2\sigma)x^2+x^3$, at most $-3/8 + x - (95/144)x^2 + x^3$. The latter increases to $-3121/294912$ at θ; $1 - 4\sigma + 2\sigma^2 \geq 15649/41472 > 3/8$. For $n \geq 6$, multiplication by $1 - z$ gives $M - z^n$, $M = A_0 + \sigma z + \sigma z^2 + \sigma z^3$, and

$$A_0^2(1+x) - |M|^2 = x[1 - 4\sigma + 2\sigma^2 + \sigma^2 x - \sigma^2 x^2] \geq (3/8)x.$$

Since $|z + z^2 + z^3|^2 = x - x^2 + x^3 \leq 1$, $|M| \leq A_0 + \sigma \leq 1$. The tail increment divided by x is at most $2\theta^2 + \theta^5 < 877/2560 < 3/8$; $x = 0$ is direct. Here $h \leq 15/128 < 49/288$, and every used index is at least four.

Q178.3.4 Continuous coefficient $17/2$ templates

Let $t=\sqrt{d}\le 1/2$, $W\ge 0$, and suppose $\mathbb{E}W\le\beta\le e^{-\gamma d}$, $W+\mathbb{E}W\le\beta(q+1+r_0t+s_0t^2)$, and $C_a\le p+(6/5)t$. The pointwise coefficient is at most

$$e^{-2\gamma t^2}(p+(6/5)t)\{q^2+2q+3+2(q+1)(r_0t+s_0t^2)+(r_0t+s_0t^2)^2\}.$$

The constant is $6+q-q^3\le 13/2$, and the exact linear coefficient is $(6/5-2r_0)q^2+(12/5+2r_0)q+18/5+4r_0$. The next four coefficients have bounds

$$2r_0^2+(9/2)s_0+(36/5)r_0,\quad 4r_0s_0+(6/5)r_0^2+(36/5)s_0,\quad 2s_0^2+(12/5)r_0s_0,\quad (6/5)s_0^2.$$

Compress higher degrees using $t\le 1/2$. The linear bounds are obtained from the concave quadratic on $[0,2]$:

r_0	s_0	γ	linear ℓ	quadratic B_*	$17\gamma-B_*$
$11/10$	0	$19/8$	$66/5$	12	$227/8$
$13/10$	0	$19/8$	$133/10$	14	$211/8$
$7/5$	0	$19/8$	$27/2$	16	$195/8$
$7/5$	$3/16$	$51/16$	$27/2$	18	$579/16$
$6/5$	0	$19/8$	$66/5$	13	$219/8$
$6/5$	1	$115/32$	$66/5$	25	$1155/32$
$13/10$	1	$145/48$	$133/10$	26	$1217/48$
$5/4$	$5/12$	$251/96$	$53/4$	18	$2539/96$.

Every quadratic $2-\ell t+(17\gamma-B_*)t^2$ has negative discriminant; compare with $(17/2)e^{2\gamma t^2}\ge 17/2+17\gamma t^2$ to obtain coefficient at most $17/2$. For example the third row has discriminant $-51/4$. The case $r_0=13/10, s_0=0, \gamma=89/32$ may use the weaker second row.

Q178.3.5 The unit large-b coefficient eight

If $\mathbb{E}W\le\beta\le e^{-(79/32)d}$ and $W+\mathbb{E}W\le\beta[q+1+(23/20)t]$, the coefficient is at most eight. The constant is at most $32/5$, since $2/(3\sqrt{3})<2/5$. At $r_0=23/20$, $s_0=0$, the linear coefficient increases on $[0,2]$ and is at most $66/5$; the compressed quadratic is $(13/5)r_0^2+(36/5)r_0<12$. Compare with $8e^{(79/16)t^2}\ge 8+(79/2)t^2$: the remaining quadratic is $8/5-(66/5)t+(55/2)t^2>0$, with discriminant $-44/25$. This coefficient is for the actual scaled W.

Q178.4 Every nonunit connector

Let $\alpha=e^{-\Gamma}$ and $h=15d/32$. The means μ_Z,μ_Y refer to unscaled original uppers. A symbol norm below is on the original spectral support and bounds $\|P(B)u\|_2/\sqrt{d}$. The exponential payments use $1-x\le e^{-x}$ and actual nonnegative messages, with the direction fixed.

Q178.4.1 Every $L\ge 5$

Retain $\bar f_0,\bar f_1,\bar f_2,\bar f_3$ and use $D\bar f_{L-1}\le DS'_{L-1}$:

$$Z=Cq+BP(B)u-h(1-d)\mathbf{1},\quad C=1+D-3h\le 1-13d/32,$$

$$P=G_L+z^{L-1}r+DG_{L-2}-h(2+z),$$

$$Q=G_{L+1}+z^Lr+DG_{L-1}-h(3+2z+z^2),\quad s=1+D-4h+hd.$$

Here $Z\ge Y_L\ge 0$. The finite shifts proved above give $\|P\|\le 1-2h+D+\rho\theta^2<11/10$ and $\|Q\|\le C+\rho\theta^{5/2}<C+1/30$. For the first use the worst bound $1+d/16+1/20\le 1+1/64+1/20<11/10$; for the

second use $\sqrt{\theta} < 13/20$ and $(3/10)\theta^{5/2} < 1/30$. The true mean loss is $\zeta = \gamma_{L+2} + D\beta_L - 2hd > 0$, so $\mu_Z \le 1 + D - 5h + 3hd - D\tau \le 1 - 127d/128$. The $G_a G_n$ products are nonnegative for $a \ge 3, n \ge 4$, and the three-term shift has receiving real part at most three. Therefore $\mathrm{Cov}(C_a, Z) \le 3hd + \rho\theta^{L/2}d \le 45d^2/32 + d/30$, and $\mathrm{Var}\, Z \le (C + 1/30)^2 d$. The row $Z \le q + (11/10)\sqrt{d}$, $\mu_Z \le 1$ satisfies the first $17/2$ template for $W = \alpha Z$. The first-term rate is at least $4(19/8) + 19/16 + 4(127/128) = 469/32$. The variance term is at most $(16337/1800)de^{-(1373/248)d} \le (16337/1800)de^{-11d/2}$; the covariance term is at most $[(2/15)d + (45/8)d^2]e^{-(1597/128)d}$. Relative to $de^{-11d/2}$ its extra coefficient is at most $2/15 + 270/893$, and $16337/1800 + 2/15 + 270/893 < 10$. Thus this entire row is bounded by $e^{-469d/32} + 10de^{-11d/2}$, which is dominated by the common scalar upper below.

Q178.4.2 $L = 4$

Retain only $\bar{f}_0, \bar{f}_1, \bar{f}_2$:

$$C = 1 + D - 2h, \quad s = 1 + D - 3h,$$
$$Q = G_5 + z^4 r + DG_3 - h(2 + z), \quad P = G_4 + z^3 r + DG_2 - h, \quad Z = Cq + BP(B)u - h\mathbf{1}.$$

The loss $\zeta = \gamma_6 + (D - h)d > 0$ gives $\mu_Z \le 1 + D - 4h - (D - h)d \le 1 - 7d/8$. Also $Cq - h \le q - 11d/32$ and $\|P\| \le 1 - h + (6/5)D + \rho\theta^{3/2} < 13/10$. Indeed $|G_4 - h| \le 1 - h$, since its squared difference quotient is $-1 + 2h - x + x^2 < 0$. The second template gives $17/2$. Moreover $\|Q\| \le 1 - 2h + D + \rho\theta^2 \le 21/20 + d/16$. The actual DG_3 exception costs $Dd/16$, so $\mathrm{Cov}(C_a, Z) \le 2hd + (D/16)d + \rho\theta^2 d \le d^2 + d/20$. The three fourth-expansion terms are at most

$$e^{-227d/16} + (7497/800)de^{-389d/84} + [(1/5)d + 4d^2]e^{-97d/8}.$$

Relative to $de^{-9d/2}$ the coefficient is at most $7497/800 + 1/5 + 12/61 < 10$.

Q178.4.3 $L = 3$**, with all three** b **ranges**

Here

$$Z = Cq + BP(B)u - h\mathbf{1}, \quad C = 1 + D - h, \quad s = 1 + D - 2h,$$
$$Q = G_4 + z^3 r + DG_2 - h, \quad P = G_3 + z^2 r + D.$$

The positive loss is $\zeta = \gamma_5 + Dd$, and $\mu_Z \le 1 + D - 3h - Dd$. Also $\mathrm{Cov}(C_a, Z) \le hd + \rho\theta^{3/2}d$ and $\|Q\| \le 1 - h + (6/5)D + \rho\theta^{3/2}$, using $|G_4 - h| \le 1 - h$. The pointwise row is $Z + \mu_Z \le q + 1 + \|P\|\sqrt{d} + 2(D - h)_+ + D - 4h$. If $D \ge h$, its last constant is $3D - 6h$; if $D < h$, retain $D - 4h$ rather than setting $q = 2$ against a negative coefficient. For $b = 4$, $\mu_Z \le 1 - 13d/32$, $\|P\| < 7/5$, and $3D - 6h = 3d/16$. The fourth template gives $17/2$. Also $\|Q\| \le 27/25 + (117/160)d$. The variance term is at most $(12393/1250)de^{-5d}$, with $12393/1250 = (17/2)(27/25)^2$. The first rate is $249/16 > 31/2$. The covariance has quadratic cost $(15/8)d^2$, rate $447/32$, and gap $287/32$ relative to de^{-5d}. Using $e > 8/3$ adds $45/574$; the total $12393/1250 + 8/25 + 45/574 < 52/5$. Thus the upper is $e^{-31d/2} + (52/5)de^{-5d}$. It is at most $F_0 = e^{-25d/2} + (103/10)de^{-9d/2}$: the extra first rate of three provides at least $3de^{-31d/2}$, whereas the extra coefficient costs at most $de^{-5d}/10$; $e^{(21/2)d} < e^3 < 27 < 30$ on the quarter-domain pays it. For $b = 5$, $\mu_Z \le 1 - 71d/96$, $\|P\| < 13/10$, and $3D - 6h < 0$. Use the second template with its stronger γ. One has $\|Q\| \le 16/15 + (53/160)d$, variance coefficient $2176/225$, and rate $1265/256 > 9/2$. The first rate is $733/48 > 25/2$; the covariance linear coefficient is $4/15$, with normalized quadratic cost at most $45/566$. Since $2176/225 + 4/15 + 45/566 < 103/10$,

this is at most F_0. For $b \ge 6$, $\mu_Z \le 1 - 169d/160$, $\|P\| < 6/5$, $D < h$ and $D - 4h < 0$. The fifth template gives $17/2$. Here $\|Q\| \le 31/30 - (39/800)d$, variance coefficient $16337/1800$ with rate greater than $9/2$, and first rate $1193/80 > 25/2$. The covariance linear coefficient is $2/15$, with normalized quadratic cost at most $225/2614$; the total is less than $103/10$.

Q178.4.4 $L = 2$: exact constant cancellation

The actual form is $Y_2 = Cq + BP(B)u + c\mathbf{1}$, with $C = 1 + D$, $P = G_2 + zr$, and $c = D_1 - d = \mathbb{E}(P(B)u)$. The centered row yields $Y_2 \le (C + c)q + \sqrt{pq\,\mathrm{Var}(P(B)u)}$, where $C + c = 1 + D - d + D_1 \le 1$. Also $\|P\| \le 6/5 + \rho\sqrt{\theta} < 7/5$ and $\mu_{Y_2} \le 1$. The third template gives $17/2$ uniformly. The full symbol is $Q = G_3 + D + z^2 r$, with $\zeta = d - D_2 > 0$. Its real and imaginary coordinates are positive: $|A_0| \le \delta_2 + x^2$, $|C_0| \le x^2$, and $D \ge 7d/20 > \theta^2 d$ give real part at least $1 - x - x^3 + D - x\delta_2 > 0$ and imaginary coordinate at least $1 - x^3 > 0$. Thus their positive uppers may be squared. For $b = 4$, the real coordinate is at most $1 + D - x$ since $A_0 \ge 0$, and the imaginary coordinate at most $1 + x^3$. The squared difference quotient is at most $-1 - 2D + x + 2x^3 + x^6 < 0$, so $\|Q\| \le 1 + D$. Then $\mathrm{Var}\, Y_2 \le (1 + d)^2 d - d^2 \le de^{5d/4}$. The mean is at most $1 - d$, with first rate $287/16$ and variance rate $41/8 > 9/2$. The covariance is at most $d(1/16 + 39/320)$, giving fourth cost $59/80$. The combined coefficient $17/2 + 59/80 < 103/10$ is dominated by F_0. For $b = 5$, real coordinate is at most $1 + D - x + x^4$, imaginary at most $1 + x^3$. The quotient is at most $-1 + x + 4x^3 + x^6 + x^7 < 0$, all D terms being nonpositive; hence $\|Q\| \le 1 + D$. The variance is at most $(1 + 2d/3)^2 d - d^2 \le de^{4d/9}$, and the mean at most $1 - 5d/4$. The first rate is $277/16 > 25/2$, the variance rate $89/16 - 4/9 > 9/2$. The covariance bound $d(1/16 + 13/128) = 21d/128$ gives cost $21/32$; $17/2 + 21/32 < 103/10$. For $b \ge 6$ use $|A_0| \le \delta_3 + x^3$, $|C_0| \le x^3$ and $\widetilde{D} = D + \theta^3 d < 21d/50$. The real coordinate is at most $1 + \widetilde{D} - x + x^4$, imaginary at most $1 + x^4 \le 1 + x^3$, so $\|Q\| \le 1 + \widetilde{D}$ as above. Since $\zeta \ge 35d/36$, the variance is at most d: $21/25 - (35/36)^2$ plus the worst cubic contribution $441/10000$ is still negative. The mean is at most $1 - 3d/2$, with first rate $267/16 > 25/2$ and variance rate $19/4 > 9/2$. The covariance bound $d(1/16 + 13/256) = 29d/256$ gives cost $29/64$; $17/2 + 29/64 < 103/10$. In these cases use $\mathrm{Re}(G_a G_3) \ge 1 - 3x + x^2 \ge -55/1024 > -1/16$; the DG_a term is nonnegative, the r remainder costs $\rho\theta$, and the mean correction $-\beta_{a+1}\zeta$ is nonpositive.

Q178.4.5 A common continuous scalar payment

For $F_0 = e^{-25d/2} + (103/10)de^{-9d/2}$, degree-four Taylor bounds give

$$\begin{aligned}\frac{e^{9d/2} - e^{-8d}}{d} &\ge 25/2 - (175/8)d + (4825/48)d^2 - (58975/384)d^3 \\ &\ge 25/2 - (175/8)d + 62d^2 \ge 25/2 - 30625/15872 > 211/20.\end{aligned}$$

Since $211/20 = 103/10 + 1/4$ and $e^{9/8} < 4$, this proves $F_0 \le 1 - d/16$ for $0 < d \le 1/4$. For the latter exponential bound use $e < 3$, from a geometric factorial tail, and $e^{1/8} \le 8/7$, so $e^{9/8} < 24/7 < 4$. Every nonunit row above is dominated by F_0, completing all $L \ge 2$ on the low domain.

Q178.5 Every unit connector

For $L = 1$, $Y_1 = q + B(u + R) + D\mathbf{1}$. Write $r = A_0 + zC_0$, $P = 1 + r$, $Q = 1 + zP$, and

$$e = \mathbb{E}R = -\int C_0\, d\nu, \quad \zeta = -\mathbb{E}[Q(B)u] = \int (1 + A_0)\, d\nu, \quad \mu = \mathbb{E}Y_1 = 1 - d + D + D_1.$$

The original centered row gives $Y_1 \le (1+e)q + (D-e)\mathbf{1} + \sqrt{pq}\sigma$, where $\sigma^2 = \mathrm{Var}((1+r)(B)u)$. Set the actual upper $W = \alpha Y_1$. For its coefficient bound use $\beta = \alpha(1+e) > 0$, retaining α as the true scale in the fourth expansion. The variance and covariance of Y_1 equal those of $Q(B)u$. The V in the finite coordinates $G_n(-z) = U - zV$ is a spectral scalar, and the receiving mean correction remains $-\beta_{a+1}\zeta$.

Q178.5.1 $b = 4, c = 5$

Here $n = 1$, $c_0 = \beta_4 - \beta_3 = 0$, $R = x^2u$, $(U, V) = (1, 0)$, and $e = 0$. One has $\Gamma = 4d - \delta_2 \ge 115d/32$, $D = d$, $\mu = 1-\delta_3 \le 1$, $Q = 1+z(1+x^2)$ and $\zeta \ge d$. Thus $|Q|^2 = 1+x+2x^3+x^5 \le 25/16$ and $\mathrm{Var}\, Y_1 \le 25d/16 - d^2$. The product has $\mathrm{Re}(G_a Q) \ge 1-2x-x^3 > 0$, so its receiving covariance is nonpositive. Since $\|1+r\| \le 1+\theta^2 < 6/5$, $Y_1 + \mu \le q + 1 + (6/5)\sqrt{d} + d$. The template $r_0 = 6/5, s_0 = 1, \gamma = 115/32$ has linear coefficient at most $66/5$ and merged quadratic less than 25. The successive contributions $801/50, 858/125, 61/50, 3/20$ sum to less than 25. The remaining quadratic $2-(66/5)t+(1155/32)t^2 > 0$ has negative discriminant, proving the coefficient throughout $0 \le t \le 1/2$. Put $H = 1+(115/16)d+(13225/512)d^2 \le e^{2\Gamma}$, $A = 1-19d/16$, and $V_* = 25d/16 - d^2$. The reference fourth is at most $A/H^2 + (17/2)V_*/H$. For $G = H^2 - A - (17/2)V_*H$, the coefficients at degrees one through three are $73/32, 8377/512, 1463835/16384$; degree four is $b_H^2 + (17/2)b_H > 0$, $b_H = 13225/512$. Hence $G \ge 73d/32$. Since $H(1/4) < 5$, $G/H^2 \ge 73d/800 > d/16$, proving the unit bound $1 - d/16$.

Q178.5.2 $b = 4, c \ge 6$

For $n \ge 2$, $r = -c_0 + x^2U - zx^2V$, $c_0 \le 0$, $|c_0| \le \delta_2$. The finite relation $U - (1-x)V \le x$ was proved in part N.22. Both coordinates of Q are positive: $Q_{\mathrm{re}} = 1 + x^3V$ and $Q_{\mathrm{im}} = 1 - c_0 + x^2U$, where the latter denotes the coefficient of z. For odd receiving index, $D_a = 1$ is the lower extremum, giving $\mathrm{Re}(G_a Q) \ge 1-2x+c_0x-x^4$. The even-index value is at least this, with difference containing $x^2Q_{\mathrm{im}} \ge 0$. Let $v = \delta_2/d \in [d, \theta]$. Then $\Gamma \ge 4d - 2\delta_2$ and $\mu = 1 + D_1 \le 1 - \kappa\delta_3 \le 1 - \kappa v^2 d =: M(v)$. Also $\mu \ge 1 - \delta_2 d - \delta_3 \ge 1 - (273/1024)d \ge 1 - 4d/15$, and $\zeta \ge d$. Keep the true mean correction to get

$$\begin{aligned}\mathrm{Cov}(C_a, Y_1) &\le -d + 2\delta_2 + \delta_2^2 + \delta_5 - \tau d \\ &\le -(1 - 2\theta - \theta^4)d - (1 - \theta - \theta^2)d^2 \\ &\le -4d/25 - 2d^2/5.\end{aligned}$$

Here $1 - 2\theta - \theta^4 = 168047/1048576 > 4/25$ and $1 - \theta - \theta^2 = 439/1024 > 2/5$. Since $\mu^3 \ge 1 - 4d/5$,

$$4\mu^3\,\mathrm{Cov}(C_a, Y_1) \le -L(d), \quad L(d) = (16d/25 + 8d^2/5)(1 - 4d/5) = 16d/25 + 136d^2/125 - 32d^3/25.$$

The signed norm satisfies

$$|Q|^2 \le 1 + x + 4x^3 + x^5 + x^6 + 2\delta_2 x(1 + x^2) + \delta_2^2 x.$$

Subtracting $\zeta^2 \ge d^2$ gives $\mathrm{Var}\, Y_1 \le V_* := 17d/10 - 3d^2/5 + 7d^3/100$, using $1+\theta+4\theta^3+\theta^5+\theta^6 < 17/10$, $2\theta^2(1+\theta^2) - 1 < -3/5$ and $\theta^3 < 7/100$. For the coefficient, $0 \le e \le \delta_3 \le d/6$, $\|1 + r\| \le 13/10$, and $\beta = \alpha(1+e) \le e^{-(145/48)d}$. Because $\mu \le 1 \le 1+e$, $W + \mathbb{E}W \le \beta[q+1+(13/10)\sqrt{d}+d]$. The template $r_0 = 13/10, s_0 = 1, \gamma = 145/48$ gives linear upper $133/10$, merged quadratic less than 26; $2-(133/10)t+(1217/48)t^2 > 0$ has negative discriminant. For the same-v endpoint replacement

define $A(d,v) = [1-2(1-v)d]M(v)^4 - L(d)$. This is positive: $M \ge 39/40$, $M^4 \ge 9/10$, $E_0 = 1-2(1-v)d \ge 1/2$ and $L \le 13/50$, giving $A \ge 19/100$. The derivative of $e^{8vd}A$, divided by $e^{8vd}d$, is

$$2M^4 + 8E_0M^3(M-\kappa v) - 8L > 0.$$

Indeed, $2M^4 \ge 9/5$, $E_0 \ge 1/2$, $M^3 \ge 7/10$, $M - \kappa v \ge 7/10$ and $8L \le 52/25$, giving $42/25 > 0$. Consequently $e^{-4\Gamma}A(d,v) \le e^{-(51/4)d}A(d,\theta)$. Since $\kappa\theta^2 > 1/12$, $M(\theta) \le 1 - d/12$; using $(1-d/12)^4 \le 1 - d/3 + d^2/24$ gives

$$A(d,\theta) \le \overline{A} = 1 - (2593/1200)d - (1301/2000)d^2 + (11813/9600)d^3.$$

Put $H = 1 + (51/8)d + (2601/128)d^2 \le e^{2\Gamma_{\min}}$. The moment is at most $\overline{A}/H^2 + (17/2)V_*/H$. For $G = H^2 - \overline{A} - (17/2)V_*H = d[g_1 + g_2d + g_3d^2 + g_4d^3 + g_5d^4]$,

$$g_1 = 553/1200 > 9/20, \quad g_2 = -5087/1000 > -51/10, \quad g_3 = -14813/3840 > -4,$$
$$g_4 = 6765201/16384 + 638979/6400, \qquad g_5 = -309519/25600.$$

For $d \le 1/4$, $g_4 + g_5 d > 509$. The bracket is at least $P(d) = 9/20 - 51d/10 - 4d^2 + 509d^3$. For $d \le 1/16$, $P \ge 37/320 > 1/10$. For $d \ge 1/16$, use $509d^3 \ge 509d^2/16$ to get $P \ge 9/20 - 51d/10 + (445/16)d^2 \ge 9/20 - 2601/11125 > 1/10$. Thus $G \ge d/10$. Since $H(1/4) = 7913/2048 < 4$, the reference fourth is at most $1 - d/160$.

Q178.5.3 $b = 5, c \ge 6$

The finite coefficients are $A_0 = -c_0 - x^3V$, $C_0 = -x^2U$, $c_0 \le 0$, $|c_0| \le \delta_2$. Here $\Gamma \ge 89d/32$, $D = 2d/3$, $0 \le e \le \delta_3 \le d/6$, and $\mu = 1 - d/3 + D_1 \le 1 - d/4$, since $D_1 \le \delta_4 \le \theta^3 d < d/12$. Also $\mu \ge 1 - d/3 - \delta_2 d \ge 1 - 7d/16$ and $\zeta = \int (1 - c_0 - x^3V)\,d\nu \ge (1-\theta^3)d \ge 11d/12$. The coefficient scale satisfies $\beta = \alpha(1+e) \le e^{-(251/96)d}$, and

$$W + \mathbb{E}W \le \beta[q + 1 + (5/4)\sqrt{d} + (5/12)d].$$

Here $\|1 + r\| \le 5/4$ and the remaining constant $D - d/4 - 2e \le 5d/12$. The corresponding template gives linear upper $53/4$ and merged quadratic less than 18; the contributions are $14, 167/48, 115/288, 5/192$. The quadratic $2 - (53/4)t + (2539/96)t^2 > 0$ has negative discriminant. The coordinates $Q_{\text{re}} = 1 + x^3U$ and $Q_{\text{im}} = 1 - c_0 - x^3V$ are positive and $|Q|^2 \le 1 + x + 2x^3 + x^6 + 2\delta_2 x + \delta_2^2 x$. After the exact negative mean square, $\operatorname{Var} Y_1 \le V_* := 25d/16 - d^2/2 + 7d^3/100$, using $1 + \theta + 2\theta^3 + \theta^6 < 25/16$, $2\theta^2 - (11/12)^2 < -1/2$ and $\theta^3 < 7/100$. Since $\operatorname{Re}(G_a Q) \ge 1 - 2x - \delta_2 x$,

$$\operatorname{Cov}(C_a, Y_1) \le -3d/16 + [\theta^2 - \kappa(11/12)]d^2 \le -3d/16 - 3d^2/8,$$

where $\kappa(11/12) - \theta^2 = 1165/3072 > 3/8$. Use $\mu^3 \ge 1 - 21d/16$, $\mathbb{E}C_a \le 1 - 19d/16$ and $\mu^4 \le (1-d/4)^4 \le 1 - d + 3d^2/8$. The mean-plus-covariance bracket is at most $\overline{A} = 1 - 47d/16 + 67d^2/64 + 195d^3/128$. Let $H = 1 + (89/16)d + (7921/512)d^2$. For $G = H^2 - \overline{A} - (17/2)V_*H = d[g_1 + g_2d + g_3d^2 + g_4d^3 + g_5d^4]$,

$$g_1 = 25/32 > 3/4, \qquad g_2 = -4501/512 > -9,$$
$$g_3 = a_H^3 - 195/128 - (17/2)[(25/16)b_H - a_H/2 + 7/100] > -12,$$
$$g_4 = b_H^2 + (17/2)(b_H/2 - (7/100)a_H),$$
$$g_5 = -(119/200)b_H = -942599/102400, \qquad a_H = 89/16, \quad b_H = 7921/512.$$

On $d \le 1/4$, $g_4 + g_5 d > 299$, so the bracket is at least $P(d) = 3/4 - 9d - 12d^2 + 299d^3$. For $d \le 1/16$, $P \ge 9/64$. For $d \ge 1/16$, $P'' = -24 + 1794d > 80$. At $d_0 = 1/9$, $P(d_0) = 35/2916$ and $P'(d_0) = -16/27$. The tangent quadratic gives $P(d) \ge 35/2916 - 8/3645 = 143/14580 > 1/200$. Thus $G \ge d/200$. Since $H(1/4) = 27505/8192 < 7/2$, $H^2 < 13$, the reference fourth is at most $1 - d/2600$, hence at most $1 - d/3000$.

Q178.5.4 Every $b \ge 6, c > b$

The actual alternating moments give $\beta_b + \beta_{b-1} \ge 2\tau$ and $\beta_c + \beta_{c-1} \ge 2\tau + \delta_3$ for $c \ge 7$. At $c = 7$ equality holds; for $c \ge 8$ each β is at least $\tau + \kappa\delta_3$, with $2\kappa > 1$. For $v = \delta_2/d$, the global inequality $\delta_3 \ge v^2 d$ yields

$$\Gamma \ge 4\tau + \delta_3 \ge (4 - 4v + v^2)d \ge (2601/1024)d > 5d/2.$$

This imposes no pointwise lower spectral bound. Also $|e| \le \delta_4 \le \theta^3 d < 7d/100$. The actual $D_1 < 67d/600$ gives $\mu \le 1 - d/2$, with loss $323/600 > 1/2$; and $\mu \ge 1 - 13d/20 - \delta_3 d - \delta_4 \ge 1 - 4d/5$, $\zeta \ge d - \delta_3 d - \delta_4 \ge 7d/8$. The scale $\beta = \alpha(1+e) > 0$ obeys $\beta \le e^{-(79/32)d}$, since $2601/1024 - \theta^3 = 81035/32768 > 79/32$. The centered noise has bound

$$\frac{\sigma}{(1+e)\sqrt{d}} \le \frac{9/8}{1 - \theta^3/4} < \frac{9/8}{393/400} = 450/393 < 23/20.$$

Moreover $\mu \le 1 - d/2 \le 1 + e$ and $D - d/2 - 2e \le 7d/20 - d/2 + 14d/100 = -d/100 < 0$. Thus $W + \mathbb{E}W \le \beta[q + 1 + (23/20)\sqrt{d}]$. For coefficient eight, use $p(q^2 + 2q + 3) = 6 + q - q^3 \le 32/5$, since $2/(3\sqrt{3}) < 2/5$. At $r_0 = 23/20, s_0 = 0$, the linear coefficient increases to at most $66/5$ on $[0, 2]$ and the merged quadratic is $(13/5)r_0^2 + (36/5)r_0 < 12$. Compare with $8e^{(79/16)t^2} \ge 8 + (79/2)t^2$; $8/5 - (66/5)t + (55/2)t^2 > 0$, with discriminant $-44/25$. The signed covariance uses $\mathrm{Re}(G_a Q) \ge 1 - 2x - (2 - x)x^4 - x\delta_3$. After the true mean correction,

$$\mathrm{Cov}(C_a, Y_1) \le -(3/16 - 2\theta^4)d + [\theta^3 - 7\kappa/8]d^2 \le -13d/100 - 2d^2/5.$$

Here $\theta^4 < 1/36$ gives $3/16 - 1/18 = 19/144 > 13/100$, and $\theta^3 < 7/100$. The exact norm bound is

$$|Q|^2 \le 1 + x + 4x^4 + x^7 + x^8 + 2\delta_3 x(1 + x^3) + \delta_3^2 x.$$

The linear coefficient is less than $61/40$, using $\theta^4 < 11/400$, $\theta^7 < 1/500$, $\theta^8 < 1/1000$. The cross quadratic contribution is at most $2\theta^2(\theta + \theta^4) \le 4/27$; after subtracting $\zeta^2 \ge 49d^2/64$, it is at most $-3d^2/5$. The cubic contribution satisfies $\delta_3^2 \delta_2 \le \theta^5 d^3 < d^3/80$. Therefore $\mathrm{Var}\, Y_1 \le V_* := 61d/40 - 3d^2/5 + d^3/80$. Use $\mu^3 \ge 1 - 12d/5$ and $\mu^4 \le (1 - d/2)^4 \le 1 - 2d + 3d^2/2$. The mean-plus-covariance bracket is at most $\overline{A} = 1 - (1483/400)d + (3523/1000)d^2 + (1647/800)d^3$. Set $H = 1 + 5d + (25/2)d^2 \le e^{2\Gamma}$. With coefficient eight, $G = H^2 - \overline{A} - 8V_* H = d[g_1 + g_2 d + g_3 d^2 + g_4 d^3 + g_5 d^4]$, where

$$g_1 = 603/400 > 3/2, \quad g_2 = -9723/1000 > -10, \quad g_3 = -4527/800 > -6, \quad g_4 = 863/4, \quad g_5 = -5/4.$$

For $d \le 1/4$, $g_4 + g_5 d > 215$, so the bracket is at least $P(d) = 3/2 - 10d - 6d^2 + 215d^3$. At $d \le 1/16$ it exceeds $1/2$. At $d \ge 1/16$, $P'' = -12 + 1290d > 64$. For $d_0 = 2/15$, $P(d_0) = 1/6 - 8/75 + 1720/3375$ and $P'(d_0) = -2/15$; hence $P(d) \ge P(d_0) - 1/7200 > 1/2$. Thus $G \ge d/2$. Since $H(1/4) = 97/32 < 13/4$, $H^2 < 11$, the reference fourth is at most $1 - d/22$, in particular

at most $1-d/32$. In all four unit cases, the mean-plus-covariance upper is nonnegative before replacing $e^{2\Gamma}$ by H. In part Q178.5.1, $A \geq 45/64$; in Q178.5.2, $A(d,v) > 0$ and $\overline{A} \geq A(d,\theta)$; in Q178.5.3, $\overline{A} \geq 1-47d/16 \geq 17/64$; in Q178.5.4, $\overline{A} \geq 1-(1483/400)d \geq 117/1600$. Each V_* is nonnegative. Therefore the replacements of α^4 by H^{-2} and α^2 by H^{-1} preserve their directions.

Q178.6 The high domain and the complete local table

For $1/4 \leq d \leq 1/3$, $c \geq 5$ gives $h_c = Tf_{c-1} \leq 2\mathbb{E}f_{c-1} \leq 2(1-2d) \leq 1$. With the same canonical direction, $T^L(qf_bh_c) \leq T^L(qf_b)$. For the actual $a \geq 3, b \geq 3, L \geq 1$, (N.5) gives

$$\mathbb{E}[f_a\{T^L(qf_b)\}^4] \leq 1 - d/3000.$$

Only that full $(1,b)$ interface is used; the short $(1,2)$ or unit h_2f_b rows are not premises of this comparison. The low-domain bounds are

original window	$0 < d \leq 1/4$ reference fourth upper
$b \geq 4, c > b, L \geq 2$	$1 - d/16$
$L = 1, b = 4, c = 5$	$1 - d/16$
$L = 1, b = 4, c \geq 6$	$1 - d/160$
$L = 1, b = 5, c \geq 6$	$1 - d/2600$
$L = 1, b \geq 6, c > b$	$1 - d/32.$

The high domain gives $1 - d/3000$ for every window. At $d = 0$ only $K\mathbf{1} = 0$ is implied; original leaf integration makes all actual messages and equal-pair products one. Combining these domains proves (Q178.T186.A) for all original integers. The common positivity restoration of $E_t, E_o \in [0,1]$ and the complete reversal of the original kernel, receiving arm, terminal and connector give the decorated and reversed versions in the same original finite measure.

The genuine receiving arm of length two

For the same actual terminal $F = qf_bh_c$, $4 \leq b < c$, and every $L \geq 1$,

$$\mathbb{E}[f_2\{T^LF\}^4] \leq 1. \tag{N.32}$$

On $0 < d \leq 1/4$, the proof gives the stronger common bound $1 - d/3000$. On the high domain the assertion in (N.32) uses only the genuine length-two reference bound by one. The strict all-domain $d/3000$ loss for receiving lengths $a \geq 3$ remains the assertion above.

Put $C_2 = f_2 = 1 - u + Bu$, $\mathbb{E}C_2 = 1 - d$, and $C_2 \leq p + \sqrt{d}$. The coefficient templates in paragraph N.22 assume only $C_a \leq p + (6/5)\sqrt{d}$. The coefficient eight in paragraph N.22 uses this same weak cap. Thus all the actual scaled coefficients hold for C_2, without a positive lower bound on β_2. The packet r, R, D, Γ, finite γ's, positive uppers, actual means, and variances remain those constructed in paragraphs N.22, N.22, and N.22. Only the receiving covariance and mass change.

For $L \geq 2$, the following unscaled covariance bounds follow from (M.21), with $a = 2, t = 0$:

L	new covariance upper	comparison budget v_0
2	$-33d/800$	$29d/256$
3	$hd - d/4$	$hd + \rho\theta^{3/2}d$
4	$2hd - 7d/25$	$d^2 + d/20$
≥ 5	$3hd - 3d/10$	$3hd + d/30,$

$h = 15d/32.$ (N.33)

The $L = 2$ comparison budget is the smallest of its three actual b rows; the other two allow larger values. To verify the new column, use the finite genuine products $\mathrm{Re}(G_2G_3) \geq 3/16$ and $\mathrm{Re}(G_2G_n) \geq \kappa^2$

for $n \geq 4$. The latter follows by the finite even/odd formulas as in paragraph N.20. At length two the full symbol $G_3 + D + z^2 r$ has product at least $3/16 - (6/5)(3/10)\theta = 33/800$. At length three, DG_2 is favorable and $\kappa^2 - (6/5)(3/10)(4/15) > 1/4$. At length four, DG_3 is favorable and $\kappa^2 - (6/5)(3/10)\theta^2 > 7/25$. For all $L \geq 5$, DG_{L-1} is favorable and $\kappa^2 - (6/5)(3/10)\theta^{5/2} > 3/10$. The retained shifts cost respectively $h, 2h, 3h$ in the last three rows. Each actual signed mean parameter is positive, as proved for these same upper functions in paragraph N.22.

The comparison mass in every nonunit row is $1 - 19d/16$, and the actual means are at most one. Each improvement in (N.33) exceeds $3d/64$. Lemma M.8 therefore applies before the actual mean substitutions. The same variances and scaled coefficients give the identical scalar upper $e^{-25d/2} + (103/10)de^{-9d/2} \leq 1 - d/16$.

At $L = 1$, retain all four actual finite configurations:

(b, c)	new covariance upper	v_0
$(4, 5)$	$-d/2$	0
$(4, c),\ c \geq 6$	$-d/2 - 4d^2/5$	$-4d/25 - 2d^2/5$
$(5, c),\ c \geq 6$	$-\kappa d - 3d^2/4$	$-3d/16 - 3d^2/8$
$b \geq 6$	$-53d/100 - 4d^2/5$	$-13d/100 - 2d^2/5.$

(N.34)

Write $r = A_0 + zC_0$. The full output symbol is $Q = (1 - xC_0) + z(1 + A_0)$, and its actual mean parameter is $\zeta = \int (1 + A_0)\, d\nu > 0$. The first row has $Q = 1 + z(1 + x^2)$, hence product at least $1 - x - x^3 > 1/2$. For $b = 4, c \geq 6$, the finite relation $U - (1 - x)V \leq x$ and $|c_0| \leq \delta_2$ give

$$\mathrm{Cov}(C_2, Y_1) \leq -(\kappa - \theta^4)d - (1 - \theta^2)d^2 \leq -d/2 - 4d^2/5.$$

For $b = 5$, the negative finite coordinates give $A_0 + C_0 \leq \delta_2$ and $\zeta \geq 11d/12$, so the upper is $-\kappa d - (11/12 - \theta^2)d^2$. For $b \geq 6$, use $|A_0| \leq \delta_3 + x^3$, $|C_0| \leq x^3$, and $\zeta \geq 7d/8$. The upper is then

$$-(\kappa - 2\theta^4)d - (7/8 - \theta^3)d^2 \leq -53d/100 - 4d^2/5.$$

These calculations retain the signed term $-d\zeta$.

Every unit-row improvement exceeds $d/4$. The comparison mass is either the same-host $1 - 2(1 - v)d$, $v = \delta_2/d$, or its conservative endpoint $1 - 19d/16$. Actual means are at most one. The mass-transfer lemma therefore pays the full change before any mean or v endpoint is taken. The four scalar computations of paragraph N.22, with their actual negative variance squares and the stronger $\Gamma \geq (2601/1024)d > 5d/2$ in the last row, now give deficits $d/16, d/160, d/2600, d/22$, respectively. When clearing a scalar denominator, use the same nonnegative comparison bracket established there; the scaled variance term retains its second-power scale. This proves the stated common low-domain deficit.

For $d \geq 1/4$, the same actual arm has $h_c \leq 1$, so $T^L(qf_b h_c) \leq T^L(qf_b)$. The genuine f_2-weighted input (L58.16.1) gives a moment at most one for every $b \geq 4, L \geq 1$. At $d = 0$, use leaf integration. Positivity restores the original equal-pair factors, and full reversal reverses the receiving arm and connector together.

N.23 A retained reference square for terminals $(1, 2, c)$

Proposition N.3. *Let $F_c = qf_2 h_c$ in the canonical direction: the unit arm points toward the terminal, the two-edge arm points outward, and the c-arm has its first edge toward the terminal and its remaining*

edges outward. The connector of length L points from terminal to receiver. For every outward receiving arm $a \ge 3$, $c \ge 4$, and $L \ge 1$,

$$\mathbb{E}[E_o f_a \{T^L(E_t F_c)\}^4] \le 1 - d/800. \tag{N.35}$$

For the genuine receiving arm of length two, the same statement holds when $c \ge 5$:

$$\mathbb{E}[E_o f_2 \{T^L(E_t F_c)\}^4] \le 1 - d/800. \tag{N.36}$$

Here E_t, E_o are the actual equal-pair factors at the two roots. For $L \ge 2$ both conclusions have the common stronger loss $d/600$.

Proof. **The original measure and the two receiving masses.**

At $d = 0$, $u = 0$ and leaf integration gives equality. Assume $d > 0$ for the proof. Use $B = -K$, $u = B\mathbf{1}$, $q = 1 + u$, $p = 1 - u$ and the spectral measure ν at u from Section M.4. It has mass d, support $[0, \theta]$, and $\int x^j \, d\nu = \delta_{j+1}$. All means and adjoints below use the original finite measure. For $a \ge 3$ take

$$C_a = \mathbf{1} - G_a(-B)u - \beta_a \mathbf{1} \ge f_a, \quad \mathbb{E}C_a \le 1 - 19d/16, \quad C_a \le p + 6\sqrt{d}/5.$$

For $a = 2$ take exactly $C_2 = f_2 = p + Bu$, of mass $1 - d$ and cap $p + \sqrt{d}$. The proof below uses the weak coefficient-nine profile of Lemma A.2.

For every actual upper $Z = c_0\mathbf{1} + Q(B)u$, put $\alpha = -\mathbb{E}Q(B)u$. The identities of Lemma M.6 read

$$\operatorname{Var} Z = \int |Q|^2 \, d\nu - \alpha^2, \qquad \operatorname{Cov}(C_a, Z) = -\int \operatorname{Re}(G_a Q) \, d\nu - \beta_{a+1}\alpha. \tag{N.37}$$

They also hold at $a = 2$, where $G_2 = 1 + z$ and $\beta_3 = d$.

One finite square and its positive continuation.

For $c \ge 5$, $d \le 1/5$, use the actual $\sigma = 9/4$ row of Lemma M.10: $m = 1 - 9d/8$, $k = m^{-1}$, $U = (a - zb)u$, $R = -kU = (-t + zs)u$ and $D = 2d/5$. The retained coefficients keep $0 \le t, s \le 1$, $t \ge k\tau + (1 - x)s$, $a, b \ge x(1 - x)$ and all signed D_i. For $c = 4$, $d \le 1/4$, use instead

$$m = 1 - d, \quad U = (d + x - xz)u, \quad t = k(d + x), \quad s = kx, \quad D = 5d/12.$$

The actual arm sum is $f_2 + h_4 = 2m - U$. Its first-row error is bounded by D in (C169.6.1). Here $D_1 = k(d^2 + \delta_2)$, $D_2 = k\delta_3$, $D_3 = -k(d\delta_2 + \delta_3)$ and $D_4 = -k\delta_4$. These give $\gamma_3 \ge 0$, $\gamma_4 \ge 3d/4$ and $\gamma_j \ge 41d/96$ for $j \ge 5$. In both families

$$T^L F_c \le m^2 Y_L, \quad Y_L = \mathbf{1} + G_{L+1}(B)u + B^L R + D\bar{f}_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar{f}_{L+1-j}. \tag{N.38}$$

This is the finite identity of Lemma M.11, with the same direct induction for the $c = 4$ first row. Every discarded loss is nonnegative.

Retain no final loss at $L = 1$, γ_3 at $L = 2$, both $g_3 q$ and $g_4\mathbf{1}$ at $L = 3$, where $g_i = \gamma_i$, and $h = 41d/96$ on each of the last two paths at $L \ge 4$. In every case the resulting upper is $W = m^2 Z \ge T^L F_c$, with the following full polynomial and unscaled mean:

L	$Z = c_0\mathbf{1} + Q(B)u$	$\mathbb{E}Z$ upper
1	$c_0 = 1 + D, \quad Q = 1 - xs + z(1 - t)$	$1 + D - d + D_1$
2	$c_0 = 1 + D - \gamma_3, \quad Q = Q_3 + D$	$1 - 7d/20$
3	$c_0 = 1 + D - g_3 - g_4, \quad Q = G_4 - tz^3 + sz^4 + DG_2 - g_3$	$1 - 31d/32$
≥ 4	$c_0 = 1 + D - 2h, \quad Q = G_{L+1} - tz^L + sz^{L+1} + DG_{L-1} - h$	$1 - 83d/96$.

(N.39)

The $L = 2$ row is used here only for $c \geq 5$; its exceptional $c = 4$ conclusion is (Q167.T175.0). In that row $Q_3 = 1 - x + tx + z(1 - sx)$. At $L = 3$, $\alpha = \gamma_5 + Dd \geq \tau + Dd$; at $L \geq 4$, $\alpha = \gamma_{L+2} + D\beta_L \geq h + D\tau$. Thus the separate mean term in (N.37) has the favorable sign in all nonunit rows. The genuine retained first loss has the uniform lower bound $g_3 \geq d/8$: for $c = 4$, $D_1 \leq kd(d + \theta) \leq 7d/8$; for $c \geq 5$, the actual coordinate cap $a < 16/25$ gives $D_1 < (40/31)(16/25)d = 128d/155 < 7d/8$. Also $0 \leq g_3 \leq d$ because $D_1 \geq 0$. Thus $\mathbb{E}Z \leq 1 + (5/12)d - d/8 - 2d/3 - \kappa d = 1 - 31d/32$. At $L = 3$ the retained polynomial subtracts g_3, while g_4 remains a constant loss. This upper follows directly from $\bar{f}_2 = \mathbf{1} + G_2(B)u$ in the exact finite continuation.

Finite polynomial bounds.

The finite alternating sums give real and normalized imaginary coordinates in $[1 - x, 1]$ for G_a, $a \geq 3$, and in $[1 - x, 1 - x + x^2]$ for G_n, $n \geq 4$. Consequently

$$\mathrm{Re}(G_a G_n) \geq 0 \ (n \geq 4), \qquad \mathrm{Re}(G_a G_3) \geq 1 - 3x + x^2 \geq -1/16, \qquad \mathrm{Re}(G_a G_2) \geq 1 - 2x \geq 0.$$

For $H_n = G_{n-1} + z^n$, the $n = 4$ coordinates are $(1 - x + x^2, 1)$; for $n \geq 5$ they belong to the second rectangle above. Thus $\mathrm{Re}(G_a H_n) \geq 1 - 3x + 2x^2 - x^3 > 0$. All these finite corners have real part at least κ. They contract: H_4, H_5 have squared norms $1 - x + 3x^2 - 2x^3 + x^4$ and $1 - x - x^2 + 3x^3 - 2x^4 + x^5$, respectively. For H_4 the residual bracket increases to $-1451/32768$; for H_5 use $x(3x - 1) \leq 91/1024 < 1/4$ and $-2x^4 + x^5 \leq 0$. For $n \geq 6$ use

$$(1 - z)H_n = 1 - z^{n-1}(1 - z + z^2), \qquad |1 - z + z^2|^2 = 1 - x + x^2 \leq 1, \quad 2\theta^{3/2} + \theta^4 < 1.$$

The corresponding finite geometric identity gives $\|G_n\| \leq 1$ for $n = 1$ and $n \geq 3$; $\|G_2\| = \sqrt{1 + \theta} < 6/5$. Hence the rectangle polynomial $P_L = G_L - tz^{L-1} + sz^L$ contracts for every $L \geq 4$.

The two shorter polynomials retain their sharper bounds

$$|Q_2|^2 \leq 8/7, \qquad |Q_3|^2 \leq 8/7 \quad (c \geq 5), \qquad Q_2 = 1 - sx + z(1 - t). \tag{N.40}$$

These are (M.30) and (M.31), proved for the actual retained coefficients and the full frequency interval. For $c = 4$, Q_3 instead has squared norm less than $7/6 < (13/12)^2$. Indeed its coordinates are affine in $k \in [1, 4/3]$. At $k = 1$ the squared norm is at most one. At $k = 4/3$ it is

$$1 - x/3 + (28/9)x^2 - (40/9)x^3 + (16/9)x^4 + (16/9)x^5.$$

Its second derivative, with $v = x - 1/4$, is $13/9 - (28/3)v + 48v^2 + (320/9)v^3 > 0$: for $v \leq 0$ use $v^3 \geq -v^2/4$; for $v \geq 0$ the quadratic has minimum $107/108$. At $x = \theta$ its coordinates are $243/256 < 19/20$, $599/768 < 4/5$, giving squared norm less than $93/80 < 7/6$.

The weak-receiver rows and the genuine-f_2 correction.

For $a \geq 3$, (N.37) gives the following unscaled covariance budgets:

L	weak budget b_0	genuine $\mathrm{Cov}(f_2, Z)$ upper	$\mathbb{E}Z$ cap for transfer
1	$-3d/16$	$-439d/1024$	$9/8$
2	0	$-3d/16$	1
3	$d/16 + d^2$	$-3d/16 + d^2$	1
≥ 4	$29d^2/64$	$-3d/16 + hd$	1.

(N.41)

For the unit weak row, $\mathrm{Re}(G_a Q) \geq 1 - 2x + x[t - (1 - x)s] \geq 1 - 2x$. At $L = 2$ the nonnegative coordinates and $a, b \geq x(1 - x)$ give $\mathrm{Re}(G_a Q) \geq 1 - 3x + 2x^2 - x^3 > 0$. At $L = 3$ the G_3 corner

costs $d/16$, and subtracting g_3 costs at most d^2. At $L \geq 4$ the subtracted h costs hd and the possible DG_3 term costs at most $Dd/16$, giving $29d^2/64$.

For the genuine row use $G_2 = 1 + z$ directly. At $L = 1$,

$$\mathrm{Re}(G_2 Q) = 1 - x + xt - sx \geq 1 - x - x^2 \geq 439/1024, \qquad \alpha = d - D_1 \geq 0.$$

At $L = 2$ its product is at least $1 - 2x \geq 3/16$ and $\alpha = d - D_2 > 0$. At $L = 3$ the corners are G_3, G_4, G_5, H_4; their genuine products are at least $3/16$. For longer connectors the G corners give at least κ^2; H_5 gives $1 - 2x + x^2 - x^3 > 3/16$, and H_n, $n \geq 6$, gives at least $\kappa^2 - \theta^3 > 3/16$. The D term and the separate $-d\alpha$ are favorable. These facts prove the genuine column of the table.

Lemma M.8, with $\eta = 19/16$, therefore replaces the genuine mass $1-d$ by the common comparison mass $1 - 19d/16$ in the mean/covariance bracket. The required fees are $27d/512$ at unit length and $3d/64$ otherwise; each is strictly smaller than the gain in the table. In particular, the unscaled unit mean may exceed one: it is at most $1 + 2d/5 \leq 27/25 < 9/8$. No change of measure or replacement of the actual receiving arm occurs.

Curvature on nonunit connectors.

For $L \geq 2$ the actual scaled upper $W = m^2 Z$ satisfies

$$W \leq (1 - 5d/8)q - d/2 + (13/12)\sqrt{d}, \qquad \mathbb{E}W \leq 1 - 3d/2.$$

Here the $L = 2$ row uses $\|Q_2\| < 15/14$ and $\gamma_3 \geq dz_0/m$, $z_0 = 19/32 - 89d/32 > 0$. The actual $L = 3$ pointwise upper is

$$W \leq m^2[(1 + D - g_3)q + (13/12 + D)\sqrt{d} - g_4].$$

Here $\|Q_3 + D\| \leq 13/12 + D$. To compare with the weak profile, drop only the nonnegative loss $g_3 q$, use $g_4 \geq 2d/3$ and $D \leq 5d/12$. This gives $W \leq m^2 \mathcal{B}$, where

$$\mathcal{B} = (1 + 5d/12)q + (13/12 + 5d/12)\sqrt{d} - 2d/3 \geq (3/4)\sqrt{d} \geq 0.$$

The lower bound uses $q \geq 0$ and $\sqrt{d} \leq 1/2$. Scale this entire nonnegative row to get $W \leq (1-d)^2 \mathcal{B}$. Its q coefficient is at most $(1 - d)^2(1 + 5d/12) \leq 1 - 5d/8$. Its remaining constant difference, with the real constant loss $-2d/3$, is exactly $dR_3(\sqrt{d})$ below. For $L \geq 4$ the row is bounded by $(1-d)^2[(1-d/96)q + (1+d/2)\sqrt{d} - h]$. After comparing the q coefficients, the remaining constant differences are $tR_2(t)$, $dR_3(t)$, $tR_4(t)$, respectively, where $t = \sqrt{d}$ and

$$\begin{aligned}
R_2(t) &= 13/12 - (15/14)(1 - 9t^2/8)^2 + t[(1 - 9t^2/8)(19/32 - 89t^2/32) - 1/2], \\
R_3(t) &= 1/6 + (7/4)t - (4/3)t^2 - t^3/4 + (2/3)t^4 - (5/12)t^5, \\
R_4(t) &= 1/12 - (7/96)t + (3/2)t^2 - (41/48)t^3 + (41/96)t^5 - t^6/2.
\end{aligned}$$

Their Bernstein coefficients have minima $1/84, 1/6, 89/1152$ on $[0, 9/20]$, $[0, 1/2]$, $[0, 1/2]$, respectively. The scaled means follow from (N.39) and $m \leq 1 - d$. Thus the weak case of Lemma A.2 gives curvature nine for both receiving types.

The exact variance remains (N.37) until bounded. For $L = 2$ it is at most

$$m^4 d[8/7 + 2D + D^2] \leq (8/7)d \exp[-15d/4 - 81d^2/32].$$

This uses $\operatorname{Re} Q_3 \le 1$, $\log m \le -9d/8 - 81d^2/128$, and $1 + (7/10)d + (7/50)d^2 \le e^{(91/125)d}$. For $L = 3$, each full corner minus g_3 has norm at most $1 - g_3/2$, since its real part is at least κ and $2\kappa - 1 \ge 3g_3/4$. Adding DG_2 gives full norm at most $1 - g_3/2 + (6/5)D \le 1 + 7d/16$. For $L \ge 4$ the analogous two-loss norm is at most $1 + 13d/64$. One may now discard the nonpositive variance mean-square to get $d(1 + 7d/16)^2$ and $d(1 + 13d/64)^2$ before scaling by m^4. For the former keep the quadratic logarithmic gain $m^4 \le (1 - d)^4 \le e^{-4d-2d^2}$, so its scaled variance is at most $de^{-25d/8-2d^2}$.

At $L = 2$ the mean is bounded more sharply by

$$\mathbb{E}W \le m[1 - (11027/5120)d + (9953/2560)d^2 + d^4], \qquad \mathbb{E}W \le m^2(1 - 7d/20).$$

Indeed $D_1 + D_2 \le k[\delta_2 + \delta_3 + (53/32)d^2 + (17/16)d\delta_2 + d^4]$; its E term is $E[d^2 - (1-d)\delta_2] \le d^4$. For $d \le 3/16$, comparison with the cubic lower Taylor polynomial of $e^{-23d/8}$ leaves an upper difference at most $d[-403/1000 + (109/50)d - (2/5)d^2 + d^3] < 0$. For $3/16 \le d \le 1/5$, the second upper has logarithm at most $-(13/5)d - (53/40)d^2 - (15/16)d^3 \le -23d/8$. Thus the three scaled mean bounds are $e^{-23d/8}$, $e^{-95d/32}$ and $e^{-275d/96}$. The common fourth expansion, with the signed bracket just established, now gives

L	moment upper
2	$\mathcal{U}_3(d)$
3	$e^{-209d/16} + 9de^{-25d/8-2d^2} + (d/4 + 4d^2)e^{-349d/32} = \mathcal{U}_4(d)$
≥ 4	$e^{-607d/48} + 9de^{-115d/32} + (29/16)d^2e^{-339d/32} \le \mathcal{U}_5(d)$.

For the last inequality use $e > 8/3$ and $\sup_{s \ge 0} se^{-cs} = 1/(ce)$. Lemma M.14 proves losses $d/160$, $d/600$ and $d/60$ on the stated low domains.

Unit curvature and the two legal mean endpoints.

For the actual unit row set $\mu = \mathbb{E}W$. The relation $t \ge (1 - x)s$ gives $|(1 - t) + zs|^2 \le 1$, hence $W \le m^2(q + \sqrt{d} + D)$. If $C_0 \le p + 6\sqrt{d}/5$, $m \le 1 - d$, $D \le 5d/12$ and $\mu \le m^2(1 + 2d/5)$, then

$$C_0[(W + \mu)^2 + 2\mu^2] \le 9. \tag{N.42}$$

To prove it, write $v = \sqrt{d} \le 1/2$ and factor m^4. The nonnegative polynomial upper is $(p + 6v/5)[(q + 1 + v + (49/60)v^2)^2 + 2(1 + (2/5)v^2)^2]$. Its coefficients are bounded by $13/2, 66/5, 65/4, 25/2, 4, 6/5$. The terms of degrees at least two are at most $24v^2$. Comparing with $9e^{4v^2}$ reduces to $P(v) = 5/2 - (66/5)v + 12v^2 + 72v^4 > 0$. On $[0, 1/4]$ it decreases to $37/160$; afterwards $P'' \ge 78$, $P'(1/4) = -27/10$, giving $P \ge 37/160 - 243/5200 > 0$. The actual unit means obey the required cap: for $c = 4$, $D_1 \le 7d/8$ gives $\mu/m^2 \le 1 + 7d/24$; for $c \ge 5$, $D_1 < d$ gives $\mu/m^2 \le 1 + 2d/5$.

In the mean/covariance bracket after the genuine-mass compensation, use $h_0 = 1 - 19d/16$ and $f(y) = h_0y^4 - (3/4)m^2dy^3$. Its maximum on $[0, M]$ is zero or $f(M)$. The following actual scaled mean bounds have $M > 1/2$ and $f(M) > 0$:

c	M	$e^{-ad} \le M$	$M \le e^{-\lambda d}$
4	$(1 - d)[1 - (113/96)d + (19/12)d^2]$	$a = 209/96$	$\lambda = 129/64$
≥ 5	$(1 - 9d/8)[1 - (211/160)d + (373/160)d^2]$	$a = 391/160$	$\lambda = 35/16$.

These follow by bounding D_1 by $k(d^2 + \delta_2)$ and $k[\delta_2 + (53/32)d^2]$, respectively. Write each cubic as $M = 1 - ad + bd^2 - hd^3$. The log-convexity numerator is

$$MM'' - (M')^2 = (2b - a^2) + (2ab - 6h)d - 2b^2d^2 + 4bhd^3 - 3h^2d^4.$$

At $(a,b,h,D)=(209/96,265/96,19/12,1/4)$ its Bernstein coefficients are

$$7199/9216,\quad 7689/8192,\quad 414635/442368,\quad 41561/49152,\quad 5753/8192;$$

at $(391/160,4883/1280,3357/1280,1/5)$ they are

$$42439/25600,\quad 3693013/2048000,\quad 215610671/122880000,$$
$$163026843/102400000,\quad 1395197823/1024000000.$$

They are positive. The tangent at zero gives the lower exponential; the endpoint chord gives the upper, using

$$M(1/4)=309/512<(2329/3840)(255/256)<e^{-129/256},$$
$$M(1/5)=51429/80000<1-7/16+(7/16)^2/2-(7/16)^3/6<e^{-7/16}.$$

The first comparison uses the degree-five lower Taylor polynomial for $e^{-1/2}$ and $e^{-1/256}\ge 255/256$. Since $h_0\ge 45/64$, $M>1/2$ and $(3/4)m^2d\le 3/16$, the positive endpoint is valid before replacing its mean factors.

For $c=4$ the nonnegative unit coordinates decrease with $k\ge 1$, so $|Q|^2\le (1-x^2)^2+x(1-x)^2\le 1+x-3x^2\le 13/12$. Moreover $\alpha=d-D_1\ge d(\kappa-2d)/m>0$. Retaining the negative square gives

$$\operatorname{Var} W\le (13/12)m^4d-m^2(\kappa-2d)^2d^2.$$

For $c\ge 5$, $a,b\ge x(1-x)$ give

$$|Q|^2\le 1+x-4x^2+5x^3-x^4-x^5+x^6\le 1+x-(13/6)x^2.$$

The last inequality follows since $4-5x+x^2+x^3-x^4\ge 2279183/1048576>13/6$. Using $\delta_3\ge\delta_2^2/d$ and maximizing the resulting quadratic gives $\int |Q|^2\,d\nu\le 29d/26$. The inequalities $m^2\ge e^{-7d/3}$ and $m^2\ge e^{-8d/3}$, respectively, together with the two valid mean bounds, yield moment uppers $\mathcal{U}_1(d)$ and $\mathcal{U}_2(d)$. In the second row the negative cubic term is at least $(3/4)de^{-10d}$, because $8/3+3(391/160)=4799/480<10$. Lemma M.14 proves losses $d/800$ and $d/16$ for the unit low domains.

A single high comparison in the same direction.

For $c\ge 5$, (A.8) gives, throughout $d\ge 1/5$,

$$h_c\le 2\mu_{c-1}\le 2-(83/16)d\le 77/80,\qquad T^LF_c\le (77/80)T^L(qf_2).$$

The genuine-f_2 reference for qf_2 is proved in Section N.2 at $L=1$, Section N.5 at $L=2,3,4,5$, and Section N.3 at $L\ge 6$. For $a\ge 3$, the actual all-length proof comprises Section N.12 at $L=1$, (Q191.3.1) at $L=2$, (Q191.11.1) at $L=3,4,5$ and (N.6) at $L\ge 6$. Each is at most one. Thus all these fourth moments are at most $(77/80)^4<1-1/2400\le 1-d/800$; the cap also covers the stated stronger nonunit loss.

For $c=4$, $d\ge 1/4$, one has $h_4\le 2(1-2d)\le 1$. At $L=2$ use (Q167.T175.0), with loss $d/16$. At $L=3,4,5$ the same-direction qf_2 bound (Q191.11.1) has loss $9d/10$; for $L\ge 6$, (N.6) has loss $3d/800$. All exceed the required $d/600$. At unit length, the actual unpropagated square gives

$$\mathbb{E}F_4\le H(d)=1-2d+[-2+\theta(1+\theta)/2]d^2+(9/2)d^3.$$

This follows from $\mathbb{E}F_4\le m^2-m(d^2+2\delta_2)+(d^3+2d\delta_2+\delta_3+\delta_4)/2$. The moment bounds imply the displayed H. Its derivative increases on $[1/4,1/3]$ and has $H'(1/3)<-3/2$, while $H(1/4)<15/32$. Consequently $TF_4<15/16$ and its actual reference fourth is less than $15/16<1-d/800$.

Finally positivity inserts $0\le E_t,E_o\le 1$ into the proved inequalities. All arm and connector directions are fixed throughout the host domain. □

N.24 The short connectors for the reference terminal pair $(1, 2)$

Q191.0 The original finite model and receiving envelope

On the original finite probability space, delete only zero-mass atoms and let $K(x, y) = -K(y, x)$ with $|K(x, y)| \le 1$. An operator kernel acts by $(Kv)(x) = \sum_y \mu(y)K(x, y)v(y)$, and $Jv = (\mathbb{E}v)\mathbf{1}$. Put

$$B = -K, \quad T = J + B, \quad A = J - B = T^*, \quad u = B\mathbf{1}, \quad p = 1 - u, \quad q = 1 + u,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad f_0 = \bar{f}_0 = \mathbf{1}, \quad h_n = Tf_{n-1}, \quad \delta_j = \|B^j\mathbf{1}\|_2^2,$$

$$d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2, \quad \alpha = 1 - d + \delta_2/2, \quad m = 1 - 2d + 3\delta_2/2.$$

The finite polynomials and alternating moments are

$$G_n(z) = \sum_{j=0}^{n-1} z^j, \quad G_0 = 0, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j \quad (n \ge 3).$$

If $G_n(z) = E_n(x) + zD_n(x)$ at $z = i\sqrt{x}$, then the same finite polynomial at $-z$ is $E_n(x) - zD_n(x)$. Bounds on the two coordinates refer to these real functions, with this sign convention. All means, rows, norms and spectral measures use the same original μ. For the receiving envelope put

$$S_a = \mathbf{1} - G_a(-B)u, \qquad C_3 = f_3, \qquad C_a = S_a - \beta_a\mathbf{1} - \beta_{a-1}p \ge f_a \ge 0 \quad (a \ge 4).$$

The finite convolution gives $\tau \le \beta_j \le d$ for $j \ge 3$.

Q191.3 Exactly two original connector edges

For every original receiving integer $a \ge 3$, every original finite probability host and every real skew kernel $|K| \le 1$, the fixed actual terminal $F = qf_2$ satisfies

$$\mathbb{E}[E_o f_a \{T^2(E_t q f_2)\}^4] \le 1 - d/1000. \tag{Q191.3.1}$$

The receiving arm points outward, the terminal unit arm points inward, its two-edge arm points outward, and both original connector edges point from terminal to receiver. The factors E_o, E_t are arbitrary actual equal-pair products in their original locations. The undecorated $a = 3$ estimate below is stronger, with deficit $d/150$. The proof uses the actual two-step message, rather than propagating the unit reference fourth moment.

Q191.4 Actual means, covariance and simultaneous rows

The finite framework gives $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, and the centered-row inequality

$$|BZ(x)| \le \sqrt{p(x)q(x)}\,\|Z\|_2 \qquad (\mathbb{E}Z = 0). \tag{Q191.4.1}$$

Subtract the original row mean $u(x)$ before Cauchy–Schwarz; the row variance is at most $1 - u(x)^2$. This leaves the measure unchanged. The spectral measure ν of $-B^2$ at u has mass d, support $[0, \theta]$, and $\int x^j\, d\nu = \delta_{j+1}$. For $d > 0$ write

$$v = \delta_2/d, \quad s = 1 - v, \quad b = 2 - 3v/2, \quad a_0 = 1 - v/2, \quad \alpha = 1 - a_0 d, \quad m = 1 - bd = \alpha - ds.$$

Here a_0 is a scalar, while a remains the receiving length. The actual moment range is $d \le v \le \theta$; no lower bound $x \ge d$ or $x \ge v$ is imposed on the spectrum. Also $1/2 \le m \le 1$ and $\alpha \ge m > 0$. With $r = Bu$, the exact terminal identity is

$$qf_2 = (1 + r/2)^2 - (-u + r/2)^2.$$

Positivity and $T(r^2) \le 2\delta_2$ give the one-step upper $q - (d - \delta_2/2)\mathbf{1} + B^2u$. Apply the same positive T once more:

$$X_2 = T^2(qf_2) \le W = \alpha q + Bu + B^3u, \qquad \mathbb{E}W = m,$$
$$V = \mathrm{Var}(W) = \alpha^2 d + \delta_2 - 2\delta_3 + \delta_4 - \tau^2. \tag{Q191.4.2}$$

The upper W is nonnegative because it dominates the actual X_2. The receiving upper is the finite-convolution expression in part N.24. For $a = 3$, $f_3 = 1 - d - u + Bu - B^2u$. For $a \ge 4$, set $V_a = G_{a-1}(-B)u$; then $\mathbb{E}V_a = \beta_a$ and

$$C_a = (1 - \beta_a - \beta_{a-1})p + B(V_a - \beta_a\mathbf{1}).$$

The exact means and covariances are

$$\mathbb{E}C_3 = 1 - 2d, \qquad \mathrm{Cov}(C_3, W) = -(d - 2\delta_2 + \delta_3) - \delta_2\tau/2, \tag{Q191.4.3}$$

$$\begin{aligned} \mathbb{E}C_a &= 1 - \beta_a - \beta_{a+1} - \beta_{a-1}, \\ \mathrm{Cov}(C_a, W) &\le -\alpha\tau + \delta_2 - 2\delta_3 + 2\delta_4 - \delta_5 - \beta_{a+1}\tau + \beta_{a-1}\alpha d. \end{aligned} \tag{Q191.4.4}$$

For the latter, write $G_a(z) = E_a(x) + zD_a(x)$ at $z = i\sqrt{x}$. Then $E_a \ge 1 - x$, and, for $a \ge 4$, $D_a \le 1 - x + x^2$. Before this comparison the covariance equals

$$-\int [\alpha E_a - x(1 - x)D_a]\, d\nu - \beta_{a+1}\tau + \beta_{a-1}\alpha d.$$

Its integral upper is exactly (Q191.4.4). The positive β_{a-1} covariance coefficient must be combined with its receiving mean. For the $a = 3$ simultaneous row put $Z_3 = 2u - Bu + B^2u - d\mathbf{1}$. Then

$$\mathbb{E}Z_3 = 0, \quad \|Z_3\|_2^2 = 4d - 3\delta_2 + \delta_3 - d^2, \quad C_3 + W + m\mathbf{1} = (3 - 4d + 2\delta_2)\mathbf{1} + (\delta_2/2)u + BZ_3.$$

The uncentered symbol is $(2 - x) - z$, with squared modulus $4 - 3x + x^2$ and mean d. Maximizing $cu + \sqrt{1 - u^2}\|Z_3\|_2$ and using $\delta_3 \le \theta\delta_2$ gives

$$C_3 + W + m\mathbf{1} \le [3 - (4 - 2v)d + \sqrt{H_0}]\mathbf{1}, \quad H_0 = [4 - (83/32)v]d - [1 - v^2/4]d^2. \tag{Q191.4.5}$$

Here $H_0 \ge 0$ because it bounds the actual squared combined row norm. The square $(\sqrt{H_0} - 1/2)^2 \ge 0$ gives the joint loss

$$C_3 + W + m\mathbf{1} \le (13/4 - \delta)\mathbf{1}, \quad \delta = (19/32)vd + (1 - v^2/4)d^2. \tag{Q191.4.6}$$

It is sharper in the middle range than either $13/4$ or $27/8 - 4d/3$, for the same actual direction. For $a \ge 4$ put $U = (I + B^2)u$ and $Z_a = V_a - \beta_a\mathbf{1} + U$. Since $\mathbb{E}U = 0$, also $\mathbb{E}Z_a = 0$. Using $\beta_a + \beta_{a-1} \ge 2\tau$ only against nonnegative p gives

$$C_a + W + m\mathbf{1} \le (3 - 5d + 4\delta_2)\mathbf{1} + (d - 3\delta_2/2)u + BZ_a.$$

For $n = a - 1 \geq 4$, both coordinates of $G_n(-z)$ lie in $[0, 1 - x + x^2]$; at $n = 3$ they are $1 - x$ and 1. Thus the squared symbol of $V_a + U$ is at most

$$\begin{aligned} P(x) &= (2 - 2x + x^2)^2 + x(1 - x + x^2)^2 \\ &= 4 - 7x + 6x^2 - x^3 - x^4 + x^5 \leq 4 - (73/16)x. \end{aligned}$$

At $n = 3$ the squared symbol $4-7x+4x^2$ is smaller, because their difference is $x^2(2-x-x^2+x^3) \geq 0$. The final comparison uses $6x \leq 39/16$ and $-x^3 - x^4 + x^5 \leq 0$. Centering subtracts $\beta_a^2 \geq \tau^2$, so

$$\begin{aligned} C_a + W + m\mathbf{1} &\leq \left[3 - 5d + 4\delta_2 + \sqrt{4d - (73/16)\delta_2 - d\delta_2 + (5/4)\delta_2^2}\right]\mathbf{1} \\ &\leq (13/4 - d)\mathbf{1}. \end{aligned} \tag{Q191.4.7}$$

For the last comparison the positive square difference is

$$1/16 - [2 - (41/16)v]d + [16 - 31v + (59/4)v^2]d^2.$$

Its leading coefficient is positive and its discriminant is $v[-5/2 + (737/256)v] \leq 0$. At $v = 0$ it is $(1/4 - 4d)^2$.

Q191.5 A common cubic coefficient with the retained joint loss

Put $C_0 = 49/16$, $A_0 = 10441/2048$ and $k_0 = 15561/2048$. For $a = 3$ use δ from (Q191.4.6); for $a \geq 4$ use $\delta_* = d$. Define

$$k_\delta = A_0 + (5/2)m^2 - \delta(C_0 + 2m^2). \tag{Q191.5.1}$$

Both choices satisfy, for the same nonnegative receiver and W,

$$C_a[(W + m)^2 + 2m^2] \leq k_\delta \mathbf{1}. \tag{Q191.5.2}$$

Indeed, for $y = W + m \geq 0$ and $C_a + y \leq 13/4 - \delta$, the left side is at most $(13/4 - \delta - y)(y^2 + 2m^2)$. First take $m = 1, \delta = 0$ and $y = 7/4 + h$. Its difference from k_0 is

$$h^3 + 2h^2 - (3/16)h + 9/2048.$$

For $h \geq 0$ this is $h^3 + 2(h - 3/64)^2$; for $-7/4 \leq h \leq 0$, $h^2(h + 2)$ and the remaining terms are nonnegative. Retaining m gives the bound $A_0 + (5/2)m^2$. At $y \leq 2$ the decrease from $m = 1$ is at least $(5/2)(1 - m^2)$. At $y = 2 + h \geq 2$ the expression is

$$5 + (5/2)m^2 + (1 - 2m^2)h - (11/4)h^2 - h^3.$$

Its excess above the constant part is at most $1/44$ since $m^2 \geq 1/4$, while $A_0 - 5 = 201/2048 > 1/44$. At $y \geq 7/4$ the additional loss is at least $\delta(C_0 + 2m^2)$. On $0 \leq y \leq 7/4$ the cubic has its maximum at an endpoint if its derivative at $7/4$ is nonnegative: the derivative is a downward quadratic, negative at zero, so its only interior local maximum lies to the right of $7/4$. The endpoint derivative is $35/16 - (7/2)\delta - 2m^2$. For $\delta_* = d$ and $m \leq 1 - 89d/64$ its lower bound is

$$3/16 + (33/16)d - (7921/2048)d^2 > 0.$$

For (Q191.4.6), use $\delta \leq (247/1024)d + d^2$; the lower bound is

$$3/16 + (9663/2048)d - (15089/2048)d^2 > 0.$$

Both quadratics are concave on $[0,1/3]$ with positive endpoint values; the second has right endpoint $4339/4608$. At $y=7/4$ the difference from (Q191.5.1) is $1033/2048-m^2/2\ge 9/2048$. At $y=0$ it is $A_0-4m^2-C_0\delta$. For $\delta_*=d$ this is at least $475/6144>0$; for the other choice, $\delta\le 1765/9216<1/5$, so it is at least $A_0-4-49/80>0$. This proves (Q191.5.2) in both cases. All coefficients are positive and at most $k_0<8$. For (Q191.4.6) put $j=5/2-2\delta>21/10$. Its derivative satisfies

$$\begin{aligned}\partial_v k_\delta &= 3jdm-[(19/32)d-vd^2/2](C_0+2m^2)\\ &\ge d[63/20-1539/512]=(369/2560)d\ge 0.\end{aligned}\tag{Q191.5.3}$$

The nonnegative coefficient derivative is retained in the continuous one-boundary payment.

Q191.6 Coupled moment payment on the original spectrum

The exact identity

$$W^4=m^4+4m^3(W-m)+(W-m)^2[(W+m)^2+2m^2]$$

and (Q191.5.2) give the two complete upper expressions

$$\begin{aligned}R_3&=(1-2d)m^4-4m^3(d-2\delta_2+\delta_3+\delta_2\tau/2)+kV,\\ R_a&=[1-\beta_a-\beta_{a+1}-\beta_{a-1}]m^4\\ &\quad+4m^3[-\alpha\tau+\delta_2-2\delta_3+2\delta_4-\delta_5-\beta_{a+1}\tau+\beta_{a-1}\alpha d]+kV,\end{aligned}$$

where k is the corresponding coefficient just proved, not an independent parameter. These are the original receiving means, covariances and variances, coupled before relaxation. For $a=3$ the combined frequency polynomial in δ_3,δ_4 is

$$-(4m^3+2k)x^2+kx^3.$$

Its second derivative is $-8m^3+k(-4+6x)<0$ on $[0,\theta]$. Jensen for ν/d therefore bounds the combined expression by substituting dv^2,dv^3 for δ_3,δ_4. This joint substitution bounds the combined polynomial for the original host. Set

$$n=s^2+dvs/2,\qquad H=\alpha^2+(v-d)s^2\ge 0.$$

The $a=3$ expression is at most

$$F_3(d,v)=(1-2d)m^4-4dm^3n+dk_\delta H.\tag{Q191.6.1}$$

For $a\ge 4$ the full covariance-plus-variance frequency polynomial is

$$(4m^3+k)x-(8m^3+2k)x^2+(8m^3+k)x^3-4m^3x^4.$$

Its second derivative is $-16m^3(1-3x+3x^2)+k(-4+6x)<0$. Jensen applies at the same first moment v, replacing $\delta_2-2\delta_3+2\delta_4-\delta_5$ by $dv(1-v)(1-v+v^2)$ jointly with the variance.

Q191.7 The $a=3$ reduction increases continuously in v

For fixed d, differentiating (Q191.6.1) gives

$$\begin{aligned}(\partial_v F_3)/d&=m^2P(d,s)+(\partial_v k_\delta)H+k_\delta H_v,\\ P(d,s)&=6+8s-d(13+17s+30s^2)+d^2(5+8s+15s^2),\\ H_v&=\alpha d+s(3s-2+2d).\end{aligned}\tag{Q191.7.1}$$

The permitted interval is $19/32 \le s \le 1-d$. The polynomial P decreases with d on $[0,1/3]$. If $s \le 2/3$, then

$$P \ge P(1/3,s) = (20+29s-75s^2)/9 \ge 2/3,$$

using $s^2 \le (2/3)s$. If $s \ge 2/3$, use $d \le 1-s$:

$$P \ge P(1-s,s) = -2+2s-9s^2+8s^3+15s^4 \ge 2/3.$$

The last polynomial increases on $[2/3,1]$: its derivative is $2+6s(4s-3)+60s^3 \ge 2-2s+60s^3 > 0$, and its value at $2/3$ is $2/3$. For $d \ge 1/8$, $H_v \ge 0$ since $3s-2+2d \ge 1/32$; together with (Q191.5.3), this makes (Q191.7.1) positive. For $d \le 1/8$ use $P \ge 43/4-60d \ge 13/4$ and $m \ge 3/4$. Also $H_v \ge s(3s-2) \ge -133/1024$ and $k_\delta < 8$, so

$$(\partial_v F_3)/d \ge 117/64 - 133/128 = 101/128 > 0.$$

Both continuous subintervals and their common endpoint are covered. Hence

$$F_3(d,v) \le F_3(d,\theta). \tag{Q191.7.2}$$

Q191.8 An exact univariate certificate for $a=3$

At $v=\theta=13/32$ put $b_0=89/64$. Then

$$\begin{aligned} m &= 1-b_0 d, \qquad n = 361/1024 + (247/2048)d, \\ H &= 37461/32768 - (1993/1024)d + (2601/4096)d^2, \\ \delta &= (247/1024)d + (3927/4096)d^2. \end{aligned} \tag{Q191.8.1}$$

Let F_{30} denote (Q191.6.1) with δ set to zero only in the analytic coefficient, and let $P_0 = (1-F_{30})/d$. Its coefficients are

$$\begin{aligned} P_0 = {} & 19214147/67108864 - (5653367/1048576)d \\ & + (4361211079/268435456)d^2 - (833648493/67108864)d^3 \\ & + (417698093/134217728)d^4. \end{aligned} \tag{Q191.8.2}$$

They give

$$P_0 \ge 2/7 - (27/5)d + (81/5)d^2 - (25/2)d^3. \tag{Q191.8.3}$$

The positive integer gaps for the first four coefficient comparisons are $281301, 44717, 62783459, 5212307$, after multiplying the two positive denominators. The fourth-degree coefficient is positive. The retained δ increases the deficit by δF, where $F = (49/16+2m^2)H$. Its first four coefficients are

$$3034341/524288, \quad -8499885/524288, \quad 1239002117/67108864, \quad -23194201/2097152,$$

and its degree-four coefficient is positive. Thus

$$F \ge 23/4 - (65/4)d + 18d^2 - (45/4)d^3. \tag{Q191.8.4}$$

The corresponding positive integer gaps are $19685, 19795, 31042565, 398759$. Combine (Q191.8.3) with δ times (Q191.8.4), where $\delta \ge 0$, and pair degrees four and five using $d \le 1/3$:

$$(1-F_3(d,\theta))/d \ge Q(d), \quad Q(d) = 2/7 - (257/64)d + (71/4)d^2 - (95/4)d^3 + (175/16)d^4. \tag{Q191.8.5}$$

Before the last simplification the coefficients of degrees one through five are

$$-27/5+5681/4096,\quad 81/5+26101/16384,\quad -388919/16384,\quad 59571/4096,\quad -176715/16384.$$

The combined degree-four coefficient is at least $538137/49152>175/16$. The other three comparison gaps are $53/20480$, $14116/327680$ and $201/16384$. To pay Q continuously, its second derivative decreases on $[0,1/3]$ to $31/12$. At $t=5/32$,

$$Q(t)=885817/117440512>3/400,\qquad Q'(t)=-5421/131072,\quad |Q'(t)|<1/24.$$

Taylor's lower quadratic with curvature $31/12$ gives

$$Q(d)\ge 3/400-1/2976>1/150. \tag{Q191.8.6}$$

Equations (Q191.6.1)–(Q191.8.6) yield $\mathbb{E}[f_3X_2^4]\le 1-d/150$ for all original hosts with $d>0$. This certificate uses a continuous quadratic lower bound.

Q191.9 Complete β payment for every $a\ge 4$

Use $k_*=A_0+(5/2)m^2-d(C_0+2m^2)$, justified by (Q191.4.7) and (Q191.5.2). In the exact fourth expansion the β_{a-1} coefficient is m^3h, where

$$h=4\alpha d-m=-1+(6-3v/2)d-(4-2v)d^2.$$

Replace β_a,β_{a+1} by τ only against their negative coefficients; replace β_{a-1} by τ if $h\le 0$ and by d if $h>0$. After the coupled Jensen reduction of part N.24, the valid upper is

$$\begin{aligned}&F_*(d,v)+dvm^3h_+,\\ F_*&=(1-3ds)m^4+4dm^3[-\alpha s(1-d)-ds^2+vs(1-v+v^2)]+dk_*H.\end{aligned} \tag{Q191.9.1}$$

Here $h_+=\max(h,0)$ is an analytic bound, with the actual direction fixed. The positive covariance part of the true β coefficient has been retained. Compare this with the same F_3 and its stronger quadratic δ. For $d>0$, $\delta<d$, $H\ge m^2$, and

$$(C_0+2m^2)H/m^3\ge C_0/m+2m\ge 81/16.$$

Subtraction gives

$$F_*+dvm^3h_+\le F_3+dm^3[S(d,v)+vh_+], \tag{Q191.9.2}$$

with

$$\begin{aligned}&S=G_0+dL+d^2M,\qquad G_0=3v-1-4v^2(1-v)^2,\\ &L=15/16-(2301/512)v+v^2/2,\qquad M=17/16+6v-(209/64)v^2.\end{aligned} \tag{Q191.9.3}$$

For the subtraction, the covariance difference divided by d is $-v^2s^2+ds(1+v)-(1-v/2)d^2s$. After division by m^3, the mean-plus-covariance part is

$$G_0+d[6-(15/2)v+v^2/2]-4(1-v/2)sd^2.$$

Subtracting $(81/16)(d-\delta)$ gives (Q191.9.3) exactly.

Q191.9.1 The entire low domain $d \le 1/5$

Since h decreases in v and $v \ge d$,

$$h \le -1 + 6d - (11/2)d^2 + 2d^3 \le -1/250 < 0.$$

The last polynomial increases up to $1/5$, so $h_+ = 0$. Also $M > 0$, hence S is convex in d. For the required endpoint bounds, G_0 increases on $[0, \theta]$ because $G_0' = 3 - 8v(1-v)(1-2v) \ge 1$, and

$$G_0(\theta) = -3665/262144 < 0.$$

If $v \le 1/5$, the largest allowed d is v, and

$$\begin{aligned} S(v,v) &= -1 + (63/16)v - (3805/512)v^2 + (29/2)v^3 - (465/64)v^4 \\ &\le -1 + 63/80 + 29/250 = -193/2000 < 0. \end{aligned}$$

If $v \ge 1/5$, use $d = 1/5$. The function $S(1/5, v)$ increases in v, since $G_0' \ge 1$, $L'/5 \ge -2301/2560$, and $M' > 0$. At θ it is negative: $G_0(\theta) < 0$ and

$$5L(\theta) + M(\theta) = 23/4 - (8433/512)\theta - (49/64)\theta^2 \le -15421/16384 < 0.$$

Convexity proves $S \le 0$ throughout the low domain. Equations (Q191.9.2) and (Q191.8.6) give deficit $d/150$ there for every $a \ge 4$.

Q191.9.2 The entire high domain $1/5 \le d \le 1/3$

Now $v \ge d \ge 1/5$; keep the possibly positive β coefficient. For fixed v, the proved starting values are $S(1/5, v) \le 0$ and $h(1/5, v) < 0$. Throughout this domain,

$$S_d = L + 2dM \le L + (2/3)M = 79/48 - (253/512)v - (161/96)v^2 < 3/2,$$

$$vh_d \le v[22/5 - (7/10)v] \le 17121/10240 < 27/16.$$

For the first inequality, at $v \ge 1/5$ the two subtracted terms exceed $3/32$ and $1/16$, whose sum exceeds $7/48$; the second quadratic increases up to θ. Integrating gives

$$S(d,v) + vh_+(d,v) \le (51/16)(d - 1/5).$$

Since $m \le 231/320$ and

$$231^3 \cdot 85 = 1047743235 < 1048576000 = 32 \cdot 320^3,$$

we have $m^3 < 32/85$. Thus (Q191.9.2) gives

$$F_* + dvm^3 h_+ \le F_3 + (6/5)d(d - 1/5). \tag{Q191.9.4}$$

The same quartic pays this excess. Put $Q_H = Q - (6/5)(d - 1/5)$. On $[1/5, 3/10]$ its second derivative is at least $73/16 > 9/2$. At $t = 1/4$,

$$Q_H(t) = 2017/716800 > 1/360, \quad Q_H'(t) = -141/1280, \quad |Q_H'(t)| < 1/9.$$

Taylor's lower quadratic yields

$$Q_H \ge 1/360 - 1/729 = 41/29160 > 1/1000.$$

On $[3/10, 1/3]$, Q_H increases because $Q_H'(3/10) = 13/64 > 0$ and $Q_H'' > 0$. Therefore $Q_H > 1/1000$ on the whole high domain. Equations (Q191.8.5) and (Q191.9.4) prove (Q191.3.1) for all $a \ge 4$.

Q191.11 The actual three-, four- and five-edge messages

Under the same original assumptions and the fixed directions in part N.24, every $a \ge 3$ and $L \in \{3, 4, 5\}$ satisfy

$$\mathbb{E}\big[E_o f_a \{T^L(E_t q f_2)\}^4\big] \le 1 - (9/10)d. \tag{Q191.11.1}$$

Thus the four lengths $2, 3, 4, 5$ have common deficit $d/1000$. We prove the three lengths by their actual finite propagation. Retain α, m, τ and set

$$\begin{aligned} M &= m - \alpha d, \qquad \rho = \alpha d - \delta_2 + \delta_3, \qquad N = M - \rho, \\ \sigma &= md - \alpha\delta_2 = d(1-v)(\alpha - d), \qquad P = N - \sigma. \end{aligned} \tag{Q191.11.2}$$

Starting with $W_2 = W$, apply the original positive T successively: $W_3 = TW_2$, $W_4 = TW_3$, $W_5 = TW_4$. Explicitly,

$$\begin{aligned} W_3 &= m\mathbf{1} + Q_3(B)u, \quad Q_3(z) = \alpha + \alpha z + z^2 + z^4, \\ W_4 &= M\mathbf{1} + Q_4(B)u, \quad Q_4(z) = m + \alpha z + \alpha z^2 + z^3 + z^5, \\ W_5 &= N\mathbf{1} + Q_5(B)u, \quad Q_5(z) = M + mz + \alpha z^2 + \alpha z^3 + z^4 + z^6. \end{aligned}$$

Their means are M, N, P, and their mean offsets from the displayed constants are $-\alpha d, -\rho, -\sigma$. Every $W_L \ge 0$ dominates the actual $T^L(qf_2)$. At $z = i\sqrt{x}$ write $Q_L = E_L + zD_L$:

L	$E_L(x)$	$D_L(x)$
3	$\alpha - x + x^2$	α
4	$m - \alpha x$	$\alpha - x + x^2$
5	$M - \alpha x + x^2 - x^3$	$m - \alpha x$

The original moment inequalities give

$$\begin{gathered} 13/18 \le \alpha \le 1 - (51/64)d \le 1 - 3d/4, \qquad 0 \le \tau \le d(1-d) \le 2/9, \\ \alpha - 2\tau = 1 - 3d + (5/2)\delta_2 \ge 5/18, \qquad (25/96)d \le \rho \le d, \qquad \sigma = d(1-v)(\alpha - d) \ge 0. \end{gathered} \tag{Q191.11.3}$$

For the lower bound on ρ, use $\rho \ge d(\alpha - v) \ge d(1 - d - v) \ge 25d/96$; the upper bound uses $\delta_3 \le \delta_2$ and $\alpha \le 1$. Also

$$M = 1 - [3 - (3/2)v]d + (1 - v/2)d^2 \le 1 - (17/8)d \le 1 - 2d.$$

The loss coefficient $3 - 3v/2 - (1 - v/2)d$ decreases in d, v, with value $17/8$ at $d = 1/3, v = \theta$. Positivity and $\rho, \sigma \ge 0$ yield

$$0 \le P \le N \le M \le 1 - 2d. \tag{Q191.11.4}$$

For these lengths use the single receiving-convolution loss, denoted by R_a to distinguish it from the double-loss upper used at $L = 2$:

$$0 \le f_a \le R_a := S_a - \beta_a \mathbf{1}, \qquad \mathbb{E}R_a = 1 - \beta_a - \beta_{a+1} \le 1 - 2\tau \le 1 - (19/16)d. \tag{Q191.11.5}$$

In particular $R_3 = f_3$. Restore all original equal pairs after proving the three moment bounds for these same R_a.

Q191.12 Positive coordinates and a common variance bound

The polynomial $\phi(x) = 1 - 3x + 2x^2 - x^3$ decreases on $[0, \theta]$, and

$$\phi(x) \geq c_0 := 1451/32768 > 0. \tag{Q191.12.1}$$

For Q_3, $E_3/\alpha = 1 - x(1-x)/\alpha$ lies between $5/8$ and $1 - x + x^2$, while the normalized imaginary coordinate is one. Hence

$$|Q_3|^2 \leq \alpha^2[(1 - x + x^2)^2 + x] = \alpha^2[1 - x\phi(x)] \leq \alpha^2. \tag{Q191.12.2}$$

Since $Q_4 = m + zQ_3$, (Q191.11.3) gives

$$|Q_4|^2 = m^2 + x|Q_3|^2 - 2m\alpha x \leq m^2 - \alpha(2m - \alpha)x \leq m^2. \tag{Q191.12.3}$$

Its real coordinate satisfies $E_4 = m - \alpha x \geq \alpha\kappa - \tau \geq 119/576 > 0$, and $D_4 = E_3 > 0$. For Q_5, $E_5' = -\alpha + 2x - 3x^2 \leq -1/3$, and at θ it is at least

$$(13/18)(25/96) - 2/9 + \theta^2\kappa = -59/1728 + 3211/32768 > 1/48 > 0.$$

Its imaginary coordinate is $E_4 > 0$. Compare its coordinates with m times the finite polynomial having coordinates

$$e_5(x) = 1 - x + x^2 - x^3 = (1-x)(1+x^2), \qquad 1 - x.$$

The imaginary comparison is $m - \alpha x \leq m(1-x)$ since $\alpha \geq m$. For the real coordinate,

$$E_5 - me_5 = -\alpha d - \tau x + (1-m)x^2(1-x) \leq d(-\alpha + 2\theta^2) \leq 0,$$

using $1 - m \leq 2d$ and $\alpha \geq 2/3 > 2\theta^2$. Therefore

$$|Q_5|^2 \leq m^2(1-x)^2[1 + x + 2x^2 + x^4] \leq m^2. \tag{Q191.12.4}$$

The last normalized squared modulus is $1 - x[1 - x + 3x^2(1-x) + x^4(2-x)] \leq 1$, with nonnegative bracketed terms. These are finite polynomials of the actual messages. Integrate their squared symbols and subtract the actual squared mean offsets:

$$\mathrm{Var}(W_3) \leq \alpha^2 d(1-d), \quad \mathrm{Var}(W_4) \leq m^2 d - \rho^2 \leq m^2 d, \quad \mathrm{Var}(W_5) \leq m^2 d - \sigma^2 \leq m^2 d.$$

For the last two, put $t = \tau/\alpha \in [0, 1]$. Since $t \geq \kappa d$,

$$2t - t^2 \geq 2\kappa d - \kappa^2 d^2 \geq d,$$

where $2\kappa - \kappa^2/3 = 1 + 215/3072 > 1$. Thus $m^2 \leq \alpha^2(1-d)$ and all three actual variances obey

$$\mathrm{Var}(W_L) \leq \alpha^2 d(1-d) \leq d(1-d)(1-3d/4)^2 \qquad (L = 3, 4, 5). \tag{Q191.12.5}$$

Q191.13 Three simultaneous original-row bounds

For every $a \geq 3$, we prove separately

$$R_a + W_L + (\mathbb{E}W_L)\mathbf{1} \leq (13/4)\mathbf{1} \qquad (L = 3, 4, 5). \tag{Q191.13.1}$$

Let $V_a = G_{a-1}(-B)u$, with $\mathbb{E}V_a = \beta_a$. Then $R_a = (1-\beta_a)p + B(V_a - \beta_a\mathbf{1})$. For $n = a-1 \geq 2$ the real and normalized imaginary coordinates of $G_n(-z)$ lie in $[1-x, 1]$; for $n \geq 3$ the real coordinate is at most $1 - x + x^2$. These are finite alternating sums; at $n = 2$ the coordinates are $1, 1$.

Q191.13.1 $L = 3$**,** $a \geq 4$

Put $Y = \alpha u + Bu + B^3 u$, whose mean is $-\tau$, and $Z = V_a - \beta_a \mathbf{1} + Y + \tau \mathbf{1}$. The centered-row inequality and $\beta_a \geq \tau$ give

$$R_a + W_3 + M\mathbf{1} \leq [3 - (6 - 4v)d + (1 - v/2)d^2]\mathbf{1} + (\alpha - 1)u + BZ.$$

The raw symbol of $V_a + Y$ has real coordinate at most $2 - x + x^2$ and absolute normalized imaginary coordinate at most x. Its squared modulus is at most

$$(2 - x + x^2)^2 + x^3 = 4 - 4x + 5x^2 - x^3 + x^4 \leq 4 - (63/32)x.$$

Centering decreases the norm. With $a_0 = 1 - v/2$ the combined squared row bound is at most $[4 - (63/32)v]d + a_0^2 d^2$. Apply $\sqrt{H} \leq H + 1/4$:

$$\begin{aligned} R_a + W_3 + M\mathbf{1} &\leq [13/4 + (-2 + (65/32)v)d + (a_0 + a_0^2)d^2]\mathbf{1} \\ &\leq [13/4 - (1561/3072)d]\mathbf{1} \leq (13/4)\mathbf{1}. \end{aligned}$$

Q191.13.2 $L = 3$**,** $a = 3$

Here $R_3 = f_3$. The centered vector is $(1+\alpha)u + B^3 u - \delta_2 \mathbf{1}$, with squared norm $(1+\alpha)^2 d + \delta_4 - \delta_2^2$; the remaining u coefficient is $-d + 3\delta_2/2$. The joint sum is at most

$$[3 - (35/8)d + \sqrt{(33/8)d + d^2}]\mathbf{1}.$$

Indeed, the deterministic part satisfies $3 - 6d + d^2 + 3\delta_2 - d\delta_2/2 \leq 3 - (35/8)d$, and the combined squared row norm is at most $(4 + \theta^3)d + d^2 \leq (33/8)d + d^2$. The positive square difference for comparison with $13/4$ is

$$1/16 - (31/16)d + (1161/64)d^2 > 0,$$

whose discriminant is $-25/32$.

Q191.13.3 $L = 4$**, every** $a \geq 3$

Use the exact centered recursion $W_4 = Mq + B(W_3 - M\mathbf{1})$. The combined centered vector is $V_a - \beta_a \mathbf{1} + W_3 - M\mathbf{1}$, with raw symbol $G_{a-1}(-z) + Q_3(z)$. Its real coordinate is at most 2; its absolute normalized imaginary coordinate is at most $\max(1 - \alpha, \alpha - (1 - x)) \leq \max(d, x) \leq \theta$. Its squared norm is therefore at most $(33/8)d$. After $\beta_a \geq \tau$, the remaining u coefficient is $M - 1 + \tau = -(1 - \alpha + \alpha d)$, with absolute value at most $2d$. The deterministic part is at most $1 - \tau + M + N \leq 3 - (147/32)d$, so

$$R_a + W_4 + N\mathbf{1} \leq [3 - (147/32)d + \sqrt{(33/8)d + 4d^2}]\mathbf{1} \leq (13/4)\mathbf{1}.$$

The positive square difference has coefficients $1/16, -117/64, 17513/1024$ and discriminant $-3824/4096 < 0$.

Q191.13.4 $L = 5$**, every** $a \geq 3$

Use $W_5 = Nq + B(W_4 - N\mathbf{1})$. The raw symbol of the combined centered vector is $G_{a-1}(-z) + Q_4(z)$. Its real coordinate is at most 2; its absolute normalized imaginary coordinate is at most

$$1 - E_3 = 1 - \alpha + x - x^2 \leq d + 1/4 \leq 7/12.$$

This covers both signs: $E_3-(1-x)=x^2-(1-\alpha)\le 1-E_3$, since $2(1-\alpha)+x-2x^2\ge 0$. The squared norm is at most $(17/4)d$, using $x(7/12)^2<1/4$ on the entire frequency interval. The remaining u coefficient is $N-1+\tau=-(1-\alpha+\alpha d+\rho)$, of absolute value at most $3d$. The deterministic part obeys

$$1-\tau+N+P\le 1-\tau+2M-2\rho\le 3-5d,$$

because $4+\kappa+2(25/96)=491/96>5$. Hence

$$R_a+W_5+P\mathbf{1}\le[3-5d+\sqrt{(17/4)d+9d^2}]\mathbf{1}\le(13/4)\mathbf{1}.$$

The square difference $1/16-(7/4)d+16d^2$ has discriminant $-15/16$. This proves all three bounds in (Q191.13.1).

Q191.14 Nonpositive actual receiving covariance

The positive coordinates in part N.24 permit the same finite receiving comparison for every $a\ge 3$. Let λ_L be $\alpha d,\rho,\sigma$, respectively. With the receiving coordinates E_a,D_a, the exact covariance and sufficient upper are

$$\begin{aligned}\operatorname{Cov}(R_a,W_L)&=-\int[E_aE_L-xD_aD_L]\,d\nu-\beta_{a+1}\lambda_L\\&\le-\int[(1-x)E_L-xD_L]\,d\nu-\tau\lambda_L.\end{aligned}\tag{Q191.14.1}$$

The negative mean-offset term is retained; $E_a\ge 1-x$ and $D_a\le 1$ are used only against positive E_L, D_L.

Q191.14.1 $L=3$

The bracket is $\alpha(1-2x)-x(1-x)^2$, decreasing on $[0,\theta]$, with endpoint $3\alpha/16-4693/32768$. Using $\alpha\ge 1-d$ gives

$$\operatorname{Cov}(R_a,W_3)\le -c_0d-(13/32)d^2+(19/32)d^3\le 0.\tag{Q191.14.2}$$

The quadratic-plus-cubic coefficient is at most $-5/24$ for $d\le 1/3$.

Q191.14.2 $L=4$

The bracket in (Q191.14.1) is

$$\alpha\phi(x)+(1-\alpha)x^2(1-x)-\tau(1-x).$$

Write $\gamma=1-\alpha$. Since $\rho=\alpha d-\delta_2+\delta_3$, the covariance upper becomes

$$-\alpha d+3\alpha\delta_2-(2-\delta_2/2)\delta_3+\delta_4+\gamma\tau d.$$

The combined polynomial $-(2-\delta_2/2)x^2+x^3$ is concave because its second derivative is $-4+\delta_2+6x<0$. Jensen for ν/d gives

$$\operatorname{Cov}(R_a,W_4)/d\le U(d,v):=-\alpha+3\alpha v-2v^2+v^3+(d/2)v^3+\gamma\tau.$$

Its v derivative is

$$3-4v+3v^2+d(-7/2+3v+3v^2/2)-d^2(3/2-v)\ge 3-4\theta-7/6-1/6=1/24>0.$$

At $v=\theta$ the upper is

$$-c_0-(9227/65536)d+(969/2048)d^2\le -7597/196608<0.$$

Use $d^2\le d/3$ and then $d\le 1/3$ for the last bound.

Q191.14.3 $L = 5$

The bracket in (Q191.14.1) is

$$M - (M + m + \alpha)x + (2\alpha + 1)x^2 - 2x^3 + x^4.$$

The combined polynomial $-(2\alpha + 1)x^2 + 2x^3 - x^4$ is concave: its second derivative is $-4\alpha - 2 + 12x - 12x^2 \le -4\alpha + 1 < 0$. Jensen and $\sigma = d(1 - v)(\alpha - d)$ yield

$$\begin{aligned} \mathrm{Cov}(R_a, W_5)/d &\le -(1 - v)J(d, v), \\ J &= (\alpha - d)(1 - 2v) - \alpha dv + (v^2 - d^2)(1 - v) \\ &= 1 - 2v + v^2 - v^3 + d(-2 + 7v/2 - v^2) + d^2(-1 + 2v - v^2/2). \end{aligned} \tag{Q191.14.3}$$

The coefficients of d, d^2 are negative on $[0, \theta]$, so J decreases in d. If $v \le 1/3$, use $d \le v$:

$$J \ge J(v, v) = 1 - 4v + (7/2)v^2 - v^4/2 \ge 4/81 > 0;$$

the last polynomial decreases up to $1/3$. If $v \ge 1/3$, use $d \le 1/3$:

$$J \ge J(1/3, v) = (4 - 11v + 11v^2 - 18v^3)/18 \ge 2291/294912 > 0.$$

Its derivative is negative and the endpoint shown is its value at θ. Thus the actual covariance is nonpositive also at $L = 5$.

Q191.15 A common scalar payment for all three lengths

The coefficient argument in part N.24, with $\delta = 0$, works for every actual mean $r \in [0, 1]$, including M, N, P:

$$R_a[(W_L + r)^2 + 2r^2] \le [A_0 + (5/2)r^2]\mathbf{1}. \tag{Q191.15.1}$$

For $y \ge 2$, replace the $1/44$ upper by $1/11$, since $1 - 2r^2 \le 1$, and use $A_0 - 5 = 201/2048 > 1/11$. The proof for $y \le 2$ is unchanged. Thus (Q191.15.1) follows from the actual joint bounds (Q191.13.1) for every such r. Use the exact fourth expansion, the nonpositive covariances in part N.24, the actual mean bounds (Q191.11.4)–(Q191.11.5), and (Q191.12.5). Since $A_0 < 41/8$, all $L = 3, 4, 5$ and $a \ge 3$ satisfy

$$\begin{aligned} \mathbb{E}[f_a\{T^L(qf_2)\}^4] &\le \mathbb{E}[R_a W_L^4] \le R(d), \\ R(d) &= [1 - (19/16)d](1 - 2d)^4 \\ &\quad + [61/8 - 10d + 10d^2]d(1 - d)(1 - 3d/4)^2. \end{aligned} \tag{Q191.15.2}$$

Every substitution acts on a nonnegative term. The covariance is first proved nonpositive for the same R_a, W_L, then discarded. The exact deficit polynomial is

$$\begin{aligned} (1 - R(d))/d &= 25/16 - (71/16)d + (1251/128)d^2 - (523/128)d^3 \\ &\quad - (29/4)d^4 + (45/8)d^5 \\ &\ge 25/16 - (71/16)d + (4381/576)d^2 \\ &\ge 25/16 - 45369/70096 > 9/10. \end{aligned} \tag{Q191.15.3}$$

The first comparison uses $d^3 \le d^2/3$, $d^4 \le d^2/9$ and drops the positive fifth term; the second completes the quadratic over all real d. For the last rational comparison, $(53/80)70096 - 45369 =$

$85568/80 > 0$. This proves (Q191.11.1) continuously on $0 < d \le 1/3$. For the $d = 0$ boundary and equal-pair decorations, use the common restoration: $B\mathbf{1} = 0$ makes every actual path and terminal message $\mathbf{1}$, while positivity and $0 \le E_o, E_t \le 1$ restore the original pairs at their own vertices. The original edge counts, same-f_a four-terminal Hölder interface, dependent joint-label KL laws and noninjective maps use the common assembly already stated in the article.

N.25 The complete two-connector coefficient certificate

We prove the pointwise premise (L37.7.C) for the same two-connector message, with the original row constraints retained. Its vector $R = r + t_3 = K(g + s)$ has $\mathbb{E}(g + s) = 0$; centering an original row gives

$$|R| \le \sqrt{pq(d - 2\delta_2 + \delta_3)} \le \sqrt{pqd[1 - (51/32)v]}, \qquad |r| \le \sqrt{pqd}.$$

Put $x = \sqrt{d}$, $t = \sqrt{d[1 - (51/32)v]}$. Then

$$a = 1 - (35/51)x^2 - (16/51)t^2, \qquad m = 1 - (18/17)x^2 - (16/17)t^2,$$

where $m = \mu_2$. The actual parameters satisfy $(19/32)x \le t \le x$ and $t^2 \le d - (51/32)d^2 \le 8/51 < 4/25$. It suffices to work on the full closed rectangle

$$0 \le x \le 1/\sqrt{3}, \qquad 0 \le t \le 2/5, \tag{L37.8.3}$$

where $a, m > 0$. Since $\sqrt{pq} \le 1$, the actual coefficient is at most

$$(p + x)\{[a(2 - p) + t]^2 + 2m[a(2 - p) + t] + 3m^2\}.$$

Writing $y = p + x \in [x, 2 + x]$ and $B_0 = a(2 + x) + t$ makes this $H(y) = y[(B_0 - ay)^2 + 2m(B_0 - ay) + 3m^2]$. At $y_0 = 7/5$, define

$$z_0 = a(3/5 + x) + t, \quad u_0 = z_0 + m, \quad H_0 = (7/5)(u_0^2 + 2m^2),$$
$$D = H'(7/5) = u_0^2 + 2m^2 - (14/5)au_0.$$

The proof will establish $|D| < 3/10$ on the entire rectangle, and the additional independent estimates needed to combine that derivative bound.

The exact Taylor square for every allowed value of the weight. For $h_0 = y - 7/5$, the cubic expansion is exactly

$$H(y) = H_0 + Dh_0 - a[2u_0 - ay]h_0^2.$$

Since $y \le 2 + x$, the square coefficient is at least $K_0 = a[a(x - 4/5) + 2t + 2m]$. Let $b = 16/51$, $A_0 = 1 - 35x^2/51$, $M_0 = 1 - 18x^2/17$, $L = x + 26/5$, $B_1 = A_0(x - 4/5) + 2M_0$. Then

$$K_0 = (A_0 - bt^2)(B_1 + 2t - bLt^2), \quad \partial_t^2 K_0 = -2b(A_0L + B_1) - 12bt + 12b^2Lt^2 < 0.$$

Indeed $A_0 > 3/4$, $L \le 29/5$, $B_1 \le 2$, $A_0L + B_1 > 4$; with $b < 1/3$, $t \le 2/5$ the last derivative is less than $-8b + (1392/125)b^2 < 0$. Dropping positive terms from the endpoint difference and using $A_0L + B_1 \le 39/5$ gives $K_0(x, 2/5) - K_0(x, 0) > 3/5 - 52/125 - 16/375 > 0$. Thus the smaller t endpoint is zero. There the exact polynomial is

$$k(x) = 6/5 + x - (122/51)x^2 - (70/51)x^3 + (2800/2601)x^4 + (1225/2601)x^5.$$

Its second derivative is convex and negative at both endpoints: $k''(0) = -244/51$ and $k''(1/\sqrt{3}) = -1244/2601 - 39760/(7803\sqrt{3}) < 0$. So k is concave. Its endpoint minimum exceeds $13/16$: $k(0) = 6/5$, and using $1/\sqrt{3} > 1/2$ gives

$$k(1/\sqrt{3}) > (118/153)[59/153 + 518/765] = 31978/39015 > 13/16.$$

Consequently $K_0 > 13/16$ on the full rectangle, without requiring $H'' \le 0$ for all y, and

$$H(y) \le H_0 + (4/13)D^2 \le H_0 + D^2/3. \tag{L37.8.6}$$

The complete anchor-value certificate. Let $F(x,t) = H_0 + 8x^2$, $b = 16/51$, $q_0 = x + 18/5$,

$$U_0 = 8/5 + x - (25/17)x^2 - (35/51)x^3, \qquad M_0 = 1 - (18/17)x^2.$$

Then $u_0 = U_0 + t - bq_0t^2$, $m = M_0 - 3bt^2$. Concavity of U_0 and its endpoint values give $U_0 \ge 3/2$; also $M_0 \ge 11/17$ and $18/5 \le q_0 \le 21/5$. The exact second derivative is

$$\frac{F_{tt}}{14/5} = 1 - 2bq_0U_0 - 12bM_0 - 6bq_0t + b^2(6q_0^2 + 108)t^2.$$

The constant part is at most $1 - 288/85 - 704/289 < -19/4$. The remaining part is at most $6/25$: the convex polynomial $R(q) = -6bq + (2/5)b^2(6q^2 + 108)$ has endpoint values $174528/325125$ and $166752/325125$, both less than $3/5$, and is multiplied by $t \le 2/5$. Hence $F_{tt} < -12$. At $t = 1/4$,

$$F_t(x, 1/4) = (14/5)N(x),$$
$$N = -643/5780 + (128/867)x + (527/2601)x^2 - (59/867)x^3 + (280/2601)x^4.$$

The numerator of N' is $384 + 1054x - 531x^2 + 1120x^3 > 0$, using $x^2 \le 1/3$. Thus F_t increases in x. Its left endpoint is $-4501/14450 > -1/3$; at the right endpoint, $1/\sqrt{3} < 3/5$ yields $N < -1/10 + 7/100 + 3/250 + 3/40 = 57/1000$ and hence $F_t < 1/3$. So $|F_t(x, 1/4)| < 1/3$, giving $F(x,t) < F(x, 1/4) + 1/216$ by strong concavity.

Set $F_0(x) = F(x, 1/4)$. Its complete derivative is

$$F_0'' = -92647/13005 - (12929/289)x + (222796/4335)x^2 + (49000/867)x^3 + (17150/867)x^4.$$

All coefficients of $F_0^{(4)}$ are positive, so F_0'' is convex. The left endpoint is below -2; the right is

$$F_0''(1/\sqrt{3}) = 476197/39015 - 67361/(2601\sqrt{3}) < 49/4 - (129/5)(4/7) = -349/140 < -2.$$

At $2/5$, $F_0 = 115038439/14450000 < 637/80$ and $F_0' = -24517/541875 \in (-1/20, 0)$. Thus $F_0 < 637/80 + 1/1600$, and the separate strong concavity bounds give

$$F < 637/80 + 1/1600 + 1/216 = 255/32 - 43/43200.$$

For the stronger combined bound, put $n(x) = F_t(x, 1/4)$. The same numerator for N' gives $0 < n'(x) < 7/5$: discarding its nonpositive quadratic term and using $x < 3/5$ bounds it by

$$\frac{14}{5}\frac{384 + 1054(3/5) + 1120(3/5)^3}{2601} = 440412/325125 < 7/5.$$

Also $n(2/5) = -12971/216750 \in (-1/16, 0)$. For $h = x - 2/5$, $z = t - 1/4$, $s = F_0'(2/5)$ and $n_0 = n(2/5)$, the two strong concavity bounds and this Lipschitz bound give

$$\begin{aligned} F(x,t) &\le F_0(2/5) + sh + n_0 z - h^2 - 6z^2 + (7/5)|hz| \\ &< 637/80 + sh + n_0 z - h^2/2 - 5z^2. \end{aligned}$$

Here $(7/5)|hz| \le h^2/2 + (49/50)z^2$. Completing both squares yields

$$H_0 + 8d < 637/80 + 1/800 + 1/5120 = 203877/25600. \tag{L37.8.9}$$

The derivative certificate on the entire rectangle. Retain the full polynomial certificate for the same $D = u_0^2 + 2m^2 - (14/5)au_0$. Write $D(x,t) = \sum_{j=0}^{4} D_j(x)t^j$. Direct expansion gives

$$\begin{aligned} D_0 &= 2/25 + (2/5)x - (191/255)x^2 - (22/17)x^3 \\ &\quad + (179/867)x^4 + (1820/2601)x^5 + (1225/2601)x^6, \\ D_1 &= 2/5 + 2x - (52/51)x^2 - (70/51)x^3, \\ D_2 &= -2309/1275 - (128/85)x + (13952/4335)x^2 + (3296/2601)x^3 + (1120/2601)x^4, \\ D_3 &= -2b(x + 11/5), \\ D_4 &= b^2\mathcal{C}(x), \quad \mathcal{C} = x^2 + (22/5)x + 522/25. \end{aligned}$$

Indeed, with $A_0 = 1 - 35x^2/51$, $M_0 = 1 - 18x^2/17$ one has $u_0 = U_0 + t - b(x + 18/5)t^2$, giving all five coefficients. These are ordinary polynomials, including the rectangle boundary.

First,

$$D_{tt} = 2D_2 - 12b(x + 11/5)t + 12b^2\mathcal{C}(x)t^2.$$

It is convex in each coordinate separately, since $D_2'' > 0$ and $\mathcal{C}'' = 2 > 0$. Its maximum is therefore at a corner. Put $X = 1/\sqrt{3}$. The exact value

$$D_2(X) = -403991/585225 - 718624/(663255\sqrt{3}) < -9/7$$

follows because the first term is below $-2/3$, the coefficient in the second exceeds $13/12$ (the integer difference is 1173), and $X > 4/7$. At $(0,0)$, $D_{tt} = -4618/1275 < -18/5$; at $(X,0)$ it is less than $-18/7$. At $(0, 2/5)$ its three terms are less than $-18/5, -33/10, 4$, summing to $-29/10$. At $(X, 2/5)$ the first is below $-18/7$; since $X > 1/2$ the negative term has absolute value greater than $(24/5)(16/51)(27/10) = 1728/425 > 81/20$. Since $b^2 < 1/10$, $X < 3/5$, $X^2 = 1/3$, $\mathcal{C}(X) < 1/3 + 66/25 + 522/25 = 1789/75$, and the positive term is below $(48/25)(1/10)(1789/75) = 85872/18750 < 23/5$. Thus the final corner is below $-18/7 - 81/20 + 23/5 = -283/140 < -2$, proving $D_{tt} < -2$ throughout.

Second, the full other derivative is

$$\begin{aligned} D_{xx} = &-382/255 - (132/17)x + (716/289)x^2 + (36400/2601)x^3 + (36750/2601)x^4 \\ &+ [-104/51 - (140/17)x]t \\ &+ [27904/4335 + (19776/2601)x + (13440/2601)x^2]t^2 + (512/2601)t^4. \end{aligned}$$

It too is separately convex: in t directly, and in x because its quadratic and higher coefficients are nonnegative, with positive quadratic coefficient. At $(0,0)$ it equals $-382/255 < -4/5$. At $(X,0)$,

$$D_{xx} = 105072/117045 - 24188/(7803\sqrt{3}) < 9/10 - 3(4/7) = -57/70 < -4/5.$$

At $(0, 2/5)$ it is

$$-118/51 + 111616/108375 + 8192/1625625.$$

The three terms are below $-23/10$, $21/20$ and $1/100$, and their sum is less than $-4/5$. For the last corner compare to $(X, 0)$. Its linear t term is less than $-(2/5)(104/51 + 70/17) = -628/255 < -12/5$. The bracket in the quadratic t term is less than $13/2 + 24/5 + 7/4 = 261/20$, using $X < 3/5$ and $X^2 = 1/3$; after multiplication by $4/25$ it is below $21/10$. The quartic term is below $1/100$. The increment is therefore below $-12/5 + 21/10 + 1/100 = -29/100 < 0$. All corners satisfy the desired bound, proving $D_{xx} < -4/5$.

For an upper bound on D, take $x_0 = 1/4$, $t_0 = 1/5$. There $a = 6423/6800$, $m = 3047/3400$, $u_0 = 258271/136000$ and

$$D(x_0, t_0) = 3516398793/18496000000 < 191/1000,$$
$$D_t(x_0, t_0) = -7468703/86700000, \quad |D_t| < 1/10.$$

The t concavity bound gives $D(1/4, t) < 191/1000 + 1/400 = 387/2000 < 1/5$ on the full vertical line. On that line the exact coefficient derivatives are

$$D_1' = 503/408 \in (6/5, 5/4), \quad D_2' = 4784/13005 \in (0, 2/5),$$
$$D_3' = -32/51 < -3/5, \quad D_4' = 6272/13005 \in (0, 1/2),$$
$$D_0' = 13/510 - 33/136 + 179/13872 + 2275/166464 + 1225/443904 \in (-1/5, -9/50).$$

For the last interval the lower bound uses $13/510 > 1/40$, $33/136 < 1/4$ and the next two positive terms each greater than $1/80$; the upper uses $13/510 < 13/500$, $33/136 > 6/25$, and the remaining positive terms less than $13/1000, 7/500, 3/1000$. Furthermore

$$\partial_t D_x(1/4, t) = D_1' + 2D_2' t + 3D_3' t^2 + 4D_4' t^3 > 6/5 - (96/51)(4/25) = 382/425 > 0.$$

Thus its lower endpoint exceeds $-1/5$, and its upper endpoint is below

$$-9/50 + (5/4)(2/5) + (2/5)(4/25) - (3/5)(8/125) + (1/2)(16/625) = 224/625 < 3/8.$$

Hence $|D_x(1/4, t)| < 3/8$. The x strong concavity bound yields

$$D(x, t) < 1/5 + (3/8)^2/(8/5) = 1/5 + 45/512 = 737/2560 < 3/10.$$

Maximizing that quadratic on the whole real line merely enlarges the allowed interval; it requires no assumption on the maximizing point.

For the lower bound, the established separate concavity puts the minimum at one of the four corners, without asserting joint concavity. At $(0, 0)$, $D = 2/25 > 0$. At $(0, 2/5)$, $a = 1211/1275 < 19/20$, $m = 361/425 > 21/25$, $u_0 = 3866/2125 \in (9/5, 11/6)$, so

$$D > 81/25 + 2(441/625) - (14/5)(19/20)(11/6) = -1691/7500 > -3/10.$$

At $(X, 0)$, $a = 118/153 < 31/40$, $m = 11/17 > 16/25$; $4/7 < X < 7/12$ gives $8303/5355 < u_0 < 7159/4590$ and thus $31/20 < u_0 < 25/16$. Therefore

$$D > (31/20)^2 + 2(16/25)^2 - (14/5)(31/40)(25/16) = -6757/40000 > -3/10.$$

At $(X,2/5)$, $a=2758/3825<73/100$, $m=211/425>49/100$; the same bounds on X give $233093/133875<u_0<401558/229500$ and $87/50<u_0<7/4$. Thus

$$D>(87/50)^2+2(49/100)^2-(14/5)(73/100)(7/4)=-173/2500>-3/10.$$

This proves $-3/10<D<3/10$ on every point of the full closed rectangle. Finally (L37.8.6) and (L37.8.9) give

$$H(y)+8d<203877/25600+(4/13)(3/10)^2=1023/128-183/332800.$$

In particular the required coefficient $C_2(d)=1023/128-8d$ in (L37.7.C) is valid, completing the two-connector premise.

N.26 The remaining short-two unit and long-connector rows

This continuation retains exactly Q5-C073, Sections 2 and 4–8, needed in the all-connector union (L77.7.table). For the unit $(2,3)$ row put $m=1-d/2$, $R=s=K^2g$. Then $h_2+f_3=2m+s\ge 0$ and the square notation (L69.10.square) gives

$$\mathbb{E}[f_2w_0]=-m^2d-m(\delta_2-\delta_3),\quad \mathbb{E}[f_2e]\le\delta_3/2,\quad \mathbb{E}w_0^2=m^4d+m^2\delta_4,$$

$$2\mathbb{E}(w_0e)=-(m^2/2)\mathbb{E}(rs^2)-(m/2)[\delta_2\delta_3+\mathbb{E}((Kt_0)s^2)].$$

The original row bounds $|r|\le\sqrt{d}$, $|Kt_0|\le\sqrt{\delta_4}<4\sqrt{d}/15$ yield $\mathbb{E}(w_0+e)^2\le m^4d+m^2\delta_4+(19/30)m\sqrt{d}\,\delta_3+\delta_3^2/4$. On $d\le 1/4$, $x=\sqrt{d}$, the row gives $Y\le m^2q+B_0$, $B_0/m\le C=5x/12+x^2/10$. Thus the coefficient divided by m^2 is bounded by $Q=(2+x-q)\{[m(q+1)+C]^2+2m^2\}$. For $c=C/m\le(8/15)x$, direct derivatives locate its maximum in $[1/2,1]$, where $Q''\le-2m^2$. The normalized derivative at $3/4$ is $D=-11/16+(7/2)x+(2x-1)c-c^2$, with $|D|\le 1$. For the lower estimate use $c+c^2\le(8/15+32/225)x<(7/2)x$. For $x\le 5/12$ use $D\le-11/16+35/24<1$; afterward $c\ge x/2$ gives $D\le-11/16+(7/2)x-x^2/4\le 1$. The locating signs, divided by m^2, are $(1+c)(3+2x-c)-2>0$ at zero, $1/4+3x+2xc-c^2>0$ at $1/2$, $-2+4x+(-2+2x)c-c^2\le 0$ at one; Q' is concave. The tangent polynomial is

$$\Phi=(5/4+x)\{[7/4+5x/12-31x^2/40]^2+2(1-x^2/2)^2\}+(1/4)(1-x^2/2)^2.$$

Its coefficients are $421/64, 661/96, -643/144, -15397/2880, 3043/3840, 1761/1600$. Absorbing $x^4\le x^2/4$, $x^5\le x^3/4$ gives $421/64+(661/96)x-(17/4)x^2-5x^3$. It increases to $7999/960<17/2$ on $[0,2/5]$; after $2/5$, $x^3\ge(2/5)x^2$ gives an increasing quadratic with endpoint $203/24<17/2$. Hence the coefficient is at most $(17/2)m^2$. With $u_2=\delta_2/d\in[d,\theta]$, the normalized actual deficit is

$$\begin{aligned}D_3\ge{}&1/2-d/2+(67/8)d^2-(3/8)d^3\\&+u_2[327/2048-(11/4)d+(101/64)d^2-(19/64)d^3]\\&-(323/60)(\theta u_2\sqrt{d}/m)-(17/8)(\theta^2u_2^2d/m^2).\end{aligned}$$

The inverse expansion used here is $[m^{-4}-m^4(1-5d)]/d\ge 9-9d+(21/2)d^2-(3/8)d^3$. For $u_2\le\theta$, $m\ge 7/8$, discard positive terms to get $D_3\ge 1/2-(51/50)x-(17/10)x^2+(33/4)x^4$. The exact relaxations are $54587/53760<51/50$ for the square-root cost, $1/2+11\theta/4+(136/49)\theta^4<17/10$, $3/8+19\theta/64=1015/2048<1/2$, and $d^3\le d^2/4$. At $x=2/5+y$, the lower polynomial is $39/1250-(67/250)y+(311/50)y^2+(66/5)y^3+(33/4)y^4$, $-2/5\le y\le 1/10$. Use

$y^3 \ge -(2/5)y^2$ to get the decreasing quadratic with coefficient $47/50$ and endpoint $69/5000$. Since $(69/5000)(7/8)^4 > 1/125$, the unit loss is $d/125$. For $d \ge 1/4$, the actual mean $1 - 2d - d^2 \le 7/16$ gives $TF_3 \le 7/8$, proving the same row.

For longer connectors use the exact propagation (L77.1.6). Put $Z_L^* = S'_L + \delta S'_{L-1} - aS'_{L-2} + kB^L R = \nu_0 + Q_L(B)u$. Its difference from that propagated upper bound is $\sum_{j=3}^{L} \gamma_j \bar{f}_{L-j}$, with

$$\begin{aligned} \gamma_3 &= d - kD_2, \quad \gamma_4 = d + \delta d - kD_3, \\ \gamma_j &= \beta_j + \delta\beta_{j-1} - a\beta_{j-2} - kD_{j-1} \ (j \ge 5). \end{aligned} \tag{L73c.4.2}$$

For $\alpha_L = \nu_0 - \mathbb{E}Z_L^*$, actual skew integration gives $\mathbb{E}(uZ_L^*) = \alpha_{L+1}$, $\mathbb{E}[r(Z_L^* - \nu_0)] = \nu_0 d - \alpha_{L+2}$. If every $\gamma_j \ge hd$ and $L \ge 4$, retain the last two actual losses to obtain $Z = Z_L^* - hd(1+q) \ge T^L F_0/m^2 \ge 0$. If the three α terms are at least hd and $\|Q_L\| \le N_0$, then exact centering yields

$$\mu \le \nu_0 - 3hd, \quad \mathbb{E}[f_2(Z-\mu)] \le (\nu_0 - 2h)d, \quad \operatorname{Var} Z \le N_0^2 d - 3h^2 d^2 + h^2 d^3. \tag{L73c.4.4}$$

If $\|Q_{L-1}\| \le N_0$ as well, the original row is $Z \le (\nu_0 - hd)q - hd + N_0\sqrt{d}$.

For $(2,3)$ over the full $d \le 1/3$, $D_0 = D_2 = D_4 = 0$, $D_1 = -\delta_2$, $D_3 = \delta_3$, with later nonzero signs alternating. Put $a = k\delta_2$, $\delta = k^2\delta_3/2$, $\eta = a - \delta$. Then $\nu_0 = 1 - \eta$, $\delta/a \le 39/160$, $0 \le \eta \le 39d/80 \le 13/80 < 1/6$, $\delta \le \eta/3$. The exact finite polynomial is $Q_n = G_n - \eta G_{n-2} + \delta z^{n-2} - kz^{n+2}$ for $n \ge 3$. For $n = 3$, $P_3 = G_3 - kz^5$ has squared norm at most $(1-\xi)^2 + \xi \le 1$. Q_3 has coordinates $1 - \xi - \eta$, $1 + \delta - k\xi^2$; its square difference is at most $\eta[-2 + (8/3)\xi + \eta(1 + \xi/9)] < 0$, bracket at most $-53/108$. For $n = 4$ the positive coordinates are bounded by those of $P_4 = G_4 - kz^6$, whose norm is at most $(1-\xi)(1+\xi/2) + k\xi^3 \le 1$ since $k\xi \le 39/80 < 1/2$. The coordinate positivity bounds are $11/36$ and $1/3$. For every finite $n \ge 5$,

$$Q_n = \frac{1 - \eta + z^{n-2}[a - \delta z - z^2 - kz^4 + kz^5]}{1 - z}.$$

The last three terms of the bracket have norm at most ξ because $(1 - k\xi)^2 + k^2\xi^3 \le 1$, implied by $k\xi(1+\xi) \le 2$. Also $a + \delta\sqrt{\xi} \le 19\eta/12$, $(19/12)\xi^{3/2} < 1$. The numerator norm is at most $1 + \xi^{5/2}$ and $2\xi^{3/2} + \xi^4 \le 13/16 < 1$ gives $\|Q_n\| \le 1$. The losses obey $\gamma_3 = d$, $\gamma_4 \ge 4d/5$; positive D_{j-1} for $j \ge 5$ first occurs at index seven and is at most $\delta_5 \le \theta^4 d$. All losses are therefore at least hd, $h = 9/16 - d/2$. After multiplying the comparison by m, its difference is $1/32 - \theta^4 + (5/64)d - d^2/4$, concave with positive endpoints $4207/1048576$ and $4207/1048576 - 1/576$. For $L \ge 4$, $\alpha_L = \gamma_{L+1} \ge hd$ and (L73c.4.4) applies with $N_0 = 1$, $\nu_0 \le 1$. The full-domain coefficient ten follows whenever $Z \le (1 - cd)q - cd + \sqrt{d}$, $\mu \le 1 - 3cd$, $c = 19/48$. Its quadratic-loss polynomial is $A(q) = (5111/6912)q^2 + (3059/864)q + 17767/3456$. It dominates $(2 - q^2/2)(q+1)^2$: a sufficient difference is $q^4/2 + q^3 - (4/5)q^2 - q/2 + 3$, at least $17/10$ on $[0,1]$ by dropping positive terms, and positive on $[1,2]$ by $q^3 \ge q^2$. Completing the square adds at most one to the base nine. Since $h \ge 19/48$, the actual mean upper bound is $v_0 = 1 - 27d/16 + 3d^2/2$, covariance at most $-d/8 + d^2$, variance at most $d - 3h^2d^2 + h^2d^3$. The upper mean endpoint is positive: $v_0 \ge 29/48$, $(1-d)v_0 \ge 29/72$, $4d^2 - d/2 \ge -1/64$. For $a_0 = 27d/16 - 3d^2/2 < 2/5$ use $v_0^4 \le 1 - 4a_0 + 6a_0^2$ and $(4d^2 - d/2)v_0^3 \le (4d^2 - d/2)(1 - 3a_0) + 12d^2a_0^2$. The full polynomial before m^8 is at most $1 + (7/4)d + (215/8)d^2 - (3579/64)d^3 + (5307/64)d^4 - (287/4)d^5 + 27d^6$. The last two terms sum nonpositively; inverse Taylor gives gap $d[9/4 - (143/8)d + (1605/32)d^2] \ge (8441/12840)d$. With

$m^8 > 1/5$ this proves deficit $d/8$ for $L \geq 4$. For $L = 3$ take the actual $Z = Z_3^* - d$. $\alpha_3 \geq 4d/5$ gives $\mu \leq 1-9d/5$, covariance at most $-d/8+d^2/5 \leq -d/20$, variance at most $d-16d^2/25$. $Q_2 = 1+\delta+z-kz^4$ has square at most $(1+\delta)^2+\xi$; $\delta \leq 3d/25 \leq 1/25$ gives $\|Q_2\| < 5/4$ and $Z \leq q-d+(5/4)\sqrt{d}$. The coefficient ten follows by square completion with loss $A(q) = (28/5)(q+1)+1037/1200$, using $(2-q^2/2)(q+1) \leq 25/8$ and $28/5 > 625/128$. The upper mean bracket is at least $1/5$ and the full polynomial is $1+(8/5)d+(533/25)d^2-(5589/125)d^3+(4374/125)d^4-(6561/625)d^5$. Its inverse gap is at least $d[12/5-13d+54d^2] > (8/5)d$, so the actual loss exceeds $d/4$.

For $(2,4), L \geq 4$, on $d \leq 1/4$ use $m = 1-d$ and the actual $R = s+t_0-dg$ of Q5-C069. The exact Q4-C128, Section 4, finite losses retained in Section I.3 give $|\varepsilon| \leq d/6$, $\delta \leq 25d/24$, $\gamma_j \geq hd$, $h = 1/2-6d/5$. Its finite P_n contraction gives $\|Q_L\|, \|Q_{L-1}\| \leq 1+d$. The shortest Q_3 cost is $1/6+(13/20)(25/24) = 27/32 < 1$; Q_4 uses $\|G_2\| \leq 6/5$, and later tails are smaller. All three actual α terms are at least hd. Thus $\mu \leq 1-4d/3+18d^2/5$, covariance at most $3d^2$, variance at most $(1+d)^2d-3h^2d^2+h^2d^3$. The row and mean sum is at most $q+B(x)$, $B(x) = 1+x-(11/6)x^2+x^3+(24/5)x^4 \leq 22/15 < 3/2$. It increases: $B' \geq 1-11/12 > 0$ up to $1/4$, and then $B' \geq 1-(11/3)x+(39/5)x^2 > 0$. With $f_2 \leq 5/2-q$, $\mu \leq 1$, the curvature cubic has $Q'' \leq -1$, $Q(9/10) = 1552/125$, $Q'(9/10) = -2/25$, so its maximum is at most $7762/625 < 13$. Using $a_0 = (4/3)d-(18/5)d^2 \geq 0$ and $(1-d)(1-a_0)^4+12d^2(1-a_0)^3 \leq 1-d-4a_0+4da_0+6a_0^2+12d^2$, the full bound is $P \leq 1+(20/3)d+(1173/20)d^2-(179/20)d^3+6d^4+(468/25)d^5$. The inverse gap is at least $d[4/3-(453/20)d+(2579/20)d^2] > d/3$, since the completion loss $205209/206320 < 1$. $m^8 > 1/10$ gives low loss $d/30$.

For every $b \geq 5, L \geq 4$, $d \leq 1/5$, the finite vector proof (L77.4.1) applies. It gives $|\varepsilon| \leq 3d/5$, $\delta \leq 6d/5$, all $\gamma_j \geq hd$, $h = 1/2-6d/5$. Every positive D_j, $j \geq 4$, is at most $\theta^2 z_0 d$ because $\theta u_0 \leq z_0$. For $j \geq 5$, compare $\kappa-d-k\theta^2 z_0$ with h; after multiplying by m its decreasing difference is $875/32768-(3241/81920)d-(9/40)d^2$, endpoint $40101/4096000 > 0$. The first two losses are stronger. The finite P_n contraction holds for all $n \geq 3$ using $ku_0 \leq 1$: for $n = 3$ coordinates are nonnegative and bounded by G_3; for $n = 4$ use $(1-\xi)(1+\xi/2)+\xi^2 \leq 1$; for $n = 5$ use $|G_5| \leq 1-\xi/2+\xi^2/2$ and $\xi^{5/2} \leq 13\xi^2/20$, whose base square difference is $\xi^2(1+2\xi-3\xi^2)/4 \geq 0$; for $n \geq 6$ use $(1+\xi^3)/\sqrt{1+\xi}+\xi^3 \leq 1$, following from $\xi(1+\sqrt{1+\xi}) \leq 143/160 < 1$. Thus $\|Q_L\|, \|Q_{L-1}\| \leq 1+7d/5$; the shortest cost is $3/5+(13/20)(6/5) = 69/50 < 7/5$, and Q_4 uses the true G_2 bound. The actual mean is at most $v_0 = 1-9d/10+18d^2/5 \leq 1$ and covariance at most $3d^2$. Its row is $Z \leq aq+b$, $a \leq 267/250 < 107/100$, $b \leq 21/40$; at $d = 1/5$ the increasing latter bound is below $(32/25)(9/20)-13/250 = 131/250 < 21/40$. The curvature cubic $(49/20-q)[((107/100)q+61/40)^2+2]$ has $Q'' \leq -91699/100000 < -9/10$, $Q(9/10) = 3967101/312500 < 127/10$ and $|Q'(9/10)| = 7819/125000 < 1/10$, so its maximum is below $127/10+1/180 < 13$. The mean part is at most $1-(23/5)d+(1743/50)d^2-(1332/25)d^3+(1944/25)d^4$; variance at most $d+(41/20)d^2+(581/100)d^3-(138/25)d^4+(36/25)d^5$. The full bound is therefore $P \leq 1+(42/5)d+(6151/100)d^2+(89/4)d^3+6d^4+(468/25)d^5$. Inverse Taylor gives gap $d[3/5-16d+147d^2] \geq (121/735)d$; $m^8 > 1/8$ and $121/5880 > 1/50$ prove low loss $d/50$.

For the high range, the precise original path inputs are $\mathbb{E}[\bar{f}_3^4/q] \leq 1-23d/36$ (L56.3.15) and $\mathbb{E}\bar{f}_4^4 \leq 1-d-4d^2$ (Q64.3.12). On positive support $q(x) \geq \mu(x) > 0$, so inverse weights are well defined. Since $f_2 q \leq pq(1+q/2) = 2q-q^3/2 \leq 8/(3\sqrt{3}) < 8/5$, the weighted bare moments are at most $(8/5)(1-23d/36)$ and $2(1-d-4d^2)$, respectively. The all-integer path means proved in

Q5-C090 give $\bar{f}_n \le 2\mu_{n-1} \le 2-5d \le 1$ for $n \ge 5, d \ge 1/5$. For $b \ge 5$, use the actual terminal-mean convolution $t_b = 1 - \beta_{b+1} - \beta_{b+2} + \beta_{b+3} - \sum_{j=3}^{b} \beta_j t_{b-j}$ with short values (L77.7.1). $t_5 \le 79/200$ for $d \ge 1/5$; $t_6 \le 1 - (121/32)d + (45/16)d^2 + d^3$ decreases and is at most $1457/4000 < 91/200$ at $1/5$. For every $b \ge 7$ retain five actual terms $t_0, \dots, t_4$, $t_4 \ge 1 - 3d \ge 0$, all beta indices at least three, to get $t_b \le 1 - d(3479 - 3287d - 608d^2)/1024$. The loss increases with derivative numerator at least 1085; at $1/5$ it is $17483/32000$. Thus all $b \ge 5$ have $t_b \le 91/200$, $TF_b \le 91/100$. For $L = 4$, the target is below $(91/100)^4(8/5)(7/8) < 35/36$, using $(91/100)^2 < 5/6$. For $L = 5$, the weighted fourth-path bound $32/25$ gives target below $8/9$; for $L \ge 6$, the actual path is at most one, giving target at most $1 - d \le 4/5$. All are stronger than $1 - d/50$. For $b = 4, d \ge 1/4$, its mean gives $TF_4 \le 4/5$. For $L = 4$ use weighted three-step bound $8/5$, for $L = 5$ use fourth-step bound two, and for $L \ge 6$ the pointwise path bound; all targets are at most $512/625$. Together with its low proof this gives loss $d/40$ for $b = 4$. Combining all finite b proves the common loss $d/50$ for every $b \ge 3, L \ge 4$.

N.27 The unit connector with terminal arms of lengths two and three

We use the auxiliary coordinates $B = -K$ and $u = B\mathbf{1} = -g$. Thus $T = J + B = A^*$, $A = J - B$, $p = 1-u$, $q = 1+u$, and $\bar{f}_n = T^n\mathbf{1}$, $h_n = Tf_{n-1}$, $\delta_i = \|B^i\mathbf{1}\|_2^2$, $d = \delta_1$, $\theta = 13/32$, and $\tau = d - \delta_2$. All expectations and adjoints use the original measure. The two residual terminal arms have lengths $(2, 3)$ and messages h_2, f_3; the unit connector points from the terminal to the receiving root and has output $X = T(h_2 f_3)$. The receiving arm has its actual outward message f_a, where $a \ge 3$. These directions depend only on the original arm lengths. We prove

$$\mathbb{E}[f_a X^4] \le 1 - d/100. \tag{R07.0.1}$$

The same conclusion holds with $\mathbb{E}[E_o f_a \{T(E_t h_2 f_3)\}^4]$ on the left, where E_o, E_t are any actual equal-arm decorations.

The finite calculus of Q3-C014 gives $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{i+1} \le \theta\delta_i$, $\|B\|^2 \le \theta$, $|BV(x)| \le \|V\|_2$, $f_n \le 2$, and $\mathbb{E}f_n \le 1 - 2d$ for $n \ge 3$. The operators A, T are positive. Equal-arm pairs lie in $[0, 1]$, and $p, q > 0$ on the original positive support. Put $G_n(z) = \sum_{j=0}^{n-1} z^j$, $\beta_1 = \beta_2 = 0$, and $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j$. The exact positive convolution (Q117.1.1) and the receiving uppers (Q117.1.2) and (Q187.T198.1) give the receiving bounds

$$\begin{gathered} S_a = 1 - G_a(-B)u, \qquad C_3 = f_3 = S_3 - d, \\ C_a = S_a - \beta_a - \beta_{a-1}p \quad (a \ge 4), \qquad 0 \le f_a \le C_a, \qquad \tau \le \beta_j \le d \quad (j \ge 3), \\ C_a \le (1-d)p + \tfrac{6}{5}\sqrt{pqd} \le (1-d)p + \tfrac{6}{5}\sqrt{d} \quad (a \ge 3). \end{gathered} \tag{R07.1.1}$$

For $a = 3$ the last bound uses the actual variance $d + \delta_2 - d^2 \le 36d/25$ of $u - Bu$. For $a \ge 4$ it retains two convolution terms, uses $\beta_a + \beta_{a-1} \ge 2\tau \ge d$, and $\|G_{a-1}(-B)u\|_2 \le \sqrt{d}$. On the original spectral measure ν of $-B^2$ at u, write $z = i\sqrt{x}$, $0 \le x \le \theta$, and $\int x^j \, d\nu = \delta_{j+1}$. If $G_a = E_a + zD_a$, then $1 - x \le E_a, D_a \le 1$ for $a \ge 3$.

Put $m = 1 - d/2$ and $s = -B^2u$. Since $h_2 + f_3 = 2m + s$,

$$0 \le X \le T(m + s/2)^2 \le Y := m^2 q - mB^3u + \delta_3/2. \tag{R07.2.1}$$

Here $\mathbb{E}s = 0$, $Ts = -B^3u$, and $T(s^2) \le 2\delta_3$; in particular, $Y \ge 0$. With $w = Y - m^2$ and $c = \delta_3/2$ the exact moments are

$$\mathbb{E}w = -m\delta_2 + c, \qquad \mathbb{E}w^2 = m^4 d + m^2\delta_4 - m\delta_2\delta_3 + \delta_3^2/4. \tag{R07.2.2}$$

The mixed product of u and B^3u vanishes; the negative constant–mean cross term in (R07.2.2) is retained. We use the identity

$$Y^4 = m^8 + 4m^6 w + w^2(Y^2 + 2m^2 Y + 3m^4). \tag{R07.2.3}$$

For $0 \le d \le 1/4$ the following continuous pointwise bound holds:

$$C_a(Y^2 + 2m^2 Y + 3m^4) \le 8m^2 \quad (a \ge 3). \tag{R07.3.1}$$

Here is its full certificate. Set $t = \sqrt{d} \in [0, 1/2]$, $\alpha = 1 - t^2$, $c_0 = 6t/5$, and $C = 5t/12 + t^2/10$. The row bound and $\delta_3 \le \theta^2 d$ give $Y \le m^2 q + mC$: indeed $\theta < 5/12$, $\theta^2/(2m) \le 4\theta^2/7 < 1/10$, and $m \ge 7/8$. After division by m^2 it suffices to bound

$$Q(q,t) = [\alpha(2-q) + c_0]\{[m(q+1) + C]^2 + 2m^2\} \le 8, \qquad 0 \le q \le 2.$$

Let $h = C/m \le 4/15$. The derivative Q_q is a concave quadratic, and

$$\frac{Q_q(0)}{m^2} = \alpha(1 + 2h - h^2) + 2c_0(1+h) > 0, \quad \frac{Q_q(1/2)}{m^2} = \alpha(1/4 - h^2) + 2c_0(3/2 + h) > 0.$$

Thus Q increases on $[0, 1/2]$. On $q \ge 1/2$,

$$Q_{qq} = 2m^2(c_0 - 3\alpha q) - 4\alpha m C \le -5/4.$$

For $t \le 2/5$ the first term is at most $-41262/31250 < -5/4$. For $t \ge 2/5$, use $m^2 \ge 49/64$, $c_0 - 3\alpha/2 \le -21/40$, $\alpha m \ge 21/32$, and $C \ge 137/750$ to obtain $-Q_{qq} \ge 1029/1280 + 2877/6000 > 5/4$. The same formula gives $Q_{qq} \le -\lambda$ with

t upper bound	$1/8$	$1/5$	$1/4$	$1/3$	$1/2$
λ	$5/2$	$9/4$	2	$3/2$	$5/4$

by inserting the respective bounds for c_0, α, m and retaining the nonpositive final term. Direct multiplication gives

$$\begin{aligned} Q(1,t) &= 6 + \tfrac{133}{15}t - \tfrac{6787}{720}t^2 - \tfrac{5357}{600}t^3 + \tfrac{21011}{3600}t^4 + \tfrac{1161}{500}t^5 - \tfrac{131}{100}t^6 \\ &\le F(t) := 6 + \tfrac{133}{15}t - \tfrac{159}{20}t^2 - \tfrac{25}{3}t^3, \\ D(t) := Q_q(1,t) &= -2 + \tfrac{119}{30}t + \tfrac{3331}{720}t^2 - \tfrac{509}{150}t^3 - \tfrac{9131}{3600}t^4 + \tfrac{56}{75}t^5 + \tfrac{41}{100}t^6. \end{aligned}$$

The first comparison uses $t^4 \le t^2/4$, $t^5 \le t^3/4$. F increases on $[0, 1/3]$, $F'(1/3) = 71/90$, and $F'' \le -159/10$, so $F \le 12577/1620 + (71/90)^2(5/159) < 39/5$ on the full interval. Discarding positive derivative terms and bounding the two negative ones at $t = 1/2$ gives $D' \ge 1105/7200 > 0$. Exact endpoint substitution gives $D(0) = -2$, $D(1/8) > -3/2$, $D(1/5) > -16/15$, $D(1/4) > -4/5$, $D(1/3) > -1/3$, and $D(1/2) < 3/5$. The strong concavity bound $Q \le Q(1,t) + D(t)^2/(2\lambda)$ therefore has the following certificates on five closed intervals:

t	$\lvert D\rvert$	λ	upper bound, strictly below 8
$[0, 1/8]$	2	$5/2$	$F(1/8) + 4/5$
$[1/8, 1/5]$	$3/2$	$9/4$	$F(1/5) + 1/2$
$[1/5, 1/4]$	$16/15$	2	$F(1/4) + 64/225$
$[1/4, 1/3]$	$4/5$	$3/2$	$F(1/3) + 16/75$
$[1/3, 1/2]$	$3/5$	$5/4$	$39/5 + 18/125$

These rational comparisons, including $12577/1620 + 16/75 < 8$ in the fourth row, prove (R07.3.1) throughout its domain.

For $a = 3$ and $0 < d \le 1/4$, put $v = \delta_2/d$, $w_3 = \delta_3/d$, $w_4 = \delta_4/d$. Then $d \le v \le \theta$, $v^2 \le w_3 \le \theta v$, $w_4 \le \theta w_3$, and the actual receiving moments are

$$\begin{aligned} \mathbb{E}C_3 &= 1 - 2d, \\ \mathbb{E}[C_3(m^2u - mB^3u)] &= -m^2(d - \delta_2) - m(1-d)\delta_2 + m\delta_3, \\ \mathbb{E}[C_3 w]/d &= -m^2 + (md/2)v + \tfrac{3}{2}(1-d)w_3. \end{aligned}$$

Equations (R07.2.2)–(R07.3.1) bound $\mathbb{E}[C_3 Y^4]$ by

$$m^8(1-6d) + 8m^6 d + 2m^7 v d^2 + 6m^6(1-d)w_3 d + 8m^4 w_4 d - 8m^3 v w_3 d^2 + 2m^2 w_3^2 d^2.$$

Writing its normalized deficit as $D_3 = (1 - \text{bound})/(m^4 d)$, the positive Taylor expansion gives

$$\frac{m^{-4} - m^4(1-6d)}{d} - 8m^2 \ge 2 - 3d + 10d^2 - \tfrac{7}{8}d^3.$$

Use the spectral bounds above and $m^2 \ge 49/64$, and discard the positive term $8vw_3 d/m$. This gives

$$\begin{aligned} D_3 &\ge 2 - 3d + 10d^2 - \tfrac{7}{8}d^3 - 2\theta d - [6m^2(1-d) + 8\theta]\theta^2 - \tfrac{128}{49}\theta^4 d \\ &\ge \tfrac{15}{32} - 2d + \tfrac{17}{2}d^2 \ge \tfrac{191}{544} > 1/3. \end{aligned}$$

For the intermediate comparison the constant is $2 - 6\theta^2 - 8\theta^3 = 1939/4096 > 15/32$, the linear coefficient is greater than -2, and, after absorbing the negative cubic using $d \le 1/4$, the quadratic coefficient is $313/32 - (57/8)\theta^2 > 17/2$. Completing the square proves the last inequality. Hence the actual deficit exceeds $m^4 d/3 > d/6$.

For $a \ge 4$ work on $0 < d \le 1/5$, set $\beta = \beta_{a-1}$, $e_a = \int E_a \, d\nu$, $l_a = \int x^2 D_a \, d\nu$. The exact finite receiving data are

$$\begin{aligned} \mathbb{E}C_a &= 1 - \beta_a - \beta_{a+1} - \beta, \qquad & \mathbb{E}[C_a u] &= -e_a + \beta d, \\ \mathbb{E}[C_a B^3 u] &= (1 - \beta_a - \beta)\delta_2 - l_a, \qquad & \tau \le e_a &\le d, \quad 0 \le l_a \le \delta_3. \end{aligned} \tag{R07.5.1}$$

In $m^8 \mathbb{E}C_a + 4m^6 \mathbb{E}[C_a w]$ the coefficients of β_a and β_{a+1} are negative: for the former use $m^2 + 2\delta_3 - 4m\delta_2 \ge m(m - 4\theta d) > 0$. The coefficient of β is $m^6 J$, where $J = -m^2(1-4d) - 2\delta_3 + 4m\delta_2$. Lowering all three betas and e_a to τ and raising l_a to δ_3 costs at most $m^6 \delta_2 J_+$ beyond the resulting bound M_0. Its normalized deficit satisfies

$$\begin{aligned} D_0 := \frac{1 - M_0}{m^4 d} &\ge B_0(d) - 3m^4 v - 2m^3 v d(1 + d - 4v) - [m^2(6 - 8d) + 8\theta]w_3 \\ &\quad + v w_3 d(8/m - 6m^2) - 2w_3^2 d/m^2, \\ B_0(d) := \frac{m^{-4} - m^4(1 - 7d + 4d^2)}{d} &- 8m^2 \\ &\ge 3 - 9d + \tfrac{39}{2}d^2 - \tfrac{59}{8}d^3 + \tfrac{67}{16}d^4. \end{aligned}$$

The right side decreases in w_3: its negative main coefficient is at least 8θ, whereas its positive correction is at most $vd(8/m) \le 13/18$. Insert $w_3 = \theta v$. The resulting expression is $B_0 - Lv + dQv^2$, with

$$L = 3m^4 + 2m^3 d(1+d) + \theta[m^2(6 - 8d) + 8\theta], \quad Q = 8m^3 + \theta(8/m - 6m^2) - 2\theta^2/m^2 < 12.$$

Its v derivative is at most $-3m^4+24d\theta<0$, because $3m^4\ge 3(9/10)^4>39/20\ge 24d\theta$. Thus $v=\theta$ gives

$$\begin{aligned}D_0 &\ge B_0-3\theta m^4-2\theta m^3 d(1+d-4\theta)-\theta^2[m^2(6-8d)+8\theta]\\ &\quad+\theta^3 d(8/m-6m^2)-2\theta^4 d/m^2\\ &\ge P(d):=\tfrac{1043}{4096}-\tfrac{11}{3}d+\tfrac{151}{10}d^2-\tfrac{49}{10}d^3.\end{aligned}$$

Indeed $m^{-2}\le 1+6d/5$ and $8/m-6m^2\ge 2+10d+d^2/2+d^3$; terms of degree at least four are nonnegative, while the first four coefficients are bounded by

$$\begin{gathered}c_0=1043/4096,\quad c_1=-9+\tfrac{29}{4}\theta+14\theta^2+2\theta^3-2\theta^4\ge -11/3,\\ c_2=\tfrac{39}{2}-\tfrac{67}{8}\theta-\tfrac{19}{2}\theta^2+10\theta^3-\tfrac{12}{5}\theta^4\ge 151/10,\\ c_3=-\tfrac{59}{8}+\tfrac{87}{16}\theta+2\theta^2+\theta^3/2\ge -49/10.\end{gathered}$$

For example $c_1+11/3=3373/1572864$ and $c_2-151/10=9145/262144$. The positive-beta cost is controlled using $\delta_3\ge\delta_2^2/d$:

$$J\le -m^2(1-4d)+(4mv-2v^2)d\le J_*(d):=-1+\tfrac{3223}{512}d-\tfrac{81}{16}d^2+d^3.$$

The second comparison uses $m>v$. The polynomial J_* increases on $[0,1/5]$, $J_*(9/50)<0$, and $J_*(1/5)=4127/64000<1/15$. For $d\le 9/50$ the cost vanishes, and

$$P(d)\ge 1043/4096-(11/3)d+(7109/500)d^2>1/64.$$

Completing the square reduces this to $1043/4096-15125/63981>1/64$, whose cross-multiplied positive difference is 685399. On $[9/50,1/5]$, P increases and $P(9/50)>1/20$, whereas the normalized extra cost is $m^2vJ_+<\theta/15<1/36$. The remaining deficit is at least $1/20-1/36=1/45>1/64$. Consequently

$$\mathbb{E}[C_aY^4]\le 1-(m^4/64)d\le 1-d/100.$$

For the remaining range use the actual terminal, whose exact mean is $\mathbb{E}[h_2f_3]=1-2d-d^2$. This follows by multiplication, or by subtracting the means of the squares in

$$h_2f_3=(m+s/2)^2-(-u+Bu-B^2u/2-d/2)^2;$$

the two square means are $m^2+\delta_3/4$ and $d+\delta_3/4+5d^2/4$. Positivity gives $X\le 2(1-2d-d^2)$. For $1/5\le d\le 1/3$ both nonnegative factors decrease, so

$$\mathbb{E}[f_aX^4]\le(1-2d)[2(1-2d-d^2)]^4\le\tfrac{3}{5}(28/25)^4=1843968/1953125<19/20\le 1-d/100.$$

This overlaps both low ranges and proves (R07.0.1). Decorations follow from $0\le E_o,E_t\le 1$ and positivity of T. At $d=0$, $u=0$ and successive leaf integration gives every original tree message equal to one; the kernel need not vanish. All original arms, decorations, and the unit connector retain their original edges. Full reversal changes the receiving arm, terminal arms, connector, and decorations together. Combining this unit row with (N.7) at $L\ge 2$ gives, for terminal $(2,3)$, every $L\ge 1$ with deficit $d/400$. Terminals $(2,t)$ with $t\ge 4$ require a separate estimate.

N.28 The remaining unit connectors with a terminal arm of length two

Use the coordinates and receiving bounds of R2-C007. The receiving arm has message f_a, $a \ge 3$. The terminal residual lengths are $(2,b)$, $b \ge 4$, with actual message $F = h_2 f_b$, and the unit connector has message $X = TF$. We prove

$$\mathbb{E}[f_a X^4] \le \begin{cases} 1 - d/200, & b = 4, \\ 1 - d/20, & b = 5, \\ 1 - d/10, & b \ge 6. \end{cases} \tag{R08.0.1}$$

Equal-arm decorations may be restored inside T and beside f_a.

In addition to (R07.1.1), the receiving bounds satisfy $\mathbb{E}C_a \le 1 - 57d/32$: for $a = 3$ use $1 - 2d$, and for $a \ge 4$ use $1 - 3\tau$, where $3\tau \ge 57d/32$. For each terminal use the actual square bound $F \le (m + R/2)^2$, $m > 0$, described below. On the original spectral measure write $R = (A_0 - zB_0)(B)u$, $z = i\sqrt{x}$, with $A_0, B_0 \ge 0$; these are finite scalar spectral coordinates. Put

$$V = \mathbb{E}R^2, \quad D_0 = \mathbb{E}R = \int B_0 \, d\nu, \quad a_1 = \mathbb{E}(KR) = \int A_0 \, d\nu,$$
$$\rho_0 = \int x B_0 \, d\nu, \quad I = \int (1 - x)(B_0 - A_0) \, d\nu.$$

Positivity and $T(R^2) \le 2V$ give

$$\begin{aligned} 0 \le X \le Y &:= m^2 q + mBR + mD_0 + V/2, \\ w_0 = m^2 u + mBR + mD_0, &\quad w = Y - m^2 = w_0 + V/2, \\ \mathbb{E}w_0 = m(D_0 - a_1), &\qquad \mathbb{E}w^2 = \mathbb{E}w_0^2 + mV(D_0 - a_1) + V^2/4. \end{aligned} \tag{R08.1.2}$$

The identity (R07.2.3) and the pointwise bound proved below yield

$$\mathbb{E}[f_a X^4] \le m^8 \mathbb{E}C_a + 4m^6 \mathbb{E}[C_a w] + 8m^2 \mathbb{E}w^2. \tag{R08.1.4}$$

For $G_a = E_a + zD_a$, $\beta = \beta_{a-1}$, $e_a = \int E_a \, d\nu$, let $I_a = D_0 - a_1 + \int x(D_a A_0 - E_a B_0) \, d\nu$. Since $1 - x \le E_a, D_a \le 1$,

$$\begin{aligned} I_a &\le I + \int x^2 B_0 \, d\nu, \\ \mathbb{E}[C_a w_0] &= -m^2 e_a + mI_a - m(\beta_a + \beta)(D_0 - a_1) - mD_0\beta_{a+1} + \beta(m^2 d + m\rho_0) \quad (a \ge 4), \\ \mathbb{E}[C_3 w_0] &= -m^2\tau + mI_3 - md(2D_0 - a_1), \qquad I_3 = I + \int x^2 B_0 \, d\nu. \end{aligned} \tag{R08.1.5}$$

Indeed $C_a = (1 - \beta_a - \beta) + (-E_a + zD_a)u + \beta u$ and $m^2 u + mBR$ has coordinates $m^2 + mxB_0 + zmA_0$. For $a = 3$ the constant is $1 - d$ and the coordinates are $-1 + x + z$. In particular, the term $-mD_0\beta_{a+1}$ is part of the actual covariance.

We give the common continuous certificate. Set $t = \sqrt{d} \in [0, 1/2]$, $\alpha = 1 - t^2$, $c = 6t/5$, and $C_0 = 3t/8 + 109t^2/64$. Then

$$Q(q,t) := [\alpha(2 - q) + c]\{[\alpha(q + 1) + C_0]^2 + 2\alpha^2\} \le 39/5 \quad (0 \le q \le 2). \tag{R08.2.1}$$

For $h = C_0/\alpha \le 157t/96 < 1$, $h^2 < 3t^2$, the derivatives at $q = 0, 1/2$, divided by α^2, are respectively $\alpha(1 + 2h - h^2) + 2c(1 + h)$ and $\alpha(1/4 - h^2) + 2c(3/2 + h)$. Both are positive, the latter being at least $\alpha/4 - 3t^2 + 18t/5 > 0$. Since Q_q is a concave quadratic, Q increases on $[0, 1/2]$. For $q \ge 1/2$,

$$Q_{qq} = 2\alpha^2[c - 3\alpha q - 2C_0] \le 2\alpha^2[-3/2 + (9/20)t - (61/32)t^2] \le -(47/16)\alpha^2;$$

the last comparison completes the square, using $81/3050 < 1/32$. The exact tangent polynomials are

$$P(t) := Q(1, t) = 6 + \tfrac{87}{10}t - \tfrac{2959}{320}t^2 - \tfrac{9957}{1280}t^3 + \tfrac{140653}{20480}t^4 + \tfrac{27939}{10240}t^5 - \tfrac{8553}{4096}t^6,$$
$$D(t) := Q_q(1, t) = -2 + \tfrac{81}{20}t + \tfrac{1073}{320}t^2 - \tfrac{6771}{1280}t^3 - \tfrac{58317}{20480}t^4 + \tfrac{1587}{1280}t^5 + \tfrac{6121}{4096}t^6.$$

Using $t^2 \le t/2$, $t^3 \le t/4$ to absorb the two positive terms in P'', and dropping its negative quartic, gives $P'' \le -147517/10240 < -14$. Also $P'(1/3) > 0$, $P(3/8) < 771/100$, and $0 < P'(3/8) < 1/8$; hence $P \le 771/100 + 1/1792 < 7711/1000$. For D', retaining the positive linear term and using the same absorptions for its negative quadratic and cubic gives $D' > 2$. Exact substitution gives

$$D(1/8) > -3/2, \quad D(1/5) > -10/9, \quad D(1/4) > -9/10,$$
$$D(3/10) > -2/3, \quad D(1/3) > -51/100, \quad D(2/5) > -1/4, \quad D(1/2) < 1/10.$$

The bound $Q \le P + D^2/(2\lambda)$ has the following complete interval certificates; all entries in the last column are below $39/5$:

t	$\lvert D\rvert$	λ	upper bound
$[0, 1/8]$	2	$11/4$	$7 + 8/11$
$[1/8, 1/5]$	$3/2$	$8/3$	$147/20 + 27/64$
$[1/5, 1/4]$	$10/9$	$5/2$	$151/20 + 20/81$
$[1/4, 3/10]$	$9/10$	$12/5$	$763/100 + 27/160$
$[3/10, 1/3]$	$2/3$	$9/4$	$77/10 + 8/81$
$[1/3, 2/5]$	$51/100$	2	$7711/1000 + 2601/40000$
$[2/5, 1/2]$	$1/4$	$3/2$	$7711/1000 + 1/48$

The first five rows use the corresponding rational endpoint bounds $P(1/8) < 7$, $P(1/5) < 147/20$, $P(1/4) < 151/20$, $P(3/10) < 763/100$, and $P(1/3) < 77/10$.

For $b = 5$, equations (L69.11.2)–(L69.11.6), and for $b \ge 6$, equations (L73.3.low)–(L73.3.5mean), give, for the same finite R, $(Y + m^2)/m \le \alpha(q+1) + C_0$ and $m \le \alpha$. In the latter case $m = 1 - 9d/8$; the original row bound contributes $3t/8 + 69d/40$, and $69/40 - 109/64 = 7/320 < 1/8$. These bounds use only V, D_0 and $e \le V/2$, and therefore apply to the constant upper bound Y in (R08.1.2). For $b = 4$ the bound is $(Y + m^2)/m \le \alpha(q + 1) + C_4$, $C_4 = t/2 + 21t^2/16$, $m = \alpha$. The positive part of $C_4 - C_0 = t/8 - 25t^2/64$ is at most $1/100$ and occurs only for $t < 8/25$. There, for $y = \alpha(q+1) + C_0$, $C_{\text{rec}} = \alpha(2 - q) + 6t/5$,

$$C_{\text{rec}} + y = 3 + \tfrac{63}{40}t - \tfrac{83}{64}t^2 \le 27/8, \quad C_{\text{rec}}y \le 729/256,$$
$$C_{\text{rec}} \le 109/50.$$

The increase in the coefficient is at most $(1/50)(729/256) + (1/10000)(109/50) < 1/10$. Thus in every case $C_a(Y^2 + 2m^2Y + 3m^4) \le 8m^2$, as required in (R08.1.4).

For $b=4$ use Q5-C069, Section 10: $m=1-d$ and $R=(d+x-xz)(B)u$, so $A_0=d+x$, $B_0=x$. The exact input is

$$D_0=\delta_2,\quad a_1=d^2+\delta_2,\quad \rho_0=\delta_3,\quad I=-d\tau,$$
$$V\le dv,\quad v=15/64+13d/16+d^2,\qquad \mathbb{E}w_0^2\le m^2dH,\quad H=57/40-54d/25+3d^2/5.$$

Since $\mathbb{E}w_0=-md^2\le 0$, $\mathbb{E}w^2\le m^2dH+d^2v^2/4$. Retaining $-mD_0\beta_{a+1}$ in (R08.1.5), using $\beta_j\le d$, $\beta_{a+1}\ge\tau$, and $I_a\le -d\tau+\delta_4$, gives

$$\mathbb{E}[C_aw_0]\le -m^2\tau+m\delta_4+md\delta_3+md^3+m\delta_2^2.$$

For $a=3$ the stronger expression is $-m^2\tau+m\delta_4-m^2d^2$. Thus for every $a\ge 3$,

$$\mathbb{E}[C_aw_0]/d\le -(19/32)m^2+m\theta^3+2m\theta^2d+md^2.$$

Let $A_*=133/32$ and $B_*(d)=[m^{-4}-m^4(1-A_*d)]/d$. On $d\le 1/5$, (R08.1.4), including $\mathbb{E}[C_aw]\le \mathbb{E}[C_aw_0]+V/2$, gives the normalized deficit

$$D_4\ge B_*-4m^3\theta^3-8m^3\theta^2d-4m^3d^2-2m^2v-8H-2dv^2/m^2.$$

The positive Taylor expansion gives $B_*\ge 389/32-101d/8+783d^2/16$, hence $B_*-8H\ge 121/160+931d/200+3531d^2/80$. Use

$$-4m^3\theta^3\ge -4\theta^3+12\theta^3d-12\theta^3d^2,\quad -8m^3\theta^2d\ge -8\theta^2d,\quad -4m^3d^2\ge -4d^2,$$
$$-2m^2v\ge -15/32-11d/16,\qquad 2v^2/m^2\le 1225/2048<3/5.$$

The penultimate bound discards the nonnegative remainder $(25/32)d^2+(19/8)d^3-2d^4$; the last uses $v\le 7/16$, $m\ge 4/5$. It follows that $D_4\ge 1/64+(5/2)d+39d^2\ge 1/64$; the actual constant is $791/40960>1/64$, and the next two coefficients are greater than $5/2$ and 39. Since $m^4\ge 256/625$, the actual deficit is at least $4d/625>d/200$. For $d\ge 1/5$, the actual mean is $\mathbb{E}F=1-3d+\delta_2-d^2+\delta_3\le 1-(3-\theta-\theta^2)d-d^2$. This decreases and is below $1/2$ at $d=1/5$, so $TF\le 1$ and $\mathbb{E}[f_aX^4]\le \mathbb{E}f_a\le 1-2d$.

For $b=5$, Q5-C069, Section 11, gives $m=1-d$ and

$$R=\sum_{j=3}^{5}K^j\mathbf{1}-(d+\delta_2)g-\delta_2Kg,\quad A_0=d+\delta_2+x-x^2,\quad B_0=\delta_2+x.$$

Its exact data are

$$D_0=(1+d)\delta_2,\quad a_1=D_0+d^2-\delta_3,\quad \rho_0=\delta_2^2+\delta_3,\quad I=-d\tau+\delta_3-\delta_4,$$
$$V\le dv,\quad v=101/800+13d/16+33d^2/16,$$
$$\mathbb{E}w_0^2\le m^2dH,\quad H=177/128-17d/10+d^2/2-19d^3/20.$$

Thus $\mathbb{E}w^2\le m^2dH+m\theta^2d^2v+d^2v^2/4$. In (R08.1.5), $I_a\le -d\tau+(1+\delta_2)\delta_3$. Retain $-mD_0\beta_{a+1}$ and bound the positive part of $-m(\beta_a+\beta)(D_0-a_1)$ by $2md^3$. This gives

$$\mathbb{E}[C_aw_0]\le -m^2\tau+m(1+d+\delta_2)\delta_3+md^3-md^2\delta_2+m(1+2d)\delta_2^2.$$

For $a=3$ the exact expression $-m^2\tau+m(1-d+\delta_2)\delta_3-m^2d^2-md^2\delta_2$ is no greater. Dropping $-md^2\delta_2$ gives

$$\mathbb{E}[C_aw_0]/d\le-(19/32)m^2+m\theta^2+m\theta^2(2+\theta)d+m(1+2\theta^2)d^2.$$

For $d\le1/6$ the full normalized deficit is at least

$$D_5\ge B_*-4m^3\theta^2-4m^3\theta^2(2+\theta)d-4m^3(1+2\theta^2)d^2-2m^2v-8H-8\theta^2vd/m-2v^2d/m^2.$$

Here $B_*-8H\ge35/32+39d/40+719d^2/16$. Since $v<1/3$, $m\ge5/6$, the last two costs are at most $(16/5)\theta^2d$ and $8d/25$. Also

$$-4m^3\theta^2\ge-4\theta^2+12\theta^2d-12\theta^2d^2,\quad -2m^2v\ge-101/400-28d/25-451d^2/400.$$

The latter discards $(53/8)d^3-(33/8)d^4\ge0$. In the remaining negative terms replace m^3 by 1. Then

$$D_5\ge9/50-(5/8)d+36d^2\ge9/50-25/9216>1/6.$$

The actual constant is $1159/6400$, the linear coefficient is $-93/200-(33/40)\theta^2>-5/8$, and the quadratic coefficient exceeds 36. Since $m^4\ge(5/6)^4$, the deficit exceeds $d/20$. For $d\ge1/6$, the exact mean $\mathbb{E}F=1-4d+2(1+d)\delta_2$ and (L69.11.8) give $TF\le283/288<1$, proving the high-range claim with the same direction.

For every finite $b\ge6$, use the finite positive convolution in (L73.3.retained). On $0<d\le1/5$ put $m=1-9d/8$, $N=b-2\ge4$,

$$\ell=\frac{9d/4-\beta_b-\beta_{b-1}}{\beta_{b-2}+(1-d)\beta_{b-3}},\quad C_1=\beta_{b-1},\ C_2=\ell\beta_{b-2},\ C_3=\ell\beta_{b-3},$$
$$C=C_1+C_2+C_3,\quad D=C_2+C_3,\quad E=C_3,$$
$$R=\sum_{j=3}^{b}K^j\mathbf{1}-Cg-DKg-EK^2g.$$

The actual finite convolution gives $0\le\ell\le1$ and $h_2+f_b\le2m+R$; no actual orientation depends on ℓ. For $G_N(i\sqrt{x})=a(x)+i\sqrt{x}\,b_0(x)$, $A_0=C+x(a-E)$, $B_0=D+xb_0$ are nonnegative. The input, valid for every finite N and both parities, is

$$\rho=3/8+9d/5,\quad v=\rho^2,\quad z_0=5/16+17d/16+d^2,\quad \kappa=19/32,$$
$$V\le dv,\quad D_0\le dz_0,\quad \rho_0\le\theta dz_0,\quad a_1\le\rho d,\quad a_1\ge\kappa D_0+\tau d,$$
$$I\le d/60-57d^2/512,\quad \mathbb{E}w_0^2\le m^2dH,$$
$$H=2685/2048-11509d/10240+124067d^2/51200-117d^3/128.$$

From (R08.1.5), $I_a\le d/60-57d^2/512+\theta^2dz_0$. Drop $-mD_0\beta_{a+1}$, bound other betas by d, and use $(a_1-D_0)_+\le\rho d$. For every $a\ge4$ this gives

$$\mathbb{E}[C_aw_0]/d\le-\kappa m^2+m(1/60+\theta^2z_0)+d[m^2+2m\rho+m\theta z_0-57m/512].$$

For $a=3$ the extra cost from its exact expression is at most $m\rho d^2$, so the same bound applies. Moreover $\mathbb{E}w_0\le m\theta D_0$, and $\mathbb{E}w^2\le m^2dH+m\theta z_0vd^2+d^2v^2/4$. For $d\le9/50$, discarding the favorable $57m^3d/128$ in the normalized deficit gives

$$D_6\ge B_*-(1/15)m^3-4\theta^2m^3z_0-4m^4d-8m^3\rho d-4\theta m^3z_0d$$
$$-2m^2v-8H-8\theta z_0vd/m-2v^2d/m^2.$$

Here the positive Taylor expansion, with the present m, gives

$$B_* \geq 421/32 - 873d/64 + 67311d^2/1024,$$
$$B_* - 8H \geq 683/256 - 5951d/1280 + \ 1186507d^2/25600.$$

Throughout $[0, 9/50]$,

$$m^3 z_0 \leq 5/16 + d/128, \quad m^3\rho \leq 3/8 + 171d/320,$$
$$-2m^2 v \geq -9/32 - (21/10)d - (11/10)d^2, \quad -4m^4 d \geq -4d + (25/2)d^2,$$
$$8\theta z_0 v/m < 6/5, \qquad 2v^2/m^2 < 5/6.$$

For the first two inequalities, the negative quadratic terms in the expansions dominate the positive higher terms. For the third square $m\rho = 3/8 + 441d/320 - 81d^2/40$; its exact negative terms after multiplication by -2 are $-9/32 - 1323d/640 - 38961d^2/51200 - 6561d^4/800$. Drop the positive cubic and use $d^4 \leq 81d^2/2500$. For the fourth use $m^4 \leq 1 - (9/2)d + (38/5)d^2$. For the last two use $z_0 < 27/50$, $v < 49/100$, $m^{-1} < 13/10$, and $m^{-2} < 17/10$. The resulting constant exceeds 2, the linear cost is below 17, and the quadratic remainder exceeds 53, explicitly

$$683/256 - 1/15 - (5/4)\theta^2 - 9/32 > 2,$$
$$5951/1280 + \theta^2/32 + 4 + 3 + (5/4)\theta + 21/10 + 6/5 + 5/6 < 17,$$
$$1186507/25600 + 25/2 - 171/40 - \theta/32 - 11/10 > 53.$$

Thus $D_6 \geq 2 - 17d + 53d^2 \geq 135/212 > 3/5$. Since $m \geq 31/40$ gives $m^4 > 1/3$, the low-range actual deficit exceeds $d/5$. For $d \geq 9/50$, M.20, using the Q4-C128 finite path, gives for all $b \geq 6$

$$\mathbb{E}F \leq 1 - L(d), \qquad L(d) = d(2871 - 1463d - 608d^2)/1024.$$

L increases on $[0, 1/3]$ and $L(9/50) = 58229118/128000000 > 9/20$. Hence $TF \leq 11/10$ and $\mathbb{E}f_a \leq 16/25$, so

$$\mathbb{E}[f_a X^4] \leq (16/25)(11/10)^4 = 14641/15625 < 19/20 \leq 1 - d/10.$$

This proves all of (R08.0.1). Decorations and $d = 0$ follow exactly as in Module R2-C007. Full reversal changes all original edges together. Combining R2-C007 with this module gives the unit row for every $b \geq 3$ with deficit at least $d/200$. Together with (N.7) for $L \geq 2$, this yields, independently for every $a, b \geq 3$ and $L \geq 1$,

$$\mathbb{E}[E_o f_a \{T^L(E_t h_2 f_b)\}^4] \leq 1 - d/400. \tag{R08.6.1}$$

N.29 The three-step bare reference estimate

Use $B = -K$, $u = -g$ and the notation of R2-C007. For every integer $a \geq 3$, orient the receiving arm outward and all three input edges toward the receiving root. We prove

$$\mathbb{E}[f_a \bar{f}_3^4] \leq 1 - 3d/40, \qquad \mathbb{E}[E_o f_a \{T^3(E_t)\}^4] \leq 1 - 3d/40 \tag{R12.1.1}$$

for actual equal-arm decorations $0 \leq E_o, E_t \leq 1$ at their original vertices.

The finite calculus gives $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_3 \le \theta\delta_2$, and $d\delta_3 \ge \delta_2^2$. For any centered real V, the original row bound is

$$|BV(x)|^2 = |\mathbb{E}_y[(B(x,y) - u(x))V(y)]|^2 \le p(x)q(x)\|V\|_2^2. \tag{R12.2.2}$$

Use the exact finite G_n, β_n, S_a, C_a of (R07.1.1); in particular

$$S_a = f_a + \sum_{j=3}^{a} \beta_j f_{a-j}, \quad 0 \le f_a \le C_a, \quad \mathbb{E}C_3 = 1 - 2d,$$
$$\mathbb{E}C_a = 1 - \beta_{a-1} - \beta_a - \beta_{a+1} \le 1 - 57d/32 \ (a \ge 4).$$

For $a \ge 4$, $V_a = G_{a-1}(-B)u$ has $\mathbb{E}V_a = \beta_a$, $\|V_a\|_2 \le \sqrt{d}$, and

$$C_a = (1 - \beta_a - \beta_{a-1})p + B(V_a - \beta_a) \le (1-d)p + \sqrt{pqd}.$$

Here $\beta_a + \beta_{a-1} \ge 2\tau \ge 19d/16 \ge d$, and the finite G norm holds for every $a - 1 \ge 3$. The case $a = 3$ instead uses the following exact centered identities:

$$\bar{f}_3 = 1 - d + u + Bu + B^2u = q - d + B(u + Bu) = (1-d)q + B(u + Bu + d),$$
$$f_3 = (1-d)p + B(u - Bu - d), \qquad \|u + Bu + d\|_2^2 = \|u - Bu - d\|_2^2 = d + \delta_2 - d^2. \tag{R12.3.3}$$

Both inputs of B are centered because $\mathbb{E}Bu = -d$.

Put $\alpha = 1 - d$, $m = \mathbb{E}\bar{f}_3 = 1 - 2d$, $\sigma^2 = d + \delta_2 - d^2 \ge d$, and $c = \sqrt{d(45/32 - d)}$. Then for every $a \ge 3$,

$$C_a \le \alpha p + \sqrt{pq}\,\sigma, \quad \bar{f}_3 \le \alpha q + \sqrt{pq}\,\sigma,$$
$$C_a + (\bar{f}_3 + m) \le 3 - 4d + 2c. \tag{R12.3.6}$$

The row errors need not have the same sign.

We record the continuous cubic inequality used here and below. If $x, y, m \ge 0$, $x + y \le D$, and $D \ge 3m$, then

$$x(y^2 + 2m^2) \le \tfrac{4}{27}D^3 + \tfrac{2}{3}Dm^2 + \tfrac{1}{2}m^3. \tag{R12.4.1}$$

For $D = 0$ this is immediate. Otherwise monotonicity in y reduces the problem to $G(x) = x[(D - x)^2 + 2m^2]$. With $z = x - D/3$,

$$G(x) = \tfrac{4}{27}D^3 + \tfrac{2}{3}Dm^2 + 2m^2 z - Dz^2 + z^3.$$

On $-D/3 \le z \le 0$ the last three terms are nonpositive. On $0 \le z \le D/3$ they are at most $2m^2 z - (2D/3)z^2 \le 3m^4/(2D) \le m^3/2$. On $2D/3 \le x \le D$, $G'' = 6x - 4D \ge 0$, so only its endpoints remain. The left endpoint is already covered, and $G(D) = 2Dm^2 \le (4/27)D^3 + (2/3)Dm^2 + m^3/2$ because $D^2 \ge 9m^2$. This also covers $m = 0$.

On $0 \le d \le 1/4$, $m \ge 1/2$ and

$$c \le (1 + 5d)/4, \quad ((1+5d)/4)^2 - c^2 = \frac{82(d - 25/164)^2 + 31/328}{32} > 0.$$

Hence (R12.3.6) gives $C_a + (\bar{f}_3 + m) \le D = (7 - 3d)/2$, $D - 3m = (1 + 9d)/2 \ge 0$. Applying (R12.4.1) yields

$$C_a[(\bar{f}_3 + m)^2 + 2m^2] \le \tfrac{248}{27} - \tfrac{43}{2}d + \tfrac{137}{6}d^2 - \tfrac{17}{2}d^3$$
$$\le \tfrac{19}{2}(1-d)^2 - \tfrac{23}{216}.$$

Indeed the difference before the last constant is $17/54 + (5/2)d - (40/3)d^2 + (17/2)d^3 \geq 17/54 - (5/6)d \geq 23/216$, using $d^2 \leq d/4$. This is a certificate on the entire continuous interval.

The spectral polynomial $G_3(z) = 1 + z + z^2$ satisfies $|G_3(i\sqrt{x})|^2 = 1 - x + x^2$, so the exact variance is

$$N := \mathrm{Var}(\bar{f}_3) = d - \delta_2 + \delta_3 - d^2 \leq d - \tfrac{19}{32}\delta_2 - d^2 \leq d - \tfrac{51}{32}d^2. \tag{R12.6.1}$$

For every finite $a \geq 4$, the alternating coordinates of $G_a(z) = E_a(x) + zD_a(x)$ satisfy $E_a \geq 1 - x$, $D_a \leq 1 - x + x^2$. For $a = 4, 5$ the latter is exactly $1 - x$; all later finite truncations obey the displayed bound since $0 \leq x \leq \theta < 1$. Consequently

$$\mathrm{Re}(G_a G_3) = (1 - x)E_a - xD_a \geq \phi(x) := 1 - 3x + 2x^2 - x^3 > 0.$$

Here $\phi' = -3 + 4x - 3x^2 < 0$ and $\phi(13/32) = 1451/32768 > 0$. Writing $\beta = \beta_{a-1}$, the exact centered covariance is

$$\mathrm{Cov}(C_a, \bar{f}_3) = -\int \mathrm{Re}(G_a G_3)\, d\nu + \beta\tau - \beta_{a+1}d \leq 0 \quad (a \geq 4). \tag{R12.6.5}$$

This uses $\mathbb{E}[uG_3(B)u] = \tau$, $\mathbb{E}G_3(B)u = -d$, $\mathbb{E}G_a(-B)u = \beta_{a+1}$, $\beta \leq d$, and $\beta_{a+1} \geq \tau$. For $a = 3$ the exact signed formula is

$$\mathrm{Cov}(f_3, \bar{f}_3) = -d + 3\delta_2 - \delta_3 - d^2 \leq d(55/1024 - d). \tag{R12.6.6}$$

For $d > 0$ the last comparison writes $v = \delta_2/d$ and uses $\delta_3/d \geq v^2$ and monotonicity of $-1 + 3v - v^2$ on $0 \leq v \leq 13/32$. The variable v is a global moment ratio.

The exact Taylor identity about the actual mean m gives, on $0 < d \leq 1/4$,

$$\mathbb{E}[C_a \bar{f}_3^4] \leq (\mathbb{E}C_a)m^4 + 4m^3\,\mathrm{Cov}(C_a, \bar{f}_3) + \tfrac{19}{2}(1 - d)^2 N. \tag{R12.7.1}$$

For $a \geq 4$ the resulting bound is $(1 - 57d/32)(1 - 2d)^4 + (19/2)(1 - d)^2[d - (51/32)d^2]$. After subtracting one and dividing by d, it becomes

$$\begin{aligned} P_4 &= -9/32 + (263/64)d - (1119/32)d^2 + (3703/64)d^3 - (57/2)d^4 \\ &\leq -9/32 + (263/64)d - (5249/256)d^2 \\ &\leq -9/32 + 69169/335936 < -3/40. \end{aligned}$$

The final rational comparison has numerator $335936 \cdot 33 - 69169 \cdot 160 = 18848 > 0$. For $a = 3$ omit the covariance temporarily; the baseline is $m^5 + (19/2)(1 - d)^2[d - (51/32)d^2]$. Its normalized excess is

$$P_3 = -1/2 + (375/64)d - (1287/32)d^2 + (4151/64)d^3 - 32d^4 \leq -1/2 + (375/64)d - (6145/256)d^2.$$

If $d \geq 55/1024$, the covariance is nonpositive and $P_3 \leq -1/2 + 140625/393280 < -1/10 < -3/40$, using $140625/393280 < 2/5$. If $0 < d \leq 55/1024$, its upper bound is nonnegative and $m^3 \leq 1$ can be used for this term only. The total normalized excess is at most

$$\begin{aligned} &-73/256 + (119/64)d - (6145/256)d^2 \\ &\leq -73/256 + 14161/393280 < -73/256 + 1/20 < -1/5 < -3/40. \end{aligned}$$

Thus the possible positive covariance is fully included.

For $1/4 \le d \le 1/3$, the actual message satisfies $0 \le \bar f_3 = T\bar f_2 \le 2\mathbb{E}\bar f_2 = 2(1-d)$. All the covariances above are now nonpositive, and for every $a \ge 3$ $\mathbb{E}C_a \le 1 - 57d/32$. Therefore

$$\mathbb{E}[f_a \bar f_3^4] \le [2(1-d)]^3 \mathbb{E}[C_a \bar f_3] \le 8(1-d)^3(1-57d/32)(1-2d) \le 1917/2048 < 39/40 \le 1-3d/40.$$

All factors used in the endpoint comparison are nonnegative and decreasing. This overlaps the low range at $d = 1/4$. Finally $T^3(E_t) \le T^3\mathbf{1} = \bar f_3$, which proves the decorated claim. At $d = 0$ all messages are one by original leaf integration; K need not vanish. Full reversal replaces $f_a, \bar f_3$ by $\bar f_a, f_3$ and reverses every decoration. The resulting three-step reference estimate uses the original tree edge count.

N.30 A unit connector with a nonunit odd terminal starting with two and three

Use the original-measure notation of Module R2-C007. Let the terminal residual list be

$$(2, 3, c, s_2, t_2, \dots, s_j, t_j), \qquad 4 \le c < s_2 < t_2 < \cdots < s_j < t_j, \quad j \ge 1.$$

The actual fixed orientation has

$$F = \bar f_2 f_3 h_c \prod_{i=2}^{j} h_{s_i} f_{t_i}, \qquad X = TF.$$

Thus the length-two arm points inward, the length-three arm outward, and the c arm and subsequent pairs have the specified h, f messages. The unit connector points from the terminal to the receiving root. For every independent finite choice of these lengths and $a \ge 3$,

$$\mathbb{E}[f_a X^4] \le 1 - d/5. \tag{R13.1.1}$$

Actual equal-arm decorations may be restored at both original roots. Only the unit connector is asserted here.

Use C_a from (R07.1.1), so $0 \le f_a \le C_a$, $\mathbb{E}C_a \le 1-57d/32$ and $C_a \le (1-d)p + (6/5)\sqrt{d}$. The centered row identities in (R12.3.3) justify this also at $a = 3$. The terminal input is precisely Q5-C181, Sections 4–6, using its actual finite envelopes rather than its root-pP conclusion.

First suppose $0 < d \le 1/8$ and $j = 1$. Put $m_c = c - 3 \ge 1$, $\beta = \beta_{c-1}$, and write $G_{m_c}(-z) = U - zV$. For $m_c = 1$ use $U = 1, V = 0$; otherwise $U, V \in [1-x, 1]$. The actual finite vector and envelope are

$$\begin{gathered} R_0 = \mathfrak{r}_0(B)u, \qquad \mathfrak{r}_0 = -\beta + x^2 V + zxU, \qquad \Gamma_0 = d + \beta_c + \beta \ge 35d/16, \\ F \le \bar f_2 e^{R_0 - \Gamma_0}, \quad e^{\Gamma_0} X \le Y = \bar f_3 + BR_0 + D, \quad D = 3d/5, \quad D_1 = \mathbb{E}BR_0 \in [-\delta_3, d^2], \\ m = \mathbb{E}Y = 1 - 2d + D + D_1 \le 1 - 51d/40. \end{gathered} \tag{R13.3.1}$$

The bound on D retains the three actual mixed means in Q5-C181's first-row calculation. In $Y = 1 - d + D + Q(B)u$, $Q = G_3 + z\mathfrak{r}_0$ has coordinates $Q_e = 1 - x - x^2U$, $Q_o = 1 - \beta + x^2V$, both positive. Hence

$$|Q|^2 \le (1 - x - x^2 + x^3)^2 + x(1 + x^2)^2 = 1 - x - x^2 + 6x^3 - x^4 - x^5 + x^6 \le 1.$$

The last difference from one is $-x[1+x-6x^2+x^3+x^4-x^5]$; $1+x-6x^2 \ge 213/512$ on $[0,\theta]$ and the last three terms are nonnegative. This covers every finite c, including $m_c = 1$, and gives $\operatorname{Var} Y \le d$. Moreover

$$\operatorname{Re}(G_a Q) \ge (1-x)(1-x-x^2) - x(1+x^2) = 1-3x \ge -7/32.$$

For $a \ge 4$, the exact covariance is

$$\operatorname{Cov}(C_a, Y) = -\int \operatorname{Re}(G_a Q)\, d\nu + \beta_{a-1} \int Q_e\, d\nu - \beta_{a+1}(d - D_1).$$

Since $\int Q_e \le \tau$ and $d - D_1 > 0$, the scalar terms are at most $d\tau - \tau(d-D_1) = \tau D_1 \le d^3$. At $a = 3$ the positive-beta term is absent and the actual mean product is $-d(d-D_1) \le 0$. Therefore

$$\operatorname{Cov}(C_a, Y) \le (7/32)d + d^3 \quad (a \ge 3). \tag{R13.3.7}$$

The finite bound $\|G_{m_c}\| \le 6/5$ and $\theta^{3/2} < 4/15$ give $\|R_0\|_2/\sqrt{d} \le d + (6/5)\theta^{3/2} < 1/8 + 8/25 = 89/200 < 9/20$. Using the correct identity $\bar{f}_3 = q - d + B(u + Bu)$ and $\|u + Bu\|_2 \le 6\sqrt{d}/5$, we get $Y \le q + (33/20)\sqrt{d} - (2/5)d$. Since $\sqrt{d} \le 3/8$ and $m \le 1$,

$$C_a[(Y+m)^2 + 2m^2] \le (p + 9/20)[(579/160 - p)^2 + 2] \le 14.$$

For a continuous certificate, the cubic on the right has $F'' = 6p - 543/40 \le -63/40 < -3/2$ on $[0,2]$, $F(6/5) = 6631977/512000 < 13$, and $F'(6/5) = -3367/25600$ of absolute value below $1/7$. Thus its strong concavity bound is $13 + 1/147 < 14$. Taylor expansion about the actual nonnegative mean m now gives

$$\mathbb{E}[C_a Y^4] \le (1 - 57d/32)(1 - 51d/40)^4 + 4(7/32 + d^2)d + 14d \le e^{-1101d/160} + (239/16)d.$$

For a negative covariance simply omit it; use $m^3 \le 1$ only when its upper bound is nonnegative. With $A_0 = 2501/160$, $B_0 = 239/16$, $C_0 = 35/4$,

$$\mathbb{E}[f_a X^4] \le e^{-A_0 d} + B_0 d e^{-C_0 d}.$$

Here $C_0 \ge A_0/2$, $A_0 - B_0 = 111/160$, and $1 - e^{-A_0 d} \ge A_0 d e^{-A_0 d/2}$ by $\sinh y \ge y$. The deficit is at least $(111/160)de^{-A_0 d/2} > (37/160)d > d/5$, because $A_0 d/2 \le 2501/2560 < 1$ and $e < 3$.

For every finite $j \ge 2$, retain all subsequent disjoint finite intervals in one mask S, and put $R = R_0 + S$. Q5-C181, Section 5, gives

$$\Gamma \ge 431d/80 + (16/5)(j-2)d, \quad \|S\|_2 < (3/20)\sqrt{d}, \quad \|R\|_2 < (3/5)\sqrt{d},$$
$$e^{\Gamma} X \le Y = \bar{f}_3 + BR + D \ge 0, \quad D = 3d/2,$$
$$D_1 = \mathbb{E}BR \le 19d/120, \quad m = \mathbb{E}Y \le 1 - 41d/120.$$

Since $\|BS\|_2 < \sqrt{d}/10$, the proved Q bound gives $\operatorname{Var} Y \le (11/10)^2 d$. The receiving centered norm is at most $(1+d)\sqrt{d} \le 9\sqrt{d}/8$; for $a \ge 4$ use its vector $[-G_a(-B) + \beta_{a-1}]u$, and for $a = 3$ use $\|G_3 u\|_2 \le \sqrt{d}$. Thus the extra covariance cost is at most $9d/80$, including all beta–mask cross terms, and

$$\operatorname{Cov}(C_a, Y) \le (53/160)d + d^3 \le (111/320)d.$$

The changed D is a constant and does not enter this covariance. The row bound gives $Y \le q + (9/5)\sqrt{d} + d/2$, and

$$C_a[(Y+m)^2 + 2m^2] \le (p + 9/20)[(299/80 - p)^2 + 2] \le 16.$$

This cubic has $F'' = 6p - 281/20 \le -41/20 < -2$, $F(5/4) = 890817/64000 < 14$ and $F'(5/4) = -1727/6400$ of absolute value below $3/10$. Its strong concavity bound $14 + 9/400 < 16$ covers the full interval. Discarding only the favorable extra scalar for $j > 2$ gives

$$\mathbb{E}[f_a X^4] \le e^{-A_1 d} + B_1 d e^{-C_1 d}, \quad A_1 = 2371/96,\ B_1 = 8299/400,\ C_1 = 431/20.$$

These constants are $A_1 = 4(431/80) + 57/32 + 4(41/120)$ and $B_1 = 16(121/100) + 4(111/320)$. Since $C_1 \ge A_1/2$ and $A_1 - B_1 = 9481/2400$, the deficit is at least $(9481/2400)de^{-A_1 d/2} > (9481/21600)d > d/5$; indeed $A_1 d/2 \le 2371/1536 < 2$ and $e^2 < 9$.

For $1/8 \le d \le 1/3$, use the argument of Q5-R2-A002, Section 7. (C181.T190.31) gives $f_3 h_c \le e^{-d}$, and every subsequent $h_s f_t \le e^{-2d}$. Thus with the same actual direction, $F \le e^{-(2j-1)d}\bar{f}_2$ and $X \le e^{-(2j-1)d}\bar{f}_3$. The variance and signed covariance formulas (R12.6.1)–(R12.6.6) follow directly from the finite path, and all these covariances are nonpositive on the present interval. Put $\mu_3 = 1 - 2d$. Since $t = \sqrt{d} \ge 1/3$, $6t/5 - 3t^2 \le 1/15$, so

$$\bar{f}_3 + \mu_3 \le q + 1 + (6/5)\sqrt{d} - 3d < 25/8 - p, \quad C_a \le p + 3/4, \quad \mu_3 \le 3/4.$$

Consequently

$$C_a[(\bar{f}_3 + \mu_3)^2 + 2\mu_3^2] \le \Phi(p) := (p + 3/4)[(25/8 - p)^2 + 9/8] \le 11.$$

Indeed $\Phi' = 3p^2 - 11p + 397/64$, $\Phi'' = 6p - 11$. On $[1,2]$ the convex quadratic derivative has endpoint values $-115/64, -243/64$, so Φ decreases. On $[0,1]$, $\Phi'' \le -5$, $\Phi(3/4) = 1299/128$, $\Phi'(3/4) = -23/64$, and the tangent bound is $1299/128 + 529/40960 = 416209/40960 < 11$. The actual centered Taylor formula therefore gives

$$\mathbb{E}[f_a \bar{f}_3^4] \le (1 - 57d/32)(1 - 2d)^4 + 11d(1 - 51d/32), \quad \mathbb{E}[f_a X^4] \le e^{-4d}\mathbb{E}[f_a \bar{f}_3^4].$$

The first term after multiplication is at most $27/160$, using $e^{-4d} \le e^{-1/2} \le 2/3$, $1 - 57d/32 \le 199/256 < 4/5$ and $(1-2d)^4 \le (3/4)^4$. Completing the square gives $1 - 11d + (1021/32)d^2 \ge 53/1021 > 0$. Thus $15d(1 - 51d/32) \le 1 + 4d + 8d^2 \le e^{4d}$ and $e^{-4d}d(1 - 51d/32) \le 1/15$. The total is at most $27/160 + 11/15 = 433/480 < 14/15 \le 1 - d/5$. This joins the low range at $d = 1/8$ and proves (R13.1.1). Decorations, $d = 0$, and full reversal follow by original positivity and leaf integration, as in Module R2-C007. The local edge count is $a + 1 + 2 + 3 + c + \sum_{i \ge 2}(s_i + t_i)$ plus every actual repeated pair. The high-range argument uses the finite identities of R2-C012.

N.31 Four bare steps and two steps from an odd terminal

In the notation of Module R2-C007 put $\kappa = 19/32$. The two assertions are

$$\mathbb{E}[f_a \bar{f}_4^4] \le 1 - d/2 \quad (a \ge 3) \tag{R14.1.1}$$

and, for the fixed terminal orientation and length list of Module R2-C013, $F = \bar{f}_2 f_3 h_c \prod_{i=2}^{j} h_{s_i} f_{t_i}$,

$$\mathbb{E}[f_a (T^2 F)^4] \le 1 - d/5. \tag{R14.1.2}$$

Both connector edges in the second assertion point from the terminal toward the receiving root. All original lengths are independent finite integers, with $a \ge 3$ and $4 \le c < s_2 < t_2 < \cdots < s_j < t_j$. The exact receiving data are (R07.1.1) and $\mathbb{E}C_a \le 1 - 57d/32$, $\sigma(C_a) \le (1+d)\sqrt{d}$. The latter follows for $a \ge 4$ from the nonconstant vector $[-G_a(-B) + \beta_{a-1}]u$, and for $a = 3$ from G_3.

The correct finite recurrences are

$$\bar{f}_3 = q - d + B(u + Bu), \qquad \bar{f}_4 = (1-d)q - d + BG_3(B)u = 1 - 2d + [G_4(B) - d]u. \quad \text{(R14.2.1)}$$

Thus $\mu = \mathbb{E}\bar{f}_4 = 1 - 3d + \delta_2 \le 1 - (5/2)d$. The exact variance supplied in Q5-C101, Section 2, is

$$\operatorname{Var} \bar{f}_4 = (1-d)^2 d - (1-2d)\delta_2 - \delta_3 + \delta_4 - \tau^2 \le v_4(d) := d - 4d^2 + (141/32)d^3 - d^4. \quad \text{(R14.2.2)}$$

Put $v = \delta_2/d \in [d, \theta]$ when $d > 0$. Using $\delta_4 \le \theta\delta_3$ and $\delta_3 \ge v^2 d$, the normalized variance is at most $(1-d)^2 - (1-2d)v - \kappa v^2 - d(1-v)^2$. Its derivative in v is at most $-1 + (45/16)d - 2d^2 \le -41/144 < 0$, so its maximum is at the legal global endpoint $v = d$, giving (R14.2.2). At $a = 3$ cancellation of actual means gives

$$\operatorname{Cov}(f_3, \bar{f}_4) = -\int (1-x)(1-2x)\, d\nu \le 0;$$

indeed $\operatorname{Re}[G_3(G_4 - d)] = (1-x)(1-2x-d)$ and the mean-product correction is $-d\tau$. For $a \ge 4$, with $\beta = \beta_{a-1}$, the exact covariance is

$$-\int \operatorname{Re}(G_a G_4)\, d\nu + d\int E_a\, d\nu + \beta(\tau - d^2) - \beta_{a+1}\tau.$$

Since $\operatorname{Re}(G_a G_4) = (1-x)(E_a - xD_a) \ge (1-x)(1-2x) \ge 57/512$, $\beta - \beta_{a+1} \le \delta_2$, $\int E_a \le d$, and $\tau \le d$,

$$\operatorname{Cov}(C_a, \bar{f}_4) \le -57d/512 + (45/32)d^2 \le (45/32)d^2. \quad \text{(R14.2.4)}$$

The final nonnegative bound also covers $a = 3$.

Here is the full continuous curvature certificate for $0 \le d \le 1/4$. Write $t = \sqrt{d}$, $\alpha = 1 - t^2$, $c = 6t/5$. The correct recurrence and $\|G_3 u\|_2 \le \sqrt{d}$ give

$$\bar{f}_4 + \mu \le \alpha(q+1) + t - (5/2)d \le \alpha(q+1) + c - 2d, \qquad 0 \le \mu \le \alpha.$$

For $h = (c - 2t^2)/\alpha$, $0 \le h \le 1/5$ because $\alpha/5 - c + 2t^2 = (3t-1)^2/5$. The curvature divided by α^2 is therefore at most $[\alpha p + c][(16/5 - p)^2 + 2]$. For fixed p, completing the square in t, or using its endpoint, gives

$$\alpha p + c \le \begin{cases} 3p/4 + 3/5, & 0 \le p \le 6/5, \\ p + 9/(25p), & 6/5 \le p \le 2. \end{cases}$$

On the first interval the resulting cubic is $F(p) = (3/4)p^3 - (21/5)p^2 + (267/50)p + 918/125$. It has $F'' \le -3$, $F(4/5) = 1164/125$, $F'(4/5) = 3/50$, and $F \le 1164/125 + (3/50)^2/6 = 46563/5000 < 19/2$. On the second interval put $H = (p + 9/(25p))[(16/5 - p)^2 + 2]$. Then

$$p^2 H' = N = 3p^4 - (64/5)p^3 + (63/5)p^2 - 2754/625, \quad N' = 12p[p^2 - (16/5)p + 21/10].$$

The bracket is convex with endpoint values $-3/10$, hence negative; $N(6/5) = -54/25 < 0$, so H decreases and $H(6/5) = 9 < 19/2$. Consequently throughout the low range

$$C_a[(\bar{f}_4 + \mu)^2 + 2\mu^2] \le (19/2)(1-d)^2. \quad \text{(R14.3.2)}$$

Taylor expansion about the actual mean μ, using the displayed variance and covariance, bounds $\mathbb{E}[C_a \bar{f}_4^4]$ by

$$(1-57d/32)(1-5d/2)^4+(45/8)d^2+(19/2)(1-d)^2[d-4d^2+(141/32)d^3-d^4].$$

Its excess above one, divided by d, is exactly

$$\begin{aligned}&-73/32+(63/16)d-(31/16)d^2+(1227/64)d^3-(4465/512)d^4-(19/2)d^5\\&\le -73/32+63/64+1227/4096=-4085/4096<-1/2.\end{aligned}$$

For $d\ge 1/4$, $\bar{f}_4=T\bar{f}_3\le 2\mathbb{E}\bar{f}_3=2(1-2d)\le 1$, and $\mathbb{E}f_a\le 1-2d$. This proves (R14.1.1) on the full range with unchanged directions.

For (R14.1.2) first take $0<d\le 1/8$, $j=1$. Use $\mathfrak{r}_0, R_0, U, V, \beta=\beta_{c-1}$ from (R13.3.1), including the $c=4$ case $U=1, V=0$. Q5-C181, Section 4, gives the two-step envelope

$$\begin{gathered}\Gamma_0=d+\beta_c+\beta\ge 35d/16,\quad D=3d/5,\quad D_1=\mathbb{E}BR_0\in[-\delta_3,d^2],\quad 0\le D_2=\mathbb{E}B^2R_0\le\delta_3,\\ e^{\Gamma_0}T^2F\le Y=(1-d+D)q+B[G_3+z\mathfrak{r}_0](B)u+D_1-d\ge 0.\end{gathered}\tag{R14.5.1}$$

The proof in Module R2-C013 gives, for $Q=G_3+z\mathfrak{r}_0$, $Q_e=1-x-x^2U$, $Q_o=1-\beta+x^2V$, and $|Q|^2\le 1$ on the entire original spectrum. The nonconstant polynomial of Y is

$$P=G_4+z^2\mathfrak{r}_0-(2/5)d,\quad P_e=1-x+\beta x-x^3V-(2/5)d,\quad P_o=1-x-x^2U.$$

Both coordinates are positive. For $x\le 2/5$, $\beta x-(2/5)d\le 0$ so $|P|^2\le(1-x)^2(1+x)\le 1$. For $2/5\le x\le\theta$, $P_e\le 3/5+d/160<5/8$, $P_o\le 3/5$, so $|P|^2<25/64+3/20<1$. Thus $\operatorname{Var}Y\le d$. The actual mean is

$$m=1-2d+D+D_1-\tau+D_2\le 1-(8723/5120)d\le 1-(17/10)d,$$

because $D_1\le d^2\le d/8$, $D_2\le\theta^2 d$ and $2-3/5-1/8+\kappa-\theta^2=8723/5120>17/10$. Using $U\ge 1-x$, we have $P_o\le(1-x)(1-x^2)$, $P_e\ge 1-x-x^3-(2/5)d$, hence $\operatorname{Re}(G_aP)\ge(1-x)[1-2x-(2/5)d]\ge 209/2560$. Also $\int P_e\le\tau+d^2/160$ and $\int P_o=\tau-D_2>0$. For $a\ge 4$, the actual covariance is

$$-\int \operatorname{Re}(G_aP)\,d\nu+\beta_{a-1}\int P_e\,d\nu-\beta_{a+1}\int P_o\,d\nu.$$

Its scalar correction is at most $d(\tau+d^2/160)-\tau(\tau-D_2)\le(\theta+\theta^2)d^2+d^3/160\le(2929/40960)d$. Since $209/2560=3344/40960$, the covariance is at most $-83d/8192<0$. For $a=3$ it is $-\int\operatorname{Re}(G_3P)\,d\nu-d\int P_o\,d\nu\le 0$. The row bound in (R14.5.1) gives $Y\le(1-2d/5)q+\sqrt{d}+D_1-d\le q+\sqrt{d}$. Thus

$$C_a[(Y+m)^2+2m^2]\le(p+9/20)[(27/8-p)^2+2]\le 12.$$

This cubic has $F''=6p-63/5\le -3/5$, $F(1)=14181/1280<111/10$, and $F'(1)=241/320<4/5$. The strong concavity bound is $111/10+8/15=349/30<12$. Taylor expansion yields

$$\mathbb{E}[f_a(T^2F)^4]\le e^{-A_0d}+12de^{-C_0d},\quad A_0=2773/160,\ C_0=35/4\ge A_0/2.$$

Here $A_0 = 4\Gamma_0/d + 57/32 + 4(17/10)$ with the lower bound for Γ_0/d. In general, for $A \ge B \ge 0$, $C \ge A/2$,

$$1 - e^{-Ad} - Bde^{-Cd} \ge (A - B)de^{-Ad/2}. \tag{R14.7.3}$$

Since $A_0 - 12 = 853/160$, $A_0 d/2 \le 2773/2560 < 2$, and $e^2 < 9$, the deficit exceeds $853d/1440 > d/5$.

For any finite $j \ge 2$ retain Q5-C181's single complete mask S, $R = R_0 + S$. Section 5 gives

$$\|S\|_2 < (3/20)\sqrt{d}, \quad \Gamma \ge 431d/80 + (16/5)(j-2)d, \quad D = 3d/2,$$
$$D_1 = \mathbb{E}BR \le 19d/120, \quad D_2 = \mathbb{E}B^2R \le d/4,$$
$$e^{\Gamma}T^2F \le Y = (1 - d + D)q + B[G_3 + z\mathfrak{r}](B)u + D_1 - d \ge 0, \quad R = \mathfrak{r}(B)u.$$

Its mean is $m \le 1 - (329/480)d \le 1 - 2d/3$. Relative to the single-terminal nonconstant vector the addition is $(9/10)du + B^2S$. Since $\|B^2S\|_2 < \sqrt{d}/16$,

$$\operatorname{Var} Y \le [1 + 1/16 + (9/10)d]^2 d \le (2209/1600)d.$$

The added q term has nonpositive covariance: for $a \ge 4$, $\operatorname{Cov}(C_a, q) = -\int E_a \, d\nu + \beta_{a-1}d \le -\tau + d^2 < 0$, and for $a = 3$ it is $-\tau$. The other added term costs at most $(9/8)\sqrt{d}\,\sqrt{d}/16 = 9d/128$ in covariance. Further $\|Q(B)u + BS\|_2 < (11/10)\sqrt{d}$ and $q \le 2$ give $Y \le q + (11/10)\sqrt{d} + (19/120)d \le q + 83/192$. Since $3 + 83/192 < 299/80$, the full curvature is at most $(p + 9/20)[(299/80 - p)^2 + 2] \le 16$, with the continuous certificate already proved in Module R2-C013. Taylor expansion gives

$$\mathbb{E}[f_a(T^2F)^4] \le e^{-A_1 d} + B_1 d e^{-C_1 d}, \quad A_1 = 12479/480,\ B_1 = 17897/800,\ C_1 = 431/20.$$

Here $A_1 = 431/20 + 57/32 + 8/3$ and $B_1 = 16(2209/1600) + 9/32$. $C_1 \ge A_1/2$, $A_1 - B_1 = 272/75$, and $A_1 d/2 \le 12479/7680 < 2$, so (R14.7.3) gives a deficit greater than $272d/675 > d/5$ for every finite mask.

On $1/8 \le d \le 1/3$, Q5-C181, Section 6, gives the same actual $F \le e^{-(2j-1)d}\bar{f}_2$ and hence $T^2F \le e^{-(2j-1)d}\bar{f}_4$. By (R14.1.1),

$$\mathbb{E}[f_a(T^2F)^4] \le e^{-4(2j-1)d}(1 - d/2) \le e^{-4d} \le 1 - \frac{4d}{1 + 4d} \le 1 - 12d/7 \le 1 - d/5.$$

This joins at $d = 1/8$. Decorations follow by positivity, including $T^4E_t \le \bar{f}_4$ in the bare assertion. At $d = 0$ original leaf integration gives equality, without requiring $K = 0$. Full reversal changes the receiving arm, terminal, both connector edges, and all decorations. The terminal edge count is $a + 2 + 2 + 3 + c + \sum_{i \ge 2}(s_i + t_i)$ plus all equal-arm edges. The estimates apply to the stated four-step bare path and two-step terminal connector.

N.32 Every bare input path of length at least five

For all independent finite integers $a \ge 3$, $n \ge 5$, the actual outward receiving arm and inward input path satisfy

$$\mathbb{E}[f_a \bar{f}_n^4] \le 1 - d/2, \qquad \mathbb{E}[E_o f_a \{T^n(E_t)\}^4] \le 1 - d/2. \tag{R17.1.1}$$

Use Module R2-C007's coordinates, receiving upper bounds, and spectral measure. In particular $\kappa = 19/32$, $\tau = d - \delta_2$, $\kappa d \le \tau \le d - d^2 \le 2/9$, $\mathbb{E}C_a \le 1 - 57d/32$, and $C_a \le (1 - d)p + (6/5)\sqrt{d}$. For

every nonnegative $Z = c + Q(B)u$, $Q = Q_e + zQ_o$, $m = \mathbb{E}Z$, the exact original-measure identities are

$$\operatorname{Cov}(C_a, Z) = -\int \operatorname{Re}(G_a Q)\, d\nu + \beta_{a-1}\int Q_e\, d\nu - \beta_{a+1}\int Q_o\, d\nu \tag{R17.2.3}$$

and

$$\mathbb{E}[C_a Z^4] = (\mathbb{E}C_a)m^4 + 4m^3 \operatorname{Cov}(C_a, Z) + \mathbb{E}[C_a(Z-m)^2\{(Z+m)^2 + 2m^2\}]. \tag{R17.2.4}$$

They include $a = 3$ by setting $\beta_{a-1} = 0$, $\beta_{a+1} = d$. The final term of (R17.2.3) is the actual mean-product correction. All covariance upper bounds used below are nonnegative. The elementary continuous cubic inequality (R12.4.1), whose full proof was given there, implies

$$C_a + Z + m \le D, \quad m \le 1, \quad D \ge 3 \implies C_a[(Z+m)^2 + 2m^2] \le 4D^3/27 + 2D/3 + 1/2. \tag{R17.3.2}$$

For every finite $n \ge 6$, the positive convolution $S'_n = 1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}$ permits retention of four original terms:

$$\begin{aligned} 0 \le \bar{f}_n \le Z = S'_n - \tau(\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3), \\ \bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3 = 4 - d + 3u + 2Bu + B^2 u, \\ Z = 1 - \tau(4-d) + Q_n(B)u, \qquad Q_n = G_n - \tau(3 + 2z + z^2). \end{aligned} \tag{R17.4.1}$$

All four convolution indices are at least three. The finite alternating coordinates of G_n belong to $[1-x, 1-x+x^2]$, including $n = 6$. Thus $Q_e \ge 1 - x - \tau(3-x)$ and $Q_o \ge 1 - x - 2\tau$, which are at least $1/108$ and $5/36$, using $x \le \theta < 5/12$, $\tau \le 2/9$. With $a_0 = 1 - 3\tau$, the coordinate upper bounds give

$$|Q_n|^2 - a_0^2 \le x\{-1 + 4\tau - 2\tau^2 + (1 - 4\tau + \tau^2)x + (1 - 2\tau)x^2 - x^3 + x^4\}.$$

The bracket's τ derivative is $4(1-x) - \tau(4-2x) - 2x^2 \ge 79/72 > 0$ on the full rectangle. At $\tau = 2/9$, it is at most $-17/81 + (13/81)x + (5/9)x^2 - x^3 + x^4 \le -181/3888 < 0$, using $-x^3 + x^4 \le 0$ and $x \le 5/12$. Hence $\|Q_n(B)u\|_2 \le (1 - 3\tau)\sqrt{d}$. The actual mean and variance satisfy

$$\begin{aligned} m = 1 - 4\tau - \beta_{n+1} + 3\tau d \le 1 - 4\kappa d =: m_0 = 1 - 19d/8, \\ \operatorname{Var} Z \le (1-3\tau)^2 d - \tau^2(1-2d)^2 \le w(d) := d - (6\kappa + \kappa^2)d^2 + 13\kappa^2 d^3 - 4\kappa^2 d^4. \end{aligned} \tag{R17.4.3}$$

Here $\mathbb{E}Q_n(B)u = -\beta_{n+1} + 2\tau d \le -\tau(1-2d)$; the variance bound decreases in τ, with derivative $-6d(1-3\tau) - 2\tau(1-2d)^2 < 0$. Thus $\tau = \kappa d$ is legal. $m \ge 0$ by positivity and $m_0 \ge 5/24$.

Put $\phi = 1 - 3x + 2x^2 - x^3$, $\psi = 3 - 6x + x^2$. Coordinate positivity gives

$$\operatorname{Re}(G_a Q_n) \ge \phi - \tau\psi, \qquad \int Q_e\, d\nu \le d - \delta_2 + \delta_3 - \tau(3d - \delta_2),$$

$$\int Q_o\, d\nu = \beta_{n+1} - 2\tau d \ge \tau(1-2d) > 0.$$

In (R17.2.3), use $\beta_{a-1} \le d$, $\beta_{a+1} \ge \tau$. The covariance is at most

$$-\int \phi\, d\nu + \tau\int \psi\, d\nu + d[d - \delta_2 + \delta_3 - \tau(3d - \delta_2)] - \tau^2(1-2d).$$

For the global ratios $v=\delta_2/d$, $w_3=\delta_3/d$, use $v\in[d,\theta]$, $w_3\ge v^2$, $\delta_4\le\theta\delta_3$ and $\tau=d(1-v)$. After division by d the bound is

$$-1+3v-(2-\theta)w_3+d[3-8v+5v^2+(2-v)w_3]-d^2(1-v^2).$$

Its w_3 coefficient is negative, so insert $w_3=v^2$. The resulting v derivative is $3-2(2-\theta)v+d(-8+14v-3v^2)+2d^2v\ge 1/3+(2/3+2\theta)v-v^2>0$. The comparison uses that the bracket multiplying d is negative and $d\le 1/3$. At $v=\theta$ the result is $[-1451+27467d-27360d^2]/32768$, increasing on $[0,1/3]$ and at most $6997/49152<3/20$. Thus $\mathrm{Cov}(C_a,Z)\le(3/20)d$ for every $a\ge 3, n\ge 6$.

Discarding only the extra nonnegative $\bar f_2,\bar f_3$ terms in (R17.4.1), and using $S_n'=q+BG_{n-1}(B)u$, gives $Z\le(1-\tau)q-\tau+\sqrt{d}$. Combining with the receiving and mean bounds yields

$$C_a+Z+m\le 3-7\kappa d+(11/5)\sqrt{d}\le 33/10.$$

The full real quadratic maximum of the increment is $968/3325<3/10$. Equation (R17.3.2) gives the curvature coefficient $1003/125<81/10$. Therefore

$$\mathbb{E}[f_a\bar f_n^4]\le(1-57d/32)(1-19d/8)^4+(3/5)d(1-19d/8)^3+(81/10)w(d).$$

Its normalized excess is exactly

$$-413/160+(30267/2048)d-(681929/10240)d^2+(551969/5120)d^3-(7428297/131072)d^4.$$

Using $d^3\le d^2/3$ gives quadratic coefficient $-941849/30720<-30$; the positive linear coefficient is below 15. Discarding the negative quartic and completing the square gives the bound $-413/160+15d-30d^2\le -113/160<-1/2$. This covers every finite $n\ge 6$ on the whole $0<d\le 1/3$.

For $n=5$ the actual finite expansion is

$$\bar f_5=1-3d+\delta_2+P_5(B)u,\quad P_5=G_5-d(2+z),\quad (P_{5,e},P_{5,o})=(1-x+x^2-2d,1-x-d). \tag{R17.7.1}$$

Both coordinates are positive. Their exact square gives

$$|P_5|^2-(1-2d)^2=x[-1+2d+d^2+(1-2d)x-x^2+x^3]\le 0.$$

The bracket increases in d and at $d=1/3$ is at most $-2/9+\theta/3<0$. Thus $\|P_5(B)u\|_2\le(1-2d)\sqrt{d}$. Its mean is $m=1-4d+2\delta_2+d^2\le 1-137d/48\le 1-11d/4=:m_5$. Since $\mathbb{E}P_5(B)u=-\tau+d^2$ and $\tau-d^2\ge d(\kappa-d)>0$,

$$\mathrm{Var}\,\bar f_5\le(1-2d)^2d-(\tau-d^2)^2\le v_5(d):=d-(4+\kappa^2)d^2+(4+2\kappa)d^3-d^4.$$

Also $\mathrm{Re}(G_aP_5)\ge(1-x)^3-d(2-3x)$, $\int P_{5,e}=\tau+\delta_3-2d^2\ge 0$, $\int P_{5,o}=\tau-d^2>0$. Equation (R17.2.3) gives

$$\mathrm{Cov}(C_a,\bar f_5)\le-\int(1-x)^3\,d\nu+d(2d-3\delta_2)+d(\tau+\delta_3-2d^2)-\tau(\tau-d^2).$$

For the same global ratios its normalized upper bound is $-1+3v-(3-\theta)w_3+d(2-2v+w_3-v^2)-d^2(1+v)$. The w_3 coefficient is negative. At $w_3=v^2$ the v derivative is $3-2(3-\theta)v-2d-d^2\ge 529/4608>0$. Insert $v=\theta$ to get $-\kappa^3+2\kappa d-(1+\theta)d^2$, increasing up to $d=1/3$ and there equal

to $2975/98304 < 1/32$. Hence the covariance is at most $d/32$. The alternative correct recurrence is $\bar{f}_5 = (1-2d)q - \tau + B(G_4 - d)u$. The squared coordinate difference for $\|G_4 - d\| \le 1-d$ is $x[-1+2d-x+x^2] \le 0$. Thus

$$\bar{f}_5 \le (1-2d)q - \tau + (1-d)\sqrt{d}, \qquad C_a + \bar{f}_5 + m \le 3 - (523/96)d + (11/5)\sqrt{d} \le 13/4.$$

The full quadratic maximum is $2904/13075 < 1/4$. Equation (R17.3.2) gives coefficient $3349/432 < 8$, so

$$\mathbb{E}[f_a \bar{f}_5^4] \le (1-57d/32)(1-11d/4)^4 + (1/8)d(1-11d/4)^3 + 8v_5(d).$$

Its normalized excess is

$$-149/32 + (3727/128)d - (30637/256)d^2 + (49861/256)d^3 - (834537/8192)d^4.$$

Using $d^3 \le d^2/3$ gives coefficient $-21025/384 < -54$, and $3727/128 < 117/4$. Completing the square gives $-149/32 + (117/4)d - 54d^2 \le -89/128 < -1/2$. This completes (R17.1.1) for $d > 0$. For $d = 0$, positivity, decorations, and simultaneous full reversal, use the original leaf-integration argument in Module R2-C007. Combining this module with the three-step Module R2-C012 and the four-step assertion (R14.1.1) gives

$$\mathbb{E}[E_o f_a \{T^n(E_t)\}^4] \le 1 - 3d/40 \quad (a \ge 3, n \ge 3). \tag{R17.10.1}$$

The union combines those two estimates with the proof for $n \ge 5$ above.

N.33 All positive connectors from an odd terminal starting with two and three

For the actual terminal $F = \bar{f}_2 f_3 h_c \prod_{i=2}^{j} h_{s_i} f_{t_i}$ with $4 \le c < s_2 < t_2 < \cdots < s_j < t_j$ and every finite $j \ge 1$, orient all L connector edges toward the receiving root and the receiving arm outward. For every independent $a \ge 3, L \ge 1$,

$$\mathbb{E}[f_a (T^L F)^4] \le 1 - d/5. \tag{R18.1.1}$$

The $L = 1$ assertion is exactly Module R2-C013, and $L = 2$ is (R14.1.2). We prove $L \ge 3$ here, using the independent $n \ge 5$ assertion of Module R2-C017 only in the high range. Equal-arm decorations may be restored at their original vertices.

Use the original finite receiving definitions of Module R2-C007, $\sigma(C_a) \le (1+d)\sqrt{d}$, and the exact covariance and Taylor identities (R17.2.3)–(R17.2.4). The continuous cubic inequality (R12.4.1) gives curvature coefficient $1081/128 < 17/2$ when $C_a + Y + m \le 27/8$, and $683/54 < 13$ when $C_a + Y + m \le 4$, provided $m = \mathbb{E}Y \le 1$. We also use (R14.7.3): if $\mathbb{E}[f_a X^4] \le e^{-Ad} + Bde^{-Cd}$, $C \ge A/2$, $A > B \ge 0$, then the deficit is at least $(A-B)de^{-Ad/2}$. All rows below have $d \le 1/8$, $Ad/2 < 2$, so this is greater than $(A-B)d/9$.

We give the complete finite terminal input from Q5-C181, Sections 4–6. On $0 < d \le 1/8$, write $m_c = c - 3 \ge 1$, $\beta = \beta_{c-1}$, $G_{m_c}(-z) = U - zV$, and $R_0 = \mathfrak{r}_0(B)u$, $\mathfrak{r}_0 = -\beta + x^2 V + zxU$. For $m_c = 1$, $(U,V) = (1,0)$; otherwise $U, V \in [1-x, 1]$. For $j = 1$,

$$\Gamma_0 = d + \beta_c + \beta \ge 35d/16, \quad F \le \bar{f}_2 e^{R_0 - \Gamma_0}, \quad \|R_0\|_2 < (9/20)\sqrt{d}, \quad D = 3d/5,$$
$$D_i = \mathbb{E}B^i R_0, \quad -\delta_3 \le D_1 \le d^2, \quad 0 \le D_2 \le \delta_3, \quad D_3 \le \delta_4, \quad D_i \le d/36 \ (i \ge 4).$$

For all finite $j \ge 2$, the complete disjoint interval mask S gives $R = R_0 + S = \mathfrak{r}(B)u$ and

$$F \le \bar{f}_2 e^{R-\Gamma}, \quad \Gamma \ge [431/80 + (16/5)(j-2)]d, \quad \|S\|_2 < (3/20)\sqrt{d}, \quad \|R\|_2 < (3/5)\sqrt{d},$$
$$D = 3d/2, \quad D_i = \mathbb{E}B^i R, \quad -19d/60 \le D_1 \le 19d/120, \quad D_2 \le d/4,$$
$$D_3 \le d/6, \quad D_i \le d/9 \ (i \ge 4).$$

In both cases finite positive propagation gives

$$e^{\Gamma} T^L F \le Y_L^{\text{old}} = \bar{f}_{L+2} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \ge 0, \tag{R18.3.1}$$

using Γ_0, R_0 when $j = 1$. The exact $L = 3$ form is

$$Y_3 = Cq + BV - \tau + D_2, \quad C = 1 - 2d + D + D_1, \quad V = [G_4 - d + D + z^2\mathfrak{r}](B)u,$$
$$m = 1 - 2d - 2\tau + D + D_1 + D_2 + D_3 + d^2 - Dd. \tag{R18.3.2}$$

For all $L \ge 4$, retain short finite convolution tails to obtain a nonnegative upper bound $Y \ge Y_L^{\text{old}}$:

$$j = 1: \quad h = 2d/5, \quad Y = (1-3h)q + BV - h(1-d),$$
$$V = [G_{L+1} + z^{L-1}\mathfrak{r}_0 - h(2+z)](B)u, \qquad \mathbb{E}BV \le -h + 2hd; \tag{R18.3.3}$$

$$j \ge 2: \quad h = d/3, \quad Y = (1-2h)q + BV + d - h,$$
$$V = [G_{L+1} + z^{L-1}\mathfrak{r} - h](B)u, \qquad \mathbb{E}BV \le -h + hd. \tag{R18.3.4}$$

In the latter case $d - D = -d/2$ is not treated as nonnegative: the positive remainder $(D-d)\bar{f}_{L-1}$ was paid using the actual $\bar{f}_{L-1} \le 2$, producing the displayed d.

For $j = 1, L = 3$, $C = 1 - 7d/5 + D_1 \le 1 - 51d/40 \le 1 - d$. The exact mean in (R18.3.2) gives $m \le 1 - 2d$: the loss is at least $49/40 + 19/16 - \theta^2 - \theta^3 > 2$, using $\theta^2 < 1/6$, $\theta^3 < 1/12$. The core $G_4 - 2d/5$ only lowers a positive real coordinate, and $\theta(9/20) = 117/640 < 3/16$, so $\|V\|_2 < (19/16)\sqrt{d}$. As $-\tau + D_2 < 0$,

$$Y_3 \le (1-d)q + (19/16)\sqrt{d}, \quad C_a + Y_3 + m \le 3 - 4d + (191/80)\sqrt{d} \le 27/8.$$

The completed-square increment is $36481/102400 < 3/8$. The exact nonconstant polynomial is $P = G_5 - a_0 - (2/5)dz + z^3\mathfrak{r}_0$, where $a_0 = 7d/5 - D_1 \in [(51/40)d, (7/5+\theta^2)d]$. Its core has positive coordinates bounded by those of G_5, hence norm at most one. The tail satisfies $\|B^3 R_0\|_2 < (3/25)\sqrt{d} < \sqrt{d}/8$, so $\operatorname{Var} Y_3 \le (81/64)d$. For the covariance, $\operatorname{Re}(G_a G_5) \ge (1-x)^3 \ge \kappa^3$; subtracting $(2/5)dz$ is favorable. The actual scalar correction is at most $d^2 - \tau^2$, because $-a_0 d^2 + (2/5)\tau d^2 \le 0$. Thus the core covariance is at most $-\kappa^3 d + (177/80)d^2$, where $7/5 + \theta^2 + 1 - \kappa^2 = 177/80$. The tail costs at most $(9/8)(3/25)d = 27d/200$. Since $\kappa^3 > 1/5$, the normalized sum is below $177/640 + 27/200 - 1/5 = 677/3200 < 1/4$. Use covariance bound $d/4$ and curvature $17/2$ in Taylor expansion. After restoring $e^{-4\Gamma_0}$ the mean exponent is $35/4 + 57/32 + 8 = 593/32$, which may be lowered to $A = 35/2$. The remaining constants are $B = (17/2)(81/64) + 1 = 1505/128$, $C = 35/4$, $A - B = 735/128$, and $Ad/2 \le 35/32 < 2$. The deficit is therefore greater than $735d/1152 > d/5$.

For every finite $j \ge 2, L = 3$, the exact data give $C \le 1 - 41d/120$ and $m \le 1 - (89/80)d - d^2/2 \le 1 - d$, with $89/80 = 1/2 + 2\kappa - 19/120 - 1/4 - 1/6$. Now $V = [G_4 + d/2 + z^2\mathfrak{r}]u$, so $\|V\|_2 \le [1 + 1/16 + (3/5)\theta]\sqrt{d} < (21/16)\sqrt{d}$. Since $-\tau + D_2 \le -11d/32 < 0$,

$$Y_3 \le q + (21/16)\sqrt{d}, \quad C_a + Y_3 + m \le 3 - d - dp + (201/80)\sqrt{d} < 4.$$

The exact polynomial is $P = G_5 - a_0 + (d/2)z + z^3\mathfrak{r}$, $a_0 = d/2 - D_1 \in [(41/120)d, (49/60)d]$. $G_5 - a_0$ lowers a positive real coordinate and has norm at most one; $(d/2)\sqrt{\theta} < 1/24$ and $(3/5)\theta^{3/2} < 4/25$. Hence $\|P(B)u\|_2 \le (721/600)\sqrt{d} < (5/4)\sqrt{d}$ and $\operatorname{Var} Y_3 \le (25/16)d$. Retaining actual scalar means, the core covariance is at most

$$-\kappa^3 d + [49/60 + \theta/2 + 1 - \kappa^2]d^2 = -\kappa^3 d + (25609/15360)d^2 \le -(449/491520)d < 0.$$

The scalar correction is $\le d^2 - \tau^2$ and the added positive imaginary coordinate costs $\le \theta d^2/2$. The complete tail costs at most $(9/8)(4/25)d = 9d/50$; use covariance bound $d/5$ and curvature 13. The exponential payment has

$$A = 431/20 + 57/32 + 4 = 4373/160, \quad B = 13(25/16) + 4/5 = 1689/80, \quad C = 431/20.$$

$C \ge A/2$, $A - B = 199/32$, $Ad/2 \le 4373/2560 < 2$, giving deficit greater than $199d/288 > d/5$.

For unbounded finite L we need two finite polynomial bounds:

$$\|G_N - h(3 + 2z + z^2)\| \le 1 - 3h \quad (N \ge 6, 0 \le h \le 2/9), \tag{R18.6.1}$$

$$\|G_N - h(2 + z)\| \le 1 - 2h \quad (N \ge 5, 0 \le h \le 2/9). \tag{R18.6.2}$$

All finite G_N coordinates lie between $1-x$ and $1-x+x^2$ in these ranges, and shifted coordinates remain positive. For (R18.6.1), after subtracting $(1 - 3h)^2$ and dividing the square bound by x, one obtains $-1+4h-2h^2+(1-4h+h^2)x+(1-2h)x^2-x^3+x^4$. Its h derivative is at least $79/72$; at $h = 2/9$, $x \le 5/12$, it is at most $-181/3888 < 0$, as in Module R2-C017. For (R18.6.2), the corresponding expression is $-1 + 2h + h^2 + (1 - 2h)(x + x^2) - x^3 + x^4$. Its derivative $2 + 2h - 2(x + x^2) > 0$; at $h = 2/9$ use $x + x^2 \le 85/144$ to get $-41/81 + (5/9)(85/144) = -231/1296 < 0$. $x = 0$ is included by continuity.

For $j = 1, L \ge 4$, use (R18.3.3), $h = 2d/5$. Its mean is $m \le 1 - 5h + 3hd \le 1 - (37/20)d$. The nonconstant polynomial is $P = G_{L+2} - h(3 + 2z + z^2) + z^L\mathfrak{r}_0$. By (R18.6.1) its core norm is at most $1 - 3h$; the tail norm is at most $(9/20)\theta^2\sqrt{d} < (3/40)\sqrt{d}$. Thus $\operatorname{Var} Y \le (1849/1600)d$. The V bound in (R18.3.3) is

$$\|V\|_2/\sqrt{d} \le 1 + (9/20)\theta^{3/2} + h\sqrt{4 + \theta} < 1 + 3/25 + 21/200 = 49/40 < 5/4.$$

Together with the receiving bound this gives $C_a + Y + m \le 3 - (21/5)d + (49/20)\sqrt{d} \le 27/8$, whose completed-square increment is $2401/6720 < 3/8$. For the covariance core, $\operatorname{Re}(G_a G_{L+2}) \ge \phi$, $\operatorname{Re}[G_a(3+2z+z^2)] \le 3$, and the actual scalar correction is at most $d^2 - \tau^2 + 2hd^2$; indeed $\int Q_e \le d$ and $\int Q_o = \beta_{L+3} - 2hd \ge \tau - 2hd > 0$. The resulting core covariance is at most

$$-\phi(\theta)d + (11/5 - \kappa^2)d^2 + (4/5)d^3 < d/5,$$

with normalized upper bound $32629/163840 < 1/5$ on the full low range. The tail costs at most $27d/320$, so use total covariance $3d/10$. Taylor expansion gives the exponential payment with

$$A = 35/2 \le 35/4 + 57/32 + 37/5 = 2869/160,$$

$$B = (17/2)(1849/1600) + 6/5 = 35273/3200, \quad C = 35/4.$$

Here $A - B = 20727/3200$ and $Ad/2 \le 35/32 < 2$, giving a deficit greater than $20727d/28800 > d/5$ for every finite $L \ge 4$.

For all finite $j \ge 2, L \ge 4$, use (R18.3.4), $h = d/3$. Its actual mean is $m \le 1 + d - 4h + hd \le 1 - (7/24)d$. The nonconstant polynomial is $P = G_{L+2} - h(2+z) + z^L\mathfrak{r}$. Its core norm is at most $1 - 2h$ by (R18.6.2); the complete tail norm is at most $(3/5)\theta^2\sqrt{d} < \sqrt{d}/10$. Thus $\operatorname{Var} Y \le (121/100)d$. $G_{L+1} - h$ only lowers a positive real coordinate, and $(3/5)\theta^{3/2} < 4/25 < 1/5$, so $\|V\|_2 \le (6/5)\sqrt{d}$. Consequently $C_a + Y + m \le 3 - (23/24)d + (12/5)\sqrt{d} < 4$. The core covariance uses $\operatorname{Re}(G_a G_{L+2}) \ge \phi$ and $\operatorname{Re}[G_a(2+z)] \le 2$. Its actual scalar correction is at most $d^2 - \tau^2$, since $-2hd^2 + h\tau d \le 0$. Including the tail cost $9d/80$, the normalized covariance is at most

$$-1451/32768 + (2/3 + 1 - \kappa^2)/8 + 9/80 = 114271/491520 < 1/4.$$

With curvature 13, the payment constants are

$$A = 431/20 + 57/32 + 7/6 = 11759/480, \quad B = 13(121/100) + 1 = 1673/100, \quad C = 431/20.$$

$C \ge A/2$, $A - B = 18643/2400$, and $Ad/2 \le 11759/7680 < 2$, so the deficit is greater than $18643d/21600 > d/5$. Every subsequent pair is included in the one finite mask, without a bound on j, L, gaps, or the largest arm.

For $1/8 \le d \le 1/3$, Q5-C181, Section 6, gives in the same direction $F \le e^{-(2j-1)d}\bar{f}_2$. For $L \ge 3$, the $n = L + 2 \ge 5$ part of Module R2-C017 yields

$$\mathbb{E}[f_a(T^L F)^4] \le e^{-4(2j-1)d}\mathbb{E}[f_a \bar{f}_{L+2}^4] \le e^{-4d}(1 - d/2) \le e^{-4d} \le 1 - \frac{4d}{1+4d} \le 1 - 12d/7 \le 1 - d/5.$$

This joins the low range at $d = 1/8$. The disjoint integer cases $L = 1$, $L = 2$, and $L \ge 3$ give all positive-connector ranges and prove (R18.1.1). Decorations follow from positivity of T^L; $d = 0$ gives equality by original leaf integration; full reversal changes every original edge, including the reference arm and all decorations, together.

N.34 All odd terminals with nonunit shortest arm

Let the original residual list be $(\ell, s_1, t_1, \dots, s_j, t_j)$, with $2 \le \ell < s_1 < t_1 < \cdots < s_j < t_j$, $s_1 \ge 4$, and finite $j \ge 1$. Fix the actual terminal $F = \bar{f}_\ell \prod_{i=1}^{j} h_{s_i} f_{t_i}$, orient all connector edges toward the receiving root, and orient the receiving arm outward. We prove, for every independent $a \ge 3, L \ge 1$,

$$\mathbb{E}[f_a(T^L F)^4] \le 1 - d/5. \tag{R20.1.1}$$

Together with R2-C018, whose second shortest arm is three, this covers every odd residual list with at least three entries and shortest entry at least two. Actual equal-arm decorations can be restored in place.

Use the original finite receiving notation, exact covariance (R17.2.3), and centered Taylor identity (R17.2.4). In particular $\mathbb{E}C_a \le 1 - 57d/32$, $C_a \le (1-d)p + (6/5)\sqrt{d}$, $\sigma(C_a) \le (1+d)\sqrt{d}$. The cubic inequality (R12.4.1) gives the following continuous coefficients for $0 \le m = \mathbb{E}Y \le 1$:

$C_a + Y + m$ upper bound	4	7/2	18/5
curvature coefficient	$683/54 < 13$	$248/27 < 19/2$	$2453/250 < 10$.

In the low range $0 < d \le 1/8$, the exponential payment (R14.7.3) will use

	A	B	C	deficit lower bound
$j = 1$	$39/2$	$53/3$	$64/5$	$(11/54)d > d/5$
$j \ge 2$	29	22	$128/5$	$(7/9)d > d/5$.

(R20.2.4)

Indeed $C \ge A/2$, $Ad/2 \le 39/32$ or $29/16$, both below two, and $e^2 < 9$.

Here is the exact finite anchored input from Q5-C181, Section 3. Let R be the sum of the disjoint original power intervals of all actual $h_s f_t$ pairs, and let Γ be its actual loss. Put $\mu_\ell = \mathbb{E}\bar{f}_\ell \in [0,1]$, $D_i = \mathbb{E}B^i R$. Then $F \le \bar{f}_\ell e^{R-\Gamma}$, $|R|^2 < d/3$, and

row	Γ/d lower bound	ρ	c_0	c_1	c_2
$I:\ j = 1, s_1 = 4$	$69/20$	$1/5$	$72/125$	$1/6$	$7/100$
$II:\ j = 1, s_1 \ge 5$	$16/5$	$3/20$	$3/5$	$1/30$	$1/12$
$III:\ j \ge 2, s_1 \ge 4$	$16j/5$	$1/4$	$5/6$	$1/5$	$1/12$

(R20.3.table)

Here $\|R\|_2 < \rho\sqrt{d}$, $\delta_+/d \le c_0$, $(D_1)_+/d \le c_1$, and $(D_i)_+/d \le c_2$ for $i \ge 2$. All bounds hold for every finite mask and the whole $0 < d \le 1/3$. The actual first row and finite propagation are

$$T(\bar{f}_\ell e^R) \le \bar{f}_{\ell+1} + \mu_\ell BR + \delta,$$
$$e^\Gamma T^L F \le \bar{f}_n + \mu_\ell B^L R + \delta \bar{f}_{L-1} + \mu_\ell \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \quad n = \ell + L. \tag{R20.3.1}$$

δ may be negative. In positive propagation we may replace it by $D = c_0 d \ge \delta$, and each D_i only by its declared actual positive-part bound. Resulting upper bounds Y remain nonnegative. In row I the original ℓ is two or three, so $\beta_{\ell+1} = d$.

For $n = 3, 4, 5$ the correct finite bare recurrences are

n	$\bar{f}_n$	mean upper bound
3	$q - d + B(u + Bu)$	$1 - 2d$
4	$(1-d)q - d + BG_3 u$	$1 - 5d/2$
5	$(1-2d)q - \tau + B(G_4 - d)u$	$1 - 11d/4$.

Their variances are at most d and, uniformly, $\bar{f}_n \le q + (6/5)\sqrt{d} - \kappa d$. The exact short covariance formulas retained in Modules R2-C012, R2-C014 and R2-C017 give

$$\mathrm{Cov}(C_a, \bar{f}_n) \le d/12 \ (n = 3, 4, 5), \qquad \mathrm{Cov}(C_a, \bar{f}_5) \le d/32.$$

For three steps use nonpositivity at $a \ge 4$ and $d(55/1024 - d)$ at $a = 3$; for four steps use zero at $a = 3$ and $-57d/512 + 45d^2/32 \le 33d/512 < d/12$ otherwise. Two favorable short covariances, directly from (R17.2.3), are

$$\mathrm{Cov}(C_a, q) \le -\tau + d^2 < 0, \qquad \mathrm{Cov}(C_a, \bar{f}_2) \le -3d/16 + \theta d^2 < 0. \tag{R20.4.3}$$

For the second use $Q = 1 + z$, $\mathrm{Re}(G_a Q) \ge 1 - 2x \ge 3/16$, and scalar correction at most $d^2 - \tau d = d\delta_2$. Also $\sigma(\bar{f}_2)^2 = d + \delta_2 - d^2 \le (36/25)d$.

Suppose $n \le 5, L = 1$. Then $Y = \bar{f}_n + \mu_\ell BR + D$ has

$$m \le 1 - kd, \quad k = 2 - c_0 - c_1, \quad \mathrm{Var}\, Y \le (1 + \sqrt{\theta}\rho)^2 d,$$
$$\mathrm{Cov}(C_a, Y) \le [1/12 + (9/8)\sqrt{\theta}\rho]d.$$

The actual nonzero mean of BR is paid by D_1. The pointwise row bound gives $Y \le q + (6/5 + \rho)\sqrt{d} + (c_0 - \kappa)d$. All rows have $c_0 - \kappa - k < 0$, so $C_a + Y + m \le 3 + (12/5 + \rho)\sqrt{d} < 4$; the largest increment is $159/160 < 1$ because $\rho \le 1/4$, $\sqrt{d} < 3/8$. For single pairs, $\sqrt{\theta}\rho < 2/15$, covariance is at most $7d/30$, and the remainder coefficient is $13(17/15)^2 + 4(7/30) = 3967/225 < 53/3$. The mean exponent $4\Gamma/d + 57/32 + 4k$ has $k_I = 943/750$, $k_{II} = 41/30$ and exceeds $39/2$ (row II gives $9623/480$). For multiple pairs $k = 29/30$, $\sqrt{\theta}\rho < 1/6$, the remainder is at most $13(7/6)^2 + 4(13/48) = 169/9 < 22$, and the mean exponent is $14999/480 > 29$. Thus (R20.2.4) applies.

If $n \le 5, L = 2$, then $n = 4, 5$, $\ell = 2, 3$, and use $Y = \bar{f}_n + \mu_\ell B^2 R + Dq + \mu_\ell (D_1)_+$. Here $m \le 1 - kd$, $k = 5/2 - c_0 - c_1 - c_2$: $k_I = 2531/1500 > 5/3$, $k_{II} = 107/60 > 5/3$, $k_{III} = 83/60$. The standard deviation is at most $[1 + \theta\rho + c_0 d]\sqrt{d}$, bounded by $7\sqrt{d}/6$, $23\sqrt{d}/20$, and $5\sqrt{d}/4$ in the three rows. For row I use $\theta/5 < 1/12$, $(72/125)d \le 9/125$; for II use $3\theta/20 < 1/16$; for III use $\theta/4 < 5/48$, $(5/6)d \le 5/48$. By (R20.4.3), covariance is at most $d/5$ for single pairs and $d/4$ for multiple pairs. The stronger short bound gives $Y \le (1 - d + D)q + (1 + \sqrt{\theta}\rho)\sqrt{d} + (c_1 - \kappa)d$. For single pairs, $c_0 \le 3/5$, $c_1 \le 1/6$, $k \ge 5/3$ give $C_a + Y + m \le 3 - (463/160)d + (7/3)\sqrt{d} \le 7/2$; the full increment maximum is $1960/4167 < 1/2$. For multiple pairs the d coefficient remains negative and the $\sqrt{d}$ coefficient is $71/30$, giving sum below four. The remainder is at most $(19/2)(49/36) + 4/5 = 4943/360 < 14 < 53/3$ for single pairs, and $13(25/16) + 1 = 341/16 < 22$ for multiple pairs. The mean exponents are at least $64/5 + 57/32 + 20/3 > 39/2$ and $128/5 + 57/32 + 83/15 > 29$, respectively.

The only remaining $n \le 5, L = 3$ case is $n = 5, \ell = 2$. Take $Y = \bar{f}_5 + \mu_\ell B^3 R + D\bar{f}_2 + \mu_\ell (D_1)_+ q + \mu_\ell (D_2)_+$. Then $m \le 1 - kd$, $k = 11/4 - c_0 - c_1 - 2c_2$: $k_I = 2801/1500 > 11/6$, $k_{II} = 39/20 > 11/6$, $k_{III} = 31/20$. Its standard deviation is at most $[1 + \theta^{3/2}\rho + (6/5)c_0 d + c_1 d]\sqrt{d}$, bounded by the same three row values $7/6, 23/20, 5/4$. The respective increments are below $4/75 + 7/80 + 1/48 < 1/6$, $1/25 + 9/100 + 1/240 < 3/20$, and $1/15 + 1/8 + 1/40 < 1/4$. The added $\bar{f}_2, q$ terms have nonpositive covariance. The remaining tail covariance costs at most $(9/8)\theta^{3/2}\rho d \le 3d/40$; with $d/32$ use total $d/8$. For single pairs, $D + (D_1)_+ \le (23/30)d < d$ and the actual five-step bound gives $C_a + Y + m \le 3 - 4d + (7/3)\sqrt{d} \le 7/2$. Here the negative coefficient is at least $2 + \kappa + 11/6 - 1/12 > 4$, the square-root coefficient is below $11/5 + \theta/5 < 7/3$, and the increment maximum is $49/144 < 1/2$. For multiple pairs the q coefficient is at most $1 - (29/30)d$, $-\tau + (D_2)_+ < 0$, so $C_a + Y + m \le 3 + (11/5 + 5/48)\sqrt{d} < 4$. The remainders are below 14 and at most $13(25/16) + 1/2 < 22$; the mean exponents are at least $64/5 + 57/32 + 22/3 > 39/2$ and $128/5 + 57/32 + 31/5 > 29$. This exhausts $n \le 5$.

For every finite $n \ge 6$, rewrite the actual propagated upper bound as

$$Y_{\text{old}} = 1 + G_n(B)u + \mu_\ell B^L R - \sum_{r=3}^{n} \gamma_r \bar{f}_{n-r}, \quad \gamma_r = \begin{cases} \beta_r, & r < \ell + 1, \\ \beta_r - D, & r = \ell + 1, \\ \beta_r - \mu_\ell D_{r-\ell-1}, & r > \ell + 1. \end{cases}$$

Use the row parameters

$$\begin{array}{c|cc} I & h = 2d/5 & E = 0 \\ II & h = d/2 & E = 81d/80 \\ III & h = 3d/8 & E = 59d/48. \end{array}$$

Except for the one index $r = \ell + 1$, all $\gamma_r \ge h$ and the actual mean coefficient $\beta_{n+1} - \mu_\ell D_L \ge h$. For I, $\beta_{\ell+1} = d$ gives $d - D \ge 53d/125 > 2d/5$, and the other tight coefficient is $\kappa d - d/6 =$

$41d/96 > 2d/5$. For II and III the nonexceptional bounds are $\kappa d - d/12 = 49d/96 > d/2$ and $\kappa d - d/5 = 63d/160 > 3d/8$. The exceptional coefficient in II and III can be raised to h at cost at most $(81d/160)\bar{f}_{L-1}$ or $(59d/96)\bar{f}_{L-1}$. The actual bound $\bar{f}_{L-1} \le 2$ pays exactly the stated E, even if the index is outside the last four terms. Retaining only those four actual terms then gives

$$e^{\Gamma}T^{L}F \le Y = 1 - h(4-d) + Q_{h,n}(B)u + \mu_{\ell}B^{L}R + E \ge 0, \quad Q_{h,n} = G_n - h(3 + 2z + z^2). \tag{R20.8.1}$$

The entire finite bound (R18.6.1) gives $\|Q_{h,n}\| \le 1 - 3h$. Retaining D_L in the actual mean yields $m \le 1 - 5h + 3hd + E \le 1 - kd$ with $k_I = 37/20$, $k_{II} = 13/10$, $k_{III} = 97/192$. The variance row bounds are $(17/15)^2 d$, $(11/10)^2 d$, and $(7/6)^2 d$, using only $L \ge 1$, $\mu_{\ell} \le 1$.

The pointwise form of (R20.8.1) is $Y = (1 - 3h)q + BV - h(1-d) + E$, $V = [G_{n-1} - h(2 + z)]u + \mu_{\ell}B^{L-1}R$. Thus $\|V\|_2 \le [1 + \rho + h\sqrt{4+\theta}]\sqrt{d}$, with row bounds $4/3$, $41/32$, $7/5$. They give respectively

I	$C_a + Y + m \le 3 - (21/5)d + (38/15)\sqrt{d} \le 7/2$	$19/2$
II	$C_a + Y + m \le 3 - (109/40)d + (5/2)\sqrt{d} \le 18/5$	10
III	$C_a + Y + m \le 3 - (77/48)d + (13/5)\sqrt{d} < 4$	$13.$

The final column is the curvature coefficient. The first two full increment maxima are $361/945 < 1/2$, $125/218 < 3/5$; the third is at most $39/40 < 1$ using $\sqrt{d} < 3/8$. For $\phi(x) = 1 - 3x + 2x^2 - x^3$, $\phi(\theta) = 1451/32768$, $\mathrm{Re}(G_a G_n) \ge \phi$ and $\mathrm{Re}[G_a(3 + 2z + z^2)] \le 3$. The shifted core has $\int Q_e \le d$ and $\int Q_o = \beta_{n+1} - 2hd \ge \tau - 2hd > 0$. Its actual scalar correction is at most $d^2 - \tau^2 + 2h\tau d$, so

$$\mathrm{Cov}(C_a, Q_{h,n}(B)u) \le -\phi(\theta)d + [3(h/d) + 1 - \kappa^2]d^2 + 2(h/d)d^3.$$

The normalized bound in I is $32629/163840 < 1/5$; in II it is $7857/32768 < 1/4$; III is smaller than I. Add the complete tail costs $3d/20$, $9d/80$, and $3d/16$, respectively. The total covariance is at most $3d/8$ in I and II, and $2d/5$ in III. The constant E has no covariance. The row mean exponents are $3677/160 > 39/2$, $633/32 > 39/2$, and $14113/480 > 29$; the remainders are at most

$$(19/2)(289/225) + 3/2 = 3083/225 < 53/3,$$
$$10(121/100) + 3/2 = 68/5 < 53/3,$$
$$13(49/36) + 8/5 = 3473/180 < 22.$$

Since $\Gamma \ge 16d/5$ in I,II and $\Gamma \ge 32d/5$ in III, (R20.2.4) proves the full long low range.

For $1/8 \le d \le 1/3$, the same actual envelope $|R|^2 < d/3$ gives $|R| \le (5/3)d$, $\Gamma \ge (16/5)jd$, and $F \le e^{-(23/15)d}\bar{f}_{\ell} \le e^{-d}\bar{f}_{\ell}$. Thus $T^L F \le e^{-d}\bar{f}_n$, $n = \ell + L \ge 3$. The three-step Module R2-C012, four-step Module R2-C014, and $n \ge 5$ Module R2-C017 give $\mathbb{E}[f_a \bar{f}_n^4] \le 1$, so

$$\mathbb{E}[f_a(T^L F)^4] \le e^{-4d} \le 1 - \frac{4d}{1+4d} \le 1 - 12d/7 \le 1 - d/5.$$

This joins at $d = 1/8$. If the second shortest residual arm is three, the shortest must be two and Module R2-C018 applies. Otherwise the family above applies, proving the announced complete nonunit odd terminal union. Directions use only this structural distinction. At $d = 0$ all actual messages are one by original leaf integration, without implying $K = 0$. Positivity restores all original equal pairs; full reversal reverses the receiving arm, every terminal arm, each connector edge, and all decorations simultaneously.

N.35 Unit shortest arm and second shortest arm at least four

For the residual list $(1, s_1, t_1, \dots, s_j, t_j)$, $4 \le s_1 < t_1 < \cdots < s_j < t_j$, fix, for every finite $j \ge 2$, $F = q \prod_{i=1}^{j} h_{s_i} f_{t_i}$. The unit arm points toward the terminal, every h_s has its first edge toward the terminal and the rest toward the leaf, and every f_t points outward. All $L \ge 1$ connector edges point toward the receiving root and its arm $a \ge 3$ points outward. We prove

$$\mathbb{E}[f_a (T^L F)^4] \le 1 - d/5. \tag{R21.1.1}$$

For $j = 1$, use the orientation $q f_s h_t$ from the short reference table Section N.43/D_{full}, with deficit $d/3000$. Choosing between these two directions by the original j gives the entire family $j \ge 1$ with bound $1 - d/3000$.

Use the receiving bounds and exact centered identities (R17.2.3)–(R17.2.4). For $m \le 1$, $C_a + Y + m \le 4$, the full continuous cubic proof in (R12.4.1) gives curvature coefficient $683/54 < 13$. In the low range $0 < d \le 1/8$ every row will satisfy

$$\mathbb{E}[f_a (T^L F)^4] \le e^{-(55/2)d} + (51/2) d e^{-(128/5)d}. \tag{R21.2.3}$$

By (R14.7.3), its deficit is at least $2de^{-(55/4)d} > 2d/9 > d/5$, since $(55/4)d \le 55/32 < 2$.

The finite mask R in Q5-C181, Section 3, depends only on the pairs, so the same actual terminal satisfies

$$F \le q e^{R-\Gamma}, \quad \Gamma \ge (16/5) j d \ge 32 d/5, \quad \|R\|_2 < (1/4)\sqrt{d}, \quad |R|^2 < d/3,$$
$$D_0 = \mathbb{E}R \le d/12, \quad (D_1)_+ \le d/5, \quad (D_i)_+ \le d/12 \ (i \ge 2), \quad D_i = \mathbb{E}B^i R.$$

The first row must be proved for the actual unit anchor q. As $|R| \le 1/3$, $e^R \le 1 + R + (3/5)R^2$. Using $T(qR) = D_0 + BR + T(uR)$, $T(uR) \le 2\mathbb{E}|uR|$, and $T(qR^2) \le 4\mathbb{E}R^2$, we obtain

$$T(qe^R) \le \bar{f}_2 + BR + D_0 + 2\sqrt{d}\|R\|_2 + (12/5)\|R\|_2^2 \le \bar{f}_2 + BR + D, \qquad D = 11d/15. \tag{R21.3.1}$$

The coefficient is $1/12 + 1/2 + 3/20 = 11/15$. Finite propagation, with the exact $\mathbb{E}q = 1$, gives $n = L + 1$ and

$$e^{\Gamma} T^L F \le \bar{f}_n + B^L R + D \bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{R21.3.2}$$

Only declared positive parts will be raised in this expression.

For the short paths the exact bounds are

n	pointwise upper bound	mean upper bound	variance
2	$q + \sqrt{d}$	$1 - d$	$(45/32)d$
3	$q + (6/5)\sqrt{d} - d$	$1 - 2d$	d
4	$(1-d)q + \sqrt{d} - d$	$1 - 5d/2$	d
5	$(1-2d)q + (1-d)\sqrt{d} - \tau$	$1 - 11d/4$	d.

The corresponding receiving covariances are at most $d/12, d/12, d/32$ for $n = 3, 4, 5$, respectively; the sharper three-step bound is $(55/1024)d$. They follow from the exact finite formulas in Modules R2-C012, R2-C014 and R2-C017. In addition,

$$\mathrm{Cov}(C_a, q) \le -\tau + d^2 < 0, \qquad \mathrm{Cov}(C_a, \bar{f}_2) \le -3d/16 + \theta d^2 \le -(5/96)d \le 0. \tag{R21.4.1}$$

The latter holds on the full domain $d \le 1/3$: use $\mathrm{Re}[G_a(1+z)] \ge 1-2x \ge 3/16$ and actual scalar correction at most $d\delta_2$. Also $\sigma(\bar f_2) \le 6\sqrt{d}/5$ and $\sigma(q) = \sqrt{d}$.

For $L = 1$, take $Y = \bar f_2 + BR + D$. Then $m \le 1 - d/15$, $\mathrm{Var}\, Y \le (6/5 + 1/6)^2 d = (1681/900)d$, and covariance at most $(9/8)(1/6)d = 3d/16$. The row bound uses $|BR| \le \|R\|_2 < \sqrt{d}/4$, giving $Y \le q + (5/4)\sqrt{d} + 11d/15$ and

$$C_a + Y + m \le 3 - dp + (49/20)\sqrt{d} + (2/3)d < 4.$$

Indeed $\sqrt{d} \le 17/48$ since $289/2304 > 1/8$, and the positive increment is at most $833/960 + 1/12 = 913/960 < 1$. The Taylor remainder is at most $13(1681/900) + 3/4 = 5632/225 < 51/2$; the mean exponent is at least $4\Gamma/d + 57/32 + 4/15 \ge 13271/480 > 55/2$. This proves (R21.2.3) in the shortest case without any bare-two-step contraction assumption.

For $L = 2$, take $Y = \bar f_3 + B^2 R + Dq + (D_1)_+$. The actual mean is at most $1 - (59/60)d$ and

$$\sigma(Y) \le [1 + \theta/4 + (11/15)d]\sqrt{d} < [1 + 5/48 + 11/120]\sqrt{d} < (6/5)\sqrt{d}.$$

The Dq covariance is nonpositive; the remainder is at most $d/12 + (9/8)(5/48)d < d/4$. Since $|B^2 R| \le \|BR\|_2 < \sqrt{d}/6$, $Y \le (1+D)q + (41/30)\sqrt{d} - (4/5)d$ and $C_a + Y + m \le 3 - (19/60)d + (77/30)\sqrt{d} < 4$, using increment $77/80 < 1$. The Taylor remainder is at most $13(36/25) + 1 = 493/25 < 51/2$; the mean exponent is at least $128/5 + 57/32 + 59/15 > 55/2$.

For $L = 3$, take $Y = \bar f_4 + B^3 R + D\bar f_2 + (D_1)_+ q + (D_2)_+$. Its mean is at most $1 - 7d/5$ and, since $\theta^{3/2} < 4/15$, $\sigma(Y) \le [1 + 1/15 + (22/25 + 1/5)d]\sqrt{d} < (5/4)\sqrt{d}$. The added $\bar f_2, q$ covariances are nonpositive, and the remainder is at most $d/12 + 3d/40 < d/5$. The q coefficient is at most $1 - d/15$, the square-root coefficient at most $1 + 11/120 + 5/48 = 287/240 < 6/5$, and the scalar remainder is at most $-11d/12$. Hence $C_a + Y + m < 3 + (12/5)\sqrt{d} < 4$. The remainder is at most $13(25/16) + 4/5 = 1689/80 < 51/2$; the mean exponent is at least $128/5 + 57/32 + 28/5 > 55/2$.

For $L = 4$, retain every short addition: $Y = \bar f_5 + B^4 R + D\bar f_3 + (D_1)_+ \bar f_2 + (D_2)_+ q + (D_3)_+$. Retaining D_4 in the actual mean gives $m \le 1 - (11/4)d + D + d/5 + 3d/12 = 1 - (47/30)d$. Also $\|B^4 R\|_2 < \sqrt{d}/24$ and

$$\sigma(Y) \le [1 + 1/24 + (11/15 + 6/25 + 1/12)d]\sqrt{d} = [1 + 1/24 + (317/300)d]\sqrt{d} < (6/5)\sqrt{d}.$$

The $\bar f_2, q$ additions have nonpositive covariance, but the $\bar f_3$ addition is paid explicitly:

$$\mathrm{Cov}(C_a, Y) \le d/32 + 3d/64 + (11d/15)(55d/1024) \le [5/64 + 605/122880]d < d/10.$$

The correct short expansions give q coefficient at most $1 - 59d/60$, square-root coefficient at most $16/15 + (2/25)d \le 323/300 < 6/5$, and scalar remainder at most $-\tau + (D_3)_+ < 0$. Thus $C_a + Y + m < 4$. The remainder is at most $13(36/25) + 2/5 < 51/2$; the mean exponent is at least $128/5 + 57/32 + 94/15 > 55/2$.

For all $L \ge 5$, $n = L + 1 \ge 6$, the positive term $D\bar f_{L-1}$ has convolution index two and must be paid separately: use the actual $\bar f_{L-1} \le 2$ at cost $E = 2D = 22d/15$. For $r \ge 3$ the remaining coefficients are $\gamma_r = \beta_r - D_{r-2}$, with $\gamma_3 \ge 4d/5$, $\gamma_4 \ge 11d/12$, and $\gamma_r \ge 49d/96 > d/2$ for $r \ge 5$. The actual mean coefficient $\gamma_{n+1} = \beta_{n+1} - D_L$ obeys the same bound. Put $h = d/2$ and retain the last four actual messages:

$$e^{\Gamma} T^L F \le Y = 1 - h(4-d) + Q_{h,n}(B)u + B^L R + E \ge 0, \quad Q_{h,n} = G_n - h(3 + 2z + z^2). \tag{R21.9.1}$$

The full finite certificate (R18.6.1) gives $\|Q_{h,n}\| \le 1-3h$ for every $n \ge 6$. The actual mean and variance satisfy $m \le 1-5h+3hd+E \le 1-(203/240)d$ and $\operatorname{Var} Y \le (49/36)d$. The pointwise form is $(1-3h)q + BV - h(1-d) + E$ with $V = [G_{n-1} - h(2+z)]u + B^{L-1}R$ and $\|V\|_2 \le [1+1/4+(d/2)\sqrt{4+\theta}]\sqrt{d} < (7/5)\sqrt{d}$. Thus $C_a + Y + m \le 3 - (109/60)d + (13/5)\sqrt{d} < 4$. As in the retained-mean calculation of Module R2-C020,

$$\operatorname{Cov}(C_a, Q_{h,n}(B)u) \le -\phi(\theta)d + (5/2-\kappa^2)d^2 + d^3 \le (7857/32768)d < d/4,$$

where $\phi = 1-3x+2x^2-x^3$. The actual scalar correction is $\le d^2 - \tau^2 + 2h\tau d$, since $\int Q_e \le d$, $\int Q_o \ge \tau - 2hd > 0$; $\operatorname{Re}(G_a G_n) \ge \phi$ and $\operatorname{Re}[G_a(3+2z+z^2)] \le 3$. The complete tail costs at most $3d/16$, so total covariance is at most $d/2$. The remainder is at most $13(49/36)+2 = 709/36 < 51/2$; the mean exponent is at least $128/5 + 57/32 + 203/60 > 55/2$. This proves (R21.2.3) for every positive L.

For $1/8 \le d \le 1/3$, the same finite mask gives $|R| \le 5d/3$, so $\Gamma - |R| \ge 71d/15 > 4d$ and $F \le e^{-4d}q$, $T^L F \le e^{-4d}\bar{f}_{L+1}$. For $L \ge 2$, the precise bare assertions of Modules R2-C012, R2-C014 and R2-C017 give $\mathbb{E}[f_a \bar{f}_{L+1}^4] \le 1$; e^{-16d} then suffices. For $L = 1$ instead use the actual $\bar{f}_2 \le 2$ and the full-domain covariance in (R21.4.1):

$$\mathbb{E}[f_a \bar{f}_2^4] \le 8\mathbb{E}[C_a \bar{f}_2] \le 8(1-57d/32)(1-d).$$

On this range the last two factors are at most $4/5, 7/8$ and $e^{-16d} \le e^{-2} \le 3/19$, since $e^2 \ge 1+2+2+4/3 = 19/3$. Hence $\mathbb{E}[f_a(TF)^4] \le 84/95 < 14/15 \le 1-d/5$. The high and low ranges meet at $d = 1/8$ with identical directions. At $d = 0$ use original leaf integration; positivity restores all equal pairs. The $j = 1$ union uses $q f_s h_t$, while $j \ge 2$ uses $q \prod h_s f_t$ as stated. Full reversal changes every original edge, receiving weight, and decoration together.

N.36 Even terminals whose shortest residual arm is at least four

Let $4 \le s_1 < t_1 < \cdots < s_j < t_j$ be any finite even residual list, $j \ge 1$. Fix $F = \prod_{i=1}^{j} h_{s_i} f_{t_i}$, all $L \ge 1$ connector edges toward the receiving root, and its arm $a \ge 3$ outward. For every independent choice of these integers,

$$\mathbb{E}[f_a(T^L F)^4] \le 1 - d/100, \qquad \mathbb{E}[E_o f_a \{T^L(E_t F)\}^4] \le 1 - d/100. \tag{R22.1.1}$$

Use the finite receiving bounds of R2-C007, the covariance and Taylor identities (R17.2.3)–(R17.2.4), and $\kappa = 19/32$. The favorable full-domain bounds are

$$\operatorname{Cov}(C_a, q) \le -\kappa d + d^2 < 0, \quad \operatorname{Cov}(C_a, \bar{f}_2) \le -3d/16 + \theta d^2 \le -5d/96.$$

For the second, $\operatorname{Re}(G_a G_2) \ge 1 - 2x \ge 3/16$ and the actual scalar correction is at most $d\delta_2$. The true G_2 norm is $\sqrt{1+x}$, rather than one. For $C_a + Y + m \le 18/5$, the continuous cubic inequality (R12.4.1) gives curvature $2453/250 < 10$ if $m \le 1$, and $162189/16000 < 41/4$ if $m \le 21/20$. The low-range exponential payments use (R14.7.3); all have $d \le 1/16$ and $Ad/2 < 1$, so the deficit exceeds $(A-B)d/3$.

(L96.2.5), in the actual stated direction, gives

$$h_4 f_t \le e^{R_{4,t} - 69d/20}, \quad h_s f_t \le e^{R_{s,t} - 16d/5} \ (s \ge 5),$$

$$R_{s,t} = \sum_{r=s+1}^{t} K^r \mathbf{1}.$$

Thus $F \le e^{R-\Gamma}$ for one complete finite disjoint-interval mask R, $\Gamma \ge 16jd/5$; retain $\Gamma \ge 69d/20$ for $j = 1, s_1 = 4$. For a finite zero–one polynomial, separating the positive and negative subsum in each real and imaginary coordinate and taking the larger absolute value gives

$$|Q(i\sqrt{x})|^2 \le \frac{1+x}{(1-x^2)^2} \le 32768/16245 < 101/50.$$

The infinite geometric sum only bounds the actual finite mask. Writing $R = BV_0$, its smallest original power is five, so

$$\begin{gathered}\|V_0\|_2^2 \le (101/50)\theta^3 d < d/7, \quad \sigma := \mathbb{E}R^2 \le (101/50)\theta^4 d < d/18, \\ |R|^2 < d/7 \le 1/21 < 1/16.\end{gathered} \tag{R22.3.3}$$

Hence $|R| < 1/4$. The positive Taylor-tail ratio is at most $1/12$, and the integral remainder treats negative R, giving $e^R \le 1 + R + (5/8)R^2$. The original signed moments $D_i = \mathbb{E}B^i R$ obey

$$D_0 \in [-d/5, d/12], \quad D_1 \in [-d/12, d/5], \quad D_i \le d/12 \ (i \ge 2). \tag{R22.3.4}$$

Indeed $\mathbb{E}K^{2r}\mathbf{1} = (-1)^r \delta_r$ and odd moments vanish. Same-sign moment indices are spaced by two, giving the sharper bounds $D_0 \in [-C_\beta\delta_3, C_\beta\delta_4]$, $D_1 \in [-C_\beta\delta_4, C_\beta\delta_3]$, $D_i \le C_\beta\delta_4$ for $i \ge 2$, where $C_\beta = 1024/855$. Here $C_\beta\theta^2 = 169/855 < 1/5$ and $C_\beta\theta^3 = 2197/27360 < 1/12$. The actual first row and propagation are

$$\begin{gathered}T(e^R) \le q + BR + D_0 + (5/4)\sigma \le q + BR + D, \quad D = 11d/72, \\ e^\Gamma T^L F \le Y_L = \bar{f}_L + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \ge 0.\end{gathered} \tag{R22.3.6}$$

Individual signed summands need not be nonnegative. The finite convolution coefficients satisfy $\gamma_r = \beta_r - D_{r-1} \ge d/2$ for $r \ge 3$ because $\kappa - 1/12 = 49/96 > 1/2$; indices one and two will be paid separately. From (R22.3.3) and $\|B\|^2 \le \theta$,

i	1	2	3	4	5	6
$\|B^i R\|_2/\sqrt{d}$ upper bound	$5/32$	$1/10$	$1/16$	$1/25$	$1/40$	$1/60$.

(R22.3.8)

These are rational-square comparisons with $\theta^i/18$; all later finite powers only reduce the bound.

On $0 < d \le 1/16$, first let $L = 1$ and $Y = q + BR + D$. Its actual mean satisfies $1 + 5d/72 \le m \le 1 + 127d/360 \le 21/20$, and $\operatorname{Var} Y \le (1369/1024)d$. The covariance is at most $-15d/32 + (9/8)(5/32)d = -75d/256$. Since $m \ge 1$, the negative Taylor term is at most $-75d/64$. The row bound $Y \le q + (6/25)\sqrt{d} + D$ gives $C_a + Y + m \le 3 + (36/25)\sqrt{d} + (91/180)d \le 18/5$; the increment is at most $9/25 + 91/2880 < 3/5$. Using curvature $41/4$, the exponential constants are

$$\begin{gathered}A = 18965/1440 = 64/5 + 57/32 - 4(127/360), \\ B = 51329/4096 = (41/4)(1369/1024) - 75/64, \quad C = 64/5.\end{gathered}$$

$C \ge A/2$, $A - B = 117715/184320 > 3/5$, and $Ad/2 < 1$, so the deficit exceeds $d/5$.

For $L = 2$, use the actual $Y = \bar{f}_2 + B^2 R + Dq + D_1$. Its mean satisfies $m \le 1 - 203d/360$. Since $|D_2| = |\langle B^2\mathbf{1}, R\rangle| \le (5/32)d$ and $D_1 \ge -d/12$, it also satisfies $m \ge 1 - (313/288)d \ge 9/10$. The

actual random vector is $(G_2+D)(B)u+B^2R$; $\|G_2\|\le\sqrt{45/32}<19/16$, $D\le 11/1152<1/100$ and (R22.3.8) give $\operatorname{Var} Y\le(169/100)d$. The Dq covariance is nonpositive, and

$$\operatorname{Cov}(C_a,\bar f_2)\le-(83/512)d,\quad |\operatorname{Cov}(C_a,B^2R)|\le(17/16)(1/10)d,$$
$$\operatorname{Cov}(C_a,Y)\le-(143/2560)d.$$

Therefore $4m^3\operatorname{Cov}(C_a,Y)\le-3d/20$, since $4(9/10)^3(143/2560)=416988/2560000>3/20$. The row bound $Y\le(1+D)q+(37/32)\sqrt{d}+d/5$ gives $C_a+Y+m\le 3-(7/120)d+(377/160)\sqrt{d}\le 3+377/640<18/5$. With curvature ten, the payment constants are

$$A=4849/288=64/5+57/32+4(203/360),\quad B=67/4=10(169/100)-3/20,\quad C=64/5.$$

$A-B=25/288$, $C\ge A/2$, $Ad/2<1$, giving deficit $>25d/864>d/100$.

For $L=3,4,5$ replace signed coefficients only before nonnegative actual path factors, obtaining the nonnegative upper bounds

$$\begin{aligned}\overline{Y}_3&=\bar f_3+B^3R+D\bar f_2+(d/5)q+d/12,\\ \overline{Y}_4&=\bar f_4+B^4R+D\bar f_3+(d/5)\bar f_2+(d/12)q+d/12,\\ \overline{Y}_5&=\bar f_5+B^5R+D\bar f_4+(d/5)\bar f_3+(d/12)\bar f_2+(d/12)q+d/12.\end{aligned}\tag{R22.6.1}$$

Their means are the actual means after these replacements. Use the correct short expansions already displayed in Module R2-C021; in this range $\mathbb{E}\bar f_4\le 1-83d/32$, $\|G_4-d\|\le 1$ and $\|P_5\|\le 1-2d$, where $P_5=G_5-d(2+z)$ and $|P_5|^2-(1-2d)^2=x[-1+2d+d^2+(1-2d)x-x^2+x^3]\le 0$. The signed mean bounds and whole-vector norm bounds give

L	$\mathbb{E}\overline{Y}_L$ upper bound	random-vector norm$/\sqrt{d}$ upper bound
3	$1-533d/360$	$1+(23/60)d+1/16<11/10$
4	$1-19d/10$	$1+(857/1800)d+1/25<11/10$
5	$1-2d$	$1+(193/360)d+1/40<11/10$.

The mean losses are respectively $2-11/72-1/5-1/6=533/360$, $83/32-11/72-1/5-1/4=2867/1440>19/10$, and $11/4-11/72-1/5-1/3=743/360>2$. The vector estimates retain the true G_2 norm bound $6/5$. Their actual pointwise bounds are

$$\begin{aligned}\overline{Y}_3&\le(1+127d/360)q+(21/16)\sqrt{d}-11d/12,\\ \overline{Y}_4&\le(1-203d/360)q+(11/10)\sqrt{d}-11d/12,\\ \overline{Y}_5&\le(1-533d/360)q+(11/10)\sqrt{d}-\tau+d/12.\end{aligned}$$

For the last two square-root coefficients use $17/16+(23/60)d<11/10$ and $1+1/25+(857/1800)d<11/10$. For $L=3$, $C_a+\overline{Y}_3+m\le 3-(203/120)d+(201/80)\sqrt{d}<18/5$: the increment increases for $t=\sqrt{d}\le 1/4$ and has endpoint $1003/1920<3/5$. For $L=4,5$ discard favorable negative terms to obtain $3+(23/10)\sqrt{d}\le 3+23/40<18/5$. All three rows therefore have curvature ten and the common bounds

$$m\le 1-7d/5,\quad \operatorname{Var}\overline{Y}_L\le 121d/100,$$
$$\operatorname{Cov}(C_a,\overline{Y}_L)\le(6/5)d.$$

The covariance upper bound is nonnegative, so $m^3 \le 1$ is legal. Payment uses $A = 20$, $B = 169/10 = 10(121/100) + 24/5$, $C = 64/5$; the actual mean exponent $64/5 + 57/32 + 28/5$ exceeds twenty. $A - B = 31/10$, $C \ge A/2$, $Ad/2 < 1$, so the deficit exceeds $31d/30$.

For every finite $L \ge 6$, indices one and two in (R22.3.6) cannot be absorbed into γ_r. Pay their actual positive cost explicitly:

$$D\bar{f}_{L-1} + D_1\bar{f}_{L-2} \le E_0 = (127/180)d,$$

using $\bar{f}_n \le 2$. All remaining $\gamma_r \ge h = d/2$. Retain the four last actual paths, whose earliest original index is $L - 3 \ge 3$:

$$\widehat{Y} = 1 - h(4 - d) + [G_L - h(3 + 2z + z^2)](B)u + B^L R + E_0. \tag{R22.7.2}$$

The actual mean coefficient $\gamma_{L+1} \ge h$ is also retained, so $m \le 1 - 5h + 3hd + E_0 \le 1 - 8d/5$; its linear loss is at least $323/180 - 3/32 = 2449/1440 > 8/5$. The full finite polynomial certificates (R18.6.1)–(R18.6.2) give random-vector norm at most $(1 - 3h + 1/60)\sqrt{d} < (11/10)\sqrt{d}$. The pointwise form is

$$\widehat{Y} = (1 - 3h)q + BV - h(1 - d) + E_0,$$
$$V = [G_{L-1} - h(2 + z)](B)u + B^{L-1}R, \quad \|V\|_2 \le (1 - 2h + 1/40)\sqrt{d}.$$

Since $dp + 3hq \ge 2d$ and $-2d - h(1-d) + E_0 \le 0$, $C_a + \widehat{Y} + m \le 3 + (89/40)\sqrt{d} \le 3 + 89/160 < 18/5$. Thus the preceding common variance, covariance and payment bounds also handle every finite $L \ge 6$.

It remains to treat $d \ge 1/16$ without changing F. For a single pair with $s_1 = 4$, its original interval and $\|G_m\| \le 6/5$ give $|R| \le (6/5)\theta^{3/2}\sqrt{d} < (8/25)\sqrt{d} \le 32d/25$, so $\Gamma - |R| > 2d$ using $\Gamma \ge 69d/20$. For $s_1 \ge 5$, $|R| \le (6/5)\theta^2\sqrt{d} < \sqrt{d}/5 \le 4d/5$ gives the same conclusion. For $j \ge 2$, $|R| < \sqrt{d/7} \le 4d/\sqrt{7} < 8d/5$ and $\Gamma \ge 32d/5$. Consequently the same actual directions satisfy

$$0 \le F \le e^{-2d}, \qquad T^L F \le e^{-2d}\bar{f}_L. \tag{R22.8.1}$$

For $L \ge 3$, the precise bare union of Modules R2-C012, R2-C014 and R2-C017 gives $\mathbb{E}[f_a \bar{f}_L^4] \le 1 - 3d/40$. The target is at most $e^{-8d} \le 1 - 8d/(1 + 8d) \le 1 - 24d/11 < 1 - d/100$.

For $L = 1$, on the full $1/16 \le d \le 1/3$, use $C_a \le p + 3/4$, $q + 1 = 3 - p$. The cubic inequality with $D = 15/4, m = 1$ gives curvature $173/16 < 11$. Since covariance with q is nonpositive, $\mathbb{E}[f_a q^4] \le 1 - 57d/32 + 11d = 1 + 295d/32$. But $e^{8d} \ge 1 + 8d + 32d^2$, whose difference from this bound is at least $d(32d - 39/32) \ge 25d/32$. After multiplying by e^{-8d} the deficit is greater than $25d/864 > d/100$, because $8d \le 8/3 < 3$ and $e^3 < 27$.

For $L = 2$, first take $1/16 \le d \le 1/8$. The actual mean and variance are $m_2 = 1 - d$, $\mathrm{Var}\,\bar{f}_2 = d + \delta_2 - d^2 \le d(45/32 - d)$, and covariance with C_a is nonpositive. Since $\bar{f}_2 = q + Bu$ and $\sqrt{d} - d \le 1/4$, $\bar{f}_2 + m_2 \le 13/4 - p$, $C_a \le p + 9/20$, $m_2 \le 15/16$. The full pointwise coefficient is at most

$$\Phi(p) = (p + 9/20)[(13/4 - p)^2 + 225/128] \le 10.$$

Indeed $\Phi'' = 6p - 121/10 \le -3$ on $[0, 3/2]$, $\Phi(1) = 25317/2560 < 99/10$, $|\Phi'(1)| = 189/640 < 3/10$, so the tangent increment is below $3/200$. After $3/2$, $\Phi'(3/2) = -1283/640 < 0$ and $\Phi'' \le$

$-1/10$, so the remainder decreases. Taylor expansion gives

$$\begin{aligned}\mathbb{E}[f_a \bar{f}_2^4] &\le P(d) := (1-57d/32)(1-d)^4 + 10d(45/32-d)\\ &= 1 + (265/32)d + (25/8)d^2 - (235/16)d^3 + (65/8)d^4 - (57/32)d^5\\ &\le 1 + (265/32)d + (25/8)d^2.\end{aligned}$$

The omitted tail is nonpositive for $d \le 1/8$. Thus $e^{8d} - P(d) \ge d[(231/8)d - 9/32] \ge 195d/128$; the resulting deficit exceeds $(65/128)d > d/100$ since $e^{-8d} > 1/3$.

Finally take $L = 2$, $1/8 \le d \le 1/3$. Now $C_a \le p + 3/4$, $m_2 \le 7/8$ and the coefficient is bounded by

$$\Psi(p) = (p + 3/4)[(13/4 - p)^2 + 49/32] \le 12.$$

On $[0,1]$, $\Psi'' \le -5$, $\Psi(3/4) = 747/64$, $\Psi'(3/4) = 9/32$, so the tangent increment is at most $81/10240$ and the value below twelve. On $[1,2]$, the convex quadratic derivative has negative endpoint values $-41/32, -121/32$, so Ψ decreases. The target is bounded by

$$e^{-8d}[(1-57d/32)(1-d)^4 + 12d(45/32-d)].$$

The first term is at most $(3/8)(4/5)(7/8)^4 = 7203/40960$, using $e \ge 8/3$. The function $h(d) = d(45/32-d)e^{-8d}$ decreases here: after removal of the positive exponential its derivative is $45/32 - 2d - (45/4)d + 8d^2$, whose derivative is at most $-95/12$ and whose value at $1/8$ is $-1/8$. Thus $h(d) \le 41/(256e) \le 123/2048 < 1/16$. The total is at most $7203/40960 + 3/4 = 37923/40960 < 14/15 \le 1 - d/5$. All high rows join the low range at $d = 1/16$. At $d = 0$, original leaf integration gives equality, without requiring $K = 0$. Positivity restores all original equal pairs and subdivision vertices. Full reversal reverses receiving arm, terminal arms, connectors, and decorations together.

N.37 Unit shortest arm and second shortest arm three

For any finite residual list $(1, 3, c, s_2, t_2, \dots, s_j, t_j)$, $4 \le c < s_2 < t_2 < \cdots < s_j < t_j$, fix the actual terminal $F = q f_3 h_c \prod_{i=2}^{j} h_{s_i} f_{t_i}$. Orient every $L \ge 1$ connector edge toward the receiving root and its arm $a \ge 3$ outward. Then

$$\mathbb{E}[f_a (T^L F)^4] \le 1 - d/800. \tag{R23.1.1}$$

The $j = 1$ row is the $(1,3,c)$ row of Section N.43/D_{full}. The argument below treats every finite $j \ge 2$ with the same directions in both parameter ranges. Decorations may be restored.

Use the finite original receiving bounds, the exact covariance (R17.2.3), and the Taylor formula (R17.2.4). G_1 has coordinates $(1,0)$; for $n \ge 2$ the finite alternating coordinates are in $[1-x, 1]$. $\|G_n\| \le 1$ for $n = 1$ or $n \ge 3$, whereas G_2 has norm $\sqrt{1+x} < 6/5$. In particular $\mathrm{Cov}(C_a, \bar{f}_2) \le -3d/16 + \theta d^2$. The full continuous cubic inequality (R12.4.1), with $D = 17/5$, $m \le 33/32$, gives

$$C_a + Y + m \le 17/5 \implies C_a[(Y+m)^2 + 2m^2] \le 9, \qquad m^3 \le 9/8. \tag{R23.2.5}$$

Indeed $D \ge 3m$, and its three terms are below $35/6$, $29/12$, $11/20$, totaling $44/5 < 9$. In the low range $0 < d \le 1/64$, each exponential payment below has $C \ge A/2$, $A > B \ge 0$, $Ad/2 < 1$. By (R14.7.3) the deficit exceeds $(A-B)d/3$.

The actual canonical first pair from Q5-C181, Section 4, has

$$\begin{gathered} f_3 h_c \le e^{R_0 - \Gamma_0}, \quad \Gamma_0 = d + \beta_c + \beta_{c-1} \ge 35d/16, \quad R_0 = \mathfrak{r}_0(B)u,\\ \mathfrak{r}_0 = -\beta - z^3 G_{c-3}(-z), \quad \beta = \beta_{c-1}. \end{gathered} \tag{R23.3.1}$$

Write $G_{c-3}(-z) = U - zV$, with $(U, V) = (1, 0)$ for $c = 4$ and $U, V \in [1 - x, 1]$ otherwise. Thus $\mathfrak{r}_0 = -\beta + x^2 V + zxU$ and the actual signed mixed mean is

$$\mathbb{E}R_0 + \mathbb{E}(uR_0) = -\beta d - \int x(U - xV)\, d\nu \le 0, \tag{R23.3.2}$$

because $U - xV \ge 1 - 2x \ge 3/16$. Every later actual pair has the envelope (L96.2.6), $h_s f_t \le e^{R_{s,t} - 16d/5}$, $R_{s,t} = \sum_{r=s+1}^{t} K^r \mathbf{1}$. Let S be their one complete finite interval mask, $R = R_0 + S$. Then

$$F \le q e^{R - \Gamma}, \qquad \Gamma \ge 431d/80 + (16/5)(j - 2)d.$$

Since $s_2 \ge 5$, the first original power of S is at least six. The four residue classes of its finite zero–one mask give

$$\|S\|_2^2 \le \theta^5 \frac{1 + \theta}{(1 - \theta^2)^2} d = \frac{371293}{16634880} d < 9d/400, \quad \mathbb{E}S + \mathbb{E}(uS) \le \frac{2\delta_4}{1 - \theta^2} < d/6. \tag{R23.3.4}$$

The positive mixed-mean contributions start with the paired original K^7, K^8 terms δ_4, with later same-sign indices spaced by two. The geometric sums bound finite masks only. The row bound gives $|S| < \sqrt{d}/4$ and

$$\|R\|_2/\sqrt{d} \le d + (6/5)\theta^{3/2} + 3/20 < 777/1600 < 1/2, \qquad |R| \le d + 2\sqrt{d} \le 17/64 < 1/2.$$

Therefore $e^R \le 1 + R + (3/5)R^2$. Expanding the actual $q = 1 + u$, using the kernel of T bounded by two and $q \le 2$, gives

$$T(qe^R) \le \bar{f}_2 + BR + \delta, \qquad \delta = \mathbb{E}R + \mathbb{E}(uR) + \sqrt{d}\|R\|_2 + (12/5)\|R\|_2^2 \le 19d/15 < D := 4d/3. \tag{R23.3.6}$$

The $B(uR)$ cost is $\mathbb{E}|uR|$ and $T(qR^2) \le 4\mathbb{E}R^2$. The signed means $D_i = \mathbb{E}B^i R$ satisfy

$$\begin{gathered} |D_i| \le \tfrac{1}{2}\theta^{(i-1)/2} d \ (i \ge 1), \quad D_1 \le d/2, \quad D_2 \le d/3, \\ D_i \le \theta d/2 \ (i \ge 3). \end{gathered} \tag{R23.3.7}$$

Finite propagation preserves the nonnegative upper bound

$$e^{\Gamma} T^L F \le Y_L = \bar{f}_{L+1} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \ge 0. \tag{R23.3.8}$$

Two joint symbols retain the actual cross terms with R_0. For $G_2 + z\mathfrak{r}_0$, the coordinates are $Q_e = 1 - x^2 U$, $Q_o = 1 - \beta + x^2 V$, both positive. Hence

$$|G_2 + z\mathfrak{r}_0|^2 \le 1 + x - 2x^2 + 4x^3 + x^4 - x^5 + x^6 \le 1 + x + x^4 \le 1 + \theta + \theta^4 < 36/25. \tag{R23.4.1}$$

Use $x < 1/2$ for the two negative pairs; the last rational comparison is $37578025 < 37748736$. For $G_3 + z^2\mathfrak{r}_0$ the coordinates are $Q_e = 1 - x + \beta x - x^3 V$, $Q_o = 1 - x^2 U$, positive with lower bounds $1 - x - x^3$, $1 - x^2$. Their square is bounded by

$$[1 - (1 - \beta)x]^2 + x = 1 + x[-1 + 2\beta + (1 - \beta)^2 x] \le 1, \tag{R23.4.2}$$

since $\beta \le 1/64$, $x \le \theta$. Both symbols include $c = 4$.

For $L=1$, $Y=\bar f_2+BR+D$ has actual mean $m=1-d+D+D_1\le 1+5d/6\le 33/32$. Its random vector is $[G_2+z\mathfrak{r}_0](B)u+BS$. Using (R23.4.1) and $\|BS\|_2<\sqrt{d}/10$ gives $\operatorname{Var} Y\le 169d/100$. Since $\sigma(C_a)\le 65\sqrt{d}/64$ and $\|BR\|_2<\sqrt{d}/3$,

$$\operatorname{Cov}(C_a,Y)\le -3d/16+\theta d^2+(65/64)(1/3)d\le 967d/6144<d/6.$$

The pointwise row uses $|BR|\le\|R\|_2\le\sqrt{d}/2$, giving $Y\le q+(3/2)\sqrt{d}+D$ and $C_a+Y+m\le 3+(27/10)\sqrt{d}+(13/6)d\le 3+713/1920<17/5$. By (R23.2.5), the remainder is at most $9(169/100)+4(9/8)(1/6)=399/25<16$.

For $L=2$, $Y=\bar f_3+B^2R+Dq+D_1$ has $m\le 1+d/6\le 1+5d/6$. Its random vector is $[G_3+z^2\mathfrak{r}_0](B)u+Du+B^2S$. By (R23.4.2), $D\le 1/48$ and $\|B^2S\|_2<\sqrt{d}/16$, its standard deviation is at most $(13/12)\sqrt{d}$, and covariance at most $(65/64)(13/12)d<9d/8$. The correct three-step recurrence gives $Y\le(1+D)q+(23/15)\sqrt{d}-d/2$ and $C_a+Y+m\le 3+(41/15)\sqrt{d}+(7/3)d\le 3+121/320<17/5$. The remainder is at most $9(169/144)+4(9/8)^2=125/8<16$. For both rows the actual mean exponent is at least $431/20+57/32-10/3=9599/480>19$, so

$$\mathbb{E}[f_a(T^LF)^4]\le e^{-19d}+16de^{-(431/20)d}. \tag{R23.5.10}$$

Its deficit exceeds $d>d/800$.

For $L=3,4$ raise coefficients only before nonnegative actual paths:

$$\overline{Y}_3=\bar f_4+B^3R+D\bar f_2+(d/2)q+d/3,$$
$$\overline{Y}_4=\bar f_5+B^4R+D\bar f_3+(d/2)\bar f_2+(d/3)q+d/4.$$

They dominate (R23.3.8) and are nonnegative. The correct bare recurrences and vector polynomials are those of Modules R2-C014 and R2-C017, with means at most $1-83d/32$ and $1-11d/4$, and norms $\|G_4-d\|\le 1$, $\|G_5-d(2+z)\|\le 1-2d$. At $L=3$, retain $\mathbb{E}B^3R\le\theta d/2$ to get $m\le 1-[83/32-4/3-1/2-1/3-\theta/2]d=1-43d/192\le 1$. The whole-vector norm is at most $[1+(21/10)d+2/15]\sqrt{d}<(6/5)\sqrt{d}$, including the true G_2 cost. The row bound is $\overline{Y}_3\le(1+5d/6)q+(5/4)\sqrt{d}-2d/3$, so $C_a+\overline{Y}_3+m\le 3+d+(49/20)\sqrt{d}<17/5$. At $L=4$, $\mathbb{E}B^4R<(2/15)d$ gives $m\le 1-[11/4-4/3-1/2-1/3-1/4-2/15]d=1-d/5$. The whole-vector norm is at most $[1+(34/15)d+1/12]\sqrt{d}<(6/5)\sqrt{d}$. The actual row bound is $\overline{Y}_4\le(1+d/6)q+(6/5)\sqrt{d}-\tau+d/4$; the scalar remainder is negative, so $C_a+\overline{Y}_4+m\le 3+d/3+(12/5)\sqrt{d}<17/5$. Both rows have variance at most $36d/25$, covariance at most $5d/4$, and remainder at most $9(36/25)+5=449/25<18$. The mean exponent is at least $431/20+57/32=3733/160>23$, giving

$$\mathbb{E}[f_a(T^LF)^4]\le e^{-23d}+18de^{-(431/20)d}. \tag{R23.6.8}$$

Its deficit exceeds $5d/3>d/800$.

For all finite $L\ge 5$, put $n=L+1\ge 6$. The term $D\bar f_{L-1}$ has index two, where $\beta_2=0$, and is paid explicitly at cost $2D=8d/3$. For $r\ge 3$, $\gamma_r=\beta_r-D_{r-2}$ satisfies $\gamma_3\ge d/2$, $\gamma_4\ge 2d/3$, and $\gamma_r\ge(\kappa-\theta/2)d=25d/64\ge h=3d/8$ for $r\ge 5$. The additional actual mean index satisfies $\beta_{n+1}-D_L\ge h$ as well. Retaining the last four actual paths gives the nonnegative bound

$$e^{\Gamma}T^LF\le\widehat{Y}=1-h(4-d)+[G_n-h(3+2z+z^2)](B)u+B^LR+8d/3.$$

The exact mean estimate is $m \le 1 - 5h + 3hd + 8d/3 = 1 + (19/24)d + (9/8)d^2 \le 1 + 5d/6 \le 33/32$. The full finite shifted-symbol certificates (R18.6.1)–(R18.6.2) apply. Since $\|B^L R\|_2 \le (1/2)\theta^{L/2}\sqrt{d} < \sqrt{d}/16$, variance is at most $289d/256$ and covariance at most $(1105/1024)d < (11/10)d$. The pointwise form is $(1-3h)q + BV - h(1-d) + 8d/3$, where $V = [G_L - h(2+z)](B)u + B^{L-1}R$ and $\|V\|_2 \le (1 - 2h + 1/12)\sqrt{d} \le (13/12)\sqrt{d}$. As $dp + 3hq \ge 2d$,

$$C_a + \widehat{Y} + m \le 3 + (9/8)d + (3/8)d^2 + (137/60)\sqrt{d} \le 3 + (6/5)d + (12/5)\sqrt{d} < 17/5.$$

The remainder is at most $9(289/256) + 4(9/8)(11/10) = 19341/1280 < 16$. Thus (R23.5.10) covers every $L \ge 5$ too, completing the entire low range.

For $d \ge 1/64$, retain the actual first three arms. Every later pair has $s \ge 5$ and its original interval satisfies $|R_{s,t}| \le (6/5)\theta^{(s-1)/2}\sqrt{d} < \sqrt{d}/5 \le 8d/5$. Its loss is at least $16d/5$, so $h_s f_t \le e^{-8d/5}$. Consequently, with unchanged directions,

$$F \le e^{-(8/5)(j-1)d} q f_3 h_c, \qquad T^L F \le e^{-(8/5)(j-1)d} T^L(q f_3 h_c).$$

The exact Module Section N.43/D_{full} row, originating in Q5-C175, gives $\mathbb{E}[f_a\{T^L(qf_3h_c)\}^4] \le 1 - d/800$ for every $a \ge 3, c \ge 4, L \ge 1$. The full target is therefore at most $e^{-(32/5)(j-1)d}(1 - d/800) \le 1 - d/800$. This joins the low range at $d = 1/64$; $j = 1$ uses the same directory row directly. At $d = 0$ original leaf integration gives equality, without $K = 0$. Positivity restores original equal pairs and full reversal reverses the receiving arm, first three arms, all subsequent pairs, connectors, and decorations together.

N.38 Unit shortest arm and second shortest arm two

For the residual list $(1, 2, c, s_2, t_2, \dots, s_j, t_j)$, $3 \le c < s_2 < t_2 < \cdots < s_j < t_j$, orient the receiving arm $a \ge 3$ outward and every $L \ge 1$ connector edge toward its root. Fix the actual terminal by the following structural rule:

$j = 1, c = 3, L = 1$	$q f_2 \bar{f}_3$
$j = 1$, all other cases	$q f_2 h_c$
$j \ge 2, c = 3, s_2 = 4$	$q f_2 h_{t_2} h_3 f_4 \prod_{i \ge 3} h_{s_i} f_{t_i}$
$j \ge 2$, all other cases	$q f_2 h_c \prod_{i \ge 2} h_{s_i} f_{t_i}$.

(R24.1.table)

The crossing in the third row exchanges roles of existing original arms only. Every direction is fixed by the original integers. For this F, every independent finite parameter choice satisfies

$$\mathbb{E}[f_a (T^L F)^4] \le 1 - d/800. \tag{R24.1.1}$$

The $j = 1$ inputs are precisely the short-directory rows; the proof below treats every $j \ge 2$. Actual equal pairs can be restored.

Use the original receiving bounds of Module R2-C007 and exact covariance and Taylor identities (R17.2.3)–(R17.2.4). The short actual path formulas and bounds of Module R2-C021 apply on $d \le 1/8$: means at most $1 - d$, $1 - 2d$, $1 - 83d/32$, $1 - 11d/4$ for $n = 2, 3, 4, 5$; receiving covariances at most zero, $d/12$, $d/12$, $d/32$, respectively. $\|G_4 - d\| \le 1$, $\|G_5 - d(2+z)\| \le 1 - 2d$; the latter squared difference is $x[-1 + 2d + d^2 + (1-2d)x - x^2 + x^3] \le 0$, here already implied by $-1 + 1/4 + 1/64 + \theta < 0$. The favorable covariances with $q, \bar{f}_2$ are nonpositive. The full continuous cubic inequality (R12.4.1) gives coefficient below nine when $C_a + Y + m \le 17/5$, $m \le 33/32$,

and below ten when the corresponding sum is at most $18/5$, $m \le 1$. We use the exact exponential comparison (R14.7.3).

The following scaled form of Taylor expansion is essential. If $X \le e^{-\Gamma}Y$, $Y \ge 0$, $W = e^{-\Gamma}Y$, and $C_a[(W + \mathbb{E}W)^2 + 2(\mathbb{E}W)^2] \le M$, then

$$\mathbb{E}[f_a X^4] \le (\mathbb{E}C_a)e^{-4\Gamma}m_Y^4 + 4e^{-4\Gamma}m_Y^3 \operatorname{Cov}(C_a, Y) + Me^{-2\Gamma} \operatorname{Var} Y. \tag{R24.2.6}$$

Here $m_Y = \mathbb{E}Y$ is the actual mean; the last exponent is -2Γ.

We give the finite terminal algebra from Q5-C177, Sections 6–8, and the report A138, Sections 8–10. For every actual $4 \le s < t$, $h_s f_t \le e^{R_{s,t}-16d/5}$, $R_{s,t} = \sum_{r=s+1}^{t} K^r \mathbf{1}$. All subsequent pairs form one finite disjoint-interval mask S. The stripped zero–one polynomial has square at most $(1+x)/(1-x^2)^2 \le 32768/16245 < 100/49$; in each of the four residue classes take the larger positive or negative finite subsum. When its first short arm is at least five,

$$\begin{gathered}\|S\|_2 < 3\sqrt{d}/20, \quad |S| < \sqrt{d}/4, \quad \mathbb{E}S + \mathbb{E}(uS) \le 2C_\beta\delta_4 < d/6, \quad C_\beta = 1024/855, \\ S = (A_S + zC_S)(B)u, \quad |A_S| \le x^3/(1-x^2), \quad |C_S| \le x^2/(1-x^2).\end{gathered} \tag{R24.3.1}$$

If the first short arm is at least six, improve the final bound to $|C_S| \le x^3/(1-x^2)$. In all cases $\|BS\|_2 < \sqrt{d}/10$, $\|B^2S\|_2 < \sqrt{d}/16$, $\|B^3S\|_2 < \sqrt{d}/25$, $\|B^4S\|_2 < \sqrt{d}/40$. For $F \le qe^{R-\Gamma}$ and $|R| \le 1/2$, the actual unit-anchor first row is

$$T(qe^R) \le \bar{f}_2 + BR + \delta, \quad \delta = \mathbb{E}R + \mathbb{E}(uR) + \sqrt{d}\|R\|_2 + (12/5)\|R\|_2^2.$$

This follows by $e^R \le 1 + R + 3R^2/5$, $T(qR) = \mathbb{E}R + \mathbb{E}(uR) + BR + B(uR)$, and $T(qR^2) \le 4\mathbb{E}R^2$. For $|R| \le 3/4$, replace $12/5$ by $8/3$. Taking $D \ge \max(\delta, 0)$ and $D_i = \mathbb{E}B^i R$, finite positive propagation gives

$$e^{\Gamma}T^L F \le Y_L = \bar{f}_{L+1} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \ge 0. \tag{R24.3.3}$$

We distinguish four exhaustive structural packages. Package II has $c \ge 4, j \ge 2$ and $d \le 1/64$. The actual first-pair convolution gives

$$\begin{gathered}\Gamma_0 = \beta_c + \beta_{c-1} \ge 19d/16, \quad R_0 = -\beta_{c-1}u - \sum_{r=3}^{c} K^r \mathbf{1}, \quad R = R_0 + S, \quad \Gamma \ge 351d/80, \\ \mathfrak{r}_0 = -\beta - xU + zxV, \quad G_{c-2}(-z) = U - zV, \quad \beta = \beta_{c-1}, \quad U, V \in [1-x, 1].\end{gathered}$$

The actual first mixed mean is at most $-2\delta_2$ if $c = 4$, and $-2\kappa\delta_2$ if $c \ge 5$. For $v = \delta_2/d$, $\|R\|_2/\sqrt{d} \le d + C\sqrt{\theta v}$, with $C = 8/5$ or $5/4$, respectively; the mask mixed positive mean is at most $2C_\beta\theta^2\delta_2$. Thus

$$\begin{gathered}\delta/d \le -\lambda v + d + C\sqrt{\theta v} + (12/5)[d + C\sqrt{\theta v}]^2, \quad \lambda_4 = 1372/855, \\ \lambda_5 = 2\kappa - 338/855.\end{gathered}$$

As a quadratic in $\sqrt{v}$, the displayed linear and quadratic coefficients are positive, and it increases with d. At $v = \theta, d = 1/64$, the two bounds are $221307/128000 - 4459/6840 < 11/10$ and $2373/2000 - 3/10 < 9/10$. Take $D = 11d/10$. The complete original symbol obeys $|\mathfrak{r}| < 2/3$ and $|R| \le d + 2\sqrt{d} < 1/2$. Its actual signed means satisfy

$$D_1 \le d^2 + \delta_2 + C_\beta\delta_4 \le d^2 + C_\beta\delta_2 \le 33d/64, \qquad D_i < d/4 \quad (i \ge 2). \tag{R24.3.5}$$

Use $C_\beta\theta = 416/855 < 1/2$ in the first line; later four residue-class positive parts are at most $\beta\delta_2 + \delta_3 + C_\beta\delta_5$, with $\theta/64 < 1/128$, $\theta^2 < 1/6$, $C_\beta\theta^4 < 1/30$. All $\gamma_r = \beta_r - D_{r-2} \geq d/3$ for $r \geq 3$, including the additional actual mean index $\beta_{L+2} - D_L$ for $L \geq 5$.

Package III has $c = 3, s_2 \geq 5, j \geq 2$ and $d \leq 1/8$. The exact identity $f_2 + h_3 = 2 - d - K^3\mathbf{1}$ gives

$$\begin{gathered} R = B^2u + S, \quad \Gamma \geq 21d/5, \quad D = 11d/10, \quad \rho = 3/5, \\ \|R\|_2 < \rho\sqrt{d}, \quad |R| < 1/2, \\ D_1 < d/2,\ D_2 < d/12,\ D_i < d/10\ (i \geq 3). \end{gathered} \tag{R24.3.6}$$

To retain the actual calculation, the mixed mean is at most $-(1 - 2C_\beta\theta^2)\delta_2$ and $\|R\|_2^2/d \leq \theta(617/448)^2 v$. The resulting quadratic in $\sqrt{v}$ increases; at $v = \theta$ its root bound is below $9/16$ and the negative mixed term exceeds $6/25$. Hence $\delta/d < 9/16 + 243/320 - 6/25 < 11/10$. The pointwise bound is $[\sqrt{\theta} + (10/7)\theta^2]\sqrt{d} < 1/2$. $D_1 \leq \delta_2 + C_\beta\delta_4 \leq 416d/855$, $D_2 \leq C_\beta\delta_4$, and later positive residue-class leading terms are $C_\beta\delta_5, C_\beta\delta_5, \delta_4 + C_\beta\delta_6, C_\beta\delta_6$, decaying every four powers. Consequently

$$\gamma_3 \geq d/2, \quad \gamma_4 \geq 11d/12, \quad \gamma_r \geq 79d/160 \geq 15d/32 \quad (r \geq 5),$$

also at the additional mean index.

Package IV has $c = 3, s_2 = 4, j = 2$, $e = t_2 \geq 5$, $d \leq 1/8$, and the actual crossing $F = qf_2h_eh_3f_4$. For $e = 5$ take $c_1 = d, c_2 = d/4 + \delta_2, c_3 = 0$. For $e \geq 6$ take

$$\begin{gathered} c_1 = \beta_{e-1}, \quad c_2 = \lambda\beta_{e-2}, \quad c_3 = \lambda\beta_{e-3}, \\ \lambda = \frac{9d/4 - \beta_e - \beta_{e-1}}{\beta_{e-2} + (1-d)\beta_{e-3}} \in [0,1]. \end{gathered}$$

The denominator is positive and the total available constants are at least $(4-d)\tau > 9d/4$. Put $C = c_1 + c_2 + c_3$, $D_* = c_2 + c_3$, $E = c_3$. The original convolution gives $f_2 + h_e \leq 2 - 9d/4 - U_e$, $U_e = (A_{\text{old}} - zB_{\text{old}})(B)u$. For $e \geq 6$, $A_{\text{old}} = C + x(a_f - E)$, $B_{\text{old}} = D_* + xb_f$, $G_{e-2}(-z) = a_f - zb_f$; for $e = 5$ the actual coordinates are $A_{\text{old}} = 5d/4 + \delta_2 + x - x^2$, $B_{\text{old}} = d/4 + \delta_2 + x$. The actual constants satisfy $C \geq 5d/4 + dE$, $d/4 \leq D_* \leq (17/16 + d)d$, $0 \leq E \leq d$. Combining with $h_3 + f_4 = 2 - 3d + K^4\mathbf{1} - dg$ gives

$$\begin{gathered} F \leq qe^{R_c - 21d/4}, \quad R_c = (-A_c + zB_c)(B)u, \quad A_c = A_{\text{old}} - d, \quad B_c = B_{\text{old}} - x, \\ 0 \leq A_c \leq 19/40, \quad -x^2 \leq B_c \leq 19d/16, \quad A_c + B_c \geq d/2 + x/16. \end{gathered} \tag{R24.3.8}$$

The last comparison uses $a_f + b_f \geq 2(1-x)$ and $1 - 2x - E \geq 1/16$; for $e = 5$ it is direct. $A_{\text{old}} \leq 3/8 + 9d/5$ follows from $x(1 - x + x^2) \leq 5/16$ and the C bound. The coordinate B_c can be negative and is retained with its sign. The actual mixed mean is $-\int (A_c + B_c)\, d\nu \leq 0$ and $\|R_c\|_2 < \sqrt{d}/2$. The row formula $R_c = (C-d)g + D_*Kg + (E-1)K^2g - \sum_{r=5}^{e} K^r\mathbf{1}$ gives $|R_c| \leq 25/256 + (19/128 + 119/125)\sqrt{1/8} < 1/2$; here $119/125 = 16/25 + 39/125$ pays the actual third-power and continuous tail, and $\sqrt{1/8} < 9/25$ completes the rational comparison. Thus

$$\Gamma = 21d/4, \quad D = 11d/10, \quad \rho = 1/2, \quad D_1 \leq 19d/40, \quad D_2 < d/16, \quad D_i < d/12\ (i \geq 3). \tag{R24.3.9}$$

In detail $D_3 \le 0$, $D_4 \le \delta_5$, $D_5 < 19\delta_3/40$, $D_6 < 19d\delta_4/16$, followed by four-class decay. Every $\gamma_r \ge d/2$, including the mean index.

Package V is the same crossing with $j \ge 3$, $d \le 1/8$. Put all subsequent actual pairs $i \ge 3$, whose short arms start at six, in one mask S. Then

$$\begin{gathered} R = R_c + S, \quad \Gamma \ge 169d/20, \quad D = 7d/4, \quad \rho = 3/5, \\ |R| < 7/12 < 3/4, \quad \|R\|_2 < \rho\sqrt{d}, \quad D_1 < 2d/3, \quad D_2 < d/3, \quad D_i < d/8 \ (i \ge 3). \end{gathered} \tag{R24.3.10}$$

Here $|S| < 13/224 < 1/12$, mixed mean at most $d/6$, and the $8/3$ first-row version gives $\delta/d \le 1/6 + 3/5 + (8/3)(9/25) < 7/4$. The mask's D_1 positive part is below $d/12$, and its later positive parts below $d/30$. Thus every $\gamma_r \ge d/3$, including the additional mean index.

All long rows use the following actual finite bound. Put $n = L + 1$. The index-two term $D\bar{f}_{L-1}$ is first paid at cost $2D$ since $\beta_2 = 0$. If $\gamma_r \ge h$ for all $r \ge 3$ and $L \ge 5$, retain the last four original path messages to get

$$e^{\Gamma} T^L F \le \widehat{Y} = 1 - h(4-d) + [G_n - h(3 + 2z + z^2)](B)u + B^L R + 2D \ge 0. \tag{R24.4.1}$$

The actual mean, retaining the extra index, is

$$\mathbb{E}\widehat{Y} = 1 - 4h + 3hd - [\beta_{n+1} - D_L] + 2D \le 1 - 5h + 3hd + 2D. \tag{R24.4.2}$$

The pointwise form is $(1-3h)q + BV - h(1-d) + 2D$, with $V = [G_L - h(2+z)](B)u + B^{L-1}R$. The full finite certificates (R18.6.1) and (R18.6.2) apply for every eligible N and $h \le 2/9$; their negative square-difference bounds are $-181/3888$ and $-231/1296$, including the shortest $N = 6, 5$ and $x = 0$.

We now pay package II on $d \le 1/64$. The joint symbol $G_2 + z\mathfrak{r}_0$ has coordinates $1 - x^2V$, $1 - \beta - xU$, both in $[0, 1]$, and norm at most $6/5$. Thus $Y_1 = \bar{f}_2 + BR + D$ has variance at most $169d/100$, actual mean $m_1 \le 1 + 197d/320$, and covariance at most $d/3$. For the covariance use the negative bare-two bound and $\|BR\|_2 < (4/9)\sqrt{d}$ to get $-3/16 + \theta/64 + 65/144 < 1/3$ after normalization. The joint symbol $G_3 + z^2\mathfrak{r}_0$ has positive coordinates $1 - x + \beta x + x^2U$, $1 - x^2V$. Its squared norm is at most $[1 - (1-\beta)x + x^2]^2 + x$; the difference from one divided by x is $-1 + 2\beta + [(1-\beta)^2 + 2]x - 2(1-\beta)x^2 + x^3$. It increases in both $\beta \in [0, 1/64]$ and $x \in [0, \theta]$; the upper-right value is $-2683/131072 < 0$. Therefore $Y_2 = \bar{f}_3 + B^2R + Dq + D_1$ has $\sigma(Y_2) < [1 + 11/640 + 1/16]\sqrt{d} < (11/10)\sqrt{d}$. Use the weaker common variance $36d/25$, covariance $5d/4$, and actual mean $m_2 \le 1 - 43d/320$. The pointwise sums for the two rows are at most

$$3 + (43/15)\sqrt{d} + (549/320)d < 17/5, \qquad 3 + (128/45)\sqrt{d} + (549/320)d < 17/5.$$

Both means are at most $33/32$, so curvature nine and positive-term $m^3 \le 9/8$ are legitimate. For $L = 3, 4$, use the actual nonnegative upper bounds

$$\begin{aligned} \overline{Y}_3 &= \bar{f}_4 + B^3R + D\bar{f}_2 + (33d/64)q + d/4, \\ \overline{Y}_4 &= \bar{f}_5 + B^4R + D\bar{f}_3 + (33d/64)\bar{f}_2 + (d/4)q + d/4. \end{aligned}$$

Their actual means, including D_L, are at most $1 - 153d/320$, $1 - 123d/320$; standard deviations at most $[1 + (33/25 + 33/64)d + 8/45]\sqrt{d} < (5/4)\sqrt{d}$ and $[1 + (63/32)d + 1/9]\sqrt{d} < (6/5)\sqrt{d}$. Use centered covariance bounds $13d/10$, $5d/4$. Their row bounds are $(1 + 197d/320)q + (13/10)\sqrt{d} - 3d/4$ and

$(1-43d/320)q+(6/5)\sqrt{d}-\tau+d/4$. Adding C_a and the actual mean gives sum below $17/5$ in both rows. For $L\ge 5$ take $h=d/3$ in (R24.4.1). Then $m\le 1+(8/15)d+d^2\le 1+3d/5$, $\|B^L R\|_2<\sqrt{d}/12$, $\sigma\le 13\sqrt{d}/12$, and covariance at most $9d/8$. Its pointwise V tail is at most $\sqrt{d}/9$, so $C_a+\widehat{Y}+m\le 3+(104/45)\sqrt{d}+(7/15)d+d^2/3<17/5$. All package-II payments therefore have $\mathbb{E}[f_a(T^L F)^4]\le e^{-Ad}+Bde^{-(351/20)d}$ with

L	m upper bound	Var $/d$	Cov $/d$	A	B
1	$1+197d/320$	$169/100$	$1/3$	$84/5$	$1671/100$
2	$1-43d/320$	$36/25$	$5/4$	19	18
3	$1-153d/320$	$25/16$	$13/10$	21	20
4	$1-123d/320$	$36/25$	$5/4$	20	18
≥ 5	$1+3d/5$	$169/144$	$9/8$	$84/5$	16.

(R24.5.table)

The actual remainder constants are $1671/100$, at most $449/25<18$, $1541/80<20$, $449/25<18$, and $125/8<16$. The exact first-term exponents are respectively $2699/160, 3179/160, 3399/160, 3339/160, 2709/160$, exceeding the listed A. In all rows $C\ge A/2$, $Ad/2<1$, and $A-B\ge 9/100$, so the deficit exceeds $3d/100>d/800$.

For packages III–V on $d\le 1/8$ we prove two scaled continuous curvature certificates, with $t=\sqrt{d}<3/8$, $W=e^{-\Gamma}Y$. Certificate I assumes $\Gamma\ge 21d/5$, $m_Y\le 1+3d/5$, and

$$Y+m_Y\le(1+\delta d)q+1+(8/5)t+(9/4)d,\qquad 0\le\delta\le 11/10.$$

Then $\mathbb{E}W\le 1$ and $C_a+W+\mathbb{E}W\le 18/5$. Indeed $e^{-\Gamma}(1+\delta d)\le 1-d$; the coefficient of q in the sum is nonpositive, so take $q=0,p=2$. After $e^{-\Gamma}\le 1/(1+21d/5)$, multiplication by the positive denominator bounds the difference from $18/5$ below by

$$3/5-(14/5)t+(647/100)t^2-(126/25)t^3+(42/5)t^4\ge 3/5-(14/5)t+(229/50)t^2>0,$$

whose discriminant is $-394/125$. Certificate II assumes $\Gamma\ge 169d/20$, $m_Y\le 1+2d$, and $Y+m_Y\le(1+3d)q+1+2t+(21/4)d$. The same conclusion holds: the maximum again occurs at $q=0$, and the positive-denominator difference is at least

$$3/5-(16/5)t+(1027/100)t^2-(507/50)t^3+(169/10)t^4\ge 3/5-(16/5)t+(2587/400)t^2>0,$$

whose discriminant is $-2641/500$. Both give $M=10$ in (R24.2.6) throughout the full continuous domain.

For package III, $R=B^2u+S$. At $L=1$ the joint core is $Q_1=G_2+z^3=1+z(1-x)$, with $|Q_1|^2=1+x(1-x)^2\le 31/27<(43/40)^2$. Adding BS gives $\sigma(Y_1)\le 6\sqrt{d}/5$ and $m_1\le 1+3d/5$. $\mathrm{Re}(G_a Q_1)\ge(1-x)^2$ and its true scalar correction is at most $d^2-\tau^2$, so core covariance is at most $-2225d/8192$. The BS cost is at most $9d/80$, hence total covariance is nonpositive. At $L=2$, the core $Q_2=G_3+z^4=(1-x+x^2)+z$ has norm at most one: the square difference divided by x is $-1+3x-2x^2+x^3$, increasing to $-1451/32768$ at θ. Adding $Du+B^2S$ gives $\sigma(Y_2)\le[1+11/80+1/16]\sqrt{d}=(6/5)\sqrt{d}$, and $m_2\le 1-19d/60$. The core real product is at least $\phi=1-3x+2x^2-x^3$; its scalar correction is at most $d\delta_3$. Since $D\,\mathrm{Cov}(C_a,q)\le 0$, $\mathrm{Cov}(C_a,Y_2)\le[-1451/32768+\theta^2/8+9/128]d<d/20$. The two row sums $Y+m_Y$ are bounded respectively by $q+1+(8/5)t+(17/10)d$ and $(1+D)q+1+(8/5)t-(49/60)d$, so Certificate I applies.

For III at $L = 3$ use $\overline{Y}_3 = \bar{f}_4 + B^3R + D\bar{f}_2 + (d/2)q + d/12$. Its actual mean is at most $1 - 389d/480$. Combining B^2u with all short paths gives exact core coordinates $1 - x + (3/5)d$, $1 - x + x^2 + (11/10)d$. Their square is at most $(1 + 3d/5)^2$, since the difference divided by x is at most $-1 + \theta + d + (121/100)d^2 < 0$. Adding B^3S gives $\sigma \le 9\sqrt{d}/8$. The $D\bar{f}_2, (d/2)q$ covariances are nonpositive; the others cost $d/12 + (9/8)(8/75 + 1/25)d < d/4$, using $\theta^{5/2} < (1/6)(16/25) = 8/75$. The row bound $(1 + 3d/5)q + (7/5)t - 11d/12$ enters Certificate I. For III at $L = 4$ take $\overline{Y}_4 = \bar{f}_5 + B^4R + D\bar{f}_3 + (d/2)\bar{f}_2 + (d/12)q + d/10$. Its mean is at most $1 - 13d/15$, and the core coordinates are exactly $1 - x + x^2 - x^3 - (19/60)d - (11/10)dx, 1 - x + (3/5)d$. The first is positive and at most the G_5 real coordinate; the second adds $3d/5$ to its imaginary coordinate, so the core norm is at most $1 + (2/5)d$. Adding B^4S gives $\sigma < 11\sqrt{d}/10$. The possibly positive $D\bar{f}_3$ covariance is retained:

$$\mathrm{Cov}(C_a, \overline{Y}_4) \le [1/32 + (9/8)(7/100 + 1/40) + 11/960]d = 359d/2400 < 3d/20.$$

Here $\|B^4B^2u\|_2 \le \theta^3\sqrt{d} < 7\sqrt{d}/100$ and the $D\bar{f}_3$ cost before $d \le 1/8$ is $(11/120)d^2$. The row bound is $q + (13/10)t$, so Certificate I applies. For III at $L \ge 5$, take $h = 15d/32$ in (R24.4.1). Then $m \le 1 - (23/160)d + (45/32)d^2 \le 1 + d/30$. The complete tail $B^LR = B^{L+2}u + B^LS$ has norm below $(3/50)\sqrt{d}$: its two costs are below $27\sqrt{d}/625$ and $2\sqrt{d}/125$, totaling $37\sqrt{d}/625 < 3\sqrt{d}/50$. Thus $\sigma \le 16\sqrt{d}/15$, covariance at most $6d/5$, and $m^3 \le 33/32$, since $m \le 1 + 1/240$ and $(1+x)^3 \le 1 + 4x$ for $x \le 1/4$. The V tail is below $\sqrt{d}/10$, giving $\widehat{Y} + m \le (1 - 3h)q + 1 + (11/10)t + (9/4)d$, again Certificate I. For short III rows, (R24.2.6) gives

$$\mathbb{E}[f_a(T^LF)^4] \le e^{-16d} + (73/5)de^{-(42/5)d}. \tag{R24.7.1}$$

The four remainder constants are at most $72/5, 73/5, 437/32, 127/10$; the smallest mean exponent is $84/5 + 57/32 - 12/5 = 2589/160 > 16$. The deficit exceeds $7d/15 > d/800$. For long III rows the remainder is at most $10(256/225) + 4(33/32)(6/5) = 2939/180 < 33/2$ and the mean exponent exceeds $84/5$, so $\mathbb{E}[f_a(T^LF)^4] \le e^{-(84/5)d} + (33/2)de^{-(42/5)d}$. $A - B = 3/10$, $Ad/2 \le 21/20 < 2$, giving deficit $> d/30 > d/800$.

For package IV retain the actual signed coordinates (R24.3.8). At $L = 1$, the joint coordinates $1 - xB_c, 1 - A_c$ are positive and have squared norm at most $(1 + x^3)^2 + x < 25/16$; use $(107/100)^2 + 13/32 < 25/16$. The actual mean is at most $1 + 23d/40$. The real product is at least $3/16 - 19d/64 \ge 77/512$, and actual scalar correction is at most $[(1 + \theta^3) - \kappa(21/40)]d^2 < 47d^2/60$. Thus covariance is nonpositive. The row sum is at most $q + 1 + (3/2)t + (67/40)d$, so Certificate I gives $M = 10$; variance is at most $25d/16$. At $L = 2$, the positive coordinates are $1 - x + xA_c, 1 - xB_c$. Their square is at most

$$(1 - 21x/40)^2 + x(1 + x^3)^2 \le 1 + (441/1600)\theta^2 + 2\theta^4 + \theta^7 < (17/16)^2.$$

After Du, $\sigma \le 17/16 + 11/80 = 6/5$ in units of $\sqrt{d}$. The actual mean is at most $1 - 29d/80$, covariance at most $27d/20$, and row sum at most $(1 + D)q + 1 + (8/5)t$, so Certificate I applies. At $L = 3, 4$ use

$$\overline{Y}_3 = \bar{f}_4 + B^3R + D\bar{f}_2 + (19d/40)q + d/16,$$
$$\overline{Y}_4 = \bar{f}_5 + B^4R + D\bar{f}_3 + (19d/40)\bar{f}_2 + (d/16)q + d/12.$$

Their actual means are at most $1 - 419d/480$, $1 - 227d/240$. Their standard deviations are at most $[1 + (359/200)d + 2/15]\sqrt{d} < (11/8)\sqrt{d}$ and, retaining the P_5 loss $-2d$, $[1 - 107d/400 + 1/12]\sqrt{d} \le$

$(13/12)\sqrt{d}$. Covariances are at most $d/4$ and $3d/20$: the first is bounded by $d/12 + (9/8)(2/15)d$ after favorable terms; the second by $d/32 + 3d/32 + 11d/960$. The actual row bounds are $(1 + 23d/40)q + (7/5)t - 15d/16$ and $q + (5/4)t$, both in Certificate I. For IV at $L \geq 5$ use $h = d/2$: actual mean at most $1 - 9d/80$, $\sigma \leq 17\sqrt{d}/16$, covariance at most $6d/5$; $\|V\|_2 \leq 13\sqrt{d}/12$ gives row sum $\widehat{Y} + m \leq q + 1 + (8/5)t + (9/4)d$. The IV $L = 1$ remainder is at most $125/8 < 16$ and its mean exponent $21 + 57/32 - 23/10 > 20$, so take $(A, B, C) = (20, 16, 21/2)$. For $L = 2, 3, 4, \geq 5$, the remainders are at most $99/5, 637/32, 4441/360, 10297/640$, all below twenty; their mean exponents exceed twenty-one. Take $(21, 20, 21/2)$. All $Ad/2 < 2$, giving deficit $> d/9 > d/800$.

For package V use the actual R, D_i in (R24.3.10). The first two bounds are (R24.3.3); take additionally

$$\overline{Y}_3 = \bar{f}_4 + B^3 R + D\bar{f}_2 + (2d/3)q + d/3, \quad \overline{Y}_4 = \bar{f}_5 + B^4 R + D\bar{f}_3 + (2d/3)\bar{f}_2 + (d/3)q + d/8,$$

and $h = d/3$ in the long row. Keeping all actual D_L means gives

L	m_Y upper bound	$\sigma(Y)/\sqrt{d}$	$\mathrm{Cov}(C_a, Y)/d$	m_Y^3 upper bound
1	$1 + 17d/12$	$8/5$	$9/20$	$5/3$
2	$1 + 3d/4$	$3/2$	$3/8$	$4/3$
3	$1 + 9d/32$	$8/5$	$4/15$	$9/8$
4	$1 + d/4$	$5/4$	$1/6$	$9/8$
≥ 5	$1 + (11/6)d + d^2 \leq 1 + 2d$	$11/10$	$5/4$	$2.$

(R24.9.table)

For the first norm use $6/5 + 2/5$, dropping only the favorable bare-two covariance and paying tail $9d/20$. For the second use $1 + \theta(3/5) + 7/32 < 3/2$ and covariance $d/12 + 9d/32 < 3d/8$, retaining the negative Dq term. For the third use $1 + (83/30)d + 4/25 < 8/5$ and covariance $d/12 + 9d/50 < 4d/15$. For the fourth retain P_5's $-2d$, giving $1 + (53/60)d + 1/10 < 5/4$, and covariance $d/32 + 9d/80 + 7d/384 < d/6$. For the long row the core norm is at most $1 - d$, complete tail norm below $\sqrt{d}/12$, and centered Cauchy–Schwarz pays the covariance. Each actual row expands to $Y + m_Y \leq (1 + 3d)q + 1 + 2t + (21/4)d$: the scalar coefficients are $19/6, 5/12$, negative, and at most $3/8$ in the four short rows; the long scalar is $5d + (4/3)d^2 \leq 31d/6 < 21d/4$. The respective q coefficients are at most $1, 1 + 7d/4, 1 + 17d/12, 1 + 3d/4, 1 - d$; every square-root coefficient is below two. Certificate II thus gives $M = 10$. For V at $L = 1$, the remainder is at most $143/5 < 29$, while the mean exponent $169/5 + 57/32 - 17/3 > 59/2$. Take $(A, B, C) = (59/2, 29, 169/10)$. The other row remainders are at most $49/2, 134/5, 131/8, 221/10$, all at most $134/5$; their mean exponents exceed twenty-seven, even the weakest long exponent $169/5 + 57/32 - 8 > 27$. Take $(A, B, C) = (27, 134/5, 169/10)$. $C \geq A/2$, $Ad/2 < 2$, and the smallest deficit is $> d/45 > d/800$. This pays the entire finite later mask once, independently of its size.

For package II's high range $d \geq 1/64$, every later short arm is at least five and $|R_{s,t}| < \sqrt{d}/5 \leq 8d/5$, so $h_s f_t \leq e^{-8d/5}$. The same F is at most $e^{-8(j-1)d/5} q f_2 h_c$. The exact D_{full} row gives the required $1 - d/800$ for the remaining $q f_2 h_c$, all $a \geq 3, c \geq 4, L \geq 1$. For III at $d \geq 1/8$, only rearrange multiplication at the same root:

$$F = (q f_2 h_{s_2})(h_3 f_{t_2}) \prod_{i \geq 3} h_{s_i} f_{t_i}, \qquad s_2 \geq 5,\ t_2 \geq 6.$$

The original directions are unchanged. For $t \geq 6$ put $\beta = \beta_{t-1}$, $\Gamma_t = d + \beta_t + \beta$, $U = \sum_{r=3}^{t-1} K^r \mathbf{1}$, and $\alpha = \beta - \mathbb{E}U = \Gamma_t - 2d \geq 3d/16$. The actual two-arm sum gives $h_3 f_t \leq e^{-\Gamma_t - R_0}$, $R_0 =$

$\alpha g - K(U - \mathbb{E}U)$. The centered row bound and $g^2 + pq = 1$ give

$$|R_0| \le \sqrt{\alpha^2 + \operatorname{Var} U}, \quad \operatorname{Var} U \le \theta^2 d < 11d^2/8 \le d^2 + 2\alpha d.$$

Thus $|R_0| \le \alpha + d$ and $\Gamma_t - |R_0| \ge d$, proving $h_3 f_t \le e^{-d}$. Pay all other pairs as above and use the same-direction D_{full} row for $qf_2h_{s_2}$. For IV,V at $d \ge 1/8$ the original expression is already $(qf_2h_e)(h_3f_4)\prod_{i\ge3} h_{s_i}f_{t_i}$. From $h_3 + f_4 = 2 - 3d + K^4\mathbf{1} - dg$, centered rows give $|K^4\mathbf{1} - dg| \le \sqrt{d^2 + \delta_3} \le 2d$, since $\delta_3 \le \theta^2 d \le 3d^2$. Hence $h_3 f_4 \le e^{-d}$; pay later pairs and use the exact directory row for qf_2h_e, $e \ge 5$. All high rows meet their low rows at the stated closed endpoints with identical original directions.

For $j \ge 2$, Packages II–V give the bounds at $L = 1, 2, 3, 4, \ge 5$. For $j = 1$, the original $(1, 2, 3), L = 1$ direction $qf_2\bar{f}_3$ has deficit $d/256$; its other connectors use qf_2h_3 with deficit $d/16$; $c \ge 4$ uses qf_2h_c with deficit $d/800$. These precise short-directory rows and the table (R24.1.table) prove (R24.1.1). At $d = 0$, original leaf integration gives equality without $K = 0$. Positivity restores equal-pair decorations in place. The crossing retains every original arm and subdivision vertex; full reversal changes every receiving, terminal, connector, and decoration edge together. This is a reference fourth-moment interface, to be used with its actual weights in the complete original-tree integral.

N.39 Even terminal lists of minimum two or three

R25.1 Original statement and the precise new theorem

We use the original finite probability space and real antisymmetric kernel with pointwise $|K| \le 1$. On the same original probability space put

$$\begin{gathered} Jv = (Ev)\mathbf{1}, A = J + K, T = J - K, B = -K, u = B\mathbf{1} = -K\mathbf{1}, \\ p = 1 - u, q = 1 + u, f_n = A^n\mathbf{1}, \bar{f}_n = T^n\mathbf{1}, h_n = Tf_{n-1}, \\ \delta_i = \|B^i\mathbf{1}\|^2, d = \delta_1, \theta = \frac{13}{32}, \kappa = \frac{19}{32}, \tau = d - \delta_2. \end{gathered} \tag{R25.L17}$$

The original receiving arm $a \ge 2$ points entirely outward. The original terminal has any finite ordered residual

$$s_1 < t_1 < s_2 < t_2 < \ldots < s_j < t_j, j \ge 2, s_1 \in \{2, 3\}. \tag{R25.L24}$$

Use the SAME actual terminal $F = \prod_{i=1}^{j} h_{s_i}f_{t_i}$ at EVERY positive original connector length L. Each h arm has its first edge toward the terminal and its remaining edges toward the leaf; each f arm points entirely toward the leaf. Every connector edge points terminal-to-receiver. None of these directions depends on d, μ, K or a spectral quantity.

Lemma N.4 (Even finite masks)**.** *All independent $a \ge 2, L \ge 1, j \ge 2$ and all these finite arm tables satisfy, on every original legal host,*

$$E[f_a(T^L F)^4] \le 1 - d/100. \tag{R25.1.1}$$

For arbitrary genuine original equal-pair decorations E_o, E_t, the same actual directions satisfy

$$E[E_o f_a T^L (E_t F)^4] \le 1 - d/100. \tag{R25.1.2}$$

For receiving lengths $a \ge 3$, this and Section N.36 cover every even terminal with $j \ge 2$ and minimum at least two, in the same canonical direction and with the same deficit.

R25.2 Original finite calculus and receiving covariance

Use the original Q3-C014 finite calculus, at its original measure:

$$\begin{gathered}0 \le d \le \frac{1}{3}, d^2 \le \delta_2 \le \theta d, \delta_{i+1} \le \theta \delta_i, \\ \|B\|^2 \le \theta, |BV(x)| \le \|V\|, \\ EK^{2i}1 = (-1)^i \delta_i, EK^{2i+1}1 = 0.\end{gathered} \tag{R25.L56}$$

A and T are positive operators. Original path messages are positive, $f_n, \bar{f}_n \le 2$ and $Ef_n = E\bar{f}_n \le 1$. Equal pairs satisfy $0 \le f_l h_l \le 1$. Positive atoms have $p, q \ge \mu(x) > 0$. The common zero-boundary argument applies when $d = 0$.

Set $G_n(z) = \sum_{r=0}^{n-1} z^r, G_0 = 0, \beta_1 = \beta_2 = 0$ and

$$\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1} \delta_i. \tag{R25.L67}$$

The exact finite positive convolution is

$$\begin{gathered}1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}, \\ \tau \le \beta_r \le d (r \ge 3), \beta_3 = \beta_4 = d.\end{gathered} \tag{R25.2.1}$$

$Atz = i\sqrt{x}, 0 \le x \le \theta$, every $n \ge 2$ has $G_n = A_n + zC_n$ with $1 - x \le A_n, C_n \le 1$. For $n \ge 3$, $A_n \le 1 - x + x^2$; for $n \ge 4$ both coordinates are $\le 1 - x + x^2$. All finite $G_n, n \ge 3$ contract; only G_2 has norm $\sqrt{1+x}$, which is strictly below $\frac{6}{5}$. In addition

$$|G_n|^2 \le 1 - 7x/15, n \ge 3. \tag{R25.2.2}$$

Here are finite proofs of the latter useful bound. For $n = 3$ use $1 - x + x^2 \le 1 - 7x/15$. For $n = 4$ use $(1 - x^2)^2/(1 + x) \le 1/(1 + x)$. For $n \ge 5$ the geometric numerator is $\le (1 + x^3)^2$. The remaining inequality is $8 - 7x - 30x^2 - 15x^5 \ge 0$, decreasing to $\frac{1311885}{33554432} > 0$ at $x = \frac{13}{32}$. Thus every finite integer is covered.

Use the weak actual receiving upper (Q117.1.2), also obtained directly from (R25.2.1) with K in place of B:

$$\begin{gathered}0 \le f_a \le C = 1 - G_a(-B)u - \beta_a, \\ EC = 1 - \beta_a - \beta_{a+1} \le 1 - d, \\ C \le p + (\frac{6}{5})\sqrt{d}.\end{gathered} \tag{R25.2.3}$$

The last bound writes $C = p - \beta_a + BG_{a-1}(-B)u$ and explicitly pays the $a = 3$ predecessor G_2. C is the nonnegative upper bound used in the analysis. For $a = 2$, the same finite formula is exactly $C = f_2$, since $\beta_2 = 0, \beta_3 = d$; the predecessor is $G_1 = 1$, and $C = p + Bu \le p + \sqrt{d}$. Thus (R25.2.3)–(R25.2.5) hold in this case too.

Let ν be the original spectral measure of $-B^2$ at u, mass d and $\int x^i d\nu = \delta_{i+1}$. If $H = Q(B)u$, $EH = -\alpha$, then exactly

$$\operatorname{Cov}(C,H)=-\int \operatorname{Re}(G_aQ)d\nu-\beta_{a+1}\alpha. \tag{R25.2.4}$$

Indeed the nonconstant receiver part is $-G_a(-B)u$, with mean $-\beta_{a+1}$. This proves the signed mean-product term in (R25.2.4).

For every nonnegative Y with its ACTUAL mean m,

$$E[CY^4]=(EC)m^4+4m^3\operatorname{Cov}(C,Y)+E[C(Y-m)^2(Y+m)^2+2m^2]. \tag{R25.2.5}$$

When $W=e^{-\lambda d}Y$, the first two terms have factor $e^{-4\lambda d}$, whereas a pointwise remainder coefficient M paid on W multiplies $e^{-2\lambda d}\operatorname{Var}Y$. The latter factor is $e^{-2\lambda d}$.

R25.3 One entire finite mask: envelope, full signed parameter packet

The actual pair-sum inputs, available in C090, Section 2 and C096, Section 2.2, are

$$\begin{aligned} h_2+f_t&\le 2+R_{2,t}-\kappa d,\\ h_3+f_t&\le 2+R_{3,t}-(2-\theta)d,\\ h_s+f_t&\le 2+R_{s,t}-16d/5,\ s\ge 4,\\ R_{s,t}&=\sum_{r=s+1}^{t}K^r\mathbf{1}. \end{aligned} \tag{R25.3.1}$$

The first two follow from the finite C014 difference, retaining the true $\beta_t f_0$ term, including the shortest t. For $s=4$ the retained negative τf_2 term and $f_2\ge p^2/2$ give loss at least $13d/4$. For $s\ge 5$, the exact finite mean bound $\mu_{s-1}\le 1-83d/32$ and positive difference cost $1664d/855$ give net loss $(\frac{16}{5}+\frac{113}{2736})d$. These are the pair estimates used below.

AM–GM is legal because each actual pair sum is nonnegative. Multiplying the resulting exponential bounds gives one original variable

$$\begin{aligned} F&\le \exp(R-\lambda d),\ R=\sum_i R_{s_i,t_i},\\ \lambda&=\frac{607}{160}\quad\text{if } s_1=2;\ \lambda=\frac{767}{160}\quad\text{if } s_1=3. \end{aligned} \tag{R25.3.2}$$

Every original pair remains in R; only its nonnegative loss was weakened. The exponent intervals are disjoint, hence $R=U(B)u$ is a finite 0–1 mask in the original K powers, whose lowest power is $K^{s_1+1}1$. No pointwise division by u occurs.

For any finite [0,1] coefficient polynomial P, separate the four residue classes at $z=i\sqrt{x}$. The positive and negative subcollections in each coordinate are bounded separately; the absolute difference is bounded by their maximum. Consequently

$$|P(z)|^2\le (1+x)/(1-x^2)^2\le C_\star=32768/16245. \tag{R25.3.3}$$

These infinite positive geometric sums only bound original finite sums. The original spectral and row bounds yield

$$\|U\|^2\le\theta^{s_1}C_\star,\ |R|^2\le\theta^{s_1-1}C_\star d. \tag{R25.L153}$$

Thus, writing $V = ER^2$,

$$\begin{aligned} s_1 = 2 : \|U\| < 1/\sqrt{3}, V < d/3, |R| < \frac{3}{5}; \\ s_1 = 3 : \|U\| < \frac{3}{8}, V < 9d/64, |R| < 1/3. \end{aligned} \tag{R25.3.4}$$

For example $\theta^2 C_\star = \frac{5408}{16245} < \frac{1}{3}$ and $\theta^3 C_\star = \frac{2197}{16245} < \frac{9}{64}$. The Taylor remainder $n! \geq 2 \cdot 3^{n-2}$ gives

$$e^R \leq 1 + R + cR^2/2, c = \frac{5}{4}(s_1 = 2), c = \frac{6}{5}(s_1 = 3). \tag{R25.3.5}$$

Define the SAME actual signed quantities

$$D_n = EB^n R, a_0 = -D_1, \rho = -D_2, D = D_0 + cV, \epsilon = D - a_0 = D + D_1. \tag{R25.3.6}$$

Their complete bounds, all in units of d, are

2 D/d: $[-\frac{1}{5}, \frac{11}{12}]$;
ϵ/d: $[-\frac{7}{10}, \frac{7}{10}]$;
a_0/d: $[0, \frac{1}{2}]$;
ρ/d: $[-\frac{1}{12}, \frac{1}{5}]$;
D_3/d: $[0, \frac{1}{5}]$;
$|D_n|/d, n \geq 4$: $\leq \frac{1}{12}$.

3 D/d: $[-\frac{1}{5}, \frac{2}{3}]$;
ϵ/d: $[-\frac{17}{60}, \frac{7}{8}]$;
a_0/d: $[-\frac{1}{5}, \frac{1}{12}]$;
ρ/d: $[-\frac{1}{12}, \frac{1}{5}]$;
D_3/d: $[-\frac{1}{12}, \frac{1}{30}]$;
$|D_n|/d, n \geq 4$: $\leq \frac{1}{12}$.

Put $C_\beta = 1/(1-\theta^2) = \frac{1024}{855}$. The even moments of a finite mask satisfy D_0 in $[-C_\beta\delta_3, C_\beta\delta_2]$, with $C_\beta\theta^2 = \frac{169}{855} < \frac{1}{5}$ and $C_\beta\theta = \frac{416}{855} < \frac{1}{2}$. For $s_1 = 2$, the mandatory $K^3\mathbf{1}$ gives

$$a_0 \geq \delta_2 - C_\beta\delta_3 \geq 439\delta_2/855 \geq 0, a_0 \leq C_\beta\delta_2 < d/2. \tag{R25.L184}$$

It likewise gives $0 \leq D_3 \leq C_\beta\delta_3 < d/5$. With $K^3\mathbf{1}$ absent at $s_1 = 3$, a_0 lies in $[-C_\beta\delta_3, C_\beta\delta_4]$ and D_3 in $[-C_\beta\delta_4, C_\beta\delta_5]$. For both minima, ρ lies in $[-C_\beta\delta_4, C_\beta\delta_3]$. For all $n \geq 4$, the largest same-sign tail begins at δ_4, so $|D_n| \leq C_\beta\delta_4 < d/12$. Here $C_\beta\theta^3 = \frac{2197}{27360} < \frac{1}{12}$ and $C_\beta\theta^4 = \frac{28561}{875520} < \frac{1}{30}$.

The bounds on D follow from (R25.3.4). For $s_1 = 3$ use the retained sharper $D_0 \leq 416d/855$, so $D \leq (\frac{416}{855} + \frac{27}{160})d < 2d/3$. For $s_1 = 2$ the mandatory first negative contribution to $D_0 - a_0$ cancels the possible $+\delta_2$; the remaining positive subcollections are at most

$$C_\beta(\delta_3+\delta_4)\le \delta_3/(1-\theta)\le 13\delta_2/19. \tag{R25.L198}$$

Hence $\epsilon\le(13\theta/19+\frac{5}{12})d<7d/10$. Its lower bound is $-d/5-d/2$. $Ats_1=3,\epsilon\ge -d/5-d/12=-17d/60$ and $\epsilon\le 2d/3+d/5=13d/15<7d/8$.

Finally write $U=\alpha+z\beta$. The actual finite coefficient signs give

$$\begin{gathered} s_1=2: \\ x-x^2/(1-x^2)\le\alpha\le x/(1-x^2), \\ -x/(1-x^2)\le\beta\le x^2/(1-x^2); \\ \text{thus } x/2\le\alpha\le 6x/5, -6x/5\le\beta\le x/2. \end{gathered} \tag{R25.3.7}$$

$$\begin{gathered} s_1=3: \\ -x^2/(1-x^2)\le\alpha\le x^3/(1-x^2), \\ -x/(1-x^2)\le\beta\le x^2/(1-x^2). \end{gathered} \tag{R25.3.8}$$

These are frequency statements for the same finite polynomial, without using the frequency lower bound $x\ge d$.

R25.4 Positive propagation, every finite convolution loss and true means

Because the original T kernel lies in $[0,2], T(R^2)\le 2V$. Equations(R25.3.2)–(R25.3.6) give a positive actual upper and its exact finite propagation:

$$\begin{gathered} e^{\lambda d}T^L F\le Y_L, \\ Y_1=q+D+BR, \\ Y_L=\bar f_L+B^L R+D\bar f_{L-1} \\ +\sum_{i=1}^{L-1} D_i\bar f_{L-1-i}. \end{gathered} \tag{R25.4.1}$$

Y_L is nonnegative because it is T^{L-1} applied to the actual first positive upper. The signed summands remain in this upper bound. The finite identity $T^rR=B^rR+\sum_{i=0}^{r-1}D_i\bar f_{r-1-i}$ proves(R25.4.1).

Define, for $n\ge 2$,

$$P_n=G_n+z^nU, Q_n=P_n+\epsilon G_{n-2}+Dz^{n-2}. \tag{R25.L233}$$

Then $Q_2=G_2+D+z^2U$ and $Q_{n+1}=1+\epsilon+zQ_n$. Direct substitution of the finite positive convolution gives

$$\begin{gathered} Y_L=1+\epsilon+Q_L(B)u-\sum_{n=3}^{L}\gamma_n\bar f_{L-n}, \\ \gamma_3=d-D_2=d+\rho, \\ \gamma_4=d+Dd-D_3, \\ \gamma_n=\beta_n+D\beta_{n-1}-a_0\beta_{n-2}-D_{n-1}, n\ge 5. \\ (R25.4.2) \end{gathered} \tag{R25.L239}$$

In particular the full original means are

$$\begin{aligned} EY_1 &= 1+\epsilon, \\ EY_2 &= 1+\epsilon-\gamma_3, \\ EY_3 &= 1+\epsilon-\gamma_3-\gamma_4. \end{aligned} \tag{R25.4.3}$$

The parameter table gives

$$\begin{aligned} s_1 = 2 : \gamma_3 \geq 11d/12, \gamma_4 \geq 11d/15, \\ \gamma_n \geq [\kappa - \frac{1}{12} - (\frac{7}{10})d]d \geq 133d/480 > d/4; \\ s_1 = 3 : \gamma_3 \geq 11d/12, \gamma_4 \geq 9d/10, \\ \gamma_n \geq [\kappa - \frac{1}{12} - (\frac{17}{60})d]d > 2d/5, n \geq 5. \end{aligned} \tag{R25.4.4}$$

For the latter long bounds use $D \geq -d/5, \beta \leq d$ and respectively $a_0 \leq d/2$ or $d/12$. Only these bounds on the β sequence are used.

Let $h = \frac{1}{4}$ in the minimum-two case, $h = \frac{2}{5}$ in the minimum-three case. For EVERY finite $L \geq 4$ retain precisely $\gamma_L \bar{f}_0$ and $\gamma_{L-1}\bar{f}_1$:

$$\begin{aligned} 0 \leq Y_L \leq Z_L = 1+\epsilon+H-hd(1+q), H = Q_L(B)u, \\ Z_L = (1+\epsilon-hd)q - hd + BQ_{L-1}(B)u. \end{aligned} \tag{R25.4.5}$$

$AtL = 4$ these are γ_4 and γ_3; no positive β_2 is fabricated. The same finite formula yields

$$\begin{aligned} EH = -\gamma_{L+1}, \langle H, u\rangle = \gamma_{L+2}, \\ EZ_L \leq 1+\epsilon-3hd, \\ \text{Var}\, Z_L = \|H\|^2 - 2hd\gamma_{L+2} + h^2d^3 - \gamma_{L+1}^2 \\ \leq \|Q_L\|^2 d. \end{aligned} \tag{R25.4.6}$$

The last inequality retains the real u cross and the mean subtraction; $\gamma_{L+2} \geq hd$ and $d \leq \frac{1}{3}$ suffice. The extra mean indices are paid by(R25.4.4).

R25.5 Complete finite symbol norms and receiving signs

For minimum two, P_3 has positive coordinates

$$1 - x + x^2\beta, 1 - x\alpha. \tag{R25.L282}$$

By(R25.3.7) its squared norm is at most

$$(1-x+x^3/2)^2 + x(1-x^2/2)^2 = 1 - x + x^2 - x^4 + x^5/4 + x^6/4 \leq 1. \tag{R25.5.1}$$

The final comparison follows from $x(1-x) \geq 0$ and $x^4[1 - x/4 - x^2/4] \geq 0$. For minimum three, the G_3 norm gap $\kappa x/2$ absorbs the perturbation $(\frac{3}{8})x^{3/2}$, since $(\frac{3}{8})\sqrt{\theta} < \kappa/2$. $Atn = 4$,

$|G_4| \le 1 - x/2$ and $x^2\|U\| < x/2$. $Atn = 5, |G_5| \le 1 - \kappa x/2$ and $x^{5/2}\|U\| < \kappa x/2$. For EVERY $n \ge 6$, (R25.2.2) gives gap $7x/30$ and $x^{n/2}\|U\| \le x^3(\frac{3}{5}) < 7x/30$. Thus both cases have

$$\|P_n\| \le 1 \quad \text{for every finite } n \ge 3. \tag{R25.5.2}$$

The actual scalar and $z-$ coordinate perturbations give

$$\begin{gathered} \|Q_3\| \le 1 + d, \\ \|Q_n\| \le 1 + 5d/4(\text{minimum two}), \\ \|Q_n\| \le 1 + 4d/3(\text{minimum three}), n \ge 3. \end{gathered} \tag{R25.5.3}$$

For Q_3 their squared costs are at most $d^2[(\frac{7}{10})^2 + \theta(\frac{11}{12})^2] < d^2$ and $d^2[(\frac{7}{8})^2 + \theta(\frac{2}{3})^2] < d^2$. $Atn = 4$ explicitly pay G_2 : $(\frac{7}{10})(\frac{6}{5}) + (\frac{11}{12})\theta < \frac{5}{4}$ and $(\frac{7}{8})(\frac{6}{5}) + (\frac{2}{3})\theta = \frac{317}{240} < \frac{4}{3}$. Every later predecessor contracts and its z power decreases.

For $a \ge 3$, the receiving signs follow directly from the finite coordinates:

$$\begin{gathered} \text{Re}(G_a G_2) \ge 1 - 2x \ge \frac{3}{16}, \\ \text{Re}(G_a G_3) \ge 1 - 3x + x^2 \ge -\frac{55}{1024} > -\frac{3}{50}, \\ \text{Re}(G_a G_4) \ge (1 - x)(1 - 2x) \ge \frac{57}{512}, \\ \text{Re}(G_a G_5) \ge (1 - x)^3 > 1/5. \end{gathered} \tag{R25.5.4}$$

For $n \ge 6$, both coordinates of G_n are $\le 1 - x + x^2$, so the weaker but sufficient lower is

$$\text{Re}(G_a G_n) \ge 1 - 3x + 2x^2 - x^3 \ge 1451/32768. \tag{R25.L323}$$

The polynomial is decreasing on $[0, \theta]$. The perturbation is at most $\theta^3/\sqrt{3} < 3\theta^3/5 = \frac{6591}{163840}$, whereas $\frac{1451}{32768} = \frac{7255}{163840}$. Therefore $\text{Re}(G_a P_n) > 0$ for ALL $n \ge 6$. $Atn = 4$ the remaining gap exceeds $\frac{63}{5120}$; at $n = 5$ the perturbation is $< 2\theta^2/5 < \frac{1}{5}$. Thus

$$\text{Re}(G_a P_n) \ge 0 \quad \text{for all } n \ge 4. \tag{R25.5.5}$$

$Atn = 3$, use $\sqrt{\theta} < \frac{16}{25}$ and $\theta^{3/2} < \frac{13}{50}$:

$$\begin{gathered} \text{Re}(G_a P_3) > -\frac{1}{4}(\text{minimum two}), > -\frac{1}{6}(\text{minimum three}). \\ (R25.5.6) \end{gathered} \tag{R25.L335}$$

Indeed $\frac{3}{50} + (\frac{3}{5})(\frac{13}{50}) < \frac{1}{4}$ and $\frac{3}{50} + (\frac{3}{8})(\frac{13}{50}) < \frac{1}{6}$. These bounds cover a_3 and every finite a.

The genuine receiving arm of length two

For $a = 2$, the same receiving factor in (R25.2.3) is $C = f_2 = 1 - u + Bu$, with $\mathbb{E}C = 1 - d$ and $C \le p + \sqrt{d}$. Its covariance is exactly (R25.2.4), with $G_a = G_2$ and $\beta_{a+1} = d$. The following checks give every receiving bound used in the remaining proof for this genuine arm as well.

At $L=1$, write $U=\alpha+z\beta$. Then

$$\mathrm{Re}[G_2(1+zU)]=1-x(\alpha+\beta)\ge\begin{cases}1-17x^2/10, & s_1=2,\\ 1-x^3/(1-x), & s_1=3.\end{cases}$$

In the first case $a_0\ge 0$, so the signed mean term is favorable. In the second case its possible fee is at most $d^2/5$, and the displayed real gap exceeds $d/5\le 1/15$. Thus both covariances are nonpositive. At $L=2$, the full coordinates are $Q_e=1+D-x\alpha$, $Q_o=1-x\beta$, both positive. Their genuine product has the positive lower bounds

$$Q_e-xQ_o\ge\begin{cases}1-x-d/5-(6/5)x^2-(6/5)x^3, & s_1=2,\\ 1-x-d/5-x^3/(1-x), & s_1=3.\end{cases}$$

The actual mean parameter is $\gamma_3>0$. Hence these covariances are nonpositive, in particular within the old bounds $d/16$ and zero.

For every finite $m\ge 2$, the original coordinates give $3/16\le\mathrm{Re}(G_2G_m)\le 1$; for $m=1$ the product is one. Since $|G_2|<6/5$ and $|U|<3/5$,

$$\mathrm{Re}(G_2P_3)>3/16-(18/25)(13/50)>0,\qquad \mathrm{Re}(G_2P_n)>3/16-(18/25)\theta^2>0\quad(n\ge 4).$$

Here the perturbations are the full z^nU, not independently chosen coordinates. Also $\mathrm{Re}(G_2z)=-x$, and $|\mathrm{Re}(G_2z^r)|=x^{\lceil r/2\rceil}\le\theta$ for every integer $r\ge 2$. The signed ϵG_{n-2}, Dz^{n-2}, and translated $-hd\,u$ terms therefore have at most the fees already used below. The remaining actual mean term $-d\gamma_{n+1}$ is favorable. All receiving covariance budgets in (R25.7.1)–(R25.8.5) follow for $a=2$, and the scaled pointwise estimates use its valid weaker cap $C\le p+6\sqrt{d}/5$. Thus the common finite mean, variance, curvature and scalar proof applies to every $a\ge 2$. At $d=0$, one has $u=0$; the exact finite formulas give $C=1$ and $Y_L=1$, so the same conclusion holds.

R25.6 Three self-contained scaled pointwise remainder lemmas

Let $t=\sqrt{d}\le\frac{3}{5}$ and $0\le q\le 2$. Suppose the SAME nonnegative W and $m=EW$ satisfy

$$W\le e^{-\lambda d}(q+at+vd),\, m\le e^{-\lambda d}(1+wd).\tag{R25.L346}$$

By(R25.2.3), the remainder coefficient of(R25.2.5), after multiplication by $e^{2\lambda d}$, is at most

$$H=(2-q+6t/5)[(q+1+at+(v+w)t^2)^2+2(1+wt^2)^2].\tag{R25.6.1}$$

For $b=v+w$, $c=\frac{6}{5}$, the five coefficients beyond $H_0=6+q-q^3\le\frac{32}{5}$ are

$$\begin{aligned}H_1&=(c-2a)q^2+(2a+2c)q+4a+3c,\\ H_2&=-2bq^2+(2b-a^2-4w+2ac)q+2a^2+4b+8w+2ac,\\ H_3&=2b(c-a)q+4ab+ca^2+2bc+4cw,\\ H_4&=(2-q)(b^2+2w^2)+2abc,\\ H_5&=c(b^2+2w^2).\end{aligned}\tag{R25.6.2}$$

All squared majorants dominate the same nonnegative $W+m$, so squaring is legal. The following full continuous bounds follow from(R25.6.2):

A a: $\frac{3}{5}$;

v: $\frac{11}{12}$;

w: $\frac{7}{10}$;

λ lower: $\frac{607}{160}$;

bounds H_1, H_2, H_3, H_4, H_5: 14,15,16,10,5;

M: 8.

B a: $\frac{7}{4}$;

v: $\frac{7}{10}$;

w: 0;

λ lower: $\frac{607}{160}$;

bounds H_1, H_2, H_3, H_4, H_5: $\frac{29}{2}, 15, 11, 4, 1$;

M: 8.

C a: $\frac{8}{5}$;

v: 2;

w: $\frac{3}{2}$;

λ lower: $\frac{19}{4}$;

bounds H_1, H_2, H_3, H_4, H_5: 14,36,42,47,21;

M: 10.

For explicit checks, A has $H_1 = 6 + 18q/5$ and $H_2 = \frac{1067}{75} + (\frac{227}{150})q - (\frac{97}{30})q^2$. Its H_3 is $\frac{1444}{125} + (\frac{97}{50})q < 16$. B has $H_1 = \frac{53}{5} + (\frac{59}{10})q - (\frac{23}{10})q^2$ and $H_2 = \frac{105}{8} + (\frac{203}{80})q - (\frac{7}{5})q^2$; its $H_3 \le \frac{2051}{200} < 11$. The remaining bounds are immediate from nonnegative $2 - q$ in(R25.6.2).

For $A, H \le \frac{32}{5} + 14t + 30t^2$, whereas $8e^{2\lambda t^2} \ge 8 + (\frac{607}{10})t^2$. Their difference is bounded below by $\frac{8}{5} - 14t + (\frac{307}{10})t^2$, with negative discriminant $-\frac{12}{25}$. For $B, H \le \frac{32}{5} + (\frac{29}{2})t + 24t^2$ and the lower difference is $\frac{8}{5} - (\frac{29}{2})t + (\frac{367}{10})t^2$, discriminant $-\frac{2463}{100}$.

Here is a complete replacement for the unavailable old coefficient-ten lemma. For C,

$$\begin{aligned} H_1 &= 10 + 28q/5 - 2q^2 \le \frac{348}{25} < 14, \\ H_2 &= \frac{874}{25} + (\frac{57}{25})q - 7q^2 < 36, \\ H_3 &= \frac{5134}{125} - (\frac{14}{5})q < 42, \\ H_4 &< 47, H_5 = \frac{201}{10} < 21. \end{aligned} \tag{R25.L387}$$

Using $t \le \frac{3}{5}, H \le \frac{32}{5} + 14t + (\frac{306}{5})t^2 + (\frac{298}{5})t^4$. Also $10e^{2\lambda t^2} \ge 10 + 95t^2 + (\frac{1805}{4})t^4$. The difference is at least

$$\frac{18}{5} - 14t + (\frac{169}{5})t^2 + (\frac{7833}{20})t^4 > 0, \tag{R25.L395}$$

because the quadratic discriminant is $-\frac{7268}{25}$. This proves $M = 10$ on the entire original domain.

R25.7 Minimum two: every positive connector length

Here $\lambda = \frac{607}{160}$. In each row set $W = e^{-\lambda d}Y$, using Y_L at $L = 1 - -3$ and Z_L from(R25.4.5) at $L \geq 4$. These are actual nonnegative uppers.

R25.7.1 $L = 1$

$Y_1 = q + D + BR$ has $m_Y \leq 1 + 7d/10$ and $\operatorname{Var} Y_1 \leq 737d/480$, since

$$\begin{aligned} \operatorname{Var}(u + BR) &= d + 2\rho + \|BR\|^2 - a_0^2 \\ &\leq [1 + \frac{2}{5} + \theta/3]d. \end{aligned} \tag{R25.L409}$$

Its receiving covariance is nonpositive: the symbol $1 + zU$ has real receiving product $\geq \kappa - \sqrt{\theta}/\sqrt{3} > 0$, and $\alpha = a_0 \geq 0$ in(R25.2.4). Pointwise $Y_1 \leq q + (\frac{3}{5})\sqrt{d} + 11d/12$, so Lemma A applies. The scaled Taylor identity gives

$$\begin{aligned} E[CW^4] &\leq e^{-107d/8} + (\frac{737}{60})de^{-607d/80} \\ &\leq e^{-13d} + (\frac{25}{2})de^{-7d}. \end{aligned} \tag{R25.7.1}$$

R25.7.2 $L = 2$**: full precursor, output variance and small positive covariance**

$Y_2 = q + \epsilon + BV_2$ with $V_2 = D\mathbf{1} + u + BR$. Its FULL precursor norm is

$$\|V_2\|^2 = d + 2\rho + \|BR\|^2 + D^2 - 2Da_0. \tag{R25.L424}$$

For $D \geq 0$ its last cross is favorable. For $D < 0$ the constant cost is at most $6d^2/25$, less than $121d^2/144$. Hence

$$\|V_2\|^2 \leq 737d/480 + 121d^2/144 < (\frac{7}{4})^2 d. \tag{R25.L429}$$

The actual mean $m_Y = 1 + \epsilon - \gamma_3 \leq 1 - d/10$. Lemma B applies. The real and imaginary coordinates of Q_2 are positive. Using (R25.3.7),

$$\begin{aligned} \operatorname{Re}(G_a Q_2) &\geq 1 - 2x - (d/5)(1 - x) - (\frac{6}{5})x^2 \\ &\geq -\frac{77}{1536} > -1/16. \end{aligned} \tag{R25.7.2}$$

The scalar $\alpha = \gamma_3$ is positive, so (R25.2.4) gives $\operatorname{Cov}(C, Y_2) \leq d/16$.

For $H = Q_2 u = (1 + D)u + Bu + B^2 R$, the output variance is

$$\operatorname{Var} Y_2 = (1 + D)^2 d + \delta_2 - 2(1 + D)D_3 + 2D_4 + \|B^2 R\|^2 - \gamma_3^2. \tag{R25.L442}$$

Here $D_3 \geq 0$ and $1 + D > 0$, so

$$\begin{aligned} \operatorname{Var} Y_2 &\le C_0 + (\frac{11}{6})d + (\frac{121}{144})d^2 d, \\ C_0 &= 1 + \theta + \frac{1}{6} + \theta^2/3 < \frac{5}{3}, \\ \operatorname{Var} Y_2 &\le (\frac{5}{3})(1 + 13d/10)d. \end{aligned} \tag{R25.7.3}$$

The signed D_4 and actual mean subtraction are not confused with V_2. The mean fourth term is $\le e^{-663d/40}$, and the covariance term is $\le (d/4)e^{-619d/40}$. Together they are at most $e^{-15d}(1+d/4) \le e^{-59d/4}$. The scaled variance contributes at most $(\frac{40}{3})de^{-6d}$, since $\frac{607}{80} - \frac{13}{10} > 6$. Therefore

$$E[CW^4] \le e^{-59d/4} + (\frac{40}{3})de^{-6d}. \tag{R25.7.4}$$

R25.7.3 $L = 3$

The actual $Y_3 = (1+\epsilon)q - \gamma_3 + BQ_2u$ has

$$\begin{aligned} m_Y \le 1 - 19d/20, \operatorname{Var} Y_3 \le (1+d)^2 d, \\ \operatorname{Cov}(C, Y_3) \le d/4 + (\frac{11}{10})d^2. \end{aligned} \tag{R25.7.5}$$

For the covariance use(R25.5.6), the actual $\alpha = \gamma_4 > 0$, and $|\epsilon| + D_+\theta \le (\frac{7}{10} + 11\theta/12)d < 11d/10$. For the pointwise predecessor use

$$\begin{aligned} \|Q_2\| \le \frac{6}{5} + \frac{11}{36} + \frac{39}{160} = \frac{2519}{1440} < \frac{7}{4}, \\ \epsilon q - \gamma_3 \le 29d/60 < 7d/10. \end{aligned} \tag{R25.L469}$$

Thus Lemma B applies with the same actual mean, which is ≤ 1. The first two Taylor terms are bounded by $e^{-15d}[1 + d + (\frac{22}{5})d^2] \le e^{-12d}$; the variance term is $\le 8de^{-5d}$. Consequently

$$E[CW^4] \le e^{-12d} + 8de^{-5d}. \tag{R25.7.6}$$

R25.7.4 Every finite $L \ge 4$

Use the exact $Z_L, h = \frac{1}{4}$, and (R25.4.6). Then

$$\begin{aligned} m_Y \le 1 - d/20, \operatorname{Var} Y \le (1 + 5d/4)^2 d, \\ \operatorname{Cov}(C, Y) \le (\frac{7}{10} + 11\theta/12 + \frac{1}{4})d^2 < 4d^2/3. \end{aligned} \tag{R25.7.7}$$

To see the covariance, (R25.5.5) pays the P_L part, the ϵG_{L-2} product has absolute real part ≤ 1, and Dz^{L-2} costs at most $|D|\theta d$. $AtL = 4$ the actual G_2 product lies in $[\frac{3}{16}, 1]$, so its exceptional norm is retained. The $-hdu$ term costs at most hd^2. The remaining signed mean product is favorable.

The predecessor norm $\le 1 + 5d/4 < \frac{7}{4}$ and $(\epsilon - hd)q - hd \le 13d/20 < 7d/10$, so Lemma B applies. The first two terms are at most $e^{-15d}(1 + 16d/9) \le e^{-12d}$. The variance exponent is $\frac{607}{80} - \frac{5}{2} = \frac{407}{80} > 5$. Hence (R25.7.6) pays every finite $L \ge 4$, without any limiting length or gap cutoff.

R25.8 Minimum three: every positive connector length

Here $\lambda = \frac{767}{160} > \frac{19}{4}$. Every actual row meets Lemma C.

$AtL = 1$ the pointwise radical is $< 3\sqrt{d}/8$ and $D_+ \le 2d/3$. $AtL = 2$ the full V_2 above has

$$\begin{aligned} \|V_2\|^2 &\le (1 + \frac{2}{5} + 9\theta/64)d + (\frac{4}{9} + \frac{4}{15})d^2 \\ &< (\frac{8}{5})^2 d. \end{aligned} \tag{R25.L505}$$

The cross $-2Da_0$ is paid, allowing either sign. $AtL = 3, \|Q_2\| \le \frac{6}{5} + \frac{2}{9} + (\frac{3}{8})\theta < \frac{8}{5}$ and $\epsilon q - \gamma_3 \le 5d/6 < 2d$. *At* every $L \ge 4, \|Q_{L-1}\| \le 1 + 4d/3 < \frac{8}{5}$ and $(\epsilon - 2d/5)q - 2d/5 \le 11d/20 < 2d$. All means below are $\le 1 + 3d/2$. These prove the literal hypotheses of the self-contained coefficient-ten lemma, not merely a similar old scope.

R25.8.1 $L = 1$

The receiving product of $1 + zU$ is at least $\kappa - (\frac{3}{8})\sqrt{\theta} > \frac{1}{3}$. Its possible unfavorable signed mean-product cost is $\le d^2/5 < d/3$, so $\mathrm{Cov}(C, Y_1) \le 0$. Variance $\le 3d/2$ and $m_Y \le 1 + 7d/8$. Thus

$$E[CW^4] \le e^{-16d} + 15de^{-8d}. \tag{R25.8.1}$$

R25.8.2 $L = 2$

Using the actual coordinate bounds(R25.3.8), both Q_2 coordinates are positive and the complete lower is

$$\begin{aligned} \mathrm{Re}(G_a Q_2) &\ge 1 - 2x - (d/5)(1 - x) \\ &-(1 - x)x^4/(1 - x^2) - x^3/(1 - x^2) \\ &> \frac{3}{16} - \frac{1}{15} - \frac{1}{30} - \frac{1}{12} = 1/240. \end{aligned} \tag{R25.8.2}$$

The bounds on the last two terms use $x^4/(1 + x) < \frac{1}{30}$ and $\theta^3/(1 - \theta^2) < \frac{1}{12}$. Since $\alpha = \gamma_3 > 0$, covariance is nonpositive. The complete output formula of(R25.7.3) now gives

$$\begin{aligned} \mathrm{Var}\, Y_2 &\le C_0 + (\frac{13}{9})d + (\frac{4}{9})d^2 d, \\ C_0 &= 1 + \theta + \frac{1}{3} + \frac{1}{40} < \frac{9}{5}, \\ \mathrm{Var}\, Y_2 &\le (\frac{9}{5})(1 + 9d/10)d. \end{aligned} \tag{R25.8.3}$$

Indeed $D_3 \ge -d/12$ contributes at most $(1 + D)d/6, D_4$ contributes $d/6$, and $\|B^2 R\|^2 < d/40$. Every signed and constant coordinate is included. The actual mean is $\le 1 - d/24$. The mean exponent exceeds 19 and the variance exponent $\frac{767}{80} - \frac{9}{10}$ exceeds $\frac{17}{2}$. Therefore

$$E[CW^4] \le e^{-19d} + 18de^{-17d/2}. \tag{R25.8.4}$$

R25.8.3 $L = 3$

Equations(R25.4.3),(R25.5.3),(R25.5.6) and $\alpha = \gamma_4 > 0$ give

$$\begin{aligned} m_Y \le 1 - 9d/10, \operatorname{Var} Y_3 \le (1+d)^2 d, \\ \operatorname{Cov}(C, Y_3) \le d/6 + (\frac{7}{6})d^2. \end{aligned} \tag{R25.L553}$$

The coordinate cost is at most $(\frac{7}{8}+2\theta/3)d < 7d/6$. The first two Taylor terms are at most $e^{-19d}[1+(\frac{2}{3})d+(\frac{14}{3})d^2] \le e^{-16d}$. The variance exponent $\frac{767}{80} - 2$ exceeds 6. Thus

$$E[CW^4] \le e^{-16d} + 10de^{-6d}. \tag{R25.8.5}$$

R25.8.4 Every finite $L \ge 4$

Use $h = \frac{2}{5}$ in(R25.4.5). All actual losses and extra mean indices have already been paid. Then

$$\begin{aligned} m_Y \le 1 - 13d/40, \operatorname{Var} Y \le (1+4d/3)^2 d, \\ \operatorname{Cov}(C, Y) \le (\frac{7}{8} + 2\theta/3 + \frac{2}{5})d^2 < 8d^2/5. \end{aligned} \tag{R25.L568}$$

The same finite receiving calculation as(R25.7.7) includes the $L = 4G_2$ case. The first two Taylor terms are at most $e^{-19d}(1 + 32d/15) \le e^{-16d}$. The variance exponent $\frac{767}{80} - \frac{8}{3} = \frac{1661}{240} > 6$, so (R25.8.5) also pays every finite long length.

R25.9 Complete continuous scalar payments: one original host interval

All rows above hold on $0 \le d \le \frac{1}{3}$ with one original direction. Elementary rational Taylor bounds give $e < \frac{11}{4} < 3$ and $e^2 < 8$. For example the terms through $n = 4$ sum to $\frac{65}{24}$ and the remaining positive tail is $< \frac{1}{100}$, which proves $e < \frac{11}{4}$.

If $A > b \ge 0$ and $B \ge A/2$, the exact identity for sinh gives

$$1 - e^{-Ad} - bde^{-Bd} \ge (A-b)de^{-Ad/2}. \tag{R25.9.1}$$

This pays (R25.7.1) by more than $d/54$, using $A = 13, b = \frac{25}{2}, B = 7$ and $e^{13/6} < e^3 < 27$. It pays(R25.8.1) by more than $d/27$ using $A = 16, b = 15, B = 8$.

For the other four rows, use e^{Bd} bounded below through degree four and $e^{-(A-B)d}$ bounded above through degree four, then divide their difference by d. The quartic Taylor upper for a negative exponent is global for nonnegative d. The resulting comparisons are:

$$\begin{aligned} &\frac{e^{6d} - e^{-35d/4}}{d} \\ &\ge \frac{59}{4} - (\frac{649}{32})d + 84d^2 > \frac{40}{3} + \frac{1}{6}; \end{aligned} \tag{R25.9.2}$$

$$\frac{e^{5d} - e^{-7d}}{d} \ge 12 - 12d + 53d^2 > 11; \tag{R25.9.3}$$

$$\frac{e^{17d/2} - e^{-21d/2}}{d} \ge 19 - 19d + 198d^2 > \frac{37}{2}; \tag{R25.9.4}$$

$$\frac{e^{6d}-e^{-10d}}{d} \geq 16-32d+81d^2 > 12. \tag{R25.9.5}$$

For full coefficient verification, the quadratic coefficients after using $d \leq \frac{1}{3}$ on the signed cubic remainders are respectively

$$\begin{aligned} \frac{56699}{384}-\frac{1168849}{18432} &> 84, \\ 78-\frac{74}{3}=\frac{160}{3} &> 53, \\ \frac{7087}{24}-\frac{6935}{72}=\frac{7163}{36} &> 198, \\ \frac{608}{3}-\frac{1088}{9}=\frac{736}{9} &> 81. \end{aligned} \tag{R25.L607}$$

The all-real minimum of the first quadratic above $\frac{40}{3}$ is $\frac{66223}{344064} > \frac{1}{6}$. The others have minima $12-\frac{36}{53} > 11, 19-\frac{361}{792} > \frac{37}{2}$, and $16-\frac{256}{81} > 12$. These are continuous polynomial certificates, not checks at sample points.

Multiplying back by the exact positive scale gives deficits:

Table 34: Even terminal lists of minimum two or three: parameter values.

Entry	Certified deficit
(R25.7.4)	$> (d/6)e^{-6d} > d/48$.
(R25.7.6)	$> 3de^{-5d} > d/3$.
(R25.8.4)	$> (d/2)e^{-17d/2} > d/54$.
(R25.8.5)	$> 2de^{-6d} > d/4$.

Every deficit in this section exceeds $d/100$. This proves Lemma N.4 on the entire positive original host domain.

N.40 Even terminal lists of minimum one, including the global third class

R28.1 Target, original objects and exact structural partition

The original problem is unchanged: for every finite original tree U with at most five vertices of original degree at least three, choose every original edge direction before the host, so that

$$Z_D = E_{\mu^{V(U)}} \prod_{v \to w} (1 + K(x_v, x_w)) \leq 1 \tag{R28.L20}$$

for every original finite probability μ and real antisymmetric $|K| \leq 1$. Every original arm, multiplicity, vertex and subdivision vertex remains.

The local theorem of this module is the following all-parameter family. After actual maximal equal-length pairing, take an even terminal residual

$$\begin{aligned} 1 = \text{basis}_{\min} < b_1 < a_2 < b_2 < \ldots < a_j < b_j, j \geq 2, \\ F = q f_{b_1} \prod_{i=2}^{j} h_{a_i} f_{b_i}. \end{aligned} \tag{R28.L29}$$

The first unit arm points toward the terminal, each f arm points entirely outward, and each h arm has its first edge toward the terminal and all later edges toward the leaf. Every one of the $L \geq 1$ connector edges points from terminal toward receiver. The actual existing receiving arm points outward and has its original length $a \geq 3$. Its factor is exactly f_a. The bound is:

$$E[E_o f_a T^L (E_t F)^4] \leq 1 - d/3000. \tag{R28.1.1}$$

E_t and E_o are arbitrary actual equal-pair products in their original locations. All finite residual lengths/gaps and all finite j are allowed. In this particular theorem one fixed F works for every L and every host. No orientation is selected according to d or another host statistic.

We divide the integer parameters into four cases:

I Exact condition: $b_1 \geq 3$;

Analytic ranges: low $d \leq \frac{1}{64}$;

same-direction high comparison.

II Exact condition: $b_1 = 2, a_2 \geq 4$;

Analytic ranges: low $d \leq \frac{1}{64}$;

same-direction high comparison.

IIIa

Exact condition: $b_1 = 2, a_2 = 3, j = 2$;

Analytic ranges: low $d \leq \frac{1}{8}$;

same-direction high comparison.

IIIb

Exact condition: $b_1 = 2, a_2 = 3, j \geq 3$;

Analytic ranges: complete global finite-mask proof.

The low/high partitions change only estimates of the same message.

R28.2 Original finite calculus and the actual receiver

Work on the positive support of the original μ. Write

$$\begin{gathered} B = -K, u = B\mathbf{1}, T = J + B, A = J - B = T^*, Jv = (Ev)\mathbf{1}, \\ p = 1 - u, q = 1 + u, f_n = A^n\mathbf{1}, \bar{f}_n = T^n\mathbf{1}, h_n = Tf_{n-1}, \\ \delta_i = \|B^i\mathbf{1}\|^2, d = \delta_1, \theta = \frac{13}{32}, \kappa = \frac{19}{32}, \tau = d - \delta_2. \end{gathered} \tag{R28.L60}$$

The original C014 finite calculus gives $0 \leq d \leq \frac{1}{3}, d^2 \leq \delta_2, \delta_{i+1} \leq \theta \cdot \delta_i, \|B\|^2 \leq \theta$ and $|BV(x)| \leq \|V\|$. A, T are positive kernels, all actual path messages lie in[0,2], and $Ef_n = E\bar{f}_n \leq 1$. On every positive-mass atom, $p, q \geq \mu(x) > 0$. The equality $f_l + h_l = 2Ef_{l-1}$ gives $0 \leq f_l h_l \leq 1$ for genuine equal pairs.

Put $G_n(z) = \sum_{r=0}^{n-1} z^r, G_0 = 0, \beta_1 = \beta_2 = 0$ and

$$\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1} \delta_i, n \geq 3. \tag{R28.L72}$$

Finite positive convolution gives

$$\begin{aligned} 1+G_n(B)u &= \bar{f}_n+\sum_{r=3}^{n}\beta_r\bar{f}_{n-r}, \\ \tau \le \beta_r \le d(r\ge 3), &\beta_3=\beta_4=d. \end{aligned} \tag{R28.2.1}$$

These are finite identities, with the actual β_1,β_2 retained as zero. For $z=i\sqrt{x}, 0\le x\le\theta$, the real/normalized-imaginary coordinates of G_n lie in $[1-x,1]$ for $n\ge 2$. Its real coordinate is $\le 1-x+x^2$ for $n\ge 3$; both coordinates obey that upper for $n\ge 4$. Every $n\ge 3$ contracts, while $\|G_2\|=\sqrt{1+x}<\frac{6}{5}$. The strengthened finite gap is

$$|G_n|^2 \le 1-7x/15, n\ge 3. \tag{R28.2.2}$$

This exact finite fact is proved in Q5-C025/E004. For example the only potentially larger tail numerator is $(1+x^3)^2$; comparison after multiplying by $1+x$ reduces to $8-7x-30x^2-15x^5\ge\frac{1311885}{33554432}>0$ at its decreasing endpoint θ. Odd tails and multiples of four satisfy the stronger respective bounds.

The same weak receiver is

$$\begin{aligned} 0\le f_a\le C &= 1-G_a(-B)u-\beta_a, \\ EC &= 1-\beta_a-\beta_{a+1}\le 1-d, \\ C &\le p+(\frac{6}{5})\sqrt{d}, a\ge 3. \end{aligned} \tag{R28.2.3}$$

Let σ denote u's original spectral measure for $-B^2$; its mass isd and $\int x^r d\sigma=\delta_{r+1}$. If $H=Q(B)u$ and $EH=-\alpha$, exact adjunction gives

$$\mathrm{Cov}(C,H)=-\int \mathrm{Re}(G_aQ)d\sigma-\beta_{a+1}\cdot\alpha. \tag{R28.2.4}$$

For every actual nonnegative Y and its actual mean m,

$$E[CY^4]=(EC)m^4+4m^3\,\mathrm{Cov}(C,Y)+E[C(Y-m)^2(Y+m)^2+2m^2]. \tag{R28.2.5}$$

Useful all-finite receiving bounds are

$$\begin{aligned} \mathrm{Re}(G_aG_2) &\ge 1-2x\ge\frac{3}{16}, \\ \mathrm{Re}(G_aG_3) &\ge 1-3x+x^2\ge-\frac{55}{1024}>-\frac{3}{50}, \\ \mathrm{Re}(G_aG_4) &\ge (1-x)(1-2x)\ge\frac{57}{512}, \\ \mathrm{Re}(G_aG_5) &\ge (1-x)^3>\frac{1}{5}, \\ \mathrm{Re}(G_aG_n) &\ge 1-3x+2x^2-x^3\ge\frac{1451}{32768}, n\ge 6. \end{aligned} \tag{R28.2.6}$$

Also $G_6=A_3(x)(1+z), A_3=1-x+x^2$, so its receiving lower is at least $\frac{57}{512}$. The last cubic in(R28.2.6) decreases on $[0,\theta]$. In all cases $a=3$ and zero frequency are included.

R28.3 Finite masks and positive propagation

For an actual pair define $R_{st} = \sum_{n=s+1}^{t} K^n \mathbf{1}$. Exact finite differences and nonnegative AM-GM give

$$\begin{aligned} qf_{b_1} &\le \exp(R_{1b_1} - \beta_{b_1}), \\ h_s f_t &\le \exp(R_{st} - 16d/5), 4 \le s < t, \end{aligned} \tag{R28.3.1}$$

$$h_3 f_t \le \exp(R_{3t} - \tau \cdot g - (3d - 2\delta_2)), t \ge 4, g = K\mathbf{1} = -u. \tag{R28.L130}$$

For the last line retain the actual $\beta_{t-1} f_1$ term in $f_t - f_3$. For the long-pair line $s = 4$ retain $-\tau \cdot f_2$ and use $f_2 \ge p^2/2$, which bounds $\delta_2(1+p) - \tau f_2$ by $3d/4$. For $s \ge 5$ the actual path mean $Ef_n \le 1 - 83d/32, n \ge 4$, combined with the finite difference positive part $4\delta_2/(1-\theta^2) \le 1664d/855$ leaves loss $\frac{83}{16} - \frac{1664}{855} > \frac{16}{5}$. These exact finite proofs are given in the actual C092§2, C067§2 and reproduced E004§2.2.

Let $R_0 = \sum_i R_{a_i, b_i}$ with $a_1 = 1$. Then $R_0 = BP_0(-B)u$, where P_0 is one finite $\frac{0}{1}$ polynomial coming from all disjoint original pair intervals. Write at $z = i\sqrt{x}$

$$P_0(-z) = U_0(x) - zV_0(x), c(x) = 1/(1-x^2), R_0 = (xV_0 + zU_0)(B)u. \tag{R28.L146}$$

Grouping the four coefficient residue classes gives

Table 35: Even terminal lists of minimum one, including the global third class: parameter values.

Entry	Mandatory coefficients	Exact useful coordinates	$\lvert P_0\rvert^2$ upper
I	$(P_0)_0 = (P_0)_1 = 1$;	$1 - xc \le U_0, V_0 \le c$;	$\frac{100}{49}$.
II	$(P_0)_0 = 1, (P_0)_1 = (P_0)_2 = 0$;	$1 - x^3 c \le U_0 \le 1 + x^2 c; -xc \le V_0 \le x^2 c$;	$\frac{25}{16}$.
III	$(P_0)_0 = (P_0)_2 = 1, (P_0)_1 = 0$;	$1 - xc \le U_0 \le 1 - x/3; -xc \le V_0 \le x^2 c$;	1.

For a general $\frac{0}{1}$ mask its square is at most $(1+x)/(1-x^2)^2 \le \frac{32768}{16245} < \frac{100}{49}$. The missing II coefficients improve this to $(1+x^3)/(1-x^2)^2 < \frac{25}{16}$. In III the compulsory negative real term gives $(1 - x/3)^2 + (\frac{16}{9})x^3 \le 1 - x/6 \le 1$. These bounds apply to each finite mask.

Thus $F \le \exp(R - \Gamma)$, where

Table 36: Even terminal lists of minimum one, including the global third class: parameter values.

Entry	Actual R	Γ lower
I	R_0;	$(\frac{607}{160})d + (\frac{16}{5})(j-2)d$.
II	R_0;	$(\frac{16}{5})(j-1)d$.
III	$R_0 + \tau \cdot u$;	$(\frac{35}{16})d + (\frac{16}{5})(j-2)d$.

Suppose $e^R \le 1 + R + (\eta/2)R^2$. Write $D_i = EB^i R$ and choose a scalar $D \ge D_0 + \eta \cdot ER^2; D$ can be an explicitly enlarged nonnegative value. Original positive T and $T(R^2) \le 2ER^2$ give the actual nonnegative upper

$$
\begin{aligned}
0 \le e^{\Gamma} T^{L} F \le Y_{L} \\
= \bar{f}_{L} + B^{L} R + D \bar{f}_{L-1} + \sum_{i=1}^{L-1} D_{i} \bar{f}_{L-1-i}.
\end{aligned}
\tag{R28.3.2}
$$

Put $\epsilon = D + D_1$ and for $n \ge 2$

$$
P_n = G_n + z^n U, R = U(B)u, Q_n = P_n + \epsilon G_{n-2} + D z^{n-2}. \tag{R28.L180}
$$

Then $Q_2 = G_2 + D + z^2 U, Q_{n+1} = 1 + \epsilon + z Q_n$. The exact finite identity is

$$
\begin{aligned}
Y_{L} = 1 + \epsilon + Q_{L} u - \sum_{n=3}^{L} \gamma_{n} \bar{f}_{L-n}, \\
\gamma_3 = d - D_2, \gamma_4 = d + Dd - D_3, \\
\gamma_n = \beta_n + D\beta_{n-1} + D_1 \beta_{n-2} - D_{n-1}, n \ge 5.
\end{aligned}
\tag{R28.3.3}
$$

In particular

$$
\begin{aligned}
Y_1 = q + D + BR, \\
Y_2 = q + \epsilon + B(D\mathbf{1} + u + BR), \\
Y_3 = (1+\epsilon)q - \gamma_3 + BQ_2 u, \\
EY_1 = 1 + \epsilon, EY_2 = 1 + \epsilon - \gamma_3, \\
EY_3 = 1 + \epsilon - \gamma_3 - \gamma_4.
\end{aligned}
\tag{R28.3.4}
$$

If all $\gamma_n \ge h \cdot d, n \ge 3$, every original $L \ge 4$ permits retaining exactly the last two true paths $\gamma_L \bar{f}_0$ and $\gamma_{L-1} \bar{f}_1$:

$$
0 \le Y_L \le Z_L = 1 + \epsilon + Q_L u - hd(1+q) = (1 + \epsilon - hd)q - hd + BQ_{L-1} u. \tag{R28.L200}
$$

If $H = Q_L u$, then $EH = -\gamma_{L+1}, \langle H, u \rangle = \gamma_{L+2}$. Hence

$$
\begin{aligned}
EZ \le 1 + \epsilon - 3hd, \\
\text{Var}\, Z = \|H\|^2 - 2hd\gamma_{L+2} + h^2 d^3 - \gamma_{L+1}^2 \\
\le \|Q_L\|^2 d.
\end{aligned}
\tag{R28.3.5}
$$

Both extra mean indices are paid by the complete γ bound. The full $L = 2$ predecessor $D\mathbf{1}+u+BR$ is distinct from its output centered variance.

R28.4 Four self-contained scaled fourth-remainder estimates

Let $t = \sqrt{d}, 0 \le q \le 2$ and suppose the nonnegative W and $m = EW$ obey $W \le e^{-\lambda d}(q + At + vd), m \le e^{-\lambda d}(1 + wd)$. Equation(R28.2.3) bounds the coefficient of $(W-m)^2$ in(R28.2.5) by $e^{-2\lambda d} H$, where

$$
H = (2 - q + (\frac{6}{5})t)[(q + 1 + At + (v+w)t^2)^2 + 2(1 + wt^2)^2]. \tag{R28.4.1}
$$

With $b = v + w, c = \frac{6}{5}$, direct multiplication gives $H_0 = 6 + q - q^3 \le \frac{32}{5}$ and

$$\begin{aligned} H_1 &= (c - 2A)q^2 + (2A + 2c)q + 4A + 3c, \\ H_2 &= -2bq^2 + (2b - A^2 - 4w + 2Ac)q + 2A^2 + 4b + 8w + 2Ac, \\ H_3 &= 2b(c - A)q + 4Ab + cA^2 + 2bc + 4cw, \\ H_4 &= (2 - q)(b^2 + 2w^2) + 2Abc, \\ H_5 &= c(b^2 + 2w^2). \end{aligned} \tag{R28.4.2}$$

The following table is proven below over the whole indicated intervals.

Table 37: Even terminal lists of minimum one, including the global third class: parameter values.

Entry	A, v, w	λ lower	t upper	$H_1, \dots, H_5$ upper	Coefficient M
$I/$ II	$\frac{8}{5}, \frac{3}{4}, \frac{3}{4}$;	$\frac{16}{5}$;	$\frac{1}{8}$;	14,22,20,13,5;	8.
IIIa $L = 1$	$\frac{7}{10}, \frac{1}{40}, \frac{9}{40}$;	$\frac{35}{16}$;	$\frac{3}{8}$;	$\frac{66}{5}, 6, \frac{7}{2}, \frac{3}{4}, \frac{1}{5}$;	8.
IIIa $L = 2$, new	$\frac{21}{20}, \frac{1}{6}, 0$;	$\frac{35}{16}$;	$\frac{3}{8}$;	$\frac{66}{5}, \frac{189}{25}, \frac{13}{5}, \frac{1}{2}, \frac{1}{20}$;	8.
IIIa later	$\frac{6}{5}, \frac{1}{6}, 0$;	$\frac{35}{16}$;	$\frac{3}{8}$;	$\frac{66}{5}, 9, 3, 1, 1$;	9.

For $I/$ II, $H_1 = 10 + 28q/5 - 2q^2 < 14, H_2 = \frac{524}{25} + (\frac{32}{25})q - 3q^2 < 22, H_3 = \frac{2484}{125} - (\frac{6}{5})q < 20; H_4 \le \frac{1251}{100} < 13, H_5 = \frac{81}{20} < 5$. With $t \le \frac{1}{8}$ this gives $H \le \frac{32}{5} + 14t + 25t^2$. The difference from $8e^{32t^2/5}$ is at least $\frac{8}{5} - 14t + (\frac{131}{5})t^2$, decreasing on $[0, \frac{1}{8}]$ to $\frac{83}{320} > 0$.

For IIIa $L = 1, H_1 = \frac{32}{5} + 19q/5 - q^2/5 \le \frac{66}{5}, H_2 = \frac{273}{50} + (\frac{79}{100})q - q^2/2 < 6$. The remaining coefficients follow directly from(R28.4.2). Thus $H \le \frac{32}{5} + (\frac{66}{5})t + (\frac{15}{2})t^2$. The lower difference from $8e^{35t^2/8}$ is $\frac{8}{5} - (\frac{66}{5})t + (\frac{55}{2})t^2$, discriminant $-\frac{44}{25}$.

For the NEW $L = 2$ row, H_1 and H_2 increase on $q \in [0, 2]$ and reach $\frac{66}{5}$ and $\frac{189}{25}$. $H_3 = \frac{2423}{1000} + q/20 < \frac{13}{5}, H_4 \le \frac{1}{18} + \frac{21}{50} < \frac{1}{2}, H_5 = \frac{1}{30} < \frac{1}{20}$. Consequently $H \le \frac{32}{5} + (\frac{66}{5})t + (\frac{44}{5})t^2$. Positive exponential Taylor gives

$$8e^{35t^2/8} - H \ge \frac{8}{5} - (\frac{66}{5})t + (\frac{131}{5})t^2 + 75t^4. \tag{R28.4.3}$$

For $t \le \frac{1}{8}$ its quadratic alone is $\ge \frac{23}{64}$. For $t \ge \frac{1}{8}$ use $75t^4 \ge 75t^2/64$: the resulting quadratic has t^2 coefficient $\frac{8759}{320}$ and discriminant $-\frac{47}{50}$. This pays the full actual range, without a scan.

For the last row, $H_1 = \frac{42}{5} + 24q/5 - 6q^2/5 \le \frac{66}{5}$ and $H_2 = \frac{482}{75} + (\frac{133}{75})q - q^2/3 < 9$; the remaining coefficients are $\le 3, 1, 1$. Thus $H \le \frac{32}{5} + (\frac{66}{5})t + 11t^2$, and the difference from $9e^{35t^2/8}$ is at least $\frac{13}{5} - (\frac{66}{5})t + (\frac{227}{8})t^2$, with negative discriminant.

All squared majorants dominate the same nonnegative $W + m$. These comparisons prove that the actual remainder coefficient is $\le M$. In particular for $W = e^{-\lambda d}Y$,

$$\begin{aligned} E[CW^4] \le e^{-4\lambda d}[(EC)(EY)^4 + 4(EY)^3 \operatorname{Cov}(C, Y)] \\ + Me^{-2\lambda d} \operatorname{Var} Y. \end{aligned} \tag{R28.4.4}$$

The variance has the second-power scale, not the fourth-power scale.

R28.5 Classes I and II: complete low-domain packet and finite bounds

Assume $0 < d \le \frac{1}{64}$. The original row bound gives $|R_0|^2 < 100d/49 < \frac{1}{25}$, so $\eta = \frac{15}{14}$ is valid (use $n! \ge 2 \cdot 3^{n-2}$). Set $D = 3d/8$ in I and $D = 0$ in II. The actual signed packet, from the finite coordinates in§3, is

Table 38: Even terminal lists of minimum one, including the global third class: parameter values.

Entry	$\mathbb{E}R_0^2$ upper	D_1/d	ϵ/d
I	$325d/392$;	$[-\frac{1}{2}, 0]$;	$[-\frac{1}{8}, \frac{3}{8}]$.
II	$325d/512$;	$[-\frac{1}{12}, \frac{1}{5}]$;	$[-\frac{1}{12}, \frac{1}{5}]$.

Indeed I has $\mathbb{E}R_0 \le -439d/855$ and $\mathbb{E}R_0 + (\frac{15}{14})\mathbb{E}R_0^2 \le (-\frac{439}{855} + \frac{4875}{5488})d < 3d/8$. II has $\mathbb{E}R_0 \le -11d/12$ and corresponding scalar $\le (-\frac{11}{12} + \frac{4875}{7168})d < 0$. For I the real/imaginary first coefficients give $D_1 \le 0$; for II use $V_0 \in [-xc, x^2c]$. In both cases

$$0 \le D_2 \le d/2, D_3 \le d/5, D_i \le d/10 \quad \text{for every } i \ge 4. \tag{R28.5.1}$$

Explicitly $D_2 = \int xU_0, D_3 = \int x^2V_0, D_4 = -\int x^2U_0 \le 0, D_5 = -\int x^3V_0 \le d/30, D_6 \le C_\beta \cdot \delta_4 < d/10, D_7 \le C_\beta \cdot \delta_5 < d/10$; each later residue class decays by θ^2. Here $C_\beta = \frac{1024}{855}$. These are upper bounds on the actual signed means, not absolute values.

Equation(R28.3.3) gives $\gamma_3 \ge d/2, \gamma_4 \ge 4d/5$. For $n \ge 5$, using $D \ge 0$ and $\beta \le d$,

$$\gamma_n \ge \kappa \cdot d - d/10 - d^2/2 \ge 3d/8. \tag{R28.L299}$$

The II lower $D_1 = -d/12$ is stronger. Thus take $h = \frac{3}{8}$ for every required index. The actual means from(R28.3.4)-(R28.3.5) obey

$$\begin{aligned} EY_1 &\le 1 + 3d/8(I), \le 1 + d/5(II), \\ EY_2 &\le 1 - d/8(I), \le 1 - 3d/10(II), \\ EY_3 &\le 1 - 37d/40, EZ_L \le 1 - 3d/4(L \ge 4). \end{aligned} \tag{R28.5.2}$$

At frequencyz the actual $U = R_0/u = zP_0(-z)$ has $|U| < 1$, since $x|P_0|^2 \le \frac{325}{392} < 1$. The symbols are

$$Q_1 = 1 + z^2P_0(-z) : (1 - xU_0, xV_0), Q_2 : (1 + D - x^2V_0, 1 - xU_0). \tag{R28.5.3}$$

All Q_2 coordinates are positive. In I its real coordinate $\le 1 + \frac{3}{512}$, in II $\le \frac{13}{12}$; both imaginary coefficient $s \le 1$. Hence

$$\|Q_2\|^2 \le \frac{13}{8}, \|Q_2\| < 4/3. \tag{R28.5.4}$$

For $Q_3, |P_3| \le 1 + \theta^{3/2} < \frac{19}{15}$; the added $\epsilon+$ Dz vector costs at most $9d/20$ in I andd/5 in II. Thus $\|Q_3\| < \frac{4891}{3840} < \frac{51}{40}$. For every $n \ge 4$,

$$\|Q_n\| \le 6/5. \tag{R28.5.5}$$

$Atn = 4$ use $1 + \theta^2 + (\frac{3}{8})(\frac{6}{5} + \theta)/64 < \frac{6}{5}$. Subsequent G predecessors contract and their perturbation powers decrease. The II fee is smaller.

Here are all receiving signs, including the genuine short cases. $AtL = 1$,I has $\mathrm{Re}(G_a Q_1) \ge 1 - x - x/(1 - x^2) \ge \frac{2933}{27360} > \frac{17}{160}$. II has lower $1 - (\frac{11}{5})x + x^2 > \frac{17}{160}$. I's mean correction is favorable; II's adverse correction is $\le d^2/5$, smaller than $17d/160$. Thus $\mathrm{Cov}(C, Y_1) \le 0$. $AtL = 2$,

$$\begin{aligned} \mathrm{Re}(G_a Q_2) &\ge (1 - x)(1 + D - x^2 V_0) - x(1 - xU_0) \\ &\ge 1 - 2x + x^2[U_0 - (1 - x)V_0]. \end{aligned} \tag{R28.5.6}$$

I gives lower $1 - 2x - x^4/(1 - x^2) \ge \frac{3}{16} - \frac{1}{30} > 0$. II gives $U_0 \ge \frac{11}{12}, V_0 \le \frac{1}{5}$, so the bracket is positive. The actual $\alpha = \gamma_3 > 0$; hence $\mathrm{Cov}(C, Y_2) \le 0$.

For every $n \ge 4, \mathrm{Re}(G_a P_n) \ge 0$. $Atn = 4$, I gives the decreasing lower $1 - 3x + 2x^2 - (\frac{6}{5})x^3$, still $\frac{1}{96}$ at $x = \frac{5}{12}$. In II, after multiplication by $1 - x^2$, the lower is $1 - 3x + x^2 + 2x^3 - 3x^4 + x^5$. Its derivative is negative on $[0, \frac{5}{12}]$, and its θ value is $\frac{326653}{33554432} > 0$. $Atn = 5$ use baseline $\frac{1}{5}$ and perturb $\theta^{5/2} < \frac{1}{5}$. $Atn = 6$ use the exact G_6 baseline $\frac{57}{512}$ and perturb $\theta^3 < \frac{57}{512}$. At every $n \ge 7$ use the last baseline in(R28.2.6) and

$$\theta^{7/2} < (\frac{16}{25})\theta^3 < 1451/32768. \tag{R28.L349}$$

Thus the general $\frac{1}{10}$ bound is unnecessary. $Atn = 3$ the lower is greater than $-\frac{3}{50} - \theta^{3/2} > -\frac{1}{3}$. Including the true ϵ and D terms,

$$\begin{aligned} \mathrm{Cov}(C, Y_3) &\le d/3 + (\frac{3}{10})d^2, \\ \mathrm{Cov}(C, Z_L) &\le d^2, L \ge 4. \end{aligned} \tag{R28.5.7}$$

For $L = 3$ the adverse fee is $\le d^2/8 + (3\theta/8)d^2 < 3d^2/10$ in I and $\le d^2/12$ in II. For long L use(R28.2.4), the paid P_L sign, the actual real G_2 product in $[\frac{3}{16}, 1]$ at $L = 4$, and the translationhd $*u$. Its fee is at most $(\frac{3}{8} + 3\theta/8 + \frac{3}{8})d^2 < d^2$; II is smaller. The mean correction is favorable.

R28.6 Classes I and II: all low lengths and final continuous payment

Every row fits the $I/$ II coefficient-eight lemma of§4. $AtL = 1$ use $\|R_0\| < \sqrt{d}$ and $D \le 3d/8$. $AtL = 2$ the FULL precursor norm is

$$\|D\mathbf{1} + u + BR\|^2 = d - 2D_2 + \|BR\|^2 + D^2 + 2DD_1 < (\frac{8}{5})^2 d. \tag{R28.6.1}$$

In I the last cross is nonpositive; in II $D = 0$. $AtL = 3$ use $\|Q_2\| < \frac{4}{3}$ and $\epsilon \cdot q - \gamma_3 \le d/4$. For $L = 4$ the predecessor $Q_3 < \frac{51}{40}$; later one $s < \frac{6}{5}$, and $(\epsilon - hd)q - hd \le 0$. All actual means are $\le 1 + 3d/4$.

The actual variances are

$$\begin{aligned}\operatorname{Var} Y_1 &= d - 2D_2 + \|BR\|^2 - D_1^2 \le 27d/20,\\ \operatorname{Var} Y_2 &\le 13d/8,\\ \operatorname{Var} Y_3 &\le (\frac{51}{40})^2 d,\\ \operatorname{Var} Z_L &\le 36d/25.\end{aligned} \tag{R28.6.2}$$

The $L = 2$ variance uses its output Q_2, not(R28.6.1). The long bound follows only after retaining the full mean/cross terms in(R28.3.5).

Use $\lambda = \frac{607}{160}$ in I and $\frac{16}{5}$ in II. Formula(R28.4.4) gives

$$\begin{aligned}L1 : E[CW^4] &\le e^{-64d/5} + (\frac{54}{5})de^{-32d/5},\\ L2 : E[CW^4] &\le e^{-15d} + 13de^{-6d},\\ L3\text{or } L \ge 4 : E[CW^4] &\le e^{-14d} + (\frac{53}{4})de^{-6d}.\end{aligned} \tag{R28.6.3}$$

For $L = 1$ the mean exponents are at least $\frac{587}{40}$ and 13; the covariance is nonpositive. $L = 2$ gives mean exponents at least $\frac{667}{40}$ and 15. For $L = 3$ use mean $\le 1-37d/40$ and(R28.5.7): its first two terms are at most $e^{-31d/2}[1 + (\frac{4}{3} + \frac{3}{160})d] \le e^{-14d}$. Its variance coefficient is $\frac{2601}{200} < \frac{53}{4}$. The long mean exponent is $\ge \frac{84}{5}$; the covariance term has exponent $\ge \frac{301}{20} > 15$, so their sum $\le e^{-15d}(1 + 4d^2) \le e^{-14d}$. Its variance coefficient $\frac{288}{25}$ is smaller than $\frac{53}{4}$. In all rows $2\lambda \ge \frac{32}{5} > 6$.

Full continuous scalar certificates on $0 \le d \le \frac{1}{64}$:

- The first row use $s2\sinh$: loss $\ge 2de^{-32d/5} > d$.
- The second uses the quadratic exponential upper and gives loss

$$\ge 2d - (\frac{69}{2})d^2 - 234d^3 > d. \tag{R28.L400}$$

- The third gives loss $\ge (\frac{3}{4})d - (\frac{37}{2})d^2 - (\frac{477}{2})d^3$

$$\ge 3299d/8192 > d/4. \tag{R28.L402}$$

All are stronger than(R28.1.1).

R28.7 Class IIIa: the true short symbols and complete low packet

Now $F = qf_2h_3f_b, b \ge 4$, with $0 < d \le \frac{1}{8}$. Put $m = b - 3 \ge 1$ and $G_m(-z) = A_0 - zC_0$, $1 - x \le A_0 \le 1, 0 \le C_0 \le 1$. The exponent is

$$R = [\tau + z + z^3 G_m(-z)]u, \Gamma = 3d - 2\delta_2 \ge 35d/16. \tag{R28.7.1}$$

Thus $R_0 = zP_0(-z)u$ with $P_0(-z) = 1 + z^2 G_m(-z)$, coordinates $U_0 = 1 - xA_0, V_0 = -xC_0 \le 0, |P_0| \le 1$. The original row gives $|R| \le \sqrt{d} + d < \frac{1}{2}$ and $\eta = \frac{6}{5}$ is legal. The full signed square obeys

$$ER^2 \le \delta_2 + \tau^2 d,$$
$$ER \le -d + \delta_2,$$
$$ER + (\frac{6}{5})ER^2 \le -d + (\frac{11}{5})\delta_2 + (\frac{6}{5})\tau^2 d \le -7d/80 < 0. \tag{R28.7.2}$$

The cross $2\tau\langle u, R_0\rangle$ is nonpositive, not discarded as an absolute value. Choose $D = 0$. Then

$$\epsilon = D_1 = -\tau \cdot d + \int x^2 C_0 \in [-d^2, \delta_3],$$
$$D_2 = \delta_2 - \int x^2 A_0 \in [\delta_2 - \delta_3, \delta_2], \tag{R28.7.3}$$
$$D_3 = \tau \cdot \delta_2 - \int x^3 C_0 \le \theta \cdot d^2.$$

For every $i \ge 4, D_i \le d/10 : D_4 \le 0, D_5 \le \delta_5, D_6 \le \delta_4, D_7 \le \tau \cdot \delta_4$, and the four signed classes subsequently decay by θ^2. Therefore

$$\gamma_3 \ge \kappa \cdot d, \gamma_4 \ge 243d/256,$$
$$\gamma_n \ge [\kappa - \frac{1}{10} - d^2]d \ge 153d/320 > 7d/16, n \ge 5. \tag{R28.7.4}$$

Take $h = \frac{7}{16}$ for all needed indices. The short symbols are

$$Q_1 = 1 + z^2 + z^4 G_m(-z) + \tau \cdot z,$$
$$Q_2 = G_2 + \tau \cdot z^2 + z^3 + z^5 G_m(-z), \tag{R28.7.5}$$
$$Q_2 coordinates : (1 - \tau \cdot x + x^3 C_0, 1 - x + x^2 A_0).$$

These expressions follow from (R28.3.4).

R28.7.1 $L = 1$

Let $Q_0 = 1 + z^2 + z^4 G_m(-z)$, with coordinates $(1 - x + x^2 A_0, -x^2 C_0)$. Then $|Q_0|^2 \le (1 - x + x^2)^2 + x^5 \le 1$ on $x \le \theta < \frac{1}{2}$, and $\langle Q_0 u, Bu\rangle = -\int x^3 C_0 \le 0$. Hence

$$\operatorname{Var} Y_1 \le d + \tau^2 \delta_2 \le d + \theta \cdot d^3. \tag{R28.7.6}$$

The receiving lower is $\operatorname{Re}(G_a Q_1) \ge (1-x)^2 - \tau \cdot x \ge \kappa^2 - \theta/8$. Its possible bad mean correction is $\le \theta^2 d^2$, so $\operatorname{Cov}(C, Y_1) \le 0$. The actual mean is $\le 1 + \theta^2 d$ and pointwise $Y_1 \le q + \sqrt{\delta_2 + \tau^2 d} \le q + (\frac{7}{10})\sqrt{d}$. These meet the IIIa $L = 1$ lemma in§4. The mean exponent $4\lambda + 1 - 4\theta^2 > 9$; the variance exponent $2\lambda - \theta/8 > 4$. Thus

$$E[CW^4] \le e^{-9d} + 8de^{-4d}. \tag{R28.7.7}$$

R28.7.2 $L = 2$**: variance, curvature and covariance**

The finite parity identity gives $C_0 - A_0 \le x$: even m has equal coordinates; odd m contributes one signed power (and $m = 1$ has difference-1). By(R28.7.3),

$$EY_2 = 1 - \tau - \tau \cdot d + \int x^2 (C_0 - A_0),$$
$$\frac{55}{64} \le 1 - d - d^2 \le EY_2 \le 1 - (\kappa - \theta^3) d \le 1 - d/2. \tag{R28.7.8}$$

The Q_2 coordinates in(R28.7.5) are positive, and

$$\begin{aligned} \operatorname{Re}(G_a Q_2) &\ge (1-x)(1-\tau \cdot x) - x(1 - x + x^2) \\ &\ge \frac{8367}{32768} > 1/4. \end{aligned} \tag{R28.7.9}$$

The polynomial decreases on $[0, \theta]$ after $\tau \le \frac{1}{8}$. The actual mean parameter is $\gamma_3 > 0$, so $\operatorname{Cov}(C, Y_2) \le -d/4$.

The correct squared symbol is bounded by

$$P(x) = (1 + x^3)^2 + x(1 - x + x^2)^2 = 1 + x - 2x^2 + 5x^3 - 2x^4 + x^5 + x^6. \tag{R28.L481}$$

$P' \ge 1 - 4x + 11x^2 > 0$ on $[0, \theta]$; the exact endpoint difference is $\frac{11}{8} - P(\theta) = \frac{2692519}{1073741824} > 0$. Keep the actual mean subtraction:

$$\operatorname{Var} Y_2 \le (\frac{11}{8}) d - \kappa^2 d^2. \tag{R28.7.10}$$

The FULL predecessor $Q_1 u$ has squared norm $\le d + \tau^2 \delta_2 < (\frac{21}{20})^2 d$ by(R28.7.6), and $\epsilon \le \theta^2 d < d/6$. Hence $Y_2 \le q + (\frac{21}{20}) \sqrt{d} + d/6$, with $EY_2 \le 1$. The NEW coefficient-eight lemma of§4 applies. From(R28.7.8)-(R28.7.9),

$$\begin{aligned} &e^{-4\lambda d} [(EC)(EY_2)^4 + 4(EY_2)^3 \operatorname{Cov}(C, Y_2)] \\ &\le e^{-4\lambda d} [e^{-3d} - (\frac{3}{5}) d] \le e^{-247 d/20}, \end{aligned} \tag{R28.7.11}$$

because $(\frac{55}{64})^3 > \frac{3}{5}$ and $e^{-3d} - e^{-18d/5} \le 3d/5$. Thus

$$E[CW^4] \le e^{-247 d/20} + [11d - (\frac{361}{128}) d^2] e^{-35 d/8}. \tag{R28.7.12}$$

The covariance sign and squared mean are retained; the argument needs no bound $\|Q_2\| \le 1$.

R28.7.3 $L = 3$

The correct symbol is $Q_3 = G_3 + \epsilon + \tau \cdot z^3 + z^4 + z^6 G_m(-z)$. Before ϵ, its positive coordinates are $1 - x + x^2 - x^3 A_0$ and $1 - \tau \cdot x + x^3 C_0$. Their squared sum is at most

$$1 - x + 3x^2 - 2x^3 + 3x^4 + x^7 < 9/8. \tag{R28.L509}$$

Indeed $[-1 + 3x - 2x^2 + 3x^3 + x^6]$ increases on $[0, \frac{5}{12}]$ to $\frac{373321}{2985984} < \frac{1}{6}$. Negative ϵ reduces the positive real coordinate; positive ϵ costs at most $\theta^2 d$. Thus $\|Q_3\| < \frac{17}{16} + \frac{1}{48} = \frac{13}{12}$. The exact mean formula gives $EY_3 \le 1 - 4d/3$.

Here $|U| \le \tau + \sqrt{\theta} < \frac{31}{40}$, and(R28.2.6) gives $\mathrm{Re}(G_a P_3) > -\frac{3}{50} - (\frac{31}{40})(\frac{4}{15}) = -\frac{4}{15}$. Positive ϵ is favorable; negative ϵ costs at most d^3; $\gamma_4 > 0$. Hence

$$\mathrm{Cov}(C, Y_3) \le 4d/15 + d^3. \tag{R28.7.13}$$

The correct predecessor satisfies $\|Q_2\| \le \sqrt{11/8} < \frac{6}{5}$, and $\epsilon \cdot q - \gamma_3 \le 2\delta_3 - \kappa \cdot d < 0$. So $Y_3 \le q + (\frac{6}{5})\sqrt{d}$ and the coefficient-nine lemma applies. The mean term $\le e^{-15d}$ and the covariance term $\le (16d/15 + d^2/2)e^{-51d/4}$. Their sum $\le e^{-13d}$: after dividing by $e^{-51d/4}$, the needed difference is $\ge 2de^{-9d/4} > 4d/3$, which exceeds $(\frac{16}{15} + \frac{1}{16})d$. Therefore

$$E[CW^4] \le e^{-13d} + (\frac{169}{16})de^{-35d/8}. \tag{R28.7.14}$$

R28.7.4 Every finite $L \ge 4$, with the preceding estimates

All $P_n, n \ge 4$ contract. $Atn = 4$ use $|G_4| \le 1 - x/2$ and perturb $\le 31x^2/40 < x/2$. $Atn = 5$ use $|G_5| \le 1 - \kappa \cdot x/2$ and perturb $\le 31x^{5/2}/40 < \kappa \cdot x/2$. For every $n \ge 6$ use(R28.2.2), $|G_n| \le 1 - 7x/30$, and $31x^3/40 < 7x/30$. As $|\epsilon| \le \theta^2 d$ and $\|G_2\| < \frac{6}{5}$,

$$\|Q_n\| \le 1 + (\frac{6}{5})\theta^2 d < 1 + d/5, n \ge 4. \tag{R28.7.15}$$

Every receiving P_n product is nonnegative. $Atn = 4$ its exact coordinates are $1 - x + \tau \cdot x^2 - x^4 C_0$ and $1 - x + x^2 - x^3 A_0$. The receiving lower is $1 - 3x + 2x^2 - x^3 - x^4 + x^5$, decreasing on $[0, \frac{5}{12}]$ to $\frac{1817}{248832} > 0$. $Atn = 5$ use baseline $\frac{1}{5}$ and its smaller perturbation. $Atn = 6$ use the exact G_6 baseline $\frac{57}{512}$, which exceeds $(\frac{31}{40})\theta^3$. For $n \ge 7$ use(R28.2.6) and

$$(\frac{31}{40})\theta^{7/2} < (\frac{62}{125})\theta^3 < 1451/32768. \tag{R28.L546}$$

With $h = \frac{7}{16}$, the true Z_L has mean $\le 1 - 9d/8$ and variance $\le (1 + d/5)^2 d$ by(R28.3.5). Its $L = 4$ predecessor is the proved $Q_3 < \frac{13}{12}$; all later ones are $\le 1 + d/5 < \frac{6}{5}$. The q excess $(\epsilon - hd)q - hd$ is nonpositive, so the same coefficient-nine lemma applies.

For the covariance, negative ϵ costs at most d^3. Positive ϵ can only be unfavorable when the predecessor is $G_3(L = 5)$, costing $3\theta^2 d^2/50$. The retained $-hd \cdot u$ costs at most hd^2, and the actual mean correction is favorable. Thus $\mathrm{Cov}(C, Z_L) < 3d^2/5$. Its first two terms are at most

$$e^{-57d/4} + (\frac{12}{5})d^2 e^{-97d/8} \le e^{-13d}. \tag{R28.L558}$$

For the last comparison divide by $e^{-97d/8}$ and use $e^{-7d/8} - e^{-17d/8} \ge (\frac{5}{4})de^{-17d/8} \ge 235d/256 > 3d/10$. The variance cost is $\le 9de^{-159d/40} \le 9de^{-15d/4}$. Hence

$$E[CW^4] \le e^{-13d} + 9de^{-15d/4}. \tag{R28.7.16}$$

R28.8 Complete low-domain scalar payments for IIIa

All statements here hold throughout $0 \le d \le \frac{1}{8}$. The positive Taylor lower of e^{Bd} and the degree-four upper of e^{-Cd} give the following exact continuous lower polynomials. After division by d, the negative cubic terms are bounded using $d \le \frac{1}{8}$.

For(R28.7.7), $\frac{e^{4d}-e^{-5d}}{d} \geq 9-(\frac{9}{2})d+29d^2 > \frac{17}{2}$. The all-real minimum of the quadratic is $9-\frac{81}{464} > \frac{17}{2}$. Subtract 8 and multiply by $e^{-4d} > \frac{1}{2}$; the deficit exceeds $d/4$.

For(R28.7.12), take $A = \frac{247}{20}, B = \frac{35}{8}, C = \frac{319}{40}$. Then

$$\begin{aligned} &\frac{e^{Bd}-e^{-Cd}}{d} - 11 + (\frac{361}{128})d \\ &\geq \frac{27}{20} - (\frac{39}{2})d + 77d^2 \geq 37/320. \end{aligned} \tag{R28.8.1}$$

The exact linear coefficient before weakening is $-\frac{62111}{3200} > -\frac{39}{2}$. The quadratic coefficient, even discarding the positive fourth Taylor term, is $\frac{38055750399}{491520000} > 77$. The final quadratic decreases on $[0, \frac{1}{8}]$ and reaches $\frac{37}{320}$. Since $e^{35/64} < 2$, the deficit is $\geq 37d/640$.

For(R28.7.14),

$$\frac{e^{35d/8}-e^{-69d/8}}{d} \geq 13 - (\frac{221}{8})d + 90d^2 \geq 701/64. \tag{R28.8.2}$$

The exact quadratic coefficient before weakening, including the positive fourth term, is $\frac{2309619}{24576} > 90$. The displayed quadratic decreases on $[0, \frac{1}{8}]$. Subtract $\frac{169}{16} = \frac{676}{64}$ and multiply by $e^{-35d/8} > \frac{1}{2}$; the deficit is $\geq 25d/128$.

For(R28.7.16),

$$\frac{e^{15d/4}-e^{-37d/4}}{d} \geq 13 - (\frac{143}{4})d + 100d^2 \geq 323/32. \tag{R28.8.3}$$

The exact quadratic coefficient even without the positive fourth term is $\frac{5041423}{49152} > 100$. Again the quadratic decreases over the interval. Subtract 9 and multiply by $e^{-15d/4} > \frac{1}{2}$; the deficit is $\geq 35d/64$.

The elementary bound $e^{2/3} < 2$ used here follows, for example, from its positive Taylor series with the factorial tail bounded geometrically. Every low IIIa row is consequently stronger than(R28.1.1).

R28.9 Same-direction high comparisons, preserving every actual arm

R28.9.1 I and II, $d \geq \frac{1}{64}$

Every actual later pair has shorter arma $_i \geq 4$. For its original finite $R_{ab} = K^a G_{b-a}(K)g$, $\|G_m\| \leq \frac{19}{16}$ for every $m \geq 1 : m = 1$ is $1, m = 2$ is $\sqrt{1+x} \leq \sqrt{45/32} < \frac{19}{16}$, and all $m \geq 3$ contract. The row bound gives

$$\begin{aligned} |R_{ab}| &\leq \theta^{(a-1)/2}(\frac{19}{16})\sqrt{d} \\ &< (\frac{39}{125})\sqrt{d} \leq (\frac{312}{125})d. \end{aligned} \tag{R28.L618}$$

Together with(R28.3.1), $h_a f_b \leq \exp(-d/2)$. Thus the SAME actual canonical $F \leq q f_{b_1}$ pointwise. The exact Section N.43 D_{full} row is

$$E[f_a T^L (q f_b)^4] \leq 1 - d/3000, a \geq 3, b \geq 2, L \geq 1. \tag{R28.9.1}$$

For $b = 2$ it has the strongerd/1000 margin. Original positivity now proves (R28.1.1) throughout this high domain. Original arms are still oriented and present; the pointwise estimate is only inside their actual $\int$.

R28.9.2 IIIa, $d \geq \frac{1}{8}$

Lemma O.6 proves the same actual $h_3 f_b \leq e^{-d}$ for every $b \geq 4$ on $d \geq 1/8$. It includes the genuine G_1 case $b = 4$ and retains the centered original row for every later finite length.

The same canonical $F \leq e^{-d} q f_2$, and(R28.9.1) proves(R28.1.1) for every L. Both low and high proofs include their common boundary $\frac{1}{8}$.

R28.10 A uniform payment for class IIIb

The compulsory coefficients of the class-III mask make it possible to pay every connector with the same estimate.

Lemma N.5 (A finite-mask receiving estimate)**.** *Let P be a finite polynomial with coefficients in $\{0,1\}$, with $P_0 = P_2 = 1$ and $P_1 = 0$. Put*

$$R = BP(-B)u + \tau u, \qquad \tau = d - \delta_2.$$

If a nonnegative function F satisfies

$$F \leq e^{R-\Gamma}, \qquad \Gamma \geq \frac{431}{80} d,$$

then, for all integers $a \geq 2$ and $L \geq 1$,

$$\mathbb{E}\big[f_a (T^L F)^4\big] \leq 1 - \frac{2d}{405}. \tag{N.43}$$

Proof. When $d = 0$, one has $u = 0$, $R = 0$ and $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$; positivity gives the assertion. Assume $d > 0$, and use the receiver C of (R28.2.3) when $a \geq 3$. When $a = 2$, set $C = f_2 = p + Bu$. In both cases

$$f_a \leq C, \qquad \mathbb{E}C \leq 1 - d, \qquad C \leq 2 - q + \frac{6}{5}\sqrt{d}.$$

All the spectral integrals below use the measure ν at u: $\nu([0,\theta]) = d$ and $\int x^r \, d\nu = \delta_{r+1}$.

The finite input. Write $P(-z) = U_0 - zV_0$, $z = i\sqrt{x}$, and $c = (1 - x^2)^{-1}$. The four coefficient residue classes, with the three compulsory coefficients retained, give

$$\begin{gathered} 1 - xc \leq U_0 \leq 1 - x + x^2 c \leq 1 - x/3, \qquad -xc \leq V_0 \leq x^2 c, \\ |P|^2 \leq (1 - x/3)^2 + \frac{16}{9} x^3 \leq 1. \end{gathered} \tag{N.44}$$

Indeed each allowed residue class is bounded by its geometric sum, $c \leq 1024/855 < 4/3$, and the last inequality follows from $2x + 32x^2 \leq 9$ on $0 \leq x \leq 1/2$. Set

$$U(z) = \tau + zP(-z) = \tau + xV_0 + zU_0, \qquad R = U(B)u.$$

Since $\tau \leq d - d^2 \leq 2/9$, one has $|U| \leq \tau + \sqrt{x} < 9/10$. For the original pointwise bound use $R = B\{P(-B)u + \tau\mathbf{1}\}$. Here $\mathbb{E}[P(-B)u] = \int V_0 \, d\nu < d/5$, whence

$$|R|^2 \leq d + \tau^2 + \frac{2\tau d}{5} \leq d + \frac{7}{5} d^2 \leq \frac{22}{45} < \frac{9}{16}.$$

Also, with $\sigma_R = \mathbb{E}R^2$,

$$\begin{aligned} \sigma_R &\le \delta_2 + \frac{\tau d}{6} + \tau^2 d \le \frac{1277}{2592} d < d/2, \\ D_0 &= -\int U_0 \, d\nu, \qquad D_1 = -\tau d - \int x V_0 \, d\nu, \\ D_2 &= \int x U_0 \, d\nu, \qquad D_3 = \tau\delta_2 + \int x^2 V_0 \, d\nu, \qquad D_i = \mathbb{E}B^i R. \end{aligned} \tag{N.45}$$

The cross term in the bound for σ_R uses $\langle u, BP(-B)u\rangle = \int x V_0 \, d\nu < d/12$. The resulting bounds are

$$D_0 \le -\frac{439}{855} d, \quad -\frac{11}{36} d \le D_1 \le \frac{1}{5} d, \quad 0 \le D_2 \le \delta_2, \quad D_3 < \frac{1}{8} d.$$

For every $r \ge 1$, the same coordinates give

$$D_{2r} = (-1)^{r+1} \int x^r U_0 \, d\nu, \qquad D_{2r+1} = (-1)^{r+1} \left(\tau\delta_{r+1} + \int x^{r+1} V_0 \, d\nu \right).$$

Thus $D_i \le d/12$ for every $i \ge 4$. For even indices the first positive bound is $\theta^3 d < d/14$. For odd indices $1 \pmod 4$ the τ term is favorable and the bound is $(1024/855)\theta^4 d < d/30$. For odd indices $3 \pmod 4$, starting at seven, it is $\{(2/9)\theta^3 + (1024/855)\theta^6\} d < d/30$. Subsequent indices in each class decrease by θ^2.

One propagated packet. Since $|R| < 3/4$, the Taylor comparison (R28.3.2) applies with $\eta = 4/3$. Choose

$$D = \frac{d}{4}, \qquad \epsilon = D + D_1, \qquad h = \frac{3}{10}, \qquad \lambda = \frac{431}{80}.$$

This choice is valid because

$$D_0 + \frac{4}{3}\sigma_R \le \left(-\frac{439}{855} + \frac{1277}{1944} \right) d < \frac{3}{20} d < D, \qquad -\frac{d}{18} \le \epsilon \le \frac{9d}{20}.$$

Use Y_L, Q_L, γ_n from (R28.3.2)–(R28.3.4), with

$$Q_1 = 1 + zU, \qquad Q_n = G_n + z^n U + \epsilon G_{n-2} + D z^{n-2} \quad (n \ge 2).$$

Their short coordinates and loss bounds are

$$\begin{aligned} Q_1 &= (1 - xU_0) + z(\tau + xV_0), \\ Q_2 &= (1 + D - \tau x - x^2 V_0) + z(1 - xU_0), \\ \gamma_3 = d - D_2 &\ge \frac{19}{32} d, \qquad \gamma_4 = d + Dd - D_3 \ge \frac{7}{8} d + Dd, \\ \gamma_n &\ge \left(\frac{19}{32} - \frac{1}{12} \right) d - \frac{11}{36} d^2 \ge \frac{353}{864} d > hd \quad (n \ge 5). \end{aligned} \tag{N.46}$$

The first two losses use $\beta_2 = 0$. For $L \ge 4$, retain the last two path terms, as in (R28.3.5), and put

$$X_L = \begin{cases} Y_L, & L = 1, 2, 3, \\ Z_L = 1 + \epsilon + Q_L u - hd(1 + q), & L \ge 4. \end{cases}$$

Then $0 \le T^L F \le e^{-\lambda d} X_L$. The identities $\mathbb{E} Q_n u = -\gamma_{n+1}$ and $\langle Q_n u, u\rangle = \gamma_{n+2}$ give

$$\operatorname{Var} Z_L = \|Q_L u\|^2 - 2hd\gamma_{L+2} + h^2 d^3 - \gamma_{L+1}^2 \le \|Q_L\|^2 d.$$

Finite spectral bounds. The coordinates of Q_2 lie respectively in $(0, 7/6)$ and $(0, 1]$, so

$$|Q_2|^2 \le \frac{509}{288} < \frac{9}{5}.$$

The pointwise predecessor at $L = 2$ is the full vector $V_2 = D\mathbf{1} + u + BR$. Its constant and its cross term give

$$\|V_2\|^2 = d - 2D_2 + \|BR\|^2 + D^2 + 2DD_1 \le \frac{1207}{960} d < \frac{64}{25} d.$$

For $Q_3 = P_3 + \epsilon + Dz$, where $P_n = G_n + z^n U$,

$$|P_3| < \frac{31}{25}, \qquad |\epsilon + Dz|^2 \le \left(\frac{81}{400} + \frac{13}{512}\right) d^2 < \frac{1}{4} d^2.$$

Consequently $\|Q_3\| < 211/150$ and $(211/150)^2 < 2$. For every $n \ge 4$, one has $\|Q_n\| < 7/5$. At $n = 4$ the bound is

$$1 + \theta^2 + d(27/50 + \theta/4) \le 1 + \frac{29099}{76800} < \frac{7}{5};$$

for $n \ge 5$ it is at most

$$1 + \frac{9}{10}\theta^{5/2} + d(9/20 + \theta^{3/2}/4) < 1 + \frac{971}{3600} < \frac{7}{5}.$$

These comparisons include the exceptional G_2 predecessor.

For $a \ge 3$, the receiving products needed below satisfy

$$\operatorname{Re}(G_a Q_1) \ge 1 - \frac{20}{9} x + \frac{4}{5} x^2 \ge \frac{23}{108},$$

$$\operatorname{Re}(G_a Q_2) \ge 1 - \frac{20}{9} x + \frac{1}{45} x^2 + \frac{1}{5} x^3 \ge \frac{2}{27}.$$

Here the Q_2 coordinates are positive; for Q_1 a negative imaginary coordinate contributes favorably. For P_3, (R28.2.6) and $|U| < 9/10$ give $\operatorname{Re}(G_a P_3) > -3/10$. For every $n \ge 4$, $\operatorname{Re}(G_a P_n) > 0$. At $n = 4$, its exact coordinates give the lower bound

$$1 - 3x + 2x^2 - x^3 - (1 - x) c x^4 > \frac{1451}{32768} - \frac{1}{30} > 0.$$

At $n = 5, 6$ the perturbations are respectively less than $39/400$ and $9/140$, below the baselines $1/5$ and $57/512$. For $n \ge 7$ the perturbation is less than $3/70 < 1451/32768$. Thus each finite index is covered by the same geometric bounds.

For the genuine receiver $a = 2$, both short products have the stronger lower bound $1 - 11x/9 - x^2/5$, which exceeds $23/108$ on $0 \le x \le \theta$. Also

$$\operatorname{Re}(G_2 P_3) > \frac{3}{16} - \frac{6}{5}\frac{9}{10}\frac{4}{15} > -\frac{3}{10}.$$

For every $n \ge 4$, the finite coordinates give $\mathrm{Re}(G_2 G_n) \ge 1 - 2x \ge 3/16$, whereas

$$|G_2 z^n U| < \frac{27}{25}\theta^2 < \frac{3}{16}.$$

Thus the same receiving bounds hold for every $a \ge 2$.

One fourth-moment payment. The preceding bounds and the exact covariance identity (R28.2.4) give the following data for $X = X_L$, with $\mu = \mathbb{E}X$.

Table 39: The data for the global class-III packet.

L	μ upper	$\mathrm{Var}\, X/d$ upper	$\mathrm{Cov}(C, X)$ upper	$X - q$ upper
1	$1 + 9d/20$	$77/64$	0	$\sqrt{d/2} + d/4$
2	$1 - 23d/160$	$509/288$	0	$8\sqrt{d}/5 + 9d/20$
3	$1 - 163d/160$	$(211/150)^2$	$3d/10 + d^2/6$	$8\sqrt{d}/5 + 49d/160$
≥ 4	$1 - 9d/20$	$49/25$	$d^2/2$	$8\sqrt{d}/5$

For the first row the covariance is at most $-23d/108 + d^2/5 < 0$ and $\mathrm{Var}\, Y_1 = d - 2D_2 + \|BR\|^2 - D_1^2 \le 77d/64$. For the second row both the receiving integral and the $-\beta_{a+1}\gamma_3$ correction are nonpositive. At $L = 3$ the two corrections $\epsilon + Dz$ cost at most $(1/18 + 13/128)d^2 < d^2/6$ beyond the $3d/10$ baseline. At $L \ge 4$, the ϵG_{L-2}, Dz^{L-2} and $-hd$ terms cost at most

$$\left(\frac{1}{18} + \frac{13}{128} + \frac{3}{10}\right) d^2 < \frac{1}{2} d^2.$$

For $a \ge 3$, this uses $-3/50 \le \mathrm{Re}(G_a G_n) \le 1$ for $n \ge 2$. For $a = 2$ the latter interval improves to $[3/16, 1]$, and $\mathrm{Re}(G_2 z) = -x$. Moreover $\mathrm{Re}(G_2 z^k) \ge -\theta$ for every $k \ge 2$, by the four finite parity classes, so the D-term has the same fee. The remaining mean corrections are $-\beta_{a+1}\gamma_{L+1} \le 0$. Finally, the pointwise rows follow from Y_1, Y_2, Y_3 and the representation $Z_L = (1 + \epsilon - hd)q - hd + BQ_{L-1}u$; its extra q term is at most zero.

In particular, every row has variance at most $2d$, and

$$X \le q + \frac{8}{5}\sqrt{d} + 2d, \qquad \mu \le 1 + \frac{3}{2}d.$$

Apply the coefficient-ten comparison with $W = e^{-\lambda d} X$. Its scalar proof uses only the displayed cap on C, so it includes $a = 2$. The fourth-order identity (R28.2.5) gives

$$\mathbb{E}[CW^4] \le e^{-4\lambda d}\{(1 - d)\mu^4 + 4\mu^3 \,\mathrm{Cov}(C, X)\} + 20d\, e^{-2\lambda d}.$$

The expression in braces is at most $e^{4d/5}$. For $L = 1$ this follows from $(1 - d)(1 + 9d/20)^4 \le e^{4d/5}$; for $L = 2$ it is at most one. For $L = 3$, replace μ by $\mu_0 = 1 - 163d/160$ after taking the nonnegative covariance upper. The result is

$$\mu_0^3 \left\{1 - \frac{131}{160}d + \frac{809}{480}d^2\right\} \le \mu_0^3(1 - 37d/144) \le 1.$$

For $L \ge 4$ the corresponding expression, with $\mu_0 = 1 - 9d/20$, is at most

$$\mu_0^3\{1 - 29d/20 + 49d^2/20\} \le \mu_0^3(1 - 19d/30) \le 1.$$

All replacements use nonnegative coefficients of μ^4 and μ^3. Since $4\lambda - 4/5 = 83/4$ and $2\lambda > 83/8$, every connector therefore satisfies the single bound

$$\mathbb{E}[CW^4] \le e^{-83d/4} + 20d\, e^{-83d/8} \le 1 - \frac{3}{4} d\, e^{-83d/8}.$$

The last step uses $2\sinh(83d/8) \ge 83d/4$. For $d \le 1/3$, one has $83d/8 < 4$ and $e^4 < 81$, so the deficit is greater than $d/108 > 2d/405$. This proves the lemma. □

For the actual class-IIIb terminal, the pair estimates (R28.3.1) give $F \le e^{R-\Gamma}$ with its single finite mask and

$$\Gamma = 3d - 2\delta_2 + \frac{16}{5}(j-2)d \ge \frac{431}{80} d \qquad (j \ge 3).$$

Lemma N.5 proves (R28.1.1) for this last class. Positivity restores the equal-pair factors at their original locations, as in the other three classes.

N.41 Every single pair of minimum three: the reference proof

R32.1 The original object and directions.

All operators, products and expectations use the original finite probability measure μ and a real skew kernel with $|K(x,y)| \le 1$. Directions are chosen from the original tree before the host. Original arm multiplicities, subdivisions and the four independent positive core lengths are retained.

Theorem A. For every receiving arm $a \ge 3$, terminal residual list exactly $(3, b)$, $b \ge 4$, and original connector length $L \ge 1$, use

L	Actual terminal F	All connector edges
1 or 2	$f_3 \bar{f}_b$	terminal to receiver
$L \ge 3$	$h_3 f_b$	terminal to receiver.

The receiver's existing arm points outward, giving f_a. An f-arm is fully outward, a $\bar{f}$-arm fully inward, and h_3 has its first edge toward the terminal and its other two edges toward the leaf. For genuine equal-arm-pair products E_o, E_t at their original locations,

$$\mathbb{E}[E_o f_a \{T^L(E_t F)\}^4] \le 1 - d/3000.$$

The stronger deficits are $d/10$ for $L = 1$, $3d/320$ for $L = 2$, and the explicit low/high deficits below for later lengths. The endpoint retains its two unequal arms.

Theorem B. The component statements listed below complete the directory for every strictly increasing positive terminal residual list of size at least two, every original $a \ge 3, L \ge 1$, with common same-f_a deficit $d/3000$. All short-length direction exceptions in those statements are retained.

Single pairs of minimum one or two and unit-minimum odd triples use Section N.43; minimum-three single pairs use Theorem A. Even lists of minimum at least four use Section N.36; even lists with at least two pairs and minimum two or three use Section N.39; and even multipair lists of minimum one use Section N.40. Odd lists with nonunit minimum use Section N.33 when the second length is three and Section N.34 otherwise. For unit-minimum odd lists with at least five entries, second length two, three or at least four uses, respectively, Sections N.38, N.37 and N.35. These alternatives cover every strict residual list, preserve the actual outward receiving arm f_a, and choose directions from the original integer data. Every cited deficit is at least $d/3000$.

The central allocations using these local reference inputs are given in Section 10.

R32.2 Original finite calculus and receiver.

Delete only zero-mass host atoms and put

$$Jv = (\mathbb{E}v)\mathbf{1}, \quad B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B = T^*,$$
$$p = 1 - u, \quad q = 1 + u, \quad f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1},$$
$$h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}, \quad \delta_i = \|B^i\mathbf{1}\|_2^2, \quad d = \delta_1,$$
$$\theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The finite calculus of Appendix D gives positive A, T and

$$0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta\delta_i, \quad \|B\|^2 \le \theta,$$
$$|Bv(x)| \le \|v\|_2, \qquad p(x), q(x) \ge \mu(x) > 0.$$

Also $d^2 \le \delta_2$ follows directly from $\mathbb{E}Bu = -d$ and Cauchy–Schwarz. The finite sums are

$$G_n(z) = \sum_{j=0}^{n-1} z^j, \quad G_0 = 0, \quad \beta_1 = \beta_2 = 0, \qquad \beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j \quad (n \ge 3).$$

The exact positive convolution and its full reverse are

$$1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r},$$
$$1 - G_n(-B)u = f_n + \sum_{r=3}^{n} \beta_r f_{n-r}, \qquad \tau \le \beta_r \le d \quad (r \ge 3).$$

The sums are empty for $n = 1, 2$. The pure path messages f_n and $\bar{f}_n$ lie in $[0, 2]$ and have means at most one. The identity $f_\ell + h_\ell = 2\mathbb{E}f_{\ell-1}$ gives $0 \le f_\ell h_\ell \le 1$; reversal gives $0 \le \bar{f}_\ell W_\ell \le 1$. These are the freely removable genuine pair decorations.

The original spectral measure ν of $-B^2$ at u has mass d, support $[0, \theta]$, and $\int x^j \, d\nu = \delta_{j+1}$. Integrals below use this measure. At $z = i\sqrt{x}$, for a real finite polynomial $Q = A_0 + zC_0$,

$$\|Q(B)u\|_2^2 = \int (A_0^2 + xC_0^2),$$
$$\mathbb{E}[Q(B)u] = -\int C_0, \qquad \langle Q(B)u, u\rangle = \int A_0.$$

Moments such as τ remain global. The ratio δ_2/d does not impose a pointwise condition $x \ge d$ on the spectral support. For $G_n = A_n + zC_n$, every finite $n \ge 1$ has $1 - x \le A_n \le 1$, $0 \le C_n \le 1$, with $(A_1, C_1) = (1, 0)$. For $n \ge 2$, both coordinates are at least $1 - x$. For $n \ge 3$, $A_n \le 1 - x + x^2$, and for $n \ge 4$, the same holds for C_n. Finite parity gives $C_n - A_n \le x$. The exact geometric formula gives $\|G_n\| \le 1$ for $n = 1$ or $n \ge 3$, whereas $\|G_2\|^2 \le 1 + \theta < (19/16)^2$. Every finite row therefore has norm less than $19/16$, with both G_1, G_2 exceptions retained. We also have the finite gaps

$$|G_4| \le 1 - x/2, \qquad |G_5| \le 1 - \kappa x/2, \qquad |G_n| \le 1 - 7x/30 \quad (n \ge 6).$$

For G_4, use $(1-x)^2(1+x)$; for G_5, its exact square $1 - x + x^2 - x^3 + x^4 \le 1 - \kappa x$. More generally, $|G_n|^2 \le 1 - 7x/15$ for $n \ge 3$. The cases $3, 4$ are direct; for $n \ge 5$, the finite numerator is at most

$(1+x^3)^2$. The polynomial $8-7x-30x^2-15x^5$ decreases on $[0,\theta]$ and equals $1311885/33554432 > 0$ at θ. Taking square roots gives the gaps, with the tails bounding the finite sums.

For every $a \geq 3$, the true weak receiving upper is

$$\begin{aligned} C &= 1 - G_a(-B)u - \beta_a \geq f_a \geq 0, \\ \mathbb{E}C &= 1 - \beta_a - \beta_{a+1} \leq 1 - 2\tau \leq 1 - 19d/16 \leq 1 - d, \\ C &= p - \beta_a + BG_{a-1}(-B)u \leq p + (6/5)\sqrt{d}. \end{aligned}$$

For $H = Q(B)u$ of actual mean $-\alpha$,

$$\operatorname{Cov}(C, H) = -\int \operatorname{Re}(G_a Q) - \beta_{a+1}\alpha.$$

Indeed $\mathbb{E}[G_a(-B)u] = \beta_{a+1}$, fixing the sign before estimating. The $a = 3$ predecessor is the paid G_2 row. Uniformly in $a \geq 3$,

n	Lower bound for $\operatorname{Re}(G_a G_n)$
2	$1 - 2x \geq 3/16$
3	$1 - 3x + x^2 \geq -55/1024 > -3/50$
4	$(1-x)(1-2x) \geq 57/512$
5	$(1-x)^3 > 1/5$
$n \geq 6$	$1 - 3x + 2x^2 - x^3 \geq 1451/32768.$

At $n = 6$, $G_6 = (1-x+x^2)(1+z)$ gives the stronger $57/512$. For $n \geq 6$, use $(1-x)^2 - x(1-x+x^2)$; for $n = 5$, use $(1-x)(1-x+x^2) - x(1-x)$. The other rows are immediate, and the polynomials decrease on the interval. The general tail row is $1451/32768$, not $1/10$. Upper real products are at most one when used: for $n = 2$, $A_a - xC_a \leq 1$; for $n \geq 3$, use the two contractions.

R32.2.1 Fourth expansion and finite positive propagation.

For nonnegative Y of mean m_Y,

$$\begin{aligned} \mathbb{E}(CY^4) = (\mathbb{E}C)m_Y^4 &+ 4m_Y^3 \operatorname{Cov}(C, Y) \\ &+ \mathbb{E}\big[C(Y - m_Y)^2\{(Y + m_Y)^2 + 2m_Y^2\}\big]. \end{aligned}$$

For $W = e^{-\Gamma}Y$, if the actual scaled curvature is paid by M, the variance contribution is $Me^{-2\Gamma} \operatorname{Var} Y$, with second-power scaling. For a canonical $F \leq e^{R-\Gamma}$, put $D_i = \mathbb{E}B^i R$ and choose $D \geq D_0 + (6/5)\mathbb{E}R^2$ when $|R| \leq 1/2$. The factorial tail for $R \geq 0$, and integral remainder for $R \leq 0$, give $e^R \leq 1 + R + (3/5)R^2$. Positivity and $T(R^2) \leq 2\mathbb{E}R^2$ imply

$$0 \leq e^{\Gamma}T^L F \leq Y_L = \bar{f}_L + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}.$$

Write $R = U(B)u$, $\epsilon = D + D_1$, and, for $n \geq 2$, $P_n = G_n + z^n U$, $Q_n = P_n + \epsilon G_{n-2} + Dz^{n-2}$. Finite substitution gives

$$\begin{aligned} Y_L &= 1 + \epsilon + Q_L(B)u - \sum_{n=3}^{L} \gamma_n \bar{f}_{L-n}, \\ \gamma_3 &= d - D_2, \qquad \gamma_4 = d + Dd - D_3, \\ \gamma_n &= \beta_n + D\beta_{n-1} + D_1\beta_{n-2} - D_{n-1} \quad (n \geq 5), \\ \mathbb{E}[Q_n(B)u] &= -\gamma_{n+1}, \qquad \langle Q_n(B)u, u\rangle = \gamma_{n+2}. \end{aligned}$$

The last identity follows by taking expectations in $Q_{n+1}(B)u = (1+\epsilon)u + BQ_n(B)u$ and using skewness; its future index differs from the output mean. In particular, $Y_3 = (1+\epsilon)q - \gamma_3 + BQ_2(B)u$. If $\gamma_n \geq hd$ for all $n \geq 3$, retain the last two genuine convolution terms for every $L \geq 4$ to obtain the positive upper

$$\begin{aligned} Z_L &= 1 + \epsilon + Q_L(B)u - hd(1+q) \\ &= (1 + \epsilon - hd)q - hd + BQ_{L-1}(B)u, \\ \mathbb{E}Z_L &\leq 1 + \epsilon - 3hd, \\ \operatorname{Var} Z_L &= \|Q_L(B)u\|_2^2 - 2hd\gamma_{L+2} + h^2d^3 - \gamma_{L+1}^2 \\ &\leq \|Q_L\|^2 d. \end{aligned}$$

All other removed terms have nonnegative original path factors and proved nonnegative coefficients. Both future indices remain paid.

R32.3 Global $L = 1$, with actual terminal $f_3\bar{f}_b$.

Fix $a \geq 3, b \geq 4$. The short arm points outward, the long arm inward, the one connector terminal-to-receiver, and the receiver is f_a. We prove $\mathbb{E}[f_a\{T(f_3\bar{f}_b)\}^4] \leq 1 - d/10$ on $0 \leq d \leq 1/3$. Use the original receiver C, measure ν and covariance formula above throughout.

R32.A1 Actual positive square and its negative constant.

Put $n = b - 3 \geq 1$, $G_n = A_0 + zC_0$. For every finite n, $1 - x \leq A_0 \leq 1$, $0 \leq C_0 \leq 1$; at $n = 1$, $(A_0, C_0) = (1, 0)$. The exact arm and positive convolution give

$$f_3 = 1 - d - u + Bu - B^2u, \qquad \bar{f}_b \leq 1 + G_b(B)u - \beta_b.$$

Consequently

$$\begin{aligned} f_3 + \bar{f}_b &\leq 2 - \Gamma + R, \qquad \Gamma = d + \beta_b \in [2d - \delta_2, 2d], \\ R &= 2Bu + B^3G_n(B)u = [x^2C_0 + zr]u, \\ r &= 2 - xA_0 \in [2 - x, 2 - x + x^2]. \end{aligned}$$

Bracket notation denotes the finite spectral polynomial applied to u. The nonnegative actual arm sum gives $m + R/2 \geq 0$, where $m = 1 - \Gamma/2 \in [2/3, 1]$, $k = 1/m \in [1, 3/2]$. Hence

$$F \leq m^2(1 + kR + k^2R^2/4), \qquad 0 \leq TF \leq m^2Y \leq e^{-\Gamma}Y,$$

where $Y = q + kBR + \delta$, $\delta = k\mathbb{E}R + (k^2/2)\mathbb{E}R^2$. This uses $T(R^2) \leq 2\mathbb{E}R^2$, with no exponential Taylor approximation of R. The actual means are

$$\mathbb{E}R = -\int r \leq -51d/32, \qquad \mathbb{E}BR = -\int x^2C_0 \leq 0.$$

The full square $\sigma = \mathbb{E}R^2$ satisfies

$$\sigma \leq \int P(x), \qquad P = x(2 - x + x^2)^2 + x^4,$$

and $P' = (2 - x + x^2)(2 - 3x + 5x^2) + 4x^3 > 0$. Its value at θ is $43080765/33554432$, smaller than $21/16$ by $959427/33554432$. Thus $\sigma < 21d/16$ and

$$\delta/k \leq -51d/32 + (3/4)(21/16)d = -39d/64.$$

In particular $\delta \le -3d/5$ and $0 \le m_Y = \mathbb{E}Y = 1 + \delta + k\mathbb{E}BR \le 1 - 3d/5$. Pointwise, $Y \le q + k\sqrt{\sigma} \le q + (7/4)\sqrt{d}$, since $k \le 3/2$ and $3\sqrt{21} < 14$. It is nonnegative because it dominates TF/m^2.

R32.A2 Complete variance contraction for $1 \le k \le 3/2$.

The random part of Y is $Q_k(B)u$, with $Q_k = (1 - kxr) + zkx^2C_0$. Its squared modulus is convex in k at fixed x, A_0, C_0, so consider $k = 1, 3/2$. At $k = 1$, the real coordinate is positive and

$$|Q_1|^2 \le (1 - x)^4 + x^5 \le 1 - 2x.$$

The last difference divided by x is $-2+6x-4x^2+x^3+x^4$. It increases on $[0, 5/12]$ to $-1907/20736 < 0$; $x = 0$ is direct. At $k = 3/2$, the squared difference from $1 - 2x$, divided by x, is at most $2 - 3r + (9/4)xr^2 + (9/4)x^4$, convex in r. At its two allowed endpoints it equals

$$\begin{aligned} J_{\text{lo}} &= -4 + 12x - 9x^2 + (9/4)x^3 + (9/4)x^4, \\ J_{\text{hi}} &= -4 + 12x - 12x^2 + (45/4)x^3 - (9/4)x^4 + (9/4)x^5. \end{aligned}$$

Both increase on $[0, 5/12]$, with derivatives at least $9/2$ and $259/192$, respectively. Their endpoint values are $-3059/9216$ and $-34183/110592$, both negative. This pays every finite n and every k, so, retaining the actual mean before weakening,

$$\operatorname{Var} Y = \|Q_k(B)u\|_2^2 - (k\mathbb{E}BR)^2 \le d - 2\delta_2.$$

Writing $G_a = A_a + zC_a$, its coordinates obey $1 - x \le A_a \le 1$ and $0 \le C_a \le 1$. Without assuming the real Q_k coordinate positive,

$$\begin{aligned} \operatorname{Re}(G_aQ_k) &\ge 1 - x - kx(2 - x + x^2) - kx^3 \\ &= 1 - (1 + 2k)x + kx^2(1 - 2x) \ge 1 - 4x. \end{aligned}$$

Its mean is $k\mathbb{E}BR \le 0$, so the receiver mean-product is favorable. Hence $\operatorname{Cov}(C, Y) \le -d + 4\delta_2$; this possibly positive upper is retained.

R32.A3 The scaled coefficient 10.

Put $t = \sqrt{d} \le 3/5$, $W = e^{-\Gamma}Y$. As $\mathbb{E}Y \le 1$ and $Y \le q + 7t/4$, the same-reference remainder coefficient is at most $e^{-2\Gamma}H$, where

$$H = (2 - q + 6t/5)\{(q + 1 + 7t/4)^2 + 2\}.$$

Its constant is $6 + q - q^3 \le 32/5$, and

$$\begin{aligned} H_1 &= 53/5 + (59/10)q - (23/10)q^2 \le 29/2, \\ H_2 &= 413/40 + (91/80)q \le 63/5, \qquad H_3 = 147/40, \quad H_4 = H_5 = 0. \end{aligned}$$

The first maximum has margin $107/920$ below $29/2$. Hence

$$\begin{aligned} H &\le 32/5 + (29/2)t + (2961/200)t^2, \\ 10e^{2\Gamma} - H &\ge 18/5 - (29/2)t + (1707/100)t^2 > 0. \end{aligned}$$

Here $\Gamma/d \geq 51/32$; the last discriminant is $-17779/500$. This proves the continuous bound, and the exact fourth expansion gives

$$\mathbb{E}(CW^4) \leq e^{-4\Gamma}[(1-d)m_Y^4 + 4m_Y^3 \operatorname{Cov}(C,Y)] + 10e^{-2\Gamma} \operatorname{Var} Y.$$

The variance term therefore has the second-power scale.

R32.A4 Coupled scalar payment on the original domain.

For $d > 0$, let $\bar{x} = \delta_2/d$, so $d \leq \bar{x} \leq \theta$. The true mean upper $m_Y \leq 1 - 3d/5$, covariance and variance bounds, and $\Gamma \geq 2d - \delta_2$, bound the moment by

$$F(d,\bar{x}) = e^{-(57/5-4\bar{x})d} + 10(1-2\bar{x})de^{-(4-2\bar{x})d} + 4(4\bar{x}-1)_+ de^{-(49/5-4\bar{x})d}.$$

The positive part bounds the covariance with either sign. For fixed $0 < d \leq 1/3$, F decreases in $\bar{x}$ on $[0,\theta]$. Below $1/4$, its derivative divided by $de^{-(4-2\bar{x})d}$ is $4e^{-(37/5-2\bar{x})d} - 20 + 20d(1-2\bar{x}) < 0$. Above $1/4$, add $16e^{-yd}(1+zd)$, where $y = 29/5 - 2\bar{x}$, $z = 4\bar{x} - 1$. We have $4 < y < 6$, $0 \leq z < 1$, $y - z > 3$, and

$$e^{-yd}(1+zd) \leq \frac{1+zd}{1+yd} \leq 1 - \frac{3d}{1+6d} \leq 1 - d.$$

The full bracket is at most $4-20+10d+16(1-d) = -6d < 0$. Continuity at $1/4$ gives $F(d,\bar{x}) \leq F(d, d)$, using $\delta_2 \geq d^2$. For $0 < d \leq 1/4$, the positive part vanishes at $\bar{x} = d$. Set $A = 57/5 - 4d$, $B = 4 - 2d$, $C_0 = A - B = 37/5 - 2d$. Positive quadratic Taylor for e^{Bd} and the quadratic upper for $e^{-C_0 d}$ give

$$\frac{e^{Bd} - e^{-C_0 d}}{d} - 10(1-2d) \geq 7/5 - (169/50)d + (34/5)d^2 \geq 111/200.$$

Since $Bd \leq 1$, $e < 3$, this gives $F(d,d) \leq 1 - 37d/200$. For $1/4 \leq d \leq 1/3$, bound its three terms separately by

$$\text{first} < e^{-5/2} < 1/8,$$

$$\text{second} \leq 5de^{-10d/3} \leq 3/(2e) < 3/5,$$

$$\text{third} < 4/(9e^2) < 4/63 < 1/15.$$

The inequalities $e > 5/2$, $e^2 > 7$ follow from positive Taylor series. The sum is less than $19/24 \leq 1 - 5d/8$. At $d = 0$, every actual message is one even if $K \neq 0$. Thus

$$\mathbb{E}[f_a\{T(f_3 \bar{f}_b)\}^4] \leq 1 - d/10 \qquad (a \geq 3,\ b \geq 4).$$

Genuine decorations restore by positivity and full reversal reverses every original edge together.

R32.4 Global $L = 2$, with the same $f_3\bar{f}_b$.

Use the actual terminal above. Put $\Gamma = d + \beta_b$, $m = 1 - \Gamma/2$, $k = 1/m$, $a_0 = 1 - 3d/5$. Then $\Gamma \ge 51d/32$, $2/3 \le m \le 1$, $1 \le k \le 3/2$, $4/5 \le a_0 \le 1$. The exponent-free envelope is

$$\begin{aligned} F &\le m^2(1 + kR + k^2R^2/4), \\ R &= 2Bu + B^3G_{b-3}(B)u = [x^2C_0 + zr]u, \\ r &= 2 - xA_0, \quad 1 - x \le A_0 \le 1, \quad 0 \le C_0 \le 1, \\ 2 - x \le r &\le 2 - x + x^2 \le 2. \end{aligned}$$

The signed means are $\mathbb{E}R = -\int r$, $D_1 = \mathbb{E}BR = -\int x^2C_0 \le 0$, $D_2 = \mathbb{E}B^2R = \int xr$. The same-resource bounds give $\mathbb{E}R \le -51d/32$, $\sigma \le 21d/16$ and $\delta = k\mathbb{E}R + (k^2/2)\sigma \le -39d/64 \le -3d/5$.

R32.E1 Positive propagation and complete mean.

Positivity first gives $TF \le m^2Y_1$, $Y_1 = q + kBR - 3d/5 \ge 0$. Apply the second actual T:

$$\begin{aligned} T^2F \le m^2Y, &\qquad Y = q + \epsilon + BV \ge 0, \\ \epsilon = -3d/5 + kD_1, &\qquad V = -(3d/5)\mathbf{1} + u + kBR, \\ Q_2 = (a_0 - kx^3C_0) + z(1 - kxr), &\qquad Y = 1 + \epsilon + Q_2(B)u. \end{aligned}$$

It is positive as the actual T-image of Y_1; no signed upper is squared. Its mean is $\mu_Y = 1 - 8d/5 + k\int(xr - x^2C_0)$. The bracket is nonnegative because $r - xC_0 \ge 2 - 2x > 0$, while $xr \le x(2 - x + x^2) \le \theta(2 - \theta + \theta^2) = 23413/32768$. Therefore

$$1 - 8d/5 \le \mu_Y \le 1 - [8/5 - 70239/65536]d \le 1 - 21d/40.$$

The last difference is $1061/327680 > 0$, and $\mu_Y \ge 7/15$.

R32.E2 Precursor, variance and covariance.

The actual precursor contains the constant vector $-(3d/5)\mathbf{1}$. Let $c = 3d/5$. R32.A2 gives $\|u + kBR\|_2^2 \le d - 2\delta_2$. Since $\mathbb{E}(u + kBR) = kD_1$, $D_1 = -\int x^2C_0$, its complete square is

$$\begin{aligned} \|V\|_2^2 &= \|u + kBR\|_2^2 + c^2 - 2ckD_1 \\ &\le d - 2\delta_2 + 9d^2/25 + (9/5)d\delta_3 \\ &\le d - (281/160)\delta_2 + 9d^2/25 \\ &\le d - (1117/800)d^2 \le d. \end{aligned}$$

This uses $\delta_3 \le \theta\delta_2$, $d \le 1/3$, $\delta_2 \ge d^2$, and retains the positive cross $-2ckD_1$. The original row now gives $Y \le q + \epsilon + \sqrt{d} \le q - 3d/5 + \sqrt{d} \le q + \sqrt{d}$. The output polynomial Q_2 is used separately for variance/covariance; the precursor remains the actual constant vector, rather than a multiple of u.

The real coordinate of Q_2 is positive and at most a_0, since $a_0 - kx^3C_0 \ge 4/5 - (3/2)\theta^3 > 0$. Put $y = kxr$. For $y \le 1$, $x(1-y)^2 \le x(1-x)^4 \le 256/3125 < 1/12$; the derivative is $(1-x)^3(1-5x)$, with maximum at $x = 1/5$. For $y \ge 1$, $y - 1 \le 4703/65536 < 3/40$, so $x(1-y)^2 < 9/3200 < 1/12$. Before the favorable squared-mean subtraction,

$$\operatorname{Var} Y \le \|Q_2(B)u\|_2^2 \le (a_0^2 + 1/12)d \le [13/12 - 27d/25]d,$$

using $9d^2/25 \le 3d/25$. For the actual receiver,

$$\operatorname{Cov}(C,Y) = -\int \operatorname{Re}(G_a Q_2) - \beta_{a+1}\alpha, \qquad \alpha = d - kD_2.$$

For every finite $a \ge 3$, $\operatorname{Re}(G_a Q_2) \ge 7/25$. If $1 - kxr \ge 0$, the nonnegative coordinates give

$$\begin{aligned}\operatorname{Re}(G_a Q_2) &\ge (1-x)(a_0 - kx^3 C_0) - x(1-kxr)\\ &\ge 4/5 - (9/5)x + 2x^2 - 2x^3 + x^4.\end{aligned}$$

Its derivative is at most $-9/5 + 4x < 0$ on $x \le 5/12$. Its value there is $29309/103680 > 7/25$, with positive cross difference 6965. If $1 - kxr < 0$, the imaginary term is favorable and the real product is at least $\kappa[4/5 - (3/2)\theta^3] > 27/80 > 7/25$. Also $\alpha \ge -4703d/65536 > -3d/40$, $\beta_{a+1} \le d$, hence

$$\operatorname{Cov}(C,Y) \le -7d/25 + 3d^2/40 \le -51d/200 \le -d/4.$$

The case $\alpha < 0$ is included in this estimate.

R32.E3 Global continuous remainder with coefficient 9.

Let $W = e^{-\Gamma}Y \ge T^2 F$, $t = \sqrt{d} \le 3/5$. We have $\mathbb{E}Y \le 1$, $Y \le q + t$, $C \le 2 - q + 6t/5$, and the scaled fourth curvature is controlled by $H = (2 - q + 6t/5)[(q + 1 + t)^2 + 2]$. On $0 \le q \le 2$,

$$\begin{aligned}&H_0 \le 32/5, \qquad H_1 = -(4/5)q^2 + (22/5)q + 38/5 \le 66/5,\\ &H_2 = (7/5)q + 22/5 \le 36/5, \qquad H_3 = 6/5, \quad H_4 = H_5 = 0.\end{aligned}$$

Thus $H \le 32/5 + (66/5)t + (198/25)t^2$. Since $2\Gamma \ge 51t^2/16$,

$$9e^{2\Gamma} - H \ge 13/5 - (66/5)t + (8307/400)t^2 > 0,$$

with discriminant $-20871/500$. The resulting actual moment is

$$\begin{aligned}\mathbb{E}(CW^4) \le e^{-4\Gamma}&[(\mathbb{E}C)\mu_Y^4 + 4\mu_Y^3 \operatorname{Cov}(C,Y)]\\ &+ 9e^{-2\Gamma}\operatorname{Var} Y.\end{aligned}$$

R32.E4 The signed cubic mean factor on its actual range.

Use $\mathbb{E}C \le 1 - 2\tau \le 1 - 19d/16$. The bracket is at most $g(\mu_Y) = (1 - 19d/16)\mu_Y^4 - d\mu_Y^3$. Its derivative is nonnegative for $\mu_Y \ge 7/15$, because

$$4(1 - 19d/16)\mu_Y - 3d \ge 4(29/48)(7/15) - 1 = 23/180 > 0.$$

It may therefore be bounded at $\mu_0 = 1 - 21d/40$, despite the negative covariance. The resulting positive expression satisfies

$$\mu_0^3[1 - (217/80)d + (399/640)d^2] \le e^{-(2611/640)d}.$$

The bracket is positive, and the exponential upper uses $d \le 1/3$. Together with $4\Gamma \ge 51d/8$, the mean/covariance exponent is $6691/640 > 52/5$. Thus

$$\mathbb{E}[f_a\{T^2(f_3\bar{f}_b)\}^4] \le F(d) := e^{-52d/5} + [(39/4)d - (243/25)d^2]e^{-51d/16}.$$

R32.E5 Exact continuous scalar payment.

Set $A = 52/5$, $B = 51/16$, $C = A - B = 577/80$, $b_0 = 39/4$, $b_1 = 243/25$. For $0 < d \le 1/8$, positive Taylor through degree three for e^{Bd}, and even Taylor through degree four for e^{-Cd}, imply

$$\frac{e^{Bd} - e^{-Cd}}{d} - b_0 + b_1 d \ge 13/20 - (1121/100)d + 53d^2 \ge 1/20.$$

The unweakened quadratic coefficient is

$$(B^3 + C^3)/6 - C^4/192 = 423382685439/7864320000 > 53,$$

with positive numerator difference 6573725439. The completed square uses $1121^2 = 1256641 < 1272000 = 10000 \cdot 4 \cdot 53 \cdot (3/5)$. Since $e^{-Bd} > 1/2$, $F(d) \le 1 - d/40$ throughout this interval. For $1/8 \le d \le 1/3$,

$$\begin{aligned} F'(d) &= e^{-Bd}[P(d) - Ae^{-Cd}], \\ P(d) &= b_0 - \alpha_0 d + \beta_0 d^2, \\ \alpha_0 &= 80829/1600 \ge 101/2, \qquad \beta_0 = 12393/400 < 31. \end{aligned}$$

Put $\Delta = Ae^{-Cd} - P$. At $d = 1/8$, $C/8 < 29/32$ and the cubic lower $e^{-t} \ge 1 - t + t^2/2 - t^3/6$ give

$$\Delta(1/8) > (52/5)(74779/196608) - 251/64 = 33148/983040 > 0.$$

For all d in the interval, $C \ge 7$, $AC < 609/8$, $\alpha_0 > 50$, $\beta_0 < 31$, so

$$\Delta'(d) \ge 50 - 62d - \frac{609/8}{1 + 7d} > 0.$$

After multiplying the positive denominator, the numerator is the concave quadratic $-209/8 + 288d - 434d^2$, whose endpoint values are $99/32$ and $1559/72$, both positive. Hence F decreases and

$$F(d) \le F(1/8) \le 1 - 1/320 \le 1 - 3d/320.$$

Combining the domains proves

$$\mathbb{E}[f_a\{T^2(f_3\bar{f}_b)\}^4] \le 1 - 3d/320 \qquad (a \ge 3, b \ge 4).$$

At $d = 0$, equality follows from $K\mathbf{1} = 0$; directions are independent of d.

R32.5 Every $L \ge 3$, with actual $h_3 f_b$.

Keep this canonical terminal for every $b \ge 4, L \ge 3$ and every host. Only the proof splits at $d = 1/8$. We reconstruct its finite packet, without using an ordinary fourth-moment theorem as a same-f_a premise.

R32.C1 Exact low-domain packet and all path constants.

Let $0 < d \le 1/8$, $m = b - 3$, and $G_m(-z) = A_0 - zC_0$, where $\kappa \le A_0 \le 1$, $0 \le C_0 \le 1$, $C_0 - A_0 \le x$. The actual finite envelope is

$$\begin{aligned} F &\le e^{R-\Gamma}, \qquad R = \tau u + B^3 G_m(-B)u, \\ \Gamma &= 3d - 2\delta_2 \ge 35d/16, \qquad U = \tau - x^2 C_0 - xzA_0. \end{aligned}$$

Retain $\beta_b f_0$ and τf_1 in the original positive convolution for f_b, use $\beta_b \ge \tau$, and add $h_3 = 1 - d + u - Bu + B^2u$. This gives $h_3 + f_b \le 2 - \Gamma + R$, and nonnegative AM–GM proves the envelope without an extra original unit arm. All coefficients are real finite polynomials. We have $|R| < 1/2$ and $\sigma = \|R\|_2^2 < 3d/20$. The actual cross $\mathbb{E}[uB^3G_m(-B)u] = -\int x^2 C_0$ is nonpositive, so

$$\sigma \le [(1+\theta)\theta^3 + 1/64]d < (55/448)d < 3d/20.$$

The row gives $|R| \le d + (19/16)\theta\sqrt{d} < 1/2$ for $d \le 1/8$. The quadratic exponential envelope with coefficient $6/5$ yields

$$\begin{aligned}
D &= D_0 + (6/5)\sigma, \qquad 0 \le D \le 47d/80,\\
D_0 &= \mathbb{E}R = \int xA_0,\\
D_1 &= \mathbb{E}BR = \int x^2C_0 - \tau d,\\
D_2 &= \mathbb{E}B^2R = -\int x^2A_0 \le 0,\\
D_3 &= \mathbb{E}B^3R = -\int x^3C_0 + \tau\delta_2 \le \theta d^2.
\end{aligned}$$

Indeed $D \le \theta d + (6/5)(55/448)d = 31d/56 < 47d/80$. The next actual means are

$$\begin{aligned}
D_4 &= \int x^3A_0, \qquad D_5 = \int x^4C_0 - \tau\delta_3,\\
D_6 &= -\int x^4A_0, \quad D_7 = -\int x^5C_0 + \tau\delta_4.
\end{aligned}$$

Each later residue class gains two powers of x. Thus all D_i, $i \ge 4$, are at most $d/14$: the first positive even term is $\delta_4 < d/14$, the positive C_0 term is at most δ_5, and the positive τ term at most $\tau\delta_4$. Put $\epsilon = D + D_1$. Since $D \ge 0$ and $D_1 \ge -\tau d$, $\epsilon \ge -d^2$. The same R gives

$$\begin{aligned}
\epsilon = &\int xA_0 + (1 - 12\tau/5)\int x^2C_0\\
&+ (6/5)\|B^3G_m(-B)u\|_2^2 + (-\tau + 6\tau^2/5)d\\
&\le \delta_2 + \delta_3 + (6/5)(1+\theta)\delta_4\\
&\le (358839/524288)d < 11d/16.
\end{aligned}$$

It also gives $D + (D_1)_+ \le 11d/16$, without separately maximizing the mean or discarding the negative cross. For the actual positive propagation, $Q_n = G_n + z^nU + \epsilon G_{n-2} + Dz^{n-2}$, and

$$\begin{aligned}
\gamma_3 &= d - D_2 \ge d,\\
\gamma_4 &= d + Dd - D_3 \ge (243/256)d > 9d/10,\\
\gamma_n &\ge \kappa d - d/14 - d^3 \ge (227/448)d > d/2 \quad (n \ge 5).
\end{aligned}$$

Thus $h = 1/2$ pays every required future index. The positive upper at $L = 3$ is

$$Y_3 = (1+\epsilon)q - \gamma_3 + BQ_2(B)u, \qquad \mathbb{E}Y_3 \le 1 - 97d/80.$$

For every $L \ge 4$, the actual positive output is bounded by

$$Z_L = 1 + \epsilon + Q_L(B)u - (d/2)(1+q), \qquad \mathbb{E}Z_L \le 1 - 13d/16.$$

The removed path sum is nonnegative. Retaining the constant cross and squared mean gives Var $Z_L \le \|Q_L\|^2 d$, as in R32.2.1.

R32.C2 Spectral norms and receiving products.

In the low domain,

$$|U| \le d + (19/16)\theta^{3/2} < 1/8 + 19/60 = 53/120 < 9/20.$$

For $P_n = G_n + z^n U$, P_3 has positive coordinates $1 - x - x^3 A_0$, $1 - \tau x + x^3 C_0$. Thus

$$|P_3|^2 \le (1-x)^2 + x(1+x^3)^2 \le 1,$$

using $-1 + x + 2x^3 + x^6 < 0$ for $x \le 1/2$. The G_4, G_5, G_n gaps $x/2, \kappa x/2, 7x/30$, respectively, pay the perturbations $(9/20)x^{n/2}$. Every finite P_n, $n \ge 3$, contracts, including $n = 4, 5$. The added $\epsilon + Dz$ in Q_3 has norm at most $4d/5$, because

$$(11/16)^2 + \theta(47/80)^2 = 125517/204800 < 16/25,$$

with difference $5555/204800$. Hence $\|Q_3\| \le 1 + 4d/5$. At $n = 4$, the added fee is $[(11/16)(6/5) + (47/80)\theta]d = (2723/2560)d < (16/15)d$. At $n \ge 5$, it is $[11/16 + (47/80)(4/15)]d = (1013/1200)d < (16/15)d$. Thus $\|Q_n\| \le 1 + (16/15)d$ for every $n \ge 4$. The Q_2 coordinates are $1 + D - \tau x + x^3 C_0$, $1 + x^2 A_0$, bounded above by $1 + 47/640 + 1/14 < 23/20$ and $7/6$. Hence

$$\|Q_2\|^2 < 529/400 + 637/1152 < 49/25, \qquad \|Q_2\| < 7/5.$$

For the actual $C \ge f_a$, $\mathbb{E}C \le 1 - d$, $C \le p + (6/5)\sqrt{d}$, the receiving products give $\mathrm{Re}(G_a P_3) > -3/50 - (9/20)(4/15) = -9/50$. For $n \ge 4$, they are nonnegative: the $n = 4$ baseline $57/512$ exceeds $(9/20)\theta^2$; the $n = 5$ baseline $1/5$ exceeds $(9/20)\theta^{5/2}$; at $n = 6$, use its exact $57/512$; for $n \ge 7$, $1451/32768$ pays the decreasing $(9/20)\theta^{n/2}$. The tail uses the displayed lower bound, below $1/10$. The actual mean-product in $\mathrm{Cov}(C, Q_n(B)u)$ is favorable, and

$$\mathrm{Cov}(C, Y_3) \le 9d/50 + (47\theta/80)d^2 + d^3 \le 9d/50 + 3d^2/8.$$

Negative ϵ costs at most d^3. For the long upper, the bad product with positive ϵ occurs only at the actual G_3 index $L = 5$, costing $(11/16)(3/50)d^2$. Negative ϵ costs d^3, the D-power costs $(47/80)\theta d^2$, and $-hdu$ costs $d^2/2$. Therefore

$$\mathrm{Cov}(C, Z_L) \le d^3 + (33/800 + 611/2560 + 1/2)d^2 < d^2.$$

R32.C3 Continuous remainders: 9 at $L = 3$, 8 at $L \ge 4$.

For $t = \sqrt{d} \le 3/8$, $Y_3 \le q + (7/5)t + (3/8)t^2$, $\mathbb{E}Y_3 \le 1$, because $\epsilon q - \gamma_3 \le (11/8 - 1)d = 3d/8$. For $W = e^{-\Gamma} Y_3$, the fourth remainder is at most $9e^{-2\Gamma}$ Var Y_3. Expand

$$H = (2 - q + 6t/5)[(q + 1 + 7t/5 + 3t^2/8)^2 + 2].$$

On $0 \le q \le 2$,

$$H_0 \le 32/5, \quad H_1 \le 27/2, \quad H_2 \le 21/2, \quad H_3 \le 11/2, \quad H_4 \le 8/5, \quad H_5 \le 1/5.$$

Using $t \le 3/8$, $H \le 32/5 + (27/2)t + 13t^2$, and

$$9e^{35t^2/8} - H \ge 13/5 - (27/2)t + (211/8)t^2 > 0,$$

with negative discriminant. The exact continuous coefficient calculations are

$$H_1 = -(8/5)q^2 + (26/5)q + 46/5, \qquad \max H_1 = 537/40 < 27/2,$$
$$H_2 = -(3/4)q^2 + (43/20)q + 439/50, \qquad \max H_2 = 12385/1200 < 21/2,$$
$$H_3 = 669/125 - (3/20)q < 11/2,$$
$$H_4 \le 1233/800 < 8/5, \qquad H_5 = 27/160 < 1/5.$$

For $L \ge 4$, $\|Q_3\| \le 1.1$, and every later norm is at most $17/15 < 6/5$. Also $(\epsilon - d/2)q - d/2 \le 2(11/16 - 1/2)d - d/2 = -d/8$, so $Z_L \le q + (6/5)t$, $\mathbb{E}Z_L \le 1$. The analogous H has

$$H_0 \le 32/5, \quad H_1 \le 66/5, \quad H_2 \le 216/25,$$
$$H_3 = 216/125, \qquad H_4 = H_5 = 0.$$

Hence $H \le 32/5 + (66/5)t + (47/5)t^2$. Retaining the positive fourth exponential term gives

$$8e^{35t^2/8} - H \ge 8/5 - (66/5)t + (128/5)t^2 + 75t^4.$$

For $t \le 1/6$, the quadratic decreases to $1/9 > 0$. For $t \ge 1/6$, $75t^4 \ge 25t^2/12$, so the quadratic coefficient becomes $1661/60$, with discriminant $-44/15$. This proves the full low-domain coefficient 8.

R32.C4 The $L = 3$ scalar payment.

The true mean term is at most $e^{-73d/5}$, and the positive covariance contribution at most $[(18/25)d + (3/2)d^2]e^{-991d/80}$. Their sum is at most e^{-13d}: after dividing the common exponential,

$$e^{-49d/80} - e^{-177d/80} \ge (8/5)de^{-177d/80} \ge 463d/400,$$

whereas $18/25 + 3d/2 \le 363/400$. For negative covariance, discard that contribution. The variance term is at most $9d(1 + 4d/5)^2 e^{-35d/8} \le 9de^{-111d/40}$. Thus the actual same-$f_a$ moment is at most $e^{-13d} + 9de^{-111d/40}$. Put $B = 111/40$, $C = 409/40$. Positive Taylor through degree four for e^{Bd}, and even Taylor through degree six for e^{-Cd}, give

$$\frac{e^{Bd} - e^{-Cd}}{d} \ge 13 - (97/2)d + 125d^2 + 700d^4.$$

Before weakening, the quadratic coefficient is $61494390880/491520000 > 125$, with numerator difference 54390880; it equals $(B^3 + C^3)/6 - (C^4 - B^4)/192$. The quartic coefficient is at least

$$\frac{C^5(1 - C/48)}{120} > \frac{(51/5)^5(47/60)}{120} = 16216186797/22500000 > 700.$$

The weaker polynomial decreases on $[0, 1/8]$, its derivative being at most $-48.5 + 250/8 + 2800/512 < 0$. Its endpoint $9279/1024$ exceeds nine by $63/1024$. Since $e^{-Bd} > 1/2$,

$$\mathbb{E}[f_a\{T^3(h_3 f_b)\}^4] \le 1 - 63d/2048.$$

R32.C5 Scalar payment for every $L \geq 4$.

The mean is at most e^{-13d}, and covariance at most $4d^2e^{-179d/16}$. Their sum is at most e^{-12d}, because

$$e^{-13d/16} - e^{-29d/16} \geq de^{-29d/16} \geq 99d/128 > 4d^2.$$

The variance part is at most $8d(1 + 16d/15)^2e^{-35d/8} \leq 8de^{-269d/120}$. Thus every actual $L \geq 4$ is bounded by $e^{-12d} + 8de^{-269d/120}$. Set $B = 269/120$, $C = 1171/120$, $v = C - B = 451/60 \in [15/2, 8]$. The continuous Taylor method through degree four gives

$$\frac{e^{Bd} - e^{-Cd}}{d} \geq 12 - (451/10)d + 109d^2.$$

The unweakened quadratic coefficient is

$$(B^3 + C^3)/6 - (C^4 - B^4)/192 = 72 + (3/2)v^2 - (9/2)v - v^3/32.$$

It increases on $[15/2, 8]$, with derivative at least twelve; its left value $28017/256$ exceeds 109. The final quadratic decreases on $[0, 1/8]$ to $2581/320$, exceeding eight by $21/320$. Since $e^{-Bd} > 1/2$,

$$\mathbb{E}[f_a\{T^L(h_3f_b)\}^4] \leq 1 - 21d/640 \qquad (L \geq 4,\ d \leq 1/8).$$

R32.C6 The same direction on the high domain.

For $d \geq 1/8$, Lemma O.6 gives $h_3f_b \leq e^{-d}$ for every $b \geq 4$ in this same canonical direction. Positivity gives $T^L(h_3f_b) \leq e^{-d}\bar{f}_L$. The actual bare-reference bounds R2-C012/A005 at $L = 3$, R2-C014/A003 at $L = 4$, and R2-C017/A006 at $L \geq 5$, give

$$\mathbb{E}(f_a\bar{f}_L^4) \leq 1 - 3d/40 \qquad (a \geq 3, L \geq 3).$$

The high-domain moment is at most $e^{-4d}(1 - 3d/40) \leq 1 - 3d/40$. All low/high deficits exceed $d/3000$. At $d = 0$, all finite tree messages equal one because $K\mathbf{1} = 0$, even when $K \neq 0$. Decorations and complete reversal use the actual positive operators and preserve all original edges. The canonical h_3f_b direction proves every $L \geq 3$; R32.3–R32.4 prove $L = 1, 2$ with $f_3\bar{f}_b$. This completes Theorem A.

N.42 Every genuine length-two reference arm: the complete directory

R34.1 Exact theorem, actual directions and exhaustive structural table

On the original finite probability space put

$$Jv = (Ev)1, \quad B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B, \quad p = 1 - u, \quad q = 1 + u,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \qquad \delta_i = \|B^i\mathbf{1}\|^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The receiver in the theorem is exactly the existing outward arm $C = f_2 = A^21$. There is no variable receiving length in the theorem. For every strictly increasing positive finite residual table Σ of size at least two and every integer $L \geq 1$, select the actual terminal product in the table below. Every connector edge points from terminal to receiver, so its actual message is $X = T^LF$. Then

$$E[f_2X^4] \leq 1. \tag{R34.1.1}$$

The directions use only Σ, L. In an f_s arm every edge points toward its leaf; in an h_s arm the first edge points toward the terminal and all remaining edges toward the leaf; in $\bar{f}_s$ every edge points toward the terminal. These statements specify each original edge, including all degree-two subdivisions. A split of the proof by d leaves them unchanged.

even, $(1, b)$

Actual terminal product, all unmentioned pairs retained: qf_b;

Proof: exact Section N.7.

even, $(2, b)$

Actual terminal product, all unmentioned pairs retained: $h_2 f_b$;

Proof: exact Section N.10.

even, $(3, b), b \geq 4$

Actual terminal product, all unmentioned pairs retained: $f_3 \bar{f}_b$ for $L = 1, 2$;

$h_3 f_b$ for $L \geq 3$;

Proof: §4.

even, $s_1 < t_1 < \cdots < s_j < t_j$**,** $s_1 \geq 4$

Actual terminal product, all unmentioned pairs retained: $\prod h_{s_i} f_{t_i}$;

Proof: §5.

even, $j \geq 2, s_1 = 2$ **or** 3

Actual terminal product, all unmentioned pairs retained: $\prod h_{s_i} f_{t_i}$;

Proof: §6.

even, $1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j$**,** $j \geq 2$

Actual terminal product, all unmentioned pairs retained: $qf_{b_1} \prod_{i=2}^{j} h_{a_i} f_{b_i}$;

Proof: §7.

odd, $2 \leq \ell < s_1 < t_1 < \cdots < s_j < t_j$**,** $s_1 \geq 4$

Actual terminal product, all unmentioned pairs retained: $\bar{f}_\ell \prod h_{s_i} f_{t_i}$;

Proof: Lemma N.6, family $\mathcal{C}$.

odd, $(2, 3, c, s_2, t_2, ...)$

Actual terminal product, all unmentioned pairs retained: $\bar{f}_2 f_3 h_c \prod_{i=2}^{j} h_{s_i} f_{t_i}$;

Proof: Lemma N.6, family $\mathcal{P}$.

odd, $(1, 2, 3)$

Actual terminal product, all unmentioned pairs retained: $qf_2 \bar{f}_3$ at $L = 1$;

$qf_2 h_3$ at $L \geq 2$;

Proof: Section N.17;

Section N.15.

odd, $(1, 2, 4)$

Actual terminal product, all unmentioned pairs retained: $qf_2 h_4$;

Proof: Section N.16.

odd, $(1,2,c), c \geq 5$

Actual terminal product, all unmentioned pairs retained: qf_2h_c;

Proof: §12.

odd, $(1,3,c), c \geq 4$

Actual terminal product, all unmentioned pairs retained: qf_3h_c;

Proof: §10.

odd, $(1,b,c), 4 \leq b < c$

Actual terminal product, all unmentioned pairs retained: qf_bh_c;

Proof: §11.

odd, $j \geq 2, s_1 \geq 4$

Actual terminal product, all unmentioned pairs retained: $q\prod h_{s_i}f_{t_i}$;

Proof: Lemma N.7, family A.

odd, $j \geq 2, s_1 = 3$

Actual terminal product, all unmentioned pairs retained: $qf_3h_{t_1}\prod_{i=2}^{j} h_{s_i}f_{t_i}$;

Proof: Lemma N.7, family B.

odd, $j \geq 2, s_1 = 2, t_1 \geq 4$

Actual terminal product, all unmentioned pairs retained: $qf_2h_{t_1}\prod_{i=2}^{j} h_{s_i}f_{t_i}$;

Proof: Lemma N.7, family II.

odd, $j \geq 2, (s_1,t_1) = (2,3), s_2 \geq 5$

Actual terminal product, all unmentioned pairs retained: $qf_2h_3\prod_{i=2}^{j} h_{s_i}f_{t_i}$;

Proof: Lemma N.7, family III.

odd, $j \geq 2, (s_1,t_1,s_2) = (2,3,4)$

Actual terminal product, all unmentioned pairs retained: $qf_2h_{t_2}h_3f_4\prod_{i=3}^{j} h_{s_i}f_{t_i}$;

Proof: Lemma N.7, families IV, V.

The last row crosses the roles of the first two pairs, as prescribed by their lengths. The table is exhaustive by parity, minimum, second minimum and, in the last odd case, the next original integer.

Arbitrary original equal-pair factors E_o, E_t satisfy

$$E[E_of_2\{T^L(E_tF)\}^4] \leq 1. \tag{R34.1.2}$$

Every positive length, arbitrary original repeats and unbounded finite numbers of pairs are retained. The complete reversal uses the same result with $-K$, on the same measure.

R34.2 Genuine f2 calculus and scalar estimates

All strict inequalities proportional to d in the proof are used for $d > 0$. At $d = 0$, $B\mathbf{1} = 0$, and successive integration of the original leaves gives every actual tree density one.

R34.2.1 Finite original identities

The supplied C014 calculus gives

$$0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta \delta_i, \quad \|B\|^2 \le \theta, \quad |BV(x)| \le \|V\|.$$

The original positive operators have kernel in $[0,2]$. The path messages $f_n, \bar{f}_n$ lie in $[0,2]$ and have mean at most one. One must not extend the mean assertion to h_n: $Eh_2 = 1 + d$.

Write

$$G_n(z) = \sum_{r=0}^{n-1} z^r, \quad G_0 = 0, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1} \delta_i.$$

The exact finite positive convolution is

$$1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}, \qquad \tau \le \beta_r \le d \ (r \ge 3), \quad \beta_3 = \beta_4 = d. \tag{R34.2.1}$$

All sums are finite; positive geometric series bound their subcollections. At $z = i\sqrt{x}$, $0 \le x \le \theta$, write $G_n = A_n + zC_n$. For $n \ge 2$, $1 - x \le A_n, C_n \le 1$; for $n \ge 3$, $A_n \le 1 - x + x^2$; for $n \ge 4$, the same holds for C_n. The exceptional $G_1 = 1$ has imaginary coordinate zero. Every finite G_n, $n = 1$ or $n \ge 3$, contracts, and $|G_2| < 19/16 < 6/5$. In particular, G_2 is not contractive.

R34.2.2 Exact receiving covariance and basic products

Directly,

$$C = f_2 = 1 - u + Bu, \quad EC = 1 - d, \quad 0 \le C \le p + \sqrt{d} \le p + \tfrac{6}{5}\sqrt{d}, \quad C \le 2 - q^2/2, \tag{R34.2.2}$$
$$\sigma(C)^2 = d + \delta_2 - d^2 \le 45d/32 < (6/5)^2 d.$$

The point cap in 2.2 is also a direct original-row fact. For the fixed row $v(y) = B(x,y)$, skewness gives $Bu(x) = \frac{1}{2} E_{y,z}[(v(y) - v(z))B(y,z)]$. Since $|v(y) - v(z)| \le 1 - v(y)v(z)$ on $[-1,1]^2$, $Bu(x) \le (1 - u(x)^2)/2$, which is exactly $C \le 2 - q^2/2$.

Let ν be the original spectral measure of $-B^2$ at u: its mass is d, support is $[0,\theta]$, and $\int x^r \, d\nu = \delta_{r+1}$. If $H = Q(B)u$ and $EH = -\alpha$, then the exact identity is

$$\mathrm{Cov}(C,H) = -\int \mathrm{Re}[(1+z)Q(z)] \, d\nu - d\alpha. \tag{R34.2.3}$$

The final signed mean product is retained in the covariance estimates.

The decisive genuine-f2 products are

n	$\mathrm{Re}(G_2 G_n)$
1	1
2	$1 - x \ge \kappa$
3	$1 - 2x \ge 3/16$
$n \ge 4$	$\ge (1-x)^2 \ge \kappa^2 = 361/1024.$

(R34.2.4)

For even n, $A_n = C_n \ge 1 - x$. For odd $n \ge 5$, $A_n = 1 - xC_n$ and $C_n \le 1 - x + x^2$, whence $A_n - xC_n \ge 1 - 2x + 2x^2 - 2x^3 \ge (1-x)^2$. Every product in 2.4 is at most one. Also

$$|\operatorname{Re}(G_2 z^r)| = x^{\lceil r/2 \rceil} \le x \quad (r \ge 2), \qquad \operatorname{Re}(G_2 z) = -x,$$
$$\operatorname{Re}(G_2(2+z)) = 2 - x \le 2, \quad \operatorname{Re}(G_2(3 + 2z + z^2)) = 3 - 3x \le 3. \tag{R34.2.5}$$

For later positive additions, the actual bare covariances are

$$\begin{aligned}
\operatorname{Cov}(C, q) &= -d, \\
\operatorname{Cov}(C, \bar{f}_2) &= -d + \delta_2 - d^2 < 0, \\
\operatorname{Cov}(C, \bar{f}_3) &= -d + 2\delta_2 - d^2 < 0, \\
\operatorname{Cov}(C, \bar{f}_4) &= -d + 2\delta_2 - \delta_3 + d\delta_2 \le -667d/3072, \\
\operatorname{Cov}(C, \bar{f}_5) &= -d + 2\delta_2 - 2\delta_3 + d^2 + d^3 \le -337d/4608.
\end{aligned} \tag{R34.2.6}$$

For the last two use $v = \delta_2/d \le \theta$, $\delta_3/d \ge v^2$, and monotonicity of $-1 + 2v - v^2 + dv$ and $-1 + 2v - 2v^2 + d + d^2$ on the stated rectangle. Their respective upper corners give the displayed exact rationals.

R34.2.3 Taylor, signed cubic means and exact bracket compensation

For nonnegative actual Y, with its actual $m = EY$,

$$E[CY^4] = (1-d)m^4 + 4m^3 \operatorname{Cov}(C, Y) + E[C(Y-m)^2\{(Y+m)^2 + 2m^2\}]. \tag{R34.2.7}$$

If $W = e^{-\Gamma} Y$, the first two terms scale by $e^{-4\Gamma}$. A pointwise coefficient M proved on W multiplies $e^{-2\Gamma} \operatorname{Var} Y$, not $e^{-4\Gamma} \operatorname{Var} Y$. If instead M is proved on Y, all terms have the fourth-power scale.

We will use the following comparison of scalar brackets. Suppose an old bracket is $(1 - \eta d)m^4 + 4vm^3$, $1 \le \eta \le 2$. If $0 \le m \le M$ and

$$\operatorname{Cov}(C, Y) \le v - (\eta - 1)Md/4, \tag{R34.2.8}$$

then the new mean-plus-covariance bracket is no larger: its difference is at most $(\eta - 1)dm^3(m - M) \le 0$. Usually $M = 1$. This compensates for the mean $1 - d$. When covariance is negative, all actual cubic means or their valid lower bounds are kept; no replacement of m^3 by an upper bound is made with the wrong sign.

R34.2.4 Pointwise curvature and scalar certificates

For $x, y, m \ge 0$, $x + y \le H$, $H \ge 3m$, the cubic estimate is

$$x(y^2 + 2m^2) \le 4H^3/27 + 2Hm^2/3 + m^3/2. \tag{R34.2.9}$$

To verify it, put $y = H - x$, then $x = H/3 + t$. For $t \le 0$ the remaining cubic terms are nonpositive; for $0 \le t \le H/3$, complete the quadratic square to get at most $3m^4/(2H) \le m^3/2$. On $x \ge 2H/3$ the polynomial is convex and its two endpoint values are covered by the same bound using $H^2 \ge 9m^2$. Thus the curvature constants used below are:

Table 40: Every genuine length-two reference arm: the complete directory: parameter values.

Entry	Bound for m	Valid coefficient
4	1;	13.
18/5	1;	10.
17/5	33/32;	9.
18/5	21/20;	41/4.

The same statement applies to W and EW. All the pointwise proofs below use the actual 2.2; the stronger old $(1-d)p$ bound is not assigned to f_2.

If $C_0 \geq A_0/2$ and $A_0 > B_0 \geq 0$, then

$$e^{-A_0 d} + B_0 d e^{-C_0 d} \leq 1 - (A_0 - B_0) d e^{-A_0 d/2}. \tag{R34.2.10}$$

This follows from $2\sinh(A_0 d/2) \geq A_0 d$. We use the elementary finite positive-series estimates $e < 3, e^2 < 9, e^4 < 81$. Other, tighter continuous certificates with their polynomial lower bounds are stated below.

R34.2.5 Finite masks and positive propagation

For an actual ordered pair put

$$R_{s,t} = \sum_{r=s+1}^{t} K^r \mathbf{1}. \tag{R34.2.11}$$

Disjoint original pairs give disjoint finite intervals and a single finite coefficient mask; no sum of separate worst-case norms is charged once per pair. For any finite coefficient polynomial P with coefficients in $[0,1]$, split its four residue classes. Positive and negative subcollections in each coordinate give

$$|P(i\sqrt{x})|^2 \leq \frac{1+x}{(1-x^2)^2} \leq \frac{32768}{16245}. \tag{R34.2.12}$$

The actual finite pair inequalities are

$$h_2 + f_t \leq 2 + R_{2,t} - \kappa d, \quad h_3 + f_t \leq 2 + R_{3,t} - (2-\theta)d, \quad h_s + f_t \leq 2 + R_{s,t} - 16d/5 \quad (s \geq 4). \tag{R34.2.13}$$

The $s = 4$ loss may be strengthened in the indicated single-pair packets. These are finite positive-convolution facts, proved in the finite-mask calculation of Section N.39 and the corresponding C020/C022 packets. AM–GM applies to their actual nonnegative left sides. A pair product is bounded by its corresponding exponential.

For $D_i = EB^i R$ the exact Duhamel identity is

$$T^r R = B^r R + \sum_{i=0}^{r-1} D_i \bar{f}_{r-1-i}. \tag{R34.2.14}$$

Every upper used below is obtained by positive propagation of a true first row. Signed D_i remain signed unless explicitly replaced by positive parts in a positive message coefficient. The displayed γ's

record every finite convolution loss, including the one needed for the actual mean. Precursor and output norms are different quantities.

R34.3 Exact genuine-f2 boundary rows

Table 41: Every genuine length-two reference arm: the complete directory: parameter values.

Entry	Actual object and exact usable scope
Section N.7	$E[f_2\{T^L(qf_b)\}^4] \le 1$, every $b \ge 2, L \ge 1$.
Section N.10	$E[f_2\{T^L(h_2f_b)\}^4] \le 1 - d/10000$, every $b \ge 3, L \ge 1$.
Section N.14	$E[f_2\bar{f}_n^4] \le 1 - d/20$, every $n \ge 3$.
Section N.17	$E[f_2\{T(qf_2\bar{f}_3)\}^4] \le 1 - d/256$.
Section N.15	$E[f_2\{T^L(qf_2h_3)\}^4] \le 1 - d/500$, every $L \ge 2$.
Section N.16	$E[f_2\{T^L(qf_2h_4)\}^4] \le 1$, every $L \ge 1$.

Section N.14's bare $n = 3, 4, 5$ proofs are stronger than the uniform $d/20$, and its exact Section N.13 input covers every $n \ge 6$. Bare $n = 1, 2$ are not contractions and will be paid separately whenever they arise. The C090/C092/Section N.14 larger-terminal thresholds enter in their stated ranges.

R34.4 The single pair (3, b) under the genuine f2 reference

This section independently changes the receiver in the C032 finite proof. Its low-length direction is $F = f_3\bar{f}_b$; its $L \ge 3$ direction is $F = h_3f_b$. Here $b \ge 4$.

R34.4.1 Length one and the full all-k precursor

Put $n = b - 3$, $G_n = A_0 + zC_0$, and

$$R = 2Bu + B^3G_n(B)u, \quad \Gamma = d + \beta_b \ge 51d/32, \quad m_0 = 1 - \Gamma/2, \quad k = m_0^{-1} \in [1, 3/2].$$

The exact coordinates of (R/u), interpreted as a polynomial and not a pointwise quotient, are $x^2C_0 + zr$, $r = 2 - xA_0$. The finite first row in C032 §§4–5, derived from the actual pair sum, gives

$$e^{\Gamma}TF \le Y_1 = q + kBR - c, \qquad c = 3d/5,$$

$$ER \le -51d/32, \quad D_1 = EBR \le 0, \quad ER^2 \le 21d/16, \quad \|u + kBR\|^2 \le d - 2\delta_2. \tag{R34.4.1}$$

For every $k \in [1, 3/2]$, not merely its endpoints, the output symbol

$$Q_k = 1 - kxr + zkx^2C_0$$

satisfies

$$\mathrm{Re}(G_2Q_k) = 1 - kxr - kx^3C_0 \ge 1 - 2kx \ge 1 - 3x \ge 1 - 4x. \tag{R34.4.2}$$

The signed mean product in 2.3 is favorable here. Equations 4.1, 4.2, the exact $EC = 1 - d$, and $C \le p + \sqrt{d}$ meet every premise of the receiver-independent C032 length-one curvature/scalar certificate (coefficient 10). Thus its $d/10$ deficit is retained. In particular no bound $|G_2| \le 1$ has been inserted.

R34.4.2 Length two: actual negative constant, covariance and cubic mean

The true precursor is

$$V = -c\,1 + u + kBR, \quad Y_2 = q + \epsilon + BV, \quad \epsilon = -c + kD_1.$$

It is not ((1-c)u+kBR). The entire constant cross is

$$\begin{aligned} \|V\|^2 &= \|u + kBR\|^2 + c^2 - 2ckD_1 \\ &\le d - 2\delta_2 + 9d^2/25 + (9/5)d\,\delta_3 \\ &\le d - (281/160)\delta_2 + 9d^2/25 \le d - (1117/800)d^2 \le d. \end{aligned} \tag{R34.4.3}$$

This uses $D_1 = -\int x^2 C_0\,d\nu$, $\delta_3 \le \theta\delta_2$, $d \le 1/3$, $\delta_2 \ge d^2$; its cross term has the positive sign it actually has. Consequently $Y_2 \le q + \epsilon + \sqrt{d} \le q + \sqrt{d}$.

Write $a_0 = 1 - c$. The output symbol, distinct from the precursor, is

$$Q_2 = (a_0 - kx^3 C_0) + z(1 - kxr).$$

Its genuine-f2 product is at least

$$4/5 - x + 2x^2 - 2x^3 > 7/12. \tag{R34.4.4}$$

The derivative is $-6(x-1/3)^2 - 1/3 < 0$, and its value at the larger endpoint $5/12$ is $2531/4320 > 7/12$. The possibly adverse signed mean product is at most $3d^2/40$. Hence

$$\mathrm{Cov}(C, Y_2) \le -7d/12 + 3d^2/40 \le -67d/120 \le -3d/10. \tag{R34.4.5}$$

The actual mean lies in ([1-8d/5, 1-21d/40]), in particular above $7/15$. The required function is now

$$g(\mu) = (1 - d)\mu^4 - (6/5)d\mu^3.$$

On this actual interval, $g'(\mu)/\mu^2 = 4(1-d)\mu - (18/5)d \ge 56/45 - 6/5 = 2/45 > 0$. It is therefore legitimate to use the upper mean endpoint in the negative cubic bracket. At that endpoint its exponential rate is at least $33/8$; combined with $4\Gamma \ge 51d/8$ it is stronger than $52/5$. The full output variance is at most ([$13/12 - 27d/25$]d). The genuine pointwise bound above gives coefficient 9. Thus the exact continuous certificate reconstructed in C032 and A016 applies:

$$e^{-52d/5} + \left(39d/4 - 243d^2/25\right) e^{-51d/16} \le 1 - 3d/320, \qquad 0 \le d \le 1/3. \tag{R34.4.6}$$

R34.4.3 Every length at least three

For $d \le 1/8$, the canonical $h_3 f_b$ packet is

$$R = \tau u + B^3 G_{b-3}(-B)u, \quad \Gamma \ge 35d/16, \quad U = \tau - x^2 C_0 - xzA_0, \quad |U| < 9/20.$$

The same finite first row has $\sigma < 55d/448$, $D \le 47d/80$, $\epsilon = D + D_1 \in [-d^2, 11d/16]$, and $D + (D_1)_+ \le 11d/16$. The exact losses include $\gamma_3 \ge d, \gamma_4 \ge 243d/256, \gamma_n \ge 227d/448$ for $n \ge 5$. The actual means and norms are

$$EY_3 \le 1 - 97d/80, \quad EZ_{L\ge 4} \le 1 - 13d/16, \quad |P_n| \le 1 \ (n \ge 3),$$
$$|Q_2| < 7/5, \quad |Q_3| \le 1 + 4d/5, \quad |Q_n| \le 1 + 16d/15 \ (n \ge 4).$$

For $P_n = G_n + z^n U$, the genuine receiving real product is positive for every $n \ge 3$: its minimum baseline is $3/16$, whereas $(6/5)(9/20)\theta^{3/2} < (6/5)(9/20)(4/15) = 18/125 < 3/16$. Positive ϵG_{n-2} is favorable; negative (epsilon) costs at most d^3, since all products 2.4 are between zero and one. The Dz^{n-2} and the retained (h(1+q)) costs remain exactly those of 2.5. The full signed mean terms therefore preserve the upper bounds $\mathrm{Cov}(C, Y_3) \le 9d/50 + 3d^2/8$ and $\mathrm{Cov}(C, Z_{L\ge 4}) \le d^2$.

The actual pointwise upper $C \le p + \sqrt{d}$ meets the coefficients 9 and 8. The complete low-domain scalar certificates are consequently

$$e^{-13d} + 9de^{-111d/40} \le 1 - 63d/2048, \quad e^{-12d} + 8de^{-269d/120} \le 1 - 21d/640. \tag{R34.4.7}$$

For $d \ge 1/8$, the same canonical actual pair satisfies $h_3 f_b \le e^{-d}$. The centered finite proof uses $\beta_* = \beta_{b-1}$; $b = 4$ is its actual short endpoint, not an unavailable geometric tail. Hence

$$E[f_2\{T^L(h_3 f_b)\}^4] \le e^{-4d} E[f_2 \bar{f}_L^4] \le e^{-4d}(1 - d/20), \quad L \ge 3.$$

This completes (3, b), with a conservative common $d/1000$ deficit.

R34.5 Every even table of minimum at least four

Let $F = \prod_{i=1}^{j} h_{s_i} f_{t_i}$, $j \ge 1, s_1 \ge 4$. The entire finite C022 mask has

$$F \le e^{R-\Gamma}, \quad \Gamma \ge 16jd/5, \quad R = BV_0, \quad \|V_0\|^2 < d/7, \quad ER^2 < d/18, \quad |R| < 1/4.$$

For the single pair starting at four, the stronger loss is (69d/20). The actual moments satisfy

$$D_0 \in [-d/5, d/12], \quad D_1 \in [-d/12, d/5], \quad D_i \le d/12 \ (i \ge 2).$$

Since $e^R \le 1 + R + 5R^2/8$, take $D = 11d/72 \ge D_0 + 5ER^2/4$. Positive propagation gives

$$e^{\Gamma} T^L F \le Y_L = \bar{f}_L + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{R34.5.1}$$

The finite loss $\gamma_r = \beta_r - D_{r-1} \ge d/2$ for every $r \ge 3$. The $r = 1, 2$ terms are paid separately in long rows. The successive spectral tail constants for $BR, B^2R, \dots, B^6R$, in units of $\sqrt{d}$, are respectively

$$5/32, \quad 1/10, \quad 1/16, \quad 1/25, \quad 1/40, \quad 1/60. \tag{R34.5.2}$$

They follow directly from $\|B\|^2 \le \theta$ and the same finite mask, so cover every subsequent finite power as well.

R34.5.1 Low domain $0 < d \le 1/16$, with all new receiving fees

At $L = 1$, the actual mean satisfies $1 + 5d/72 \le m \le 1 + 127d/360 \le 21/20$, and $\operatorname{Var} Y \le 1369d/1024$. By 2.3 or Cauchy on the tail,

$$\operatorname{Cov}(C, Y) \le -d + (6/5)(5/32)d = -13d/16.$$

The negative cubic therefore contributes at most (-13d/4), since $m \ge 1$. The actual pointwise sum is

$$C + Y + m \le 3 + (31/25)\sqrt{d} + (91/180)d < 18/5,$$

so coefficient $41/4$ is legal. The mean exponent is $223/18 > 12$, and the combined variance/covariance coefficient is

$$(41/4)(1369/1024) - 13/4 = 42817/4096 < 11.$$

Thus 2.10 applies with $(A_0, B_0, C_0) = (12, 11, 64/5)$.

At $L = 2$,

$$m \le 1 - 203d/360, \quad m \ge 1 - 313d/288 \ge 9/10, \quad \operatorname{Var} Y \le 169d/100,$$

$$\operatorname{Cov}(C, Y) \le -\kappa d + (6/5)(1/10)d = -379d/800 \le -9d/20.$$

Hence $4m^3 \operatorname{Cov}(C, Y) \le -6561d/5000 \le -5d/4$, with the lower, not upper, mean used. The genuine pointwise sum is

$$C + Y + m \le 3 + (69/32)\sqrt{d} - 7d/120 < 18/5.$$

The coefficient is 10, and the scalar triple is

$$(A_0, B_0, C_0) = (289/18, 313/20, 64/5).$$

Its gap is $73/180$; since $A_0 d/2 < 1$, the deficit exceeds $73d/540 > d/100$.

For $L = 3, 4, 5$, use the actual positive upper from the full finite convolution, with all negative coefficients treated as in 5.1. It has

$$m \le 1 - 7d/5, \quad \operatorname{Var} Y \le 121d/100, \quad \operatorname{Cov}(C, Y) \le (6/5)(11/10)d = 33d/25.$$

The pointwise sums are bounded by $3 + (37/16)\sqrt{d} - (609/360)d$ at $L = 3$, and by $3 + (21/10)\sqrt{d}$ at $L = 4, 5$, all below $18/5$. For $L \ge 6$, pay the two exceptional finite coefficients by $E_0 = 127d/180$, retain the loss $h = d/2$, and use the same shifted finite polynomial as C022 §7. Then

$$m \le 1 - 8d/5, \quad \operatorname{Var} Y \le 121d/100, \quad C + Y + m \le 3 + (81/40)\sqrt{d} + (37/180)d + d^2/2 < 18/5.$$

The same Cauchy covariance bound applies. All $L \ge 3$ therefore have coefficient 10 and common scalar triple $(19, 18, 64/5)$; the actual remainder coefficient is only $10(121/100) + 4(33/25) = 869/50 < 18$. Every low row has deficit at least $d/100$.

R34.5.2 High domain, including the two bare short obstructions

For $d \ge 1/16$, the same actual entire pair product satisfies $F \le e^{-2d}$, by the finite centered C022 pair bound. For $L \ge 3$ use the exact bare genuine-f2 theorem. At $L = 1$, the direct coefficient 11 with $C \le p + 3/4$, $Eq = 1$, $\mathrm{Cov}(C, q) = -d$, gives

$$E[Cq^4] \le 1 + 6d, \quad e^{-8d}E[Cq^4] \le 1 - 2de^{-8d} < 1 - 2d/27.$$

At $L = 2, d \in [1/16, 1/8]$, coefficient 10 is valid because $C \le p + 9/20$; 2.6 gives

$$E[C\bar{f}_2^4] \le (1-d)^5 + 10d(45/32 - d) - (19/8)(1-d)^3 d \le 1 + (30603/4096)d < 1 + 8d. \tag{R34.5.3}$$

For the second inequality use $(1-d)^5 \le 1 - 5d + 10d^2$ and $(1-d)^3 \ge (7/8)^3$; the d^2 terms cancel. Comparison with $e^{8d} \ge 1 + 8d$ leaves more than $d/100$ after scaling.

At $L = 2, d \ge 1/8$, coefficient 12 is valid under $C \le p + 3/4$. Discarding the negative covariance,

$$e^{-8d}E[C\bar{f}_2^4] \le e^{-8d}(1-d)^5 + 12d(45/32 - d)e^{-8d} \le 239349/262144 < 14/15 \le 1 - d/5. \tag{R34.5.4}$$

Indeed the first term is at most $50421/262144$, using $e^{-1} < 3/8$. The derivative of $d(45/32 - d)e^{-8d}$ is negative on ($[1/8, 1/3]$): the derivative of its bracket $(45/32 - 2d - 8d(45/32 - d))$ is negative there, and its value at $1/8$ is negative. Its second term is at most $1476/2048$. Thus every minimum-at-least-four even table has deficit at least $d/100$, including all single pairs and all positive L.

For clarity, the two bare $\bar{f}_2$ curvature constants above have complete one-variable certificates. The actual bound $\bar{f}_2 + Eb_2 \le 13/4 - p$ follows from $\sqrt{d} - d \le 1/4$. On the lower strip the polynomial is

$$\Phi(p) = (p + 9/20)[(13/4 - p)^2 + 225/128] \le 10.$$

On $0 \le p \le 3/2$, $\Phi'' \le -3$, $\Phi(1) = 25317/2560 < 99/10$, and $|\Phi'(1)| = 189/640 < 3/10$; its quadratic-tangent excess is below $3/200$. Beyond $3/2$, its derivative is negative and decreasing. On the upper strip use

$$\Psi(p) = (p + 3/4)[(13/4 - p)^2 + 49/32] \le 12.$$

For $p \le 1$, $\Psi'' \le -5$, $\Psi(3/4) = 747/64$, $\Psi'(3/4) = 9/32$; its excess is at most $81/10240$. On $[1, 2]$, the convex quadratic derivative has two negative endpoint values, so it remains negative. These are continuous proofs of the exact coefficients used in 5.3–5.4.

R34.6 Even multipairs of minimum two or three

Let $j \ge 2$, $s_1 \in \{2, 3\}$, and retain the actual canonical terminal $F = \prod_{i=1}^{j} h_{s_i} f_{t_i}$, with its strict original arm order. Lemma N.4, at the genuine receiving length $a = 2$, gives

$$\mathbb{E}[f_2(T^L F)^4] \le 1 - d/100 \qquad \text{for every integer } L \ge 1.$$

The shared proof retains the full finite mask, signed first constants, the constant vector in the length-two precursor, and all actual long mean indices. Its genuine G_2 receiving branch is proved in paragraph N.39. The original directions and the decoration bound are therefore exactly those in the reference directory.

R34.7 All minimum-one even multipair tables

The fixed product is

$$F = qf_{b_1}\prod_{i=2}^{j} h_{a_i}f_{b_i}, \quad 1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j, \quad j \geq 2. \tag{R34.7.1}$$

For classes I, II and IIIa, Section N.40 supplies the finite masks, positive first rows, norms and scalar certificates; the genuine f_2 receiving products are checked below. For class IIIb, Lemma N.5 applies with the actual receiver length $a = 2$, retaining the stronger deficit $2d/405$.

Every higher-index D_i bound in this section is ONE-SIDED. Negative D_i values are retained in the finite formulas.

Throughout this section, $D_i = EB^iR$ belongs to its one actual finite mask, and positive propagation is exactly

$$Y_L = \bar{f}_L + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}.$$

The symbols $P_n = G_n + z^n U$, $Q_n = P_n + \epsilon G_{n-2} + Dz^{n-2}$, $\epsilon = D + D_1$, and every finite γ_n use the algebraic identities in Section N.39, with this same U, R, D, D_i; the mask bounds are those of this section. Thus all displayed mean bounds retain the extra finite index rather than setting a spectral mean to zero.

R34.7.1 Classes I and II

Class I is $b_1 \geq 3$; class II is $b_1 = 2, a_2 \geq 4$. On $d \leq 1/64$, write the one actual finite mask as

$$R = zP_0(-z)u = Uu, \quad P_0(-z) = U_0 - zV_0, \quad c_x = (1-x^2)^{-1}, \quad |U| < 1.$$

The actual mask, before coordinate bounds, is

$$P_0(w) = G_{b_1-1}(w) + \sum_{i=2}^{j}\sum_{r=a_i+1}^{b_i} w^{r-2}.$$

Its exponent intervals are disjoint. Together with $R = zP_0(-z)u$, this specifies both classes. Its complete coordinate bounds are

	U_0	V_0
I	$[1 - xc_x, c_x]$	$[1 - xc_x, c_x]$
II	$[1 - x^3c_x, 1 + x^2c_x]$	$[-xc_x, x^2c_x]$.

The losses satisfy $\Gamma_{\mathrm{I}} \geq 607d/160 + 16(j-2)d/5$ and $\Gamma_{\mathrm{II}} \geq 16(j-1)d/5$. The true first-row constants are $D = 3d/8$ and $D = 0$, respectively; $\epsilon = D + D_1 \in [-d/8, 3d/8]$ or ([-d/12, d/5]). The finite moments obey $0 \leq D_2 \leq d/2, D_3 \leq d/5$, $D_i \leq d/10$ for $i \geq 4$, and every retained $\gamma \geq 3d/8$. The resulting exact mean/variance table is

Table 42: Every genuine length-two reference arm: the complete directory: parameter values.

Entry	Mean upper (I / II where different)	Variance upper
1	$(1 + 3d/8)/(1 + d/5)$;	$(27d/20)$.

Entry	Mean upper (I / II where different)	Variance upper
2	$(1-d/8)/(1-3d/10)$;	$(13d/8)$.
3	$(1-37d/40)$;	$(51/40)^2 d$.
≥ 4	$(1-3d/4)$;	$(36d/25)$.

The new length-one product has $Q_1 = (1 - xU_0) + zxV_0$. Its genuine real product is at least $1 - x/(1-x) \geq 6/19$ in I, and at least $\kappa - \theta^3/\kappa > 0$ in II. The latter's possible signed mean cost is at most $d^2/5$; both covariances are therefore nonpositive. At length two both full coordinates are positive, so the old $(1-x)Q_e - xQ_o$ lower bound improves to $Q_e - xQ_o$, with favorable signed mean. Its covariance remains nonpositive.

For $P_n = G_n + z^n U$, $n \geq 4$, the product is at least $\kappa^2 - (6/5)\theta^2 > 0$. For $n = 3$ it is at least $3/16 - (6/5)(4/15) = -53/400 > -1/3$. Equation 2.4 pays every signed (epsilon) term, and 2.5 every D term. The actual translated mean products then preserve

$$\mathrm{Cov}(C, Y_3) \leq d/3 + 3d^2/10, \qquad \mathrm{Cov}(C, Z_{L\geq 4}) \leq d^2.$$

The original coefficient 8 is valid with 2.2. The full continuous low-domain scalar rows are

$$e^{-64d/5} + (54/5)de^{-32d/5}, \quad e^{-15d} + 13de^{-6d}, \quad e^{-14d} + (53/4)de^{-6d}; \tag{R34.7.2}$$

Each is at most $1 - d/4$ on $0 \leq d \leq 1/64$, by the continuous payments following (R28.6.3) in Section N.40. For $d \geq 1/64$, every later actual pair is at most $e^{-d/2}$. Preserve that factor:

$$F \leq e^{-(j-1)d/2} q f_{b_1}, \qquad E[C(T^L F)^4] \leq e^{-2d} \leq 1 - 6d/5. \tag{R34.7.3}$$

The last inequality follows from $1 - e^{-2d} \geq 2d/(1+2d) \geq 6d/5$. Merely discarding these pairs would prove a weaker conclusion and would not justify the strict margin in 7.3.

R34.7.2 Class III, two pairs

Here $F = qf_2 h_3 f_b$, $b \geq 4$. On $d \leq 1/8$,

$$R = [\tau + z + z^3 G_{b-3}(-z)]u, \quad \Gamma \geq 35d/16, \quad D = 0.$$

Writing $G_{b-3} = A_0 + zC_0$, the actual short packet is

$$\epsilon = D_1 \in [-d^2, \delta_3], \quad \delta_2 - \delta_3 \leq D_2 \leq \delta_2, \quad D_3 \leq \theta d^2, \quad D_i \leq d/10 \ (i \geq 4),$$
$$\gamma_3 \geq \kappa d, \quad \gamma_4 \geq 243d/256, \quad \gamma_n \geq 7d/16 \quad (n \geq 5).$$

The two actual output symbols are

$$Q_1 = (1 - x + x^2 A_0) + z(\tau - x^2 C_0), \quad Q_2 = (1 - \tau x + x^3 C_0) + z(1 - x + x^2 A_0). \tag{R34.7.4}$$

The first genuine product is at least $\kappa - \theta/8$; its only adverse mean cost is at most $\theta^2 d^2$. Thus its covariance is nonpositive. The second has positive coordinates, so its old lower product $1/4$ is preserved, giving covariance at most (-d/4). This is the sign needed in the tight second-length cubic bracket. The actual mean is between $55/64$ and (1-d/2), the output variance is at most $11d/8 - \kappa^2 d^2$, and the entire constant-containing precursor is at most $21\sqrt{d}/20$. These are distinct bounds.

For the later P_n, use $|U| < 31/40$:

$$\mathrm{Re}(G_2P_3) \geq 3/16 - (6/5)(31/40)(4/15) > -4/15,$$
$$\mathrm{Re}(G_2P_n) \geq \kappa^2 - (6/5)(31/40)\theta^2 > 0 \quad (n \geq 4).$$

Positive (epsilon) is favorable; its negative part costs at most d^3. Full translated covariance bounds remain $4d/15 + d^3$ at $L = 3$, and $3d^2/5$ for longer lengths. The other actual data are

L	mean upper	variance upper
1	$1 + D_1 \leq 1 + \delta_3 \leq 1 + \theta^2 d$	$d + \theta d^3$
3	$1 - 4d/3$	$169d/144$
≥ 4	$1 - 9d/8$	$(1 + d/5)^2 d.$

The genuine C satisfies the exact pointwise premises for coefficients 8 at lengths 1, 2 and 9 later. All low-domain scalar certificates are

$$e^{-9d} + 8de^{-4d},$$
$$e^{-247d/20} + (11d - 361d^2/128)e^{-35d/8},$$
$$e^{-13d} + (169/16)de^{-35d/8}, \qquad e^{-13d} + 9de^{-15d/4}. \tag{R34.7.5}$$

Their respective R2-C028 continuous deficits are at least (d/4, 37d/640, 25d/128, 35d/64). For $d \geq 1/8$, retain the same actual inequality $h_3 f_b \leq e^{-d}$ and compare with the exact Section N.7 qf_2 row. This gives $E[C(T^L F)^4] \leq e^{-4d}$ for every L.

R34.7.3 Class III, at least three pairs, on the whole domain

The actual terminal is

$$F = qf_2h_3f_{b_2}\prod_{i=3}^{j} h_{a_i}f_{b_i}, \qquad 4 \leq b_2 < a_3 < b_3 < \cdots < a_j < b_j, \quad j \geq 3.$$

Its finite polynomial is

$$P_0(w) = 1 + w^2 G_{b_2-3}(w) + \sum_{i=3}^{j}\sum_{r=a_i+1}^{b_i} w^{r-2}.$$

The original intervals are disjoint, so every coefficient is zero or one, with $(P_0)_0 = (P_0)_2 = 1$ and $(P_0)_1 = 0$. The actual pair inequalities give

$$F \leq e^{R-\Gamma}, \qquad R = BP_0(-B)u + \tau u, \qquad \Gamma = 3d - 2\delta_2 + \frac{16}{5}(j-2)d \geq \frac{431}{80}d.$$

Lemma N.5, in its genuine $a = 2$ case, yields $\mathbb{E}[f_2(T^L F)^4] \leq 1 - 2d/405$ for every integer $L \geq 1$, throughout $0 \leq d \leq 1/3$. Together with the preceding classes, this proves the minimum-one even multipair family with deficit $d/3000$.

R34.8 Nonunit odd terminals

Lemma N.6. *For every integer $L \ge 1$, the following actual terminals satisfy $\mathbb{E}[f_2(T^L F)^4] \le 1 - d/100$:*

	original residual lengths	F
$\mathcal{C}$	$2 \le \ell < s_1 < t_1 < \cdots < s_j < t_j, \quad s_1 \ge 4$	$\bar{f}_\ell \prod_{i=1}^{j} h_{s_i} f_{t_i}$
$\mathcal{P}$	$(2, 3, c, s_2, t_2, \dots, s_j, t_j), \quad c \ge 4$	$\bar{f}_2 f_3 h_c \prod_{i=2}^{j} h_{s_i} f_{t_i}$.

Here $j \ge 1$, all listed lengths increase strictly, every connector edge points toward the receiver, and the existing receiving arm has length two and points outward.

Proof. Use $C = f_2$, with mass $1 - d$, pointwise bound $C \le p + \sqrt{d}$, and $\sigma(C) < (6/5)\sqrt{d}$. Leaf integration handles $d = 0$. First suppose $0 < d \le 1/8$.

The actual finite packets. For $\mathcal{C}$, put $\mu_\ell = \mathbb{E}\bar{f}_\ell \in [0, 1]$. The finite anchored calculation (R20.3.table) gives $F \le \bar{f}_\ell e^{R-\Gamma}$ and the positive upper

$$e^{\Gamma} T^L F \le Y_L = \bar{f}_n + \mu_\ell B^L R + D\bar{f}_{L-1} + \mu_\ell \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \quad n = \ell + L, \quad D_i = \mathbb{E}B^i R. \tag{N.47}$$

Write $D = c_0 d$. Its linked parameters are

	$\Gamma/d \ge g$	$\|R\|/\sqrt{d} < \rho$	c_0	$(D_1)_+/d \le c_1$	$(D_i)_+/d \le c_2,\ i \ge 2$
$I:\ j = 1, s_1 = 4$	$69/20$	$1/5$	$72/125$	$1/6$	$7/100$
$II:\ j = 1, s_1 \ge 5$	$16/5$	$3/20$	$3/5$	$1/30$	$1/12$
$III:\ j \ge 2$	$16j/5$	$1/4$	$5/6$	$1/5$	$1/12.$

In row I, necessarily $\ell = 2, 3$, and the actual $\beta_{\ell+1} = d$ is retained. In every row D is the upper for the anchored first constant, including the actual $\bar{f}_\ell$ mixed terms; no larger anchor variance is combined with a different row's smaller mask. For $n \le 5$, raise coefficients only before nonnegative paths in (N.47). The mean losses at $L = 1, 2, 3$ are respectively

$$k = 2 - c_0 - c_1, \qquad k = 5/2 - c_0 - c_1 - c_2, \qquad k = 11/4 - c_0 - c_1 - 2c_2,$$

and $\mathbb{E}Y_L \le 1 - kd$. The norm bounds are $1 + \sqrt{\theta}\rho$, $1 + \theta\rho + c_0 d$, and $1 + \theta^{3/2}\rho + (6c_0/5 + c_1)d$, in units of $\sqrt{d}$. Thus the standard deviation is at most $7\sqrt{d}/6$ for single pairs and $5\sqrt{d}/4$ for multipairs.

For $n \ge 6$, the same finite convolution gives (R20.8.1), with

$$Y = 1 - h(4 - d) + [G_n - h(3 + 2z + z^2)](B)u + \mu_\ell B^L R + E,$$
$$(h/d, E/d) = (2/5, 0), (1/2, 81/80), (3/8, 59/48).$$

The exceptional original coefficient at $\ell + 1$ is paid by E, and the extra actual mean coefficient is retained. Hence $\mathbb{E}Y \le 1 - 5h + 3hd + E$, giving mean losses $37/20, 13/10, 97/192$. The finite shifted-symbol inequalities (R18.6.1)–(R18.6.2) apply to $n \ge 6$ and $n - 1 \ge 5$. The long standard deviation is at most $(1 + \rho\sqrt{\theta})\sqrt{d}$. Consequently all single rows have $k \ge 5/4$, variance at most $49d/36$; all multiple rows have $k \ge 97/192$, variance at most $25d/16$.

For $\mathcal{P}$, the actual packets (R13.3.1), (R14.5.1), and (R18.3.1) use

$$R_0 = r_0(B)u, \quad r_0 = -\beta + x^2 V + zxU, \quad \beta = \beta_{c-1}, \quad U \in [1 - x, 1], \quad V \in [0, 1].$$

The endpoint $c = 4$ is $(U, V) = (1, 0)$. The signed real coordinate gives the useful common improvement

$$|r_0|^2 = (-\beta + x^2V)^2 + x^3U^2 \le d^2 + x^4 + x^3 \le 1/64 + \theta^4 + \theta^3 < 1/9.$$

Thus $\|R_0\| < \sqrt{d}/3$. For $j \ge 2$, add the one complete later mask S, with $\|S\| < 3\sqrt{d}/20$, so $\|R_0 + S\| < (29/60)\sqrt{d}$. The first constants remain $D = 3d/5$ and $3d/2$, and the losses remain $g = 35/16$ and $431/80$, respectively. Their finite propagation is (N.47) with $n = L + 2$ and coefficient one before $B^L R, D_i$. The actual $\bar{f}_2 = 1 + u + Bu$ is retained in this first row. The same finite identities give the following mean/variance data:

	$j = 1$		$j \ge 2$	
L	k	$\operatorname{Var} Y/d$	k	$\operatorname{Var} Y/d$
1	$51/40$	1	$41/120$	$121/100$
2	$17/10$	1	$2/3$	$2209/1600$
3	2	$81/64$	1	$25/16$
≥ 4	$37/20$	$1849/1600$	$7/24$	$121/100.$

For $L \ge 4$, use the distinct positive bounds (R18.3.3)–(R18.3.4), with $h = 2d/5, d/3$. The multirow's negative coefficient $\gamma_3 = d - D = -d/2$ is paid by $\bar{f}_{L-1} \le 2$, giving the actual extra constant d, rather than deleting it.

Genuine covariances. For $\mathcal{C}$ with $n \le 5$, the bare and every positively added short path have nonpositive covariance by 2.6. Only the full tail is charged, at most $(6/5)\rho\theta^{L/2}d$; this is at most $4d/25$ for single pairs and $d/5$ for multipairs. In the long rows the core product is at least $\kappa^2 - 3h$, its signed mean parameter $\beta_{n+1} - 2hd$ is positive, and

$$\kappa^2 - 3(h/d)/8 > (6/5)\rho\sqrt{\theta}$$

in each of the three linked rows. Thus all long covariances are nonpositive, even when $L = 1$ and the anchor is long.

For $\mathcal{P}$ at $L = 1$, the joint core $G_3 + zr_0$ has genuine product at least $1 - 2x - x^2 - x^3 > -1/20$ and positive mean parameter. Its covariance is at most $d/20$; adding BS costs at most $3d/25$. At $L = 2$, the single core $G_4 + z^2 r_0 - 2d/5$ has positive imaginary coordinate and genuine real product at least $(1 - x)^2 - 2d/5 - x^4$. Its gap exceeds the whole later fee $3d/40$. The added $(9/10)du$ has negative covariance, so both single and multiple covariances are nonpositive. At $L = 3$, keep the signed constants

$$P = G_5 - a_0 + \epsilon z + z^L r, \qquad (a_0, \epsilon) = \begin{cases} (7d/5 - D_1, -2d/5), & j = 1, \\ (d/2 - D_1, d/2), & j \ge 2. \end{cases}$$

The retained cores $G_5 - a_0 + \epsilon z$ have mean parameters $\tau - 2d^2/5$ and $\tau + d^2/2$, both positive; the complete $z^L r$ tails are paid separately in the following gaps:

$$\kappa^2 - (7/5 + \theta^2)/8 - 18/125 > 0, \qquad \kappa^2 - (49/60 + \theta/2)/8 - 24/125 > 0.$$

Here D_1 keeps its sign; its range gives the displayed upper bound for a_0, not an unjustified favorable $D_1 q$ term. For $L \ge 4$, the genuine products of the retained four-path or two-path core exceed their whole tail fees $9d/100, 3d/25$, respectively, and their actual signed mean parameters are positive. Thus every $\mathcal{P}$ covariance at $L \ge 2$ is nonpositive.

One curvature coefficient. Every positive upper above satisfies $C + Y + \mathbb{E}Y < 4$ and $\mathbb{E}Y \le 1$. For $\mathcal{C}, n \le 5$, the bound is $3 + (5/2)\sqrt{d} < 4$. In the long rows the pointwise precursor norm is at most $(1+\rho)\sqrt{d}$, by the second shifted-symbol inequality, giving the common upper $3+(9/4)\sqrt{d}+19d/48 < 4$. For $\mathcal{P}, L = 1$, the improved R_0 norm gives sums at most

$$3 + (38/15)\sqrt{d}, \qquad 3 + (161/60)\sqrt{d} + 19d/120$$

for single and multiple rows; both are below four using $\sqrt{1/8} < 17/48$. At $L = 2$, the actual upper is at most $q + \sqrt{d}$ or $q + 83/192$; their sums are below four. The finite pointwise forms (R18.3.2), (R18.3.3), and (R18.3.4) give, for $L \ge 3$, the common upper $3 + (12/5)\sqrt{d} + 3d/8 < 4$. By 2.9, the one unscaled coefficient is therefore $K = 64/5 > 683/54$.

One payment. If $\mathbb{E}Y \le 1 - kd$, $\operatorname{Var} Y \le vd$, and $\operatorname{Cov}(C, Y) \le bd$ with $b \ge 0$, the exact fourth identity gives

$$\mathbb{E}[C(T^L F)^4] \le e^{-Ad} + Bd\, e^{-4gd}, \qquad A \le 4g + 1 + 4k, \quad B \ge (64/5)v + 4b.$$

The receiving mass $1 - d$ and every actual negative mean loss are retained. The whole nonunit odd family is paid by five rows:

	L	g	k	v	b	A	B
$\mathcal{C}, j = 1$	≥ 1	$16/5$	$5/4$	$49/36$	$4/25$	$94/5$	$91/5$
$\mathcal{C}, j \ge 2$	≥ 1	$32/5$	$97/192$	$25/16$	$1/5$	28	21
$\mathcal{P}, j = 1$	1	$35/16$	$51/40$	1	$1/20$	$74/5$	13
$\mathcal{P}, j = 1$	≥ 2	$35/16$	$17/10$	$81/64$	0	$33/2$	$81/5$
$\mathcal{P}, j \ge 2$	≥ 1	$431/80$	$7/24$	$25/16$	$1/5$	23	$21.$

Each has $4g \ge A/2$, $A - B \ge 3/10$, and $Ad/2 < 2$. By 2.10 and $e^2 < 9$, the low-domain deficit exceeds $d/30 > d/100$.

For $d \ge 1/8$, the same actual pair comparisons give $F \le e^{-d} \bar{f}_\ell$ in $\mathcal{C}$, and $F \le e^{-(2j-1)d} \bar{f}_2$ in $\mathcal{P}$. The propagated bare index is at least three in both cases, so (Q101.0.2) yields at most $e^{-4d}(1 - d/20)$. Since $1 - e^{-4d} \ge 4d/(1 + 4d) \ge 12d/7$, the same margin holds in the high domain. Both closed endpoints and every original direction agree across the split. □

R34.10 Unit odd triple (1, 3, c), every c and every length

The genuine length-two extension (N.29) gives, in the actual direction $F = q f_3 h_c$,

$$\mathbb{E}[f_2\{T^L(q f_3 h_c)\}^4] \le 1 - d/800 \qquad (c \ge 4,\ L \ge 1).$$

Its proof uses the genuine G_2 product and receiving mass $1 - d$, all five connector windows, and the high-domain path comparison. Equal-pair decorations follow by positivity.

R34.11 Unit odd triples (1, b, c), $4 \le b < c$

Apply (N.32) with the same terminal $F = q f_b h_c$, the same outward receiving arm of length two, and the same terminal-to-receiver connector of length $L \ge 1$. The moment is at most $1 - d/3000$ on $0 < d \le 1/4$, and at most one on $1/4 \le d \le 1/3$. The upstream proof retains the three nonunit b ranges and all four unit configurations, including the stronger actual loss for $b \ge 6$. At $d = 0$, use leaf integration; decorations use positivity.

R34.12 Unit odd triples (1, 2, c), $c \geq 5$

Use (N.36) with the actual terminal $F = qf_2h_c$, the same receiving arm f_2, and the original terminal-to-receiver connector T^L, for every $L \geq 1$. It gives $1 - d/800$ throughout the host domain, with loss $d/600$ for nonunit connectors. On $d \leq 1/5$, the stronger unit and connector-dependent bounds are those of Lemma M.14. The cases $c = 3, 4$ retain their separate inputs in §3, including the $d/50$ low-domain bound of (Q105.0.3).

R34.13 A common estimate for unit-leading multipairs

Lemma N.7. *Let all arm lengths be strictly increasing in their original order, and let $j \geq 2$. The following six terminals satisfy $\mathbb{E}[f_2(T^LF)^4] \leq 1 - d/800$ for every integer $L \geq 1$:*

family	*original lengths*	F
A	$s_1 \geq 4$	$q\prod_{i=1}^{j} h_{s_i}f_{t_i}$
B	$s_1 = 3, c = t_1 \geq 4$	$qf_3h_c\prod_{i=2}^{j} h_{s_i}f_{t_i}$
II	$s_1 = 2, c = t_1 \geq 4$	$qf_2h_c\prod_{i=2}^{j} h_{s_i}f_{t_i}$
III	$(s_1, t_1) = (2, 3),\ s_2 \geq 5$	$qf_2h_3\prod_{i=2}^{j} h_{s_i}f_{t_i}$
IV	$(s_1, t_1, s_2) = (2, 3, 4),\ j = 2$	$qf_2h_eh_3f_4, \quad e = t_2 \geq 5$
V	$(s_1, t_1, s_2) = (2, 3, 4),\ j \geq 3$	$qf_2h_eh_3f_4\prod_{i=3}^{j} h_{s_i}f_{t_i}$.

Family A satisfies the stronger bound $1 - d/100$.

Proof. Write $C = f_2$, so $\mathbb{E}C = 1 - d$, $C \leq p + \sqrt{d}$ and $\sigma(C) < (6/5)\sqrt{d}$. At $d = 0$, integrate the original leaves. Suppose henceforth $d > 0$.

The finite packets. The positive first rows for families A, B are (R21.3.1) and (R23.3.6). For families II–V, use (R24.3.1)–(R24.3.10). Together these give

$$F \leq qe^{R-\Gamma}, \qquad T(qe^R) \leq \bar{f}_2 + BR + D, \qquad D_i = \mathbb{E}B^iR.$$

These are inequalities between actual messages, before a receiving factor is inserted. In the following table $D = \delta d$, $h = \eta d$, and $(D_i)_+ \leq \alpha_i d$ for $i = 1, 2, 3$.

	d_0	$\Gamma/d \geq g$	$\|R\|/\sqrt{d} \leq \rho$	δ	η	$(\alpha_1, \alpha_2, \alpha_3)$
A	$1/8$	$32/5$	$1/4$	$11/15$	$1/2$	$(1/5, 1/12, 1/12)$
B	$1/64$	$431/80$	$1/2$	$4/3$	$3/8$	$(1/2, 1/3, 1/4)$
II	$1/64$	$351/80$	$2/3$	$11/10$	$1/3$	$(33/64, 1/4, 1/4)$
III	$1/8$	$21/5$	$14/25$	$11/10$	$15/32$	$(1/2, 1/12, 1/10)$
IV	$1/8$	$21/4$	$1/2$	$11/10$	$1/2$	$(19/40, 1/16, 1/12)$
V	$1/8$	$169/20$	$3/5$	$7/4$	$1/3$	$(2/3, 1/3, 1/8)$.

For III, $R = B^2u + S$ and $\|S\| < 3\sqrt{d}/20$, whence $\rho \leq \theta + 3/20 = 89/160 < 14/25$. For IV, retain the signed crossing coordinates (R24.3.8): $R = (-A_c + zB_c)(B)u$, $0 \leq A_c \leq 19/40$, $-x^2 \leq B_c \leq 19d/16$, and $A_c + B_c \geq d/2 + x/16$. Family V adds its one complete later mask to this same crossing. Thus the table preserves every original interval and, in particular, a possibly negative B_c.

For $L = 1, 2$, take

$$Y_1 = \bar{f}_2 + BR + D, \qquad Y_2 = \bar{f}_3 + B^2R + Dq + D_1.$$

For $L = 3, 4$, raise only coefficients of nonnegative paths and take

$$Y_3 = \bar{f}_4 + B^3 R + D\bar{f}_2 + \alpha_1 dq + \alpha_2 d,$$
$$Y_4 = \bar{f}_5 + B^4 R + D\bar{f}_3 + \alpha_1 d\bar{f}_2 + \alpha_2 dq + \alpha_3 d.$$

For every finite $L \geq 5$, take the retained four-path upper (R24.4.1), also used in (R21.9.1):

$$Y_L = 1 - h(4 - d) + [G_{L+1} - h(3 + 2z + z^2)](B)u + B^L R + 2D.$$

In each case $0 \leq e^{\Gamma} T^L F \leq Y_L$. The finite packets give $\beta_r - D_{r-2} \geq h$ at all retained indices $r \geq 3$, including the additional mean index. Consequently $\mathbb{E}Y_L \leq 1 - 5h + 3hd + 2D$ for $L \geq 5$.

Genuine receiving covariances. Apply (M.21) with $a = 2, t = 0$. For a joint core $Q(B)u$ with mean $-\zeta$, this is

$$\operatorname{Cov}(C, Q(B)u) = -\int \operatorname{Re}(G_2 Q)\, d\nu - d\zeta.$$

For B, II, write respectively $r_0 = -\beta + x^2 V + zxU$ and $r_0 = -\beta - xU + zxV$, with $U \in [1 - x, 1]$, $V \in [0, 1]$. The latter interval includes the actual B-endpoint $c = 4$, where $(U, V) = (1, 0)$. For III, use $R = B^2 u + S$; for IV, use the signed coordinates above. At $L = 1, 2$, the joint cores $G_{L+1} + z^L r_0$ have positive imaginary coordinates, and hence $\zeta > 0$. Their real products with G_2 are at least

	$L = 1$	$L = 2$	later-mask covariance fees
B	$1 - x - x^2 - x^3$	$1 - 2x - x^4$	$(3d/25, 3d/40)$
II	$1 - x - x^3$	$1 - 2x$	$(3d/25, 3d/40)$
III	$1 - x + x^2$	$(1 - x)^2$	$(3d/25, 3d/40)$
IV	$1 - x - 19dx/16$	$1 - 2x - x^4$	$(0, 0)$.

Every displayed gap exceeds its fee throughout its stated domain. The term Dq is favorable. These eight covariances are therefore nonpositive. For A, V at $L = 1$, the bare gap κd exceeds $(6/5)\rho\sqrt{\theta}\, d$; for A, V at $L = 2$, only the tail fee $(6/5)\rho\theta d$ need be retained.

At $L = 3, 4$, all added path covariances are nonpositive by 2.6, so only $B^L R$ is charged. Uniformly its fee is $(6/5)\rho\theta^{L/2} d$. Use $\theta^{3/2} < 13/50$ and $\theta^2 < 1/6$. For $L \geq 5$, the core has real product at least $\kappa^2 - 3h$; its actual mean parameter is $\beta_{L+2} - 2hd > 0$. The full tail costs at most $(6/5)\rho\theta^{5/2} d$, with $\theta^{5/2} < 8/75$. The six parameter rows give $\kappa^2 - 3\eta d_0 > (6/5)\rho(8/75)$. Thus the long covariances, too, are nonpositive.

One payment. Put $\mu = \mathbb{E}Y_L$, and suppose that

$$\mu \leq 1 + ad, \quad \operatorname{Var} Y_L \leq vd, \quad \operatorname{Cov}(C, Y_L) \leq bd, \quad b \geq 0.$$

If the scaled curvature coefficient on $W = e^{-\Gamma} Y_L$ is K, the exact fourth identity gives

$$\mathbb{E}[C(T^L F)^4] \leq e^{-Ad} + Bd\, e^{-2gd} \quad \text{when} \quad A \leq 4g - \max\{0, 4a - 1\}, \quad B \geq Kv + 4br, \quad \text{(N.48)}$$

where $r \geq \mu^3$ is needed only when $b > 0$. Indeed $(1 - d)\mu^4 \leq e^{(4a-1)d}$, while the nonnegative covariance fee and the variance fee have the common upper scale e^{-2gd}. No mean above one is discarded.

Here is sufficient finite data for this single payment. The mean and variance columns follow by substituting the displayed positive uppers in (R21.3.2), (R23.3.8), and (R24.3.3), using their joint short

norms and finite shifted-symbol bounds. The covariance column was proved above.

	L	a	v	b	K	r	A	B
A	≥ 1	0	$1681/900$	$1/5$	13	1	$128/5$	$251/10$
B	≥ 1	$5/6$	$169/100$	$4/25$	9	$9/8$	19	16
II	1	$197/320$	$169/100$	0	9	1	16	$61/4$
II	≥ 2	$3/5$	$25/16$	$16/75$	9	$9/8$	16	$61/4$
III	≥ 1	$3/5$	$36/25$	$22/125$	10	$5/4$	$77/5$	$153/10$
IV	1	$23/40$	$25/16$	0	10	1	19	16
IV	≥ 2	0	$121/64$	$4/25$	10	1	21	20
V	$1 \leq L \leq 4$	$17/12$	$64/25$	$3/10$	10	$5/3$	29	28
V	≥ 5	2	$121/100$	0	10	1	26	$13.$

Here are direct norm checks for the variance column. At $L = 1, 2$, the joint cores for B, II, III, IV have norm bounds respectively $(6/5, 1), (6/5, 1), (43/40, 1), (5/4, 17/16)$; add the single $B^L S$ tail and, at $L = 2$, Du. For A, V, add the full tail to the bare G_2, G_3 cores. The ordinary triangle bounds on $\sigma(Y_L)/\sqrt{d}$ at $L = 3, 4$ are respectively

$$1 + (6\delta/5 + \alpha_1)d + \rho\theta^{3/2}, \qquad 1 + (-2 + \delta + 6\alpha_1/5 + \alpha_2)d + \rho\theta^2.$$

These use the bare centered norms $6/5, 1, 1, 1 - 2d$ for indices $2, 3, 4, 5$. Only III needs its tighter joint cores: at $L = 3$ the coordinates are $(1 - x + 3d/5, 1 - x + x^2 + 11d/10)$, of norm at most $1 + 3d/5$; the square difference divided by x is at most $-1 + \theta + d + 121d^2/100 < 0$. At $L = 4$, its positive real coordinate is at most the real coordinate of G_5, and its imaginary coordinate exceeds that of G_5 by $3d/5$; the norm is therefore at most $1 + (3/5)\sqrt{\theta}d < 1 + 2d/5$. Add $\|B^3 S\| < \sqrt{d}/25$, $\|B^4 S\| < \sqrt{d}/40$ in these two rows. Finally the long norm is at most $1 + \rho\theta^{5/2}$, by the retained finite shifted-symbol inequality. Substitution of the six parameter rows proves every listed variance bound; precursor and output norms have not been identified. For the four positive-covariance groups with $a > 0$, the displayed r's follow from $(1 + ad_0)^3$; all other charged groups have $\mu \leq 1$. Every listed A, B satisfies (N.48).

Curvature. In A, B, II, direct path-row bounds give respectively

$$C + Y_L + \mu \leq 3 + \tfrac{12}{5}\sqrt{d} + \tfrac{2}{3}d < 4, \quad 3 + \tfrac{38}{15}\sqrt{d} + \tfrac{13}{3}d < \tfrac{17}{5}, \quad 3 + \tfrac{8}{3}\sqrt{d} + \tfrac{1189}{320}d < \tfrac{17}{5}.$$

Also $\mu \leq 1, 33/32, 33/32$, respectively. Thus 2.9 gives $K = 13, 9, 9$, on Y_L and hence on its contraction W. For III, IV, the positive rows satisfy

$$Y_L + \mu \leq (1 + \delta d)q + 1 + \tfrac{8}{5}\sqrt{d} + \tfrac{9}{4}d, \quad 0 \leq \delta \leq 11/10, \quad \mu \leq 1 + 3d/5.$$

For V, they satisfy $Y_L + \mu \leq (1 + 3d)q + 1 + 2\sqrt{d} + 21d/4$, $\mu \leq 1 + 2d$. These bounds follow from the same positive coefficients displayed above, including the long excess $2D$. In both groups $\mathbb{E}W \leq 1$, and the coefficient of q in $C + W + \mathbb{E}W$, using $C \leq 2 - q + \sqrt{d}$, is nonpositive. Put $t = \sqrt{d} < 3/8$, and use $e^{-gd} \leq (1 + gt^2)^{-1}$. The numerator of the gap to $18/5$ is at least

$$\begin{array}{c|l} III, IV & \tfrac{3}{5} - \tfrac{13}{5}t + \tfrac{447}{100}t^2 - \tfrac{21}{5}t^3 \;\geq\; \tfrac{3}{5} - \tfrac{13}{5}t + \tfrac{579}{200}t^2 \\ V & \tfrac{3}{5} - 3t + \tfrac{827}{100}t^2 - \tfrac{169}{20}t^3 \;\geq\; \tfrac{3}{5} - 3t + \tfrac{4081}{800}t^2. \end{array}$$

The last two quadratics have negative discriminants $-47/250, -3243/1000$. Therefore $C + W + \mathbb{E}W < 18/5$, and 2.9 gives $K = 10$ with the variance scale $e^{-2\Gamma}$.

For every payment row, $A - B \geq 1/10$, $2g \geq A/2$, and $Ad_0/2 < 2$. Applying 2.10 and $e^2 < 9$ therefore gives the common low-domain bound $1 - d/100$.

The high domains use the same directed terminals. For A, $d \geq 1/8$ gives $F \leq e^{-4d}q$. At $L \geq 2$, use (Q101.0.2), whose index is $L + 1 \geq 3$. At $L = 1$,

$$\mathbb{E}[C\bar{f}_2^4] \leq 8\mathbb{E}[C\bar{f}_2] \leq 8(1-d)^2 \leq 49/8, \qquad e^{-16d}\mathbb{E}[C\bar{f}_2^4] \leq 147/152 < 1 - d/100,$$

using $e^{-2} < 3/19$ and the nonpositive bare covariance. For B, II at $d \geq 1/64$, every later pair is at most $e^{-8d/5}$. Use (N.29) for B, and Section N.16 or paragraph N.42 for II, in their same actual directions. For III at $d \geq 1/8$, regroup the same product as

$$F = (qf_2h_{s_2})(h_3f_{t_2})\prod_{i=3}^{j} h_{s_i}f_{t_i}.$$

Here $s_2 \geq 5$, $h_3f_{t_2} \leq e^{-d}$, and the first factor is the genuine base of paragraph N.42. For IV, V, the fixed crossing is already $(qf_2h_e)(h_3f_4)$ times the later pairs; $h_3f_4 \leq e^{-d}$. The two pair inequalities are the actual pointwise comparisons at the end of Section N.38. These last three families have bound at most e^{-4d}, and all later factors are at most one. Thus the high domains retain the claimed margins, including both closed splitting endpoints. □

R34.19 Supplementary finite certificates used in §§11–12

R34.19.1 The stronger actual unit loss when $b \geq 6$

Paragraph N.22 proves, from the actual finite alternating moments and $\delta_3 \geq \delta_2^2/d$,

$$\Gamma \geq 4\tau + \delta_3 \geq (4 - 4v + v^2)d \geq (2601/1024)d > 5d/2, \qquad v = \delta_2/d.$$

This is the stronger unit-row loss used in (N.32).

R34.19.3 The weak-receiver coefficient nine

Lemma A.2, case w, applies whenever $C_0, Z \geq 0$ and

$$C_0 \leq p + (6/5)\sqrt{d}, \quad Z \leq (1 - 5d/8)q - d/2 + (13/12)\sqrt{d}, \quad \mathbb{E}Z \leq 1 - 3d/2.$$

It gives $C_0[(Z + \mathbb{E}Z)^2 + 2(\mathbb{E}Z)^2] \leq 9$ throughout $0 \leq d \leq 1/3$. The actual f_2 satisfies this pointwise receiver condition by (R34.2.2).

R34.19.5 The high-domain path mean

The stronger finite estimate (A.8), with $83/32 > 5/2$, gives

$$\mu_n = \mathbb{E}f_n = \mathbb{E}\bar{f}_n \leq 1 - 5d/2 \quad (n \geq 4). \tag{R34.19.2}$$

Its proof covers every finite short and long index on $0 \leq d \leq 1/3$. Consequently $h_c \leq 2\mu_{c-1} \leq 2 - 5d$ for $c \geq 5$, as used in the same-direction high comparisons above. At $d = 0$ the bound follows directly from leaf integration.

N.43 The short-terminal reference catalogue and its row proofs

Let f_a be an actual outward receiving arm, with $a \geq 3$, and let E_o, E_t be the products of equal-arm factors at the receiving and terminal roots. Each such factor is $f_l h_l \leq 1$ or its full reversal $\bar{f}_l W_l \leq 1$. For every row below, with the indicated actual terminal factor F and connector length L, put $X = T^L(E_t F)$. Then

$$\mathbb{E}[E_o f_a X^4] \leq 1 - \epsilon d. \tag{R9.1.ref}$$

All operators use the given finite measure. The connector points from the terminal toward the receiving root; its L edges and all equal-arm pairs remain in the oriented tree.

$(1,2)$**;** $F = qf_2$**;** $L \geq 1$**;** $\epsilon = 1/1000$
The unit case is proved in Section N.12 with loss $d/50$. Section N.24 gives loss $d/1000$ at $L = 2$ and at least $9d/10$ at $L = 3, 4, 5$. For $L \geq 6$, (N.6) gives the actual receiving bound with loss $3d/800$. These use the same factor qf_2.

$(1,b)$**,** $b \geq 3$**;** $F = qf_b$**;** $L \geq 1$**;** $\epsilon = 1/3000$
The short and long connectors are proved together in (N.5), for every actual $a, b \geq 3$ and $L \geq 1$.

$(2,b)$**,** $b \geq 3$**;** $F = h_2 f_b$**;** $L \geq 1$**;** $\epsilon = 1/400$
For $L = 1$, Section N.27 gives $d/100$ at $b = 3$, and Section N.28 gives $d/200$, $d/20$ and $d/10$ at $b = 4$, $b = 5$ and $b \geq 6$, respectively. For $L \geq 2$, (N.7) gives the full receiving estimate with common loss $d/400$. The terminal factor is $h_2 f_b$ in all these ranges.

$(1,2,3)$**;** $F = qf_2\bar{f}_3$**;** $L = 1$**;** $\epsilon = 1/256$
This is the all-receiver fourth-moment proof in Section N.18.

$(1,2,3)$**;** $F = qf_2h_3$**;** $L \geq 2$**;** $\epsilon = 1/16$
Section N.21 proves every short connector case and the full finite tail with this direction.

$(1,2,c)$**,** $c \geq 4$**;** $F = qf_2h_c$**;** $L \geq 1$**;** $\epsilon = 1/800$
Proposition N.3 proves the whole row. Its $c = 4, L = 2$ case uses the actual independent bound (Q167.T175.0).

$(1,3,c)$**,** $c \geq 4$**;** $F = qf_3h_c$**;** $L \geq 1$**;** $\epsilon = 1/800$
The complete proof, including the unit connector and its finite-length cases, is Section N.20.

$(1,b,c)$**,** $4 \leq b < c$**;** $F = qf_bh_c$**;** $L \geq 1$**;** $\epsilon = 1/3000$
Section N.22 proves this family for every receiving arm $a \geq 3$ and every positive connector length.

We denote this eight-row catalogue by D_{full}. Its common deficit is $1/3000$. In the $(1,2,3)$ rows the choice of $\bar{f}_3$ or h_3 is determined by L. All other displayed terminal directions are unchanged across the connector ranges. Positivity inserts E_t, E_o into the undecorated inequalities without increasing a moment. At $d = 0$, leaf integration gives equality. Applying the same rule to $-K$ gives the fully reversed catalogue, with its receiving arm reversed as well.

O Ordinary fourth and eighth moments

This family supplies Proposition 9.2. Post blocks retain their stated order and their actual arm factors. The unit-arm ordinary catalogue is distinguished from the nonunit catalogue.

O.1 The shortest-two root: one finite packet and nine payments

Let $W_2 = Aq = p - r$, $r = K^2\mathbf{1}$, $s = K^3\mathbf{1}$, and $t = K^4\mathbf{1}$. With the notation and actual finite packet of Lemma M.4, we prove

$$\mathbb{E}[W_2 T^L(qf_b)]^8 \le e^{-d/10} \le 1 - d/12 \le 1 - d/256 \qquad (b \ge 2,\ L \ge 2). \tag{Q71.0.2}$$

The terminal unit arm points inward, its b-arm outward, and the whole connector points from the terminal to the root. The first edge of W_2 points outward and its second inward. These directions are fixed before choosing the host. Actual equal-pair decorations $E_t, E_r \in [0,1]$ give

$$0 \le E_r W_2 T^L(E_t q f_b) \le W_2 T^L(q f_b),$$

so the same bounds hold with such decorations. At $d = 0$, every actual tree message is one by leaf integration; below $d > 0$.

The common square and its complete exponent.

Write $X = T^L(qf_b)$, $Y = W_2 X$, and use the $m, k, R, \gamma, a_1, B_*$ of (M.6). For $b = 2$ set

$$\gamma = d - \delta_2/2, \quad B_2 = 1 - \gamma, \qquad D_4 = B_2^2 d + \delta_2 - 2\delta_3 + \delta_4 - \tau^2.$$

All nine rows below have an actual positive upper bound

$$0 < X \le C[Bq + Br + Z - a], \qquad 0 \le Y \le CB[1 + (Z-a)/(2B)]^2, \qquad Z = K^j H. \tag{O.1}$$

The square base is half the sum of W_2 and the positive terminal upper divided by CB. Put $c_* = (1+\theta)^2 = 2025/1024$. Lemma M.5 gives

$$\log \mathbb{E}Y^8 \le \mathcal{E}_{b,L} := 8\log C + 8\log B - \frac{8a}{B} + \frac{8M}{B} + \begin{cases} 6D/B^2, & j = 2, \\ 8c_* D/B^2, & j = 1, \end{cases} \tag{O.2}$$

where $\|H\|_2^2 \le D$ and $M \ge \max(\mathbb{E}Z, 0)$. In the tables $z = -K$, P_n is the actual polynomial with the same b, k, and

$$A_5(\tau) = \theta - 2\theta\tau + \tau^2, \qquad A_6(\tau) = \theta - \tfrac{7}{15}\theta^2 - 2\theta\kappa\tau + \tau^2.$$

Table 43: Actual short-one terminal masks for the shortest-two root.

Row	(C, B, a)	(j, H)
$b=2, L=2$	$1, B_2, 0$	$2,\ \gamma\mathbf{1} + r$
$b=2, L=3$	$1, B_2, \tau$	$2,\ -(g+s)$
$b=2, L=4$	$1, 1-\gamma-\tau, d(1-\gamma)$	$2,\ \tau\mathbf{1} - B_2 g + r + t$
$b=2, L=5$	$1, 1-2\tau, \tau$	$1,\ [\tau + zP_3(z)]g$
$b=2, L\ge 6$	$1, 1-2\tau, \tau$	$1,\ [\tau + zP_{L-2}(z)]g$
$b\ge 3, L=2$	$m^2, 1-\gamma, a_1$	$2,\ \gamma\mathbf{1} + kR$
$b\ge 3, L=3$	$m^2, 1-B_*, d-k\rho$	$1,\ [a_1 + zP_1(z)]g$
$b\ge 3, L=4$	$m^2, 1-41d/80, \kappa d$	$1,\ [41d/80 + zP_2(z)]g$
$b\ge 3, L\ge 5$	$1-d/2, 1-d/2, d/2$	$1,\ [d/2 + zP_{L-2}(z)]g$

Table 44: Moments of the same fluctuations and their paid exponents: $\mathcal{E}_{b,L} \le -r_{b,L} d$.

Row	D	M	$r_{b,L}$
$2,2$	δ_2	$\delta_2 - \gamma d$	$1/8$
$2,3$	$d - 2\delta_2 + \delta_3$	0	$3/4$
$2,4$	D_4	$(\theta\kappa - \tau)d$	1
$2,5$	$A_5(\tau)d$	$(\theta - \tau)d$	$2/3$
$2,\ge 6$	$A_6(\tau)d$	$(\theta - \tau)d$	$4/5$
$\ge 3,2$	$k^2 V$	$\max(k\rho - \gamma d, 0)$	$1/8$
$\ge 3,3$	$[351/1280 + a_1^2]d$	$(\theta - a_1)d$	$1/10$
$\ge 3,4$	$[\theta(71/63) + (41d/80)^2]d$	$(\theta - 41d/80)d$	$8/25$
$\ge 3,\ge 5$	$(\theta + d^2/4)d$	$(\theta - d/2)d$	$2/5$

Derivation of all masks and moment data.

For every b, the first two propagated masks are

$$m^{-2} X_{b,1} \le q + kB_0 R - \gamma, \qquad m^{-2} X_{b,2} \le (1-\gamma)q + r + kK^2 R - a_1.$$

Thus the $L = 2$ rows follow by AM–GM. Their full vector includes its constant term: with $v_0 = kt_0,\ v_2 = k^2 V$,

$$\|\gamma \mathbf{1} + kR\|_2^2 = v_2 - v_0^2 + v_2^2/4 \le k^2 V. \tag{Q71.8.2}$$

Indeed $\gamma = v_0 - v_2/2 \ge 0$. Its actual K^2-mean is $k\rho - \gamma d$. At $b = 2$ this is nonnegative because $\delta_2 \ge d^2$; its exact vector norm is $\delta_2 - d^2 + \delta_2^2/4$.

At $b = 2$, $\varepsilon_3 = \tau, \varepsilon_4 = d$. Keeping these terms in (M.7) gives

$$X_{2,3} \le B_2(q + r) - s - K^5 \mathbf{1} - \tau,$$

and

$$X_{2,4} \le (1 - \gamma - \tau)(q + r) + \tau r - B_2 s + t + K^6 \mathbf{1} - dB_2.$$

These are the displayed $L = 3, 4$ masks. Their two-row vector norms are $d - 2\delta_2 + \delta_3$ and D_4. The latter obeys $D_4 \le [B_2^2 + 4/27]d$, since $x(1-x)^2 \le 4/27$. Its mean is $\delta_2 - \delta_3 - \tau d \le (\theta\kappa - \tau)d$, using $x(1-x) \le \theta\kappa$. For $L \ge 5$, keep the last three losses, all at least τ:

$$X_{2,L} \le (1 - 2\tau)q + R_L - \tau r - \tau.$$

Consequently $Z = K[\tau + zP_{L-2}]g$. At $L = 5$, $\Re(zP_3) = -x$ and $|P_3| \le 1$, giving $D = A_5 d$. At $L \ge 6$, (M.10) and (M.12) give

$$|\tau + zP_{L-2}|^2 \le \tau^2 + (1 - 2\tau)x + (2\tau - 7/15)x^2 \le A_6(\tau).$$

This polynomial increases on $0 \le x \le \theta$: its derivative is at least $127/720 > 0$.

For $b \ge 3, L = 3$, keep the three actual losses:

$$X_{b,3} \le m^2[V_3 - \gamma r - (d - k\rho) - B_* q].$$

This gives the listed $Z = R_3 - r + a_1 r$. For $L = 4$, retain only $\varepsilon_4 \ge \kappa d$ and $\varepsilon_3 \ge 41d/80$. For $L \ge 5$, retain the last two losses, each at least $d/2$, then enlarge m^2 to $1 - d/2$ after the remaining

bracket is positive. These operations give exactly the last three mask rows. In every one-row case $Z = K[\delta + zP_n]g$,

$$\Re(zP_n) \le 0, \qquad \mathbb{E}Z = -\alpha_L + (1-\delta)d \le (\theta - \delta)d.$$

Use $|P_1|^2 \le 1 - 4x/5$, $|P_2|^2 \le 71/63$, or $|P_n|^2 \le 1$ as appropriate to obtain the three displayed D's. The endpoint $x - 4x^2/5$ is $351/1280$. All M's are nonnegative because $\tau \le 2/9 < \theta\kappa$, $a_1 \le 39d/80 \le 13/80$, and $d/2 < \theta$. All B's are positive: $B_2 \ge 13/18$, $1 - \gamma - \tau \ge 1/2$, $1 - 2\tau \ge 5/9$, $1 - \gamma \ge 17/25$, and $1 - B_* \ge 1/2$.

The two shortest terminal-two connectors.

At $b = 2, L = 2$, let $v = \delta_2/d \le \theta$, $B = B_2$. The complete exponent in (O.2), using $D = \delta_2$ and $\log B \le -\gamma$, is at most $8dF/B^2$, where

$$F = \tfrac{v}{2}B^2 - (1-v)B + \tfrac{3}{4}v.$$

It increases in v and decreases in B, since $vB - (1-v) \le \theta - \kappa < 0$. Hence

$$\mathcal{E}_{2,2} \le \frac{8dF}{B_2^2}, \qquad F \le \tfrac{13}{64}(\tfrac{13}{18})^2 - \tfrac{19}{32}\tfrac{13}{18} + \tfrac{39}{128} = -\tfrac{377}{20736} < -\tfrac{1}{64}. \tag{O.3}$$

In particular $\mathcal{E}_{2,2} \le -d/8$. For $L = 3$, use $D \le d$, $\gamma \ge (51/64)d$, $\tau \ge \kappa d$, and $B_2 \ge 13/18$:

$$\mathcal{E}_{2,3} \le \frac{8d}{B_2^2}\left[-\frac{51}{64}B_2^2 - \kappa B_2 + \frac{3}{4}\right] \le -\frac{653}{864}d \le -\frac{3}{4}d. \tag{Q71.5.2}$$

The bracket decreases in B_2, and its value at $13/18$ is $-653/6912$.

Terminal two at length four.

Put $B = 1 - \gamma - \tau$, $v = (\gamma + \tau)/d$. Using the vector and mean bounds already proved, the exponent is at most $8dF/B^2$, where

$$F = -(v + 1/4)B^2 + (\theta\kappa - \tau/2)B + 3\tau^2/4 + 1/9.$$

Here $v \ge 89/64$, $B \ge 1/2$, $\tau \le 2/9$, so

$$F \le -\tfrac{105}{64}B^2 + \tfrac{247}{1024}B + \tfrac{4}{27} \le -\tfrac{7819}{55296} < -\tfrac{1}{8}.$$

The quadratic decreases on $B \ge 1/2$. Thus

$$\mathcal{E}_{2,4} \le \frac{8dF}{B^2} \le -\frac{d}{B^2} \le -d. \tag{O.4}$$

The entire terminal-two tail.

For $j = 5, 6$, set $(A_0, \sigma) = (\theta, 1)$ or $(\theta - 7\theta^2/15, \kappa)$, respectively, and $B = 1 - 2\tau$. The complete exponent is at most $8dF_j(d, \tau)/B^2$, where

$$F_j = -\frac{2\tau}{d}(1-2\tau)^2 + (1-2\tau)(\theta - \tau - \tau/d) + c_*[A_0 - 2\theta\sigma\tau + \tau^2]. \tag{O.5}$$

Table 45: All coefficients for the two terminal-two tail payments.

(j,e)	D	$D(b_0,\dots,b_n)$
$(5,\ell)$	9437184	$(4608096, 2256333, 190121)$
$(5,u)$	39813120	$(67962645, 42533343, 26218242,$ $15176862, 7537011, 2084065)$
$(6,\ell)$	1509949440	$(942104736, 468288531, 40160486)$
$(6,u)$	1274019840	$(2347612551, 1478434437, 910150830,$ $519866370, 247670913, 54696491)$

Its second τ-derivative is $(20-48\tau)/d+4+2c_*>0$. It therefore suffices to check $\tau=\kappa d$ and $\tau=d-d^2$, throughout $0<d\le 1/3$. With $a_5=1/12, a_6=1/10$, the four polynomials are

$$P_{j,\ell}(x)=-F_j(x/3,\kappa x/3)-a_j,\qquad P_{j,u}(x)=-F_j(x/3,x/3-x^2/9)-a_j. \tag{O.6}$$

Their complete Bernstein coefficients on $[0,1]$ are listed below; the entries are the integer vector divided by D. Lemma A.1 gives $F_5<-1/12, F_6<-1/10$. Consequently the actual complete exponents satisfy

$$\mathcal{E}_{2,5}\le\frac{8dF_5}{B^2}\le-\frac{2d}{3B^2}\le-\frac{2d}{3}, \tag{O.7}$$

$$\mathcal{E}_{2,L}\le\frac{8dF_6}{B^2}\le-\frac{4d}{5B^2}\le-\frac{4d}{5}\qquad(L\ge 6). \tag{O.8}$$

Every longer terminal at connector length two.

The same complete exponent, including its nonzero constant vector, is

$$\mathcal{E}_{b,2}=16\log m+8\log B+\frac{8}{B}[\max(k\rho-\gamma d,0)-ke]+\frac{6k^2V}{B^2},\qquad B=1-\gamma\ge 17/25.$$

If $\rho\le e$, the bracket is nonpositive; otherwise it is at most $k(\rho-e)$. At $b=3$, $\beta_b=d$, $V\le(585/1024)d<(3/5)d$, and $\gamma\ge 7d/10$. Using $16\log m\le -8d$ and $\log B\le-\gamma$, the exponent after division by $8d/B^2$ is at most

$$-\tfrac{17}{10}B^2+\tfrac{3}{4}k^2(585/1024)\le-\tfrac{34}{45}+\tfrac{81}{125}=-\tfrac{121}{1125}<-\tfrac{1}{10}.$$

Thus $\mathcal{E}_{3,2}\le -4d/5$, and $\mathbb{E}Y_{3,2}^8\le 1-d/2$. For $b\ge 4$, $V\le\delta_2$, $\beta_b\ge\kappa d$, and $\gamma\ge(k\kappa-k^2\theta/2)d\ge 25d/64$. At nonpositive end sign this gives

$$\frac{B^2}{8d}\mathcal{E}_{b,2}\le-\tfrac{63}{64}B^2+\tfrac{3}{4}k^2\theta\le-\tfrac{657}{40000}<-\tfrac{1}{64}.$$

At positive end sign, necessarily $b\equiv 2\pmod 4$, $b\ge 6$. The exact $\Delta=\rho-e=\delta_{b/2+1}$ satisfies $\Delta\le\theta\delta_3$. The same finite sums and the G_{b-1} loss in the proof of Lemma M.4 give

$$t_0\ge\tau+\kappa\delta_3,\quad V\le\delta_2-\tfrac{7}{15}\delta_3,\quad \gamma\ge\tfrac{25}{64}d+(k\kappa+7k^2/30)\delta_3.$$

This improvement pays the positive sign before any independent relaxation of moments, because

$$-B^2(k\kappa+7k^2/30)\delta_3+Bk\Delta\le 0,\qquad B(\kappa+7k/30)\ge 6749/12000>\theta.$$

We have therefore proved the complete scalar bound

$$\mathcal{E}_{b,2}\le -d/8\qquad(b\ge 3). \tag{O.9}$$

All remaining longer-terminal rows.

Set $h_0 = 209/384$. For $L = 3$, let $v = B_*/d \in [5/8, 3/2]$, $B = 1 - vd$, and $\delta = a_1 \le 13/80$. The exponent is at most $8dF/B^2$, where

$$F \le -(h_0 + v)B^2 + B(-17/160 - \delta) + c_*(351/1280 + \delta^2).$$

As $c_*\delta < B$, the two δ-terms are nonpositive. The function $(13/24 + v)(1 - v/3)^2$ has at most one interior maximum and endpoint values $2527/3456, 49/96$. Hence

$$(h_0 + v)B^2 \ge 49/96, \qquad c_*351/1280 < 11/20,$$
$$F \le -49/96 - 17/320 + 11/20 = -13/960 < -1/80.$$

Thus

$$\mathcal{E}_{b,3} \le -d/10 \qquad (b \ge 3). \tag{O.10}$$

For $L = 4$, $u = 41/80$, $B = 1 - ud$, and

$$F = -(h_0 + u)B^2 + B(\theta - \kappa - ud) + c_*[\theta(71/63) + u^2d^2].$$

Since $h_0 + u \ge 25/24,\ u \ge 1/2,\ 71/63 \le 181/160$, put $B' = 1 - 33d/64 \le B$ to obtain

$$F \le -\tfrac{25}{24}(B')^2 + B'(-3/16 - d/2) + c_*[\theta(181/160) + (33d/64)^2] = Q(d).$$

Both its linear coefficient $2061/3072$ and its quadratic coefficient $(33/64)/2 + (33/64)^2(c_* - 25/24)$ are positive. Thus $Q(d) \le Q(1/3) = -2543205/62914560 < -1/25$, proving

$$\mathcal{E}_{b,4} \le -8d/25 \qquad (b \ge 3). \tag{O.11}$$

For $L \ge 5$, $B = 1 - d/2$, and the scalar after division by $8d/B^2$ is at most

$$F = -B^2 + B(\theta - 1/2 - d/2) + c_*(\theta + d^2/4) = \theta - \tfrac{3}{2} + c_*\theta + (\tfrac{3}{4} - \tfrac{\theta}{2})d + \tfrac{c_*}{4}d^2.$$

It increases to $F(1/3) = -5225/98304 < -1/20$, so

$$\mathcal{E}_{b,L} \le -2d/5 \qquad (b \ge 3,\ L \ge 5). \tag{O.12}$$

These are all rows in Table 44, proving (Q71.0.2). The corresponding linear bounds are

row	$b = 2, L \ge 3$	$b \ge 3, L = 2$	$b \ge 3, L = 3$	$b \ge 3, L = 4$	$b \ge 3, L \ge 5$
$\mathbb{E}Y^8 \le$	$1 - d/2$	$1 - d/10$	$1 - d/12$	$1 - d/4$	$1 - d/3.$

They follow from $e^{-x} \le (1 + x)^{-1}$ and $d \le 1/3$. For $b = 2, L = 4$, the same argument gives the stronger $1 - 3d/4$.

O.2 The nonunit-inner ordinary eighth moment

Q75.0 Exact new theorem and its original scope

Let the original finite probability space and original real skew kernel satisfy the conventions in part Q75.1. Put

$$W_s = A\bar{f}_{s-1}, \qquad X_{b,L} = T^L(qf_b).$$

The claim is

$$\boxed{E(W_s X_{b,L})^8 \le e^{-3d/20} \le 1 - d/7, \qquad s \ge 3, \quad b \ge 2, \quad L \ge 2.} \tag{Q75.T080.1}$$

In fact the shortest new arm has the stronger bound

$$E(W_3 X_{b,L})^8 \le e^{-d} \qquad (b \ge 2, L \ge 2). \tag{Q75.T080.2}$$

At an actual root with distinct residual arms (s, a), $1 \le s < a$, the negative canonical factor is $W_s \bar{f}_a$. The short arm's first edge points away from the root, and all its remaining edges point toward the root. The long arm points entirely toward the root. At the terminal with residual arms (1, b), the unit arm points toward the terminal and its b-arm points away. The inner connector of length L points from terminal to root. These descriptions are literal directions of every original subdivision edge.

After multiplication by the genuine equal-pair decorations E_r, E_t, the actual input and actual receiving-root output are

$$Y_D = E_r W_s T^L (E_t q f_b), \qquad U_D = T^M(\bar{f}_a Y_D), \qquad M \ge 1. \tag{Q75.0.1}$$

The exact C066 map, valid for every real Y and every $a \ge 2, M \ge 1$, is

$$I_q(T^M(\bar{f}_a Y))^3 \le (1 - d/72) E|Y|^8, \quad I_q(U) = E[q^{-1/3}|U|^{8/3}]. \tag{Q75.0.2}$$

Consequently the new $s \ge 3$ family satisfies

$$\boxed{I_q(U_D)^3 \le (1 - d/72) e^{-3d/20} \quad (3 \le s < a,\ b \ge 2, L \ge 2, M \ge 1).} \tag{Q75.T080.4}$$

The two connector lengths are independent. Section 12 gives the exact arbitrary-joint-law and original P5 applications, with no assumption of independent branch labels.

This module does not cover unit inner L for $s \ge 2$, arbitrary terminal tables, multiple unequal residual pairs at the root, or every central role. Other work may supply separate directions in some of those cases; none is imported here.

Q75.1 Original objects, positive convolution, and row bounds

Delete zero-mass atoms from the finite probability space and use its expectation and L2 inner product. Set

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad K(x,y) = -K(y,x), \quad |K(x,y)| \le 1,$$

$$g = K1,\ p = 1 + g,\ q = 1 - g,\ r = Kg,\ s_0 = Kr,\ t = Ks_0, \quad \delta_j = \|K^j \mathbf{1}\|_2^2, \quad d = \delta_1,$$

$$f_n = A^n \mathbf{1}, \quad \bar{f}_n = T^n \mathbf{1}, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The symbol s is an arm length; $s_0 = K^3\, 1$ is a function. This avoids confusing a scalar length with the third score.

The C014 calculus supplies

$$\|K\|^2 \le \theta, \quad 0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta \delta_j,$$

$$\kappa d \le \tau \le d - d^2 \le 2/9, \quad Er = -d, \quad Es_0 = 0, \quad Et = \delta_2, \quad Egr = 0, \quad Egs_0 = -\delta_2, \quad Ert = -\delta_3. \tag{Q75.1.1}$$

All actual path and tree messages are positive on the positive-mass atoms. In particular p, $q \ge \mu(x) > 0$. A and T preserve order.

Define the finite coefficients and finite polynomials

$$\beta_1 = \beta_2 = 0, \qquad \beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1}\delta_i \quad (j \ge 3), \qquad G_n(z) = \sum_{i=0}^{n-1} z^i.$$

The original positive finite convolution is

$$\sum_{j=0}^{n} K^j \mathbf{1} = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad \tau \le \beta_j \le d \quad (j \ge 3). \tag{Q75.1.2}$$

The same finite identity holds after K is replaced everywhere by -K.

For every original vector v, row Cauchy gives $|Kv(x)| \le \|v\|_2$. The useful two-row estimate is

$$|K^2 v(x)|^2 \le \|v\|_2^2/3. \tag{Q75.1.3}$$

This is (M.15); its proof uses conjugated legal skew kernels and the original row measure.

At $d = 0, g = 0$ and every actual tree message here is one by leaf integration, even when K is nonzero. All theorem boundaries hold directly. Subsequent divisions by d are restricted to $d > 0$.

Q75.2 The exact scalar input and its positive majorant interpretation

Use the positive-mean exponential inequality (M.16). If $|Z| \le U$ and $EZ^2 \le hU^2, 0 \le h < 1$, then, for $\lambda \ge 0$,

$$Ee^{\lambda Z} \le \exp\{\lambda \max(EZ, 0) + \lambda^2(1+h)^2U^2/8\}. \tag{Q75.2.1}$$

Here all moments belong to the same original finite variable. A nonnegative upper bound $M \ge EZ$ may replace max(EZ, 0).

The full proof, allowing either sign of the actual mean, is Lemma M.5.

At $\lambda = 8/B$ the two forms needed below are:

$$Z = KV,\ \|V\|^2 \le D \quad \Longrightarrow \quad \log Ee^{8Z/B} \le 8M/B + 8c_*D/B^2, \quad c_* = (1+\theta)^2 = 2025/1024; \tag{Q75.2.2}$$

$$Z = K^2V,\ \|V\|^2 \le D \quad \Longrightarrow \quad \log Ee^{8Z/B} \le 8M/B + 6D/B^2. \tag{Q75.2.3}$$

For (Q75.2.2), the range squared is D and the raw second is at most theta D. For (Q75.2.3), (Q75.1.3) gives range squared D/3 and $\theta^2 < 1/6$ gives raw second at most D/6, so $h = 1/2$ is legitimate.

Apply AMGM to the following positive factors. With $W_2 = Aq = p - r$ and $X = T^L(qf_b)$, there is an actual positive upper bound

$$0 < X \le C[Bq + Br + Z - a], \qquad C, B > 0. \tag{Q75.2.4}$$

Thus AMGM between W_2 and the bracket divided by B gives

$$W_2 X \le CB\left(1 + \frac{Z-a}{2B}\right)^2. \tag{Q75.2.5}$$

The base is half a sum of nonnegative quantities. Consequently its eighth power is at most C^8B^8 exp(8(Z-a)/B), including at a zero base.

For a specified nonnegative mean upper M and vector bound D, call the right-hand logarithmic estimate

$$\mathcal{E} = 8\log C + 8\log B - 8a/B + 8M/B + \begin{cases} 6D/B^2, & Z = K^2V, \\ 8c_*D/B^2, & Z = KV. \end{cases} \tag{Q75.2.6}$$

Section O.1 proves every displayed actual mask and its complete exponent, including the $b = 2, L = 2$ row. These complete expressions are used below for the longer root factors.

Q75.3 The compatible scalar table

Q75.3.1 Exact parameters for every b, L

For $b = 2$ set $\gamma = d - \delta_2/2$ and $B_2 = 1 - \gamma$. The same original terminal has $TF \le q - \gamma - s_0$ and $T^2F \le B_2 q + r + t$. The C071 finite masks yield:

Table 46: Finite masks and paid scalar bounds for $b = 2$.

L	C	B	a	Z and its row form	Paid scalar upper
$L = 2$	1;	B_2;	0;	$K^2(\gamma 1 + r)$;	$\le$-d/8 by part Q75.3.2.
$L = 3$	1;	B_2;	τ;	$-K^2(g + s_0)$;	$\le -3d/4$, by (Q71.5.2).
$L = 4$	1;	$1 - \gamma - \tau$;	$d(1-\gamma)$;	$K^2[\tau 1 - (1-\gamma)g + r + t]$;	$\le -d$, by (O.4).
$L = 5$	1;	$1 - 2\tau$;	τ;	$K[\tau + zP_3(z)]g$, $z = -K$;	$\le -2d/3$, by (O.7).
$L \ge 6$	1;	$1 - 2\tau$;	τ;	$K[\tau + zP_{L-2}(z)]g$;	$\le -4d/5$, by (O.8).

Here $P_n(z) = G_n(z) + z^{n+1}$. For $L = 2$ one may take $D = \delta_2$ and $M = \delta_2 - \gamma d \ge 0$. For $L = 3$ the exact vector norm is $d - 2\delta_2 + \delta_3$ and $M = 0$. For $L = 4$ its exact norm is $(1-\gamma)^2 d + \delta_2 - 2\delta_3 + \delta_4 - \tau^2$, and the used upper mean is $(\theta\,\kappa - \tau)d \ge 0$. The scalar proof uses their stated upper bounds. For $L \ge 5$ the old one-row vector is Qg with $Q = \delta + zP_n, \delta = \tau$. Its old bound $D = A(\tau)d$ and mean $M = (\theta - \tau)d$ are those in Table 44, with

$$A(\tau) = A_0 - 2\theta\sigma\tau + \tau^2, \quad (A_0, \sigma) = \begin{cases} (\theta, 1), & L = 5, \\ (\theta - 7\theta^2/15, \kappa), & L \ge 6. \end{cases} \tag{Q75.3.1}$$

The convex endpoint payment there covers the full region kappa $d \le \tau \le d - d^2$, not a finite sample of moments.

For every $b \ge 3$ use the exact C058/C071 same-direction finite toolkit:

$$R = \sum_{j=2}^{b} K^j \mathbf{1} = KG_{b-1}(K)g, \quad m_b = 1 - \beta_b/2, \quad k = 1/m_b, \quad t_0 = \beta_{b+1},$$

$$e = d - \beta_{b+2}, \quad V = ER^2, \quad \rho = E(rR), \quad \gamma = kt_0 - k^2V/2, \quad a_1 = ke, \quad B_* = \gamma + a_1. \tag{Q75.3.2}$$

In particular $ER = -t_0, E(gR) = -e \le 0, 1 \le k \le 6/5, 3d/10 \le \gamma \le 6d/5, 5d/8 \le B_* \le 3d/2$, $0 \le a_1 \le 39d/80$, and $m_b^2 \le 1 - 209d/384$. Define

$$P_n(z) = G_n(z) + kz^{n+1}G_{b-1}(-z), \quad z = -K. \tag{Q75.3.3}$$

Every actual upper mask is a finite positive convolution. Specifically, with $V_L = \sum_{j=0}^{L} z^j 1 + kz^L$ R, its upper bracket before discarding any nonnegative loss is

$$X_{b,L} \le m_b^2 \left[V_L - \gamma \bar{f}_{L-1} - a_1 \bar{f}_{L-2} - \sum_{j=3}^{L} \varepsilon_j \bar{f}_{L-j} \right], \tag{Q75.3.4}$$

where $\epsilon_3 = d - k\rho \ge 41d/80$ and $\epsilon_j \ge \tau$ for $j \ge 4$. Only existing indices are used. These exact masks produce (Q75.2.4) with:

Table 47: Finite masks and paid scalar bounds for $b \ge 3$.

L	C	B	a	Z and its row form	Paid scalar upper
$L = 2$	m_b^2;	$1 - \gamma$;	a_1;	$K^2(\gamma 1 + kR)$;	$\le -d/8$, (O.9).
$L = 3$	m_b^2;	$1 - B_*$;	$d - k\rho$;	$K[\delta + zP_1(z)]g$, $\delta = a_1$;	$\le -d/10$, (O.10).
$L = 4$	m_b^2;	$1 - 41d/80$;	κd;	$K[\delta + zP_2(z)]g$, $\delta = 41d/80$;	$\le -8d/25$, (O.11).
$L \ge 5$	$1 - d/2$;	$1 - d/2$;	$d/2$;	$K[\delta + zP_{L-2}(z)]g$, $\delta = d/2$;	$\le -2d/5$, (O.12).

In the last row m_b^2 is replaced by the larger positive factor 1-d/2 only after the actual upper bracket is known to be nonnegative. For $L = 2$ the vector squared norm is at most $k^2 V$, its actual mean is k rho-gamma d, and $M = max(k$ rho-gamma d, 0). In the three other rows $M = (\theta - \delta)d \ge 0$ and the old squared vector bounds D/d are, respectively,

$$351/1280 + \delta^2, \qquad \theta(71/63) + \delta^2, \qquad \theta + d^2/4. \tag{Q75.3.5}$$

The nonpositive real cross term in those masks is retained in the original scalar bounds. The finite end sign in the $L = 2$ proof can be positive for $b = 2$ mod 4; C071 pays it with its same δ_3 improvement. We use that complete paid formula, with its actual b, m_b, R, not a sign-free relaxation of it.

Every B in the two tables is in (0, 1]. The necessary explicit lower bounds are $B_2 \ge 13/18$, $1 - \gamma - \tau \ge 1/2, 1 - 2\tau \ge 5/9, 1 - \gamma \ge 17/25$ in (Q75.3.2), and $1 - B_* \ge 1/2$.

Q75.3.2 The terminal-two, length-two payment

For this actual row $B = B_2$, $Z = K^2(\gamma \mathbf{1} + r)$, $D = \delta_2$, and $M = \delta_2 - \gamma d$. Equation (O.3) proves its complete scalar upper bound $\mathcal{E} \le -d/8$. In particular every row in the two tables has $\mathcal{E} \le -d/10$.

Q75.4 Finite coefficients

For $z = i\sqrt{x}$, $0 \le x \le \theta$, the finite coordinates are

$$G_N(z) = A_N(x) + zB_N(x), \quad 1 - x \le A_N, B_N \le 1 \quad (N \ge 2). \tag{Q75.4.1}$$

At $N = 1$, $(A_N, B_N) = (1, 0)$. For $P_n = G_n + z^{n+1}$, or the actual polynomial (Q75.3.3), write $P_n = A_P + zB_P$. Equation (M.11) gives

$$1 - x \le A_P, B_P \le 1 \quad (n \ge 3). \tag{Q75.4.2}$$

Its short rows are

$$\begin{aligned} n = 1: &\quad A_P = 1 - kxA_N \ge 1 - 6x/5, \quad 0 \le B_P = kxB_N \le 6x/5; \\ n = 2: &\quad A_P = 1 - kx^2 B_N \ge 1 - x, \quad 1 - 6x/5 \le B_P = 1 - kxA_N \le 1. \end{aligned} \tag{Q75.4.3}$$

Thus $\Re P_n \ge 1/2$ at every index used below. The same finite spectral calculus, Lemma M.1, also gives

$$\|G_m(-K)\|^2 \le A_m, \quad A_2 = 1 + \theta, \quad A_m = 1 \ (m \ge 3). \tag{Q75.4.4}$$

The proof in Lemma M.4 includes every finite endpoint and both $N = 1$ and $N \ge 2$.

Q75.5 The actual shortest-three root

The exact root identity is

$$W_3 = A\bar{f}_2 = p - r + s_0 - d = W_2 + s_0 - d. \tag{Q75.5.1}$$

In each positive factorization (Q75.2.4), AMGM with W_3 therefore gives

$$W_3 X \le CB\left(1 + \frac{Z + Bs_0 - a - Bd}{2B}\right)^2. \tag{Q75.5.2}$$

The base remains nonnegative because the root is actual and the terminal upper is positive. The new fluctuation has the same mean, and its deterministic loss adds -8d to (Q75.2.6).

For $b = 2, L = 2$, the new two-row vector is gamma1+Bg+r. Its squared norm is $B^2 d + \delta_2 - d^2 + \delta_2^2/4 \le B^2 d + \delta_2$. Thus its cost over the paid expression of part Q75.3.2 increases by at most 6d. The added -8d leaves a bound at most $-d/8 - 2d \le -d$. For $b \ge 3, L = 2$, the new two-row vector is gamma1+Bg+kR. Since $E(gR) = -e \le 0$ and $Eg = 0$, its squared norm is at most $k^2 V + B^2 d$. The same argument gives at most $-d/8 - 2d \le -d$. This reuses the explicit scalar formula with its positive mean, not just the old eighth bound.

For $b = 2, L = 3$, the new vector is -(gamma $g + s_0$). On the original skew spectrum,

$$(\gamma - x)^2 \le (1 - x)^2, \quad (1 - x)^2 - (\gamma - x)^2 = (1 - \gamma)(1 + \gamma - 2x) \ge 0.$$

The two-row norm bound still applies. At $b = 2, L = 4$ the new vector is tau1-tau g+r+t. Orthogonality of g to 1, r, t and $\tau < 1 - \gamma$ show its squared norm is no larger than the old one. Means are unchanged in both cases, and their added -8d is more than required.

In every remaining table row, $Z = K[\delta + zP_n(z)]$g with $z = -K$. The new vector is $[\delta + z(P_n - B)]$g. For every original frequency,

$$|\delta + z(P_n - B)|^2 - |\delta + zP_n|^2 = Bx[B - 2\Re P_n]. \tag{Q75.5.3}$$

This is nonpositive by part Q75.4 and $0 < B \le 1$. Thus both the original row range and the theta raw-second estimate use no larger vector norm. Again the mean is unchanged, and the exponent improves by -8d. The complete old table therefore proves (Q75.T080.2) for all $b \ge 2, L \ge 2$. No step asserted $W_3 \le W_2$ pointwise; that assertion would be false.

Q75.6 Every longer shortest arm: an actual loss and a centered perturbation

Fix the root arm length $s \geq 4$ and put

$$m = s - 2 \geq 2, \quad c = 1 - \tau \geq 7/9, \quad U = K^2 G_m(-K)g, \quad \beta = \beta_s, \quad H = G_m(-K)g - \tau 1. \tag{Q75.6.1}$$

The letter m here is a root-arm polynomial length; it is not m_b from the terminal toolkit. Their values are independent.

Apply positive A to the negative-sign version of (Q75.1.2) for $\bar{f}_{s-1}$. The left side becomes $p - r + U - \beta_s$, while the last convolution term on the right is $\beta_{s-1}p$. Every other term is nonnegative. Since $s - 1 \geq 3$, retaining $\beta_{s-1}p \geq \tau$ p gives

$$0 < W_s \leq cp - r + U - \beta. \tag{Q75.6.2}$$

The upper side is nonnegative because it is the actual W_s plus the remaining nonnegative convolution terms. In particular, it is not an arbitrarily signed formal series.

Combine (Q75.6.2) with the positive terminal upper (Q75.2.4). AMGM between the root upper divided by c and the terminal upper divided by CB gives, with $h = B/c$,

$$W_s X \leq CBc \left(1 + \frac{Z + hK^2 H - a - h\beta}{2B}\right)^2. \tag{Q75.6.3}$$

To check the algebra, the new fluctuation is Z+(B/c)(U-tau r), and $K^2 H = U - \tau$ r. The deterministic addition is -(B/c)beta; the constant prefactor is CBc. Both AMGM factors are nonnegative. This scaling is needed: it retains the actual loss tau p without introducing a fictitious root factor.

Q75.6.1 The same original perturbation has a nonpositive mean

Let beta'=E $G_m(-K)g$. The finite skew identities give

$$\beta' = \beta_{m+1}, \quad \tau \leq \beta' \leq d, \quad 0 \leq EH = \beta' - \tau \leq \delta_2,$$

$$\|H\|^2 = \|G_m(-K)g\|^2 - 2\tau\beta' + \tau^2 \leq A_m d - \tau^2 \leq A_m d. \tag{Q75.6.4}$$

To control its K^2 mean without separating incompatible moment extrema, apply the original norm bound to the centered vector r+d1:

$$\delta_3 - 2d\delta_2 + d^3 = \|K(r + \delta_1)\|^2 \leq \theta(\delta_2 - d^2).$$

It follows that

$$\delta_2 - \delta_3 - \tau d \geq (\kappa - d)(\delta_2 - d^2) \geq 0. \tag{Q75.6.5}$$

For every $m \geq 2$, the finite alternating mean of U begins -δ_2 and is at most -$\delta_2 + \delta_3$. Hence the same H satisfies

$$EK^2 H = EU + \tau d \leq 0. \tag{Q75.6.6}$$

Thus adding $hK^2 H$ never increases any nonnegative mean upper M used in the old scalar table. This is also valid when an actual mean was already negative.

Q75.6.2 A nonpositive covariance for every terminal at $L = 2$

The old two-row vector for $L = 2$ is $V_0 = gamma1 + kR$. For $b \geq 3$ its parameters are (Q75.3.2). For $b = 2$ use $R = r, k = 1, t_0 = d, V = \delta_2$ and the same identity $\gamma = kt_0 - k^2V/2$. In both cases

$$EV_0 = -k^2V/2 \leq 0.$$

We prove

$$\mathrm{Cov}(R, G_m(-K)g) \leq 0, \qquad E(V_0H) \leq 0. \tag{Q75.6.7}$$

The second follows from the first and $EH \geq 0$.

For $b \geq 3$ put $N = b - 1 \geq 2$, and at an original skew rotation block write

$$G_N(z) = A + zB_c, \qquad G_m(z) = C + zD, \qquad z = i\sqrt{x}.$$

All four real coefficients belong to [1-x, 1], hence to [kappa, 1]. Let ν_1 be the spectral measure of the original vector 1 for -K^2. It has mass one, including any zero block. The corresponding measure of $g = K1$ is x ν_1. This is spectral bookkeeping in the original finite Hilbert space; it is not a changed host or a new row bound. Skew adjunction gives the exact three quantities

$$t_0 = \int xA\, d\nu_1, \qquad \beta' = \int xD\, d\nu_1, \quad E[RG_m(-K)g] = -\int x^2(AD + B_cC)\, d\nu_1. \tag{Q75.6.8}$$

Put $J_0 = \int x^2AD\, d\nu_1$. Pointwise $B_cC \geq \kappa^2 AD$, and A/D lies in [kappa, 1/kappa]. By $2uv \leq u^2 + v^2$ and Cauchy with the probability measure ν_1,

$$t_0\beta' \leq \tfrac{1}{2}(t_0^2 + \beta'^2) \leq \tfrac{1}{2}\int x^2(A^2 + D^2)\, d\nu_1 \leq \frac{\kappa + \kappa^{-1}}{2} J_0 \leq (1 + \kappa^2)J_0. \tag{Q75.6.9}$$

The final comparison is precisely $2\kappa \geq 1$. Combining (Q75.6.8)-(Q75.6.9) proves the first inequality of (Q75.6.7).

Here the covariance in (Q75.6.7) equals

$$-\mathrm{Cov}_{\nu_1}(x, xD(x)).$$

The function xD(x) is a finite alternating sum $x - x^2 + x^3 - \ldots$. It is increasing on [0, theta]. Indeed its derivative begins $1 - 2x > 0$; each subsequent positive/negative pair $(2j+1)x^{2j} - (2j+2)x^{2j+1}$, $j \geq 1$, is nonnegative, and an unpaired last positive term is nonnegative. For the constant $D = 1$ the assertion is immediate. The finite double-integral identity

$$2\,\mathrm{Cov}_{\nu_1}(u, v) = \iint [u(x) - u(y)][v(x) - v(y)]\, d\nu_1(x) d\nu_1(y)$$

therefore proves the needed sign. It includes the zero block.

Finally, for both terminal ranges,

$$E(V_0H) = k\,\mathrm{Cov}(R, G_m(-K)g) + (EV_0)(EH) \leq 0.$$

Thus the new two-row vector $V_0 + hH$ obeys

$$\|V_0 + hH\|^2 \leq \|V_0\|^2 + h^2A_md. \tag{Q75.6.10}$$

The old relaxed norm D from part Q75.3 may replace $\|V_0\|^2$ on the right. The underlying vector and its cross term remain actual.

Q75.6.3 The other two-row cases $b = 2, L = 3$, 4

For $L = 3, V_0 = -(g + s_0), EV_0 = 0$. Its product with $G_m(-K)g$ has spectral real part -(1-x)C, so $E(V_0H) \le 0$.

For $L = 4, V_0 = tau1 - (1-\gamma)g + r + t$ and $EV_0 = 0$. The product with H, using the original spectral measure ν_g, is exactly

$$E(V_0H) = \tau\beta' - (1-\gamma)\int C\,d\nu_g - \int x(1-x)D\,d\nu_g \le [\tau - \kappa(1-\gamma)]d \le 0. \qquad \text{(Q75.6.11)}$$

For the last sign use $\tau \le 2/9$ and $\kappa(1-\gamma) \ge \kappa(13/18) = 247/576 > 2/9$. The same bound (Q75.6.10) therefore holds in every old two-row case.

Q75.6.4 Every remaining one-row case

The old vector is $V_0 = Q(-K)g$ with $Q = \delta + zP_n(z)$. The added vector in the one-row representation is hKH, since $Z + hK^2H = K(V_0 + hKH)$. At $z = -K$ we have

$$KH = -(\tau + zG_m(z))g.$$

We prove its cross with V_0 is nonpositive at every original frequency. Write $G_m = C + zD$ and $P_n = A_P + zB_P$ as in part Q75.4. Direct multiplication gives

$$\Re\{Q\overline{\tau + zG_m}\} = \delta\tau + x[\Re(P_n\overline{G_m}) - \delta D - \tau B_P]. \qquad \text{(Q75.6.12)}$$

Here $\delta \ge 0$. Every table row with $n \ge 3$ has $\delta \le 2/9$. By (Q75.4.1)-(Q75.4.2) the square bracket is at least

$$(1+x)(1-x)^2 - \delta - \tau \ge 16245/32768 - 4/9 > 0. \qquad \text{(Q75.6.13)}$$

The function $(1+x)(1-x)^2$ decreases on [0, theta]; its endpoint is $(45/32)(19/32)^2$. The last rational comparison is $146205 > 131072$. This covers every $b = 2, L \ge 5$ and $b \ge 3, L \ge 5$.

For $b \ge 3, L = 4, n = 2$ and $\delta = 41d/80 \le 2/9$. Use (Q75.4.3) to bound the square bracket by

$$(1-x)(1-6x^2/5) - \delta - \tau \ge 78014/163840 - 4/9 > 0. \qquad \text{(Q75.6.14)}$$

The displayed frequency function decreases throughout the interval: its derivative is $-1 - 12x/5 + 18x^2/5 < 0$ for $0 \le x \le \theta$. At theta it has the stated value; the final comparison is $702126 > 655360$.

For $b \ge 3, L = 3, n = 1$ and $\delta = a_1 \le 13/80$. Again (Q75.4.3) gives

$$\Re(P_n\overline{G_m}) - \delta D - \tau B_P \ge (1-6x/5)(1-x) - \delta - (6/5)\tau x \ge 779/2560 - 13/80 - 13/120 = 257/7680 > 0. \qquad \text{(Q75.6.15)}$$

Both negative replacements are legitimate since $D \le 1$ and $B_P \le 6x/5$. The first product decreases on [0, theta].

These are the full possible n-ranges, including $n = 1, 2$. Equations (Q75.6.12)-(Q75.6.15) show $E(V_0KH) \le 0$. Therefore

$$\|V_0 + hKH\|^2 \le \|V_0\|^2 + h^2\|KH\|^2 \le \|V_0\|^2 + h^2\theta A_m d. \qquad \text{(Q75.6.16)}$$

This norm bound uses $\|H\|^2$ from (Q75.6.4). Row Cauchy and the spectral theta bound apply to this vector.

Q75.7 Payment of every longer root, on the whole continuous domain

All old mean uppers stay valid by (Q75.6.6). In a two-row case, (Q75.6.10) adds at most $6h^2A_md/B^2 = 6A_md/c^2$ to its exponential cost. In a one-row case, (Q75.6.16) adds at most $8c_*\theta A_md/c^2$. Since $6 < 8c_*\theta = 26325/4096$, both are paid by the common upper increment

$$\Delta \le 8\log c - 8\beta/c + 8c_*\theta A_m d/c^2. \tag{Q75.7.1}$$

The first two terms come from the actual factor c and actual loss beta in (Q75.6.3).

For $m \ge 3$, equivalently $s \ge 5, A_m = 1$ and $\beta \ge \tau$. Using $\log c \le -\tau, \tau/d \ge \kappa$ and $c \ge 7/9$, divide by $8d/c^2$ to get

$$\frac{c^2\Delta}{8d} \le -\kappa(c^2+c) + c_*\theta \le -\frac{133}{162} + \frac{26325}{32768} < -\frac{1}{64}. \tag{Q75.7.2}$$

For the last comparison, $133/162 - 26837/32768$ equals $10550/5308416 > 0$. Thus $\Delta \le -d/8$ in this range.

For $m = 2$, equivalently $s = 4$, the actual β_4 is d and $A_m = 1+\theta$. Consequently

$$\frac{c^2\Delta}{8d} \le -\kappa c^2 - c + c_*\theta(1+\theta) \le -\frac{2947}{2592} + \frac{1184625}{1048576} < -\frac{1}{160}. \tag{Q75.7.3}$$

Indeed $1184625/1048576 < 113/100$ and $2947/2592 - 113/100 = 451/64800 > 1/160$. These rational comparisons are by hand; no continuous-domain or numerical computation is implicit. Thus in every $s \ge 4$ case $\Delta \le -d/20$.

The complete old table, including part Q75.3.2, has scalar exponent at most -d/10. Adding the fully paid increment proves

$$E(W_sX_{b,L})^8 \le e^{-3d/20} \qquad (s \ge 4, b \ge 2, L \ge 2).$$

part Q75.5 supplies $s = 3$. Finally

$$e^{-3d/20} \le \frac{1}{1+3d/20} = 1 - \frac{3d/20}{1+3d/20} \le 1 - d/7 \qquad (0 \le d \le 1/3).$$

This proves (Q75.T080.1) in every integer and continuous parameter, with no exceptional intermediate d interval.

O.3 Unit receivers with negative root pairs

Let K be a real skew kernel with $|K| \le 1$ on the given finite probability space, after removal of zero-mass atoms, and put $Jv = (\mathbb{E}v)1$. We use $A = J + K, T = J - K, g = K1, p = 1 + g, q = 1 - g$, $f_n = A^n 1$, $\bar{f}_n = T^n 1$, $h_n = Tf_{n-1}$, and $W_n = A\bar{f}_{n-1}$. The spectral notation is

$$d = \|g\|^2, \qquad \delta_j = \|K^j 1\|^2, \qquad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

All expectations and norms use the given finite probability space. The finite spectral calculus gives

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j, \quad |KV(x)| \le \|V\|.$$

Thus a spectral bound for a polynomial at $z = i\sqrt{x}$, $0 \le x \le \theta$, integrates against the spectral measure of g, of mass d and moments $\int x^j\, d\nu = \delta_{j+1}$.

The two directed interfaces.

Family A has residual order

$$2 \le s < a < u_1 < v_1 < \cdots < u_r < v_r, \qquad r \ge 1. \tag{Q78.0.1}$$

Its s-arm is inward, its a-arm outward, and each later pair has message $W_{u_i}\bar{f}_{v_i}$. Its terminal is $p\bar{f}_b$, with a unit inner edge from the terminal to the root. Thus

$$P^- = \prod_{i=1}^{r} W_{u_i}\bar{f}_{v_i}. \tag{Q78.0.2}$$

$$J_A = f_a P^- T(p\bar{f}_b). \tag{Q78.0.3}$$

Proposition M.39 gives

$$\mathbb{E}J_A^8 \le e^{-5d}. \tag{Q78.0.4}$$

$$I_q(T^M[\bar{f}_s J_A])^3 \le (1 - d/72)e^{-5d} \quad (M \ge 1). \tag{Q78.0.5}$$

Indeed its post parameter is r, its ordering requires $a < u_1$, and its exponent is at least $2005/192 > 5$.

Family B instead has

$$1 < t < u_1 < v_1 < \cdots < u_r < v_r, \qquad r \ge 1. \tag{Q78.0.6}$$

The root unit arm is outward, the reserved t-arm inward, and the terminal has its unit arm inward and its b-arm outward. Every inner edge points from the terminal to the root. Its ordinary input and outer output are

$$J_B = pP^- T^L(qf_b), \qquad I_q(T^M[\bar{f}_t J_B])^3 \le (1 - d/72)\mathbb{E}J_B^8, \quad b \ge 2, \quad L, M \ge 1. \tag{Q78.0.7}$$

The last inequality is the all-real multiplier bound of Section M.16; the reserved arm is multiplied before the outer propagation. The proposition below proves

$$\mathbb{E}J_B^8 \le e^{-25d/24 - 24(r-1)d} \le 1 - d/12000. \tag{Q78.0.8}$$

Consequently the former outer deficit is preserved:

$$I_q(T^M[\bar{f}_t J_B])^3 \le (1 - d/72)(1 - d/12000). \tag{Q78.0.9}$$

For the center-reference application in Section O.27, the excluded t-arm is attached outward as f_t. The ordinary input J_B is unchanged. If $u_1 = 3$, the displayed order forces $t = 2$.

Proposition O.1 (Negative root pairs at a unit receiver)**.** *Let $r \ge 1$, $3 \le u_1 < v_1 < \cdots < u_r < v_r$, $b \ge 2$, and $L \ge 1$ be integers. For the fixed directed message*

$$Y = p\left(\prod_{i=1}^{r} W_{u_i}\bar{f}_{v_i}\right) T^L(qf_b) \tag{O.13}$$

one has, on the entire host domain $0 \le d \le 1/3$,

$$\mathbb{E}Y^8 \le e^{-25d/24 - 24(r-1)d}, \qquad \mathbb{E}Y^4 \le e^{-705d/128}. \tag{O.14}$$

In addition, the proof supplies a pointwise certificate

$$0 \le Y \le e^{a + KZ}, \qquad \|Z\|^2 \le 9d/4, \qquad 4a + 4\mathbb{E}(KZ) + (17/4)\|Z\|^2 \le -705d/128. \tag{O.15}$$

The vector Z here is the full, uncentered vector specified below. The length b and the root arm lengths may be interleaved. Equal-arm factors can be restored by positivity.

Proof. At $d = 0$, successive leaf integration gives every tree message the value one. Hence suppose $d > 0$, and put $\zeta = \delta_2/d \le \theta$.

A square and its signed mean.

For any real vector V, the two signed estimates (M.43), applied to $-V$ since their operator is $-K$, state that

$$\mathbb{E}e^{\nu KV} \le \exp\{\nu\mathbb{E}(KV) + (17/64)\nu^2\|V\|^2\}, \qquad \nu \in \{4, 8\}. \tag{O.16}$$

If $0 \le H \le C(c + N/2)^2$, $N = KV$, the square base is nonnegative, and $P^- \le e^{S-\Gamma}$, $S = KU$, then

$$P^-H \le C\exp\{2(c-1) - \Gamma + K(V+U)\}. \tag{O.17}$$

This follows from $x^2 \le e^{2(x-1)}$ for $x \ge 0$. In particular, for $\nu = 4$ and $\nu = 8$,

$$\log \mathbb{E}(P^-H)^\nu \le \nu \log C + 2\nu(c-1) - \nu\Gamma + \nu\mathbb{E}(N+S) + \frac{17\nu^2}{64}\|V+U\|^2. \tag{O.18}$$

The same vector $V + U$ occurs in both moments and in the pointwise certificate.

The finite root-post mask.

The pair bound from Section O.15, including $u = 4$, is

$$h_u f_v \le \exp\left\{\sum_{j=u}^{v-1} K^j g - 3d\right\} \quad (4 \le u < v). \tag{Q78.1.6}$$

For $u = 3$, finite positive convolution gives $f_v - f_3 \le \sum_{j=3}^{v-1} K^j g + \delta_2$. Together with $h_3 + f_3 = 2 - 2d$, AM–GM therefore gives

$$h_3 f_v \le \exp\left\{\sum_{j=3}^{v-1} K^j g - (2d - \delta_2)\right\}. \tag{Q78.1.7}$$

After full reversal these inequalities give

$$\begin{gathered} P^- \le e^{S-\Gamma}, \qquad S = K\mathcal{U}(-K)g, \qquad \mathcal{U}(z) = \sum_{i=1}^{r}\sum_{j=u_i}^{v_i-1} z^{j-1}, \\ \Gamma = \begin{cases} 3rd, & u_1 \ge 4, \\ 2d - \delta_2 + 3(r-1)d, & u_1 = 3. \end{cases} \end{gathered} \tag{O.19}$$

The intervals are disjoint, so $\mathcal{U}$ is a finite zero-one mask. For any such mask Q,

$$|Q(i\sqrt{x})|^2 \le \frac{1+x}{(1-x^2)^2} \le \frac{32768}{16245} < \frac{100}{49} \qquad (0 \le x \le \theta).$$

It follows that

$$\|\mathcal{U}(-K)g\| \le (3/8)\sqrt{d}, \qquad \mathbb{E}S \le d/12 \quad (u_1 \ge 4). \tag{O.20}$$

For the norm, factor out z^3 and use $(10/7)\theta^{3/2} < 585/1568 < 3/8$. Only the mean powers $j = 3, 7, 11, \ldots$ are positive. If $u_1 \ge 4$, their total is at most $\delta_4/(1-\theta^2) \le 2197d/27360 < d/12$. If $u_1 = 3$, retain the compulsory z^2 and write $\mathcal{U}(z) = u - zv$, $z = i\sqrt{x}$. The four residue classes give

$$-\frac{6}{5}x \le u \le -\frac{1}{2}x, \qquad -\frac{1}{2}x \le v \le \frac{6}{5}x, \qquad \mathbb{E}S \le \delta_2 + d/12. \tag{O.21}$$

For example $-x/(1-x^2) \le u \le -x + x^2/(1-x^2)$; the other two bounds follow from the odd classes, using $1/(1-\theta^2) < 6/5$ and $\theta/(1-\theta^2) < 1/2$. For a single pair starting at three the finite alternating sums give

$$\mathcal{U} = z^2 G_{v_1-3}(z), \qquad -x \le u \le -x(1-x), \quad 0 \le v \le x, \quad \mathbb{E}S \le \delta_2. \tag{O.22}$$

Nine actual square envelopes.

Put $H = pT^L(qf_b)$ and $z = -K$ in vector polynomials. For $b \ge 3$, use the finite short-one packet (M.6)–(M.13):

$$R = \sum_{j=2}^{b} K^j 1, \quad m = 1 - \beta_b/2, \quad k = 1/m, \quad t_0 = \beta_{b+1}, \quad e = d - \beta_{b+2},$$
$$V_R = \mathbb{E}R^2, \quad \rho = \mathbb{E}[(K^2 1)R], \quad \gamma = kt_0 - k^2 V_R/2, \quad a_1 = ke, \quad B_* = \gamma + a_1,$$
$$P_n(z) = G_n(z) + kz^{n+1} G_{b-1}(-z), \qquad B = 1 - \gamma.$$

In particular

$$3d/10 \le \gamma \le 6d/5, \quad B \ge 17/25,$$
$$B_* \ge 5d/8, \quad B_* \ge d - 29\delta_2/32,$$
$$m \ge 1 - d/2, \quad 1 \le k \le 6/5, \quad 0 \le \rho \le \delta_2,$$
$$\varepsilon_3 = d - k\rho, \quad \varepsilon_j \ge \tau \ (j \ge 4),$$
$$\|P_n(-K)g\|^2 \le d \ (n \ge 3), \qquad \|P_2(-K)g\|^2 \le (71/63)d.$$

For $b = 2$, set

$$m = k = 1, \quad R = K^2 1, \quad \gamma = d - \delta_2/2, \quad B = 1 - \gamma, \quad \mathcal{P}_n(z) = G_n(z) + z^{n+1}.$$

The short-one packet treats this case separately: its P_2 bound is $31/27$, not $71/63$. Below $\mathcal{P}_2$ is used only through its explicit coordinates.

In Table 48, each row means the actual nonnegative envelope $H \le C(c + N/2)^2$, with $N = KV$. An upper value for c enlarges the same nonnegative square base. Set $\delta = \kappa d$ and $B_0 = 1 - 2\delta$ in its last two rows.

Table 48: Actual positive squares at the unit receiver.

(b, L)	C	c, or an upper bound	Full vector V; actual mean $\mathbb{E}N$
$b \ge 3, 1$	m^2	$1 - \gamma/2$	$P_0 g$, $P_0 = kzG_{b-1}(-z)$; $-ke$.
$b \ge 3, 2$	1	$m\sqrt{B} - ma_1/(2\sqrt{B})$ $\le 1 - 37d/64$	$(m/\sqrt{B})P_1 g$; $-(md - \rho)/\sqrt{B}$.
$b \ge 3, 3$	1	$1 - 4d/5$	$(P_2 - \gamma)g$; $\le -(4/5 - 6d/5)d$.
$b \ge 3, L \ge 4$	1	$1 - 3d/4$	$P_{L-1} g$; $\le -\tau$.
$2, 1$	1	$1 - \gamma/2$	zg; 0.
$2, 2$	1	$\sqrt{B}$	$(1 + z^2)g/\sqrt{B}$; $-\tau/\sqrt{B}$.
$2, 3$	1	$\sqrt{B} - \tau/(2\sqrt{B})$ $\le 1 - d + 3\delta_2/4$	$(B + z + z^3)g/\sqrt{B}$; $-\sqrt{B}\, d$.
$2, 4$	1	$\sqrt{B_0} - \delta(1 - d)/(2\sqrt{B_0})$ $\le 1 - \delta(3 - d)/2$	$[\mathcal{P}_3 - \tau(1 + z)]g/\sqrt{B_0}$; $\le -\tau(1 - d)/\sqrt{B_0}$.
$2, L \ge 5$	1	$\sqrt{B_0} - \delta/(2\sqrt{B_0})$ $\le 1 - 3\delta/2$	$(\mathcal{P}_{L-1} - \tau)g/\sqrt{B_0}$; $\le -\tau(1 - d)/\sqrt{B_0}$.

The finite derivation of the short exceptions is as follows. The propagation (M.7) gives the first four rows by retaining, respectively, γ, then γ, a_1, then expanding $\bar{f}_2 = q + K^2 1$, and finally retaining the last two losses. The exact $L = 2$ scale and mean are proved in (Q58O.6.1)–(Q58O.6.4); in particular

$$\|(m/\sqrt{B})P_1 g\|^2 \le d, \qquad m^2/B \le 9/8, \qquad m^2 B = m^2 - mt_0 + V_R/2.$$

This identity keeps the common arm remainder in both factors. For $L = 3$, $\Re P_2 \ge 1-(6/5)\theta^2 > 4/5$ and $\gamma \le 2/5$, so subtracting γ reduces its modulus. The actual center and mean then follow from (Q58O.5.1)–(Q58O.5.2). For $L \ge 4$, (Q58O.4.1) gives the stated square.

For $b = 2$, the common finite propagation, beginning with $qf_2 \le (1 + K^2 1/2)^2$, is

$$T^L(qf_2) \le \sum_{j=0}^{L}(-K)^j 1 + (-K)^{L+2}1 - \gamma \bar{f}_{L-1} - \sum_{j=3}^{L} \eta_j \bar{f}_{L-j}, \quad \eta_j = \begin{cases} \beta_j, & 2 \mid j, \\ \beta_{j+2}, & 2 \nmid j. \end{cases}$$

Only nonnegative indices occur. All $\eta_j \ge \tau$. Expanding the first three lengths gives the three short rows. For $L = 4$, retain the τ multiples of $\bar{f}_0, \bar{f}_1, \bar{f}_3$; for $L \ge 5$, retain $\bar{f}_0, \bar{f}_1, \bar{f}_2$. This gives

$$\begin{aligned} T^4(qf_2) &\le (1-2\tau)q + K[\mathcal{P}_3 - \tau(1+z)]g - \tau(1-d), \\ T^L(qf_2) &\le (1-2\tau)q + K[\mathcal{P}_{L-1} - \tau]g - \tau \qquad (L \ge 5). \end{aligned} \tag{O.23}$$

The remainder means are at most $-\tau(1-d)$. Enlarge only the nonnegative q coefficient and the scalar loss by replacing τ there by $\delta \le \tau$, leaving the remainders unchanged. AM–GM with $B_0 q$ and p proves the last two rows.

Full-domain base and joint norms.

For each row of Table 49, let $\lambda\sqrt{d}$ bound $\|V\|$, and let Jd bound $\|V + \mathcal{U}(-K)g\|^2$ when $u_1 = 3$. We first prove these bounds over the full domain.

For $b \ge 3, L = 1$, $\|P_0 g\|^2/d \le (6/5)^2\theta(1+\theta) < (11/12)^2$. At $L = 2$, put $c_0 = m/\sqrt{B}$. The actual scale satisfies

$$c_0^2 \ge \frac{(1-d/2)^2}{1-3d/10} \ge \frac{125}{162} > \frac{49}{64}, \qquad c_0^2 \le 9/8, \qquad c_0 k = 1/\sqrt{B} \le 5/\sqrt{17} < 5/4.$$

For $b = 2, L = 2$, the exact norm is

$$\|(1+z^2)g/\sqrt{B}\|^2 = \frac{d - 2\delta_2 + \delta_3}{B} \le d.$$

The difference after multiplication by B is $-d^2 + (2 + d/2)\delta_2 - \delta_3 \ge 0$. For $b = 2, L = 3$,

$$|(B + z + z^3)/\sqrt{B}|^2 = B + x(1-x)^2/B \le 1 + 4/(27B) \le 47/39 < (10/9)^2,$$

where $B \ge 13/18$.

For $b = 2, L = 4$, direct expansion yields

$$|\mathcal{P}_3 - \tau(1+z)|^2 = (1-\tau)^2 - xH(x,\tau), \quad H = 1 - 3x + 2x^2 - x^3 + 2\tau x - \tau^2.$$

The function H is concave in $\tau \in [0, 2/9]$. At both endpoints it is decreasing in x, with positive value at θ, respectively $1451/32768$ and $465691/2654208$. Hence it is nonnegative on the whole rectangle. For $b = 2, L \ge 5$, the finite packet gives $|\mathcal{P}_n|^2 \le 1 - 7x/15$ and $\Re\mathcal{P}_n \ge 1 - x$, where $n = L - 1 \ge 4$. Therefore

$$|\mathcal{P}_n - \tau|^2 \le (1-\tau)^2 + x(2\tau - 7/15) \le (1-\tau)^2.$$

Both long base norms are at most

$$\frac{1-\tau}{\sqrt{B_0}} \le \frac{1-\delta}{\sqrt{1-2\delta}} \le \sqrt{\frac{5929}{5568}} < \frac{17}{16}.$$

All remaining base norms follow from the finite P_n bounds above.

For the joint norms assume $u_1 = 3$ and use (O.21). If $Q = A + zC$, $0 \le A, C \le 1$, then

$$|Q + \mathcal{U}|^2 \le (1 - x/2)^2 + x(1 + x/2)^2 = 1 + 5x^2/4 + x^3/4 < 5/4.$$

The negative real magnitude is at most $6x/5 \le 1 - x/2$. The finite packet puts P_n, $n \ge 3$, and $P_2 - \gamma$ in this coordinate square. This proves the $b \ge 3, L \ge 3$ entries. For $P_0 = kxb + zka$, $1 - x \le a, b \le 1$, the real magnitude of $P_0 + \mathcal{U}$ is at most $7x/10$, and its imaginary coordinate lies between zero and $6/5 + x/2$. Hence

$$|P_0 + \mathcal{U}|^2 \le 36x/25 + 169x^2/100 + x^3/4 < 9/10.$$

For $c_0 P_1$, the real magnitude after addition is at most $c_0 - x$: the upper side uses $c_0\kappa + 1/2 \ge 1$, and the lower side uses $(c_0 k + 11/5)x \le 2c_0$. Its imaginary magnitude is at most $7x^{3/2}/4$. Thus

$$|c_0 P_1 + \mathcal{U}|^2 \le (c_0 - x)^2 + 49x^3/16 \le c_0^2 \le 9/8,$$

because $-2c_0 + x + 49x^2/16 < 0$ for $c_0 \ge 7/8$.

For $b = 2, L = 1$, direct coordinates give

$$|z + \mathcal{U}|^2 \le x + 61x^2/25 + x^3/4 < 5/6.$$

For a single pair, (O.22) improves this to

$$|z + \mathcal{U}|^2 \le x + x^2, \qquad \|(z + \mathcal{U})g\|^2 \le \delta_2 + \delta_3. \tag{O.24}$$

Indeed $G_h = a + zc$ has $0 \le a, c \le 1$, also for $h = 1$; the squared modulus is $x^2a^2 + x(1 - xc)^2 \le x^2 + x$. At $L = 2$, put $D = (1 - x)/\sqrt{B} \ge 1 - x$. The real magnitude is at most $D - x/2$, and the imaginary magnitude at most $6x^{3/2}/5$. Consequently

$$|D + \mathcal{U}|^2 \le (D - x/2)^2 + 36x^3/25 \le D^2;$$

the lower real side uses $2D \ge 17x/10$. Integrating the base norm proves $J = 1$. At $L = 3$, $\sqrt{B} \ge 5/6$, $1/\sqrt{B} \le 6/5$. The real magnitude is at most $1 - x/2$; the imaginary coordinate is positive and at most $6/5 - 7x/10$. Thus

$$\begin{aligned} |(B + z + z^3)/\sqrt{B} + \mathcal{U}|^2 &\le 1 + 11x/25 - 143x^2/100 + 49x^3/100 \\ &\le 1 + 11x/25 - 6x^2/5 \le 1 + 121/3000 < 9/8. \end{aligned}$$

Finally let $A_0 = 17/16$ and $C_0 = 35/32 = \kappa + 1/2$. For $L = 4$, the unscaled coordinates are $1 - \tau - x + x^2, 1 - \tau$. For $L \ge 5$, both coordinates of $\mathcal{P}_{L-1}$ lie in $[1 - x, 1 - x + x^2]$; only the real coordinate loses τ. After division by $\sqrt{B_0}$ and addition of $\mathcal{U}$, the real magnitude in both cases is at most $A_0 - C_0 x$. Indeed its upper side uses $x(1 - x)/\sqrt{B_0} \ge \kappa x$; its negative side is at most $6x/5 \le A_0 - C_0 x$. The imaginary coordinate remains positive. For $L = 4$ it is at most $A_0 + x/2$; for $L \ge 5$ it is at most $(4/3)(1 - x + x^2) + x/2 \le 4/3$. These yield

$$\begin{aligned} |Q_4 + \mathcal{U}|^2 &\le (A_0 - C_0 x)^2 + x(A_0 + x/2)^2 \le A_0^2, \\ |Q_L + \mathcal{U}|^2 &\le (A_0 - C_0 x)^2 + (16/9)x \le A_0^2 \quad (L \ge 5). \end{aligned}$$

For the two final inequalities, subtract from A_0^2 and divide by x; the resulting decreasing polynomials have positive value at θ, respectively $7747/32768$ and $17827/294912$. All lower-coordinate checks use $(C_0 + 6/5)\theta < A_0$, $1 - 11\theta/5 > 0$, and $7/9 - 6\theta/5 > 0$. In particular the joint bound $289/256 < 5/4$ holds for every d.

One fourth/eighth payment table.

Define the normalized base cost

$$D = \frac{8\log C + 16(c-1) + 8\mathbb{E}N}{d}.$$

Table 49 lists an upper bound $D_i(\zeta)$, the base norm λ_i, and the proved joint bound J_i. Every coefficient multiplying ζ is nonnegative.

Table 49: Full-domain base costs and the same-vector norm bounds.

(b, L)	$D_i(\zeta)$	λ_i	J_i, when $u_1 = 3$
$b \ge 3, 1$	$-16 + 61\zeta/4$	$11/12$	$9/10$
$b \ge 3, 2$	$-191/12 + 8\zeta$	1	$9/8$
$b \ge 3, 3$	-16	$13/12$	$5/4$
$b \ge 3, L \ge 4$	$-20 + 8\zeta$	1	$5/4$
$2, 1$	$-8 + 4\zeta$	$2/3$	$5/6$
$2, 2$	$-16 + 12\zeta$	1	1
$2, 3$	$-68/3 + 12\zeta$	$10/9$	$9/8$
$2, 4$	$-95/6$	$17/16$	$5/4$
$2, L \ge 5$	$-209/12$	$17/16$	$5/4$

For the first row, $8\log m^2 - 8B_* \le -8(\beta_b + B_*) \le -16d + 61\delta_2/4$. For the second use $m \ge 5/6$, $\rho \le \delta_2$, $B \le 1$, and $md - \rho > 0$, giving $\mathbb{E}N \le -(5/6 - \zeta)d$. At the third row its mean is at most $-2d/5$. The long $b \ge 3$ row uses $\mathbb{E}N \le -\tau$. For $b = 2, L = 2$, the actual $C = 1, c = \sqrt{B}$ gives $16(c-1) = 16(\sqrt{1-\gamma} - 1) \le -8\gamma$ by concavity, and $\mathbb{E}N = -\tau/\sqrt{B} \le -\tau$. For $L = 3$, use $\sqrt{B} \ge 5/6$. The last two rows follow from

$$16(c-1) + 8\mathbb{E}N \le -8\delta(3-d) - 8\delta(1-d) \le -95d/6 \quad (L = 4),$$
$$16(c-1) + 8\mathbb{E}N \le -24\delta - 8\delta(1-d) \le -209d/12 \qquad (L \ge 5).$$

Thus each row includes the actual scale and the actual signed mean.

When $u_1 \ge 4$, (O.18) at $\nu = 8$ has normalized cost, after the extra $-24(r-1)$,

$$B_i = D_i(\theta) - 24 + 2/3 + 17(\lambda_i + 3/8)^2 \le -2.$$

When $u_1 = 3$, the corresponding costs are

$$A_i = D_i(\theta) - 16 + 16\theta + 2/3 + 17J_i \le -25/24.$$

For clarity, their exact values in table order are

i	B_i	A_i
1	$-5501/1152$	$-6409/1920$
2	$-247/64$	$-19/8$
3	$-1831/576$	$-43/12$
4	$-1525/192$	$-13/3$
5	$-6487/576$	$-25/24$
6	$-445/192$	$-71/24$
7	$-18559/5184$	$-15/2$
8	$-3101/768$	$-41/12$
9	$-1439/256$	-5

This proves the eighth moment, with the stated extra loss for every additional pair. The finite mask bounds are independent of its degree.

For the fourth moment, apply the $\nu = 4$ instance of the same formula. If $u_1 \geq 4$, its cost is

$$\frac{1}{2}B_i - \frac{17}{4}(\lambda_i + 3/8)^2 \leq -1 - \frac{17}{4}(25/24)^2 < -705/128.$$

The nine direct expressions before this last relaxation have maximum $-7061/768 < -7$, so the stronger $u_1 = 4$ fourth rate e^{-7d} also follows from the same table. If $u_1 = 3$ and $r \geq 2$, its cost is at most

$$\frac{1}{2}A_i - \frac{17}{4}J_i - 12(r-1) \leq -\frac{65}{16} - 12(r-1) < -705/128.$$

For $u_1 = 3, r = 1$, use the sharper mean in (O.22). Except for rows $1, 5$, one has $A_i \leq -19/8$ and $J_i \geq 1$, so the fourth cost is at most $-19/16 - 17/4 - 1/3 = -277/48 < -705/128$. Row 1 gives the direct fourth cost

$$-487/40 + (125/8)\zeta \leq -7459/1280 < -705/128.$$

For row 5, use (O.24) before integrating. Its full fourth cost is

$$-12d + \frac{57}{4}\delta_2 + \frac{17}{4}\delta_3 \leq -\frac{22567}{4096}d < -\frac{705}{128}d.$$

These are direct fourth payments, not interpolation from the eighth moment.

In (O.17) take

$$a = \log C + 2(c-1) - \Gamma, \qquad Z = V + \mathcal{U}(-K)g.$$

If $u_1 = 3$, the joint bounds give $\|Z\|^2 \leq (5/4)d$; if $u_1 \geq 4$, they give $\|Z\|^2 \leq (10/9 + 3/8)^2 d = (11449/5184)d < 9d/4$. The preceding direct fourth calculation proves (O.15). □

Bare inputs and decorations.

For later comparisons we retain the two distinct bare statements

$$\mathbb{E}[pT^L(qf_b)]^8 \leq 1 - d/12000, \qquad \mathbb{E}[f_2\{T^L(qf_b)\}^4] \leq 1 \quad (b \geq 2, L \geq 1). \tag{Q78.1.5}$$

Their respective proofs are (Q58O.0.2) and (L58.16.1). All directions in the proposition depend only on the arm lengths. The positive operators A, T restore every equal-arm factor $f_\ell h_\ell \leq 1$ or its full reversal without altering a remaining edge. Finally $e^{-25d/24} \leq 1 - d/12000$ on $[0, 1/3]$, for instance by $e^{-d} \leq 1 - d + d^2/2$.

O.4 Negative odd roots at every nonunit inner length

Q80.0 Original objective and exact new claims

The ambient problem includes the P5, T5, and K1, 4 skeletons of Q5. The following estimates treat the specified root roles.

Fix the finite probability space and kernel throughout. Write

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n \mathbf{1}, \quad \bar{f}_n = T^n \mathbf{1}, \quad h_n = T f_{n-1}, \quad W_n = A\bar{f}_{n-1}, \quad f_0 = \bar{f}_0 = 1.$$

Take the strictly ordered original odd residual root table

$$a < u_1 < v_1 < \cdots < u_k < v_k, \qquad k \geq 1, \qquad P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}. \tag{Q80.0.1}$$

The terminal has original residual arms (1, b), with every integer $b \geq 2$. For every nonunit inner length $L \geq 2$, the actual ordinary input is

$$X_{b,L} = T^L(qf_b), \qquad Y_{a,b,L} = W_a P X_{b,L}. \tag{Q80.0.2}$$

Theorem T085.1. If $a \geq 3$, all original hosts and all the independent finite lengths in (Q80.0.1)–(Q80.0.2) satisfy

$$\boxed{EY_{a,b,L}^8 \leq e^{-97d/12}, \qquad I_q(T^M Y_{a,b,L})^3 \leq e^{-49d/228} \quad (M \geq 1),} \tag{Q80.0.3}$$

where $d = \|K1\|_2^2$ and $I_q(U) = \mathbb{E}[q^{-1/3}|U|^{8/3}]$. For $a = 3$ the ordinary exponent improves to $403/48$, and the all-M weighted exponent improves to $481/912$.

Theorem T085.2. If $a = 2$ and $u_1 \geq 4$, every $b \geq 2, L \geq 2, M \geq 1$ satisfies

$$\boxed{EY_{2,b,L}^8 \leq e^{-479d/60}, \qquad I_q(T^M Y_{2,b,L})^3 \leq e^{-131d/1140}.} \tag{Q80.0.4}$$

The factors in (Q80.0.2) are the complete actual root input, and T^M is its original outer path.

Theorem T085.3, exact all-length union. For every odd root table (Q80.0.1) with $a \geq 3$, terminal residual (1, b), $b \geq 2$, and all independent positive original L, M, choose the following directions from L alone:

Table 50: Negative odd roots at every nonunit inner length: parameter values.

Entry	Complete ordinary root input before the outer path
$L = 1$	$Y = f_a PT(p\bar{f}_b)$.
$L \geq 2$	$Y = W_a PT^L(qf_b)$.

Then, including equal-arm factors at their roots by positivity,

$$\boxed{I_q(T^M Y)^3 \leq e^{-25d/912} \leq 1.} \tag{Q80.0.5}$$

All claimed root and terminal directions are literal original directions:

$Root a - arm$

Nonunit row of (Q80.0.2): First edge away from root, all later edges toward its first neighbor;

factor W_a.

$Root u_i - arm$

Nonunit row of (Q80.0.2): Same actual reversed bent pattern;

factor W_{u_i}.

$Rootv_i - arm$

Nonunit row of (Q80.0.2): $Entirelytowardroot; factor\bar{f}_{v_i}$.

Terminal unit arm

Nonunit row of (Q80.0.2): Toward terminal;

factor q.

Terminal b-arm

Nonunit row of (Q80.0.2): Entirely away from terminal;

factor f_b.

Inner connector

Nonunit row of (Q80.0.2): All original L edges from terminal toward root.

Outer connector

Nonunit row of (Q80.0.2): All original M edges from root toward receiving root.

For $L = 1$, replace the a-arm by its entirely outward orientation f_a, and the terminal by the actual unit-outward, b-arm-inward pattern $p\bar{f}_b$. The extra pairs P and both core-path directions remain as displayed. The original integer L, not any host parameter, determines this choice.

Q80.1 Original calculus and one finite mask for all extra pairs

Delete zero-mass atoms. Set

$$r = Kg, \quad s_0 = Kr, \quad t = Ks_0, \quad \delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The C014 calculus gives

$$0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \|K\|^2 \le \theta, \quad \delta_{j+1} \le \theta\delta_j, \quad |Kv(x)| \le \|v\|_2. \tag{Q80.1.1}$$

The original two-row estimate is $|K^2v(x)|^2 \le \|v\|_2^2/3$. It is not an operator-norm bound $\|K\|^2 \le 1/3$. The path convolution uses the finite quantities

$$\beta_1 = \beta_2 = 0, \qquad \beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j, \quad \tau \le \beta_n \le d \quad (n \ge 3),$$

$$\sum_{j=0}^{n} K^j\mathbf{1} = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}. \tag{Q80.1.2}$$

The same identity holds after complete reversal $K- > -K$. Every path message here is nonnegative. In fact p, $q \ge \mu(x) > 0$ on each retained atom. Actual equal pairs obey $f_l h_l \le 1$ and, after complete reversal, $W_l\bar{f}_l \le 1$.

At $d = 0, g = 0$ while K may be nonzero. Nevertheless $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, and successive leaf integrations make every tree message here equal to one. Every displayed final bound holds directly. All later divisions by d are restricted to $d > 0$.

Q80.1.1 The exact actual pair loss and original finite mask

C067 proves, including the separate finite $u = 4$ case,

$$h_u f_v \le \exp\left(\sum_{j=u}^{v-1} K^j g - 3d\right) \qquad (4 \le u < v). \tag{Q80.1.3}$$

We use this only with $u \ge 4$. Full reversal on the same original probability space gives

$$P \le e^{S-3kd}, \qquad S = -\sum_{i=1}^{k}\sum_{j=u_i}^{v_i-1}(-K)^j g = KV_S,$$

$$V_S = U(-K)g, \qquad U(z) = \sum_{i=1}^{k}\sum_{j=u_i}^{v_i-1} z^{j-1}. \tag{Q80.1.4}$$

The intervals are disjoint. Thus U is a finite polynomial whose coefficients are zero or one, independently of the number of arms, the gaps, and the largest original length.

For every finite [0, 1]-coefficient polynomial Q and $z = i\sqrt{x}$, $0 \le x \le \theta$, separate the four sign classes. Their finite sums give

$$|\Re Q(z)| \le (1-x^2)^{-1}, \qquad |\Im Q(z)| \le \sqrt{x}(1-x^2)^{-1},$$

$$\|Q(K)\|^2 \le \frac{1+\theta}{(1-\theta^2)^2} = \frac{32768}{16245} < \frac{100}{49}. \tag{Q80.1.5}$$

The comparison is $32768 * 49 = 1605632 < 1624500 = 16245 * 100$. Only a finite polynomial is evaluated; the geometric sums are upper bounds for finite positive subcollections.

If $u_1 \ge 4$, U begins in degree at least three. If $u_1 \ge 5$ it begins in degree at least four. Consequently

$$\|V_S\| \le \begin{cases} (3/8)\sqrt{d}, & u_1 \ge 4, \\ (1/4)\sqrt{d}, & u_1 \ge 5. \end{cases} \tag{Q80.1.6}$$

For the first, $(10/7)\theta^{3/2} < 585/1568 < 3/8$, using $\sqrt{\theta} < 9/14$. For the second, $(10/7)\theta^2 = 845/3584 < 1/4$. Original row Cauchy applies only to the outermost K in $S = KV_S$, not to a compressed or spectral kernel.

The positive contributions to ES have $j = 3, 7, 11,...$ and values $\delta_2, \delta_4, \delta_6, \ldots$. Since $u_1 \ge 4$ excludes the first such term,

$$ES \le \frac{\delta_4}{1-\theta^2} \le \frac{2197}{27360} d < \frac{d}{12}. \tag{Q80.1.7}$$

Equations (Q80.1.4)–(Q80.1.7) hold for every finite k and every original set of gaps. Unequal pairs are paid by these inequalities, rather than deleted from the tree or assumed pointwise at most one.

Q80.2 A precise envelope extension with the same original fluctuation

Let $c_* = (1+\theta)^2 = 2025/1024 < 2$. The exact C071/C075 exponential tool says that if $Z = KV$, $\|V\|^2 \le D$, and $M \ge max(EZ, 0)$, then

$$\log E e^{8Z/B} \le \frac{8M}{B} + \frac{8c_* D}{B^2} \qquad (B > 0). \tag{Q80.2.1}$$

Its hypotheses are the original range $|Z|^2 \le D$ and original raw second moment $EZ^2 \le \theta$ D. It covers both signs of EZ. For $Z = K^2 V_2$ with $\|V_2\|^2 \le D_2$, the original two-row version is

$$\log E e^{8Z/B} \le \frac{8M}{B} + \frac{6D_2}{B^2}. \tag{Q80.2.2}$$

Writing the same two-row variable as $K(KV_2)$ gives a one-row primitive with squared norm at most theta D_2. Its scalar cost increases by at most

$$(8c_*\theta - 6)D_2/B^2 = \frac{1749}{4096} D_2/B^2 < \frac{7}{16} D_2/B^2. \tag{Q80.2.3}$$

Apply these scalar lemmas with their stated hypotheses.

Lemma T085.4. Suppose an actual nonnegative base Y_0 has specified $C_0, B > 0$, a real loss ell, a vector V, $M \ge max(EKV, 0)$, and $D \ge \|V\|^2$, with

$$Y_0 \le C_0 \left(1 + \frac{KV - \ell}{2B}\right)^2, \qquad 1 + \frac{KV - \ell}{2B} \ge 0, \quad D \le \eta^2 B^2 d. \tag{Q80.2.4}$$

Let its complete specified one-row scalar expression be

$$\mathcal{E}_0 = 8 \log C_0 - 8\ell/B + 8M/B + 8c_* D/B^2. \tag{Q80.2.5}$$

For the actual P in (Q80.1.4), if $\|V_S\| \le \epsilon\sqrt{d}$, then

$$\boxed{E(PY_0)^8 \le \exp\left[\mathcal{E}_0 + \left\{16(2\eta\epsilon + \epsilon^2) + \frac{2}{3} - 24k\right\} d\right].} \tag{Q80.2.6}$$

Proof. Positivity of the square base and $\log(1 + x) \le x$ give

$$(PY_0)^8 \le C_0^8 \exp\{8(KV - \ell)/B + 8S - 24kd\}.$$

At a zero square base this follows by continuity. Combine the two remainders before taking expectations:

$$KV + BS = K(V + BV_S).$$

Its mean is bounded by the nonnegative M'=M+Bd/12. Its specified primitive norm can be bounded by

$$D' = D + 2\sqrt{D}\, B\epsilon\sqrt{d} + B^2\epsilon^2 d \le D + B^2(2\eta\epsilon + \epsilon^2)d.$$

Use (Q80.2.1) on this same original function, and retain (Q80.2.5) with its own originally paid D. The extra mean cost is 2d/3; the extra norm cost is at most $8c_*(2\eta\epsilon+\epsilon^2)d$, hence at most 16(2eta $\epsilon+\epsilon^2)d$. This proves (Q80.2.6). In particular, a looser norm normalization is used only for the increment; it does not replace D inside an already paid scalar expression. QED.

Q80.3 The complete actual base table and its exact normalizations

We use the actual shortest-two construction of C071, with C075's additional all-domain $b = 2$, $L = 2$ payment and its shortest-three comparison. C076/T082 records the following full normalization. We reproduce all rows needed for the new estimates.

Since $W_2 = Aq = p - r$, the exact positive terminal masks give

$$0 < X_{b,L} \le C[Bq + Br + Z - \ell], \qquad W_2 X_{b,L} \le CB\left(1 + \frac{Z - \ell}{2B}\right)^2. \tag{Q80.3.1}$$

The bracket is half the sum of two actual nonnegative upper factors. Thus the sign condition in (Q80.2.4) is satisfied, even before subsequent root comparisons.

For $b = 2$ put $\gamma = d - \delta_2/2$ and $B_2 = 1 - \gamma$. For $b \ge 3$ put

$$R = KG_{b-1}(K)g, \quad G_n(z) = \sum_{j=0}^{n-1} z^j, \quad m_b = 1 - \beta_b/2, \quad k_b = 1/m_b,$$

$$t_0 = \beta_{b+1}, \quad e = d - \beta_{b+2}, \quad V_R = ER^2, \quad \rho = E(rR), \quad \gamma = k_b t_0 - k_b^2 V_R/2, \quad a_1 = k_b e, \quad B_* = \gamma + a_1. \tag{Q80.3.2}$$

The subscript on k_b distinguishes it from the number k of root pairs. The exact terminal toolkit gives $1 \le k_b \le 6/5, ER = -t_0, E(gR) = -e \le 0, V_R = \delta_2 + \delta_3$ for $b = 3$ and $V_R \le \delta_2$ for $b \ge 4$, $0 \le a_1 \le 39d/80, \gamma \le 6d/5$, and $B_* \le 3d/2$.

For the finite polynomials in the table set $z = -K$ and

$$P_n(z) = \begin{cases} G_n(z) + z^{n+1}, & b = 2, \\ G_n(z) + k_b z^{n+1} G_{b-1}(-z), & b \ge 3. \end{cases} \tag{Q80.3.3}$$

The entire finite table, including every short connector, is:

Table 51: Finite masks and row-form scalar bounds for the terminal parameters (b, L).

(b, L)	C, B and ell	Same original fluctuation Z	Original row-form scalar upper
2, 2	$C = 1; B = B_2; \ell = 0;$	$K^2(\gamma 1 + r);$	$-d/8.$
2, 3	$C = 1; B = B_2; \ell = \tau;$	$-K^2(g + s_0);$	$-3d/4.$
2, 4	$C = 1; B = 1 - \gamma - \tau; \ell = d(1 - \gamma);$	$K^2[\tau 1 - (1 - \gamma)g + r + t];$	$-d.$
2, 5	$C = 1; B = 1 - 2\tau; \ell = \tau;$	$K[\tau + zP_3(z)]g;$	$-2d/3.$
$2, L \ge 6$	$C = 1; B = 1 - 2\tau; \ell = \tau;$	$K[\tau + zP_{L-2}(z)]g;$	$-4d/5.$
$b \ge 3, 2$	$C = m_b^2; B = 1 - \gamma; \ell = a_1;$	$K^2(\gamma 1 + k_b R);$	$-d/8.$
$b \ge 3, 3$	$C = m_b^2; B = 1 - B_*; \ell = d - k_b \rho;$	$K[a_1 + zP_1(z)]g;$	$-d/10.$
$b \ge 3, 4$	$C = m_b^2; B = 1 - 41d/80; \ell = \kappa d;$	$K[41d/80 + zP_2(z)]g;$	$-8d/25.$
$b \ge 3, L \ge 5$	$C = B = 1 - d/2; \ell = d/2;$	$K[d/2 + zP_{L-2}(z)]g;$	$-2d/5.$

Here the original row-form scalar is

$$8\log(CB) - 8\ell/B + 8M/B + \begin{cases} 6D_2/B^2, & Z = K^2V_2, \\ 8c_*D_1/B^2, & Z = KV_1. \end{cases} \tag{Q80.3.4}$$

The D_j in (Q80.3.4) are the specified bounds for those particular vectors. Each paid scalar expression retains its own shared terminal parameters. These bounds give (Q80.3.4) directly.

Nonnegative mean uppers M in the displayed order are $\delta_2 - \gamma$ d, 0, (theta kappa-tau)d, (theta-tau)d for both $b = 2$ tails, $max(k_b$ rho-gamma d, 0), and (theta-delta)d in the last three rows, where $\delta = a_1$, 41d/80, d/2 respectively. They bound the original Z in each row and satisfy $M \geq 0$ over the entire original domain.

Q80.3.1 Two-row vectors and conversion costs

The four two-row vector squares have the following whole-domain normalizations. These upper bounds are used for increments and norm comparisons, not substituted into a scalar payment that used a sharper same-parameter expression:

(b, L)	$D_2/(B^2d)$
$(2, 2)$	$\theta\, 324/169 < 1$
$(2, 3)$	$324/169 < 2$
$(2, 4)$	$217/81 < 25/9$
$(b \geq 3, 2)$	$540/289 < 2.$

(Q80.3.5)

The original lower bounds are $B_2 \geq 13/18, 1 - \gamma - \tau \geq 1/2$, and $1 - \gamma \geq 17/25$ for $b \geq 3$. The vector squares are bounded respectively by $\delta_2, d - 2\delta_2 + \delta_3 \leq d$, [$(1 - \gamma)^2 + 4/27$]d, and $k_b^2 V_R$. For $b = 2, L = 4$, use $1 - \gamma = B + \tau$ and $\tau/B \leq 4/9$, giving $(1 + 4/9)^2 + 16/27 = 217/81$. For $b \geq 3$, $L = 2, \theta(1 + \theta) = 585/1024 < 3/5$ gives the upper $540/289$.

After the exact two-row to one-row conversion (Q80.2.3), the squared primitive normalization is multiplied by theta, while the specified scalar upper increases by less than $(7/16)D_2/B^2$.

Q80.3.2 Every one-row tail

For $b = 2, L \geq 5$ the old squared primitive is bounded by A(tau)d, where

$$A(\tau) = A_0 - 2\theta\sigma\tau + \tau^2, \quad (A_0, \sigma) = \begin{cases} (\theta, 1), & L = 5, \\ (\theta - 7\theta^2/15, \kappa), & L \geq 6. \end{cases}$$

Both are bounded above by theta-2theta kappa $\tau + \tau^2$. Its quotient by $(1 - 2\tau)^2$ increases for $0 \leq \tau \leq 2/9$: after multiplying the derivative by $(1 - 2\tau)^3$, its numerator is 4theta-2theta kappa+(2-4theta $\kappa)\tau > 0$. Its endpoint is $289/256$. The other three one-row normalizations are

$$\frac{D_1}{B^2 d} \leq \begin{cases} 289/256, & b = 2, L \geq 5, \\ 481/400, & b \geq 3, L = 3, \\ 27/32, & b \geq 3, L = 4, \\ 18/25, & b \geq 3, L \geq 5. \end{cases} \tag{Q80.3.6}$$

The latter three follow from the actual bounds $D_1/d \leq 351/1280 + \delta^2, \theta(71/63) + \delta^2, \theta + d^2/4$, and the respective $B \geq 1/2, 4/5, 5/6$. All four normalizations are at most $5/4$. Thus (Q80.3.5)–(Q80.3.6) include every $b \geq 2, L \geq 2$, including the finite cases immediately before each tail.

Q80.3.3 The uniform base for $a \geq 4$

C076/T082 Lemma T082.2 proves the following particular one-row certificate: for each $a \geq 2$, $b \geq 2, L \geq 2$, the actual Y_0=W_$aX_{b,L}$ admits (Q80.2.4) with

$$D \leq (9/4)B^2 d, \qquad \mathcal{E}_0 \leq (9/4)d. \tag{Q80.3.7}$$

Its exact construction for $a \geq 4$ starts from the rows above. Put

$$m = a - 2, \quad c = 1 - \tau \geq 7/9, \quad H = G_m(-K)g - \tau 1, \quad h = B/c,$$
$$A_m = 45/32 \ (m = 2), \qquad A_m = 1 \ (m \geq 3).$$

The retained actual positive loss tau p gives

$$W_a X \leq CBc \left(1 + \frac{Z + hK^2 H - \ell - h\beta_a}{2B}\right)^2. \tag{Q80.3.8}$$

The bracket is an actual nonnegative arithmetic mean. The exact Q5-C075 same-kernel comparisons are $EK^2 H \leq 0, \|H\|^2 \leq A_m$ d, and

$$\|V_2 + hH\|^2 \leq D_2 + h^2 A_m d, \qquad \|V_1 + hKH\|^2 \leq D_1 + h^2 \theta A_m d. \tag{Q80.3.9}$$

In its original row form, (Q80.3.8) has scalar upper at most -3d/20. The largest normalized two-row upper is

$$25/9 + 3645/1568 < 36/7,$$

with positive difference $571/14112$. Conversion adds less than $(7/16)(36/7)d = 9d/4$, while its new one-row norm is at most $\theta(36/7) = 117/56 < 9/4$. In an already one-row case its norm is at most $5/4 + 47385/50176 < 9/4$ and no conversion is needed. The mean upper remains valid by $EK^2 H \leq 0$. Set $C_0 = CBc$, replace ell by ell+h β_a, and use these actual vectors. This explains exactly the uniform certificate (Q80.3.7) used below. The finite covariance proof and all actual coefficient end cases are the specified C075 parts Q80.4 and 6 / C076 part Q80.2 inputs.

Q80.4 A sharper certificate for the actual shortest-three base

The generic normalization (Q80.3.7) loses too much for the first extra short arm four. We therefore retain the precise shortest-three identity

$$W_3 = W_2 + s_0 - d.$$

In every original row (Q80.3.1), replace Z by Z+B s_0 and ell by ell+Bd:

$$W_3 X \leq CB \left(1 + \frac{Z + Bs_0 - \ell - Bd}{2B}\right)^2. \tag{Q80.4.1}$$

The base remains nonnegative, being the arithmetic mean of the actual W_3 and the positive terminal upper factor. Since $Es_0 = 0$, the old nonnegative mean upper M still applies. The deterministic improvement in the scalar expression is exactly -8d.

Lemma T085.5. For every $b \geq 2, L \geq 2$, (Q80.4.1) has a specified one-row certificate satisfying

$$\boxed{D \leq (81/64)B^2 d, \qquad \mathcal{E}_0 \leq -13d/16.} \tag{Q80.4.2}$$

Proof, both $L = 2$ terminal rows. The new two-row vector is $V_2 + B$ g. When $b = 2$, g is orthogonal to 1 and r. When $b \geq 3$, the additional cross term uses $E(gR) = -e \leq 0$. Thus one may take the new vector upper $D_2 + B^2 d$. The old ratio in (Q80.3.5) is below two, so the new ratio is below three. Its original row-form scalar is at most

$$-d/8 + 6d - 8d = -17d/8.$$

Conversion to one row costs less than 21d/16, leaving -13d/16. The new one-row squared primitive is at most 3theta $B^2 d$, and $3\theta = 39/32 = 78/64 < 81/64$.

Proof, $b = 2$ and $L = 3$ or 4. For $L = 3$ the vector changes from -($g + s_0$) to -(gamma $g + s_0$). The squared-norm bound follows from the pointwise inequality

$$(1 - x)^2 - (\gamma - x)^2 = (1 - \gamma)(1 + \gamma - 2x) \geq 0.$$

For $L = 4$ the new vector is tau 1-tau g+r+t. Orthogonality of g to 1, r, t and $\tau < 1 - \gamma$ again makes its norm no larger. The normalized old square is at most $25/9 < 3$. The original scalar is at most -8d, before conversion. After conversion it is at most $-8d + 21d/16 = -107d/16$, stronger than required. The squared one-row norm is at most $\theta(25/9) < 81/64$.

Proof, all remaining one-row cases. The old primitive is [$\delta + zP_n(z)$]g and the new one is [$\delta + z(P_n(z) - B)$]g, with $z = -K$. At frequency $z = i\sqrt{x}$, the difference of squared moduli is exactly

$$Bx[B - 2\Re P_n(z)].$$

The complete C075 finite coefficient lemma gives Re $P_n \geq 1/2$ for every relevant finite n, including $n = 1, 2$ where needed. Since $0 < B \leq 1$, the difference is nonpositive. The norm therefore retains (Q80.3.6), at most $5/4 = 80/64 < 81/64$. Its original scalar retains the full extra -8d and is at most -8d. These cases exhaust all b, L and prove (Q80.4.2). QED.

Q80.5 Minimum at least three, every nonunit inner connector

Take the actual input (Q80.0.2), $k \geq 1$.

If $a \geq 4$, the first extra short arm satisfies $u_1 \geq a + 1 \geq 5$. Use the precise certificate (Q80.3.7) in Lemma T085.4 with $\eta = 3/2, \epsilon = 1/4$. The full exponent is at most

$$\left[\frac{9}{4} + 16\left(2 \cdot \frac{3}{2} \cdot \frac{1}{4} + \frac{1}{16}\right) + \frac{2}{3} - 24k\right] d \leq \left(\frac{9}{4} + 13 + \frac{2}{3} - 24\right) d = -\frac{97}{12} d. \qquad \text{(Q80.5.1)}$$

If $a = 3$, the first extra short arm satisfies $u_1 \geq 4$. Use the sharper (Q80.4.2), $\eta = 9/8, \epsilon = 3/8$. The full exponent is

$$\left[-\frac{13}{16} + 16\left(2 \cdot \frac{9}{8} \cdot \frac{3}{8} + \frac{9}{64}\right) + \frac{2}{3} - 24k\right] d \leq \left(-\frac{13}{16} + \frac{63}{4} + \frac{2}{3} - 24\right) d = -\frac{403}{48} d. \quad \text{(Q80.5.2)}$$

Since $403/48 > 97/12$, these prove the ordinary part of Theorem T085.1 over the full original host domain. Every additional pair beyond the first contributes another -24d; its number never increases the finite mask bound. All original lengths and gaps remain arbitrary.

Q80.6 Minimum two with first extra short arm at least four

Now take $a = 2, u_1 \geq 4$. The extra mask has $\epsilon = 3/8$. Each actual shortest-two row of (Q80.3.4)–(Q80.3.6) retains its own paid scalar cost. For a normalized primitive upper eta, the increment in (Q80.2.6) is

$$16(2\eta \cdot 3/8 + 9/64)d + (2/3)d - 24kd. \tag{Q80.6.1}$$

The following six rows exhaust every $b \geq 2, L \geq 2$. The column E_0/d is an upper for the complete specified one-row scalar after any necessary two-row conversion.

Table 52: Negative odd roots at every nonunit inner length: parameter values.

Entry	eta	E_0/d upper	Extra norm cost divided by d	Final logarithmic upper divided by d at $k = 1$
Every $L = 2$; also $b = 2, L = 3$	$11/12$;	$3/4$;	$53/4$;	$-28/3$.
$b = 2, L = 4$	$17/16$;	$1/4$;	15;	$-97/12$.
$b = 2, L \geq 5$	$17/16$;	0;	15;	$-25/3$.
$b \geq 3, L = 3$	$11/10$;	$-1/10$;	$309/20$;	$-479/60$.
$b \geq 3, L = 4$	$15/16$;	0;	$27/2$;	$-59/6$.
$b \geq 3, L \geq 5$	$6/7$;	0;	$351/28$;	$-907/84$.

The scalar and norm calculations for the table are as follows.

For the first row, (Q80.3.5) has $D_2/(B^2 d) < 2$. Conversion gives a one-row norm at most $2\theta = 13/16 < (11/12)^2$. The weakest original scalar is -d/8; conversion costs less than 7d/8. Their sum is at most 3d/4. The extra norm cost is $16(11/16+9/64)d = 53d/4$. The full sum is $3/4+53/4+2/3-24 = -28/3$.

For $b = 2, L = 4$, (Q80.3.5) gives $D_2/(B^2 d) \leq 217/81$. The one-row norm is at most $\theta(217/81) = 2821/2592 < (17/16)^2$. Conversion costs at most 1519d/1296, while the original scalar is at most -d. Thus its scalar upper is $223d/1296 < d/4$. The extra norm cost is $16(51/64 + 9/64)d = 15d$. This gives $1/4 + 15 + 2/3 - 24 = -97/12$.

For $b = 2, L \geq 5$, (Q80.3.6) is exactly at most $(17/16)^2$ and its original one-row scalar is nonpositive. No conversion cost is added. The same extra norm cost 15d gives $15 + 2/3 - 24 = -25/3$.

For $b \geq 3, L = 3$, (Q80.3.6) gives $481/400 < (11/10)^2$, with the complete original scalar at most -d/10. The extra norm cost is $16(33/40+9/64)d = 309d/20$. The resulting sum is $-1/10+309/20+2/3-24 = -479/60$.

For $b \geq 3, L = 4$, the squared norm is at most $27/32 < (15/16)^2$ and the original scalar is nonpositive. The extra norm cost $16(45/64 + 9/64)d = 27d/2$ gives $27/2 + 2/3 - 24 = -59/6$.

For $b \geq 3, L \geq 5$, the squared norm is at most $18/25 < (6/7)^2$ and the original scalar is nonpositive. The extra norm cost $16(9/14 + 9/64)d = 351d/28$ gives $351/28 + 2/3 - 24 = -907/84$.

Every table entry is at most $-479/60$. In particular, the close second row is $-97/12 = -485/60$, with surplus $1/10$. The first and third give losses $560/60$ and $500/60$; the fifth gives $590/60$. For the last comparison, $907/84 - 479/60 = 1182/420 > 0$. All $k > 1$ are stronger by 24(k-1)d. This proves the ordinary part of Theorem T085.2 throughout $0 \leq d \leq 1/3$.

The short-three exception $u_1 = 3$ is excluded here: its first-pair loss is not $3d$, so (Q80.1.3) and this table do not establish that boundary. No conclusion is drawn from this sufficient payment about whether that boundary is true or false.

O.5 The unit ordinary terminal catalogue

Q106.1 Original model and what is proved

Let (X, mu) be any nonempty finite probability space and let K be real with $K(x,y) = -K(y,x)$ and $|K(x,y)| \le 1$. All expectations and operators below use mu:

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g, \quad d = Eg^2 \in [0, 1/3],$$

$$f_n = A^n \mathbf{1}, \qquad \bar{f}_n = T^n \mathbf{1}, \qquad h_n = T f_{n-1}.$$

An f_n arm points entirely away from its root. A $\bar{f}_n$ arm points entirely toward its root. An h_n arm has its first edge toward the root and all remaining edges toward its leaf. In particular $f_1 = p$ and $h_1 = q$.

At each root, pair equal-length original arms as far as possible using f_ℓ and h_ℓ on each pair. Their actual product E_0 satisfies $0 < E_0 \le 1$ after zero-mass atoms are removed. Entire pairs may instead be reversed when the complete construction calls for it. The residual lengths are distinct. The original equal pairs remain in the tree and edge ledger.

Write

$$Q(d) = 1 - d/20000, \qquad I_q(U) = E[q^{-1/3}|U|^{8/3}]. \tag{Q106.1.1}$$

Theorem A (complete ordinary terminal supply). Let Sigma be any finite original terminal arm multiset with at least two distinct residual arms after maximum equal pairing. For every positive original connector length L, the tables in parts Q106.2–3 select a terminal orientation $F^{\mathrm{ord}}_{\Sigma,L}$, depending only on Sigma and L, such that

$$\boxed{E[(E_r p T^L (E_t F^{ord}_{Sigma,L}))^8] \le Q(d).} \tag{Q106.1.2}$$

Here E_t and E_r contain arbitrary genuine equal pairs at the terminal and receiving root. The receiving root has an original outward unit arm, whose factor is p. All L connector edges point toward the receiving root. The statement covers residual size at least two; the same-direction PW4 and an f2-reference fourth for the chosen F are separate assertions.

Theorem B (root pair (1, a), all terminals and both lengths). At an original internal root with residual exactly (1, a), $a \ge 2$, orient the unit arm outward and the a-arm entirely inward. With any Sigma as in A and any independent positive L, M, put

$$Y = E_r p T^L (E_t F^{ord}_{Sigma,L}), \qquad U = T^M (\bar{f}_a Y).$$

Then

$$\boxed{I_q(U)^3 \le (1 - d/72) Q(d) \le 1.} \tag{Q106.1.3}$$

All M outer edges point from the internal root toward the output root. The $\bar{f}_a$ factor is multiplied before T^M; it is not commuted through T.

Theorem C (unit internal root, all terminals, outer length at least three). If the internal root instead has residual exactly one outward unit arm, the same Y satisfies, for every $M \geq 3$,

$$\boxed{I_q(T^M Y)^3 \leq (1-d)Q(d) \leq 1.} \tag{Q106.1.4}$$

The applications of these ordinary roles are given in Sections 11 and 10.

Q106.2 Every even residual terminal, with every direction exception

Write the distinct residual table as

$$a_1 < b_1 < \cdots < a_j < b_j, \qquad j \geq 1.$$

All rows below use every $L \geq 1$ except where explicitly split by L. Each table entry omits the common genuine E_t only. The displayed factor holds in the displayed range.

Table 53: The unit ordinary terminal catalogue: parameter values.

Entry	Connector	F	Proof
$j=1, a_1=1, b_1 \geq 2$	every L;	qf_{b_1};	Section O.29.
$j=1, a_1=2, b_1 \geq 3$	$L=1$;	$h_2 f_{b_1}$;	Sections O.28 and O.10.
$j=1, (a_1,b_1)=(2,3)$	$L=2$;	$h_2 f_3$;	Section O.17.
$j=1, a_1=2, b_1 \geq 4$	$L=2$;	$f_2 h_{b_1}$;	Section O.17.
$j=1, a_1=2, b_1 \geq 3$	$L \geq 3$;	$h_2 f_{b_1}$;	Section O.17/Section O.18 exact union.
$j=1, a_1=3, b_1 \geq 4$	$L=1$;	$f_3 h_{b_1}$;	Section O.19.
$j=1, (a_1,b_1)=(3,4)$	$L=2$;	$h_3 f_4$;	Section O.19.
$j=1, a_1=3, b_1 \geq 5$	$L=2$;	$f_3 h_{b_1}$;	Section O.19.
$j=1, a_1=3, b_1 \geq 4$	$L \geq 3$;	$h_3 f_{b_1}$;	Section O.19.
$j=1, a_1 \geq 4$	every L;	$h_{a_1} f_{b_1}$;	Section O.30.
$j \geq 2, a_1=1$	every L;	$qf_{b_1} \prod_{i=2}^{j} h_{a_i} f_{b_i}$;	Section O.20.
$j \geq 2, a_1 \geq 2$	every L;	$\prod_{i=1}^{j} h_{a_i} f_{b_i}$;	Section O.30.

Section O.29 supplies 1-d/12000. The Section O.17/Section O.18 union supplies Q(d). Section O.19 supplies 1-d/500. Section O.30 supplies exp(-d/6) for its multiple-pair row; for $a_1 \geq 4$, $j \geq 1$ its stronger expression is exp(-(128j/5-24)d), whose coefficient is at least $8/5$. Section O.20 supplies $\exp(-(1/4+16(j-2))d)$. Each is at most Q(d): for the exponential rows use $\exp(-ad) \leq 1 - ad/(1+ad)$, with $d \leq 1/3$ and the stated $a \geq 1/6$. The latter fraction is at least d/7, which is more than d/20000.

Exhaustion is structural. A positive even residual has either $j=1$ or $j \geq 2$. With $j=1$ the shorter length is 1, 2, 3 or at least 4. The cases shorter 2 and 3 are partitioned by $L=1, L=2$ and $L \geq 3$, retaining the actual shorter-pair exceptions. With $j \geq 2$ the shorter length is 1 or at least 2. These cases are disjoint and cover all even residual lists. No direction is chosen using K, mu, or d.

Q106.3 Every odd residual terminal

Write

$$a < b_1 < c_1 < \cdots < b_k < c_k, \qquad k \geq 1.$$

Q106.3.1 Minimum $a \geq 2$

The catalogue in Section O.23, with the unit and adjacent proofs in Sections O.16, O.21 and O.22, gives the following table:

Table 54: The unit ordinary terminal catalogue: parameter values.

Entry	Connector	Actual F
$b_1 \geq 5$, any $a \geq 2, k \geq 1$	every L;	$\bar{f}_a \prod_{i=1}^{k} h_{b_i} f_{c_i}$.
$b_1 = 4, a = 2$ or 3, any $k \geq 1$	every L;	$\bar{f}_a \prod_{i=1}^{k} h_{b_i} f_{c_i}$.
$a = 2, b_1 = 3, k \geq 2$	every L;	$\bar{f}_2 f_3 h_{c_1} \prod_{i=2}^{k} h_{b_i} f_{c_i}$.
exact (2, 3, 4)	every L;	$\bar{f}_2 f_3 h_4$.
exact (2, 3, c), $c \geq 5$	$L = 1$;	$\bar{f}_2 h_3 f_c$.
exact (2, 3, c), $c \geq 5$	$L \geq 2$;	$\bar{f}_2 f_3 h_c$.

Each row gives at most $\exp(-d/27) \leq 1 - d/28 \leq Q(d)$. Since $b_1 > a \geq 2$, b_1 is either 3, 4 or at least 5; the value 3 forces $a = 2$. The last three rows split the remaining k and c cases. Thus all minima at least two and all positive L are covered. In particular the $L = 1$ direction for (2, 3, $c \geq 5$) is not changed to the $L \geq 2$ direction.

Q106.3.2 Minimum $a = 1$

In the following table, later pairs mean the displayed increasing pairs, and every row uses all $L \geq 1$ unless a connector is specified.

Table 55: The unit ordinary terminal catalogue: parameter values.

Entry	Connector	Actual F	Reference
$b_1 \geq 4$, any $k \geq 1$	every L;	$q \prod_{i=1}^{k} h_{b_i} f_{c_i}$;	Section O.31.
$b_1 = 3, k \geq 2$	every L;	$q f_3 h_{c_1} \prod_{i=2}^{k} h_{b_i} f_{c_i}$;	Section O.31.
exact (1, 3, 4)	every L;	$q f_3 h_4$;	Section O.31.
exact (1, 3, c), $c \geq 5$	every L;	$q f_3 h_c$;	Section O.25.
$b_1 = 2, k \geq 3$	every L;	$q f_2 h_{c_1} \prod_{i=2}^{k} h_{b_i} f_{c_i}$;	Section O.31.
exact (1, 2, c, b, e), $4 \leq c < b < e$	every L;	$q f_2 h_c h_b f_e$;	Section O.25.
exact (1, 2, 3, b, e), $5 \leq b < e$	every L;	$q f_2 h_3 h_b f_e$;	Section O.25.
exact (1, 2, 3, 4, e), $e \geq 5$	every L;	$q f_2 h_e h_3 f_4$;	Section O.25.
exact (1, 2, c), $c \geq 5$	every L;	$q f_2 h_c$;	Section O.25.
exact (1, 2, 4)	every L;	$q f_2 h_4$;	Section O.26.
exact (1, 2, 3)	$L = 2$;	$q f_2 \bar{f}_3$;	Section O.26.
exact (1, 2, 3)	$L = 1$ or $L \geq 3$;	$q f_2 h_3$;	Section O.26.

The Section O.31 rows give exp(-d), exp(-d/2), or 1-d/601, all at most Q(d). Section O.25 gives 1-d/601, 1-d/12000, exp(-d/2), or exp(-8d), as recorded in its local proof. Section O.26 gives 1-d/601 for the complete 123 table, and 1-d/12000 for the complete 124 table. Thus every row supplies (Q106.1.2).

For exhaustion, b_1 is 2, 3 or at least 4. When it is 3, k is 1 or at least 2; $k = 1$ has $c_1 = 4$ or $c_1 \geq 5$. When $b_1 = 2$, first split $k \geq 3$, $k = 2$, and $k = 1$. With $k = 2$, either $c_1 \geq 4$; or $c_1 = 3$ and $b_2 \geq 5$; or $c_1 = 3$, $b_2 = 4$. With $k = 1$, c_1 is 3, 4 or at least 5. All lengths in these partitions are original integers. In the crossed five-arm row each of the lengths 1, 2, 3, 4, e is used exactly once. There is no artificial pair or omitted arm. The Section O.26 $L = 2$ row uses $\bar{f}_3$, not h_3.

Q106.3.3 Genuine decorations, reversal, and degenerate kernels

The moment statements include actual equal pairs. Alternatively positivity gives $E_r pT^L(E_t F) \leq pT^L F$, and hence (Q106.1.2) from the undecorated row. This comparison retains all original pair edges in the orientation. After removal of zero-mass atoms, p, $q \geq \mu(x) > 0$, so inverse q powers below are defined. If $d = 0, g = 0$ but K need not be zero; original leaf integration gives every actual message one. All bounds then hold with equality.

Applying the same fixed rule to -K reverses every original arm, connector and pair and exchanges A/T and p/q. This gives the full reversed version.

parts Q106.2 and 3 prove Theorem A from exact local statements.

Q106.4 The complete internal-root transfers

The theorem of Section M.16 applies to every real Y on the original space:

$$I_q(T^M(\bar{f}_a Y))^3 \leq (1 - d/72)E|Y|^8, \qquad a \geq 2, \quad M \geq 1. \tag{Q106.4.1}$$

Substitute the actual Y from (Q106.1.2). This proves Theorem B with no restriction on Sigma besides its at least two residual arms. The root's original a-arm provides exactly $\bar{f}_a$, and its original unit arm provides p. Thus (Q106.4.1) is applied in its actual operator order and with no added reference arm. All original L and M are independent.

For Theorem C, the all-real map of Section M.19 for every $M \geq 3$ is

$$I_q(T^M Y)^3 \leq (1 - d)E|Y|^8. \tag{Q106.4.2}$$

Substituting (Q106.1.2) proves (Q106.1.4). The proof covers $M \geq 3$. For $M = 1, 2$ the Section M.19 costs need not leave enough loss in $Q(d)$ for a contraction.

For a half-tree of type B put $B_p air = (1 - d/72)Q(d)$, and for type C put $B_u nit = (1 - d)Q(d)$. Both outputs satisfy $I_q(U)^3 \leq B_t ype \leq 1$.

O.6 The ordinary eighth moment of the unit short-one connector

On any nonempty finite probability space (X, μ), let K be real and antisymmetric with $|K(x, y)| \leq 1$. Delete zero-mass atoms and use the same μ for all operators and inner products. Set

$$Jv = (\mathbb{E}v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*,$$
$$g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \quad r = Kg, \quad s = K^2 g,$$
$$d = \mathbb{E}g^2, \qquad \delta_2 = \mathbb{E}r^2, \qquad \delta_3 = \mathbb{E}s^2, \qquad \theta = 13/32.$$

The finite spectral input in Section M.1 gives

$$0 \leq d \leq 1/3, \qquad \delta_2 \leq \theta d, \qquad \delta_3 \leq \theta\delta_2 \leq \theta^2 d. \tag{Q49.1.1}$$

The actual terminal and unit-connector messages are

$$f_2 = A^2\mathbf{1} = p + r, \qquad F = qf_2, \qquad H = pTF = pT(qf_2). \tag{Q49.1.2}$$

The terminal unit arm points inward, its two-edge arm points outward, the connector points from the terminal to the receiving root, and the receiving unit arm points outward. These directions are fixed by the tree. We prove, for every such host,

$$\mathbb{E}H^8 \le 1 - d/500. \tag{Q49.1.3}$$

Complete reversal gives

$$\mathbb{E}[\{qA(p\bar{f}_2)\}^8] \le 1 - d/500, \qquad \bar{f}_2 = T^2\mathbf{1}. \tag{Q49.1.4}$$

Genuine equal pairs can be restored at their roots using $f_\ell h_\ell \le 1$ and positivity.

Q49.2 A row bound for K^2

For every real v with $|v| \le 1$,

$$\|Kv\|_2^2 \le 1/3. \tag{Q49.2.1}$$

To see this, for a legal antisymmetric kernel L, put $a = L(x,y)$, $b = L(y,z)$, $c = L(z,x)$. Pointwise,

$$1 + ab + bc + ca = \frac{(1+a)(1+b)(1+c) + (1-a)(1-b)(1-c)}{2} \ge 0.$$

Integrating over three independent labels with measure μ, each quadratic term equals $-\mathbb{E}(L\mathbf{1})^2$. Hence

$$\mathbb{E}(L\mathbf{1})^2 \le 1/3. \tag{Q49.2.2}$$

For $v \in \{-1, 1\}$, the kernel $L(x,y) = v(x)K(x,y)v(y)$ is legal and $L\mathbf{1}(x) = v(x)Kv(x)$, proving (Q49.2.1). For general $|v| \le 1$, choose independent signs $\sigma(x)$ at the finitely many atoms, with $\mathbb{E}_\sigma\sigma(x) = v(x)$. Convexity gives

$$\|Kv\|_2^2 = \|\mathbb{E}_\sigma K\sigma\|_2^2 \le \mathbb{E}_\sigma\|K\sigma\|_2^2 \le 1/3.$$

For a fixed x, let $k_x(y) = K(x,y)$. Since $|k_x| \le 1$, for every real w,

$$K^2w(x) = \langle k_x, Kw\rangle = -\langle Kk_x, w\rangle, \qquad |K^2w(x)|^2 \le \|Kk_x\|_2^2\|w\|_2^2 \le \|w\|_2^2/3. \tag{Q49.2.3}$$

In particular,

$$|s(x)|^2 \le d/3. \tag{Q49.2.4}$$

Equation (Q49.2.1) is an L^∞-to-L^2 estimate, while (Q49.2.3) is a row estimate for K^2. The spectral bound used below remains $\theta = 13/32$.

Q49.3 A positive square envelope

Since $q, f_2 \ge 0$ and $q + f_2 = 2 + r$, AM–GM gives $F = qf_2 \le (1 + r/2)^2 = 1 + r + r^2/4$. Using $\mathbb{E}r = -d$, $Kr = s$, and $0 \le T(r^2) \le 2\mathbb{E}r^2 = 2\delta_2$,

$$TF \le q - d - s + \delta_2/2 \le q - (51/64)d - s. \tag{Q49.3.1}$$

The last inequality uses $\delta_2 \le 13d/32$, and its right side is nonnegative because it dominates TF. Put $a = 51/128$ and $c = 1 - ad$. AM–GM for p and the nonnegative upper in (Q49.3.1) gives

$$0 \le H = pTF \le [1 - (51/128)d - s/2]^2 = (c - s/2)^2. \tag{Q49.3.2}$$

Moreover,

$$c \ge 111/128 > 0, \qquad |s|/2 \le \sqrt{d/12} \le 1/6 < c. \tag{Q49.3.3}$$

Skewness gives $\mathbb{E}s = \langle \mathbf{1}, K^3\mathbf{1}\rangle = 0$, and

$$\mathbb{E}s^2 = \delta_3 \le \theta^2 d. \tag{Q49.3.4}$$

Q49.4 The scalar sixteenth-moment estimate

At $d = 0$, $g = r = s = 0$ and the actual messages p, q, f_2, F, H are one. For $d > 0$, set

$$x = \frac{\sqrt{d/12}}{c}, \qquad \eta = -\frac{s}{2c}, \qquad \alpha = 3\theta^2 = \frac{507}{1024}, \qquad k = 12a = \frac{153}{32}, \qquad w = \frac{1}{c}.$$

The function $\sqrt{d}/(1 - ad)$ increases, so

$$0 < x \le \frac{1/6}{111/128} = \frac{64}{333} < \frac{1}{5}. \tag{Q49.4.1}$$

The row and second-moment bounds give

$$|\eta| \le x, \qquad \mathbb{E}\eta = 0, \qquad \mathbb{E}\eta^2 \le \alpha x^2. \tag{Q49.4.2}$$

For every integer $j \ge 2$, $\eta^j \le x^{j-2}\eta^2$: for even j use absolute values; for odd j and negative η, the left side is nonpositive. The binomial expansion therefore gives

$$\mathbb{E}(1 + \eta)^{16} \le 1 + \alpha[(1 + x)^{16} - 1 - 16x]. \tag{Q49.4.3}$$

The exact relation $1 - c = ad$ implies

$$w^2 - w = kx^2. \tag{Q49.4.4}$$

Put $v = w - 1 \ge 0$ and $t = kx^2$. Then $w - v = 1$, $wv = t$, and the two roots $w, -v$ satisfy $z^2 = z + t$. Thus

$$\begin{aligned} w^{16} + v^{16} = 1 + 16t + 104t^2 + 352t^3 + 660t^4 \\ + 672t^5 + 336t^6 + 64t^7 + 2t^8. \end{aligned} \tag{Q49.4.5}$$

For example, the finite identity follows from $S_0 = 2$, $S_1 = 1$, $S_n = S_{n-1} + tS_{n-2}$, where $S_n = w^n + (-v)^n$. Since $v^2 \le wv = t$, we have $v^{16} \le t^8$; retaining the first three positive terms gives

$$w^{16} \ge 1 + 16kx^2 + 104k^2x^4 + 352k^3x^6. \tag{Q49.4.6}$$

For $0 \le x \le 1/5$, the ratio of consecutive binomial terms after degree five is at most $((16-5)/(5+1))x \le 11/30$. Hence

$$\sum_{j=5}^{16} \binom{16}{j} x^j \le \frac{4368}{1 - 11/30} x^5 = \frac{131040}{19} x^5 < 7000x^5. \tag{Q49.4.7}$$

Combining (Q49.4.6) and (Q49.4.7) yields

$$w^{16} - 1 - \alpha[(1+x)^{16} - 1 - 16x] \geq x^2[A - Bx + Cx^2 - Dx^3 + Ex^4]. \tag{Q49.4.8}$$

Here A, B, C, D, E are scalar coefficients:

$$\begin{aligned} A &= 16k - 120\alpha = 2187/128 > 17, \\ B &= 560\alpha = 17745/64 < 278, \\ C &= 104k^2 - 1820\alpha = 377949/256 > 1476, \\ D &= 7000\alpha = 443625/128 < 3500, \\ E &= 352k^3 = 39397347/1024 > 38000. \end{aligned}$$

Therefore

$$w^{16} - 1 - \alpha[(1+x)^{16} - 1 - 16x] \geq x^2[17 - 278x + 1476x^2 - 3500x^3 + 38000x^4]. \tag{Q49.4.9}$$

The polynomial in brackets is

$$(17 - 278x + 1152x^2) + x^2(324 - 3500x + 38000x^2).$$

Completing the first square gives

$$17 - 278x + 1152x^2 = 1152(x - 139/1152)^2 + 263/1152 \geq 263/1152. \tag{Q49.4.10}$$

The second bracket is positive for every real x, since

$$3500^2 = 12250000 < 49248000 = 4 \cdot 324 \cdot 38000. \tag{Q49.4.11}$$

Consequently, on the entire interval,

$$w^{16} - 1 - \alpha[(1+x)^{16} - 1 - 16x] \geq (263/1152)x^2. \tag{Q49.4.12}$$

Q49.5 The uniform deficit

Using the positive square envelope,

$$\mathbb{E}H^8 \leq c^{16}\{w^{16} - (263/1152)x^2\} = 1 - (263/13824)c^{14}d. \tag{Q49.5.1}$$

Since $111 \cdot 15 = 1665 > 1664 = 128 \cdot 13$, we have $c > 13/15$. Furthermore,

$$(13/15)^7 = \frac{62748517}{170859375} > \frac{4}{11}, \qquad 11 \cdot 62748517 = 690233687 > 683437500 = 4 \cdot 170859375.$$

Thus $c^{14} > 16/121 > 1/8$, and

$$\mathbb{E}H^8 \leq 1 - (263/110592)d \leq 1 - d/500, \tag{Q49.5.2}$$

because $263 \cdot 500 = 131500 > 110592$. This proves (Q49.1.3). The conclusion allows one-atom and unequal-mass spaces, zero edge factors, and nonzero balanced kernels at $d = 0$. At positive-mass atoms, $p, q > 0$; the argument uses division only by the positive scalar c.

O.7 Unit-root eighth moments for the terminal $(1, 2)$

Use the original finite probability measure, $A = J + K$, $T = J - K$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $r = Kg$, $s = Kr$, $t = Ks$, and $d = \|g\|^2$, $\delta_j = \|K^j\mathbf{1}\|^2$. Write $\theta = 13/32$, $\kappa = 19/32$, $\tau = d - \delta_2$, and $\gamma = d - \delta_2/2$. Lemma D.1, skew-adjointness, and the row bound give

$$0 \le d \le 1/3, \qquad d^2 \le \delta_2 \le \theta d, \qquad \delta_{j+1} \le \theta\delta_j, \qquad |Kv| \le \|v\|.$$

All connector edges point from the terminal toward the receiving root, whose original unit arm points outward. The terminal unit arm points inward and its two-edge arm outward. Thus the actual message is

$$F = qf_2, \qquad X_L = T^L F, \qquad H_L = pX_L. \tag{Q50.0.1}$$

We shall prove the stronger estimate $\mathbb{E}H_L^8 \le 1 - d$ for $L \ge 2$; in particular,

$$\mathbb{E}H_L^8 \le 1 - d/50 \qquad (L \ge 2). \tag{Q50.0.2}$$

The case $L = 1$ is Section O.6. When $d = 0$, leaf integration makes every tree message one. Assume $d > 0$ below.

A common unit-root payment.

Suppose a nonnegative message satisfies

$$H \le B\left(1 + \frac{Z - a}{2B}\right)^2, \qquad |Z| \le U, \quad \mathbb{E}Z \le -\alpha, \quad \mathbb{E}Z^2 \le v. \tag{O.25}$$

Here $B > 0$, $a, \alpha \ge 0$, $\alpha^2 \le v \le U^2$, and $2B - a > U > 0$. Apply Lemma P.1 with exponent eight, the variable $Z/\sqrt{B}$, and center $(B - a/2)/\sqrt{B}$. The compatible double contact and upper probability are

$$\ell = -\frac{v + \alpha U}{U + \alpha}, \qquad \pi = \frac{v - \alpha^2}{U^2 + 2\alpha U + v}.$$

Consequently,

$$\mathbb{E}H^8 \le \frac{(1 - \pi)(B - a/2 + \ell/2)^{16} + \pi(B - a/2 + U/2)^{16}}{B^8}. \tag{O.26}$$

At $v = U^2$, the same Hermite quadratic has contacts $-U, U$, with upper probability $(U - \alpha)/(2U)$, so the displayed formula also holds. This uses the moments of the same Z.

Here are five continuous payments for this formula. Put $c = 51/64$, $\beta = 89/64$, $k = 19/36$, and

$$v_3(d) = \theta(1 - cd)^2 + \theta^2\kappa^2 - 2\kappa^2\theta d + \kappa^2 d^2.$$

The table writes $U = tu$, $\alpha = da_0$, $v = dv_0$, where $d = t^2$. Every row implies $\mathbb{E}H^8 \le 1 - d$.

The single polynomial certificate for each row can be reconstructed as follows:

$$C = B - a/2, \quad D = u + ta_0, \quad E = u^2 + 2tua_0 + v_0, \quad N = v_0 + ta_0u, \quad R = v_0 - t^2a_0^2,$$
$$\mathcal{G}(t) = \frac{2^{16}D^{14}EB^8(1 - t^2) - (2CD - tN)^{16} - RD^{14}(2C + tu)^{16}}{t^2}.$$

The numerator is divisible by t^2, and $\mathcal{G}(0) > 1$ in every row. The denominator of the bound (O.26) is $2^{16}D^{14}EB^8 > 0$. On the certificate intervals in the next table, every Bernstein coefficient of $\mathcal{G}/\mathcal{G}(0)$ exceeds $1/4000$. The six guard polynomials

$$B, \quad u, \quad a_0, \quad v_0 - t^2a_0^2, \quad u^2 - v_0, \quad 2B - a - tu$$

Table 56: Joint moment data for the unit $(1,2)$ terminal.

Row	B	a	a_0	u	v_0	Actual d
I	$1-cd$	0	κ	$1-cd$	$4/27$	$[0,1/4]$
II	$1-\beta d$	0	$1-cd$	$\frac{43}{40}(1-\frac{4}{5}d)$	$v_3(d)$	$[0,1/3]$
III	$1-\beta d$	$d(1-d)$	$\frac{777}{1024}-d$	1	$\theta(1-\frac{153}{128}d)$	$[0,1/3]$
IV	$1-3kd$	$kd(1-d)$	$k(1-2d)$	1	$1/4$	$[0,1/3]$
V	$1-3\kappa d$	$\kappa d(1-d)$	$\kappa(1-2d)$	1	$1/3$	$[0,1/3]$

each have Bernstein coefficients greater than $1/32$ on the same interval. In particular all the Hermite hypotheses hold strictly when $t>0$. The coefficients follow from (A.18). Since $(7/12)^2>1/3$, these

Table 57: Whole-interval certificates for Table 56.

Row	Interval for t	Degree of $\mathcal{G}$	Coefficient count
I	$[0,1/2]$	76	77
II	$[0,7/12]$	93	94
III–V	$[0,7/12]$	107	108 each

rational intervals cover every actual d-domain. Lemma A.1 now proves the five payments, including the limiting endpoint $d=0$.

The finite terminal envelope.

The arm identity and positivity give

$$qf_2=(1+r/2)^2-(g+r/2)^2,\qquad TF\le q-\gamma-s.$$

Put $B_0=-K$, $t_j=B_0^j\mathbf{1}$, $S'_L=\sum_{j=0}^{L}t_j$, and $\bar f_n=T^n\mathbf{1}$. Let $\beta_1=\beta_2=0$ and $\beta_j=\sum_{i=1}^{\lfloor (j-1)/2\rfloor}(-1)^{i-1}\delta_i$ for $j\ge 3$; alternating sums give $\tau\le\beta_j\le d$. Apply T^{L-1} to the one-step bound above. The finite convolution (M.2), or direct finite Duhamel expansion, gives for every $L\ge 1$

$$\begin{aligned}X_L\le \mathcal{W}_L&:=S'_L+t_{L+2}-\gamma\bar f_{L-1}-\sum_{j=3}^{L}\omega_j\bar f_{L-j},\\ \omega_j&=\begin{cases}\beta_j, & j\text{ even},\\ \beta_{j+2}, & j\text{ odd},\end{cases}\qquad \omega_j\ge\tau.\end{aligned}\tag{O.27}$$

Only nonnegative indices occur. Indeed $T^Lr=t_{L+2}-d\bar f_{L-1}+\delta_2\bar f_{L-3}-\delta_3\bar f_{L-5}+\cdots$, a finite identity. All retained path losses are nonnegative.

Two and three connector edges.

For $L=2$, $X_2\le(1-\gamma)q+r+t$. At $d\le 1/4$, weaken $\gamma\ge cd$ and use $Z=r+t$. Its mean is $-d+\delta_2\le-\kappa d$, and

$$\mathbb{E}Z^2=\delta_2-2\delta_3+\delta_4\le 4d/27,\qquad |Z|^2\le d-2\delta_2+\delta_3\le d(1-2cd).$$

The first bound follows from $4/27-x(1-x)^2=(x-1/3)^2(4/3-x)\ge 0$. For the row bound use $Z=K(g+s)$, center the row, and note that $g+s$ is centered. Since $\sqrt{1-2cd}\le 1-cd$, these are row I. For $d\ge 1/4$, $TF\le 2\mathbb{E}F=2-4d\le 1$, hence $H_2\le(2-4d)pq\le 1$ and $\mathbb{E}H_2^8\le\mathbb{E}(pq)^8\le 1-d$.

For $L = 3$, expand (O.27) to obtain

$$X_3 \le (1-\beta d)q + Z_3, \qquad Z_3 = (1-cd)r - s - K^4 g - \kappa d g, \qquad \mathbb{E}Z_3 = -d(1-cd).$$

Writing $A_0 = 1 - cd$, the row and spectral bounds are

$$|Z_3|^2 \le dD_3(d), \quad D_3 = A_0^2 + 4/27 - \kappa^2 d, \qquad \mathbb{E}Z_3^2 \le dv_3(d).$$

For the row estimate use $Z_3 = (-K)(-A_0 g + r + t + \kappa d)$: its squared predecessor norm is at most $A_0^2 d + 4d/27 - \kappa^2 d^2$, since $\mathbb{E}(r+t) = -\tau$ and $\tau \ge \kappa d$. For its second moment the exact spectral integrand is $A_0^2 x + [x(1-x) - \kappa d]^2$. Its derivative is at least $A_0^2 - 2\kappa d > 0$, so its value at $x = \theta$ gives v_3. Finally $D_3 \le (43/40)^2(1-4d/5)^2$: the three coefficients of the difference are $1849/1600 - 31/27$, $1993/1024 - 1849/1000$, and $1849/2500 - 2601/4096$, all positive. Thus row II applies.

Four connector edges.

Set $R_L = \sum_{j=2}^{L} t_j + t_{L+2}$. Expansion of the same finite envelope gives

$$X_4 \le (1-\tau-\gamma)q + Z_4 - d(1-\gamma), \qquad Z_4 = R_4 - \gamma r + \gamma s.$$

Here $Z_4 = B_0 Q(B_0) g'$, $g' = B_0 \mathbf{1}$, $Q = P_3 - \gamma(1+z)$, $P_3 = 1 + z + z^2 + z^4$. At $z = i\sqrt{x}$,

$$|P_3|^2 = 1 - x + x^2(3 - 2x + x^2) \le 1, \quad |Q|^2 \le 1 - 2\gamma + (1+\theta)\gamma^2 \le 1.$$

The first inequality uses $1 - 3x + 2x^2 - x^3 \ge 1451/32768 > 0$. Consequently $|Z_4| \le \sqrt{d}$, and

$$\mathbb{E}Z_4^2 \le \theta d(1 - 153d/128), \qquad \mathbb{E}Z_4 \le -(777/1024)d + d^2.$$

For the variance use $\gamma \ge cd$, $(1+\theta)cd \le 1/2$; for the mean use $\tau + \delta_3 \ge d[1 - u + u^2] \ge (777/1024)d$, where $u = \delta_2/d \le \theta$ and $\delta_3 \ge \delta_2^2/d$. Weakening only positive path losses gives $X_4 \le (1-\beta d)q + Z_4 - d(1-d)$, namely row III.

Five edges and all longer connectors.

At $L = 5$, retain the four losses already present in (O.27). Expanding $\bar{f}_4 = (1-d)q - d + r - s + t$ gives

$$X_5 \le [1 - (3-d)\tau]q + Z_5 - \tau(1-d), \qquad Z_5 = R_5 - 2\tau r + \tau s - \tau t.$$

Its spectral polynomial is $Q = P_4 - \tau(2 + z + z^2)$, $P_4 = 1 + z + z^2 + z^3 + z^5$. The identities

$$|P_4|^2 \le 1 - x, \quad \Re[P_4 \overline{(2 + z + z^2)}] = 2 - 2x + x^3 \ge 19/16, \quad |2 + z + z^2|^2 \le 4$$

and $\tau \le 2/9$ imply $|Q|^2 \le |P_4|^2$. Thus $|Z_5| \le \sqrt{d}$, $\mathbb{E}Z_5^2 \le d/4$, and $\mathbb{E}Z_5 = -\tau + 2\tau d - \tau\delta_2 \le -\tau(1-2d)$. Since $(3-d)\tau \ge 3kd$, row IV applies.

For $L \ge 6$, retain the last four losses, whose indices are all at least three. They give

$$X_L \le (1-3\tau)q + Z_L - \tau(1-d), \qquad Z_L = R_L - 2\tau r + \tau s. \tag{O.28}$$

For $m = L - 1 \ge 5$, put $P_m = G_m + z^{m+1}$, $G_m = \sum_{j=0}^{m-1} z^j$, and $Q_m = P_m - \tau(2+z)$. Direct multiplication gives

$$|P_m|^2 = \frac{1 + x^m(1 - x + x^2) - 2\Re[z^m(1 - x - z)]}{1 + x}.$$

For $m \equiv 0,1 \pmod 4$, the subtracted terms are respectively $2x^{m/2}(1-x)$ and $2x^{(m+1)/2}$; both dominate $x^m(1-x+x^2)$. For the other residues the numerator is at most $1+2x^3+x^6$ or $1+2x^4+x^7$. These estimates yield $|P_m| \le 1$, $x|P_m|^2 \le 1/3$. For the latter bound in the last two cases it suffices to use $2x+6x^4+3x^7 \le 2269/2304 < 1$. Also $\Re[P_m\overline{(2+z)}] \ge 2-3x/(1-x^2) \ge 154/285$. Together with $\tau(4+x) \le 47/48 < 308/285$, this proves $|Q_m|^2 \le |P_m|^2$. Therefore

$$|Z_L| \le \sqrt{d}, \qquad \mathbb{E}Z_L^2 \le d/3, \qquad \mathbb{E}Z_L \le -\tau(1-2d).$$

For the last bound $\mathbb{E}R_L = -\beta_{L+3}$ when L is even, and $-\beta_{L+1}$ when L is odd. Replacing τ by κd only on the positive path factors in (O.28) gives row V. In each of the five cases, multiplying the nonnegative envelope $X_L \le Bq + Z - a$ by p and applying AM–GM to Bp and $Bq + Z - a$ proves (O.25). This completes the proof.

A scalar estimate used by the neighbouring terminal family.

We retain the elementary bound

$$(1+x)^{16} \le 1 + 16x + 256x^2 \qquad (|x| \le 1/7). \tag{Q50.2.1}$$

On the negative half-interval Taylor's quadratic coefficient is at most 120; on the positive half the quotient is increasing, and at $1/7$ the assertion is $8^{16} \le 417\,7^{14}$, whose margin is $1344044667377 > 0$.

Finally, actual equal-pair factors $E_r, E_t \le 1$ satisfy $0 \le E_r p T^L (E_t q f_2) \le H_L$ by positivity. Full reversal replaces K by $-K$ and reverses every arm and connector. Together with Section O.6, the catalogue for all positive lengths retains

$$\mathbb{E}H_L^8 \le 1 - d/500 \qquad (L \ge 1). \tag{Q50.8.1}$$

O.8 Unit connectors for every short-one terminal

Work on the original finite probability space with a real antisymmetric kernel $|K| \le 1$. Put

$$A = J + K, \quad T = J - K, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g,$$

$$r = Kg, \quad s = K^2 g, \quad t = K^3 g, \quad d = \mathbb{E}g^2, \quad \delta_j = \|K^j \mathbf{1}\|_2^2, \quad \theta = 13/32, \quad f_b = A^b \mathbf{1}.$$

For each integer $b \ge 2$, use $F_b = q f_b$ and $H_b = pTF_b$. The terminal unit arm points inward, its b-edge arm outward, the unit connector toward the receiving root, and the receiving unit arm outward. We prove

$$\mathbb{E}H_b^8 \le 1 - d/500. \tag{Q51.1.1}$$

Genuine equal-length pairs may be restored at their roots by positivity. The case $b = 2$ is Section O.6. Its row estimate (Q49.2.3) applies to K^2 on every real input; the spectral input remains $\|K\|^2 \le \theta$.

Q51.2 The signed-mean scalar input

Let $0 \le d \le 1/3$, and suppose the real function Z satisfies

$$\mathbb{E}Z \ge 0, \qquad |Z|^2 \le d/3, \qquad \mathbb{E}Z^2 \le \theta^2 d. \tag{Q51.2.1}$$

For $c = 1 - 51d/128$,

$$\mathbb{E}(c - Z/2)^{16} \le 1 - d/500. \tag{Q51.2.2}$$

At $d = 0$, $Z = 0$. For $d > 0$, take $x = \sqrt{d/12}/c$, $\eta = -Z/(2c)$, and $\alpha = 507/1024$, as in Section O.6. Now $\mathbb{E}\eta \le 0$, so the linear term $16\mathbb{E}\eta$ in the binomial expansion is nonpositive. The same termwise estimate gives

$$\mathbb{E}(1+\eta)^{16} \le 1 + \alpha[(1+x)^{16} - 1 - 16x]. \tag{Q51.2.3}$$

The scalar comparisons (Q49.4.4)–(Q49.5.2) use only c, d, x. In particular,

$$x \le 64/333 < 1/5, \qquad w = 1/c, \qquad w^2 - w = (153/32)x^2,$$
$$w^{16} - 1 - \alpha[(1+x)^{16} - 1 - 16x] \ge (263/1152)x^2, \qquad c^{14} > 1/8,$$
$$(263/1152)c^{16}x^2 \ge (263/110592)d \ge d/500.$$

Together with (Q51.2.3), this proves (Q51.2.2) with the stated one-sided mean condition.

Q51.3 Finite positive convolution

Define

$$\beta_1 = \beta_2 = 0, \qquad \beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1}\delta_i \quad (j \ge 3), \qquad S_b = \sum_{j=0}^{b} K^j \mathbf{1}.$$

Apply Lemma M.1 after full reversal, $(B, u, T, \bar{f}_n) \mapsto (K, g, A, f_n)$. Its recursion gives

$$1 - \mathbb{E}S_{b-1} = \beta_b, \qquad S_b = AS_{b-1} + \beta_b. \tag{Q51.3.1}$$

and (M.2) gives

$$S_b = f_b + \sum_{j=3}^{b} \beta_j f_{b-j}. \tag{Q51.3.2}$$

with

$$\tau := d - \delta_2 \le \beta_j \le d \qquad (j \ge 3). \tag{Q51.3.3}$$

Retaining the $j = b$ term, since $f_\ell \ge 0$, yields

$$f_b \le S_b - \beta_b. \tag{Q51.3.4}$$

Q51.4 Every $b \ge 4$

Fix $b \ge 4$ and set

$$R = \sum_{j=2}^{b} K^j \mathbf{1}, \qquad a = 1 - \beta_b/2, \qquad t_0 = \beta_{b+1} = -\mathbb{E}R, \qquad V = \mathbb{E}R^2, \qquad Z = KR.$$

Equation (Q51.3.4) gives $q + f_b \le 2 - \beta_b + R = 2a + R$. AM–GM therefore yields $F_b = qf_b \le (a + R/2)^2$. Using positivity and $T(R^2) \le 2V$, we obtain

$$TF_b \le a^2 q - at_0 - aZ + V/2. \tag{Q51.4.1}$$

Its right side is nonnegative. Apply AM–GM to ap and that right side divided by a:

$$H_b \le [a - t_0/2 - Z/2 + V/(4a)]^2. \tag{Q51.4.2}$$

The bracket is half the sum of these two nonnegative quantities.

For $G_N(z)=\sum_{j=0}^{N-1} z^j$, every $N\ge 3$ satisfies

$$|G_N(i\sqrt{x})|^2\le 1 \qquad (0\le x\le \theta). \tag{Q51.4.3}$$

At $x=0$, this is direct. For odd N, the ratio is $(1+x^N)/(1+x)\le 1$. For $4\mid N$, the numerator is $(1-x^{N/2})^2\le 1$, with denominator $1+x$. For $N\equiv 2 \pmod 4$, $N\ge 6$, the required inequality is $2x^{N/2-1}+x^{N-1}\le 1$. Its left side is at most $2\theta^2+\theta^5<1/2<1$, because $\theta^2=169/1024<1/6$ and $\theta^5<\theta^2<1/6$. The finite-dimensional spectral theorem gives $\|G_N(K)\|\le 1$.

Since $R=KG_{b-1}(K)g$ and $b-1\ge 3$,

$$V\le \theta d, \qquad \mathbb{E}Z^2=\|KR\|_2^2\le \theta^2 d, \qquad |Z(x)|^2=|K^2G_{b-1}(K)g(x)|^2\le d/3. \tag{Q51.4.4}$$

The final bound uses (Q49.2.3). Moreover, $\mathbb{E}Z=\mathbb{E}KR$ is the nonempty alternating sum beginning with δ_2, so

$$\delta_2-\delta_3\le \mathbb{E}Z\le \delta_2, \qquad \mathbb{E}Z\ge 0. \tag{Q51.4.5}$$

Using $\beta_b, t_0\ge \tau\ge 19d/32$, $a\ge 1-d/2\ge 5/6$, and (Q51.4.4),

$$a-t_0/2+V/(4a)\le 1-\tau+(3/10)\theta d\le 1-(151/320)d\le 1-(51/128)d=:c. \tag{Q51.4.6}$$

The last coefficient difference is $151/320-51/128=47/640>0$. Enlarging the nonnegative bracket in (Q51.4.2) gives

$$0\le H_b\le (c-Z/2)^2. \tag{Q51.4.7}$$

The conditions (Q51.4.4)–(Q51.4.5) allow (Q51.2.2), proving (Q51.1.1) for every $b\ge 4$.

Q51.5 The remaining arm length $b=3$

Here $f_3=p-d+r+s$, so

$$q+f_3=2-d+r+s=2a+R, \qquad a=1-d/2, \quad R=r+s, \quad V=\delta_2+\delta_3, \quad Z=KR=s+t.$$

Positivity and the same AM–GM step give

$$H_3\le [a-d/2-Z/2+V/(4a)]^2. \tag{Q51.5.1}$$

We have $\mathbb{E}R=-d$, $\mathbb{E}(rs)=0$, $a\ge 5/6$, and $V\le \theta(1+\theta)d$. Thus

$$a-d/2+V/(4a)\le 1-d+(3/10)\theta(1+\theta)d=1-(1697/2048)d\le 1-(4/5)d=:c. \tag{Q51.5.2}$$

The last coefficient difference is $1697/2048-4/5=293/10240>0$. Since the original bracket is nonnegative,

$$0\le H_3\le (c-Z/2)^2. \tag{Q51.5.3}$$

Skewness gives $\mathbb{E}(gr)=0$ and $\mathbb{E}(st)=0$. The row bound yields

$$\begin{gathered}\mathbb{E}Z=\delta_2\ge 0, \qquad |Z(x)|^2=|K^2(g+r)(x)|^2\le (d+\delta_2)/3\le Bd,\\ \mathbb{E}Z^2=\delta_3+\delta_4\le \theta^2(1+\theta)d=\alpha Bd,\\ B=(1+\theta)/3=15/32, \qquad \alpha=3\theta^2=507/1024.\end{gathered} \tag{Q51.5.4}$$

The case $d=0$ is equality. For $d>0$, put

$$x=\frac{\sqrt{Bd}}{2c}, \qquad \eta=-\frac{Z}{2c}, \qquad w=1/c, \qquad k=\frac{4(4/5)}{B}=\frac{512}{75}.$$

Then $|\eta| \le x$, $\mathbb{E}\eta \le 0$, $\mathbb{E}\eta^2 \le \alpha x^2$, and

$$w^2 - w = kx^2. \tag{Q51.5.5}$$

The variable x increases with d. At $d = 1/3$, $c = 11/15$, giving

$$x^2 \le 1125/15488 < 1152/15488 = (3/11)^2. \tag{Q51.5.6}$$

The binomial terms after degree five have consecutive ratio at most $(11/6)(3/11) = 1/2$. Hence

$$\sum_{j=5}^{16} \binom{16}{j} x^j \le 8736x^5 < 9000x^5. \tag{Q51.5.7}$$

The Lucas identity (Q49.4.5) gives, for this k,

$$w^{16} \ge 1 + 16kx^2 + 104k^2x^4 + 352k^3x^6. \tag{Q51.5.8}$$

Since $k > 34/5$ and $\alpha < 1/2$,

$$\begin{aligned} 16k - 120\alpha &> 16(34/5) - 60 = 244/5 > 48, \\ 560\alpha &< 280, \\ 104k^2 - 1820\alpha &> 104(34/5)^2 - 910 = 97474/25 > 3800, \\ 9000\alpha &< 4500, \\ 352k^3 &> 352(34/5)^3 = 13835008/125 > 110000. \end{aligned}$$

Combining the tail estimate and the Lucas lower bound,

$$w^{16} - 1 - \alpha[(1+x)^{16} - 1 - 16x] \ge x^2[48 - 280x + 3800x^2 - 4500x^3 + 110000x^4]. \tag{Q51.5.9}$$

The bracket equals

$$(48 - 280x + 1000x^2) + x^2(2800 - 4500x + 110000x^2).$$

The first quadratic has minimum $48 - 280^2/4000 = 142/5 > 28$. The second is positive for every real x, because

$$4500^2 = 20250000 < 1232000000 = 4 \cdot 2800 \cdot 110000.$$

Thus the difference in (Q51.5.9) is at least $28x^2$. The binomial expectation bound (Q51.2.3) applies with $\mathbb{E}\eta \le 0$, giving

$$\mathbb{E}H_3^8 \le 1 - 28c^{16}x^2 = 1 - 7Bc^{14}d. \tag{Q51.5.10}$$

Finally $c \ge 11/15 > 2/3$, and $(2/3)^{14} = 16384/4782969 > 1/300$, since $16384 \cdot 300 = 4915200 > 4782969$. Therefore $7Bc^{14} > 7(15/32)/300 = 7/640 > 1/500$. This proves (Q51.1.1) for $b = 3$, and completes the unit-connector catalogue together with $b = 2$ and $b \ge 4$.

O.9 The unit eighth moment for terminal pairs $(2, b)$ on the low domain

Let $A = J + K$, $T = J - K = A^*$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $r = Kg$, $s = Kr$, $t = Ks$, $d = \mathbb{E}g^2$, $\delta_j = \|K^j\mathbf{1}\|_2^2$ and $\theta = 13/32$. The finite calculus gives $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{j+1} \le \theta\delta_j$ and $|Kv(x)| \le \|v\|_2$. For

$$F_b = h_2 f_b, \qquad H_b = pTF_b, \qquad b \ge 4,$$

we prove

$$\mathbb{E}H_b^8 \le 1 - d/500 \quad \begin{cases} b \ge 4, & 0 \le d \le 1/4, \\ b = 4, & 0 \le d \le 1/3. \end{cases} \tag{Q55.0.1}$$

The two-edge arm has message $h_2 = q - r$, the b-arm points outward, the unit connector points from the terminal to the receiving root, and the receiving unit arm points outward. Thus this direction uses $b + 4$ edges. All expectations and operators use the given finite measure. Zero-mass atoms may be deleted; on the remaining atoms $p, q > 0$. At $d = 0$, leaf integration gives every tree message equal to one.

Q55.2 Two scalar estimates

For $|z| \le 1/7$,

$$(1 + z)^{16} \le 1 + 16z + 256z^2. \tag{Q55.2.1}$$

This is (Q50.2.1).

Suppose $C, U > 0$, $C > U/2$, and a real function W satisfies $|W| \le U$, $\mathbb{E}W \le 0$ and $\mathbb{E}W^2 \le \theta U^2$. Then

$$\mathbb{E}(C + W/2)^{16} \le \frac{32}{45}(C - \theta U/2)^{16} + \frac{13}{45}(C + U/2)^{16}. \tag{Q55.2.2}$$

Apply (M.46) with $h = \theta = 13/32$ and $f(z) = (C + z/2)^{16}$. The hypothesis $C > U/2$ gives $f', f'', f''' \ge 0$ on the full interval, and the probabilities are $1/(1 + \theta) = 32/45$, $\theta/(1 + \theta) = 13/45$.

Q55.3 The propagated square

The retained envelope in Section I.3 is $0 \le F_b \le (m + R/2)^2$, where $m = 1 - d > 0$. Write $D_0 = \mathbb{E}R$, $V = \mathbb{E}R^2$, $Z = KR$. Expansion gives

$$T(m + R/2)^2 = m^2 q + m(D_0 - Z) + \tfrac{1}{4}(V - KR^2).$$

Using $|K(R^2)| \le V$, positivity and arithmetic–geometric mean yields

$$0 \le H_b \le p\{m^2 q + m(D_0 - Z) + V/2\} \le \{m + D_0/2 - Z/2 + V/(4m)\}^2. \tag{Q55.3.1}$$

The second factor in braces before the last comparison bounds a positive operator output. After division by m^2, its sum with p is $2 + D_0/m - Z/m + V/(2m^2) \ge 0$, which also establishes the sign of the square base.

Q55.4 Every $b \ge 4$ on $0 < d \le 1/4$

The same retained envelope satisfies

$$D_0 \ge 0, \quad \mathbb{E}Z \ge 0, \quad V \le d(1/2 + d)^2, \quad D_0 \le d(\theta + 11d/16). \tag{Q55.4.1}$$

Hence, with $U = (1/2 + d)\sqrt{d}$,

$$|Z| \le U, \qquad \mathbb{E}Z^2 \le \theta U^2. \tag{Q55.4.2}$$

Put $C = 1 - (47/64)d + (27/32)d^2$. For $0 \le d \le 1/4$,

$$\frac{(1/2+d)^2}{4(1-d)} \le \frac{1}{16} + \frac{d}{2}, \tag{Q55.4.4}$$

because multiplication by $16(1-d)$ leaves the difference $3d(1-4d)$. It follows that $m + D_0/2 + V/(4m) \le C$. The function C decreases, $C \ge 445/512$, and $U \le 3/8$; thus $C > U/2$ and

$$0 \le H_b \le (C - Z/2)^2. \tag{Q55.4.5}$$

If $0 < d \le 1/16$, then $C \ge 15/16$, $U \le 9/64$, and $|Z|/(2C) \le 3/40 < 1/7$. Also,

$$C \le 1 - ad, \quad a = 349/512, \qquad \mathbb{E}Z^2 \le (1053/8192)d.$$

Applying (Q55.2.1) to $-Z/(2C)$ and using $\mathbb{E}Z \ge 0$ gives

$$\mathbb{E}H_b^8 \le C^{16} + 64C^{14}\mathbb{E}Z^2 \le C^{14}(1 + 1053d/128).$$

Since $C^{-14} \ge (1-ad)^{-14} \ge 1 + 14ad$,

$$\mathbb{E}H_b^8 \le 1 - (14a - 1053/128)C^{14}d = 1 - (337/256)C^{14}d.$$

Bernoulli's inequality gives $C^{14} \ge (15/16)^{14} \ge 1/8$, so

$$\mathbb{E}H_b^8 \le 1 - 337d/2048 \le 1 - d/500. \tag{Q55.4.6}$$

For $1/16 \le d \le 1/4$, put $x = \sqrt{d} \in [1/4, 1/2]$. The positive node is

$$G(x) = C + U/2 = 1 + x/4 - 47x^2/64 + x^3/2 + 27x^4/32.$$

Its derivative is $1/4 - 47x/32 + 3x^2/2 + 27x^3/8 > 0$. For $x \le 1/5$, deletion of the cubic term leaves a decreasing quadratic whose value at $1/5$ is $13/800$. For $x \ge 1/5$, the cubic term is at least $27x^2/40$, and the resulting quadratic has minimum $91/44544$. Thus

$$C + U/2 \le 541/512 < 37/35, \qquad (C + U/2)^{16} < 625/256. \tag{Q55.4.7}$$

Here $(37/35)^4 < 5/4$ follows from $4(37^4 - 35^4) = 1494144 < 1500625 = 35^4$. The negative node $C - \theta U/2$ decreases; at $d = 1/16$ it is $7609/8192 < 15/16$. Since $(15/16)^4 < 7/9$ and $(7/9)^4 < 3/8$, using $455625 < 458752$ and $19208 < 19683$, we obtain

$$(C - \theta U/2)^{16} < 3/8. \tag{Q55.4.8}$$

Equations (Q55.2.2), (Q55.4.7) and (Q55.4.8) imply

$$\mathbb{E}H_b^8 < \frac{32}{45}\frac{3}{8} + \frac{13}{45}\frac{625}{256} = \frac{11197}{11520} = 1 - \frac{323}{11520} \le 1 - d/500, \tag{Q55.4.9}$$

where $d \le 1/4$ and $323/11520 > 1/2000$.

Q55.5 The pair $(2,4)$ on the full domain

The exact identities $f_4 = 1+g+r+s+t-d-dp$ and $h_2 = q-r$ give

$$h_2 + f_4 = 2(1-d) + R, \qquad R = s+t-dg.$$

The two factors are nonnegative, so $F_4 \le (m+R/2)^2$ throughout the domain. In this case,

$$\begin{aligned} D_0 = \delta_2, \qquad \mathbb{E}(KR) = \delta_2 + d^2 \ge 0, \\ V = \delta_3 + \delta_4 + 2d\delta_2 + d^3 \le d\{(d+\theta)^2 + \theta^3\} \le d(1/2+d)^2. \end{aligned} \tag{Q55.5.1}$$

The last step uses $\theta \le 1/2$ and $\theta^2(1+\theta) = 7605/32768 < 1/4$. Thus (Q55.4.2) still holds and $D_0 \le \theta d$. Put $C_4 = 1-(47/64)d+(19/32)d^2$. The full-domain chord bound is

$$\frac{(1/2+d)^2}{4(1-d)} \le \frac{1}{16} + \frac{19}{32}d; \tag{Q55.5.3}$$

after multiplication by $16(1-d)$ the difference is $(9/2)d(1-3d)$. Consequently $m+D_0/2+V/(4m) \le C_4$ and

$$H_4 \le (C_4 - Z/2)^2. \tag{Q55.5.4}$$

The function C_4 decreases, $C_4 \ge 473/576$, and $U \le 5/(6\sqrt{3}) < 1/2$, so the base is positive. The range $d \le 1/16$ is covered above.

For $x = \sqrt{d} \in [1/4, 1/\sqrt{3}]$, write

$$G_4(x) = C_4 + U/2 = 1 + x/4 - 47x^2/64 + x^3/2 + 19x^4/32.$$

Let $h = G_4'$. Then $h'' = 3 + 57x/4 \ge 3$, while $h(1/4) = 7/512$ and $h'(1/4) = -35/128$. Hence

$$h(x) \ge 7/512 - (35/128)(x-1/4) + (3/2)(x-1/4)^2 \ge 119/98304 > 0.$$

It follows that

$$G_4(1/\sqrt{3}) = 1 - 103/576 + 5/(12\sqrt{3}) \le 17/16. \tag{Q55.5.5}$$

The endpoint comparison is $80\sqrt{3} \le 139$, which follows from $19200 < 19321$. Moreover $(17/16)^4 < 51/40$ and $(51/40)^4 < 8/3$, using $3340840 < 3342336$ and $20295603 < 20480000$. Thus

$$(C_4 + U/2)^{16} < 8/3. \tag{Q55.5.6}$$

For $d \ge 1/16$, the negative node is at most $7601/8192 < 13/14$, using $106414 < 106496$. Since $(13/14)^4 < 3/4$ by $114244 < 115248$,

$$(C_4 - \theta U/2)^{16} < 81/256. \tag{Q55.5.7}$$

The Hermite bound now gives

$$\mathbb{E}H_4^8 < \frac{32}{45}\frac{81}{256} + \frac{13}{45}\frac{8}{3} = \frac{9}{40} + \frac{104}{135} = \frac{215}{216} \le 1 - d/500, \tag{Q55.5.8}$$

because $d \le 1/3$ and $1/216 > 1/1500$. This proves (Q55.0.1). Positivity permits equal-arm factors at both roots, and replacing K by $-K$ gives the fully reversed statement.

O.10 High-domain unit eighth moments for every single pair $(2, b)$

Work on a nonempty finite probability space with a real antisymmetric kernel $|K| \le 1$, and use

$$A = J + K,\ T = J - K,\ g = K1,\ p = 1 + g,\ q = 1 - g,\ d = Eg^2,$$
$$\theta = 13/32, \quad \kappa = 19/32, \quad \delta_j = \|K^j 1\|_2^2.$$

Write $f_n = A^n \mathbf{1}$ and $h_n = T f_{n-1}$. The fixed actual direction is

$$F_b = h_2 f_b, \qquad H_b = pTF_b, \qquad b \ge 4.$$

We prove

$$\boxed{EH_b^8 \le 199/200 < 1 - d/500 \quad \text{for every integer } b \ge 5, \quad 1/4 \le d \le 1/3.} \tag{Q57.0.1}$$

The finite spectral input is that of Section M.1. The finite retained-envelope construction and its high-domain estimates are given below. The low-domain bound for every $b \ge 4$, and the full-domain $b = 4$ bound, are proved in Section O.9. They are used only in the final union.

Q57.1 The retained finite construction on the full domain

Let $0 < d \le 1/3$, $m = 1 - d$ and $\tau = d - \delta_2$. Define

$$\beta_1 = \beta_2 = 0, \qquad \beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1} \delta_i.$$

The finite positive convolution and its alternating terms give $\tau \le \beta_j \le d$ for $j \ge 3$. For $j \ge 5$, retain

$$\beta_j \le d - \delta_2 + \delta_3 \le d - \kappa\delta_2. \tag{Q57.1.1}$$

If the finite sum ends earlier the first inequality still holds; each later pair begins with a negative term and has nonpositive sum. Put $R_{\text{base}} = \sum_{j=3}^{b} K^j \mathbf{1}$. Since $h_2 = 2 - f_2$, positive path convolution gives

$$h_2 + f_b = 2 + R_{\text{base}} - \sum_{j=3}^{b} \beta_j f_{b-j}.$$

For $b = 5$, take $C_1 = d, C_2 = \delta_2, C_3 = 0$. For $b \ge 6$, take

$$\lambda = \frac{2d - \beta_b - \beta_{b-1}}{\beta_{b-2} + (1-d)\beta_{b-3}}, \quad C_1 = \beta_{b-1},\ C_2 = \lambda\beta_{b-2},\ C_3 = \lambda\beta_{b-3}.$$

The denominator is positive and the numerator is nonnegative. The full constant from these four losses is at least $(4 - d)\tau \ge (11/3)\kappa d = 209d/96 > 2d$, so $\lambda \le 1$. Thus the retained portion is valid throughout $d \le 1/3$, and

$$0 \le C_i \le d, \quad C_1 \ge \tau, \quad \beta_b + C_1 + C_2 + (1-d)C_3 = 2d.$$

Set

$$C_R = C_1 + C_2 + C_3, \quad D_R = C_2 + C_3, \quad E_R = C_3, \quad R = R_{\text{base}} - C_R g - D_R r - E_R s.$$

Here $r = Kg, s = Kr$. Retaining the actual f_1, f_2, f_3 terms in the positive convolution gives

$$0 \le h_2 + f_b \le 2m + R, \qquad 0 \le F_b \le (m + R/2)^2. \tag{Q57.1.2}$$

The base before squaring is nonnegative. Moreover,

$$C_R = 2d - \beta_b + dE_R, \qquad D_R = 2d - \beta_b - \beta_{b-1} + dE_R, \qquad 0 \le E_R \le d. \tag{Q57.1.3}$$

Q57.2 The same finite spectral representation and its positive mean

Let $N = b - 2$. On $0 \le x \le \theta$, write

$$\sum_{j=0}^{N-1} (i\sqrt{x})^j = a(x) + i\sqrt{x}\, b(x).$$

Finite alternating sums give $1 - x \le a, b \le 1$; for $N \ge 4$, also $a, b \le 1 - x + x^2$. Let ν be the positive spectral measure relative to g, with mass d, first moment δ_2 and second moment δ_3. The multiplier of R is

$$-A(x) - i\sqrt{x}\, B(x), \quad A = C_R + x(a - E_R), \quad B = D_R + xb.$$

Since $a \ge \kappa > d \ge E_R$, both A, B are nonnegative. Put $V = \mathbb{E}R^2$, $D_0 = \mathbb{E}R$, $Z = KR$ and $\alpha_{\text{actual}} = \mathbb{E}Z$. Skew adjointness and the spectral representation give

$$V = \int (A^2 + xB^2)\, d\nu, \quad D_0 = \int B\, d\nu, \quad \alpha_{\text{actual}} = \int A\, d\nu. \tag{Q57.2.1}$$

The mean and variance estimates below apply to this same R. By (Q57.1.1) and (Q57.1.3), $C_R \ge d + \kappa\delta_2 + dE_R$. Since $a \ge 1 - x$,

$$\begin{aligned} \alpha_{\text{actual}} &\ge d^2 + \kappa(1 + d)\delta_2 - E_R(\delta_2 - d^2) \\ &\ge d^2 + (\kappa - \theta d)\delta_2 + d^3 \\ &\ge (1 + \kappa + \kappa d)d^2 =: \alpha. \end{aligned} \tag{Q57.2.2}$$

The second step uses $E_R \le d, \delta_2 \ge d^2$; the last uses $\kappa - \theta d > 0$.

Q57.3 A high-domain spectral envelope for every $b \ge 5$

Set

$$a_0(d) = 3/8 + 3d/2, \qquad z_0(d) = 79/256 + 13d/16 + d^2, \qquad U = a_0\sqrt{d}.$$

For every $b \ge 5$ and $1/4 \le d \le 1/3$, we show

$$V \le da_0^2, \qquad D_0 \le dz_0, \qquad |Z| \le U, \qquad EZ^2 \le \theta U^2. \tag{Q57.3.1}$$

The last two bounds follow from $|KR| \le \|R\|_2$ and the spectral bound once the first is proved.

For $b = 5$, $C_R = d + \delta_2$, $D_R = \delta_2$, $E_R = 0$, and $A \le \theta - \theta^2 + (1 + \theta)d$, $B \le \theta + \theta d$. The triangle inequality in two coordinates yields

$$\sqrt{A^2 + xB^2} \le 3/8 + 3d/2.$$

The constants are

$$(\theta - \theta^2)^2 + \theta^3 = 131313/1048576 < 9/64,$$

$$(1 + \theta)^2 + \theta^3 = 66997/32768 < 9/4.$$

This vector estimate holds on the full legal domain. Also $B \le \theta(1 + d) \le z_0$ when $d \ge 1/4$: the difference is increasing, and at $d = 1/4$, replacing $79/256$ by the smaller $u = \theta - \theta^2 + \theta^3$ already leaves $2165/32768 > 0$.

For $b \ge 6$, put $u = \theta - \theta^2 + \theta^3 = 10101/32768 < 79/256$. By (Q57.1.3) and $a, b \le 1 - x + x^2$, the nonnegative upper bounds for A, B increase on the spectral interval: their derivatives are at least $2/3 - E_R > 0$ and $2/3$. Hence

$$A^2 + xB^2 \le \{u + (1+\theta)d + E_R(d-\theta)\}^2 + \theta\{u + 2\theta d + dE_R\}^2. \tag{Q57.3.2}$$

The right side is a convex quadratic in $E_R \in [0, d]$. At $E_R = 0$, the triangle inequality gives

$$\sqrt{\text{right side}} \le u\sqrt{1+\theta} + d\sqrt{(1+\theta)^2 + 4\theta^3} < 3/8 + 3d/2.$$

Here $u < 5/16$, $\sqrt{1+\theta} \le 6/5$, and the second squared norm is $18397/8192 < 9/4$, with difference $35/8192$.

At $E_R = d$, enlarge u to $u_* = 79/256$. It suffices to prove

$$(u_* + d + d^2)^2 + \theta(u_* + 2\theta d + d^2)^2 \le (3/8 + 3d/2)^2, \quad 0 \le d \le 1/3. \tag{Q57.3.3}$$

The right side minus the left is $P(d) = A_0 + B_0 d + C_0 d^2 - D_0^* d^3 - E_0 d^4$, where

$$A_0 = 9/64 - (1+\theta)u_*^2, \quad B_0 = 9/8 - 2u_*(1 + 2\theta^2),$$

$$C_0 = 5/4 - 4\theta^3 - 2(1+\theta)u_*, \quad D_0^* = 2 + 4\theta^2, \quad E_0 = 1 + \theta.$$

Since $d^3 \le d^2/3, d^4 \le d^2/9$, $P(d) \ge L(d) = A_0 + B_0 d + (C_0 - D_0^*/3 - E_0/9)d^2$. The quadratic coefficient is negative: using just $u_* > 1/4, D_0^* \ge 2, E_0 \ge 1$, it is at most $5/4 - 1/2 - 2/3 - 1/9 = -1/36$. Thus L is concave and its minimum on the interval is at an endpoint. The exact endpoint differences are

$$L(0) = 14067/2097152 > 0,$$

$$L(1/3) = P(1/3) = 822339/169869312 > 0.$$

The second denominator is $32 \cdot 2304^2$. The two squared bases are $1735/2304$ and $1591/2304$; their weighted numerator is 129233853, whereas the target $(7/8)^2$ has numerator 130056192. This proves (Q57.3.3) on the entire interval. Finally $B \le u + 2\theta d + d^2 \le z_0$, and (Q57.2.1) gives $D_0 \le d z_0$.

Q57.4 Positive propagation and Hermite domination

Positive propagation and AM–GM for the two nonnegative factors give

$$H_b \le \{m + D_0/2 - Z/2 + V/(4m)\}^2 \le (C - Z/2)^2,$$

$$C(d) = 1 - d + \frac{d}{2} z_0(d) + \frac{d a_0(d)^2}{4(1-d)}. \tag{Q57.4.1}$$

Since $C \ge m \ge 2/3$ and $U \le 7/8$, $C > U/2$, so the square and the increase of its base are valid. Let $W = -Z$. Then $\mathbb{E}W \le -\alpha, \mathbb{E}W^2 \le \theta U^2, |W| \le U$, and $\alpha^2 \le \theta U^2$. For $f(w) = (C + w/2)^{16}$, use (M.45) with $a = \alpha$, $v_0 = \theta U^2$, and $b_{-\alpha} \le f(U)/(U+\alpha)^2$. The positive base follows from $C > U/2$, and the mean coefficient is $f'(-\alpha) + 2\alpha b_{-\alpha} \ge 0$.

$$EH_b^8 \le (C - \alpha/2)^{16} + \mathcal{R}(k)(C + U/2)^{16}, \quad k = \alpha/U, \quad \mathcal{R}(k) = \frac{\theta - k^2}{(1+k)^2}. \tag{Q57.4.2}$$

The denominator is positive and $\theta - k^2 \geq 0$. On the legal range $k \geq 0$, $\mathcal{R}'(k) = -2(k+\theta)/(1+k)^3 < 0$. Precisely,

$$k(d) = \frac{d^{3/2}(51 + 19d)}{12(1 + 4d)}$$

This increases on the high domain, since its logarithmic derivative is $3/(2d) + 19/(51 + 19d) - 4/(1 + 4d) > 0$. The first and third terms already have positive difference $(3 + 4d)/(2d(1 + 4d))$.

The other needed monotone function is $G(d) = C + U/2$. For $d \geq 1/4$, $C' \geq -1 + z_0/2$ and $z_0 \geq 1/2$. Writing $x = \sqrt{d}$, $(U/2)' = (9/8)x + (3/32)/x \geq 3/4$: this last function increases for $x \geq 1/2$ and equals $3/4$ at $1/2$. Thus $G' \geq 0$, with positive terms still retained.

The function C is convex. The second derivative of $dz_0/2$ is nonnegative, and $da_0^2/(1 - d)$ is a product of two nonnegative functions with nonnegative first and second derivatives; the remaining $1 - d$ is affine. Hence C is bounded by the larger endpoint value on $[1/4, 1/3]$:

$$C(1/4) = 1779/2048 < 113/128, \qquad C(1/3) = 6065/6912 < 113/128.$$

Also α increases and $\alpha(1/4) = 223/2048$. Therefore

$$C - \alpha/2 \leq 3393/4096 < 5/6, \qquad (C - \alpha/2)^{16} < 1/16. \tag{Q57.4.3}$$

The power bound follows from $(5/6)^4 < 1/2$. For $d \geq 5/16$, $\alpha \geq 22775/131072$, giving

$$C - \alpha/2 \leq 208649/262144 < 4/5, \qquad (C - \alpha/2)^{16} < 3/100. \tag{Q57.4.4}$$

For the last power bound, use $(4/5)^8 < 17/100$ and $289/10000 < 3/100$; the required cross comparison is $6553600 < 6640625$. All these bases are positive, since $\alpha \leq 43/216 < 1/5$ and $C \geq 2/3$.

Q57.5 Four rational interval payments

Use the four closed intervals

$$[1/4, 2/7], \quad [2/7, 5/16], \quad [5/16, 21/64], \quad [21/64, 1/3].$$

The endpoint estimates below apply throughout these intervals by the proved monotonicity of k and G. The exact lower endpoints give

$$k(1/4) = 223/768 \geq 2/7, \qquad k(2/7) = 79\sqrt{14}/882 \geq 1/3,$$

$$k(5/16) = 4555\sqrt{5}/27648 \geq 11/30, \qquad k(21/64) = 693\sqrt{21}/8192 \geq 31/80. \tag{Q57.5.1}$$

The first cross difference is 25. The second follows from $87374 > 86436$. For the third, use $\sqrt{5} \geq 67/30$, since $4500 > 4489$, then $305185 > 304128$. For the fourth, use $\sqrt{21} \geq 197/43$, since $38829 > 38809$, then $682605 > 682496$. Thus the respective upper bounds for $\mathcal{R}(k)$ are

$$\frac{509}{2592}, \quad \frac{85}{512}, \quad \frac{1957}{13448}, \quad \frac{1639}{12321}. \tag{Q57.5.2}$$

At $d = 2/7$, $a_0 = 45/56$; using $\sqrt{2/7} \leq 15/28$,

$$G(2/7) \leq \frac{5}{7} + \frac{79}{1792} + \frac{13}{392} + \frac{4}{343} + \frac{1755}{6272} = \frac{95097}{87808} < \frac{13}{12}. \tag{Q57.5.3}$$

The final cross difference is 340; the square-root comparison is $1568 < 1575$. At $d = 5/16$,

$$C(5/16) = \frac{78537}{90112} < \frac{447}{512}, \qquad U/2 \le \frac{243}{1024},$$

The second bound uses $\sqrt{5} \le 9/4$. Hence $G(5/16) \le 1137/1024 < 10/9$, with cross difference 7. The same-denominator difference for the first C comparison is 135. At $d = 21/64$,

$$C(21/64) = \frac{19742627}{22544384} < \frac{897}{1024}, \qquad U/2 \le \frac{2035}{8192}.$$

These comparisons use the same-denominator difference 5725 and $\sqrt{21} \le 55/12$, since $3024 < 3025$. Consequently $G(21/64) \le 9211/8192 < 9/8$. At $d = 1/3$, $U/2 = 7\sqrt{3}/48$; using $\sqrt{3} \le 26/15$, since $675 < 676$, gives

$$G(1/3) \le \frac{6065}{6912} + \frac{91}{360} = \frac{39061}{34560} < \frac{181}{160}. \tag{Q57.5.4}$$

The final same-denominator difference is 35. The respective sixteenth-power upper bounds for the positive nodes are

$$4, \qquad 11/2, \qquad 53/8, \qquad 29/4. \tag{Q57.5.5}$$

Indeed, $(13/12)^4 < 7/5$ by $142805 < 145152$, so its sixteenth power is less than $(7/5)^4 = 2401/625 < 4$. Also $(10/9)^8 < 7/3$ by $300000000 < 301327047$, so the sixteenth power is less than $49/9 < 11/2$. Next $(9/8)^8 < 18/7$ by $301327047 < 301989888$, giving $324/49 < 53/8$. Finally $(181/160)^4 < 41/25$ by $1073283121 < 1074790400$, and $(41/25)^4 < 29/4$ follows from $11303044 < 11328125$.

Substitute (Q57.4.3)–(Q57.4.4), (Q57.5.2) and (Q57.5.5) into (Q57.4.2). The four interval bounds are

$$\begin{aligned}
\frac{1}{16} + 4\frac{509}{2592} &= \frac{1099}{1296}, \\
\frac{1}{16} + \frac{11}{2}\frac{85}{512} &= \frac{999}{1024}, \\
\frac{3}{100} + \frac{53}{8}\frac{1957}{13448} &= 1 - \frac{15887}{2689600}, \\
\frac{3}{100} + \frac{29}{4}\frac{1639}{12321} &= 1 - \frac{3431}{616050}.
\end{aligned} \tag{Q57.5.6}$$

Each deficit exceeds $1/200$. For the last two, the exact comparisons are $3177400 > 2689600$ and $686200 > 616050$; the first two are immediate. Thus $\mathbb{E}H_b^8 \le 199/200$ for every $b \ge 5$ on the whole high domain. Since $d/500 \le 1/1500 < 1/200$, this proves (Q57.0.1).

Q57.6 The unit catalogue and its fixed direction

Combining the high-domain proof with Section O.9's bound for every $b \ge 4, d \le 1/4$ and its full-domain $b = 4$ bound yields

$$\boxed{EH_b^8 \le 1 - d/500 \quad \text{for every integer } b \ge 4, \text{every finite admissible kernel.}} \tag{Q57.6.1}$$

Here $H_b = pT(h_2 f_b)$ is the same actual function in both domains. At $d = 0$, leaf messages equal one.

The long terminal arm points toward its leaf. The intermediate vertex of the two-edge short arm points toward both the terminal and its leaf. The unit connector points from the terminal to the receiving root, whose original unit arm points outward. The undecorated component has $b + 4$ edges. Genuine equal-arm pairs retain their directions $f_\ell h_\ell \le 1$; positivity restores any number and multiplicity of them. Complete reversal holds with the same original probability measure, kernel and adjoint.

O.11 Unit-connector ordinary eighth moments for even terminals of minimum one

Q61.0 The terminal and its directions.

Use the fixed finite model of Appendix M.15, with $Jv = (\mathbb{E}v)\mathbf{1}$, $A = J + K$, $T = J - K = A^*$, $K^* = -K$, $|K(x,y)| \le 1$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $d = \mathbb{E}g^2$, $\delta_j = \|K^j\mathbf{1}\|_2^2$, $\theta = 13/32$, $\kappa = 19/32$, $f_n = A^n\mathbf{1}$, $h_n = Tf_{n-1}$, and $h_1 = q$. After maximal equal pairing, suppose $1 = a_1 < b_1 < a_2 < b_2 < \cdots < a_k < b_k$, $k \ge 2$. For the genuine equal-pair product $E_0 = \prod_\ell f_\ell h_\ell \le 1$, set

$$\boxed{F = E_0 q f_{b_1} \prod_{i=2}^{k} h_{a_i} f_{b_i}, \qquad H = pTF.} \tag{Q61.0.1}$$

Each f_n-arm points outward; each h_n-arm has only its first edge inward. The terminal unit arm points inward. A single connector edge points from the terminal to the new root, whose unit arm points outward. This choice depends only on the arm list. For all finite integer lengths, $k \ge 2$, legal kernels and equal-pair multiplicities,

$$\boxed{\mathbb{E}[(pTF)^8] \le e^{-63d/50} \le 1 - d/2.} \tag{Q61.0.2}$$

The proof includes $(1,2,a,c)$ with $a \ge 4$, $(1,3,a,c)$ with $a \ge 4$, every $b_1 \ge 4$, and $(1,2,3,c)$, in the ordered direction (Q61.0.1).

Q61.1 Finite identities and boundary cases.

After deletion of zero-mass atoms, skewness gives $|g(x)| \le 1 - \mu(x)$ and $p(x), q(x) \ge \mu(x) > 0$. The finite calculus gives

$$0 \le d \le 1/3, \quad \|K\|_{2\to 2}^2 \le \theta, \quad \delta_{j+1} \le \theta\delta_j, \quad |Kv(x)| \le \|v\|_2, \tag{Q61.1.1}$$

and

$$\mathbb{E}K^{2j}\mathbf{1} = (-1)^j\delta_j, \quad \mathbb{E}K^{2j+1}\mathbf{1} = 0, \quad 0 \le f_n, h_n \le 2, \quad f_n h_n \le 1. \tag{Q61.1.2}$$

These are identities in $L^2(\mu)$. When $d = 0$, $g = 0$ and $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$; integration from the leaves gives every tree message equal to one even if $K \ne 0$. Below take $d > 0$ where needed.

Q61.2 The K^2 row input.

The finite three-label and random-sign argument in (Q49.2.1)–(Q49.2.3) proves

$$\|Kv\|_2^2 \le 1/3 \quad (|v| \le 1), \qquad |K^2 w(x)|^2 \le \|w\|_2^2/3 \quad (w \text{ real}).$$

The first estimate is from L^∞ to L^2, and the second is a row estimate for K^2. The spectral estimate used here is $\|K\|^2 \le \theta$.

Q61.3.1 A centered variable in a bounded interval.

Let a finite real variable Y have $\mathbb{E}Y = 0$ and take values in an interval of length ℓ. For $\psi(s) = \log \mathbb{E}e^{sY}$, the exponentially tilted law has the same support, and $\psi''(s) = \mathrm{Var}_s(Y) \le \ell^2/4$. Indeed, if c is the interval midpoint, then $\mathrm{Var}_s(Y) \le \mathbb{E}_s(Y - c)^2 \le \ell^2/4$. Since $\psi(0) = \psi'(0) = 0$, two integrations give, for every $s \ge 0$,

$$\mathbb{E}e^{sY} \le e^{s^2\ell^2/8}. \tag{Q61.3.1}$$

Q61.3.2 Hermite comparison with either sign of the mean.

Suppose the same finite real variable W satisfies $|W| \le U$, $\mathbb{E}W^2 \le U^2/2$, $U \ge 0$. For $s \ge 0$, put $m = \mathbb{E}W$, $m_+ = \max(m, 0)$. We prove

$$\boxed{\mathbb{E}e^{sW} \le \exp\{sm_+ + 9s^2U^2/32\}.} \tag{Q61.3.2}$$

The cases $U = 0$ or $s = 0$ are immediate. For $U, s > 0$, let $f(w) = e^{sw}$. Its Hermite majorant is

$$Q(w) = f(a)+f'(a)(w-a)+B(w-a)^2, \qquad B = \frac{f(U) - f(a) - f'(a)(U-a)}{(U-a)^2} \ge 0. \tag{Q61.3.3}$$

Lemma M.24 proves $f \le Q$ for $-U \le a < U$. Its remainder argument also covers $a < -U$, since $f''' \ge 0$ throughout the contact interval. Thus the stated majorant holds for every $a < U$, with $B \ge 0$.

If $m \le 0$, (M.46) with $h = 1/2$ gives $\mathbb{E}e^{sW} \le (2/3)e^{-sU/2} + (1/3)e^{sU}$. This centered two-point variable has interval length $3U/2$, so (Q61.3.1) proves (Q61.3.2) in this case.

If $m \ge 0$, keep the variance $v = \mathbb{E}(W-m)^2 = \mathbb{E}W^2 - m^2 \le U^2/2 - m^2$. Then $0 \le m \le U/\sqrt{2} < U$. Put $M = U - m > 0$ and $a = m - v/M$. Take the centered two-point variable $Y = \ell(Z - \pi)$ of (M.47), where $\ell = M + v/M$. Lemma M.24 places a in $[-U, U]$ and matches the first two moments, so

$$\mathbb{E}e^{sW} \le \mathbb{E}Q(W) = \mathbb{E}Q(m + Y) = e^{sm}\mathbb{E}e^{sY}. \tag{Q61.3.4}$$

Its interval length obeys

$$M + \frac{v}{M} \le U - m + \frac{U^2/2 - m^2}{U - m} = \frac{3U^2/2 - 2Um}{U - m} \le \frac{3U}{2}. \tag{Q61.3.5}$$

After multiplication by $U - m > 0$, the last inequality is $-2Um \le -3Um/2$, valid for $m \ge 0$. Apply (Q61.3.1) in (Q61.3.4) to complete the proof.

Q61.4 Positive propagation and a uniform exponential payment.

The scalar estimate on the whole interval $|u| \le 1$ is

$$e^u \le 1 + u + \frac{3}{4}u^2. \tag{Q61.4.1}$$

For $u \le 0$, the integral remainder coefficient is at most $1/2$. For $0 \le u \le 1$, successive terms starting at degree two have ratio at most $1/3$, so their sum is at most $(u^2/2)/(1 - 1/3) = 3u^2/4$. Suppose a nonnegative message F_0 has

$$F_0 \le e^{R - \Lambda d}, \qquad R = KP(K)g, \qquad \|P(K)\|^2 \le B, \qquad Bd \le 1, \tag{Q61.4.2}$$

where P is its finite arm polynomial. Set $V = \mathbb{E}R^2$, $Z = KR$, $U^2 = Bd/3$. The row, spectral and K^2 row estimates give

$$\begin{gathered} |R|^2 \le Bd \le 1, \quad V \le \theta Bd, \quad |Z|^2 \le U^2, \\ \mathbb{E}Z^2 \le \theta^2 Bd = 3\theta^2 U^2 \le U^2/2, \end{gathered} \tag{Q61.4.3}$$

using $3\theta^2 = 507/1024 < 1/2$. The kernel of T is nonnegative and at most two, so

$$Te^R \le q + \mathbb{E}R - Z + \tfrac{3}{2}V =: B_0. \tag{Q61.4.4}$$

Since $B_0 \geq Te^R \geq 0$, AM–GM yields

$$0 \leq pTF_0 \leq e^{-\Lambda d} pB_0 \leq e^{-\Lambda d}\left(1 + \frac{\mathbb{E}R}{2} + \frac{3}{4}V - \frac{Z}{2}\right)^2. \tag{Q61.4.5}$$

The base equals $(p + B_0)/2 \geq 0$. Applying $x \leq e^{x-1}$, taking the eighth power and integrating gives

$$\mathbb{E}[(pTF_0)^8] \leq e^{-8\Lambda d + 8\mathbb{E}R + 12V}\mathbb{E}e^{-8Z}. \tag{Q61.4.6}$$

Apply (Q61.3.2) to the same $W = -Z$, $s = 8$: $\mathbb{E}e^{-8Z} \leq e^{8(-\mathbb{E}Z)_+ + 18U^2}$. Since $12\theta + 6 = 87/8$, this gives

$$\boxed{\mathbb{E}[(pTF_0)^8] \leq \exp\left[-8\Lambda d + 8\mathbb{E}R + \frac{87}{8}Bd + 8(-\mathbb{E}Z)_+\right].} \tag{Q61.4.7}$$

Q61.5.1 Finite pair envelopes.

For $a < b$, put $R_{ab} = \sum_{j=a+1}^{b} K^j\mathbf{1}$, $\beta_1 = \beta_2 = 0$, and $\beta_b = \sum_{j=1}^{\lfloor (b-1)/2 \rfloor}(-1)^{j-1}\delta_j$. Positive convolution gives the pair envelopes

$$h_a f_b \leq \begin{cases} e^{R_{ab}-\beta_b}, & a = 1, 2, \\ e^{R_{ab}-(2d-\delta_2)}, & a = 3, \\ e^{R_{ab}-2d}, & a \geq 4. \end{cases} \tag{Q61.5.1}$$

For the first two short lengths, the pair-sum inequalities are $q + f_b \leq 2 - \beta_b + R_{1b}$ and $h_2 + f_b \leq 2 - \beta_b + R_{2b}$. For $a = 3$, the sum is at most $2 + R_{3b} - (2d - \delta_2)$. These are the finite short sums; the longer-pair estimate follows also from the stronger bound (L90.2.3). AM–GM has a nonnegative base in each case. For all $b \geq 3$, $\kappa d \leq d - \delta_2 \leq \beta_b \leq d$. Removing E_0 from (Q61.0.1), write the remaining message as F_0. Its exponent variable is

$$R = \sum_{i=1}^{k} R_{a_i b_i} = KP(K)g. \tag{Q61.5.2}$$

Disjoint ordered intervals give a finite zero-one mask: the term $K^j\mathbf{1}$ corresponds to the coefficient of z^{j-2} in P.

Q61.5.2 An arbitrary finite zero-one mask.

At $z = i\sqrt{x}$, $0 \leq x \leq \theta$, the positive and negative real subsums are each at most $1/(1 - x^2)$ in modulus; the imaginary part is at most $\sqrt{x}/(1 - x^2)$. Hence

$$|P(i\sqrt{x})|^2 \leq \frac{1 + x}{(1 - x^2)^2} \leq \frac{54}{25} =: B_{\mathrm{I}}. \tag{Q61.5.3}$$

Here $1 + \theta < 3/2$, $\theta^2 = 169/1024 < 1/6$, and the denominator exceeds $(5/6)^2$. This includes $x = 0$. The finite skew spectral theorem gives the corresponding operator norm.

Q61.5.3 The mask $P_0 = 1, P_1 = P_2 = 0$.

Its positive real subsum is at most $1/(1 - x^2)$, the negative one at most $x^3/(1 - x^2)$, and the imaginary part at most $x\sqrt{x}/(1 - x^2)$. Thus

$$|P(i\sqrt{x})|^2 \leq \frac{1 + x^3}{(1 - x^2)^2} \leq \frac{54}{35} < \frac{25}{16} =: B_{\mathrm{II}}. \tag{Q61.5.4}$$

Indeed, $\theta^2 < 1/6$, $\theta^3 = 2197/32768 < 1/14$, so the ratio is below $(15/14)(36/25) = 54/35$; the last comparison is $864 < 875$.

Q61.5.4 The mask $P_0 = P_2 = 1, P_1 = 0$.

Here

$$1 - \frac{x}{1-x^2} \le \Re P \le 1 - x + \frac{x^2}{1-x^2}.$$

For $0 \le x \le 1/2$, the lower bound is at least $1/3$, and the upper bound at most $1 - x/3$. The imaginary part is again at most $x\sqrt{x}/(1-x^2)$, so

$$|P|^2 \le (1-x/3)^2 + \frac{16}{9}x^3 \le (1-x/3)^2 + \frac{4}{9}x \le 1 - x/6 \le 1 =: B_{\mathrm{III}}. \tag{Q61.5.5}$$

The steps use $x^2 \le 1/4$ and $x^2/9 \le x/18$. These forced coefficients and gaps come from the ordered arm list. The geometric sums bound finite subsums. All three constants are at most $54/25$, hence

$$Bd \le (54/25)(1/3) = 18/25 < 1. \tag{Q61.5.6}$$

Thus the exponential interval in Q61.4 holds on the whole domain.

Q61.6.1 The class $b_1 \ge 3$.

All later $a_i \ge 4$. Equation (Q61.5.1) gives $F_0 \le e^{R - \Lambda d}$, $\Lambda = \kappa + 2(k-1)$. The terms $K^2\mathbf{1}, K^3\mathbf{1}$ both have coefficient one in R. Retain the $-d$ contribution of the first and drop the other negative mean terms:

$$\mathbb{E}R \le -d + \frac{\delta_2}{1-\theta^2} \le -\frac{439}{855}d \le -d/2. \tag{Q61.6.1}$$

Retain the $+\delta_2$ contribution of the second to $\mathbb{E}Z = \mathbb{E}KR$ and drop the other positive terms:

$$\mathbb{E}Z \ge \delta_2 - \frac{\delta_3}{1-\theta^2} \ge \frac{439}{855}\delta_2 \ge 0. \tag{Q61.6.2}$$

The coefficient used is $\theta/(1-\theta^2) = 416/855$. Insert $B = B_{\mathrm{I}} = 54/25$ into (Q61.4.7):

$$\mathbb{E}[(pTF_0)^8] \le e^{-[16(k-2)+63/50]d}. \tag{Q61.6.3}$$

The constant calculation is

$$8\kappa + 16 + 4 - \frac{87}{8}\frac{54}{25} = \frac{99}{4} - \frac{2349}{100} = \frac{63}{50}.$$

This holds for all $b_1 = 3$ or $b_1 \ge 4$, all later finite lengths and $0 < d \le 1/3$.

Q61.6.2 The class $b_1 = 2, a_2 \ge 4$.

Every later short arm is at least four, so

$$F_0 \le e^{R - 2(k-1)d}, \qquad R = K^2\mathbf{1} + \sum_{i=2}^{k} \sum_{j=a_i+1}^{b_i} K^j\mathbf{1}.$$

The latter sum starts at $j \ge 5$, so $P_0 = 1, P_1 = P_2 = 0$, and $B = B_{\mathrm{II}} = 25/16$. The first possible positive mean term has $j = 8$; hence

$$\mathbb{E}R \le -d + \frac{\delta_4}{1-\theta^2} \le -11d/12. \tag{Q61.6.4}$$

Here $\theta^3/(1-\theta^2) = 2197/27360 < 1/12$, by $26364 < 27360$. The first possible negative term of $\mathbb{E}Z$ has $j = 5$, giving

$$\mathbb{E}Z \geq -\frac{\delta_3}{1-\theta^2} \geq -\frac{169}{855}d \geq -d/5, \tag{Q61.6.5}$$

where $845 < 855$. Equation (Q61.4.7) yields

$$\mathbb{E}[(pTF_0)^8] \leq e^{-[16(k-2)+9103/1920]d}. \tag{Q61.6.6}$$

The exact constant is

$$16 + \frac{22}{3} - \frac{8}{5} - \frac{87}{8}\frac{25}{16} = \frac{9103}{1920} > 4 > \frac{63}{50}.$$

The $8/5$ term retains the possible negative mean of Z.

Q61.6.3 The class $b_1 = 2, a_2 = 3$.

The direction remains $qf_2h_3f_{b_2}\prod_{i\geq 3} h_{a_i}f_{b_i}$. The second pair has loss $2d - \delta_2$, and each later pair loss $2d$, so $\Lambda = 2(k-1) - \theta$. The terms $K^2\mathbf{1}, K^4\mathbf{1}$ are present, and $K^3\mathbf{1}$ is absent. Thus $P_0 = P_2 = 1, P_1 = 0$, and $B = B_{\mathrm{III}} = 1$. The same first-sign estimates give

$$\mathbb{E}R \leq -d/2, \qquad \mathbb{E}Z \geq -d/5. \tag{Q61.6.7}$$

Consequently

$$\mathbb{E}[(pTF_0)^8] \leq e^{-[16(k-2)+171/40]d}, \tag{Q61.6.8}$$

since

$$16 - 8\theta + 4 - \frac{8}{5} - \frac{87}{8} = \frac{171}{40} > 4 > \frac{63}{50}.$$

This includes $k = 2$ and every $(1, 2, 3, c)$ in this direction.

Q61.6.4 Completion of the three classes.

For $k \geq 2$, either $b_1 \geq 3$, or $b_1 = 2$ with $a_2 = 3$ or $a_2 \geq 4$. Each class has the stronger bound $\mathbb{E}[(pTF_0)^8] \leq e^{-[16(k-2)+63/50]d}$, and hence $\mathbb{E}[(pTF_0)^8] \leq e^{-63d/50}$. Using $e^x \geq 1 + x$ and $d \leq 1/3$,

$$e^{-63d/50} \leq \frac{1}{1+63d/50} = 1 - \frac{63d}{50+63d} \leq 1 - \frac{63}{71}d \leq 1 - d/2. \tag{Q61.6.9}$$

Restore $E_0 \leq 1$ by $0 \leq pT(E_0F_0) \leq pTF_0$, proving (Q61.0.2).

Q61.7 Decorations, extreme entries and full reversal.

Genuine equal pairs at both terminal and receiving root multiply nonnegative messages by factors at most one. Positivity restores every finite multiplicity and pair length. Entries $|K(x, y)| = 1$ are allowed; the proof divides by neither p nor q. The one-atom space has $d = 0$; a zero message is also compatible with AM–GM and the exponential upper bounds. If $\overline{F}$ is obtained by reversing every edge of F, replace K by $-K$ in the whole formula. Then p becomes q, T becomes A, and d, δ_j stay unchanged. Thus

$$\mathbb{E}[(qA\overline{F})^8] \leq e^{-63d/50} \leq 1 - d/2. \tag{Q61.7.1}$$

All terminal, connector, receiving-unit and decoration edges reverse together. The ordered terminal direction is also that of Appendix I.6, which independently proves

$$\boxed{\mathbb{E}[p(T^nF)^4] \leq 1 \qquad (n \geq 1).} \tag{Q61.7.2}$$

O.12 The unit ordinary eighth moment for terminal $(1,2,3)$

Q62.1 Model, directions and local bound

Let (X,μ) be a nonempty finite probability space and let K be real and skew with $|K(x,y)|\le 1$. All integrals, adjoints, norms and spectral measures use μ. Put

$$Jv=(Ev)1,\quad A=J+K,\quad T=A^*=J-K,\quad g=K1,\ p=1+g,\ q=1-g,\ d=Eg^2.$$

Write $f_n=A^n\mathbf{1}$, $\bar{f}_n=T^n\mathbf{1}$, $h_n=Tf_{n-1}$, and

$$r=Kg,\quad s=K^2g,\quad \delta_j=\|K^j1\|_2^2,\quad \theta=13/32.$$

The finite calculus of Section D gives $0\le d\le 1/3$, $d^2\le\delta_2\le\theta d$, $\|K\|^2\le\theta$, $|Kv(x)|\le\|v\|_2$, path positivity and the equal-arm bound $0\le f_\ell h_\ell\le 1$.

The terminal residual lengths are $(1,2,3)$, with messages q,f_2,h_3, respectively. The unit arm points toward the terminal, the two-edge arm points outward, and only the first edge of the three-edge arm points toward the terminal. Allow any finite number of equal-arm pairs and put

$$\begin{aligned} E_0&=\prod_j f_{\ell_j}h_{\ell_j},\\ F=E_0qf_2h_3,\qquad X&=TF,\qquad H=pX. \end{aligned}\tag{Q62.1.1}$$

The connector has one edge, directed from the terminal to the receiving root, whose unit arm points outward. These choices depend only on the arm lengths and pair multiplicities.

For every such structure and every finite host,

$$\boxed{EH^8\le e^{-d/200}\le 1-d/601.}\tag{Q62.1.2}$$

The bound also permits equal-arm decorations at the receiving root. Full reversal gives the corresponding ordinary eighth-moment bound for qAF^-.

Q62.2 Positive envelope with the same kernel parameters

By $E_0\le 1$ and positivity of T, it suffices to consider $P=qf_2h_3$. Define

$$t=1-d/2,\qquad k=t^{-1},\qquad a=t^2,\qquad 5/6\le t\le 1.$$

The path identities are

$$f_2=p+r,\quad h_3=q-d-r-s,\quad \bar{f}_3=q-d+r-s,\quad f_2+h_3=2-d-s.$$

Arithmetic–geometric mean for the two nonnegative factors gives

$$0\le P\le q(t-s/2)^2.\tag{Q62.2.1}$$

To establish positivity of $H_0=q-ks$, use $h_3,\bar{f}_3\ge 0$, which imply $q-s\ge d+|r|$, together with $|s|\le\sqrt{\delta_2}\le\sqrt{13/96}<3/8$. Thus

$$H_0=(q-s)-(k-1)s\ge d-\frac{d}{2t}|s|\ge 31d/40\ge 0.$$

Since $Es = 0$, we have $EH_0 = 1$. The kernel of T lies in $[0, 2]$, so

$$B = TH_0 = q + R, \qquad R = r + kK^3 g, \qquad 0 \le B \le 2. \tag{Q62.2.2}$$

Expanding (Q62.2.1) gives

$$\begin{aligned} TP &\le a(q + r) + tK^3 g - tK(gs) - t\delta_2 + \tfrac{1}{4}T(qs^2) \\ &\le aB + t(|K(gs)| - \delta_2) + \tfrac{1}{2}E(qs^2). \end{aligned}$$

Here $T(qs) = \delta_2 - K^3 g + K(gs)$. The row bound gives $|K(gs)| \le \sqrt{d\delta_3} \le \sqrt{\theta d\delta_2}$. For $d > 0$, put $v = \delta_2/d$. Completing the square in $\sqrt{\theta v} - v \le \theta/4$ retains the negative term $-t\delta_2$. Also $E(qs^2)/2 \le \delta_3 \le \theta^2 d$. Hence

$$0 \le TP \le t^2(q + R) + \gamma d, \qquad \gamma = t\theta/4 + \theta^2. \tag{Q62.2.3}$$

Taking the arithmetic mean of the nonnegative quantities tp and $t(q + R) + \gamma d/t$ yields

$$0 \le pTP \le (c + N/2)^2, \quad N = tR = K(tg + s), \quad c = t + \gamma d/(2t). \tag{Q62.2.4}$$

The base of this square is nonnegative, being their arithmetic mean. As $t \ge 5/6$,

$$c \le 1 - \frac{1793}{5120}d \le 1 - \frac{7}{20}d =: c_0. \tag{Q62.2.5}$$

The last comparison has slack $d/5120$. Enlarging the base preserves its nonnegativity.

Retain the same parameter $v = \delta_2/d$ throughout. Then

$$d \le v \le \theta, \qquad m = t - v > 0, \qquad EN = -md. \tag{Q62.2.6}$$

The spectral measure σ of g has mass d, support $[0, \theta]$, and first moment δ_2. The second derivative of $x(t - x)^2$ is $-4t + 6x < 0$, since $-4(5/6) + 6(13/32) < 0$. Jensen gives

$$EN^2 = \int x(t - x)^2 \, d\sigma(x) \le v(t - v)^2 d. \tag{Q62.2.7}$$

Here Jensen is applied to the probability measure σ/d, with mean v. The row bound and $\delta_3 \le \theta\delta_2$ also give

$$\begin{aligned} |N|^2 &\le \|tg + s\|_2^2 \\ &= t^2 d - 2t\delta_2 + \delta_3 \\ &\le d\{t^2 - (2t - \theta)v\} =: U^2. \end{aligned} \tag{Q62.2.8}$$

Since $2t - \theta > 0$ and $v \ge d$,

$$U^2 \le f(d) := d - \frac{83}{32}d^2 + \frac{5}{4}d^3 < \frac{1}{9}. \tag{Q62.2.9}$$

To verify the last comparison, first take $d \le 1/4$. Then $f(d) \le d - (73/32)d^2 \le 8/73 < 1/9$. For $1/4 \le d \le 1/3$, $f''(d) = -83/16 + (15/2)d \le -43/16 < 0$ and $f'(1/4) = -1/16$. Thus f decreases on $[1/4, 1/3]$ and has maximum $f(1/4) = 55/512 < 1/9$. Take U to be the nonnegative square root; at zero use continuity.

Q62.3 A scalar exponential bound

For $U \geq 0$, define

$$C(U) = \begin{cases} (e^{8U} - 1 - 8U)/U^2, & U > 0, \\ 32, & U = 0. \end{cases}$$

The integral form of the Taylor remainder gives $C(U) = 64 \int_0^1 (1-z) e^{8zU}\, dz$. This representation holds for every real argument and is increasing in that argument. Consequently, for $n \leq U$,

$$e^{8n} \leq 1 + 8n + C(U) n^2. \tag{Q62.3.1}$$

For $0 \leq U \leq 1/3$, we claim

$$\boxed{C(U) \leq 195/2, \qquad C(U) \leq 48 + 448U^2.} \tag{Q62.3.2}$$

The terms of orders zero through four in $e^{4/3}$ sum to $911/243$. The fifth term is $128/3645$, and every subsequent ratio is at most $2/9$. Therefore

$$\begin{aligned} e^{4/3} &\leq \frac{911}{243} + \frac{128}{3645}\frac{9}{7} \\ &= \frac{32269}{8505} < \frac{19}{5}. \end{aligned}$$

The final numerator difference is 50. It follows that $e^{8/3} < (19/5)^2 = 361/25 < 29/2$, so $C(1/3) < 195/2$; monotonicity proves the first bound.

Separate the first two terms of $C(U)$. The remaining coefficients are nonnegative and their degrees are at least two. For $U \leq 1/3$,

$$C(U) \leq 32 + \frac{256}{3} U + \frac{667}{2} U^2.$$

The quadratic coefficient is $9\{195/2 - 32 - 256/9\} = 667/2$. The nonnegative square $(32U/3 - 4)^2$ gives

$$\frac{256}{3} U \leq 16 + \frac{1024}{9} U^2.$$

Hence $C(U) \leq 48 + (8051/18) U^2 \leq 48 + 448U^2$, proving (Q62.3.2).

Q62.4 Reduction to two parameters of the same kernel

By (Q62.2.4)–(Q62.2.5) and $1 + x \leq e^x$,

$$E(pTP)^8 \leq E(c_0 + N/2)^{16} \leq e^{-28d/5} E e^{8N}.$$

Now apply (Q62.3.1), (Q62.2.6)–(Q62.2.8), and $1 + x \leq e^x$:

$$\boxed{\begin{aligned} E(pTP)^8 &\leq \exp\{d\, \mathcal{E}(d, v)\}, \\ \mathcal{E}(d, v) &= -\frac{28}{5} - 8(t - v) + C(U) v (t - v)^2. \end{aligned}} \tag{Q62.4.1}$$

The integrated polynomial upper bound is positive because it dominates $Ee^{8N} > 0$. It can therefore be exponentiated. We prove, for $0 < d \le 1/3$ and $d \le v \le 13/32$, that

$$\mathcal{E}(d, v) \le -1/200. \tag{Q62.4.2}$$

Q62.5 The range $v \le 1/4$

Here $d \le 1/4$. Since $t \ge 7/8$, the function $v(t-v)^2$ increases on $[0, 1/4]$, and

$$\begin{aligned} v(t-v)^2 &\le \frac{1}{4}(t-1/4)^2 \\ &= \frac{9}{64} - \frac{3}{16}d + \frac{1}{16}d^2, \\ -\frac{28}{5} - 8(t-v) &\le -\frac{58}{5} + 4d. \end{aligned} \tag{Q62.5.1}$$

Q62.5.1 The range $0 < d \le 3/16$

Using $d^3 \le (3/16)d^2$ in (Q62.2.9) gives

$$U^2 \le d - \frac{151}{64}d^2, \qquad C(U) \le 48 + 448d - 1057d^2.$$

Multiplying by the bounds in (Q62.5.1) and expanding gives

$$\begin{aligned} \mathcal{E} \le &-\frac{97}{20} + 58d - \frac{14697}{64}d^2 \\ &+\frac{3619}{16}d^3 - \frac{1057}{16}d^4. \end{aligned}$$

Drop the final nonpositive term and use $d^3 \le (3/16)d^2$ to obtain

$$\begin{aligned} \mathcal{E} &\le -\frac{97}{20} + 58d - \frac{47931}{256}d^2 \\ &\le -\frac{97}{20} + \frac{215296}{47931} < -\frac{1}{4}. \end{aligned} \tag{Q62.5.2}$$

The last comparison follows from $5 \cdot 215296 = 1076480 < 23 \cdot 47931 = 1102413$. The quadratic upper bound is obtained by completing the square and holds for every real d.

Q62.5.2 The range $3/16 \le d \le 1/4$

The variance upper bound in (Q62.5.1) decreases with d, so it is at most $441/4096$. Also $C(U) \le 195/2$ and $-58/5 + 4d \le -53/5$. Thus

$$\begin{aligned} \mathcal{E} &\le \frac{195}{2}\frac{441}{4096} - \frac{53}{5} \\ &= -\frac{4201}{40960} < -\frac{1}{10}. \end{aligned} \tag{Q62.5.3}$$

Q62.6 The range $1/4 \le v \le 5/16$

Now $d \le 5/16$. On $0 \le v \le t$, the maximum of $v(t-v)^2$ is $4t^3/27$. Moreover,

$$-\frac{28}{5}-8(t-v)\le -\frac{111}{10}+4d.$$

Using $v\ge 1/4$ in (Q62.2.8) gives

$$\begin{aligned} U^2 &\le d\left(\frac{77}{128}-\frac{3}{4}d+\frac{1}{4}d^2\right)\\ &\le \frac{77}{128}d-\frac{43}{64}d^2, \\ C(U) &\le 48+\frac{539}{2}d-301d^2. \end{aligned} \tag{Q62.6.1}$$

The second step uses $d\le 5/16$. For $0\le d\le 1/3$,

$$t^3=(1-d/2)^3\le 1-\frac{5}{4}d, \tag{Q62.6.2}$$

because $(3/4)d^2\le d/4$ and the remaining cubic term is nonpositive. The bound in (Q62.6.1) dominates the positive quantity $C(U)$ and can be multiplied by (Q62.6.2). This yields

$$\begin{aligned} \mathcal{E} &\le -\frac{359}{90}+\frac{946}{27}d-\frac{189}{2}d^2+\frac{1505}{27}d^3\\ &\le -\frac{359}{90}+\frac{946}{27}d-\frac{33299}{432}d^2 \\ &\le -\frac{359}{90}+\frac{3579664}{899073}<-\frac{1}{200}. \end{aligned} \tag{Q62.6.3}$$

The second line uses $d^3\le (5/16)d^2$; the third completes the square. The final rational comparison follows from

$$\begin{aligned} 3579664\cdot 1800 &= 6443395200\\ &< 899073\cdot 7171 = 6447252483,\\ \frac{359}{90}-\frac{1}{200} &= \frac{7171}{1800}. \end{aligned}$$

Q62.7 The range $5/16\le v\le \theta$

By (Q62.2.8) and $d\le 1/3$,

$$\begin{aligned} U^2 &\le d\left(\frac{257}{512}-\frac{11}{16}d+\frac{1}{4}d^2\right)\le \frac{257}{512}d-\frac{29}{48}d^2,\\ C(U) &\le 48+\frac{1799}{8}d-\frac{812}{3}d^2. \end{aligned} \tag{Q62.7.1}$$

For the mean term, use $v\le\theta$ to obtain $-28/5-8(t-v)\le -207/20+4d$. For the variance use $4t^3/27$ and (Q62.6.2). Expanding and absorbing the cubic term gives

$$\begin{aligned}\mathcal{E} &\le -\frac{583}{180}+\frac{1535}{54}d-\frac{52969}{648}d^2+\frac{4060}{81}d^3\\ &\le -\frac{583}{180}+\frac{1535}{54}d-\frac{126427}{1944}d^2\\ &\le -\frac{583}{180}+\frac{4712450}{1517124}<-\frac{1}{10}.\end{aligned}\tag{Q62.7.2}$$

The second line uses $d^3 \le d^2/3$; the third completes the square. The last comparison follows from

$$\begin{gathered}4712450\cdot 36 = 169648200\\ < 1517124\cdot 113 = 171435012,\\ \frac{583}{180}-\frac{1}{10}=\frac{113}{36}\end{gathered}$$

The bounds (Q62.5.2), (Q62.5.3), (Q62.6.3), and (Q62.7.2) cover the allowed parameter domain. Together they prove (Q62.4.2), and hence $EH^8 \le e^{-d/200}$. Finally,

$$e^{-d/200} \le \frac{1}{1+d/200} = 1-\frac{d}{200+d} \le 1-\frac{d}{601}$$

since $200+d \le 601$.

Q62.8 Boundaries, decorations and reversal

If $d=0$, then $g=0$ and $A\mathbf{1}=T\mathbf{1}=\mathbf{1}$. Leaf integration gives every path message equal to one, so the conclusion holds with equality. Divisions by v or U above are taken for positive parameters or by the stated continuous extension. Omit zero-mass atoms. On the remaining support, $p, q \ge \mu(x) > 0$.

As $E_0 \le 1$ and T is positive, the decorated message is bounded by the undecorated one. Equal-arm factors at the receiving root can be restored in the same way. Replacing K by $-K$ reverses all edges and gives the bound for qAF^- with the same d.

O.13 Unit ordinary eighth moments for terminals $(1,2,c)$

Q63.1 Statement and directions

Let (X,μ) be a nonempty finite probability space and let K be real, skew, and bounded by one in absolute value. All expectations, adjoints, inner products and spectral measures use μ. Write

$$Jv=(Ev)1,\quad A=J+K,\quad T=J-K=A^*,\quad g=K1,\quad p=1+g,\ q=1-g,\ d=Eg^2,$$

$$\begin{gathered}f_n=A^n1,\quad \bar{f}_n=T^n1,\quad h_n=Tf_{n-1},\\ \delta_j=\|K^j1\|_2^2,\quad \theta=13/32,\quad \kappa=19/32.\end{gathered}$$

The calculus of Section D gives $d\le 1/3$, $\|K\|^2\le\theta$, $d^2\le\delta_2\le\theta d$, $\delta_{j+1}\le\theta\delta_j$, $|Kv(x)|\le\|v\|_2$, path positivity, finite positive convolution and the equal-arm bound. The spectral support is $[0,\theta]$.

The three distinct residual arm lengths are $(1,2,c)$ with integer $c\ge 4$. Their messages are q, f_2, h_c: the unit arm points inward, the two-edge arm outward, and only the first edge of the c-edge arm

points inward. The unit connector points toward the receiving root, whose unit arm points outward. For any equal-arm decorations put

$$\begin{aligned} E_0 &= \prod_j f_{\ell_j} h_{\ell_j}, \\ F_c = E_0 q f_2 h_c, &\quad X_c = T F_c, \quad H_c = p X_c. \end{aligned} \tag{Q63.1.1}$$

For every integer $c \geq 4$, the same direction satisfies

$$\boxed{EH_c^8 \leq 1 - d/500.} \tag{Q63.1.2}$$

More precisely, the undecorated message satisfies the following low-range bounds:

$$EH_c^8 \leq \begin{cases} e^{-5d/63}, & c = 4,\ 0 \leq d \leq 1/4, \\ e^{-d/4}, & c \geq 5,\ 0 \leq d \leq 1/5. \end{cases} \tag{Q63.1.3}$$

Equal-arm decorations at either root and full reversal are permitted. The parameter intervals below use the single stated direction. The connector length in this result is one.

Q63.2 Finite path bounds and the high range

Put $r = Kg$, $s = K^2 g$, and define the finite alternating sum

$$\begin{aligned} \beta_1 &= \beta_2 = 0, \\ \beta_n &= \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1} \delta_j \quad (n \geq 3). \end{aligned}$$

Let $\tau = d - \delta_2$. Finite positive convolution gives

$$\begin{aligned} S_n := \sum_{j=0}^{n} K^j 1 &= f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \\ \tau \leq \beta_j &\leq d \quad (j \geq 3). \end{aligned} \tag{Q63.2.1}$$

It also gives $0 \leq f_n, \bar{f}_n, h_n \leq 2$, $\mu_n = E f_n = E \bar{f}_n \leq 1$, $f_n + h_n = 2\mu_{n-1}$, $0 \leq f_n h_n \leq 1$.
The alternating sum further yields $\beta_j \geq \tau + \kappa \delta_3$ for $j \geq 7$.
We need the following path-mean bound for all $n \geq 4$:

$$\mu_n \leq 1 - 83d/32 \qquad (n \geq 4). \tag{Q63.2.2}$$

Here is its finite proof. Positive convolution gives

$$\begin{aligned} \mu_2 &= 1 - d, & \mu_3 &= 1 - 2d, \\ \mu_4 &= 1 - 3d + \delta_2, & \mu_5 &= 1 - 4d + 2\delta_2 + d^2, \\ \mu_6 &= 1 - 5d + 3\delta_2 + 3d^2 - \delta_3, \\ \mu_7 &= 1 - 6d + 4\delta_2 + 6d^2 - 2\delta_3 - 2d\delta_2. \end{aligned}$$

For $n = 4, 5, 6, 7$, the respective lower bounds on $(1 - \mu_n)/d$ are $83/32$, $137/48$, $89/32$, and $127/48$, each at least $83/32$. For $n \geq 8$, retain the terms $f_0, \dots, f_5$ in (Q63.2.1) and lower their coefficients to τ. The leading β_{n+1} and the coefficients of f_0, f_1 each retain an additional $\kappa\delta_3$. Thus

$$\begin{aligned} 1 - \mu_n &\geq \tau(7 - 10d + 3\delta_2 + d^2) + 3\kappa\delta_3 \\ &\geq \kappa d(7 - 10d + 7d^2) \geq 95d/36 > 83d/32. \end{aligned}$$

Use $\delta_2 \geq d^2$ and $\delta_3 \geq d^3$. The final bracket is decreasing and positive on $[0, 1/3]$.

If $c = 4$ and $d \geq 1/4$, then $h_4 \leq 2\mu_3 = 2(1-2d) \leq 1$. If $c \geq 5$ and $d \geq 1/5$, then $h_c \leq 2\mu_{c-1} \leq 77/80 < 1$. Positivity therefore gives, in these ranges,

$$0 \leq pT(qf_2h_c) \leq pT(qf_2). \tag{Q63.2.3}$$

The unit ordinary-eighth bound for qf_2 from Q5-C051 bounds the eighth moment on the right by $1 - d/500$. It remains to prove the two low-range bounds in (Q63.1.3).

Q63.3 Positive input, error and coupled parameters

First omit E_0 by positivity. Set

$$\begin{gathered} \alpha = \beta_c, \quad B = \beta_{c-1}, \quad m = 1 - (\alpha + B)/2, \\ k = m^{-1}, \quad W = \sum_{j=3}^{c} K^j 1 = K^2 G_{c-2}(K) g, \\ U_0 = Bg - W, \quad H_0 = q + kU_0, \\ D_1 = d - \beta_{c+1}, \quad D_2 = d - \beta_{c+2}, \\ G_n(z) = \sum_{j=0}^{n-1} z^j. \end{gathered}$$

Retain the Bq term in the positive convolution for h_c to obtain

$$\begin{gathered} h_c \leq (1 - B)q - r - W - \alpha, \\ f_2 + h_c \leq 2m + U_0, \\ 0 \leq qf_2h_c \leq q(m + U_0/2)^2. \end{gathered} \tag{Q63.3.1}$$

To prove $H_0 \geq 0$, put $V = S_{c-3} \geq 0$ and $a_0 = \beta_{c-2}$. Then $EV = 1 - a_0$, $E(gV) = B$, and $E(rV) = -\alpha$, so

$$T^3V = (1 - a_0 + B)q + (1 - a_0)r - \alpha - W \geq 0.$$

Another nonnegative upper bound is $Z = (1 - B)q - r - \alpha - W \geq h_c \geq 0$. Add $(1 - a_0)Z$ to T^3V and divide by $2 - a_0$:

$$q - W \geq \alpha + \frac{a_0(1 - B)}{2 - a_0} q \geq \alpha. \tag{Q63.3.2}$$

Since $\alpha, B \in [\tau, d]$, we have $kB \leq 1/2$. If $g \geq 1/4$, (Q63.3.2) gives

$$H_0 \ge \frac{k}{2}\{\alpha - B + (\alpha + 3B)g\} \ge \frac{k}{2}(d - 2\delta_2) \ge 0.$$

Here $\alpha - B \ge -\delta_2$ and $\alpha + 3B \ge 4\tau$. If $g \le 1/4$, then $q + kBg \ge 3/4$. Also $|W|^2 \le \theta(1+\theta)d \le 195/1024 < 1/4$ and $k \le 3/2$, whence $k|W| < 3/4$. Thus $H_0 \ge 0$ in this case as well. Finally, $EH_0 = 1 - kD_1 \le 1$, and hence $0 \le TH_0 \le 2$.

Applying T to (Q63.3.1) gives

$$T(qf_2h_c) \le m^2 TH_0 - mE(gU_0) + mK(gU_0) + \tfrac{1}{4}T(qU_0^2). \tag{Q63.3.3}$$

If $J = EU_0^2$ and $v_0 = E(gU_0)$, the final three terms are at most $m(\sqrt{dJ} - v_0) + J$. Put $t = m - B$ and $N = K(tg + W)$. Then

$$TH_0 = q - kD_1 + kN, \qquad EN = -td + D_2. \tag{Q63.3.4}$$

Bound the final three terms of (Q63.3.3) by a nonnegative δd. Arithmetic–geometric mean for the nonnegative quantities mp and $mq + N - D_1 + \delta d/m$ gives

$$pT(qf_2h_c) \le (c_* + N/2)^2, \qquad c_* = m - D_1/2 + \delta d/(2m). \tag{Q63.3.5}$$

The second quantity dominates a nonnegative message by (Q63.3.3). Thus the base of the square is nonnegative.

Q63.4 A scalar remainder bound on $U \le 7/25$

Define $C(U) = (e^{8U} - 1 - 8U)/U^2$, with its continuous value $C(0) = 32$. The integral form of the Taylor remainder, $C(U) = 64\int_0^1 (1-z)e^{8zU}\,dz$ gives, for every real $n \le U$,

$$e^{8n} \le 1 + 8n + C(U)n^2. \tag{Q63.4.1}$$

For $0 \le U \le 7/25$,

$$\boxed{C(U) \le 42 + 483U^2.} \tag{Q63.4.2}$$

To check the constants, put $x = 28/25$. Its Taylor terms of orders two through five are less than $63/100$, $47/200$, $33/500$, and $3/200$, respectively; every subsequent ratio is at most $14/75$. Consequently,

$$\begin{aligned} e^{28/25} &< 1 + \frac{28}{25} + \frac{63}{100} + \frac{47}{200} \\ &\quad + \frac{33}{500} + \frac{3}{200}\frac{5}{4} \\ &= \frac{12279}{4000} < \frac{123}{40}. \end{aligned}$$

using $75/61 < 5/4$. Hence $e^{56/25} < 15129/1600$. At $U_0 = 7/25$ this gives $C(U_0) < 248625/3136$. After separating $32 + (256/3)U$, the remaining coefficients of $C(U)$ are nonnegative and have degree at least two. Throughout $[0, U_0]$ they are bounded by a quadratic coefficient 300, since

$$\frac{248625/3136 - 4192/75}{49/625} < 300,$$

the cross products are $3437976875 < 3457440000$. Moreover, $C(U) \le 32 + (256/3)U + 300U^2$. Young's inequality gives

$$\frac{256}{3}U \le 10 + \frac{8192}{45}U^2,$$

Thus $C(U) \le 42 + (21692/45)U^2 \le 42 + 483U^2$ throughout the stated interval.

Q63.5 Finite-tail bounds for $c \ge 5$

Fix $0 < d \le 1/5$ and $c \ge 5$. Put $v = \delta_2/d \in [d, \theta]$ and $y = \sqrt{\theta v}$. Since $\alpha, B \in [\tau, d]$, we have $m \ge 4/5$ and $t = m - B \ge 3/5$.

Q63.5.1 A common spectral deficit for every G_n, $n \ge 3$

For $z = i\sqrt{x}$ and $0 \le x \le \theta$, the finite geometric identity gives

$$|G_n(z)|^2 \le \frac{(1+x^3)^2}{1+x} \le 1 - \eta x, \qquad \eta = 7/16, \quad n \ge 3. \tag{Q63.5.1}$$

For odd $n \ge 3$, its numerator is $1+x^n \le 1+x^3$. If $n \equiv 0 \pmod 4$, the numerator is $(1-x^{n/2})^2 \le 1$. If $n \equiv 2 \pmod 4$ and $n \ge 6$, it is $(1+x^{n/2})^2 \le (1+x^3)^2$. These cases prove the first comparison. The second comparison reduces to $9 - 7x - 32x^2 - 16x^5 \ge 0$. This polynomial decreases on $[0, \theta]$, and $9 - 7\theta - 32\theta^2 = 7/8$ whereas $16\theta^5 < 1/2$. The real coordinate of the same G_n also belongs to $[1 - x, 1]$.

By (Q63.5.1), the same $N = K(tg + W)$ satisfies

$$|N|^2 \le dA(t,v), \qquad EN^2 \le d\,vA(t,v), \qquad A(t,v) = t^2 - (2\kappa t - \theta)v - \eta v^3. \tag{Q63.5.2}$$

Both bounds come from the spectral expression $|t - xG|^2 \le t^2 - 2tx + (2t+1)x^2 - \eta x^3$. Integration gives $t^2 d - 2t\delta_2 + (2t+1)\delta_3 - \eta\delta_4$. Use $\delta_3 \le \theta\delta_2$ and $\delta_4 \ge v^3 d$. After multiplication by one spectral factor x, integration gives $t^2\delta_2 - 2t\delta_3 + (2t+1)\delta_4 - \eta\delta_5$. Use $\delta_4 \le \theta\delta_3$, $\delta_3 \ge v^2 d$, and $\delta_5 \ge v^4 d$. The coefficient $2\kappa t - \theta \ge 49/160 > 0$ fixes the direction of the comparison.

On the subsequent parameter rectangle, $A > 1/5$: it increases with t and decreases with v, so take $t = 3/5$, $v = \theta$. Then $(2\kappa t - \theta)\theta = 637/5120 < 1/8$ and $\eta\theta^3 < 3/100$, giving $A > 9/25 - 1/8 - 3/100 = 41/200 > 1/5$.

Q63.5.2 Coupled error and alternating-tail deficit

As $\|G_{c-2}\| \le 1$, we have $\sqrt{J} \le \sqrt{d}(B+y)$, while $v_0 = Bd + D_2 \ge (B + \kappa v)d$. Thus in (Q63.3.3) one can take

$$\delta = m(y - \kappa v) + (B+y)^2 \ge 0. \tag{Q63.5.3}$$

Keep the dependence between $D_2 - D_1$ and α. For $c = 5, 6$, $\alpha = \tau$ and $D_2 - D_1 \le 0$. For $c \ge 7$, the alternating tail gives $D_2 - D_1 \le \delta_4$, while $\alpha - \tau \ge \kappa\delta_3$ and $\delta_4 \le \theta\delta_3$. Consequently, uniformly for $c \ge 5$,

$$D_2 - D_1 \le h(\alpha - \tau), \qquad h = \theta/\kappa = 13/19. \tag{Q63.5.4}$$

Q63.6 The row bounds and parameter relaxation for $c \ge 5$

Q63.6.1 The bound $U \le 7/25$

The function $A(t, v)$ increases with t. As $\alpha, B \ge \tau$, $t \le \bar{t} = 1 - 2(1 - v)d$. Put $a = 1 - v$ and define

$$\begin{aligned} \mathcal{A}(d, v) &= A(\bar{t}, v) \\ &= 1 - \frac{25}{32}v - \eta v^3 - 4a(1 - \kappa v)d + 4a^2d^2. \end{aligned} \tag{Q63.6.1}$$

Its second derivative in v is $-6\eta v - 8\kappa d + 8d^2 < 0$. For $v \ge d$, its first derivative is at most

$$Q(d) = -25/32 + (51/8)d - (225/16)d^2 + 8d^3 < 0.$$

Indeed $Q'' = -225/8 + 48d < 0$ and $Q'(1/5) = 171/100 > 0$. Thus Q increases on $[0, 1/5]$ and has maximum $Q(1/5) = -19/4000 < 0$. It follows that $\mathcal{A}$ decreases for $v \ge d$. Therefore $U^2 = dA(t, v)$ is at most

$$\begin{aligned} f(d) &:= d\mathcal{A}(d, d) \\ &= d - \frac{153}{32}d^2 + \frac{83}{8}d^3 - \frac{173}{16}d^4 + 4d^5. \end{aligned}$$

$f'''(d) \ge 207/20 > 0$, $f''(1/5) = -133/80 < 0$, and $f'(1/5) = 37/2000 > 0$. Hence f increases on $[0, 1/5]$, with maximum

$$f(1/5) = 7573/100000 < 49/625 < 1/12. \tag{Q63.6.2}$$

The two strict comparisons are $7573 < 7840$ and $588 < 625$. Thus $U \le 7/25$, so (Q63.4.2) applies. We also have $d\mathcal{A} < 1/12$.

Q63.6.2 Joint monotonicity in α and B

By (Q63.3.5), (Q63.4.1), (Q63.5.2)–(Q63.5.4), and $1 + x \le e^x$,

$$E\{[pT(qf_2h_c)]^8\} \le e^{d\Phi(\alpha, B; d, v)},$$

where

$$\begin{aligned} \Phi = &-\frac{8}{d}\{(1 - h)\alpha + B + h\tau\} - 8t + 8(y - \kappa v) + 8(B + y)^2/m \\ &+ vA(t, v)\{42 + 483dA(t, v)\}. \end{aligned} \tag{Q63.6.3}$$

This expression comes directly from $16(c^* - 1) + 8EN$, retaining both $-D_1$ and D_2. The integrated polynomial bound dominates $Ee^{8N} > 0$, so a further use of $1 + x \le e^x$ is legitimate.

Fix d, v and allow α, B to range independently in $[\tau, d]$. On this rectangle $A > 1/5$ and $A_t > 0$. The last term increases with t, whose derivatives in α, B are $-1/2, -3/2$. Also $(B+y)/m \le 97/128 < 1$. Discarding the nonpositive derivatives of the last term gives

$$\partial_\alpha \Phi \le -\frac{48}{19d} + 4 + 4(B+y)^2/m^2 < -240/19 + 8 < 0,$$

$$\partial_B \Phi \le -8/d + 12 + 8\{2(B+y)/m + (B+y)^2/(2m^2)\} < -40 + 12 + 20 < 0.$$

Thus the upper bound is maximized at $\alpha = B = \tau$. Then $m = 1 - ad$, $t = 1 - 2ad$, and the upper bound in (Q63.5.4) is zero. It remains to bound

$$\begin{aligned} \Psi(d, v) = &-16a - 8t + 8(y - \kappa v) + 8(ad + y)^2/m \\ &+ v\mathcal{A}(d, v)\{42 + 483d\mathcal{A}(d, v)\}. \end{aligned} \tag{Q63.6.4}$$

Q63.7 Monotonicity throughout the v interval

Put $\mathcal{B} = \mathcal{A} + v\mathcal{A}_v$ and $\mathcal{D} = \mathcal{A} + 2v\mathcal{A}_v$. For fixed d, both functions are concave in v: their second derivatives are $-24\eta v - 24d(\kappa - d)$ and $-42\eta v - 40d(\kappa - d)$, respectively, and are nonpositive. Their minima on $[0, \theta]$ occur at the endpoints. At $v = 0$, both equal $(1 - 2d)^2 \ge 9/25$.

At $v = \theta$, expansion gives

$$\begin{aligned} \mathcal{B} &= \frac{32493}{131072} + \frac{31}{8192}d - \frac{133}{256}d^2 \\ &> \frac{15}{64} - \frac{3}{128} > \frac{1}{5}, \end{aligned}$$

Here $133/6400 < 3/128$. Also,

$$\begin{aligned} \mathcal{D} &= -\frac{82565}{524288} + \frac{14825}{8192}d - \frac{627}{256}d^2 \\ &> -\frac{1}{6}. \end{aligned}$$

Since $14825/8192 > 1$ and $(627/256)d \le 627/1280 < 1$, the combined d terms are nonnegative; also $6 \cdot 82565 < 524288$. Hence $\mathcal{B} \ge 1/5$ and $\mathcal{D} \ge -1/6$ on the full rectangle.

Let $G = v\mathcal{A}(42 + 483d\mathcal{A})$. For $v \ge d$, (Q63.6.2) gives

$$\begin{aligned} G_v &= 42\mathcal{B} + 483d\mathcal{A}\mathcal{D} \\ &\ge 42/5 - 483/72 = 203/120 > 0. \end{aligned} \tag{Q63.7.1}$$

The other part of (Q63.6.4) also increases strictly. Specifically, $y' \ge 1/2$ and $z = ad + y$ give $z' = y' - d$, $m' = d$, and $z/m < 1$. Thus

$$(z^2/m)' = (z/m)\{2y' - 2d - dz/m\} > 0.$$

The other explicit derivatives total at least $16 - 16d - 8\kappa \ge 161/20 > 0$. Therefore Ψ increases on $d \le v \le \theta$, and it suffices to take $v = \theta$.

Q63.8 The final one-variable estimate for $c \geq 5$

At $v = \theta$, put $a = \kappa$ and $y = \theta$. Write

$$M(d) = 1 - \kappa d, \quad T_0(d) = 1 - 2\kappa d,$$
$$P(d) = \frac{342509}{524288} - \frac{14763}{8192}d + \frac{361}{256}d^2.$$

The one-variable function to be bounded is

$$\begin{aligned} F(d) = &-16\kappa - 8T_0(d) + 8\theta^2 \\ &+\frac{8(\kappa d + \theta)^2}{M(d)} \\ &+\theta\{42P(d) + 483dP(d)^2\}. \end{aligned} \tag{Q63.8.1}$$

It equals $\Psi(d, \theta)$; the alternating-tail term has already been accounted for in (Q63.5.4). We show $F \leq -1/4$ throughout $[0, 1/5]$.

Q63.8.1 Curvature on the full interval

Write $P = a_0 - b_0 d + c_0 d^2$, so $P' < 0$. The function $P + dP' = a_0 - 2b_0 d + 3c_0 d^2$ decreases and is positive on $[0, 1/5]$: $a_0 > 13/20$, $b_0 < 181/100$, and $c_0 > 7/5$ give a right-endpoint value greater than $47/500 > 0$; $b_0 > 9/5$ and $c_0 < 3/2$ make its derivative negative.

Put $S = 4PP' + 2d((P')^2 + PP'')$. Then

$$S' = 6\{P'^2 + P''(P + dP')\} > 0.$$

whereas

$$F'' = \theta\{42P'' + 483S\} + \frac{16\kappa^2(1+\theta)^2}{M^3}. \tag{Q63.8.2}$$

The second term increases with d, so F'' also increases and is maximized at $d = 1/5$. The exact endpoint values are

$$P = 4577893/13107200 \in (349/1000, 7/20),$$
$$-P' = 50711/40960 \in (619/500, 5/4), \quad P'' = 361/128 < 113/40.$$

Put $h = -P'$, so $S = -4Ph + (2/5)(h^2 + PP'')$. This upper bound decreases with both P, h on their stated rectangle. Consequently,

$$\begin{aligned} S(1/5) &\leq -4\frac{349}{1000}\frac{619}{500} \\ &+\frac{2}{5}\left\{\left(\frac{619}{500}\right)^2 + \frac{349}{1000}\frac{113}{40}\right\} \\ &= -\frac{1802051}{2500000} < -18/25. \end{aligned}$$

The second term in (Q63.8.2) has endpoint value $91378125/5606442 < 17$. Therefore

$$F'' \le \frac{13}{32}\left(\frac{7581}{64} - \frac{8694}{25}\right) + 17 < -75. \tag{Q63.8.3}$$

The final comparison has positive slack $59183/51200$.

Q63.8.2 A rational value and slope bound

At $d_0 = 1/8$ the exact values are

$$P(d_0) = 235957/524288, \quad P'(d_0) = -11875/8192,$$
$$P(d_0) + P'(d_0)/4 = 45957/524288.$$

Put $G(d) = \theta(42P + 483dP^2)$. Its central value is

$$G(d_0) = \frac{889950519380559}{70368744177664} < 253/20.$$

The cross-product difference in the last comparison is $4281889337812 > 0$. The other terms of (Q63.8.1) total $-261/16 + 169/128 + 5043/2528$ at d_0, and $5043/2528 < 399/200$. Thus

$$F(d_0) < 253/20 - 261/16 + 169/128 + 399/200 = -1111/3200 < -1/3. \tag{Q63.8.4}$$

The derivative of those other terms at d_0 is $392179/24964 \in (15, 16)$. $G'(d_0) = \theta[42P' + 483P(P + P'/4)]$. Here $-29/20 < P' < -7/5$ and $39/1000 < P(P + P'/4) < 1/25$. The latter follows from $9/20 < P < 451/1000$ and $87/1000 < P + P'/4 < 88/1000$. Hence $-18 < G'(d_0) < -15$, giving $|F'(d_0)| < 3$.

Apply the full-interval quadratic upper bound from (Q63.8.3) and complete the square:

$$\begin{aligned} F(d) &\le F(d_0) + F'(d_0)(d - d_0) - \frac{75}{2}(d - d_0)^2 \\ &< -\frac{1}{3} + \frac{9}{150} = -\frac{41}{150} < -\frac{1}{4}. \end{aligned} \tag{Q63.8.5}$$

Thus (Q63.6.3)–(Q63.8.5) prove the second row of (Q63.1.3) for every $c \ge 5$ and $0 < d \le 1/5$.

Q63.9 The short case $c = 4$

Let $0 < d \le 1/4$. The finite coefficients are $\alpha = B = d$, $m = 1 - d$, $t = 1 - 2d$, $D_1 = D_2 = \delta_2$, and $W = s + K^3 g = K^2(1 + K)g$. Put

$$D = \theta(1 + \theta) = 585/1024 < 3/5, \quad v = \delta_2/d, \quad y = \sqrt{Dv} < 1/2.$$

Here $D\theta = 7605/32768 < 1/4$. As in (Q63.5.3), the exact error can be bounded by $\delta = m(y - v) + (d + y)^2 \ge 0$, since $y \ge v$. The spectral identity $|t - x(1 + i\sqrt{x})|^2 = t^2 - 2tx + x^2 + x^3$ yields

$$|N|^2 \le dA, \quad EN^2 \le dvA, \quad A = t^2 - (2t - D)v > 0. \tag{Q63.9.1}$$

Here $2t - D \ge 1 - D > 0$; for the variance use $\delta_3 \ge v^2 d$. Also $A > 0$ since $t \ge 1/2$ and $v \le \theta$: its minimum is at least $(1/2 - \theta)^2 + \theta^3 > 0$. The terms $D_1 = D_2$ cancel exactly. Exponentiating (Q63.3.5) gives

$$E\{[pT(qf_2h_4)]^8\} \le$$
$$\exp\{d[-16-8t+8(y-v)+8(d+y)^2/m+C(U)vA]\}, \tag{Q63.9.2}$$
$$U^2 = dA.$$

The function $d(1-2d)^2$ is maximized at $d=1/6$, so $U^2 \le dt^2 \le 2/27 < 49/625$. By (Q63.4.2), $C(U) \le 42 + 483(2/27) = 700/9$.

Furthermore $m \ge 3/4$, $y < 1/2$, and $d+y \le 3/4$, so the first three scalar terms of (Q63.9.2) total at most $-6-8t$. Complete the square in v to get

$$vA \le \frac{t^4}{4(2t-D)} \le \frac{t^4}{4(2t-3/5)}.$$

The function $t^4/[(2t-3/5)(6+8t)]$ increases on $[1/2,1]$: the numerator controlling its logarithmic derivative is $32t^2+(108/5)t-72/5$, which is already $22/5>0$ at the left endpoint. Therefore

$$\frac{700}{9}\frac{t^4}{4(2t-3/5)} \le \frac{875}{882}(6+8t).$$

The exponent coefficient in (Q63.9.2) is at most $-(6+8t)/126 \le -5/63$, proving the first row of (Q63.1.3).

Q63.10 Boundaries, decorations and the same-direction fourth moment

Using $e^{-xd} \le 1/(1+xd)$, the two low-range bounds in (Q63.1.3) imply $1-d/13$ and $1-d/5$, respectively, and hence $1-d/500$. In the high ranges, (Q63.2.3) and Q5-C051 give the same bound. This proves (Q63.1.2) for every $c \ge 4$ and $0 \le d \le 1/3$.

At $d=0$, we have $g=0$ and leaf integration gives every message equal to one. The differentiations involving division by d or v use $d>0$ and extend continuously to zero. On positive-mass atoms p, $q \ge \mu(x) > 0$; zero-mass atoms can be omitted. Positivity restores equal-arm factors at both roots. Replacing K by $-K$ reverses every edge and gives the corresponding bound for qAF^-.

This is also the direction qf_2h_c of the four-branch terminal catalogue. Its same-direction fourth-moment bound $E[p(T^LF_c)^4] \le 1$, for all positive L, is supplied by Theorem 5.1.

O.14 The unit-connector eighth moment for every triple $(1,b,c)$, $3 \le b < c$

Q65.1 Directions and statement.

Use the finite model, operators and arm messages of Appendix M.15: K is real and skew, $|K| \le 1$, $A = J+K$, $T = J-K = A^*$, $g = K\mathbf{1}$, $p = 1+g$, $q = 1-g$, $f_n = A^n\mathbf{1}$, $\bar{f}_n = T^n\mathbf{1}$, and $h_n = Tf_{n-1}$. Write $\delta_j = \|K^j\mathbf{1}\|_2^2$, $d = \delta_1$, $\theta = 13/32$, $\kappa = 19/32$, and $\tau = d-\delta_2$. All expectations, norms, adjoints and spectral measures use the same finite probability measure after zero-mass atoms have been removed. The finite skew-kernel bounds are

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j, \quad |Kv(x)| \le \|v\|_2. \tag{Q65.1.1}$$

The spectral support is $[0,\theta]$, and $p(x), q(x) \ge \mu(x) > 0$ on positive-mass atoms. For the terminal residual $(1,b,c)$, $3 \le b < c$, direct the unit arm inward, the b-arm outward, and the first edge of the

c-arm inward with its remaining edges outward. The unit connector points from the terminal to a new root, whose unit arm points outward. For genuine equal-pair products at the terminal and root, set

$$E_0 = \prod_j f_{\ell_j} h_{\ell_j}, \qquad F = E_0 q f_b h_c, \quad H = pTF, \quad H_D = E_r pTF. \tag{Q65.1.2}$$

Each equal-pair factor belongs to $[0,1]$. For all these finite integer parameters and decorations, the fixed directions satisfy

$$\boxed{\mathbb{E}H_D^8 \le 1 - d/500.} \tag{Q65.1.3}$$

The low-domain estimates are stronger:

$$\mathbb{E}[(pT(qf_bh_c))^8] \le e^{-d/30} \quad (c \ge 5,\ d \le 1/5), \tag{Q65.1.4}$$

$$\mathbb{E}[(pT(qf_3h_4))^8] \le e^{-5d} \quad (d \le 1/4). \tag{Q65.1.5}$$

Full reversal gives the corresponding bounds with every edge reversed.

Q65.2 A bounded exponential estimate retaining a negative mean.

Let B take values zero and one, with $\Pr(B=1) = \pi$, $0 < \pi < 1/2$, and put $a = \frac{1}{2}\log((1-\pi)/\pi) > 0$. The centered Bernoulli inequality (Q81.2.5), including every real t, gives

$$\mathbb{E}e^{t(B-\pi)} \le \exp\left\{\frac{1-2\pi}{4\log((1-\pi)/\pi)}t^2\right\}. \tag{Q65.2.1}$$

The coefficient $\tanh(a)/(8a)$ increases with π. For $\pi \le 13/45$, it is at most $19/[180\log(32/13)]$, and

$$\frac{19}{180\log(32/13)} < \frac{47}{400}. \tag{Q65.2.2}$$

To verify the constant, put $x = 19/45$. The positive integral or geometric series gives

$$\log(32/13) = 2\operatorname{arctanh} x \ge 2(x + x^3/3 + x^5/5) > 38/45 + 13718/273375 + 64/15625 > 380/423.$$

The first strict substitution uses $x > 2/5$. The first two terms sum to $244568/273375$, whose difference from $380/423$ is $430236/115637625$; the final comparison is

$$64 \cdot 115637625 = 7400808000 > 430236 \cdot 15625 = 6722437500.$$

At $\pi = 0$, the centered variable is identically zero.

Now let one finite real variable W satisfy

$$\mathbb{E}W = \mu \le 0, \quad |W| \le U, \quad \mathbb{E}W^2 \le \theta U^2.$$

The case $U = 0$ is direct. For $U > 0$, put $M = U - \mu > 0$, $v = \mathbb{E}(W-\mu)^2 \ge 0$, and use the centered two-point variable Y in (M.47): its values are $-v/M, M$, with probabilities $M^2/(M^2+v), v/(M^2+v)$. For $f(w) = e^{sw}$, $s \ge 0$, take $Q = Q_{\mu - v/M}$ from (M.44). Equation (M.48) gives

$$\mathbb{E}e^{sW} \le \mathbb{E}Q(W) = e^{s\mu}\mathbb{E}e^{sY}.$$

Put $L = M + v/M$. Its upper probability is $\pi = v/(M^2+v) \le \theta/(1+\theta) = 13/45$, because $v \le \theta U^2 - \mu^2 \le \theta(U-\mu)^2$. Apply (Q65.2.1)–(Q65.2.2) to $Y = L(B-\pi)$:

$$\boxed{\mathbb{E}e^{sW} \le \exp\{s\mu + (47/400)s^2L^2\}.} \tag{Q65.2.3}$$

At $v = 0$, $Y = 0$ and the same formula holds. If, further, $U^2 = A_0 d$, $|\mu| \le \sqrt{A_0 d}$, $A_0 > 0$, then $a_1 = |\mu|/U \le \sqrt{d}$, and

$$L \le U\left(2 - \frac{\kappa}{1+a_1}\right) \le \sqrt{A_0 d}\left(2 - \frac{\kappa}{1+\sqrt{d}}\right).$$

For $s = 8$, this gives

$$\boxed{\mathbb{E}e^{8W} \le \exp\{8\mathbb{E}W + \lambda(\sqrt{d})A_0 d\}, \qquad \lambda(x) = \frac{188}{25}\left(2 - \frac{\kappa}{1+x}\right)^2.} \tag{Q65.2.4}$$

For the continuous interval $0 \le x \le 9/20$,

$$\boxed{\lambda(x) \le 149/10 + (63/5)x - 4x^2.} \tag{Q65.2.5}$$

Indeed, with $Q(x) = (2 - \kappa/(1+x))^2$,

$$Q(0) = 2025/1024, \quad Q'(0) = 855/512, \quad Q''(x)/2 = \frac{\kappa(3\kappa - 4 - 4x)}{(1+x)^4}.$$

Since $Q'''(x) = 24\kappa(1 + x - \kappa)/(1+x)^5 > 0$, throughout the interval $Q''/2$ is at most its right endpoint value $-1527125/2829124 \le -25/47$. The cross-products are $1527125 \cdot 47 = 71774875 > 2829124 \cdot 25 = 70728100$. Integrating twice gives

$$\lambda(x) \le 3807/256 + (8037/640)x - 4x^2.$$

Now $3807/256 < 149/10$, $8037/640 < 63/5$ imply (Q65.2.5). Also λ increases, so

$$\lambda(x) \le \frac{188}{25}(77/48)^2 = 278663/14400 < 20 \qquad (0 \le x \le 1/2). \tag{Q65.2.6}$$

Q65.3 Finite paths, the high domain and a positive envelope.

Set $\beta_1 = \beta_2 = 0$, $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j$, $S_n = \sum_{j=0}^{n} K^j \mathbf{1}$, and $G_s(z) = \sum_{j=0}^{s-1} z^j$. The finite positive convolution is

$$S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad \tau \le \beta_j \le d \quad (j \ge 3). \tag{Q65.3.1}$$

Also $\mathbb{E}S_n = 1 - \beta_{n+1}$, $\mathbb{E}(gS_n) = \beta_{n+2}$, $\beta_3 = \beta_4 = d$, $\beta_5 = \beta_6 = \tau$, with the finite alternating-sum bounds

$$\beta_j \in [\tau + \kappa\delta_3, \tau + \delta_3] \quad (j \ge 7),$$
$$\beta_j \in [\tau + \delta_3 - \delta_4, \tau + \delta_3 - \delta_4 + \delta_5] \quad (j \ge 9).$$

The all-index mean bound $\mathbb{E}f_n \le 1 - 83d/32$, $n \ge 4$, is proved in Appendix M.17, following Q4-C134, §2.3. Together with $h_c \le 2\mathbb{E}f_{c-1}$ and the short paths, it gives

$$h_c \le 77/80 \quad (c \ge 5, d \ge 1/5), \qquad h_4 \le 1 \quad (d \ge 1/4). \tag{Q65.3.2}$$

The first bound follows from $2(1 - 83d/32)$. For the second, $h_4 = 2 - S_4 - d - dq$ and $S_4 \ge d(1+p)$ give $h_4 \le 2 - 4d \le 1$. In these high domains the message is bounded by qf_b. Q5-C051 supplies the

same-direction unit bound $\mathbb{E}[(pT(qf_b))^8] \le 1 - d/500$. It remains to consider $c \ge 5, d \le 1/5$, and the short case $(b,c) = (3,4)$ on $d \le 1/4$. For fixed b, c, set

$$\alpha = \beta_c, \quad B = \beta_{c-1}, \quad \Delta = \beta_b + \alpha + B, \quad m = 1 - \Delta/2, \quad t = m - B,$$
$$W = S_c - S_b = K^b G_{c-b}(K)g, \quad U = Bg - W, \quad H_0 = q + U/m = 1 - (tg + W)/m. \tag{Q65.3.3}$$

Retaining β_b in the positive convolution for f_b, and $\alpha + Bq$ in that for h_c, gives

$$f_b + h_c \le 2 - \Delta + Bg - W, \qquad P := qf_b h_c \le q(m + U/2)^2. \tag{Q65.3.4}$$

For $c \ge 5, d \le 1/5$, one has $m \ge 7/10$, $t \ge 1/2$. To prove $H_0 \ge 0$, put $v_0 = K^{b-1}G_{c-b}(K)g$, so $W = Kv_0$, $\mathbb{E}v_0 = \beta_b - \alpha \le \delta_2 \le B$. Every finite G_s has modulus at most $\sqrt{1+\theta}$, and the row bound gives

$$|v_0|^2 \le (1+\theta)\delta_{b-1} \le \theta(1+\theta)d \le 117/1024 < 1/4.$$

Thus $z = t + v_0 \ge 0$ and $mH_0 = Tz + B - \mathbb{E}v_0 \ge 0$. For $(b,c) = (3,4), d \le 1/4$, one has $v_0 = K^2 g$, $\mathbb{E}v_0 = 0$, and $|v_0| \le \sqrt{\theta d} < 3/8 \le t$, proving positivity there too. Apply T to (Q65.3.4), and put $J_0 = \mathbb{E}U^2$:

$$TP \le m^2 TH_0 - m\mathbb{E}(gU) + mK(gU) + \tfrac{1}{4}T(qU^2).$$

The bounds $|K(gU)| \le \sqrt{dJ_0}$, $T(qU^2)/4 \le J_0$ show that any nonnegative δ satisfying

$$\delta d \ge m\{\sqrt{dJ_0} - \mathbb{E}(gU)\} + J_0 \tag{Q65.3.5}$$

gives $TP \le m^2 TH_0 + \delta d$. Define $D = \beta_{b+2} - \beta_{c+2} = -\mathbb{E}(gW)$ and $N = K(tg + W)$. Then

$$TH_0 = q - \mathbb{E}W/m + N/m, \qquad \mathbb{E}N = -td + D. \tag{Q65.3.6}$$

Both mp and $mTH_0 + \delta d/m$ are nonnegative. AM–GM gives

$$0 \le pTP \le \left(m - \frac{\mathbb{E}W}{2} + \frac{\delta d}{2m} + \frac{N}{2}\right)^2. \tag{Q65.3.7}$$

Suppose that this same N has an $A_0 > 0$ with

$$|N|^2 \le A_0 d, \quad \mathbb{E}N^2 \le \theta A_0 d, \quad \mathbb{E}N \le 0, \quad |\mathbb{E}N| \le \sqrt{A_0}d. \tag{Q65.3.8}$$

Equation (Q65.2.4), the nonnegative base in (Q65.3.7), and $\log z \le z - 1$ yield $\mathbb{E}[(pTP)^8] \le e^{d\Phi}$, where

$$\boxed{\Phi = -8\{\Delta - (D - \mathbb{E}W)\}/d - 8t + 8\delta/m + \lambda(\sqrt{d})A_0.} \tag{Q65.3.9}$$

At a zero base, the comparison holds directly.

Q65.4 Bounds for all finite tails.

At $z = i\sqrt{x}$, $0 \le x \le \theta$, write $G_\infty = (1+z)/(1+x)$. Finite summation gives

$$G_\infty = \frac{G_3 + xG_1}{1+x}, \qquad G_{4h+r} = (1 - x^{2h})G_\infty + x^{2h}G_r \quad (r = 1,2,3,4). \tag{Q65.4.1}$$

Every finite G_s lies in the real convex hull of G_1, G_2, G_3, G_4. Thus $|A + BG_s|^2$, for fixed complex A, B, is bounded by its four vertex values. For $b = 3$, the vertices of $t + z^3 G_s$ have real parts between t

and $t + x^2$, with imaginary magnitude at most $x^{3/2}$; the second vertex dominates. For $b = 4$, the real parts lie between $t+x^2-x^3$ and $t+x^2$, and the imaginary magnitude is at most $x^{5/2}$; again the second vertex dominates. For $b = 5$, $t \geq 1/2 > x^3$, so the real parts are nonnegative and at most t, and the imaginary magnitude is at most $x^{5/2}$; the first vertex dominates. Row and spectral bounds therefore give the following choices in (Q65.3.8):

b	$A_0(t)$
3	$(t+\theta^2)^2+\theta^3$
4	$(t+\theta^2)^2+\theta^5$
5	$t^2+1/64$
≥ 6	$(t+7/250)^2+4/625.$

(Q65.4.2)

The first two follow from the vertex bounds; the third uses $\theta^5 < 1/64$. For $b \geq 6$, $(-D)_+ \leq \delta_5$, $D_+ \leq \delta_4$, and

$$\|W\|^2/d \leq (1+\theta)\theta^b < 4/625, \qquad \|tg+W\|^2/d \leq t^2 + 2\theta^4 t + 4/625.$$

Use $\theta^4 < 7/250$ for the last row. In every case $\|K(tg+W)\|^2 \leq \theta\|tg+W\|^2$. For the mean conditions, $b = 3, 4$ have $-\delta_3 \leq D \leq 0$, so $\mathbb{E}N \leq 0$, $|\mathbb{E}N| \leq (t+\theta^2)d$. For $b = 5$, $0 \leq D \leq \delta_4$, giving $\mathbb{E}N < 0$, $|\mathbb{E}N| \leq td$. For $b \geq 6$, $-\delta_5 \leq D \leq \delta_4$, giving $\mathbb{E}N < 0$, $|\mathbb{E}N| \leq (t+\theta^4)d$; here $t \geq 1/2 > \theta^3$. Each upper bound is at most $\sqrt{A_0}d$. For the all-index interval when $b \geq 6$, at $b = 6$, $\beta_{b+2} = \beta_8 = \tau + \delta_3$ is the upper endpoint of its alternating layer, hence $D \geq 0$. For $b \geq 7$, both indices are at least nine, with interval width at most δ_5. The common interval for all indices at least eight has width at most δ_4, giving $D_+ \leq \delta_4$.

For $c \geq 5, d \leq 1/5$, the error in (Q65.3.5) may be chosen as

b	δ
3	$(143/300)m + d^2 + 19/200$
4	$(11/30)m + (d+1/5)^2$
5	$m/8 + (d+1/8)^2$
≥ 6	$(3/20)m + (d+2/25)^2.$

(Q65.4.3)

Indeed $J_0 = \|Bg - W\|^2$, $\mathbb{E}(gU) = Bd - \mathbb{E}(gW)$, $B \leq d$. For $b = 3$, $\mathbb{E}(gW) \geq 0$ and

$$J_0/d \leq B^2 + C_0, \quad C_0 = (1+\theta)\theta^3 = 98865/1048576 < 19/200, \quad \sqrt{C_0} < 31/100.$$

Also $\mathbb{E}(gW)/d \leq \theta^2 < 1/6$, so $\sqrt{J_0/d} - B + \mathbb{E}(gW)/d \leq 31/100 + 1/6 = 143/300$, proving the first row. For $b = 4$, $\|W\|/\sqrt{d} \leq \theta^2\sqrt{1+\theta} < 1/5$, and $\mathbb{E}(gW)/d \leq \theta^2 < 1/6$; Minkowski gives the second row. For $b = 5$, $\mathbb{E}(gW) \leq 0$, and $(1+\theta)\theta^5 = 16708185/1073741824 < 1/64$, hence $\|W\|/\sqrt{d} < 1/8$, proving the third. For $b \geq 6$, $\|W\|/\sqrt{d} \leq \theta^3\sqrt{1+\theta} < 2/25$, and $\mathbb{E}(gW)/d \leq \theta^3 < 7/100$, proving the final row. For the strict $2/25$ comparison, use $\theta^3 < 269/4000$, $\sqrt{1+\theta} < 19/16$, and $5111/64000 < 2/25$. Each δ is nonnegative and uses the same m.

The coupled mean and convolution loss satisfy the finite identity

$$D - \mathbb{E}W = \mathbb{E}K^{c+1}\mathbf{1} - \mathbb{E}K^{b+1}\mathbf{1}. \tag{Q65.4.4}$$

For $b = 3, c \geq 5$, the right side is at most $-\delta_2$ when $c = 5, 6$, and at most $\delta_4 - \delta_2$ when $c \geq 7$. In the latter case $\alpha \geq \tau + \kappa\delta_3$, and $\delta_4 \leq \theta\delta_3 < \kappa\delta_3$. Using $\beta_b = d$, $B \geq \tau$, gives

$$\Delta - (D - \mathbb{E}W) \geq 3d - \delta_2. \tag{Q65.4.5}$$

For $b = 4, c \geq 5$, $D - \mathbb{E}W \leq 0$ at $c = 5, 6$, and $D - \mathbb{E}W \leq \delta_4$ at $c \geq 7$. The extra $\kappa\delta_3$ in the same α pays this, giving

$$\Delta - (D - \mathbb{E}W) \geq 3d - 2\delta_2. \tag{Q65.4.6}$$

For $b = 5$, $D - \mathbb{E}W \leq \delta_3 + \delta_4$, so

$$\Delta - (D - \mathbb{E}W) \geq 3\tau - (\delta_3 + \delta_4). \tag{Q65.4.7}$$

For every $b \geq 6$, $D - \mathbb{E}W \leq \delta_4$, giving

$$\Delta - (D - \mathbb{E}W) \geq 3\tau - \delta_4. \tag{Q65.4.8}$$

For this last bound, if b is even the second term in (Q65.4.4) is zero, and the positive part of the first is at most δ_4. If b is odd and its second term is nonnegative, discard it. In the remaining class, $b + 1 \equiv 2 \pmod 4$ and $b + 1 \geq 10$, so the difference is at most $\delta_5 + \delta_6 \leq \theta(1 + \theta)\delta_4 < \delta_4$.

Q65.5 The continuous low-domain estimates for $c \geq 5$.

Take $0 < d \leq 1/5$, $v = \delta_2/d \in [d, \theta]$, and $x = \sqrt{d} < 9/20$. In dividing (Q65.4.3) by m, use $m \geq 7/10$. The bounds on t are

$$\begin{aligned} b = 3, 4: &\quad t \leq 1 - (5/2 - 2v)d, \\ b \geq 5: &\quad t \leq 1 - (5/2)(1 - v)d. \end{aligned} \tag{Q65.5.1}$$

For every A_0 in (Q65.4.2), the function $-8t + \lambda(x)A_0(t)$ increases on $t \geq 1/2$, since $\lambda(x) \geq 3807/256 > 14$ and $A_0'(t) \geq 2t \geq 1$. Substitute (Q65.5.1). The resulting bound also increases in v: its explicit linear term has positive derivative, the upper bound on t increases with v, and its derivative in t is positive. Thus take $v = \theta$. For $b = 3, 4$, put $a = \theta^2$, $k = 27/16$, $t_0 = 1 - kx^2$. For $b = 5$ and $b \geq 6$, use $k = 95/64$ and $t_0 = 1 - kx^2$.

Q65.5.1 The class $b = 3, c \geq 5$.

Equations (Q65.3.9), (Q65.4.3), and (Q65.4.5) bound the exponent coefficient by

$$-24 + 8\theta - 8t_0 + 286/75 + (80/7)(x^4 + 19/200) + \lambda(x)\{(t_0 + \theta^2)^2 + \theta^3\}. \tag{Q65.5.2}$$

Write $A_0 = A_{00} - A_{02}x^2 + k^2x^4$, where

$$7/5 < A_{00} = 1493553/1048576 < 57/40, \quad A_{02} = 32211/8192,$$
$$39/10 < A_{02} < 63/16, \qquad k^2 < 57/20.$$

Expansion with (Q65.2.5) gives the coefficientwise upper bound

$$-2 + 18x - 50x^2 - 49x^3 + 70x^4 + 36x^5 - 4k^2x^6. \tag{Q65.5.3}$$

For the constant, its part outside λ satisfies

$$-115/4 + 286/75 + 38/35 < -119/5.$$

Adding $(149/10)(57/40) = 8493/400$ leaves a value below -2. The linear coefficient is less than $(63/5)(57/40) < 18$. The magnitude of the negative quadratic coefficient is at least $(149/10)(39/10) + 4(7/5) - 27/2 > 50$; the negative cubic coefficient has magnitude at least $(63/5)(39/10) > 49$. The positive quartic coefficient is at most $(149/10)(57/20) + 63/4 + 80/7 < 70$, and the quintic coefficient is less than $(63/5)(57/20) < 36$. The sixth-degree term is nonpositive. Since $x \leq 9/20$, the cubic through quintic terms sum to at most $x^3[-49 + 70(9/20) + 36(9/20)^2] < 0$. Completing the square gives

$$\Phi \leq -2 + 18x - 50x^2 \leq -2 + 81/50 = -19/50. \tag{Q65.5.4}$$

Q65.5.2 The class $b = 4, c \geq 5$.

Using (Q65.4.6), the exponent coefficient is at most

$$-24 + 16\theta - 8t_0 + 44/15 + (80/7)(x^2 + 1/5)^2 + \lambda(x)\{(t_0 + \theta^2)^2 + \theta^5\}. \tag{Q65.5.5}$$

Here A_{02} is unchanged, while $A_{00} = 45915261/33554432 < 1369/1000$ and $A_{00} > 27/20$. The first cross-product comparison is $45915261000 < 45936017408$. The constant outside λ is exactly $-4643/210$, so the total constant is at most $-4643/210 + 203981/10000 < -17/10$, using $46406010 < 46430000$. The linear coefficient is less than $(63/5)(1369/1000) < 173/10$. The negative quadratic coefficient has magnitude greater than

$$(149/10)(39/10) + 4(27/20) - 27/2 - 32/7 > 45.$$

The cubic through sixth-degree bounds are those in (Q65.5.3), and their sum is nonpositive. Thus

$$\Phi \leq -17/10 + (173/10)x - 45x^2 \leq -671/18000 < -1/30. \tag{Q65.5.6}$$

This uses the coupled alternating loss in the same α from (Q65.4.6).

Q65.5.3 The class $b = 5, c \geq 6$.

Put $k = 95/64$, $t_0 = 1 - kx^2$. By (Q65.4.7), the exponent coefficient is at most

$$-24\kappa - 8t_0 + 8\theta^2(1 + \theta) + 1 + (80/7)(x^2 + 1/8)^2 + \lambda(x)(t_0^2 + 1/64). \tag{Q65.5.7}$$

Here $\delta_3 + \delta_4 \leq \theta(1 + \theta)\delta_2$, and the same increasing comparison in v has already set $v = \theta$. Using $\theta^2(1 + \theta) < 7/30$, the constant is at most

$$-57/4 - 7 + 28/15 + 5/28 + (149/10)(65/64) < -4.$$

The linear coefficient is less than 13, and the magnitude of the negative quadratic coefficient is

$$(109/5)k + 65/16 - 20/7 = 15037/448 > 33.$$

The negative cubic coefficient has magnitude $(126/5)k = 1197/32 > 37$. The positive quartic coefficient is less than $(149/10)(9/4) + 12 + 80/7 < 57$; the quintic coefficient is $(63/5)k^2 = 568575/20480 < 28$, and the sixth-degree term is nonpositive. Therefore

$$\Phi \leq -4 + 13x - 33x^2 + x^3(-37 + 57x + 28x^2) \leq -4 + 169/132 < -2. \tag{Q65.5.8}$$

The parenthesis is negative throughout $x \leq 9/20$.

Q65.5.4 The class $b \geq 6, c > b$.

With the same $k = 95/64$, $t_0 = 1 - kx^2$, (Q65.4.8) gives

$$-24\kappa - 8t_0 + 8\theta^3 + 6/5 + (80/7)(x^2 + 2/25)^2 + \lambda(x)\{(t_0 + 7/250)^2 + 4/625\} \tag{Q65.5.9}$$

as an upper bound for Φ. Here $A_{00} = 66449/62500$, $1 < A_{00} < 107/100$, and $A_{02} = 24415/8000$, $61/20 < A_{02} < 31/10$. Using $\theta^3 < 7/100$, the constant is at most

$$-57/4 - 8 + 14/25 + 6/5 + 64/875 + (149/10)(107/100) < -4.$$

The linear coefficient is less than 14. The magnitude of the negative quadratic coefficient is greater than

$$(149/10)(61/20) + 4 - 95/8 - 64/35 > 35.$$

The negative cubic coefficient has magnitude greater than $(63/5)(61/20) > 38$, the positive quartic coefficient is less than $(149/10)(9/4) + 4(31/10) + 80/7 < 58$, the quintic coefficient remains less than 28, and the sixth-degree term is nonpositive. Thus

$$\Phi \le -4 + 14x - 35x^2 + x^3(-38 + 58x + 28x^2) \le -4 + 196/140 = -13/5 < -2. \quad \text{(Q65.5.10)}$$

The cubic through quintic parenthesis is negative on $x \le 9/20$. The four classes cover every $c \ge 5$, proving (Q65.1.4).

Q65.6 The short case $(1, 3, 4)$ on $d \le 1/4$.

Here $\Delta = 3d$, $m = 1 - 3d/2$, $t = 1 - 5d/2 \ge 3/8$, $B = d$, $W = K^3 g$. Exactly, $\mathbb{E}W = \delta_2$, $\mathbb{E}(gW) = 0$, $D = 0$, and $J_0 = d^3 + \delta_4$. Since $\sqrt{dJ_0} - d^2 \le \sqrt{d\delta_4}$, $\theta^3 < (13/50)^2$, and $1/16 + \theta^3 = 4245/32768 < 13/100$, one may take $\delta = 2/5$ in (Q65.3.5). The positivity proof of H_0 in part Q65.3 applies on this range. For $N = K(tg + W)$, the row and spectral bounds give

$$|N|^2 \le d(t^2 + \theta^3), \quad \mathbb{E}N^2 \le \theta d(t^2 + \theta^3), \quad \mathbb{E}N = -td.$$

Thus (Q65.2.4) applies with $A_0 = t^2 + \theta^3$. Using (Q65.2.6), $m \ge 5/8$, and discarding $-8\mathbb{E}W/d \le 0$ in (Q65.3.9),

$$\Phi \le -24 - 8t + 128/25 + 20(t^2 + \theta^3).$$

The function $-8t + 20(t^2 + \theta^3)$ increases on $[3/8, 1]$. Since $\theta^3 < 7/100$,

$$\Phi \le -24 - 8 + 128/25 + 20(1 + 7/100) = -137/25 < -5. \quad \text{(Q65.6.1)}$$

This proves (Q65.1.5), including $d = 1/4$.

Q65.7 Completion, decorations and reversal.

The common low-domain estimate satisfies

$$e^{-d/30} \le (1 + d/30)^{-1} = 1 - \frac{d}{30 + d} \le 1 - d/31 \le 1 - d/500.$$

The high domains follow from (Q65.3.2) and Q5-C051. At $d = 0$, $g = 0$ while K may be nonzero; leaf integration makes all tree messages one. Every division by d above applies on $d > 0$. Finally $0 \le E_0, E_r \le 1$ and positivity of T give

$$0 \le H_D = E_r pT(E_0 q f_b h_c) \le pT(q f_b h_c).$$

Taking eighth powers and integrating proves the decorated bound. Apply the result to $-K$, reversing the terminal, connector, root unit arm and all equal pairs together. This gives the corresponding ordinary eighth moment for qAF^-. The chosen $q f_b h_c$ direction also has the all-positive-length fourth-moment bound from Q4-C134, proved in Section J.3:

$$\mathbb{E}[p(T^L F)^4] \le 1 \qquad (L \ge 1).$$

O.15 The unit-connector eighth moment for odd terminals with a unit arm and multiple pairs

Q67.1 Directions and bounds.

Use the finite probability space, real skew kernel K, and operators and messages of Appendix M.15: $A = J + K$, $T = J - K = A^*$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $f_n = A^n\mathbf{1}$, $\bar{f}_n = T^n\mathbf{1}$, $h_n = Tf_{n-1}$, $\delta_j = \|K^j\mathbf{1}\|_2^2$, $d = \delta_1$, $\theta = 13/32$, $\kappa = 19/32$, and $\tau = d - \delta_2$. Fix any finite $k \geq 2$ and strictly increasing list $1 < b_1 < c_1 < b_2 < c_2 < \cdots < b_k < c_k$. Let E_0, E_r be arbitrary finite genuine equal-pair products, with any lengths and multiplicities; the empty product is one. Set

$$F = E_0 q f_{b_1} h_{c_1} \prod_{i=2}^{k} h_{b_i} f_{c_i}, \qquad H_D = E_r p T F. \tag{Q67.1.1}$$

The terminal unit arm points inward. In the first pair the short arm is outward, while the long arm has its first edge inward and its remaining edges outward. Every later pair has the short arm first inward and then outward, and the long arm outward. The unit connector points from terminal to root; the root unit arm points outward. Each equal pair consists of an outward arm and an arm first inward and then outward. These directions depend only on the arm list. For all such parameters,

$$\boxed{\mathbb{E}H_D^8 \leq e^{-d/2} \leq 1 - 2d/5.} \tag{Q67.1.2}$$

On $d \geq 1/24$, the stronger bound is e^{-9d}. Combining the multiple-pair result with the triple bounds of Q5-C062, Q5-C063 and Appendix O.14 gives, for every odd distinct residual list of at least three arms and minimum one, in these same directions,

$$\boxed{\mathbb{E}H_D^8 \leq 1 - d/601.} \tag{Q67.1.3}$$

Three residual arms use the triple bounds; at least five use (Q67.1.2). Residual zero and one follow the reduction to fewer branch vertices.

The finite convolution, the all-index mean bound $\mu_n \leq 1 - 83d/32$ for $n \geq 4$, and the finite-mask norm are proved in Appendix M.17. Its long-pair estimate (L90.2.3) supplies in particular $h_a f_c \leq e^{R_{ac} - 3d}$ for $4 \leq a < c$, where $R_{ac} = \sum_{j=a}^{c-1} K^j g$. We use the finite identity $S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}$, with $S_n = \sum_{j=0}^{n} K^j\mathbf{1}$, $\beta_1 = \beta_2 = 0$, $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j$, and $\tau \leq \beta_n \leq d$ for $n \geq 3$.

Q67.3 Unit propagation retaining the negative mean.

Suppose $0 < d \leq 1/24$ and a nonnegative message F satisfies

$$F \leq e^{R - \Gamma}, \quad |R| \leq 5/4, \quad \mathbb{E}R^2 \leq A_0 d, \quad \mathbb{E}(gR) \leq 0, \qquad A_0 > 0. \tag{Q67.3.1}$$

Put $N = -KR$. Row and spectral bounds, followed by Cauchy in the same inner product, give

$$|N| \leq \sqrt{A_0 d}, \quad \mathbb{E}N^2 \leq \theta A_0 d, \quad \mathbb{E}N = \mathbb{E}(gR) \leq 0, \quad |\mathbb{E}N| \leq \sqrt{A_0} d. \tag{Q67.3.2}$$

The negative-mean lemma (Q65.2.4), including its two-point comparison, Hermite interpolation and degenerate cases, gives

$$\mathbb{E}e^{8N} \leq \exp\{8\mathbb{E}N + \lambda(\sqrt{d}) A_0 d\}, \qquad \lambda(x) = \frac{188}{25}\left(2 - \frac{\kappa}{1 + x}\right)^2. \tag{Q67.3.3}$$

Since λ increases and $\sqrt{d} \le 5/24$,

$$\lambda(\sqrt{d}) \le \lambda(5/24) = 57575/3364 < 69/4, \tag{Q67.3.4}$$

where $230300 < 232116$. For every real $u \le 5/4$, one has $e^u \le 1 + u + (4/5)u^2$. Indeed, $(e^u - 1 - u)/u^2 = \int_0^1 (1-t)e^{tu}\,dt$ increases with u, so it suffices that $e^{5/4} < 7/2$. The exponential series through degree four sums to $21249/6144$; starting with degree five, adjacent tail terms have ratio at most $5/24$. Thus the total is at most

$$21249/6144 + 3750/116736 = 407481/116736 < 7/2.$$

Positivity of T and $|K(R^2)| \le \mathbb{E}R^2$ give

$$Te^R \le q + \mathbb{E}R - KR + (8/5)\mathbb{E}R^2 = q + C + N, \qquad C = \mathbb{E}R + (8/5)\mathbb{E}R^2. \tag{Q67.3.5}$$

The right side is nonnegative. Since $p + q = 2$, AM–GM gives $p(q + C + N) \le [1 + (C + N)/2]^2$ with nonnegative base. Use $(1 + v)^{16} \le e^{16v}$ for $v \ge -1$, obtaining

$$\boxed{\mathbb{E}[(pTF)^8] \le \exp\{-8\Gamma + 8\mathbb{E}R + 8\mathbb{E}(gR) + (601/20)A_0 d\}.} \tag{Q67.3.6}$$

Here $601/20 = 64/5 + 69/4$. A zero base satisfies the comparison directly. Both signed means remain in this bound.

Q67.4 The odd-table envelope and high domain.

First take $E_0 = E_r = 1$. Write $b = b_1$, $c = c_1$, $B = \beta_{c-1}$, and define

$$W = \sum_{j=b}^{c-1} K^j g, \qquad S = \sum_{i=2}^{k} \sum_{j=b_i}^{c_i - 1} K^j g, \tag{Q67.4.1}$$

$$\Delta = \beta_b + \beta_c + B, \quad \alpha = 1 - B, \quad R = -\alpha g - W + S, \quad \Gamma = \Delta + 3(k-1)d.$$

Finite convolution and $h_c = Tf_{c-1}$ give $f_b + h_c \le 2 - \Delta + Bg - W$. Apply AM–GM, $q \le e^{-g}$, and the long-pair bound to obtain

$$q f_b h_c \prod_{i=2}^{k} h_{b_i} f_{c_i} \le e^{R - \Gamma}. \tag{Q67.4.2}$$

The later exponent intervals are disjoint, so $S = K^{c+1}Q(K)g$, where Q has coefficients zero or one and may have initial zero coefficients. The finite-mask bound is

$$\|Q(K)\|^2 \le \frac{1+\theta}{(1-\theta^2)^2} = 32768/16245 < 101/50 < (10/7)^2. \tag{Q67.4.3}$$

For $c \ge 3$, this implies $|S| \le (10/7)\theta^{3/2}\sqrt{d} < (3/8)\sqrt{d}$; the squared comparison is $14060800 < 14450688$. If $d \ge 1/24$, $\sqrt{d} \le \sqrt{24}d < 5d$, so

$$\prod_{i=2}^{k} h_{b_i} f_{c_i} \le e^{S - 3(k-1)d} \le e^{-9d/8}. \tag{Q67.4.4}$$

Positivity and the same-direction unit eighth moments of Q5-C062 for $(b, c) = (2, 3)$, Q5-C063 for $b = 2, c \ge 4$, and Appendix O.14 for $b \ge 3$, give

$$\mathbb{E}[(pTF)^8] \le e^{-9d}\mathbb{E}[(pT(qf_b h_c))^8] \le e^{-9d}. \tag{Q67.4.5}$$

Q67.5 Signs, pointwise bounds and finite frequencies on the low domain.

Now let $0 < d \le 1/24$, so $23/24 \le \alpha \le 1$. Every finite G_m satisfies $\|G_m(K)\| \le \sqrt{1+\theta} < 6/5$. Apply the original row bound to the outermost K in W and S:

$$|R| \le 1 + \left((6/5)\sqrt{\theta} + (10/7)\theta^{3/2}\right)\sqrt{d} < 1 + (6/5)\sqrt{d} \le 5/4. \tag{Q67.5.1}$$

The middle coefficient is bounded using $\sqrt{\theta} < 2/3$, which gives $997/840 < 6/5$. For the single interval W, alternating sums give $-\mathbb{E}(gW) \le \delta_2$. The positive mixed-mean terms of S begin at $K^4 g$, and their sum is at most $\delta_3/(1-\theta^2) < \delta_2/2$. Therefore

$$\mathbb{E}(gR) \le -\alpha d + (3/2)\delta_2 \le -67d/192 < 0. \tag{Q67.5.2}$$

We next prove, for all finite intervals and masks,

$$\mathbb{E}R^2 \le A_0 d, \qquad A_0 = \begin{cases} 1, & b = 2, \\ 33/20, & b = 3, \\ 19/12, & b = 4, \\ 5/4, & b \ge 5. \end{cases} \tag{Q67.5.3}$$

On a skew-operator rotation block, put $z = i\sqrt{x}$, $0 \le x \le \theta$, and write $W = W_0(K)g$, $S = U(K)g$. The polynomial representing $-R$ is $P(z) = \alpha + W_0(z) - U(z)$, where $W_0 = \sum_{j=b}^{c-1} z^j$, and the zero-one coefficients of U begin at degree at least $c+1$. Finite sums in the four residue classes give the coordinate bounds below. We use

$$\begin{gathered} \frac{x^2}{1-x^2} \le 169/855 < 1/5, \quad \frac{x}{1-x^2} \le 416/855 < 1/2, \quad \frac{x^3}{1-x^2} < 1/12, \\ x^2 < 1/6, \quad x^3 < 1/14, \quad x^4 < 1/32, \quad x^5 < 1/90, \quad \frac{x^4}{1-x^2} < 1/30. \end{gathered} \tag{Q67.5.4}$$

The third fraction is at most $2197/27360 < 1/12$. For x^3 and x^5, the integer comparisons are $30758 < 32768$ and $33416370 < 33554432$; the remaining bounds follow by cross multiplication.

Q67.5.1 The class $b = 2, c \ge 3$.

At $c = 3$, $B = 0$, $P = 1 + z^2 - U$, with U starting at degree four. The real part is positive and at most $1 - 4x/5$, and the imaginary magnitude is at most $x^{5/2}/(1-x^2)$. Hence

$$\frac{|P|^2 - 1}{x} \le -8/5 + 16x/25 + \frac{x^4}{(1-x^2)^2} \le -8/5 + 8/25 + 1/25 < 0.$$

At $c = 4$, $W_0 = z^2 + z^3$, with U starting at degree five. The real part is positive and at most $1 - 4x/5$; the imaginary part is negative, with magnitude at most $(3/2)x^{3/2}$. Thus

$$\frac{|P|^2 - 1}{x} \le -8/5 + 16x/25 + 9x^2/4 \le -8/5 + 8/25 + 3/8 < 0.$$

For $c \ge 5$, the real part of W_0 lies in $[-x, -x + x^2]$, and its imaginary part in $[-x^{3/2}, -x^{3/2}(1-x)]$. The mask U starts at least at degree six, with positive imaginary terms starting at nine and negative imaginary terms at seven. The real part of P is positive and

$$\operatorname{Re} P \le 1 - x + x^2 + \frac{x^3}{1-x^2} \le 1 - 3x/8.$$

Here $1 - x - x^2/(1-x^2) \geq 1 - \theta - 1/5 = 63/160 > 3/8$. Its imaginary part remains nonpositive because $1 - x > x^2/(1-x^2)$, and

$$|\operatorname{Im} P| \leq x^{3/2}\left(1 + \frac{x^3}{1-x^2}\right) \leq (13/12)x^{3/2}.$$

Consequently

$$\frac{|P|^2 - 1}{x} \leq -3/4 + 9x/64 + 169x^2/144 \leq -3/4 + 9/128 + 169/864 = -1673/3456 < 0.$$

The real-part positivity used above follows from $1-x-x^2/(1-x^2) > 0$ or $23/24-x-x^4/(1-x^2) > 0$, respectively. At zero the bound is direct. This proves $A_0 = 1$ for every finite c.

Q67.5.2 The class $b = 3, c \geq 4$.

At $c = 4$, $W_0 = z^3$, and U starts at degree five. The real part is positive and at most $13/12$, and the imaginary magnitude is at most $(3/2)x^{3/2}$. Thus $|P|^2 \leq 169/144+9/56 = 1345/1008 < 33/20$. For $c \geq 5$, $\operatorname{Re} W_0 \in [0, x^2]$, $\operatorname{Im} W_0 \in [-x^{3/2}, -x^{3/2}(1-x)]$. The mask starts at least at degree six, with negative real terms starting at six and positive imaginary terms at nine. The real part of P is positive and at most $1 + x^2 + x^3/(1-x^2) \leq 5/4$. As above its imaginary part is nonpositive, of magnitude at most $(13/12)x^{3/2}$. Therefore

$$|P|^2 \leq 25/16 + 169/2016 = 3319/2016 < 33/20,$$

where $66380 < 66528$.

Q67.5.3 The class $b = 4, c \geq 5$.

Here $\operatorname{Re} W_0 \in [x^2(1-x), x^2]$ and $\operatorname{Im} W_0 \in [0, x^{5/2}]$; the latter is zero at $c = 5$. The mask begins at least at degree six. The real part is at least $23/24 - x^4/(1-x^2) > 0$, and at most

$$1 + x^2 + \frac{x^3}{1-x^2} \leq 319/256.$$

This uses $\theta^2 = 169/1024$ and $2197/27360 < 83/1024$, with $2249728 < 2270880$. The imaginary magnitude is at most $x^{5/2}[1 + x/(1-x^2)] \leq (3/2)x^{5/2}$. Thus

$$|P|^2 \leq 101761/65536 + 1/40 < 19/12, \tag{Q67.5.5}$$

where the final comparison is $12211320 < 12255232$.

Q67.5.4 The class $b \geq 5, c > b$.

The contiguous W_0 has real part in $[-x^3, x^4]$ and imaginary magnitude at most $x^{5/2}$. The mask begins at least at degree seven: its negative real terms start at ten, its positive real terms at eight, and its imaginary magnitude is at most $x^{7/2}/(1-x^2)$. Thus the real part of P is positive and

$$\operatorname{Re} P \leq 1 + x^4 + \frac{x^5}{1-x^2} < 1 + 1/32 + 1/70 < 21/20.$$

The imaginary magnitude is at most $(3/2)x^{5/2}$. For positivity use $23/24 - x^3 - x^4/(1-x^2) > 0$. It follows that $|P|^2 \leq 441/400 + 1/40 = 451/400 < 5/4$. These four classes prove (Q67.5.3) for arbitrary finite degree, pair count and gaps.

Q67.6 The common linear functional and continuous low-domain bounds.

Write $a_j = \mathbb{E}K^j\mathbf{1}$, so $a_{2j} = (-1)^j\delta_j$ and every odd a_j is zero. Skew adjunction gives the telescoping identity

$$\mathbb{E}W + \mathbb{E}(gW) = a_{b+1} - a_{c+1}. \tag{Q67.6.1}$$

Put $V = a_{c+1} - a_{b+1}$. Then

$$\mathbb{E}R + \mathbb{E}(gR) = -\alpha d + V + \mathbb{E}S + \mathbb{E}(gS). \tag{Q67.6.2}$$

For $(b, c) = (2, 3)$, S starts at degree four. The positive $\mathbb{E}S$ terms begin at degree seven with δ_4, while the positive $\mathbb{E}(gS)$ terms begin at degree four with δ_3. Therefore

$$\mathbb{E}S + \mathbb{E}(gS) \le \frac{\delta_3 + \delta_4}{1 - \theta^2} \le \frac{\theta^2}{\kappa} d = 169d/608 < 7d/25. \tag{Q67.6.3}$$

For the other branches $c \ge 4$, the mask starts at least at degree five. The positive $\mathbb{E}S$ terms start at seven with δ_4, and the positive $\mathbb{E}(gS)$ terms at eight with δ_5. Thus

$$\mathbb{E}S + \mathbb{E}(gS) \le \frac{\delta_4 + \delta_5}{1 - \theta^2} \le \frac{\theta^3}{\kappa} d = 2197d/19456 < 23d/200. \tag{Q67.6.4}$$

The integer comparisons are $4225 < 4256$ and $439400 < 447488$. Discarding the negative terms of the finite mask gives these upper bounds for the common moments. Equations (Q67.3.6), (Q67.4.1) and (Q67.6.2) give

$$\begin{aligned} \log \mathbb{E}[(pTF)^8] \le & -8(\Delta - V) - 8d + 8Bd - 24(k-1)d \\ & + 8(\mathbb{E}S + \mathbb{E}(gS)) + (601/20)A_0 d. \end{aligned} \tag{Q67.6.5}$$

At zero expectation, take the logarithm to be $-\infty$; the tree messages on positive-mass atoms are positive. All following bounds hold on $0 < d \le 1/24$, using $k \ge 2$.

For $(b, c) = (2, 3)$, $\Delta = d$, $B = 0$, $V = \delta_2$. Use (Q67.6.3) and $A_0 = 1$. Dividing the upper exponent by d gives

$$-40 + 8\theta + 56/25 + 601/20 = -223/50 < -1/2. \tag{Q67.6.6}$$

For $b = 2, c \ge 4$, $\Delta \ge 2\tau$ and $V = a_{c+1} \le \delta_4$, since the first positive even moment for $c + 1 \ge 5$ is at $c + 1 = 8$. Thus $\Delta - V \ge 2\tau - \delta_4$. With (Q67.6.4), $8B \le 1/3$, and $A_0 = 1$, the coefficient is at most

$$-48 + 16\theta + 8\theta^3 + 1/3 + 23/25 + 601/20 < -9 < -1/2. \tag{Q67.6.7}$$

Use $16\theta = 13/2$ and $8\theta^3 < 4/7$ for the last bound. For $b = 3$, $\Delta \ge d + 2\tau$ and $V = a_{c+1} - \delta_2 \le \delta_4 - \delta_2$. Hence $\Delta - V \ge 3d - \delta_2 - \delta_4$, and $A_0 = 33/20$ gives

$$-56 + 8\theta + 8\theta^3 + 1/3 + 23/25 + 19833/400 < -11279/8400 < -1 < -1/2. \tag{Q67.6.8}$$

Substituting $8\theta = 13/4$, $8\theta^3 < 4/7$ gives the stated $-11279/8400$. For $b = 4$, $V = a_{c+1}$. At $c = 5$, 6, $V \le 0$ and $\Delta \ge 3d - 2\delta_2$. For $c \ge 7$, $\beta_c \ge \tau + \kappa\delta_3$, $\beta_{c-1} \ge \tau$, and $V \le \delta_4 \le \theta\delta_3$. Since $\kappa > \theta$, this gives throughout

$$\Delta - V \ge 3d - 2\delta_2. \tag{Q67.6.9}$$

The extra term in β_c is the alternating tail $\delta_3 - \delta_4 + \cdots \geq \delta_3 - \delta_4$, with precisely $+\delta_3$ at $c = 7$. This preserves the common-moment relation. With $A_0 = 19/12$, the coefficient is at most

$$-56 + 16\theta + 1/3 + 23/25 + 11419/240 = -267/400 < -1/2. \tag{Q67.6.10}$$

Its margin below $-1/2$ is $67/400$. For $b \geq 5$, $\Delta \geq 3\tau$, $V \leq \delta_3 + \delta_4$: the first positive contribution of $-a_{b+1}$ is δ_3, and that of a_{c+1} is δ_4. Using $A_0 = 5/4$, the coefficient is at most

$$-56 + 24\theta + 8\theta^2 + 8\theta^3 + 1/3 + 23/25 + 601/16 < -5 < -1/2. \tag{Q67.6.11}$$

For this bound use $\theta^2 < 1/6$, $\theta^3 < 1/14$. Equations (Q67.6.6)–(Q67.6.11) prove $\mathbb{E}[(pTF)^8] \leq e^{-d/2}$ on the low domain.

Q67.7 Completion and the full odd unit-arm list.

The high-domain bound (Q67.4.5) and the low-domain proof cover $d > 0$ with the same directions. At $d = 0$, $g = 0$ and K may be nonzero; leaf integration makes all path, spider and tree messages one. Since $0 \leq E_0, E_r \leq 1$ and T is positive,

$$0 \leq E_r pT \left(E_0 q f_b h_c \prod_{i=2}^{k} h_{b_i} f_{c_i} \right) \leq pT \left(q f_b h_c \prod_{i=2}^{k} h_{b_i} f_{c_i} \right).$$

Thus all genuine equal-pair decorations satisfy the bound. Moreover,

$$e^{-d/2} \leq (1 + d/2)^{-1} \leq 1 - 3d/7 \leq 1 - 2d/5 \qquad (0 \leq d \leq 1/3). \tag{Q67.7.1}$$

Apply the theorem to $-K$, exchanging A/T, p/q and every edge together, to obtain the reversed qAF^- ordinary eighth moment. The same multiple-pair direction is that of Q4-C138, whose all-positive-length fourth moment is proved in Section J.6:

$$\mathbb{E}[p(T^L F)^4] \leq 1 \qquad (L \geq 1).$$

The triple directions have the fourth moments of Q4-C125, Q4-C129 and Q4-C134. For their unit eighth moments, Q5-C062 gives $1 - d/601$, and Q5-C063 and Q5-C065 give $1 - d/500$. Together with $1 - 2d/5$ for every list of at least five arms, these yield (Q67.1.3).

O.16 The unit-connector eighth moment for every triple $(2, b, c)$, $2 < b < c$

Q70.0 Statement and directions.

For every finite probability space and real skew kernel $|K| \leq 1$, and all finite integers $2 < b < c$, consider

$$F_D = E_0 \bar{f}_2 h_b f_c, \qquad H_D = E_r pT(F_D), \tag{Q70.0.1}$$

where E_0, E_r are arbitrary finite genuine equal-pair products. Then the directions specified below satisfy

$$\boxed{\mathbb{E} H_D^8 \leq e^{-d/27} \leq 1 - d/28.} \tag{Q70.0.2}$$

The entire two-edge arm points toward the terminal, giving $\bar{f}_2$. The b-arm has its first edge inward and its remaining edges outward, giving h_b; the c-arm is outward, giving f_c. The unit connector points from terminal to root, whose unit arm points outward. Each equal pair uses an outward arm and an arm first inward and then outward. Any finite number of pairs and repeated lengths is allowed, with empty product one. For $b = 3, 4$, this direction differs from the corresponding positive-step weighted-fourth direction of Q4-C143.

Q70.1 Finite model and short paths.

Use the finite-model definitions in Appendix M.15:

$$\begin{gathered} Jv = (\mathbb{E}v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \\ f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad r = Kg, \quad s = Kr, \\ \delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2. \end{gathered} \tag{Q70.1.1}$$

After removing zero-mass atoms, all expectations, inner products, spectra and adjoints use the same measure. The kernel is real, skew, and bounded in absolute value by one. Its bounds are

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad |Kv(x)| \le \|v\|_2, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j. \tag{Q70.1.2}$$

For $-K^2$, the spectral measure ν of $\mathbf{1}$ is supported on $[0, \theta]$, with $\delta_j = \int x^j \, d\nu(x)$. Both A, T preserve order and $p(x), q(x) \ge \mu(x) > 0$. The exact short identities are

$$\mathbb{E}r = -d, \quad \mathbb{E}s = 0, \quad \mathbb{E}(gr) = 0, \quad \bar{f}_2 = q + r, \qquad \bar{f}_3 = q + r - s - d. \tag{Q70.1.3}$$

The arm and equal-pair bounds are

$$0 \le f_n, \bar{f}_n, h_n \le 2 \quad (n \ge 1), \qquad 0 \le f_\ell h_\ell \le 1. \tag{Q70.1.4}$$

Thus $0 \le E_0, E_r \le 1$.

Q70.2 A finite exponential bound for the long-arm pair.

Put

$$S_n = \sum_{j=0}^{n} K^j\mathbf{1}, \qquad G_m(z) = \sum_{j=0}^{m-1} z^j, \qquad G_0 = 0,$$

and $\beta_1 = \beta_2 = 0$, $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j$. The finite convolution and geometric bounds are

$$S_n = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad \tau \le \beta_n \le d \quad (n \ge 3), \tag{Q70.2.1}$$

$$\|G_m(K)\| \le 1 \quad (m = 0, 1 \text{ or } m \ge 3), \qquad \|G_2(K)\|^2 \le 1 + \theta. \tag{Q70.2.2}$$

For $3 \le b < c$, set

$$R = R_{bc} = \sum_{j=b}^{c-1} K^j g = K^b G_{c-b}(K) g, \qquad \Gamma = \begin{cases} 2d - \delta_2, & b = 3, \\ 3d, & b \ge 4. \end{cases} \tag{Q70.2.3}$$

The pair satisfies

$$0 \le h_b f_c \le e^{R - \Gamma}. \tag{Q70.2.4}$$

The case $b = 3$ is the finite short-pair inequality of Q3-C014, Lemma 6.1, also displayed as (Q78.1.7) in Appendix O.3. For $b \ge 4$, subtract the finite convolutions:

$$f_c - f_b = R + \sum_{i=0}^{b-3} (\beta_{b-i} - \beta_{c-i}) f_i - \sum_{i=b-2}^{c-3} \beta_{c-i} f_i. \tag{Q70.2.5}$$

The positive part of the first sum is at most $4\delta_2/(1-\theta^2) \le 2d$. For every $n \ge 4$, the finite path-mean bound is $\mathbb{E}f_n \le 1 - 83d/32 \le 1 - 5d/2$, proved in Appendix M.17. Hence for $b \ge 5$, $h_b + f_c = 2\mathbb{E}f_{b-1} + f_c - f_b \le 2 + R - 3d$. For $b = 4$, retain the $i = 2$ term in the last sum, including the shortest case $c = 5$:

$$f_c - f_4 \le R + \delta_2(1+p) - \tau f_2.$$

The row bound $|r| \le pq/2$ gives $f_2 = p + r \ge p^2/2$. Completing the square for $p \ge 0$, and using $\delta_2 \le d/2$, $\tau \ge d/2$, gives

$$\delta_2(1+p) - \tau f_2 \le \delta_2 + \delta_2 p - \tau p^2/2 \le \delta_2 + \frac{\delta_2^2}{2\tau} \le 3d/4.$$

Thus $h_4 + f_c \le 2 + R - 13d/4 \le 2 + R - 3d$. For $d > 0$, $\tau > 0$; the zero boundary is treated below. The arm sum is nonnegative, so AM–GM gives $h_b f_c \le [(h_b + f_c)/2]^2 \le e^{R-\Gamma}$. Let $J_R = \mathbb{E}R^2$. Equation (Q70.2.2) gives

$$|R|^2 \le (1+\theta)\theta^{b-1}d, \qquad J_R \le (1+\theta)\theta^b d. \tag{Q70.2.6}$$

In particular,

$$|R|^2 \le 7605d/32768 \le 7605/98304 < 9/100, \qquad |R| < 3/10, \tag{Q70.2.7}$$

where $760500 < 884736$.

Q70.3 Propagation retaining the two-edge arm.

The negative-mean lemma (Q65.2.4) applies whenever $\mathbb{E}N \le 0$, $|N| \le \sqrt{A_0 d}$, $\mathbb{E}N^2 \le \theta A_0 d$, and $|\mathbb{E}N| \le \sqrt{A_0}d$:

$$\mathbb{E}e^{8N} \le \exp\{8\mathbb{E}N + \lambda(\sqrt{d})A_0 d\}, \qquad \lambda(t) = \frac{188}{25}\left(2 - \frac{\kappa}{1+t}\right)^2. \tag{Q70.3.1}$$

Since λ increases and $\sqrt{d} \le 1/\sqrt{3} < 3/5$,

$$\lambda(\sqrt{d}) \le \lambda(3/5) = 32691132/1638400 < 20. \tag{Q70.3.2}$$

Here $2 - \kappa/(8/5) = 417/256$, and $188 \cdot 417^2 = 32691132 < 32768000 = 20 \cdot 1638400$. For every real $u \le 3/10$,

$$e^u \le 1 + u + (3/5)u^2. \tag{Q70.3.3}$$

Indeed, $(e^u - 1 - u)/u^2 = \int_0^1 (1-t)e^{tu}\,dt$ increases with u. At $u = 3/10$, the exponential series and $j! \ge 2 \cdot 3^{j-2}$ give

$$e^u \le 1 + u + \frac{u^2}{2(1-u/3)} = 27/20 < 677/500 = 1 + 3/10 + (3/5)(3/10)^2.$$

Use the continuous extension at zero. Set

$$\begin{aligned} G &= g - r, \quad \bar{f}_2 = 1 - G, \quad \mathbb{E}G^2 = d + \delta_2, \\ V &= g - r - R, \quad N = KV = r - s - KR, \\ C &= -d + \mathbb{E}R - \mathbb{E}(GR) + \sqrt{(d+\delta_2)J_R} + (12/5)J_R. \end{aligned} \tag{Q70.3.4}$$

Positivity of T, $\bar{f}_2 \geq 0$, and (Q70.3.3) give

$$T(\bar{f}_2 e^R) \leq \bar{f}_3 + T(\bar{f}_2 R) + (3/5)T(\bar{f}_2 R^2).$$

The exact linear term is $T(\bar{f}_2 R) = \mathbb{E}R - \mathbb{E}(GR) - KR + K(GR)$, and

$$K(GR) \leq \mathbb{E}|GR| \leq \sqrt{(d+\delta_2)J_R}, \qquad T(\bar{f}_2 R^2) \leq 2\mathbb{E}(\bar{f}_2 R^2) \leq 4J_R.$$

Therefore

$$T(\bar{f}_2 e^R) \leq q + C + N. \tag{Q70.3.5}$$

The right side is nonnegative. By $p + q = 2$ and AM–GM, $p(q + C + N) \leq [1 + (C + N)/2]^2 \leq e^{C+N}$, with nonnegative base. For the undecorated $H = pT(\bar{f}_2 h_b f_c)$,

$$0 \leq H \leq e^{-\Gamma+C+N}. \tag{Q70.3.6}$$

The coupled mean identities are

$$\begin{aligned} \mathbb{E}N &= -d + \mathbb{E}(gR), \\ C + \mathbb{E}N &= -2d + \mathbb{E}R + \mathbb{E}(rR) + \sqrt{(d+\delta_2)J_R} + (12/5)J_R. \end{aligned} \tag{Q70.3.7}$$

Thus $\mathbb{E}(gR)$ cancels between the terms. If $\|V\|^2 \leq A_0 d$ and $\mathbb{E}N \leq 0$, the row and spectral bounds and Cauchy give all hypotheses of (Q70.3.1). Using (Q70.3.2) then gives

$$\boxed{\begin{aligned} \log \mathbb{E}H^8 \leq\ & -8\Gamma - 16d + 8\{\mathbb{E}R + \mathbb{E}(rR)\} \\ & + 8\sqrt{(d+\delta_2)J_R} + (96/5)J_R + 20A_0 d. \end{aligned}} \tag{Q70.3.8}$$

Q70.4 Mixed moments and the negative mean.

Put $a_j = \mathbb{E}K^j\mathbf{1}$, so $a_{2j} = (-1)^j\delta_j$ and odd a_j vanish. Skewness gives

$$\mathbb{E}(rK^j g) = a_{j+3}, \qquad \mathbb{E}(K^j g) = a_{j+1}. \tag{Q70.4.1}$$

Write $e_n = \delta_n - \delta_{n+1}$. The spectral representation gives

$$e_n = \int x^n(1-x)\,d\nu(x) \geq 0, \qquad e_{n+1} \leq \theta e_n. \tag{Q70.4.2}$$

For each odd $j = 2n-1$, the sum in (Q70.4.1) is $(-1)^n e_n$; for even j, it is zero. The contiguous finite interval in R therefore gives a finite alternating sum of decreasing nonnegative terms, and

$$\mathbb{E}R + \mathbb{E}(rR) \leq \begin{cases} \delta_2 - \delta_3, & b = 3, \\ 0, & b = 4, \\ \delta_4, & b \geq 5. \end{cases} \tag{Q70.4.3}$$

The first class begins with e_2, and equals $\delta_2 - \delta_3$ for $(b,c) = (3,4), (3,5)$. The second, when nonempty, begins with $-e_3$. In the third, a positive first term has index at least four and is at most $e_4 \leq \delta_4$; a negative first term gives upper bound zero. Similarly, only even j contribute to $\mathbb{E}(gK^j g)$. For $j \geq 3$, the first possible positive term is $j = 4$, with value δ_3, followed by decreasing alternating terms. Every finite truncation gives

$$\mathbb{E}(gR) \leq \delta_3 \leq \theta^2 d < d \qquad (d > 0). \tag{Q70.4.4}$$

Hence $\mathbb{E}N < 0$ in every class.

Q70.5 Norms for every finite b, c.

On a rotation block write $z = i\sqrt{x}$, $0 \le x \le \theta$. Then $V = P(K)g$, with

$$P(z) = 1 - z - \sum_{j=b}^{c-1} z^j. \tag{Q70.5.1}$$

For $(b, c) = (3, 4)$, $P = 1 - z - z^3$, so

$$|P(i\sqrt{x})|^2 = 1 + x(1-x)^2 \le 1 + 4/27 = 31/27. \tag{Q70.5.2}$$

On $[0, 1]$, the derivative of $x(1-x)^2$ is $(1-x)(1-3x)$, giving maximum $4/27$. Here $J_R = \delta_4$. For $(b, c) = (3, 5)$, $P = 1 - z - z^3 - z^4$, and

$$|P|^2 = (1-x^2)^2 + x(1-x)^2 = 1 + x - 4x^2 + x^3 + x^4.$$

Since $x + x^2 \le \theta + \theta^2 < 1$,

$$|P|^2 \le 1 + x - 3x^2 \le 13/12. \tag{Q70.5.3}$$

This case has the exact second moment $J_R = \delta_4 + \delta_5$, retaining the exceptional G_2 contribution. For $b = 3, c \ge 6$, the finite sums satisfy

$$x^2 - x^3 \le \mathrm{Re} \sum_{j=3}^{c-1} z^j \le x^2, \qquad -x^{3/2} \le \mathrm{Im} \sum_{j=3}^{c-1} z^j \le -x^{3/2} + x^{5/2}.$$

Thus $\mathrm{Re}\, P > 0$, $\mathrm{Im}\, P \le 0$, and

$$\begin{aligned} |P|^2 &\le (1 - x^2 + x^3)^2 + x(1 - x + x^2)^2 \\ &= 1 + x - 4x^2 + 5x^3 - x^4 - x^5 + x^6 \\ &\le 1 + x - (63/32)x^2 \le 1 + 8/63 = 71/63. \end{aligned} \tag{Q70.5.4}$$

Use $-x^4 - x^5 + x^6 \le 0$, $4 - 5\theta = 63/32$. Here $c - 3 \ge 3$, so (Q70.2.2) gives $J_R \le \delta_4$. For $b = 4$, $c > 4$, the contiguous sum has real part in $[x^2 - x^3, x^2]$ and imaginary part in $[0, x^{5/2}]$. Therefore

$$|P|^2 \le (1 - x^2 + x^3)^2 + x(1 + x^2)^2 = 1 + x - 2x^2 + 4x^3 + x^4 - x^5 + x^6.$$

As $x \le 1/2$, both $-2x^2 + 4x^3$ and $-x^5 + x^6$ are nonpositive, so

$$|P|^2 \le 1 + x + x^4 < 1 + 13/32 + 1/32 = 23/16 < 3/2. \tag{Q70.5.5}$$

Here $\theta^4 = 28561/1048576 < 1/32$, and (Q70.2.6) also gives $J_R < (3/64)d$. For $b \ge 5, c > b$, the finite alternating sums give

$$-x^3 \le \mathrm{Re} \sum_{j=b}^{c-1} z^j \le x^4, \qquad \left| \mathrm{Im} \sum_{j=b}^{c-1} z^j \right| \le x^{5/2}.$$

Since $\mathrm{Re}\, P > 0$,

$$|P|^2 \le (1 + x^3)^2 + x(1 + x^2)^2 = 1 + x + 4x^3 + x^5 + x^6 < 7/4. \tag{Q70.5.6}$$

For the rational comparison use $x \le \theta < 5/12$, $4\theta^3 < 3/10$, and $\theta^5 + \theta^6 < (3/2)/90 < 1/30$; these sum to less than $3/4$. The cross-products for $\theta^5 < 1/90$ and $4\theta^3 < 3/10$ are $33416370 < 33554432$ and $87880 < 98304$. Also

$$J_R \le (1 + \theta)\theta^5 d < d/60. \tag{Q70.5.7}$$

Integrating the block bounds gives $\mathbb{E}V^2 \le A_0 d$ in each class.

Q70.6 The eighth-moment bounds on $0 < d \le 1/3$.

Apply (Q70.3.8), retaining the common contribution $\mathbb{E}R + \mathbb{E}(rR) + (12/5)J_R$. For $(b,c) = (3,4)$, $J_R = \delta_4 \le \theta\delta_3$, and

$$\mathbb{E}R + \mathbb{E}(rR) + (12/5)J_R \le \delta_2 - \delta_3 + (12/5)\delta_4 \le \delta_2 - \delta_3/40 \le \delta_2. \tag{Q70.6.1}$$

Moreover,

$$\sqrt{(d+\delta_2)J_R} \le \sqrt{(1+\theta)\theta^3}\, d < 5d/16. \tag{Q70.6.2}$$

The squared comparison is $98865 < 102400$, with denominator 1048576. Use $\Gamma = 2d - \delta_2$, $A_0 = 31/27$:

$$\log \mathbb{E}H^8 \le (-32 + 16\theta + 5/2 + 20 \cdot 31/27)d = -d/27. \tag{Q70.6.3}$$

The final equality is $-23 + 620/27 = -1/27$. For $(b,c) = (3,5)$, use the exact $J_R = \delta_4 + \delta_5$:

$$\begin{aligned} \mathbb{E}R + \mathbb{E}(rR) + (12/5)J_R &\le \delta_2 + [-1 + (12/5)\theta(1+\theta)]\delta_3 \\ &= \delta_2 + (95/256)\delta_3 < \delta_2 + d/16. \end{aligned} \tag{Q70.6.4}$$

The last comparison is $16055 < 16384$, with denominator 262144. Also

$$\sqrt{(d+\delta_2)J_R} \le (1+\theta)\theta^{3/2} d < 3d/8, \tag{Q70.6.5}$$

where the squared comparison is $4448925 < 4718592$, with denominator 33554432. Using $A_0 = 13/12$,

$$\log \mathbb{E}H^8 \le (-32 + 16\theta + 1/2 + 3 + 20 \cdot 13/12)d = -d/3. \tag{Q70.6.6}$$

For $b = 3, c \ge 6$, $J_R \le \delta_4$, so (Q70.6.1)–(Q70.6.2) apply. With $A_0 = 71/63$,

$$\log \mathbb{E}H^8 \le (-23 + 20 \cdot 71/63)d = -(29/63)d. \tag{Q70.6.7}$$

For $b = 4, c > 4$, $\Gamma = 3d$, $\mathbb{E}R + \mathbb{E}(rR) \le 0$, $A_0 = 3/2$. From $J_R < 3d/64$, $d + \delta_2 < (3/2)d$,

$$\sqrt{(d+\delta_2)J_R} < d/3, \qquad (12/5)J_R < d/8.$$

The comparisons are $9/128 < 1/9$ and $9/80 < 1/8$. Hence

$$\log \mathbb{E}H^8 \le (-40 + 30 + 8/3 + 1)d = -(19/3)d. \tag{Q70.6.8}$$

For $b \ge 5, c > b$, $\Gamma = 3d$, $A_0 = 7/4$, and $\mathbb{E}R + \mathbb{E}(rR) \le \delta_4 < d/14$, using $\theta^3 < 1/14$ and $30758 < 32768$. Since $J_R < d/60$,

$$\sqrt{(d+\delta_2)J_R} < d/6, \qquad (12/5)J_R < d/25.$$

Now $1/14 + 1/6 + 1/25 = 146/525 < 3/10$, giving

$$\log \mathbb{E}H^8 \le (-40 + 35 + 12/5)d = -13d/5. \tag{Q70.6.9}$$

The five classes cover every $3 \le b < c$. Their common bound is $\mathbb{E}H^8 \le e^{-d/27}$.

Q70.7 Boundary, decorations and full reversal.

At $d = 0$, $g = 0$, $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, and leaf integration makes every rooted tree message one, although K may be nonzero. For the decorations, positivity gives

$$0 \le E_r pT(E_0 \bar{f}_2 h_b f_c) \le pT(\bar{f}_2 h_b f_c).$$

Full reversal replaces K by $-K$, exchanges A/T, p/q, and changes $\bar{f}_2$ to f_2, h_b to $W_b = A\bar{f}_{b-1}$, and f_c to $\bar{f}_c$, with every decoration reversed as well. The value of d is unchanged. Finally,

$$e^{-d/27} \le \frac{1}{1+d/27} = 1 - \frac{d}{27+d} \le 1 - d/28 \qquad (d \le 1/3).$$

This proves (Q70.0.2). For the direction comparison, Q4-C143, §5.3, uses $\bar{f}_2 f_3 h_c$ at $b = 3$ and $\bar{f}_2 f_4 h_c$ at $b = 4$, whereas the present input is $\bar{f}_2 h_b f_c$. For $b \ge 5$, the directions agree, so the existing all-positive-length p-weighted fourth and the present unit eighth can be used for the same input.

O.17 Short connectors for every terminal pair $(2, b)$

On the original nonempty finite probability space, let K be real and antisymmetric with $|K| \le 1$, and put

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n 1, \quad \bar{f}_n = T^n 1, \quad h_n = T f_{n-1}, \qquad \delta_j = \|K^j 1\|_2^2, \quad d = \delta_1,$$

$$\theta = 13/32, \quad \kappa = 19/32, \quad r = Kg, \quad s = Kr, \quad t = Ks.$$

All adjoints, norms, spectral measures and expectations use the same original μ. Choose the terminal directions by the original integer lengths:

terminal and connector	$F_{b,L}$
$b \ge 3,\ L = 1$	$h_2 f_b$
$b = 3,\ L = 2$	$h_2 f_3$
$b \ge 4,\ L = 2$	$f_2 h_b$
$b \ge 3,\ L \ge 3$	$h_2 f_b$

Each f_n arm points entirely away from the terminal. Each h_n arm has its first edge toward the terminal and its remaining edges away from its first neighbour toward the leaf. All L connector edges point from the terminal toward the receiving root, whose original unit arm points outward. The actual message is $H = pT^L F_{b,L}$; these directions are fixed before the host is given.

The short-connector bounds are

$$E[pT^2(h_2 f_3)]^8 \le e^{-d/25}, \tag{Q85.0.1}$$

$$E[pT^2(f_2 h_4)]^8 \le e^{-d/3}, \qquad E[pT^3(h_2 f_4)]^8 \le e^{-d/3}, \tag{Q85.0.2}$$

$$E[pT^2(f_2 h_b)]^8 \le e^{-d/5}, \qquad E[pT^3(h_2 f_b)]^8 \le e^{-d/5} \quad (b \ge 5). \tag{Q85.0.3}$$

The high-domain part of the second bound in (Q85.0.3) uses the terminal-mean theorem in Section O.18; its use is stated below. Together with the unit and long-connector rows, these estimates give

$$EH^8 \le Q(d) := 1 - d/20000. \tag{Q85.0.4}$$

The ordinary row $F_{b,2} = f_2 h_b$ for $b \ge 4$ has its stated direction. The f_2-reference fourth bound for $h_2 f_b$ is a separate input on that actual branch.

Q85.1 Finite framework and local inputs

The finite calculus in Section M.1 gives

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta \delta_j,$$

$$|Kv(x)| \le \|v\|_2, \quad EK^{2j}1 = (-1)^j \delta_j, \quad EK^{2j+1}1 = 0. \tag{Q85.1.1}$$

After deleting zero-mass atoms, $p(x), q(x) \ge \mu(x) > 0$. Actual tree messages are positive: conditional on the root label, assigning every other label that same value has positive mass and makes every edge factor one. At $d = 0$, $g = 0$ and successive leaf integration gives every actual message one, so all moment bounds hold directly. Subsequent divisions by d use $d > 0$.

The finite positive convolution is

$$S_n := \sum_{j=0}^{n} K^j 1 = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \quad \beta_1 = \beta_2 = 0,$$

$$\beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1} \delta_i, \qquad \tau := d - \delta_2 \le \beta_j \le d \quad (j \ge 3). \tag{Q85.1.2}$$

Its complete reversal uses $K' = -K, g' = -g$. All finite path messages in discarded convolution terms are nonnegative.

We use the signed coefficient-17 exponential estimate in Section M.1, the retained-envelope formulas and full-frequency bounds of Section N.10, and the corresponding $b = 4$ parameter bounds of Section N.9. The arithmetic-mean input for $h_4 + f_v$, for every finite $v \ge 5$, is the one in Section O.15; its needed formula is restated in (Q85.6.4)'s derivation. The ordinary unit row is $\mathbb{E}[pT(h_2 f_b)]^8 \le 1 - d/500$ for every $b \ge 3$. The remaining long rows and the high-domain actual terminal mean are supplied by Section O.18.

Q85.2 A variance-sensitive exponential comparison

Let W be finite and real with $|W| \le U$ and $\mathbb{E}W^2 \le \theta U^2$. For every $s \ge 0$, retaining the actual mean of either sign,

$$Ee^{sW} \le \exp\left\{ sEW + \frac{s^2}{8}(U^2 + 3EW^2) \right\}. \tag{Q85.2.1}$$

In particular, for every real original vector V,

$$Ee^{8KV} \le \exp\{8E(KV) + 8\|V\|^2 + 24\|KV\|^2\}. \tag{Q85.2.2}$$

The cases $U = 0$ and constant-variable endpoints are immediate. Otherwise set $h = \mathbb{E}W^2/U^2$, $m = \mathbb{E}W$, $v = \mathbb{E}(W - m)^2$, and $M = U - m$. Apply (M.48) to $f(w) = e^{sw}$:

$$Ee^{sW} \le e^{sm} Ee^{sL(B-\pi)}, \qquad L = M + v/M, \quad \pi = \frac{v}{M^2 + v}, \tag{Q85.2.3}$$

where B is Bernoulli with mean π.

When $m \ge 0$, write $w = m/U$. Then

$$L/U = \frac{1 + h - 2w}{1 - w} \le 1 + h.$$

The centered Bernoulli logarithmic moment has tilted second derivative at most $L^2/4$. Integrating twice gives coefficient $L^2/8$, at most $(1 + 3h)U^2/8$, since $(1 + h)^2 \le 1 + 3h$ for $0 \le h \le 1$.

When $m \le 0$, put $u = -m/U, t = 1 + u$ and

$$a = \frac{h - u^2}{(1+u)^2} \in [0, h], \qquad L/U = t(1+a), \quad y = 1 - 2\pi = \frac{1-a}{1+a}. \tag{Q85.2.4}$$

The centered Bernoulli estimate is

$$Ee^{z(B-\pi)} \le \exp\left\{ \frac{1-2\pi}{4\log((1-\pi)/\pi)} z^2 \right\}. \tag{Q85.2.5}$$

This is (Q81.2.5), including its continuous endpoints.

Use $\log((1+y)/(1-y)) \ge 2(y + y^3/3)$. It suffices to bound $L^2/[8(1+y^2/3)]$. After division by t^2, the desired inequality becomes

$$\frac{3(1+a)^4}{4(1+a+a^2)} \le \frac{1+3u^2}{(1+u)^2} + 3a. \tag{Q85.2.6}$$

Subtracting $3a$ on the left yields

$$\frac{3(1+a^2)^2}{4(1+a+a^2)} \le \frac{3}{4},$$

because $a^3 + a \le \theta^3 + \theta < 1$. On the right,

$$\frac{1+3u^2}{(1+u)^2} - \frac{3}{4} = \frac{(3u-1)^2}{4(1+u)^2} \ge 0.$$

This proves (Q85.2.1). The original row and spectral bounds, with $U = \|V\|$ and $W = KV$, prove (Q85.2.2). Degenerate Bernoulli endpoints follow directly or by continuity. The coefficient-17 estimate and (Q85.2.2) can both be applied to the same primitive.

Q85.3 The exceptional row $b = 3, L = 2$

Use $F = h_2 f_3$. Its arithmetic mean is

$$h_2 + f_3 = 2 - d + s = 2m + s, \quad m = 1 - d/2, \quad k = 1/m \in [1, 6/5].$$

Put $K' = -K, g' = K'\mathbf{1}, a = k\delta_2, \delta = k^2\delta_3/2$ and $\eta = a - \delta$. Positive two-step propagation of $F \le (m + s/2)^2$ gives

$$0 < X = T^2 F \le m^2 Z, \quad Z = 1 - \eta + (1 + \delta + K' - kK'^4)g'. \tag{Q85.3.1}$$

This is the two-step envelope obtained by $T(s^2) \le 2\delta_3$. Set

$$V = \delta 1 + g' - kK'^3 g', \qquad N = K'V.$$

Then $p + Z = 2 - \eta + N$, with $p, Z \ge 0$, so

$$(pX)^8 \le m^{16}[1 + (N - \eta)/2]^{16} \le m^{16} \exp(8N - 8\eta). \tag{Q85.3.2}$$

The base is the actual arithmetic mean. Its zero endpoint is included by $(1+x)^{16} \le e^{16x}$ for $x \ge -1$. Skew orthogonality gives

$$EN = -d,$$

$$\|V\|^2 = d + k^2\delta_4 - 2\delta k\delta_2 + \delta^2 \le d + k^2\delta_4,$$
$$\|K'V\|^2 = \delta_2 + k^2\delta_5 - 2\delta k\delta_3 + \delta^2 d \le \delta_2 + k^2\delta_5. \tag{Q85.3.3}$$

The discarded terms are nonpositive because $\delta \le 2k\delta_2$ and $\delta d \le 2k\delta_3$. Both follow from $\delta = k^2\delta_3/2$, $k \le 6/5, d \le 1/3$ and $\delta_3 \le \theta\delta_2$.

Using the coefficient-17 estimate in (Q85.3.2), and $16\log m \le -8d$, yields

$$\log E(pX)^8 \le d - 8k\delta_2 + 4k^2\delta_3 + 17k^2\delta_4 \le d - (78/25)\delta_2. \tag{Q85.3.4}$$

Indeed $4\theta + 17\theta^2 = 4537/1024 < 9/2$, and $-8k + (9/2)k^2$ increases on $[1, 6/5]$, with maximum $-78/25$. Using (Q85.2.2) instead gives

$$\log E(pX)^8 \le -8d + (24 - 8k)\delta_2 + 4k^2\delta_3 + 8k^2\delta_4 + 24k^2\delta_5 \le -8d + (108/5)\delta_2. \tag{Q85.3.5}$$

Here $4\theta + 8\theta^2 + 24\theta^3 < 5$: replacing θ by $5/12$ already gives $115/24 < 5$. The quadratic $24 - 8k + 5k^2$ increases on $[1, 6/5]$ and has maximum $108/5$. If $\delta_2/d \ge 1/3$, (Q85.3.4) is at most $-d/25$. If $\delta_2/d \le 1/3$, (Q85.3.5) is at most $-4d/5$. Both include equality, proving (Q85.0.1) on the full domain.

Q85.4 Two actual signs of the retained square envelope

For $b = 4$, let $m = 1 - d, R = s + t - dg$. The finite formulas give

$$h_2 + f_4 = 2m + R, \qquad f_2 + h_4 = 2m - R. \tag{Q85.4.1}$$

For every $b \ge 5$ on $0 < d \le 1/5$, use the retained coefficients with $m = 1 - 9d/8$. For $b = 5$, take

$$c_1 = d, \quad c_2 = d/4 + \delta_2, \quad c_3 = 0.$$

For $b \ge 6$, set

$$\lambda = \frac{9d/4 - \beta_b - \beta_{b-1}}{\beta_{b-2} + (1-d)\beta_{b-3}}, \quad c_1 = \beta_{b-1},\ c_2 = \lambda\beta_{b-2},\ c_3 = \lambda\beta_{b-3}. \tag{Q85.4.2}$$

The denominator is positive and $0 \le \lambda \le 1$: the full retained constant is at least $(4 - d)\tau \ge 361d/160 > 9d/4$. Thus every c_i is bounded by its actual positive-convolution coefficient. Put

$$C = c_1 + c_2 + c_3, \quad D = c_2 + c_3, \quad E_* = c_3,$$

$$R = \sum_{j=3}^{b} K^j 1 - Cg - Dr - E_* s.$$

In the positive convolution for f_b, retain $\beta_b, c_1 f_1, c_2 f_2, c_3 f_3$. This gives $h_2 + f_b \le 2m + R$. For the opposite pair, apply T to the positive convolution for f_{b-1}, retaining $c_1 q, c_2 h_2, c_3 h_3$. Since

$$h_3 = (1 - d) - g - r - s,$$

the constant loss is again $\beta_b + c_1 + c_2 + (1-d)c_3 = 9d/4$, and the remaining polynomial is $-R$. Thus

$$0 < h_2 + f_b \le 2m + R, \qquad 0 < f_2 + h_b \le 2m - R. \tag{Q85.4.3}$$

The actual products are bounded by $(m + R/2)^2$ and $(m - R/2)^2$, respectively.

Write $K' = -K, g' = -g$. On its rotation blocks $z = i\sqrt{x}$, $R = (\mathsf{A}(x) - \mathsf{B}(x)z)g'$. All unlabelled integrals below use the finite positive spectral measure of $-K'^2$ at g', with total mass d. For $b = 5$,

$$\mathsf{A} = d + D + x - x^2, \qquad \mathsf{B} = D + x, \quad D = d/4 + \delta_2. \tag{Q85.4.4}$$

For $b \geq 6$, let $N = b - 2 \geq 4$ and $G_N(z) = a_N + zb_N$. Then

$$\mathsf{A} = C + x(a_N - E_*), \quad \mathsf{B} = D + xb_N, \tag{Q85.4.5}$$

$$1 - x \leq a_N, b_N \leq 1 - x + x^2, \quad a_N \geq (1 - x)b_N.$$

These are finite alternating geometric sums. The coefficients and the global δ_2 in c_i stay fixed while x varies. Put $k = 1/m$ and

$$V_R = ER^2 = \int (\mathsf{A}^2 + x\mathsf{B}^2), \quad D_0 = ER = \int \mathsf{B}, \quad a_1 = EKR = \int \mathsf{A},$$

$$\rho = -E(rR) = \int x\mathsf{B}, \quad U_R = \|KR\|^2 = \int x(\mathsf{A}^2 + x\mathsf{B}^2),$$

$$\delta = kD_0 + k^2 V_R/2, \quad a = ka_1, \quad \varepsilon = \delta - a. \tag{Q85.4.6}$$

The retained-envelope estimates give

$$\mathsf{A}, \mathsf{B} \geq 0, \quad \mathsf{A} - (1 - x)\mathsf{B} \geq c_1 + c_2 x, \quad c_1 \geq \tau,$$

$$\sqrt{\mathsf{A}^2 + x\mathsf{B}^2} \leq u_0 = 3/8 + 9d/5, \quad \mathsf{B} \leq z_0 = \theta + 13d/16, \tag{Q85.4.7}$$

$$|\varepsilon| \leq 3d/5, \quad \delta \leq 6d/5, \quad ku_0 \leq 1, \quad a_1 - D_0 \leq d^2 + \theta^3 d. \tag{Q85.4.8}$$

For the last inequality, (Q85.4.5) gives $\mathsf{A} - \mathsf{B} = c_1 - E_* x + x(a_N - b_N)$; the first possible positive finite end term is x^3. For $b = 5$, the difference integrates to $d^2 - \delta_3$. Multiplying the same finite observation by x also gives

$$D_3 - \rho := \int x(\mathsf{A} - \mathsf{B}) \leq d\delta_2 + \delta_5. \tag{Q85.4.9}$$

For $b = 4$ on $0 < d \leq 1/4$, the corresponding exact quantities are

$$D_0 = \delta_2, \quad a_1 = d^2 + \delta_2, \quad \rho = \delta_3,$$

$$V_R = d^3 + 2d\delta_2 + \delta_3 + \delta_4, \quad U_R = d^2\delta_2 + 2d\delta_3 + \delta_4 + \delta_5,$$

$$\delta = k\delta_2 + k^2 V_R/2, \quad \varepsilon = -kd^2 + k^2 V_R/2, \quad |\varepsilon| \leq d/6, \quad \delta \leq 25d/24. \tag{Q85.4.10}$$

The last two estimates apply to this same R, m.

Q85.5 Full-frequency contraction of the retained envelope

For every $b \ge 5, 0 < d \le 1/5$ and every original frequency $0 \le x \le \theta$,

$$\boxed{\mathsf{A}(x)^2 + x\mathsf{B}(x)^2 \le 2m\mathsf{B}(x).} \tag{Q85.5.1}$$

Consequently,

$$V_R \le 2mD_0, \qquad kU_R \le 2\rho. \tag{Q85.5.2}$$

First suppose $b \ge 6$, and let $f = x - x^2 + x^3, \ell = x - x^2$. Equations (Q85.4.2)–(Q85.4.5) give

$$\mathsf{A} \le D + d + f, \quad D + \ell \le \mathsf{B} \le D + f,$$

$$d/4 \le D \le (17/16)d + d^2. \tag{Q85.5.3}$$

Indeed $D = 9d/4 - \beta_b - \beta_{b-1} + dE_*$, and these two global bounds follow from $\tau \le \beta_j \le d, E_* \le d$. It is enough to prove

$$Q = (1+x)(D+f)^2 + 2d(D+f) + d^2 - 2m(D+\ell) \le 0. \tag{Q85.5.4}$$

For fixed d, Q is convex in D. It is also convex in x. To see the latter, use

$$(1+x)f = x + x^4,$$

$$(1+x)f^2 = x^2 - x^3 + x^4 + x^5 - x^6 + x^7 =: P(x).$$

Here $P'' \ge 5/4$, since $2 - 6x + 12x^2 \ge 5/4$, $20x^3 - 30x^4 \ge 0$, and the remaining terms are nonnegative. The potentially negative term $2df'' \ge -4d$ is paid by the contribution $4m$. Thus $Q_{xx} > 0$ on the whole rectangle. It suffices to use $x = 0, \theta$ and the two D endpoints while retaining the full interval of d.

At $x = 0, D = d/4$,

$$Q = d[(17/8)d - 1/2] \le -3d/40.$$

At $x = 0, D = (17/16)d + d^2$,

$$Q/d = -17/8 + (1189/256)d + (51/8)d^2 + d^3 < 0.$$

For the last inequality, increase the three positive coefficients to $5, 7, 1$ and use $d \le 1/5$. At $x = \theta$, use $f(\theta) < 5/16$, $6/25 \le \ell(\theta) \le 1/4$ and $1 + \theta < 3/2$. For $D = d/4$,

$$Q \le 75/512 - 12/25 + (59/64)d + (69/32)d^2 \le -161/2560. \tag{Q85.5.5}$$

For $D = (17/16)d + d^2$, the linear coefficient is at most $-15/4096$. The quadratic, cubic and quartic coefficients are less than $6, 15/2, 3/2$, respectively, so

$$Q \le 75/512 - 12/25 + 6d^2 + (15/2)d^3 + (3/2)d^4 \le -9957/320000. \tag{Q85.5.6}$$

Before enlargement the quadratic coefficient is $49005/8192 < 6$, and the cubic coefficient is $1853/256 < 15/2$. The linear comparison is $(1277/256)(5/16) - 17/8 + 9/16 = -15/4096$. Every remaining positive polynomial was bounded at $d = 1/5$. Convexity proves the intermediate frequencies and D values.

For $b = 5$, $\mathsf{A} = D + d + x - x^2$, $\mathsf{B} = D + x$, with $d/4 \le D \le (21/32)d$. Because $\mathsf{A} \ge 0$, discard its $-x^2$ inside the positive square. It suffices to prove

$$Q = (1 + x)(D + x)^2 + 2d(D + x) + d^2 - 2m(D + x) \le 0. \tag{Q85.5.7}$$

This Q is convex in D and x; its second x-derivative is $2(1 + 2D) + 6x > 0$. At $x = 0$, the lower D endpoint was proved above, and the upper gives $Q/d = -21/16 + (4321/1024)d < 0$. At $x = \theta$, write $D = t_0 d$, $t_0 = 1/4$ or $21/32$. The constant is $-19019/32768 \le -9/16$, the linear coefficient is at most $13/8$, and the quadratic coefficient is at most $9/2$. For the quadratic bound, use $1 + \theta < 3/2$, $t_0 < 2/3$; for the linear bound, use $2\theta + 2\theta^2 - 2 < -3/4$, $t_0 \ge 1/4$. Therefore

$$Q \le -9/16 + (13/8)d + (9/2)d^2 \le -23/400.$$

This proves (Q85.5.1) for every finite $b \ge 5$. Integrate against the original spectral measure, first directly and then after multiplication by x, to obtain (Q85.5.2). The global δ_2 is constant in this frequency calculation.

Q85.6 Every $b \ge 5$, $L = 2$, in the reversed-pair direction

For $0 < d \le 1/5$, take $F = f_2 h_b$ and use its actual square $(m - R/2)^2$ from (Q85.4.3). In the two-step Duhamel calculation write

$$\delta_- = -kD_0 + k^2 V_R/2, \quad a_- = -ka_1, \quad \varepsilon_- = \delta_- - a_-.$$

Positive propagation gives

$$0 < T^2 F/m^2 \le Z = \bar{f}_2 - kK'^2 R + ka_1 + \delta_- q.$$

This follows by expanding that square and using $T(R^2) \le 2V_R$. In particular its right side is nonnegative. It satisfies

$$p + Z = 2 + \varepsilon_- + N, \quad N = K' V_-, \quad V_- = \delta_- 1 + g' - kK' R.$$

The exact moments are

$$EN = -d + k\rho,$$

$$\|V_-\|^2 = d - 2k\rho + k^2 U_R + \delta_-^2 - 2\delta_- a_-. \tag{Q85.6.1}$$

Equation (Q85.5.2) gives $\delta_- \le 0$ and $kU_R \le 2\rho$. Also $a_1 \ge \kappa D_0 + \tau d$ implies $D_0 \le 2a_1$, so $\delta_- \ge 2a_-$. Thus the entire constant-coordinate term in (Q85.6.1) is nonpositive and

$$\|V_-\|^2 \le d.$$

AM–GM and the coefficient-17 estimate yield

$$\log E[pT^2 F]^8 \le 16 \log m + 9d + 8(\varepsilon_- + k\rho). \tag{Q85.6.2}$$

For the same R,

$$\varepsilon_- + k\rho \le \{k(d + \theta^3) + (k^2/2)u_0^2 + k\theta z_0\}d.$$

All three terms increase with d on $[0, 1/5]$, and their endpoint comparisons are

$$43753/126976 < 7/20, \quad 21609/48050 < 9/20, \quad 1183/3968 < 3/10.$$

Their sum is therefore less than $11/10$. Since $16\log m \le -18d$, (Q85.6.2) is at most $-d/5$. This proves the low-domain row for every finite $b \ge 5$.

For $1/5 \le d \le 1/3$, original adjunction gives

$$EF = E(f_2 h_b) = E(f_3 f_{b-1}) = E(h_4 f_{b-2}). \tag{Q85.6.3}$$

For $b = 5$, this is $\mathbb{E}[f_3 A f_3] = \mathbb{E}[f_3 T f_3] = (\mathbb{E} f_3)^2 = (1-2d)^2$. For $b = 6$, it is $\mathbb{E}(h_4 f_4) \le \mu_3^2 = (1-2d)^2$, since $h_4 + f_4 = 2\mu_3$ pointwise. For $b \ge 7$, set $v = b - 2 \ge 5$. The arithmetic-mean input gives

$$h_4 + f_v \le 2 - 13d/4 + R_{4v}, \quad R_{4v} = \sum_{j=5}^{v} K^j 1.$$

The finite alternating means start with a zero odd term and a negative even term, so $\mathbb{E} R_{4v} \le 0$. Its norm satisfies

$$E R_{4v}^2 \le (1+\theta)\delta_5 \le (1+\theta)\theta^4 d.$$

The finite G_N norm is at most one except at $N = 2$, where its square is at most $1 + \theta$; $N = 1$ is included. Since $1 - 13d/8 > 0$,

$$E(h_4 f_v) \le (1 - 13d/8)^2 + \frac{(1+\theta)\theta^4}{4} d \le (1 - 13d/8)^2 + d/80. \tag{Q85.6.4}$$

The coefficient comparison already follows from $\theta < 5/12$. The last quadratic decreases on $[1/5, 1/3]$, with maximum $733/1600 < 11/24$. The $b = 5, 6$ maximum $9/25$ is smaller. Thus $TF \le 2\mathbb{E}F \le 11/12$ at every atom. Positivity gives $T^2 F \le (11/12)q$, whence

$$E[pT^2 F]^8 \le (11/12)^8 \le e^{-2/3} \le e^{-d/5}. \tag{Q85.6.5}$$

The same direction applies on both domains, proving the first inequality in (Q85.0.3).

Q85.7 The two short $b = 4$ rows

For $L = 2, F = f_2 h_4, 0 < d \le 1/4$, use (Q85.4.10). In the reversed square, $\delta_- = -k\delta_2 + k^2 V_R/2 \le 0$, because

$$V_R/\delta_2 \le 3d + \theta + \theta^2 \le 2(1-d).$$

Also $\varepsilon_- = kd^2 + k^2 V_R/2 \ge 0$, so $\delta_- \ge a_-$ and the constant norm contribution is nonpositive. The two-step calculation in (Q85.6.1) gives

$$\begin{aligned} \log EH^8 &\le 16\log m + 9d + 8kd^2 \\ &\quad + k^2[4d^3 + (8d + 17d^2)\delta_2 + (4 + 34d)\delta_3 + 21\delta_4 + 17\delta_5] - 26k\delta_3. \end{aligned} \tag{Q85.7.1}$$

Group $\delta_3, \delta_4, \delta_5$ before estimating their signs. Since $4 + 21\theta + 17\theta^2 < 31/2$, the total coefficient of δ_3 is at most $k^2(31/2 + 34d) - 26k \le 24k^2 - 26k \le 8$. Bound the remaining positive terms at $d = 1/4, k = 4/3$. Using $16\log m \le -16d$, the result divided by d is

$$-7 + 8/3 + 4/9 + 49\theta/9 + 8\theta^2 = -137/384 < -1/3. \tag{Q85.7.2}$$

For $L = 3, F = h_2 f_4$, use the canonical parameters in (Q85.4.10). The three-step arithmetic-mean primitive is

$$V_3 = \varepsilon 1 + (1+\delta)g' + K'g' + kK'^2 R.$$

Its nonconstant spectral coordinates are

$$1+\delta-kdx-kx^2, \qquad 1+kx^2.$$

The scalar constant plus the signed mean of $K'V_3$ is $-2d-kd^2+kV_R/2$. Thus

$$\log EH^8 \le 16\log m - 16d - 8kd^2 + 4kV_R + 17\|V_3\|^2. \tag{Q85.7.3}$$

In the nonconstant norm, group $\delta_3,\delta_4,\delta_5,\delta_6$. Their coefficient is at most

$$k[-2\kappa + k((d+\theta)^2+\theta^3)] < 0,$$

because $(d+\theta)^2+\theta^3 \le 16309/32768 < 1/2$ and $k \le 4/3$. The constant-vector contribution is $\varepsilon^2 - 2\varepsilon(d+k\delta_3) \le (1001/2304)d^2 < d^2/2$. The parameter bounds on δ,ε give

$$\|V_3\|^2/d \le 1+\theta+(85/48)d+(625/576)d^2. \tag{Q85.7.4}$$

The coefficient $1-2k(1+\delta)d$ of δ_2 is nonnegative: its minimum from those parameter bounds is $23/144$. Thus $\delta_2 \le \theta d$ can be used in this term. Use $16\log m \le -16d-8d^2, k\ge 1$ and $4kV_R/d \le 16309/6144$, and bound the remaining positive powers at $d=1/4$. Equation (Q85.7.3), divided by d, is at most

$$-259/32+677/192+10625/9216+16309/6144 = -14015/18432 < -3/4. \tag{Q85.7.5}$$

For the actual bare path, Corollary P.2 gives

$$\mathbb{E}(p\bar{f}_2)^8 \le 82/15 \qquad (1/4 \le d \le 1/3). \tag{Q85.7.6}$$

On the adjacent interval it gives

$$\mathbb{E}(p\bar{f}_2)^8 \le 35/8 \qquad (1/5 \le d \le 1/4). \tag{Q85.7.7}$$

Both statements use the same actual $r = K^2\mathbf{1}$ and its compatible mean and second moment, with the proof now shared at the unit root.

For the $b=4$ reversed pair on $d \ge 1/4$,

$$E(f_2h_4) = Ef_3^2 = 1-3d+3d^2-\delta_2+\delta_3 \le 1-3d+(77/32)d^2 \le 205/512 \tag{Q85.7.8}$$

The bounding quadratic decreases on $[1/4,1/3]$. Thus $T(f_2h_4) \le 205/256$ and $pT^2(f_2h_4) \le 205/256$. Its eighth power is smaller than $e^{-d/3}$: $(205/256)^2 < 13/20$ gives an eighth power below $9/50$, whereas $e^{-d/3} \ge 8/9$. For the canonical pair, its exact mean gives

$$E(h_2f_4) = 1-3d+\delta_2-d^2+\delta_3 \le 1-(2487/1024)d-d^2 \le 1353/4096 < 1/3. \tag{Q85.7.9}$$

Hence $pT^3(h_2f_4) \le (2/3)p\bar{f}_2$. Equation (Q85.7.6) gives an eighth moment at most $(2/3)^8(82/15) < 1/2 < e^{-d/3}$. Together with (Q85.7.2) and (Q85.7.5), this proves (Q85.0.2) in its fixed directions.

Q85.8 Every $b \geq 5, L = 3$ on the low domain

Let $0 < d \leq 1/5, F = h_2 f_b$. Three-step Duhamel propagation of the actual square gives

$$p + Z = 2 + \varepsilon - d - k\rho + N,$$

$$N = K'V_3, \quad V_3 = \varepsilon 1 + (1+\delta)g' + K'g' + kK'^2 R,$$

$$EN = -(1+\delta)d + kD_3, \qquad D_3 = \int x\mathsf{A}.$$

Consequently,

$$\log EH^8 \leq 16 \log m - 16d + 8\varepsilon - 8\delta d + 8k(D_3 - \rho) + 17\|V_3\|^2. \tag{Q85.8.1}$$

The nonconstant coordinates of V_3 are $1 + \delta - kx\mathsf{A}$ and $1 + kx\mathsf{B}$. For every $b \geq 5$, their squared norm satisfies

$$(1 + \delta - kx\mathsf{A})^2 + x(1 + kx\mathsf{B})^2 \leq (1+\delta)^2 + x. \tag{Q85.8.2}$$

Here is the full finite certificate. Using $\mathsf{A}^2 + x\mathsf{B}^2 \leq u_0\mathsf{A} + xz_0\mathsf{B}$ and $\mathsf{A} \geq (1-x)\mathsf{B}$, the unshifted squared difference is at most

$$kx\mathsf{B}[-2 + 4x + kx\{u_0(1-x) + xz_0\}].$$

The substitution is valid because $2 - kxu_0 > 0$. The bracket is negative on the full rectangle: $x[u_0(1-x) + xz_0]$ increases with x, with derivative at least $u_0(1-2x) > 0$, and all parameters increase with d. At $d = 1/5, x = 13/32$, its negative margin is $79745/3174400$. A nonnegative δ shift adds $-2\delta kx\mathsf{A}$ to the difference. This proves (Q85.8.2), including $b = 5$. The constant coordinate then gives

$$\|V_3\|^2 \leq (1+\delta)^2 d + \delta_2 + \varepsilon^2 - 2\varepsilon(d + k\rho). \tag{Q85.8.3}$$

Besides $\varepsilon \leq 3d/5$, the parameter and endpoint bounds give

$$-\varepsilon \leq k(a_1 - D_0) < 7d/20, \qquad k\rho < 3d/10.$$

The full ε contribution satisfies

$$8\varepsilon + 17\varepsilon^2 - 34\varepsilon(d + k\rho) \leq (24/5)d - (357/25)d^2. \tag{Q85.8.4}$$

For $\varepsilon \geq 0$, discard the nonpositive ρ term. The convex quadratic on $[0, 3d/5]$ has maximum at an endpoint; $8 - 34d > 0$ makes the upper endpoint larger, giving exactly the right side. For $\varepsilon \leq 0$, put $w = -\varepsilon \leq 7d/20$. The upper bound is a convex quadratic in w, with endpoints zero and

$$-(14/5)d + (7021/400)d^2.$$

The right side of (Q85.8.4) is positive and exceeds this endpoint: their difference is $d[38/5 - (12733/400)d] > 0$ for $d \leq 1/5$. Thus the actual mean of either sign and its constant-vector cross term remain included. Use $\delta \leq 6d/5$, (Q85.4.9), and

$$16 \log m \leq -18d - (81/8)d^2, \quad 8k\theta^4 < 3/10, \quad 8k\theta < 17/4.$$

Substitution into (Q85.8.1)–(Q85.8.4) gives

$$\log EH^8 \leq -(799/160)d + (2209/200)d^2 + (612/25)d^3 \leq -(36111/20000)d < -(9/5)d. \tag{Q85.8.5}$$

All remaining positive coefficients were bounded at $d = 1/5$. This proves the canonical low-domain row for every finite $b \geq 5$.

Q85.10 Completion using the actual long-connector inputs

For the canonical $F = h_2 f_b$, Section O.18 proves $\mathbb{E}[pT^L F]^8 \le 1 - d/20000$ when $b = 3, L \ge 3$, or $b \ge 4, L \ge 4$. It also proves, for every finite $b \ge 5$ and $1/5 \le d \le 1/3$, $\mathbb{E}(h_2 f_b) \le (1-d)/2$.

For the remaining $b \ge 5, L = 3, d \ge 1/5$ row, the kernel $0 \le 1 - K(x,y) \le 2$ gives $TF \le 1 - d$. Positivity yields

$$pT^3 F \le (1-d) p \bar{f}_2.$$

The actual path estimates (Q85.7.6)–(Q85.7.7) now give

$$E[pT^3F]^8 \le \begin{cases} (4/5)^8(35/8) = 286720/390625 < 3/4, & 1/5 \le d \le 1/4, \\ (3/4)^8(82/15) = 538002/983040 < 3/4, & 1/4 \le d \le 1/3. \end{cases} \tag{Q85.10.1}$$

Since $e^{-d/5} \ge 1 - d/5 \ge 14/15$, this proves the second inequality in (Q85.0.3) on the remaining domain. Together with (Q85.8.5), it proves that row in the same direction.

The full ordinary catalogue has the following rows:

integer row	$F_{b,L}$	ordinary eighth upper bound
$b \ge 3,\ L = 1$	$h_2 f_b$	$1 - d/500$
$b = 3,\ L = 2$	$h_2 f_3$	$e^{-d/25}$
$b = 3,\ L \ge 3$	$h_2 f_3$	$1 - d/20000$
$b = 4,\ L = 2$	$f_2 h_4$	$e^{-d/3}$
$b = 4,\ L = 3$	$h_2 f_4$	$e^{-d/3}$
$b \ge 5,\ L = 2$	$f_2 h_b$	$e^{-d/5}$
$b \ge 5,\ L = 3$	$h_2 f_b$	$e^{-d/5}$
$b \ge 4,\ L \ge 4$	$h_2 f_b$	$1 - d/20000$

The unit row uses the $(2,3)$ unit input of Section O.28 and Section O.10 for $b \ge 4$. The $b = 3, L \ge 3$ and $b \ge 4, L \ge 4$ rows use Section O.18; all other rows were proved above. For $0 \le d \le 1/3$,

$$e^{-d/25} \le \frac{1}{1 + d/25} = 1 - \frac{d}{25 + d} \le 1 - \frac{3d}{76} \le 1 - d/20000.$$

The stronger short-row exponentials satisfy the same bound. Therefore

$$\boxed{E[pT^L F_{b,L}]^8 \le Q(d) = 1 - d/20000 \quad (b \ge 3, L \ge 1).} \tag{Q85.10.2}$$

Q85.11 Decorations and complete reversal

At the terminal, let E_t be the product of its original equal pairs. One member of each pair points entirely outward; the other has its first edge inward and its remaining edges outward. At the receiving root, let E_r be the corresponding product of actual pairs or their complete reversals, as specified by the structure. Each pair factor lies in $[0,1]$. The decorated actual message is

$$H_D = E_r p T^L (E_t F_{b,L}). \tag{Q85.11.1}$$

Positivity and (Q85.10.2) imply $\mathbb{E}H_D^8 \le Q(d)$; the decorations are inserted before their outgoing propagation.

For complete reversal, replace K by $-K$ on the entire fixed structure. Then p, q exchange, A, T exchange, $f_n, \bar{f}_n$ exchange, and h_n becomes $A\bar{f}_{n-1}$. All terminal and root decorations reverse together. In particular, the $L = 2, b \ge 4$ row reverses $f_2 h_b$ as a whole. The reversed ordinary bound has the same $Q(d)$ and the same original edge and vertex counts.

O.18 Long connectors for every terminal pair $(2, b)$

Fix the original finite probability space and a real antisymmetric kernel $|K| \le 1$. The terminal has residual arms of lengths $2, b$, $b \ge 3$. The internal vertex of the two-edge arm points toward both the terminal and its leaf; the b-edge arm points entirely toward its leaf. All L connector edges point from the terminal toward the receiving root, whose genuine original unit arm points outward. The undecorated message is

$$F_b = h_2 f_b, \qquad X_{b,L} = T^L F_b, \qquad H_{b,L} = pX_{b,L}. \tag{Q86.0.1}$$

Let $E_t, E_r \in [0,1]$ be the actual equal-pair factors at the terminal and root. Each pair uses one fully outward f_ℓ arm and one h_ℓ arm with first edge inward and remaining edges outward, or the full reversal of that pair, fixed by the application. The full message is $H_D = E_r p T^L (E_t F_b)$. All directions are fixed from the tree, its lengths and pairings before the host is given.

For every finite admissible model,

$$\boxed{EH_D^8 \le 1 - d/20000} \tag{Q86.0.2}$$

for $b = 3, L \ge 3$, and for every $b \ge 4, L \ge 4$. We also prove the two local bounds

$$\boxed{E[(p\bar f_n)^8] \le e^{253d/32} \quad (n \ge 3),} \tag{Q86.0.3}$$

$$\boxed{E[h_2 f_b] \le (1-d)/2 \quad (b \ge 5,\ 1/5 \le d \le 1/3).} \tag{Q86.0.4}$$

The first is an upper cost for actual bare paths. The second supplies $TF_b \le 1 - d$ in its specified high domain.

Q86.1 Original finite model

Delete zero-mass atoms, and write

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad K^* = -K, \quad |K(x,y)| \le 1,$$

$g = K1,\ p = 1+g,\ q = 1-g,\ r = Kg,\ s = Kr,\ t = Ks, \quad f_n = A^n 1,\ \bar f_n = T^n 1,\ h_n = T f_{n-1}.$

Put $\delta_j = \|K^j \mathbf{1}\|_2^2, d = \delta_1, \theta = 13/32, \kappa = 19/32, \tau = d - \delta_2$. All expectations, inner products, adjoints, spectral quantities and norms use $L^2(\mu)$. The shared finite calculus in Section M.1 gives

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta \delta_j, \quad |Kv(x)| \le \|v\|_2. \tag{Q86.1.1}$$

Cauchy–Schwarz in the spectral measure of $-K^2$ also gives $d\delta_3 \ge \delta_2^2$. The operators A, T are positive, $0 \le f_n, \bar f_n, h_n \le 2$, and $\mathbb{E} f_n = \mathbb{E}\bar f_n \le 1$. Actual equal pairs and their full reversals satisfy $f_\ell h_\ell \le 1$. Since $K(x,x) = 0$, $p(x), q(x) \ge \mu(x) > 0$. Every finite rooted-tree message is positive by assigning all vertices the root label.

Let $\beta_1 = \beta_2 = 0$, and

$$\beta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1} \delta_i \quad (j \ge 3), \qquad \tau \le \beta_j \le d.$$

Finite Duhamel expansion gives

$$S_n := \sum_{j=0}^{n} K^j 1 = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}, \qquad S_n' := \sum_{j=0}^{n} (-K)^j 1 = \bar f_n + \sum_{j=3}^{n} \beta_j \bar f_{n-j}. \tag{Q86.1.2}$$

Every retained or discarded path loss is nonnegative. For $G_n(z) = 1 + z + \cdots + z^{n-1}$, $G_0 = 0$,

$$\|G_n(K)\| \le 1 \quad (n = 0, 1 \text{ or } n \ge 3), \qquad \|G_2(K)\|^2 \le 1 + \theta. \tag{Q86.1.3}$$

The signed exponential estimate from Section M.1 states that $|W| \le U, \mathbb{E}W^2 \le \theta U^2$ imply, for every $s \ge 0$,

$$Ee^{sW} \le \exp\{sEW + (17/64)s^2U^2\}.$$

In particular, for every real original function V,

$$\boxed{Ee^{8KV} \le \exp\{8E(KV) + 17\|V\|_2^2\}.} \tag{Q86.1.4}$$

At $d = 0$, $g = 0$, $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, and leaf integration gives every actual message one. All divisions below use $d > 0$.

Q86.2 Square envelopes and their low-domain data

For $b = 3$, $h_2 + f_3 = 2 - d + s$, so

$$F_3 \le (m + R/2)^2, \qquad m = 1 - d/2, \quad R = s \quad (0 < d \le 1/3). \tag{Q86.2.1}$$

For $b = 4$, $h_2 + f_4 = 2 - 2d + s + t - dg$, so

$$F_4 \le (m + R/2)^2, \qquad m = 1 - d, \quad R = s + t - dg \quad (0 < d \le 1/4). \tag{Q86.2.2}$$

Before squaring, $2m + R$ is a sum of nonnegative original messages.

For $b \ge 5$, work on $0 < d \le 1/5$, with $m = 1 - 9d/8, k = 1/m$. For $b = 5$, set $C_1 = d, C_2 = d/4 + \delta_2, C_3 = 0$. Then $C_2 \le 21d/32 \le \beta_3$ and $\beta_5 + C_1 + C_2 = 9d/4$. For $b \ge 6$, set

$$\lambda = \frac{9d/4 - \beta_b - \beta_{b-1}}{\beta_{b-2} + (1-d)\beta_{b-3}}, \quad C_1 = \beta_{b-1}, \quad C_2 = \lambda\beta_{b-2}, \quad C_3 = \lambda\beta_{b-3}.$$

The denominator is positive, the numerator is nonnegative, and the full four-term constant is at least $(4-d)\tau \ge 361d/160 > 9d/4$. Thus $0 \le \lambda \le 1$, each C_i is no larger than its actual β coefficient, $0 \le C_i \le d$, and $C_1 \ge \tau$. Set $C = C_1 + C_2 + C_3, D = C_2 + C_3, E_0 = C_3$, and

$$R = \sum_{j=3}^{b} K^j 1 - Cg - Dr - E_0 s. \tag{Q86.2.3}$$

Retain $\beta_b, C_1 f_1, C_2 f_2, E_0 f_3$ in (Q86.1.2). Using $f_1 = 1 + g$, $f_2 = 1 + g + r$, $f_3 = 1 - d + g + r + s$, their constant loss is $\beta_b + C_1 + C_2 + (1-d)E_0 = 9d/4$. Hence

$$0 \le h_2 + f_b \le 2m + R, \qquad F_b \le F_0 := (m + R/2)^2. \tag{Q86.2.4}$$

For $b = 5$, $E_0 = 0$, so no positive amount is retained from $\beta_2 = 0$.

On the original spectral interval $\xi \in [0, \theta]$, the multiplier of R relative to g is $-A(\xi) - i\sqrt{\xi}B(\xi)$, with $A, B \ge 0$. For $b = 5$,

$$A = d + C_2 + \xi - \xi^2, \qquad B = C_2 + \xi.$$

For $b \ge 6$, let $N = b - 2 \ge 4$ and $G_N(i\sqrt{\xi}) = a_f + i\sqrt{\xi}b_f$. Then

$$A = C + \xi(a_f - E_0), \quad B = D + \xi b_f, \quad 1 - \xi \le a_f, b_f \le 1 - \xi + \xi^2, \quad a_f \ge (1-\xi)b_f. \tag{Q86.2.5}$$

These alternating finite-sum bounds include $N = 4, 5$. For $N = 2j$, $a_f = b_f = \sum_{i=0}^{j-1}(-\xi)^i$. For $N = 2j+1$, $a_f - (1-\xi)b_f = [\xi + (-\xi)^j]/(1+\xi) \geq 0$, $j \geq 2$. It follows that

$$A - (1-\xi)B \geq C_1 + C_2\xi, \qquad C_1 \geq \tau. \tag{Q86.2.6}$$

The uniform frequency bounds are

$$\boxed{\sqrt{A^2 + \xi B^2} \leq u_0 := 3/8 + 9d/5, \qquad B \leq z_0 := \theta + 13d/16.} \tag{Q86.2.7}$$

For $b = 5$, with the global δ_2 fixed, A, B increase nonnegatively with ξ. First use $\xi = \theta$, then $\delta_2 \leq \theta d$. The endpoint vector is bounded by $(\theta - \theta^2, \sqrt{\theta}\,\theta) + d(53/32, \sqrt{\theta}\,21/32)$. The squared norms of these two vectors are $131313/1048576 < 9/64$ and $95621/32768 < (9/5)^2$. Also $B \leq \theta + 21d/32 \leq z_0$.

For $b \geq 6$, the exact retained constant yields

$$C = 9d/4 - \beta_b + dE_0, \qquad D = 9d/4 - \beta_b - \beta_{b-1} + dE_0.$$

Let $u(\xi) = \xi - \xi^2 + \xi^3$. The frequency bounds are

$$A \leq \overline{A}(\xi) = u(\xi) + 5d/4 + \delta_2 + E_0(d - \xi), \quad B \leq \overline{B}(\xi) = u(\xi) + d/4 + 2\delta_2 + dE_0. \tag{Q86.2.8}$$

Throughout the frequency interval, d, δ_2, E_0 are fixed. Since $u' \geq 2/3$ and $E_0 \leq d \leq 1/5$, both nonnegative upper bounds increase, as does $\overline{A}^2 + \xi\overline{B}^2$. Use $\xi = \theta$ before $\delta_2 \leq \theta d$, obtaining

$$A_* = u + (5/4 + \theta)d + E_0(d - \theta), \quad B_* = u + (1/4 + 2\theta)d + dE_0, \quad u = 10101/32768 < 5/16.$$

The expression $A_*^2 + \theta B_*^2$ is convex in $E_0 \in [0, d]$. At $E_0 = 0$, the constant-vector norm is less than $3/8$, and the squared d-vector norm is $26229/8192 < (9/5)^2$. At $E_0 = d$, the squared d-vector norm is $16557/8192 < (10/7)^2$, and the d^2-vector norm is below $6/5$. The latter endpoint norm is therefore at most $3/8 + (10/7)d + (6/5)d^2 \leq 3/8 + (9/5)d$, since $10/7 + 6/25 < 9/5$. This bounds every E_0. Also $B \leq 5/16 + 17d/16 + d^2 \leq z_0$, because the difference $-3/32 + d/4 + d^2$ increases on $[0, 1/5]$ and has right endpoint $-3/800$.

Write $B_0 = -K, g' = -g, k = 1/m, V = \mathbb{E}R^2, D_j = \mathbb{E}(B_0^j R)$, and

$$a_1 = EKR = -E(gR), \quad \delta = kD_0 + k^2V/2, \quad a = ka_1, \quad \varepsilon = \delta - a, \quad \nu = 1 + \varepsilon. \tag{Q86.2.9}$$

The operator B_0 is distinct from the positive scalar B used below. For $b \geq 4$, use the spectral measure of g, with mass d:

$$V = \int (A^2 + \xi B^2), \quad D_0 = \int B, \quad a_1 = \int A, \quad D_2 = -\int \xi B,$$

$$D_{4j+1} = -\int \xi^{2j} A, \quad D_{4j+2} = -\int \xi^{2j+1} B, \quad D_{4j+3} = \int \xi^{2j+1} A, \quad D_{4j+4} = \int \xi^{2j+2} B. \tag{Q86.2.10}$$

For $b \geq 5$ on its low domain, (Q86.2.6) gives $a_1 \geq \kappa D_0 + \tau d$. Also $a_1 - D_0 \leq d^2 + \theta^3 d$. For $b = 5$, this difference is $d^2 - \delta_3$. For $b \geq 6$, use $A - B = C_1 - E_0\xi + \xi(a_f - b_f)$, whose first possible positive end term has degree at least three. Consequently,

$$|\varepsilon| \leq 3d/5, \quad \delta \leq 6d/5, \quad ku_0 \leq 1 \qquad (b \geq 5,\ d \leq 1/5). \tag{Q86.2.11}$$

$\delta/d \le z_0/m + u_0^2/(2m^2)$. After multiplication by $2m^2$, the required positive difference is $463/320 - (955/128)d + (2601/1600)d^2$, decreasing on the interval with right endpoint $1577/80000$. For the upper ε bound, use $a_1 \ge \kappa D_0 + \tau d$; the positive difference, multiplied by $2m^2$, is $1867/2560 - (64539/20480)d - (118503/51200)d^2$, with right endpoint $16519/2560000$. For its lower bound, $-\varepsilon/d \le k(d+\theta^3) < 3/5$. Finally u_0/m increases, and $ku_0 \le 1$ at the right endpoint.

For $b = 4$, $A = d + \xi$, $B = \xi$, $D_0 = \delta_2$, $a_1 = d^2 + \delta_2$, and $3d^3 \le V \le d(1/2+d)^2$. These give

$$|\varepsilon| \le d/6, \quad \delta \le 25d/24 \qquad (b = 4,\ d \le 1/4). \tag{Q86.2.12}$$

Indeed $\varepsilon = -kd^2 + k^2V/2$. For its upper bound, the positive difference multiplied by m^2 is $(1-4d)(1+8d)/24 \ge 0$. For the lower bound let $z = d/m \le 1/3$, so $\varepsilon/d \ge -z + 3z^2/2 \ge -1/6$. The upper bound $\delta/d \le \theta/m + (1/2+d)^2/(2m^2)$ increases to $25/24$.

For $b = 3$, the parameters are

$$\begin{gathered} D_0 = D_2 = D_4 = 0, \quad D_1 = -\delta_2, \quad D_3 = \delta_3, \\ a = k\delta_2, \quad \delta = k^2\delta_3/2, \\ \eta = a - \delta \ge 0, \quad \varepsilon = -\eta. \end{gathered} \tag{Q86.2.13}$$

Since $k \le 6/5, \delta_3 \le \theta\delta_2$, $\delta/a \le 39/160, \eta \le 13/80 < 1/6, \delta \le \eta/3, \delta \le 3d/25$. The last uses $\delta/d \le (18/25)\theta^2 < 3/25$.

Q86.3 Finite propagation, actual losses and weighted AM–GM

For every $L \ge 2$, positivity, $T(R^2) \le 2V$ and finite Duhamel expansion give

$$\frac{T^L F_0}{m^2} \le \bar f_L + kB_0^L R + \delta \bar f_{L-1} - a\bar f_{L-2} + k\sum_{j=2}^{L-1} D_j \bar f_{L-1-j}. \tag{Q86.3.1}$$

The exact formula for any original R is $T^L R = B_0^L R + \sum_{j=0}^{L-1} D_j \bar f_{L-1-j}$. Insert $F_0 = m^2 + mR + R^2/4$ and use $T^L(R^2) \le 2V\bar f_{L-1}$ to obtain (Q86.3.1) term by term. Define

$$Z_L^* = S_L' + \delta S_{L-1}' - aS_{L-2}' + kB_0^L R = \nu + Q_L(B_0)g', \qquad \alpha_L = \nu - EZ_L^*.$$

Expanding by (Q86.1.2), the difference between Z_L^* and the right side of (Q86.3.1) is $\sum_{j=3}^{L} \gamma_j \bar f_{L-j}$, where

$$\gamma_3 = d - kD_2, \quad \gamma_4 = d + \delta d - kD_3, \quad \gamma_j = \beta_j + \delta\beta_{j-1} - a\beta_{j-2} - kD_{j-1} (j \ge 5). \tag{Q86.3.2}$$

Only existing nonnegative indices occur. For $L \ge 4$, $\alpha_L = \gamma_{L+1}$. The estimates used at every length are

range	$\gamma_j \ge hd$ $(j \ge 3)$	$\|Q_n\|$ $(n \ge 3)$
$b = 3,\ d \le 1/3$	$h = 9/16 - d/2$	≤ 1
$b = 4,\ d \le 1/4$	$h = 1/2 - 6d/5$	$\le 1 + d$
$b \ge 5,\ d \le 1/5$	$h = 1/2 - 6d/5$	$\le 1 + 7d/5$

Here are the finite certificates for all indices.

For $b = 3$,

$$Q_n(z) = G_n(z) - \eta G_{n-2}(z) + \delta z^{n-2} - kz^{n+2} \quad (n \ge 3). \tag{Q86.3.3}$$

At $z = i\sqrt{\xi}$, the real and normalized imaginary parts for $n = 3$ are $1 - \xi - \eta, 1 + \delta - k\xi^2$. Compared with $P_3 = G_3 - kz^5$, the increase in squared modulus is at most $\eta[-2 + (8/3)\xi + \eta(1 + \xi/9)] \le -53\eta/108$, where $|P_3| \le 1$. For $n = 4$, the two components are nonnegative and bounded by those of $P_4 = G_4 - kz^6$; their lower bounds are $11/36, 1/3$. Moreover, $|P_4| \le (1 - \xi)(1 + \xi/2) + k\xi^3 \le 1$, because $k\xi \le 39/80 < 1/2$.

For every $n \ge 5$,

$$Q_n(z) = \frac{1 - \eta + z^{n-2}[a - \delta z - z^2 - kz^4 + kz^5]}{1 - z}.$$

The last three terms have modulus at most ξ, since $(1-k\xi)^2+k^2\xi^3 \le 1$, which follows from $k\xi(1+\xi) \le 2$. Also $a + \delta\sqrt{\xi} \le 19\eta/12$ and $(19/12)\xi^{3/2} < 1$, so the numerator has modulus at most $1 + \xi^{5/2}$. Finally $2\xi^{3/2} + \xi^4 \le 13/16 < 1$, proving $|Q_n| \le 1$.

For the losses, $\gamma_3 = d$ and $\gamma_4 \ge (1 - k\theta^2)d \ge 4d/5$. For $j \ge 5$, a positive D_{j-1} first occurs at $D_7 = \delta_5$, so $\gamma_j \ge [\kappa - k\theta d - k\theta^4]d$. Subtracting $h = 9/16 - d/2$ and multiplying by m gives $1/32 - \theta^4 + (5/64)d - d^2/4$. This is concave, and its two endpoint values are $4207/1048576$ and $4207/1048576 - 1/576$, both positive.

For $b \ge 4$, on the B_0 frequency blocks write $R = (A - Bz)g'$, and

$$P_n = G_n + kz^n(A - Bz), \qquad Q_n = P_n + \varepsilon G_{n-2} + \delta z^{n-2}. \tag{Q86.3.4}$$

For $b = 4$, use $u_0 = 1/2 + d$; for $b \ge 5$, use (Q86.2.7). In both cases $ku_0 \le 1$. For every $n \ge 3$, $\|P_n\| \le 1$: at $n = 3$, the components $1 - \xi - k\xi^2 B, 1 - k\xi A$ are nonnegative and no larger than those of G_3. For $n = 4$, the modulus is at most $(1 - \xi)(1 + \xi/2) + \xi^2 \le 1$. For $n = 5$, use $|G_5| \le 1-\xi/2+\xi^2/2$, then add $\xi^{5/2} \le 13\xi^2/20$. The squared comparison has difference $\xi^2(1+2\xi-3\xi^2)/4 \ge 0$. For $n \ge 6$, the modulus is at most $(1 + \xi^3)/\sqrt{1 + \xi} + \xi^3 \le 1$. The last inequality is equivalent to $\xi(1 + \sqrt{1 + \xi}) \le 1$, and this quantity is at most $143/160$.

For Q_3, use $\|G_1\| = 1$, $\sqrt{\theta} < 13/20$. The added cost for $b = 4$ is at most

$$[1/6 + (13/20)(25/24)]d = 27d/32 < d;$$

for $b \ge 5$, it is at most

$$[3/5 + (13/20)(6/5)]d = 69d/50 < 7d/5.$$

For Q_4, also use $\|G_2\| \le 6/5$. For larger n, the geometric-sum norm is at most one and the power tail is smaller. This proves the stated Q_n bounds.

For the losses, $\gamma_3 \ge d, \gamma_4 \ge \kappa d$. When $b = 4$, the same spectral estimate gives $\gamma_j \ge [\kappa - ku_0 d - k\theta^2 z_0]d$, $j \ge 5$, with $z_0 = \theta+11d/16$. Subtract h and multiply by m. The resulting concave quadratic is $875/32768+(40369/81920)d-(11/5)d^2$, whose endpoint values are $875/32768, 4063/327680 > 0$. When $b \ge 5$, every positive D_j, $j \ge 4$, is at most $\theta^2 z_0 d$, and $a \le d$. Thus $\gamma_j \ge [\kappa - d - k\theta^2 z_0]d$. After subtracting h and multiplying by m, the difference is $875/32768 - (3241/81920)d - 9d^2/40$; it decreases to $40101/4096000 > 0$.

For each $L \ge 4$, retain the $j = L, L - 1$ losses in (Q86.3.2). This gives

$$0 \le X_{b,L}/m^2 \le Z := Z_L^* - hd(1 + q). \tag{Q86.3.5}$$

The finite polynomials satisfy $Q_L = \nu + zQ_{L-1}$, hence

$$Z = Bq - hd + N_0, \quad B = \nu - hd, \quad N_0 = B_0 Q_{L-1}(B_0)g', \quad EN_0 = -\alpha_L \le -hd. \tag{Q86.3.6}$$

In particular, the actual constant coordinate $\nu - hd$ is retained. The estimates below prove $B > 0$ on each domain. Weighted AM–GM between p and the nonnegative upper bound in (Q86.3.5) gives

$$H_{b,L} \le m^2 pZ \le (c_* + N/2)^2, \quad c_* = m\sqrt{B} - \frac{mhd}{2\sqrt{B}}, \quad N = \frac{m}{\sqrt{B}} N_0. \tag{Q86.3.7}$$

The base is exactly $m(\sqrt{B}\,p + Z/\sqrt{B})/2 \ge 0$. If $c_* \le c$, then $c + N/2$ remains nonnegative. All row, second-moment and mean bounds for N come from this same original function.

For $b = 3, L = 3$, the only retained loss is $\gamma_3 = d$. The exact short-length formula is

$$Z = (1-\eta)q - d + N_0, \quad N_0 = B_0 Q_2(B_0) g', \quad Q_2 = 1 + \delta + z - kz^4. \tag{Q86.3.8}$$

Here $B = 1 - \eta$, and hd in (Q86.3.7) is replaced by d; $\mathbb{E}N_0 = -\alpha_3$. The polynomial Q_2 is estimated separately below.

Q86.4 Fixed-node Hermite–Lucas payment

Let $A, M > 0, \beta \ge 0$, and let the same real N satisfy

$$|N|^2 \le Md, \quad EN^2 \le \theta M d, \quad EN \le -\beta d, \quad 0 \le H \le (c + N/2)^2, \quad c = 1 - Ad > 0, \tag{Q86.4.1}$$

Assume also $c + N/2 \ge 0$. Put $u = \sqrt{Md}, x = u/(2c) < 1, k_0 = 4A/M$ and $\ell(d) = 2c\beta/M$. For $f(z) = (1 + z/(2c))^{16}$, use quadratic Hermite interpolation with double contact at $-\theta u$ and single contact at u. Since $f''' \ge 0$ on $[-u, u]$, its remainder has sign $(z + \theta u)^2(z - u) \le 0$. Explicitly the majorant is $f(-\theta u) + f'(-\theta u)(z + \theta u) + B_H(z + \theta u)^2$, where

$$B_H = \frac{f(u) - f(-\theta u) - f'(-\theta u)(1+\theta)u}{[(1+\theta)u]^2} \ge 0.$$

Its linear coefficient $f'(-\theta u) + 2\theta u B_H$ is also nonnegative. Substituting the actual mean and second-moment bounds yields

$$E(1 + N/(2c))^{16} \le M_\theta(x) - \ell(d) x D_\theta(x), \tag{Q86.4.2}$$

$$M_h(x) = \frac{h(1+x)^{16} + (1-hx)^{16}}{1+h},$$

$$D_h(x) = \frac{1-h}{1+h} 16x(1-hx)^{15} + \frac{2h}{(1+h)^2}\{(1+x)^{16} - (1-hx)^{16}\}.$$

Write the degree-j coefficients as $\binom{16}{j} M_j, \binom{16}{j} D_j$, respectively. Then

$$M_j = \frac{h + (-h)^j}{1+h}, \qquad D_j = \frac{1-h}{1+h} j(-h)^{j-1} + \frac{2h}{(1+h)^2}(1 - (-h)^j). \tag{Q86.4.3}$$

These M_j, D_j are scalar coefficients local to this paragraph, distinct from the original moments D_j in the propagation formulas. Here $D_0 = 0$. At $h = \theta$, the required low-degree data are

$$M_0 = 1, \quad M_1 = 0, \quad M_2 = \theta,$$

$$560 M_3 = 8645/64 < 136, \quad M_j \le M_4 = 10101/32768 < 5/16 \ (j \ge 4),$$

$$D_1 = 1, \quad D_2 = 0, \quad D_3 = 663/1024 > 16/25, \quad 560 D_3 > 350, \quad D_j \ge 0 \ (j \ge 0). \tag{Q86.4.4}$$

To see $D_j \ge 0$, both terms are nonnegative when j is odd. For even $j \ge 4$, factor out $h(1-h)/(1+h)^2$. The remainder is $2\sum_{i=0}^{j-1} h^i - jh^{j-2}(1+h) > 0$, since $0 < h \le 1/2$ and $jh^{j-2} \le 1$. The cases $j = 0, 1, 2$ are direct.

Let $w = 1/c, v = w - 1 \ge 0$, so $wv = k_0x^2$. The finite Lucas identity is

$$w^{16} + v^{16} = 1 + 16t + 104t^2 + 352t^3 + 660t^4 + 672t^5 + 336t^6 + 64t^7 + 2t^8, \quad t = wv.$$

It follows from $w-v = 1$ and the recursion $S_0 = 2, S_1 = 1, S_n = S_{n-1}+tS_{n-2}$ for $S_n = w^n+(-v)^n$. Since $v^{16} \le t^8$,

$$c^{-16} \ge 1 + 16k_0x^2 + 104k_0^2x^4 + 352k_0^3x^6 + 660k_0^4x^8 + 672k_0^5x^{10} + 336k_0^6x^{12} + 64k_0^7x^{14} + k_0^8x^{16}. \tag{Q86.4.5}$$

If these coefficient bounds show

$$c^{-16} - M_\theta(x) + \ell(d)xD_\theta(x) \ge r_*x^2, \tag{Q86.4.6}$$

then (Q86.4.2) gives

$$\boxed{EH^8 \le 1 - (r_*M/4)c^{14}d.} \tag{Q86.4.7}$$

The three finite polynomials needed below are as follows. Except for the first x^2, x^4 coefficients, every even payment uses (Q86.4.5). For the x^3 term, use $560M_3 < 136$; for other odd terms, use $M_j \le 5/16$, discarding positive D_j terms. When $k_0 \ge 3$, after division of the difference by x^2, the $x^4, x^6, x^8, x^{10}, x^{12}, x^{14}$ coefficients are at least $7000, 49000, 160000, 244000, 139000, 6500$. The first is $352 \cdot 3^3 - 8008 \cdot 5/16 = 14003/2 > 7000$. The others follow from $660 \cdot 3^4 - 12870 \cdot 5/16$, $672 \cdot 3^5 - 8008 \cdot 5/16$, $336 \cdot 3^6 - 1820 \cdot 5/16$, $64 \cdot 3^7 - 120 \cdot 5/16$, and $3^8 - 5/16$.

If the first two coefficients are at least $45/4, 629$, the difference divided by x^2 is at least

$$\begin{aligned} P_A(x) = {} & 45/4 - 136x + 629x^2 - 1365x^3 + 7000x^4 - 3575x^5 + 49000x^6 - 3575x^7 \\ & + 160000x^8 - 1365x^9 + 244000x^{10} - 175x^{11} + 139000x^{12} - 5x^{13} + 6500x^{14}. \end{aligned}$$

Its full positive-quadratic decomposition is

$$\begin{aligned} P_A = {} & (45/4 - 136x + 420x^2) + x^2(209 - 1365x + 2400x^2) \\ & + x^4(4600 - 3575x + 1000x^2) + x^6(48000 - 3575x + 100x^2) \\ & + x^8(159900 - 1365x + 10x^2) + x^{10}(243990 - 175x + x^2) \\ & + x^{12}(138999 - 5x + x^2) + 6499x^{14}. \end{aligned} \tag{Q86.4.8}$$

Each quadratic has positive leading coefficient and negative discriminant. For example, $1365^2 < 4 \cdot 209 \cdot 2400$ and $3575^2 < 4 \cdot 4600 \cdot 1000$; the last four comparisons follow directly in the same way. The first bracket has minimum $101/420$, so $P_A \ge 101/420$.

If $k_0 \ge 17/6$, the conservative high-even coefficient bounds are instead $5500, 38000, 120000, 170000, 90000, 4000$. The first exceeds 5500 by $209/54$; the others follow from

$$\frac{4593655}{108} - \frac{32175}{8} > 38000, \quad \frac{9938999}{81} - \frac{5005}{2} > 120000,$$

$$\frac{168962983}{972} - \frac{2275}{4} > 170000, \quad \frac{410338673}{4374} - \frac{75}{2} > 90000, \quad \frac{6975757441}{1679616} - \frac{5}{16} > 4000.$$

If the first two coefficients are at least $15,500$, use

$$\begin{aligned} P_B &= 15 - 136x + 500x^2 - 1365x^3 + 5500x^4 - 3575x^5 + 38000x^6 - 3575x^7 \\ &\quad + 120000x^8 - 1365x^9 + 170000x^{10} - 175x^{11} + 90000x^{12} - 5x^{13} + 4000x^{14} \\ &= (15 - 136x + 310x^2) + x^2(190 - 1365x + 3000x^2) \\ &\quad + x^4(2500 - 3575x + 2000x^2) + x^6(36000 - 3575x + 100x^2) \\ &\quad + x^8(119900 - 1365x + 10x^2) + x^{10}(169990 - 175x + x^2) \\ &\quad + x^{12}(89999 - 5x + x^2) + 3999x^{14}. \end{aligned} \tag{Q86.4.9}$$

Every bracket is positive; the first has minimum $13/155$. Thus $P_B \geq 13/155$.

Finally, if $k_0 \geq 3$ and the first two coefficients are at least $35/4,685$, use

$$\begin{aligned} P_C &= (35/4 - 136x + 530x^2) + x^2(155 - 1365x + 3100x^2) \\ &\quad + x^4(3900 - 3575x + 1000x^2) + x^6(48000 - 3575x + 100x^2) \\ &\quad + x^8(159900 - 1365x + 10x^2) + x^{10}(243990 - 175x + x^2) \\ &\quad + x^{12}(138999 - 5x + x^2) + 6499x^{14}. \end{aligned} \tag{Q86.4.10}$$

Its expansion changes only the constant and x^2 coefficients of P_A, to $35/4,685$. The second and third discriminants satisfy $1365^2 < 4 \cdot 155 \cdot 3100$ and $3575^2 < 4 \cdot 3900 \cdot 1000$; the remaining brackets are as in (Q86.4.8). The first has minimum $27/1060$, so $P_C \geq 27/1060$. These decompositions hold for all real x.

Q86.5 $b = 3$ **and every** $L \geq 4$

Use $m = 1 - d/2, \eta = k\delta_2 - k^2\delta_3/2, h = 9/16 - d/2$, and $B = 1 - \eta - hd$. From $d\delta_3 \geq \delta_2^2$, set $u = \delta_2/d, z = ku \leq 39/80$. Then

$$\eta/d \leq z - z^2/2 \leq 4719/12800 < 3/8.$$

Also $\delta/a \leq 39/160$ gives

$$\eta \geq (121/160)\delta_2 \geq (121/160)d^2. \tag{Q86.5.1}$$

Therefore $\eta + hd \leq 15d/16 - d^2/2 \leq 1 - m^2$, so $m^2 \leq B \leq 1$. By (Q86.3.6) and $\|Q_{L-1}\| \leq 1$, the same N in (Q86.3.7) satisfies

$$|N|^2 \leq d, \qquad EN^2 \leq \theta d, \qquad EN \leq -\frac{5}{6}\frac{19}{48}d = -\frac{95}{288}d. \tag{Q86.5.2}$$

The mean estimate uses $m/\sqrt{B} \geq m \geq 5/6$ and $h \geq 19/48$. Since $B - hd/2 > 0$ and $m/\sqrt{B} \leq 1$, use the whole expression

$$c_* = (m/\sqrt{B})(B - hd/2) \leq B - hd/2 = 1 - \eta - 3hd/2 \leq 1 - 27d/32 - d^2/160 \leq c := 1 - 27d/32. \tag{Q86.5.3}$$

In the scalar lemma take $M = 1, A = 27/32, \beta = 95/288$. Then $c \geq 23/32 > 2/3, k_0 = 27/8$, and $\ell \geq 2185/4608 > 7/15$. The first coefficients satisfy

$$16(27/8) - 120\theta + 16(7/15) > 45/4,$$

$$104(27/8)^2 - 1820(5/16) + 350(7/15) > 629.$$

Since $k_0 \geq 3$, P_A pays all remaining terms. By (Q86.4.7) and $(2/3)^{14} > 1/300$,

$$EH_{3,L}^8 \leq 1 - (101/1680)c^{14}d \leq 1 - d/6000 \qquad (L \geq 4). \tag{Q86.5.4}$$

Q86.6 The short boundary $b = 3, L = 3$

Use (Q86.3.8), with $B = 1 - \eta \geq 1 - 3d/8$, so again $m^2 \leq B \leq 1$. More precisely,

$$m^2/B \leq 1 - d/2, \tag{Q86.6.1}$$

because $(1 - d/2)(1 - 3d/8) - m^2 = d/8 - d^2/16 \geq 0$. The exact $Q_2 = 1 + \delta + z - kz^4$ needs its own estimate. For $\xi \in [0, \theta]$, $1 - k\xi^2 \geq 0$, and

$$|1 + z - kz^4|^2 = (1 - k\xi^2)^2 + \xi \leq 1 + \xi - 2\xi^2 + \xi^4 \leq 1 + \xi - (11/6)\xi^2 \leq 1 + 3/22 < 8/7.$$

Here $\theta^2 < 1/6$. Since $\delta \leq 3d/25$, $2\delta + \delta^2 \leq 153d/625 < d/4$, whence

$$|Q_2|^2 \leq 8/7 + d/4, \qquad \frac{m^2}{B}\|Q_2\|^2 \leq (1 - d/2)(8/7 + d/4) \leq 8/7 < 6/5. \tag{Q86.6.2}$$

On the other hand, $\alpha_3 = d + \delta d - k\delta_3 \geq 4d/5$. Thus the same $N = (m/\sqrt{B})B_0 Q_2 g'$ satisfies

$$|N|^2 \leq (6/5)d, \quad EN^2 \leq \theta(6/5)d, \quad EN \leq -(5/6)(4/5)d = -2d/3. \tag{Q86.6.3}$$

Because $B \leq 1$ and $\sqrt{B} - d/(2\sqrt{B})$ increases,

$$c_* \leq m(1 - d/2) = m^2 \leq 1 - 7d/8 =: c. \tag{Q86.6.4}$$

Take $M = 6/5, A = 7/8, \beta = 2/3$. Then $c \geq 17/24 > 2/3, k_0 = 35/12 > 17/6$, and

$$\ell = (10/9)c \geq 85/108 > 39/50, \qquad \ell \geq 10/9 - (175/54)x^2, \tag{Q86.6.5}$$

The last inequality uses $d = 4c^2x^2/M \leq (10/3)x^2$. Apply the second bound in (Q86.6.5) to D_1, its first bound to D_3, and discard the other nonnegative D_j. The first two coefficients are bounded below by

$$16(35/12) - 120\theta + 16(10/9) = 565/36 > 15,$$

$$104(35/12)^2 - 1820(5/16) + 560(16/25)(39/50) - 16(175/54) > 500.$$

The polynomial P_B therefore pays all higher terms, giving

$$EH_{3,3}^8 \leq 1 - (39/1550)c^{14}d \leq 1 - d/20000. \tag{Q86.6.6}$$

Here again $c^{14} > (2/3)^{14} > 1/300$. This proves every $b = 3, L \geq 3$ row on the full domain.

Q86.7 $b = 4, L \geq 4$ **on** $0 < d \leq 1/4$

Now $m = 1 - d, h = 1/2 - 6d/5, B = 1 + \varepsilon - hd$, with $|\varepsilon| \leq d/6$. Hence

$$1 - 2d/3 + 6d^2/5 \leq B \leq 1, \qquad B - m^2 \geq 4d/3 + d^2/5 > 0. \tag{Q86.7.1}$$

Use $\|Q_{L-1}\| \leq 1 + d$ and $m^2(1 + d)^2 = (1 - d^2)^2$. We have

$$\frac{17}{16}(1 - 2d/3 + 6d^2/5) - (1 - d^2)^2 \geq \frac{1}{16} - \frac{17}{24}d + \frac{257}{80}d^2 > 0. \tag{Q86.7.2}$$

The estimate uses $d^4 \leq d^2/16$. The final quadratic has negative discriminant because $289/576 < 257/320$. Thus the actual row and second moment of N can use $M = 17/16$. Also $m/\sqrt{B} \geq m \geq 3/4, h \geq 1/5$, so $\mathbb{E}N \leq -3d/20$.

For the center, $m\sqrt{B} \le (m^2+B)/2$ and $m/\sqrt{B} \ge 3/4$ give

$$c_* \le 1-d+d^2/2+\varepsilon/2-(7/8)hd \le 1-(65/48)d+(31/20)d^2 \le 1-(29/30)d =: c. \quad \text{(Q86.7.3)}$$

Take $A = 29/30, \beta = 3/20$. Then $c \ge 91/120 > 3/4, k_0 = 928/255 > 18/5$, and $\ell \ge 91/425 > 1/5$. The first coefficients obey

$$16(18/5) - 120\theta + 16/5 = 241/20 > 45/4,$$

$$104(18/5)^2 - 1820(5/16) + 350/5 > 629.$$

The polynomial P_A and (Q86.4.7) imply

$$EH^8_{4,L} \le 1 - \frac{1717}{26880}c^{14}d \le 1 - d/2000 \qquad (L \ge 4,\ d \le 1/4). \quad \text{(Q86.7.4)}$$

Use $(3/4)^{14} > 1/64$. The interval includes $d = 1/4$.

Q86.8 Every $b \ge 5, L \ge 4$ on $0 < d \le 1/5$

Here $m = 1-9d/8, h = 1/2-6d/5 \ge 13/50, B = 1+\varepsilon-hd$. Equation (Q86.2.11) gives

$$B_{\text{low}} := 1 - 11d/10 + 6d^2/5 \le B \le 1 + d/10 + 6d^2/5 \le 267/250,$$

$$B - m^2 \ge (23/20)d - (21/320)d^2 > 0. \quad \text{(Q86.8.1)}$$

All these B are positive. Using $\|Q_{L-1}\| \le 1+7d/5$, calculate

$$m^2(1+7d/5)^2 = (1+11d/40-63d^2/40)^2 \le 1+11d/20-(4919/1600)d^2.$$

The remaining cubic and quartic terms are $d^3[-693/800+(3969/1600)d] \le 0$. Therefore

$$\frac{5}{4}B_{\text{low}} - m^2(1+7d/5)^2 \ge \frac{1}{4} - \frac{77}{40}d + \frac{7319}{1600}d^2 > 0, \quad \text{(Q86.8.2)}$$

The final discriminant comparison is $5929 < 7319$, so take $M = 5/4$. For the mean, use the precise ratio

$$\frac{m}{\sqrt{B}} \ge \frac{149}{200}, \quad \text{(Q86.8.3)}$$

Indeed $m \ge 31/40, B \le 267/250$, and $961/1600 - (149/200)^2(267/250) = 78583/10000000 > 0$. Thus $\mathbb{E}N \le -(149/200)(13/50)d \le -19d/100$.

Using $\sqrt{B} \le 1+(B-1)/2$ and (Q86.8.3), the center satisfies

$$\begin{aligned} c_* &\le (1-9d/8)(1+d/20+3d^2/5) - (149/400)hd \\ &\le 1-(1009/800)d+(3963/4000)d^2 \le 1-(21/20)d =: c. \end{aligned} \quad \text{(Q86.8.4)}$$

The last comparison is $1009/800 - 3963/20000 = 10631/10000 > 21/20$. Take $A = 21/20, \beta = 19/100$. Then $c \ge 79/100 > 3/4, k_0 = 84/25$, and $\ell \ge 3002/12500 > 6/25$. The first coefficients satisfy

$$16(84/25) - 120\theta + 16(6/25) = 177/20 > 35/4,$$

$$104(84/25)^2 - 1820(5/16) + 350(6/25) = 1723421/2500 > 685.$$

The polynomial P_C pays the remaining terms, giving

$$EH^8_{b,L} \le 1 - (27/3392)c^{14}d \le 1 - d/10000 \quad (b \ge 5, L \ge 4, d \le 1/5). \quad \text{(Q86.8.5)}$$

Again use $(3/4)^{14} > 1/64$. The scalar condition $x < 1$ holds in all four applications: $c > 2/3, Md \le 5/12$, so $x^2 < 15/64$. The nonnegative square base follows from the actual AM–GM comparison and $c \ge c_*$.

Q86.9 Actual terminal means and bare-path costs

Let $t_b = \mathbb{E}(h_2 f_b)$. Adjunction gives $t_b = \mathbb{E}(p f_{b+1}) = 2\mathbb{E}f_{b+1} - \mathbb{E}f_{b+2}$. Integrating (Q86.1.2) also gives

$$t_b = 1 - \beta_{b+1} - \beta_{b+2} + \beta_{b+3} - \sum_{j=3}^{b} \beta_j t_{b-j}. \tag{Q86.9.1}$$

All $t_j \geq 0$. The exact initial values are

$$t_0 = 1 + d, \quad t_1 = 1, \quad t_2 = 1 - d - \delta_2, \quad t_3 = 1 - 2d - d^2,$$

$$t_4 = 1 - 3d + \delta_2 - d^2 + \delta_3, \qquad t_5 = 1 - 4d + 2(1+d)\delta_2,$$

$$t_6 = 1 - 5d + 2d^2 + d^3 + (3 + 2d)\delta_2 - \delta_3 - \delta_4. \tag{Q86.9.2}$$

Two further finite expansions give

$$t_7 = 1 - 6d + 5d^2 + 2d^3 + 4\delta_2 - \delta_2^2 - 2(1+d)\delta_3, \tag{Q86.9.3}$$

$$t_8 = 1 - 7d + 9d^2 + 2d^3 + (5 - 4d - 3d^2)\delta_2 - \delta_2^2 - (3 + 2d)\delta_3 + \delta_4 + \delta_5. \tag{Q86.9.4}$$

For $b = 7$, the base term is $1 - d + \delta_2 - \delta_3$, and the loss is $d(t_4 + t_3) + \tau(t_2 + t_1 + t_0) + \delta_3 t_0$. Its exact expansion is $5d - 5d^2 - 2d^3 - 3\delta_2 + \delta_2^2 + \delta_3 + 2d\delta_3$. For $b = 8$, the base is $1 - d + \delta_2 - \delta_3 + \delta_4 + \delta_5$, and the loss is $d(t_5 + t_4) + \tau(t_3 + t_2 + t_1 + t_0) + \delta_3(t_1 + t_0)$. It expands to $6d - 9d^2 - 2d^3 + (4d + 3d^2 - 4)\delta_2 + \delta_2^2 + 2\delta_3 + 2d\delta_3$. Subtracting gives (Q86.9.3)–(Q86.9.4).

Now assume $1/5 \leq d \leq 1/3$. For $b = 5$, use $\delta_2 \leq \theta d$:

$$t_5 - (1-d)/2 \leq 1/2 - (43/16)d + (13/16)d^2 \leq -1/200. \tag{Q86.9.5}$$

The quadratic decreases on the whole interval, with left endpoint $-1/200$. For $b = 6$, (Q86.9.2) similarly gives

$$t_6 - (1-d)/2 \leq 1/2 - (105/32)d + (45/16)d^2 + d^3 \leq -143/4000. \tag{Q86.9.6}$$

Its derivative at the right endpoint is still at most $-103/96$, so it decreases throughout. For $b = 7$, discard two nonpositive terms and use $\delta_2 \leq \theta d$, obtaining the convex upper bound

$$t_7 - (1-d)/2 \leq 1/2 - (31/8)d + 5d^2 + 2d^3.$$

Its endpoints are $-59/1000, -35/216$. For $b = 8$, $\delta_4 + \delta_5 \leq (\theta + \theta^2)\delta_3 \leq (3 + 2d)\delta_3$, and $5 - 4d - 3d^2 > 0$. Thus

$$t_8 - (1-d)/2 \leq 1/2 - (143/32)d + (59/8)d^2 + (25/32)d^3.$$

This is also convex, with endpoint values $-37/400, -61/432$.

For every $b \geq 9$, all β indices corresponding to $t_0, \ldots, t_6$ in (Q86.9.1) are at least three. Retain all seven nonnegative terms. The finite alternating sums bound the base by $1 - \beta_{b+1} - \beta_{b+2} + \beta_{b+3} \leq 1 - \tau + \delta_3$, hence

$$t_b \leq 1 - \tau(1 + t_0 + \cdots + t_6) + \delta_3, \quad t_0 + \cdots + t_6 = 7 - 14d + d^3 + (5 + 4d)\delta_2 - \delta_4. \tag{Q86.9.7}$$

Let P be the quantity subtracted from one in (Q86.9.7). Use $\delta_4 \le \theta^2\delta_2, \delta_3 \le \theta\delta_2$, retaining the global $u = \delta_2/d \in [d, \theta]$:

$$P \ge d[(1-u)(A + Cud) - \theta u], \quad A = 8 - 14d + d^3, \quad C = 5 + 4d - \theta^2. \tag{Q86.9.8}$$

The bracket decreases with u, since its derivative is $-A + Cd(1-2u) - \theta < 0$, using $A \ge 10/3$, $Cd \le 19/9$. Thus the global endpoint $u = \theta$ gives

$$P - (1+d)/2 \ge -1/2 + (8\kappa - \theta^2 - 1/2)d - \kappa(14 - 5\theta + \theta^3)d^2. \tag{Q86.9.9}$$

Only nonnegative d^3, d^4 terms were discarded here. The coefficients satisfy $8\kappa - \theta^2 - 1/2 = 4183/1024 > 4$ and $\kappa(14 - 5\theta + \theta^3) = 7493391/1048576 < 29/4$. Therefore (Q86.9.9) is at least $-1/2 + 4d - 29d^2/4$. This is concave, with endpoint values $1/100, 1/36$, so it is at least $1/100$. Consequently $t_b \le (1-d)/2 - 1/100$ for every $b \ge 9$. Together with the $b = 5, 6, 7, 8$ estimates, this proves (Q86.0.4). Since the original kernel of T is at most two,

$$\boxed{0 \le TF_b \le 2t_b \le 1 - d \quad (b \ge 5, d \ge 1/5).} \tag{Q86.9.10}$$

For the bare-path cost, reserve the notation $\mathcal{V}_n$ for the finite sum

$$\mathcal{V}_n = \sum_{j=2}^{n} (-K)^j 1 = B_0 G_{n-1}(B_0) g'.$$

Retaining $\beta_n \bar{f}_0$ in (Q86.1.2) gives $0 \le \bar{f}_n \le q + \mathcal{V}_n - \beta_n$, and $\mathbb{E}\mathcal{V}_n = -\beta_{n+1}$. AM–GM for p and this nonnegative upper bound yields

$$p\bar{f}_n \le [1 + (\mathcal{V}_n - \beta_n)/2]^2, \qquad E[(p\bar{f}_n)^8] \le e^{-8\beta_n} E e^{8\mathcal{V}_n}. \tag{Q86.9.11}$$

The square base is nonnegative, so use $y \le e^{y-1}$ for $y \ge 0$. For $n \ge 4$, $\|G_{n-1}g'\|^2 \le d$. By (Q86.1.4) and $\beta_n, \beta_{n+1} \ge \kappa d$,

$$E[(p\bar{f}_n)^8] \le \exp[-8\beta_n - 8\beta_{n+1} + 17d] \le e^{(15/2)d}. \tag{Q86.9.12}$$

For $n = 3$, $\mathcal{V}_3 = r - s, \mathbb{E}\mathcal{V}_3 = -d, \beta_3 = d$. Its predecessor is $g' + B_0 g'$, with squared norm exactly $d + \delta_2 \le (1+\theta)d$; skew adjointness cancels the cross term. Therefore

$$E[(p\bar{f}_3)^8] \le \exp[(17(1+\theta) - 16)d] = e^{253d/32}. \tag{Q86.9.13}$$

Since $15/2 \le 253/32$, these prove (Q86.0.3). The $n = 3$ calculation uses the actual G_2 bound separately.

Q86.10 High domains and completion

For $b \ge 5, d \ge 1/5, L \ge 4$, the actual mean bound (Q86.9.10) and positivity give

$$0 \le X_{b,L} \le (1-d)\bar{f}_{L-1}.$$

Since $L - 1 \ge 3$, (Q86.0.3) yields

$$EH_{b,L}^8 \le (1-d)^8 e^{253d/32} \le e^{-3d/32} \le 1 - d/11. \tag{Q86.10.1}$$

The last comparison uses $e^{-3d/32} \le (1+3d/32)^{-1}$ and $d \le 1/3$. This joins (Q86.8.5) at $d = 1/5$.

For $b = 4, d \ge 1/4$, return to the actual t_4 in (Q86.9.2):

$$t_4 \le 1 - (2487/1024)d - d^2 \le 1353/4096 < 1/3, \quad TF_4 \le 2/3.$$

For every $L \ge 4$, it follows that

$$\log EH_{4,L}^8 \le 8\log(2/3) + (253/32)d \le -28/9 + 253/96 = -137/288 \le -d. \tag{Q86.10.2}$$

Here $\log(2/3) \le -7/18$ follows from $\log(1-x) \le -x - x^2/2$ at $x = 1/3$. Thus $\mathbb{E}H_{4,L}^8 \le e^{-d} \le 1 - 3d/4$, joining (Q86.7.4) at $d = 1/4$. Sections Q86.5–Q86.6 give every $b = 3, L \ge 3$ row on the full domain. This proves (Q86.0.2), including $d = 0$ and every stated integer boundary.

For decorations, positivity gives $0 \le E_r p T^L(E_t F_b) \le p T^L F_b$. Replace K by $-K$ to reverse the terminal, connector, receiving unit arm and all decorations together. Then

$$E\left[\{\overline{E}_r q A^L(\overline{E}_t\, \overline{h_2 f_b})\}^8\right] \le 1 - d/20000.$$

The overline denotes the full reversal of the corresponding original edges.

O.19 Unit-root eighth moments for terminal pairs $(3, b)$

Keep the finite model and notation of Section O.7, and write $f_n = A^n\mathbf{1}$, $h_n = Tf_{n-1}$, $\bar{f}_n = T^n\mathbf{1}$. Choose the terminal direction from the integer lengths before choosing the host:

$$\begin{array}{l|l} b \ge 4, L = 1 & f_3 h_b \\ b = 4, L = 2 & h_3 f_4 \\ b \ge 5, L = 2 & f_3 h_b \\ b \ge 4, L \ge 3 & h_3 f_b. \end{array} \tag{O.29}$$

An f_n arm points outward throughout. In an h_n arm only its first edge points toward the terminal. The connector points from the terminal to the receiving root, whose original unit arm supplies p. For actual equal-pair factors $0 \le E_t, E_r \le 1$, set

$$H_D = E_r p T^L(E_t F_{b,L}). \tag{Q89.0.2}$$

We prove

$$\mathbb{E}H_D^8 \le 1 - d/500 \qquad (b \ge 4, L \ge 1). \tag{Q89.0.3}$$

The $L = 1$ row is Section O.28. Positivity allows us to set $E_t = E_r = 1$ until the end. At $d = 0$, all tree messages are one by leaf integration.

Finite square propagation at the original unit root.

Put $K' = -K$, $g' = K'\mathbf{1}$, $G_n(z) = \sum_{j=0}^{n-1} z^j$, $G_0 = 0$. The finite convolution (M.2) gives

$$\tau \le \beta_j \le d \ (j \ge 3), \qquad \|G_n(K')\| \le 1 \ (n = 0, 1 \text{ or } n \ge 3), \quad \|G_2(K')\| \le \sqrt{1+\theta} < 6/5.$$

Suppose $0 \le F \le (m + R/2)^2$, where $m > 0$, and put

$$k = m^{-1}, \quad D_j = \mathbb{E}(K')^j R, \quad V = \mathbb{E}R^2, \quad a_1 = -D_1, \quad \rho = -D_2,$$
$$\delta = kD_0 + k^2 V/2, \quad a = ka_1, \quad \varepsilon = \delta - a.$$

Expanding the square, using $T(R^2) \le 2V$, and applying finite Duhamel gives

$$\frac{T^L F}{m^2} \le \bar f_L + k(K')^L R + \delta \bar f_{L-1} - a \bar f_{L-2} + k \sum_{j=2}^{L-1} D_j \bar f_{L-1-j} \quad (L \ge 2). \tag{O.30}$$

For $R = (A(x) - B(x)z)g'$, $z = i\sqrt{x}$, use the same spectral measure ν of g', supported on $[0, \theta]$ and of mass d. Thus

$$\begin{gathered} V = \int (A^2 + xB^2)\, d\nu, \quad D_0 = \int B\, d\nu, \quad a_1 = \int A\, d\nu, \quad \rho = \int xB\, d\nu, \\ D_{4j+1} = -\int x^{2j} A\, d\nu, \quad D_{4j+2} = -\int x^{2j+1} B\, d\nu, \\ D_{4j+3} = \int x^{2j+1} A\, d\nu, \quad D_{4j+4} = \int x^{2j+2} B\, d\nu. \end{gathered} \tag{O.31}$$

Let $S'_L = \sum_{j=0}^{L} (K')^j \mathbf{1}$, and define

$$\begin{gathered} P_n = G_n + kz^n(A - Bz), \quad Q_n = P_n + \varepsilon G_{n-2} + \delta z^{n-2} \quad (n \ge 2), \\ Z^*_L = S'_L + \delta S'_{L-1} - a S'_{L-2} + k(K')^L R = 1 + \varepsilon + Q_L(K')g'. \end{gathered}$$

The exact difference between Z^*_L and the right side of (O.30) is

$$\begin{gathered} \sum_{j=3}^{L} \gamma_j \bar f_{L-j}, \qquad \gamma_3 = d + k\rho, \quad \gamma_4 = d + \delta d - kD_3, \\ \gamma_j = \beta_j + \delta\beta_{j-1} - a\beta_{j-2} - kD_{j-1} \qquad (j \ge 5). \end{gathered} \tag{O.32}$$

Also $Q_L = 1 + \varepsilon + zQ_{L-1}$ for $L \ge 3$, and $\mathbb{E} Z^*_L = 1 + \varepsilon - \gamma_{L+1}$ for $L \ge 4$. These are finite identities, including the constant coordinate.

Whenever $N = K'W$, $0 \le H \le (c + N/2)^2$, and the square base is nonnegative, the signed estimate of Section M.1 and (M.49) give

$$\mathbb{E} H^8 \le \exp\{16(c-1) + 8\mathbb{E} N + 17\|W\|^2\}. \tag{O.33}$$

For a normalized square with prefactor m^{16} this exponent becomes $16 \log m + 8\eta + 8\mathbb{E} N + 17\|W\|^2$, when its center before rescaling is $1 + \eta/2$.

Two terminal packets, with all finite arm lengths.

For this paragraph assume $0 < d \le 1/5$. At $b = 4$, the exact identity $h_3 + f_4 = 2 - 3d + t - dg$ gives

$$\begin{gathered} h_3 + f_4 = 2m + R, \qquad m = 1 - 3d/2, \\ R = t - dg, \qquad A = d, \quad B = x. \end{gathered} \tag{O.34}$$

In particular

$$\begin{gathered} D_0 = \delta_2, \quad a_1 = d^2, \quad \rho = \delta_3, \quad V = d^3 + \delta_4, \\ U := \|KR\|^2 = d^2\delta_2 + \delta_5, \qquad W := \|K^2 R\|^2 = d^2\delta_3 + \delta_6. \end{gathered} \tag{O.35}$$

Moreover $0 \le \varepsilon \le \delta \le 7d/10$, as will also follow from the scalar certificates below.

For every $b \ge 5$, retain three terms of each positive convolution. At $b = 5$ take $c_1 = d$, $c_2 = d/4 + \delta_2$, $c_3 = 0$. For $b \ge 6$, take

$$\lambda = \frac{9d/4 - \beta_b - \beta_{b-1}}{\beta_{b-2} + (1-d)\beta_{b-3}}, \quad c_1 = \beta_{b-1}, \quad c_2 = \lambda\beta_{b-2}, \quad c_3 = \lambda\beta_{b-3}. \tag{O.36}$$

The denominator is at least $(2-d)\tau > 0$, the numerator is nonnegative, and the available constant loss is at least $(4-d)\tau \ge 361d/160 > 9d/4$; hence $0 \le \lambda \le 1$. For $b = 5$, $c_2 \le 21d/32 \le \beta_3$, without using the zero coefficient β_2. Put $C = c_1 + c_2 + c_3$, $D = c_2 + c_3$, $E = c_3$, and

$$R_{\text{old}} = \sum_{j=3}^{b} K^j\mathbf{1} - Cg - Dr - Es, \quad R = R_{\text{old}} - s, \qquad m = 1 - 13d/8.$$

The two positive convolutions yield the two actual arm sums

$$\begin{gathered} h_2 + f_b \le 2 - 9d/4 + R_{\text{old}}, \quad f_2 + h_b \le 2 - 9d/4 - R_{\text{old}}, \\ h_3 + f_b \le 2m + R, \qquad f_3 + h_b \le 2m - R. \end{gathered} \tag{O.37}$$

Both signs therefore give positive square envelopes. For the second sign retain the actual q, h_2, h_3 factors after applying T. Their constant loss in either direction is $\beta_b + c_1 + c_2 + (1-d)c_3 = 9d/4$.

Here is the complete finite coordinate description. At $b = 5$, $A = C - x^2$, $B = D + x$. At $b \ge 6$, put $N = b - 2 \ge 4$ and $G_N = a_f + zb_f$. Then

$$A = C + x(a_f - E - 1), \quad B = D + xb_f, \quad a_f, b_f \in [1-x, 1-x+x^2], \quad a_f \ge (1-x)b_f. \tag{O.38}$$

These finite geometric bounds include $N = 4, 5$. The retained constants satisfy

$$\begin{gathered} C = 9d/4 - \beta_b + dE, \quad D = 9d/4 - \beta_b - \beta_{b-1} + dE, \quad E \le d, \\ \tau \le c_1 \le d, \quad C \le d + D, \quad d/4 \le D \le 85d/64, \quad C \le C_0 := (53/32 + d)d. \end{gathered}$$

Consequently $-x^2 \le A \le C$, $D + (1-x)x \le B \le D + x$, and $A + x - (1-x)B \ge c_1 \ge \tau$. Using $D \le d/4 + 2\delta_2 + dE$ in the upper bound for B gives

$$B \le x - x^2 + x^3 + d/4 + 2\delta_2 + dE \le z_0 := \theta + 13d/16.$$

Indeed the difference after setting $x = \theta$, $\delta_2 \le \theta d$, $E \le d$ is at most $-3/32 + d/4 + d^2 \le -3/800$; the $b = 5$ bound is smaller. Thus, with

$$v_0 = C_0^2 + \theta^4 + \theta z_0^2, \quad s_0 = kz_0 + \tfrac{1}{2}k^2 v_0, \quad e_0 = k[\theta(1+z_0) - \kappa d] + \tfrac{1}{2}k^2 v_0, \tag{O.39}$$

we have $V \le v_0 d$, $\delta \le s_0 d$, and $\varepsilon \le e_0 d$. The last bound uses the same spectral measure and $D_0 - a_1 \le [\theta(1+z_0) - \kappa d]d$. For the opposite bound, $a_f - b_f$ is zero or an alternating end term; its first positive possibility is x^2. Hence

$$a_1 - D_0 \le d^2 - \delta_2 + \delta_4 \le \theta^3 d, \qquad \varepsilon \ge -k\theta^3 d > -d/10.$$

At $b = 5$, the left side is exactly $d^2 - \delta_2 - \delta_3$.

One additional pointwise estimate is needed for the exchanged two-step direction:

$$A^2 + xB^2 \le (5/4)B < 2mB. \tag{O.40}$$

Indeed its ratio to B is at most $(C^2 + x^4)/(D + \kappa x) + x(D + x)$, a convex function of x. At $x = 0$ it is at most $d(1 + D/d)^2/(D/d) \le 5/4$. At $x = \theta$, replace C by $d + D$; the resulting function is convex in D, and then in d on each endpoint $D/d \in \{1/4, 85/64\}$. The corner bounds are $169/608$, $8/15$, and $13/16$, all below $5/4$. Also $2m \ge 27/20 > 5/4$.

A single continuous payment for each short connector.

For the canonical square, direct AM–GM gives the following two complete vector formulas:

$$\begin{aligned} V_2 &= \delta\mathbf{1} + g' + kK'R, \qquad \mathbb{E}(K'V_2) = -d - k\rho, \\ \|V_2\|^2 &= d + 2k\rho + k^2U + \delta^2 - 2\delta a, \\ V_3 &= \varepsilon\mathbf{1} + (1+\delta)g' + K'g' + k(K')^2R, \\ \mathbb{E}(K'V_3) &= -(1+\delta)d + kD_3, \\ \|V_3\|^2 &= (1+\delta)^2 d + \delta_2 - 2k(1+\delta)D_3 \\ &\quad + 2kD_4 + k^2W + \varepsilon^2 - 2\varepsilon(d+k\rho). \end{aligned} \tag{O.41}$$

For $L = 3$ the one retained path loss is $\gamma_3 = d + k\rho$. The square bases are halves of sums of nonnegative factors. Applying (O.33) yields

$$\begin{aligned} \log \mathbb{E}(pT^2F)^8 &\le 16\log m + 9d + 8\varepsilon + 26k\rho + 17k^2U + 17(\delta^2 - 2\delta a), \\ \log \mathbb{E}(pT^3F)^8 &\le 16\log m + (253/32)d + 26\delta d + 17\delta^2 d \\ &\quad - (26 + 34\delta)kD_3 + 34kD_4 - 8k\rho + 17k^2W + \mathcal{E}, \\ \mathcal{E} &= 8\varepsilon + 17\varepsilon^2 - 34\varepsilon(d + k\rho). \end{aligned} \tag{O.42}$$

In particular the full constant-vector cross terms remain in both rows.

Here are explicit rational functions which pay these bounds on the single interval $0 \le d \le 1/5$. Write $\mathcal{L}_q = -16\sum_{j=1}^{4}(qd)^j/j \ge 16\log(1-qd)$. For $b = 4$, let $m_4 = 1 - 3d/2$, $k_4 = 1/m_4$, and $s_4 = k_4\theta + \frac{1}{2}k_4^2(d^2 + \theta^3)$. Define

$$\begin{aligned} F_{42} &= \mathcal{L}_{3/2} + 9d + 8s_4 d - 8k_4 d^2 \\ &\quad + 26k_4\theta^2 d + 17k_4^2(d^2\theta + \theta^4)d + 17s_4^2 d^2 - 34s_4 k_4 d^3, \\ F_{43} &= \mathcal{L}_{3/2} + (253/32)d + (91/5)d^2 + (833/100)d^3 \\ &\quad + 34k_4\theta^3 d + 17k_4^2(d^2\theta^2 + \theta^5)d + (28/5)d - (1547/100)d^2. \end{aligned}$$

The two bounds are $F_{42} \le -d$, $F_{43} \le -6d$. For the first, the dependence on δ is increasing and $\delta \le s_4 d$; for the second use $0 \le \varepsilon \le \delta \le 7d/10$, $D_3 \ge 0$, and the data (O.35).

For $b \ge 5$, use $m = 1 - 13d/8$, $k = 1/m$, and the joint functions (O.39). Put

$$l = k\theta^3, \quad j_0 = 1 + k\theta z_0, \quad E_+ = 8e_0 d + (17e_0^2 - 34e_0)d^2, \quad E_- = -8ld + (17l^2 + 34j_0 l)d^2,$$

$$\begin{aligned} F_{\ge 5,3} &= \mathcal{L}_{13/8} + (253/32)d + 26s_0 d^2 + 17s_0^2 d^3 \\ &\quad + (26 + 34s_0 d)k\theta^3 d + 34k\theta^2 z_0 d + 17k^2\theta^2 v_0 d + E_+. \end{aligned}$$

Convexity in ε, retaining its sign, gives $\mathcal{E} \le \max(0, E_+, E_-) = E_+$. Indeed $-ld \le \varepsilon \le e_0 d$, $d \le d + k\rho \le j_0 d$, and the certificates below show $E_+ \ge 0, E_-$. The remaining substitutions are $D_3 \ge -\theta^3 d$, $D_4 \le \theta^2 z_0 d$, $W \le \theta^2 v_0 d$. They prove the whole-domain bound $F_{\ge 5,3} \le -d/50$.

For the actual exchanged $L = 2$ packet replace R by $-R$. Equation (O.40) gives $\delta_- = -kD_0 + k^2V/2 \le 0$ and $\|g' - kK'R\|^2 = d - 2k\rho + k^2U \le d$, since $U \le (5/4)\rho$, $(5/4)k < 2$. The actual mean is $-d + k\rho$, and

$$|\delta_-| \le 6d/7, \quad |a_-| \le 4d/7, \quad \varepsilon_- \le d/2, \quad k\rho \le 7d/20, \quad \delta_-^2 - 2\delta_- a_- \le 12d^2/7.$$

Thus its exponent is at most $[-26+9+4+14/5+204/35]d=-153d/35<-4d$.

The continuous certificates just used have only 78 rational Bernstein coefficients in total. Each polynomial below has all its coefficients greater than $1/200$ on $[0,1/5]$. The positive factors in the second column clear all denominators. At $d=0$, quotients by d mean their polynomial continuations. These are direct applications of (A.18); the exact coefficients and inverse expansions accompany the

Table 58: Continuous short-connector payments and their moment guards.

Function	Positive factor	Degree
$(-F_{\geq 5,3}-d/50)/d$	m^4	10
$6/5-s_0,\ 9/8-e_0,\ e_0$	m^2	4 each
$E_+/d,\ (E_+-E_-)/d$	m^4	9 each
$81/256-v_0$	1	4
$4/7-kC_0$	m	2
$7/20-k\theta z_0$	m	1
$1/2-k\theta^3-k^2v_0/2$	m^2	4
$(-F_{42}-d)/d$	m_4^4	7
$(-F_{43}-6d)/d$	m_4^2	5
$7/10-s_4$	m_4^2	2

manuscript. Also $k\theta^2<4/7$ controls the negative part of a_1, and $kz_0<6/7$ controls δ_-. The table therefore proves all the moment guards used in both directions, including the endpoints.

Every low-domain connector of length at least four.

For both terminal packets, $\|P_n\|\leq 1$ for all $n\geq 3$. Here is a finite-index proof. For $b\geq 5, n=3$, the real and normalized imaginary parts are $1-x-kx^2B$, $1-kxA$. They are nonnegative and bounded above by $1-x$, $1+kx^3$. Hence

$$|P_3|^2\leq 1-x[1-x-2kx^3-k^2x^6]\leq 1;$$

the bracket is at least $23/256$ on $x\leq 1/2, k\leq 3/2$. For $b=4$, those two components are $1-x-kx^3$, $1-kdx$, between zero and the components of G_3. In both cases $k\sqrt{A^2+xB^2}\leq 1$. At $n=4$, use $|P_4|\leq(1-x)\sqrt{1+x}+x^2\leq 1$. At $n=5$, use $|P_5|\leq 1-x/2+(23/20)x^2\leq 1$. For $n\geq 6$, the finite geometric formula gives $|P_n|\leq(1+x^3)/\sqrt{1+x}+x^3\leq 1$, since $x(1+\sqrt{1+x})\leq 143/160<1$.

The same data imply the following complete loss and predecessor table:

	h	$\gamma_j\ (j\geq 3)$	$\|Q_n\|\ (n\geq 3)$	ε/d
$b=4$	$2/5$	$\geq hd$	$\leq 1+6d/5$	$[0,7/10]$
$b\geq 5$	$1/3$	$\geq hd$	$\leq 1+2d$	$[-1/10,9/8]$

(O.43)

Indeed $\gamma_3\geq d$, $\gamma_4>3d/4$. For $j\geq 5$, (O.32) gives respectively

$$\gamma_j/d\geq\kappa-kd^2-k\theta^3\geq 2/5,\qquad \gamma_j/d\geq\kappa-(4/7)d-k\theta^2z_0\geq 1/3.$$

For the latter bound $(D_j)_+\leq\theta^2z_0d$ for every $j\geq 4$; the four residue formulas in (O.31) prove this also when $A<0$. For Q_n, use its defining sum and separate $n=3,4$ from $n\geq 5$, so the sole G_2 factor is charged $6/5$. The largest extra coefficients are respectively $231/200, 1799/1600, 63/64$ for $b=4$, and $381/200, 147/80, 129/80$ for $b\geq 5$.

Retain the last two losses in (O.32). With $B_*=1+\varepsilon-hd$, the finite recursion gives

$$0\leq T^LF/m^2\leq Z=B_*q-hd+N_0,\quad N_0=K'Q_{L-1}(K')g',\quad \mathbb{E}N_0\leq -hd.$$

Weighted AM–GM at the genuine unit root gives

$$pT^{L}F \le (c_{*}+N/2)^{2}, \quad c_{*}=m\sqrt{B_{*}}-\frac{mhd}{2\sqrt{B_{*}}}, \quad N=\frac{m}{\sqrt{B_{*}}}N_{0}.$$

The square base is half a sum of nonnegative factors. The following data concern this same complete vector, including its mean:

	B_{*}	$m/\sqrt{B_{*}}$	$\|\frac{m}{\sqrt{B_{*}}}Q_{L-1}g'\|^{2}$	$\mathbb{E}N$	c_{*}
$b=4$	$[1-2d/5, 1+3d/10]$	$\ge 2/3$	$\le d$	$\le -4d/15$	$\le 1-5d/4$
$b\ge 5$	$[1-13d/30, 1+19d/24]$	$\ge 5/8$	$\le 9d/8$	$\le -5d/24$	$\le 1-4d/3$

(O.44)

For the norm column, use respectively

$$m^{2}(1+6d/5)^{2} \le 1-3d/5 \le B_{*}; \quad m^{2}(1+2d)^{2} \le 1+3d/4-(407/64)d^{2} \le (9/8)(1-13d/30).$$

The latter difference is bounded below by $1/8-(99/80)d+(407/64)d^{2} > 0$. The other columns follow from these B_{*}-intervals and $\sqrt{1+x} \le 1+x/2$. Substitution in (O.33) proves the rates $77/15$ and $31/8$ in the final table below.

High-domain terminal means.

Assume $1/5 \le d \le 1/3$. The actual canonical mean is $s_{b}=\mathbb{E}(h_{3}f_{b})=\mathbb{E}(f_{2}f_{b+1})$. Let $u_{j}=\mathbb{E}K^{j}\mathbf{1}$. The finite convolution gives

$$s_{b}=1-(2-d)\beta_{b+1}+u_{b+1}+u_{b+3}-\sum_{j=3}^{b}\beta_{j}s_{b-j}.$$

Direct expansion yields

$$s_{0}=1,\ s_{1}=1-d+\delta_{2},\ s_{2}=(1-d)^{2}, \quad s_{3}=1-3d+d^{2}+\delta_{2}-\delta_{3},$$
$$s_{4}=1-4d+2d^{2}+2(1-d)\delta_{2},$$
$$s_{5}=1-5d+4d^{2}-d^{3}+(3-2d)\delta_{2}-\delta_{3}+\delta_{4},$$
$$s_{6}=1-6d+7d^{2}-2d^{3}+4(1-d)\delta_{2}+\delta_{2}^{2}-2(1-d)\delta_{3}.$$

Consequently

$$s_{4} \le M_{4} := 1-(51/16)d+(19/16)d^{2}, \quad 2M_{4} \le 1-9d/10, \quad s_{b} \le (1-d)/2 \quad (b\ge 5). \tag{O.45}$$

For the last bound at $b=5$, use $s_{5}-(1-d)/2 \le 1/2-(105/32)d+(51/16)d^{2}-d^{3} \le -147/4000$. At $b=6$, Cauchy–Schwarz in the spectral measure, together with $\delta_{2} \ge d^{2}$, gives $\delta_{3} \ge d\delta_{2}$, and hence $s_{6}-(1-d)/2 \le 1/2-(31/8)d+(19/4)d^{2} \le -17/200$. For every $b \ge 7$, the base in the finite convolution is at most $1-(2-d)\tau+\delta_{4}$. Retaining its five actual nonnegative terms $s_{0}, \dots, s_{4}$ gives

$$s_{b} \le 1-\tau\{7-11d+4d^{2}+(4-2d)\delta_{2}-\delta_{3}\}+\delta_{4}.$$

Keep the global ratio $w=\delta_{2}/d \in [d,\theta]$. The subtracted quantity is at least $d\{(1-w)[7-11d+4d^{2}+(4-2d-\theta)wd]-\theta^{2}w\}$. Its bracket decreases in w, since $7-11d+4d^{2} \ge 34/9$ and $(4-2d-\theta)d \le 4/3$. At $w=\theta$ the full subtracted quantity exceeds $(1+d)/2+7/1500$: after retaining the cubic contribution

with $d \ge 1/5$, it is enough to check $-1/2 + (43/12)d - (53/10)d^2 \ge 7/1500$, a concave quadratic with endpoint values $7/1500, 19/180$. This proves all finite b, without monotonicity of path means.

For the exchanged direction, adjunction instead gives

$$\tilde{s}_b = \mathbb{E}(f_3 h_b) = \mathbb{E}(f_4 f_{b-1}), \quad \mathbb{E}f_4^2 \le U_4 := 1 - (147/32)d + (83/16)d^2 + d^3.$$

Here the exact square is $1 - 5d + 6d^2 + d^3 + (1-2d)\delta_2 - \delta_3 + \delta_4$; use $\delta_2 \le \theta d$, $\delta_4 \le \delta_3$. The polynomial U_4 decreases on this domain, with maximum $U_4(1/5) = 1187/4000$. The actual finite path-norm bounds of Section M.15 give, for every $n \ge 0$, $\|A^n\| \le 1 + 32d/19$; they include $n = 0, 1, 2$ separately. Thus

$$\tilde{s}_b \le (1 + 32d/19)U_4 \le 105643/228000 < 15/32 \quad (b \ge 5). \tag{O.46}$$

Propagation, strong rates, and restoration.

Corollary P.2 and the independent bare-path estimate (Q86.0.3) give

$$\mathbb{E}(p\bar{f}_2)^8 \le \begin{cases} 35/8, & 1/5 \le d \le 1/4, \\ 82/15, & 1/4 \le d \le 1/3, \end{cases} \qquad \mathbb{E}(p\bar{f}_n)^8 \le e^{253d/32} \quad (n \ge 3).$$

Since the kernel of T is at most two, $pT^L F \le 2(\mathbb{E}F)p\bar{f}_{L-1}$. Together with (O.45) and (O.46), this yields the following full set of rates. Entries without an expectation are pointwise bounds on $H = pT^L F$. For $L = 3, b \ge 5$, the high-domain costs are $(4/5)^8(35/8) < 3/4$ and

Table 59: Actual directions and complete eighth-moment payments.

Actual parameters and terminal	$0 < d \le 1/5$	$1/5 \le d \le 1/3$
$b = 4, L = 2, h_3 f_4$	$\mathbb{E}H^8 \le e^{-d}$	$H \le 1 - 9d/10$
$b \ge 5, L = 2, f_3 h_b$	$\mathbb{E}H^8 \le e^{-4d}$	$H \le 15/16$
$b = 4, L = 3, h_3 f_4$	$\mathbb{E}H^8 \le e^{-6d}$	$\mathbb{E}H^8 \le 15/16$
$b \ge 5, L = 3, h_3 f_b$	$\mathbb{E}H^8 \le e^{-d/50}$	$\mathbb{E}H^8 \le 3/4$
$b = 4, L \ge 4, h_3 f_4$	$\mathbb{E}H^8 \le e^{-77d/15}$	$\mathbb{E}H^8 \le e^{-d/100}$
$b \ge 5, L \ge 4, h_3 f_b$	$\mathbb{E}H^8 \le e^{-31d/8}$	$\mathbb{E}H^8 \le e^{-3d/32}$

$(3/4)^8(82/15) < 3/4$. For $b = 4$, use $2M_4(1/5) = 41/50$, $2M_4(1/4) = 71/128 < 3/5$, giving respectively $(41/50)^8(35/8) < 15/16$ and $(3/5)^8(82/15) < 3/4$. For $L \ge 4, b \ge 5$, use $8\log(1-d) + (253/32)d \le -3d/32$. For $b = 4$, retaining three logarithm terms gives

$$\frac{8\log(1 - 9d/10)}{d} + 253/32 \le 113/160 - (81/25)d - (243/125)d^2 \le -1951/100000 < -1/100.$$

These inequalities include both closed intervals at $d = 1/4$. Finally $e^{-d/100} \le 1 - 3d/301 \le 1 - d/500$, and the pointwise and constant bounds are stronger still. This proves (Q89.0.3). Positivity restores both actual equal-pair factors. Replacing K by $-K$ gives the full reversal, with message $\overline{E}_r q A^L(\overline{E}_t\, \overline{F}_{b,L})$.

O.20 Ordinary eighth moments for even terminals of minimum one with multiple pairs

Q92.0 The terminal and the ordinary moment.

Work on the finite probability space of Appendix M.15, with $K(x,y) = -K(y,x)$, $|K(x,y)| \le 1$, and

$$\begin{gathered} Jv = (\mathbb{E}v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \\ f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad \delta_i = \|K^i\mathbf{1}\|_2^2, \quad d = \delta_1, \\ \theta = 13/32, \qquad \kappa = 19/32, \qquad \tau = d - \delta_2. \end{gathered} \tag{Q92.0.1}$$

After maximal equal pairing, suppose the distinct residual arms satisfy

$$1 = a_1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j, \qquad j \ge 2. \tag{Q92.0.2}$$

Every integer in this list is finite and arbitrary. Set

$$F_\Sigma = qf_{b_1}\prod_{i=2}^{j} h_{a_i}f_{b_i}, \qquad H_D = E_r pT^L(E_t F_\Sigma), \quad L \ge 1. \tag{Q92.0.3}$$

Here E_t, E_r are products of genuine equal pairs, each in $[0,1]$, and the receiving root has an outward unit arm. Each f_n-arm points toward its leaf. Each h_n-arm has its first edge toward its terminal and all later edges toward the leaf; in particular, $q = h_1$ points toward the terminal. The connector points from terminal to root. These directions are chosen from the arm lengths and hold throughout the parameter domains below. We prove

$$\boxed{\mathbb{E}H_D^8 \le e^{-[1/4+16(j-2)]d} \le e^{-d/4} \le 1 - d/5.} \tag{Q92.0.4}$$

The same assertion holds with every edge reversed.

Q92.1.1 Finite identities and the zero boundary.

The finite calculus in Appendix M.15 gives

$$\begin{aligned} &0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta\delta_i, \quad |Kv(x)| \le \|v\|_2, \\ &\mathbb{E}K^{2i}\mathbf{1} = (-1)^i\delta_i, \quad \mathbb{E}K^{2i+1}\mathbf{1} = 0, \quad 0 \le f_n, \bar{f}_n, h_n \le 2, \quad \mathbb{E}f_n = \mathbb{E}\bar{f}_n \le 1. \end{aligned} \tag{Q92.1.1}$$

All expectations, adjoints and row estimates use the fixed measure. With $\beta_1 = \beta_2 = 0$, finite positive convolution gives

$$\begin{gathered} \beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i \quad (n \ge 3), \\ S_n = \sum_{i=0}^{n} K^i\mathbf{1} = f_n + \sum_{i=3}^{n} \beta_i f_{n-i}, \\ S'_n = \sum_{i=0}^{n} K'^i\mathbf{1} = \bar{f}_n + \sum_{i=3}^{n} \beta_i \bar{f}_{n-i}, \qquad K' = -K, \quad g' = -g, \\ \tau \le \beta_n \le d \quad (n \ge 3), \qquad 0 \le \tau \le d - d^2. \end{gathered} \tag{Q92.1.2}$$

The index two has $\beta_2 = 0$. For $G_n(z) = \sum_{i=0}^{n-1} z^i$,

$$\|G_n(K)\| \le 1 \quad (n = 0, 1 \text{ or } n \ge 3), \qquad \|G_2(K)\|^2 \le 1 + \theta. \tag{Q92.1.3}$$

Genuine equal pairs satisfy

$$f_\ell + h_\ell = 2\mathbb{E} f_{\ell-1}, \qquad 0 \le f_\ell h_\ell \le 1. \tag{Q92.1.4}$$

After zero-mass atoms are removed, $p(x), q(x) \ge \mu(x) > 0$. When $d = 0$, $g = 0$, and integration from the leaves gives one for every tree message, including when $K \neq 0$. Hence divisions below are made for $d > 0$. Kernel entries may attain ± 1.

Q92.1.2 Finite masks and three coefficient patterns.

Write

$$R_0 = \sum_{i=1}^{j} \sum_{n=a_i+1}^{b_i} K^n \mathbf{1} = KP(K)g. \tag{Q92.1.5}$$

The disjoint intervals make P a finite polynomial with coefficients zero or one and $P_0 = 1$. At $z = i\sqrt{x}$, $0 \le x \le \theta$, put $P(-z) = U(x) - zV(x)$. The residue-class estimate (L90.2.6) gives

$$|P(z)|^2 \le \frac{1+x}{(1-x^2)^2} \le \frac{32768}{16245} < \frac{101}{50} < \frac{100}{49}. \tag{Q92.1.6}$$

Thus $\|P(K)\| < 10/7$, and

$$\frac{x}{1-x^2} \le \frac{416}{855} < \frac{1}{2}, \quad \frac{x^2}{1-x^2} < \frac{1}{5}, \quad \frac{x^3}{1-x^2} < \frac{1}{12}, \quad \frac{x^4}{1-x^2} < \frac{1}{30}. \tag{Q92.1.7}$$

The last two comparisons use $2197/27360 < 1/12$ and $\theta^4/(1-\theta^2) = 28561/875520 < 1/30$. Also $\theta^2 < 1/6$, $\theta^3 < 1/14$, $\theta^4 < 1/32$, and $\sqrt{\theta} < 2/3$. Every mask has $U \ge 1 - x/(1-x^2) > 1/2$. The three patterns are

class	arm conditions	forced coefficients	$\lvert P\rvert^2$ bound
I	$b_1 \ge 3$	$P_0 = P_1 = 1$	$100/49$
II	$b_1 = 2,\ a_2 \ge 4$	$P_0 = 1,\ P_1 = P_2 = 0$	$25/16$
III	$b_1 = 2,\ a_2 = 3$	$P_0 = P_2 = 1,\ P_1 = 0$	1

For II the first possible real and imaginary terms give

$$|P(z)|^2 \le \frac{1+x^3}{(1-x^2)^2} < \frac{54}{35} < \frac{25}{16}. \tag{Q92.1.8}$$

For III,

$$\begin{gathered} 1 - \frac{x}{1-x^2} \le U \le 1 - x + \frac{x^2}{1-x^2} \le 1 - x/3, \\ -\frac{x}{1-x^2} \le V \le \frac{x^2}{1-x^2}. \end{gathered} \tag{Q92.1.9}$$

The real coordinate is positive, and $x \le 1/2$ gives

$$|P(z)|^2 \le (1-x/3)^2 + \frac{16}{9}x^3 \le (1-x/3)^2 + \frac{4}{9}x \le 1 - x/6 \le 1. \tag{Q92.1.10}$$

Q92.1.3 Moment inputs.

The unit-connector estimate of Q5-C061, in these same directions, is

$$\mathbb{E}[(pTF_{\Sigma})^{8}] \le e^{-[16(j-2)+63/50]d}. \tag{Q92.1.11}$$

The single-pair inputs (Q58O.0.2) and (L58.16.1) give, for $b \ge 2, L \ge 1$ and a genuine equal-pair factor E,

$$\mathbb{E}[(pT^{L}(Eqf_{b}))^{8}] \le 1 - d/12000, \qquad \mathbb{E}[f_{2}\{T^{L}(Eqf_{b})\}^{4}] \le 1. \tag{Q92.1.12}$$

The full-mean inequality of Appendix M.1 is

$$\mathbb{E}e^{8K'W} \le \exp\{8\mathbb{E}(K'W) + 17\|W\|_{2}^{2}\} \tag{Q92.1.13}$$

for every real W, with its signed mean.

Q92.2.1 Stronger losses for pairs of minimum four.

Let $R_{ab} = \sum_{n=a+1}^{b} K^{n}\mathbf{1}$. The all-index mean estimate in Appendix M.17 is

$$\mathbb{E}f_{n} \le 1 - \frac{83}{32}d \qquad (n \ge 4). \tag{Q92.2.1}$$

Its proof treats $n = 4, 5, 6, 7$ separately. For each $n \ge 8$, the finite convolution retains $f_{0}, \dots, f_{5}$ and three additional losses to give

$$1 - \mathbb{E}f_{n} \ge \tau(7 - 10d + 3\delta_{2} + d^{2}) + 3\kappa\delta_{3} \ge \kappa d(7 - 10d + 7d^{2}) \ge 95d/36. \tag{Q92.2.2}$$

Here $\delta_{3} \ge d^{3}$. The loss divided by d for the first four lengths is at least $83/32, 137/48, 89/32, 127/48$, respectively. The finite difference formula is

$$f_{b} - f_{a} = R_{ab} + \sum_{i=0}^{a-3}(\beta_{a-i} - \beta_{b-i})f_{i} - \sum_{i=a-2}^{b-3}\beta_{b-i}f_{i}. \tag{Q92.2.3}$$

The positive part of the middle sum is at most $4\delta_{2}/(1-\theta^{2})$. For $a \ge 5$, this yields

$$\begin{aligned} h_{a} + f_{b} &= 2\mathbb{E}f_{a-1} + f_{b} - f_{a} \\ &\le 2 + R_{ab} - \left(\frac{83}{16} - \frac{1664}{855}\right)d \le 2 + R_{ab} - 16d/5. \end{aligned} \tag{Q92.2.4}$$

The final positive margin is $2825/68400$. At $a = 4$, retain the negative term in the convolution:

$$f_{b} - f_{4} \le R_{4b} + \delta_{2}(1+p) - \tau f_{2}. \tag{Q92.2.5}$$

The row estimate $|Kg| \le (1-g^{2})/2$ gives $f_{2} \ge p^{2}/2$. Completing the square in $p \ge 0$, using $\delta_{2} \le d/2, \tau \ge d/2$, gives

$$\delta_{2}(1+p) - \tau f_{2} \le \delta_{2} + \delta_{2}^{2}/(2\tau) \le 3d/4. \tag{Q92.2.6}$$

Since $\mathbb{E}f_{3} = 1 - 2d$, $h_{4} + f_{b} \le 2 + R_{4b} - 13d/4$. AM–GM on the nonnegative factors therefore gives, including $(a,b) = (4,5)$,

$$\boxed{h_{a}f_{b} \le e^{R_{ab} - 16d/5} \qquad (4 \le a < b).} \tag{Q92.2.7}$$

Q92.2.2 The retained negative term for minimum three.

For every $b \geq 4$, (Q92.2.3) gives

$$f_b - f_3 = R_{3b} + (d - \beta_b) - \sum_{i=1}^{b-3} \beta_{b-i} f_i. \tag{Q92.2.8}$$

The term $i = 1$ is present, $\beta_{b-1} \geq \tau$, and $d - \beta_b \leq \delta_2$. Thus

$$h_3 + f_b \leq 2 - 2d + R_{3b} + \delta_2 - \tau p = 2 - 3d + 2\delta_2 + R_{3b} - \tau g, \tag{Q92.2.9}$$

and

$$\boxed{h_3 f_b \leq e^{R_{3b} - \tau g - (3d - 2\delta_2)}.} \tag{Q92.2.10}$$

The term $-\tau p$ comes from the displayed finite convolution. We also use the coarser estimate

$$h_3 f_b \leq e^{R_{3b} - (2d - \delta_2)}. \tag{Q92.2.11}$$

Q92.2.3 The three exponential envelopes.

Positive convolution gives $q + f_{b_1} \leq 2 - \beta_{b_1} + R_{1b_1}$, so

$$q f_{b_1} \leq e^{R_{1b_1} - \beta_{b_1}}. \tag{Q92.2.12}$$

Apply (Q92.2.7) to all remaining pairs except the second pair of III, where (Q92.2.10) is used. Then

$$F_\Sigma \leq e^{R - \Gamma}, \tag{Q92.2.13}$$

with the following functions and losses:

class	R	Γ
I	R_0	$\beta_{b_1} + 16(j-1)d/5$
II	R_0	$16(j-1)d/5$
III	$R_0 - \tau g$	$3d - 2\delta_2 + 16(j-2)d/5$

Their losses are at least $607d/160 + 16(j-2)d/5$, $16d/5 + 16(j-2)d/5$, and $35d/16 + 16(j-2)d/5$, respectively.

Q92.3 Finite propagation with a signed mean.

For the function R in (Q92.2.13), put

$$e^R \leq 1 + R + (\eta/2)R^2, \quad V_R = \mathbb{E}R^2, \quad D_i = \mathbb{E}K'^i R, \quad \delta = D_0 + \eta V_R. \tag{Q92.3.1}$$

One may take $\eta = 15/14$ if $|R| \leq 1/5$, and $\eta = 6/5$ if $|R| \leq 1/2$. Indeed, positive Taylor terms starting at degree two have successive ratio at most $|R|/3$, giving remainder $R^2/[2(1 - |R|/3)]$; for negative R, the integral remainder is at most $R^2/2$. Positivity of T and $T(R^2) \leq 2V_R$ imply

$$\begin{gathered} 0 \leq T^L F_\Sigma \leq e^{-\Gamma} Z_L, \\ Z_L = \bar{f}_L + K'^L R + \delta \bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \geq 0. \end{gathered} \tag{Q92.3.2}$$

Here the exact finite identity is

$$T^L R = K'^L R + \sum_{i=0}^{L-1} D_i \bar{f}_{L-1-i}. \tag{Q92.3.3}$$

For $L \geq 2$, set $\gamma_\ell = \beta_\ell - D_{\ell-1}$ for $\ell \geq 3$. Then

$$Z_L = S'_L + K'^L R + \delta \bar{f}_{L-1} + D_1 \bar{f}_{L-2} - \sum_{\ell=3}^{L} \gamma_\ell \bar{f}_{L-\ell}. \tag{Q92.3.4}$$

Empty sums contribute zero. Write $R = R_0 + \xi g'$, with $\xi = 0$ for I and II and $\xi = \tau$ for III, and define

$$Q_n(z) = G_n(z) + z^{n+1} P(-z) + \xi z^n. \tag{Q92.3.5}$$

The finite skew rotation blocks apply the frequency bounds to g'. The quantities δ_i, τ remain global moments of the fixed model. We have

$$\begin{gathered} p + S'_L + K'^L R = 2 + N_L, \quad N_L = K' Q_{L-1}(K') g', \\ \mathbb{E} N_L = -\beta_{L+1} + D_L. \end{gathered} \tag{Q92.3.6}$$

If all $\gamma_\ell \geq 0$, the bound $0 \leq \bar{f}_n \leq 2$ gives

$$c_2 = 2\delta_+ + (D_1)_+. \tag{Q92.3.7}$$

For $L \geq 3$, retain $\bar{f}_0 = 1$ when $\gamma_L \geq h_L d$ and take

$$c_L = 2\delta_+ + 2(D_1)_+ - h_L d. \tag{Q92.3.8}$$

These choices give $p + Z_L \leq 2 + c_L + N_L$, a nonnegative upper bound. AM–GM and (Q92.1.13) give

$$\boxed{\mathbb{E}[(pT^L F_\Sigma)^8] \leq e^{-8\Gamma + 8c_L + 8\mathbb{E}N_L + 17\|Q_{L-1}(K')g'\|_2^2}.} \tag{Q92.3.9}$$

Both c_L and $\mathbb{E}N_L$ retain their signs.

Q92.4.1 Classes I and II for $0 < d \leq 1/64$.

In both classes, $|R_0|^2 \leq 100d/49 < 1/25$, so $\eta = 15/14$. For I the mandatory terms $K^2\mathbf{1}, K^3\mathbf{1}$ give

$$\mathbb{E}R_0 \leq -d + \frac{\delta_2}{1-\theta^2} \leq -\frac{439}{855} d, \qquad D_1 = -\mathbb{E}KR_0 \leq -\delta_2 + \frac{\delta_3}{1-\theta^2} \leq 0. \tag{Q92.4.1}$$

Since $V_R \leq 325d/392$,

$$\delta \leq \left(-\frac{439}{855} + \frac{4875}{5488} \right) d < 3d/8. \tag{Q92.4.2}$$

The last comparison is $2817 \cdot 855 = 2408535 < 439 \cdot 5488 = 2409232$. For II, $K^3\mathbf{1}, K^4\mathbf{1}$ are absent, and the first possible positive mean term is $K^8\mathbf{1}$. Consequently

$$\begin{gathered} \mathbb{E}R_0 \leq -d + \frac{\delta_4}{1-\theta^2} \leq -11d/12, \qquad V_R \leq \frac{325}{512} d, \\ \delta \leq \left(-11/12 + \frac{4875}{7168} \right) d < 0, \qquad D_1 \leq \frac{\delta_3}{1-\theta^2} < d/5. \end{gathered} \tag{Q92.4.3}$$

For I, $D_2 \le \delta_2/(1-\theta^2) < d/2$. For II, the missing $K^4\mathbf{1}$ gives

$$D_2 \le \delta_2 + \frac{\delta_4}{1-\theta^2} < (\theta + 1/12)d < d/2. \tag{Q92.4.4}$$

In both classes $D_3 \le \delta_3/(1-\theta^2) < d/5$. For every $i \ge 4$, positive mean terms of $K'^i R_0$ lie in one residue class modulo four, with first possible moment index at least three:

$$D_i \le \frac{\delta_3}{1-\theta^2} < d/5 \qquad (i \ge 4). \tag{Q92.4.5}$$

Thus

$$\begin{aligned} &\gamma_3 \ge d/2, \quad \gamma_4 \ge 4d/5, \quad \gamma_\ell \ge (\kappa - 1/5)d \ge 3d/8 \quad (\ell \ge 5), \\ &\mathbb{E}N_2 \le -d/2, \quad \mathbb{E}N_3 \le -4d/5, \quad \mathbb{E}N_L \le -3d/8 \quad (L \ge 4). \end{aligned} \tag{Q92.4.6}$$

Q92.4.2 All finite frequency predecessors for I and II.

Here $\xi = 0$. Since $U \ge 1/2$,

$$|Q_1|^2 = |1 - xP(-z)|^2 \le 1 - x + (100/49)x^2 \le 1, \tag{Q92.4.7}$$

using $(100/49)\theta = 325/392 < 1$. For I, $U, V \ge 1 - x/(1-x^2) > 1/2$, and both are at most $1/(1-x^2)$. Both coordinates of $Q_2 = 1 - x^2 V + z(1 - xU)$ are nonnegative, so

$$|Q_2|^2 \le (1 - x^2/2)^2 + x(1 - x/2)^2 \le 1 + x - \frac{29}{16}x^2 \le 1 + \frac{4}{29} < \frac{8}{7}. \tag{Q92.4.8}$$

The middle comparison uses $x \le 1/2$. For II, $P_1 = 0$ gives $-x/(1-x^2) \le V \le x^2/(1-x^2)$. The real coordinate is positive and at most $13/12$; the imaginary coordinate has modulus at most $\sqrt{x}(1-x/2)$. The function $x(1-x/2)^2$ increases on $[0, 1/2]$ to $9/32$, whence

$$|Q_2|^2 \le 169/144 + 9/32 = 419/288 < 3/2. \tag{Q92.4.9}$$

For both classes,

$$\boxed{\|Q_n(K')\| \le 6/5 \qquad (n \ge 3).} \tag{Q92.4.10}$$

At $n = 3$,

$$|G_3| \le 1 - x/2 + x^2/2, \qquad |Q_3| \le 1 - x/2 + (27/14)x^2.$$

The latter convex quadratic has values at most one and $1 + 1651/14336 < 6/5$ at $0, \theta$. At $n = 4$, $|G_4| \le 1$ and the remainder is at most $(10/7)x^{5/2} \le 845/5376 < 1/5$. For each $n \ge 5$, $|G_n| \le 1$, and the remainder is at most $(10/7)\theta^3 < 5/49 < 1/5$.

Q92.4.3 The low-domain moment bounds for I and II.

First take $j = 2$. Every additional pair subtracts $(128/5)d$ from the exponent of (Q92.3.9). For I the bounds are

L	c_L/d	$\mathbb{E}N_L/d$	$\|Q_{L-1}g'\|^2/d$	exponent$/d$
2	3/4	−1/2	1	−227/20
3	1/4	−4/5	8/7	−429/28
$L \ge 4$	3/8	−3/8	36/25	−587/100

(Q92.4.11)

The three exponent calculations are $-607/20+6-4+17$, $-607/20+2-32/5+136/7$, and $-607/20+612/25=-587/100$. For II, $\delta_+=0$ and $(D_1)_+\le d/5$, giving

L	c_L/d	$\mathbb{E}N_L/d$	$\|Q_{L-1}g'\|^2/d$	exponent$/d$
2	$1/5$	$-1/2$	1	-11
3	$-1/10$	$-4/5$	$3/2$	$-73/10$
$L\ge 4$	$1/40$	$-3/8$	$36/25$	$-98/25$

(Q92.4.12)

The calculations are $-128/5+8/5-4+17$, $-128/5-4/5-32/5+51/2$, and $-128/5+1/5-3+612/25$. All these bounds hold throughout $0<d\le 1/64$.

Q92.5.1 Class III for $0<d\le 1/6$: the same variable and its variance.

Here the direction is $(1,2,3,b_2,...)$ and $R=R_0-\tau g$. The III mask has $\|P(K)\|\le 1$; its first odd coefficient is absent, so

$$\mathbb{E}[P(K)g]\ge -\frac{\delta_3}{1-\theta^2}>-d/5. \tag{Q92.5.1}$$

Apply the row bound to $R=K(P(K)g-\tau\mathbf{1})$:

$$|R|^2\le d+\tau^2+(2\tau/5)d\le d+(7/5)d^2\le 37/180<1/4. \tag{Q92.5.2}$$

Thus $\eta=6/5$ throughout this domain. The terms $K^2\mathbf{1}, K^4\mathbf{1}$ are present in R_0, whereas $K^3\mathbf{1}$ is absent. Retaining their signs gives

$$\mathbb{E}R_0\le -d+\frac{\delta_2}{1-\theta^2}\le -\frac{439}{855}d,\qquad \mathbb{E}(gR_0)\ge -\frac{\delta_4}{1-\theta^2}>-d/12. \tag{Q92.5.3}$$

Since $\tau\le d-d^2\le 5/36$,

$$\begin{aligned} V_R&=\mathbb{E}R_0^2-2\tau\mathbb{E}(gR_0)+\tau^2 d\\ &\le \delta_2+(\tau/6)d+\tau^2 d\le \frac{1163}{2592}d. \end{aligned} \tag{Q92.5.4}$$

Consequently

$$\boxed{\delta=\mathbb{E}R+(6/5)V_R\le(-439/855+1163/2160)\,d=\frac{205}{8208}d<d/40.} \tag{Q92.5.5}$$

The final comparison uses $205\cdot 40=8200<8208$.

Q92.5.2 All signed means and convolution losses for III.

We have

$$D_1=\mathbb{E}(gR_0)-\tau d\le\frac{\delta_3}{1-\theta^2}-\tau d<d/5. \tag{Q92.5.6}$$

Retaining the negative contribution from $K^4\mathbf{1}$ gives

$$D_2\le\delta_2-\delta_3+\frac{\delta_4}{1-\theta^2}\le\delta_2. \tag{Q92.5.7}$$

At the third step, the first possible positive contribution from R_0 is $K^7\mathbf{1}$, and $\tau g'$ contributes $+\tau\delta_2$. Hence

$$D_3\le\frac{\delta_5}{1-\theta^2}+\tau\delta_2<\left(\frac{13}{384}+\frac{65}{1152}\right)d=\frac{13}{144}d<d/10. \tag{Q92.5.8}$$

For every $i \geq 4$, the positive contribution from R_0 is at most $\delta_3/(1-\theta^2)$. The first positive contribution of $\tau g'$ in this range is $\tau\delta_4$ at $i = 7$. Therefore

$$D_i \leq \frac{\delta_3}{1-\theta^2} + \tau\delta_4 < \left(\frac{1}{5} + \frac{5}{504}\right) d < d/4 \qquad (i \geq 4). \tag{Q92.5.9}$$

The complete loss and mean bounds are

$$\begin{gathered} \gamma_3 \geq \kappa d, \quad \gamma_4 \geq 9d/10, \quad \gamma_\ell \geq 11d/32 \quad (\ell \geq 5), \\ \mathbb{E}N_2 \leq -\kappa d, \quad \mathbb{E}N_3 \leq -9d/10, \quad \mathbb{E}N_L \leq -11d/32 \quad (L \geq 4). \end{gathered} \tag{Q92.5.10}$$

Q92.5.3 The two short frequency predecessors for III.

Now $Q_n = G_n + z^{n+1}P(-z) + \tau z^n$. For $Q_1 = 1 - xU + z(\tau + xV)$, the real coordinate lies in $[0, 1 - x/2]$. By (Q92.1.9), $-1/5 < \tau + xV \leq 5/36 + 1/12 = 2/9 < 1/4$, so

$$|Q_1|^2 \leq (1 - x/2)^2 + x/16 \leq 1. \tag{Q92.5.11}$$

The real coordinate of $Q_2 = 1 - \tau x - x^2V + z(1 - xU)$ is greater than $1 - 1/16 - 1/30 > 0$ and less than $13/12$. Its imaginary coordinate has modulus at most $\sqrt{x}(1 - x/2)$. Thus

$$|Q_2|^2 \leq 419/288 < 3/2. \tag{Q92.5.12}$$

Q92.5.4 All long frequency predecessors for III.

First put $\widetilde{Q}_n = G_n + z^{n+1}P(-z)$. We show

$$|\widetilde{Q}_n(z)| \leq 1 \qquad (n \geq 3,\ 0 \leq x \leq \theta), \tag{Q92.5.13}$$

using both forced coefficients of III. At $n = 3$, both coordinates are nonnegative, and

$$\Re\widetilde{Q}_3 \leq \frac{1 - x + x^5}{1 - x^2}, \qquad \Im\widetilde{Q}_3 \leq \frac{\sqrt{x}(1 - x^2 + x^3)}{1 - x^2}. \tag{Q92.5.14}$$

The sum of their squared upper bounds is at most one precisely when

$$G(x) = 1 - 3x + 2x^2 - x^3 - 3x^4 + 4x^5 - x^6 - x^9 \geq 0. \tag{Q92.5.15}$$

Since $x + x^4 < 7/16 < 1/2$, $G(x) \geq G_*(x) = 1 - 3x + 2x^2 - x^3 - 3x^4 + (7/2)x^5$. On $[0, \theta]$,

$$G_*'(x) \leq -3 + 4\theta + (35/2)(1/32) = -53/64 < 0, \qquad G_*(\theta) = 86987/67108864 > 0. \tag{Q92.5.16}$$

At $n = 4$,

$$|\widetilde{Q}_4| \leq (1 - x)\sqrt{1 + x} + x^{5/2} \leq 1 - x/2 + x^2/6 \leq 1. \tag{Q92.5.17}$$

At $n = 5$,

$$|G_5| \leq 1 - x/2 + x^2/2, \qquad |\widetilde{Q}_5| \leq 1 - x/2 + x^2/2 + x^3 \leq 1. \tag{Q92.5.18}$$

The first squared comparison has positive difference $x^2(1 + 2x - 3x^2)/4$; the second uses $x \leq 1/2$. For every $n \geq 6$, the finite geometric formula gives

$$|\widetilde{Q}_n| \leq \frac{1 + x^3}{\sqrt{1 + x}} + x^3 \leq 1. \tag{Q92.5.19}$$

The final comparison is equivalent to $x(1 + \sqrt{1 + x}) \leq 1$, whose left side is at most $(13/32)(11/5) = 143/160 < 1$. At $x = 0$, equality holds directly. Restoring the τ term, for all $n \geq 3$,

$$|Q_n| \leq 1 + \tau\theta^{3/2} < 1 + 65/1728 < 21/20, \tag{Q92.5.20}$$

using $\tau \leq 5/36$ and $\sqrt{\theta} < 2/3$.

Q92.5.5 The low-domain moment bounds for III.

For $j = 2$, use $\Gamma \ge 35d/16$ in (Q92.3.9):

L	c_L/d	$\mathbb{E}N_L/d$	$\|Q_{L-1}g'\|^2/d$	exponent$/d$
2	$1/4$	$-19/32$	1	$-13/4$
3	$-23/160$	$-9/10$	$3/2$	$-7/20$
$L \ge 4$	$17/160$	$-11/32$	$441/400$	$-263/400$

(Q92.5.21)

These are the calculations

$$-35/2 + 2 - 19/4 + 17 = -13/4,$$
$$-35/2 - 23/20 - 36/5 + 51/2 = -7/20,$$
$$-35/2 + 17/20 - 11/4 + 7497/400 = -263/400.$$

Each additional pair subtracts $(128/5)d$ from the exponent. This covers every $b_2 \ge 4$, every later pair and $L \ge 2$ on $0 < d \le 1/6$.

Q92.6 Pointwise tail comparison on the remaining domains.

Compare with the same-direction single-pair input qf_{b_1} in (Q92.1.12). Put

$$S = \sum_{i=2}^{j} R_{a_i b_i}. \tag{Q92.6.1}$$

The intervals remain disjoint, so the row and mask bounds imply

$$|S| \le \frac{10}{7}\theta^{(a_2-1)/2}\sqrt{d}. \tag{Q92.6.2}$$

For I and II, $a_2 \ge 4$, so $|S| < (3/8)\sqrt{d}$: indeed, $\sqrt{\theta} < 9/14$ gives $(10/7)\theta^{3/2} < 585/1568 < 3/8$. For $d \ge 1/64$, $\sqrt{d} \le 8d$, and

$$\prod_{i=2}^{j} h_{a_i} f_{b_i} \le e^{S-(16/5)(j-1)d} \le e^{-[1/5+(16/5)(j-2)]d}. \tag{Q92.6.3}$$

For III, $|S| \le (65/112)\sqrt{d}$. For $d \ge 1/6$, $\sqrt{d} \le \sqrt{6}\, d < (5/2)d$, so

$$\prod_{i=2}^{j} h_{a_i} f_{b_i} \le e^{S-[2-\theta+(16/5)(j-2)]d} \le e^{-[1/7+(16/5)(j-2)]d}, \tag{Q92.6.4}$$

because $2 - \theta - 325/224 = 51/32 - 325/224 = 1/7$. Positivity of T^L and (Q92.1.12) now give, for all $L \ge 1$,

$$\mathbb{E}[(pT^L F_\Sigma)^8] \le \begin{cases} e^{-[8/5+(128/5)(j-2)]d}, & \text{I and II, } d \ge 1/64, \\ e^{-[8/7+(128/5)(j-2)]d}, & \text{III, } d \ge 1/6. \end{cases} \tag{Q92.6.5}$$

Both adjacent domains include their common endpoint and use the directions of (Q92.0.3).

Q92.7 Completion and genuine equal-pair factors.

Classes I–III cover (Q92.0.2). Use (Q92.1.11) for $L = 1$, and the preceding bounds for $L = 2, 3$ and $L \geq 4$. The weakest new basic exponent is $7/20 > 1/4$, and $128/5 > 16$, hence

$$\mathbb{E}[(pT^L F_\Sigma)^8] \leq e^{-[1/4+16(j-2)]d}. \tag{Q92.7.1}$$

Since $d \leq 1/3$,

$$e^{-d/4} \leq \frac{1}{1+d/4} = 1 - \frac{d}{4+d} \leq 1 - \frac{3}{13}d \leq 1 - d/5. \tag{Q92.7.2}$$

Restore $E_t, E_r \leq 1$ by positive comparison to obtain (Q92.0.4). Replacing K by $-K$ reverses all terminal, connector and equal-pair edges together, interchanges p/q and A/T, and preserves d, δ_i. The same directions also have the separate weighted fourth estimate proved in Appendix I.6:

$$\mathbb{E}[p\{T^L(E_t F_\Sigma)\}^4] \leq 1 \qquad (L \geq 1). \tag{Q92.7.3}$$

O.21 Ordinary eighth moments for odd terminals with an inward arm of length at least two

Q93.0 Structures and directions.

After maximal equal pairing, let the distinct residual arms form

$$a < b_1 < c_1 < \cdots < b_k < c_k, \qquad a \geq 2, \quad k \geq 1. \tag{Q93.0.1}$$

Every edge of the a-arm points toward the terminal. Each b_i-arm has only its first edge inward and all remaining edges outward; each c_i-arm points outward. All connector edges point from the terminal to the receiving root, which has an outward unit arm. With genuine equal-pair products E_t, E_r, set

$$F = \bar{f}_a \prod_{i=1}^{k} h_{b_i} f_{c_i}, \qquad X_D = T^L(E_t F), \qquad H_D = E_r p X_D. \tag{Q93.0.2}$$

These directions depend only on the arm list. For every $L \geq 1$, all finite lengths and gaps, and all genuine decorations, we prove

$$\boxed{\mathbb{E}H_D^8 \leq Q(d) \leq e^{-d} \leq 1 - 3d/4,} \tag{Q93.0.3}$$

The three ranges and bounds are listed in Table 60.

Table 60: The three ordinary eighth-moment ranges in (Q93.0.3).

First short arm	Pair count	$Q(d)$
$b_1 = 3,\ a = 2$	$k \geq 3$	$e^{-[10+(128/5)(k-3)]d}$
$b_1 = 4,\ a \in \{2, 3\}$	$k \geq 2$	$e^{-[17+(128/5)(k-2)]d}$
$b_1 \geq 5,\ 2 \leq a < b_1$	$k \geq 1$	$e^{-[1+(128/5)(k-1)]d}$

Q93.1 The finite calculus.

Use the fixed finite probability space and operators of Appendix M.15, with

$$\begin{gathered} Jv = (\mathbb{E}v)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad g = K\mathbf{1}, \quad p = 1 + g, \quad q = 1 - g, \\ f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad \delta_i = \|K^i\mathbf{1}\|_2^2, \quad d = \delta_1, \\ \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2. \end{gathered} \tag{Q93.1.1}$$

The finite skew and path identities give

$$\begin{gathered} 0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta\delta_i, \quad \|K\|^2 \le \theta, \quad |Kv(x)| \le \|v\|_2, \\ 0 \le f_n, \bar{f}_n, h_n \le 2, \qquad \mu_n := \mathbb{E}f_n = \mathbb{E}\bar{f}_n \in [0,1], \\ \mathbb{E}K^{2i}\mathbf{1} = (-1)^i\delta_i, \qquad \mathbb{E}K^{2i+1}\mathbf{1} = 0. \end{gathered} \tag{Q93.1.2}$$

After zero-mass atoms are removed, $p(x), q(x) \ge \mu(x) > 0$. Each tree message is positive: conditional on the root label, the all-equal labelling has positive mass and all diagonal edge factors are one. At $d = 0$, $g = 0$ and leaf integration gives every message equal to one. We henceforth take $d > 0$ where division is used. For every genuine equal pair,

$$f_\ell + h_\ell = 2\mu_{\ell-1}, \qquad 0 \le f_\ell h_\ell \le \mu_{\ell-1}^2 \le 1. \tag{Q93.1.3}$$

With $\beta_1 = \beta_2 = 0$,

$$\begin{gathered} \beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i \quad (n \ge 3), \qquad G_n(z) = \sum_{i=0}^{n-1} z^i, \\ S_n = \sum_{i=0}^{n} K^i\mathbf{1} = f_n + \sum_{i=3}^{n} \beta_i f_{n-i}, \quad \kappa d \le \tau \le \beta_n \le d \quad (n \ge 3), \\ \beta_3 = \beta_4 = d, \quad \|G_n(K)\| \le 1 \quad (n = 0, 1 \text{ or } n \ge 3), \quad \|G_2(K)\| \le \sqrt{1+\theta} < 6/5. \end{gathered} \tag{Q93.1.4}$$

Under reversal put $K' = -K$, $g' = -g$, and $S_n' = \sum_{i=0}^n K'^i\mathbf{1}$; the same convolution uses $\bar{f}_n$. Appendix M.1 proves, for every real V,

$$\boxed{\mathbb{E}e^{8K'V} \le e^{8\mathbb{E}(K'V) + 17\|V\|_2^2}.} \tag{Q93.1.5}$$

Q93.2.1 Pair losses at every arm length.

Let $R_{st} = \sum_{r=s+1}^t K^r\mathbf{1}$, $s < t$. The finite difference formula gives, for $s = 3$,

$$h_3 + f_t \le 2 + R_{3t} - (2d - \delta_2) \le 2 + R_{3t} - (2 - \theta)d. \tag{Q93.2.1}$$

For $s = 4$, retain the negative convolution term:

$$f_t - f_4 \le R_{4t} + \delta_2(1 + p) - \tau f_2, \qquad f_2 \ge p^2/2.$$

Completing the square in $p \ge 0$, with $\delta_2 \le d/2$ and $\tau \ge d/2$, gives $\delta_2(1+p) - \tau f_2 \le \delta_2 + \delta_2^2/(2\tau) \le 3d/4$. Since $h_4 + f_4 = 2 - 4d$, this yields $h_4 + f_t \le 2 + R_{4t} - 13d/4$, including $t = 5$. For $s \ge 5$, the all-index mean proof in Appendix M.17 gives $\mu_n \le 1 - 83d/32$ for $n \ge 4$. It treats $n = 4, 5, 6, 7$ explicitly and retains $f_0, \dots, f_5$ and three higher losses in the convolution for $n \ge 8$, giving loss at

least $95d/36$. The positive part of the finite difference is at most $4\delta_2/(1-\theta^2) \le 1664d/855$, so the pair-sum loss is

$$83d/16 - 1664d/855 = (16/5 + 113/2736)d > 16d/5.$$

The sums are nonnegative. AM–GM and $1+u \le e^u$ give

$$h_s f_t \le \begin{cases} e^{R_{3t}-(51/32)d}, & s=3,\\ e^{R_{4t}-(13/4)d}, & s=4,\\ e^{R_{st}-(16/5)d}, & s\ge 5. \end{cases} \tag{Q93.2.2}$$

Later pairs of minimum at least four may use the weaker loss $16d/5$. Thus

$$\begin{aligned} F &\le \bar f_a e^{R-\Gamma}, \qquad R = \sum_{i=1}^{k}\sum_{r=b_i+1}^{c_i} K^r\mathbf{1},\\ \Gamma/d &= \begin{cases} 51/32 + (16/5)(k-1), & b_1=3,\\ 13/4 + (16/5)(k-1), & b_1=4,\\ 16k/5, & b_1\ge 5. \end{cases} \end{aligned} \tag{Q93.2.3}$$

Q93.2.2 Finite masks and the exponential remainder.

The intervals are disjoint, so $R = K^{b_1}P(K)g$, where P is a finite zero-one polynomial with arbitrary gaps. The finite mask proof in (L90.2.6) gives, at $z = i\sqrt{x}$, $0 \le x \le \theta$,

$$|P(z)|^2 \le \frac{1+x}{(1-x^2)^2} \le C_* := \frac{32768}{16245} < \frac{100}{49}. \tag{Q93.2.4}$$

The spectral and row bounds respectively imply

$$\|R\|^2 \le \theta^{b_1} C_* d, \qquad |R|^2 \le \theta^{b_1-1} C_* d. \tag{Q93.2.5}$$

Since $b_1 \ge 3$ and $\theta^2 C_* = 5408/16245 < 1/3$, $|R|^2 < d/3 \le 1/9$. The factorial bound $n! \ge 2\cdot 3^{n-2}$ bounds the exponential remainder by $9R^2/16$ for $|R| \le 1/3$. Hence throughout the domain

$$e^R \le 1 + R + 3R^2/5. \tag{Q93.2.6}$$

Use the scalar norm constants

$$\|R\| \le \varrho\sqrt{d}, \qquad \varrho = 3/8\ (b_1=3),\quad 1/4\ (b_1=4),\quad 3/20\ (b_1\ge 5). \tag{Q93.2.7}$$

Indeed,

$$\theta^3 C_* = 2197/16245 < 9/64,$$
$$\theta^4 C_* = 28561/519840 < 1/16,$$
$$\theta^5 C_* = 371293/16634880 < 9/400.$$

Q93.2.3 Mean and centered part of the inward arm.

Put $\mu = \mathbb{E}\bar f_a = \mu_a \in [0,1]$, $v = \bar f_a - \mu$, $\Pi = I - J$, and $C_0 = \Pi K \Pi$. The centered path identity is

$$\bar f_a - \mu_a = -\sum_{i=0}^{a-1} \mu_{a-1-i}(-C_0)^i g. \tag{Q93.2.8}$$

Every coefficient is in $[0,1]$. Applying the same finite mask spectral bound to the skew compression gives $\sigma(\bar{f}_a) \le \sqrt{C_*}\sqrt{d} < (3/2)\sqrt{d}$. For $b_1 = 3,4$, only $a = 2,3$ are possible. Direct expansion gives $\sigma(\bar{f}_2)^2 = d + \delta_2 - d^2 \le 36d/25$. For $\bar{f}_3$, its centered expression $-(g + K^3\mathbf{1}) + (Kg + d)$ has orthogonal summands, so $\sigma(\bar{f}_3)^2 = d - \delta_2 + \delta_3 - d^2 \le d$. Consequently

$$\sigma(\bar{f}_a) \le s\sqrt{d}, \qquad s = 6/5 \ (b_1 = 3,4), \quad s = 3/2 \ (b_1 \ge 5). \tag{Q93.2.9}$$

Q93.3 Positive parts of all finite propagation means.

Let $D_i = \mathbb{E}K'^i R$ and write $R = \sum_{r \ge b_1+1} c_r K^r \mathbf{1}$, a finite sum with $c_r \in \{0,1\}$. A term with odd $i+r$ has zero mean; otherwise its contribution is $(-1)^{i+(i+r)/2} c_r \delta_{(i+r)/2}$. Positive terms can therefore occur precisely when $r \equiv i \pmod 4$. For $i = 1,2,3,4$, their first possible moments are Adding four

Table 61: First possible positive moments in $D_i = \mathbb{E}B^i R$. Adding four to either index raises the moment index by two.

b_1	$i=1$	$i=2$	$i=3$	$i=4$
3	δ_3	δ_4	δ_5	δ_4
4	δ_3	δ_4	δ_5	δ_6
≥ 5	δ_5	δ_4	δ_5	δ_6

to either index increases the moment index by two. Bounding both finite tails by their geometric upper sums yields

$$\sum_{i\ge 1}(D_i)_+ \le \begin{cases} \dfrac{\theta^2}{(1-\theta)^2} d = \dfrac{169}{361} d < d/2, & b_1 = 3, \\ \dfrac{\theta^2(1+\theta+\theta^2+\theta^3)}{(1-\theta^2)^2} d = \dfrac{201617}{519840} d < 2d/5, & b_1 = 4, \\ \dfrac{\theta^3}{(1-\theta)^2} d = \dfrac{2197}{11552} d < d/5, & b_1 \ge 5. \end{cases} \tag{Q93.3.1}$$

The bound applies to each finite initial sum used by Duhamel. The same sign calculation gives For

Table 62: Upper bounds for individual propagation means.

b_1	D_0	D_1	$D_i,\ i \ge 2$
3	$d/2$	$d/5$	$d/12$
4	$d/12$	$d/5$	$d/12$
≥ 5	$d/12$	$d/30$	$d/12$

D_0, positive terms start at δ_2 or δ_4; for D_1, at δ_3 or δ_5; for $i \ge 2$, their first possible moment index is at least four. The needed bounds are

$$\theta/(1-\theta^2) = 416/855 < 1/2, \quad \theta^2/(1-\theta^2) = 169/855 < 1/5,$$
$$\theta^3/(1-\theta^2) = 2197/27360 < 1/12, \quad \theta^4/(1-\theta^2) = 28561/875520 < 1/30.$$

Deleting an absent positive term only improves these upper bounds.

Q93.4 The common inward-arm propagation lemma.

Apply Lemma M.23 with $B = K' = -K$, $\mu = \mathbb{E}\bar{f}_a$, and the values of s, ϱ in (Q93.2.7) and (Q93.2.9). Its first-step constant $c_0 = 2s\varrho + (12/5)\varrho^2$ equals $99/80$, $3/4$, and $63/125$ in the three

rows. The individual mean bounds in Table 62 therefore give

$$\delta_+ \le \begin{cases} 7d/4, & b_1 = 3, \\ 5d/6, & b_1 = 4, \\ 3d/5, & b_1 \ge 5. \end{cases} \tag{Q93.4.2}$$

For the last row, $(D_0)_+ + 63d/125 \le 881d/1500 < 3d/5$. The lemma retains the coefficient μ of every perturbation and gives the exact expressions (Q93.4.3)–(Q93.4.6).

Q93.5 Every nonunit connector.

For $L \ge 2$, $n \ge 4$, so G_{n-1} is a contraction. Using $\mu \le 1$ and $\sqrt{\theta} < 13/20$,

$$\|V\| \le (1 + \sqrt{\theta}\,\varrho)\sqrt{d}. \tag{Q93.5.1}$$

The three coefficients are smaller than $5/4, 7/6, 11/10$; the middle one is $1 + 13/80 = 93/80 < 7/6$. Equations (Q93.3.1), (Q93.4.2) and (Q93.4.5) give the complete cost table The first row consists of costs $163d/6$ and $425d/16$, totaling $2579d/48$; the second, $109d/10$ and $833d/36$, totaling $6127d/180$; the third, $119d/30$ and $2057d/100$. At the minimum allowed pair counts, the net losses are

Table 63: Nonunit connector costs in (Q93.4.6). The last column excludes -8Γ. All entries are upper bounds.

b_1	c_L/d	$\mathbb{E}N/d$	$\|V\|^2/d$	Cost divided by d
3	$9/2-\kappa$	$-\kappa+1/12$	$25/16$	$2579/48$
4	$37/15-\kappa$	$-\kappa+1/12$	$49/36$	$6127/180$
≥ 5	$8/5-\kappa$	$-\kappa+1/12$	$121/100$	$119/30+2057/100$

$$\begin{aligned} b_1 = 3, k = 3: &\quad 1279d/20 - 2579d/48 = 2453d/240 > 10d, \\ b_1 = 4, k = 2: &\quad 258d/5 - 6127d/180 = 3161d/180 > 17d, \\ b_1 \ge 5, k = 1: &\quad 128d/5 - 119d/30 - 2057d/100 = 319d/300 > d. \end{aligned} \tag{Q93.5.2}$$

Each additional pair adds $16d/5$ to Γ without increasing the mask costs, and therefore adds $128d/5$ to the eighth-moment loss.

Q93.6 Unit connectors and the G_2 exception.

For $L = 1$, retain the exact value $\bar{f}_0 = 1$. Then

$$c_1 + \mathbb{E}N \le \delta + \mu D_1 - \beta_{a+1} - \beta_{a+2}, \qquad \|V\| \le (\|G_a\| + \varrho)\sqrt{d}. \tag{Q93.6.1}$$

For $a = 2$, the two beta coefficients in (Q93.6.1) are both d; for $a \ge 3$, use $\beta \ge \kappa d$. The sole G_2 exception is $a = 2$, when $\|G_2\| \le 6/5$; all G_a with $a \ge 3$ are contractions. These observations and the individual means in Table 62 give Table 64. Its mean bounds are $(7/4 + 1/5 - 2)d = -d/20$, $(5/6 + 1/5 - 2\kappa)d = -37d/240$, $(3/5 + 1/30 - 2)d = -41d/30$, and $(3/5 + 1/30 - 2\kappa)d = -133d/240$, respectively. The first two costs are $-2/5 + 17(63/40)^2 = 66833/1600$ and $-37/30 + 17(29/20)^2 = 41411/1200$. In the third row the loss is $128/5 + 164/15 - 17(27/20)^2 = 6661/1200 > 5$. For $a \ge 3$ the last-row loss is

$$[128/5 + 133/30 - 17(23/20)^2]d = 9061d/1200 > d. \tag{Q93.6.2}$$

Table 64: Unit connectors. The mean, norm and cost columns are upper bounds. The final column is the lower loss at the minimum permitted pair count; each additional pair adds $128d/5$.

Range	$k_{\min}$	$(c_1+\mathbb{E}N)/d$	$\|V\|/\sqrt{d}$	Cost$/d$	Loss$/d$
$b_1=3,\ a=2$	3	$-1/20$	$63/40$	$66833/1600$	$35487/1600$
$b_1=4,\ a\in\{2,3\}$	2	$-37/240$	$29/20$	$41411/1200$	$20509/1200$
$b_1\ge 5,\ a=2$	1	$-41/30$	$27/20$	$-164/15+17(27/20)^2$	$6661/1200$
$b_1\ge 5,\ a\ge 3$	1	$-133/240$	$23/20$	$-133/30+17(23/20)^2$	$9061/1200$

Every further pair adds the same $128d/5$ loss in all these cases. Together with the nonunit calculation, this proves all three rows of (Q93.0.3). Positive comparison restores E_t, E_r. Finally $e^{-d}\le 1/(1+d)\le 1-3d/4$ because $d\le 1/3$. Replacing K by $-K$ reverses the receiving unit arm, inward arm, all pairs, connector and decorations together while preserving the measure and every δ_i.

O.22 Short-first-pair ordinary eighth moments for odd terminals of minimum two

Q95.0 Structures and bounds.

The receiving root has an outward unit arm, all connector edges point from the terminal to this root, and $L\ge 1$. For genuine equal-pair products E_t, E_r, put $H=E_rpT^L(E_tF)$. We prove the following three rows, for every finite arm list and every positive connector length:

distinct residual arms	F	upper bound for $\mathbb{E}H^8$
$(a,4,c),\ a=2,3,\ c\ge 5$	$\bar f_a h_4 f_c$	e^{-d}
$(2,3,4)$	$\bar f_2 f_3 h_4$	$e^{-3d/32}$
$2<3<c_1<b_2<c_2<\cdots<b_k<c_k,\ k\ge 2$	$\bar f_2 f_3 h_{c_1}\prod_{i=2}^{k} h_{b_i} f_{c_i}$	$e^{-[1+(128/5)(k-2)]d}$

The last row with $c_1=4$ has the stronger bound $e^{-[5+(128/5)(k-2)]d}$. The $\bar f_a$-arm is fully inward, each f_n-arm fully outward, and each h_n-arm has its first edge inward and all later edges outward. In the last two rows the first pair is $f_3h_{c_1}$. Directions are fixed by the integer arm list and are the same in the low and high parameter domains.

Q95.1 The finite inputs.

Use $A=J+K, T=J-K=A^*, g=K\mathbf{1}, p=1+g, q=1-g, f_n=A^n\mathbf{1}, \bar f_n=T^n\mathbf{1}, h_n=Tf_{n-1}$, and $\delta_i=\|K^i\mathbf{1}\|^2, d=\delta_1$, as in Appendix M.15. In the fixed measure,

$$\begin{gathered}0\le d\le 1/3,\quad d^2\le\delta_2\le\theta d,\quad \delta_{i+1}\le\theta\delta_i,\quad \|K\|^2\le\theta,\\ |Kv(x)|\le\|v\|_2,\quad \theta=13/32,\quad \kappa=19/32,\\ 0\le f_n,\bar f_n,h_n\le 2,\quad \mu_n=\mathbb{E}f_n=\mathbb{E}\bar f_n\in[0,1],\\ \mathbb{E}K^{2j}\mathbf{1}=(-1)^j\delta_j,\quad \mathbb{E}K^{2j+1}\mathbf{1}=0.\end{gathered}\tag{Q95.1.1}$$

Each genuine pair satisfies $f_\ell+h_\ell=2\mu_{\ell-1}$ and $0\le f_\ell h_\ell\le 1$, and can be restored by positivity. After removal of zero-mass atoms, $p(x),q(x)\ge\mu(x)>0$. The all-equal labelling makes every tree message positive. For $d=0$, $g=0$ and leaf integration gives one, including when $K\ne 0$; divisions

below are restricted to $d > 0$. Set $\tau = d - \delta_2$, $\beta_1 = \beta_2 = 0$, and

$$\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1}\delta_j \quad (n \ge 3), \quad \tau \le \beta_n \le d, \quad \tau \ge \kappa d, \quad \beta_3 = \beta_4 = d,$$
$$S_n = \sum_{j=0}^{n} K^j \mathbf{1} = f_n + \sum_{i=3}^{n} \beta_i f_{n-i}. \tag{Q95.1.2}$$

Under $K' = -K$, $g' = -g$, the convolution for S'_n uses $\bar{f}_n$. With $G_m(z) = \sum_{j=0}^{m-1} z^j$, $\|G_m(K)\| \le 1$ for $m = 0, 1$ or $m \ge 3$, whereas $\|G_2\| \le \sqrt{1+\theta} < 6/5$. The full-mean exponential inequality of Appendix M.1 is

$$\mathbb{E}e^{8K'V} \le e^{8\mathbb{E}(K'V) + 17\|V\|_2^2} \tag{Q95.1.3}$$

for each real V. For a fixed row, write $k(y) = K(x, y)$. Skewness gives

$$r(x) = Kg(x) = \frac{1}{2}\mathbb{E}_{y,z}[(k(y) - k(z))K(y, z)].$$

Since $|u - v| \le 1 - uv$ for $u, v \in [-1, 1]$, $|r(x)| \le (1 - g(x)^2)/2 = pq/2$; hence $f_2 = p + r \ge p^2/2$. For every $5 \le s < t$, the all-index path mean $\mu_n \le 1 - 83d/32$ for $n \ge 4$ and the positive finite-difference bound $4\delta_2/(1 - \theta^2) \le 1664d/855$ give pair-sum loss $[83/16 - 1664/855]d = [16/5 + 113/2736]d$. AM–GM therefore gives $h_s f_t \le e^{R_{st} - 16d/5}$, where $R_{st} = \sum_{r=s+1}^{t} K^r \mathbf{1}$; see Appendix M.17.

Q95.2 Finite propagation retaining the inward arm.

Here $a = 2$ or 3. Direct expansion gives $\bar{f}_2 = q + Kg$, $\mathbb{E}\bar{f}_2 = 1 - d$, and $\sigma(\bar{f}_2)^2 = d + \delta_2 - d^2 \le 36d/25$. The orthogonal odd/even decomposition $\bar{f}_3 - \mathbb{E}\bar{f}_3 = -(g + K^3\mathbf{1}) + (Kg + d)$ gives $\sigma(\bar{f}_3)^2 = d - \delta_2 + \delta_3 - d^2 \le d$. Thus $\sigma(\bar{f}_a) \le 6\sqrt{d}/5$. Suppose $0 \le F \le \bar{f}_a e^{R - \Gamma}$ and $|R| \le 1/2$. Set $\mu = \mathbb{E}\bar{f}_a$ and $D_i = \mathbb{E}B^i R$, with $B = K' = -K$ and $u = B\mathbf{1} = g'$. Apply Lemma M.23 with $s = \sigma(\bar{f}_a)/\sqrt{d}$ and $\varrho = \|R\|/\sqrt{d}$. The first step and its finite continuation are

$$T(\bar{f}_a e^R) \le \bar{f}_{a+1} + \mu BR + \delta, \qquad \delta = \mu D_0 + 2\sigma(\bar{f}_a)\|R\| + \tfrac{12}{5}\|R\|^2, \tag{Q95.2.1}$$

$$0 < e^{\Gamma} T^L F \le Y_L = \bar{f}_{a+L} + \mu B^L R + \delta \bar{f}_{L-1} + \mu \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{Q95.2.2}$$

For an identically zero input replace the initial strict inequality by a weak one. The moment bound follows by continuity. Use the common parameters, here written explicitly with $n = a + L$:

$$V = G_{n-1}(B)u + \mu B^{L-1} R, \quad N = BV, \quad \mathbb{E}N = -\beta_{n+1} + \mu D_L,$$
$$c_1 = \delta - \beta_n, \qquad c_L = 2\delta_+ + 2\mu \sum_{i=1}^{L-1} (D_i)_+ - \beta_n \quad (L \ge 2).$$

Thus

$$\log \mathbb{E}[(pT^L F)^8] \le -8\Gamma + 8c_L + 8\mathbb{E}N + 17\|V\|^2. \tag{Q95.2.3}$$

The actual messages are positive. At $d = 0$ use leaf integration rather than the displayed ratios; for a zero input the logarithm is $-\infty$.

Q95.3 Triples $(a,4,c)$, $a=2,3$, $c\ge 5$.

Take $F=\bar f_a h_4 f_c$. Retaining the negative f_2 term gives

$$h_4+f_c\le 2+R-4d+\delta_2(1+p)-\tau f_2,\qquad R=\sum_{r=5}^{c}K^r\mathbf{1},\quad f_2\ge p^2/2.$$

Completing the square bounds the excess by $\delta_2+\delta_2^2/(2\tau)$. For $t=\delta_2/d\le\theta$, the function $t+t^2/[2(1-t)]$ increases on $[0,\theta]$, so this is at most $663d/1216$. The pair loss is at least $4201d/1216>69d/20$, with difference $29d/6080$. Hence $F\le \bar f_a e^{R-\Gamma}$, $\Gamma=69d/20$. The interval is consecutive: $R=K^4G_{c-4}(K)g$. Therefore

$$\|R\|\le(6/5)\theta^2\sqrt d=507\sqrt d/2560<\sqrt d/5,\qquad |R|<\sqrt d/3\le 1/3.$$

The Taylor bound holds on the whole domain. Alternating interval moments give $D_0\le 0$, $D_1\le\delta_3<d/6$, and $D_i\le\delta_4<7d/100$ for each $i\ge 2$. Positive classes are $i\equiv 1,2\pmod 4$, and

$$\sum_{i\ge1}(D_i)_+\le\frac{\delta_3+\delta_4}{1-\theta^2}\le\frac{\theta^2}{1-\theta}d=169d/608<7d/25.$$

These bound the finite Duhamel sums. With $\sigma(\bar f_a)\le 6\sqrt d/5$ and $\|R\|\le\sqrt d/5$, (Q95.2.1) gives $\delta_+\le 72d/125$. Use (Q95.2.2) with $n=a+L$ and the same V,N,c_L to apply (Q95.2.3). For $L\ge2$, $\|V\|\le(113/100)\sqrt d$, and

$$c_L+\mathbb{E}N\le(144/125+14/25+7/100-19/16)d=1189d/2000.$$

The total cost is $264633d/10000$, whereas $8\Gamma=276000d/10000$, leaving loss $11367d/10000>d$. For $L=1$, $\beta_{a+1}+\beta_{a+2}\ge 51d/32$, $\|V\|\le(7/5)\sqrt d$, and $c_1+\mathbb{E}N\le(72/125+1/6-51/32)d$. The cost is $39767d/1500$, while $8\Gamma=41400d/1500$, leaving $1633d/1500>d$. Thus $\mathbb{E}H^8\le e^{-d}$ for all these triples.

Q95.4 The triple $(2,3,4)$.

Use $F=\bar f_2 f_3 h_4$. Its finite expansion is

$$f_3+h_4=2-3d+dg-K^4\mathbf{1},\qquad R=dg-K^4\mathbf{1},\qquad F\le\bar f_2 e^{R-3d}.$$

Q95.4.1 The domain $1/6\le d\le 1/3$.

The row and spectral bounds give $|K^4\mathbf{1}|\le\theta\sqrt d\le d$ because $\theta^2<1/6$. Thus $R\le 2d$, $f_3h_4\le e^{-d}$, and $H\le e^{-d}p\bar f_{L+2}$. Finite convolution and (Q95.1.3) give $\mathbb{E}[(p\bar f_n)^8]\le e^{253d/32}$ for $n\ge3$: at $n=3$, the exponent is at most $-16d+17(d+\delta_2)\le 253d/32$, and at $n\ge4$, it is at most $-16\kappa d+17d=15d/2\le 253d/32$. Consequently $\mathbb{E}H^8\le e^{-3d/32}$ throughout this high domain.

Q95.4.2 The first step for $0<d\le 1/6$.

Put $\mu=\mathbb{E}\bar f_2=1-d$. Parity gives $\mathbb{E}R=-\delta_2$ and $\mathbb{E}R^2=\delta_4+d^3$. The pointwise range $|R|\le d+\theta\sqrt d<1/3$ permits the same Taylor estimate. For $t=\delta_2/d\in[d,\theta]$, $\sigma(\bar f_2)^2=(1+t-d)d$ and $\mathbb{E}R^2\le(\theta^2t+d^2)d$, so

$$\delta/d\le-(1-d)t+2\sqrt{1+t-d}\sqrt{\theta^2t+d^2}+(12/5)(\theta^2t+d^2).$$

Apply $2uv \le u^2/3 + 3v^2$. The resulting bound is

$$1/3 - d/3 + (27/5)d^2 + t[-2/3 + d + (27/5)\theta^2].$$

The coefficient of t is positive, its value at $d = 0$ being $3449/15360$. At $t = \theta$, the expression increases in d because $\theta - 1/3 = 7/96 > 0$. At $d = 1/6$, it is $1/16 + 59319/163840 + 7/576 + 3/20 < 3/5$, with positive margin $19601/1474560$. Hence $\delta_+ \le 3d/5$.

Q95.4.3 The signed propagation means.

For $D_i = \mathbb{E}K'^i R$, the first values are

$$D_0 = -\delta_2, \quad D_1 = d^2, \quad D_2 = \delta_3, \quad D_3 = -d\delta_2, \quad D_4 = -\delta_4,$$
$$D_5 = d\delta_3, \qquad D_6 = \delta_5.$$

Positive contributions continue in precisely two classes: $i = 1 + 4j$ contributes $d\delta_{1+2j}$, and $i = 2 + 4j$ contributes δ_{3+2j}. Therefore

$$\sum_{i \ge 1} (D_i)_+ \le \frac{d^2 + \delta_3}{1 - \theta^2} \le 1019d/2565 < 2d/5.$$

For $i \ge 5$, positive terms are at most $\max(d\delta_3, \delta_5) < d/36$, since $d \le 1/6$ and $\theta^2 < 1/6$. Also $D_2 < d/6$, while $D_3, D_4 \le 0$. The positive partial sum used for $L = 2$ is $d^2 \le d/6$, and for $L = 3, 4$ it is $d^2 + \delta_3 \le d/3$.

Q95.4.4 Finite predecessors for every $L \ge 2$.

Here $n = L + 2$, and

$$V = Q_L(K')g', \qquad Q_L(z) = G_{L+1}(z) - \mu d z^{L-1} - \mu z^{L+2}.$$

Write $x = -z^2 \in [0, \theta]$, $\gamma = \mu d \in [0, 5/36]$, and $\mu \in [5/6, 1]$. The real coordinates below are nonnegative. For $L = 2$, $\Re Q = 1 - x - \mu x^2 \in [0, 1 - x]$ and $\Im Q = (1 - \gamma)\sqrt{x}$; hence $|Q|^2 \le (1 - x)^2 + x \le 1$. For $L = 3$,

$$\Re Q = 1 - (1 - \gamma)x \le 1 - 31x/36,$$
$$\Im Q = \sqrt{x}(1 - x - \mu x^2) \in [0, \sqrt{x}(1 - x)],$$
$$|Q|^2 - 1 \le -13x/18 - 1631x^2/1296 + x^3 \le 0 \quad (x \le 1/2).$$

For $L = 4$, put $R_* = 1 - x + x^2 + x^3$. Then $\Re Q = 1 - x + x^2 + \mu x^3 \le R_*$ and $\Im Q \le \sqrt{x}(1 - 31x/36)$. Since $R_*^2 = 1 - 2x + 3x^2 - x^4 + 2x^5 + x^6$,

$$|Q|^2 - 1 \le -x + (23/18)x^2 + (961/1296)x^3 - x^4 + 2x^5 + x^6.$$

Drop $-x^4$ and divide by x. On $x \le 1/2$, the positive terms sum to at most $23/36 + 961/5184 + 1/8 + 1/32 = 5083/5184 < 1$. For $L = 5$,

$$\Re Q = 1 - x + (1 - \gamma)x^2 \le 1 - x + x^2,$$
$$\Im Q = \sqrt{x}(1 - x + x^2 + \mu x^3) \le \sqrt{x} R_*,$$
$$|Q|^2 - 1 \le -x + x^2 + x^3 + x^4 - x^5 + 2x^6 + x^7.$$

After $-x^5$ is dropped and division by x, the positive terms are at most $1/2+1/4+1/8+1/16+1/64 = 61/64 < 1$. For every $L \geq 6$, finite geometric summation and $(1+x)^{-1/2} \leq 1 - x/2 + 3x^2/8$ give

$$|Q_L| \leq 1 - x/2 + 3x^2/8 + x^{7/2} + (5/36)x^{5/2} + x^4.$$

Divide the positive error by x and use $x \leq 5/12$, $\sqrt{x} \leq 2/3$. Its four terms are at most $5/32+25/216+25/648+125/1728 = 1985/5184 < 2/5$. Thus $|Q_L| \leq 1 - x/10 \leq 1$. At $x = 0$, equality holds. For $L = 1$, separately, $Q_1 = 1 - \gamma + z - \mu z^3$, and $|Q_1|^2 \leq 1 + x(1+x)^2 \leq 1 + 26325/32768 < 29/16$.

Q95.4.5 The low-domain moment bounds.

Use $\Gamma = 3d$, the constants in Q95.2 and $\|V\|^2 \leq d$ for $L \geq 2$. For $L = 2$, $c_L + \mathbb{E}N \leq (6/5 + 1/3 + 1/6 - 19/16)d = 41d/80$, giving exponent at most $-29d/10$. For $L = 3, 4$, the positive partial sum is at most $d/3$ and $D_L \leq 0$, so $c_L + \mathbb{E}N \leq (6/5 + 2/3 - 19/16)d = 163d/240$, giving $-47d/30$. For every $L \geq 5$, $c_L + \mathbb{E}N \leq (6/5 + 4/5 + 1/36 - 19/16)d = 121d/144$, giving $-5d/18$. For $L = 1$, $\beta_3 + \beta_4 = 2d$, $D_1 = d^2$, $c_1 + \mathbb{E}N \leq -37d/30$, and the variance cost is at most $493d/16$. The resulting exponent is at most $-733d/240$. The low domain therefore gives $\mathbb{E}H^8 \leq e^{-5d/18}$. It meets the high domain at $d = 1/6$ with the same $F = \bar{f}_2 f_3 h_4$. The zero boundary and positive restoration of decorations give $\mathbb{E}H^8 \leq e^{-3d/32}$ for every $L \geq 1$.

Q95.5.1 Finite tails for the first short pair three.

Let $2 < 3 < c_1 < b_2 < c_2 < \cdots < b_k < c_k$, $k \geq 2$, and

$$F = \bar{f}_2 f_3 h_{c_1} \prod_{i=2}^{k} h_{b_i} f_{c_i}, \qquad S = \sum_{i=2}^{k} \sum_{r=b_i+1}^{c_i} K^r \mathbf{1}.$$

Every later $b_i \geq 5$, so its pair has loss $16d/5$. The finite zero-one mask S has first possible exponent $c_1 + 2$, and $\|S\|^2 \leq \theta^{b_2} C_* d$, $|S|^2 \leq \theta^{b_2 - 1} C_* d$, where $C_* = 32768/16245 < 100/49$ is (L90.2.6). Since $\theta^3 < 7/100$, $\sqrt{\theta} < 13/20$, and $\theta^5 C_* < 9/400$, we have

c_1	b_2 lower	$\|S\|/\sqrt{d}$ upper	$\|S\|/\sqrt{d}$ upper
4	5	3/20	1/4
5	6	1/10	3/20
≥ 6	≥ 7	7/100	1/10

For instance $\theta^6 C_* < (7/100)^2(100/49) = 1/100$, and the next norm is below $13/200 < 7/100$; pointwise bounds shift the power by one. In all cases $\mathbb{E}S \leq \delta_4/(1-\theta^2) < d/12$ and $D_1(S) < d/30$. The complete positive $D_i(S)$ tails are bounded respectively by $d/5, d/8, d/12$. For exponents $r \geq 6$, the first four positive moment indices are $\delta_5, \delta_4, \delta_5, \delta_6$, giving $\theta^3 d/(1-\theta)^2 = 2197d/11552 < d/5$. For $r \geq 7$, they are $\delta_5, \delta_6, \delta_5, \delta_6$, giving $2\theta^4 C_* d < (1/18)(100/49)d = 50d/441 < d/8$. For $r \geq 8$, they are $\delta_5, \delta_6, \delta_7, \delta_6$, giving $\theta^4 d/(1-\theta)^2 = 28561d/369664 < d/12$. For each $i \geq 2$, the individual bounds are $d/12, d/30, d/72$, respectively. The middle bound follows from $\delta_5/(1-\theta^2) < (1/36)/(5/6)d = d/30$; the last from $\delta_6/(1-\theta^2) < (1/90)/(5/6)d = d/75 < d/72$.

Q95.5.2 The first long arm $c_1 = 4$.

Take $R_0 = dg - K^4\mathbf{1}$, $R = R_0 + S$, and $\Gamma = [3 + (16/5)(k-1)]d$. First let $d \leq 1/6$. The pointwise range is less than $1/3 + (1/4)\sqrt{1/6} < 1/2$, so the same Taylor bound applies. Adding S

increases the first-step constant by at most

$$(\mathbb{E}S)_+ + 2\sigma(\bar{f}_2)\|S\| + (12/5)(2\|R_0\|\|S\| + \|S\|^2).$$

Since $\|R_0\|^2 < d/10 < d/9$ and $\sigma(\bar{f}_2) \le 6\sqrt{d}/5$,

$$\delta_+/d \le 3/5 + 1/12 + 9/25 + (12/5)(1/10 + 9/400) = 1003/750 < 27/20.$$

The positive D_i tail is at most $3d/5$, and each $D_L \le d/4$ for $L \ge 2$. The base Q_L of Q95.4.4 is a contraction for $L \ge 2$; adding the propagated S gives $\|V\| \le (11/10)\sqrt{d}$. For $k = 2, L \ge 2$, $c_L + \mathbb{E}N \le (27/10 + 6/5 - 19/16 + 1/4)d = 237d/80$. The cost apart from -8Γ is $4427d/100$, whereas $8\Gamma = 4960d/100$; the net loss is $533d/100 > 5d$. For $L = 1$, $\|Q_1 g'\| \le (27/20)\sqrt{d}$; adding S gives $\|V\| \le 3\sqrt{d}/2$. Also $D_1 \le d^2 + d/30 \le d/5$, $\beta_3 + \beta_4 = 2d$, and $c_1 + \mathbb{E}N \le (27/20 + 1/5 - 2)d = -9d/20$. The net loss for $k = 2$ is $299d/20 > 5d$. For $d \ge 1/6$, the first pair is at most e^{-d}, while $|S| \le (1/4)\sqrt{d} \le 5d/8$. The pair product is at most $e^{-[3/8+(16/5)(k-1)]d}$. The bare-path cost $253d/32$ leaves loss $3311d/160 > 5d$ at $k = 2$. In all cases an additional pair changes only Γ. Therefore

$$\mathbb{E}H^8 \le e^{-[5+(128/5)(k-2)]d} \qquad (c_1 = 4,\ k \ge 2,\ L \ge 1).$$

Q95.5.3 The exact base for $c = c_1 \ge 5$ and $d \le 1/32$.

Set $m = c - 3 \ge 2$, $B = \beta_{c-1}$, $\mu = 1 - d$, and

$$W = \sum_{r=4}^{c} K^r \mathbf{1} = K^3 G_m(K) g, \qquad R_0 = Bg - W, \qquad \Gamma_0 = d + \beta_c + B.$$

The finite difference gives $f_3 + h_c \le 2 - \Gamma_0 + R_0$, so $F \le \bar{f}_2 e^{R_0 + S - \Gamma_0 - (16/5)(k-1)d}$. For $c = 5$, $B = d$ and $\Gamma_0 = 3d - \delta_2 \ge 83d/32$. For $c \ge 6$, $\Gamma_0 \ge d + 2\tau \ge 35d/16$. Consecutive alternating moments give $\mathbb{E}W \ge 0$ and $\mathbb{E}(gW) \ge 0$. More precisely $\mathbb{E}W = \delta_2$ for $c = 5$, while $\mathbb{E}W \ge \delta_2 - \delta_3 \ge \kappa\delta_2$ for $c \ge 6$. Thus $\mathbb{E}R_0 = -\delta_2$ or at most $-\kappa\delta_2$, respectively, and $\mathbb{E}R_0^2 \le B^2 d + \mathbb{E}W^2$. For $t = \delta_2/d$,

$$\mathbb{E}R_0^2/d \le \begin{cases} \theta^2(1+\theta)t + d^2, & c = 5, \\ \theta^2 t + d^2, & c \ge 6. \end{cases}$$

For $c = 5$, $\mathbb{E}R_0^2 < d/10$, since $\theta^3(1+\theta) + 1/1024 = 99889/1048576 < 1/10$. Also $\|W\|^2 = \delta_4 + \delta_5 \le 98865d/1048576 < 25d/256$, so $\|W\| < 5\sqrt{d}/16$. For $c \ge 6$, the variance is below $9d/100$, using $\theta^3 < 7/100$ and $1/1024 < 1/100$. Including S, the pointwise range is at most $d + (51/80)\sqrt{d} < 1/32 + 51/400 = 127/800 < 1/3$.

Both base first-step constants have $(\delta_0)_+ \le 3d/5$. For $c \ge 6$, apply $2uv \le u^2/3 + 3v^2$. The bound divided by d is

$$1/3 - d/3 + (27/5)d^2 + t[1/3 - (1-d)\kappa + (27/5)\theta^2].$$

Its t-coefficient is positive. At $t = \theta$, the linear d-coefficient $\kappa\theta - 1/3$ is negative; dropping it and using $d^2 \le 1/1024$ yields $292389/491520 < 3/5$, with margin $2523/491520$. For $c = 5$, the corresponding expression is

$$1/3 - d/3 + (27/5)d^2 + t[-2/3 + d + (27/5)\theta^2(1+\theta)].$$

It increases in t. At $t = \theta$, the linear coefficient is $7/96 > 0$; at $d = 1/32$, the bound is $9109889/15728640 < 3/5$, with margin $327295/15728640$. For the base signed means, $-W$

contributes positive classes $i \equiv 2,3 \pmod 4$, totaling at most $(\delta_3+\delta_4)/(1-\theta^2)$. The Bg term contributes class $i \equiv 1 \pmod 4$, at most $Bd/(1-\theta^2)$. Hence

$$\sum_{i\ge 1}(D_i(R_0))_+ \le 169d/608 + 32d/855 < d/3.$$

Also $D_1(R_0) \le Bd \le d^2$, and every $D_i(R_0) \le \delta_3 < d/6$ for $i \ge 2$. At $i=3$, the Bg term is negative; at $i=5$, dropping the negative W term leaves $B\delta_3 \le \delta_3$.

Q95.5.4 A finite predecessor for all $c \ge 5, L \ge 2$ on the small domain.

Put $z = K'$ and $\gamma = \mu B \le 1/32$. The base predecessor is

$$Q_{L,c}(z) = G_{L+1}(z) - \gamma z^{L-1} - \mu z^{L+2} G_m(-z).$$

Write $G_m(-z) = U - zV$. For $m \ge 2$, both U, V are finite alternating sums in $[1-x,1]$; also $x \le \theta < 1/2$ and $\mu \ge 31/32$. For $L=2$, the real coordinate is positive and $1-x-\mu x^2 U \le 1-x-(31/64)x^2$. The imaginary coordinate $\sqrt{x}[1-\gamma+\mu x^2 V]$ is positive and at most $\sqrt{x}(1+x^2)$. Thus

$$|Q|^2 - 1 \le -x + (1/32)x^2 + (95/32)x^3 + (1/4)x^4 + x^5.$$

After division by x, the positive terms sum to at most $109/128 < 1$ on $x \le 1/2$. For $L=3$, $|G_4| \le (1-x)(1+x/2) = 1 - x/2 - x^2/2$. Adding γx and $(6/5)x^{5/2}$ gives $|Q| \le 1 - 15x/32 + 3x^2/10 \le 1 - 51x/160 \le 1$. For $L=4$, $|G_5| \le 1 - x/2 + x^2/2$, since the squared difference is $x^2(1+2x-3x^2)/4 \ge 0$. Adding $\gamma x^{3/2}$ and $(6/5)x^3$ gives $|Q| \le 1 - 23x/48 + x^2/2 + (6/5)x^3 \le 1 - x/16$, using $x/2 + (6/5)x^2 \le 5/12$ for $x \le 5/12$. For every $L \ge 5$, finite geometric summation and the negative third derivative of $(1+x)^{-1/2}$ give

$$|G_{L+1}| \le \frac{1+x^3}{\sqrt{1+x}} \le 1 - x/2 + 3x^2/8 + x^3.$$

The extra terms are at most $x^2/32 + (6/5)x^{7/2}$. After division by x, use $x \le 5/12$, $\sqrt{x} \le 2/3$ to bound $(13/32)x + x^2 + (6/5)x^{5/2} \le 185/384 < 1/2$. Therefore $|Q| \le 1 - 7x/384 \le 1$; equality at $x=0$ is direct. For $L=1$, the safe norm bounds are $31/20$ for $c=5$ and $3/2$ for $c \ge 6$. These use $\|G_2\| \le 6/5$, $\gamma \le 1/32$, and $\|W\|/\sqrt{d} < 5/16$ or $\theta^{3/2} < (13/32)(13/20)$, respectively.

Q95.5.5 Small-domain moment bounds.

For $c=5$, $\|R_0\| < \sqrt{d}/3$, $\|S\| \le \sqrt{d}/10$. The increase of δ/d is at most $1/12 + 6/25 + (12/5)(1/15 + 1/100) = 761/1500$, giving $\delta_+ \le 9d/8$. The positive D_i tail is at most $d/3 + d/8 = 11d/24$, and $D_L \le d/6 + d/30 = d/5$ for $L \ge 2$. The predecessor bound is $\|V\| \le (16/15)\sqrt{d}$, so $c_L + \mathbb{E}N \le 523d/240$. The cost is $523d/30 + 4352d/225$; for $k=2$, $8\Gamma \ge 927d/20$, leaving $17234d/1800 > 9d$. At $L=1$, $\|V\| \le 33\sqrt{d}/20$ and $c_1 + \mathbb{E}N \le (9/8 + 1/32 + 1/30 - 2)d = -389d/480$; the net loss is $7861d/1200 > 6d$.

For $c \ge 6$, $\|R_0\| < 3\sqrt{d}/10$, $\|S\| \le 7\sqrt{d}/100$, and

$$\delta \le [3/5 + 1/12 + 21/125 + (12/5)(21/500 + 49/10000)]d < d.$$

The positive tail is at most $d/3 + d/12 = 5d/12$, while $D_L \le d/6 + d/72 < d/5$ for $L \ge 2$. Now $\|V\| \le 21\sqrt{d}/20$ and $c_L + \mathbb{E}N \le 443d/240$, with total cost $443d/30 + 7497d/400$. Since $8\Gamma \ge 431d/10$, the net loss is $11509d/1200 > 9d$. At $L=1$, $\|V\| \le 8\sqrt{d}/5$ and $c_1 + \mathbb{E}N \le (1 + 1/32 + 1/30 - 2)d = -449d/480$, leaving $2119d/300 > 7d$. Additional pairs increase only 8Γ, by $128d/5$ for each pair.

Q95.5.6 The domain $d \ge 1/32$.

Here $\sqrt{d} \le \sqrt{32}\, d < (23/4)d$. For $c = 5$,

$$|R_0 + S| \le d + [(6/5)\theta + 3/20]\sqrt{d} \le d + (51/80)(23/4)d = 1493d/320.$$

At $k = 2$, $\Gamma \ge 927d/160$, so the pair product is at most $e^{-361d/320}$. Multiplying the bare path $p\bar{f}_{L+2}$ and using its $253d/32$ cost leaves $179d/160 > d$, for all $L \ge 1$. For $c \ge 6$, use $\|G_m\| \le 1$, obtaining $|R_0+S| \le d+(\theta+1/10)\sqrt{d}$. Even $\sqrt{d} < 6d$ gives $|R_0+S| \le 323d/80$. At $k = 2, \Gamma \ge 431d/80$, so the pair product is at most $e^{-27d/20}$; the bare-path cost leaves $463d/160 > 2d$. Both domains include $d = 1/32$ and keep the same direction. Thus

$$\boxed{\mathbb{E}H^8 \le e^{-[1+(128/5)(k-2)]d} \qquad (c_1 \ge 5,\ k \ge 2,\ L \ge 1).}$$

Together with Q95.5.2, the weaker bound holds for every $c_1 \ge 4$.

Q95.6 Relation to the ordinary rows.

Appendix O.21 proves the direction $\bar{f}_a \prod_i h_{b_i} f_{c_i}$, with bounds $e^{-[10+(128/5)(k-3)]d}$ for $a = 2, b_1 = 3, k \ge 3$, $e^{-[17+(128/5)(k-2)]d}$ for $a = 2, 3, b_1 = 4, k \ge 2$, and $e^{-[1+(128/5)(k-1)]d}$ for $a \ge 2, b_1 \ge 5$, $k \ge 1$. Q95.3 supplies $b_1 = 4, k = 1$, Q95.4 supplies $(2, 3, 4)$, and Q95.5 supplies $b_1 = 3, k \ge 2$ with first pair $f_3 h_{c_1}$. These integer ranges leave only the triple $(2, 3, c)$, $c \ge 5$, for the calculation of Q5-C097. For a list belonging to more than one row, choose one row from its integer structure and retain that row's full direction and moment bound. Positive comparison restores every genuine equal pair at its vertex. Replacing K by $-K$ reverses all arms, connector, receiving unit arm and decorations together, with the fixed measure and all δ_i preserved.

O.23 The remaining odd triple and two short bare-path eighth moments

Q97.0 The triples and their directions.

Work on a nonempty finite probability space with a real skew kernel $|K| \le 1$. The distinct residual terminal arms are $(2, 3, c)$, $c \ge 5$; genuine equal pairs may occur at the terminal and receiving root. The root has an outward unit arm. Every connector edge points from terminal to root. Choose the terminal direction from its integer connector length:

L	F	ordinary eighth bound	
1	$f_2 h_3 f_c$	$e^{-d/27}$	(Appendix O.16)
≥ 2	$\bar{f}_2 f_3 h_c$	e^{-d}	

For $H = E_r p T^L (E_t F)$, with genuine equal-pair factors E_t, E_r, we prove the second row:

$$\boxed{\mathbb{E}[\{E_r p T^L (E_t \bar{f}_2 f_3 h_c)\}^8] \le e^{-d} \qquad (c \ge 5,\ L \ge 2).} \tag{Q97.0.1}$$

Here f_n is fully outward, $\bar{f}_n$ fully inward, and h_n has only its first edge inward. The two directions are selected by L, before the kernel is given; the low and high domains use the same nonunit direction. With the first row they yield $\mathbb{E}H^8 \le e^{-d/27} \le 1 - d/28$ for all $L \ge 1$. We also prove, on the whole parameter domain,

$$\mathbb{E}[(p\bar{f}_4)^8] \le e^{3d}, \qquad \boxed{\mathbb{E}[(p\bar{f}_5)^8] \le e^{-425d/864}.} \tag{Q97.0.2}$$

Q97.1 Finite inputs and boundary cases.

Use the finite definitions of Appendix M.15: $A = J + K$, $T = J - K = A^*$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $f_n = A^n\mathbf{1}$, $\bar{f}_n = T^n\mathbf{1}$, $h_n = Tf_{n-1}$, $\delta_j = \|K^j\mathbf{1}\|_2^2$, $d = \delta_1$, $\theta = 13/32$, $\kappa = 19/32$, and $\tau = d - \delta_2$. All expectations, norms, rows and adjoints use the fixed measure. We have

$$\begin{aligned} &0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j, \quad \|K\|^2 \le \theta, \quad |Kv(x)| \le \|v\|_2, \\ &\mathbb{E}K^{2j}\mathbf{1} = (-1)^j\delta_j, \quad \mathbb{E}K^{2j+1}\mathbf{1} = 0, \quad 0 \le f_n, \bar{f}_n, h_n \le 2, \quad \mathbb{E}f_n = \mathbb{E}\bar{f}_n \le 1. \end{aligned} \tag{Q97.1.1}$$

The same spectral measure gives $\delta_3 \ge \delta_2^2/d$ for $d > 0$. For a mean-zero vector v, the variance of the row $K(x,\cdot) - g(x)$ is at most $1 - g(x)^2$, hence

$$|Kv(x)| \le \sqrt{p(x)q(x)}\|v\|_2. \tag{Q97.1.2}$$

Set $\beta_1 = \beta_2 = 0$ and $\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor}(-1)^{j-1}\delta_j$ for $n \ge 3$. Then $\tau \le \beta_n \le d$, $\beta_3 = \beta_4 = d$, $\beta_5 = \beta_6 = \tau$, and

$$S_n = \sum_{j=0}^{n} K^j\mathbf{1} = f_n + \sum_{j=3}^{n}\beta_j f_{n-j}. \tag{Q97.1.3}$$

Put $K' = -K$, $z = K'$, $g' = -g$; the reversed convolution uses $\bar{f}_n$. With $G_m(z) = \sum_{j=0}^{m-1} z^j$, $G_0 = 0$, $\|G_m\| \le 1$ for $m = 0, 1$ or $m \ge 3$, whereas $\|G_2\| < 6/5$. Frequencies have $z = i\sqrt{x}$, $0 \le x \le \theta$. The full-mean inequality in Appendix M.1 gives

$$\boxed{\mathbb{E}e^{8K'V} \le e^{8\mathbb{E}(K'V) + 17\|V\|_2^2}} \tag{Q97.1.4}$$

for every real V, including V/C for any scalar $C > 0$. The bare-path bound of (Q72.bare) is

$$\mathbb{E}[(p\bar{f}_n)^8] \le 1 - d/50 \qquad (n \ge 6). \tag{Q97.1.5}$$

After zero-mass atoms are removed, $p, q \ge \mu(x) > 0$. The all-equal labelling gives positive tree messages. If $d = 0$, $g = 0$, and leaf integration gives every message equal to one even when $K \ne 0$. Divisions below use $d > 0$. Genuine equal pairs satisfy $f_\ell + h_\ell = 2\mathbb{E}f_{\ell-1}$ and $0 \le f_\ell h_\ell \le 1$, and are restored by positive comparison.

Q97.2.1 The bare fourth path: cost at most $3d$.

Finite convolution gives the exact identity

$$\bar{f}_4 = (1 - d)q + zG_3(z)g' - d. \tag{Q97.2.1}$$

Put $C = 1 - d \ge 2/3$, $N = zG_3g'$. AM–GM on Cp and $\bar{f}_4$ gives

$$p\bar{f}_4 \le C\left(1 + \frac{N - d}{2C}\right)^2.$$

The base is $(Cp + \bar{f}_4)/(2C) \ge 0$. Using $y^{16} \le e^{16(y-1)}$ and (Q97.1.4),

$$\log \mathbb{E}[(p\bar{f}_4)^8] \le 8\log C + \frac{8}{C}(-d - \tau) + \frac{17}{C^2}(d - \delta_2 + \delta_3). \tag{Q97.2.2}$$

Here $\mathbb{E}N = -\tau$ and $\|G_3 g'\|^2 = d - \delta_2 + \delta_3$. Let $t = \delta_2/d \geq d$. Since $\delta_3 \leq \theta\delta_2$, the squared norm divided by d is at most $1 - \kappa t$. Use $\log C \leq -d$ and multiply the bound by C^2/d:

$$-7 + 32d - 8d^2 - (67/32 + 8d)t \leq -7 + (957/32)d - 16d^2. \tag{Q97.2.3}$$

Subtracting $3C^2$ from the last expression gives $-10 + (1149/32)d - 19d^2$. This increases on $[0, 1/3]$, since its derivative is at least $2231/96 > 0$, and its endpoint value is $-41/288 < 0$. Hence (Q97.2.2) is at most $3d$.

Q97.2.2 The bare fifth path: contraction.

Retaining three convolution terms gives

$$\bar{f}_5 = (1 - 2d)q + z(G_4(z) - d)g' - \tau. \tag{Q97.2.4}$$

Put $C = 1 - 2d \geq 1/3$, $V = (G_4 - d)g'$, $N = zV$. Then $\mathbb{E}N = -\tau + d^2$, and the spectral blocks give

$$\|V\|^2 = (1 - 2d + d^2)d - (1 - 2d)\delta_2 - \delta_3 + \delta_4.$$

Using $t = \delta_2/d \geq d$, $\delta_4 \leq \theta\delta_3$, and $\delta_3 \geq \delta_2^2/d$,

$$\|V\|^2/d \leq 1 - 2d + d^2 - Ct - \kappa t^2. \tag{Q97.2.5}$$

The same weighted AM–GM and exponential inequality yield

$$\begin{aligned} \log \mathbb{E}[(p\bar{f}_5)^8] \leq 8 \log C + \frac{8d}{C}(-2 + 2t + d) \\ + \frac{17d}{C^2}[1 - 2d + d^2 - Ct - \kappa t^2]. \end{aligned} \tag{Q97.2.6}$$

The inequality $\log C \leq -2d - 2d^2$, initially using only $-2d$, gives after multiplication by C^2/d $-15 + 70d - 63d^2 - Ct - 17\kappa t^2$, decreasing for $t \geq d$. Set $t = d$ and include the remaining logarithmic term $-16dC^2$. The result is

$$P(d) = -15 + 53d - (227/32)d^2 - 64d^3. \tag{Q97.2.7}$$

On $[0, 1/3]$, $P'(d) \geq 431/16 > 0$, and $P(1/3) = -425/864$. Thus the exponent is at most $-425d/(864C^2) \leq -425d/864$, proving (Q97.0.2).

Q97.3 The triple envelope and the exact first step.

Fix $c \geq 5, L \geq 2$, $F = \bar{f}_2 f_3 h_c$, and set

$$\begin{aligned} B = \beta_{c-1}, \quad m = c - 3 \geq 2, \quad W = \sum_{r=4}^{c} K^r \mathbf{1} = K^3 G_m(K) g, \\ R = Bg - W, \qquad \Gamma = d + \beta_c + B. \end{aligned} \tag{Q97.3.1}$$

Since $f_{c-1} \leq S_{c-1} - B$, positivity gives $h_c \leq TS_{c-1} - Bq$. Expanding and adding f_3 yields

$$f_3 + h_c \leq 2 - \Gamma + R, \qquad 0 < F \leq \bar{f}_2 e^{R-\Gamma}. \tag{Q97.3.2}$$

At $c = 5$, $B = d$ and $\Gamma = 3d - \delta_2 = 2d + \tau \geq 83d/32$. For $c \geq 6$, $B \geq \tau$ and $\Gamma \geq d + 2\tau \geq 35d/16$. The exponential envelope follows by AM–GM on the two nonnegative arms.

First suppose $0 < d \le 1/8$. The row estimate and G_2 bound give $|R| \le d + (6/5)\theta\sqrt{d} < 1/8 + (39/80)(3/8) < 1/2$. The factorial bound $n! \ge 2 \cdot 3^{n-2}$, $n \ge 2$, bounds the remainder by $R^2/[2(1-|R|/3)] \le 3R^2/5$. Write $r = Kg$, $G = g - r$, $\bar{f}_2 = 1 - G$, so $\mathbb{E}G^2 = d + \delta_2$. Keeping the complete mixed mean in the first row gives

$$T(\bar{f}_2 e^R) \le \bar{f}_3 + zR + \delta,$$
$$\boxed{\delta = \mathbb{E}R - \mathbb{E}(gR) + \mathbb{E}(rR) + \sqrt{(d+\delta_2)\mathbb{E}R^2} + \frac{12}{5}\mathbb{E}R^2.} \tag{Q97.3.3}$$

Indeed, $T(\bar{f}_2 R) = \mathbb{E}R - \mathbb{E}(GR) - KR + K(GR)$, $K(GR) \le \mathbb{E}|GR| \le \sqrt{(d+\delta_2)\mathbb{E}R^2}$, and $T(\bar{f}_2 R^2) \le 4\mathbb{E}R^2$. For $D_i = \mathbb{E}z^i R$, finite propagation is

$$e^{\Gamma} T^L F \le Y_L = \bar{f}_{L+2} + z^L R + \delta \bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{Q97.3.4}$$

The first perturbation is zR, with coefficient one.

Q97.4.1 The first-step constant for $c = 5$.

Put $t = \delta_2/d$, $u = \delta_3/d$. The exact mixed moments are

$$\mathbb{E}R - \mathbb{E}(gR) + \mathbb{E}(rR) = -\delta_2 - d^2 + 2\delta_3, \qquad \mathbb{E}R^2 = d^3 + \delta_4 + \delta_5 - 2d\delta_3. \tag{Q97.4.1}$$

Since $\theta(1+\theta) - 2d > 0$ and $u \le \theta t$, $\mathbb{E}R^2/d \le J = d^2 + [\theta^2(1+\theta) - 2\theta d]t$. Insert $\sqrt{(1+t)J} \le (1+t)/6 + (3/2)J$ into (Q97.3.3):

$$\begin{aligned} \delta/d &\le 1/6 - d + (39/10)d^2 \\ &\quad + t[-1/48 + (39/10)\theta^2(1+\theta) - (39/5)\theta d]. \end{aligned} \tag{Q97.4.2}$$

The t-coefficient is positive for $d \le 1/8$. A lower bound, using $\theta^2 > 4/25$, $1 + \theta > 7/5$, $\theta < 5/12$, is $-1/48 + (39/10)(4/25)(7/5) - (39/5)(5/12)(1/8) > 0$. Set $t \le \theta$. The resulting polynomial has derivative at most $-1 - (39/5)\theta^2 + 39/40 < 0$. Its value at zero is

$$81/512 + (39/10)(98865/1048576) = 5514615/10485760 < 17/32, \tag{Q97.4.3}$$

with margin $55945/10485760$. Thus $\delta_+ \le 17d/32$.

Q97.4.2 The first-step constant for $c \ge 6$.

Let $e_j = \delta_j - \delta_{j+1}$. Then $e_j \ge 0$ and $e_{j+1} \le \theta e_j$. Consecutive alternating sums give $\mathbb{E}W + \mathbb{E}(rW) \ge \kappa(\delta_2 - \delta_3)$ and $\kappa\delta_3 \le \mathbb{E}(gW) \le \delta_3$. Together with $B \ge \tau \ge \kappa d$,

$$\begin{aligned} \mathbb{E}R - \mathbb{E}(gR) + \mathbb{E}(rR) &\le -\kappa\delta_2 + (1+\kappa)\delta_3 - Bd, \\ \mathbb{E}R^2 &\le d^3 + (\theta - 2\kappa^2 d)\delta_3. \end{aligned} \tag{Q97.4.4}$$

Here $\theta - 2\kappa^2 d > 0$. Using $u \le \theta t$, the mixed mean divided by d is at most $(55/1024)t - \kappa d$, and $J = \mathbb{E}R^2/d \le d^2 + [\theta^2 - 2\kappa^2\theta d]t$. Apply $\sqrt{(1+t)J} \le (1+t)/8 + 2J$:

$$\begin{aligned} \delta/d &\le 1/8 - \kappa d + (22/5)d^2 \\ &\quad + t[1/8 + 55/1024 + (22/5)\theta^2 - (44/5)\kappa^2\theta d]. \end{aligned} \tag{Q97.4.5}$$

The t-coefficient is positive: dropping its first two positive terms and using $d \le 1/8$ leaves $(22/5)\theta^2 - (11/10)\kappa^2\theta = (11/10)\theta(4\theta - \kappa^2) > 0$. After $t \le \theta$, the derivative in d is at most $11/10 - \kappa - (44/5)\kappa^2\theta^2 < 0$. The exact comparison is $81/160 = 2654208/5242880 < 2684396/5242880$. The value at zero is $80709/163840 < 1/2$, with margin $1211/163840$. Hence $\delta_+ \le d/2$ for every $c \ge 6$.

Q97.4.3 All signed propagation tails.

The positive means from $-W$ have classes $i \equiv 2, 3 \pmod 4$, with first terms at most δ_3, δ_4. Those from Bg have class $i \equiv 1$, first term Bd. Increasing i by four increases the moment index by two. Thus every finite positive sum obeys

$$\sum_{i \ge 1} (D_i)_+ \le \frac{\delta_3 + \delta_4 + Bd}{1 - \theta^2} \le \frac{169}{608} d + \frac{1024}{855} d^2. \tag{Q97.4.6}$$

The individual bounds are

$$\begin{aligned} -\delta_3 \le D_1 \le d^2, \quad 0 \le D_2 \le \delta_3, \quad D_3 \le \delta_4, \\ D_i \le \max(\delta_5, B\delta_3) \le \theta^4 d < d/36 \qquad (i \ge 4). \end{aligned} \tag{Q97.4.7}$$

For the last bound, $d \le 1/8 < \theta^2$. At $i \equiv 0$, both parts are nonpositive; class $i \equiv 1$ starts at $i = 5$ with $B\delta_3$ and a nonpositive other part; classes $i \equiv 2, 3$ start at $i = 6, 7$ with δ_5, δ_6 and nonpositive Bg part.

Q97.5 Four retained path losses for $L \ge 4$, $d \le 1/8$.

Here $n = L + 2 \ge 6$. Retain the four nonnegative convolution terms $\tau\bar{f}_0, \tau\bar{f}_1, \tau\bar{f}_2, \tau\bar{f}_3$:

$$\bar{f}_n \le (1 - 3\tau)q + z[G_{L+1}(z) - \tau(2 + z)]g' - \tau(1 - d). \tag{Q97.5.1}$$

Indeed, $\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3 = 1 - d + 3q + 2Kg - K^3\mathbf{1}$. Set $C = 1 - 3\tau > 0$, $a = 1 - 2\tau$, and

$$P_{L,c}(z) = G_{L+1}(z) - \tau(2 + z) - Bz^{L-1} - z^{L+2}G_m(-z). \tag{Q97.5.2}$$

As $R = -Bg' - z^3 G_m(-z)g'$, the term $z^L R$ in (Q97.3.4) combines with (Q97.5.1) as $zP_{L,c}g'$. We prove

$$\boxed{\|P_{L,c}(K')\| \le 1 - 2\tau = a.} \tag{Q97.5.3}$$

Q97.5.1 The frequency certificate for $L = 4$.

At $z = i\sqrt{x}$, $0 \le x \le \theta$, write $G_m(-z) = U - zV$, where $1 - x \le U, V \le 1$. With $v = 1 - \tau$, the coordinates are

$$\Re P = a - x + x^2 + x^3 U, \qquad \Im P/\sqrt{x} = v - (1 - B)x - x^3 V.$$

They are nonnegative since $a \ge 3/4$, $v \ge 7/8$, and $x + x^3 < 1/2$, and are at most $a - x + x^2 + x^3$ and $v - 7x/8$, respectively. Squaring gives

$$\begin{aligned} |P|^2 \le a^2 + (-1 + 2\tau + \tau^2)x + (5/4 - 9\tau/4)x^2 \\ + (49/64 - 4\tau)x^3 - x^4 + 2x^5 + x^6. \end{aligned}$$

Since $\tau \le 1/8$, the first coefficient is at most $-47/64$, and the next two positive coefficients are at most $5/4, 49/64$. Drop $-x^4$ and divide by x, treating $x = 0$ by equality. For $x \le 5/12$, the total positive cost is at most

$$\frac{25}{48} + \frac{1225}{9216} + \frac{625}{10368} + \frac{3125}{248832} = \frac{180800}{248832} < \frac{182736}{248832} = \frac{47}{64}.$$

Thus $|P|^2 \le a^2$ at every frequency and for every c.

Q97.5.2 The frequency certificate for $L = 5$.

The real coordinate is $a - x + (1 - B)x^2 + x^4V$, and the imaginary coordinate divided by $\sqrt{x}$ is $v - x + x^2 + x^3U$, both nonnegative. Bound these by $a - x + x^2 + x^4$ and $v - x + x^2 + x^3$, respectively:

$$\begin{aligned}|P|^2 - a^2 \le x[&-1 + 2\tau + \tau^2 + (1 - 2\tau)x + (1 - 2\tau)x^2 \\ &+ (3 - 6\tau)x^3 - 3x^4 + 4x^5 + x^6 + x^7].\end{aligned}$$

The derivative of the bracket in τ is at least $2 - 2x - 2x^2 - 6x^3 \ge 37/96 > 0$ for $x \le 5/12$. Take $\tau = 1/8$ and drop $-3x^4$. The remaining positive terms are $(3/4)x + (3/4)x^2 + (9/4)x^3 + 4x^5 + x^6 + x^7$. Using $x \le 5/12$, $x^2 < 7/40$, $x^3 < 3/40$, $x^5 < 1/72$, $x^6 < 1/180$, $x^7 < 1/400$, their sum is below

$$49/80 + 1/18 + 1/180 + 1/400 = 1217/1800 < 47/64.$$

This proves $|P|^2 \le a^2$.

Q97.5.3 Every $L \ge 6$.

For comparison with the finite geometric remainder, define the scalar rational function $P_\infty(z) = 1/(1 - z) - \tau(2 + z)$. Directly,

$$a^2 - |P_\infty|^2 = x\left[\frac{a}{1 + x} - \tau^2\right] \ge \frac{559}{1088}x.$$

The last coefficient follows from $a \ge 3/4$, $x \le 5/12$, $\tau \le 1/8$, giving $9/17 - 1/64$. Also $a + |P_\infty| \le 2a \le 2$, so $|P_\infty| \le a - 559x/2176 \le a - x/4$. The finite G_{L+1} remainder is at most $x^{(L+1)/2}/\sqrt{1 + x} \le x^{7/2}$. The other two terms in (Q97.5.2) are at most $x^{5/2}/8 + (6/5)x^4$. After division by x, their combined bound, using $x \le 5/12$, $\sqrt{x} \le 2/3$, is

$$25/216 + 5/144 + 25/288 = 205/864 < 1/4.$$

Hence $|P_{L,c}| \le a - 11x/864 \le a$, with equality at $x = 0$. This completes (Q97.5.3) for every finite $L \ge 4, c \ge 5$.

Q97.5.4 Weighted AM–GM and the cost.

Put $P_{\text{cost}} = 2\delta_+ + 2\sum_{i \ge 1}(D_i)_+$, which bounds the cost of every finite propagation. From (Q97.3.4), (Q97.5.1) and $\bar{f}_j \le 2$,

$$0 < e^\Gamma T^L F \le Cq + N + P_{\text{cost}} - \tau(1 - d), \qquad N = zP_{L,c}(z)g'.$$

Here $C = 1 - 3\tau \ge 5/8$, $\mathbb{E}N = -\beta_{n+1} + 2\tau d + D_L$, and $\|P_{L,c}g'\|^2 \le (1 - 2\tau)^2 d$. AM–GM on Cp and this nonnegative upper bound, followed by (Q97.1.4) applied to $P_{L,c}g'/C$, gives

$$\begin{aligned}\log \mathbb{E}[(pT^L F)^8] \le\ &-8\Gamma + 8\log C \\ &+ \frac{8}{C}[P_{\text{cost}} - \tau(1 - d) - \beta_{n+1} + 2\tau d + D_L] \\ &+ \frac{17}{C^2}(1 - 2\tau)^2 d.\end{aligned} \tag{Q97.5.4}$$

Q97.5.5 The two-variable payment for $c \ge 6$.

The preceding bounds give $P_{\text{cost}} \le (473/304)d + (2048/855)d^2$ and $D_L \le d/36$. Put $b = 1024/855$, $A_0 = 4333/2736$. The bracket in (Q97.5.4) is at most $A_0 d + 2bd^2 - 2\tau + 3\tau d$, and $\Gamma \ge d + 2\tau$. Use $\log C \le -3\tau$, multiply by C^2/d, and set $s = \tau/d \in [\kappa, 1]$. Bound the positive costs by $A = 8/5$, $b = 6/5$, obtaining

$$\begin{aligned} F(s,d) &= 9 + 8A - 56s + d[288s^2 + (4 - 24A)s + 16b] \\ &\quad - d^2[360s^3 + 76s^2 + 48bs]. \end{aligned} \tag{Q97.5.5}$$

For $d \le 1/8$, the derivative of F_s in d is at least $538s - 270s^2 + 4 - 24A - 12b \ge 268s + 4 - 24A - 12b > 0$. Its maximum in d is thus at $1/8$; there it increases in $s \le 1$ and has value $-11/4 - 3A - 3b/4 < 0$ at $s = 1$. Consequently F decreases in s, with maximum at $s = \kappa$. At this s, the linear d-coefficient exceeds 75, and the absolute negative quadratic coefficient is below 140, using $\kappa^2 > 1/3$, $\kappa < 3/5$. Hence the d-derivative exceeds $75 - 140/4 = 40$, and the maximum is at $d = 1/8$. Its exact value is

$$\begin{aligned} F(\kappa, 1/8) &= 121/5 - (306/5)\kappa + (557/16)\kappa^2 - (45/8)\kappa^3 \\ &= -\frac{1365979}{1310720} < -1. \end{aligned} \tag{Q97.5.6}$$

Thus (Q97.5.4) is at most $-d/C^2 \le -d$.

Q97.5.6 The payment for $c = 5$.

Here P_{cost} increases by $d/16$, so $A_0 = 563/342 < 5/3$, and $\Gamma = 2d + \tau$. With $A = 5/3$, $b = 6/5$, the scaled expression is

$$\begin{aligned} F_5(s,d) &= 1 + 8A - 48s + d[240s^2 + (52 - 24A)s + 16b] \\ &\quad - d^2[288s^3 + 148s^2 + 48bs]. \end{aligned} \tag{Q97.5.7}$$

Its s-derivative increases in d, because the corresponding derivative is bounded below by $406s - 216s^2 + 52 - 24A - 12b \ge 190s + 52 - 24A - 12b > 0$. At $d = 1/8$, $(F_5)_s$ increases in $s \le 1$, with value $3/8 - 3A - 3b/4 < 0$ at $s = 1$. The maximum is again at $s = \kappa$. There the linear coefficient exceeds 80 and the absolute negative quadratic coefficient is below 152; the d-derivative is thus greater than $80 - 152/4 > 0$. At the endpoint,

$$\begin{aligned} F_5(\kappa, 1/8) &= 251/15 - (237/5)\kappa + (443/16)\kappa^2 - (9/2)\kappa^3 \\ &= -\frac{2547481}{983040} < -2. \end{aligned} \tag{Q97.5.8}$$

This row gives $\mathbb{E}H^8 \le e^{-2d}$. Together the last two calculations prove e^{-d} for all $c \ge 5, L \ge 4, d \le 1/8$.

Q97.6 The two short connectors on the low domain.

Fix $\bar{\delta} = 17d/32 \ge \delta$ and $\lambda = d - \bar{\delta} = 15d/32$. Replacing δ by $\bar{\delta}$ increases the same positive upper bound. Put

$$Q_L(z) = G_{L+1}(z) - Bz^{L-1} - z^{L+2}G_m(-z). \tag{Q97.6.1}$$

Q97.6.1 The connector $L = 2$.

The same $U, V \in [1-x, 1]$ give $\Re Q_2 = 1 - x - x^2 U \in [0, 1 - x - x^2/2]$ and $\Im Q_2/\sqrt{x} = 1 - B + x^2 V \in [0, 1 + x^2]$. Thus $|Q_2|^2 \le 1 - x + 3x^3 + x^4/4 + x^5 \le 1$, since after division by x, the positive cost is at most $27/32$ for $x \le 1/2$. The exact convolution $\bar{f}_4 = S_4' - d(1+q)$ and finite propagation give $p + Y \le 2 + c + zV_{\text{tot}}$, with

$$c = \bar{\delta} + D_1 - 2d, \qquad V_{\text{tot}} = Q_2(z) g' - \lambda \mathbf{1}, \qquad \mathbb{E}(zV_{\text{tot}}) = -\tau + D_2.$$

By (Q97.4.7), $d \le 1/8$ and $\delta_3 < d/6$,

$$c + \mathbb{E}N \le [17/32 + 1/8 + 1/6 - 2 - 19/32]d = -85d/48 \le -7d/4.$$

Since $\mathbb{E}(Q_2 g') = -d + D_1$,

$$\|V_{\text{tot}}\|^2 \le d + \lambda^2 + 2\lambda(d - D_1) \le d + (10/3)d^2 \le 17d/12 < 3d/2.$$

AM–GM and (Q97.1.4) give exponent at most $-8(35/16)d - 14d + (51/2)d = -6d$. This proves the low-domain $L = 2$ bound e^{-6d} for every $c \ge 5$.

Q97.6.2 The connector $L = 3$.

The bound $|G_4| = (1-x)\sqrt{1+x} \le 1 - x/2 - x^2/2$, together with $B \le 1/8$, $\|G_m\| \le 6/5$ and $\sqrt{x} \le 13/20$, gives $|Q_3| \le 1 - 3x/8 + (7/25)x^2 \le 1$. Also $\Re Q_3 = 1 - x + Bx - x^3 V \ge 1 - x - x^3 > 1/2$. Since $\lambda \le 1/8$, $|Q_3 - \lambda|^2 \le |Q_3|^2 - 2\lambda \Re Q_3 + \lambda^2 \le 1$. Use the exact convolution $\bar{f}_5 = S_5' - \tau - dq - d\bar{f}_2$. Then

$$c = \bar{\delta} + D_1 + D_2 - 2d - \tau, \quad V_{\text{tot}} = (Q_3 - \lambda) g' - \eta \mathbf{1},$$
$$\eta = 2d - \bar{\delta} - D_1 > 0, \qquad \mathbb{E}(zV_{\text{tot}}) = -\tau + \lambda d + D_3.$$

As $D_1 \le d^2$, $D_2 \le \delta_3 < d/6$, $D_3 \le \delta_4 < d/14$,

$$c + \mathbb{E}N \le [17/32 + 1/4 + 1/6 + 1/14 - 2 - 19/16]d = -1457d/672 \le -2d.$$

On the other hand, $D_1 \ge -\delta_3$ gives $\eta \le 157d/96 < 5d/3$. Also $\mathbb{E}[(Q_3 - \lambda)g'] = -\tau + D_2 \in [-\tau, 0]$, since $\tau \ge \kappa d > \delta_3$. Therefore

$$\|V_{\text{tot}}\|^2 \le d + \eta^2 + 2\eta\tau \le d + (55/9)d^2 \le 127d/72 < 9d/5.$$

The total exponent is at most $-35d/2 - 16d + 153d/5 = -29d/10 < -2d$. Thus every $c \ge 5$ has the low-domain $L = 3$ bound e^{-2d}.

Q97.7 The high domain with the row mean retained.

Let $1/8 \le d \le 1/3$, keeping $F = \bar{f}_2 f_3 h_c$. Set

$$U = \sum_{r=3}^{c-1} K^r \mathbf{1} = K^2 G_m(K) g, \qquad W = KU.$$

The finite alternating sum gives $\mathbb{E}U = d - \beta_c$, so $B - \mathbb{E}U = \Gamma - 2d \in [0, d]$. The centered row estimate (Q97.1.2) gives

$$R = (\Gamma - 2d)g - K(U - \mathbb{E}U),$$
$$|R|^2 \le (\Gamma - 2d)^2 + \operatorname{Var}(U) \le d^2 + \|U\|^2. \tag{Q97.7.1}$$

The middle inequality is Cauchy on the two coordinates $g, \sqrt{1-g^2}$. For $c \geq 6$, $m \geq 3$, and $\|U\|^2 \leq \delta_3 \leq \theta^2 d$. Thus $|R|/d \leq \sqrt{1+8\theta^2} = \sqrt{297/128} < 25/16$. Since $\Gamma/d \geq 35/16$, $\Gamma - R \geq 5d/8$. For $c = 5$, $\|U\|^2 = \delta_3 + \delta_4 \leq \theta^2(1+\theta)d$, so $|R|/d \leq \sqrt{11701/4096} < 7/4$. Now $\Gamma/d \geq 83/32$ yields $\Gamma - R \geq 27d/32 \geq 5d/8$. Equation (Q97.3.2) therefore gives $f_3 h_c \leq e^{-5d/8}$ for every $c \geq 5$. Positivity implies $H \leq e^{-5d/8} p \bar{f}_{L+2}$. Use the new $n = 4$ cost for $L = 2$, the new $n = 5$ contraction for $L = 3$, and (Q97.1.5) for $L \geq 4$. Uniformly,

$$\mathbb{E}H^8 \leq e^{-2d} \qquad (L \geq 2). \tag{Q97.7.2}$$

At $d = 1/8$ this meets the low-domain proof with the same direction. The weakest low-domain bound is e^{-d}, proving (Q97.0.1). The zero boundary, genuine decorations and full reversal are included by the finite positive comparison. Full reversal replaces K by $-K$ and reverses all arms, connector and receiving unit arm together while preserving the measure and δ_j.

Q97.8 The ordinary catalogue for minimum arm at least two.

For $a < b_1 < c_1 < \cdots < b_k < c_k$, $a \geq 2, k \geq 1$, the following choices depend on the integer structure:

range	F	$Q(d)$	proof
$b_1 \geq 5$	$f_a \prod_i h_{b_i} f_{c_i}$	e^{-d}	$O.21$
$b_1 = 4,\ a = 2,3$	$\bar{f}_a \prod_i h_{b_i} f_{c_i}$	e^{-d}	$k = 1:\ O.22;\ k \geq 2:\ O.21$
$b_1 = 3,\ a = 2,\ k \geq 2$	$\bar{f}_2 f_3 h_{c_1} \prod_{i=2}^{k} h_{b_i} f_{c_i}$	e^{-d}	$O.22$
$(a, b_1, c_1) = (2,3,4),\ k = 1$	$\bar{f}_2 f_3 h_4$	$e^{-3d/32}$	$O.22$
$a = 2,\ k = 1,\ b_1 = 3,\ c_1 \geq 5,\ L = 1$	$\bar{f}_2 h_3 f_{c_1}$	$e^{-d/27}$	$O.16$
$a = 2,\ k = 1,\ b_1 = 3,\ c_1 \geq 5,\ L \geq 2$	$\bar{f}_2 f_3 h_{c_1}$	e^{-d}	(Q97.0.1)

Except for the last two rows, the listed ranges have every $L \geq 1$. Strict integer order forces $b_1 = 3, 4$ or $b_1 \geq 5$, with $a = 2$ when $b_1 = 3$. The rows therefore give

$$\boxed{\mathbb{E}[\{E_r p T^L (E_t F)\}^8] \leq e^{-d/27} \leq 1 - d/28.} \tag{Q97.8.1}$$

The last inequality follows from $e^x \geq 1 + x$ and $27 + d \leq 28$. Each row also retains its displayed stronger bound in its own direction.

O.24 Ordinary moments at an actual unit root

Throughout this subsection the receiving vertex has one existing outward unit arm. Its message is p. The inner path has length $L \geq 1$ and contributes T^L. At its other endpoint let F be the terminal message specified below. We estimate

$$H = pT^L F, \qquad \mathcal{J}(U) = \mathbb{E}[q^{-1/3}|U|^{8/3}].$$

The measure, operators and finite spectral measure are those of (M.1). Write $B = -K$, $u = B\mathbf{1} = -g$, $z = i\sqrt{x}$, $0 \leq x \leq \theta = 13/32$, $\kappa = 19/32$, and $\tau = d - \delta_2$. Thus $\int x^j \, d\nu = \delta_{j+1}$ and $\nu([0,\theta]) = d$. All polynomials and masks in this proof are finite. The proof works with $d > 0$; its last paragraph treats $d = 0$ directly.

Theorem O.2 (The unit-root ordinary catalogue)**.** *Terminal residual lists are strictly increasing:* $(a, b_1, c_1, \ldots, b_k, c_k)$*, with* $1 \leq a < b_1 < c_1 < \cdots < b_k < c_k$ *and* $k \geq 1$*. The directions in Table 65*

depend only on the original residual lengths. They satisfy the fourth and eighth moment bounds proved in the corresponding paragraphs below. Every fourth-moment entry in Table 65 has loss at least $15d/4$. Every eighth-moment row with terminal minimum one has bound $1 - d/12000$; the sharper rates in the tables are retained. These assertions allow every positive L, all genuine equal-pair decorations, and complete reversal.

Table 65: Terminal messages at the unit root. An absent eighth entry carries no eighth-moment estimate for that direction.

Residual lengths	Actual F	Moments proved
$(a, b_1, c_1, \dots, b_k, c_k),\ b_1 \geq 4$	$\bar{f}_a \prod_i h_{b_i} f_{c_i}$	4; also 8 if $a = 1$
$b_1 = 3,\ k \geq 2,\ a = 1, 2$	$\bar{f}_a f_3 h_{c_1} \prod_{i \geq 2} h_{b_i} f_{c_i}$	4; also 8 if $a = 1$
$(a, 3, c),\ a = 1, 2,\ c \geq 4$	$\bar{f}_a f_3 h_c$	4; also 8 if $a = 1$
$(1, 2, c_1, \dots, b_k, c_k),\ k \geq 3$	$q f_2 h_{c_1} \prod_{i \geq 2} h_{b_i} f_{c_i}$	4, 8
$(1, 2, c, b, e),\ 4 \leq c < b < e$	$q f_2 h_c h_b f_e$	4, 8
$(1, 2, 3, b, e),\ 5 \leq b < e$	$q f_2 h_3 h_b f_e$	4, 8
$(1, 2, 3, 4, e),\ e \geq 5$	$q f_2 h_e h_3 f_4$	4, 8
$(1, 2, c),\ c \geq 4$	$q f_2 h_c$	8
$(1, 2, 3),\ L \neq 2$	$q f_2 h_3$	8
$(1, 2, 3),\ L = 2$	$q f_2 \bar{f}_3$	8

O.24.1 One finite ledger and its two moment payments.

We use Theorem M.18 on the original positive row. The four first-row forms needed here are

F upper	s	r	λ
$e^{R-\Gamma}$	$e^{-\Gamma}$	1	1
$\bar{f}_a e^{R-\Gamma}$	$e^{-\Gamma}$	$a+1$	$\mu_a := \mathbb{E}\bar{f}_a$
$q e^{R-\Gamma}$	$e^{-\Gamma}$	2	1
$q(m - U/2)^2$	m^2	2	$1, \quad R = -U/m.$

(O.47)

For the exponential inputs $|R| \leq 1/2$, except for the explicitly larger interval in the first-pair-two multipair case. The respective admissible first-row constants are as follows. The coefficient $6/5$ in the first line is replaced by 2 on the larger interval $|R| \leq 3/2$.

$$\begin{aligned} D_E &\geq \mathbb{E}R + \tfrac{6}{5}\mathbb{E}R^2, \\ D_A &\geq \mu_a \mathbb{E}R + 2\sigma(\bar{f}_a)\|R\| + \tfrac{12}{5}\mathbb{E}R^2, \\ D_Q &\geq \mathbb{E}(qR) + \sqrt{d\mathbb{E}R^2} + \tfrac{12}{5}\mathbb{E}R^2, \\ D_{QS} &\geq -\frac{\mathbb{E}U - \mathbb{E}(gU)}{m} + \frac{\sqrt{d\mathbb{E}U^2}}{m} + \frac{\mathbb{E}U^2}{m^2}. \end{aligned} \tag{O.48}$$

We always enlarge to a nonnegative D. In the first two lines the linear vector has coefficient λ; in the last two it has coefficient one. In particular the $a = 2$ uncentred mode used below has coefficient one and squared distance $d + \delta_2$, rather than coefficient $1 - d$ and centered variance $d + \delta_2 - d^2$. Here R denotes the seed before multiplication by λ; the normalized linear vector in the transport theorem is λR.

Put $D_i = \mathbb{E}B^i R$. Positive continuation gives the actual finite upper

$$T^L F \leq s Y_L, \qquad Y_L = \bar{f}_{r+L-1} + \lambda B^L R + D\bar{f}_{L-1} + \lambda \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{O.49}$$

For a q-weighted input, $\lambda = 1, r = 2$, this is equivalently

$$Y_L = S'_{L+1} + B^L R + D\bar{f}_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar{f}_{L+1-j}, \qquad \gamma_j = \beta_j - D_{j-2}. \tag{O.50}$$

Only after proving $\gamma_j \geq 0$ do we drop a loss.

We record the complete comparison used to pay both moments:

$$0 \leq Y_L \leq Cq + BV + c, \qquad C > 0. \tag{O.51}$$

The upper is positive because it majorizes the actual message. The unit pairing and the signed-exponential estimate give, for $\nu = 4, 8$,

$$\log \mathbb{E}H^\nu \leq \nu \log(sC) + \frac{\nu}{C}(c + \mathbb{E}BV) + \frac{17\nu^2}{64C^2}\|V\|^2. \tag{O.52}$$

Indeed $pY_L \leq C[1 + (BV + c)/(2C)]^2$, with nonnegative base, and $\mathbb{E}e^{tBV} \leq \exp(t\mathbb{E}BV + 17t^2\|V\|^2/64)$ for $t \geq 0$. For one specified norm upper N, the fourth payment equals one half of its eighth payment minus $17N/(4C^2)$. This relation concerns the two proof expenses for this same vector.

In short rows $V = P(B)u + e\mathbf{1}$, its complete norm is

$$\|V\|^2 = \int |P|^2 \, d\nu + 2e\,\mathbb{E}[P(B)u] + e^2. \tag{O.53}$$

We retain the mixed term. The subsequent tables specify either this complete vector or a proved positive enlargement of it.

Lemma O.3 (Short-ledger normalization)**.** *Write $R = P_R(B)u$, with P_R its finite polynomial, and put $Q_L = G_{r+L-2} + z^{L-1}P_R$. For $r = 2, 3$, the exact upper Y_L of (O.49), with $\lambda = 1$, has the form $C_0 q + BP_0(B)u + c_0$ with the following complete data:*

r	L	C_0	P_0	c_0
2	1	1	Q_1	D
2	2	$1 + D$	Q_2	$D_1 - d$
2	3	$1 - d + D + D_1$	$Q_3 + D$	$D_2 - d$
2	4	$1 - 2d + D + D_1 + D_2$	$Q_4 - d + DG_2 + D_1$	$D_3 - \tau - Dd$
3	1	1	$G_2 + P_R$	$D - d$
3	2	$1 - d + D$	Q_2	$D_1 - d$
3	3	$1 - 2d + D + D_1$	$Q_3 - d + D$	$D_2 - \tau$.

(O.54)

For every real e and $C > 0$ with $C \geq C_0 - e$, the same table gives the positive comparison

$$Y_L \leq Cq + B(P_0 u + e\mathbf{1}) + (c_0 + e). \tag{O.55}$$

Proof. Insert

$$\bar{f}_2 = q + Bu, \quad \bar{f}_3 = q + BG_2 u - d, \quad \bar{f}_4 = (1-d)q + BG_3 u - d, \quad \bar{f}_5 = (1-2d)q + B(G_4 - d)u - \tau$$

in the finite ledger. This gives each row, including every constant coordinate. Since $B\mathbf{1} = u = q - \mathbf{1}$, the difference between the upper in (O.55) and the exact table is $(C - C_0 + e)q \geq 0$. The norm is then exactly (O.53). □

O.24.2 Centered inward arms and the minimum-four packet.

Lemma O.4. *For every $a \ge 1$, $\sigma(\bar f_a) < 3\sqrt{d}/2$; for $a = 1, 2, 3$ one may use $6\sqrt{d}/5$, and $\sigma(q) = \sqrt{d}$. The first-row comparison in (O.49) for $\bar f_a e^{R-\Gamma}$ has the actual coefficient $\lambda = \mu_a$, with D_A from (O.48). More precisely, for every real R with $|R| \le 1/2$,*

$$T(\bar f_a e^R) \le \bar f_{a+1} + \mu_a BR + \delta_A, \qquad \delta_A = \mu_a \mathbb{E}R + 2\sigma(\bar f_a)\|R\| + (12/5)\|R\|^2.$$

This formula allows the actual signed δ_A. The catalogue payments enlarge it to a nonnegative D_A.

Proof. With $C_0 = (I - J)B(I - J)$, finite iteration gives

$$\bar f_a - \mu_a = \sum_{i=0}^{a-1} \mu_{a-1-i} C_0^i u.$$

The coefficients belong to $[0, 1]$. The finite-mask squared norm bound $C_* = 32768/16245 < 9/4$, applied on the centered subspace, gives the first assertion. This uses its operator norm, not an original pointwise row assertion on C_0. The three short variances are

$$d, \qquad d + \delta_2 - d^2, \qquad d - \delta_2 + \delta_3 - d^2,$$

each below $36d/25$. The centered original row in Theorem M.18 gives D_A. □

Lemma O.5. *Let*

$$R = \sum_{i=1}^{k} \sum_{j=b_i+1}^{c_i} K^j \mathbf{1}, \qquad a < b_1 < c_1 < \cdots < b_k < c_k, \quad b_1 \ge 4.$$

Then

$$\bar f_a \prod_i h_{b_i} f_{c_i} \le \bar f_a e^{R-\Gamma},$$

where a pair beginning at four contributes $69d/20$ to Γ, and every other pair contributes $16d/5$. The following finite bounds hold on all $0 \le d \le 1/3$.

b_1, k	r_0	D_0/d	D_1/d	$D_{i\ge2}/d$	$\sum_{i\ge1}(D_i)_+/d$	D_q/d	D_{gen}/d
$4, 1$	$1/5$	0	$1/6$	$7/100$	$7/25$	$62/125$	$72/125$
$4, k \ge 2$	$1/4$	$1/12$	$1/5$	$1/12$	$2/5$	$11/15$	$5/6$
$\ge 5, k \ge 1$	$3/20$	$1/12$	$1/30$	$1/12$	$1/5$	$164/375$	$3/5$

(O.56)

Here $\|R\| \le r_0\sqrt{d}$, D_q is an admissible constant when $a = 1$, and D_{gen} is admissible for every listed a.

Proof. The full finite path estimate (A.8), $\mu_n \le 1 - 83d/32$, $n \ge 4$, and the positive convolution give, for $t > 4$,

$$h_4 + f_t \le 2 + R_{4t} - 4d + \delta_2(1 + p) - \tau f_2.$$

The original row identity

$$Kg(x) = \tfrac{1}{2}\mathbb{E}_{y,z}[(K(x, y) - K(x, z))K(y, z)]$$

and $|a-b| \le 1-ab$ for $a,b \in [-1,1]$ give $|Kg| \le pq/2$, hence $f_2 \ge p^2/2$. Completing the square in $p \ge 0$ yields

$$\delta_2(1+p) - \tau f_2 \le \delta_2 + \delta_2^2/(2\tau) \le 663d/1216.$$

The resulting loss $4201/1216$ exceeds $69/20$. For $s \ge 5$, the same convolution gives

$$h_s + f_t \le 2 + R_{st} - (83/16)d + 4\delta_2/(1-\theta^2).$$

Its loss is at least $83/16 - 1664/855 > 16/5$. Positive AM–GM proves the pair bounds.

The disjoint finite coefficient masks start at powers five and six, respectively. The mask bound gives $r_0 = 1/4$ or $3/20$; the latter uses $\theta^5 C_* < 9/400$. A single first interval is $K^4 G_{c_1-4}(K)g$, giving $(6/5)\theta^2 = 507/2560 < 1/5$. Removing one original K also proves $|R| < 1/3$. For every finite mask the exact signed formula is

$$D_i = \sum_j (-1)^i e_j \mathbb{E} K^{i+j}\mathbf{1}, \quad \mathbb{E}K^{2r}\mathbf{1} = (-1)^r \delta_r, \quad \mathbb{E}K^{2r+1}\mathbf{1} = 0.$$

Its positive terms have $j \equiv i \pmod 4$; increasing either index by four multiplies the upper tail by at most θ^2. For the single interval the positive classes begin with δ_3, δ_4, and their total is at most $169d/608 < 7d/25$. For a mask beginning at five the total is $201617d/519840 < 2d/5$; when it begins at six it is $2197d/11552 < d/5$. The same first signed terms give the individual columns of (O.56). The q-row bound $D_0 + 2\sqrt{d}\|R\| + (12/5)\|R\|^2$ gives D_q. For a single pair beginning at four $a \le 3$, as also for several pairs beginning at four; use Lemma O.4 with $6/5$. At minimum at least five use $3/2$. These give precisely the D_{gen} column. □

O.24.3 Both moments of the minimum-four family.

Put $n = a + L$. For (O.51) take $C = 1$,

$$V = G_{n-1}(B)u + \mu_a B^{L-1} R.$$

For $L = 1$, its constant-plus-mean is at most $D - \beta_n - \beta_{n+1} + D_1$. For $L \ge 2$, replace the first two terms by $2D + 2\sum_{i=1}^{L-1}(D_i)_+$; this gives a positive upper because all discarded path factors lie in $[0,2]$. Use $\beta_n + \beta_{n+1} \ge d$ at $L = 1$, the exact sum $2d$ at $(a,L) = (1,2)$, and $2\kappa d$ in all other $L \ge 2$ rows, where $n \ge 4$. The norm bounds are $6/5 + r_0$ at $L = 1$, $6/5 + 13r_0/20$ at $a = 1, L = 2$, and $1 + 13r_0/20$ at $L \ge 2, n \ge 4$. For each of the three seed rows, the cases are $L = 1$, $(a,L) = (1,2)$, and $L \ge 2, n \ge 4$. For $D = D_{\text{gen}}$, substitution in (O.52) gives fourth losses greater than five in these nine rows.

When $a = 1$, use D_q, $1 + r_0$ at $L = 1$, and $6/5 + 13r_0/20$ only at $L = 2$. At $L \ge 3$ use $1 + 13r_0/20$. The exact smallest eighth losses are

b_1, k	loss$/d$
$4, 1$	$70981/30000 > 2$
$4, k \ge 2$	$121127/6400 + 128(k-2)/5 > 18 + 128(k-2)/5$
$b_1 \ge 5$	$601503/160000 + 128(k-1)/5 > 3 + 128(k-1)/5.$

(O.57)

Each entry is a direct payment for the same positive comparison; the fourth conclusion is not derived by interpolating the eighth.

O.24.4 A same-direction high-domain comparison.

Lemma O.6.

$$f_3 h_c \le e^{-d}, \qquad h_3 f_c \le e^{-d} \quad (c \ge 4,\ d \ge 1/8). \tag{O.58}$$

Moreover

$$\log \mathbb{E}[(p\bar{f}_n)^8] \le 253d/32 \quad (n \ge 3), \qquad \log \mathbb{E}[(p\bar{f}_2)^8] \le 9d.$$

Proof. Let $\eta = \beta_{c-1}$, $\Gamma_0 = d + \beta_c + \eta$, $U = \sum_{j=3}^{c-1} K^j \mathbf{1}$, and $R = \eta g - KU$. Finite positive convolution gives the actual arm sums

$$f_3 + h_c \le 2 - \Gamma_0 + R, \qquad h_3 + f_c \le 2 - \Gamma_0 - R.$$

Since $\mathbb{E}U = d - \beta_c$, put $\alpha = \eta - \mathbb{E}U = \Gamma_0 - 2d$. Centering the original row and using $g^2 + pq = 1$ gives

$$|R| \le \sqrt{\alpha^2 + \operatorname{Var} U}.$$

For $c = 4, 5, \ge 6$, respectively, the lower bounds for α/d are $1, \kappa, 3/16$. The finite $G_1, G_2, G_{n\ge 3}$ bounds give

$$\operatorname{Var} U/d^2 \le 169/128, \quad 7605/4096, \quad 169/128.$$

They are smaller than $3, 35/16, 11/8$, respectively, hence $\operatorname{Var} U \le d^2 + 2\alpha d$. Thus $\Gamma_0 - |R| \ge d$, and positive AM–GM proves both inequalities without changing a direction.

For a bare path, retain β_n in its finite convolution and apply (O.52):

$$\log \mathbb{E}[(p\bar{f}_n)^8] \le -8\beta_n - 8\beta_{n+1} + 17\|G_{n-1}(B)u\|^2.$$

At $n = 3$ the actual norm is $d + \delta_2$, giving $253d/32$. For $n \ge 4$ it is at most d, giving $15d/2$. At $n = 2$, $p + \bar{f}_2 = 2 + Bu$ gives $-8d + 17d = 9d$. □

O.24.5 First pair three with a full later mask.

Suppose $a = 1, 2$, $k \ge 2$, and

$$F = \bar{f}_a f_3 h_c \prod_{i\ge 2} h_{b_i} f_{c_i}.$$

The one finite seed is

$$\eta = \beta_{c-1}, \quad W = \sum_{j=4}^{c} K^j \mathbf{1}, \quad S = \sum_{i\ge 2} R_{b_i c_i}, \quad R = \eta g - W + S,$$

$$\Gamma = d + \beta_c + \eta + (16/5)(k-1)d. \tag{O.59}$$

On $d \le 1/6$ it has $|R| \le d + (2/3)\sqrt{d} < 1/2$. The complete low-domain data are

c	$\|R\|^2/d$	r_0	D_2/d	D_3/d	T	D_q/d	D_{gen}/d
4	$79/400$	$9/20$	$1/4$	$1/30$	$3/5$	$687/500$	$777/500$
5	$3/20$	$2/5$	$1/5$	$1/6$	$17/25$	$29/25$	$33/25$
≥ 6	$3/25$	$7/20$	$1/5$	$1/6$	$17/25$	$247/250$	$141/125$

(O.60)

Here $D_0 \le 0$, $D_1 \le d/5$, $\sum_{i\ge 1}(D_i)_+ \le Td$, and $\|R\| \le r_0\sqrt{d}$. At $k = 2$ the corresponding lower Γ/d values are $31/5, 927/160, 431/80$.

We give the finite justification of all columns. Relative to g, the multiplier is $\eta-\sum_{j=3}^{c-1} z^j+P_S(z)$, with the zero-one later mask starting at degree $c+1$. For $c=4$ its real part is in $[-1/12,1/5]$ and its nonnegative imaginary part is at most $(3/2)x^{3/2}$. For $c=5$ use $[-1/4,1/6]$ and $(13/12)x^{3/2}$. For $c\ge 6$ use real absolute bound $9/50$ and the same latter imaginary bound. Indeed the continuous real interval is between $x^2(1-x)$ and x^2, its imaginary interval between $-x^{3/2}$ and $-x^{3/2}(1-x)$, and the disjoint later tail is smaller than the retained leading terms. Squaring gives $79/400, 3/20, 3/25$, respectively.

The exact signed formula

$$D_i=(-1)^i\left[\eta\mathbb{E}K^{i+1}\mathbf{1}-\sum_{j=4}^{c}\mathbb{E}K^{i+j}\mathbf{1}+\sum_{j\in S}\mathbb{E}K^{i+j}\mathbf{1}\right]$$

has $D_0\le 0$, since $\mathbb{E}W\ge\kappa\delta_2$ while $\mathbb{E}S\le\delta_4/(1-\theta^2)<\delta_2/5$. The positive mixed tail in D_1 starts at δ_5. The positive S contribution to D_2 starts at δ_4 when $c=4$, and at δ_6 otherwise. At D_3, the ηg term is nonpositive; the continuous contribution is zero at $c=4$ and at most δ_4 otherwise. Continuing the four residue classes gives exactly the D_2, D_3, T columns. For every $L\ge 4$, original Cauchy–Schwarz also gives

$$D_L/d\le(4/15)r_0,\qquad \|B^{L-1}R\|/\sqrt{d}\le(4/15)r_0.$$

Finally $\sigma(q)=\sqrt{d}$ gives D_q, whereas $\sigma(\bar{f}_2)<6\sqrt{d}/5$ gives D_{gen}.

For the fourth moment use D_{gen} and the following complete expenses; $M=(c+\mathbb{E}BV)/d$ and $\|V\|\le N\sqrt{d}$:

c	L	M	N
4	1	377/500	33/20
4	2	5141/2000	597/400
4	≥ 3	6741/2000	517/400
5	1	13/25	8/5
5	2	821/400	73/50
5	≥ 3	241/80	63/50
≥ 6	1	41/125	31/20
≥ 6	2	3337/2000	571/400
≥ 6	≥ 3	5257/2000	491/400

(O.61)

The two beta sums are at least d at $L=1$ and $2\kappa d$ thereafter. At $L=2$ the positive partial sum contains only D_1; the contracting G applies at $L\ge 3$. These facts and (O.60) derive the table. Every direct fourth loss exceeds four.

For $a=1$ use the smaller D_q. At $L=1$ use $M=D_q/d-1+1/5$, $N=1+r_0$. At $L=2$ use

$$M=2D_q/d+2/5-2+D_2/d,\qquad N=6/5+13r_0/20.$$

The smallest eighth loss there is $87607/160000>1/2$. At $L=3$, only D_1, D_2 enter the partial sum and $\beta_4+\beta_5\ge 51d/32$; this gives $M\le 21/10, 17/10, 27/20$, and $N\le 6/5$. For all $L\ge 4$ use

$$M=2D_q/d+2T-2\kappa+(4/15)r_0,\qquad N=1+(4/15)r_0. \tag{O.62}$$

Substitution gives eighth loss at least $d/2$ in every row. Additional pairs increase the fourth loss by $64d/5$ and the eighth loss by $128d/5$, without increasing the finite mask.

For $d \ge 1/6$, Lemma O.6 and $|S| \le \sqrt{d}/4 \le 5d/8$ give, in the same actual direction,

$$F \le e^{-[143/40+(16/5)(k-2)]d}\,\bar{f}_a.$$

The bare fourth cost is at most $d/4$, and at $a = 1$ the bare eighth cost is at most $9d$. Thus this complementary interval preserves both claimed rates, including $L = 1$, and meets the low interval at its endpoint.

O.24.6 The three-arm packet, including its two inward minima.

For $F = \bar{f}_a f_3 h_c$, $a \in \{1, 2\}$, $c \ge 4$, put

$$\eta = \beta_{c-1}, \quad W = \sum_{j=4}^{c} K^j \mathbf{1}, \quad R = \eta g - W, \quad \Gamma = d + \beta_c + \eta. \tag{O.63}$$

The two arm sums in Lemma O.6 prove this exponential upper. Work first on $d \le 1/8$. Write $j = c-3 \ge 1$, so $W = K^3 G_j(K) g$. The original point range is below $d+(6/5)\theta\sqrt{d} < 1/2$. The available losses are $3d$ at $c = 4$, $2d + \tau$ at $c = 5$, and at least $d + 2\tau$ at $c \ge 6$. In particular $\Gamma \ge 35d/16$; at both $c = 4, 5$ one may use the lower bound $2d + \tau$.

There are two first rows for this one seed. At $a = 1$, use the Q row of (O.48). At $a = 2$, expand the actual $\bar{f}_2 = 1 - g + Kg$ with coefficient one:

$$T(\bar{f}_2 e^R) \le \bar{f}_3 + BR + D, \quad D \ge \mathbb{E}R - \mathbb{E}(gR) + \mathbb{E}[(Kg)R] + \sqrt{(d+\delta_2)\mathbb{E}R^2} + \tfrac{12}{5}\mathbb{E}R^2. \tag{O.64}$$

The following choices are valid:

$$D^{(1)} = 7d/20, \qquad D^{(2)} = \begin{cases} 17d/32, & c = 4, 5, \\ d/2, & c \ge 6. \end{cases} \tag{O.65}$$

Here is a direct same-moment verification. At $c = 4$,

$$R = dg - K^4\mathbf{1}, \quad \Gamma = 3d, \quad \mathbb{E}R^2 = d^3 + \delta_4, \quad \mathbb{E}R - \mathbb{E}(gR) + \mathbb{E}[(Kg)R] = -\delta_2 - d^2 + \delta_3.$$

Put $t = \delta_2/d$. The Q-row error is at most

$$\theta\sqrt{t} - (773/1280)t + (12/5)d^2 \le 845/12368 + 3/80 < 7/20.$$

For the uncentred B row use $\sqrt{(1+t)(d^2+\theta^2 t)} \le (1+t)/6 + (3/2)(d^2+\theta^2 t)$. Its error divided by d is at most

$$1/6 - d + (39/10)d^2 + t[-\kappa + 1/6 + (39/10)\theta^2] \le 1/6 + \theta[-\kappa + 1/6 + (39/10)\theta^2] < 17/32.$$

The t coefficient is positive and $-d + (39/10)d^2 \le 0$ throughout this domain. Thus the same two first-row constants cover the genuine G_1. For $c = 5$, set $t = \delta_2/d$, $J = d^2 + [\theta^2(1+\theta) - 2\theta d]t$. Then

$$\begin{aligned} \mathbb{E}R - \mathbb{E}(gR) &= -\delta_2 - d^2 + \delta_3, \\ \mathbb{E}R - \mathbb{E}(gR) + \mathbb{E}[(Kg)R] &= -\delta_2 - d^2 + 2\delta_3, \\ \mathbb{E}R^2 &= d^3 + \delta_4 + \delta_5 - 2d\delta_3 \le dJ. \end{aligned}$$

Use $\sqrt{J} \le 1/8 + 2J$ for $a = 1$, and $\sqrt{(1+t)J} \le (1+t)/6 + 3J/2$ for $a = 2$. The resulting bounds are $156575/524288 < 3/10$ and $5514615/10485760 < 17/32$. For $c \ge 6$, the finite alternating sums give

$$\mathbb{E}W - \mathbb{E}(gW) \ge \delta_2 - 2\delta_3, \quad \kappa\delta_3 \le \mathbb{E}(gW) \le \delta_3, \quad \|W\|^2 \le \theta\delta_3,$$
$$\mathbb{E}W + \mathbb{E}[(Kg)W] \ge \kappa(\delta_2 - \delta_3).$$

The last assertion is the alternating sum beginning $(\delta_2 - \delta_3) - (\delta_3 - \delta_4)$; its successive differences contract by θ. Now $\mathbb{E}R^2/d \le d^2 + [\theta^2 - 2\kappa^2\theta d]t$. The two Young bounds yield

$$\frac{1}{8} - \kappa d + \frac{22}{5}d^2 + t\left[b + \frac{22}{5}\theta^2 - \frac{44}{5}\kappa^2\theta d\right], \quad b = -3/16, \quad b = 1/8 + 55/1024,$$

respectively. Both t coefficients are positive. At $t = \theta$ the d-derivative is at most $11/10 - \kappa - (44/5)\kappa^2\theta^2 < 0$. At zero the values are $28167/81920 < 7/20$ and $80709/163840 < 1/2$. This proves (O.65).

The one signed sequence satisfies

$$\begin{aligned} &-\delta_3 \le D_1 \le d^2, \quad 0 \le D_2 \le \delta_3, \quad -\eta\delta_2 \le D_3 \le \delta_4, \quad D_4 \le 0, \\ &D_i \le \theta^4 d < d/36 \quad (i \ge 4), \qquad \sum_{i \ge 1} (D_i)_+ \le (169/608)d + (1024/855)d^2. \end{aligned} \tag{O.66}$$

Indeed its exact formula is

$$D_i = (-1)^i \left[\eta\mathbb{E}K^{i+1}\mathbf{1} - \sum_{r=4}^{c} \mathbb{E}K^{i+r}\mathbf{1}\right].$$

The positive ηg class has $i = 1 \bmod 4$, whereas the positive continuous classes have $i = 2, 3 \bmod 4$. Thus their separate maxima are not charged in the same class. At $c = 4$, the initial values are exactly $-\delta_2, d^2, \delta_3, -d\delta_2, -\delta_4$. The two later positive classes give the same bounds in (O.66), using $d \le 1/8 < \theta^2$.

For either a, let

$$Q_L = G_{L+a-1} - \eta z^{L-1} - z^{L+2}G_j(-z), \qquad G_j(-z) = U - zV, \quad 1 - x \le U \le 1, \quad 0 \le V \le 1. \tag{O.67}$$

These coordinate intervals include $j = 1$, with $(U, V) = (1, 0)$. The norm arguments below require no positive lower bound on V. At $L = 1$, the positive comparison has $C = 1$ and

a	V	$(c + \mathbb{E}BV)/d$	$\|V\|^2/d$
1	$(1 - \eta - z^3G_j(-z))u$	$-21/40$	$3/2$
2	$(G_2 - \eta - z^3G_j(-z))u$	$-43/32$	$9/4.$

In the second row the scalar is $c = D^{(2)} - d$; in the first it is $D^{(1)}$. The squared coordinate upper bounds are $(1 + x^2)^2 + x^3 < 3/2$ and $1 + x + 4x^2 + x^3 + x^4 \le 45901/20736 < 9/4$, respectively. The direct fourth rates are $179/40$ and $73/16$, both larger than four.

At $a = 1$ the three remaining short complete vectors are

L	c	V_{tot}	$\mathbb{E}BV_{\text{tot}}$
2	$D + D_1 - d$	$Q_2u + D\mathbf{1}$	$-d + D_2$
3	$D + D_1 + D_2 - 2d$	$(Q_3 + D)u - \lambda\mathbf{1}$	$-\tau + D_3 - Dd$
4	$-\eta_0 - \tau - Dd + D_3$	$(Q_4 - \lambda + Dz)u - \eta_0\mathbf{1}$	$-\tau + D_4 + \lambda d$

(O.68)

Here $D = 7d/20$, $\lambda = d - D - D_1$, and $\eta_0 = 2d - D - D_1 - D_2$. This table is obtained by expanding the actual $\bar{f}_3, \bar{f}_4, \bar{f}_5$ in the finite ledger.

At $L = 2$, the two coordinates of Q_2 are $1 - x^2U$, $1 - \eta + x^2V$, positive, and

$$|Q_2|^2 \le 1 + x - 2x^2 + 4x^3 + x^4 - x^5 + x^6 < 11/8.$$

Adding $D\mathbf{1}$ decreases the norm, because $\mathbb{E}Q_2u = -d + D_1$ and $D \le 2(d - D_1)$. At $L = 3$, its coordinates are $1-(1-\eta)x-x^3V$, $1-x^2U$, proving $|Q_3| \le 1$. Here $0 < \lambda \le d$ and $\mathbb{E}[(Q_3+D)u] = -d + D_2 \in [-d, 0]$. The complete norm is at most

$$(1 + D)^2 d + \lambda^2 + 2\lambda d \le d + (15/4)d^2 \le 47d/32. \tag{O.69}$$

At $L = 4$, the squared excess of Q_4, divided by x, is at most $-1 - 3x/4 + (177/64)x^2 - 2x^3 + x^5 \le -71/256$. Thus $|Q_4| \le 1$, $\Re Q_4 \ge \kappa$, and $|Q_4 - \lambda + Dz|^2 \le 1 + d/2$. Since $0 < \eta_0 < 11d/6$ and the mean of this last vector is in $[-9d/8, 0]$, (O.53) gives

$$\|V_{\text{tot}}\|^2 \le d + (575/72)d^2 \le 1151d/576 < 2d.$$

These ranges follow from $0 < \lambda \le d$, $D_1 \in [-\delta_3, d^2]$, $0 \le D_2 < d/6$, and $\mathbb{E}[(Q_4 - \lambda + Dz)u] = -\tau + D_3 - Dd$; the lower mean bound uses $D_3 \ge -\eta\delta_2$. Consequently the short $a = 1$ payments are

L	$(c + \mathbb{E}BV_{\text{tot}})/d$	$\|V_{\text{tot}}\|^2/d$	eighth exponent$/d$
2	$-163/120$	$11/8$	$-599/120$
3	$-15/8$	$47/32$	$-241/32$
4	$-7/3$	2	$-13/6$.

(O.70)

All direct fourth payments in these same rows are less than $-4d$.

For $a = 2$, enlarge D to $17d/32$, and put $\lambda = d - D = 15d/32$. At $L = 2$, Q_2 has positive coordinates $1 - x - x^2U$, $1 - \eta + x^2V$ and $|Q_2| \le 1$. The complete data are

$$c = D + D_1 - 2d, \quad V_{\text{tot}} = Q_2u - \lambda\mathbf{1}, \quad \mathbb{E}BV_{\text{tot}} = -\tau + D_2.$$

Its mean sum is at most $-85d/48$, its norm squared below $3d/2$, and the eighth expense at most $-6d$. Indeed $\mathbb{E}Q_2u = -d + D_1$, so the entire constant-vector cost is at most $(10/3)d^2$, using $0 < \lambda \le d$, $D_1 \ge -\delta_3$, and $\delta_3 < d/6$. At $L = 3$,

$$|Q_3| \le 1 - 3x/8 + (7/25)x^2 \le 1, \qquad \Re Q_3 > 1/2.$$

Here $|G_4| \le 1 - x/2 - x^2/2$, $\eta \le 1/8$, and $|G_j(-z)| < 6/5$, including G_1; also $(6/5)\sqrt{x} - 1/2 \le 7/25$. The real lower bound is $1-x-x^3 > 1/2$. Consequently $|Q_3-\lambda| \le 1$. With $\eta_0 = 2d-D-D_1$ the complete data are

$$c = D + D_1 + D_2 - 2d - \tau, \quad V_{\text{tot}} = (Q_3 - \lambda)u - \eta_0\mathbf{1},$$
$$\mathbb{E}BV_{\text{tot}} = -\tau + \lambda d + D_3.$$

The mean sum is below $-2d$, the complete norm is below $9d/5$, and the eighth expense at most $-29d/10$. More explicitly its mean sum divided by d is at most $17/32 + 1/8 + 1/6 + 1/14 + 1/8 - 2 - 19/16 = -1457/672$. Also $0 < \eta_0 < 5d/3$ and $\mathbb{E}[(Q_3 - \lambda)u] = -\tau + D_2 \in [-\tau, 0]$; thus the complete norm is at most $d + (55/9)d^2 < 9d/5$. The two corresponding direct fourth payments are below $-4d$.

O.24.7 One whole finite tail for both inward minima.

For $a = 1, L \geq 5$ or $a = 2, L \geq 4$, put $n = a + L \geq 6$. Retaining the four genuine convolution terms at indices $0, 1, 2, 3$ gives

$$\bar{f}_n \leq Cq + B[G_{n-1} - \tau(2 + z)]u - \tau(1 - d), \quad C = 1 - 3\tau, \quad a_0 = 1 - 2\tau.$$

The polynomial for this same complete upper is

$$P = G_{L+a-1} - \tau(2 + z) - \eta z^{L-1} - z^{L+2} G_j(-z).$$

It satisfies $|P| \leq a_0$ for every finite index. At $a = 2, L = 4$ its two coordinates are at most $a_0 - x + x^2 + x^3$ and $1 - \tau - 7x/8$. Their squared excess divided by x is at most

$$-1 + 2\tau + \tau^2 + (5/4 - 9\tau/4)x + (49/64 - 4\tau)x^2 - x^3 + 2x^4 + x^5.$$

The leading negative magnitude is at least $47/64$. The remaining positive part satisfies

$$180800/248832 < 47/64.$$

At $L = 5$ the common coordinate bounds for both a are

$$a_0 - x + (1 - \eta)x^2 + x^4 V, \qquad 1 - \tau - x + x^2 + x^3.$$

The resulting quotient is at most

$$-1 + 2\tau + \tau^2 + (1 - 2\tau)x + (1 - 2\tau)x^2 + (3 - 6\tau)x^3 - 3x^4 + 4x^5 + x^6 + x^7.$$

Its derivative in τ is positive. At $\tau = 1/8$ its positive part after deleting $-3x^4$ is at most $1217/1800 < 47/64$. For all $L \geq 6$ use only the scalar comparator $P_\infty = (1-z)^{-1} - \tau(2+z)$. Direct algebra gives

$$a_0^2 - |P_\infty|^2 = x[a_0/(1 + x) - \tau^2], \qquad a_0 - |P_\infty| > x/3.$$

The actual finite remainder divided by x is at most

$$x^2 + x^{3/2}/8 + (6/5)x^3 \leq 85/288 < 1/3.$$

This proves every finite tail; $x = 0$ is equality.

Let $P_{\text{cost}} = 2D^{(a)} + 2\sum_{i \geq 1}(D_i)_+$. The positive comparison is

$$Y_L \leq Cq + BPu + P_{\text{cost}} - \tau(1 - d), \qquad \mathbb{E}BPu = -\beta_{n+1} + 2\tau d + D_L.$$

By (O.66), for either a,

$$P_{\text{cost}} + D_L \leq Ad + 2bd^2, \qquad b = 6/5, \quad A = \begin{cases} 5/3, & c = 4, 5, \\ 8/5, & c \geq 6. \end{cases} \tag{O.71}$$

Here $D_1 \leq d^2$, $D_2 \leq \delta_3$, and the finite tail bound are combined before enlarging constants; these bounds are also valid for the smaller $a = 1$ first-row error. For $c \geq 6$ the unrounded linear constant is $4333/2736 < 8/5$; for $c = 4, 5$ it is $563/342 < 5/3$.

Set $t = \tau/d \in [\kappa, 1]$, a global moment ratio, not the frequency x. Using $\log C \le -3\tau$ in (O.52), the eighth expense times C^2/d is bounded by

$$\begin{aligned} F &= 9 + 8A - 56t + d[288t^2 + (4 - 24A)t + 16b] \\ &\quad - d^2[360t^3 + 76t^2 + 48bt] \qquad (c \ge 6), \\ F_5 &= 1 + 8A - 48t + d[240t^2 + (52 - 24A)t + 16b] \\ &\quad - d^2[288t^3 + 148t^2 + 48bt] \qquad (c = 4, 5). \end{aligned} \tag{O.72}$$

On the whole rectangle $0 \le d \le 1/8,\ \kappa \le t \le 1$, the degree-$(2,3)$ Bernstein coefficients of $-F - (1 - 3td)^2$ have minimum $573959/1310720 > 0$. Those of $-F_5 - 2(1 - 3td)^2$ have minimum $1359451/983040 > 0$. These are the exact coefficients from (A.18). Lemma A.1 therefore gives eighth expenses $-d$ and $-2d$, respectively, simultaneously for both a. Since $a_0 \ge C > 0$, their direct fourth expenses are at most $-19d/4$.

For $d \ge 1/8$, the same direction satisfies

$$pT^L(\bar f_a f_3 h_c) \le e^{-d} p \bar f_{a+L}.$$

It has fourth rate four, except at $a = L = 1$ where the bare fourth cost $d/4$ gives $15/4$. At $a = 1, L \ge 2$ its eighth rate is $3/32$. The unit-connector eighth input $\mathbb{E}[(pT(qf_3h_c))^8] \le 1 - d/500$ is the actual one-step theorem of Section O.14. Thus the common eighth bound for qf_3h_c, $c \ge 4$, is $1 - d/601$.

O.24.8 First pair two and at least three pairs.

Here

$$F = qf_2h_c \prod_{i \ge 2} h_{b_i} f_{c_i}, \quad k \ge 3.$$

Put $\eta = \beta_{c-1}$, $\alpha = 1 - \eta$, $W = \sum_{j=2}^{c-1} K^j g$, and $S = \sum_{i \ge 2} R_{b_i c_i}$. The actual arm sum and $q \le e^{-g}$ give the single unweighted seed

$$F \le e^{R - \Gamma}, \quad R = -\alpha g - W + S, \qquad \Gamma = \beta_c + \eta + (16/5)(k-1)d \ge [1 + (16/5)(k-1)]d. \tag{O.73}$$

The q is absorbed only into this upper bound; it is still the terminal's original unit arm.

On $d \le 1/6$ the finite polynomial $P_0(z) = \alpha + \sum_{j=2}^{c-1} z^j - P_S(z)$ has the following coordinate bounds. The later mask starts at degree $c + 1$.

c	positive real part, upper	|imaginary part|
3	$1 - 4x/5$	$x^{5/2}/(1 - x^2)$
4	$1 - 4x/5$	$(3/2)x^{3/2}$
≥ 5	$1 - 3x/8$	$(13/12)x^{3/2}$.

(O.74)

They follow by summing the four finite signed residue intervals. The positive real lower bounds are $1 - x - x^2/(1 - x^2)$ at $c = 3$, and $5/6 - x - x^4/(1 - x^2)$ otherwise. The squared excess quotients are bounded by

$$-8/5 + 8/25 + 1/25, \quad -8/5 + 8/25 + 3/8, \quad -1673/3456,$$

all negative. Hence $\|R\|^2 \le d$. The original row gives $|R| < 1 + (6/5)\sqrt{d} < 3/2$. Use $e^R \le 1 + R + R^2$, obtaining $D = 25d/12$.

The exact signed classes give

$$D_0 \le d/12, \quad D_1 \le -\alpha d + (3/2)\delta_2 \le -43d/192 < 0, \quad D_2 < d/4, \quad |D_i| \le \theta^{(i-1)/2} d \ (i \ge 3).$$

Thus the unweighted convolution losses $\gamma_j = \beta_j - D_{j-1}$ satisfy $\gamma_3 \ge 3d/4, \gamma_4 \ge \kappa d, \gamma_j > d/4$ for $j \ge 5$. Retain D_1 until its sign is known, then discard it and the other nonpositive path terms, keeping the last gamma loss. If $P = P_0(-z) = A + zV$, the coordinates also give

$$0 < A \le 1, \quad |P| \le 1, \quad |V| \le 3x/2, \quad V \ge -x^2/(1-x^2).$$

For $Q_n = G_n + z^n P$,

$$|Q_1|^2 \le 455/288 < 8/5, \quad |Q_2|^2 \le 249/128 < 2, \quad |Q_n| \le 1 + \theta^{3/2} < 19/15 \quad (n \ge 3).$$

These prove, with $C = 1$, the four complete comparison rows:

L	$(c + \mathbb{E}BV)/d$	$\|V\|^2/d$	fourth loss$/d$	eighth loss$/d$
1	$119/64$	1	$1433/80$	
2	$41/12$	$8/5$	$137/15$	$14/3$
3	$271/96$	2	$1177/120$	$157/60$
≥ 4	$11/3$	$361/225$	$7303/900$	$583/225$.

(O.75)

At $L = 1$, the direct eighth payment also exceeds $5/2$. For $L = 2$, only the two indicated short terms occur; for larger L the finite gamma bound pays the whole tail. Each extra pair adds $64d/5$ or $128d/5$ to the fourth or eighth loss.

For $d \ge 1/6$, $h_c \le 2$, $|S| \le (3/8)\sqrt{d} \le 15d/16$, and $2 \le e^{9d/2}$. The same actual factors satisfy

$$F \le e^{-[77/80 + (16/5)(k-3)]d} q f_2.$$

The all-length ordinary eighth theorem for the actual qf_2 input bounds its fourth as well by one. Consequently the global fourth rate is at least $77/20$, and the eighth rate exceeds $1/2$, for every L.

O.24.9 One q-weighted square for the shortest-two triples.

For $F = qf_2 h_c$, $c \ge 3$, use the following actual arm sums:

c, direction	m	$U = (A - Bz)u$	$d_{\max}$	D
3, h_3	$1 - d/2$	$\mathsf{X}u$	$1/3$	$3d/8$
3, $\bar{f}_3$, $L = 2$	$1 - d/2$	$(\mathsf{X} - 2B)u$	$1/3$	specified below
4, h_4	$1 - d$	$(d + \mathsf{X} - \mathsf{X}B)u$	$1/4$	$d/2$
$c \ge 5$, h_c	$1 - 9d/8$	retained U_c	$1/5$	$2d/5$.

(O.76)

Here $\mathsf{X} = -B^2$, and the B's inside the vector column are the skew operator; $A(x), B(x)$ in U's coordinate description are scalar coordinates. In each row the sum of the two actual nonnegative arms is bounded by $2m - U$. Thus $F \le q(m - U/2)^2$ with nonnegative square base. The exceptional row uses $qf_2\bar{f}_3$, so it has no asserted canonical h_3 fourth counterpart.

Let $k = 1/m$, $R = -kU$. The QS first row of (O.48) and (O.54) are the only propagation used. For the canonical $c = 3$ row,

$$D_{\text{actual}} = -k\delta_2 + k\sqrt{d\delta_3} + k^2\delta_3, \qquad D_{\text{actual}}/d \le \frac{k\theta}{4(1 - k\theta)} \le 39/164 < 1/4.$$

We choose $D = 3d/8$ to retain one convenient long normalization. Its exact sequence has

$$D_1 = k\delta_2, \quad D_2 = 0, \quad D_3 = -k\delta_3, \quad D_4 = 0, \quad \gamma_j \geq 49d/96 \quad (j \geq 3). \tag{O.77}$$

The last statement follows from $\gamma_3 \geq 41d/80$, $\gamma_4 = d$, and the first remaining positive class $k\delta_4 < d/12$, followed by factors θ^2.

For $c = 4$ the actual raw moments are

$$\mathbb{E}U^2 = d^3 + 2d\delta_2 + \delta_3 + \delta_4, \qquad \mathbb{E}U - \mathbb{E}(gU) = d^2 + 2\delta_2.$$

Writing $t = \delta_2/d$, Young's inequality gives

$$D_{\text{actual}}/(kd) \leq 3/8 - d + 2d^2 + t[-2 + 4d + 2\theta(1+\theta)] \leq 3/8.$$

If the bracket is negative discard it; otherwise take $t = \theta$, where the residual increases to $-1099/16384 < 0$. As $k \leq 4/3$, $D = d/2$ is admissible. The exact signed classes are

$$D_{2j-1} = (-1)^{j+1}k(d\delta_j + \delta_{j+1}), \qquad D_{2j} = (-1)^{j+1}k\delta_{j+2}.$$

Their first two upper bounds are $7d/8, d/4$. Every positive term from index three on is below $d/6$: the first are $k(d\delta_3 + \delta_4) \leq 1183d/8192$ and $k\delta_5$, followed by θ^2. Hence

$$\gamma_3 \geq d/8, \quad \gamma_4 \geq 3d/4, \quad \gamma_j \geq 41d/96 \quad (j \geq 5). \tag{O.78}$$

For $c \geq 5$, Lemma M.10, at $\sigma = 9/4$, proves the actual QS row $D = 2d/5$ and all its finite signed classes. It retains

$$D_1, D_2 \geq 0, \quad D_3, D_4 \leq 0, \quad D_1 + D_2 < 5d/4, \quad \gamma_3, \gamma_4 \geq 0, \quad \gamma_{j\geq 5} \geq 41d/96.$$

The real retained coefficients also give

$$0 \leq kA, kB < 1, \quad \|1 - kBx + z(1-kA)\|^2 \leq 8/7, \quad \|1 - x + kAx + z(1-kBx)\|^2 \leq 8/7. \tag{O.79}$$

These are the two actual short-polynomial inequalities (M.30) and (M.31), not an arbitrary independent rectangle substituted for the retained coefficients.

Lemma O.7 (Two finite geometric comparisons)**.** *For $0 \leq x \leq \theta$, $z = i\sqrt{x}$,*

$$|G_{n-1}(z) + z^n| \leq 1 \qquad (n \geq 4), \tag{O.80}$$

and, for $0 \leq h \leq 3/16$,

$$|G_n(z) - h(2+z)| \leq 1 - 2h \qquad (n \geq 3). \tag{Q102.5.6}$$

Proof. For the first assertion, at $n = 4$ the excess is $x[-1 + 3x - 2x^2 + x^3]$; its bracket increases to $-1451/32768$. At $n = 5$ it is $x[-1 - x + 3x^2 - 2x^3 + x^4] < 0$. For every $n \geq 6$,

$$(1-z)(G_{n-1} + z^n) = 1 - z^{n-1}(1 - z + z^2), \qquad |1 - z + z^2|^2 = 1 - x + x^2 \leq 1.$$

The remaining numerator expense, divided by x, is at most $2\theta^{3/2} + \theta^4 < 889/1600 < 1$.

For the second assertion put $a = 1 - 2h$. At $n = 3, 4$ the squared excess is respectively

$$x[-1 + 2h + h^2 + x], \qquad x[-1 + 2h + h^2 + (-1+2h)x + x^2],$$

both nonpositive. For $n \geq 5$, the numerator after multiplication by $1-z$ is $M - z^n$, $M = a + hz + hz^2$, and $|M| \leq 1$. Its exact available margin is

$$a^2(1+x) - |M|^2 = x[1 - 2h - h^2(1+x)] \geq (4715/8192)x.$$

The finite tail costs at most $2x^{n/2} + x^n$. After division by x, this is below $889/1600 < 9/16 < 4715/8192$. This proves every finite index; both assertions are direct at $x = 0$. □

O.24.10 Canonical 123: two short rows and one retained tail.

Here $m = 1 - d/2,\ k \in [1, 6/5],\ D = 3d/8$. For $L = 3$, the short-ledger table gives

$$C_0 = 1 - d + D + k\delta_2, \quad P = 1 + D + z + (1 - k\mathsf{X})z^2, \quad c_0 = -d.$$

Enlarge C_0 to $C = 1 - d/8$. The squared frequency modulus of P, minus $(1 + D)^2$, divided by x, is

$$-5/2 + 3/(2k) + (7k/2 - 1/2)x - 2kx^2 + k^2x^3.$$

It is convex in k. At $k = 1, 6/5$ it increases with x; the endpoint values are

$$-1451/32768, \qquad -9507/204800.$$

Thus $\|Pu\|^2 \le (1 + D)^2 d$. Its constant plus signed mean is at most $-(7/4 + D)d$. Using the actual scale m^2 in (O.52), the eighth expense divided by d is at most

$$-23 - 3d + 17\left(\frac{1 + 3d/8}{1 - d/8}\right)^2 \le -1/2. \tag{O.81}$$

The last continuous inequality is one of the exact polynomial certificates recorded below.

At $L = 4$, write

$$C_0 = 1 - 2d + D + k\delta_2, \quad A_0 = C_0 + d, \quad P = A_0 + (1 + D)z + z^2 + (1 - k\mathsf{X})z^3.$$

The frequency bracket in

$$|P|^2 = A_0^2 + x[(1 + D - x + kx^2)^2 + x - 2A_0]$$

is convex in x. At $x = 0$ it is below $-1/4$; at $x = \theta$ use $1 + D - \theta + k\theta^2 \le 2347/2560 < 15/16$ and $A_0 \ge 19/24$, obtaining $-229/768 < -1/4$. Thus $\|Pu\|^2 \le A_0^2 d - \delta_2/4$. Enlarge to $C = 1 - 9d/8$. The actual same-moment bounds are

$$\|Pu\|^2/d \le 1 - d/2 + d^2/64, \qquad (c_0 + \mathbb{E}BPu)/d \le -19/16 + d/4 - 2d^2. \tag{O.82}$$

For the mean bound use $\delta_3/d \ge (\delta_2/d)^2$ and $d \le \delta_2/d \le \theta$; these are integrated moment constraints, not a lower bound on spectral frequencies. The exact cleared polynomial below proves eighth loss $d/8$.

For every $L \ge 5$, put $h = 49d/96$. By (O.77) retain the last four nonnegative losses in (O.50), and enlarge $D\bar{f}_{L-1} \le DS'_{L-1}$. The complete comparison has

$$C = 1 + D - 3h = 1 - 37d/32, \quad c = -h(1 - d),$$
$$P = G_L - k\mathsf{X}z^{L-1} + DG_{L-2} - h(2 + z).$$

The first two terms are the convex combination $(1 - kx)G_L + kxG_{L-1}$, with $kx \le 39/80$. Lemma O.7 therefore gives

$$\|P\| \le 1 - 2h + D = 1 - 31d/48, \qquad c + \mathbb{E}BPu \le d[-49/48 + (335/256)d]. \tag{O.83}$$

Here $\mathbb{E}BPu = -\beta_{L+2} + D_L - D\beta_L + 2hd$; the retained γ_{L+2} and $\beta_L \ge \kappa d$ are used before any enlargement. Truncating the negative logarithm expansions at degree three, the cleared polynomial below gives eighth loss $d/24$ on all $0 \le d \le 1/3$. The $L = 1$ direction is the actual one-step input of Section O.12. At $L = 2$ use the different row next.

O.24.11 The actual $123, L = 2$ square.

The existing three-edge arm points entirely inward. Its exact sum is $f_2 + \bar{f}_3 = 2 - d + 2Kg - K^2 g$. Thus $m = 1 - d/2,\ U = (\mathsf{X} - 2B)u$ and

$$\mathbb{E}U^2 = 4\delta_2 + \delta_3, \quad D_{\text{actual}} = -k(2d + \delta_2) + k\sqrt{d(4\delta_2 + \delta_3)} + k^2(4\delta_2 + \delta_3).$$

For $t = \delta_2/d$, the bound on its positive part is

$$(D_{\text{actual}}/d)_+ \le k[-2 - t + \sqrt{(141/32)t} + k(141/32)t]_+.$$

At $t \le 1/4$ it is at most $117/800 < 1/6$, by taking $k = 6/5, t = 1/4$ and $\sqrt{141/128} < 21/20$. For $t \ge 1/4$, divide by t. The positive expression increases in k and t; at $k = 6/5, t = \theta$ it is below $8313/2600 < 16/5$. Use $D = d/6$ or $D = (16/5)td$, respectively.

The same exact short-ledger row is

$$C = 1 + D, \quad c = k\delta_2 - d, \quad P = (1 - 2k\mathsf{X}) + (1 - k\mathsf{X})z.$$

The actual mean and raw norm satisfy

$$(c + \mathbb{E}BPu)/d = -2 + 3kt, \qquad \|Pu\|^2 \le d(1 - 2t).$$

Indeed the squared-frequency excess quotient increases with x and is convex in k; its two endpoint values at $x = \theta$ are $-2071/1024, -14063/6400 < -2$. On $t \le 1/4$, direct substitution gives eighth expense $[-275/57 - (638/95)t]d < -4d$. On the whole remaining rectangle $0 \le d \le 1/3,\ 1/4 \le t \le \theta$, set $C_t = 1 + (16/5)td$ and $m = 1 - d/2$. The actual logarithmic truncations yield the numerator

$$\begin{aligned} P_2(d,t) = {} & mC_t^2[-8 - 2d + (128/5)t - (1024/25)t^2 d + (32768/375)t^3 d^2] \\ & + 8(-2m + 3t)C_t + 17(1 - 2t)m. \end{aligned} \tag{O.84}$$

The eighth expense divided by d is at most $P_2/(mC_t^2)$. The exact degree-$(5,5)$ Bernstein coefficients of $-P_2 - \frac{1}{4}mC_t^2$ have minimum $33/80 > 0$. This proves the whole two-moment domain directly, including its finite boundary, and gives the eighth loss $d/4$. The canonical fourth direction is unchanged at this integer length.

O.24.12 The canonical 124 and retained $12c$ rows.

First take $c = 4$, $d \le 1/4$, $m = 1 - d, D = d/2$. At $L = 2$, the actual polynomial from the short ledger is

$$P = 1 - k\mathsf{X}^2 + [1 - k(d + \mathsf{X})]z, \quad C = 1 + D.$$

Its two coordinates are nonnegative, and

$$|P|^2 \le (1 - x^2)^2 + x(1 - x)^2 \le 1 + x - 3x^2 \le 13/12.$$

The constant plus true signed mean is at most $-695d/768$. With $1 \le C \le 9/8$, the eighth expense is

$$[-16 + 4 - 695/108 + 221/12]d = -d/54. \tag{O.85}$$

At $L = 3$, use

$$P = 1 + D + z + [1 - k(d + \mathsf{X})]z^2 + k\mathsf{X}z^3, \quad C = 1 + 3d/8 \ge C_0.$$

Its real coordinate lies in $[0, 1 + D]$, and its imaginary coefficient in $[0, 1]$, so $|P|^2 \le (1 + D)^2 + x$. Consequently its normalized squared norm is at most $45d/32$: after multiplication by C^2, the scalar margin over $(1 + d/2)^2 + \theta$ is $d(112 - 107d)/2048$. The $k\delta_3$ terms cancel in the exact mean, leaving $c_0 + \mathbb{E}BPu \le -51d/32$. Its eighth expense is at most $-841d/1120 < -3d/4$.

For $c \ge 5$, take $m = 1 - 9d/8, D = 2d/5$ on $d \le 1/5$. At $L = 2$ use the real short-ledger normalization $C = 1, e = D$. By (O.79) the complete data give

$$c + \mathbb{E}BV \le -7d/20, \qquad \|V\|^2 \le 8d/7 + D^2 < 6d/5,$$

and the eighth expense is at most $-2d/5$. At $L = 3$ set $\lambda = d - D - D_1,\ P = Q_3 + D, C = 1, e = -\lambda$. The exact mean sum is at most $-151d/160$, and $\mathbb{E}Pu = -d + D_2 < 0$. If $\lambda \le 0$, adding $-\lambda \mathbf{1}$ decreases the norm, since $0 \le -\lambda < D < 2(d - D_2)$. If $\lambda > 0$, write the actual retained coordinate as $A = C_0 + x(a_f - E_0)$, where $C_0 \ge 5d/4 + dE_0,\ a_f \ge \kappa,\ E_0 \le d$. Integration with $\delta_2 \ge d^2$ gives

$$D_1 = k \int A \, d\nu \ge (59/32)kd^2 \ge 3d^2/2.$$

The E_0 contributions cancel in this lower bound. Thus $\lambda/d \le 3/5 - 3d/2$. Its entire extra constant-vector norm cost divided by d is below $7/50$, since

$$d[(3/5 - 3d/2)^2 + 2(3/5 - 3d/2)] \le (39/25)d - (87/20)d^2 < 7/50.$$

Also $\|Pu\|^2/d \le 8/7 + (6/7)d + (4/25)d^2$. Together these give $\|Pu - \lambda \mathbf{1}\|^2 < 3d/2$. The full eighth expense is therefore at most $-d/20$.

It remains to prove the entire finite tail for both 124 and retained $12c$. In each case

$$Q_L = G_L - kAz^{L-1} + kBz^L.$$

For 124, $A = d + x, B = x$, so $0 \le kB \le kA \le 7/8$. For retained $12c$, the actual coordinates have $0 \le kA, kB < 1$. Thus Q_L is a convex combination of $G_{L-1}, G_L, G_{L-1} + z^L, G_{L+1}$. Lemma O.7 proves $|Q_L| \le 1$ for every finite $L \ge 4$. For 124 retain the last gamma loss and $D\bar{f}_{L-1} \le DS'_{L-1}$. Then

$$C = 1 + D, \quad P = Q_L + DG_{L-2}, \quad c = -\gamma_{L+1},$$

$$c + \mathbb{E}BPu \le -41d/48, \quad \|P\|/C \le (1 + 3d/5)/(1 + d/2) \le 46/45.$$

The G_2 exception at $L = 4$ is the reason for $3d/5$. The eighth expense is at most $-628d/2025 < -d/4$. For retained $12c$, simply use $C = 1,\ c = 2D - \gamma_{L+1},\ V = Q_L u$. Its mean sum is at most $-13d/240$; the scale m^2 gives eighth expense at most $-43d/30 < -d$.

The low-domain bounds meet the same-direction high comparisons. At $c = 4, d \ge 1/4$, $h_4 \le 2(1 - 2d) \le 1$; at $c \ge 5, d \ge 1/5$, $h_c \le 2\mu_{c-1} \le 77/80 < 1$. Therefore $pT^L(qf_2h_c) \le pT^L(qf_2)$. The actual short-one ordinary theorem (Q58O.0.2) supplies $1 - d/12000$, at every L. The $L = 1$ low rows are the same-direction one-step input of Section O.13, with deficit $d/500$. For canonical 123 the $L = 1$ input is Section O.12.

O.24.13 The six complete continuous payments.

For clarity we give the unexpanded defining polynomials for the remaining certificates. Put

$$C_3 = 1 - d/8, \qquad C_4 = 1 - 9d/8, \qquad \xi = 37/32, \quad C_5 = 1 - \xi d,$$
$$P_4 = [-17 - (113/16)d - (243/64)d^2]C_4^2 + [-19/2 + 2d - 16d^2]C_4 + 17(1 - d/2 + d^2/64),$$
$$P_5 = [-8 - 8\xi - (2 + 4\xi^2)d - (2/3 + 8\xi^3/3)d^2]C_5^2$$
$$+[-49/6 + (335/32)d]C_5 + 17(1 - 31d/48)^2.$$

These are obtained from (O.82) and (O.83) after using the signed means and the degree-three negative logarithm bounds. In particular the factor m^{16} of the ordinary eighth moment has been included. Define

$$\begin{aligned} G_3 &= (45/2 + 3d)C_3^2 - 17(1 + 3d/8)^2, \\ G_4 &= -P_4 - 1/8, \qquad G_5 = -P_5 - 1/24, \\ G_2(d,t) &= -P_2(d,t) - \tfrac{1}{4}(1 - d/2)[1 + (16/5)td]^2, \\ G_{13}(d,t) &= -F(d,t) - (1 - 3td)^2, \\ G_{13,5}(d,t) &= -F_5(d,t) - 2(1 - 3td)^2. \end{aligned} \tag{O.86}$$

Here $t = \delta_2/d$ in G_2, whereas $t = \tau/d$ in the two G_{13}'s. For each row below map its complete interval or rectangle affinely to the unit interval or square. Formula (A.18) gives the indicated minimum among all coefficients, with exact reconstruction of the displayed polynomial:

polynomial	complete domain	degree	minimum coefficient
G_3	$0 \le d \le 1/3$	3	$77/1152$
G_4	$0 \le d \le 1/3$	4	$1319/36864$
G_5	$0 \le d \le 1/3$	4	$2038583/339738624$
G_2	$0 \le d \le 1/3,\ 1/4 \le t \le \theta$	$(5,5)$	$33/80$
G_{13}	$0 \le d \le 1/8,\ \kappa \le t \le 1$	$(2,3)$	$573959/1310720$
$G_{13,5}$	$0 \le d \le 1/8,\ \kappa \le t \le 1$	$(2,3)$	$1359451/983040.$

(O.87)

Each positive minimum proves the claimed continuous inequality by Lemma A.1.

Corollary O.8 (An ordinary-fourth resource for the canonical shortest-two square)**.** *The same actual $pT^L(qf_2h_c)$ has ordinary fourth at most $e^{-18d/5}$ on the following domains:*

$$c = 3, \quad L \ge 3, \quad 0 \le d \le 1/3; \qquad c = 4, \quad L \ge 2, \quad 0 \le d \le 1/4;$$
$$c \ge 5, \quad L \ge 2, \quad 0 \le d \le 1/5.$$

This assertion concerns no inverse-fourth output. It does not replace the canonical $123, L = 2$ fourth row.

Proof. Use the fourth expression in (O.52) for each already specified complete vector. For canonical 123, its normalized norm upper is at least d: this is direct at $L = 3$ and $L \ge 5$; at $L = 4$ the difference from $C_4^2 d$ is $(7d/4 - 5d^2/4)d \ge 0$. Thus the eighth rates $1/2, 1/8, 1/24$ give direct fourth rates at least $9/2, 69/16, 205/48$, respectively. For $124, L = 2$, the normalized upper $13d/(12C^2)$ is at least $208d/243$, so its fourth rate is at least $1/108 + 884/243 > 18/5$. The $L = 3$ and finite tail rows are stronger. For retained $12c$, the specified norm uppers are $6d/5, 3d/2, d$; combined with the stated eighth expenses, they give the claimed rate directly. This is the same-comparison expense identity, not a consequence of ordinary eighth-to-fourth interpolation. □

O.24.14 The three finite five-arm packets.

The actual directions are those in Table 65. All three use $F \le qe^{R-\Gamma}$, the Q-row, and $\gamma_j = \beta_j - D_{j-2}$. Their data on the stated low domains are

row	$d_{\max}$	Γ/d	D/d	D_1/d	D_2/d	$\gamma_{j\ge5}/d$
$A:\ c\ge4$	$1/64$	$351/80$	$9/10$	$1/2$	$1/4$	$11/32$
$B:\ c=3,\ b\ge5$	$1/8$	$21/5$	$5/6$	$3/5$	$1/14$	$79/160$
$C:$ crossed	$1/8$	$21/4$	$11/10$	$19/40$	$1/16$	$49/96.$

(O.88)

In row A retain $\gamma_3 \ge d/2, \gamma_4 \ge 3d/4$. In B retain $2d/5, 13d/14$, and in C retain $d/2, 15d/16$. The following finite construction proves all these columns.

For A set

$$\eta = \beta_{c-1}, \quad W = \sum_{j=3}^{c} K^j\mathbf{1}, \quad S = \sum_{j=b+1}^{e} K^j\mathbf{1}, \quad R = \eta g - W + S.$$

Here $|R| \le d + \sqrt{d} \le 9/64$. With $t = \delta_2/d$, at $c = 4$ the raw norm is at most $[d + (3/2)\sqrt{\theta t}]^2 d$ and the signed mixed mean is at most $-2(1-\theta^2)td$. At $c \ge 5$ replace $3/2$ by $6/5$ and the last coefficient by $2\kappa - 2\theta^2$. The first-row error increases in $\sqrt{t}$ and d. At $t = \theta, d = 1/64$ it is at most $3/4 + 2\theta^3 < 9/10$ in the first case and $-1/3 + 51/100 + (12/5)(51/100)^2 < 9/10$ in the second. This retains the two consecutive positive δ_4 terms in $\mathbb{E}S - \mathbb{E}(gS)$. The finite sign formula gives

$$D_1 \le d^2 + \delta_2 + \delta_4, \quad D_2 \le \delta_3 + \delta_4, \quad D_{i\ge3} < d/4.$$

The four first tail classes are bounded by $\delta_5, \delta_6, \delta_4 + \eta\delta_3 + \delta_7, \delta_5 + \delta_6$; every further block advances all moment indices by two.

For the polynomial norm in A put

$$P = G_{c-2}(z) - z^{b-2}G_{e-b}(z), \qquad P(-z) = U - zV.$$

Its opposite-sign finite intervals are disjoint and

$$(1-2x)/(1-x) \le U, V \le 1/(1-x), \qquad |P| \le 3/2.$$

The $c = 4$ estimate is $\sqrt{1+\theta} + 39/125 < 3/2$; the first block contracts when $c \ge 5$, with a smaller tail. Now $Q_L = G_L - \eta z^{L-1} + z^{L+1}P(-z)$. At $L = 2$ both coordinates are in $[0, 1]$, so $|Q_2|^2 \le 45/32$. At $L = 3$, $|Q_3| \le 1 + 273/2048 < 6/5$. At $L = 4$ the perturbation of G_4 is at most $11/64$, and at every $L \ge 5$ at most $1/384 + 21/200 < 1/5$. Thus $|Q_L| \le 6/5$ for all $L \ge 3$.

For B take $R = -K^3\mathbf{1} + S$, $S = R_{be}$. The point row is below $(119/125)\sqrt{d} < 1/2$. At $b = 5$ its coordinates are $(-A + Cz)u$, where $0 \le A = x - x^3V \le x,\ 0 \le C = x^2U \le x^2$. Writing $w = \mathbb{E}R^2/d$ and $v = d^{-1}\int (A + C)\, d\nu$, we have $w \le \theta v,\ w < 1/4$; hence the Q error is at most $\sqrt{w} - (4/65)w < 1/2$. For $b \ge 6$, its mixed mean is at most $-\delta_2 + 2\delta_4$ and

$$\mathbb{E}R^2/d \le \theta[1 + (6/5)\theta^2]^2 t.$$

The error increases in t, and at $t = \theta$ is below $49/100 + (12/5)(49/100)^2 - 4/15 < 4/5$. Thus $D = 5d/6$ suffices. Its signed tails give $D_1 \le \delta_2 + \delta_3,\ D_2 \le \delta_4,\ D_{i\ge3} \le \delta_4 + \delta_5$. The polynomial

is $Q_L = G_L + z^{L+1} + z^{L-1}(S/u)$. The base $G_L + z^{L+1}$ contracts for all $L \geq 3$: at $L = 3$ its excess is $x[-1 + 3x - 2x^2 + x^3] < 0$; at $L = 4$ it is at most $-3x/4$. For every $L \geq 5$, use

$$(1 - z)(G_L + z^{L+1}) = 1 - z^L(1 - z + z^2), \qquad |1 - z + z^2|^2 = 1 - x + x^2 \leq 1.$$

The tail is paid by $2x^{3/2} + x^4 < 1$. The additional propagated S costs less than $1/10$. At $L = 2$ the base has squared norm at most $31/27$ and the tail at most $2/15$, so $|Q_2| < 5/4$.

For C take the actual retained $U_e = (A - Bz)u$ of Lemma M.9, $\sigma = 9/4$, and

$$R = -U_e + K^4\mathbf{1} - dg = (-\bar{A} + \bar{B}z)u, \quad \bar{A} = A - d, \quad \bar{B} = B - x.$$

The exact $e = 5$ coordinates are $A = 5d/4 + \delta_2 + x - x^2$, $B = d/4 + \delta_2 + x$; for $e \geq 6$ use (M.25). Their retained constants give

$$0 \leq \bar{A} \leq 19/40, \quad -x^2 \leq \bar{B} \leq 19d/16, \quad |\bar{B}| < 1/6, \quad \bar{A} + \bar{B} \geq d/2 + x/16.$$

For the $\bar{A}$ upper, if $x \geq d$, the retained $E_0(d - x)$ term is nonpositive and $\bar{A} \leq 21d/32 + x - x^2 + x^3 \leq 101/256 < 19/40$. If $x \leq d$, use $\bar{A} \leq 53d/32 + d^2 \leq 57/256 < 19/40$. Thus the actual mixed mean is nonpositive and $\mathbb{E}R^2/d < (19/40)^2 + \theta/36 < 1/4$. The original expression

$$R = (C_0 - d)g + D_0Kg + (E_0 - 1)K^2g - \sum_{j=5}^{e} K^j\mathbf{1}$$

has point bound $25/256 + (19/128 + 119/125)\sqrt{1/8} < 1/2$. The Q error is therefore at most $11d/10$. The signed classes are

$$D_{4j+1} = \int x^{2j}\bar{A}, \quad D_{4j+2} = \int x^{2j+1}\bar{B}, \quad D_{4j+3} = -\int x^{2j+1}\bar{A}, \quad D_{4j+4} = -\int x^{2j+2}\bar{B}.$$

They include the possible positive fourth class when $\bar{B} < 0$. It is at most $\theta^4 d$; the fifth class is at most $(19/40)\theta^2 d < d/12$. At $L = 1$, $\|u + R\|^2 \leq 81d/80$. At $L = 2, 3$ the complete coordinates of Q_L give squared norms below $8/5, 3/2$, respectively. For every $L \geq 4$,

$$|Q_L| \leq 1 + (19/40)\theta^{3/2} + \theta^2/6 < 6/5.$$

Here are all the remaining low-domain payments, with the actual first-row constants of (O.88):

row	L	$(c + \mathbb{E}BV)/d$	$\|V\|^2/d$	eighth exponent$/d$
A	1	$2/5$	$169/64$	one-step input
A	2	$11/20$	$45/32$	< -6
A	≥ 3	$89/80$	$36/25$	$-43/25$
B	1	$13/30$	$64/25$	one-step input
B	2	$71/210$	$25/16$	< -4
B	≥ 3	$163/240$	$36/25$	< -3
C	1	$23/40$	$81/80$	$-323/16$
C	2	$61/80$	$8/5$	$-87/10$
C	3	$107/160$	$3/2$	< -11
C	≥ 4	$283/240$	$36/25$	< -8.

(O.89)

At $L \geq 2$, take $c = 2D - \gamma_{L+1}$, $\mathbb{E}BV = -\beta_{L+2} + D_L$, with the finite $Q_L u$ just proved. At $L = 1$, take $V = u + R$, preserving its full mean. The direct fourth payments at A1 and B1 are $-6051d/1280$ and

$-314d/75$. Every other row has direct fourth loss greater than four by (O.52). For A1/B1 the ordinary eighth bound $e^{-d/2}$ is exactly the same-direction one-step theorem of Section O.15. The crossed C1 row is proved here and needs no crossed extension of that one-step theorem.

For A on $d \geq 1/64$, regroup the same factors as $(qf_2h_b)(h_cf_e)$. If $c = 4$, the already proved short-four loss gives $h_4f_e \leq e^{-477d/500}$, since

$$|R_{4e}| < (39/125)\sqrt{d} \leq (312/125)d.$$

If $c \geq 5$, its row tail is below $\sqrt{d}/5$, giving the stronger loss $8d/5$. The actual all-length eighth base qf_2h_b, $b \geq 5$, proved in the square-packet argument above, is at most one. It follows that the fourth rate is $477/125 > 15/4$ and the eighth rate exceeds $1/2$. For B and C on $d \geq 1/8$, use (O.58) to regroup respectively $(qf_2h_b)(h_3f_e)$ and $(qf_2h_e)(h_3f_4)$. This gives fourth loss four and eighth loss eight.

O.24.15 Completeness and original decorations.

For a residual beginning with one, split its first pair minimum into at least four, three, or two. The first two cases were proved above, including their single-triple boundaries. At first minimum two, $k \geq 3$ is the finite-mask family, $k = 2$ is exactly the three five-arm rows, and $k = 1$ is the square family. Its $c = 3, L = 2$ entry directs only the existing three-edge arm differently. Every d-split changes the estimate, not this integer direction. For nonunit terminal minimum, the fourth rows state exactly their additional ranges; no corresponding eighth direction is inferred.

Restore the actual pair products at their own vertices:

$$0 \leq E_rpT^L(E_tF) \leq pT^LF, \qquad 0 \leq E_t, E_r \leq 1.$$

All original edges and subdivision vertices remain. Full reversal replaces K by $-K$ throughout, including the unit root, connector, terminal and every decoration. At $d = 0$, $K\mathbf{1} = 0$ although K need not vanish; original leaf integration makes all actual tree messages one.

For the R37 two-edge application the actual output is $U = T^2[E_rpT^L(E_tF)]$, with its original two intermediate edges and the same original q. Its common ordinary-fourth loss $15d/4$, followed by Corollary M.19, gives

$$\mathcal{J}(U)^3 \leq e^{(24427/3420)d-5d^2}(\mathbb{E}H_D^4)^2 \leq e^{-1223d/3420-5d^2}.$$

The separate R26 inverse-fourth resources and their directions have not entered this ordinary-moment proof.

O.25 Three further unit-terminal ordinary-eighth families

Let the residual list be $\Sigma = (1, b_1, c_1, \dots, b_k, c_k)$, with $1 < b_1 < c_1 < \cdots < b_k < c_k$. The receiving root has an outward unit arm. Every connector edge points from the terminal to that root, so for any positive integer L,

$$H = E_rpT^L(E_tF). \tag{Q98.0.1}$$

Here E_t, E_r are any genuine equal-pair products at their actual roots. The following choices depend only on the residual lengths:

Σ	F	$\mathbb{E}H^8$ upper
$(1,3,c),\ c\ge 5$	qf_3h_c	$1-d/601$
$(1,2,c),\ c\ge 5$	qf_2h_c	$1-d/12000$
$(1,2,c,b,e),\ 4\le c<b<e$	$qf_2h_ch_bf_e$	$e^{-d/2}$
$(1,2,3,b,e),\ 5\le b<e$	$qf_2h_3h_bf_e$	$e^{-d/2}$
$(1,2,3,4,e),\ e\ge 5$	$qf_2h_eh_3f_4$	e^{-8d}.

The crossed last row is used for every L, including one. The first two rows retain the stronger continuation estimates

Σ	d-domain	$L=2$	$L=3$	$L=4$	$L\ge 5$
$(1,3,c),\ c\ge 5$	$[0,1/8]$	e^{-4d}	e^{-7d}	e^{-2d}	$e^{-3d/4}$
$(1,2,c),\ c\ge 5$	$[0,1/5]$	$e^{-2d/5}$	$e^{-d/20}$	e^{-d}	e^{-d}.

For $d\ge 1/8, L\ge 2$, the first row has the same-direction bound $e^{-3d/32}$. For $d\ge 1/5$, the second row compares to the actual ordinary-eighth qf_2 input at every L.

Proof. Sections O.24.6 and O.24.7 prove the first row with its complete signed means and all finite short vectors. Section O.24.12, using the actual $9/4$-retained packet, proves the second row and its strong continuation rates. In these two rows the unit-connector inputs are Sections O.14 and O.13.

Section O.24.14 proves the last three rows. The first two one-step directions in that paragraph are the actual multipair input of Section O.15; its crossed one-step direction is proved directly. The high-domain regrouping uses the shortest-two square family, already proved in Section O.24.9, with its original terminal q and root p.

Every comparison uses the same finite host and its original measure. Restore decorations and full reversal by Section O.24.15; at $d=0$ original leaf integration gives equality. □

O.26 Unit-root ordinary eighth moments for 123 and 124

The terminal has the distinct residual lengths $(1,2,c)$, where $c=3$ or 4. Its connector has any positive integer length L, all edges pointing toward the receiving root, which has an existing outward unit arm. The terminal and receiver may carry arbitrary genuine equal-pair products.

Choose the direction from the original integers c, L. The following bounds hold for the undecorated message:

c	L	d-domain	F	$\mathbb{E}[(pT^LF)^8]$ upper
3	1	$[0,1/3]$	qf_2h_3	$e^{-d/200}$
3	2	$[0,1/3]$	$qf_2\bar{f}_3$	$e^{-d/4}$
3	3	$[0,1/3]$	qf_2h_3	$e^{-d/2}$
3	4	$[0,1/3]$	qf_2h_3	$e^{-d/8}$
3	≥ 5	$[0,1/3]$	qf_2h_3	$e^{-d/24}$
4	1	$[0,1/3]$	qf_2h_4	$1-d/500$
4	2	$[0,1/4]$	qf_2h_4	$e^{-d/54}$
4	3	$[0,1/4]$	qf_2h_4	$e^{-3d/4}$
4	≥ 4	$[0,1/4]$	qf_2h_4	$e^{-d/4}$
4	≥ 1	$[1/4,1/3]$	qf_2h_4	$1-d/12000$.

Proposition O.9. *For every nonempty finite probability space and every real antisymmetric kernel $|K| \le 1$, these fixed directions give*

$$H_D = E_o p T^L (E_t F), \qquad \mathbb{E} H_D^8 \le 1 - d/12000 \le 1. \tag{Q102.0.1}$$

For $c = 3$, the stronger uniform bound is $1 - d/601$. The assertion includes positive atom masses, all genuine equal-pair products, complete reversal and $d = 0$.

Proof. The $L = 1$ directions are the actual one-step inputs of Sections O.12 and O.13. The canonical $123, L \ge 3$ proof is Section O.24.10. Its finite long comparison retains the four convolution losses and the factor $(1 - d/2)^{16}$. Section O.24.11 proves $123, L = 2$ for its actual terminal $q f_2 \bar{f}_3$, on the whole moment rectangle. This direction is distinct from the canonical $q f_2 h_3$ used in separate reference or fourth estimates. Section O.24.12 proves all 124 rows and their common high-domain comparison.

The continuous scalar payments are the six full polynomials in Section O.24.13; the finite geometric inequality (Q102.5.6) is proved at Lemma O.7. The smallest 123 exponential rate is $1/200$, so $e^{-d/200} \le 1 - d/601$ on $0 \le d \le 1/3$. The 124 low rows are stronger than $1 - d/12000$; its high rows use (Q58O.0.2). Finally positivity restores E_t, E_o, and Section O.24.15 treats reversal and $d = 0$. □

O.27 One reference arm at a nonempty even center

Let the distinct residual arm lengths at the center be $a_1 < \cdots < a_{2j}$, $j \ge 1$. Select the reference length by

$$a_* = \begin{cases} a_2, & a_1 = 1, \\ a_1, & a_1 \ge 2, \end{cases} \qquad a_* \ge 2. \tag{Q113.1.2}$$

Orient this arm outward, with message f_{a_*}. Fix a simple terminal with residual $(1, b)$, $b \ge 2$, and connector length $L \ge 1$. There is a direction on the other center arms, this terminal and its connector such that their message H at the center satisfies

$$\mathbb{E} H^8 \le e^{-d/12000}. \tag{Q113.1.3}$$

The factor H contains all equal-arm decorations at the center and terminal, and excludes the selected reference arm.

Use $A = J + K$, $T = J - K$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $f_n = A^n \mathbf{1}$, $\bar{f}_n = T^n \mathbf{1}$, $h_n = T f_{n-1}$ and $W_n = A \bar{f}_{n-1}$ on the same finite probability space. An h_n arm has its first edge toward the root and the rest toward the leaf; a W_n arm has its first edge away from the root and the rest toward the first interior vertex. Each T^L connector points from terminal to center. Equal-arm products $E_c, E_e \le 1$ stay at their respective vertices.

Q113.2 The second moment of the reference

The path-power estimate of Section M.15 gives

$$\|A^n\| \le (1 - d/8)^{\lfloor n/3 \rfloor} \le 1 \qquad (n \ge 3).$$

For $n = 2$, use

$$f_2 = 1 + g + Kg, \qquad \mathbb{E}[g(Kg)] = 0, \quad \mathbb{E}(Kg) = -d, \qquad \mathbb{E} f_2^2 = 1 - d + \delta_2 \le 1,$$

where $\delta_2 \le (13/32)d$. Applying the path-power estimate to $\mathbf{1}$ for $n \ge 3$ therefore proves

$$\mathbb{E} f_a^2 \le 1 \qquad (a \ge 2). \tag{Q113.2.3}$$

Q113.3 Directions at every even center

If $a_1 = 1$, write the center list as

$$(1, a_*, u_1, v_1, \dots, u_k, v_k), \qquad 1 < a_* < u_1 < v_1 < \cdots < u_k < v_k, \quad k = j - 1 \geq 0.$$

Put $P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}$. The center unit arm points outward; the other center arms use P. At the terminal, the unit arm points inward and its b-edge arm outward. Thus

$$H = E_c p P T^L (E_e q f_b). \tag{Q113.3.1}$$

For $k = 0$, the ordinary-eighth estimate is the full qf_b directory in Section O.29. For $k \geq 1$, Proposition O.1 applies to the ordinary input excluding the reference arm. Its first root pair starts above the retained length $t = a_*$, so the Family B order of Section O.3 holds. In particular $u_1 = 3$ forces $a_* = 2$, which is included in that family. Positivity restores E_c, E_e, giving

$$\mathbb{E}H^8 \leq 1 - d/12000 \leq e^{-d/12000}.$$

If $a_1 \geq 2$, write the list as

$$(a_*, a, u_1, v_1, \dots, u_k, v_k), \qquad 2 \leq a_* < a < u_1 < v_1 < \cdots < u_k < v_k, \quad k = j - 1 \geq 0.$$

In particular $a \geq 3$. Keep the reference arm outward and again put $P = \prod_i W_{u_i} \bar{f}_{v_i}$. Use the following two directions, chosen by the integer L:

$$H = \begin{cases} E_c f_a P T (E_e p \bar{f}_b), & L = 1, \\ E_c W_a P T^L (E_e q f_b), & L \geq 2. \end{cases} \tag{Q113.dir}$$

For $L = 1$, the center a-arm and terminal unit arm point outward, and the terminal b-arm points inward. For $L \geq 2$, the center a-arm has message W_a, and the terminal uses qf_b. All connector edges point toward the center in both cases.

For $k = 0$, Proposition M.39 and Section O.2 give, respectively,

$$\mathbb{E}H^8 \leq e^{-9d/4} \quad (L = 1), \qquad \mathbb{E}H^8 \leq e^{-3d/20} \quad (L \geq 2).$$

For $k \geq 1$, the residual list outside the reference arm is $(a, u_1, v_1, \dots, u_k, v_k)$ with minimum $a \geq 3$. Proposition M.39 and Section O.4 give

$$\mathbb{E}H^8 \leq e^{-(2005/192)d} \quad (L = 1), \qquad \mathbb{E}H^8 \leq e^{-97d/12} \quad (L \geq 2).$$

These inputs exclude the reference arm a_*. All four coefficients are at least $1/12000$, proving (Q113.1.3) in every case.

The message H and its selected reference f_{a_*} enter the central allocations of Table 9.

O.28 The unit eighth moments for minimum-three even lists and the pair $(2, 3)$

Use $A = J + K$, $T = J - K$, $g = K\mathbf{1}$, $p = 1 + g$, $q = 1 - g$, $d = \mathbb{E}g^2$, $\delta_j = \|K^j \mathbf{1}\|^2$, $\theta = 13/32$, $\kappa = 19/32$, $f_n = A^n \mathbf{1}$ and $h_n = T f_{n-1}$. For $3 = b_1 < c_1 < b_2 < c_2 < \cdots < b_j < c_j$, set

$$F = f_3 h_{c_1} \prod_{i=2}^{j} h_{b_i} f_{c_i}.$$

We prove

$$\mathbb{E}(pTF)^8 \le e^{-(16j-61/4)d} \le e^{-3d/4}. \tag{Q53.1.4}$$

We also prove

$$\mathbb{E}[pT(h_2 f_3)]^8 \le 1 - d/500. \tag{Q53.1.3}$$

In these formulas the eighth power is inside the expectation. All directions are determined by the listed arm lengths. The receiving unit arm points outward, and the unit connector points from the terminal toward that root.

Q53.2 Exponential transfer at a unit connector

For $|u| \le 8/5$,

$$e^u \le 1 + u + u^2. \tag{Q53.2.1}$$

Divide the absolute Taylor tail by u^2. Its quadratic term is $1/2$, the cubic term is at most $4/15$, and the fourth and later terms sum to at most $(8/75)/(1 - 8/25) = 8/51$. Their total is $157/170 < 1$. Suppose $F \ge 0$, $F \le e^{R-\Lambda d}$, $|R| \le 8/5$ and $|8KR| \le 8/5$. Write $V = \mathbb{E}R^2$ and $Z = KR$. Since the kernel of T is at most two,

$$Te^R \le q + \mathbb{E}R - Z + 2V =: U.$$

The function U is nonnegative. Arithmetic–geometric mean gives

$$pTF \le e^{-\Lambda d} pU \le e^{-\Lambda d}[1 + \mathbb{E}R/2 + V - Z/2]^2.$$

The base is $(p + U)/2 \ge 0$. Raising to the eighth power and using $1 + y \le e^y$, followed by (Q53.2.1) for $-8Z$, yields

$$\mathbb{E}(pTF)^8 \le e^{-8\Lambda d + 8\mathbb{E}R + 16V - 8\mathbb{E}Z + 64\mathbb{E}Z^2}. \tag{Q53.2.2}$$

Because $\mathbb{E}Z^2 \le \theta V$ and $16 + 64\theta = 42$, the conditions $\mathbb{E}R \le md$, $V \le vd$, $-\mathbb{E}Z \le ad$ imply

$$\mathbb{E}(pTF)^8 \le e^{-(8\Lambda - 8m - 42v - 8a)d}. \tag{Q53.2.4}$$

If instead $F \le e^{-cd}$ for $c \ge 0$, positivity and $pq \le 1$ give

$$\mathbb{E}(pTF)^8 \le e^{-8cd}. \tag{Q53.2.5}$$

Q53.5 The minimum-three list

The retained envelope of Section I.5, with the same direction $f_3 h_b$, gives a function R_0 such that

$$f_3 h_b \le e^{R_0 - 3d}, \qquad \mathbb{E}R_0 \le -\kappa\delta_2. \tag{Q53.5.1}$$

Its explicit estimates (I.10) and (I.11) are

$$\begin{cases} \|R_0\| \le (31/100)\sqrt{d}, & |R_0| \le 1/4, \quad 0 < d \le 1/9, \\ |R_0| \le (69/32 - 1/4000)d, & 1/9 \le d \le 1/3. \end{cases} \tag{Q53.5.2}$$

For the mean, the spectral representation there is $\mathbb{E}R_0 = -\int B(x)\, d\nu_g$, where $B = D + xb_x$, $D \ge 0$ and $b_x \ge 1 - x \ge \kappa$. Set $S = \sum_{i=2}^{j} R_{b_i c_i}$, with $S = 0$ for $j = 1$. Each later shorter arm is at least five, and each of its exponents begins at six. The finite-mask bound (L90.2.5) and the pair estimate give

$$F \le e^{R - (2j+1)d}, \qquad R = R_0 + S. \tag{Q53.5.3}$$

Writing $S = K^5 P(K) g$ with a finite zero–one polynomial gives

$$\|S\| \le (3/2)\theta^{5/2}\sqrt{d} < \sqrt{d}/6, \qquad |S| \le (3/2)\theta^2\sqrt{d} < \sqrt{d}/4. \tag{Q53.5.4}$$

Here $\theta^2 < 1/6$ and $\sqrt{\theta} < 2/3$. Its possible positive mean starts at δ_4, so

$$\mathbb{E}S \le \frac{\delta_4}{1-\theta^2} \le \frac{\theta^2}{1-\theta^2}\delta_2 < \delta_2/5, \qquad \mathbb{E}R \le -(\kappa - 1/5)\delta_2 \le 0. \tag{Q53.5.5}$$

For $0 < d \le 1/9$,

$$\|R\| \le (143/300)\sqrt{d} < \sqrt{d}/2,$$
$$V \le d/4, \quad |R| \le 1/4 + \sqrt{d}/4 \le 1/3,$$
$$|8KR| \le 4\sqrt{d} \le 4/3 < 8/5.$$

Skew-adjointness gives $-\mathbb{E}(KR) = \mathbb{E}(gR) \le d/2$. Taking $\Lambda = 2j+1$, $m = 0$, $v = 1/4$ and $a = 1/2$ in (Q53.2.4) gives

$$\mathbb{E}(pTF)^8 \le e^{-(16j-13/2)d}. \tag{Q53.5.6}$$

For $1/9 \le d \le 1/3$, use $\sqrt{d} \le 3d$ to obtain $R \le (93/32)d$, and hence

$$F \le e^{-(2j-61/32)d}. \tag{Q53.5.7}$$

The loss is at least $3/32$ when $j \ge 1$, so (Q53.2.5) gives

$$\mathbb{E}(pTF)^8 \le e^{-(16j-61/4)d}. \tag{Q53.5.8}$$

The small-domain bound is stronger, proving (Q53.1.4).

Q53.7 The pair $(2,3)$

Put $r = Kg$, $s = Kr$, $t = Ks$ and $a = 1 - d/2 \ge 5/6$. The identities $h_2 = q - r$, $f_3 = p - d + r + s$ give $h_2 + f_3 = 2 - d + s$, so $h_2 f_3 \le (a + s/2)^2$. Since $\mathbb{E}s = 0$, $Ts = -t$ and $T(s^2) \le 2\delta_3$,

$$T(h_2 f_3) \le a^2 q - at + \delta_3/2 =: U \ge 0.$$

Arithmetic–geometric mean for ap and U/a gives

$$0 \le pT(h_2 f_3) \le [a - t/2 + \delta_3/(4a)]^2. \tag{Q53.7.1}$$

The base is half the sum of the two nonnegative factors. Moreover,

$$a + \delta_3/(4a) \le 1 - d/2 + (3/10)\theta^2 d = 1 - (4613/10240)d \le 1 - (51/128)d =: c.$$

The two loss coefficients differ by $533/10240 > 0$. Thus

$$0 \le pT(h_2 f_3) \le (c - t/2)^2. \tag{Q53.7.3}$$

The row estimate of Section O.6, applied to $t = K^2 r$, gives

$$\mathbb{E}t = \delta_2 \ge 0, \qquad |t|^2 \le \delta_2/3 \le d/3, \qquad \mathbb{E}t^2 = \delta_4 \le \theta^3 d \le \theta^2 d. \tag{Q53.7.4}$$

These are precisely the hypotheses of (Q51.2.2); with $Z = t$ it proves (Q53.1.3). Positivity restores all equal-arm decorations. At $d = 0$ the bounds follow by leaf integration, and replacing K by $-K$ gives their full reversals.

O.29 Ordinary eighth moments for every short-one terminal

Let K be a real antisymmetric kernel with $|K| \le 1$ on a nonempty finite probability space (X, μ). Use the finite notation, square envelope and signed coefficients of Section N.7, in particular (L58.1.6), (L58.1.7) and (L58.1.9). In this calculation write $B_0 = -K$ and $g' = B_0\mathbf{1} = -g$. Put

$$F_b = qf_b, \qquad X_{b,L} = T^L(qf_b), \qquad H_{b,L} = pX_{b,L}. \tag{Q58O.0.1}$$

The terminal unit arm points inward, the b-edge arm points outward, every connector edge points toward the receiving root, and the receiving unit arm points outward. These directions are fixed by the original structure.

We prove the ordinary-eighth bound for every $b \ge 3, L \ge 2$, with deficits $d/2000$ for $L \ge 4$, $d/12000$ for $L = 3$, and $d/3000$ for $L = 2$. The $b = 2, L \ge 2$ rows follow from Section O.7; the $L = 1$ rows for every $b \ge 2$ follow from Section O.8, using the scalar estimate of Section O.6. Their combination gives

$$\boxed{\mathbb{E}H_{b,L}^8 \le 1 - d/12000 \qquad (b \ge 2,\ L \ge 1).} \tag{Q58O.0.2}$$

All calculations below use μ. At $d = 0$, leaf integration gives every message equal to one; divisions by d are used only for $d > 0$.

Q58O.2 Fixed-node Hermite comparison and the Lucas polynomial

Suppose N is real with $|N| \le u$, $\mathbb{E}N \le -a$ and $\mathbb{E}N^2 \le hu^2$, where $0 < h \le 1/2$ and $a \ge 0$. Take $c > 0$ and $x = u/(2c) < 1$. For $f(z) = (1 + z/(2c))^{16}$, quadratic Hermite interpolation with double contact at $-hu$ and single contact at u is an upper bound on $[-u, u]$, since $f''' \ge 0$ and $z - u \le 0$. Both the quadratic and the mean coefficients are nonnegative. Consequently

$$E(1 + N/(2c))^{16} \le M_h(x) - \frac{a}{u} D_h(x), \tag{Q58O.2.1}$$

Here

$$M_h(x) = \frac{h(1+x)^{16} + (1-hx)^{16}}{1+h},$$
$$D_h(x) = \frac{1-h}{1+h}\, 16x(1-hx)^{15} + \frac{2h}{(1+h)^2}\big[(1+x)^{16} - (1-hx)^{16}\big].$$

In this scalar comparison, write the degree-j coefficients as $\binom{16}{j}M_j$ and $\binom{16}{j}D_j$. Then

$$M_j = \frac{h + (-h)^j}{1+h}, \quad D_j = \frac{1-h}{1+h} j(-h)^{j-1} + \frac{2h}{(1+h)^2}(1 - (-h)^j). \tag{Q58O.2.2}$$

We have $M_0 = 1$, $M_1 = 0$, $M_2 = h$, $M_3 = h(1-h)$, and $M_j \le M_4 = h(1 - h + h^2)$ for $j \ge 4$. Every $D_j \ge 0$. For even $j \ge 4$, after factoring $h(1-h)/(1+h)^2$, the remainder is

$$2\sum_{i=0}^{j-1} h^i - jh^{j-2}(1+h) > 0,$$

because $jh^{j-2} \le 1$. The low-degree values used below are

$$D_1 = 1,\ D_2 = 0,\ D_3 = h(2-h),\ D_4 = 2h(1-h)^2, \quad D_5 = 2h - 4h^2 + 6h^3 - 3h^4. \tag{Q58O.2.3}$$

If $c = 1 - Ad$, $u^2 = Md$ and $w = 1/c$, then $w^2 - w = k_0 x^2$, where $k_0 = 4A/M$. Put $v = w - 1 \geq 0$ and $t = k_0 x^2 = wv$. The finite Lucas identity gives

$$w^{16} + v^{16} = 1 + 16t + 104t^2 + 352t^3 + 660t^4 + 672t^5 + 336t^6 + 64t^7 + 2t^8.$$

Since $v^{16} \leq t^8$,

$$w^{16} \geq 1 + 16t + 104t^2 + 352t^3 + 660t^4 + 672t^5 + 336t^6 + 64t^7 + t^8. \tag{Q58O.2.4}$$

The identity follows from $S_0 = 2$, $S_1 = 1$ and $S_n = S_{n-1} + tS_{n-2}$.

If $\mathbb{E}N \leq -b(d)d$, then $a/u \geq [2b(d)c/M]x$. Denote the coefficient in brackets by $\ell(d)$. Equations (Q58O.2.1) and (Q58O.2.4) reduce the estimate to a positive lower bound for $w^{16} - M_h(x) + \ell(d)xD_h(x)$, using the mean, second moment and row bound of the same N.

Q58O.3 Three scalar estimates on the full continuous domains

Scalar I. Assume $|N|^2 \leq d$, $\mathbb{E}N \leq -d/2$, $\mathbb{E}N^2 \leq \theta d$, and set $c = 1 - 3d/4$. Then $h = \theta$, $k_0 = 3$, $c \geq 3/4$, $x < 1$, and $\ell(d) = c \geq 3/4$. Also $M_4 < 5/16$, $560M_3 < 136$, and $560D_3 > 350$. After division by x^2, the difference from (Q58O.2.4) is at least

$$\begin{aligned} Q_I(x) = {} & 45/4 - 136x + 629x^2 - 1365x^3 + 7000x^4 \\ & - 3575x^5 + 49000x^6 - 3575x^7 + 160000x^8 \\ & - 1365x^9 + 244000x^{10} - 175x^{11} + 139000x^{12} - 5x^{13} + 6500x^{14}. \end{aligned}$$

The quartic coefficient is at least $936 - 1820(5/16) + (3/4)350 = 629.75$. From degree six onward, retain the positive Lucas terms and use $M_j \leq 5/16$. The polynomial decomposes as

$$\begin{aligned} Q_I = {} & (45/4 - 136x + 420x^2) + x^2(209 - 1365x + 2400x^2) \\ & + x^4(4600 - 3575x + 1000x^2) + x^6(48000 - 3575x + 100x^2) \\ & + x^8(159900 - 1365x + 10x^2) + x^{10}(243990 - 175x + x^2) \\ & + x^{12}(138999 - 5x + x^2) + 6499x^{14}. \end{aligned} \tag{Q58O.3.1}$$

Every bracket has negative discriminant; the first has minimum $101/420$. Thus every nonnegative $H \leq (c + N/2)^2$ satisfies

$$EH^8 \leq 1 - (101/1680)c^{14}d \leq 1 - d/2000. \tag{Q58O.3.2}$$

Here $(3/4)^{14} > 1/64$.

Scalar II. Let $M = 71/63$, and suppose

$$|N|^2 \leq Md, \qquad \mathbb{E}N^2 \leq \theta Md, \qquad \mathbb{E}N \leq -(4/5 - 6d/5)d, \qquad c = 1 - 4d/5.$$

Then $k_0 = 1008/355 > 17/6$, $c \geq 11/15$, $x < 1$, and

$$\ell(d) = \frac{2c(4/5 - 6d/5)}{M} \geq \frac{44}{75M} > 13/25.$$

Since $d = 4c^2x^2/M \leq 4x^2/M$, we also have

$$\ell(d) \geq \frac{8}{5M} - \frac{368}{25M^2}x^2. \tag{Q58O.3.3}$$

Use (Q58O.3.3) for D_1, and $\ell > 13/25$ for D_3, D_4, D_5; discard the remaining positive terms. The difference divided by x^2 is at least

$$\begin{aligned} Q_{II}(x) = {} & 155/8 - (1351/10)x + 274x^2 - 960x^3 + 6500x^4 \\ & - 3575x^5 + 38000x^6 - 3575x^7 + 120000x^8 \\ & - 1365x^9 + 170000x^{10} - 175x^{11} + 90000x^{12} - 5x^{13} + 4000x^{14}. \end{aligned}$$

The quadratic coefficient is $384/(5M) - 195/4 > 155/8$, and $560M_3 = 8645/64 < 1351/10$. The quartic coefficient is at least

$$20736/(25M^2) - 1820M_4 + 560D_3 \cdot 44/(75M) > 274,$$

using $M^2 < 1271/1000$, $20736/(25M^2) > 652$, $M_4 < 31/100$, and $D_3 > 16/25$. The magnitude of the negative quintic coefficient is below

$$4368(9/32) - 1820(2/7)(13/25) = 958.1 < 960,$$

because $M_5 < 9/32$ and $D_4 > 2/7$. The sextic coefficient exceeds $8000 - 2483 + 1067 = 6584 > 6500$, by $352k_0^3 > 8000$, $M_6 < 31/100$, and $D_5 = 495885/1048576 > 47/100$. The remaining even coefficients, using $k_0 > 17/6$, exceed $38000, 120000, 170000, 90000, 4000$, respectively. The magnitudes of the remaining negative odd coefficients, using $M_j < 31/100$, are below $3575, 3575, 1365, 175, 5$.

The decomposition is

$$\begin{aligned} Q_{II} = {} & (155/8 - (1351/10)x + 237x^2) + x^2(37 - 960x + 6400x^2) \\ & + x^4(100 - 3575x + 32000x^2) + x^6(6000 - 3575x + 600x^2) \\ & + x^8(119400 - 1365x + 10x^2) + x^{10}(169990 - 175x + x^2) \\ & + x^{12}(89999 - 5x + x^2) + 3999x^{14}. \end{aligned} \tag{Q58O.3.4}$$

Every bracket is positive. The first has minimum $11549/94800 > 1/10$, and the second has minimum one. For $0 \le H \le (c + N/2)^2$, this gives

$$EH^8 \le 1 - (M/40)c^{14}d \le 1 - d/12000. \tag{Q58O.3.5}$$

Use $c \ge 11/15 > 2/3$ and $(2/3)^{14} > 1/300$.

Scalar III. Assume

$$|N|^2 \le d, \qquad \mathbb{E}N \le -41d/96, \qquad \mathbb{E}N^2 \le 3d/10, \qquad c = 1 - 37d/64.$$

Then $h = 3/10$, $k_0 = 37/16$, $c \ge 155/192$, $x < 1$, and $\ell(d) \ge 6355/9216 > 11/16$. The difference divided by x^2 is at least

$$\begin{aligned} Q_{III}(x) = {} & 12 - 118x + 321x^2 - 632x^3 + 3500x^4 \\ & - 2720x^5 + 15800x^6 - 2720x^7 + 42000x^8 \\ & - 1040x^9 + 50000x^{10} - 133x^{11} + 22000x^{12} - 4x^{13} + 800x^{14}. \end{aligned}$$

Here

$$M_3 = 21/100, \quad M_4 = 237/1000, \quad M_5 = 2289/10000,$$
$$D_3 = 51/100, \quad D_4 = 147/500, \quad D_5 = 3777/10000 > 3/8.$$

The quadratic coefficient exceeds $37 - 36 + 11 = 12$. The quartic coefficient exceeds $17797/32 - 431.34 + 196.35 = 321.16625$. The magnitude of the negative quintic coefficient is below

$$4368(2289/10000) - (11/16)1820(147/500) = 631.9677 < 632.$$

The sextic coefficient exceeds $557183/128 - 1898 + 1126 > 3500$. Using $k_0^4 > 200/7$ and $k_0^2 > 53/10$, the subsequent even coefficients exceed $15800, 42000, 50000, 22000, 800$; the remaining odd terms use $M_j \le 237/1000$.

We have the decomposition

$$\begin{aligned} Q_{III} &= (12 - 118x + 291x^2) + x^2(30 - 632x + 3330x^2) \\ &\quad + x^4(170 - 2720x + 11000x^2) + x^6(4800 - 2720x + 400x^2) \\ &\quad + x^8(41600 - 1040x + 10x^2) + x^{10}(49990 - 133x + x^2) \\ &\quad + x^{12}(21999 - 4x + x^2) + 799x^{14}. \end{aligned} \tag{Q58O.3.6}$$

Every bracket is positive, and the first has minimum $11/291$. Thus $0 \le H \le (c + N/2)^2$ implies

$$EH^8 \le 1 - (11/1164)c^{14}d \le 1 - d/3000. \tag{Q58O.3.7}$$

Use $c > 4/5$ and $(4/5)^{14} > 1/25$. The displayed terminating decimals denote exact finite rational numbers.

Q58O.4 Every $b \ge 3$ and $L \ge 4$

In the finite envelope (L58.1.7), retain $\varepsilon_L \bar{f}_0$ and $\varepsilon_{L-1}\bar{f}_1$. Both coefficients are at least $d/2$. Moreover, $R_L = V_L - q$ satisfies

$$\mathbb{E}R_L \le -\tau \le -d/2, \qquad |R_L|^2 \le d, \qquad \mathbb{E}R_L^2 \le \theta d.$$

Hence

$$X_{b,L} \le m^2[(1 - d/2)q + R_L - d/2].$$

From $\beta_b \le d$ and $\beta_b \ge \kappa d$,

$$m^2 = 1 - \beta_b + \beta_b^2/4 \le 1 - (11\kappa/12)d \le 1 - d/2.$$

AM–GM gives

$$H_{b,L} \le (1 - 3d/4 + R_L/2)^2. \tag{Q58O.4.1}$$

The square base is the average of two nonnegative factors, and $m^2/(1 - d/2) \le 1$. Scalar I now gives $\mathbb{E}H_{b,L}^8 \le 1 - d/2000$.

Q58O.5 Every $b \geq 3$ at $L = 3$

Expanding $\bar{f}_2 = q + r$ in (L58.1.7) gives

$$X_{b,3} \leq m^2[V_3 - \gamma r - a - B_* q], \quad a = d - k\rho \geq 41d/80.$$

Put $N = V_3 - \gamma r - q$. Its row polynomial is $P_2 - \gamma$. The real part of P_2 is at least $1 - (6/5)\theta^2 > 4/5$, while $\gamma \leq 2/5$; subtracting γ therefore does not increase the modulus. Thus $|N|^2 \leq Md$ and $\mathbb{E}N^2 \leq \theta M d$, where $M = 71/63$. Also

$$-EN = d - kD_3 - \gamma d \geq (4/5 - 6d/5)d. \tag{Q58O.5.1}$$

Here $D_3 \leq \delta_3 \leq \theta^2 d$ and $1 - (6/5)\theta^2 > 4/5$. Let $B = 1 - (209/384)d$. The bound above gives $m^2 \leq B$, and $B_* \geq 5d/8 > (209/384)d$. The positive envelope and AM–GM give

$$H_{b,3} \leq [B - (41/160)d + N/2]^2 \leq (1 - 4d/5 + N/2)^2, \tag{Q58O.5.2}$$

Indeed $209/384 + 41/160 = 1537/1920 > 4/5$. Scalar II proves $\mathbb{E}H_{b,3}^8 \leq 1 - d/12000$.

Q58O.6 Every $b \geq 3$ at $L = 2$

Keep the dependence between m, γ and the finite arm tail. Let $B = 1 - \gamma > 0$ be a scalar, distinct from the operator $B_0 = -K$, and put $R_2 = r + kK^2 R$. The envelope is

$$X_{b,2} \leq m^2[Bq + R_2 - a_1].$$

AM–GM gives $H_{b,2} \leq (c_* + N/2)^2$, where

$$N = \frac{m}{\sqrt{B}} R_2, \quad c_* = m\sqrt{B} - \frac{m a_1}{2\sqrt{B}} \leq m(1 - B_*/2) \leq 1 - \frac{1777}{3072} d \leq 1 - 37d/64. \tag{Q58O.6.1}$$

This uses $m \leq 1 - \kappa d/2$, $B_* \geq 5d/8$, and $d \leq 1/3$.

The row bound. Write $R_2 = Kw$, $w = g + kKR$, so $\|w\|_2^2 = d - 2k\rho + k^2 U$. For all $b \geq 3$, $2m\rho - U \geq 7\delta_2/12$: when $b \geq 4$, use $V \leq \delta_2$ and $\rho \geq \kappa\delta_2$; when $b = 3$, use $\rho = \delta_2$ and $U \leq \theta(1 + \theta)\delta_2$, giving a stronger bound. Also $t_0 - \beta_b \leq \delta_3 \leq \theta\delta_2$ and $t_0 \leq d$, hence

$$1 - m^2 - \gamma \geq -\beta_b^2/4 - \theta\delta_2 - (3/5)\beta_b d.$$

Therefore

$$Bd - m^2\|w\|^2 \geq [7/12 - (17/20 + \theta)d]\delta_2 \geq 79\delta_2/480 \geq 0. \tag{Q58O.6.2}$$

The row Cauchy–Schwarz inequality gives $|N|^2 \leq d$.

The second-order filter and its coupled ratio. At $z = i\sqrt{x}$, write $G_{b-1}(-z) = a_x - i\sqrt{x}\, b_x$, with $\kappa \leq 1 - x \leq a_x, b_x \leq 1$. Since $P_1 = 1 + kz^2 G_{b-1}(-z)$,

$$|P_1|^2 \leq [1 - kx(1 - x)]^2 + k^2 x^3 \leq (1 - x + x^2)^2 + x^3.$$

The middle expression decreases for $1 \leq k \leq 6/5$, because $(6/5)x \leq 1 - x$. Multiplying the last expression by x gives $x - 2x^2 + 3x^3 - x^4 + x^5$, whose derivative is $(1 - 2x)^2 + x^2(5 - 4x + 5x^2) > 0$. Consequently

$$ER_2^2 \leq (4/15)d, \tag{Q58O.6.3}$$

The endpoint value is $8762429/33554432 < 4/15$, with cross-product difference $2781293 > 0$.

To bound the ratio m^2/B, set $C = m^2B = m^2 - mt_0 + V/2$. For $b = 3, 4$, $\beta_b = d$, $m = 1 - d/2$, and $C \geq m^2 - md + d^2/2$. For $b = 3$, use $V \geq \delta_2 \geq d^2$. For $b = 4$, $t_0 = d - \delta_2$, so the difference between $-mt_0$ and $-md$ is $m\delta_2 \geq d^2/2$. Thus

$$C - (8/9)m^4 \geq 1/9 - (2/9)d - d^2/12 + (4/9)d^3 - d^4/18 \geq 79/2916 > 0.$$

For $b \geq 5$, $\beta_b \leq d - \kappa\delta_2 \leq 77/288$, so $m \geq 499/576 > 6/7$. The function $f(m) = 3m^2 - 2m - (8/9)m^4$ increases on $[6/7, 1]$, with $f(6/7) = 24/2401$. If $t_0 \leq \beta_b$, then $C - (8/9)m^4 \geq f(m) > 0$. If $t_0 > \beta_b$, then $b \equiv 2 \pmod 4$, $b \geq 6$, and $t_0 - \beta_b \leq \theta\delta_2$. Now $b - 1$ is odd, and

$$|G_{b-1}(i\sqrt{x})|^2 = \frac{1 + x^{b-1}}{1 + x} \geq 32/45,$$

so $V \geq (32/45)\delta_2$. Therefore

$$C - (8/9)m^4 \geq 24/2401 - (73/1440)\delta_2 \geq 24/2401 - 949/138240 > 0. \tag{Q58O.6.4}$$

This proves $m^2/B \leq 9/8$ for every b. Together with (Q58O.6.3), it gives $\mathbb{E}N^2 \leq (9/8)(4/15)d = 3d/10$. The signed mean is $\mathbb{E}N = -(md - \rho)/\sqrt{B} \leq -41d/96$. Scalar III applies to this same N; using (Q58O.6.1), we obtain $\mathbb{E}H^8_{b,2} \leq 1 - d/3000$.

Q58O.7 The ordinary-eighth catalogue

The estimates for $b \geq 3, L \geq 2$ have common deficit $d/12000$. Combine them with the $b = 2$ positive-connector bounds of Section O.7 and the unit-connector bounds of Section O.8 to obtain (Q58O.0.2). The eight-copy root construction has edge cover one, nonroot cost one and root cost $1/8$, with the joint-law relative-entropy interface of the corresponding ordinary component. Restore genuine equal pairs by positivity, using their fixed $f_\ell h_\ell \leq 1$ directions. Complete reversal replaces K by $-K$ and reverses the receiving arm, terminal arms, connector and equal pairs together.

O.30 Ordinary eighth moments for even terminals with multiple pairs at every connector length

Q90.0 The terminal and its bounds.

After maximal equal pairing, let the distinct residual arms be

$$a_1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j, \qquad j \geq 1. \tag{Q90.0.1}$$

Every a_i-arm has its first edge inward and the remaining edges outward; every b_i-arm is outward. All $L \geq 1$ connector edges point from terminal to receiving root, whose unit arm points outward. For arbitrary genuine equal-pair products E_t, E_r, define

$$F = \prod_{i=1}^{j} h_{a_i} f_{b_i}, \quad X_D = T^L(E_t F), \qquad H_D = E_r p X_D. \tag{Q90.0.2}$$

For $a_1 \geq 2, j \geq 2$, the same directions give

$$\boxed{\mathbb{E}H^8_D \leq e^{-d/6} \qquad (L \geq 1).} \tag{Q90.0.3}$$

The stronger rows are

$$a_1 = 3,\ j \ge 2: \qquad \mathbb{E}H_D^8 \le e^{-d}, \tag{Q90.0.4}$$

$$a_1 \ge 4,\ j \ge 1: \qquad \mathbb{E}H_D^8 \le e^{-(128j/5-24)d}. \tag{Q90.0.5}$$

All integer lengths, gaps and finite pair counts in these ranges are allowed.

Use the finite model, path definitions, skew moments, row bounds and finite convolution in Appendix M.15, and the pair and finite-mask bounds in Appendix M.17. Thus $K' = -K$, $g' = -g$, $T = J + K'$, $\delta_n = \|K^n\mathbf{1}\|_2^2$, $d = \delta_1$, $\theta = 13/32$, $\kappa = 19/32$, and $\tau = d - \delta_2$. All expectations and adjoints use the fixed finite probability measure. In particular $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{n+1} \le \theta\delta_n$, $d\delta_3 \ge \delta_2^2$, and $\delta_3 \ge d^3$. Write $G_n(z) = \sum_{i=0}^{n-1} z^i$, $G_0 = 0$, $\beta_1 = \beta_2 = 0$, $\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i$. The finite G_n norm is at most one for $n = 0, 1$ or $n \ge 3$, while $\|G_2\| \le \sqrt{1+\theta} < 19/16$. For the actual finite mask set

$$R = \sum_{i=1}^{j} \sum_{r=a_i+1}^{b_i} K^r\mathbf{1}, \qquad F \le e^{R-\Gamma},$$

$$\Gamma/d = \begin{cases} \kappa + 16(j-1)/5, & a_1 = 2, \\ 2 - \theta + 16(j-1)/5, & a_1 = 3, \\ 16j/5, & a_1 \ge 4. \end{cases}$$

These are precisely the pair losses of (L90.2.3). The mask has $R = U(K')g'$, with the frequency and propagated norm bounds (L90.2.5)–(L90.2.6); $C_* = 32768/16245 < 100/49$. For $a_1 \ge 4$, the pointwise bound gives $|R|^2 < d/7 \le 1/21 < 1/16$, and for $a_1 = 3$, $|R|^2 < d/3 \le 1/9$. Taking Taylor range $1/4$ or $1/3$ gives $e^R \le 1 + R + (3/5)R^2$. For $a_1 = 2$, $|R|^2 < 5d/6 \le 5/18 < 9/25$, and Taylor range $3/5$ gives $e^R \le 1 + R + (5/8)R^2$. These follow from the finite remainder bound $e^R \le 1 + R + R^2/[2(1-t/3)]$ for $|R| \le t < 3$, proved in Appendix M.17. The full-mean exponential inequality of Appendix M.1 is $\mathbb{E}e^{8K'V} \le e^{8\mathbb{E}(K'V)+17\|V\|_2^2}$. Consequently, if $H \le (c + N/2)^2$, $N = K'V$, with nonnegative square base, then $\log \mathbb{E}H^8 \le 16(c-1) + 8\mathbb{E}N + 17\|V\|^2$.

Q90.3 Finite propagation of the common mask.

Let $V = \mathbb{E}R^2$, $D_n = \mathbb{E}K'^n R$, $a_0 = \mathbb{E}KR = -D_1$, and $\rho = -D_2$. Define

$$\delta = D_0 + cV, \quad \varepsilon = \delta - a_0, \qquad c = 6/5\ (a_1 \ge 3), \quad c = 5/4\ (a_1 = 2). \tag{Q90.3.1}$$

With $B = K'$, $u = g'$, $\sigma = V$, and $D_1 = -a_0$, (R29.5.3) gives the positive upper $0 < T^L F \le e^{-\Gamma} Y_L$:

$$\begin{aligned} Y_1 &= q + \delta + K'R, \\ Y_L &= \bar{f}_L + \delta\bar{f}_{L-1} - a_0\bar{f}_{L-2} + K'^L R + \sum_{n=2}^{L-1} D_n \bar{f}_{L-1-n} \quad (L \ge 2). \end{aligned} \tag{Q90.3.2}$$

The sum is empty at $L = 2$. Writing $p + Y_L = 2 + c_L + N_L$, $N_L = K'V_L$, use (M.50) at $r = 8$ to obtain

$$\log \mathbb{E}[(pT^L F)^8] \le -8\Gamma + 8c_L + 8\mathbb{E}N_L + 17\|V_L\|^2. \tag{Q90.3.3}$$

The three short lengths have the exact forms

$$L = 1: \qquad c_1 = \delta, \quad V_1 = R, \quad \mathbb{E}N_1 = -a_0; \tag{Q90.3.4}$$

$$\begin{aligned} L=2:\quad & c_2=\varepsilon,\quad V_2=\delta\mathbf{1}+g'+K'R,\quad \mathbb{E}N_2=-d-\rho,\\ & \|V_2\|^2=d+2\rho+\|K'R\|^2+\delta^2-2\delta a_0; \end{aligned} \tag{Q90.3.5}$$

$$\begin{aligned} L=3:\quad & c_3=\varepsilon-d-\rho,\quad V_3=\varepsilon\mathbf{1}+(1+\delta)g'+K'g'+K'^2R,\\ & \mathbb{E}N_3=-(1+\delta)d+D_3, \end{aligned} \tag{Q90.3.6}$$

$$\|V_3\|^2=(1+\delta)^2d+\delta_2-2(1+\delta)D_3+2D_4+\|K'^2R\|^2+\varepsilon^2-2\varepsilon(d+\rho).$$

In particular, these retain the constant coordinates and their cross terms.

Q90.3.2 Finite convolution losses.

Put $S'_n=\sum_{i=0}^{n}K'^i\mathbf{1}$, $\nu=1+\varepsilon$, and define

$$\begin{aligned} Z_L^*&=S'_L+\delta S'_{L-1}-a_0S'_{L-2}+K'^LR,\\ P_n(z)=G_n(z)+z^nU(z),&\qquad Q_n(z)=P_n(z)+\varepsilon G_{n-2}(z)+\delta z^{n-2}\quad (n\ge 2). \end{aligned} \tag{Q90.3.7}$$

Subtracting (Q90.3.2) term by term gives

$$\begin{aligned} Z_L^*-Y_L&=\sum_{n=3}^{L}\gamma_n\bar f_{L-n},\\ \gamma_3=d-D_2&=d+\rho,\quad \gamma_4=d+\delta d-D_3,\\ \gamma_n&=\beta_n+\delta\beta_{n-1}-a_0\beta_{n-2}-D_{n-1}\quad (n\ge 5). \end{aligned} \tag{Q90.3.8}$$

The finite polynomials also give $Z_L^*=\nu+Q_L(K')g'$, $Q_L=\nu+zQ_{L-1}$ for $L\ge 3$, and $\mathbb{E}Z_L^*=\nu-\gamma_{L+1}$ for $L\ge 4$. Suppose every $\gamma_n\ge hd$, $n\ge 3$, and $B=1+\varepsilon-hd>0$. For $L\ge 4$, keep the two distinct last terms $\gamma_L\bar f_0$, $\gamma_{L-1}\bar f_1$:

$$\begin{aligned} 0<e^{\Gamma}T^LF\le Y_L\le Z&:=Z_L^*-hd(1+q)=Bq-hd+N_0,\\ N_0=K'Q_{L-1}(K')g',&\qquad \mathbb{E}N_0=-\gamma_{L+1}\le -hd. \end{aligned} \tag{Q90.3.9}$$

Let $m=e^{-\Gamma/2}$. Weighted AM–GM for $p,Z\ge 0$ gives

$$\begin{aligned} pT^LF\le(c_*+N/2)^2,&\qquad c_*=m\sqrt{B}-\frac{mhd}{2\sqrt{B}},\\ N=K'V_*,&\qquad V_*=\frac{m}{\sqrt{B}}Q_{L-1}(K')g'. \end{aligned} \tag{Q90.3.10}$$

Its nonnegative square base is $m(\sqrt{B}p+Z/\sqrt{B})/2$. Alternatively, retain $m^2=e^{-\Gamma}$ outside the square and use the exponential form (Q90.3.3).

Q90.3.3 Finite contraction when U is small.

If $|U(i\sqrt{x})|\le 3/8$, then $|P_n|\le 1$ for every $n\ge 3$. For $n=3,5$, $|G_n|^2\le 1-\kappa x$, hence $|G_n|\le 1-\kappa x/2$. The perturbation is at most $(3/8)x^{3/2}<39x/160<\kappa x/2$, and is smaller for $n=5$. For $n=4$, $|G_4|=(1-x)\sqrt{1+x}\le 1-x/2$, while the perturbation is at most $3\theta x/8<x/2$. For $n\ge 6$, the finite geometric formula and $|U|\le 1$ give

$$|P_n|\le(1+x^3)/\sqrt{1+x}+x^3\le 1. \tag{Q90.3.11}$$

The last comparison is equivalent to $x(1+\sqrt{1+x})\le 1$, whose left side is at most $143/160$. The zero frequency is included.

Q90.4 Minimum arm at least four, every pair count and length.

Let $a_1 \ge 4$, $j \ge 1$, $\Gamma = 16jd/5$. Summing the positive and negative even-moment subcollections of the finite mask separately gives

$$D_0, a_0 \in [-d/5, d/12], \quad \rho, D_3 \in [-d/12, d/30], \quad |D_n| \le d/30 \quad (n \ge 4). \tag{Q90.4.1}$$

The geometric constants are $\theta^2/(1-\theta^2) = 169/855 < 1/5$, $\theta^3/(1-\theta^2) = 2197/27360 < 1/12$, and $\theta^4/(1-\theta^2) = 28561/875520 < 1/30$. The first possible negative even term of D_0 is $-\delta_3$ and the first positive one is δ_4, since R starts at least at $K^5\mathbf{1}$; the classes for ρ, D_3 start with $-\delta_4$ or δ_5. The mask bound further gives

$$V \le 28561d/519840 < d/18, \quad \|K'R\|^2 < d/44, \quad \|K'^2R\|^2 < d/108, \quad \|U\|^2 < 1/18 < 1/16.$$

Thus

$$\delta \in [-d/5, 3d/20], \quad a_0 \in [-d/5, d/12], \quad \varepsilon \in [-17d/60, 7d/20], \quad d + \rho \in [11d/12, 31d/30]. \tag{Q90.4.2}$$

At $L = 1$, the cost beyond -8Γ is $8\varepsilon + 17V \le 337d/90 < 4d$. At $L = 2$, the constant square block is at most $3d^2/25 \le d/25$, so

$$\|V_2\|^2 \le (1 + 1/15 + 1/44 + 1/25)d < 8d/7, \qquad \mathbb{E}N_2 \le -11d/12.$$

The extra cost is at most $(14/5 - 22/3 + 136/7)d = 1564d/105 < 15d$. At $L = 3$, (Q90.3.6) gives

$$c_3 + \mathbb{E}N_3 \le (7/20 - 2 + 1/15 + 7/60)d = -22d/15.$$

The constant block $\varepsilon^2 - 2\varepsilon(d + \rho)$ is at most $203d^2/240$. Taking $1 + \delta \le 21/20$,

$$\|V_3\|^2/d \le 441/400 + 13/32 + 7/40 + 1/15 + 1/108 + 203/720 = 44099/21600 < 21/10.$$

Here $7/40$ comes from the negative endpoint of D_3, and $1/15$ from the positive endpoint of D_4. The extra cost is at most $(-176/15 + 357/10)d = 719d/30 < 24d$. Therefore

$$\mathbb{E}[(pT^L F)^8] \le e^{-(128j/5 - 24)d} \qquad (L = 1, 2, 3). \tag{Q90.4.3}$$

For the longer lengths, (Q90.3.8) gives $\gamma_3 \ge 11d/12$, $\gamma_4 \ge 9d/10$, and for $n \ge 5$,

$$\gamma_n \ge [19/32 - 1/30 - (17/60)d]d \ge (9/20 + 23/1440)d.$$

Take $h = 9/20$. Part Q90.3.3 gives $\|P_n\| \le 1$. The three finite ranges for Q_n are

$$\begin{aligned} n = 3: &\quad \|Q_n\| \le 1 + (7/20 + 13/100)d < 1 + d/2, \\ n = 4: &\quad \|Q_n\| \le 1 + (133/320 + 13/160)d = 1 + (159/320)d < 1 + d/2. \end{aligned}$$

For $n \ge 5$, contraction of G_{n-2} and the smaller last power give the same bound. The $n = 4$ row uses $\sqrt{1+\theta} < 19/16$ for its G_2 term. Equation (Q90.3.9) now has

$$1 - 11d/15 \le B \le 1 - d/10, \qquad \mathbb{E}N_0 \le -9d/20. \tag{Q90.4.4}$$

Keep $m^2 = e^{-\Gamma}$ outside the square. The unscaled precursor norm is

$$\frac{(1 + d/2)^2}{1 - 11d/15} d \le (245/136)d. \tag{Q90.4.5}$$

The ratio increases on the full domain and equals $245/136$ at $d = 1/3$. Its unscaled mean is at most $-9d/20$, since $B \le 1$, and its constant satisfies

$$\sqrt{B} - \frac{hd}{2\sqrt{B}} \le (1+B)/2 - hd/2 \le 1 - 11d/40.$$

The full exponential cost is therefore

$$-8\Gamma + [-22/5 - 18/5 + 17(245/136)]d = -(128j/5 - 181/8)d. \tag{Q90.4.6}$$

As $181/8 < 24$, this and (Q90.4.3) prove (Q90.0.5), including the two retained terms at $L = 4$.

Q90.5 Minimum arm three and at least two pairs.

Let $a_1 = 3$, $j \ge 2$, and $\Gamma = [2 - \theta + 16(j-1)/5]d \ge 767d/160$. The mask begins at $K^4\mathbf{1}$. Its finite signed subcollections give

$$\begin{aligned} D_0 \in [-d/5, d/2], \quad a_0 \in [-d/5, d/12], \quad \rho \in [-d/12, d/5], \\ D_3 \in [-d/12, d/30], \qquad |D_n| \le d/12 \quad (n \ge 4). \end{aligned} \tag{Q90.5.1}$$

In particular $D_3 = -\mathbb{E}K^3R$ has first possible negative term $-\delta_4$. The finite mask bounds give

$$V \le 2197d/16245 < 5d/36, \quad \|K'R\|^2 < 3d/50, \quad \|K'^2R\|^2 < d/40, \quad |U|^2 < 5/36 < 9/64.$$

Consequently

$$\delta \in [-d/5, 2d/3], \quad \varepsilon \in [-17d/60, 7d/8], \quad d + \rho \in [11d/12, 6d/5]. \tag{Q90.5.2}$$

The upper δ uses $D_0 \le d/2$, $6V/5 \le d/6$; the upper ε first gives $13d/15$, then $7d/8$. The bound on U holds at every frequency. At $L = 1$, the extra cost is $7d + 85d/36 = 337d/36 < 10d$. At $L = 2$, the constant block is at most $32d^2/45$, so

$$\|V_2\|^2/d \le 1 + 2/5 + 3/50 + 32/135 < 7/4, \qquad \mathbb{E}N_2 \le -11d/12.$$

Its extra cost is at most $(7 - 22/3 + 119/4)d = 353d/12 < 30d$. At $L = 3$,

$$c_3 + \mathbb{E}N_3 \le (7/8 - 2 + 1/12 + 1/15 + 1/30)d = -113d/120. \tag{Q90.5.3}$$

If $\varepsilon \ge 0$, then $\varepsilon \le 7d/8 < 2(d + \rho)$, so its constant block is nonpositive. If $\varepsilon < 0$, the block is at most $2737d^2/3600$, using $|\varepsilon| \le 17d/60$. Take $1 + \delta \le 11/9$ in (Q90.3.6); then

$$\begin{aligned} \|V_3\|^2/d &\le 121/81 + 13/32 + 11/54 + 1/6 + 1/40 + 2737/10800 \\ &< 3/2 + 13/32 + 5/24 + 1/6 + 1/40 + 13/50 = 6159/2400 < 13/5. \end{aligned} \tag{Q90.5.4}$$

The extra cost is at most $(-113/15 + 221/5)d = 110d/3$. Since $8\Gamma \ge 767d/20$, the gap is $101d/60 > d$. Thus each short length has bound e^{-d}. For $L \ge 4$, $\gamma_3 \ge 11d/12$, $\gamma_4 \ge 9d/10$, and

$$\gamma_n \ge [19/32 - 1/12 - (17/60)d]d \ge (2/5 + 23/1440)d \quad (n \ge 5). \tag{Q90.5.5}$$

Take $h = 2/5$. Part Q90.3.3 gives $\|P_n\| \le 1$. The extra coefficients for Q_3, Q_4 are at most $157d/120$ and $503d/384$; for $n \ge 5$ they are no larger than the $n = 3$ bound. Hence $\|Q_n\| \le 1 + 4d/3$ for every $n \ge 3$, and

$$1 - 41d/60 \le B \le 1 + 19d/40, \qquad \mathbb{E}N_0 \le -2d/5. \tag{Q90.5.6}$$

Let $\lambda = \Gamma/d \geq 767/160$, $m = e^{-\Gamma/2}$, retaining the scale in V_*. Then

$$\|V_*\|^2 \leq e^{-\lambda d}\frac{(1+4d/3)^2}{1-41d/60}d \leq d. \tag{Q90.5.7}$$

The logarithmic derivative of the ratio is at most $-767/160 + 8/3 + 123/139 < 0$, and its value at zero is one. The constant $c_* \leq m\sqrt{B}$ satisfies

$$c_* \leq e^{-691d/320} \leq (1+691d/320)^{-1} \leq 1-(2073/1651)d \leq 1-5d/4. \tag{Q90.5.8}$$

The last step uses $d \leq 1/3$ and $2073/1651 > 5/4$. The actual mean $\mathbb{E}N = (m/\sqrt{B})\mathbb{E}N_0$ is nonpositive. The full-mean exponential bound gives $\mathbb{E}[(pT^L F)^8] \leq e^{-3d}$. Together with the short lengths this proves (Q90.0.4).

Q90.6 Minimum arm two and at least two pairs.

Let $a_1 = 2$, $j \geq 2$, and $\Gamma = [\kappa + 16(j-1)/5]d \geq 607d/160$. The coefficient of $K^3\mathbf{1}$ in R is exactly one; all later coefficients are zero or one. The signed even-moment subcollections and the mask norm give

$$\begin{gathered} D_0 \in [-d/5, d/2], \quad a_0 \leq d/2, \quad \rho \in [-d/12, d/5], \quad D_3 \leq d/5, \\ |D_n| \leq d/12 \quad (n \geq 4), \quad V \leq 5408d/16245 < d/3, \\ \|K'R\|^2 \leq 13d/96, \quad \|K'^2R\|^2 < d/18, \quad \delta = D_0 + 5V/4 \in [-d/5, 11d/12]. \end{gathered} \tag{Q90.6.1}$$

The required $K^3\mathbf{1}$ term contributes $+\delta_2$ to $a_0 = \mathbb{E}KR$ and $+\delta_3$ to $D_3 = -\mathbb{E}K^3R$. The possible negative tails are bounded by $\delta_3/(1-\theta^2)$ and $\delta_4/(1-\theta^2)$, respectively, so

$$\begin{aligned} a_0 &\geq \delta_2 - \frac{\delta_3}{1-\theta^2} \geq (439/855)\delta_2 \geq 0, \\ D_3 &\geq \delta_3 - \frac{\delta_4}{1-\theta^2} \geq (439/855)\delta_3 > \delta_3/2 \geq d^3/2. \end{aligned} \tag{Q90.6.2}$$

Here $\theta/(1-\theta^2) = 416/855 < 1/2$; the sign of D_3 uses $K' = -K$. The positive terms of D_0 sum to at most $\delta_2/(1-\theta^2)$. Thus

$$D_0 - a_0 \leq \frac{\delta_2+\delta_3}{1-\theta^2} - \delta_2 \leq \frac{\theta}{1-\theta}\delta_2 = (13/19)\delta_2.$$

With $V \leq d/3$ and $\delta_2 \leq \theta d$, this yields $\varepsilon \leq (169/608 + 5/12)d = (1267/1824)d < 7d/10$. The lower bound follows from $\delta \geq -d/5$, $a_0 \leq d/2$, hence

$$|\varepsilon| \leq 7d/10, \qquad d + \rho \in [11d/12, 6d/5]. \tag{Q90.6.3}$$

At $L = 1$, the extra cost is at most $28d/5 + 17d/3 < 12d < 8\Gamma$. At $L = 2$, if $\delta \geq 0$, then $a_0 \geq 0$ bounds the constant block by $\delta^2 \leq 121d^2/144$. If $\delta < 0$, it is at most $d^2/25 + d^2/5 = 6d^2/25 < 121d^2/144$. Therefore

$$\|V_2\|^2/d \leq 1 + 2/5 + 13/96 + 121/432 = 7843/4320 < 11/6.$$

Since $\mathbb{E}N_2 \leq -11d/12$, $c_2 \leq 7d/10$, the extra cost is

$$8c_2 + 8\mathbb{E}N_2 + 17\|V_2\|^2 \leq (28/5 - 22/3 + 187/6)d = 883d/30.$$

As $8\Gamma \geq 607d/20$, and $607/20 - 883/30 = 11/12$,

$$\mathbb{E}[(pT^2F)^8] \leq e^{-11d/12}. \tag{Q90.6.4}$$

Q90.6.3 The common-variable cancellation at $L = 3$.

Equations (Q90.3.6) and (Q90.6.1)–(Q90.6.3) give

$$c_3 + \mathbb{E}N_3 \le (7/10 - 2 + 1/12 + 1/15 + 1/5)d = -19d/20. \tag{Q90.6.5}$$

Use the exact $\delta = \varepsilon + a_0$ to rewrite the full norm as

$$\begin{aligned}\|V_3\|^2 &= d + \delta_2 + 2a_0 d + \delta^2 d + \varepsilon^2 - 2\varepsilon\rho \\ &\quad + \|K'^2 R\|^2 - 2(1+\delta)D_3 + 2D_4.\end{aligned} \tag{Q90.6.6}$$

The term $2\varepsilon d$ has canceled with the constant-coordinate cross term. The required coefficient in (Q90.6.2), and $1 + \delta \ge 14/15$, imply

$$\delta^2 d - 2(1+\delta)D_3 \le (121/144 - 14/15)d^3 \le 0. \tag{Q90.6.7}$$

The remaining quadratic terms satisfy $2a_0 d \le d^2$, $\varepsilon^2 \le 49d^2/100$, $-2\varepsilon\rho \le 7d^2/25$. Hence

$$\begin{aligned}\|V_3\|^2 &\le (1 + 13/32 + 1/18 + 1/6)d + (177/100)d^2 \\ &\le (469/288 + 59/100)d = (15973/7200)d.\end{aligned} \tag{Q90.6.8}$$

Substitute (Q90.6.5) and (Q90.6.8) in (Q90.3.3). The cost beyond -8Γ is at most $216821d/7200$; its gap from $607d/20$ is $1699d/7200 > d/5$. Thus

$$\mathbb{E}[(pT^3F)^8] \le e^{-d/5}. \tag{Q90.6.9}$$

This includes the shortest table $(2, 3, 4, 5)$ and every larger finite gap and pair count.

Q90.6.4 The required coefficient and contraction of every P_n.

At $z = i\sqrt{x}$, write $U(z) = A_x + zB_x$. The required $K^3\mathbf{1}$ term becomes $-z^2$ in the K' notation. Separately bounding the positive and negative finite tails gives

$$\begin{aligned}x - \frac{x^2}{1-x^2} \le A_x \le \frac{x}{1-x^2}, \qquad A_x \ge 0, \\ -\frac{x}{1-x^2} \le B_x \le \frac{x^2}{1-x^2}.\end{aligned} \tag{Q90.6.10}$$

The coordinate B_x is distinct from the scalar B in (Q90.3.9). For $n = 3$, the real and normalized imaginary coordinates are $1 - x + x^2 B_x$, $1 - xA_x$. Their lower bounds are $(1 - x - x^2)/(1 - x^2) > 0$ and $(1 - 2x^2)/(1 - x^2) > 0$, respectively. Since $\theta^3/(1 - \theta^2) < 1/12$, their upper bounds are $1 - 11x/12$ and one. Consequently

$$|P_3|^2 \le (1 - 11x/12)^2 + x \le 1. \tag{Q90.6.11}$$

For $n = 4$, the coordinates are $1 - x + x^2 A_x$, $1 - x + x^2 B_x$, both nonnegative and at most $1 - 4x/5$, using $\theta^2/(1 - \theta^2) < 1/5$. Thus

$$|P_4|^2 \le (1 + x)(1 - 4x/5)^2 \le 1. \tag{Q90.6.12}$$

For $n = 5$, the coordinates are $1 - x + x^2 - x^3 B_x$, $1 - x + x^2 A_x$, again nonnegative. Using $\theta/(1-\theta^2) < 1/2$, they are at most $1 - x/2$ and $1 - 4x/5$. Therefore

$$|P_5|^2 \le (1 - x/2)^2 + x(1 - 4x/5)^2 = 1 - 27x^2/20 + 16x^3/25 \le 1. \tag{Q90.6.13}$$

For every $n \ge 6$, the mask's frequency bound gives $|U(z)|^2 \le x^2 C_* < 1/3$, so the finite estimate (Q90.3.11) applies. Thus $\|P_n\| \le 1$ for all $n \ge 3$, including the P_3 used at $L = 4$.

Q90.6.5 Every finite $L \geq 4$.

The convolution coefficients satisfy $\gamma_3 \geq 11d/12$, $\gamma_4 \geq 11d/15$. For $n \geq 5$, use $\delta \geq -d/5$, $a_0 \leq d/2$, and $D_{n-1} \leq d/12$ to obtain

$$\gamma_n \geq [19/32 - 1/12 - (7/10)d]d \geq 133d/480 > d/4. \tag{Q90.6.14}$$

Take $h = 1/4$. The extra coefficient for Q_3 is $7/10 + (11/12)(13/20) = 311/240 < 4/3$. For Q_4, it is $(7/10)(19/16) + (11/12)(13/32) < 4/3$; for $n \geq 5$, it is no larger than for $n = 3$. Hence $\|Q_n\| \leq 1 + 4d/3$ for every $n \geq 3$. Equation (Q90.3.9) gives $B \in [1 - 19d/20, 1 + 9d/20]$, with positive lower endpoint, and $\mathbb{E}N_0 \leq -d/4$. Let $\lambda = \Gamma/d \geq 607/160$, $m = e^{-\lambda d/2}$, in (Q90.3.10). Then

$$\|V_*\|^2 \leq e^{-\lambda d}\frac{(1 + 4d/3)^2}{1 - 19d/20}d \leq d. \tag{Q90.6.15}$$

For the last inequality, the logarithmic derivative of the ratio is at most

$$-607/160 + \frac{8/3}{1 + 4d/3} + \frac{19/20}{1 - 19d/20}.$$

The sum of the two fractions is convex in d, so its maximum on $[0, 1/3]$ occurs at an endpoint. The endpoint values are $217/60$, $1725/533$, both less than $607/160$. The ratio therefore decreases from one at zero. The constant satisfies

$$c_* \leq m\sqrt{B} \leq e^{-107d/64} \leq (1 + 107d/64)^{-1} \leq 1 - (321/299)d \leq 1 - (103/96)d. \tag{Q90.6.16}$$

This uses $d \leq 1/3$; the cross-product difference in the last comparison is 19. The actual mean $\mathbb{E}N$ is nonpositive and the AM–GM square base is nonnegative. The exponential bound yields

$$\log \mathbb{E}[(pT^L F)^8] \leq -[16(103/96) - 17]d = -d/6. \tag{Q90.6.17}$$

Together with the $L = 1$ bound, (Q90.6.4), and (Q90.6.9), this proves the minimum-two case for all $j \geq 2, L \geq 1$. Parts Q90.4–Q90.5 complete (Q90.0.3)–(Q90.0.5).

Q90.8 Decorations, zero boundary and full reversal.

By positivity and the equal-pair comparison,

$$0 < H_D = E_r pT^L(E_t F) \leq pT^L F, \qquad 0 < X_D = T^L(E_t F) \leq T^L F. \tag{Q90.8.1}$$

The terminal product stays before propagation and the root product at the receiving root. At $d = 0$, $g = 0$; leaf integration makes all tree messages one even if $K \neq 0$. All divisions by d above concern $d > 0$. Full reversal reverses terminal, connector, root unit arm and every equal-pair decoration together. It replaces K by $-K$, exchanges A/T, p/q, $f_n/\bar{f}_n$, and h_n/W_n, where $W_n = A\bar{f}_{n-1}$. The spectral moments and probability measure are unchanged, giving the reversed ordinary eighth bounds.

O.31 Four unit-root ordinary-eighth families

After maximal genuine equal pairing let the residual lengths be

$$\Sigma = (1, b_1, c_1, \dots, b_k, c_k), \qquad 1 < b_1 < c_1 < \cdots < b_k < c_k, \quad k \geq 1. \tag{Q96.0.1}$$

All $L \ge 1$ connector edges point from terminal to receiver, whose existing unit arm is outward. For genuine equal-pair products at the two roots, including empty products, put

$$H = E_r p T^L (E_t F). \tag{Q96.0.2}$$

The following directions are fixed by the residual lengths:

range	F	$\mathbb{E}H^8$ upper
$b_1 \ge 4,\ k \ge 1$	$q \prod_i h_{b_i} f_{c_i}$	e^{-d}
$b_1 = 3,\ k \ge 2$	$q f_3 h_{c_1} \prod_{i \ge 2} h_{b_i} f_{c_i}$	$e^{-d/2}$
$b_1 = 2,\ k \ge 3$	$q f_2 h_{c_1} \prod_{i \ge 2} h_{b_i} f_{c_i}$	$e^{-d/2}$
$\Sigma = (1,3,4)$	$q f_3 h_4$	$1 - d/601.$

The first row retains the stronger all-domain rates

$$\begin{aligned} &b_1 = 4, k = 1: \quad \mathbb{E}H^8 \le e^{-2d}, \\ &b_1 \ge 4, k \ge 2: \quad \mathbb{E}H^8 \le e^{-[18+128(k-2)/5]d}, \\ &b_1 \ge 5, k \ge 1: \quad \mathbb{E}H^8 \le e^{-[3+128(k-1)/5]d}. \end{aligned} \tag{Q96.0.3}$$

Every instance of the first row keeps its displayed $h_b f_c$ direction even at $L = 1$. In each of the other rows the first pair has direction $f_b h_c$.

Proof. Apply Theorem O.2 with the actual terminal minimum $a = 1$. Lemma O.5 and Section O.24.3 give the three strong rates in (Q96.0.3). Section O.24.5 gives the second row for every finite L, on the intervals meeting at $d = 1/6$. Section O.24.8 gives the third row in the same way. The fourth row is the genuine G_1 case of Section O.24.6; its low-domain finite continuation and the same-direction high comparison meet at $d = 1/8$.

The original measure, terminal factor q and connector T^L are retained throughout. Positivity restores E_t, E_r; complete reversal and $d = 0$ follow as in Section O.24.15. □

O.32 All nonunit odd terminals: choosing the actual eighth-moment direction

Let the strict odd residual list at the terminal be

$$\Sigma = (\ell, r_1, t_1, \dots, r_j, t_j), \qquad 2 \le \ell < r_1 < t_1 < \cdots < r_j < t_j, \quad j \ge 1. \tag{R30.1.1}$$

Keep its genuine equal-pair product E_t. For the existing receiving arm $m \ge 3$ and connector $L \ge 1$, put $n = \ell + L$ and choose

$$F_\Sigma = \begin{cases} \bar{f}_\ell \prod_{i=1}^{j} h_{r_i} f_{t_i}, & r_1 \ge 4, \\ \bar{f}_2 f_3 h_c \prod_{i=2}^{j} h_{r_i} f_{t_i}, & r_1 = 3,\ \ell = 2,\ c = t_1 \ge 4. \end{cases} \tag{R30.1.2}$$

Empty products equal one. If $m \le n$, the receiving arm is $W_m = A \bar{f}_{m-1}$ and the connector is T^L. If $m > n$, the receiving arm is $\bar{f}_m = T^m \mathbf{1}$ and the connector is $A T^{L-1}$. Thus the first connector edge changes only according to the actual integers. When $L = 1$ the latter operator is A on the one existing edge. Denote the chosen receiving arm by $R_{m,n}$ and the connector by $\mathcal{P}_{m,n,L}$. We prove

$$\mathbb{E}[\{R_{m,n} \mathcal{P}_{m,n,L} (E_t F_\Sigma)\}^8] \le e^{-3d}. \tag{R30.1.3}$$

The same bound holds after a genuine equal-pair product E_c and any finite ordered negative post block $P_c^- = \prod_{\nu=1}^{k} W_{u_\nu} \bar{f}_{v_\nu}$, $4 \le u_1 < v_1 < \cdots < u_k < v_k$, $k \ge 0$, are restored at the receiver. The terminal and receiver length intervals may overlap. This is a construction for the stated direction.

R30.2 Finite tools

Use $B=-K$, $u=B\mathbf{1}=-g$, $T=J+B$, $A=J-B$, $p=1-u$, $q=1+u$, $\delta_i=\|B^i\mathbf{1}\|_2^2$, $d=\delta_1$, $\theta=13/32$, $\kappa=19/32$ and $\tau=d-\delta_2$. The finite spectral and row bounds give

$$d^2\le\delta_2\le\theta d,\quad d\le 1/3,\quad \delta_{i+1}\le\theta\delta_i,\quad \|B\|^2\le\theta,\quad |BV(x)|\le\|V\|_2.$$

For $G_s(z)=\sum_{i=0}^{s-1}z^i$, $G_0=0$, one has $\|G_s(B)\|\le 1$ for $s=0,1$ or $s\ge 3$, while $\|G_2(B)\|^2\le 1+\theta$. Define the finite sums

$$\beta_1=\beta_2=0,\qquad \beta_n=\sum_{i=1}^{\lfloor(n-1)/2\rfloor}(-1)^{i-1}\delta_i,\qquad \tau\le\beta_n\le d\quad(n\ge 3).$$

The spectral measure ν at u has mass d, support $[0,\theta]$ and moments $\int x^i\,d\nu=\delta_{i+1}$. Writing $z=i\sqrt{x}$, a real polynomial $P=P_e+zP_o$ satisfies $\mathbb{E}P(B)u=-\int P_o\,d\nu$. All four path messages lie in $[0,2]$; their means satisfy $\mu_a=\mathbb{E}f_a=\mathbb{E}\bar f_a\le 1$, $\mu_2=1-d$, $\mu_a\le 1-2d$ for $a\ge 3$, and $W_a+\bar f_a=2\mu_{a-1}$. Only genuine equal pairs are bounded by one pointwise.

The full-mean exponential lemma of Section M.1 is

$$\mathbb{E}e^{8BV}\le\exp\{8\mathbb{E}(BV)+17\|V\|_2^2\}.\tag{R30.2.5}$$

It applies to every real V, including its constant component. The actual pair-sum estimates of Section M.17, after full reversal, are

$$W_h+\bar f_H\le 2+\sum_{a=h}^{H-1}B^au-\gamma_h,\quad 3\le h<H,\qquad \gamma_h=\begin{cases}2d-\delta_2, & h=3,\\ 16d/5, & h\ge 4.\end{cases}$$

For $4\le r<t$, the actual positive terminal pair satisfies $h_rf_t\le\exp\{\sum_{a=r}^{t-1}K^ag-16d/5\}$. A finite disjoint-interval mask with least power a has bounds

$$\|R\|_2^2\le\theta^aC_*d,\quad |R|^2\le\theta^{a-1}C_*d,\qquad C_*=32768/16245.$$

These bounds apply to the finite sums in the path expansions.

R30.3 Matching the two actual lengths

For $3\le h\le H$, put

$$Q_{h,H}=W_h\bar f_H,\quad R_*=\sum_{a=h}^{H-1}B^au=BZ_*,\quad Z_*=B^{h-1}G_{H-h}(B)u.$$

At $h=H$ these sums vanish. Uniformly,

$$W_h+\bar f_H\le 2-\lambda d+R_*,\quad \lambda=51/32,\quad \mathbb{E}R_*\le\theta d,\quad \|Z_*\|_2\le\tfrac12\sqrt{d}.\tag{R30.3.2}$$

For $h<H$ the first assertion follows from the pair-sum estimate, since $2d-\delta_2\ge 51d/32$. At equality use $2\mu_{h-1}\le 2-2d$, including $h=3$. In the finite alternating mean sum the first possible positive term is δ_2, so $\mathbb{E}R_*\le\delta_2\le\theta d$. Furthermore

$$\|Z_*\|_2^2\le\theta^{h-1}(1+\theta)d\le 7605d/32768<d/4,$$

including the G_2 exception. Both actual arms are nonnegative. Their arithmetic-geometric mean bound and $1+x\le e^x$ give $Q_{h,H}\le e^{R_*-\lambda d}$. Applying (R30.2.5) once yields

$$\mathbb{E}Q_{h,H}^8\le\exp\{(-8\lambda+8\theta+17/4)d\}=e^{-21d/4}.\tag{R30.3.4}$$

R30.4 Both first-edge choices and every finite propagation length

For $\sigma \in \{+1, -1\}$ let

$$S_\sigma = J + \sigma B, \quad P_L^\sigma = S_\sigma T^{L-1}, \quad w_0^\sigma = 1, \quad w_j^+ = \bar{f}_j, \quad w_j^- = W_j \ (j \geq 1).$$

Choose $\sigma = +$ when $m \leq n$, and $-$ otherwise. Suppose $F \leq \bar{f}_\ell e^{R-\Gamma}$ and write $D_i = \mathbb{E}B^i R$. For the actual range $|R| \leq 1/2$ verified below, (R29.5.1) with $t = 1/2$, $c = 6/5$ gives $e^R \leq 1 + R + 3R^2/5$.

If $r_1 \geq 4$, the finite first row is

$$S_\sigma(\bar{f}_\ell e^R) \leq w_{\ell+1}^\sigma + \sigma\chi BR + \delta, \quad \chi = \mu_\ell \in [0,1],$$
$$\delta = \mu_\ell D_0 + 2\,\mathrm{sd}(\bar{f}_\ell)\|R\|_2 + \tfrac{12}{5}\|R\|_2^2. \tag{R30.4.2}$$

Indeed split $S_\sigma(\bar{f}_\ell R)$ into $\mu_\ell D_0 + \sigma\mu_\ell BR$ and the centered part, which is at most $2\mathbb{E}|(\bar{f}_\ell - \mu_\ell)R|$. The two positive kernels have the same absolute row bound, and the quadratic fee is $12\mathbb{E}R^2/5$. Thus the same signed δ works for both first edges.

If $r_1 = 3$, the actual first row instead has $\chi = 1$:

$$S_\sigma(\bar{f}_2 e^R) \leq w_3^\sigma + \sigma BR + \delta,$$
$$\delta = \mathbb{E}R + \mathbb{E}(uR) + \mathbb{E}((Bu)R) + \sqrt{(d+\delta_2)\mathbb{E}R^2} + \tfrac{12}{5}\mathbb{E}R^2. \tag{R30.4.4}$$

Here $\bar{f}_2 = 1 + u + Bu$; the displayed three means are precisely $J(\bar{f}_2 R)$, and the remaining row term is bounded by $\mathbb{E}|(u + Bu)R| \leq \sqrt{(d+\delta_2)\mathbb{E}R^2}$. This proves sign compatibility without identifying the two distinct first-row coefficients.

For any $D \geq \delta$ and every finite $L \geq 1$, with $n = \ell + L$,

$$e^\Gamma P_L^\sigma F \leq Y_L^\sigma = w_n^\sigma + \sigma\chi B^L R + D w_{L-1}^\sigma + \chi \sum_{i=1}^{L-1} D_i w_{L-1-i}^\sigma. \tag{R30.4.5}$$

For $L = 1$ this is the corresponding S_σ first row. For $L \geq 2$, apply $S_\sigma T^{L-2}$ to the T first row and use (M.33). Its final summand is $\chi D_{L-1} w_0^\sigma$; all indices are nonnegative and positivity gives $Y_L^\sigma \geq 0$. Replacing signed coefficients by positive parts only when they multiply a nonnegative message gives the scalar upper bounds

L	C bounding $Dw_{L-1}^\sigma + \chi\sum_{i=1}^{L-1}(D_i)_+ w_{L-1-i}^\sigma$
1	D
2	$2D + (D_1)_+$
3	$2D + 2(D_1)_+ + (D_2)_+$
$L \geq 4$	$2D + 2\sum_{i\geq 1}(D_i)_+$

(R30.4.6)

The last geometric bound dominates each finite positive subsum.

The product of the actual receiver and w_n^σ is always $Q_{h,H}$ with $h = \min(m,n)$ and $H = \max(m,n)$. Consequently

$$R_{m,n} + Y_L^\sigma \leq 2 - \lambda d + C + BV, \qquad V = Z_* + \sigma\chi B^{L-1} R.$$

If $\|B^{L-1}R\|_2 \leq \rho_L\sqrt{d}$, then $|D_L| \leq \rho_L d$, and hence $\mathbb{E}BV \leq (\theta + \rho_L)d$ and $\|V\|_2 \leq (1/2 + \rho_L)\sqrt{d}$. Arithmetic-geometric mean followed by (R30.2.5) gives the complete same-kernel payment

$$\log \mathbb{E}[\{R_{m,n} P_L^\sigma F\}^8] \leq -8\Gamma - \tfrac{51}{4}d + 8C + 8(\theta + \rho_L)d + 17(1/2 + \rho_L)^2 d. \tag{R30.4.9}$$

If the moment is zero use the equivalent exponential inequality. The terminal mask and root interval need not be disjoint: this is one exponential expectation, with every scalar cost displayed.

R30.5 Second-shortest terminal length at least four

Section Q.11 gives, for the first row of (R30.1.2),

$$R=\sum_{i=1}^{j}\sum_{a=r_i}^{t_i-1}K^a g,\quad F_\Sigma\le \bar f_\ell e^{R-\Gamma},$$
$$\Gamma\ge 16jd/5,\quad \|R\|_2<\tfrac14\sqrt{d},\quad |R|^2<d/3,$$
$$\delta\le 5d/6,\qquad \sum_{i\ge1}(D_i)_+<2d/5. \tag{R30.5.1}$$

For $j=1,r_1=4$ the stronger three entries are $\sqrt{d}/5,72d/125,7d/25$; for $j=1,r_1\ge5$ they are $3\sqrt{d}/20,3d/5,d/5$. For $j\ge2$ use the joint row, without combining incompatible extrema. The finite mask of least original power four can be written $R=(U_e+zU_o)u$ with

$$-\frac{x^2}{1-x^2}\le U_e\le\frac{x^3}{1-x^2},\qquad -\frac{x^3}{1-x^2}\le U_o\le\frac{x^2}{1-x^2}.$$

Its four residue classes give the positive-tail bound

$$\frac{x^2(1+x)(1+x^2)}{(1-x^2)^2}=\frac{x^2(1+x^2)}{(1-x)^2(1+x)}.$$

It has nonnegative power-series coefficients; its value at θ is $201617/519840<2/5$. Integration proves the stated finite-tail bound. If $r_1=4$ then $\ell=2$ or 3, so the larger mask and the small path variance belong to the same row; the $r_1\ge5$ row has the smaller mask.

Take $D=5d/6$, $C=37d/15$ and $\rho_L=1/4$ for every finite L. Substitution in (R30.4.9) gives

$$\begin{aligned}\log\mathbb{E}[\{R_{m,n}P_L^\sigma F\}^8]&\le\left[-\frac{128j}{5}-\frac{51}{4}+\frac{296}{15}+\frac{21}{4}+\frac{153}{16}\right]d\\&=-\left(\frac{128j}{5}-\frac{5231}{240}\right)d\le-\frac{913}{240}d<-3d.\end{aligned}$$

The actual $\chi=\mu_\ell$ was retained before using $0\le\chi\le1$. This proof also retains $\|V\|_2\le3\sqrt{d}/4$ on the whole domain.

R30.6 The exact second-shortest length three

Strict order forces $\ell=2,r_1=3,c=t_1\ge4$. Set

$$\beta_*=\beta_{c-1},\quad R_0=-\beta_*u-B^3G_{c-3}(-B)u,\quad \Gamma_0=d+\beta_c+\beta_*.$$

The actual pair estimate in Section Q.11 gives $f_3+h_c\le2-\Gamma_0+R_0$ and $\bar f_2f_3h_c\le\bar f_2e^{R_0-\Gamma_0}$, with $\Gamma_0\ge35d/16$. The $c=4,5$ values are respectively $3d$ and $3d-\delta_2$; for $c\ge6$ use $d+2\tau\ge35d/16$. Let

$$S=\sum_{i=2}^{j}\sum_{a=r_i}^{t_i-1}K^a g,\quad R=R_0+S,\quad \Gamma=\Gamma_0+16(j-1)d/5.$$

At $j=1$, $S=0$. Otherwise $r_2\ge c+1\ge5$, and the finite terminal mask gives

$$\|S\|_2<3\sqrt{d}/20,\quad |S|<\sqrt{d}/4,\quad \sum_{i\ge1}(\mathbb{E}B^iS)_+<d/5.$$

The actual shortest arm $\bar{f}_2$ is still present.

For $0 < d \le 1/8$ the full first row (R30.4.4) has the following joint bounds from Section Q.11:

	D	$(D_1)_+$	$(D_2)_+$
$j = 1$	$3d/5$	d^2	δ_3
$j \ge 2$	$3d/2$	$19d/120$	$d/4$

These bound the complete δ, including its correlated means. They use coefficient one on BR. Also

$$\|R_0\|_2 \le (d + \tfrac{6}{5}\theta^{3/2})\sqrt{d} < 89\sqrt{d}/200 < 9\sqrt{d}/20,$$

since $\sqrt{1+\theta} < 6/5$ and $\theta^{3/2} < 4/15$. Thus $\|R\|_2 < 3\sqrt{d}/5$ for $j \ge 2$. The Taylor conditions follow from $|R_0| \le d + 39\sqrt{d}/80 < 1/3$ and $|R_0 + S| \le d + 59\sqrt{d}/80 < 1/2$; $\sqrt{d} < 3/8$ gives $197/640 < 1/3$ and $257/640 < 1/2$.

For the necessary finite positive-tail sum, write $G_{c-3}(-z) = U - zV$, so $r_0 = -\beta_* + x^2 V + zxU$. At $c = 4$, $(U, V) = (1, 0)$; otherwise $0 \le U, V \le 1$. For every integer $k \ge 0$, the four residue classes are

$$D_{4k}(R_0) = -\int x^{2k+1} U\, d\nu, \quad D_{4k+1}(R_0) = \beta_* \delta_{2k+1} - \int x^{2k+2} V\, d\nu,$$
$$D_{4k+2}(R_0) = \int x^{2k+2} U\, d\nu, \quad D_{4k+3}(R_0) = -\beta_* \delta_{2k+2} + \int x^{2k+3} V\, d\nu.$$

Consequently every finite positive subsum is bounded by

$$\begin{aligned}\sum_{i \ge 1} (D_i(R_0))_+ &\le \frac{\beta_* d + \delta_3 + \delta_4}{1 - \theta^2} \\ &\le \frac{d^2 + \theta^2(1+\theta)d}{1 - \theta^2} \\ &\le \frac{11701}{27360} d < \frac{3}{7} d.\end{aligned}$$

The last comparison uses $d \le 1/8$ and $7 \cdot 11701 = 81907 < 82080 = 3 \cdot 27360$. For S the corresponding four-class bound is

$$\frac{x^3 + 2x^4 + x^5}{(1 - x^2)^2} = \frac{x^3}{(1 - x)^2} \le \frac{2197}{11552} < \frac{1}{5}.$$

Thus $\sum_{i \ge 1} (D_i(R))_+ < 22d/35$ by $(a + b)_+ \le a_+ + b_+$. There is no independence assumption.

R30.7 The complete second-shortest-three payment

For $j = 1$ use $D = 3d/5$ and the following finite-length data in (R30.4.9):

L	C/d	ρ_L	loss coefficient
1	$3/5$	$9/20$	$1303/400$
2	$53/40$	$3/10$	$78/25$
3	$97/60$	$3/16$	$17401/3840$
≥ 4	$72/35$	$3/25$	$53341/17500$

The norm bounds use 1, $\sqrt{\theta} < 2/3$, $\theta < 5/12$, $\theta^{3/2} < 4/15$. The scalar bounds use $(D_1)_+ \le d/8$, $(D_2)_+ \le \theta^2 d < d/6$ and the finite-tail bound above. The negative base is $8\Gamma_0 + 51d/4 \ge 121d/4$; the four positive costs are respectively

$$24/5 + 137/20 + 6137/400, \quad 53/5 + 113/20 + 272/25,$$
$$194/15 + 19/4 + 2057/256, \quad 576/35 + 421/100 + 16337/2500.$$

Subtracting gives the table, all strictly greater than three. The tight long row has margin $841/17500$ above three. This includes $c = 4, L = 1, m = n = 3$.

For $j \geq 2$ use $D = 3d/2$ and

L	C/d	ρ_L	loss at $j = 2$
1	$3/2$	$3/5$	$1523/100$
2	$379/120$	$2/5$	$3109/300$
3	$107/30$	$1/4$	$3001/240$
≥ 4	$149/35$	$4/25$	$172509/17500$

Here $8\Gamma + 51d/4 \geq 1117d/20$ at $j = 2$, and the positive costs are

$$12 + 161/20 + 2057/100, \quad 379/15 + 129/20 + 1377/100,$$
$$428/15 + 21/4 + 153/16, \quad 1192/35 + 453/100 + 18513/2500.$$

All losses exceed nine. Each further pair adds $16d/5$ to Γ, while the joint mask, D and finite positive-tail bounds remain the same. Both low-domain families retain $\|V\|_2 \leq 11\sqrt{d}/10$.

For $1/8 \leq d \leq 1/3$ keep exactly the same direction. The row and finite G bounds give

$$|R_0| \leq d + 39\sqrt{d}/80 \leq 197d/80, \quad \Gamma_0 \geq 175d/80, \quad |S| \leq \tfrac{3}{4}(j-1)d.$$

Since $\sqrt{d} \leq 3d$ on this domain,

$$F \leq \bar{f}_2 \exp\{[11/40 - 49(j-1)/20]d\}.$$

The positive connector sends $\bar{f}_2$ to w_n^σ; the remaining pair is $Q_{h,H}$. By (R30.3.4) the resulting loss is $61/20 + 98(j-1)/5 \geq 3$. The domains overlap at $d = 1/8$. Together with the structural case above this proves (R30.1.3).

R30.8 Receiver posts and genuine equal pairs

For $k \geq 1$, the actual negative post block has

$$P_c^- \leq e^{R_c - \Gamma_c}, \quad R_c = \sum_{\nu=1}^{k} \sum_{a=u_\nu}^{v_\nu - 1} B^a u = BZ_c, \quad \Gamma_c \geq 16kd/5.$$

Here extracting a factor B is a polynomial identity, not inversion. The finite mask gives

$$\|Z_c\|_2^2 \leq \theta^3 C_* d = 2197d/16245 < 4d/25, \quad \mathbb{E}R_c \leq \frac{\delta_4}{1-\theta^2} \leq 2197d/27360 < d/12.$$

The first possible positive mean comes from original power seven. On the whole domain of part R30.5 and the low domain of R30.7, combine $V + Z_c$ before applying the exponential lemma once. The extra cost, including all cross terms, is at most

$$\begin{aligned} 8\mathbb{E}R_c + 34\|V\|_2\|Z_c\|_2 + 17\|Z_c\|_2^2 - 8\Gamma_c &\leq [2/3 + 374/25 + 68/25 - 128k/5]d \\ &= [1376/75 - 128k/5]d \leq -544d/75 < 0. \end{aligned}$$

The terminal and receiver masks may overlap. On the high domain of R30.7, $|R_c| \leq \|Z_c\|_2 < 2\sqrt{d}/5 \leq 6d/5$ gives $P_c^- \leq \exp[-(16k/5 - 6/5)d] \leq 1$. For $k = 0$ no additional cost occurs. Finally restore $E_c, E_t \leq 1$ using positivity of the actual operators. This proves the decorated bound with every original edge retained.

O.33 Higher even terminal lists with a nonunit ordinary anchor

R36.1 Original object and exact ordinary theorem

The original problem quantifies every finite tree U with at most five vertices of original degree ≥ 3. Every original edge direction must be fixed from U before every finite original probability μ and real skew pointwise $|K| \leq 1$. The target is $Z_D = E_\mu^V \prod(1+K) \leq 1$. Every original arm, repeated arm, subdivision, all four independent positive core lengths, correlated joint law and noninjective tournament map remain in the problem.

At an original terminal t let the strictly ordered residual after genuine equal pairing be $a_1 < b_1 < ... < a_j < b_j, a_1 \geq 4, j \geq 1$. Let its actual F be $\prod_i h_{a_i} f_{b_i}$. *At* receiver r there is an actual arm $m \geq 3$ and an optional original root block $P = \prod_i W_{u_i} \bar{f}_{v_i}$ with $4 \leq u_1 < v_1 < ... < u_k < v_k, k \geq 0$. The block's length intervals, the terminal intervals and m may overlap. Within each block the written intervals are disjoint because the original residual table is strictly ordered. Equal-arm products E_t, E_r stay at their original vertices and satisfy $0 \leq E_t, E_r \leq 1$.

For each original connector length $L \geq 1$ fix

$$H_D = \begin{cases} E_r P W_m T^L (E_t F), & m \leq L, \\ E_r P \bar{f}_m A T^{L-1} (E_t F), & m > L. \end{cases} \tag{R36.1.1}$$

Every terminal h arm has its first edge toward t, remaining edges toward the leaf; its f arm is entirely outward. W_m has its first edge outward and remaining edges inward toward that first internal point; $\bar{f}_m$ is entirely inward. In the second row only the root-adjacent connector edge is A, from r toward t; all remaining $L-1$ connector edges are T. The first row is all T. Root P and equal pairs use their actual displayed directions.

Theorem A. Every original legal host satisfies

$$\boxed{EH_D^8 \leq e^{-2d}, \qquad d = \|K\mathbf{1}\|_2^2.} \tag{R36.1.2}$$

The direction depends only on m, L and original arm tables. In particular $L = 1, 2$ always use the second row. No low/high host split selects directions. This theorem is about an ordinary eighth moment, not an arbitrary input or a same-direction reference-fourth or I_q assertion.

R36.2 Exact original finite tools

All operators and inner products use the same μ. Put $B = -K, u = B\mathbf{1}, T = J + B, A = J - B = T^*, p = 1 - u, q = 1 + u, f_n = A^n \mathbf{1}, \bar{f}_n = T^n \mathbf{1}, h_n = T f_{n-1}, W_n = A \bar{f}_{n-1}, \delta_i = \|B^i \mathbf{1}\|^2, d = \delta_1, \theta = \frac{13}{32}, \kappa = \frac{19}{32}, \tau = d - \delta_2$. C014 gives $d \leq \frac{1}{3}, \|B\|^2 \leq \theta, d^2 \leq \delta_2 \leq \theta \cdot d, \delta_{i+1} \leq \theta \cdot \delta_i$ and the original row $|Bv(x)| \leq \|v\|_2$. The $f_n, \bar{f}_n$ means μ_n lie in[0,1]; all four path types lie in[0,2]. The means $Eh_2 = EW_2 = 1 + d$ exceed one.

$G_n(z) = \sum_{r=0}^{n-1} z^r, G_0 = 0, \beta_1 = \beta_2 = 0$ and $\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1} \delta_i$. The actual finite identity $1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}$ gives $\tau \leq \beta_n \leq d$ for $n \geq 3$. Finite geometry gives $\|G_n\| \leq 1$ for $n = 1$ or $n \geq 3$; G_2 has square bound $1 + \theta = \frac{45}{32}$.

We use the exact signed exponential theorem, fully reconstructed in actual A013: for every original real V,

$$Ee^{8BV} \leq e^{8EBV + 17\|V\|^2}. \tag{R36.2.1}$$

It is the all-real fourth-exponent lemma applied to $2V$; constant-vector parts and either sign of EBV are allowed. C064, equation (9.21), gives $\|T^n\| \le 1$ for every $n \ge 3$. Exact C094's original T^2 bound gives

$$\|T^2\| \le (1 + 1024\delta_2/855)^2 \le (\frac{2981}{2565})^2 < (\frac{7}{6})^2 < 7/5. \tag{R36.2.2}$$

The rational comparison use $s_{17886} < 17955$ and $245 < 252$. Thus for $L \ge 3, \|T^{L-1}\| \le \frac{7}{5}$, including the genuine $L = 3$ exception.

The strong ACTUAL unequal-pair envelopes, from C096, Section 2.2, are $h_4 f_b \le \exp(R_{4b} - 69d/20), h_a f_b \le \exp(R_{ab} - 16d/5)$ for $a \ge 5$, where $R_{ab} = \sum_{n=a+1}^{b} K^n \mathbf{1}$. The $a = 4$ retained negative term $-\tau f_2$ and $f_2 \ge p^2/2$ give a positive error at most $\delta_2 + \delta_2^2/(2\tau) \le 663d/1216$; subtract it from $4d$ to get $4201d/1216 > 69d/20$. For $a \ge 5$ the actual finite mean $\mu_{a-1} \le 1 - 83d/32$ and positive difference tail $\le 4\delta_2/(1-\theta^2)$ give loss $\frac{83}{16} - \frac{1664}{855} > \frac{16}{5}$. These are finite unequal-pair bounds on the actual arm messages.

R36.3 One terminal mask and its signed moments

Write $F \le \exp(R - \Gamma)$, where R is the one finite union of the terminal intervals. Always $\Gamma \ge 16jd/5$; if $a_1 = 4$ then $\Gamma \ge 69d/20 + 16(j-1)d/5$, an extra $d/4$ beyond $16jd/5$. There is a finite 0-1 polynomial Q with $R = -B^4 Q(-B)u$. $Atz = i\sqrt{x}$, write $Q(-z) = U(x) - zV(x)$. Finite alternating subseries give

$$\begin{aligned} -\frac{x}{1-x^2} \le U, V \le \frac{1}{1-x^2}, \\ |Q|^2 \le \frac{1+x}{(1-x^2)^2} \le C_* := 32768/16245. \end{aligned} \tag{R36.3.1}$$

Only finite subseries occur; geometric tails bound them without changing an original path. With $C_\beta = \frac{1024}{855} < \frac{6}{5}$ and ν the original u spectral measure, R has coordinates $-x^2U + zx^2V$. Let $D_i = EB^iR, \sigma = ER^2$. The exact identities and useful bounds are

$$D_0 = -\int x^2 V, \quad D_1 = \int x^2 U, \quad D_2 = \int x^3 V, \tag{R36.3.2}$$

$$\begin{aligned} \sigma \le C_* \theta^4 d = 28561d/519840 < d/18, \\ |R|^2 \le C_* \theta^3 d < 9d/64 < \frac{1}{16}, \end{aligned} \tag{R36.3.3}$$

$$\begin{aligned} |D_0|, |D_1| < d/5, \qquad D_0 \le C_\beta \delta_4 < d/12, \\ D_0 - D_1 \le 2C_\beta \delta_4 < d/6. \end{aligned} \tag{R36.3.4}$$

Here $C_\beta \theta^2 = \frac{169}{855} < \frac{1}{5}$ and $C_\beta \theta^3 = \frac{2197}{27360} < \frac{1}{12}$. The last inequality in(R36.3.4) bounds $U + V$ together. Also

$$\begin{aligned} D_0 + D_1 - D_2 \le C_\beta(2\delta_3 + \delta_4) < d/2, \\ \|TR\|^2 = D_0^2 + 2D_0 D_1 + \|BR\|^2 \\ \le (\frac{1}{25} + \frac{5}{216})d < (\frac{4}{15})^2 d. \end{aligned} \tag{R36.3.5}$$

For the first use $C_\beta < \frac{6}{5}, \theta^2 < \frac{1}{6}, 2+\theta < \frac{5}{2}$. For the second retain the actual constant cross, then use $d \le \frac{1}{3}, |D_0|, |D_1| \le d/5$ and $\theta\sigma < 5d/216$. The final rational inequality is $\frac{341}{5400} < \frac{16}{225}$. No constant vector has been replaced by u.

If the terminal minimum is at least five, a sharper finite factorization is available: $R = B^5Q(-B)u = x^3V + zx^2U$. Then $\sigma \le C_*\theta^5 d < d/40$. The same bounds(R36.3.4) still hold, and

$$-\langle G_2(B)u, R\rangle = -\int x^3(U+V) \le 2C_\beta\delta_5 < d/15. \tag{R36.3.6}$$

The final bound uses $C_\beta < \frac{6}{5}, \theta^4 < \frac{1}{36}$. This improvement is needed only in the $L=1$ proof when the first terminal arm is not four.

R36.4 True propagated perturbation, all original lengths

Since $|R| < \frac{1}{4}$, the scalar inequality $e^R \le 1 + R + (\frac{3}{5})R^2$ is valid; the complete $|R| \le \frac{1}{2}$ factorial/integral certificate appears in actual A016, Section 4. The upper polynomial is positive. If S_L is T^L or AT^{L-1}, positivity gives

$$\begin{aligned} 0 \le S_L F &\le e^{-\Gamma}\{S_L 1 + S_L R + \omega_L\}, \\ \omega_1 &= (\frac{6}{5})\sigma < d/15, \\ \omega_L &= (\frac{12}{5})\sigma < 2d/15 \ (L \ge 2). \end{aligned} \tag{R36.4.1}$$

Indeed the last positive A/T row is at most twice the preceding mean; for $L \ge 2$ that mean is $E[f_{L-1}R^2] \le 2\sigma$. $AtL = 1$ it is σ. The braced expression is a nonnegative upper for the complete sum; its summands can have either sign.

Set $V_R = T^{L-1}R$ and $a_R = EV_R$. Then $S_LR = a_R + BV_R$ for $S_L = T^L$, and $a_R - BV_R$ for $S_L = AT^{L-1}$. For $L=1, EAR = D_0 - D_1 < d/6$. For $L=2, EATR = D_0 + D_1 - D_2 < d/2$ and $\|V_R\| < 4\sqrt{d}/15$ by(R36.3.5).

For all $L \ge 3$, (R36.2.2) and(R36.3.3) give $\|V_R\| < 7\sqrt{d}/20$. There is also a uniform true mean bound. The exact finite Duhamel expansion writes, for every $k \ge 1$,

$$f_k = \mu_{k-1}1 + \sum_{j=1}^{k-1} \mu_{k-1-j}(-B)^j\mathbf{1} + (-B)^k\mathbf{1}. \tag{R36.L248}$$

Every coefficient is in[0,1]. Consequently $\|f_k - \mu_{k-1}1\| \le \sqrt{d}\sum_{j=0}^{k-1}\theta^{j/2} < 3\sqrt{d}$. Hence $a_R = \langle f_{L-1}, R\rangle \le d/12 + 3\sqrt{d\sigma} < 5d/6$. Skewness gives $|EBV_R| \le \sqrt{d}\|V_R\|$. Both possible last-edge signs therefore satisfy

$$ES_LR < 71d/60 \qquad (L \ge 3). \tag{R36.4.2}$$

R36.5 Actual bare sums and one eighth-moment payment formula

$AtL = 1$ the bare sum is $\bar{f}_m + p, m \ge 3$. The exact finite convolution gives

$$\bar{f}_m + p \le 2 - \beta_m + BV_b, V_b = G_{m-1}(B)u,$$
$$EBV_b = -\beta_{m+1}, \beta_m + \beta_{m+1} \ge 2\tau \ge 19d/16. \tag{R36.L265}$$

$AtL = 2$ it is $\bar{f}_m + W_2$. Here

$$\bar{f}_m + W_2 \le 2 - \beta_m + BV_b, V_b = BG_{m-2}(B)u,$$
$$EBV_b \le \delta_2, \|V_b\| \le 4\sqrt{d}/5, \tag{R36.5.1}$$
$$-8\beta_m + 8EBV_b \le -3d/2.$$

The G_2 exception is paid because $\theta(1+\theta) = \frac{585}{1024} < \frac{16}{25}$. For $L \ge 3$ the structure-only direction in(R36.1.1) makes the sum $W_h + \bar{f}_H, h = \min(m, L) \ge 3, H = \max(m, L)$. Its exact finite pair estimate, reconstructed in R2-C030/A014, is

$$W_h + \bar{f}_H \le 2 - (\frac{51}{32})d + BV_b,$$
$$EBV_b \le \theta d, \quad \|V_b\| \le \tfrac{1}{2}\sqrt{d}. \tag{R36.5.2}$$

For $h < H, V_b = B^{h-1}G_{H-h}(B)u; h = H$ has $V_b = 0$ and $W_h + \bar{f}_h = 2\mu_{h-1} \le 2 - 2d$. The finite G_2 exception gives $\|V_b\|^2 \le \theta^2(1+\theta)d < d/4$; the first positive finite mean is δ_2. Thus (R36.5.2) includes every finite m, L and equality $m = L$.

Combine the appropriate bare sum, the same true perturbation, and(R36.4.1). If its bare loss is $\lambda \cdot d($ or $\beta_m)$, actual nonnegative AM–GM and (R36.2.1) give the unpowered-root-pair estimate

$$\log EH^8 \le -8\Gamma - 8\lambda d + 8EBV_b$$
$$+8\omega_L + 8ES_L R + 17\|V_b \pm V_R\|^2. \tag{R36.5.3}$$

The sign is minus in the A row and plus in the T row. The right side comes from a pointwise positive upper for the actual sum.

R36.6 All L without root post-pairs

R36.6.1 $L = 1$, terminal minimum exactly four

For $m = 3, V_b = G_2 u$. From $R = -B^4 Q(-B)u$,

$$-\langle G_2 u, R\rangle = \int [x^2 U - x^3 V] \le C_\beta(\delta_3 + \delta_5) < 7d/30. \tag{R36.L308}$$

Therefore

$$\|V_b - R\|^2 \le (\frac{45}{32} + \frac{7}{15} + \frac{1}{18})d = (\frac{2777}{1440})d. \tag{R36.6.1}$$

For $m \ge 4, G_{m-1}$ contracts, so the weaker bound $\|V_b - R\|^2 < (\frac{5}{4})^2 d$ is even smaller than(R36.6.1). Using only $\Gamma \ge 16d/5$ first, (R36.5.3) gives a loss coefficient at least

$$\frac{128}{5} + \frac{19}{2} - 17(\frac{2777}{1440}) - \frac{8}{15} - \frac{4}{3} = 647/1440. \tag{R36.L320}$$

At actual $a_1 = 4$ the retained $\frac{69}{20}$ first-pair loss adds 2 to this coefficient. Thus the full $L = 1$ loss exceeds 2, for all $m \geq 3$ and all j.

R36.6.2 $L = 1$**, terminal minimum at least five**

For $m = 3$ use(R36.3.6), yielding $\|V_b - R\|^2 \leq (\frac{45}{32} + \frac{2}{15} + \frac{1}{40})d = (\frac{751}{480})d$. For $m \geq 4$ the bound $25d/16 = 750d/480$ suffices. Even keeping the weaker $\omega_1 < d/15$ and mean $ES_1R < d/6$, the coefficient is at least

$$\frac{128}{5} + \frac{19}{2} - 17(\frac{751}{480}) - \frac{8}{15} - \frac{4}{3} = \frac{637}{96} > 2. \tag{R36.6.2}$$

The two structural cases give the bound for $a_1 \geq 4$.

R36.6.3 $L = 2$

By(R36.3.5) and(R36.5.1), $\|V_b - V_R\| < 16\sqrt{d}/15$. Use $\Gamma \geq 16d/5, \omega_2 < 2d/15$ and $ES_2R < d/2$. Formula(R36.5.3) gives

$$\frac{128}{5} + \frac{3}{2} - 17(\frac{16}{15})^2 - \frac{16}{15} - 4 = \frac{1211}{450} > 2. \tag{R36.6.3}$$

This uses the genuine constant-vector square $\|TR\|^2$ in(R36.3.5).

R36.6.4 Every $L \geq 3$

Equations(R36.4.2),(R36.5.2) give $\|V_b \pm V_R\| < 17\sqrt{d}/20$. The norm cost $17(\frac{17}{20})^2 < 13$, and the combined $\delta/$ mean fee is $\frac{16}{15} + \frac{142}{15} = \frac{158}{15} < 11$. The loss before those fees is $\frac{128}{5} + 8(\frac{51}{32} - \theta) = \frac{351}{10}$. Therefore the remaining coefficient exceeds $\frac{351}{10} - 13 - 11 = \frac{111}{10} > 2$. Every finite L is covered by the actual power bound and finite path identity, including $L = 3$ separately in(R36.2.2).

R36.7 All finite negative root post-pairs, including overlapping masks

For $k \geq 1$, the actual $P = \prod W_{u_i} \bar{f}_{v_i}$ has the reversed same pair envelope $P \leq \exp(S - \Gamma_r)$, $\Gamma_r \geq 16kd/5$. Write $S = BV_p = B^4 Q_p(B)u$, where Q_p is one finite 0-1 mask. The same finite-coordinate proof as(R36.3.1) gives

$$ES < d/12, \qquad \|V_p\| < 3\sqrt{d}/8. \tag{R36.7.1}$$

The ordinary exponent for H now contains the SINGLE actual vector $V_b \pm V_R + V_p$, with additional loss $8\Gamma_r$ and mean $8ES$. No assertion that terminal and root masks are disjoint has been used. Their cross term is paid by the vector triangle bound below.

$L = 1$: (R36.6.1) gives $\|V_b - R\| < 7\sqrt{d}/5$, so the combined norm is less than $71\sqrt{d}/40$. Its 17-times square is < 54. The δ, perturbation mean and post mean fees sum to $\frac{8}{15} + \frac{4}{3} + \frac{2}{3} = \frac{38}{15} < 3$. Even at $j = k = 1$, the loss $8\Gamma + 8\Gamma_r$ plus the bare signed loss is at least $607d/10$. The final coefficient therefore exceeds $\frac{607}{10} - 54 - 3 = \frac{37}{10} > 2$.

$L = 2$: the combined norm is at most $173\sqrt{d}/120$, whose 17-times square is < 36. The other fees $\frac{16}{15} + 4 + \frac{2}{3} = \frac{86}{15}$ are < 6. The starting loss is $\frac{256}{5} + \frac{3}{2} = \frac{527}{10}$. The remaining coefficient exceeds $\frac{107}{10} > 2$.

$L \ge 3$: the combined norm is at most $49\sqrt{d}/40$, whose 17-times square is < 26. Other fees $\frac{16}{15} + \frac{142}{15} + \frac{2}{3} = \frac{56}{5} < 12$. Starting loss $\frac{607}{10}$ leaves a coefficient greater than $\frac{227}{10} > 2$.

Extra terminal or root pairs only increase Γ or Γ_r; the finite mask estimates already hold for every finite pair count. Thus all $k \ge 1$ and $j \ge 1$ are paid with the SAME bound e^{-2d}. Restore E_t, E_r by the actual positive original operators and multiplications to prove(R36.1.2).

O.34 Short even multipairs with a nonunit ordinary anchor

R38.1 Actual objects and directions

Let the original receiving arm have length $m \ge 3$, and let the terminal's strict residual list be

$$a_1 < b_1 < \cdots < a_j < b_j, \qquad j \ge 2, \qquad a_1 \in \{2, 3\}, \qquad F = \prod_{i=1}^{j} h_{a_i} f_{b_i}. \tag{R38.L74}$$

An optional negative root-post block is

$$P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}, \qquad 4 \le u_1 < v_1 < \cdots < u_k < v_k, \qquad k \ge 0. \tag{R38.L76}$$

All counts and lengths are finite. The two masks may overlap each other and the anchor length. Genuine equal-pair products $E_t, E_r \in [0, 1]$ remain at their original vertices. Define the decorated message by inserting $E_t F$ in the following connector and multiplying by $E_r P$ at the receiving root:

$$H_D = E_r P \begin{cases} \bar{f}_m A(E_t F), & a_1 = 2,\ L = 1, \\ W_m T^L (E_t F), & a_1 = 2,\ L \ge 2, \\ W_m T^L (E_t F), & a_1 = 3,\ m \le L, \\ \bar{f}_m A T^{L-1} (E_t F), & a_1 = 3,\ m > L. \end{cases} \tag{R38.L78}$$

Every f arm points outward, every $\bar{f}$ arm inward. An h arm has its first edge toward the root and all later edges toward the leaf; W reverses that entire arm. A T connector points toward the receiving root. In an AT^{L-1} row, only its root-adjacent edge is A. These choices depend only on the original integers, and satisfy

$$\boxed{E H_D^8 \le e^{-2d}, \qquad d = \|K\mathbf{1}\|_2^2.} \tag{R38.L87}$$

R38.2 Finite inputs and one actual terminal mask

Use the finite packet of R2-C025 and the unequal-pair envelopes of Q5-C090. The first pair has loss κd at minimum two and $(2-\theta)d = 51d/32$ at minimum three; every later shorter arm has length at least four and pays $16d/5$. Thus

$$F \le e^{R - \lambda d}, \qquad \lambda_2 = 607/160, \quad \lambda_3 = 767/160, \qquad R = \sum_i \sum_{r=a_i+1}^{b_i} K^r \mathbf{1}. \tag{R38.L104}$$

Extra pairs only enlarge the available loss. The intervals within this single finite mask are disjoint. Work throughout on the original measure, with

$$\begin{gathered} B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B, \quad p = 1 - u, \quad q = 1 + u, \\ \delta_i = \|B^i \mathbf{1}\|^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2, \\ f_n = A^n \mathbf{1}, \quad \bar{f}_n = T^n \mathbf{1}, \quad h_n = T f_{n-1}, \quad W_n = A \bar{f}_{n-1}. \end{gathered} \tag{R38.L106}$$

The finite calculus gives $0 \le d \le 1/3$, $d^2 \le \delta_2 \le \theta d$, $\delta_{i+1} \le \theta\delta_i$, $\|B\|^2 \le \theta$, and $|BV(x)| \le \|V\|$. Only $f_n, \bar{f}_n$ have means at most one; $Eh_2 = EW_2 = 1 + d$. All four path types lie in $[0, 2]$. The common zero-boundary argument handles $d = 0$; below take $d > 0$.

At $z = i\sqrt{x}$, let ν be the original spectral measure of $-B^2$ at u, with mass d and $\int x^i \, d\nu = \delta_{i+1}$. If $R = U(B)u$, $U = \alpha + z\beta$, skew adjunction gives

$$ER = -\int \beta \, d\nu, \qquad \|R\|^2 = \int (\alpha^2 + x\beta^2) \, d\nu. \tag{R38.L115}$$

All masks are finite; positive geometric tails only bound them. Set

$$G_0 = 0, \quad G_n(z) = \sum_{r=0}^{n-1} z^r, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i. \tag{R38.L124}$$

The exact finite positive convolution, equivalently $S_n^- = \sum_{r=0}^{n} B^r \mathbf{1} = 1 + G_n(B)u$, is

$$1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}, \qquad \tau \le \beta_r \le d \quad (r \ge 3). \tag{R38.L126}$$

Four alternating residue classes give $|Q(i\sqrt{x})|^2 \le (1+x)/(1-x^2)^2 \le 32768/16245$ for every finite zero-one mask Q. This supplies the norms below. At minimum two the mandatory term is $K^3\mathbf{1}$. The short pair retains the actual $\beta_b f_0$ term; $\beta_2 = 0$ is never charged as a positive loss.

For minimum two, the full actual packet is

$$\begin{gathered}
\sigma = ER^2 < d/3, \quad |R| < 3/5, \quad e^R \le 1 + R + 5R^2/8, \\
R = U(B)u, \quad U = \alpha + z\beta, \quad x/2 \le \alpha \le 6x/5, \quad -6x/5 \le \beta \le x/2, \\
D_i = EB^i R, \quad a_0 = -D_1 \in [0, d/2], \quad \rho = -D_2 \in [-d/12, d/5], \\
D_3 \in [0, d/5], \quad |D_i| \le d/12 \ (i \ge 4), \quad D_0 \in [-d/5, d/2], \\
D = D_0 + 5\sigma/4 \in [-d/5, 11d/12], \qquad \epsilon = D + D_1 \in [-7d/10, 7d/10].
\end{gathered} \tag{R38.L134}$$

These are correlated quantities from the same R. Positive propagation gives

$$\begin{gathered}
e^{\lambda d} T^L F \le Y_L, \quad Y_1 = q + D + BR, \\
Y_L = \bar{f}_L + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \\
Q_n = G_n + z^n U + \epsilon G_{n-2} + Dz^{n-2} \quad (n \ge 2), \\
Q_2 = G_2 + D + z^2 U, \qquad Q_{n+1} = 1 + \epsilon + zQ_n, \\
Y_L = 1 + \epsilon + Q_L(B)u - \sum_{n=3}^{L} \gamma_n \bar{f}_{L-n}.
\end{gathered} \tag{R38.L142}$$

The exact signed coefficients satisfy

$$\begin{gathered}
\gamma_3 = d - D_2 \ge 11d/12, \qquad \gamma_4 = d + Dd - D_3 \ge 11d/15, \\
\gamma_n = \beta_n + D\beta_{n-1} - a_0\beta_{n-2} - D_{n-1} \ge d/4 \quad (n \ge 5).
\end{gathered} \tag{R38.L148}$$

The long bound uses $\gamma_n/d \ge \kappa - 1/12 - (7/10)d \ge 133/480 > 1/4$. The short γ_4 bound retains $D \ge -d/5$, $D_3 \le d/5$, $d \le 1/3$. Moreover $EQ_n(B)u = -\gamma_{n+1}$ for all $n \ge 2$. At every $L \ge 4$, retaining the last two actual losses with $h = 1/4$ yields

$$Y_L \le (1 + \epsilon - hd)q - hd + BQ_{L-1}(B)u. \tag{R38.L154}$$

Every discarded loss has its proved nonnegative sign at every finite index.

Positive convolution followed by A gives, for every $m \ge 3$,

$$W_m \le p - \beta_m - BG_{m-1}(B)u, \qquad \beta_m \in [\kappa d, d]. \tag{R38.L158}$$

The actual increment $\Delta_m = \beta_{m+1} - \beta_m \le \delta_3 < d/6$: odd m gives zero, $m = 4$ gives $-\delta_2$, and positive even increments first occur at $m = 6$, with δ_3 or later $\delta_5, \ldots$. Every G_n contracts except G_2, whose squared bound is $1+\theta$, with $\sqrt{1+\theta} < 6/5$. The finite difference identity $G_n - G_r = \pm z^{\min(n,r)} G_{|n-r|}$, or zero, gives

$$\|(G_n - G_r)u\| < \sqrt{d}/2 \quad (n, r \ge 2), \qquad \theta\sqrt{1+\theta} < 1/2. \tag{R38.L163}$$

R38.3 Full-mean eighth-exponent payment

If the actual anchor plus its nonnegative propagated upper is at most $2 + c + BV$, then AM–GM and the all-real signed exponential lemma give

$$\log EH^8 \le -8\lambda d + 8(c + EBV) + 17\|V\|^2, \qquad Ee^{8BV} \le e^{8EBV + 17\|V\|^2}. \tag{R38.L169}$$

The second inequality is the full fourth-exponent lemma applied to $2V$. The actual nonnegative average also satisfies $y^{16} \le e^{16(y-1)}$. The precursor and its signed mean are retained. A root-post block adds $-8\Gamma_r + 8ES$ and the single vector V_p, as paid in part R38.8.

R38.4 Minimum two, unit connector

Use $\bar{f}_m AF$. The positive Taylor error and actual mean are

$$S_1 R = AR, \quad \omega_1 = 5\sigma/4 < 5d/12, \quad EAR = D_0 - D_1 \le d, \tag{R38.L180}$$

using $D_0, a_0 \le d/2$. The bare sum has $\bar{f}_m + p \le 2 - \beta_m + BV_b$, $V_b = G_{m-1}u$, $EBV_b = -\beta_{m+1}$. At $m = 3$, $\beta_3 + \beta_4 = 2d$; at $m \ge 4$ that sum is at least $2\tau \ge 19d/16$.

For $m = 3$,

$$\langle G_2 u, R\rangle = \int (\alpha + x\beta)\, d\nu \ge 0, \qquad \alpha + x\beta \ge x - \frac{2x^2}{1 - x^2} \ge 0. \tag{R38.L187}$$

Here $2\theta < 1 - \theta^2$ is the rational comparison $832 < 855$ in units of 1024. Hence $\|G_2 u - R\|^2 \le (45/32 + 1/3)d = 167d/96$. For $m \ge 4$, $\operatorname{Re} G \ge \kappa$, the normalized imaginary coefficient has modulus at most one, and the finite coordinate bounds imply

$$\langle G_{m-1}u, R\rangle \ge -d/12. \tag{R38.L191}$$

Indeed the concave function $\kappa x/2 - (6/5)x^2$ has its minimum at zero or θ, and its latter value is $-793/10240 > -1/12$; $\beta \ge 0$ only improves the dot product. Thus $\|V_b - R\|^2 \le d + d/3 + d/6 = 3d/2$. The two complete loss payments are

$$\begin{aligned} &607/20 + 16 - 17(167/96) - 10/3 - 8 > 14/3 > 2 \quad (m = 3), \\ &607/20 + 19/2 - 51/2 - 10/3 - 8 = 181/60 > 2 \quad (m \ge 4). \end{aligned} \tag{R38.L197}$$

For the first, $607/20 > 30$ and $17(167/96) < 30$. All $j \ge 2$ are already included in the one finite mask.

R38.5 Minimum two, length-two connector

Keep the actual $W_m T^2 F$ direction. From the genuine Y_2,

$$\begin{aligned} W_m + Y_2 &\le 2 + c + BV, \quad c = \epsilon - \beta_m, \\ V = D\mathbf{1} - B(G_{m-2}u - R), &\quad V' = B(G_{m-2}u - R), \\ EBV' &= d - \beta_{m+1} - D_2, \\ c + EBV &= \epsilon - d + \Delta_m + D_2 \le \epsilon - 3d/4 \le -d/20. \end{aligned} \tag{R38.L204}$$

For $n = m-2 \ge 1$, including $n = 1, 2$, each G_n has real coordinate in $[\kappa, 1]$ and normalized imaginary coordinate in $[0, 1]$, whereas $\alpha \le 6\theta/5 = 39/80 < \kappa$. Thus

$$|G_n - U|^2 \le (1 - x/2)^2 + x(1 + 6x/5)^2 = 1 + (53/20)x^2 + (36/25)x^3 < 8/5. \tag{R38.L213}$$

This uses $\theta^2 < 1/6$, $\theta^3 < 7/100$, and $53/120 + 63/625 < 3/5$. Hence $\|V'\|^2 \le 13d/20$, while $EV' = -\beta_m + a_0$ and $\beta_m - a_0 \ge 3d/32 > 0$. Retain the full constant cross:

$$\|V\|^2 = \|V'\|^2 + D^2 + 2D(\beta_m - a_0). \tag{R38.L220}$$

If $D < 0$, the cross is nonpositive and $D^2 \le d/75$. If $D \ge 0$, use $D \le 11d/12$, $\beta_m - a_0 \le d$, $d \le 1/3$, so the extra cost is at most $385d/432$. Consequently

$$\begin{aligned} \|V\|^2 &\le (13/20 + 385/432)d = 3329d/2160 < 25d/16, \\ 607/20 + 2/5 - 17(3329/2160) &> 17/4 > 2, \end{aligned} \tag{R38.L225}$$

where $17(3329/2160) < 53/2$.

R38.6 Minimum two, every longer connector

At $L = 3$,

$$\begin{aligned} Y_3 &= (1 + \epsilon)q + BQ_2u - \gamma_3, \quad c = \epsilon - \beta_m - \gamma_3, \\ V = V_0 + \epsilon\mathbf{1}, &\qquad V_0 = (G_2 - G_{m-1})u + Du + B^2R, \\ \|V_0\| &< (1/2 + 11/36 + 6/25)\sqrt{d} < 21\sqrt{d}/20. \end{aligned} \tag{R38.L233}$$

The B^2R term uses $\theta^2/3 < 36/625$. The mean $EV_0 = \beta_m - \gamma_3 \in [-2d/3, d/12]$, with $\gamma_3 \in [11d/12, 6d/5]$. For $\epsilon \ge 0$, its constant correction is at most $((7/10)^2 + (7/10)/6)d^2 \le 91d/450$; for $\epsilon \le 0$, it is at most $((7/10)^2 + (4/3)(7/10))d^2 \le 427d/900$. Therefore

$$\begin{aligned} \|V\|^2 &< [(21/20)^2 + 427/900]d < 169d/100, \\ c + EBV &= \epsilon - \gamma_3 - \gamma_4 + \Delta_m \le \epsilon - 89d/60 \le -47d/60, \\ 607/20 + 94/15 - 17(169/100) &> 457/60 > 2. \end{aligned} \tag{R38.L243}$$

The last comparison uses $17(169/100) < 29$.

For every $L \ge 4$, put $n = L - 1 \ge 3$, $h = 1/4$, and retain

$$c = \epsilon - 2hd - \beta_m, \quad V = V_0 + (\epsilon - hd)\mathbf{1}, \quad V_0 = (Q_n - G_{m-1})u. \tag{R38.L251}$$

The four terms in V_0 are the geometric difference, B^nR, $\epsilon G_{n-2}u$, and $DB^{n-2}u$. Their respective bounds are

$$\theta\sqrt{1+\theta}<1/2,\quad \theta^3/3<9/400,$$
$$(7/10)(1/3)(6/5)=7/25,\quad (11/12)(1/3)(13/20)=143/720<1/5. \tag{R38.L256}$$

The actual G_2 exception is paid by $6/5$. Hence $\|V_0\|<113\sqrt{d}/100$. Also $EV_0=\beta_m-\gamma_L\in[-d,3d/4]$: at $L=4$, $\gamma_4\le d+11d^2/12$ because $D_3\ge 0$; at $L\ge 5$, $\gamma_L\le 13d/12+11d^2/12\le 25d/18$. Combining with $\beta_m\ge\kappa d$ gives the lower mean bound, and $\gamma_L\ge hd$ gives the upper. The signed exponent mean is

$$c+EBV=\epsilon-2hd-\gamma_{L+1}+\Delta_m\le\epsilon-7d/12. \tag{R38.L263}$$

Put $t=\epsilon/d-1/4\in[-19/20,9/20]$ and $C=113/100$. The full squared norm obeys

$$\frac{\|V\|^2}{d}\le N(t)=\begin{cases}C^2+(t^2-2t)/3, & t\le 0,\\ C^2+(t^2+3t/2)/3, & t\ge 0.\end{cases} \tag{R38.L270}$$

The fee $8(t-1/3)+17N(t)$ is convex on each closed half. At the negative, zero and positive endpoints, the pairs $(N,\text{mean fee})$ are

$$(C^2+1121/1200,-154/15),\quad (C^2,-8/3),\quad (C^2+117/400,14/15). \tag{R38.L276}$$

Every fee is less than $607/20-2$. Indeed $17C^2=217073/10000<87/4$; the positive endpoint uses $17(117/400)=1989/400<5$, $14/15<1$, giving fee $<111/4$ and loss $>13/5$. The negative endpoint uses $17(1121/1200)<16$, $154/15>41/4$, giving fee $<55/2$ and loss $>57/20$. At zero the fee is $<87/4-8/3<20$, giving loss $>207/20$. Convexity covers every allowable correlated (ϵ,D). This proves all finite $L\ge 4$ for minimum two in the same W_mT^L direction.

R38.7 Minimum three, every connector length

The actual R2-C025 packet gives

$$\sigma<9d/64,\quad |R|<1/3,\quad e^R\le 1+R+3R^2/5,$$
$$D_0\in[-d/5,d/2],\quad D_1\in[-d/12,d/5],\quad -D_2\in[-d/12,d/5]. \tag{R38.L293}$$

with $\lambda=767/160$. Keep the structural comparison $m\le L$ versus $m>L$. The ordinary power bounds are $\|T^n\|\le 1$ for $n\ge 3$ and

$$\|T^2\|\le(1+1024\delta_2/855)^2\le(2981/2565)^2<(7/6)^2<7/5. \tag{R38.L297}$$

The genuine positive Taylor errors have the exact scopes

$$\omega_1=(6/5)\sigma<27d/160,\qquad \omega_L<27d/80\quad (L\ge 2). \tag{R38.L301}$$

Moreover $EAR\le 7d/12$, $EATR=D_0+D_1-D_2\le 9d/10$, and

$$\|TR\|^2=D_0^2+2D_0D_1+\|BR\|^2\le(3/20+117/2048)d<(11/24)^2d. \tag{R38.L302}$$

For $L\ge 3$, the power bounds give $\|T^{L-1}R\|<21\sqrt{d}/40$. The finite Duhamel expansion

$$f_k=\mu_{k-1}\mathbf{1}+\sum_{r=1}^{k-1}\mu_{k-1-r}(-B)^r\mathbf{1}+(-B)^k\mathbf{1} \tag{R38.L310}$$

has coefficients in $[0,1]$, whence $\|f_k-\mu_{k-1}\mathbf{1}\|\le\sqrt{d}\sum_{r=0}^{k-1}\theta^{r/2}<3\sqrt{d}$. It gives $a_R=Ef_{L-1}R\le d/2+3\sqrt{d\sigma}<13d/8$, and thus, for either actual last-edge sign,

$$ES_LR\le 13d/8+21d/40=43d/20\quad(L\ge 3). \tag{R38.L305}$$

Positivity and the last row bounded by two give $S_LF\le e^{-\lambda d}(S_L\mathbf{1}+S_LR+\omega_L)$, nonnegative as a whole. At $L\ge 2$, the error uses $E[f_{L-1}R^2]\le 2\sigma$; at $L=1$, it uses σ. Retain $S_LR=a_R\pm BT^{L-1}R$, including its true constant and the displayed constant cross in TR.

The exact bare sums are:

$$\begin{gathered}
\bar f_m+p\le 2-\beta_m+BG_{m-1}u\quad(L=1),\\
8(\beta_m+\beta_{m+1})=16d\ (m=3),\qquad 8(\beta_m+\beta_{m+1})\ge 19d/2\ (m\ge 4),\\
\bar f_m+W_2\le 2-\beta_m+BV_b,\quad V_b=BG_{m-2}u,\quad \|V_b\|\le 4\sqrt{d}/5,\\
-8\beta_m+8EBV_b\le -3d/2\quad(L=2),\\
W_h+\bar f_H\le 2-(51/32)d+BV_b,\quad EBV_b\le\theta d,\quad \|V_b\|\le\sqrt{d}/2\quad(L\ge 3),\\
h=\min(m,L)\ge 3,\qquad H=\max(m,L).
\end{gathered} \tag{R38.L321}$$

For $h=H$, the exact sum is $2\mu_{h-1}\le 2-2d$; otherwise the finite vector is $B^{h-1}G_{H-h}u$. The G_2 exception uses $\theta^2(1+\theta)<1/4$, and the signed mean uses $\delta_2\le\theta d$.

The four complete payments, in order $(L=1,m=3)$, $(L=1,m\ge 4)$, $L=2$, $L\ge 3$, are

$$\begin{gathered}
\|G_2u-R\|<(6/5+3/8)\sqrt{d}=63\sqrt{d}/40,\\
767/20+16-17(63/40)^2-27/20-14/3>2,\\
\|G_{m-1}u-R\|<11\sqrt{d}/8,\quad 767/20+19/2-17(11/8)^2-27/20-14/3>2,\\
\|V\|\le(4/5+11/24)\sqrt{d}=(151/120)\sqrt{d},\\
767/20+3/2-17(151/120)^2-27/10-36/5>2,\\
\|V\|\le(1/2+21/40)\sqrt{d}=(41/40)\sqrt{d},\\
767/20+19/2-17(41/40)^2-27/10-86/5>2.
\end{gathered} \tag{R38.L333}$$

Their norm costs are less than $43,33,27,18$, and their other fees are less than $7,7,10,20$. The resulting weaker losses are $87/20,157/20,57/20,197/20$, each greater than two. Equality $m=L$ and every finite connector are included.

R38.8 Every finite negative root-post block

For $k\ge 1$, the strong actual pair envelope and its one finite reversed mask give

$$\begin{gathered}
P\le e^{S-\Gamma_r},\quad \Gamma_r\ge 16kd/5,\quad S=BV_p=B^4Q_p(B)u,\quad V_p=B^3Q_p(B)u,\\
C_\star=32768/16245,\quad C_\beta=1024/855,\\
\|V_p\|^2\le C_\star\theta^3d=2197d/16245<9d/64,\qquad ES\le C_\beta\delta_4<d/12.
\end{gathered} \tag{R38.L347}$$

Add V_p to the same actual V and pay every cross by the triangle inequality. Its intervals may overlap the terminal intervals and m.

For minimum two, the short rows use base norm bounds $4/3, 5/4, 13/10$, in units of $\sqrt{d}$, and combined bounds $41/24, 13/8, 67/40$. The full losses are

$$\begin{aligned} 607/20 + 128/5 + 19/2 - 17(41/24)^2 - (10/3 + 8 + 2/3) &> 69/20 > 2, \\ 607/20 + 128/5 + 2/5 - 17(13/8)^2 - 2/3 &> 207/20 > 2, \\ 607/20 + 128/5 + 94/15 - 17(67/40)^2 - 2/3 &> 793/60 > 2. \end{aligned} \tag{R38.L357}$$

The respective norm costs are below $50, 45, 48$; the first other fees total twelve, and the last two use $2/3 < 1$.

At minimum two and $L \geq 4$, complete the squares:

$$N = C^2 - 1/3 + (t-1)^2/3 \quad (t \leq 0), \qquad N = C^2 - 3/16 + (t + 3/4)^2/3 \quad (t \geq 0). \tag{R38.L367}$$

Both residual constants are positive, so $\sqrt{N}$ and $(\sqrt{N} + 3/8)^2$ are convex on each half. Hence so is the full fee $8(t - 1/3) + 17(\sqrt{N} + 3/8)^2 + 2/3$. At the negative, positive and zero endpoints, respectively, the combined norms are less than $15/8, 327/200, 301/200$, using $\sqrt{N} < 3/2$ and $\sqrt{N} < 63/50$ at the first two. Their fees are bounded by

$$\begin{aligned} &17(15/8)^2 - 154/15 + 2/3, \\ &17(327/200)^2 + 14/15 + 2/3, \\ &17(301/200)^2 - 8/3 + 2/3. \end{aligned} \tag{R38.L375}$$

The norm costs are below $60, 46, 39$, so the complete fees are below $252/5, 238/5, 37$. Subtracting from $607/20 + 128/5 = 1119/20$ leaves losses greater than $111/20, 167/20, 379/20$, all greater than two. Composition with the increasing convex square on nonnegative arguments justifies the convexity and covers every actual ϵ.

For minimum three, the four combined norms are $39/20, 7/4, 49/30, 7/5$, in the same order as part R38.7. The complete payments are

$$\begin{aligned} 767/20 + 128/5 + 16 - 17(39/20)^2 - 27/20 - 14/3 - 2/3 &> 2, \\ 767/20 + 128/5 + 19/2 - 17(7/4)^2 - 27/20 - 14/3 - 2/3 &> 2, \\ 767/20 + 128/5 + 3/2 - 17(49/30)^2 - 27/10 - 36/5 - 2/3 &> 2, \\ 767/20 + 128/5 + 19/2 - 17(7/5)^2 - 27/10 - 86/5 - 2/3 &> 2. \end{aligned} \tag{R38.L385}$$

Their norm costs are below $65, 53, 46, 34$, and the remaining fees below $7, 7, 11, 21$. The starting losses and resulting net losses satisfy

$$\begin{aligned} &\text{starting losses: } 1599/20,\ 1469/20,\ 1309/20,\ 1469/20, \\ &\text{net losses: } >159/20,\ >269/20,\ >169/20,\ >369/20. \end{aligned} \tag{R38.L393}$$

Extra terminal or root pairs increase the available loss. Restore the actual $E_r, E_t \leq 1$ by positive operators to finish the theorem.

O.35 Higher even lists: both ordinary moments and outer outlets

R39.1 Original problem and complete new local statements

The original Q_5 problem asks, for every finite original tree U with at most five vertices of original degree at least three, for a direction of every original edge fixed from U before any host, such that

every finite original probability μ and real skew pointwise $|K| \le 1$ satisfy $Z_D = E_\mu^V \prod_{v\to w}(1+K(x_v, x_w)) \le 1$. Every original arm, repetition, subdivision, four independent positive core lengths, arbitrary correlated original joint-label law and noninjective tournament map remains. We prove the following families.

Use on the same original $L = 2(\mu)$

$$\begin{gathered} B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B = T^*, \\ p = 1 - u, \quad q = 1 + u, \quad \delta_i = \|B^i\mathbf{1}\|^2, \quad d = \delta_1, \end{gathered} \tag{R39.L80}$$

$$\begin{gathered} f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}, \\ I_q(U) = E[q^{-\frac{1}{3}}|U|^{8/3}]. \end{gathered} \tag{R39.L84}$$

After maximal equal-arm pairing, let the terminal residual be $4 \le a_1 < b_1 < \cdots < a_j < b_j$, $j \ge 1$, with actual message $F = \prod_i h_{a_i} f_{b_i}$. At the receiver there is an existing arm of length $m \ge 3$ and an optional negative post block $P = \prod_i W_{u_i} \bar{f}_{v_i}$, where $4 \le u_1 < v_1 < \cdots < u_k < v_k$ and $k \ge 0$. The two length masks and m may overlap. Let E_t, E_r be the equal-pair products at the terminal and receiver.

For every original inner $L \ge 1$ choose exactly

$$H_D = E_r P \begin{cases} W_m T^L(E_t F), & m \le L, \\ \bar{f}_m A T^{L-1}(E_t F), & m > L. \end{cases} \tag{R39.1.1}$$

In an AT^{L-1} row, only the root-adjacent connector edge is A; all remaining $L-1$ edges are T. The m arm is the actual inward $\bar{f}_m$ or its specified W_m direction. Every f arm is outward, every b arm inward, h has its first edge toward the root and all remaining edges away from that first neighbor, and W is its full reversal. These are directions on the original edges, not substitute arms.

Theorem A. For every stated table, every $m \ge 3$, every $L \ge 1$ and every finite host,

$$\boxed{EH_D^8 \le e^{-2d}, \qquad EH_D^4 \le e^{-6d}.} \tag{R39.1.2}$$

The eighth moment is the estimate of Section O.33. The fourth moment is proved below for the same direction.

Theorem B. For every independent $M \ge 1$, append M outer T edges and put $U_D = T^M H_D$. Then

$$\boxed{I_q(U_D)^3 \le e^{-24d/19}.} \tag{R39.1.3}$$

The stronger exact statements are

$$I_q(U_D)^3 \le \begin{cases} e^{-24d/19}, & M = 1, \\ e^{-16613d/3420 - 5d^2}, & M = 2, \\ e^{-(3+\lfloor M/3 \rfloor/4)d}, & M \ge 3. \end{cases} \tag{R39.1.4}$$

The same H_D is used for every M. The direction depends only on the original integer data.

R39.2 Precisely finite data and actual higher-even masks

The original finite calculus gives $\theta = \frac{13}{32}, \kappa = \frac{19}{32}, 0 \le d \le \frac{1}{3}, d^2 \le \delta_2 \le \theta d, \delta_{i+1} \le \theta\delta_i, \|B\|^2 \le \theta$ and $|Bv(x)| \le \|v\|_2$. Only $f_n, \bar{f}_n$ have means μ_n in[0,1]. All four path types are nonnegative and at most 2; h_2, W_2 have mean $1 + d$ and are not assigned the f/b mean property. The genuine equal-pair products $f_n h_n$ and $\bar{f}_n W_n$ lie in[0,1].

For $G_n(z) = \sum_{r=0}^{n-1} z^r, G_0 = 0$, put $\beta_1 = \beta_2 = 0$ and $\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i$. The finite identity

$$\begin{aligned} 1 + G_n(B)u &= \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}, \\ \tau := d - \delta_2 \le \beta_n &\le d \quad (n \ge 3) \end{aligned} \tag{R39.2.1}$$

has actual nonnegative remainder terms. The finite G_n norm is at most one for $n = 1$ or $n \ge 3$; G_2 alone has squared norm at most $1+\theta = \frac{45}{32}$. These finite statements hold without imposing a maximum arm length.

Use the actual intermediate finite certificates in parts R36.3–R36.7, and their English reconstruction in E006, Sections 3–8.

The true terminal product has one finite mask R and loss Γ:

$$\begin{gathered} F \le e^{R-\Gamma}, \\ R = \sum_i \sum_{n=a_i+1}^{b_i} K^n \mathbf{1} = -B^4 Q(-B)u, \\ \Gamma \ge 16jd/5. \end{gathered} \tag{R39.2.2}$$

Q has finite zero-one coefficients because intervals within the actual strict terminal table are disjoint. The stronger first loss $69d/20$ at $a_1 = 4$ is valid but unnecessary for the fourth-moment estimate. On the original spectral measure ν of $-B^2$ at u, write $z = i\sqrt{x}, Q(-z) = U(x) - zV(x)$. The actual bounds are

$$\begin{aligned} -x/(1-x^2) \le U, V &\le 1/(1-x^2), \\ |Q|^2 \le C_* := \frac{32768}{16245}, &\quad 0 \le x \le \theta. \end{aligned} \tag{R39.L194}$$

Let $\sigma = ER^2$ and $D_i = EB^i R$. The same finite coordinates give

$$\begin{gathered} D_0 = -\int x^2 V, \quad D_1 = \int x^2 U, \quad D_2 = \int x^3 V, \\ \sigma < d/18, \quad |R| < \frac{1}{4}, \end{gathered} \tag{R39.2.3}$$

$$\begin{gathered} |D_0|, |D_1| < d/5, \quad D_0 < d/12, \quad D_0 - D_1 < d/6, \\ D_0 + D_1 - D_2 < d/2. \end{gathered} \tag{R39.2.4}$$

$Atd = 0$ we use the exact original boundary separately. The crucial genuine propagated vector is $TR = D_0 \cdot 1 + BR$. Its complete square, including the possibly positive constant cross, has already been proved as

$$\|TR\|^2 = D_0^2 + 2D_0D_1 + \|BR\|^2 < (\frac{341}{5400})d < (\frac{4}{15})^2 d. \tag{R39.2.5}$$

It is not permissible to replace $D_0 \cdot 1$ by $D_0 \cdot u$. We retain (R39.2.5). The needed original power bounds are $\|T^n\| \le 1$ for $n \ge 3$ and $\|T^2\| < \frac{7}{5}$; the latter covers the genuine inner $L = 3$ exception.

For either actual $S_L = T^L$ or AT^{L-1}, positivity and $e^R \le 1+R+3R^2/5$ give the true nonnegative upper

$$0 \le S_L F \le e^{-\Gamma} Y_L, \quad Y_L = S_L 1 + S_L R + \omega_L \ge 0, \tag{R39.2.6}$$

where $\omega_1 < d/15$ and $\omega_L < 2d/15$ for $L \ge 2$. These follow by applying the last original positive row, bounded by twice the preceding mean; $E[f_{L-1}R^2] \le 2\sigma$. For $L = 1$ the preceding input is just R^2. The upper is nonnegative as a whole; its summands need not be.

Put $V_R = T^{L-1}R$ and $a_R = EV_R$. Then $S_L R = a_R + BV_R$ in a T row and $a_R - BV_R$ in an A row. $AtL = 1, EAR < d/6$; at $L = 2, EATR < d/2$ and $\|V_R\| < 4\sqrt{d}/15$. For every $L \ge 3$ the exact finite Duhamel identity with coefficients μ_i in[0,1] gives

$$a_R = \langle f_{L-1}, R\rangle \le d/12 + 3\sqrt{d\sigma} < 5d/6. \tag{R39.2.7}$$

The finite identity is $f_k = \mu_{k-1}1 + \sum_{j=1}^{k-1} \mu_{k-1-j}(-B)^j \mathbf{1} + (-B)^k \mathbf{1}$; its nonconstant norm is at most $\sqrt{d}\sum_{j=0}^{k-1} \theta^{j/2} < 3\sqrt{d}$. A geometric tail bounds this finite path on the original measure.

Two convenient valid consequences for all $L \ge 3$ are

$$\|V_R\| < (\frac{7}{20})\sqrt{d}, \quad ES_L R < 71d/60, \tag{R39.2.8}$$

and the newly sharpened, same-direction bounds

$$\|V_R\| < (\frac{7}{5})\sqrt{d/18} < \sqrt{d}/3, \qquad ES_L R < 7d/6. \tag{R39.2.9}$$

The first strict comparison in(R39.2.9) use $s_{49}/450 < \frac{1}{9}$; the mean uses $|EBV_R| \le \sqrt{d}\|V_R\|$. It holds for either last-edge sign.

Finally the actual negative root posts have their own finite mask:

$$\begin{gathered} P \le e^{S-\Gamma_r}, \quad \Gamma_r \ge 16kd/5, \quad S = BV_p, \\ ES < d/12, \quad \|V_p\| < (\frac{3}{8})\sqrt{d}. \end{gathered} \tag{R39.2.10}$$

For $k0$ take $\Gamma_r = S = V_p = 0$. All cross terms below are paid in the square of one combined vector on the same original μ.

R39.3 One actual positive upper gives both ordinary moment certificates

The supplied all-real signed exponential statement is

$$\begin{aligned} Ee^{4BV} &\le e^{4EBV+(\frac{17}{4})\|V\|^2}, \\ Ee^{8BV} &\le e^{8EBV+17\|V\|^2}. \end{aligned} \tag{R39.3.1}$$

The second follows by applying the first to $2V$. Both allow nonzero mean, either mean sign, and a constant-vector coordinate.

Temporarily omit only the genuine equal pairs. Let X be the actual anchor $\bar{f}_m$ or W_m, and suppose its true bare sum obeys

$$X + S_L 1 \le 2 + c_b + BV_b. \tag{R39.3.2}$$

Equation(R39.2.6) gives $H \le Pe^{-\Gamma} XY_L$ with $X, Y_L \ge 0$. Their actual sum is at most $2 + c_b + a_R + \omega_L + B(V_b + / - V_R)$. That upper is nonnegative because it dominates the nonnegative sum. Thus AM-GM and $(1+t)^2 \le e^{2t}$ for $1 + t \ge 0$ give a pointwise bound

$$\begin{aligned} H \le \exp[&-\Gamma - \Gamma_r + c_b + a_R + \omega_L \\ &+B(V_b \pm V_R + V_p)]. \end{aligned} \tag{R39.3.3}$$

The sign is minus in an A row and plus in a T row. Applying(R39.3.1) once, on the same combined vector, yields for $r = 4$ or 8

$$\begin{aligned} \log EH^r \le &-r(\Gamma + \Gamma_r) \\ &+r(c_b + EBV_b) + r\omega_L + rES_L R + rES \\ &+C_r \|V_b \pm V_R + V_p\|^2, \end{aligned} \tag{R39.3.4}$$

where $C_4 = \frac{17}{4}$ and $C_8 = 17$. The fourth moment is proved directly from this formula.

R39.4 Every $m \ge 3$: new ordinary fourth at the unchanged C036 direction

C036/E006 supply the following actual bare-sum data in(R39.3.2):

$L = 1$

Bare sum: $\bar{f}_m + p$;

$c_b + EBV_b$ upper: $-19d/16$;

Norm/cross data: $Atm = 3$ retain G_2;

combined square with $-R$ at most $2777d/1440$ for terminal minimum $4,751d/480$ for minimum $\ge 5; m \ge 4$ is smaller.

$L = 2$

Bare sum: $\bar{f}_m + W_2$;

$c_b + EBV_b$ upper: $-3d/16$;

Norm/cross data: $\|V_b\| < 4\sqrt{d}/5$;

genuine $\|TR\| < 4\sqrt{d}/15$.

$L \ge 3$

Bare sum: $W_{\min(m,L)} + \bar{f}_{\max(m,L)}$;

$c_b + EBV_b$ upper: $-19d/16$;

Norm/cross data: $\|V_b\| \le \sqrt{d}/2$, including $m = L$ and the finite G_2 exception.

For the first line the bound is actually $c_b + EBV_b = -\beta_m - \beta_{m+1} \le -2\tau$. For the second it follows from $-\beta_m + \delta_2 \le -d + 2\delta_2$. For the third the actual loss $51d/32$ and mean allowance $13d/32$ give $-19d/16$. These are all the original structural rows, not selected from the host's $d-$ domain.

With no root posts, put $r = 4$ in(R39.3.4). $AtL = 1$ the combined norm square is less than $2d$, and the two other fees are at most $14d/15$. Thus the fourth loss is greater than

$$\frac{64}{5} + \frac{19}{4} - \frac{14}{15} - \frac{17}{2} = \frac{487}{60} > 8. \tag{R39.4.1}$$

$AtL = 2$ use $\|V_b - V_R\| < 16\sqrt{d}/15$ and both actual short means:

$$\frac{64}{5} + \frac{3}{4} - \frac{8}{15} - 2 - (\frac{17}{4})(\frac{16}{15})^2 = \frac{5563}{900} > 6. \tag{R39.4.2}$$

For every $L \ge 3$, the weaker proved bounds(R39.2.8) suffice. The vector norm is at most $17\sqrt{d}/20$, the other fee $79d/15$, so

$$\frac{64}{5} + \frac{19}{4} - \frac{79}{15} - \frac{4913}{1600} > \frac{551}{60} > 9, \tag{R39.4.3}$$

where $\frac{4913}{1600} < \frac{31}{10}$ gives the displayed weaker rational coefficient. These estimates include every finite terminal pair count; extra pairs only enlarge Γ while the finite-mask estimates are uniform.

With at least one root post pair, C036/E006 paid the full combined vectors without a cross-block disjointness assumption. Their norm bounds are $71\sqrt{d}/40$ at $L = 1, 173\sqrt{d}/120$ at $L = 2$, and $49\sqrt{d}/40$ at $L \ge 3$. The corresponding eighth norm costs are less than $54d, 36d, 26d$, so the fourth costs are less than $27d/2, 9d, 13d/2$ respectively. The fourth other fees, including $4ES$, are $19d/15$, $43d/15, 28d/5$. The exact resulting weaker losses are

$$\frac{128}{5} + \frac{19}{4} - \frac{19}{15} - \frac{27}{2} = \frac{187}{12} > 15 \quad (L1), \tag{R39.L370}$$

$$\frac{128}{5} + \frac{3}{4} - \frac{43}{15} - 9 = \frac{869}{60} > 14 \quad (L2), \tag{R39.L373}$$

$$\frac{128}{5} + \frac{19}{4} - \frac{28}{5} - \frac{13}{2} = \frac{73}{4} > 18 \quad (L \ge 3). \tag{R39.4.4}$$

All larger root pair counts improve Γ_r. Thus the same message of Section O.33 has $EH_D^4 \le e^{-6d}$ for every $m \ge 3$ and $L \ge 1$, with all its stated posts. That section gives $EH_D^8 \le e^{-2d}$ for the same direction. This proves Theorem A.

R39.6 A fully paid one-edge bridge and all original outer lengths

R39.6.1 General real ordinary-fourth transfer on the original measure

Let P_{op} be any original positive operator with positive row mass $h = P_o p1$ and let Y be a real original function. Original row Cauchy gives $|P_{\text{op}}Y|^2 \le hP_{\text{op}}(|Y|^2)$. Raise this to $\frac{4}{3}$, multiply by $q^{-\frac{1}{3}}$, then use original Hölder with exponents 3 and $\frac{3}{2}$:

$$
\begin{aligned}
&I_q(P_{op}Y)\\
&\le E[(h^4/q)^{1/3}(P_{\mathrm{op}}|Y|^2)^{4/3}]\\
&\le [E(h^4/q)]^{1/3}[E(P_{\mathrm{op}}|Y|^2)^2]^{2/3}.
\end{aligned}
\tag{R39.L560}
$$

Cubing and applying the ordinary original $L = 2(\mu)$ norm gives

$$
\boxed{\begin{aligned}
&I_q(P_{op}Y)^3 \le E(h^4/q)\|P_{\mathrm{op}}\|^4\\
&(E|Y|^4)^2.
\end{aligned}}
\tag{R39.6.1}
$$

In the middle display $E(P_{\mathrm{op}}|Y|^2)^2$ means the expectation of the square of that actual function, not the square of its expectation. This reproduces exact C094(2.5), fixing its operator order, measure and input exponent. The input Y may be any real vector.

For one original T edge, $h = q$, and the exact finite moments give

$$
Eq^3 = E(1+u)^3 = 1 + 3d + Eu^3 \le 1 + 4d, \tag{R39.6.2}
$$

since $Eu = 0$ and $|u| \le 1$ implies $u^3 \le u^2$. The supplied original ordinary norm is $\|T\| \le 1 + 32d/19$. Therefore(R39.6.1) proves the new useful same-input inequality for every real Y:

$$
\begin{aligned}
I_q(TY)^3 &\le (1+4d)(1+32d/19)^4(E|Y|^4)^2\\
&\le e^{(\frac{204}{19})d}(E|Y|^4)^2.
\end{aligned}
\tag{R39.6.3}
$$

Apply (R39.6.3) to the actual H_D of Theorem A:

$$
I_q(TH_D)^3 \le e^{(\frac{204}{19}-12)d} = e^{-24d/19}. \tag{R39.6.4}
$$

The outer edge is retained when the inner connector has length one.

R39.6.2 Exactly two original outer edges

The exact C094 all-real bridge, on the unchanged original measure, is

$$
\begin{aligned}
I_q(T^2Y)^3 &\le [1 - 2d^2 + (8-2d)\delta_2]\\
&\quad (1 + 1024\delta_2/855)^8(E|Y|^4)^2\\
&\le e^{(\frac{24427}{3420})d - 5d^2}(E|Y|^4)^2.
\end{aligned}
\tag{R39.6.5}
$$

The actual T^2 kernel retains both outer edges and their middle vertex. For the same H_D, Theorem A and (R39.6.5) give

$$
I_q(T^2H_D)^3 \le e^{-(\frac{16613}{3420})d - 5d^2}. \tag{R39.6.6}
$$

This is the existing $m \ge 3$ two-edge output, and its rate is stronger than the common rate in Theorem B.

R39.6.3 Every finite outer $M \geq 3$

The exact original C079 positive-operator eighth transfer is

$$I_q(T^M Y)^3 \leq \|T^M\|^2 E|Y|^8 \{E[\bar{f}_M^3/\sqrt{q}]\}^2. \tag{R39.6.7}$$

Its complete actual row proof gives the last brace squared at most $1-d$ for every $M \geq 3$. For $M = 3$ it uses the actual inverse fourth and actual second of $\bar{f}_3$; $M = 4, 5$ use their respective supplied ordinary fourths, and every $M \geq 6$ uses the finite tail.

The power bound is $\|T^M\| \leq (1-d/8)^{\lfloor M/3 \rfloor}$. Combining (R39.6.7) with Theorem A gives

$$\begin{aligned} I_q(T^M H_D)^3 &\leq (1-d)(1-d/8)^{2\lfloor M/3 \rfloor} e^{-2d} \\ &\leq e^{-(3+\lfloor M/3 \rfloor/4)d}. \end{aligned} \tag{R39.6.8}$$

Together with (R39.6.4) and (R39.6.6), this proves Theorem B. The direction is H_D followed by the M outer T edges.

O.36 Ordinary moments with a unit terminal arm and a nonunit receiver

Messages and directions.

Use $B = -K$, $u = B1$, $T = J + B$, $A = T^* = J - B$, $q = 1 + u$, $p = 1 - u$, and

$$f_n = A^n 1, \qquad \bar{f}_n = T^n 1, \qquad h_n = T f_{n-1}, \qquad W_n = A \bar{f}_{n-1}.$$

Let $(1, r_1, t_1, \dots, r_j, t_j)$ be a strict terminal list, $1 < r_1 < t_1 < \cdots < r_j < t_j$, with $j \geq 1$. The receiver has an existing arm of length $m \geq 3$, and the connector has length $L \geq 1$. Its negative post block is

$$P = \prod_{\ell=1}^{k} W_{u_\ell} \bar{f}_{v_\ell}, \qquad 4 \leq u_1 < v_1 < \cdots < u_k < v_k, \qquad k \geq 0. \tag{R40.1.3}$$

There is no restriction between a terminal interval and a post interval. Genuine equal pairs are oriented as $f_\ell h_\ell$ at the terminal and $\bar{f}_\ell W_\ell$ at the receiver, giving factors $E_t, E_r \leq 1$. The following two tables specify every original edge before the host is chosen; $S_\sigma = J + \sigma B$ and $Q_L^\sigma = S_\sigma T^{L-1}$.

Table 66: Terminal messages for the ordinary unit-minimum list.

Terminal list	F
$r_1 \geq 4$	$q \prod_i h_{r_i} f_{t_i}$
$r_1 = 3$	$q f_3 h_{t_1} \prod_{i \geq 2} h_{r_i} f_{t_i}$
$(1, 2, 3, 4, e)$, exactly five arms	$q f_2 h_e h_3 f_4$
All other $r_1 = 2$	$q f_2 h_{t_1} \prod_{i \geq 2} h_{r_i} f_{t_i}$

Table 67: Receiver and connector directions.

Condition	a_m	Q_L^σ
$L = 1, r_1 = 2$	$\bar{f}_m$	A
$L = 1, r_1 \geq 3$	W_m	T
$L \geq 2, m \leq L + 1$	W_m	T^L

Condition	a_m	Q_L^σ
$L \ge 2, m > L+1$	$\bar{f}_m$	AT^{L-1}

In AT^{L-1}, the A-edge is adjacent to the receiver. In particular qf_2h_3 is the exact 123 message also at $L = 2$; the five-arm crossing is not applied inside a longer list.

Proposition O.10. *For these directions the actual message*

$$H = E_r P\, a_m Q_L^\sigma(E_t F) \tag{R40.1.5}$$

satisfies $\mathbb{E}H^8 \le e^{-\gamma_8 d}$ and $\mathbb{E}H^4 \le e^{-\gamma_4 d}$, with the following unposted losses.

Table 68: Simultaneous ordinary losses in the selected directions.

Terminal family	γ_8	γ_4
$r_1 \ge 4$	$6 + \frac{128}{5}(j-1)$	$5 + \frac{64}{5}(j-1)$
$r_1 = 3, j = 1$	3	$9/2$
$r_1 = 3, j \ge 2$	$10 + \frac{128}{5}(j-2)$	$7 + \frac{64}{5}(j-2)$
Exact 123	2	3
Exact 124	$1/4$	1
Exact $12c, c \ge 5$	$57/400$	1
$(1,2,c,b,e), c \ge 4$	4	3
$(1,2,3,b,e), b \ge 5$	7	5
Crossed $(1,2,3,4,e)$	8	5
$r_1 = 2, j \ge 3$	$8 + \frac{128}{5}(j-3)$	$8 + \frac{64}{5}(j-3)$

For $k \ge 1$, the losses improve by

$$\Delta_8(k) = 128k/5 - 24, \qquad \Delta_4(k) = 64k/5 - 12. \tag{R40.1.7}$$

Consequently, for every listed terminal and every post block,

$$\mathbb{E}H^8 \le e^{-57d/400} \le e^{-d/8}, \qquad \mathbb{E}H^4 \le e^{-d}. \tag{R40.1.6}$$

The proof separates the exact three-arm packets from seven remaining terminal profiles, then uses one finite-tail comparison for every receiver and connector. All domains include their analytic boundaries. When $d = 0$, $u = 0$, so every path factor and the actual message is one. In what follows $d > 0$ whenever a ratio by d occurs.

R40.2 Basic estimates and finite algebra

The exact finite calculus gives

$$0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta\delta_i, \quad \|B\|^2 \le \theta, \quad |BV(x)| \le \|V\|_2. \tag{R40.2.1}$$

All four path types are nonnegative and at most two. Only $f_n, \bar{f}_n$ have the common means $\mu_n \le 1$. In particular $Eh_2 = EW_2 = 1 + d$ is not replaced by a false mean-at-most-one assertion. We use

$$\mu_2 = 1 - d, \quad \mu_3 = 1 - 2d, \quad \mu_n \le 1 - 2d\ (n \ge 3), \quad \mu_n \le 1 - \tfrac{5}{2}d\ (n \ge 4). \tag{R40.2.2}$$

The last bound follows from the stronger finite estimate (A.8). The identities

$$f_n + h_n = \bar{f}_n + W_n = 2\mu_{n-1} \tag{R40.2.3}$$

give $0 \le f_n h_n, \bar{f}_n W_n \le 1$, hence $0 \le E_t, E_r \le 1$.

The signed exponential estimate (M.43) gives

$$Ee^{8BV} \le e^{8E(BV)+17\|V\|^2}, \qquad Ee^{4BV} \le e^{4E(BV)+(17/4)\|V\|^2}. \tag{R40.2.4}$$

It allows arbitrary real V, either sign of $E(BV)$, and genuine constant-vector coordinates. Its hypotheses follow from (R40.2.1). Only these hypotheses are used below.

For every finite n define

$$G_n(z) = \sum_{a=0}^{n-1} z^a, \quad G_0 = 0, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i. \tag{R40.2.5}$$

The exact finite positive convolution is

$$1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}, \qquad \tau \le \beta_n \le d \quad (n \ge 3). \tag{R40.2.6}$$

This is (M.2) in the unchanged B, u convention. We use $\beta_3 = \beta_4 = d$, $\beta_5 = \beta_6 = \tau$, and the valid alternating bounds

$$\beta_n \le d - \kappa\delta_2 \ (n \ge 5), \qquad \beta_n \ge \tau + \kappa\delta_3 \ (n \ge 7). \tag{R40.2.7}$$

On a rotation block put $z = i\sqrt{x}$, $0 \le x \le \theta$. Finite geometry gives

$$\|G_n(B)\| \le 1 \quad (n = 0, 1 \text{ or } n \ge 3), \qquad \|G_2(B)\|^2 \le 1 + \theta. \tag{R40.2.8}$$

Lemma M.1 proves these bounds, including the separate G_2 formula. Writing $G_n(z) = U_n + zV_n$, both finite coordinates U_n, V_n lie in $[0, 1]$. For $n \ge 1$, $U_n \ge 1 - x$; for $n \ge 2$, also $V_n \ge 1 - x$. The separately defined $G_0 = 0$ has both coordinates zero.

Let ν be the spectral measure of the actual vector u for $-B^2$. Then

$$\int d\nu = d, \quad \int x^i \, d\nu = \delta_{i+1}, \quad E[(P_e(-B^2) + BP_o(-B^2))u] = -\int P_o \, d\nu. \tag{R40.2.9}$$

This identity follows from skewness. Cauchy also gives $\delta_3/d \ge (\delta_2/d)^2$ when $d > 0$.

For the finite zero-one masks used below, whose first power on u is $a \ge 1$, (M.4) and the row bound give

$$\|R\|^2 \le C_*\theta^a d, \qquad |R|^2 \le C_*\theta^{a-1}d, \qquad C_* = \frac{32768}{16245} < \frac{100}{49}. \tag{R40.2.10}$$

The actual unequal-arm finite difference gives, for $4 \le r < t$,

$$h_r + f_t \le 2 - \frac{16}{5}d + R_{rt}, \qquad R_{rt} = \sum_{n=r+1}^{t} K^n \mathbf{1}. \tag{R40.2.11}$$

This is an actual arm-sum inequality, so AM–GM gives $h_r f_t \le e^{R_{rt} - 16d/5}$. Its full reversal is also valid. The pair-loss calculation is also given in (L90.2.3).

A positive q-row and the prescribed last edge.

Let $B = -K$, $u = B1$, $q = 1 + u$, $S_\sigma = J + \sigma B$, and $Q_L^\sigma = S_\sigma T^{L-1}$. Suppose, on the given finite probability space,

$$0 \le F \le s_0 q(1 + R + \chi R^2), \qquad s_0 > 0, \quad \chi \ge 0.$$

Write $D_i = \mathbb{E}B^i R$ and

$$\delta = D_0 - D_1 + \sqrt{d\,\mathbb{E}R^2} + 4\chi \mathbb{E}R^2. \tag{O.90}$$

For any $D \ge \delta$, the actual first row and its continuation obey

$$0 \le s_0^{-1} Q_L^\sigma F \le Y_L^\sigma,$$

$$Y_L^\sigma = w_{L+1}^\sigma + \sigma B^L R + D w_{L-1}^\sigma + \sum_{i=1}^{L-1} D_i w_{L-1-i}^\sigma, \qquad w_0^\sigma = 1, \quad w_n^+ = \bar{f}_n, \quad w_n^- = W_n \ (n \ge 1). \tag{O.91}$$

Indeed $\mathbb{E}(qR) = D_0 - D_1$. The remaining linear row has absolute value at most $\mathbb{E}|uR| \le \sqrt{d\,\mathbb{E}R^2}$, and $S_\sigma(qR^2) \le 4\mathbb{E}R^2$. This proves $L = 1$. For $L \ge 2$, apply T in the first row, continue it by (M.33), and apply S_σ in the last step. In particular

$$S_\sigma B^i R = D_i 1 + \sigma B^{i+1} R.$$

This proves (O.91) in the stated operator order. It is the $w = q, \lambda = 1$ instance of the positive first-row calculation in Theorem M.18. If $F \le q e^{R-\Gamma}$, $|R| \le 1/2$, take $s_0 = e^{-\Gamma}$, $\chi = 3/5$. The choice follows from Lemma M.16.

Two receiver geometries, with all finite tails.

The bare receiver estimates used below, and by the triple calculation, are as follows. If $3 \le h \le H$, the actual sum has an upper bound

$$W_h + \bar{f}_H \le 2 - (51/32)d + BZ, \qquad \mathbb{E}(BZ) \le \theta d, \quad \|Z\| \le \sqrt{d}/2. \tag{R40.3.1}$$

At $h < H$ take $Z = B^{h-1} G_{H-h}(B) u$. The arm-sum loss is $2d - \delta_2$ at $h = 3$, and $16d/5$ at $h \ge 4$. Its finite mean is at most δ_2. The G_2 norm is retained in $\theta^2(1+\theta) = 7605/32768 < 1/4$. At equality take $Z = 0$ and use $W_h + \bar{f}_h = 2\mu_{h-1}$. Thus in the notation $2 - \lambda d + BZ$, the combined loss $\lambda - \mathbb{E}(BZ)/d$ is at least $19/16$. AM–GM and the signed exponential bound therefore give

$$\mathbb{E}(W_h \bar{f}_H)^8 \le e^{-21d/4}, \qquad \mathbb{E}(W_h \bar{f}_H)^4 \le e^{-59d/16}. \tag{R40.3.2}$$

For $h \ge 4$, the same finite interval has

$$\lambda = 16/5, \qquad \mathbb{E}(BZ) \le \delta_4 < d/14, \qquad \|Z\| \le \sqrt{d}/3, \quad \lambda - \mathbb{E}(BZ)/d \ge 219/70. \tag{R40.3.3}$$

The first positive mean term is δ_4, and $\theta^3(1+\theta) < 1/9$; this proves the stronger data.

At $L = 1$ the effective bare index is two instead. The two actual sums, obtained by applying A to the finite path convolution, are

$$W_m + \bar{f}_2 \le 2 - \beta_m + BZ_m, \qquad Z_m = -B G_{m-2}(B) u, \qquad \mathbb{E}(BZ_m) = \beta_{m+1} - d, \tag{R40.3.4}$$

$$-\beta_m + \mathbb{E}(BZ_m) \le -5d/6, \qquad \|Z_m\|^2 \le \theta(1+\theta)d < (16/25)d. \tag{R40.3.5}$$

The possible positive increment $\beta_{m+1}-\beta_m$ begins at $\delta_3 < d/6$. The norm includes the G_2 value $\theta(1+\theta) < 16/25$. The opposite sum has $\widetilde{Z}_m = -Z_m$ and

$$\bar{f}_m + W_2 \le 2 - \beta_m + B\widetilde{Z}_m, \tag{R40.3.6}$$

$$-\beta_m + \mathbb{E}(B\widetilde{Z}_m) = d - \beta_m - \beta_{m+1} \le -d + 2\delta_2. \tag{R40.3.7}$$

Lemma O.11 (One finite-tail bound)**.** *If $\|R\| \le r\sqrt{d}$, $0 \le r \le 2/3$, then*

$$\begin{aligned} |D_i| &\le r\theta^{(i-1)/2}d \ (i \ge 1), \qquad \sum_{i\ge 1}(D_i)_+ \le 3rd, \\ \gamma_s &:= \beta_s - D_{s-2} \ge (\kappa - \theta r)d \ge 31d/96 \quad (s \ge 3). \end{aligned} \tag{O.92}$$

Each displayed sum dominates every actual finite positive subsum. For a nonnegative first-row bound $D \ge \max(\delta, 0)$ and $L \ge 3, m = 3$, retain the final path coefficient in (O.91)*. With*

$$V = (G_L - G_2)u + B^{L-1}R, \qquad M \le 2D/d - \gamma_{L+1}/d - \gamma_{L+2}/d, \tag{O.93}$$

one has the uniform full-vector bound $\|V\| < (7/10)\sqrt{d}$.

Proof. Skew adjunction gives $|D_i| = |\langle u, B^{i-1}R\rangle| \le r\theta^{(i-1)/2}d$. Use $\sqrt{\theta} < 2/3$ for the geometric bound. At $s = 3, 4$, the coefficient $\beta_s = d$ gives lower bounds $1-r$ and $1-2r/3$; both exceed $\kappa - \theta r$. For all $s \ge 5$, use $\beta_s \ge \tau \ge \kappa d$ and $|D_{s-2}| \le \theta r d$. This proves every finite index at once.

For $m = 3$ the selected direction is all T, and finite convolution gives

$$Y_L^+ = 1 + G_{L+1}u + B^L R + D\bar{f}_{L-1} - \sum_{s=3}^{L+1}\gamma_s \bar{f}_{L+1-s}.$$

Use $W_3 = 1 - d - u - Bu - B^2u$, retain $\gamma_{L+1}\bar{f}_0$, and bound $D\bar{f}_{L-1} \le 2D$. The signed mean of the remaining full vector is

$$\mathbb{E}BV = -\beta_{L+2} + d + D_L.$$

It supplies the second loss in (O.93); it is not an additional unavailable convolution term. At $L = 3$, its norm is at most $\theta(1+r)\sqrt{d} \le (25/36)\sqrt{d} < (7/10)\sqrt{d}$. At $L \ge 4$, the bare part has norm at most $\sqrt{d}/2$, including $L = 4$'s G_2. The remaining part has norm at most $(4/15)r\sqrt{d}$, so the total is below $61\sqrt{d}/90$. □

A single moment payment.

In any positive finite comparison

$$a_m + Y_L^\sigma \le 2 + c + BV, \qquad M = (c + \mathbb{E}BV)/d, \quad N^2 = \|V\|^2/d,$$

AM–GM and (M.43) give

$$\log \mathbb{E}(a_m Q_L^\sigma F)^\nu \le \nu \log s_0 + [\nu M + (17\nu^2/64)N^2]d, \qquad \nu \in \{4, 8\}. \tag{O.94}$$

Its pointwise certificate is $a_m Q_L^\sigma F \le s_0 e^{c+BV}$, with this same full vector. For $D \ge 0$, let $\alpha = D/d$ and let b_1 bound $(D_1)_+/d$. The receiver bounds and Lemma O.11 yield:

mode	M upper	N^2 upper
$L = 1$	M_1	N_1^2
$L = 2$	$-19/16 + 2\alpha + b_1 + 2r/3$	$(1/2 + 2r/3)^2$
$L \ge 3,\ m \ge 4$	$-219/70 + 2\alpha + 6r + 5r/12$	$(1/3 + 5r/12)^2$
$L \ge 3,\ m = 3$	$2\alpha - 2(\kappa - \theta r)$	$49/100$

(O.95)

At $L=2$, keep $Dw_1^\sigma + D_1 w_0^\sigma$. At $L \ge 3, m \ge 4$, the smaller bare index is at least four. After positive propagation, its remaining path cost is bounded by $2D + 2\sum (D_i)_+ < 2D + 6rd$. The terminal vector has norm at most $\theta r\sqrt{d} < (5/12) r\sqrt{d}$.

R40.6–8 Ordinary moments of the unit-minimum triples

Let $F_c = q f_2 h_c$, $c \ge 3$, and retain the directions of the direction table: at $L=1$ put $H = \bar{f}_m A F_c$; at $L \ge 2$ put

$$H = \begin{cases} W_m T^L F_c, & 3 \le m \le L+1, \\ \bar{f}_m A T^{L-1} F_c, & m > L+1. \end{cases}$$

We prove, in these same directions,

$$\mathbb{E}H^8 \le e^{-g_8(c)d}, \qquad \mathbb{E}H^4 \le e^{-d}, \qquad g_8(3)=2, \quad g_8(4)=\tfrac{1}{4}, \quad g_8(c)=\tfrac{57}{400}\ (c \ge 5). \tag{O.96}$$

The fourth-moment data below are retained for a subsequent outer path. All expectations are in the original finite measure. In particular the eighth moment here is $\mathbb{E}(H^8)$. During the normalized calculations assume $d > 0$; the case $d = 0$ is settled at the end.

Finite inputs and their retained coefficients

On a spectral block write $z = i\sqrt{x}$, $0 \le x \le \theta$, and set $u_0(x) = x - x^2 + x^3$, $v_0(x) = x - x^2$. For each row of the following table the actual arm sum satisfies $f_2 + h_c \le 2a_0 - U$, so

$$F_c \le a_0^2 q(1 + R + R^2/4), \qquad R = -U/a_0 = (-a + zb)u. \tag{O.97}$$

The base $2a_0 - U$ is nonnegative, since it bounds $f_2 + h_c$.

c	domain	a_0	U
3	$d \le 1/3$	$1 - d/2$	$-B^2 u$
4	$d \le 1/3$	$1 - d$	$(d + x - xz)u$
≥ 5	$d \le 1/4$	$1 - 9d/8$	$(A_0 - zB_0)u$

(O.98)

For $c = 3, 4$ these are the exact identities $f_2 + h_3 = 2 - d + B^2 u$ and $f_2 + h_4 = 2 - 2d - du + B^2 u - B^3 u$.

For $c \ge 5$, retain the following actual positive convolution terms:

$$\begin{aligned} &c = 5: \quad c_1 = d, \quad c_2 = d/4 + \delta_2, \quad c_3 = 0, \\ &c \ge 6: \quad c_1 = \beta_{c-1}, \quad \lambda = \frac{9d/4 - \beta_c - \beta_{c-1}}{\beta_{c-2} + (1-d)\beta_{c-3}}, \quad (c_2, c_3) = \lambda(\beta_{c-2}, \beta_{c-3}). \end{aligned} \tag{O.99}$$

For $d > 0$ the denominator is positive and the numerator nonnegative. The full available constant is at least $9d/4$: for $c = 6$ it is $2\tau + (2-d)d$, for $c = 7$ it is $3\tau + \delta_3 + (1-d)d$, and for $c \ge 8$ it is at least $(4-d)\tau + 2\kappa\delta_3$. For the last case, $t = \delta_2/d \le \theta$ and $\delta_3/d \ge t^2$ give

$$(4-d)(1-t) + 2\kappa t^2 \ge \tfrac{15}{4}(1-t) + 2\kappa t^2 \ge \tfrac{39691}{16384} > \tfrac{9}{4}.$$

The last quadratic decreases on $[0, \theta]$. Thus $0 \le \lambda \le 1$. At $c = 5$, $c_2 \le 21d/32 < \beta_3 = d$. No coefficient is taken from $\beta_2 = 0$. Put $C = c_1 + c_2 + c_3$, $D_* = c_2 + c_3$, $E_* = c_3$. Keeping these terms in the convolution for f_{c-1} and applying T gives the subtractions $c_1 q + c_2 h_2 + c_3 h_3$. Use $h_2 = q - Bu$,

$h_3 = q - d - Bu + B^2u$, and write $G_{c-2}(-z) = a_f - zb_f$. Expanding gives $f_2 + h_c \le 2 - 9d/4 - U$, with

$$\beta_c + c_1 + c_2 + (1-d)c_3 = 9d/4, \qquad C = 9d/4 - \beta_c + dE_*, \quad D_* = C - c_1, \quad 0 \le E_* \le d,$$
$$5d/4 + dE_* \le C \le 53d/32 + dE_*, \qquad d/4 \le D_* \le 17d/16 + dE_*,$$
$$(A_0, B_0) = \begin{cases} (5d/4 + \delta_2 + x - x^2, d/4 + \delta_2 + x), & c = 5, \\ (C + x(a_f - E_*), D_* + xb_f), & c \ge 6, \end{cases}$$
$$1 - x \le a_f, b_f \le 1 - x + x^2 \quad (c \ge 6), \qquad A_0 - c_1 - (1-x)B_0 = x[c_2 + a_f - (1-x)b_f] \ge 0. \tag{O.100}$$

In the last identity the $c = 5$ coordinates are $a_f = 1 - x, b_f = 1$. In particular $A_0, B_0 \ge x(1-x)$.

One propagation and two ordinary moments

Write $D_i = \mathbb{E}B^iR$. Expanding (O.97) in one positive row gives, for either $S_\sigma = J + \sigma B$,

$$a_0^{-2} S_\sigma F_c \le w_2^\sigma + \sigma BR + \delta, \qquad \delta = -\int (a+b)\, d\nu + \sqrt{d\mathbb{E}R^2} + \mathbb{E}R^2. \tag{O.101}$$

Indeed $\mathbb{E}(qR) = D_0 - D_1 = -\int (a+b) d\nu$, the remaining linear row is bounded by $\mathbb{E}|uR|$, and $S_\sigma(qR^2)/4 \le \mathbb{E}R^2$. Here $w_n^+ = \bar{f}_n, w_n^- = W_n$ for $n \ge 1$, and $w_0^\sigma = 1$. For any $D \ge \max(\delta, 0)$, finite Duhamel gives

$$0 \le a_0^{-2} Q_L^\sigma F_c \le Y_L^\sigma = w_{L+1}^\sigma + \sigma B^L R + D w_{L-1}^\sigma + \sum_{i=1}^{L-1} D_i w_{L-1-i}^\sigma. \tag{O.102}$$

For $L = 1$ the actual δ, even when negative, can be retained. All signed terms in (O.102) are propagated before any negative coefficient is discarded. If the sum with the actual receiver is bounded by $a_m + Y_L^\sigma \le 2 + c_0 + BV$, put $M = (c_0 + \mathbb{E}BV)/d$, $N^2 = \|V\|^2/d$. AM–GM and (M.43) give both moments at once:

$$\frac{\log \mathbb{E}H^s}{d} \le \frac{2s \log a_0}{d} + sM + c_s N^2, \qquad (s, c_s) = (4, 17/4), (8, 17). \tag{O.103}$$

The vector V includes every constant coordinate.

Lemma O.12 (Two fractional final coefficients)**.** *For $n \ge 1$, $0 \le a, b \le 1$, and $z = i\sqrt{x}$, $0 \le x \le \theta$, the real and z-coordinates of $G_n(z) - az^{n-1} + bz^n$ belong to $[0, 1]$. Consequently its squared modulus is at most $1 + x$. Also $|G_n(z) - z| \le 1$ for every $n \ge 2$.*

Proof. The first polynomial is affine in (a, b). Its four corner polynomials are $G_n, G_{n-1}, G_{n+1}, G_{n-1} + z^n$. Finite alternating sums give coordinates in $[0, 1]$ for all four, including $G_0 = 0$. For the last polynomial they are consecutive partial sums of $1 - x + x^2 - \cdots$, with a possible zero coordinate at $n = 1, 2$. Convex combinations preserve these coordinate bounds. For the second assertion, $G_2 - z = 1$. For $n \ge 3$, the actual coordinates give

$$|G_n - z|^2 \le (1 - x + x^2)^2 + x^3 \le 1, \tag{O.104}$$

since $2 - 3x + x^2 - x^3 \ge 2 - 3\theta - \theta^3 > 0$. □

Continuous first-row certificates

The following bounds hold on the stated full intervals:

c	d range	$a \le$	$b \le$	D/d	$\sum_{i\ge1}(D_i)_+/d$	γ_s/d $(s \ge 5)$
3	$[0, 1/3]$	$39/80$	0	$1/4$	$3/5$	$49/96$
4	$[0, 1/4]$	$7/8$	$13/24$	$5/12$	$7/5$	$41/96$
≥ 5	$[0, 1/5]$	$5/6$	$3/4$	$2/5$	$3/2$	$41/96$

(O.105)

Here $\gamma_s = \beta_s - D_{s-2}$. For $c = 3$, maximizing $k[\sqrt{\theta t} - (1 - k\theta)t]$, $k = (1 - d/2)^{-1} \le 6/5$, gives $\delta/d \le 39/164 < 1/4$. For $c = 4$, $k = (1 - d)^{-1}$, $J = \mathbb{E}U^2/d = d^2 + 2dt + \delta_3/d + \delta_4/d$, and Young's inequality $\sqrt{J} \le 3/8 + 2J/3$ give

$$\frac{\delta}{kd} \le \frac{3}{8} - d + 2d^2 + t\left(-\frac{439}{512} + 4d\right). \tag{O.106}$$

If the coefficient of t is negative, subtraction of $(5/12)(1-d)$ leaves at most $-1/24-7d/12+2d^2 < 0$. Otherwise take $t = \theta$; the resulting convex quadratic is negative at zero and equals $-75/16384$ at $d = 1/4$. On $d \le 1/5$, using $t \ge d$ in the negative-coefficient branch and subtracting $(3/8)(1 - d)$ instead gives $d(-759/512 + 6d) \le -723d/2560$. Thus one may use

$$D = 3d/8 \quad (c = 4, d \le 1/5). \tag{O.107}$$

For $c \ge 5, d \le 1/5$, the retained data imply $B_0 \le \theta + 13d/16 \le 91/160$. At $c = 5$ this is immediate. At $c \ge 6$, use $B_0 \le (17/16 + d)d + u_0(\theta)$ and $d/4 + d^2 \le 9/100 < \kappa\theta^2 = \theta - u_0(\theta)$. We also have

$$\tfrac{15}{7}(A_0^2 + xB_0^2) \le A_0 + B_0. \tag{O.108}$$

Here is a finite certificate on the whole rectangle $[0, 1/5]\times[0, \theta]$. For $\phi(A, B) = \frac{15}{7}(A^2+xB^2)-A-B$, $\partial_B\phi < 0$ on the relevant range, since $2(15/7)\theta(91/160) = 35490/35840 < 1$. Convexity in A reduces to $A = 0$, where $\phi \le 0$, or $A = C + u_0 - xE_*$; decrease B to $C - d + v_0$. Convexity first in C, then in E_*, reduces to the four polynomials

$$\begin{gathered} P_{\alpha,e}(d, x) = A + B - \tfrac{15}{7}(A^2 + xB^2), \\ A = \alpha d + ed^2 + u_0 - edx, \qquad B = (\alpha - 1)d + ed^2 + v_0, \\ \alpha \in \{5/4, 53/32\}, \qquad e \in \{0, 1\}. \end{gathered} \tag{O.109}$$

Their bidegrees are $(2, 6)$ for $e = 0$ and $(4, 6)$ for $e = 1$. For an exact small certificate, let $r_i = \min_{0\le j\le 6} b_{ij}$, where b_{ij} is given by (A.18) after $d = y_1/5, x = 13y_2/32$, $0 \le y_1, y_2 \le 1$. The row minima are

(α, e)	$(r_0, \dots, r_n)$
$(5/4, 0)$	$(0,\ \frac{3}{20},\ \frac{4051919789}{37580963840})$
$(5/4, 1)$	$(0,\ \frac{3}{40},\ \frac{2369}{16800},\ \frac{1049}{5600},\ \frac{33811712609}{187904819200})$
$(53/32, 0)$	$(0,\ \frac{37}{160},\ \frac{79653291}{5368709120})$
$(53/32, 1)$	$(0,\ \frac{37}{320},\ \frac{220841}{1075200},\ \frac{39423284321}{187904819200},\ \frac{17975757409}{187904819200})$.

(O.110)

All 112 coefficients are therefore nonnegative. Lemma A.1 proves (O.108). Now $k = a_0^{-1}$, $\sqrt{d\mathbb{E}U^2} \le 3d/10 + 5\mathbb{E}U^2/6$, and $k + 5/6 \le 15/7$ give $\delta \le 3kd/10 \le 12d/31 < 2d/5$. If $x \le d$, $a \le 85/124$; if $x \ge d$, $a \le 104777/126976 < 5/6$. Also $b \le 91/124 < 3/4$, proving the last row of (O.105).

All finite signed moments in that table follow from the one formula

$$D_{4r+1} = \int x^{2r} a, \qquad D_{4r+2} = \int x^{2r+1} b,$$
$$D_{4r+3} = -\int x^{2r+1} a, \qquad D_{4r+4} = -\int x^{2r+2} b \qquad (r \geq 0). \tag{O.111}$$

Thus $\sum (D_i)_+ \leq \int (a + xb)/(1 - x^2)$. The later positive classes are at most $d/12$ for $c = 3$, and $d/6$ for the other rows. This gives the stated γ_s; $\gamma_3 \geq d/8$, $\gamma_4 \geq 2d/3$ also hold in every row. Each sum here bounds every finite positive sub-sum in the connector.

Every connector of length at least two

For $L = 2, m > 3$, take the complete vector $V = B^2 G_{m-3} u - BR - D\mathbf{1}$. Its nonconstant part has coordinates $-x(E_{m-3} - b) + z(a - xC_{m-3})$, hence squared norm at most $\theta(1 + \theta)d$; its mean is $d - \beta_m - D_1 \geq -D_1$. At $m = 3$, take $V = BR + D\mathbf{1}$ instead. In both directions

$$N^2 \leq \theta(1 + \theta) + (D^2 + 2DD_1)/d, \quad M \leq -2\kappa + D/d + D_1/d - D_2/d \tag{O.112}$$

for $m > 3$; the $m = 3$ mean is the stronger $M = -2 + D/d + (D_1 + D_2)/d$. For $c = 3$ improve the first term to $4/15$: the nonconstant vector is $B^2(G_{m-3} - kB)u$, $k \leq 6/5$. The indices $m - 3 = 1, 2$ give respectively $\theta^2(1 + 36\theta/25)$ and $\theta^2(1 + \theta/25)$; all later indices give $\theta^2(17/15)^2 < 4/15$ by (O.104). At $m = 3$, use $kB^3 u$. The constant contributes at most $49d^2/160$, giving $M \leq -9/20$, $N^2 \leq 59/160$.

For $L \geq 3, m \geq 4$, the matched bare arm has smaller index at least four. Its scalar plus signed mean is at most $-219d/70$ and its vector norm at most $\sqrt{d}/3$, by (R40.3.3). Add $\sigma B^{L-1} R$ to that same vector. At $L = 3$ the remaining scalar is at most $2D + 2D_1 + D_2$; at $L \geq 4$ it is at most $2D + 2\sum (D_i)_+$. The G_2 exception remains in the bare bound. For $m = 3, L \geq 3$, retain the final actual subtraction $\gamma_{L+1} \bar{f}_0$. Adding the signed mean of the remaining finite vector gives the two indexed losses in

$$V = (G_L - G_2)u + B^{L-1} R = B^2 [G_{L-2} - aB^{L-3} + bB^{L-2}]u, \quad M \leq 2D/d - \gamma_{L+1}/d - \gamma_{L+2}/d. \tag{O.113}$$

Indeed its mean is $-\beta_{L+2} + d + D_L$; the extra index $L + 2$ comes from this mean, rather than an additional convolution term. Lemma O.12 yields $N^2 < 1/4$ at every one of these lengths, rather than separate short and long bounds. The resulting simultaneous payment table is

c	indices	M upper	N^2 upper
3	$L = 2$	$-9/20$	$59/160$
3	$L \geq 3, m \geq 4$	$-219/70 + 17/10 + 1/5$	$64/225$
3	$L \geq 3, m = 3$	$-25/48$	$1/4$
4	$L = 2$	$5/48$	$4/5$
4	$L = 3, m \geq 4$	$17/140$	$9/16$
4	$L \geq 4, m \geq 4$	$27/35$	$9/25$
4	$L \geq 3, m = 3$	$-1/48$	$1/4$
≥ 5	$L = 2$	$11/240$	$3/4$
≥ 5	$L = 3, m \geq 4$	$37/420$	$9/16$
≥ 5	$L \geq 4, m \geq 4$	$197/210$	$9/25$
≥ 5	$L \geq 3, m = 3$	$-13/240$	$1/4$

(O.114)

All rows are on their domains in (O.105). For example the bounds on $\|B^{L-1} R\|/\sqrt{d}$ are $1/5$ in the first family, and $5/12, 4/15$ at $L = 3, L \geq 4$ in the other families. At $c = 4, L \geq 4, m \geq 4, d \leq 1/5$,

(O.107) decreases M by $1/12$. Using $-2\log a_0/d \geq 1, 2, 9/4$ in the three families proves all low-domain eighth rates in (O.96). Every fourth rate exceeds $29/8$, except the $c = 4$ long matched row on $1/5 \leq d \leq 1/4$. Its actual fourth rate is

$$8 - 4(27/35) - (17/4)(9/25) = 2369/700; \tag{O.115}$$

on $d \leq 1/5$ the improved value is $7807/2100 > 37/10$. The complete minimum fourth rates for $c = 3$ are greater than four.

The unit connector

At $c = 3, L = 1$, use the exponential input $F_3 \leq qe^{B^2u-d}$, for which $\delta = -\delta_2 + \sqrt{d\delta_3} + (12/5)\delta_3 \leq 2d/5$. After division by d, the upper $-t + \sqrt{\theta t} + (12/5)\theta t$ increases on $[0, \theta]$ and is at most $(12/5)\theta^2 < 2/5$. The vector is $B(G_{m-2} - B)u$. At $m = 3$, its norm square is $\delta_2 + \delta_3$ and $M \leq -3/5 - \delta_2/d$. At $m = 4$, $M \leq -3/5$ and $N^2 \leq \theta$. For $m \geq 5$,

$$|z(G_{m-2} - z)|^2 \leq F(x) = x - 2x^2 + 3x^3 - x^4 + x^5.$$

Since $F' = (1 - 2x)^2 + x^2(5 - 4x + 5x^2) > 0$ and $F(\theta) = 8762429/33554432 < 17/64$, use $M \leq -\kappa + 2/5$, $N^2 \leq 17/64$. The fourth rate is at least $4667/1280 > 29/8$, and the eighth rate exceeds two, for all three ranges of m.

For $c = 4, L = 1$, retain the actual δ and the norm cross. With $k = (1 - d)^{-1}$, $t = \delta_2/d$, $s_3 = \delta_3/d$, $J = d^2 + 2dt + s_3 + \delta_4/d$, the full vector is $V = [a - xC_{m-2} + z(E_{m-2} - b)]u$ and $N^2 \leq k^2J + t - 2ks_3$. For $(s, c_s) = (4, 17/4), (8, 17)$ its complete moment upper is

$$\frac{2s\log(1-d)}{d} - s - 2skd + (2s + c_s - 3sk)t + sk\sqrt{J} + (s + c_s)k^2J - 2c_sks_3. \tag{O.116}$$

First use $\delta_4 \leq \theta\delta_3$, then $s_3 \leq \theta t$, then $t \leq \theta$. The respective coefficients after each step are nonnegative in both moments. For the last step, multiplication by $(1-d)^2$ gives $9-9\theta+25\theta^2+(8+34\theta)d+33d^2$ for $s = 8$, and $(1 - \theta + 33\theta^2)/4 + (4 + 17\theta/2)d + 49d^2/4$ for $s = 4$. The convex chord yields $\sqrt{J} \leq 1/2 + 9d/10$. Using $\log(1 - d) \leq -d - d^2/2 - d^3/3$ at eighth order and $\log(1 - d) \leq -d$ at fourth order gives upper polynomials

$$\begin{aligned} P_8(d) &= -529299/32768 + (92317/2560)d + (16259/480)d^2 + (8/3)d^3 - (16/3)d^4, \\ P_4(d) &= -(899/128)(1 - d)^2 + [-8761/2048 - (22/5)d](1 - d) \\ &\quad + (33/4)[d^2 + (13/16)d + 7605/32768]. \end{aligned} \tag{O.117}$$

These bound $(1 - d)^2 \log \mathbb{E}H^s/d$. The Bernstein coefficients of $-P_8 - (1 - d)^2/4$ on $[0, 1/3]$ are the five positive numbers

$$\tfrac{521107}{32768}, \quad \tfrac{6360013}{491520}, \quad \tfrac{124006319}{13271040}, \quad \tfrac{67578559}{13271040}, \quad \tfrac{8949613}{39813120}.$$

For $-P_4 - (39/10)(1 - d)^2$ they are $3595671/655360, 6583589/1966080, 897167/5898240$. Thus the eighth rate is $1/4$ and the same-direction fourth rate is $39/10$. The full vector satisfies $N^2 < 4/3$: its pointwise bound increases with d, and at $d = 1/3$ is $1/4 + (5/2)x - (3/4)x^2 + (9/4)x^3 \leq 169437/131072 < 4/3$. The derivative has positive numerator $2[d(1 + x + x^2) + x + x^3]$.

For $c \geq 5, L = 1, d \leq 1/5, V = [a-xC_{m-2}+z(E_{m-2}-b)]u$ satisfies $N^2 \leq 4/5$. Indeed $a \geq x/2$ and $b \leq 1-x/2$ give the pointwise bound $a^2+x(1-b)^2$. On $x \leq d$ this is at most $9/16+1/5 = 61/80$.

On $x \ge d$, increase A_0 to $C + u_0 - xE_*$ and decrease B_0 to $C - d + v_0$. The resulting squared norm increases with C; its derivative has the sign of $k[(1+x)C - xE_* - xd] + (k-1)x > 0$, since the bracket is at least $d(5/4 - 3x/4) > 0$. After $C = 53d/32 + dE_*$, it decreases with E_* on this region. It therefore suffices that

$$\tfrac{4}{5}(1 - 9d/8)^2 - (53d/32 + u_0)^2 - x(1 - 9d/8 - 21d/32 - v_0)^2 \ge 0. \tag{O.118}$$

This bidegree $(2,6)$ polynomial has positive Bernstein row minima

$$\frac{2529059971}{5368709120}, \qquad \frac{8020330127}{26843545600}, \qquad \frac{765259979}{134217728000}$$

on the full rectangle $[0, 1/5] \times [0, \theta]$. Equation (A.18) gives this finite certificate without imposing $x \ge d$ on the measure. Now $M \le -3/16 + 2/5 = 17/80$; the eighth and fourth rates are $27/10$ and $19/4$.

The same unit connector on the middle interval

For $c \ge 5$, the retained construction remains valid on $d \le 1/4$. It gives $A_0 \le 11/16$, $B_0 \le 13/20$, and hence

$$0 \le a \le 22/23, \qquad 0 \le b \le 104/115 < 10/11, \qquad \sigma := \mathbb{E}R^2/d < 81/64. \tag{O.119}$$

To check all finite lengths, on $x \le d$ use $A_0 \le 85d/32 \le 85/128$. On $x \ge d$, the upper bounds for A_0 at $c = 5, 6$, $c = 7, 8$, and $c \ge 9$ are, respectively,

$$\tfrac{85}{128}, \qquad \tfrac{1527}{4096} + u_0(\theta) = \tfrac{22317}{32768}, \qquad \tfrac{51061}{131072} + u_0(\theta) - \kappa\theta^4 < \tfrac{137}{200}.$$

Here $\beta_7 = \beta_8 = \tau + \delta_3$, $\beta_c \ge \tau + \kappa\delta_3$ for $c \ge 9$, $\delta_3/d \ge (\delta_2/d)^2$, and the finite coordinate for $c \ge 9$ satisfies $xa_f \le u_0(x) - \kappa x^4$. Also $B_0 \le 73/128$ for $c = 5$ and $B_0 \le 21/64 + u_0(\theta) < 13/20$ otherwise.

Retain δ rather than extending a low-domain error bound. The actual receiver coordinates give

$$N^2 \le \sigma + t - \tfrac{2}{d}\int xb\, d\nu + \tfrac{1}{25}, \qquad N^2 < 36/25. \tag{O.120}$$

The extra term bounds $x^2(2b - 2 + x)_+ \le 133679/3768320 < 1/25$. Since $b^2 - 2b \le 0$, the same estimate gives $N^2 \le (22/23)^2 + \theta + 1/25 < 36/25$. Substitution into (O.103) gives the one joint expression, for both values of s,

$$\begin{aligned}\frac{\log \mathbb{E}H^s}{d} \le{}& \frac{2s\log a_0}{d} - s + (2s + c_s)t + s\sqrt{\sigma} + \frac{c_s}{25} \\ &+ \frac{1}{d}\int [(s + c_s)a^2 - 2sa + (s + c_s)xb^2 - (s + 2c_s x)b]\, d\nu.\end{aligned} \tag{O.121}$$

At eighth order, $25a^2 - 16a \le 4004/529 < 38/5$, and the remaining integrand is at most $-(8 + 9x)b$. For $c \ge 6$, the retained coefficients give $B_0 \ge d/4 + 2\kappa\delta_2 + \kappa x$, so on $d, t \ge 1/5$

$$\frac{1}{d}\int (8 + 9x)b\, d\nu \ge k[2d + 16\kappa t d + 8\kappa t + 9\kappa t^2] \ge 311/124 > 5/2.$$

For $c = 5$ the lower bound is $536/155 > 5/2$. Thus the nonlogarithmic cost is at most $16149/800$, whereas $16\log a_0/d \le -18 - (81/8)d - (243/32)d^2 \le -16263/800$. This proves the eighth rate $57/400$. At fourth order both integrand terms in (O.121) are nonpositive, since $a < 32/33$ and $b \le 1$. The cost is less than $125/128 + 9/2 + 17/100 < 6$. Together with $8\log a_0/d \le -9 - (81/16)d$, this gives fourth rate $321/80 > 4$ throughout $1/5 \le d \le 1/4$.

The remaining host interval

For $c \ge 5, d \ge 1/4, L = 1$, positivity and the actual mean give $AF_c \le 2(2-5d)(1-2d) \le 3/4$. For $m = 3$, $\bar{f}_3 \le 3/2$ and $\mathbb{E}\bar{f}_3^2 = 1-3d+3d^2-\delta_2+\delta_3 \le 7/16$. For $m \ge 4$, $\bar{f}_m \le 1$ and $\mathbb{E}\bar{f}_m \le 1/2$. Consequently

$$\mathbb{E}H^8 \le 3^{15}/2^{24} < 7/8 \le e^{-3d/8}, \qquad \mathbb{E}H^4 \le 5103/16384 < 5/16. \tag{O.122}$$

For $L \ge 2$, the remaining intervals are $d \ge 1/4$ at $c = 4$ and $d \ge 1/5$ at $c \ge 5$. The same terminal pair obeys

$$f_2 h_c \le e^{3d/5} \quad (c = 4,\ d \ge 1/4;\ \text{or } c \ge 5,\ d \ge 1/5). \tag{O.123}$$

For $c = 4$, center $Bu - B^2u$: its mean is $-d$, and the perturbation in the arm sum is at most $\sqrt{3d^2+\delta_2+\delta_3} \le 13d/5$. For $c \ge 5$, the unretained positive convolution gives

$$f_2 + h_c \le 2 - \Gamma_0 + R_0, \quad \Gamma_0 = \beta_c + \beta_{c-1}, \quad R_0 = -\beta_{c-1}u + B^2 G_{c-2}(-B)u. \tag{O.124}$$

Put $V_* = BG_{c-2}(-B)u$. Then $\mathbb{E}V_* = -\beta_c$, $\operatorname{Var} V_* \le \delta_2 - \tau^2$, and $R_0 = -\Gamma_0 u + B(V_* + \beta_c)$. The centered row bound is at most $\Gamma_0 + 3d/5$, since

$$t \le d[(1-t)^2 + (12/5)(1-t) + 9/25], \qquad t = \delta_2/d \le \theta, \quad d \ge 1/5.$$

The right minus left decreases with t, increases with d, and at $(d,t) = (1/5, \theta)$ is $2721/128000 > 0$. The matched bare moments (R40.3.2) now give eighth rate $9/20$ and fourth rate $103/80$. This closes (O.96).

All exponential full vectors used here have norm at most $3\sqrt{d}/2$; the largest special bounds are $N^2 < 4/3$ in (O.116) and $N^2 < 36/25$ in (O.120). Thus the same actual negative posts are restored by the full-vector post calculation, with the increments (R40.1.7). Equal pairs are restored by positivity. If $d = 0$, then $u = 0$, all four path types and every displayed tree message are one; all bounds hold with equality.

Seven terminal profiles.

The adopted exact-triple proof handles all qf_2h_c. The remaining terminals have the seven profiles below. In each row $F \le qe^{R-\Gamma}$, $|R| < 1/2$, $\Gamma \ge gd$, $\|R\| \le r\sqrt{d}$, and $D = \alpha d$ is a valid first-row error. The integer j contributes the displayed extra loss without increasing a norm.

profile	d range	g	r	α
$H:\ r_1 \ge 4$	$[0, 1/3]$	$16j/5$	$1/4$	$11/15$
$3_1:\ r_1 = 3, j = 1$	$[0, 1/8]$	$35/16$	$1/3$	$1/2$
$3_+:\ r_1 = 3, j \ge 2$	$[0, 1/8]$	$431/80 + 16(j-2)/5$	$9/20$	$10/9$
$A:\ (1,2,c,b,e), c \ge 4$	$[0, 1/64]$	$351/80$	$5/8$	$11/12$
$B:\ (1,2,3,b,e), b \ge 5$	$[0, 1/8]$	$21/5$	$1/2$	$7/10$
$C:\ (1,2,3,4,e)$, crossed	$[0, 1/8]$	$21/4$	$1/2$	$11/10$
$N:\ r_1 = 2, j \ge 3$	$[0, 1/64]$	$37/5 + 16(j-3)/5$	$2/3$	$19/10$

The actual terminal factors remain those in the direction table; in particular profile N is canonical at every prefix.

Here are complete finite proofs of these parameters. For a continuous pair put

$$R_{ab} = \sum_{n=a+1}^{b} K^n 1, \qquad h_a f_b \le e^{R_{ab} - 16d/5} \quad (4 \le a < b),$$

using (L90.2.3). A finite union of disjoint intervals with first degree a on g has squared norm at most $C_*\theta^a d$ and original row square at most $C_*\theta^{a-1}d$, $C_* < 100/49$. Four residue classes give the following bounds for a later mask S starting at degree at least four:

$$\mathbb{E}S - \mathbb{E}BS \le \frac{2\delta_4}{1-\theta^2} < d/6, \qquad \mathbb{E}BS \ge -\frac{\delta_4}{1-\theta^2} > -d/12. \tag{O.125}$$

For first degree at least five the positive terms in $\mathbb{E}BS$ start only at δ_5. These follow by summing the possible positive terms $\delta_4, \delta_6, ...$; they bound every finite mask. For H, the first degree is four:

$$\|R\| \le \sqrt{d}/4, \quad |R| \le 3\sqrt{d}/8, \quad D_0 \le d/12, \quad (D_1)_+ \le d/5,$$
$$\delta/d \le 1/12 + 1/4 + 1/4 + (12/5)(1/16) = 11/15.$$

For 3_1, put $c = t_1 \ge 4$, $\eta = \beta_{c-1}$, and

$$R_0 = -\eta u - B^3 G_{c-3}(-B)u, \qquad \Gamma_0 = d + \beta_c + \eta \ge 35d/16.$$

In the reversed positive convolution for f_{c-1}, retain its last coefficient η before applying T. For f_c, retain the last two coefficients β_c, η. Together with the exact f_3, h_3, these give the two actual sums

$$f_3 + h_c \le 2 - \Gamma_0 + R_0, \qquad h_3 + f_c \le 2 - \Gamma_0 - R_0.$$

Their nonnegative square bases imply $f_3 h_c \le e^{R_0 - \Gamma_0}$ and $h_3 f_c \le e^{-R_0-\Gamma_0}$. Write $G_{c-3}(-z) = a - zb$, where $1 - x \le a \le 1$, $0 \le b \le 1$, including G_1. If $t = \delta_2/d$ and $J = \mathbb{E}R_0^2/d$, these same coefficients give

$$J \le d^2 + \theta^2(1+\theta)t, \qquad (D_0 - D_1)/d \le -\kappa d - (3/16)t.$$

Indeed $R_0 = (-\eta + x^2 b + zxa)u$, and its negative cross can be discarded because $\eta, b \ge 0$. Also $a - xb \ge 1 - 2x \ge 3/16$. The G_2 factor $1+\theta$ is kept in J. Using $\sqrt{J} \le 1/8 + 2J$, the q-row error is at most

$$Q(d) = 1/8 - \kappa d + (22/5)d^2 + \theta[-3/16 + (22/5)\theta^2(1+\theta)] < 1/2.$$

The coefficient of t is positive, so $t \le \theta$ is valid; Q is convex on $[0, 1/8]$ and both endpoint values are below $1/2$. Moreover $J \le 1/64 + \theta^3(1+\theta) < 1/9$, proving the common $r = 1/3$.

For 3_+, add the one finite mask $S = \sum_{i\ge 2} R_{r_i t_i}$, whose first degree is at least $c+1$. At $c = 4, 5, \ge 6$, respectively, the norm bounds for R_0 are $29/100, 7/20, 29/100$, in units of $\sqrt{d}$. Their squared bounds are $d^2 + \theta^3$, $d^2 + \theta^3(1+\theta)$, and $d^2 + \theta^3$. Here only $c = 5$ has a G_2 norm. The later mask bounds are $4/25, 1/10, 1/10$, so their sums are at most $9/20$. The mean of R_0 in the q-row is nonpositive; hence

$$\delta/d \le 1/6 + 9/20 + (12/5)(9/20)^2 < 10/9.$$

The original row bound is $|R_0 + S| \le d + (3/4)\sqrt{d} < 1/2$. Indeed, after taking one B-row outside, the first term has norm at most $\sqrt{d}/2$, and the later mask has norm at most $\sqrt{d}/4$. Finally $(D_1)_+ \le d^2 + \delta_5/(1-\theta^2) < d/5$.

For A, use the actual canonical factors

$$F = q f_2 h_c h_b f_e, \quad R = -\eta u + B^2 G_{c-2}(-B)u + S, \quad S = R_{be}, \quad \Gamma = \beta_c + \eta + 16d/5 \ge 351d/80.$$

The initial f_2+h_c upper is (O.124); at $c=4$ it is the exact identity in (O.98). Writing $R=(-a+zb)u$, finite coordinates give

$$a=\eta+xU,\quad b=xV,\qquad \ell(x):=\frac{x(1-2x)}{1-x}\le xU,xV\le h(x):=\frac{x}{1-x},\qquad 0\le\eta\le 1/64.$$

The constant and linear coefficients of the first finite interval are one, while the negative later interval is disjoint; summing the absolute coordinate tails gives precisely these bounds. They imply

$$a^2+xb^2\le(3/5)(a+b). \tag{O.126}$$

For a complete small certificate, the derivative of $a^2+xb^2-(3/5)(a+b)$ in b is negative on this range. Set $b=\ell$, then use convexity in a and η. The four resulting polynomials are

$$P_{\eta,e}(x)=(1-x)^2\{(3/5)(\eta+e+\ell)-(\eta+e)^2-x\ell^2\},\quad \eta\in\{0,1/64\},\quad e\in\{\ell,h\}.$$

Their full Bernstein coefficients on $[0,\theta]$, in that order, are supplied in Table 69. Thus Lemma A.1 proves (O.126). For the norm, $c=4$ retains G_2, giving $\|R\|/\sqrt{d}\le d+(3/2)\theta\le 5/8$. Here the two continuous intervals contribute at most $\theta\sqrt{1+\theta}(1+\theta^{3/2})<(3/2)\theta$. For $c\ge 5$, finite contraction and the later continuous tail give the bound $d+\theta+1/10<5/8$. Also $|R|\le d+\sqrt{d}<1/2$. For this last row estimate, at $c=4$ the two terms after removing one B have norms less than $4\sqrt{d}/5$ and $\sqrt{d}/5$. At $c\ge 5$ their bounds are $2\sqrt{d}/3$ and $\sqrt{d}/5$.

For B, $R=B^2u+S$, $S=R_{be}$, and $\Gamma=21d/5$. At $b=5$, the actual coefficients are

$$a=x-x^3V,\quad b_0=x^2U,\qquad 0\le U,V\le 1.$$

At every $b\ge 6$, they satisfy

$$x-x^3\le a\le x+x^3,\qquad |b_0|\le x^3.$$

Here $R=(-a+zb_0)u$. These finite bounds prove (O.126) also for this profile. At $b=5$ decrease b_0 to zero and use $a\le x<3/5$. For $b\ge 6$, decrease b_0 to $-x^3$ and use convexity in a. After division by x, the two nonnegative gaps are

$$3/5-x-2x^3-x^5-x^6,\qquad 3/5-x-(6/5)x^2+2x^3-x^5-x^6.$$

Their complete coefficients are in the same table. The raw norm is below $1/4$: use x^2+x^5 for $b=5$, and $(x+x^3)^2+x^7$ thereafter. The original row bound gives $|R|\le(13/15)\sqrt{d}<1/2$. To see the latter, remove one B; the leading term has norm at most $2\sqrt{d}/3$, and every continuous later interval has norm at most $\theta^2\sqrt{1+\theta}\sqrt{d}<\sqrt{d}/5$.

For both A,B, write $b_0=b$ in profile A, and put $w=\mathbb{E}R^2/d$ and $v=d^{-1}\int(a+b_0)\,d\nu$. The coupled inequality gives $w\le(3/5)v$; hence

$$\delta/d=-v+\sqrt{w}+(12/5)w\le\sqrt{w}+(11/15)w.$$

At $r=5/8$ this is $175/192<11/12$; at $r=1/2$ it is $41/60<7/10$. This is the compatible signed q-mean calculation underlying both rows.

Table 69: Complete Bernstein coefficients on $0 \le x \le 13/32$. The first row certifies the unit-connector norm; the four A rows are $P_{\eta,e}$, and the last rows are the two B gaps.

Polynomial	Coefficients in increasing index
$2-5x+x^2-x^3$	$\left(2, \frac{127}{96}, \frac{2153}{3072}, \frac{2187}{32768}\right)$
$A:0,\ell$	$\left(0, \frac{39}{400}, \frac{6097}{51200}, \frac{165399}{1638400}, \frac{32487}{409600}, \frac{1934127}{41943040}\right)$
$A:0,h$	$\left(0, \frac{39}{400}, \frac{7111}{51200}, \frac{205621}{1638400}, \frac{527397}{6553600}, \frac{187343}{41943040}\right)$
$A:1/64,\ell$	$\left(\frac{187}{20480}, \frac{168113}{1638400}, \frac{25556979}{209715200}, \frac{21535377}{209715200}, \frac{1695861}{20971520}, \frac{2009861}{41943040}\right)$
$A:1/64,h$	$\left(\frac{187}{20480}, \frac{168113}{1638400}, \frac{29494003}{209715200}, \frac{26122713}{209715200}, \frac{1625557}{20971520}, \frac{6197}{41943040}\right)$
$B:+$	$\left(\frac{3}{5}, \frac{511}{960}, \frac{223}{480}, \frac{127851}{327680}, \frac{14861}{49152}, \frac{64614757}{335544320}, \frac{236733507}{5368709120}\right)$
$B:-$	$\left(\frac{3}{5}, \frac{511}{960}, \frac{17333}{38400}, \frac{596329}{1638400}, \frac{340091}{1228800}, \frac{65306981}{335544320}, \frac{613303363}{5368709120}\right)$

All denominators suppressed in $P_{\eta,e}$ are positive: $1-x \ge 19/32$. Thus these are closed-interval certificates, including $x=0$, by (A.18).

For C, retain the already proved coefficients (O.100) of the actual f_2+h_e sum:

$$R = -U_e + B^3u + du = (-\bar{a} + z\bar{b})u, \qquad \bar{a} = A_0 - d, \quad \bar{b} = B_0 - x, \quad \Gamma = 21d/4.$$

Here $h_3 + f_4 = 2 - 3d + du + B^3u$ is exact. Combining its nonnegative square with $f_2 + h_e \le 2 - 9d/4 - U_e$ explains both the crossed factors and the loss $21d/4$. Those coefficients give

$$0 \le \bar{a} \le 19/40, \quad -x^2 \le \bar{b} \le 19d/16, \quad |\bar{b}| < 1/6, \quad \bar{a} + \bar{b} \ge d/2 + x/16.$$

For $e \ge 6$, the lower sum follows from $C + D_* - d \ge d/2$ and $a_f + b_f - E_* - 1 \ge 1 - 2x - d \ge 1/16$; at $e = 5$, the exact pair is $\bar{a} = d/4 + \delta_2 + x - x^2$, $\bar{b} = d/4 + \delta_2$. For the real upper split at $x = d$: $\bar{a} \le 53d/32 + d^2$ when $x \le d$, and $\bar{a} \le 21d/32 + x - x^2 + x^3$ when $x \ge d$. Both are below $19/40$. For $\bar{b}$, use $D_* \ge 0$ and $D_* \le 17d/16 + d^2$; the exact $e = 5$ value satisfies the same inequalities. Thus $\mathbb{E}R^2 < d/4$ and $D_0 - D_1 \le 0$, proving $\delta \le 11d/10$. The actual row form

$$R = -(C-d)u + B[z(a_f - E_*) + D_* + z^2(1-b_f)]u$$

has bracket modulus below one, with $C-d \le 25/256$. Indeed $D_* \le 19/128 < \theta^2$, $|a_f - E_*| \le 1$, and $0 \le 1 - b_f \le x$; the bracket square is at most $\theta^4 + \theta < 1$. Consequently $|R| \le 25/256 + \sqrt{d} < 1/2$.

For N, take the canonical product at every prefix and set

$$R = -\eta u + B^2 G_{c-2}(-B)u + S, \qquad S = \sum_{i \ge 2} R_{r_i t_i}, \qquad \Gamma = \beta_c + \eta + (16/5)(j-1)d.$$

At $c = 3, 4, \ge 5$, the norm bounds in units of $\sqrt{d}$ are

$$\theta + (10/7)\theta^2, \quad 1/64 + (6/5)\theta + 4/25, \quad 1/64 + \theta + 1/10,$$

all below $2/3$. The middle row keeps G_2. Also $|R| \le d + (6/5)\sqrt{d} < 1/2$. After removing one B-row, the three cases have norm bounds $2/3 + 3/8$, $4/5 + 1/4$, and $2/3 + 4/25$, respectively, in units of $\sqrt{d}$; each is below $6/5$. The first interval has nonpositive q-mean, so $\delta/d \le 1/6 + 2/3 + (12/5)(2/3)^2 = 19/10$. Its D_1 is nonnegative; the later mask gives $D_1 > -d/12$. The scalar loss is at least $1 + (16/5)(j-1)$, including $c = 3$.

The unit connector and one finite payment table.

For $L = 1$, the following are the actual bounds in (O.95); the b_1 column is also used at $L = 2$:

profile	σ	b_1	M_1	N_1^2
H	$+$	$1/5$	$1/10$	$441/400$
3_1	$+$	$1/8$	$-5/24$	$16/25$
3_+	$+$	$1/5$	$43/90$	$25/16$
A	$-$	$5/8$	$35/48$	$9/10$
B	$-$	$1/2$	$41/80$	$9/10$
C	$-$	$1/2$	$73/80$	$4/5$
N	$-$	$2/3$	$431/240$	$484/225$

For the plus sign use $M_1 \le -5/6 + \alpha + (D_1)_+/d$; for the minus sign use $M_1 \le -3/16 + \alpha - D_1/d$. The $H, 3_+, N$ norms are the actual triangle bounds $4/5 + r$. For 3_1, the two contiguous finite coordinates give

$$\|Z_m + R_0\|^2 \le \delta_2 + \delta_3 + \eta^2 d < (16/25)d.$$

Indeed the coordinate square is at most $(x + x^2)^2 + x(1 - x + x^2)^2 + \eta^2 \le x(1 + x) + \eta^2$; the difference after division by x^2 is $-2 + 5x - x^2 + x^3 < 0$. For A, $0 \le a, b \le 7/10$, and its opposite-edge vector has real magnitude at most $7/10$ and imaginary coefficient magnitude at most one. For B, $a \le 1/2, |b| \le 1/5$ give the corresponding norm bound $1/4 + (36/25)\theta < 9/10$. For C, use $(19/40)^2 + (49/36)\theta < 4/5$. All three have $D_1 = \int a \, d\nu \ge 0$. The other b_1 values follow above, or simply from $|D_1| \le rd$.

Insert these two parameter tables into (O.94) and (O.95). All four modes give the following low-domain losses at the smallest pair count:

profile	eighth loss at least	fourth loss at least
H	6	5
3_1	3	$9/2$
3_+	10	7
A	4	5
B	7	5
C	8	5
N	8	8

Every entry is a direct rational substitution of the printed M, N^2.

Each added terminal pair contributes $128/5$ at order eight and $64/5$ at order four. Every vector in these comparisons has norm at most $3\sqrt{d}/2$.

Completion of the same directions.

Profile H already holds on the whole domain. For $3_1, 3_+$, the centered-pair inequality (O.58) gives $f_3h_c, h_3f_c \le e^{-d}$ when $d \ge 1/8$. The same later mask has size at most $\sqrt{d}/4 \le 3d/4$. At $L = 1$, the actual bare product has eighth cost at most $316d/75$ and fourth loss at least $46d/75$, by (R40.3.5). At $L \ge 2$ use (R40.3.2). Thus the global 3_1 losses are at least 3 and $9/2$; the 3_+ losses are at least the stated $10, 7$, with their terminal-pair increments. In particular, in every original m, L, d direction,

$$\mathbb{E}\big[a_m Q_L^{\sigma}(qf_3h_c)\big]^8 \le e^{-3d}, \qquad \mathbb{E}\big[a_m Q_L^{\sigma}(qf_3h_c)\big]^4 \le e^{-9d/2} \quad (c \ge 4). \tag{O.127}$$

For A, at $d \ge 1/64$ regroup its unchanged product as

$$F = (qf_2h_b)(h_cf_e).$$

The actual continuous pair satisfies

$$h_af_b \le e^{R_{ab}-16d/5}, \qquad |R_{ab}| \le (39/125)\sqrt{d} \quad (4 \le a < b). \tag{O.128}$$

Its original row bound follows, including G_2, from $\theta^3(1+\theta) < (39/125)^2$. As $\sqrt{d} \le 8d$, its scalar loss $16d/5$ gives $h_cf_e \le e^{-d/2}$. The adopted exact $12b$ theorem applies with the same receiver and connector directions. Hence the global A losses are at least $4, 3$. For B, C, at $d \ge 1/8$ use respectively

$$F = (qf_2h_b)(h_3f_e), \qquad F = (qf_2h_e)(h_3f_4).$$

The second factor is at most e^{-d}, so the global losses are at least $7, 5$ for B and $8, 5$ for C. For N, at $d \ge 1/64$, $|S| \le 3\sqrt{d}/8 \le 3d$. The same actual later product is at most $\exp[-((16/5)(j-1)-3)d]$. Combine with the unchanged exact triple, including its $c = 3, L = 2$ direction, to obtain the stated $8, 8$ and all terminal-pair increments. All analytic boundaries retain one integer direction and are included in the adjacent proofs.

R40.12 Every finite negative root post and every overlap

For the actual negative post block (R40.1.3), full reversal of (R40.2.11) gives

$$P \le e^{S_p-\Gamma_p}, \quad \Gamma_p \ge (16/5)kd, \quad S_p = \sum_{\nu=1}^{k}\sum_{a=u_\nu}^{v_\nu-1} B^a u = BZ_p. \tag{R40.12.1}$$

It is one finite zero-one mask. Its first power is at least four. The actual positive mean begins with δ_4, and (R40.2.10) gives

$$ES_p < d/12, \qquad \|Z_p\| < \tfrac{3}{8}\sqrt{d}. \tag{R40.12.2}$$

These bounds are uniform in the number of post pairs and allow overlap with the terminal and bare receiver masks.

In the seven-profile payment and in every low- or middle-domain exact-triple comparison, the unposted full vector satisfies

$$\|V\| \le \tfrac{3}{2}\sqrt{d}. \tag{R40.12.3}$$

The largest bound in (O.95) and its unit-connector table is $22/15$. The exact 124 vector is bounded by $\sqrt{4/3}$, and the middle $12c$ vector by $6/5$. Thus (R40.12.3) includes the joint first-row arguments, not only the tabular estimates.

Multiply the actual moment upper by P^s before taking its expectation, and apply (R40.2.4) to the single vector $V + Z_p$. At eighth order the added cost is at most

$$\begin{aligned} &8ES_p + 34\|V\|\|Z_p\| + 17\|Z_p\|^2 - 8\Gamma_p \\ &\le \left(\frac{4259}{192} - \frac{128}{5}k\right) d \le -\frac{3281}{960}d \quad (k \ge 1). \end{aligned} \tag{R40.12.4}$$

The constant is exactly $2/3 + 153/8 + 153/64 = 4259/192$. At fourth order the same actual vector gives

$$4ES_p + \frac{17}{2}\|V\|\|Z_p\| + \frac{17}{4}\|Z_p\|^2 - 4\Gamma_p \le \left(\frac{4387}{768} - \frac{64}{5}k\right) d. \tag{R40.12.5}$$

The constant is $1/3 + 153/32 + 153/256 = 4387/768$. Every cross with the terminal, bare receiver, and genuine constant coordinates has therefore been paid on the original kernel.

Every high-domain comparison above starts at $d \ge 1/64$. On that whole domain,

$$|S_p| \le (3/8)\sqrt{d} \le 3d, \quad P \le e^{-((16/5)k-3)d}. \tag{R40.12.6}$$

It can be multiplied directly into each same-direction pointwise comparison. Since $4259/192 < 24$ and $4387/768 < 12$, (R40.12.4)–(R40.12.6) prove the common post improvements (R40.1.7). There is no Hölder splitting into unrelated mask expectations.

Finally positivity gives

$$0 \le H = E_r P\, a_m Q_L^{\sigma}(E_t F) \le P\, a_m Q_L^{\sigma} F, \tag{R40.12.7}$$

so all genuine equal pairs are restored at their original vertices.

O.37 Short even lists: the fourth moment and all outer lengths

R41.1 Exact theorem and actual integer direction table

All expectations and operators are on the unchanged original finite probability space. Let K be real skew, pointwise bounded by one, and put

$$B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B = T^*, \quad p = 1 - u, \quad q = 1 + u, \tag{R41.L60}$$

$$\begin{gathered} f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}, \\ \delta_i = \|B^i\mathbf{1}\|^2, \quad d = \delta_1, \quad \theta = 13/32. \end{gathered} \tag{R41.L64}$$

The terminal has an actual strict even residual

$$\begin{gathered} a_1 < b_1 < \cdots < a_j < b_j, \qquad j \ge 2, \quad a_1 \in \{2, 3\}, \\ F = \prod_{i=1}^{j} h_{a_i} f_{b_i}. \end{gathered} \tag{R41.L69}$$

There is an existing original anchor $m \geq 3$ at the receiving vertex. Optional original negative posts and genuine equal pairs are

$$P = \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}, \qquad 4 \leq u_1 < v_1 < \cdots < u_k < v_k, \quad k \geq 0; \qquad E_r, E_t. \tag{R41.L75}$$

The blocks may overlap and be dependent.

Choose the actual anchor X and actual connector S_L as follows.

$a_1 = 2$**,** $L = 1$
Use $XS_L = \bar{f}_m A$.

$a_1 = 2$**,** $L \geq 2$
Use $XS_L = W_m T^L$.

$a_1 = 3$**,** $m \leq L$
Use $XS_L = W_m T^L$.

$a_1 = 3$**,** $m > L$
Use $XS_L = \bar{f}_m A T^{L-1}$.

Set

$$H_D = E_r P X S_L (E_t F). \tag{R41.1.1}$$

Every direction is fixed solely from original integers. In an AT^{L-1} row only the receiver-adjacent connector edge is A; all remaining connector edges are T. The b, W, h and f factors are the actual paths defined above, with every original subdivision edge retained.

Theorem A. For every stated table and every finite host,

$$E H_D^4 \leq e^{-cd}, \qquad c = 6. \tag{R41.1.2}$$

For every independent original $M \geq 1$ append exactly M actual T edges, $U_D = T^M H_D$, and write $I_q(U) = E[q^{-\frac{1}{3}} |U|^{8/3}]$.

Theorem B. The SAME H_D satisfies

$$I_q(U_D)^3 \leq \begin{cases} e^{-(2c - \frac{204}{19})d}, & M = 1, \\ e^{-(2c - \frac{24427}{3420})d - 5d^2}, & M = 2, \\ e^{-(2c + \frac{23}{36})d}, & M = 3, \\ e^{-[2c - 11 + \lfloor M/3 \rfloor/2]d}, & M \geq 4. \end{cases} \tag{R41.1.3}$$

In particular, every row is at most $e^{-24d/19}$. The same input direction applies for every independent positive M, with all equal-pair factors restored. The fully reversed statement follows by reversing the complete configuration.

R41.2 Exact finite inputs, without a moment substitution

The original C014 calculus gives

$$\begin{gathered} 0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \\ \delta_{i+1} \le \theta \delta_i, \quad \|B\|^2 \le \theta, \\ |BV(x)| \le \|V\|. \end{gathered} \tag{R41.2.1}$$

Only $f_n, \bar{f}_n$ have means μ_n in [0,1]. In particular $Eh_2 = EW_2 = 1 + d$. All four actual arm messages are nonnegative and at most two. Genuine original pairs satisfy $f_n h_n \le 1$ and $\bar{f}_n W_n \le 1$.

Let $G_n(z) = \sum_{i=0}^{n-1} z^i, G_0 = 0, \beta_1 = \beta_2 = 0$ and

$$\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1} \delta_i. \tag{R41.L147}$$

The actual finite positive convolution and geometric bounds are

$$\begin{gathered} 1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}, \\ (1-\theta)d \le \beta_n \le d \quad (n \ge 3), \end{gathered} \tag{R41.2.2}$$

$$\begin{gathered} \|G_n(B)\| \le 1 \quad (n = 0, 1 \text{ or } n \ge 3), \\ \|G_2(B)\|^2 \le 1 + \theta = 45/32. \end{gathered} \tag{R41.2.3}$$

The G_2 exception is not discarded. The true operator powers satisfy

$$\begin{gathered} \|T\| \le 1 + 32d/19, \quad \|T^2\| < \frac{7}{5}, \\ \|T^n\| \le (1 - d/8)^{\lfloor n/3 \rfloor} \quad (n \ge 3). \end{gathered} \tag{R41.2.4}$$

The last bound uses actual blocks of lengths 3,4,5, not monotonicity of operator norms with length.

The supplied actual R2-C038/A019 terminal packets are independent of the receiving anchor. They give

$$\begin{gathered} F \le e^{R - \lambda d}, \\ \lambda_2 = \frac{607}{160}, \quad \lambda_3 = \frac{767}{160}, \\ R = \sum_i \sum_{r=a_i+1}^{b_i} K^r \mathbf{1}. \end{gathered} \tag{R41.2.5}$$

Each mask is one finite zero-one coefficient mask. The disjoint intervals are only those within the terminal, not the union with root posts. Every extra terminal pair increases the available loss. The stated minimum losses already include $j = 2$.

For minimum 2 put $\sigma = ER^2, R = U(B)u, U = \alpha + z\beta$ at $z = i\sqrt{x}$. The actual original spectral measure ν at u has mass d, support $[0, \theta]$, and $\int x^r = \delta_{r+1}$. The finite packet is

$$\sigma < d/3, \quad |R| < \frac{3}{5}, \quad e^R \le 1 + R + 5R^2/8, \tag{R41.L186}$$

$$x/2 \le \alpha \le 6x/5, \qquad -6x/5 \le \beta \le x/2. \tag{R41.2.6}$$

For $D_i = EB^iR, a_0 = -D_1$ and $D = D_0 + 5\sigma/4$, put $\epsilon = D + D_1$. Then

$$0 \le a_0 \le d/2, \quad -d/5 \le D \le 11d/12, \quad |\epsilon| \le 7d/10, \tag{R41.L195}$$

$$\begin{gathered} -d/5 \le D_2 \le d/12, \quad 0 \le D_3 \le d/5, \\ |D_i| \le d/12 \ (i \ge 4), \quad D_0 \le d/2. \end{gathered} \tag{R41.2.7}$$

These are actual joint quantities, not independently chosen parameters. The positive upper and complete finite losses are

$$\begin{gathered} e^{\lambda_2 d} T^L F \le Y_L \\ = \bar{f}_L + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \end{gathered} \tag{R41.2.8}$$

$$\begin{gathered} Y_1 = q + D + BR, \\ Y_L = 1 + \epsilon + Q_L(B)u - \sum_{r=3}^{L} \gamma_r \bar{f}_{L-r} \quad (L \ge 2), \end{gathered} \tag{R41.L211}$$

$$\begin{gathered} Q_n = G_n + z^n U + \epsilon G_{n-2} + Dz^{n-2} \quad (n \ge 2), \\ Q_{n+1} = 1 + \epsilon + zQ_n, \quad EQ_n(B)u = -\gamma_{n+1}, \end{gathered} \tag{R41.2.9}$$

$$\gamma_n = \beta_n + D\beta_{n-1} - a_0\beta_{n-2} - D_{n-1} \quad (n \ge 3). \tag{R41.2.10}$$

Thus $\gamma_3 = d - D_2, \gamma_4 = d + Dd - D_3$. The actual uniform bounds are

$$\begin{gathered} 11d/12 \le \gamma_3 \le 6d/5, \\ \gamma_4 \ge 11d/15, \\ d/4 \le \gamma_n \le 25d/18 \quad (n \ge 4). \end{gathered} \tag{R41.2.11}$$

The last upper includes γ_4, since $\gamma_4 \le d + 11d^2/12$. For $n \ge 5$ it follows from $\beta \le d, D \le 11d/12$ and $|D_{n-1}| \le d/12$. Every lower bound retains $\beta_1 = \beta_2 = 0$ at its actual short endpoint.

For minimum 3 the supplied packet gives

$$\sigma < 9d/64, \ |R| < \frac{1}{3}, \quad e^R \le 1 + R + 3R^2/5, \tag{R41.L236}$$

$$D_0 \in [-d/5, d/2], \quad D_1 \in [-d/12, d/5], \quad -D_2 \in [-d/12, d/5]. \tag{R41.2.12}$$

For $S_L = T^L$ or AT^{L-1}, the actual positive upper is

$$0 \le S_L F \le e^{-\lambda_3 d}$$
$$(S_L 1 + S_L R + \omega_L), \tag{R41.2.13}$$
$$\omega_1 < 27d/160, \quad \omega_{L\ge 2} < 27d/80.$$

Here $EAR \le 7d/12, EATR \le 9d/10$, and the TRUE propagated square gives

$$TR = D_0\, 1 + BR,$$
$$\|TR\|^2 = D_0^2 + 2D_0 D_1 + \|BR\|^2 < (\frac{11}{24})^2 d, \tag{R41.2.14}$$
$$\|T^{L-1}R\| < (\frac{21}{40})\sqrt{d} \quad (L \ge 3).$$

The genuine constant cross in TR is part of this bound.

All negative root posts have one actual finite envelope

$$P \le e^{S-\Gamma_r},$$
$$\Gamma_r \ge 16kd/5, \quad S = BV_p, \tag{R41.2.15}$$
$$ES < d/12, \quad \|V_p\| < 3\sqrt{d}/8 \quad (k \ge 1).$$

It follows from the actual finite negative pair mask $B^4 Q_p(B)u$. The squared norm of V_p is at most $(\frac{2197}{16245})d$, and its mean is bounded by $(\frac{1024}{855})\delta_4 < d/12$. Overlap with the terminal mask is allowed and is paid below by adding the vectors.

R41.3 A separate fourth-exponent payment

The supplied all-real signed lemma gives, for EVERY original real V,

$$Ee^{4BV} \le \exp\{4EBV + (\frac{17}{4})\|V\|^2\}. \tag{R41.3.1}$$

It includes actual constant coordinates and either sign of EBV. Suppose X is the actual nonnegative anchor and $S_L F \le e^{-\lambda d} Y$ with $Y \ge 0$, and the actual sum has upper

$$X + Y \le 2 + c_0 + BV. \tag{R41.L282}$$

AM–GM and $1 + t \le e^t$ applied to the nonnegative average imply

$$\log E(XS_L F)^4 \le -4\lambda d + 4(c_0 + EBV) + (\frac{17}{4})\|V\|^2. \tag{R41.3.2}$$

For posts, the SAME calculation gives

$$\log E(PXS_L F)^4 \le$$
$$-4\lambda d - 4\Gamma_r + 4(c_0 + EBV) + 4ES \tag{R41.3.3}$$
$$+(\frac{17}{4})\|V + V_p\|^2.$$

Genuine equal pairs can then be restored by positivity. These formulas, not an eighth-to-fourth upgrade, prove all new fourth bounds.

R41.4 All $m \geq 3$: complete fourth payment for the R2-C038 directions

The positive sums in R2-C038/A019 give the following fourth-moment estimates. The coefficients track the change of exponent. All entries are in units of d. Minimum 2 starts with $4\lambda_2 = \frac{607}{40}$.

Table 70: Short even lists: the fourth moment and all outer lengths: parameter values.

Entry	Bare fourth loss	Other scalar fourth fee	Full squared-norm cost	Resulting loss
$L = 1, m = 3$	8;	$\frac{17}{3}$;	$< \frac{15}{2}$;	$> \frac{1201}{120}$.
$L = 1, m \geq 4$	$\frac{19}{4}$;	$\frac{17}{3}$;	$\frac{51}{8}$;	$\frac{473}{60}$.
$L = 2$	$\frac{1}{5}$;	0;	$< \frac{53}{8}$;	$> \frac{35}{4}$.
$L = 3$	$\frac{47}{15}$;	0;	$< \frac{29}{4}$;	$> \frac{1327}{120}$.

$AtL = 1$ the actual vectors are $G_{m-1}u - R$. Their squared norms are $\leq 167d/96$ for $m = 3$ and $\leq 3d/2$ for $m \geq 4$; the full signed scalar fee is $4(\omega_1 + EAR) \leq 17d/3$. $AtL = 2$ the true vector is $D\mathbf{1} - B(G_{m-2}u - R)$, with squared norm $\leq 3329d/2160$ and total signed mean $\leq -d/20$. $AtL = 3$ the true vector is $(G_2 - G_{m-1})u + Du + B^2R + \epsilon\mathbf{1}$, with squared norm $< 169d/100$ and total signed mean $\leq -47d/60$. Thus every table entry follows directly from (R41.3.2) with its actual constant cross.

For every $L \geq 4$ put $t = \epsilon/d - \frac{1}{4}$ in $[-\frac{19}{20}, \frac{9}{20}]$ and $C = \frac{113}{100}$. The full actual mean and norm satisfy

$$(c_0 + EBV)/d \leq t - \frac{1}{3}, \quad \|V\|^2/d \leq N(t) = \begin{cases} C^2 + (t^2 - 2t)/3, & t \leq 0, \\ C^2 + (t^2 + 3t/2)/3, & t \geq 0. \end{cases} \tag{R41.4.1}$$

These follow from the exact $Q_{L-1} - G_{m-1}$ vector, whose norm is $< (\frac{1}{2} + \frac{3}{20} + \frac{7}{25} + \frac{1}{5})\sqrt{d}$, and whose mean lies $[-d, 3d/4]$. The fourth fee $4(t - \frac{1}{3}) + (\frac{17}{4})N(t)$ is convex on each whole half. At the negative endpoint, zero and positive endpoint respectively,

$$\begin{aligned} (\frac{17}{4})C^2 &< \frac{87}{16}, \\ (\frac{17}{4})(\frac{1121}{1200}) &< 4, \\ (\frac{17}{4})(\frac{117}{400}) &< 5/4. \end{aligned} \tag{R41.L340}$$

The fourth mean fees at those endpoints are $-\frac{77}{15}, -\frac{4}{3}, \frac{7}{15}$. The resulting fees are respectively $< \frac{71}{16}$, $< 5, < \frac{115}{16}$. Subtracting from $\frac{607}{40}$ leaves more than 6 in every case. Convexity proves every actual ϵ.

For minimum 3 the actual R2-C038 bare sums and propagated norms give:

Table 71: Short even lists: the fourth moment and all outer lengths: parameter values.

Entry	Start including bare loss	Norm cost	Other scalar fee	Net loss
$L = 1, m = 3$	$\frac{767}{40} + 8$;	$< \frac{43}{4}$;	$< \frac{7}{2}$;	$> \frac{517}{40}$.

Entry	Start including bare loss	Norm cost	Other scalar fee	Net loss
$L=1, m\ge 4$	$\frac{767}{40}+\frac{19}{4}$;	$<\frac{33}{4}$;	$<\frac{7}{2}$;	$>\frac{487}{40}$.
$L=2$	$\frac{767}{40}+\frac{3}{4}$;	$<\frac{27}{4}$;	<5;	$>\frac{327}{40}$.
$L\ge 3$	$\frac{767}{40}+\frac{19}{4}$;	$<\frac{9}{2}$;	<10;	$>\frac{377}{40}$.

The respective vector norms are $\frac{63}{40}$, $\frac{11}{8}$, $\frac{151}{120}$, $\frac{41}{40}$ times $\sqrt{d}$. The last row uses the actual $W_{\min(m,L)}+\bar{f}_{\max(m,L)}$ sum. These rows include equality $m=L$ and the G_2 and T^2 exceptions.

For $k\ge 1$ posts add $\frac{64}{5}$ to the starting fourth loss and at most $\frac{1}{3}$ to the mean fee. In minimum 2's $L=1, L=2, L=3$ rows the full vector norms are respectively $<\frac{41}{24}$, $\frac{13}{8}$, $\frac{67}{40}$ times $\sqrt{d}$; their costs are $<\frac{25}{2}, <\frac{45}{4}, <12$. The resulting losses exceed $\frac{569}{40}, 6, \frac{751}{40}$. The $L=2$ explicit lower bound is $\frac{1119}{40}+\frac{1}{5}-\frac{45}{4}-\frac{1}{3}>6$.

For minimum $2L\ge 4$, complete the squares in (R41.4.1):

$$\begin{aligned} N &= C^2-\frac{1}{3}+(t-1)^2/3\ (t\le 0),\\ N &= C^2-\frac{3}{16}+(t+\frac{3}{4})^2/3\ (t\ge 0). \end{aligned} \tag{R41.L369}$$

Both residual constants are positive. Hence $\sqrt{N}$, and then $(\sqrt{N}+\frac{3}{8})^2$, are convex on each half. At the negative, zero, positive endpoints the fourth norm costs are $<15, <\frac{39}{4}, <\frac{23}{2}$. Including means and ES, the full fees are $<\frac{51}{5}, <\frac{35}{4}, <\frac{123}{10}$. The starting loss is $\frac{1119}{40}$, so all leave more than 6.

For minimum 3 posts the full vector norms are $\frac{39}{20}$, $\frac{7}{4}$, $\frac{49}{30}$, $\frac{7}{5}$ in the same four rows. The starting losses including bare terms are $\frac{1599}{40}$, $\frac{1469}{40}$, $\frac{1309}{40}$, $\frac{1469}{40}$; norm costs are respectively $<\frac{65}{4}, <\frac{53}{4}, <\frac{23}{2}$, $<\frac{17}{2}$, and other fees $<\frac{7}{2}, <\frac{7}{2}, <\frac{11}{2}, <\frac{21}{2}$. Their differences all exceed 6. Extra pairs only improve Γ. This proves (R41.1.2) for every $m\ge 3$.

R41.8 Every independent outer length from the same fourth input

For any original positive kernel operator P, put $h=P_1$. Original row Cauchy and Hölder with exponents 3 and $\frac{3}{2}$ give, for every real Y,

$$\begin{aligned} |PY|^2 &\le hP(|Y|^2),\\ I_q(PY)^3 &\le E(h^4/q)\|P\|^4(E|Y|^4)^2. \end{aligned} \tag{R41.8.1}$$

All quantities remain on $L=2(\mu)$, with actual adjoint $T^*=A$.

$AtM=1, h=q$. Since $Eu=0, Eu^2=d$ and $|u|\le 1$,

$$Eq^3=1+3d+Eu^3\le 1+4d. \tag{R41.L671}$$

Together with (R41.2.4), this pays the complete cost $4+\frac{128}{19}=\frac{204}{19}$ in (R41.8.1). The ordinary-fourth bridge combines the displayed row and norm costs with the unweighted fourth moment of the input. Its input bound is distinct from the $p-$ or $q-$ weighted moments.

For $M=2$, the exact C094 whole-host calculation gives

$$I_q(T^2Y)^3\le e^{(\frac{24427}{3420})d-5d^2}(E|Y|^4)^2. \tag{R41.8.2}$$

Its negative curvature combines the $\bar{f}_2$ inverse-fourth row with the same δ_2.

For $M = 3$, C094, Section 2.3 gives the actual row $E(\bar{f}_3^4/q) \le 1 - 23d/36$ and $\|T^3\| \le 1$. Hence

$$I_q(T^3Y)^3 \le (1 - 23d/36)(E|Y|^4)^2. \tag{R41.8.3}$$

For every $M \ge 4$, the exact C084 row used in the same C094 section is $E(\bar{f}_M^4/q) \le e^{11d}$. The fourth power of the true operator bound retains the floor:

$$I_q(T^MY)^3 \le e^{[11-\lfloor M/3\rfloor/2]d}(E|Y|^4)^2. \tag{R41.8.4}$$

The $M = 3$ and $M \ge 4$ rows are distinct.

Substitute the actual H_D of Theorem A into (R41.8.1)–(R41.8.4). For $c = 6$, the losses are $24/19$ at $M = 1$, $16613/3420$ with the additional term $5d^2$ at $M = 2$, $455/36$ at $M = 3$, and $1+\lfloor M/3\rfloor/2 \ge 3/2$ at $M \ge 4$. These are the bounds in (R41.1.3), each at least $24/19$.

O.38 Short terminal pairs and unit-minimum even masks

The messages and their bounds.

Use the original finite model of Lemma M.1: $B = -K$, $u = B\mathbf{1}$, $T = J + B$, $A = J - B$, $p = 1 - u$, $q = 1 + u$, and $\theta = 13/32$, $\kappa = 19/32$, $\tau = d - \delta_2$. Use the finite convolution (M.2) and the spectral coordinates (M.3); in particular $G_n(z) = \sum_{r=0}^{n-1} z^r$ and the spectral measure of u has mass d and moments $\int x^r\, d\nu = \delta_{r+1}$. The equal-pair products $E_t, E_r \in [0, 1]$ remain at the terminal and receiving root. Let

$$P = \prod_{i=1}^{k} W_{u_i}\bar{f}_{v_i}, \qquad 4 \le u_1 < v_1 < \cdots < u_k < v_k,$$

where the product may be empty. For $m \ge 3$, use the following actual messages, multiplied by E_rP; restore E_t inside the displayed connector.

Table 72: The remaining even terminal directions.

Terminal lengths	Connector	Anchor and terminal message
$(2, b),\ b \ge 3$	$L = 1$	$f_mT(f_2\bar{f}_b)$
	$L \ge 2$	$W_mT^L(f_2W_b)$
$(3, b),\ b \ge 4$	$L = 1$	$f_mT(f_3h_b)$
	$L \ge 2$	$W_mT^L(f_3h_b)$
Minimum one, $j \ge 2$	$L = 1$	$f_mT(\overline{F})$
	$L \ge 2$	W_mT^LF

In the last two rows,

$$F = qf_{b_1}\prod_{i=2}^{j} h_{a_i}f_{b_i}, \qquad 1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j, \quad j \ge 2.$$

Only these rows require $m < u_1$ when $P \ne 1$. The single-pair rows allow every $m \ge 3$, with no order against the posts. In either case terminal and post lengths may overlap. The bar in the last unit-connector row reverses every terminal edge. All choices depend only on the listed integers.

Proposition O.13 (The short-terminal ordinary moments)**.** *For every finite choice of the indicated lengths, the actual message H satisfies*

$$\mathbb{E}H^8 \le e^{-d/4}, \qquad P \ne 1 \implies \mathbb{E}H^8 \le e^{-8d}.$$

In the same directions the single-pair fourth moments satisfy

$$\begin{array}{l|l} (2,b),\ b\ge 3 & \mathbb{E}H^4 \le e^{-(326249/86400)d} \\ (3,b),\ b\ge 4 & \mathbb{E}H^4 \le e^{-(1901/300)d}. \end{array}$$

For the minimum-one even rows, put $r = j-2$. Their bounds include

$$\mathbb{E}H^8 \le e^{-[8/7+1/1024+(128/5)r]d},$$
$$\mathbb{E}H^4 \le e^{-[4/7+1/2048+(64/5)r]d}.$$

With nonempty posts their eighth and fourth rates are at least $3037/336 + (128/5)r$ and $3037/672 + (64/5)r$, respectively.

Proof. **A single propagation and moment calculation.**

For any actual terminal envelope $F \le e^{R-\Gamma}$, put

$$D_i = \mathbb{E}B^i R, \qquad \sigma = \mathbb{E}R^2, \qquad \delta = D_0 + \eta\sigma, \qquad \epsilon = \delta + D_1,$$

where $e^R \le 1 + R + (\eta/2)R^2$ on its actual range. Positive propagation and (M.33) give

$$0 \le e^{\Gamma} T^L F \le Y_L = \bar{f}_L + B^L R + \delta \bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{O.129}$$

Use $\gamma_l = \beta_l - D_{l-1}$. The existing receiving arm obeys $W_m \le p - \beta_m - BG_{m-1}(B)u$, and $\Delta_m = \beta_{m+1} - \beta_m \le \delta_3 < d/6$. The following complete vectors are therefore sufficient for $W_m + Y_L \le 2 + c + BV$. The last column is $S = -\Gamma + c + \mathbb{E}BV$.

L	V	S
2	$\delta\mathbf{1} - B(G_{m-2}u - R)$	$-\Gamma + \epsilon - d + \Delta_m + D_2$
3	$V_0 + \epsilon\mathbf{1}, \quad V_0 = (G_2 - G_{m-1})u + \delta u + B^2 R$	$-\Gamma + \epsilon - 2d + \Delta_m + D_2 + D_3 - \delta d.$

(O.130)

Their unshifted means are

$$\mathbb{E}[-B(G_{m-2}u - R)] = \beta_m + D_1, \qquad \mathbb{E}V_0 = \beta_m - d + D_2. \tag{O.131}$$

These follow by adding the actual path formulas; in particular the constant coordinates in (O.130) are not orthogonal to the remaining vector.

There is a second bound for every finite $L \ge 3$. If all required γ_l are nonnegative, pay the positive δ and D_1 coefficients by $\bar{f}_s \le 2$, retain the last actual loss $\gamma_L \bar{f}_0$, and discard the remaining negative terms. This gives

$$\begin{aligned} V &= (G_{L-1} - G_{m-1})u + B^{L-1}R, \\ S &\le -\Gamma + 2\delta_+ + 2(D_1)_+ - \gamma_L - \gamma_{L+1} + \Delta_m. \end{aligned} \tag{O.132}$$

The mean index $L+1$ is included. The finite difference satisfies $\|(G_s - G_t)u\| \le \theta\sqrt{1+\theta}\sqrt{d} < \sqrt{d}/2$ for $s, t \ge 2$.

For $L = 1$ two direct sums will be used. The actual message $f_m TF$ has

$$V = G_{m-1}(-B)u + R, \qquad S = -\Gamma + \epsilon - \beta_m - \beta_{m+1}. \tag{O.133}$$

The complete reversal of the prescribed minimum-one message is $\bar{f}_m AF$, with its posts reversed too. Its corresponding sum is

$$V = G_{m-1}(B)u - R, \qquad S = -\Gamma + \delta - D_1 - \beta_m - \beta_{m+1}. \tag{O.134}$$

If $S \le -ad$, $\|V\|^2 \le Nd$, then (M.50) and (M.43) prove the two estimates at once:

$$\log \mathbb{E}H^8 \le -(8a - 17N)d, \qquad \log \mathbb{E}H^4 \le -(4a - 17N/4)d. \tag{O.135}$$

Here $H \le e^{-\Gamma} XY$, where $X + Y \le 2 + c + BV$. The square in AM–GM has a nonnegative base because it bounds an actual nonnegative arm sum. A scalar prefactor s^2 is handled by replacing $-\Gamma$ in S with $2 \log s$.

The whole finite post block has one pointwise envelope $P \le e^{BV_P - \Gamma_P}$, with

$$\Gamma_P \ge 16kd/5, \qquad \mathbb{E}BV_P \le d/12, \qquad \|V_P\| \le 3\sqrt{d}/8.$$

For each actual pair starting at four, use (L96.2.5) or (L96.2.6), followed by complete reversal. Its exponent is the sum of $B^r \mathbf{1}$ on its original finite interval; the union of all these intervals starts at $r = 5$. Lemma M.2 therefore gives $\|V_P\|^2 \le (101/50)\theta^3 d < 9d/64$. The first possible positive even mean is δ_4, so $\mathbb{E}BV_P \le \delta_4/(1-\theta^2) < d/12$. These arguments also apply after complete reversal. If H_0 is the message without posts and $\|V\| \le v\sqrt{d}$, then

$$\begin{aligned} \log \mathbb{E}(PH_0)^8 &\le [-8a - 128k/5 + 2/3 + 17(v + 3/8)^2]d, \\ \log \mathbb{E}(PH_0)^4 &\le [-4a - 64k/5 + 1/3 + (17/4)(v + 3/8)^2]d. \end{aligned} \tag{O.136}$$

In particular, if $v \le 51/40$, adding the first post improves the eighth and fourth rates in (O.135) by at least

$$\begin{aligned} z_8 &= 128/5 - 2/3 - 17\{2(51/40)(3/8) + 9/64\} = 1207/192, \\ z_4 &= 64/5 - 1/3 - (17/4)\{2(51/40)(3/8) + 9/64\} = 29971/3840. \end{aligned} \tag{O.137}$$

Each further post adds $128/5$ or $64/5$, since the norm bound is for the whole block. This pays every cross term, including overlap between terminal, receiving and post lengths.

Two terminal packets.

For the terminal $F = \bar{f}_2 W_b$, $b \ge 4$, use the actual packet of Lemma M.36, with $\eta = 3/2$. Thus, with $r = b - 2$, $G_r = e + zf$,

$$R = (\gamma + xe + zxf)u, \qquad \gamma = \beta_{b-1}, \qquad \Gamma = \beta_b + \gamma \ge 19d/16.$$

Use $\delta \le 7d/12$ and retain the joint loss (M.81) together with its signed moments. One additional finite contraction is useful:

$$\|(G_n(B)u - R)\|^2 \le d \qquad (n \ge 1). \tag{O.138}$$

Indeed before subtracting γ, the real coordinate $E_n - xe$ is at least $1 - 2x \ge 3/16$. Since $\gamma \le 1/3 < 2(3/16)$, subtraction does not increase its absolute value. Both resulting coordinate magnitudes are at

most $h = 1 - x + x^2$; for $n = 1$ the imaginary magnitude is $xf \le x \le h$. Consequently the squared modulus is at most $(1 + x)h^2 \le 1$, because $\theta + \theta^2 + \theta^4 < 1$. This proof includes G_1.

At $L = 2$, let $V_0 = -B(G_{m-2}u - R)$. Then $\|V_0\|^2 \le \delta_2$, and $\mathbb{E}V_0 = \beta_m + D_1 = \beta_m + \epsilon - \delta$. For $\delta \ge 0$, keep this identity before estimating the constant:

$$\delta^2 + 2\delta\mathbb{E}V_0 \le 2\delta d - \delta^2 + 2\delta\epsilon \le 119d^2/144.$$

For $\delta < 0$, use $\delta \ge -\delta_2$ and $\mathbb{E}V_0 \ge -7d/48$; its correction is at most $871d^2/3072 < 119d^2/144$. Hence

$$\|\delta\mathbf{1} + V_0\|^2 \le (\theta + 119/432)d = 589d/864. \tag{O.139}$$

This coupling is why the public $7d/12$ cap suffices. For $L = 3$, $m \ge 4$, $V_0 = \delta u - B^2(G_{m-3}u - R)$, so its norm is at most $(\theta + 7/36)\sqrt{d} = (173/288)\sqrt{d}$. At $m = 3$ the absent G_0 term is handled using $\|R\| < 4\sqrt{d}/5$ and gives a smaller bound. Using $\epsilon \in [-13d/16, 0]$ and $\mathbb{E}V_0 \in [-\theta d, \theta^2 d]$, its complete square is at most $(66433/82944)d < 13d/16$. The first two scalar bounds in (O.130) are $S \le -89d/48$, $S \le -29d/12$: use $\epsilon \le 0$, $D_2 \le \delta_3$, $D_3 \le d\delta_2 + \delta_3$, and $\delta \ge -\delta_2$. For all $L \ge 4$, the finite residue classes give $\gamma_l \ge 17d/32$; (O.132) and (M.81) give $S \le -83d/48$, while the norm is below $(1/2 + 16/75)\sqrt{d} < 3\sqrt{d}/4$.

For $b = 3$, keep its actual G_1 packet separately:

$$R = -B^2u, \quad \Gamma = d, \quad \sigma = \delta_3, \quad D_0 = 0, \quad D_1 = -\delta_2, \quad D_2 = 0, \quad D_3 = \delta_3.$$

Use $\eta = 3/2$, so $\delta = 3\delta_3/2 < d/4$ and $\epsilon \in [-\delta_2, 0]$. The finite coordinates give $\|(G_n - x)u\| \le \sqrt{d}$. At $L = 2$, the constant correction is at most $9d^2/16$, giving $N = 19/32$. At $L = 3$, the nonconstant norm is at most $\sqrt{d}/2$, and its ϵ correction is at most $3\theta^2 d^2 < d/6$, giving $N = 5/12$. Their scalar losses are $11/6$, $8/3$. Here $\gamma_3 = d$, $\gamma_4 = d - \delta_3$; every later positive moment is at most δ_5, and $\kappa - \theta^4 > 9/16$. Thus every finite $\gamma_l \ge 9d/16$, and the long row gives $a = 35/24$, $N = 25/64$.

For $F = f_3 h_b$, $b \ge 4$, put $n = b - 3$, $G_n(-z) = U - zV$, so $U \in [1 - x, 1]$, $V \in [0, 1]$, including $(U, V) = (1, 0)$ at $b = 4$. The actual swapped arm sum is

$$f_3 + h_b \le 2 - \Gamma + R, \qquad \Gamma = 3d - 2\delta_2, \qquad R = (-\tau + x^2V + zxU)u.$$

To obtain it, apply T to the finite positive convolution for f_{b-1}, retaining its τq term. This is a direct bound for the swapped arms, not a sign change of the canonical pair inequality. Its square satisfies $\sigma \le (1 + \theta)\delta_4 + \tau^2 d < 3d/20$, the cross term being nonpositive. Its predecessor has square below $1/12 + 1/18 < 1/4$, so $\eta = 6/5$ is valid on the full domain. The exact signed moments are

$$D_0 = -\int xU, \quad D_1 = \tau d - \int x^2V,$$
$$D_2 = \int x^2U, \quad D_3 = -\tau\delta_2 + \int x^3V.$$

Each further four steps multiplies the spectral powers by x^2. Thus

$$\begin{gathered} -\theta d \le \delta \le 9d/50, \quad -3d/5 \le \epsilon < d/4, \quad -\theta^2 d \le D_1 \le 2d/9, \\ 0 \le D_2 \le \theta^2 d, \quad D_3 \le \theta^3 d, \quad \gamma_3 \ge 5d/6, \quad \gamma_4 \ge 9d/10, \quad \gamma_l \ge d/2 \ (l \ge 5). \end{gathered} \tag{O.140}$$

For example, retain the negative D_0 when bounding $\epsilon \le \tau d - \kappa\delta_2 + 9d/50 \le (\theta^2/4 + 9/50)d < d/4$. The later positive moments are at most $\max(\tau\delta_3, \delta_5) < d/24$, which proves every finite tail bound in (O.140).

For $L = 2$, the nonconstant squared norm is below $4d/5$, since its coordinate bounds are $11/9$ and one, including G_1. Its mean is positive and at most $11d/9$. The true constant correction is at most $(81/2500 + 11/25)d^2 < 4d/25$, so $N = 24/25$. For $L = 3$, the nonconstant norm is below $(1/2 + 13/96 + 13/80)\sqrt{d} < 4\sqrt{d}/5$. Its mean is in $[-\theta d, \theta^2 d]$; the two signs of ϵ give constant corrections at most $113d/400$, $7d/144$, respectively. Hence its complete square is below d. Equations (O.130) and (O.140) give $a = 125/48, 10/3$ in these rows. For every $L \geq 4$, use (O.132): the positive cost is at most $181d/225$, the two final losses total at least d, and $\Delta_m < d/6$. Thus $a = 35/16$; the norm is below $(1/2 + 8/75)\sqrt{d} < 5\sqrt{d}/8$.

The actual unit connectors.

For $(2, b)$, $L = 1$, the terminal is the different $f_2 \bar{f}_b$. Set $G_{b-2} = e + zf$, $s = 1 - \beta_b/2$, $k = s^{-1} \in [1, 6/5]$. The finite arm sum is

$$0 \leq f_2 + \bar{f}_b \leq 2s + R, \qquad R = (-xe + z(2 - xf))u.$$

Its actual nonnegative square gives $TF \leq s^2(q + \delta + kBR)$, where $\delta = k\mathbb{E}R + (k^2/2)\mathbb{E}R^2$, $\epsilon = \delta + k\mathbb{E}BR$. For $m \geq 4$, retain $\alpha = \beta_{m-1}$ in the receiving f_m convolution. The complete vector and scalar are

$$V = G_{m-1}(-B)u + kR + \alpha\mathbf{1}, \qquad S = 2\log s + \epsilon - \beta_m - \beta_{m+1} - \alpha.$$

At $m = 3$, set $\alpha = 0$. If $b \geq 4$, the epsilon integrand increases in e, f; at $e = f = 1$ it is convex in k. Its two endpoint polynomials increase in x, and their values at θ are $-38603/65536$, $-230691/409600$, both below $-9/16$. Also $\mathbb{E}[V - \alpha\mathbf{1}] \leq -\kappa d$, so $\alpha^2 + 2\alpha\mathbb{E}[V - \alpha\mathbf{1}] \leq 0$. At $m = 3$ the two positive coordinate bounds are $1 - x + x^2$ and $7/5 - 6x/5 + 6x^2/5$, with square below $6/5$. At $m \geq 4$ they are $1 - 2x + 2x^2$ and $7/5 - x/5 + 6x^2/5$. Their square is convex on $[0, \theta]$, with endpoint values below $5/4$. For $m \geq 4$, retain all three receiving losses $\beta_m, \beta_{m+1}, \alpha \geq \kappa d$, as well as $2\log s \leq -\beta_b \leq -\kappa d$. At $m = 3$, the two receiving losses total $2d$. Thus every m gives $a = 47/16$, $N = 5/4$.

At $b = 3$, retain its actual G_1:

$$\epsilon = -2kd + (2k^2 + k)\delta_2 + (k^2/2)\delta_3 \leq -3d/5.$$

The upper is convex in k, and its two endpoint values at $x = \theta$ are below $-3/5$. The constant correction is again nonpositive. For $m = 3$ the squared norm is bounded by $(1-x)^2 + (49/25)x < 7/6$; for $m \geq 4$ it is bounded by $(1-x)^4 + x(7/5+x)^2 < 29/20$. The latter polynomial is convex and its upper endpoint is below $29/20$. Here $2\log s \leq -d$; the same receiving losses give $a = 1 + 3\kappa + 3/5 = 541/160$ for $m \geq 4$, and a larger value at $m = 3$. Thus $a = 541/160$, $N = 29/20$ suffice for every m.

For $(3, b)$, use (O.133) with its actual packet above. At $m = 3$, the squared norm is bounded by $(1+x^2)^2 + x(1-x+x^2)^2 < 13/8$; at $m \geq 4$ it is bounded by $(1-x+2x^2)^2 + x(1-x+x^2)^2 < 9/8$. The coordinate bounds are nonnegative, including the $b = 4$ endpoint. Equation (O.140) gives the scalar losses $63/16$, $25/8$, respectively.

Minimum-one even masks.

Let $F = qf_{b_1} \prod_{i=2}^{j} h_{a_i} f_{b_i}$, as in Table 72. Put $a_1 = 1$ and form its one finite interval mask

$$R_0 = \sum_{i=1}^{j} \sum_{r=a_i+1}^{b_i} K^r\mathbf{1} = BP(-B)u, \qquad P(-z) = U - zV, \qquad C_\beta = (1 - \theta^2)^{-1} = 1024/855.$$

The pair envelopes follow from finite positive convolution. For the unit pair this is $f_b - f_1 = R_{1b} - \sum_{i=0}^{b-3} \beta_{b-i} f_i$. Each later shorter arm at least four supplies loss $16d/5$. When the next short arm is three, retain τf_1 in that pair, giving the same $3d - 2\delta_2$ loss used above. Thus the following packets bound the actual F by $e^{R-\Gamma}$; every extra pair after the second adds $16d/5$ to Γ.

Table 73: The three finite-mask packets. Here $\Gamma \ge gd + 16(j - 2)d/5$.

Class	Condition	Low domain	R	g	η
I	$b_1 \ge 3$	$d \le 1/64$	R_0	$607/160$	$15/14$
II	$b_1 = 2,\ a_2 \ge 4$	$d \le 1/64$	R_0	$16/5$	$15/14$
III	$b_1 = 2,\ a_2 = 3$	$d \le 1/6$	$R_0 + \tau u$	$35/16$	$6/5$

At the lowest degrees the forced coefficients of P are $(P_0, P_1) = (1, 1)$, $(P_0, P_1, P_2) = (1, 0, 0)$, and $(P_0, P_1, P_2) = (1, 0, 1)$, respectively. Split the four finite residue classes to obtain

	U	V
I	$1 - C_\beta x \le U \le C_\beta$	$1 - C_\beta x \le V \le C_\beta$
II	$11/12 \le U \le C_\beta$	$-C_\beta x \le V \le C_\beta x^2$
III	$1 - C_\beta x \le U \le 1 - x/3$	$-C_\beta x \le V \le C_\beta x^2$.

(O.141)

In I, $U, V > 1/2$. In II, $|P|^2 < 25/16$; in III, $|P|^2 \le (1 - x/3)^2 + (16/9)x^3 \le 1$. The whole mask has $|P|^2 \le (1 + x)/(1 - x^2)^2 < 100/49$. In I/II the row bound gives $|R| < 1/5$; in III its actual predecessor $P(-B)u + \tau\mathbf{1}$ has squared norm at most $d + (7/5)d^2 \le 37/180 < 1/4$. This verifies the Taylor ranges in Table 73.

The same coordinates and signed spectral means give the finite data below. The column $D_{i\ge4}$ bounds every positive part with that index, not a truncated list.

	σ/d	δ/d	D_1/d	D_2/d	D_3/d	$D_{i>4}/d$
I	$325/392$	$[-6/5, 3/8]$	$[-1/2, 0]$	$1/2$	$1/5$	$1/5$
II	$325/512$	$[-6/5, 0]$	$[-1/12, 1/5]$	$1/2$	$1/5$	$1/5$
III	$1163/2592$	$[-1, 1/40]$	$[-2/9, 1/5]$	θ	$1/10$	$1/4$.

(O.142)

Here is a verification of the signs and the full ranges. Since $R_0 = (xV + zU)u$, $D_0(R_0) = -\int U$, $D_1(R_0) = -\int xV$. The bounds on σ in I/II are $\theta(100/49)d$, $\theta(25/16)d$, respectively. The upper delta bounds follow from

$$-439/855 + (15/14)(325/392) < 3/8, \qquad -11/12 + (15/14)(325/512) < 0.$$

In III, $\tau \le 5/36$ and the actual constant cross term gives $\sigma/d \le \theta + \tau/6 + \tau^2 \le 1163/2592$, so $\delta/d \le -439/855 + 1163/2160 < 1/40$. Its D_1 lower bound is $-d/12 - \tau d \ge -2d/9$. The forced fourth power gives $D_2 \le \delta_2 - \delta_3 + C_\beta\delta_4 \le \delta_2$; $D_3 \le C_\beta\delta_5 + \tau\delta_2 < d/10$. For the later positive classes, retain $\tau = d - \delta_2$ before estimating: if $v = \delta_2/d$, then

$$(C_\beta\delta_3 + \tau\delta_2)/d \le C_\beta\theta v + d\, v(1 - v) < 1/4 \quad (0 \le d \le 1/6,\ 0 \le v \le \theta).$$

Every further four powers advance the moment index by two. The same four classes give the I/II later bounds in (O.142). Consequently, including the next mean index,

	γ_3/d	γ_4/d	γ_l/d $(l \ge 5)$
I, II	$1/2$	$4/5$	$3/8$
III	κ	$9/10$	$11/32$.

(O.143)

At $L = 1$, use the complete reversed vector in (O.134). The real and imaginary bounds for I are $[0, 1 - x/2]$, $[-5/8, 5/8]$, so $N \le 1$. In II they are bounded in absolute value by $6/5, 5/8$, giving $N < 8/5$. In III with $m = 3$ use $6/5, 1/2$, again $N < 8/5$; with $m \ge 4$ the real coordinate lies in $[107/288, 1]$, the imaginary one in $[-1/2, 1/2]$, and $N \le 141/128 < 9/8$. The scalar bound is $\delta - D_1 - \beta_m - \beta_{m+1}$; retain $\beta_3 + \beta_4 = 2d$ when $m = 3$.

At $L = 2$, retain $\delta\mathbf{1}$ in the exact vector (O.130). Its remaining mean $\beta_m + D_1$ is nonnegative in all three classes. The G_1 nonconstant squared bounds before applying B are $5/4, 81/40, 15/8$; all higher indices have smaller sufficient bounds from the preceding coordinates. After B, use $65/128, 5/6, 25/32$. In I/II the complete constant correction is at most $36d^2/25 \le 9d/400$; in III it is at most $d^2 \le d/6$. This gives $N = 17/32, 7/8, 91/96$. The scalar in (O.130) uses the actual $\epsilon = \delta + D_1$.

For $L \ge 3$, use (O.132) with (O.143). The norms of R are below $11/12, 4/5, 27/40$ times $\sqrt{d}$; the finite difference then gives the norms in Table 74. For example, the III scalar contributions before $-\Gamma$ at $L = 3$ and $L \ge 4$ are $1/20 + 2/5 - \kappa - 9/10 + 1/6 = -421/480$ and $1/20 + 2/5 - 22/32 + 1/6 = -17/240$. All other entries follow from the same formula and the displayed finite data; the original final path and its mean index are retained.

The joint parameter table.

Table 74 collects the complete-vector bounds proved above. An entry (a, N) means $S \le -ad$, $\|V\|^2 \le Nd$, so both moments follow from (O.135). In the last thirteen rows add $16(j-2)/5$ to a.

Table 74: One payment table for both ordinary moments.

Terminal packet	Range	a	N
$(2, b \ge 4)$	$L = 1$	$47/16$	$5/4$
$(2, 3)$	$L = 1$	$541/160$	$29/20$
$(2, b \ge 4)$	$L = 2$	$89/48$	$589/864$
	$L = 3$	$29/12$	$13/16$
	$L \ge 4$	$83/48$	$9/16$
$(2, 3)$	$L = 2$	$11/6$	$19/32$
	$L = 3$	$8/3$	$5/12$
	$L \ge 4$	$35/24$	$25/64$
$(3, b)$	$L = 1, m = 3$	$63/16$	$13/8$
	$L = 1, m \ge 4$	$25/8$	$9/8$
	$L = 2$	$125/48$	$24/25$
	$L = 3$	$10/3$	1
	$L \ge 4$	$35/16$	$25/64$
I	$L = 1$	$657/160$	1
II	$L = 1$	$1033/240$	$8/5$
III	$L = 1, m = 3$	$2837/720$	$8/5$
	$L = 1, m \ge 4$	$563/180$	$9/8$
I	$L = 2$	$1801/480$	$17/32$
II	$L = 2$	$10/3$	$7/8$
III	$L = 2$	$1147/480$	$91/96$
I	$L = 3$	$2005/480$	$49/64$
II	$L = 3$	$59/15$	$25/36$
III	$L = 3$	$1471/480$	$16/25$

Terminal packet	Range	a	N
I	$L \ge 4$	$1741/480$	$9/16$
II	$L \ge 4$	$203/60$	$9/16$
III	$L \ge 4$	$271/120$	$49/100$

Every row has $N \le (51/40)^2$ and eighth rate at least $9/4$. The minimum-one rows have eighth and fourth rates at least $3 + 128(j-2)/5$ and $11/2 + 64(j-2)/5$. The single-pair fourth rates dominate those in Proposition O.13; these are direct fourth payments, not consequences of the common eighth bound. Equation (O.137) gives a nonempty-post eighth rate at least $9/4 + 1207/192 > 8$, with a positive gain in its same-direction fourth rate. These statements are exact rational substitutions into the one payment formula.

The high domains of the mask packets.

The same actual directions are used. The entire later terminal mask in I/II has pointwise size below $(3/8)\sqrt{d}$; at $d \ge 1/64$ this is at most $3d$. Its actual pair losses give

$$\prod_{i=2}^{j} h_{a_i} f_{b_i} \le e^{-[1/5+16(j-2)/5]d}.$$

In III its pointwise size is below $(65/112)\sqrt{d}$. At $d \ge 1/6$, use $\sqrt{d} < (5/2)d$ and the actual shorter-three loss $(2-\theta)d$, giving

$$\prod_{i=2}^{j} h_{a_i} f_{b_i} \le e^{-[1/7+16(j-2)/5]d}.$$

For the remaining $(1, b_1)$ terminal the actual base directions are precisely $f_m T(p\bar{f}_{b_1})$ at $L = 1$ and $W_m T^L(q f_{b_1})$ at $L \ge 2$. Proposition M.39 supplies eighth rates $9/4, 2005/192$ for the first, with empty and nonempty posts. Sections O.2 and O.4 supply $3/20, 97/12$ for the second. The nonempty cases use exactly $m < u_1$; they are bounds for the complete post block. It follows that throughout all the high domains

$$\begin{array}{l|l} P = 1 & \text{eighth rate } 181/140 + (128/5)(j-2) \\ P \ne 1 & \text{eighth rate } 775/84 + (128/5)(j-2). \end{array}$$

These exceed the rates stated in Proposition O.13. Cauchy gives the weaker fourth high-domain bounds there, with half the eighth rate. Each high comparison preserves the entire terminal product and agrees with its low-domain direction at the closed endpoint.

Equal-pair decorations are restored by positivity at their original vertices. At $d = 0$, leaf integration gives one. □

If a distinct central unit arm contributes p, ordinary Cauchy gives

$$\mathbb{E}[pH^4] \le \sqrt{\mathbb{E}p^2}\sqrt{\mathbb{E}H^8} \le e^{-(\gamma-1)d/2} \quad \text{whenever } \mathbb{E}H^8 \le e^{-\gamma d},$$

since $\mathbb{E}p^2 = 1 + d \le e^d$. This uses the same direction and does not supply an extra arm to the catalogue.

O.39 Ordinary eighth moments and the nonunit-pair outlet

R46.1 Local statements and conventions

Use the original finite probability space and operators of Section 2. A terminal residual has at least two distinct lengths. An ordinary component consists of that terminal, its connector of length $L \geq 1$, one existing receiving arm of length $m \geq 3$, and the ordered negative post block specified in part R46.3. The message includes its root and terminal equal-pair factors. The directions in that directory satisfy, for every finite host,

$$EH^{8} \leq e^{-d/1024}, \qquad d = \|K1\|^{2}. \tag{R46.1.2}$$

For a root whose strict residual is (s, m), $2 \leq s < m$, every nonreducible terminal and every pair of independent positive lengths L, M admit the output constructed below, with

$$I_{q}(U)^{3} \leq e^{-\eta_{\text{pair}} d}, \qquad \eta_{\text{pair}} = \frac{1}{72} + \frac{1}{1024} = \frac{137}{9216}. \tag{R46.1.3}$$

R46.2 Equal pairs

Equal-pair factors satisfy (2.5) and its complete reversal. They may be restored inside their positive terminal connector or at the receiving root, where they only decrease the message. Unequal post pairs retain their displayed directions and ordering.

R46.3 Exhaustive actual ordinary-eighth directory

Let $m \geq 3$ be one existing receiving arm, $L \geq 1$ its original connector to a terminal with strict residual size ≥ 2, and

$$P = \prod_{\nu=1}^{k} W_{u_{\nu}} \bar{f}_{v_{\nu}}, \qquad 4 \leq u_{1} < v_{1} < \cdots < u_{k} < v_{k}. \tag{R46.3.1}$$

Impose $m < u_{1}$ when $k \geq 1$. The restriction is imposed uniformly, even in rows where a weaker one would suffice. There is no restriction between the terminal lengths and the other blocks. Every post is an actual original pair of unequal arms.

The complete message is $H = E_{r} P$ times the displayed anchor/connector function, with the terminal equal pairs restored inside its actual positive connector. Empty products equal one. Canonical pairs mean $h_{a} f_{b}$ for adjacent increasing residual lengths; their negatives are $W_{a} \bar{f}_{b}$.

R46.3.1 Even terminal lists

Write the strict terminal list as $a_{1} < b_{1} < \cdots < a_{j} < b_{j}$.

Table 75: Ordinary eighth moments and the nonunit-pair outlet: parameter values.

Entry	L/anchor condition	Unpaired anchor/connector factor H_{0}	Eighth loss
$j = 1, a_{1} = 1$	$L = 1$;	$f_{m} T(p \bar{f}_{b_{1}})$;	$1/1024$.
$j = 1, a_{1} = 1$	$L \geq 2$;	$W_{m} T^{L}(q f_{b_{1}})$;	$3/20$.
$j = 1, a_{1} = 2$	$L = 1$;	$f_{m} T(f_{2} \bar{f}_{b_{1}})$;	at least $1/4$.
$j = 1, a_{1} = 2$	$L \geq 2$;	$W_{m} T^{L}(\bar{f}_{2} W_{b_{1}})$;	at least $1/4$.
$j = 1, a_{1} = 3$	$L = 1$;	$f_{m} T(f_{3} h_{b_{1}})$;	at least $1/4$.

Entry	L/anchor condition	Unpaired anchor/connector factor H_0		Eighth loss
$j=1, a_1=3$	$L\ge 2$;	$W_m T^L(f_3 h_{b_1})$;		at least $1/4$.
$j\ge 2, a_1=1$	$L=1$;	$f_m T(\overline{F}), F$ $q f_{b_1}\prod_{i\ge 2} h_{a_i} f_{b_i}$;	$=$	at least $8/7+1/1024$.
$j\ge 2, a_1=1$	$L\ge 2$;	$W_m T^L(F)$, same F;		at least $8/7+1/1024$.
$j\ge 2, a_1=2$	$L=1$;	$\bar{f}_m A(F), F$ $\prod_i h_{a_i} f_{b_i}$;	$=$	2.
$j\ge 2, a_1=2$	$L\ge 2$;	$W_m T^L(F)$, same F;		2.
$j\ge 2, a_1=3$	$m\le L$;	$W_m T^L(F), F$ $\prod_i h_{a_i} f_{b_i}$;	$=$	2.
$j\ge 2, a_1=3$	$m> L$;	$\bar{f}_m A T^{L-1}(F)$, same F;		2.
$a_1\ge 4$, any $j\ge 1$	$m\le L$;	$W_m T^L(F), F$ $\prod_i h_{a_i} f_{b_i}$;	$=$	2.
$a_1\ge 4$, any $j\ge 1$	$m> L$;	$\bar{f}_m A T^{L-1}(F)$, same F;		2.

The bar in the minimum-one $L=1$ row reverses every terminal residual edge. The proof reverses the entire configuration, including posts and pairs, when applying its analytic comparison; the actual output in this table retains the prescribed negative P. Its condition $m<u_1$ is preserved. Genuine equal pairs can either follow that convention or be restored by (2.5) and its complete reversal.

The first two rows use the center input of Section O.27. For nonempty later-pair lists one may use rates $2005/192$ and $97/12$, with $m<u_1$. Rows 3–8 follow Proposition O.13, rows 9–12 follow Section O.34, and rows 13–14 follow Section O.35, with the directions in their respective tables.

For $j=1$, a_1 is exactly 1, 2, 3, or at least 4. For $j\ge 2$, the same four-way split applies. Thus every even list is covered.

R46.3.2 Odd terminal lists of minimum at least two

Write $(\ell, r_1, t_1, \dots, r_j, t_j)$, $\ell\ge 2, j\ge 1$. Set

$$F=\begin{cases}\bar{f}_\ell \prod_i h_{r_i} f_{t_i}, & r_1\ge 4,\\ \bar{f}_2 f_3 h_{t_1}\prod_{i\ge 2} h_{r_i} f_{t_i}, & r_1=3.\end{cases} \tag{R46.3.2}$$

The latter case forces $\ell=2$. Use

$$H_0=\begin{cases}W_m T^L F, & m\le \ell+L,\\ \bar{f}_m A T^{L-1} F, & m>\ell+L.\end{cases} \tag{R46.3.3}$$

R2-C030/A014 gives eighth loss 3, including every permitted P and arbitrary cross-block overlap. The comparison is with $\ell+L$, not merely L. Every strict odd list with minimum at least 2 has second entry 3 or at least 4, so (R46.3.2) exhausts it.

R46.3.3 Odd terminal lists of minimum one

Write $(1, r_1, t_1, \dots, r_j, t_j)$, $j\ge 1$. Choose

Table 76: Ordinary eighth moments and the nonunit-pair outlet: parameter values.

Entry	Actual F
$r_1 \geq 4$	$q \prod_i h_{r_i} f_{t_i}$.
$r_1 = 3$	$q f_3 h_{t_1} \prod_{i \geq 2} h_{r_i} f_{t_i}$.
$r_1 = 2, j = 1$	$q f_2 h_{t_1}$.
exact $j = 2$ list $(1, 2, 3, 4, e)$	$q f_2 h_e h_3 f_4$.
every other $r_1 = 2$	$q f_2 h_{t_1} \prod_{i \geq 2} h_{r_i} f_{t_i}$.

The actual anchor/connector is

Table 77: Ordinary eighth moments and the nonunit-pair outlet: parameter values.

Entry	Unpaired anchor/connector factor H_0
$L = 1, r_1 = 2$	$\bar{f}_m AF$.
$L = 1, r_1 \geq 3$	$W_m TF$.
$L \geq 2, m \leq L + 1$	$W_m T^L F$.
$L \geq 2, m > L + 1$	$\bar{f}_m A T^{L-1} F$.

Proposition O.10 supplies ordinary moment bounds of orders 8 and 4 for these same directions, with eighth loss at least $57/400$. The direction for $(1, 2, 3)$ is $q f_2 h_3$ even at $L = 2$; it must not be replaced by C106's distinct ordinary-unit exception. The crossed five-arm row uses exactly its existing arms. Every unit-minimum odd list has second entry 2, 3, or at least 4; the displayed cardinality/prefix splits are exhaustive.

R46.3.4 Proof of the common theorem and exact post scope

Each cited local theorem holds on all original finite hosts, all integers in its row and the stated complete finite post block. All its displayed eighth losses are at least $1/1024$. Genuine equal pairs only decrease the same positive message by (2.5) and its complete reversal. This proves (R46.1.2) for the complete union.

In every later central application, the anchor is selected before all post lengths in the strict central list, so $m < u_1$ and $u_1 \geq 4$ hold automatically. At a nonunit-pair root there is no unequal post.

This theorem has $m \geq 3$. It gives no general eighth-moment bound for $m = 2$ and no standalone $M = 2$ outlet for a singleton residual by itself.

R46.4 Every nonunit pair, all inner and outer lengths

At an actual internal root with strict residual (s, m), $2 \leq s < m$, orient its s-arm inward as $\bar{f}_s$. Put its distinct genuine m-arm, terminal, L connector and all its root/terminal equal pairs in the actual H of part R46.3, with $P = 1$. Then the exact outgoing message along an outer connector of length $M \geq 1$ directed toward the receiving center is

$$U = T^M(\bar{f}_s H). \tag{R46.4.1}$$

The m-arm has length at least 3. The s-arm is a different original arm, even when its length coincides with a length in the other terminal's table. Each actual edge is used once.

The full-multiplier inequality of Section M.16 holds for every real Y:

$$I_q(T^M(\bar{f}_a Y))^3 \le (1-d/72)E|Y|^8, \qquad a \ge 2, M \ge 1. \tag{R46.4.2}$$

The multiplier $T^M M_{\bar{f}_a}$ retains the inward arm in this operator order; its proof uses the row $\bar{f}_{M+a}$. Thus Y may be the AT^{L-1} component of the preceding directory.

Apply (R46.4.2) with $a = s, Y = H$ and use (R46.1.2):

$$I_q(U)^3 \le (1-d/72)e^{-d/1024} \le e^{-137d/9216}. \tag{R46.4.3}$$

The full multiplier includes every $M \ge 1$, independently of L. The retained inward arm is the existing s-arm, distinct from the m-arm used in H.

The path construction combines this local output with the outlet on the other side using (11.1). The tournament and entropy consequences are proved in Section 12.

P Unit-arm two-step outlets

These estimates cover the unit singleton with receiving length $M = 2$. The separate short terminal directions are retained. The unit singleton with $M = 1$ is treated by the whole-tree alternatives in Section 11.

P.1 Compatible square moments at a unit root

Lemma P.1 (A square with compatible moments)**.** *Suppose* $0 \le H \le (c+X/2)^2$, $|X| \le U$, $c > U/2$, *and*

$$\mathbb{E}X \le -\alpha, \qquad \mathbb{E}X^2 \le hU^2, \qquad 0 \le \alpha \le \sqrt{h}\,U, \quad U > 0, \quad 0 \le h < 1.$$

Put $k = \alpha/U$ *and*

$$x_- = -U\frac{k+h}{1+k}, \qquad \pi = \frac{h-k^2}{1+2k+h}.$$

Then, for every integer $j \ge 2$,

$$\mathbb{E}H^j \le (1-\pi)(c+x_-/2)^{2j} + \pi(c+U/2)^{2j}. \tag{P.1}$$

In particular,

$$\mathbb{E}H^4 \le (1-\pi)(c+x_-/2)^8 + \pi(c+U/2)^8. \tag{P.2}$$

If $U = 0$, *the bound for the* j*th moment is* c^{2j}.

Proof. The quadratic tangent to $f(x) = (c+x/2)^{2j}$ at x_- and agreeing with f at U majorizes f on $[-U, U]$. Its Hermite remainder has the sign of $(x-x_-)^2(x-U)$, since $f''' \ge 0$. The quadratic coefficient is nonnegative by convexity, and the linear coefficient is nonnegative because $x_- \le 0$. Hence its expectation is at most its value at the moment bounds $\mathbb{E}X = -\alpha$, $\mathbb{E}X^2 = hU^2$. The two points x_-, U, with upper probability π, have exactly these moments; $x_- \in [-U, 0]$ and $0 \le \pi \le 1$. This proves the general bound; $j = 4$ gives the displayed eighth-power formula. The zero case is direct. □

Corollary P.2 (The high-domain two-step path cost)**.** *In the finite skew-kernel model, put* $p = 1 + K\mathbf{1}$ *and* $\bar{f}_2 = T^2\mathbf{1}$*. Then*

$$\mathbb{E}(p\bar{f}_2)^8 \le \begin{cases} 35/8, & 1/5 \le d \le 1/4, \\ 82/15, & 1/4 \le d \le 1/3. \end{cases} \tag{P.3}$$

Proof. Let $r = K^2\mathbf{1}$. The nonnegative factors satisfy $p + \bar{f}_2 = 2 + r$, so $p\bar{f}_2 \le (1 + r/2)^2$. The same actual r has $|r| \le \sqrt{d}$, $\mathbb{E}r = -d$, and $\mathbb{E}r^2 \le \theta d$, where $\theta = 13/32$. Apply (P.1) with $j = 8, c = 1, U = \sqrt{d}, \alpha = d, h = \theta$. The negative atom contributes at most one, and $\pi = (\theta - d)/(1 + 2\sqrt{d} + \theta)$. For $d \ge 1/4$, $\pi \le 5/77 < 1/15$ and $(1+\sqrt{d}/2)^{16} < (13/10)^{16} < 67$, giving $82/15$. For $1/5 \le d \le 1/4$, use $\pi \le 33/353 < 3/32$ and $(1 + \sqrt{d}/2)^{16} \le (5/4)^{16} < 36$, giving $35/8$. The two last power comparisons follow from $(13/10)^4 < 20/7$, $(20/7)^4 < 67$, and $(5/4)^8 < 6$. □

The following three scalar payments apply whenever the square in Lemma P.1 has the displayed moment data, on $0 \le d \le 1/3$. Write $U = t\,u(d)$, $\alpha = t^2 a(d)$, and $c = C/D$, with $D = 1$.

Table 78: Compatible square-moment data.

Row	d	u	a	c	h
I	t^2	1	$1/2$	$1 - 3d/4$	$13/32$
II	$63t^2/71$	1	$(63/71)(4/5 - d)$	$1 - 4d/5$	$13/32$
III	t^2	1	$41/96$	$1 - 37d/64$	$3/10$

The targets are

$$\mathcal{T}_{\mathrm{I}} = (1 - 15d/32)^8, \quad \mathcal{T}_{\mathrm{II}} = (1 - 9d/20)^8, \quad \mathcal{T}_{\mathrm{III}} = (1 - 15d/32 + 3d^2/8)^8.$$

Here is one polynomial certificate for all three targets. Define

$$\begin{aligned} P &= u + ta, \qquad & Q &= (1 + h)u^2 + 2tau, \\ L &= CP - \tfrac{1}{2}Dtu(ta + hu), \qquad & B &= C + \tfrac{1}{2}Dtu, \\ N &= L^8 + (hu^2 - t^2a^2)B^8P^6, \qquad & \mathcal{D} &= D^8QP^6. \end{aligned}$$

Substitution into (P.2) gives exactly $N/\mathcal{D}$. The factors P, Q, D are positive on each actual interval. Set $\mathcal{G} = (\mathcal{T}\mathcal{D} - N)/t^2$, dividing out its exact endpoint factor. Let a_* be the left endpoint of the certificate interval and $\widehat{\mathcal{G}}(t) = \mathcal{G}(t)/\mathcal{G}(a_*)$.

Table 79: Continuous polynomial certificates for Table 78.

Row	Certificate interval for t	Degree	Bernstein bound
I	$[0, 577351/10^6]$	21	$B_j(\widehat{\mathcal{G}}) \ge 1/10$
II	$[0, 612913/10^6]$	37	$B_j(\widehat{\mathcal{G}}) \ge 1/10$
III	$[0, 577351/10^6]$	37	$B_j(\widehat{\mathcal{G}}) \ge 1/10$

In every row $\mathcal{G}(a_*) > 1/2$. The coefficient formula is (A.18), with one variable. Thus Lemma A.1 proves $\mathbb{E}H^4 \le \mathcal{T}$ throughout the stated actual domains. The rational certificate intervals include those domains, as is seen by squaring their positive endpoints. At $d = 0$, rows I–III have $X = 0, c = 1$ and give equality.

P.2 Whole-domain retained square moments

Lemma P.3 (Whole-domain data for $\sigma = 2$)**.** *Take the actual U of Lemma M.9 with $\sigma = 2$, and put $Z = KU$. The following data hold for the complete interval $0 \le d \le 1/3$:*

$$\|U\|^2 \le d\, v_n(d)^2, \quad \mathbb{E}U \le d\, z_n(d), \quad \mathbb{E}Z \ge \alpha_n(d), \quad |Z| \le v_n\sqrt{d}, \quad \mathbb{E}Z^2 \le \theta d\, v_n^2,$$

where

n	v_n	z_n	α_n
5	$3/8 + 23d/16$	$\theta(1+d)$	$(51/32+d)d^2$
≥ 6	$3/8 + 3d/2$	$79/256 + 13d/16 + d^2$	$(51+19d)d^2/32.$

Proof. The original spectral measure at u has mass d and $\int x = \delta_2$. Skew adjunction gives $\|U\|^2 = \int (a^2 + xb^2)$, $\mathbb{E}U = \int b$, and $\mathbb{E}KU = \int a$. For $n = 5$, $a' = 1 - 2x > 0$ and $b' = 1$. First maximize the coordinates in x and then use $\delta_2 \le \theta d$. The constant-vector squared norm is $131313/1048576 < 9/64$ and the linear-vector squared norm is $66997/32768 < (23/16)^2$; the latter margin is $715/32768$. Also $b \le \theta(1+d)$ and

$$\mathbb{E}Z = d^2 + (1+d)\delta_2 - \delta_3 \ge (51/32 + d)d^2.$$

For $n \ge 6$ put $u_\theta = 10101/32768$. The exact coordinates are bounded above by

$$x - x^2 + x^3 + d + \delta_2 + E_0(d - x), \qquad x - x^2 + x^3 + 2\delta_2 + dE_0.$$

They increase in x, with derivatives at least $1/3$ and $2/3$. At $x = \theta$, then $\delta_2 \le \theta d$, the bounding vector is

$$(u_\theta + (1+\theta)d + E_0(d - \theta), \sqrt{\theta}[u_\theta + 2\theta d + dE_0]).$$

Its squared norm is convex in $E_0 \in [0, d]$. At $E_0 = 0$, the constant norm is below $3/8$ and the linear squared norm is $18397/8192 < 9/4$. At $E_0 = d$, increase u_θ to $u_* = 79/256$. The difference

$$(3/8 + 3d/2)^2 - (u_* + d + d^2)^2 - \theta(u_* + 2\theta d + d^2)^2$$

has degree four and Bernstein coefficients on $[0, 1/3]$

$$\frac{14067}{2097152}, \quad \frac{67211}{2097152}, \quad \frac{1123003}{18874368}, \quad \frac{1216019}{18874368}, \quad \frac{30457}{6291456}.$$

All are positive. This proves the norm bound at both E_0 endpoints and hence for the complete interval. Further $b \le u_\theta + 2\theta d + d^2 < z_n$. Using $\beta_n \le d - \kappa\delta_2$, $a_N \ge 1 - x$, $E_0 \le d$, and $\delta_3 \le \theta\delta_2$ gives

$$\begin{aligned} \int a &\ge d^2 + \kappa(1+d)\delta_2 - E_0(\delta_2 - d^2) \\ &\ge d^2 + (\kappa - \theta d)\delta_2 + d^3 \ge (1 + \kappa + \kappa d)d^2. \end{aligned}$$

The last coefficient is positive throughout the full domain. Finally apply the original row and spectral bounds to KU. At $d = 0$ every vector appearing here vanishes. □

Proposition P.4 (One host-domain payment for every $(2, n)$, $n \ge 5$)**.** *On $0 \le d \le 1/3$, in the same canonical direction,*

$$H = pT(h_2 f_n), \quad U_{\text{out}} = T^2 H \qquad \Longrightarrow \qquad I_q(U_{\text{out}})^3 \le e^{-d/500}.$$

Proof. Put $m = 1 - d$, $v = v_n$, $z = z_n$, $\alpha = \alpha_n$. The actual square in (M.24) gives

$$T(h_2 f_n) \le m^2 q + m(\mathbb{E}U - KU) + \|U\|^2/2.$$

AM–GM between the two nonnegative original factors gives

$$0 \le H \le (c - Z/2)^2, \qquad c = 1 - d + dz/2 + dv^2/[4(1-d)].$$

Here $c \ge 2/3 > 7/16 \ge v\sqrt{d}/2$. Lemma P.1 therefore applies to the actual $X = -Z$, with cap $v\sqrt{d}$, mean upper bound $-\alpha$, and raw second bound $\theta d v^2$. Its moment compatibility follows from these same actual moments.

Let $d = t^2$, $a = \alpha/d$, $u = v$, and use the polynomial definitions $P, Q, L, B, N, \mathcal{D}$ in Section P.1. Take

$$\xi = \tfrac{1}{2}(24427/3420 + 1/500)d - \tfrac{5}{2}d^2, \qquad \mathcal{T} = \sum_{j=0}^{7} (-\xi)^j/j!.$$

The exact factor in both rows is t^2: $\mathcal{G} = (\mathcal{T}\mathcal{D} - N)/t^2$. For $n = 5$ take

$$C = 1024 - 1804d + 1300d^2 + 321d^3, \qquad D = 1024(1-d);$$

for $n \ge 6$ take

$$C = 512 - 927d + 785d^2 + 336d^3 - 256d^4, \qquad D = 512(1-d).$$

These are the actual $c = C/D$ above. On the single interval $0 \le t \le 577351/10^6$, the two polynomials have degrees $85, 99$, respectively. Their left values exceed $1/2$; after division by their left values, every one of their $86 + 100 = 186$ Bernstein coefficients is at least $9/10$.

Lemma A.1 proves $0 \le \mathbb{E}H^4 \le \mathcal{T} \le e^{-\xi}$; the last inequality is the odd Taylor lower bound, since $\xi \ge 0$. With the ordinary-fourth output factor of paragraph P.5, $B_2 \le e^{(24427/3420)d - 5d^2}$, this gives $I_q(U_{\text{out}})^3 \le B_2(\mathbb{E}H^4)^2 \le e^{-d/500}$. At $d = 0$ equality is evaluated directly. □

P.3 Parameters for the unit-root catalogue

In the table below, Z is the finite terminal seed, c is its exponential-envelope constant, and $D_i = \mathbb{E}B^i R$ is the continuation sequence after normalization. In an exponential row the scalar $e^{-\Gamma}$ is kept outside the first comparison.

Table 80: First-row parameters for all unit-root families.

Packet	Terminal upper bound and parameters	Normalized seed and scalar error
E	$e^{Z-\Gamma}$; $w = 1$, $r = 1$, $\lambda = 1$	$R = Z$, $D = \mathbb{E}Z + c\|Z\|^2$.
Q	$qe^{Z-\Gamma}$; $w = q$, $r = 2$, $\lambda = 1$	$R = Z$, $D = \mathbb{E}(qZ) + \sqrt{d}\|Z\| + 2c\|Z\|^2$.
A	$\bar{f}_a e^{Z-\Gamma}$; $r = a+1$, $\lambda = \mu_a = \mathbb{E}\bar{f}_a$	$R = \mu_a Z$, $D = \mu_a \mathbb{E}Z + 2\|\bar{f}_a - \mu_a \mathbf{1}\|\|Z\| + 2c\|Z\|^2$.
B	$\bar{f}_2 e^{Z-\Gamma}$; $r = 3$, $\lambda = 1$	$R = Z$, $D = \mathbb{E}(\bar{f}_2 Z) + \sqrt{d + \delta_2}\|Z\| + 2c\|Z\|^2$.
S	$(m + Z/2)^2$; $w = 1$, $r = 1$, $\lambda = 1$, $s_0 = m^2$	$R = Z/m$, $D = \mathbb{E}Z/m + \|Z\|^2/(2m^2)$.
QS	$q(m - Z/2)^2$; $w = q$, $r = 2$, $\lambda = 1$, $s_0 = m^2$	$R = -Z/m$, $D = -\mathbb{E}(qZ)/m + \sqrt{d}\|Z\|/m + \|Z\|^2/m^2$.

E applies to Sections P.5 and P.6 and the multipair low-domain case in Section P.9, where $q \le e^{-g}$ is absorbed into the seed. The inward-anchor cases of that section use A; its precise unit-anchor and five-arm cases use Q. The exact $(2,3,c)$ family with $c \ge 5$ uses B. For the centered A row at $a = 2$, $\lambda = 1-d$ and the variance is $d+\delta_2-d^2$; the B row has $\lambda = 1$ and squared distance $\|\bar{f}_2-\mathbf{1}\|^2 = d+\delta_2$.

The retained squares in Sections P.8 and P.7 use S and QS, respectively. For the reversed short directions in Section P.4, the first-square corollary uses inner kernel $-K$; the outer T^2 and its weight q stay original. That section proves the stronger inverse-fourth exit separately. Its remaining $(1,b)$ inputs with $b \ge 3$, $L \ge 2$ keep the finite data of Section I.1 and the three compatible square payments of Section P.1.

The local cases check the signed coefficients of (M.35) before discarding paths. Their G_2 exceptions and the stronger retained inequality in paragraph P.7 retain their stated ranges.

P.4 All terminal pairs with minimum one at a unit root and a two-edge outlet

For every terminal residual $(1,b)$, $b \ge 2$, every inner length $L \ge 1$, an internal root with exactly one residual unit side arm and an outer path of exactly two edges, the direction table below gives an actual output U with $I_q(U)^3 \le 1$, where $I_q(V) = \mathbb{E}[q^{-1/3}|V|^{8/3}]$. Equal pairs remain at their original vertices.

R26.2 Original finite facts and the cubic certificate

R26.2.1 The actual probability space and same-kernel moments

Throughout this proof, ER^2 means E[R^2]; the square of a mean is written $(ER)^2$. All such expectations are on the same original mu.

Delete zero-mass atoms only. Then p(x), $q(x) \ge \mu(x) > 0$ because the diagonal contribution has K(x, x) $= 0$ and all off-diagonal factors 1+/-K are nonnegative. Thus inverse weights are legitimate without a uniform lower bound on p, q.

The C014 calculus gives, on the original L2(mu),

$$0 \le d \le \tfrac{1}{3}, \quad d^2 \le \delta_2,$$
$$\|K\|^2 \le \theta = \tfrac{13}{32},$$
$$\delta_{i+1} \le \theta\delta_i.$$

Write $k_i = K^i 1, r = k_2, s = k_3, t = k_4, v = k_5$. Skew-adjointness gives

$$Er = -d, \quad Et = \delta_2, \quad Eg = Es = Ev = 0,$$
$$E(gr) = E(gt) = E(rv) = 0, \quad E(gs) = -\delta_2, \quad E(gv) = \delta_3.$$

The three further inequalities used below follow directly:

$$\delta_4 \ge \delta_2^2, \qquad d\delta_3 \ge \delta_2^2, \qquad \delta_3 \ge d\delta_2.$$

The last follows from the second and $\delta_2 \ge d^2$ when $d > 0$; $d = 0$ is immediate.

For $EV = 0$, subtract the row mean g(x) before Cauchy:

$$\begin{aligned} &|KV(x)|^2 \\ &\le E_y(K(x,y) - g(x))^2\, EV^2 \\ &\le (1-g(x)^2)EV^2 = p(x)q(x)EV^2. \end{aligned} \tag{R26.A.1}$$

For any original real R, a direct calculation gives the useful identity

$$T^2R = (ER)q + E(gR) + K^2R, \qquad T^21 = q + r = \bar{f}_2. \tag{R26.A.2}$$

In particular, a constant is propagated to $\bar{f}_2$,. Also, for any nonnegative R^2,

$$T^2(R^2) \le 2E(R^2)q, \tag{R26.A.3}$$

by first using $T(R^2) \le 2E(R^2)$, then applying the positive T.

R26.2.2 Derivation of the entire cubic expansion

Suppose $h = aq + j \ge 0, a > 0, j = KV, EV = 0$, and write

$$Ej = -\beta d, \quad C_1 = E(gj) \ge 0, \quad N = EV^2, \quad J_2 = Ej^2.$$

The parameters a and beta need not agree. From (R26.A.1),

$$E(j^4/q) \le NE(pj^2) \le 2NJ_2. \tag{R26.A.4}$$

Put

$$W = 2j - ag = K(2V - a), \qquad z = a + 2N/a.$$

The original row norm yields $|W| \le \sqrt{a^2 + 4N} \le z$, while

$$EW = -2\beta d, \qquad EW^2 = a^2d + 4J_2 - 4aC_1.$$

The pointwise polynomial (W+2 beta $d)^2(W - z) \le 0$ therefore gives

$$EW^3 \le (z - 4\beta d)EW^2 - 4z\beta^2d^2 + 8\beta^3d^3.$$

All cubic terms in the exact h^4/q expansion combine as

$$-a^4Eg^3 + 4a^3E(g^2j) - 6a^2E(gj^2) + 4aEj^3 = \frac{a}{2}\{EW^3 + a^2E(g^2W)\}.$$

Using $E(g^2W) \le zd$ and (R26.A.4), we obtain

$$\begin{aligned} E(h^4/q) \le \mathcal{A} &+ (8a^2 + 6N - 8a\beta d)J_2 \\ &-2a^2(4a + z - 4\beta d)C_1, \end{aligned} \tag{R26.A.5}$$

where

$$\begin{aligned} \mathcal{A} &= a^4 + 4a^3(a - \beta)d + 2a^2Nd - 2a^3\beta d^2 - 2a^2\beta^2d^2 \\ &\quad - 4N\beta^2d^2 + 4a\beta^3d^3 \\ &= a^4 + 2a^2d(N - \beta^2d) + 4a^3(a - \beta)d - 4N\beta^2d^2 \\ &\quad - 2a\beta d^2(a^2 - 2\beta^2d). \end{aligned} \tag{R26.A.6}$$

No fourth term or unfavorable cubic remainder is omitted, and the signed C_1 term is kept.

Cauchy applied to $Ej = - < g, V >$ gives $N \ge \beta^2$ d. Consequently

$$z - 4\beta d \ge a\{1 - 2d + 2d(\beta/a - 1)^2\} > 0.$$

Under $a \le \beta \le 6a/5$ and $N \le 4a$ beta d/3, every remainder discarded from (R26.A.5)-(R26.A.6) is nonpositive: in particular

$$a^2 - 2\beta^2 d \ge a^2/25, \qquad 6N - 8a\beta d \le 0.$$

This gives

$$E(h^4/q) \le a^4 + 8a^2 J_2 + 2a^2 d(N - \beta^2 d). \tag{R26.A.7}$$

If $a \le \beta$ is not available, retain $4a^3(a-\beta)_+ d$; this is precisely the positive-part variant of the displayed certificate.

For the broader variant, assume only $0 < a \le 1, 0 < \beta \le 1, C_1 \ge 0$ and

$$(a-\beta)_+ \le w, \qquad N \le nd, \qquad J_2 \le jd.$$

Dropping only the explicit negative terms of (R26.A.6), and retaining the positive cubic term as $4ad^3$, gives

$$E(h^4/q) \le a^4 + (4wa^3 + 8ja^2)d + (2na^2 + 6nj)d^2 + 4ad^3. \tag{R26.A.8}$$

This is the broader variant above.

The following coefficient formula provides an independent check on the later displayed polynomials. Substitution $a = 1 - \rho$ d in (R26.A.8) has coefficients, after the constant 1,

$$\begin{array}{c|l} d & -4\rho + 4w + 8j \\ d^2 & 6\rho^2 - 12w\rho - 16j\rho + 2n + 6nj \\ d^3 & -4\rho^3 + 12w\rho^2 + 8j\rho^2 - 4n\rho + 4 \\ d^4 & \rho^4 - 4w\rho^3 + 2n\rho^2 - 4\rho. \end{array} \tag{R26.A.9}$$

The comparisons in the sequel substitute into the monotone expression (R26.A.8) before using the resulting signed polynomial.

R26.3 New row $b = 2, L = 1$

Set $\gamma = d - \delta_2/2, c = 1 - \gamma/2$ and $\beta = c^2$. Since $p + \bar{f}_2 = 2 + r$,

$$p\bar{f}_2 \le (1 + r/2)^2, \qquad A(p\bar{f}_2) \le p - \gamma + s.$$

Apply Corollary M.20 to the inner operator $A = J + K$ (inner kernel $-K$ in the common notation). It gives

$$qA(p\bar{f}_2) \le (c + s/2)^2.$$

Using $T^2 s = -\delta_2 + v$, (R26.A.2)-(R26.A.3) give

$$\begin{aligned} 0 < U_* \le h &= aq + j, \\ a &= \beta - c\delta_2 + \delta_3/2, \\ V &= \beta g + c(t - \delta_2), \\ j = KV &= \beta r + c(v - \delta_2 g). \end{aligned} \tag{R26.A.10}$$

The exact covariance data are

$$Ej = -\beta d, \quad C_1 = c(\delta_3 - d\delta_2) \ge 0,$$

$$N = \beta^2 d + c^2(\delta_4 - \delta_2^2),$$

$$\begin{aligned} J_2 &= \beta^2\delta_2 + c^2(\delta_5 - 2\delta_2\delta_3 + \delta_2^2 d) \\ &\le \beta^2\delta_2 + c^2\delta_5. \end{aligned}$$

Both omitted cross terms vanish by skew parity, rather than by an independence assumption.

The certificate hypotheses are paid as follows:

$$\begin{gathered} \tfrac{5}{6} \le c \le 1, \quad \tfrac{25}{36} \le \beta \le 1, \\ \tfrac{a}{\beta} \ge \tfrac{67}{80} > \tfrac{5}{6}, \quad a \le \beta, \end{gathered}$$

$$\begin{aligned} &\frac{N}{a\beta d} \\ &\le \frac{6}{5}\left(1 + \frac{36}{25}\theta^3\right) \\ &< \frac{831}{625} < \frac{4}{3}. \end{aligned}$$

The divisions are used only for $d > 0$. Also $N \ge \beta^2 d$ is immediate from its exact expression.

Applying (R26.A.7), $\delta_5 \le \theta^3\delta_2$ and $\delta_4 \le \theta^3 d$ yields

$$\begin{aligned} &E(h^4/q) \le \\ &a^4 + a^2\left(\frac{559}{160}d + \frac{3}{20}d^2\right). \end{aligned} \tag{R26.A.11}$$

The exact scalar is

$$a = 1 - d - \frac{1}{2}\delta_2 + \frac{1}{4}d^2 + \frac{1}{4}d\delta_2 - \frac{3}{16}\delta_2^2 + \frac{1}{2}\delta_3.$$

Using $d^2 \le \delta_2$ and $d \le 1/3$ gives

$$a \le 1 - d + \frac{7}{192}\delta_2 \le 1 - \frac{6053}{6144}d \le 1 - \frac{63}{64}d.$$

For $0 \le z < 1$,

$$(1 - z)^{-2} \ge 1 + 2z + 3z^2.$$

With $\rho = 63/64, C = 559/160, D = 3/20$, this proves

$$\begin{aligned} &1 - (1 - \rho d)^2\{(1 - \rho d)^2 + Cd + Dd^2\} \\ &\ge (1 - \rho d)^2\{(4\rho - C)d + (2\rho^2 - D)d^2\}. \end{aligned}$$

Here $4\rho - C = 71/160, 2\rho^2 - D > 0$ and 1-rho $d \ge 43/64$. Thus the loss is at least

$$\frac{131279}{655360}d > \frac{1}{5}d.$$

It follows that $E(U_*^4/q) \le 1 - d/5$ on the entire legal host domain.

R26.4 The new finite inner lengths $L = 2$, 3, 4

R26.4.1 Two same-measure difference-moment facts

Since $E(r + d) = 0$,

$$\delta_3 - 2d\delta_2 + d^3 = \|K(r + d)\|^2 \le \theta(\delta_2 - d^2).$$

Subtracting this bound from the desired upper expression leaves $(1 - \theta - d)(\delta_2 - d^2) \ge 0$. Hence

$$\begin{aligned} \delta_3 &\le (1 + d)\delta_2 - d^2, \\ \delta_2 - \delta_3 &\ge d(d - \delta_2). \end{aligned} \tag{R26.A.12}$$

Let sigma be the spectral measure of g for -K^2. Cauchy for the nonnegative measure (1-x)d sigma gives

$$(\delta_2 - \delta_3)^2 \le (d - \delta_2)(\delta_3 - \delta_4).$$

For $d > 0$, (R26.A.12) then gives

$$\delta_3 - \delta_4 \ge d(\delta_2 - \delta_3). \tag{R26.A.13}$$

These are actual moment inequalities. They are not the false assertion that every original spectral point satisfies $x \ge d$.

R26.4.2 $L = 2$

Put

$$B = 1 - d + \delta_2/2, \quad \tau = d - \delta_2, \quad R = r + t, \quad V_R = ER^2 = \delta_2 - 2\delta_3 + \delta_4.$$

Positive propagation from the preceding row yields

$$A^2(p\bar{f}_2) \le Bp + R, \qquad qA^2(p\bar{f}_2) \le B(1 + R/(2B))^2.$$

Using $T^2 r = -dq + t$ and $T^2 t = \delta_2 q + k_6$ gives the actual upper bound

$$\begin{aligned} U_* &\le aq + j, \quad a = B - \tau + V_R/(2B), \\ V &= Bg + s + v, \quad j = KV = Br + t + k_6. \end{aligned} \tag{R26.A.14}$$

For $d > 0$ write $x = \delta_2/d, y = \delta_3/d$. The odd filter is $phi(z) = B - z + z^2$, and $\beta = B - x + y$. On $0 \le z \le \theta, 0 < phi(z) \le B \le 1$. Therefore

$$Ej = -\beta d, \quad C_1 = 0,$$

$$N = \int \phi^2 \, d\sigma, \quad J_2 = \int z\phi^2 \, d\sigma,$$

$$\beta^2 d \le N \le d, \qquad J_2 \le \theta d.$$

Equation (R26.A.12) gives $a \ge \beta > 0$. Also $V_R \le \delta_2$ and $2Btau > \delta_2$ show $0 \le a - \beta \le x \le \theta$.

To verify the nontrivial certificate condition, set $\kappa = 19/32$ and $m = B - \theta + \theta^2 \ge B/2$. Since $B - phi(z) \ge \kappa$ z,

$$N \le B\beta d - \kappa m\delta_2 \le \beta d(B - \kappa x/2) \le \beta d(B - \kappa d/2).$$

The exact sufficient comparison is

$$4a - 3B + \tfrac{3}{2}\kappa d \geq 1 - \frac{263}{64}d + \frac{9}{2}d^2 \geq \frac{25}{192} > 0.$$

It follows that $N \leq 4a$ beta d/3; $\beta/a \leq 1$ is already known.

For the scalar loss, log-convexity and $\delta_4 \leq \theta\delta_3$ give

$$V_R \leq dx\{1 - (2-\theta)x\}, \qquad B \geq (4+x)/6,$$

so

$$a \leq 1 - d\left[2 - \frac{3}{2}x - \frac{3x\{1-(2-\theta)x\}}{4+x}\right] \leq 1 - \frac{5}{4}d.$$

The final inequality is equivalent to $264x - 105x^2 \leq 96$. Its left side is increasing on [0, theta], and the endpoint margin is $6225/1024$.

The retained-remainder form of (R26.A.7) now gives

$$E(h^4/q) \leq a^4 + a^2\left(\frac{39}{8}d + 2d^2\right).$$

The same reciprocal-square comparison as in part R26.3, now with $\rho = 5/4, C = 39/8, D = 2$, supplies loss

$$(7/12)^2(1/8)d = \frac{49}{1152}d > \frac{1}{24}d.$$

Thus $E(U_*^4/q) \leq 1 - d/24$.

R26.4.3 $L = 3$, **including the propagated constant**

Let $\omega_2 = \delta_2 - \delta_3$ and retain B, tau. This time

$$A^3(p\bar{f}_2) \leq Bp + R - \tau, \qquad R = Br + s + v.$$

The root upper envelope is $B(1 + (R-\tau)/(2B))^2$. With

$$V_R = E(R-\tau)^2 = B^2\delta_2 + \delta_3 - 2\delta_4 + \delta_5 + 2\tau Bd + \tau^2,$$

the exact propagated upper bound is

$$\begin{aligned} a &= B(1-d) - \tau - \omega_2 + V_R/(2B), \\ j &= (B-\tau)r + Bt + v + k_7 - \omega_2 g, \\ V &= (B-\tau)g + Bs + (t + k_6 - \omega_2), \qquad j = KV. \end{aligned} \tag{R26.A.15}$$

The coefficient of r is B-tau, as it must be because the constant -tau propagates to -tau $\bar{f}_2$. Replacing it by -tau q would change this row.

The odd filter $phi(z) = B(1-z) - \tau$ is between $1/16$ and 1. Its mean coefficient is $\beta = (B - d)(1-x) > 0$, and

$$C_1 = \delta_3 - \delta_4 - d(\delta_2 - \delta_3) \geq 0$$

by (R26.A.13). The odd and centered even components of V are orthogonal. Their squared norms give

$$N \leq d + \delta_4 \leq \frac{43}{40}d, \qquad J_2 \leq \frac{559}{1280}d.$$

Since $h \geq U_{*} > 0$ and $Eh = a - \beta$ d, $a > 0$.

The exact scalar expansion is

$$a = 1 - 3d + 2d^{2} + (1 - 2d + \delta_{2}/4)\delta_{2} + \delta_{3} + \frac{\delta_{3} - 2\delta_{4} + \delta_{5} + \tau^{2}}{2B}.$$

Use $\delta_{3} - 2\delta_{4} + \delta_{5} \leq 4\delta_{2}/27$ and $B \geq 2/3$. After collecting the d terms before substituting $d \leq 1/3$, one obtains

$$\begin{aligned} &\frac{a - 1 + 3d}{d} \\ &\leq \frac{11}{12} + (\theta - 1/18)x + \frac{1}{3}x^{2} \\ &\leq \frac{325}{288} < \frac{5}{4}. \end{aligned}$$

Thus $a \leq 1 - 7d/4$. Also

$$a - \beta \leq Bx + d(1/4 - 3x/2).$$

For $x \geq 1/6$ this is at most x; for $x < 1/6, d \leq x$ makes it at most 5x/4. Hence $(a - \beta)_{+} \leq \theta$.

Apply (R26.A.8) with

$$\rho = \tfrac{7}{4}, \quad w = \theta, \quad n = \tfrac{43}{40}, \quad j = \tfrac{559}{1280}.$$

Formula (R26.A.9) gives exactly

$$\begin{aligned} &E(h^{4}/q) \leq \\ &1 - \frac{301}{160}d + \frac{66111}{25600}d^{2} \\ &+ \frac{1707}{2560}d^{3} + \frac{651}{2560}d^{4}. \end{aligned}$$

The three positive coefficients are respectively less than 3, 1, 1. Consequently the loss is at least

$$\begin{aligned} &\left(\frac{301}{160} - 1 - \frac{1}{9} - \frac{1}{27}\right) d \\ &= \frac{3167}{4320}d > \frac{7}{10}d. \end{aligned}$$

This proves the $L = 3$ row, retaining every cubic and quartic term.

R26.4.4 $L = 4$

Set $C = B - \tau = 1 - 2d + 3\delta_{2}/2 \geq 1/2$ and

$$R = B(r + s) + t + k_{6}.$$

The true propagated envelope is

$$A^{4}(p\bar{f}_{2}) \leq Cp + R - Bd, \qquad qA^{4}(p\bar{f}_{2}) \leq C(1 + (R - Bd)/(2C))^{2}.$$

With $V_{R} = E(R - Bd)^{2}$, (R26.A.2) gives

$$\begin{aligned} a &= C - 2Bd + (1 - B)\delta_{2} - \delta_{3} + V_{R}/(2C), \\ j &= (C - Bd)r + Bt + k_{6} + k_{8} + B(v - \delta_{2}g), \\ V &= (C - Bd)g + Bs + v + k_{7} + B(t - \delta_{2}). \end{aligned} \tag{R26.A.16}$$

Again, the constant -Bd contributes -Bd $\bar{f}_2$, including its r term.

The odd filter

$$\phi(z) = C - Bd - Bz + z^2 - z^3$$

decreases on [0, theta], because phi'≤-$B + 1/3 < 0$. Its endpoint has the strict original-parameter lower bound

$$\phi(\theta) \geq$$
$$(1-\theta) - (3-\theta)d + 2d^2 + \theta^2(1-\theta)$$
$$\geq \frac{14563}{294912} > 0.$$

It is also at most 1. The centered even component gives

$$0 < \beta \leq 1,$$
$$C_1 = B(\delta_3 - d\delta_2) \geq 0,$$
$$N \leq \frac{43}{40}d, \quad J_2 \leq \frac{559}{1280}d.$$

The even filter in R is $B - z + z^2$, between 0 and B, so

$$ER^2 \leq B^2(\delta_2 + \delta_3), \quad ER = -Bd + \delta_2 - \delta_3,$$

$$V_R \leq B^2(\delta_2 + \delta_3) + 3B^2d^2 - 2Bd(\delta_2 - \delta_3).$$

Since $B^2 - C \leq d^2/2$ and $C \geq 1/2, B^2/C \leq 1 + d^2 \leq 10/9$. Substitution gives

$$a \leq 1 - 4d + \frac{11}{3}d^2 + \frac{37}{18}\delta_2$$
$$-\frac{4}{9}\delta_3 - \frac{1}{2}\delta_2^2$$
$$\leq 1 - \frac{373}{192}d \leq 1 - \frac{7}{4}d.$$

Writing $\beta = C - Bd - Bx + y - z$ with $y = \delta_3/d, z = \delta_4/d$ and retaining $z \leq \theta$ y,

$$a - \beta \leq x + d(5/4 - 3x/2) \leq \frac{9}{4}x - \frac{3}{2}x^2 \leq \frac{1365}{2048} < \frac{2}{3}.$$

Apply (R26.A.8) with the previous n, j, rho and $w = 2/3$. The exact polynomial is

$$1 - \frac{403}{480}d - \frac{73889}{25600}d^2$$
$$+\frac{26207}{2560}d^3 - \frac{40922}{7680}d^4.$$

The last three coefficients are bounded above by -2, 11, 0. The convex function $-2d + 11d^2$ has maximum $5/9$ on [0, $1/3$]. Therefore

$$E(h^4/q) \leq 1 - \frac{409}{1440}d \leq 1 - \frac{1}{4}d.$$

This completes all finite inner lengths $L = 1, 2, 3, 4$ without a change of host-dependent direction.

R26.5 New row $b=3, L=1$

The genuine three-edge inward arm is

$$\bar{f}_3 = q + r - s - d.$$

Both the scalar -d here and both propagated δ_2 terms below are retained. Put $m = 1 - d/2, S = s - t$, $D = \delta_2 + \delta_3$. From $p + \bar{f}_3 = 2m + r - s$ and positivity,

$$p\bar{f}_3 \le (m + (r-s)/2)^2, \qquad A(p\bar{f}_3) \le m^2 p - md + mS + D/2.$$

The same first-square corollary, with inner kernel $-K$, gives

$$qA(p\bar{f}_3) \le (c + S/2)^2, \qquad c = 1 - d + \frac{\delta_2 + \delta_3}{4m}.$$

The correct two-step identity is

$$T^2 S = -\delta_2 - \delta_2 q + v - k_6.$$

Consequently

$$\begin{aligned} a &= c^2 - 2c\delta_2 + (\delta_3 + \delta_4)/2, \\ V &= c^2 g - cv + c(t - \delta_2), \\ j &= c^2 r - ck_6 + c(v - \delta_2 g). \end{aligned} \tag{R26.A.17}$$

The odd filter is $c^2 - cz^2$. The bounds

$$\begin{aligned} 1 - d \le c \le 1 - d + \frac{27}{64}\delta_2 \le 1, \\ c \ge 1 - d + (d^2 + d^3)/4 \ge \frac{19}{27} > \frac{7}{10} \end{aligned}$$

make it positive. Thus

$$\begin{aligned} \beta &= c^2 - c\delta_3/d > 0, \\ C_1 &= c(\delta_3 - d\delta_2) \ge 0, \\ N &\le \frac{43}{40}d, \quad J_2 \le \frac{559}{1280}d. \end{aligned}$$

Also $a \le c^2$, $(a - \beta)_+ \le \theta^2 \le \theta$, and positivity of h gives $a > 0$.

Using $\lambda = 1 - c$, the exact available scalar losses yield

$$a \le 1 - 2d + d^2 - \frac{2771}{10240}\delta_2 \le 1 - \frac{53971}{30720}d \le 1 - \frac{7}{4}d.$$

The parameters of (R26.A.8) are therefore exactly those of part R26.4.3. Its already checked polynomial proves

$$E(U_*^4/q) \le 1 - \frac{7}{10}d.$$

This is a separate actual-message calculation for terminal (1, 3), not a reuse of the terminal (1, 2), $L = 3$ message.

R26.6 Every original inner length $L \geq 5$ with terminal (1, 2)

R26.6.1 Finite convolution and the actual two losses

Define

$$\zeta_1 = \zeta_2 = 0,$$
$$\zeta_j = \sum_{i=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{i-1} \delta_i,$$
$$S_L = \sum_{j=0}^{L} k_j.$$

The identity $S_j = AS_{j-1}+\zeta_j$ follows by taking the actual mean of S_{j-1}. Iteration gives

$$f_L = S_L - \sum_{j=3}^{L} \zeta_j f_{L-j}, \qquad \tau \leq \zeta_j \leq d.$$

The latter bounds are finite alternating-sum bounds, with $\tau = d - \delta_2$. The finite Duhamel formula applied to r gives

$$A^L r = k_{L+2} - d f_{L-1} + \delta_2 f_{L-3} - \delta_3 f_{L-5} + \cdots,$$

with nonnegative indices only. Combining this with $p\bar{f}_2 \leq (1 + r/2)^2$ and $A(r^2) \leq 2\delta_2$ gives

$$\begin{aligned} A^L(p\bar{f}_2) \leq S_L + k_{L+2} - \gamma f_{L-1} \\ - \sum_{j=3}^{L} \omega_j f_{L-j}, \end{aligned} \tag{R26.A.18}$$

where $\omega_j = \zeta_j$ for even j and $\omega_j = \zeta_{j+2}$ for odd j. Every ω_j is at least τ. This verifies the truncation-end signs.

For $L \geq 5$ the $j = L$ and $j = L-1$ terms exist. Retain their tau multiples of f_0 and f_1 and discard only other nonnegative subtractions. With

$$B = 1 - \tau, \qquad R = \sum_{j=2}^{L} k_j + k_{L+2},$$

the resulting genuine nonnegative upper bound is

$$\begin{aligned} A^L(p\bar{f}_2) &\leq Bp + R - \tau, \\ H_* &\leq B(1 + (R - \tau)/(2B))^2. \end{aligned} \tag{R26.A.19}$$

These are two losses already present in the original path expansion; they are not additional tree arms.

R26.6.2 Every finite mask, rather than a length cutoff

Let $m = L - 1 \geq 4$ and $P_m(z) = G_m(z) + z^{m+1}$, where $G_m(z) = 1 + ... + z^{m-1}$. Then $R = KP_m(K)g$. For $z = i$ sqrt(x),

$$|P_m(z)|^2 = \frac{1 + x^m(1 - x + x^2) - 2\Re[z^m(1 - x - z)]}{1 + x}. \tag{R26.A.20}$$

This identity is obtained by putting the whole finite numerator over 1-z; it includes the final extra monomial.

The four residue classes of m give a complete proof:

Table 81: All terminal pairs with minimum one at a unit root and a two-edge outlet: parameter values.

Entry	Numerator bound in (R26.A.20)
0	At most 1; the subtraction is $2x^{m/2}(1-x)$..
1	At most 1; the subtraction is $2x^{(m+1)/2}$..
2	At most $1+2x^3+x^6$, because $m \geq 6$..
3	At most $1+2x^4+x^7$, because $m \geq 7$..

For the last two rows, $2\theta^2+\theta^5 < 1$ proves $|P_m| \leq 1$. For the second weighted bound the only larger required expression is

$$2x+6x^4+3x^7 \leq \frac{13}{16}+\frac{1}{6}+\frac{13}{2304} = \frac{2269}{2304} < 1.$$

The other bad-residue expression is smaller. The first two rows use $x/(1+x) \leq 13/45 < 1/3$. Thus every finite $m \geq 4$ satisfies

$$\begin{aligned} |P_m(i\sqrt{x})|^2 &\leq 1, \\ x|P_m(i\sqrt{x})|^2 &\leq \tfrac{1}{3}. \end{aligned} \tag{R26.A.21}$$

For $A_k(x) = \sum_{i=0}^{k-1}(-x)^i$,

$$1-x \leq A_k(x) \leq 1 \quad (k \geq 1),$$

and the exact real parts are

m	$\Re P_m$	$\Re(i\sqrt{x}P_m)$
$2h$	A_h	$-xA_{h+1}$
$2h+1$	A_{h+2}	$-xA_h$.

The relevant indices are at least two. Therefore, on the same sigma,

$$\begin{gathered} -d \leq ER \leq -\tau, \quad E(gR) \leq -(\delta_2-\delta_3), \\ ER^2 \leq d/3. \end{gathered} \tag{R26.A.22}$$

R26.6.3 The covariance is on the spectral measure of 1

It is necessary to retain the correct spectral measure in the sign argument. For every finite k,

$$A_k'(x) \geq -\frac{1}{(1-x)^2},$$

so, on [0, theta],

$$\begin{aligned} &\frac{d}{dx}(x^2A_k(x)) \\ &\geq x\left(2\kappa-\frac{\theta}{\kappa^2}\right) \geq 0, \\ &2\kappa^3-\theta = \frac{406}{32768} > 0. \end{aligned} \tag{R26.A.23}$$

Let nu be the original probability spectral measure of 1 for -K^2. Then d $\sigma = x$ d nu and integral x d $\nu = d$. For increasing F,

$$\begin{aligned} &\int (x-d)F(x)\,d\nu \\ &= \tfrac{1}{2}\iint (x-y)(F(x)-F(y))\,d\nu(x)d\nu(y) \geq 0. \end{aligned} \tag{R26.A.24}$$

The legitimate choice is $F = x^2 A_k$. This proves an integrated covariance sign even when nu has spectral mass below d.

R26.6.4 Actual T^2 propagation and its scalar payment

Put $\alpha = B - \tau = 1 - 2\tau$ and $V_R = E(R-\tau)^2$. Since $\tau \leq 2/9, \alpha \geq 5/9$. Expanding (R26.A.19) with (R26.A.2) gives

$$\begin{aligned} a &= \alpha + ER + E(gR) + V_R/(2B), \\ j &= \alpha r + K^2 R + E(gR)g, \\ V &= \alpha g + KR + E(gR), \qquad EV = 0. \end{aligned} \tag{R26.A.25}$$

With $\beta = \alpha - d^{-1}$ integral x $Re(P_m)d$ sigma,

$$Ej = -\beta d, \quad 0 < 5/9 - \theta \leq \beta \leq 1.$$

The exact covariance is

$$C_1 = E[(s+dg)R] = \int x^2(x-d)A_\ell(x)\,d\nu(x) \geq 0$$

by (R26.A.23)-(R26.A.24).

The vector V is the centered version of $(\alpha + K^2 P_m(K))g$. By $Re(P_m) \geq \kappa > 1/2$ and (R26.A.21),

$$\begin{aligned} N &\leq \int |\alpha - xP_m|^2\,d\sigma \\ &\leq \int (\alpha^2 - \alpha x + x^2)\,d\sigma \leq \alpha^2 d \leq d, \qquad J_2 \leq \theta d, \end{aligned}$$

because $x \leq \theta < \alpha$. Positivity of h again gives $a > 0$.

Combining the same ER terms before discarding signs is essential:

$$a = \alpha + \frac{\alpha}{B}ER + E(gR) + \frac{ER^2 + \tau^2}{2B}.$$

Equation (R26.A.22) gives

$$\begin{aligned} a &\leq 1 - 3\tau - (\delta_2 - \delta_3) \\ &\quad + \frac{d/6 + (3/2)\tau^2}{B}. \end{aligned} \tag{R26.A.26}$$

In particular,

$$a - \alpha \leq \frac{-\tau + (5/2)\tau^2 + d/6}{B} \leq -\frac{7d}{72B} \leq 0,$$

so $(a-\beta)_+ \leq \theta$.

For $x = \delta_2/d, y = \delta_3/d$, (R26.A.26) also gives

$$a \le 1 - d\left[3 - 2x - y - \frac{1/6 + (3/2)d(1-x)^2}{1 - d(1-x)}\right] \le 1 - \frac{3}{2}d.$$

To verify the last inequality over the entire domain, use $y \le \theta$ x, the monotonicity of the displayed fraction in d, and $d \le 1/3$. After multiplication by 2+x, the remaining expression is

$$1 - \frac{5}{16}x - \frac{125}{32}x^2 \ge \frac{7483}{32768} > 0.$$

In (R26.A.8), take $\rho = 3/2, w = \theta, n = 1, j = \theta$. The full polynomial is

$$E(h^4/q) \le 1 - \frac{9}{8}d + \frac{7}{8}d^2 + \frac{89}{32}d^3 - \frac{123}{64}d^4.$$

Discard the negative quartic term and use $d^2 \le d/3, d^3 \le d/9$. The loss is at least $151d/288 > d/2$. Thus, for every original integer $L \ge 5$,

$$E(U_*^4/q) \le 1 - \frac{1}{2}d.$$

R26.7 Every original terminal length $b \ge 4$ with $L = 1$

R26.7.1 The finite arm and its genuine envelope

From the finite convolution for T, retaining only its last nonnegative loss,

$$\bar{f}_b \le \sum_{j=0}^{b} (-K)^j 1 - \zeta_b.$$

Let

$$n = b - 1 \ge 3, \quad m = 1 - \zeta_b/2, \quad R = KG_n(-K)g = \sum_{j=2}^{b} (-K)^j 1,$$

$$t_0 = -ER = \zeta_{b+1}, \quad V_R = ER^2, \quad S = KR, \quad c = m - t_0/2 + V_R/(4m).$$

The sum $p + \bar{f}_b$ is bounded by 2m+R, which is nonnegative because it dominates that nonnegative sum. Thus

$$p\bar{f}_b \le (m + R/2)^2, \quad A(p\bar{f}_b) \le m^2 p - mt_0 + mS + V_R/2,$$

$$0 < H_* \le (c + S/2)^2. \tag{R26.A.27}$$

For every $n \ge 3$ the finite geometric sum has $|G_n(i\sqrt{x})| \le 1$ on [0, theta]. Odd n gives numerator $1 + x^n$; n divisible by four gives $(1 - x^{n/2})^2$; the remaining $n \ge 6$ satisfies $2x^{n/2-1} + x^{n-1} \le 2\theta^2 + \theta^5 < 1$. Therefore

$$ER^2 \le \delta_2, \qquad ES^2 \le \delta_3. \tag{R26.A.28}$$

Writing

$$G_n(-i\sqrt{x}) = B_0(x) - i\sqrt{x}A_0(x),$$

$$B_0 = A_{\lceil n/2 \rceil}, \quad A_0 = A_{\lfloor n/2 \rfloor},$$

both A_0, B_0 lie in [1-x, 1]. The exact means are

$$e = -ES = E(gR) = \int xA_0\, d\sigma, \quad \ell = -E(gS) = \int xB_0\, d\sigma,$$

$$\delta_2 - \delta_3 \le e, \ell \le \delta_2. \tag{R26.A.29}$$

In this row E(gR) is positive, unlike the long-inner-path row. The calculation retains this sign change.

The same finite arm supplies

$$1 - d + \delta_2/2 \le c \le 1 - d + \frac{13}{10}\delta_2 \le 1 - \frac{151}{320}d.$$

For the lower bound, when $b = 4$ the two zeta terms sum to $2d - \delta_2$. For $b \ge 5$ their sum is at most $2d - 2\delta_2 + 2\delta_3 \le 2d - \delta_2$. Thus $c \ge 13/18$; in particular $c > \theta^2$.

R26.7.2 The necessary variance improvement survives

Use $T^2S = -eq - \ell + K^2S$ in (R26.A.27). The actual nonnegative upper bound is

$$\begin{aligned} a &= c^2 - c(e + \ell) + ES^2/2, \\ j &= c^2 r + cK^2S - c\ell g, \\ V &= c^2 g + cKS - c\ell, \qquad EV = 0, \quad j = KV. \end{aligned} \tag{R26.A.30}$$

Its odd Krylov component is $(c^2 - cx^2A_0(x))g$. Its centered even component is $c(K^3B_0(-K^2)g - \ell)$. They are orthogonal. Consequently, with $\xi = d^{-1}$ integral x^2A_0 d sigma,

$$Ej = -\beta d, \quad \beta = c^2 - c\xi, \quad 0 < \beta \le 1.$$

The same original spectral covariance proof (R26.A.23)-(R26.A.24) gives

$$C_1 = c \int x^2(x - d)B_0(x)\, d\nu(x) \ge 0.$$

The stronger variance bound is not optional. Put $z = x^2A_0(x)$, so $0 \le z \le \theta^2 < c$. Then

$$(c^2 - cz)^2 \le c^4 - c^3 z.$$

Using $A_0 \ge \kappa$ and the centered even variance at most $c^2\delta_4$,

$$N \le c^4 d - c^3\kappa\delta_3 + c^2\delta_4 \le c^4 d + c^2(\theta - c\kappa)\delta_3 \le c^4 d,$$

since c $\kappa \ge 247/576 > \theta = 234/576$. For $\lambda_0 = 151/320$, the complete fourth expansion, with $d \le 1/3$, gives

$$c^4 \le (1 - \lambda_0 d)^4 \le 1 - (4\lambda_0 - 2\lambda_0^2)d \le 1 - \frac{7}{5}d.$$

The cubic-plus-quartic remainder discarded here is -$\lambda_0^3 d^3(4 - \lambda_0 d) \le 0$, and $4\lambda_0 - 2\lambda_0^2 = 73839/51200 > 7/5$. Therefore

$$\begin{aligned} N &\le d(1 - 7d/5), \\ J_2 &\le \theta d(1 - 7d/5). \end{aligned} \tag{R26.A.31}$$

Equations (R26.A.28)-(R26.A.29) also give $a \le c^2$, because 2c $\kappa > \theta/2$. Hence $(a - \beta)_+ \le c\xi \le \theta^2$. As in the other rows, positivity of h and $Eh = a - \beta$ d give $a > 0$.

R26.7.3 Scalar loss and the retained second-order budget

Let $\lambda = 1 - c$. Then $d - 13\delta_2/10 \le \lambda \le d$, and

$$a \le (1-\lambda)^2 - 2(1-\lambda)(\delta_2 - \delta_3) + \delta_3/2.$$

This expression decreases with lambda because $\lambda + (\delta_2 - \delta_3) \le d + \delta_2 < 1$. Substitute its lower bound. The resulting δ_3 coefficient is positive, allowing $\delta_3/d \le \theta$ x. The exact result is

$$a \le 1 - d\left[2 - \frac{517}{320}x - d(1-13x/10)(1-9x/80)\right].$$

Both factors in the last product are nonnegative. With $d \le 1/3$, the bracket is at least

$$\begin{aligned}&\frac{5}{3} - \frac{1099}{960}x - \frac{39}{800}x^2\\ &\ge \frac{2933267}{2457600} > \frac{19}{16}.\end{aligned}$$

Thus $a \le 1 - 19d/16$.

In (R26.A.5)-(R26.A.6), retain (R26.A.31) specifically in $8a^2J_2$ and use the coarser $N \le d$, $J_2 \le \theta$ d only elsewhere. This gives

$$\begin{aligned}&E(h^4/q) \le\\ &a^4 + 4\theta^2 a^3 d\\ &+a^2\left(8\theta d - \frac{51}{20}d^2\right)\\ &+6\theta d^2 + 4ad^3.\end{aligned} \tag{R26.A.32}$$

The coefficient in parentheses is at least 12d/5, so substitution of the upper bound for a is legitimate. The complete resulting polynomial is

$$\begin{aligned}&1 - \frac{215}{256}d - \frac{35269}{20480}d^2\\ &+\frac{3517247}{327680}d^3 - \frac{39126719}{5242880}d^4.\end{aligned}$$

Direct expansion gives these coefficients. The second, third and fourth coefficients are respectively at most -3/2, 11,-7. The function -$3d/2 + 11d^2 - 7d^3$ is convex on [0, $1/3$], since its second derivative is $22 - 42d \ge 8$. Its endpoint maximum is $25/54$. Therefore

$$\begin{aligned}&E(h^4/q) \le 1-\\ &\left(\frac{215}{256} - \frac{25}{54}\right)d\\ &= 1 - \frac{2605}{6912}d\\ &\le 1 - \frac{3}{8}d.\end{aligned}$$

This proves every finite $b \ge 4$. It would not be justified to replace (R26.A.31) by $N \le d$ throughout: the retained negative quadratic coefficient in (R26.A.32) is part of the proof.

R26.8 The exact old C094 row and its three ordinary-fourth inputs

The following argument proves the required bridge and its scalar estimates.

R26.8.1 The unweighted fourth-to-Iq bridge

For any original positive kernel operator B, $h = B1$, and any original real Y, row Cauchy and original-measure Holder give

$$|BY|^2 \le hB(|Y|^2),$$

$$I_q(BY)^3 \le E(h^4/q)\,\|B\|^4 (E|Y|^4)^2. \tag{R26.A.33}$$

This is an ordinary L2(mu) operator norm and an ordinary fourth input. There is no change of adjoint or input weight.

For $B = T^2, h = \bar{f}_2 = q + r$. Taking $W = 2r - g$ and $z = 1 + 2d$ in the cubic argument, and retaining the actual row relation $r^2 \le pqd$, gives

$$E(\bar{f}_2^4/q) \le 1 - 2d^2 + (8 - 2d)\delta_2. \tag{R26.A.34}$$

More explicitly,

$$E(\bar{f}_2^4/q) = 1 - d + 6\delta_2 + \mathcal{C} + E(r^4/q),$$

$$\mathcal{C} \le d - 2d^2 + 2(1 - 2d)\delta_2, \quad E(r^4/q) \le 2d\delta_2.$$

These reproduce C094 (2.1)-(2.4).

The exact C064 part Q64.7.2 norm proof is compatible with this original space. On the cyclic subspace generated by 1 under K, split even and odd Krylov vectors. Multiplication of T^2 by the orthogonal sign operator $\mathrm{diag}(I_{\text{even}}, -I_{\text{odd}})$ gives the self-adjoint block matrix

$$\begin{pmatrix} J - S_{\text{even}} & 1 \otimes g \\ g \otimes 1 & S_{\text{odd}} \end{pmatrix}, \qquad S = -K^2.$$

The orthogonal complement has $T^2 = K^2$ and norm at most theta. For $\lambda \ge 1$, the positive and negative Schur complements reduce the only nonautomatic condition to

$$\lambda h_\lambda t_\lambda \le 1,$$

$$h_\lambda = \int \frac{d\nu(x)}{\lambda - x},$$

$$t_\lambda = \int \frac{d\nu(x)}{\lambda + x}.$$

Writing $\alpha = intx^2/(\lambda^2 - x^2)d$ nu and $\eta = intx/(\lambda^2 - x^2)d$ nu,

$$\lambda h_\lambda t_\lambda$$

$$= \frac{(1+\alpha)^2}{\lambda} - \lambda\eta^2$$

$$\le \frac{(1+\alpha)^2}{\lambda}.$$

For $k_2 = (1 - \theta^2)^{-1} = 1024/855$ and $\lambda = (1 + k_2\delta_2)^2, \alpha \le k_2\delta_2$. Thus

$$\|T^2\| \le (1 + 1024\delta_2/855)^2.$$

Together with (R26.A.33)-(R26.A.34), this gives exactly

$$I_q(T^2Y)^3 \le B_2(E|Y|^4)^2,$$

$$B_2 = \{1 - 2d^2 + (8 - 2d)\delta_2\}(1 + 1024\delta_2/855)^8. \tag{R26.A.35}$$

For the claimed logarithmic cost, the first factor is at most $R = 1 + (13/4)d - (45/16)d^2$. If $u = R - 1$, then $0 \le u \le 37/48$ and $u \ge (37/16)d$. The elementary integral remainder inequality $\log(1 + u) \le u - u^2/[2(1+u)]$ gives

$$\log R \le \frac{13}{4}d - \frac{11757}{2720}d^2.$$

For $\lambda_2 = 416/855, 12/25 < \lambda_2 < 1/2$ and

$$8\log(1 + \lambda_2 d) \le 8\lambda_2 d - \frac{3}{4}d^2.$$

Since $11757/2720 + 3/4 > 5$, the precise result is

$$B_2 \le \exp\left(\frac{24427}{3420}d - 5d^2\right). \tag{R26.A.36}$$

R26.8.2 Which actual messages enter this bridge

The old row uses, for all $b \ge 3, L \ge 2$,

$$Y = pT^L(qf_b).$$

C058 fixes this same actual direction: terminal unit arm toward the terminal, long arm outward, inner connector toward the internal root, internal-root unit arm outward. The two outer edges remain directed from that root to the receiver.

The actual square-envelope estimates of Q5-C058 supply the following same-input data to Q5-C094:

Table 82: All terminal pairs with minimum one at a unit root and a two-edge outlet: parameter values.

Entry	Actual upper envelope $Y \le (c + N/2)^2$
$L \ge 4$	$c = 1-3d/4, N = R_L$; Pointwise and moment data: $N^2 \le d, EN \le -d/2, EN^2 \le \theta d$.
$L = 3$	$c = 1 - 4d/5, N = V_3 - \gamma r - q$; Pointwise and moment data: $N^2 \le (71/63)d, EN^2 \le \theta(71/63)d, EN \le -(4/5 - d)d$.
$L = 2$	$c = 1 - 37d/64, N = mR_2/\sqrt{B}, R_2 = r + kK^2R, B = 1 - \gamma$; Pointwise and moment data: $N^2 \le d, EN \le -41d/96, EN^2 \le 3d/10$.

The scalar envelope base is nonnegative because it arises from a sum of actual nonnegative factors. The L3 mean improvement uses the same R: $V \ge t_0^2, u = kt_0 \le d/(1 - d/2)$, and

$$\gamma \le u - u^2/2 \le d(1 - d)/(1 - d/2)^2 \le d.$$

For $L = 2$, the original ratio $m^2/B \le 9/8$ and $ER_2^2 \le 4d/15$ are retained, giving $EN^2 \le 3d/10$. These are the exact finite-envelope premises of Q5-C058 for the stated direction.

R26.8.3 Three compatible square moments

Use the three actual envelopes and their same-input moment data from paragraph R26.8.2. They are rows I, II and III of Table 78, for $L \geq 4$, $L = 3$ and $L = 2$, respectively. Their bases satisfy $c > U/2$; the compatibility $\alpha^2 \leq hU^2$ also follows from the actual first and second moments. Lemma P.1 and the three continuous certificates in Table 79 give

$$\mathbb{E}Y^4 \leq (1 - 15d/32)^8, \qquad L \geq 4, \tag{R26.A.39}$$

$$\mathbb{E}Y^4 \leq (1 - 9d/20)^8, \qquad L = 3, \tag{R26.A.40}$$

$$\mathbb{E}Y^4 \leq (1 - 15d/32 + 3d^2/8)^8, \qquad L = 2. \tag{R26.A.41}$$

These bounds hold on $0 \leq d \leq 1/3$ for the original direction $Y = pT^L(qf_b)$, $b \geq 3$. At $d = 0$, its actual leaf integration gives equality.

R26.8.7 Final exponents and the literal old union

Use only (R26.A.39)-(R26.A.41) in (R26.A.35)-(R26.A.36). For $L = 2$,

$$\log I_q(T^2Y)^3 \leq -\frac{1223}{3420}d + d^2 \leq -\frac{83}{3420}d.$$

For $L = 3$ the linear coefficient is -197/3420, and for $L \geq 4$ it is -1223/3420; both also retain the nonpositive $-5d^2$ term. Thus the old row uniformly supplies

$$I_q(T^2[pT^L(qf_b)])^3 \leq e^{-83d/3420} \qquad (b \geq 3, L \geq 2).$$

The transcription of B_2, all three ordinary-fourth inputs, the inner direction and the final exponent is exact. No inverse-fourth conclusion for this old direction follows from this argument, and none is included in this derivation.

R26.9 Complete local directions, original pairs and reversal

The complete direction table is:

Table 83: All terminal pairs with minimum one at a unit root and a two-edge outlet: parameter values.

Entry	Inner L
(1, 2)	1; Actual bare root H: $qA(p\bar{f}_2)$; New inverse fourth: $\leq 1-d/5$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-d/5)^2$.
(1, 2)	2; Actual bare root H: $qA^2(p\bar{f}_2)$; New inverse fourth: $\leq 1 - d/24$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-d/24)^2$.
(1, 2)	3; Actual bare root H: $qA^3(p\bar{f}_2)$; New inverse fourth: $\leq 1 - 7d/10$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-7d/10)^2$.
(1, 2)	4; Actual bare root H: $qA^4(p\bar{f}_2)$; New inverse fourth: $\leq 1 - d/4$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-d/4)^2$.
(1, 2)	every $L \geq 5$; Actual bare root H: $qA^L(p\bar{f}_2)$; New inverse fourth: $\leq 1 - d/2$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-d/2)^2$.
(1, 3)	1; Actual bare root H: $qA(p\bar{f}_3)$; New inverse fourth: $\leq 1 - 7d/10$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-7d/10)^2$.
(1, b), every $b \geq 4$	1; Actual bare root H: $qA(p\bar{f}_b)$; New inverse fourth: $\leq 1 - 3d/8$; $Iq(U)^3$ bound, $U = T^2H$: $\leq (1-3d/8)^2$.

Entry	Inner L
(1, b), every $b \geq 3$	every $L \geq 2$; Actual bare root H: $pT^L(qf_b)$; New inverse fourth: not bounded; $Iq(U)^3$ bound, $U = T^2H$: $\leq \exp(-83d/3420)$.

The integer partition is exhaustive: $b = 2$ uses the first five rows; if $b \geq 3$, either $L = 1$ uses one of the next two rows or $L \geq 2$ uses the last row. All tests in this direction selection depend only on original integer lengths. The analytic d intervals estimate this fixed direction.

For the new rows the literal directions are:

- The terminal's original unit arm points away from the terminal.
- Its complete original b-edge arm points toward the terminal.
- Every one of the L inner edges points from the internal root toward the terminal.
- The internal root's one residual unit arm points toward that root.
- Both original outer edges point from the internal root toward the receiver.

At a root, a genuine equal pair contributes $f_l h_l \leq 1$. Full reversal of the pair contributes $\bar{f}_l W_l \leq 1$, $W_l = A\bar{f}_{l-1}$. These are actual two-arm products, not arbitrary functions chosen to satisfy a budget. Thus the new decorated message is exactly

$$H_D = E_r^- qA^L(E_t^- p\bar{f}_b), \qquad U_D = T^2 H_D.$$

The terminal factor stays inside A^L; the root factor stays at that root. Original nonnegativity gives $0 < U_D \leq U_*$ and therefore preserves every new inverse-fourth bound. The old row has, instead,

$$H_{o,D} = E_r^+ pT^L(E_t^+ qf_b), \qquad U_{o,D} = T^2 H_{o,D},$$

with the exact same positivity comparison for its ordinary-fourth and Iq bounds.

The outer operator is the actual original two-edge kernel:

$$T^2H(x) = \sum_{z,y} \mu(z)\mu(y)(1 - K(x,z))(1 - K(z,y))H(y).$$

By antisymmetry these factors are exactly the directed edges $y- > z- > x$. No independent row choices or tilted measures have been introduced.

Complete component reversal substitutes -K simultaneously in every factor. It swaps p, q and A, T, leaves d unchanged, reverses every decoration, and changes the outer T^2 to A^2. Thus the new reversed message is

$$A^2[E_r^+ pT^L(E_t^+ qf_b)]$$

with denominator p and the corresponding I_p budget. The reversed old row is $A^2[E_r^- qA^L(E_t^- p\bar{f}_b)]$ and has the corresponding old I_p statement only. Reversing just the inner component does not effect this global weight swap.

P.5 All even terminal lists with at least two unequal pairs at a unit root

Let $a_1 < b_1 < \cdots < a_j < b_j$ with $j \geq 2$. Put $F = \prod_i h_{a_i} f_{b_i}$ and $H = pT^L F$, where $L \geq 1$. Then

$$\mathbb{E}H^4 \leq \exp[-(15/4 + 64(j-2)/5)d], \quad I_q(T^2H)^3 \leq \exp[-(1223/3420 + 128(j-2)/5)d - 5d^2].$$

The same statements hold with genuine equal-pair factors at the terminal and unit internal root. The following common calculus, including the ordinary fourth-moment transfer, is also used in Section P.6.

R29.2 Actual messages and finite calculus

All operators, inner products, adjoints and expectations use one unchanged original finite probability measure. Write

$$Jv = (Ev)\mathbf{1}, \quad A = J + K, \quad T = J - K = A^*, \quad B = -K, \quad u = B\mathbf{1} = -g,$$
$$p = 1 - u, \quad q = 1 + u, \quad f_n = A^n\mathbf{1},$$
$$\bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}.$$

An f arm points entirely outward. An h arm has its first edge toward the terminal and its remaining edges outward. The actual root unit arm points outward and supplies p. Every inner edge points from terminal to root and supplies T^L. The two outer edges point from root to receiver and supply T^2. Thus the bare actual messages are

$$F = \prod_i h_{a_i} f_{b_i}, \qquad H = pT^L F, \qquad U = T^2 H. \tag{R29.2.1}$$

For R2-C031 the product has exactly one factor, with its stated restrictions. The root unit arm supplies the factor p. Additional root-arm factors are treated in Section Q.12.

Put

$$\delta_i = \|B^i\mathbf{1}\|_2^2, \quad d = \delta_1,$$
$$\theta = \frac{13}{32}, \quad \kappa = \frac{19}{32}, \quad \tau = d - \delta_2.$$

The finite calculus gives

$$0 \le d \le \frac{1}{3}, \quad d^2 \le \delta_2 \le \theta d,$$
$$\|B\|^2 \le \theta, \quad \delta_{i+1} \le \theta\delta_i, \tag{R29.2.2}$$
$$|Bv(x)| \le \|v\|_2.$$

The lower bound $\delta_2 \ge d^2$ also follows from $EB^2 1 = -d$ and Cauchy. The spectral measure of -B^2 at 1 is a probability measure, so $\delta_3 \ge d^3$.

After zero-mass atoms are removed, antisymmetry gives K(x, x) = 0, whence p(x), $q(x) \ge \mu(x) > 0$. Every actual rooted tree message is positive: assigning all integrated vertices the root's label produces a positive term with every diagonal factor equal to one. The finite path calculus also gives $0 < f_n, \bar{f}_n, h_n \le 2$ and $Ef_n = E\bar{f}_n \le 1$.

The spectral measure nu of -B^2 at u has mass d and $\int x^r\, d\nu = \delta_{r+1}$, supported on [0, theta]. At $z = i$ sqrt(x), if a real polynomial Q has coordinates $Q(z) = E(x) + zD(x)$, then

$$E[Q(B)u] = -\int D(x)\, d\nu(x),$$
$$\|Q(B)u\|^2 = \int (E(x)^2 + xD(x)^2)\, d\nu(x). \tag{R29.2.3}$$

R29.3 All finite path and unequal-pair estimates

R29.3.1 Finite convolution and a uniform mean loss

Define $\beta_1 = \beta_2 = 0$ and

$$\beta_n = \sum_{r=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{r-1}\delta_r \quad (n \geq 3).$$

The exact finite convolution is

$$\begin{aligned} S_n &= \sum_{r=0}^{n} K^r 1 \\ &= f_n + \sum_{r=3}^{n} \beta_r f_{n-r}, \\ &\tau \leq \beta_r \leq d. \end{aligned} \tag{R29.3.1}$$

The same identity holds after full reversal for $\bar{f}_n$ and B. The beta coefficients are nonnegative by decreasing alternating moments.

Writing $\mu_n = Ef_n$, the exact initial means are

$$\begin{aligned} &\mu_2 = 1 - d, && \mu_3 = 1 - 2d, \\ &\mu_4 = 1 - 3d + \delta_2, && \mu_5 = 1 - 4d + 2\delta_2 + d^2, \\ &\mu_6 = 1 - 5d + 3\delta_2 + 3d^2 - \delta_3, && \mu_7 = 1 - 6d + 4\delta_2 + 6d^2 - 2\delta_3 - 2d\delta_2. \end{aligned}$$

Their losses divided by d, for $n = 4, 5, 6, 7$, are at least respectively $83/32, 137/48, 89/32, 127/48$. For $n = 7$, for example, discard the positive $2\delta_3$ and use

$$6d - (4 - 2d)\delta_2 - 6d^2 \geq [6 - 4\theta - (6 - 2\theta)d]d \geq \frac{127}{48}d.$$

For every $n \geq 8$, retain the six actual terms corresponding to $\mu_0, ..., \mu_5$ in (R29.3.1). The three coefficients $\beta_{n+1}, \beta_n, \beta_{n-1}$ are at least tau+kappa δ_3. Consequently

$$\begin{aligned} 1 - \mu_n &\geq \tau(7 - 10d + 3\delta_2 + d^2) + 3\kappa\delta_3 \\ &\geq \kappa d(7 - 10d + 7d^2) \geq \frac{95}{36}d > \frac{83}{32}d. \end{aligned} \tag{R29.3.2}$$

This proves $\mu_n \leq 1 - 83d/32$ for every $n \geq 4$.

R29.3.2 The difference formula and every unequal pair

For $a < b$ let $R_{ab} = \sum_{r=a+1}^{b} K^r 1$. Subtracting two finite convolutions gives exactly

$$\begin{aligned} f_b - f_a &= R_{ab} \\ &+ \sum_{i=0}^{a-3} (\beta_{a-i} - \beta_{b-i}) f_i \\ &- \sum_{i=\max(0,a-2)}^{b-3} \beta_{b-i} f_i. \end{aligned} \tag{R29.3.3}$$

Empty sums are zero. At $a = 1$ the identity reads

$$f_b - f_1 = R_{1b} - \sum_{i=0}^{b-3} \beta_{b-i} f_i. \tag{R29.3.3a}$$

Since $h_1 + f_1 = 2$, retaining the genuine $f_0 = 1$ term gives $h_1 + f_b \le 2 + R_{1b} - \beta_b$ for $b \ge 3$. For $b = 2$ the sum is empty and $\beta_2 = 0$, yielding exactly the same bound. AM–GM and $\log t \le t - 1$ then give $h_1 f_b \le \exp(R_{1b} - \beta_b)$. This proves the $a = 1$ row.

In the first sum of (R29.3.3) a positive coefficient requires an odd starting alternating-moment index; its first omitted even-index moment is δ_{2r}. Each such first moment occurs at most twice, and $f_i \le 2$. Therefore the positive part of that entire sum is at most $4\delta_2/(1 - \theta^2)$. For $a \ge 5$, (R29.3.2) and $f_a + h_a = 2\mu_{a-1}$ yield

$$\begin{aligned} h_a + f_b &\le 2 + R_{ab} - \left(\frac{83}{16} - \frac{1664}{855}\right) d, \\ \frac{83}{16} - \frac{1664}{855} &= \frac{16}{5} + \frac{113}{2736}. \end{aligned} \tag{R29.3.4}$$

For $a = 4$ retain the actual negative term -tau f_2 in (R29.3.3):

$$f_b - f_4 \le R_{4b} + \delta_2(1 + p) - \tau f_2.$$

The original row identity

$$2Kg(x) = E_{y,z}[(K(x, y) - K(x, z))K(y, z)]$$

and $|a - b| \le 1 - ab$ for a, b in [-1, 1] give $|Kg| \le pq/2$. Hence $f_2 = p + Kg \ge p^2/2$, and completing this real quadratic gives

$$\begin{aligned} &\delta_2(1 + p) - \tau f_2 \\ &\quad \le \delta_2 + \frac{\delta_2^2}{2\tau} \\ &\le \frac{663}{1216} d < \frac{3}{4} d. \end{aligned} \tag{R29.3.5}$$

The $a = 4$ pair sum therefore loses at least 13d/4. For any positive pair X, Y, $XY \le ((X+Y)/2)^2$ and $\log t \le t - 1$ give $\log(XY) \le X + Y - 2$. Equations (R29.3.4)-(R29.3.5) establish the common strong pair bound

$$h_a f_b \le \exp\left(R_{ab} - \frac{16}{5} d\right) \qquad (4 \le a < b). \tag{R29.3.6}$$

The short finite differences give, with no additional arm,

$$\begin{aligned} h_1 f_b &\le e^{R_{1b} - \beta_b}, \\ h_2 f_b &\le e^{R_{2b} - \beta_b}, \\ h_3 f_b &\le e^{R_{3b} - (2d - \delta_2)}. \end{aligned} \tag{R29.3.7}$$

For $a = 3, b \ge 4$, retain -$\beta_{b-1} f_1 \le -\tau$ p:

$$h_3 f_b \le \exp\{R_{3b} + \tau u - (3d - 2\delta_2)\}. \tag{R29.3.8}$$

For the exact pair (2, 3),

$$h_2 + f_3 = 2 - d + K^3 1, \qquad h_2 f_3 \le e^{-d+K^3 1}. \tag{R29.3.9}$$

The retained tau u in (R29.3.8) is the scalar correction for the same kernel.

R29.4 The signed-mean exponential certificate

The full signed-mean lemma of Section M.1, paragraph M.1, gives

$$\mathbb{E}e^{4BV} \le e^{4\mathbb{E}BV + (17/4)\|V\|^2}. \tag{R29.4.1}$$

Its scalar statement is

$$\mathbb{E}e^{sW} \le e^{s\mathbb{E}W + (17/64)s^2U^2} \quad (|W| \le U,\ \mathbb{E}W^2 \le \theta U^2,\ s \ge 0). \tag{R29.4.2}$$

These are the signed-mean inequalities proved in paragraph M.1. The complete-vector payment is (M.38).

Use Lemmas M.15 and M.22.

R29.6 R2-C029: exhaustive finite masks and all later pairs

R29.6.1 One mask for every finite number of pairs

Put

$$C_\beta = \frac{1}{1-\theta^2} = \frac{1024}{855}, \qquad C_* = \frac{32768}{16245}.$$

For every finite polynomial P with coefficients in {0, 1}, separate the positive and negative terms of each real/imaginary coordinate at i sqrt(x). Each coordinate's absolute value is bounded by the larger one-sided geometric tail, not the sum of both tails. This gives

$$\begin{aligned} |P(i\sqrt{x})|^2 &\le \frac{1+x}{(1-x^2)^2} \\ &\le C_* < \frac{101}{50} < \frac{100}{49}. \end{aligned} \tag{R29.6.1}$$

The exponent intervals $[a_i + 1, b_i]$ are disjoint because the actual arm lengths are strictly ordered. Their union therefore makes one finite 0-1 mask, regardless of j or any gap. The geometric tails bound these finite sums with a norm bound independent of j.

Useful exact constants are

$$\begin{aligned} C_\beta\theta &= \frac{416}{855} < \frac{1}{2}, \\ C_\beta\theta^2 &= \frac{169}{855} < \frac{1}{5}, \\ C_\beta\theta^3 &= \frac{2197}{27360} < \frac{1}{12}, \\ C_\beta\theta^4 &= \frac{28561}{875520} < \frac{1}{30}. \end{aligned} \tag{R29.6.2}$$

Also $\theta^2 < 1/6, \theta^3 < 1/14$, and $\theta^{3/2} < 4/15$. Let $\Delta = (16/5)(j-2)d$. The following five cases are a structural partition of every $j \ge 2$ table:

Table 84: The five finite even-terminal packets, with $R_0 = \sum_i R_{a_i b_i}$ and $\Delta = (16/5)(j-2)d$.

Packet	Arm condition	R	Lower Γ
E2	$a_1 = 2$	R_0	$(607/160)d + \Delta$
E3	$a_1 \ge 3$	R_0	$(767/160)d + \Delta$
I	$a_1 = 1,\ b_1 \ge 3$	R_0	$(607/160)d + \Delta$
II	$a_1 = 1,\ b_1 = 2,\ a_2 \ge 4$	R_0	$(16/5)d + \Delta$
III	$a_1 = 1,\ b_1 = 2,\ a_2 = 3$	$R_0 + \tau u$	$3d - 2\delta_2 + \Delta$

For E2 and I the first pair loses at least kappa d and the next at least 16d/5. For E3 the first loses at least $2d - \delta_2$ and the next at least 16d/5; $a_1 \ge 4$ only improves that. Case II discards the nonnegative first-pair loss and keeps the next strong pair. Case III uses (R29.3.8) on its actual $a_2 = 3$ pair. Every pair after the second has shorter length at least four and supplies exactly the extra Delta loss used here. This justifies the linear increment in j without discarding any later arm.

R29.6.2 Complete scalar packets

The following table records proved bounds, with entries normalized by d. The row D_2 for III retains its actual sharper value δ_2.

Table 85: Scalar packet bounds for Lemma M.22. Interval rows contain δ/d and ϵ/d; other rows are upper bounds. The D_2 entry in III retains δ_2, a moment of the same host.

Quantity	E2	E3	I	II	III
c	$5/4$	$6/5$	$3/2$	$3/2$	$4/3$
σ/d	$1/3$	$9/64$	$325/392$	$325/512$	$1277/2592$
δ/d	$[-1/5, 91/100]$	$[-1/5, 2/3]$	$[-6/5, 3/4]$	$[-6/5, 1/25]$	$[-1, 3/20]$
ϵ/d	$[-7/10, 7/10]$	$[-7/8, 7/8]$	$[-17/10, 3/4]$	$[-13/10, 6/25]$	$[-4/3, 7/20]$
D_1/d	0	$1/5$	0	$1/5$	$1/5$
D_2	$d/12$	$d/12$	$d/2$	$d/2$	δ_2
D_3/d	$1/5$	$1/30$	$1/5$	$1/5$	$1/8$
$D_k/d,\ k \ge 4$	$1/12$	$1/12$	$1/5$	$1/5$	$1/12$
h	$1/2$	$1/2$	$3/8$	$3/8$	$1/2$

Here is a reconstruction of each packet, including the infinite index quantifier on D_k.

E2. Write $R = -B^2$ P(-B)u, where $P_0 = 1$, and $P(-z) = U - zV$. The coordinates of R are xU,-xV. One-sided alternating tails give

$$\frac{1}{2} < U < \frac{6}{5}, \qquad -C_\beta x \le V \le C_\beta.$$

Therefore

$$D_0 = \int xV \in [-d/5, d/2], \qquad D_1 = -\int xU \in [-d/2, 0].$$

The row predecessor and the spectral norm give

$$|R|^2 \le C_*\theta d < \frac{9}{25},$$
$$\sigma \le C_*\theta^2 d = \frac{5408}{16245}d < \frac{d}{3}.$$

Thus $c = 5/4$ is valid. The positive delta bound is

$$\delta \le$$
$$\left(\frac{416}{855} + \frac{5}{4}\frac{5408}{16245}\right)d$$
$$= \frac{4888}{5415}d < \frac{91}{100}d.$$

The useful epsilon bound keeps $D_0 + D_1$ together. Since $V - U \le x/(1-x)$, one has

$$D_0 + D_1 \le \frac{13}{19}\delta_2,$$
$$\epsilon \le \frac{13}{19}\delta_2 + \frac{5}{4}\sigma < \frac{7}{10}d.$$

The negative bound is $D_0 + D_1 \ge -7d/10$. Multiplying the two coordinates successively by z shows the positive part of D_2 begins at δ_4, that of D_3 at δ_3, and those of every $k \ge 4$ no earlier than δ_4. Within each of the four power classes, subsequent positive even-moment indices increase by two, so (R29.6.2) bounds the entire finite tail. In particular $\gamma_3 \ge 11d/12, \gamma_4 \ge 4d/5$, and every γ_k for $k \ge 5$ is at least $(\kappa - 1/12)d = 49d/96$. Hence $h = 1/2$.

E3. Write $R = B^3P(-B)u$, allowing $P_0 = 0$. The general coordinates U, V lie between -C_β x and C_β; those of R are -$x^2V, -xU$. Consequently

$$\sigma \le C_*\theta^3 d = \frac{2197}{16245}d < \frac{9}{64}d,$$
$$|R|^2 \le C_*\theta^2 d < \frac{1}{3}d \le \frac{1}{9}.$$

Use $c = 6/5$. The signed means are

$$D_0 \in [-C_\beta\delta_3, C_\beta\delta_2], \qquad D_1 \in [-C_\beta\delta_4, C_\beta\delta_3].$$

Thus $\delta \ge -d/5$ and $\delta \le 52702d/81225 < 2d/3$. Epsilon lies between -17d/60 and 13d/15, which is inside the stated [-7/8, 7/8] interval. The positive D_2 tail begins at δ_4, D_3 at δ_5, and every later positive tail no earlier than δ_4. Four-power continuation and (R29.6.2) give the table. In particular $\gamma_3 \ge 11d/12, \gamma_4 \ge 29d/30$, and the remaining $\gamma_k \ge 49d/96$; $h = 1/2$.

I. Here $R = BP(-B)u$ and $P_0 = P_1 = 1$. Its coordinates are xV, U, where

$$1 - C_\beta x \le U, V \le C_\beta, \qquad \frac{1}{2} < U, V < \frac{6}{5}.$$

The row bound gives $|R| < 1$; (R29.6.1) gives $\sigma \le (325/392)d$. Use $c = 3/2$. The actual first two moments are

$$D_0 = -\int U \in [-6d/5, -439d/855], \qquad D_1 = -\int xV \in [-d/2, 0].$$

Hence $\delta \le (-439/855 + (3/2)(325/392))d < 3d/4$, and the other delta/epsilon bounds follow without replacing signs. $D_2 \le C_\beta\delta_2 < d/2, D_3 \le C_\beta\delta_3 < d/5$, and each $k \ge 4$ has positive part bounded by d/5 through the same four-power continuation. Thus $\gamma_3 \ge d/2, \gamma_4 \ge 4d/5$, and all later $\gamma_k \ge (\kappa - 1/5)d = 63d/160 > 3d/8$.

II. The actual missing exponents impose $P_0 = 1, P_1 = P_2 = 0$. Consequently

$$1 - C_\beta x^3 \le U \le C_\beta,$$
$$-C_\beta x \le V \le C_\beta x^2,$$
$$U \ge \frac{11}{12}.$$

The sharper mask bound is

$$|P|^2 \le \frac{1 + x^3}{(1 - x^2)^2}$$
$$\le \frac{24864}{16245} < \frac{54}{35} < \frac{25}{16}.$$

This yields $\sigma \le (325/512)d$ and $|R| < 1$; take $c = 3/2.D_0$ is in [-6d/5,-11d/12] and D_1 is in [-d/12, d/5]. Therefore

$$\delta \le -\frac{11}{12}d + \frac{975}{1024}d = \frac{109}{3072}d < \frac{1}{25}d.$$

The lower epsilon bound is -13d/10 and its upper bound is 6d/25. Also $D_2 \le \delta_2 + C_\beta\delta_4 < d/2$. The remaining means have the same sufficient upper bounds as I. The full gamma tail has the same $h = 3/8$. Missing original powers are used only as mask information; no shorter path replaces the finite original path.

III. Now $P_0 = P_2 = 1$ and $P_1 = 0$. Its finite coordinates satisfy

$$1 - C_\beta x \le U \le 1 - \frac{x}{3} \le 1,$$
$$-C_\beta x \le V \le C_\beta x^2, \quad |V| \le \frac{4}{3}x.$$

Indeed $U \le 1 - x + C_\beta x^2 \le 1 - x/3$ on the full range. Therefore

$$|P|^2 \le (1 - x/3)^2 + \frac{16}{9}x^3 \le 1 - \frac{x}{6} \le 1 \qquad \text{(R29.6.3)}$$
$$(0 \le x \le 1/2).$$

Here $R = R_0 + \tau$ u. The complete row predecessor is P(-B)u+tau 1. Its constant-vector cross term is retained:

$$
\begin{aligned}
|R|^2 &\le d + \tau^2 + 2\tau E[P(-B)u] \\
&\le d + \tau^2 + \frac{2\tau}{5} d \\
&\le \frac{22}{45} < \frac{9}{16}.
\end{aligned}
\tag{R29.6.4}
$$

The bound on the cross mean follows from E[P(-B)u]=int $V \le C_\beta \delta_3 < d/5$. Thus the globally valid Taylor choice is $c = 4/3$. In the squared norm of R itself, $E(uR_0) = intxV \le C_\beta \delta_4 < d/12$, so

$$
\begin{aligned}
\sigma &\le \delta_2 + \frac{\tau}{6} d + \tau^2 d \\
&\le \left(\frac{13}{32} + \frac{1}{27} + \frac{4}{81} \right) d \\
&= \frac{1277}{2592} d.
\end{aligned}
\tag{R29.6.5}
$$

Since $D_0 = -int$ U lies in [-d,-439d/855],

$$
-d \le \delta \le \left(-\frac{439}{855} + \frac{1277}{1944} \right) d < \frac{3}{20} d.
$$

$D_1 = -E(uR_0) - \tau$ d lies above -d/12-tau d and below d/5. The resulting epsilon lower bound is $-47d/36 > -4d/3$ and the upper bound is 7d/20. The retained finite moments give

$$
\begin{aligned}
D_2 &\le \delta_2 - \delta_3 + C_\beta \delta_4 \le \delta_2, \\
D_3 &\le C_\beta \delta_5 + \tau \delta_2 \\
&< \left(\frac{1}{30} + \frac{13}{144} \right) d \\
&= \frac{89}{720} d < \frac{1}{8} d.
\end{aligned}
$$

The all-k tail is essential. For every even $k \ge 4$, tau u has zero mean after B^k and the positive R_0 tail starts at δ_4. For $k = 5$ mod 4, the tau term is nonpositive and the positive R_0 tail starts at δ_5. For $k = 7$ mod 4, the first possible positive contributions are $C_\beta \delta_7$ and tau δ_4; together they are less than $(1/12 + 2/9)\delta_4 < 11d/504 < d/30$. Moving four powers onward shifts these even-moment indices by two and only reduces the tail bound. Thus $D_k \le d/12$ for every $k \ge 4$. Accordingly $\gamma_3 \ge \kappa$ d, $\gamma_4 \ge 7d/8$ and all later $\gamma_k \ge 49d/96$; $h = 1/2$ is valid for every integer index.

This establishes the complete packet table. In particular, the dangerous III row retains both its actual constant-vector cross and its stronger long-tail gamma value.

R29.7 R2-C029: all four inner-length ranges

In this section every displayed lower loss coefficient omits the additional 4Delta, which is restored at the end.

R29.7.1 $L = 1$

Substitute the packets into (R29.5.6). Five valid loss coefficients are

Table 86: Lower loss coefficients for $L = 1$.

Packet	Lower $-\log \mathbb{E}H^4/d$
E2	$607/40 - 14/5 - 17/12$
E3	$767/40 - 7/2 - 153/256$
I	$607/40 - 3 - 119/32$
II	$64/5 - 24/25 - 17/6$
III	$35/4 - 7/5 - 17/8 = 209/40$

Each exceeds $15/4$. The weaker sigma bounds 7d/8, 2d/3, d/2 used in the last three rows follow from the already proved packets. All signed epsilon terms have their proper upper bounds.

R29.7.2 $L = 2$

Let $V_2 = u + BR$. For E2, $\|BR\|^2 \le 2197d/16245 < (37/100)^2 d$, so $\|V_2\| \le 11\sqrt{d}/8$. For E3, $\|BR\| \le \sqrt{d}/4$, so $\|V_2\| \le 5\sqrt{d}/4$.

For I and II its coordinates are 1-xU, xV. Since $U > 1/2$ and $|P|^2 < 100/49$,

$$|1 - xP(-z)|^2 \le 1 - x + \frac{100}{49}x^2 \le 1 \quad (0 \le x \le \theta).$$

For III its coordinates are 1-xU, tau+xV. The first lies between zero and 1-x/2; the absolute value of the second is below $1/3$, using $\tau \le 2/9$, -$C_\beta x^2 \le xV \le C_\beta x^3$. Hence its squared modulus is at most $(1 - x/2)^2 + x/9 \le 1$. These estimates keep tau as its actual scalar.

Apply (R29.5.7), using respectively $|\delta| \le d$, 2d/3, 6d/5, 6d/5, d:

Table 87: Lower loss coefficients for $L = 2$.

Packet	Lower $-\log \mathbb{E}H^4/d$
E2	$607/40 - 14/5 + 4 - 1/3 - 2057/256 - 10/9$
E3	$767/40 - 7/2 + 4 - 1/3 - 425/64 - 40/81$
I	$607/40 - 3 + 4 - 2 - 17/4 - 8/5$
II	$64/5 - 24/25 + 4 - 2 - 17/4 - 8/5$
III	$35/4 - 7/5 + 4 - 13/8 - 17/4 - 10/9 = 1571/360$

All exceed $15/4$; III alone has the explicit margin $221/360$. The -4d mean and D_2 terms in (R29.5.7) are retained in every row.

R29.7.3 $L = 3$, **including the continuous III comparison**

Write $V_3 = (1 + B)u + \delta u + B^2 R$. The mean part $4\epsilon + 4D_2 + 4D_3 - 4\delta$ d has the following upper coefficients, and the curvature term $(10/3)\epsilon^2$ has the listed upper coefficients after using $d^2 \le d/3$:

Table 88: Upper signed-mean and curvature costs at $L = 3$.

Packet	Mean cost divided by d	Curvature cost divided by d
E2	$21/5$	$49/90$
E3	$127/30$	$245/288$
I	$37/5$	$289/90$
II	$134/25$	$169/90$
III	$583/120$	$160/81$

The negative lower bounds on delta are needed for the term -4delta d. They have not been dropped.

For E2, triangle bounds give

$$\frac{\|V_3\|}{\sqrt{d}} < \frac{19}{16} + \frac{91}{300} + \frac{1}{4} < \frac{7}{4}.$$

For E3 the corresponding sum is $19/16 + 2/9 + 1/6 < 8/5$. The exceptional G_2 norm is explicitly $\sqrt{1+\theta} < 19/16$; it is not bounded by one.

For I and II the coordinates are

$$A_3 = 1 + \delta - x^2 V, \qquad C_3 = 1 - xU.$$

In I, both are positive, $A_3 \le 5/4$ and $C_3 \le 1$, so $\|V_3\|^2 \le (63/32)d$. In II, $0 < A_3 \le 329/300$ and $0 < C_3 \le 1 - x/2$. Since $x(1-x/2)^2 \le 9/32$ on [0, 1/2],

$$\|V_3\|^2 \le [(329/300)^2 + 9/32]d < \frac{3}{2}d.$$

For III the exact coordinates are

$$A_3 = 1 + \delta - \tau x - x^2 V, \qquad C_3 = 1 - xU. \tag{R29.7.1}$$

The lower bounds give $A_3 > 1/2$ and $C_3 \ge \kappa$. Their upper bounds, with the same original delta and tau, are

$$A_3 \le 1 + d(3/20 - \kappa x) + x^2/2, \qquad C_3 \le 1 - x/2. \tag{R29.7.2}$$

For $0 \le x \le 1/4$, (R29.7.2) gives

$$A_3^2 + xC_3^2 \le (173/160)^2 + \frac{1}{4} = \frac{36329}{25600} < \frac{23}{16},$$

with difference $471/25600$. For $1/4 \le x \le \theta, 3/20 - \kappa x \le 1/640$, so set $\epsilon_0 = 1/1920$ and obtain

$$A_3^2 + xC_3^2 \le 1 + x + x^3/4 + x^4/4 + 2\epsilon_0 + \epsilon_0 x^2 + \epsilon_0^2.$$

The last three terms are less than $1/800$. Thus the entire expression is less than $1 + 13/32 + 1/56 + 1/128 + 1/800$, which is below $23/16$ by $97/22400$. This is a proof on the entire continuous spectral interval, not checking selected frequencies or assuming $x \ge d$. It gives $\|V_3\|^2 \le (23/16)d$.

Substitution into (R29.5.8) gives

Table 89: Lower loss coefficients for $L = 3$.

Packet	Lower $-\log \mathbb{E}H^4/d$
E2	$607/40 + 8 - 21/5 - 49/90 - 833/64 > 5$
E3	$767/40 + 8 - 127/30 - 245/288 - 272/25 > 11$
I	$607/40 + 8 - 37/5 - 289/90 - 1071/128 = 24173/5760$
II	$64/5 + 8 - 134/25 - 169/90 - 51/8 > 7$
III	$35/4 + 8 - 583/120 - 160/81 - 391/64 = 98677/25920$

The I margin over $15/4$ is $2573/5760$; the III margin is $1477/25920$. These are positive exact rational margins.

R29.7.4 Every finite $L \geq 4$

For every finite $m \geq 3$, $\|G_m(B)\| \leq 1$. To see the complete length scope, at $z = i$ sqrt(x), an odd m gives $|G_m|^2 = (1 + x^m)/(1 + x) \leq 1$. For m divisible by four the numerator is $(1 - x^{m/2})^2$. For $m = 2$ mod 4 and $m \geq 6$ it is at most $(1 + x^3)^2 \leq 1 + x$ for $x \leq 1/2$. The omitted $m = 2$ is the genuine exception already used correctly in $L = 3$.

Every packet has $\sigma \leq 7d/8$. Hence

$$\begin{aligned} \|B^{L-1}R\|^2 &\leq \theta^3\sigma < d/16, \\ \|V_L\| &\leq \frac{5}{4}\sqrt{d}. \end{aligned} \tag{R29.7.3}$$

The gamma hypotheses were proved for every index in part R29.6, not merely for $k \leq L$. Equation (R29.5.10) now has norm cost 425d/64 and curvature cost at most $10h^2d/9$. Its five lower loss coefficients are

Table 90: Lower loss coefficients for every $L \geq 4$.

Packet	Lower $-\log \mathbb{E}H^4/d$
E2	$607/40 + 6 - 182/25 - 425/64 - 5/18$
E3	$767/40 + 6 - 16/3 - 8/5 - 425/64 - 5/18$
I	$607/40 + 9/2 - 6 - 425/64 - 5/32$
II	$64/5 + 9/2 - 8/25 - 8/5 - 425/64 - 5/32$
III	$35/4 + 6 - 6/5 - 8/5 - 425/64 - 5/18$

Each is greater than 5, and therefore greater than $15/4$. These are one finite-polynomial proof for all $L \geq 4$, with no cutoff or unproved limiting passage.

Restoring the omitted 4Delta in all four length ranges proves

$$\begin{aligned} E[(pT^L F)^4] \leq \\ \exp\left[-\left(\frac{15}{4} + \frac{64}{5}(j-2)\right)d\right]. \end{aligned} \tag{R29.7.4}$$

This completes R2-C029's ordinary-fourth claim for all five structural classes, every $j \geq 2$ and every original $L \geq 1$.

R29.11 The exact C094 ordinary-fourth bridge

The input to the Q5-C094 Theorem A, Sections 2.1–2.2, transfer is every original real Y with its ordinary fourth moment $E|Y|^4$. Its same-kernel row and norm argument is reproduced here.

R29.11.1 Same-kernel inverse fourth of the genuine $\bar{f}_2$

For this subsection use the original $g = K1$ and $r = Kg = K^2\ 1$. Then $\bar{f}_2 = q + r, Er = -d$, $Egr = 0, Er^2 = \delta_2$. The original centered-row Cauchy inequality, valid for every mean-zero v, is

$$|Kv(x)|^2 \leq p(x)q(x)E[v^2]. \tag{R29.11.1}$$

Indeed subtract g(x) from the row and use E[(K(x, $y) - g(x))^2$]≤$1 - g(x)^2$. Applying this to $v = g$ gives $r^2 \leq pqd$. This controls the actual small-q denominator:

$$E[r^4/q] \leq dE[pr^2] \leq 2d\delta_2. \tag{R29.11.2}$$

Let $W = 2r-g = K(2g-1)$ and $t_0 = 1+2d$. The original row bound gives $|W| \le \sqrt{1+4d} \le t_0$, $EW = -2d$ and $EW^2 = d + 4\delta_2$. The pointwise polynomial $(W+2d)^2(W-t_0) \le 0$ implies

$$EW^3 \le (t_0 - 4d)(d + 4\delta_2) - 4t_0 d^2 + 8d^3.$$

The exact cubic combination in $\bar{f}_2^4/q$ is

$$\mathcal{C} = -Eg^3 + 4E(g^2 r) - 6E(gr^2) + 4Er^3 = \frac{EW^3 + E(g^2 W)}{2}.$$

Using $E(g^2W) \le t_0 d$ gives

$$\mathcal{C} \le d - 2d^2 + 2(1-2d)\delta_2.$$

The full expansion, with $Egr = 0$ and the same δ_2 retained, is

$$E[\bar{f}_2^4/q] = 1 - d + 6\delta_2 + \mathcal{C} + E[r^4/q] \le 1 - 2d^2 + (8-2d)\delta_2. \tag{R29.11.3}$$

There is no independent choice of moment extrema in this expansion and no host-independent lower bound on q.

R29.11.2 Exact transfer exponents and the original norm

For any positive original kernel operator $\mathcal{B}$, let $h = \mathcal{B}1$. Original-row Cauchy and then Hölder with exponents 3 and $3/2$ give

$$\begin{aligned} |\mathcal{B}Y|^2 &\le h\mathcal{B}(|Y|^2), \\ I_q(\mathcal{B}Y) &\le E[(h^4/q)^{1/3}(\mathcal{B}(|Y|^2)^2)^{2/3}] \\ &\le E[h^4/q]^{1/3} E[\mathcal{B}(|Y|^2)^2]^{2/3}. \end{aligned}$$

Hence, in the unchanged ordinary L2(mu) space,

$$I_q(\mathcal{B}Y)^3 \le E[h^4/q]\|\mathcal{B}\|^4 (E|Y|^4)^2. \tag{R29.11.4}$$

For $\mathcal{B} = T^2$, the actual h is $\bar{f}_2$, not q. The exact norm input recorded in C094 is $\|T^2\| \le (1 + (1024/855)\delta_2)^2$. This is an accepted original ordinary norm bound; neither its adjoint nor its input is changed by a q-tilt. Together with (R29.11.3),

$$I_q(T^2Y)^3 \le B_2(d,\delta_2)(E|Y|^4)^2,$$

$$B_2 = \{1 - 2d^2 + (8-2d)\delta_2\}\left(1 + \frac{1024}{855}\delta_2\right)^8. \tag{R29.11.5}$$

This transfer uses the stated original norm estimate for T^2.

The negative curvature is also retained. Since $8 - 2d > 0$, the first factor is at most

$$R = 1 + \frac{13}{4}d - \frac{45}{16}d^2.$$

If $v = R - 1$, then $0 \le v \le 37/48$ and $v \ge 37d/16$. Integrating 1/(1+t), or differentiating the difference, gives $\log(1+v) \le v - v^2/[2(1+v)]$. It follows that

$$\log R \le \frac{13}{4}d - \frac{11757}{2720}d^2.$$

Let $\lambda_0 = (1024/855)\theta = 416/855$, with $12/25 < \lambda_0 < 1/2$. Applying the same logarithm inequality,

$$\begin{aligned} &8\log(1+\lambda_0 d) \\ &\le 8\lambda_0 d - \frac{4\lambda_0^2}{1+\lambda_0/3}d^2 \\ &\le 8\lambda_0 d - \frac{3}{4}d^2. \end{aligned}$$

Here $(24/7)(12/25)^2 = 3456/4375 > 3/4$, while $11757/2720 + 3/4 = 13797/2720 > 5$. Therefore

$$B_2(d,\delta_2) \le \exp\left(\frac{24427}{3420}d - 5d^2\right). \tag{R29.11.6}$$

For R2-C029 use (R29.7.4); for R2-C031 use the ordinary fourth bound in Section P.6. Since $15/2 - 24427/3420 = 1223/3420$, they give respectively

$$I_q(T^2H)^3 \le \exp\left[-\left(\frac{1223}{3420} + \frac{128}{5}(j-2)\right)d - 5d^2\right], \tag{R29.11.7}$$

$$I_q(T^2H)^3 \le \exp\left[-\frac{1223}{3420}d - 5d^2\right]. \tag{R29.11.8}$$

The transfer uses the ordinary fourth moment of H in the original $L^2(\mu)$ space.

R29.12 Genuine decorations, full reversal, and boundary cases

R29.12.1 Every original equal pair remains

At each original terminal or internal root, pair equal-length arms as far as possible. Each pair has its two actual directions f_l, h_l, and

$$0 < f_l h_l \le (Ef_{l-1})^2 \le 1.$$

Let E_t and E_r be the actual products of all these pairs at their own vertices. The decorated component is exactly

$$H_D = E_r p T^L (E_t F), \qquad U_D = T^2 H_D. \tag{R29.12.1}$$

Positivity proves $H_D \le pT^L F$ and $U_D \le T^2(pT^L F)$, so both ordinary-fourth and Iq estimates persist. The terminal pairs remain inside T^L, the root pairs remain outside it at the actual root, and every pair's edges remain in the tree.

R29.12.2 Complete reversal

Replacing K by -K reverses every actual edge simultaneously. It interchanges A, T; p, q; $f, \bar{f}$; and h, W. Thus the fully reversed component is

$$A^2\left[E_r^- qA^L\left(E_t^-\prod_i W_{a_i}\bar{f}_{b_i}\right)\right], \tag{R29.12.2}$$

with a single pair in R2-C031, and the corresponding budget is I_p. The decorations E_t^-, E_r^- are the actual $\bar{f}_l W_l$ products. The norm moments δ_i remain the same under reversal. This second direction is fixed by the structure before observing K. Reversing only the inner component would not justify keeping the same Iq claim. Both outer edges in (R29.12.2) are reversed as well.

R29.12.3 Small masses, extreme kernels and $d = 0$

All estimates allow arbitrary finite positive atom masses, including arbitrarily small ones, and pointwise K values at either endpoint of [-1, 1]. The positive diagonal contribution guarantees that the actual messages and p, q are positive on the retained support. No estimate requires a mass lower bound uniform over hosts. The only inverse q expression used for the output bridge is paid by the actual centered-row inequality (R29.11.1)-(R29.11.2).

When $d = 0$, only $K1 = 0$ follows. Then $p = q = 1$ and all actual directed tree messages equal one by recursively integrating their leaves. This works for a nonzero skew K with $K1 = 0$. All displayed actual moment and Iq estimates are equalities at $d = 0$. The preceding divisions by d or tau were used only when $d > 0$, where $\tau \geq \kappa d > 0$.

R29.12.4 Original edge and vertex counts

Let S_pair be the sum of one arm length from each genuine equal pair at the two vertices. Include the receiver c in the component. Then

$$e_B = L + 3 + \sum_{i=1}^{j}(a_i + b_i) + 2S_{\text{pair}} \quad \text{for R2-C029,} \tag{R29.12.3}$$

$$e_B = L + 3 + a + b + 2S_{\text{pair}} \quad \text{for R2-C031.} \tag{R29.12.4}$$

The three extra edges are the original root unit arm and the two outer edges. Since this is the original tree component, $|V_B| = e_B + 1$ and it has e_B nonreceiver vertices. All path subdivision vertices are counted, as are all paired-arm vertices. No skeleton edge count replaces e_B or the final e(U).

P.6 Single terminal pairs with minimum at least two

For $(a, b) = (2, 3)$ or $3 \leq a < b$, put $F = h_a f_b$ and $H = pT^L F$, $L \geq 1$. The inner connector points from terminal to root, the root unit arm points outward, and the two outer edges point from root to receiver. We prove

$$\mathbb{E}H^4 \leq e^{-15d/4}, \qquad I_q(T^2 H)^3 \leq e^{-1223d/3420 - 5d^2}.$$

Use the finite calculus of Section P.5 and Lemma M.22. The transfer is (R29.11.5)–(R29.11.6). The separate case $a = 2, b \geq 4$ is Section P.8.

A first arm of length three.

Let $a = 3$, $n = b - 3 \geq 1$, and set

$$R_0 = B^3 G_n(-B)u, \quad R = R_0 + \tau u, \quad \Gamma = 3d - 2\delta_2 \geq \tfrac{35}{16}d. \tag{R31.8.1}$$

This is the actual envelope (R29.3.8). At $z = i\sqrt{x}$, write $G_n(-z) = U - zV$. Finite alternating sums give

$$1 - x \le U \le 1, \quad 0 \le V \le 1, \quad V - U \le x, \quad |G_n(-z)|^2 \le 1 + \theta. \tag{R31.8.2}$$

For $n = 1$ the coordinates are $(1, 0)$; for even n they agree; for odd $n \ge 3$ they differ by one alternating terminal power. The sole exception to the unit norm bound is $n = 2$, whose squared norm is $1 + x$.

The coordinates of R_0 are $(-x^2V, -xU)$, giving $\mathbb{E}(uR_0) = -\int x^2V \le 0$. The row predecessor is $B^2G_n(-B)u + \tau\mathbf{1}$, whose cross mean is $-\int xV \le 0$. Consequently

$$|R|^2 \le (1+\theta)\delta_3 + \tau^2 < \tfrac{1}{12} + \tfrac{1}{18} < \tfrac{1}{4},$$

$$\sigma \le (1+\theta)\delta_4 + \tau^2 d \le \left(\tfrac{45}{448} + \tfrac{4}{81}\right) d = \tfrac{5437}{36288} d < \tfrac{3}{20} d. \tag{R31.8.3}$$

Here $\tau \le 2/9$, so the remainder constant is $c = 6/5$. The spectral rule (R29.2.3) gives

$$\begin{aligned} D_0 &= \textstyle\int xU, & D_1 &= \textstyle\int x^2V - \tau d, \\ D_2 &= -\textstyle\int x^2U, & D_3 &= -\textstyle\int x^3V + \tau\delta_2, \\ D_4 &= \textstyle\int x^3U, & D_5 &= \textstyle\int x^4V - \tau\delta_3, \\ D_6 &= -\textstyle\int x^4U, & D_7 &= -\textstyle\int x^5V + \tau\delta_4. \end{aligned} \tag{R31.8.4}$$

Increasing an index by four raises each moment index by two. Thus

$$D_2 \le 0, \quad D_3 \le \tau\delta_2 \le \tfrac{13}{144} d < \tfrac{d}{10}, \qquad D_k \le \tfrac{d}{14} \quad (k \ge 4). \tag{R31.8.5}$$

For the last bound, the first positive terms in residue classes $k = 0, 1, 3 \pmod 4$, $k \ge 4$, are respectively $\delta_4, \delta_5, \tau\delta_4$; class 2 is nonpositive. Hence $\gamma_3 \ge d$, $\gamma_4 > 9d/10$, and $\gamma_k \ge (\kappa - 1/14)d = 117d/224 > d/2$ thereafter. The long formula uses $h = 1/2$.

The short scalar bounds are

$$0 \le \delta \le \left(\tfrac{13}{32} + \tfrac{9}{50}\right) d = \tfrac{469}{800} d < \tfrac{47}{80} d, \qquad -\tfrac{2}{9} d \le \epsilon \le \tfrac{181}{240} d. \tag{R31.8.6}$$

Combining terms before estimating, $V - U \le x$ gives

$$\begin{aligned} \epsilon + D_2 &\le \delta + \delta_4 - \tau d, \\ \epsilon + D_2 + D_3 - \delta d &\le \delta + \delta_4 - \tau^2 - \delta d \le \delta + \delta_4. \end{aligned} \tag{R31.8.7}$$

For long connectors, retain the negative cross term in $\sigma = \|R_0\|^2 - 2\tau \int x^2V + \tau^2 d$. It gives

$$\epsilon = D_0 + (1 - 12\tau/5) \int x^2V + \tfrac{6}{5}\|R_0\|^2 + (-\tau + 6\tau^2/5)d.$$

Since $0 \le 1 - 12\tau/5 \le 1$ and the final term is nonpositive,

$$\epsilon \le \delta_2 + \delta_3 + \tfrac{6}{5}(1+\theta)\delta_4 \le \tfrac{358839}{524288} d < \tfrac{11}{16} d. \tag{R31.8.8}$$

The last margin is $1609/524288$. Together with $\delta \ge 0$,

$$\delta + (D_1)_+ = \max(\delta, \epsilon) \le \tfrac{11}{16} d. \tag{R31.8.9}$$

For $L = 2$, the coordinates of $u + BR$ are $(1 + x^2U, \tau - x^2V)$, bounded in absolute value by $7/6, 2/9$. Thus

$$\|u + BR\|^2 \le \left(\tfrac{49}{36} + \tfrac{13}{648}\right) d = \tfrac{895}{648} d < \tfrac{7}{5} d. \tag{R31.8.11}$$

For $L = 3$, the coordinates of the vector in (R29.5.8) are

$$(1 + \delta - \tau x + x^3 V,\ 1 + x^2 U). \tag{R31.8.13}$$

The first is positive and at most $51/40$, since $1 + 47/240 + 1/14 < 51/40$; the second is at most $7/6$. Hence

$$\|V_3\|^2 < \left(\tfrac{2601}{1600} + \tfrac{637}{1152}\right) d < \tfrac{9}{4} d, \qquad \tfrac{10}{3}\epsilon^2 \le \tfrac{32761}{51840} d.$$

For $L \ge 4$, the finite G_{L-1} contraction gives

$$\|B^{L-1} R\|^2 \le \theta^3 \sigma < \tfrac{3}{280} d < (21/200)^2 d, \qquad \|V_L\| < \tfrac{221}{200}\sqrt{d}.$$

Substitution in Lemma M.22, using (R31.8.7) at $L = 2, 3$ and (R31.8.9) for $L \ge 4$, yields the following lower bounds for $-\log \mathbb{E} H^4/d$:

$$L = 1: \quad \tfrac{35}{4} - \tfrac{181}{60} - \tfrac{51}{80} = \tfrac{1223}{240} > \tfrac{15}{4}, \tag{R31.8.10}$$

$$L = 2: \quad \tfrac{35}{4} + 4 - 4(\tfrac{47}{80} + \tfrac{1}{14}) - \tfrac{119}{20} - \tfrac{10}{9}(\tfrac{47}{80})^2 = \tfrac{15}{4} + \tfrac{1241}{40320}, \tag{R31.8.12}$$

$$L = 3: \quad \tfrac{35}{4} + 8 - 4(\tfrac{47}{80} + \tfrac{1}{14}) - \tfrac{153}{16} - \tfrac{32761}{51840} = \tfrac{15}{4} + \tfrac{12325}{72576}, \tag{R31.8.14}$$

$$L \ge 4: \quad \tfrac{35}{4} + 6 - \tfrac{11}{2} - \tfrac{830297}{160000} - \tfrac{5}{18} = \tfrac{15}{4} + \tfrac{47327}{1440000}. \tag{R31.8.15}$$

A first arm of length at least four.

For $4 \le a < b$, use (R29.3.6) with $\Gamma = 16d/5$ and $R = R_{ab} = B^3 P(-B)u$. The finite zero-one mask may have $P_0 = 0$. Writing $P(-z) = U - zV$, its coordinates satisfy $U, V \in [-C_\beta x, C_\beta]$; the coordinates of R are $(-x^2 V, -xU)$. The mask bound (R29.6.1) gives

$$\sigma \le \tfrac{2197}{16245} d < \tfrac{9}{64} d, \quad |R| < \tfrac{1}{3}, \quad c = \tfrac{6}{5},$$

and $D_0 \in [-C_\beta \delta_3, C_\beta \delta_2]$, $D_1 \in [-C_\beta \delta_4, C_\beta \delta_3]$. Consequently

$$-\tfrac{1}{5} d \le \delta \le \tfrac{2}{3} d, \qquad -\tfrac{17}{60} d \le \epsilon \le \tfrac{13}{15} d, \qquad D_1 \le \tfrac{1}{5} d. \tag{R31.9.1}$$

The positive tails of D_2, D_3, D_k ($k \ge 4$) start at $\delta_4, \delta_5, \delta_4$, respectively, and four-step continuation decreases each by θ^2. Thus

$$D_2 \le \tfrac{d}{12}, \quad D_3 \le \tfrac{d}{30}, \quad D_k \le \tfrac{d}{12}\ (k \ge 4), \quad \gamma_k \ge \tfrac{d}{2}\ (k \ge 3). \tag{R31.9.2}$$

The vectors in Lemma M.22 satisfy

$$\|u + BR\| \le \tfrac{5}{4}\sqrt{d}, \qquad \|V_3\| < (\tfrac{19}{16} + \tfrac{2}{9} + \tfrac{1}{6})\sqrt{d} < \tfrac{8}{5}\sqrt{d}, \qquad \|V_L\| \le \tfrac{5}{4}\sqrt{d} \quad (L \ge 4).$$

The $L = 3$ mean and curvature costs are at most $127d/30$ and $245d/288$. The lower loss coefficients are

L	$-\log \mathbb{E} H^4/d$ is at least
1	$64/5 - 7/2 - 153/256$
2	$64/5 - 7/2 + 4 - 1/3 - 425/64 - 40/81$
3	$64/5 + 8 - 127/30 - 245/288 - 272/25$
≥ 4	$64/5 + 6 - 16/3 - 8/5 - 425/64 - 5/18$

(R31.9.3)

All exceed $15/4$; the last two exceed $24/5$ and $49/10$.

The pair $(2,3)$.

The exact envelope (R29.3.9) is

$$R = K^3\mathbf{1} = -B^2u, \qquad \Gamma = d. \tag{R31.10.1}$$

Its row predecessor has squared norm $\delta_2 \le \theta/3 < 9/25$, so $c = 5/4$. Its moments are

$$\begin{aligned} \sigma = \delta_3, \quad D_0 = D_2 = D_4 = D_6 = 0, \\ D_1 = -\delta_2, \quad D_3 = \delta_3, \quad D_5 = -\delta_4, \quad D_7 = \delta_5. \end{aligned} \tag{R31.10.2}$$

Even moments vanish; odd signs alternate, with moment index rising by two under four-step continuation. Therefore

$$\delta = \tfrac{5}{4}\delta_3, \qquad \epsilon = \tfrac{5}{4}\delta_3 - \delta_2 \le 0. \tag{R31.10.3}$$

Here $5\theta/4 < 1$. Also $\gamma_3 = d$, $\gamma_4 = d - \delta_3$, and for $k \ge 5$ the first positive D_{k-1} is δ_5. Since $\kappa - \theta^4 > 1/2$, all $\gamma_k \ge d/2$.

At $L = 1$, formula (R29.5.6) gives

$$\log \mathbb{E}H^4 \le -4d - 4\delta_2 + \tfrac{37}{4}\delta_3 \le -4d - \tfrac{31}{128}\delta_2 \le -\tfrac{15}{4}d. \tag{R31.10.4}$$

At $L = 2$, $V_2 = u - B^3u$ has squared norm $d + \delta_4$ by skewness. Keep the quadratic fee to obtain

$$\log \mathbb{E}H^4 \le -\tfrac{15}{4}d - 4\delta_2 + 5\delta_3 + \tfrac{17}{4}\delta_4 + \tfrac{125}{24}\delta_3^2. \tag{R31.10.5}$$

Since $\delta_3^2 \le \theta^3 d\delta_2 \le \theta^3\delta_2/3$,

$$5\theta + \tfrac{17}{4}\theta^2 + \tfrac{125}{72}\theta^3 < \tfrac{366625}{124416} < 3. \tag{R31.10.6}$$

The first comparison uses $\theta < 5/12$; the required loss has an additional δ_2 to spare.

At $L = 3$, the coordinates of V_3 are $(1 + \delta - x^2, 1)$. Since $\delta \le 5d/24 \le 5/72$, the first coordinate is positive and

$$\|V_3\|^2 \le [(77/72)^2 + \theta]d = \tfrac{8035}{5184}d < \tfrac{25}{16}d. \tag{R31.10.7}$$

Discard $4\epsilon \le 0$ and $-4\delta d \le 0$, and use $4D_3 < 2d/3$, $\epsilon^2 \le \delta_2^2 \le d/18$. The loss is at least

$$12 - \tfrac{2}{3} - \tfrac{425}{64} - \tfrac{5}{27} > \tfrac{15}{4}. \tag{R31.10.8}$$

For $L \ge 4$, set $m = L - 1 \ge 3$. The exact vector is

$$V_L = [G_m(B) - B^{m+2}]u. \tag{R31.10.9}$$

It satisfies $\|V_L\|^2 \le d$. At $m = 3$, its squared multiplier is

$$(1-x)^2 + x(1-x^2)^2 = 1 - x + x^2 - 2x^3 + x^5 \le 1.$$

At $m = 4$, the nonnegative coordinates $(1 - x + x^3, 1 - x)$ are bounded by those of G_5, namely $(1 - x + x^2, 1 - x)$. For every $m \ge 5$,

$$(1-z)(G_m(z) - z^{m+2}) = 1 - z^m(1 + z^2 - z^3), \qquad |1 + z^2 - z^3|^2 = (1-x)^2 + x^3 \le 1.$$

Thus the squared numerator is at most $1 + 2x^{m/2} + x^m$. For $x > 0$,

$$2x^{m/2-1} + x^{m-1} \le 2\theta^{3/2} + \theta^4 < \tfrac{8}{15} + \tfrac{1}{30} < 1;$$

the numerator is at most $1 + x = |1 - z|^2$. At $x = 0$ the multiplier is one. This proves the norm bound for every finite m. Now $\delta \le 5d/24$, $D_1 \le 0$, and $h = 1/2$ give

$$4 + 6 - \tfrac{5}{3} - \tfrac{17}{4} - \tfrac{5}{18} = \tfrac{137}{36} = \tfrac{15}{4} + \tfrac{1}{18}. \tag{R31.10.10}$$

This proves the ordinary fourth bound in all stated cases. The transfer (R29.11.5)–(R29.11.6) gives the asserted I_q bound. Genuine equal pairs, full reversal, arbitrary positive atom masses, the case $d = 0$, and the original edge count are handled exactly as in R29.12: specifically, $H_D = E_r p T^L(E_t F)$, $0 < E_t, E_r \le 1$, and $e_B = L + 3 + a + b + 2S_{\text{pair}}$.

P.7 Odd terminal triples $(1, 2, c)$ at a unit root

R33.1 Exact object, direction and domains

On the one original finite probability space let

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n 1, \quad \bar{f}_n = T^n 1, \quad h_n = T f_{n-1},$$
$$W_n = A\bar{f}_{n-1}, \qquad I_q(V) = E[q^{-1/3}|V|^{8/3}].$$

The terminal residual is exactly $(1, 2, c)$, $c \ge 3$. The adjacent internal root has exactly one residual unit arm. Its original inner connector has length $L \ge 1$, and its original outer connector has length two. Genuine equal pairs at either root are allowed with arbitrary finite lengths and multiplicities.

The fixed actual message is

$$H_D = E_r^- q A^L(E_t^- p \bar{f}_2 W_c), \qquad U_D = T^2 H_D. \tag{R33.A1}$$

The factors E_t^-, E_r^- are the actual products of $\bar{f}_\ell W_\ell$ on original paired arms. The literal orientation is:

Table 91: Odd terminal triples $(1, 2, c)$ at a unit root: parameter values.

Entry	Direction/message
Terminal unit arm	Away from terminal, p.
Terminal two-edge arm	Both edges toward terminal, $\bar{f}_2$.
Terminal c-edge arm	First edge away from terminal; remaining edges toward its first vertex, W_c.
Internal root unit arm	Toward the root, q.
All L inner edges	From internal root toward terminal, A^L.
Both outer edges	From internal root toward receiver, T^2.
Each genuine equal pair	Entire reversal of its $f_\ell h_\ell$ pair.

The parameters K, μ, d and the spectral moments vary under this fixed rule.

The ordinary-fourth auxiliary is

$$F_c = q f_2 h_c, \qquad H_{c,L} = p T^L F_c.$$

The exact intermediate assertion proved below is

$$EH_{c,L}^4 \le e^{-18d/5} \quad \begin{cases} 0 \le d \le 1/3, & c = 3, \\ 0 \le d \le 1/4, & c = 4, \\ 0 \le d \le 1/5, & c \ge 5, \end{cases} \quad L \ge 1. \tag{R33.A2}$$

On the remaining high domains, compare W_c with one and apply C026 in the same direction.

The final result is

$$\boxed{I_q(U_D)^3 \le e^{-197d/3420} \le 1.} \tag{R33.A3}$$

R33.2 Original finite calculus and path means

R33.2.1 Finite notation and the short spectral exception

Use $B = -K$, $u = -g$, $\mathsf{X} = -B^2$, $\tau = d - \delta_2$ and (M.1), with Lemma M.1. Its original spectral measure ν at u has mass d, support $[0, \theta]$, and $\int x^j \, d\nu = \delta_{j+1}$. Write $S_n^- = \mathbf{1} + G_n(B)u$ and $\mu_n = \mathbb{E} f_n = \mathbb{E}\bar{f}_n$. For $Q(B) = E_0(\mathsf{X}) + BD_0(\mathsf{X})$ its two formulas are

$$E[Q(B)u] = -\int D_0 \, d\nu, \qquad \|Q(B)u\|^2 = \int (E_0^2 + xD_0^2) \, d\nu. \tag{R33.A5}$$

For $v = \delta_2/d$, Jensen gives $v \ge d$, $\delta_3/d \ge v^2$ and $\delta_4/d \ge v^3$. These are integrated moment restrictions, with no restriction $x \ge d$. The positive convolution is (M.2); the genuine message and equal-pair bounds are (M.1). The spectral exception used below is

$$|G_n(i\sqrt{x})| \le 1 \quad (n = 1 \text{ or } n \ge 3), \qquad |G_2(i\sqrt{x})|^2 = 1 + x. \tag{R33.A7}$$

The real coordinate for every $n \ge 3$ lies in $[1 - x, 1]$.

R33.2.2 The all-length mean used in the high domain

The stronger finite path estimate (A.8) gives, on the same original measure,

$$\boxed{\mu_n \le 1 - 83d/32 \quad (n \ge 4).} \tag{R33.A9}$$

Its proof includes the four short indices $n = 4, 5, 6, 7$ and every $n \ge 8$, with no monotonicity assumption on the integer n.

R33.3 Signed-mean moment payments

The common square payments (M.39) and (M.40) give, for every real vector V,

$$Ee^{4BV} \le e^{4EBV + (17/4)\|V\|^2}, \tag{R33.A13}$$

$$Ee^{4BV} \le e^{4EBV + 2\|V\|^2 + 6\|BV\|^2}. \tag{R33.A14}$$

Both also hold with K in place of B.

R33.4 Exact positive propagation and the ordinary-fourth outlet

R33.4.1 The retained q square and all finite Duhamel coefficients

Suppose a true nonnegative F obeys

$$F \le q(m - U/2)^2, \qquad m > 0, \quad k = 1/m, \quad V = EU^2, \quad R = -kU.$$

Every application below obtains this square from the sum of two nonnegative original arms; the square base is nonnegative. Use the QS row of Table 80: $w = q$, $r = 2$, $s_0 = m^2$, $s_1 = -m$, $s_2 = 1/4$ and $\lambda = 1$. The first row is

$$TF/m^2 \le \bar{f}_2 + BR + \delta, \qquad \delta = -k(EU - EgU) + k\sqrt{dV} + k^2 V. \tag{R33.A15}$$

Choose $D \ge \max(\delta, 0)$ and put $D_i = EB^i R$. The common finite continuation gives

$$0 \le T^L F/m^2 \le Y_L = \bar{f}_{L+1} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}, \tag{R33.A16}$$

$$Y_L = S^-_{L+1} + B^L R + D\bar{f}_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar{f}_{L+1-j}. \tag{R33.A17}$$

Here $\gamma_j = \beta_j - D_{j-2}$. The index is $j-2$ because the initial path is $\bar{f}_2$. Each local case below proves the signs it uses before discarding a path term.

R33.4.2 The two fourth-moment payment formulas

If $Y_L \le Cq + BPu + c$, $C > 0$, apply (M.37) and the log-scale square payment with scale $m^2 C$:

$$\log E(pT^L F)^4 \le 8 \log m + 4 \log C + \frac{4}{C}(c + EBPu) + \frac{17}{4C^2}\|Pu\|^2. \tag{R33.A18}$$

Increasing C to a specified larger positive value keeps the comparison because $q > 0$. If instead $p + Y_L \le 2 + c + BV$, the same payment with scale m^2 gives

$$\log E(pT^L F)^4 \le 8 \log m + 4(c + EBV) + (17/4)\|V\|^2. \tag{R33.A19}$$

R33.4.3 The ordinary-fourth outlet with its exact curvature

For an original positive kernel operator M, row Cauchy and Hölder with exponents $3, 3/2$ give

$$I_q(MY)^3 \le \mathbb{E}[(M\mathbf{1})^4/q]\,\|M\|^4 (\mathbb{E}|Y|^4)^2. \tag{R33.A20}$$

Apply the complete same-measure calculation in paragraph P.5. Its inverse row and ordinary norm use $\bar{f}_2 = T^2\mathbf{1}$ and $\|T^2\| \le (1 + 1024\delta_2/855)^2$; the latter's Schur proof is also given in Section P.4, paragraph R26.8.1. Thus, for every original real Y,

$$I_q(T^2 Y)^3 \le B_2 (\mathbb{E}|Y|^4)^2, \qquad B_2 = [1 - 2d^2 + (8 - 2d)\delta_2](1 + 1024\delta_2/855)^8,$$

$$B_2 \le \exp\{(24427/3420)d - 5d^2\}. \tag{R33.A21}$$

Consequently the ordinary-fourth input $\mathbb{E}|Y|^4 \le e^{-18d/5}$ gives

$$I_q(T^2 Y)^3 \le e^{-(197/3420)d - 5d^2}. \tag{R33.A22}$$

The input is unweighted and real, and the outer operator and weight remain the original T^2 and q.

R33.5 Terminal (1, 2, 3): every L and the whole original d domain

R33.5.1 $L = 1$ **: true positive square, actual moments and both continuous domains**

Let $r = Kg,\ s = K^2 g,\ t = 1 - d/2,\ k = 1/t$. The original paths are

$$f_2 = p + r, \qquad h_3 = q - d - r - s, \qquad \bar{f}_3 = q - d + r - s.$$

Thus $f_2 + h_3 = 2 - d - s$ and

$$F_3 \le q(t - s/2)^2.$$

Apply Corollary M.20 with $m = t$, $a = 0$, $W = s$, and $\Delta = \gamma$:

$$TF_3 \le t^2 q + tK(tg + s) + \gamma, \qquad \gamma = t(\sqrt{d\delta_3} - \delta_2) + \delta_3 \ge 0. \tag{R33.A23}$$

Here $d\delta_3 \ge \delta_2^2$. The resulting positive square is

$$H_{3,1} \le (c + N/2)^2, \qquad c = t + \gamma/(2t), \quad N = K(tg + s). \tag{R33.A24}$$

The complete upper bound dominates $TF_3 > 0$; the corollary therefore supplies the nonnegative square base.

The complete-square inequality $\sqrt{\theta v} - v \le \theta/4$, $v = \delta_2/d$, yields

$$c \le 1 - \left(\frac{1}{2} - \frac{\theta}{8} - \frac{3}{5}\theta^2\right) d = 1 - \frac{1793}{5120} d \le 1 - \frac{7}{20} d.$$

For $w = \delta_3/d$,

$$EN = -(t - v)d, \quad \|tg + s\|^2 = d(t^2 - 2tv + w) \le d[t^2 - (2t - \theta)v].$$

Furthermore

$$EN^2 = \int x(t - x)^2 \, d\nu \le dv(t - v)^2. \tag{R33.A25}$$

This is Jensen for the concave function $x(t - x)^2$: its second derivative is $-4t + 6x < 0$ for $t \ge 5/6$, $x \le 13/32$.

For $v \le 1/5$, (R33.A14) and the nonnegative base in (R33.A24) give

$$\frac{\log EH_{3,1}^4}{d} \le -\frac{14}{5} - 4t + 2t^2 + (4 - 4t + 2\theta)v + 6v(t - v)^2.$$

The constant term is at most $-427/90$; $4 - 4t + 2\theta \le 71/48$; and $v(1 - v)^2$ is increasing on $[0, 1/5]$, with endpoint $16/125$. Therefore

$$\begin{aligned} &\frac{\log EH_{3,1}^4}{d} \\ &\le -\frac{427}{90} + \frac{71}{240} + \frac{96}{125} \\ &= -\frac{18}{5} - \frac{1451}{18000}. \end{aligned} \tag{R33.A26}$$

For $v \ge 1/5$, retaining the exact γ and using (R33.A13) first gives

$$\begin{aligned} &-4 - 4t + \frac{17}{4}t^2 + 4\sqrt{w} \\ &-\frac{17}{2}tv + \left(\frac{4}{t} + \frac{17}{4}\right) w. \end{aligned}$$

Only now use $w \le \theta v$. The resulting $\Phi(t,v)$ is convex in t, with second derivative $17/2 + 8\theta v/t^3 > 0$. At its two endpoints:

$$\Phi(1,v) = -15/4 + 4\sqrt{\theta v} - (659/128)v,$$

$$\Phi(5/6,v) = -631/144 + 4\sqrt{\theta v} - (6541/1920)v.$$

Both decrease on $v \ge 1/5$, since $2\sqrt{\theta/v} < 3$ while both subtracted coefficients exceed three. The rational comparison $\sqrt{13/160} < 2/7$ follows from $637 < 640$. It gives

$$\begin{aligned} \Phi(1,1/5) &< -\frac{18}{5} - \frac{33}{896}, \\ \Phi(5/6,1/5) &< -\frac{31}{8} < -\frac{18}{5}. \end{aligned} \tag{R33.A27}$$

Thus every actual $v \in [d,\theta]$ is covered, with no spectral support restriction beyond $[0,\theta]$.

R33.5.2 All $L \ge 2$: the complete coefficient signs

In (R33.A15) use

$$m = 1 - d/2, \quad k = 1/m, \quad U = s = \mathsf{X}u, \quad R = -k\mathsf{X}u = kB^2u.$$

Then $EU - EgU = \delta_2$, $EU^2 = \delta_3$, and

$$\begin{aligned} \delta &= -k\delta_2 + k\sqrt{d\delta_3} + k^2\delta_3, \\ \frac{\delta}{d} &\le \frac{k\theta}{4(1-k\theta)} \\ &\le \frac{39}{164} < \frac{1}{4}. \end{aligned} \tag{R33.A28}$$

The first inequality completes the square in $k[\sqrt{\theta v} - (1-k\theta)v]$. The exact finite coefficients are

$$D_i = kEB^{i+3}1, \quad D_1 = k\delta_2,\ D_2 = 0,\ D_3 = -k\delta_3,\ D_4 = 0.$$

Every positive D_i, $i \ge 3$, is at most $k\delta_4 \le k\theta^3 d < d/12$. Consequently

$$\begin{aligned} \gamma_3 &\ge 41d/80, \quad \gamma_4 = d, \\ \gamma_j &\ge 49d/96 =: \sigma \quad (j \ge 5). \end{aligned} \tag{R33.A29}$$

The first bound exceeds σ, since $41 \cdot 96 - 49 \cdot 80 = 16 > 0$. For $L = 2$ I take $D = d/4$; for all $L \ge 3$ I take $D = 3d/8$. Both dominate the actual δ.

R33.5.3 $L = 2$ in the canonical h3 direction

The exact (R33.A16) reads

$$Y_2 = Cq + BQu + c, \quad C = 1 + D, \quad c = k\delta_2 - d, \quad Q = 1 + B + kB^3, \quad D = d/4.$$

Thus

$$c + EBQu = k\delta_2 - 2d \le -121d/80.$$

At a frequency, $Q = 1 + z(1 - kx)$ and $0 \le 1 - kx \le 1 - x$. Hence

$$\|Qu\|^2 \le d\left[1 + \max_{0\le x\le 1} x(1-x)^2\right] = 31d/27.$$

Use $C \le 13/12$, $\log m \le -d/2$, $\log C \le d/4$ in (R33.A18). The full payment is

$$\begin{aligned} &\frac{\log EH_{3,2}^4}{d} \\ &\le -4 + 1 - \frac{363}{65} + \frac{527}{108} \\ &= -\frac{26009}{7020} < -\frac{37}{10} < -\frac{18}{5}. \end{aligned} \tag{R33.A30}$$

This gives the canonical qf_2h_3 fourth estimate. The C106/C102 $L = 2$ eighth row concerns the direction $qf_2\bar{f}_3$.

R33.5.4 $L = 3$

The exact finite form is

$$Y_3 = Cq + BPu - d, \quad C = 1 - d + D + k\delta_2 \le \bar{C} = 1 - d/8,$$

$$P = 1 + D + B + (1 - k\mathsf{X})B^2, \qquad D = 3d/8.$$

After using $d = 2(1 - 1/k)$, the frequency difference $(|P|^2 - (1 + D)^2)/x$ equals

$$-5/2 + 3/(2k) + (7k/2 - 1/2)x - 2kx^2 + k^2x^3.$$

It is convex in $k \in [1, 6/5]$. Each endpoint is increasing in $x \in [0, \theta]$, and its final value is respectively

$$-1451/32768, \qquad -9507/204800.$$

Thus $\|Pu\|^2 \le (1 + D)^2 d$. The exact signed mean is

$$-d + EBPu = -(2 + D)d + \delta_2 - k\delta_3 \le -(7/4 + D)d,$$

using $\delta_3/d \ge v^2$ and $v - kv^2 \le 1/(4k) \le 1/4$. The ratio $(1+D)/\bar{C}$ increases to $27/23 < 6/5$. Since $\bar{C} \le 1$, division strengthens the negative mean, and

$$\begin{aligned} &\frac{\log EH_{3,3}^4}{d} \\ &\le -4 - \frac{1}{2} - 7 - 4D + \frac{17}{4}\left(\frac{6}{5}\right)^2 \\ &\le -\frac{269}{50} < -\frac{18}{5}. \end{aligned} \tag{R33.A31}$$

R33.5.5 $L = 4$

The actual finite expression is

$$Y_4 = Cq + BPu + c, \quad C = 1 - 2d + D + k\delta_2, \quad A_0 = C + d,$$
$$P = A_0 + (1 + D)B + B^2 + (1 - k\mathsf{X})B^3,$$
$$c = -\tau - Dd - k\delta_3, \qquad D = 3d/8.$$

At a frequency,

$$|P|^2 = A_0^2 + x[(1 + D - x + kx^2)^2 + x - 2A_0].$$

The bracket is convex in x; its second derivative is $2(-1 + 2kx)^2 + 4k(1 + D - x + kx^2) > 0$. At zero it is at most $-1 + 2d + D^2 \le -1/3 + 1/64 < -1/4$. At θ, use

$$1 + D - \theta + k\theta^2 \le 2347/2560 < 15/16, \quad A_0 \ge 19/24$$

to obtain at most $-229/768 < -1/4$. Thus

$$\|Pu\|^2 \le A_0^2 d - \delta_2/4.$$

The legal enlarged coefficient is $\bar{C} = 1 - 9d/8 \ge C$, and $A_0 \le 1 - d/8$. The exact mean plus constant is

$$c + EBPu = -2d + 2\delta_2 + d^2 - 2Dd - kd\delta_2 - k\delta_3.$$

Using the same $v = \delta_2/d$, $\delta_3/d \ge v^2$, $v \ge d$ and $v \le \theta$, it is at most

$$d[-19/16 + d/4 - 2d^2] \le -53d/48.$$

Also

$$\|Pu\|^2/d \le 1 - d/2 + d^2/64, \qquad \frac{1 - d/2 + d^2/64}{(1 - 9d/8)^2} \le 481/225.$$

The last ratio increases, because the derivative numerator has the sign of $7/4 - 17d/32 > 0$, and its value at $d = 1/3$ is $481/225$. Therefore

$$\begin{aligned} &\frac{\log EH_{3,4}^4}{d} \\ &\le -4 - \frac{9}{2} - \frac{53}{12} + \frac{8177}{900} \\ &= -\frac{862}{225} < -\frac{18}{5}. \end{aligned} \tag{R33.A32}$$

R33.5.6 Every finite $L \ge 5$, with no length cutoff

The needed shifted finite-geometric assertion is

$$\begin{aligned} &|G_n(i\sqrt{x}) - s_0(2 + i\sqrt{x})| \le 1 - 2s_0 \\ &(0 \le s_0 \le 3/16,\ n \ge 3). \end{aligned} \tag{R33.A33}$$

This finite two-term shift, including zero frequency and every $n \ge 3$, is proved in paragraph N.22.

Retain the final four actual losses $\bar{f}_0, \bar{f}_1, \bar{f}_2, \bar{f}_3$ in (R33.A17), using (R33.A29), and replace $D\bar{f}_{L-1}$ by DS^{-}_{L-1}. The identity

$$\bar{f}_0 + \bar{f}_1 + \bar{f}_2 + \bar{f}_3 = 1 + 3q + B(2+B)u - d$$

gives

$$C = 1 + D - 3\sigma = 1 - 37d/32, \quad c = -\sigma(1-d),$$

$$P = G_L - k\mathsf{X}B^{L-1} + DG_{L-2} - \sigma(2+B).$$

At every frequency $0 \le kx \le 39/80$, and

$$G_L - kxz^{L-1} = (1-kx)G_L + kxG_{L-1}.$$

Apply (R33.A33) with $s_0 = \sigma = 49d/96 \le 49/288 < 3/16$ to both terms; G_{L-2} contracts because $L - 2 \ge 3$. It follows that

$$\|Pu\|^2 \le a^2 d, \qquad a = 1 - 2\sigma + D = 1 - 31d/48.$$

The actual signed mean is

$$EBPu = -\beta_{L+2} + D_L - D\beta_L + 2\sigma d,$$

so

$$c + EBPu \le d[-49/48 + (335/256)d] \le -449d/768.$$

Here $-\beta_{L+2} + D_L = -\gamma_{L+2} \le -\sigma$; every discarded loss has already been paid. Finally $0 < C \le 1$, $a/C \le 226/177 < 13/10$, and

$$\begin{aligned} &\frac{\log EH^4_{3,L}}{d} \\ &\le -4 - \frac{37}{8} - \frac{449}{192} \\ &\quad + \frac{17}{4}\left(\frac{13}{10}\right)^2 \\ &= -\frac{18149}{4800} < -\frac{18}{5}. \end{aligned} \tag{R33.A34}$$

Equations (R33.A26)-(R33.A34) complete every $c = 3$ row on $0 < d \le 1/3$.

R33.6 Terminal (1, 2, 4): every L on $0 < d \le 1/4$

R33.6.1 $L = 1$

Let $r = Kg,\ s = K^2 g,\ w = K^3 g,\ W = s + w, m = 1 - d,\ k = 1/m,\ t = 1 - 2d$. The actual sum is

$$f_2 + h_4 = 2 - 2d + dg - W.$$

Thus $F_4 \le q(m + U_0/2)^2, U_0 = dg - W$. Put

$$\mathcal{D} = \theta(1+\theta) = 585/1024 < 3/5, \qquad v = \delta_2/d, \quad y = \sqrt{\mathcal{D}v} < 1/2.$$

The actual moments are $EW = \delta_2$, $E(gU_0) = d^2 + \delta_2$, and $\|U_0\| \le \sqrt{d}(d+y)$. Thus the q-square corollary permits

$$\Delta = \delta d, \qquad \delta = m(y - v) + (d+y)^2 \ge 0,$$

since $\sqrt{d\|U_0\|^2} \le d(d+y)$. For $N = K(tg+W)$ it gives directly

$$TF_4 \le m^2 q + mN - m\delta_2 + \delta d, \qquad EN = -td + \delta_2,$$
$$H_{4,1} \le (c_* + N/2)^2, \qquad c_* = m - \delta_2/2 + \delta d/(2m),$$

with the unchanged signed-vector bound

$$\|tg+W\|^2 \le d[t^2 - (2t - \mathcal{D})v] \le dt^2.$$

Its base is the arithmetic mean of mp and the complete positive upper factor.

Using (R33.A13), the two linear δ_2 terms cancel exactly:

$$\frac{\log EH_{4,1}^4}{d}$$
$$\le -8 - 4t + 4(y-v) + 4(d+y)^2/m$$
$$+\frac{17}{4}[t^2 - (2t-\mathcal{D})v].$$

Because $y - v \le \mathcal{D}/4 < 3/20$, $d + y < 3/4$, $m \ge 3/4$, this is at most

$$-22/5 - 4t + (17/4)t^2.$$

This convex quadratic has maximum $-83/20 < -18/5$ on $t \in [1/2, 1]$. This proves the unit row without using the old eighth bound as a fourth budget.

R33.6.2 The normalized input, quadratic error and every coefficient

For $L \ge 2$ use

$$U = s + w - dg = (d + \mathsf{X} - \mathsf{X}B)u, \quad R = -kU,$$
$$EU^2 = d^3 + 2d\delta_2 + \delta_3 + \delta_4, \qquad EU - EgU = d^2 + 2\delta_2.$$

The symbols d^3 and δ_3 are distinct. Let $J_0 = EU^2/d$, $v = \delta_2/d$. Then

$$J_0 \le d^2 + [2d + \theta(1+\theta)]v.$$

The scalar inequality $\sqrt{J_0} \le 3/8 + (2/3)J_0$, together with $k \le 4/3$, gives

$$\delta/(kd) \le 3/8 - d + 2d^2 + v[-2 + 4d + 2\theta(1+\theta)] \le 3/8.$$

If the coefficient of v is nonpositive, discard it. Otherwise use $v \le \theta$; the remaining quadratic increases in d and its value at $d = 1/4$ is $-1099/16384 < 0$. Thus $D = d/2$ dominates δ.

For every $j \ge 1$, direct skew moments give

$$D_{2j-1} = (-1)^{j+1}k(d\delta_j + \delta_{j+1}), \qquad D_{2j} = (-1)^{j+1}k\delta_{j+2}.$$

In particular

$$D_1 \le 7d/8, \quad D_2 < d/4, \quad D_3 = -k(d\delta_2 + \delta_3), \quad D_4 = -k\delta_4.$$

The first later positive odd term is at most

$$D_5 \le (4/3)(1/4 + \theta)\theta^2 d = 1183d/8192 < d/6,$$

and $D_6 \le (4/3)\theta^4 d < d/6$. Each subsequent member of either positive class contracts by at least θ^2 every four indices. Therefore

$$\gamma_3 \ge d/8, \quad \gamma_4 \ge 3d/4,$$
$$\gamma_j \ge 41d/96 =: \sigma \quad (j \ge 5). \tag{R33.A35}$$

R33.6.3 $L = 2$ **and** $L = 3$

For $L = 2$,

$$Y_2 = Cq + BQu + c, \quad C = 1 + D, \quad c = k(d^2 + \delta_2) - d,$$

$$Q = (1 - k\mathsf{X}^2) + [1 - k(d + \mathsf{X})]B.$$

Both frequency coordinates are nonnegative: $k(d + x) \le 7/8$ and $kx^2 \le 169/768$. Hence

$$|Q|^2 \le (1 - x^2)^2 + x(1 - x)^2 = 1 + x - 4x^2 + x^3 + x^4 \le 1 + x - 3x^2 \le 13/12.$$

The actual mean is

$$c + EBQu = -2d + k(d^2 + \delta_2 + \delta_3) \le -695d/768.$$

Use $C \le 9/8,\ \log m \le -d,\ \log C \le d/2$ in (R33.A18):

$$\begin{aligned} &\frac{\log EH_{4,2}^4}{d} \\ &\le -8 + 2 - \frac{695}{216} + \frac{221}{48} \\ &= -\frac{1993}{432} < -\frac{18}{5}. \end{aligned} \tag{R33.A36}$$

For $L = 3$,

$$Y_3 = Cq + BPu + c, \quad C = 1 - d + D + D_1 \le \bar{C} = 1 + 3d/8, \quad c = -d + k\delta_3,$$

$$P = 1 + D + B + [1 - k(d + \mathsf{X})]B^2 + k\mathsf{X}B^3.$$

Its real coordinate is in $[0, 1 + D]$, and its imaginary coefficient is $1 - kx^2 \in [0, 1]$. Thus $\|Pu\|^2 \le d[(1 + D)^2 + \theta]$. The exact rational comparison

$$\frac{(1 + d/2)^2 + \theta}{(1 + 3d/8)^2} \le 45/32$$

has positive cleared difference $d(112 - 107d)/2048$. The $k\delta_3$ terms cancel in the signed mean:

$$c + EBPu = -(2 + D)d + (1 - kd)\delta_2 \le -51d/32.$$

With $\bar{C} \le 35/32$,

$$\begin{aligned} &\frac{\log EH_{4,3}^4}{d} \\ &\le -8 + \frac{3}{2} - \frac{204}{35} + \frac{765}{128} \\ &= -\frac{28457}{4480} < -\frac{18}{5}. \end{aligned} \tag{R33.A37}$$

R33.6.4 Every finite $L \ge 4$

The finite filter is

$$Q_L = G_L - k(d + \mathsf{X})B^{L-1} + k\mathsf{X}B^L.$$

At a frequency set $a = k(d + x)$, $b = kx$, so $0 \le b \le a \le 7/8$. Then

$$\begin{aligned} Q_L &= (1-a)G_L + (a-b)G_{L-1} \\ &\quad + b[G_{L-1} + z^L]. \end{aligned} \tag{R33.A38}$$

The coefficients are nonnegative and sum to one. The extra finite polynomial $H_L = G_{L-1} + z^L$ is P_{L-1} of Section N.3. For $L \ge 5$, the estimate following (L33.3.1) gives $|H_L|^2 \le 1 - 7x/15 \le 1$. At $L = 4$, its squared modulus minus one is

$$x[-1 + 3x - 2x^2 + x^3] \le 0;$$

the bracket increases to $-1451/32768$ at $x = \theta$. Thus $|H_L| \le 1$, hence $|Q_L| \le 1$, for every finite $L \ge 4$.

Retain the final $\gamma_{L+1}\bar{f}_0$ loss in (R33.A17), and use $D\bar{f}_{L-1} \le DS^-_{L-1}$. The resulting true upper is

$$Y_L \le (1 + D)q + BPu - \gamma_{L+1}, \qquad P = Q_L + DG_{L-2}.$$

By (R33.A35),

$$EBPu = -\beta_{L+2} + D_L - D\beta_L \le -\sigma, \quad c + EBPu \le -2\sigma.$$

The exceptional G_2 occurs only at $L = 4$, and $\|G_2\| < 6/5$. Uniformly,

$$\frac{\|Pu\|}{(1+D)\sqrt{d}} \le \frac{1 + 6D/5}{1 + D} \le 46/45.$$

Substitution into (R33.A18) gives

$$\begin{aligned} &\frac{\log EH^4_{4,L}}{d} \\ &\le -8 + 2 - \frac{82}{27} \\ &\quad + \frac{17}{4}\left(\frac{46}{45}\right)^2 \\ &= -\frac{9307}{2025} < -\frac{18}{5}. \end{aligned} \tag{R33.A39}$$

This completes the entire $c = 4$ low domain.

R33.7 Every $c \ge 5, L = 1$: the full continuous moment rectangle

R33.7.1 Actual finite endpoints and positive auxiliary function

Put

$$\alpha = \beta_c, \quad a = \beta_{c-1},$$
$$m = 1 - (\alpha + a)/2, \quad k = 1/m, \quad t = m - a,$$

$$W=\sum_{j=3}^{c}K^{j}1=K^{2}G_{c-2}(K)g,$$
$$U_{0}=ag-W,$$
$$e_{1}=d-\beta_{c+1},\quad e_{2}=d-\beta_{c+2}.$$

Here e_{1},e_{2} denote the short means called D_{1},D_{2} in R2-C033 part R33.5; they are not the later Duhamel sequence. The actual positive convolution, retaining aq, gives

$$\begin{aligned}h_{c}&\le(1-a)q-r-W-\alpha,\\ f_{2}+h_{c}&\le 2m+U_{0},\\ F_{c}&\le q(m+U_{0}/2)^{2}.\end{aligned}\tag{R33.A40}$$

The first-connector square now follows from Corollary M.20; its actual moments are evaluated next.

R33.7.2 Same-moment errors, finite spectral loss and endpoint coupling

Let $v=\delta_{2}/d\in[d,\theta]$, $y=\sqrt{\theta v}$. Since $c-2\ge3$, (R33.A7) gives

$$\|U_{0}\|\le\sqrt{d}(a+y),\qquad E(gU_{0})=ad+e_{2}\ge(a+\kappa v)d.$$

The same expansion as in the c4 unit row therefore permits

$$\begin{aligned}\delta&=m(y-\kappa v)+(a+y)^{2}\ge0,\\ N=K(tg+W),\quad TF_{c}&\le m^{2}q+mN-me_{1}+\delta d,\quad EN=-td+e_{2},\\ H_{c,1}&\le(c_{*}+N/2)^{2},\\ c_{*}&=m-e_{1}/2+\delta d/(2m).\end{aligned}\tag{R33.A42}$$

Again the square is a nonnegative arithmetic mean of the actual positive factors.

For every finite $n\ge3$,

$$\begin{aligned}&|G_{n}(i\sqrt{x})|^{2}\\ &\le\frac{(1+x^{3})^{2}}{1+x}\le1-\eta x,\qquad\eta=7/16.\end{aligned}\tag{R33.A43}$$

The first bound follows separately from the three finite parity formulas in part R33.2.1. For the second, clear $1+x$: the remaining inequality is $9-7x-32x^{2}-16x^{5}\ge0$. It decreases on $[0,\theta]$, and

$$9-7\theta-32\theta^{2}=7/8,\qquad16\theta^{5}<1/2.$$

Thus it holds on the complete continuous interval. Using also $\Re G_{n}\ge1-x$,

$$|t-xG_{n}|^{2}\le t^{2}-2tx+(2t+1)x^{2}-\eta x^{3}.$$

Integration with the same ν, then $\delta_{3}\le\theta\delta_{2}$ and $\delta_{4}/d\ge v^{3}$, gives

$$\begin{aligned}\|tg+W\|^{2}&\le dA(t,v),\\ A&=t^{2}-(2\kappa t-\theta)v-\eta v^{3}.\end{aligned}\tag{R33.A44}$$

Here $m \ge 4/5,\ t \ge 3/5$, so $2\kappa t - \theta > 0$. The bound is positive on the entire enlarged rectangle: it increases in t, decreases in v, and at $t = 3/5, v = \theta$ exceeds

$$9/25 - 1/8 - 3/100 = 41/200.$$

The short endpoint difference must be coupled to α:

$$e_2 - e_1 \le h(\alpha - \tau), \qquad h = \theta/\kappa = 13/19. \tag{R33.A45}$$

For $c = 5$, 6, $\alpha = \tau$ and the difference is nonpositive. For $c \ge 7$, the alternating endpoints give $e_2 - e_1 \le \delta_4$, while $\alpha - \tau \ge \kappa\delta_3$ and $\delta_4 \le \theta\delta_3$. This verifies (R33.A45) for every finite c and preserves the same-kernel relation rather than separately maximizing its two terms.

R33.7.3 Monotonicity in both structural parameters and the whole v interval

By (R33.A13) and (R33.A42)-(R33.A45), the logarithmic fourth bound divided by d is at most

$$\begin{gathered}\mathcal{E} = \\ -4(\alpha + a)/d - 4t + 4h(\alpha - \tau)/d \\ +4(y - \kappa v) + 4(a + y)^2/m + (17/4)A(t, v).\end{gathered} \tag{R33.A46}$$

Enlarge to the full rectangle $\alpha, a \in [\tau, d]$, with the original d, v fixed. There $m \ge 4/5,\ t \ge 3/5$, $(a + y)/m < 4/5$, and $t - \kappa v > 0$. Direct differentiation gives

$$\mathcal{E}_\alpha = -4(1 - h)/d + 2 + 2[(a + y)/m]^2 - (17/4)(t - \kappa v) < 0,$$

$$\mathcal{E}_a = -4/d + 6 + 8(a + y)/m + 2[(a + y)/m]^2 - (51/4)(t - \kappa v) < 0.$$

The first is bounded above, after discarding its negative final term, by $-120/19 + 2 + 32/25 < 0$; the second by $-20 + 6 + 32/5 + 32/25 < 0$. Hence the single upper bound (R33.A46) is maximized at $\alpha = a = \tau = (1 - v)d$.

Now

$$m = 1 - (1 - v)d, \quad t = 1 - 2(1 - v)d, \quad z = (1 - v)d + y.$$

With $y' = \frac{1}{2}\sqrt{\theta/v} \ge 1/2$,

$$(z^2/m)' = (z/m)[2y' - 2d - dz/m] > 0.$$

Also

$$A' = 4d(t - \kappa v) - (2\kappa t - \theta) - 3\eta v^2 > -1.$$

Indeed $2\kappa - \theta = 25/32$ and $3\eta\theta^2 < 7/32$. The total derivative of (R33.A46) is therefore bounded below by

$$8 - \frac{8}{5} - \frac{3}{8} - \frac{17}{4} = \frac{71}{40} > 0.$$

This pays the entire v interval, so it is enough to set $v = \theta,\ \alpha = a = \kappa d,\ y = \theta$.

R33.7.4 The final continuous d interval

At those endpoints, (R33.A44) becomes

$$P(d) = 342509/524288 - (14763/8192)d + (361/256)d^2.$$

The remaining bound is exactly

$$\begin{aligned}\mathcal{E}_*(d) = \\ -8\kappa - 4(1-2\kappa d) + 4\theta^2 \\ +\frac{4(\kappa d+\theta)^2}{1-\kappa d} \\ +(17/4)P(d).\end{aligned} \tag{R33.A47}$$

It is convex on $[0, 1/5]$: $P'' > 0$, and the square of an affine function divided by a positive affine function has nonnegative second derivative here. At zero, $P(0) < 2/3$, $8\theta^2 < 4/3$, so

$$\mathcal{E}_*(0) < -35/4 + 25/6 = -55/12.$$

At $d = 1/5$,

$$t = 61/80, \quad m = 141/160, \quad z = 21/40, \quad P(1/5) = 4577893/13107200 < 7/20,$$

$$4z^2/m = 294/235 < 13/10, \qquad 4\theta^2 < 2/3.$$

Thus

$$\begin{aligned}\mathcal{E}_*(1/5) \\ < -39/5 + 2/3 + 13/10 + 119/80 \\ = -1043/240 < -18/5.\end{aligned} \tag{R33.A48}$$

Convexity proves the entire d interval. Together with the preceding monotonicities and finite-tail arguments, this completes every $c \geq 5$ unit-inner row in the exact low domain.

R33.8 Every $c \geq 5, L \geq 2$: all retained masks and signed propagation

Throughout this section $0 < d \leq 1/5$, $m = 1 - 9d/8$, and $k = 1/m \leq 40/31$.

R33.8.1 Genuine retained coefficients for every finite c

Use Lemma M.9 with $n = c$ and $\sigma = 9/4$. Its genuine coefficients a_1, a_2, a_3, sums C_0, D_0, E_0, and actual vector U are used here. It includes $c = 5$ with $a_3 = 0$, and every $c \geq 6$ with $0 \leq \lambda \leq 1$. The second actual arm sum gives

$$f_2 + h_c \leq 2m - U, \qquad F_c \leq q(m - U/2)^2. \tag{R33.A49}$$

Its exact constant loss and coefficient ranges are

$$\begin{aligned}\beta_c + a_1 + a_2 + (1-d)a_3 = 9d/4, \\ C_0 = 9d/4 - \beta_c + dE_0, \quad D_0 = 9d/4 - \beta_c - \beta_{c-1} + dE_0, \\ 5d/4 + dE_0 \leq C_0 \leq (53/32)d + dE_0, \\ d/4 \leq D_0 \leq (17/16 + d)d, \qquad 0 \leq E_0 \leq d.\end{aligned} \tag{R33.A50}$$

The ranges follow from $\tau \leq \beta_j \leq d$ and $E_0 \leq d$.

R33.8.2 Every finite frequency mask, including $c = 5$

Write $U = [a(\mathsf{X}) - b(\mathsf{X})B]u$. The same lemma gives

$$a = 5d/4 + \delta_2 + x - x^2, \qquad b = d/4 + \delta_2 + x \quad (c = 5), \tag{R33.A51}$$

and, for $c \geq 6$, with $G_{c-2}(i\sqrt{x}) = a_f + i\sqrt{x}\, b_f$,

$$a = C_0 + x(a_f - E_0), \qquad b = D_0 + x b_f. \tag{R33.A52}$$

The finite alternating sums obey $1 - x \leq a_f, b_f \leq 1 - x + x^2$ and $a_f \geq (1 - x)b_f$. In both cases,

$$a, b \geq 0, \qquad a - (1 - x)b \geq a_1 + a_2 x \geq \tau. \tag{R33.A53}$$

The following two paragraphs prove the full-frequency bounds and the quadratic error for these same actual coefficients, with the original global δ_2 fixed.

R33.8.3 Strong retained norms

Lemma M.12 applies to this same actual $U = [a(\mathsf{X}) - b(\mathsf{X})B]u$, with the original global d, δ_2 fixed during every frequency maximization. It gives

$$\sqrt{a^2 + xb^2} \leq u_0 := 3/8 + 9d/5 \leq 147/200, \qquad b \leq z_0 := \theta + 13d/16 \leq 91/160. \tag{R33.A54}$$

R33.8.4 The retained first-row allowance

The $\sigma = 9/4$ row of Lemma M.10 proves, for these actual coordinates and the whole frequency interval,

$$\frac{15}{7}(a^2 + xb^2) \leq a + b. \tag{R33.A57}$$

With $V = \mathbb{E}U^2$, the actual first-row error is

$$\delta = -k \int (a + b)\, d\nu + k\sqrt{dV} + k^2 V \leq 3kd/10 \leq 12d/31 < 2d/5.$$

Thus the same positive continuation uses

$$D = 2d/5. \tag{R33.A59}$$

R33.8.5 Signed finite path losses

Use the exact four sign classes in (M.27) for this same $R = -kU$. The coordinate bounds of that lemma imply the original strong estimates

$$D_1 \leq (147/155)d < d, \quad D_2 \leq (1183/3968)d < d/3, \quad D_1 + D_2 \leq (24731/19840)d < 5d/4. \tag{R33.A61}$$

Every subsequent positive class is below $d/6$, so

$$\gamma_3 > 0, \quad \gamma_4 \geq 2d/3, \quad \gamma_j \geq 41d/96 \quad (j \geq 5). \tag{R33.A62}$$

R33.8.6 Every finite $L \geq 4$

The filter is

$$Q_L = G_L - kaB^{L-1} + kbB^L.$$

At each frequency $0 \leq ka \leq 147/155 < 1$ and $0 \leq kb \leq 91/124 < 1$. It is exactly the convex combination of

$$G_{L-1}, \quad G_L, \quad G_{L-1} + z^L, \quad G_{L+1}$$

with respective weights

$$(ka)(1-kb), \quad (1-ka)(1-kb), \quad (ka)(kb), \quad (1-ka)(kb).$$

The four weights sum to one. All four finite polynomials contract: the G polynomials by (R33.A7), the missing-term polynomial by the complete proof in part R33.6.4. Therefore $\|Q_L u\|^2 \leq d$ for every finite $L \geq 4$.

Retain the final loss in (R33.A17) and use $D\bar{f}_{L-1} \leq 2D$:

$$p + Y_L \leq 2 + c + BQ_L u, \qquad c = 2D - \gamma_{L+1}.$$

The exact mean is

$$EBQ_L u = -\beta_{L+2} + D_L = -\gamma_{L+2} \leq -\sigma.$$

Hence

$$c + EBQ_L u \leq 4d/5 - 82d/96 = -13d/240.$$

The right side is a true nonnegative upper sum. Since $8 \log m \leq -9d$, (R33.A19) gives

$$\begin{aligned} &\frac{\log EH_{c,L}^4}{d} \\ &\leq -9 - \frac{13}{60} + \frac{17}{4} \\ &= -\frac{149}{30} < -\frac{18}{5}. \end{aligned} \tag{R33.A63}$$

R33.8.7 The two short connectors

The retained bounds give $0 \leq ka \leq 147/155 < 1$ and $0 \leq kb \leq 91/124 < 1$. At $z = i\sqrt{x}$, the two filters are

$$Q_2 = 1 - kbx + z(1-ka), \qquad Q_3 = 1 - x + kax + z(1-kbx).$$

Their real and normalized imaginary coordinates lie in $[0, 1]$. Consequently

$$|Q_j|^2 \leq 1 + x \leq 45/32, \qquad \|Q_j u\|^2 \leq (45/32)d \quad (j = 2, 3). \tag{R33.A64}$$

For $L = 2$, the original finite expansion is

$$\begin{gathered} p + Y_2 = 2 + c + BV, \qquad c = D + D_1 - d, \qquad V = Q_2 u + D\,1_{\text{const}}, \\ EBV = -d + D_2, \qquad EQ_2 u = -d + D_1 \leq 0. \end{gathered} \tag{R33.A65}$$

Thus $c + EBV \leq -7d/20$, and the constant-vector cross term is nonpositive:

$$\frac{\|V\|^2}{d} \leq \frac{45}{32} + \frac{4}{25}d \leq \frac{5753}{4000} < \frac{3}{2}.$$

Here and below $D = 2d/5$. Formula (R33.A19), with $8 \log m \leq -9d$, gives

$$\frac{\log EH_{c,2}^4}{d} \leq -9 - \frac{7}{5} + \frac{51}{8} = -\frac{161}{40} < -\frac{18}{5}. \tag{R33.A66}$$

R33.8.8 The connector $L = 3$

The exact expansion, including its constant vector, is

$$\begin{aligned} p + Y_3 &= 2 + c + BV, \quad \lambda_0 = d - D - D_1, \quad P = Q_3 + D, \\ c &= D + D_1 + D_2 - 2d, \qquad V = Pu - \lambda_0 1_{\text{const}}, \\ EBV &= -\tau + D_3 - Dd, \qquad EPu = -d + D_2. \end{aligned} \tag{R33.A70}$$

The signed-moment bounds give

$$c + EBV \le [2/5 + 5/4 - 2 - \kappa]d = -151d/160. \tag{R33.A71}$$

Since $0 \le D_1 < d$ and $0 \le D_2 < d/3$, $|\lambda_0| \le 3d/5$ and $|EPu| \le d$. The constant-vector correction therefore satisfies

$$-2\lambda_0 EPu + \lambda_0^2 \le (39/25)d^2.$$

Using $\sqrt{45/32} < 6/5$ for the additional Du term gives

$$\begin{aligned} \|V\|^2/d &\le 45/32 + (24/25)d + (4/25)d^2 + (39/25)d \\ &\le 38333/20000 < 2. \end{aligned}$$

Finally (R33.A19) gives

$$\frac{\log EH_{c,3}^4}{d} \le -9 - \frac{151}{40} + \frac{17}{2} = -\frac{171}{40} < -\frac{18}{5}. \tag{R33.A73}$$

Together with part R33.7 and the preceding long rows, these two estimates prove every $c \ge 5$ low-domain case.

R33.9 The same actual direction on the low and high domains

R33.9.1 Low-domain inner reversal followed by the original outer T^2

parts R33.5-8 prove (R33.A2) for the actual canonical input. Apply that statement to the legal kernel $-K$. Then

$$p \leftrightarrow q, \quad A \leftrightarrow T, \quad f_2 \mapsto \bar{f}_2, \quad h_c \mapsto W_c,$$

and all original moments δ_i stay unchanged. Thus

$$E[qA^L(p\bar{f}_2 W_c)]^4 \le e^{-18d/5} \tag{R33.A74}$$

in precisely the stated low domains. This step reverses the inner component only. The actual output is then obtained by applying the original T^2, with the original receiver weight q. For the genuine decorations,

$$0 < H_D = E_r^- qA^L(E_t^- p\bar{f}_2 W_c) \le qA^L(p\bar{f}_2 W_c),$$

because $0 < E_t^-, E_r^- \le 1$ and A^L is positive. Hence $EH_D^4 \le e^{-18d/5}$. Apply the original ordinary-fourth bridge (R33.A21)-(R33.A22) directly to this actual H_D:

$$I_q(U_D)^3 \le e^{-(197/3420)d - 5d^2}. \tag{R33.A75}$$

This proves all $c = 3$ hosts, the $c = 4$ hosts with $d \le 1/4$, and all $c \ge 5$ hosts with $d \le 1/5$.

R33.9.2 Actual high-domain path comparison

For $c = 4$ and $d \geq 1/4$,

$$W_4 = A\bar{f}_3 \leq 2E\bar{f}_3 = 2(1-2d) \leq 1.$$

For $c \geq 5$ and $d \geq 1/5$, the all-finite mean bound (R33.A9) gives

$$\begin{aligned} W_c = A\bar{f}_{c-1} &\leq 2\mu_{c-1} \\ &\leq 2(1-83d/32) \leq 77/80 < 1. \end{aligned} \tag{R33.A76}$$

Consequently the very same direction satisfies

$$0 < U_D \leq U_{12,L} := T^2[qA^L(p\bar{f}_2)]. \tag{R33.A77}$$

This is a comparison inside a nonnegative integral. The actual long arm and all its original edges have not been removed from the oriented object or its final edge count.

The right side of (R33.A77) is the actual terminal $(1,2)$ direction of Section P.4, with the same original q and outer T^2.

The high-domain comparison table is:

$L = 1$**.**
$\mathbb{E}[U_{12,L}^4/q] \leq 1 - d/5$, by R26.3.

$L = 2$**.**
$\mathbb{E}[U_{12,L}^4/q] \leq 1 - d/24$, by R26.4.2.

$L = 3$**.**
$\mathbb{E}[U_{12,L}^4/q] \leq 1 - 7d/10$, by R26.4.3.

$L = 4$**.**
$\mathbb{E}[U_{12,L}^4/q] \leq 1 - d/4$, by R26.4.4.

$L \geq 5$**.**
$\mathbb{E}[U_{12,L}^4/q] \leq 1 - d/2$, by R26.6.

The proofs at $L = 3, 4$ propagate constants as $T^2\mathbf{1} = \bar{f}_2$. The proof for every $L \geq 5$ retains its finite final monomial and the covariance sign on the spectral measure of $\mathbf{1}$. Only these five actual terminal $(1,2)$ entries are used here.

All entries are at most $1 - d/24$. Positivity, $q > 0$ and (R33.A77) therefore give

$$E(U_D^4/q) \leq 1 - d/24.$$

Hölder with exponents $3/2, 3$, using $Eq = 1$, is exactly

$$I_q(U_D) = E[(U_D^4/q)^{2/3} q^{1/3}] \leq [E(U_D^4/q)]^{2/3}.$$

Thus the high-domain payment is

$$\begin{aligned} I_q(U_D)^3 \\ \leq (1 - d/24)^2 \leq e^{-d/12} \\ \leq e^{-197d/3420}. \end{aligned} \tag{R33.A78}$$

The final coefficient margin is $1/12 - 197/3420 = 22/855 > 0$. Both split endpoints $d = 1/4$ and $d = 1/5$ belong to both domains. Combining (R33.A75) and (R33.A78) proves (R33.A3).

R33.10 Original decorations, reversal, KL and exact gluing responsibilities

R33.10.1 Original edges and all repeated arms

Each genuine pair is formed from two actually present equal-length arms. The pointwise inequality

$$\bar{f}_\ell W_\ell \le 1$$

is the complete reversal of $f_\ell h_\ell \le 1$. It permits comparison at the same root label, followed by positive propagation. The terminal pair factor remains inside A^L, and the internal-root pair factor remains outside A^L and before T^2.

If repeated arms need a tie-breaking choice, any fixed labeling or structure-based choice may be made before the host is supplied. Every original arm in each pair is still oriented, and every original subdivision vertex is still integrated over the original μ.

Without paired decorations the component has

$$e_B = (1+2+c) + L + 1 + 2 = L + c + 6$$

original edges. If S_{pair} is the sum of one arm length from each actual pair at both roots, then

$$e_B = L + c + 6 + 2S_{\text{pair}}, \qquad |V_B| - 1 = e_B. \tag{R33.A79}$$

Every two-edge path, every internal connector edge and every long-arm subdivision edge appears exactly once in (R33.A1).

R33.10.2 Complete component reversal is a separate operation

Reversing every original edge means substituting $-K$ throughout the complete component, including both outer edges and every decoration. The fully reversed output is

$$U_D^{\text{rev}} = A^2[E_r^+ pT^L(E_t^+ q f_2 h_c)],$$

and its budget is

$$I_p(U_D^{\text{rev}})^3 \le e^{-197d/3420}, \qquad I_p(V) = E[p^{-1/3}|V|^{8/3}].$$

This is different from the inner reversal in part R33.9.1. The outer operator and the receiver weight must both swap in a complete reversal. Keeping T^2 while changing only the weight, or keeping q while changing only the outer path, is not approved.

P.8 The terminal pairs $(2, b)$ for every $b \ge 4$ at a unit root

For every $b \ge 4$ and $L \ge 1$, the fixed actual direction

$$H_D = E_r pT^L(E_t h_2 f_b), \qquad U_D = T^2 H_D$$

at a unit internal root and an outer path of exactly two edges satisfies $I_q(U_D)^3 \le e^{-d/500}$. Here E_t, E_r are the genuine equal-pair products at the indicated original vertices. The direction is the same on every analytic subdomain below. For $L \ge 2$, the ordinary fourth-moment estimate is $\mathbb{E}[pT^L(h_2 f_b)]^4 \le e^{-15d/4}$ on the full domain $0 \le d \le 1/3$.

R35.2 Original finite notation

Use the original-kernel notation and bounds of (M.1) and Lemma M.1:

$$B=-K,\quad u=-g,\quad X=-B^2,\quad \tau=d-\delta_2,\quad \theta=13/32,\quad \kappa=19/32.$$

Put $S_n^- = \mathbf{1}+G_n(B)u$. Its positive convolution is (M.2). The original X-spectral measure at u is denoted by ν_{sp}, and its coordinate formulas are (M.3). In particular a constant-vector component retains its cross term with the same spectral mean. The genuine message means obey $\mathbb{E}f_n=\mathbb{E}\bar{f}_n\le 1$; the other two arm messages can have larger means, since $\mathbb{E}h_2=\mathbb{E}W_2=1+d$. For the undecorated terminal write $F_b=h_2f_b$, $H_{b,L}=pT^LF_b$ and $I_q(Y)=\mathbb{E}[q^{-1/3}|Y|^{8/3}]$.

R35.3 Signed-mean payments and bare paths

The common signed-mean bound gives

$$\mathbb{E}e^{4BV}\le e^{4\mathbb{E}BV+(17/4)\|V\|^2}. \tag{R35.A9}$$

Corollary M.21 gives

$$E(p\bar{f}_2)^4\le e^{d/4}, \tag{R35.A10}$$

$$E(p\bar{f}_n)^4\le 1\quad (n\ge 3),\qquad pq\le 1. \tag{R35.A11}$$

The $n=3$ proof retains $\|G_2u\|^2=d+\delta_2$.

R35.4 Positive square transport and signed constant-vector terms

Suppose the actual F satisfies $0\le F\le (m+R/2)^2, m>0$ and $2m+R\ge 0$. Set

$$\begin{gathered}k=1/m,\quad V=ER^2,\quad D_i=EB^iR,\quad a_1=-D_1=EKR,\quad a=ka_1,\\ \delta=kD_0+k^2V/2,\quad \epsilon=\delta-a.\end{gathered} \tag{R35.A12}$$

Use the S row of Table 80, with $w=1$, $r=1$, $s_0=m^2$, $s_1=m$, $s_2=1/4$ and $\lambda=1$. The common finite continuation gives

$$\begin{aligned}&0\le T^LF/m^2\le Y_L\\ &=\bar{f}_L+kB^LR+\delta\bar{f}_{L-1}-a\bar{f}_{L-2}\\ &\quad +k\sum_{j=2}^{L-1}D_j\bar{f}_{L-1-j}\quad (L\ge 2).\end{aligned} \tag{R35.A13}$$

This upper bound is nonnegative as the result of the positive-square comparison, despite its signed expansion.

For L1, $TF/m^2\le q+kBR+\delta$. The sum with p is 2+delta+kBR, yielding

$$\log E(pTF)^4\le 8\log m+4\epsilon+(17/4)k^2V. \tag{R35.A14}$$

Suppose $R=[a_f(X)-b_f(X)B]u$ and $\rho=\int xb_f$. For $L=2$,

$$p+Y_2=2+\epsilon+BV_2,\qquad V_2=\delta\mathbf{1}+u+kBR,$$

$$EBV_2=-d-k\rho,\quad \|V_2\|^2=d+2k\rho+k^2\|BR\|^2+\delta^2-2\delta a.$$

Indeed $\langle u, BR\rangle = \rho$ and $\mathbb{E}BR = -a_1$. The last contribution is exactly $\epsilon^2 - a^2$, not δ^2 alone. Dropping only -a^2 gives

$$\begin{aligned}\log E(pT^2F)^4 &\le 8\log m + d/4 + 4\epsilon + (9/2)k\rho \\ &+(17/4)k^2\|BR\|^2 + (17/4)\epsilon^2.\end{aligned} \tag{R35.A15}$$

This is the canonical R calculation, without reversing the terminal pair.

For L3, direct expansion gives

$$p + Y_3 = 2 + \epsilon - d - k\rho + BV_3, \qquad V_3 = \epsilon\mathbf{1} + (1+\delta)u + Bu + kB^2R,$$
$$EBV_3 = -(1+\delta)d + kD_3.$$

Therefore

$$\begin{aligned}\log E(pT^3F)^4 &\le 8\log m - 8d + 4\epsilon - 4\delta d \\ &+4k(D_3 - \rho) + (17/4)\|V_3\|^2.\end{aligned} \tag{R35.A16}$$

The nonconstant part of V3 has mean -d-k rho, so its cross term with epsilon1 is -2epsilon(d+k rho), not zero.

For the first connector, Corollary M.20 with $Z = R$ gives directly

$$pTF \le \{m + D_0/2 - KR/2 + V/(4m)\}^2. \tag{R35.A17}$$

The square base is nonnegative.

R35.5 Actual low-domain envelopes and whole-frequency bounds

R35.5.1 $b = 4$

Let $m = 1 - d$ and $R = K^2g + K^3g - dg$. Actual expansion gives $h_2 + f_4 = 2m + R$ and

$$\begin{aligned}R &= (d + X - XB)u, \quad a_f = d + x, \quad b_f = x, \\ D_0 &= \delta_2, \quad a_1 = d^2 + \delta_2, \quad \rho = \delta_3, \\ V &= d^3 + 2d\delta_2 + \delta_3 + \delta_4, \\ \|BR\|^2 &= d^2\delta_2 + 2d\delta_3 + \delta_4 + \delta_5.\end{aligned} \tag{R35.A18}$$

For $d \le 1/4$, $\theta^2(1+\theta) = 7605/32768 < 1/4$ gives $\sqrt{a_f^2 + xb_f^2} \le u_0 = 1/2 + d$ and $ku_0 \le 1$. The padded $z_0 = \theta + 11d/16$ is valid; $b_f \le \theta$ is also available. Exactly,

$$\begin{aligned}\epsilon &= -kd^2 + k^2V/2, \\ |\epsilon| &\le d/6, \\ \delta &\le 25d/24.\end{aligned} \tag{R35.A19}$$

For epsilon's upper bound, $m^2/6 + dm - u0^2/2 = (1-4d)(1+8d)/24 \ge 0$. For its lower bound, $V \ge 3d^3$; put $z = d/m \le 1/3$ and use $\epsilon/d \ge -z + 3z^2/2 \ge -1/6$. Finally $\delta/d \le \theta/m + u0^2/(2m^2)$, increasing to $25/24$.

R35.5.2 Every finite $b \ge 5$, $d \le 1/5$

Use Lemma M.9 with $n = b$, $\sigma = 9/4$, and $m = 1 - 9d/8$. Denote its retained coefficients by c_i, put $C = c_1 + c_2 + c_3$, $D = c_2 + c_3$, $E = c_3$, and write $R = U$. Its first actual arm sum gives

$$h_2 + f_b \le 2m + R, \qquad h_2 f_b \le (m + R/2)^2.$$

At $b = 5$ the coordinates of $R = (a_f - b_f B)u$ are $a_f = 5d/4 + \delta_2 + x - x^2$, $b_f = d/4 + \delta_2 + x$. For $b \ge 6$, write $N = b - 2$ and $G_N = a_N + Bb_N$. Then

$$\begin{gathered} a_f = C + x(a_N - E), \qquad b_f = D + xb_N, \\ a_f - (1 - x)b_f \ge c_1 + c_2 x \ge \tau. \end{gathered} \tag{R35.A21}$$

These are exactly the retained coordinates of R33.8.1–R33.8.3, whose continuous norm proof keeps the same global d, δ_2, E. It therefore gives

$$\sqrt{a_f^2 + xb_f^2} \le u_0 = 3/8 + 9d/5, \qquad b_f \le z_0 = \theta + 13d/16. \tag{R35.A23}$$

The finite propagation uses the seed $+kU$; the signed moments below are computed for $R = U$, before normalization.

The exact signed spectral moments are

$$\begin{gathered} V = \int (a_f^2 + xb_f^2), \quad D_0 = \int b_f, \quad a_1 = \int a_f, \\ \rho = \int xb_f, \quad D_3 = \int xa_f, \\ D_{4j+1} = -\int x^{2j} a_f, \quad D_{4j+2} = -\int x^{2j+1} b_f, \\ D_{4j+3} = \int x^{2j+1} a_f, \quad D_{4j+4} = \int x^{2j+2} b_f. \end{gathered} \tag{R35.A24}$$

From (R35.A21), $a1 \ge \kappa$ D0+tau d. The finite difference $a_f - b_f = c1 - Ex + x(a_N - b_N)$ gives

$$a_1 - D_0 \le d^2 + \theta^3 d, \qquad D_3 - \rho \le d\delta_2 + \delta_5. \tag{R35.A25}$$

The first possible positive end term is x^3, or x^4 after the extra x. At b5 the first integrated difference is exactly $d^2 - d3$.

The complete parameter bounds are

$$\begin{gathered} |\epsilon| \le 3d/5, \quad -\epsilon < 7d/20, \quad \delta \le 6d/5, \\ ku_0 \le 147/155 < 1, \quad k\rho < 3d/10, \quad k^2\theta u_0^2 < 3/8. \end{gathered} \tag{R35.A26}$$

For delta, the target difference after multiplication by $2m^2$ is $463/320 - (955/128)d + (2601/1600)d^2$, decreasing on [0, 1/5], with endpoint $1577/80000$. For epsilon use $a1 \ge \kappa$ D0+tau d. The difference is $1867/2560 - (64539/20480)d - (118503/51200)d^2$, decreasing with endpoint $16519/2560000$. The other exact endpoint bounds are $k(d + \theta^3) \le 43753/126976 < 7/20$, k theta $z0 \le 1183/3968 < 3/10$, and $k^2\theta u0^2 \le 280917/768800 < 3/8$. All belong to the same actual R.

R35.5.3 L3 frequency inequality and its certificate

In both low domains,

$$(1+\delta-kxa_f)^2+x(1+kxb_f)^2\le(1+\delta)^2+x. \tag{R35.A27}$$

Use $a_f^2+xb_f^2\le u0a_f+x$ z0 b_f and $a_f\ge(1-x)b_f$. Since 2-kx $u0>0$, the unshifted difference is at most

$$kxb_f[-2+4x+kx\{u_0(1-x)+xz_0\}].$$

The quantity x[u0(1-x)+xz0] increases with x, since its derivative is $\ge u0(1-2x)>0$, and the parameters increase with d. At the upper corners, the negative-bracket margins are

$$323/49152>0\quad(b4,\ z_0=\theta+11d/16),\qquad 79745/3174400>0\quad(b\ge5).$$

For $b=4$, the sharper input $b_f=x\le\theta$ gives margin $1091/24576$. With the padded input $z_0=\theta+11d/16$, the valid margin is $323/49152$. Both are positive. The scalar shift contributes $-2\delta kxa_f\le0$.

The actual constant coordinate consequently gives

$$\|V_3\|^2\le(1+\delta)^2d+\delta_2+\epsilon^2-2\epsilon(d+k\rho). \tag{R35.A28}$$

R35.6 All low-domain short connectors

For b4, $d\le1/4$, (R35.A14) and exact epsilon yield

$$\log EH^4_{4,1}/d\le-8-8d+\frac{25}{4}\left(\frac{1/2+d}{1-d}\right)^2. \tag{R35.A29}$$

Here $8\log(1-d)\le-8d-4d^2$ and $k\ge1$. The right side is convex, with endpoint values -103/16, $-15/4$. For $b\ge5$, the complete-host $L=1$ conclusion is Proposition P.4.

For canonical L2, $b\ge5$, use $\|BR\|^2\le\theta$ V in (R35.A15), keeping the negative quadratic logarithm term:

$$\begin{aligned}&\log EH^4_{b,2}/d\\&\le-127/20-(1413/400)d+(9/2)\theta z_0/m\\&\quad+(17/4)\theta(u_0/m)^2.\end{aligned} \tag{R35.A31}$$

This is convex. At zero the two positive terms sum to less than one, giving below -107/20. At $d1/5$, (R35.A26) gives below -16451/4000 $<-15/4$. The coefficient -1413/400 is exactly -81/16 + $(17/4)(9/25)$, retaining the signed constant-vector budget.

For b4, L2,

$$\|BR\|^2/d\le\theta/16+\theta^2/2+\theta^3+\theta^4=212017/1048576<13/64.$$

Use $k\le4/3,\epsilon^2/d\le1/144$ and $\rho/d\le\theta^2<1/6$ in (R35.A15):

$$\log EH^4_{4,2}/d\le-85/12+1+221/144+17/576=-2603/576<-15/4. \tag{R35.A32}$$

For b4, L3, the scalar shift plus the mean in (R35.A16) simplifies exactly to -$2d - kd^2 + kV/2$. By (R35.A19), k $\rho < 2d/9$ and (R35.A28),

$$\|V_3\|^2/d \le (1 + 25d/24)^2 + \theta + (47/108)d \le 58163/27648 < 17/8.$$

The coefficient $47/108 = 1/36+11/27$ pays ϵ^2 and its possible positive cross term. Also $2kV/d \le 3/2$, so

$$\log EH^4_{4,3}/d \le -16 + 3/2 + (17/4)(17/8) = -175/32. \tag{R35.A33}$$

For $b \ge 5$, L3, if $\epsilon \ge 0$ its cross term in (R35.A28) is nonpositive; if $\epsilon < 0$ it is <$91d^2/100$. Hence

$$\|V_3\|^2/d \le 1 + \theta + (367/100)d + (36/25)d^2 < 9/4.$$

Also $4k(D3 - \rho)/d \le 4k(\theta d + \theta^4) < 3/5$, since 4k $\theta < 17/8$, 4k $\theta^4 < 3/20$ and $d \le 1/5$. Therefore (R35.A16) is at most

$$[-9 - 8 + 12/5 + 3/5 + (17/4)(9/4)]d = -71d/16. \tag{R35.A34}$$

This completes the low-domain $L = 2, 3$ ordinary-fourth rows with $\mathbb{E}H^4 \le e^{-15d/4}$, together with the $b = 4, L = 1$ row. For $b \ge 5, L = 1$, Proposition P.4 provides the required whole-domain outlet.

R35.7 Every finite $L \ge 4$, including the real constant coordinate

Rewrite (R35.A13) using the positive convolution (M.2):

$$\begin{aligned} Z^*_L &= S^-_L + \delta S^-_{L-1} - aS^-_{L-2} + kB^L R \\ &= \nu_0 + Q_L(B)u, \quad \nu_0 = 1 + \epsilon. \end{aligned}$$

Its difference from Y_L is exactly $\sum_{j=3}^{L} \gamma_j \bar{f}_{L-j}$, where

$$\begin{aligned} \gamma_3 &= d - kD_2, \quad \gamma_4 = d + \delta d - kD_3, \\ \gamma_j &= \beta_j + \delta\beta_{j-1} - a\beta_{j-2} - kD_{j-1} \ (j \ge 5). \end{aligned} \tag{R35.A35}$$

Taking means gives $\alpha_L := nu0 - EZ^*_L = \gamma_{L+1}$ for $L \ge 4$. This uses $ES^-_n = 1 - \beta_{n+1}$, not a mean-zero assumption.

The exact finite polynomials are

$$\begin{aligned} P_n &= G_n + kB^n(a_f - b_f B), \\ Q_n &= P_n + \epsilon G_{n-2} + \delta B^{n-2} \quad (n \ge 3). \end{aligned} \tag{R35.A36}$$

For every such n, $\|P_n\| \le 1$. At n3 the coordinates are $1 - x - kx^2 b_f$, 1-kx a_f, nonnegative and no larger than the G3 coordinates, because $ku0 \le 1$ and $kz0 \le 37/48$ for b4 or $91/124$ for $b \ge 5$; $1 - x - x^2 > 0$. At n4, $|P4| \le (1 - x)(1 + x/2) + x^2 \le 1$. At n5, $|G_5| \le 1 - x/2 + x^2/2$ and $x^{5/2} \le 13x^2/20$, giving $|P5| \le 1 - x/2 + 23x^2/20 \le 1$ since $\theta < 10/23$. The squared comparison for G5 has exact difference $x^2(1 + 2x - 3x^2)/4 \ge 0$. For every $n \ge 6$,

$$|P_n| \le (1 + x^3)/\sqrt{1 + x} + x^3 \le 1.$$

For $x > 0$ the last condition is equivalent to $x(1+\sqrt{1+x}) \le 1$, which follows from its bound $(13/32)(11/5) = 143/160$. The zero frequency is direct.

Using (R35.A19),(R35.A26), $\sqrt{\theta} < 13/20$ and $\|G_2\| < 6/5$ gives

$$\begin{aligned} \|Q_n\| &\le 1+d \quad (b4), \\ \|Q_n\| &\le 1+7d/5 \quad (b \ge 5), \qquad n \ge 3. \end{aligned} \tag{R35.A37}$$

At n3 the added costs are at most 27d/32 and 69d/50. At n4 use $(6/5)|\epsilon| + \theta$ delta; at $n \ge 5$ use |epsilon|+$\theta^{3/2}$delta. Both are smaller than the stated respective budgets. These bounds cover the finite-length tails.

Let $h = 1/2 - 6d/5 > 0$. Every $\gamma_j \ge h$ d. $D2 \le 0$ gives $gamma3 \ge d$. Since $ka1 \le ku0d \le d$ and $D3 \le \theta$ a1, $gamma4 \ge \kappa$ d. For $j \ge 5$, each positive D_{j-1} is $\le \theta^2 z0$ d. The first positive even class is D4; the first relevant positive odd class is $D7 \le \theta^3 u0d \le \theta^2 z0$ d, and later powers decrease. The needed theta $u0 \le z0$ holds throughout both domains.

For b4 the lower coefficient is kappa-ku0 d-k $\theta^2 z0$. Subtract h and multiply by m; the result is

$$875/32768 + (40369/81920)d - (11/5)d^2,$$

a concave quadratic with positive endpoints $875/32768$ and $4063/327680$. For $b \ge 5$ use $a \le d$. The corresponding difference is

$$875/32768 - (3241/81920)d - (9/40)d^2,$$

decreasing with positive endpoint $40101/4096000$. Therefore $\gamma_j \ge hd$ for every $j \ge 3$ and $\alpha_L \ge hd$.

Retain the last two actual terms $j = L$, L-1:

$$0 \le T^L F/m^2 \le Z = Z_L^* - hd(1+q).$$

The finite recurrence $Q_L = nu0 + BQ_{L-1}$ gives

$$\begin{aligned} Z &= Cq - hd + N_0, \\ C &= \nu_0 - hd, \\ N_0 &= BQ_{L-1}u, \\ EN_0 &= -\alpha_L \le -hd. \end{aligned} \tag{R35.A38}$$

This displays the real constant C. Once $C > 0$, weighted AM–GM between actual nonnegative p, Z/C yields

$$\begin{aligned} H_{b,L} &\le (c_* + N/2)^2, \\ c_* &= m\sqrt{C} - \frac{mhd}{2\sqrt{C}}, \quad N = \frac{m}{\sqrt{C}} N_0. \end{aligned} \tag{R35.A39}$$

The base equals $m(\sqrt{C}p + Z/\sqrt{C})/2 \ge 0$. Writing $N = BV_*$, (R35.A9) gives fourth exponent

$$8(c_* - 1) + 4EN + (17/4)\|V_*\|^2. \tag{R35.A40}$$

For b4, $1 - 2d/3 + 6d^2/5 \le C \le 1$ and $C - m^2 \ge 4d/3 + d^2/5 > 0$; in particular $C > 0$. By (R35.A37), $\|V_*\|^2 \le (1-d^2)^2 d/C \le 17d/16$. The full comparison is

$$(17/16)(1 - 2d/3 + 6d^2/5) - (1-d^2)^2 \ge 1/16 - (17/24)d + (257/80)d^2 > 0.$$

Here $d^4 \le d^2/16$ and the discriminant is negative because $289/576 < 257/320$. Also $m/\sqrt{C} \ge 3/4$ and $h \ge 1/5$, so $EN \le -3d/20$. Using m $\sqrt{C} \le (m^2 + C)/2$,

$$c_* \le 1 - (65/48)d + (31/20)d^2 \le 1 - 29d/30.$$

Thus (R35.A40) is at most [-$116/15 - 3/5 + 289/64$]$d = -733d/192 < -15d/4$.

For $b \ge 5$,

$$C_{\text{low}} = 1 - 11d/10 + 6d^2/5 \le C \le 1 + d/10 + 6d^2/5 \le 267/250,$$
$$C - m^2 \ge (23/20)d - (21/320)d^2 > 0.$$

Again $C > 0$. Expanding,

$$m^2(1 + 7d/5)^2 = (1 + 11d/40 - 63d^2/40)^2 \le 1 + 11d/20 - (4919/1600)d^2.$$

The discarded tail is d^3[-$693/800 + (3969/1600)d$]≤ 0. Subtracting this upper bound from $(5/4)C_low$ leaves $1/4 - (77/40)d + (7319/1600)d^2 > 0$, whose discriminant comparison is $5929 < 7319$. Thus $\|V_*\|^2 \le 5d/4$. Moreover $m/\sqrt{C} \ge 149/200$, since

$$961/1600 - (149/200)^2(267/250) = 78583/10000000 > 0.$$

With $h \ge 13/50, EN \le -(149/200)(13/50)d < -19d/100$. Using $\sqrt{C} \le 1 + (C-1)/2$ and dropping only a nonpositive cubic,

$$c_* \le 1 - (1009/800)d + (3963/4000)d^2 \le 1 - 21d/20.$$

The endpoint coefficient is $10631/10000 > 21/20$. Hence (R35.A40) is at most [-$42/5 - 19/25 + 85/16$]$d = -1539d/400 < -15d/4$. Every finite low-domain $L \ge 4$ is now paid.

R35.8 Actual high-domain terminal means and every $L \ge 2$

R35.8.1 $b = 4$, including the remaining L1 domain

Original adjunction and finite convolution give

$$t_4 := E(h_2 f_4) = 1 - 3d + \delta_2 - d^2 + \delta_3 \le 1 - (2487/1024)d - d^2.$$

For $d \ge 1/4$ this upper bound decreases, with value $1353/4096 < 1/3$ at $1/4$. Thus $TF4 \le 2/3$. Positivity gives $H_{4,2} \le (2/3)pq, H_{4,3} \le (2/3)p\bar{f}_2$ and $H_{4,L} \le (2/3)p\bar{f}_{L-1}$ for $L \ge 4$. Equations (R35.A10)-(R35.A11), $4\log(2/3) \le -4/3$ and $d \le 1/3$ prove all these ordinary-fourth bounds $\le$exp(-15d/4). In the only cost-bearing row L3, $-4/3 + d/4 \le -15d/4$ is exactly $d \le 1/3$.

For b4, L1 use (R35.A17) with exact $D0 = \delta_2$ and $EZ = d^2 + \delta_2$. Applying (R35.A9) to -R cancels the linear d2 terms and gives

$$\log EH_{4,1}^4 \le -8d - 4d^2 + \{2/(1-d) + 17/4\}V. \qquad \text{(R35.A41)}$$

The bound $V/d \le d^2+2\theta d+\theta^2+\theta^3 < 5/8$ holds throughout $d \le 1/3$: its left side increases, and at $1/3$ the positive cross difference is $35*32768-7605*144 = 51760$. For $d \ge 1/4$ the coefficient in (R35.A41) is $\le 29/4$, giving $\log(\mathbb{E}H_{4,1}^4)/d \le -9+145/32 = -143/32 < -15/4$. Combined with part R35.6 this proves the b4, L1 full-domain row.

R35.8.2 Every finite $b \ge 5$ on $1/5 \le d \le 1/3$

Let $t_b = E(h_2 f_b)$, an actual nonnegative mean. Adjunction gives $t_b = E[\mathrm{p}\ f_{b+1}]$=2$\mu_{b+1}$-$\mu_{b+2}$. Multiply the original convolution by $h_2 = 1-g-Kg$ to obtain

$$\begin{aligned} t_b &= 1-\beta_{b+1}-\beta_{b+2}+\beta_{b+3} \\ &\quad -\sum_{j=3}^{b}\beta_j t_{b-j}. \end{aligned} \tag{R35.A42}$$

In particular actual $t0 = 1+d$ is retained. Direct finite expansion gives

$$\begin{aligned} &t_0 = 1+d, && t_1 = 1, \quad t_2 = 1-d-\delta_2, \\ &t_3 = 1-2d-d^2, \\ &t_4 = 1-3d+\delta_2-d^2+\delta_3, \\ &t_5 = 1-4d+2(1+d)\delta_2, \\ &t_6 = 1-5d+2d^2+d^3+(3+2d)\delta_2-\delta_3-\delta_4, \\ &t_7 = 1-6d+5d^2+2d^3+4\delta_2-\delta_2^2-2(1+d)\delta_3, \\ &t_8 = 1-7d+9d^2+2d^3+(5-4d-3d^2)\delta_2-\delta_2^2 \\ &\qquad -(3+2d)\delta_3+\delta_4+\delta_5. \end{aligned} \tag{R35.A43}$$

For explicit finite-end checks, b7 has base 1-d+d2-d3 and loss

$$d(t_4+t_3)+\tau(t_2+t_1+t_0)+\delta_3 t_0 = 5d-5d^2-2d^3-3\delta_2+\delta_2^2+\delta_3+2d\delta_3.$$

At b8 the base is 1-d+d2-d3+d4+d5 and loss is

$$\begin{aligned} &d(t_5+t_4)+\tau(t_3+t_2+t_1+t_0)+\delta_3(t_1+t_0) \\ &= 6d-9d^2-2d^3+(4d+3d^2-4)\delta_2+\delta_2^2+2\delta_3+2d\delta_3. \end{aligned}$$

The four short cases have the following upper polynomials for $t_b-(1-d)/2$:

Table 92: The terminal pairs $(2,b)$ for every $b \ge 4$ at a unit root: parameter values.

Entry	Upper polynomial
5	$1/2-(43/16)d+(13/16)d^2$; Complete continuous payment: decreasing; left value -1/200.
6	$1/2-(105/32)d+(45/16)d^2+d^3$; Complete continuous payment: derivative $\le$-103/96; left value -143/4000.

Entry	Upper polynomial
7	$1/2-(31/8)d+5d^2+2d^3$; Complete continuous payment: convex; endpoints -$59/1000$, $-35/216$.
8	$1/2-(143/32)d+(59/8)d^2+(25/32)d^3$; Complete continuous payment: convex; endpoints -$37/400$, $-61/432$.

For b8, first group $d4+d5 \le (\theta+\theta^2)d3 \le (3+2d)d3$, then use the positive coefficient $5-4d-3d^2$ with $\delta_2 \le \theta$ d. At b7 the omitted terms are explicitly nonpositive.

For every finite $b \ge 9$, the seven actual coefficients of t0,..., t6 in (R35.A42) have indices at least three. The base is $\le$1-tau+d3, from the finite beta increments. Their exact sum is

$$\sum_{j=0}^{6} t_j = 7 - 14d + d^3 + (5+4d)\delta_2 - \delta_4.$$

Thus

$$t_b \le 1 - \tau[8 - 14d + d^3 + (5+4d)\delta_2 - \delta_4] + \delta_3. \tag{R35.A44}$$

Let P be the amount subtracted from one. For the original global $v = \delta_2/d$ in [d, theta],

$$P \ge d[(1-v)(A+Cvd) - \theta v], \qquad A = 8 - 14d + d^3, \quad C = 5 + 4d - \theta^2.$$

The bracket decreases in v because its derivative is -$A+Cd(1-2v)-\theta < 0$, using $A \ge 10/3$ and $Cd \le 19/9$. Take the legitimate global endpoint $v = \theta$, and discard only nonnegative d^3, d^4 terms. Then

$$\begin{aligned} P - (1+d)/2 &\ge -1/2 + (4183/1024)d - (7493391/1048576)d^2 \\ &\ge -1/2 + 4d - (29/4)d^2 > 0. \end{aligned} \tag{R35.A45}$$

The last concave polynomial has endpoints $1/100, 1/36$. Hence every finite $b \ge 5$ satisfies

$$t_b \le (1-d)/2, \qquad 0 \le TF_b \le 1 - d. \tag{R35.A46}$$

All finite b cases are exhausted.

For $L \ge 2$, positivity gives $H_{b,L}$$\le$(1-d)$p\bar{f}_{L-1}$. Use $pq \le 1$ at L2, (R35.A10) at L3, and (R35.A11) at $L \ge 4$. Since $4\log(1-d) \le -4d$, the exponents are respectively -4d,-15d/4,-4d. Combined with parts R35.6-7 and b4 high,

$$EH_{b,L}^4 \le e^{-15d/4} \qquad (b \ge 4,\ L \ge 2,\ 0 \le d \le 1/3). \tag{R35.A47}$$

R35.9 Every $b \ge 5$, $L = 1$, on the complete host domain

Apply Proposition P.4 to the actual $H_{b,1} = pT(h_2 f_b)$. It proves

$$I_q(T^2 H_{b,1})^3 \le e^{-d/500} \qquad (0 \le d \le 1/3).$$

The same $\sigma = 2$ retained vector and the same canonical direction are used throughout this interval. Its $b = 5$ and $b \ge 6$ data are the two integer cases in Lemma P.3.

R35.11 Exact C094 bridge, curvature and logarithm payment

R35.11.1 The original outer output

Use Corollary M.19 and paragraph P.5. On the original measure, every real Y satisfies

$$I_q(T^2Y)^3 \le B_2(E|Y|^4)^2, \tag{R35.A61}$$

$$B_2 = [1-2d^2+(8-2d)\delta_2](1+1024\delta_2/855)^8 \le e^{c_2 d-5d^2}, \qquad c_2 = 24427/3420. \tag{R35.A62}$$

R35.11.2 Completion of the ordinary-fourth rows

Every row with $\mathbb{E}H^4 \le e^{-15d/4}$ yields

$$\log I_q(T^2H)^3 \le (c_2-15/2)d-5d^2 = -(1223/3420)d-5d^2 \le -d/500. \tag{R35.A63}$$

Equation (R35.A47) covers all $L \ge 2$ and $b \ge 4$. Paragraph R35.8.1 and the low-domain bound in R35.6 cover $b=4, L=1$. For $b \ge 5, L=1$ use Proposition P.4 on its complete host domain. Hence

$$I_q(T^2[pT^L(h_2f_b)])^3 \le e^{-d/500} \le 1 \qquad (b \ge 4,\ L \ge 1). \tag{R35.A68}$$

R35.12 Original-tree realization, correlated KL, full reversal and all maps

The direction is fixed before (mu, K): the terminal b-arm points outward; the two-arm has first edge toward the terminal and second toward its leaf; the internal root's unit arm points outward; all L inner edges point from terminal to internal root; both outer edges point from internal root to receiving root. Every genuine equal pair uses f_l, h_l at its original vertex. Hence the exact original output is

$$H_D = E_r pT^L(E_t h_2 f_b), \qquad U_D = T^2 H_D.$$

No analytic case changes this direction, including L2. Since $0 < E_t, E_r \le 1$ on the positive support and T is positive,

$$0 < H_D \le pT^L(h_2 f_b), \qquad 0 < U_D \le T^2[pT^L(h_2 f_b)].$$

The positive I_q weight makes (R35.A68) apply to the actual decorated output. The multiplier is the specified product of equal-pair factors. Without decorations the component has exactly L+b+5 original edges: terminal 2+b, inner L, root side 1, outer 2. Each equal pair adds its actual 2l edges and original interior vertices. All subdivision vertices and independently specified core lengths remain in the integral. The local outer length is two; other independently specified outer lengths retain their original values.

The outer kernel remains

$$T^2(x,y) = E_z[(1-K(x,z))(1-K(z,y))].$$

Adjoints and Holder use the original mu, $T* = A$. Paragraph P.5 controls the inverse weight through the same-kernel centered-row bound as q becomes small. Constants are independent of min mu; q may be arbitrarily small across finite models.

At $d = 0, K1 = 0$ on the positive support but K need not be zero. $A1 = T1 = 1$, so leaf integration gives every actual original tree message one. Then $H_D = U_D = q = 1$ and the claimed result is equality. All divisions by d were confined to $d > 0$. Scalar zero-variance cases and frequency $x = 0$ were also included.

Complete reversal is substitution $K \mapsto$ -K in the same fixed structure. The terminal becomes W2 $\bar{f}_b$; inner T^L becomes A^L; unit p becomes q; all equal pairs become $\bar{f}_l W_l$; outer T^2 becomes A^2; I_q becomes I_p. The same loss follows because d is unchanged. Swapping only one weight or one connector while retaining other orientations is not justified.

P.9 Odd terminal lists at a unit root with a two-edge outlet

Let the original distinct residual lengths be

$$(a, b_1, c_1, \dots, b_k, c_k), \qquad 1 \le a < b_1 < c_1 < \cdots < b_k < c_k, \quad k \ge 1,$$

excluding the exact list $(1, 2, c)$. The receiving root has one outward unit arm, the inner connector has any positive integer length L, and the outer connector consists of two original edges. Genuine equal-pair products at the receiving root and terminal are denoted by E_r, E_t. The actual messages are

$$H_D = E_r p T^L (E_t F), \qquad U_D = T^2 H_D.$$

The directions and their stronger ordinary-fourth rates are listed below. Each direction is chosen from the original lengths, before the host kernel is given. In particular, with the original reference

Table 93: Unit-root directions and fourth rates: $\mathbb{E}H_D^4 \le e^{-\rho d}$.

Residual range	Actual F	ρ
$b_1 \ge 4$, all $a < b_1$, $k \ge 1$	$\bar{f}_a \prod_i h_{b_i} f_{c_i}$	5
$b_1 = 3$, $a = 1, 2$, $k \ge 2$	$\bar{f}_a f_3 h_{c_1} \prod_{i \ge 2} h_{b_i} f_{c_i}$	4
$(1, 3, c)$, $c \ge 4$	$q f_3 h_c$	15/4
$(2, 3, c)$, $c \ge 4$	$f_2 f_3 h_c$	4
$b_1 = 2$, $k \ge 3$	$q f_2 h_{c_1} \prod_{i \ge 2} h_{b_i} f_{c_i}$	77/20
$(1, 2, c, b, e)$, $4 \le c < b < e$	$q f_2 h_c h_b f_e$	477/125
$(1, 2, 3, b, e)$, $5 \le b < e$	$q f_2 h_3 h_b f_e$	4
$(1, 2, 3, 4, e)$, $e \ge 5$	$q f_2 h_e h_3 f_4$	4

factor q,

$$\mathbb{E}H_D^4 \le e^{-15d/4}, \qquad I_q(U_D)^3 \le e^{-1223d/3420 - 5d^2} \le 1.$$

Proof. The finite-message construction and all its moment payments are Theorem O.2. Its minimum-four packet is Lemma O.5; the next two families are Sections O.24.5 and O.24.6. The first-pair-two multipair family and the three five-arm directions are proved in Sections O.24.8 and O.24.14, respectively. These ranges partition every displayed increasing odd list.

Restore the genuine products by positivity as in Section O.24.15. Corollary M.19 then applies to the actual $T^2 H_D$:

$$I_q(U_D)^3 \le e^{(24427/3420)d - 5d^2} (\mathbb{E}H_D^4)^2 \le e^{-1223d/3420 - 5d^2}.$$

This uses the ordinary fourth resource, the two original outer edges and the unchanged reference factor q. Equal lengths, complete reversal and $d = 0$ are included by the same positive comparison and original leaf integration. □

Q Odd-root fourth moments and outlets

This family treats odd root residuals with at least three entries. The full signed-vector estimates and the curved-tail estimates state their distinct hypotheses; short first rows are evaluated before the general finite continuation is applied.

Q.1 Minimum-one odd-root bridges and the complete fourth transfer

Q84.0 Exact completed scope and the new unanchored fourth interface

The original finite host, measure, adjoint, vertices, and edges are retained. Write

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n 1, \quad \bar{f}_n = T^n 1, \quad h_n = T f_{n-1}, \quad W_n = A\bar{f}_{n-1}, \quad I_q(H) = \mathbb{E}[q^{-1/3}|H|^{8/3}]. \tag{Q84.0.1}$$

The root has the original odd residual list

$$(1, u_1, v_1, \dots, u_j, v_j), \qquad 1 < u_1 < v_1 < \cdots < u_j < v_j, \qquad j \geq 1. \tag{Q84.0.2}$$

Its terminal has residual arms $(1, b)$, with every integer $b \geq 2$. The inner and outer lengths $L, M \geq 1$ are independent original lengths.

The root configurations considered here are:

* $u_1 = 2$, every finite j, every remaining arm, and all b, L, M. * $u_1 = 3$ or 4, $j = 1$, every $v_1 > u_1$, and all b, L, M.

Put $X_{b,L} = T^L(qf_b)$. For the second block use

$$Y = pW_{u_1}\bar{f}_{v_1}X_{b,L}. \tag{Q84.0.3}$$

For the first block put $v = v_1$, $P = \prod_{i=2}^{j} W_{u_i}\bar{f}_{v_i}$, and use

$$Y = pP \begin{cases} h_2 f_v\, T(qf_b), & L = 1,\ b \geq 3,\ v \geq 4, \\ \bar{f}_2 W_v\, X_{b,L}, & \text{all other original integer cases.} \end{cases} \tag{Q84.0.4}$$

The empty product is one. The choice depends on b, v, L, independently of μ or K.

Fourth-moment bound. Every message (Q84.0.3) or (Q84.0.4), including all original equal-pair decorations, satisfies

$$\boxed{EY^4 \leq \exp(-705d/128), \qquad d = \mathbb{E}g^2.} \tag{Q84.0.5}$$

Unanchored fourth-moment transfer. For every real original function H, every legal original kernel, and every integer $M \geq 1$,

$$\boxed{I_q(T^M H)^3 \leq e^{11d}(\mathbb{E}|H|^4)^2.} \tag{Q84.0.6}$$

There is no reserved arm in (Q84.0.6). In particular no $\bar{f}_a$ is inserted into the tree. The stronger coefficients for $M = 1, 2, 3$ are $e^{204d/19}, e^{10d}, 1$, respectively.

The new all-length path ingredient is

$$E(\bar{f}_M^4/q) \leq e^{11d} \quad (M \geq 4). \tag{Q84.0.7}$$

It is proved here from the actual finite path convolution and a truncated quartic majorant. It is not ordinary path-mean monotonicity, a general weighted contraction, or a conclusion drawn from an arbitrary fourth budget.

The exact final bridge for every listed original half-tree is therefore

$$\boxed{I_q(T^M Y)^3 \le e^{-d/64} \le 1 \qquad (M \ge 1).} \tag{Q84.0.8}$$

This single bound covers the full continuous host domain and all integer tails. Its margin follows from $2(705/128) - 11 = 1/64$.

Equal-arm factors and full reversal are handled below. The whole-tree assembly is given in Section 11. A union with other exact root catalogues is explicitly separated from this theorem. No complete Q5, arbitrary terminal table, singleton-root role, or general branching recursion is claimed.

Q84.1 Original finite calculus and finite spectral bookkeeping

Delete only atoms of zero original mass. All expectations, inner products, adjoints, and rotation blocks below use the original μ. Since $K(x,x) = 0$, $p(x), q(x) \ge \mu(x) > 0$. Every actual tree message is positive: the all-equal labeling already has positive weight. A and T are positive operators.

Use the C014 calculus and the exact identities:

$$r = Kg, \quad \sigma = Kr, \quad t = K\sigma, \delta_i = \|K^i 1\|_2^2, \quad d = \delta_1, \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2,$$

$$0 \le d \le 1/3, \quad \|K\|^2 \le \theta, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{i+1} \le \theta \delta_i, \quad |KV(x)| \le \|V\|_2, \quad r \le pq/2. \tag{Q84.1.1}$$

Also $\mathbb{E}r = -d$, $E\sigma = 0$, $\mathbb{E}t = \delta_2$, and $\mathbb{E}(gr) = 0$. The lower spectral-moment inequality used below is

$$\delta_3 \ge \delta_2^2/d \qquad (d > 0), \tag{Q84.1.2}$$

which is Cauchy on the same spectral measure of g.

Define finite sums only:

$$G_n(z) = \sum_{a=0}^{n-1} z^a, \quad \beta_1 = \beta_2 = 0, \quad \beta_n = \sum_{a=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{a-1} \delta_a.$$

Positive finite convolution gives

$$\sum_{a=0}^{n} (\pm K)^a 1 = f_n^{\pm} + \sum_{a=3}^{n} \beta_a f_{n-a}^{\pm}, \qquad f_n^{+} = f_n, \; f_n^{-} = \bar{f}_n. \tag{Q84.1.3}$$

Its coefficients obey

$$\tau \le \beta_n \le d \; (n \ge 3), \qquad \beta_n \le d - \kappa\delta_2 \; (n \ge 5). \tag{Q84.1.4}$$

The second bound retains $-\delta_2 + \delta_3$ and bounds the remaining finite alternating tail by its first positive end. No path mean is declared monotone in its length.

At a frequency $z = i\sqrt{x}$, $0 \le x \le \theta$, write

$$G_n(z) = a_n + z c_n.$$

For $n \ge 2$, both coordinates lie in $[1 - x, 1]$; for $n = 1$ they are $1, 0$. For $n \ge 3$, $a_n \le 1 - x + x^2$ and $|G_n| \le 1$. For $n \ge 4$, both coordinates are at most $1 - x + x^2$. Always $|G_n|^2 \le 1 + x$. These

are finite alternating-sum statements. The geometric formula gives squared numerator $1+x^n$ for odd n and $(1\pm x^{n/2})^2$ for even n, over $1+x$. In the positive-ending even case $n\ge 6$, use $2x^2+x^5\le 1$.

Let ν be the original spectral measure of g. Then

$$\nu([0,\theta])=d, \qquad \int x^a\,d\nu=\delta_{a+1}.$$

For $Q(z)=A(x)+zB(x)$ and $z=-K$, the exact signs are

$$\mathbb{E}[Q(-K)g]=\int B\,d\nu, \qquad \mathbb{E}[KQ(-K)g]=-\int A\,d\nu. \tag{Q84.1.5}$$

Every spectral inequality below is applied only in this original spectral theorem.

If $d=0$, g vanishes, and successive leaf integration makes all actual messages equal to one. K need not vanish. All final conclusions include this equality case. Divisions by d below are used only when d is positive.

Q84.1.1 The full-mean exponential lemma

The full-mean bound in Q5-C081, §2, holds for every real original vector V and every $s\ge 0$

$$\mathbb{E}e^{sKV}\le \exp\{sE(KV)+(17/64)s^2\|V\|_2^2\}. \tag{Q84.1.6}$$

This full signed estimate is proved in Section M.1, paragraph M.1. Its fourth instance is also (M.43) with the vector sign changed because its operator is $-K$. In particular the fourth-moment application is

$$\mathbb{E}e^{4KV}\le e^{4E(KV)+(17/4)\|V\|^2}. \tag{Q84.1.7}$$

Q84.1.2 Later negative pairs

The exact C067 pair-loss lemma and full reversal give

$$P=\prod_i W_{u_i}\bar f_{v_i}\le e^{S-3nd}, \quad S=KU(-K)g, \quad U(z)=\sum_i\sum_{a=u_i}^{v_i-1}z^{a-1}, \tag{Q84.1.8}$$

when all the displayed $u_i\ge 4$; n is their number. The intervals are disjoint finite intervals. Equation (O.20) gives

$$\|U(-K)g\|\le (3/8)\sqrt{d}, \qquad ES\le d/12. \tag{Q84.1.9}$$

For example a finite zero-one mask Q satisfies $|Q(i\sqrt{x})|^2\le (1+x)/(1-x^2)^2$, whose maximum is $32768/16245<100/49$. Multiplying by the first compulsory power and summing the possible positive mean powers proves (Q84.1.9). If the first short is at least five the sharper norm is $\sqrt{d}/4$, and the common restoration uses the preceding bound.

For a single canonical pair starting at three,

$$W_3\bar f_v\le e^{S-(2d-\delta_2)}, \qquad S=K[z^2G_{v-3}(z)]g, \quad ES\le\delta_2. \tag{Q84.1.10}$$

Its loss is exactly $2d-\delta_2$, not $3d$.

Q84.2 An unanchored ordinary-fourth bridge for every outer length

Let B be any positive original kernel operator, with row $h = B1$. Pointwise Cauchy and then Holder give

$$|BH|^2 \le hB(|H|^2),$$

$$I_q(BH)^3 \le E(h^4/q)\left[E((B(|H|^2))^2)\right]^2 \le E(h^4/q)\,\|B\|_{2\to 2}^4(\mathbb{E}|H|^4)^2. \tag{Q84.2.1}$$

The Holder exponents are 3 and $3/2$; its first factor cubed is exactly h^4/q. This proves (Q84.2.1) for arbitrary real H without an added arm.

The exact C064 operator estimates are

$$\|T\| \le 1 + 32d/19, \quad \|T^2\|^2 \le (1 + d/2)^4, \quad \|T^M\| \le 1 \ (M \ge 3). \tag{Q84.2.2}$$

The middle expression contains d/2, not δ_2.

For $M = 1$ the row is q and $\mathbb{E}q^3 \le 1 + 4d$, because $|\mathbb{E}g^3| \le d$. Thus the coefficient in (Q84.2.1) is at most $e^{204d/19}$.

For $M = 2$ write $\bar{f}_2 = q + r$ and $t = r/q$. Positivity and (Q84.1.1) give $-1 \le t \le p/2 \le 1$. On this interval,

$$(1 + t)^4 \le 1 + 4t + (6 + 2p + p^2/4)t^2.$$

Since $q(6 + 2p + p^2/4) \le 12$, integration yields

$$E(\bar{f}_2^4/q) \le 1 - d - \mathbb{E}g^3 + 4E(g^2 r) + 12\delta_2 \le 1 + (9/8)d + 12\delta_2 \le 1 + 6d. \tag{Q84.2.3}$$

The middle step uses $r \le (1 - g^2)/2$ and $-g - 2g^2 \le 1/8$. To check the coefficient 12, expand it as $33/4 - 23g/4 - 9g^2/4 - g^3/4$; it decreases on [-1, 1] and equals 12 at -1. Combining (Q84.2.3) and (Q84.2.2) gives coefficient e^{10d}.

For $M = 3$, the C056 actual path theorem gives $E(\bar{f}_3^4/q) \le 1 - 23d/36 \le 1$. Together with (Q84.2.2) this gives coefficient at most one.

Q84.2.1 Every remaining path row: a finite quartic proof

Fix $M \ge 4$. From (Q84.1.3),

$$0 < \bar{f}_M \le U = q + R - \beta_M, \quad R = KG_{M-1}(-K)g. \tag{Q84.2.4}$$

The upper U is nonnegative because it dominates an actual path. All $n \ge 3$ also satisfy $\bar{f}_n \le 5/2$. Indeed their finite upper gives

$$\bar{f}_n \le 2 + \sqrt{(1 + \theta)d} - \kappa d \le 5/2.$$

The scalar middle function increases on $[0, 1/3]$; at $1/3$ the last comparison is $\sqrt{15/32} \le 67/96$, or $4320 \le 4489$. Thus positivity gives the second actual upper $\bar{f}_M \le (5/2)q$.

For every real $s \ge -1$,

$$\min(1 + s, 5/2)^4 \le 1 + 4s + (57/4)s^2. \tag{Q84.2.5}$$

On [-1, $3/2$] this is the quartic expansion with $6 + 4s + s^2 \le 57/4$. Beyond $3/2$, the quadratic is increasing and already majorizes $(5/2)^4$. Apply (Q84.2.5) to $s = (R - \beta_M)/q$, which is at least -1 by (Q84.2.4):

$$\bar{f}_M^4/q \le q^3 + 4q^2(R - \beta_M) + (57/4)q(R - \beta_M)^2. \tag{Q84.2.6}$$

Apply the truncated quartic inequality directly to U.

The same finite R obeys

$$-d \le \mathbb{E}R \le -\tau, \quad \kappa\delta_2 \le E(gR) \le \delta_2, \quad \mathbb{E}R^2 \le \delta_2, \quad |R| \le \sqrt{d}. \tag{Q84.2.7}$$

Here $M-1 \ge 3$ permits $|G_{M-1}| \le 1$. The sign of E(gR) follows from its positive finite imaginary coordinate. In particular

$$\begin{aligned} Eq^2(R-\beta_M) &\le -2\tau - 2\kappa\delta_2 + d\sqrt{d} - \tau d, \\ Eq(R-\beta_M)^2 &\le \delta_2 + 3d^2 + d\sqrt{\delta_2} + 2d\delta_2. \end{aligned} \tag{Q84.2.8}$$

For the last inequality use $|E(gR^2)| \le d\sqrt{\delta_2}$, from Cauchy and $\mathbb{E}R^4 \le d\,\mathbb{E}R^2$, and retain the same $E(gR)$.

Substitution into (Q84.2.6), with $\sqrt{\theta} < 2/3$, gives

$$E(\bar{f}_M^4/q) \le 1 + \frac{199}{64}d + \frac{27}{2}d\sqrt{d} + \frac{3325}{64}d^2 \le 1 + 11d + \frac{121}{2}d^2 \le e^{11d}. \tag{Q84.2.9}$$

The middle scalar inequality is valid on the full domain. After dividing its difference by d and multiplying by 64, it is

$$505 - 864x + 547x^2 = 547(x - 432/547)^2 + 89611/547 > 0, \qquad x = \sqrt{d}.$$

This proves (Q84.0.7) for every $M \ge 4$ and then (Q84.0.6) by (Q84.2.1)–(Q84.2.3).

Q84.3 Same-direction terminal envelopes and a stronger joint loss

For $b \ge 3$ define the actual finite data

$$\begin{gathered} R_b = KG_{b-1}(K)g, \quad m = 1 - \beta_b/2, \\ \lambda = 1/m, \quad t_0 = \beta_{b+1}, \\ e = d - \beta_{b+2}, \quad V_b = \mathbb{E}R_b^2, \quad \rho = E(rR_b), \\ \gamma = \lambda t_0 - \lambda^2 V_b/2, \\ a_1 = \lambda e, \quad B_* = \gamma + a_1, \quad B = 1 - \gamma, \\ P_n(z) = G_n(z) + \lambda z^{n+1} G_{b-1}(-z). \end{gathered} \tag{Q84.3.1}$$

The exact C058/C075 finite proof supplies

$$\begin{gathered} 1 \le \lambda \le 6/5, \quad 3d/10 \le \gamma \le 6d/5, \quad \kappa\delta_2 \le e, \rho \le \delta_2, \\ V_b = \delta_2 + \delta_3 \ (b=3), \quad V_b \le \delta_2 \ (b \ge 4), \quad 5d/8 \le B_* \le 3d/2. \end{gathered} \tag{Q84.3.2}$$

For $b = 3, t_0 = d, e = \rho = \delta_2, \beta_b = d$. For $b = 4, t_0 = \tau, e = \delta_2, \beta_b = d$. When $b \ge 5, V_b \ge \kappa^2\delta_2$, by the finite coordinate lower bound.

The exact finite positive envelope is

$$X_{b,L} \le m^2 \left[V_L - \gamma \bar{f}_{L-1} - a_1 \bar{f}_{L-2} - \sum_{i=3}^{L} \varepsilon_i \bar{f}_{L-i} \right],$$

$$V_L = \sum_{i=0}^{L} (-K)^i 1 + \lambda(-K)^L R_b, \quad \varepsilon_3 = d - \lambda\rho \ge 41d/80, \quad \varepsilon_i \ge \tau \ (i \ge 4). \tag{Q84.3.3}$$

Only existing indices occur. This is an actual envelope for the one terminal qf_b; it is not obtained by swapping a different terminal into an old moment conclusion.

Q84.3.1 A new stronger common bound for B_*

We shall need the whole-integer bound

$$\boxed{B_* \geq d - 3\delta_2/4.} \tag{Q84.3.4}$$

In addition $B_* \geq d$ for $b = 3$, $B_* \geq d - \delta_2/2$ for $b = 4$, and on all $b \geq 3$

$$B_* - d \leq 11\delta_2/10. \tag{Q84.3.5}$$

Put $S = t_0 + e$. For fixed actual S, V_b the expression $\lambda S - \lambda^2 V_b/2$ increases for $1 \leq \lambda \leq 6/5$. For $b \geq 4, S \geq d - \delta_3$, so its derivative is at least $d - (\theta + 6/5)\delta_2 > 0$; the $b = 3$ derivative is positive using $S = d + \delta_2, V_b = \delta_2 + \delta_3$. It therefore suffices to put $\lambda = 1$ for a lower bound.

For even b, $S = d$ and $V_b \leq \delta_2$. For $b = 3$ the bound is $d + (\delta_2 - \delta_3)/2 \geq d$. For odd $b \geq 5$ write $b - 1 = 2a, a \geq 2$. On each frequency, put $c = 1 - (-1)^a x^a$. The exact finite formulas give the integrands

$$S : \ c, \qquad V_b : \ xc^2/(1 + x).$$

The function $c - xc^2/[2(1 + x)]$ increases throughout the allowed interval. Since $c \geq 1 - x^2$, its value is at least

$$1 - x^2 - \frac{x(1 - x^2)^2}{2(1 + x)} \geq 1 - 3x/4.$$

The last difference divided by x is $1/4 - x/2 + x^2/2 - x^3/2 \geq 1/4 - \theta/2 > 0$. This proves (Q84.3.4), every finite end sign included.

For (Q84.3.5), $\lambda - 1 = \lambda\beta_b/2 \leq 3d/5$, hence $(\lambda - 1)d \leq 3\delta_2/5$. For $b = 3$ use $\lambda - \lambda^2/2 \leq 1/2$. For $b \geq 4$ use $S \leq d + \delta_3$ and drop the negative quadratic term; this gives $B_* - d \leq (3/5 + 39/80)\delta_2 < 11\delta_2/10$.

Q84.3.2 All finite coordinate ranges

The exact C075 coefficient proof gives

$$P_n = A_n^P + zC_n^P, \quad 1 - x \leq A_n^P, C_n^P \leq 1 \qquad (n \geq 3). \tag{Q84.3.6}$$

The same statement holds for $\mathcal{P}_n = G_n + z^{n+1}$, $n \geq 3$. Split n into its four residue classes. With $S_a = \sum_{i=0}^{a} (-x)^i$, the base pairs and corrections for P_n are

n	base coordinates	corrections
$4a$	(S_{2a-1}, S_{2a-1})	$(\lambda x^{2a+1} c_N, \lambda x^{2a} a_N)$
$4a + 1$	(S_{2a}, S_{2a-1})	$(-\lambda x^{2a+1} a_N, \lambda x^{2a+1} c_N)$
$4a + 2$	(S_{2a}, S_{2a})	$(-\lambda x^{2a+2} c_N, -\lambda x^{2a+1} a_N)$
$4a + 3$	(S_{2a+1}, S_{2a})	$(\lambda x^{2a+2} a_N, -\lambda x^{2a+2} c_N)$.

Every correction fits the adjacent last alternating term, since $\lambda x < 1$ and $\lambda x^2 < x$. The case $n = 3$ in the final imaginary coordinate is $1 - \lambda x^2 c_N \geq 1 - x$ directly. This proves every integer n. The two short polynomials are

$$P_1 = (1 - \lambda x a_N) + z\lambda x c_N, \quad P_2 = (1 - \lambda x^2 c_N) + z(1 - \lambda x a_N). \tag{Q84.3.7}$$

Q84.3.3 Raw envelopes for use with the same first pair

Write an actual nonnegative upper as $X_{b,L} \le C[(1-c)q + KQ(-K)g - a]$, and put $D = c + a$. Here is the full table used in parts Q84.5–8:

Table 94: Raw terminal masks indexed by (b, L).

(b, L)	C	c	a	Q
2, 1	1;	0;	γ;	z.
2, 2	1;	γ;	0;	$1 + z^2$.
2, 3	1;	γ;	τ;	$B + z + z^3$.
$2, L \ge 4$	1;	τ;	τ;	$\mathcal{P}_{L-1}$.
$b \ge 3, 1$	m^2;	0;	γ;	$\lambda z G_{b-1}(-z)$.
$b \ge 3, 2$	m^2;	γ;	a_1;	P_1.
$b \ge 3, 3$	m^2;	B_*;	ϵ_3;	$P_2 - \gamma$.
$b \ge 3, 4$	m^2;	$\delta = 41d/80$;	τ;	P_3.
$b \ge 3, L \ge 5$	m^2;	τ;	τ;	P_{L-1}.

In the $b = 2$ rows $\gamma = d - \delta_2/2$ and $B = 1 - \gamma$. Their finite envelope is

$$X_{2,L} \le \sum_{i=0}^{L} z^i 1 + z^{L+2} 1 - \gamma \bar{f}_{L-1} - \sum_{i=3}^{L} \eta_i \bar{f}_{L-i}, \quad \eta_i \ge \tau. \tag{Q84.3.8}$$

At $L \ge 4$ retain the actual last two losses $\tau\bar{f}_1, \tau\bar{f}_0$. For $b \ge 3, L = 4$ retain $\delta\bar{f}_1, \tau\bar{f}_0$; for $L \ge 5$ retain $\tau\bar{f}_1, \tau\bar{f}_0$. The short rows follow by applying T once to the preceding actual row. Every upper remains nonnegative because the discarded terms are positive.

Q84.4 The exact first-pair envelopes and the common fourth bookkeeping

Applying A to the negative finite convolution, retaining its last constant term, gives for $v > 2$ the literal bound

$$W_v \le 2 - \beta_v - S'_v - \beta_{v-1} p, \qquad S'_v = \sum_{a=0}^{v} (-K)^a 1.$$

Here $\bar{f}_2 = S'_2$ and $S'_v - S'_2 = -z^2 G_{v-2}(z) g$, with $z = -K$. Consequently

$$\bar{f}_2 + W_v \le 2 - \Gamma + K[-zG_{v-2}(z)g - \beta\mathbf{1}], \quad \beta = \beta_{v-1}, \quad \Gamma = \beta_v + \beta. \tag{Q84.4.1}$$

In particular for $v = 3, \beta = 0, \Gamma = d$, and the remainder is sigma. For $v \ge 4$ put $e_v = d - \beta_{v+1}$; then

$$\kappa\delta_2 \le e_v \le \delta_2, \quad \mathbb{E}[-zG_{v-2}(-K)g] = -\beta_v, \quad EK[-zG_{v-2}(-K)g - \beta\mathbf{1}] = -e_v. \tag{Q84.4.2}$$

For $v = 3$ take $e_v = 0$. The exact short losses are

$$\Gamma = d \ (v = 3), \quad 2d \ (v = 4), \quad 2d - \delta_2 \ (v = 5), \quad \Gamma \ge 2\tau \ (v \ge 6). \tag{Q84.4.3}$$

Combine (Q84.4.1) with a raw row of part Q84.3.3 and $p = 1 + g$. AM–GM, or $\log u \le u - 1$ on the four positive actual factors, proves

$$H = p\bar{f}_2 W_v X_{b,L} \le C \exp\{-D - \Gamma + KN\},$$

$$N = [Q(z) - zG_{v-2}(z)]g + (c - \beta)1, \qquad E(KN) = \mathbb{E}[KQ(z)g] - e_v. \tag{Q84.4.4}$$

The constant $(c-\beta)1$ is part of the same original vector. It cannot be discarded or bounded independently before squaring its norm. For $b \geq 3, \log C = 2\log m \leq -\beta_b$. Applying (Q84.1.7) gives the single-variable certificate

$$\log \mathbb{E}H^4 \leq 4\log C - 4(D+\Gamma) + 4E(KN) + (17/4)\|N\|^2. \tag{Q84.4.5}$$

All the following tables are explicit payments of (Q84.4.5), using that same actual N. Write $\zeta = \delta_2/d$ only when $d > 0$; then $d \leq \zeta \leq \theta$.

The one structural unit-inner exception in (Q84.0.4) has instead

$$h_2 + f_v \leq 2 - \beta_v + K[-zG_{v-2}(-z)]g. \tag{Q84.4.6}$$

Since $h_2 = q - r$, this is the positive finite convolution after the actual r terms cancel. Its mean is $+e_v$. part Q84.7 proves its whole-parameter unit-inner payment separately.

Q84.5 The entire first pair (2, 3), all terminals and all inner lengths

Here $\beta = 0, \Gamma = d$, and N in (Q84.4.4) is $[Q-z]g + c_1$. The following table lists complete bounds for its squared norm and for $\mathcal{L} = -\log C + D + \Gamma - E(KN)$.

Table 95: The first pair $(2,3)$: lower bounds for the loss $\mathcal{L}$ and upper bounds for the full squared norm.

b	L	$\mathcal{L}$ lower	$\|N\|^2$ upper
2	1	$2d - \delta_2/2$	0
2	2	$3d - 3\delta_2/2$	$d - \kappa\delta_2$
2	3	$4d - 3\delta_2/2 - \gamma d$	$d + \theta^2\delta_2$
2	≥ 4	$4d - 3\delta_2$	$d + \delta_2/3$
≥ 3	1	$3d - 7\delta_2/4$	$d/3$
≥ 3	2	$4d - 59\delta_2/20$	d
≥ 3	3	$22d/5 - 59\delta_2/20$	d
≥ 3	4	$361d/80 - 3\delta_2$	d
≥ 3	≥ 5	$5d - 4\delta_2$	$d + \delta_2/2$

Here and below notation in tables denotes the displayed mathematical quantities.

We prove all norm entries. For $b = 2, L = 1$ the joint primitive is zero, by the preceding primitive formula. For $L = 2$,

$$\|[1 + z^2 - z]g + \gamma\mathbf{1}\|^2 = d - \delta_2 + \delta_3 - d^2 + \delta_2^2/4 \leq d - \kappa\delta_2. \tag{Q84.5.1}$$

For $L = 3$ it is $(B + z^3)g + \gamma\mathbf{1}$, with squared norm $B^2 d + \delta_4 - 2\gamma\delta_2 + \gamma^2 \leq d + \theta^2\delta_2$. Indeed $B^2 d + \gamma^2 \leq d$, since $\gamma \leq d \leq 1/3$. For $L \geq 4$, the coordinates of $\mathcal{P}_n - z$, $n \geq 3$, give

$$|\mathcal{P}_n - z|^2 \leq (1 - x + x^2)^2 + x^3 \leq 1 - 2x/3. \tag{Q84.5.2}$$

The last deficit follows from $2 - 3\theta - \theta^3 > 2/3$. Its mean as a primitive is nonpositive, so adding $\tau\mathbf{1}$ increases its squared norm by at most $\tau^2 \leq \delta_2$. This proves that row.

For $b \geq 3, L = 1$,

$$|\lambda z G_{b-1}(-z) - z|^2 \leq (36/25)x^2 + x^3 + x/25 < 1/3 \tag{Q84.5.3}$$

on the whole frequency interval. The imaginary coordinate is bounded by $\max(\lambda - 1, x)$; substitution of θ in the nonnegative polynomial proves the last strict comparison.

For $L=2, |P_1-z|^2 \le (1+x)(1-x+x^2)^2 \le 1-2x/5$. The last scalar deficit follows from $1-\theta-\theta^2-\theta^4 > 2/5$. Also $\mathbb{E}[(P_1-z)g] = a_1 - d$. Its constant contribution is at most

$$\gamma^2 + 2\gamma(a_1-d) \le (21/100)d^2 \le (21/100)\delta_2. \tag{Q84.5.4}$$

To see this, use $\gamma \le 6d/5$ and $a_1 \le 6\delta_2/5$; the possible positive bracket is at most $7d/40$. Thus the norm is at most d.

For $L=3$ the polynomial after the subtraction is $B + \lambda z^3 G_{b-1}(-z)$. Its squared modulus is at most B^2. In fact the positive tail term is paid by its real cross term because

$$\lambda x|G_{b-1}|^2 \le 351/512 < 57/80 \le 2Bc_{b-1}.$$

The primitive mean is $-\lambda\rho$. Consequently

$$\|N\|^2 \le B^2 d - 2B_*\lambda\rho + B_*^2 \le d. \tag{Q84.5.5}$$

For the final inequality, the quadratic in B_* is increasing above $\lambda\kappa\delta_2$, and $B_* \ge 5d/8$ lies above that point. Replace B_* by $\gamma + \lambda\delta_2$, and ρ by kappa δ_2. The difference from d is at most

$$\gamma[\gamma(1+d) - 2d + 2\lambda\theta\delta_2] \le -\gamma d/256 \le 0. \tag{Q84.5.6}$$

Here $(6/5)(4/3) + (12/5)\theta^2 = 2 - 1/256$. This retains the constant cross term; a triangle bound on B_*1 would not give (Q84.5.5).

For every $n \ge 3, |P_n - z|^2 \le 1 - 2x/5$; for $n \ge 4$ the stronger $1 - x/2$ holds. For $n=3$ use $(1 - x + 6x^2/5)^2 + (36/25)x^5$. Its excess divided by x is at most $-2 + (17/5)\theta + (36/25)\theta^3 + (36/25)\theta^4 < -2/5$. For $n \ge 4$ its real coordinate is at most $1 - x + x^2 + (6/5)x^3 \le 1 - 3x/8$, since $1 - \theta - (6/5)\theta^2 = 1013/2560 > 3/8$. Its imaginary coordinate after subtraction has magnitude at most x. Hence use $(1 - 3x/8)^2 + x^3 \le 1 - x/2$. The means of these primitive polynomials are nonpositive. Adding delta1 at $L=4$ costs at most $\delta^2 < (2/5)\delta_2$; adding $\tau\mathbf{1}$ later costs at most δ_2. This proves the last two norm entries.

The loss column follows directly from (Q84.3.3), (Q84.3.4), and the exact means in (Q84.1.5). In particular at $L=3$, $EK(P_2-\gamma)g \le -(4/5 - 6d/5)d \le -2d/5$. All table entries in (Q84.4.5) are at most -$6d$. The two tight $b=2$ entries are

$$\begin{aligned} -\frac{31}{4}d + \frac{445}{128}\delta_2 &\le -\frac{25959}{4096}d \quad (L=2), \\ -\frac{47}{4}d + \frac{161}{12}\delta_2 &\le -\frac{2419}{384}d \quad (L \ge 4). \end{aligned} \tag{Q84.5.7}$$

For $b=2, L=3$ use gamma $d \le \delta_2$ and $\theta^2 < 1/6$ to get an upper $-5683d/768 < -6d$. The other entries are stronger by direct use of $\delta_2 \le \theta d$. Thus

$$\mathbb{E}[p\bar{f}_2 W_3 X_{b,L}]^4 \le e^{-6d} \qquad \text{for every } b \ge 2, L \ge 1. \tag{Q84.5.8}$$

Every same primitive in this proof has squared norm at most $9d/4$.

Q84.6 Terminal two and every longer first pair (2, v), $v \ge 4$

Use the same $\bar{f}_2 W_v$ direction for every L here. Let $h = v-2$ and $G_h = a + zc$. Write $D_0 = Q - zG_h$ and $s = c_{\text{row}} - \beta$; then the norm in (Q84.4.5) is exactly

$$\|D_0 g + s_1\|^2 = \|D_0 g\|^2 + 2sE(D_0 g) + s^2. \tag{Q84.6.1}$$

This section gives the complete $v=4$, $v=5$, and $v \ge 6$ payments.

Q84.6.1 First and second inner lengths

For $L = 1, D_0 = -z^2 G_{v-3}$, its primitive mean is nonnegative, and $s = -\beta$. Therefore

$$\|N\|^2 \le \delta_2 + (1+\theta)\delta_3 \le (8/5)\delta_2. \tag{Q84.6.2}$$

Using $\Gamma \ge 2\tau$ and $e_v \ge \kappa\delta_2$, (Q84.4.5) is at most

$$-12d + (577/40)\delta_2 \le -7859d/1280 < -6d. \tag{Q84.6.3}$$

For $L = 2, \gamma = d - \delta_2/2$ and $s = \gamma - \beta$.

* $v = 4 : D_0 = 1 - z$, $s = -\delta_2/2$, and $\|N\|^2 = d + \delta_2 + d\delta_2 + \delta_2^2/4 \le (8/5)d$. The total loss is $\mathcal{L} = 4d - \delta_2/2$, so the exponent is at most $-16d + 2\delta_2 + (34/5)d < -6d$. * $v = 5 : D_0 = 1 - z - z^3$, $s = -\delta_2/2$, and

$$\|N\|^2 \le d + \delta_2 - (2-\theta)\delta_3 + d\delta_2 - 3\delta_2^2/4 \le (4/3)d.$$

Indeed $d \le \zeta$ and (Q84.1.2) bound its ratio by $1+\zeta-\kappa\zeta^2 \le 1+\theta-\kappa\theta^2 < 4/3$. Here $\mathcal{L} = 4d - 3\delta_2/2$, giving an upper $-379d/48 < -6d$. * $v \ge 6$: both coordinates of G_h are at most $1 - x + x^2$. The polynomial has squared modulus at most

$$(1 - x^2 + x^3)^2 + x(1 - x + x^2)^2 \le 1 + x - (19/10)x^2.$$

The deficit uses $4 - 5\theta - \theta^4 > 19/10$. Now $0 \le s \le \delta_2/2$, $E(D_0 g) = -\beta_v$, and its constant contribution is nonpositive. Hence

$$\|N\|^2 \le d + \delta_2 - (19/10)\delta_3 \le (23/20)d.$$

The last bound follows from $1+\zeta-(19/10)\zeta^2 \le 43/38 < 23/20$. The loss is at least $4d-(7/2-\kappa)\delta_2$, so (Q84.4.5) is at most $-16d + (93/8)\delta_2 + (391/80)d < -6d$.

The polynomial estimates are full interval inequalities. For the first $v = 4$ bound use $d \le 1/3$ and $\delta_2 \le \theta$ d; its ratio is at most $1 + (175/128)\theta < 8/5$.

Q84.6.2 The third inner length

Here $Q = B + z + z^3$. The three exact/upper norm formulas are

v	$\|N\|^2$ upper
4	$B^2 d + 2B\delta_2 + \delta_3 + \delta_4 + 5\delta_2^2/4$
5	$B^2 d + 2B\delta_2 + \delta_3 + \delta_2^2/4$
≥ 6	$d + 2\delta_2 - (2/5)\delta_3 - 2\gamma d + \gamma^2 d + \delta_2^2/4.$

(Q84.6.4)

For $v = 4$ the imaginary coordinate is -x; for $v = 5$ it is zero. For $v \ge 6$ the real coordinate is at most $B + x - x^2 + x^3$, and the imaginary coordinate has magnitude at most x^2. The inequality

$$(1 + x - x^2 + x^3)^2 + x^5 \le 1 + 2x - (2/5)x^2$$

follows from $1 - 3\theta^2 - \theta^4 > 2/5$. Its constant s is nonnegative and at most $\delta_2/2$, with nonpositive primitive mean. These facts prove all entries of (Q84.6.4).

Substitute them in (Q84.4.5). All gamma-dependent terms are nonpositive after the following common estimate:

$$-9\gamma/2 + (17/4)\gamma^2 + (85/16)\zeta^2 d \le 0. \tag{Q84.6.5}$$

Indeed $\gamma \ge 51d/64, \gamma \le 1/3$, and $(37/12)(51/64) > (85/16)\theta^2$. Writing $C_0 = \theta(1+\theta)$, the remaining fourth exponents are bounded by

$$\begin{array}{r|l} 4 & [-63/4 + (21/2 + 17C_0/4)\zeta]d \\ 5 & [-63/4 + (29/2 + 17\theta/4)\zeta]d \\ \ge 6 & [-63/4 + (161/8)\zeta - (17/10)\zeta^2]d. \end{array} \tag{Q84.6.6}$$

They are all below -6d on $[0,\theta]$; even dropping the last negative quadratic leaves the third at most $-1939d/256 < -6d$. Each norm in (Q84.6.4) is at most $9d/4$, using $\delta_2 \le \theta$ d and $d \le 1/3$.

Q84.6.3 Fourth and every longer inner length

Now $Q = \mathcal{P}_n$, $n = L-1 \ge 3$, and $s = \tau - \beta \le 0$. Its real coordinate is at most $1 - x + x^2$, and both coordinates lie in $[1-x, 1]$. For $v = 4$ or 5,

$$|Q - zG_h|^2 \le (1+x^2)^2 + x^3 \le 1 + (13/5)x^2. \tag{Q84.6.7}$$

For $v = 4$ the constant contribution is at most $3\delta_2^2$; for $v = 5$ its cross term is nonpositive and only δ_2^2 is needed. It follows that

$$\|N\|^2 \le d + (3/2)\delta_2 \quad (v=4), \qquad \|N\|^2 \le d + (6/5)\delta_2 \quad (v=5). \tag{Q84.6.8}$$

For $v \ge 6, c_h \le 1 - x + x^2$ improves the real coordinate to $1 + x^3$, while the imaginary difference is at most x. Hence

$$|Q - zG_h|^2 \le 1 + 3x^3 + x^6.$$

Moreover $\mathbb{E}[(Q - zG_h)g] \ge -\theta\delta_2$, since $\mathbb{E}(Qg) \ge \tau$ and $\beta_v \le d - \kappa\delta_2$. Thus

$$\|N\|^2 \le d + (3\theta^2 + \theta^5)\delta_2 + (1+2\theta)\delta_2^2 \le d + (4/5)\delta_2. \tag{Q84.6.9}$$

The last scalar inequality uses $\delta_2 \le \theta/3$ and $3\theta^2 + \theta^5 + (1+2\theta)\theta/3 < 4/5$.

The loss lower bounds for $v = 4, 5, \ge 6$ are respectively $5d - 2\delta_2,\ 5d - 3\delta_2,\ 5d - 141\delta_2/32$. Using (Q84.6.8)–(Q84.6.9), the fourth exponents are at most

$$-2537d/256, \qquad -2817d/320, \qquad -9227d/1280, \tag{Q84.6.10}$$

respectively. This covers every $L \ge 4$. Combining the whole section,

$$\mathbb{E}[p\bar{f}_2 W_v X_{2,L}]^4 \le e^{-6d} \qquad (v \ge 4, L \ge 1). \tag{Q84.6.11}$$

Every primitive used here has squared norm at most $9d/4$.

Q84.7 Unit inner connector, all $b \ge 3$ and $v \ge 4$: the new actual direction

Use $H = ph_2 f_v T(qf_b)$, exactly as prescribed in (Q84.0.4). Combining (Q84.4.6) with the first $b \ge 3$ row gives

$$H \le m^2 e^{-\gamma - \beta_v + KN}, \qquad N = z[\lambda G_{b-1}(-z) - G_{v-2}(-z)]g,$$

$$E(KN) = -\lambda e + e_v. \tag{Q84.7.1}$$

Consequently its total loss is $\beta_b + B_* + \beta_v - e_v$. The opposite signs that caused the old unit-inner norm difficulty have disappeared in this actual direction.

For all n, $h \ge 2$, putting $\Delta = \lambda - 1 \le 1/5$, finite coordinate differences give

$$|z[\lambda G_n(z) - G_h(z)]|^2 \le x(1+x)(\Delta + x)^2 \le 17/80. \tag{Q84.7.2}$$

Indeed each coordinate difference of two G's is at most x, and each coordinate of ΔG_n is at most Delta. The endpoint upper is $\theta(1+\theta)(1/5+\theta)^2 < 17/80$.

This bound suffices when $b = 3, b = 4$, or $v = 4$. Their loss lower bounds are respectively

$$3d - 2\delta_2, \qquad 3d - 5\delta_2/2, \qquad 3d - 11\delta_2/4. \tag{Q84.7.3}$$

For $b = 3$ use $\beta_b = d$ and $B_* \ge d$. For $b = 4$ use $\beta_b = d$ and $B_* \ge d - \delta_2/2$. For $v = 4$ use $\beta_v - e_v = \tau$. With norm at most $17d/80$, every resulting fourth exponent is below -6d.

It remains to cover $b \ge 5$ and $v \ge 5$, with no upper bound on either integer. Here

$$\boxed{\|N\|^2 \le 3d/32.} \tag{Q84.7.4}$$

For $v = 5$, write $n = b - 1 \ge 4, h = 3$, and $G_n = G_3 + z^3 G_{n-3}$. Then

$$z[\lambda G_n - G_3] = \Delta z G_3 + \lambda z^4 G_{n-3}.$$

Their real cross term is nonpositive: apart from positive factors it is $(1-x)c_{n-3} - a_{n-3} \le 0$. This also holds for $n - 3 = 1$, whose coordinates are 1, 0. Thus the squared modulus is at most

$$\Delta^2 x + \lambda^2(1+x)x^4,$$

whose uniform upper is $\theta/25 + (36/25)(1+\theta)\theta^4 < 3/32$. For $v \ge 6$, both n and h are at least four. Their difference begins at degree at least four. Since $|G_n| \le 1$, its norm is bounded by

$$\theta[1/5 + (6/5)\theta^2]^2 = 13498693/209715200 < 3/32. \tag{Q84.7.5}$$

Here $\sqrt{1+\theta} < 6/5$ bounds the finite difference tail. This proves every finite n, h and all their relative orders; no length scan is being substituted for the finite factorization.

Finally (Q84.3.4) and $\beta_v - e_v \ge d - 2\delta_2$ give

$$\beta_b + B_* + \beta_v - e_v \ge 3d - 15\delta_2/4.$$

Using (Q84.7.4) in (Q84.1.7), the fourth exponent is at most

$$-12d + 15\delta_2 + (51/128)d \le -705d/128. \tag{Q84.7.6}$$

This is the tight common constant in Theorem A. This proof supplies all $b \ge 3, v \ge 4$ for the actual unit-inner direction, with no score-dependent choice and no lost original arm. Its primitive is also bounded by $9d/4$.

Q84.8 All $b \ge 3, v \ge 4$, and every nonunit inner connector

Return to the original $H = p\bar{f}_2 W_v X_{b,L}$ direction, $L \ge 2$. We give the remaining full-domain raw-norm certificates. Define

$$D_0 = Q - zG_{v-2}, \quad s = c - \beta, \quad C_s = 2sE(D_0 g) + s^2. \tag{Q84.8.1}$$

All bounds in this section retain C_s with its actual constant and mean.

Q84.8.1 The complete second-connector norm table

Here $Q = P_1, c = \gamma$. Put $h = v - 2$ and write $G_{b-1} = a_N + zc_N$. Then

$$D_0 = (1 - \lambda x a_N + x c_h) + z(\lambda x c_N - a_h), \quad E(D_0 g) = a_1 - \beta_v < 0. \tag{Q84.8.2}$$

The last sign follows from $a_1 \le 6\delta_2/5 < \tau$. The complete norm table is

range	$\|N\|^2$ upper
$b = 3$	$3d/2$
$b = 4$	$2d$
$b \ge 5, v = 4$	$9d/4$
$b \ge 5, v = 5$	$2d$
$b \ge 5, v \ge 6$	$8d/5$.

(Q84.8.3)

$b = 3$. The polynomial in (Q84.8.2) has squared modulus at most $1 + x - 2x^2 + x^3$. Let $m_0 = \beta_v - \lambda\delta_2 > 0$, so $C_s = s^2 - 2m_0 s$. When $s \ge 0$, this is nonpositive. Indeed

$$\gamma + 2\lambda\delta_2 \le 3\tau,$$

because after division by d the difference is at least

$$9/5 - (117/25)\zeta + (18/25)\zeta^2 \ge 9/512 > 0.$$

This uses (Q84.1.2), the exact $b = 3$ gamma, and monotonicity in lambda and zeta on their stated intervals. If $s < 0$, then $|s| \le \lambda^2(\delta_2 + \delta_3)/2$, whence

$$C_s \le \lambda^2(1 + \theta)d\delta_2 + (\lambda^4/4)(1 + \theta)^2\delta_2^2 \le (5/6)\delta_2. \tag{Q84.8.4}$$

The coefficient before the last relaxation is $27/40 + 28431/204800 = 166671/204800 < 5/6$. Thus

$$\|N\|^2 \le d + (11/6)\delta_2 - (51/32)\delta_3 \le 3d/2.$$

The final quadratic in zeta increases to θ and its endpoint is less than $3/2$. This proves the first whole-parameter row.

$b = 4$. Since $a_N = 1 - x, c_N = 1$, the polynomial norm is at most $1 + x + \lambda^2 x^3(1 + x)$. If $s \ge 0$ then $s \le d/5 < 2(\tau - 6\delta_2/5)$, so $C_s \le 0$. If $s < 0$, use $|s| \le 43\delta_2/25$ and $m_0 \le \tau$. This yields

$$C_s/\delta_2 \le (86/25)d - (301/625)\delta_2 \le 6149/5625 < 11/10. \tag{Q84.8.5}$$

Here $\delta_2 \ge d^2$ and monotonicity of the remaining quadratic put its maximum at $d = 1/3$. Also $(36/25)\theta^2(1 + \theta) < 27/80$. Therefore $\|N\|^2 \le d + (39/16)\delta_2 < 2d$, proving the second row.

The constant bound just used, and reused below, follows on every $b \ge 4$ from $\gamma \ge \lambda\tau - \lambda^2\delta_2/2$ and $\beta \le d$:

$$(\beta - \gamma)_+ \le [\lambda + \lambda^2/2 - (\lambda - 1)/\theta]\delta_2 \le (3/2)\delta_2 < (43/25)\delta_2.$$

The bracket decreases for $1 \le \lambda \le 6/5$, since $1 + \lambda - 1/\theta < 0$. For $v \ge 6$, the further bound $\beta \le d - \kappa\delta_2$ improves this to $(\beta - \gamma)_+ \le (29/32)\delta_2 < (901/800)\delta_2$.

$b \ge 5, v = 4$. The polynomial is bounded by $1 + x + (8/5)x^2$. To verify it, use real upper $1 + \lambda x^2$ and imaginary magnitude at most $1 - \kappa x$. The excess quadratic coefficient is at most $97/80 + 23/160 + 6/25 < 8/5$. Positive s again gives $C_s \le 0$. For negative s, $|s| \le 43\delta_2/25$ and $m_0 \le d - \kappa\delta_2$, so

$$C_s/\delta_2 \le 86/75 + (9159/10000)(13/96) < 13/10.$$

Hence $\|N\|^2 \le d + (59/20)\delta_2 < 9d/4$.

$b \ge 5, v = 5$. Now $a_h = 1 - x, c_h = 1$. The polynomial norm is at most $1 + x + x^2/2$: use real upper $1 + \lambda x^2$ and imaginary magnitude at most $1 - (1 + \kappa)x$. The excess coefficient is bounded by $-63/80 + 663/640 + 6/25 = 7815/16000 < 1/2$. When $s \ge 0$ the constant term is nonpositive. Otherwise $m_0 \le d - (1 + \kappa)\delta_2$, giving

$$C_s/\delta_2 \le (86/25)d - (25241/10000)\delta_2 \le 77959/90000 < 7/8. \tag{Q84.8.6}$$

Thus $\|N\|^2 \le d + (15/8 + \theta/2)\delta_2 < 2d$.

$b \ge 5, v \ge 6$. Both finite coordinate pairs have the stronger upper $1 - x + x^2$. The polynomial norm is at most

$$(1 + x^3)^2 + x(1 - 2x + 2x^2)^2 \le 1 + x - (9/10)x^2. \tag{Q84.8.7}$$

The deficit coefficient is $4 - 10x + 8x^2 - 4x^3 - x^4$; it decreases to $1009135/1048576 > 9/10$. If $s < 0$, (Q84.1.4) and (Q84.3.2) give

$$|s| \le (901/800)\delta_2, \quad m_0 \le d - 2\kappa\delta_2,$$

$$C_s/\delta_2 \le (901/400)d - (900099/640000)\delta_2 \le 3424701/5760000 < 3/5. \tag{Q84.8.8}$$

For $s \ge 0$ use $t_0 \le d - \kappa\delta_2$ and $V_b \ge \kappa^2\delta_2$. The possible positive factor in C_s is at most

$$\gamma + 2a_1 - 3\tau \le [-9/5 + (9239/2048)\zeta]d \le d/30.$$

Since $s \le 6d/5$, it follows that $C_s \le d^2/25 \le (3/5)\delta_2$ in this case as well. Combining with (Q84.8.7),

$$\|N\|^2 \le d + (8/5)\delta_2 - (9/10)\delta_3 \le (3075/2048)d < (8/5)d. \tag{Q84.8.9}$$

This proves every row of (Q84.8.3), including all terminal and first-pair integer tails and the whole continuous domain.

The corresponding lower losses are, in the same row order,

$$5d - 697\delta_2/160, \quad 5d - 497\delta_2/160, \quad 5d - 39\delta_2/20, \quad 5d - 59\delta_2/20, \quad 5d - 697\delta_2/160. \tag{Q84.8.10}$$

For the first we use a weaker common bound. For $b = 4$ keep $\beta_b = d$ and $B_* \ge d - \delta_2/2$. For $v = 4$ or5 keep its exact Γ and e_v. Equation (Q84.4.5) gives upper fourth exponents, respectively,

$$-8379d/1280, \ -8259d/1280, \ -1163d/160, \ -1073d/160, \ -1567d/256. \tag{Q84.8.11}$$

Every one is below -6d.

Q84.8.2 The complete third-connector norm and loss table

Here $Q = P_2 - \gamma, c = B_*$, and

$$E(D_0 g) = \varepsilon_3 - \beta_v, \quad -6\delta_2/5 \le E(D_0 g) \le \theta\delta_2, \quad -3\delta_2/4 \le s \le 21\delta_2/10. \tag{Q84.8.12}$$

These follow from (Q84.3.4)–(Q84.3.5). In particular $C_s \le (49/8)\delta_2^2$.

For $v \ge 6$ its real coordinate is at most $1 + x - (1 + \kappa)x^2 + x^3$, and the imaginary magnitude is at most $x/5 + x^2$. Squaring gives

$$|D_0|^2 \le 1 + 2x - x^2. \tag{Q84.8.13}$$

For a direct continuous check, the two discarded negative terms have degrees three and five; the remaining coefficient comparison is $((51/32)^2+12/5)\theta^2+\theta^4<19/16$. Then (Q84.1.2), $d\le 1/3$, and (Q84.8.12) imply

$$\|N\|^2\le d+2\delta_2-\delta_3+(49/8)\delta_2^2\le 2d,$$

because $2\theta+(25/24)\theta^2=24193/24576<1$.

For $v=4$, the polynomial is bounded by $(1+x)^2+(36/25)x^3$, hence by $1+(8/3)x$. Now $s=B_*-d$ and $E(D_0g)=-\lambda\rho$. Positive s costs nothing, since $s\le 11\delta_2/10<2\lambda\rho$; negative s costs at most $(189/80)\delta_2^2\le\delta_2/3$. Thus $\|N\|^2\le d+3\delta_2<9d/4$. For $v=5$ the polynomial is bounded by $(1+x)^2+\theta^2x^3\le 1+(5/2)x$. Its mean is $\delta_2-\lambda\rho\in[-\delta_2/5,\theta\delta_2]$, and the constant costs at most $(17/8)\delta_2^2<3\delta_2/10$. Again $\|N\|^2<9d/4$.

The loss lower bounds for $v=4,5,\ge 6$ are

$$27d/5-39\delta_2/20,\quad 27d/5-59\delta_2/20,\quad 27d/5-697\delta_2/160. \tag{Q84.8.14}$$

They use the actual mean $EK(P_2-\gamma)g\le -2d/5$, and the exact first-pair loss where available. With the norm bounds $9d/4, 9d/4, 2d$, (Q84.4.5) is at most

$$-1419d/160,\qquad -1159d/160,\qquad -7707d/1280. \tag{Q84.8.15}$$

Thus this whole connector also has a fourth exponent below -6d.

Q84.8.3 Fourth and every longer connector, all first-pair lengths

At $L=4, Q=P_3$ and $c=\delta=41d/80$. The polynomial has real coordinate at most $1+\lambda x^2$ and imaginary difference of magnitude at most x. Hence

$$|D_0|^2\le 1+(31/10)x^2.$$

Here $s=\delta-\beta<0$, with $|s|\le 39d/80$, and $E(D_0g)\ge-\lambda\delta_3$. Thus

$$C_s\le 2(39d/80)\lambda\delta_3+(39d/80)^2\le(507/1280)\delta_2,$$

$$\|N\|^2\le d+(2119/1280)\delta_2\le 2d. \tag{Q84.8.16}$$

The loss is at least $441d/80-141\delta_2/32$. Its fourth payment is $-8179d/1280<-6d$.

For every $L\ge 5$, both coordinates of $Q=P_{L-1}$ lie in [1-x, 1]. Thus the real coordinate of D_0 is at most 1+x and its imaginary magnitude at most x. Since $s=\tau-\beta$ lies in [-δ_2, 0] and $E(D_0g)\ge-\delta_2$,

$$\|N\|^2\le d+2\delta_2+\delta_3+\delta_4+3\delta_2^2\le d+(3049/1024)\delta_2<9d/4. \tag{Q84.8.17}$$

The loss is at least $6d-173\delta_2/32$. Therefore

$$\log\mathbb{E}H^4\le -24d+(173/8)\delta_2+(153/16)d\le -1447d/256<-705d/128. \tag{Q84.8.18}$$

All finite long connectors are covered by the one coefficient lemma, not by a numerical tail or an unproved monotonicity assertion.

parts Q84.5–8 now prove (Q84.0.5) for every bare first pair $u_1=2$, with every b, L and all $v>2$. Every chosen same primitive has squared norm at most $9d/4$. The further original pairs will be restored in part Q84.10 without changing any chosen edge direction.

Q84.9 Canonical single pairs with $u = 3$ or 4

For the actual message $Y = pW_u\bar{f}_v T^L(qf_b)$, apply Proposition O.1 with one pair (u, v). It covers every $b \geq 2$, $L \geq 1$, and $v > u$, without requiring a reserved root arm. Thus

$$\mathbb{E}Y^4 \leq e^{-705d/128}.$$

The same proposition supplies the stronger interface (O.15): the pointwise exponent has its actual signed mean and its full vector has norm at most $3\sqrt{d}/2$. In particular its tight $b = 2, L = 1, u = 3$ payment retains the joint norm $\delta_2 + \delta_3$, giving

$$-12d + (57/4)\delta_2 + (17/4)\delta_3 \leq -22567d/4096 < -705d/128.$$

The proof for $u = 2$ in parts Q84.5–8 is separate, with its stated integer choice of the first-pair direction. These two inputs prove the ordinary-fourth assertion for all listed root configurations.

Q84.10 Restore every later residual pair and every equal pair

For $u_1 = 2$ every bare proof in parts Q84.5–8 has an explicit pointwise exponential certificate, with the same vector N satisfying $\|N\| \leq 3\sqrt{d}/2$. Its paid scalar fourth expression is at most $-705d/128$. The later disjoint pairs start at least at four.

If there are $n = j - 1 \geq 1$ later pairs, multiply this actual exponential upper by (Q84.1.8). Its remainder becomes the one original variable $K[N + U(-K)g]$. The increase in the squared norm is at most

$$2(3/2)(3/8)d + (3/8)^2 d = 81d/64.$$

Its mean increases by at most d/12. The additional scalar fourth cost is therefore at most

$$-12nd + d/3 + (17/4)(81/64)d = -4829d/768 - 12(n-1)d < 0. \tag{Q84.10.1}$$

This operates on the proved same-variable certificate, not on an isolated numerical bound for EY^4. It proves every finite number of later pairs, all their gaps and lengths, and preserves every original factor.

Now restore the original maximum equal-length pairing. On a root pair of length a choose $\bar{f}_a W_a \leq 1$, or its full reversal; on a terminal pair choose $f_a h_a \leq 1$, or its full reversal. These are the actual C014/C143 equal-pair inequalities. Let E_r, E_t be the full products. The actual message is explicitly

$$Y_D = E_r\,[\text{chosen original root residual factors}]T^L(E_t q f_b). \tag{Q84.10.2}$$

It is nonnegative and at most its corresponding bare residual message. Thus both the ordinary fourth budget and the final T/I_q target follow by positivity. No unequal factor is deleted pointwise. All multiplicities and all original equal-pair edges remain present in the oriented tree.

Applying the new full-real bridge (Q84.0.6) to the actual Y_D proves (Q84.0.8). This proves the T089 bound throughout the stated parameter ranges.

Q.2 Short terminal pairs at a root beginning (1,2,v)

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\,d\nu$, $\beta_v = \int E\,d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges.

The root residual list is $(1,2,v)$, $v \geq 3$. Both original connector lengths $L, M \geq 1$ are independent. Write $H = pPT^L F$, with outer message $T^M H$. The unit arm has actual factor p; an f arm points entirely outward, a $\bar{f}$ arm inward, an h arm first inward then outward, and a W arm first outward then inward. Every inner edge points from terminal to root; every outer edge from root to receiver. Fix the following original directions:

U2. Terminal $(2,b)$, $b \geq 4$, $L = 1$: $F = \bar{f}_2 W_b$; take $P = \bar{f}_2 W_3$ at $v = 3$, and $P = h_2 f_v$ at $v \geq 4$.

U3. Terminal $(3,b)$, $b \geq 5$, $L = 1$: $F = h_3 f_b$, $P = \bar{f}_2 W_v$.

24. Terminal $(2,4)$: $F = \bar{f}_2 W_4$; take $P = h_2 f_v$ at $L = 2$, and $P = \bar{f}_2 W_v$ at $L \geq 3$.

L2B. Terminal $(2,b)$, $b \geq 4$: $F = \bar{f}_2 W_b$; take $P = \bar{f}_2 W_v$ at $L \geq 3$, and $P = h_2 f_3$ at $L = 2, v = 3$.

Auxiliary L2B. At $L = 2, v \geq 4$, use the same $F = \bar{f}_2 W_b$, $P = h_2 f_v$. The present auxiliary calculation requires $d \leq 1/8$; the remaining domain is handled by Q5-C163 in the same direction.

L3B. Terminal $(3,b)$, $b \geq 5$: $F = \bar{f}_3 W_b$; take $P = h_2 f_v$ at $L = 2$, and $P = \bar{f}_2 W_v$ at $L \geq 3$.

The direction uses only original integers. The overlapping $(2,4)$ rows agree. Each complete row satisfies $EH^4 \leq e^{-45d/8}$. The ordinary-fourth outlet in Q5-C084, Section 0, gives $I_q(T^M H)^3 \leq e^{11d}(EH^4)^2 \leq e^{-d/4}$ for every $M \geq 1$. Genuine equal-pair decorations are restored by positivity.

For later reference, write $m = v - 2$, $G_m = E + zD$, $\beta = \int D$ and $\beta_v = \int E$. The standard root $\bar{f}_2 W_v$ has vector $-BG_m(B)u + \beta\mathbf{1}$ and scalar $-\beta_v - \beta$; the alternative root $h_2 f_v$ has vector $-BG_m(-B)u + \beta\mathbf{1}$ with the same scalar. Their scalar-plus-mean losses, for $v \geq 4$, are at least $1463d/1024$ and $969d/1024$, respectively. Dropping the retained βf_1 term gives weaker losses $5d/6$ and $3d/16$, without changing the actual direction. At $v = 3$ both roots have exactly vector $-Bu$, scalar-plus-mean $-d$ and $\beta_2 = 0$. The full finite-coordinate proof is Lemma M.28; the weaker standard bound uses $1 - \theta^2 = 855/1024 > 5/6$ in even parity, while the odd case has loss d.

For $n \geq 3, m \geq 1$, the full joint bound of C180.3.3 is $\|(G_n - BG_m)u\| < (6/5)\sqrt{d}$: its frequency square is at most $1 + 2\theta^2 + \theta^3 + \theta^4 = 1493553/1048576 < 36/25$, with comparison $37338825 < 37748736$. This includes the exact $G_1 = (1,0)$ endpoint. In the following calculations the signed exponential estimate is applied once to each full vector, including all its displayed constants and means.

C160.3.2 One, two and three positive steps for the same remainder

Suppose $F \leq e^{R-\Gamma}$ and $e^R \leq 1 + R + \eta R^2/2$. Define $\sigma = ER^2$, $D_j = EB^j R$, $\delta = D_0 + \eta\sigma$, and $\epsilon = \delta + D_1$. Since $T(R^2) \leq 2\sigma$,

$$TF \leq e^{-\Gamma}(q + \delta + BR).$$

Apply positive T again. Use $\log y \leq y - 1$ on the complete strictly positive upper bound and $\log p \leq -u$. The resulting vector and scalar-plus-mean $S = a + EBV$ are

Table 96: Short-connector vectors and scalar-plus-mean terms for $\log(pT^L F)$.

L	V
1	R; S: $-\Gamma + \epsilon$.

L	V
2	$\delta\mathbf{1}+u+BR$; S: $-\Gamma+\epsilon-d+D_2$.
3	$\epsilon\mathbf{1}+G_2u+\delta u+B^2R$; S: $-\Gamma+\epsilon-2d+D_2+D_3-\delta d$.

For two steps the exponent scalar is $-\Gamma+\epsilon$, while $EBV=-d+D_2$. For three steps these are $-\Gamma+\epsilon-d+D_2$ and $-d-\delta d+D_3$. Thus the actual mean remains, including the shortest finite G_2.

C160.3.3 Every longer connector and the terminal gamma index

Use finite continuation (M.33) for the same first-row data δ, D_i. If $h\ge 0$ and $\gamma_j=\beta_j-D_{j-1}\ge hd$ for every $j\ge 3$, then (M.56) gives, for $L\ge 4$,

$$V=G_{L-1}(B)u+B^{L-1}R-hd\mathbf{1},\qquad a=-\Gamma+2\delta_{+}+2(D_1)_{+}-2hd.$$

Thus $\log(pT^LF)\le a+BV$, with $\mathbb{E}BV=-\gamma_{L+1}$ and $S\le-\Gamma+2\delta_{+}+2(D_1)_{+}-3hd$. The periodic moment calculations below establish the index $L+1$ as well; the constant $-hd\mathbf{1}$ stays in the full vector.

C160.4 The three actual terminal remainders and their pointwise Taylor domains

Write $k=b-2$, $G_k=A+zC$, and $\gamma=\beta_{b-1}=\int C$. Each row uses its same actual R in every σ, D_j and propagated vector.

Table 97: Short terminal pairs at a root beginning (1,2,v): parameter values.

Entry	Gamma
$\bar{f}_2W_b, b\ge 4$	$\beta_b+\gamma=\int(A+C)$; Spectral coordinates of R: $(\gamma+xA+zxC)u$.
$h_3f_b, b\ge 5$	$d+\beta_b+\gamma$; Spectral coordinates of R: $(\gamma+x(A-1)-zxC)u$.
$\bar{f}_3W_b, b\ge 5$	$d+\beta_b+\gamma$; Spectral coordinates of R: $(\gamma-x(1-A)+zxC)u$.

The first row is the original arm sum in the root-pair envelope. For the next two use $h_3=h_2-d+B^2u$ and $\bar{f}_3=\bar{f}_2-d+B^2u$, retaining the actual βf_1 term. Each actual arm sum proves $F\le e^{R-\Gamma}$.

For the first row, the original row inequality gives $|R|\le d+\sqrt{\delta_2+\delta_3}<1/3+7/16<1$, since $\delta_2+\delta_3\le 195/1024<49/256$. The other two tails have squared row bound $\delta_3+\delta_4\le 2535/32768=7605/98304$; the shortest G_2 retains its factor $1+x$. Thus the second row has $|R|<2/3$, and the third has $|R|<1/3+7/24=5/8$.

On $[-1,1]$, the positive Taylor remainder coefficient is at most $1/2+1/6+5/96=23/32<3/4$; the negative remainder is at most $R^2/2$. Hence the first two rows permit $\eta=3/2$. On $[0,5/8]$, consecutive Taylor-tail ratios from degree four are at most $y/5$, and $1/2+y/6+y^2/[24(1-y/5)]\le 279/448<5/8$. The negative coefficient remains at most $1/2$, so the third row permits $\eta=5/4$ throughout the pointwise domain.

For the first row the exact same-vector moments are

$$\sigma=\gamma^2d+2\gamma\int xA+\int(x^2A^2+x^3C^2),$$

$$D_0=-\int xC,\quad D_1=-\gamma d-\int xA.$$

For the second row, $D_0 = \int xC$, $D_1 = -\gamma d + \int x(1-A)$, and $\sigma \le \gamma^2 d + \delta_4 + \delta_5$. For the third, $D_0 = -\int xC$, with the same D_1 and variance bound. Their signs come from their full original terminal directions.

C160.5 The unit connector for terminal (2, b)

C160.5.1 The shortest root $v = 3$

The root vector is exactly $-Bu$. The full squared norm is $\sigma + \delta_2 - 2\int x^2 C$. Apply the signed exponential estimate at four. Put $c = 41/4 = 4\eta + 17/4$, the total coefficient of σ. The complete exponent is

$$-4\Gamma - 4d - 4\gamma d + \int [17x/4 - 4x(A+C) - 17x^2 C/2] + c\sigma.$$

Combine all γ terms before estimating: $P(\gamma) = -4\gamma - 4\gamma d + c\gamma^2 d + 2c\gamma \int xA$. Its derivative is at most $-4 + (277/64)d + (41/2)d^2 \le -161/576 < 0$. Therefore use the complete block $P(\kappa d)$. Its contribution $-4\kappa d^2 + c\kappa^2 d^3$ is nonpositive because $c\kappa/3 = 779/384 < 4$; the remaining mixed term is at most $779\delta_2/192$. With $A, C \ge \kappa$ and $\beta_b \ge \kappa d$,

$$\log EH^4 \le -35d/4 + (683/192)\delta_2 + (333/64)\delta_3 + (41/4)\delta_4.$$

The last three costs, divided by d, are at most $8879/6144 < 3/2$, $56277/65536 < 7/8$, and $90077/131072 < 3/4$. Their sum is below $25/8$, yielding loss greater than $45/8$.

C160.5.2 Every v at least four and both finite root parities

Use the entire actual root $h_2 f_v$, with strong vector $-BG_m(-B)u + \beta\mathbf{1}$. The complete squared norm is

$$\sigma + \int (x^2 D^2 + xE^2) - 2\gamma \int xD$$
$$-2\int x^2(AD + CE) + \beta^2 + 2\beta \int E - 2\beta \int xC.$$

The γ block now includes $-(17/2)\gamma \int xD$, so its derivative remains negative. Replace the whole block by its value at $\gamma = \tau$, also using $\beta_b \ge \tau$. The term $-4\tau d + c\tau^2 d$ is nonpositive since $\tau \le 2/9$. Retain the negative constant cross term, weakening it to $-(17\kappa/2)\tau \int xC$, because $\beta \ge \tau \ge \kappa\tau$.

For even m, use $E = D$ and $3\beta^2 \le 3d \int D^2 \le \int D^2$. For odd m,

$$\beta^2 + 2\beta \int E$$
$$\le d \int [D^2 + 2D - 2\kappa x D]$$
$$\le \tfrac{1}{3} \int [D^2 + 2D - 2\kappa x D].$$

The bracket is positive. The negative term was first paid with the actual $\beta \ge \kappa d$, before using $d \le 1/3$. This gives the two complete integral polynomials

$$F_e = -8 + 8x - 4x(A + C) - (8 - 4x)D$$
$$+\tfrac{17}{4}(1 + x + x^2)D^2 - \tfrac{17}{2}x^2(A + C)D$$
$$+c(x^2A^2 + x^3C^2)$$
$$+\tau x[\tfrac{41}{2}A - \tfrac{17}{2}D - \tfrac{17\kappa}{2}C],$$
$$F_o = -12 + 8x - 4x(A + C) + c(x^2A^2 + x^3C^2)$$
$$+\tfrac{17}{4}x - \tfrac{17}{2}x^2C$$
$$+[-\tfrac{7}{6} + (8 - \tfrac{17\kappa}{6})x - \tfrac{17}{2}x^2(1 + A)$$
$$+\tfrac{17}{2}x^3C]D$$
$$+[\tfrac{17}{12} + \tfrac{17}{4}x^2(1 + x)]D^2$$
$$+\tau x[\tfrac{41}{2}A - \tfrac{17}{2}D - \tfrac{17\kappa}{2}C].$$

Both expressions are convex separately in A, D and affine in τ. The C derivative divided by x is at most $-4 - (17/2)x + 29x^2$, negative at zero and at θ, where it equals $-2731/1024$. Set $C = 1 - x$. At $\tau = 2/9$, the A derivative divided by x is at least $5/9 + 12x - (41/2)x^2 > 0$. At $A = 1$, the τ coefficient is at least $445x/64 > 0$. Thus the complete continuous rectangle reduces to I: $(A, C, \tau) = (1, 1 - x, 2/9)$, and II: $(1 - x, 1 - x, 0)$, each with $D = 1$ or $D = 1 - x$.

All eight endpoint polynomials, with every finite tail term retained, are

Table 98: Short terminal pairs at a root beginning (1,2,v): parameter values.

Entry	Endpoint polynomial
even I 1	$-47/4 + (2821/288)x + (755/288)x^2 + (75/4)x^3 - (41/2)x^4 + (41/4)x^5$.
even I 1-x	$-47/4 + (2677/288)x - (359/96)x^2 + (63/2)x^3 - (99/4)x^4 + (41/4)x^5$.
even II 1	$-47/4 + (33/4)x + (11/2)x^2 + (27/4)x^3 - (41/4)x^4 + (41/4)x^5$.
even II 1-x	$-47/4 + (31/4)x - (11/4)x^2 + (39/2)x^3 - 23x^4 + (41/4)x^5$.
odd I 1	$-47/4 + (6977/576)x - (1693/288)x^2 + (63/2)x^3 - 29x^4 + (41/4)x^5$.
odd I 1-x	$-47/4 + (6017/576)x - (569/64)x^2 + 40x^3 - (167/4)x^4 + 23x^5$.
odd II 1	$-47/4 + (2029/192)x - 3x^2 + (39/2)x^3 - (75/4)x^4 + (41/4)x^5$.
odd II 1-x	$-47/4 + (1709/192)x - (1517/192)x^2 + 28x^3 - 40x^4 + 23x^5$.

For example, even I with $D = 1$ has τ contribution $445x/288 + 323x^2/288$, constant $-47/4$, quadratic coefficient $3/2 + 323/288 = 755/288$, and cubic coefficient $75/4$. Odd I with $D = 1$ has linear coefficient $17/4 + 1213/192 + 445/288 = 6977/576$. The other rows expand D, D^2 with $1 - x$, $(1 - x)^2$, retaining the finite parity relations.

For the four even rows, drop nonpositive final terms. Their positive costs are paid respectively by $4 + 1/2 + 4/3$, $19/5 + 17/8$, $27/8 + 1 + 1/2$, and $13/4 + 4/3$. The resulting upper bounds are $-71/12$, $-233/40, -55/8, -43/6$, all below $-45/8$. In the first row the exact endpoint costs $36673/9216$, $127595/294912, 164775/131072$ are below $4, 1/2, 4/3$. In the second, $34801/9216 < 19/5$ and $138411/65536 < 17/8$.

For the four odd rows, use $x \le \theta$ to fold the cubic into the quadratic. The resulting positive quadratic coefficients are $3985/576, 471/64, 315/64, 667/192$. After dropping nonpositive final terms the upper

bounds are convex. Their endpoint positive costs are respectively below $493/100 + 23/20 = 152/25$, $17/4 + 5/4$, $13/3 + 5/6$, and $11/3 + 3/5$, all below $49/8$. The tightest first row uses $90701/18432 < 493/100$ and $673465/589824 < 23/20$, leaving margin at least $9/200$. These rational comparisons cover the full continuous interval and include the shortest even root G_2. Together with the preceding $v = 3$ proof, this covers all $b \geq 4, v \geq 3, L = 1$.

C160.6 The unit connector for the whole terminal h3 f_b

C160.6.1 The root $v = 3$

Here $R = (\gamma + x(A-1) - zxC)u$, $\sigma \leq \gamma^2 d + \delta_4 + \delta_5$, $D_0 \leq \delta_2$, and $D_1 \leq -\gamma d + \delta_3$. The root cross term with $-Bu$ is positive; the full norm is at most $\sigma + \delta_2 + 2\delta_3$. Thus the complete exponent is

$$-8d - 4\beta_b - 4\gamma - 4\gamma d + (33/4)\delta_2 + (25/2)\delta_3 + c\sigma.$$

The block $P = -4\gamma - 4\gamma d + c\gamma^2 d$ has negative derivative. Set the whole block to $\gamma = \tau$, use $\beta_b \geq \tau$, and drop $-4\tau d + c\tau^2 d \leq 0$. The result is $-16d + (65/4)\delta_2 + (25/2)\delta_3 + c(\delta_4 + \delta_5)$. The three costs per d are below $53/8, 17/8, 1$, with sum $39/4$, giving $EH^4 \leq e^{-25d/4}$.

C160.6.2 Every v at least four

Use the entire root $\bar{f}_2 W_v$, strong vector $-BG_m(B)u + \beta\mathbf{1}$. Its full squared norm is

$$\sigma + \int (x^2 D^2 + xE^2) + 2\gamma \int xD$$
$$+2\int x^2[(A-1)D + CE]$$
$$+\beta^2 + 2\beta(\int E + \int xC).$$

The γ derivative is at most $-4 - (35/64)d + (41/2)d^2 < 0$. Set its whole block to τ. Drop the nonpositive $2c\tau \int x(A-1)$ and $-4\tau d + c\tau^2 d$, retaining positive τ terms until $\tau \leq 2/9$. The remaining C coefficient is nonnegative, so set $C = 1$. In $s = A - 1 \in [-x, 0]$, convexity reduces to its two endpoints, and $s = -x$ exceeds zero by $4x^2 + cx^4 - (17/2)x^3 D \geq 0$. Thus set $A = 1 - x$.

The complete even and odd integral polynomials are respectively

$$F_e = -12 + (89/6)x + 4x^2 + cx^3 + cx^4$$
$$+[-8 - (19/9)x + (17/2)x^2 - (17/2)x^3]D$$
$$+(17/4)(1 + x + x^2)D^2,$$

$$F_o = -16 + (229/12)x + (25/2)x^2 + cx^3 + cx^4$$
$$+[-7/6 + (119/576)x - (17/2)x^2 - 17x^3]D$$
$$+[17/12 + (17/4)x^2 + (17/4)x^3]D^2.$$

Handle the actual constant first. For even m, $3\beta^2 + 2\beta \int xC \leq 3d \int D^2 + 2d \int xC$. For odd m, first use $\beta^2 + 2\beta \int E \leq d \int [D^2 + 2D - 2\kappa xD]$, then pay the terminal positive cross term $2\beta \int xC$. Only afterwards use $d \leq 1/3$. Convexity in D gives all four endpoints:

Table 99: Short terminal pairs at a root beginning (1,2,v): parameter values.

Entry	Expanded polynomial
even 1	$-63/4 + (611/36)x + (67/4)x^2 + (7/4)x^3 + (41/4)x^4$.
even 1-x	$-63/4 + (593/36)x + (263/18)x^2 - 11x^3 + 23x^4$.
odd 1	$-63/4 + (11111/576)x + (33/4)x^2 - (5/2)x^3 + (41/4)x^4$.
odd 1-x	$-63/4 + (10151/576)x + (5449/576)x^2 - (5/2)x^3 + 23x^4 + (17/4)x^5$.

For example, the even $1-x$ linear and quadratic coefficients are $89/6+53/9-17/4 = 593/36$ and $4+19/9+17/2 = 263/18$. The odd $1-x$ quadratic is $25/2-17/2-119/576+17/3 = 5449/576$. The four endpoint positive costs are below $1007/100, 199/20, 1911/200, 193/20$, all below $81/8$. The tightest first row uses $7943/1152 < 69/10$, $11323/4096 < 277/100$, $15379/131072 < 3/25$, and $1171001/4194304 < 7/25$, summing to $1007/100$. Drop the other negative cubic terms. This covers all $v \geq 4$, with $v = 3$ covered separately.

C160.7 Every nonunit inner connector for terminal (2, 4)

C160.7.1 Exact same-vector data

At $b = 4$, $G_2 = 1 + z$, $\gamma = d$, and $\Gamma = 2d$. Thus

$$R = du - B^2u - B^3u = (d + x + zx)u, \quad \sigma = d^3 + 2d\delta_2 + \delta_3 + \delta_4,$$
$$D_0 = -\delta_2, \quad D_1 = -d^2 - \delta_2, \quad D_2 = \delta_3, \quad D_3 = d\delta_2 + \delta_3.$$

The actual expressions give

$$\sigma < 5d/8, \quad -\delta_2 \leq \delta < 8d/15,$$
$$-13d/16 \leq \epsilon \leq -\tfrac{1}{2}d^2 - \tfrac{293}{2048}\delta_2 \leq 0.$$

The four terms of σ/d are at most $1/9, 13/48, 169/1024, 2197/32768$, respectively below $9/80$, $11/40, 1/6, 11/160$, with total $299/480 < 5/8$. Also $\delta/d \leq 1/6 + (3/2)\theta^2(1+\theta) < 31/60 < 8/15$, using $(3d-1)\delta_2 \leq 0$. The ϵ coefficient uses $(3/2)\theta(1+\theta) = 1755/2048$. In the other direction, $|\epsilon|/d \leq d - (3/2)d^2 + (2-3d)(\delta_2/d) \leq 2\theta + (1-3\theta)d - (3/2)d^2 \leq 13/16$.

C160.7.2 Two inner edges

The terminal vector is $\delta\mathbf{1} + u + BR$, with $S_2 = -3d + \epsilon + \delta_3 \leq -3d + \delta_3$. Use the weaker envelope for the entire root $h_2 f_v$. The full vector is $Q + \delta\mathbf{1}$, where

$$Q = [1 - xD - x^2 + z(d + x - E)]u.$$

For $v \geq 4$, the real coordinate lies in $[0, 1-x]$, and the imaginary coefficient has absolute value at most $1-x$, using $d + 3x < 2$ on its positive side. Hence $\|Q\|^2 \leq \int (1+x)(1-x)^2 \leq d$. For $v = 3$, the exact $(E, D) = (1, 0)$ gives $\|Q\|^2 \leq \int [1 + x - 4x^2 + x^3 + x^4] \leq 13d/12$, since $x + x^2 < 1$ and $x - 3x^2 \leq 1/12$. In both cases $EQ = \int E - d^2 - \delta_2$, so $|EQ| \leq d$. The full constant cost is $\delta^2 + 2\delta EQ \leq (304/225)d^2 \leq 304d/675$. The resulting norms are below $3d/2, 5d/3$, respectively; the root losses are $3d/16$ and exactly d. The fourth losses are $1463/256, 6341/768$, both greater than $45/8$.

C160.7.3 Three inner edges

The exact terminal scalar-plus-mean is $S_3 = -4d + \epsilon + 2\delta_3 + d\delta_2 - \delta d \le -4d + 2\delta_3 + 2d\delta_2$. Use the weaker standard root. The full vector is $Q + \epsilon\mathbf{1}$, where

$$Q = [1 + \delta + xD - dx - x^2 + z(1 - E - x^2)]u.$$

The real coordinate is greater than $1 - 2\theta/3 - \theta^2 > 1/2$, and is at most $1 + x - x^2 + (8/15 - x)d \le 53/45 + (2/3)x - x^2 \le 58/45$. The imaginary coefficient has absolute value at most x, including the exact $v = 3$ endpoint. Thus $\|Q\|^2 < 7d/4$. Since $EQ = \int E - d + \delta_3 \ge -\delta_2$ and $\epsilon \le 0$, the constant cost is at most $\epsilon^2 + 2|\epsilon|\delta_2 \le 169d/384$. The full norm is below $9d/4$. With root loss $5d/6$, the fourth loss is $943/128 > 45/8$.

C160.7.4 Every L at least four, including the last gamma index

The same remainder has all the periodic moments

$$\begin{aligned} D_{4j} &= -\delta_{2j+2}, \\ D_{4j+1} &= -d\,\delta_{2j+1} - \delta_{2j+2}, \\ D_{4j+2} &= \delta_{2j+3}, \\ D_{4j+3} &= d\,\delta_{2j+2} + \delta_{2j+3}. \end{aligned}$$

Here $\gamma_3 = d - \delta_3$, and $\gamma_4 = d - d\delta_2 - \delta_3 \ge (1 - 39/128)d$. For later indices, the positive part is at most $(3/4)\delta_4 < d/16$, because $d + \theta < 3/4$ and $\theta^3 < 1/12$. Thus every $\gamma_l \ge 17d/32$ for $l \ge 3$, including $L + 1$. Set $h = 17/32$, $n = L - 1 \ge 3$. Propagation gives $S \le -1213d/480$, with $V = G_n(B)u + B^n R - hd\mathbf{1}$. Since $\|R\| < (4/5)\sqrt{d}$ and $\theta^{3/2} < 4/15$, the actual tail has norm below $16\sqrt{d}/75$.

For $v \ge 4$, use the strong standard root. Put $U = (G_n - BG_m)u + B^n R$; then $\|U\| < 106\sqrt{d}/75$, and $EU \le \delta_2 + (D_n)_+ \le 7\delta_2/4$. The complete constant $c = \beta - hd$ lies in $[0, 15d/32]$, and

$$\|U + c1\|^2 = \|U\|^2 + c^2 + 2cEU < [2 + 3/40 + 9/40]d = 23d/10.$$

The three comparisons are $(106/75)^2 < 2$, $75/1024 < 3/40$, and $455/2048 < 9/40$. Add the strong root loss $1463d/1024$; the fourth loss is $4645/768 > 45/8$.

For $v = 3$, use the exact root vector $-Bu$ and loss d. The base $G_n(B)u - Bu$ has coordinates $(E_n, D_n - 1)$, with norm below $21\sqrt{d}/20$. Adding the tail gives $379\sqrt{d}/300$, with square below $8d/5$. Its mean satisfies $EU = d - \beta_{n+1} + D_n \ge -(3/4)\delta_3$: the negative moments start at D_4, D_5, controlled by $\delta_4, (d + \theta)\delta_3$, and later terms are smaller. Retain the actual negative constant:

$$\|U - hd\mathbf{1}\|^2 < [8/5 + h^2/3 + (h/2)\theta^2]d < 7d/4.$$

The resulting fourth loss is $1601/240 > 45/8$. This proves every v for $L = 2, 3$ and all $L \ge 4$ at terminal $(2, 4)$.

C160.8 Common bounds for a general terminal (2, b)

Here $F = \bar{f}_2 W_b$, $k = b - 2 \ge 2$. Use the same first-row R, σ, D_0, D_1 of part C160.4 throughout. In particular $\gamma = \int C$ retains its relation to A, C.

C160.8.1 Variance, epsilon and delta

From $\gamma \le d$ and $A, C \le 1$, $\sigma \le d^3 + 2d\delta_2 + \delta_3 + \delta_4 < 5d/8$.

The exact expression for epsilon is

$$\epsilon = d[-\gamma + (3/2)\gamma^2] + \int [(3\gamma - 1)xA - xC + (3/2)x^2A^2 + (3/2)x^3C^2].$$

Fix the actual A, C. The expression is convex in $\gamma \in [0, 1/3]$. At zero, the negative terms absorb the square terms separately. At $1/3$, set $A = 1, C = 1 - x$; the scalar is $-d/6$, and the integral is bounded by $J = -x + (5/2)x^2 + (3/2)x^3 - 3x^4 + (3/2)x^5$. After dropping the nonpositive final term the bound is convex, with endpoint $7007/65536 < 1/6$. Thus $\epsilon \le 0$. Inserting the first two actual variance terms in the opposite direction gives

$$|\epsilon| \le d[2\theta + \gamma(1 - 3\theta) - (3/2)\gamma^2] \le 13d/16.$$

The lower bound is $\delta \ge -\delta_2$. For its upper bound retain $\gamma = \int C$ and split the domain.

If $d \le 9/32$, enlarge the actual expression using $A = 1, C = 1 - x$, obtaining $\delta/d \le 243/2048 + \int f/d$, where $f = -(5/32)x + (5/2)x^2 + (3/2)x^3 - 3x^4 + (3/2)x^5$. Since $f' = 5 + 9x - 36x^2 + 30x^3 \ge 5 - 9x > 0$, its maximum lies at an endpoint. The terms at θ are respectively below $1/8, -1/16, 5/12, 1/9, -1/16$; the last negative comparison is $4369833 > 4194304$. Their total is below $19/36 < 8/15$.

If $d \ge 9/32$, put $r = d - \gamma = \int (1 - C)$, so $0 \le r \le \delta_2$. Compare with $\omega_{24} = -\delta_2 + (3/2)(d^3 + 2d\delta_2 + \delta_3 + \delta_4)$:

$$\begin{aligned} &\delta - \omega_{24} \\ &\le r[\theta - 3d^2 - 3\delta_2 + (3/2)dr] \\ &\le r[\theta - 6d^2 + (3/2)d^3] \le 0. \end{aligned}$$

The bracket decreases and equals $-2293/65536$ at $9/32$. Hence again $\delta < 8d/15$. Together the two domains give $|\delta| \le 8d/15$.

C160.8.2 The joint scalar payment

We prove $\Gamma - 2\delta_+ \ge 5d/6$. If $\delta \le 0$, use $\Gamma \ge 2\kappa d$. For $\delta > 0$ retain the actual finite parity relation.

For even k, $A = C$. Put $X = \int xC$ and $Y = \int x^2(1 + x)C^2$. Then

$$\Gamma - 2\delta = 2\gamma - 3\gamma^2 d + (2 - 6\gamma)X - 3Y.$$

Use $Y \le \theta(1 + \theta)X$ and set $c = 2 - 6\gamma - 3\theta(1 + \theta)$. If $c \ge 0$, the bound is $\gamma(2 - 3\gamma d) \ge 95d/96$. If $c < 0$, use the actual $X \le \theta\gamma$ and $t = \gamma/d \in [\kappa, 1]$:

$$(\Gamma - 2\delta)/d \ge C_0 t - (55/48)t^2, \quad C_0 = 69345/32768 > 21/10.$$

The concave quadratic reduces to its two continuous endpoints. At one the lower bound is $229/240$; at κ, it is $207157/245760 > 204800/245760 = 5/6$. The coefficient $55/48$ comes from $d(3d + 6\theta) \le 55/48$, retaining the actual scalar relation.

For odd k, $A = 1 - xC$. Put $X = \int xC$, $Z = \int x^2C$. The actual variance is

$$\sigma = \gamma^2 d + 2\gamma\delta_2 - 2\gamma Z + \delta_3 - \int x^3 C[2 - (1+x)C].$$

Use $Z \geq \kappa\delta_3$ and $C[2-(1+x)C] \geq \kappa^2$, obtaining

$$\Gamma - 2\delta \geq d + \gamma - 3\gamma^2 d + \int J_\gamma(x),$$

$$J_\gamma = (\kappa - 6\gamma)x + (6\gamma\kappa - 3)x^2 + 3\kappa^2 x^3.$$

Here $J_\gamma'' \leq 4\kappa - 6 + 18\kappa^2\theta < 0$; $\kappa < 3/5, \theta < 5/12$ give the stronger bound $-9/10$. Also $J_\gamma(0) = 0$ and $J_\gamma(\theta) < 0$: its γ coefficient is negative, and at $\gamma = 0$ use $\kappa - 3\theta = -5/8$ and $3\kappa^2\theta^2 < 3/16$. Thus the full interval has lower bound $J_\gamma(\theta)$. With $d \leq 1/3$, $t = \gamma/d$,

$$(\Gamma - 2\delta)/d \geq A_0 + B_0 t - t^2/3, \quad A_0 > 13/16, \quad B_0 = 6283/16384 > 3/8.$$

The exact constant is $A_0 = 191/256 + 2379351/33554432$; its comparison uses $2379351 > 2228224$. The two concave-quadratic endpoints have lower bounds $41/48, 233/256$, both above $5/6$. This includes the shortest odd $k = 3$. Thus the joint payment holds for every original $b \geq 4$.

C160.8.3 All moment orders and every long connector

The same actual remainder has all the moments

$$D_{4j} = -\int x^{2j+1} C, \quad D_{4j+1} = -\gamma\delta_{2j+1} - \int x^{2j+1} A,$$

$$D_{4j+2} = \int x^{2j+2} C, \quad D_{4j+3} = \gamma\delta_{2j+2} + \int x^{2j+2} A.$$

The corresponding positive parts are bounded by those of the actual $(2,4)$ remainder. Thus every $\gamma_l \geq 17d/32$ for $l \geq 3$, including $L+1$; also $D_1 < 0$. The joint payment gives $S \leq -(5/6 + 51/32)d = -233d/96$ for $L \geq 4$. The identical full norm, mean and root-constant calculations in part C160.7.4 give, for $v \geq 4$, norm below $23d/10$ and strong root loss $1463d/1024$; for $v = 3$, norm below $7d/4$ and exact root loss d. The fourth losses are $21689/3840, 301/48$, respectively. The first exceeds $45/8$ by $89/3840$. The joint scalar payment remains essential here.

C160.9 Short connectors for general (2, b)

C160.9.1 Three inner edges and every root

Put $X_A = \int xA, Z_A = \int x^2 A, Z_C = \int x^2 C$. Exact propagation gives

$$S_3 = -\Gamma + (1-d)\delta - 2d - \gamma(d - \delta_2) - X_A + Z_A + Z_C.$$

If $\delta \geq 0$, the joint payment gives $\Gamma - (1-d)\delta \geq 5d/6$; if $\delta < 0$, use $\Gamma \geq 2\kappa d$. Since $X_A \geq \kappa\delta_2$ and $Z_A, Z_C \leq \delta_3$, $S_3 \leq -17d/6 + (7/32)\delta_2 \leq -(17/6 - 91/1024)d$.

For $v \geq 4$, use the strong standard root. Its complete vector is $Q + c\mathbf{1}$, where

$$Q = [1 + \delta + xD - \gamma x - x^2 A + z(1 - E - x^2 C)]u,$$

$$c = \epsilon + \beta, \quad -7d/32 \leq c \leq d.$$

The real coordinate is above $1/2$, and at most $53/45 + (77/96)x - x^2 + x^3 < 45/32$. This polynomial has derivative $3(x-1/3)^2 + 15/32 > 0$; the exact endpoint margin is $911/1474560$. The imaginary coefficient is bounded by x, so $\|Q\|^2 \le 66997d/32768$. Its mean is $EQ = \int E - d + Z_C \in [-\delta_2, \delta_3]$. If $c \ge 0$, $2cEQ \le 2d\delta_3$; if $c < 0$, it is at most $7d\delta_2/16$. Both are at most $(2/3)\theta^2 d$, and $c^2 \le d/3$. The full norm is at most $81525d/32768 < 5d/2$, giving fourth loss $1165/192 > 45/8$.

For $v = 3$, use $E = 1, D = 0$. The real coordinate is at most $53/45$, the imaginary coefficient is $-x^2 C$, and $EQ = Z_C \ge 0$. Since $\epsilon \le 0$, its actual cross term is nonpositive. The comparisons $(53/45)^2 < 25/18$, $\theta^5 < 1/90$ give $\|Q\|^2 < 7d/5$. Adding $\epsilon^2 \le 169d/768$ gives full norm below $5d/3$; the exact root loss d yields fourth loss $2021/256 > 45/8$.

C160.9.2 Two inner edges and $v = 3$ on the whole domain

The entire root is $h_2 f_3$, selected using only L, v. The exact full vector is $Q + \delta\mathbf{1}$, where

$$Q = [1 - x^2 C + z(\gamma + xA - 1)]u, \quad EQ = d - \gamma d - X_A.$$

The real coordinate lies in $[0, 1 - x^2 + x^3]$; the negative imaginary coefficient has absolute value at most $1 - x + x^2$. Its squared polynomial satisfies $1 + x - 4x^2 + 5x^3 - x^4 - x^5 + x^6 \le 1 + x - 2x^2 \le 9/8$, since $5x - x^2 \le 1911/1024 < 2$ and $-x^5 + x^6 \le 0$. Thus $\|Q\|^2 \le 9d/8$, and $25d/96 \le EQ \le d$. If $\delta \le 0$, then $|\delta| \le \theta d < 2EQ$, and the constant cost is nonpositive. If $\delta \ge 0$,

$$\begin{aligned} &\tfrac{17}{4}(\delta^2 + 2\delta EQ) \\ &\le \tfrac{17}{4}\tfrac{38}{15} d\delta \\ &\le \tfrac{323}{90}\delta < 4\delta. \end{aligned}$$

Consequently $4(-\Gamma + \delta)$ plus the constant cost is at most $-4(\Gamma - 2\delta_+) \le -10d/3$. The remaining exact scalar is $-2d - \gamma d - X_A + Z_C \le -2d$, using $\kappa - \theta > 0$. The complete exponent is at most $-34d/3 + (17/4)(9/8)d = -629d/96 < -45d/8$. This covers all hosts and all $b \ge 4$.

C160.9.3 The same direction for $L = 2$ and v at least four on the auxiliary domain

Assume $0 < d \le 1/8$. Retain the entire root $h_2 f_v$, strong loss $969d/1024$, and vector $-BG_m(-B)u + \beta\mathbf{1}$. The improved exact bound is $\epsilon \le -c_* \delta_2$, where $c_* = 13663/65536 > 1/5$. Indeed $\gamma \le 1/8$, its scalar term is nonpositive, $(3\gamma - 1)xA \le -(5/8)\kappa x$, and the remaining two blocks give $c_* = \kappa(13/8 - (3/2)\theta^2) - (3/2)\theta$. Also $\delta/d \le 18527/65536 < 2/7$, with $\delta \ge -\delta_2$. The complete constant $c = \delta + \beta$ lies in $(0, 9d/7]$.

$$Q = [1 - xD - x^2 C + z(\gamma + xA - E)]u, \quad EQ \le d.$$

The real coordinate is positive and at most $1 - x + x^3$, while the imaginary coefficient is bounded by $1 - x + x^2$, using $\gamma + 3x - x^2 < 2$ on the positive side. The squared polynomial $1 - x - x^2 + 5x^3 - 4x^4 + x^5 + x^6$ is at most one, since $5\theta^2 + \theta^4 + \theta^5 < 7/8$. The complete norm is therefore at most $d + (207/49)d^2 \le 599d/392$. For the scalar, first combine $\epsilon + D_2 \le (\theta - 1/5)\delta_2$, then use $\delta_2 \le \theta d$, giving $S_2 \le -[35/16 - \theta^2 + \theta/5]d$. Adding the strong alternative-root loss gives exactly $61d/20$. The fourth loss is $44733/7840 > 45/8$, with margin $633/7840$.

This auxiliary calculation is restricted to $d \le 1/8$. The fixed direction on the remaining domain is treated separately by Q5-C163.

C160.10 Every nonunit inner connector for terminal (3, b)

C160.10.1 The actual terminal and its common bounds

Here $k = b - 2 \ge 3$, $t = 1 - A \in [0, x]$, $R = (\gamma - xt + zxC)u$, and $\Gamma = d + \int(A + C) \ge 3d - 2\delta_2 \ge 35d/16$. Use $\eta = 5/4$. The actual envelope $\bar{f}_3 W_b \le e^{R-\Gamma}$ is the third row of part C160.4 and holds independently of L, including $L = 1$.

$$\sigma = \gamma^2 d - 2\gamma \int xt + \int (x^2 t^2 + x^3 C^2) \le d^3 + \delta_4 + \delta_5 < 5d/24,$$

$$D_0 = -\int xC, \quad D_1 = -\gamma d + \int xt,$$

$$D_2 = \int x^2 C, \quad D_3 = \gamma\delta_2 - \int x^2 t.$$

The variance bound uses $\theta^3(1+\theta) < 7/72$. Set $c_0 = \kappa - (5/4)\theta^2(1+\theta) = 39799/131072 > 3/10$. Then $-\delta_2 \le \delta \le (5/4)d^3 - c_0\delta_2 \le 7d/180 < d/24$. This follows by comparing both endpoints of the convex quadratic δ/d on the whole $d \in [0, 1/3]$.

The exact epsilon expression is

$$d[-\gamma + (5/4)\gamma^2] - (5/2)\gamma \int xt + \int x[t - C + (5/4)xt^2 + (5/4)x^2 C^2].$$

The first two terms are nonpositive. The bracket increases with t and decreases with C; at $t = x$, $C = 1 - x$, it is bounded by $f = -1 + 2x + (5/4)x^2 - (5/4)x^3 + (5/4)x^4$. This polynomial is increasing and $f(\theta) = -129867/4194304 < 0$, proving $\epsilon \le 0$. In the other direction, $\epsilon \ge -\gamma d - \int xC \ge -(d^2 + \delta_2) \ge -71d/96 > -3d/4$. The moment D_1 can be positive, so retain $(D_1)_+ \le \delta_3$.

C160.10.2 Two inner edges and the entire alternative root

The terminal has $S_2 \le -4d + 2\delta_2 + \delta_3$. The weaker alternative-root loss is $3d/16$ for $v \ge 4$, or exactly d for $v = 3$. The full vector is $Q + \delta\mathbf{1}$, where

$$Q = [1 - xD - x^2 C + z(\gamma - xt - E)]u.$$

For $v \ge 4$, the real coordinate is positive and at most $1 - x + x^3$, while the negative imaginary coefficient is bounded by $1 + x^2$, since $\gamma < E$. The squared upper bound is $1 - x + x^2 + 4x^3 - 2x^4 + x^5 + x^6$. Subtract one and divide by x: $g = -1 + x + 4x^2 - 2x^3 + x^4 + x^5$ is increasing, with derivative at least $1 - 6\theta^2 > 0$, and $g(\theta) = -985987/33554432 < 0$. Hence $\|Q\|^2 \le d$.

For both root ranges, $25d/96 \le EQ \le (1 + \theta^2)d$. If $\delta \le 0$, the constant cost is nonpositive; if positive it is below $19d^2/192 \le 19d/576 < d/30$. The full norm for $v \ge 4$ is below $31d/30$. At the exact $v = 3$ endpoint, $\|Q\|^2 \le d[1 + \theta(1 + \theta^2)^2] < 8d/5$, so its full norm is below $49d/30$. The fourth losses are $1093/120 - 169/256$ and $1177/120 - 169/256$, both above eight. This includes every finite root index.

C160.10.3 Three inner edges and the entire standard root

The terminal satisfies $S_3 \le -5d + 2\delta_2 + \delta_3 + 2d\delta_2$. The full vector is $Q + \epsilon\mathbf{1}$, where

$$Q = [1 + \delta + xD - \gamma x + x^2 t + z(1 - E - x^2 C)]u.$$

The real coordinate is positive and at most $1 + 1/72 + \theta + \theta^3 < 3/2$; the imaginary coefficient is bounded by x. Thus $\|Q\|^2 < 7d/3$, and $EQ = \int E - d + \int x^2 C \geq -\delta_2$. Retain the epsilon constant and its mean cross term to obtain full norm at most $[7/3 + 25/64]d < 11d/4$. The root loss $5d/6$ yields fourth loss $1703/256 > 45/8$. The exact $v = 3$ coordinates satisfy the same bounds.

C160.10.4 Every L at least four, all gamma indices and the constant cross term

Choose $h = 9/16$. Here $\gamma_3 \geq d - \delta_3, \gamma_4 \geq d - d\delta_2$. For $l \geq 5$, the positive D_{l-1} is at most δ_5 or $d\delta_4$, each below $d/32$. Therefore $\gamma_l \geq \kappa d - d/32 = 9d/16$ for all $l \geq 3$. Explicitly, $D_4 = -\int x^3 C \leq 0$, $D_5 = -\gamma\delta_3 + \int x^3 t \leq \delta_5$, $D_6 = \int x^4 C \leq \delta_5$, and $D_7 = \gamma\delta_4 - \int x^4 t \leq d\delta_4$. Every later four-step cycle only increases the powers, including the index $L + 1$.

Long propagation gives $S \leq -[91/24 - 2\theta^2]d$, retaining $2(D_1)_+ \leq 2\delta_3$. Since $\|R\| < \sqrt{d}/2$, its actual tail is below $2\sqrt{d}/15$. The joint $6/5$ estimate gives $U = (G_n - BG_m)u + B^n R$ with $\|U\| < 4\sqrt{d}/3$. Its actual mean satisfies the needed lower bound $EU \geq -\delta_2 - (3/4)\delta_3$: the base mean is at least $-\delta_2$, and negative D_n for $n \geq 3$ are controlled by δ_4 or $d\delta_3$. Retain the full constant $-hd\mathbf{1}$:

$$\begin{aligned} &\|U - hd\mathbf{1}\|^2 \\ &\leq [16/9 + h^2/3 + (2h/3)(\theta + (3/4)\theta^2)]d \\ &= [16/9 + 9969/32768]d < 17d/8. \end{aligned}$$

Add the standard-root loss $5d/6$. The fourth loss is $1043/128 > 45/8$. The lower bound on the mean has the required sign for the negative constant. Together the cases $L = 2, 3$ and $L \geq 4$ cover every $b \geq 5, v \geq 3, L \geq 2$ in this fixed terminal direction.

Q.3 The actual unit-arm curvature for the short-two connector

Q163.0 Original goal, exact new scope, and fixed directions

The surrounding original Q5 problem is the following. For every nonempty finite original tree U having at most five vertices of original degree at least three, choose all original edge directions solely from U, before the host, such that for every original finite probability measure mu and every real skew kernel K with pointwise $|K| \leq 1$,

$$Z_D = \mathbb{E}_{\mu^{V(U)}} \prod_{i \to j} (1 + K(x_i, x_j)) \leq 1.$$

Original arm lengths, equal multiplicities, all subdivision vertices, all four independent positive core lengths, the original edge count, arbitrary original joint-label laws in entropy applications, and noninjective maps are retained. We prove the half-tree interface below.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. All expectations, adjoints, norms, and finite spectral integrals use the same original mu. Until the separate $d = 0$ case in part Q163.7, the proof assumes $d > 0$.

The original half-tree has root residual arms (1, 2, v), $v \geq 4$, and terminal residual arms (2, b), $b \geq 5$. Its inner connector has exactly two original edges. Use the actual, structure-fixed messages

$$F = \bar{f}_2 W_b, \qquad P = h_2 f_v, \qquad H = pP T^2 F. \tag{Q163.0.1}$$

The root unit arm is outward, giving $p = f_1$. The root length-two arm has its first edge toward the root and its other edge away, giving h_2; its length-v arm is all outward, giving f_v. The terminal length-two arm is all inward, giving $\bar{f}_2$; its length-b arm has its first edge outward and remaining edges toward the terminal, giving W_b. The inner connector is directed from the terminal toward the root, giving T^2. The independent outer connector is directed from this root toward its receiving center, giving T^M for every original integer $M \geq 1$.

Author theorem T171.1. For every original $b \geq 5, v \geq 4$, and $M \geq 1$, the single direction (Q163.0.1), including every genuine equal-pair decoration at the two original vertices, satisfies

$$\boxed{EH^4 \leq e^{-25d/4}, \qquad I_q(T^M H)^3 \leq e^{-3d/2}, \qquad I_q(Y) = E[q^{-1/3}|Y|^{8/3}].} \tag{Q163.0.2}$$

The proof actually gives $EH^4 \leq \exp(-6929d/1080)$. In particular it pays the previously requested ordinary-fourth loss 45d/8 and outer loss d/4.

Q163.1 Precisely used inputs and finite notation

Use the finite-space bounds (M.1), the spectral identities (M.3), and the finite convolution (M.2). The fourth signed-mean bound is (M.43), proved in Section M.1; it applies to every original real vector, with its signed mean. Section Q.1 gives the ordinary, unanchored outlet

$$I_q(T^M H)^3 \leq e^{11d}(\mathbb{E}|H|^4)^2 \qquad (M \geq 1). \tag{Q163.1.2}$$

For $G_n = E_n + zD_n$, the finite coordinate ranges and $|G_n| \leq 1$ at $n \geq 3$ are Lemma M.1. Here the root index is at least two and the terminal index at least three, so G_1 does not occur. The genuine G_2 square remains $1 + x$. The following proof uses its own Taylor coefficient and its own ϵ, δ and joint-norm estimates.

Q163.2 The same actual terminal and a sharper whole-domain Taylor coefficient

Let $k = b - 2 \geq 3$ and write

$$G_k = A + zC, \quad \gamma = \beta_{b-1} = \int C, \quad \Gamma = \beta_b + \gamma = \int (A + C).$$

Thus A, C are in [1-x, 1], gamma lies in [kappa d, d], and $\Gamma \geq 2\kappa d$.

To verify the actual envelope, apply A to the $\bar{f}_{b-1}$ convolution while retaining its original $\beta_{b-1}\bar{f}_0$ term. Since $A1 = p$ and $E(1 + G_{b-1}(B)u) = 1 - \beta_b$,

$$W_b \leq 1 - \beta_b - u - BG_{b-1}(B)u - \gamma p.$$

As $\bar{f}_2 = 1 + u + Bu$ and $G_{b-1} = 1 + zG_k$, their sum satisfies

$$\bar{f}_2 + W_b \leq 2 - \Gamma + R, \qquad R = \gamma u - B^2 G_k(B)u = (\gamma + xA + zxC)u.$$

The two messages are genuine and nonnegative. AM-GM and $1 + t \leq e^t$ therefore give

$$\boxed{F = \bar{f}_2 W_b \leq e^{R-\Gamma}.} \tag{Q163.2.1}$$

The improvement specific to $b \geq 5$ is the actual finite contraction $|G_k| \leq 1$. The original row bound yields

$$|R| \leq d + \|BG_k(B)u\| \leq d + \sqrt{\delta_2} \leq 1/3 + \sqrt{13/96} < 1/3 + 3/8 = 17/24 < 3/4. \tag{Q163.2.2}$$

The strict square comparison is $13/96 < 9/64$.

For $0 \le t \le 3/4$, the positive Taylor series gives

$$\frac{e^t - 1 - t}{t^2} \le \frac{1}{2} + \frac{t}{6} + \frac{t^2}{24(1 - t/5)} \le \frac{1}{2} + \frac{1}{8} + \frac{15}{544} = \frac{355}{544} < \frac{2}{3}. \tag{Q163.2.3}$$

Here the $n \ge 4$ remainder is bounded using its successive ratio at most t/5. The last rational comparison is $1065 < 1088$. At $t = 0$ use continuity. For $t \le 0$, the integral Taylor remainder is at most $t^2/2$. Hence the same actual R, everywhere on every original host, obeys

$$e^R \le 1 + R + (2/3)R^2. \tag{Q163.2.4}$$

This is $\eta = 4/3$, not C160's $\eta = 3/2$. Only an analysis bound changes; the original terminal F and all its edge directions remain the same.

Q163.3 Actual moments, epsilon loss, and the real constant

For this same R define

$$\sigma = ER^2, \quad D_j = EB^j R, \quad \delta = D_0 + (4/3)\sigma, \quad \epsilon = \delta + D_1.$$

The exact finite identities are

$$\begin{aligned} \sigma &= \gamma^2 d + 2\gamma \int xA + \int x^2(A^2 + xC^2), \\ D_0 &= -\int xC, \qquad D_1 = -\gamma d - \int xA, \qquad D_2 = \int x^2 C. \end{aligned} \tag{Q163.3.1}$$

Since $|G_k|^2 = A^2 + xC^2 \le 1$,

$$\sigma \le \gamma^2 d + 2\gamma \int xA + \delta_3. \tag{Q163.3.2}$$

No gamma, delta, or sigma is being substituted from a different vector.

Q163.3.1 A uniform negative epsilon

Let $X_A = \int xA$, $X_C = \int xC$. Using (Q163.3.1)–(Q163.3.2),

$$\epsilon \le d[-\gamma + (4/3)\gamma^2] + [(8/3)\gamma - 1]X_A - X_C + (4/3)\delta_3.$$

The first term is nonpositive. Since $\gamma \le 1/3$, the coefficient of X_A is at most -1/9. Also $X_A, X_C \ge \kappa\delta_2$ and $\delta_3 \le \theta\delta_2$. Therefore

$$\boxed{\epsilon \le -\left(\frac{10\kappa}{9} - \frac{4\theta}{3}\right)\delta_2 = -\frac{17}{144}\delta_2 \le 0.} \tag{Q163.3.3}$$

This is valid on the entire host domain, without a split at $d = 1/8$.

Q163.3.2 A sufficient delta bound with a short continuous proof

The lower bound is immediate: $\delta \ge -X_C \ge -\delta_2$. For the upper bound,

$$\delta \le \frac{4}{3}d^3 + \left(\frac{8}{3}d - \kappa + \frac{4\theta}{3}\right)\delta_2.$$

The parenthesized coefficient is first bounded from above by the positive constant $241/288$; only then is $\delta_2 \le \theta$ d used. Since $d^2 \le 1/9$,

$$\frac{\delta}{d} \le \frac{4}{27} + \frac{241\theta}{288} = \frac{4}{27} + \frac{3133}{9216} < \frac{3}{20} + \frac{7}{20} = \frac{1}{2}. \tag{Q163.3.4}$$

The two last comparisons are $80 < 81$ and $62660 < 64512$. Consequently

$$\boxed{-\delta_2 \le \delta < d/2.} \tag{Q163.3.5}$$

Q163.4 Exact two-edge propagation and the entire original root pair

The positive T kernel is at most two, so $T(R^2) \le 2\sigma$. Applying T to (Q163.2.1) and then using (Q163.2.4),

$$TF \le e^{-\Gamma}(q + \delta + BR).$$

A second actual application of T gives

$$T^2F \le e^{-\Gamma}Y, \qquad Y = 1 + u + \epsilon + BV_2, \qquad V_2 = \delta\mathbf{1} + u + BR. \tag{Q163.4.1}$$

Indeed $T\bar{f}_1 = \bar{f}_2 = 1 + u + Bu$ and $TBR = D_1 + B^2R$. The right side is strictly positive on every positive-mass atom because it dominates the actual positive message T^2F. In particular log Y is legal.

Let $m = v - 2 \ge 2, G_m = E + zD$, and

$$\beta = \beta_{v-1} = \int D, \qquad \beta_v = \int E.$$

For the root, retain the actual $\beta_v f_0$ and $\beta_{v-1} f_1$ terms in the positive convolution for f_v. Since $h_2 = 1 + u - Bu$ and $G_v(-B) = 1 - B + B^2 G_m(-B)$, one obtains

$$h_2 + f_v \le 2 - \beta_v - \beta - B^2 G_m(-B)u + \beta u.$$

Thus the entire actual root pair obeys

$$\boxed{P = h_2 f_v \le \exp\{-\beta_v - \beta + BV_{\text{root}}\}, \quad V_{\text{root}} = -BG_m(-B)u + \beta\mathbf{1}.} \tag{Q163.4.2}$$

The actual root unit p remains separate.

Its true scalar-plus-mean loss is

$$\ell_{\text{root}} = \beta_v + \beta - \int xD.$$

If m is even then $E = D$, so

$$\ell_{\text{root}} = \int (2 - x)D \ge (2 - \theta)(1 - \theta)d = \frac{969}{1024}d.$$

If m is odd then $E = 1 - xD$, so

$$\ell_{\text{root}} = d + \int (1 - 2x)D \ge [1 + (1 - 2\theta)(1 - \theta)]d = \frac{569}{512}d > \frac{969}{1024}d.$$

Therefore, for every actual finite $v \geq 4$,

$$\boxed{\ell_{\text{root}} \geq (969/1024)d.} \tag{Q163.4.3}$$

The even minimum $m = 2$, namely $v = 4$, is included literally.

Combining the vectors in (Q163.4.1)–(Q163.4.2), define the genuine joint polynomial vector Q and real scalar c by

$$Q = [1 - xD - x^2C + z(\gamma + xA - E)]u, \qquad c = \delta + \beta, \qquad V_2 + V_{\text{root}} = Q + c1. \tag{Q163.4.4}$$

From (Q163.3.5), $\beta \geq \kappa$ d, $\beta \leq d$, and $\delta_2 \leq \theta$ d,

$$\boxed{(3/16)d \leq c < (3/2)d \quad (d > 0).} \tag{Q163.4.5}$$

In particular c is not zero, and no orthogonality between the constant vector and Q may be assumed.

The two means are

$$EQ = \int E - \gamma d - \int xA, \qquad E(BQ) = -d + \int xD + \int x^2C. \tag{Q163.4.6}$$

Since $Eu = 0, E[B(Q + c1)] = E(BQ)$, even though EQ need not vanish. The full squared norm remains exactly

$$\|Q + c1\|^2 = \|Q\|^2 + c^2 + 2c\,EQ. \tag{Q163.4.7}$$

The next section pays the pointwise contribution c B1 in full before applying the exponential lemma.

Q163.5 The original unit arm pays the constant contribution

Apply (R29.5.4) to the actual p. Using $\log Y \leq Y - 1$ in (Q163.4.1)–(Q163.4.2) and the exact vector (Q163.4.4), set $a = -\Gamma + \epsilon - \beta_v - \beta$. Then

$$\log H \leq a + BQ + cu - (3/10)u^2. \tag{Q163.5.3}$$

Lemma M.27, with its real coefficient c, gives

$$\boxed{\log \mathbb{E}H^4 \leq 4S + (17/4)\|Q\|^2 + (10/3)c^2,} \qquad S = a + \mathbb{E}(BQ). \tag{Q163.5.5}$$

The pointwise square (R29.5.5) pays the entire cu term before the exponential estimate is applied to Q. This keeps the signed mean from (Q163.4.6),

$$S = -\Gamma + \epsilon - d + D_2 - \ell_{\text{root}}.$$

Thus (Q163.3.3), $D_2 \leq \delta_3 \leq \theta^2 d$, (Q163.4.3), and $\Gamma \geq 2\kappa d$ give

$$\boxed{S \leq -\frac{95}{32}d - \frac{17}{144}\delta_2.} \tag{Q163.5.6}$$

Here $2\kappa + 1 - \theta^2 + 969/1024 = 95/32$.

Q163.6 A strengthened joint norm, on the full continuous frequency interval

Write the two coordinates of Q as

$$R_0 = 1 - xD - x^2C, \qquad I_0 = \gamma + xA - E.$$

The actual finite-coordinate ranges imply

$$0 < 1 - x - x^2 \le R_0 \le 1 - x + x^3.$$

Also $I_0 \ge x(1-x) - 1 = -(1 - x + x^2)$. On the other side, $I_0 \le 1/3 + 2x - 1 \le 7/48$, whereas $1 - x + x^2 \ge 3/4$. Consequently

$$|I_0| \le 1 - x + x^2.$$

These inequalities use $\gamma \le 1/3$ and are valid on the entire original domain, not just on $d \le 1/8$. Thus

$$\|Q\|^2 \le \int [1 - x - x^2 + 5x^3 - 4x^4 + x^5 + x^6]. \tag{Q163.6.1}$$

The following single continuous polynomial estimate pays a useful loss:

$$1 - x - x^2 + 5x^3 - 4x^4 + x^5 + x^6 \le 1 - (4/5)x \quad (0 \le x \le 13/32). \tag{Q163.6.2}$$

After subtracting the right side and dividing by $x > 0$, it is enough to check

$$h(x) = -1/5 - x + 5x^2 - 4x^3 + x^4 + x^5 \le 0.$$

Now

$$h''(x) = 10 - 24x + 12x^2 + 20x^3 \ge 10 - 24\theta = 1/4 > 0.$$

A convex function is bounded above by its endpoint chord. The two actual endpoint values are

$$h(0) = -1/5, \qquad h(13/32) = -\frac{1835407}{167772160} < 0.$$

This proves (Q163.6.2) throughout the interval; $x = 0$ follows directly. Therefore

$$\boxed{\|Q\|^2 \le d - (4/5)\delta_2.} \tag{Q163.6.3}$$

The vector Q contains the root and terminal polynomials together.

Q163.7 Complete scalar payment and all outer lengths

Insert (Q163.4.5), (Q163.5.6), and (Q163.6.3) into (Q163.5.5):

$$\begin{aligned} \log EH^4 &\le -\frac{95}{8}d - \frac{17}{36}\delta_2 + \frac{17}{4}d - \frac{17}{5}\delta_2 + \frac{15}{2}d^2 \\ &= -\frac{61}{8}d - \frac{697}{180}\delta_2 + \frac{15}{2}d^2 \\ &\le -\frac{61}{8}d + \frac{653}{180}d^2 \\ &\le -\frac{6929}{1080}d < -\frac{25}{4}d \quad (d > 0). \end{aligned} \tag{Q163.7.1}$$

The last two steps use only the original moment inequality $\delta_2 \ge d^2$ and the original $d \le 1/3$. The exact final excess over $25/4$ is $179/1080 > 0$. Hence the ordinary-fourth claim in (Q163.0.2) is proved.

By the exact C084 interface (Q163.1.2), for every independent original outer length $M \ge 1$,

$$I_q(T^M H)^3 \le e^{11d}(EH^4)^2 \le e^{(11-25/2)d} = e^{-3d/2}. \tag{Q163.7.2}$$

This is the ordinary-fourth-to-Iq outlet.

At $d = 0, u = B\mathbf{1} = 0$ on every positive-mass atom. Original leaf integration makes all actual tree messages equal to one, even when B is not zero. Thus $H = 1, q = 1$ and both (Q163.0.2) inequalities hold with equality. No part of the direction changes at that boundary.

Q.4 The terminal pair (2,3) at a root (1,2,v)

Let $v \ge 3$ and let the original inner and outer connector lengths $L, M \ge 1$ be independent. Either terminal direction may be fixed:

$$F = \bar{f}_2 W_3 \quad \text{or} \quad F = h_2 f_3, \qquad P_{v,L} = \begin{cases} h_2 f_v, & L = 2, \\ \bar{f}_2 W_v, & L = 1 \text{ or } L \ge 3. \end{cases} \tag{Q.1}$$

Use the outward unit arm p, direct inner edges from terminal to root and outer edges from root to receiver, and put

$$H = pP_{v,L}T^L F. \qquad \mathbb{E}H^4 \le e^{-25d/4}, \qquad I_q(T^M H)^3 \le e^{-3d/2}. \tag{Q.2}$$

The directions depend only on the original lengths. Arbitrary genuine equal pairs and subdivision vertices are allowed. We use the notation and path inputs of Section M.2.

Proof. The arm sums for both terminal directions are $2 - d + R$, with $R = -B^2 u$:

$$\bar{f}_2 = 1+u+Bu, \quad W_3 = 1-d-u-Bu-B^2u, \quad h_2 = 1+u-Bu, \quad f_3 = 1-d-u+Bu-B^2u.$$

Thus $F \le e^{R-d}$. The row bound gives $|R| \le \sqrt{\delta_2} \le \sqrt{13/96} < 3/8$. On $[0, 3/8]$, the Taylor remainder coefficient is at most

$$\tfrac{1}{2} + \tfrac{t}{6} + \frac{t^2}{24(1-t/5)} \le \tfrac{1347}{2368} < \tfrac{4}{7};$$

on negative arguments it is at most $1/2$. Use the positive majorant (R29.5.3), with the coefficient $c = \eta = 8/7$ permitted at $t = 3/8$, and data

$$\begin{gathered} \Gamma = d, \quad \eta = 8/7, \quad \sigma = \delta_3, \quad D_{2k} = 0, \quad D_{2k-1} = (-1)^k \delta_{k+1} \ (k \ge 1), \\ \delta = 8\delta_3/7 \le 169d/896 < d/5, \quad \delta \le 13\delta_2/28, \quad \epsilon = \delta - \delta_2 < 0. \end{gathered} \tag{Q.3}$$

Write $m = v-2$, $G_m = E + zC$, and $\beta = \int C\, d\nu$. The standard and alternative root losses are ℓ_s, ℓ_a from Lemma M.28.

For the three short lengths, combine their positive majorants with the indicated root pair and the actual p curvature. The resulting parameters (Q_L, c_L, S_L), where $S_L = a_L + \mathbb{E}BQ_L$, are

$$\begin{aligned} Q_1 &= [x(1+C) - zE]u, \qquad c_1 = \beta, \\ S_1 &= -d + \delta - \delta_2 - \ell_s, \\ Q_2 &= [1 - xC + z(x-E)]u, \qquad c_2 = \delta + \beta, \\ S_2 &= -2d - \delta_2 + \delta - \ell_a, \\ Q_3 &= [1 + xC - x^2 + \delta + z(1-E)]u, \qquad c_3 = \beta + \epsilon, \\ S_3 &= -3d - \delta_2 + \delta + \delta_3 - \delta d - \ell_s. \end{aligned} \tag{Q.4}$$

Here $z = i\sqrt{x}$; the coordinate notation defines vectors by functional calculus, not pointwise functions of the original labels. In every row (M.55) gives $\log \mathbb{E}H^4 \le 4S_L + (17/4)\|Q_L\|^2 + (10/3)c_L^2$.

Length one. For $v = 3$, $(E, C, \beta) = (1, 0, 0)$, giving

$$\log \mathbb{E}H^4 \le -8d + \tfrac{1}{4}\delta_2 + \tfrac{247}{28}\delta_3 < -8d + \tfrac{529}{336}d < -25d/4.$$

For $v \ge 4$, pay $\beta^2 \le d \int C^2 \, d\nu$ first. The complete frequency integrands, for even and odd m, are

$$\begin{aligned} F_e &= -4 - 4x + \tfrac{32}{7}x^2 - (8+4x)C + \tfrac{17}{4}[x^2(1+C)^2 + xC^2] + \tfrac{10}{9}C^2, \\ F_o &= -8 + \tfrac{x}{4} + \tfrac{247}{28}x^2 - 4C + [\tfrac{17}{4}x^2(1+x) + \tfrac{10}{9}]C^2. \end{aligned}$$

They are convex in $C \in [1-x, 1]$. All four endpoint polynomials and their upper bounds on $[0, \theta]$ are

(m, C)	Endpoint polynomial	Upper bound
Even, 1	$-\frac{98}{9} - \frac{15}{4}x + \frac{151}{7}x^2$	$-\frac{919}{126}$
Even, $1-x$	$-\frac{98}{9} + \frac{73}{36}x + \frac{2291}{126}x^2$ $-\frac{51}{4}x^3 + \frac{17}{4}x^4$	$-\frac{10349}{1512}$
Odd, 1	$-\frac{98}{9} + \frac{x}{4} + \frac{183}{14}x^2 + \frac{17}{4}x^3$	$-\frac{8297}{1008}$
Odd, $1-x$	$-\frac{98}{9} + \frac{73}{36}x + \frac{1787}{126}x^2$ $-\frac{17}{4}x^3 - \frac{17}{4}x^4 + \frac{17}{4}x^5$	$-\frac{11357}{1512}$

The bounds use $x \le 1/2$, $x^2 < 1/6$, $x^3 < 1/12$. In the second and fourth rows the final displayed tails are nonpositive. Every bound is below $-25/4$.

Length two. For $v \ge 4$, the actual alternative root gives

$$\|Q_2\|^2 \le \int [(1 - x + x^2)^2 + x(1-x)^2] \, d\nu = \int [1 - x + x^2 - x^3 + x^4] \, d\nu \le d.$$

Indeed the last polynomial minus one is $-x(1-x)(1+x^2) \le 0$. Moreover $0 \le c_2 \le 6d/5$ and $4\delta - 4\delta_2 \le -15\delta_2/7$. Therefore

$$\log \mathbb{E}H^4 \le -\tfrac{1929}{256}d - \tfrac{15}{7}\delta_2 + \tfrac{24}{5}d^2 \le -(\tfrac{1929}{256} - \tfrac{31}{35})d < -13d/2.$$

For $v = 3$, keep the exact norm $\|Q_2\|^2 = d + \delta_2 - 2\delta_3 + \delta_4$ and $c_2 = \delta$:

$$\log \mathbb{E}H^4 \le -\tfrac{31}{4}d + \tfrac{1}{4}\delta_2 - \tfrac{55}{14}\delta_3 + \tfrac{17}{4}\delta_4 + \tfrac{10}{3}\delta^2 < -\tfrac{303}{40}d.$$

Here $\delta_4 \le \theta\delta_3$ makes the middle pair nonpositive, and the remaining costs are less than $d/8 + d/20$.

Length three. For $v \ge 4$, $0 \le c_3 \le d$; for $v = 3$, $c_3 = \epsilon < 0$ and $|c_3| \le \delta_2 \le d$. Removing δu only for the norm comparison gives

$$\|Q_3 - \delta u\|^2 \le \int [(1 + x - x^2)^2 + x^3]\, d\nu \le d + 2\delta_2 - \delta_3.$$

Restoring its square and cross term adds at most $(5/2)\delta d + \delta^2 d \le 77d^2/150$. Drop the favorable $-4\delta d$ from the scalar term. The full bound is

$$\log \mathbb{E}H^4 \le -\tfrac{47}{4}d + \tfrac{9}{2}\delta_2 + \tfrac{121}{28}\delta_3 + \tfrac{1103}{200}d^2 < -\tfrac{27}{4}d.$$

The three positive terms are less than $9d/4, 3d/4, 2d$, respectively. This includes the negative constant at $v = 3$.

Every longer connector. For $\gamma_j = \beta_j - D_{j-1}$, the finite moments in (Q.3) give

$$\gamma_3 = d, \quad \gamma_4 = d - \delta_3 \ge d/2, \qquad \gamma_j \ge \kappa d - \delta_4 > \tfrac{117}{224}d > d/2 \quad (j \ge 5). \tag{Q.5}$$

In the last inequality $\delta_4 \le \theta^3 d < d/14$. Apply (M.56) with $h = 1/2$, $D_1 < 0$, $n = L - 1 \ge 3$, and $R = -B^2 u$. The retained $\gamma_L \bar{f}_0$ and $\gamma_{L-1}\bar{f}_1$ give a terminal scalar-plus-mean bound

$$S_t \le -\tfrac{5}{2}d + \tfrac{16}{7}\delta_3 \le -\tfrac{951}{448}d < -\tfrac{21}{10}d. \tag{Q.6}$$

The next mean uses γ_{L+1}, covered by the same finite bounds. Adding the standard root leaves the curvature parameters

$$Q = (G_n(B) - BG_m(B) - B^{n+2})u, \qquad c = \beta - d/2, \qquad S \le S_t - \ell_s. \tag{Q.7}$$

Thus $|c| \le d/2$, whose curvature cost is at most $5d/18$. Also $\|B^{n+2}u\| < 11\sqrt{d}/100$, since $\theta^5 < 121/10000$. For $v \ge 4$, Lemma M.3 gives $\|Q\| < 131\sqrt{d}/100$, hence $\|Q\|^2 < 7d/4$, while $S < -7d/2$. Consequently

$$\log \mathbb{E}H^4 < -14d + \tfrac{119}{16}d + \tfrac{5}{18}d = -\tfrac{905}{144}d < -25d/4.$$

For $v = 3$, the stronger $m = 1$ bound gives $\|Q\| < 111\sqrt{d}/100$, $\|Q\|^2 < 5d/4$, $S < -31d/10$, and hence $\log \mathbb{E}H^4 < -4903d/720 < -25d/4$.

All $L \ge 1$ are now covered. The uniform cost 11 in (M.66) gives $I_q(T^M H)^3 \le e^{[11-2(25/4)]d} = e^{-3d/2}$ for every independent $M \ge 1$. Genuine equal pairs are restored by positivity and their pointwise bound in (M.1). Complete reversal applies the same proof to $-K$, exchanging the corresponding directions and weights. The $d = 0$ boundary follows by leaf integration. □

Q.5 Terminal triples (1,2,w) at a root (1,2,v)

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges. Here $n = v-2$, $\ell_+ = \ell_a$ and $\ell_- = \ell_s$. The ordinary-fourth bridge of Q5-C084, Section 2, has cost $11d$ for every $M \ge 1$. Its same-direction terminal $(1, 2)$ row, Sections 3–6 and equations (0.4)–(0.5), states $\mathbb{E}[p\bar{f}_2 W_v T^L(qf_2)]^4 \le e^{-705d/128}$ for every $v \ge 3, L \ge 1$. This exact actual message supplies the high-domain comparison below.

C169.0 The actual root and terminal triples

All vertices, arm multiplicities, subdivisions, independent original connector lengths and label laws retain their original meanings.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The root residual is $(1, 2, v)$, $v \geq 3$, and the terminal residual is $(1, 2, w)$, $w \geq 3$. Genuine equal pairs are restored by positivity. The original positive lengths L, M are independent. Always take the actual canonical terminal $F = qf_2h_w$. Choose the root using only the original integer w:

$$P_{v,w} = \begin{cases} h_2 f_v, & w = 3, \\ \bar{f}_2 W_v, & w \geq 4, \end{cases} \qquad H = pP_{v,w}T^L F. \tag{C169.0.1}$$

For every original v, w at least three and positive independent L, M,

$$\boxed{\begin{aligned} EH^4 &\leq \exp(-2817d/512), \\ I_q(T^M H)^3 &\leq \exp(-d/256), \quad I_q(Y) = E[q^{-1/3}|Y|^{8/3}]. \end{aligned}} \tag{C169.0.2}$$

The more precise same-direction rates are

Terminal	Root pair	Fourth loss λ; $EH^4 \leq e^{-\lambda d}$
$w = 3$	$h_2 f_v$	$11/2 + 3581/921600$
$w = 4$	$\bar{f}_2 W_v$	$705/128$
$w \geq 5$	$\bar{f}_2 W_v$	$705/128$.

(C169.0.3)

For $w = 4$, the low-domain rate on $d \leq 1/4$ is at least $239821/43200$; for $w \geq 5$, the low-domain rate on $d \leq 1/5$ is at least six. The corresponding high domains use the exact same-direction Q5-C084 ordinary fourth input. The common rate follows from $3581/921600 - 1/512 = 1781/921600 > 0$ and $705/128 - 2817/512 = 3/512 > 0$. The Q5-C084 all-length cost $11d$ leaves the stated $1/256$ output loss.

C169.0.1 Literal directions

Use the four genuine arm types. The root unit arm points outward, giving p. At the terminal the unit arm points inward, the two-edge arm outward, and the w-arm first inward and then outward. All inner edges point from terminal to root, and outer edges from root to receiver, giving T^L, T^M.

C169.3 The actual q-square first row and every finite propagation

For the same actual terminal $F \leq q(m + S/2)^2$, $m > 0$, put $k = 1/m$, $R = kS = r(B)u$. Then

$$\begin{aligned} TF/m^2 &\leq \bar{f}_2 + BR + \delta, \\ \delta &= \mathbb{E}R + \mathbb{E}(uR) + \sqrt{d\mathbb{E}R^2} + \mathbb{E}R^2. \end{aligned} \tag{C169.3.1}$$

Indeed $T(qR) = \mathbb{E}R + \mathbb{E}(uR) + BR + B(uR)$, with the last term at most $\sqrt{d\mathbb{E}R^2}$, and $T(qR^2/4) \leq \mathbb{E}R^2$. Choose the proved upper bound $D_0 \geq \max(\delta, 0)$, and write $D_i = \mathbb{E}B^i R$ for $i \geq 1$. Apply Lemma M.32 with $\alpha = 2\log m$ and its $D = D_0$. Thus $\gamma_j = \beta_j - D_{j-2}$, $Q_L = G_L + z^{L-1}r$, $c_L = 2D_0 - \gamma_{L+1}$, and $\mathbb{E}BQ_L(B)u = -\gamma_{L+2}$ for $L \geq 2$, after the required γ-signs have been proved.

For the complete root pair, set $n = v - 2$ and use (M.61) with $C_0 = 1$, $c_0 = c_L$:

$$\begin{aligned}\log \mathbb{E}H^4 &\le 8\log m + 4(2D_0 - \gamma_{L+1} - \gamma_{L+2}) \\ &-4\ell_\pm + (17/4)\|[Q_L - zG_n(\mp z)]u\|^2 + (10/3)\beta^2.\end{aligned} \tag{C169.3.4}$$

The positive root uses $G_n(-z)$, and the negative root $G_n(z)$. At $L = 1$, use the exact majorant $1 + u + D_0 + B(u + R)$, replacing the scalar gamma expression by $D_0 - d + D_1$. In particular D_0 is not doubled there.

C169.4 Terminal (1, 2, 3) on the full kernel domain

The actual arm sum is $f_2 + h_3 = 2 - d + B^2u$. Thus $m = 1 - d/2$, $S = B^2u$, $k \le 6/5$, and $F \le q(m + S/2)^2$. Here $R = -kxu$, $ER^2 = k^2\delta_3$, $ER = 0$, $E(uR) = -k\delta_2$. The first row gives

$$\delta = -k\delta_2 + k\sqrt{d\delta_3} + k^2\delta_3 \le d/4. \tag{C169.4.1}$$

For the full payment put $a = \delta_2/d$, $b = 1 - k\theta \ge 41/80$. Completing a square gives $\delta/d \le k[-ba + \sqrt{\theta a}] \le k\theta/(4b) \le 39/164 < 1/4$. Take $D_0 = d/4$. The signed moments are $D_1 = k\delta_2, D_2 = 0$, $D_3 = -k\delta_3, D_4 = 0$; positive classes have $D_{4j+1} = k\delta_{2j+2}$. Hence $\gamma_3 \ge 41d/80 > d/2$, $\gamma_4 = d$, and every $j \ge 5$ has $\gamma_j \ge [\kappa - (6/5)\theta^3]d = 42049d/81920 > d/2$.

Always use $h_2 f_v$ here. Put $Q_{\text{joint}} = G_L - kxz^{L-1} - zG_n(-z)$, $Q_{\text{base}} = G_L - zG_n(-z)$. For $L \ge 3$, the base coordinates are $U - xD, V - E$, with real part in $[1 - 2x, 1 - x + x^2]$ and imaginary coefficient of absolute value at most x. Thus

$$|Q_{base}|^2 \le (1 - x + x^2)^2 + x^3 \le 1 - (7/8)x. \tag{C169.4.2}$$

The final difference divided by x is $2 - 3x + x^2 - x^3 - 7/8$, decreasing, with θ-endpoint $139/32768 > 0$.

C169.4.1 Every L at least five

The perturbation modulus is at most $kx^{(L+1)/2}$, giving squared-frequency bound $1 - 7x/8 + 12x^3/5 + 36x^6/25$. Since $\theta^5 < 1/90$, by $33416370 < 33554432$, and $12\theta^2/5 + 36\theta^5/25 < 507/1280 + 2/125 < 83/200$,

$$\|Q_{joint}u\|^2 \le d - (23/50)\delta_2. \tag{C169.4.3}$$

At most one of adjacent D_{L-1}, D_L is positive, of size at most $k\delta_4$. Thus their adjacent gamma sum is at least $[2\kappa - (6/5)\theta^3]d$. Using $8\log m \le -4d$, $\ell_+ \ge 969d/1024$, the common payment gives

$$\begin{aligned}\log EH^4 &\le -\frac{209169}{20480}d + \frac{17}{4}d \\ &+ \left(\frac{10}{3} - \frac{391}{200}\right)d^2 \\ &\le -\left(\frac{11}{2} + \frac{3581}{921600}\right)d.\end{aligned} \tag{C169.4.4}$$

Use $\delta_2 \ge d^2$, $d \le 1/3$ in the last step. The coefficient $209169/20480$ is $4[1 - 1/2 + 2\kappa - (6/5)\theta^3 + 969/1024]$.

C169.4.2 Lengths two, three and four

At $L = 2$, the joint coordinates are $1 - xD, 1 - kx - E$, lying in $[0, 1]$ and $[-kx, 0]$. Thus the squared norm is at most $d + k^2\theta^2\delta_2$, and $\gamma_3 + \gamma_4 = 2d - k\delta_2$. With $\theta^2 < 1/6$,

$$\begin{aligned} \log EH^4 &\le -(2441/256)d + (1373/150)\delta_2 \\ &\le -(111679/19200)d < -(23/4)d. \end{aligned} \tag{C169.4.5}$$

The full curvature cost $10\beta^2/3$ is paid using $d^2 \le \delta_2$. At $L = 3$, the coordinates are $1-x+kx^2-xD$, $1 - E$, with square at most $1 - 3x/4$: bound it by $(1 - x + 6x^2/5)^2 + x^3$, whose difference divided by x decreases to the positive margin $667/204800$. The gamma sum is at least $(1 + \kappa)d$, yielding loss $18113/2304 > 7$. At $L = 4$, the coordinates are $1 - x - xD, 1 - x + kx^2 - E$; their square is at most $(1 - x)^2 + x^3 \le 1 - 7x/5$. With gamma sum at least $2\kappa d$, the loss is $1609/256 > 6$.

C169.4.3 The unit connector in the same canonical direction

The joint vector is [1-kx-xD-zE]u, with exact payment

$$\begin{aligned} \log EH^4 &\le 8\log m + 4D_0 - 4d + 4k\delta_2 - 4\ell_+ \\ &\quad +(17/4)\|Q_{joint}u\|^2 + (10/3)\beta^2. \end{aligned} \tag{C169.4.6}$$

At $v = 3$, $D = 0, E = 1, \beta = 0, \ell_+ = d$. The joint δ_2 coefficient $4k+(17/4)(1-2k+k^2\theta)$ decreases for $1 \le k \le 6/5$, with maximum $189/128$. The loss is at least $27/4-2457/4096 = 25191/4096 > 6$. For $v \ge 4$, combine root loss and joint norm at one frequency: $J = -4E-4(1-x)D+(17/4)[(1-kx-xD)^2 + xE^2]$. The real coordinate is positive, and J decreases in D, E. Set both to $1 - x$, then bound with $k = 1$. This gives $J \le -15/4 - 3x/4 + 13x^2 - 51x^3/4 + 17x^4/4 \le -5/2$. Its second derivative $26 - 153x/2 + 51x^2$ decreases but is positive at θ, with value $3419/1024$. Both endpoints lie below $-5/2$, the last margin being $-621887/4194304$. Using $8\log m \le -4d - d^2$ yields loss at least $1219/180 > 6$.

C169.5 Joint-frequency payments for the negative root

For $w \ge 4$, use $\bar{f}_2 W_v$. Write $G_n = E + zD$, with $(D, E) = (0, 1)$ at $n = 1$, and both in $[1 - x, 1]$ later. The root-loss integrand is $l = E + (1 + x)D$. A propagated polynomial $U + zV$ has joint square $(U + xD)^2 + x(V - E)^2$. For the stated ranges $U \le 1, V \ge 1 - x$,

$$\begin{aligned} \partial_D[-4l + (17/4)\{(U + xD)^2 + x(V - E)^2\}] < 0, \\ \partial_E[\cdots] < 0, \end{aligned} \tag{C169.5.1}$$

The first derivative is at most $(1 + x)(17x/2 - 4) < 0$, the second at most $-4 + 17x^2/2 < 0$. Thus for $n \ge 2$, take $D = E = 1 - x$; retain $n = 1$ separately.

Let $0 \le t, s \le 1$. The following joint estimates include the propagation means and root loss.

For the first-row type, $U = 1 - t, V = s$, and the actual $D_1 = \int t$ gives

$$4t - 4l + (17/4)[(1 - t + xD)^2 + x(s - E)^2] \le 2. \tag{C169.5.2}$$

At $n = 1$, $4t + 17(1 - t)^2/4 \le 17/4$, so the left side is at most $1/4 + 17x/4 < 2$. For $n \ge 2$, set $D = E = 1 - x$, then $s = 0$, and use convex endpoint values of t. The $t = 0$ value is larger by

$1/4 + 17(x - x^2)/2 > 0$. The resulting $B(x) = -15/4 + 67x/4 - 35x^2/4 - 17x^3/4 + 17x^4/4$ increases, with $B(\theta) - 3/2 = -245759/4194304 < 0$.

For the two-step type, $U = 1 - sx, V = 1 - t$, with actual mean $D_1 + D_2 = \int (t + sx)$,

$$4t + 4sx - 4l + (17/4)[(1 + x(D - s))^2 + x(1 - t - E)^2] \le 6. \tag{C169.5.3}$$

After the same coordinate derivatives, the s-derivative is $x[4 - (17/2)(1 + x(D - s))] < 0$. The expression is convex in t, and its value at one exceeds that at zero by $4 + 17x(2E - 1)/4 > 0$. At $n = 1$, the result is $17(1 + x)/4 < 6$; later it is $4 + B(x) \le 11/2$.

For the three-step type, $U = 1 - x + tx, V = 1 - sx$,

$$-4l + (17/4)[(U + xD)^2 + x(V - E)^2] \le 5/4. \tag{C169.5.4}$$

For $n \ge 2$, take $D = E = 1 - x, t = 1, s = 0$. The result $C(x) = -15/4 + 25x/2 - x^2/4 - 17x^3/4 + 17x^4/4$ increases, and $C(\theta) - 5/4 = -555007/4194304 < 0$. At $n = 1$, the square is at most $1 + x^3$, yielding the same claim.

For $L = 4$, take $U = 1 - x + sx^2, V = 1 - x + tx$; for the long base take the two coordinates of G_L, $L \ge 3$, with $U \le 1 - x + x^2, V \in [1 - x, 1]$. Then

$$-4l + (17/4)[(U + xD)^2 + x(V - E)^2] \le 1/4. \tag{C169.5.5}$$

For $n \ge 2$, the extreme real coordinate is at most one and imaginary coefficient at most x; the bound is $-15/4 + 4x + 4x^2 + 17x^3/4 \le -7/32 < 1/4$, using $x \le 1/2$. At $n = 1$, use $(1 - x + x^2)^2 + x^3 \le 1$. All derivative uses have the stated U, V scopes; those in the first two types include the displayed t terms directly.

The long-base modulus also satisfies

$$|G_L - zG_n(z)| \le 23/16, \quad L \ge 3, \tag{C169.5.6}$$

Indeed its square is at most $(1 + x)^2 + x^3$, whose endpoint is $66997/32768$, while $(23/16)^2 = 67712/32768$.

C169.6 Terminal (1, 2, 4) on its full low domain

The actual sum is $f_2 + h_4 = 2 - 2d - du + B^2u - B^3u$. Put $m = 1 - d$, $k = 1/m$, $S = -du + B^2u - B^3u$, $R = [-k(d + x) + kxz]u$. On $0 < d \le 1/4$, $k \le 4/3$, $t = k(d + x) \le 7/8$, $s = kx \le 13/24$. Put $a = \delta_2/d$, $V = ES^2 = d^3 + 2d\delta_2 + \delta_3 + \delta_4$, $J = V/d \le d^2 + (2d + \theta + \theta^2)a$. The first row gives

$$\delta = -k(d^2 + 2\delta_2) + k\sqrt{dV} + k^2V \le (5/12)d. \tag{C169.6.1}$$

Use $\sqrt{J} \le 3/8 + 2J/3$, $k + 2/3 \le 2$. Then $\delta/(kd) \le 3/8 - d + 2d^2 + a[-2 + 4d + 2\theta + 2\theta^2]$. If the coefficient of a is nonpositive, subtracting $(5/12)(1 - d)$ leaves at most $-1/24 - 7d/12 + 2d^2 < 0$. Otherwise take $a = \theta$, obtaining a convex quadratic in d, negative at both endpoints; the $d = 1/4$ value is $-75/16384$. Thus the full interval is covered; take $D_0 = 5d/12$.

The same R has signed moments

$$\begin{gathered} D_1 = k(d^2 + \delta_2),\ D_2 = k\delta_3,\ D_3 = -k(d\delta_2 + \delta_3),\ D_4 = -k\delta_4, \\ D_5 = k(d\delta_3 + \delta_4),\ D_6 = k\delta_5. \end{gathered} \tag{C169.6.2}$$

Here $D_1 \le 7d/8, D_2 < d/4$; positive D_i, $i \ge 3$, are below $d/6$. Hence $\gamma_3 \ge d/8$, $\gamma_4 \ge 3d/4$, and $\gamma_j \ge 41d/96$ for every $j \ge 5$. For $L \ge 5$, the positive adjacent-moment sum is at most $k\theta^2(d+\theta+\theta^2)d < 5d/27$, so

$$\gamma_{L+1} + \gamma_{L+2} \ge (433/432)d. \tag{C169.6.3}$$

This uses $\theta < 5/12$, $\theta^2 < 1/6$, on the entire integer tail.

Combining the common fourth payment with the four joint estimates gives the following explicit short rows.

At $L = 1$, use $8\log m \le -8d$, $4D_0 \le 5d/3$, the actual $-4d$ term, the joint cost $2d$, and curvature cost at most $5d/6$, giving loss at least $15/2$. At $L = 2$, the terms are $-8d + 10d/3 - 8d + 6d + 5d/6$, giving loss at least $35/6$. At $L = 3$, $2D_0 - 2d + \delta_2 + D_2 + D_3 \le -49d/96$, and the third joint estimate gives loss $191/24$. At $L = 4$, the adjacent gamma sum is at least $2\kappa d$, so the scalar is at most $-17d/48$; the fourth joint estimate gives loss $25/3$.

For $L \ge 5$, the polynomial is $G_L + z^{L-1}(-t + sz)$. The perturbation primitive has square at most $k^2[(d+x)^2 + x^3] \le 16309/18432 < (19/20)^2$, so its modulus is at most $19x^2/20$. The long-base bound gives additional joint square at most

$$(437/160)x^2 + (361/400)x^4,$$

After integration this is at most $1729d/3600$, using $\theta^2 < 1/6, \theta^4 < 1/36$; its norm cost is $29393d/14400$. The adjacent gamma sum, $8\log m \le -8d$ and curvature payment give loss

$$8 + 73/108 - 1/4 - 29393/14400 - 5/6 = 239821/43200 > 705/128. \tag{C169.6.4}$$

The margin above $705/128$ is $3767/86400 > 0$, completing every low-domain connector.

C169.7 Every terminal (1, 2, w), w at least five, on the low domain

On $0 < d \le 1/5$, use Lemma M.9 with $\sigma = 9/4$. Lemmas M.12 and M.10 prove the strong continuous norm and first-row error for this same actual retained arm sum. In the present $B = -K$ notation they give

$$\begin{gathered} f_2 + h_w \le 2m - U, \quad m = 1 - 9d/8,\ k = 1/m \le 40/31, \\ U = [a(x) - zb(x)]u. \end{gathered} \tag{C169.7.1}$$

Thus $F \le q(m - U/2)^2$, $R = -kU = (-t + sz)u$, with $t = ka$, $s = kb$, and the complete retained-coordinate properties are

$$\begin{gathered} a, b \ge 0, \quad a \ge \tau + (1-x)b, \\ a^2 + xb^2 \le u_0^2, \quad u_0 = 3/8 + 9d/5 \le 147/200, \\ b \le z_0 = \theta + 13d/16 \le 91/160. \end{gathered} \tag{C169.7.2}$$

Hence $0 \le t \le 147/155 < 1$, $0 \le s \le 91/124 < 1$, and the primitive modulus is at most $147/155$. At $w = 5$, the retained coefficients are $c_1 = d, c_2 = d/4 + \delta_2, c_3 = 0$. For $w \ge 6$, take $c_1 = \beta_{w-1}$, $c_2 = \lambda\beta_{w-2}$, $c_3 = \lambda\beta_{w-3}$, where $\lambda = (9d/4 - \beta_w - \beta_{w-1})/[\beta_{w-2} + (1-d)\beta_{w-3}] \in [0,1]$. Set $C = c_1 + c_2 + c_3, D = c_2 + c_3, E = c_3$. Then $a = C + x(a_f - E)$, $b = D + xb_f$, with a_f, b_f the finite coordinates of G_{w-2}. Retaining $c_1 q + c_2 h_2 + c_3 h_3$ after applying the genuine T to the positive convolution for f_{w-1} gives exact constant loss $\beta_w + c_1 + c_2 + (1-d)c_3 = 9d/4$. Its remaining

polynomial is exactly U, with the stated sign. The global δ_2 is constant. The full continuous coefficient inequality is (M.26), for this same $\sigma = 9/4$ vector.

That quadratic-error certificate gives

$$\begin{aligned}\delta = -k\int (a+b) + k\sqrt{d\int (a^2+xb^2)} \\ +k^2\int (a^2+xb^2) \le 2d/5.\end{aligned} \tag{C169.7.3}$$

Use $\sqrt{dV} \le 3d/10 + 5V/6$, $k + 5/6 \le 15/7$, and $3k/10 \le 12/31 < 2/5$. Take $D_0 = 2d/5$. The moments are $D_1 = \int t, D_2 = \int xs, D_3 = -\int xt, D_4 = -\int x^2 s$. Every later positive D_i is at most $d/6$; thus the first gamma values are positive, and $\gamma_j \ge 41d/96$ for $j \ge 5$. Also

$$\begin{aligned}D_1 + D_2 \le k(u_0 + \theta z_0)d \le 24731d/19840 < 5d/4, \\ D_2 \le 1183d/3968 < d/3.\end{aligned} \tag{C169.7.4}$$

The positive-tail bound uses $ku_0 \le 147/155$, $\theta^2 < 1/6$; the b-coordinate terms are smaller.

The actual negative root and joint vector remain unchanged.

The four short losses are respectively $9-8/5+4-2-2/3 = 131/15 > 8$, $9-16/5+8-6-2/3 = 107/15 > 7$, $357/40 > 8$, using $2D_0 - 2d + \delta_2 + D_2 + D_3 \le -221d/480$, and $83/10 > 8$, using $2D_0 - \gamma_5 - \gamma_6 \le -13d/240$. Each uses its corresponding full joint-frequency bound.

For $L \ge 5$, the long joint base costs at most $d/4$. The perturbation modulus is at most x^2, while the base modulus is at most $3/2$, so the added square is at most $3x^2 + x^4$. Its integrated norm cost is at most $323d/144$. Thus the fourth loss is at least

$$9 + 13/60 - 1/4 - 323/144 - 2/3 = 4361/720 > 6. \tag{C169.7.5}$$

Every row pays the complete βu curvature cost, at most $2d/3$ on $d \le 1/5$. This proves the rate six for every low-domain inner length.

C169.8 The same actual direction on the high domain and every outer length

The exact finite-mean proof in part R29.3.1 of Section P.5 gives the stronger $\mu_n \le 1 - 83d/32$ for every $n \ge 4$, hence in particular

$$\mu_n = Ef_n = E\bar{f}_n \le 1 - (5/2)d \quad (n \ge 4). \tag{C169.8.1}$$

It uses the exact four initial means and, for $n \ge 8$, retains $\mu_0, \dots, \mu_5$ and the actual $\beta_{n+1}, \beta_n, \beta_{n-1}$ improvements. The resulting bound is $1-\mu_n \ge \tau(7-10d+3\delta_2+d^2)+3\kappa\delta_3 \ge \kappa d(7-10d+7d^2) \ge 95d/36$, with $\delta_3 \ge d^3$.

$$\begin{aligned}1 - \mu_n \ge \tau(7 - 10d + 3\delta_2 + d^2) + 3\kappa\delta_3 \\ \ge \kappa d(7 - 10d + 7d^2) \ge 95d/36 > 5d/2.\end{aligned}$$

For $w = 4, d \ge 1/4$, $h_4 \le 2\mu_3 = 2 - 4d \le 1$. For $w \ge 5, d \ge 1/5$, $h_w \le 2\mu_{w-1} \le 2 - 5d \le 1$. Thus on each corresponding high domain, with the same actual direction,

$$0 \le p\bar{f}_2 W_v T^L(qf_2 h_w) \le p\bar{f}_2 W_v T^L(qf_2). \tag{C169.8.2}$$

The right side is exactly the Q5-C084 ordinary fourth input, with original p, root pair and L retained, giving loss $705/128$. The positive comparison only bounds the actual extra arm factor; original edge and vertex counts remain. Together with the low-domain proofs this covers every $w \geq 4, d, L$ in the same fixed orientation.

For $w = 3$, use part C169.4; otherwise use the two-domain proof. The precise rates give the uniform ordinary fourth conclusion. The Q5-C084 all-length bridge then gives

$$I_q(T^M H)^3 \leq e^{11d}(EH^4)^2 \leq e^{-d/256}. \tag{C169.8.3}$$

At $d = 0$, $u = 0$ even when $B \neq 0$, and original leaf integration makes every actual message one.

Q.6 The exceptional terminal pair (3,4) at a root (1,2,v)

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges.

C170.0 The actual terminal (3, 4)

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The genuine arms are $f_n, \bar{f}_n, h_n, W_n$. The internal residual is $(1, 2, v)$, $v \geq 3$, and the terminal residual is $(3, 4)$. Original positive L, M are independent, and all equal pairs, multiplicities and subdivisions remain.

Fix the following actual directions using only the original integers:

$$F = h_3 f_4, \qquad P_{v,L} = \begin{cases} h_2 f_v, & L = 2, \\ \bar{f}_2 W_v, & L = 1 \text{ or } L \geq 3, \end{cases} \qquad H = pP_{v,L}T^L F.$$

All inner edges point from terminal to root, and outer edges from root to receiver; the root unit arm supplies its original p.

For every listed direction and legal original host,

$$\boxed{\begin{array}{c} EH^4 \leq e^{-13d/2}, \\ I_q(T^M H)^3 \leq e^{-2d} \\ (M \geq 1), \end{array}} \tag{C170.T178.1}$$

Use $I_q(X) = E[q^{-1/3}|X|^{8/3}]$. Restore genuine decorations and reversal by the common positivity argument.

C170.3 The same terminal polynomial and finite propagation

The actual expansions are

$$h_3 = 1 - d + u - Bu + B^2 u, \quad f_4 = 1 - 2d - (1 - d)u + Bu - B^2 u + B^3 u.$$

Their sum is $2 - 3d + R$, with $R = du + B^3 u$, so

$$F \leq e^{-3d+R}. \tag{C170.T178.8}$$

The row bound gives $|R| \le d + \sqrt{\delta_3} < 1/3 + 1/4 = 7/12 < 3/5$. For $0 \le t \le 3/5$, the Taylor-tail ratio is at most $t/5$, yielding

$$\frac{e^t - 1 - t}{t^2} \le \frac{1}{2} + \frac{t}{6} + \frac{t^2}{24(1 - t/5)} \le 543/880 < 5/8.$$

For negative t, the integral remainder is at most $t^2/2$. Hence $e^R \le 1 + R + 5R^2/8$ pointwise.

Put $\sigma = ER^2$, $D_j = EB^j R$, $\delta = D_0 + 5\sigma/4$, $\epsilon = \delta + D_1$. The exact same-polynomial data are

$$\begin{gathered} \sigma = d^3 + \delta_4, \quad D_0 = \delta_2, \quad D_1 = -d^2, \\ D_2 = -\delta_3, \quad D_3 = d\delta_2, \\ \delta = \delta_2 + (5/4)(d^3 + \delta_4), \quad \epsilon = \delta - d^2. \end{gathered} \tag{C170.T178.9}$$

Using $\theta^3 < 27/400$, $d^2 \le 1/9$,

$$\begin{gathered} \sigma < (643/3600)d < (9/50)d, \\ 0 \le \epsilon \le \delta < (101/160)d, \quad \delta^2 < (2/5)d^2. \end{gathered} \tag{C170.T178.10}$$

The nonnegative ϵ follows from $\delta_2 \ge d^2$; the squared-delta comparison is $10201 < 10240$.

The first positive propagation, with kernel bounded by two, is $TF \le e^{-3d}(q + \delta + BR)$. Iterating the actual T gives

$$\begin{gathered} T^L F \le e^{-3d} Z_L, \\ Z_L = \bar{f}_L + B^L R + \delta \bar{f}_{L-1} + \sum_{j=1}^{L-1} D_j \bar{f}_{L-1-j} > 0. \end{gathered} \tag{C170.T178.11}$$

The right side dominates the actual positive message. Every step is finite original propagation.

C170.4 The unit connector

Use the standard pair at $L = 1$. The full parameters in the curvature formula are

$$\begin{gathered} Q = [d + xC + z(-x - E)]u, \quad c = \beta, \\ S = -3d + \delta - d^2 - \ell_s. \end{gathered} \tag{C170.T178.12}$$

For every $m \ge 1$, $0 \le C, E \le 1$, hence

$$\|Q\|^2 \le d^3 + (1 + 2d)\delta_2 + 3\delta_3 + \delta_4 < (7/5)d.$$

The final coefficient is at most $1/9 + 25/36 + 1/2 + 1/14 = 347/252 < 7/5$, using $\theta < 5/12$, $\theta^2 < 1/6$, $\theta^3 < 1/14$. Also $S \le -539d/160 - d^2$, $c^2 \le d^2$. Retain the common negative quadratic term to obtain

$$\begin{aligned} \log EH^4 &\le -(539/40)d - 4d^2 + (119/20)d + (10/3)d^2 \\ &\le -(301/40)d < -13d/2. \end{aligned} \tag{C170.T178.13}$$

The $v = 3, G_1$ and $v = 4, G_2$ boundaries are both included.

C170.5 Inner length two

Use the complete alternative pair. The exact second row is $Z_2 = q + \epsilon + B(\delta\mathbf{1} + u + BR)$, giving

$$\begin{aligned} Q &= [1 - xC + x^2 + z(d - E)]u, \quad c = \delta + \beta, \\ S &= -4d + \delta - d^2 - \delta_3 - \ell_a. \end{aligned} \tag{C170.T178.14}$$

For $v \geq 4$, $0 < 1 - xC + x^2 \leq 1 - x + 2x^2 \leq 1$. Since $E \geq 1 - x \geq 19/32 > d$, $|d - E| \leq 1 - d$, so $\|Q\|^2 \leq d + (1 - d)^2\delta_2 \leq 45d/32 < 9d/5$. At $v = 3$, $E = 1, C = 0$, and the exact squared norm $d + 2\delta_3 + \delta_5 + (1 - d)^2\delta_2$ is at most

$$(1 + \theta + 2\theta^2 + \theta^4)d < (1 + 5/12 + 1/3 + 1/36)d = (16/9)d < 9d/5.$$

This separately includes the genuine G1 case and keeps the actual G2 coordinates.

For every v, $0 \leq c < 261d/160$, so

$$(10/3)c^2 < (7569/2560)d < 3d,$$

Here $d^2 \leq d/3$; the final margin is $111/2560$. The root and terminal scalar bounds give $S \leq -689d/160$. Combining all terms,

$$\log EH^4 < -(689/40)d + (153/20)d + 3d = -(263/40)d < -13d/2. \tag{C170.T178.15}$$

C170.6 Inner length three

Return to the standard pair. Finite propagation gives the complete parameters

$$\begin{aligned} Q &= [1 + xC - dx + \delta + z(1 - E + x^2)]u, \quad c = \beta + \epsilon, \\ S &= -5d + \delta - d^2 - \delta_3 + d\delta_2 - \delta d - \ell_s. \end{aligned} \tag{C170.T178.16}$$

Since $\delta_2 < d$, $-d^2 + d\delta_2 \leq 0$, and the other favorable negative terms may be dropped. Hence $S \leq -859d/160$. Remove only δu to form Q_0. For every $m \geq 1$, $0 < 1 + xC - dx \leq 1 + x$, $0 \leq 1 - E + x^2 \leq x + x^2$, giving

$$\|Q_0\|^2 \leq (1 + \theta)^2(1 + \theta^3)d < 17d/8.$$

The last comparison has cross-product difference $3992344 > 0$, namely $8 \cdot 70804125 < 17 \cdot 33554432$. Retain the full square and cross term of δu:

$$\|Q\|^2 < 17d/8 + (9/5)d^2 + (2/5)d^3 \leq (997/360)d < 14d/5. \tag{C170.T178.17}$$

Indeed $2\delta(1 + \theta)d < 9d^2/5$, $\delta^2 d < 2d^3/5$; the final margin is $11/360$. Since $0 \leq c \leq d + \delta$, its curvature cost is still less than $3d$. Therefore

$$\log EH^4 < -(859/40)d + (119/10)d + 3d = -(263/40)d < -13d/2. \tag{C170.T178.18}$$

All terms associated with the positive epsilon remain paid.

C170.7 Every finite longer connector

Use the complete finite-difference proof in Lemma M.3. It proves, for all $n \ge 3, m \ge 1$,

$$|G_n(z) - zG_m(z)| \le 6/5. \tag{C170.T178.19}$$

For the actual present R, the full moment formulas are

$$D_{2k} = (-1)^k \delta_{k+2}\ (k \ge 0), \qquad D_{2k-1} = (-1)^k d\delta_k\ (k \ge 1).$$

Put $\gamma_j = \beta_j - D_{j-1}$. Then $\gamma_3 = d + \delta_3 \ge d$, $\gamma_4 = d - d\delta_2 \ge d/2$. For every $j \ge 5$, a nonzero D_{j-1} is δ_k, $k \ge 4$, or $d\delta_k$, $k \ge 3$; its absolute value is at most $\theta^3 d < d/14$, using $d \le 1/3 < \theta$. Thus $\gamma_j \ge \kappa d - d/14 = 117d/224 > d/2$ for every finite $j \ge 3$.

For $L \ge 4$, put $n = L - 1 \ge 3$ and group the exact positive convolution:

$$Z_L = S_L + B^L R + \delta \bar{f}_{L-1} + D_1 \bar{f}_{L-2} - \sum_{j=3}^{L} \gamma_j \bar{f}_{L-j}.$$

Retain $\gamma_L \bar{f}_0$, $\gamma_{L-1}\bar{f}_1$, use $\delta \ge 0$, $\bar{f}_{L-1} \le 2$, and drop the negative D_1 term and other nonnegative subtracted terms:

$$Z_L \le 1 + u + 2\delta - d + B[G_n(B)u + B^n R - (d/2)1].$$

This remains a positive upper bound; the exact vector B-mean is $-\gamma_{L+1} \le -d/2$. Adding the actual p and standard root pair gives

$$\begin{aligned} Q &= (G_n - BG_m)u + B^n R, \quad c = \beta - d/2, \\ S &\le -9d/2 + 2\delta - \ell_s \le -339d/80. \end{aligned} \tag{C170.T178.20}$$

At $L = 4$, the retained terms are exactly $\gamma_4 \bar{f}_0$ and $\gamma_3 \bar{f}_1$.

For n at least three the same-polynomial variance bound gives

$$\|B^n R\|^2 \le \theta^3 \sigma < (243/20000)d < (9/625)d.$$

Hence $\|Q\| < (6/5 + 3/25)\sqrt{d} = 33\sqrt{d}/25 < 4\sqrt{d}/3$. Also $|c| \le d/2$, with curvature cost at most $5d^2/6 \le 5d/18$. Therefore

$$\log EH^4 < -(339/20)d + (68/9)d + (5/18)d = -(547/60)d < -13d/2. \tag{C170.T178.21}$$

Together with the three short rows, this covers every positive inner length.

Q.7 Terminal triples (1,3,w) at a root (1,2,v)

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges. Here $n = v - 2$, and D_r denotes the imaginary coordinate D. The ordinary-fourth bridge of Q5-C084, Section 2, has cost $11d$ for every positive outer length. Its exact terminal $(1, 3)$ row is $\mathbb{E}[pP_{v,L}T^L(qf_3)]^4 \le e^{-705d/128}$, with precisely the same $P_{v,L}$ as below. It supplies the high-domain comparison.

C172.0 The terminal family (1, 3, w)

The construction retains the original tree, every arm, subdivision and equal-pair multiplicity, all independent positive connector lengths, the original edge count and arbitrary original label laws.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The root residual is $(1, 2, v)$, $v \geq 3$, and the terminal residual $(1, 3, w)$, $w \geq 4$. The inner and outer lengths are independent positive integers. Fix

$$F = qf_3h_w.$$

Choose the root direction by v and L:

$$P = P_{v,L} = \begin{cases} h_2 f_v, & L = 1,\ v \geq 4, \\ \bar{f}_2 W_v, & \text{all other cases,} \end{cases} \qquad H = pP\,T^L F. \tag{C172.T179.1}$$

For every listed original integer and legal kernel,

$$\boxed{\begin{array}{c} EH^4 \leq e^{-705d/128}, \\ I_q(T^M H)^3 \leq e^{-d/64}, \quad I_q(Y) = E[q^{-1/3}|Y|^{8/3}]. \end{array}} \tag{C172.T179.2}$$

The rate $705/128 = 11/2 + 1/128$ is the ordinary fourth rate of the message containing the original unit arm and complete root pair. Restore genuine equal pairs and complete reversal by the common argument.

The root unit arm points outward, giving p. The terminal unit arm points inward, its three-edge arm entirely outward, and its w-arm first inward then outward. Inner edges point from terminal to root and outer edges from root to receiver. On the entire low domain $0 < d \leq 1/4$, the more precise rates are

$$\log EH^4 \leq \begin{cases} -421d/60, & L = 1, \\ -427d/60, & L = 2, \\ -337d/60, & L \geq 3. \end{cases} \tag{C172.T179.3}$$

The common low-domain margin is $337/60 - 705/128 = 209/1920 > 0$. The same root direction on the high domain compares with the exact Q5-C084 qf_3 row.

C172.2 The actual terminal envelope and uniform low domain

Until the final high-domain comparison, take $0 < d \leq 1/4$. Set

$$m_0 = w - 3 \geq 1, \quad B_0 = \beta_{w-1}, \quad \Gamma = d + \beta_w + B_0 \geq (35/16)d,$$

$$R = -B_0 u - B^3 G_{m_0}(-B)u. \tag{C172.T179.8}$$

Retain the actual constant B_0 in the finite convolution of f_{w-1}, then apply positive T:

$$h_w \leq TS_{w-1} - B_0 q, \quad S_n = \sum_{j=0}^{n} K^j 1,\ K = -B.$$

Add the exact f_3. Cancellation of the first three original K-powers gives

$$f_3 + h_w \leq 2 - \Gamma + R. \tag{C172.T179.9}$$

At $w = 4$, $\beta_3 = \beta_4 = d$, $\Gamma = 3d$, and $R = -du - B^3u$; this is the exact short identity. Nonnegativity and AM–GM yield

$$F = qf_3h_w \le q\exp(R - \Gamma). \tag{C172.T179.10}$$

The original row and spectral bounds, retaining the G_2 cost at $m_0 = 2$, give

$$|R| \le d + (6/5)\theta\sqrt{d} \le 1/4 + 39/160 = 79/160 < 1/2.$$

Thus $e^R \le 1 + R + 3R^2/5$. The integral Taylor coefficient increases for $R \le 1/2$, and $e^{1/2} < 33/20$ makes it less than $3/5$; negative R is included. Use $q \le 2$, the original kernel bound two, and $|B(uR)| \le \sqrt{dER^2}$:

$$\begin{aligned} T(qe^R) &\le \bar{f}_2 + BR + \delta, \\ \delta &= ER + E(uR) + \sqrt{dER^2} + (12/5)ER^2. \end{aligned} \tag{C172.T179.11}$$

The actual mixed mean remains signed.

C172.2.1 The case $w = 4$

Here $ER + E(uR) = -\delta_2 - d^2$, $ER^2 = d^3 + \delta_4$. Put $v = \delta_2/d$, $J = ER^2/d \le d^2 + \theta^2 v$. Young gives $\sqrt{J} \le 1/8 + 2J$, hence

$$\delta/d \le 1/8 - d + (22/5)d^2 + v[-1 + (22/5)\theta^2].$$

The v coefficient is negative, since $\theta^2 < 1/6$. Dropping it leaves a convex quadratic in d, with endpoint values $1/8, 3/20$ on $[0, 1/4]$. Hence $\delta \le 3d/20 < 7d/20$.

C172.2.2 The case $w = 5$ on the entire low domain

Now $B_0 = d$, $ER + E(uR) = -\delta_2 - d^2 + \delta_3$, and $ER^2 = d^3 + \delta_4 + \delta_5 - 2d\delta_3$. Thus $J \le d^2 + [\theta^2(1 + \theta) - 2\theta d]v$, with a positive middle coefficient. The same Young bound gives

$$\begin{aligned} \delta/d &\le 1/8 - d + (22/5)d^2 \\ &+ v[-\kappa + (22/5)\theta^2(1 + \theta) - (44/5)\theta d]. \end{aligned} \tag{C172.T179.12}$$

For $d \le 1/8$, drop the negative vd term and $-d + 22d^2/5 \le 0$, then bound a positive v coefficient by $v \le \theta$. The result is $156575/524288 < 3/10$. On $[1/8, 1/4]$, that coefficient decreases from $-1593/81920 < 0$. Dropping it leaves the convex quadratic with endpoint values $11/160, 3/20$. Thus the whole low domain has $\delta \le 7d/20$.

C172.2.3 Every w at least six

In the original K notation let $V_0 = \sum_{j=4}^{w} K^j\mathbf{1}$, $g = -u$, so $R = B_0g - V_0$. Finite alternating moments give $EV_0 - E(gV_0) \ge \delta_2 - 2\delta_3$, $E(gV_0) \ge \kappa\delta_3$, and the finite contracting G_{w-3} gives $\|V_0\|^2 \le \delta_4$. Therefore

$$ER + E(uR) \le -(3/16)\delta_2 - \kappa d^2, \quad J \le d^2 + [\theta^2 - 2\kappa^2\theta d]v.$$

The coefficient $\theta - 2\kappa^2 d$ is positive, so $\delta_3 \le \theta\delta_2$ is used in its valid direction. The same Young inequality gives

$$\begin{aligned} \delta/d &\le 1/8 - \kappa d + (22/5)d^2 \\ &+ v[-3/16 + (22/5)\theta^2 - (44/5)\kappa^2\theta d]. \end{aligned} \tag{C172.T179.13}$$

The v coefficient stays positive for $d \le 1/4$; using $2/5 < \theta < 5/12$, $\kappa < 3/5$, a lower bound is $373/2000$. Set $v = \theta$. The resulting convex quadratic has exact endpoints

$$P(0) = 28167/81920, \qquad P(1/4) = 1795141/5242880,$$

Both are less than $7/20$, whose numerator over 5242880 is 1835008. Convexity covers the whole domain and every $w \ge 6$.

Thus use the uniform actual first-row bound

$$\boxed{D = 7d/20 \ge \max(\delta, 0)}. \tag{C172.T179.14}$$

This uniform certificate keeps the same original direction.

C172.3 Every finite signed moment and positive propagation

For i at least one, let $D_i = EB^i R$. Finite alternating sums give

$$\begin{aligned} D_{2r-1} &= (-1)^{r+1}[B_0\delta_r - H_{r+2}], \\ D_{2r} &= (-1)^{r+1}\widetilde{H}_{r+2}, \end{aligned} \tag{C172.T179.15}$$

Here $H, \widetilde{H}$ have respectively $\lfloor m_0/2 \rfloor, \lceil m_0/2 \rceil$ terms, starting at the indicated δ_{r+2}, and empty sums are zero. Hence $0 \le H, \widetilde{H} \le \delta_{r+2}$, including $m_0 = 1$. Consequently

$$\begin{aligned} D_1 \le d^2, \quad 0 \le D_2 \le \delta_3, \quad D_3 \le \delta_4, \\ D_4 \le 0, \quad (D_i)_+ \le d/24 \quad (i \ge 4). \end{aligned} \tag{C172.T179.16}$$

For odd $i \ge 5$, the two summands have opposite signs, so the positive part is at most $\max(B_0\delta_3, \delta_5)$. Since $B_0 \le d \le 1/4$, $\theta^2 < 1/6$, the first is less than $d/24$; the second is at most $\theta^4 d < d/36$. Positive even indices start with $D_6 \le \delta_5$.

Put $\gamma_j = \beta_j - D_{j-2}$. For every actual j at least three,

$$\boxed{\gamma_j \ge d/2}. \tag{C172.T179.17}$$

The precise short bounds are $\gamma_3 \ge 3d/4$, $\gamma_4 \ge (1-\theta^2)d$, $\gamma_5 \ge (\kappa-\theta^3)d > 49d/96$, and $\gamma_j \ge 53d/96$, $j \ge 6$.

Original positive propagation gives, for every L at least one,

$$\begin{aligned} T^L F &\le e^{-\Gamma} Y_L, \\ Y_L &= \bar{f}_{L+1} + B^L R + D\bar{f}_{L-1} \\ &\quad + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \end{aligned} \tag{C172.T179.18}$$

Y_L dominates a strictly positive actual message. For L at least two, the finite convolution is

$$Y_L = 1 + G_{L+1}(B)u + B^L R + D\bar{f}_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar{f}_{L+1-j}.$$

Every gamma is nonnegative. Keep the final original $\bar{f}_0$ term and use $\bar{f}_j \le 2$ for the positive D term:

$$\begin{aligned} Y_L &\le 1 + u + c_L + BQ_L(B)u, \\ c_L &= 2D - \gamma_{L+1}, \\ Q_L &= G_L + z^{L-1}r(z), \quad R = r(B)u. \end{aligned} \tag{C172.T179.19}$$

The exact mean is $EBQ_L(B)u = -\beta_{L+2} + D_L = -\gamma_{L+2}$. At $L = 1$, retain $Y_1 = 1+u+D+B(u+R)$, using D only once.

Combine with the complete root-pair envelope and the curvature payment of the original p. For L at least two this gives

$$\begin{aligned} \log EH^4 &\le -4\Gamma + 4(2D - \gamma_{L+1} - \gamma_{L+2}) \\ -4\ell_\sigma &+ (17/4)\|[Q_L - zG_n(\mp z)]u\|^2 + (10/3)\beta^2. \end{aligned} \tag{C172.T179.23}$$

The positive root uses $G_n(-z)$, the negative one $G_n(z)$; every cross term belongs to the same actual kernel. At $L = 1$, replace the scalar bracket by $D - d + D_1$ and Q_L by $1 + r$. Since $\beta \le d \le 1/4$, the curvature cost is at most $5d/6$.

C172.5 The two short-connector joint-frequency payments

Write $r = -t + zs$. At $w = 4$, $t = d, s = x$. For $w \ge 5$, write $G_{w-3}(-z) = U - zV$, with $1 - x \le U, V \le 1$, so

$$t = B_0 - x^2 V, \qquad s = xU.$$

The common range is

$$-x^2 \le t \le d \le 1/4, \qquad 0 \le s \le x. \tag{C172.T179.24}$$

The coordinate t may be negative here. The actual means are $D_1 = \int t$, $D_2 = \int xs$.

C172.5.1 The first connector, including its actual mean

For the negative root the frequency expression is

$$\begin{aligned} J_1^- &= 4t - 4[E + (1+x)D_r] \\ &+ (17/4)[(1 - t + xD_r)^2 + x(s - E)^2]. \end{aligned} \tag{C172.T179.25}$$

For $n \ge 2$, both root-coordinate derivatives are negative. The D_r-derivative is bounded by $-4(1+x) + (17/2)x(1+x+x^2)$, increasing to $-13051/65536 < 0$ at θ; the E-derivative is at most $-4 + 17\theta/2 = -35/64$. Thus take $D_r = E = 1 - x$. Since $s \le x < 1 - x$, take $s = 0$. Convexity in t, and derivative at most $4 - (17/2)(3/4) < 0$ on $[0, 1/4]$, reduce to $t = 0, -x^2$. At zero the expression is

$$\mathcal{B}(x) = -15/4 + (67/4)x - (35/4)x^2 - (17/4)x^3 + (17/4)x^4 < 3/2.$$

This increases with derivative at least $77/16$ for $x \le 1/2$, and endpoint margin $-245759/4194304$ below $3/2$. At $t = -x^2$, it is

$$-15/4 + (67/4)x - (17/4)x^2 + (17/4)x^3 \le -15/4 + 871/128 + 17/48 = 1309/384 < 7/2.$$

This drops a negative quadratic term and uses $\theta^3 < 1/12$. At $n = 1$, retain $(E, D_r) = (1, 0)$. For $t \ge 0$, the maximum at zero is $1/4 + 17x/4 < 2$; at $t = -x^2$, it is at most $1/4 + 9x^2/2 + 17x^4/4 + 17x/4 < 26/9 < 7/2$. Hence every negative-root first row costs at most $7/2$.

For the actual $L = 1, v \geq 4$ positive root, replace the real coordinate by $1 - t - xD_r$, and root loss by $E + (1-x)D_r$. The real coordinate is at least $1 - 1/4 - \theta > 0$, so both root derivatives are negative. Take $D_r = E = 1 - x$. Throughout the actual t range, $0 < 1 - t - x(1-x) \leq 1 - x + 2x^2 \leq 1$, and $4t \leq 1$. Set $s = 0$, obtaining

$$1 - 969/256 + (17/4)(1 + \theta) = 817/256 < 7/2. \tag{C172.T179.26}$$

This covers precisely the actual first-row directions.

C172.5.2 The second connector, including both actual means

Use the negative root, with $Q2 = 1 - xs + z(1-t)$. The joint expression is

$$\begin{aligned} J_2 &= 4t + 4xs - 4[E + (1+x)D_r] \\ &+ (17/4)[(1 + x(D_r - s))^2 + x(1 - t - E)^2]. \end{aligned} \tag{C172.T179.27}$$

The D_r-derivative is at most $(1+x)(17x/2-4) < 0$; the E-derivative is $-4+(17/2)x(E+t-1) < 0$. For $n \geq 2$, set $D_r = E = 1-x$; at one retain $(0, 1)$. The s-derivative is $x[4-(17/2)(1+x(D_r-s))] < 0$. Indeed even enlarging to $s \leq 1$ leaves the real coordinate at least $1 - x > 8/17$.

For nonnegative t, enlarge to $[0, 1]$, keeping the E-derivative negative. Convexity reduces to endpoints; one exceeds zero by $4 + 17x(2E - 1)/4 > 0$. At $s = 0, t = 1$, the $n = 1$ value is $17(1 + x)/4 \leq 765/128 < 6$, and later values are $4 + \mathcal{B}(x) < 11/2$.

For negative t, only the endpoint $-x^2$ remains. The real coordinate is at most $1 + x$, and the imaginary coefficient $x + x^2$ is at most one; this also bounds $n = 1$. Root loss is at least one, so

$$\begin{aligned} J_2 &\leq -4x^2 - 4 + (17/4)[(1+x)^2 + x] \\ &= 1/4 + (51/4)x + x^2/4 \leq 22409/4096 < 6. \end{aligned} \tag{C172.T179.28}$$

Thus the full second-row cost is at most six, including negative t.

C172.6 Every finite connector of length at least three

The root is negative. Write $G_L = U + zV$, with both coordinates in $[1-x, 1]$ and $U \leq 1 - x + x^2$. The joint base satisfies

$$-4[E + (1+x)D_r] + (17/4)[(U + xD_r)^2 + x(V - E)^2] \leq 1/4. \tag{C172.T179.29}$$

At $n = 1$, its square is at most $(1 - x + x^2)^2 + x^3 \leq 1$, since $2 - 3x + x^2 - x^3 > 0$. At $n \geq 2$, both root derivatives are negative. Set $D_r = E = 1 - x$; the real coordinate is at most one, imaginary coefficient has absolute value at most x, giving $-15/4 + 4x + 4x^2 + 17x^3/4 \leq -7/32 < 1/4$.

Uniformly for every finite n at least one and L at least three,

$$\begin{aligned} |G_L - zG_n(z)|^2 &\leq (1+x)^2 + x^3 \\ &\leq 66997/32768 < 67712/32768 = (23/16)^2. \end{aligned} \tag{C172.T179.30}$$

The actual r is finite. Retaining every G_{m_0} bound $6/5$ and $\sqrt{\theta} < 13/20$,

$$|r(z)| \leq d + (6/5)x^{3/2} \leq 1/4 + (6/5)\theta^{3/2} < 907/1600 < 3/5.$$

The perturbation modulus is at most $3x/5$, for every finite $L \geq 3$. Its additional square is at most $69x/40 + 9x^2/25$. After integration, the norm cost is

$$\begin{aligned} &(17/4)[(69/40)\theta + (9/25)\theta^2]d \\ &= 330837d/102400 < 13d/4. \end{aligned} \tag{C172.T179.31}$$

This one certificate covers lengths three, four and every finite longer length.

C172.7 Continuous low-domain payments and the same-direction high domain

For $L = 1$, the exact common payment is $-4\Gamma + 4D - 4d + \int J_1 + 10\beta^2/3$. Use $\Gamma \geq 35d/16$, $D = 7d/20$, $J_1 \leq 7/2$, and curvature cost $5d/6$:

$$\log EH^4 \leq [-35/4 + 7/5 - 4 + 7/2 + 5/6]d = -421d/60.$$

For $L = 2$, the adjacent gamma sum is $2d - D_1 - D_2$, so

$$\log EH^4 \leq [-35/4 + 14/5 - 8 + 6 + 5/6]d = -427d/60.$$

For every $L \geq 3$, the adjacent gamma sum is at least d, with base cost $d/4$ and additional norm cost $13d/4$:

$$\log EH^4 \leq [-35/4 + 14/5 - 4 + 1/4 + 13/4 + 5/6]d = -337d/60.$$

Every low-domain row therefore has ordinary fourth loss greater than $705/128$.

For $d \geq 1/4$, keep the same F, P. Since $w - 1 \geq 3$,

$$h_w \leq 2Ef_{w-1} \leq 2(1 - 2d) \leq 1.$$

Positivity gives $0 \leq H \leq pPT^L(qf_3)$. The root is exactly the Q5-C084 direction: positive at $L = 1$, $v \geq 4$, negative otherwise. Its same-direction fourth rate is $705/128$. At $d = 1/4$, both analytic domains include the endpoint; at $d = 0$, original leaf integration gives every message one even if $B \neq 0$.

For every independent positive outer length use the Q5-C084 ordinary fourth bridge:

$$I_q(T^M H)^3 \leq e^{11d}(EH^4)^2 \leq \exp\{[11 - 2(705/128)]d\} = e^{-d/64}.$$

The input is the actual ordinary fourth moment of the complete message.

Q.8 All terminal triples (1,a,b) with a at least four

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges. Here the root coordinates are represented by $G_{v-2} = E_0 + zD_0$. The ordinary-fourth bridge of Q5-C084, Section 2, has cost $11d$ for every positive outer length. Its terminal $(1, a)$ row, $a \geq 4$, uses the identical $P_{v,L}$ below and has rate $705/128$. This is the actual same-direction high-domain comparison.

C173.0 Fixed original object and direction

Every arm length and multiplicity, subdivision vertex, independent positive core length, original edge count and original joint label law is retained.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The root residual is $(1, 2, v)$, $v \geq 3$, and the terminal residual $(1, a, b)$, $4 \leq a < b$. Original positive L, M are independent. Always use the same actual terminal and the integer-selected root pair

$$F_{ab} = qf_a h_b, \qquad P_{v,L} = \begin{cases} h_2 f_v, & L = 1,\ v \geq 4, \\ \bar{f}_2 W_v, & \text{otherwise}, \end{cases} \qquad H = pP_{v,L}T^L F_{ab}. \tag{C173.T183.1}$$

For all listed original integers and legal kernels,

$$\boxed{\begin{aligned} &EH^4 \leq e^{-705d/128}, \\ &I_q(T^M H)^3 \leq e^{-d/64}, \\ &I_q(Y) = E[q^{-1/3}|Y|^{8/3}]. \end{aligned}} \tag{C173.T183.2}$$

Genuine equal-pair decorations and simultaneous reversal are included. The moment is the ordinary fourth moment with the actual unit arm p and complete pair P.

The four arm types have their fixed directions. Root and terminal unit arms give p, q; inner edges point terminal to root, outer edges root to receiver. All choices precede μ, K, d.

The entire low domain $0 < d \leq 1/4$ has common rate at least $349/60$, exceeding $705/128$ by $593/1920$. The high domain compares the same direction with the Q5-C084 qf_a row.

C173.2 Four actual beta constants and every terminal pair

Through the low-domain payment take $0 < d \leq 1/4$. Let $n_0 = b - a \geq 1$, and define

$$c = \beta_{b-1} - \beta_{a-1}, \quad \Gamma = \beta_a + \beta_{a-1} + \beta_b + \beta_{b-1},$$

$$R = r(B)u, \qquad r(z) = -c + (-1)^a z^a G_{n_0}(-z). \tag{C173.T183.7}$$

Retain $\beta_a f_0, \beta_{a-1} f_1$ in the finite convolution for f_a; retain $\beta_{b-1} f_0$ for f_{b-1} and apply positive T. With $K = -B$, $S_n = \sum_{j=0}^{n} K^j\mathbf{1}$,

$$f_a \leq S_a - \beta_a - \beta_{a-1}p, \qquad h_b \leq TS_{b-1} - \beta_{b-1}q,$$

$$TS_{b-1} = 1 - \beta_b - \sum_{j=1}^{b} K^j 1.$$

Adding gives the exact envelope

$$f_a + h_b \leq 2 - \Gamma + R, \qquad F_{ab} \leq qe^{R-\Gamma}. \tag{C173.T183.8}$$

AM–GM acts on the two actual nonnegative arms; their upper sum remains nonnegative, and $(1 + s/2)^2 \leq e^s$.

The beta difference is a finite decreasing alternating sum beginning at index $\lfloor a/2 \rfloor$, so

$$|c| \leq \delta_{\lfloor a/2 \rfloor}.$$

For $a = 4, 5$, $\beta_{a-1} = d$, hence $c \le 0, |c| \le \delta_2$; for $a \ge 6$, $|c| \le \delta_3$. The corresponding scalar lower bounds are

$$\Gamma \ge \begin{cases} (51/16)d, & a = 4, \\ (89/32)d, & a = 5, \\ (19/8)d, & a \ge 6. \end{cases} \tag{C173.T183.9}$$

They retain respectively $2d + 2\tau, d + 3\tau, 4\tau$, preserving the extra short-arm loss.

The original row bound gives

$$\begin{aligned} |R| &\le \theta d + (6/5)\theta^{3/2}\sqrt{d} \\ &< 13/128 + 507/3200 = 13/50 < 1/2. \end{aligned} \tag{C173.T183.10}$$

Use $\sqrt{\theta} < 13/20$. On $|t| \le 1/2$, $e^t \le 1 + t + 3t^2/5$, since its integral remainder coefficient increases in t, and $e^{1/2} \le 79/48 + 5/1728 = 2849/1728 < 33/20$. Keeping the actual mean, $q \le 2$, the kernel bound two and row Cauchy give

$$\begin{aligned} T(qe^R) &\le \bar{f}_2 + BR + \delta, \\ \delta &= ER + E(uR) + \sqrt{dER^2} + (12/5)ER^2. \end{aligned} \tag{C173.T183.11}$$

Expand $T(qR) = ER + E(uR) + BR + B(uR)$; bound only the final term by $\sqrt{dER^2}$. The quadratic remainder is at most $(12/5)ER^2$.

C173.3 Three first-row bounds on the full low domain

Write $G_{b-a}(-z) = U - zV$, with $0 \le U, V \le 1$ for every $n_0 \ge 1$, including the actual $V = 0$ at one. Write $r = A_0 + zC_0$, and let ρ bound the spectral modulus of this finite polynomial.

C173.3.1 The case $a = 4$

$$A_0 = -c + x^2U, \quad C_0 = -x^2V, \quad \rho \le \theta/4 + (6/5)\theta^2 = 767/2560 < 3/10.$$

The actual means obey $ER \le d3$ and $E(uR) \le |c|d + d3$, with $ER^2 \le \rho^2$ d. Thus

$$\begin{aligned} \delta/d &\le 2\theta^2 + \theta/4 + 3/10 + 27/125 \\ &< 1/3 + 5/48 + 3/10 + 27/125 = 1907/2000 < 1. \end{aligned} \tag{C173.T183.12}$$

Take $D = d$. Also retain the exact signs $D_1 \le 0, D_2 \le 0$.

C173.3.2 The case $a = 5$

$$A_0 = -c - x^3V, \quad C_0 = -x^2U.$$

$$\rho \le \theta/4 + (6/5)\theta^{5/2} < 5/48 + 13/100 = 281/1200 < 1/4.$$

Now $ER \le d3$ and $E(uR) \le |c|d$, whence

$$\begin{aligned} \delta/d &\le \theta^2 + \theta/4 + 1/4 + 3/20 \\ &= 273/1024 + 2/5 < 2/3. \end{aligned} \tag{C173.T183.13}$$

The final margin is $1/15360$. Take $D = 2d/3$, retaining $D_2 \le 0, D_1 \le \delta_4$.

C173.3.3 Every a at least six, both parities and every finite gap

For $a = 2h$, h at least three,

$$A_0 = -c + (-1)^h x^h U, \quad C_0 = -(-1)^h x^h V.$$

For $a = 2h + 1$, h at least three,

$$A_0 = -c + (-1)^{h+1} x^{h+1} V, \quad C_0 = (-1)^{h+1} x^h U.$$

Thus $|A_0| \le |c| + x^3$, $|C_0| \le x^3$, and

$$\rho \le \theta^2/4 + (6/5)\theta^3 = 9971/81920 < 1/8.$$

The actual mean sum is at most $\delta_3 d + 2\delta_4$. Using $\theta^2 < 1/6$, $\theta^3 < 7/100$,

$$\begin{aligned} \delta/d &\le \theta^2/4 + 2\theta^3 + 1/8 + 3/80 \\ &< 1/24 + 7/50 + 13/80 = 413/1200 < 7/20. \end{aligned} \tag{C173.T183.14}$$

Take $D = 7d/20$. This covers all integer lengths and the whole low domain with the same directions.

C173.4 Every finite signed mean and gamma sign

For i at least one define $D_i = EB^i R$. The exact signs are

$$\begin{aligned} D_{2j-1} &= (-1)^j \int x^{j-1} A_0, \\ D_{2j} &= (-1)^{j+1} \int x^j C_0. \end{aligned} \tag{C173.T183.15}$$

In particular $D_1 = -\int A_0$, $D_2 = \int x C_0$. Uniformly in $a \ge 4$, $|A_0| \le \delta_2 + x^2$, $|C_0| \le x^2$, so

$$\begin{aligned} |D_1| &\le \delta_2 d + \delta_3 \le (\theta/4 + \theta^2)d = (273/1024)d < d/2, \\ |D_2| &\le \delta_4 \le \theta^3 d < d/12, \\ |D_i| &\le \delta_2^2 + \delta_4 < d/8 \quad (i \ge 3). \end{aligned} \tag{C173.T183.16}$$

For odd $i \ge 3$, the first bound is $|c|\delta_2 + \delta_4$; for even $i \ge 4$, it starts at δ_5. All later moments decrease since $x \le \theta < 1$. Use $\theta^2/4 < 1/24$, $\theta^3 < 1/12$. Put $\gamma_j = \beta_j - D_{j-2}$; since $\beta_3 = \beta_4 = d$,

$$\begin{aligned} \gamma_3 &\ge d/2, \quad \gamma_4 \ge 11d/12, \\ &\boxed{\gamma_j \ge 15d/32 \quad (j \ge 3).} \end{aligned} \tag{C173.T183.17}$$

For $j \ge 5$, use $\kappa - 1/8 = 15/32$. Thus every discarded path term has its proved nonnegative coefficient.

Keep these stronger short-length signs:

	$4D_1$	$4(D_1 + D_2)$
$a = 4$	≤ 0	≤ 0
$a = 5$	$< d/3$	$< d/3$
$a \ge 6$	$< d/2$	$< 3d/5$

(C173.T183.18)

For $a = 4$, use $A_0 \ge 0, C_0 \le 0$. For $a = 5$, $D_1 = cd + \int x^3 V \le \delta_4$, $D_2 \le 0$, and $4\theta^3 < 1/3$. For $a \ge 6$,

$$|D_1|/d \le \theta^2/4 + \theta^3 < 67/600, \quad |D_2|/d \le \theta^4 < 1/36,$$

Hence $4|D_1| < 67d/150 < d/2$, and $4(|D_1| + |D_2|) < 251d/450 < 3d/5$, using $\theta^4 < 1/36$.

C173.5 Actual positive propagation and the full root payment

Finite induction using the actual positive T gives

$$\begin{aligned} T^L F_{ab} &\le e^{-\Gamma} Y_L, \\ Y_L &= \bar f_{L+1} + B^L R + D\bar f_{L-1} \\ &\quad + \sum_{i=1}^{L-1} D_i \bar f_{L-1-i}. \end{aligned} \tag{C173.T183.19}$$

This follows step by step from $T(B^i R) = D_i + B^{i+1} R$. The right side dominates a positive actual message. For $L \ge 2$, exact convolution gives

$$Y_L = 1 + G_{L+1}(B)u + B^L R + D\bar f_{L-1} - \sum_{j=3}^{L+1} \gamma_j \bar f_{L+1-j}.$$

Retain the final original $\bar f_0$ and use $\bar f_{L-1} \le 2$:

$$\begin{aligned} Y_L &\le 1 + u + 2D - \gamma_{L+1} + BQ_L(B)u, \\ Q_L &= G_L + z^{L-1} r, \quad E[BQ_L(B)u] = -\gamma_{L+2}. \end{aligned} \tag{C173.T183.20}$$

At $L = 1$, use exactly $Y_1 = 1 + u + D + B(u + R)$, with D once.

Use the complete root envelopes of Lemma M.28 and the actual p curvature in (M.55). For L at least two the full joint vector gives

$$\begin{aligned} \log E H^4 &\le -4\Gamma + 8D \\ &\quad -4(\gamma_{L+1} + \gamma_{L+2}) - 4\ell_- \\ &\quad +(17/4)\|[Q_L - zG_n(z)]u\|^2 + (10/3)\beta^2. \end{aligned} \tag{C173.T183.24}$$

At $L = 1$ the exact corresponding expression is

$$\begin{aligned} \log E H^4 &\le -4\Gamma + 4D - 4d + 4D_1 - 4\ell_\sigma \\ &\quad +(17/4)\|[1 + r - zG_n(\mp z)]u\|^2 + (10/3)\beta^2. \end{aligned} \tag{C173.T183.25}$$

The positive root uses $G_n(-z)$, only when $v \ge 4$; the negative-root first row occurs only at $v = 3$. At $L = 2$, retain $\gamma_3 + \gamma_4 = 2d - D_1 - D_2$. The curvature cost is at most $5d/6$, and exactly zero at $v = 3$.

C173.6 Three root bases on the entire finite-frequency interval

Temporarily omit r, and let $\ell_\sigma(x)$ be the same-frequency root-loss integrand. Each inequality below holds on $[0, \theta]$, before integration against the original ν.

C173.6.1 The first connector in its actual two root directions

At $v = 3$ the negative-root base is 1-z, with loss integrand one and beta zero. Thus

$$\begin{aligned} -4\ell_- + (17/4)|1 - z|^2 &\le 253/128 < 2, \\ |1 - z| &< 6/5. \end{aligned} \tag{C173.T183.26}$$

For $v \ge 4$, the positive-root base is $1 - zG_n(-z) = 1 - xD_0 - zE_0$. Since $0 \le 1 - xD_0$, $E_0 \le 1$, $D_0 \ge 1 - x$,

$$|1 - zG_n(-z)|^2 \le (1 - x + x^2)^2 + x \le 1.$$

The difference divided by x is $-1+3x-2x^2+x^3$, increasing with derivative $3-4x+3x^2>0$; its endpoint is $-1451/32768<0$. Also $\ell_+\ge(1-x)(2-x)\ge 969/1024$, hence

$$-4\ell_+ + (17/4)|1-zG_n(-z)|^2 \le 119/256 < 1/2. \tag{C173.T183.27}$$

Only the actual positive-root range n at least two is used.

C173.6.2 The second connector with the negative root

The base is $G_2 - zG_n = 1 + xD_0 + z(1-E_0)$. Its joint derivative in D_0 is at most $(1+x)(17x/2-4)<0$, and in E_0 at most -4. For $n\ge 2$, set both to $1-x$, obtaining

$$\begin{aligned} -15/4 + (25/2)x - x^2/4 - (17/4)x^3 + (17/4)x^4 \\ \le -15/4 + (25/2)\theta = 85/64 < 3/2. \end{aligned} \tag{C173.T183.28}$$

At $n=1$, the actual base is one and the joint expression $1/4$. Uniformly in n,

$$\begin{aligned} |G2 - zG_n|^2 \le (1+x)^2 + x^3 \\ \le 66997/32768 < (23/16)^2. \end{aligned} \tag{C173.T183.29}$$

C173.6.3 Every finite length at least three with the negative root

Write $G_L = U_L + zV_L$. Its joint base is

$$-4[E_0 + (1+x)D_0] + (17/4)[(U_L + xD_0)^2 + x(V_L - E_0)^2].$$

The D_0-derivative is negative as above, and the E_0-derivative at most $-4+17x^2/2<0$. At $n\ge 2$, set both to $1-x$, use $U_L\le 1-x+x^2$, $|V_L-(1-x)|\le x$, giving

$$-15/4 + 4x + 4x^2 + (17/4)x^3 \le -7/32 < 1/4.$$

At $n=1$, the square is at most $(1-x+x^2)^2+x^3\le 1$, because $2-3x+x^2-x^3>0$; the joint bound is again $1/4$. The norm coefficient $23/16$ above holds for every finite $n, L\ge 3$, retaining G_1, G_2.

C173.7 Three complete perturbation payments

At the same frequency use $|B+\varepsilon|^2\le |B|^2+2|B||\varepsilon|+|\varepsilon|^2$. Take $\rho=3/10, 1/4, 1/8$ for $a=4$, $5, \ge 6$. The following are costs after multiplication by $17/4$, per unit d.

The first-row base has modulus below $6/5$ for the negative root and at most one for the positive root:

	$L1$ negative root	$L1$ positive root
$a4$	$1377/400 < 7/2$	$1173/400 < 3$
$a5$	$901/320 < 3$	$153/64 < 5/2$
$a\ge 6$	$1717/1280 < 7/5$	$289/256 < 6/5$

(C173.T183.30)

The formulas are $(17/4)[(12/5)\rho+\rho^2]$ and $(17/4)[2\rho+\rho^2]$, respectively.

At $L=2$, the perturbation is zr, of modulus at most $\sqrt{\theta}\rho<(13/20)\rho$. With base bound $23/16$,

	$\lvert\epsilon\rvert$ upper bound	$(17/4)[(23/8)\lvert\epsilon\rvert+\lvert\epsilon\rvert^2]$
$a4$	$39/200$	$203541/80000 < 13/5$
$a5$	$13/80$	$53703/25600 < 21/10$
$a\ge 6$	$13/160$	$104533/102400 < 21/20$

(C173.T183.31)

The second-row rational margin is $57/25600$.

For every finite $L \geq 3$, the perturbation $z^{L-1}r$ has modulus at most ρx. After integration its full cost is

$$(17/4)[(23/8)\rho\theta + \rho^2\theta^2]d \leq \begin{cases} (635817/409600)d < (8/5)d, & a4, \\ (84201/65536)d < (13/10)d, & a5, \\ (165529/262144)d < (13/20)d, & a \geq 6. \end{cases} \tag{C173.T183.32}$$

The three comparisons are $635817 < 655360$, $842010 < 851968$, and $3310580 < 3407872$.

C173.8 All short connectors and every finite tail

Substitute the actual Γ, D, D_i, root beta and full joint base into the two common payment formulas. The table gives positive fourth losses λ, with $\log EH^4 \leq -\lambda d$, on $d \leq 1/4$.

Table 100: All terminal triples (1,a,b) with a at least four: parameter values.

Entry	$L = 1, v = 3$, negative root
$a = 4$	$29/4$; $L = 1, v \geq 4$, positive root: $101/12$; $L = 2$, all v: $469/60$; $L \geq 3$, all v: $349/60$.
$a = 5$	$57/8$; $L = 1, v \geq 4$, positive root: $199/24$; $L = 2$, all v: $361/40$; $L \geq 3$, all v: $859/120$.
$a \geq 6$	$41/5$; $L = 1, v \geq 4$, positive root: $136/15$; $L = 2$, all v: $643/60$; $L \geq 3$, all v: $523/60$.

The following expressions give every rational entry.

For $a = 4, L = 1$, $4D - 4d = 0, 4D_1 \leq 0$, and the negative root has $\beta = 0$:

$$51/4 - 2 - 7/2 = 29/4, \qquad 51/4 - 1/2 - 3 - 5/6 = 101/12.$$

At $L = 2$, $8D - 8d = 0$, $4(D_1 + D_2) \leq 0$:

$$51/4 - 3/2 - 13/5 - 5/6 = 469/60.$$

For $L \geq 3$, the adjacent gamma sum is at least $15d/16$:

$$51/4 - 8 + 15/4 - 1/4 - 8/5 - 5/6 = 349/60.$$

For $a = 5, L = 1$, $4d - 4D = 4d/3$, $4D_1 < d/3$:

$$89/8 + 4/3 - 1/3 - 2 - 3 = 57/8,$$

$$89/8 + 4/3 - 1/3 - 1/2 - 5/2 - 5/6 = 199/24.$$

At $L = 2$, retain $4(D_1 + D_2) < d/3$:

$$89/8 + 8/3 - 1/3 - 3/2 - 21/10 - 5/6 = 361/40.$$

For longer L:

$$89/8 - 16/3 + 15/4 - 1/4 - 13/10 - 5/6 = 859/120.$$

For $a \geq 6, L = 1$, $4d - 4D = 13d/5$:

$$19/2 + 13/5 - 1/2 - 2 - 7/5 = 41/5,$$

$$19/2 + 13/5 - 1/2 - 1/2 - 6/5 - 5/6 = 136/15.$$

At $L = 2$ and at longer lengths the payments are respectively

$$19/2 + 26/5 - 3/5 - 3/2 - 21/20 - 5/6 = 643/60,$$

$$19/2 - 14/5 + 15/4 - 1/4 - 13/20 - 5/6 = 523/60.$$

Thus the entire low domain and every original integer satisfy

$$\boxed{EH^4 \le e^{-349d/60} < e^{-705d/128}.} \tag{C173.T183.33}$$

The difference from $705/128$ is $593/1920$.

C173.9 The same-direction high domain and every outer length

For $1/4 \le d \le 1/3$, $b \ge 5$, so the exact original path-mean bound gives

$$0 \le h_b \le 2Ef_{b-1} \le 2(1-2d) \le 1.$$

With the same actual F and P, positivity yields

$$0 \le H \le pP_{v,L}T^L(qf_a).$$

The right side is exactly the Q5-C084 same-direction row for $a \ge 3$, giving rate $705/128$. Both domains include $d = 1/4$. At zero, leaf integration makes every message one, with no requirement that $B = 0$. The ordinary fourth bound therefore holds on the full original host domain.

For every independent positive outer length, the ordinary fourth bridge gives

$$\begin{aligned} &I_q(T^M H)^3 \le \\ &\exp\{[11 - 2(705/128)]d\} = e^{-d/64}. \end{aligned} \tag{C173.T183.34}$$

Its input is the actual ordinary fourth moment.

Q.9 All odd multipair terminals at a root (1,2,v)

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges. The three complete finite base-vector proofs used below are C173.6.1–C173.6.3 of Section Q.8; the negative-root joint payment is C169.5 of Section Q.5. The ordinary-fourth outer bridge is Q5-C084, Section 2, with cost $11d$ for every $M \ge 1$. The high-domain regrouping uses exactly the fixed-direction terminal triples proved in Sections Q.5, Q.7 and Q.8.

C177.0 Fixed original data and direction

The construction retains all original arms, equal-arm multiplicities, subdivisions, independent positive core lengths, edge counts and arbitrary original label laws.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All

means, norms, adjoints and spectral integrals use the same original probability measure. The root residual is (1, 2, v), v at least three; the terminal residual is

$$(1, a_1, b_1, \dots, a_j, b_j), \quad 1 < a_1 < b_1 < \cdots < a_j < b_j, \quad j \geq 2.$$

Original positive L, M are independent.

Set the terminal direction as follows:

$$F = \begin{cases} q f_2 h_{b_2} h_3 f_4 \prod_{i=3}^{j} h_{a_i} f_{b_i}, & (a_1, b_1, a_2) = (2, 3, 4), \\ q f_{a_1} h_{b_1} \prod_{i=2}^{j} h_{a_i} f_{b_i}, & \text{otherwise.} \end{cases} \tag{C177.T184.1}$$

An empty product is one. The same crossed rule is used at $j = 2$ and every larger pair count. The actual root unit arm gives p; fix

$$P = \begin{cases} \bar{f}_2 W_v, & a_1 = 2, \\ h_2 f_v, & a_1 \geq 3,\ L = 1,\ v \geq 4, \\ \bar{f}_2 W_v, & a_1 \geq 3; \text{ otherwise,} \end{cases} \qquad H = pP\,T^L F. \tag{C177.T184.2}$$

Inner edges point terminal to root, and outer edges root to receiver. The four arm types retain their specified literal directions.

For every listed integer, legal finite original host and genuine equal-pair decoration,

$$\boxed{\begin{array}{c} EH^4 \leq e^{-705d/128}, \\ I_q(T^M H)^3 \leq e^{-d/64}, \quad I_q(Y) = E[q^{-1/3}|Y|^{8/3}]. \end{array}} \tag{C177.T184.3}$$

The rate $705/128 = 11/2 + 1/128$ includes the actual unit root arm and complete pair.

The original finite calculus gives positive A, T with kernels in $[0, 2]$, $0 \leq d \leq 1/3$, $|u| \leq 1$, $\|B\|^2 \leq \theta$, $|BV(x)| \leq \|V\|$, $d^2 \leq \delta_2$, and $\delta_{r+1} \leq \theta\delta_r$. Consistently directed path means are at most one, and for $n \geq 3$ at most $1 - 2d$. Genuine equal pairs lie in $[0, 1]$. Delete zero masses only; then p, $q \geq \mu(x) > 0$, and constant labels give strict positivity.

The common full-vector exponential and all-length ordinary fourth interfaces are

$$\begin{array}{c} Ee^{4BV} \leq e^{4E(BV)+(17/4)\|V\|^2} \quad (V \text{any original real vector}), \\ I_q(T^M Y)^3 \leq e^{11d}(E|Y|^4)^2 \quad (M \geq 1). \end{array} \tag{C177.T184.4}$$

Use the signed mean.

For each actual pair with $4 \leq s < t$, the Q5-C096 envelope is

$$\begin{array}{c} h_s f_t \leq \exp\{R_{st} - 16d/5\}, \\ R_{st} = \sum_{r=s+1}^{t} K^r 1. \end{array} \tag{C177.T184.5}$$

Its $16/5$ loss is the finite unequal-pair payment.

C177.2 One finite zero-one mask for every later pair

Put $G_n(z) = \sum_{r=0}^{n-1} z^r$, $\beta_1 = \beta_2 = 0$, and for $n \geq 3$,

$$\beta_n = \sum_{r=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{r-1} \delta_r,$$
$$\tau = d - \delta_2 \leq \beta_n \leq d.$$

The exact finite convolution is

$$1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}. \tag{C177.T184.6}$$

The spectral measure has mass d, support [0, theta], and moments integral $x^r = \delta_{r+1}$. At $z = i$ sqrt(x),

$$\begin{aligned} \|(A_0 + zC_0)(B)u\|^2 &= \int (A_0^2 + xC_0^2), \\ E[(A_0 + zC_0)(B)u] &= -\int C_0. \end{aligned} \tag{C177.T184.7}$$

For $n \geq 2$, both coordinates of G_n lie in $[1 - x, 1]$, while $n = 1$ gives $(1, 0)$. For $n \geq 3$, the real coordinate is at most $1 - x + x^2$ and the modulus at most one. Retain $|G_2| \leq \sqrt{1 + \theta} < 6/5$.

For any finite group of later pairs, their exponent intervals are disjoint; their sum has form

$$S = -P(-B)u, \qquad P(z) = \sum_i \sum_{k=s_i}^{t_i - 1} z^k$$

Thus P is a finite zero-one mask, with minimum shorter length s. Four residue classes give

$$\begin{aligned} |P(i\sqrt{x})|^2 &\leq \frac{1 + x}{(1 - x^2)^2} \\ &\leq \frac{32768}{16245} < \frac{101}{50} < \frac{100}{49}. \end{aligned} \tag{C177.T184.8}$$

After removing the lowest z^s, its remaining mask has norm less than $10/7$, independently of pair count. Row and spectral bounds give

$$\begin{aligned} |S| &\leq (10/7)\theta^{(s-1)/2}\sqrt{d}, \\ \|S\| &\leq (10/7)\theta^{(s-2)/2}\sqrt{\delta_3}, \\ |S/u|_{\text{spectral}} &\leq (10/7)x^{s/2}. \end{aligned} \tag{C177.T184.9}$$

The final notation is a finite spectral polynomial.

For s at least five, its real and imaginary coordinates satisfy

$$\begin{aligned} |A_S| &\leq \frac{x^3}{1 - x^2}, \\ |C_S| &\leq \frac{x^2}{1 - x^2}. \end{aligned} \tag{C177.T184.10}$$

For $s \ge 6$, improve the second bound to $x^3/(1-x^2)$. In each residue class bound the positive and negative subseries separately. The positive original mean terms arise from paired powers K^7, K^8, then K^{11}, K^{12}, and so on. Hence

$$\begin{aligned} ES - E(gS) &\le 2\sum_{r\ge 0}\delta_{4+2r} \\ &\le 2C_\beta\delta_4 < d/6, \quad g=-u,\ C_\beta = 1024/855. \end{aligned} \tag{C177.T184.11}$$

The geometric tail bounds only the actual finite subseries. We shall also use the exact rational inequalities

$$\begin{gathered} \sqrt{\theta} < 16/25 < 13/20, \quad \theta^2 < 1/6, \quad \theta^3 < 7/100 < 1/12, \\ C_\beta\theta = 416/855 < 1/2, \\ C_\beta\theta^2 = 169/855 < 1/5, \\ C_\beta\theta^3 = 2197/27360 < 1/12, \\ C_\beta\theta^4 < 1/30. \end{gathered} \tag{C177.T184.12}$$

These bounds include zero frequency, every finite highest power and all pair counts.

C177.3 Actual first-row propagation and the full root payment

If $F \le qe^{R-\Gamma}$ and $|R| \le 1/2$, use $e^R \le 1 + R + 3R^2/5$, $q \le 2$ and the original kernel bound two:

$$\begin{aligned} TF &\le e^{-\Gamma}(\bar f_2 + BR + \delta), \\ \delta &= ER + E(uR) + \sqrt{dER^2} + (12/5)ER^2. \end{aligned} \tag{C177.T184.13}$$

Expand $T(qR) = ER + E(uR) + BR + B(uR)$, applying Cauchy only to the last row term. If only $|R| \le 3/4$ is available, use $e^R \le 1 + R + 2R^2/3$, replacing $12/5$ by $8/3$. The positive Taylor-tail coefficient is at most $355/544 < 2/3$, and the negative integral coefficient at most $1/2$.

Take $D \ge \max(\delta, 0)$, $D_i = EB^iR$, $\gamma_r = \beta_r - D_{r-2}$. Actual positive finite propagation gives

$$\begin{aligned} T^LF &\le e^{-\Gamma}Y_L, \\ Y_L &= \bar f_{L+1} + B^LR + D\bar f_{L-1} \\ &\quad + \sum_{i=1}^{L-1} D_i\bar f_{L-1-i}. \end{aligned} \tag{C177.T184.14}$$

After proving every gamma sign, retain the final original $\bar f_0$ term for $L \ge 2$:

$$\begin{aligned} Y_L &\le 1 + u + 2D - \gamma_{L+1} + BQ_L(B)u, \\ Q_L &= G_L + z^{L-1}r(z), \quad R = r(B)u, \\ E[BQ_L(B)u] &= -\gamma_{L+2}. \end{aligned} \tag{C177.T184.15}$$

At $L = 1$, use exactly $Y_1 = 1 + u + D + B(u + R)$. Combine the complete root envelopes of Lemma M.28 with (M.55), retaining the original $\beta\mathbf{1}$ component.

For L at least two, the complete payment is

$$\begin{aligned} \log EH^4 &\le -4\Gamma + 8D - 4(\gamma_{L+1} + \gamma_{L+2}) \\ &-4\ell_- + (17/4)\|[Q_L - zG_n(z)]u\|^2 + (10/3)\beta^2. \end{aligned} \tag{C177.T184.18}$$

All these actual nonunit rows use the negative root. At $L = 1$, replace $8D$ minus four times the gamma sum by $4D - 4d + 4D_1$, using the prescribed positive or negative root. At $L = 2$, retain $\gamma_3 + \gamma_4 = 2d - D_1 - D_2$. The curvature cost is at most $5d/96$ on $d \le 1/64$, and $5d/12$ on $d \le 1/8$, and zero at $v = 3$.

C177.4 Root bases, continuous short rectangles and all finite longer lengths

Use the full base proofs in C173.6 and C169.5. The negative first-row base has joint cost at most two and modulus below $8/5$, or $6/5$ at $n = 1$. The positive first-row base, used for $n \ge 2$, has modulus at most one and joint cost $119/256 < 1/2$. The negative second-row base has modulus below $23/16$, joint cost at most $3/2$; every longer finite base has the same modulus bound and joint cost at most $1/4$. These whole-frequency proofs retain G_1, G_2.

If $r = -t + zs$ lies in the continuous rectangle

$$-1/16 \le t \le 2/3, \qquad -1/5 \le s \le 1/2, \tag{C177.T184.19}$$

the negative-root short rows, including their actual means, satisfy

$$\begin{aligned} J_1 &= 4t - 4[E_0 + (1+x)D_0] \\ &+(17/4)[(1 - t + xD_0)^2 + x(s - E_0)^2] \le 4, \end{aligned} \tag{C177.T184.20}$$

$$\begin{aligned} J_2 &= 4t + 4xs - 4[E_0 + (1+x)D_0] \\ &+(17/4)[(1 + x(D_0 - s))^2 + x(1 - t - E_0)^2] \le 6. \end{aligned} \tag{C177.T184.21}$$

The D_0-derivative of J_1 is at most $-4(1+x)+(17/2)x(17/16+x)$, whose endpoint is $-1133/2048 < 0$. The square is dominated by $s = -1/5$. Convexity in t reduces to endpoints; the difference bracket between $2/3$ and $-1/16$ is at most $-371/192$, so the negative endpoint is larger. For $n \ge 2$, set $D_0 = 1 - x$, use $1 - x \le E_0 \le 1$ and $x - x^2 \le 1/4$, giving $J_1 \le 98373/25600 < 4$. At $n = 1$, direct substitution gives $77673/25600 < 4$. This covers the whole continuous rectangle.

The expression J_2 decreases in D_0, E_0, s; the first derivative has endpoint upper bound $-2501/5120 < 0$. For $n \ge 2$, set both root coordinates to $1 - x$, and $s = -1/5$. Convexity in t selects $2/3$, since its difference bracket is $4 + (17/4)x[29/48 + 2(E_0 - 1)] > 0$. The real coordinate is at most $4/3$, using $\theta(6/5 - \theta) = 1651/5120 < 1/3$, and the imaginary coefficient has absolute value at most $2/3$. Thus $J_2 \le 19007/3840 < 6$. At $n = 1$, the square is below $7/5$ and scalar at most $-4/3$, giving $J_2 < 277/60 < 6$. If $0 \le t, s \le 2/3$, enlarge to $[0,1]^2$: $s = 0$ dominates, the first-row $t = 0$ and second-row $t = 1$ endpoints give $J_1 \le 2, J_2 \le 6$.

For every longer L, a modulus bound ρ on r bounds the perturbation by ρx; its full added squared-norm cost is

$$\mathcal{E}(\rho)d = (17/4)[(23/8)\rho\theta + \rho^2\theta^2]d. \tag{C177.T184.22}$$

For $\rho = 1/2, 2/3, 3/5$, use respectively $8/3, 11/3, 13/4$. This controls every finite longer length.

C177.5 Case I: minimum first pair at least three

Take the first canonical pair $qf_{a_1}h_{b_1} \le qe^{R_0 - \Gamma_0}$. At $a_1 = 3$, $\Gamma_0 = d + \beta_{b_1} + \beta_{b_1 - 1} \ge 35d/16$, $R_0 = -\beta_{b_1-1}u - B^3 G_{b_1-3}(-B)u$, with mixed mean nonpositive. For $b_1 = 4, 5$, the means are $-\delta_2 - d^2$, $-\delta_2 - d^2 + \delta_3$; for $b_1 \ge 6$, at most $-3\delta_2/16 - \kappa d^2$. For $a_1 \ge 4$, the full four-beta envelope

is

$$\Gamma_0 = \beta_{a_1} + \beta_{a_1-1} + \beta_{b_1} + \beta_{b_1-1} \ge 19d/8,$$
$$r_0 = -c + (-1)^{a_1} z^{a_1} G_{b_1-a_1}(-z),$$
$$c = \beta_{b_1-1} - \beta_{a_1-1},\ |c| \le \delta_2.$$

This comes from the actual finite convolution retaining f_0, f_1 and the q term. Its mixed mean is at most $\delta_2 d + 2\delta_3 \le 7d/20$ on the present low domain.

Let S be all later pair intervals, starting at $s = a2$ at least five. The actual pair losses give

$$F \le qe^{R-\Gamma}, \quad R = R_0 + S,$$
$$\Gamma \ge (35/16 + 16/5)d = (431/80)d. \tag{C177.T184.23}$$

Each additional pair increases the scalar loss while the same single mask bounds S. At $a_1 = 3$, $|r_0| \le 1/64 + 507/1600$, while the later mask modulus is below $13/80$, giving $|r| < 99/200 < 1/2$. At $a_1 \ge 4$, $|r_0| \le \theta/64 + (6/5)\theta^2$, and the later mask starts at six, again giving modulus below $1/2$. The original row bound gives $|R| \le d + 2\sqrt{d} \le 17/64 < 1/2$. The mixed mean is at most $7d/20 + d/6 = 31d/60$, and $ER^2 \le d/4$. Therefore

$$\delta/d \le 31/60 + 1/2 + 3/5 = 97/60 < 5/3.$$

Take $D = 5d/3$. Spectral Cauchy gives $|D_i| \le \rho\theta^{(i-1)/2}d$; hence $\gamma_3 \ge d/2$, $\gamma_4 \ge 27d/40$, and later gammas at least $25d/64 > d/3$.

At $L = 1$, only $v = 3$ uses the negative root. Base cost at most two, modulus at most $6/5$, and $\rho = 1/2$ add norm cost $493d/80$; including $4D_1 \le 2d$, the total is below $21d/2$. At $L = 2$, perturbation modulus is below $13/40$, added cost below $9d/2$; with base $3d/2$, mean cost below $33d/10$, the total is at most $93d/10$. Longer lengths add $8d/3$.

C177.6 Case II: first pair (2, b1), b1 at least four

The actual first-pair convolution gives

$$\Gamma_0 = \beta_{b_1} + B_0 \ge 2\kappa d,$$
$$R_0 = B_0 g - \sum_{r=3}^{b_1} K^r 1,$$
$$B_0 = \beta_{b_1-1}.$$

The later mask starts at $a_2 \ge 5$, and $\Gamma \ge 351d/80$. The mixed mean of R_0 is at most $-2\delta_2$ at $b_1 = 4$, and $-2\kappa\delta_2$ otherwise. Retaining the finite G_2 exception gives, with $v_0 = \delta_2/d$,

$$\sqrt{ER^2/d} \le d + C\sqrt{\theta v_0},$$
$$C = 8/5\ (b1 = 4), \quad C = 5/4\ (b1 \ge 5). \tag{C177.T184.24}$$

For the first coefficient use $\sqrt{1+\theta} + (10/7)\theta^{3/2} < 6/5 + 169/448 < 8/5$. For the second use $1 + (10/7)\theta^2 < 5/4$, since $a_2 \ge 6$. The positive mixed-mean tail of S is at most $2C_\beta\theta^2\delta_2$, so

$$\delta/d \le -\lambda v_0 + d + C\sqrt{\theta v_0} + (12/5)[d + C\sqrt{\theta v_0}]^2,$$

Take $\lambda_4 = 1372/855$, $\lambda_5 = 2\kappa - 338/855$. In the variable $\sqrt{v_0}$, the quadratic and linear coefficients are positive, so the bound increases on $[0, \theta]$; it also increases in d. At $b_1 = 4$, the endpoint square root is $213/320$, and the value $221307/128000 - 4459/6840 < 11/10$, with cross-product margin 20084120. At $b_1 \geq 5$, the square root is $67/128 < 21/40$; use $\lambda_5 > 3/4, \theta > 2/5$ to get $2373/2000 - 3/10 = 1773/2000 < 9/10$. Uniformly take $D = 11d/10$.

The first-pair polynomial is $r_0 = -B_0 - xU + zxV$, with $G_{b_1-2}(-z) = U - zV$. The later mask gives $|r| < 1/64 + 39/80 + 13/80 = 213/320 < 2/3$, and $|R| \leq d + 2\sqrt{d} < 1/2$. Its coordinates satisfy

$$0 \leq t \leq 2/3, \qquad 0 \leq s \leq 2/3. \tag{C177.T184.25}$$

The lower bounds follow from $x(1-x) - C_\beta x^3 > 0$, $x(1-x) - C_\beta x^2 > 0$; upper bounds from $1/64 + C_\beta\theta < 2/3$, $\theta + C_\beta\theta^2 < 2/3$. Thus the full short-window bounds apply.

The finite moments obey $|D_1| \leq d^2 + \delta_2 + C_\beta\delta_4 < 2d/3$, $D_2 \leq \delta_3 + C_\beta\delta_4 < d/4$, and every later positive part is at most $B_0\delta_2 + \delta_3 + C_\beta\delta_5 < d/4$. Use $\theta/64 < 1/128, \theta^2 < 1/6, C_\beta\theta^4 < 1/30$. Hence $\gamma_3 \geq d/3, \gamma_4 \geq 3d/4$, and later gammas are at least $11d/32 > d/3$. The long cost is $11d/3$.

C177.7 Case III: first pair (2, 3), next shorter length at least five

The actual terminal is $qf_2h_3\prod_{i\geq 2} h_{a_i}f_{b_i}$. The exact first-pair sum is $f_2 + h_3 = 2 - d - K^3\mathbf{1}$, so $\Gamma_0 = d, R_0 = -K^3\mathbf{1}$. Later pairs give

$$\Gamma \geq 21d/5, \quad R = -K^3 1 + S, \quad S \text{ first shorter length } s = a2 \geq 5.$$

The mixed mean is at most $-(1 - 2C_\beta\theta^2)\delta_2$. For $v_0 = \delta_2/d$,

$$ER^2/d \leq \theta(617/448)^2 v_0,$$

Here $1 + (10/7)\theta^{3/2} < 617/448$. The first-row bound increases in $\sqrt{v_0}$, with positive quadratic and linear coefficients. At $v_0 = \theta$, the square root is below $9/16$, and the mean deduction exceeds $6d/25$. Thus

$$\delta/d < 9/16 + 243/320 - 6/25 = 8655/8000 < 11/10.$$

Take $D = 11d/10$, $\rho = 3/5$. The original pointwise row bound is still below $1/2$.

The finite coordinates $r = -x + A_S + zC_S$ give $0 \leq t \leq 416/855 < 2/3$, $|s| \leq 169/855 < 1/5$. Also $D_1 \leq \delta_2 + C_\beta\delta_4 < d/2$, $D_2 \leq C_\beta\delta_4 < d/12$. Across the four residue classes, later positive parts start at $C_\beta\delta_5, C_\beta\delta_5, \delta_4 + C_\beta\delta_6, C_\beta\delta_6$, then decrease by θ^2 every four indices. They are at most $\delta_4 + C_\beta\delta_5 < d/10$, using $\theta^3 < 43/640, C_\beta\theta^4 < 21/640$. Thus $\gamma_3 \geq d/2, \gamma_4 \geq 11d/12$, and later gammas at least $79d/160 \geq 15d/32$. Use short costs four and six; the long adjacent-gamma sum is at least $15d/16$, with norm increment $13d/4$.

C177.8 Cases IV and V: one fixed crossed direction

Let $e = b_2 \geq 5$. The first two pairs are always $qf_2h_eh_3f_4$; later pairs retain $h_{a_i}f_{b_i}$. The retained coefficients are those of Lemma M.9 with $\sigma = 9/4$; Lemmas M.12 and M.10 give their actual continuous frequency and error bounds, here on the smaller domain $d \leq 1/8$. At $e = 5$, take $c_1 = d$, $c_2 = d/4 + \delta_2, c_3 = 0$; otherwise use

$$c_1 = \beta_{e-1}, \quad c_2 = \lambda\beta_{e-2}, \quad c_3 = \lambda\beta_{e-3},$$
$$\lambda = \frac{9d/4 - \beta_e - \beta_{e-1}}{\beta_{e-2} + (1-d)\beta_{e-3}} \in [0, 1].$$

Each retained coefficient lies within the actual positive convolution, with positive denominator and total available constant exceeding $9d/4$. Put $C = c_1 + c_2 + c_3, D_* = c_2 + c_3, E = c_3$. Then $f_2 + h_e \le 2 - 9d/4 - U_e$, where $U_e = (A_{\text{old}} - zB_{\text{old}})u$, $A_{\text{old}} \le 3/8 + 9d/5$. For $e \ge 6$, its coordinates are $C + x(a_f - E), D_* + xb_f$; for $e = 5$, they are $5d/4 + \delta_2 + x - x^2, d/4 + \delta_2 + x$. Retain $C \ge 5d/4 + dE, D_* \ge d/4, D_* \le (17/16 + d)d, 0 \le E \le d$.

Combine this with the exact $h_3 + f_4 = 2 - 3d + K^4\mathbf{1} - dg$, yielding

$$F_{\text{cross}} \le qe^{R_c - 21d/4},$$
$$R_c = (-A_c + zB_c)u, \quad A_c = A_{old} - d, \quad B_c = B_{old} - x.$$

The second coordinate may be negative. Its complete range is

$$\begin{gathered} 0 \le A_c \le 19/40, \quad -x^2 \le B_c \le 19d/16 \le 19/128, \\ A_c + B_c \ge d/2 + x/16. \end{gathered} \tag{C177.T184.26}$$

The final inequality uses $a_f + b_f \ge 2(1 - x), 1 - 2x - E \ge 1/16$ for $e \ge 6$; substitute directly at five. Hence $ER_c + E(uR_c) = -\int (A_c + B_c) \le 0$, and $\rho_c^2 < (19/40)^2 + \theta/36 < 1/4$. The original row formula is $R_c = (C - d)g + D_* Kg + (E - 1)K^2 g - \sum_{r=5}^{e} K^r \mathbf{1}$, so

$$|R_c| \le 25/256 + (19/128 + 119/125)\sqrt{1/8} < 1/2.$$

Using $\sqrt{1/8} < 9/25$, its second term is below $158463/400000$, still giving total below $1/2$. Thus $\delta_c < d/2 + 3d/5 = 11d/10$. The moments obey $D_1 \le 19d/40, D_2 < d/16$, and later positive parts below $d/12$: $D_3 \le 0, D_4 \le \delta_5, D_5 < 19\delta_3/40, D_6 < 19d\delta_4/16$, with each residue class then decreasing. Hence $\gamma_3 \ge d/2, \gamma_4 \ge 15d/16$, and later gammas at least $49d/96$.

In IV, $j = 2$, use $\Gamma = 21d/4, D = 11d/10, \rho = 1/2$. The coordinates satisfy the full short rectangle, giving costs four and six, with long increment $8d/3$.

In V, $j \ge 3$, the later-pair mask starts at $a_3 \ge 6$:

$$R = R_c + S, \qquad \Gamma \ge (21/4 + 16/5)d = 169d/20.$$

The later mask has $|S| < 13/224 < 1/12$, so $|R| < 7/12 < 3/4$. Use the Taylor coefficient $2/3$. Its spectral modulus is less than $1/2 + (10/7)\theta^3 < 3/5$, and mixed mean at most $d/6$. Therefore

$$\delta/d \le 1/6 + 3/5 + (8/3)(9/25) = 259/150 < 7/4.$$

Take $D = 7d/4$. The minimum-six coordinate bounds give $D_1 < 19d/40 + d/12 < 2d/3$, $D_2 < d/16 + d/30 < d/3$, and later positive parts below $d/12 + d/30 < d/8$. Hence $\gamma_3 \ge d/3, \gamma_4 \ge 2d/3$, and later gammas at least $15d/32$; the long adjacent sum remains at least $15d/16$.

For V at the short rows use the complete coarse joint payment. At $L = 1$, base cost at most two and modulus below $8/5$, with $\rho = 3/5$, add at most $969d/100$; including $4D_1 < 8d/3$, the total is below $29d/2$. At $L = 2$, additional norm cost is below $11d/2$, base $3d/2$, and mean cost below $4d$, giving total at most $11d$. Longer lengths add $13d/4$. One finite mask controls every later pair count.

C177.9 Complete rational payments on every low domain

The table lists positive fourth losses. Cases I–II use $d \le 1/64$; III–V use $d \le 1/8$.

Table 101: All odd multipair terminals at a root (1,2,v): parameter values.

Entry	Lower Gamma/d
$I, a_1 \geq 3$	$431/80$; D/d: $5/3$; Curvature cost/d: $5/96$; $L = 1$ loss: $1333/160$; $L = 2$ loss: $659/96$; Every $L \geq 3$ loss: $3799/480$.
$II, a_1 = 2, b_1 \geq 4$	$351/80$; D/d: $11/10$; Curvature cost/d: $5/96$; $L = 1$ loss: $6287/480$; $L = 2$ loss: $1027/96$; Every $L \geq 3$ loss: $715/96$.
$III, a_1 = 2, b_1 = 3, a_2 \geq 5$	$21/5$; D/d: $11/10$; Curvature cost/d: $5/12$; $L = 1$ loss: $719/60$; $L = 2$ loss: $115/12$; Every $L \geq 3$ loss: $47/6$.
IV, cross, $j = 2$	$21/4$; D/d: $11/10$; Curvature cost/d: $5/12$; $L = 1$ loss: $971/60$; $L = 2$ loss: $827/60$; Every $L \geq 3$ loss: $757/60$.
V, cross, $j \geq 3$	$169/20$; D/d: $7/4$; Curvature cost/d: $5/12$; $L = 1$ loss: $953/60$; $L = 2$ loss: $983/60$; Every $L \geq 3$ loss: $589/30$.

The common full-vector payment gives every entry by the following rational arithmetic.

$$431/20 - 20/3 + 4 - 21/2 - 5/96 = 1333/160,$$

$$431/20 - 40/3 + 8 - 93/10 - 5/96 = 659/96,$$

$$431/20 - 40/3 + 8/3 - 1/4 - 8/3 - 5/96 = 3799/480.$$

For II:

$$351/20 - 22/5 + 4 - 4 - 5/96 = 6287/480,$$

$$351/20 - 44/5 + 8 - 6 - 5/96 = 1027/96,$$

$$351/20 - 44/5 + 8/3 - 1/4 - 11/3 - 5/96 = 715/96.$$

For III:

$$84/5 - 22/5 + 4 - 4 - 5/12 = 719/60, \quad 84/5 - 44/5 + 8 - 6 - 5/12 = 115/12,$$

$$84/5 - 44/5 + 15/4 - 1/4 - 13/4 - 5/12 = 47/6.$$

For IV:

$$21 - 22/5 + 4 - 4 - 5/12 = 971/60, \quad 21 - 44/5 + 8 - 6 - 5/12 = 827/60,$$

$$21 - 44/5 + 15/4 - 1/4 - 8/3 - 5/12 = 757/60.$$

For V:

$$169/5 - 7 + 4 - 29/2 - 5/12 = 953/60, \quad 169/5 - 14 + 8 - 11 - 5/12 = 983/60,$$

$$169/5 - 14 + 15/4 - 1/4 - 13/4 - 5/12 = 589/30.$$

The weakest low-domain loss is $659/96$, exceeding $705/128$ by $521/384$. Every bound is a continuous-frequency finite-mask estimate.

C177.10 Same-direction regrouping on the high domains

For every actual pair with s at least four, the original row bound gives

$$\begin{aligned} h_s f_t &\le \exp\{(39/125)\sqrt{d} - 16d/5\} \\ &\le e^{-88d/125} \le 1 \quad (d \ge 1/64). \end{aligned} \tag{C177.T184.27}$$

Its tail is at most $(6/5)\theta^{(s-1)/2}\sqrt{d} < 39\sqrt{d}/125$, and $\sqrt{d} \le 8d$ when $d \ge 1/64$.

For I and II on $d \ge 1/64$, drop only the paid later pair factors from the upper bound, obtaining $H \le pPT^L(qf_{a_1}h_{b_1})$. Case I uses C172 or C173 according as $a_1 = 3$ or at least four; case II uses C169 with its fixed negative root. Every rate is $705/128$.

For III on d at least one eighth, retain every arm direction and regroup the same factors as

$$F = (qf_2h_{a_2})(h_3f_{b_2})\prod_{i=3}^{j} h_{a_i}f_{b_i}. \tag{C177.T184.28}$$

This only changes the multiplication order at one root. Here $b_2 \ge 6$, and the actual centered-row estimate below gives $h_3f_{b_2} \le e^{-d}$; other later pair factors are at most one. Compare with $qf_2h_{a_2}$, $a_2 \ge 5$, with the same negative root, and use C169.

For IV and V on the same high domain, the original product is already

$$F = (qf_2h_{b_2})(h_3f_4)\prod_{i=3}^{j} h_{a_i}f_{b_i}. \tag{C177.T184.29}$$

The factor $h_3f_4 \le e^{-d}$, and later pairs start at six, hence are at most one. Compare with $qf_2h_{b_2}$, $b_2 \ge 5$, and use the same C169 negative-root row. The crossed orientation is identical in IV and V.

The actual centered row is as follows. At $c \ge 5$, put $B_0 = \beta_{c-1}, \Gamma_c = d+\beta_c+B_0, U = \sum_{r=3}^{c-1} K^r\mathbf{1}$, $a = B_0 - EU = \Gamma_c - 2d \ge 0$. The arm sum gives $h_3f_c \le e^{-\Gamma_c - R_0}$, $R_0 = ag - K(U-EU)$. Centered-row Cauchy and $g^2+pq=1$ give $|R_0| \le \sqrt{a^2+\operatorname{Var}U}$. For $c \ge 6$, $\operatorname{Var}U \le \theta^2 d$, $a \ge 3d/16$; when $d \ge 1/8$, this variance is less than $11d^2/8 \le d^2+2ad$. At $c=5$, retain $\operatorname{Var}U \le \theta^2(1+\theta)d$, $a \ge \kappa d$, and $8\theta^2(1+\theta) = 7605/4096 < 35/16 \le 1+2a/d$. Thus $|R_0| \le a+d$, leaving loss at least d. At four, use $h_3+f_4 = 2-3d+K^4\mathbf{1}-dg$; the centered row bounds its remainder by $\sqrt{d^2+\delta_3} \le 2d$ on the same domain. This proves $h_3f_4 \le e^{-d}$ as well.

The low and high intervals share their closed endpoints, with the same fixed directions. At zero, original leaf integration gives equality, allowing nonzero kernel. Apply the Q5-C084 all-length ordinary fourth bridge:

$$I_q(T^MH)^3 \le e^{11d}(EH^4)^2 \le \exp\{[11-2(705/128)]d\} = e^{-d/64}.$$

Q.10 Large-minimum even terminal packets at a root (1,2,v)

C180.0 Fixed directions and statement

The internal root residual is $(1,2,v)$, $v \ge 3$, and the terminal residual satisfies

$$4 \le a_1 < b_1 < \cdots < a_j < b_j, \qquad j \ge 1, \tag{C180.1}$$

The finite positive pair count j, lengths and length gaps are unbounded. The original $L, M \ge 1$ are independent. Genuine equal pairs may be attached at both roots.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. Fix the actual terminal and root pair as

$$F = \prod_{i=1}^{j} h_{a_i} f_{b_i}, \qquad P = \bar{f}_2 W_v, \qquad H = pPT^L F. \tag{C180.2}$$

The four arm types have their specified original directions. The unit root arm points outward and gives p; inner edges point from terminal to root and outer edges from root to receiver. The direction is the same for every L and host.

Set

$$\lambda_j = \frac{29}{5} + \frac{64}{5}(j-1).$$

For every listed integer, legal original kernel and genuine equal-pair decoration,

$$\boxed{EH^4 \le e^{-\lambda_j d} \le e^{-29d/5}.} \tag{C180.3}$$

This fourth moment includes the genuine original unit root arm and complete root pair.

The outer output satisfies

$$I_q(T^M H)^3 \le e^{-[2\lambda_j - c(M)]d} \le e^{-82d/95}, \qquad I_q(X) = E[q^{-1/3}|X|^{8/3}], \tag{C180.4}$$

with the all-length Q5-C164 costs

$$c(1) = 204/19, \quad c(2) = 10, \quad c(3) = 0, \qquad c(4) = 7, \quad c(M) = 6 \ (M \ge 5). \tag{C180.5}$$

For two such sides on an original five-branch path and the compatible negative zero-step central table, the four positive core lengths remain independent and

$$\boxed{Z_D \le e^{-41d/190} \le 1.} \tag{C180.6}$$

The stronger exponent is $[2\lambda_- + 2\lambda_+ - c(M_-) - c(M_+)]/8$.

C180.1 Finite calculus and the actual unequal-pair losses

Delete only zero-mass atoms. Antisymmetry gives $K(x,x) = 0$, so $p, q \ge \mu(x) > 0$. The kernels of A, T lie in $[0,2]$. Constant labels prove strict positivity of every actual tree message. Every norm, mean, spectral measure and Hölder application uses the original μ.

The finite calculus gives

$$0 \le d \le 1/3, \quad \theta = 13/32, \quad \|B\|^2 \le \theta, \qquad |BV(x)| \le \|V\|_2, \quad \delta_{r+1} \le \theta\delta_r.$$

Put $G_n(z) = \sum_{k=0}^{n-1} z^k$, $G_0 = 0$, $\beta_1 = \beta_2 = 0$, and

$$\beta_n = \sum_{r=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{r-1}\delta_r.$$

The exact finite positive convolution is

$$S_n = 1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r},$$
$$d - \delta_2 \le \beta_r \le d \quad (r \ge 3). \tag{C180.7}$$

Full reversal gives the f_n identity. The four path types lie in $[0, 2]$; $f_n, \bar{f}_n$ have means at most one. The genuine paired products $f_l h_l, \bar{f}_l W_l$ lie in $[0, 1]$.

The spectral measure of u is supported on $[0, \theta]$, has mass d, and moments $\int x^r \, d\nu = \delta_{r+1}$. At $z = i\sqrt{x}$, write $G_n = E_n + zD_n$. For $n \ge 2$ both coordinates lie in $[1 - x, 1]$, while $n = 1$ gives $(1, 0)$. For even n, $E_n = D_n$; for odd n, $E_n = 1 - xD_n$. For $n \ge 3$, $E_n \le 1 - x + x^2$, $|G_n| \le 1$, while $|G_2|^2 = 1 + x$.

For a real polynomial $Q = E + zD$, the actual mean and norm are

$$E[Q(B)u] = -\int D \, d\nu,$$
$$\|Q(B)u\|_2^2 = \int (E^2 + xD^2) d\nu. \tag{C180.8}$$

The global moments are integrated restrictions on the spectral measure.

The actual unequal-pair bound (L96.2.6) is

$$h_a f_b \le \exp\left(\sum_{r=a+1}^{b} K^r 1 - 16d/5 \right)$$
$$(4 \le a < b). \tag{C180.9}$$

Consequently one common finite vector gives

$$R = \sum_{i=1}^{j} \sum_{r=a_i+1}^{b_i} K^r 1,$$
$$\Gamma = (16j/5)d, \qquad F \le e^{R-\Gamma}. \tag{C180.10}$$

Every unequal pair retains its actual scalar loss and original arms.

C180.2 One mask for every finite pair count

The original exponent intervals are disjoint. Their finite zero-one mask, with either kernel sign, satisfies (M.4) by Lemma M.2.

Write $R = BV_0$. Its lowest original power is five, so V_0 acting on u starts at B^3, and R at B^4. Thus

$$\|V_0\|^2 \le (101/50)\theta^3 d < d/7,$$
$$\sigma := ER^2 \le (101/50)\theta^4 d < d/18. \tag{C180.12}$$

The first comparison has numerator 221897, denominator 1638400, and $1553279 < 1638400$ after multiplication by seven. The second has numerator 2884661, denominator 52428800, and $51923898 < 52428800$ after multiplication by eighteen. The row bound gives $|R|^2 < d/7 \le 1/21 < 1/16$, hence $|R| < 1/4$. For positive R, consecutive Taylor-tail ratios after the quadratic term are at most $1/12$,

so their coefficient is at most $6/11 < 5/8$. For negative R, the integral coefficient is at most $1/2$. Therefore

$$e^R \le 1 + R + (5/8)R^2. \tag{C180.13}$$

Put $D_k = EB^kR$. The exact skew moments are $EK^{2r}\mathbf{1} = (-1)^r\delta_r$, with odd moments zero. Since original powers start at five, the negative D_0 subseries starts at δ_3, its positive subseries at δ_4. The positive $D_1 = -EKR$ subseries starts at δ_3, its negative one at δ_4. For every fixed $k \ge 2$, the positive subseries starts no earlier than δ_4; same-sign moment indices differ by two. With $C_\beta = (1-\theta^2)^{-1} = 1024/855$,

$$\begin{gathered} -Cbeta\,\delta_3 \le D_0 \le Cbeta\,\delta_4, \\ -Cbeta\,\delta_4 \le D_1 \le Cbeta\,\delta_3, \\ D_k \le Cbeta\,\delta_4 \quad (k \ge 2). \end{gathered} \tag{C180.14}$$

This sign classification covers every finite k. Even when the positive-start parity changes, it starts no earlier than δ_4, with the same two-index spacing. Since $C_\beta\theta^2 = 169/855 < 1/5$, $C_\beta\theta^3 = 2197/27360 < 1/12$,

$$\begin{gathered} D_0 \in [-d/5, d/12], \quad D_1 \in [-d/12, d/5], \\ D_k \le d/12 \ (k \ge 2). \end{gathered} \tag{C180.15}$$

The positive bound for D1 is d/5.

Define

$$\delta = D_0 + (5/4)\sigma, \qquad \epsilon = \delta + D_1.$$

The same quantities satisfy

$$\begin{aligned} \delta &\le 11d/72, \quad |\delta| \le d/5, \\ \epsilon &\le 127d/360, \quad |\epsilon| \le 3d/8. \end{aligned} \tag{C180.16}$$

For example $-17d/60 \le \epsilon \le 127d/360$, both within $[-3d/8, 3d/8]$. Put $\gamma_k = \beta_k - D_{k-1}$. For every finite $k \ge 3$,

$$\gamma_k \ge (19/32 - 1/12)d = 49d/96 \ge d/2. \tag{C180.17}$$

Only the finite alternating-moment bounds have been used.

C180.3 A root envelope and three complete finite bases

Let $n = v - 2 \ge 1$, $G_n = E + zD$, $\beta_v = \int E$, $\beta_{v-1} = \int D$. Add the actual $\bar{f}_2$ and W_v, retaining their original negative terms:

$$\bar{f}_2 + W_v \le 2 - \beta_v - \beta_{v-1} - B^2G_n(B)u + \beta_{v-1}u.$$

For nonnegative x, y, $xy \le e^{x+y-2}$. Since $u \le 1$, dropping $\beta_{v-1}(u-1) \le 0$ gives the valid weaker envelope

$$P \le \exp\{-\beta_v + B[-BG_n(B)u]\}. \tag{C180.18}$$

Thus root scalar plus mean is

$$-\beta_v + E[-B^2G_n(B)u] = -\int (E + xD)d\nu \le -5d/6. \tag{C180.19}$$

For odd n, $E + xD = 1$; for even n, $E = D \ge 1 - x$, so $E + xD \ge 1 - x^2 > 5/6$. At $n = 1$, the root vector is exactly $-Bu$, with loss d. Retaining the G_2 exception gives

$$\|BG_n(B)u\|^2 \le \theta(1+\theta)d = (585/1024)d < (16/25)d. \tag{C180.20}$$

The two short bases below pay their same-frequency root loss and norm jointly.

C180.3.1 The two-step base

Let $A_n(z) = 1 - zG_n(z)$. Then

$$|A_n|^2 = (1 + xD)^2 + xE^2 \le (1 + x)^2 + x < (8/5)^2. \tag{C180.21}$$

The joint inequality is

$$\int \{-4 - 4E - 4xD + (17/4)|A_n|^2\} d\nu \le (17/16)d. \tag{C180.22}$$

At $n = 1$, the bracket is $-15/4 + 17x/4$. For $n \ge 2$, its derivative in E is $-4 + 17xE/2 < 0$, and in D is $x(9/2 + 17xD/2) \ge 0$. The coordinate rectangle therefore reduces to $E = 1 - x, D = 1$, giving

$$-15/4 + 51x/4 - 17x^2/4 + 17x^3/4.$$

This polynomial increases on $[0, \theta]$, with right endpoint $132805/131072 < 17/16$.

C180.3.2 The three-step base

Let $B_n(z) = G_2(z) - zG_n(z)$. Its squared modulus is

$$\begin{aligned} (1 + xD)^2 + x(1 - E)^2 &\le (1 + x)^2 + x^3 \\ &\le 66997/32768 < (23/16)^2. \end{aligned} \tag{C180.23}$$

The full joint estimate is

$$\int \{-8 - 4E - 4xD + (17/4)|B_n|^2\} d\nu \le -3d. \tag{C180.24}$$

At $n = 1$, the bracket is $-31/4$. For $n \ge 2$, it decreases in E and increases in D, giving the endpoint polynomial

$$-31/4 + 17x/2 + 17x^2/4 + 17x^3/4$$

This increases, with value $-433915/131072 < -3$ at θ.

C180.3.3 Every longer finite base

For $m \ge 3, n \ge 1$, write $G_m = U + zV$. The finite coordinates give $U + xD \le 1 + x^2$, $|V - E| \le x$, including $n = 1$. Hence

$$\begin{aligned} |G_m - zG_n|^2 &\le (1 + x^2)^2 + x^3 \\ &\le 1493553/1048576 < 36/25. \end{aligned} \tag{C180.25}$$

The final comparison is $37338825 < 37748736$, valid for every finite m, n.

C180.4 Actual positive propagation and the short connectors

Use the signed full-real-vector estimate

$$Ee^{4BV} \le \exp\{4EBV + (17/4)\|V\|^2\}. \tag{C180.26}$$

Its mean may have either sign; each application below retains the complete vector.

The terminal envelope, Taylor bound and $T(R^2) \le 2\sigma$ give

$$TF \le e^{-\Gamma}(1 + u + BR + \delta).$$

The bracket dominates the actual positive message. Repeated application of the same original positive operator gives

$$\begin{aligned} T^L F &\le e^{-\Gamma} Y_L, \\ Y_L &= \bar{f}_L + \delta \bar{f}_{L-1} + B^L R \\ &+ \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} > 0. \end{aligned} \tag{C180.27}$$

This is finite Duhamel induction. Combining $\log p \le -u$, $\log Y_L \le Y_L - 1$ and the root envelope gives the following actual exponential bounds.

C180.4.1 The unit connector

The full vector is $V = R - BG_n(B)u$, and root scalar plus mean gives

$$S \le -\Gamma + \epsilon - 5d/6 \le -\Gamma - 173d/360.$$

The packet and root bounds give $\|V\| \le 21\sqrt{d}/20$, so

$$\begin{aligned} \log EH^4 &\le -4\Gamma - 173d/90 + 7497d/1600 \\ &\le -[10 + 64(j-1)/5]d. \end{aligned} \tag{C180.28}$$

At $j = 1$, the exact margin over loss ten is $527/14400$.

C180.4.2 The two-step connector

Retaining the actual second, first and zeroth paths gives the full vector

$$V = A_n(B)u + BR + \delta \mathbf{1}.$$

After the jointly paid base is removed, additional scalar plus mean is $\epsilon + D_2$. Since $\sqrt{d} \le 3/5$, $d \le 1/3$,

$$\|BR\|^2 \le (13/576)d < (16/625)d, \quad |\delta| \le (3/25)\sqrt{d},$$

Thus $\|BR + \delta \mathbf{1}\| \le 7\sqrt{d}/25$. This retains the constant coordinate and all cross terms by the triangle inequality. Its norm increment costs

$$(17/4)\{2(8/5)(7/25) + (7/25)^2\}d = 10353d/2500.$$

The actual additional scalar cost is $4\epsilon + 4D_2 \le 157d/90$. Therefore

$$\begin{aligned} \log EH^4 &\le -4\Gamma + \\ &[17/16 + 157/90 + 10353/2500]d \\ &< -4\Gamma + 7d = -\lambda_j d. \end{aligned} \tag{C180.29}$$

The bracket is less than seven because $10353 \cdot 720 = 7454160 < 7547500 = 3019 \cdot 2500$, and $7 - 17/16 - 157/90 = 3019/720$. This includes the actual positive D_1 cost $d/5$.

C180.4.3 The three-step connector

The exact $\bar{f}_3 = 1 - d + u + Bu + B^2u$ gives

$$V = B_n(B)u + \delta u + B^2R + \epsilon\mathbf{1}.$$

Outside the jointly paid base, scalar plus mean is $\epsilon + D_2 + D_3 - \delta d$. The packet and $d \le 1/3$, $\sqrt{d} \le 3/5$ bound the perturbation norm by

$$[1/15 + 1/10 + 9/40]\sqrt{d} < (5/12)\sqrt{d}.$$

Here $\|B^2R\|^2 \le \theta^2 d/18 < d/100$. The base gives norm increment at most $6715d/1152 < 6d$, while additional scalar cost is at most

$$4\epsilon + 4D_2 + 4D_3 - 4\delta d \le (211/90)d < (12/5)d.$$

Thus

$$\begin{aligned} \log EH^4 &\le -4\Gamma - 3d + 6d + (12/5)d \\ &\le -[7 + 64(j-1)/5]d. \end{aligned} \tag{C180.30}$$

C180.5 Every finite length at least four

Substitution of the exact positive convolution gives

$$\begin{aligned} Y_L = S_L + \delta\bar{f}_{L-1} + D_1\bar{f}_{L-2} + B^LR \\ -\sum_{k=3}^{L}\gamma_k\bar{f}_{L-k}. \end{aligned} \tag{C180.31}$$

Both $k = L, L-1$ occur in the actual finite sum when $L \ge 4$, giving the original $\bar{f}_0, \bar{f}_1$. Retain them using the proved gamma signs, drop only other negative terms, and use $\bar{f}_n \le 2$ on positive-coefficient terms:

$$Y_L \le S_L + B^LR + 2\delta_+ + 2(D_1)_+ - (d/2)(2+u).$$

The right side dominates positive Y_L. After multiplying by the actual p and root pair, the full exponent vector is

$$V = [G_{L-1}(B) - BG_n(B)]u + B^{L-1}R - (d/2)1. \tag{C180.32}$$

The exact B-mean is $-\beta_{L+1} + D_L = -\gamma_{L+1} \le -d/2$; the extra endpoint index is covered. Combining with the root mean gives

$$\begin{aligned} S &\le -\Gamma + 2\delta_+ + 2(D_1)_+ - 3d/2 - 5d/6 \\ &\le -\Gamma - 293d/180. \end{aligned} \tag{C180.33}$$

Indeed $2\delta_+ + 2(D_1)_+ \le 11d/36 + 2d/5 = 127d/180$.

The packet gives $\|B^{L-1}R\|^2 \le \theta^3 d/18 < d/256$; the finite base has norm coefficient $6/5$. The genuine constant $(d/2)\mathbf{1}$ has norm at most $3\sqrt{d}/10$. The triangle bound retains all cross terms:

$$\|V\| \le [6/5 + 1/16 + 3/10]\sqrt{d} = (25/16)\sqrt{d}.$$

The signed full-vector estimate now gives

$$\begin{aligned} \log EH^4 &\le -4\Gamma - 293d/45 + 10625d/1024 \\ &\le -[8 + 64(j-1)/5]d. \end{aligned} \tag{C180.34}$$

For example $10625/1024 < 21/2$, while $64/5 + 293/45 - 21/2 > 8$. Every arbitrarily large finite connector is covered.

Comparing the four length ranges, the smallest uniform loss is the two-step value λ_j. The same direction works on the entire domain and every adjacent or unit-length boundary.

Q.11 All nonunit-minimum odd terminals with their actual inward anchors

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges. Also $\ell = \ell_-$, with $\ell \geq d$. The complete finite joint-base proofs, including the shortest G_2, are C180.3.1–C180.3.3 of Section Q.10. Lemma O.4 proves the centered inward-path variance and the full first-anchor comparison. Lemma O.5 proves every minimum-four finite packet used here, with its actual inward anchor.

C181.0 Parameters and fixed directions

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The internal root residual is (1, 2, v), v at least three, and the terminal residual is

$$\begin{gathered}(a, b_1, c_1, \dots, b_j, c_j),\\ 2 \leq a < b_1 < c_1 < \cdots < b_j < c_j, \quad j \geq 1.\end{gathered} \tag{C181.T190.1}$$

The finite pair count and all original arm lengths are unbounded. The inner and outer positive lengths L, M are independent. The root unit arm points outward, its two-edge arm entirely inward, and its v-arm first outward then inward, giving

$$P = \bar{f}_2 W_v.$$

The actual terminal direction is fixed by the original integers:

Table 102: All nonunit-minimum odd terminals with their actual inward anchors: parameter values.

Entry	Actual F
$b_1 = 3$, necessarily $a = 2$	$\bar{f}_2 f_3 h_{c_1} \prod_{i\geq 2} h_{b_i} f_{c_i}$.
$b_1 \geq 4$	$\bar{f}_a \prod_{i=1}^{j} h_{b_i} f_{c_i}$.

All inner edges point terminal to root, all outer edges root to receiver. These fixed choices retain every genuine equal-pair decoration E_t, E_r. The actual complete message and output are

$$H = E_r p P\, T^L(E_t F), \qquad X = T^M H. \tag{C181.T190.2}$$

For every listed original parameter and legal kernel,

$$\boxed{EH^4 \le e^{-\lambda d}, \qquad \lambda = 705/128,} \tag{C181.T190.A}$$

$$\boxed{\begin{aligned} I_q(X)^3 &\le e^{-[2\lambda - c(M)]d} \le e^{-339d/1216}, \\ I_q(X) &= E[q^{-1/3}|X|^{8/3}], \end{aligned}} \tag{C181.T190.B}$$

The Q5-C164 costs are $c(1) = 204/19, c(2) = 10, c(3) = 0, c(4) = 7, c(M) = 6$ for $M \ge 5$. Here $\lambda > 102/19$, and $2\lambda - 204/19 = 339/1216 > 0$. For two such sides with the compatible negative zero-step center and four independent original core lengths,

$$Z_D \le e^{-339d/4864}. \tag{C181.T190.C}$$

The fourth moment includes the actual p and complete original root pair.

For an actual positive comparison $T^L F \le e^{-\Gamma}[Cq + BV + c]$, apply (M.61) with $\alpha = -\Gamma, C_0 = C$, $c_0 = c, m = v - 2$, and the standard root $P = \bar{f}_2 W_v$. Its loss is $\ell_s = \int [E + (1+x)D] \ge d$; the entire vector V and the residual coefficient $\beta + C - 1$ are kept in that certificate.

C181.2.4 The full bare-root fourth bound

For every original $n \ge 3$, retain $\beta_n \bar{f}_0$ in the positive convolution: $\bar{f}_n \le q + BG_{n-1}u - \beta_n$. In the complete root formula take $\Gamma = 0, C = 1, c = -\beta_n$. For $n \ge 4$, the full finite base bound $6/5$ proved in C180.3.3 and $\beta_n + \beta_{n+1} \ge 2\kappa d$ give

$$\log E[(pP\bar{f}_n)^4] \le -[35/4 - 153/25 - 10/9]d = -(1367/900)d. \tag{C181.T190.14}$$

The curvature cost is at most $10d^2/3 \le 10d/9$ on the full domain. At $n = 3$, the two beta terms equal $2d$, and the joint variance cost is below $9d$, so the loss is at least $12 - 9 - 10/9 = 17/9 > 1367/900$. Thus the bound holds for every $n \ge 3$.

C181.3 First pair minimum at least four: the actual inward anchor

C181.3.1 One finite mask, the real anchor and its first row

Take $F = \bar{f}_a \prod h_{b_i} f_{c_i}$, $R = \sum_i R_{b_i c_i}$, and the sum Γ of actual pair losses. Their disjoint exponent intervals form one finite zero-one mask:

$$|Q(i\sqrt{x})|^2 \le \frac{1+x}{(1-x^2)^2} \le 32768/16245 < 100/49. \tag{C181.T190.15}$$

The full finite coordinate and signed-tail proofs are Lemma O.5, using the same original mask and anchor. Their data on $0 < d \le 1/3$ are

Table 103: All nonunit-minimum odd terminals with their actual inward anchors: parameter values.

Entry	Gamma/d
$j = 1, b_1 = 4$	$69/20$; $\|R\|/\sqrt{d}$: $< 1/5$; δ_+/d: $72/125$; Sum positive $D_i/d, i \ge 1$: $< 7/25$; D_1/d: $< 1/6$; $D_i/d, i \ge 2$: $< 7/100$.
$j = 1, b_1 \ge 5$	$16/5$; $\|R\|/\sqrt{d}$: $< 3/20$; δ_+/d: $3/5$; Sum positive $D_i/d, i \ge 1$: $< 1/5$; D_1/d: $< 1/30$; $D_i/d, i \ge 2$: $< 1/12$.

Entry	Gamma/d
$j \ge 2, b_1 \ge 4$	$\ge 16j/5$; $\|R\|/\sqrt{d}$: $< 1/4$; δ_+/d: $5/6$; Sum positive $D_i/d, i \ge 1$: $< 2/5$; D_1/d: $< 1/5$; $D_i/d, i \ge 2$: $< 1/12$.

The scalar delta is the actual first-row bound below. Let $\mu_a = E\bar{f}_a$, $\sigma_a^2 = \operatorname{Var} \bar{f}_a$. The full centered finite-path proof in Lemma O.4 gives $\sigma_a < 3\sqrt{d}/2$; for $a = 2, 3$, use $6\sqrt{d}/5$. A single pair starting at four has $a = 2, 3$, a continuous interval mask with $\|R\| < \sqrt{d}/5, D_0 \le 0$, and positive moments in two residue classes with total at most $(\delta_3 + \delta_4)/(1-\theta^2) < 7d/25$. At minimum at least five, the mask norm is below $3\sqrt{d}/20$, $D_0 \le d/12$, positive tail below $d/5$. Several pairs starting at four use $\sqrt{d}/4$, $d/12, 2d/5$. Every individual positive-moment bound comes from its actual four residue classes.

All these masks have $|R|^2 < d/3 \le 1/9$, permitting $e^R \le 1 + R + 3R^2/5$. Retaining the complete actual $\bar{f}_a$ in the first positive row gives

$$\begin{aligned} T(\bar{f}_a e^R) &\le \bar{f}_{a+1} + \mu_a BR + \delta, \\ \delta &= \mu_a D_0 + 2\sigma_a \|R\| + (12/5)\|R\|^2. \end{aligned} \tag{C181.T190.16}$$

Indeed $T((\bar{f}_a - \mu_a)R) \le 2E|(\bar{f}_a - \mu_a)R|$, and the second-order term is at most $12ER^2/5$. The three delta coefficients are $2(6/5)(1/5) + (12/5)(1/25) = 72/125$; $1/12 + 2(3/2)(3/20) + (12/5)(9/400) < 3/5$; and $1/12 + 2(6/5)(1/4) + (12/5)(1/16) = 5/6$. For multipair minima at least five use their stronger mask bound with its matching anchor variance.

Finite Duhamel propagation, with $n = a + L$, gives the same actual upper bound

$$\begin{aligned} e^{\Gamma} T^L F \le Y_L &= \bar{f}_n + \mu_a B^L R + \delta \bar{f}_{L-1} \\ &\quad + \mu_a \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \end{aligned} \tag{C181.T190.17}$$

The genuine anchor remains $\bar{f}_{a+L}$. Retain its actual beta term and put

$$V = G_{n-1}(B)u + \mu_a B^{L-1} R, \qquad EBV = -\beta_{n+1} + \mu_a D_L.$$

At $L = 1$, take $C = 1, c = \delta - \beta_n$. At $L \ge 2$, take $C = 1, c = 2\delta_+ + 2\mu_a \sum_{i=1}^{L-1} (D_i)_+ - \beta_n$. Every index is original and nonnegative, and the resulting upper bound remains positive.

C181.3.2 Every length, anchor and pair count

Use the complete-root formula. With $C = 1$, curvature costs at most $10d/9$. For $L \ge 2$, the joint norm coefficient is at most $6/5 + \sqrt{\theta}\rho$; at $L = 1$, at most $23/16 + \rho$, retaining the actual G_2.

For a single pair starting at four and L at least two, full scalar per d is at most

$$-69/20 + 144/125 + 14/25 - 19/16 + 7/100 - 1 = -7711/2000.$$

The joint coefficient is at most $133/100$, giving loss $7711/500 - (17/4)(133/100)^2 - 10/9 > 13/2$. At $L = 1$, the beta sum for $a = 2, 3$ is at least $51d/32$, and the joint coefficient at most $131/80$, giving loss $4(69/20 + 51/32 + 1 - 72/125 - 1/6) - (17/4)(131/80)^2 - 10/9 > 13/2$.

For a single pair with minimum at least five and $L \ge 2$, scalar is at most $-889d/240$, with norm coefficient below $13/10$, so the loss is

$$889/60 - 2873/400 - 10/9 = 23483/3600 > 13/2.$$

At $L = 1$, scalar is at most $-[16/5 + 19/16 + 1 - 3/5 - 1/30]d$, and norm coefficient at most $127/80$, again giving loss greater than $13/2$.

For $j \geq 2, L \geq 2$, scalar is at most $-\Gamma + 29d/80$, norm coefficient below $11/8$, giving

$$\lambda_{j,L\geq 2} \geq 64j/5 - 29/20 - 2057/256 - 10/9 > 15 + 64(j-2)/5.$$

At $L = 1$, scalar is at most $-\Gamma - 277d/240$, norm coefficient at most $27/16$, giving

$$\lambda_{j,L1} \geq 64j/5 + 277/60 - 12393/1024 - 10/9 > 15 + 64(j-2)/5.$$

Thus the entire first-pair-minimum-at-least-four family has the stronger full-domain rate $13/2$, for all original a, j, L.

C181.4 First pair minimum three: the genuine canonical first row

The strict order forces $a = 2$. Put $c = c_1 \geq 4, m_c = c - 3 \geq 1, B_0 = \beta_{c-1}$,

$$R_0 = B_0 g - \sum_{r=4}^{c} K^r 1 = r_0(B)u, \quad r_0(z) = -B_0 - z^3 G_{m_c}(-z),$$

$$\begin{aligned} \Gamma_0 &= d + \beta_c + B_0 \geq 35d/16, \\ f_3 + h_c &\leq 2 - \Gamma_0 + R_0, \\ \bar{f}_2 f_3 h_c &\leq \bar{f}_2 e^{R_0 - \Gamma_0}. \end{aligned} \tag{C181.T190.18}$$

At $c = 4$, $\Gamma_0 = 3d$; at five, $3d - \delta_2 \geq 83d/32$; later use $d + 2\tau \geq 35d/16$. Retain both G_1, G_2.

First take $0 < d \leq 1/8$. The row bound gives $|R_0| \leq d + (6/5)\theta\sqrt{d} < 1/3$. Write $G_{m_c}(-z) = U - zV$; at one its coordinates are $(1, 0)$, and otherwise lie in $[1-x, 1]$. Its exact frequency polynomial is

$$r_0 = -B_0 + x^2 V + zxU. \tag{C181.T190.19}$$

Retain the actual $\bar{f}_2 = 1 + u + Bu$:

$$\begin{aligned} T(\bar{f}_2 e^R) &\leq \bar{f}_3 + BR + \delta, \\ \delta &= ER + E(uR) + E((Bu)R) \\ &+ \sqrt{(d + \delta_2)ER^2} + (12/5)ER^2. \end{aligned} \tag{C181.T190.20}$$

Apply the original row bound to $B((u + Bu)R)$, using $E(u + Bu)^2 = d + \delta_2$. All three mixed means belong to this same R.

C181.4.1 Every c: the genuine BR first row and all signed moments

At $c = 4$, recompute the actual first row. Here $R_0 = dg - K^4\mathbf{1}$, $ER_0 = -\delta_2$, $ER_0^2 = \delta_4 + d^3$, and the mixed-mean sum is $-\delta_2 - d^2 + \delta_3$. Put $t = \delta_2/d$, $J = d^2 + \theta^2 t$. Young gives $\sqrt{(1+t)J} \leq (1+t)/6 + 3J/2$, hence

$$\delta_0/d \le 1/6 - d + (39/10)d^2$$
$$+t[1/6 - \kappa + (39/10)\theta^2]$$
$$< 1/6 + (5/12)(49/60) = 73/144 < 3/5.$$

Drop only $-d + 39d^2/10 \le 0$; use $t \le \theta < 5/12, \theta^2 < 1/6$, and the negative $-\kappa t$. This is the genuine BR first row throughout $d \le 1/8$. At $c = 5$, the mixed mean is $-\delta_2 - d^2 + 2\delta_3$, and variance $d^3 + \delta_4 + \delta_5 - 2d\delta_3$. The same Young bound gives

$$\delta_0/d \le 1/6 - d + (39/10)d^2 + t[-1/48 + (39/10)\theta^2(1+\theta) - (39/5)\theta d].$$

The t coefficient is positive; taking $t = \theta$ leaves a decreasing function of d, whose value at zero is $5514615/10485760 < 17/32$. For $c \ge 6$, finite alternating intervals give mixed mean at most $-\kappa\delta_2 + (1+\kappa)\delta_3 - B_0 d$, and variance at most $d^3 + (\theta - 2\kappa^2 d)\delta_3$. Young gives

$$\delta_0/d \le 1/8 - \kappa d + (22/5)d^2$$
$$+t[1/8 + 55/1024 + (22/5)\theta^2 - (44/5)\kappa^2\theta d].$$

The t coefficient is positive; taking θ gives a decreasing function of d, with zero endpoint $80709/163840 < 1/2$. Thus uniformly for all $c \ge 4$,

$$D = 3d/5 \ge \max(\delta_0, 0). \tag{C181.T190.21}$$

The finite spectral means directly give

$$\begin{aligned} -\delta_3 \le D_1 \le d^2, \quad 0 \le D_2 \le \delta_3, \quad D_3 \le \delta_4, \\ D_i \le d/36 \ (i \ge 4), \qquad D_i = EB^i R_0. \end{aligned} \tag{C181.T190.22}$$

The four residue classes are explicit: the negative interval has positive terms only for indices congruent to two and three, starting at δ_3, δ_4; the $B_0 g$ term is positive in class one, starting at $B_0 d$. For every $i \ge 4$, the positive part is at most $\max(\delta_5, B_0\delta_3) \le \theta^4 d < d/36$, using $d \le 1/8 < \theta^2$. At four, the exact initial terms are $D_0 = -\delta_2, D_1 = d^2, D_2 = \delta_3, D_3 = -d\delta_2, D_4 = -\delta_4$.

Actual finite positive propagation yields

$$\begin{aligned} e^{\Gamma_0} T^L(\bar{f}_2 f_3 h_c) \le \\ Y_L = \bar{f}_{L+2} + B^L R_0 + D\bar{f}_{L-1} \\ + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \ge 0. \end{aligned} \tag{C181.T190.23}$$

C181.4.2 The unit connector with its full joint spectrum

Here $Y_1 = q + B(G_2 + r_0)u + D - d$. In the full root formula, take $C = 1, c = D - d$, $V = (G_2 + r_0)u$. The scalar is at most

$$-\Gamma_0 + D + D_1 - 3d \le -(35/16 + 91/40)d.$$

The real coordinate of $G_2 + r_0 - zG_m$ is positive and at most $1 + x + x^2$; its imaginary coefficient lies in $[0, 2x]$. Hence

$$|G_2 + r_0 - zG_m|^2 \le (1 + x + x^2)^2 + 4x^3 \le 2870097/1048576 < 25/9.$$

Curvature costs at most $5d/12$; the fourth loss is at least

$$35/4 + 91/10 - 425/36 - 5/12 = 1013/180 > 705/128. \tag{C181.T190.24}$$

C181.4.3 The second connector and the exact fourth path

The exact $\bar{f}_4 = (1-d)q + BG_3u - d$ gives

$$Y_2 = (1 - d + D)q + B[G_3 + zr_0]u + D_1 - d.$$

The actual mean is $EBV = -\tau + D_2$, and full scalar at most $-\Gamma_0 + D + D_1 + D_2 - 2d - \tau \le -1867d/480$. The real joint coordinate lies in $(0, 1]$, imaginary coefficient in $[-B_0, x+x^2]$; its square is at most $1 + \theta/64 + \theta^3(1+\theta)^2 < 36/25$. The actual curvature coefficient $\alpha = \beta + D - d$ has absolute value at most $3d/5$, costing $3d/20$. The loss is at least

$$1867/120 - 153/25 - 3/20 = 5573/600 > 705/128. \tag{C181.T190.25}$$

C181.4.4 The third connector and the exact fifth path

The identity $\bar{f}_5 = (1-2d)q + B(G_4 - d)u - \tau$ gives

$$Y_3 = (1 - 2d + D + D_1)q + B[G_4 - d + z^2r_0 + D]u - \tau + D_2.$$

The exact mean is $-\tau + d^2 + D_3 - Dd$, and the full scalar is at most

$$-\Gamma_0 - 3d - 2\tau + D + D_1 + D_2 + D_3 + d^2 - Dd \le -211d/40.$$

Put $Q_3 = G_4 + z^2r_0$. Its joint real coordinate lies in $[1-x-x^3, 1+B_0x]$, and imaginary coefficient in $[-x-x^2, 0]$. Subtracting $d - D = 2d/5$ leaves the real coordinate positive, so the joint square is at most $(1+\theta/8)^2 + \theta^3(1+\theta)^2 < 25/16$. The actual $\alpha = \beta - 7d/5 + D_1$ is nonpositive with absolute value at most $47d/30$, costing less than $21d/20$. The loss is at least $211/10 - 425/64 - 21/20 = 4291/320 > 13$.

C181.4.5 Every finite longer length with the actual D retained

In the same positive convolution define

$$\gamma_3 = d - D = 2d/5, \qquad \gamma_j = \beta_j - D_{j-3} \quad (j \ge 4).$$

The signed moments give $\gamma_4 \ge 7d/8$, $\gamma_5 \ge \tau - \delta_3 > 2d/5$, $\gamma_6 \ge \tau - \delta_4 > 2d/5$, and later $\gamma_j \ge \tau - d/36 > 2d/5$. Thus all $j \ge 3$, including $L+3$, satisfy $\gamma_j \ge h = 2d/5$. Retain the final four original messages $\bar{f}_0, \bar{f}_1, \bar{f}_2, \bar{f}_3$:

$$Y_L \le Cq + BV - h(1-d), \quad C = 1 - 3h,$$
$$V = [G_{L+1} + z^{L-1}r_0 - h(2+z)](B)u, \quad EBV = -\gamma_{L+3} + 2hd \le -h + 2hd.$$

The complete scalar is at most $-\Gamma_0 - 5h + 3hd - \ell \le -403d/80$. Since $|r_0| < 9/20$, the joint norm coefficient is less than $6/5 + (9/20)\theta^{3/2} + h\sqrt{4+\theta} < 3/2$, using $\theta^{3/2} < 4/15, h \le 1/20$, $\sqrt{4+\theta} < 21/10$. The actual $|\alpha| = |\beta - 3h| \le 6d/5$ costs $3d/5$. The loss is at least

$$403/20 - 153/16 - 3/5 = 799/80 > 705/128. \tag{C181.T190.26}$$

C181.5 First pair minimum three with every later-pair count

Still take $d \le 1/8$. Let $S = \sum_{i\ge 2} R_{b_i c_i}$, whose first shorter length is at least $c_1 + 1 \ge 5$. The actual terminal obeys $F \le \bar{f}_2 e^{R-\Gamma}$, where

$$\begin{aligned} R = R_0 + S, \quad \Gamma = \Gamma_0 + (16/5)(j-1)d \\ \ge [431/80 + (16/5)(j-2)]d. \end{aligned} \tag{C181.T190.27}$$

One finite mask covers all later intervals, independently of pair count. The row and four-residue-class bounds give

$$\begin{aligned} &\|S\| < (3/20)\sqrt{d}, \quad |S| < (1/4)\sqrt{d}, \quad ES + E(uS) \le d/6, \\ &D_1(S) < d/30, \quad D_i(S) < d/12 \quad (i \ge 2). \end{aligned} \tag{C181.T190.28}$$

The positive mixed-mean contributions start at original powers K^7, K^8, then K^{11}, K^{12}; each contributes at most δ_4, so $ES + E(uS) \le 2\delta_4/(1-\theta^2) < d/6$. Also $|D_1(S)| \le \sqrt{d}\|S\| < 3d/20$.

We have $\|R_0\| < 9\sqrt{d}/20$, $\|R\| < 3\sqrt{d}/5$, and $|R| \le d + [(6/5)\theta + 1/4]\sqrt{d} < 1/2$. Adding S to the genuine first row costs, per unit d, at most

$$1/6 + 39/400 + 9/50 + 189/500 < 9/10.$$

These four terms pay the mixed mean, $E(BuS)$, square-root increment, and quadratic variance increment. The total is below $9/10$, so take $D = 3d/2$. Every D_i now belongs to the same new R; its actual bounds are

$$\begin{aligned} D_1 \le 19d/120, \quad D_2 \le d/4, \quad D_3 \le d/6, \\ D_i \le d/9 \ (i \ge 4), \quad D_1 \ge -19d/60. \end{aligned} \tag{C181.T190.29}$$

The finite propagation remains the displayed actual first-row propagation with these new R, Gamma, D.

At $L = 1$, joint norm coefficient is below $5/3 + 3/20 < 11/6$, curvature cost at most $5d/12$, scalar at most $-\Gamma - 161d/120$. At the weakest pair count $j = 2$, loss is at least

$$431/20 + 161/30 - 2057/144 - 5/12 = 1759/144 > 12.$$

At $L = 2$, joint coefficient is below $6/5 + (3/20)\sqrt{\theta} < 13/10$, curvature cost at most $15d/16$, and scalar at most $-\Gamma - 329d/480$, giving loss $9703/600 > 16$.

At $L = 3$, add $(D-d)u = (d/2)u$ and $B^2 S$. The joint norm coefficient is below $5/4 + 1/16 + 1/16 < 3/2$. The actual $\alpha = \beta - d/2 + D_1$ has absolute value at most $5d/6$, costing below $d/3$. Scalar is at most $-\Gamma - 159d/80$, giving loss $941/48 > 19$.

For $L \ge 4$, retain the negative $\gamma_3 = d - D = -d/2$. All later $\gamma_j = \beta_j - D_{j-3}$ are at least $h = d/3$. Pay the actual positive remainder $(D-d)\bar{f}_{L-1} \le d$, then retain only the final original $\bar{f}_0, \bar{f}_1, \bar{f}_2$:

$$Y_L \le Cq + BV + d - h, \quad C = 1 - 2h, \quad V = [G_{L+1} + z^{L-1} r - h](B)u.$$

The actual mean is $EBV = -\gamma_{L+3} + hd \le -h + hd$, and complete scalar at most $-\Gamma - 31d/24$. Joint norm coefficient is below $6/5 + (3/5)\theta^{3/2} + h < 3/2$. The actual $|\alpha| = |\beta - 2h| \le 2d/3$ costs at most $5d/27$. At $j = 2$, the loss is $431/20 + 31/6 - 153/16 - 5/27 = 36653/2160 > 16$. Each additional pair increases Γ, so the whole low-domain family has loss at least $12 + 64(j-2)/5$.

C181.6 The same direction on the full high domain

For $1/8 \le d \le 1/3$, keep the original direction. Put $U = \sum_{r=3}^{c-1} K^r \mathbf{1}$, so $R_0 = B_0 g - KU$, $EU = d - \beta_c$. Let $t_0 = B_0 - EU = \Gamma_0 - 2d \ge 0$. Centered-row Cauchy gives

$$R_0 = t_0 g - K(U - EU), \qquad |R_0|^2 \le t_0^2 + \operatorname{Var} U. \tag{C181.T190.30}$$

This combines the two same-point coordinates $g, \sqrt{1-g^2}$. For $c \ge 6$, $\|U\|^2 \le \delta_3 \le \theta^2 d \le 8\theta^2 d^2 < 11d^2/8$, while $t_0 \ge 3d/16$, so $\operatorname{Var} U \le d^2 + 2t_0 d$. At $c = 5$, retain $\theta^2(1+\theta)d$ and $8\theta^2(1+\theta) = 7605/4096 < 35/16 \le 1 + 2t_0/d$. At four, $t_0 = d, U = K^3\mathbf{1}$, with variance $\delta_3 \le \theta^2 d < 3d^2$. Thus every $c \ge 4$ has $|R_0| \le t_0 + d = \Gamma_0 - d$, and the actual two arms give

$$f_3 h_c \le e^{-d}. \tag{C181.T190.31}$$

Every later shorter length is at least five. Its original row remainder is less than $\sqrt{d}/5 < 3d/5$, so the pair envelope gives $h_s f_t \le e^{-2d}$. The same actual message therefore satisfies

$$H \le e^{-[1+2(j-1)]d} p P \bar{f}_{L+2}.$$

The full bare-root bound then gives the high-domain rate

$$4 + 1367/900 + 8(j-1) = 4967/900 + 8(j-1) > 705/128. \tag{C181.T190.32}$$

At the weakest $j = 1$, the margin above $705/128$ is $319/28800 > 0$. The low and high domains share $d = 1/8$, with the same direction. Together with the full-domain anchor family this proves the claimed rows.

Q.12 Five even multipair packets with the actual two-step root exception

Use the original-measure calculus of Section M.2 and the complete root-pair certificates of Section M.10. For a root $(1, 2, v)$, write $m = v - 2$, $G_m = E + zD$, $\beta = \beta_{v-1} = \int D\, d\nu$, $\beta_v = \int E\, d\nu$, $P_+ = h_2 f_v$, $P_- = \bar{f}_2 W_v$, $\ell_+ = \ell_a$, and $\ell_- = \ell_s$. The two vectors, their signed means and full constants are (M.57)–(M.59). At $v = 3$, $(E, D) = (1, 0)$ and $\beta = 0$. All subsequent direction choices and packet estimates retain their stated integer and parameter ranges. Here $n = v - 2$. The complete finite unequal-pair mask proofs and all five terminal packets are in R29.3.2, R29.6.1–R29.6.2 of Section P.5. The following payments retain the genuine positive root P_+ precisely at $L = 2$; the negative root P_- is used at all other lengths.

C182.0 Original parameters and the genuine two-step exception

All original arms, equal-pair multiplicities, subdivisions, independent positive core lengths, edge counts and arbitrary label laws remain.

The internal root residual is (1, 2, v), v at least three. The terminal residual is

$$1 \le a_1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j, \qquad j \ge 2. \tag{C182.1}$$

The positive original inner and outer lengths L, M are independent, and arbitrary genuine equal pairs may remain at both roots.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. Fix the actual directions as

$$F = \prod_{i=1}^{j} h_{a_i} f_{b_i}, \qquad P = \begin{cases} h_2 f_v, & L = 2, \\ \bar{f}_2 W_v, & L \neq 2, \end{cases} \qquad H = pPT^L F. \tag{C182.2}$$

The root unit arm points outward and gives p. Inner edges point terminal to root, outer edges root to receiver. If $a_1 = 1$, its $h_1 = q$ is an actual terminal unit arm. The two root pairs correspond to $L = 2$ and $L \neq 2$, respectively.

Let

$$\lambda_j = \frac{27}{5} + \frac{64}{5}(j-2).$$

For every listed original parameter and genuine equal-pair decoration,

$$\boxed{EH^4 \le e^{-\lambda_j d}.} \tag{C182.3}$$

For $a_1 = 1$, strengthen the basic rate $27/5$ to $13/2$. Apply the Q5-C164 all-length outlet to obtain

$$\begin{gathered} I_q(T^M H)^3 \le e^{-[2\lambda_j - c(M)]d} \le e^{-6d/95}, \\ I_q(X) = E[q^{-1/3}|X|^{8/3}], \end{gathered} \tag{C182.4}$$

Here $c(1) = 204/19, c(2) = 10, c(3) = 0, c(4) = 7$, and $c(M) = 6$ for $M \ge 5$.

For two such sides and the compatible negative zero-step center on an original five-branch path, all four positive core lengths remain independent and

$$\boxed{Z_D \le e^{-3d/190} \le 1.} \tag{C182.5}$$

If both terminal minima are one, the stronger exponent is $43/76$.

C182.2 One actual exponent and every finite propagation

Suppose $F \le e^{R-\Gamma}$, and the same pointwise domain has $e^R \le 1 + R + cR^2/2$. Put

$$\sigma = ER^2, \quad D_k = EB^k R, \quad \delta = D_0 + c\sigma, \quad \epsilon = \delta + D_1.$$

For $|R| \le t < 3$, the positive Taylor bound $n! \ge 2 \cdot 3^{n-2}$ permits $c = (1 - t/3)^{-1}$; negative R has integral coefficient at most one. Thus the bounds $t = 3/5, 1/3, 1, 3/4$ allow $c = 5/4, 6/5, 3/2, 4/3$. Positivity and $T(R^2) \le 2\sigma$ give

$$\begin{gathered} T^L F \le e^{-\Gamma} Y_L, \\ Y_L = \bar{f}_L + \delta \bar{f}_{L-1} + B^L R + \\ \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} > 0. \end{gathered} \tag{C182.14}$$

This is finite Duhamel induction. The scalar δ may be negative, but Y_{L} dominates the actual positive message. Combine $\log Y_{L} \leq Y_{L} - 1$ with the complete root envelope and the original p curvature. The curvature coefficient c_{0} is distinct from the Taylor coefficient.

At $L = 1$, use $V = R - BG_{n}(B)u$, $c_{0} = \beta$, giving

$$\begin{aligned} \log EH^{4} &\leq -4\Gamma + 4\epsilon - 4\ell_{-} \\ &+(17/4)\|V\|^{2} + (10/3)\beta^{2}. \end{aligned} \tag{C182.15}$$

At $L = 2$, retain the actual positive-root exception. Its vector is $V = [1 - BG_{n}(-B)]u + BR$, and its full curvature coefficient is $c_{0} = \beta + \delta$:

$$\begin{aligned} \log EH^{4} &\leq -4\Gamma + 4\epsilon - 4d + 4D_{2} - 4\ell_{+} \\ &+(17/4)\|V\|^{2} + (10/3)(\beta + \delta)^{2}. \end{aligned} \tag{C182.16}$$

At $L = 3$, use the negative root, $V = [G_{2} - BG_{n}]u + \delta u + B^{2}R$, with full $c_{0} = \beta + \epsilon$:

$$\begin{aligned} \log EH^{4} &\leq -4\Gamma + 4\epsilon + 4D_{2} + 4D_{3} - 4\delta d \\ &+J_{3} + (10/3)(\beta + \epsilon)^{2}, \end{aligned} \tag{C182.17}$$

Here $J_{3} = -8d - 4\ell_{-} + 17\|V\|^{2}/4$, retaining this same actual δ, R.

For $L \geq 4$, put $\gamma_{k} = \beta_{k} - D_{k-1}$. If every $k \geq 3$ has $\gamma_{k} \geq hd$, finite convolution retains the original final $\bar{f}_{0}, \bar{f}_{1}$ terms:

$$Y_{L} \leq S_{L} + B^{L}R + 2\delta_{+} + 2(D_{1})_{+} - hd(2 + u).$$

The actual mean also contains $-\beta_{L+1} + D_{L} = -\gamma_{L+1} \leq -hd$. Set $V = [G_{L-1} - BG_{n}]u + B^{L-1}R$, $c_{0} = \beta - hd$, obtaining

$$\begin{aligned} \log EH^{4} &\leq -4\Gamma + 8\delta_{+} + 8(D_{1})_{+} - 12hd - 4\ell_{-} \\ &+(17/4)\|V\|^{2} + (10/3)(\beta - hd)^{2}. \end{aligned} \tag{C182.18}$$

The gamma proof must include every finite index, especially $L + 1$. Every discarded path message has a proved nonnegative subtracted coefficient.

C182.2.1 Three finite common bases

For every n at least one, the two-step positive-root base satisfies

$$|1 - zG_{n}(-z)|^{2} = (1 - xD)^{2} + xE^{2} \leq 1 + x < (19/16)^{2}. \tag{C182.19}$$

The negative-root three-step base $Q = G_{2} - zG_{n}$ has modulus below $23/16$, and the stronger joint payment

$$-8d - 4\ell_{-} + (17/4)\|Q(B)u\|^{2} \leq -27d/4. \tag{C182.20}$$

At $n = 1$, the left side equals $-31d/4$. For $n \geq 2$, the frequency expression decreases in both E, D: the D-derivative is at most $-4 + 9x/2 + 17x^{2}/2 < 0$, the E-derivative at most -4. Set both to $1 - x$, obtaining

$$-47/4 + (25/2)x - x^{2}/4 - (17/4)x^{3} + (17/4)x^{4}.$$

This increases on $[0, \theta]$, with endpoint $-28866559/4194304 < -27/4$, a margin of $555007/4194304$.

For every m at least three and n at least one, the long base satisfies

$$\begin{aligned} |G_m - zG_n|^2 &\le (1+x^2)^2 + x^3 \\ &\le 1493553/1048576 < 36/25. \end{aligned} \tag{C182.21}$$

The real coordinate of G_m is at most $1 - x + x^2$, and the imaginary-coordinate difference has absolute value at most x, including the actual $(1,0)$ endpoint. Every finite length is covered.

C182.4 Packet E2: terminal minimum two

Take the actual finite mask $R = \sum_i R_{a_i b_i} = -B^2 P(-B)u$, with $P_0 = 1$, and $\Gamma \ge 607d/160 + \Delta$, $\Delta = 16(j-2)d/5$. Its complete zero-one-mask, coordinate, signed-mean and all-index gamma proofs are parts R29.6.1–R29.6.2 of Section P.5. In particular,

$$\begin{aligned} \sigma &\le (5408/16245)d < d/3, \\ \|BR\|^2 &\le (2197/16245)d < (37/100)^2 d. \end{aligned} \tag{C182.27}$$

The row bound gives $|R|^2 \le C_* \theta d < 9/25$; use $c = 5/4$. For $R = (\alpha + z\beta_0)u$, the actual coordinates satisfy

$$x/2 \le \alpha \le 6x/5, \qquad -6x/5 \le beta0 \le x/2.$$

The forced original K^3 term gives $D_1 \le 0$; the complete same-sign moment tails give

$$\begin{aligned} -d/5 \le \delta \le (4888/5415)d < (91/100)d, \\ |\epsilon| \le 7d/10, \end{aligned} \tag{C182.28}$$

$$D_2 \le d/12, \quad D_3 \le d/5, \quad D_k \le d/12 \ (k \ge 4).$$

The positive delta coefficient is exactly $416/855 + (5/4)(5408/16245) = 4888/5415$. Epsilon retains the same-model inequality $D_0 + D_1 \le 13\delta_2/19$, giving its positive bound; its lower bound uses $D_0 \ge -d/5, D_1 \ge -d/2$. All later positive moment tails start at δ_4, with same-sign indices separated by two. Hence $\gamma_3 \ge 11d/12, \gamma_4 \ge 4d/5$, and all later gammas at least $49d/96$; take $h = 1/2$.

At $L = 1$, the coordinates of V are $\alpha + xD, \beta_0 - E$, with square at most $(11x/5)^2 + x(1+6x/5)^2 \le 16/9$, using $x \le 5/12$. The complete first-row payment, with $\beta^2 \le d^2$, gives basic loss

$$607/40 - 14/5 + 4 - 68/9 - 10/9 = 185/24. \tag{C182.29}$$

At $L = 2$, the positive-root base and BR bound give $\|V\| < 623\sqrt{d}/400$. The actual coefficient obeys $|\beta + \delta| \le 191d/100$. With $d \le 1/3$, the loss is at least

$$\begin{aligned} &\frac{385}{24} + \frac{969}{256} \\ &-\frac{6598193}{640000} - \frac{36481}{9000} \\ &> \frac{27}{5}. \end{aligned} \tag{C182.30}$$

The left side has numerator 31470923 over 5760000, while $27/5$ has numerator 31104000; the margin is 366923. The actual delta constant is paid through its original curvature.

At $L = 3$, the jointly paid base has modulus bound $23/16$. The perturbation $\delta u + B^2 R$ has norm coefficient at most $91/300 + 1/4 = 83/150$, adding norm cost

$$(17/4)[(23/8)(83/150) + (83/150)^2] = 2902427/360000 < 81/10.$$

The additional mean cost is at most $21d/5$, and $|\beta + \epsilon| \le 17d/10$ costs at most $289d/90$ in curvature. Thus the basic loss is

$$607/40 + 27/4 - 81/10 - 21/5 - 289/90 = 2309/360 > 32/5. \tag{C182.31}$$

For $L \ge 4$, $\|B^{L-1}R\| < 3\sqrt{d}/20$, since $\theta^3/3 < 9/400$. The full vector has norm below $27\sqrt{d}/20$, $|\beta - d/2| \le d/2$, and $2\delta_+ + 2(D_1)_+ \le 91d/50$. The basic loss is at least

$$607/40 + 68/25 - 12393/1600 - 5/18 > 49/5. \tag{C182.32}$$

All four lengths pay $27/5$, with an additional $64/5$ for each later terminal pair.

C182.5 Packet E3: terminal minimum at least three

Take the same finite interval mask, now with $\Gamma \ge 767d/160 + \Delta$. Its full derivation is the E3 packet in R29.6.2, including every finite signed moment. The lowest original power is four, giving

$$\|R\| < (3/8)\sqrt{d}, \quad |R| < 1/3,$$

Use $c = 6/5$ on the full domain. The same-sign moment tails give

$$-d/5 \le \delta \le 2d/3, \quad |\epsilon| \le 7d/8, \quad D_1 \le d/5,$$

$$D_2 \le d/12, \quad D_3 \le d/30, \quad D_k \le d/12 \ (k \ge 4). \tag{C182.33}$$

Explicitly $\sigma \le 2197d/16245$, and $\delta \le 52702d/81225 < 2d/3$. Since $D_1 \ge -d/12$, $\epsilon \ge -17d/60$, while its upper bound is $13d/15 < 7d/8$. The gamma bounds are $11d/12, 29d/30$ at the first two indices and $49d/96$ thereafter, allowing $h = 1/2$.

At $L = 1$, $\|BG_n(B)u\| < 4\sqrt{d}/5$ gives $\|V\| < 47\sqrt{d}/40$, with basic loss $787/40 - 37553/6400 - 10/9 > 12$. At $L = 2$, $\|BR\| < \sqrt{d}/4$ and the positive-root base give $\|V\| < 23\sqrt{d}/16$, $|\beta + \delta| \le 5d/3$, and loss $767/40 - 7/2 + 4 - 1/3 + 969/256 - 8993/1024 - 250/81 > 11$.

At $L = 3$, the perturbation coefficient is at most $2/9 + 1/6 = 7/18$, adding norm cost $27965d/5184 < 11d/2$. The actual additional mean costs at most $127d/30$; $|\beta + \epsilon| \le 15d/8$ gives curvature cost at most $125d/32$. Hence the basic loss is $767/40 + 27/4 - 11/2 - 127/30 - 125/32 > 12$.

For $L \ge 4$, the tail norm is below $\sqrt{d}/10$, so the full vector coefficient is below $13/10$. The positive scalar terms satisfy $2\delta_+ + 2(D_1)_+ \le 26d/15$, and the curvature cost is at most $5d/18$. The loss is $767/40 + 46/15 - 2873/400 - 5/18 > 14$. Thus every finite parameter in E3 pays the common rate and pair-count increment.

C182.7 All unit and two-step connectors for minimum one

The three packets I–III are completely derived, including their forced missing exponents, masks, scalar losses and signed moments, in R29.6.1–R29.6.2. They are I: $a_1 = 1, b_1 \ge 3$; II: $a_1 = 1, b_1 = 2$, $a_2 \ge 4$; III: $a_1 = 1, b_1 = 2, a_2 = 3$. Their basic gamma lower bounds are $607d/160, 16d/5, 3d - 2\delta_2$, respectively, plus Δ. Write $R = (xV + \xi) + zU$, with $\xi = 0$ in I–II and the original global $\xi = \tau$ in III. All use the negative root at one step and the actual positive root at two.

C182.7.1 Packet I

At $L = 1$, the coordinates are $x(V + D), U - E$, with absolute values at most $11x/5, 5/8$. The square is below $121/144+325/2048 = 18413/18432 < 1$. The basic loss is $607/40-3+4-17/4-10/9 > 10$.

At $L = 2$, the coordinates are $1-xD-xU, -E+xV$. The first is positive and the second negative, since $xV \le 416/855 < 19/32 \le E$. Both absolute values are at most $1 - x/2$, so their square is at most $(1+x)(1-x/2)^2 \le 1$. The actual curvature coefficient has absolute value at most $7d/4$, costing $245d/72$. The basic loss is $607/40 - 3 + 4 - 2 + 969/256 - 17/4 - 245/72 > 10$.

C182.7.2 Packet II

At $L = 1$, the real-coordinate absolute value is below $1/2$, and imaginary coefficient below $5/8$; the squared modulus is below $5/12$. The basic loss is $396/25 - 85/48 - 10/9 > 12$.

At $L = 2$, the real coordinate is positive and at most one, and the negative imaginary coefficient has absolute value at most $6/5$. The square is at most $1 + 36\theta/25 = 317/200 < 8/5$. The actual curvature coefficient has absolute value at most $6d/5$, costing $8d/5$. The loss is $64/5 - 24/25 + 4 - 2 + 969/256 - 34/5 - 8/5 > 9$.

C182.7.3 Packet III

At $L = 1$, the real coordinate $\tau + x(V + D)$ has positive bound $205/288 < 5/7$ and negative absolute bound $C_\beta x^2 < 1/5$. The imaginary coefficient $U - E$ has absolute value at most $1/2$. Thus the square is below $5/8$, giving loss

$$227/20 - 85/32 - 10/9 > 15/2. \tag{C182.41}$$

At $L = 2$, the real coordinate lies in $[1 - 2x, 1 - x/2]$; the imaginary coefficient $-E + \tau + xV$ is negative, since $\tau + xV \le 2/9 + 1/12 < 19/32$, and has absolute value at most $6/5$. Its square is less than $(1 - x/2)^2 + 36x/25 < 5/4$, by $1 + 143/800 + 169/4096 < 5/4$. With actual constant coefficient at most $23d/20$, curvature costs $529d/360$, so the basic loss is

$$35/4 - 7/5 + 4 - 13/8 + 969/256 - 85/16 - 529/360 > 13/2. \tag{C182.42}$$

The actual $D_2 \le \delta_2 \le \theta d$ is used; the global δ_2 is constant.

C182.8 The three-step and every longer connector for minimum one

C182.8.1 The full three-step continuous-frequency comparison

The coordinates of the same full vector are

$$A_3 = 1 + xD - x\xi - x^2V + \delta, \qquad C_3 = 1 - E - xU,$$

$$J_3 = \int [-8 - 4E - 4(1 + x)D + (17/4)(A_3^2 + xC_3^2)]d\nu. \tag{C182.43}$$

The scalars ξ, either zero or global τ, and δ retain their actual global values. The real coordinate is positive in all three cases, so its proved delta upper bound may be used.

In I, use $\delta \le 1/4, V \ge 1/2$. The D-derivative is at most $-4 + 53x/8 + 17x^2/2 - 17x^3/4$, increasing to $-493/6912 < 0$ at $5/12$. In II and III use $\delta \le 1/20, V \ge -6x/5, \xi \ge 0$. The derivative is at most $-4 + 197x/40 + 17x^2/2 + 51x^4/5$, with numerator -17085 over 103680 at $5/12$. The E-derivative is at most $-4 + 51x^2/5 < 0$ in every packet. Thus set $E = D = 1 - x$ for $n \ge 2$, retaining $(1, 0)$ separately at one.

In I, the later-root real coordinate is at most $17/12$, imaginary coefficient at most $1/5$, and square below $49/24$, giving $J_3 \le -3511d/256 + 833d/96 < -5d$. At one, the square is below $5/3$, giving $J_3 \le -59d/12$; uniformly use $-19d/4$.

In II, the later-root real bound is $101/75$, imaginary bound $1/12$, square below $11/6$, giving $-3511d/256 + 187d/24 < -23d/4$. At one, the corresponding bounds are $11/10, 1/2, 4/3$, with the same joint conclusion.

In III, the later-root bounds $83/60, 1/5, 39/20$ give $J_3 \le -3511d/256 + 663d/80 = -6947d/1280 < -27d/5$. At one, bounds $17/15, x, 11/8$ give $-197d/32 < -27d/5$. These coordinate comparisons use $x - x^2 \le 1/4, C_\beta x^3 < 1/12$, and additionally $x - 3x^2/2 \le 1/6$ in I, over the whole domain.

C182.8.2 Actual three-step mean and curvature costs

In I, additional mean cost is at most $37d/5$; the actual curvature coefficient is at most $7d/4$, costing $245d/72$. Hence loss $607/40 + 19/4 - 37/5 - 245/72 = 821/90 > 9$.

In II, additional mean cost is at most $134d/25$; coefficient $13d/10$ costs $169d/90$, giving $64/5 + 23/4 - 134/25 - 169/90 = 10181/900 > 11$.

In III, additional mean cost is at most $583d/120$; coefficient $27d/20$ costs $81d/40$, giving

$$35/4 + 27/5 - 583/120 - 81/40 = 109/15. \tag{C182.44}$$

The negative-delta cost has been retained.

C182.8.3 Every finite length at least four

In I and II, the tail norm is below $6\sqrt{d}/25$, giving full vector norm below $36\sqrt{d}/25$. Take $h = 3/8$; $|\beta - hd| \le 5d/8$, so curvature costs at most $125d/288$. In I, $2\delta_+ + 2(D_1)_+ \le 3d/2$, giving loss

$$707/40 - 5508/625 - 125/288 > 42/5. \tag{C182.45}$$

One may use $5508/625 < 53/6$, $125/288 < 7/16$, leaving $2017/240 > 42/5$. In II, $2\delta_+ + 2(D_1)_+ \le 12d/25$, giving loss $969/50 - 5508/625 - 125/288 > 10$.

In III retain its exact all-index gamma bound and take $h = 1/2$. The tail satisfies

$$\begin{aligned} \|B^{L-1}R\| &\le [\theta^2 + (2/9)\theta^{3/2}]\sqrt{d} \\ &< (61/270)\sqrt{d} < (23/100)\sqrt{d}. \end{aligned}$$

The full vector norm is below $143\sqrt{d}/100$, with norm cost $347633d/40000 < 87d/10$. The positive scalar terms satisfy $2\delta_+ + 2(D_1)_+ \le 7d/10$, and curvature costs $5d/18 < 7d/25$. The basic loss is

$$319/20 - 87/10 - 7/25 = 697/100 > 69/10. \tag{C182.46}$$

This covers every finite connector, gamma index and terminal pair count.

C182.8.4 All five packets and every length range

Table 104: Five even multipair packets with the actual two-step root exception: parameter values.

Entry	$L=1$ basic loss
$a_1=2$	$185/24$; $L=2$ basic loss: $27/5$; $L=3$ basic loss: $32/5$; Every $L\ge 4$ basic loss: $49/5$.
$a_1\ge 3$	12; $L=2$ basic loss: 11; $L=3$ basic loss: 12; Every $L\ge 4$ basic loss: 14.
$a_1=1, b_1\ge 3$	10; $L=2$ basic loss: 10; $L=3$ basic loss: 9; Every $L\ge 4$ basic loss: $42/5$.
$a_1=1, b_1=2, a_2\ge 4$	12; $L=2$ basic loss: 9; $L=3$ basic loss: 11; Every $L\ge 4$ basic loss: 10.
$a_1=1, b_1=2, a_2=3$	$15/2$; $L=2$ basic loss: $13/2$; $L=3$ basic loss: $109/15$; Every $L\ge 4$ basic loss: $69/10$.

Each displayed value is a valid weaker paid rate, with the same additional $64(j-2)/5$ in every cell. The five rows partition all original terminal lists, and the four columns cover every positive L in the same preselected direction. The minimum is $27/5$, or $13/2$ for the three minimum-one rows.

Q.13 Odd root lists beginning (1,2,v): lifting all later pairs

Q185.0 The original half-tree and structural direction rule

The original target fixes every original edge direction before the finite host and kernel, for a tree with at most five original vertices of degree at least three. The original skeleton, arm lengths, equal-arm multiplicities, subdivision vertices, four independent positive core lengths, edge count, correlated-law KL and all-map interpretation are retained.

After maximal genuine equal pairing, the internal-root residual is

$$\begin{gathered}(1,2,v,s_1,t_1,\dots,s_k,t_k),\\ 3\le v<s_1<t_1<\cdots<s_k<t_k,\quad k\ge 0.\end{gathered}\tag{Q185.1}$$

The empty later-pair product is allowed. The terminal residual has at least two arms; the original lengths $L,M\ge 1$ are independent, and every genuine equal-pair decoration is allowed. Fix all additional unequal root pairs as

$$Q_k=\prod_{i=1}^{k} W_{s_i}\bar f_{t_i}.\tag{Q185.2}$$

Each pair has shorter arm length at least four. The first three root arms and the terminal use the integer table below.

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K=-B$ and $g=K\mathbf{1}=-u$; in either notation $\delta_j=\|B^j\mathbf{1}\|_2^2=\|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The four arm directions are those of Section Q.25. Every inner edge points from terminal to root; every outer edge from root toward the receiving center. The root unit arm points outward and supplies the actual p.

For every listed root and every terminal residual of size at least two, the structural table below selects an actual F and first root pair P so that the full bare message

$$H_k=pPQ_kT^LF$$

satisfies

$$EH_k^4\le e^{-\lambda d},\qquad \lambda\ge 27/5.\tag{Q185.3}$$

The exact λ is given in the direction table. The same statement holds after genuine equal-pair decorations are restored. For every independent $M \ge 1$,

$$I_q(T^M H_k)^3 \le e^{-[2\lambda - c(M)]d} \le e^{-6d/95}, \qquad I_q(X) = E[q^{-1/3}|X|^{8/3}], \tag{Q185.4}$$

Here $c(1) = 204/19$, $c(2) = 10$, $c(3) = 0$, $c(4) = 7$, and $c(M) = 6$ for $M \ge 5$. The even multipair row uses exactly the Q5-C182 premise recorded below.

Q185.1 One finite mask for all later unequal pairs

Use Section M.2, retaining the exact G_1, G_2 coordinates. Thus $|G_n| \le 6/5$ at every finite index and $|zG_n(\pm z)| < 4/5$. Complete reversal of (L96.2.6) gives the same actual pair bound

$$W_s \bar{f}_t \le \exp\left\{ \sum_{r=s+1}^{t} B^r \mathbf{1} - \frac{16}{5} d \right\}, \qquad 4 \le s < t. \tag{Q185.6}$$

The disjoint original intervals therefore give

$$R_* = BV_* = \sum_{i=1}^{k} \sum_{r=s_i+1}^{t_i} B^r \mathbf{1}, \qquad V_* = \sum_{i=1}^{k} \sum_{r=s_i+1}^{t_i} B^{r-2} u, \qquad Q_k \le e^{R_* - \Gamma_*}, \quad \Gamma_* = (16k/5)d. \tag{Q185.7}$$

The lowest power in V_* is at least B^3. Lemma M.2 yields

$$\|V_*\|^2 \le (101/50)\theta^3 d < 9d/64. \tag{Q185.9}$$

The first positive mean term of R_* is no earlier than δ_4, with subsequent positive moment indices two apart. Thus

$$\mathbb{E}R_* \le \frac{\delta_4}{1-\theta^2} \le \frac{2197}{27360} d < d/12. \tag{Q185.10}$$

For $d \ge 1/64$, the original row bound gives

$$|R_*| \le (3/8)\sqrt{d} \le 3d, \qquad Q_k \le e^{-(16k/5-3)d} \quad (k \ge 1). \tag{Q185.11}$$

Q185.2 Lifting a full exponential certificate

The low-domain input is the same actual message with a complete bound $H \le e^{a+BV+W}$, $4W \le C$, satisfying

$$4a + 4\mathbb{E}BV + (17/4)\|V\|^2 + C \le -\lambda d, \qquad \|V\| \le (7/2)\sqrt{d} \qquad (0 < d \le 1/64). \tag{Q185.12}$$

Here $W = C = 0$, or $W = cu - 3u^2/10$ and $C = 10c^2/3$ come from the same actual unit arm. Keep all constants and signed means in V. Suppose also that the same direction has $\mathbb{E}H^4 \le e^{-\lambda d}$ on the full domain. The actual Q_k above meets Lemma M.31:

$$\mathbb{E}(HQ_k)^4 \le e^{-[\lambda + \zeta_k]d}, \qquad \zeta_k = \frac{2737}{3840} + \frac{64}{5}(k-1) > 0 \quad (k \ge 1). \tag{Q185.13}$$

Its proof keeps the full $V + V_*$ and the existing curvature payment; the same Q_k gives the high-domain gain. At $d = 0$, use leaf integration.

Q185.3 Matching the full low-domain certificates

The matching below retains each exponential certificate, including its curvature payment. The lifting adds precisely the vector cost just displayed. Q5-C160 supplies the auxiliary $L = 2, v \geq 4$ certificate on $0 < d \leq 1/8$; Q5-C163 supplies the same-direction full-domain fourth estimate. Both are used with the weaker common loss $\lambda = 45/8$.

Q185.3.1 Minimum-two and minimum-three single pairs

Every row has $H \leq e^{a+BV}$, $W = C = 0$, retaining all constant components. At the unit connector the terminal polynomials are

$$R = \gamma u - B^2 G_{b-2}(B)u \quad (2b),$$
$$R = \gamma u - B^2[G_{b-2}(-B) - 1]u \quad (3b),$$

Here $0 \leq \gamma \leq d$. The actual root vector is $-BG_{v-2}(\pm B)u + \beta\mathbf{1}$, with $0 \leq \beta \leq d$, and $\beta = 0$ at $v = 3$. Both terminal tails are bounded by $(6/5)\theta\sqrt{d}$. On $d \leq 1/64$,

$$\|V\|/\sqrt{d} \leq 1/64 + 39/80 + 4/5 + 1/8 = 457/320 < 3/2. \tag{Q185.14}$$

The norm of $\beta\mathbf{1}$, at most d, is explicitly paid as $\sqrt{d}/8$. This is the full vector at the unit connector.

The nonunit-connector certificates have the following full-vector forms and squared-norm bounds.

Table 105: Odd root lists beginning (1,2,v): lifting all later pairs: parameter values.

Entry	Full vector, including its constant
$(2,4), L = 2$	original Q + $\delta\mathbf{1}$; Squared bound divided by d: $\leq 5/3$.
$(2,4), L = 3$	original Q + $\epsilon\mathbf{1}$; Squared bound divided by d: $< 9/4$.
$(2,4), L \geq 4$	original finite base + tail + $(\beta - hd)\mathbf{1}$; at $v = 3$ use -$hd\mathbf{1}$; Squared bound divided by d: $< 23/10$.
general (2, b), $L = 3$	original Q + $\epsilon\mathbf{1}$; Squared bound divided by d: $< 5/2$.
general (2, b), $L \geq 4$	original finite base + tail + $(\beta - hd)\mathbf{1}$; Squared bound divided by d: $< 23/10$.
general (2, b), $L = 2$, $v = 3$	original Q + $\delta\mathbf{1}$; $\|Q\|^2 \leq 9d/8, \|\delta\| \leq 8d/15$; Squared bound divided by d: <4 on the low domain.
general (2, b), $L = 2$, $v \geq 4$	original Q + $(\delta + \beta)\mathbf{1}$; Squared bound divided by d: $\leq 599/392$ for $d \leq 1/8$.
general (3, b), $L = 2$	original Q + $\delta\mathbf{1}$; Squared bound divided by d: $\leq 49/30$.
general (3, b), $L = 3$	original Q + $\epsilon\mathbf{1}$; Squared bound divided by d: $< 11/4$.
general (3, b), $L \geq 4$	original base + tail - $hd\mathbf{1}$, using the stated weaker root envelope; Squared bound divided by d: $< 17/8$.

For $v = 3, L = 2$, $\|V\|/\sqrt{d} \leq \sqrt{9/8} + 1/15 < 2$ on the present low domain. Every other squared bound in the table is less than four as well, so all vectors have norm less than $2\sqrt{d}$. Their original combined scalar, signed-mean and norm payments give $45/8$.

Q185.3.2 The exceptional short pairs (2, 3) and (3, 4)

For Q5-C168, the short and long data are respectively (Q.4) and (Q.7). They give the complete bound $H \leq \exp(a + BQ + cu - 3u^2/10)$, with $a + \mathbb{E}BQ = S$. The proof of (Q.2) bounds its full certificate $4S + (17/4)\|Q\|^2 + (10/3)c^2$ by $-25d/4$, using (M.55). Thus this is an input to (Q185.12),

with $V = Q$, $W = cu - 3u^2/10$, and $C = 10c^2/3$. Q5-C170 uses the same curvature convention in T178.12, 14, 16, 20, with full-domain loss $13/2$.

In Q5-C168, the $L = 1$ coordinate $[x(1 + C) - zE]u$ has squared-frequency bound less than four; at $L = 2$ its original coordinate bounds give norm less than $2\sqrt{d}$; at $L = 3$ its full squared norm is at most $d + 2\delta_2 - \delta_3 + 77d^2/150 < 2d$; for later lengths its norm is at most $131\sqrt{d}/100$. In Q5-C170, the squared bounds for $L = 1, 2, 3$ are respectively $7d/5, 9d/5, 14d/5$, and for later lengths $16d/9$. All full vectors have norm less than $2\sqrt{d}$. The $v = 3, G_1$ and $v = 4, G_2$ boundaries are retained in those original formulas.

Q185.3.3 Every minimum-one terminal triple

On the common low domain $d \le 1/64$, Q5-C169, Q5-C172 and Q5-C173 use the full vector

$$V = [G_L + z^{L-1}r - zG_n(\pm z)]u, \quad n = v - 2, \tag{Q185.15}$$

Here $W = \beta u - 3u^2/10$, $C = 10\beta^2/3$. At $L = 1$, retain $G_1 = 1$, giving the exact $u + R$ first row. The notation $R = r(B)u$ means a finite polynomial, with no pointwise division by u. The full-frequency bounds are as follows; $\xi = 1/m$ in the first row is the reciprocal scalar of the original square envelope.

Table 106: Odd root lists beginning (1,2,v): lifting all later pairs: parameter values.

Entry	Reference and domain
(1, 2, 3)	Q5-C169 Section 4, full d; $r = -\xi \cdot$ x, $\xi \le 6/5$; Upper bound for the modulus of r: $\le 39/80$.
(1, 2, 4)	Q5-C169 Section 6, $d \le 1/4$; Upper bound for the modulus of r: $< 19/20$.
$(1, 2, w), w \ge 5$	Q5-C169 Section 7, $d \le 1/5$; Upper bound for the modulus of r: $\le 147/155 < 1$.
$(1, 3, w), w \ge 4$	Q5-C172 Sections 3–7, $d \le 1/4$; Upper bound for the modulus of r: $< 3/5$.
$(1, a, b), a \ge 4$	Q5-C173 Sections 2–6, $d \le 1/4$; Upper bound for the modulus of r: $< 3/10$.

The finite geometric bounds give $\|V\|/\sqrt{d} \le 6/5 + 4/5 + \sup|r| < 3$. The original vector, mean, curvature term and joint-frequency payment remain unchanged. The common loss is $2817/512$ for $(1, 2, 3)$, and $705/128$ for every other row. The same-direction full-domain statements come from the corresponding original results.

Q185.3.4 Minimum-one odd lists with several pairs

Q5-C177, equations T184.13–18, uses the same vector and original β curvature. The low domains of its five structural blocks I–V all contain $[0, 1/64]$; their actual modulus bounds for r are $1/2, 2/3, 3/5, 1/2, 3/5$. Hence every pair count and L has $\|V\|/\sqrt{d} < 2 + 2/3 < 3$. The original low-domain loss is at least $659/96 > 705/128$. The crossed row keeps exactly its fixed permutation of arms $2, 3, 4, \bar{f}_2$. Its high-domain statement uses the same crossed or ordinary direction together with the pointwise mask bound.

Q185.3.5 Even terminal lists with large minimum

Q5-C180 has $W = C = 0$; the full vectors in equations (28)–(34) include $\delta\mathbf{1}$, $\epsilon\mathbf{1}$, or $-(d/2)\mathbf{1}$. Their norm coefficients for the four length ranges are at most $21/20, 47/25, 89/48, 25/16$, all below two. Its $L = 2$ joint Φ payment retains this actual vector. Its full-domain loss is $29/5 + 64(j - 1)/5$.

Q185.3.6 All nonunit-minimum odd terminal lists

For Q5-C181, apply (M.61) to its actual positive comparison with $\alpha = -\Gamma$, retaining C_0, c_0, and the propagated vector. The actual p supplies the curvature in (M.55), and the full vector is $V_{\text{actual}} = V_{\text{propagated}} - BG_n(B)u$, where $n = v-2$. Thus $W = (\beta+C_0-1)u-3u^2/10$, $C = 10(\beta+C_0-1)^2/3$. On our low domain, the full norm coefficient is at most $27/16$ whenever $b_1 \geq 4$. For a single pair with $b_1 = 3$, the four length-range coefficients are at most $5/3, 6/5, 5/4, 3/2$; with multiple pairs they are at most $11/6, 13/10, 3/2, 3/2$. All are below two. Every original γ, the actual $\bar{f}_a$, and the genuine BR first row are retained. The original full-domain ordinary fourth loss is $705/128$.

Q185.3.7 The five even multipair blocks

This step uses the four complete propagation estimates (C182.15)–(C182.18) from Section Q.12. Their vectors are

$$\begin{gathered} R - BG_n(B)u, \quad (1 - BG_n(-B))u + BR, \\ (G_2 - BG_n)u + \delta u + B^2 R, \\ (G_{L-1} - BG_n)u + B^{L-1}R. \end{gathered} \tag{Q185.16}$$

The original β and δ, ϵ constant terms have been paid with the same pointwise p curvature, so these are the actual vectors used.

For blocks E2, E3, UI, UII and UIII, σ/d is at most $1/3, 9/64, 325/392, 325/512, 1277/2592$, all below one, and $|\delta| \leq 6d/5$. Thus $\|R\| < \sqrt{d}$ uniformly in the pair count. Retaining the joint-base bounds $19/16, 23/16, 6/5$, on $d \leq 1/64$ we obtain

L	$\|V\|/\sqrt{d}$ common upper bound
1	$1 + 4/5 = 9/5$
2	$19/16 + 2/3 = 89/48$
3	$23/16 + 13/32 + 3/160 = 149/80$
≥ 4	$6/5 + 4/15 = 22/15$

(Q185.17)

All four are below two. At $L = 3$, δ is the original global scalar, bounded by $6d/5$, so the joint polynomial optimizations remain valid for the actual vector. Retaining its full $L = 2$ positive-root exception, this supplies loss $27/5 + 64(j-2)/5$.

Q185.3.8 The exact full-root interface for a terminal pair (1, b)

This row uses the full-root statement Q5-C084, Section 0, equations (0.4)–(0.5). It already covers $(1, 2, v, s_1, t_1, \dots, s_k, t_k)$, every finite k, terminal qf_b, and exactly the later-pair product Q_k used here. Choose $P = h_2 f_v$ precisely when $L = 1, b \geq 3, v \geq 4$; otherwise take $P = \bar{f}_2 W_v$. Its ordinary fourth loss is $705/128$ for all independent L, M.

Every other terminal row therefore has the required full exponential certificate, with the stronger bound $\|V\| < 3\sqrt{d}$ on the common low domain. The lifting uses the uniform allowance $7\sqrt{d}/2$.

Q185.4 The complete terminal direction algorithm

Put $P_- = \bar{f}_2 W_v$, $P_+ = h_2 f_v$, $H_0 = pPT^L F$, and $H_k = H_0 Q_k$. Every original length is positive, with $v \geq 3$.

Table 107: Odd root lists beginning (1,2,v): lifting all later pairs: parameter values.

Entry	Actual F
$(1,b), b \geq 2$	qf_b; First root pair P: P_+ if$L = 1, b \geq 3, v \geq 4$; otherwise P_-; Lambda used: $705/128$.
$(2,b), b \geq 4$	$\bar{f}_2 W_b$; First root pair P: P_+ if$L = 2$ or $L = 1, v \geq 4$; otherwise P_-; Lambda used: $45/8$.
(2, 3)	$h_2 f_3$; First root pair P: P_+ if$L = 2$; otherwise P_-; Lambda used: $25/4$.
$(3,b), b \geq 5, L = 1$	$h_3 f_b$; First root pair P: P_-; Lambda used: $45/8$.
$(3,b), b \geq 5, L = 2$	$\bar{f}_3 W_b$; First root pair P: P_+; Lambda used: $45/8$.
$(3,b), b \geq 5, L \geq 3$	$\bar{f}_3 W_b$; First root pair P: P_-; Lambda used: $45/8$.
(3, 4)	$h_3 f_4$; First root pair P: P_+ if$L = 2$; otherwise P_-; Lambda used: $13/2$.
even $4 \leq a_1 < b_1 < \cdots < a_j < b_j, j \geq 1$	$\prod h_{a_i} f_{b_i}$; First root pair P: P_-; Lambda used: $29/5 + 64(j-1)/5$.
even multipair $a_1 \leq 3$, $j \geq 2$	$\prod h_{a_i} f_{b_i}$; First root pair P: P_+ if$L = 2$; otherwise P_-; Lambda used: $27/5 + 64(j-2)/5$, exact C182 premise.
(1, 2, 3)	$qf_2 h_3$; First root pair P: P_+ for every L; Lambda used: $2817/512$.
$(1,2,w), w \geq 4$	$qf_2 h_w$; First root pair P: P_- for every L; Lambda used: $705/128$.
$(1,a,b), a \geq 3$	$qf_a h_b$; First root pair P: P_+ if$L = 1, v \geq 4$; otherwise P_-; Lambda used: $705/128$.
odd minimum one, $j \geq 2, (a_1, b_1, a_2) = (2, 3, 4)$	$qf_2 h_{b_2} h_3 f_4 \prod_{i \geq 3} h_{a_i} f_{b_i}$; First root pair P: P_-; Lambda used: $705/128$.
odd minimum one, $j \geq 2, a_1 = 2$, excluding the preceding row	$qf_{a_1} h_{b_1} \prod_{i \geq 2} h_{a_i} f_{b_i}$; First root pair P: P_-; Lambda used: $705/128$.
odd minimum one, $j \geq 2, a_1 \geq 3$	same ordinary $\prod$; First root pair P: P_+ if$L = 1, v \geq 4$; otherwise P_-; Lambda used: $705/128$.
$(a, b_1, c_1, \cdots, b_j, c_j)$, $a \geq 2, b_1 = 3, soa = 2$	$\bar{f}_2 f_3 h_{c_1} \prod_{i \geq 2} h_{b_i} f_{c_i}$; First root pair P: P_-; Lambda used: $705/128$.
nonunit odd residual, $b_1 \geq 4$	$\bar{f}_a \prod h_{b_i} f_{c_i}$; First root pair P: P_-; Lambda used: $705/128$.

The residual lengths are strictly ordered. A single pair has minimum one, two, three, or at least four. Even multipair lists have minimum at most three or at least four. Odd lists split by minimum one or at least two and the stated short exceptions. Thus every finite terminal residual of size at least two appears. The low- and high-domain proofs use the same row, and M never chooses a direction.

For $k = 0$, the exact original interface gives the ordinary fourth bound directly. For $k \geq 1$, the first row uses the full-root Q5-C084 statement; every other row uses the lifting lemma, with the stronger loss $\lambda + \zeta_k$. The terminal $qf_2 h_3$ and the fixed crossed permutation are retained exactly. The smallest tabulated λ is $27/5$: the differences for $2817/512, 705/128, 45/8$ are respectively $261/2560, 69/640, 9/40$.

For the genuine equal-pair products at the root and terminal, positivity gives

$$0 < \widetilde{H}_k = E_r p P Q_k T^L (E_t F) \leq H_k.$$

Only genuine equal pairs are bounded by one; each additional unequal pair has been paid by the finite-mask argument. The ordinary fourth bound is order preserving. The Q5-C164 outlet applies to every original real input and every independent M, yielding the outlet statement since $54/5 - 204/19 = 6/95$.

The whole-tree path alternatives and the join of compatible outlets are proved in Section 11. Their kernel, tournament and entropy consequences are given in Section 12.

Q.14 Comparing complete certificates at the remaining odd roots

All operators and vectors in this section use the original finite probability measure. Write $z = i\sqrt{x}$, $\theta = 13/32$, $\kappa = 19/32$, $\tau = d - \delta_2$, and let ν have mass d. We compare the same actual positive terminal majorant

$$X \le e^{\alpha}[Cq + c + BQ(B)u], \qquad \gamma = C - 1, \quad k_0 = C + c - 1. \tag{Q.8}$$

The coefficients C, c and all moments of Q retain their actual values. In particular $\gamma\mathbf{1}$ is a vector of norm $|\gamma|$.

The finite bare-exponent packet.

For an actual $F \le e^{R-\Gamma}$, with $R = r(B)u$, put $D_i = \mathbb{E}B^i R$, $\delta = \mathbb{E}R + c_T\mathbb{E}R^2$ and $\epsilon = \delta + D_1$, using the pointwise Taylor coefficient c_T proved for that terminal. The positive first row and its finite continuation give

$$e^{\Gamma}T^L F \le \bar{f}_L + \delta\bar{f}_{L-1} + B^L R + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}.$$

Substituting the exact short paths into this same majorant gives the parameters of (Q.8), with $\alpha = -\Gamma$:

L	C	c	Q
1	1	δ	r
2	$1+\delta$	D_1	$1 + zr$
3	$1+\epsilon$	$-d + D_2$	$G_2 + \delta + z^2 r$
≥ 4	$1 - hd$	$2\delta_+ + 2(D_1)_+ - hd$	$G_{L-1} + z^{L-1} r.$

(Q.9)

The last row assumes $\gamma_i = \beta_i - D_{i-1} \ge hd$ also at the mean index $i = L + 1$, and retains the final two paths $\bar{f}_0, \bar{f}_1$ from that positive convolution. The signed means in the four windows are exactly

$$D_1, \qquad -d + D_2, \qquad -(1+\delta)d + D_3, \qquad -\gamma_{L+1}. \tag{Q.10}$$

Both the constant $C - 1$ and these moments belong to the same R.

Lemma Q.1 (One certificate with an explicit constant split)**.** *Suppose an actual root pair satisfies $P \le \exp(BS - \zeta u - \Gamma_r)$, and its separate unit arm is p. For any real η put*

$$V = Q(B)u + S + (\gamma - \zeta - \eta)\mathbf{1}.$$

Then $H = pPX$ has the pointwise certificate

$$\log H \le \alpha + k_0 - \Gamma_r + BV + \eta u - 3u^2/10, \tag{Q.11}$$

and its complete fourth exponent is at most

$$4(\alpha + k_0 - \Gamma_r + \mathbb{E}BV) + (17/4)\|V\|^2 + (10/3)\eta^2.$$

For the auxiliary pair $P_0 \le e^{-d-B^2u}$, the choices $\eta = \gamma$ and $\eta = 0$ give, respectively, the curved and full-constant baselines

$$J_0^c = 4(\alpha + k_0 + \mathbb{E}BQu - d) + (17/4)\|(Q - B)u\|^2 + (10/3)\gamma^2, \tag{Q.12}$$

$$J_0^f = 4(\alpha + k_0 + \mathbb{E}BQu - d) + (17/4)\|(Q - B)u + \gamma\mathbf{1}\|^2. \tag{Q.13}$$

Proof. Apply $\log y \le y-1$ to the positive bracket in (Q.8), and add the actual pair envelope and $\log p \le -u - 3u^2/10$. Since $B\mathbf{1} = u$, this gives (Q.11) exactly. Apply Lemma M.27 to these complete vectors. □

Lemma Q.2 (Minimum-one root transfers)**.** *Take $3 \le a < b$ and the actual root $pW_a\bar{f}_b$. Write $Q = A + zD$. Suppose $A \ge 0$, $D \le \bar{D}$, $\bar{D} \ge 1$, $\|Q(B)u\| \le R\sqrt{d}$, and $\gamma \ge -\chi d$ on $0 < d \le d_*$, with $0 \le \chi \le 17/10$, $d_* \le 1/3$. Relative to the curved baseline, the complete exponent increments are at most*

$$J_3 - J_0^c \le d\Phi(\bar{D}, \chi, d_*), \tag{Q.14}$$

$$J_{a\ge 4} - J_0^c \le d\Psi(\bar{D}, R), \tag{Q.15}$$

where

$$\Phi(\bar{D}, \chi, d_*) = -8 + [12 + (17/4)(2\bar{D} - 1 + \theta)]\theta + (10/3)d_*[\kappa^2 + 2\chi\kappa],$$
$$\Psi(\bar{D}, R) = -127/15 + (17/4)[(2\bar{D} - 1)\theta + 3R/4 + 9/64].$$

For a full-constant baseline with $\gamma \ge -\chi d$, use instead $\Phi(\bar{D}, 0, d_*)$ or $\Psi(\bar{D}, R)$, and add at most

$$(17/6)\chi d \quad (a = 3), \qquad (17/6)\chi(1 + \theta^2)d \quad (a \ge 4). \tag{Q.16}$$

If $A \ge -\rho_0$ replaces $A \ge 0$, replace $2\bar{D} - 1$ by $2\bar{D} - 1 + 2\rho_0$ in Φ.

Proof. The actual reversed length-three pair, including its constant loss, is Lemma M.26 in the form

$$W_3\bar{f}_b \le e^{BS - \tau u - (3d - 2\delta_2)}, \qquad S = -x(E + zF)u, \quad 0 \le E, F \le 1.$$

The scalar plus signed-mean increment is at most $-8d + 12\delta_2$. The nonconstant norm differences satisfy

$$(A - xE)^2 - A^2 \le x^2, \qquad x[(D - xF)^2 - (D - 1)^2] = x(1 - xF)(2D - 1 - xF) \le (2\bar{D} - 1)x.$$

The curved choice of Lemma Q.1 changes γ to $\gamma - \tau$, costing at most $(10/3)(\tau^2 + 2\chi d\tau)$. Only after integrating set $t = \delta_2/d$. The resulting upper

$$-8 + [12 + (17/4)(2\bar{D} - 1 + \theta)]t + (10/3)d_*[(1 - t)^2 + 2\chi(1 - t)]$$

increases on $[0, \theta]$: its derivative is at least $12 + (17/4)(1 + \theta) - (20/9)(1 + 17/10) > 0$. This proves Φ without replacing the same global moment twice. For $A \ge -\rho_0$, its real norm difference adds $2\rho_0 x$.

For $a \ge 4$, the actual pair estimate (L96.2.6) gives

$$W_a\bar{f}_b \le e^{BS - 16d/5}, \qquad S = \sum_{j=a+1}^{b} B^{j-2}u.$$

Its contiguous finite mask gives $\|S\| < 3\sqrt{d}/8$, $\mathbb{E}BS \le d/12$, and $-\delta_3 \le \mathbb{E}S \le \delta_2$. Thus the scalar plus mean improves by at least $127d/15$. The base norm difference is $\|Qu\|^2 - \|(Q - B)u\|^2 = \int (2D - 1)x\, d\nu$; the further norm increment is at most $2R(3/8)d + 9d/64$. The curvature coefficient is unchanged, giving Ψ.

For the full-constant convention choose $\eta = -\tau$ at $a = 3$ and $\eta = 0$ at $a \ge 4$. The additional exact cross term is $2\gamma\mathbb{E}(S + Bu)$, while $\mathbb{E}(S + Bu) \in [-d, -\tau]$ or $[-d - \delta_3, -\tau]$, respectively. A nonnegative γ helps; otherwise $\gamma \ge -\chi d$ and $d \le 1/3$ give (Q.16), with the constant's signed mean retained inside both vectors. □

Lemma Q.3 (Positive-root transfers, with signed overshoots)**.** *Take the actual root $f_a h_b f_c$, $3 \le a < b < c$. Write $G_{a-1}(-z) = E_a - zD_a$ and $S_{bc} = (r+zs)u = \sum_{j=b+1}^{c} (-1)^j B^{j-2} u$. Put $A_* = E_a + r$, $\phi = 1 - D_a + s$, and $V_* = (Q + G_{a-1}(-B))u + S_{bc}$. For $Q = U + zV$ define*

$$\mathcal{T}(Q) = \frac{17}{2} \int [(1+x^2)(U-1)_+ + (x^2+x^3)(V-1)_+ + x^3(-V)_+] \, d\nu.$$

The complete exponent differences are bounded by

$$J_+ - J_0^f \le -11d/10 + \mathcal{T}(Q) - (17/2)\gamma \int \phi \, d\nu, \tag{Q.17}$$

$$J_+ - J_0^c \le -11d/10 + \mathcal{T}(Q) + (17/2)\gamma \mathbb{E} V_* + (11/12)\gamma^2. \tag{Q.18}$$

Alternatively, if $Q = Q_0 + \Delta Q$ with both normalized coordinates of Q_0 in $[0,1]$, replace $-11d/10 + \mathcal{T}(Q)$ by

$$-d + (34/3)\sqrt{d}\, \|\Delta Q(B) u\|. \tag{Q.19}$$

The same full-constant or curved correction still applies.

Proof. Subtract the actual exponents in Lemma M.30. This gives its exact density j_Q, together with the displayed two distinct constant corrections. On $0 \le U, V \le 1$ the three actual finite root blocks have upper polynomials

(a,b)	j_Q upper
$(3,4)$	$-81/20 + 13x^2 + (17/4)x^3 + (17/4)x^4$,
$(3, b \ge 5)$	$-81/20 + 13x^2 + (17/2)x^3 + (17/4)x^4 + (17/4)x^5$,
$a \ge 4$	$-81/20 - 5x + 22x^2 + (17/2)x^3 + (17/4)x^4 + (17/4)x^5$.

For the first block $E_a = D_a = 1$, $r = x^2 F$, $s = xE$, with $0 \le E, F \le 1$, including the finite gap one. For $b \ge 5$ the four starting residues give $|r|, |s| \le x^2$. Convexity selects $r = x^2$, then $U = 1$; the imaginary part is affine in V and convex in $D_a - s$. For $a \ge 4$, $E_a \le 1 - x + x^2$, $D_a \ge 1 - x$; decreasing E_a from one saves at least $9x(1-x)$, while relaxing $-4D_a$ costs at most $4x$. These give the three displayed polynomials. The first two are below $-11/10$ using $\theta^2 < 1/6$, $\theta^3 < 1/14$, $\theta^4 < 1/36$. The third is convex on $[0, 5/12]$; at its right endpoint its excess over $-3/2$ is $-87307/4976640$.

All finite blocks also give $0 < A_* \le 1 + x^2$, $-x^2 \le \phi \le x + x^2$. The dependence of the exact density on the terminal coordinates is

$$\Delta j = (17/2)(A_* \Delta U + x\phi \Delta V).$$

Clipping Q into the unit rectangle therefore gives the three signed charges in $\mathcal{T}$. A negative real coordinate only helps. For the norm version the coefficient is at most $(17/2)\sqrt{A_*^2 + x\phi^2} < 34/3$; use $|\phi| \le 2x$ and $(1+\theta^2)^2 + 4\theta^3 < 16/9$, then Cauchy with the same spectral measure of mass d. The L^2 tail is used in this integral Cauchy inequality. □

Corollary Q.4 (Bare comparisons in the same directions)**.** *For $n \ge 3$ the actual $pW_a \bar{f}_b \bar{f}_n$, $3 \le a < b$, has fourth loss at least $29/10$. For the positive root $f_a h_b f_c$, $3 \le a < b < c$, the actual bare input $\bar{f}_n$ has respective losses*

$$419/128 \ (n=1), \quad 387/128 \ (n=2), \quad 35/4 \ (n=3), \quad 11/2 \ (n \ge 4).$$

Proof. Retain $\beta_n \bar{f}_0$ in the positive convolution: $\bar{f}_n \le q - \beta_n + BG_{n-1}(B)u$. At the auxiliary root the actual exponents have losses $291/128$, $259/128$, $31/4$, and at least $9/2$, respectively. For $n \ge 4$ this uses $\beta_n + \beta_{n+1} \ge 2\kappa d$ and $|G_{n-1} - z|^2 \le (1 - x + x^2)^2 + x^3 \le 1$. Every G_{n-1} is in the coordinate rectangle. The positive transfer saves at least d. For the minimum-one root, $\gamma = 0$, $\bar{D} = 1$, $R \le 6/5$; both $\Phi(1, 0, 1/3)$ and $\Psi(1, 6/5)$ are negative. The weaker claimed $29/10$ follows. □

Q.15 The remaining short terminal directions

The positive-root convention for a terminal $(1, t)$ is separate from that for the other terminal lists. Write $P_-^{(N)} = \prod_{i=1}^{N} W_{b_i} \bar{f}_{c_i}$, $a < b_1 < c_1 < \cdots < b_N < c_N$, $a \ge 3$, $N \ge 1$. The actual messages are

$$H_N = \begin{cases} f_a P_-^{(N)} T(p\bar{f}_t), & L = 1, \\ W_a P_-^{(N)} T^L(q f_t), & L \ge 2. \end{cases} \tag{Q.20}$$

The correlated square packets of Sections M.19 and O.4 give respective fourth losses

$$33337/3840 + 64(N-1)/5, \qquad 22897/3840 + 64(N-1)/5. \tag{Q.21}$$

To see the complete-vector payment, at $L = 1$ the actual square for $f_a T(p\bar{f}_t)$ has an unscaled real precursor V with $\|V\|^2 \le d$ and already paid exponent $E_0 = 8 \log C - 8\ell + 8\mathbb{E}KV + 20\|V\|^2 \le 0$. The full finite mask of $P_-^{(N)}$ contributes KV_S, $\|V_S\| < 3\sqrt{d}/8$, $\mathbb{E}KV_S \le d/12$, and scalar $-16Nd/5$. At fourth order, $E_0/2$ pays the base exponent; the extra mean and norm payments are $d/3 + (17/4)(57/64)d$. Subtracting these from $64Nd/5$ gives the first displayed loss. For $L \ge 2$ the actual W_a square has normalized precursor norm at most $3\sqrt{d}/2$ and exponent $E_0 \le 9d/4$. The added mask has the same norm and signed mean, so the payment is $9d/8 + d/3 + (17/4)(81/64)d$, giving the second loss. At $a = 3$ its separate packet has norm at most $9\sqrt{d}/8$ and $E_0 \le -13d/16$, hence is stronger. The finite square packets include $t = 2$, $L = 2, 3, 4$ and the full finite tail.

For every other positive-root terminal the root is $f_a h_b f_c$, $3 \le a < b < c$. Two remaining short terminals use the actual bare-exponent rows of Section Q.2 or Section Q.4. We give their complete payments.

The terminal $(2, t)$, $t \ge 4$.

Take the actual $F = \bar{f}_2 W_t$ and its remainder $R = (\beta_{t-1} + xA + zxC)u$. The actual constants satisfy

$$\Gamma = \beta_t + \beta_{t-1}, \quad \sigma < 5d/8, \quad |\delta| \le 8d/15, \quad -13d/16 \le \epsilon \le 0, \quad D_1 < 0, \quad \Gamma - 2\delta_+ \ge 5d/6.$$

All finite losses $\beta_i - D_{i-1}$ are at least $17d/32$, including the mean index. The four actual C, c, Q rows are (Q.9), with $h = 17/32$, interpreted in the full-constant convention. Their auxiliary losses are $45/8$, $629/96$, $2021/256$ and $301/48$.

At $L = 1$, $Q = r$ is rectangular and $\gamma = 0$, saving d. At $L = 2$ the coordinates are $1 - x^2 C$, $\beta_{t-1} + xA$ and remain rectangular. The full $\gamma = \delta$ cross correction is less than $3d^2 \le d$, paid by that saving. At $L = 3$ the coordinates are $1 + \delta - \beta_{t-1} x - x^2 A$, $1 - x^2 C$. Only real overshoot δ_+ occurs; its fee is less than $16d^2/3$. The signed $\epsilon \mathbf{1}$ correction costs less than $4d^2$. The full increment is therefore at most $-d + 28d^2/3 \le 19d/9$. The residual loss satisfies

$$2021/256 - 19/9 = 23/4 + 77/2304.$$

For the long row $n = L - 1 = 3$ the actual coordinates are $1 - x + x^3 C$, $1 - x\beta_{t-1} - x^2 A$; both lie in $[0, 1]$. For $n \geq 4$ the finite tail is at most $x^2(1/3 + x)$, absorbed by the adjacent alternating-prefix margins since $(4/3)x + x^2 < 1$. The constant $-(17d/32)\mathbf{1}$ costs less than $3d^2 \leq d$. Thus every L, t has loss at least $23/4$.

The terminal $(2, 3)$**.**

Take the actual $F = h_2 f_3$. Its packet is $R = xu$, $\Gamma = d$, $\delta = 8\delta_3/7 < d/5$ and $\epsilon = \delta - \delta_2 \in [-\delta_2, 0]$. All finite losses are at least $d/2$. The curved auxiliary losses are $25/4$, $15/2$, $27/4$ and $4903/720$. At $L = 1$ the polynomial is rectangular and $\gamma = 0$. At $L = 2$, $Q = 1 + zx$ is rectangular and the actual new virtual coordinate lies in $[-1, 0]$; therefore $0 \leq \mathbb{E}V_* \leq d$. The curved $\delta \geq 0$ correction is at most $(521/300)d^2 < d$. At $L = 3$ the real overshoot and the actual $\epsilon\mathbf{1}$ correction total less than $(767/180)d^2$. Its increment, including the saved d, is at most $227d/540$. For the long row, $Q = G_n - z^{n+2}$ is rectangular, including its separate $n = 3$ coordinates $1 - x$, $1 - x^2$. The curved coefficient $-d/2$ has correction less than $11d^2/4 \leq 11d/12$, paid by the saved d. Every resulting loss exceeds six.

These short proofs retain their original signed means and their full or curved convention. At $d = 0$ the same finite messages are one.

Q.16 All terminals beginning at three, at both remaining roots

Theorem Q.5. *Let $3 \leq a < b$ or $3 \leq a < b < c$. The internal root is, respectively, $pW_a \bar{f}_b$ or $f_a h_b f_c$. For a terminal residual $(3, t)$, $t \geq 4$, use the actual direction*

$$F_{t,L} = \begin{cases} h_3 f_t, & L = 1 \text{ or } t = 4, \\ \bar{f}_3 W_t, & t \geq 5,\ L \geq 2. \end{cases}$$

For every finite t, every $L \geq 1$, and the whole host domain, the resulting complete message H_0 has

$$\mathbb{E}H_0^4 \leq e^{-6d}. \tag{Q.22}$$

On $0 < d \leq 1/64$, its proof supplies a pointwise certificate

$$\log H_0 \leq a_0 + BV_0 + W_0, \qquad \|V_0\| < \tfrac{7}{2}\sqrt{d}, \tag{Q.23}$$

with $W_0 = 0$ or an actual already paid p-curvature term. All later ordered root pairs, genuine equal pairs and every independent outer length are therefore covered by Lemmas M.31 and (M.66).

Proof. The separate terminal directions are fixed by the original t, L. All intermediate means, constants and vectors below use the original measure. We give one actual terminal packet and one four-row payment for both root classes.

The packet for $t \geq 5$**.**

Write $G_{t-2} = A + zC$, $w = 1 - A \in [0, x]$, and $b_0 = \beta_{t-1} = \int C\, d\nu$. The actual two directions have

$$F_{t,1} \leq e^{R_+ - \Gamma}, \quad F_{t,L} \leq e^{R_- - \Gamma}\ (L \geq 2), \qquad R_\pm = (b_0 - xw \mp zxC)u, \quad \Gamma = d + \beta_t + b_0 \geq 3d - 2\delta_2.$$

These are the arm-sum identities of part Q.2. Their first positive rows use the same actual remainders. For R_{-}, the complete finite calculation in Section Q.2 gives the following data on the entire domain:

$$\|R_{-}\| < \tfrac{1}{2}\sqrt{d}, \quad -\delta_2 \le \delta < d/24, \quad -3d/4 \le \epsilon = \delta + D_1 \le 0,$$
$$(D_1)_{+} \le \delta_3, \qquad \beta_j - D_{j-1} \ge 9d/16 \ (j \ge 3).$$

Here $D_i = \mathbb{E}B^i R_{-}$ and the Taylor coefficient is $5/4$. For R_{+} at $L = 1$, use coefficient $3/2$ and its actual unit certificate. The finite propagation is the bare exponent packet (R29.5.3).

Put $n = L - 1$ in the long row and $h = 9/16$. In the full-constant convention its propagated coordinates are

L	Q	$\gamma\mathbf{1}$ retained in the full vector
1	r_{+}	0
2	$1 + zr_{-}$	$\delta\mathbf{1}$
3	$G_2 + \delta + z^2 r_{-}$	$\epsilon\mathbf{1}$
≥ 4	$G_n + z^n r_{-}$	$-hd\mathbf{1}$.

(Q.24)

The mean and scalar in each row are obtained from this actual upper, including the same D_i; no alternate full polynomial is substituted. At the auxiliary root P_0, the complete baseline losses are

$$25/4, \qquad 8, \qquad 13, \qquad 3385/384. \tag{Q.25}$$

The first two are the actual unit and two-step calculations of Section Q.2. For the long row its complete calculation has loss $1043/128$ after using the weaker root loss $5d/6$. At this same $v = 3$ root the loss is exactly d and its constant β_2 is zero, saving $2d/3$ without changing the full vector.

For the third row a direct specialization is substantially shorter. The nonconstant old vector $(Q - B)u$ has real coordinate $1 + \delta - b_0 x + x^2 w$ and imaginary coefficient $-x^2 C$. The real part is positive and at most $1 + 1/72 + \theta^3 < 13/12$, so its squared norm is below $6d/5$. Its actual mean $\int x^2 C \, d\nu$ is nonnegative. Since $\epsilon \le 0$, the cross with $\epsilon\mathbf{1}$ helps; its square costs at most $3d/16$. The complete norm is thus below $7d/5$. The scalar plus mean is at most $[-6 + 2\theta + \theta^2 + 2\theta/3]d$. Consequently its loss exceeds

$$24 - 8\theta - 4\theta^2 - 8\theta/3 - 119/20 = 50137/3840 > 13.$$

The two root transfers for this same packet.

For the minimum-one root, use Lemma Q.2 in the full-constant convention. At $L = 1$ the signed imaginary part gives a stronger direct comparison: its nonconstant norm increment is at most

$$\int (-x + x^2 + 3x^3) \, d\nu \le 0.$$

Indeed the real increment is at most $x^2 + 2x^3$, while the imaginary increment is at most $-x + x^3$. The scalar gain is $25d/8$ and the new curvature costs at most $10d/9$, giving a further improvement $145d/72$. For $a \ge 4$, $\Psi(1, 1/2) < 0$ suffices.

At $L = 2$, $A_Q \ge 0$, $D_Q \le 1$, $\|Qu\| \le (6/5)\sqrt{d}$, and $\delta \ge -\theta d$. The worst full-constant expense is

$$\Phi(1, 0, 1/3) + (17/6)\theta = 31177/36864 < 17/20;$$

the $a \ge 4$ expense is negative. At $L = 3$, take $\bar{D} = 1$, $R = 7/5$, $\epsilon \ge -3d/4$; the two expenses are below $2d$ and d, respectively. For the long row, $\bar{D} = 5/4$, $R = 6/5$, $\gamma = -9d/16$; the two expenses are below $(11/5)d$ and d. These bounds keep the complete constant vector in every row.

For the positive root use Lemma Q.3, again in its full-constant convention. At $L = 1$ the only adverse overshoot is the negative imaginary part of size at most x, costing $17d/72 < d$. At $L = 2$ its size is at most x^2, costing $17d/168$. The signed constant correction is at most $(17/8)d^2$, since $\delta \geq -\delta_2$ and $\delta_+ < d/24$. Together with the saved d, this row improves.

At $L = 3$, the real overshoot is at most $d/24+x^3$ and the imaginary coordinate is in $[0, 1]$. Its charge is below $(119/288)d^2 + (119/168)d < 17d/20$. The complete $\epsilon\mathbf{1}$ correction is below $(15/4)d^2 \leq 5d/4$. The increment is therefore below $11d/10$. For the long row $n = 3$ has coordinates $1-x+x^3C$, $1-b_0x+x^2w$, with only imaginary overshoot x^3. Its charge is below $17d/112 < d/5$. For every $n \geq 4$ the coordinate perturbations are at most $x^2(1/3 + x)$; they lie between the adjacent finite alternating prefixes, because $(4/3)x + x^2 < 1$. Thus they stay in $[0, 1]$. The negative constant $-hd\mathbf{1}$ costs less than $3d^2 \leq d$, paid by the root saving. The long increment is below $d/5$.

The resulting simultaneous payment table is

L	actual old full loss	minimum-one expense$/d$	positive-root expense$/d$
1	$25/4$	0	0
2	8	$17/20$	0
3	13	2	$11/10$
≥ 4	$3385/384$	$11/5$	$1/5$.

Every loss minus its indicated expense is greater than six.

The adjacent terminal $t = 4$.

Keep the actual $F = h_3f_4$ for every L. Its one packet is

$$\Gamma = 3d, \quad R = (d - zx)u, \quad \delta = \delta_2 + (5/4)(d^3 + \delta_4),$$
$$\epsilon = \delta - d^2 \geq 0, \quad \delta < 101d/160, \quad \delta^2 < 2d^2/5.$$

Every finite loss is at least $d/2$, by the signed moments of Section Q.6. Use the same four rows as (Q.24), with $r = d - zx$, $h = 1/2$, but with the actual curved convention. The auxiliary complete baseline losses can be taken

$$301/40, \qquad 9, \qquad 14, \qquad 547/60.$$

Here the unit and long rows are the original complete calculations. For $L = 2$, the actual $v = 3$ scalar plus mean is at most $-699d/160$; its nonconstant squared norm is below $16d/9$, and its curvature costs at most $4d/9$. The loss is thus greater than $699/40 - 68/9 - 4/9 = 379/40 > 9$. At $L = 3$ the nonconstant coordinates are $1-dx+\delta$, x^2, with squared norm below $3d/2$. The scalar plus mean is at most $-859d/160$ and the actual ϵ curvature costs at most $4d/9$. This gives loss $859/40-51/8-4/9 = 1319/90 > 14$.

For the minimum-one root, the four curved expenses are at most $0, 0, 3/5, 21/10$, using respectively $(\bar{D}, R) = (1, 6/5), (1, 6/5), (5/4, 3/2), (3/2, 8/5)$. All resulting losses exceed six on the full domain. For the positive root first take $d \leq 1/8$. The first row improves as above. In the second row, real overshoot x^2 costs at most $119d/72$, and its actual curved correction is less than d. After the saved d, its increment is less than $2d$. In the third row the two overshoots cost less than $9d/8$; its new virtual coordinate is nonnegative, so $\epsilon\mathbb{E}V_* \leq 0$. Only $11\epsilon^2/12 \leq 11d/240$ remains from the curved correction. The increment is below $2d$. The long tail has norm below $(8/75)\sqrt{d}$; the integral comparison costs $47d/225$. Its curved correction is below $9d^2 \leq 9d/8$, again leaving an increment below $2d$. The baseline losses therefore give at least six in all four windows.

On $d \ge 1/8$, the actual arm sum is $h_3 + f_4 = 2 - 3d + du + B^3u$. The centered-row inequality gives $|du + B^3u| \le \sqrt{d^2 + \delta_3} \le 2d$, since $\delta_3 \le \theta^2 d \le 8\theta^2 d^2 < 3d^2$. Thus this same actual $F = h_3 f_4$ satisfies $F \le e^{-d}$. Thus $T^L F \le e^{-d} \bar{f}_L$, and Corollary Q.4 gives positive-root loss at least $4 + 387/128 > 6$ with the same directions.

Actual full vectors and later arms.

On $d \le 1/64$, the packet rows above have $\|Q(B)u\| \le (3/2)\sqrt{d}$ and $|\gamma|/\sqrt{d} \le 3/32$. The minimum-one root adds a nonconstant precursor of norm at most $\sqrt{d}/2$; its constant or curve is the one used in the proof. The positive root adds its actual anchor precursor of norm below $(6/5)\sqrt{d}$ and its pair precursor below $(3/8)\sqrt{d}$. Consequently the complete vector is below $(3/2+6/5+3/8+3/32)\sqrt{d} < (7/2)\sqrt{d}$. This proves (Q.23) and permits the existing paid restoration of every later ordered root pair and independent outer connector. Equal pairs are restored by positivity. At $d = 0$ leaf integration gives one, with no requirement that $B = 0$. □

Q.17 The odd terminal packets with a unit arm

Let F have the literal directions in Table 108. The two root classes considered here are $pW_a \bar{f}_b$, $3 \le a < b$, and $f_a h_b f_c$, $3 \le a < b < c$. Their terminal messages and their inner connector L are the same. We prove the respective global fourth losses $705/128$ and $11/2$ by comparing their complete certificates with the auxiliary root pP_0 of Lemma Q.1.

The propagated packet.

Every terminal in this section has an actual first-row upper

$$TF \le e^{\alpha}(\bar{f}_2 + BR + D), \qquad R = r(B)u, \quad D \ge 0. \tag{Q.26}$$

For a retained square $\alpha = 2\log m$; for an exponential terminal $\alpha = -\Gamma$. Put $D_i = \mathbb{E}B^i R$ and $\gamma_j = \beta_j - D_{j-2}$. The finite identity (M.68), applied before discarding any term, gives the parameters of (Q.8):

L	C	c	Q	$\mathbb{E}BQ(B)u$
1	1	D	$1 + r$	$-d + D_1$
$L \ge 2$	1	$2D - \gamma_{L+1}$	$G_L + z^{L-1} r$	$-\gamma_{L+2}$.

(Q.27)

The long row uses the proved nonnegative losses $\gamma_3, \dots, \gamma_{L+1}$, and its mean uses the additional actual index $L + 2$. Its auxiliary-root exponent is exactly

$$J_0 = 4(\alpha + c + \mathbb{E}BQu - d) + (17/4)\|(Q - B)u\|^2. \tag{Q.28}$$

In particular the constant split is $\gamma = C - 1 = 0$ throughout.

If $r = -t + zs$ and $0 \le t, s \le 1$, both coordinates of every finite Q_L lie in $[0, 1]$. Indeed $Q_L = G_{L-1} + (1 - t)z^{L-1} + sz^L$: each parity coordinate is an alternating finite partial sum with its final coefficient between zero and one. Convexity in t, s reduces to the four adjacent partial sums. This includes $Q_1 = (1 - t) + zs$ and $Q_2 = (1 - xs) + z(1 - t)$.

Canonical triples $(1,2,w)$**.**

Use the actual square $F = qf_2h_w$. Its three retained packets are

w	$d_{\max}$	m	r	D
3	$1/3$	$1-d/2$	$-kx$	$d/4$
4	$1/4$	$1-d$	$-k(d+x)+zkx$	$5d/12$
$w\ge 5$	$1/5$	$1-9d/8$	$-ka+zkb$	$2d/5$.

$k = m^{-1}$.

The first two rows are the actual sums and signed moments in Section Q.5. The last is Lemma M.10 with $\sigma = 9/4$; its actual a, b, U and all four residue classes of D_i are retained. In all three rows $0 \le t, s \le 1$, and (Q.28) has respective uniform losses $2817/512$, $705/128$ and 6. These are complete $v = 3$ certificates: the auxiliary pair is P_0 in either original direction, and its β_2 curvature coefficient is zero.

At a minimum-one root use $\Phi(1,0,1/3) = -11255/36864 < 0$ and $\Psi(1,6/5) = -8899/3840 < 0$. Thus every low-domain row has loss greater than $705/128$. On the complementary domain, the same extra arm satisfies $h_4 \le 1$ for $d \ge 1/4$, and $h_w \le 1$ for $w \ge 5$, $d \ge 1/5$. The complete message is then bounded by its actual qf_2 terminal. Proposition O.1, applied to the one pair (a,b) and terminal parameter two, supplies $705/128$.

For a positive root the $w \ge 5$ low domain must extend to $1/4$. Use instead the $\sigma = 2$ row of Lemma M.10: $m = 1-d$, $R = (-ka+zkb)u$, $D = d/2$, $a < 7/10$, $b < 3/5$. Its own signed moments give $\gamma_3 \ge 0$, $\gamma_4 \ge 2d/3$, $\gamma_j \ge 41d/96$ for $j \ge 5$, including the mean index. The actual short-window values of (Q.28) have losses at least $8, 6, 187/24, 47/6$ for $L = 1,2,3,4$. For $L \ge 5$ its base has modulus at most $23/16$ and its perturbation at most $(6/5)x^2$. The latter adds less than $2091d/800$ to the complete norm payment; the adjacent actual losses satisfy $\gamma_{L+1} + \gamma_{L+2} \ge 41d/48$. Consequently J_0 has loss at least $43/6 - 2091/800$. The positive-root comparison of Lemma Q.3 saves at least d on every one of these rectangles. The weakest resulting loss is

$$49/6 - 2091/800 = 13327/2400 = 11/2 + 127/2400. \tag{Q.29}$$

For $w = 3, 4$ the same comparison gives $3329/512$ and $833/128$.

For the positive-root high domain use the actual reference estimate of Proposition N.3, $\mathbb{E}[f_a X^4] \le 1$ for $X = T^L(qf_2h_w)$, $a \ge 3$, $w \ge 4$. The remaining factor is $f_a^3(h_bf_c)^4$. For $(a,b) = (3,4)$ the actual pair loss $69d/20$ and precursor norm $3\sqrt{d}/8$ give $(h_bf_c)^4 \le e^{-54d/5}$ on $d \ge 1/4$. For $b \ge 5$ use $16d/5$ and $\sqrt{d}/4$, with the same result. If $a = 3$, then $f_3 \le 3/2$ and $3\log(3/2) < 5/4 \le 5d$; if $a \ge 4$, $f_a \le 1$. The complete high-domain loss is therefore at least $29/5$. This argument also applies to the canonical qf_A reference estimate (N.5) for $A \ge 3$ in the following triple rows.

The other single triples.

For $F = qf_Ah_C$, the actual exponential first row is (Q.26) with $\alpha = -\Gamma$ and $d \le 1/4$. For $A = 3$,

$$r = -\beta_{C-1} + x^2V + zxU, \qquad 0 \le U, V \le 1, \quad |r| < 3/5, \quad D = 7d/20,$$

with the true $G_1 = (1,0)$ retained at $C = 4$. For $A \ge 4$ use the actual four-beta packet

$$r = -c + (-1)^A z^A G_{C-A}(-z), \qquad c = \beta_{C-1} - \beta_{A-1}.$$

Its $(\Gamma/d, D/d, \rho)$ are $(51/16, 1, 3/10)$ at $A = 4$, $(89/32, 2/3, 1/4)$ at $A = 5$ and $(19/8, 7/20, 1/8)$ at $A \ge 6$. These are the same actual packets and all signed moment indices proved in Sections Q.7

and Q.8. Substitution in (Q.28), with zero auxiliary curvature, gives the following losses and positive-root increments. An increment already includes the saved root loss.

	auxiliary loss			positive-root increment upper		
A	$L=1$	$L=2$	$L\ge 3$	$L=1$	$L=2$	$L\ge 3$
3	$157/20$	$159/20$	$129/20$	$2/3$	0	0
4	$29/4$	$173/20$	$133/20$	$7/4$	$121/100$	$61/160$
5	$57/8$	$1183/120$	$959/120$	1	$121/100$	$61/160$
$A\ge 6$	$41/5$	$231/20$	$191/20$	$5/12$	$121/100$	$61/160$.

The $A=4, L=1$ increment is strict. Here are the signed coordinate charges that prove this table directly. At $A=3$, Q_1 has real overshoot at most x^2; its fee is below $119d/72$, before the saved d. At $L=2$ only the imaginary coordinate may exceed one, by x^2; its fee is $(17/2)(\theta^4+\theta^5)d<d$. Lengths three and four are in the rectangle. Every later tail is at most $x^2(1/4+x)$, below the finite partial-sum margins since $x(5/4+x)<1$ on $[0,\theta]$. At $A=4, L=1$, the real and negative imaginary overshoots are at most δ_2+x^2 and x^2, respectively. Keeping the stronger $-11d/10$ gives the strict increment

$$-11/10+1547/576+17/168=34009/20160<7/4.$$

At $A=5$ the real overshoot is only δ_2, giving an increment below d. At $A\ge 6$ use the integral norm version with $\rho=1/8$, giving $5d/12$. At $L=2$ and $L\ge 3$ the largest respective perturbations $39/200$ and $39/320$ give $121d/100$ and $61d/160$.

For a minimum-one root the $A=3$ polynomials have real coordinate nonnegative, imaginary coordinate at most $5/4$ and norm at most $3\sqrt{d}/2$. Their Φ cost is below $3d/5$ and their Ψ cost is negative. For $A\ge 4$ use instead $\bar{D}=4/3$, $R=3/2$ and $\Phi(4/3,0,1/4)=3063/4096<3/4$; $\Psi(4/3,3/2)<0$. The low-domain losses exceed $705/128$. In the high domain $h_C\le 1$ in the unchanged direction. At a minimum-one root apply Proposition O.1 with terminal parameter A. At a positive root use (N.5) and the preceding remaining-factor payment. The respective losses are $705/128$ and $29/5$.

Every odd multipair terminal of minimum one.

Write the ordered residual list as $(1,A_1,C_1,\dots,A_j,C_j)$, $j\ge 2$. The canonical direction is $qf_{A_1}h_{C_1}\prod_{i\ge 2}h_{A_i}f_{C_i}$. When $(A_1,C_1,A_2)=(2,3,4)$ use the fixed cross $qf_2h_{C_2}h_3f_4\prod_{i\ge 3}h_{A_i}f_{C_i}$. The single finite mask and its actual first-row moments are those of Section Q.9. The five records below give d_*, Γ/d, D/d, a modulus bound ρ for r, and the complete auxiliary losses at $L=1,2,\ge 3$.

	d_*	Γ/d	D/d	ρ	1	2	≥ 3
I : $A_1\ge 3$	$1/64$	$431/80$	$5/3$	$1/2$	$1333/160$	$659/96$	$3799/480$
II : $A_1=2, C_1\ge 4$	$1/64$	$351/80$	$11/10$	$2/3$	$6287/480$	$1027/96$	$715/96$
III : $A_1=2, C_1=3, A_2\ge 5$	$1/8$	$21/5$	$11/10$	$3/5$	$719/60$	$115/12$	$47/6$
IV : cross, $j=2$	$1/8$	$21/4$	$11/10$	$1/2$	$971/60$	$827/60$	$757/60$
V : cross, $j\ge 3$	$1/8$	$169/20$	$7/4$	$3/5$	$953/60$	$983/60$	$589/30$.

The table conservatively keeps the original uniform root-curvature fees $5d/96$ or $5d/12$; their actual auxiliary values are zero. The same four residue classes prove all finite γ signs.

For the minimum-one transfer every Q_L has nonnegative real coordinate, imaginary coordinate at most $3/2$ and norm at most $8\sqrt{d}/5$. Indeed the actual $r=A_r+zD_r$ satisfy $-2/3\le A_r\le 1/2$, $|D_r|\le 2/3$. For $L=1,2$ use the explicit rows $(1+A_r,D_r)$ and $(1-xD_r,1+A_r)$; for $L\ge 3$ use $1-x-\rho x>0$, $1+\rho\sqrt{x}<3/2$ and $|Q_L|\le 1+\rho x<8/5$. Thus Φ costs less than $11d/10$ in I–II

and $6d/5$ in III–V, and Ψ costs less than $7d/10$ throughout. Every residual loss exceeds $705/128$, for example $659/96 - 11/10 = 2767/480$.

The positive-root increments are controlled by the same signed overshoot functional, as follows:

	$L = 1$	$L = 2$	$L \ge 3$
I	$1547/600 + 17/112 - 1$	$4/5$	$125/96$
II	-1	-1	-1
III	0	0	$141/80$
IV	0	0	$125/96$
V	$29/5$	$171/50$	$141/80$.

For I, $A_r \le 13/50$ and $D_r \in [-1/4, 2/3]$. At $L = 1$ these give the two fees $1547d/600$ and $17d/112$; at $L = 2$ the real and imaginary overshoots are at most $x/4$ and $13/50$, giving a fee below $9d/5$ before saving d. For II, $r = -t + zs$ lies in $[0, 2/3]^2$ in its signed coordinates, so every finite row is in the rectangle. For III, $0 \le t \le 2/3$, $|s| \le 1/5$; the first two fees are $17d/140$ and $119d/144$, both below the saved d. For IV, $0 \le t \le 19/40$, $-x^2 \le s \le 19/128$; the corresponding fees are $17d/168$ and $17d/24$. For longer rows of I, III and IV use the integral norm estimate with ρx. For V use that estimate at every length with $\rho = 3/5$, $\sqrt{\theta} < 13/20$ and $\theta = 13/32$. Each entry subtracted from its auxiliary loss leaves more than six.

For $d \ge 1/64$ every later actual pair with minimum at least four is at most $e^{-88d/125} \le 1$. Hence I–II compare with their first triple. On $d \ge 1/8$, III can be written as the same product

$$F = (q f_2 h_{A_2})(h_3 f_{C_2}) \prod_{i \ge 3} h_{A_i} f_{C_i};$$

IV–V already have the analogous factors $q f_2 h_{C_2}$ and $h_3 f_4$. The actual centered-row proof gives, on this domain,

$$h_3 f_t \le e^{-d}, \qquad f_3 h_t \le e^{-d}, \qquad t \ge 4. \tag{Q.30}$$

For $t \ge 5$ its centered fluctuation has absolute value at most $\sqrt{a_0^2 + \operatorname{Var} U} \le a_0 + d$; here $U = \sum_{r=3}^{t-1} K^r \mathbf{1}$ and $a_0 = \beta_{t-1} - \mathbb{E}U = \Gamma_t - 2d \ge 0$, with $\Gamma_t = d + \beta_t + \beta_{t-1}$. The centered original row gives this bound for $a_0 g - K(U - \mathbb{E}U)$, $g = -u$, since its two row coordinates satisfy $g^2 + pq = 1$. The finite $t = 5$ row uses $8\theta^2(1+\theta) < 35/16$, and the $t \ge 6$ tail uses $\operatorname{Var} U \le \theta^2 d \le 11d^2/8$. At four use $\sqrt{d^2 + \delta_3} \le 2d$ in the exact arm sum. The two literal sums have opposite signs of this same centered fluctuation, so its absolute-value bound proves both inequalities. The resulting actual comparison is with the canonical triple already proved, in the same arm directions. This gives the claimed global losses in both root classes.

All parameter intervals include their nonzero endpoints. At $d = 0$, every vector here vanishes and leaf integration gives the value one.

Q.18 Even terminal packets at the two remaining root types

Use the root and input conventions of Section Q.14. The actual even terminal is $F = \prod_{i=1}^{j} h_{a_i} f_{b_i}$, with its original strict ordering. Every pair keeps this direction in every host interval.

Proposition Q.6. *For either root $p W_a \bar{f}_b$, $3 \le a < b$, or $f_a h_b f_c$, $3 \le a < b < c$, and every even terminal residual of size at least four, the actual complete message has fourth loss at least $11d/2$. The same holds for a terminal single pair whose minimum is at least four. On $0 < d \le 1/64$ the proof retains a pointwise complete certificate whose full vector has norm below $(7/2)\sqrt{d}$.*

Proof. **One finite propagation, including its constants.**

Use the bare-exponent parameters (Q.9)–(Q.10) for the actual mask $R = r(B)u$ of each terminal. The scalar, complete constant and signed means are kept in the chosen convention.

The five smaller-minimum finite packets.

Use the actual masks of Section Q.12. Their original scalar is increased by $16(j-2)d/5$ for every extra terminal pair, while the same disjoint mask bounds all coordinates. The data needed here are

packet	Γ/d lower	δ/d range	ϵ/d range	h	actual finite coordinates of r
$E2$	$607/160$	$[-1/5, 91/100]$	$[-7/10, 7/10]$	$1/2$	$x/2 \le A_r \le 6x/5$, $-6x/5 \le D_r \le x/2$
$E3$	$767/160$	$[-1/5, 2/3]$	$[-7/8, 7/8]$	$1/2$	$\lvert A_r\rvert \le C_\beta x^2$, $\lvert D_r\rvert \le C_\beta x$
I	$607/160$	$[-6/5, 3/4]$	$[-17/10, 3/4]$	$3/8$	$r = xV + zU$, $1/2 < U, V < 6/5$
II	$16/5$	$[-6/5, 1/25]$	$[-13/10, 6/25]$	$3/8$	$r = xV + zU$, $11/12 \le U \le 6/5$
III	$3 - 2\delta_2/d$	$[-1, 3/20]$	$[-4/3, 7/20]$	$1/2$	$r = \tau + xV + zU$, $1 - C_\beta x \le U \le 1 - x/3$.

Here $C_\beta = (1-\theta^2)^{-1} = 1024/855$; II and III retain $-C_\beta x \le V \le C_\beta x^2$. The packets are respectively minimum two, minimum at least three, $(1, b_1)$ with $b_1 \ge 3$, $(1, 2, a_2)$ with $a_2 \ge 4$, and $(1, 2, 3, ...)$. Their forced missing exponents, original means and all-index losses are those of the actual masks, not freely chosen parameters. In III, τ is the original global moment.

At the auxiliary root use the curved convention. Complete baseline losses, in the four windows of (Q.9), are

	1	2	3	≥ 4
$E2$	$185/24$	$101/12$	12	$49/5$
$E3$	12	11	12	14
I	10	10	$821/90$	$42/5$
II	12	9	$10181/900$	10
III	$15/2$	7	$109/15$	$697/100$.

(Q.31)

All but three entries are the same complete calculations in Section Q.12, with the actual $v = 3$ pair envelope. The three specializations are short.

For E2 at $L = 2$, the actual nonconstant vector is $(1 - B)u + BR$, with squared norm below $5d/2$ since $\|BR\| < (37/100)\sqrt{d}$. Its curvature coefficient is δ, not $\beta + \delta$; $(10/3)\delta^2 < d$ on the entire domain. Keeping $\epsilon \le 7d/10$ and $D_2 \le d/12$ gives the full loss

$$607/40 + 8 - 14/5 - 1/3 - 85/8 - 1 = 101/12.$$

For E2 at $L = 3$, the old unperturbed vector is exactly u, whose jointly paid base is $-31d/4$. The actual perturbation $\delta u + B^2 R$ has norm coefficient at most $83/150$. Its added norm cost is $(17/4)[2(83/150) + (83/150)^2]d$. The additional signed mean costs at most $21d/5$, and the actual ϵ curvature costs at most $49d/90$. Their substitution gives loss greater than twelve on the whole domain. For III at $L = 2$, retain the actual zero β_2 and $|\delta| \le d$: its complete loss is at least

$$35/4 - 7/5 + 8 - 13/8 - 85/16 - 10/9 > 7.$$

These estimates specialize the full vectors before taking moments.

Minimum-one roots: one parameter payment.

The complete curved expenses from Lemma Q.2, divided by d, are bounded by the following table for all $0 < d \le 1/3$:

	1	2	3	≥ 4
$E2$	$11/10$	0	$3/2$	$21/10$
$E3$	$11/10$	0	$7/4$	$21/10$
I	$11/10$	$13/10$	2	$21/10$
II	$11/10$	$13/10$	$3/2$	$21/10$
III	$11/10$	$21/20$	$3/2$	$5/4$.

For the first window $A_Q \ge -1/5$, $D_Q \le 6/5$; the signed real-coordinate variant of Φ gives the common first column. For the second, $A_Q > 0$, $D_Q \le 1$, and $\gamma = \delta$; insert the actual lower endpoints of δ. For the third, $\gamma = \epsilon$, $D_Q \le 5/4$ in E2/E3 and $D_Q \le 1$ in I–III. The actual lower endpoints of ϵ give the third column. All these Q have norm at most $(8/5)\sqrt{d}$, and the $a \ge 4$ expense is smaller by Ψ.

For the long window all real coordinates are positive. In the first four packets use $D_Q \le 3/2$, $R = 8/5$ and $\gamma = -hd$, $h \le 1/2$. For III the actual coordinate is smaller: at $L - 1 = 3$ it is $1 - x\tau - x^2 V \le 1 + C_\beta x^3 < 5/4$. For later finite lengths its upper is at most $1 - x + 2x^2 + \tau x^{3/2} \le 1$, since $2\theta + (2/9)\sqrt{\theta} < 1$. Thus $\Phi(5/4, 1/2, 1/3) < 5/4$ gives the last entry. Subtracting this table from (Q.31) leaves more than $11/2$ in every cell; the smallest displayed margins are $109/15 - 3/2$ and $697/100 - 5/4$.

Positive roots: the same rows with complete constant charges.

Use the actual $\gamma = 0, \delta, \epsilon, -hd$ in the four windows, now in (Q.18). For E2/E3 first take $d \le 1/8$; for I–III take $d \le 1/4$. The following are upper bounds for the full exponent increments, including the signed constant cross and square:

	1	2	3	≥ 4
$E2$	0	$67/30$	$37/28$	$145/72$
$E3$	$13/4$	$261/80$	$23/6$	$151/120$
I	0	$5/2$	3	0
II	0	$423/140$	1	0
III	0	0	$17/10$	0.

(Q.32)

Here is the common calculation of these entries. E2's second polynomial increment, including the saved d, is at most $59d/60$; its curved correction is less than $10d^2$, giving $67d/30$. Its third charges are at most $10d^2 + 17d/42$ and $(16/3)d^2$; the saved d gives $37d/28$. The long charges are below $8d/9$ and $9d^2$, giving $145d/72$. E3's polynomial charges are $13d/4$, $141d/80$, $11d/6$, $2d/15$; the last three constant corrections are $3d/2$, $2d$, $9d/8$. These use the actual D_i in the signed mean, not a frequencywise interpretation of $\|R\|$.

For I, the second curved correction is below $14d^2$ and its polynomial is rectangular. At $L = 3$, real overshoot costs $119d^2/16$. The signed virtual interval is $[-x^2 - 6x/5, x/2 + x^2]$; splitting ϵ by sign bounds its cross and square by $(867/160 + 3179/1200)d^2$. The total is below $16d^2$. This proves the second and third entries after the saved d. For II, the second polynomial charge is $17d/140$ and its curved correction at most $(78/5)d^2$. At $L = 3$ the virtual interval is $[-x^2 - 6x/5, x/12 + x^2]$; its signed curved correction is less than $4d^2$, and the polynomial charge is less than d. For III, the second virtual coordinate is negative, so $\delta < 0$ helps; the whole curved correction is less than $2d^2$. At $L = 3$

real overshoot costs less than $6d/5$. The new virtual interval is $[-x-x^2,(11/5)x^2]$. For negative ϵ its cross plus square is at most $(187/45+44/27)d^2<6d^2$; positive ϵ costs less. After the saved d this is $17d/10$.

In the long I–III windows, the actual finite alternating coordinates are in $[0,1]$ for every $L-1\ge 4$. At $L-1=3$, the possible virtual overshoot is at most $C_\beta x^3$, and is absent in I. Its charge is $85d/504$. The respective signed curved corrections are below $2d^2$, $(5/2)d^2$, $(49/16)d^2$. Each total is less than the saved d on its stated interval. This proves the zero final entries, including the global τ in III and every finite highest exponent. Subtracting (Q.32) from (Q.31) leaves at least $11/2$; the tight III third row is $109/15-17/10=167/30>11/2$.

On the complementary E2/E3 domains the original finite row bounds give the same actual terminal a pointwise scalar surplus. For E2 it is at least $127d/160$; the bare-root bound gives $387/128+127/40>6$. E3 has larger surplus. For I, the paid later pairs leave qf_{b_1}. For II/III they leave the canonical $qf_2h_{a_2}$; $f_{b_2}\le 2(1-2d)\le 1$ there. The same-direction bare reference or canonical odd-triple payment below gives $11/2$. These are bounds on the same products; no terminal arm is reversed at a host-domain boundary.

Larger-minimum even lists.

For $a_1\ge 4$, use the independent actual mask of Section Q.10, with $\Gamma=16jd/5$, $\|R\|<\sqrt{d}/4$, $|\delta|\le d/5$, $|\epsilon|\le 3d/8$ and all losses at least $d/2$. The same four rows (Q.9) apply. At a minimum-one root their direct complete vectors have norm coefficients $3/4,5/3,28/15,25/16$ and curve coefficients at most $d,6d/5,11d/8,3d/2$. Their scalar-plus-mean bounds, before $-\Gamma$, are respectively $-45d/32,-223d/96,-1523d/480,-3709d/1440$. The direct certificate therefore gives the original loss $8+64(j-1)/5$ on the entire domain.

For a positive root take $d\le 1/64$. The auxiliary full-constant baselines are $10,29/5+395/128,47/4,8$. The second and third actual $v=3$ base savings are $395/128$ and $19/4$ relative to their general-root bounds. The full-constant increments are bounded by $11d/6,7d/8,d/3,0$. These follow from the same norm comparison with tail coefficients $1/4,13/80,1/9,1/15$, retaining $\delta\mathbf{1}$, $\epsilon\mathbf{1}$ and $-(d/2)\mathbf{1}$ inside the full vectors. Every resulting loss exceeds eight. For $d\ge 1/64$ the actual pair estimate gives $F\le e^{-88jd/125}$; the paid bare root gives $387/128+352j/125>11/2$.

Finally, on $d\le 1/64$, the four rows in (Q.9) obey $\|Qu\|+|\gamma|<(19/10)\sqrt{d}$. For the short windows this follows from $\rho\le 1$, $|\delta|\le 6d/5$, $|\epsilon|\le 17d/10$; the third window has bound $147\sqrt{d}/80$. The long bound is $(167/120)\sqrt{d}$. Adding the actual anchor and pair precursors gives at most $(19/10+6/5+3/8)\sqrt{d}<(7/2)\sqrt{d}$. The complete later-root certificate is therefore available in both directions. Equal pairs and the independent outer connector are restored by the existing common lemmas. At zero, leaf integration gives equality. □

Q.19 Odd terminals with their actual inward path

Write the strictly increasing terminal list as $(A,b_1,c_1,\dots,b_j,c_j)$, $A\ge 2$, $j\ge 1$. If $b_1\ge 4$ use $F=\bar f_A\prod_i h_{b_i}f_{c_i}$. If $b_1=3$, the order forces $A=2$; use $F=\bar f_2 f_3 h_{c_1}\prod_{i\ge 2}h_{b_i}f_{c_i}$. We prove fourth losses $13/2$ at $pW_a\bar f_b$, $3\le a<b$, and 6 at $f_ah_bf_c$, $3\le a<b<c$, keeping these directions for every positive inner connector L.

The first pair has minimum at least four.

Let $R = r(B)u$ be the one finite sum of the actual pair intervals, and Γ their total scalar loss. Put $\mu_A = \mathbb{E}\bar{f}_A$ and $\sigma_A^2 = \operatorname{Var}\bar{f}_A$. The actual centered first row of Section Q.11 is

$$\begin{aligned} T(\bar{f}_A e^R) &\le \bar{f}_{A+1} + \mu_A BR + \delta, \\ \delta &= \mu_A \mathbb{E}R + 2\sigma_A \|R\| + (12/5)\|R\|^2. \end{aligned} \tag{Q.33}$$

The centered product is bounded in the original measure by $2\mathbb{E}|(\bar{f}_A - \mu_A)R|$. The mixed mean and variance use the same R. The three finite-mask records are

	Γ/d	$\rho : \|R\| < \rho\sqrt{d}$	δ_+/d	$d^{-1}\sum_{i\ge1}(D_i)_+$
$j = 1, b_1 = 4$	$69/20$	$1/5$	$72/125$	$7/25$
$j = 1, b_1 \ge 5$	$16/5$	$3/20$	$3/5$	$1/5$
$j \ge 2$	$16j/5$	$1/4$	$5/6$	$2/5$.

Here $D_i = \mathbb{E}B^i R$. The first row uses $A = 2, 3$ and $\sigma_A < 6\sqrt{d}/5$; at higher first-pair minimum use $\sigma_A < 3\sqrt{d}/2$ and its matching stronger mask.

Finite propagation retains the actual inward anchor:

$$e^{\Gamma} T^L F \le \bar{f}_{A+L} + \mu_A B^L R + \delta \bar{f}_{L-1} + \mu_A \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{Q.34}$$

Thus (Q.8) has $\alpha = -\Gamma$, $C = 1$,

$$Q = G_{A+L-1} + \mu_A z^{L-1} r, \qquad \mathbb{E}BQu = -\beta_{A+L+1} + \mu_A D_L,$$

and $c = \delta - \beta_{A+L}$ at $L = 1$, or $c = 2\delta_+ + 2\mu_A \sum_{i<L}(D_i)_+ - \beta_{A+L}$ at $L \ge 2$. The true anchor mean μ_A remains in both terms. Substituting these data into the auxiliary exponent (Q.12) gives losses greater than $13/2 + 10/9$ for one pair and $15 + 10/9 + 64(j-2)/5$ for several pairs. To verify these numbers directly, the complete scalar and joint-norm bounds are

	$-(\alpha + c + \mathbb{E}BQu - d)/d$		$\|(Q-B)u\|/\sqrt{d}$	
	$L = 1$	$L \ge 2$	$L = 1$	$L \ge 2$
$j = 1, b_1 = 4$	$69/20 + 51/32 + 1 - 72/125 - 1/6$	$7711/2000$	$131/80$	$133/100$
$j = 1, b_1 \ge 5$	$16/5 + 19/16 + 1 - 3/5 - 1/30$	$889/240$	$127/80$	$13/10$
$j \ge 2$	$16j/5 + 277/240$	$16j/5 - 29/80$	$27/16$	$11/8$.

There is no auxiliary β_2 curvature cost; the $10/9$ just reclaims the isolated uniform fee in the corresponding general root certificate of Section Q.11.

For a minimum-one root these polynomials have nonnegative real coordinate, imaginary coordinate at most $5/4$ and norm at most $3\sqrt{d}/2$. For the coordinate assertions use the actual mask, whose first degree is at least four: $|\operatorname{Re} r|, |\operatorname{Im} r/z| \le (1024/855)x^2 < 1/5$. The real coordinate is at least $\kappa - 1/5$, while the imaginary coordinate is at most $6/5$. The norm follows from $\|R\| < \sqrt{d}/4$, $|G_2| < 6/5$ and $|G_n| \le 1$ for $n \ge 3$, keeping $\mu_A \le 1$. Their Φ increment is below $3d/5$; $\Psi(5/4, 3/2) < 0$. The full-domain loss exceeds $13/2$. For a positive root write $Q = G_{A+L-1} + \Delta Q$ and use the integral form (Q.19). The actual tail norms are at most $\rho\sqrt{d}$, including $L = 1$ and every finite A, L. The increments are at most $19d/15$, $7d/10$ and $11d/6$ in the three rows. The weakest resulting loss is $13/2 + 10/9 - 19/15 = 571/90 > 6$.

The first pair has minimum three.

The actual short anchor is $\bar{f}_2 = 1 + u + Bu$. Let $c = c_1 \geq 4$, $B_0 = \beta_{c-1}$ and

$$r_0 = -B_0 - z^3 G_{c-3}(-z) = -B_0 + x^2 V + zxU, \qquad \Gamma_0 = d + \beta_c + B_0 \geq 35d/16.$$

At $c = 4$, $(U, V) = (1, 0)$; thereafter the coordinates of the same finite $G_{c-3}(-z)$ are used. For one pair take $R = r_0(B)u$, $\Gamma = \Gamma_0$. For several pairs add the one actual finite later mask S to R and its true scalar losses to Γ. On $d \leq 1/8$, both cases have the centered row

$$\begin{aligned} &T(\bar{f}_2 e^R) \leq \bar{f}_3 + BR + D, \\ &\qquad D \geq \max\{\mathbb{E}R + \mathbb{E}(uR) + \mathbb{E}((Bu)R) + \sqrt{(d+\delta_2)\mathbb{E}R^2} + (12/5)\mathbb{E}R^2, 0\}. \end{aligned} \tag{Q.35}$$

The linear coefficient is one, rather than μ_2; all three mixed means are retained. The actual finite heads $c = 4, 5$ and the remaining contracting tail give the following records:

	Γ/d	D/d	D_1/d	D_2/d	D_i/d $(i \geq 4)$
$j = 1$	$35/16$	$3/5$	$[-\delta_3/d, d]$	$[0, \delta_3/d]$	$\leq 1/36$
$j \geq 2$	$431/80 + 16(j-2)/5$	$3/2$	$[-19/60, 19/120]$	$\leq 1/4$	$\leq 1/9$.

Also $D_3 \leq \delta_4$ in the first row and $D_3 \leq d/6$ in the second. These are the genuine first-row and signed-tail certificates of Section Q.11 for the displayed short anchor.

The finite propagation is

$$e^{\Gamma} T^L F \leq Y_L := \bar{f}_{L+2} + B^L R + D\bar{f}_{L-1} + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \tag{Q.36}$$

Expanding its first three actual paths gives the exact rows below for (Q.8), with $\alpha = -\Gamma$:

L	C	c	Q
1	1	$D - d$	$G_2 + r$
2	$1 - d + D$	$D_1 - d$	$G_3 + zr$
3	$1 - 2d + D + D_1$	$-\tau + D_2$	$G_4 - d + D + z^2 r$.

(Q.37)

For example Y_3 retains both losses in $\bar{f}_5 = \mathbf{1} + G_5 u - d\bar{f}_2 - dq - \tau\mathbf{1}$; this gives the displayed coefficient of q and the separate constant $-\tau + D_2$. Its actual mean is $-\tau + d^2 + D_3 - Dd$.

For $j = 1$ put $h = 2d/5$. The loss coefficients in the same convolution are $\gamma_3 = d - D = h$ and $\gamma_i = \beta_i - D_{i-3} \geq h$, $i \geq 4$. At $L \geq 4$ retain the final four paths $\bar{f}_0, \ldots, \bar{f}_3$. Their sum is $4 - d + 3u + 2Bu + B^2 u$, giving

$$C = 1 - 3h, \qquad c = -h(1-d), \qquad Q = G_{L+1} + z^{L-1} r - h(2+z), \quad \mathbb{E}BQu = -\gamma_{L+3} + 2hd. \tag{Q.38}$$

For $j \geq 2$, the actual first coefficient is instead $\gamma_3 = d - D = -d/2$. Keep its positive contribution $(D-d)\bar{f}_{L-1} \leq d$, and take $h = d/3$ for every later loss, including the mean index. Retaining the final three paths $\bar{f}_0, \bar{f}_1, \bar{f}_2$ gives

$$C = 1 - 2h, \qquad c = d - h, \qquad Q = G_{L+1} + z^{L-1} r - h, \quad \mathbb{E}BQu = -\gamma_{L+3} + hd. \tag{Q.39}$$

Both rows follow from (Q.36); their different constants are kept in the subsequent full vectors.

One complete payment for both root classes.

The exact curved baseline (Q.12) and the two transfer lemmas give the following sufficient ledger. An entry is a loss for J_0, followed by an upper bound on its increment at the stated root. For multiple pairs add $64(j-2)/5$ to every baseline.

	$L=1$	$L=2$	$L=3$	$L\geq 4$
$j=1:\ -J_0/d$	$1173/100$	$5573/600$	$4291/320$	$799/80$
minimum-one increment	2	2	2	2
positive increment	2	0	$49/12$	$10/3$
$j\geq 2:\ -J_0/d$	12	16	19	16
minimum-one increment	$7/2$	$7/2$	$7/2$	$15/4$
positive increment	$29/5$	$171/50+9/8$	$25/9+15/8$	$11/6+3/2.$

Here is a direct check of the entries that differ from an unweighted path input. For $j=1, L=1$, the actual auxiliary joint vector has squared norm at most $(1+x^2)^2+x^3<36/25$. The scalar is at most $-(35/16+91/40)d$, and $C-1=0$. This gives the first loss $35/4+91/10-153/25=1173/100$. For the other single-pair rows, the respective scalar upper bounds are $-1867d/480$, $-211d/40$ and $-403d/80$; their joint squared-norm caps are $36d/25$, $25d/16$ and $9d/4$. Keeping the actual curved coefficients from (Q.37)–(Q.38) gives the displayed weaker baseline losses. For several pairs the same calculation uses scalar upper bounds $-\Gamma-161d/120$, $-\Gamma-329d/480$, $-\Gamma-159d/80$ and $-\Gamma-31d/24$.

The single-pair polynomials have real coordinate nonnegative, imaginary coordinate at most $3/2$, norm at most $8\sqrt{d}/5$, and $\gamma\geq -8d/5$. Thus $\Phi(3/2,8/5,1/8)<2$ and $\Psi(3/2,8/5)<7/10$. The multiple-pair values fit $\bar{D}=2$, $R=2$, $\gamma\geq -d$; their respective costs are below $7/2$ and $15/4$.

For the positive root apply (Q.18) to the same C, c, Q. The single-pair first row has real overshoot at most x^2 and imaginary overshoot at most x. Its increment is below $2d$. The second row has only imaginary overshoot x^2, costing $85d/252$; $\gamma=-2d/5$ contributes less than $3d^2$, so the saved d covers both on $d\leq 1/8$. At the third row, the integral norm increment is at most $11d/6$, while $|\gamma|\leq 47d/30$ and $|\mathbb{E}V_*|\leq 7d/6$ give a curved correction below $18d^2$. At longer rows use the same $11d/6$ norm increment, $\gamma=-6d/5$, $|\mathbb{E}V_*|\leq d$ and correction below $12d^2$. For the latter mean bound, the finite base and root coordinates differ by at most x each, the actual propagated tail contributes at most $\max(x/8,x^3)<1/14$, and $h\leq 1/20$; $2\theta+1/14+1/20<1$. For multiple pairs the corresponding norm increments are $29d/5$, $171d/50$, $25d/9$ and $11d/6$; the first curved coefficient is zero and the other corrections are below $9d^2$, $15d^2$ and $12d^2$. These are comparisons with the complete curved baseline; $\gamma\mathbf{1}$ and its signed cross term occur in the positive root's vector. The table leaves more than $13/2$ at a minimum-one root and more than six at a positive root.

The same short anchor on the high domain.

For $d\geq 1/8$, the actual inequality (Q.30) gives $f_3 h_{c_1}\leq e^{-d}$. Each later pair has minimum at least five and is at most e^{-2d}. Hence the same propagated terminal satisfies

$$T^L F\leq e^{-[1+2(j-1)]d}\bar{f}_{L+2}.$$

The bare comparisons of Corollary Q.4 give rates at least $4+29/10+8(j-1)>13/2$ and $4+11/2+8(j-1)>6$, respectively. The closed endpoint $d=1/8$ is shared with the low domain. At $d=0$ the actual vectors vanish and leaf integration gives one.

Q.20 Restoring the complete odd root

The terminal directions in Table 108 are shared by the two root classes, apart from the short-one direction (Q.20). The preceding complete-certificate proofs give the following global rates:

terminal	$pW_a\bar{f}_b$	$f_a h_b f_c$	proof
$(1,t)$	$705/128$	$11/2$	whole-message short-one packets
$(2,t),\ t\ge 4$	6	$23/4$	two-arm theorem and short exceptions
$(2,3)$	6	6	two-arm theorem and short exceptions
$(3,t),\ t\ge 4$	6	6	Theorem Q.5
even lists of minimum at least four	8	$11/2$	Proposition Q.6
other even multipair lists	$11/2$	$11/2$	Proposition Q.6
odd lists of minimum one	$705/128$	$11/2$	Section Q.17
odd lists of nonunit minimum	$13/2$	6	Section Q.19.

(Q.40)

The positive-root entry in the first row means its actual special direction (Q.20), including all later negative pairs. Every other positive root uses $f_a h_b f_c$.

Lemma Q.7 (The full low-domain vector)**.** *On $0 < d \le 1/64$, each base certificate used in* (Q.40) *has its original pointwise form*

$$\log H_0 \le a_0 + BV_0 + W_0, \qquad \|V_0\| < (7/2)\sqrt{d},$$

with its already paid complete fourth exponent at most $-27d/5$ at a minimum-one root and $-11d/2$ at a positive root. Here $W_0 = 0$ or the same paid unit-arm curvature term.

Proof. The minimum-one short-one bound is (Q.42), with vector norm $3\sqrt{d}/2$. Its two-arm-minimum-two counterpart is supplied by Theorem M.37, with norm at most $2\sqrt{d}$. For the other minimum-one rows the transferred nonconstant root precursor has norm at most $\sqrt{d}/2$; the original constant or curved coefficient is the one used in its proof.

For the positive root outside its short-one special direction the complete vector is exactly

$$V_0 = Q(B)u + G_{a-1}(-B)u + V_{bc} + \gamma\mathbf{1}.$$

The two root precursors have norms below $(6/5)\sqrt{d}$ and $(3/8)\sqrt{d}$. The following bounds for the core $(\|Qu\| + |\gamma|)/\sqrt{d}$ include the actual constant:

q-first-row and retained-square packets	$5/3$
bare-exponent packets, first three windows	$1,\ 9/5,\ 147/80$
bare-exponent packets, long window	$167/120$
general inward anchor	$29/20$
short inward anchor, first three windows	$9/5,\quad 1+(13/20)(3/5)+1/16,$ $1+1/128+(13/32)(3/5)+1/4$
short inward anchor, long window	$7/5.$

For the exponent rows use $\rho \le 1$, $|\delta| \le 6d/5$, $|\epsilon| \le 17d/10$ and $h \le 1$; at $L = 3$ the genuine G_2 contribution is $6/5$, the scalar term in Q costs at most $3/160$, the attenuated remainder at most $13/32$, and the constant vector at most $17/80$. The general anchor keeps $\mu_A \le 1$ and its G_2 exception. The short anchor uses its two actual D values and (Q.37)–(Q.39); its $|D-d|u$ term is distinct from $\gamma\mathbf{1}$. All entries are less than $19/10$, so $\|V_0\| < (19/10+6/5+3/8)\sqrt{d} = (139/40)\sqrt{d}$. The special short-one direction was paid directly for all its root pairs in (Q.21). □

Let $k \geq 1$ further ordered unequal root pairs follow the base root triple. At a minimum-one root they are $W_s \bar{f}_t$; at a positive root they are $h_s f_t$. Their disjoint original exponent intervals give one finite mask, with scalar loss $16kd/5$, vector norm below $3\sqrt{d}/8$ and signed mean at most $d/12$. In the positive direction its positive mean powers first occur at original power eight, giving $\delta_4/(1-\theta^2) < d/12$; complete reversal gives the minimum-one direction. Applying Lemma M.31 to the full vector above gives the additional rate

$$\zeta_k = 2737/3840 + 64(k-1)/5. \tag{Q.41}$$

The high-domain estimate in that lemma uses these same products. For the short-one positive root, the directly proved increment $64k/5$ in (Q.21) is stronger.

Genuine equal pairs are oriented as $f_\ell h_\ell$ or their complete reverse. Their products are at most one. Positivity therefore restores all root and terminal decorations in the actual message. Finally apply the unanchored ordinary outlet (M.66), independently of the inner L. The base losses $27/5$ and $11/2$ give the respective output losses $6/95$ and $5/19$ for all $M \geq 1$, and each later root pair supplies twice its additional fourth loss.

Q.21 Odd internal roots of minimum one

Take ordered root lengths

$$3 \leq a < b < s_1 < t_1 < \cdots < s_k < t_k, \qquad k \geq 0, \tag{Q186.0.2}$$

and a terminal residual list Σ of at least two distinct lengths. Orient its arms by Table 108. For genuine equal-pair decorations E_o, E_t, set

$$H_k = E_o p W_a \bar{f}_b \prod_{i=1}^{k} W_{s_i} \bar{f}_{t_i}\, T^L(E_t F_{\Sigma,L}). \tag{Q186.0.3}$$

All means and operators use the original finite probability measure and the original skew kernel; $d = \mathbb{E}u^2$.

Proposition Q.8. *For every such residual list, every finite k and all independent positive integers L, M,*

$$\mathbb{E}H_0^4 \leq e^{-27d/5}, \tag{Q186.0.4}$$

$$\mathbb{E}H_k^4 \leq e^{-[27/5+\zeta_k]d} \quad (k \geq 1), \qquad \zeta_k = 2737/3840 + 64(k-1)/5. \tag{Q186.0.5}$$

Writing $\lambda_0 = 27/5$ and $\lambda_k = 27/5 + \zeta_k$ for $k \geq 1$, the outgoing message satisfies

$$I_q(T^M H_k)^3 \leq e^{-[2\lambda_k - c(M)]d} \leq e^{-6d/95}, \qquad I_q(X) = \mathbb{E}[q^{-1/3}|X|^{8/3}], \tag{Q186.0.6}$$

where $c(1) = 204/19$, $c(2) = 10$, $c(3) = 0$, $c(4) = 7$ and $c(M) = 6$ for $M \geq 5$.

Q186.1 Literal directions and exhaustive terminal table

An f_n arm has every edge away from its root; $\bar{f}_n$ has every edge toward its root; h_n has its first edge toward the root and its remaining edges away from the first neighbor; W_n has its first edge away from the root and its remaining edges toward that neighbor. T^L means every original inner edge points from the terminal toward the internal root. T^M means every outer edge points from that root toward the receiving center. The original unit arm at the internal root points away from the root, giving p.

In the table, every displayed residual list is strictly increasing. The indices used for terminal lengths are independent of the root a, b.

Table 108: Terminal directions at a minimum-one internal root. All residual lists are strictly increasing; their indices are independent of the internal root lengths $3 \le a < b$. The root pair is always $W_a \bar{f}_b$.

Terminal residual and length condition	Actual terminal message F
$(1,t)$, $t \ge 2$	qf_t
$(2,t)$, $t \ge 4$	$\bar{f}_2 W_t$
$(2,3)$	$h_2 f_3$
$(3,t)$, $t \ge 5$, $L = 1$	$h_3 f_t$
$(3,t)$, $t \ge 5$, $L \ge 2$	$\bar{f}_3 W_t$
$(3,4)$	$h_3 f_4$
Even $(\alpha_1, \beta_1, \dots, \alpha_j, \beta_j)$, $\alpha_1 \ge 4$, $j \ge 1$	$\prod_{i=1}^{j} h_{\alpha_i} f_{\beta_i}$
The same even list, $\alpha_1 \le 3$, $j \ge 2$	$\prod_{i=1}^{j} h_{\alpha_i} f_{\beta_i}$
$(1,s,t)$, $1 < s < t$	$qf_s h_t$
$(1, \alpha_1, \beta_1, \dots, \alpha_j, \beta_j)$, $j \ge 2$, $(\alpha_1, \beta_1, \alpha_2) = (2,3,4)$	$qf_2 h_{\beta_2} h_3 f_4 \cdot \prod_{i=3}^{j} h_{\alpha_i} f_{\beta_i}$
The same minimum-one odd list, all other cases	$qf_{\alpha_1} h_{\beta_1} \prod_{i=2}^{j} h_{\alpha_i} f_{\beta_i}$
Nonunit odd $(s, \beta_1, \gamma_1, \dots, \beta_j, \gamma_j)$, $s \ge 2$, $\beta_1 = 3$ (hence $s = 2$)	$\bar{f}_2 f_3 h_{\gamma_1} \prod_{i=2}^{j} h_{\beta_i} f_{\gamma_i}$
The same nonunit odd list, $\beta_1 \ge 4$	$\bar{f}_s \prod_{i=1}^{j} h_{\beta_i} f_{\gamma_i}$

The fixed cross in the tenth row is part of the original direction, not an analytical switch. Every list with at least two residual arms occurs in exactly one row after the displayed conditions are applied. The separate (3, t) rows depend only on whether $L = 1$. No direction depends on d, μ, K or a frequency. For all of them the new internal root direction is exactly $W_a \bar{f}_b$.

Proof. **Complete root and terminal certificates**

Use the full constant split and the minimum-one root transfer in Section Q.14. The positive finite propagation in each terminal family provides its own C, c, Q; the transfer is applied to those complete parameters.

Q186.5 Two-arm terminals $(1,t)$

For every $3 \le a < b$, $t \ge 2$ and $L \ge 1$, the actual message is $H_0 = pW_a \bar{f}_b T^L(qf_t)$. Apply Proposition O.1 with one root pair $(u_1, v_1) = (a, b)$ and terminal parameter t. It gives

$$\mathbb{E} H_0^4 \le e^{-705d/128}. \tag{Q186.5.1}$$

The root and terminal lengths are independent in that proposition. Its pointwise conclusion (O.15) also supplies the whole vector required for later root pairs. In the present $B = -K$ notation, set $V = -Z$. Then

$$\begin{aligned} H_0 &\le e^{a_0 + BV}, \qquad \|V\|^2 \le 9d/4, \\ 4a_0 + 4\mathbb{E}BV + (17/4)\|V\|^2 &\le -705d/128. \end{aligned} \tag{Q.42}$$

There is no separate curvature term in this certificate. It holds on the entire host domain, including all short inner lengths, and therefore applies to the low-domain restoration in part Q.21. For $a \ge 4$, the direct fourth payment in part O.3 also gives the stronger loss $7d$.

Q186.6 All remaining two-arm terminals of minimum two or three

Q186.6.1 Every terminal $(2,t)$, $t \geq 4$

Use the same actual root $pW_a\bar{f}_b$ and terminal $F = \bar{f}_2 W_t$. The minimum-one instance of Theorem M.37 proves, for all $3 \leq a < b$ and every inner length,

$$\mathbb{E}[pW_a\bar{f}_b T^L F]^4 \leq e^{-6d}. \tag{Q.43}$$

Its complete vectors have norm at most $2\sqrt{d}$, with any used p-curvature payment kept separate. Thus this is also an input for the paid restoration of later root pairs below.

Q186.6.2 The actual terminal $(2,3)$

Choose $F = h_2 f_3$, as in the direction table. Its actual arm sum is $2 - d - B^2 u$, equal to the arm sum for $\bar{f}_2 W_3$. Theorem M.37 treats this genuine alternative with the same $6d$ fourth loss and the same complete-vector bound. It uses this common positive envelope, not an equality of the two different terminal products.

The remaining terminal families

Theorem Q.5 proves loss six for the actual $(3,t)$ directions, including the fixed $(3,4)$ direction. Proposition Q.6 proves loss eight for even lists of minimum at least four and at least $11/2$ for the other even multipair lists. The q-first-row packets of Section Q.17 give $705/128$ for every minimum-one odd terminal, with the fixed cross prescribed in the table. Section Q.19 gives $13/2$ for every nonunit-minimum odd terminal. All these are complete fourth certificates at the same root $pW_a\bar{f}_b$.

Complete ordered root and outgoing connector

The losses in (Q.40) are at least $27/5$, proving (Q186.0.4). Lemma Q.7 supplies the pointwise low-domain certificate with its full vector norm at most $(7/2)\sqrt{d}$. The paid restoration in Section Q.20 gives the additional ζ_k and restores the genuine equal pairs. Finally (M.66) applies to the actual real message with the independent outgoing length M. Since $54/5 - 204/19 = 6/95$, it gives (Q186.0.6). □

Q.22 The minimum-two odd root: unit and minimum-one odd terminals

Q188.1 Original model and the actual orientation table

Use the original finite-space model, path messages and spectral coordinates of Section M.2. In particular $B = -K$, $u = B\mathbf{1}$, $T = J + B$, $A = T^*$, $p = 1 - u$, $q = 1 + u$, and $\tau = d - \delta_2$. The two signed means and the complete-vector square are (M.3); the finite positive convolution is (M.2). At $n = 1$, $G_n = (1,0)$, whereas $|G_2|^2 = 1 + x$; the contraction $|G_n| \leq 1$ is used only for $n \geq 3$. Every certificate $H \leq e^{a+BV}$ below uses the first convention of Lemma M.27, with the actual signed mean and all constant components retained in V. Only the genuine equal-length decorations in (M.1) may be dropped by positivity. For terminal (1, a), $F = qf_a$. For the odd catalogue, normally

$$F = qf_{a_1} h_{b_1} \prod_{i \geq 2} h_{a_i} f_{b_i}. \tag{Q188.L63}$$

In the sole cross pattern,

$$F = qf_2 h_{b_2} h_3 f_4 \prod_{i \geq 3} h_{a_i} f_{b_i}. \tag{Q188.L67}$$

Its complete reversal $\bar{F}$ replaces q/f/h by p/b/W. The actual root input is

Table 109: The minimum-two odd root: unit and minimum-one odd terminals: parameter values.

Entry	Actual H
$L=1$	$f_2 h_b f_c \prod h_{s_i} f_{t_i} * T(\bar{F})$.
$L \geq 2, b=3$	$W_2 \bar{f}_3 W_c \prod W_{s_i} \bar{f}_{t_i} * T^L(F)$.
$L \geq 2, b \geq 4$	$W_2 W_b \bar{f}_c \prod W_{s_i} \bar{f}_{t_i} * T^L(F)$.

The inner path points from terminal to root and the outer path from root to receiver. f is an entirely outward original arm, b entirely inward, h first inward then outward, and W first outward then toward the first neighbour. All decisions use only original integer lengths and the cross pattern. No direction is chosen after d, μ or K is known.

Q188.2 The actual root3 pair, full constants, and one mask for all later pairs

Write $n = c-3 \geq 1$, $G_n = E + zD$, and $\beta = \beta_{c-1}$. Use the positive pair (M.51) for $h_3 f_c$, and the distinct exchanged pair (M.53) for $\bar{f}_3 W_c$. Thus $V_+ = (-xE + zxD)u + \tau \mathbf{1}$, whereas $V_- = (xE + zxD)u + \beta \mathbf{1}$ and $\Gamma_- = d + \beta_c + \beta$. Their nonconstant norm and signed means are (M.54), including $n = 1, D = 0$. Each displayed constant stays in the complete vector used below. For every actual ordered pair with short length at least four, C096's retained finite difference supplies the stronger loss 16d/5. The union of all disjoint exponent intervals is one finite 0–1 mask. Four residue classes give

$$|P(i\sqrt{x})|^2 \leq (1+x)/(1-x^2)^2 < 100/49. \tag{Q188.L103}$$

Consequently the complete primitive U of all such positive or negative pairs has

$$\|U\| \leq 3\sqrt{d}/8, E(BU) \leq d/12. \tag{Q188.L107}$$

If its first short length is at least five, $\|U\| \leq \sqrt{d}/4$. The mean bound follows by retaining only positive even-power contributions: the first possible one is at most $\delta_4/(1-\theta^2) < d/12$. These are original-measure statements, with no independence between pairs and no multiplication of separate moment estimates.

For a paid root3 certificate $H \leq \exp(a + BV)$, with complete $\|V\| \leq 2\sqrt{d}$, adding all $k \geq 1$ later original root pairs at once changes its fourth exponent by at most

$$\begin{aligned} &[1/3 + (17/4)(2 \cdot 2 \cdot (1/4) + 1/16) - 64k/5]d \\ &\quad = [931/192 - 64k/5]d \leq -7633d/960. \end{aligned} \tag{Q188.L113}$$

Here the full constant term of V is included. The complete norm cap is checked in §8 below. If $b \geq 4$, all root pairs are instead paid in the one mask of norm $3\sqrt{d}/8$; their total loss is 16(k+1)d/5. Thus arbitrary pair counts are covered.

Q188.3 New all-host $L = 1$ core for every terminal (1, a)

The preceding fourth-moment estimate uses the positive root-pair direction. C081 Theorem A uses a distinct unit direction.

For $a = 2, p + \bar{f}_2 = 2 + Bu$, so

$$p\bar{f}_2 \le (1+Bu/2)^2,$$
$$T(p\bar{f}_2) \le q-\gamma+B^2u, \gamma = d-\delta_2/2. \tag{Q188.L124}$$

The latter follows by expanding the same square, using $EBu=-d$ and $T((Bu)^2)\le 2\delta_2$. Since $f_2=1-u+Bu$, the arithmetic mean of the two actual positive factors gives

$$f_2\cdot T(p\bar{f}_2) \le \exp[-\gamma+B(1+B)u]. \tag{Q188.L129}$$

For $a\ge 3$ set $N=a-1, R_a=BG_N(B)u, m=1-\beta_a/2, k_0=1/m\le 6/5$,

$$t_0=\beta_{a+1}, e=d-\beta_{a+2}, V_0=ER_a^2,$$
$$\gamma = k_0\cdot t_0-k_0^2\cdot V_0/2. \tag{Q188.L133}$$

The actual finite positive convolution gives

$$p\bar{f}_a \le (m+R_a/2)^2, T(p\bar{f}_a)\le m^2[q-\gamma+k_0BR_a]. \tag{Q188.L138}$$

Here $ER_a=-t_0$, $E(uR_a)=-e$, $0\le e\le\delta_2$; $V_0=\delta_2+\delta_3$ for $a=3$ and $V_0\le\delta_2$ for $a\ge 4$. Positive AM–GM with the actual f_2 therefore gives

$$f_2\cdot T(p\bar{f}_a) \le \exp[2\log m-\gamma+BQ(B)u],$$
$$Q=1+k_0zG_N, E[BQ(B)u]=-d+k_0e. \tag{Q188.L143}$$

For $a\ge 4, t_0-e\ge d-2\delta_2\ge 0$, $k_0\ge 1$ and $k_0^2/2\le 18/25$ prove

$$\gamma-k_0e\ge d-68\delta_2/25. \tag{Q188.L148}$$

For $a=3$ the direct values $t_0=d, e=\delta_2$ and $V_0=\delta_2+\delta_3$ give the stronger coefficient $161/80$, hence the same stated bound. This checks the whole finite a-tail and its end signs.

Now $b=3$. In the $a=2$ row the combined nonconstant primitive has coordinates

$$(1-xE, 1+xD). \tag{Q188.L154}$$

Its squared modulus is at most $(1-x+x^2)^2+x(1+x)^2\le 3/2$: its second derivative is positive, so the two endpoints suffice, and the θ endpoint is $1446129/1048576<3/2$. Its imaginary coefficient is at least 1. The full τ constant therefore has nonpositive incremental square $\tau^2+2\tau EV_{\text{non}}\le 0$. Including the actual mean and the full-mean fourth exponential lemma gives

$$\log EH^4\le -20d+14\delta_2+51d/8\le -127d/16. \tag{Q188.L158}$$

For $a\ge 3$ the coordinates are

$$(1-k_0xD_N-xE, k_0E_N+xD). \tag{Q188.L162}$$

The real coordinate is at least $17/160>0$, and both coordinates can therefore be safely upper-bounded before squaring. Their square is at most

$$(1-2\kappa\cdot x)^2+x(6/5+x)^2\le 3/2. \tag{Q188.L166}$$

The displayed polynomial is convex in x; at θ its value is $17249442/13107200 < 3/2$. The imaginary coefficient is at least $\kappa > 1/2$, so the τ cross is again nonpositive. Since $\beta_a \geq \tau$,

$$\begin{aligned} \log EH^4 &\leq -24d + (672/25)\delta_2 + 51d/8 \\ &\leq -1341d/200 < -6d. \end{aligned} \tag{Q188.L170}$$

For $b \geq 4$, use the single complete root mask. In the $a = 2$ row the base norm is below $6\sqrt{d}/5$, scalar-plus-mean fourth loss is at least $115/16$, and the resulting loss is at least

$$115/16 + 64/5 - 1/3 - 67473/6400 > 6. \tag{Q188.L175}$$

For $a \geq 3$, the exact Q norm is below $9\sqrt{d}/8$. One direct check starts from

$$\begin{aligned} |Q|^2 &\leq 1 - (24/25)x + (96/25)x^2 \\ &\leq 1 + 3x/200 + (36/25)x^2 < 997/800 < 81/64. \end{aligned} \tag{Q188.L179}$$

The scalar-plus-mean fourth loss is at least $1191/200$, because it is at least $12 - (372/25)\theta$. After the same whole root mask, the loss is at least

$$1191/200 + 64/5 - 1/3 - 153/16 > 6. \tag{Q188.L184}$$

Further root pairs only improve this payment. Thus all $a \geq 2$, all hosts, and every original positive root table have the required actual $L = 1$ fourth loss 6.

Correlated first rows and the actual root cancellation

Use Lemma M.32 with the terminal's own R, Γ, D, taking $\alpha = -\Gamma$ in an exponential row. Its first row is (M.67), where

$$D \geq \max\{\delta, 0\}, \qquad \delta = \mathbb{E}R + \mathbb{E}(uR) + \sqrt{d\mathbb{E}R^2} + c_T\mathbb{E}R^2.$$

Here $c_T = 12/5$ or $8/3$, as proved for the packet; the square-envelope row uses $c_T = 1$ and $\alpha = 2\log m - \Gamma$, retaining its factor m^2 at every length. Keep these actual first-row data. For $L = 1$, use the reversed terminal and the first row of (M.71); for $L \geq 2$, use the second row. Each following packet proves the required nonnegative $\gamma_j = \beta_j - D_{j-2}$, including the next mean index $L + 2$. Its root vector retains the full $\tau\mathbf{1}$ or $\beta\mathbf{1}$ component from the actual pair envelope.

Q188.5 Canonical123: signed mean and all-length loss $13/2$

Put $m = 1 - d/2$, $k_0 = 1/m$. The actual sums are

$$f_2 + h_3 = 2 - d + B^2u, \bar{f}_2 + W_3 = 2 - d - B^2u. \tag{Q188.L239}$$

The same positive squares have first-row error

$$\delta = -k_0 \cdot \delta_2 + k_0 \cdot \sqrt{d \cdot \delta_3} + k_0^2 \cdot \delta_3 \leq 39d/164 < d/4. \tag{Q188.L243}$$

To check the whole domain, put $a = \delta_2/d$ and $\mathfrak{b}_0 = 1 - k_0\theta \geq 41/80$; then

$$\begin{aligned} \delta/d &\leq k_0[-\mathfrak{b}_0 \cdot a + \sqrt{\theta \cdot a}] \\ &\leq k_0\theta/(4\mathfrak{b}_0) \leq 39/164. \end{aligned} \tag{Q188.L247}$$

This is completion of a square. Take $D = d/4$. The actual signed coefficients are $D_1 = k_0\delta_2$, $D_2 = 0, D_3 = -k_0\delta_3, D_4 = 0$, and all $\gamma_j \geq d/2$ for $j \geq 3$. Thus

$$Q_L = G_L - k_0 \cdot x \cdot z^{L-1}, c_L = d/2 - \gamma_{L+1}. \tag{Q188.L252}$$

For $L = 1, b = 3$, the full nonconstant primitive is $[x(k_0 - E) + zxD]u$; its mean after B is nonpositive, norm squared is below 9d/64, and adding the actual τ constant still leaves norm below $\sqrt{d}$. For example its nonconstant square is at most $x^2(1/5+x)^2 + x^3$; the stated weaker $9/64$ is valid. The scalar fourth exponent is at most -47d/4. Hence the fourth loss is at least $15/2$. For $b \geq 4$ the nonconstant base norm is below $\sqrt{d}/2$ and the full mask norm gives $7\sqrt{d}/8$; the resulting fourth loss also exceeds $15/2$.

For $L \geq 2, b = 3$ the nonconstant primitive is

$$[Q_L - 1 + xE + zxD]u, \tag{Q188.L258}$$

and the full constant is $\beta_{c-1}\mathbf{1}$. Its imaginary coefficient is at least $41/80$ at L2, at least 1 at L3, at least κ at L4, and at least $\kappa - 1/12 = 49/96 > 1/2$ for every $L \geq 5$. This last lower bound remains valid when the root gap is one and $D = 0$. Therefore the entire β-square increment is nonpositive.

Table 110: The $b = 3$ short and long fourth-moment ledger. Norms concern the nonconstant vector; its actual constant is retained and has a nonpositive square increment by the preceding argument.

L	$\|V_{\text{non}}\|^2/d \leq$	$4a/d \leq$	$4\mathbb{E}BV_{\text{non}}/d \leq$	Loss $\geq$
2	$3/5$	$-64/5$	0	$41/4$
3	1	$-59/4$	0	$21/2$
4	1	$-105/8$	$2/3$	$197/24$
≥ 5	$6/5$	$-51/4$	$106/105$	$2789/420$

The short coordinates are respectively

$$\begin{aligned} L2 &: (xE, 1 - k_0 \cdot x + xD), \\ L3 &: (x(E-1) + k_0 \cdot x^2, 1 + xD), \\ L4 &: (x(E-1), 1 - x + k_0 \cdot x^2 + xD). \end{aligned} \tag{Q188.L271}$$

Their displayed norms follow, for instance, from $x + x^2 < 3/5$ at L2; at L3 use $x + 2x^2 + x^3 + (36/25)x^4 < 1$; at L4 use $x+(12/5)x^3+x^4+(36/25)x^5 < 1$. The mean at L3 is at most $\delta_3 - k_0\delta_3 \leq 0$, and at L4 at most δ_3.

For every $L \geq 5$ the unperturbed square is at most $x+3x^2+x^3 \leq 31733/32768 < 1$. The terminal perturbation is at most $k_0 \cdot x^3 < 1/12$; thus the whole square is below $(13/12)^2 < 6/5$. The actual mean is not always nonpositive. Precisely,

$$\begin{aligned} EBV_{\text{non}} &\leq \delta_3 + (6/5)\delta_4, \\ 4EBV_{\text{non}} &< 2d/3 + 12d/35 = 106d/105. \end{aligned} \tag{Q188.L279}$$

This uses $\beta_{L+2} \geq d - \delta_2$ and $\int xE \geq \delta_2 - \delta_3$, and the largest positive actual D_L is at most k0δ_4. The final loss is therefore $51/4 - 51/10 - 106/105 = 2789/420 > 13/2$.

For $b \geq 4$ and $L \geq 2$, $\|(Q_L - 1)u\| < 9\sqrt{d}/10$: L2–4 are bounded by $x + x^2 < 3/5$, and the long tail by $\sqrt{3/5} + 1/12 < 53/60 < 9/10$. The complete mask gives fourth loss at least

$$217/15 - 44217/6400 = 145109/19200 > 15/2. \tag{Q188.L286}$$

The canonical123 conclusion $13/2$ is thus uniform in all L, all root gaps and all hosts, before later root pairs, which improve it.

Q188.6 Canonical12c: new actual2d retention and all low lengths

Here $0 < d \leq 1/4$, $m = 1 - d$ and $k_0 \leq 4/3$. The actual $c = 4$ retained square has

$$\begin{gathered} R = (-t + zs)u, t = k_0(d + x), s = k_0 \cdot x, D = 5d/12, \\ \gamma_3 \geq d/8, \gamma_4 \geq 3d/4, \gamma_j \geq 41d/96 (j \geq 5). \end{gathered} \tag{Q188.L294}$$

For $c \geq 5$, use the actual $\sigma = 2$ row of Lemma M.10, throughout $0 < d \leq 1/4$. It has the same canonical $F = qf_2h_c$, $m = 1 - d$, $R = (-t + zs)u = -m^{-1}(A_0 - zB_0)u$, and $D = d/2$. The true retained construction gives

$$A_0 \geq \tau + (1-x)B_0, \quad A_0, B_0 \geq x(1-x), \quad A_0 < 7/10, \quad B_0 < 3/5, \quad 2(A_0^2 + xB_0^2) \leq A_0 + B_0.$$

Hence $0 \leq t < 14/15$, $0 \leq s < 4/5$, and $t \geq x(1 - x)$. The signed moments in (M.27) give $\gamma_3 \geq 0$, $\gamma_4 \geq 2d/3$, $\gamma_5 \geq \tau$, and $\gamma_j \geq 41d/96$ for all $j \geq 5$. The root-fourth calculation below uses this same vector and its signs.

For $L \geq 3$, both coordinates of $Q_L = G_L + z^{L-1}(-t + zs)$ lie in [1-x, 1]. This can also be seen directly by residues mod four: the last two terms are fractional truncations of the final alternating interval, because t, s lie in [0, 1]. L2 instead has coordinates $(1 - sx, 1 - t)$ and is treated separately.

For $b = 3, L = 1$ the complete nonconstant primitive has coordinates

$$(t - xE, s + xD). \tag{Q188.L314}$$

The plus sign $s + xD$ is necessary: $\bar{R}$ and the actual positive root pair have the same imaginary sign here. In $c = 4$ the real coordinate is nonnegative and below $2/3$, and the imaginary coefficient is between 0 and $7x/3 < 1$. Thus the nonconstant square is below d; its B mean is nonpositive. The τ cross is nonpositive and its remaining square is at most $d^2 \leq d/4$. The complete fourth exponent is at most

$$-181d/12 + (17/4)(5d/4) = -469d/48 < -6d. \tag{Q188.L318}$$

For $c \geq 5$ the real coordinate is in [-x^2, $14/15$] and the imaginary coefficient is at most $4/5 + x$. Their square is below $3/2$, since

$$196/225 + (5/12)(73/60)^2 = 64277/43200 < 3/2. \tag{Q188.L322}$$

Again the constant cross is nonpositive and its square is at most d^2, so complete squared norm $\leq 7d/4$. The actual B mean is at most δ_3. With scalar fourth exponent at most -59d/4 the loss is at least

$$59/4 - 2/3 - 119/16 = 319/48 > 6. \tag{Q188.L326}$$

For $b = 3, L = 2$ the coordinates are $(x(E-s), 1-t+xD)$. They have square below 1, imaginary coefficient nonnegative, and actual B mean at most δ_3. The constant square costs at most d^2, giving complete squared norm $\le 5d/4$. The weakest scalar is -51d/4, leaving loss

$$51/4 - 2/3 - 85/16 = 325/48 > 6. \tag{Q188.L330}$$

The $c = 4$ row is stronger: E-$s \ge 5/96$, the imaginary coefficient is at most 1, nonconstant square $< 3/5$ and scalar $\le -167d/12$. For every $L \ge 3$ the rectangle gives nonconstant square < 1 and imaginary coefficient $\ge \kappa$, so the full β cross is nonpositive. The signed mean is at most δ_3. Scalar fourth bounds are -121d/8 for $c = 4$ and -347d/24 for $c \ge 5$; after mean and $17/4$ norm payment both leave loss > 6.

For $b \ge 4, L = 1$ the base norm is below $6\sqrt{d}/5$ and B mean is nonpositive; after the full root mask the mean is at most d/12 and norm below $(6/5 + 3/8)\sqrt{d}$. Its weakest scalar, -8d+4D-64d/5, still pays loss > 6. For $L \ge 2$ the base norm is below $4\sqrt{d}/5$. Adjacent γ sums are at least 41d/48 in $c = 4$ and 2d/3 for $c \ge 5$. In the exact scalar-plus-mean formula of §4 the resulting losses exceed 10 and 9 respectively. All later root pairs are already in this same mask.

Q188.7 All remaining triples: actual envelopes, G_1 and weakest $2427/400$

For $F = qf_a h_c$, $c > a \ge 3$, and $0 < d \le 1/4$, the actual first-row packages extracted from C172/C173 are:

Table 111: First-row parameters for $F = qf_a h_c$, $3 \le a < c$ and $0 < d \le 1/4$.

a	$g = \Gamma/d$ lower	$D* = D/d$ upper	ρ with $\lvert r\rvert \le \rho$	h with all $\gamma_j \ge h \cdot d$
3	$35/16$;	$7/20$;	$3/5$;	$1/2$.
4	$51/16$;	1;	$3/10$;	$15/32$.
5	$89/32$;	$2/3$;	$1/4$;	$15/32$.
≥ 6	$19/8$;	$7/20$;	$1/8$;	$15/32$.

For $a = 3$ the exact polynomial is $r = -\beta_{c-1} - z^3 G_{c-3}(-z)$. For $a \ge 4$ it is

$$\begin{aligned} r &= -c_0 + (-1)^a z^a G_{c-a}(-z), \\ c_0 &= \beta_{c-1} - \beta_{a-1}. \end{aligned} \tag{Q188.L349}$$

The cited finite constructions supply these signs, the finite G_2 cost, the actual δ and all γ indices. In particular c_0 is a finite alternating difference, not a monotone-β increment. These actual envelopes, not p-root moment conclusions, are the inputs to the new W_2/f_2 root proof.

For $a \ge 4$ and $b = 3, L = 1$, complete norm is at most $(\rho + 1)\sqrt{d}$, since it includes $\bar{R}$, the root nonconstant norm $< \sqrt{d}/2$ and $\tau \le d \le \sqrt{d}/2$. The actual B mean is at most $(\rho + \theta)d$. The fourth loss is consequently at least

$$4g - 4D* + 57/8 - 4\rho - (17/4)(\rho + 1)^2. \tag{Q188.L356}$$

The $a = 4$, $a = 5$ and $a \ge 6$ rows respectively exceed 7, 7 and 9.

For $L \ge 2$ the unperturbed primitive $G_L - 1 + xE + zxD$ has norm below $\sqrt{d}$, since its real coordinate has absolute value at most x and its imaginary coefficient at most 1+x. The terminal perturbation has norm below $2\rho/3$, uniformly in L. Its imaginary coefficient is nonnegative after addition:

the L2 base is at least 1 and the perturbation is at most ρ, the L3 base is at least 1 and the perturbation at most sqrt(θ)*ρ, and from L4 onward the base is at least κ and perturbation at most θ*ρ. This argument explicitly allows root $n = 1$ and $D = 0$. Hence the β cross is nonpositive, but the β square is still paid by $\beta^2 \le d^2 \le d/4$.

The fourth loss is therefore at least

$$4g + 35/4 - 4(2D* + 1 - 2h) - (17/4)[(1 + 2\rho/3)^2 + 1/4]. \tag{Q188.L364}$$

For $a = 4$ this is exactly $53/4-(17/4)(169/100) = 2427/400 > 6$. For $a = 5$ it equals $67/9 > 7$, and for $a \ge 6$ it is greater than 9. This is the weakest general-triple payment; it contains the actual β square.

The $a = 3$ unit row benefits from its actual correlation. Its nonconstant coordinates are

$$(\beta_t - x^2 D_t - xE, x(E_t + D)), t = c - 3. \tag{Q188.L371}$$

Their square is at most $d^2 + x^2 + 6x^3 + x^4 < 2/3$. After the τ square the complete squared norm is at most 11d/12, and the signed mean after B is at most δ_2+δ_3. The scalar fourth exponent is at most -161d/10. Using $4(\delta_2 + \delta_3) \le 585d/256$ leaves loss greater than $793/80 > 6$.

For $a = 3$, L2 the nonconstant coordinates are $(xE - x^2E_t, 1 - \beta_t + x^2D_t + xD)$. Their square is below $6/5$; the θ certificate is $39193245/33554432 < 6/5$. The complete constant adds at most d^2. For $L \ge 3$, the unperturbed norm is below 1 and the perturbation below $1/4$, so complete squared norm $\le 29d/16$. The exact scalar-plus-mean loss is at least $147/10$. Even the latter weaker norm leaves

$$147/10 - 493/64 > 6. \tag{Q188.L377}$$

For $b \ge 4$ the actual unit and nonunit rows use the same mask as before. Their respective lower losses are

$$\begin{aligned} &4g - 4D* + 64/5 - 4\rho - 1/3 - (17/4)(\rho + 3/8)^2, \\ &4g + 64/5 - 4(2D* + 1 - 2h) - 1/3 \\ &-(17/4)(4/5 + 2\rho/3 + 3/8)^2. \end{aligned} \tag{Q188.L381}$$

All four table rows exceed 6; the weakest nonunit one is already the $a = 3$ row, exceeding 7. Thus all triples and every later root pair are paid in the full low domain.

Q188.8 All odd multipairs, all finite j, and complete root-norm control

For $j \ge 2$, use the five exhaustive terminal blocks and actual finite first-row constructions of C177. Their exact packages are:

I Original structure: $a_1 \ge 3$;

g: $431/80$;

D*: $5/3$;

ρ: $1/2$;

h: $1/3$;

Δ: $1/64$.

II Original structure: $a_1 = 2, b_1 \geq 4$;

g: $351/80$;

D*: $11/10$;

ρ: $2/3$;

h: $1/3$;

Δ: $1/64$.

III

Original structure: $a_1 = 2, b_1 = 3, a_2 \geq 5$;

g: $21/5$;

D*: $11/10$;

ρ: $3/5$;

h: $15/32$;

Δ: $1/8$.

IV

Original structure: cross, $j = 2$;

g: $21/4$;

D*: $11/10$;

ρ: $1/2$;

h: $1/2$;

Δ: $1/8$.

V Original structure: cross, $j \geq 3$;

g: $169/20$;

D*: $7/4$;

ρ: $3/5$;

h: $1/3$;

Δ: $1/8$.

All later terminal pairs belong to the same actual R and finite disjoint mask, so ρ and D do not increase with j. The Γ lower bounds come from the actual first package plus 16d/5 for each later pair. The β coefficients enter through their finite identities. In III the actual $\gamma_3 \geq d/2, \gamma_4 \geq 11d/12$, later $\gamma \geq 79d/160$ justify $h = 15/32$. IV later $\gamma \geq 49d/96$ justify $h = 1/2$ only with its short bounds. V is bounded using $h = 1/3$ rather than a nonexistent universal $1/2$.

For $b = 3, L1$ the complete root certificate, including τ, gives fourth loss at least

$$U = 4g - 4D* + 57/8 - 4\rho - (17/4)(\rho + 1/2 + \sqrt{\Delta})^2. \tag{Q188.L403}$$

Use $\sqrt{\Delta} = 1/8$ in I/II and the upper $3/8$ in III/IV/V. The five lower bounds respectively exceed 14, 10, 7, 13 and 22. For example III has $U = 137/8 - 59177/6400 > 7$. This pays the possibly positive actual B mean of $\bar{R}$ by 4ρ*d.

For $b = 3, L \geq 2$, the same base and perturbation proof as §7 gives nonnegative imaginary coefficient even when the root gap is one. The constant increment is at most $\beta^2 \leq \Delta \cdot d$. Hence the fourth loss is at least

$$N = 4g + 35/4 - 4(2D* + 1 - 2h) - (17/4)[(1 + 2\rho/3)^2 + \Delta]. \tag{Q188.L410}$$

The five rows respectively exceed 8, 7, 7, 12 and 18. The closest exact comparisons are

$$\begin{aligned} I : N &= 469/30 - 17561/2304 = 8 + 131/11520 > 8, \\ II : N &= 97/6 - 185249/20736 > 7, \\ III : N &= 33/2 - 7089/800 > 7. \end{aligned} \tag{Q188.L415}$$

Thus no finite-j or short-L gap is hidden by these envelopes. The full complete vector is still being used, not a product of expectations.

For $b \geq 4$ the respective unit/nonunit losses are

$$\begin{aligned} U_4 &= 4g - 4D* + 64/5 - 4\rho - 1/3 - (17/4)(\rho + 3/8)^2, \\ N4 &= 4g + 64/5 - 4(2D* + 1 - 2h) - 1/3 \\ &\quad -(17/4)(4/5 + 2\rho/3 + 3/8)^2. \end{aligned} \tag{Q188.L423}$$

All five U_4 rows exceed 18, and all five N4 rows exceed 8. These are direct substitutions into positive-denominator rational expressions; every parameter family is already covered by the finite-mask and finite-propagation arguments.

The norm premise needed to add later root pairs in part Q188.2 holds as follows. In the general nonunit rows the unperturbed norm is below 1 and terminal perturbation below $2\rho/3 \leq 4/9$, while the complete constant has norm at most $\sqrt{d}/2$ (at most $3\sqrt{d}/8$ in the multipair low domains). Thus complete norm is below $2\sqrt{d}$. In unit rows, $\bar{R}$ plus root nonconstant and constant is at most $(\rho + 1/2 + \sqrt{\Delta})\sqrt{d}$ in the low domains, also below $2\sqrt{d}$. The special canonical12c unit bound is at most $(\sqrt{3/2} + 1/2)\sqrt{d} < 2\sqrt{d}$. The all-host (1, a) unit proof has nonconstant norm at most $\sqrt{3d/2}$, and $\tau \leq d$ gives norm at most $(\sqrt{3/2} + 1/\sqrt{3})\sqrt{d} < 2\sqrt{d}$. Canonical123 is stronger. Therefore the whole-root-pair lifting increment is valid in every row where it is used.

Q188.9 Closed high domains and exact paid exterior interfaces

For $c = 4$ and $d \geq 1/4, h_4 \leq 2Ef_3 = 2 - 4d \leq 1$. For $c \geq 5$, the finite path-mean proof in Q5-C169, Section 8 gives $Ef_n \leq 1 - 5d/2$ for all $n \geq 4$, so $h_c \leq 2 - 5d \leq 1$ already for $d \geq 1/5$ and in particular for $d \geq 1/4$. Complete reversal gives $W_c \leq 1$ in the same domains. Thus each single triple compares in the same direction to terminal (1, a): L1 to $T(p\bar{f}_a)$ and $L \geq 2$ to $T^L(qf_a)$.

For multipair blocks I/II and $d \geq 1/64$, C177's actual later-pair estimate is

$$\begin{aligned} h_s f_t &\leq \exp[(39/125)\sqrt{d} - 16d/5] \\ &\leq \exp(-88d/125) \leq 1, s \geq 4. \end{aligned} \tag{Q188.L437}$$

Thus remove only already-paid factors inside the same positive T^L; the resulting single triple preserves every surviving arm's direction. For III and $d \geq 1/8$, use the exact multiplication identity

$$F = (qf_2h_{a_2})(h_3f_{b_2})\prod_{i\geq 3} h_{a_i}f_{b_i}. \tag{Q188.L442}$$

Here $h_3f_{b_2} \leq \exp(-d)$, with $b_2 \geq 6$, and later pairs are ≤ 1. For IV/V the original cross already is

$$F = (qf_2h_{b_2})(h_3f_4)\prod_{i\geq 3} h_{a_i}f_{b_i}, \tag{Q188.L446}$$

with $h_3f_4 \leq e^{-d}$. These actual high-domain bounds are proved by centred original rows, including the finite G_2 exception. Complete reversal gives the corresponding $\bar{F}$ comparisons. These operations change no actual direction; they are comparisons after the original factors were oriented. Low and high intervals overlap or meet at included endpoints.

Every paid fourth-loss-6 domain uses exact C164's all-M real-input outlet

$$\begin{gathered} I_q(T^MH)^3 \leq \exp(c(M)d)(EH^4)^2, \\ c(1) = 204/19, c(2) = 10, c(3) = 0, c(4) = 7, c(M) = 6 \quad \text{for } M \geq 5. \end{gathered} \tag{Q188.L452}$$

Its worst cost leaves $12 - 204/19 = 24/19 > 3/95$. Canonical123 is stronger.

For all remaining nonunit high-domain comparisons, the exact actual C081 Theorem B input for $b = 3$ is

$$W_2\bar{f}_3W_c\prod W_{s_i}\bar{f}_{t_i}T^L(qf_a), L \geq 2, \tag{Q188.L459}$$

with ordinary eighth moment $\leq e^{-79d/10}$. This is exactly the new root direction, including every later pair. For $b \geq 4$ the exact C080 Theorem T085.2 input is

$$W_2W_b\bar{f}_c\prod W_{s_i}\bar{f}_{t_i}T^L(qf_a), L \geq 2, \tag{Q188.L463}$$

with ordinary eighth moment $\leq e^{-479d/60}$. These exact ordinary-eighth premises concern the same actual messages.

C079's all-real, unanchored outlet has maximum cost 299d/38 over all M, with separate costs 6d at M2 and factor 1-d at $M \geq 3$. Therefore the exact losses are

$$79/10 - 299/38 = 3/95, 479/60 - 299/38 = 131/1140 > 3/95. \tag{Q188.L469}$$

This pays the exterior with the same actual ordinary-eight function. L1 high domains use the new positive-pair fourth proof of §3 instead, so the old C081 unit special direction is irrelevant. Every original host is covered with the same orientation. This proves the stated all-host output bound.

Q.23 Odd internal roots of minimum at least three

Take a root residual list $(a, b, c, s_1, t_1, \ldots, s_k, t_k)$ with $3 \leq a < b < c < s_1 < t_1 < \cdots < s_k < t_k$ and $k \geq 0$. Let the terminal residual list contain at least two distinct positive lengths. For a list other than $(1, t)$, use the same terminal directions as Table 108 and set

$$H_k = E_rf_ah_bf_c\prod_{i=1}^{k} h_{s_i}f_{t_i}\,T^L(E_tF).$$

For the short-one terminal use (Q.20) with $N = k+1$ and ordered pairs $(b,c),(s_1,t_1),\dots,(s_k,t_k)$, then restore its genuine equal-pair decorations. The root and terminal decorations E_r, E_t retain their actual arm vertices.

Proposition Q.9. *For all the stated residual lists and every independent $L, M \geq 1$,*

$$\mathbb{E}H_k^4 \leq e^{-\lambda_k d}, \qquad I_q(T^M H_k)^3 \leq e^{-[2\lambda_k - c(M)]d} \leq e^{-5d/19},$$

where $\lambda_0 = 11/2$, $\lambda_k = 11/2 + \zeta_k$ for $k \geq 1$, ζ_k is (Q.41), and $c(M)$ is the unanchored ordinary outlet cost (M.66).

Proof. The complete positive-root transfer in Lemma Q.3 keeps the actual constant vector $\gamma\mathbf{1}$ and its signed mean cross. Sections Q.15, Q.16, Q.18, Q.17 and Q.19 give all the terminal families, with rates at least $11/2$ as summarized in (Q.40). Their full low-domain vectors satisfy Lemma Q.7. Section Q.20 therefore restores all later ordered pairs and genuine equal pairs in their specified directions. In the short-one family those root pairs are already paid together by (Q.21), with a stronger increment. Apply (M.66) to the resulting actual message; $11 - 204/19 = 5/19$. At $d = 0$ the finite leaf integrations give equality. □

Q.24 The terminal pair at a minimum-two odd root

Use the original finite-space model and messages of Section M.2. The residual lists are

$$(2,a,b,s_1,t_1,\dots,s_k,t_k), \qquad 3 \leq a < b < s_1 < t_1 < \cdots < s_k < t_k, \qquad (2,t), \quad t \geq 3.$$

All lists are finite; the inner and outer lengths $L, M \geq 1$ are independent. Take exactly the minimum-two directions of Theorem M.37: for $L = 1$, the root factor is $f_2 h_a f_b \prod h_{s_i} f_{t_i}$ and the terminal is $W_2 \bar{f}_t$. For $L \geq 2$, the terminal is $\bar{f}_2 W_t$; the root factor is $W_2 \bar{f}_3 W_b \prod W_{s_i} \bar{f}_{t_i}$ at $a = 3$, and $W_2 W_a \bar{f}_b \prod W_{s_i} \bar{f}_{t_i}$ at $a \geq 4$. Every inner edge points terminal-to-root and every outer edge root-to-receiver. Thus these directions are fixed before the original host is chosen.

Let C_r, F_t be those actual factors, and let E_r, E_t be arbitrary compatible equal-pair products in their original positions. Define $H_{\text{full}} = E_r C_r T^L (E_t F_t)$ and $I_q(X) = \mathbb{E}[q^{-1/3}|X|^{8/3}]$. Theorem M.37 gives the complete all-length conclusion

$$\mathbb{E}H_{\text{full}}^4 \leq e^{-6d}, \qquad I_q(T^M H_{\text{full}})^3 \leq e^{-24d/19}. \tag{Q192.A.1}$$

Its proof keeps both finite G_1 boundaries, the full constant vectors, and the distinct unit and nonunit terminal directions. The ordinary fourth moment is followed by the paid unanchored outlet with the original denominator q; no reference arm is added.

Section 11 assembles these local interfaces with the appropriate central estimate. The common theorem concerns the actual local half-tree; its use leaves every root direction, original edge and decoration in place. At $d = 0$, original leaf integration gives equality; the kernel itself need not be zero.

Q.25 Minimum-two odd roots with every large-minimum even terminal

R2.0 Statement and fixed directions

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All

means, norms, adjoints and spectral integrals use the same original probability measure. The distinct residual arm list at the internal root is

$$(2, b, c, s_1, t_1, \cdots, s_k, t_k),$$
$$3 \le b < c < s_1 < t_1 < \cdots < s_k < t_k, k \ge 0$$

The distinct residual list at the distant terminal is

$$(a_1, z_1, \cdots, a_j, z_j), 4 \le a_1 < z_1 < \cdots < a_j < z_j, j \ge 1$$

The inner length $L \ge 1$ and outer length $M \ge 1$ are independent. Arbitrary genuine equal-pair products E_r, E_t are allowed at their original roots.

Fix $F = \prod_{i=1}^{j} h_{a_i} f_{z_i}$, and choose the root pair product as follows:

$$P = \bar{f}_3 W_c \prod_{i=1}^{k} W_{s_i} \bar{f}_{t_i}, \quad \text{if} b = 3;$$
$$P = W_b \bar{f}_c \prod_{i=1}^{k} W_{s_i} \bar{f}_{t_i}, \quad \text{if} b \ge 4$$

The shortest root arm, of length two, has the actual message $W_2 = Aq$. Every inner edge points from the terminal toward the internal root; every outer edge points from that root toward the receiver. Thus the actual half-tree message is

$$H = E_r W_2 P T^L (E_t F)$$

An f arm points entirely outward; a $\bar{f}$ arm entirely toward its root; an h arm has its first edge toward the root and all remaining edges toward the leaf; a W arm has its first edge away from the root and all remaining edges toward the first neighbour. The direction depends only on the original lengths and on whether $b = 3$ or $b \ge 4$. For all these parameters and hosts, put

$$\lambda_j = 15/2 + 64(j-1)/5$$

Then

$$EH^4 \le \exp(-\lambda_j d), \tag{R2.0.1}$$

$$\begin{aligned} I_q(T^M H)^3 &\le \exp[-(2\lambda_j - c(M))d] \\ &\le \exp[-(81/19 + 128(j-1)/5)d], \end{aligned} \tag{R2.0.2}$$

$$I_q(X) = E[q^{-1/3}|X|^{8/3}]$$

The all-positive-length outlet of Q5-C164 gives $c(1) = 204/19$, $c(2) = 10$, $c(3) = 0$, $c(4) = 7$, and $c(M) = 6$ for $M \ge 5$.

R2.1 Finite identities and root-pair inputs

Use the original finite-space bounds of Section M.2, the positive convolution (M.2), and the signed mean and complete-vector square (M.3). At $n = 1$, $G_n = (1, 0)$; $|G_2|^2 = 1 + x$, and the contraction is used only at $n \ge 3$. Every full-vector fourth estimate below is (M.43), including its constant component.

R2.1.1 The root pair when $b = 3$

Q5-C188, Section 1, equation (10), gives the following estimate for the same actual pair $\bar{f}_3 W_c$, $c \geq 4$:

$$\begin{gathered} \bar{f}_3 W_c \leq \exp[-\Gamma_r + B(U_r + \beta \mathbf{1})], \\ \Gamma_r = d + \beta_c + \beta_{c-1} \geq 3d - 2\delta_2 \geq 35d/16, \\ \beta = \beta_{c-1}, \tau \leq \beta \leq d, \\ U_r = (xE + zxD)(B)u, G_{c-3} = E + zD \end{gathered} \tag{R2.1.4}$$

This is the direction $\bar{f}_3 W_c$. For all later root pairs, whose shorter lengths satisfy $s_i \geq 5$, Q5-C188, Sections 1–2, gives

$$\begin{gathered} \prod W_{s_i} \bar{f}_{t_i} \leq \exp[-16kd/5 + BU_+], \\ \|U_+\| \leq \sqrt{d}/4, EBU_+ \leq d/12, \quad \text{if} k \geq 1 \end{gathered} \tag{R2.1.5}$$

R2.1.2 The root pairs when b is at least four

Combining the first and all later pairs in one actual finite mask U, Q5-C096 and Q5-C188 give

$$\begin{gathered} P \leq \exp[-16(k+1)d/5 + BU], \\ \|U\| \leq 3\sqrt{d}/8, EBU \leq d/12 \end{gathered} \tag{R2.1.6}$$

Each unequal pair retains its negative scalar loss and its finite mask.

R2.2 One finite terminal packet for every pair count

Q5-C180, Sections 1–2, gives, for the same actual terminal F,

$$\begin{gathered} F \leq \exp(R - \Gamma), \Gamma = 16jd/5, \\ R = \sum_{i=1}^{j} \sum_{r=a_i+1}^{z_i} K^r 1 \end{gathered}$$

Define $D_i = EB^i R$, $\sigma = ER^2$, $\delta = D_0 + 5\sigma/4$, and $\epsilon = \delta + D_1$. The complete finite-mask estimates are

$$\begin{gathered} \sigma < d/18, \|R\| < \sqrt{d}/4, |R| < 1/4, \\ -d/5 \leq D_0 \leq d/12, -d/12 \leq D_1 \leq d/5, \\ D_i \leq d/12 (alli \geq 2), \\ |\delta| \leq d/5, \delta \leq 11d/72, \\ |\epsilon| \leq 3d/8, \epsilon \leq 127d/360, \\ \gamma_l := \beta_l - D_{l-1} \geq d/2 (allfinitel \geq 3) \end{gathered} \tag{R2.2.1}$$

The index range for γ_l includes $L+1$. The upper bound for D_1 is $d/5$. The bounds are independent of j because the original power intervals are disjoint. These finite R bounds give the intermediate estimates of Q5-C180.

The actual Taylor bound $e^R \le 1 + R + 5R^2/8$ and positivity give the finite majorant (R29.5.3), with $c = 5/4$. At $L = 1$ it is $q + BR + \delta$. Every bracket is positive because it dominates the same positive actual message.

R2.3 The complete exponential table

Apply Lemma M.34 to the actual terminal packet, with $c_T = 5/4$ and $h = 1/2$. The four rows of (M.74) give $\log(W_2 T^L F) \le a + B(V_0 + c_0\mathbf{1})$ for $L = 1, 2, 3$ and $L \ge 4$, respectively. In the long row $n = L - 1$, its exact mean is

$$\mathbb{E}BV_0 = d - \beta_{L+1} + D_L = d - \gamma_{L+1} \le d/2. \tag{R2.3.1}$$

The constant $c_0 = -d/2$ stays in the full vector. Every row uses the fixed directions stated above.

R2.4 The root $b = 3$ for every inner length

First take $k = 0$. Add the scalar and full vector of the root-pair bound to the same exponent in part R2.3 and apply the full-vector exponential bound once. Write $S = a - \Gamma_r + EBV_{\text{full}}$ and $N = \|V_{\text{full}}\|^2$. We give the weakest scalar case $j = 1$. Each additional terminal pair subtracts $16d/5$ from the scalar, without changing the vector or mean estimates, and hence adds $64d/5$ to the fourth-moment loss.

R2.4.1 Three root norms and a finite path norm

The spectral identities and $1 - x \le E \le 1$, $0 \le D \le 1$, including G_1, imply

$$\begin{aligned} \|U_r\|^2 &\le \theta^2(1+\theta)d < d/4, \\ \|-u + U_r\|^2 &\le d, \\ \|Bu + U_r\|^2 \le d,\ EBU_r \le 0,\ EU_r &\le 0 \end{aligned} \tag{R2.4.1}$$

For the second inequality, the frequency bound is $(1-x+x^2)^2+x^3 = 1+x(-2+3x-x^2+x^3) \le 1$, since $-2 + 3x - x^2 + x^3 \le -2 + 3\theta + \theta^3 < 0$. For the third it is $x + 3x^2 + x^3 \le 31733/32768 < 1$. The first uses $\theta^2(1 + \theta) = 7605/32768 < 1/4$. The real coordinate one and imaginary coordinate zero of G_1 are both included.

For $n \ge 3$, $G_n - 1 = zG_{n-1}$. At $n = 3$, its squared modulus is $x + x^2 \le 585/1024 < 16/25$; for $n \ge 4$, use $|G_{n-1}| \le 1$. Therefore

$$\|(G_n - 1)u\| \le 4\sqrt{d}/5$$

Also $\|BR\| < \sqrt{d}/6$, $\|B^2R\| < 5\sqrt{d}/48$, and $\|B^nR\| < \sqrt{d}/15$ for $n \ge 3$. These follow respectively from $\theta/16 < 1/36$, $\theta/4 < 5/48$, and $\theta^3/16 < 1/225$, which are rational comparisons on the whole interval.

R2.4.2 The unit inner length and its constant component

Here $V_{\text{full}} = R - u + U_r + \beta\mathbf{1}$. The nonconstant part has norm at most $5\sqrt{d}/4$ and mean $E(R - u + U_r) = D_0 + EU_r \le d/12$. Thus its constant square and cross term give

$$\begin{aligned} N &\le 25d/16 + 2\beta(d/12) + \beta^2 \\ &\le 25d/16 + 7d^2/6 \le 281d/144 \end{aligned} \tag{R2.4.2}$$

The actual B-mean is at most $D_1 + d + EBU_r \le 6d/5$, so

$$S \le (-16/5 - 35/16 + 11/72 + 6/5)d = -581d/144$$

Consequently

$$\begin{aligned} \log EH^4 &\le -[581/36 - (17/4)(281/144)]d \\ &= -4519d/576 < -15d/2 \end{aligned} \tag{R2.4.3}$$

R2.4.3 Inner length two

Now $V_{\text{full}} = BR + U_r + (\beta + \delta)\mathbf{1}$. We have $\|BR + U_r\| \le 2\sqrt{d}/3$ and $|\beta + \delta| \le 6d/5$. Using $\sqrt{d} \le 3/5$ gives $\|V_{\text{full}}\| \le 7\sqrt{d}/5$. The constant has zero B-mean, hence $EBV_{\text{full}} \le D_2 \le d/12$. Therefore

$$\begin{aligned} S &\le (-16/5 - 35/16 + 3/8 + 1/12)d = -1183d/240, \\ \log EH^4 &\le -[1183/60 - 833/100]d = -854d/75 < -15d/2 \end{aligned} \tag{R2.4.4}$$

R2.4.4 Inner length three and the signed constant cross term

Write $V_{\text{full}} = V_0 + c_0\mathbf{1}$, where $V_0 = (B + \delta)u + B^2R + U_r$ and $c_0 = \beta + \epsilon$. The terminal packet gives $7d/32 \le c_0 \le 11d/8$, while

$$EV_0 = -d + D_2 + EU_r \le -11d/12$$

Thus $2c_0EV_0 + c_0^2 \le c_0(c_0 - 11d/6) \le 0$. This pays the actual constant square and cross term. The preceding norms give

$$\begin{aligned} \|V_0\| &\le (1 + 1/15 + 5/48)\sqrt{d} < 6\sqrt{d}/5, \\ N &\le 36d/25 \end{aligned}$$

Also $EBV_{\text{full}} = -\delta d + D_3 + EBU_r \le d^2/5 + d/12$. Combining this with the actual scalar D_2, we obtain

$$\begin{aligned} S &\le (-16/5 - 35/16 + 3/8 - 1 + 1/6 + 1/15)d = -1387d/240, \\ \log EH^4 &\le -[1387/60 - 153/25]d = -5099d/300 < -15d/2 \end{aligned} \tag{R2.4.5}$$

R2.4.5 Every finite inner length at least four

Write $V_{\text{full}} = V_0 + c_0\mathbf{1}$, where $V_0 = (G_n - 1)u + B^nR + U_r$, $c_0 = \beta - d/2$, and $n = L - 1 \ge 3$. Then $3d/32 \le c_0 \le d/2$, and

$$EV_0 = -\beta_{n+1} + D_n + EU_r \le -49d/96$$

Hence $2c_0EV_0 + c_0^2 \le 0$ and $N \le \|V_0\|^2 \le (41/30)^2 d$. The exact terminal mean and $EBU_r \le 0$ give

$$S \le (-16/5 - 35/16 + 11/36 + 2/5 - 1 + 1/2)d = -3731d/720$$

It follows that

$$\begin{aligned}\log EH^4 &\le -[3731/180 - 28577/3600]d \\ &= -46043d/3600 < -15d/2\end{aligned} \tag{R2.4.6}$$

The terminal packet supplies γ_{L+1}; the terminal mean at the end of the finite path is retained.

R2.5 The root b at least four, including all later pairs

Add the complete root-pair mask to the table in part R2.3. The weakest scalar is $j = 1, k = 0$; arbitrary j, k subtract a further $16(j-1)d/5 + 16kd/5$. Every vector in the following table includes its displayed constant. Direct triangle bounds suffice.

Table 112: Complete-vector bounds for roots with $b \ge 4$, indexed by inner length L.

L	full V_{full}
1	$R - u + U$; upper bound for S/d: $-1787/360$; upper bound for $\|V_{\text{full}}\|/\sqrt{d}$: $13/8$.
2	$BR + U + \delta\mathbf{1}$; upper bound for S/d: $-703/120$; upper bound for $\|V_{\text{full}}\|/\sqrt{d}$: $2/3$.
3	$(B+\delta)u + B^2R + U + \epsilon\mathbf{1}$; upper bound for S/d: $-161/24$; upper bound for $\|V_{\text{full}}\|/\sqrt{d}$: $3/2$.
≥ 4	$(G_n - 1)u + B^nR + U - (d/2)\mathbf{1}$; upper bound for S/d: $-55/9$; upper bound for $\|V_{\text{full}}\|/\sqrt{d}$: $8/5$.

Here are the four scalar and norm calculations.

At $L = 1$, scalar plus mean is $-32d/5 + \delta + D_1 + d + EBU$, at most $(-32/5 + 11/72 + 1/5 + 1 + 1/12)d = -1787d/360$. The norm coefficient is $1/4 + 1 + 3/8 = 13/8$.

At $L = 2$, it is $-32d/5 + \epsilon + D_2 + EBU$, at most $(-32/5 + 3/8 + 1/6)d = -703d/120$. The norm coefficient is $1/6 + 3/8 + 3/25 < 2/3$.

At $L = 3$, it is $-32d/5 + \epsilon - d + D_2 - \delta d + D_3 + EBU$, at most $(-32/5 + 3/8 - 1 + 1/4 + 1/15)d = -161d/24$. Its norm coefficient is $13/20 + 1/15 + 5/48 + 3/8 + 9/40 = 341/240 < 3/2$. Here $\sqrt{\theta} < 13/20$, and the full constant $|\epsilon| \le 3d/8$ has been included.

For $L \ge 4$, the expression is $-32d/5 + 2\delta_+ + 2(D_1)_+ - d + EBV_{\text{full}}$. The exact terminal mean and $EBU \le d/12$ give coefficient $-32/5 + 11/36 + 2/5 - 1 + 1/2 + 1/12 = -55/9$. The norm coefficient is $4/5 + 1/15 + 3/8 + 3/10 = 37/24 < 8/5$.

The full-vector exponential bound therefore gives fourth-moment losses at least

$$99451/11520, 1939/90, 829/48, 3052/225,$$

All four exceed $15/2$. This treats all $b \ge 4$, all root-pair counts and every L.

R2.6 Arbitrarily many later root pairs when $b = 3$

Every full vector in part R2.4 has norm at most $2\sqrt{d}$: use respectively the bound $281d/144$, coefficient $7/5$, coefficient $6/5$ after the cross term is retained, and coefficient $41/30$. Adding the one mask U_+ for all later pairs gives the additional fourth exponent

$$\begin{aligned}&-64kd/5 + 4EBU_+ + (17/4)(2\|V_{\text{full}}\|\|U_+\| + \|U_+\|^2) \\ &\le [931/192 - 64k/5]d < 0, k \ge 1\end{aligned} \tag{R2.6.1}$$

The bound uses the full old vector and therefore includes its constant cross terms. Together with parts R2.4–R2.5, it proves the stated ordinary fourth bound and the complete λ_j formula.

R2.7 Decorations, outer propagation and the path-skeleton consequence

Genuine equal pairs have their Q3-C014 directions and actual factors in $[0,1]$. Positivity gives $E_r W_2 P T^L (E_t F) \le W_2 P T^L F$, retaining every original edge and vertex in the actual tree. Apply the Q5-C164 outlet to this actual H. Its worst cost gives $2\lambda_1 - 204/19 = 15 - 204/19 = 81/19$.

Suppose the two sides satisfy this theorem, with independent original core lengths L_-, M_-, M_+, L_+. Let the central residual list be empty or have at least two arms. The exact negative zero-step table of Q4-C143 gives an actual center C with $E[qC^4] \le 1$. Write $X_i = T^{M_i} H_i$.

Leaf integration gives $Z_D = E[CX_-X_+]$, and Hölder with the same original q gives

$$\begin{aligned} Z_D &\le (E[qC^4])^{1/4} I_q(X_-)^{3/8} I_q(X_+)^{3/8} \\ &\le \exp[-(2\lambda_{j_-} + 2\lambda_{j_+} - c(M_-) - c(M_+))d/8] \\ &\le \exp(-81d/76) \end{aligned} \tag{R2.7.1}$$

The center direction follows the rules of Q5-C180: an empty residual uses reversed equal pairs; an even residual uses the reversed pair product; an odd residual with minimum one uses $p\bar{f}_s W_t$ and later reversed pairs; an odd residual with minimum at least two uses f_a and later $h_s f_t$ pairs.

R2.8 Original counts, boundaries and entropy

The original half-tree, before adding its outer connector, has edge count

$$e_h alf = 2 + b + c + sum_i(s_i + t_i) + L + sum_i(a_i + z_i) + 2S_r + 2S_t$$

Here S_r, S_t are the sums of one length from each original equal pair. The outer connector adds M original edges. For the full tree, the four core lengths plus all original side-arm lengths, with their multiplicities, give $e(U)$, and $|V(U)| = e(U) + 1$. Branch vertices are counted by their original degrees.

At $d = 0$, $B\mathbf{1} = 0$, hence $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$, and leaf integration makes all actual messages one. This allows nonzero K. Every positive atom, the one-atom host and arbitrarily small atom masses are included. Complete reversal reverses all original edges, arms, connectors, weights and decorations simultaneously by replacing K with $-K$; it leaves d unchanged.

For any correlated original joint label law ρ, the actual edge product W_D and $P_0 = \mu^{V(U)}$, Gibbs gives $E_\rho \log W_D \le D(\rho \| P_0) + \log Z_D$. If $W_D = 0$ on positive ρ-mass, the left side is minus infinity; lack of absolute continuity gives infinite KL. Constant labels ensure $Z_D > 0$ on positive support. For a loopless tournament on N vertices, apply $1_{\text{arc}} \le (1+K)/2$ to each original edge and sum over all original maps to obtain $\hom(D, \mathcal{T}) \le 2^{-e(U)} N^{e(U)+1} Z_D$. Nonadjacent original vertices may have the same image.

Q.26 Minimum-two odd roots with every even multipair terminal

R3.0 Parameters, actual directions and conclusion

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The internal

root has distinct residual list $(2, b, c, s_1, t_1, \dots, s_k, t_k)$, with $3 \le b < c < \cdots$, $k \ge 0$. The terminal residual is $(a_1, z_1, \dots, a_j, z_j)$, with $1 \le a_1 < z_1 < \cdots < a_j < z_j$, $j \ge 2$. The original lengths L, $M \ge 1$ are independent. Equal pairs, subdivisions and multiplicities are arbitrary.

Always use $F = \prod_i h_{a_i} f_{z_i}$, with $h_1 = q$. Every inner edge points from terminal to internal root, and every outer edge from that root to the receiver. The usual root direction is

$$C_r = W_2 \bar{f}_3 W_c \prod W_{s_i} \bar{f}_{t_i}, b = 3;$$
$$C_r = W_2 W_b \bar{f}_c \prod W_{s_i} \bar{f}_{t_i}, b \ge 4$$

There is one structural exception. If $(a_1, z_1, a_2) = (1, 2, 3)$, $L = 1$ and $b \ge 4$, use

$$C_r = f_2 h_b f_c \prod h_{s_i} f_{t_i} \tag{R3.0.1}$$

This exception depends only on the three original terminal lengths, b and L, before the host is known. The literal arm directions are those of Section Q.25. The decorated actual message is $H = E_r C_r T^L (E_t F)$.

For all these original parameters and hosts,

$$I_q (T^M H)^3 \le \exp(-3d/95), all independent L, M \ge 1 \tag{R3.0.2}$$

More precisely, for $a_1 \ge 2$, the full-domain ordinary fourth bound is

$$EH^4 \le \exp[-(15/2 + 64(j-2)/5)d] \tag{R3.0.3}$$

For $a_1 = 1$, on the whole d domain when $L \ne 3$, and on $0 \le d \le 1/8$ when $L = 3$,

$$EH^4 \le \exp[-(11/2 + 64(j-2)/5)d] \tag{R3.0.4}$$

When $L = 3$ and $d \ge 1/8$, the same actual direction is compared directly with the Q5-C188 outlet, giving the stated outlet bound.

R3.1 Common finite inputs and propagation

Use the finite convolution (M.2) and the complete-vector estimate (M.43), with the original signed mean.

For $b = 3$, the actual root pair $\bar{f}_3 W_c$ satisfies

$$\begin{gathered} \bar{f}_3 W_c \le \exp[-\Gamma_r + B(U_r + \beta \mathbf{1})], \\ \Gamma_r = d + \beta_c + \beta_{c-1} \ge 3d - 2\delta_2, \\ \beta = \beta_{c-1} \in [\tau, d], U_r = (xE + zxD)(B)u, \\ G_{c-3} = E + zD, z = i\sqrt{x} \end{gathered} \tag{R3.1.2}$$

The coordinates satisfy $E \ge 1 - x$, $0 \le D \le 1$, including $G_1 = (1, 0)$. All later root pairs, whose shorter lengths are at least five, combine into one U_+ with $\|U_+\| \le \sqrt{d}/4$, $EBU_+ \le d/12$, and scalar loss $16kd/5$. If $b \ge 4$, the first and all later pairs combine into one U with $\|U\| \le 3\sqrt{d}/8$, $EBU \le d/12$, and scalar loss $16(k+1)d/5$. Both the positive $h_s f_t$ and negative $W_s \bar{f}_t$ directions have these finite-mask bounds.

For $b \geq 4$ we also need the original mean bound

$$|EU| \leq \delta_2/(1-\theta^2) < d/2 \tag{R3.1.3}$$

The lowest power acting on u in U is B^3. The absolute same-sign sum of its nonzero constant means starts at δ_2, with moment indices separated by two. Each of the positive and negative tails is at most $\delta_2/(1-\theta^2)$; take the larger tail, rather than their sum. It is less than $d/2$ because $C_\beta\theta = 416/855 < 1/2$.

The three root norms are

$$\begin{gathered}\|U_r\| < \sqrt{d}/2, \| - u + U_r\| \leq \sqrt{d}, \\ \|Bu + U_r\| \leq \sqrt{d}, EBU_r \leq 0, EU_r \leq 0\end{gathered} \tag{R3.1.4}$$

The frequency proof is in part R2.4.1, or follows by substitution in the displayed root coordinates. Also $\|(G_n - 1)u\| \leq 4\sqrt{d}/5$ for every $n \geq 3$.

R3.1.1 The exponential table for a genuine terminal certificate

For the same actual $F \leq e^{R-\Gamma}$, use its proved Taylor coefficient c_T, and write $D_i = \mathbb{E}B^i R$, $\delta = D_0 + c_T\mathbb{E}R^2$, $\epsilon = \delta + D_1$. Once $h \geq 0$ and $\gamma_l = \beta_l - D_{l-1} \geq hd$ for all $l \geq 3$, Lemma M.34 gives all four length rows. For $L \geq 4$, $n = L - 1$, its mean is

$$\mathbb{E}BV_0 = d - \beta_{L+1} + D_L \leq (1-h)d. \tag{R3.1.5}$$

For the structural $L = 1$ exception, use the actual f_2 row: $V_0 = R + u$. After adding the actual pairs, set $V = V_0 + U_r + (\beta + c_0)\mathbf{1}$ at $b = 3$, and $V = V_0 + U + c_0\mathbf{1}$ at $b \geq 4$. With $S = a - \Gamma_{\text{root}} + \mathbb{E}BV$, $N = \|V\|^2$, (M.76) gives $\log \mathbb{E}H^4 \leq 4S + 17N/4$. Every subsequent N includes all constant squares and cross terms.

R3.2 Five finite terminal packets

Use the actual R, Γ and finite estimates in Q5-C182, Sections 3–6. Put $\Delta = 16(j-2)d/5$; add Δ to each basic Γ in the table.

Table 113: Minimum-two odd roots with every even multipair terminal: parameter values.

Entry	Original terminal condition
E2	$a_1 = 2$; Basic lower bound for Gamma/d: $607/160$; $\|R\|/\sqrt{d}$: $1/\sqrt{3}$; delta/d: $[-1/5, 91/100]$; epsilon/d: $[-7/10, 7/10]$; h: $1/2$.
E3	$a_1 \geq 3$; Basic lower bound for Gamma/d: $767/160$; $\|R\|/\sqrt{d}$: $3/8$; delta/d: $[-1/5, 2/3]$; epsilon/d: $[-17/60, 7/8]$; h: $1/2$.
I	$a_1 = 1, z_1 \geq 3$; Basic lower bound for Gamma/d: $607/160$; $\|R\|/\sqrt{d}$: $\sqrt{325/392}$; delta/d: $[-6/5, 3/4]$; epsilon/d: $[-17/10, 3/4]$; h: $3/8$.
II	$a_1 = 1, z_1 = 2, a_2 \geq 4$; Basic lower bound for Gamma/d: $16/5$; $\|R\|/\sqrt{d}$: $\sqrt{325/512}$; delta/d: $[-6/5, 1/25]$; epsilon/d: $[-13/10, 6/25]$; h: $3/8$.
III	$a_1 = 1, z_1 = 2, a_2 = 3$; Basic lower bound for Gamma/d: $3 - 2\delta_2/d$; $\|R\|/\sqrt{d}$: $\sqrt{1277/2592}$; delta/d: $[-1, 3/20]$; epsilon/d: $[-4/3, 7/20]$; h: $1/2$.

In the final row use the undivided expression $\Gamma = 3d - 2\delta_2$ at $d = 0$. The sharper E3 lower endpoint $\epsilon \geq -17d/60$ is supplied in Q5-C182, Section 5. All packet estimates hold throughout $0 \leq d \leq 1/3$; only the new $L = 3$ payment below is restricted to $d \leq 1/8$.

Table 114: Minimum-two odd roots with every even multipair terminal: parameter values.

Entry	upper bound for D_1
E2	0; upper bound for D_2: $d/12$; upper bound for D_3: $d/5$; upper bound for every $D_n, n \geq 4$: $d/12$.
E3	$d/5$; upper bound for D_2: $d/12$; upper bound for D_3: $d/30$; upper bound for every D_n, $n \geq 4$: $d/12$.
I	0; upper bound for D_2: $d/2$; upper bound for D_3: $d/5$; upper bound for every $D_n, n \geq 4$: $d/5$.
II	$d/5$; upper bound for D_2: $d/2$; upper bound for D_3: $d/5$; upper bound for every D_n, $n \geq 4$: $d/5$.
III	$d/5$; upper bound for D_2: δ_2; upper bound for D_3: $d/8$; upper bound for every $D_n, n \geq 4$: $d/12$.

Every finite D_i and γ_l index, including $L+1$, is covered by the proof. All bounds concern one finite zero-one mask, so the norms do not grow with j.

R3.2.1 The retained finite coordinates

For E2 write $R = (\alpha + z\beta_0)(B)u$. Then

$$x/2 \leq \alpha \leq 6x/5, -6x/5 \leq \beta_0 \leq x/2 \tag{R3.2.1}$$

For I, II and III, write $R = (\xi + xV + zU)(B)u$. Here U, V denote terminal spectral coordinates only; the root vector denoted U above is distinct.

Table 115: Minimum-two odd roots with every even multipair terminal: parameter values.

Entry	xi
I	0; U: $[1/2, 6/5]$; V: $[1/2, 6/5]$.
II	0; U: $[11/12, 6/5]$; V: $[-C_\beta x, C_\beta x^2]$.
III	τ; U: $[1 - C_\beta x, 1 - x/3]$; V: $[-C_\beta x, C_\beta x^2]$.

Here $C_\beta = 1024/855 < 6/5$ and $0 \leq \tau \leq 2/9$. Thus $D_2 = \int xU \geq 0$ in all three cases. The global scalars $\xi = \tau, \delta, \epsilon$ are constant in the frequency.

R3.3 E2 and E3: eight full-domain fourth-moment payments

First take $j = 2, k = 0$, with either root class. Restore the increments below. In each cell, $-S/d \geq s$ and $N/d \leq n$; the loss is $4s - 17n/4$.

Table 116: Minimum-two odd roots with every even multipair terminal: parameter values.

Entry	L1
$E2, b = 3$	(s, n): $(4, 5/3)$; L2: (s, n): $(5, 7/3)$; L3: (s, n): $(23/4, 8/3)$; $L \geq 4$: (s, n): $(9/2, 17/8)$.
$E2, b \geq 4$	(s, n): $(5, 121/64)$; L2: (s, n): $(6, 16/9)$; L3: (s, n): $(27/4, 4)$; $L \geq 4$: (s, n): $(11/2, 169/64)$.
$E3, b = 3$	(s, n): $(5, 11/4)$; L2: (s, n): $(6, 7/4)$; L3: (s, n): $(27/4, 2)$; $L \geq 4$: (s, n): $(11/2, 49/25)$.

Entry	L1
$E3, b \geq 4$	(s, n): $(6, 49/16)$; L2: (s, n): $(27/4, 441/400)$; L3: (s, n): $(31/4, 4)$; $L \geq 4$: (s, n): $(13/2, 64/25)$.

Every cell gives a loss greater than $15/2$; the smallest listed sufficient value is $133/16$, in E3 with $b = 3, L = 1$. We prove all cells on their continuous domains.

R3.3.1 The E2 unit-connector vector

The real coordinate of $R - u + U_r$ is $-1 + xE + \alpha < 0$, with absolute value at most $1 - 3x/2 + x^2$; the imaginary coefficient has absolute value at most $3x/2$. Its squared norm is at most d, since

$$(1 - 3x/2 + x^2)^2 + (9/4)x^3$$
$$= 1 + x[-3 + (17/4)x - (3/4)x^2 + x^3] \leq 1$$

The bracket is nonpositive because $-3 + 17\theta/4 + \theta^3 < 0$. The nonconstant mean is at most $6\delta_2/5 \leq 39d/80$; adding the real β constant costs at most $79d^2/40 \leq 79d/120$. Hence $N < 5d/3$. Without U_r, $\|R - u\| \leq \sqrt{d}$, by $(1 - x/2)^2 + 36x^3/25 \leq 1$. Adding the complete $b \geq 4$ root mask yields $N \leq 121d/64$.

R3.3.2 E2 at lengths two, three and every later length

At $L = 2$, the real coordinate of $BR + U_r$ is nonnegative and at most $x + 6x^2/5$; its imaginary coefficient is at most $11x/5$. Thus its squared-frequency bound is $x^2 + 181x^3/25 + 36x^4/25 < 1$, using $x^2 < 1/6$, $x^3 < 1/14$, $x^4 < 1/36$. The constant $c = \beta + \delta$ lies in $[(19/32 - 1/5)d, 191d/100]$, while $E(BR + U_r) \leq D_1 \leq 0$. Hence $N \leq d + 4d^2 \leq 7d/3$. For $b \geq 4$, use $\|BR\| < 3\sqrt{d}/8$, $|\delta| \leq 91d/100$: the full norm coefficient is less than $3/8 + 3/8 + 273/500 < 4/3$.

At $L = 3$, $V_0 = (B + \delta)u + B^2R + U_r$ has norm at most $(1 + 91/300 + 1/4)\sqrt{d} = 233\sqrt{d}/150$. Its mean is at most $-11d/12$ and has absolute value at most $2d$, using $|D_2| \leq 3d/8$, $|EU_r| \leq \delta_2$. The positive and negative parts of $c = \beta + \epsilon$ are at most $17d/10$ and $17d/160$. For $c \geq 0$ the constant increment is nonpositive; for $c < 0$ it is at most $[17/40 + 289/25600]d^2$. Thus $N < 8d/3$. For $b \geq 4$, the full triangle coefficient is $13/20 + 91/300 + 1/4 + 3/8 + 21/50 = 1199/600 < 2$.

For $L \geq 4$, $\|B^nR\| < 3\sqrt{d}/20$, $n \geq 3$. For $b = 3$, the nonconstant norm coefficient is at most $29/20$, while $c = \beta - d/2 \in [3d/32, d/2]$ and $EV_0 \leq -19d/32 + d/5 = -63d/160$; the constant increment is nonpositive, giving $N < 17d/8$. For $b \geq 4$, the full norm coefficient is at most $4/5 + 3/20 + 3/8 + 3/10 = 13/8$.

The four exact scalar lower bounds for $b = 3$ are respectively $137/32$, $607/160 + 35/16 - 7/10 - 1/12$, $607/160 + 35/16 + 1 - 7/10 - 1/12 - 1/5 - 1/15$, and $607/160 + 35/16 - 33/25$, all exceeding the table entries. For $b \geq 4$, use $\Gamma_{\text{root}} = 16d/5$, $EBU \leq d/12$ to obtain successively bounds exceeding $5, 6, 27/4, 11/2$. The $L = 3$ term $-\delta d$ is paid using $\delta \geq -d/5$.

R3.3.3 E3 at all four length ranges

Here $\|R\| < 3\sqrt{d}/8$, $\|BR\| < \sqrt{d}/4$, $\|B^2R\| < \sqrt{d}/6$, and $\|B^nR\| < \sqrt{d}/10$ for $n \geq 3$. For $b = 3, L = 1$, the nonconstant norm is at most $11\sqrt{d}/8$, with mean at most $D_0 \leq \delta \leq 2d/3$; hence $N \leq 121d/64 + 7d^2/3 < 11d/4$. At $L = 2$, the nonconstant norm is at most $3\sqrt{d}/4$; $0 \leq c = \beta + \delta \leq 5d/3$, and the nonconstant mean is at most $d/5$, giving $N \leq 9d/16 + 31d^2/9 < 7d/4$. At $L = 3$, the

nonconstant norm is at most $25\sqrt{d}/18$, with mean at most $-11d/12$. The exact $\epsilon \geq -17d/60$ ensures $0 \leq c = \beta + \epsilon \leq 15d/8$; the constant increment is at most $5d^2/64$, so $N \leq 625d/324 + 5d^2/64 < 2d$. For later lengths the nonconstant norm is at most $7\sqrt{d}/5$, while $c = \beta - d/2 \geq 0$ and the mean is at most $-49d/96$, so the constant increment is nonpositive.

For $b \geq 4$, the four full norm coefficients are $7/4, 21/20, 2, 8/5$; the third uses $13/20 + 2/9 + 1/6 + 3/8 + 21/40 < 2$, and the others are direct sums including the actual constants. Substitution into S gives the table. The weakest scalar cell is $b = 3, L = 1$, where $\Gamma + \Gamma_r - 7d/8 - d = 817d/160 > 5d$. The other seven lower bounds are $6, 27/4, 11/2, 6, 27/4, 31/4, 13/2$. This proves the basic $j = 2$ fourth estimate.

R3.4 The three minimum-one packets

Continue with $j = 2, k = 0$. The $L = 1, 2$ and $L \geq 4$ rows use $0 \leq d \leq 1/3$; the $L = 3$ rows use $d \leq 1/8$.

Table 117: Minimum-two odd roots with every even multipair terminal: parameter values.

Entry	L1
$I, b = 3$	(s, n): $(4, 3/2)$; L2: (s, n): $(9/2, 3/2)$; L3 on the low domain: (s, n): $(21/4, 23/8)$; $L \geq 4$: (s, n): $(9/2, 12/5)$.
$I, b \geq 4$	(s, n): $(5, 9/4)$; L2: (s, n): $(11/2, 49/16)$; L3 on the low domain: (s, n): $(25/4, 7/2)$; $L \geq 4$: (s, n): $(11/2, 81/25)$.
$II, b = 3$	(s, n): (4, 2); L2: (s, n): $(9/2, 6/5)$; L3 on the low domain: (s, n): $(21/4, 5/2)$; $L \geq 4$: (s, n): $(5, 12/5)$.
$II, b \geq 4$	(s, n): $(5, 841/256)$; L2: (s, n): $(11/2, 25/9)$; L3 on the low domain: (s, n): (6, 4); $L \geq 4$: (s, n): $(23/4, 81/25)$.
$III, b = 3$	(s, n): $(93/20 - 1417/1024, 7/4)$; L2: (s, n): $(7/2, 7/5)$; L3 on the low domain: (s, n): $(17/4, 7/3)$; $L \geq 4$: (s, n): $(33/8, 12/5)$.
$III, b \geq 4$	(s, n): $(23/4, 225/64)$; L2: (s, n): $(9/2, 9/4)$; L3 on the low domain: (s, n): $(21/4, 3)$; $L \geq 4$: (s, n): $(5, 49/16)$.

The III, $b \geq 4, L = 1$ cell uses the sole positive-root exception. Every loss $4s - 17n/4$ exceeds $11/2$. The tightest sufficient value, in III with $b = 3, L = 1$, is $7203/1280$, exceeding $11/2$ by $163/1280$. The next is $6183/1024$, in II with $b \geq 4, L = 1$.

R3.5 All minimum-one vectors and constant terms

R3.5.1 Packet I at lengths one and two

For $L = 1, b = 3$, the real coordinate is negative with absolute value at most $1 - 3x/2 + x^2$, and the imaginary coefficient is at most $x + 6/5$. Its squared-frequency bound is

$$1 - (39/25)x + (133/20)x^2 - 2x^3 + x^4 \leq 3/2$$

After dropping the negative cubic term, this polynomial is convex. At zero and θ it is less than $3/2$. At θ, retain $\theta^2 = 169/1024$ and use $\theta^4 < 1/36$ to get $1 - 507/800 + 22477/20480 + 1/36 < 3/2$. The nonconstant mean is at most $-d/2$, and $c = \beta \leq d$, so the constant increment is nonpositive. For $b \geq 4$, $\|R - u\|^2 < 5d/4$, using the upper bound $1 + 11x/25 + x^2/4$. Adding the root mask gives norm at most $3\sqrt{d}/2$.

For $L = 2, b = 3$, the coordinates of $BR + U_r$ are $x(E - U)$ and $x(D + V)$. Since $|E - U| \le 5/8$, $D + V \le 11/5$, its squared norm is less than $2d/5$, by $25\theta^2/64 + 121\theta^3/25 < 2/5$. The constant lies in $[-97d/160, 7d/4]$; the nonconstant mean is nonpositive with absolute value at most $143d/160$. For nonnegative c the increment is at most $49d^2/16$; for negative c, it is at most $[2(97/160)(143/160) + (97/160)^2]d^2 < 3d^2/2$. Thus $N \le 2d/5 + 49d^2/16 < 3d/2$. For $b \ge 4$, use $\|BR\| < 3\sqrt{d}/5$, $|\delta| \le 6d/5$, giving full coefficient $3/5 + 3/8 + 18/25 < 7/4$.

R3.5.2 Packet II at lengths one and two

For $L = 1, b = 3$, the real coordinate is negative, with absolute value at most $1 - x + 11x^2/5$, and the imaginary coefficient is at most $x + 6/5$. The squared-frequency bound is

$$P(x) = 1 - (14/25)x + (39/5)x^2 - (17/5)x^3 + (121/25)x^4$$

This is convex on $[0, \theta]$, since $P'' \ge 78/5 - 102\theta/5 > 0$. Also $P(0) = 1$. Using $\theta \ge 2/5$, $\theta^2 < 1/6$, $\theta^3 > 1/15$, $\theta^4 < 1/36$, we get $P(\theta) < 8927/4500 < 2$. The nonconstant norm squared is at most $2d$; its mean is at most $-11d/12$, and $c = \beta \le d$, so the constant increment is nonpositive. For $b \ge 4$, $\|R - u\|^2 < 51d/25 < (23/16)^2 d$, since $(1 + 6x^2/5)^2 + 36x/25 < 36/25 + 3/5$. The full norm coefficient is at most $29/16$.

For $L = 2, b = 3$, the coordinate absolute values are at most $5x/8$ and $x + 6x^3/5$, giving nonconstant norm squared less than $d/5$ and mean absolute value less than $5d/8$. Since $|c| = |\beta + \delta| \le 6d/5$, the increment is at most $[2(6/5)(5/8) + 36/25]d^2 = 147d^2/50$, and $N < 6d/5$. For $b \ge 4$, the coefficients $11/20, 3/8, 18/25$ sum to less than $5/3$.

R3.5.3 Packet III and the positive-root exception

For $b = 3, L = 1$, the real coordinate $-1 + \tau + xE + xV$ is negative. Since $\tau \ge 0$ reduces its absolute value, the bound remains $1 - x + 11x^2/5$. The imaginary coefficient is at most $x + 1$, giving

$$1 - x + (37/5)x^2 - (17/5)x^3 + (121/25)x^4$$

This is again convex and its θ endpoint is less than $1567/900 < 7/4$. The nonconstant mean is at most $-(1 - C_\beta\theta)d < -d/2$, so the constant increment is nonpositive. Keep the scalar and mean together: $EBU_r = -\int xE \le -(1 - \theta)\delta_2$, whence

$$\begin{aligned} S &\le -6d + 4\delta_2 + (7/20)d + d - (1 - \theta)\delta_2 \\ &\le -[93/20 - \theta(3 + \theta)]d \\ &= -[93/20 - 1417/1024]d \end{aligned} \tag{R3.5.1}$$

This is the joint scalar bound in the table.

For $b \ge 4, L = 1$, the actual f_2 and positive root pairs give vector $R + u + U_{\text{root}}$. The real coordinate of $R + u$ is at most $1 + 2/9 + 1/12 = 47/36$, and the imaginary coefficient at most one; the squared-frequency bound $(47/36)^2 + \theta < 9/4$ gives full norm at most $15\sqrt{d}/8$. Its actual B-mean contains $-d$, yielding $-S/d \ge 35/16 + 16/5 - 7/20 + 1 - 1/12 > 23/4$. The shortest root arm still consists of its two original edges.

R3.5.4 Packet III at length two

For $b = 3$, the real coordinate $x(E - U)$ has absolute value at most $6x^2/5 < 1/5$, while the imaginary coefficient $xD+\tau+xV$ has absolute value less than $3/4$. Hence the norm squared is less than $d/25+9\theta d/16 < 3d/10$. Its mean has absolute value less than $3d/4$, using $EV_0 = -\tau d-\int x(V+D)$, $\tau \le 2/9$, $C_\beta\theta^3 < 1/12$. Since $|\beta + \delta| \le 6d/5$, the increment is at most $81d^2/25$, giving $N < 7d/5$. For $b \ge 4$, use coefficients $1/2, 3/8, 3/5$, giving full norm less than $3\sqrt{d}/2$.

R3.5.5 All three packets at length three on the low domain

Write $V_0 = (B + \delta)u + B^2R + U_r$, $c = \beta + \epsilon$ for $b = 3$. Retain every constant. Throughout this part $d \le 1/8$, $\sqrt{d} < 3/8$.

In I, $\|V_0\| \le (1 + 3/8 + 3/20)\sqrt{d} = 61\sqrt{d}/40$. Since $EV_0 = -d + \int x(U - D)$, it lies in $[-77d/64, -41d/80]$. The constant lies in $[-177d/160, 7d/4]$. For either sign, $2cEV_0 + c^2 \le 4d^2$, so $N \le 3721d/1600 + 4d^2 < 23d/8$.

In II, $\|V_0\| \le (1+1/3+3/20)\sqrt{d} = 89\sqrt{d}/60$, with mean in $[-397d/384, -41d/80]$ and constant in $[-113d/160, 31d/25]$. The increment is less than $2d^2$ for either sign, giving $N < 5d/2$.

In III, $\|V_0\| \le (1 + 3/10 + 1/8)\sqrt{d} = 57\sqrt{d}/40$, with mean in $[-6d/5, -19d/32]$ and constant in $[-71d/96, 27d/20]$. For negative c, the increment is at most $189d^2/80$; for nonnegative c it is smaller. Hence $N \le 3249d/1600 + 189d^2/80 < 7d/3$.

For $b \ge 4$, replace U_r by the full root mask and take $c = \epsilon$. In I and III the nonconstant mean lies in $[-3d/2, 0]$, using the original mean bound and $D_2 = \int xU \ge 0$, together with $D_2 \le 6\delta_2/5 < d/2$ in I and $D_2 \le \delta_2 < d/2$ in III. In I the nonconstant norm is at most $31\sqrt{d}/20$, the increment at most $799d^2/100$, and $N \le 961d/400 + 799d^2/100 < 7d/2$. In III the corresponding bounds are $29\sqrt{d}/20$, $52d^2/9$, and $N \le 841d/400 + 52d^2/9 < 3d$. In II the full triangle coefficient is $13/20 + 1/3 + 3/8 + 3/20 + 39/80 = 479/240 < 2$, so $N < 4d$.

R3.5.6 Every finite length at least four

For every $n \ge 3$, $\|B^nR\| < 6\sqrt{d}/25$ in I and II, and less than $23\sqrt{d}/100$ in III. For $b = 3$, the nonconstant coefficients are at most $77/50$ and $153/100$, respectively, with squares less than $12/5$. In I and II, $c = \beta - 3d/8 \in [7d/32, 5d/8]$, and $EV_0 \le -19d/32 + d/5 = -63d/160$, so the increment is nonpositive. In III, $c = \beta - d/2 \in [3d/32, d/2]$, and $EV_0 \le -19d/32 + d/8 = -15d/32$, with the same sign conclusion. For $b \ge 4$, the full coefficients are $4/5 + 6/25 + 3/8 + 3/8 < 9/5$ in I and II, and $4/5 + 23/100 + 3/8 + 3/10 < 7/4$ in III.

R3.5.7 Scalar formulas for every cell

Except for the joint scalar bound and positive-root exception above, every scalar entry follows from the same formulas:

Table 118: Lower bounds for $-S$ from the finite propagation formulas.

L	lower bound for -S
1	$\Gamma + \Gamma_r oot - \epsilon - d - m_r.$
2	$\Gamma + \Gamma_r oot - \epsilon - D_2 - m_r.$
3	$\Gamma + \Gamma_r oot - \epsilon + d - D_2 - D_3 + \delta \cdot d - m_r.$
≥ 4	$\Gamma + \Gamma_r oot - 2\delta_+ - 2(D_1)_+ - (1 - 3h)d - m_r.$

Take $m_r = 0$ for $b = 3$, and $m_r = d/12$ for $b \geq 4$. At $L = 3$, the lower bounds for δd are $-3d/20, -3d/20, -d/8$, respectively. Use $\Gamma_{\text{root}} = 35d/16$ or $16d/5$, and basic $\Gamma = 35d/16$ in III. For example, II with $b \geq 4, L = 3$ gives $16/5 + 16/5 - 6/25 + 1 - 1/2 - 1/5 - 3/20 - 1/12 > 6$; III gives $35/16 + 16/5 - 7/20 + 1 - 13/32 - 1/8 - 1/8 - 1/12 > 21/4$. These expressions retain every $D_2, D_3, \delta d$ term and the root mean, proving the fourth estimate on every stated domain.

R3.6 The same direction at length three on the high domain

The same-direction finite-pair estimate in Section Q.9, (C177.T184.27), gives for every actual pair with $4 \leq s < t$,

$$h_s f_t \leq \exp(-88d/125) \leq 1, d \geq 1/64 \tag{R3.6.1}$$

The centered-row calculation proving (O.58) gives

$$h_3 f_t \leq \exp(-d) \leq 1, t \geq 4, d \geq 1/8 \tag{R3.6.2}$$

These are pointwise comparisons in the actual directions. In I and II the second and later pair minima are at least four; in III the second is three and the later ones at least five. Thus, throughout $d \geq 1/8$,

$$F \leq q f_{z_1} \tag{R3.6.3}$$

At $L = 3$, our root is exactly the Q5-C188 nonunit-root direction: $W_2 \bar{f}_3 W_c$ and later reversed pairs for $b = 3$, or $W_2 W_b \bar{f}_c$ and later reversed pairs for $b \geq 4$. Both connectors use T, with the same original q. Positivity and monotonicity of I_q compare all decorations and later pairs directly with the actual terminal $(1, z_1)$ half-tree of Q5-C188, Theorem A. Hence

$$I_q(T^M H)^3 \leq \exp(-3d/95), M \geq 1 \tag{R3.6.4}$$

R3.7 All later root pairs, terminal-pair counts and outer lengths

For $b = 3$, every full paid vector has norm at most $2\sqrt{d}$, because all table entries have $n < 4$. Adding the one mask U_+ for every later root pair gives additional exponent at most

$$[931/192 - 64k/5]d < 0, k \geq 1,$$

This is the calculation in part R2.6, retaining all constant cross terms. For $b \geq 4$, every root pair is already in one mask, including the positive-root exception with its corresponding sign. Each additional terminal pair adds $16d/5$ to Γ, leaving packet bounds unchanged, and thus adds $64d/5$ to the ordinary fourth loss. The high-domain comparison allows arbitrary j, k.

Actual equal-pair products lie in $[0, 1]$; positivity restores E_r, E_t. In the ordinary-fourth domains Q5-C164 gives outlet losses at least $15 - 204/19 = 81/19$ in E2/E3, or $11 - 204/19 = 5/19$ in I/II/III. Both exceed $3/95$. Combining with the high-domain outlet proves the conclusion for all L, M, d.

R3.8 The path-skeleton interface and original boundaries

For two sides from this theorem and the Q4-C143 negative zero-step center with the same q, apply Hölder once with exponents $4, 8/3, 8/3$ to obtain

$$Z_D \leq \exp[-(3/95 + 3/95)d/8] = \exp(-3d/380) \tag{R3.8.1}$$

The four original core lengths remain independent. A mixed pair of side constructions uses its separately stated actual directions and compatible central role. The original edge count is the sum of all four core lengths and every side-arm length with its multiplicity; $|V| = e + 1$. Branch vertices use original degrees. At $d = 0$, leaf integration gives all messages one, allowing nonzero B. Complete reversal reverses arms, connectors, weights and decorations together. The Gibbs/KL and all-map tournament conclusions follow from the original edge-product argument in part R2.8, without an additional independence assumption.

Q.27 Minimum-two odd roots with every terminal pair (3,t)

R4.1 Statement and exact directions

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The internal root residual is $(2, b, c, s_1, t_1, \dots, s_k, t_k)$, $3 \le b < c < \cdots$, $k \ge 0$. The distant terminal residual is $(3, t)$, $t \ge 4$. Original lengths $L, M \ge 1$ are independent and all genuine equal-pair decorations are allowed. Fix

$$\begin{aligned} F &= h_3 f_4, t = 4; F = \bar{f}_3 W_t, t \ge 5; \\ C_r &= W_2 \bar{f}_3 W_c \prod W_{s_i} \bar{f}_{t_i}, b = 3; \\ C_r &= W_2 W_b \bar{f}_c \prod W_{s_i} \bar{f}_{t_i}, b \ge 4 \end{aligned}$$

These directions are unchanged for every L and host. Inner edges point from the terminal to the root; outer edges from the root to the receiver. The arm directions are those of Section Q.25. The actual message is $H = E_r C_r T^L (E_t F)$, with equal-pair factors E_r, E_t.

For all these original parameters and hosts,

$$\begin{aligned} EH^4 &\le \exp(-13d/2), \\ I_q(T^M H)^3 &\le \exp[-(13 - c(M))d] \le \exp(-43d/19) \end{aligned} \tag{R4.1.1}$$

Here I_q and $c(M)$ are the same original-weight outlet as in Q5-C164.

R4.2 Full root and finite propagation

The Q5-C188 root input for $b = 3$ is

$$\begin{aligned} \bar{f}_3 W_c &\le \exp[-\Gamma_r + B(U_r + \beta\mathbf{1})], \\ \Gamma_r &= d + \beta_c + \beta_{c-1} \ge 3d - 2\delta_2 \ge 35d/16, \\ \beta &= \beta_{c-1} \in [\tau, d], U_r = (xE + zxD)(B)u, \\ G_{c-3} &= E + zD, E \ge 1 - x, 0 \le D \le 1 \end{aligned}$$

Consequently $\|U_r\| < \sqrt{d}/2$, $\|-u + U_r\| \le \sqrt{d}$, $\|Bu + U_r\| \le \sqrt{d}$, $EBU_r \le 0$, and $EU_r \le 0$. For $b \ge 4$, all root pairs combine into one vector with scalar loss $16(k+1)d/5$, norm at most $3\sqrt{d}/8$, and B-mean at most $d/12$. The mask for all later pairs with $s_i \ge 5$ has norm at most $\sqrt{d}/4$. Their complete proofs are in parts R2.1 and R2.4.1.

For the actual $F \le e^{R-\Gamma}$, put $D_i = \mathbb{E}B^i R$, $\sigma = \mathbb{E}R^2$, $\delta = D_0 + 5\sigma/4$, and $\epsilon = \delta + D_1$. Its Taylor bound gives (R29.5.3), with $c_T = 5/4$. If $h \ge 0$ and $\gamma_l = \beta_l - D_{l-1} \ge hd$ for every $l \ge 3$, use the four rows of (M.74). For $L \ge 4$, $n = L-1$, retain both endpoint means $\mathbb{E}BV_0 = d - \gamma_{L+1} \le (1-h)d$ and $\mathbb{E}V_0 = -\gamma_{n+1} \le -hd$.

The complete pair vector is $V = V_0 + U_r + (\beta + c_0)\mathbf{1}$ for $b = 3$, or $V = V_0 + U + c_0\mathbf{1}$ for $b \ge 4$. With $S = a - \Gamma_{\text{root}} + \mathbb{E}BV$, $N = \|V\|^2$, (M.76) gives

$$\log \mathbb{E}H^4 \le 4S + (17/4)N. \tag{R4.2.1}$$

R4.3 The terminal pair (3, 4)

Q5-C170, Section 3, gives the following actual two-arm envelope:

$$\begin{gathered} h_3 f_4 \le \exp(R - 3d), R = d \cdot u + B^3 u, \sigma = d^3 + \delta_4, \\ D_0 = \delta_2, D_1 = -d^2, D_2 = -\delta_3, D_3 = d \cdot \delta_2, \\ \delta = \delta_2 + (5/4)(d^3 + \delta_4), \epsilon = \delta - d^2, \\ 0 \le \epsilon \le \delta < 101d/160, all \gamma_l \ge d/2 \end{gathered}$$

The Taylor coefficient $5/4$ is proved on the same pointwise domain, and its finite-tail calculation in Section Q.6 covers every finite γ index. Convenient weaker norm bounds are

$$\begin{gathered} \|R\| < 3\sqrt{d}/5, \|BR\| < 2\sqrt{d}/5, \\ \|B^2 R\| < \sqrt{d}/4, \|B^n R\| < 4\sqrt{d}/25 (n \ge 3) \end{gathered} \tag{R4.3.1}$$

The first uses $d + \theta^{3/2} < 1/3 + 4/15 = 3/5$; the others follow from the original operator norm.

First take $k = 0$. The table gives $-S/d \ge s$, $N/d \le n$.

Table 119: Minimum-two odd roots with every terminal pair (3,t): parameter values.

Entry	L1(s, n)
b3	$(569/160, 77/48)$; L2(s, n): $(729/160, 3/2)$; L3(s, n): $(5, 9/4)$; $L \ge 4(s,$ n): $(177/40, 9/4)$.
$b \ge 4$	$(17/4, 2)$; L2(s, n): $(5, 36/25)$; L3(s, n): $(25/4, 225/64)$; $L \ge 4(s,$ n): $(21/4, 25/9)$.

Every loss $4s - 17n/4$ exceeds $13/2$. For $b = 3$, $L = 1$, the sufficient value is $569/40 - 1309/192 > 13/2$.

R4.3.1 The unit connector

For $b = 3$, the real coordinate of $V_0 + U_r$ is $d - 1 + xE < 0$, with absolute value at most $1 - (1-\theta)x$. The imaginary coefficient is $x(D-1)$, of absolute value at most x. Thus $(1 - (1-\theta)x)^2 + x^3 \le 1$, because $-2(1-\theta) + (1-\theta)^2\theta + \theta^2 < 0$. Its mean is at most $D_0 \le \delta_2$. Adding β yields $N \le d + (1 + 2\theta)d^2 \le 77d/48$. Also $S \le (-3 - 35/16 + 101/160 + 1)d = -569d/160$.

For $b \ge 4$, $\|R - u\|^2 < 15d/14$, by $(1-d)^2 + x^3 \le 1 + \theta^3 < 15/14$. Since $\sqrt{15/14} < 83/80$, adding the root mask gives norm less than $113\sqrt{d}/80$, with square less than $2d$. The scalar is at most $(-3 - 16/5 + 101/160 + 1 + 1/12)d < -17d/4$.

R4.3.2 Inner length two

For $b = 3$, the nonconstant coordinates of $BR + U_r$ are $x^2 + xE$ and $d + xD$. The squared norm is at most $[(\theta + \theta^2)^2 + \theta(\theta + 1/3)^2]d < 9d/16$. The constant $c = \beta + \delta$ lies in $[0, 261d/160]$, and the nonconstant mean $-d^2 + EU_r$ is nonpositive. Hence $N \le 9d/16 + (261d/160)^2 < 3d/2$. Keeping $D_2 = -\delta_3 \le 0$ gives $S \le (-3-35/16+101/160)d = -729d/160$. For $b \ge 4$, the full norm coefficient is less than $2/5+3/8+303/800 < 6/5$, and $S \le (-3-16/5+101/160+1/12)d < -5d$.

R4.3.3 Inner length three

For $b = 3$, the nonconstant norm coefficient is at most $1 + 1/4 + 101/480 < 3/2$, and its mean $-d - \delta_3 + EU_r$ is at most $-d$. Since $c = \beta + \epsilon \in [0, 261d/160]$, $2cEV_0 + c^2 \le 0$, giving $N < 9d/4$. Scalar plus mean is $-3d - \Gamma_r + \epsilon - d - \delta_3 - \delta d + d\delta_2 + EBU_r$. Keeping the negative terms in their valid direction gives $S \le (-3-35/16+101/160-1+\theta/3)d < -5d$. For $b \ge 4$, the complete norm coefficient is $13/20 + 1/4 + 101/480 + 3/8 + 303/800 < 15/8$, and $S/d \le -3 - 16/5 + 101/160 - 1 + \theta/3 + 1/12 = -127/20 < -25/4$.

R4.3.4 Every finite connector of length at least four

Here $h = 1/2$, $D_1 < 0$. For $b = 3$, the nonconstant norm coefficient is at most $4/5+4/25+1/2 < 3/2$. The constant lies in $[3d/32, d/2]$, and the nonconstant mean is at most $-d/2$; its increment is nonpositive. Thus $S/d \le -3-35/16+101/80-1+1/2 = -177/40$. For $b \ge 4$, the full coefficient is at most $4/5+4/25+3/8+3/10 < 5/3$, and $S/d \le -3-16/5+101/80-1+1/2+1/12 < -21/4$. Every γ_l, including $L + 1$, is included.

R4.4 Every terminal (3, t) with t at least five

The actual terminal envelope in part Q.2 of Section Q.2 is

$$
\begin{gathered}
F = \bar{f}_3 W_t \le \exp(R - \Gamma), \\
R = (\gamma - xv + zxC)(B)u, \Gamma = d + \beta_t + \gamma, \\
\gamma = \beta_{t-1} \in [\tau, d], G_{t-2} = A + zC, v = 1 - A, \\
0 \le v \le x, 1 - x \le C \le 1, \Gamma \ge 3d - 2\delta_2 \ge 35d/16
\end{gathered}
$$

The pointwise envelope and its Taylor certificate apply both for $L \ge 2$ and at the unit connector with the present root. We have $e^R \le 1 + R + 5R^2/8$ on the entire original domain and gives, for the same R,

$$
\begin{gathered}
\sigma < 5d/24, \|R\| < \sqrt{d}/2, \\
D_0 = -\int xC, D_1 = -\gamma \cdot d + \int xv, \\
D_2 = \int x^2 C \in [0, \delta_3], D_3 = \gamma \cdot \delta_2 - \int x^2 v \le d \cdot \delta_2, \\
-\delta_2 \le \delta < d/24, -3d/4 \le \epsilon \le 0, \\
(D_1)_+ \le \delta_3, all \gamma_l \ge 9d/16
\end{gathered}
\tag{R4.4.1}
$$

The sign of D_1 is not assumed negative; all long-connector bounds retain its positive part. We may use

$$\|BR\| < 13\sqrt{d}/40, \|B^2R\| < 5\sqrt{d}/24,$$
$$\|B^nR\| < 2\sqrt{d}/15 (n \geq 3) \tag{R4.4.2}$$

The following cells again give s, n, and every $4s - 17n/4$ exceeds $13/2$.

Table 120: Minimum-two odd roots with every terminal pair (3,t): parameter values.

Entry	L1(s, n)
b3	$(27/8, 19/12)$; L2(s, n): $(101/24, 9/10)$; L3(s, n): $(79/16, 2)$; $L \geq 4$(s, n): $(223/48, 1849/900)$.
$b \geq 4$	$(1033/240, 3969/1600)$; L2(s, n): (5, 1); L3(s, n): $(88/15, 121/36)$; $L \geq 4$(s, n): $(223/40, 25/9)$.

The tightest cell, $b \geq 4, L = 1$, exceeds $13/2$ by $3341/19200$.

R4.4.1 The unit connector

For $b = 3$, the nonconstant real coordinate $-1+\gamma-xv+xE$ is negative, of absolute value at most $1 - x + 2x^2$; the imaginary coefficient $x(C + D)$ is at most $2x$. A squared-frequency upper bound is

$$1 - 2x + 5x^2 + 4x^4 \leq 5/4$$

This is convex on $[0, \theta]$. Using $\theta \geq 2/5$, $\theta^2 < 1/6$, $\theta^4 < 1/36$, its endpoint is less than $1 - 4/5 + 5/6 + 1/9 < 5/4$. The nonconstant mean $-\int x(C + D)$ is nonpositive, and $c = \beta \leq d$, yielding $N \leq 5d/4 + d^2 \leq 19d/12$. Since $\epsilon \leq 0$, $S \leq (-35/8 + 1)d = -27d/8$.

For $b \geq 4$, $\|R-u\|^2 \leq \int[(1+x^2)^2+x^3] < 36d/25$, the same finite rational frequency comparison used in Q5-C180. Adding the root mask gives full norm at most $63\sqrt{d}/40$, while $S/d \leq -35/16 - 16/5 + 1 + 1/12 = -1033/240$. In particular,

$$1033/60 - (17/4)(3969/1600) - 13/2 = 3341/19200 > 0$$

R4.4.2 Inner length two

For $b = 3$, the real coordinate $xE - x^2C$ lies in $[0, x]$; the imaginary coefficient $xD + \gamma - xv$ has absolute value at most $x + \gamma$. The nonconstant norm squared is less than $2d/5$. The constant $c = \beta + \delta$ lies in $[3d/16, 25d/24]$; the nonconstant mean is at most $D_1 \leq \delta_3 < d/6$. Thus $N \leq 2d/5 + 275d^2/192 < 9d/10$, and $S \leq (-35/8 + 1/6)d = -101d/24$. For $b \geq 4$, the full coefficient is $13/40 + 3/8 + 39/160 < 1$, and $S/d \leq -35/16 - 16/5 + 1/6 + 1/12 < -5$.

R4.4.3 Inner length three

For $b = 3$, the nonconstant norm coefficient is at most $1 + 5/24 + 11/80 < 27/20$, using $|\delta| \leq \delta_2$. Its mean lies in $[-3d/2, -5d/6]$, and $c = \beta + \epsilon \in [-5d/32, d]$. For nonnegative c the increment is nonpositive; for negative c, at most $505d^2/1024$. Hence $N \leq 729d/400 + 505d^2/1024 < 2d$. The scalar terms $D_3 - \delta d$ sum to at most $2d\delta_2$, so

$$-S/d \geq 35/8 + 1 - 1/6 - 2\theta/3 = 79/16$$

For $b \geq 4$, the complete norm coefficient is at most $13/20 + 5/24 + 11/80 + 3/8 + 9/20 < 11/6$, and $-S/d \geq 35/16 + 16/5 + 1 - 1/6 - 2\theta/3 - 1/12 = 88/15$.

R4.4.4 Every finite connector of length at least four

Here $h = 9/16$, $2\delta_+ \leq d/12$, and $2(D_1)_+ \leq 2\delta_3 < d/3$. For $b = 3$, the nonconstant coefficient is at most $4/5 + 2/15 + 1/2 = 43/30$; the constant lies in $[d/32, 7d/16]$, with nonconstant mean at most $-9d/16$. Its square and cross term therefore sum to a nonpositive number. The scalar lower bound is $-S/d \geq 35/8 + 11/16 - 1/12 - 1/3 = 223/48$. For $b \geq 4$, the full coefficient is at most $4/5+2/15+3/8+27/80 < 5/3$, and $-S/d \geq 35/16+16/5+11/16-1/12-1/3-1/12 = 223/40$. These estimates cover every finite $n = L - 1$.

R4.5 Later pairs, the outer outlet and the original tree

For $b = 3$, every complete vector in the two tables has norm at most $2\sqrt{d}$. Adding all later root pairs uses the full-vector increment $[931/192 - 64k/5]d < 0$ for $k \geq 1$, as in part R2.6, preserving the basic loss $13/2$. For $b \geq 4$, all root pairs remain in one mask and extra scalar loss only helps. Positivity restores all genuine equal-pair decorations. Finally Q5-C164 gives $13 - 204/19 = 43/19$, proving the outlet statement.

If both sides of an original five-branch path use this construction and the center uses the Q4-C143 negative zero-step table with the same q, Hölder gives $Z_D \leq \exp[-(43/19 + 43/19)d/8] = e^{-43d/76}$. A mixed-side construction requires the stated directions, weight and central role on each side.

The edge count is the sum of all four independent positive core lengths and every original side-arm length, counting multiplicity; $|V| = e + 1$. Branch vertices use original degrees. At $d = 0$, leaf integration gives every actual message one without requiring $K = 0$. Full reversal reverses all arms, connectors, weights and decorations. The correlated-law Gibbs/KL and all-map tournament conclusions use the original product argument of part R2.8 on the same measure.

Q.28 Minimum-two odd roots with every nonunit-minimum odd terminal

R5.0 Parameters, fixed directions and statement

Use the original finite-space model and arm messages of Section M.2. When the original K-notation is used, $K = -B$ and $g = K\mathbf{1} = -u$; in either notation $\delta_j = \|B^j\mathbf{1}\|_2^2 = \|K^j\mathbf{1}\|_2^2$. All means, norms, adjoints and spectral integrals use the same original probability measure. The distinct internal-root residual is $(2, r, s, s_1, t_1, \dots, s_k, t_k)$, with $3 \leq r < s < s_1 < t_1 < \cdots$, $k \geq 0$. The terminal residual is

$$(a, b_1, c_1, \cdots, b_j, c_j), 2 \leq a < b_1 < c_1 < \cdots < b_j < c_j, j \geq 1$$

The original inner and outer lengths $L, M \geq 1$ are independent. Before the host is known, choose

$$C_r = W_2 \bar{f}_3 W_s \prod W_{s_i} \bar{f}_{t_i}, r = 3;$$
$$C_r = W_2 W_r \bar{f}_s \prod W_{s_i} \bar{f}_{t_i}, r \geq 4;$$
$$F = \bar{f}_a \prod h_{b_i} f_{c_i}, b_1 \geq 4;$$
$$F = \bar{f}_2 f_3 h_{c_1} \prod_{i \geq 2} h_{b_i} f_{c_i}, b_1 = 3$$

The strict ordering in the last row forces $a = 2$. Arm directions are those of Section Q.25; all

inner edges point from terminal to root, and all outer edges from root to receiver. Arbitrary genuine equal-pair products at the two roots are denoted E_r, E_t. The actual half-tree and its output are

$$H = E_r C_r T^L (E_t F), X = T^M H$$

For all these original integer parameters and legal hosts,

$$EH^4 \le \exp(-6d), \tag{R5.0.1}$$

$$I_q(X)^3 \le \exp[-(12 - c(M))d] \le \exp(-24d/19), \tag{R5.0.2}$$

$$I_q(X) = E[q^{-1/3}|X|^{8/3}]$$

Q5-C164 gives $c(1) = 204/19$, $c(2) = 10$, $c(3) = 0$, $c(4) = 7$, $c(M) = 6$ for $M \ge 5$.

R5.1 Finite inputs and the complete root exponent

Use Section M.2, the finite convolution (M.2), and spectral identities (M.3). Keep $G_1 = (1, 0)$ and the noncontractive G_2 boundary. For $n \ge 2$, $|G_n - 1|^2 \le x^2 + x < (4/5)^2$; the exact square at $n = 4$ is $x - x^2 + x^3 \le x$.

For $r = 3$, the actual exchanged root pair is (M.53), with $G_{s-3} = E + zD$, $U_r = (xE + zxD)u$, $\beta = \beta_{s-1}$, and $\Gamma_r = d + \beta_s + \beta \ge 35d/16$. Its signed means and three norm bounds are (M.54); in particular $\mathbb{E}U_r \in [-\theta d, 0]$. For $r \ge 4$, all actual root pairs form one finite mask:

$$\prod \text{ root pairs} \le e^{-16(k+1)d/5 + BU}, \qquad \|U\| \le 3\sqrt{d}/8, \quad \mathbb{E}BU \le d/12. \tag{R5.1.4}$$

For $r = 3$, the $k \ge 1$ later pairs have loss $16kd/5$ and one vector U_+ with $\|U_+\| \le \sqrt{d}/4$, $\mathbb{E}BU_+ \le d/12$. First take $r = 3, k = 0$, restoring these pairs in Section Q.28.

The actual W_2 root uses (M.73). After adding the root pairs, retain the complete scalar a, vector V_{full}, $S = a + \mathbb{E}BV_{\text{full}}$, and $N = \|V_{\text{full}}\|^2$. Lemma M.27 gives $\log \mathbb{E}H^4 \le 4S + 17N/4$, including every β and $C - 1$ constant and cross term.

R5.2 Every first terminal-pair minimum at least four

R5.2.1 Finite data with the same actual anchor

Use the actual $F = \bar{f}_a \prod_i h_{b_i} f_{c_i}$ of Q5-C181, Section 3. Put $R = \sum_i \sum_{l=b_i+1}^{c_i} K^l \mathbf{1}$, let Γ be the sum of actual pair losses, and put $D_i = EB^i R$, $\mu_a = E\bar{f}_a \le 1$. Its genuine first row and positive finite propagation give

$$\begin{aligned} T(\bar{f}_a e^R) &\le \bar{f}_{a+1} + \mu_a BR + \delta, \\ e^\Gamma T^L F \le Y_L &= \bar{f}_{a+L} + \mu_a B^L R + \delta \bar{f}_{L-1} \\ &\quad + \mu_a \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i} \end{aligned} \tag{R5.2.1}$$

Here δ bounds the actual mixed mean. The complete finite data are

Table 121: Minimum-two odd roots with every nonunit-minimum odd terminal: parameter values.

Entry	Gamma/d
$j1, b_1 = 4$	$69/20$; $\rho = \|R\|/\sqrt{d}$: $< 1/5$; δ_+/d: $\le 72/125$; $\sum_{i\ge1}(D_i)_+/d$: $< 7/25$; D_1/d: $< 1/6$; $D_i/d, i \ge 2$: $< 7/100$.
$j1, b_1 \ge 5$	$16/5$; $\rho = \|R\|/\sqrt{d}$: $< 3/20$; δ_+/d: $\le 3/5$; $\sum_{i\ge1}(D_i)_+/d$: $< 1/5$; D_1/d: $< 1/30$; $D_i/d, i \ge 2$: $< 1/12$.
$j \ge 2, b_1 \ge 4$	$\ge 16j/5$; $\rho = \|R\|/\sqrt{d}$: $< 1/4$; δ_+/d: $\le 5/6$; $\sum_{i\ge1}(D_i)_+/d$: $< 2/5$; D_1/d: $< 1/5$; $D_i/d, i \ge 2$: $< 1/12$.

In the first row $D_0 \le 0$; in the other rows $D_0 \le d/12$. The finite masks and positive tails are proved in Q5-C181, Section 3.1. These are compatible data for the same R.

R5.2.2 Payment with the actual W2 root

Let $n = a + L \ge 3$, retaining the actual $\beta_n \bar{f}_0$ term:

$$V0 = (G_{n-1} - 1)(B)u + \mu_a B^{L-1} R,$$

$$a0 = -\Gamma + cost - \beta_n,$$

$$cost = \delta(L1),$$

$$cost = 2\delta_+ + 2\mu_a \sum_{i=1}^{L-1} (D_i)_+ (L \ge 2)$$

The root exponent gives $\log(W_2 T^L F) \le a_0 + BV_0$. Its exact means are $EV_0 = -\beta_n + \mu_a D_{L-1}$ and $EBV_0 = d - \beta_{n+1} + \mu_a D_L$. Thus the scalar loss is at least

$$\Gamma + \Gamma_r oot + 3d/16 - cost - \mu_a D_L - EB(rootvector) \tag{R5.2.2}$$

For $r = 3$, the full vector is $V = V_0 + U_r + \beta \mathbf{1}$. At $L = 1$, its nonconstant mean is at most $-49d/96$, so the β increment is nonpositive; the norm coefficient is less than $4/5 + 1/4 + 1/2$, giving $N < 5d/2$. For $L \ge 2$, the mean is at most $-63d/160$, and the increment is at most $17d^2/80 \le 17d/240$. The nonconstant norm coefficient is at most $4/5 + 1/2 + (13/20)(1/4) = 117/80$, giving $N < 9d/4$. With a single pair and $b_1 \ge 5$, the sharper mean $-49d/96$ and norm coefficient less than $7/5$ give $N < 2d$. We used $\sqrt{\theta} < 13/20$; norm coefficients are in units of $\sqrt{d}$.

For $r \ge 4$, there is no β constant. The complete norm coefficients are less than $3/2$ at $L = 1$ and $27/20$ at $L \ge 2$. The scalar formula gives the following s, n table.

Table 122: Minimum-two odd roots with every nonunit-minimum odd terminal: parameter values.

Entry	L1(s, n)
$r3, j1, b_1 = 4$	$(19/4, 5/2)$; $L \ge 2(s,$ n$)$: $(4043/1000, 9/4)$.
$r3, j1, b_1 \ge 5$	$(19/4, 5/2)$; $L \ge 2(s,$ n$)$: $(467/120, 2)$.
$r3, j \ge 2$	$(19/4, 5/2)$; $L \ge 2(s,$ n$)$: $(249/40, 9/4)$.
$r \ge 4$, all preceding terminals	$(23/4, 9/4)$; $L \ge 2(s,$ n$)$: $(19/4, 729/400)$.

For multiple pairs, $j = 2$ is the weakest case; every additional pair contributes scalar loss $16d/5$. The tightest row, $r = 3, b_1 = 4, L \geq 2$, gives

$$4s - 17n/4 = 4043/250 - 153/16 > 6$$

Every other cell also gives a loss greater than six, proving the stated ordinary fourth bound for this complete packet.

R5.3 First pair minimum three with one pair on the low domain

Now $a = 2$, $c = c_1 \geq 4$, and the actual terminal is $F = \bar{f}_2 f_3 h_c$. Q5-C181, Section 4, supplies the same finite polynomial

$$R0 = (-\gamma + x^2 V + zxU)(B)u, \gamma = \beta_{c-1},$$
$$G_{c-3}(-z) = U - zV, Gamma0 = d + \beta_c + \gamma \geq 35d/16$$

The G_1 boundary is $(U, V) = (1, 0)$; all later coordinates lie in $[1 - x, 1]$, and $0 \leq \gamma \leq d$. The original pair gives $F \leq \bar{f}_2 e^{R_0 - \Gamma_0}$. The complete actual first row permits the uniform choice $D = 3d/5$:

$$T(\bar{f}_2 e^{R_0}) \leq \bar{f}_3 + BR_0 + D, \qquad D \geq \max(\delta_0, 0). \tag{R5.3.1}$$
$$D_i = EB^i R0, -\delta_3 \leq D_1 \leq d^2, 0 \leq D_2 \leq \delta_3,$$
$$D_3 \leq \delta_4, D_i \leq d/36 (i \geq 4), \|R0\| < 9\sqrt{d}/20$$

Apply (M.72) with $s = 3$, $\mu = 1$, $\alpha = -\Gamma_0$, and this same D. It gives the finite Y_L with base $\bar{f}_{L+2}$, used below. This D is specific to the actual BR_0 first row. Every subsequent expression uses this same full mean.

R5.3.1 The unit connector

Here $Y_1 = q + B(G_2 + r_0)u + D - d$. In the root exponent take $a_0 = -\Gamma_0 + D - d$, $V_0 = Bu + R_0$, $c_0 = 0$. For $r = 3$, the nonconstant coordinates of $V_0 + U_r$ are

$$real = -\gamma + x^2 V + xE, imag = 1 + xU + xD_r$$

The squared-frequency bound is $x + 5x^2 + 6x^3 + x^4 + d^2$; on $x \leq \theta, d \leq 1/8$, it is at most $1758033/1048576 < 27/16$. The nonconstant mean is at most $-d$, so the full β increment is nonpositive. The scalar is $S \leq -\Gamma_0 - \Gamma_r + D - d + D_1$, giving $s \geq 93/20$, $n \leq 27/16$. For $r \geq 4$, the full coefficient is less than $13/20 + 9/20 + 3/8 < 3/2$, and $s > 11/2$.

R5.3.2 Inner length two

The exact fourth-path identity gives

$$Y2 = (1 - d + D)q + B[G3 + zr0]u + D_1 - d,$$
$$a0 = -Gamma0 + D + D_1 - 2d, V0 = (G3 - 1 + zr0)u, c0 = -2d/5$$

For $r = 3$, the nonconstant coordinates are $-x(1-E) - x^2 U$ and $1 - \gamma + x^2 V + xD_r$. Their squared-frequency bound is $x + 2x^2 + 3x^3 + 6x^4 + x^5$, whose endpoint is $37311261/33554432 < 9/8$. The mean is at most $-d + D_1 \leq -7d/8$, and $c = \beta - 2d/5 \in [31d/160, 3d/5]$, so the constant increment is nonpositive. The scalar satisfies

$$-S/d \geq 35/8 + 2 - 3/5 - 1/8 - \theta - \theta^2 > 5$$

For $r \geq 4$, the full coefficient is less than $4/5 + 3/10 + 3/8 + 3/20 = 13/8$, and the scalar loss gains $16/5 - 35/16 - 1/12$, giving $s > 6$.

R5.3.3 Inner length three

The exact fifth-path finite identity gives

$$Y3 = (1 - 2d + D + D_1)q + B[G4 - d + z^2 r0 + D]u - \tau + D_2,$$
$$a0 = -Gamma0 - 2d + D + D_1 - \tau + D_2,$$
$$V0 = (G4 - 1 - d + D + z^2 r0)u, c0 = -7d/5 + D_1$$

For $r = 3$, the nonconstant coordinates are $-x(1-E)+\gamma x - x^3 V - 2d/5$ and $1 - x - x^2 U + x D_r$. The real coordinate has absolute value less than $3/10$, and the imaginary coefficient lies in $[0, 1]$, giving nonconstant norm squared less than $d/2$ and mean in $[-d, 0]$. The constant $c = \beta - 7d/5 + D_1$ is negative with $|c| < d$; its increment is at most $3d^2 \leq 3d/8$, yielding $N < 7d/8$. The actual scalar retains every term:

$$S = -Gamma0 - \Gamma_r - 3d + 2\delta_2 + D + D_1 + D_2 + D_3 + (d - D)d + EBU_r$$

The finite packet gives $s > 5$. For $r \geq 4$, use $\|(G_4 - 1)u\| \leq \sqrt{\theta d}$. The full coefficient is less than $13/20 + 1/20 + 1/5 + 3/8 + 3/5 < 2$, and the scalar gives $s > 51/8$.

R5.3.4 Every finite inner length at least four

Define $\gamma_3 = d - D = 2d/5$, and $\gamma_l = \beta_l - D_{l-3}$ for $l \geq 4$. Q5-C181, Section 4.5, proves at each initial index and throughout the finite tail that $\gamma_l \geq h = 2d/5$, including $L + 3$. Keeping the final four original terms gives the positive comparison

$$Y_L \leq Cq + BV - h(1 - d), C = 1 - 3h,$$
$$V = [G_{L+1} + z^{L-1} r0 - h(2 + z)]u$$

Thus $a_0 = -\Gamma_0 - 4h + hd$, $V_0 = V - u$, $c_0 = -3h$, and $EBV_0 \leq d - h + 2hd$. For $r = 3$, the scalar loss has $s \geq 209/40$; the nonconstant norm coefficient is at most $4/5 + 3/25 + 21/200 + 1/2 = 61/40$. Here $\|B^{L-1} R_0\| < 3\sqrt{d}/25$, and $h\sqrt{4 + \theta} \leq 21/200$. The constant $c = \beta - 6d/5 < 0$ has absolute value at most $97d/160$, and the nonconstant mean is at least $-13d/8$. Its increment is less than $5d^2/2 \leq 5d/16$, yielding $N < 8d/3$. For $r \geq 4$, the full coefficient is at most $37/20 < 15/8$, and $s > 6$.

All four original length ranges therefore have the following full bounds:

Table 123: Minimum-two odd roots with every nonunit-minimum odd terminal: parameter values.

Entry	L1(s, n)
r3	$(93/20, 27/16)$; L2(s, n): $(5, 9/8)$; L3(s, n): $(5, 7/8)$; $L \geq 4(s$, n): $(209/40, 8/3)$.
$r \geq 4$	$(11/2, 9/4)$; L2(s, n): $(6, 169/64)$; L3(s, n): $(51/8, 4)$; $L \geq 4(s$, n): $(6, 225/64)$.

Every loss $4s - 17n/4$ exceeds six. Even the coarse $L = 3, r \geq 4$ cell gives $51/2 - 17 = 17/2 > 6$.

R5.4 First pair minimum three with at least two pairs on the low domain

Keep $F = \bar{f}_2 f_3 h_{c_1} \prod_{i\geq 2} h_{b_i} f_{c_i}$. The actual later-pair mask S_t of Q5-C181, Section 5, gives

$$R = R0 + S_t, \Gamma = Gamma0 + 16(j-1)d/5,$$
$$\Gamma \geq 431d/80 + 16(j-2)d/5, \|S_t\| < 3\sqrt{d}/20,$$
$$ES_t \leq d/12, \|R\| < 3\sqrt{d}/5$$

The mean bound for S_t comes from the finite four-residue-class mask with minimum at least five, part R5.3.1. Its actual first row permits $D = 3d/2$. The finite propagation of part R5.3 remains valid with this same R, Γ, D. Its complete data are

$$-19d/60 \leq D_1 \leq 19d/120, D_2 \leq d/4, D_3 \leq d/6,$$
$$D_i \leq d/9 (i \geq 4) \tag{R5.4.1}$$

R5.4.1 The unit connector

Here $V_0 = Bu + R$, $a_0 = -\Gamma + D - d$. For $r = 3$, $\|Bu + U_r\| \leq \sqrt{d}$ gives nonconstant coefficient less than $8/5$, and mean at most $-d + ES_t \leq -11d/12$. The full β increment is nonpositive, hence $N \leq 64d/25$, while $s \geq 83/12$. For $r \geq 4$, the full coefficient is less than $13/8$, and $s > 15/2$.

R5.4.2 Inner length two

Use the same exact form as in part R5.3.2, now with $D = 3d/2$, $R = R_0 + S_t$. The joint base $(G_3 - 1)u + U_r$ has real-coordinate absolute value at most x^2 and imaginary coefficient at most $1 + x$. Since $x^4 + x(1+x)^2 < 1$, the nonconstant norm coefficient is less than $1 + 2/5 = 7/5$. The constant $c = \beta + d/2 \in [0, 3d/2]$, with nonconstant mean at most $-101d/120$, has nonpositive increment. Thus $s > 7$, $N \leq 49d/25$. For $r \geq 4$, the full coefficient is less than $4/5 + 2/5 + 3/8 + 3/16 < 9/5$, and $s > 8$.

R5.4.3 Inner length three

Here $V_0 = (G_4 - 1 + d/2 + z^2 r)u$, $c_0 = -d/2 + D_1$. The base $(G_4 - 1)u + U_r$ has real-coordinate absolute value at most x^2, imaginary coefficient in $[0, 1]$, and norm coefficient less than $3/4$. Add $d/2 \leq 1/16$ and $\|B^2 R\| < \sqrt{d}/4$ to obtain nonconstant coefficient less than $17/16$. Its mean is at most $-\tau + D_2 \leq -11d/32$, and at least $-8d/5$, using $D_2(R_0) \geq 0$, $D_2(S_t) > -d/10$, $EU_r \geq -\theta d$. The full constant lies in $[-107d/480, 79d/120]$. For nonnegative c, the increment is nonpositive; for negative c, $|c| < d/4$ gives increment at most $69d^2/80$. Therefore $N/d < (17/16)^2 + 69/640 < 5/4$. The full scalar formula of part R5.3.3 retains $(d - D)d = -d^2/2 \leq 0$, giving $s \geq 123/16$. For $r \geq 4$, the full coefficient is less than $13/20 + 1/16 + 1/4 + 3/8 + 49/160 < 5/3$, and $s > 17/2$.

R5.4.4 Every finite later length, with the negative third coefficient retained

Now $\gamma_3 = d - D = -d/2$, and this term must be retained. For every $l \geq 4$, including $L + 3$, $\gamma_l = \beta_l - D_{l-3} \geq h = d/3$. The actual positive remainder $(D - d)\bar{f}_{L-1} \leq d$, together with the final three terms, gives

$$Y_L \leq Cq + BV + d - h, C = 1 - 2h,$$
$$V = [G_{L+1} + z^{L-1} r - h]u$$

Thus $a_0 = -\Gamma + d - 3h = -\Gamma$, $V_0 = V - u$, $c_0 = -2h$, and $EBV_0 \le 2d/3 + d^2/3$. For $r = 3$, the weakest scalar loss is $s \ge 103/15 > 27/4$; the nonconstant coefficient is less than $4/5 + 4/25 + 1/24 + 1/2 < 61/40$. The constant lies in $[-7d/96, d/3]$ and the nonconstant mean in $[-5d/3, -41d/96]$. The upper endpoint uses $D_3 \le d/6$, $D_n \le d/9$, $n \ge 4$; the lower uses spectral Cauchy. For nonnegative c, the increment is nonpositive; for negative c, it is at most $[2(7/96)(5/3) + (7/96)^2]d^2 < d^2/4$. Hence $N/d < (61/40)^2 + 1/32 < 12/5$. For $r \ge 4$, the full coefficient is less than $4/5 + 4/25 + 1/24 + 3/8 + 1/4 < 5/3$, and $s > 15/2$.

The weakest case $j = 2$ has the complete table

Table 124: Minimum-two odd roots with every nonunit-minimum odd terminal: parameter values.

Entry	L1(s, n)
r3	$(83/12, 64/25)$; L2(s, n): $(7, 49/25)$; L3(s, n): $(123/16, 5/4)$; $L \ge 4$(s, n): $(27/4, 12/5)$.
$r \ge 4$	$(15/2, 169/64)$; L2(s, n): $(8, 81/25)$; L3(s, n): $(17/2, 25/9)$; $L \ge 4$(s, n): $(15/2, 25/9)$.

Every $4s - 17n/4$ exceeds six. Each additional pair gives a further $64d/5$ loss.

R5.5 The complete bare root for every inward path of length at least three

To pay the bare $\bar{f}_n$ in the same direction on the high domain, retain the genuine $\beta_n \bar{f}_0$ term:

$$\bar{f}_n \le q + BG_{n-1}u - \beta_n$$

The root exponent has $a_0 = -\beta_n$, $V_0 = (G_{n-1} - 1)u$. For $r = 3$, the nonconstant $V_0 + U_r$ has norm at most $13\sqrt{d}/10$ and mean at most $-\kappa d$. For $\beta \in [0, d]$, its square and cross term sum to a nonpositive quantity, giving $N \le 169d/100$. The full scalar is

$$S \le -\Gamma_r - \beta_n - \beta_{n+1} + d \le -19d/8$$

The loss is therefore at least $19/2 - 2873/400 = 927/400 > 9/4$. For $r \ge 4$, $N < (6/5)^2 d$, and $s \ge 16/5 + 3/16 - 1/12 = 793/240$, also giving more than $9/4$. Thus every $n \ge 3$ and every legal d satisfy

$$E[(C_r \bar{f}_n)^4] \le \exp(-9d/4) \tag{R5.5.1}$$

The later-pair increment below applies to this full root as well.

R5.6 First pair minimum three on the entire high domain

For $1/8 \le d \le 1/3$, Q5-C181, Section 6, gives, for every $c \ge 4$,

$$f3h_c \le e^{-d},$$
$$h_s f_t \le e^{-2d}, s \ge 5$$

These pointwise bounds use the stated arms and retain the G_1, G_2 boundaries. Every later-pair shorter arm is at least five, so the same directions give

$$F \le \exp[-(1+2(j-1))d]b2,$$
$$C_r T^L F \le \exp[-(1+2(j-1))d]C_r \bar{f}_{L+2}$$

The preceding bare-root bound gives fourth loss at least $25/4+8(j-1)>6$. Combining with the low-domain parts covers every d, j, c and every positive L.

R5.7 Later root pairs, genuine decorations and every outer length

For $r=3$, all basic cases, including the bare-root estimate, have full norm at most $2\sqrt{d}$. Add the finite $k \ge 1$ later-pair mask to the same full vector:

$$4EBU_+ + (17/4)(2<V_{\text{full}}, U_+> + \|U_+\|^2) - 64kd/5$$
$$\le [931/192 - 64k/5]d < 0$$

This retains the cross term with $\beta\mathbf{1}$. For $r \ge 4$, all root pairs stay in the one mask of part R5.1 and further scalar loss helps. Genuine equal pairs lie in $[0,1]$, so positivity restores E_r, E_t and preserves the ordinary fourth bound. Finally Q5-C164 gives $12 - 204/19 = 24/19$, proving the outlet bound.

If both sides use this construction and the center uses the Q4-C143 negative zero-step table with the same q, one original-measure Hölder inequality gives $Z_D \le \exp[-(24/19+24/19)d/8] = e^{-6d/19}$. Other combinations retain their separately stated actual side and central directions.

R5.8 Original counts and boundaries

The four original positive core lengths remain independent. Every side-arm length, multiplicity, subdivision vertex and edge is retained. Sum all lengths for $e(U)$; $|V| = e+1$, and branch vertices use original degrees at least three. At $d=0$, original leaf integration gives every actual message one, allowing nonzero K. Complete reversal reverses every actual arm, connector, weight and decoration. Gibbs applies to arbitrary correlated original joint laws, and the tournament consequence counts all maps, allowing nonadjacent vertices to coincide.

R Remaining outlets and mixed estimates

This family completes the singleton and even-root outlet rows and the mixed bound (9.1). The compound reference-second estimate has its proof in Proposition 9.3.

R.1 Weighted transfer for a nonunit singleton

The square-root multiplier.

Write

$$A = J + K, \qquad T = A^* = J - K, \qquad g = K\mathbf{1}, \qquad p = 1+g, \quad q = 1-g,$$

$$f_n = A^n\mathbf{1}, \qquad \bar{f}_n = T^n\mathbf{1}, \qquad \delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1.$$

Let M_v denote multiplication by v, and set $I_q(Y) = \mathbb{E}[q^{-1/3}|Y|^{8/3}]$. The square-root estimate (C.3), including every $a \ge 2$, is

$$\|M_{\sqrt{f_a}} T\|_{2\to 2} \le 1.$$

Replace K by $-K$ in the entire construction to obtain $\|M_{\sqrt{\bar f_a}} A\|_{2\to 2} \le 1$. Taking the ordinary adjoint on the original measure gives

$$\boxed{\|T M_{\sqrt{\bar f_a}}\|_{2\to 2} \le 1, \qquad a \ge 2.} \tag{E8.10.1}$$

The displayed multiplier is a square root.

Full reversal of the actual reference input.

Use the complete outward-f_a reference directions in Sections N.41 and N.42. They give the reference fourth moment for every nonreducible terminal and every original inner length $L \ge 1$, with bound $1 - d/3000$ for $a \ge 3$ and bound one for $a = 2$. Their actual direction tables are used with all their exceptions. In particular, the unit-$(1,2,3)$ reference exception is at $L = 1$; the ordinary unit-anchor eighth-moment exception at $L = 2$ has a different role.

Let F_{ref} denote the actual terminal selected by those reference directions. Reverse the entire reference construction: the actual reference arm, terminal arms, all L connector edges and both decoration products. A superscript minus denotes this complete reversal. Put

$$X_0 = A^L(E_t^- F_{\mathrm{ref}}^-), \qquad X = E_r^- X_0,$$
$$Q_a(d) = \begin{cases} 1, & a = 2, \\ 1 - d/3000, & a \ge 3. \end{cases}$$

Since $0 < E_r^- \le 1$, the fully reversed reference bound gives

$$\mathbb{E}[\bar f_a X^4] \le \mathbb{E}[E_r^- \bar f_a X_0^4] \le Q_a(d). \tag{E8.10.2}$$

Here each root equal pair occurs once, and every terminal pair remains inside its actual A^L connector. The sole nonunit residual root arm points inward and gives $\bar f_a$. The actual final component is

$$U = T^M(\bar f_a X). \tag{E8.10.3}$$

Thus the inner connector points from the root toward its terminal, the root arm toward the root, and the outer connector from the root toward the center. These directions are fixed by the original structure. Every arm in F_{ref}^- is reversed together with the reference and connector: for example, $q f_b$ becomes $p \bar f_b$, and $f_3 \bar f_b$ becomes $\bar f_3 f_b$. The output weight q belongs to the actual outer T-path.

The weighted fourth-moment bridge.

For any real X, $a \ge 2$ and $M \ge 1$, consider the positive operator $P_0 = T^M M_{\bar f_a}$. Its row is $P_0 \mathbf{1} = \bar f_{M+a}$. Row Cauchy–Schwarz gives

$$|P_0 X|^2 \le \bar f_{M+a}\, T^M(\bar f_a |X|^2).$$

Apply Hölder with exponents 3 and $3/2$, then cube, to obtain

$$I_q(P_0 X)^3 \le \mathbb{E}[\bar f_{M+a}^4/q] \left\{ \mathbb{E}\left[(T^M(\bar f_a |X|^2))^2 \right] \right\}^2.$$

In L^2, factor the vector as

$$T^M(\bar f_a |X|^2) = [T^M M_{\sqrt{\bar f_a}}]\, [\sqrt{\bar f_a} |X|^2].$$

Consequently the exact weighted bridge is

$$\boxed{I_q(T^M(\bar f_a X))^3 \le \mathbb{E}[\bar f_{M+a}^4/q]\,\|T^M M_{\sqrt{\bar f_a}}\|_{2\to2}^4 \cdot (\mathbb{E}[\bar f_a |X|^4])^2.} \tag{E8.10.4}$$

Both the norm and input weight follow from this factorization. The factorization uses the one $\bar f_a$ arm as two square-root multipliers.

To keep every operator coefficient explicit, write

$$N_0 = 1, \qquad N_1 = 1 + (32/19)d, \qquad N_2 = (1 + (1024/855)\delta_2)^2,$$
$$N_n = (1 - d/8)^{\lfloor n/3 \rfloor} \qquad (n \ge 3).$$

Section M.15 proves $\|T^n\| \le N_n$. These N_n bound the norm itself. The bases are the separate estimates for powers three, four and five; writing lengths as $3m, 3m+1, 3m+2$ uses respectively m cubes, one fourth power with $m-1$ cubes, or one fifth power with $m-1$ cubes. Use (E8.10.1) in the actual order

$$T^M M_{\sqrt{\bar f_a}} = T^{M-1}(T M_{\sqrt{\bar f_a}}), \qquad \|T^M M_{\sqrt{\bar f_a}}\| \le N_{M-1}.$$

The bare inverse-row estimates in Sections M.15 and M.20 give $\mathbb{E}[\bar f_n^4/q] \le 1 - d/2$ for every $n \ge 3$. Their $n = 3$ bound is the separate $1 - 23d/36$ estimate; the $n \ge 4$ row uses the complete longer-path proof. Combining these with (E8.10.1), (E8.10.2) and (E8.10.4) yields

$$\boxed{I_q(U)^3 \le (1 - d/2) N_{M-1}^4 Q_a(d)^2.} \tag{E8.10.5}$$

For $M = 1$, the multiplier norm is at most one. For $M \ge 4$, the remaining exponent $M - 1 \ge 3$ is covered by the separate power bases. Thus

$$I_q(U)^3 \le 1 - d/2 \le e^{-d/2} \qquad (M = 1), \tag{E8.10.6}$$

$$I_q(U)^3 \le e^{-[1/2 + \lfloor (M-1)/3 \rfloor/2]d} \qquad (M \ge 4). \tag{E8.10.7}$$

For $a \ge 3$, the additional factor $Q_a(d)^2$ may be retained. The displayed common coefficients cover every nonunit singleton, nonreducible terminal and inner length L in these two outer ranges. At $a = 2$, $M = 1$, the actual row is $\bar f_3$ and uses its separate bound; every other relevant row has length at least four. At $d = 0$, the operator proof still applies, and actual leaf integration gives equality.

R.2 Ordinary fourth moments for genuine singleton-root inputs

Q87.0 Singleton-root fourth moments and their outer transfer

Work on the finite probability space with a real skew kernel. Put

$$A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$
$$f_n = A^n 1, \quad \bar f_n = T^n 1, \quad W_n = A \bar f_{n-1},$$
$$I_q(U) = E[q^{-1/3} |U|^{8/3}].$$

The original offcenter root has precisely one residual arm of length a, after maximal actual equal-length pairing. The terminal has precisely the two residual arms $(1, b)$, $b \ge 2$. Their inner connector

has length L, and the independently specified connector from that root to the receiving root has length M. Equal-arm factors are restored by positivity in the common tree assembly.

Local bound. For every

$$a \geq 3, \qquad b \geq 2, \qquad L \geq 1, \qquad M \geq 1,$$

the following orientation, determined by the lengths, gives a receiving-root message U satisfies $I_q(U) \leq 1$.

Table 125: Ordinary fourth moments for genuine singleton-root inputs: parameter values.

Entry	Actual input Y at the singleton root
$L = 1$, every $a \geq 3$	$f_a T(p\bar{f}_b)$; Outer message: $T^M Y$.
$a = 3, L = 2, M = 1, 2$	$f_3 TA(p\bar{f}_b)$; Outer message: $T^M Y$.
$a = 3, L = 2, M \geq 3$	$W_3 T^2(qf_b)$; Outer message: $T^M Y$.
$a = 3, L \geq 3$	$W_3 T^L(qf_b)$; Outer message: $T^M Y$.
$a \geq 4, L \geq 2$	$W_a T^L(qf_b)$; Outer message: $T^M Y$.

For the first, second, fourth, and the $a = 4$ part of the last row, the new proof gives

$$EY^4 \leq e^{-11d/2}. \tag{Q87.0.1}$$

For every $a \geq 5, b \geq 2, L \geq 2$, it gives

$$E\{[W_a T^L(qf_b)]^4\} \leq e^{-3487d/640}. \tag{Q87.0.2}$$

Here $d = Eg^2$. The new fourth-input outer bridges are

$$\begin{aligned} I_q(TY)^3 &\leq e^{204d/19}(E|Y|^4)^2, \\ I_q(T^2Y)^3 &\leq e^{87d/8}(E|Y|^4)^2. \end{aligned} \tag{Q87.0.3}$$

Consequently every short-outer row of the local bound has the common bound

$$I_q(U)^3 \leq e^{-7d/320} \qquad (M = 1, 2). \tag{Q87.0.4}$$

In (Q87.0.1) rows the stronger bounds are $e^{-5d/19}$ for $M = 1$ and $e^{-d/8}$ for $M = 2$. For $M \geq 3$, the exact Proposition M.39, Section O.2, and the unanchored transfer of Section M.19 apply to the actual rows specified above. In particular,

$$I_q(U)^3 \leq \begin{cases} (1-d)e^{-d/1024}, & L = 1, \\ (1-d)e^{-3d/20}, & L \geq 2. \end{cases} \tag{Q87.0.5}$$

For $a = 3, L = 2$, the direction is selected from the outer length M.

Q87.1 Finite identities and inputs

Delete only zero-mass atoms. On every retained atom, $p(x), q(x) \geq \mu(x) > 0$, because $K(x, x) = 0$. Every finite tree message is positive: conditional on its root, the labeling putting all other original vertices at that same atom has positive mass and all diagonal edge factors one. All norms,

expectations, adjoints and rotation blocks use μ. Throughout, EY^j means the moment $E[Y^j]$, never the power $(EY)^j$ of a mean; displayed explicit norm squares are interpreted in the usual original Hilbert space.

Write

$$r = Kg, \quad s = Kr, \quad t = Ks, \quad \delta_j = \|K^j 1\|_2^2,$$
$$\theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The finite calculus supplies

$$\begin{gathered} 0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \\ \delta_{j+1} \le \theta \delta_j, \quad \|K\|^2 \le \theta, \\ |Kv(x)| \le \|v\|_2, \quad \kappa d \le \tau \le 2/9, \end{gathered} \tag{Q87.1.1}$$

$$\begin{gathered} Er = -d, \quad Es = 0, \quad Et = \delta_2, \quad Egr = Ers = 0, \\ Egs = -\delta_2, \quad Ert = -\delta_3. \end{gathered} \tag{Q87.1.2}$$

Original row centering and antisymmetrization give, respectively,

$$r^2 \le pqd, \qquad |r| \le pq/2. \tag{Q87.1.3}$$

For the second inequality, use $r(x) = \frac{1}{2} E_{y,z}[(K(x,y) - K(x,z))K(y,z)]$ and $|u - v| \le 1 - uv$ on $[-1,1]^2$. In particular $|r| \le 1/2$.

Define finite polynomials and finite alternating sums by

$$G_n(z) = \sum_{j=0}^{n-1} z^j,$$
$$\beta_1 = \beta_2 = 0,$$
$$\beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1} \delta_j.$$

The exact finite positive convolution, for either sign, is

$$\begin{gathered} \sum_{j=0}^{n} (\pm K)^j 1 \\ = f_n^{\pm} + \sum_{j=3}^{n} \beta_j f_{n-j}^{\pm}, \\ \tau \le \beta_j \le d \quad (j \ge 3), \end{gathered} \tag{Q87.1.4}$$

where $f_n^+ = f_n, f_n^- = \bar{f}_n$. Every removed term is a nonnegative actual path. The finite sum identity is proved in the common finite calculus.

At $z = i\sqrt{x}$, $0 \le x \le \theta$, write $G_n(z) = A_n + zB_n$. For $n \ge 2$, both coordinates lie in $[1 - x, 1]$. For $n \ge 4$, both are at most $1 - x + x^2$. Also $\|G_n(K)\| \le 1$ for $n \ge 3$, and $|G_n(i\sqrt{x})|^2 \le 1 - 7x/15$ for $n \ge 3$. The coordinate estimates are Lemma M.1; the finite decay bound is (Q81.1.5) of Section M.1. they hold at the short endpoints and for every finite integer length.

We use the following inputs.

1. Section M.1, the signed-mean exponential lemma proves, for every original real vector V and every $u \ge 0$,

$$Ee^{uKV} \le \exp\{uE(KV) + (17/64)u^2\|V\|^2\}. \tag{Q87.1.5}$$

It retains the actual mean of either sign. We use it at $u = 4$ and $u = 8$, including scaled vectors. The Hermite–Bernoulli proof covers the full continuous domain. 2. Section M.15, the short-operator and all-power estimates give

$$\begin{aligned} \|T\|^2 &\le (1 + 32d/19)^2, \\ \|T^2\|^2 &\le (1 + d/2)^4, \\ \|T^M\| &\le (1 - d/8)^{\lfloor M/3 \rfloor} \quad (M \ge 3). \end{aligned} \tag{Q87.1.6}$$

The Section M.14 bound $E[\bar{f}_3^4/q] \le 1 - 23d/36$. 3. Section M.19, the eighth-input transfer estimates prove the eighth-input map and its row budgets:

$$\begin{aligned} I_q(T^M Y)^3 &\le \|T^M\|^2 E|Y|^8 R_M^2, \\ R_M = E[\bar{f}_M^3/\sqrt{q}], &\quad R_M^2 \le 1 - d(M \ge 3), \end{aligned} \tag{Q87.1.7}$$

with short costs $e^{299d/38}$, e^{6d}. 4. Lemma M.38 and Proposition M.39 give the actual unit-inner envelopes and ordinary eighth for $f_a T(p\bar{f}_b)$, all $a \ge 3, b \ge 2$. Section O.2 gives the actual nonunit-inner envelopes, their paid scalar expressions, their covariance signs, and the ordinary eighth for $W_a T^L(qf_b)$, all $a \ge 3, b \ge 2, L \ge 2$. 5. Section O.29 gives $E[pT^L(qf_b)]^8 \le 1 - d/12000$ for all $b \ge 2, L \ge 1$. Section O.1 gives $E[W_2 T^L(qf_b)]^8 \le 1 - d/256$ for $b, L \ge 2$.

At $d = 0$, $g = 0$ but K need not vanish. Leaf integration makes every actual message one. The bounds for arbitrary real inputs also follow from their displayed norm/Jensen proofs. Every later division by d assumes $d > 0$.

Q87.2 Fourth-moment transfer along short outer paths

Let B be any positive original operator with row $h = B1 > 0$. Row Cauchy, then Holder with exponents 3 and $3/2$, give

$$\begin{aligned} |BY|^2 &\le hB(|Y|^2), \\ I_q(BY) &\le \\ \{E[h^4/q]\}^{1/3}&\{E[B(|Y|^2)^2]\}^{2/3}. \end{aligned}$$

Therefore, for every real Y,

$$I_q(BY)^3 \le E[h^4/q]\,\|B\|^4 (E|Y|^4)^2. \tag{Q87.2.1}$$

For $B = T$, $h = q$, and $Eq^3 = 1 + 3d - Eg^3 \le 1 + 4d$. Together with (Q87.1.6), this proves the first inequality of (Q87.0.3), since $4 + 128/19 = 204/19$.

For $B = T^2$, the actual row is $\bar{f}_2 = q + r$. Expand the same row:

$$E[\bar{f}_2^4/q] = E[q^3 + 4q^2 r] + E[(6q + 4r + r^2/q)r^2].$$

The first term is at most $1+2d$: indeed it equals $1-d-Eg^3+4Eg^2r$, and (Q87.1.3) gives $4Eg^2r \le 2E[g^2pq] \le 2d$, while $-Eg^3 \le d$. For the second, (Q87.1.3) gives

$$6q+4r+r^2/q \le q(6+2p+p^2/4) \le 12.$$

The last polynomial is $12-2p-3p^2/2-p^3/4$ for $0 \le p \le 2$. Consequently

$$E[\bar{f}_2^4/q] \le 1+2d+12\delta_2 \le 1+55d/8. \tag{Q87.2.2}$$

Now $\|T^2\|^4 \le (1+d/2)^8 \le e^{4d}$; (Q87.2.1) yields the second inequality of (Q87.0.3).

One further exact consequence will be used later:

$$I_q(T^3Y)^3 \le (1-23d/36)(E|Y|^4)^2. \tag{Q87.2.3}$$

This follows from (Q87.2.1), $\|T^3\| \le 1$, and the actual inverse-fourth row.

Q87.3 Every unit inner connector with a singleton arm at least three

Use the actual reversed terminal $F_b^- = p\bar{f}_b$. For $b=2$,

$$0 < TF_2^- \le q-\gamma-s, \qquad \gamma = d-\delta_2/2. \tag{Q87.3.1}$$

This follows from $p\bar{f}_2 \le (1+r/2)^2$, $Tr=-d-s$, and $T(r^2) \le 2\delta_2$. For $b \ge 3$, put

$$\bar{R} = KG_{b-1}(-K)g, \quad m = 1-\beta_b/2, \quad k = 1/m,$$
$$t_0 = \beta_{b+1}, \quad e = d-\beta_{b+2}, \quad V = E\bar{R}^2,$$
$$\gamma = kt_0 - k^2V/2.$$

The exact positive comparison (M.86) is

$$\begin{aligned} 0 < TF_b^- &\le m^2[q-\gamma-kK\bar{R}], \\ 1 \le k \le 6/5, \quad \gamma - ke &\ge d - 68\delta_2/25. \end{aligned} \tag{Q87.3.2}$$

Here $E\bar{R} = -t_0$, $E(g\bar{R}) = e$, $V = \delta_2+\delta_3$ for $b=3$, and $V \le \delta_2$ for $b \ge 4$. Both upper factors are nonnegative by their positive-kernel derivation.

Write $R_a = KG_{a-1}(K)g$. The root envelope $f_a \le p+R_a-\beta_a$ and AM–GM give

$$\begin{aligned} f_aT(p\bar{f}_b) &\le \\ m^2[1-(\beta_a+\gamma)/2+Z/2]^2, \\ Z = K[G_{a-1}(K) - kKG_{b-1}(-K)]g, \end{aligned} \tag{Q87.3.3}$$

with $EZ = -\beta_{a+1}+ke$. For $b=2$, set $m=k=1$, $Z = K[G_{a-1}(K)-K]g$, $EZ = -\beta_{a+1}$. Every square base is half a sum of two nonnegative upper factors. Thus raising to the eighth power and using $1+x \le e^x$ is legal. Equation (Q87.1.5) at $u=4$ applies to this same fluctuation.

Q87.3.1 Terminal lengths at least three, root arms three and four

The exact correlated estimate (M.89) gives

$$\|[G_{a-1}(K) - kKG_{b-1}(-K)]g\|^2 \le d - 17\delta_2/20 \qquad (a, b \ge 3).$$

Consequently

$$\begin{gathered} \log EY^4 \le \\ -4(\beta_a + \beta_{a+1} + \beta_b + \gamma - ke) \\ +(17/4)d - (289/80)\delta_2. \end{gathered} \tag{Q87.3.4}$$

For $a = 3$, $\beta_3 + \beta_4 = 2d$; for $a = 4$, $\beta_4 + \beta_5 = 2d - \delta_2$. Using $\beta_b \ge \tau$ gives, respectively,

$$\log EY^4 \le -47d/4 + (4507/400)\delta_2, \qquad \log EY^4 \le -47d/4 + (6107/400)\delta_2.$$

The second is the weaker one, and its upper bound is $-71009d/12800 < -11d/2$, because $150400 - 79391 = 71009 > 70400$. This proves (Q87.0.1) in both rows.

Q87.3.2 A stronger finite frequency bound for every root arm at least five

For $m_0 \ge 4, n \ge 2, 1 \le k \le 6/5$, one has

$$\begin{gathered} |G_{m_0}(i\sqrt{x}) - ki\sqrt{x}G_n(-i\sqrt{x})|^2 \\ \le 1 - 3x/2, \qquad 0 \le x \le \theta. \end{gathered} \tag{Q87.3.5}$$

To prove it write $G_{m_0} = \alpha_0 + i\sqrt{x}\beta_0$, $G_n(-i\sqrt{x}) = \gamma_0 - i\sqrt{x}\omega_0$. The first two coordinates lie in $[1 - x, 1 - x + x^2]$, the last two in $[1 - x, 1]$. The real part of the difference is positive, at least $1 - (1 + k)x$, and at most $1 - (1 + k)x + (1 + k)x^2$. Its imaginary coefficient has magnitude at most $k - 1 + x$. Squaring gives $1 + xP(k, x)$, where

$$P = k^2 - 4k - 1 + (k^2 + 6k + 1)x - (2k^2 + 4k + 1)x^2 + (k + 1)^2x^3.$$

Its derivative in x is at least $8 - (434/25)\theta = 379/400 > 0$. Its second derivative in k is $2(1 + x - 2x^2 + x^3) > 0$. The two endpoint maxima at $x = \theta$ are

$$P(1, \theta) = -13411/8192 < -3/2, \qquad P(6/5, \theta) = -1271219/819200 < -3/2.$$

These prove the entire continuous rectangle and all integer lengths.

For $a \ge 5$, retain also the actual term $\beta_{a-1}p \ge \tau p$ in (Q87.1.4):

$$f_a \le (1 - \tau)p + R_a - \beta_a.$$

In (Q87.3.3), the new deterministic loss is $\beta_a + \gamma + \tau$ and the fluctuation is $Z' = K(V_0 - \tau 1)$, where $V_0 = [G_{a-1}(K) - kKG_{b-1}(-K)]g$. Its mean is unchanged. Moreover

$$EV_0 = -\beta_a + kt_0 \ge -\delta_2, \qquad \|V_0 - \tau 1\|^2 \le d - 3\delta_2/2 + d^2 - \delta_2^2.$$

Using (Q87.1.5), (Q87.3.2), and $\beta_a, \beta_{a+1}, \beta_b \ge \tau$,

$$\begin{aligned}\log EY^4 &\le -63d/4 + (4101/200)\delta_2 + (17/4)(d^2 - \delta_2^2)\\ &\le -115261d/19200 < -6d.\end{aligned} \tag{Q87.3.6}$$

The last step only uses $\delta_2 \le \theta d$, $d^2 \le d/3$, and discards the negative δ_2^2. Thus every $a \ge 5, b \ge 3$ is covered without a remaining score interval.

Q87.3.3 Terminal two, every root arm

For $b = 2$, (Q87.3.3) and (Q87.1.5) give

$$\begin{aligned}\log EY^4 &\le -4(\beta_a + \beta_{a+1} + \gamma)\\ &+(17/4)\|[G_{a-1}(K) - K]g\|^2.\end{aligned} \tag{Q87.3.7}$$

At $a = 3$, the last polynomial is 1; this yields $-31d/4 + 2\delta_2 \le -111d/16$. At $a = 4$, its squared modulus is $(1 - x)^2$, and (Q87.3.7) is at most $-31d/4 - (5/2)\delta_2 + (17/4)\delta_3 \le -31d/4$.

At $a = 5$, the exact squared modulus is $1-2x+x^2+x^3$, while $\beta_5+\beta_6 = 2\tau$. The upper coefficient is

$$-31/4 + (3/2)\theta + (17/4)\theta^2(1 + \theta) < -91/16.$$

For an explicit comparison, the last positive term is $129285/131072 < 1$; the whole expression is less than $-31/4 + 39/64 + 1 = -393/64 < -91/16$.

At $a = 6$, the exact squared modulus is $1 - 2x + 3x^2 - x^3 + x^4$, and $\beta_6 + \beta_7 = 2\tau + \delta_3$. Hence the bound is at most $-31/4 + (3/2)\theta + (35/4)\theta^2 = -23333/4096 < -91/16$.

For every $a \ge 7$, the same upper squared polynomial $1-2x+3x^2-x^3+x^4$ follows from the finite coordinate bounds. Also $\beta_a, \beta_{a+1} \ge \tau + \kappa\delta_3$, by the finite alternating tails. Substitution in (Q87.3.7) gives

$$\log EY^4 \le -31d/4 + (3/2)\delta_2 + 8\delta_3 - (17/4)\delta_4 + (17/4)\delta_5 \le -745d/128 < -91d/16.$$

All terminal-two cases therefore satisfy (Q87.0.1). This completes every $a \ge 3, b \ge 2, L = 1$, with the single orientation in the local bound.

Q87.4 The nonunit-inner scalar table

The remaining proof uses the scalar estimates in Section O.2, the scalar-input and row-table estimates. Set $X = T^L(qf_b)$, $L \ge 2$, and $W_2 = p - r$. The positive envelope has the form

$$0 < X \le C[Bq + Br + Z - a_0], \qquad W_2X \le CB[1 + (Z - a_0)/(2B)]^2. \tag{Q87.4.1}$$

Every B below is positive. If $Z = K^2V_0$, the paid expression is

$$S_8 = 8\log C + 8\log B - 8a_0/B + 8M_0/B + 6D/B^2. \tag{Q87.4.2}$$

Here $EZ \le M_0$, $\|V_0\|^2 \le D$. For $Z = KV_0$, replace its last coefficient 6 by $8c_*$, $c_* = (1 + \theta)^2 = 2025/1024$. The exact rows are:

Table 126: Ordinary fourth moments for genuine singleton-root inputs: parameter values.

Entry	C, B, a_0
$b = 2, L = 2$	$1, 1-\gamma, 0$; Row vector and D: two-row $V_0 = \gamma 1 + r$, $D = \delta_2$; Accepted upper for S_8: $-d/8$.
$b = 2, L = 3$	$1, 1-\gamma, \tau$; Row vector and D: two-row $V_0 = -(g+s)$, $D = d - 2\delta_2 + \delta_3$; Accepted upper for S_8: $-3d/4$.
$b = 2, L = 4$	$1, 1-\gamma-\tau, d(1-\gamma)$; Row vector and D: two-row $V_0 = \tau 1 - (1-\gamma)g + r + t$; Accepted upper for S_8: $-d$.
$b = 2, L = 5$	$1, 1-2\tau, \tau$; Row vector and D: one-row $[\tau + zP_3(z)]g$, $D/d = \theta - 2\theta\tau + \tau^2$; Accepted upper for S_8: $-2d/3$.
$b = 2, L \geq 6$	$1, 1-2\tau, \tau$; Row vector and D: one-row $[\tau + zP_{L-2}(z)]g$, $D/d = \theta - 7\theta^2/15 - 2\theta\kappa\tau + \tau^2$; Accepted upper for S_8: $-4d/5$.
$b \geq 3, L = 2$	$m^2, 1-\gamma, ke$; Row vector and D: two-row $V_0 = \gamma 1 + kR$, $D = k^2 V$; Accepted upper for S_8: $-d/8$.
$b \geq 3, L = 3$	$m^2, 1 - B_*, d - k\rho$; Row vector and D: one-row $[\delta + zP_1(z)]g$, $D/d = 351/1280 + \delta^2$, $\delta = ke$; Accepted upper for S_8: $-d/10$.
$b \geq 3, L = 4$	$m^2, 1 - 41d/80, \kappa d$; Row vector and D: one-row $[\delta + zP_2(z)]g$, $D/d = \theta(71/63) + \delta^2$, $\delta = 41d/80$; Accepted upper for S_8: $-8d/25$.
$b \geq 3, L \geq 5$	$1 - d/2, 1 - d/2, d/2$; Row vector and D: one-row $[\delta + zP_{L-2}(z)]g$, $D/d = \theta + d^2/4$, $\delta = d/2$; Accepted upper for S_8: $-2d/5$.

For $b = 2$, $\gamma = d - \delta_2/2$, $z = -K$, $P_n = G_n + z^{n+1}$. In its $L = 4$ row, $D = (1-\gamma)^2 d + \delta_2 - 2\delta_3 + \delta_4 - \tau^2$. For $b \geq 3$, use the actual positive terminal remainder $R = KG_{b-1}(K)g$ and the same even-moment quantities m, k, t_0, e, V, γ as in (Q87.3.2), with $\rho = ErR$, $B_* = \gamma + ke$, and $P_n = G_n + kz^{n+1}G_{b-1}(-z)$.

The same-parameter bounds are

$$\begin{gathered} 3d/10 \leq \gamma \leq 6d/5, \\ 5d/8 \leq B_* \leq 3d/2, \\ d - k\rho \geq 41d/80, \quad 0 \leq ke \leq 39d/80, \\ \kappa\delta_2 \leq \rho \leq \delta_2, \\ m^2 \leq 1 - 209d/384, \quad 1 - \gamma \geq 17/25. \end{gathered} \tag{Q87.4.3}$$

For $b = 2$, the bounds needed here are $1 - \gamma \geq 13/18$, $1 - \gamma - \tau \geq 1/2$, $1 - 2\tau \geq 5/9$. In the three $b \geq 3, L \geq 3$ rows, $M_0 = (\theta - \delta)d$. In the two $L = 2$ rows, $M_0 = \max(k\rho - \gamma d, 0)$, with $k = 1, \rho = \delta_2$ for $b = 2$; that latter mean is already nonnegative. For $b = 2, L = 3$, $M_0 = 0$; for $L = 4$, $M_0 = (\theta\kappa - \tau)d$.

All row-vector ranges, raw second moments, and means belong to these same actual fluctuations. The $b \geq 3, L = 2$ scalar includes the terminal end-sign term from Section O.1/Section O.2. Every row has $S_8 \leq -d/10$.

For later covariance arguments, write $P_n = A_P + zB_P$, $z = i\sqrt{x}$. The finite coefficient lemma says

$$1 - x \leq A_P, B_P \leq 1 \quad (n \geq 3),$$

$$
\begin{aligned}
n=1 &: A_P \ge 1-6x/5, 0 \le B_P \le 6x/5;\\
n=2 &: A_P \ge 1-x, 1-6x/5 \le B_P \le 1.
\end{aligned}
\tag{Q87.4.4}
$$

These include every finite tail. In particular $\Re P_n \ge 1/2$ in all required rows.

Q87.5 Every nonunit inner connector with a singleton arm at least five

Let $a \ge 5$, $m_0 = a-2 \ge 3$, $U_0 = K^2 G_{m_0}(-K)g$, and $\beta = \beta_a$. Apply positive A to (Q87.1.4) for $\bar{f}_{a-1}$. Its last two nonconstant convolution terms are the actual messages $\beta_{a-1}p$ and $\beta_{a-2}W_2$. Retaining both, and using their actual lower coefficients τ, gives

$$
0 < W_a \le cp - (1-\tau)r + U_0 - \beta, \qquad c = 1-2\tau. \tag{Q87.5.1}
$$

This upper side is nonnegative because it equals the actual path plus omitted nonnegative path terms and nonnegative coefficient remainders.

Set $H = G_{m_0}(-K)g - \tau 1$. Scaled AM–GM with (Q87.4.1) gives

$$
W_a X \le CBc[1 + (Z + (B/c)K^2 H - a_0 - (B/c)\beta)/(2B)]^2. \tag{Q87.5.2}
$$

Indeed $K^2 H = U_0 - \tau r$; the coefficient of r after dividing the two factors by c and CB is exactly $-\tau/c$. Both scaled factors are nonnegative.

The exact Section O.2 centered-covariance argument gives

$$
EK^2 H \le 0, \qquad \|H\|^2 \le d - \tau^2, \tag{Q87.5.3}
$$

and the nonpositive original inner products $\langle V_0, H\rangle \le 0$ in every two-row case, or $\langle V_0, KH\rangle \le 0$ in every one-row case. These statements were proved in Section O.2, the centered-covariance estimates for this same H, all $m_0 \ge 2$, and every row of part Q87.4. They are independent of c and therefore apply in (Q87.5.2). In particular the Section O.2 proof retains $\|H\|^2 \le d - \tau^2$, rather than its later weaker $\le d$.

For clarity, the mean uses $\delta_2 - \delta_3 - \tau d \ge (\kappa - d)(\delta_2 - d^2) \ge 0$. The latter follows by applying the original spectral bound to $r + d1$. The covariance at $L = 2$ is the actual covariance of the two finite arm remainders; the other one-row covariances are the finite coefficient inequalities in Section O.2.

Put $\alpha = 17\theta/4 = 221/128$. Apply (Q87.1.5) at $u = 4/B$ to (Q87.5.2). In a two-row case its variance coefficient is at most $\alpha\|V_0 + (B/c)H\|^2/B^2$. In a one-row case it is at most $(17/4)\|V_0 + (B/c)KH\|^2/B^2$. The covariance signs, (Q87.5.3), and $\|K\|^2 \le \theta$ imply

$$
\begin{aligned}
\log E[(W_a X)^4] &\le S_8/2 + F(d,\tau),\\
F(d,\tau) &= 4\log(1-2\tau) - \frac{4\tau}{1-2\tau}\\
&\quad + \alpha\frac{d-\tau^2}{(1-2\tau)^2}.
\end{aligned}
\tag{Q87.5.4}
$$

Here $\alpha < 3$ and $17/4 < 4c_*$ pay the old vector costs. The old mean upper stays valid because $EK^2 H \le 0$.

The following pays (Q87.5.4) on the entire original domain:

$$
F(d,\tau) \le -691d/128. \tag{Q87.5.5}
$$

For fixed d, its τ-derivative, multiplied by $(1-2\tau)^3$, is

$$-12+(40-2\alpha)\tau-32\tau^2+4\alpha d\le -227/144<0$$

using $\tau\le 2/9, d\le 1/3$. Thus it suffices to put $\tau=\kappa d$. Let $h=12\kappa-\alpha=691/128$. Using $\log(1-2\tau)\le -2\tau-2\tau^2$, the assertion $F+hd\le 0$, after multiplication by $(1-2\tau)^2$, is bounded above by

$$\tau^2[32-\alpha-4h/\kappa+4hd-32\kappa^2d^2].$$

The bracket increases on $[0,1/3]$, since its derivative is at least $1351/96>0$. Its endpoint is

$$3875/128-691/19+691/96-361/288=-3295/21888<0.$$

This proves (Q87.5.5). Since $S_8\le -d/10$, (Q87.5.4) now gives precisely (Q87.0.2).

Q87.6 Every nonunit inner connector with a singleton arm four

Here the exact coefficient retained from (Q87.1.4) is $\beta_3=d$, not merely its lower bound τ. The root identity is

$$W_4=(1-d)p-r+(s-t)-d.$$

Put $c=1-d$, $H=g-r-d1$. Formula (Q87.5.2) holds with $\beta=d$, and

$$EH=0,\quad \|H\|^2=d+\delta_2-d^2,\quad EK^2H=-\delta_2+d^2\le 0. \tag{Q87.6.1}$$

The required two-row covariances remain nonpositive. At $L=2$, the covariance proof in Section O.2, the two-row covariance calculation has no centering constant inside its covariance, and $EH=0$. At $b=2, L=3,4$, $EV_0=0$, so replacing $\tau 1$ by $d1$ leaves the covariance unchanged.

Every one-row covariance is also nonpositive with this new centering. At $z=-K=i\sqrt{x}$, $G_2(z)=1+z$, and the spectral product controlling its negative is

$$\delta d+x[A_P-\delta+(x-d)B_P]. \tag{Q87.6.2}$$

For $n\ge 3$, and also $n=2$, the square bracket is at least $\kappa-2/9-1/3=11/288>0$, by (Q87.4.4). For $n=1$, it is at least $41/80-13/80-13/80=15/80>0$. Thus the same comparison to $S_8/2$ gives the increment

$$F_4=4\log(1-d)-4d/(1-d)+\alpha[(1+\theta)d-d^2]/(1-d)^2.$$

Let $h_4=8-\alpha(1+\theta)=22823/4096$. Using $\log(1-d)\le -d-d^2/2$, multiplication by $(1-d)^2$ bounds F_4+h_4d by

$$d^2[10-\alpha-2h_4+h_4d-2d^2]\le d^2[10-\alpha-5h_4/3]=-12451d^2/12288\le 0.$$

Hence $\log E[(W_4X)^4]\le -(h_4+1/20)d<-11d/2$. This covers every terminal length and every $L\ge 2$.

Q87.7 Singleton arm three and every inner length at least three

The exact root is $W_3 = W_2 + s - d$. Combining with (Q87.4.1),

$$W_3 X \le CB[1 + (Z + Bs - a_0 - Bd)/(2B)]^2. \tag{Q87.7.1}$$

The new fluctuation has the same mean as Z; its deterministic loss adds $-4d$ to the ordinary-fourth exponent. For one-row cases its row vector becomes $[\delta + z(P_n - B)]g$. At every original frequency,

$$|\delta + z(P_n - B)|^2 - |\delta + zP_n|^2 = Bx(B - 2\Re P_n) \le 0. \tag{Q87.7.2}$$

Put $\Delta = 4c_* - 17/4 = 937/256$. Except for the separately treated $b \ge 3, L = 3$ row, (Q87.1.5), (Q87.7.2), and the paid table yield

$$\log E[(W_3 X)^4] \le S_8/2 - 4d - \Delta D/B^2 \tag{Q87.7.3}$$

in every remaining one-row case.

For $b = 2, L = 5$, $D/B^2 \ge \theta d$. For $b = 2, L \ge 6$, it is at least $(\theta - 7\theta^2/15)d$. To verify both entire domains, put $A(t) = A_0 - 2\theta\sigma t + t^2$. The derivative of $A(t)/(1-2t)^2$ has positive numerator $4A_0 - 2\theta\sigma + 2t(1 - 2\theta\sigma)$, for $(A_0, \sigma) = (\theta, 1)$ or $(\theta - 7\theta^2/15, \kappa)$. Its minimum is therefore at zero. For $b \ge 3, L = 4$, $D/B^2 \ge \theta(71/63)d > \theta d$; for $b \ge 3, L \ge 5$, $D/B^2 \ge \theta d$.

The exact lower constants used in (Q87.7.3) are

$$\Delta\theta = 12181/8192 > 67/50,$$
$$\Delta(\theta - 7\theta^2/15) = 4738409/3932160 > 11/10.$$

With the corresponding $S_8/2$ values, each coefficient is strictly more negative than $-11d/2$. In particular the weakest long terminal-two row has loss at least $4 + 2/5 + 11/10 = 11/2$.

For $b = 2, L = 3$, the new two-row vector is $-(\gamma g + s)$. Since $\gamma d \le \delta_2$, its square norm is at most δ_3. Its mean is zero, and direct use of (Q87.1.5) gives

$$\begin{aligned} &\log E[(W_3 X)^4] \\ &\le 4\log(1-\gamma) - 4\tau/(1-\gamma) - 4d \\ &\quad + \alpha\delta_3/(1-\gamma)^2 \\ &\le [-153/16 + 17901/32768]d < -9d. \end{aligned} \tag{Q87.7.4}$$

For $b = 2, L = 4$, the new vector is $\tau 1 - \tau g + r + t$, whose norm square is $\delta_2 - 2\delta_3 + \delta_4 - \tau^2 + \tau^2 d \le \delta_2 + \tau^2 d$. Using the same table parameters gives

$$\begin{aligned} &\log E[(W_3 X)^4] \le \\ &-(89/16)d - 4d - 4d + 8\theta\kappa d \\ &\quad + 4\alpha(\theta + 4/81)d < -8d. \end{aligned} \tag{Q87.7.5}$$

For instance $8\theta\kappa < 2$, while $4\alpha(\theta + 4/81) = 2873/1024 + 221/648 < 13/4$, so the last strict comparison follows on the entire domain.

Finally let $b \ge 3, L = 3$, $t_* = B_* \in [0, 1/2]$, $B = 1 - t_*$. The new one-row vector is $[\delta + z(P_1 - B)]g$. Writing $G_{b-1}(-z) = A - zD_0$, its squared modulus is

$$
\begin{aligned}
&(\delta - kx^2 D_0)^2 + x(t_* - kxA)^2 \\
&\quad \le \delta^2 + \theta t_*^2 + c_0, \\
&c_0 = (36/25)\theta^3(1+\theta) < 7/50.
\end{aligned}
$$

The last rational bound follows from $\theta^3 < 27/400$, $1+\theta < 10/7$, and $(36/25)(27/400)(10/7) = 243/1750 < 7/50$. The old row has $D/d = 351/1280 + \delta^2$. Therefore its saved cost, divided by d, is at least

$$
\frac{3/2 - \alpha t_*^2}{(1-t_*)^2} \ge 3/2.
$$

Indeed

$$
4c_*(351/1280) - (17/4)c_0 > (79/10)(27/100) - (17/4)(7/50) = 769/500 > 3/2,
$$

and the displayed quotient increases on $[0, 1/2]$, since its derivative has numerator $2(3/2 - \alpha t_*) > 0$. Consequently its fourth exponent is at most $-d/20 - 4d - 3d/2 = -111d/20 < -11d/2$. This finishes every $L \ge 3$ at $a = 3$.

Q87.8 Singleton arm three, two original inner edges, and a real turn

For the remaining short-outer row use

$$
Y = f_3 TA(p\bar{f}_b). \tag{Q87.8.1}
$$

The two inner edges meet at their original midpoint and are both directed away from that midpoint. Thus at the singleton root the last-edge operator is T, and at the midpoint the terminal-edge operator is A. The root arm is the outward three-path $f_3 = p + r + s - d$.

The positive terminal square used in part Q87.3 gives

$$
\begin{aligned}
A(p\bar{f}_b) &\le m^2[p - \gamma + kK\bar{R}], \\
0 < TA(p\bar{f}_b) &\le m^2[Bq - r - kK^2\bar{R} - ke], \qquad B = 1 - \gamma.
\end{aligned} \tag{Q87.8.2}
$$

For $b = 2$, use $m = k = 1, \bar{R} = r, e = 0$. Set $V_0 = \gamma 1 + k\bar{R}$, $\rho = Er\bar{R}$. Scaled AM–GM yields

$$
Y \le m^2 B[1 - d/2 - ke/(2B) + Z/2]^2, \qquad Z = K^2[g - V_0/B]. \tag{Q87.8.3}
$$

Its exact mean and row-vector norm satisfy

$$
\begin{aligned}
EZ &= (\gamma d - k\rho)/B, \\
&\quad \|g - V_0/B\|^2 \\
&\le d - 2ke/B + k^2 V/B^2.
\end{aligned} \tag{Q87.8.4}
$$

Here $E(g\bar{R}) = e$, while $\|V_0\|^2 \le k^2 V$ follows from the identity (Q71.8.2). The sign of the mean is retained; (Q87.1.5) allows either sign. Thus

$$
\begin{aligned}
\log EY^4 &\le 4\log m^2 + 4\log B - 4d + \alpha d \\
&\quad + [4\gamma d - 4k\rho - (4 + 2\alpha)ke]/B \\
&\quad + \alpha k^2 V/B^2.
\end{aligned} \tag{Q87.8.5}
$$

Q87.8.1 Terminal two

Here $\rho = \delta_2$, $B \geq 13/18$, and the exact norm is

$$\|V_0\|^2 = \delta_2 - d^2 + \delta_2^2/4.$$

Put $c_0 = \alpha/(13/18)^2 = 1377/416$. Use $\delta_2 - \gamma d \geq 0$, $B \leq 1$, and $\log B \leq -\gamma$ in (Q87.8.3)–(Q87.8.5). After collecting the same moments,

$$\log EY^4 \leq -(8-\alpha)d + (c_0 - 2)\delta_2 + (4 - c_0)d^2 - 2d\delta_2 + (c_0/4)\delta_2^2.$$

The last two terms sum to a nonpositive number because $(c_0/4)\delta_2 \leq (c_0\theta/4)d < 2d$. Since $c_0 > 2$ and $4 - c_0 > 0$, the full bound is at most

$$[-8 + \alpha + (c_0 - 2)\theta + (4 - c_0)/3]d = -220097d/39936 < -11d/2. \tag{Q87.8.6}$$

The final numerator comparison is $220097 > 219648$.

Q87.8.2 Every terminal length at least four

Use the same δ_2 throughout, and put

$$C_0 = 1989/800, \quad N_0 = 13927/2048, \quad \gamma_0 = d - 43\delta_2/25 > 0.$$

The finite alternating sums imply $\beta_b \geq \tau$, $e, \rho \geq \kappa\delta_2$, $V \leq \delta_2$, and $\gamma \geq \gamma_0$. Thus (Q87.8.5) is bounded by

$$-4\tau - 4d + \alpha d +$$
$$\Phi(B),$$
$$\Phi(B) = 4\log B + 4(1 - B)d/B + C_0\delta_2/B^2 - N_0\delta_2/B.$$

On $B \geq 17/25$, its derivative is positive. After multiplying by B^3, the derivative is $4B^2 - 4dB + N_0\delta_2 B - 2C_0\delta_2$. The first two terms are at least $1768/1875 > 9/10$; $2C_0\delta_2 < 5(7/50) = 7/10$; the omitted term is nonnegative. Hence we may replace B by $B_0 = 1 - \gamma_0 \geq B$. Now $2/3 \leq B_0 \leq 1$.

For $1 \leq w \leq 3/2$, convexity and $N_0 \geq 5C_0/2$ imply $C_0 w^2 - N_0 w \leq C_0 - N_0$. Also $4\gamma_0 d/B_0 \leq 6d^2$. Using $\log B_0 \leq -\gamma_0$ gives

$$\begin{aligned} \log EY^4 &\leq (-10 + \alpha)d + \\ &[272/25 + C_0 - N_0]\delta_2 \\ &\leq -9184899d/1638400 < -28d/5. \end{aligned} \tag{Q87.8.7}$$

The final comparison is $9184899 > 9175040$.

Q87.8.3 Terminal three

Here $\beta_b = d$, $e = \rho = \delta_2$, $V = \delta_2 + \delta_3$. Repeat the preceding argument with

$$C_0 = \alpha(36/25)(1 + \theta) < 7/2,$$
$$N_0 = 8 + 2\alpha > 11,$$
$$\gamma_0 = d - 81\delta_2/80.$$

Monotonicity now follows immediately from $N_0B - 2C_0 > 11(17/25) - 7 > 0$ and $B > d$. The convex endpoint bound still holds since $N_0 \geq 5C_0/2$. The final upper is

$$(-10 + \alpha)d + [81/20 + C_0 - N_0]\delta_2 \leq (-10 + \alpha)d < -11d/2.$$

This completes (Q87.0.1) for (Q87.8.1). For $M \geq 3$, the direction table instead uses $W_3T^2(qf_b)$ and its eighth-moment estimate.

R.3 The retained multiplier and every even internal root

R42.1 Original model and precise claims

For a real skew kernel K with pointwise absolute value at most one on one original finite probability space, put

$$B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B = T^*, \quad p = 1 - u, \quad q = 1 + u, \quad d = Eu^2 \in [0, 1/3].$$

All adjoints, norms and expectations use this unchanged original measure. Write $f_n = A^n\mathbf{1}, \bar{f}_n = T^n\mathbf{1}, h_n = Tf_{n-1}, W_n = A\bar{f}_{n-1}.M_v$ denotes multiplication by the actual function v. After zero-mass atoms are removed, p(x), $q(x) \geq \mu(x) > 0$; there is no uniform cutoff. Define $I_q(V) = \mathbb{E}[q^{-1/3}|V|^{8/3}]$.

Theorem A — arbitrary real fourth input with an actual arm. For every real original Y, every $a \geq 2$ and every $M \geq 1$,

$$I_q(T^M(\bar{f}_a Y))^3 \leq (1 - d/2)(1 - d/72)^2\, N_{M-1}^4 (E|Y|^4)^2, \tag{R42.1.1}$$

where valid original norm bounds are

$$N_0 = 1, \quad N_1 = 1 + 32d/19, \quad N_2 = (1 + (1024/855)\delta_2)^2,$$

$$N_n = (1 - d/8)^{\lfloor n/3 \rfloor} \quad (n \geq 3), \qquad \delta_2 = \|B^2 1\|^2 \leq (13/32)d. \tag{R42.1.2}$$

Consequently a useful all-length exponential form is

$$I_q(T^M(\bar{f}_a Y))^3 \leq e^{c_M d}(E|Y|^4)^2, \quad c_M = \begin{cases} -19/36, & M = 1, \\ 4247/684, & M = 2, \\ 11507/3420, & M = 3, \\ -19/36 - \lfloor (M-1)/3 \rfloor/2, & M \geq 4. \end{cases} \tag{R42.1.3}$$

In particular $c_M \leq 4247/684$ for EVERY M. Thus any actual H with $EH^4 \leq \exp(-\lambda\ \mathrm{d})$ has the same-direction output

$$I_q(T^M(\bar{f}_a H))^3 \leq \exp[-(2\lambda - c_M)d] \leq \exp[-(2\lambda - 4247/684)d]. \tag{R42.1.4}$$

No ordinary eighth hypothesis occurs in this theorem.

Theorem B — every even root with at least four residual arms. At an original internal root let the maximally unpaired side-arm lengths be strictly increasing $r_1 < \ldots < r_{2j}, j \geq 2$. Its terminal has any finite strict residual list of size at least two. All original equal-pair decorations, all independent

$innerL \geq 1$ and $outerM \geq 1$ are allowed. Section3 specifies all original directions from these integers alone. The actual output U_D satisfies

$$I_q(U_D)^3 \leq \begin{cases} e^{-15701d/3420}, & r_1 = 1, \\ e^{-3277d/684}, & r_1 \geq 2. \end{cases} \tag{R42.1.5}$$

The common first coefficient applies to the whole family.

Theorem C — complete unit-minimum even roots and original P5. Every unit-minimum even residual, including an exact pair(1, t), has, for every nonreducible terminal, an all-L/M actual output satisfying $Iq(U)^3 \leq \exp(-d/72)$. For a P5 whose two noncentral internal roots each either have an even residual of size at least four, or have an exact pair(1, t), the two terminal and central multisets may be arbitrary, and all four positive core lengths independent: there is a structure-only direction with $Z \leq 1$. In the nonreducible-end, central-R branch, the common deficit is d/288. If both internal residuals have size at least four, it improves to 15701d/13680. Section5 gives precisely qualified mixed-side consequences.

R42.2 The complete analytic proof: retain and pay the multiplier

R42.2.1 Exact original ingredients

The actual ordinary multiplier statement is

$$\|TM_{\bar{f}_a}\|_{2\to 2}^2 \leq 1 - d/72 \quad (a \geq 2). \tag{R42.2.1}$$

This is C064(1.6)/C066(1.4). Q4/C115(8.3) states $\|M_{f_a}T\|^2 \leq 1 - d/72$. Replacing K by -K changes that operator to $M_{\bar{f}_a}A$; taking its ordinary adjoint gives T $M_{\bar{f}_a}$, exactly (R42.2.1). Reversal and adjunction use the same probability measure. This estimate uses the full multiplier.

C100 TheoremA (A068) gives

$$E[\bar{f}_n^4/q] \leq 1 - (13639/23040)d \leq 1 - d/2 \quad (n \geq 4). \tag{R42.2.2}$$

The distinct n3 bound is the C056 inequality E[$\bar{f}_3^4/q$]$\leq 1 - 23d/36 \leq 1 - d/2$. Thus (R42.2.2)'s weaker final bound holds for EVERY $n \geq 3$. The C100 proof in Sections2-3 uses two retained finite convolution losses, a truncated quartic majorant and the true correlated cubic. The derivation uses these finite bounds directly.

The ordinary T power bounds are exactly (R42.1.2). N2 is the separate two-step bound; the $n \geq 3$ bound uses $separate3/4/5$ bases and finite blocks. In particular no assumption $\|T^{n+1}\| \leq \|T^n\|$ is introduced.

R42.2.2 Every real input, every actual positive path length

Let P be any original positive-kernel operator and $h = P1$. Pointwise row Cauchy and original Holder with exponents3 $and3/2$ give

$$|PY|^2 \leq hP(|Y|^2),$$

$$I_q(PY) \leq \{E[h^4/q]\}^{1/3}\{E[(P(|Y|^2))^2]\}^{2/3}.$$

The ordinary L2 norm then yields

$$I_q(PY)^3 \leq E[h^4/q] \cdot \|P\|^4 (E|Y|^4)^2. \tag{R42.2.3}$$

This calculation uses $|Y|^2$, so it allows any real Y, without a sign or actual-tree assumption on Y. The original positive row h is unchanged.

Now put $P = T^M M_{\bar{f}_a}$. Its EXACT row and factorization are

$$h = T^M \bar{f}_a = \bar{f}_{M+a}, \qquad P = T^{M-1}(T M_{\bar{f}_a}). \tag{R42.2.4}$$

Because $M + a \geq 3$, the bare-row bound applies at every allowed boundary, including a2, M1. The actual norm satisfies

$$\|P\|^4 \leq \|T^{M-1}\|^4 (1 - d/72)^2. \tag{R42.2.5}$$

Inserting (R42.2.2),(R42.2.5) into (R42.2.3) proves (R42.1.1). In particular we did NOT replace $\|T^M M_{\bar{f}_a}\|$ by $\|T^M\|$. The actual multiplier is paid before the remaining M-1 original edges; all operator orders are retained.

For (R42.1.3), $\log(1 - d/2) + 2\log(1 - d/72) \leq -(19/36)d$. When M2 the remaining norm contributes at most128d/19, and $128/19 - 19/36 = 4247/684$. When M3 it contributes at most $8(1024/855)\delta_2 \leq 3328d/855$, and $3328/855 - 19/36 = 11507/3420$. For $M \geq 4$, (R42.1.2) contributes at most -floor((M-1)/3)d/2. M1 has no remaining norm cost. These four exact cases cover every positive integer M and prove the common maximum.

At d0, the actual $\bar{f}_a$ and q equal1 although B need not vanish. The same norm and positive-operator proof gives (R42.1.1) for all realY. Leaf integration is used only later for actual tree messages, not for an arbitrary real function. No division by d occurs in the transfer proof.

R42.3 Actual orientation algorithm for every even residual of $size \geq 4$

Fix an original ordering of equal-length arms to choose their maximal pairing. Orient each real equal pair as a valid $f_l h_l$ pair or its complete reversal. Assign ALL root equal pairs to the inner factor H; terminal pairs remain in E_t at the original terminal. Positivity permits comparison while leaving the original edges and $e(U)$ unchanged.

R42.3.1 Minimum one: omit only the second arm from H

Write the actual root list as

$$(1, t, a, b, s_1, v_1, \ldots, s_k, v_k), \quad 1 < t < a < b < s_1 < v_1 < \cdots, \qquad k = j - 2 \geq 0. \tag{R42.3.1}$$

Retain the actual t arm fully inward, as $\bar{f}_t$. Every remaining root arm forms the actual odd list(1, a, b, s1, v1,...), with $a \geq 3$. Use EXACTLY C186 Section1 at that root and terminal. The root factor is

$$H = E_r p W_a \bar{f}_b \prod_i W_{s_i} \bar{f}_{v_i} T^L (E_t F^{186}_{\Sigma,L}), \qquad U = T^M(\bar{f}_t H). \tag{R42.3.2}$$

The C186 ordinary fourth gives $EH4 \leq \exp(-27d/5)$ for allk, L. Its later-pair improvement is optional; the base suffices. Apply (R42.1.4):

$$2(27/5) - 4247/684 = 15701/3420 > 0. \tag{R42.3.3}$$

This includes t2, a3, b4 and every finite post list. No minimum-four post restriction is imposed on the first actual pair $W_a \bar{f}_b$: C186 itself includes a3. The $\bar{f}_t$ arm is not inside H and is used exactly once outside.

R42.3.2 Minimum at least two: omit only the shortest arm from H

Write the original root list as

$$(s, a, b, c, s_1, v_1, \dots, s_k, v_k), \quad 2 \le s < a < b < c < s_1 < v_1 < \cdots, \quad k = j - 2 \ge 0. \quad \text{(R42.3.4)}$$

Retain the actual s arm fully inward, as $\bar{f}_s$. The remaining actual odd root(a, b, c, s1, v1,...) has $a \ge 3$ and falls exactly in C189 TheoremA. Define $Pplus = h_b f_c$ product $h_s i f_{v_i}$ and $Pminus = W_b \bar{f}_c$ product $W_s i \bar{f}_{v_i}$. The actual root/terminal choices are:

Table 127: The retained multiplier and every even internal root: parameter values.

Entry	Actual root factor inside H	Actual F
Exact(1, z), L1	$f_a Pminus$;	$p\bar{f}_z$.
Exact(1, z), $L \ge 2$	$W_a Pminus$;	qf_z.
Every other terminal	$f_a Pplus$;	The exact C189 table below.

In all cases $H = E_r(rootfactor)T^L(E_t F), U = T^M(\bar{f}_s H)$. C189's ordinary fourth is $EH4 \le \exp(-11d/2)$, including both special(1, z) rows and all equal pairs. The new output coefficient is

$$11 - 4247/684 = 3277/684 > 15701/3420. \quad \text{(R42.3.5)}$$

The special first root arm W_a is not changed to f_a by notation. The negative pairs in the two short-one rows are not changed to Pplus. The s arm is distinct from all displayed H arms and appears only as $\bar{f}_s$.

R42.3.3 The complete inherited terminal directions used in both rows

The following table specifies the terminal directions. All lists are strictly increasing and later products contain only displayed original pairs. The two ordinary-root bounds agree here except for the exact(1, z), L1 row described above.

Table 128: The retained multiplier and every even internal root: parameter values.

Entry	C186 F	C189 F
Exact(1, z), $z \ge 2$, L1	qf_z;	$p\bar{f}_z$.
Exact(1, z), $z \ge 2, L \ge 2$	qf_z;	qf_z.
Exact(2, 3)	h2 f3;	h2 f3.
Exact(2, z), $z \ge 4$	$\bar{f}_2 W_z$;	$\bar{f}_2 W_z$.
Exact(3, 4)	h3 f4;	h3 f4.
Exact(3, z), $z \ge 5$, L1	$h_3 f_z$;	$h_3 f_z$.
Exact(3, z), $z \ge 5, L \ge 2$	$\bar{f}_3 W_z$;	$\bar{f}_3 W_z$.
Single pair $minimum \ge 4$	$h_\alpha f_\beta$;	same.
Every even list with at least two pairs	$\prod h_{\alpha_i} f_{\beta_i}$;	same.
Exact(1, A, C), $2 \le A < C$	$qf_A h_C$;	same.

Entry	C186 F	C189 F
Unit-minimum odd multipair with(A1, C1, $A2) = (2, 3, 4)$	$qf_2h_{C_2}h_3f_4\prod_{i\geq 3}h_{A_i}f_{C_i}$;	same.
All other unit-minimum odd multipairs	$qf_{A_1}h_{C_1}\prod_{i\geq 2}h_{A_i}f_{C_i}$;	same.
Nonunit-minimum odd list, first paired $shortB1 = 3$	$\bar{f}_2f_3h_{C_1}\prod_{i\geq 2}h_{B_i}f_{C_i}$;	same.
Nonunit-minimum odd list, first paired $shortB1 \geq 4$	$\bar{f}_A\prod_i h_{B_i}f_{C_i}$;	same.

The first paired $shortB1 = 3$ forces $A = 2$. The crossed unit row requires at least five residual arms. Exactly three unit-minimum arms use the triple row. This gives the stated estimate for every finite nonreducible terminal. All inner connector edges are T and all outer connector edges T; the arm turns are exactly the h/W definitions.

R42.4 Complete unit-minimum even roots, including the pair boundary

An even residual starting at one either has at least four entries and uses (R42.3.1)-(R42.3.3), or is exactly(1, t), $t \geq 2$. In the latter case use the complete Q5-C106 ordinary table F_ord for its actual nonreducible terminal:

$$H = E_rpT^L(E_tF_{ord}), \quad EH8 \leq 1 - d/20000, \quad U = T^M(\bar{f}_tH).$$

The original C066 arbitrary-real eighth theorem yields

$$I_q(U)^3 \leq (1 - d/72)(1 - d/20000) \leq e^{-d/72}. \tag{R42.4.1}$$

The exact-pair boundary uses the eighth-moment theorem for the C106 rows. The structural branches have their stated directions. Together they prove the first assertion of TheoremC. TheoremB's nonunit minimum rows also have a much stronger output than exp(-d/72).

R.4 Path second moments and the mixed central estimate

R44.1 The local central estimates

All expectations use the finite measure and kernel of Section 2. This module proves the path second moment and the mixed estimate used at a center whose residual begins with a unit arm.

At each branching vertex, maximally pair equal side-arm lengths. A residual list consists of the remaining distinct lengths in increasing order. Each paired arm is still directed and retained. The central families treated here are exactly

$$(1, s), \quad s \geq 2, \tag{R44.1.2}$$

and

$$(1, s, a, b, u_1, v_1, \dots, u_k, v_k), \qquad 2 \leq s < a < b < u_1 < v_1 < \cdots < u_k < v_k, \quad k \geq 0. \tag{R44.1.3}$$

Together they are all nonempty even residual lists of minimum one.

For every $s \geq 2$, the actual outward path message satisfies

$$Ef_s^2 \leq 1.$$

For the central residual in (R44.1.3), set

$$Q = W_a \bar{f}_b \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}.$$

The same-label mixed estimate is

$$E[f_s^2 Q^8] \le e^{-17d/64}.$$

For (R44.1.2), take $Q = 1$ and use the path second moment. The strict ordering in (R44.1.3) includes $s = 2$ and $a = 3$. These are the local inputs to Table 9; the compound reference-second construction is proved in Proposition 9.3.

R44.2 Original finite calculus and genuine path moments

All operators are on the same original real L2(mu). Delete only zero-mass host atoms, which leaves every original integral unchanged. Put

$$B = -K, \quad u = B\mathbf{1}, \quad T = J + B, \quad A = J - B = T^*, \quad p = 1 - u, \quad q = 1 + u, \quad Jv = (Ev)1,$$

$$f_n = A^n \mathbf{1}, \quad \bar{f}_n = T^n \mathbf{1}, \quad h_n = T f_{n-1}, \quad W_n = A \bar{f}_{n-1}, \quad f_0 = \bar{f}_0 = 1. \tag{R44.2.1}$$

An f_n arm points entirely away from its root; $\bar{f}_n$ points entirely toward it; h_n has its first edge toward the root and every remaining edge away along the arm; W_n has its first edge away from the root and every remaining edge toward that first neighbor. In particular $f_1 = W_1 = p$ and $\bar{f}_1 = h_1 = q$. An incoming connector of length L has operator T^L, with every original edge directed toward the receiver.

Write

$$\delta_j = \|B^j \mathbf{1}\|_2^2, \quad d = \delta_1, \quad \theta = 13/32, \quad \kappa = 19/32, \quad \tau = d - \delta_2.$$

The exact C014 finite calculus gives

$$0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta \delta_j, \quad \|B\|^2 \le \theta, \quad |BV(x)| \le \|V\|_2, \quad Eu = 0, \tag{R44.2.2}$$

$$EB^{2j}1 = (-1)^j \delta_j, \qquad EB^{2j+1}1 = 0. \tag{R44.2.3}$$

In particular δ_2 is a spectral moment, whereas d^2 is the square of the first moment; they are never identified.

For $G_n(z) = \sum_{j=0}^{n-1} z^j$, with $G_0 = 0$, define

$$\beta_1 = \beta_2 = 0, \qquad \beta_n = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{j-1} \delta_j \quad (n \ge 3).$$

The exact finite positive convolution and its complete reverse are

$$S_n^+ = 1 + G_n(B)u = \bar{f}_n + \sum_{j=3}^{n} \beta_j \bar{f}_{n-j},$$

$$S_n^- = 1 - G_n(-B)u = f_n + \sum_{j=3}^{n} \beta_j f_{n-j}. \tag{R44.2.4}$$

Every coefficient $\beta_j, j \ge 3$, lies in [tau, d], and $\beta_3 = \beta_4 = d$. The sums are finite and empty at $n = 1$, 2. Their means are

$$ES_n^+ = ES_n^- = 1 - \beta_{n+1}. \tag{R44.2.5}$$

The positive path messages $f_n, \bar{f}_n$ lie in [0, 2] and have means at most one. The mean assertion is not extended to h_n or $W_n : Eh_2 = EW_2 = 1 + d$.

The finite geometric estimate needed below is

$$\|G_n(B)\| \le 1 \quad (n = 0, 1 \text{ or } n \ge 3), \qquad \|G_2(B)\|^2 \le 1 + \theta. \tag{R44.2.6}$$

Here is the finite calculation, including every integer n. On an original real skew rotation block put $z = i\sqrt{x}$, $0 \le x \le \theta < 1/2$. The exact finite geometric identity gives

$$|G_n(z)|^2 = \frac{1 - 2\,\mathrm{Re}(z^n) + x^n}{1 + x}.$$

For odd $n \ge 3, Re(z^n) = 0$ and $x^n \le x$. For n divisible by four, $x^n - 2x^{n/2} \le 0 \le x$. For n congruent to two modulo four and $n \ge 6$,

$$x^n + 2x^{n/2} \le x^6 + 2x^3 = x(x^5 + 2x^2) \le (17/32)x \le x.$$

The estimate uses only $x \le 1/2$; it therefore covers every original rotation block and the zero block. The remaining exact cases are $G_0 = 0, G_1 = 1$ and $|G_2(z)|^2 = 1 + x$. Orthogonal summation over the original finite blocks proves (R44.2.6), with no spectral limit.

Write $G_n(z) = E_n(x) + zD_n(x)$. For every $n \ge 1$,

$$0 \le E_n \le 1, \quad 0 \le D_n \le 1; \qquad E_1 = 1,\ D_1 = 0. \tag{R44.2.7}$$

For $n \ge 2$ the two coordinates lie in [1-x, 1]. These are finite alternating partial sums. No restriction $x \ge d$ is made.

Let nu be the original spectral measure of -B^2 at u. Its mass is d, its support is [0, theta], and int $x^j d\nu = \delta_{j+1}$. For a real polynomial $Q(z) = E(x) + zD(x)$,

$$\|Q(B)u\|^2 = \int (E^2 + xD^2)d\nu, \qquad E[Q(B)u] = -\int D\,d\nu, \qquad EB[Q(B)u] = -\int E\,d\nu. \tag{R44.2.8}$$

Every constant-vector cross below is computed on this same original space before any norm estimate.

R44.2.1 Complete ordinary second moments of the actual reference arms

For $s = 2$ the exact actual formula is

$$f_2 = 1 - u + Bu, \quad Ef_2 = 1 - d, \quad Ef_2^2 = 1 - d + \delta_2 \le 1 - \kappa d. \tag{R44.2.9}$$

Indeed, $\mathbb{E}Bu = -d$ and $\langle u, Bu\rangle = 0$.

For every $s \ge 3$, (R44.2.4) gives $0 \le f_s \le S_s^-$ and (R44.2.6) gives

$$E(S_s^- - 1)^2 = \|G_s(-B)u\|^2 \le d.$$

Consequently

$$Ef_s^2 \le E(S_s^-)^2 = 1 - 2\beta_{s+1} + \|G_s(-B)u\|^2 \le 1 - d + 2\delta_2 \le 1 - 3d/16. \tag{R44.2.10}$$

Thus $Ef_s^2 \le 1$ for every genuine $s \ge 2$, directly from the finite G and positive-convolution calculus. Only these finite bounds enter this estimate. The unit arm is different: $Ep^2 = 1 + d$.

R44.2.2 A uniform fourth bound for every genuine $s \geq 2$

For -1 $\leq y \leq 1$,

$$(1+y)^4 \leq 1 + 4y + 11y^2, \tag{R44.2.11}$$

because the difference is $y^2(1-y)(5+y) \geq 0$.

At $s = 2$, f_2 in [0, 2], $E(f_2 - 1) = -d$ and $E(f_2 - 1)^2 = d + \delta_2$. Therefore

$$Ef_2^4 \leq 1 + 7d + 11\delta_2 \leq 1 + \frac{367}{32}d \leq e^{367d/32}. \tag{R44.2.12}$$

At $s \geq 3$, put $g_s = min(2, S_s^-)$. This is only an analytical nonnegative majorant: the actual arm remains f_s. Positivity in (R44.2.4) and $f_s \leq 2$ give $0 \leq f_s \leq g_s \leq 2$. Moreover

$$E(g_s - 1) \leq -\beta_{s+1} \leq 0, \qquad E(g_s - 1)^2 \leq E(S_s^- - 1)^2 \leq d.$$

The second inequality is pointwise before integration: clipping S_s^- at two can only decrease its squared distance from one. Using (R44.2.11),

$$Ef_s^4 \leq Eg_s^4 \leq 1 - 4\beta_{s+1} + 11d \leq 1 + 11d \leq e^{367d/32}. \tag{R44.2.13}$$

This proves the uniform actual-path estimate

$$\boxed{Ef_s^4 \leq e^{367d/32} \quad (s \geq 2).} \tag{R44.2.14}$$

It holds for every finite s, including the genuine $s = 3/G_3$ case. There is no large-length limit and no arbitrary positive-function replacement of f_s.

R44.2.3 Equal original pairs

For each actual equal-length pair,

$$f_\ell + h_\ell = 2Ef_{\ell-1}, \qquad 0 \leq f_\ell h_\ell \leq (Ef_{\ell-1})^2 \leq 1. \tag{R44.2.15}$$

Complete reversal gives $0 \leq \bar{f}_\ell W_\ell \leq 1$. Products of these actual factors are denoted E_v at their original vertices. They can be compared with one only inside a nonnegative original integral or positive propagation.

R44.3 The full signed exponential estimate at exponent sixteen

On the original probability space, apply the signed-exponential theorem with $W = K(-V) = BV$, $U = \|V\|_2$, and $s = t$. The bounds in (R44.2.2) give $|W| \leq U$ and $EW^2 = \|BV\|_2^2 \leq \theta U^2$. Thus every real complete vector V, with either sign of its actual mean $E(BV)$, satisfies

$$Ee^{tBV} \leq \exp\{tEBV + (17/64)t^2\|V\|^2\}, \qquad V \text{ any real original vector}, \quad t \geq 0. \tag{R44.3.1}$$

Taking $t = 16$ gives $(17/64)16^2 = 68$, hence

$$\boxed{Ee^{16BV} \leq e^{16EBV + 68\|V\|^2}.} \tag{R44.3.4}$$

R44.4 New central mixed moment, with every short and finite-pair case

Let the genuine central lengths satisfy (R44.1.3), and put

$$Q = W_a \bar{f}_b \prod_{i=1}^{k} W_{u_i} \bar{f}_{v_i}. \tag{R44.4.1}$$

This directs each of the original central arms other than 1, s exactly once. The central unit arm will not be used anywhere in this section.

Central lemma. For the same actual f_s and Q in (R44.4.1),

$$\boxed{E[f_s^2 Q^8] \le e^{-17d/64}.} \tag{R44.4.2}$$

Arbitrary actual equal-pair factors may be restored by positivity, though the final assembly assigns all central equal pairs to its separate C106 component.

R44.4.1 The finite mask and the higher-pair envelope

For $4 \le c < e$ the full reversal of (L90.2.3) gives

$$W_c \bar{f}_e \le \exp\{BV_{c,e} - 16d/5\}, \qquad V_{c,e} = \sum_{j=c+1}^{e} B^{j-2} u. \tag{R44.4.3}$$

This is an actual original pair bound, also explicitly proved in C096 part R44.2.2 and preserved under complete reversal. Only this exact pair interface is needed from C186's larger terminal theorem.

For any finite polynomial P with coefficients zero or one, the four residue classes of its powers at i $\sqrt{x}$ give

$$|P(i\sqrt{x})|^2 \le \frac{1+x}{(1-x^2)^2} \le \frac{32768}{16245} < \frac{101}{50} =: C_*. \tag{R44.4.4}$$

For each coordinate, bound its positive and negative subcollections and take the larger absolute bound; do not add the two. Consequently a finite union of disjoint exponent intervals beginning at degree h has original vector norm squared at most $C_* \theta^h$ d. All intervals used below are disjoint because the original arm lengths are strictly increasing.

If V is such a union beginning at degree at least three, its positive terms in EBV begin at δ_4. Hence

$$EBV \le \frac{\delta_4}{1-\theta^2} < \frac{d}{12}, \qquad \|V\|^2 \le C_* \theta^3 d < \frac{9}{64} d. \tag{R44.4.5}$$

The first rational comparison is $\theta^3/(1-\theta^2) = 2197/27360 < 1/12$. The norm comparison is $221897 < 230400$ after the common denominator 1638400. These bounds do not grow with the number of intervals.

R44.4.2 Every $a \ge 4$, with every finite later pair block

There are $r = k+1$ actual pairs. Let V be their single full finite mask. Equations (R44.4.3)-(R44.4.5), followed by (R44.3.4), give

$$EQ^{16} \le \exp\left[-\left\{ \frac{256r}{5} - \frac{4}{3} - \frac{153}{16} \right\} d \right]. \tag{R44.4.6}$$

At $r = 1$ the coefficient is $9673/240 > 40$. Each further original pair adds $256/5$ to this coefficient, while the full-vector estimate already pays every cross term. Thus, by Cauchy on the original mu and the actual path bound (R44.2.14),

$$E[f_s^2 Q^8] \le (Ef_s^4)^{1/2} (EQ^{16})^{1/2} \le e^{-(40-367/32)d/2} \le e^{-17d/64}. \tag{R44.4.7}$$

R44.4.3 The actual $a = 3$ envelope, derived before any moment estimate

Here the strict order forces $s = 2$. Write $n = b - 3 \geq 1$ and

$$V_0 = B^2 G_n(B)u, \qquad \Gamma = 3d - 2\delta_2 = d + 2\tau.$$

The exact $W_3 + \bar{f}_3 = 2E\bar{f}_2 = 2 - 2d$ and (R44.2.4) give

$$W_3 + \bar{f}_b = 2 - d + \sum_{j=4}^{b} B^j \mathbf{1} - \sum_{j=3}^{b} \beta_j \bar{f}_{b-j}.$$

Retain the two genuine last terms $\beta_b \bar{f}_0$ and $\beta_{b-1}\bar{f}_1$, both with coefficient at least tau. Since $b \geq 4$, both indices are at least three. Their sum is at least tau(2+u). Therefore

$$W_3 + \bar{f}_b \leq 2 + BV_0 - \tau u - \Gamma.$$

The left side is the sum of two actual nonnegative messages. AM-GM and $1 + t \leq e^t$ imply

$$Q_0 := W_3 \bar{f}_b \leq \exp\{B(V_0 - \tau 1) - \Gamma\}. \tag{R44.4.8}$$

The vector is V_0 minus tau times the constant vector 1. It is not $V_0 - \tau$ u.

In (R44.2.7), write $G_n = E_n + zD_n$. Then V_0 has coordinates -$xE_n - zxD_n$. From (R44.2.8),

$$0 \leq EV_0 = \int xD_n \, d\nu \leq \delta_2, \qquad EBV_0 = \int xE_n \, d\nu \leq \delta_2. \tag{R44.4.9}$$

Thus the full constant square is

$$\|V_0 - \tau 1\|^2 = \|V_0\|^2 + \tau^2 - 2\tau EV_0. \tag{R44.4.10}$$

Every following estimate starts with (R44.4.10). Its cross is not discarded when needed in the genuine $n = 2$ exception.

R44.4.4 No later pairs, low domain $0 < d \leq 1/16$: n not equal to two

If $n = 1$ or $n \geq 3$, (R44.2.6) gives $\|V_0\|^2 \leq \delta_3$. By (R44.4.9)-(R44.4.10),

$$\|V_0 - \tau 1\|^2 \leq \theta\delta_2 + \tau^2.$$

The signed mean in (R44.3.4) satisfies $EB(V_0 - tau1) = EBV_0 \leq \delta_2$, since $Eu = 0$. Put $t = \delta_2/d$ only after these original integrated inequalities. Then

$$\frac{\log EQ_0^{16}}{d} \leq -48 + (48 + 68\theta)t + 68d(1-t)^2. \tag{R44.4.11}$$

For every fixed $d \leq 1/16$ its derivative in t is at least $48+68\theta-136/16 > 0$ on [0, theta]. Its maximum is therefore at $t = \theta$. The remaining d coefficient is nonnegative, so use $d = 1/16$. The resulting loss is exactly

$$48\kappa - 68\theta^2 - \frac{17}{4}\kappa^2 = \frac{64631}{4096} > 15. \tag{R44.4.12}$$

This handles $b = 4$ and every $b \geq 6$.

R44.4.5 No later pairs, low domain: the genuine $n = 2, b = 5$ exception

Now $G_2 = 1 + B$, so it must not be treated as a contraction:

$$V_0 = B^2u + B^3u, \quad EV_0 = \delta_2, \quad \|V_0\|^2 = \delta_3 + \delta_4.$$

The cross between consecutive B powers vanishes by skewness. The nonzero constant cross in (R44.4.10) is exactly -2tau δ_2. Hence

$$\|V_0 - \tau 1\|^2 \le \theta(1+\theta)\delta_2 + (d - \delta_2)(d - 3\delta_2). \tag{R44.4.13}$$

This expression comes from the same actual vector; a potentially negative second summand is kept with its sign. Equations (R44.3.4) and (R44.4.8) give

$$\frac{\log EQ_0^{16}}{d} \le -48 + \{48 + 68\theta(1+\theta)\}t + 68d(1-t)(1-3t). \tag{R44.4.14}$$

For fixed $d \le 1/16$ the derivative in t is at least $48 + 68\theta(1+\theta) - 17 > 0$. At $t = \theta$ the last term is nonpositive, because $1 - 3\theta = -7/32$. Dropping only that nonpositive term yields the loss

$$48\kappa - 68\theta^2(1+\theta) = \frac{104187}{8192} > 12. \tag{R44.4.15}$$

Consequently (R44.4.11)-(R44.4.15) prove, for every $b \ge 4$,

$$EQ_0^{16} \le e^{-12d}, \qquad 0 < d \le 1/16. \tag{R44.4.16}$$

Since the reference is the actual f_2, Cauchy and (R44.2.12) give

$$E[f_2^2 Q_0^8] \le e^{-(12-367/32)d/2} = e^{-17d/64}. \tag{R44.4.17}$$

R44.4.6 No later pairs, high domain $1/16 \le d \le 1/3$

The actual direction (R44.4.8) is unchanged. From (R44.2.6), (R44.4.9) and (R44.4.10),

$$\|V_0 - \tau 1\|^2 \le (1+\theta)\delta_3 + \tau^2 \le (1+\theta)\theta^2 d + d^2. \tag{R44.4.18}$$

Also $\Gamma \ge 35d/16$. For $d \ge 1/16$,

$$(1+\theta)\theta^2 = \frac{7605}{32768} < \frac{7752}{32768} = \frac{969}{256}\frac{1}{16} \le \frac{969}{256}d.$$

Adding d^2 in (R44.4.18) therefore gives

$$\|V_0 - \tau 1\|^2 \le (35d/16)^2 \le \Gamma^2.$$

The original row bound in (R44.2.2), applied to the full real vector, shows $B(V_0 - tau1) \le \Gamma$ pointwise. Thus

$$Q_0 \le 1, \qquad E[f_2^2 Q_0^8] \le Ef_2^2 \le 1 - \kappa d \le e^{-\kappa d} \le e^{-17d/64}. \tag{R44.4.19}$$

This pointwise conclusion is proved only on its stated high domain. Both proofs include $d = 1/16$ and use the same literal $W_3 \bar{f}_b$ direction.

R44.4.7 $a = 3$ **with every nonempty finite later pair block**

Let $k \geq 1$. The original order gives $u_1 > b \geq 4$, hence $u_1 \geq 5$. Let

$$V_* = \sum_{i=1}^{k} \sum_{j=u_i+1}^{v_i} B^{j-2}u, \qquad V = V_0 + V_*.$$

The entire set of powers in V is one disjoint finite zero-one mask beginning at degree two. Therefore

$$\|V\|^2 \leq C_*\theta^2 d = \frac{17069}{51200}d < \frac{9}{25}d. \tag{R44.4.20}$$

For the later vector alone, the positive terms in EBV_* begin at δ_4, while the negative terms in EV_* begin at -δ_3. Thus, for the actual finite subcollections,

$$EBV_* \leq \frac{\delta_4}{1-\theta^2} < d/12, \qquad EV_* \geq -\frac{\delta_3}{1-\theta^2} > -d/5. \tag{R44.4.21}$$

The second rational comparison is $\theta^2/(1-\theta^2)$ =$169/855 < 1/5$. Combined with $EV_0 \geq 0$, these imply

$$\|V - \tau 1\|^2 = \|V\|^2 + \tau^2 - 2\tau EV \leq \frac{9}{25}d + \frac{7}{5}d^2 \leq \frac{62}{75}d. \tag{R44.4.22}$$

The bound includes the root-pair cross terms and the constant-vector cross term.

Using (R44.4.3), (R44.4.8) and (R44.4.21),

$$Q \leq \exp\{B(V - \tau 1) - \Gamma - 16kd/5\},$$

$$\Gamma + 16kd/5 - EB(V - \tau 1) \geq \{57/32 + 16k/5 - 1/12\}d. \tag{R44.4.23}$$

At $k = 1$, sixteen times the last coefficient is $2351/30$. The entire variance cost in (R44.3.4) is at most $68(62/75) = 4216/75$. Their exact difference is

$$\frac{2351}{30} - \frac{4216}{75} = \frac{3323}{150} > 22.$$

Every further original pair adds $256/5$. Hence, on the whole original domain without a d split,

$$EQ^{16} \leq \exp[-\{3323/150 + 256(k-1)/5\}d] \leq e^{-22d}. \tag{R44.4.24}$$

Cauchy with (R44.2.12) again proves (R44.4.2), with substantial room. There is no upper bound on k or on any original length.

Parts R44.4.2 and 4.3-4.7 treat $a \geq 4$ and $a = 3$, respectively. At $d = 0$ the actual messages all equal one, by leaf integration. This completes the new central lemma.

R.5 Finite geometric cancellation and every nonunit singleton at three outer edges

R45.1 Original model and exact statements

Work on the original finite probability space after deleting zero-mass atoms, with a real skew kernel K satisfying $|K(x, y)| \leq 1$. All norms, inner products, adjoints and expectations use the same original measure. Write

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}, \quad \delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1.$$

In particular p(x), $q(x) \geq \mu(x) > 0$. A finite actual tree message is positive because the all-equal labeling contributes positive original mass. Define

$$I_q(U) = E[q^{-1/3}|U|^{8/3}], \qquad B_n = E[\bar{f}_n^4/q],$$

and let M_v be multiplication by the actual function v. The supplied finite calculus gives

$$\theta = 13/32, \quad \kappa = 19/32, \quad 0 \leq d \leq 1/3, \quad d^2 \leq \delta_2 \leq \theta d, \quad \delta_{j+1} \leq \theta\delta_j, \quad \|K\|^2 \leq \theta. \tag{R45.1.1}$$

Theorem A: exact improved bare-row bounds. For every integer $n \geq 4$,

$$B_n \leq \begin{cases} 1 - 7d/4, & 0 \leq d \leq 2/15, \\ 1 - 3d/2, & 0 \leq d \leq 7/40, \\ 1 - 29d/20, & 0 \leq d \leq 1/5. \end{cases} \tag{R45.1.2}$$

The overlapping bounds are all valid on their stated domains for the same actual n-edge path. Section3 also proves a single explicit continuous bound on the entire original domain, before these convenient relaxations. For $n \geq 5$ there is the separate all-domain estimate

$$B_n \leq 8(1 - 2d)(1 - 5d/2)^3. \tag{R45.1.3}$$

The restriction $n \geq 5$ in (R45.1.3) is essential to the present proof.

Theorem B: every nonunit singleton, outer length three. Suppose the actual internal vertex has, after maximal genuine equal-length pairing, exactly one residual arm of length $a \geq 2$. Its actual terminal has any strict residual list of size at least two. Let $L \geq 1$ be the inner original connector length. Include every original root and terminal equal pair. Section5 prescribes a direction depending only on these integers. For outer length $M = 3$, its actual message into the original center satisfies

$$\boxed{I_q(U)^3 \leq e^{-d/80}.} \tag{R45.1.4}$$

There is no relation required between a, L and the terminal lengths and no ordinary-eighth input assumption.

Theorem C: precise union and original P5 consequence. The E008 construction and Theorem B together give, for every such half-tree,

$$I_q(U)^3 \leq e^{-\eta_M d}, \qquad \eta_M = \begin{cases} 1/2, & M = 1, \\ 1/80, & M = 3, \\ 1/2 + \lfloor (M-1)/3 \rfloor/2, & M \geq 4. \end{cases} \tag{R45.1.5}$$

A P5 with both noncentral internal residuals nonunit singletons and both outer lengths different from two therefore has an original structure-only orientation with $Z \leq 1$ for arbitrary terminal and central multisets, and both independent positive inner lengths. When both ends are nonreducible and the center has a nonempty R residual, the common strict bound is $Z \leq \exp(-d/320)$. Sections 11 and 12 give the comparison branches and the separately qualified mixed use.

R45.2 The finite geometric identity which sharpens the C100 row

Define

$$G_m(z) = \sum_{j=0}^{m-1} z^j, \qquad E_r(x) = \sum_{j=0}^{r-1} (-x)^j \quad (r \geq 1).$$

For x in [0, theta], $1-x \leq E_r(x) \leq 1$. All sums here are finite. At the original spectral block $z = i\sqrt{x}$, write

$$G_m(z) = A_m(x) + zD_m(x).$$

The letter D_m is a scalar polynomial here, not an orientation. Fix $n \geq 4$ and $m = n-1 \geq 3$. In the actual C100 notation set

$$R = KG_m(-K)g, \quad h = E[gR], \quad V = ER^2.$$

Let ν_g be the actual finite spectral measure of -K^2 at g. Its mass is d and its first moment is δ_2. The exact identities are

$$h = \int x D_m(x)\, d\nu_g(x), \quad V = \int x\{A_m(x)^2 + xD_m(x)^2\}\, d\nu_g(x). \tag{R45.2.1}$$

This is the same actual R and h as in C100; it is not an independent replacement vector. We now retain cancellation in V-h.

If $m = 2r$ is even, then $r \geq 2$ and $A_m = D_m = E_r$. Hence

$$x(A_m^2 + xD_m^2 - D_m) = (-1)^{r+1} x^{r+1} E_r. \tag{R45.2.2}$$

Indeed $(1+x)E_r = 1 - (-x)^r$. When the right side is positive, r is odd and $r \geq 3$. It is then at most x^4, which is at most $x^3(1-x+x^2)$, since $1 - x + x^2 - x = (1-x)^2 \geq 0$. Negative cases already satisfy the same upper bound.

If $m = 2r+1$ is odd, $r \geq 1$, then $D_m = E_r$ and $A_m = E_{r+1} = 1 - xE_r$. Direct substitution gives

$$A_m^2 + xD_m^2 - D_m = (-x)^r A_m. \tag{R45.2.3}$$

Only even $r \geq 2$ gives a positive right side. In that case

$$A_m = \frac{1+x^{r+1}}{1+x} \leq \frac{1+x^3}{1+x} = 1 - x + x^2,$$

and $x^{r+1} \leq x^3$. Thus every $m \geq 3$, including m3 and m4, obeys

$$x(A_m^2 + xD_m^2 - D_m) \leq x^3(1 - x + x^2). \tag{R45.2.4}$$

The derivative of $x^2(1-x+x^2)$ is $x(2-3x+4x^2) \geq 0$: the quadratic has negative discriminant and positive leading coefficient. Consequently

$$\boxed{V - h \leq r_* \delta_2, \qquad r_* = \theta^2(1 - \theta + \theta^2) = \frac{131313}{1048576}.} \tag{R45.2.5}$$

This replaces the weaker C100 bound $V - h \leq \theta\delta_2$. The estimate is uniform over all finite $m \geq 3$. It makes no assumption $x \geq d$ and uses no infinite geometric series. Useful exact scalar comparisons are

$$\frac{1}{8} < r_* < \frac{13}{100}, \qquad \theta r_* = \frac{1707069}{33554432} < \frac{51}{1000}. \tag{R45.2.6}$$

For the last comparison, $1707069000 < 1711276032$. Put

$$\delta_* := \theta^2 - 51/1000 = \frac{14597}{128000} > \frac{9}{80}. \tag{R45.2.7}$$

It follows that $\theta(\theta - r_*) > \delta_*$.

R45.3 Same-row fourth expansion and a continuous explicit bound

This section reuses the exact C100 calculation only through its given identities. To avoid confusing an arm length with a scalar, put $\alpha = \kappa$ d, $C = 1-\alpha, t = 1+d/2$. The two positive finite-convolution upper bounds on the SAME actual path are

$$\bar{f}_n \le Cq + R - \alpha, \qquad \bar{f}_n \le 2Cq. \tag{R45.3.1}$$

They hold for every $n \ge 4$. The second retains the actual mean of $\bar{f}_{n-2}$. For $s \ge -1$,

$$\min(1+s,2)^4 \le 1 + 4s + 7s^2 + 4s^3.$$

Applying this at $s = (R-\alpha)/(Cq)$, then retaining the correlated cubic through $W = 2R - Cg$, gives the exact C100(2.10):

$$\begin{aligned} B_n \le\ & C^4(1+3d) + C^3[td - 6\alpha d - 8\alpha] + C\alpha^2[19 - d - 33\alpha] \\ & + CB(d)(V-h) + C[-1 + (13+d)\alpha - 30\alpha^2]h - C^2E(gR^2), \end{aligned} \tag{R45.3.2}$$

where

$$B(d) = 9 - \frac{481}{32}d > 0.$$

Here B(d) is a scalar coefficient; it is unrelated to the index in B_n. The beta monotonicity, cubic signs and coefficient collection behind (R45.3.2) are unchanged Q5-C100 steps, given.

The genuine h bounds are kappa $\delta_2 \le h \le \delta_2$. Separating the three terms of its coefficient by sign, as C100 does, gives

$$\begin{aligned} [-1 + (13+d)\alpha - 30\alpha^2]h &\le H(d)\delta_2, \\ H(d) &= -\kappa + 13\kappa d + \kappa(1 - 30\kappa^2)d^2. \end{aligned} \tag{R45.3.3}$$

After (R45.2.5), the combined coefficient of C δ_2 is now $S_n ew(d) = r_* B(d) + H(d)$. It is positive throughout [0, 1/3]. Indeed

$$9r_* - \kappa > 17/32, \quad 13\kappa - (481/32)r_* > 5, \quad \kappa(1-30\kappa^2) = -93157/16384 > -6,$$

so $S_n ew(d) > 17/32 + 5d - 6d^2 > 0$. Only AFTER this sign check may we replace the combined δ_2 by theta d.

The exact old C100 polynomial was

$$\begin{aligned} P(d) = 1 &- \frac{963}{512}d + \frac{171113}{32768}d^2 - \frac{7275575}{1048576}d^3 \\ &+ \frac{14103187}{16777216}d^4 + \frac{1063145}{1048576}d^5. \end{aligned}$$

The only changed term is therefore exactly -C B(d) $\theta(\theta - r_*)d$. The unchanged actual correlation satisfies $|E(gR^2)| \le (2/3)d^{3/2}$. Thus

$$B_n \le P(d) + \frac{2}{3}C^2 d^{3/2} - CB(d)\theta(\theta - r_*)d. \tag{R45.3.4}$$

The already proved C100 rational comparison, valid on all $d \le 1/3$, is

$$(P(d) - 1)/d \le -963/512 + (21/4)d - (32/5)d^2. \tag{R45.3.5}$$

It follows from $A2 < 21/4, A3 < -69/10, A4 < 1, A5 < 33/32$ and $d^3 + (33/32)d^4 \le (43/96)d^2$.

Combining (R45.2.7),(R45.3.4),(R45.3.5), for $d > 0$ we obtain the claimed continuous same-row bound

$$\boxed{B_n \le 1 - s(d)d,}$$

$$s(d) = \frac{963}{512} - \frac{21}{4}d + \frac{32}{5}d^2 - \frac{2}{3}(1-\kappa d)^2\sqrt{d} + \delta_*(1-\kappa d)\left(9 - \frac{481}{32}d\right). \tag{R45.3.6}$$

It extends continuously to $d = 0$. This is an analytic formula, not a finite scan. It holds for every $n \ge 4$ without changing the actual path.

The function s is decreasing on [0, $1/3$]. The first polynomial's derivative is at most -$59/60$. The product C B(d) is decreasing with both factors positive. Finally $C^2\sqrt{d}$ is increasing: for $d > 0$ its derivative has the sign of C(1-5kappa d), and $1 - 5\kappa/3 = 1/96 > 0$. These three signs prove monotonicity on the whole interval, including the continuous endpoint at zero.

R45.4 Exact endpoint payments and the higher-domain row

It now suffices to lower-bound s at three exact rational endpoints. Evaluate the monotone analytic formula at those endpoints. Write $F(d) = 963/512 - (21/4)d + (32/5)d^2$.

R45.4.1 Endpoint $d = 2/15$

Here $C = 221/240 > 23/25$ and $B = 1679/240 > 69/10$. Therefore

$$\delta_* CB > \frac{9}{80}\frac{23}{25}\frac{69}{10} = \frac{14283}{20000} > 7/10.$$

Also $F(2/15) = 963/512 - 7/10 + 128/1125 > 129/100$: use $963/512 > 47/25$ and $128/1125 > 11/100$. Since $C^2 = 48841/57600 < 17/20$ and $\sqrt{2/15} < 11/30$,

$$\frac{2}{3}C^2\sqrt{2/15} < \frac{187}{900}.$$

Consequently

$$s(2/15) > 129/100 + 7/10 - 187/900 = 401/225 > 7/4. \tag{R45.4.1}$$

R45.4.2 Endpoint $d = 7/40$

Here $C = 1147/1280 > 89/100$ and $B = 8153/1280 > 63/10$. Hence

$$\delta_* CB > \frac{9}{80}\frac{89}{100}\frac{63}{10} = \frac{50463}{80000} > 5/8.$$

The exact F is $963/512 - 147/160 + 49/250 > 23/20$, by $963/512 > 47/25, 147/160 < 23/25$, $49/250 > 19/100$. Furthermore $C < 9/10$ and $\sqrt{7/40} < 21/50$, so

$$\frac{2}{3}C^2\sqrt{7/40} < \frac{567}{2500} < 11/40.$$

It follows that

$$s(7/40) > 23/20 + 5/8 - 567/2500 > 3/2. \tag{R45.4.2}$$

R45.4.3 Endpoint $d = 1/5$

Now $C = 141/160, B = 959/160$ and $CB = 135219/25600$. The retained gain is greater than $3/5$, because

$$14597 \cdot 135219 = 1973791743 > 1966080000 = \frac{3}{5}(128000 \cdot 25600).$$

The remaining exact comparisons are

$$F(1/5) = \frac{69559}{64000}, \quad \sqrt{1/5} < 9/20, \quad \frac{2}{3}C^2\sqrt{1/5} < \frac{59643}{256000}.$$

Thus

$$s(1/5) > \frac{372193}{256000} > \frac{371200}{256000} = 29/20. \tag{R45.4.3}$$

Equations (R45.4.1)-(R45.4.3) and monotonicity prove every line of (R45.1.2).

R45.4.4 The original finite path-mean bound

For clarity, the exact finite argument used on the other domain is included. Define $\beta_1 = \beta_2 = 0$ and

$$\beta_j = \sum_{r=1}^{\lfloor (j-1)/2 \rfloor} (-1)^{r-1}\delta_r, \qquad \tau = d - \delta_2.$$

The actual finite positive convolution is

$$\sum_{r=0}^{n}(-K)^r 1 = \bar{f}_n + \sum_{j=3}^{n}\beta_j \bar{f}_{n-j}, \qquad \tau \le \beta_j \le d \quad (j \ge 3). \tag{R45.4.4}$$

Its exact means $\mu_n = E\bar{f}_n = Ef_n$ satisfy

$$\begin{aligned}
&\mu_2 = 1 - d, && \mu_3 = 1 - 2d,\\
&\mu_4 = 1 - 3d + \delta_2, && \mu_5 = 1 - 4d + 2\delta_2 + d^2,\\
&\mu_6 = 1 - 5d + 3\delta_2 + 3d^2 - \delta_3,\\
&\mu_7 = 1 - 6d + 4\delta_2 + 6d^2 - 2\delta_3 - 2d\delta_2.
\end{aligned}$$

For n4, 5, 6, 7 their losses divided by d are at least respectively $83/32, 137/48, 89/32, 127/48$. These follow from $d \le 1/3$ and $\delta_2 \le \theta$ d, retaining the displayed positive losses. For example,

$$6d - (4 - 2d)\delta_2 - 6d^2 \ge [6 - 4\theta - (6 - 2\theta)d]d \ge 127d/48.$$

For $n \ge 8$, take the expectation in (R45.4.4) and retain the six actual terms corresponding to $\mu_0, ..., \mu_5$, together with β_{n+1}. The three coefficients $\beta_{n+1}, \beta_n, \beta_{n-1}$ are at least tau+kappa δ_3. This gives

$$\begin{aligned}
1 - \mu_n &\ge \tau(7 - 10d + 3\delta_2 + d^2) + 3\kappa\delta_3\\
&\ge \kappa d(7 - 10d + 7d^2) \ge 95d/36 > 83d/32.
\end{aligned} \tag{R45.4.5}$$

For the middle step use $\delta_2 \ge d^2$ and $\delta_3 \ge d\delta_2 \ge d^3$; the latter is the Cauchy moment inequality $\delta_2^2 \le d\delta_3$ combined with $\delta_2 \ge d^2$. The bracket is positive. The final quadratic decreases on [0, $1/3$]. Hence

$$\mu_j \le 1 - 2d \ (j \ge 3), \qquad \mu_j \le 1 - 83d/32 \le 1 - 5d/2 \ (j \ge 4). \tag{R45.4.6}$$

If $n \geq 5$, the positive original kernel gives $Tv \leq 2Ev$ and $T^2v \leq 2(Ev)q$ for $v \geq 0$. Thus, for the SAME path,

$$\bar{f}_n \leq 2(1-2d)q, \quad \bar{f}_n \leq 2(1-5d/2), \quad E\bar{f}_n \leq 1-5d/2.$$

All right sides are nonnegative for $d \leq 1/3$. Consequently

$$B_n \leq 2(1-2d)E\bar{f}_n^3 \leq 8(1-2d)(1-5d/2)^3,$$

which is (R45.1.3). The first mean is μ_{n-2}; it is exactly mu3 when $n = 5$. The latter two require indices at least four. No n4 endpoint is hidden in this argument.

R45.5 Every actual terminal and the original weighted input

Theorem B uses the E008 Section10 construction. The complete outward-reference table is reproduced to make its integer-only choice explicit. Canonical pairs mean $h_s f_t$ with $s < t$; $h_1 = q$. Every genuine terminal equal-pair decoration stays inside its original connector. The full outward connector is T^L.

Table 129: Finite geometric cancellation and every nonunit singleton at three outer edges: parameter values.

Entry	Actual outward-reference $F_r ef$
Exact (1, b)	qf_b.
Exact (2, b)	$h_2 f_b$.
Exact (3, b), $L = 1$ or2	$f_3\bar{f}_b$.
Exact (3, b), $L \geq 3$	$h_3 f_b$.
Exact pair of $minimum \geq 4$; every even multipair	Product of all canonical $h_s f_t$.
Odd $minimum \geq 2$, second $length \geq 4$	$\bar{f}_a$ times all later canonical pairs.
Odd prefix (2, 3, c)	$\bar{f}_2 f_3 h_c times later canonical pairs.$
Exact (1, s, t), $s \geq 4$	$qf_s h_t$.
Unit-minimum odd, $second \geq 4$, at least five arms	q times all later canonical pairs.
Unit-minimum odd prefix (1, 3, c)	$qf_3 h_c times later canonical pairs.$
Exact (1, 2, 3), $L = 1$	$qf_2\bar{f}_3$.
Exact (1, 2, 3), $L \geq 2$	$qf_2 h_3$.
Exact (1, 2, c), $c \geq 4$	$qf_2 h_c$.
Odd prefix (1, 2, 3, 4, t2), at least five arms	$qf_2 h_{t_2} h_3 f_4 times later pairs.$
Other odd prefix (1, 2, c), at least five arms	$qf_2 h_c times later pairs.$

In the odd $minimum \geq 2$ row, a denotes the terminal minimum; the singleton arm length is a separate parameter. The first pair in a five-or-more-arm unit prefix is displayed explicitly; later pairs are the consecutive strict residual pairs left after those displayed arms. The rows are exhaustive by parity, minimum, next integer and the indicated length exception. The exact unit123 exception is L1 for this reference role; it is not C106's different ordinary-eighth L2 exception.

Fully reverse this reference construction, including the actual arm, the L connector, every terminal arm, and both equal-pair decorations. Superscript minus means precisely this full reversal. Put

$$X_0 = A^L(E_t^- F_{\text{ref}}^-), \qquad X = E_r^- X_0,$$

$$Q_a(d) = \begin{cases} 1, & a = 2, \\ 1 - d/3000, & a \geq 3. \end{cases}$$

The reference theorem and $0 < E_r^- \leq 1$ give

$$E[\bar{f}_a X^4] \leq E[E_r^- \bar{f}_a X_0^4] \leq Q_a(d) \leq 1. \tag{R45.5.1}$$

In particular, qf_b becomes $p\bar{f}_b$, and the short $f_3\bar{f}_b$ row becomes $\bar{f}_3 f_b$. The actual singleton arm points toward its internal root and has message $\bar{f}_a$. The inner connector points from that root toward its terminal. The outer connector has $M = 3$ edges from the root toward the central receiver. Its message is exactly

$$U = T^3(\bar{f}_a X). \tag{R45.5.2}$$

The description retains every original edge in its displayed operator order.

R45.5.1 The weighted bridge and positive interpolation

For every real X and every M, actual row Cauchy gives

$$|T^M(\bar{f}_a X)|^2 \leq \bar{f}_{M+a}\, T^M(\bar{f}_a |X|^2).$$

Original-measure Holder, then ordinary L2 norm factorization, yield

$$\begin{aligned} I_q(T^M(\bar{f}_a X))^3 &\leq B_{M+a}\{E[T^M(\bar{f}_a |X|^2)^2]\}^2 \\ &\leq B_{M+a}\|T^M M_{\sqrt{\bar{f}_a}}\|^4 (E[\bar{f}_a |X|^4])^2. \end{aligned} \tag{R45.5.3}$$

The vector inside the norm is $(T^M M_{\sqrt{\bar{f}_a}})(\sqrt{\bar{f}_a}|X|^2)$. This is the weighted input and its matching operator, not an ordinary-fourth substitution into a different multiplier theorem.

For any positive original operator P and $w \geq 0$, row Cauchy gives $|P(\sqrt{w}v)|^2 \leq (P|v|)(P(w|v|))$. Integration and L2 Cauchy therefore prove

$$\|PM_{\sqrt{w}}\|^2 \leq \|P\|\,\|PM_w\|. \tag{R45.5.4}$$

Use $P = T^3, w = \bar{f}_a$. The full-multiplier theorem is $\|\text{T } M_{\bar{f}_a}\|^2 \leq 1 - d/72$ for every $a \geq 2$. Factor it in the displayed order: $T^3 M_{\bar{f}_a} = T^2(T M_{\bar{f}_a})$. The separate original norm bounds are

$$\|T^3\| \leq 1 - d/8, \qquad \|T^2\| \leq (1 + (1024/855)\delta_2)^2.$$

Squaring (R45.5.4) consequently gives

$$\|T^3 M_{\sqrt{\bar{f}_a}}\|^4 \leq (1 - d/8)^2 (1 + (1024/855)\delta_2)^4 (1 - d/72). \tag{R45.5.5}$$

Set $c = 416/855$, so $(1024/855)\delta_2 \leq c$ d and $12/25 < c < 1/2$. For the actual $n = a + 3 \geq 5$, (R45.5.1)-(R45.5.5) imply

$$I_q(U)^3 \leq B_n (1 - d/8)^2 (1 + cd)^4 (1 - d/72) Q_a(d)^2. \tag{R45.5.6}$$

We keep the genuine δ_2 until this justified reduction. The remaining proof pays exactly (R45.5.6), with $Q_a \leq 1$.

R45.6 Complete continuous payment for every M3 output

The basic logarithmic inequalities, valid on the indicated positive domains, are

$$\log(1-u) \le -u - u^2/2 \quad (0 \le u < 1),$$

$$\log(1+z) \le z - \frac{z^2}{2(1+z)} \quad (z \ge 0). \tag{R45.6.1}$$

The second follows by writing $z - \log(1+z) = \int_0^z$ t/(1+t)dt and replacing the denominator by 1+z. Since $d \le 1/3$ and $c < 1/2$,

$$4\log(1+cd) \le 4cd - \frac{3}{8}d^2. \tag{R45.6.2}$$

Indeed the exact quadratic coefficient is at least $2c^2/(1+c/3) > (12/7)(12/25)^2 = 1728/4375 > 3/8$.

R45.6.1 Domain $0 < d \le 2/15$

Use $B_n \le 1-7d/4$ and the linear logarithmic bounds. The logarithm of the right side of (R45.5.6), with $Q_a \le 1$, is at most

$$(-7/4 - 1/4 - 1/72 + 4c)d \le -d/72 \le -d/80. \tag{R45.6.3}$$

This uses $4c < 2$. All factors inside the logarithms are positive.

R45.6.2 Domain $2/15 \le d \le 7/40$

Use $B_n \le 1 - 3d/2$, retain its quadratic term $9d^2/8$, and use (R45.6.2). The linear coefficient satisfies

$$4c - 7/4 - 1/72 < 3/16,$$

since $1664 * 144 = 239616 < 240255 = 281 * 855$. The logarithm is at most

$$\frac{3}{16}d - \frac{3}{2}d^2 \le (3/16 - 1/5)d = -d/80. \tag{R45.6.4}$$

R45.6.3 Domain $7/40 \le d \le 1/5$

Use $B_n \le 1-29d/20$. Its quadratic contribution is $841d^2/800$. Together with (R45.6.2) this gives total quadratic loss $1141d^2/800$. The linear coefficient satisfies

$$4c - 1/4 - 1/72 - 29/20 < 15/64,$$

because $1/4 + 1/72 + 29/20 = 617/360$ and $1664 * 2880 = 4792320 < 4797405 = 5611 * 855$. Thus the logarithm is at most

$$\left(\frac{15}{64} - \frac{1141}{800}\frac{7}{40}\right)d = -\frac{487}{32000}d \le -d/80. \tag{R45.6.5}$$

R45.6.4 Domain $1/5 \le d \le 1/3$

Here use (R45.1.3) and $c < 1/2$ in (R45.5.6), and discard only factors already bounded above by one. The right side is at most

$$F_*(d) = 8(1-2d)(1-5d/2)^3(1+d/2)^4. \tag{R45.6.6}$$

Every factor is positive. Its logarithmic derivative is

$$-\frac{2}{1-2d} - \frac{15/2}{1-5d/2} + \frac{2}{1+d/2} < 0.$$

The first and last terms have nonpositive sum, and the middle term is negative. Hence

$$F_*(d) \le F_*(1/5) = \frac{43923}{50000} < \frac{9}{10} < \frac{239}{240} \le 1 - d/80 \le e^{-d/80}. \tag{R45.6.7}$$

The four domains are exhaustive. They all use the same X, the same outer T^3, the same singleton arm, and the same original measure. This proves Theorem B for every $d > 0$.

At $d = 0, K1 = 0$ on positive support; it does not follow that $K = 0$. Original leaf integration gives every actual tree message one, so $U = q = 1$ and $Iq(U)^3 = 1$. Thus the zero endpoint is also proved.

R.6 A genuine two-edge singleton at two outer edges

R47.1 Original model and exact new statements

The original target is an orientation of every edge of every finite tree with at most five vertices of original degree at least three, chosen from the tree before the host, with original partition at most one for every finite original probability and every real skew kernel bounded by one. This section proves the following half-tree estimate, retaining all original vertices, subdivisions, arms and their repetitions, and the independent original core lengths.

Work on the unchanged original finite probability space, after deleting only zero-mass host atoms. Put

$$Jv = (Ev)1, \quad A = J + K, \quad T = J - K = A^*, \quad g = K1, \quad p = 1 + g, \quad q = 1 - g,$$

$$f_n = A^n\mathbf{1}, \quad \bar{f}_n = T^n\mathbf{1}, \quad h_n = Tf_{n-1}, \quad W_n = A\bar{f}_{n-1}, \qquad \delta_j = \|K^j\mathbf{1}\|_2^2, \quad d = \delta_1.$$

All expectations, norms, adjoints and spectral measures use this same μ. The finite calculus is

$$\theta = 13/32, \quad \kappa = 19/32, \quad 0 \le d \le 1/3, \quad d^2 \le \delta_2 \le \theta d, \quad \delta_{j+1} \le \theta\delta_j, \quad \|K\|^2 \le \theta, \quad |Kv(x)| \le \|v\|_2. \tag{R47.1.1}$$

Moment Cauchy also gives $\delta_2^2 \le d\delta_3$. Define

$$I_q(U) = E[q^{-1/3}|U|^{8/3}], \qquad B_n = E[\bar{f}_n^4/q], \qquad N(d) = (1 + d/2)^4(1 + 32d/19)^2. \tag{R47.1.2}$$

At each retained atom, $p, q \ge \mu(x) > 0$. An actual finite rooted-tree message is positive, because the all-equal labeling contributes positive mass.

Theorem A — exact original weighted two-step transfer. On every such original host,

$$\boxed{B_4 N(d) \le 1 - d/100.} \tag{R47.1.3}$$

Consequently every real original function X satisfies

$$\boxed{I_q(T^2(\bar{f}_2 X))^3 \le (1-d/100)(E[\bar{f}_2|X|^4])^2.} \tag{R47.1.4}$$

The weight and operator in (R47.1.4) are part of the statement. It is not a claim about an ordinary fourth input to a different operator.

Theorem B — every actual $a = 2, M = 2$ singleton. Let an original internal vertex r have, after maximal equal-length pairing, exactly one residual arm of length two. Its original terminal has any strict residual list of size at least two, its inner connector has any positive length L, and both vertices may have arbitrarily many genuine equal pairs. Give every edge the direction in part R47.2. At outer length $M = 2$ the actual message into the original center obeys

$$\boxed{I_q(U)^3 \le 1 - d/100 \le e^{-d/100}.} \tag{R47.1.5}$$

The same direction table applies on both sides of $d = 6/25$.

Theorem C — exact original assembly. A side from Theorem B can be combined with any separately supported actual opposite output on the same q. If that output has $I_q^3 \le 1$, the nonreducible-end, nonempty R-center P5 branch has $Z \le \exp(-d/800)$. If both sides are from Theorem B, the corresponding deficit is $d/400$. With the separate original Q4 and central-singleton constructions, the latter gives $Z \le 1$ for arbitrary original terminal and central multisets, both inner lengths independently positive, both actual outer lengths two, and both residual internal arms of length two. No other original length is identified with these twos.

Appendix A gives a separate optional high-domain theorem for $a \ge 3$.

R47.2 Complete actual original direction and reference input

For an arm, f_n means every edge outward from its attachment, $\bar{f}_n$ every edge inward, h_n the first edge inward and the remaining $n-1$ outward, and W_n the first outward and the remaining $n-1$ inward. These definitions also specify $n = 1 : h_1 = q$ and $W_1 = p$.

Pair equal arm lengths maximally using a fixed ordering of the original tree. A genuine pair can be oriented as $f_l h_l$ in the positive table, or $\bar{f}_l W_l$ in its full reverse. Its factor is in $(0, 1]$. The original two arms are retained, with their actual location and edge count. The unpaired residual lengths are strictly increasing.

The complete C034 theorem supplies $E[f_2(T^L F_{\text{ref}})^4] \le 1$ and its genuine-decoration version for all terminal residuals below. In the table, a canonical pair is $h_s f_t$ with $s < t$. Later pairs are the remaining consecutive strict residual pairs, each retained once. Each row gives the outward-reference construction followed by the actual terminal factor used here after full reversal.

Exact $(1, t)$

$F_{\text{ref}} = q f_t$; its full reverse is $p\bar{f}_t$.

Exact $(2, t)$

$F_{\text{ref}} = h_2 f_t$; its full reverse is $W_2\bar{f}_t$.

Exact $(3, t)$**,** $L = 1$ **or** 2

$F_{\text{ref}} = f_3\bar{f}_t$; its full reverse is $\bar{f}_3 f_t$.

Exact $(3, t)$**,** $L \ge 3$

$F_{\text{ref}} = h_3 f_t$; its full reverse is $W_3\bar{f}_t$.

Single pair with minimum at least four; every even list with at least four arms

The product of the canonical pairs $h_s f_t$; its full reverse is the product of $W_s \bar{f}_t$.

Odd minimum $\ell \geq 2$, second length at least four

$\bar{f}_\ell$ times the later canonical pairs; its full reverse is f_ℓ times the later negative pairs.

Odd prefix $(2,3,t)$

$\bar{f}_2 f_3 h_t$ times the later canonical pairs; its full reverse is $f_2 \bar{f}_3 W_t$ times the later negative pairs.

Exact $(1,s,t)$, $s \geq 4$

$F_{\text{ref}} = q f_s h_t$; its full reverse is $p \bar{f}_s W_t$.

Unit minimum, odd list, second length at least four, at least five arms

q times the later canonical pairs; its full reverse is p times the later negative pairs.

Odd prefix $(1,3,t)$

$q f_3 h_t$ times the later canonical pairs; its full reverse is $p \bar{f}_3 W_t$ times the later negative pairs.

Exact $(1,2,3)$, $L = 1$

$F_{\text{ref}} = q f_2 \bar{f}_3$; its full reverse is $p \bar{f}_2 f_3$.

Exact $(1,2,3)$, $L \geq 2$

$F_{\text{ref}} = q f_2 h_3$; its full reverse is $p \bar{f}_2 W_3$.

Exact $(1,2,t)$, $t \geq 4$

$F_{\text{ref}} = q f_2 h_t$; its full reverse is $p \bar{f}_2 W_t$.

Odd prefix $(1,2,3,4,t_2)$, at least five arms

$q f_2 h_{t_2} h_3 f_4$ times the canonical pairs from index three; its full reverse is $p \bar{f}_2 W_{t_2} W_3 \bar{f}_4$ times the negative pairs from index three.

Other odd prefix $(1,2,t)$, at least five arms

$q f_2 h_t$ times the later canonical pairs; its full reverse is $p \bar{f}_2 W_t$ times the later negative pairs.

The table is indexed by parity, minimum, second minimum and the displayed short exceptions. In particular the reference $(1,2,3)$ exception is at $L = 1$. It is not the different C106 ordinary-eighth $L = 2$ exception. A predetermined integer priority resolves overlapping labels.

Let E_r^- and E_t^- be the genuine negative pair decorations at r and the terminal. Define

$$X_0 = A^L(E_t^- F_{\text{ref}}^-), \qquad X = E_r^- X_0, \qquad U = T^2(\bar{f}_2 X). \tag{R47.2.1}$$

Thus the sole two-edge arm points toward r, the inner L-edge connector points from r toward its terminal, and both outer edges point from r toward the original center. All terminal directions listed as full reversals, all pair decorations and the genuine reference arm were reversed together. This is a legal fixed orientation of the complete original half-tree.

The fully reversed C034 reference theorem gives

$$E[E_r^- \bar{f}_2 X_0^4] \leq 1.$$

Since $(E_r^-)^4 \leq E_r^-$ for $0 < E_r^- \leq 1$,

$$\boxed{E[\bar{f}_2 X^4] \leq 1.} \tag{R47.2.2}$$

There is one original $\bar{f}_2$ arm in (R47.2.1). The square-root factorization below uses that arm.

R47.3 Original weighted bridge and its exact norm cost

For any real X, positivity and row Cauchy for $T^2M_{\bar f_2}$ give

$$|T^2(\bar f_2X)|^2 \le \bar f_4\,T^2(\bar f_2|X|^2).$$

Hölder on the original measure μ, with exponents 3 and $3/2$, followed by the ordinary L^2 bound, gives

$$\begin{aligned} I_q(T^2(\bar f_2X))^3 &\le B_4\{E[T^2(\bar f_2|X|^2)^2]\}^2 \\ &\le B_4\|T^2M_{\sqrt{\bar f_2}}\|^4(E[\bar f_2|X|^4])^2. \end{aligned} \tag{R47.3.1}$$

The vector acted on is $\sqrt{\bar f_2}|X|^2$. The weight and the square-root operator are changed together in this exact factorization.

For every positive original operator P and $w\ge 0$, row Cauchy implies

$$|P(\sqrt{w}\,v)|^2 \le (P|v|)(P(w|v|)).$$

After integration and ordinary L^2 Cauchy,

$$\|PM_{\sqrt{w}}\|^2 \le \|P\|\,\|PM_w\|. \tag{R47.3.2}$$

The actual full-multiplier interface is

$$\|TM_{\bar f_2}\|^2 \le 1-d/72.$$

Its order follows from full reversal and ordinary adjunction of the original outward-arm interface, as reconstructed in actual E008 §3.1. Use $P=T^2, w=\bar f_2$ and $T^2M_{\bar f_2}=T(TM_{\bar f_2})$. The actual C064 norms give

$$\begin{aligned} \|T^2M_{\sqrt{\bar f_2}}\|^4 &\le \|T^2\|^2\|T\|^2(1-d/72) \\ &\le (1+(1024/855)\delta_2)^4(1+32d/19)^2(1-d/72) \\ &\le N(d). \end{aligned} \tag{R47.3.3}$$

The last step uses $\delta_2\le 13d/32$, so $(1024/855)\delta_2\le 416d/855<d/2$, and drops only the positive factor $1-d/72\le 1$. Equations (R47.3.1)-(R47.3.3) reduce Theorem A exactly to (R47.1.3).

R47.4 Exact four-edge row and correlated cubic

Write $\tau=d-\delta_2, C=1-d$, and

$$R=KG_3(-K)g=Kg-K^2g+K^3g, \qquad G_3(z)=1+z+z^2.$$

Direct finite multiplication gives

$$\boxed{\bar f_4 = Cq+R-d.} \tag{R47.4.1}$$

Also $\bar f_4=T^2\bar f_2\le 2(\mathbb{E}\bar f_2)q=2Cq$. The exact same-vector moments are

$$ER=-\tau, \qquad h:=E(gR)=\delta_2, \qquad V:=ER^2=\delta_2-\delta_3+\delta_4. \tag{R47.4.2}$$

For example $E(Kg) = -d, E(K^2g) = 0, E(K^3g) = \delta_2$; the odd cross inner products vanish by skewness. These are finite identities.

On the original finite spectral block $z = i\sqrt{x}$, $|G_3(z)|^2 = 1 - x + x^2 \le 1$ for $0 \le x \le \theta$. Hence

$$|R| \le \sqrt{d}, \quad V \le \delta_2, \quad |E(gR^2)| \le d\sqrt{V} \le \tfrac{2}{3}d^{3/2}. \tag{R47.4.3}$$

The last inequality uses original Cauchy, $R^2 \le d$ and $\theta < 4/9$.

Set $W = 2R - Cg$. Its actual precursor is 2G3(-K)g-C1, with E[G3(-K)g]=d. The original row bound therefore proves

$$|W|^2 \le C^2 + 4d - 4Cd = 1 - 2d + 5d^2 \le 1 \quad (0 \le d \le 1/3). \tag{R47.4.4}$$

This improves the looser generic correlated cap specifically at the actual n4 row. No arbitrary n is relabeled as four. The genuine mean and second moment are

$$EW = -2\tau, \qquad EW^2 = C^2d + 4V - 4Ch.$$

Since $(W + 2\tau)^2(W - 1) \le 0$ pointwise,

$$EW^3 \le (1 - 4\tau)EW^2 - 4\tau^2 + 8\tau^3, \qquad E(g^2W) \le d. \tag{R47.4.5}$$

For $s \ge -1$,

$$\min(1 + s, 2)^4 \le 1 + 4s + 7s^2 + 4s^3. \tag{R47.4.6}$$

For -1 $\le s \le$ 1 the difference from $(1 + s)^4$ is $s^2(1 - s^2) \ge 0$. For $s \ge 1$ the right side is16 at1 and has positive derivative. Apply (R47.4.6) at $s = (R-d)/(Cq)$, which is >-1 by (R47.4.1), keeping the cap $\bar{f}_4 \le 2Cq$. This yields

$$B_4 \le C^4Eq^3 + 4C^3E[q^2(R - d)] + 7C^2E[q(R - d)^2] + 4CE(R - d)^3. \tag{R47.4.7}$$

The correlated cubic identity used in expanding it is exactly

$$\begin{aligned} &- C^4Eg^3 + 4C^3E(g^2R) - 6C^2E(gR^2) + 4CER^3 \\ &\qquad = \frac{C}{2}\{EW^3 + C^2E(g^2W)\}. \end{aligned} \tag{R47.4.8}$$

The coefficient -7 in (R47.4.7) leaves the additional -$C^2E(gR^2)$. Substitute (R47.4.2),(R47.4.5) into (R47.4.7)-(R47.4.8). Direct collection gives

$$\begin{aligned} B_4 \le K_0 &+ C(9 - 27d + 8\delta_2)(V - h) \\ &+ C(-1 + 13d - 30d^2 + 8d\delta_2)h - C^2E(gR^2), \end{aligned} \tag{R47.4.9}$$

where the complete constant part is

$$\begin{aligned} K_0 = C^4(1 + 3d) &+ C^3[(1 - 2\tau - 4d)d - 4(d + \tau)] \\ &+ 7C^2d^2 + 14C^2d\tau - 12Cd^2\tau - 4Cd^3 - 2C\tau^2 + 4C\tau^3. \end{aligned} \tag{R47.4.10}$$

Every coefficient here comes from the exact n4 identities. In particular the cap in (R47.4.4) is1, so this is not a transcription of the generic C100 expression with $t = 1 + d/2$ or with beta lowered to kappa d.

R47.5 Complete low-domain row payment, $0 < d \le 6/25$

Put $x = \delta_2/d$, so $d \le x \le \theta$. This is one global moment ratio. Moment Cauchy and (R47.1.1) give

$$V - h = -\delta_3 + \delta_4 \le -\kappa\delta_3 \le -\kappa x^2 d.$$

The coefficient 9-27d+8d2 is positive on this domain. Use this bound, the exact $h = xd$, and (R47.4.3) in (R47.4.9). Then

$$B_4 \le C\{1 - A(x)d + B(x)d^2 - D(x)d^3 - E(x)d^4\} + \tfrac{2}{3}C^2 d^{3/2}, \tag{R47.5.1}$$

with

$$\begin{aligned} A(x) &= 7 - 3x + 9\kappa x^2, \\ B(x) &= 21 - 3x + (27\kappa - 2)x^2 - 8\kappa x^3, \\ D(x) &= 20 + 16x - 20x^2 + 4x^3, \\ E(x) &= 9 - 2x. \end{aligned} \tag{R47.5.2}$$

One direct check of the collection is to write $v = \tau/d = 1 - x$. The constant part alone satisfies

$$K_0/C = 1 - (3 + 4v)d + (3 + 20v - 2v^2)d^2 + (2 - 26v + 4v^3)d^3 - (7 + 2v)d^4.$$

Adding the exact h term and the signed V-h term gives precisely (R47.5.2).

The following uniform rational reductions retain enough curvature:

$$A(x) \ge 125/19, \quad B(x) \le 22, \quad D(x) \ge 20 + (63/8)d, \quad E(x) \ge 8. \tag{R47.5.3}$$

For A, complete its square:

$$A(x) = 9\kappa(x - 1/(6\kappa))^2 + 7 - 1/(4\kappa) = 9\kappa(x - 1/(6\kappa))^2 + 125/19.$$

For B, its second derivative on [0, theta] is at least $1055/64 > 0$. Its maximum is therefore at an endpoint; $B(0) = 21$ and

$$B(\theta) = 2854549/131072 < 22.$$

For D use $16x - 20x^2 \ge x(16 - 20\theta) = 63x/8 \ge 63d/8$ and $4x^3 \ge 0$. Finally $E \ge 131/16 > 8$. Thus (R47.5.1) gives

$$B_4 \le (1 - d)[1 - (125/19)d + 22d^2 - 20d^3 - (127/8)d^4] + \tfrac{2}{3}(1 - d)^2 d^{3/2}.$$

Since $d \le 6/25 < 1/4$, $\sqrt{d} \le 1/2$. Therefore

$$\boxed{B_4 \le (1 - d)R(d),} \qquad R(d) = 1 - \frac{356}{57}d + \frac{65}{3}d^2 - 20d^3 - \frac{127}{8}d^4. \tag{R47.5.4}$$

R47.5.1 Exact continuous polynomial certificate

Let $P(d) = (1 - d)R(d)N(d)$. This is a fixed polynomial. The coefficients of (1-d)R(d), in degrees0 through5, are

$$(1, -413/57, 1591/57, -125/3, 33/8, 127/8). \tag{R47.5.5}$$

The coefficients of N(d), in degrees0 through6, are

$$(1, 102/19, 7995/722, 8105/722, 34665/5776, 588/361, 64/361). \tag{R47.5.6}$$

Write $P = 1 + \sum_{j=1}^{11} c_j d^j$. Manual convolution gives the following exact coefficients and bounds. Equations (R47.5.5)-(R47.5.6) specify every coefficient even where a positive expression is more useful than a large single numerator.

Table 130: A genuine two-edge singleton at two outer edges: parameter values.

Entry	Exact coefficient or exact positive description	Bound used
1	$-107/57$;	retained at the endpoint through P.
2	$191/2166$;	> 0.
3	$1612016/41154$;	> 39.
4	$4671427/329232$;	> 0.
5	$-50007635/329232$;	> -153.
6	$-59573162/329232$;	> -182.
7	$5337151/329232$;	> 0.
8	$(127/8)n_3 + (33/8)n_4 - (125/3)n_5 + (1591/57)n_6$;	> 0.
9	$(127/8)n_4 + (33/8)n_5 - (125/3)n_6$;	> 0.
10	$(127/8)n_5 + (33/8)n_6$;	> 0.
11	$(127/8)n_6$;	> 0.

Here n_j denotes the actual coefficient in (R47.5.6), not a path length. For c8, the first two terms exceed165+24 and the sole negative term has magnitude less than70, since $n_3 > 11, n_4 > 6, n_5 < 5/3$. For c9 the first term exceeds90 and the negative term has magnitude less than $42/5$, since $n_6 < 1/5$. The last two rows are positive termwise. The listed integer fractions also show $c3 < 40, c4 < 15$, $c5 < -151$ and $c6 < -180$, providing further coefficient bounds.

Define $Q(d) = (P(d)-1)/d$ for $d > 0$ and extend it polynomially at zero. For every $0 \le d \le 6/25$, its derivative is bounded below by

$$\begin{aligned} Q'(d) &\ge \frac{191}{2166} + 78d - 612d^3 - 910d^4 \\ &\ge \frac{191}{2166} + d\left[78 - 612(6/25)^2 - 910(6/25)^3\right] \\ &= \frac{191}{2166} + \frac{471390}{15625}d > 0. \end{aligned} \tag{R47.5.7}$$

This proves monotonicity on the entire continuous interval. It is not a statement inferred from checking finitely many d values.

R47.5.2 Exact upper endpoint

At $d* = 6/25$, the exact values from (R47.5.4) are

$$R(d_*) = 3116469/7421875, \qquad (1 - d_*)R(d_*) = 3116469/9765625.$$

Also

$$N(d_*) = (28/25)^4(667/475)^2 < 25/8. \tag{R47.5.8}$$

For an explicit integer check, after multiplying by the positive denominators (R47.5.8) is

$$2187629545472 < 2203369140625.$$

Thus

$$P(d_*) < 3116469/3125000 = 1 - 8531/3125000.$$

By (R47.5.7), $Q(d) \le Q(d*) < -8531/750000 < -1/100$. Combining with (R47.5.4) proves

$$\boxed{B_4 N(d) \le 1 - d/100 \qquad (0 < d \le 6/25).} \tag{R47.5.9}$$

Every expression is for the same actual four-edge path.

R47.6 Complete high-domain row payment, $6/25 \le d \le 1/3$

Here a different estimate of that SAME path is more efficient. Its exact second moment, from (R47.4.1)-(R47.4.2), is

$$E\bar{f}_4^2 = 1 - 5d + 6d^2 + d^3 + (1-2d)\delta_2 - \delta_3 + \delta_4. \tag{R47.6.1}$$

For example the cross term in the direct square is 2C E[q(R-d)]=-4Cd, since $\tau + h = d$. This retains the true mean. As in Section5, with $x = \delta_2/d$,

$$(1-2d)x - \kappa x^2 \le (1-2d)^2/(4\kappa).$$

Consequently

$$\boxed{E\bar{f}_4^2 \le F(d) := 1 - \frac{87}{19}d + \frac{82}{19}d^2 + \frac{51}{19}d^3.} \tag{R47.6.2}$$

No maximizer of this scalar square is imposed on an actual spectral frequency or on the global moment ratio.

Positivity gives the two simultaneous bounds

$$b_4 \le 2(1-d)q, \qquad b_4 \le 2(1-2d).$$

The second uses the exact $Eb3 = 1 - 2d$. Thus

$$B_4 \le 4(1-d)(1-2d)E\bar{f}_4^2 \le 4(1-d)(1-2d)F(d). \tag{R47.6.3}$$

Let $H(d) = 4(1-d)(1-2d)F(d)N(d)$. The polynomial F is positive on the entire original domain: its derivative is

$$F'(d) = (-87 + 164d + 153d^2)/19 \le -46/57 < 0, \qquad F(1/3) = 1/19.$$

In particular $0 < F \le 1$ and F'/$F \le -46/57$. On $d \ge 6/25$ the logarithmic derivative of H is therefore at most

$$-25/19 - 50/13 - 46/57 + 2 + 64/19 = -445/741 < 0. \tag{R47.6.4}$$

All logarithmic factors are positive, so this is a valid continuous monotonicity proof through $d = 1/3$.

At $d* = 6/25$,

$$F(d_*) = 55441/296875, \qquad 4(1-d_*)(1-2d_*)F(d_*) = 2882932/9765625.$$

Using the same $N(d*) < 25/8$ from (R47.5.8),

$$H(d) \le H(d_*) < 2882932/3125000 < 19/20 < 299/300 \le 1 - d/100. \tag{R47.6.5}$$

Together (R47.5.9) and (R47.6.5) prove (R47.1.3) for all $d > 0$. The estimates apply at their shared endpoint.

At $d = 0, K1 = 0$ on the positive support, $A\mathbf{1} = T\mathbf{1} = \mathbf{1}$ and $\bar{f}_4 = q = 1$. Thus (R47.1.3) is equality. The arbitrary-real (R47.1.4) follows directly from (R47.3.1)-(R47.3.3), also at zero; an arbitrary X need not equal one. For the actual tree in (R47.2.1), original successive leaf integration does give $X = U = q = 1$, even when the balanced skew kernel K is nonzero. This proves Theorems A and B.

R.7 Every longer singleton at two outer edges

R48.0 Complete bounded statement

Work with the original finite probability space and the original real skew kernel, pointwise bounded in absolute value by one. At an internal root, maximal genuine equal-arm pairing leaves precisely one original arm of length $m \geq 3$. At the other end of an original inner connector of length $L \geq 1$, the strict residual terminal list has size at least two. Both original vertices may have arbitrary genuine equal-pair decorations. Append exactly two original outer edges from the internal root to the receiving vertex.

Theorem $T20$. The complete integer direction table in part R48.2 gives an actual original message H at the internal root, and $U = T^2 H$ at the receiver, such that

$$\boxed{I_q(U)^3 \leq \exp(-d/20)}. \tag{R48.0.1}$$

The directions depend only on the original terminal integers, m and L. The same table applies across the d domains and spectral frequencies. Every finite terminal cardinality, every finite arm length, every L, every original equal pair and the boundary $d = 0$ is included.

Alongside the $m = 2$ case, the $m \geq 3$ proof uses same-direction ordinary fourths, finite signed coefficients and the ordinary-fourth two-edge bridge with its two T steps.

The following new ingredients complete the finite catalogue:

1. The previously exceptional actual $W_3 T^2(qf_b)$ direction receives its own fourth estimate, with the true constant-vector cross retained.
2. The actual opposite first-edge family $\bar{f}_m AT^{L-1}(qf_2), m > L+1$, receives complete $L = 2$, $L = 3$ and every $L \geq 4$ fourth estimates.
3. The weak common fourths in C040 are strengthened where needed; its original directions remain unchanged in every analytic domain.
4. For the nonunit odd O terminal, the $m = 3$, long $-L$ row retains the complete finite convolution and its last two negative coefficients.
5. The C043 high-domain unit-minimum even directions retain their actual full tail cap and use the correct actual U fourth, including $m = 3, L = 2$.

Theorem (R48.0.1) is an original half-tree interface.

R48.1 Original calculus and the paid two-edge bridge

R48.1.1 Notation, finite identities and actual positivity

On the original $L^2(\mu)$, put

$$\begin{gathered} B = -K, \quad u = B\mathbf{1}, \quad Jv = (Ev)\mathbf{1}, \quad T = J + B, \quad A = J - B = T^*, \\ q = 1 + u, \quad p = 1 - u, \end{gathered} \tag{R48.L230}$$

$$\begin{gathered} f_n = A^n \mathbf{1}, \quad \bar{f}_n = T^n \mathbf{1}, \quad h_n = T f_{n-1}, \quad W_n = A \bar{f}_{n-1}, \\ \delta_j = \|B^j \mathbf{1}\|^2, \quad d = \delta_1, \end{gathered} \tag{R48.L234}$$

$$\theta = \frac{13}{32}, \qquad \kappa = \frac{19}{32}, \qquad \tau = d - \delta_2,$$
$$I_q(V) = E[q^{-\frac{1}{3}}|V|^{8/3}]. \tag{R48.1.1}$$

Here $f_0 = \bar{f}_0 = 1$, and h_n, W_n are defined for $n \geq 1$. In finite propagated sums $W_0 = 1$ denotes the genuine zero-edge path.

The original Q3-C014 and Q4-C143 finite calculus gives

$$0 \leq d \leq 1/3, \quad d^2 \leq \delta_2 \leq \theta d,$$
$$\delta_{j+1} \leq \theta\delta_j, \quad \|B\|^2 \leq \theta, \tag{R48.1.2}$$
$$|Bv(x)| \leq \|v\|_2.$$

All A and T kernels are original nonnegative kernels. All actual finite tree messages are positive on the positive-mass support. Indeed the all-equal labeling at any retained root atom has positive mass and all its diagonal edge factors equal one. In particular $p, q > 0$ there; the bounds retain the atomwise dependence of p and q.

Let

$$G_n(z) = \sum_{i=0}^{n-1} z^i, \quad G_0 = 0,$$
$$\beta_1 = \beta_2 = 0, \tag{R48.1.3}$$
$$\beta_n = \sum_{i=1}^{\lfloor (n-1)/2 \rfloor} (-1)^{i-1}\delta_i.$$

Thus $\beta_3 = \beta_4 = d, \beta_5 = \beta_6 = \tau$, and $\tau \leq \beta_n \leq d$ for $n \geq 3$. These β values are the actual finite alternating moments; they are not $1 - E\bar{f}_{n-1}$. The original finite positive convolution is

$$1 + G_n(B)u = \bar{f}_n + \sum_{r=3}^{n} \beta_r \bar{f}_{n-r}. \tag{R48.1.4}$$

Its completely reversed form holds as well. Every removed term in (R48.1.4) is a nonnegative actual path term. We use this finite identity below.

On the original spectral measure ν of u, write $z = i\sqrt{x}, 0 \leq x \leq \theta$. Its mass is d and $\int x^i d\nu = \delta_{i+1}$. For a real polynomial $P = E_P(x) + zC_P(x)$,

$$E[P(B)u] = -\int C_P \, d\nu, \qquad E[BP(B)u] = -\int E_P \, d\nu. \tag{R48.1.5}$$

Write $G_n = E_n + zC_n$. For $n \geq 2$ both coordinates lie in $[1 - x, 1]$; for $n \geq 3, E_n \leq h(x) := 1 - x + x^2$. Also

$$\|G_n(B)\| \leq 1 \quad (n = 0, 1 \text{ or } n \geq 3), \qquad \|G_2(B)\|^2 \leq 1 + \theta. \tag{R48.1.6}$$

Lemma O.12, (O.104), proves $|G_n(z) - z| \leq 1$ for every finite $n \geq 2$, including the short indices.

For pure paths $\mu_n := Ef_n = E\bar{f}_n \leq 1$, with $\mu_2 = 1 - d, \mu_3 = 1 - 2d$, and $\mu_n \leq 1 - 2d$ for every $n \geq 3$. Every $f_n, \bar{f}_n, h_n, W_n$ lies in [0,2], and

$$f_n + h_n = \bar{f}_n + W_n = 2\mu_{n-1}, \qquad f_n h_n \le 1, \quad \bar{f}_n W_n \le 1. \tag{R48.1.7}$$

The means of h_2 and W_2 are $1 + d$; use their exact low-index formulas.

The all-real signed exponential input in the supplied finite calculus is

$$Ee^{4BV} \le \exp\{4E(BV) + (\frac{17}{4})\|V\|^2\}. \tag{R48.1.8}$$

It applies to every real V with either sign of the actual mean, including a genuine constant component.

If actual nonnegative factors X, Y have $X + Y \le 2 + c + BV$, AM-GM therefore gives

$$\log E(XY)^4 \le 4(c + EBV) + (\frac{17}{4})\|V\|^2. \tag{R48.1.9}$$

The upper base is nonnegative because it bounds an actual nonnegative sum. If the second factor has a prefactor $\exp(-\Gamma)$, add -4Γ; if its prefactor is a_0^2, add $8 \log a_0$. This fixes the fourth convention used throughout the manuscript. Every expression such as $E[X]^4$ below denotes the expectation $E(X^4)$, never the fourth power of the mean.

R48.1.2 The exact original two-edge transfer

The actual C094, Section 2, gives for every real original H

$$\begin{aligned} I_q(T^2H)^3 &\le [1 - 2d^2 + (8 - 2d)\delta_2](1 + 1024\delta_2/855)^8 (E|H|^4)^2 \\ &\le \exp(c_2 d - 5d^2)(E|H|^4)^2, \qquad c_2 = \frac{24427}{3420} < 36/5. \end{aligned} \tag{R48.1.10}$$

For clarity the positive-operator step is exact. If P is positive with row $r = P\mathbf{1}$, row Cauchy and Hölder give

$$\begin{aligned} I_q(PH)^3 &\le E(r^4/q)\{E[P(H^2)^2]\}^2 \\ &\le E(r^4/q)\|P\|^4 (E|H|^4)^2. \end{aligned} \tag{R48.1.11}$$

Here $P(H^2)^2$ denotes the square of $P(H^2)$, not $P(H^4)$. For $P = T^2$ the actual row is $\bar{f}_2 = q + Bu$. C094's correlated cubic row calculation gives $E(\bar{f}_2^4/q) \le 1 - 2d^2 + (8 - 2d)\delta_2$. Its actual ordinary norm is $\|T^2\| \le (1 + 1024\delta_2/855)^2$. Hence the power in (R48.1.10) is eight. The variance $\delta_2 = \|B^2\mathbf{1}\|^2$ has not been changed to d^2.

For the curvature, the same calculation proves on the whole interval $0 \le d \le \frac{1}{3}$

$$\log(1 + 13d/4 - 45d^2/16) \le 13d/4 - (\frac{11757}{2720})d^2, \tag{R48.L370}$$

$$8 \log(1 + 416d/855) \le (\frac{3328}{855})d - 3d^2/4. \tag{R48.1.12}$$

Their linear sum is c_2 and their quadratic sum exceeds five. The complete inverse-fourth row and the ordinary norm occur in the same original measure.

In particular every same-direction fourth bound $EH^4 \le \exp(-29d/8)$ implies (R48.0.1), since

$$2(\frac{29}{8}) - \frac{36}{5} = 1/20. \tag{R48.1.13}$$

Some rows below instead use their explicitly retained negative curvature.

R48.2 The complete structural direction table

The following defines the actual unpaired H. Genuine equal pairs are restored at their original vertices by the common restoration argument. A bar means reversal of every arm in that displayed terminal configuration. This specifies the terminal reversal in the direction table.

R48.2.1 Even terminal residuals

Write an even strict terminal as $a_1 < b_1 < ... < a_j < b_j$.

Table 131: Every longer singleton at two outer edges: parameter values.

Entry	Actual H
Exact $(1,b), L = 1$	$f_m T(p\bar{f}_b)$.
Exact $(1,b), m = 3, L = 2$	$f_3 TA(p\bar{f}_b)$.
Exact $(1,b)$, all remaining $L \geq 2$	$W_m T^L(qf_b)$.
Exact $(2,b), b \geq 3, L = 1$	$f_m T(f_2\bar{f}_b)$.
Exact $(2,b), b \geq 3, L \geq 2$	$W_m T^L(\bar{f}_2 W_b)$.
Exact $(3,b), b \geq 4, L = 1$	$f_m T(f_3 h_b)$.
Exact $(3,b), b \geq 4, L \geq 2$	$W_m T^L(f_3 h_b)$.
$a_1 \geq 4$, every $j \geq 1; m \leq L$	$W_m T^L(F), F = \prod h_{a_i} f_{b_i}$.
$a_1 \geq 4$, every $j \geq 1; m > L$	$\bar{f}_m AT^{L-1}(F)$, same F.
$a_1 = 2, j \geq 2, L = 1$	$\bar{f}_m A(F), F = \prod h_{a_i} f_{b_i}$.
$a_1 = 2, j \geq 2, L \geq 2$	$W_m T^L(F)$, same F.
$a_1 = 3, j \geq 2, m \leq L$	$W_m T^L(F), F = \prod h_{a_i} f_{b_i}$.
$a_1 = 3, j \geq 2, m > L$	$\bar{f}_m AT^{L-1}(F)$, same F.
$a_1 = 1, j \geq 2, L = 1$	$f_m T(\overline{F}), F = qf_{b_1} \prod_{i \geq 2} h_{a_i} f_{b_i}$.
$a_1 = 1, j \geq 2, L \geq 2$	$W_m T^L(F)$, same F.

The exact $-(1,b)m = 3, L = 2$ turn is the old Q5-C087 short-outer direction. It is used here because the outer length is two. The unit-minimum multipair $m = 3, L = 2$ row instead keeps its actual C043 all $-T$ direction; part R48.3 pays that different message explicitly.

R48.2.2 Odd terminal residuals of nonunit minimum

Write the strict list as $(\ell, r_1, t_1, ..., r_j, t_j)$, where $2 \leq \ell < r_1 < t_1 < ... < r_j < t_j$ and $j \geq 1$. Put

$$F = \begin{cases} \bar{f}_\ell \prod_{i=1}^{j} h_{r_i} f_{t_i}, & r_1 \geq 4, \\ \bar{f}_2 f_3 h_{t_1} \prod_{i=2}^{j} h_{r_i} f_{t_i}, & r_1 = 3. \end{cases} \tag{R48.2.1}$$

The second row necessarily has $\ell = 2$. Set $n = \ell + L$ and use

$$H = \begin{cases} W_m T^L F, & m \le n, \\ \bar{f}_m A T^{L-1} F, & m > n. \end{cases} \tag{R48.2.2}$$

These are exactly the R2-C030 actual O directions.

R48.2.3 Odd terminal residuals of unit minimum

Write the strict list as $(1, r_1, t_1, \dots, r_j, t_j), j \ge 1$. The actual terminal is the C040 terminal:

$$F = \begin{cases} q \prod_{i=1}^{j} h_{r_i} f_{t_i}, & r_1 \ge 4, \\ q f_3 h_{t_1} \prod_{i=2}^{j} h_{r_i} f_{t_i}, & r_1 = 3, \\ q f_2 h_e h_3 f_4, & \text{exact list } (1,2,3,4,e),\ j = 2, \\ q f_2 h_{t_1} \prod_{i=2}^{j} h_{r_i} f_{t_i}, & \text{all other } r_1 = 2. \end{cases} \tag{R48.2.3}$$

For $L = 1$ use $H = W_m T F$ when $r_1 \ge 3$, and $H = \bar{f}_m A F$ when $r_1 = 2$. For every $L \ge 2$ use

$$H = \begin{cases} W_m T^L F, & m \le L + 1, \\ \bar{f}_m A T^{L-1} F, & m > L + 1. \end{cases} \tag{R48.2.4}$$

The crossing in (R48.2.3) is only the exact five-arm list. The same prefix inside a list with at least seven residual arms stays canonical, exactly as in C040. The equality $m = L + 1$ belongs to the first row of (R48.2.4). Each analytic split below estimates the direction fixed by this table.

In every AT^{L-1} row the receiver-adjacent original connector edge is A and the other $L - 1$ edges are T. Every $f, \bar{f}, h, W$ is its actual path of the displayed original length. The special T A row contains the two original inner edges and their actual common midpoint. Finally every row appends the same two original outer T edges, giving $U = T^2 H$.

R48.3 The actual $W_3 T^2(q f_b)$ boundary

The turn $f_3 T A(p \bar{f}_b)$ differs from the C043 all $-T$ message estimated here.

R48.3.1 Terminal two: the real constant cross and a quadratic reserve

The actual identities are $f_2 = p + Bu$ and $q + f_2 = 2 + Bu$. Therefore

$$\begin{aligned} q f_2 &\le (1 + Bu/2)^2, \\ T(q f_2) &\le q - \gamma + B^2 u, \\ \gamma &= d - \delta_2/2. \end{aligned} \tag{R48.3.1}$$

Indeed $T(Bu) = -d + B^2 u$ and $T((Bu)^2) \le 2\delta_2$. Applying the actual positive T once more gives

$$T^2(q f_2) \le \bar{f}_2 - \gamma q + B^3 u. \tag{R48.3.2}$$

The exact W_3 is $1 - d - u - Bu - B^2 u$, so $W_3 + \bar{f}_2 = 2 - d - B^2 u$. For the sum of W_3 and the nonnegative upper in (R48.3.2), use

$$c = -d - \gamma, \qquad V = -\gamma\, 1 - Bu + B^2 u, \qquad E(BV) = \delta_2. \tag{R48.3.3}$$

The vector is not centered. Its genuine constant cross is

$$\begin{aligned}\|V\|^2 &= \delta_2 + \delta_3 + \gamma^2 - 2\gamma d \\ &= \delta_2 + \delta_3 - d^2 + \delta_2^2/4.\end{aligned} \tag{R48.3.4}$$

Equation (R48.1.9) gives

$$\begin{aligned}&\log E[W_3 T^2(qf_2)]^4 \\ &\le -8d + (\frac{41}{4})\delta_2 + (\frac{17}{4})\delta_3 - (\frac{17}{4})d^2 + (\frac{17}{16})\delta_2^2\end{aligned} \tag{R48.L527}$$

$$\begin{aligned}&\le -\frac{12839}{4096}d \\ &-\frac{17}{4}(1-\theta^2/4)d^2 \\ &\le -\frac{25}{8}d - 4d^2.\end{aligned} \tag{R48.3.5}$$

The last comparisons are exact: $12839 > 12800$, and $\theta^2 < \frac{1}{6}$ gives $(\frac{17}{4})(1-\theta^2/4) > \frac{391}{96} > 4$. Thus every original host is included, not only the high domain in which this estimate will later be used.

R48.3.2 Every $b \ge 3$: one finite terminal polynomial

Set

$$\begin{aligned}R = BG_{b-1}(-B)u, \quad a = 1-\beta_b/2, \quad k = a^{-1}, \\ V_R = ER^2, \quad t_0 = \beta_{b+1},\end{aligned} \tag{R48.L546}$$

$$e = d - \beta_{b+2}, \quad \rho = E[(Bu)R], \qquad \gamma = kt_0 - k^2 V_R/2. \tag{R48.3.6}$$

The actual finite positive terminal square, also used in C087 parts R48.3–R48.4, is $qf_b \le a^2(1 + kR/2)^2$. Its first and second rows give

$$\begin{aligned}T(qf_b) &\le a^2[q - \gamma + kBR], \\ T^2(qf_b) &\le a^2[\bar{f}_2 - \gamma q - ke + kB^2R].\end{aligned} \tag{R48.3.7}$$

Both displayed upper factors come from actual positive kernels. The exact same-parameter bounds in C087 (4.3), at every finite b, are

$$\begin{aligned}1 \le k \le 6/5, \quad 3d/10 \le \gamma \le 6d/5, \\ 0 \le ke \le 39d/80, \quad \beta_b \ge \tau.\end{aligned} \tag{R48.3.8}$$

We use simultaneous consequences of the definitions in (R48.3.6). For $b \ge 4$ the finite G contraction gives $\|BR\|^2 \le \delta_3$. For $b = 3$ the actual G_2 gives $\|BR\|^2 = \delta_3 + \delta_4$.

Write $G_{b-1}(-z) = E - zC$, so $R = (xC + zE)u$. Then

$$\begin{aligned}e = \int xC\, d\nu, \quad \rho = \int xE\, d\nu, \\ \langle Bu, BR\rangle = \int x^2 C\, d\nu \ge 0.\end{aligned} \tag{R48.3.9}$$

The entire finite difference $\rho - e$ is controlled without a limiting series. If $b-1$ is even, $E = C$. If $b - 1 = 2r + 1$ is odd, $E - C = (-x)^r$; here $r \geq 1$. A positive difference requires even $r \geq 2$, whence

$$\rho - e \leq \delta_4. \tag{R48.3.10}$$

For $b = 3$ specifically $\rho = e = \delta_2$.

Apply AM-GM to W_3 and the bracket in the second row of (R48.3.7). The scalar and vector are

$$c = -d - \gamma - ke, \qquad V = -\gamma\,1 - Bu + kBR, \quad c + EBV = -d - \gamma + k(\rho - e). \tag{R48.3.11}$$

For $V_0 = -Bu + kBR$, its true mean is $EV_0 = d-$ ke $\geq 41d/80$. Consequently its constant correction has the whole-domain bound

$$\begin{aligned} &\gamma^2 - 2\gamma EV_0 \\ &\leq \max_{0 \leq y \leq 6d/5} \{y^2 - (\frac{41}{40})dy\} \\ &\leq 21d^2/100 \leq 7d/100. \end{aligned} \tag{R48.3.12}$$

The maximum is at an endpoint of this convex quadratic. At the nonzero endpoint its coefficient is $\frac{36}{25} - \frac{123}{100} = \frac{21}{100}$. Using the nonnegative cross in (R48.3.9), for $b \geq 4$ we obtain

$$\begin{aligned} \|V\|^2 &\leq \delta_2 + (\frac{36}{25})\delta_3 + 7d/100 \\ &\leq (\frac{4569}{6400})d < 3d/4. \end{aligned} \tag{R48.3.13}$$

For $b = 3$ the same argument gives a norm below d: indeed $\theta^2(1+\theta) < \frac{1}{4}$ and $\theta + (\frac{36}{25})(\frac{1}{4}) + \frac{7}{100} < 1$.

Now $8 \log a \leq -4\beta_b$. For $b \geq 4$, equations (R48.1.9) and (R48.3.8)-(R48.3.13) give a fourth loss strictly larger than

$$\begin{aligned} &4\kappa + 4 + \frac{6}{5} - (\frac{24}{5})\theta^3 - \frac{51}{16} \\ &> \frac{19}{8} + \frac{26}{5} - \frac{3}{8} - \frac{51}{16} = \frac{321}{80} > 4. \end{aligned} \tag{R48.3.14}$$

Here $\theta^3 < \frac{5}{64}$ pays the penultimate strict comparison. $Atb = 3, \beta_b = d$ and $\rho = e$; the weaker norm bound d gives loss at least $4 + 4 + \frac{6}{5} - \frac{17}{4} = \frac{99}{20} > 4$. We have therefore proved

$$\boxed{E[W_3 T^2(qf_b)]^4 \leq e^{-4d} \quad (b \geq 3).} \tag{R48.3.15}$$

Every finite b, both parity classes, the true G_2 exception and the actual constant cross are included in this unchanged all $-T$ direction.

R48.4 The actual $\bar{f}_m AT^{L-1}(qf_2)$ family

Assume $L \geq 2$ and $m > L + 1$, precisely the second branch of (R48.2.4). Write $J_{m,L} = \bar{f}_m AT^{L-1}(qf_2)$. We prove

$$EJ_{m,L}^{4} \le \begin{cases} e^{-5d}, & L = 2, \\ e^{-167d/48}, & L = 3, \\ e^{-6d}, & L \ge 4. \end{cases} \tag{R48.4.1}$$

The positive-message calculation uses the stated single-root orientation; the forward U family is different.

R48.4.1 $L = 2$ **and every** $m \ge 4$

Apply A to (R48.3.1):

$$AT(qf_2) \le W_2 - \gamma p - B^3 u. \tag{R48.4.2}$$

Keep the last two positive coefficients of (R48.1.4) : $\bar{f}_m \le 1 + G_m u - \beta_m - \beta_{m-1} q$. Together with $W_2 = p - Bu$, the sum of $\bar{f}_m$ and the upper in (R48.4.2) is at most $2 + c + BV$, where

$$c = -(\beta_m + \beta_{m-1} + \gamma), \quad V = B(G_{m-2}(B) - B)u + (\gamma - \beta_{m-1})1. \tag{R48.4.3}$$

For $n = m - 2 \ge 2$ the finite coordinate $C_n \le 1$ gives

$$EBV = \int x(C_n - 1)\, d\nu \le 0. \tag{R48.4.4}$$

Moreover $|\gamma - \beta_{m-1}| \le \delta_2/2$, because both actual β values lie in $[d - \delta_2, d]$. The finite inequality $|G_n - z| \le 1$ yields

$$\|V\| \le \sqrt{\delta_2} + \delta_2/2 < (\frac{19}{24})\sqrt{d}. \tag{R48.4.5}$$

For the rational comparison use $\sqrt{\theta} < \frac{2}{3}, \theta < \frac{5}{12}$ and $\sqrt{d} < \frac{3}{5}$, so the second normalized term is less than $\frac{1}{8}$. The scalar obeys $c \le -[3d - (\frac{5}{2})\delta_2] \le -127d/64$. Equation (R48.1.9) now gives fourth loss at least

$$\frac{127}{16} - (\frac{17}{4})(\frac{19}{24})^2 = \frac{12151}{2304} > 5. \tag{R48.4.6}$$

The constant in (R48.4.3) was included in the norm, not discarded.

R48.4.2 $L = 3$ **and every** $m \ge 5$

From (R48.3.1),

$$AT^2(qf_2) \le W_3 - \gamma W_2 + \delta_2 - B^4 u. \tag{R48.4.7}$$

The actual finite unequal-arm sum for $h = 3, H = m$ is

$$W_3 + \bar{f}_m \le 2 - (2d - \delta_2) + B^3 G_{m-3}(B)u. \tag{R48.4.8}$$

Set $n = m - 3 \ge 2$. Combining (R48.4.7)-(R48.4.8) gives

$$\begin{aligned} c &= -3d + (\frac{5}{2})\delta_2, \\ V &= \gamma \mathbf{1} + V_0, \quad V_0 = \gamma u + B^2(G_n(B) - B)u. \end{aligned} \tag{R48.4.9}$$

The exact finite coordinates of V_0 are $\gamma - xE_n + zx(1 - C_n)$. Thus

$$EV_0 = \int x(C_n - 1)\, d\nu \le 0, \qquad \|V_0\|^2 \le \gamma^2 d + \delta_3 + \delta_6. \tag{R48.4.10}$$

The omitted real cross is $-2\gamma \int xE_n$ and is nonpositive. Since $0 \le \gamma \le d$, the full constant cross is also nonpositive, and

$$\begin{aligned} \|V\|^2 &\le \gamma^2(1 + d) + \delta_3 + \delta_6 \\ &\le [\frac{4}{9} + \theta^2 + \theta^5]d < 2d/3. \end{aligned} \tag{R48.4.11}$$

One entirely rational last check is $\theta^2 < \frac{1}{6}, \theta^5 < \frac{1}{32}$, and $\frac{4}{9} + \frac{1}{6} + \frac{1}{32} = \frac{185}{288} < \frac{2}{3}$. The actual mean in the exponent satisfies

$$\begin{aligned} EBV &= -\gamma d + E[B^3(G_n - B)u] \le \delta_2, \\ c + EBV &\le -3d + (\frac{7}{2})\delta_2 \le -101d/64. \end{aligned} \tag{R48.4.12}$$

Here $E[B^3(G_n - B)u] = \int xE_n \le \delta_2$. The fourth loss is therefore at least

$$\frac{101}{16} - \frac{17}{6} = 167/48. \tag{R48.4.13}$$

In particular no assertion stronger than the proved (R48.4.13) is needed. On $d \ge \frac{1}{5}$, (R48.1.10) already gives an $M = 2$ loss at least

$$2(\frac{167}{48}) + 1 - \frac{36}{5} = \frac{91}{120} > 1/20. \tag{R48.4.14}$$

R48.4.3 Every finite $L \ge 4$

Finite Duhamel applied to the first row (R48.3.1) gives exactly

$$\begin{aligned} AT^{L-1}(qf_2) &\le W_L - \gamma W_{L-1} - B^{L+1}u \\ &\quad + \sum_{i=0}^{L-2} (EB^{i+2}u) W_{L-2-i}. \end{aligned} \tag{R48.4.15}$$

The last term has $W_0 = 1$. The $i = 0$ coefficient is zero. The positive classes of the remaining finite sum are $\delta_2, \delta_4, \delta_6$,...; hence, using $0 \le W_j \le 2$, their entire finite contribution is at most

$$C \le \frac{2\delta_2}{1 - \theta^2} \le \frac{832}{855} d < d. \tag{R48.4.16}$$

We may discard the negative $-\gamma W_{L-1}$. This is after actual positive propagation, not an unsupported sign change of an input. Since $4 \le L < m$, (R40.3.3) gives a vector Z with

$$c_{\text{bare}} + E(BZ) \le -219d/70, \qquad \|Z\| \le \sqrt{d}/3. \tag{R48.4.17}$$

The new tail vector is $-B^L u$. Its norm is at most $\theta^2\sqrt{d} < \sqrt{d}/6$ and its exponent mean is at most $d/6$ in absolute value. Combining it with Z and (R48.4.16), the complete sum has

$$c + EBV \le (-\frac{219}{70} + 1 + \frac{1}{6})d = -206d/105, \qquad \|V\| \le \sqrt{d}/2. \tag{R48.4.18}$$

The fourth loss is at least $\frac{824}{105} - \frac{17}{16} = \frac{11399}{1680} > 6$. This proves the last row of (R48.4.1) for every finite $L \ge 4$. In particular the last scalar in (R48.4.15), the G_2 boundaries in the bare vector and all later residue classes remain.

R48.4.4 High-score consequence in the exact C040 directions

For $c \ge 5$ and $d \ge \frac{1}{5}$, actual path means give

$$h_c \le 2\mu_{c-1} \le 2 - 5d \le 1. \tag{R48.4.19}$$

The slightly stronger supplied $\mu_{c-1} \le 1 - 83d/32$ for $c-1 \ge 4$ justifies the weaker $1 - 5d/2$ used here. For $c = 4$ and $d \ge \frac{1}{4}$, $h_4 \le 2\mu_3 = 2 - 4d \le 1$. Thus in those exact domains

$$qf_2h_c \le qf_2. \tag{R48.4.20}$$

If $m \le L+1$, positive propagation compares the C040 message to $W_m T^L(qf_2)$. Except at $m = 3$, $L = 2$, its actual C087 fourth loss is at least $\frac{3487}{640} > \frac{29}{8}$. $Atm = 3, L = 2,$ (R48.3.5) and (R48.1.10) give, on $d \ge \frac{1}{5}$,

$$\log I_q(T^2H)^3 \le (\frac{36}{5} - \frac{25}{4})d - 13d^2 \le -33d/20. \tag{R48.4.21}$$

If $m > L + 1$, the actual comparator is $J_{m,L}$, exactly (R48.4.1). Its $L = 2$ and all $L \ge 4$ fourths pay (R48.1.13); $L = 3$ is paid by (R48.4.14). Consequently all original $L \ge 2$, all $m \ge 3$ and both entire high domains in (R48.4.19)-(R48.4.20) satisfy (R48.0.1) with their unchanged C040 directions.

R48.5 Every unit-minimum odd terminal

Use Proposition O.10 and the exact-triple calculation in the ordinary triple construction, with the fixed direction table (R48.2.3)–(R48.2.4).

R48.5.1 First pair three and exactly one pair

For $F = qf_3h_c$, $c \ge 4$, the receiver and connector table is exactly the one in (O.127). That full-domain same-direction calculation gives

$$\mathbb{E}H^4 \le e^{-9d/2} \le e^{-39d/10}. \tag{R48.5.3}$$

Genuine equal-pair factors are restored by positivity. Since $9/2 > 29/8$, the ordinary-fourth two-edge bridge (R48.1.10)–(R48.1.13) proves the required rate $1/20$ for the actual $U = T^2H$. Its $q^{-1/3}$ denominator is unchanged.

R48.5.2–5.5 Unit-minimum triples with an outer path

The messages $F_c = qf_2h_c$, $c \ge 3$, and their actual m, L direction table coincide with the ordinary triple construction. Its simultaneous fourth and eighth calculation includes both short connector directions and every $m = 3$ row; no terminal is reoriented here.

The transfer from a same-direction ordinary fourth has a formulation with an arbitrary outer length $r \geq 1$. Set $P_r = T^r$, $\rho_r = P_r\mathbf{1}$, with the ordinary L^2 operator norm in the original measure. Positive-row Cauchy and Hölder give

$$\begin{aligned} I_q(P_rH)^3 &\leq \mathbb{E}(\rho_r^4/q)\{\mathbb{E}[(P_r(H^2))^2]\}^2 \\ &\leq \mathbb{E}(\rho_r^4/q)\|P_r\|^4(\mathbb{E}H^4)^2. \end{aligned} \tag{R.1}$$

Indeed $|P_rH|^2 \leq \rho_r P_r(H^2)$, and the exponents $3, 3/2$ separate $(\rho_r^4/q)^{1/3}$ and $(P_r(H^2)^2)^{2/3}$. Thus the formula retains the inverse row weight at the receiving vertex; it does not turn the ordinary moment of H into a weighted eighth moment. At $r = 2$, the previously established actual row and norm estimates (R48.1.10) give

$$\mathbb{E}H^4 \leq e^{-\eta d} \quad \Longrightarrow \quad \log I_q(T^2H)^3 \leq (c_2 - 2\eta)d - 5d^2, \qquad c_2 < 36/5. \tag{R.2}$$

The theorem below uses this explicit two-edge value, rather than an unproved contraction at arbitrary outer lengths.

For $c = 3$, the complete ordinary proof gives fourth rate $4667/1280 > 29/8$ at $L = 1$, and rate greater than four for every $L \geq 2$. Hence (R.2) proves the desired rate $1/20$ for every m, L, d. For $c = 4$, the same proof gives rate $39/10$ at $L = 1$. At $L \geq 2, d \leq 1/4$, every row of (O.114) pays $29/8$, except $L \geq 4, m \geq 4, 1/5 \leq d \leq 1/4$. On that remaining interval retain the actual fourth rate $2369/700$ from (O.115); then

$$2(2369/700) + 5d - 36/5 \geq 199/350 > 1/20.$$

On $d \leq 1/5$, the smaller first-row error in (O.107) instead gives fourth rate $7807/2100 > 37/10$. On $d \geq 1/4, L \geq 2$, the actual cap $h_4 \leq 1$ permits the same-direction comparison in the high-score calculation, including its curved W_3T^2 and opposite-first-edge $L = 3$ cases.

Every exact $12c$ with $c \geq 5$

On $d \leq 1/5$, the unit-connector rate is $19/4$ and every nonunit-connector fourth rate exceeds $37/10$, by (O.114) and its unit-connector calculation. On $1/5 \leq d \leq 1/4, L = 1$, the single joint calculation (O.121) gives rate $321/80 > 4$. These rates all pay (R.2). On $d \geq 1/4, L = 1$, use the actual pointwise and mean estimate (O.122): $\mathbb{E}H^4 < 5/16$, while $c_2 d - 5d^2 < 2$. Thus

$$I_q(T^2H)^3 < 8(5/16)^2 = 25/32 < 5/6 \leq e^{-d/2}. \tag{R.3}$$

Finally, on $d \geq 1/5, L \geq 2$, the actual cap $h_c \leq 1$ gives exactly the forward or opposite-first-edge comparator in the high-score calculation; it proves the same rate $1/20$. These intervals meet at their endpoints and retain the prescribed integer direction throughout. Positivity restores all original equal pairs before both outer T steps. At $d = 0$ the message and both outer rows are one.

R48.5.6 The remaining complete unit-minimum odd catalogue

The following rows are the same-direction fourth entries of Table 68:

Table 132: Every longer singleton at two outer edges: parameter values.

Entry	Existing fourth loss
$r_1 \geq 4, j \geq 1$	$5 + (\frac{64}{5})(j-1)$.

Entry	Existing fourth loss
$r_1 = 3, j \geq 2$	$7 + (\frac{64}{5})(j-2)$.
exact $(1,2,3,b,e), b \geq 5$	5.
exact crossed $(1,2,3,4,e)$	5.
$r_1 = 2, j \geq 3$, canonical throughout	$8 + (\frac{64}{5})(j-3)$.

The remaining five-arm class is $(1,2,c,b,e), c \geq 4$. Its complete low domain $d \leq \frac{1}{64}$ has fourth loss greater than five for every m, L. On $d \geq \frac{1}{64}$ the same actual terminal regroups as

$$F = (qf_2h_b)(h_cf_e), \qquad b \geq 5. \tag{R48.5.27}$$

The actual full finite continuous pair tail in (O.128) has size at most $(\frac{39}{125})\sqrt{d}$. Its scalar loss is $16d/5$, and $\sqrt{d} \leq 8d$ here. Consequently

$$h_cf_e \leq \exp[-(\frac{16}{5} - \frac{312}{125})d] \leq e^{-d/2}. \tag{R48.5.28}$$

The base triple in (R48.5.27) has exactly the same receiver and connector choice as part R48.5.5. Positive propagation, including both outer T edges, therefore multiplies the actual U by at most $e^{-d/2}$. Since $I_q(U)^3$ is homogeneous of degree eight, its bound gains e^{-4d}. This bound covers the entire high domain for the fixed direction.

Every unit-minimum odd terminal has now been covered: $r_1 \geq 4, r_1 = 3$, or $r_1 = 2$; the last case has $j = 1, j = 2$ or $j \geq 3$, with $j = 2$ exhausted by the three explicit classes. The analytic boundaries $\frac{1}{64}, \frac{1}{8}, \frac{1}{5}, \frac{1}{4}$ are included on both sides of their estimates.

R48.6 Every odd terminal of nonunit minimum

We now prove the stronger same-direction statement

$$\boxed{EH^4 \leq e^{-4d}} \quad \text{for every O direction in (R48.2.1)-(R48.2.2).} \tag{R48.6.1}$$

Use the full finite packet in parts R30.2–R30.8, and A014, Sections 3–7, to prove (R48.6.1).

R48.6.1 Second-shortest terminal length at least four

R2-C030 gives $F \leq \bar{f}_\ell \exp(R - \Gamma), \Gamma \geq 16jd/5$. Its two actual first-edge signs have the same valid $D \leq 5d/6$, while the one finite terminal mask satisfies $\|R\| \leq \sqrt{d}/4$ and $\sum_{i \geq 1}(D_i)_+ < 2d/5$. The matched integer choice m versus $\ell + L$ makes the bare pair $W_h\bar{f}_H$ with $3 \leq h \leq H$. Thus for every finite L the complete scalar/vector bounds are

$$M \leq -\frac{19}{16} + \frac{37}{15} + \frac{1}{4}, \qquad N \leq 3/4. \tag{R48.6.2}$$

Here $C = 2D + 2\sum(D_i)_+ \leq 37d/15$ bounds each finite propagated positive sub-sum, not an infinite actual connector. Applying (R48.1.9) to this same combined vector gives

$$\text{fourth loss} \geq \frac{4121}{960} + (\frac{64}{5})(j-1) > 4. \tag{R48.6.3}$$

The numerator 4121 follows from $\frac{64}{5} - \frac{367}{60} - \frac{153}{64} = \frac{4121}{960}$. This covers every ℓ, m, L and j in this entire structural class without an additional d split.

R48.6.2 Second-shortest length three: the complete low packet

Now $\ell = 2, r_1 = 3$ and $c = t_1 \geq 4$. Put $\eta = \beta_{c-1}$ and

$$R_0 = -\eta u - B^3 G_{c-3}(-B)u, \quad \Gamma_0 = d + \beta_c + \eta \geq 35d/16. \tag{R48.6.4}$$

For $j = 1, F = \bar{f}_2 f_3 h_c \leq \bar{f}_2 \exp(R_0 - \Gamma_0)$. On $d \leq \frac{1}{8}$ the actual $\bar{f}_2-$ anchored first row has coefficient one on $BR0$, and its supplied whole signed δ allows $D = 3d/5$. The full packet is

$$\begin{gathered} \|R_0\| \leq (\frac{9}{20})\sqrt{d}, \\ (D_1)_+ \leq d^2, \quad (D_2)_+ \leq \delta_3, \\ \sum_{i \geq 1} (D_i)_+ < 3d/7. \end{gathered} \tag{R48.6.5}$$

This $\bar{f}_2$ first row is not the q first row of part R48.5. Its actual δ includes the additional mean $E[(Bu)R_0]$ and the original $\sqrt{(d+\delta_2)ER_0^2}$ cost, as explicitly paid in A014, Section 6.

The exact short $-L$ constants are

Table 133: Every longer singleton at two outer edges: parameter values.

Entry	C/d	ρ with $\|B^{L-1}R_0\| \leq \rho\sqrt{d}$
1	$\frac{3}{5}$;	$\frac{9}{20}$.
2	$\frac{53}{40}$;	$\frac{3}{10}$.
3	$\frac{97}{60}$;	$\frac{3}{16}$.
every $L \geq 4$	$\frac{72}{35}$;	$\frac{3}{25}$.

For $L = 1, L = 2, L = 3$ the matched bare estimate gives $M \leq -\frac{19}{16} + C/d + \rho$ and $N \leq \frac{1}{2} + \rho$. Substitution at exponent four gives respectively

$$\frac{8743}{1600} > 4, \qquad 107/25 > 4, \qquad 377/60 - \frac{2057}{1024} > 4. \tag{R48.6.6}$$

The last comparison even follows from $\frac{2057}{1024} < \frac{21}{10}$.

For every $L \geq 4$ and $m \geq 4$, the matched smaller index is at least four. Use the actual stronger bare pair instead:

$$\begin{gathered} M \leq -\frac{219}{70} + \frac{72}{35} + \frac{3}{25} = -\frac{333}{350} < 0, \\ N \leq 1/3 + \frac{3}{25} = \frac{34}{75} < 1/2. \end{gathered} \tag{R48.6.7}$$

The fourth loss is then at least $\frac{35}{4} - \frac{17}{16} = \frac{123}{16} > 4$. Only $m = 3$ and all $L \geq 4$ remain in the $j = 1$ low domain. They are handled by the complete finite coefficient argument next.

R48.6.3 $m = 3$ **and every** $L \geq 4$**: retain the last two finite coefficients**

The structural rule (R48.2.2) selects $W_3 T^L$ because $n = L + 2 > 3$. Write $G_{c-3}(-z) = U - zV$. $Atc = 4$ the genuine coordinates are $U_1 = 1, V_1 = 0$; at every other finite $c, 0 \leq U, V \leq 1$. The four signed moment classes are

$$\begin{array}{ll} D_{4a} = -\int x^{2a+1} U \, d\nu, & a \geq 0, \\ D_{4a+1} = \eta\delta_{2a+1} - \int x^{2a+2} V \, d\nu, & a \geq 0, \\ D_{4a+2} = \int x^{2a+2} U \, d\nu, & a \geq 0, \\ D_{4a+3} = -\eta\delta_{2a+2} + \int x^{2a+3} V \, d\nu, & a \geq 0. \end{array} \tag{R48.6.8}$$

Thus every $i \geq 2$ satisfies

$$\begin{aligned} (D_i)_+ &\leq \delta_3, \\ |D_i| &\leq \delta_3 + \eta\delta_2 \\ &\leq (\frac{221}{1024})d < d/4, \end{aligned} \tag{R48.6.9}$$

where $d \leq \frac{1}{8}$ was used. These are bounds for all four residue classes, not checks of finitely many connector lengths.

The actual finite first-row propagation is

$$\begin{aligned} e^{\Gamma_0} T^L F &\leq Y_L \\ &= \bar{f}_{L+2} + B^L R_0 + D\bar{f}_{L-1} \\ &\quad + \sum_{i=1}^{L-1} D_i \bar{f}_{L-1-i}. \end{aligned} \tag{R48.6.10}$$

Combine it with the complete identity (R48.1.4) at $L + 2$. Define

$$\Lambda_3 = d - D, \qquad \Lambda_r = \beta_r - D_{r-3} \quad (4 \leq r \leq L + 2). \tag{R48.L1331}$$

Then (R48.6.10) becomes exactly

$$Y_L = 1 + G_{L+2}(B)u + B^L R_0 - \sum_{r=3}^{L+2} \Lambda_r \bar{f}_{L+2-r}. \tag{R48.6.11}$$

In particular the actual scalar D is absorbed only into Λ_3; it has not been confused with D_3. By (R48.6.5),(R48.6.9),

$$\begin{aligned} \Lambda_3 = 2d/5, \quad \Lambda_4 \geq 7d/8, \\ \Lambda_r \geq (\kappa - \theta^2)d > 2d/5 \ (r \geq 5), \\ \Lambda_r \leq 5d/4 \ (r \geq 5). \end{aligned} \tag{R48.6.12}$$

Every coefficient is nonnegative. We may therefore retain the last two actual terms $\Lambda_{L+2}\bar{f}_0$ and $\Lambda_{L+1}\bar{f}_1$ and discard the other nonnegative subtractions. With the exact $W_3 = 1 - d - G_3 u$, the resulting sum $W_3 + Y_L$ is at most $2 + c + BV$, with

$$
\begin{aligned}
c &= -d - \Lambda_{L+1} - \Lambda_{L+2},\\
V &= B^2 G_{L-1}(B)u + B^{L-1}R_0 - \Lambda_{L+1}1.
\end{aligned}
\tag{R48.6.13}
$$

The two future coefficients retain both $L+1$ losses. The finite odd-power mean $E[B^3 G_{L-1}u]$ is at most δ_2. Since $L \geq 4$, (R48.6.5) gives $\|B^{L-1}R_0\| \leq 3\sqrt{d}/25$ and $|D_L| \leq 3d/25$. Hence

$$
(c + EBV)/d \leq -\frac{9}{5} + \theta + \frac{3}{25} < -5/4. \tag{R48.6.14}
$$

The complete norm, including the constant, has the valid triangle bound

$$
\begin{aligned}
&\|V\|/\sqrt{d}\\
&\leq \theta + \frac{3}{25} + (\frac{5}{4})\sqrt{d}\\
&< \frac{5}{12} + \frac{3}{25} + \frac{15}{32} = \frac{2413}{2400} < 21/20.
\end{aligned}
\tag{R48.6.15}
$$

Here $\sqrt{d} < \frac{3}{8}$ on the actual low domain. The fourth loss is at least

$$
\frac{35}{4} + 5 - (\frac{17}{4})(\frac{21}{20})^2 = \frac{14503}{1600} > 4. \tag{R48.6.16}
$$

This proves the entire previously weak long $m = 3$ row while retaining the original finite signed propagation and all original directions.

R48.6.4 All later pairs and the entire high domain

For $j \geq 2$ and $d \leq \frac{1}{8}$, the complete actual R2-C030/A014 $j = 2$ table has ordinary-eighth loss greater than nine for every L, and each later terminal pair only improves it. Ordinary Cauchy on the same H gives ordinary-fourth loss at least $\frac{9}{2} > 4$. No weighted moment is involved.

For $d \geq \frac{1}{8}$, use the actual centered-pair inequalities (O.58). The complete later terminal mask S has $|S| \leq \sqrt{d}/4 \leq 3d/4$; $S = 0$ when $j = 1$. Therefore, uniformly for every $j \geq 1$,

$$
F \leq \bar{f}_2 \exp[-(\frac{1}{4} + (\frac{16}{5})(j-1))d]. \tag{R48.6.17}
$$

The original selected connector maps $\bar{f}_2$ to $\bar{f}_{L+2}$ or W_{L+2}. Multiplication by the selected original arm gives exactly $W_h \bar{f}_H, h = \min(m, L+2) \geq 3$. Its same-direction fourth loss is at least $\frac{59}{16}$, as follows immediately from the actual bare sum used in R2-C030 and (R48.1.8). Thus (R48.6.17) gives total fourth loss at least

$$
\frac{59}{16} + 1 + (\frac{64}{5})(j-1) \geq 75/16 > 4. \tag{R48.6.18}
$$

Equations (R48.6.3),(R48.6.6)-(R48.6.7),(R48.6.16),(R48.6.18) and the complete $j \geq 2$ low table prove (R48.6.1). The analytic boundary $d = \frac{1}{8}$ is shared, and the integer comparison m versus $\ell + L$ is unchanged throughout.

R48.7 Every even terminal

R48.7.1 Exact pairs and nonunit even multipairs

Use the theorem bodies and exact direction scopes of R2-C039/A021 and R2-C041/A022. They give $EH^4 \le \exp(-6d)$ for all $m \ge 3$ in the E and S rows of part R48.2.1. E includes a single pair of minimum at least four and every larger even list of that minimum. S includes every even list of at least four arms of minimum two or three. Their already complete $M = 2$ output is retained with its exact direction; the proof uses those stated estimates.

For exact $(1, b)$, C087's actual fourth catalogue at outer length two has rate at least $\frac{3487}{640}$, including its prescribed $m = 3, L = 2$ turn. For exact $(2, b)$, the complete C043/A026 gives same-direction fourth rate $\frac{326249}{86400}$ at every $m \ge 3$ and $L \ge 1$; for exact $(3, b)$, the corresponding rate is $\frac{1901}{300}$. All these rates exceed $\frac{29}{8}$. In particular the true short 23 and its finite G_1/G_2 boundaries are included in the exact $-(2, b)$ theorem, with the actual table in part R48.2.

R48.7.2 Unit-minimum even multipairs: the actual full tail

Let $j \ge 2$ and

$$F = q f_{b_1} \prod_{i=2}^{j} h_{a_i} f_{b_i}, \qquad 1 < b_1 < a_2 < b_2 < \cdots < a_j < b_j. \tag{R48.7.1}$$

The complete C043/A026 finite mask proof has the following exhaustive classes and analytic domains:

Table 134: Every longer singleton at two outer edges: parameter values.

Entry	Original integers	Low domain	High-domain full tail bound
I	$b_1 \ge 3$;	$d \le \frac{1}{64}$;	$\exp[-(\frac{1}{5} + (\frac{16}{5})(j-2))d]$.
II	$b_1 = 2, a_2 \ge 4$;	$d \le \frac{1}{64}$;	same bound.
III	$b_1 = 2, a_2 = 3$;	$d \le \frac{1}{6}$;	$\exp[-(\frac{1}{7} + (\frac{16}{5})(j-2))d]$.

In every low-domain row its actual ordinary fourth is at most $\exp(-11d/2)$, for every m, L and every finite j. These fourth-moment bounds include all signed constants.

Each high-domain tail bound in the table concerns the entire original product after the first pair. It is a pointwise cap for one finite mask. It survives complete reversal. For $L = 1$ the exact comparator is $f_m T(p\bar{f}_{b_1})$, because the actual terminal in part R48.2 is $\overline{F}$. For $L \ge 2$ it is exactly $W_m T^L(q f_{b_1})$. The selected directions thus coincide with the actual U directions in C087, except for $m = 3, L = 2$, which is handled by part R48.3 above rather than by a turn.

Outside $m = 3, L = 2$ the U fourth loss is at least $\frac{3487}{640}$. $Atm = 3, L = 2$ with $b_1 \ge 3$, (R48.3.15) gives at least four. $Atm = 3, L = 2$ with $b_1 = 2$, (R48.3.5) gives at least $\frac{25}{8}$, with an additional negative $4d^2$. The weakest fourth loss after multiplying the true tail is therefore

$$\frac{25}{8} + \frac{4}{7} = 207/56. \tag{R48.7.2}$$

Class II instead gives $\frac{25}{8} + \frac{4}{5}$; class I gives at least $4 + \frac{4}{5}$. Every later pair improves the fourth loss by $\frac{64}{5}$. Including the stronger low-domain table, the entire actual unit-minimum even family satisfies

$$\boxed{EH^4 \le e^{-207d/56}}. \tag{R48.7.3}$$

In particular $2(\frac{207}{56}) - \frac{36}{5} = \frac{27}{140} > \frac{1}{20}$. The same-direction all $-Tm = 3, L = 2$ exception is now fully paid for all terminal lengths, all finite pair counts and the whole continuous host domain.

R48.8 Exhaustion, uniform rate and absence of a host-selected direction

All strict terminal lists of size at least two are even or odd. At even cardinality two, the minimum is one, two, three or at least four. At larger even cardinality, its minimum is one, two, three or at least four. Odd lists have minimum one or at least two; the corresponding subtables are given in Sections 5 and 6.

The complete minimum payment ledger is:

E **and** S**, all** $m \ge 3$

Actual proved payment sufficient for $M = 2$: same-direction fourth loss 6.

Exact $(1, b)$

Actual proved payment sufficient for $M = 2$: C087 actual short-outer fourth loss at least $\frac{3487}{640}$.

Exact $(2, b), (3, b)$

Actual proved payment sufficient for $M = 2$: fourth losses $\frac{326249}{86400}$ and $\frac{1901}{300}$.

Unit-minimum even multipairs

Actual proved payment sufficient for $M = 2$: fourth loss $\frac{207}{56}$.

All nonunit odd O

Actual proved payment sufficient for $M = 2$: fourth loss 4.

Unit-minimum odd first pair at least four

Actual proved payment sufficient for $M = 2$: fourth loss at least 5.

Unit-minimum odd first pair three, one pair

Actual proved payment sufficient for $M = 2$: fourth loss $\frac{39}{10}$.

Unit-minimum odd first pair three, multiple pairs

Actual proved payment sufficient for $M = 2$: fourth loss at least 7.

Exact 123

Actual proved payment sufficient for $M = 2$: fourth loss $\frac{29}{8}$.

Exact $124, L = 1$

Actual proved payment sufficient for $M = 2$: fourth loss $\frac{39}{10}$.

Exact 124, low nonunit L

Actual proved payment sufficient for $M = 2$: fourth at least $\frac{29}{8}$, or the explicit $\frac{199}{350}$ curved output loss.

Exact $12c, c \ge 5$**, low/middle** $L = 1$ **and low nonunit** L

Actual proved payment sufficient for $M = 2$: fourth greater than $\frac{29}{8}$.

Exact $12c, c \ge 5$**, high** $L = 1$

Actual proved payment sufficient for $M = 2$: actual output loss at least $\frac{1}{2}$.

Exact 124 or $12c$, high nonunit L

Actual proved payment sufficient for $M = 2$: part R48.4.4, including exact curved W_3 and opposite-first-edge $L = 3$.

Five-arm class $(1, 2, c, b, e), c \geq 4$

Actual proved payment sufficient for $M = 2$: low fourth > 5;

high same-direction triple output plus $4d$.

Other five-arm unit-minimum odd classes

Actual proved payment sufficient for $M = 2$: fourth loss at least 5.

Unit-minimum odd, $r_1 = 2, j \geq 3$

Actual proved payment sufficient for $M = 2$: fourth loss at least 8.

Every fourth rate at least $\frac{29}{8}$ pays the uniform output loss $\frac{1}{20}$ by (R48.1.10)-(R48.1.13). The explicitly curved rows have larger output losses: $\frac{199}{350}$, $\frac{91}{120}$ or $\frac{33}{20}$ in the displayed domains. The high $L = 1$ direct estimate is stronger as well. Thus the common coefficient $\frac{1}{20}$ applies to every entry and proves Theorem $T20$.

No threshold in d appears in part R48.2. Each analytic split estimates the same original message on both sides and includes its endpoints. In particular no C087 turn is substituted into the C043 $m = 3$, $L = 2$ row; no opposite-first-edge message is renamed as a forward U message; no O comparison $\ell + L$ is replaced by L or $L + 1$; and the unit-minimum odd crossing is used only in its exact five-arm class.